The Foreman and the serif family

Handbook of the general definitions, constructions and theorems

29 October 2026

How this handbook is written

This handbook states the general theory of the Foreman and the serif family: the members of the serif family and their relations; the tables and families they evaluate; the Foreman's classification of a family and his tree; the generalized integrals the Foreman sets; the constructions and compositions of families; the motive of a family and the arithmetic of its values; rays of data and the laws their values obey; the relations among values; and the family read as a model of data. Every definition, construction, theorem and procedure of the program is stated here, once, at the place where it is first needed.

The order of the book. The serif family comes first (Part I), because every value the theory produces is closed in it. Then the path of a computation: a list of marks and the tables and families built on it (Part II); the classification of a family by its realizations, its reading, its bunches and the Foreman tree (Part III); the integrals the Foreman procedure sets, and their evaluation along the tree (Part IV); the constructions that build families from cells, holds and bunches, and price them (Part V); the motive of a family and the coaction along its tree, positivity and the gap (Part VI); rays of data, their modules, their two ends and the reading of their values (Part VII); relations among values and the content they live in (Part VIII); and the family as a model of data (Part IX). No statement and no proof uses a statement of a later chapter, and a later chapter is mentioned only as a pointer; inside a chapter a statement may point ahead to a result proved below it, and then says so.

Proofs. A statement that is more general than its counterpart in the reference *The Foreman and the serif family* (16 October 2026), or that has no counterpart there, is proved in Part X, one section per chapter in the order of the chapters; the other statements are proved in the reference. In the source file every statement is preceded by a comment that says which: either the label and line of the statement in the reference, or that it is proved in the last part.

Exercises. Computations on particular lists, tables and rays are the exercises of Part XI. Each is a guided example with its solution worked in full: every step names the statement or procedure it uses, and every quantity the next step needs is displayed in exact arithmetic.

Names. The text uses the program's own names. The classical names of its objects, and the classical names of results that the program reaches by its own route, are collected in the Aliases at the end.

One governing definition per object. Each object of the theory is defined once, with all of its slots. Every later appearance of the object is that definition with particular data in its slots, written through the full signature of its governing symbol: a specialization displays the data it chooses, and it never receives a symbol of its own. Thus a Breta on one scale is a Breta with one bunch, a Trailmoment on a line is a Trailmoment whose cut region is a half-line, and a value of interest at particular exponents is the interest with those exponents in its slots.

Abbreviations are local. Inside one statement an abbreviation may be declared, once and in words, for data that the statement holds fixed; the declaration displays the full symbol it

abbreviates, and it lapses at the end of the statement.

Relations are stated at their height. A relation holds on the values of the slots on which its proof establishes it (Definition 5.6). A hypothesis that restricts a slot is written into the statement that needs it, and nothing outside that statement restricts the object.

Final answers are written with exp, log, the stacked Gamma and the serif family (Definition 1.1(vi)).

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Part I

The serif family

Chapter 1

Conventions, the stacked Gamma and the finite part

Definition 1.1 (Conventions)**.** (i) $\mathcal{E}$ and B are finite nonempty sets and $n = |\mathcal{E}|$. For $x \in \mathbb{R}^{\mathcal{E}}_{>0}$ and $v \in \mathbb{C}^{\mathcal{E}}$, $x^v = \prod_e x_e^{v_e}$ with $x_e^{v_e} = \exp(v_e \log x_e)$. For a vector v, $|v| = \sum_e v_e$. $\mathbf{1}$ is the all-ones vector, $\mathbf{1}_S$ the indicator of S, $x^S = x^{\mathbf{1}_S}$, $v_S = \sum_{e \in S} v_e$, $v|_S$ the restriction of v to S, and $x^{-1} = (x_e^{-1})_e$. For a set X and elements a, b, $X - a = X \setminus \{a\}$ and $X + b = X \cup \{b\}$. $\mathbb{N} = \{0, 1, 2, \dots\}$.

(ii) For $\alpha \in \mathbb{R}^{\mathcal{E}}_{>0}$ the *logarithmic coordinates* are $y = \log \alpha$ and the *signed coordinates* are $z = \frac{1}{2} y$, coordinatewise, so $\alpha = e^y = e^{2z}$.

(iii) $\Delta_B = \{u \in \mathbb{R}^B_{>0} : |u| = 1\}$ is the *shape simplex* of B, and $d\omega_B(u)$ is Lebesgue measure in the coordinates $(u_e)_{e \neq e_0}$ for a fixed $e_0 \in B$; a change of e_0 is affine of determinant ± 1. For $|B| = 1$, ω_B is the Dirac measure at the point of Δ_B.

(iv) An integral *converges* when it converges absolutely.

(v) $(a)_j = \Gamma(a + j)/\Gamma(a)$ for $j \in \mathbb{N}$; $d_n = \operatorname{lcm}(1, \dots, n)$ with $d_0 = 1$; e_k is the elementary symmetric polynomial of degree k; for a coordinate x of the unit interval, $\bar{x} = 1 - x$; $[X]\, Y$ is the coefficient of X in Y.

(vi) *Final answers* are written with exp, log, the stacked Gamma and the serif family: the Gramma G and the Zamma Z (Chapter 2), the Breta B (Chapter 3) and the Trailmoment T (Chapter 5). A value of interest is left in another form only with a proof that it has no such expression. A *word* is an expression built from rationals, the data and the variables by sums, products, powers with rational exponents or exponents among the data, these functions, integrals over simplices, half-lines and fibres, and the iterate $\mathcal{Q}^M$ of an integral operator whose kernel is a word. A finite sequence $(w_1, \dots, w_k)$ is a *string* of length k. An *insertion* is a function multiplied into an integrand.

(vii) For masses $\mathbf{m}$, $d\alpha_{\mathbf{m}} := \alpha^{\mathbf{m}-\mathbf{1}} d\alpha$ is a measure on the orthant, and a form $f\, d\alpha_{\mathbf{m}}$ unchanged by $\alpha \mapsto \lambda\alpha$, f homogeneous of degree $-m_{\mathcal{E}}$, is integrated over the rays of the orthant, that is over any section (Chapter 16), as the form $W\, d\alpha_{\mathbf{m}}$ of a family word W (Chapter 8) is on its balance wall; a mark or a set of marks as a subscript, as in $d\alpha_B$, denotes coordinates. The law of a random variable, or of a measure under a map, is its image measure; an identity between members of the serif family is a relation (Chapter 5). An expectation names its random variable and that variable's law, $\mathbb{E}_{U \sim P}[g(U)] = \int g(u)\, P(du)$; inside a statement that has named them once, the subscript may be dropped for the rest of that

statement. The *Gamma law* of index $\nu > 0$ is $\mathrm{Gamma}_\nu(dx) = x^{\nu-1}e^{-x}dx/\Gamma(\nu)$ on $(0,\infty)$, and for $\mathbf{m} \in \mathbb{R}^B_{>0}$, $\mathrm{Gamma}_{\mathbf{m}} = \bigotimes_{e\in B} \mathrm{Gamma}_{m_e}$ is the law on $\mathbb{R}^B_{>0}$ of independent Gamma coordinates of indices m_e. The generalized integrals of Chapter 16 are unnormalized, and their expectations divide them by the period of the centre: on a connected family every expectation of a form is taken under the law of its centre, $\mathbb{P}_{\mathrm{c}}$, of the departure that its evaluation sets.

(viii) The *root rate* of terms a_η indexed by sizes $\eta \to \infty$, as the terms of a lattice by the real parts of their exponents or the terms of a closing by its steps, is $\limsup |a_\eta|^{1/\eta}$. When the number of terms of size at most N grows at most polynomially in N, terms of root rate below one are absolutely summable, and terms of root rate above one do not tend to zero.

1.1 The stacked Gamma

Definition 1.2 (The stacked Gamma)**.** For $\alpha \in \mathbb{C}^p$ and $\beta \in \mathbb{C}^q$,

$$\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1,\ \dots,\ \alpha_p\\ \beta_1,\ \dots,\ \beta_q\end{matrix}\right] := \frac{\Gamma(\alpha_1)\cdots\Gamma(\alpha_p)}{\Gamma(\beta_1)\cdots\Gamma(\beta_q)},$$

wherever no α_i is a pole of Γ; it is a meromorphic function of its entries. The stacked Gamma is *balanced* when $\sum_i \alpha_i = \sum_j \beta_j$.

The subscripts of the bracket record the numbers of its upper and lower entries, so a quotient of Gammas is read off the symbol with no translation. The stacked Gamma $\Gamma_{\substack{|B|\\1}}\!\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right] = \prod_{e\in B}\Gamma(m_e)/\Gamma(m_B)$ of masses $\mathbf{m} \in \mathbb{C}^B$ over their total m_B is, for $\mathrm{Re}\, m_e > 0$, the integral of $u^{\mathbf{m}-\mathbf{1}}$ over the shape simplex Δ_B (Theorem 3.9).

Proposition 1.3 (Relations of the stacked Gamma)**.** *(i)* Concatenation. $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]\Gamma_{\substack{p'\\q'}}\!\left[\begin{matrix}\boldsymbol{\alpha}'\\ \boldsymbol{\beta}'\end{matrix}\right] = \Gamma_{\substack{p+p'\\q+q'}}\!\left[\begin{matrix}\boldsymbol{\alpha},\ \boldsymbol{\alpha}'\\ \boldsymbol{\beta},\ \boldsymbol{\beta}'\end{matrix}\right]$.

(ii) Cancellation. $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha},\ c\\ \boldsymbol{\beta},\ c\end{matrix}\right] = \Gamma_{\substack{p-1\\q-1}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]$.

(iii) Shift. $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1+1,\ \boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right] = \alpha_1\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1,\ \boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]$ *and* $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \beta_1+1,\ \boldsymbol{\beta}\end{matrix}\right] = \beta_1^{-1}\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \beta_1,\ \boldsymbol{\beta}\end{matrix}\right]$.

(iv) Residue. *For* $j \in \mathbb{N}$, $\operatorname*{Res}_{\alpha_1=-j}\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1,\ \boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right] = \frac{(-1)^j}{j!}\Gamma_{\substack{p-1\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]\Big|_{\alpha_1=-j}$.

(v) Reflection. $\Gamma_{\substack{2\\0}}\big[a,\ 1-a\big] = \pi/\sin\pi a$.

(vi) Laurent data at a pole. *For* $j \in \mathbb{N}$,

$$\Gamma(-j+\varepsilon) = \frac{(-1)^j}{j!}\Big(\frac{1}{\varepsilon} + (\log\Gamma)'(j+1) + O(\varepsilon)\Big), \qquad (\log\Gamma)'(\nu+1) = (\log\Gamma)'(\nu) + \frac{1}{\nu}.$$

Remark 1.4 (Balance is a property of the written stacked Gamma). Balance is not a property of the value of a stacked Gamma: the shift (iii) and the reflection (v) change the sums of the entries,

multiplying the value by an algebraic number when the entries are rational. For rational $a \notin \mathbb{Z}$, (v) at a and at $\frac{1}{2}$, with $\Gamma(1) = 1$, gives

$$\Gamma_{\substack{2\\0}}\big[a,\ 1-a\big] = \frac{1}{\sin \pi a}\,\Gamma_{\substack{2\\1}}\left[\begin{matrix}\frac{1}{2},\ \frac{1}{2}\\ 1\end{matrix}\right],$$

the left side with the sums 1 and 0, the right side balanced, and $\sin \pi a$ algebraic.

1.2 The finite part

Definition 1.5 (The finite part)**.** A function h on $(0,\infty)$ has a *log-power expansion* at 0 if for every N there are finitely many coefficients $c_{k,j}$, with exponents $k \in \mathbb{C}$ of real part below N and $j \in \mathbb{N}$, such that $h(x) - \sum c_{k,j} x^k \log^j x = O(x^N)$ as $x \to 0$; at ∞ the same holds with x^{-k} and $O(x^{-N})$. The coefficients are determined by h. Then $\operatorname{Reg}_{x\to 0} h$ and $\operatorname{Reg}_{x\to\infty} h$ are the coefficients of $x^0 \log^0 x$ in these expansions, zero if absent. If f has a primitive h with log-power expansions at both ends,

$$\operatorname{Reg}\int_0^\infty f\,dx := \operatorname{Reg}_{x\to\infty} h - \operatorname{Reg}_{x\to 0} h,$$

which does not depend on the primitive; on $(0,b]$, $\operatorname{Reg}\int_0^b f := h(b) - \operatorname{Reg}_{x\to 0} h$. A sum is regularized the same way, $\operatorname{Reg}_{N\to\infty}$ of its partial sums. The coordinate is part of the data of a finite part, and for iterated finite parts so is the order.

Proposition 1.6 (A convergent integral is its own finite part)**.** *If $\int_0^\infty f\,dx$ converges and f has a primitive with log-power expansions at both ends, then $\operatorname{Reg}\int_0^\infty f\,dx = \int_0^\infty f\,dx$.*

Theorem 1.7 (A finite part is a constant term)**.** *If f has log-power expansions at 0 and at ∞, then*

$$\operatorname{Reg}\int_0^\infty f\,dx = \operatorname*{CT}_{\varepsilon=0}\int_0^\infty x^\varepsilon f(x)\,dx,$$

the right side continued meromorphically in ε from $\int_0^1$ and from $\int_1^\infty$ separately. Termwise, $x^b \log^j x$ in the expansion at 0 contributes $\int_0^1 x^{b+\varepsilon}\log^j x\,dx = (-1)^j j!/(b+1+\varepsilon)^{j+1}$, whose constant term is $(-1)^j j!/(b+1)^{j+1}$ for $b \neq -1$ and 0 for $b = -1$; the end at ∞ is the same after $x \mapsto 1/x$.

Corollary 1.8 (Finite parts at a pole of a Mellin integral)**.** *Let f have a log-power expansion at 0 and be integrable on $[\delta,1]$ for every $\delta > 0$, and let $I(a) = \int_0^1 x^{a-1} f(x)\,dx$, continued meromorphically. Then $\operatorname{Reg}\int_0^1 x^{a-1} f(x)\,dx = \operatorname{CT}_{a'=a} I(a')$ for every $a \in \mathbb{C}$: it is $I(a)$ at every a that is not a pole, and the poles lie at the points $a = -k$ for the exponents k of the expansion, a term $x^k \log^j x$ giving a pole of order at most $j+1$.*

Lemma 1.9 (A finite part depends on the coordinate)**.** *Let h have a log-power expansion at ∞, and let a_j be its coefficient of $x^0 \log^j x$, $j \in \mathbb{N}$. For $c,p > 0$, $h(cx^p)$ has a log-power expansion at ∞, and*

$$\operatorname{Reg}_{x\to\infty} h(cx^p) = \sum_{j\geq 0} a_j\,(\log c)^j;$$

the same holds at 0. Consequently, if f has log-power expansions at 0 and at ∞, with coefficients f_j^0 and f_j^∞ of $x^{-1}\log^j x$, then for $c,p > 0$

$$\operatorname{Reg}\int_0^\infty f(cx^p)\,c\,p\,x^{p-1}\,dx = \operatorname{Reg}\int_0^\infty f(x)\,dx + \sum_{j\geq 0}\frac{(\log c)^{j+1}}{j+1}\big(f_j^\infty - f_j^0\big)\,:$$

a power of the coordinate leaves a finite part unchanged, and a dilation moves it by a polynomial in $\log c$ *read from the terms* $x^{-1}\log^j x$ *of the integrand at its two ends.*

A finite part is the difference of the regularized limits of one primitive at the two ends, not the integral of a measure. It is linear, it moves with the coordinate (Lemma 1.9), and iterated in several variables it moves with the order. The generalized integrals of Chapter 16 are built from it: a quasi-integral is an iterated finite part along a chart and an order, and the pseudointegral is the value of the native completion (Definition 16.29).

Chapter 2

The Gramma and the Zamma

2.1 The Gramma

Definition 2.1 (Gramma)**.** Let $\mathcal{E}$ be a finite set of *scales*. A *configuration* on $\mathcal{E}$ is a finite family $t = (t_1, \dots, t_N)$ of *exponents* $t_k \in \mathbb{R}^{\mathcal{E}}$; t also denotes the $\mathcal{E} \times N$ matrix with these columns. For *coefficients* $\boldsymbol{\xi} \in (0,\infty)^N$ and an *index* $\nu \in \mathbb{C}^{\mathcal{E}}$, the *Gramma* is

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) := \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\nu-\mathbf{1}} \exp\Big(-\sum_{k=1}^{N} \xi_k\, x^{t_k} \Big)\, dx$$

wherever the integral converges. Put $\operatorname{cone}(t) = \{\sum_k c_k t_k :\ c \in [0,\infty)^N\}$ and $\mathcal{T}_t = \{\nu : \operatorname{Re}\nu \in \operatorname{int}\operatorname{cone}(t)\}$. In $y = \log x$,

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \int_{\mathbb{R}^{\mathcal{E}}} e^{\langle \nu, y\rangle} \exp\Big(-\sum_{k} \xi_k e^{\langle t_k, y\rangle} \Big)\, dy,$$

the moment generating function, in the index, of the density $\exp\big(-\sum_k \xi_k e^{\langle t_k,y\rangle}\big)$. Exponents and coefficients combine by concatenation, $t \oplus t'$ and $\boldsymbol{\xi} \oplus \boldsymbol{\xi}'$ (configurations on two sets of scales extended by zero to their union), and a shape u dresses coefficients by $u^t\boldsymbol{\xi} := (u^{t_k}\xi_k)_k$. The *unit configuration* on $\mathcal{E}$ is $I_{\mathcal{E}} = (\mathbf{e}_e)_{e\in\mathcal{E}}$, one unit column per scale, and on one scale the configuration of the single exponent 1 is written (1).

The three slots of the Gramma (Definition 5.6) are its index, its configuration and its coefficients. Every structural relation of the Gramma is a statement about how these slots move: the index moves along the exponents, the coefficients are rescaled by the torus of the scales, and the configuration passes to its facets at its walls.

Theorem 2.2 (Domain)**.** $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ *converges if and only if* $\nu \in \mathcal{T}_t$*, and it is holomorphic on* $\mathcal{T}_t$*. If* $\operatorname{cone}(t) = \mathbb{R}^{\mathcal{E}}$ *it is entire; if* $\operatorname{cone}(t)$ *lies in a hyperplane it converges nowhere.*

Proposition 2.3 (Affine changes of the logarithmic coordinates)**.** *For* $A \in GL_{\mathcal{E}}(\mathbb{R})$ *and* $\mathbf{h} \in \mathbb{R}^{\mathcal{E}}$*:* $\mathcal{T}_{A^{\mathsf{T}}t} = A^{\mathsf{T}}\mathcal{T}_t$*, and for* $\nu \in \mathcal{T}_t$

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det A|\, e^{\langle \nu, \mathbf{h}\rangle}\, \mathsf{G}_{A^{\mathsf{T}}\nu;A^{\mathsf{T}}t}\big((e^{\langle t_k,\mathbf{h}\rangle}\xi_k)_k\big).$$

At $A = -I_{\mathcal{E}}$ *and* $\mathbf{h} = 0$ *this is the reflection,* $\mathcal{T}_{-t} = -\mathcal{T}_t$ *and* $\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \mathsf{G}_{-\nu;-t}(\boldsymbol{\xi})$*.*

Theorem 2.4 (Relations of the Gramma)**.** *Let* $\nu \in \mathcal{T}_t$*; then* $\nu + t_k \in \mathcal{T}_t$ *for every* k*.*

(i) Ladder. $\partial_{\xi_k}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = -\mathsf{G}_{\nu+t_k;t}(\boldsymbol{\xi})$*; for* $\mathbf{w}\in\mathbb{N}^N$, $\partial^{\mathbf{w}}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = (-1)^{|\mathbf{w}|}\mathsf{G}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi})$.

(ii) Exchange, one relation per scale. *For each* $e\in\mathcal{E}$: $\nu_e\,\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \sum_k t_{k,e}\,\xi_k\,\mathsf{G}_{\nu+t_k;t}(\boldsymbol{\xi})$.

(iii) Torus homogeneity. *For* $\lambda\in\mathbb{R}^{\mathcal{E}}_{>0}$: $\mathsf{G}_{\nu;t}\big((\lambda^{t_k}\xi_k)_k\big) = \lambda^{-\nu}\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ *(Proposition 2.3 at* $A = I_{\mathcal{E}}$, $\mathbf{h} = \log\lambda$*).*

(iv) Toric relation. *For every* $\mathbf{v}\in\mathbb{Z}^N$ *with* $t\mathbf{v} = 0$*, written* $\mathbf{v} = \mathbf{v}^+ - \mathbf{v}^-$ *with* $\mathbf{v}^{\pm}\in\mathbb{N}^N$ *of disjoint supports,*

$$\partial^{\mathbf{v}^+}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = (-1)^{|\mathbf{v}^+|-|\mathbf{v}^-|}\,\partial^{\mathbf{v}^-}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}).$$

The ladder moves the index along one exponent; the exchange relation, one for each scale, is linear in the Grammas at the index and at its N neighbours $\nu + t_k$; the toric relation records the linear dependencies among the exponents themselves. On one scale with integer exponents the exchange relation is a recurrence on the indices $\nu + \mathbb{Z}$ linking the indices from $\nu + \min(0, \min_k t_k)$ to $\nu + \max(0, \max_k t_k)$.

Theorem 2.5 (Product)**.** *If* $\mathcal{E} = \mathcal{E}_1 \sqcup \cdots \sqcup \mathcal{E}_q$ *and each* t_k *is supported in one* $\mathcal{E}_i$ *(zero vectors assigned to* $\mathcal{E}_1$*), then* $\mathcal{T}_t = \mathcal{T}_{t^{(1)}} \times \cdots \times \mathcal{T}_{t^{(q)}}$ *and*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \prod_{i=1}^{q} \mathsf{G}_{\nu|_{\mathcal{E}_i};t^{(i)}}(\boldsymbol{\xi}^{(i)}),$$

with $t^{(i)}$ *the exponents supported in* $\mathcal{E}_i$ *and* $\boldsymbol{\xi}^{(i)}$ *their coefficients.*

Corollary 2.6 (Square configurations)**.** *Let* $N = |\mathcal{E}|$ *and let the matrix* t *be invertible, and put* $\mathbf{b} = t^{-1}\nu$*. Then* $\mathcal{T}_t = \{\nu : \operatorname{Re}\mathbf{b}\in(0,\infty)^N\}$ *and*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det t|^{-1}\prod_{k=1}^{N}\Gamma(b_k)\,\xi_k^{-b_k}.$$

In particular $\mathsf{G}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) = \prod_e \Gamma(\nu_e)\,\xi_e^{-\nu_e}$*; on one scale* $\mathsf{G}_{\nu;(t_1)}(\xi) = t_1^{-1}\Gamma(\nu/t_1)\,\xi^{-\nu/t_1}$ *for* $t_1 > 0$*; and* $\Gamma(\nu) = \mathsf{G}_{\nu;(1)}(1)$*, whose exchange relation is* $\nu\Gamma(\nu) = \Gamma(\nu+1)$.

The Gamma function is thus the Gramma of one scale and one unit exponent, and at every invertible square configuration the Gramma is a stacked Gamma times powers, divided by $|\det t|$. A configuration with more exponents than scales is where the Gramma becomes more than a stacked Gamma times powers: the difference $N - |\mathcal{E}|$ is the dimension of the share side of Proposition 4.2.

Lemma 2.7 (Scale and shape)**.** *The map* $(0,\infty)\times\Delta_B\to\mathbb{R}^B_{>0}$, $(\rho,u)\mapsto\rho u$*, is a bijection with inverse* $x\mapsto(|x|, x/|x|)$*, and* $dx = \rho^{|B|-1}\,d\rho\,d\omega_B(u)$*. Consequently* $(\rho u)^{\pi} = \rho^{\langle\pi,\mathbf{1}_B\rangle}u^{\pi}$ *for* $\pi\in\mathbb{R}^B$*, and* $x^{\mathbf{m}-\mathbf{1}}dx = \rho^{m_B-1}u^{\mathbf{m}-\mathbf{1}}\,d\rho\,d\omega_B(u)$ *for* $\mathbf{m}\in\mathbb{C}^B$.

Theorem 2.8 (Scale and shape along sets)**.** *Let* $B_1,\dots,B_q$ *be pairwise disjoint nonempty subsets of* $\mathcal{E}$*,* $R = \mathcal{E}\setminus(B_1\cup\cdots\cup B_q)$ *and* $\nu\in\mathcal{T}_t$*. With the* levels $\ell_{k,i} = \langle t_k, \mathbf{1}_{B_i}\rangle$*, the configuration* $\ell = (\ell_1,\dots,\ell_N)$ *on* q *scales and the coefficients* $\mathbf{b}(u,x_R) = \big(\xi_k\prod_i u_i^{t_{k|B_i}}x_R^{t_{k|R}}\big)_k$ *for* $u = (u_i)_i\in\prod_i\Delta_{B_i}$*,*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \int_{\mathbb{R}^R_{>0}} x_R^{\nu|_R-\mathbf{1}}\int_{\prod_i\Delta_{B_i}}\prod_i u_i^{\nu|_{B_i}-\mathbf{1}}\;\mathsf{G}_{(\nu_{B_i})_i;\ell}\big(\mathbf{b}(u,x_R)\big)\,d\omega(u)\,dx_R,$$

with $d\omega = \prod_i d\omega_{B_i}$*, the inner Gramma, on* q *scales at the index* $(\nu_{B_i})_i$*,* $\nu_{B_i} = \sum_{e\in B_i}\nu_e$*, converging for almost every* (u,x_R)*; when* R *is empty there is no outer integral.*

Passing to scale and shape on disjoint sets of scales replaces each set by one scale, its total, carrying the levels of the exponents on the set; what is left of the exponents dresses the coefficients by the shapes. The Breta of Chapter 3 is this passage made into a symbol.

Theorem 2.9 (The walls of a Gramma)**.** *Let $\mathcal{E} = \mathcal{E}' \sqcup \{e\}$, write $t_k = (t'_k, t_{k,e})$ and $\nu = (\nu', \nu_e)$, and suppose $t_{k,e} \geq 0$ for every k, with some $t_{k,e} > 0$. Let $H = \{k : t_{k,e} = 0\}$ be the* facet terms*, suppose* $\operatorname{cone}(t'_H)$ *has nonempty interior in $\mathbb{R}^{\mathcal{E}'}$ and contains t'_k for every $k \notin H$, and let $\operatorname{Re}\nu' \in \operatorname{int}\operatorname{cone}(t'_H)$. Then $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ converges for $0 < \operatorname{Re}\nu_e < c$, for some $c \in (0, \infty]$, and extends to a meromorphic function of $\nu_e \in \mathbb{C}$ with at most simple poles at $\nu_e = -p$, $p \in P = \{\sum_{k\notin H} n_k t_{k,e} : \mathbf{n} \in \mathbb{N}^{H^c}\}$, where*

$$\operatorname*{Res}_{\nu_e=-p} \mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \sum_{\mathbf{n}\in\mathbb{N}^{H^c}:\ \sum_k n_k t_{k,e} = p} \prod_{k\notin H} \frac{(-\xi_k)^{n_k}}{n_k!}\ \mathsf{G}_{\nu' + \sum_{k\notin H} n_k t'_k;\ t'_H}(\boldsymbol{\xi}_H).$$

At $p = 0$ the residue is the Gramma of the facet terms: at its first wall a Gramma factors onto its facet. A linear change of the logarithmic coordinates, $y = Ay'$ with $A \in GL_{\mathcal{E}}(\mathbb{R})$, gives $\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det A|\, \mathsf{G}_{A^{\mathsf{T}}\nu;A^{\mathsf{T}}t}(\boldsymbol{\xi})$ (Proposition 2.3 at $\mathbf{h} = 0$), and with the last column of A the inward normal of a facet of $\operatorname{cone}(t)$ *it puts that facet in the position of H.*

The poles of a Gramma sit on the walls of its cone, and each residue is a Gramma of the facet, of one scale fewer, at an index moved by the exponents that leave the facet. On one scale with positive exponents this is the continuation of the Gramma to the whole index line, with at most simple poles at the points $-p$, $p = \langle \mathbf{n}, t\rangle$ for $\mathbf{n} \in \mathbb{N}^N$, the residue at $-p$ being $\sum_{\langle \mathbf{n},t\rangle = p} \prod_k (-\xi_k)^{n_k}/n_k!$.

Proposition 2.10 (The expansion lattice)**.** *Let S be a set of terms such that the integral with the signs of those terms flipped converges,*

$$\int_{\mathbb{R}^{\mathcal{E}}} \exp\Big(\langle \operatorname{Re}\nu, y\rangle + \sum_{k\in S} \xi_k e^{\langle t_k, y\rangle} - \sum_{k\notin S} \xi_k e^{\langle t_k, y\rangle}\Big) dy < \infty.$$

Then, absolutely,

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \sum_{\mathbf{n}\in\mathbb{N}^S} \prod_{k\in S} \frac{(-\xi_k)^{n_k}}{n_k!}\ \mathsf{G}_{\nu + \sum_{k\in S} n_k t_k;\ t_{S^c}}(\boldsymbol{\xi}_{S^c}).$$

Theorem 2.11 (The decay polytope)**.** *Let $\nu_0 \in \operatorname{int}\operatorname{cone}(t)$ be real, $\eta \in \mathbb{R}^{\mathcal{E}}$, and $D_t = \{y \in \mathbb{R}^{\mathcal{E}} : |\langle t_k, y\rangle| < \frac{\pi}{2}\ \forall k\}$. For every $y \in D_t$,*

$$\big|\mathsf{G}_{\nu_0+i\eta;t}(\boldsymbol{\xi})\big| \leq e^{-\langle \eta, y\rangle}\ \mathsf{G}_{\nu_0;t}\big((\xi_k \cos\langle t_k, y\rangle)_k\big).$$

Consequently, with $h_t(\eta) = \sup_{y\in D_t}\langle \eta, y\rangle$, for every $\varepsilon \in (0,1)$

$$\big|\mathsf{G}_{\nu_0+i\eta;t}(\boldsymbol{\xi})\big| \leq \mathsf{G}_{\nu_0;t}\big(\sin(\tfrac{\pi\varepsilon}{2})\boldsymbol{\xi}\big)\, e^{-(1-\varepsilon)h_t(\eta)},$$

and h_t is a norm when the t_k span $\mathbb{R}^{\mathcal{E}}$.

2.2 The Zamma

Definition 2.12 (Zamma)**.** Wherever $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ converges and no ν_e lies in $-\mathbb{N}$, the *Zamma* is the Gramma divided by the Gammas of its own index,

$$\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) := \frac{\mathsf{G}_{\nu;t}(\boldsymbol{\xi})}{\prod_{e\in\mathcal{E}} \Gamma(\nu_e)}.$$

At the unit configuration it is $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) := \prod_e \xi_e^{-\nu_e}$ for every $\nu \in \mathbb{C}^{\mathcal{E}}$; where $\operatorname{Re}\nu_e > 0$ for every e this is the quotient $\mathsf{G}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi})/\prod_e \Gamma(\nu_e)$. The Zamma carries the continuations of the Gramma wherever the Gramma has them.

The Zamma divides out of the Gramma what is first order in its index, the Gammas that the unit configuration produces on its own; what remains is the content of the configuration. It has the three slots of the Gramma.

Proposition 2.13 (First facts about the Zamma)**.** *(i)* First order. *At the unit configuration the Zamma is the monomial of Definition 2.12; at an invertible square configuration t, $N = |\mathcal{E}|$, with $\mathbf{b} = t^{-1}\nu$,*

$$\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) = |\det t|^{-1}\,\Gamma_{\!N}\!\begin{bmatrix}\mathbf{b}\\ \nu\end{bmatrix}\prod_k \xi_k^{-b_k}.$$

(ii) A power of a sum is a Zamma. *With N unit exponents on one scale, $\mathsf{Z}_{\nu;(1,\dots,1)}(\boldsymbol{\xi}) = \big(\sum_k \xi_k\big)^{-\nu}$.*

(iii) Relations. *Where the terms converge, with quotients of Gammas in place of the factors the index moves,*

$$\partial_{\xi_k}\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) = -\prod_e \Gamma_1\!\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix}\mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}),$$

$$\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) = \sum_k t_{k,e}\,\xi_k\,\Gamma_1\!\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e + 1\end{bmatrix}\prod_{f\neq e}\Gamma_1\!\begin{bmatrix}\nu_f + t_{k,f}\\ \nu_f\end{bmatrix}\mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}) \qquad (e \in \mathcal{E}).$$

The other relations of Theorem 2.4 are read with the same normalization, and identities persist on the meromorphic continuations.

(iv) As an expectation. *If a configuration on $\mathcal{E}$ contains the unit configuration $I_{\mathcal{E}}$ with unit coefficients, then for real $\nu \in \mathbb{R}^{\mathcal{E}}_{>0}$*

$$\mathsf{Z}_{\nu;I_{\mathcal{E}}\oplus t'}(\mathbf{1}\oplus\boldsymbol{\xi}') = \mathbb{E}_{U\sim\mathrm{Gamma}_\nu}\Big[\exp\Big(-\sum_k \xi'_k U^{t'_k}\Big)\Big];$$

it lies in $(0,1]$, decreases in every coefficient, and, where every $\nu + t'_k$ lies in $\mathbb{R}^{\mathcal{E}}_{>0}$,

$$\mathsf{Z}_{\nu;I_{\mathcal{E}}\oplus t'}(\mathbf{1}\oplus\boldsymbol{\xi}') \;\geq\; \exp\Big(-\sum_k \xi'_k \prod_{e\in\mathcal{E}}\Gamma_1\!\begin{bmatrix}\nu_e + t'_{k,e}\\ \nu_e\end{bmatrix}\Big).$$

Proposition 2.14 (Powers of a polynomial in the scales)**.** *Let t be a configuration and $\nu \in \mathcal{T}_t$ with no ν_e in $-\mathbb{N}$, or $t = I_{\mathcal{E}}$ and $\nu \in \mathbb{C}^{\mathcal{E}}$; let $\xi_k(\rho) = \sum_{l\in\mathbb{N}^d}\rho^l\xi_k^{(l)}$ be polynomials in $\rho \in \mathbb{R}^d$ with real coefficients and $\boldsymbol{\xi}^{(0)} \in (0,\infty)^N$, and let $c_q^\nu = [\rho^q]\,\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho))$ be the Taylor coefficients at $\rho = 0$. At the unit configuration, where $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) = \boldsymbol{\xi}^{-\nu}$ is read with the principal branch near $\boldsymbol{\xi}^{(0)}$, the coefficients $\xi_k^{(l)}$ with $l \neq 0$ may be complex and $\rho \in \mathbb{C}^d$.*

(i) $c_q^\nu = \sum_{\mathbf{n}}(-1)^{|\mathbf{n}|}\prod_{e\in\mathcal{E}}\Gamma_1\!\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}\mathsf{Z}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi}^{(0)})\prod_{l\neq0,\,k}\frac{(\xi_k^{(l)})^{n_{l,k}}}{n_{l,k}!}$, *the sum over the arrays $\mathbf{n} = (n_{l,k})_{l\neq0,\,k}$ in $\mathbb{N}$ with $\sum_{l,k} l\,n_{l,k} = q$, where $\mathbf{w} = (\sum_l n_{l,k})_k$; at the unit configuration the factor $\prod_e \Gamma_1\!\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}$ is the polynomial $\prod_e(\nu_e)_{w_e}$, and away from it each product $\prod_e \Gamma_1\!\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}\mathsf{Z}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi}^{(0)})$ is $\mathsf{G}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi}^{(0)})/\prod_e\Gamma(\nu_e)$.*

(ii) *At one unit exponent, $t = (1)$, with $\boldsymbol{\xi}(\rho) = L(\rho)$ and the index λ: for every i, $\sum_l L^{(l)}\,(q_i - l_i + 1 + \lambda l_i)\,c^{\lambda}_{q-l+\mathbf{e}_i} = 0$, coefficients with a negative index being 0.*

(iii) $\partial_{\xi_k^{(l)}} c^{\nu}_{q} = -\prod_e \Gamma_1\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix} c^{\nu+t_k}_{q-l}$, *read in the same way.*

(iv) *For a configuration t, an index $\nu \in \mathbb{C}^{\mathcal{E}}$ and coefficients $\xi_k(\rho)$ depending smoothly on ρ (Proposition 2.13(iii) and the chain rule),*

$$\rho_i \partial_{\rho_i} \mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho)) = -\sum_k \prod_e \Gamma_1\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix} \big(\rho_i \partial_{\rho_i} \xi_k\big) \mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}(\rho)).$$

Chapter 3

The Breta

3.1 Definition and the tower

Definition 3.1 (Breta)**.** The data of a Breta are:

- a finite set $\mathcal{E}$ of scales and a partition $\mathcal{P}$ of $\mathcal{E}$ into *bunches* Q; each bunch is read in scale and shape, $x_Q = \rho_Q u_Q$ with $\rho_Q > 0$ and $u_Q \in \Delta_Q$;
- a *shape index* $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ and a *scale index* $\boldsymbol{\nu} = (\nu_Q)_{Q\in\mathcal{P}}$;
- terms $k = 1, \dots, N$, each with a *shape exponent* $\pi_k \in \mathbb{R}^{\mathcal{E}}$, a *level* $\ell_k = (\ell_{k,Q})_Q \in \mathbb{R}^{\mathcal{P}}$ and a coefficient $\xi_k > 0$.

With $u^{\pi_k} = \prod_Q u_Q^{\pi_k|_Q}$ and $d\omega = \prod_Q d\omega_Q$, the *Breta* is

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) := \int_{\prod_Q \Delta_Q} u^{\mathbf{m}-\mathbf{1}}\ \mathsf{Z}_{\boldsymbol{\nu};\ell}\big((u^{\pi_k}\xi_k)_k\big)\, d\omega(u),$$

the Zamma on the bunch scales with configuration ℓ, wherever the integral converges. The data are *homogeneous* when every level is the degree of its shape exponent on each bunch, $\ell_{k,Q} = \langle \pi_k, \mathbf{1}_Q\rangle$; then the shape exponents form a configuration $t = (\pi_k)_k$ on $\mathcal{E}$ and the Breta is written $\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};t}(\boldsymbol{\xi})$. When $\mathcal{P} = \{B\}$ has one bunch, the scale index has one entry and is written as a number, $\mathsf{B}^{\{B\}}_{\mathbf{m},\nu;t}(\boldsymbol{\xi})$.

The Breta has five slots: the partition, the shape index, the scale index, the terms and the coefficients. It is the Gramma read in scale and shape on every bunch at once (Theorem 2.8), with the scale integrals closed into a Zamma and the shapes left as an integral; the scale index is free, and for homogeneous data fixing it at the bunch totals of the shape index, $\nu_Q = m_Q$, makes the Breta the Gramma itself up to Gammas.

Theorem 3.2 (The tower)**.** *Let the data be homogeneous and* $\mathbf{m} \in \mathcal{T}_t$*. For every partition* $\mathcal{P}$ *of the scales,*

$$\mathsf{G}_{\mathbf{m};t}(\boldsymbol{\xi}) = \prod_{Q\in\mathcal{P}} \Gamma(m_Q)\ \mathsf{B}^{\mathcal{P}}_{\mathbf{m},(m_Q)_Q;t}(\boldsymbol{\xi}).$$

With the finest partition the Breta is $\mathsf{Z}_{\mathbf{m};t}(\boldsymbol{\xi})$*; with the coarsest,* $\Gamma(m_{\mathcal{E}})$ *times the Breta is the integral of Theorem 2.8 with* $R = \varnothing$*. At the unit configuration* I_B *of a bunch* B *and the coarsest partition* $\{B\}$*, the Zamma inside is* $\big(\sum_e \xi_e u_e\big)^{-m_B}$*, and*

$$\prod_{e\in B} \Gamma(m_e)\, \xi_e^{-m_e} = \Gamma(m_B)\, \mathsf{B}^{\{B\}}_{\mathbf{m},m_B;I_B}(\boldsymbol{\xi}) \qquad (\operatorname{Re} m_e > 0).$$

Corollary 3.3 (Coarsening)**.** *Let the data be homogeneous, and let $\mathcal{P}$ refine $\mathcal{Q}$. Then*

$$\mathsf{B}^{\mathcal{Q}}_{\mathbf{m},(m_Q)_Q;t}(\boldsymbol{\xi}) = \prod_{Q\in\mathcal{Q}} \Gamma_{1}^{k_Q}\begin{bmatrix}(m_P)_{P\subseteq Q}\\ m_Q\end{bmatrix}\ \mathsf{B}^{\mathcal{P}}_{\mathbf{m},(m_P)_P;t}(\boldsymbol{\xi}),$$

with k_Q the number of bunches of $\mathcal{P}$ inside Q.

Proposition 3.4 (The Breta on the orthant)**.** *For any data,*

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \frac{1}{\prod_Q \Gamma(\nu_Q)} \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{m}-\mathbf{1}} \prod_Q |x_Q|^{\nu_Q - m_Q} \exp\Big(-\sum_k \xi_k\, x^{\pi_k} \prod_Q |x_Q|^{\ell_{k,Q}-\langle \pi_k,\mathbf{1}_Q\rangle}\Big)\, dx,$$

either side converging when the other does: a free scale index is one more power of each bunch's total, a level that is not the degree of its shape exponent on a bunch is one more power of that bunch's total inside its term, and for homogeneous data at $\nu_Q = m_Q$ this is Theorem 3.2.

Theorem 3.5 (Relations of the Breta)**.** *Each identity holds wherever every Breta in it converges. The terms are (π,ℓ); homogeneous data have $t_k = \pi_k$ and $\ell_{k,Q} = \langle t_k, \mathbf{1}_Q\rangle$. In this statement write $g_k = \prod_Q \Gamma_1\begin{bmatrix}\nu_Q+\ell_{k,Q}\\ \nu_Q\end{bmatrix}$.*

(i) Shape split*, per bunch Q:*

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \sum_{e\in Q} \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}).$$

(ii) Ladder*, per term k:*

$$\partial_{\xi_k} \mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = -g_k\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}).$$

(iii) Scale exchange*, per bunch Q:*

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \sum_k \ell_{k,Q}\,\xi_k\,\Gamma_1\begin{bmatrix}\nu_Q+\ell_{k,Q}\\ \nu_Q+1\end{bmatrix} \prod_{Q'\neq Q} \Gamma_1\begin{bmatrix}\nu_{Q'}+\ell_{k,Q'}\\ \nu_{Q'}\end{bmatrix}\ \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}).$$

(iv) Shape exchange*, for scales $e\neq f$ of one bunch, with $\operatorname{Re} m_e, \operatorname{Re} m_f > 0$:*

$$m_e\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_f,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) - m_f\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \sum_k \xi_k\, g_k \Big[\pi_{k,e}\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_f,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}) - \pi_{k,f}\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_e,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})\Big].$$

(v) Shape exchange on the orthant*, for scales $e\neq f$ of one bunch Q:*

$$\begin{aligned}(m_e - m_f)\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) + (\nu_Q - m_Q)\Big(\mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) - \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_f,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi})\Big)\\ = \sum_k \xi_k\, g_k \Big[(\pi_{k,e}-\pi_{k,f})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})\\ + \big(\ell_{k,Q} - \langle \pi_k,\mathbf{1}_Q\rangle\big)\Big(\mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_e,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}) - \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_f,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})\Big)\Big];\end{aligned}$$

for homogeneous data the second line of the right side is absent.

For homogeneous data a term raises the shape total of each bunch and its scale index by the same level, so the ladder, the scale exchange and the shape exchange (v) keep the condition $\nu_Q = m_Q$; in (v) the shifts of the shape index alone carry the factor $\nu_Q - m_Q$, which vanishes there. The shape split and the shape exchange (iv) shift the shape index alone.

3.2 Letters

A *letter* is a function that enters an integrand through a power P^{-s}, s its *letter exponent*. On bunches read in scale and shape, a polynomial letter with positive coefficients is a sum of patterns, each a monomial in the scales and the shapes; the letter presentation writes the integral of a kernel against a product of letters as one Breta, by giving every letter a scale of its own.

Proposition 3.6 (Letter presentation)**.** *Let the scales B carry a partition $\mathcal{Q}$ into bunches, each read in scale and shape, $x_Q = \rho_Q u_Q$, with $\rho = (\rho_Q)_{Q\in\mathcal{Q}}$ and $u = (u_Q)_{Q\in\mathcal{Q}}$; let* kernel terms *$g_j \rho^{\lambda_j} u^{\pi_j}$ ($g_j > 0$, $\lambda_j \in \mathbb{R}^{\mathcal{Q}}$, $\pi_j \in \mathbb{R}^B$) and letters*

$$P_i(\rho, u) = \sum_p C_{i,p}\, \rho^{\ell_{i,p}} u^{\pi_{i,p}}, \qquad C_{i,p} > 0, \qquad i = 1, \dots, N,$$

be given, the sum over the patterns *p of the letter P_i, with $\ell_{i,p} \in \mathbb{R}^{\mathcal{Q}}$, $\pi_{i,p} \in \mathbb{R}^B$ and letter exponents s_i, $\operatorname{Re} s_i > 0$. The array $(C_{i,p})$ lists the coefficients of the patterns. Introduce one scale v_i per letter, put $V = \{v_1, \dots, v_N\}$, and take the partition of the scales $B \sqcup V$*

$$\mathcal{P} = \mathcal{Q} \cup \{\{v_1\}, \dots, \{v_N\}\}.$$

On $B \sqcup V$ the shape index is $(\mathbf{m}, \mathbf{s})$, and on the bunches the scale index is $(\boldsymbol{\nu}, \mathbf{s})$, $\boldsymbol{\nu} \in \mathbb{C}^{\mathcal{Q}}$. The term arrays $(\widetilde{\pi}, \widetilde{\ell})$ have one column per kernel term and one per pattern:

column	*shape exponent on $B \sqcup V$*	*level on $\mathcal{P}$*	*coefficient*
j	$(\pi_j, 0_V)$	$(\lambda_j, 0_V)$	g_j
(i,p)	$(\pi_{i,p}, \mathbf{e}_i)$	$(\ell_{i,p}, \mathbf{e}_i)$	$C_{i,p}$

with $\mathbf{e}_i$ the unit vector at v_i in the shape column and at the singleton bunch $\{v_i\}$ in the level column, and $\mathbf{g} \oplus \mathbf{C}$ lists the coefficients in this column order. Then

$$\int_{\prod_Q \Delta_Q} u^{\mathbf{m}-\mathbf{1}} \int_{\mathbb{R}^{\mathcal{Q}}_{>0}} \rho^{\boldsymbol{\nu}-\mathbf{1}} \exp\Big(-\sum_j g_j \rho^{\lambda_j} u^{\pi_j}\Big) \prod_i P_i(\rho, u)^{-s_i}\, d\rho\, d\omega(u) = \prod_{Q\in\mathcal{Q}} \Gamma(\nu_Q)\, \mathsf{B}^{\mathcal{P}}_{(\mathbf{m},\mathbf{s}),(\boldsymbol{\nu},\mathbf{s});(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g} \oplus \mathbf{C}).$$

The shapes of the singleton bunches are points, so the entries of the shape index on V do not enter the integral; they move with the scale entries on V, which keeps the relations of Theorem 3.5 literal. In the rest of this statement the partition, the term arrays and the coefficients are fixed, and

$$b(\mathbf{m}, \boldsymbol{\nu}; \mathbf{s}) \quad \textit{stands for} \quad \mathsf{B}^{\mathcal{P}}_{(\mathbf{m},\mathbf{s}),(\boldsymbol{\nu},\mathbf{s});(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g} \oplus \mathbf{C}).$$

The relations of Theorem 3.5 read: the scale exchange at $\{v_i\}$ is the letter split

$$b(\mathbf{m}, \boldsymbol{\nu}; \mathbf{s}) = \sum_p C_{i,p} \prod_{Q\in\mathcal{Q}} \Gamma_1\begin{bmatrix} \nu_Q + \ell_{i,p,Q} \\ \nu_Q \end{bmatrix} b(\mathbf{m} + \pi_{i,p}, \boldsymbol{\nu} + \ell_{i,p}; \mathbf{s} + \mathbf{e}_i);$$

the ladders are

$$\partial_{g_j} b(\mathbf{m}, \boldsymbol{\nu}; \mathbf{s}) = -\prod_{Q\in\mathcal{Q}} \Gamma_1\begin{bmatrix} \nu_Q + \lambda_{j,Q} \\ \nu_Q \end{bmatrix} b(\mathbf{m} + \pi_j, \boldsymbol{\nu} + \lambda_j; \mathbf{s}),$$

$$\partial_{C_{i,p}} b(\mathbf{m},\boldsymbol{\nu};\mathbf{s}) = -s_i \prod_{Q\in\mathcal{Q}} \Gamma_1\!\begin{bmatrix}\nu_Q+\ell_{i,p,Q}\\ \nu_Q\end{bmatrix}\, b(\mathbf{m}+\pi_{i,p},\boldsymbol{\nu}+\ell_{i,p};\mathbf{s}+\mathbf{e}_i);$$

and the scale exchange on each $Q\in\mathcal{Q}$ is

$$\nu_Q\, b(\mathbf{m},\boldsymbol{\nu};\mathbf{s}) = \sum_j \lambda_{j,Q}\, g_j \prod_{Q'\in\mathcal{Q}} \Gamma_1\!\begin{bmatrix}\nu_{Q'}+\lambda_{j,Q'}\\ \nu_{Q'}\end{bmatrix}\, b(\mathbf{m}+\pi_j,\boldsymbol{\nu}+\lambda_j;\mathbf{s}) + \sum_{i,p} s_i\, \ell_{i,p,Q}\, C_{i,p} \prod_{Q'\in\mathcal{Q}} \Gamma_1\!\begin{bmatrix}\nu_{Q'}+\ell_{i,p,Q'}\\ \nu_{Q'}\end{bmatrix}\, b(\mathbf{m}+\pi_{i,p},\boldsymbol{\nu}+\ell_{i,p};\mathbf{s}+\mathbf{e}_i).$$

Proposition 3.7 (Letter data on the orthant)**.** *Use the partition and the term arrays of Proposition 3.6, and for $\alpha\in\mathbb{R}^B_{>0}$ write $P_i(\alpha)$ for the letter at $\rho_Q=|\alpha_Q|$, $u_Q=\alpha_Q/|\alpha_Q|$:*

$$P_i(\alpha) = \sum_p C_{i,p}\,\alpha^{\pi_{i,p}} \prod_{Q\in\mathcal{Q}} |\alpha_Q|^{\ell_{i,p,Q}-\langle\pi_{i,p},\mathbf{1}_Q\rangle}.$$

Then, either side converging when the other does,

$$\prod_{Q\in\mathcal{Q}} \Gamma(\nu_Q)\, \mathsf{B}^{\mathcal{P}}_{(\mathbf{m},\mathbf{s}),(\boldsymbol{\nu},\mathbf{s});(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g}\oplus\mathbf{C}) = \int_{\mathbb{R}^B_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} \prod_{Q\in\mathcal{Q}} |\alpha_Q|^{\nu_Q-m_Q} \exp\Big(-\sum_j g_j\,\alpha^{\pi_j} \prod_{Q\in\mathcal{Q}} |\alpha_Q|^{\lambda_{j,Q}-\langle\pi_j,\mathbf{1}_Q\rangle}\Big) \prod_i P_i(\alpha)^{-s_i}\, d\alpha :$$

each bunch total $|\alpha_Q|$ is one more letter, at the exponent $m_Q-\nu_Q$, and it enters every kernel term and pattern whose level on Q is not its degree there. For homogeneous letter data, $\lambda_{j,Q}=\langle\pi_j,\mathbf{1}_Q\rangle$ and $\ell_{i,p,Q}=\langle\pi_{i,p},\mathbf{1}_Q\rangle$, the letters are $P_i(\alpha)=\sum_p C_{i,p}\alpha^{\pi_{i,p}}$, and at $\nu_Q=m_Q$ the totals are absent.

Proposition 3.8 (Equal levels collapse)**.** *In a Breta the dressed coefficients of all terms with one level add: the Zamma inside has one term per distinct level, whose coefficient is that level's polynomial in the shape. For homogeneous letters of a bunch the level of a pattern is its degree.*

3.3 The shape integral

Theorem 3.9 (The shape integral)**.** *For $\operatorname{Re} m_e>0$ ($e\in B$),*

$$\int_{\Delta_B} u^{\mathbf{m}-\mathbf{1}}\, d\omega_B(u) = \mathsf{B}^{\{B\}}_{\mathbf{m},m_B;I_B}(\mathbf{1}) = \Gamma_{|B|}\!\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix} :$$

the tower (Theorem 3.2) at the unit configuration I_B with unit coefficients, where the Zamma inside the Breta is $\big(\sum_e u_e\big)^{-m_B}=1$.

The stacked Gamma of the masses over their total is therefore the integral of the shape density $u^{\mathbf{m}-\mathbf{1}}$ on the simplex, and the Dirichlet law is that density normalised.

Definition 3.10 (Dirichlet law)**.** For $\mathbf{m}\in\mathbb{R}^B_{>0}$, $\operatorname{Dir}_{\mathbf{m}}(du) := u^{\mathbf{m}-\mathbf{1}}\, d\omega_B(u)/\Gamma_{|B|}\!\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}$, a probability measure on Δ_B.

Theorem 3.11 (Moments and steps)**.** *For* $\mathbf{m} \in \mathbb{R}^B_{>0}$ *and* $\mathbf{k} \in \mathbb{C}^B$ *with* $\mathrm{Re}(m_e + k_e) > 0$ *for every* $e \in B$*,*

$$\mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}\big[U^{\mathbf{k}}\big] = \frac{\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{k}\\ m_B+|\mathbf{k}|\end{matrix}\right]}{\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right]},$$

which for $\mathbf{k} \in \mathbb{N}^B$ *is* $\prod_e (m_e)_{k_e}/(m_B)_{|\mathbf{k}|}$*. For every* $\mathbf{m} \in \mathbb{C}^B$ *off the poles the stacked Gamma* $\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right]$ *satisfies the shape split and the shape exchange of its Breta* $\mathsf{B}^{\{B\}}_{\mathbf{m},m_B;I_B}(\mathbf{1})$*:*

$$\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right] = \sum_{e\in B}\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{e}_e\\ m_B+1\end{matrix}\right], \qquad m_e\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{e}_f\\ m_B+1\end{matrix}\right] = m_f\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{e}_e\\ m_B+1\end{matrix}\right],$$

$$\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{e}_e\\ m_B+1\end{matrix}\right] = \frac{m_e}{m_B}\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right].$$

Theorem 3.12 (Aggregation)**.** *Let* $B = B_1 \sqcup \cdots \sqcup B_q$ *with nonempty blocks and* $J = \{1, \dots, q\}$*. The map* $(w, v^{(1)}, \dots, v^{(q)}) \mapsto u$*,* $u|_{B_j} = w_j v^{(j)}$*, is a bijection* $\Delta_J \times \prod_j \Delta_{B_j} \to \Delta_B$*, and*

$$u^{\mathbf{m}-\mathbf{1}} d\omega_B = \prod_j w_j^{m_{B_j}-1}\, d\omega_J(w) \prod_j (v^{(j)})^{\mathbf{m}|_{B_j}-\mathbf{1}} d\omega_{B_j}(v^{(j)}).$$

Consequently $\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right] = \Gamma_{\substack{q\\1}}\left[\begin{matrix}(m_{B_j})_j\\ m_B\end{matrix}\right] \prod_j \Gamma_{\substack{|B_j|\\1}}\left[\begin{matrix}\mathbf{m}|_{B_j}\\ m_{B_j}\end{matrix}\right]$*, and under* $\mathrm{Dir}_{\mathbf{m}}$ *the block totals and the block shapes are independent, with laws* $\mathrm{Dir}_{(m_{B_j})_j}$ *and* $\mathrm{Dir}_{\mathbf{m}|_{B_j}}$*.*

3.4 Products of shape integrals

Theorem 3.13 (The Shruffle)**.** *For a bunch* Q *and a total order* ς *on* Q *let* $\Delta^{\varsigma}_Q = \{u \in \Delta_Q : u_{\varsigma(1)} > u_{\varsigma(2)} > \cdots\}$ *and* $I^{\varsigma}_Q(f; \mathbf{m}) = \int_{\Delta^{\varsigma}_Q} u^{\mathbf{m}-\mathbf{1}} f(u)\, d\omega_Q$*. For pairwise disjoint bunches* $A_1, \dots, A_q$ *with total orders* ς_i *on* A_i*, functions* f_i *on* Δ_{A_i}*,* $A = A_1 \sqcup \cdots \sqcup A_q$ *and* $\bigotimes_i f_i(w) = \prod_i f_i(w_{A_i}/|w_{A_i}|)$ *on* Δ_A*, with* $\mathrm{Re}\, m_{A_i} > 0$ *and the integrals absolutely convergent,*

$$\prod_{i=1}^{q} I^{\varsigma_i}_{A_i}(f_i; \mathbf{m}|_{A_i}) = \Gamma_{\substack{q\\1}}\left[\begin{matrix} & m_A & \\ m_{A_1}, & \dots, & m_{A_q}\end{matrix}\right] \sum_{\varsigma} I^{\varsigma}_A\Big(\bigotimes_i f_i; \mathbf{m}\Big),$$

the sum over the orders ς *of* A *that restrict to every* ς_i*.*

The product of shape integrals on disjoint bunches $A_1, \dots, A_q$ is $\Gamma_{\substack{q\\1}}\left[\begin{matrix} & m_A & \\ m_{A_1}, & \dots, & m_{A_q}\end{matrix}\right]$ times the sum of the shape integrals on the joined bunch over the interleavings of their orders: this is the Shruffle, the product rule of the shape side, whose counterpart for Trailmoments at a common cut is the Struffle (Theorem 5.25).

3.5 The Beta port

Definition 3.14 (Beta port)**.** For $q \in \mathbb{N}$ and $a, b \in \mathbb{C}$,

$$\Pi^{a,b}_q(x) = \frac{(-1)^q x^{1-a}(1-x)^{1-b}}{q!}\, \partial^q_x\big[x^{a+q-1}(1-x)^{b+q-1}\big]$$

$$= \sum_{k=0}^{q} (-1)^{q+k} \binom{a+q-1}{q-k} \binom{b+q-1}{k} x^k (1-x)^{q-k},$$

a polynomial of degree at most q. For positive integers a, b its coefficients are integers; for fixed q and a, b affine in an integer parameter, its coefficients are integers of polynomial growth in that parameter.

Theorem 3.15 (Relations of the Beta port)**.** *(i)* Action on a monomial. *For* $\operatorname{Re} c > 0$ *and* $\operatorname{Re} b > 0$*:* $\int_0^1 x^{c-1}(1-x)^{b-1}\,\Pi_q^{a,b}(x)\,dx = \binom{c-a}{q}\Gamma\,{}_2{}_1\!\left[\begin{matrix} c,\ b+q \\ c+b+q \end{matrix}\right]$*. With* $c = a + j$*,* $j \in \mathbb{N}$*, the binomial vanishes for* $j < q$*: the Beta port is orthogonal, for the density* $x^{a-1}(1-x)^{b-1}$*, to every polynomial of degree below* q*.*

(ii) Integration by parts. *For* $\operatorname{Re} a, \operatorname{Re} b > 0$ *and* g *with* q *continuous derivatives on* $[0,1]$*,*

$$\int_0^1 x^{a-1}(1-x)^{b-1}\,\Pi_q^{a,b}(x)\,g(x)\,dx = \frac{1}{q!}\int_0^1 x^{a+q-1}(1-x)^{b+q-1}\,g^{(q)}(x)\,dx;$$

with one Beta port per scale, of orders $\mathbf{n} = (n_e)_{e\in\mathcal{E}}$*,*

$$\int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{a_e-1}(1-x_e)^{b_e-1}\Pi_{n_e}^{a_e,b_e}(x_e)\,g(\mathbf{x})\,d\mathbf{x} \\ = \frac{1}{\prod_e n_e!}\int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{a_e+n_e-1}(1-x_e)^{b_e+n_e-1}\,\partial^{\mathbf{n}} g(\mathbf{x})\,d\mathbf{x}.$$

(iii) Letter ladder. *For letters* $L_i = c_i + d_i x$ *positive on* $[0,1]$ *(*$c_i > 0$ *and* $c_i + d_i > 0$*) and exponents* $w_i \in \mathbb{C}$*,*

$$\frac{(-1)^q}{q!}\,\partial_x^q \prod_i L_i^{-w_i} = \prod_i L_i^{-w_i} \sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!}\Big(\frac{d_i}{L_i}\Big)^{v_i},$$

so that, for $\operatorname{Re} a, \operatorname{Re} b > 0$*,*

$$\int_0^1 x^{a-1}(1-x)^{b-1}\,(-1)^q \Pi_q^{a,b} \prod_i L_i^{-w_i}\,dx \\ = \sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!}\int_0^1 x^{a-1}(1-x)^{b-1}\prod_i L_i^{-w_i}\prod_i \Big(\frac{x(1-x)d_i}{L_i}\Big)^{v_i}dx.$$

Let $a, b > 0$ *and every* $w_i \geq 0$*. If every* $d_i \geq 0$ *the value is nonnegative, and if every* $d_i \leq 0$ *it is* $(-1)^q$ *times a nonnegative number; whatever the signs of the* d_i*, if* $x(1-x)|d_i|/L_i \leq \varepsilon$ *on* $[0,1]$ *for every* i*, its modulus is at most* $\frac{(|\mathbf{w}|)_q}{q!}\varepsilon^q \int_0^1 x^{a-1}(1-x)^{b-1}\prod_i L_i^{-w_i}dx$*.*

(iv) Reflection. $\Pi_q^{a,b}(1-x) = (-1)^q \Pi_q^{b,a}(x)$*.*

(v) Three-term and derivative relations. *For* $q \geq 1$ *and* $a, b \in \mathbb{C}$ *with* $a + b + 2q - i \neq 0$ *for* $i = 0, 1, 2$*,*

$$x\,\Pi_q^{a,b} = \frac{(q+1)(a+b+q-1)}{(a+b+2q)(a+b+2q-1)}\,\Pi_{q+1}^{a,b} + \Big(\frac{(q+1)(a+q)}{a+b+2q} - \frac{q(a+q-1)}{a+b+2q-2}\Big)\Pi_q^{a,b}$$

$$+\frac{(a+q-1)(b+q-1)}{(a+b+2q-1)(a+b+2q-2)}\,\Pi^{a,b}_{q-1},$$

$$x(1-x)\,\partial_x\Pi^{a,b}_q=\Big(\frac{q(b+q-1)}{a+b+2q-2}-q\,x\Big)\Pi^{a,b}_q+\frac{(a+q-1)(b+q-1)}{a+b+2q-2}\,\Pi^{a,b}_{q-1};$$

for every $q\in\mathbb{N}$ the coefficient of x^j in $\Pi^{a,b}_q$ is $(-1)^{q+j}\binom{a+q-1}{q-j}\binom{a+b+q+j-2}{j}$, and that of x^q is $\binom{a+b+2q-2}{q}$.

The Beta port moves derivatives of an integrand onto its shape density: integrating against $\Pi^{a,b}_q$ is integrating the q-th derivative against the density with both exponents raised by q. Against a product of linear letters whose slopes have one sign it keeps the sign of the integrand, up to $(-1)^q$ when the slopes are negative, which is what the arithmetic of Part VI uses. The Beta port of order q is orthogonal, for the density $x^{a-1}(1-x)^{b-1}$, to every polynomial of lower degree (Theorem 3.15(i)), and at every pair of exponents multiplication by x and the operator $x(1-x)\partial_x$ move each Beta port by at most one step in its order (Theorem 3.15(v)): finite coefficient rules, which is what turns a mark relation (Theorem 35.10) into a recurrence of bounded width.

Chapter 4

Fibre integrals and closing

Definition 4.1 (Fibre integral)**.** Let $\mathbf{b} \in \mathbb{C}^N$ be *shares*, pinned by $P\mathbf{b} = \boldsymbol{\pi}$ with P a real $q \times N$ matrix with linearly independent rows and $\boldsymbol{\pi} \in \mathbb{C}^q$. Let f be holomorphic on a tube $\{\operatorname{Re}\mathbf{b} \in U\}$, and let $\mathbf{c} \in U$ be an *admissible real point*, $P\mathbf{c} = \operatorname{Re}\boldsymbol{\pi}$. The *fibre integral* is

$$\int_{P\mathbf{b}=\boldsymbol{\pi}} f := (2\pi)^{q-N} \int_{\mathbb{R}^N} f(\mathbf{c}+i\mathbf{y})\,\delta^q\big(P\mathbf{y} - \operatorname{Im}\boldsymbol{\pi}\big)\,d\mathbf{y} = \frac{(2\pi)^{q-N}}{\sqrt{\det PP^{\mathsf{T}}}} \int_{P\mathbf{y}=\operatorname{Im}\boldsymbol{\pi}} f(\mathbf{c}+i\mathbf{y})\,d\mathrm{vol}_{N-q}(\mathbf{y})$$

whenever it converges, vol_{N-q} being the $(N-q)$-dimensional volume on the affine subspace; by Cauchy's theorem it does not depend on $\mathbf{c}$ within one connected set of admissible real points. Its *dimension* is $N-q$. With one pin, $P = (1,\dots,1)$ and $\boldsymbol{\pi} = s$, it is the *share integral*

$$\int_{a_1+\cdots+a_k=s} f(\mathbf{a}) = \frac{1}{(2\pi i)^{k-1}} \int_{c_1+i\mathbb{R}} \cdots \int_{c_{k-1}+i\mathbb{R}} f\Big(a_1,\dots,a_{k-1},\, s - \sum_{i<k} a_i\Big)\,da_1\cdots da_{k-1}.$$

At dimension zero, $q = N$, the value is $|\det P|^{-1} f(P^{-1}\boldsymbol{\pi})$. When $\boldsymbol{\pi}$ is real, with $V = \{\mathbf{y} :\ P\mathbf{y} = 0\}$ and the measure $d\boldsymbol{\vartheta} = (\det PP^{\mathsf{T}})^{-1/2}\,d\mathrm{vol}_{N-q}$ on it, the fibre integral is written $\int_{\mathbf{c}+iV} f = (2\pi)^{q-N}\int_V f(\mathbf{c}+i\boldsymbol{\vartheta})\,d\boldsymbol{\vartheta}$, its admissible real point shown.

Proposition 4.2 (The share side of the Gramma)**.** *For $\nu \in \mathcal{T}_t$,*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \int_{t\mathbf{b}=\nu} \prod_{k=1}^{N} \Gamma(b_k)\,\xi_k^{-b_k},$$

the fibre integral with one share per term, pinned by the configuration ($|\mathcal{E}|$ pins), of dimension $N - |\mathcal{E}|$; its admissible real points are the $\mathbf{c} \in (0,\infty)^N$ with $t\mathbf{c} = \operatorname{Re}\nu$, and they exist exactly on the domain. At dimension zero it is the value of Corollary 2.6.

On the share side the configuration is the matrix of the pins and the index their values, one pin per scale; each term carries one share, with its Gamma and its coefficient's power as factors.

Theorem 4.3 (The native gauge of a configuration)**.** *Let the t_k span $\mathbb{R}^{\mathcal{E}}$, $\nu \in \operatorname{int}\operatorname{cone}(t)$ real and $\boldsymbol{\xi} > 0$.*

(i) $\Phi_{\boldsymbol{\xi}}(y) = \sum_k \xi_k e^{\langle t_k, y\rangle} - \langle \nu, y\rangle$ is strictly convex and tends to $+\infty$ along every ray; its unique critical point $y^\star$ solves $\sum_k \xi_k e^{\langle t_k, y^\star\rangle} t_k = \nu$, and $e^{y^\star}$ is the native gauge *of the configuration.*

(ii) The shares $w_k^\star = \xi_k e^{\langle t_k, y^\star\rangle}$ are an admissible real point of the share side (Proposition 4.2). On the real fibre $\{\mathbf{w} > 0 :\ t\mathbf{w} = \nu\}$ the function $S(\mathbf{w}) = \sum_k \big(w_k - w_k \log(w_k/\xi_k)\big)$ is strictly concave with unique critical point $\mathbf{w}^\star$, and $S(\mathbf{w}^\star) = \Phi_{\boldsymbol{\xi}}(y^\star)$.

(iii) For $\nu' \in \mathbb{C}^{\mathcal{E}}$, as $\lambda \to \infty$,

$$\mathsf{G}_{\lambda\nu+\nu';t}(\lambda\boldsymbol{\xi}) = e^{\langle \nu', y^\star\rangle}\, e^{-\lambda\Phi_{\boldsymbol{\xi}}(y^\star)} \Big(\frac{2\pi}{\lambda}\Big)^{|\mathcal{E}|/2} \det\Big(\sum_k w_k^\star t_k t_k^{\mathsf{T}}\Big)^{-1/2} \big(1 + O(\lambda^{-1})\big).$$

The native gauge of a configuration is read on both of its sides at once: on the orthant side its logarithmic coordinate $y^\star$ minimizes the convex function $\Phi_{\boldsymbol{\xi}}$, on the share side the shares $\mathbf{w}^\star$ it determines maximize the entropy-like function S on the real fibre, and the minimum equals the maximum.

Corollary 4.4 (Splitting a coefficient)**.** *Let $k \geq 2$, $X \in (0,\infty)^k$, $\operatorname{Re} s > 0$ and $\mathbf{c} \in (0,\infty)^{k-1}$ with $\sum_i c_i < \operatorname{Re} s$. On this contour, every share having positive real part,*

$$\mathsf{Z}_{s;(1)}(X_1 + \cdots + X_k) = \Big(\sum_{i=1}^k X_i\Big)^{-s} = \int_{a_1+\cdots+a_k=s} \Gamma_{\substack{k\\1}}\!\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix} \prod_{i=1}^k X_i^{-a_i},$$

equivalently $\Gamma(s)(X_1+\cdots+X_k)^{-s} = \int_{|\mathbf{a}|=s} \prod_i \Gamma(a_i) X_i^{-a_i}$: the share side of the one-scale Gamma $\mathsf{G}_{s;(1,\dots,1)}(X)$ (Proposition 4.2). The stacked Gamma $\Gamma_{\substack{k\\1}}\!\left[\begin{smallmatrix}\mathbf{a}\\ s\end{smallmatrix}\right]$ is the share kernel *of the letter $X_1 + \cdots + X_k$ at the exponent s.*

Theorem 4.5 (Closing)**.** *Let a fibre integral have the form $\int_{P\mathbf{b}=\boldsymbol{\pi}} \Gamma\big(\ell(\mathbf{b})\big) G(\mathbf{b})$ with ℓ affine and not constant on the fibre. Suppose that, moving the contour along a direction of the fibre in which ℓ decreases, $\Gamma(\ell)$ is the only factor with poles crossed, and the shifted contour's contribution tends to zero. Then*

$$\int_{P\mathbf{b}=\boldsymbol{\pi}} \Gamma\big(\ell(\mathbf{b})\big)\, G(\mathbf{b}) = \sum_{j\geq 0} \frac{(-1)^j}{j!} \int_{P\mathbf{b}=\boldsymbol{\pi},\ \ell(\mathbf{b})=-j} G(\mathbf{b}),$$

each term a fibre integral with one more pin. When several factors have poles crossed, each contributes its own lattice; where several share a pole hyperplane, the residue is a Laurent coefficient of the Gamma factors (Chapter 5).

Closing trades one dimension of the fibre for a sum over the poles crossed: the Gamma whose argument falls is replaced by its residues. Where closing can be repeated until the fibre has dimension zero, every term is the value $|\det P|^{-1} G(P^{-1}\boldsymbol{\pi})$ of the remaining factors at one point.

Theorem 4.6 (The share kernels form a semigroup)**.** *For a letter $\sum_{i=1}^k X_i$, shares $\mathbf{a}, \mathbf{b}, \mathbf{g} \in \mathbb{C}^k$ and $|\mathbf{g}| = s + s'$, on admissible contours,*

$$\int_{\mathbf{a}+\mathbf{b}=\mathbf{g},\ |\mathbf{a}|=s} \Gamma_{\substack{k\\1}}\!\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix} \Gamma_{\substack{k\\1}}\!\begin{bmatrix}\mathbf{b}\\ s'\end{bmatrix} = \Gamma_{\substack{k\\1}}\!\begin{bmatrix}\mathbf{g}\\ s+s'\end{bmatrix}.$$

Splitting the letter at the exponent s and at the exponent s' and merging the shares is splitting it at $s + s'$: the share kernels of Corollary 4.4 compose like the powers they open.

Corollary 4.7 (Four Gammas on a line)**.** *On a contour separating the poles of $\Gamma(a+z)\Gamma(b+z)$ from those of $\Gamma(c-z)\Gamma(d-z)$,*

$$\frac{1}{2\pi i}\int \Gamma(a+z)\,\Gamma(b+z)\,\Gamma(c-z)\,\Gamma(d-z)\,dz = \Gamma_{\substack{4\\1}}\!\begin{bmatrix} a+c,\ b+d,\ a+d,\ b+c \\ a+b+c+d \end{bmatrix}:$$

Theorem 4.6 at $k=2$, multiplied by $\Gamma(a+c)\Gamma(b+d)$, with the shares $(a+z,\ c-z)$ at the exponent $a+c$ and $(d-z,\ b+z)$ at the exponent $b+d$, merged to $(a+d,\ b+c)$.

The four Gammas are the Gamma factors of the four shares of a Gramma on three scales, pinned by the three scales, and the one free share is the contour variable; integrating the orthant side scale by scale leaves $\Gamma(a+c)\Gamma(b+d)$ times the Beta corner ${}_{2}\Gamma_{1}\left[\begin{matrix} a+d,\ b+c \\ a+b+c+d \end{matrix}\right]$, the shape integral of two scales (Theorem 3.9).

Chapter 5

The Trailmoment

5.1 Definition

Definition 5.1 (Trailmoment)**.** The data of a Trailmoment are an index ν and a configuration t, those of a Zamma; a measure ϕ, the *input*, on a space Y, the *source*; a measurable *coefficient function* $\mathbf{c} : Y \to (0,\infty)^N$; and a measurable *cut region* $Y' \subseteq Y$. The *Trailmoment* is

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \int_{Y'} \mathsf{Z}_{\nu;t}\big(\mathbf{c}(y)\big)\,\phi(dy)$$

wherever it converges: a normalized Zamma kernel read against the input over the cut region.

The Trailmoment has five slots: its index and configuration, which are the kernel's, its coefficient function, its input and its cut region. On a line, $Y = \mathbb{R}$, the cut region (N,∞) is the *cut* at N, and the region $(N, N']$ between two cuts is a *head*; on $\mathbb{R}^d$ the region $X_N = \{N < y_1 < \cdots < y_d\}$ is a *trail region*, the nested tails that give the Trailmoment its name; the trails of Section 5.8 are read over it. The input is written as a density with its domain marked by an indicator, so that the measure is part of the data.

Definition 5.2 (Lattice inputs; the index-one value)**.** The *lattice input* on the coset $b+q\mathbb{Z}$, $b \in \mathbb{R}$, of step $q > 0$ and frequency $a \in \mathbb{C}$ is $e^{2\pi i a x}\,\mathbb{I}_{b+q\mathbb{Z}} = \sum_{m\in b+q\mathbb{Z}} e^{2\pi i a m}\,\delta_m$; at frequency 0 it is counting measure on $b+q\mathbb{Z}$, written $\mathbb{I}_{b+q\mathbb{Z}}$, and for $\operatorname{Im} a > 0$ the values $e^{2\pi i a m}$ decay geometrically toward $+\infty$. The input $f\,\mathbb{I}_{b+q\mathbb{Z}}$ has the density f on the lattice, and $\mathbb{I}_{\mathbb{Z}}$ is the *counting input*. At one unit exponent and the coefficient function x the kernel is $\mathsf{Z}_{\nu;(1)}(x) = x^{-\nu}$, so at a cut beyond which every lattice point is positive

$$(\mathsf{T}_{\nu;(1)}[x]\,f\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) = \sum_{m\in b+\mathbb{Z},\ m>N} f(m)\,m^{-\nu}.$$

At index one the lattice input is regularized at its upper end:

$$(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) := \lim_{M\to\infty}\Big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,N+M]) - \log M\Big),$$

which exists, the heads being $\log M + (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) + O(M^{-1})$: the regularized value is the coefficient of $M^0 \log^0 M$ in the heads.

Proposition 5.3 (The cut relation)**.** *The Trailmoment is additive in its cut region: for disjoint measurable $Y_1', Y_2' \subseteq Y$, $(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y_1' \sqcup Y_2') = (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y_1') + (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y_2')$, either side converging when the other does; on a line, for $N < N'$,*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N,\infty)) = (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N,N']) + (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N',\infty)),$$

and for a lattice input at index one it holds with the regularized values.

Definition 5.4 (Inputs)**.** An input is a nonnegative σ-finite measure μ, or such a measure with a complex density, $\phi = f\mu$ with $|\phi| = |f|\mu$; a Trailmoment converges when $\int |\mathsf{Z}_{\nu;t}(\mathbf{c})|\, d|\phi|$ over the cut region is finite. The image $\tau_*\phi$ of an input with a complex density under a measurable map τ is used only in integrals $\int g\, d(\tau_*\phi) := \int g \circ \tau\, d\phi$ with $\int |g \circ \tau|\, d|\phi| < \infty$; $|\tau_*\phi|$ denotes $\tau_*|\phi|$, and tameness and convergence of $\tau_*\phi$ are meant with it. The *Lebesgue input* $\mathbb{I}_{\mathbb{R}}$ is Lebesgue measure on $\mathbb{R}$, $f\,\mathbb{I}_{\mathbb{R}}$ is the input with density f, $\mathbb{I}_D$ is Lebesgue measure on an interval D, and $\mathbb{I}_{\mathbb{R}^{\mathcal{E}}}$ is Lebesgue measure on $\mathbb{R}^{\mathcal{E}}$; the lattice inputs are those of Definition 5.2, and at integer frequency the lattice input on $\mathbb{Z}$ is the counting input.

An input is *tame beyond the cut N* when its part beyond N is carried by $[x_0,\infty)$ for some $x_0 > 0$ and $\int_{(N,\infty)} x^{-\beta}\, |\phi|(dx) < \infty$ for some real β. Its Laplace transform beyond the cut is

$$\mathcal{F}_N(u) = \int_{(N,\infty)} e^{-ux}\, \phi(dx),$$

which converges for $\operatorname{Re} u > 0$, with $|\mathcal{F}_N(u)| \le e^{-(u-1)x_0} \int_{(N,\infty)} e^{-x}\, |\phi|(dx)$ for $u \ge 1$. When $\mathcal{F}_N$ is moreover meromorphic at $u = 0$, $(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ denotes, for every ν off its poles, the continuation in which $\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ is the meromorphic continuation of $\int_0^\infty u^{\nu-1}\mathcal{F}_N(u)\, du$, the Laplace side of Theorem 5.11 below at one unit exponent. At index one on a lattice input it is the regularized value of Definition 5.2, which is the constant term of the continuation at $\nu = 1$.

Proposition 5.5 (The continued Trailmoment on a line)**.** *Let ϕ be tame beyond the cut N with $\mathcal{F}_N(u) = \sum_{k \ge -k_0} c_k u^k$ near $u = 0$. Then $\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ is meromorphic in ν with at most simple poles, at $\nu = -k$, $k \ge -k_0$, with residue c_k, and at the nonpositive integers*

$$(\mathsf{T}_{-j;(1)}[x]\,\phi)((N,\infty)) = (-1)^j j!\, c_j \qquad (j \in \mathbb{N}).$$

At a nonpositive integer index the Trailmoment of a tame input is a Taylor coefficient of its Laplace transform, so a cut that releases a head changes these values by the head read at the same index.

5.2 Slots, corners and heights

Definition 5.6 (Slots, corners, height)**.** A *slot* is a place of a symbol of the serif family where one of its data stands. A Gramma or a Zamma has three slots: its index, its configuration and its coefficients. A Breta has five: its partition into bunches, its shape index, its scale index, its terms (shape exponents and levels) and its coefficients. A Trailmoment has five: its index ν, its configuration t, its coefficient function $\mathbf{c}$, its input ϕ and its cut region. A *corner* of a member of the serif family is that member with some of its slots given particular values. A *relation* is an identity between members of the serif family. Its *height* is the set of values of the slots on which one proof establishes it; a slot is *free* in a relation when the height does not restrict it and *pinned* otherwise. For a pinned slot, the *mechanism* is the step of the proof that uses the pinned value.

A relation recited at a corner is understood once its height is known: which slots the proof leaves free, which it uses, and at which step. A relation whose height is every value of every slot is a property of the operator. A pin records what a proof uses; it does not say that the relation fails at the other values of the slot, and where the converse is proved, so that the relation is a property of the pinned value, it is stated with the relation.

5.3 Relations at free slots

Theorem 5.7 (The gauge of the coefficient function)**.** *Let $(\nu, t, \mathbf{c}, \phi, Y')$ be the data of a Trailmoment on a source Y and let $\ell : Y \to (0,\infty)^{\mathcal{E}}$ be measurable. Then*

$$\big(\mathsf{T}_{\nu;t}[(\ell^{t_k} c_k)_k]\,\phi\big)(Y') = \big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\ell^{-\nu}\phi\big)(Y'),$$

either side converging when the other does. If the columns t_k, $k \in S$, form a basis of $\mathbb{R}^{\mathcal{E}}$, the choice $\log \ell = -(t_S^{\mathsf{T}})^{-1} \log \mathbf{c}_S$ makes $\ell^{t_k} c_k = 1$ for every $k \in S$: the gauge moves $|\mathcal{E}|$ of the N coefficient functions into the input, and the kernel then depends on the source point only through the $N - |\mathcal{E}|$ functions $\ell^{t_k} c_k$, $k \notin S$. At the unit configuration, $(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = \int_{Y'} \mathbf{c}^{-\nu}\, d\phi$.

At the unit configuration the kernel is a density on the input: index, coefficient function and input enter only through the measure $\mathbf{c}^{-\nu}\phi$. With $N > |\mathcal{E}|$ terms the kernel depends on the source point through $N - |\mathcal{E}|$ gauge-invariant functions as well as through a density, and this count is the dimension of the share side (Theorem 5.10).

Theorem 5.8 (Exchange and ladder of the Trailmoment)**.** *Each identity holds wherever every Trailmoment in it converges.*

(i) Exchange, one relation per scale. *For each $e \in \mathcal{E}$,*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \sum_{k=1}^{N} t_{k,e}\ \Gamma_1^{\,1}\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e + 1\end{bmatrix} \prod_{f \neq e} \Gamma_1^{\,1}\begin{bmatrix}\nu_f + t_{k,f}\\ \nu_f\end{bmatrix}\ \big(\mathsf{T}_{\nu + t_k;t}[\mathbf{c}]\,c_k\phi\big)(Y').$$

(ii) Ladder. *If $\mathbf{c} = \mathbf{c}_\zeta$ depends differentiably on a real parameter ζ and the differentiated integrand is dominated by an integrable function locally uniformly in ζ,*

$$\partial_\zeta(\mathsf{T}_{\nu;t}[\mathbf{c}_\zeta]\,\phi)(Y') = -\sum_{k=1}^{N} \prod_{e} \Gamma_1^{\,1}\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix}\ \big(\mathsf{T}_{\nu + t_k;t}[\mathbf{c}_\zeta]\,(\partial_\zeta c_k)\,\phi\big)(Y').$$

(iii) The unit configuration. *At $t = I_{\mathcal{E}}$ the exchange relation at each scale has one term, and on the interior of the set of indices where the Trailmoment converges the index is differentiated through the input: for every $e \in \mathcal{E}$,*

$$(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = (\mathsf{T}_{\nu + \mathbf{e}_e;I_{\mathcal{E}}}[\mathbf{c}]\,c_e\phi)(Y'), \qquad \partial_{\nu_e}(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = -(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,(\log c_e)\,\phi)(Y').$$

At the unit configuration a step of the index at a scale is a factor of that coordinate of the coefficient function on the input, in both directions. With several terms on a scale the exchange relation is linear in $N + 1$ Trailmoments whose inputs differ by the factors c_k, one relation per scale, and a step of the index is no longer a density.

Theorem 5.9 (Transport)**.** *For a measurable map $\tau : Y \to Y_2$, an input ϕ on Y, a coefficient function $\mathbf{c}$ on Y_2 and a cut region $Y_2' \subseteq Y_2$,*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\tau_*\phi)(Y_2') = (\mathsf{T}_{\nu;t}[\mathbf{c}\circ\tau]\,\phi)(\tau^{-1}Y_2'),$$

whenever the right side converges; for a nonnegative input either side converges when the other does. In particular, transporting by the coefficient function itself, a convergent Trailmoment is

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \big(\mathsf{T}_{\nu;t}[\mathrm{id}]\,\mathbf{c}_*(\phi|_{Y'})\big)\big((0,\infty)^N\big)\,:$$

coefficient function, input and cut region enter only through the image $\mathbf{c}_*(\phi|_{Y'})$ of the input on the orthant of coefficients.

A Trailmoment of a nonnegative input is therefore a function of three things: an index, a configuration, and a measure on the orthant of coefficients. Two descriptions of that measure, by different coefficient functions, inputs and cut regions, give the same member of the serif family. An increasing bijection of the line carries half-lines to half-lines: with $\tau(x) = x + h$ it is the *shift* of the cut, and with $\tau(x) = qx$ at one unit exponent it gives $(\mathsf{T}_{\nu;(1)}[x]\,\tau_*\phi)((qN,\infty)) = q^{-\nu}(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$.

Theorem 5.10 (The share side of the Trailmoment)**.** *Let $\operatorname{Re}\nu \in \operatorname{int}\operatorname{cone}(t)$, and let $\mathbf{s} \in (0,\infty)^N$ be an admissible real point, $t\mathbf{s} = \operatorname{Re}\nu$, with $\int_{Y'}\prod_k c_k(y)^{-s_k}\,|\phi|(dy) < \infty$. On the contour through $\mathbf{s}$,*

$$\prod_{e\in\mathcal{E}}\Gamma(\nu_e)\ (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \int_{t\mathbf{b}=\nu}\ \prod_{k=1}^{N}\Gamma(b_k)\ (\mathsf{T}_{\mathbf{b};I_N}[\mathbf{c}]\,\phi)(Y'),$$

$$(\mathsf{T}_{\mathbf{b};I_N}[\mathbf{c}]\,\phi)(Y') = \int_{Y'}\prod_{k=1}^{N} c_k(y)^{-b_k}\,\phi(dy),$$

a fibre integral of dimension $N - |\mathcal{E}|$ of Trailmoments at the unit configuration I_N, with one scale per term of t. Where some ν_e lies in $-\mathbb{N}$ the left side is read as $\int_{Y'}\mathsf{G}_{\nu;t}(\mathbf{c})\,d\phi$.

Every configuration is written through the unit configuration: a Trailmoment at any configuration is a superposition of Trailmoments at the unit configuration over the fibre of its configuration, each multiplied by the Gammas of its shares. Closing the fibre integral (Theorem 4.5) replaces the superposition by residues, and the residues are Trailmoments at the unit configuration at shifted indices; that is why their values appear in the closings of every configuration.

Theorem 5.11 (The Laplace side of the Trailmoment)**.** *The joint Laplace transform of the data $(\mathbf{c},\phi,Y')$ is $\mathcal{F}(v) = \int_{Y'} e^{-\langle v,\mathbf{c}(y)\rangle}\,\phi(dy)$ for $v \in (0,\infty)^N$. For $\nu \in \mathcal{T}_t$, wherever the double integral converges,*

$$\prod_{e\in\mathcal{E}}\Gamma(\nu_e)\ (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\nu-\mathbf{1}}\,\mathcal{F}\big(x^{t_1},\dots,x^{t_N}\big)\,dx = \big(\mathsf{T}_{\mathbf{1}-\nu;I_{\mathcal{E}}}[x]\ \mathcal{F}(x^t)\,\mathbb{I}_{\mathbb{R}^{\mathcal{E}}}\big)\big(\mathbb{R}^{\mathcal{E}}_{>0}\big),$$

where $\mathcal{F}(x^t) = \mathcal{F}(x^{t_1},\dots,x^{t_N})$ and the kernel of the last Trailmoment is $\prod_e x_e^{\nu_e-1}$, the Zamma of the unit configuration at the index $\mathbf{1}-\nu$; where some ν_e lies in $-\mathbb{N}$ the left side is read as $\int_{Y'}\mathsf{G}_{\nu;t}(\mathbf{c})\,d\phi$. The exponential is the Zamma of a term at exponent zero, $\mathsf{Z}_{1;(1,0)}(1,\xi) = e^{-\xi}$, so the joint Laplace transform is a Trailmoment of the same input: $\mathcal{F}(v) = (\mathsf{T}_{1;(1,0)}[(1,\langle v,\mathbf{c}\rangle)]\,\phi)(Y')$.

Index ν on the input side is index $\mathbf{1}-\nu$ on the Laplace side. Besides the input itself, a Trailmoment has two sides: the share side, which resolves the configuration into unit configurations, and the Laplace side, which replaces the input by its Laplace transform; the reflections of the index (Theorems 5.18 and 5.19), the Shruffle (Theorem 5.26) and the Cremona (Theorem 5.29) are read on the Laplace side, and the lattice relations and the Struffle on the input itself.

5.4 Lattices

Theorem 5.12 (The lattice relations)**.** *Let $b\in\mathbb{R}$, let $q\geq 1$ be an integer, let $(\nu,t,\mathbf{c})$ be the data of a kernel on $\mathbb{R}$, f a density and N a cut.*

(i) Cosets. *Wherever the right side converges,*

$$\sum_{r=0}^{q-1}\big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,f\,\mathbb{I}_{\frac{b+r}{q}+\mathbb{Z}}\big)((N,\infty))=\big(\mathsf{T}_{\nu;t}[\mathbf{c}(\,\cdot\,/q)]\,f(\,\cdot\,/q)\,\mathbb{I}_{b+\mathbb{Z}}\big)((qN,\infty)).$$

(ii) Frequencies. *For $a\in\mathbb{R}$, wherever the Trailmoments on the left converge,*

$$\sum_{r=0}^{q-1}\big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,e^{2\pi i(a+\frac{r}{q})(x-b)}f\,\mathbb{I}_{b+\mathbb{Z}}\big)((N,\infty))=q\,\big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,e^{2\pi ia(x-b)}f\,\mathbb{I}_{b+q\mathbb{Z}}\big)((N,\infty)),$$

the right side summing over the lattice of step q.

A lattice is assembled from a sublattice in two ways: by cosets, the q cosets of $\mathbb{Z}$ in $q^{-1}\mathbb{Z}$; by frequencies, the q exponentials $e^{2\pi irx/q}$, $0\leq r<q$. Both statements are free in the index, the configuration, the coefficient function, the density, the coset and the cut. At one unit exponent with the coefficient function x the rescaled coefficient function of (i) multiplies the kernel by the scalar q^{ν}, $\mathsf{Z}_{\nu;(1)}(x/q)=q^{\nu}x^{-\nu}$; at index one, with the regularized values, the same assembly carries the further term $q\log q$, the finite part read in two coordinates (Lemma 1.9).

5.5 The Gamma read by Trailmoments

Proposition 5.13 (Heads and higher orders at lattice data)**.** *For real $b>-1$ and $|z|<1+b$,*

$$\log\Gamma(1+b-z)=\log\Gamma(1+b)+\sum_{i\geq 1}\frac{z^i}{i}\,(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((b,\infty)),$$

the index-one Trailmoment regularized; equivalently, for $j\in\mathbb{N}$,

$$(\log\Gamma)^{(j+1)}(1+b)=(-1)^{j+1}j!\,(\mathsf{T}_{j+1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((b,\infty)).$$

Further $\Gamma(-k+z)=\dfrac{(-1)^k\,\Gamma(1+z)\,\Gamma(1-z)}{z\,\Gamma(k+1-z)}$ for $k\in\mathbb{N}$. In this statement write $h=\prod_a\Gamma(1+b_a-z_a)/\prod_c\Gamma(1+b'_c-z'_c)$, a stacked Gamma whose constants b_a,b'_c lie in one translate $b_0+\mathbb{N}$ with $b_0>-1$, with $\sum_a z_a=\sum_c z'_c$, $|z_a|<1+b_a$ and $|z'_c|<1+b'_c$. Then

$$\log\frac{h}{h|_{z=0}}=\sum_c z'_c\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b_0+\mathbb{Z}})((b_0,b'_c])$$

$$- \sum_a z_a \, (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{b_0+\mathbb{Z}})((b_0, b_a])$$
$$+ \sum_{i\geq 2} \frac{1}{i} \Big(\sum_a z_a^i \, (\mathsf{T}_{i;(1)}[x]\, \mathbb{I}_{b_0+\mathbb{Z}})((b_a, \infty))$$
$$- \sum_c z_c'^i \, (\mathsf{T}_{i;(1)}[x]\, \mathbb{I}_{b_0+\mathbb{Z}})((b_c', \infty))\Big) :$$

the regularized index-one values cancel, the first order is a finite sum over a head of the lattice, rational when b_0 is, and the orders $i \geq 2$ are Trailmoments of the lattice input cut at the data. At integer data, $b_0 = 0$, the first order is rational and the higher orders are Trailmoments of the counting input.

Proposition 5.14 (Laurent data of the Gamma at its poles)**.** *Let $j \in \mathbb{N}$. For $|\varepsilon| < 1$, with $(-1)^j \varepsilon\, \Gamma(-j+\varepsilon) = \Gamma(1+\varepsilon)\,\Gamma(1-\varepsilon)/\Gamma(1+j-\varepsilon)$,*

$$\log\big((-1)^j \varepsilon\, \Gamma(-j+\varepsilon)\big) = -\log j! - (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((j,\infty))\, \varepsilon$$
$$- \sum_{k\geq 2} \frac{(\mathsf{T}_{k;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((j,\infty))}{k}\, \varepsilon^k + \sum_{k\geq 1} \frac{(\mathsf{T}_{2k;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))}{k}\, \varepsilon^{2k},$$

and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((j,\infty))$. Let h be holomorphic near $b = -j$, and let $k_1, \ldots, k_p$ be integers with $j_i = j - k_i \in \mathbb{N}$, so that every $\Gamma(b+k_i)$ has a pole at $-j$. Then

$$\operatorname*{Res}_{b=-j} \Big(h(b) \prod_{i=1}^{p} \Gamma(b+k_i) \Big) = (-1)^{j_1+\cdots+j_p} \sum_{l=0}^{p-1} \frac{h^{(l)}(-j)}{l!}\, [\varepsilon^{p-1-l}] \prod_{i=1}^{p} \big((-1)^{j_i} \varepsilon\, \Gamma(-j_i+\varepsilon)\big),$$

each factor of the product being the exponential of the display at j_i; at $p = 2$ and $k_1 = k_2 = 0$ it is $\frac{1}{(j!)^2}\big(h'(-j) + 2\,(\log\Gamma)'(j+1)\, h(-j)\big)$. Every Laurent coefficient of $\prod_i \Gamma(b+k_i)$ at $-j$ is $\prod_i (j_i!)^{-1}$ times a polynomial in Trailmoments of the counting input cut at the j_i and at 0.

Proposition 5.15 (Selecting an index inside a word)**.** *Let $w \geq 2$ be an integer and $h(z) = \prod_i \Gamma(1 + b_i - a_i z)^{n_i}$, a finite product with real $b_i > -1$, $a_i \in \mathbb{C}$ and $n_i \in \mathbb{Z}$. Its w-th multiplicative difference satisfies*

$$\prod_{j=0}^{w} h(jz)^{(-1)^{w-j}\binom{w}{j}} = 1 + (w-1)! \Big[\sum_i n_i a_i^w \, (\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{b_i+\mathbb{Z}})((b_i,\infty)) \Big] z^w + O(z^{w+1}).$$

At integer values $b_i = k_i \in \mathbb{N}$ with rational a_i, an integer $\ell \geq 1$ with every $\ell a_i \in \mathbb{Z}$ and $k_* = \max_i k_i$, the coefficient of z^w is $A\,(\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - B$ with $A = (w-1)! \sum_i n_i a_i^w$ and $B = (w-1)! \sum_i n_i a_i^w \, (\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,k_i])$, both cleared by $\ell^w d_{k_*}^w$. For the one factor $\Gamma(k+1-z)$, the coefficient of z^w in $\prod_{j=0}^{w} \Gamma(k+1-jz)^{(-1)^{w-j}\binom{w}{j}}$ is $(w-1)!\,(\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((k,\infty))$, and passing from k to $k+1$ multiplies this product by $\prod_{j=0}^{w}(k+1-jz)^{(-1)^{w-j}\binom{w}{j}}$. The difference acts multiplicatively on one word, not on the sums of a closing; along a ray from the origin whose law of $[1]$ is reflexive of order two, the coefficient of the coordinate is read without any sum, from the closings of the ray (Chapter 43).

The w-th difference annihilates every power of z below the w-th in the logarithm of a Gamma word, so it removes the regularized index-one values together with all Trailmoments of lower index and keeps exactly those of index w: from a word in the Gammas of the data it selects one Trailmoment index, with a rational head.

Lemma 5.16 (Two Gammas and their reflections). *The stacked Gamma* $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big] = \pi^2/\sin^2\pi a$ *has at every integer* m *the principal part* $(a-m)^{-2}$ *and no simple part, it equals* $\sum_{m\in\mathbb{Z}}(a+m)^{-2}$*, and on the line* $a=\frac12+iy$ *it equals* $4\pi^2/(e^{\pi y}+e^{-\pi y})^2$*.*

(i) Right closing. *For* $0<c<1$ *and* g *rational, holomorphic on* $\{\operatorname{Re} a\ge c\}$*, with* $g(a)=O(|a|^{-2})$*:* $\dfrac{1}{2\pi i}\displaystyle\int_{(c)}\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\ g(a)\,da = -\sum_{m\ge1} g'(m)$.

(ii) Left closing. *For* $0<X<1$*:* $\dfrac{1}{2\pi i}\displaystyle\int_{(1/2)}\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\ X^{-a}\,da = \frac{\log X}{X-1}$.

(iii) Fourier read. *For real* t*,*

$$\frac{1}{2\pi}\int_{\mathbb{R}}\Gamma_{\substack{4\\0}}\big[\tfrac12+iy,\ \tfrac12+iy,\ \tfrac12-iy,\ \tfrac12-iy\big]\ e^{ity}\,dy = \Gamma_{\substack{2\\0}}\big[1+\tfrac{it}{2\pi},\ 1-\tfrac{it}{2\pi}\big] = \frac{t}{e^{t/2}-e^{-t/2}}.$$

A closing whose poles coincide (Theorem 4.5) takes Laurent coefficients of its Gammas, and a balanced word near lattice data has heads and higher orders (Proposition 5.13): in both cases the Gammas contribute words in Trailmoments of lattice inputs, read at the cuts that the data set. The stacked Gamma of Lemma 5.16 is the simplest coincidence, two Gammas and their two reflections meeting at every integer: its double poles have no simple parts, so its right closing reads the derivative of the rest of the integrand at the positive integers, and on the line $\operatorname{Re} a=\frac12$ it is 2π times the density of the critical-line law (Chapter 36).

Theorem 5.17 (Reflected pairs and matched parity). *(i)* Cosets. *For* $0<b<1$ *and every integer* $n\ge1$*,*

$$(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((b-1,\infty)) + (-1)^n\,(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{-b+\mathbb{Z}})((-b,\infty)) = \frac{(-1)^{n-1}}{(n-1)!}\,\partial_b^{n-1}\big(\pi\cot\pi b\big),$$

the index-one values regularized. The left side is $\sum_{m\in b+\mathbb{Z}} m^{-n}$ *over the whole lattice, in symmetric order at* $n=1$*, and the right side is* π^n *times a polynomial of degree* n *in* $\cot\pi b$ *with rational coefficients.*

(ii) Frequencies. *For* $0<b\le1$ *and every integer* $n\ge2$*, the Trailmoment* $(\mathsf{T}_{n;(1)}[x]\,\sin\big(2\pi bx-\frac{\pi(n-1)}{2}\big)\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ *is* $\frac{(-1)^{n-1}(2\pi)^n}{2}$ *times the coefficient of* u^{n-1} *in* $e^{-bu}/(1-e^{-u})$*. The density is* $(-1)^{n/2}\cos(2\pi bx)$ *at even* n *and* $(-1)^{(n-1)/2}\sin(2\pi bx)$ *at odd* n*, and the value is* π^n *times a polynomial of degree* n *in* b *with rational coefficients.*

(iii) Periodic densities. *For integers* $0<r<q$ *and* $n\ge1$*,*

$$(\mathsf{T}_{n;(1)}[x]\,\big(\mathbb{I}_{r+q\mathbb{Z}}+(-1)^n\,\mathbb{I}_{-r+q\mathbb{Z}}\big))((0,\infty)) = \frac{(-1)^{n-1}}{q^n(n-1)!}\,\partial_b^{n-1}\big(\pi\cot\pi b\big)\Big|_{b=r/q},$$

at $n=1$ *with the points of the two lattices taken in increasing order.*

(iv) Folding. *At* $b=\frac12$ *the two lattices of (i) are one, and (i) reads*

$$\big(1+(-1)^n\big)(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty)) = \frac{(-1)^{n-1}}{(n-1)!}\,\partial_b^{n-1}(\pi\cot\pi b)\Big|_{b=1/2}:$$

at odd n *both sides vanish, and at even* $n=2j$*, with the lattice relation at the step* 2 *(Theorem 5.12(i)),*

$$(\mathsf{T}_{2j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \frac{-1}{2\,(2^{2j}-1)\,(2j-1)!}\ \partial_b^{2j-1}\big(\pi\cot\pi b\big)\Big|_{b=1/2}.$$

At $b = 1$, (ii) gives $(\mathsf{T}_{2j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (-1)^{j+1}\frac{(2\pi)^{2j}}{2}$ times the coefficient of u^{2j-1} in $1/(e^u-1)$, and the identity $0=0$ at odd n.

At an integer index n the kernel $\mathsf{Z}_{n;(1)}(x) = x^{-n}$ has the parity of n under $x \mapsto -x$. What closes, for every coset $b+\mathbb{Z}$ and every integer index, is the value of the whole coset, points of both signs: the reflection of the Gamma evaluates it, and at $n=2$ and $0<b<1$ it is $\Gamma_{\substack{4\\0}}\big[b,\ b,\ 1-b,\ 1-b\big]$ (Lemma 5.16). A Trailmoment reads the points beyond its cut only, so this relation gives it when the input, continued to the whole lattice, has the parity of the index: an even periodic density closes at even indices and an odd one at odd indices, by (iii), and on the coset $q\mathbb{Z}$ by (iv). On a lattice equal to its own reflection the pair of (i) folds onto one Trailmoment with the factor $1+(-1)^n$; at even n that Trailmoment closes, and at odd n the relation is the identity $0=0$ and carries no information. Statement (ii) is the same rule read on frequencies. The mismatched parities, the sine at an even index and the cosine at an odd index, are reached by neither statement on any coset or at any frequency; the counting input at an odd index is the cosine at frequency zero, and its Trailmoments are the ones that the positive words of Part VI carry.

5.6 Reflections of the index

Theorem 5.18 (Reflection through the poles of the Laplace transform)**.** *Let $\mathcal{F}$ be a function on $(0,\infty)$ that extends to a meromorphic function on $\mathbb{C}$ with no pole on $(0,\infty)$, such that*

(a) its poles other than $u=0$ are simple, with residue r_p at the pole p, and its pole at 0 has order $k_0 \geq 0$, with $k_0 = 0$ when there is none;

(b) there are radii $R_m \to \infty$ and a real A with $\sup_{|u|=R_m} |\mathcal{F}(u)| = O(R_m^A)$.

Write $\Sigma_m(\nu) = \sum_{0<|p|<R_m} r_p\,(-p)^{\nu-1}$, with $(-p)^{\nu-1} = \exp\big((\nu-1)\log(-p)\big)$ and $|\arg(-p)| < \pi$.

(i) On the strip. *If moreover $\mathcal{F}(u) = O(u^A)$ as $u \to +\infty$, then for $k_0 < \operatorname{Re}\nu < -A$ with $\nu \notin \mathbb{Z}$,*

$$\int_0^\infty u^{\nu-1}\mathcal{F}(u)\,du = \frac{\pi}{\sin\pi\nu}\lim_{m\to\infty}\Sigma_m(\nu).$$

(ii) Beyond the strip. *If $\mathcal{F} = \mathcal{F}_N$ is the Laplace transform of an input ϕ tame beyond the cut N, its poles lie in $\operatorname{Re} u \leq 0$, and for $\operatorname{Re}\nu < -A$ with $\nu \notin \{1,2,3,\dots\}$,*

$$(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty)) = \Gamma(1-\nu)\lim_{m\to\infty}\Sigma_m(\nu).$$

Reflection of the index replaces an input by the poles of its Laplace transform: each simple pole p becomes the point measure $r_p\,\delta_{-p}$, read by the kernel $\mathsf{Z}_{1-\nu;(1)}$. The relation is free in the input within (a) and (b), and pinned at one unit exponent with the coefficient function x; the mechanism is the Laplace side of the kernel $\mathsf{Z}_{\nu;(1)}$, where the index is the exponent of the scale and a full turn of the scale multiplies the integrand by $e^{2\pi i\nu}$. For the input $e^{-x}\,\mathbb{I}_{\mathbb{R}}$ cut at 0, whose Laplace transform $1/(1+u)$ has the one pole -1, statement (i) is the reflection of the Gamma (Proposition 1.3(v)); for a lattice input the strip of (i) is empty, and the reflection is a statement about the continued Trailmoment, which is (ii).

Theorem 5.19 (Reflection by inversion of the Laplace transform)**.** *Let ϕ and ψ be tame beyond the cut 0, with Laplace transforms $\mathcal{F}$ and $\mathcal{G}$, and suppose that for some $k \in \mathbb{R}$, $\lambda > 0$, $C \neq 0$ and $f_0, g_0 \in \mathbb{C}$,*

$$\mathcal{F}(u) + f_0 = C\,u^{-k}\Big(\mathcal{G}\big(\tfrac{\lambda}{u}\big) + g_0\Big) \qquad (u>0).$$

Then $\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty))$*, defined for large* $\operatorname{Re}\nu$ *by the Laplace side, continues meromorphically to* $\mathbb{C}$*,*

$$\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty)) = \int_{\sqrt{\lambda}}^{\infty} u^{\nu-1}\mathcal{F}(u)\,du + C\lambda^{\nu-k}\int_{\sqrt{\lambda}}^{\infty} u^{k-\nu-1}\mathcal{G}(u)\,du + \frac{C\,g_0\,\lambda^{(\nu-k)/2}}{\nu-k} - \frac{f_0\,\lambda^{\nu/2}}{\nu},$$

whose only possible poles are simple, at $\nu = k$ *and* $\nu = 0$*. The same holds with* ψ *in the place of* ϕ*, and*

$$\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty)) = C\,\lambda^{\nu-k}\,\Gamma(k-\nu)\,(\mathsf{T}_{k-\nu;(1)}[x]\,\psi)((0,\infty)).$$

The inversion $u \mapsto \lambda/u$ of the scale is the reflection of the Gramma (Proposition 2.3) on the Laplace side. The relation is free in the two inputs, which need not be lattices and whose Laplace transforms need not be meromorphic; it uses one property of the pair, that the inversion of the scale carries the Laplace transform of one to that of the other. The constants f_0 and g_0 are the multiples of δ_0 that the two inputs would carry at the origin, where the kernel $\mathsf{Z}_{\nu;(1)}$ is not defined, and for $k \neq 0$ they give the residues of the continuation, $-f_0$ at $\nu = 0$ and Cg_0 at $\nu = k$.

Theorem 5.20 (Reflection at the coefficient function $x+\mu$)**.** *Let* ϕ*,* ψ*,* $\mathcal{F}$*,* $\mathcal{G}$*,* k*,* λ*,* C*,* f_0 *and* g_0 *be as in Theorem 5.19, and let* $\mu > 0$*. For* $\operatorname{Re} s > \max(k,0)$ *with* $\int_{(0,\infty)}(x+\mu)^{-\operatorname{Re} s}\,|\phi|(dx) < \infty$*,*

$$\Gamma(s)\Big[(\mathsf{T}_{s;(1)}[x+\mu]\,\phi)((0,\infty)) + f_0\,\mu^{-s}\Big] = C\,\lambda^{s-k}\int_{(0,\infty)}\mathsf{G}_{k-s;(1,-1)}(y,\lambda\mu)\,\psi(dy) + C\,g_0\,\Gamma(s-k)\,\mu^{k-s}.$$

The integral on the right converges for every $s \in \mathbb{C}$ *and is entire in* s*.*

5.7 Joint sources

Theorem 5.21 (Closure with the source kept joint)**.** *(i)* $\mathsf{Z}_{\nu;t}(\mathbf{c})\,\mathsf{Z}_{\eta;s}(\mathbf{d}) = \mathsf{Z}_{(\nu,\eta);\,t\oplus s}(\mathbf{c}\oplus\mathbf{d})$*, the configuration* $t \oplus s$ *acting on the disjoint union of the two sets of scales.*

(ii) A finite expression built from Zammas with positive indices by products and by nested reads against nonnegative measures is one Trailmoment on a source: its configuration is the concatenation $\oplus$ *of the factors' configurations, and its input is the joint measure* $\phi(dy)\,\psi_y(dw)$ *of the source variables, every shared variable, conditional dependence and cut retained. The same holds for complex data where every integral converges absolutely.*

Proposition 5.22 (The joint Laplace transform of Dirichlet inputs)**.** *Let* $U = (U_Q)_{Q\in\mathcal{P}}$ *have independent blocks* $U_Q \sim \operatorname{Dir}_{\mathbf{m}|_Q}$ *on a partition* $\mathcal{P}$ *of the scales, let* $P_1,\dots,P_q$ *be finite sums* $P_i(u) = \sum_{\mathbf{a}}[u^{\mathbf{a}}]P_i\,u^{\mathbf{a}}$ *over exponents* $\mathbf{a} \in [0,\infty)^{\mathcal{E}}$*, polynomials when the exponents are integers, and let* t *be a configuration of* q *columns on* d *scales and* $\nu \in \mathbb{C}^d$*. Then* $\mathcal{F}(v) = \mathbb{E}_{U\sim\bigotimes_Q \operatorname{Dir}_{\mathbf{m}|_Q}}\big[\exp\big(-\sum_i v_i P_i(U)\big)\big]$ *is entire and*

$$\mathcal{F}(v) = \sum_{\mathbf{k}\in\mathbb{N}^q}\frac{(-v)^{\mathbf{k}}}{\mathbf{k}!}M(\mathbf{k}),$$

$$M(\mathbf{k}) = \sum_{\mathbf{a}}\big[u^{\mathbf{a}}\big]\big(P_1(u)^{k_1}\cdots P_q(u)^{k_q}\big)\prod_e \Gamma_1\begin{bmatrix} m_e + a_e \\ m_e\end{bmatrix}\prod_Q \Gamma_1\begin{bmatrix} m_Q \\ m_Q + a_Q\end{bmatrix}, \qquad a_Q = \sum_{e\in Q} a_e;$$

for integer exponents the stacked Gammas are $\prod_e (m_e)_{a_e} / \prod_Q (m_Q)_{a_Q}$. *If every* P_i *is positive on the interior of the source and* $\operatorname{Re}\nu_j > 0$, *then, wherever the read converges,*

$$\mathbb{E}_{U\sim\bigotimes_Q \mathrm{Dir}_{\mathbf{m}|_Q}}\big[\mathsf{Z}_{\nu;t}\big(P_1(U),\dots,P_q(U)\big)\big] = \frac{1}{\prod_j \Gamma(\nu_j)} \int_{(0,\infty)^d} x^{\nu-\mathbf{1}}\, \mathcal{F}\big(x^{t_1},\dots,x^{t_q}\big)\, dx,$$

the Laplace side of the Trailmoment $(\mathsf{T}_{\nu;t}[(P_i)_i] \otimes_Q \mathrm{Dir}_{\mathbf{m}|_Q})\big(\prod_Q \Delta_Q\big)$. *For real coefficients, two choices of the data with equal joint moment arrays* M *give the vectors* $(P_i(U))_i$ *equal laws, hence equal reads of every kernel.*

5.8 Trails

Definition 5.23 (Trail data)**.** Take d scales, an index $\boldsymbol{\nu} \in \mathbb{C}^d$, the *trail columns* $I_d = (\mathbf{e}_1,\dots,\mathbf{e}_d)$, and *constant columns* $(t_k)_{k\in C}$ with constant coefficients $\lambda_k > 0$ (a constant column of coefficient zero is absent). Put

$$t = I_d \oplus (t_k)_{k\in C}, \qquad \mathbf{c}(y) = (y_1,\dots,y_d,(\lambda_k)_{k\in C}), \qquad X_N = \{N < y_1 < \cdots < y_d\}.$$

The *trail* at the cut N of an input ϕ on $\mathbb{R}^d$, carried where every trail coefficient is positive, is the Trailmoment of Definition 5.1 with this coefficient function and cut region,

$$(\mathsf{T}_{\boldsymbol{\nu};\, I_d\oplus(t_k)_{k\in C}}[(y_1,\dots,y_d,(\lambda_k)_{k\in C})]\,\phi)(X_N);$$

with no constant columns it is $(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\,\phi)(X_N)$. For lattices $\mathbb{L}_1,\dots,\mathbb{L}_d \subset \mathbb{R}$ the input $\mathbb{I}_{\mathbb{L}_1\times\cdots\times\mathbb{L}_d}$ is counting measure on their product, of which the cut region selects the points in X_N. The input is *given by increments* $\psi_1,\dots,\psi_d$, measures on $(0,\infty)$, when it is the image ϕ_N of $\bigotimes_j \psi_j$ under $s \mapsto y$, $y_i = N + s_1 + \cdots + s_i$; their Laplace transforms are $L_j(U) = \int e^{-sU}\psi_j(ds)$. By transport (Theorem 5.9) the trail of such an input is the Trailmoment on the increment source $(0,\infty)^d$ with the input $\bigotimes_j \psi_j$ and the coefficient function $s \mapsto (N+s_1,\dots,N+s_1+\cdots+s_d,(\lambda_k)_{k\in C})$, over the whole source: there the cut N is carried by the coefficient function.

Theorem 5.24 (The trail by its increments)**.** *Let the input be given by increments, put* $t = I_d \oplus (t_k)_{k\in C}$, *and let* $U_i = x_i + \cdots + x_d$ *be the sums of the scales from the* i*-th on,* $U_{d+1} = 0$. *Wherever the right side converges absolutely, with* $x_i = U_i - U_{i+1}$,

$$\begin{aligned}&(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N)\\ &\qquad = \int_{U_1>\cdots>U_d>0} \prod_{i=1}^{d} \frac{(U_i-U_{i+1})^{\nu_i-1}}{\Gamma(\nu_i)}\, \exp\Big(-\sum_{k\in C}\lambda_k x^{t_k}\Big) e^{-NU_1} \prod_{j=1}^{d} L_j(U_j)\, dU.\end{aligned}$$

Its relations hold wherever the reads in them converge, the derivatives under the integral being dominated:

(i) Moving the cut, *with the increments fixed:*

$$\partial_N(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N) = -\sum_i \nu_i\,(\mathsf{T}_{\boldsymbol{\nu}+\mathbf{e}_i;t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N).$$

(ii) Ladder.

$$\partial_{\lambda_k}(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_h)_{h\in C})]\,\phi_N)(X_N) = -\Gamma_d\begin{bmatrix}\boldsymbol{\nu}+t_k\\ \boldsymbol{\nu}\end{bmatrix}\,(\mathsf{T}_{\boldsymbol{\nu}+t_k;t}[(y,(\lambda_h)_{h\in C})]\,\phi_N)(X_N).$$

(iii) Exchange, per scale.

$$(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N) = (\mathsf{T}_{\boldsymbol{\nu}+\mathbf{e}_i;t}[(y,(\lambda_k)_{k\in C})]\,y_i\phi_N)(X_N) + \frac{1}{\nu_i}\sum_{k\in C} t_{k,i}\lambda_k \Gamma_{\substack{d\\ d}}\begin{bmatrix}\boldsymbol{\nu}+t_k\\ \boldsymbol{\nu}\end{bmatrix}\,(\mathsf{T}_{\boldsymbol{\nu}+t_k;t}[(y,(\lambda_h)_{h\in C})]\,\phi_N)(X_N).$$

(iv) Composition. *If every constant column is supported on one scale, partition C into sets C_i with $\mathrm{supp}(t_k)\subseteq\{i\}$ for $k\in C_i$, assigning every zero column to one scale, and put $t^{(i)} = (1)\oplus(t_{k,i})_{k\in C_i}$. Then*

$$\mathsf{Z}_{\boldsymbol{\nu};t}(y,(\lambda_k)_{k\in C}) = \prod_i \mathsf{Z}_{\nu_i;t^{(i)}}(y_i,(\lambda_k)_{k\in C_i}),$$

and the trail is the iterated read over $s_1,\dots,s_d>0$, in that order, of $\prod_i \mathsf{Z}_{\nu_i;t^{(i)}}(N+s_1+\cdots+s_i,(\lambda_k)_{k\in C_i})$ against $\bigotimes_i\psi_i$: each one-scale Trailmoment has the next read in its input. With counting increments, every ψ_j the counting input on the positive integers, no constant columns and $N\in b+\mathbb{Z}$, this is the nest of one-scale Trailmoments at successive running cuts, with indices $\nu_1,\dots,\nu_d$ and lattice input $\mathbb{I}_{b+\mathbb{Z}}$, and the trail is $(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\,\mathbb{I}_{(b+\mathbb{Z})^d})(\{N<y_1<\cdots<y_d\})$.

5.9 Products of Trailmoments

Theorem 5.25 (The Struffle)**.** *(i) Let ϕ,ψ be inputs on $\mathbb{R}$ with any indices, configurations and coefficient functions. If the product below converges with $|\phi|$, $|\psi|$ and the moduli of the kernels, then*

$$\begin{aligned}&(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N,\infty))\,(\mathsf{T}_{\nu';t'}[\mathbf{c}']\,\psi)((N,\infty))\\ &\quad=\Big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\big[x\mapsto(\mathsf{T}_{\nu';t'}[\mathbf{c}']\,\psi)((x,\infty))\big]\phi\Big)((N,\infty))\\ &\qquad+\Big(\mathsf{T}_{\nu';t'}[\mathbf{c}']\,\big[x\mapsto(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((x,\infty))\big]\psi\Big)((N,\infty))\\ &\qquad+\big(\mathsf{T}_{(\nu,\nu');\,t\oplus t'}[\mathbf{c}\oplus\mathbf{c}']\,\phi\wedge\psi\big)((N,\infty)),\end{aligned}$$

where the diagonal input *is $\phi\wedge\psi=\sum_x\phi(\{x\})\psi(\{x\})\delta_x$, the sum over the points that both inputs charge, and the diagonal kernel is the Zamma of the concatenation $t\oplus t'$ (Theorem 5.21(i)).*

(ii) For inputs $\phi_1,\dots,\phi_r$ on $\mathbb{R}$ with kernel data $(\nu_i,t_i,\mathbf{c}_i)$, if $\prod_i(\mathsf{T}_{\nu_i;t_i}[\mathbf{c}_i]\,\phi_i)((N,\infty))$ converges with the total variations and the moduli of the kernels, it is the sum over the ordered partitions $(B_1,\dots,B_s)$ of $\{1,\dots,r\}$ into nonempty blocks of

$$\int_{N<x_1<\cdots<x_s}\prod_{j=1}^{s}\Big(\prod_{i\in B_j}\mathsf{Z}_{\nu_i;t_i}(\mathbf{c}_i(x_j))\Big)\Big(\bigwedge_{i\in B_j}\phi_i\Big)(dx_j),$$

with $\bigwedge_{i\in B}\phi_i=\sum_x\prod_{i\in B}\phi_i(\{x\})\delta_x$ for a block of at least two elements and ϕ_i for the block $\{i\}$.

The Struffle orders the points of the inputs along the line; the diagonal is there exactly when the inputs charge common points beyond the cut. It is free in the index, the configuration and

the coefficient function of both factors and in their inputs on the line; the cut region is pinned at one common cut. At one unit exponent with one coefficient function on both factors the diagonal kernel is $\mathsf{Z}_{\nu+\nu';(1)}$.

Theorem 5.26 (The Shruffle on a line)**.** *For a string $w = (w_1, \dots, w_n)$ of measurable functions on $(0,\infty)$ put $\operatorname{chain}(w) = \int_{x_1>\cdots>x_n>0} \prod_i w_i(x_i)\,dx$, defined when $\operatorname{chain}(|w|) < \infty$.*

(i) If $\operatorname{chain}(|w^{(1)}|), \dots, \operatorname{chain}(|w^{(r)}|)$ are finite, $\prod_{i=1}^r \operatorname{chain}(w^{(i)}) = \sum_\varsigma \operatorname{chain}(\varsigma)$, the sum over the $(n_1+\cdots+n_r)!/(n_1!\cdots n_r!)$ interleavings of the r strings, n_i the length of $w^{(i)}$.

(ii) Let a trail have its input given by nonnegative increments, with Laplace transforms $L_1, \dots, L_d$, and a cut $N \in \mathbb{R}$ such that the image of the first increment ψ_1 under $s \mapsto s+N$ is carried by $(0,\infty)$, as it is when $N \geq 0$; let it have no constant columns and positive integer indices $a_1, \dots, a_d$. Then, both sides finite or infinite together, with 1^a for a copies of the constant function 1, the trail is the chain integral of its string:

$$(\mathsf{T}_{\mathbf{a};I_d}[y]\,\phi_N)(X_N) = \operatorname{chain}\big(e^{-N\cdot}L_1,\ 1^{a_1-1},\ L_2,\ 1^{a_2-1},\ \dots,\ L_d,\ 1^{a_d-1}\big).$$

(iii) The product of finitely many such trails is the sum, over the interleavings of their strings, of trails of the same kind at the cut 0: every interleaving begins with a Laplace transform and is the string of the trail whose increments have the Laplace transforms of the interleaving in their order, each index being one more than the number of constant functions 1 that follow its Laplace transform, $e^{-N\cdot}L$ being the Laplace transform of the increment moved by N.

The Shruffle orders the scales on the Laplace side. It is free in the nonnegative increments, in the number of scales, and in the cut as long as the first point beyond it is positive; it is pinned at positive integer indices without constant columns, where $(U_i - U_{i+1})^{a-1}/\Gamma(a)$ is the volume of a chain of points. Where a nest of Trailmoments on the input side is a trail given by increments all equal to one measure at the cut 0, as for the counting input cut at 0, the Struffle and the Shruffle expand one product into the same members, and comparing the two expansions gives linear relations among the trails.

Theorem 5.27 (Trails at frequencies are iterated integrals)**.** *Let $d \geq 1$, $\boldsymbol{\nu} \in \{1,2,\dots\}^d$ and $\mathbf{a} \in \mathbb{R}^d$, put $\zeta_i = e^{2\pi i a_i}$, and let $\zeta_d \neq 1$ when $\nu_d = 1$. Put $c_i = \zeta_i\zeta_{i+1}\cdots\zeta_d$, $w = |\boldsymbol{\nu}|$, and let $(\ell_1, \dots, \ell_w)$ be the word $c_1^{-1}, 0^{\nu_1-1}, c_2^{-1}, 0^{\nu_2-1}, \dots, c_d^{-1}, 0^{\nu_d-1}$, 0^r standing for r letters 0. Then the heads of the trail over $\{0 < y_1 < \cdots < y_d \leq N\}$ of the lattice input $e^{2\pi i\langle\mathbf{a},y\rangle}\,\mathbb{I}_{\mathbb{Z}^d}$ at the indices $\boldsymbol{\nu}$ converge as $N \to \infty$, to the trail itself when $\nu_d \geq 2$, and with the trail read as this limit*

$$\big(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\ e^{2\pi i\langle\mathbf{a},y\rangle}\,\mathbb{I}_{\mathbb{Z}^d}\big)\big(\{0 < y_1 < \cdots < y_d\}\big) = (-1)^d \int_{0<t_1<\cdots<t_w<1} \prod_{j=1}^{w} \frac{dt_j}{t_j - \ell_j},$$

an absolutely convergent integral. For $|z| < 1$ the trail with the factor z^{y_d} in its density is $(-1)^d$ times the same integral over $0 < t_1 < \cdots < t_w < z$ along the segment from 0 to z.

On the Laplace side the input of the trail, given by its increments, has the letters $c_i z/(1-c_i z)$ in the variable $z = e^{-U}$, singular at the c_i^{-1}, and the index of each scale adds the letters 0 of the factor 1^{ν_i-1} of the Shruffle (Theorem 5.26(ii)): the theorem is the chain integral of that string, read in z, for inputs with complex densities.

Theorem 5.28 (The sum over the splittings of an index)**.** *Let $b \in \mathbb{R}$, let N be a cut, and let $m_0 > 0$ be the first point of $b + \mathbb{Z}$ beyond N. For an integer $w \geq 3$,*

$$\sum_{a=1}^{w-2} (\mathsf{T}_{(a,w-a);I_2}[y]\, \mathbb{I}_{(b+\mathbb{Z})^2})(\{N < y_1 < \cdots < y_2\}) = (\mathsf{T}_{w-1;(1)}[x]\, \eta\, \mathbb{I}_{b+\mathbb{Z}})((N,\infty)),$$

$$\eta(m) = (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m-m_0,\infty)) - (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{b+\mathbb{Z}})((m,\infty)) = \sum_{k\geq 1} \Big(\frac{1}{m-m_0+k} - \frac{1}{m+k}\Big),$$

the index-one values regularized. At every point m of the lattice beyond the cut,

$$\eta(m) - \frac{1}{m} = (\log\Gamma)'(m) - (\log\Gamma)'(m-m_0+1),$$

which is positive for $m_0 > 1$, zero for $m_0 = 1$ and negative for $m_0 < 1$. So the sum over the splittings is $(\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{b+\mathbb{Z}})((N,\infty))$ if and only if $m_0 = 1$, that is, for the counting input at a cut in $[0,1)$; it is larger for $m_0 > 1$ and smaller for $m_0 < 1$. For the counting input cut at an integer $c \geq 0$, $m_0 = c+1$ and $\eta(m) = (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m-c-1,m])$, a head of $c+1$ terms. For a lattice $b + q\mathbb{Z}$ of step $q > 0$ with a positive first point m_0 beyond the cut, the sum over $1 \leq a \leq w-2$ of the trails with the indices $(a, w-a)$ on its points is $(\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{b+q\mathbb{Z}})((N,\infty))$ if and only if $m_0 = q$.

Summing the trails on two scales over the splittings of an index telescopes in the inner variable, and what is left is one Trailmoment of index one less, with the density η. The index rises back by one exactly when $\eta(m) = \frac{1}{m}$ at every point, which is exactly when the first point beyond the cut is 1.

Theorem 5.29 (The Cremona of an increment)**.** *Let $L > 0$ be continuous on $(0,\infty)$ with $\int_x^\infty L < \infty$ for $x > 0$ and $\int_0^\infty L = \infty$, and let $\mathrm{tail}_L(x) = \int_x^\infty L(u)\, du$, the* tail function*, a decreasing bijection of $(0,\infty)$. For a string w whose entries are $L(x)\, dx$ or dx, let $\mathrm{chain}_L(w)$ be its chain integral over $x_1 > \cdots > x_n > 0$, largest variable first, and w^* the reversed string with $L(x)\, dx$ and dx swapped. Then $\mathrm{tail}_L \circ \mathrm{tail}_L = \mathrm{id}$ if and only if $L(\mathrm{tail}_L(x))\, L(x) = 1$ for every x, and in that case $\mathrm{chain}_L(w) = \mathrm{chain}_L(w^*)$ for every string, finite or infinite together.*

The Cremona uses one property of the increment: its tail function tail_L is an involution of the scale, so that the substitution $y = \mathrm{tail}_L(x)$ carries dx to $L(y)\, dy$ as well as $L(x)\, dx$ to dy. For the counting increments, $L(u) = 1/(e^u - 1)$ and the involution is the curve $e^{-x} + e^{-y} = 1$.

5.10 Convolutions of inputs

Definition 5.30 (The two convolutions)**.** Let ϕ and ψ be inputs carried by $[0,\infty)$, and $\phi \otimes \psi$ their product on $[0,\infty)^2$. The *additive convolution* $\phi \boxplus \psi$ is the image of $\phi \otimes \psi$ under $(x,y) \mapsto x + y$. When both inputs are carried by $(0,\infty)$, the *multiplicative convolution* $\phi \boxtimes \psi$ is the image of $\phi \otimes \psi$ under $(x,y) \mapsto xy$. Both are images in the sense of Definition 5.4: for inputs with complex densities they are used only in integrals that converge absolutely on the product $\phi \otimes \psi$. The multiplicative convolution of two σ-finite measures need not be σ-finite; it is used only where an integral of Theorem 5.31(ii) below converges absolutely, and there it is σ-finite, its kernel being positive and finite on $(0,\infty)$. For lattice inputs with densities, $(f\, \mathbb{I}_{\mathbb{Z}}) \boxplus (g\, \mathbb{I}_{\mathbb{Z}})$ and $(f\, \mathbb{I}_{\mathbb{Z}}) \boxtimes (g\, \mathbb{I}_{\mathbb{Z}})$ denote the convolutions of the parts beyond the cut 0: for densities f and g on the positive integers, the first has the density $\sum_{j+k=n} f(j)g(k)$ at n and the second the density $\sum_{jk=n} f(j)g(k)$; likewise for lattice inputs on other cosets. For an input carried by $[0,\infty)$ write $\mathcal{F}_\phi(u) = \int_{[0,\infty)} e^{-ux}\, \phi(dx)$, the kernel being 1 at the point $x = 0$: it is the Laplace transform beyond every cut $N < 0$, and for an input carried by $(0,\infty)$ the Laplace transform beyond the cut 0.

Theorem 5.31 (Two convolutions)**.** *Let ϕ on Y_1 and ψ on Y_2 be inputs with coefficient functions c and c' and cut regions Y_1' and Y_2', and write $c+c'$ and $c\,c'$ for the functions $(y_1, y_2) \mapsto c(y_1)+c'(y_2)$ and $(y_1, y_2) \mapsto c(y_1)\,c'(y_2)$ on $Y_1 \times Y_2$.*

(i) The kernel $\mathsf{Z}_{1;(1,0)}(1, \cdot)$ on a sum. *If the two Trailmoments on the right converge,*

$$\big(\mathsf{T}_{1;(1,0)}[(1, c+c')]\ \phi \otimes \psi\big)(Y_1' \times Y_2') = (\mathsf{T}_{1;(1,0)}[(1, c)]\,\phi)(Y_1')\ (\mathsf{T}_{1;(1,0)}[(1, c')]\,\psi)(Y_2').$$

For inputs carried by $[0, \infty)$ and $u \in \mathbb{C}$ such that $e^{-x \operatorname{Re} u}$ is integrable for $|\phi|$ and for $|\psi|$, $\mathcal{F}_{\phi \boxplus \psi}(u) = \mathcal{F}_\phi(u)\,\mathcal{F}_\psi(u)$.

(ii) The kernel $\mathsf{Z}_{\nu;I_\mathcal{E}}$ on a product. *For coefficient functions $\mathbf{c}$ and $\mathbf{c}'$ with values in $(0, \infty)^\mathcal{E}$, $\mathbf{c}\,\mathbf{c}'$ their coordinatewise product and $\nu \in \mathbb{C}^\mathcal{E}$, if the two Trailmoments on the right converge,*

$$\big(\mathsf{T}_{\nu;I_\mathcal{E}}[\mathbf{c}\,\mathbf{c}']\ \phi \otimes \psi\big)(Y_1' \times Y_2') = (\mathsf{T}_{\nu;I_\mathcal{E}}[\mathbf{c}]\,\phi)(Y_1')\ (\mathsf{T}_{\nu;I_\mathcal{E}}[\mathbf{c}']\,\psi)(Y_2').$$

For inputs carried by $(0, \infty)$,

$$(\mathsf{T}_{\nu;(1)}[x]\,\phi \boxtimes \psi)((0,\infty)) = (\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty))\ (\mathsf{T}_{\nu;(1)}[x]\,\psi)((0,\infty)),$$
$$\mathcal{F}_{\phi \boxtimes \psi}(u) = \int_{(0,\infty)} \mathcal{F}_\psi(ux)\,\phi(dx),$$

the first when its right side converges and the second when $\iint e^{-xy \operatorname{Re} u}\,|\phi|(dx)\,|\psi|(dy) < \infty$.

(iii) Converses. *Let g be a continuous complex-valued function on $(0, \infty)^\mathcal{E}$, not identically zero, with $g(s+s') = g(s)\,g(s')$ for all $s, s' \in (0, \infty)^\mathcal{E}$. Then $g(s) = e^{-\langle u, s\rangle}$ for one $u \in \mathbb{C}^\mathcal{E}$. Let h be a continuous complex-valued function on $(0, \infty)^\mathcal{E}$, not identically zero, with $h(x\,x') = h(x)\,h(x')$ for the coordinatewise product. Then $h = \mathsf{Z}_{\nu;I_\mathcal{E}}$ for one $\nu \in \mathbb{C}^\mathcal{E}$. The two hypotheses are the product relations of (i) and of (ii), for a kernel g or h in the place of the kernel there, on every pair of Dirac measures.*

The two convolutions add and multiply the points of two inputs, and each has one kernel that turns it into a product: on sums the kernel $\mathsf{Z}_{1;(1,0)}(1, \cdot)$ of a term at exponent zero beside a unit term, which is e^{-c}; on products the kernel $\mathsf{Z}_{\nu;I_\mathcal{E}}$ of the unit configuration, the power $\mathbf{c} \mapsto \mathbf{c}^{-\nu}$. Both statements are free in the two inputs, the two coefficient functions and the two cut regions, and each is pinned in its kernel; (iii) is the converse of that pin.

Chapter 6

Rational words and live inputs

A rational word acting on a Trailmoment through its input is moved, by partial fractions and the shift of the cut, onto Trailmoments of shifted inputs at the unit configuration; the pole columns of the word then separate what is a Trailmoment from what is a finite difference. An input is *live* when the values of its density are themselves read from the serif family, a Breta or a Gramma at a shifting slot, and the relations of that member give the input a recurrence of finite order. Every identity for rational words and live inputs is a finite identity first, at finite cuts; its infinite form uses the displayed convergence, or the common finite upper cut of the closing it comes from.

Definition 6.1 (Rational words and their transport)**.** A *rational word* is a function

$$R(x) = \sum_{k=0}^{k_{\max}} \sum_{j=1}^{J_k} c_{k,j}(x+k)^{-j},$$

kept with its partial-fraction presentation. For the data $(\nu, t, \mathbf{c}, \phi)$ of a Trailmoment on a line and the cut region (N, ∞), the rational word acts through the input,

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,(R\phi))((N,\infty)) = \int_{(N,\infty)} R(x)\,\mathsf{Z}_{\nu;t}(\mathbf{c}(x))\,\phi(dx).$$

Write $(\mathrm{Sh}^h f)(m) = f(m+h)$, $\Delta H(m) = H(m+1) - H(m)$, $\tau_k(x) = x + k$ and $\phi_k = (\tau_k)_*\phi$. On the counting input at index zero, where $\mathsf{Z}_{0;(1)}(x) = 1$, this is $(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} R(m)f(m)$ for an integer cut $N \geq 0$.

Theorem 6.2 (Shifted inputs)**.** *With* $g_k(y) = \mathsf{Z}_{\nu;t}(\mathbf{c}(y-k))f(y-k)$,

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,(Rf\phi))((N,\infty)) = \sum_{k,j} c_{k,j}\,(\mathsf{T}_{j;(1)}[x]\,(g_k\phi_k))((N+k,\infty)).$$

On the counting input at integer cuts $M > N \geq 0$*,*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,M]) = \sum_{k,j} c_{k,j}\,(\mathsf{T}_{j;(1)}[x]\,(\mathrm{Sh}^{-k} f)\,\mathbb{I}_{\mathbb{Z}})((N+k,M+k]),$$

and when the Trailmoments converge individually

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{k,j} c_{k,j}\,(\mathsf{T}_{j;(1)}[x]\,(\mathrm{Sh}^{-k} f)\,\mathbb{I}_{\mathbb{Z}})((N+k,\infty)).$$

When only the combination converges, the common finite upper cut is kept and the limit is taken in the combination. For $f = \mathbf{1}$, $\sum_k c_{k,1} = 0$ and an integer $m_0 \geq 1$, with the index-one Trailmoments regularized,

$$\begin{aligned}(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((m_0-1,\infty)) &= \sum_{k,s} c_{k,s}\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m_0-1+k,\infty)) \\ &= \sum_{s\geq 2}\sum_{k} c_{k,s}\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m_0-1+k,\infty)) - \sum_{k} c_{k,1}\, (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0, m_0-1+k]) \\ &= \int_0^\infty \Big[\sum_{k,s} \frac{c_{k,s}}{\Gamma(s)} u^{s-1} e^{-ku}\Big] \frac{e^{-(m_0-1)u}}{e^u-1}\, du.\end{aligned}$$

Theorem 6.3 (Common cut and heads)**.** *For a common cut $N' \geq N + k_{\max}$,*

$$\begin{aligned}(\mathsf{T}_{j;(1)}[x]\, (\mathrm{Sh}^{-k} f)\mathbb{I}_{\mathbb{Z}})((N+k,\infty)) &= (\mathsf{T}_{j;(1)}[x]\, (\mathrm{Sh}^{-k} f)\mathbb{I}_{\mathbb{Z}})((N',\infty)) \\ &\quad + (\mathsf{T}_{j;(1)}[x]\, (\mathrm{Sh}^{-k} f)\mathbb{I}_{\mathbb{Z}})((N+k,N']),\end{aligned}$$

the head being $\sum_{N+k<\ell\leq N'} \ell^{-j} f(\ell-k)$ (Proposition 5.3). For $R = -g'$ with $g(a) = \sum_{h,s} c_{h,s}(a+h)^{-s} = O(a^{-2})$ and $C_s = \sum_h c_{h,s}$, under individual convergence,

$$\begin{aligned}(\mathsf{T}_{0;(1)}[x]\, (-g')f\, \mathbb{I}_{\mathbb{Z}})((N,\infty)) &= \sum_{s} s\, C_s\, (\mathsf{T}_{s+1;(1)}[x]\, f\, \mathbb{I}_{\mathbb{Z}})((N',\infty)) \\ &\quad + \sum_{h,s} s\, c_{h,s}\, (\mathsf{T}_{s+1;(1)}[x]\, (\mathrm{Sh}^{-h} f - f)\, \mathbb{I}_{\mathbb{Z}})((N',\infty)) \\ &\quad + \sum_{h,s} s\, c_{h,s}\, (\mathsf{T}_{s+1;(1)}[x]\, (\mathrm{Sh}^{-h} f)\, \mathbb{I}_{\mathbb{Z}})((N+h,N']).\end{aligned}$$

Theorem 6.4 (Pole columns are a rational difference quotient)**.** *Let $R(t) = \sum_{k=0}^{k_{\max}} \sum_{s=1}^{w} c_{k,s}(t+k)^{-s}$ with coefficients in a subfield $\mathbb{K}$ of $\mathbb{C}$, $C_s = \sum_k c_{k,s}$, and $H(t) = \sum_{k,s} c_{k,s} \sum_{j=0}^{k-1} (t+j)^{-s}$, a rational function with coefficients in $\mathbb{K}$. Then*

$$R(t) = \sum_{s=1}^{w} C_s t^{-s} + \Delta H(t),$$

and $R(t) = O(t^{-2})$ exactly when $C_1 = 0$. Conversely a difference ΔH of a proper rational function with poles on the nonpositive integer lattice has every column sum zero; modulo such differences, the rational words of maximum order w have the basis $t^{-1}, \ldots, t^{-w}$, and those that are $O(t^{-2})$ the basis $t^{-2}, \ldots, t^{-w}$. If the poles fall in several cosets of $\mathbb{Z}$, each coset has its own columns.

Theorem 6.5 (Summation by parts with a live input)**.** *Let H be a function on the integers above N and f a density on them.*

(i) $\Delta(Hf)(m) = \Delta H(m) f(m) + H(m+1)\Delta f(m)$, *so at finite cuts $M > N$*

$$\sum_{N<m\leq M} \Delta H(m) f(m) = H(M+1)f(M+1) - H(N+1)f(N+1) - \sum_{N<m\leq M} H(m+1)\Delta f(m).$$

(ii) If the upper boundary vanishes and the series converges, $\sum_{m>N} \Delta H(m) f(m) = -H(N+1)f(N+1) - \sum_{m>N} H(m+1)\Delta f(m)$.

(iii) For a Trailmoment read as a function of its cut, $\Delta\,(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((m,\infty)) = -(m+1)^{-p}g(m+1)$ *(Proposition 5.3), and hence*

$$\sum_{m>N}\Delta H(m)\,(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((m,\infty)) \\ = -H(N+1)\,(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((N+1,\infty)) + (\mathsf{T}_{p;(1)}[x]\,Hg\,\mathbb{I}_{\mathbb{Z}})((N+1,\infty)),$$

with one fewer Trailmoment in the input.

(iv) With R, C_s, H *as in Theorem 6.4 and the index-one Trailmoment regularized (Definition 5.2),*

$$\lim_{M\to\infty}\Big((\mathsf{T}_{0;(1)}[x]\,R\,\mathbb{I}_{\mathbb{Z}})((N,N+M]) - C_1\log M\Big) = \sum_{s=1}^{w} C_s\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) - H(N+1);$$

when $C_1 = 0$*, that is when* $R = O(t^{-2})$*, the left side is* $(\mathsf{T}_{0;(1)}[x]\,R\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$ *and the sum starts at* $s = 2$*. For a general density* f*, when the terms converge and the upper boundary vanishes,*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{s=1}^{w} C_s\,(\mathsf{T}_{s;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) \\ - H(N+1)f(N+1) - \sum_{m>N} H(m+1)\Delta f(m).$$

Theorem 6.6 (Translation and the terminating descent)**.** *For the counting input, each inner Trailmoment below is read at its displayed running cut.*

(i) $(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((y-k,\infty)) = (\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((y,\infty)) + (\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((y-k,y])$*, the head being* $\sum_{a=0}^{k-1}(y-a)^{-p}g(y-a)$ *(Proposition 5.3).*

(ii)

$$(\mathsf{T}_{j;(1)}[x]\,\big[x\mapsto(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((x-k,\infty))\big]\,\mathbb{I}_{\mathbb{Z}})((N+k,\infty)) \\ = (\mathsf{T}_{j;(1)}[x]\,\big[x\mapsto(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((x,\infty))\big]\,\mathbb{I}_{\mathbb{Z}})((N+k,\infty)) \\ + \sum_{a=0}^{k-1}(\mathsf{T}_{p;(1)}[x]\,(x+a)^{-j}g\,\mathbb{I}_{\mathbb{Z}})((N+k-a,\infty)).$$

(iii) For positive integers p, j*, each product* $z^{-p}(z+a)^{-j}$ *has finite rational partial fractions with poles at* 0 *and* $-a$ *only. The recursion therefore terminates on every finite nest of Trailmoments of the counting input, each taking the next as a function of its cut: it returns finitely many such nests, finite shifted copies of* g *and their heads, keeps the selected cuts, lowers at each correction the number of Trailmoments nested in the remaining input, and uses only rational operations on the pole data.*

(iv) If $R(-L-t) = -R(t)$*, then* $\sum_k c_{k,j} = 0$ *for every even* j*, and for a density* f *that is a nest of* d *Trailmoments of the counting input, at a common cut* C*,*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{j}\Big(\sum_{k} c_{k,j}\Big)(\mathsf{T}_{j;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((C,\infty)) + E_d,$$

with E_d a sum of terms with at most d nested Trailmoments, so the terms with $d+1$ nested Trailmoments carry only odd outer indices, starting at three when $\sum_k c_{k,1} = 0$. For $f = \mathbf{1}$, an integer $m_0 \geq 1$ and $R(-k_{\max} - t) = -R(t)$, $c_{k_{\max}-k,s} = -(-1)^s c_{k,s}$, and at every even index s the Trailmoments combine into the finite sum

$$\sum_k c_{k,s}\Big[(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m_0 - 1 + k, \infty)) - (\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m_0 - 1, \infty))\Big] \\ = -\sum_k c_{k,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m_0 - 1, m_0 - 1 + k]).$$

Proposition 6.7 (Zamma kernels and general measures)**.** *For counting measure the product rule applies to the kernel times the density, $g(m) = \mathsf{Z}_{\nu;t}(\mathbf{c}(m))f(m)$, with $\Delta g(m) = \mathsf{Z}_{\nu;t}(\mathbf{c}(m+1))\Delta f(m) + f(m)\Delta[\mathsf{Z}_{\nu;t}(\mathbf{c}(m))]$. For a general measure μ restricted to $(N, M]$, $\int \Delta H\, g\, d\mu_{N,M} = \int H\, d\big((\tau_1)_*(g\mu_{N,M}) - g\mu_{N,M}\big)$.*

6.1 Live inputs

Theorem 6.8 (A Breta input with a recurrence of finite order)**.** *For $\operatorname{Re} a, \operatorname{Re} b > 0$, real $q_i < 1$ with $q_i \neq 0$, and $s_i \in \mathbb{C}$, let*

$$f(m) = \int_0^1 x^{m+a-1}(1-x)^{b-1} \prod_{i=1}^{r} (1 - q_i x)^{-s_i}\, dx \qquad (m \geq 0).$$

For $\operatorname{Re} s_i > 0$ this is a Breta in which m sits in the shape index: on the bunch $\{1, 2\}$ with the shape $u = (x, 1-x)$, the kernel terms ρu_1 and ρu_2 with unit coefficients and the letters $P_i(\rho, u) = \rho\big((1-q_i)u_1 + u_2\big)$, Proposition 3.6 at the scale index $1 + |\mathbf{s}|$ gives

$$f(m) = \Gamma(1 + |\mathbf{s}|)\, \mathsf{B}^{\mathcal{P}}_{((m+a,\,b),\,\mathbf{s}),\,(1+|\mathbf{s}|,\,\mathbf{s});\,(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g} \oplus \mathbf{C}).$$

Then $f(m) = O(m^{-\operatorname{Re} b})$, and with $Q = \prod_i (1 - q_i x)$, $Q_i = Q/(1 - q_i x)$ and

$$B_m(x) = \big[(m+a)(1-x) - bx\big]Q(x) + x(1-x)\sum_i (s_i - 1) q_i Q_i(x) = \sum_{\ell=0}^{r+1} B_\ell(m) x^\ell,$$

the density obeys $\sum_{\ell=0}^{r+1} B_\ell(m) f(m+\ell) = 0$, with $B_0(m) = m + a$ and $B_{r+1}(m) = (-1)^{r+1}(\prod_i q_i)(m + a + b + r - \sum_i s_i)$. If $a + b + r - \sum_i s_i \notin -\mathbb{N}$, both extreme coefficients are nonzero for every $m \geq 0$, and all shifts reduce to $(f(m), \ldots, f(m+r))$ with coefficients rational in m.

Theorem 6.9 (A fixed set of input-dependent evaluations)**.** *With f, a, b, r, q_i and s_i as in Theorem 6.8, write in this statement $Q(x) = \prod_i (1 - q_i x)$, $C(x) = (1-x)Q(x)$, $\eta(x) = x^{a-1}(1-x)^{b-1}\prod_i (1 - q_i x)^{-s_i}$ and*

$$B(x) = \big[a(1-x) - bx\big]Q(x) + x(1-x)\sum_i (s_i - 1)\, q_i\, \frac{Q(x)}{1 - q_i x},$$

so that $(xC\eta)' = B\eta$; let B_ℓ, C_ℓ and Q_ℓ be the coefficients of x^ℓ.

(i) For each maximum index A, every shift reduces to $f(N+1),\dots,f(N+r+1)$ and the Trailmoments $(\mathsf{T}_{p;(1)}[x+k]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} f(m)(m+k)^{-p}$, $1\le p\le A$, $0\le k\le r$, together with $(\mathsf{T}_{p;(1)}[x+a]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$ when a is an integer at least $r+1$.

(ii) For $k\ge r+1$ and $p\ge 2$,

$$\sum_{\ell=0}^{r+1}(B_\ell - kC_\ell)\,(\mathsf{T}_{p;(1)}[x+k-\ell]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) + \sum_{\ell=0}^{r+1} C_\ell\,(\mathsf{T}_{p-1;(1)}[x+k-\ell]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$$
$$= \sum_{\ell=0}^{r+1}\sum_{v=1}^{\ell}\Big[\frac{B_\ell - kC_\ell}{(N+v+k-\ell)^p} + \frac{C_\ell}{(N+v+k-\ell)^{p-1}}\Big] f(N+v),$$

and at $p=1$,

$$\sum_{\ell=0}^{r+1}(B_\ell - kC_\ell)\,(\mathsf{T}_{1;(1)}[x+k-\ell]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$$
$$= \sum_{\ell=0}^{r+1}\sum_{v=1}^{\ell}\frac{B_\ell - kC_\ell}{N+v+k-\ell} f(N+v) - \sum_{\ell=0}^{r} Q_\ell f(N+1+\ell).$$

(iii) Cuts move by $(\mathsf{T}_{p;(1)}[x+k]\,f\,\mathbb{I}_{\mathbb{Z}})((N+1,\infty)) = (\mathsf{T}_{p;(1)}[x+k]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) - f(N+1)/(N+1+k)^p$ (Proposition 5.3).

(iv) For rational data, if c_* clears the coefficients of B, C and Q, then $d^A_{N+J}\prod_{r+1\le k\le J,\ k\ne a}|c_*(a-k)|^A$ clears all shifts through J and indices through A.

Theorem 6.10 (The row of a rational function against an input with a recurrence)**.** *Let f be a density on the integers $m>N$, $N\in\mathbb{Z}$, with $\sum_{m>N}|f(m)|(1+|m|)^d<\infty$, that satisfies $\sum_{i=0}^{\tau} p_i(m)\,f(m+i)=0$ for every integer $m>N$, with $\tau\ge 1$ and $p_i\in\mathbb{Z}[m]$ of degree at most d. Let $u\ge 1$ be an integer and $h_*\in\frac{1}{u}\mathbb{Z}$ with $N+1+h_*>0$. In this statement write, for integers $s\ge 1$, $r\ge 0$ and $h\in h_*+\mathbb{N}$,*

$$T_s(h) = (\mathsf{T}_{s;(1)}[x+h]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N}\frac{f(m)}{(m+h)^s},$$
$$M_r = (\mathsf{T}_{-r;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} m^r f(m).$$

(i) For a rational function $R(m) = P(m) + \sum_{h,s} c_{h,s}(m+h)^{-s}$, P a polynomial of degree at most d and finitely many $h\in h_*+\mathbb{N}$, $\sum_{m>N} R(m)f(m) = \sum_r([m^r]P)\,M_r + \sum_{h,s} c_{h,s}\,T_s(h)$: one coordinate for each index and shift.

(ii) Let $J\ge\tau$ with $p_0(-h_*-j)\ne 0$ for $\tau\le j\le J$. Then for $s\ge 1$ and $\tau\le j\le J$, $T_s(h_*+j)$ is a rational combination of the $T_{s'}(h_*+i)$, $1\le s'\le s$, $0\le i<\tau$, of the moments M_r, $0\le r\le d-1$, and of the values $f(N+1),\dots,f(N+\tau)$.

(iii) For every $w\ge 1$ the coefficients of (ii) of all the $T_s(h_*+j)$, $1\le s\le w$, $\tau\le j\le J$, are cleared by

$$d^w_{u(N+h_*+J)}\prod_{j=\tau}^{J}\big|u^d p_0(-h_*-j)\big|^w.$$

A rational function with poles in several cosets of $\mathbb{Z}$ takes one such reduction for each coset.

Theorem 6.9 is the case of the Breta input of Theorem 6.8, read through its differential equation; here the input is any density with a recurrence, the live inputs of the closings along a ray among them (Theorem 43.1). The clearing has two parts: the heads released by the moves of the cut, cleared by a power of d_n on the scale u of the coset, and the values of the first coefficient of the recurrence at the shifts used, raised to the power of the index, which along a ray is the frame of the recurrence (Proposition 38.11).

Theorem 6.11 (A Gramma input with a recurrence of finite order)**.** *Let a one-scale configuration t have positive integer exponents with greatest exponent L, the term at exponent L present. For $\lambda, c > 0$ put $g(m) = \mathsf{G}_{\lambda+m;t}(\boldsymbol{\xi})/\Gamma(c+m)$. Then*

$$(m+\lambda)g(m) = \sum_k t_k \xi_k (m+c)_{t_k}\, g(m+t_k),$$

a shift relation of order L, the exchange relation of the Gramma (Theorem 2.4(ii)) divided by $\Gamma(c+m)$; for $L \geq 2$, $\log g(m) \leq -(1-1/L)m\log m + O(m)$, so g decays faster than every inverse power.

Part II

Tables and families

Chapter 7

Tables

7.1 Characters, tables, lists

Definition 7.1 (Characters, tables, lists)**.** The *characters* are a finite set $\mathcal{E}$, whose elements are the *marks*; each mark e carries a positive variable α_e, and its name carries nothing. A *table* on $\mathcal{E}$ is a nonempty family $\mathcal{B}$ of r-element subsets of $\mathcal{E}$, the *bases*, with the *exchange property*: for bases T, T' and $a \in T \setminus T'$ there is $b \in T' \setminus T$ with $(T \setminus \{a\}) \cup \{b\}$ a basis. Each basis carries a weight $\theta_T > 0$; r is the *rank* and $r^* = n - r$ the *corank*. A *list* is a table whose weights are all 1; it is specified by its characters and its bases, $(\mathcal{E}, \mathcal{B})$, and by nothing else; for a set S and $0 \le k \le |S|$, $\binom{S}{k}$ is the list on S whose bases are the k-subsets of S. A mark in no basis is a *loop*, a mark in every basis a *coloop*. The *dual table* has the bases $\mathcal{E} \setminus T$ with the weights θ_T. The *Foreman polynomials* of a table are its basis and cobasis polynomials,

$$K(\alpha) = \sum_{T \in \mathcal{B}} \theta_T \, \alpha^T, \qquad \Psi(\alpha) = \sum_{T \in \mathcal{B}} \theta_T \, \alpha^{\mathcal{E} \setminus T} \qquad (\alpha \in \mathbb{R}^{\mathcal{E}}_{>0});$$

$K(\mathbf{1}) = \Psi(\mathbf{1}) = \sum_T \theta_T$ is the total weight. K and Ψ are the two letters of the table (Section 3.2): they enter a family word through powers (Chapter 8). The *table's gauge* is $\alpha = \mathbf{1}$.

A table carries two descriptors of one object, what is there (the bases, read by K) and what is not (their complements, read by Ψ), and the marks on which both sit; any two of these determine the third, and the theory keeps both polynomials at every step.

Proposition 7.2 (One function read twice)**.** *For every table, $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1})$, and the Foreman polynomials of the dual table are $K^* = \Psi$, $\Psi^* = K$. Inverting the gauge swaps K and Ψ up to the factor $\alpha^{\mathcal{E}}$, and up to scale the gauges fixed by $\alpha \mapsto \alpha^{-1}$ are $\alpha \propto \mathbf{1}$.*

Definition 7.3 (Deletion and contraction)**.** For a mark e and $R = \mathcal{E} \setminus \{e\}$, the *deletion* has the bases $T \not\ni e$ and the *contraction* the bases $T \setminus \{e\}$, $T \ni e$, both on R with the inherited weights; the deletion is a table when e is not a coloop, and the contraction when e is not a loop. With $K_{\setminus e}, \Psi_{\setminus e}$ and $K_{/e}, \Psi_{/e}$ the sums over these bases, a sum without terms being 0,

$$K = K_{\setminus e} + \alpha_e K_{/e}, \qquad \Psi = \alpha_e \Psi_{\setminus e} + \Psi_{/e}.$$

7.2 Circuits and molecules

Definition 7.4 (Independence, circuits, linking, molecules)**.** A set $S \subseteq \mathcal{E}$ is *independent* in a table if it lies in a basis; its *rank* $r(S)$ is the largest size of an independent subset. A *circuit* is a minimal

set that is not independent, and a *cocircuit* is a minimal set that meets every basis. Two marks are *linked* if they lie in a common circuit of some table of a family on $\mathcal{E}$, a finite sequence of tables on $\mathcal{E}$ (Chapter 8). The *molecules* are the smallest nonempty sets of marks that contain, with each mark, every mark linked to it; a molecule with one mark is a *unit*, and the set of molecules is the *polycule*. A family whose marks form more than one molecule is a *compound*.

Definition 7.5 (Minors, restriction, contraction)**.** For disjoint $A, D \subseteq \mathcal{E}$ such that some basis contains A and avoids D, the *minor* contracting A and deleting D is the table on $\mathcal{E} \setminus (A \cup D)$ with the bases $T \setminus A$, $T \supseteq A$, $T \cap D = \varnothing$, and the weights θ_T; deletion and contraction of one mark are the minors with $D = \{e\}$ and with $A = \{e\}$. For $X \subseteq \mathcal{E}$ the *restriction* $\mathcal{B}|X$ is the table on X with the bases $T \cap X$, $|T \cap X| = r(X)$, and the *contraction* $\mathcal{B}/X$ is the table on $\mathcal{E} \setminus X$ with the bases $T \setminus X$ of the same T; when the weights of these T factor as $\theta_T = \theta'_{T\cap X}\theta''_{T\setminus X}$, $\mathcal{B}|X$ and $\mathcal{B}/X$ carry θ' and θ''. When e is not a loop, $\mathcal{B}/\{e\}$ is the contraction of e (Definition 7.3); at a loop $\mathcal{B}/\{e\}$ is the deletion of e, while $K_{/e} = \Psi_{/e} = 0$.

Proposition 7.6 (Duality of minors)**.** *For a table $(\mathcal{B}, \theta)$ on $\mathcal{E}$ with the dual table $\mathcal{B}^*$ and $X \subseteq \mathcal{E}$, $X' = \mathcal{E} \setminus X$,*

$$(\mathcal{B}|X)^* = \mathcal{B}^*/X', \qquad (\mathcal{B}/X)^* = \mathcal{B}^*|X',$$

the duals taken on X and on X': a basis is at the top of X exactly when its complement is at the top of X' in the dual table. When the weights of the bases at the top of X factor, $\theta_T = \theta'_{T\cap X}\theta''_{T\setminus X}$, the minors of $\mathcal{B}^*$ carry the same weights, $\mathcal{B}^*/X'$ those of $\mathcal{B}|X$ and $\mathcal{B}^*|X'$ those of $\mathcal{B}/X$ on the complemented bases. So $K_{\mathcal{B}^*/X'} = \Psi_{\mathcal{B}|X}$, $\Psi_{\mathcal{B}^*/X'} = K_{\mathcal{B}|X}$, $K_{\mathcal{B}^*|X'} = \Psi_{\mathcal{B}/X}$ and $\Psi_{\mathcal{B}^*|X'} = K_{\mathcal{B}/X}$.

Lemma 7.7 (Linking read on the bases)**.** *Marks $e \neq f$ lie in a common circuit of a table exactly when some basis T has $e \in T$, $f \notin T$ and $(T \setminus \{e\}) \cup \{f\}$ a basis.*

Lemma 7.8 (The circuit lemma)**.** *If $w \in \mathbb{R}^{\mathcal{E}}$ has $\langle w, \mathbf{1}_T\rangle$ constant over the bases of a table, then $w_a = w_b$ for any two marks a, b linked by a circuit of that table. If $\langle w, \mathbf{1}_T\rangle$ is constant over the bases of every table of a family, w is constant on every molecule.*

Proposition 7.9 (Molecules split every table)**.** *Each molecule is a union of molecules of every table, those of a table being the molecules of that table alone, and every table is the* direct sum *of its restrictions to the molecules: its bases are exactly the unions of one basis of each restriction. When the weights factor across the molecules, $\theta_T = \prod_M \theta_M(T \cap \mathcal{E}_M)$, as they do for a list, the Foreman polynomials factor, $K(\alpha) = \prod_M K_M(\alpha_{\mathcal{E}_M})$ and $\Psi(\alpha) = \prod_M \Psi_M(\alpha_{\mathcal{E}_M})$.*

Lemma 7.10 (Unions of molecules)**.** *For a table on $\mathcal{E}$ of rank r and $S \subseteq \mathcal{E}$ the following are equivalent: (a) $r(S) + r(\mathcal{E} \setminus S) = r$; (b) every basis meets S in the same number of marks; (c) S is a union of molecules of the table (Definition 7.4).*

Theorem 7.11 (The Foreman polynomials of a linked table do not factor)**.** *If the circuits of a table link all of its $n \geq 2$ marks, then K and Ψ are irreducible polynomials.*

7.3 The Skeleton

Definition 7.12 (Skeleton counts)**.** The *cocircuit layer* of a table is its set of cocircuits and the *circuit layer* its set of circuits; the two layers together are the table's *Skeleton*. For $S \subseteq \mathcal{E}$, with $r^*(S) = |S| + r(\mathcal{E} \setminus S) - r$ the dual rank,

$$d(S) = |S| - r^*(S), \qquad d^*(S) = |S| - r(S), \qquad c(S) = |S| - d(S) - d^*(S), \qquad \varepsilon(S) = d(S) - d^*(S) :$$

$c(S)$ is the *width* and $\varepsilon(S)$ the *slant* of S.

Theorem 7.13 (The window)**.** *For every $S \subseteq \mathcal{E}$: $d(S) = \min_T |T \cap S|$, $d^*(S) = \min_T |S \setminus T|$, $r(S) = \max_T |T \cap S|$, and $\{|T \cap S| : T \in \mathcal{B}\} = \{d(S), d(S)+1, \dots, r(S)\}$. Hence $c(S) = r(S) - d(S)$ is the width of this window. Moreover $c(\mathcal{E} \setminus S) = c(S)$ and $\varepsilon(S) + \varepsilon(\mathcal{E} \setminus S) = 2r - n$.*

The number of marks of S that a basis occupies ranges over a window of consecutive integers; its floor is $d(S)$, its width is the width $c(S)$, and the same window read on the complements has the floor $d^*(S)$. Every count that the integrals of a family need is read from these two layers.

Definition 7.14 (Signed indicators and the centred polynomial)**.** For $T \subseteq \mathcal{E}$ the *signed indicator* is $\chi_T = \mathbf{1}_T - \mathbf{1}_{\mathcal{E}\setminus T}$. For $\alpha = e^{2z}$ the *centred polynomial* is

$$\Lambda(z) = \sum_{T \in \mathcal{B}} \theta_T \, e^{\langle z, \chi_T \rangle}.$$

Theorem 7.15 (One function, two Foreman polynomials)**.** *For $\alpha = e^{2z}$:*

(i) $K(\alpha) = e^{\langle \mathbf{1}, z\rangle}\Lambda(z)$ *and* $\Psi(\alpha) = e^{\langle \mathbf{1}, z\rangle}\Lambda(-z)$*;*

(ii) the dual table has the centred polynomial $\Lambda^*(z) = \Lambda(-z)$: duality and inversion of the gauge are $z \mapsto -z$;

(iii) for every $v \in \mathbb{R}^{\mathcal{E}}$ constant on each molecule of the table alone, with value v_M on the molecule M, $\langle v, \chi_T \rangle = \sum_M v_M(2r(M) - |M|)$ for every basis and

$$\Lambda(z + v) = \exp\Big(\sum_M v_M\big(2r(M) - |M|\big)\Big)\Lambda(z);$$

these are the only v for which $\Lambda(z + \lambda v)/\Lambda(z)$ does not depend on z for every λ, and at $v = \lambda\mathbf{1}$ the factor is $e^{\lambda(2r-n)}$.

Theorem 7.16 (The inner moment generating function)**.** *For a gauge $\alpha = e^{2z}$ the* basis law *and the* cobasis law *at α are the laws on $\mathcal{B}$*

$$\mathbb{P}_\alpha(T) = \frac{\theta_T \alpha^T}{K(\alpha)} = \frac{\theta_T e^{\langle z, \chi_T\rangle}}{\Lambda(z)}, \qquad \mathbb{P}^*_\alpha(T) = \frac{\theta_T \alpha^{\mathcal{E}\setminus T}}{\Psi(\alpha)} = \mathbb{P}_{\alpha^{-1}}(T).$$

Then

$$\frac{\Lambda(z+w)}{\Lambda(z)} = \mathbb{E}_{T\sim\mathbb{P}_\alpha}\big[e^{\langle w, \chi_T\rangle}\big], \qquad \nabla \log \Lambda(z) = \mathbb{E}_{T\sim\mathbb{P}_\alpha}[\chi_T] = 2R(\alpha) - \mathbf{1},$$

where $R_e(\alpha) = \alpha_e \partial_e \log K(\alpha) = \mathbb{P}_\alpha(e \in T)$ is the occupation*, and the derivatives of $\log\Lambda$ of order k at z are the joint cumulants of order k of χ_T under $\mathbb{P}_\alpha$. The* even *and* odd statistics *$E(z) = \log\Lambda(z) + \log\Lambda(-z)$ and $O(z) = \log\Lambda(z) - \log\Lambda(-z)$ satisfy $e^{E(z)} = K\Psi/\alpha^{\mathcal{E}}$, $e^{O(z)} = K/\Psi$, and, for every $w \in \mathbb{R}^{\mathcal{E}}$,*

$$e^{E(z+w)} = e^{E(z)}\,\mathbb{E}_{(T,T')\sim\mathbb{P}_\alpha\otimes\mathbb{P}^*_\alpha}\big[e^{\langle 2w, \mathbf{1}_T - \mathbf{1}_{T'}\rangle}\big] :$$

$E(z + \cdot) - E(z)$ is the cumulant generating function of $\chi_T - \chi_{T'}$ for independent $T \sim \mathbb{P}_\alpha$ and $T' \sim \mathbb{P}^*_\alpha$; at the table's gauge $e^{E(0)} = K(\mathbf{1})^2$. For table j of a family the two laws at α are $\mathbb{P}_{j,\alpha}$ and $\mathbb{P}^*_{j,\alpha} = \mathbb{P}_{j,\alpha^{-1}}$, the occupation is $R_j(\alpha)$, and the dual occupation *$R^*_j(\alpha)$, with $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T) = 1 - R_{j,e}(\alpha^{-1})$, is the occupation of the dual table. At the table's gauge the two laws coincide, $\mathbb{P}^*_{j,\mathbf{1}} = \mathbb{P}_{j,\mathbf{1}}$.*

The centred polynomial is a moment generating function: its terms at z, divided by $\Lambda(z)$, are the law of a random basis, and its logarithm generates the cumulants of the signed indicator. The basis law and the cobasis law at a gauge are one law read at α and at α^{-1}, and at every gauge the product of the two evaluations generates the difference of a basis drawn from the basis law and an independent basis drawn from the cobasis law.

Definition 7.17 (Weight law, indicators, levels, cumulants)**.** The *weight law* of table j is its basis law at the table's gauge, $\mathbb{P}_{j,\mathbf{1}}(T) = \theta_{j,T}/K_j(\mathbf{1})$ on $\mathcal{B}_j$; for a list it is uniform. For a random basis T the *indicators* are $X_e = \mathbf{1}[e \in T]$, and for a set C of marks the *level* is $N_C = |T \cap C| = \sum_{e \in C} X_e$. For functions $Y_1, \dots, Y_k$ of T, $\kappa_{j,\alpha}(Y_1, \dots, Y_k)$ is their joint cumulant under $\mathbb{P}_{j,\alpha}$, and $Y^{\times a}$ in its argument lists Y a times; along a string $w = (e_1, \dots, e_k)$ of marks, $\kappa_{j,\alpha}(w) = \kappa_{j,\alpha}(X_{e_1}, \dots, X_{e_k}) = \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \log K_j(e^y)$ at $y = \log \alpha$. With one table the index j is dropped: $\mathbb{P}_\alpha$, $R(\alpha)$, κ_α.

Theorem 7.18 (The Skeleton is the Newton data of Λ)**.** *For every $S \subseteq \mathcal{E}$ the values $\langle \mathbf{1}_S, \chi_T \rangle$, $T \in \mathcal{B}$, fill $\{2j - |S| : \ d(S) \le j \le r(S)\}$. The Newton polytope of Λ is $2P - \mathbf{1}$ with $P = \operatorname{conv}\{\mathbf{1}_T\}$ the base polytope; its width in the direction $\mathbf{1}_S$ is $2c(S)$ and the midpoint of its extent in that direction is $\varepsilon(S)$.*

Proposition 7.19 (Loops and coloops)**.** *If e is a loop, $\Lambda(z) = e^{-z_e} \Lambda'(z|_{\mathcal{E} \setminus e})$ with Λ' the centred polynomial of the same bases on $\mathcal{E} \setminus e$; if e is a coloop, $\Lambda(z) = e^{z_e} \Lambda'(z|_{\mathcal{E} \setminus e})$ with Λ' that of the bases $T \setminus e$.*

7.4 Multivectors of any tensor rank

Definition 7.20 (Multivectors and tensor rank)**.** Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$. A *multivector* of a table of rank r is an $\boldsymbol{\omega} \in \bigwedge^r \mathbb{F}^{\mathcal{E}}$ with $|\boldsymbol{\omega}_T|^2 = \theta_T$ for $T \in \mathcal{B}$ and $\boldsymbol{\omega}_T = 0$ otherwise; $\boldsymbol{\omega}_T = \sqrt{\theta_T}$ is one. Its *tensor rank* $\operatorname{rk}(\boldsymbol{\omega})$ is the least number of decomposable elements of $\bigwedge^r \mathbb{F}^{\mathcal{E}}$ with sum $\boldsymbol{\omega}$, written $\boldsymbol{\omega} = \sum_{k=1}^{\operatorname{rk}(\boldsymbol{\omega})} \boldsymbol{\omega}^{(k)}$, $\boldsymbol{\omega}^{(k)}_T = \det A^{(k)}_T$ with *slices* $A^{(k)} \in \mathbb{F}^{r \times n}$. The table is *tensorminental of tensor rank q* over $\mathbb{F}$ when q is the least tensor rank of its multivectors, and *determinantal* when $q = 1$. The *dual multivector* is the Hodge complement $\boldsymbol{\omega}^* \in \bigwedge^{n-r} \mathbb{F}^{\mathcal{E}}$, $|\boldsymbol{\omega}^*_{\mathcal{E} \setminus T}| = |\boldsymbol{\omega}_T|$, and $\operatorname{rk}(\boldsymbol{\omega}^*) = \operatorname{rk}(\boldsymbol{\omega})$.

Every table has multivectors, and every multivector has a tensor rank; tensor rank one, the determinantal case, is where a single matrix carries the table, and it is one case of the statements on multivectors, which hold at every tensor rank.

Proposition 7.21 (The letters of a tensorminental table)**.** *For a decomposition $\boldsymbol{\omega} = \sum_k \boldsymbol{\omega}^{(k)}$ into decomposable elements with slices $A^{(k)}$ and $D = \operatorname{diag}(\alpha)$,*

$$K(\alpha) = \sum_{k,l} \det\left(A^{(k)} D A^{(l)\mathsf{H}}\right) = \mathbf{1}^{\mathsf{H}} \mathbf{G}(\alpha) \mathbf{1}, \qquad \mathbf{G}_{kl}(\alpha) = \det\left(A^{(k)} D A^{(l)\mathsf{H}}\right),$$

and Ψ is the same expression for the slices of the dual multivector. $\mathbf{G}(\alpha)$ is the Gram matrix of the $\boldsymbol{\omega}^{(k)}$ for $\langle \eta, \xi \rangle_\alpha = \sum_T \alpha^T \eta_T \overline{\xi_T}$, Hermitian and positive semidefinite.

Theorem 7.22 (Tensor rank is hereditary)**.** *For disjoint $A, D \subseteq \mathcal{E}$ with A extending to a basis avoiding D, $\boldsymbol{\omega}_{A|D} := (\mathbf{e}_A \lrcorner \boldsymbol{\omega})|_{\mathcal{E} \setminus (A \cup D)}$, the restriction of the interior product of $\boldsymbol{\omega}$ with the wedge $\mathbf{e}_A$ of the unit vectors of the marks of A, $(\mathbf{e}_A \lrcorner \boldsymbol{\omega})_X = \pm \boldsymbol{\omega}_{X \cup A}$, is a multivector of the minor that contracts A and deletes D, with the weight $\theta_{X \cup A}$ on each basis X of the minor, and $\operatorname{rk}(\boldsymbol{\omega}_{A|D}) \le \operatorname{rk}(\boldsymbol{\omega})$.*

Theorem 7.23 (The pattern law of a tensorminental table is a finite interference)**.** *Let the table have a multivector of tensor rank q with slices $A^{(1)},\dots,A^{(q)}$, let $B\subseteq\mathcal{E}$ and $R=\mathcal{E}\setminus B$, and let α be a gauge at which every $M_{kl}=A^{(k)}D_\alpha A^{(l)\mathsf{H}}$, $D_\alpha=\operatorname{diag}(\alpha)$, with $\mathbf{G}_{kl}\not\equiv 0$ is invertible; pairs with $\mathbf{G}_{kl}\equiv 0$ as a polynomial in the gauge are omitted (Proposition 7.21). For $x\in\mathbb{C}^B$, with $x\circ\alpha_B=(x_e\alpha_e)_{e\in B}$,*

$$\mathbb{E}_{T\sim\mathbb{P}_\alpha}\Big[\prod_{e\in B}x_e^{\mathbf{1}[e\in T]}\Big]=\frac{K(x\circ\alpha_B,\alpha_R)}{K(\alpha)}=\sum_{k,l}w_{kl}\det\big(I+(D_x-I)\Pi^{(kl)}_{BB}\big),$$

where $\mathbb{P}_\alpha$ is the basis law at α, $D_x=\operatorname{diag}(x)$, $w_{kl}=\mathbf{G}_{kl}(\alpha)/K(\alpha)$ and $\Pi^{(kl)}=A^{(l)\mathsf{H}}M_{kl}^{-1}A^{(k)}D_\alpha$. Here $\sum_{k,l}w_{kl}=1$, $w_{lk}=\overline{w_{kl}}$, every $\Pi^{(kl)}$ is idempotent, and $\Pi^{(kk)}$ is the D_α-orthogonal projection onto the row space of $A^{(k)}$. The hypothesis holds off a proper algebraic set of gauges.

Corollary 7.24 (Moments and cumulants)**.** *In the setting of Theorem 7.23:*

(i) $\mathbb{P}_\alpha(S\subseteq T)=\sum_{k,l}w_{kl}\det\Pi^{(kl)}_{SS}$*;*

(ii) the law of $T\cap B$ is the w-mixture of q^2 determinantal components, whose cumulants are $\kappa^{(kl)}(e)=\Pi^{(kl)}_{ee}$ and, for $p\geq 2$ distinct marks,

$$\kappa^{(kl)}(e_1,\dots,e_p)=(-1)^{p-1}\sum_{\varsigma}\prod_{i=1}^{p}\Pi^{(kl)}_{e_{\varsigma(i)}e_{\varsigma(i+1)}},$$

the sum over the cyclic orders ς of $\{1,\dots,p\}$, with $\varsigma(p+1)=\varsigma(1)$; the joint cumulants κ_α of the indicators $X_e=\mathbf{1}[e\in T]$ under $\mathbb{P}_\alpha$ (Definition 7.17) are their total cumulants over the mixing index: for distinct marks $e_1,\dots,e_p$ of B,

$$\kappa_\alpha(e_1,\dots,e_p)=\sum_{\mathfrak{p}}\kappa_w\Big(\big(\kappa^{(kl)}(e_C)\big)_{C\in\mathfrak{p}}\Big),$$

the sum over the partitions $\mathfrak{p}$ of $\{1,\dots,p\}$ into nonempty blocks, $e_C=(e_c)_{c\in C}$, where for finitely many functions $(Y_i)_{i\in I}$ of the index (k,l)

$$\kappa_w\big((Y_i)_{i\in I}\big)=\sum_{\mathfrak{q}}(-1)^{|\mathfrak{q}|-1}(|\mathfrak{q}|-1)!\prod_{D\in\mathfrak{q}}\sum_{k,l}w_{kl}\prod_{i\in D}Y_i(k,l),$$

the sum over the partitions $\mathfrak{q}$ of I, is their joint cumulant over the index with the mixing coefficients w_{kl}; at one mark this is the occupation $R_e(\alpha)=\sum_{k,l}w_{kl}\Pi^{(kl)}_{ee}$. This is an identity of generating functions, valid for complex mixing coefficients summing to one.

A table with a multivector of tensor rank q reads the law of its patterns as an interference of q^2 determinantal components, mixed with complex coefficients that sum to one. At tensor rank one there is a single component and the mixture is a determinantal law; at every higher tensor rank the joint cumulants over the mixing index of the components' cumulants are the part that no single component carries.

Chapter 8

Families

8.1 The interest and its walls

Definition 8.1 (Family, interest, walls)**.** A *family* on $\mathcal{E}$ is a finite sequence $F = \big((\mathcal{B}_j, \theta_j);\, s_j, t_j\big)_{j=1}^{N}$ of tables on $\mathcal{E}$ with exponents $s_j, t_j \in \mathbb{C}$; $K_j, \Psi_j, \Lambda_j, E_j, O_j$, r_j and $d_j, d_j^*, c_j, \varepsilon_j$ belong to table j. A table is *active* when its exponents are not both zero, and a letter when its exponent is not zero. For *masses* $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ the *interest* is

$$J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j}\, d\alpha,$$

with its slots: the family's tables, the masses $\mathbf{m}$ and the exponents $\mathbf{s} = (s_j)_j$, $\mathbf{t} = (t_j)_j$. The *family word* $W = \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ is homogeneous of degree $-\sum_j (s_j r_j + t_j r_j^*)$, and with the measure $d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m}-\mathbf{1}} d\alpha$ (Definition 1.1(vii)), $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int W\, e^{-|\alpha|} d\alpha_{\mathbf{m}}$. The *pair coordinates* are the *strengths* $\sigma_j = \frac{s_j + t_j}{2}$ and the *leans* $\delta_j = \frac{s_j - t_j}{2}$, with the total strength $\sigma = \sum_j \sigma_j$, so that $s_j = \sigma_j + \delta_j$ and $t_j = \sigma_j - \delta_j$; the *offsets* are $\boldsymbol{\mu} = \mathbf{m} - \sigma\mathbf{1}$. The *walls* and *top walls* are

$$\kappa_S = m_S - \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big),$$

$$\kappa_S^{\mathrm{top}} = \sum_j \big(s_j r_j(S) + t_j(|S| - d_j(S))\big) - m_S \qquad (\varnothing \neq S \subseteq \mathcal{E}).$$

The *imbalance* is $\kappa_{\mathcal{E}} = m_{\mathcal{E}} - \sum_j (s_j r_j + t_j r_j^*)$, and the *balance wall* is the hyperplane $\kappa_{\mathcal{E}} = 0$. On it the data are fixed by their pair coordinates $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$, with $\boldsymbol{\mu}_\perp = \boldsymbol{\mu} - \frac{\mu_{\mathcal{E}}}{n}\mathbf{1}$ the projection of the offsets to $\mathbf{1}^\perp$: the exponents are $(\boldsymbol{\sigma} + \boldsymbol{\delta}, \boldsymbol{\sigma} - \boldsymbol{\delta})$ and the masses are the *balanced masses*

$$\mathbf{m}_F(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta}) = \boldsymbol{\mu}_\perp + \Big(\sigma + \frac{1}{n}\sum_j \delta_j(2r_j - n)\Big)\mathbf{1},$$

linear in $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$. F is *connected* if every nonempty proper S has $c_j(S) > 0$ for some active table j. A family of one table is named by its table: for a table $\mathcal{A}$ with its weights $\theta^{\mathcal{A}}$, every object of the family $((\mathcal{A}, \theta^{\mathcal{A}}); s, t)$ carries $\mathcal{A}$ in place of F, as $J_{\mathcal{A}}(\mathbf{m}; s, t)$. A table may carry two systems of positive weights, θ^K for its letter K and θ^Ψ for its letter Ψ; at (s, t) it names the family of the two tables $((\mathcal{B}, \theta^K); s, 0)$ and $((\mathcal{B}, \theta^\Psi); 0, t)$, whose word is $K_{\theta^K}^{-s} \Psi_{\theta^\Psi}^{-t}$.

The exponents of a table enter in two coordinates: σ_j, how much the family weighs the table, and δ_j, which side it leans to, toward the bases (K) or toward their complements (Ψ). The pair $(K_j \Psi_j, K_j/\Psi_j)$ carries σ_j and δ_j separately, and the walls are read in either coordinates.

Theorem 8.2 (The interest is a partition function in the pair coordinates)**.** *(i) In the offsets,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\boldsymbol{\mu}-\mathbf{1}} e^{-|\alpha|} \prod_j \Lambda_j(\tfrac{1}{2}\log\alpha)^{-s_j}\Lambda_j(-\tfrac{1}{2}\log\alpha)^{-t_j}\,d\alpha.$$

(ii) In signed coordinates,

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = 2^n \int_{\mathbb{R}^{\mathcal{E}}} \exp\Big(2\langle\boldsymbol{\mu},z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big) - \sum_e e^{2z_e}\Big)\,dz,$$

the partition function of the exponential family with natural parameters $(\boldsymbol{\mu};\sigma_j,\delta_j)$ and sufficient statistics $(2z;-E_j,-O_j)$.

(iii) $\kappa_S = \mu_S + \sum_j \big(\sigma_j\, c_j(S) - \delta_j\, \varepsilon_j(S)\big)$ *and* $\kappa_{\mathcal{E}} = \mu_{\mathcal{E}} - \sum_j \delta_j(2r_j - n)$.

Lemma 8.3 (Greedy bases on a sector)**.** *Let π be a total order of $\mathcal{E}$ with tails $U_k = \{\pi(k),\dots,\pi(n)\}$. There are bases $T^{\pm}_{\pi}$ with $|T^{+}_{\pi}\cap U_k| = d(U_k)$ and $|U_k \setminus T^{-}_{\pi}| = d^*(U_k)$ for all k, and on the sector $H_\pi = \{\alpha_{\pi(1)} \ge \cdots \ge \alpha_{\pi(n)}\}$ every basis has $\alpha^T \le \alpha^{T^+_\pi}$ and $\alpha^{\mathcal{E}\setminus T} \le \alpha^{\mathcal{E}\setminus T^-_\pi}$. For a nonempty set of r-element subsets of $\mathcal{E}$ this conclusion holds exactly when the set has the exchange property.*

The sectors are where the exchange property is used: on each sector one basis dominates every monomial of K and one cobasis every monomial of Ψ, and that is what gives the walls their form. What does not use the exchange property is the opening of every monomial by its shares (Corollary 4.4).

Lemma 8.4 (Cube integrals)**.** *Let $h(q,\lambda)$ be smooth in q near $[0,1]^m$, with every q-derivative continuous in (q,λ) and holomorphic in λ on an open set. Then* $\mathrm{cube}(x,\lambda;h) = \int_{[0,1]^m}\prod_k q_k^{x_k-1}h(q,\lambda)\,dq$*, defined for* $\operatorname{Re} x_k > 0$*, extends meromorphically to $\mathbb{C}^m$, jointly in λ, with poles on the hyperplanes $x_k = -j$, $j\in\mathbb{N}$: with $h = \sum_{j<M} h_j(q',\lambda)q_1^j + q_1^M r_M$,*

$$\mathrm{cube}(x,\lambda;h) = \sum_{j<M}\frac{\mathrm{cube}(x',\lambda;h_j)}{x_1+j} + \mathrm{cube}\big((x_1+M,x'),\lambda;r_M\big),$$

repeated in each coordinate.

Theorem 8.5 (Existence and continuation)**.** *(i) $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converges if and only if* $\operatorname{Re}\kappa_S > 0$ *for every nonempty $S\subseteq\mathcal{E}$, whatever $\mathbf{s},\mathbf{t}\in\mathbb{C}^N$; it is holomorphic there.*

(ii) On that region

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_{\mathcal{E}})\sum_\pi \int_{[0,1]^{n-1}} \prod_{k=2}^{n} q_k^{\kappa_{U_k}-1} h_\pi(q)\,dq, \qquad h_\pi = L_\pi^{-\kappa_{\mathcal{E}}}\prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j},$$

where in the sector coordinates $\alpha_{\pi(1)} = \rho$, $\alpha_{\pi(k)} = \rho q_2\cdots q_k$ one has

$$K_j = \rho^{r_j}\prod_k q_k^{d_j(U_k)} k_{j,\pi}(q), \qquad \Psi_j = \rho^{n-r_j}\prod_k q_k^{d_j^*(U_k)}\psi_{j,\pi}(q), \qquad |\alpha| = \rho L_\pi(q);$$

the sector letters *$k_{j,\pi},\psi_{j,\pi},L_\pi$ have nonnegative coefficients and positive constant terms $\theta_{j,T^+_{j,\pi}}$, $\theta_{j,T^-_{j,\pi}}$, $T^{\pm}_{j,\pi}$ the greedy bases of table j (Lemma 8.3), and* 1*, and $L_\pi\in[1,n]$.*

(iii) $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *extends to a meromorphic function of* $(\mathbf{m},\mathbf{s},\mathbf{t})$ *with poles on the hyperplanes* $\kappa_S=-j$, $S\neq\varnothing$, $j\in\mathbb{N}$*; identities valid on an open subset of the region persist.*

Corollary 8.6 (Past several walls at once)**.** *For an order* π *write* $\kappa_k=\kappa_{U_k}$ *and let*

$$I_\pi=\int_{[0,1]^{n-1}}\prod_k q_k^{\kappa_k-1}h_\pi(q)\,dq=\int_{[0,\infty)^{n-1}}e^{-\sum_k\kappa_k u_k}h_\pi(e^{-u})\,du$$

be the term of Theorem 8.5(ii), written also in the gaps $u_k=-\log q_k$*; in the log-ratios* $\eta_k=\log(\alpha_{\pi(1)}/\alpha_{\pi(k)})=u_2+\cdots+u_k$ *the sector is the trail region* $0<\eta_2<\cdots<\eta_n$*. Let no* κ_k *be a nonpositive integer and* $L_k\in\mathbb{N}$ *with* $\operatorname{Re}\kappa_k+L_k>0$*. Then the continuation of* I_π *in the masses is*

$$\sum_{A\subseteq\{2,\dots,n\}}\ \sum_{\mathbf{l}:\ l_k<L_k\ (k\in A)}\ \prod_{k\in A}\frac{1}{\kappa_k+l_k}\int_{[0,1]^{A^c}}\prod_{k\notin A}q_k^{\kappa_k-1}\Big[\prod_{k\notin A}(1-\mathrm{Tay}_k)\,h_\pi^{(A,\mathbf{l})}\Big]dq_{A^c},$$

where $h_\pi^{(A,\mathbf{l})}=\partial_{q_A}^{\mathbf{l}}h_\pi/\mathbf{l}!$ *at* $q_A=0$ *and* Tay_k *takes the Taylor polynomial in* q_k *of degree below* L_k*: the term* A *collapses the tails* U_k*,* $k\in A$*, together, at the levels* $\mathbf{l}$*.*

Theorem 8.7 (Projective form; the period)**.** *Put*

$$\begin{aligned}P_F(\mathbf{m};\mathbf{s},\mathbf{t})&=\int_{\Delta_\mathcal{E}}u^{\boldsymbol{\mu}-\mathbf{1}}\prod_j\Lambda_j(\tfrac12\log u)^{-s_j}\Lambda_j(-\tfrac12\log u)^{-t_j}\,d\omega_\mathcal{E}(u)\\&=\int_{\Delta_\mathcal{E}}u^{\mathbf{m}-\mathbf{1}}\prod_j K_j(u)^{-s_j}\Psi_j(u)^{-t_j}\,d\omega_\mathcal{E}(u).\end{aligned}$$

It converges if and only if $\operatorname{Re}\kappa_S>0$ *for every nonempty proper* S*, and* $J_F(\mathbf{m};\mathbf{s},\mathbf{t})=\Gamma(\kappa_\mathcal{E})P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *as meromorphic functions;* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *is the* period*. On the balance wall the period is the residue of* $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *there; for every positive* h *homogeneous of degree one, the integral of* $\alpha^{\mathbf{m}-\mathbf{1}}\prod_j K_j^{-s_j}\Psi_j^{-t_j}d\alpha$ *over* $\{h=1\}$ *(Definition 1.1(vii)) is the period.*

Definition 8.8 (The continued interest)**.** The continuation of Theorem 8.5(iii) keeps the symbol $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ away from its poles. Among them is the whole balance wall, where $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ has the residue $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 8.7). The factor $e^{-|\alpha|}$ damps the scaling mode, so no completion enters the continued interest.

Theorem 8.9 (The interest is a Gramma; its letter and mark relations)**.** *For* $\operatorname{Re}s_j,\operatorname{Re}t_j>0$*, on the scales* $\mathcal{E}\sqcup\{v_j\}\sqcup\{w_j\}$ *let* t_F *be the configuration of the columns* $(\mathbf{e}_e;0;0)$ *with coefficient* 1*, and* $(\chi_T/2;\mathbf{e}_j;0)$*,* $(-\chi_T/2;0;\mathbf{e}_j)$ *with coefficient* $\theta_{j,T}$ *for* $T\in\mathcal{B}_j$*. Then*

$$\prod_j\Gamma(s_j)\Gamma(t_j)\,J_F(\mathbf{m};\mathbf{s},\mathbf{t})=\mathsf{G}_{(\boldsymbol{\mu};\mathbf{s};\mathbf{t});t_F}(\mathbf{1}\oplus\boldsymbol{\theta}\oplus\boldsymbol{\theta}),$$

either side converging with the other. As identities of meromorphic functions:

(i) (letter relations) for each j*,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t})=\sum_{T\in\mathcal{B}_j}\theta_{j,T}J_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t})=\sum_{T\in\mathcal{B}_j}\theta_{j,T}J_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{s},\mathbf{t}+\mathbf{e}_j);$$

(ii) (mark relations) for each e,

$$m_e J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = J_F(\mathbf{m}+\mathbf{e}_e;\mathbf{s},\mathbf{t}) + \sum_j s_j \sum_{T\ni e} \theta_{j,T} J_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t}) + \sum_j t_j \sum_{T\not\ni e} \theta_{j,T} J_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{s},\mathbf{t}+\mathbf{e}_j);$$

(iii) the same for the period $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ of Theorem 8.7, with $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_{\mathcal{E}})P_F(\mathbf{m};\mathbf{s},\mathbf{t})$: the letter relations hold verbatim, and in the mark relation the pure mass shift carries the factor $\kappa_{\mathcal{E}}$, $m_e P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \kappa_{\mathcal{E}} P_F(\mathbf{m}+\mathbf{e}_e;\mathbf{s},\mathbf{t}) + \cdots$.

The interest is not a new kind of object: it is one Gramma, on the marks and one scale for each letter, and its relations are that Gramma's exchange relations. The letter relation is the exchange relation on a letter's scale, and the mark relation the exchange relation on a mark's scale, rewritten with the letter relations.

Definition 8.10 (Breta shape law; Breta law)**.** For real data with every proper wall positive, the *Breta shape law* $\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ is the probability measure

$$u^{\mathbf{m}-\mathbf{1}} \prod_j K_j(u)^{-s_j} \Psi_j(u)^{-t_j}\, d\omega_{\mathcal{E}}(u) / P_F(\mathbf{m};\mathbf{s},\mathbf{t}) \quad \text{on } \Delta_{\mathcal{E}}.$$

With every wall positive, the *Breta law* $\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ is the law on $\mathbb{R}^{\mathcal{E}}_{>0}$ with density $\alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j} / J_F(\mathbf{m};\mathbf{s},\mathbf{t})$.

Proposition 8.11 (The interest over latent vectors)**.** *Let every table j of a family be determinantal over $\mathbb{F}_j \in \{\mathbb{R},\mathbb{C}\}$ (Definition 7.20), put $\beta_j = \dim_{\mathbb{R}} \mathbb{F}_j$, and let $2s_j/\beta_j$ and $2t_j/\beta_j$ be nonnegative integers. Let $V_j \subseteq \mathbb{F}_j^{\mathcal{E}}$ be the row space of a slice of table j and $V_j^{\perp}$ its orthogonal complement, each with its Lebesgue measure, and for the* latent vectors *$x_{j,i} \in V_j$, $1 \le i \le 2s_j/\beta_j$, and $x^*_{j,i} \in V_j^{\perp}$, $1 \le i \le 2t_j/\beta_j$, put $\mathfrak{q}_e = \sum_j \big(\sum_i |(x_{j,i})_e|^2 + \sum_i |(x^*_{j,i})_e|^2 \big)$. Then*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \frac{\prod_e \Gamma(m_e)}{\prod_j K_j(\mathbf{1})^{s_j+t_j} \pi^{r_j s_j + r_j^* t_j}} \int \prod_{e\in\mathcal{E}} \big(1+\mathfrak{q}_e\big)^{-m_e} \prod_{j,i} dx_{j,i}\, dx^*_{j,i},$$

both sides finite or infinite together. Where they are finite, the Breta law $\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ (Definition 8.10) is the law of α under the joint law of α and the latent vectors with density proportional to $\alpha^{\mathbf{m}-\mathbf{1}} \exp\big(-\sum_e \alpha_e(1+\mathfrak{q}_e)\big)$, under which, given the latent vectors, the scales α_e are independent and $\alpha_e(1+\mathfrak{q}_e)$ has the law Gamma_{m_e}. Over $\mathbb{R}$ with $s_j = t_j$ the sums $u_{j,i} = x_{j,i} + x^*_{j,i}$ run over $\mathbb{R}^{\mathcal{E}}$, with $|(x_{j,i})_e|^2 + |(x^*_{j,i})_e|^2 = (\mathrm{pr}_j\, u_{j,i})_e^2 + \big((I-\mathrm{pr}_j)u_{j,i}\big)_e^2$, pr_j the orthogonal projection onto V_j.

The letters of a determinantal table open as Gaussian integrals: its basis polynomial is a Gram determinant, whose inverse powers are Gaussian integrals over the row space of its slice, real or complex, and its cobasis polynomial is the same over the orthogonal complement, each letter with as many latent vectors as its exponent asks; over $\mathbb{R}$ at equal exponents the two pair into Gaussian vectors on $\mathbb{R}^{\mathcal{E}}$ read through the two projections. The integrals over the scales are then Gamma integrals, and the interest becomes a rational integral in the latent vectors.

Proposition 8.12 (The interest under Gamma draws)**.** *Let the data be real with $\mathbf{m} > 0$.*

(i) Where every proper wall is positive, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\,\mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}[W(U)]$. *With* $\phi_{\mathbf{m}}$ *the joint law of* $\big(K_1(U),\dots,K_N(U),\Psi_1(U),\dots,\Psi_N(U)\big)$ *for* $U\sim \mathrm{Dir}_{\mathbf{m}}$, *an input on the source* $(0,\infty)^{2N}$, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *is a Trailmoment at the unit configuration* I_{2N}*:*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\,\big(\mathsf{T}_{(\mathbf{s},\mathbf{t});I_{2N}}[\mathrm{id}]\,\phi_{\mathbf{m}}\big)\big((0,\infty)^{2N}\big).$$

(ii) Where moreover $\kappa_{\mathcal{E}}>0$, *so that every wall is positive, let* $\alpha\sim \mathrm{Gamma}_{\mathbf{m}}$, *with total* $|\alpha|$ *and shape* $U=\alpha/|\alpha|$. *The scale and the shape separate:* $|\alpha|\sim \mathrm{Gamma}_{m_{\mathcal{E}}}$ *and* $U\sim \mathrm{Dir}_{\mathbf{m}}$ *are independent, and*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_e \Gamma(m_e)\,\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}[W(\alpha)],$$

$$\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}[W(\alpha)] = \frac{\Gamma(\kappa_{\mathcal{E}})}{\Gamma(m_{\mathcal{E}})}\,\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)].$$

(iii) In the exponents, $J_F(\mathbf{m};\mathbf{s},\mathbf{t})/\prod_e\Gamma(m_e)$ *is the moment generating function of the letters' logarithms* $-\log K_j(\alpha)$, $-\log\Psi_j(\alpha)$ *for* $\alpha\sim\mathrm{Gamma}_{\mathbf{m}}$, *and* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ *that of* $-\log K_j(U)$, $-\log\Psi_j(U)$ *for* $U\sim\mathrm{Dir}_{\mathbf{m}}$. *Their ratio* $\Gamma(\kappa_{\mathcal{E}})/\Gamma(m_{\mathcal{E}})$ *has its pole on the balance wall, where the moment generating function under* $\mathrm{Gamma}_{\mathbf{m}}$ *diverges and that under* $\mathrm{Dir}_{\mathbf{m}}$ *is the period over* $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$.

Theorem 8.13 (Walls under Cremona)**.** *As functions of* $(\boldsymbol{\mu};\boldsymbol{\sigma},\boldsymbol{\delta})$: $\kappa_S^{\mathrm{top}}(\boldsymbol{\mu};\boldsymbol{\sigma},\boldsymbol{\delta}) = \kappa_S(-\boldsymbol{\mu};\boldsymbol{\sigma},-\boldsymbol{\delta})$, $\kappa_S+\kappa_S^{\mathrm{top}} = 2\sum_j\sigma_j c_j(S)$, *and* $\kappa_{\mathcal{E}\setminus S}=\kappa_{\mathcal{E}}+\kappa_S^{\mathrm{top}}$. *In pair coordinates* $\kappa_S^{\mathrm{top}} = \sum_j\big(\sigma_j c_j(S)+\delta_j\varepsilon_j(S)\big)-\mu_S$.

Theorem 8.14 (Cremona on the balance wall)**.** *For a connected family, on the hyperplane* $\kappa_{\mathcal{E}}=0$,

$$P_F(\boldsymbol{\mu}+\sigma\mathbf{1};\boldsymbol{\sigma}+\boldsymbol{\delta},\boldsymbol{\sigma}-\boldsymbol{\delta}) = P_F(-\boldsymbol{\mu}+\sigma\mathbf{1};\boldsymbol{\sigma}-\boldsymbol{\delta},\boldsymbol{\sigma}+\boldsymbol{\delta}),$$

the walls of the right side at S *being those of the left at* $\mathcal{E}\setminus S$*; the identity continues meromorphically along the hyperplane.*

On the balance wall the period is unchanged when the offsets and the leans change sign together: Cremona (Definition 9.1) swaps the two exponents of each table, $\delta_j\mapsto-\delta_j$, negates the offsets, and swaps every wall with the wall of the complement.

Proposition 8.15 (Dual tables)**.** *Replacing every table by its dual, with the same exponents, gives the interest at the exponents* (t_j,s_j)*: in pair coordinates* $\delta_j\mapsto-\delta_j$.

Proposition 8.16 (The wall of a compound)**.** *For every nonempty proper* S, $\kappa_S+\kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}} + 2\sum_j\sigma_j c_j(S)$. *So if* $c_j(S)=0$ *in every table with* $\sigma_j\neq0$, *then* $\kappa_S+\kappa_{\mathcal{E}\setminus S}=\kappa_{\mathcal{E}}$, *and on the balance wall a compound split along* S *has a wall with real part at most zero, and its period diverges.*

8.2 The native gauge

Definition 8.17 (Target; native gauge)**.** For data with $m_{\mathcal{E}} \neq 0$ the *target* is $\boldsymbol{\tau} = \frac{\sum_j (s_j r_j + t_j r_j^*)}{m_{\mathcal{E}}} \mathbf{m}$; on the balance wall it is $\mathbf{m}$. A point $\alpha^\star \in (\mathbb{C}^\times)^{\mathcal{E}}$ with $K_j(\alpha^\star)\Psi_j(\alpha^\star) \neq 0$ for every j is *native* when its *usage* equals the target,

$$\sum_j \big(s_j\, R_j(\alpha^\star) + t_j\, R_j^*(\alpha^\star)\big) = \boldsymbol{\tau},$$

where $R_{j,e} = \alpha_e \partial_e K_j / K_j$ is the occupation of table j and $R^*_{j,e} = \alpha_e \partial_e \Psi_j / \Psi_j$ that of its dual (Theorem 7.16), rational functions homogeneous of degree zero; so nativity is a property of the point up to scale. A native point in $\mathbb{R}^{\mathcal{E}}_{>0}$, written $\alpha^\star = e^{2z^\star}$, is a *native gauge*; the other native points are the *native gauges at complex points.*

Definition 8.18 (Potential; its even and odd parts)**.** In logarithmic coordinates the *potential* of a family with its exponent pairs is

$$\Phi(y) = \sum_j \Big(s_j \log K_j(e^y) + t_j \log \Psi_j(e^y)\Big) = \Phi_{\mathbf{E}}(y) + \Phi_{\mathbf{O}}(y),$$

$$\Phi_{\mathbf{E}}(y) = \sum_j \sigma_j \log \mathbf{E}_j(e^y), \qquad \Phi_{\mathbf{O}}(y) = \sum_j \delta_j \log \mathbf{O}_j(e^y), \qquad \mathbf{E}_j = K_j\Psi_j, \quad \mathbf{O}_j = K_j/\Psi_j,$$

the *even* and *odd letters* of table j, so that $\mathbf{E}_j(e^{2z}) = e^{2\langle \mathbf{1}, z\rangle} e^{E_j(z)}$ and $\mathbf{O}_j(e^{2z}) = e^{O_j(z)}$. The pair $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$, either part written $\Phi_{\mathbf{X}}$ with $\mathbf{X} \in \{\mathbf{E}, \mathbf{O}\}$, is read whole: neither part is switched off at any lean, and the two are never added before they are read.

Theorem 8.19 (The native gauge is the saddle)**.** *The potential satisfies* $\partial_{y_e}\Phi = \sum_j \big(s_j R_{j,e} + t_j R^*_{j,e}\big)$*, the usage at* $\alpha = e^y$*, and the integrand of the interest is* $e^{\langle \mathbf{m}, y\rangle - \Phi(y) - |\alpha|}$ *in* y*. In the signed coordinates* $z = \frac{1}{2}y$*,* $\Phi(y) = \sigma\langle \mathbf{1}, y\rangle + \Phi^\circ(z)$ *with the* centred potential $\Phi^\circ(z) = \sum_j \big(s_j \log \Lambda_j(z) + t_j \log \Lambda_j(-z)\big) = \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)$*.*

(i) A gauge $\alpha^\star$ *is native iff* $\nabla\Phi(\log \alpha^\star) = \boldsymbol{\tau}$*; on the balance wall, iff* $\log \alpha^\star$ *is a critical point of* $\langle \mathbf{m}, y\rangle - \Phi(y)$*.*

(ii) For real data with $s_j, t_j \geq 0$*, not all zero, and* $\mathbf{m} > 0$*,* Φ *is convex, and a native gauge exists iff* $\boldsymbol{\tau}$ *lies in the relative interior of the Newton sum* $\mathcal{N}_F = \sum_j \big(s_j P_j + t_j P_j^*\big)$*, with* P_j *the base polytope of table* j *(Theorem 7.18) and* P_j^* *that of its dual; it is then unique modulo the space of directions along which* Φ *is affine.*

(iii) For such data $\mathcal{N}_F = \{x :\ x_{\mathcal{E}} = g(\mathcal{E}),\ x_S \geq g(S)\ \forall S\}$ *with the set function* $g = \sum_j (s_j d_j + t_j d_j^*)$*. For a connected family that space is* $\mathbb{R}\mathbf{1}$*, and the condition is* $\tau_S - g(S) > 0$ *for every nonempty proper* S*: the walls at the target are positive. The native gauge is then the unique point, up to scale, at which* $\alpha^{\boldsymbol{\tau}} W(\alpha)$*, constant along each ray, is largest; on the balance wall* $\boldsymbol{\tau} = \mathbf{m}$*.*

A native gauge is a gauge at which the family uses each mark exactly as much as the target asks, and the same equation at a complex point gives the native gauges at complex points that the rays of Chapter 38 use. With real data, nonnegative exponents and positive masses, for a connected family it is unique up to scale and exists exactly when every proper wall at the target is positive (Theorem 8.19(iii)); on a compound it exists exactly when every molecule of the active family is balanced at the target with its proper walls positive there, with one free scale for each such molecule (Proposition 8.20). It is a point at which $\alpha^{\boldsymbol{\tau}} W(\alpha)$ is largest, the saddle of the potential, and it is the measuring stick of every computation on the family.

Proposition 8.20 (The native gauges of a compound)**.** *Let F have real data, $s_j, t_j \geq 0$ not all zero, $\mathbf{m} > 0$, and target $\boldsymbol{\tau}$, and let $\mathcal{E}_1, \ldots, \mathcal{E}_q$ be the molecules of its active family. A native gauge exists exactly when every $\mathcal{E}_i$ is balanced at the target, $\tau_{\mathcal{E}_i} = \sum_j \big(s_j r_j(\mathcal{E}_i) + t_j r_j^*(\mathcal{E}_i)\big)$, and every proper wall of every $\mathcal{E}_i$ is positive there, $\tau_S > \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big)$ for $\varnothing \neq S \subsetneq \mathcal{E}_i$. The native gauges are then the gauges β with $\beta/\alpha^\star$ constant on every $\mathcal{E}_i$, $\alpha^\star$ any one of them: one free scale for each molecule. When the weights factor across the $\mathcal{E}_i$, the native gauges are the products of native gauges of the molecules. A mark that is a loop or a coloop in every table is a unit, whose letters are monomials.*

Proposition 8.21 (The centre of an arbitrary family)**.** *Take tables on $\mathcal{E}$, $s_j = t_j = \sigma_j \geq 0$ with $\sigma = \sum_j \sigma_j > 0$, and $\mathbf{m} = \sigma\mathbf{1}$. Then $\delta_j = 0$, $\boldsymbol{\mu} = 0$, the centred potential $\Phi^\circ(z) = \sum_j \sigma_j E_j(z)$ is even, and:*

(i) the gauge $\mathbf{1}$ is native, unique modulo one scale for each molecule of the active family (Proposition 8.20);

(ii) the walls are $\kappa_S = \sum_j \sigma_j c_j(S)$, positive on every proper nonempty S exactly when the active family is connected;

(iii) every odd derivative of Φ° vanishes at $z = 0$, so every odd derivative of order at least three of the potential Φ vanishes at $y = 0$.

The family at the centre, every $\delta_j = 0$ (Definition 9.1), with the masses $\sigma\mathbf{1}$ has its offsets zero. It exists for every family, its native gauge is the table's gauge, and its walls are the combinations $\kappa_S = \sum_j \sigma_j c_j(S)$ of the widths; it is where the family is read before any lean is asked of it.

Proposition 8.22 (Prescribing a gauge)**.** *For any positive gauge a and positive exponent pairs, the usage $\tau_e = \sum_j \big(s_j R_{j,e}(a) + t_j R_{j,e}^*(a)\big)$ has total $\sum_j (s_j r_j + t_j r_j^*)$, and every mass vector on the ray $c\boldsymbol{\tau}$, $c > 0$, makes a native. Fixing values $K_j^0 > 0$ and $\Psi_j^0 > 0$ of the letters at a, the constraints*

$$\sum_T \theta_{j,T} a^T = K_j^0, \qquad \sum_T \theta_{j,T} a^{\mathcal{E}\setminus T} = \Psi_j^0, \qquad \sum_{j,T} \theta_{j,T} \Big[\frac{s_j a^T}{K_j^0} \mathbf{1}_T + \frac{t_j a^{\mathcal{E}\setminus T}}{\Psi_j^0} \mathbf{1}_{\mathcal{E}\setminus T} \Big] = \boldsymbol{\tau}$$

are linear in the weights, and every positive deformation in their homogeneous kernel preserves them exactly.

Part III

Classification

Chapter 9

Realizations and the still

9.1 Realizations

Definition 9.1 (Realization, lean, carrier)**.** Let a family carry tables j on its marks $\mathcal{E}$, $n = |\mathcal{E}|$, its potential read as the pair $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ of Definition 8.18. A *realization* of the family is $\Upsilon = \big((s_j, t_j)_j;\, \alpha\big)$: exponents $s_j, t_j \geq 0$ with $s_j + t_j > 0$, and a gauge $\alpha \in \mathbb{R}^{\mathcal{E}}_{>0}$. Its pair coordinates are the strengths $\sigma_j = \frac{1}{2}(s_j + t_j) > 0$ and the leans $\delta_j = \frac{1}{2}(s_j - t_j)$ of Definition 8.1, $|\delta_j| \leq \sigma_j$, and in them the realization is written

$$\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha} = \big((\sigma_j + \delta_j,\ \sigma_j - \delta_j)_j;\ \alpha\big).$$

$\boldsymbol{\delta} = (\delta_j)_j$ is the *lean*, and the *centre* is $\boldsymbol{\delta} = 0$; at the table's gauge the realization at the centre has the masses $\sigma\mathbf{1}$ (Proposition 8.21), and a family at the centre is taken with these masses. Table j has a *basis lean* when $\delta_j > 0$, a *cobasis lean* when $\delta_j < 0$, and *rides the line* when $\delta_j = 0$; its lean is *full* when $|\delta_j| = \sigma_j$ (then the factor of table j in the family word is $K_j^{-s_j}$ alone, or $\Psi_j^{-t_j}$ alone) and *partial* when $0 < |\delta_j| < \sigma_j$. The family has a basis lean when every $\delta_j \geq 0$ and some $\delta_j > 0$, a cobasis lean when every $\delta_j \leq 0$ and some $\delta_j < 0$, and otherwise, off the centre, a mixed lean. The realization is *at one pair* (σ, δ), $\sigma = \sum_j \sigma_j$ and $\delta = \sum_j \delta_j$, when its pairs lie on one ray, $(\sigma_j, \delta_j) = \frac{\sigma_j}{\sigma}(\sigma, \delta)$ for every j; its *lean sign* is the sign of δ. A family with one table is at one pair. The *masses* of Υ are its usage,

$$\tau_e(\Upsilon) = \sum_j \Big(s_j \, \frac{\alpha_e \partial_e K_j}{K_j}(\alpha) + t_j \, \frac{\alpha_e \partial_e \Psi_j}{\Psi_j}(\alpha) \Big),$$

for which α is the native gauge (Proposition 8.22). The *carrier* of Υ is $(\alpha;\, K_j, \Psi_j;\, \boldsymbol{\tau};\, s_j, t_j)$. A gauge is written as a deviation from the table's gauge, $\alpha = e^{y_0}$. Inverting the gauge and swapping the two exponents of every table, $\big((s_j, t_j)_j; \alpha\big) \mapsto \big((t_j, s_j)_j; \alpha^{-1}\big)$, sends the offsets and the lean to $-\boldsymbol{\mu}$ and $-\boldsymbol{\delta}$, the change of data of Theorem 8.14; so *Cremona* swaps basis and cobasis leans, and the table's gauge is its fixed point up to scale (Proposition 7.2).

A realization is read off its exponents: the masses are the usage at its gauge, so the gauge is native by definition, and a realization sits on a wall only along a union of molecules, so that a realization of a molecule never does (Proposition 9.2). A molecule is read at all of its realizations, and a particular integral is one of them, fixed by its exponents and its native gauge.

Proposition 9.2 (A realization sits on a wall only along its molecules)**.** *Let a family carry real exponents $s_j, t_j \geq 0$, $s_j + t_j > 0$, and a gauge α, with masses its usage $\mathbf{m} = \sum_j \big(s_j R_j(\alpha) + t_j R_j^*(\alpha)\big)$, that is, a realization (Definition 9.1). With $N_S = |T \cap S|$,*

$$\kappa_S = \sum_j \Big(s_j \big(\mathbb{E}_{T \sim \mathbb{P}_{j,\alpha}}[N_S] - d_j(S) \big) + t_j \big(r_j(S) - \mathbb{E}_{T \sim \mathbb{P}^*_{j,\alpha}}[N_S] \big) \Big) \;\geq\; 0 \qquad (\varnothing \neq S \subseteq \mathcal{E}),$$

with equality exactly when S is a union of molecules (Definition 7.4). In particular $\kappa_{\mathcal{E}} = 0$, and every proper wall of a connected family is positive.

Proposition 9.3 (Masses at a gauge)**.** *At $\alpha = e^{y_0}$, in this statement write $R^{\pm}_{j,e} = \frac{1}{2}\big(R_{j,e}(e^{y_0}) \pm R_{j,e}(e^{-y_0})\big)$; then*

$$\tau_e = \sum_j \big(s_j R_{j,e}(e^{y_0}) + t_j (1 - R_{j,e}(e^{-y_0})) \big) = \sum_j \big(\sigma_j (1 + 2R^-_{j,e}) + \delta_j (2R^+_{j,e} - 1) \big),$$

and at $\alpha = \mathbf{1}$, where $R^-_{j,e} = 0$ and $R^+_{j,e} = R_{j,e}(\mathbf{1})$, $\tau_e = \sum_j \big(\sigma_j + \delta_j (2R_{j,e}(\mathbf{1}) - 1) \big)$.

9.2 The potential and the still

Definition 9.4 (Still, rungs, tallies, line functions)**.** The *still* of Υ is the sequence of symmetric tensors

$$\kappa_\Upsilon(e_1, \ldots, e_k) = 2^k \, \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \Phi(\log \alpha), \qquad k \geq 2,$$

the Taylor tensor of Φ in the signed coordinates $z = \frac{1}{2} y$ from order two on; its entries of order k form *rung k*. *Rung one* is the gradient $\partial_{y_e} \Phi(\log \alpha) = \tau_e(\Upsilon)$, the masses (Theorem 8.19), taken without the factor 2. Each part has its own still, $\kappa^{\mathbf{E}}_\Upsilon$ and $\kappa^{\mathbf{O}}_\Upsilon$, with $\kappa_\Upsilon = \kappa^{\mathbf{E}}_\Upsilon + \kappa^{\mathbf{O}}_\Upsilon$. The *tally* of a mark e at rung k is the multiset $\mathrm{tally}^{\mathbf{X}}_k(e) = \{\!\{ \kappa^{\mathbf{X}}_\Upsilon(e, w) : \; w \in \mathcal{E}^{k-1} \}\!\}$. The *self-line* and *partner-line functions* of e are

$$\varphi^{\mathbf{X}}_e(v) = \partial_{y_e} \Phi_{\mathbf{X}}\big(\log \alpha + v \mathbf{1}_e \big), \qquad \varphi^{\mathbf{X}}_{e,g}(v) = \partial_{y_e} \Phi_{\mathbf{X}}\big(\log \alpha + v \mathbf{1}_g \big) \quad (g \neq e).$$

The still is the Taylor tensor of the potential at the realization from rung two on, one object, recording how the marks covary under the two laws; beside it, rung one is the masses. The letters are affine in every variable, so they have finitely many nontrivial derivatives, and every rung is read from those through the logarithm; the readings of Chapter 10 take the potential as one function, of which the still is the expansion.

Proposition 9.5 (The potential of a compound splits by molecules)**.** *Let the weights factor across the molecules. Then $\Phi_{\mathbf{X}}(y) = \sum_M \Phi_{\mathbf{X},M}(y_{\mathcal{E}_M})$ with $\Phi_{\mathbf{X},M}$ the potential of the restriction to M, and every entry of the still on a string that meets two molecules is 0. A mark that is a loop or a coloop in every table is a unit whose letters are monomials (Proposition 8.20): every entry of the still on a string through it is 0, and its rung one is its mass.*

Proposition 9.6 (A deviation reweights the bases)**.** *At the realization $\Upsilon = \big((s_j, t_j)_j; e^{y_0}\big)$, for $k \geq 2$ and every string w of length k,*

$$\kappa_\Upsilon(w) = 2^k \sum_j \Big(s_j \, \kappa_{j, e^{y_0}}(w) + t_j (-1)^k \kappa_{j, e^{-y_0}}(w) \Big).$$

In this statement write $\kappa_j^{\pm}(w) = \frac{1}{2}\big(\kappa_{j,e^{y_0}}(w) \pm \kappa_{j,e^{-y_0}}(w)\big)$*; then*

$$\kappa_\Upsilon(w) = \begin{cases} 2^{k+1}\sum_j \big(\sigma_j\kappa_j^+(w) + \delta_j\kappa_j^-(w)\big), & k \text{ even}, \\ 2^{k+1}\sum_j \big(\sigma_j\kappa_j^-(w) + \delta_j\kappa_j^+(w)\big), & k \text{ odd}, \end{cases}$$

and the cumulants at the deviation are read from those at the table's gauge, $\kappa_{j,e^{y_0}}(w) = \sum_{i\geq 0}\frac{1}{i!}\sum_{u\in\mathcal{E}^i}\kappa_{j,\mathbf{1}}(w,u)\prod_{l=1}^i y_{0,u_l}$*, for* y_0 *in the domain of convergence of the Taylor series of* $y \mapsto \log K_j(e^y)$ *at* 0*.*

Proposition 9.7 (The still at the table's gauge)**.** *At every realization* $\Upsilon = \big((s_j,t_j)_j;\mathbf{1}\big)$*, for* $k \geq 2$ *and every string* w *of length* k*,*

$$\kappa_\Upsilon(w) = 2^k\sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w) = \begin{cases} 2^{k+1}\sum_j \sigma_j\,\kappa_{j,\mathbf{1}}(w), & k \text{ even}, \\ 2^{k+1}\sum_j \delta_j\,\kappa_{j,\mathbf{1}}(w), & k \text{ odd}. \end{cases}$$

This is Proposition 9.6 at $y_0 = 0$*. At the table's gauge* $\Phi_{\mathbf{E}}$ *is even in* y *and* $\Phi_{\mathbf{O}}$ *is odd, each up to a linear term; so the even rungs are those of* $\kappa_\Upsilon^{\mathbf{E}}$*, independent of the lean, and the odd rungs are those of* $\kappa_\Upsilon^{\mathbf{O}}$*, linear in the lean and zero at the centre.*

Proposition 9.8 (Balanced tallies)**.** *At every realization* Υ*, for each part, every rung* $k \geq 2$ *and every mark* e*, the tally of* e *sums to* 0*:* $\sum_{w\in\mathcal{E}^{k-1}}\kappa_\Upsilon^{\mathbf{X}}(e,w) = 0$*.*

Proposition 9.9 (The two occupations of a mark)**.** *For a mark* e *of table* j*, neither a loop nor a coloop, with room* $R = \mathcal{E}-e$*, let* $K_{j,/e}(\alpha_R)$ *and* $K_{j,\backslash e}(\alpha_R)$ *be the basis polynomials of its contraction and deletion at* e *(Definition 7.3), and*

$$\mathrm{D}_{j,e}(y_R) = \log K_{j,/e}(e^{y_R}) - \log K_{j,\backslash e}(e^{y_R}), \qquad \mathrm{D}_{j,e}^{\pm}(y_R) = \tfrac{1}{2}\big(\mathrm{D}_{j,e}(y_R) \pm \mathrm{D}_{j,e}(-y_R)\big).$$

At the gauge e^y *the occupation and the dual occupation of* e *(Theorem 7.16) are*

$$\frac{\alpha_e\partial_e K_j}{K_j} = \frac{1}{1+e^{-\lambda_{j,e}-\eta_{j,e}}}, \qquad \frac{\alpha_e\partial_e \Psi_j}{\Psi_j} = \frac{1}{1+e^{-\lambda_{j,e}+\eta_{j,e}}},$$
$$\lambda_{j,e} = y_e + \mathrm{D}_{j,e}^-(y_R), \qquad \eta_{j,e} = \mathrm{D}_{j,e}^+(y_R),$$

and at the table's gauge $\lambda_{j,e} = 0$*: the two occupations are* $(1+e^{-\eta_{j,e}})^{-1}$ *and* $(1+e^{\eta_{j,e}})^{-1}$*.*

Proposition 9.10 (Occupations at a gauge)**.** *At* $\alpha = e^{y_0}$*, with* $R_{j,e}^{\pm}$ *as in Proposition 9.3,* $\varphi_e^{\mathbf{E}}(0) = \sum_j \sigma_j(1+2R_{j,e}^-)$ *and* $\varphi_e^{\mathbf{O}}(0) = \sum_j \delta_j(2R_{j,e}^+ - 1)$*. At* $\alpha = \mathbf{1}$*,* $\varphi_e^{\mathbf{E}}(0) = \sum_j \sigma_j$ *at every mark and* $\varphi_e^{\mathbf{O}}(0) = \sum_j \delta_j(2R_{j,e}(\mathbf{1}) - 1)$*.*

Chapter 10

The reading, classes, atoms

10.1 Views and the reading

Definition 10.1 (Views; the reading; classes; atoms)**.** Let M be a molecule with n marks carrying tables j, and Υ a realization of M (Definition 9.1). The room of a mark e is $\mathcal{E} \setminus \{e\}$ (Chapter 12 calls it the room of the bunch $\{e\}$), and an *order* of a set is a listing of its elements. For a mark e and distinct marks $\mathbf{g} = (g_1, \dots, g_k)$ of its room, $0 \le k \le n-1$, the *view* of e with the partners $\mathbf{g}$ is the pair $(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}})$,

$$\Phi^{\mathbf{X}}_{e,\mathbf{g}}(v_0, v_1, \dots, v_k) = \Phi_{\mathbf{X}}\Big(\log\alpha + v_0 \mathbf{1}_e + \sum_{i=1}^{k} v_i \mathbf{1}_{g_i} \Big),$$

the potential on the scales of e and its partners with every other mark at the gauge; the variable v_i sits at *position* i. The *reading of* e *with* k *partners* is the multiset of its views over all k-tuples of distinct partners,

$$\mathrm{read}^{(k)}_{\Upsilon}(e) = \{\{ (\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}}) : \ \mathbf{g} \in (\mathcal{E} \setminus e)^{(k)} \}\} :$$

the functions are kept and the names of the partners are not. With $k = n-1$, $\mathbf{g}$ is an order of the room and the pair $(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}})$ is the potential *written from* e in that order; $\mathrm{read}^{(n-1)}_{\Upsilon}(e)$ is the *reading* of e. The *classes* of Υ are the level sets of $e \mapsto \mathrm{read}^{(n-1)}_{\Upsilon}(e)$; they form the partition $\mathrm{Cl}(\Upsilon)$. The realization is *atomic*, an *atom*, when $\mathrm{Cl}(\Upsilon) = \{\mathcal{E}\}$. For a family whose weights factor across its molecules, the classes of a realization are, by definition, the classes of the realizations of its molecules, each read on its own marks. With the gauge α and $\boldsymbol{\sigma}$ fixed, the partition at the lean $\boldsymbol{\delta}$ is $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$.

The reading of a mark is the whole potential seen from that mark, its partners unnamed, and the reading is a function on the marks whose values are functions. Classes are its level sets, and an atom is a realization at which it is constant, so that the integral cannot tell its marks apart at any order.

Proposition 10.2 (A view through the patterns)**.** *For table j and positions $0, \dots, k$ held by $e, g_1, \dots, g_k$, let the pattern of a basis be the set $p \subseteq \{0, \dots, k\}$ of positions it holds (its trace $T \cap \{e, g_1, \dots, g_k\}$ read in positions, the pattern of the basis on the bunch $\{e, g_1, \dots, g_k\}$ in Chapter 12), $\bar{p}$ its complement, and*

$$K_{j;e,\mathbf{g}}(x) = \sum_{p} \vartheta^{K}_{j,p}\, x^{p}, \qquad \Psi_{j;e,\mathbf{g}}(x) = \sum_{p} \vartheta^{\Psi}_{j,p}\, x^{\bar{p}}, \qquad \vartheta^{K}_{j,p} = \sum_{T:\, p} \theta_{j,T} \alpha^{T}, \quad \vartheta^{\Psi}_{j,p} = \sum_{T:\, p} \theta_{j,T} \alpha^{\mathcal{E}\setminus T},$$

sums over the bases of pattern p. With $x_i = e^{v_i}$,

$$\Phi^{\mathbf{E}}_{e,\mathbf{g}} = \sum_j \sigma_j \log\big(K_{j;e,\mathbf{g}}\Psi_{j;e,\mathbf{g}}\big), \qquad \Phi^{\mathbf{O}}_{e,\mathbf{g}} = \sum_j \delta_j \log\big(K_{j;e,\mathbf{g}}/\Psi_{j;e,\mathbf{g}}\big).$$

At every gauge

$$K_{j;e,\mathbf{g}}(\mathbf{1}+x) = K_j(\alpha) \sum_{A\subseteq\{0,\dots,k\}} \mathbb{P}_{j,\alpha}\big(p(T)\supseteq A\big)\, x^A,$$
$$\Psi_{j;e,\mathbf{g}}(\mathbf{1}+x) = \Psi_j(\alpha) \sum_{A\subseteq\{0,\dots,k\}} \mathbb{P}^*_{j,\alpha}\big(p(T)\cap A=\varnothing\big)\, x^A,$$

with $p(T)$ the pattern of T, so the view with k partners is read from the joint inclusion laws of $e, g_1, \dots, g_k$ under the basis and cobasis laws. At the table's gauge $\vartheta^K_{j,p} = \vartheta^\Psi_{j,p} = \vartheta_{j,p}$, the weight of the bases of pattern p, both laws are the weight law $\mathbb{P}_{j,\mathbf{1}}$, and $\Psi_{j;e,\mathbf{g}}(x) = x^{\{0,\dots,k\}}K_{j;e,\mathbf{g}}(x^{-1})$. For one table, two views of one realization agree at the centre exactly when the polynomials $K_{e,\mathbf{g}}\Psi_{e,\mathbf{g}}$ agree, and at a lean exactly when $K_{e,\mathbf{g}}$ and $\Psi_{e,\mathbf{g}}$ both agree; at the table's gauge $K_{e,\mathbf{g}}$ alone decides at a lean.

Proposition 10.3 (More partners refine; the tallies are inside)**.** *(i) $\mathrm{read}^{(k)}_\Upsilon(e)$ determines $\mathrm{read}^{(k')}_\Upsilon(e)$ for every $k' \le k$; so the level sets of the reading with k partners refine those with k'.*

(ii) For $k \ge 1$, $\mathrm{read}^{(k)}_\Upsilon(e)$ determines the self-line functions $\varphi^{\mathbf{X}}_e$, the multisets $\{\{\varphi^{\mathbf{X}}_{e,g}\}\}_{g\ne e}$ of partner-line functions, and for every rung the multiset of the entries $\kappa^{\mathbf{X}}_\Upsilon(e,w)$ over the strings w of marks with at most k distinct marks other than e; in particular the tallies of e at the rungs $2,\dots,k+1$.

(iii) The reading determines the tallies of e at every rung $k \ge 2$, its self-line and partner-line functions for both parts, and its mass $\tau_e(\Upsilon)$.

Proposition 10.4 (One writing determines the reading)**.** *The reading of e is determined by any one writing of the potential from e. So the reading is constant on a set C of marks exactly when one pair of functions is the potential written from every mark of C: there are orders $\mathbf{g}_e$ of the rooms, $e \in C$, with $\Phi^{\mathbf{X}}_{e,\mathbf{g}_e} = \Phi^{\mathbf{X}}_{f,\mathbf{g}_f}$ for all $e, f \in C$ and both parts. In particular Υ is an atom exactly when one pair of functions is the potential written from every mark.*

10.2 Identifying writings

Theorem 10.5 (One table)**.** *Let M carry one table, $n \ge 2$, at the table's gauge, and let C be a set of marks. The reading is constant on C*

(i) at a lean, exactly when one polynomial is K written from every mark of C;

(ii) at the centre, exactly when one polynomial is, from every mark of C, either K or Ψ written from it.

Here K written from e in the order $\mathbf{g}$ is $K_{e,\mathbf{g}}$ with $k = n-1$, a polynomial in the position variables.

Theorem 10.6 (Several tables: the potential is read factor by factor)**.** *At a realization Υ write every $K_j(\alpha\circ x)$ and $\Psi_j(\alpha\circ x)$ as a positive constant times a product of irreducible polynomials in*

x, each positive on the orthant and equal to 1 at $x = \mathbf{1}$. Give each distinct such polynomial L the exponents

$$w_{\mathbf{E}}(L) = \sum \sigma_j > 0, \qquad w_{\mathbf{O}}(L) = \sum \pm\delta_j,$$

summed over its occurrences with multiplicity, with $+$ in a K_j and $-$ in a Ψ_j. Then $\Phi_{\mathbf{X}}(\log\alpha + \log x) = \Phi_{\mathbf{X}}(\log\alpha) + \sum_L w_{\mathbf{X}}(L)\log L(x)$, and two pairs $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ of this form, in the same variables, are equal exactly when they have the same polynomials with the same pairs of exponents $(w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))$. So the reading is constant on C exactly when one set of polynomials with exponents $\{(L, w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))\}$, with the variables placed in positions, is the family written from every mark of C. For tables whose weights factor across their molecules the polynomials L are the normalized Foreman polynomials of their restrictions to those molecules.

Corollary 10.7 (Moving blocks of partners reads alike)**.** *Let $\mathbf{g}, \mathbf{g}'$ be orders of the rooms of e and f with $(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}}) = (\Phi^{\mathbf{E}}_{f,\mathbf{g}'}, \Phi^{\mathbf{O}}_{f,\mathbf{g}'})$, and let β be the bijection of $\mathcal{E}$ with $\beta(e) = f$ and $\beta(g_i) = g'_i$. Then for all disjoint sets $S_1, \dots, S_m$ of the room of e,*

$$\Phi_{\mathbf{X}}\Big(\log\alpha + v_0\mathbf{1}_e + \sum_{i=1}^{m} v_i\mathbf{1}_{S_i}\Big) = \Phi_{\mathbf{X}}\Big(\log\alpha + v_0\mathbf{1}_f + \sum_{i=1}^{m} v_i\mathbf{1}_{\beta(S_i)}\Big).$$

So on a class every move of the gauge that shifts a mark and several sets of partners, each by one amount, reads alike: for every partition $S_1, \dots, S_m$ of the room, the potential moved by v_0 on the mark and by v_i on S_i reads alike from every mark of the class.

Proposition 10.8 (Computing the classes)**.** *Read every mark with $k = 0, 1, 2, \dots$ partners and group the marks by value. The level sets only refine as k grows (Proposition 10.3). A level set is a class as soon as one pair of functions is written from each of its marks (Proposition 10.4); the orders are built one partner at a time, keeping every view equal to the restriction of that one function, which loses nothing since a view with fewer partners is a restriction of a writing. Where no one function serves a whole level set, the marks from which a given function can be written form one class and the rest are read again. The computation ends by $k = n - 1$, and at a gauge α every step is an identity of polynomials with coefficients in the field generated by the weights and the coordinates of α; at the table's gauge that is the field generated by the weights, rational for rational weights and integers for lists (Proposition 10.2, Theorem 10.6).*

10.3 What a class holds

Proposition 10.9 (Masses and occupations on a class)**.** *At every realization Υ the masses $\tau_e(\Upsilon)$ and each of their two parts $\varphi^{\mathbf{E}}_e(0)$ and $\varphi^{\mathbf{O}}_e(0)$ (Proposition 9.10) are constant on every class. At the table's gauge:*

(i) $\sum_j \delta_j(2R_{j,e}(\mathbf{1}) - 1)$ is constant on every class, and $\kappa^{\mathbf{E}}_{\Upsilon}(e, e) = 8\sum_j \sigma_j R_{j,e}(\mathbf{1})(1 - R_{j,e}(\mathbf{1}))$;

(ii) with one table, the occupation $R_e(\mathbf{1})$ is constant on every class at a lean, and at the centre two marks e, f of one class have $R_f(\mathbf{1}) = R_e(\mathbf{1})$ or $R_f(\mathbf{1}) = 1 - R_e(\mathbf{1})$.

Chapter 11

Lean, gauge and label

11.1 The lean

Theorem 11.1 (The lean only splits)**.** *At every gauge and every $\boldsymbol{\sigma}$, the partition at a lean $\boldsymbol{\delta}$ refines the partition at the centre. An atom at a lean is an atom at the centre, and a realization that is not an atom at the centre is an atom at no lean.*

Proposition 11.2 (What of the lean is read)**.** *With the gauge α fixed, $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$ depends on $(\boldsymbol{\sigma},\boldsymbol{\delta})$ only through the rays $\{c\boldsymbol{\sigma}\}$ and $\{c'\boldsymbol{\delta}\}$, $c,c'>0$, and $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\alpha})=\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$. So at one pair, $\boldsymbol{\delta}=\frac{\delta}{\sigma}\boldsymbol{\sigma}$, there are two partitions: $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},0;\alpha})$ at the centre, and $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$, the same for every $\delta\neq 0$ and every $\boldsymbol{\sigma}$ on its ray, which refines it.*

Proposition 11.3 (A lean reads one side)**.** *Let Υ be at one pair (σ,δ), at the table's gauge, and put*

$$\Phi_K(y)=\frac{1}{\sigma}\sum_j \sigma_j \log K_j(e^y), \qquad \Phi_\Psi(y)=\frac{1}{\sigma}\sum_j \sigma_j \log \Psi_j(e^y)=\Phi_K(-y)+\langle \mathbf{1},y\rangle.$$

Then $\Phi_{\mathbf{E}}=\sigma(\Phi_K+\Phi_\Psi)$ and $\Phi_{\mathbf{O}}=\delta(\Phi_K-\Phi_\Psi)$, and:

(i) at every lean the reading of a mark determines, and is determined by, the multiset of writings of Φ_K from it; so for $\delta\neq 0$, $\mathrm{Cl}(\Upsilon)$ is the partition of the marks into the level sets of the multiset of writings of Φ_K, and equally of Φ_Ψ;

(ii) with the tensors $\mathcal{K}_k(w)=\frac{1}{\sigma}\sum_j \sigma_j\kappa_{j,\mathbf{1}}(w)$ on the strings of length k, rung k is $2^{k+1}\sigma\mathcal{K}_k$ for even k and $2^{k+1}\delta\mathcal{K}_k$ for odd k: at a basis lean every rung is a positive multiple of the cumulant sequence $(\mathcal{K}_k)_k$ of the bases, at a cobasis lean of the cumulant sequence $((-1)^k\mathcal{K}_k)_k$ of the cobases, and at the centre only the even rungs, common to the two sequences, are read.

A lean reads the molecule with a preference: at a basis lean it reads the cumulant sequence of the bases, at a cobasis lean that of the cobases, and at the centre only what the two sequences share. The lean can only split the classes of the centre, never join them.

11.2 The gauge

Theorem 11.4 (The gauge step)**.** *Let $\Upsilon=((s_j,t_j)_j;\alpha)$ be a realization of a molecule. If the self-line functions $\varphi_e^{\mathbf{E}}$ of all marks agree up to additive constants, in particular if Υ is an atom, then $\alpha\propto\mathbf{1}$.*

Corollary 11.5 (Scale changes nothing)**.** *At $\alpha = c\mathbf{1}$ every writing is the writing at $\mathbf{1}$ plus a constant that depends on neither the mark nor the order. So $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};c\mathbf{1}}) = \mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}})$, and every atom is an atom at the table's gauge.*

11.3 Answers and the label

Proposition 11.6 (Assembly from arbitrary seed tables)**.** *Let H be a set of bijections of $\mathcal{E}$ closed under composition and inverses that takes every mark to every other. Start from seed tables on $\mathcal{E}$, and for each seed include its image under every $h \in H$, weights transported with the bases and repetitions kept. Give every image of seed a the exponents (s_a, t_a); let q_a be the number of its images with multiplicity, r_a its rank, $D = \sum_a q_a(s_a r_a + t_a(n - r_a))$, $\mathbf{m} = \frac{D}{n}\mathbf{1}$, and suppose the active assembled family is connected. Then for every positive exponent pairs the gauge $\mathbf{1}$ is native, every proper wall is positive, and the realization $((s_a, t_a)_a; \mathbf{1})$ of the assembled family is an atom.*

(i) At $\mathbf{1}$ the occupations summed over the images of seed a are the same at every mark, $q_a r_a/n$, and the dual occupations summed likewise are $q_a(n - r_a)/n$; their sum with the coefficients s_a and t_a is D/n.

(ii) The images under all permutations of a seed with at least two bases form a connected family.

(iii) The seed weights may vary over any positive values before assembly, giving atoms at the one gauge $\mathbf{1}$.

Definition 11.7 (Answers and the label)**.** A molecule *answers yes* when some realization of it is atomic, and *no* otherwise. Let k be the number of molecules of a family with at least two marks and i the number of them that answer yes; units are atomic at every realization and are not counted. The family is *Foreman k-symmetric* when $i = k$ and *Foreman asymmetric* when $i < k$; an asymmetric family with $i > 0$ is *sub-symmetric (i of k)*. The *label* of the family is the pair (i, k).

Theorem 11.8 (The answer is read at the centre)**.** *A molecule answers yes if and only if some centre $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ is atomic. With one table this holds if and only if one polynomial is, from every mark, either K or Ψ written from it; the condition does not involve σ.*

Corollary 11.9 (The label is read at the centres)**.** *By Theorem 11.8 the label of a family is read at the centres $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ of the table's gauge of its molecules, and a molecule carrying one table is decided by reading it once, at any one such centre.*

Corollary 11.10 (The leaning realizations at one pair)**.** *Let a molecule be read at one pair at the table's gauge, with the strengths $\boldsymbol{\sigma}$. Its leaning realizations $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$, $\boldsymbol{\delta} = \frac{\delta}{\sigma}\boldsymbol{\sigma}$, $\delta \neq 0$, are atomic all together or not at all; if they are, so is the centre, and an atomic centre need not make them atomic.*

Whether a molecule can be atomic and whether a particular realization of it is atomic are read differently. The first is read at the centres of the table's gauge, at any one centre when the molecule carries one table; the second at the realization itself, and a lean may split what the centre held together.

Lemma 11.11 (No degenerate pair)**.** *At every realization Υ of a family and for marks $e \neq f$, $\kappa^{\mathbf{E}}_{\Upsilon}(e, f)^2 \leq \kappa^{\mathbf{E}}_{\Upsilon}(e, e)\,\kappa^{\mathbf{E}}_{\Upsilon}(f, f)$, with equality exactly when e or f is a unit or $\{e, f\}$ is a molecule. In particular the inequality is strict for any two marks of a molecule with at least three marks.*

Lemma 11.12 (Equal diagonals on a class)**.** *At every realization Υ, for each part and every $k \geq 2$, the diagonal entry $\kappa^{\mathbf{X}}_{\Upsilon}(e,\dots,e) = 2^k(\varphi^{\mathbf{X}}_e)^{(k-1)}(0)$ of rung k is the same at every mark of a class, the reading determining the self-line functions (Proposition 10.3(iii)); in particular $\kappa^{\mathbf{E}}_{\Upsilon}(e,e) = 4\,(\varphi^{\mathbf{E}}_e)'(0)$ is constant on every class, and at an atom it is the same at every mark.*

11.4 Maps as certificates

Definition 11.13 (Carrying and exchanging)**.** Let F and F' be families on $\mathcal{E}$ and $\mathcal{E}'$ with tables indexed by $\{1,\dots,N\}$ and $\{1,\dots,N'\}$. A bijection $g : \mathcal{E} \to \mathcal{E}'$ *carries* F to F' when some bijection $h : \{1,\dots,N\} \to \{1,\dots,N'\}$ has $g(\mathcal{B}_j) = \mathcal{B}'_{h(j)}$, $\theta'_{h(j),g(T)} = \theta_{j,T}$ and $(\sigma'_{h(j)}, \delta'_{h(j)}) = (\sigma_j, \delta_j)$; it *exchanges* F with F' when instead $T \mapsto \mathcal{E}' \setminus g(T)$ maps $\mathcal{B}_j$ onto $\mathcal{B}'_{h(j)}$ with $\theta'_{h(j),\mathcal{E}'\setminus g(T)} = \theta_{j,T}$ and the same condition on the pairs. With $F' = F$, g carries or exchanges the family. Write $(g^*y)_e = y_{g(e)}$ for $y \in \mathbb{R}^{\mathcal{E}'}$.

Proposition 11.14 (What a map certifies)**.** *At gauges α of F and α' of F' with $g^* \log \alpha' = \log \alpha$, the tables' gauges in particular, and in this statement writing $\Phi^{\boldsymbol{\delta}}_{\mathbf{O}}$ for the odd potential at the lean $\boldsymbol{\delta}$: if g carries F to F' then $\Phi_{\mathbf{X}} \circ g^* = \Phi'_{\mathbf{X}}$, and the reading of $g(e)$ in F' equals that of e in F; if g exchanges F with F' then $\Phi_{\mathbf{E}} \circ g^* = \Phi'_{\mathbf{E}}$ and $\Phi^{-\boldsymbol{\delta}}_{\mathbf{O}} \circ g^* = \Phi'^{\boldsymbol{\delta}}_{\mathbf{O}}$, and the reading of $g(e)$ in F' at $-\boldsymbol{\delta}$ equals that of e in F at $\boldsymbol{\delta}$. So a map carrying the family takes classes onto classes, and an exchange of the family takes the classes at $\boldsymbol{\delta}$ onto those at $-\boldsymbol{\delta}$ and the classes of the centre onto the classes of the centre.*

A map is a certificate, not a definition: classes are the level sets of the reading, and a map carrying or exchanging the family is one way to exhibit that two marks read alike. With one table the writings and the maps say the same thing (Corollary 11.16); with several tables the reading is the definition.

Corollary 11.15 (An exchange negates odd cumulants)**.** *If g exchanges a family, with h the bijection of its tables (Definition 11.13), then under the weight laws $\kappa_{h(j),\mathbf{1}}(g(w)) = (-1)^k \kappa_{j,\mathbf{1}}(w)$ for every table j and every string w of length $k \geq 2$; so the stills at the table's gauge satisfy $\kappa_{\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}}(g(w)) = \kappa_{\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}}(w)$, and g maps every class of the centre onto itself.*

Corollary 11.16 (With one table the writings are certificates)**.** *In a one-table molecule at the table's gauge, e and f lie in one class at a lean exactly when some bijection taking e to f carries the table, and in one class at the centre exactly when some bijection taking e to f carries or exchanges it.*

Theorem 11.17 (A class across the centre)**.** *In a one-table molecule at the table's gauge, a class C of the centre is, at every lean, one class or two classes with equally many marks. In the second case the two are C_K and C_Ψ, the marks of C from which K, respectively Ψ, is written as one and the same polynomial, and an exchange β built from two such writings has $\beta(C) = C$ and $\beta(C_K) = C_\Psi$.*

Leaning away from the centre, a class of a one-table molecule either stays whole or splits into the half from which the basis polynomial is written and the half from which the cobasis polynomial is: the lean sign decides which half the Foreman meets first (Corollary 13.11).

Chapter 12

Restriction to a bunch

12.1 Patterns and the Taylor cube

Definition 12.1 (Patterns and children)**.** Let $(\mathcal{B},\theta)$ be a table of rank r on $\mathcal{E}$. A bunch (Definition 3.1) is here any nonempty set $B\subseteq\mathcal{E}$, and $R=\mathcal{E}\setminus B$ is its *room*; for a family the objects of a bunch in table j carry the index j. The *patterns* of B are the traces of the bases on it, $\mathrm{pat}(B)=\{T\cap B:\ T\in\mathcal{B}\}$, and $\mathrm{pat}_j(B)$ for table j of a family. For $p\in\mathrm{pat}(B)$ the *signed pattern* is $\chi_p=\mathbf{1}_p-\mathbf{1}_{B\setminus p}\in\mathbb{R}^B$ and the *signed level* is $\ell_p=|p|-\frac{|B|}{2}$. The *child* at p is the table on R with the bases $\mathcal{B}_{(p)}=\{T\setminus p:\ T\cap B=p\}$ and the weights $\theta_{(p),X}=\theta_{X\cup p}$; its Foreman polynomials and its centred polynomial are $K_{(p)}$, $\Psi_{(p)}$ and $\Lambda_{(p)}$.

Lemma 12.2 (Children are tables)**.** *$\mathcal{B}_{(p)}$ is the set of bases of the minor contracting p and deleting $B\setminus p$ (Definition 7.5); it has the exchange property and rank $r-|p|$.*

Lemma 12.3 (Children do not widen)**.** *Let B be a bunch of a table with the room R, p a pattern of B, $A\subseteq R$ and $W\subseteq B$. The width of A in the child at p is at most the width of $A\cup W$ in the table: $c_{(p)}(A)\le c(A\cup W)$.*

Definition 12.4 (Communicatroid; interchange)**.** The *communicatroid* of B is the indexed family of its children $(\mathcal{B}_{(p)},\theta_{(p)})_{p\in\mathrm{pat}(B)}$, tables on the room indexed by their patterns; its *gluing* is $\{U\sqcup p:\ p\in\mathrm{pat}(B),\ U\in\mathcal{B}_{(p)}\}$ with the weights $\theta_{(p),U}$. An indexed family $(\mathcal{A}_p)_p$ of tables on R, indexed by sets $p\subseteq B$ with $\mathcal{A}_p$ of rank $r-|p|$, has *interchange* when, for $U\sqcup p$ and $U'\sqcup p'$ in its gluing and $x\in(U\sqcup p)\setminus(U'\sqcup p')$, there is $y\in(U'\sqcup p')\setminus(U\sqcup p)$ with

$$U-x+y\in\mathcal{A}_p\ \ (x,y\in R),\qquad U+y\in\mathcal{A}_{p-x}\ \ (x\in B,\ y\in R),$$
$$U-x\in\mathcal{A}_{p+y}\ \ (x\in R,\ y\in B),\qquad U\in\mathcal{A}_{p-x+y}\ \ (x,y\in B):$$

a step of the exchange property inside a child, one level down, one level up, or along a level.

Proposition 12.5 (Interchange is the parent's exchange property)**.** *The gluing of the communicatroid of B is the table, weights included, and the communicatroid has interchange. Conversely, an indexed family of tables on R with interchange is the communicatroid of B in its gluing, which is a table: the children remember the parent because the parent is the glued children.*

Proposition 12.6 (Adjacent siblings)**.** *If p and $p+b$ are patterns of B, the children at p and at $p+b$ are the deletion and the contraction of b in one table on $R+b$, the minor contracting p and deleting $B\setminus(p+b)$. So the child at $p+b$ is a quotient of the child at p of rank one less: every*

set of room marks that is maximal for its rank in the child at $p+b$ is maximal for its rank in the child at p, every basis of the child at $p+b$ extends by one room mark to a basis of the child at p, and every basis of the child at p less one suitable mark is a basis of the child at $p+b$.

Theorem 12.7 (The Taylor cube)**.** *For $\alpha=(\alpha_B,\alpha_R)$ and $z=(z_B,z_R)$,*

$$K(\alpha)=\sum_{p\in\mathrm{pat}(B)}\alpha_B^{p}K_{(p)}(\alpha_R),\qquad \Psi(\alpha)=\sum_{p\in\mathrm{pat}(B)}\alpha_B^{B\setminus p}\Psi_{(p)}(\alpha_R),$$
$$\Lambda(z)=\sum_{p\in\mathrm{pat}(B)}e^{\langle z_B,\chi_p\rangle}\Lambda_{(p)}(z_R),$$

with $K_{(p)}=(\partial^p K)|_{\alpha_B=0}$. Along the scale of B, $\alpha_B=e^{2z_B}=\rho u$ with $u\in\Delta_B$, the factor of p in Λ is $e^{\langle z_B,\chi_p\rangle}=\rho^{\ell_p}u^{\chi_p/2}$; the values $|p|$ fill $\{d(B),\dots,r(B)\}$, and the signed levels fill a window of width $c(B)$ centred at $\varepsilon(B)/2$. For $B=\{g\}$ this is the expansion $K=K_{\setminus g}+\alpha_g K_{/g}$, $\Psi=\alpha_g\Psi_{\setminus g}+\Psi_{/g}$ of Definition 7.3.

The Taylor cube is the bunch read on its own coordinates: each pattern contributes a monomial on the bunch, at the level $|p|$, times the polynomial of its child on the room. The levels that occur fill the window of the bunch (Theorem 7.13), and nothing about the room is lost, since the children glued back are the table.

Theorem 12.8 (The pattern tensor)**.** *Under the basis law at α, $\mathbb{P}_\alpha(T\cap B=p)=\alpha_B^{p}K_{(p)}(\alpha_R)/K(\alpha)$, and under the cobasis law $\mathbb{P}^*_\alpha(T\cap B=p)=\alpha_B^{B\setminus p}\Psi_{(p)}(\alpha_R)/\Psi(\alpha)$. The moments of the bunch are the upper sums $\mathbb{P}_\alpha(T\cap B\supseteq S)=\sum_{p\supseteq S}\mathbb{P}_\alpha(T\cap B=p)$, and the joint cumulants $\kappa_\alpha(w)$ on the strings w of marks of B are their image under the moment–cumulant relation. Scaling α_B multiplies the entry of p by its own monomial, and inverting the gauge swaps the two laws pattern by pattern (Proposition 12.28(ii) below).*

Proposition 12.9 (Tinkers along a bunch; selecting patterns)**.** *For a* tinker *$v=(v_p)_{p\in\mathrm{pat}(B)}$ of positive numbers, the weights $v_{T\cap B}\theta_T$ define the* tinkered table*, with the same bases and the Foreman polynomials*

$$K_v(\alpha)=\sum_p v_p\,\alpha_B^{p}K_{(p)}(\alpha_R),\qquad \Psi_v(\alpha)=\sum_p v_p\,\alpha_B^{B\setminus p}\Psi_{(p)}(\alpha_R).$$

Selecting the pattern p on B, that is keeping the bases with $T\cap B=p$, and then p' on B' selects $p\cup p'$ on $B\cup B'$ when $p\cap B'=p'\cap B$, and nothing otherwise; so the selections on a table with n marks are at most the 3^n conditions that require each mark present, absent, or nothing. At the centre every tinkered table keeps **1** *native (Proposition 8.21).*

12.2 The Skeleton polynomials of a bunch

Definition 12.10 (Skeleton polynomials, values and coins)**.** At a gauge α, with the level $N_B=|T\cap B|$ and the *colevel* $N_B^*=|B\setminus T|$, the *Skeleton polynomials* of B are

$$\mathsf{Q}_B^K(x;\alpha)=\mathbb{E}_{T\sim\mathbb{P}_\alpha}\big[x^{N_B-d(B)}\big]=x^{-d(B)}\frac{K(x\alpha_B,\alpha_R)}{K(\alpha)},$$
$$\mathsf{Q}_B^\Psi(x;\alpha)=\mathbb{E}_{T\sim\mathbb{P}^*_\alpha}\big[x^{N_B^*-d^*(B)}\big]=x^{-d^*(B)}\frac{\Psi(x\alpha_B,\alpha_R)}{\Psi(\alpha)},$$

and $\mathsf{Q}^K_{j,B}$, $\mathsf{Q}^\Psi_{j,B}$ for table j of a family. Each has degree $c(B)$, positive coefficients and value 1 at $x=1$, so its roots avoid $[0,\infty)$; writing them $-\xi_1,\dots,-\xi_{c(B)}$,

$$\mathsf{Q}^K_B(x;\alpha)=\prod_{i=1}^{c(B)}\frac{x+\xi_i}{1+\xi_i}=\prod_{i=1}^{c(B)}(1-p_i+p_ix),\qquad p_i=\frac{1}{1+\xi_i}.$$

The ξ_i are the *Skeleton values*, their multiset $\operatorname{spec}^K_B(\alpha)\subset\mathbb{C}\setminus(-\infty,0]$ is the *Skeleton spectrum*, and the p_i are the *coins*; likewise $\operatorname{spec}^\Psi_B(\alpha)$ and the coins of Q^Ψ_B. The *cumulants of a coin* are $\varkappa_1(p)=p$ and $\varkappa_{k+1}(p)=p(1-p)\,\varkappa_k'(p)$.

The Skeleton polynomial is the law of the level of the bunch, shifted to start at its floor: its degree is the width of the bunch, and its roots are the negatives of the Skeleton values. When the spectrum is positive the level less its floor $d(B)$ is a sum of $c(B)$ independent $\{0,1\}$-valued variables whose means are the coins.

Proposition 12.11 (Relations of the Skeleton polynomials)**.** *(i)* Scaling. *Scaling α_B by $\lambda>0$ divides the spectrum by λ:*

$$\mathsf{Q}^K_B\big(x;(\lambda\alpha_B,\alpha_R)\big)=\mathsf{Q}^K_B(\lambda x;\alpha)/\mathsf{Q}^K_B(\lambda;\alpha);$$

likewise for Ψ; scaling all of α changes nothing.

(ii) Complement. $\mathsf{Q}^K_R(x;\alpha)=x^{c(B)}\mathsf{Q}^K_B(x^{-1};\alpha)$*: the spectrum of the room is the reciprocal of that of the bunch, and each coin p_i becomes $1-p_i$; likewise for Ψ.*

(iii) Cremona. $\mathsf{Q}^\Psi_B(x;\alpha)=x^{c(B)}\mathsf{Q}^K_B(x^{-1};\alpha^{-1})$*; at the table's gauge,* $\mathsf{Q}^\Psi_B(x;\mathbf{1})=x^{c(B)}\mathsf{Q}^K_B(x^{-1};\mathbf{1})$.

(iv) Coins. $\log\mathsf{Q}^K_B(e^v;\alpha)=\sum_i\log(1-p_i+p_ie^v)$*; under $\mathbb{P}_\alpha$ the level N_B has mean $d(B)+\sum_i p_i=\sum_{e\in B}R_e(\alpha)$ and k-th cumulant $\sum_i\varkappa_k(p_i)$ for $k\ge2$. When the spectrum is positive the coins lie in $(0,1)$ and $N_B-d(B)$ is distributed as a sum of independent $\{0,1\}$-valued variables whose means are the coins.*

Proposition 12.12 (The still read along a bunch)**.** *Let a family be at a gauge α, let $p_{j,i}$ be the coins of $\mathsf{Q}^K_{j,B}(\cdot;\alpha)$ and $p^*_{j,i}$ those of $\mathsf{Q}^\Psi_{j,B}(\cdot;\alpha)$, and put $\partial_B=\sum_{e\in B}\partial_{y_e}$. Then*

$$\sum_j\Big(s_j\sum_i p_{j,i}+t_j\sum_i p^*_{j,i}\Big)=\partial_B\Phi(\log\alpha)-\sum_j\big(s_jd_j(B)+t_jd^*_j(B)\big),$$
$$\partial_B^k\Phi(\log\alpha)=\sum_j\Big(s_j\sum_i\varkappa_k(p_{j,i})+t_j\sum_i\varkappa_k(p^*_{j,i})\Big)\qquad(k\ge2).$$

At a native gauge the first total is the wall κ_B at the masses $\boldsymbol{\tau}$, and on the balance wall it is κ_B; the entries of rung k of the still of $\big((s_j,t_j)_j;\alpha\big)$, summed over B^k, are 2^k times the second total.

Proposition 12.13 (Coins of a tensorminental table)**.** *In the setting of Theorem 7.23, let $p_i^{(kl)}$, $1\le i\le|B|$, be the eigenvalues of $\Pi^{(kl)}_{BB}$. Then*

$$x^{d(B)}\mathsf{Q}^K_B(x;\alpha)=\sum_{k,l}w_{kl}\prod_{i=1}^{|B|}\big(1-p_i^{(kl)}+p_i^{(kl)}x\big)\,:$$

the Skeleton polynomial is the w-mixture of the coin products of the components. At tensor rank one over $\mathbb{F}\in\{\mathbb{R},\mathbb{C}\}$ there is one component, and $\Pi^{(11)}_{BB}$ is similar to the compression to $\mathbb{F}^B$ of the

orthogonal projection onto the row space of $AD_\alpha^{1/2}$, A the slice; its eigenvalues are $\cos^2\vartheta_i$ for the principal angles ϑ_i between $\mathbb{F}^B$ and that row space. Of these, $d(B)$ are 1 and $d^*(B)$ are 0, and the $c(B)$ coins are the values $\cos^2\vartheta_i \in (0,1)$, with the positive Skeleton values $\tan^2\vartheta_i$.

Definition 12.14 (The pair of Skeleton functions)**.** The *even* and *odd Skeleton functions* of B are $\mathsf{Q}_B^{\mathbf{E}}(x;\alpha) = \mathsf{Q}_B^K(x;\alpha)\,\mathsf{Q}_B^\Psi(x;\alpha)$ and $\mathsf{Q}_B^{\mathbf{O}}(x;\alpha) = \mathsf{Q}_B^K(x;\alpha)/\mathsf{Q}_B^\Psi(x;\alpha)$. So

$$\mathsf{Q}_B^{\mathbf{E}}(x;\alpha) = \mathbb{E}_{(T,T')\sim\mathbb{P}_\alpha\otimes\mathbb{P}_\alpha^*}\Big[x^{|T\cap B|-d(B)+|B\setminus T'|-d^*(B)}\Big],$$

and its roots are the negatives of the *joint spectrum* $\operatorname{spec}_B^K(\alpha)\cup\operatorname{spec}_B^\Psi(\alpha)$.

Proposition 12.15 (A set on its own scale is read by its Skeleton functions)**.** *For a family at a gauge α, a set C of marks and $x = e^v$,*

$$\Phi_{\mathbf{E}}(\log\alpha + v\mathbf{1}_C) - \Phi_{\mathbf{E}}(\log\alpha) = \sum_j \sigma_j\Big(\log\mathsf{Q}_{j,C}^{\mathbf{E}}(x;\alpha) + \big(d_j(C)+d_j^*(C)\big)v\Big),$$
$$\Phi_{\mathbf{O}}(\log\alpha + v\mathbf{1}_C) - \Phi_{\mathbf{O}}(\log\alpha) = \sum_j \delta_j\Big(\log\mathsf{Q}_{j,C}^{\mathbf{O}}(x;\alpha) + \big(d_j(C)-d_j^*(C)\big)v\Big).$$

Along the scale of a set the K-half of each table's joint spectrum carries the exponent $s_j = \sigma_j + \delta_j$ and the Ψ-half carries $t_j = \sigma_j - \delta_j$.

12.3 Levels

Definition 12.16 (Levels)**.** For $d(B) \le k \le r(B)$ the *level-k patterns* are $\operatorname{pat}^{(k)}(B) = \{p \in \operatorname{pat}(B) :\ |p| = k\}$, the *level-$k$ bases* are $\mathcal{B}_B^{(k)} = \{T \in \mathcal{B} :\ |T\cap B| = k\}$, and the *level letter* is

$$K_B^{(k)}(\alpha) = \sum_{T\in\mathcal{B}_B^{(k)}} \theta_T\alpha^T = \sum_{p\in\operatorname{pat}^{(k)}(B)} \alpha_B^p K_{(p)}(\alpha_R).$$

For $d^*(B) \le k \le |B| - d(B)$ the *colevel letter* is $\Psi_B^{(k)}(\alpha) = \sum_{T:\,|B\setminus T|=k}\theta_T\alpha^{\mathcal{E}\setminus T}$. The *excess* of the level k, of its patterns and of its bases is $k - d(B)$, and that of the colevel k is $k - d^*(B)$. Then $K = \sum_k K_B^{(k)}$, $\Psi = \sum_k \Psi_B^{(k)}$,

$$x^{d(B)}\mathsf{Q}_B^K(x;\alpha) = \sum_k x^k\,\frac{K_B^{(k)}(\alpha)}{K(\alpha)}, \qquad x^{d^*(B)}\mathsf{Q}_B^\Psi(x;\alpha) = \sum_k x^k\,\frac{\Psi_B^{(k)}(\alpha)}{\Psi(\alpha)}.$$

Proposition 12.17 (Pattern levels are lifts)**.** *Let $\mathcal{B}|B$ be the restriction to B and $\mathcal{B}/R$ the contraction of the room, a table on B with rank function $r_{\mathcal{B}/R}(X) = r(X\cup R) - r(R)$ and rank $d(B)$.*

(i) $p \subseteq B$ is a pattern if and only if p is independent and $r_{\mathcal{B}/R}(p) = d(B)$.

(ii) For $i = k - d(B)$, the independent sets $X \subseteq B$ with $|X| - r_{\mathcal{B}/R}(X) \le i$ are the independent sets of a table on B of rank k, the i-th lift of $\mathcal{B}/R$ toward $\mathcal{B}|B$, whose bases are the level-k patterns. So every pattern level $\operatorname{pat}^{(k)}(B)$ is a table on B.

(iii) The room sides $T\cap R$ of the level-k bases are the bases of the $(r(B)-k)$-th lift of $\mathcal{B}/B$ toward $\mathcal{B}|R$.

Definition 12.18 (Interaction)**.** Let a table have a bunch B with room R and $\mathrm{pat}(B) = 2^B$: every $p \subseteq B$ is a pattern, equivalently $c(B) = |B|$, that is B is independent and some basis avoids it (Proposition 12.17(i)). The *interaction* and the *dual interaction* of B are

$$\iota_B(\alpha_R) = \sum_{p\subseteq B} (-1)^{|B\setminus p|} \log K_{(p)}(\alpha_R), \qquad \iota^*_B(\alpha_R) = \sum_{p\subseteq B} (-1)^{|p|} \log \Psi_{(p)}(\alpha_R),$$

the second the interaction of the dual table, and for a split of B into blocks $B_1, \dots, B_k$,

$$\iota_{(B_1,\dots,B_k)}(\alpha_R) = \sum_{A\subseteq\{1,\dots,k\}} (-1)^{k-|A|} \log K_{(\cup_{a\in A} B_a)}(\alpha_R).$$

For two marks, $\iota_{\{e,f\}} = \log \frac{K_{(\{e,f\})} K_{(\varnothing)}}{K_{(\{e\})} K_{(\{f\})}}$, the logarithm of the odds ratio of the presence of the two marks. For a family with $\mathrm{pat}_j(B) = 2^B$ in every active table j, the *interaction* of B is the pair

$$\iota^{\mathbf{E}}_B = \sum_j \sigma_j\big(\iota_{j,B} + \iota^*_{j,B}\big), \qquad \iota^{\mathbf{O}}_B = \sum_j \delta_j\big(\iota_{j,B} - \iota^*_{j,B}\big), \qquad \iota^{\mathbf{E}}_B + \iota^{\mathbf{O}}_B = \sum_j \big(s_j \iota_{j,B} + t_j \iota^*_{j,B}\big),$$

the interactions of the even and the odd part of the potential (Definition 8.18).

Theorem 12.19 (Levels are sums of children)**.** $\mathcal{B}^{(k)}_B = \bigsqcup_{p\in\mathrm{pat}^{(k)}(B)} \mathcal{B}_{[p]}$, *where* $\mathcal{B}_{[p]} = \{T \in \mathcal{B} : T \cap B = p\}$ *is the set of bases of the table on* $\mathcal{E}$ *that has* p *as coloops,* $B \setminus p$ *as loops and the child at* p *on* R*, with the parent's weights; its basis polynomial is* $\alpha^p_B K_{(p)}(\alpha_R)$*. A bijection* g *of* $\mathcal{E}$ *that carries the table and keeps* B *takes* $\mathcal{B}_{[p]}$ *to* $\mathcal{B}_{[g(p)]}$*. The extreme levels have constant children:*

$$\mathcal{B}^{(d(B))}_B = \{p \sqcup X :\ p \in \mathcal{B}/R,\ X \in \mathcal{B}|R\}, \qquad \mathcal{B}^{(r(B))}_B = \{p \sqcup X :\ p \in \mathcal{B}|B,\ X \in \mathcal{B}/B\};$$

for a table determinantal over $\mathbb{R}$ *or* $\mathbb{C}$ *the weights of the extreme levels factor along these products, and with them the extreme letters,* $K^{(d(B))}_B(\alpha) = K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}|R}(\alpha_R)$ *and* $K^{(r(B))}_B(\alpha) = K_{\mathcal{B}|B}(\alpha_B)\, K_{\mathcal{B}/B}(\alpha_R)$*.*

The bases of one level need not form a table: a step of the exchange property may leave the level, and it then moves to an adjacent one (Proposition 12.5). What a level keeps is its patterns, which form a table (Proposition 12.17); its bases are the glued children of those patterns, and children on adjacent levels are linked by Proposition 12.6. The level is the coordinate the bunch keeps for itself, and the children hand the rest down to the room.

Proposition 12.20 (The bottom of a chain factors over its steps)**.** *Let* $(\mathcal{B}, \theta)$ *be a table on* $\mathcal{E}$ *and* $\varnothing = S_0 \subsetneq S_1 \subsetneq \cdots \subsetneq S_\ell \subsetneq S_{\ell+1} = \mathcal{E}$*,* $\ell \geq 1$*, a chain with the steps* $Z_i = S_i \setminus S_{i-1}$*,* $1 \leq i \leq \ell+1$*. Some bases are at the bottom of every* S_i*,* $|T \cap S_i| = d(S_i)$ *for* $1 \leq i \leq \ell$*. For any one of them,* T_0*, and every* i*, the traces on* Z_i *of these bases are the bases of the child* $\mathcal{B}_{(T_0\setminus Z_i)}$ *at the pattern* $T_0 \setminus Z_i$ *of* $\mathcal{E} \setminus Z_i$ *(Definition 12.1), the* step table at the bottom *at* Z_i*, which depends on* S_{i-1} *and* S_i *alone; and the bases at the bottom of every* S_i *are exactly the unions* $\bigsqcup_i X_i$ *of one basis* X_i *of each step table. Likewise the bases at the top of every* S_i*,* $|T \cap S_i| = r(S_i)$*, with one of them* T_1*, give the* step tables at the top $\mathcal{B}_{(T_1\setminus Z_i)}$*.*

For a list, the terms of K *of the least degree in* α_{S_i} *for every* i *at once are the product* $\prod_i K_{\mathcal{B}_{(T_0\setminus Z_i)}}(\alpha_{Z_i})$*, and those of* Ψ *the product* $\prod_i \Psi_{\mathcal{B}_{(T_1\setminus Z_i)}}(\alpha_{Z_i})$*; the factors are the* step letters *of* K *and of* Ψ*, the same products holding for a table whose weights on these bases are products of one factor for each step. For the linear letter* α_J*, the basis polynomial of* $\binom{J}{1}$ *read on* $\mathcal{E}$ *with the marks outside* J *as loops, the step letter is* $\alpha_{J\cap Z_i}$ *at the first step* Z_i *with* $J \subseteq S_i$ *and* 1 *at the other*

steps. Along a maximal chain, every step one mark, exactly one basis is at the bottom of every S_i, and for a list the terms of K of the least degree in every α_{S_i} at once are one monomial with the coefficient 1.

Theorem 12.21 (Shared walls)**.** *Let $L_i = \sum_h K_{i,h}$, $i = 1, \dots, q$, each a sum of the basis polynomials $K_{i,h}$ of tables $\mathcal{B}_{i,h}$ on $\mathcal{E}$, with exponents $w_i > 0$, and put $d_{i,h}(S) = \min_{T \in \mathcal{B}_{i,h}} |T \cap S|$. Then*

$$\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} \prod_i L_i(\alpha)^{-w_i}\, d\alpha$$

converges if and only if there are $w_{i,h} > 0$ with $\sum_h w_{i,h} = w_i$ and $m_S > \sum_{i,h} w_{i,h}\, d_{i,h}(S)$ for every nonempty $S \subseteq \mathcal{E}$; equivalently $\mathbf{m} \in \operatorname{int}\big(\sum_i w_i \operatorname{conv} \bigcup_h P_{i,h} + \mathbb{R}^{\mathcal{E}}_{\geq 0}\big)$, with $P_{i,h}$ the base polytope of $\mathcal{B}_{i,h}$.

Proposition 12.22 (The walls of one level are exact)**.** *With P the base polytope, $P_B^{(k)} = \operatorname{conv}\{\mathbf{1}_T : T \in \mathcal{B}_B^{(k)}\}$ and $d_B^{(k)}(S) = \min_{T \in \mathcal{B}_B^{(k)}} |T \cap S|$:*

$$P_B^{(k)} = P \cap \{x_B = k\}, \qquad P_B^{(k)} + \mathbb{R}^{\mathcal{E}}_{\geq 0} = \{x :\ x_S \geq d_B^{(k)}(S) \text{ for all } S\}.$$

So for $w > 0$ the integral $\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} K_B^{(k)}(\alpha)^{-w} d\alpha$ converges exactly when $m_S > w\, d_B^{(k)}(S)$ for every nonempty S. The one-dimensional faces of $P_B^{(k)}$ that do not join two bases differing by a single step of the exchange property are diagonals of square two-dimensional faces of P, one step into B and one out of it.

For several levels together the 0/1 walls need not be exact: the facets of $\sum_k w_k P_B^{(k)} + \mathbb{R}^{\mathcal{E}}_{\geq 0}$ are normal to rays of the common refinement of the normal fans of the $P_B^{(k)}$, which can lie outside $\{0,1\}^{\mathcal{E}}$, and the exact domain of several levels together is then the one of Theorem 12.21.

Corollary 12.23 (The native level split is interior)**.** *Let a family with real data, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, have a native gauge $\alpha^\star$, and give the level letters $K_{j,B}^{(k)}$ of table j the exponents $s_j\, \mathbb{P}_{j,\alpha^\star}(N_B = k)$ and its colevel letters $\Psi_{j,B}^{(k)}$ the exponents $t_j\, \mathbb{P}^*_{j,\alpha^\star}(N_B^* = k)$. Then $\boldsymbol{\tau}$ lies in the relative interior of the Minkowski sum of the level and colevel polytopes, each multiplied by the exponent of its letter, and every wall of these letters at the masses $\boldsymbol{\tau}$,*

$$\tau_S - \sum_{j,k} \Big(s_j\, \mathbb{P}_{j,\alpha^\star}(N_B = k)\, d_{j,B}^{(k)}(S) + t_j\, \mathbb{P}^*_{j,\alpha^\star}(N_B^* = k)\, d_{j,B}^{*(k)}(S) \Big),$$

$d_{j,B}^{(k)}$ the level counts of Proposition 12.22 and $d_{j,B}^{*(k)}(S)$ the least of $|S \setminus T|$ over the colevel-k bases, is positive except on the sets S on which x_S is constant over every level and colevel polytope, among them B and R. If the marks of B are joined to one another, and the marks of R to one another, by chains in which consecutive marks lie in a common circuit of a table with a positive exponent, as they are when such a table is connected, then the directions v with $\langle v, x\rangle$ constant on that Minkowski sum are spanned by $\mathbf{1}_B$ and $\mathbf{1}_R$, there are no exceptions besides these and $\varnothing, \mathcal{E}$, and

$$\int_{\Delta_B \times \Delta_R} u^{\boldsymbol{\tau}|_B - \mathbf{1}} v^{\boldsymbol{\tau}|_R - \mathbf{1}} \prod_{j,k} K_{j,B}^{(k)}(u,v)^{-s_j \mathbb{P}_{j,\alpha^\star}(N_B = k)} \Psi_{j,B}^{(k)}(u,v)^{-t_j \mathbb{P}^*_{j,\alpha^\star}(N_B^* = k)}\, d\omega_B(u)\, d\omega_R(v)$$

converges.

12.4 Shares and the native split

Definition 12.24 (Shares, loads, children of a family)**.** For a family and a bunch B, *shares* are $\mathbf{a}_j = (a_{j,p})_{p\in \mathrm{pat}_j(B)}$ with $|\mathbf{a}_j| = s_j$ and $\mathbf{b}_j = (b_{j,p})_{p\in \mathrm{pat}_j(B)}$ with $|\mathbf{b}_j| = t_j$; the *pair shares* are $\sigma_{j,p} = \frac{a_{j,p}+b_{j,p}}{2}$ and $\delta_{j,p} = \frac{a_{j,p}-b_{j,p}}{2}$. A K-share of pattern p loads the marks of p, and a Ψ-share loads those of $B \setminus p$: the *load* and the *odd load* on $e \in B$ are

$$\mathrm{load}_e = \sum_j \Big(\sum_{p\ni e} a_{j,p} + \sum_{p\not\ni e} b_{j,p} \Big) = \sigma + \langle \boldsymbol{\delta} \rangle_e, \qquad \langle \boldsymbol{\delta} \rangle_e = \sum_{j,p} \delta_{j,p} \chi_p(e).$$

The *children* are the family

$$F_B(\mathbf{a}, \mathbf{b}) = \big((\mathcal{B}_{j,(p)}, \theta_{j,(p)});\, a_{j,p}, b_{j,p} \big)_{j,p} \quad \text{on } R.$$

Definition 12.25 (Pattern probabilities; the native split)**.** At a gauge β the *pattern probabilities* of table j are $\pi_{j,p}(\beta) = \mathbb{P}_{j,\beta}(T \cap B = p)$ and $\pi^*_{j,p}(\beta) = \mathbb{P}^*_{j,\beta}(T \cap B = p)$. At a native gauge $\alpha^\star$ the *native split* is the choice of shares $\mathbf{a}^\star$, $\mathbf{b}^\star$ with $a^\star_{j,p} = s_j \pi_{j,p}(\alpha^\star)$ and $b^\star_{j,p} = t_j \pi^*_{j,p}(\alpha^\star)$, and $F_B(\mathbf{a}^\star, \mathbf{b}^\star)$ are the children at the native split.

Theorem 12.26 (Nativity is hereditary)**.** *$\alpha^\star|_R$ is a native gauge for the children $F_B(\mathbf{a}^\star, \mathbf{b}^\star)$ with target $\boldsymbol{\tau}|_R$, and they lie on their own balance wall at the masses $\boldsymbol{\tau}|_R$; every set $S \subseteq R$ with $c_{j,(p)}(S) = 0$ for every child table is balanced, $\tau_S = \sum_{j,p} \big(a^\star_{j,p} r_{j,(p)}(S) + b^\star_{j,p} r^*_{j,(p)}(S) \big)$.*

Proposition 12.27 (The centre is hereditary)**.** *Let $\Upsilon = ((s_j, t_j)_j; \mathbf{1})$ be at the table's gauge, with its masses. The native split gives the child (j,p) the pair*

$$\sigma_{j,p} = \pi_{j,p}(\mathbf{1})\, \sigma_j, \qquad \delta_{j,p} = \pi_{j,p}(\mathbf{1})\, \delta_j, \qquad \pi_{j,p}(\mathbf{1}) = \mathbb{P}_{j,\mathbf{1}}(T \cap B = p) > 0,$$

and the gauge $\mathbf{1}$ on R, the children's own table's gauge, native for them. At a centre every child sits at a centre; at one pair the children are at the same pair; a full lean is handed down full.

Proposition 12.28 (The split is forced)**.** *(i) For every gauge β and every probability vector w on $\mathrm{pat}_j(B)$ with every $w_p > 0$,*

$$K_j(\beta) \;\geq\; \prod_p \Big(\frac{\beta_B^p K_{j,(p)}(\beta_R)}{w_p} \Big)^{w_p},$$

with equality if and only if $\pi_{j,p}(\beta) = w_p$ for every p; likewise for Ψ_j with the pattern terms $\beta_B^{B\setminus p} \Psi_{j,(p)}(\beta_R)$ and $\pi^*_{j,p}(\beta)$. So the shares $a_{j,p} = s_j w_p$ replace $K_j^{-s_j}$ by the product of its pattern terms exactly at the gauges where the basis law puts the proportions w on the patterns; at the gauge of a realization the only such shares are its native split.

(ii) $\beta_B^{-p} K_{j,(p)}(\beta_R^{-1}) = \beta^{-\mathcal{E}} \beta_B^{B\setminus p} \Psi_{j,(p)}(\beta_R)$ and $\pi_{j,p}(\beta^{-1}) = \pi^*_{j,p}(\beta)$: Cremona turns the K-term of each pattern into its Ψ-term, acting on each child as the child's own Cremona. At the table's gauge $\pi^*_{j,p}(\mathbf{1}) = \pi_{j,p}(\mathbf{1})$: the forced shares split s_j and t_j in the same proportions.

A bunch hands its room a family of children, one for each pattern of each table, and the native split is the one choice of shares under which the room inherits everything: the gauge stays native, the masses restrict, and a centre hands down centres. The inequality of (i) is why: at a gauge, the pattern terms of a letter can stand in for the letter with no loss only in the proportions in which the law at that gauge already uses them.

Chapter 13

The order of classes and the Foreman bunch

13.1 The rises of a class

Definition 13.1 (The rises of a class)**.** Let a molecule M with n marks be read at realizations $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ at the table's gauge with $\boldsymbol{\sigma}$ fixed, and let C be a class of the realization at hand, with m marks, and R its room. For a function f of $v=(v_0,\dots,v_k)$ analytic at 0, $\mathring{f}(v)=f(v)-f(0)-\sum_i v_i\,\partial_{v_i}f(0)$ is f less its value and its tangent.

(i) For $0\le k\le n-m$ and distinct partners $\mathbf{g}=(g_1,\dots,g_k)$ in R, the *rise of C with the partners* $\mathbf{g}$ is the increase of the potential along the class, each partner on its own scale:

$$V^{\mathbf{X}}_{C,\mathbf{g}}(v_0,\dots,v_k)=\Phi_{\mathbf{X}}\Big(v_0\mathbf{1}_C+\sum_{i=1}^{k}v_i\mathbf{1}_{g_i}\Big)-\Phi_{\mathbf{X}}\Big(\sum_{i=1}^{k}v_i\mathbf{1}_{g_i}\Big).$$

Stage k is the multiset $\mathcal{V}_k(C)$ of the pairs $(\mathring{V}^{\mathbf{E}}_{C,\mathbf{g}},\mathring{V}^{\mathbf{O}}_{C,\mathbf{g}})$ over the k-tuples of distinct marks of R; stage 0 is the class on its own scale.

(ii) For an order $w=(c_1,\dots,c_m;r_1,\dots,r_{n-m})$ of the marks that lists those of C first and then those of R, the *rise from the class* is

$$\overline{V}^{\mathbf{X}}_{C,w}(u_1,\dots,u_m,v_1,\dots,v_{n-m})=\Phi_{\mathbf{X}}\Big(\sum_i u_i\mathbf{1}_{c_i}+\sum_j v_j\mathbf{1}_{r_j}\Big)-\Phi_{\mathbf{X}}(0),$$

with positions $0,\dots,n-1$ in the order of w, and the *last stage* is the multiset $\mathcal{V}_*(C)$ of the pairs $(\overline{V}^{\mathbf{E}}_{C,w},\overline{V}^{\mathbf{O}}_{C,w})$ over the $m!\,(n-m)!$ such orders.

$\mathcal{V}_k(C)$ has $(n-m)!/(n-m-k)!$ elements; $v_k=0$ turns the rise with $(g_1,\dots,g_k)$ into the rise with $(g_1,\dots,g_{k-1})$; for a single mark an element of $\mathcal{V}_*(\{e\})$ is a writing of the potential from e, less $\Phi_{\mathbf{X}}(0)$.

A class is first read on its own scale, then with one partner of the room at a time, then with more, and last from the class with every mark on its own scale. Every stage before the last is taken less its value and tangent; the last stage holds the whole potential.

Proposition 13.2 (Every rise is read by the Skeleton functions of the class)**.** *With $x=e^{v_0}$ and $\alpha=e^{\sum_i v_i\mathbf{1}_{g_i}}$, the gauge the partners move to,*

$$V^{\mathbf{E}}_{C,\mathbf{g}}(v_0,\dots,v_k)=\sum_j\sigma_j\Big(\log\mathtt{Q}^{\mathbf{E}}_{j,C}(x;\alpha)+\big(d_j(C)+d_j^*(C)\big)v_0\Big),$$

$$V^{\mathbf{O}}_{C,\mathbf{g}}(v_0,\dots,v_k)=\sum_j \delta_j\Big(\log \mathsf{Q}^{\mathbf{O}}_{j,C}(x;\alpha)+\big(d_j(C)-d^*_j(C)\big)v_0\Big).$$

Stage 0 reads the Skeleton functions of C at the table's gauge, and stage k reads them at every gauge that k partners move to.

Proposition 13.3 (The rises hold the cumulants of the class with its partners)**.** *For $\mathbf{a}=(a_0,\dots,a_k)\in\mathbb{N}^{k+1}$ write, in this statement, $\kappa_j(\mathbf{a})=\kappa_{j,\mathbf{1}}\big(N_C^{\times a_0},X_{g_1}^{\times a_1},\dots,X_{g_k}^{\times a_k}\big)$, the joint cumulant, under the weight law $\mathbb{P}_{j,\mathbf{1}}$, of the sequence of variables in which the level N_C appears a_0 times and the indicator X_{g_i} appears a_i times. For $a_0\geq 1$ and $|\mathbf{a}|\geq 2$,*

$$[v^{\mathbf{a}}]\,V^{\mathbf{E}}_{C,\mathbf{g}}=\frac{1+(-1)^{|\mathbf{a}|}}{\mathbf{a}!}\sum_j\sigma_j\kappa_j(\mathbf{a}),\qquad [v^{\mathbf{a}}]\,V^{\mathbf{O}}_{C,\mathbf{g}}=\frac{1-(-1)^{|\mathbf{a}|}}{\mathbf{a}!}\sum_j\delta_j\kappa_j(\mathbf{a}),$$

and the coefficients with $a_0=0$ vanish. By multilinearity $\kappa_j(\mathbf{a})$ is the sum, over the strings $u\in C^{a_0}$, of the cumulants $\kappa_{j,\mathbf{1}}$ along the strings that continue u with each g_i taken a_i times: the entries of the still on those strings, summed over the class and read table by table.

Corollary 13.4 (The first two stages in coins)**.** *(i) For $a\geq 2$, $[v_0^a]\,\mathring{V}^{\mathbf{E}}_{C,()}=\frac{2}{a!}\sum_j\sigma_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ for even a and $[v_0^a]\,\mathring{V}^{\mathbf{O}}_{C,()}=\frac{2}{a!}\sum_j\delta_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ for odd a, the other coefficients being zero; and $\kappa_{j,\mathbf{1}}(N_C^{\times a})=\sum_i\varkappa_a(p_{j,i})$ over the coins $p_{j,i}$ of $\mathsf{Q}^K_{j,C}(\cdot;\mathbf{1})$.*

(ii) For a partner g and $q\geq 1$, with $R_{j,g}(e^{\pm v_0\mathbf{1}_C})$ the occupation of g in table j at the gauge $e^{\pm v_0\mathbf{1}_C}$,

$$[v_1^q]\,V^{\mathbf{E}}_{C,(g)}=\frac{1}{q!}\sum_j\sigma_j\Big(\varkappa_q\big(R_{j,g}(e^{v_0\mathbf{1}_C})\big)+(-1)^q\varkappa_q\big(R_{j,g}(e^{-v_0\mathbf{1}_C})\big)-\big(1+(-1)^q\big)\varkappa_q\big(R_{j,g}(\mathbf{1})\big)\Big),$$

$$[v_1^q]\,V^{\mathbf{O}}_{C,(g)}=\frac{1}{q!}\sum_j\delta_j\Big(\varkappa_q\big(R_{j,g}(e^{v_0\mathbf{1}_C})\big)-(-1)^q\varkappa_q\big(R_{j,g}(e^{-v_0\mathbf{1}_C})\big)-\big(1-(-1)^q\big)\varkappa_q\big(R_{j,g}(\mathbf{1})\big)\Big).$$

13.2 The order

Definition 13.5 (Comparing)**.** (i) *Exponents.* For $\mathbf{a},\mathbf{a}'\in\mathbb{N}^{k+1}$, $\mathbf{a}\sqsubset\mathbf{a}'$ when $a_i<a'_i$ at the largest position i with $a_i\neq a'_i$.

(ii) *Functions.* For functions $f\neq f'$ of the positions $0,\dots,k$, analytic at 0: $f\lhd f'$ when $[v^{\mathbf{a}}]f<[v^{\mathbf{a}}]f'$ at the $\sqsubset$-least exponent at which their coefficients differ.

(iii) *Pairs.* $(f^{\mathbf{E}},f^{\mathbf{O}})\lhd(f'^{\mathbf{E}},f'^{\mathbf{O}})$ when $f^{\mathbf{E}}\lhd f'^{\mathbf{E}}$, or $f^{\mathbf{E}}=f'^{\mathbf{E}}$ and $f^{\mathbf{O}}\lhd f'^{\mathbf{O}}$.

(iv) *Multisets.* Two different multisets of pairs with equally many elements are read from the least element upward in $\lhd$, and the one that first holds the smaller element comes first.

(v) *Classes.* For classes $C\neq D$ of one realization with m marks each, $C\prec D$ when $\mathcal{V}_s(C)\lhd\mathcal{V}_s(D)$ at the first s in $0,1,\dots,n-m,*$ with $\mathcal{V}_s(C)\neq\mathcal{V}_s(D)$.

Definition 13.6 (The Foreman bunch)**.** At the centre, the *Foreman bunch* $B(M;\boldsymbol{\sigma},0)$ is the class of $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ with the fewest marks and, among those, the $\prec$-least. At a lean $\boldsymbol{\delta}$ each class C of $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ lies in one class C° of the centre (Theorem 11.1), and $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ is the class with the least

$$\mathrm{key}_{\boldsymbol{\sigma},\boldsymbol{\delta}}(C)=\big(|C|,\ |C^\circ|,\ C^\circ,\ C\big),$$

compared entry by entry: sizes as integers, C° by $\prec$ at the centre, C by $\prec$ at the lean. At every realization the Foreman bunch is a class with the fewest marks among the classes of that realization; the classes of the centre and the order only choose among those.

Theorem 13.7 (What the order is)**.** *(i)* Total. *For every m, $\prec$ is a strict total order on the classes with m marks; $B(M;\boldsymbol{\sigma},0)$ and $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ are each one class.*

(ii) Directions. *With the classes given, $\prec$ depends on the realization only through the rays of $\boldsymbol{\sigma}$ and $\boldsymbol{\delta}$; at one pair only through the lean sign.*

(iii) Mirror. *If g exchanges the family (Definition 11.13), every stage of $g(C)$ at $-\boldsymbol{\delta}$ equals that stage of C at $\boldsymbol{\delta}$; so $C \prec D$ at $\boldsymbol{\delta}$ exactly when $g(C) \prec g(D)$ at $-\boldsymbol{\delta}$, and $B(M;\boldsymbol{\sigma},-\boldsymbol{\delta}) = g\big(B(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big)$. If g carries the family, the same holds at one lean.*

(iv) Nesting. *When some class at $\boldsymbol{\delta}$ inside $B(M;\boldsymbol{\sigma},0)$ has the fewest marks among the classes at $\boldsymbol{\delta}$, $B(M;\boldsymbol{\sigma},\boldsymbol{\delta}) \subseteq B(M;\boldsymbol{\sigma},0)$, and it is the class at $\boldsymbol{\delta}$ inside $B(M;\boldsymbol{\sigma},0)$ with the fewest marks and, among those, the $\prec$-least. In a one-table molecule this is so whenever $B(M;\boldsymbol{\sigma},0)$ splits at $\boldsymbol{\delta}$. Otherwise $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ need not lie in $B(M;\boldsymbol{\sigma},0)$.*

Every rise is a linear combination of logarithms of polynomials in $e^{v_0},\dots,e^{v_k}$, less the same combination at $v_0 = 0$ (Proposition 13.2), so equality of two rises is an identity of polynomials with exponents (Theorem 10.6). When two rises differ, the first differing exponent is found column by column: $v_k = 0$ is the rise with one partner fewer, the coefficient of v_k^b is a rational function of the scales at the other positions (Corollary 13.4(ii) for $k = 1$), each column is certified equal by a polynomial identity, and the deciding coefficient is a joint cumulant of Proposition 13.3, multiplied by the σ_j or the δ_j, and rational when the weights of the tables are.

Proposition 13.8 (Rung one enters only at the last stage)**.** *Every stage before the last reads the still from rung two on; the masses, rung one, enter only at the last stage, and at a centre they are the same at every mark.*

13.3 What the first stages read

Proposition 13.9 (At the centre the first number is how far the class is from a scale of its own)**.** $\mathrm{Var}_{T\sim\mathbb{P}_{j,\mathbf{1}}}|T\cap U| = 0$ *in every table exactly when U is a union of molecules. So for a class C of a molecule other than the whole,*

$$[v_0^2]\,\mathring{V}^{\mathbf{E}}_{C,()} = \sum_j \sigma_j \,\mathrm{Var}_{T\sim\mathbb{P}_{j,\mathbf{1}}}|T\cap C| > 0.$$

Proposition 13.10 (Mirror classes)**.** *Let C, D be classes with equally many marks whose levels are mirrored across the tables: there are a bijection h of the tables with $(\sigma_{h(j)},\delta_{h(j)}) = (\sigma_j,\delta_j)$ and integers k_j with $\mathbb{P}_{h(j),\mathbf{1}}(|T\cap D| = a) = \mathbb{P}_{j,\mathbf{1}}(|T\cap C| = k_j - a)$ for every j and every a. This holds with h the identity for two classes that together make up the molecule ($k_j = r_j$), and for C and $g(C)$ under an exchange g of the family, h the bijection of its tables ($k_j = |C|$). Then:*

(i) $\kappa_{h(j),\mathbf{1}}(N_D^{\times a}) = (-1)^a\kappa_{j,\mathbf{1}}(N_C^{\times a})$ *for $a \geq 2$, so $\sum_j \sigma_j\kappa_{j,\mathbf{1}}(N_D^{\times a}) = (-1)^a\sum_j \sigma_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$, and likewise with the δ_j: the $\mathbf{E}$ parts of $\mathcal{V}_0(C)$ and $\mathcal{V}_0(D)$ are equal and their $\mathbf{O}$ parts are negatives;*

(ii) at a lean, if some odd $a \geq 3$ has $\sum_j \delta_j\kappa_{j,\mathbf{1}}(N_C^{\times a}) \neq 0$ and a is the least such, then $C \prec D$ exactly when $\sum_j \delta_j\kappa_{j,\mathbf{1}}(N_C^{\times a}) < 0$, and at the opposite lean the order is the other one;

(iii) at the centre stage 0 *does not separate* C *and* D*, and a later stage does.*

Corollary 13.11 (A bunch across the centre)**.** *In a one-table molecule, if* $B(M;\sigma,0)$ *splits at a lean* δ *into* C_K *and* C_Ψ *(Theorem 11.17), then* $B(M;\sigma,\delta) \in \{C_K, C_\Psi\}$ *and* $B(M;\sigma,-\delta) = \beta\big(B(M;\sigma,\delta)\big)$ *is the other one,* β *the exchange of Theorem 11.17; the two are mirror classes, ordered at a lean by Proposition 13.10(ii).*

The Foreman bunch is the least of the smallest classes, and the order that chooses it reads the class first on its own scale. At the centre the first number read is the spread of the level of the class under the weight laws, which vanishes only for a union of molecules; a lean adds the odd cumulants of that level, and two mirror classes are told apart by the first of them that does not vanish, the opposite lean reversing the choice.

Chapter 14

Peel, children and the Foreman tree

14.1 The peel and the tree

Definition 14.1 (Peel; childcules)**.** Families are read at realizations at the table's gauge, and for a molecule M of a family F, $F|M$ is the family of the restrictions of its tables to M. The *peel* of a family F at a bunch B is the family of its children at the native split of the table's gauge (Proposition 12.27),

$$F_B(\mathbf{a}^\star, \mathbf{b}^\star), \qquad a^\star_{j,p} = \pi_{j,p}(\mathbf{1})\, s_j, \quad b^\star_{j,p} = \pi_{j,p}(\mathbf{1})\, t_j,$$

on the room R; children with the same bases and weights are one table, with their exponents added. A *childcule* is a molecule of the peel.

Definition 14.2 (The record of a peel)**.** The *record* of the peel of F at B is the bunch B with, for every table j, the communicatroid of B (Definition 12.4) and the pattern probabilities $\pi_{j,p}(\mathbf{1})$: the indexed children $\big(j,p;\ \mathcal{B}_{j,(p)}, \theta_{j,(p)};\ \pi_{j,p}(\mathbf{1})s_j,\ \pi_{j,p}(\mathbf{1})t_j\big)$. The peel is the record with the indices (j,p) forgotten and equal children merged.

Proposition 14.3 (The record keeps the parent)**.** *The record determines F: table j is the gluing of its communicatroid (Proposition 12.5), and its exponents are the sums of its children's,* $\sum_p \pi_{j,p}(\mathbf{1})(s_j, t_j) = (s_j, t_j)$.

Merging equal children forgets which pattern each share came from, and the Gammas $\Gamma\big(\mu_e - \langle\boldsymbol{\delta}\rangle_e\big)$ of the port (Theorem 15.15) depend on it. The record keeps it. A bunch of width one whose communicatroid is that of one mark, two children on two levels, may instead be pooled into one mark of the room, its shape integrated and its scale kept (Theorem 19.8).

Definition 14.4 (The Foreman tree)**.** For a family F read at $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$, the *Foreman tree* $\mathrm{Tree}(F; \boldsymbol{\sigma}, \boldsymbol{\delta})$ splits F into molecules. A molecule with one mark is a unit (Definition 7.4), and a molecule M whose realization is atomic is an atom (Definition 10.1); otherwise the tree records the Foreman bunch $B(M; \boldsymbol{\sigma}_M, \boldsymbol{\delta}_M)$, $(\boldsymbol{\sigma}_M, \boldsymbol{\delta}_M)$ the pair coordinates of the tables of M, and continues with the peel of $F|M$ at that bunch, on its room.

The tree asks each molecule whether it is an atom; where it is not, the tree takes the least of its smallest classes as the bunch, hands the room the children of the bunch at the native split, and asks again. Nothing is lost on the way down: the record keeps the parent, and the native split keeps the gauge native and the centre a centre.

Proposition 14.5 (Peeling in two steps is peeling at once)**.** *For $B = B_1 \sqcup B_2$, the peel at B_2 of the peel of F at B_1 is the peel of F at B: the child of F at (j,p) and the child at $\big((j, p\cap B_1), p\cap B_2\big)$ of the peel at B_1 are one table on $\mathcal{E}\setminus B$ with the same exponents. So if a molecule is peeled at B_1, its room is one childcule and the bunch there is B_2, the family reached is the peel at $B_1 \sqcup B_2$.*

Theorem 14.6 (Three trees)**.** $\mathrm{Tree}(F;\boldsymbol{\sigma},\boldsymbol{\delta})$ *depends on $(\boldsymbol{\sigma},\boldsymbol{\delta})$ only through their rays; at one pair only through the lean sign. So a molecule at one pair has three Foreman trees,* $\mathrm{Tree}(F;\boldsymbol{\sigma},0)$ *and* $\mathrm{Tree}(F;\boldsymbol{\sigma},\pm\boldsymbol{\delta})$ *for a basis lean $\boldsymbol{\delta}$, fixed by its tables.*

Lemma 14.7 (Maps descend)**.** *Let $g:\mathcal{E}\to\mathcal{E}'$ be a bijection, $B\subseteq\mathcal{E}$ with room R, and $B'=g(B)$.*

(i) If g carries F to F', then $g|_R$ carries the peel of F at B to the peel of F' at B', the child (j,p) going to $(h(j),g(p))$, h the bijection of the tables of Definition 11.13.

(ii) If g exchanges F with F', then $g|_R$ exchanges the peel of F at B with the peel of F' at B', the child (j,p) going to $(h(j),B'\setminus g(p))$.

(iii) In both cases g maps the molecules of F onto those of F', and on a molecule M its restriction carries, respectively exchanges, $F|M$ with $F'|g(M)$.

Theorem 14.8 (The mirror)**.** *If g exchanges F with F', then $g\big(\mathrm{Tree}(F;\boldsymbol{\sigma},\boldsymbol{\delta})\big)=\mathrm{Tree}(F';\boldsymbol{\sigma},-\boldsymbol{\delta})$: molecules go onto molecules, units onto units, atoms onto atoms, each recorded bunch onto the bunch recorded at the same place, rooms onto rooms. If g carries F to F' the same holds at the same lean.*

Corollary 14.9 (A molecule with an exchange)**.** *If g exchanges a molecule M (Definition 11.13), then g maps* $\mathrm{Tree}(M;\boldsymbol{\sigma},0)$ *onto itself, every class of the centre onto itself (Corollary 11.15), and* $\mathrm{Tree}(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ *onto* $\mathrm{Tree}(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$*: $B(M;\boldsymbol{\sigma},-\boldsymbol{\delta})=g\big(B(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big)$ and $g\big(B(M;\boldsymbol{\sigma},0)\big)=B(M;\boldsymbol{\sigma},0)$. If g carries M, it maps every class of $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ onto itself and each of the trees at 0, $\boldsymbol{\delta}$ and $-\boldsymbol{\delta}$ onto itself.*

Lemma 14.10 (An exchange turns a window over)**.** *If g exchanges a family F with F', with the bijection h of the tables (Definition 11.13), and $S'=g(S)$, then for every table j the window of S' in table $h(j)$ of F' is $\{|S|-r_j(S),\dots,|S|-d_j(S)\}$: S in table j and S' in table $h(j)$ have the same width, and the bases of table j at level k on S correspond to those of table $h(j)$ at level $|S|-k$ on S'.*

Corollary 14.11 (Below a no)**.** *A molecule that answers no is peeled at its Foreman bunch read at a centre of the table's gauge; each childcule is a molecule at a centre of the table's gauge, and whether it answers yes is read there in turn.*

14.2 Patterns of one kind

Proposition 14.12 (Level form)**.** *A function of the patterns of a bunch B has* level form *when it depends on $|p|$ alone, and B is in level form in a table when its children are, and in a family when it is in level form in every table. B is in level form in table j, $K_{j,(p)}=K_{j,(p')}$ whenever $|p|=|p'|$, if and only if every transposition of two marks of B, fixing every other mark, maps the bases of table j onto themselves with their weights; so B is in level form in a family exactly when every such transposition carries the family with the identity on its tables.*

Definition 14.13 (Kinds of pattern)**.** Let a one-table molecule be read at the table's gauge and B a set of marks. A *writing from a pattern p* is K written in an order whose first $|p|$ positions hold

p, the next $|B \setminus p|$ hold $B \setminus p$, and the rest the room; a writing of Ψ from $B \setminus p$ likewise. Patterns p, p' are of one *kind at a lean* when one polynomial is written from p and from p', and of one *kind at the centre* when moreover a writing from p may equal a writing of Ψ from $B \setminus p'$.

Theorem 14.14 (One child per kind)**.** *(i) p, p' are of one kind at a lean exactly when a bijection keeping B and carrying the table maps p to p', and of one kind at the centre exactly when, in addition, an exchanging bijection keeping B may map p to $B \setminus p'$. Being of one kind is reflexive, symmetric and transitive.*

(ii) Patterns of one kind have one child between them: the child at p' is the image of the child at p under the bijection restricted to the room, or the dual of that image when the bijection exchanges; their pattern probabilities are equal.

(iii) So the level letters $K_B^{(k)}(\alpha) = \sum_{p \in \mathrm{pat}^{(k)}(B)} \alpha_B^p K_{(p)}(\alpha_R)$ are determined by one child per kind, the patterns of each kind, and the bijections.

Corollary 14.15 (A family with maps in hand)**.** *Let $\mathcal{G}$ be any set of bijections keeping B that carry or exchange a family. Patterns reached from one another by the moves $p \mapsto g(p)$, $g \in \mathcal{G}$ carrying, and $p \mapsto B \setminus g(p)$, $g \in \mathcal{G}$ exchanging, have children that are one table up to the bijections on the room and duality, with equal pattern probabilities.*

Kinds are read on the writings of the two polynomials of one table; with several tables there is no single pair of polynomials to write, and the corollary takes the bijections as given.

Theorem 14.16 (The pattern tensor of a class)**.** *Let B be a class of a one-table molecule at a realization at the table's gauge, $e, f \in B$, and β the bijection taking an order in which the class's one function is written from e to one in which it is written from f (Theorem 10.5). Then:*

(i) β carries the table, or at the centre possibly exchanges it, and maps every class onto itself; in particular $\beta(B) = B$ and $\beta(R) = R$;

(ii) the child at $\beta(p)$ is the child at p with the room mapped by β, or, when β exchanges, the child at $B \setminus \beta(p)$ is the dual of that image; so p and $\beta(p)$, respectively p and $B \setminus \beta(p)$, are of one kind;

(iii) when β carries, the patterns through e correspond to those through f; when it exchanges, the patterns through e correspond to those avoiding f; in both cases with equal pattern probabilities and with children equal up to the bijection on the room and duality.

So the pattern tensor of a class is the same from each of its marks, at the centre up to complements.

14.3 What the room keeps

Definition 14.17 (Keeping the potential; twins; descents; copies)**.** Let a family be read at a realization $\Upsilon = ((s_j, t_j)_j; \alpha)$, with the potential $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ of Definition 8.18 and $(g^* y)_e = y_{g(e)}$ as in Definition 11.13. A bijection g of the marks of a molecule M *keeps the potential* when $\Phi_{\mathbf{X}}(\log \alpha + g^* y) = \Phi_{\mathbf{X}}(\log \alpha + y)$ for every y and both parts; at the table's gauge this is $\Phi_{\mathbf{X}} \circ g^* = \Phi_{\mathbf{X}}$. A bijection $g : M \to M'$ onto the marks of a molecule M', read at a realization with the gauge α', *takes the potential* of M to that of M' when $\Phi_{\mathbf{X}}(\log \alpha + g^* y) = \Phi'_{\mathbf{X}}(\log \alpha' + y) + c_{\mathbf{X}}$ for every y and both parts, with constants $c_{\mathbf{X}}$. Twins, descents and copies are read at the table's gauge, $\alpha = \mathbf{1}$: two childcules of the peel of M are *twins* when a bijection keeping the potential of M maps one onto the other. A *descent* of a Foreman tree is a sequence $M_0 \supset M_1 \supset \cdots \supset M_d$ of its nodes from the molecule M_0 to a leaf M_d, each M_{k+1} a childcule of the peel of M_k at its bunch.

On a descent, with B_i the bunch of M_i, the *copies inside* M_k of $Y = M_i$ or $Y = B_i$, $k \le i$, are the sets $g_k(g_{k+1}(\cdots g_{i-1}(Y)\cdots))$, each g_l a bijection keeping the potential of M_l; for $k = i$ they are Y alone.

Lemma 14.18 (The writings certify the potential)**.** *Let a molecule be read at a realization* $\Upsilon = ((s_j, t_j)_j; \alpha)$.

(i) If e and f read alike, the bijection β that takes, position by position, an order with e first to an order with f first in which one pair of functions is written from both (Proposition 10.4) keeps the potential, and $\beta(e) = f$.

(ii) A bijection keeping the potential maps every class onto itself, and a bijection taking the potential of M to that of M' maps the classes of M onto those of M', at the centre and at the realization's lean.

So two marks read alike exactly when a bijection keeping the potential takes one to the other.

Lemma 14.19 (The peel reads the factors of the letters)**.** *Write the potential of a family by its irreducible factors with their exponents $(L, w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))$, each normalized by $L(\mathbf{1}) = 1$ (Theorem 10.6), and for a bunch B let $L_{[q]}$ be the coefficient of α_B^q in L, a polynomial on the room R. In this statement write $\Phi_{\mathbf{X}}^{\mathrm{peel}}$ for the potential of the peel $F_B(\mathbf{a}^\star, \mathbf{b}^\star)$. Up to a constant,*

$$\Phi_{\mathbf{X}}^{\mathrm{peel}}(y_R) = \sum_L w_{\mathbf{X}}(L) \sum_q L_{[q]}(\mathbf{1}) \log L_{[q]}\big(e^{y_R}\big),$$

the inner sum over the q with $L_{[q]} \neq 0$, whose values $L_{[q]}(\mathbf{1})$ add to 1.

Theorem 14.20 (The room keeps its classes)**.** *Let a molecule M be read at a realization at the table's gauge, let B be a union of its classes, and let the children's weights factor across the childcules of the peel. Every bijection g keeping the potential of M keeps B; its restriction to the room keeps the potential of the peel, maps each childcule Y onto a childcule $g(Y)$, and takes the potential of Y to that of $g(Y)$, the constant being 0 when $g(Y) = Y$. Consequently, for a class C of M inside the room:*

(i) the childcules that C meets are twins of one another, and C meets each of them in equally many marks;

(ii) inside each childcule, C lies in one class.

Proposition 14.21 (The tree is read off the potential)**.** *Let a molecule M of F and a molecule M' of F' be read at realizations at the table's gauge, with pair coordinates $(\boldsymbol{\sigma}, \boldsymbol{\delta})$ and $(\boldsymbol{\sigma}', \boldsymbol{\delta}')$, the children's weights factoring across the childcules at every node of their trees, and let $g : M \to M'$ take the potential of M to that of M'. Then g maps* $\mathrm{Tree}(F|M; \boldsymbol{\sigma}, \boldsymbol{\delta})$ *onto* $\mathrm{Tree}(F'|M'; \boldsymbol{\sigma}', \boldsymbol{\delta}')$*: classes onto classes, units onto units, atoms onto atoms, each bunch onto the bunch recorded at the same place, and each childcule onto a childcule, its potential to that of its image.*

Theorem 14.22 (The leaf bounds the bunches)**.** *Let a molecule F be read at a realization at the table's gauge, the centre or a lean, the children's weights factoring across the childcules at every node, and let $M_0 \supset M_1 \supset \cdots \supset M_d$ be a descent of its Foreman tree at that realization, B_k the bunch of M_k and $R_k = M_k \setminus B_k$. Let h_k be the number of twins of M_{k+1} among the childcules of M_k, M_{k+1} itself included, and $h_{<k} = h_0 h_1 \cdots h_{k-1}$.*

(i) A class of M_k inside R_k that meets M_{k+1} meets the twins of M_{k+1} and no other childcule, each in equally many marks, and lies in one class of each. The copies inside M_k of M_i,

$k \le i \le d$, respectively of B_i, $k \le i < d$, are $h_{<i}/h_{<k}$ disjoint nodes, respectively bunches, of the subtree at M_k, leaves when $i = d$; and M_d together with its copies inside M_k is a union of whole classes of M_k.

(ii) A class C of M_k that meets M_{k+1} has at least $|B_k|$ marks and meets M_{k+1} in $|C|/h_k \ge |B_k|/h_k$ of them, all in one class of M_{k+1}; the bunch B_{k+1}, and the leaf when $k+1 = d$, contains such a part. So

$$|B_0| \le h_{<1}|B_1| \le h_{<2}|B_2| \le \cdots \le h_{<d-1}|B_{d-1}| \le h_{<d}|M_d|,$$

the k-th term being the number of marks of the copies of B_k inside M_0, and the last that of the copies of M_d: down the descent the bunches with their copies never shrink, and the leaf with its copies has at least as many marks as every bunch with its copies.

Theorem 14.22 rests on two things: the peel splits a class only across twin childcules, each part inside one class (Theorem 14.20), and the bunch is a class with the fewest marks of the realization at which its node is read. Peeled at another bunch, a molecule can end at a leaf smaller than the bunch above it; at a lean the count is read on the classes of the lean, which can split a class of the centre into parts smaller than the least class of the centre; and the copies are needed when a room falls into twins.

Part IV

Evaluation

Chapter 15

The integral at a bunch

15.1 The scale of a bunch

Proposition 15.1 (A scale integral is a Breta of one bunch)**.** *Let $J = \{0, \dots, L\}$ carry coefficients $\boldsymbol{\xi} = (\xi_\ell)_{\ell \in J}$ and letter exponents $\mathbf{w} = (w_\ell)_{\ell \in J}$ with total $|\mathbf{w}| = \sum_\ell w_\ell$, and let I_J be the unit configuration on J. For $\boldsymbol{\xi} > 0$, $\operatorname{Re} w_\ell > 0$ and $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$,*

$$\int_0^\infty x^{\kappa-1} \prod_{\ell \in J} (x + \xi_\ell)^{-w_\ell}\, dx = \Gamma_{\,|J|}^{\,2}\begin{bmatrix} \kappa, & |\mathbf{w}| - \kappa \\ & \mathbf{w} \end{bmatrix} \mathsf{B}^{\{J\}}_{\mathbf{w},\, |\mathbf{w}|-\kappa;\, I_J}(\boldsymbol{\xi}) :$$

the Breta of Definition 3.1 with the one bunch J, the shape index $\mathbf{w}$, the scale index $|\mathbf{w}| - \kappa$, the unit configuration as its terms and $\boldsymbol{\xi}$ as its coefficients. For $\xi_\ell \in \mathbb{C} \setminus (-\infty, 0]$ the left side, with principal branches, converges exactly when $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$ and is holomorphic in its data there, and it continues the right side from positive coefficients and from letter exponents with positive real part; wherever the left side converges, the product on the right denotes its value. Coincident coefficients merge into one with the summed letter exponent.

The scale of a bunch is integrated against a product of linear letters, one for each point of the joint spectrum, and that integral is a Breta: the points of the spectrum are the coefficients, the letter exponents the shape index, and the index of the scale integral sits in the scale index as $|\mathbf{w}| - \kappa$. Nothing beyond the general Breta is needed to write it. With one coefficient the Breta is the power $\xi_0^{\kappa - w_0}$, and the identity is the integral of a linear letter: for $c_0, c_1 > 0$ and $0 < \operatorname{Re}\nu < \operatorname{Re} s$, $\int_0^\infty \rho^{\nu-1}(c_0 + c_1\rho)^{-s}\, d\rho = \Gamma_{\,1}^{\,2}\begin{bmatrix} \nu, & s - \nu \\ & s \end{bmatrix} c_0^{\nu - s} c_1^{-\nu}$. A stacked Gamma $\Gamma_{\,1}^{\,2}\begin{bmatrix} a, & b \\ & a+b \end{bmatrix}$ is a *Beta corner*.

Proposition 15.2 (Where the Breta of one bunch converges)**.** *In Proposition 15.1 let $\operatorname{Re} w_\ell > 0$ for every ℓ and $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$, and let Ξ_J be the set of the $\boldsymbol{\xi} \in (\mathbb{C} \setminus (-\infty, 0])^J$ whose convex hull avoids $(-\infty, 0]$. It is open, star-shaped about $\mathbf{1}$, and contains every $\boldsymbol{\xi}$ whose coordinates have positive real parts. On Ξ_J the integral that defines the Breta at positive coefficients (Definition 3.1),*

$$\int_{\Delta_J} u^{\mathbf{w}-\mathbf{1}} \Big(\sum_{\ell \in J} \xi_\ell u_\ell \Big)^{-(|\mathbf{w}|-\kappa)} d\omega_J(u),$$

with the principal branch, converges, is holomorphic in $\boldsymbol{\xi}$, and continues the Breta $\mathsf{B}^{\{J\}}_{\mathbf{w},\, |\mathbf{w}|-\kappa;\, I_J}(\boldsymbol{\xi})$ from positive coefficients; so the identity of Proposition 15.1 holds on Ξ_J with both sides convergent integrals.

Theorem 15.3 (The scale of a bunch is one Breta)**.** *Let a family F on $\mathcal{E}$ be at a gauge α with masses $\mathbf{m}$, let B be a proper bunch with room R, its Skeleton polynomials, values and coins those of Definition 12.10, and let $\xi_0, \dots, \xi_L$ be the distinct points of the joint spectrum of the letters with nonzero exponent, the union of the spectra $\operatorname{spec}^K_{j,B}(\alpha)$ with $s_j \neq 0$ and $\operatorname{spec}^\Psi_{j,B}(\alpha)$ with $t_j \neq 0$, with the letter exponents*

$$w_\ell = \sum_j \big(s_j \operatorname{mult}^K_{j,\ell} + t_j \operatorname{mult}^\Psi_{j,\ell}\big) = \sum_j \big(\sigma_j(\operatorname{mult}^K_{j,\ell} + \operatorname{mult}^\Psi_{j,\ell}) + \delta_j(\operatorname{mult}^K_{j,\ell} - \operatorname{mult}^\Psi_{j,\ell})\big),$$

$\operatorname{mult}^K_{j,\ell}$ and $\operatorname{mult}^\Psi_{j,\ell}$ the multiplicities of ξ_ℓ in $\operatorname{spec}^K_{j,B}(\alpha)$ and $\operatorname{spec}^\Psi_{j,B}(\alpha)$, and $J = \{0, \dots, L\}$. The scale integral

$$\operatorname{Sc}_B(\alpha; \mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_0^\infty x^{m_B - 1} \prod_j K_j(x\alpha_B, \alpha_R)^{-s_j} \Psi_j(x\alpha_B, \alpha_R)^{-t_j}\, dx$$

converges exactly when $\operatorname{Re}\kappa_B > 0$ and $\operatorname{Re}\kappa_B^{\text{top}} > 0$, and then

$$\operatorname{Sc}_B(\alpha; \mathbf{m}; \mathbf{s}, \mathbf{t}) = \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j} \prod_\ell (1 + \xi_\ell)^{w_\ell}\ \Gamma_{\,2,\,|J|}\begin{bmatrix} \kappa_B, & \kappa_B^{\text{top}} \\ & \mathbf{w} \end{bmatrix} \mathsf{B}^{\{J\}}_{\mathbf{w},\, \kappa_B^{\text{top}};\, I_J}(\boldsymbol{\xi}),$$

with $|\mathbf{w}| = \sum_j (s_j + t_j) c_j(B) = \kappa_B + \kappa_B^{\text{top}}$, so that the scale index $|\mathbf{w}| - \kappa_B$ of Proposition 15.1 is the top wall; the Breta is read by that proposition at complex spectra.

Corollary 15.4 (The scale of a bunch of width one)**.** *Let B be a bunch with room R. Along B each letter of a table j with $c_j(B) = 1$ has one Skeleton value, the level letter at the floor over the level letter at the largest level, colevel letters for Ψ_j (Definition 12.16), and its coin is the probability of the largest level, respectively colevel; the letters of a table with $c_j(B) = 0$ have none:*

$$\operatorname{spec}^K_{j,B}(\alpha) = \Big\{ \frac{K^{(d_j(B))}_{j,B}(\alpha)}{K^{(d_j(B)+1)}_{j,B}(\alpha)} \Big\},\ \text{coin } \mathbb{P}_{j,\alpha}\big(N_B = d_j(B) + 1\big);$$

$$\operatorname{spec}^\Psi_{j,B}(\alpha) = \Big\{ \frac{\Psi^{(d^*_j(B))}_{j,B}(\alpha)}{\Psi^{(d^*_j(B)+1)}_{j,B}(\alpha)} \Big\},\ \text{coin } \mathbb{P}^*_{j,\alpha}\big(N^*_B = d^*_j(B) + 1\big).$$

At $B = \{e\}$, e neither a loop nor a coloop of table j, the two levels are $K_{j,\setminus e}(\alpha_R)$ and $\alpha_e K_{j,/e}(\alpha_R)$ and the two colevels $\Psi_{j,/e}(\alpha_R)$ and $\alpha_e \Psi_{j,\setminus e}(\alpha_R)$: each Skeleton value is a deletion–contraction ratio, and the coins are the occupation $R_{j,e}(\alpha)$ and the dual occupation $R^*_{j,e}(\alpha)$.

Corollary 15.5 (One share per coefficient beyond the first)**.** *Let $\operatorname{Re} w_\ell > 0$ for all ℓ, $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$, and choose $c_\ell \in (0, \operatorname{Re} w_\ell)$, $\ell = 1, \dots, L$, with $\operatorname{Re}\kappa - \operatorname{Re} w_0 < \sum_\ell c_\ell < \operatorname{Re}\kappa$, possible exactly under these conditions. With the share integrals on $\operatorname{Re} a_\ell = c_\ell$ and $|\mathbf{a}| = a_1 + \cdots + a_L$,*

$$\Gamma_{\,2,\,|J|}\begin{bmatrix} \kappa, & |\mathbf{w}| - \kappa \\ & \mathbf{w} \end{bmatrix} \mathsf{B}^{\{J\}}_{\mathbf{w},\, |\mathbf{w}| - \kappa;\, I_J}(\boldsymbol{\xi})$$
$$= \int_{a_1 + b_1 = w_1} \cdots \int_{a_L + b_L = w_L} \prod_{\ell=1}^{L} \Gamma_{\,2,\,1}\begin{bmatrix} a_\ell, & b_\ell \\ & w_\ell \end{bmatrix} \xi_\ell^{-b_\ell} \cdot \xi_0^{\kappa - |\mathbf{a}| - w_0}\, \Gamma_{\,2,\,1}\begin{bmatrix} \kappa - |\mathbf{a}|, & w_0 - \kappa + |\mathbf{a}| \\ & w_0 \end{bmatrix}.$$

Any coefficient may serve as ξ_0; $L + 1$ distinct values need L shares.

Theorem 15.6 (The scale in its spectral gaps)**.** *Let $\xi_\ell \in \mathbb{C}\setminus(-\infty,0]$ and $w_\ell \in \mathbb{C}$, $\ell \in J$, with $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$, and let $r>0$ have $|\xi_\ell - r| < r$ for every ℓ; such an r exists exactly when every ξ_ℓ has positive real part, since $|\xi_\ell - r| < r$ reads $\operatorname{Re}\xi_\ell > |\xi_\ell|^2/(2r)$. Put $g_\ell = 1-\xi_\ell/r$, the* spectral gaps *at r, so that $|g_\ell|<1$. Then*

$$\Gamma_{\,2\,|J|}\!\begin{bmatrix}\kappa,\ |\mathbf{w}|-\kappa\\ \mathbf{w}\end{bmatrix}\,\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi}) = r^{\kappa-|\mathbf{w}|}\sum_{\mathbf{n}\in\mathbb{N}^J}\prod_{\ell\in J}\frac{(w_\ell)_{n_\ell}}{n_\ell!}g_\ell^{\,n_\ell}\;\Gamma_{\,2\,1}\!\begin{bmatrix}\kappa,\ |\mathbf{w}|+|\mathbf{n}|-\kappa\\ |\mathbf{w}|+|\mathbf{n}|\end{bmatrix},$$

a convergent series whose terms, grouped by the total order $|\mathbf{n}|$, have root rate at most $\max_\ell|g_\ell|$. A coefficient equal to r has spectral gap zero and drops out. For $0<\xi_\ell\le r$, $w_\ell>0$ and $0<\kappa<|\mathbf{w}|$ the gaps lie in $[0,1)$ and every term is positive. For positive coefficients the bound $\max_\ell|g_\ell|$ is least at $r=\frac12(\max_\ell\xi_\ell+\min_\ell\xi_\ell)$, where it is $\frac{\max_\ell\xi_\ell-\min_\ell\xi_\ell}{\max_\ell\xi_\ell+\min_\ell\xi_\ell}$, against $1-\frac{\min_\ell\xi_\ell}{\max_\ell\xi_\ell}$ at $r=\max_\ell\xi_\ell$.

Corollary 15.7 (Inversion is Cremona on the scale)**.** *In Theorem 15.3:*

$$\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi}) = \prod_\ell \xi_\ell^{-w_\ell}\,\mathsf{B}^{\{J\}}_{\mathbf{w},\,\kappa;\,I_J}(\boldsymbol{\xi}^{-1}),$$

$$\mathrm{Sc}_B(\alpha;\mathbf{m};\mathbf{s},\mathbf{t}) = (\alpha^{\mathcal{E}})^{-2\sigma}\;\mathrm{Sc}_B(\alpha^{-1};2\sigma\mathbf{1}-\mathbf{m};\mathbf{t},\mathbf{s})\,:$$

the joint spectrum at $(\alpha^{-1};\mathbf{t},\mathbf{s})$ is the reciprocal of that at $(\alpha;\mathbf{s},\mathbf{t})$ with the same letter exponents, and the index of the scale integral moves from κ_B to $\kappa_B^{\mathrm{top}}=|\mathbf{w}|-\kappa_B$.

15.2 Closing

Definition 15.8 (Exceedance polynomials)**.** For a bunch B of a table the *exceedance polynomials* are $\mathsf{Q}^K_{B,>}(x;\alpha)=\sum_{k\ge0}\mathbb{P}_\alpha(N_B-d(B)>k)\,x^k$ and $\mathsf{Q}^\Psi_{B,>}(x;\alpha)=\sum_{k\ge0}\mathbb{P}^*_\alpha(N^*_B-d^*(B)>k)\,x^k$. They have nonnegative coefficients and degree $c(B)-1$, and $1-\mathsf{Q}^K_B=(1-x)\mathsf{Q}^K_{B,>}$, $1-\mathsf{Q}^\Psi_B=(1-x)\mathsf{Q}^\Psi_{B,>}$.

Theorem 15.9 (The scale closes on its two sectors)**.** *Let B be a proper bunch of a family with room R, and let the data be real with $s_j,t_j\ge0$, $\kappa_B>0$ and $\kappa_B^{\mathrm{top}}>0$. For $S\in\{B,R\}$ write the polynomials $\mathsf{Q}^K_{j,S}(\cdot;\alpha)$, with the exponents s_j, and $\mathsf{Q}^\Psi_{j,S}(\cdot;\alpha)$, with the exponents t_j, as $\mathsf{Q}_{S,i}(\cdot;\alpha)$ with exponents w_i, $1\le i\le 2N$ for a family of N tables, $\mathbf{w}=(w_i)_i$, and their exceedance polynomials as $\mathsf{Q}_{S,i,>}$.*

(i) The scale integral is the sum of the Trailmoments of its two sectors, with the coefficient functions $x\mapsto(\mathsf{Q}_{S,i}(x;\alpha))_i$ and the inputs $x^{\kappa-1}\mathbb{I}_{(0,1)}$:

$$\mathrm{Sc}_B(\alpha;\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}\Big(\big(\mathsf{T}_{\mathbf{w};I_{2N}}[(\mathsf{Q}_{B,i}(x;\alpha))_i]\,x^{\kappa_B-1}\mathbb{I}_{(0,1)}\big)\big((0,1)\big) + \big(\mathsf{T}_{\mathbf{w};I_{2N}}[(\mathsf{Q}_{R,i}(x;\alpha))_i]\,x^{\kappa_B^{\mathrm{top}}-1}\mathbb{I}_{(0,1)}\big)\big((0,1)\big)\Big).$$

(ii) With $\lambda_S=\max_i\big(1-\mathsf{Q}_{S,i}(0;\alpha)\big)<1$ and $\kappa>0$,

$$\big(\mathsf{T}_{\mathbf{w};I_{2N}}[(\mathsf{Q}_{S,i}(x;\alpha))_i]\,x^{\kappa-1}\mathbb{I}_{(0,1)}\big)\big((0,1)\big) = \int_0^1 x^{\kappa-1}\prod_i\mathsf{Q}_{S,i}(x;\alpha)^{-w_i}\,dx = \sum_{\mathbf{n}\in\mathbb{N}^{2N}}\prod_i\frac{(w_i)_{n_i}}{n_i!}\sum_{k\ge0}\Big([x^k]\prod_i\mathsf{Q}_{S,i,>}(x;\alpha)^{n_i}\Big)\Gamma_{\,2\,1}\!\begin{bmatrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{bmatrix},$$

a series of nonnegative terms, those with $|\mathbf{n}|=M$ summing to at most $\frac{(|\mathbf{w}|)_M}{M!}\lambda_S^M\kappa^{-1}$.

For complex exponents with $\operatorname{Re}\kappa > 0$ *the series converges absolutely to the same value. No Skeleton value is used, and a complex spectrum closes as a real one does.*

Proposition 15.10 (Closing around a point of the axis)**.** *The expansion* $x+\xi = (x+r)\big(1-\frac{r-\xi}{x+r}\big)$, $r>0$, *converges uniformly on* $x \ge 0$ *exactly when* $|\xi - r| < r$*; one* r *serves every value of a finite spectrum exactly when every value has positive real part, that is, when every irreducible real factor of every Skeleton polynomial has positive coefficients.*

Corollary 15.11 (The period closes on the ordering sectors)**.** *Let the data be complex with* $\kappa_{\mathcal{E}} = 0$ *and* $\operatorname{Re}\kappa_S > 0$ *for every nonempty proper* S*. For an order* π *with tails* U_k *and sector letters* $k_{j,\pi}, \psi_{j,\pi}$ *(Theorem 8.5), with* $N_S = |T \cap S|$*, on* $[0,1]^{n-1}$

$$1 - \frac{k_{j,\pi}(q)}{K_j(\mathbf{1})} = \sum_{k=2}^{n} (1-q_k)\, \mathbb{E}_{T\sim\mathbb{P}_{j,\mathbf{1}}}\Big[\prod_{k'<k} q_{k'}^{\,N_{U_{k'}} - d_j(U_{k'})} \sum_{0 \le i < N_{U_k} - d_j(U_k)} q_k^{\,i} \Big],$$

the same for $\psi_{j,\pi}$ *with the colevels and* d_j^**; both sides lie in* $[0,\lambda]$*,* $\lambda < 1$ *the largest of the numbers* $1 - \theta_{j,T^{\pm}_{j,\pi}}/K_j(\mathbf{1})$*. Consequently*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j K_j(\mathbf{1})^{-s_j-t_j} \sum_{\pi} \sum_{\mathbf{n},\mathbf{n}'} \prod_j \frac{(s_j)_{n_j}(t_j)_{n_j'}}{n_j!\, n_j'!} \times \int_{[0,1]^{n-1}} \prod_k q_k^{\,\kappa_{U_k}-1} \prod_j \Big(1 - \frac{k_{j,\pi}(q)}{K_j(\mathbf{1})}\Big)^{n_j} \Big(1 - \frac{\psi_{j,\pi}(q)}{K_j(\mathbf{1})}\Big)^{n_j'} dq,$$

a convergent series, each integral a finite positive combination of products

$$\prod_k \Gamma \begin{bmatrix} \kappa_{U_k} + \beta_k,\ \gamma_k + 1 \\ \kappa_{U_k} + \beta_k + \gamma_k + 1 \end{bmatrix}, \qquad \beta_k, \gamma_k \in \mathbb{N};$$

for real data with $s_j, t_j \ge 0$ *every term is nonnegative. With* $w = \sum_j (|s_j| + |t_j|)$*, the sum of the moduli of the exponents, the terms with* $|\mathbf{n}| + |\mathbf{n}'| = M$ *have moduli summing to at most* $\frac{(w)_M}{M!}\lambda^M \prod_j K_j(\mathbf{1})^{-\operatorname{Re}(s_j+t_j)} \sum_\pi \prod_k (\operatorname{Re}\kappa_{U_k})^{-1}$*.*

Proposition 15.12 (An enclosure of the period in the field of its data)**.** *In Corollary 15.11 let the data be real, and let the weights, masses and exponents lie in a subfield* k *of* $\mathbb{R}$*. For an order* π *put* $a_{j,\pi} = 1 - k_{j,\pi}/K_j(\mathbf{1})$ *and* $b_{j,\pi} = 1 - \psi_{j,\pi}/K_j(\mathbf{1})$*, let* $\lambda_\pi < 1$ *be the largest of the numbers* $1 - \theta_{j,T^{\pm}_{j,\pi}}/K_j(\mathbf{1})$*, and let* $f_{\pi,M}(q)$ *be the coefficient of* ζ^M *in the power series* $\prod_j (1-\zeta a_{j,\pi})^{-s_j}(1-\zeta b_{j,\pi})^{-t_j}$*.*

(i) $\prod_j K_j(\mathbf{1})^{s_j+t_j}\, P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_\pi \sum_{M\ge 0} I_{\pi,M}$ *with* $I_{\pi,M} = \int_{[0,1]^{n-1}} \prod_k q_k^{\,\kappa_{U_k}-1} f_{\pi,M}(q)\, dq$*, an element of* k*, nonnegative when* $s_j, t_j \ge 0$*, by* $\int \prod_k q_k^{\,\kappa_{U_k}-1+\beta_k} dq = \prod_k (\kappa_{U_k}+\beta_k)^{-1}$*.*

(ii) With $w = \sum_j (|s_j| + |t_j|)$*, if* $r_{\pi,Q} = \lambda_\pi \max\big(1, \frac{w+Q+1}{Q+2}\big) < 1$*, then* $\big|\sum_{M>Q} I_{\pi,M}\big| \le \operatorname{err}_{\pi,Q} = \prod_k \kappa_{U_k}^{-1} \frac{(w)_{Q+1}}{(Q+1)!} \frac{\lambda_\pi^{Q+1}}{1-r_{\pi,Q}}$*, the sum being nonnegative when* $s_j, t_j \ge 0$*, and* $\operatorname{err}_{\pi,Q} \to 0$ *as* $Q \to \infty$*: the partial sums minus and plus* $\sum_\pi \operatorname{err}_{\pi,Q}$*, and for* $s_j, t_j \ge 0$ *the partial sums and the partial sums plus* $\sum_\pi \operatorname{err}_{\pi,Q}$*, are elements of* k *enclosing the left side of (i) within any prescribed width.*

(iii) The coefficients $f_{\pi,M}$ satisfy a linear recurrence in M with coefficients polynomial in M and in the $a_{j,\pi}, b_{j,\pi}$: the power series $f(\zeta)=\prod_i(1-\zeta A_i)^{-w_i}$, with (A_i,w_i) running over the pairs $(a_{j,\pi},s_j)$ and $(b_{j,\pi},t_j)$, satisfies $\prod_i(1-\zeta A_i)\,f'(\zeta)=\big(\sum_i w_iA_i\prod_{i'\neq i}(1-\zeta A_{i'})\big)f(\zeta)$.

Theorem 15.13 (The peak closing of a period)**.** *Let the data be real with $s_j,t_j\geq 0$, $\kappa_{\mathcal{E}}=0$ and every proper wall positive. Write the letters K_j (exponent s_j, degree r_j, counts d_j) and Ψ_j (exponent t_j, degree $n-r_j$, counts d_j^*) as L_i with exponents w_i, degrees g_i and counts $d_i(S)$, choose references $M_i\geq\max_{\Delta_{\mathcal{E}}}L_i$, and put $b_{\mathbf{n}}=\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m-1}}\prod_i(M_i-L_i(u))^{n_i}d\omega_{\mathcal{E}}(u)\geq 0$.*

(i) $P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\sum_{\mathbf{n}}\prod_i\frac{(w_i)_{n_i}}{n_i!}M_i^{-w_i-n_i}b_{\mathbf{n}}$, a series of nonnegative terms. With the references $M_i=\max_T\theta_{j,T}/g_i!$, j the table of the letter L_i, every $b_{\mathbf{n}}$ is a finite positive combination of the stacked Gammas $\Gamma^{\,n}_{\,1}\!\left[\begin{matrix}\mathbf{m}+\boldsymbol{\gamma}\\ m_{\mathcal{E}}+|\boldsymbol{\gamma}|\end{matrix}\right]$, $\boldsymbol{\gamma}\in\mathbb{N}^{\mathcal{E}}$; the same series, with every $b_{\mathbf{n}}$ such a finite positive combination, holds for $\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m-1}}\prod_i L_i^{-w_i}\,d\omega_{\mathcal{E}}$ with any letters L_i homogeneous of degree g_i with nonnegative coefficients and squarefree monomials, M_i being the largest coefficient of L_i over $g_i!$.

(ii) For each letter let ϖ_i be the least of $\kappa_S/d_i(S)$ over nonempty proper S with $d_i(S)>0$, and k_i the largest length of a chain $S_1\subsetneq\cdots\subsetneq S_{k_i}$ with $\kappa_{S_l}=\varpi_i d_i(S_l)$; put $\varpi=\min_i\varpi_i$ and $k=\max\{k_i:\varpi_i=\varpi\}$. The partial sums $P^{\leq Q}$ of (i) over $\max_i n_i\leq Q$ satisfy $P_F(\mathbf{m};\mathbf{s},\mathbf{t})-P^{\leq Q}=\Theta\big(Q^{-\varpi}(\log Q)^{k-1}\big)$, and if a single letter attains ϖ,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})-P^{\leq Q}\sim\frac{c_i\,\Gamma(w_i+\varpi)}{\varpi\,\Gamma(k)\,\Gamma(w_i)}\,Q^{-\varpi}(\log Q)^{k-1},$$

with $c_i>0$ the coefficient of $(z+\varpi_i)^{-k_i}$ in $M_i^{-z}\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m-1}}L_i^z\prod_{i'}L_{i'}^{-w_{i'}}d\omega_{\mathcal{E}}$, whose pole at $-\varpi_i$ has order exactly k_i. A letter that vanishes on no proper face closes geometrically.

(iii) The fastest start is the sharp reference $M_i=\max_{\Delta_{\mathcal{E}}}L_i$; where it is attained inside the simplex, $u_e^\star\partial_e\log L_i(u^\star)=g_iu_e^\star$ for every mark. It can be attained on a proper face of the simplex.

(iv) At the target of a molecule with a native gauge every proper wall is positive (Theorem 8.19(iii)), so the series applies there and its rate ϖ is positive.

Theorem 15.14 (Continued Beta corners)**.** *In Theorem 15.9 and Corollary 15.11 let the data be complex and let no index κ, respectively no κ_{U_k}, be a nonpositive integer; for an index with $\operatorname{Re}\kappa\leq 0$ let $h\in\mathbb{N}$ have $-h-1<\operatorname{Re}\kappa\leq -h$. Then the series converge, with each Beta corner $\Gamma^{\,2}_{\,1}\!\left[\begin{matrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{matrix}\right]$ in Theorem 15.9 and $\Gamma^{\,2}_{\,1}\!\left[\begin{matrix}\kappa_{U_k}+\beta_k,\ \gamma_k+1\\ \kappa_{U_k}+\beta_k+\gamma_k+1\end{matrix}\right]$ in Corollary 15.11 read as the meromorphic stacked Gamma, to the* Reg *of the integral, its continuation in the indices; the terms of total order M are at most $\frac{(w)_M}{M!}\lambda^M$ times a polynomial in M whose degree is the sum of the integers $h+1$ over the indices with real part at most 0 (over the proper walls with $\operatorname{Re}\kappa_S\leq 0$ in Corollary 15.11), with $w=\sum_j(|s_j|+|t_j|)$ the sum of the moduli of the exponents. The sector series needs no positivity of the walls and computes the continued value at every point off the pole hyperplanes.*

The closings expand each letter around a reference and integrate term by term: on the two sectors of a bunch's scale the reference is the letter's value at the end of the sector, on the ordering sectors of the simplex it is the total weight, and at the peak it is a bound on the simplex. For real data with nonnegative exponents every term is a positive combination of stacked Gammas, nonnegative where the walls are positive, and the rate at which they fall is read from the walls.

15.3 The port

Theorem 15.15 (The port)**.** *Let B be a bunch of a family F on $\mathcal{E}$, with room R, and suppose there are reals $c_{j,p}, c'_{j,p} > 0$, one for each pattern p of B in table j, with $\sum_p c_{j,p} = \operatorname{Re} s_j$ and $\sum_p c'_{j,p} = \operatorname{Re} t_j$, such that (a) $\operatorname{Re}\mu_e - \sum_{j,p} \frac{c_{j,p}-c'_{j,p}}{2}\chi_p(e) > 0$ for $e \in B$, and (b) the children $F_B(\mathbf{c},\mathbf{c}')$ converge at the masses $\operatorname{Re}\mathbf{m}|_R$. Then $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converges and, with the share integrals on $\operatorname{Re} a_{j,p} = c_{j,p}$, $\operatorname{Re} b_{j,p} = c'_{j,p}$,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j} \int_{|\mathbf{b}_j|=t_j} \prod_j \Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix} \times \prod_{e\in B} \Gamma\big(\mu_e - \langle\boldsymbol{\delta}\rangle_e\big)\, J_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}),$$

with the odd loads $\langle\boldsymbol{\delta}\rangle_e$ of the shares (Definition 12.24). For $B = \mathcal{E}$ this is the atom formula*: every child is $\{\varnothing\}$ with the weight $\theta_{j,T}$, and $J_{F_{\mathcal{E}}(\mathbf{a},\mathbf{b})}(\varnothing;\mathbf{a},\mathbf{b}) = \prod_{j,T} \theta_{j,T}^{-a_{j,T}-b_{j,T}}$.*

The port opens every letter over the patterns of the bunch, one share for each pattern and letter. On the bunch the shares leave a product of Gammas, one per mark, whose arguments are the offsets less the odd loads; on the room they leave the interest of the children at the shares as their exponents.

Corollary 15.16 (A loop or coloop in every table)**.** *Let every mark of a set A be in every table a loop or a coloop, with $\eta_{j,e} = -1$ for a loop and $\eta_{j,e} = +1$ for a coloop. Then $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_{e\in A}\Gamma\big(\mu_e - \sum_j \eta_{j,e}\delta_j\big)\, J_{F'}(\mathbf{m}|_{\mathcal{E}\setminus A};\mathbf{s},\mathbf{t})$, with F' the family on $\mathcal{E}\setminus A$ of the deletions of its tables at the loops and the contractions at the coloops (Definition 7.3).*

Corollary 15.17 (The damped bunch integral is a Breta)**.** *In Theorem 15.15 fix α_R, let $\operatorname{Re} s_j, \operatorname{Re} t_j > 0$ (a letter with exponent zero is 1 and gets no scale), put $z = \frac{1}{2}\log\alpha$, introduce one singleton scale per letter, $V = \{v_{j,+}, v_{j,-}\}_j$, put $\mathbf{q} = (\mathbf{s},\mathbf{t})$ on V, and take the partition $\mathcal{P}_{\text{port}} = \{B\} \cup \{\{v\} : v \in V\}$. The term arrays $(\pi^{\text{port}}, \ell^{\text{port}})$ of the letter presentation (Proposition 3.6) have the columns:*

- *for each $e \in B$, the kernel column with shape exponent $(\mathbf{e}_e, 0_V)$, level $(1, 0_V)$ and coefficient 1;*
- *for every (j,p), a column with shape exponent $(\chi_p/2, \mathbf{e}_{j,+})$, level $(\ell_p, \mathbf{e}_{j,+})$ and coefficient $\Lambda_{j,(p)}(z_R)$;*
- *for every (j,p), a column with shape exponent $(-\chi_p/2, \mathbf{e}_{j,-})$, level $(-\ell_p, \mathbf{e}_{j,-})$ and coefficient $\Lambda_{j,(p)}(-z_R)$;*

and $\boldsymbol{\xi}_{\text{port}}(\alpha_R)$ lists the coefficients in this order. Then, either side converging with the other,

$$\int_{\mathbb{R}^B_{>0}} \alpha_B^{\boldsymbol{\mu}|_B - \mathbf{1}} e^{-|\alpha_B|} \prod_j \Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j}\, d\alpha_B = \Gamma(\mu_B)\, \mathsf{B}^{\mathcal{P}_{\text{port}}}_{(\boldsymbol{\mu}|_B,\mathbf{q}),\,(\mu_B,\mathbf{q});\,(\pi^{\text{port}},\ell^{\text{port}})}\big(\boldsymbol{\xi}_{\text{port}}(\alpha_R)\big).$$

The scale of Theorem 15.3 keeps the letters closed and integrates the scale alone, one share per distinct Skeleton value beyond the first (Corollary 15.5).

Theorem 15.18 (The native split is an admissible contour)**.** *For real data with $s_j, t_j \geq 0$, not all zero, $\mathbf{m} > 0$ and $\kappa_{\mathcal{E}} > 0$, a native gauge $\alpha^\star$ and a bunch B, the real parts $c_{j,p} = s_j\pi_{j,p}(\alpha^\star)$, $c'_{j,p} = t_j\pi^*_{j,p}(\alpha^\star)$ of the native split satisfy the hypotheses of Theorem 15.15; a letter with exponent zero is 1 and is not opened.*

Corollary 15.19 (The port decides convergence)**.** *For real data with $s_j, t_j \geq 0$ and any bunch B, $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ converges if and only if some real parts satisfy the hypotheses of Theorem 15.15 at B.*

15.4 Signed letters

Definition 15.20 (Signed letters)**.** A *signed letter* is $L = L_+ - L_-$ with $L_\pm$ positive combinations of monomials of one degree g; adjoining $\zeta|u|^g$ gives $L+\zeta$ on the simplex, positive for $\zeta > -\min_{\Delta_\mathcal{E}} L$.

Proposition 15.21 (A signed letter is a boundary value)**.** *Let a signed letter L enter a form with exponent s beside positive letters L_i with exponents w_i, and for $\zeta > -\min_{\Delta_\mathcal{E}} L$ put $G(\zeta) = \int_{\Delta_\mathcal{E}} u^{\mathbf{m-1}} (L+\zeta)^{-s} \prod_i L_i^{-w_i}\, d\omega_\mathcal{E}$.*

(i) With ν the image of $u^{\mathbf{m-1}} \prod_i L_i^{-w_i} d\omega_\mathcal{E}$ under $u \mapsto L(u)$, $G(\zeta) = \int (y+\zeta)^{-s}\, \nu(dy)$ is holomorphic off $(-\infty, -\min_{\Delta_\mathcal{E}} L]$, continues across that half-line except at the negatives of the critical values of L on the simplex and its faces, and satisfies a linear differential equation in ζ with polynomial coefficients.

(ii) $G(0 \pm i0) = \int_{L>0} u^{\mathbf{m-1}} L^{-s} \prod_i L_i^{-w_i} d\omega_\mathcal{E} + e^{\mp i\pi s} \int_{L<0} u^{\mathbf{m-1}} |L|^{-s} \prod_i L_i^{-w_i} d\omega_\mathcal{E}$, the region integrals convergent for $\operatorname{Re} s < 1$ near regular zeros and taken by Lemma 8.4 in the coordinate $y = L(u)$ otherwise.

(iii) $G(0+i0) - G(0-i0) = -2i \sin(\pi s) \int_{L<0} u^{\mathbf{m-1}} |L|^{-s} \prod_i L_i^{-w_i} d\omega_\mathcal{E}$.

On the scale of a bunch a zero of a signed letter is a split point: on each of the intervals it bounds, rescaled to $[0,1]$, the zero is at an endpoint, where it is a wall of the integral over that interval and Theorem 15.14 applies, and the phase $e^{\mp i\pi s}$ attaches to the interval on which the letter is negative.

Chapter 16

Integrals, expectations and the Foreman procedure

16.1 Forms, sections, completions

Definition 16.1 (Differential forms, charts, schedules)**.** The *differential form* of a family at masses $\mathbf{m}$ is

$$\omega = W\, d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m-1}} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j} d\alpha,$$

the family word against the measure $d\alpha_{\mathbf{m}}$ (Definition 8.1). It is *projective* when $\kappa_{\mathcal{E}} = 0$, and then it is integrated over the rays of the orthant, over any section (Lemma 16.2, next). The *chart* at a mark g is the section $\alpha_g = 1$ with the coordinates α_e, $e \neq g$, and the chart integral is $\int_{\alpha_g=1} \omega = \int_{\mathbb{R}^{\mathcal{E}-g}_{>0}} (\cdots)|_{\alpha_g=1} \prod_{e\neq g} d\alpha_e$. Passing to the chart at g is the *snip* at g. A *schedule* $\mathfrak{s} = (g; \varsigma)$ is the chart at a mark g and an order $\varsigma(1), \dots, \varsigma(n-1)$ of the other marks.

Lemma 16.2 (Every section gives the period)**.** *If ω is projective with convergent period, f its integrand, $h > 0$ homogeneous of degree one and φ a function on $(0,\infty)$ with $\int_0^\infty |\varphi(\rho)|\, d\rho/\rho < \infty$, then $\int_{\mathbb{R}^{\mathcal{E}}_{>0}} f\, \varphi(h)\, d\alpha = \big(\int_0^\infty \varphi(\rho) \frac{d\rho}{\rho}\big) P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$. In particular every chart integral converges and equals $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$.*

Proposition 16.3 (A section is a table)**.** *Let $W = \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ be a family word, in this statement of the degree $-d_W$, $d_W = \sum_j (s_j r_j + t_j r_j^*)$, and $\mathbf{m}$ masses with every proper wall positive. Then $\int_{\Delta_{\mathcal{E}}} u^{\mathbf{m-1}} W\, d\omega_{\mathcal{E}}$ is the value on its balance wall of the family with the list $\binom{\mathcal{E}}{1}$, whose bases are the singletons and whose letter is $K_0 = |\alpha|$, added at the exponents $(m_{\mathcal{E}} - d_W, 0)$; its proper walls are those of the family. In particular, for the family F_0 of the one table $\binom{\mathcal{E}}{1}$, $\mathrm{Dir}_{\mathbf{m}} = \mathrm{Breta}^\circ(F_0; \mathbf{m}; m_{\mathcal{E}}, 0)$, the Breta shape law of $\binom{\mathcal{E}}{1}$ at a full basis lean; and for two families F_1, F_2 on $\mathcal{E}$ with words W_1, W_2, the logarithms of the letters of F_2 under the law on $\Delta_{\mathcal{E}}$ with density proportional to $u^{\mathbf{m-1}} W_1$ have the Laplace transform*

$$\int_{\Delta_{\mathcal{E}}} u^{\mathbf{m-1}} W_1 W_2\, d\omega_{\mathcal{E}} \Big/ \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m-1}} W_1\, d\omega_{\mathcal{E}}$$

in the exponents of W_2, a ratio of two values of the family $F_1 \cup F_2 \cup F_0$.

Definition 16.4 (Completions)**.** For $w \in \mathbb{R}^{\mathcal{E}}$ with $w_{\mathcal{E}} = 1$ the *completion* of $\omega = W\, d\alpha_{\mathbf{m}}$ along w is $\omega_{\mathbf{m}'} = \alpha^{-\kappa_{\mathcal{E}} w} \omega = W\, d\alpha_{\mathbf{m}'}$, $\mathbf{m}' = \mathbf{m} - \kappa_{\mathcal{E}} w$, which is projective: a completion acts on the measure

and leaves the word. The *chart completion* at g has $w = \mathbf{e}_g$ and the masses $\mathbf{m}^{(g)} = \mathbf{m} - \kappa_{\mathcal{E}}\mathbf{e}_g$. For real data the *native completion* has $w = \mathbf{m}/m_{\mathcal{E}}$; its measure is the differential

$$d\alpha_{\boldsymbol{\tau}} = \alpha^{-(\kappa_{\mathcal{E}}/m_{\mathcal{E}})\mathbf{m}}\, d\alpha_{\mathbf{m}}, \qquad \boldsymbol{\tau} = \frac{n\sigma + \sum_j \delta_j(2r_j - n)}{n\sigma + \mu_{\mathcal{E}}}\,\mathbf{m},$$

at the target, with $\kappa_{\mathcal{E}} = \mu_{\mathcal{E}} - \sum_j \delta_j(2r_j - n)$ and $m_{\mathcal{E}} = n\sigma + \mu_{\mathcal{E}}$ in pair coordinates (Theorem 8.2(iii)), and the completed form is $\omega_{\boldsymbol{\tau}} = W\, d\alpha_{\boldsymbol{\tau}}$.

Proposition 16.5 (Snips are completions)**.** *(i)* $\int_{\alpha_g=1}\omega = \int_{\alpha_g=1}\omega_{\mathbf{m}^{(g)}}$*, the period* $P_F(\mathbf{m}^{(g)};\mathbf{s},\mathbf{t})$ *wherever it converges.*

(ii) $K_j|_{\alpha_g=1} = K_{j,\backslash g} + K_{j,/g}$ *and* $\Psi_j|_{\alpha_g=1} = \Psi_{j,\backslash g} + \Psi_{j,/g}$*, and for* $\operatorname{Re} s_j, \operatorname{Re} t_j > 0$*, on the contours of Corollary 4.4,*

$$\prod_j K_j^{-s_j}\Psi_j^{-t_j}\Big|_{\alpha_g=1} = \int\cdots\int \prod_j \Gamma_1^2\begin{bmatrix} a_j,\ a_j' \\ s_j \end{bmatrix}\Gamma_1^2\begin{bmatrix} b_j,\ b_j' \\ t_j \end{bmatrix} K_{j,\backslash g}^{-a_j}K_{j,/g}^{-a_j'}\Psi_{j,\backslash g}^{-b_j}\Psi_{j,/g}^{-b_j'}$$

over $a_j + a_j' = s_j$*,* $b_j + b_j' = t_j$*: a snip opens every letter into its deletion and its contraction, with stacked-Gamma shares.*

(iii) For a bunch B*, the integral of* ω *over the bunch section* $|\alpha_B| = 1$ *is the integral over the rays of the projective form* $|\alpha_B|^{-\kappa_{\mathcal{E}}}\omega$*. In this statement write* F' *for* F *with one table appended at the exponents* $(\kappa_{\mathcal{E}}, 0)$*: the list* $\binom{B}{1}$ *read on* $\mathcal{E}$*, whose bases are the singletons of* B*, with the marks outside* B *as loops, and whose letter is* $|\alpha_B|$*. Then* $|\alpha_B|^{-\kappa_{\mathcal{E}}}\omega$ *is the form of* F' *at the masses* $\mathbf{m}$*,* F' *is balanced there, its proper walls are* $\kappa_S - \kappa_{\mathcal{E}}\,\mathbf{1}[B \subseteq S]$*, and*

$$\int_{|\alpha_B|=1}\omega = P_{F'}\big(\mathbf{m};(\mathbf{s},\kappa_{\mathcal{E}}),(\mathbf{t},0)\big)$$

wherever $\operatorname{Re}\kappa_S - \operatorname{Re}\kappa_{\mathcal{E}}\,\mathbf{1}[B \subseteq S] > 0$ *for every nonempty proper* S*. At* $B = \{g\}$ *the appended letter is the monomial* α_g*, and this is (i); at* $B = \mathcal{E}$ *the appended table is the list* $\binom{\mathcal{E}}{1}$ *(Proposition 16.3). For* $k = -\kappa_{\mathcal{E}} \in \mathbb{N}$*,* $k \geq 1$*,* $|\alpha_B|^k = \sum_{|\mathbf{n}|=k}\binom{k}{\mathbf{n}}\alpha_B^{\mathbf{n}}$*, so* $\int_{|\alpha_B|=1}\omega = \sum_{\mathbf{n}\in\mathbb{N}^B,\,|\mathbf{n}|=k}\binom{k}{\mathbf{n}}\int_{|\alpha_B|=1}\omega_{\mathbf{m}+\mathbf{n}}$*, each term the period* $P_F(\mathbf{m}+\mathbf{n};\mathbf{s},\mathbf{t})$ *wherever it converges.*

Corollary 16.6 (The period fibres over the scale of a bunch)**.** *Let* B *be a proper bunch of a family* F *on* $\mathcal{E}$*, with room* R*, and let* $h > 0$ *be a function of* α_R *homogeneous of degree one. Wherever the integral of* $W\, d\alpha_{\mathbf{m}}$ *over the section* $\{h(\alpha_R) = 1\}$ *of the orthant converges, both sections taken with the measures of Lemma 16.2,*

$$\int_{h(\alpha_R)=1} W\, d\alpha_{\mathbf{m}} = \int_{\{\alpha_R\in\mathbb{R}^R_{>0}:\ h(\alpha_R)=1\}}\int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}}\alpha_R^{\mathbf{m}|_R - \mathbf{1}}\ \mathrm{Sc}_B(u,\alpha_R;\mathbf{m};\mathbf{s},\mathbf{t})\, d\omega_B(u),$$

and at every point Sc_B *is the product of Theorem 15.3, a stacked Gamma times a Breta of one bunch. On the balance wall the left side is* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*. At the chart* $h = \alpha_g$*,* $g \in R$*, the left side is* $P_F(\mathbf{m}^{(g)};\mathbf{s},\mathbf{t})$ *(Proposition 16.5(i)) and the outer integral is over* α_{R-g} *at* $\alpha_g = 1$*; at* $h = |\alpha_R|$ *it is over* Δ_R *against* $d\omega_R$*.*

Proposition 16.7 (The scaling mode)**.** *Along $z \mapsto z + c\mathbf{1}$, $E_j(z + c\mathbf{1}) = E_j(z)$, $O_j(z + c\mathbf{1}) = O_j(z) + 2c(2r_j - n)$, and the exponent $2\langle \boldsymbol{\mu}, z\rangle - \sum_j (\sigma_j E_j + \delta_j O_j)$ of the pair form moves by $2c\kappa_{\mathcal{E}}$. Off the balance wall the form is not integrable along this direction; it is* damped *by $e^{-|\alpha|}$ (the interest, $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_{\mathcal{E}})P_F(\mathbf{m};\mathbf{s},\mathbf{t})$),* conditioned *by a section (at a chart a completion, Proposition 16.5; in general another projective form, Proposition 16.32 below), or* slid *by moving the masses along their ray to the balance wall (the native completion).*

Theorem 16.8 (The slide)**.** *Let ω have real data on a connected family, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, native gauge $\alpha^\star$ and target $\boldsymbol{\tau}$.*

(i) If $\kappa_{\mathcal{E}} \neq 0$, exactly one completion of ω has $\alpha^\star$ as a native gauge: $\omega_{\boldsymbol{\tau}}$. If $\kappa_{\mathcal{E}} = 0$, every completion is ω and $\boldsymbol{\tau} = \mathbf{m}$.

(ii) As the masses slide along their ray, $\mathbf{m} \mapsto \lambda\mathbf{m}$, the data $(\lambda\mathbf{m};\mathbf{s},\mathbf{t})$ lie on the balance wall exactly when $\lambda\mathbf{m} = \boldsymbol{\tau}$.

(iii) Every proper wall of $\omega_{\boldsymbol{\tau}}$ is positive, so its period converges.

(iv) $\boldsymbol{\tau}$, $\omega_{\boldsymbol{\tau}}$ and $\alpha^\star$ are the same at every point of the ray of $\mathbf{m}$.

Each of the three treatments of the free scaling mode (Proposition 16.7) is an evaluation, and sliding keeps the native gauge.

16.2 Three integrals and their expectations

Lemma 16.9 (The period is meromorphic in the data)**.** *For every family, P_F is meromorphic in the masses and the exponents, with poles only where a proper wall is a nonpositive integer, and so is its restriction to the balance wall.*

Definition 16.10 (Three integrals and their expectations)**.** Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, so that W is homogeneous of degree $-\tau_{\mathcal{E}}$, $\tau_{\mathcal{E}} = n\sigma + \sum_j \delta_j(2r_j - n)$ with $\sigma = \sum_j \sigma_j$. On a connected family the centre of ω is the form of the family at the centre (Definition 9.1), at the exponents $(\boldsymbol{\sigma}, \boldsymbol{\sigma})$ and the masses $\sigma\mathbf{1}$,

$$\omega_{\mathrm{c}} = \prod_j \big(K_j \Psi_j\big)^{-\sigma_j} d\alpha_{\sigma\mathbf{1}}.$$

It is projective, its proper walls $\sum_j \sigma_j c_j(S)$ are positive and the table's gauge is native for it (Proposition 8.21), so its period $P_{\mathrm{c}} = P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$ is a positive convergent integral, and $\mathbb{P}_{\mathrm{c}} = \mathrm{Breta}^\circ(F;\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, the measure $\omega_{\mathrm{c}}/P_{\mathrm{c}}$ read on the rays, is the *law of the centre*. For masses $\mathbf{m}'$ of total $\tau_{\mathcal{E}}$ the *departure* of $W\, d\alpha_{\mathbf{m}'}$ from the centre is the ratio of the two forms,

$$\frac{W\, d\alpha_{\mathbf{m}'}}{\omega_{\mathrm{c}}}(\alpha) = \alpha^{\mathbf{m}' - \sigma\mathbf{1}} \prod_j \Big(\frac{K_j(\alpha)}{\Psi_j(\alpha)}\Big)^{-\delta_j},$$

homogeneous of degree 0: an *offset factor* $\alpha^{\mathbf{m}'-\sigma\mathbf{1}}$ and a *lean factor* $\prod_j (K_j/\Psi_j)^{-\delta_j}$. Each integral below is unnormalized. On a connected family its expectation divides it by P_{c}: it is the expectation, under the law of the centre, of the departure that its evaluation sets.

(i) The *integral* $\int W\, d\alpha_{\mathbf{m}}$ is defined when ω is projective with convergent period: it is the integral of ω over any section, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$, and $\int W\, d\alpha_{\mathbf{m}} = P_{\mathrm{c}}\, \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\big[(\omega/\omega_{\mathrm{c}})(U)\big]$.

(ii) For a schedule $\mathfrak{s} = (g; \varsigma)$ the *quasi-integral* is the iterated finite part

$$\operatorname{Reg}^{\mathfrak{s}}\!\int W\, d\alpha_{\mathbf{m}} = \operatorname{Reg}\!\int_0^\infty dx_{n-1} \cdots \operatorname{Reg}\!\int_0^\infty dx_1 \Big(\alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\Big)\Big|_{\alpha_g = 1}, \qquad x_k = \alpha_{\varsigma(k)},$$

defined whenever the integrand of every one of these iterated integrals has log-power expansions at 0 and at ∞. It sets the departure of the chart completion, and the *quasi-expectation* is $\mathfrak{E}^{\mathfrak{s}}_{U\sim\mathbb{P}_{\mathrm{c}}}\big[(\omega_{\mathbf{m}^{(g)}}/\omega_{\mathrm{c}})(U)\big] = \operatorname{Reg}^{\mathfrak{s}}\!\int W\, d\alpha_{\mathbf{m}} \big/ P_{\mathrm{c}}$.

(iii) For a connected family of at least two marks, the *pseudointegral* is

$$\Big\backslash W\, d\alpha_{\mathbf{m}} = \operatorname*{CT}_{\eta=0} P_F(\boldsymbol{\tau} + \eta\mathbf{m}; \mathbf{s}, \mathbf{t}),$$

the constant term at the target along the ray of the masses, P_F being meromorphic in the masses with poles only where a proper wall is a nonpositive integer, on the *pole hyperplanes* $\kappa_S = -q$, S proper, $q \in \mathbb{N}$ (Lemma 16.9). Off those hyperplanes it is $P_F(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$, the value at the target of the meromorphic function P_F on the balance wall. With a native gauge every proper wall is positive at $\boldsymbol{\tau}$ and $\backslash W\, d\alpha_{\mathbf{m}} = \int W\, d\alpha_{\boldsymbol{\tau}}$: the Foreman sets the measure to the native completion. It sets the departure of the native completion, and the *pseudoexpectation* is

$$\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big] = \Big\backslash W\, d\alpha_{\mathbf{m}} \Big/ P_{\mathrm{c}};$$

with a native gauge it is the expectation $\mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$: the Foreman sets the departure and the centre sets the law. On a family whose weights factor across its molecules, the pseudointegral and the pseudoexpectation are products over the molecules, each at its own target and with its own centre; a one-mark factor contributes the value of its integrand at $\alpha = \mathbf{1}$ to the pseudointegral and 1 to the pseudoexpectation, its integrand and its centre's agreeing there.

A point $(\boldsymbol{\mu}; \boldsymbol{\sigma}, \boldsymbol{\delta})$ splits into an even part, the strengths σ_j, and an odd part, the offsets and the leans; the fold (Theorem 16.18) negates the odd part and keeps the even part. The centre is the point with the same even part and no odd part. It lies on the balance wall, it is inside its walls at every strength on a connected family, and its native gauge is the table's gauge, so its period is a convergent integral of the family itself, the same for every evaluation. Dividing by it prices the odd part of a point against its even part: with a native gauge the pseudoexpectation is the exponential of a divergence from the law of the centre, at least 1, equal to 1 exactly when the native completion is the centre's form, and unchanged by the fold (Proposition 16.34).

Theorem 16.11 (Quasi-integrals extend the integral)**.** *Each* $\operatorname{Reg}^{\mathfrak{s}}\!\int$ *is linear in the integrand, and* $\operatorname{Reg}^{\mathfrak{s}}\!\int W\, d\alpha_{\mathbf{m}} = \int W\, d\alpha_{\mathbf{m}}$ *for every schedule wherever* $\int W\, d\alpha_{\mathbf{m}}$ *is defined; there every chart completion is* ω*,* $\mathbf{m}^{(g)} = \mathbf{m}$*, and on a connected family* $\mathfrak{E}^{\mathfrak{s}}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\mathbf{m}^{(g)}}/\omega_{\mathrm{c}})(U)] = \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega/\omega_{\mathrm{c}})(U)]$*. Elsewhere the value can depend on the chart, on the order and on the coordinates (Lemma 1.9).*

Proposition 16.12 (The form of a compound at one point)**.** *Let* F *be the direct sum of families* $F_1, \dots, F_q$ *on* $\mathcal{E}_1, \dots, \mathcal{E}_q$*,* $q \geq 2$*,* $\mathcal{E} = \mathcal{E}_1 \sqcup \cdots \sqcup \mathcal{E}_q$*, every table a direct sum across the split with factoring weights, so that* $K_j = \prod_i K_{j,i}$ *and* $\Psi_j = \prod_i \Psi_{j,i}$*; let the data be real, the masses* $\mathbf{m} = (\mathbf{m}_1, \dots, \mathbf{m}_q)$ *on the balance wall of* F*, and* κ_i *the imbalance of* F_i *at* $\mathbf{m}_i$*, so* $\sum_i \kappa_i = 0$*. On the section* $|\alpha_q| = 1$ *with* $\alpha_i = \rho_i u_i$*,* $u_i \in \Delta_{\mathcal{E}_i}$*, for* $i < q$ *and* $u_q = \alpha_q$*, the form of* F *is*

$$\prod_{i<q} \rho_i^{\kappa_i - 1} d\rho_i \;\cdot\; \prod_{i=1}^{q} u_i^{\mathbf{m}_i - \mathbf{1}} \prod_j K_{j,i}^{-s_j} \Psi_{j,i}^{-t_j}\, d\omega_{\mathcal{E}_i}.$$

(i) *Its integral diverges for all* $\kappa_1, \dots, \kappa_q$.

(ii) *The finite part of each* ρ_i*-integral is* 0 *for every* κ_i.

(iii) *On the masses* $\mathbf{m}(\boldsymbol{\eta}) = (\mathbf{m}_i^0 + i\eta_i w_i)_{i=1}^q$, $\boldsymbol{\eta} \in \mathbb{R}^q$ *with* $\sum_i \eta_i = 0$, *with* $\mathbf{m}_i^0$ *balanced for* F_i *inside its walls and* $w_i \in \mathbb{R}^{\mathcal{E}_i}$ *of total* 1, *so that* $\kappa_i = i\eta_i$, *the integral of the form is, as a distribution in* $(\eta_1, \dots, \eta_{q-1})$, $(2\pi)^{q-1} \prod_{i=1}^q P_{F_i}(\mathbf{m}_i^0; \mathbf{s}, \mathbf{t})$ *times the Dirac measure at* 0.

So the compound carries nothing beyond the points of its parts, and only where each molecule is balanced on its own.

Theorem 16.13 (Presentations)**.** *Let* $\omega = W\, d\alpha_{\mathbf{m}}$ *have real data on a connected family,* $s_j, t_j \geq 0$ *not all zero and* $\mathbf{m} > 0$. *A* presentation *of* ω *is a bijection* φ *of the marks together with coordinates* $x = \lambda^{-1}\alpha$, $\lambda \in \mathbb{R}^{\mathcal{E}}_{>0}$, *in which the letters are* $K_j(\lambda x)$, $\Psi_j(\lambda x)$ *and the form carries the constant* $\lambda^{\mathbf{m}}$. *The split, the targets, the native gauges as points of the torus, the stills, the classes, the native splits and the children computed from a presentation are the images under* φ *of those computed from* ω, *and, when the target lies on no pole hyperplane, the pseudointegral and the pseudoexpectation of the presentation, its centre presented with it, are* $(\lambda^{\mathbf{m}})^{\kappa\varepsilon/m_{\mathcal{E}}}$ *times those of* ω, *the factor being* 1 *on the balance wall; on a pole hyperplane the constant term moves with the coordinates as well (Proposition 16.31 below). In the coordinates* x *the table's gauge is* $x = \lambda^{-1}$, *and the Foreman tree is the same. On a family whose weights factor across its molecules the factor is the product of the molecules' factors.*

Definition 16.14 (The Foreman procedure)**.** *Classification*, from the tables alone:

(1) *Split* the marks into molecules and units (Proposition 9.5).

(2) *Read* each molecule M at a centre $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ of the table's gauge (Chapter 10).

(3) *Answer*: a constant reading is an atom; otherwise take the Foreman bunch $B(M; \boldsymbol{\sigma}, 0)$ (Definition 13.6).

(4) *Peel* at $B(M; \boldsymbol{\sigma}, 0)$; the childcules sit at the centres they inherit (Proposition 12.27).

(5) *Recurse* to atoms and units: the output is the Foreman tree and, at the root, the label.

Evaluation at a real point $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$:

(6) *Native gauge* of each molecule, and its target $\boldsymbol{\tau}$ (Theorem 8.19).

(7) *Slide* the masses to the target (Theorem 16.8).

(8) *Ports* along the tree, every share contour through the native split (Theorems 15.15, 15.18 and 12.26); on the balance wall the port of Chapter 18.

(9) *Atoms*: the atom formula (Theorem 15.15 with $B = \mathcal{E}$; on the balance wall, Chapter 18).

On the orthant side the ports integrate the Foreman bunch of each node inside the integral over its room, all its marks at once in scale and shape (Corollary 15.17): the root's bunch first and the atoms last. This is the *Foreman's order*.

Theorem 16.15 (The Foreman computes the pseudointegral)**.** *Let* $\omega = W\, d\alpha_{\mathbf{m}}$ *have real data on a connected family, with* $s_j, t_j \geq 0$, *not all zero, and* $\mathbf{m} > 0$, *and a native gauge. For every* $\eta > 0$ *the procedure at the masses* $\boldsymbol{\tau} + \eta\mathbf{m}$ *runs on admissible contours and computes* $J_F(\boldsymbol{\tau} + \eta\mathbf{m}; \mathbf{s}, \mathbf{t})$, *and*

$$\int W\, d\alpha_{\mathbf{m}} = \lim_{\eta \to 0^+} \frac{J_F(\boldsymbol{\tau} + \eta\mathbf{m}; \mathbf{s}, \mathbf{t})}{\Gamma(\eta m_{\mathcal{E}})}.$$

No finite part, chart, order or closing rule enters: the regulator η *leaves by a limit.*

Corollary 16.16 (One member among the generalized integrals)**.** *Wherever $\int W\, d\alpha_{\mathbf{m}}$ is defined for real data on a connected family, $⨍ W\, d\alpha_{\mathbf{m}} = \int W\, d\alpha_{\mathbf{m}}$, and with Theorem 16.11 every evaluation sets the one departure $\omega/\omega_{\mathrm{c}}$ on the balance wall. Off the balance wall, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, the period at the target converges exactly when the family has a native gauge (Theorem 8.19(iii)), and otherwise $⨍ W\, d\alpha_{\mathbf{m}}$ is its continuation in the masses, on a pole hyperplane the constant term of the continuation along the ray.*

Theorem 16.17 (The pseudointegral off the domain)**.** *For every family, if some nonempty proper S has $c_j(S) = 0$ in every active table, then, first where every proper wall has positive real part and then as meromorphic functions (Lemma 16.9),*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma{\scriptstyle\begin{bmatrix}\kappa_S,\ \kappa_{\mathcal{E}\setminus S}\\ \kappa_{\mathcal{E}}\end{bmatrix}}_{\!1}^{\!2} \int_{\Delta_S\times\Delta_{\mathcal{E}\setminus S}} \alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}\, d\omega_S\, d\omega_{\mathcal{E}\setminus S},$$

and the restriction of P_F to the balance wall is 0. Let now F be connected with real data, $s_j, t_j \geq 0$ not all zero; on the balance wall the value of P_F at a target depends on no path.

(i) The continuation in the masses of $f \mapsto \int_{\Delta_{\mathcal{E}}} f\, u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j}\Psi_j^{-t_j} d\omega_{\mathcal{E}}$, taken by Lemma 8.4 on each ordering sector, is a distribution on $\Delta_{\mathcal{E}}$ with normaliser $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ and monomial moments $P_F(\mathbf{m}+\boldsymbol{\gamma};\mathbf{s},\mathbf{t})/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$; where every proper wall is positive it is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ times the Breta shape law. Just past the first wall crossed it is not a positive measure: P_F passes from $+\infty$ to $-\infty$ there, while $P_F(\mathbf{m}+\boldsymbol{\gamma};\mathbf{s},\mathbf{t})$ is a positive convergent integral for γ_S large.

(ii) Theorem 8.14 and the recurrences of Theorem 8.9 hold for these values as identities of meromorphic functions.

Theorem 16.18 (The fold: Cremona off the balance wall)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$, with the exponents $(\mathbf{s},\mathbf{t})$, have real data on a connected family, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, native gauge $\alpha^\star$ and target $\boldsymbol{\tau}$. Then $2\sigma\mathbf{1} - \boldsymbol{\tau}$ is positive and balanced for the exponents $(\mathbf{t},\mathbf{s})$, with native gauge $(\alpha^\star)^{-1}$, and for every $\lambda > 0$*

$$⨍ \prod_j K_j^{-s_j}\Psi_j^{-t_j}\, d\alpha_{\mathbf{m}} = ⨍ \prod_j K_j^{-t_j}\Psi_j^{-s_j}\, d\alpha_{\lambda(2\sigma\mathbf{1}-\boldsymbol{\tau})}.$$

The map from the ray of $\mathbf{m}$ to the ray of $2\sigma\mathbf{1} - \boldsymbol{\tau}$ is an involution on rays.

The fold swaps the two letters of every table and reflects the target through $\sigma\mathbf{1}$, negating the offsets and the leans; it keeps the strengths, so the two forms have one centre, and it carries uniform masses to uniform masses.

Proposition 16.19 (Points and realizations)**.** *A real point $(\mathbf{m};\mathbf{s},\mathbf{t})$ of a connected family, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$ and native gauge $\alpha^\star$, determines the realization $((s_j,t_j)_j;\alpha^\star)$, whose masses are the target; every realization arises so (Proposition 8.22). The reading of a point is the reading of its realization and may differ from point to point; the label is read at centres of the table's gauge, the Foreman tree at the realizations of the table's gauge at the centre and at the leans (Theorem 14.6), and both are functions of the tables. An atomic point is at the table's gauge up to scale (Theorem 11.4), and its centre is atomic (Theorem 11.1).*

Theorem 16.20 (Parity)**.** *Let g be an exchange of a connected family, with the bijection h of its tables (Definition 11.13); then $r_{h(j)} = n - r_j$ for every j. With $(g^*z)_e = z_{g(e)}$ (Definition 11.13), let Φ° be the centred potential (Theorem 8.19).*

(i) $\Lambda_j(g^*z) = \Lambda_{h(j)}(-z)$*; so* $E_j(g^*z) = E_{h(j)}(z)$*,* $O_j(g^*z) = -O_{h(j)}(z)$ *and* $\Phi^\circ(-g^*z) = \Phi^\circ(z)$*.*

(ii) If the data are real with $s_j, t_j \geq 0$*, not all zero, and* $\mathbf{m} > 0$*, have a native gauge, and* $\tau_{g(e)} = 2\sigma - \tau_e$ *for every* e*, the native gauge* $e^{2z^\star}$ *with* $\langle \mathbf{1}, z^\star \rangle = 0$ *has* $-g^*z^\star = z^\star$*, and the still of the realization* $\Upsilon = ((s_j, t_j)_j; e^{2z^\star})$ *has* $\kappa_\Upsilon(g(e_1), \dots, g(e_k)) = (-1)^k \kappa_\Upsilon(e_1, \dots, e_k)$ *for every* k*.*

(iii) Under (ii) even rungs never separate e *from* $g(e)$*, and at odd rungs their tallies (Definition 9.4) are negatives of each other.*

(iv) If moreover $\boldsymbol{\delta} = 0$ *and* $\boldsymbol{\tau} = \sigma\mathbf{1}$*, then* $z^\star = 0$*, every odd rung vanishes, and no rung separates* e *from* $g(e)$*.*

(v) At $(\mathbf{t}, \mathbf{s})$ *and the target* $2\sigma\mathbf{1} - \boldsymbol{\tau}$ *the native gauge is* $e^{-2z^\star}$ *and rung* k *is* $(-1)^k$ *times rung* k *at* $(\mathbf{s}, \mathbf{t})$*; under (ii) this is the still at* $(\mathbf{s}, \mathbf{t})$ *carried by* g*, and the Foreman bunch at* $(\mathbf{t}, \mathbf{s})$ *is the image under* g^{-1} *of the Foreman bunch at* $(\mathbf{s}, \mathbf{t})$*.*

16.3 Quasi-integrals as Laurent data

Definition 16.21 (Regular schedules)**.** For a schedule $\mathfrak{s} = (g; \varsigma)$ put $x_k = \alpha_{\varsigma(k)}$ and $v_k = \mathbf{e}_{\varsigma(k)} - \mathbf{e}_g$. Let $\mathcal{J}_0$ be the chart integrand and $\mathcal{J}_k(\varepsilon_1, \dots, \varepsilon_k; x_{>k}) = \int_0^\infty x_k^{\varepsilon_k} \mathcal{J}_{k-1}\, dx_k$, with $\int_0^1$ and $\int_1^\infty$ each continued meromorphically in the *regulator* ε_k. The schedule is *regular* when for every k, as $x_k \to 0$ and as $x_k \to \infty$, $\mathcal{J}_{k-1}$ is a finite sum of terms $c(\varepsilon, x_{>k})\, x_k^{a(\varepsilon)}$ plus $O(x_k^{\pm M})$ for every M, the exponents affine in $\varepsilon_{<k}$ and independent of $x_{>k}$, uniformly for the regulators in a small polydisc away from their poles.

Theorem 16.22 (Bridge)**.** *For real data on a connected family and a regular schedule,* $\mathrm{Reg}^{\mathfrak{s}} \int W\, d\alpha_{\mathbf{m}}$ *is defined and*

$$\mathrm{Reg}^{\mathfrak{s}} \int W\, d\alpha_{\mathbf{m}} = \operatorname*{CT}_{\varepsilon_{n-1}} \cdots \operatorname*{CT}_{\varepsilon_1} P_F\Big(\mathbf{m}^{(g)} + \sum_{k=1}^{n-1} \varepsilon_k v_k; \mathbf{s}, \mathbf{t}\Big),$$

the constant terms taken innermost first, $|\varepsilon_1| \ll \cdots \ll |\varepsilon_{n-1}|$*.*

Theorem 16.23 (A wall is closed by its last variable)**.** *For a schedule* $\mathfrak{s} = (g; \varsigma)$ *put* $\mathbf{p} = \mathbf{m}^{(g)}$ *and* $I(\varepsilon) = P_F\big(\mathbf{p} + \sum_k \varepsilon_k v_k; \mathbf{s}, \mathbf{t}\big)$*. A proper* U *is* polar *when* $\kappa_U(\mathbf{p}) \in -\mathbb{N}$*; its* members *are the* k *with* $\varsigma(k) \in U$ *if* $g \notin U$*, and with* $\varsigma(k) \notin U$ *if* $g \in U$*;* $\mathrm{last}(U)$ *is its last member.*

(i) Near 0*,* $I(\varepsilon) = \sum_C f_C(\varepsilon) \big/ \prod_{U \in C} \ell_U(\varepsilon) + H(\varepsilon)$*,* $\ell_U = \pm \sum_{k\ member} \varepsilon_k$*, over the nonempty sets* C *of polar walls that are tails of one order, with* f_C, H *holomorphic; the member sets in one* C *are nested or disjoint.*

(ii) Under $\mathrm{CT}_{\varepsilon_k}$*,* $1/\ell_U$ *is holomorphic at* $\varepsilon_k = 0$ *unless* $k = \mathrm{last}(U)$*; for one polar wall with numerator* f_C *the iterated constant term of* f_C/ℓ_U *is* $\pm\partial_{\varepsilon_a} f_C(0)$*,* $a = \mathrm{last}(U)$*.*

(iii) For a fixed chart, the quasi-integral of a regular schedule depends on the order only through $U \mapsto \mathrm{last}(U)$ *on the polar walls.*

With one regulator ε_D for each class D of a realization, taken in some order of the classes, $\ell_U = \pm \sum_D |U \cap D|\, \varepsilon_D$; a wall is closed by the last class it meets, which contributes $|U \cap D|^{-1} \partial_{\varepsilon_D}$, and the value depends only on the order of the classes, which can be taken from the order of classes (Theorem 13.7).

Proposition 16.24 (The coordinates set the log-scale)**.** *In the coordinates $u = \alpha/\beta$, $\beta_g = 1$, with the regulators ε_k entering as $u_{\varsigma(k)}^{\varepsilon_k}$, the regulated function is $e^{-\sum_k \varepsilon_k \lambda_k} I(\varepsilon)$, $\lambda_k = \log \beta_{\varsigma(k)}$, and for one polar wall with numerator f_C and last member a the iterated constant term becomes $\pm\big(\partial_{\varepsilon_a} f_C(0) - \lambda_a f_C(0)\big)$: changing coordinates shifts the finite part by $-\log \beta_{\varsigma(a)}$ times the signed residue of each wall closed by a mark whose gauge differs from the chart's.*

Theorem 16.25 (Residue at a wall)**.** *Let ω be projective, S a proper bunch with room R, $g \in R$ and $\alpha_S = \rho u$. In this statement write $K_j^{[l]} = K_{j,S}^{(d_j(S)+l)}(u, \alpha_R)$ and $\Psi_j^{[l]} = \Psi_{j,S}^{(d_j^*(S)+l)}(u, \alpha_R)$ for the level and colevel letters of Definition 12.16 counted from their floors, so that $K_j(\rho u, \alpha_R) = \rho^{d_j(S)} \sum_{l \geq 0} \rho^l K_j^{[l]}$ and $\Psi_j(\rho u, \alpha_R) = \rho^{d_j^*(S)} \sum_{l \geq 0} \rho^l \Psi_j^{[l]}$. At points of $\kappa_S = -q$, $q \in \mathbb{N}$, where every other proper wall is positive,*

$$\operatorname*{Res}_{\kappa_S = -q} P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\alpha_g = 1} \int_{\Delta_S} u^{\mathbf{m}|_S - \mathbf{1}} \alpha_R^{\mathbf{m}|_R - \mathbf{1}} \prod_j \big(K_j^{[0]}\big)^{-s_j} \big(\Psi_j^{[0]}\big)^{-t_j} [\rho^q] h \; d\omega_S \, d\alpha_{R-g},$$

where

$$h = \prod_j \Big(\sum_l \rho^l K_j^{[l]} / K_j^{[0]} \Big)^{-s_j} \Big(\sum_l \rho^l \Psi_j^{[l]} / \Psi_j^{[0]} \Big)^{-t_j},$$

$$[\rho^q] \Big(1 + \sum_{l \geq 1} \rho^l X_l\Big)^{-s} = \sum_{\sum_l l n_l = q} (-1)^{|\mathbf{n}|} \frac{\Gamma(s + |\mathbf{n}|)}{\Gamma(s)} \prod_l \frac{X_l^{n_l}}{n_l!}.$$

If the weights factor on the bottom levels, then with the faces *of the wall, the families*

$$F^S = \big((\mathcal{B}_j/R;\, s_j, 0),\ (\mathcal{B}_j|S;\, 0, t_j)\big)_j \text{ on } S, \qquad F^R = \big((\mathcal{B}_j|R;\, s_j, 0),\ (\mathcal{B}_j/S;\, 0, t_j)\big)_j \text{ on } R$$

(Definition 7.5), the residue at $\kappa_S = 0$ is the product of their periods at the masses $\mathbf{m}|_S$ and $\mathbf{m}|_R$; both faces are balanced, the walls of F^S are the κ_U, $U \subseteq S$, and those of F^R are the $\kappa_{S \cup U}$, $U \subseteq R$, on the wall. Here P_F is the meromorphic function of Lemma 16.9; on a connected family with real data and $s_j, t_j \geq 0$, not all zero, it is the normaliser of the distribution of Theorem 16.17(i).

Proposition 16.26 (A molecule at a nonpositive wall)**.** *Let a molecule carry real data with $s_j, t_j \geq 0$ and target $\boldsymbol{\tau}$, and let S be a proper bunch with room R, $\alpha_S = \rho u$ on a chart $\alpha_g = 1$, $g \in R$, with $H(\rho) = H(\rho; u, \alpha_R)$ the native completion $W \, d\alpha_{\boldsymbol{\tau}}$ divided by $\rho^{\kappa_S(\boldsymbol{\tau}) - 1} d\rho \, d\omega_S \, d\alpha_{R-g}$ and $H_l = [\rho^l] H$ as in Theorem 16.25.*

(i) Along the collapse of S, $\Phi(y_0 - \lambda \mathbf{1}_S) - \langle \boldsymbol{\tau}, y_0 - \lambda \mathbf{1}_S \rangle = \lambda \, \kappa_S(\boldsymbol{\tau}) + O(1)$ as $\lambda \to \infty$, the left side being the negative logarithm of the density in $y = \log \alpha$ (Theorem 8.19); so the density does not decay along the collapse exactly when $\kappa_S(\boldsymbol{\tau}) \leq 0$, and then the form is not integrable at $\rho = 0$.

(ii) With $\langle \cdot \rangle$ the integral over u and α_R, if every other proper wall at $\boldsymbol{\tau}$ is positive, $\kappa_S(\boldsymbol{\tau})$ is not a nonpositive integer, and $L \in \mathbb{N}$ has $\kappa_S(\boldsymbol{\tau}) + L > 0$,

$$P_F(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t}) = \sum_{l < L} \frac{\langle H_l \rangle}{\kappa_S(\boldsymbol{\tau}) + l} + \int_0^1 \rho^{\kappa_S(\boldsymbol{\tau}) - 1} \Big(\langle H \rangle - \sum_{l < L} \rho^l \langle H_l \rangle \Big) d\rho + \int_1^\infty \rho^{\kappa_S(\boldsymbol{\tau}) - 1} \langle H \rangle \, d\rho,$$

the continuation of the period past the wall of S; its poles at $\kappa_S = -q$ have the residues $\langle H_q \rangle$ of Theorem 16.25. When the weights factor on the bottom levels, $\langle H_0 \rangle$ is the product of the integrals of the faces F^S at $\boldsymbol{\tau}|_S$ and F^R at $\boldsymbol{\tau}|_R$, whose imbalances are $\kappa_S(\boldsymbol{\tau})$ and $-\kappa_S(\boldsymbol{\tau})$: the leading term is the compound of Proposition 16.12.

Corollary 16.27 (The vertex)**.** *At $S = \mathcal{E} - e$, with e neither a loop nor a coloop in any table, the face F^R is the single mark e and contributes* 1*, and*

$$\langle H_0 \rangle = \int_{\Delta_{\mathcal{E}-e}} u^{\boldsymbol{\tau}|_{\mathcal{E}-e}-\mathbf{1}} \prod_j K_{j,/e}(u)^{-s_j} \Psi_{j,\backslash e}(u)^{-t_j} \, d\omega_{\mathcal{E}-e}(u),$$

the integral of the vertex child*: the contractions at e on the basis side and the deletions on the cobasis side. Proposition 16.26(ii) at $S = \mathcal{E} - e$ removes the leading terms of the vertex from the period exactly.*

16.4 The pseudointegral

Definition 16.28 (Period datum; moves)**.** Let a family of tables with positive rational weights have rational exponents and masses and lie on the balance wall. Its *period datum* is the positive orthant of $\mathbb{R}^{\mathcal{E}}$, read projectively, with the form $W \, d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} d\alpha$, algebraic over $\mathbb{Q}$ and homogeneous of degree zero, the hypersurface $\{\prod_j K_j \Psi_j = 0\}$ and the coordinate hyperplanes; the face $\{\alpha_S = 0\}$ carries the wall κ_S (Theorem 8.5). A *move* between period data is a change of variables by a birational map defined over $\mathbb{Q}$ carrying the open domain bijectively onto the open domain, faces to faces and form to form; a product of data; a linear combination; or the boundary term of an exact form.

Definition 16.29 (The pseudointegral over a region)**.** Let F be a family with real data, $s_j, t_j \geq 0$ not all zero, masses $\mathbf{m} > 0$ and weights factoring across its molecules, with form $\omega = W \, d\alpha_{\mathbf{m}}$.

(i) A *region* of ω is a domain $\mathcal{R}$ in coordinates x with a change of variables mapping it bijectively onto a section $\{h = 1\}$ of the orthant, $h > 0$ homogeneous of degree one: a chart $\alpha_g = 1$, the simplex, a bunch section $|\alpha_B| = 1$. The *density* of ω on $\mathcal{R}$ is the function f for which $f(x)\,dx$ is the form that ω induces on the section (Lemma 16.2). A region of the damped form $e^{-|\alpha|}\omega$ is the same with the orthant in place of the section. For a family of tables with positive rational weights and rational data on the balance wall the regions reached from these by moves (Definition 16.28) are domains cut out by polynomial inequalities over $\mathbb{Q}$.

(ii) The *pseudointegral* of ω over a region is

$$\int\!\!\cdots\!\!\int_{\mathcal{R}} f(x)\,dx := \int W \, d\alpha_{\mathbf{m}}$$

(Definition 16.10(iii)), and over a region of the damped form, away from the poles of J_F, it is the continued interest, $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int W \, e^{-|\alpha|} d\alpha_{\mathbf{m}}$ (Definition 8.8), written $\int \alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} \prod_j K_j^{-s_j} \Psi_j^{-t_j} d\alpha$ when the integrand is displayed. The value belongs to ω: the native completion and the ray of Definition 16.10(iii) are taken in the table's coordinates α, and a presentation (Theorem 16.13) declares them anew and is another form; the region and its coordinates say how the integration is presented, and a density determines ω only together with the family, a chart $\alpha_g = 1$ not seeing the mass m_g and the simplex not seeing a power of $|\alpha|$, the letter of the list $\binom{\mathcal{E}}{1}$ (Proposition 16.3). Divided by the period of the centre, P_{c} on a molecule, it is the pseudoexpectation $\mathsf{E}_{U \sim \mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$.

(iii) The pseudointegral of ω is read in the Foreman's order (Definition 16.14). The tree is drawn from the tables at the centre of the table's gauge; on each molecule the measure $d\alpha_{\mathbf{m}}$ is set to $d\alpha_{\boldsymbol{\tau}}$; the bunches are integrated in the Foreman's order, each at once in scale and shape; where the target lies outside a wall the value is the continuation in the masses, and on a

pole hyperplane its constant term along the ray of the masses. The damped form takes no completion, and the Foreman computes it on the ray beyond the target (Theorem 16.15).

Theorem 16.30 (The pseudointegral and the native completion)**.** *Let $\omega = W\,d\alpha_{\mathbf{m}}$ have real data on a connected family, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, with target $\boldsymbol{\tau}$.*

(i) With a native gauge $\alpha^\star$ the pseudointegral is read at the realization $\Upsilon = ((s_j,t_j)_j; \alpha^\star)$, whose masses are $\boldsymbol{\tau}$ and whose proper walls are positive. The form $W\,d\alpha_{\boldsymbol{\tau}}$ is positive with convergent period, and every iterated integral of it over every section, in any coordinates, in the Foreman's order and in every other, converges absolutely to

$$\Big\backslash W\,d\alpha_{\mathbf{m}} = \int W\,d\alpha_{\boldsymbol{\tau}} = P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t}) :$$

no finite part is taken, and the pseudoexpectation is the expectation of a positive function, $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)] = \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$.

(ii) If $\boldsymbol{\tau}$ lies on no pole hyperplane, every regular schedule of the native completion returns the pseudointegral,

$$\mathrm{Reg}^{\mathfrak{s}}\int W\,d\alpha_{\boldsymbol{\tau}} = \Big\backslash W\,d\alpha_{\mathbf{m}} = P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t}),$$

each iterated integral a finite part (Definition 1.5), and the value does not depend on the order; its quasi-expectation is the pseudoexpectation, $\mathfrak{E}^{\mathfrak{s}}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)] = \mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$.

(iii) On a pole hyperplane $\Big\backslash W\,d\alpha_{\mathbf{m}} = \mathrm{CT}_{\eta=0}\,P_F(\boldsymbol{\tau}+\eta\mathbf{m};\mathbf{s},\mathbf{t})$ exists and depends on $\mathbf{m}$ only through its ray. It takes no chart and no order: its regulator is η, along the ray of the masses, and its section is the simplex that the damping singles out, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = J_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma(\kappa_{\mathcal{E}})$ (Theorem 8.7).

Proposition 16.31 (A pole target in other coordinates)**.** *In Theorem 16.30(iii) let $x = \lambda^{-1}\alpha$ be the coordinates of a presentation (Theorem 16.13), $W^\lambda(x) = W(\lambda x)$, and*

$$G_k(\eta) = \int_{\Delta_{\mathcal{E}}} \big(\log|\lambda u|\big)^k\, u^{\boldsymbol{\tau}+\eta\mathbf{m}-\mathbf{1}} W^\lambda(u)\,d\omega_{\mathcal{E}}(u) \qquad (k\in\mathbb{N}),$$

continued in η, so that $G_0(\eta)$ is the integral over the simplex of the presented word W^λ at the masses $\boldsymbol{\tau}+\eta\mathbf{m}$. Each G_k has at $\eta = 0$ a pole of order at most $n-1$, and the presentation has pseudointegral

$$(\lambda^{\mathbf{m}})^{\kappa_{\mathcal{E}}/m_{\mathcal{E}}}\Big(\Big\backslash W\,d\alpha_{\mathbf{m}} - \lambda^{\boldsymbol{\tau}}\sum_{1\leq i+k\leq n-1}\frac{\langle\mathbf{m},\log\lambda\rangle^i}{i!}\,\frac{(-m_{\mathcal{E}})^k}{k!}\,[\eta^{-i-k}]\,G_k\Big),$$

$[\eta^{-i-k}]$ the Laurent coefficient at $\eta = 0$. Off the pole hyperplanes every G_k is holomorphic at 0 and this is Theorem 16.13; on them the finite part is taken in the coordinates of the table, as every finite part is taken in its own coordinate (Lemma 1.9).

Proposition 16.32 (Sections off the balance wall)**.** *Let $\omega = W\,d\alpha_{\mathbf{m}}$ have real data, $s_j, t_j \geq 0$ not all zero, $\mathbf{m} > 0$ and imbalance $\kappa_{\mathcal{E}} \neq 0$.*

(i) The integral of ω over a section $\{h = 1\}$ is the integral over every section of the projective form $h^{-\kappa_{\mathcal{E}}}\omega$, wherever it converges. Over a chart it is the period at the chart completion (Proposition 16.5(i)), over a bunch section $|\alpha_B| = 1$ the period of the family with the list $\binom{B}{1}$, the marks outside B as loops, appended at the exponents $(\kappa_{\mathcal{E}}, 0)$, a multinomial sum of

completions when $-\kappa_{\mathcal{E}} \in \mathbb{N}$ (Proposition 16.5(iii)), and over the simplex, where the proper walls at $\mathbf{m}$ are positive, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}[W(U)]$ (Proposition 8.12(i)). Different sections integrate different projective forms; the pseudointegral integrates the native completion over all of them.

(ii) The walls of the chart completion at g and of the native completion are

$$\kappa_S(\mathbf{m}^{(g)}) = \kappa_S(\mathbf{m}) - \kappa_{\mathcal{E}}\,\mathbf{1}_S(g), \qquad \kappa_S(\boldsymbol{\tau}) = \kappa_S(\mathbf{m}) - \frac{\kappa_{\mathcal{E}}}{m_{\mathcal{E}}}\, m_S :$$

the snip moves the walls through g by the whole imbalance and leaves the others, and the native completion moves each wall by its share. For $\kappa_{\mathcal{E}} > 0$ a proper wall through g is lower at the chart completion than at the target by $\kappa_{\mathcal{E}}(1 - m_S/m_{\mathcal{E}})$, and one avoiding g is higher by $\kappa_{\mathcal{E}} m_S/m_{\mathcal{E}}$. So a snip can put the walls through g on pole hyperplanes, or past them, while the target is inside every wall.

Theorem 16.33 (Integrals over sections)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, and for $h > 0$ homogeneous of degree one let $\int_{h=1}\omega$ denote the integral over the section $\{h = 1\}$ with the measure of Lemma 16.2.*

(i) By Proposition 16.32(i) the integral of ω over a section $\{h = 1\}$ is the integral over the rays of $h^{-\kappa_{\mathcal{E}}}\omega$, and for $\kappa_{\mathcal{E}} \neq 0$, $h \mapsto h^{-\kappa_{\mathcal{E}}}$ carries the sections bijectively onto the positive multipliers of degree $-\kappa_{\mathcal{E}}$. The monomial sections $\{\alpha^w = 1\}$, $w_{\mathcal{E}} = 1$, are the completions, $(\alpha^w)^{-\kappa_{\mathcal{E}}}\omega = W\, d\alpha_{\mathbf{m}-\kappa_{\mathcal{E}} w}$: they act on the measure and keep the word.

(ii) The native completion is the section $\{\alpha^{\mathbf{m}} = 1\}$, the hyperplane $\langle \mathbf{m}, y\rangle = 0$ in $y = \log\alpha$, where the density $e^{\langle \mathbf{m}, y\rangle}$ of $d\alpha_{\mathbf{m}}$ against dy is 1; it is the one monomial section with this property. With a native gauge on a connected family of at least two marks, for every mark g,

$$\Big\backslash W\, d\alpha_{\mathbf{m}} = \frac{m_{\mathcal{E}}}{m_g} \int_{\mathbb{R}^{\mathcal{E}-g}} W(e^y)\, dy_{\mathcal{E}-g}, \qquad y_g = -\frac{1}{m_g}\sum_{e\neq g} m_e y_e,$$

and $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$ is this integral over the integral of ω_{c} over the same section, which is P_{c}.

(iii) The simplex $\{|\alpha| = 1\}$ is not a monomial section for $n \geq 2$: its multiplier is $|\alpha|^{-\kappa_{\mathcal{E}}}$, which keeps the measure and multiplies the word by a power of the letter $K_0 = |\alpha|$ of the list $\binom{\mathcal{E}}{1}$. The integral over it is the integral of the projective form $K_0^{-\kappa_{\mathcal{E}}} W\, d\alpha_{\mathbf{m}}$, the family with that table added at the exponents $(\kappa_{\mathcal{E}}, 0)$ (Proposition 16.3, Proposition 16.5(iii) at $B = \mathcal{E}$), and wherever every proper wall at $\mathbf{m}$ is positive it is the period $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ of Proposition 8.12(i), the law $\mathrm{Dir}_{\mathbf{m}}$ there being the law of $\binom{\mathcal{E}}{1}$ at a full basis lean.

(iv) On the balance wall every section reads $\int W\, d\alpha_{\mathbf{m}}$ (Theorem 8.7).

Proposition 16.34 (The centre sets the law)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data on a connected family of $n \geq 2$ marks, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, with target $\boldsymbol{\tau}$, centre ω_{c} and law of the centre $\mathbb{P}_{\mathrm{c}}$ (Definition 16.10). For laws $\mathbb{P}, \mathbb{P}'$ with $\mathbb{P}$ absolutely continuous with respect to $\mathbb{P}'$ write $\mathrm{KL}(\mathbb{P}\,\|\,\mathbb{P}') = \mathbb{E}_{U\sim\mathbb{P}}\big[\log \frac{d\mathbb{P}}{d\mathbb{P}'}(U)\big]$, the Kullback–Leibler divergence.*

(i) The native completion is the centre times the departure, $\omega_{\boldsymbol{\tau}} = \frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}\,\omega_{\mathrm{c}}$, and with a native gauge the Foreman's law, the shape law $\mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ of the native completion, is the law of the centre reweighted by the departure:

$$\mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})(du) = \frac{(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(u)}{\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]}\,\mathbb{P}_{\mathrm{c}}(du).$$

(ii) With a native gauge,

$$\log \mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big] = \mathrm{KL}\big(\mathbb{P}_{\mathrm{c}} \,\big\|\, \mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})\big) \;\geq\; 0,$$

with equality exactly when $\mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t}) = \mathbb{P}_{\mathrm{c}}$. *So every pseudointegral with a native gauge is at least the period of its centre,* $\int W\, d\alpha_{\mathbf{m}} \geq P_{\mathrm{c}}$, *and among the forms of given strengths* σ_j *with a native gauge the centre has the least value.*

(iii) The pseudoexpectation is kept by the fold: the two forms of Theorem 16.18, at the masses $\mathbf{m}$ *with the exponents* $(\mathbf{s},\mathbf{t})$ *and at the masses* $\lambda(2\sigma\mathbf{1}-\boldsymbol{\tau})$ *with* $(\mathbf{t},\mathbf{s})$, *have one centre and one pseudoexpectation, and the same holds for continued values off the pole hyperplanes where* $2\sigma\mathbf{1}-\boldsymbol{\tau} > 0$.

(iv) Let ω *be the form of a realization at the table's gauge,* $\alpha^{\star} = \mathbf{1}$ *(Definition 9.1). In signed coordinates on* $\mathbf{1}^{\perp}$ *the departure is*

$$\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(e^{2z}) = \exp\Big(-\sum_j \delta_j \tilde{O}_j(z)\Big), \qquad \tilde{O}_j(z) = O_j(z) - \langle \nabla O_j(0), z\rangle,$$

each $\tilde{O}_j$ *odd with value, gradient and Hessian* 0 *at* 0. *The Foreman's law and the law of the centre have their mode at the table's gauge and the same Hessian of the logarithm of the density there, and the logarithm of their ratio is, up to a constant,* $-\sum_j \delta_j \tilde{O}_j$*: minus the odd part of the centred potential less its linear term, whose Taylor coefficients at the table's gauge are minus the odd rungs of order at least three of the potential, the rungs that vanish at the centre (Proposition 8.21(iii)).*

Proposition 16.35 (The list $\binom{\mathcal{E}}{1}$ as a normaliser)**.** *Let* $\omega = W\, d\alpha_{\mathbf{m}}$ *have real data on a connected family of* $n \geq 2$ *marks,* $s_j, t_j \geq 0$ *not all zero and* $\mathbf{m} > 0$, *so that* W *is homogeneous of degree* $-\tau_{\mathcal{E}}$*; let* $K_0 = |\alpha|$ *be the letter of the list* $\binom{\mathcal{E}}{1}$, *whose bases are the singletons.*

(i) Every completion $W\, d\alpha_{\mathbf{m}'}$, $\mathbf{m}' = \mathbf{m} - \kappa_{\mathcal{E}} w$, *has total mass* $m'_{\mathcal{E}} = \tau_{\mathcal{E}}$. *The native completion is the one whose masses keep the proportions of* $\mathbf{m}$, $\boldsymbol{\tau}/\tau_{\mathcal{E}} = \mathbf{m}/m_{\mathcal{E}}$, *and they are positive, while the chart completion at* g *has the mass* $m_g - \kappa_{\mathcal{E}}$ *at* g.

(ii) For $\mathbf{m}' > 0$ *with* $m'_{\mathcal{E}} = \tau_{\mathcal{E}}$ *the form* $K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}'}$ *is projective, with integral* $\Gamma_1^n \begin{bmatrix} \mathbf{m}' \\ \tau_{\mathcal{E}} \end{bmatrix}$ *over every section: the period of* $\binom{\mathcal{E}}{1}$ *at the full basis lean* $(\tau_{\mathcal{E}}, 0)$, *whose law is* $\mathrm{Dir}_{\mathbf{m}'}$ *(Proposition 16.3). So* $\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix} = \int K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}}$*; for a schedule with* $\mathbf{m}^{(g)} > 0$, $\Gamma_1^n \begin{bmatrix} \mathbf{m}^{(g)} \\ \tau_{\mathcal{E}} \end{bmatrix} = \mathrm{Reg}^{\mathfrak{s}} \int K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}}$, *a convergent integral; and with a native gauge*

$$\int W\, d\alpha_{\mathbf{m}} \Big/ \int K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}} = \mathbb{E}_{U\sim\mathrm{Dir}_{\boldsymbol{\tau}}}[W(U)].$$

(iii) With a native gauge, and for continued values off the pole hyperplanes where $2\sigma\mathbf{1}-\boldsymbol{\tau} > 0$, *the fold keeps the pseudointegral and carries the target* $\boldsymbol{\tau}$ *to* $2\sigma\mathbf{1}-\boldsymbol{\tau}$ *(Theorem 16.18), so the quotients of (ii) for* ω *and for its image are* $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ *divided by* $\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix}$ *and by* $\Gamma_1^n \begin{bmatrix} 2\sigma\mathbf{1}-\boldsymbol{\tau} \\ \tau^*_{\mathcal{E}} \end{bmatrix}$, $\tau^*_{\mathcal{E}} = 2n\sigma - \tau_{\mathcal{E}}$. *The second quotient over the first is* $\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix} / \Gamma_1^n \begin{bmatrix} 2\sigma\mathbf{1}-\boldsymbol{\tau} \\ \tau^*_{\mathcal{E}} \end{bmatrix}$, *which is not* 1 *in general. The period of* $\binom{\mathcal{E}}{1}$ *is that of a full basis lean of a table outside the family, and it prices a lean and its mirror differently; the centre, which the fold fixes, prices them alike (Proposition 16.34(iii)).*

(iv) Along the ray of the masses, wherever the proper walls at $\lambda\mathbf{m}$ are positive, the integral of the projective form $K_0^{-(\lambda m_{\mathcal{E}}-\tau_{\mathcal{E}})}W\,d\alpha_{\lambda\mathbf{m}}$ is $\Gamma_1^n\begin{bmatrix}\lambda\mathbf{m}\\ \lambda m_{\mathcal{E}}\end{bmatrix}\,\mathbb{E}_{U\sim\mathrm{Dir}_{\lambda\mathbf{m}}}[W(U)]$ (Theorem 16.33(iii), Proposition 8.12(i)). The power of K_0 vanishes exactly at $\lambda=\tau_{\mathcal{E}}/m_{\mathcal{E}}$, where the form is the native completion $W\,d\alpha_{\boldsymbol{\tau}}$.

Corollary 16.36 (The pseudointegral is the quasi-integral that always exists)**.** *For real data with $s_j,t_j\geq 0$ not all zero, $\mathbf{m}>0$ and weights factoring across the molecules, the pseudointegral $\int W\,d\alpha_{\mathbf{m}}$ and the pseudoexpectation $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$ are defined. On a molecule whose target lies on no pole hyperplane the pseudointegral is the quasi-integral of the native completion and the pseudoexpectation its quasi-expectation, the same along every regular schedule; on a pole hyperplane, where the regular schedules of the native completion can depend on the order (Theorem 16.23), it is the one constant term along the ray of the masses. On each molecule every generalized expectation divides by the one period of its centre, a positive convergent integral (Proposition 8.21); when the molecule has a native gauge, the native completion is the one completion that keeps it (Theorem 16.8(i)). The quasi-integrals of ω itself are Laurent data at its chart completions (Theorem 16.22), which off the balance wall can depend on the chart and on the order.*

The pseudointegral names a procedure as well as a value. Written for a family whose realization is specified later, $\int\cdots\int_{\mathcal{R}}$ commits to the Foreman's order at every realization: the tree is drawn once from the tables, and the realization $((s_j,t_j)_j;\alpha^\star)$ of the point fixes the measure $d\alpha_{\boldsymbol{\tau}}$ and where the contours of the ports run. An iterated integral in another order, or after a snip, computes a quasi-integral and not the pseudointegral, even where the numbers agree; when they agree, that is Theorem 16.30(i)–(ii), not the order that was followed.

On a pole hyperplane the regulator η of the pseudointegral enters as $u^{\eta\mathbf{m}}$, as in Theorem 1.7. Past a wall the pseudointegral normalizes the distribution of Theorem 16.17(i).

Chapter 17

The family as one Laplace transform

On its balance wall a family is one Laplace transform. In the pair coordinates of its data and the signed coordinates of the shape, its period is the Laplace transform, over the tilts, of the even and odd statistics against the offsets (Definition 17.1; for the interest, Theorem 8.2); the Breta shape laws at the points of the balance wall form an exponential family in these coordinates, and every ratio of periods is a moment generating function of the statistics. On the sector of every order the exponent of a question is minus twice its wall at the tilt, the walls of the tails weighted by the differences of the tilt along the order, less the logarithms of the sector letters, which are bounded. The Laplace transform is read through that one reading of the walls: where the exponent is largest and how it falls, how the period behaves at low and at high strength, how far a question can lean, and how fast the tails of the statistics fall.

17.1 Questions and the law of the tilt

Definition 17.1 (Tilts; questions)**.** The space of *tilts* is $\mathcal{T} = \mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$, with the Lebesgue measure dz of a chart $z_g = 0$, which does not depend on g. The *tilt* of a shape $u \in \Delta_{\mathcal{E}}$ is the image in $\mathcal{T}$ of its signed coordinates $\frac{1}{2}\log u$, and a tilt is read at its representative in $\mathbf{1}^{\perp}$ when an odd statistic is applied to it, since $O_j(z + c\mathbf{1}) = O_j(z) + 2c(2r_j - n)$ (Theorem 7.15(iii)). A *question* to a family F is a point of its balance wall with real data, written in its pair coordinates $(\boldsymbol{\mu}_{\perp}; \boldsymbol{\sigma}, \boldsymbol{\delta})$: the data $(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \big(\mathbf{m}_F(\boldsymbol{\mu}_{\perp}; \boldsymbol{\sigma}, \boldsymbol{\delta});\ \boldsymbol{\sigma} + \boldsymbol{\delta},\ \boldsymbol{\sigma} - \boldsymbol{\delta}\big)$ (Definition 8.1). In the coordinates $z = \frac{1}{2}\log u$ of Theorem 8.7,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{n-1} \int_{\mathcal{T}} \exp\Big(2\langle \boldsymbol{\mu}_{\perp}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)\Big)\, dz.$$

When every proper wall is positive, the tilt Z of $U \sim \mathrm{Breta}^{\circ}(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ has the density

$$2^{n-1} \exp\Big(2\langle \boldsymbol{\mu}_{\perp}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)\Big) \Big/ P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$$

with respect to dz: this is the *law of the question*. A direction $v = (\dot{\mathbf{m}}; \dot{\mathbf{s}}, \dot{\mathbf{t}})$ of the data is *balanced* when $\dot{m}_{\mathcal{E}} = \sum_j (\dot{s}_j r_j + \dot{t}_j r_j^*)$; its pair coordinates $(c; \mathbf{a}, \mathbf{b})$ have $\dot{\mathbf{m}} = \mathbf{m}_F(c; \mathbf{a}, \mathbf{b})$, $\dot{\mathbf{s}} = \mathbf{a} + \mathbf{b}$ and $\dot{\mathbf{t}} = \mathbf{a} - \mathbf{b}$, and D_v is the derivative along v. The derivatives ∂_{σ_j}, ∂_{δ_j} and $\nabla_{\boldsymbol{\mu}_{\perp}}$ of a function of the question are those along the balanced directions with the pair coordinates $(0; \mathbf{e}_j, \mathbf{0})$, $(0; \mathbf{0}, \mathbf{e}_j)$ and $(\mathbf{e}_e - \frac{1}{n}\mathbf{1}; \mathbf{0}, \mathbf{0})$, that is $(\mathbf{1}; \mathbf{e}_j, \mathbf{e}_j)$, $\big(\frac{2r_j - n}{n}\mathbf{1}; \mathbf{e}_j, -\mathbf{e}_j\big)$ and $\big(\mathbf{e}_e - \frac{1}{n}\mathbf{1}; \mathbf{0}, \mathbf{0}\big)$. The centre of a question is the question $(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$ with the same strengths, the data of the family at the centre

(Definition 9.1), of pair coordinates $(0;\boldsymbol{\sigma},\mathbf{0})$; on a connected family its law is the law of the centre $\mathbb{P}_{\mathrm{c}}$ of Definition 16.10 read on the tilts.

Corollary 17.2 (The Laplace transform of the tilt and the statistics)**.** *Let a question* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *have every proper wall positive. In this statement* $\mathbb{E}$ *is the expectation of* $U\sim\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ *and* Z *is the tilt of* U*. For* $c\in\mathbf{1}^{\perp}\otimes\mathbb{C}$ *and* $\mathbf{a},\mathbf{b}\in\mathbb{C}^N$ *such that the real parts of the shifted data* $\big(\mathbf{m}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)$ *have every proper wall positive,*

$$\mathbb{E}\Big[e^{2\langle c,Z\rangle-\sum_j(a_jE_j(Z)+b_jO_j(Z))}\Big]=\frac{P_F\big(\mathbf{m}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$

analytic in $(c,\mathbf{a},\mathbf{b})$ *there: the shift moves the pair coordinates of the question by* $(c;\mathbf{a},\mathbf{b})$*. So the joint cumulants of order* k *of* $(E_j(Z),O_j(Z))_j$ *are* $(-1)^k$ *times the derivatives of order* k *of* $\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *in* $(\sigma_j,\delta_j)_j$*, and their characteristic function is the same ratio at* $c=0$ *and purely imaginary* $(\mathbf{a},\mathbf{b})$*: the family on the fibre of imaginary exponents over the question.*

Definition 17.3 (The wall at a tilt)**.** For an order π of $\mathcal{E}$ with tails $U_k=\{\pi(k),\dots,\pi(n)\}$, the sector H_π of Lemma 8.3 is read in the tilts as $\{z\in\mathcal{T}:\ z_{\pi(1)}\geq\cdots\geq z_{\pi(n)}\}$. On it the measure dz of Definition 17.1 is the Lebesgue measure in the differences $z_{\pi(k-1)}-z_{\pi(k)}\geq 0$, $2\leq k\leq n$, which at the gauge $\alpha=e^{2z}$ are half the gaps $\log(\alpha_{\pi(k-1)}/\alpha_{\pi(k)})$ of Corollary 8.6. For data on the balance wall the *wall at a tilt* $z\in H_\pi$ is

$$\kappa(z)=\sum_{k=2}^{n}\kappa_{U_k}\big(z_{\pi(k-1)}-z_{\pi(k)}\big),$$

the walls of the tails of π weighted by the differences of z along π, half the walls weighted by the gaps. It does not depend on the order whose sector holds z, the tails with a positive difference being the sets $\{e:\ z_e<z_f\}$ for the $f\in\mathcal{E}$ with $z_f>\min_e z_e$; it is positively homogeneous of degree one and linear on every sector, and $\kappa(-\mathbf{1}_S)=\kappa_S$ for every nonempty proper S, with $-\mathbf{1}_S$ read in $\mathcal{T}$, the direction along which S collapses (Proposition 16.26(i)). In pair coordinates $\kappa(z)$ is linear in the question.

Proposition 17.4 (The exponent of a question reads its walls; the even statistic is least at the table's gauge)**.** *Let a family carry a question* $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$*, its exponents of any signs, and in this statement write* $X(z)=2\langle\boldsymbol{\mu}_\perp,z\rangle-\sum_j\big(\sigma_jE_j(z)+\delta_jO_j(z)\big)$ *for the exponent of the law of the question (Definition 17.1) and* $\kappa(z)$ *for its wall at the tilt* z *(Definition 17.3).*

(i) On the sector H_π *of every order* π*, with the sector letters* $k_{j,\pi},\psi_{j,\pi}$ *of Theorem 8.5(ii) read at the sector coordinates* $q_k=\alpha_{\pi(k)}/\alpha_{\pi(k-1)}=e^{-2(z_{\pi(k-1)}-z_{\pi(k)})}$ *of* $\alpha=e^{2z}$*,*

$$X(z)=-2\kappa(z)-\sum_j\big(s_j\log k_{j,\pi}(q)+t_j\log\psi_{j,\pi}(q)\big).$$

With $h_j(z)=\max_{T\in\mathcal{B}_j}\langle\chi_T,z\rangle$*, the support function of the Newton polytope* $2P_j-\mathbf{1}$ *of* Λ_j *(Theorem 7.18),* $2\kappa(z)=\sum_j\big(s_jh_j(z)+t_jh_j(-z)\big)-2\langle\boldsymbol{\mu}_\perp,z\rangle$ *on* $\mathbf{1}^{\perp}$*, and for* $s_j,t_j\geq 0$ *the wall at a tilt is the support function of* $\mathcal{N}_F-\mathbf{m}$*,* $\mathcal{N}_F$ *the Newton sum of Theorem 8.19(ii). The sector letters are polynomials with nonnegative coefficients, with the constant terms* $\theta_{j,T^+_{j,\pi}}$ *and* $\theta_{j,T^-_{j,\pi}}$ *and the value* $K_j(\mathbf{1})$ *at* $q=\mathbf{1}$ *(Theorem 8.5(ii)), so on the cube they lie in* $[\theta_{j,T^+_{j,\pi}},K_j(\mathbf{1})]$ *and* $[\theta_{j,T^-_{j,\pi}},K_j(\mathbf{1})]$*; with* $\theta_{j,\min}=\min_T\theta_{j,T}$*,* $|X+2\kappa|\leq\sum_j(|s_j|+|t_j|)\max\big(|\log\theta_{j,\min}|,|\log K_j(\mathbf{1})|\big)$*, and for* $s_j,t_j\geq 0$

$$2\kappa(z)+\sum_j(s_j+t_j)\log\theta_{j,\min}\leq-X(z)\leq 2\kappa(z)+\sum_j(s_j+t_j)\log K_j(\mathbf{1}).$$

(ii) $\kappa > 0$ *on* $\mathcal{T} \setminus 0$ *exactly when every proper wall is positive, and then* $\kappa(z) \geq \kappa_{\min}\big(\max_e z_e - \min_e z_e\big)$, $\kappa_{\min}$ *the least proper wall.*

(iii) At the centre of strengths $\boldsymbol{\sigma}$, $\sigma_j \geq 0$, *where* $\boldsymbol{\mu}_\perp = 0$ *and* $\boldsymbol{\delta} = 0$, *the exponent is minus the centred potential* $\sum_j \sigma_j E_j$ *(Proposition 8.21) and* $2\kappa(z) = \sum_j \sigma_j \operatorname{wd}_j(z)$, *with* $\operatorname{wd}_j(z) = h_j(z) + h_j(-z)$ *the width of* $2P_j - \mathbf{1}$ *in the direction* z, *twice the width of* P_j, *so that* $\operatorname{wd}_j(\mathbf{1}_S) = 2c_j(S)$. *Every* E_j *is convex and even,* $E_j(z) \geq E_j(0) = 2\log K_j(\mathbf{1})$, *and*

$$\sum_j \sigma_j \operatorname{wd}_j(z) + 2\sum_j \sigma_j \log \theta_{j,\min} \leq \sum_j \sigma_j E_j(z) \leq \sum_j \sigma_j \operatorname{wd}_j(z) + 2\sum_j \sigma_j \log K_j(\mathbf{1}).$$

If the family is connected, $\sum_j \sigma_j E_j$ *is strictly convex on* $\mathcal{T}$ *and* $\sum_j \sigma_j \operatorname{wd}_j > 0$ *on* $\mathcal{T} \setminus 0$.

Theorem 17.5 (The two ends of strength)**.** *Let a question* $q = (\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ *of a family have every proper wall positive, and for* $\lambda \in \mathbb{C}$ *with* $\operatorname{Re}\lambda > 0$ *read the data* λq *of pair coordinates* $(\lambda\boldsymbol{\mu}_\perp; \lambda\boldsymbol{\sigma}, \lambda\boldsymbol{\delta})$, *a question when* λ *is real, whose walls* $\lambda\kappa_S$ *have positive real parts; in this statement* $P_F(\lambda q)$ *is its period,* $2^{n-1}\int_{\mathcal{T}} e^{\lambda X(z)} dz$ *by the change of variables of Definition 17.1 with the powers of the positive letters principal (Theorem 8.7), and* X *and* κ *are those of* q *(Proposition 17.4).*

(i) Low strength. *As* $\lambda \downarrow 0$,

$$\lambda^{n-1} P_F(\lambda q) \to \sum_\pi \prod_{k=2}^{n} \kappa_{U_k}^{-1} = (n-1)!\ \operatorname{vol}\{z \in \mathcal{T} :\ \kappa(z) \leq 1\},$$

and

$$\lambda^{n-1} P_F(\lambda q) = \sum_\pi \prod_{k=2}^{n} \kappa_{U_k}^{-1}\Big(1 - \lambda \sum_j \big(s_j \log \theta_{j,T^+_{j,\pi}} + t_j \log \theta_{j,T^-_{j,\pi}}\big)\Big) + o(\lambda),$$

with the greedy bases $T^\pm_{j,\pi}$ *of Lemma 8.3: the walls set the period at low strength, and the weights enter at the next order, through the greedy bases.*

(ii) High strength, in the half-plane. *Let* $s_j, t_j \geq 0$. *Then* $-X$ *is strictly convex on* $\mathcal{T}$, *its unique critical point* $z^\star$ *is the native gauge of every question* λq, $\lambda > 0$, *and as* $|\lambda| \to \infty$ *with* $\operatorname{Re}\lambda \geq \lambda_0$ *for a fixed* $\lambda_0 > 0$, *in particular as* $\lambda \to +\infty$ *and along* $\lambda = \lambda_0 + i\eta$, $|\eta| \to \infty$,

$$P_F(\lambda q) = 2^{n-1} e^{\lambda X(z^\star)} \Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}} \det\big(-\nabla^2 X(z^\star)\big)^{-\frac{1}{2}} \big(1 + O(|\lambda|^{-1})\big),$$

the power principal and the Hessian taken on a chart $z_g = 0$, *where it is rung two of the still of the realization* $\big((s_j, t_j)_j; e^{2z^\star}\big)$ *of the active tables (Definition 9.4) with the row and the column of* g *deleted. At the centre of strengths* $\boldsymbol{\sigma}$, $z^\star = 0$, $X(0) = -2\sum_j \sigma_j \log K_j(\mathbf{1})$, *and on the chart* $z_g = 0$, $-\nabla^2 X(0) = 8\sum_j \sigma_j \big(\kappa_{j,\mathbf{1}}(X_e, X_f)\big)_{e,f\neq g}$.

Remark 17.6 (The volume of the walls is the volume of a polar body). At the centre of strengths $\boldsymbol{\sigma}$, $\sigma_j \geq 0$, Proposition 17.4(iii) gives $2\kappa(z) = \sum_j \sigma_j \operatorname{wd}_j(z)$, and $\operatorname{wd}_j(z) = h_j(z) + h_j(-z)$, with h_j the support function of $2P_j - \mathbf{1}$ and P_j the base polytope of table j (Theorem 7.18), is twice the support function of $P_j - P_j$ at z. So κ is the support function of the centrally symmetric polytope $\mathcal{Q} = \sum_j \sigma_j (P_j - P_j)$, a Minkowski sum in $\mathbf{1}^\perp$, read on $\mathcal{T} = \mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$, and $\{z \in \mathcal{T} :\ \kappa(z) \leq 1\}$ is its polar body $\mathcal{Q}^\circ = \{z :\ \langle x, z\rangle \leq 1 \text{ for every } x \in \mathcal{Q}\}$, bounded when every proper wall is positive (Proposition 17.4(ii)). By Theorem 17.5(i), $\sum_\pi \prod_{k=2}^n \kappa_{U_k}^{-1} = (n-1)!\operatorname{vol}(\mathcal{Q}^\circ)$: the volume V_1 of the direction of the centre ray at the strengths $\boldsymbol{\sigma}$ (Proposition 39.30), whose walls are those of this question, is $(n-1)!$ times the volume of the polar body of $\sum_j \sigma_j (P_j - P_j)$.

At low strength the walls alone set the period, through the volume of the region where the wall at a tilt is at most one, and the weights enter at the next order through the greedy bases. At high strength the period concentrates at the native gauge, where it is a Gaussian integral whose Hessian is rung two of the still.

Proposition 17.7 (How far a question can lean; the values of the statistics)**.** *For every family and every nonempty proper S put $\varepsilon_j^{\perp}(S) = \varepsilon_j(S) - |S|\frac{2r_j - n}{n}$, so that $\varepsilon_j^{\perp}(\mathcal{E} \setminus S) = -\varepsilon_j^{\perp}(S)$, and for a question $(\boldsymbol{\mu}_{\perp}; \boldsymbol{\sigma}, \boldsymbol{\delta})$ let*

$$\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}) = \sum_j \sigma_j c_j(S), \qquad \kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_{\perp}, \boldsymbol{\delta}) = \mu_{\perp,S} - \sum_j \delta_j \varepsilon_j^{\perp}(S),$$

the even and the odd part of the wall at S: on the balance wall $\kappa_S = \kappa_S^{\mathbf{E}} + \kappa_S^{\mathbf{O}}$ and $\kappa_{\mathcal{E}\setminus S} = \kappa_S^{\mathbf{E}} - \kappa_S^{\mathbf{O}}$, and the fold, which negates the odd part (Theorem 16.18), swaps the two. So a question is inside its walls exactly when $|\kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_{\perp}, \boldsymbol{\delta})| < \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})$ for every S. When the centre of strengths $\boldsymbol{\sigma}$ is inside its walls, the slope *of an odd direction $(c, \mathbf{b}) \in \mathbf{1}^{\perp} \times \mathbb{R}^N$ at $\boldsymbol{\sigma}$ is*

$$\ell_{\boldsymbol{\sigma}}^*(c, \mathbf{b}) = \min\Big\{ \frac{\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})}{|\kappa_S^{\mathbf{O}}(c, \mathbf{b})|} : \ \kappa_S^{\mathbf{O}}(c, \mathbf{b}) \neq 0 \Big\},$$

$\ell_{\boldsymbol{\sigma}}^* = \infty$ if every $\kappa_S^{\mathbf{O}}(c, \mathbf{b})$ vanishes; it is homogeneous of degree one in $\boldsymbol{\sigma}$, and even and positively homogeneous of degree -1 in $(c, \mathbf{b})$. The questions of strengths $\boldsymbol{\sigma}$ inside their walls are those whose odd part has $\ell_{\boldsymbol{\sigma}}^*(\boldsymbol{\mu}_{\perp}, \boldsymbol{\delta}) > 1$, an open convex polyhedron symmetric under the fold, bounded exactly when no nonzero odd direction has $\kappa_S^{\mathbf{O}}(c, \mathbf{b}) = 0$ for every S. For one connected table at offsets zero they form the wedge $\sigma > 0$, $|\delta| < \ell_1^*(0,1)\,\sigma$, with $\ell_1^*(0,1) = \min\{c(S)/|\varepsilon^{\perp}(S)| : \ \varepsilon^{\perp}(S) \neq 0\}$.

(i) For the tilt Z of $U \sim \mathrm{Breta}^{\circ}(F; \sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$, the centre of strengths $\boldsymbol{\sigma}$, and real $c \in \mathbf{1}^{\perp}$, $\mathbf{a}, \mathbf{b} \in \mathbb{R}^N$, the expectation of $e^{2\langle c, Z\rangle - \sum_j (a_j E_j(Z) + b_j O_j(Z))}$ is finite exactly when every proper wall of the question $(c; \boldsymbol{\sigma} + \mathbf{a}, \mathbf{b})$ is positive, $|\kappa_S^{\mathbf{O}}(c, \mathbf{b})| < \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma} + \mathbf{a})$ for every S, and it is then the ratio of the periods of that question and of the centre (Corollary 17.2); in particular $\mathbb{E}_{U \sim \mathrm{Breta}^{\circ}(F; \sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})}\big[e^{2\langle c, Z\rangle - \sum_j b_j O_j(Z)}\big]$ is finite exactly when $\ell_{\boldsymbol{\sigma}}^*(c, \mathbf{b}) > 1$.

(ii) For a question inside its walls with $|\delta_j| \leq \sigma_j$ the native gauge $z^{\star}$ minimizes $-2\langle \boldsymbol{\mu}_{\perp}, z\rangle + \sum_j (\sigma_j E_j + \delta_j O_j)$ on $\mathbf{1}^{\perp}$, so $\big(z^{\star}; E_j(z^{\star}), O_j(z^{\star})\big)_j$ is the point of the set of values $\{(z; E_j(z), O_j(z))_j : \ z \in \mathbf{1}^{\perp}\}$ of the statistics at which $-2\langle \boldsymbol{\mu}_{\perp}, x\rangle + \sum_j (\sigma_j y_j + \delta_j w_j)$ is least; at the centre it is $\big(0; 2\log K_j(\mathbf{1}), 0\big)_j$.

Proposition 17.8 (The tails of the statistics are read at the walls)**.** *Let a question q of a family have every proper wall positive, let Z be the tilt of $U \sim \mathrm{Breta}^{\circ}(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ at its data, and for a real balanced direction v of pair coordinates $(c; \mathbf{a}, \mathbf{b})$ (Definition 17.1) let $Y = 2\langle c, Z\rangle - \sum_j \big(a_j E_j(Z) + b_j O_j(Z)\big)$. On the balance wall the walls are linear in the pair coordinates, $\kappa_S(q + \lambda v) = \kappa_S(q) + \lambda\kappa_S(v)$ with $\kappa_S(v) = \kappa_S^{\mathbf{E}}(\mathbf{a}) + \kappa_S^{\mathbf{O}}(c, \mathbf{b})$ (Proposition 17.7). Let*

$$\lambda^* = \min\Big\{ \frac{\kappa_S(q)}{-\kappa_S(v)} : \ \kappa_S(v) < 0 \Big\},$$

the value of λ at which the ray $q + \lambda v$, $\lambda > 0$, first meets a wall, with $\lambda^* = \infty$ if no $\kappa_S(v)$ is negative. For real $\lambda \geq 0$, $\mathbb{E}[e^{\lambda Y}]$ is finite exactly for $\lambda < \lambda^*$, where it is the ratio of the periods of the questions $q + \lambda v$ and q (Corollary 17.2); if $\lambda^* < \infty$,

$$\mathbb{P}(Y > x) = e^{-\lambda^* x + o(x)} \qquad (x \to \infty),$$

and if $\lambda^* = \infty$, Y is bounded above. Under the law of the centre of strengths $\boldsymbol{\sigma}$ an odd direction, of pair coordinates $(c; \mathbf{0}, \mathbf{b})$, has $\lambda^* = \ell_{\boldsymbol{\sigma}}^*(c, \mathbf{b})$ for Y and for $-Y$ alike.

The period of a question is a Laplace transform in the tilts, so the shape laws form an exponential family whose natural parameters are the offsets and the pair coordinates of the exponents and whose statistics are the tilt and the even and odd statistics E_j, O_j. On the sector of each order the exponent of a question is minus twice its wall at the tilt, the walls of the tails weighted by the differences of the tilt along the order, up to the bounded sector letters. So the walls alone fix how the period blows up as the strength goes to zero, the weights entering only at the next order and only through the greedy bases; at high strength, real or complex with its real part bounded below, the period is read at the native gauge of the ray. At a centre the exponent is minus the centred potential $\sum_j \sigma_j E_j$, which is convex, smallest at the table's gauge and within a constant of the widths of the Newton polytopes. A question can lean only as far as its strengths allow: the odd part of every wall is bounded by its even part, the slope of its odd part exceeds one, and the same slope bounds the exponential moments of the odd statistics at the centre; along every direction the tail of a statistic falls at the rate of the first wall the direction meets. Integrating over the simplex is integrating against the list $\binom{\mathcal{E}}{1}$ at a full basis lean, so every Dirichlet expectation of letters is a ratio of periods (Proposition 16.3).

Chapter 18

The Breta law and the port on the balance wall

18.1 The laws

Theorem 18.1 (The pseudointegral is a normalizer)**.** *Let F be connected on $n \geq 2$ marks with real data, $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, a native gauge $\alpha^\star$, target $\boldsymbol{\tau}$ and form $\omega = W\, d\alpha_{\mathbf{m}}$.*

(i) $\int W\, d\alpha_{\mathbf{m}} = P_F(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ is the normalizer of $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$*, and the pseudoexpectation $\mathsf{E}_{U \sim \mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)] = \int W\, d\alpha_{\mathbf{m}}/P_{\mathrm{c}}$ is the normalizer of its density against the law of the centre (Proposition 16.34(i)). With $\phi_{\mathbf{m}'}$ the joint law of $\big(K_1(U), \dots, K_N(U), \Psi_1(U), \dots, \Psi_N(U)\big)$ for $U \sim \mathrm{Dir}_{\mathbf{m}'}$, both periods are stacked Gammas of masses over their total times Trailmoments (Proposition 8.12(i)):*

$$\mathsf{E}_{U \sim \mathbb{P}_{\mathrm{c}}}\big[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)\big] = \frac{\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix} \big(\mathsf{T}_{(\mathbf{s},\mathbf{t});I_{2N}}[\mathrm{id}]\, \phi_{\boldsymbol{\tau}}\big)\big((0,\infty)^{2N}\big)}{\Gamma_1^n \begin{bmatrix} \sigma\mathbf{1} \\ n\sigma \end{bmatrix} \big(\mathsf{T}_{(\boldsymbol{\sigma},\boldsymbol{\sigma});I_{2N}}[\mathrm{id}]\, \phi_{\sigma\mathbf{1}}\big)\big((0,\infty)^{2N}\big)}.$$

(ii) In the coordinates $y = \log u$ modulo $\mathbb{R}\mathbf{1}$ the law $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ *has density proportional to $e^{\langle \boldsymbol{\tau}, y\rangle - \Phi(y)}$, whose unique mode is $\log \alpha^\star$.*

(iii) (Cremona) If $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ on the balance wall, then $U^{-1}/|U^{-1}| \sim \mathrm{Breta}^\circ(F; 2\sigma\mathbf{1} - \mathbf{m}; \mathbf{t}, \mathbf{s})$.

(iv) (The even–odd pair) On the balance wall the inversion of (iii) carries $\mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ *at the data $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ to the law at $(-\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, -\boldsymbol{\delta})$; at $(0; \boldsymbol{\sigma}, \mathbf{0})$ it fixes the law, and with Z the signed coordinate of U the odd statistics $(O_j(Z))_j$ have a law symmetric under $O \mapsto -O$. The joint Laplace transform of the even and odd statistics is Corollary 17.2.*

Proposition 18.2 (The mark relation of the Breta shape law)**.** *Let F have real data, masses $\mathbf{m}$ on the balance wall and every proper wall positive, and in this statement write $\mathbf{u}(\alpha) = \sum_j \big(s_j R_j(\alpha) + t_j R_j^*(\alpha)\big)$ for the usage at α and $\mathbb{E}$ for the expectation of $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$. For $Y = \log U$ modulo $\mathbb{R}\mathbf{1}$, every mark e and every observable g, a smooth function of degree 0 in α such that g and $\partial_{y_e} g$ grow at most exponentially at a rate below the walls,*

$$\mathbb{E}\big[(m_e - \mathbf{u}_e(U))\, g(U) + \partial_{y_e} g(U)\big] = 0.$$

At $g=1$ the masses are the mean usage, $\mathbb{E}[\mathbf{u}(U)]=\mathbf{m}$; with a native gauge $\alpha^\star$ the usage equals the target $\boldsymbol{\tau}=\mathbf{m}$ at the mode $\log\alpha^\star$ (Theorem 18.1(ii)).

Proposition 18.3 (Scale and shapes along a bunch)**.** *For a proper bunch B with room R, masses $\mathbf{m}>0$ and every proper wall of positive real part,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\Gamma_{1}^{|B|}\begin{bmatrix}\mathbf{m}|_B\\ m_B\end{bmatrix}\Gamma_{1}^{|R|}\begin{bmatrix}\mathbf{m}|_R\\ m_R\end{bmatrix}$$
$$\times\,\mathbb{E}_{(U'_B,U'_R)\sim \mathrm{Dir}_{\mathbf{m}|_B}\otimes \mathrm{Dir}_{\mathbf{m}|_R}}\Big[\int_0^\infty x^{m_B-1}(1+x)^{-\kappa_{\mathcal{E}}}\prod_j K_j(xU'_B,U'_R)^{-s_j}\Psi_j(xU'_B,U'_R)^{-t_j}\,dx\Big],$$

the inner integral being the scale integral of the family with the list $\binom{\mathcal{E}}{1}$ appended at the exponents $(\kappa_{\mathcal{E}},0)$, whose letter at (xU'_B,U'_R) is $1+x$ (Proposition 16.3); on the balance wall it is $\mathrm{Sc}_B(U'_B,U'_R;\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 15.3), and the display is Corollary 16.6 at $h=|\alpha_R|$. For real data with every proper wall positive and $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$, with the internal shapes $V_B=U_B/|U_B|$ and $V_R=U_R/|U_R|$ and the totals $|U_B|+|U_R|=1$ (Theorem 3.12), the scale $\rho=|U_B|/|U_R|$ of B given the internal shapes has density proportional to $\rho^{m_B-1}(1+\rho)^{-\kappa_{\mathcal{E}}}\prod_j K_j(\rho V_B,V_R)^{-s_j}\Psi_j(\rho V_B,V_R)^{-t_j}$.

Theorem 18.4 (Structure of the Breta law)**.** *Let $X\sim\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$; in this statement $\mathbb{E}$ is the expectation of X.*

(i) $|X|\sim\mathrm{Gamma}_{\kappa_{\mathcal{E}}}$ *and* $X/|X|\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, independently.*

(ii) The laws form an exponential family with respect to $e^{-|\alpha|}d\alpha/\alpha^{\mathbf{1}}$, with statistics $(\log X_e)_e$ and $(-\log K_j(X),-\log\Psi_j(X))_j$, natural parameters $(\mathbf{m},\mathbf{s},\mathbf{t})$ and log-partition $\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$: $\mathbb{E}\log X_e=\partial_{m_e}\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$, $\mathbb{E}\log(K_j\Psi_j)(X)=-D_{(\mathbf{0};\mathbf{e}_j,\mathbf{e}_j)}\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$, $\mathbb{E}\log(K_j/\Psi_j)(X)=-D_{(\mathbf{0};\mathbf{e}_j,-\mathbf{e}_j)}\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$, D_v the derivative along the direction v of the data, the masses fixed, and the covariances are second derivatives.

(iii) $\mathbb{E}[X^{\mathbf{k}}]=J_F(\mathbf{m}+\mathbf{k};\mathbf{s},\mathbf{t})/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $\mathbb{E}\big[\prod_j K_j(X)^{-a_j}\Psi_j(X)^{-b_j}\big]=J_F(\mathbf{m};\mathbf{s}+\mathbf{a},\mathbf{t}+\mathbf{b})/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$.

Theorem 18.5 (The race representation)**.** *Let $X\sim\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$. Given $X=\alpha$, run for each table j a* basis race*, the basis T ringing with intensity $\theta_{j,T}\alpha^T$, and a* cobasis race*, with intensity $\theta_{j,T}\alpha^{\mathcal{E}\setminus T}$, all clocks independent; let Y_j and Y_j^* be the first ringing times and T_j, T'_j the winners. In this statement $\mathbb{E}$ and $\mathbb{P}$ are taken over the clocks of the races at α.*

(i) In each race the time and the winner are independent: $K_j(\alpha)Y_j\sim\mathrm{Gamma}_1$ and $T_j\sim\mathbb{P}_{j,\alpha}$; $\Psi_j(\alpha)Y_j^*\sim\mathrm{Gamma}_1$ and $T'_j\sim\mathbb{P}^*_{j,\alpha}$.

(ii) $K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}=\mathbb{E}\big[Y_j^{s_j}(Y_j^*)^{t_j}\big]/\big(\Gamma(1+s_j)\Gamma(1+t_j)\big)$.

(iii) With $\mathrm{g}_j=-\log(K_jY_j)$ and $\mathrm{g}'_j=-\log(\Psi_jY_j^*)$, independent, $-\log(Y_jY_j^*)=\log(K_j\Psi_j)+\mathrm{g}_j+\mathrm{g}'_j$ and $-\log(Y_j/Y_j^*)=\log(K_j/\Psi_j)+\mathrm{g}_j-\mathrm{g}'_j$, and, for real a,b with $1+a\pm b>0$, $\mathbb{E}\big[e^{-a(\mathrm{g}_j+\mathrm{g}'_j)-b(\mathrm{g}_j-\mathrm{g}'_j)}\big]=\Gamma(1+a+b)\,\Gamma(1+a-b)$.

(iv) $\mathbb{P}(Y_j<Y_j^*)=K_j/(K_j+\Psi_j)=1/(1+e^{-O_j(z)})$, $\alpha=e^{2z}$.

(v) $D=\chi_{T_j}-\chi_{T'_j}$ has $\mathbb{E}[e^{\langle w,D\rangle}]=e^{E_j(z+w)-E_j(z)}$: the even statistic E_j is the cumulant function of the difference of the two winners (by (i), Theorem 7.16 for table j).

The two exponents of a table are the two clocks of its races: its even letter reads the sum $\mathrm{g}_j+\mathrm{g}'_j$ of the two logarithmic clocks of (iii) and its odd letter their difference, so that the pair coordinates (σ_j,δ_j) are the exponents of $\mathrm{g}_j+\mathrm{g}'_j$ and $\mathrm{g}_j-\mathrm{g}'_j$.

Theorem 18.6 (Conjugacy)**.** *Let* $X \sim \mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$.

(i) Given the winner T *of the* j*-th basis race, its time unobserved,* $X \sim \mathrm{Breta}(F;\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t})$*; given the winner* T' *of the* j*-th cobasis race,* $X \sim \mathrm{Breta}(F;\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T'};\mathbf{s},\mathbf{t}+\mathbf{e}_j)$*. Both keep* $\kappa_{\mathcal{E}}$*.*

(ii) Under the joint law of X *and the races of Theorem 18.5,* $\mathbb{P}(T_j=T)=\theta_{j,T}P_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t})/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $\mathbb{P}(T'_j=T')=\theta_{j,T'}P_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T'};\mathbf{s},\mathbf{t}+\mathbf{e}_j)/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*, each summing to one.*

(iii) Given the time $Y_j=\lambda$ *as well,* X *has the density of* $\mathrm{Breta}(F;\mathbf{m}+\mathbf{1}_T;\mathbf{s},\mathbf{t})$ *multiplied by* $e^{-\lambda K_j(\alpha)}$ *and renormalized: the kernel terms* $\lambda\theta_{j,T}\alpha^T$ *join those of* $e^{-|\alpha|}$*, and the law exists only on the orthant.*

Corollary 18.7 (Conjugacy for the continued period)**.** *For every family and every table* j *of it, with its bases* $\mathcal{B}_j$ *and weights* θ_j*,*

$$\sum_{T\in\mathcal{B}_j}\theta_{j,T}\,P_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t})=P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\sum_{T'\in\mathcal{B}_j}\theta_{j,T'}\,P_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T'};\mathbf{s},\mathbf{t}+\mathbf{e}_j)$$

as identities of meromorphic functions of the masses and the exponents (Lemma 16.9), and so on the balance wall: the identities of Theorem 18.6(ii) hold for the continued values of Theorem 16.17.

Theorem 18.8 (The winners of a tensorminental table)**.** *If table* j *is tensorminental of tensor rank* q *with slices* $A^{(k)}$*, then for a bunch* B *with room* R *and at a gauge as in Theorem 7.23, the pattern* $T_j\cap B$ *of the basis winner has the generating function* $\sum_{k,l}w_{kl}\det\big(I+(D_x-I)\Pi^{(kl)}_{BB}\big)$ *of that theorem, and the cobasis winner's is the same with the slices of the dual multivector. Conditioning on the winner's pattern leaves tables with multivectors of tensor rank at most* q *on the room.*

Proposition 18.9 (The period along a ray of data)**.** *Let* F *be connected, let* $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ *be real data on its balance wall with* $\boldsymbol{\tau}>0$*,* $s_j,t_j\geq 0$ *not all zero, and a native gauge* $\alpha^\star$*, let* $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ *be complex data on the balance wall, and in this statement write* $W_0=\prod_j K_j^{-s_{0,j}}\Psi_j^{-t_{0,j}}$*. As* $\lambda\to+\infty$ *the data* $(\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$ *have every proper wall of positive real part, and, with* Φ *the potential at* $(\mathbf{s},\mathbf{t})$ *and a chart* g*,*

$$P_F(\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})\\ \sim(\alpha^\star)^{\mathbf{m}_0}W_0(\alpha^\star)\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(\nabla^2\Phi(\log\alpha^\star)\big)^{-1/2}_{(g)}\,e^{-\lambda(\Phi(\log\alpha^\star)-\langle\boldsymbol{\tau},\log\alpha^\star\rangle)},$$

(g*) deleting the row and the column of* g*, where* $\nabla^2\Phi$ *is* $\frac14$ *times rung two of the still and* $(\alpha^\star)^{\mathbf{m}_0}W_0(\alpha^\star)$ *does not depend on the scale of* $\alpha^\star$*. At* $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$ *the display is the leading term of Theorem 17.5(ii) at real* λ *for the question of the data* $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$*. For real* $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$*, under* $\mathrm{Breta}^\circ(F;\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$*,* $\sqrt{\lambda}(Y-\log\alpha^\star)$ *tends to the centred normal law on* $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$ *with covariance* $(\nabla^2\Phi)^{-1}$*,* $Y=\log U$*.*

A base point added to the data moves the ray off the origin: along $(\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$, with complex $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ on the balance wall, the leading term of Theorem 17.5(ii) gains the factor $(\alpha^\star)^{\mathbf{m}_0}\prod_j K_j(\alpha^\star)^{-s_{0,j}}\Psi_j(\alpha^\star)^{-t_{0,j}}$ at the native gauge $\alpha^\star=e^{2z^\star}$.

Definition 18.10 (Skeleton law)**.** For coefficients $\xi_\ell\in\mathbb{C}\setminus(-\infty,0]$, $\ell\in J=\{0,\dots,L\}$, closed under conjugation, real nonzero letter exponents w_ℓ, equal on conjugate coefficients, with total

$|\mathbf{w}|$, and $0<\kappa<|\mathbf{w}|$, the *Skeleton law* $\mathrm{Skel}_{\mathbf{w},\kappa}(\boldsymbol{\xi})$ is the law on $(0,\infty)$ with density

$$y^{\kappa-1}\prod_{\ell}(y+\xi_\ell)^{-w_\ell}\Big/\Gamma_{\,|J|}^{\,2}\begin{bmatrix}\kappa,\ |\mathbf{w}|-\kappa\\ \mathbf{w}\end{bmatrix}\ \mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi}),$$

its normalizer being Proposition 15.1; the density is positive, and integrable exactly when $0<\kappa<|\mathbf{w}|$.

Theorem 18.11 (The Skeleton law)**.** *(i) Under* $\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, the scale of a proper bunch* B *given the internal shapes of* B *and of its room has the law* $\mathrm{Skel}_{\mathbf{w},\kappa_B}(\boldsymbol{\xi})$*:* $\boldsymbol{\xi}$ *the joint spectrum of* B *at that point and* $\mathbf{w}$ *its letter exponents of Theorem 15.3, and off the balance wall the coefficient* 1 *besides, with the letter exponent* $\kappa_{\mathcal{E}}$*, added to the letter exponent of* 1 *when* 1 *is a point of the joint spectrum; coefficients of letter exponent zero are discarded.*

(ii) For $Y\sim\mathrm{Skel}_{\mathbf{w},\kappa}(\boldsymbol{\xi})$ *and* $-\kappa<\operatorname{Re}k<|\mathbf{w}|-\kappa$*,*

$$\mathbb{E}_{Y\sim\mathrm{Skel}_{\mathbf{w},\kappa}(\boldsymbol{\xi})}[Y^k]=\Gamma_{\,2}^{\,2}\begin{bmatrix}\kappa+k,\ |\mathbf{w}|-\kappa-k\\ \kappa,\ |\mathbf{w}|-\kappa\end{bmatrix}\frac{\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa-k;\,I_J}(\boldsymbol{\xi})}{\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi})},$$

the quotient of the values of Proposition 15.1 at the indices $\kappa+k$ *and* κ*, read as that quotient when a letter exponent is not positive; and* $1/Y\sim\mathrm{Skel}_{\mathbf{w},|\mathbf{w}|-\kappa}(\boldsymbol{\xi}^{-1})$*.*

(iii) With real coefficients, positive letter exponents and a point $r\geq\max_\ell\xi_\ell$ *of the axis,* Y *is the mixture over* $\mathbf{n}\in\mathbb{N}^J$ *of* $rY_{\mathbf{n}}$*,* $Y_{\mathbf{n}}\sim\mathrm{Skel}_{|\mathbf{w}|+|\mathbf{n}|,\kappa}(1)$ *the law with the one coefficient* 1*, mixed in proportion to* $\prod_{\ell\in J}\frac{(w_\ell)_{n_\ell}}{n_\ell!}g_\ell^{n_\ell}\Gamma_{\,1}^{\,2}\begin{bmatrix}\kappa,\ |\mathbf{w}|+|\mathbf{n}|-\kappa\\ |\mathbf{w}|+|\mathbf{n}|\end{bmatrix}$*,* $g_\ell=1-\xi_\ell/r$ *the spectral gaps at* r *of Theorem 15.6; at* $r=\max_\ell\xi_\ell$ *the largest coefficient drops out.*

(iv) In the situation of (i) on the balance wall, with $s_j,t_j\geq0$*: given* $Y\leq1$*,* Y *is the mixture of the first coordinates of the laws* $\mathrm{Dir}_{(\kappa_B+k,|\mathbf{n}|+1)}$*, and given* $Y\geq1$*,* $1/Y$ *is the mixture of the first coordinates of* $\mathrm{Dir}_{(\kappa_B^{\mathrm{top}}+k,|\mathbf{n}|+1)}$*, mixed in proportion to the terms of Theorem 15.9(ii); the two sectors carry probabilities proportional to the sector Trailmoments of that theorem at the indices* κ_B *and* κ_B^{top}*.*

18.2 The port on the balance wall

Definition 18.12 (Wall contour; joint period)**.** On the balance wall, with $\operatorname{Re}s_j>0$ or $s_j=0$, and $\operatorname{Re}t_j>0$ or $t_j=0$, for every j, a *wall contour* at a bunch B puts the shares of Theorem 15.15 of the letters with nonzero exponent on real parts $c_{j,p},c'_{j,p}>0$ with $\sum_p c_{j,p}=\operatorname{Re}s_j$, $\sum_p c'_{j,p}=\operatorname{Re}t_j$ and real odd load equal to the real part of the offset on the bunch, $\sum_{j,p}\frac{c_{j,p}-c'_{j,p}}{2}\chi_p(e)=\operatorname{Re}\mu_e$; equivalently the real part of every mass of the bunch is carried exactly, $\operatorname{Re}\mathrm{load}_e=\operatorname{Re}m_e$. The *fibre integral at* B is the fibre integral (Definition 4.1) with the pins $|\mathbf{a}_j|=s_j$, $|\mathbf{b}_j|=t_j$ and $\langle\boldsymbol{\delta}\rangle_e=\mu_e$, $e\in B$. For a family $G=((\mathcal{B}_i,\theta_i);a_i,b_i)_i$ whose molecules are $R_1,\dots,R_q$, the *joint period* $P_G^{\otimes}(\mathbf{m};\mathbf{a},\mathbf{b})$ is the integral of $\alpha^{\mathbf{m}-\mathbf{1}}\prod_i K_i(\alpha)^{-a_i}\Psi_i(\alpha)^{-b_i}$ over $\alpha\in\prod_{k=1}^q\Delta_{R_k}$ against $\prod_k d\omega_{R_k}$, wherever it converges; when the weights factor across the molecules it is the product of the molecules' periods.

Theorem 18.13 (The port on the balance wall)**.** *Let* F *have data on the balance wall with* $\operatorname{Re}s_j>0$ *or* $s_j=0$*, and* $\operatorname{Re}t_j>0$ *or* $t_j=0$*, for every* j*, let* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *converge, and let* B *be a proper bunch with room* R*; put* $z=\frac{1}{2}\log\alpha$*.*

(i) The bunch factor $\Xi_B(\alpha_R;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t}) = \int_{\mathbb{R}^B_{>0}} \alpha_B^{\boldsymbol{\mu}|_B-\mathbf{1}} \prod_j \Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j}\, d\alpha_B$ *converges for every* α_R, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_R} v^{\boldsymbol{\mu}|_R-\mathbf{1}}\, \Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})\, d\omega_R(v)$, *and on every wall contour*

$$\Xi_B(\alpha_R;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j,\ \langle\boldsymbol{\delta}\rangle=\boldsymbol{\mu}|_B} \prod_j \Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix} \times \prod_{j,p}\Lambda_{j,(p)}(z_R)^{-a_{j,p}}\Lambda_{j,(p)}(-z_R)^{-b_{j,p}}.$$

(ii) Let $R_1,\dots,R_q$ *be the molecules of the children and* $\kappa_{R_i}(\mathbf{a},\mathbf{b})$ *the wall at* R_i *of the children* $F_B(\mathbf{a},\mathbf{b})$ *with the masses* $\mathbf{m}|_R$ *(Definition 8.1); as* R_i *is a molecule,* $\kappa_{R_i}(\mathbf{a},\mathbf{b}) = m_{R_i} - \sum_{j,p}\big(a_{j,p}r_{j,(p)}(R_i) + b_{j,p}(|R_i| - r_{j,(p)}(R_i))\big)$, *with* $r_{j,(p)}$ *the rank function of the child* (j,p). *Let* $\mathcal{X}$ *be the fibre of the shares cut by the pins* $|\mathbf{a}_j| = s_j$, $|\mathbf{b}_j| = t_j$, $\langle\boldsymbol{\delta}\rangle = \boldsymbol{\mu}|_B$ *and* $\kappa_{R_1} = \cdots = \kappa_{R_{q-1}} = 0$, *with* $|B|+q-1$ *pins beyond the share sums;* $\kappa_{R_q} = 0$ *holds on it. On a wall contour with* $\operatorname{Re}\kappa_{R_i} = 0$ *for* $i < q$ *at which* $P^{\otimes}_{F_B(\mathbf{c},\mathbf{c}')}(\operatorname{Re}\mathbf{m}|_R;\mathbf{c},\mathbf{c}')$ *converges,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\mathcal{X}} \prod_j \Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\, P^{\otimes}_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}).$$

(iii) The native split of the real parts $(\operatorname{Re}\mathbf{m};\operatorname{Re}\mathbf{s},\operatorname{Re}\mathbf{t})$, *which have a native gauge, is a wall contour whose real parts are those of the points of* $\mathcal{X}$, *and* $P^{\otimes}$ *converges there; for real data it is the native split, and it lies on the fibre.*

The bunch is integrated on its orthant, scale and shape at once: writing $\alpha_B = \rho u$ before opening the letters would split the $|B|$ pins into a pin on the scale and a stacked Gamma over the shape whose indices sum to zero, carrying $1/\Gamma(0) = 0$, and neither has a value alone. For the same reason $B = \mathcal{E}$ is excluded: the imaginary part of the odd load then lies in the hyperplane $\sum_{e\in\mathcal{E}}\operatorname{Im}\langle\boldsymbol{\delta}\rangle_e = \operatorname{Im}\mu_{\mathcal{E}}$, the scaling mode, and one mark stays in the room.

Lemma 18.14 (Connected tables have connected children)**.** *If some table with* $s_j + t_j > 0$ *is connected, then at every bunch the children at a split with positive shares form a connected family.*

At the masses $\boldsymbol{\tau}+\eta\mathbf{m}$ the port on the orthant carries the stacked Gamma with the upper entries $\nu_e = \mu_e - \langle\boldsymbol{\delta}\rangle_e$, $e \in B$, and κ_{R_i}, and the lower entry $\eta m_{\mathcal{E}}$; its $|B|+q$ upper entries sum to $\eta m_{\mathcal{E}}$ and have real parts positive multiples of η on the wall contour of the native split (Theorem 18.13(iii)); as $\eta \to 0$ it tends to $(2\pi)^{|B|+q-1}$ times the delta of the pins. So the port on the balance wall is the limit of the Foreman procedure on the ray (Theorem 16.15): the Gammas are the pins.

Proposition 18.15 (Peeling a molecule)**.** *In Theorem 18.13 with a native gauge, let* R_1 *be a molecule of the children and* $g \in R_1$. *Each child letter is* $\sum_X \alpha_{R_1}^X K_{[X]}(\alpha_{R\setminus R_1})$ *over the bases* X *of its restriction to* R_1, *with* $K_{[X]}$ *the letter on* $R\setminus R_1$ *with the bases* Y *and the weights* $\theta_{X\cup Y}$ *(for* Ψ, $\alpha_{R_1}^{R_1\setminus X}\Psi_{[X]}$*). Opening these sums with one share per part,* $P^{\otimes}$ *is the fibre integral with the* $|R_1|-1$ *pins* $\operatorname{load}_e = m_e$, $e \in R_1 - g$, *of the stacked Gammas of these shares against the joint period of the new children on the remaining molecules; the pin at* g *holds by the balance of* R_1, *the native split lies on the fibre, and the joint period converges there. With weights multiplicative across* R_1 *every* $K_{[X]}$ *is a multiple of one letter and this is the factorization of Theorem 18.13(ii).*

Corollary 18.16 (Atoms on the balance wall)**.** *In Theorem 18.13, fix g. On a wall contour for $\mathcal{E}-g$, with one share per basis and letter,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j,\ \langle\boldsymbol{\delta}\rangle=\boldsymbol{\mu}|_{\mathcal{E}-g}} \prod_j \Gamma_{|\mathcal{B}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\mathcal{B}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\prod_{j,T}\theta_{j,T}^{-a_{j,T}-b_{j,T}},$$

independent of g; the pin at g holds by itself.

Running the ports on the balance wall down to rooms of one mark writes $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ as an iterated fibre integral with $n-1$ pins in all; with one share per pattern and letter its dimension is $\sum(|\mathcal{B}_j|-1)-(n-1)$, one term for each active letter, j its table, independent of the tree when no letter is split across molecules.

Proposition 18.17 (The pins reach every integer point)**.** *In Corollary 18.16 take one share for each basis and each active letter, the K_j with $s_j\neq 0$ and the Ψ_j with $t_j\neq 0$, and write the pins in the shares as the share sums $|\mathbf{a}_j|=s_j$, $|\mathbf{b}_j|=t_j$ of the active letters and the loads $\mathrm{load}_e=m_e$, $e\in\mathcal{E}-g$ (Definition 12.24). Let every two marks be joined by a sequence of marks in which any two consecutive marks lie in a common circuit of a table with an active letter. Then the pins map the integer points of the share space onto the integer points of their target. So at integer exponents and masses the fibre has integer points, its integer points are $x_0+\Lambda$ for one of them x_0 and the lattice Λ of the integer points of its directions, which has the rank of the dimension of the fibre, and a basis of Λ gives integral coordinates on the fibre in which the integer points are exactly the points with integer coordinates.*

Corollary 18.18 (Closure)**.** *If the weights of a form $W\,d\alpha_{\mathbf{m}}$ with real data, $s_j,t_j\geq 0$ not all zero, factor across its molecules and every molecule has a native gauge, $\int W\,d\alpha_{\mathbf{m}}$ is a product over the molecules of iterated fibre integrals over affine fibres of share space, with integrands products of stacked Gammas of the shares and monomials in the weights, $n-1$ pins on a molecule of n marks. The period of each molecule at its target is also the nonnegative series of Corollary 15.11 and of Theorem 15.13. Every quasi-integral is an iterated constant term of the continuation of the share integral of Theorem 15.15 at its chart completion.*

Corollary 18.19 (The marginal law of the shape)**.** *In Theorem 18.13 let $U\sim\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$, with the internal shapes $V_B=U_B/|U_B|$ and $V_R=U_R/|U_R|$ of B and R (Proposition 18.3).*

(i) $\mathbb{E}_{U'\sim\mathrm{Dir}_{\mathbf{m}|_B}}\big[\mathrm{Sc}_B(U',v;\mathbf{m};\mathbf{s},\mathbf{t})\big]=v^{-\sigma\mathbf{1}}\,\Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})/\Gamma_{|B|}\begin{bmatrix}\mathbf{m}|_B\\ m_B\end{bmatrix}$.

(ii) Given $V_R=v$, for $\mathbf{k}\in\mathbb{C}^B$ with $|\mathbf{k}|=0$, $\mathbb{E}_{U\sim\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})}[V_B^{\mathbf{k}}\mid V_R=v]=\Xi_B(v;\boldsymbol{\mu}|_B+\mathbf{k};\mathbf{s},\mathbf{t})/\Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})$, the bunch factor at shifted offsets, whose fibre is $\langle\boldsymbol{\delta}\rangle=\boldsymbol{\mu}|_B+\mathbf{k}$.

(iii) V_R has density proportional to $v^{\boldsymbol{\mu}|_R-\mathbf{1}}\,\Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})$, the fibre mixture of the children's shape laws in proportion to the stacked Gammas times $P_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$ when the children are connected (Lemma 18.14).

(iv) Given both shapes, the scale of B has the Skeleton law of Theorem 18.11(i).

18.3 Cycles

Definition 18.20 (Cycle of a family)**.** A *cycle* of a family is a bunch C that is a circuit of one of its tables. That table's cobasis polynomial has floor one on C, and on its bottom colevel the letter is linear on C; a letter linear on a bunch gives the bunch the *kernel* e^{-h}, h the letter divided by the sum of its coefficients (Theorem 18.21(ii) below).

Theorem 18.21 (A cycle carries its own kernel)**.** *Let F have real data with $s_j, t_j \geq 0$, $\kappa_{\mathcal{E}} = 0$, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ convergent, and $C \neq \mathcal{E}$ a circuit of table j_0 with $t_{j_0} > 0$ and room R.*

(i) The patterns of table j_0 on C of colevel one are exactly $p_e = C - e$, $e \in C$, and the child at p_e is the contraction $\mathcal{B}_{j_0}/C$ on R with the weights $\theta_{j_0,p_e\cup X}$. So $\Psi_{j_0} = \Psi^\flat + \Psi^\sharp$ with $\Psi^\flat = \sum_{e\in C} \alpha_e \Psi_{j_0,(p_e)}(\alpha_R)$ linear in α_C. The circuits are the only bunches whose colevel-one patterns are one per mark.

(ii) With $G(\alpha_R) = \sum_{e\in C} \Psi_{j_0,(p_e)}(\alpha_R)$ and $h = \Psi^\flat/G$, if the family with $\Psi_{j_0}^{-t_{j_0}}$ replaced by $(\Psi^\flat)^{-c}(\Psi^\sharp)^{-(t_{j_0}-c)}$ converges, then on $\operatorname{Re}\beta = c$

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \frac{1}{\Gamma(t_{j_0})} \int_{\beta+\beta'=t_{j_0}} \Gamma(\beta') \times \int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} e^{-h(\alpha)} G(\alpha_R)^{-\beta} \Psi^\sharp(\alpha)^{-\beta'} K_{j_0}^{-s_{j_0}} \prod_{j\neq j_0} K_j^{-s_j} \Psi_j^{-t_j}\, d\alpha.$$

When the colevel-one children are multiples ϑ_e of one letter, as for unit weights and for determinantal tables, the kernel is $e^{-\langle\vartheta,\alpha_C\rangle/|\vartheta|}$.

(iii) On a wall contour at C the pins at the marks of C hold exactly when $b_{j_0,p_e} = m_e - \lambda_e$, λ_e the load on e of all the other shares, and $\sum_{j,p}\big(a_{j,p}(|p| - d_j(C)) + b_{j,p}(|C\setminus p| - d_j^*(C))\big) = \kappa_C$, the coefficients being the excesses of the level and the colevel of p in table j (Definition 12.16); the fibre integral at C is over the other shares with the single pin that their total excess is the wall of the cycle, the Gammas $\Gamma(m_e - \lambda_e)$ of the cycle standing in the places of the $\Gamma(b_{j_0,p_e})$.

Dually, a cocircuit has floor one in the basis polynomial, and its bottom level gives its kernel.

Corollary 18.22 (A table that is one circuit)**.** *Let $F = ((\mathcal{B},\theta);s,t)$ with $s,t > 0$, where the bases of $\mathcal{B}$ are the sets $\mathcal{E} - e$, $e \in \mathcal{E}$, with the weights θ_e; let the masses lie on the balance wall, $m_{\mathcal{E}} = (n-1)s + t$, with $m_S > s(|S|-1)$ for every nonempty proper S, and put $\nu_e = m_e - s$. Then*

$$\begin{aligned} P_F(\mathbf{m};s,t) &= \frac{\prod_e \theta_e^{-\nu_e}}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{s-1} \prod_{e\in\mathcal{E}} \mathsf{G}_{\nu_e;(1,-1)}\big(1,\theta_e^2 y\big)\, dy \\ &= \frac{\prod_e \theta_e^{-\nu_e}}{\Gamma(s)\Gamma(t)} \int_{\sum_e a_e = s} \prod_{e\in\mathcal{E}} \Gamma(a_e)\,\Gamma(a_e+\nu_e)\,\theta_e^{-2a_e} : \end{aligned}$$

the cobasis polynomial $\Psi(\alpha) = \langle\theta,\alpha\rangle$ is linear on the whole set, which so carries its own kernel (Definition 18.20), and the fibre of the atom formula (Corollary 18.16), of dimension $n-1$, closes to one scale.

Chapter 19

Bunches of width one: pooling and cleaning

A bunch whose bases meet it on two levels only is seen by its room as one mark: the room reads the bunch through two ratios of polynomials on its marks, one for each Foreman polynomial, and the bunch is a table on its own marks and one more. A bunch of width one is read as a mark of its room, its shape integrated while its scale is kept, its walls glued, and its scale closed at a wall, at the gauge and on the totals of the bunch and the room. Cleaning, the repeated pooling of such bunches, is a move between period data and has one result.

19.1 Bunches of width one

Definition 19.1 (Parallel and series bunches)**.** A *parallel bunch* of a table (a parallel class) is a set of at least two marks, none a loop, no two of which lie in a common basis; a *series bunch* (a series class) is a set of at least two marks, none a coloop, every basis containing all but at most one of them. A bunch B of width $c(B) = r(B) - d(B)$ (Definition 7.12, Theorem 7.13) has its patterns on the $c(B)+1$ levels $d(B), \dots, r(B)$ (Theorem 12.7), and its room has the same width.

Definition 19.2 (Hinge values; their ratio)**.** Let a bunch B of a table $(\mathcal{B}, \theta)$ have width one, the weights factoring across B and its room R, $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$, so that $\mathcal{B}|B$ and $\mathcal{B}/R$ carry θ' and $\mathcal{B}|R$ and $\mathcal{B}/B$ carry θ'' (Definition 7.5). The *hinge value of B by K* and its *hinge value by Ψ* are

$$\gamma_B^K = \frac{K_{\mathcal{B}|B}}{K_{\mathcal{B}/R}}, \qquad \gamma_B^\Psi = \frac{\Psi_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|B}},$$

functions of α_B homogeneous of degree one, the values at which K and Ψ see the bunch replaced by one mark, and their *ratio*

$$\varrho_B = \frac{\gamma_B^\Psi}{\gamma_B^K} = \frac{\mathbf{E}_{\mathcal{B}/R}}{\mathbf{E}_{\mathcal{B}|B}},$$

the quotient of the even letters $\mathbf{E} = K\Psi$ (Definition 8.18) of the two tables on B, is homogeneous of degree zero, a function of the shape $u = \alpha_B/|\alpha_B|$ alone. The room has width one as well, and its hinge values are those of R as a bunch: $\gamma_R^K = K_{\mathcal{B}|R}/K_{\mathcal{B}/B}$, $\gamma_R^\Psi = \Psi_{\mathcal{B}/B}/\Psi_{\mathcal{B}|R}$ and $\varrho_R = \mathbf{E}_{\mathcal{B}/B}/\mathbf{E}_{\mathcal{B}|R}$. Statements about hinge values and their ratios are read at one fixed factorization. For table j of a family they are $\gamma_{j,B}^K$, $\gamma_{j,B}^\Psi$ and $\varrho_{j,B}$, each table with its own factorization.

Proposition 19.3 (Hinge values of parallel and series bunches)**.** *A parallel or series bunch has width at most one, and width zero exactly when it is a union of molecules. Let a parallel bunch B of width one have its weights factoring across B and R (Definition 19.2); then $\mathcal{B}/R$ has the single basis $\varnothing$, with the weight $\theta'_{\varnothing}$, and $\mathcal{B}|B$ the singletons, with the weights $\theta'_{\{e\}}$, and in this statement write $w_e = \theta'_{\{e\}}/\theta'_{\varnothing}$. So $K_{\mathcal{B}/R} = \theta'_{\varnothing}$, $K_{\mathcal{B}|B} = \theta'_{\varnothing} \sum_{e\in B} w_e\alpha_e$, $\Psi_{\mathcal{B}|B} = \theta'_{\varnothing}\, e_{|B|}(\alpha_B) \sum_{e\in B} w_e\alpha_e^{-1}$ and $\Psi_{\mathcal{B}/R} = \theta'_{\varnothing}\, e_{|B|}(\alpha_B)$, and*

$$\gamma_B^K = \sum_{e\in B} w_e\alpha_e, \qquad \gamma_B^\Psi = \Big(\sum_{e\in B} w_e\alpha_e^{-1}\Big)^{-1} :$$

K adds the marks of the class with the factors w_e and Ψ combines them as reciprocals with the same factors. A series bunch B of width one, its weights factoring, swaps the two roles: $\mathcal{B}|B$ has the single basis B, with the weight θ'_B, and $\mathcal{B}/R$ the sets $B-e$, with the weights θ'_{B-e}; with $w_e = \theta'_{B-e}/\theta'_B$, $K_{\mathcal{B}|B} = \theta'_B\, e_{|B|}(\alpha_B)$, $K_{\mathcal{B}/R} = \theta'_B\, e_{|B|}(\alpha_B) \sum_e w_e\alpha_e^{-1}$, $\Psi_{\mathcal{B}|B} = \theta'_B$ and $\Psi_{\mathcal{B}/R} = \theta'_B \sum_e w_e\alpha_e$, so $\gamma_B^\Psi = \sum_e w_e\alpha_e$ and $\gamma_B^K = \big(\sum_e w_e\alpha_e^{-1}\big)^{-1}$. In a list, the case used for cleaning (Definition 19.35 below), with the factorization $\theta' = \theta'' = 1$, every w_e is 1, and a pair with the shape $(x, 1-x)$ has $\varrho_B = x(1-x)$ when it is parallel and $\varrho_B = 1/(x(1-x))$ when it is in series.

Theorem 19.4 (A bunch of width one is one mark to its room)**.** *Let a bunch B of a table $(\mathcal{B},\theta)$ have width one, the weights factoring across B and R, $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$, and let g be a new mark. The* cell of the bunch $\mathcal{B}_B^+$ *is the table on $B+g$ whose bases avoiding g are those of $\mathcal{B}|B$ and whose bases through g are the $X+g$, $X\in\mathcal{B}/R$, with the weights θ'; the* cell of the room $\mathcal{B}_R^+$ *is the table on $R+g$ whose bases avoiding g are those of $\mathcal{B}|R$ and whose bases through g are the $Y+g$, $Y\in\mathcal{B}/B$, with the weights θ''. Both are tables, lists when $(\mathcal{B},\theta)$ is one, in which g, the* hinge*, is neither a loop nor a coloop, and the bases of $\mathcal{B}$ are theirs glued along g with g on exactly one side:*

$$\mathcal{B} = \{X\sqcup Y:\ X\in\mathcal{B}|B,\ Y\in\mathcal{B}/B\}\ \sqcup\ \{X\sqcup Y:\ X\in\mathcal{B}/R,\ Y\in\mathcal{B}|R\}.$$

With the first argument of the polynomials of a cell the variable of its hinge and the hinge values of Definition 19.2,

$$K(\alpha) = K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}_R^+}\big(\gamma_B^K(\alpha_B),\alpha_R\big), \qquad \Psi(\alpha) = \Psi_{\mathcal{B}|B}(\alpha_B)\, \Psi_{\mathcal{B}_R^+}\big(\gamma_B^\Psi(\alpha_B),\alpha_R\big).$$

Conversely, if $K(\alpha) = \sum_{l=0}^k A_l(\alpha_B)\, C_l(\alpha_R)$ with every C_l homogeneous, then $c(B)\le k$; in particular $K(\alpha) = A(\alpha_B)\big(C_0(\alpha_R) + G(\alpha_B)\, C_1(\alpha_R)\big)$ with C_0 and C_1 homogeneous gives $c(B)\le 1$.

The deletions and the contractions of the hinge are the four minors $\mathcal{B}_B^+\setminus g = \mathcal{B}|B$ and $\mathcal{B}_B^+/g = \mathcal{B}/R$, $\mathcal{B}_R^+\setminus g = \mathcal{B}|R$ and $\mathcal{B}_R^+/g = \mathcal{B}/B$ of the table. So the hinge values of B are those of B in its own cell, and across a bunch of width one the bunch and the room see each other as one mark: the room reads the hinge at $\alpha_g = \gamma_B^K(\alpha_B)$ in K and at $\alpha_g = \gamma_B^\Psi(\alpha_B)$ in Ψ.

Corollary 19.5 (The scale of a bunch of width one is that of one mark)**.** *Along the scale of a bunch B of width one of a table whose weights factor across B and R, each Foreman polynomial has one Skeleton value, the room's hinge value over the bunch's:*

$$\operatorname{spec}_B^K(\alpha) = \Big\{\frac{\gamma_R^K(\alpha_R)}{\gamma_B^K(\alpha_B)}\Big\} = \Big\{\frac{K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}|R}(\alpha_R)}{K_{\mathcal{B}|B}(\alpha_B)\, K_{\mathcal{B}/B}(\alpha_R)}\Big\},$$
$$\operatorname{spec}_B^\Psi(\alpha) = \Big\{\frac{\gamma_R^\Psi(\alpha_R)}{\gamma_B^\Psi(\alpha_B)}\Big\} = \Big\{\frac{\Psi_{\mathcal{B}|B}(\alpha_B)\, \Psi_{\mathcal{B}/B}(\alpha_R)}{\Psi_{\mathcal{B}/R}(\alpha_B)\, \Psi_{\mathcal{B}|R}(\alpha_R)}\Big\},$$

the Skeleton value of the hinge as a one-mark bunch of the cell of the room (Corollary 15.4), read at $\alpha_g = \gamma_B^K(\alpha_B)$, respectively at $\alpha_g = \gamma_B^\Psi(\alpha_B)$. The value of K over that of Ψ is $\varrho_B(\alpha_B)/\varrho_R(\alpha_R)$, the even letters of the minors of the level $d(B)$ over those of the level $r(B)$, and their product is the same quotient of the odd letters.

Lemma 19.6 (One hinge value of any degree is width at most one)**.** *Let B be a bunch of a table $(\mathcal{B}, \theta)$ with the room R, and suppose that on the open orthant*

$$K(\alpha) = A(\alpha_B)\, P\big(\alpha_R, \gamma(\alpha_B)\big),$$

with A and γ positive functions of α_B, homogeneous, γ of degree one, and P a polynomial in α_R and one more variable with real coefficients, of any degree in that variable. Then $c(B) \le 1$. Conversely, when the weights factor across B and R and $c(B) \le 1$, K has this form: at width zero with P free of its last variable, and at width one with $A = K_{\mathcal{B}/R}$, $\gamma = \gamma_B^K$ and $P(\alpha_R, y) = K_{\mathcal{B}_R^+}(y, \alpha_R)$ (Theorem 19.4). The same holds for Ψ, the basis polynomial of the dual table, in which B has the same width.

Proposition 19.7 (The freedom of the factorization)**.** *Let a bunch B of a table $(\mathcal{B}, \theta)$ have width one, the weights factoring across B and R, $\theta_T = \theta'_{T\cap B}\, \theta''_{T\cap R}$. The factorization is fixed up to one positive constant on each of the two levels: multiplying θ' by λ_d on the patterns of the level $d(B)$ and by λ_r on those of the level $r(B)$, and dividing θ'' by λ_d and by λ_r on the room sides of the bases of these levels, multiplies the hinge values γ_B^K and γ_R^K by $c = \lambda_r/\lambda_d$, γ_B^Ψ and γ_R^Ψ by c^{-1}, and the ratios ϱ_B and ϱ_R by c^{-2} (Definition 19.2), and leaves γ_R^K/γ_B^K, $\gamma_R^\Psi/\gamma_B^\Psi$ and ϱ_B/ϱ_R unchanged. In a family each table has its own two constants.*

19.2 Pooling

Theorem 19.8 (Pooling integrates the shape and keeps the scale)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family and B a bunch of width one in every table, the weights of every table factoring across B and R, and read $\alpha_B = \rho u$, $\rho > 0$, $u \in \Delta_B$ (Lemma 2.7). For $u \in \Delta_B$ the* pooled family *$F_{B\to g}(u)$ is the family on $R + g$ with, for every j, two tables with the bases of the cell of the room $\mathcal{B}_{j,R}^+$ (Theorem 19.4): the first with its weights multiplied by $\gamma_{j,B}^K(u)$ on its bases through g and the exponents $(s_j, 0)$, the second with its weights multiplied by $\gamma_{j,B}^\Psi(u)$ on its bases avoiding g and the exponents $(0, t_j)$. Its letters are $K_{\mathcal{B}_{j,R}^+}\big(\gamma_{j,B}^K(u)\alpha_g, \alpha_R\big)^{-s_j}$ and $\Psi_{\mathcal{B}_{j,R}^+}\big(\gamma_{j,B}^\Psi(u)\alpha_g, \alpha_R\big)^{-t_j}$; listing the N first tables and then the N second ones, its exponents are $(\mathbf{s}, \mathbf{0})$ and $(\mathbf{0}, \mathbf{t})$. To* pool *B is to pass from F to its pooled families $F_{B\to g}(u)$, keeping the letters $K_{\mathcal{B}_j/R}$ and $\Psi_{\mathcal{B}_j|B}$ of the shape of B, its cell kernel (Chapter 22), and the hinge values of every table. Give the hinge the mass κ_B, the wall of B. Then $F_{B\to g}(u)$ has the imbalance of F, and*

$$J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{B}_j/R}(u)^{-s_j} \Psi_{\mathcal{B}_j|B}(u)^{-t_j} \times J_{F_{B\to g}(u)}\big((\kappa_B, \mathbf{m}|_R); (\mathbf{s}, \mathbf{0}), (\mathbf{0}, \mathbf{t})\big)\, d\omega_B(u),$$

both sides converging together; the same holds with P for J, on the balance wall as well.

The port (Theorem 15.15) integrates the scale and the shape of a bunch together; pooling integrates the shape of a bunch of width one and keeps its scale, which becomes the hinge of the room. For a list the pooled families again have weights that factor across every bunch of R and

across every bunch containing g, so poolings can be repeated. Proposition 18.3 is the same split with the scale integral left whole. Read backwards, pooling is gluing: the table is its cell of the bunch substituted into its cell of the room at the hinge (Proposition 21.3), and every cell of a substitution is pooled at once by Theorem 22.3.

19.3 Walls glue

Theorem 19.9 (Walls glue)**.** *Let a bunch B of a table $(\mathcal{B},\theta)$ have width one, the weights factoring across B and R, with the cells $\mathcal{B}_B^+$ and $\mathcal{B}_R^+$ of Theorem 19.4. For $S_B \subseteq B$ and $S_R \subseteq R$ the counts of the table glue from those of the cells, the hinge counted on exactly one side, as in the gluing of the bases:*

$$d(S_B \sqcup S_R) = \min\left\{d_{\mathcal{B}_B^+}(S_B+g) + d_{\mathcal{B}_R^+}(S_R),\ d_{\mathcal{B}_B^+}(S_B) + d_{\mathcal{B}_R^+}(S_R+g)\right\},$$
$$d^*(S_B \sqcup S_R) = \min\left\{d^*_{\mathcal{B}_B^+}(S_B+g) + d^*_{\mathcal{B}_R^+}(S_R),\ d^*_{\mathcal{B}_B^+}(S_B) + d^*_{\mathcal{B}_R^+}(S_R+g)\right\}.$$

For a family $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ with masses $\mathbf{m}$, B having width one in every table and the weights of every table factoring across B and R:

(i) the walls of the pooled family $F_{B\to g}(u)$ at the masses $(\kappa_B,\mathbf{m}|_R)$ are walls of F: its wall at S_R is κ_{S_R} and its wall at S_R+g is $\kappa_{B\sqcup S_R}$;

(ii) the density $u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{B}_j/R}(u)^{-s_j}\Psi_{\mathcal{B}_j|B}(u)^{-t_j}$ of the cell kernel of B has at the face $u_{S_B}=0$ of Δ_B the wall of F, $m_{S_B} - \sum_j \left(s_j\, d_{\mathcal{B}_j/R}(S_B) + t_j\, d^*_{\mathcal{B}_j|B}(S_B)\right) = \kappa_{S_B}$;

(iii) on the balance wall, with the cell families $G_B = ((\mathcal{B}_{j,B}^+,\theta_j');s_j,t_j)_j$ at the masses $(\mathbf{m}|_B, 2\sigma-\kappa_B)$ and $G_R = ((\mathcal{B}_{j,R}^+,\theta_j'');s_j,t_j)_j$ at the masses $(\kappa_B,\mathbf{m}|_R)$, σ the total strength, whose walls are written $\kappa_S^{G_B}$ and $\kappa_S^{G_R}$, both cell families lie on their balance walls and every wall of a cell family is a wall of F,

$$\kappa_{S_B}^{G_B} = \kappa_{S_B},\qquad \kappa_{S_B+g}^{G_B} = \kappa_{S_B\sqcup R},\qquad \kappa_{S_R}^{G_R} = \kappa_{S_R},\qquad \kappa_{S_R+g}^{G_R} = \kappa_{B\sqcup S_R};$$

if moreover $s_j,t_j\geq 0$ and every active letter reads B alike, its hinge value a constant multiple of one function γ on Δ_B, as when one letter is active, every wall of F is glued from walls of the cell families,

$$\kappa_{S_B\sqcup S_R} = \min\left\{\kappa_{S_B}^{G_B} + \kappa_{S_R}^{G_R},\ \kappa_{S_B+g}^{G_B} + \kappa_{S_R+g}^{G_R}\right\},$$

and F has every proper wall positive exactly when both cell families do.

19.4 Full leans

Theorem 19.10 (Pooling factors at one hinge value)**.** *Let $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ have real exponents and be on the balance wall with every proper wall positive, and let a bunch B have width one in every table, the weights of every table factoring across B and R. Suppose that every active letter reads B alike, its hinge value a constant multiple of one function γ on Δ_B, the factorizations being chosen so that every constant is 1: every active letter has the one hinge value γ. (A change of the factorization of table j multiplies $\gamma_{j,B}^K$ by a positive constant and $\gamma_{j,B}^\Psi$ by its inverse (Proposition 19.7). So a table with one active letter can always be so normalized, and when every table is at a full lean the hypothesis is a condition across the tables alone; one table with both letters*

active can be so normalized exactly when ϱ_B is constant.) Then $0 < \kappa_B < 2\sigma$, σ the total strength (Definition 8.1), and, with the cell families $G_B = ((\mathcal{B}^+_{j,B}, \theta'_j); s_j, t_j)_j$ and $G_R = ((\mathcal{B}^+_{j,R}, \theta''_j); s_j, t_j)_j$,

$$\Gamma_{\!1}^{\,2}\begin{bmatrix}\kappa_B, & 2\sigma-\kappa_B\\ & 2\sigma\end{bmatrix} P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{G_B}\big((\mathbf{m}|_B, 2\sigma-\kappa_B);\mathbf{s},\mathbf{t}\big)\, P_{G_R}\big((\kappa_B, \mathbf{m}|_R);\mathbf{s},\mathbf{t}\big),$$

both cell families on their balance walls with every proper wall positive. One table at the exponents $(0,t)$ or $(s,0)$ has one active letter, and then 2σ is t, respectively s.

When every active letter has one hinge value, as for one table at a full lean, where one letter is active, the hinge carries the mass κ_B in the room and $2\sigma - \kappa_B$ in the bunch, and the period of the family is the product of the periods of its two cell families divided by the Beta corner of the hinge.

Proposition 19.11 (Pooling a realization)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family, B a bunch of width one in every table, the weights of every table factoring across B and R, and let $\Upsilon = ((s_j, t_j)_j; \alpha)$ be a realization of F with masses $\boldsymbol{\tau}$ (Definition 9.1); write $\alpha_B = \rho u$, $\rho > 0$, $u \in \Delta_B$ (Lemma 2.7), and read the two tables of index j of the pooled family $F_{B\to g}(u)$ of Theorem 19.8 as one table with two systems of weights at the exponents (s_j, t_j) (Definition 8.1). Then $((s_j, t_j)_j; (\rho, \alpha_R))$ is a realization of $F_{B\to g}(u)$, with the masses $\big(\tau_B - \sum_j (s_j d_j(B) + t_j d^*_j(B)),\ \boldsymbol{\tau}|_R\big)$, the first at the hinge: pooling sends a realization to a realization, and the mass of the hinge is the wall κ_B at the masses $\boldsymbol{\tau}$. With one table at the exponents $(0,t)$ the letter Ψ of $F_{B\to g}(u)$ at (ρ, α_R) is that of the cell of the room $(\mathcal{B}^+_R, \theta'')$ at $(\gamma^\Psi_B(\alpha_B), \alpha_R)$, and $((0,t); (\gamma^\Psi_B(\alpha_B), \alpha_R))$ is a realization of $(\mathcal{B}^+_R, \theta'')$ with the masses $(\tau_B - t\, d^*(B),\ \boldsymbol{\tau}|_R)$.*

Proposition 19.12 (Hinge values that agree)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real exponents and be on the balance wall with every proper wall positive, and let a bunch B have width one in every table, the weights of every table factoring across B and R. Suppose that every active letter reads B alike, its hinge value a constant multiple of one function γ on Δ_B: $\gamma^K_{j,B} = c_j\gamma$ whenever $s_j \neq 0$ and $\gamma^\Psi_{j,B} = c^*_j\gamma$ whenever $t_j \neq 0$, with constants $c_j, c^*_j > 0$; put $c_j = 1$ when $s_j = 0$ and $c^*_j = 1$ when $t_j = 0$. Let $G^{\mathbf{c}}_R$ be the pooled family of Theorem 19.8 with these constants in place of the hinge values: for every j the table $\mathcal{B}^+_{j,R}$ with its weights multiplied by c_j on its bases through g, at the exponents $(s_j, 0)$, and with its weights multiplied by c^*_j on its bases avoiding g, at the exponents $(0, t_j)$. Then*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{G^{\mathbf{c}}_R}\big((\kappa_B, \mathbf{m}|_R); (\mathbf{s},\mathbf{0}), (\mathbf{0},\mathbf{t})\big) \times \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{B}_j/R}(u)^{-s_j}\, \Psi_{\mathcal{B}_j|B}(u)^{-t_j}\, \gamma(u)^{-\kappa_B}\, d\omega_B(u) :$$

the shape and the room separate at every pair of exponents. For one table with ϱ_B constant, a factorization with $\varrho_B = 1$ (Proposition 19.7) gives $c = c^* = 1$ and $\gamma = \gamma^K_B$, and $G^{\mathbf{c}}_R$ is then $((\mathcal{B}^+_R, \theta''); s, t)$.

19.5 Partial leans

Theorem 19.13 (Gluing across a bunch of width one)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real data with $s_j, t_j \geq 0$, $s_j + t_j > 0$, and masses $\mathbf{m}$ on the balance wall with every proper wall positive, and let a bunch B have width one in every table, the weights factoring across B and R. In table j call*

$d_j(B)$ *the* level d *and* $r_j(B) = d_j(B) + 1$ *the* level r. Level shares $\mathbf{a} = (a_j^d, a_j^r)_j$ *and* $\mathbf{b} = (b_j^d, b_j^r)_j$ *split the exponents over the two levels,* $a_j^d + a_j^r = s_j$ *and* $b_j^d + b_j^r = t_j$, *a letter of exponent zero being opened by nothing, its two level shares* 0*; they are the shares of Definition 12.24 summed over the patterns of each level, and their pair shares (Definition 12.24) are the* even share $\sigma_j^\ell = \frac{1}{2}(a_j^\ell + b_j^\ell)$ *and the* odd share $\delta_j^\ell = \frac{1}{2}(a_j^\ell - b_j^\ell)$, $\ell \in \{d, r\}$. *The* bunch family *and the* room family *are*

$$F^{\text{bunch}}(\mathbf{a}, \mathbf{b}) = \big((\mathcal{B}_j/R;\, a_j^d, b_j^d),\ (\mathcal{B}_j|B;\, a_j^r, b_j^r)\big)_j \quad \text{on } B,$$
$$F^{\text{room}}(\mathbf{a}, \mathbf{b}) = \big((\mathcal{B}_j|R;\, a_j^d, b_j^d),\ (\mathcal{B}_j/B;\, a_j^r, b_j^r)\big)_j \quad \text{on } R:$$

each level carries one table of the bunch and one of the room, at the same shares. The scale of B *pins the shares of the minors it multiplies,*

$$\sum_j \big(a_j^r + b_j^d\big) = \kappa_B, \qquad \textit{equivalently} \qquad \sum_j \delta_j^d = \tfrac{1}{2}\Big(\sum_j s_j - \kappa_B\Big),$$

and, for pattern shares with these level sums, equivalently $\sum_{e\in B}\langle\boldsymbol{\delta}\rangle_e = \mu_B$, *the sum over the bunch of the pins* $\langle\boldsymbol{\delta}\rangle_e = \mu_e$ *of a wall contour (Definition 18.12). On this pin both families lie on their balance walls at* $\mathbf{m}|_B$ *and* $\mathbf{m}|_R$, *and on every contour on which the stacked Gammas below have arguments of positive real part and both periods converge absolutely,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{N-1} \int_{\sum_j \delta_j^d = \frac{1}{2}(\sum_j s_j - \kappa_B)} \prod_j \Gamma_{\substack{4\\2}}\begin{bmatrix} a_j^d,\ b_j^d,\ a_j^r,\ b_j^r \\ s_j,\ t_j \end{bmatrix} \times P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

the fibre integral (Definition 4.1) in the pair coordinates $(\sigma_j^d, \delta_j^d)_j$, *with* $\sigma_j^r = \sigma_j - \sigma_j^d$ *and* $\delta_j^r = \delta_j - \delta_j^d$*: its dimension is* $2N - 1$, *and its* N *even shares* σ_j^d *are free. This display needs every* s_j *and* t_j *positive; in general, with* p *the number of active letters,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\sum_j (a_j^r + b_j^d) = \kappa_B} \prod_{s_j \neq 0} \Gamma_{\substack{2\\1}}\begin{bmatrix} a_j^d,\ a_j^r \\ s_j \end{bmatrix} \prod_{t_j \neq 0} \Gamma_{\substack{2\\1}}\begin{bmatrix} b_j^d,\ b_j^r \\ t_j \end{bmatrix} \times P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

the share integral (Definition 4.1) over the shares a_j^r, $s_j \neq 0$, *and* b_j^d, $t_j \neq 0$, *of the minors that the scale of* B *multiplies, pinned by their sum* κ_B, *of dimension* $p - 1$*; with every exponent positive it is the display above, and at* $p = 1$ *it is the value at the one admissible point. For one table with* $s, t > 0$ *the odd share is fixed,* $2\delta^d = s - \kappa_B = \delta\,(1 + \varepsilon(B)) - \mu_B$, *and the even share alone is free: on a line* $\operatorname{Re}\sigma^d = \sigma_*^d$ *of such a contour,*

$$P_F(\mathbf{m}; s, t) = \frac{1}{2\pi i} \int_{\operatorname{Re}\sigma^d = \sigma_*^d} \Gamma_{\substack{4\\2}}\begin{bmatrix} a^d,\ b^d,\ a^r,\ b^r \\ s,\ t \end{bmatrix} P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b})\, d\sigma^d.$$

Swapping B *and* R *swaps the two levels and leaves the formula unchanged.*

The scale of the bunch is opened letter by letter, one share on each level of each letter; the scale integrates to the pin, and what is left is the form of the bunch family on its simplex times the form of the room family on its own. The level shares are the pattern shares of the port (Theorem 15.15) summed over each level: at width one the children of the level d are all $\mathcal{B}_j|R$ and those of the level

r all $\mathcal{B}_j/B$, up to constant factors of their weights (Theorem 12.19), so the room family is the children of the port with equal children merged. At the level shares $(a_j^d, b_j^d, a_j^r, b_j^r) = (s_j, 0, 0, t_j)$, on the wall $\kappa_B = 0$, the two families are the faces of Theorem 16.25. Left unopened on the room, as in Theorem 19.8, the shape enters the pooled family only through the hinge values of B by its letters, and a rescaling of the hinge variable absorbs one of them: the shares that separate the shape from the room are one fewer than the hinge values of B by the active letters counted up to constant factors, none when every active letter reads B alike (Proposition 19.12), as for one table at a full lean (Theorem 19.10).

Corollary 19.14 (One Skeleton value pins the share)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real exponents and be on the balance wall with every proper wall positive, and let a bunch B have width one in every table, the weights factoring across B and R. Suppose that every active letter reads B alike, its hinge value a constant multiple of one function γ on Δ_B, and reads R alike with the same constant, its hinge value of R that constant times one function γ_R on Δ_R: the active letters have the one Skeleton value $\gamma_R(v)/\gamma(u)$ on the scale of B at every gauge $(u, v) \in \Delta_B \times \Delta_R$ (Corollary 19.5). Then $0 < \kappa_B < 2\sigma$, σ the total strength, and for all level shares $(\mathbf{a}, \mathbf{b})$ of Theorem 19.13, those of the inactive letters 0, with $a_j^d + a_j^r = s_j$, $b_j^d + b_j^r = t_j$ and $\sum_j (a_j^r + b_j^d) = \kappa_B$,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\kappa_B, & 2\sigma - \kappa_B\\ & 2\sigma\end{bmatrix} P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

the product of the two periods not depending on the level shares: the Beta corner of the hinge times the periods of the bunch family and the room family. One active letter has one Skeleton value: for one table at $(0, t)$, at the level shares $(a^d, b^d, a^r, b^r) = (0, \kappa_B, 0, t - \kappa_B)$, the two periods are those of the forms $u^{\mathbf{m}|_B - \mathbf{1}} \Psi_{\mathcal{B}/R}^{-\kappa_B} \Psi_{\mathcal{B}|B}^{-(t-\kappa_B)}$ on Δ_B and $v^{\mathbf{m}|_R - \mathbf{1}} \Psi_{\mathcal{B}|R}^{-\kappa_B} \Psi_{\mathcal{B}/B}^{-(t-\kappa_B)}$ on Δ_R, and at $(s, 0)$, at the level shares $(s - \kappa_B, 0, \kappa_B, 0)$, the same holds with the letters of K.

Lemma 19.15 (Slopes and the range of the ratios)**.** *In Theorem 19.13 with every s_j and t_j positive, the even share σ_j^d of table j enters the two families only through $\big(\varrho_{j,B}(u)/\varrho_{j,R}(v)\big)^{-\sigma_j^d}$, $(u, v) \in \Delta_B \times \Delta_R$, the ratio of the hinge values of the bunch over that of the room in table j (Definition 19.2). The wall at S of either family is affine in the shares, with the* slope *in σ_j^d the width of S in the table of the level d of table j less its width in the table of the level r of table j: with r_j^+ the rank function of the cell of table j on the side of S,*

$$\begin{aligned} &\big[r_j^+(S + g) - r_j^+(S)\big] + \big[r_j^+(B \setminus S + g) - r_j^+(B \setminus S)\big] - 1 \qquad (S \subseteq B),\\ &1 - \big[r_j^+(S + g) - r_j^+(S)\big] - \big[r_j^+(R \setminus S + g) - r_j^+(R \setminus S)\big] \qquad (S \subseteq R). \end{aligned}$$

$\sup_{\Delta_B} \varrho_{j,B} < \infty$ exactly when no slope of table j on B is positive and $\inf_{\Delta_B} \varrho_{j,B} > 0$ exactly when none on B is negative; $\inf_{\Delta_R} \varrho_{j,R} > 0$ exactly when no slope of table j on R is positive and $\sup_{\Delta_R} \varrho_{j,R} < \infty$ exactly when none on R is negative. So a wall of either family falls as σ_j^d moves toward the level d only if $\sup \varrho_{j,B} / \inf \varrho_{j,R} = \infty$, and as it moves toward the level r only if $\inf \varrho_{j,B} / \sup \varrho_{j,R} = 0$. These conditions, like the quotient $\varrho_{j,B}/\varrho_{j,R}$, do not depend on the factorization. For one table the index j is omitted.

A wall that falls as the even share moves toward the level being closed on is therefore the same thing as the two levels coming to overlap at that wall (Proposition 19.25): the quotient ϱ_B/ϱ_R of the even letters of the two levels, below one where the closing starts, is unbounded near the face where the wall sits.

19.6 Closing at a wall and closing at the gauge

A share between the levels of a letter can be closed in two places. *At the gauge* every letter is expanded about its value at a point of the scale; the terms are nonnegative and nothing is asked of the Skeleton values. That closing is Theorem 15.9, and it closes every letter on the scale at once. *At a wall* a letter is expanded about one of its two extreme coefficients, the other coefficients entering as integer powers; the terms then separate over the bunch and the room, and the expansion holds as far as the letters are separated along the scale. The closing at a wall holds for any number of letters and any width; across a bunch of width one it is a lattice of products of periods of the bunch family and the room family, and it stops where the values no longer stand in one order, which for one table is where the two levels overlap (Proposition 19.25).

Definition 19.16 (Letters on the scale)**.** Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ have real masses and real exponents of either sign, let B be a proper bunch with room R, and read $\alpha_B = \rho u$ with $u \in \Delta_B$. At the gauge (u, α_R) the *letters on the scale* of B are the $2N$ polynomials in ρ

$$\rho^{-d_j(B)} K_j(\rho u, \alpha_R) = \sum_{l=0}^{c_j(B)} \rho^l \, K_{j,B}^{(d_j(B)+l)}(u, \alpha_R), \qquad \rho^{-d_j^*(B)} \Psi_j(\rho u, \alpha_R) = \sum_{l=0}^{c_j(B)} \rho^l \, \Psi_{j,B}^{(d_j^*(B)+l)}(u, \alpha_R),$$

with the letter exponents s_j and t_j: their coefficients are the level and colevel letters of Definition 12.16 counted from their floors, and divided by their values at $\rho = 1$ they are the Skeleton polynomials $\mathsf{Q}_{j,B}^{K}(\rho; (u, \alpha_R))$ and $\mathsf{Q}_{j,B}^{\Psi}(\rho; (u, \alpha_R))$ (Definition 12.10). Listed as $L_i(\rho) = \sum_{l=0}^{c_i} \rho^l L_i^{(l)}$, $i = 1, \dots, 2N$, with the letter exponent w_i and the width $c_i = c_j(B)$ of their table, every coefficient is positive (Theorem 12.7), a letter of width zero is constant, and the scale integral of Theorem 15.3 at the gauge (u, α_R) is

$$\mathrm{Sc}_B(u, \alpha_R; \mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_0^\infty \rho^{\kappa_B - 1} \prod_i L_i(\rho)^{-w_i} \, d\rho,$$

convergent exactly when $0 < \kappa_B < \sum_i w_i c_i$, where $\sum_i w_i c_i = \kappa_B + \kappa_B^{\mathrm{top}}$.

Definition 19.17 (Letters read from an end of the scale; separated groups)**.** Fix $\lambda \in (0, 1)$. A letter $L(\rho) = \sum_{l=0}^{c} \rho^l L^{(l)}$ with positive coefficients is *read from the bottom up to* a when $\sum_{l \ge 1} a^l L^{(l)} \le \lambda \, L^{(0)}$, and *read from the top down to* a when $\sum_{l < c} a^l L^{(l)} \le \lambda \, a^c L^{(c)}$. A partition of the letters on the scale of B into *groups* $\mathcal{G}_1, \dots, \mathcal{G}_q$ is *separated* by the scales $0 < a_1 < \cdots < a_{q-1}$ when every letter of $\mathcal{G}_k$ is read from the bottom up to a_{k-1} (for $k \ge 2$) and from the top down to a_k (for $k \le q-1$); a letter whose exponent is a nonpositive integer is a polynomial factor of the integrand, may be put in any group, and nothing is asked of it. A group $\mathcal{G}$ has the product $f_{\mathcal{G}} = \prod_{i \in \mathcal{G}} L_i^{-w_i}$, the *scale degree* $w_{\mathcal{G}} = \sum_{i \in \mathcal{G}} w_i c_i$, and the *bottom* and *top coefficients* $\beta_{\mathcal{G},n}^{\downarrow}$ and $\beta_{\mathcal{G},n}^{\uparrow}$,

$$f_{\mathcal{G}}(\rho) = \sum_{n \ge 0} \beta_{\mathcal{G},n}^{\downarrow} \, \rho^n \quad (\rho \text{ small}), \qquad f_{\mathcal{G}}(\rho) = \rho^{-w_{\mathcal{G}}} \sum_{n \ge 0} \beta_{\mathcal{G},n}^{\uparrow} \, \rho^{-n} \quad (\rho \text{ large}).$$

Below the scale a a letter read from the bottom is its lowest coefficient up to the factor $1 + \lambda$, and above a a letter read from the top is its top term $\rho^c L^{(c)}$ up to the same factor. At width one, $L(\rho) = L^{(0)} + \rho L^{(1)}$ has the single Skeleton value $\xi = L^{(0)}/L^{(1)}$ (Corollary 19.5); it is read from the bottom up to a exactly when $a \le \lambda \xi$ and from the top down to a exactly when $a \ge \xi/\lambda$. So single letters of width one with the values $\xi_1, \dots, \xi_q$ are separated, by $a_k = \sqrt{\xi_k \xi_{k+1}}$, exactly when $\xi_k \le \lambda^2 \xi_{k+1}$ for every k: their values stand in one order with room between them.

Theorem 19.18 (The scale closes between separated groups)**.** *At the gauge (u, α_R) let the letters on the scale of B be separated into groups $\mathcal{G}_1, \dots, \mathcal{G}_q$, let $0 < \kappa < \sum_i w_i c_i$, and suppose $\kappa - \sum_{k' \le k} w_{\mathcal{G}_{k'}} \notin \mathbb{Z}$ for $1 \le k < q$. For $1 \le k \le q$ and $\mathbf{n} = (n_{k'})_{k' \ne k} \in \mathbb{N}^{q-1}$ put*

$$\nu_k(\mathbf{n}) = \kappa - \sum_{k'<k} \left(w_{\mathcal{G}_{k'}} + n_{k'} \right) + \sum_{k'>k} n_{k'}.$$

(i)

$$\int_0^\infty \rho^{\kappa-1} \prod_i L_i(\rho)^{-w_i} \, d\rho = \sum_{k=1}^{q} \sum_{\mathbf{n}} \prod_{k'<k} \beta^{\uparrow}_{\mathcal{G}_{k'}, n_{k'}} \prod_{k'>k} \beta^{\downarrow}_{\mathcal{G}_{k'}, n_{k'}} \ \mathrm{Reg} \int_0^\infty \rho^{\nu_k(\mathbf{n})-1} f_{\mathcal{G}_k}(\rho) \, d\rho;$$

at $\kappa = \kappa_B$ the left side is $\mathrm{Sc}_B(u, \alpha_R; \mathbf{m}; \mathbf{s}, \mathbf{t})$.

(ii) The series converges absolutely, and the sum of the absolute values of its terms is at most C times the left side, where C depends only on λ, the letter exponents, the widths, κ and the distances of the numbers $\kappa - \sum_{k' \le k} w_{\mathcal{G}_{k'}}$ from $\mathbb{Z}$.

(iii) $\beta^{\downarrow}_{\mathcal{G},n} = \prod_{i \in \mathcal{G}} \left(L_i^{(0)} \right)^{-w_i} [\rho^n] \prod_{i \in \mathcal{G}} \left(1 + \sum_{l \ge 1} \rho^l L_i^{(l)} / L_i^{(0)} \right)^{-w_i}$: the bottom coefficients of a group are the integrand of Theorem 16.25 for the residue at $\kappa_B = -n$, taken for the letters of the group alone; the top coefficients are the same with the coefficients $L_i^{(c_i)}$ in place of $L_i^{(0)}$, for the residue at $\kappa_B^{\mathrm{top}} = -n$.

Each group in turn keeps the scale. The groups that lie above it on the scale are read from the bottom, the groups below it from the top, and what is left to integrate is the keeping group alone, at an index shifted by the integers of the others, and as a finite part when that index has left the interval $(0, w_{\mathcal{G}})$ in which the group's own integral converges. The coefficients are the wall residues the family already has: for each letter read, a power of its extreme coefficient times a polynomial in its other coefficients. Every coefficient of K_j is a sum of the pattern terms $u^p K_{j,(p)}(\alpha_R)$ and every coefficient of Ψ_j of the terms $u^{B \setminus p} \Psi_{j,(p)}(\alpha_R)$, so the children enter with their pattern index, as in the record of a peel (Definition 14.2).

Corollary 19.19 (Two groups: the walls of one against the continuation of the other)**.** *Let the letters be separated into a lower group $\mathcal{L}$ and an upper group $\mathcal{U}$, with $0 < \kappa < \sum_i w_i c_i$ and $\kappa - w_{\mathcal{L}} \notin \mathbb{Z}$. Then*

$$\int_0^\infty \rho^{\kappa-1} f_{\mathcal{L}} f_{\mathcal{U}} \, d\rho = \sum_{n \ge 0} \beta^{\downarrow}_{\mathcal{U},n} \, \mathrm{Reg} \int_0^\infty \rho^{\kappa+n-1} f_{\mathcal{L}} \, d\rho + \sum_{n \ge 0} \beta^{\uparrow}_{\mathcal{L},n} \, \mathrm{Reg} \int_0^\infty \rho^{\kappa - w_{\mathcal{L}} - n - 1} f_{\mathcal{U}} \, d\rho.$$

For any group, $\mathrm{Reg} \int_0^\infty \rho^{\nu-1} f_{\mathcal{G}} \, d\rho$ is the value at ν of a meromorphic function, and at a pole of that function its constant term there (Theorem 1.7); the poles are at most simple and lie among the points $\nu = -n$, of residue $\beta^{\downarrow}_{\mathcal{G},n}$, and $\nu = w_{\mathcal{G}} + n$, of residue $-\beta^{\uparrow}_{\mathcal{G},n}$, $n \in \mathbb{N}$; the two families of points meet only when $w_{\mathcal{G}} \in -\mathbb{N}$, and at a common point the two residues add.

The scale integral of a product closes on the walls of each factor. The residues of the upper group at the wall of B are paired with the scale integral of the lower group continued past its top, and the residues of the lower group at the top wall with the scale integral of the upper group continued past its bottom. No bound on a moved contour is used: both expansions hold on the stretch of the scale between the groups, and the scale is divided there.

Theorem 19.20 (Joint closing across a bunch of width one)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ have real data, its masses on the balance wall with $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ convergent, and let a bunch B have width at most one in every table, the weights factoring across B and R. A table in which B has width zero is the direct sum of its restrictions to B and to R (Proposition 7.9); its two letters on the scale are constants, and it enters the bunch family and the room family of Theorem 19.13 as those restrictions with its own exponents. For the N' tables in which B has width one the letters on the scale (Definition 19.16) are, on $\Delta_B \times \Delta_R$,*

$$K_{\mathcal{B}_j/R}(u)K_{\mathcal{B}_j|R}(v) + \rho\, K_{\mathcal{B}_j|B}(u)K_{\mathcal{B}_j/B}(v), \qquad \Psi_{\mathcal{B}_j|B}(u)\Psi_{\mathcal{B}_j/B}(v) + \rho\, \Psi_{\mathcal{B}_j/R}(u)\Psi_{\mathcal{B}_j|R}(v),$$

of the letter exponents s_j and t_j, with the Skeleton values of Corollary 19.5; in each the coefficient multiplying ρ is its scale coefficient*, the level letter of the level r for K_j and the colevel letter of the level d for Ψ_j, and the other its* constant coefficient*. Suppose the $2N'$ letters can be listed as $\ell = 1, \dots, 2N'$, with letter exponents w_ℓ and Skeleton values ξ_ℓ, so that for some $\vartheta < 1$*

$$\xi_\ell \le \vartheta\, \xi_{\ell+1} \quad \text{on } \Delta_B \times \Delta_R \qquad (1 \le \ell < 2N'),$$

and that $\kappa_B - \sum_{\ell' \le \ell} w_{\ell'} \notin \mathbb{Z}$ for $1 \le \ell < 2N'$. For a letter i and $\mathbf{n} = (n_\ell)_{\ell \ne i} \in \mathbb{N}^{2N'-1}$ put $\nu_i(\mathbf{n}) = \kappa_B - \sum_{\ell<i}(w_\ell + n_\ell) + \sum_{\ell>i} n_\ell$, and let $(\mathbf{a}, \mathbf{b})_{i,\mathbf{n}}$ be the level shares in which

- *a letter $\ell < i$ has its constant coefficient at $-n_\ell$ and its scale coefficient at $w_\ell + n_\ell$,*
- *a letter $\ell > i$ has its scale coefficient at $-n_\ell$ and its constant coefficient at $w_\ell + n_\ell$,*
- *the letter i has its scale coefficient at $\nu_i(\mathbf{n})$ and its constant coefficient at $w_i - \nu_i(\mathbf{n})$.*

Then:

(i) at every $(\mathbf{a}, \mathbf{b})_{i,\mathbf{n}}$ the bunch family and the room family lie on their balance walls and their periods converge;

(ii) with both periods at $(\mathbf{a}, \mathbf{b}) = (\mathbf{a}, \mathbf{b})_{i,\mathbf{n}}$,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \sum_{i=1}^{2N'} \sum_{\mathbf{n} \in \mathbb{N}^{2N'-1}} \prod_{\ell \ne i} \frac{(-1)^{n_\ell}(w_\ell)_{n_\ell}}{n_\ell!}\, \Gamma^{\,2}_{\,1}\!\begin{bmatrix} \nu_i(\mathbf{n}), & w_i - \nu_i(\mathbf{n}) \\ & w_i \end{bmatrix} \times P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

where the stacked Gamma is 0 when $w_i \in -\mathbb{N}$; when no letter exponent lies in $-\mathbb{N}$ the coefficient is $\prod_{\ell \ne i} \frac{(-1)^{n_\ell}}{n_\ell!}\, \Gamma_{2N'}^{\,2N'+1}\!\begin{bmatrix} (w_\ell + n_\ell)_{\ell \ne i}, & \nu_i(\mathbf{n}), & w_i - \nu_i(\mathbf{n}) \\ & (w_\ell)_\ell & \end{bmatrix}$;

(iii) the series converges absolutely, the absolute values of all its terms add up to at most $C\, P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$, and those with $|\mathbf{n}| = M$ to at most $C'(1+M)^D \vartheta^M\, P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$, where C, C' and D depend only on ϑ, the exponents and κ_B.

The $2N'$ Skeleton values stand in one order on the whole shape domain, and each letter in turn keeps the scale. A letter whose value lies below the keeper's is read from the top of the scale, so its constant coefficient is opened; a letter whose value lies above is read from the bottom, so its scale coefficient is opened: each letter opens the one of its two terms that is the smaller at the keeper's place on the scale. An opened coefficient sits at the share $-n$, which puts the n-th power of that coefficient into the forms of the two families, a moment; the keeper carries the Beta corner $\Gamma^{\,2}_{\,1}\!\begin{bmatrix} \nu, & w - \nu \\ & w \end{bmatrix}$ of a linear letter (Proposition 15.1 with one coefficient), continued when ν

has left $(0, w)$. So closing across the bunch pairs moments of the bunch family with moments of the room family on $2N'$ lattices of dimension $2N'-1$, the free shares of Theorem 19.13 closed at once. Nothing is asked of the walls of the two families, (i) being a conclusion, and nothing uniform in $\mathbf{n}$ is needed beyond (iii), which bounds the whole lattice by the period itself.

The values are quotients of hinge values, the room's over the bunch's by each letter (Corollary 19.5). The value of K_j over that of Ψ_j is $\varrho_{j,B}(u)/\varrho_{j,R}(v)$, the even letters of its two levels, and their product is the corresponding quotient of the odd letters: the two values of a table sit on either side of the geometric mean set by its odd letters, at the distance set by its even letters. The N' quotients of even letters and the $N'-1$ quotients of the products of one table by those of another are what the N' even shares and the $N'-1$ free odd shares of Theorem 19.13 carry; for one table only ϱ_B/ϱ_R is left.

Corollary 19.21 (Closing the even share)**.** *For one table in Theorem 19.13 let $f(\sigma^d)$ be the integrand of its one-table display.*

(i) If $\sup_{\Delta_B}\varrho_B < \inf_{\Delta_R}\varrho_R$, then $P_F(\mathbf{m};s,t) = \sum \operatorname{Res} f$ over the poles of the Gamma pair $\Gamma(a^d)\Gamma(b^d)$ of the level d, at $\sigma^d = \delta^d - n$ and $\sigma^d = -\delta^d - n$, $n \in \mathbb{N}$, the n-th terms of the size $n^{s+t-2}\big(\sup\varrho_B/\inf\varrho_R\big)^n$; every period at these points converges, and when $2\delta^d \notin \mathbb{Z}$ the n-th terms are at most $C(1+n)^D\big(\sup\varrho_B/\inf\varrho_R\big)^n P_F(\mathbf{m};s,t)$, with C and D depending only on that ratio, the exponents and κ_B (Theorem 19.20(i),(iii) with one table, its letters listed as K, then Ψ). If $\inf_{\Delta_B}\varrho_B > \sup_{\Delta_R}\varrho_R$, then $P_F(\mathbf{m};s,t) = -\sum \operatorname{Res} f$ over the poles of the Gamma pair of the level r, at $\sigma^r = \pm\delta^r - n$. The share closes toward the level whose even letter is the smaller.

(ii) At $\sigma^d = \delta^d - n$ the level shares are $(a^d, b^d, a^r, b^r) = (2\delta^d - n,\ -n,\ s - 2\delta^d + n,\ t + n)$, and the residue is

$$\frac{(-1)^n}{n!}\Gamma^{\,3}_{\,2}\!\left[\begin{matrix}2\delta^d-n,\ s-2\delta^d+n,\ t+n\\ s,\ t\end{matrix}\right] P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}),$$

where $P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})} = \int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}} K_{\mathcal{B}/R}^{-2\delta^d} K_{\mathcal{B}|B}^{-(s-2\delta^d)} \Psi_{\mathcal{B}|B}^{-t}\, \varrho_B^n\, d\omega_B$ is the n-th moment of ϱ_B against a form that does not depend on n, and $P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}$ likewise the n-th moment of ϱ_R^{-1}. At $\sigma^d = -\delta^d - n$ the level shares are $(-n,\ -2\delta^d - n,\ s+n,\ t+2\delta^d+n)$, and the same holds with the form $\Psi_{\mathcal{B}/R}^{2\delta^d} K_{\mathcal{B}|B}^{-s} \Psi_{\mathcal{B}|B}^{-t-2\delta^d}$ on B and its counterpart on R. Closing pairs the moments of the ratios ϱ_B and ϱ_R of the bunch and of the room. The two lattices of a level meet exactly when $2\delta^\ell \in \mathbb{Z}$, at the centre exactly when $\mu_B \in \mathbb{Z}$. At a common point, where $\Gamma(a^d)$ sits at $-k$ and $\Gamma(b^d)$ at $-n$, the pole is double: with $h = \Gamma^{\,2}_{\,2}\!\left[\begin{matrix}s+k,\ t+n\\ s,\ t\end{matrix}\right] P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})} P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}$,

$$\operatorname{Res} f = \frac{(-1)^{n+k}}{n!\,k!}\, h\,\Big((\log\Gamma)'(n+1) - (\log\Gamma)'(t+n) + (\log\Gamma)'(k+1) - (\log\Gamma)'(s+k) + \partial_{\sigma^d}\log\big(P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})} P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}\big)\Big).$$

At integer exponents the two differences of $(\log\Gamma)'$ are heads, finite sums of reciprocals (Proposition 5.13), and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))$ for $j \in \mathbb{N}$ (Proposition 5.14).

Closed on a lattice, the even share writes the period as a row over finitely many data of the room whenever the room's periods along the lattice satisfy a recurrence of finite order: the room's initial periods, and at a meeting of the two lattices their derivatives, against sums along the lattice of the bunch's data and the room's fundamental solutions (Proposition 43.10).

Off resonance every pole is simple and the closed value is a sum over two lattices of products of moments of the two ratios ϱ_B and ϱ_R, with stacked Gamma coefficients: no heads. Each lattice's series has a pole where the two lattices of the level meet, from $\Gamma(2\delta^d - n)$ and $\Gamma(-2\delta^d - n)$, and their sum tends to the double-pole residues of (ii); the heads at the centre are this meeting. When the ranges of the even letters of the two levels overlap, the quotient $\varrho_B(u)/\varrho_R(v)$ taking values on both sides of 1, neither closing of (i) applies; the residues of a family at one of its walls are those of Theorem 16.25, periods of the family's faces.

Corollary 19.22 (Joint closing, nested)**.** *In Theorem 19.20 let B' be a bunch of the bunch family, of width at most one in each of its tables, with factoring weights, and let the letters of the bunch family on the scale of B' have their Skeleton values in one order with a ratio $\vartheta' < 1$. Then at every $(\mathbf{a},\mathbf{b})_{i,\mathbf{n}}$ at which the non-integrality condition of Theorem 19.20 holds for the bunch family at B', $P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})$ is given by Theorem 19.20 for the family $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ at B'; likewise for the room family at a bunch of R. Continuing down a tree of splits of width one, each node splitting its family at a bunch of width one into the bunch family and the room family, with the values ordered at every node, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is a nested lattice sum of stacked Gammas and of the periods of the families at the leaves, each leaf at exponents shifted by the integers of the lattices above it. The sum is taken from the leaves up, and at each node it converges absolutely at that node's own root rate.*

The tree of the nesting is a tree of splits of width one and not the Foreman tree of Definition 14.4: a node may be any bunch of width one at which the values are ordered. At a lattice point of the node above, a letter of the family below sits at an exponent $-n$ or $w+n$; the letters below are the same letters at shifted powers, and the next split applies to them because whether the values are in one order is read off the tables. Nothing uniform in the outer integers is used: at each lattice point the inner sum is an identity. The constant C of Theorem 19.20(iii) at an inner node depends on the exponents there and grows at most geometrically in the outer integers; that matters only for summing in another order. The multiple series converges absolutely when the outer ratio ϑ beats that growth, and otherwise the sum is the iterated one.

Proposition 19.23 (Values in one order without room)**.** *In Theorem 19.20 let $\xi_\ell < \xi_{\ell+1}$ on $\Delta_B \times \Delta_R$ for every ℓ, with no $\vartheta < 1$. For $0 < \zeta < 1$ let $P^{(\zeta)}$ be the integral obtained from $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ by multiplying the constant coefficient of the ℓ-th letter by $\zeta^{-\ell}$; it is the period of the family in which K_j and Ψ_j are read from two copies of table j whose level weights are scaled accordingly. Then every period of Theorem 19.20(i) converges, and with $A_{i,\mathbf{n}}$ the term of index $(i,\mathbf{n})$ of the series of Theorem 19.20(ii) and $x_i = \sum_{\ell>i} \ell w_\ell + i\sum_{\ell\le i} w_\ell - i\kappa_B$,*

$$P^{(\zeta)} = \sum_{i,\mathbf{n}} \zeta^{\,x_i + \sum_\ell |\ell - i|\, n_\ell} A_{i,\mathbf{n}} \quad \textit{absolutely,} \qquad P^{(\zeta)} \to P_F(\mathbf{m};\mathbf{s},\mathbf{t}) \ \ (\zeta \uparrow 1).$$

So the lattice sum of Theorem 19.20(ii) is summable to the period by scaling the level weights, and equals the period whenever it converges absolutely.

Proposition 19.24 (The same shares closed at the gauge)**.** *Let $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ have complex exponents and masses $\mathbf{m}$ and let a bunch B have width one in every table, the weights factoring across B and R, with $\operatorname{Re}\kappa_B > 0$ and $\operatorname{Re}\kappa_B^{\mathrm{top}} > 0$. Write the Skeleton polynomials at the gauge $(u,v) \in \Delta_B \times \Delta_R$ as*

$$\mathsf{Q}^K_{j,B}\big(x;(u,v)\big) = \pi_{j,d} + \pi_{j,r}\,x, \qquad \mathsf{Q}^\Psi_{j,B}\big(x;(u,v)\big) = \pi^*_{j,r} + \pi^*_{j,d}\,x,$$

the chances of the two levels under the basis law and the cobasis law at (u,v), each the sum over its level of the pattern probabilities of Definition 12.25. Then, with $M=|\mathbf{n}|+|\mathbf{n}'|$,

$$\mathrm{Sc}_B(u,v;\mathbf{m};\mathbf{s},\mathbf{t})=\prod_j K_j(u,v)^{-s_j}\Psi_j(u,v)^{-t_j}\sum_{\mathbf{n},\mathbf{n}'\in\mathbb{N}^N}\prod_j\frac{(s_j)_{n_j}(t_j)_{n'_j}}{n_j!\,n'_j!}$$
$$\times\Big(\prod_j \pi_{j,r}^{n_j}\pi_{j,d}^{*\,n'_j}\ \Gamma_1^2\begin{bmatrix}\kappa_B,\ M+1\\ \kappa_B+M+1\end{bmatrix}+\prod_j \pi_{j,d}^{n_j}\pi_{j,r}^{*\,n'_j}\ \Gamma_1^2\begin{bmatrix}\kappa_B^{\mathrm{top}},\ M+1\\ \kappa_B^{\mathrm{top}}+M+1\end{bmatrix}\Big),$$

a series converging absolutely, of nonnegative terms when the data are real with $s_j,t_j\ge 0$, whatever the order of the Skeleton values: it is Theorem 15.9 at width one, whose exceedance polynomials are the constants $\pi_{j,r}$, $\pi^*_{j,d}$ on the bunch and $\pi_{j,d}$, $\pi^*_{j,r}$ on the room. Without the prefactor the terms with M fixed add up in absolute value to at most $\frac{(\sum_j(|s_j|+|t_j|))_M}{M!}\big(\lambda_B^M/\operatorname{Re}\kappa_B+\lambda_R^M/\operatorname{Re}\kappa_B^{\mathrm{top}}\big)$, with $\lambda_B=\max_j(\pi_{j,r},\pi^*_{j,d})<1$ and $\lambda_R=\max_j(\pi_{j,d},\pi^*_{j,r})<1$.

At the gauge every letter is closed about its value at one point of the scale, both levels together. The lattice has one index for each letter, and what is raised to the powers are the chances of the levels at that point. Nothing is asked of the order of the values, and the root rate is the largest chance of a level. The price is that the closed letters $K_j(u,v)$ and $\Psi_j(u,v)$ stay in front, so the terms do not separate over the bunch and the room; integrated over the shapes on their ordering sectors the same closing is Corollary 15.11, with Theorem 15.14 past the walls.

Proposition 19.25 (Overlapping levels)**.** *In Theorem 19.13 with one table let $s,t>0$ and $2\delta^d=s-\kappa_B\notin\mathbb{Z}$, and let $\mathcal{M}(x)$ be the integral over $\Delta_B\times\Delta_R$ of*

$$u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}\big(K_{\mathcal{B}/R}(u)K_{\mathcal{B}|R}(v)\big)^{\kappa_B-s}\big(K_{\mathcal{B}|B}(u)K_{\mathcal{B}/B}(v)\big)^{-\kappa_B}\big(\Psi_{\mathcal{B}|B}(u)\Psi_{\mathcal{B}/B}(v)\big)^{-t}\Big(\frac{\varrho_B(u)}{\varrho_R(v)}\Big)^x.$$

The terms of the two lattices of the level d in Corollary 19.21(ii) are $A_n=a_n\mathcal{M}(n)$ and $A'_n=a'_n\mathcal{M}(n+2\delta^d)$, with

$$a_n=\Gamma_2^2\begin{bmatrix}2\delta^d,\ 1-2\delta^d\\ s,\ t\end{bmatrix}\Gamma_2^2\begin{bmatrix}t+n,\ \kappa_B+n\\ 1+n,\ 1-2\delta^d+n\end{bmatrix},$$
$$a'_n=-\Gamma_2^2\begin{bmatrix}2\delta^d,\ 1-2\delta^d\\ s,\ t\end{bmatrix}\Gamma_2^2\begin{bmatrix}s+n,\ t+2\delta^d+n\\ 1+n,\ 1+2\delta^d+n\end{bmatrix}.$$

Suppose these integrals converge for every n. If $\varrho_B(u)>\varrho_R(v)$ on a set of positive measure, then $|A_n+A'_n|\to\infty$; likewise for the level r if $\varrho_B(u)<\varrho_R(v)$ on a set of positive measure. So when the two levels overlap*, both sets having positive measure, the lattices of neither level sum to the period by convergent periods.*

The two lattices of a level have opposite signs and coefficients whose ratio tends to -1. When $\sup\varrho_B<\inf\varrho_R$ each of them converges by itself. When $\varrho_B(u)<\varrho_R(v)$ throughout with $\sup\varrho_B=\inf\varrho_R$ neither need converge; the moments at n and at $n+2\delta^d$ then agree to leading order, so the two lattices cancel to that order, and Proposition 19.23 sums them to the period by scaling the level weights. Past that, the two lattices part and the series diverges. By Lemma 19.15 the overlap reaches a wall when ϱ_B/ϱ_R is unbounded, and then the periods themselves stop converging along the lattice. An overlap is closed on the totals of the bunch and the room (Section 19.7), and, read in one scale, on the lattice of continued periods of a level whenever the reach of the bunch on its side lies below the decay of the other (Theorem 20.39).

Proposition 19.26 (Between groups at the wall, inside a group at the gauge)**.** *In Theorem 19.18 fix a group $\mathcal{G}$, an index ν with $\nu \notin -\mathbb{N}$ and $\nu - w_{\mathcal{G}} \notin \mathbb{N}$, and a point ρ_0 of the scale. For the letter i of K_j in $\mathcal{G}$ put $\mathsf{Q}_i = \mathsf{Q}^K_{j,B}(\cdot;(\rho_0 u, \alpha_R))$ and $\mathsf{Q}_i^\top = \mathsf{Q}^K_{j,R}(\cdot;(\rho_0 u, \alpha_R))$, and for that of Ψ_j the same with Ψ; then $\mathsf{Q}_i(x) = L_i(x\rho_0)/L_i(\rho_0)$ and $\mathsf{Q}_i^\top(x) = x^{c_i}\mathsf{Q}_i(1/x)$ (Proposition 12.11(i),(ii)), and*

$$\operatorname{Reg}\int_0^\infty \rho^{\nu-1} f_{\mathcal{G}}(\rho)\,d\rho = \rho_0^\nu f_{\mathcal{G}}(\rho_0)\Big(\operatorname{Reg}\int_0^1 x^{\nu-1}\prod_{i\in\mathcal{G}}\mathsf{Q}_i(x)^{-w_i}dx + \operatorname{Reg}\int_0^1 x^{w_{\mathcal{G}}-\nu-1}\prod_{i\in\mathcal{G}}\mathsf{Q}_i^\top(x)^{-w_i}dx\Big).$$

Each of the two finite parts is the series of Theorem 15.9(ii) for these polynomials, absolutely convergent, with its Beta corners continued as in Theorem 15.14. So for every separated partition every share of the scale is closed: between the groups on the lattices of Theorem 19.18, inside a group at the gauge. With one group this is Theorem 15.9; with single letters of width one in one order it is Theorem 19.20.

A group that cannot be split, because the values of its letters do not stand in one order, is kept as one Breta of one bunch (Proposition 15.1); it is closed about a point of the scale, where no order is needed, and the letters outside it are closed about its ends. The two closings do not compete: the wall closing does as much as the scale separates, and the gauge closing does the rest. When its index lies in $(0, w_{\mathcal{G}})$ and its letter exponents are nonnegative, a group read on the totals of the bunch and the room closes about constant references, its terms pairing the bunch with the room at width one (Section 19.7).

19.7 The scale on the totals

The scale of a bunch is the quotient of two totals, that of the bunch and that of the room, and the two totals lie on the simplex $\Delta_{\{B,R\}}$ of the two blocks (Theorem 3.12). Read there, a letter on the scale is homogeneous in the totals, with its level or colevel letters as coefficients. At width one it is linear, with its constant coefficient at one vertex of the simplex and its scale coefficient at the other, and its Skeleton value is the point of the scale where its two terms are equal. The free shares of Theorem 19.13 are the share side of this integral over the totals, the pin being the shape integral over the totals (Theorem 3.9), and the lattices of Section 19.6 close them. The integral over the totals can also be closed as it stands. On the simplex a linear letter is at most the larger of its two coefficients, whatever their ratio, so it closes about a constant reference with nonnegative terms, which pair moments of the bunch with moments of the room. No order of the values is asked, and the overlap that stops the lattices of a level (Proposition 19.25) is not seen.

Definition 19.27 (The totals and their letters)**.** The *totals* of a bunch B and its room R are the point $x = (x_B, x_R)$ of the shape simplex $\Delta_{\{B,R\}}$ of the two blocks. Let $\mathbf{m}^{\mathrm{tot}} = (m_B^{\mathrm{tot}}, m_R^{\mathrm{tot}})$ have positive entries, let ϕ be a nonnegative measure on $\Delta_B \times \Delta_R$, and let finitely many *letters on the totals*

$$\widehat{L}_i(x) = \sum_{l=0}^{c_i} x_B^{\,l}\, x_R^{\,c_i-l} L_i^{(l)}, \qquad i = 1, \dots, q,$$

with letter exponents $w_i \in \mathbb{R}$, have coefficients $L_i^{(l)} \geq 0$ that are measurable functions of the shapes $(u, v) \in \Delta_B \times \Delta_R$, each letter positive on the interior of $\Delta_{\{B,R\}}$ at ϕ-almost every shape. The

integral over the totals at the shapes (u,v) is the inner integral below, over $\Delta_{\{B,R\}}$, and its integral against ϕ is the Trailmoment on the source $\Delta_{\{B,R\}}\times\Delta_B\times\Delta_R$ (Definition 5.1) with the coefficient function $(x,u,v)\mapsto(\widehat{L}_i(x))_i$ and the input $x^{\mathbf{m}^{\mathrm{tot}}-\mathbf{1}}d\omega(x)\,\phi(du,dv)$, read over the whole source,

$$\begin{aligned}\big(\mathsf{T}_{\mathbf{w};I_q}[(\widehat{L}_i)_i]\;x^{\mathbf{m}^{\mathrm{tot}}-\mathbf{1}}d\omega(x)\,\phi(du,dv)\big)\big(\Delta_{\{B,R\}}\times\Delta_B\times\Delta_R\big)&\\ =\int_{\Delta_B\times\Delta_R}\int_{\Delta_{\{B,R\}}}x^{\mathbf{m}^{\mathrm{tot}}-\mathbf{1}}\prod_i\widehat{L}_i(x)^{-w_i}\,d\omega(x)\,\phi(du,dv),&\end{aligned}$$

wherever it converges; its kernel $\mathsf{Z}_{\mathbf{w};I_q}((\widehat{L}_i)_i)=\prod_i\widehat{L}_i^{-w_i}$ is the product of the one-scale kernels $\mathsf{Z}_{w_i;(1)}(\widehat{L}_i)$ (Definition 2.12, Theorem 5.21(i)). For one linear letter $x_RL^{(0)}+x_BL^{(1)}$ the integral over the totals is the Breta with the one bunch $\{B,R\}$, the shape index $\mathbf{m}^{\mathrm{tot}}$, the scale index w and the unit configuration (Definition 3.1):

$$\int_{\Delta_{\{B,R\}}}x^{\mathbf{m}^{\mathrm{tot}}-\mathbf{1}}\big(x_BL^{(1)}+x_RL^{(0)}\big)^{-w}\,d\omega(x)=\mathsf{B}^{\{\{B,R\}\}}_{\mathbf{m}^{\mathrm{tot}},\,w;\,I_{\{B,R\}}}\big(L^{(1)},L^{(0)}\big).$$

Theorem 19.28 (The scale on the totals)**.** *At the gauge (u,α_R) let $\kappa_B>0$ and $\kappa_B^{\mathrm{top}}>0$, and read every letter on the scale of B (Definition 19.16) on the totals,*

$$\widehat{L}_i(x)=x_R^{c_i}L_i(x_B/x_R)=\sum_{l=0}^{c_i}x_B^{\,l}\,x_R^{\,c_i-l}L_i^{(l)},$$

a letter of width zero being a constant on the totals.

(i) At every point $\rho_0>0$ of the scale, which may depend on the shapes,

$$\mathrm{Sc}_B(u,\alpha_R;\mathbf{m};\mathbf{s},\mathbf{t})=\rho_0^{\kappa_B}\int_{\Delta_{\{B,R\}}}x_B^{\kappa_B-1}x_R^{\kappa_B^{\mathrm{top}}-1}\prod_i\widehat{L}_i(\rho_0x_B,\;x_R)^{-w_i}\,d\omega(x),$$

both sides converging together. At $\rho_0=1$ and $\alpha_R=v\in\Delta_R$ the point $(\rho u,v)$ is projectively $(x_Bu,x_Rv)\in\Delta_{\mathcal{E}}$, whose totals on the blocks B and R are x_B and x_R (Theorem 3.12), with $\rho=x_B/x_R$. On the balance wall $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the Trailmoment of Definition 19.27 for the letters $\widehat{L}_i$ at $\mathbf{m}^{\mathrm{tot}}=(\kappa_B,\kappa_B^{\mathrm{top}})$ with $\phi=u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}d\omega_B(u)\,d\omega_R(v)$; both sides converge together.

(ii) Let B have width at most one in every table, the weights factoring across B and R; write the letters of width one $L_i(\rho)=L_i^{(0)}+\rho L_i^{(1)}$, with their constant and scale coefficients and their Skeleton values $\xi_i=L_i^{(0)}/L_i^{(1)}$ (Theorem 19.20), and fix one of them, i_0. At the point $\rho_0=\xi_{i_0}$ the two terms of i_0 are equal and its letter on the totals is the constant $L_{i_0}^{(0)}$. The products in the displays run over the letters of width one, and both right sides carry the further factor $\prod L_i^{-w_i}$ over the constant letters of the tables of width zero:

$$\begin{aligned}\mathrm{Sc}_B(u,v;\mathbf{m};\mathbf{s},\mathbf{t})&=\big(L_{i_0}^{(0)}\big)^{\kappa_B-w_{i_0}}\big(L_{i_0}^{(1)}\big)^{\kappa_B^{\mathrm{top}}-w_{i_0}}\\&\quad\times\int_{\Delta_{\{B,R\}}}x_B^{\kappa_B-1}x_R^{\kappa_B^{\mathrm{top}}-1}\prod_{i\neq i_0}\big(x_RL_i^{(0)}L_{i_0}^{(1)}+x_BL_{i_0}^{(0)}L_i^{(1)}\big)^{-w_i}\,d\omega(x)\\&=\xi_{i_0}^{\kappa_B}\prod_i\big(L_i^{(0)}\big)^{-w_i}\int_{\Delta_{\{B,R\}}}x_B^{\kappa_B-1}x_R^{\kappa_B^{\mathrm{top}}-1}\prod_{i\neq i_0}\Big(x_R+x_B\,\frac{\xi_{i_0}}{\xi_i}\Big)^{-w_i}\,d\omega(x).\end{aligned}$$

On the balance wall $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the Trailmoment of Definition 19.27 for these linear letters at $\mathbf{m}^{\rm tot}=(\kappa_B,\kappa_B^{\rm top})$, with ϕ of (i) multiplied by $(L_{i_0}^{(0)})^{\kappa_B-w_{i_0}}(L_{i_0}^{(1)})^{\kappa_B^{\rm top}-w_{i_0}}$ and by the further factor of the constant letters: the coefficients of each linear letter are the constant coefficient of one letter times the scale coefficient of the other, and their ratio is the ratio of the two Skeleton values.

(iii) In (ii) let every $w_i>0$, and open each linear letter by Corollary 4.4, with the share ν_i on $x_BL_{i_0}^{(0)}L_i^{(1)}$ and $w_i-\nu_i$ on $x_RL_i^{(0)}L_{i_0}^{(1)}$, on a contour with $0<\operatorname{Re}\nu_i<w_i$ and $\kappa_B-w_{i_0}<\sum_{i\neq i_0}\operatorname{Re}\nu_i<\kappa_B$. The integral over the totals is then the Beta corner $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\nu_{i_0},&w_{i_0}-\nu_{i_0}\\&w_{i_0}\end{smallmatrix}\right]$ of the letter i_0 at $\nu_{i_0}=\kappa_B-\sum_{i\neq i_0}\nu_i$, so that the shares on the scale coefficients add to κ_B identically, and, with the further factor of (ii) on the right,

$$\mathrm{Sc}_B(u,v;\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\sum_i\nu_i=\kappa_B}\prod_i\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\nu_i,&w_i-\nu_i\\&w_i\end{smallmatrix}\right]\big(L_i^{(0)}\big)^{-(w_i-\nu_i)}\big(L_i^{(1)}\big)^{-\nu_i},$$

the share side of the scale integral with one share per letter (Corollary 15.5). Integrated over the shapes it is Theorem 19.13: the pin of the shares is the shape integral over the totals.

The scale integral is the integral over the totals, and the shares of Theorem 19.13 are its share side: opening every letter but one leaves over the totals the Beta corner of the last, and the pin is what that integral asks of the shares. The point ρ_0 is free. At $\rho_0=1$ the totals are those of Theorem 3.12, the coordinates of a node in Theorem 20.15; at the value of a letter that letter leaves the integral, and for one table a single linear letter is left.

Corollary 19.29 (One table: the even share on the totals)**.** *For one table in Theorem 19.13 write, on $\Delta_B\times\Delta_R$,*

$$\mathbf{E}^d=\mathbf{E}_{\mathcal{B}/R}(u)\,\mathbf{E}_{\mathcal{B}|R}(v),\qquad \mathbf{E}^r=\mathbf{E}_{\mathcal{B}|B}(u)\,\mathbf{E}_{\mathcal{B}/B}(v)$$

for the even letters of the two levels, so that $\mathbf{E}^d/\mathbf{E}^r=\varrho_B(u)/\varrho_R(v)$, and let $2\delta^d=s-\kappa_B$ and $2\delta^r=\kappa_B-t$; here $\kappa_B^{\rm top}=s+t-\kappa_B$. Then

$$\begin{aligned}\mathrm{Sc}_B(u,v;\mathbf{m};s,t)&=\big(K_{\mathcal{B}/R}(u)K_{\mathcal{B}|R}(v)\big)^{-2\delta^d}\big(K_{\mathcal{B}|B}(u)K_{\mathcal{B}/B}(v)\big)^{-2\delta^r}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\rm top}),\,t;\,I_{\{B,R\}}}\big(\mathbf{E}^d,\mathbf{E}^r\big)\\&=\big(\Psi_{\mathcal{B}/R}(u)\Psi_{\mathcal{B}|R}(v)\big)^{2\delta^d}\big(\Psi_{\mathcal{B}|B}(u)\Psi_{\mathcal{B}/B}(v)\big)^{2\delta^r}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\rm top}),\,s;\,I_{\{B,R\}}}\big(\mathbf{E}^r,\mathbf{E}^d\big),\end{aligned}$$

and $\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\rm top}),t;I_{\{B,R\}}}(\mathbf{E}^d,\mathbf{E}^r)=(\mathbf{E}^r)^{-t}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\rm top}),t;I_{\{B,R\}}}\big(\varrho_B(u)/\varrho_R(v),1\big)$. So $P_F(\mathbf{m};s,t)$ is the Trailmoment of Definition 19.27 for the single linear letter $x_R\mathbf{E}^r+x_B\mathbf{E}^d$ at the exponent t and at $\mathbf{m}^{\rm tot}=(\kappa_B,\kappa_B^{\rm top})$, with $\phi=\big(K_{\mathcal{B}/R}K_{\mathcal{B}|R}\big)^{-2\delta^d}\big(K_{\mathcal{B}|B}K_{\mathcal{B}/B}\big)^{-2\delta^r}u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}d\omega_B\,d\omega_R$, and the shapes enter the Breta only through the ratio $\varrho_B(u)/\varrho_R(v)$ of its coefficients. Opened by Corollary 4.4, the linear letter has the share $b^d=\sigma^d-\delta^d$ on $x_B\mathbf{E}^d$, and the integrand is that of Theorem 19.13: the free even share is the share of this letter, and the integral over the totals is the even share integrated.

Theorem 19.30 (Closing at the peaks, one reference per coefficient)**.** *In the Trailmoment of Definition 19.27 let every $w_i>0$, and choose constants $\bar L_{i,l}\geq L_i^{(l)}$, one for each coefficient, at ϕ-almost every shape; the* reference letters *are $\bar L_i(x)=\sum_{l=0}^{c_i}x_B^lx_R^{c_i-l}\bar L_{i,l}$. For an array $\mathbf{k}=(k_{i,l})$ with the rows $\mathbf{k}_i=(k_{i,l})_{0\leq l\leq c_i}$ put*

$$\boldsymbol{\beta}(\mathbf{k})=\big(\beta_B(\mathbf{k}),\beta_R(\mathbf{k})\big)=\Big(m_B^{\rm tot}+\sum_{i,l}l\,k_{i,l},\ m_R^{\rm tot}+\sum_{i,l}(c_i-l)\,k_{i,l}\Big),\qquad \mathbf{w}(\mathbf{k})=\big(w_i+|\mathbf{k}_i|\big)_i.$$

Introduce one singleton scale $\mathfrak{v}_i$ per letter, $\mathfrak{V} = \{\mathfrak{v}_1, \dots, \mathfrak{v}_q\}$, take the partition $\mathcal{P}_{\mathrm{tot}} = \{\{B, R\}\} \cup \{\{\mathfrak{v}_i\} : 1 \le i \le q\}$, and let $(\pi^{\mathrm{tot}}, \ell^{\mathrm{tot}})$ be the term arrays of the letter presentation (Proposition 3.6) with the kernel column of shape exponent 0, level $(1, 0_{\mathfrak{V}})$ and coefficient 1, and for every (i, l) with $\bar{L}_{i,l} > 0$ a column of shape exponent $\big((l, c_i - l), \mathbf{e}_i\big)$, level $(0, \mathbf{e}_i)$ and coefficient $\bar{L}_{i,l}$; $1 \oplus \bar{\mathbf{L}}$ lists the coefficients in this order. Then, with the arrays $\mathbf{k}$ supported on these columns,

$$\int_{\Delta_B \times \Delta_R} \int_{\Delta_{\{B,R\}}} x^{\mathbf{m}^{\mathrm{tot}} - \mathbf{1}} \prod_i \widehat{L}_i(x)^{-w_i} \, d\omega(x) \, \phi(du, dv)$$
$$= \sum_{\mathbf{k}} \prod_i \frac{(w_i)_{|\mathbf{k}_i|}}{\mathbf{k}_i!} \; \mathsf{B}^{\mathcal{P}_{\mathrm{tot}}}_{(\boldsymbol{\beta}(\mathbf{k}), \mathbf{w}(\mathbf{k})), \, (1, \mathbf{w}(\mathbf{k})); \, (\pi^{\mathrm{tot}}, \ell^{\mathrm{tot}})}\big(1 \oplus \bar{\mathbf{L}}\big) \int \prod_{i,l} \big(\bar{L}_{i,l} - L_i^{(l)}\big)^{k_{i,l}} \, d\phi,$$

a series of nonnegative terms, both sides finite or infinite together. The Breta coefficient is a number: the kernel integrates its scale to 1, and

$$\mathsf{B}^{\mathcal{P}_{\mathrm{tot}}}_{(\boldsymbol{\beta}(\mathbf{k}), \mathbf{w}(\mathbf{k})), \, (1, \mathbf{w}(\mathbf{k})); \, (\pi^{\mathrm{tot}}, \ell^{\mathrm{tot}})}\big(1 \oplus \bar{\mathbf{L}}\big) = \int_{\Delta_{\{B,R\}}} x^{\boldsymbol{\beta}(\mathbf{k}) - \mathbf{1}} \prod_i \bar{L}_i(x)^{-w_i - |\mathbf{k}_i|} \, d\omega(x).$$

Each moment is a read of ϕ against the one-scale kernels $\mathsf{Z}_{-k_{i,l};(1)}\big(\bar{L}_{i,l} - L_i^{(l)}\big) = \big(\bar{L}_{i,l} - L_i^{(l)}\big)^{k_{i,l}}$.

Corollary 19.31 (Closing at the peak on the totals)**.** *In Theorem 19.30 take the references $\bar{L}_{i,l} = \binom{c_i}{l} \bar{L}_i$ with constants $\bar{L}_i > 0$ such that the deviations $h_{i,l} = \binom{c_i}{l} \bar{L}_i - L_i^{(l)}$ are nonnegative. The reference letter is then the constant $\bar{L}_i (x_B + x_R)^{c_i} = \bar{L}_i$ on the totals, the Breta coefficient is $\prod_i \bar{L}_i^{-w_i - |\mathbf{k}_i|}$ times the Beta corner $\Gamma^{\,2}_{\,1}\!\begin{bmatrix} \beta_B(\mathbf{k}), & \beta_R(\mathbf{k}) \\ \beta_B(\mathbf{k}) + \beta_R(\mathbf{k}) & \end{bmatrix}$, and that Trailmoment is*

$$\sum_{\mathbf{k}} \prod_i \frac{(w_i)_{|\mathbf{k}_i|}}{\mathbf{k}_i!} \, \bar{L}_i^{-w_i - |\mathbf{k}_i|} \; \Gamma^{\,2}_{\,1}\!\begin{bmatrix} \beta_B(\mathbf{k}), & \beta_R(\mathbf{k}) \\ \beta_B(\mathbf{k}) + \beta_R(\mathbf{k}) & \end{bmatrix} \int \prod_{i,l} h_{i,l}^{k_{i,l}} \, d\phi.$$

For linear letters $\widehat{L}_i(x) = x_R L_i^{(0)} + x_B L_i^{(1)}$ with $\bar{L}_i \ge L_i^{(0)}, L_i^{(1)}$, with n_i counting the constant coefficient and n_i' the scale coefficient of the letter i, it is

$$\sum_{\mathbf{n}, \mathbf{n}'} \prod_i \frac{(w_i)_{n_i + n_i'}}{n_i! \, n_i'!} \, \bar{L}_i^{-w_i - n_i - n_i'} \; \Gamma^{\,2}_{\,1}\!\begin{bmatrix} m_B^{\mathrm{tot}} + |\mathbf{n}'|, & m_R^{\mathrm{tot}} + |\mathbf{n}| \\ |\mathbf{m}^{\mathrm{tot}}| + |\mathbf{n}| + |\mathbf{n}'| & \end{bmatrix} \int \prod_i \big(\bar{L}_i - L_i^{(0)}\big)^{n_i} \big(\bar{L}_i - L_i^{(1)}\big)^{n_i'} \, d\phi.$$

If a coefficient is a product $f'(u) f''(v)$ and its reference a product $\bar{f}' \bar{f}''$ of constants $\bar{f}' \ge f'$ and $\bar{f}'' \ge f''$, then $\bar{f}' \bar{f}'' - f' f'' = \bar{f}'(\bar{f}'' - f'') + f''(\bar{f}' - f')$, both terms nonnegative and each a function on the bunch times a function on the room; so when every coefficient, every reference and ϕ are products, every moment is a finite nonnegative combination of products of moments over Δ_B and over Δ_R.

No order of the values is used, only bounds on the coefficients, and the terms are nonnegative. Applied to every letter on the whole simplex with the references $M_i = \max_T \theta_{j,T} / g_i!$, Theorem 15.13 is the draw series (Proposition 20.12); applied on the totals it closes the scale of the bunch alone, and the shapes of the bunch and the room enter only through the moments of the deviations of the coefficients from their references. Across a bunch of width one every coefficient is a letter of a minor on the bunch times a letter of a minor on the room (Theorem 19.20), so with references that are products the moments pair the bunch with the room, as the lattices of Theorem 19.20 do, with nonnegative terms. If every $\widehat{L}_i$ is at least $(1 - \lambda) \bar{L}_i$ on the totals at every shape, with $\lambda < 1$,

the terms of total order $\sum_i |\mathbf{k}_i| = k$ add up to at most $\frac{(|\mathbf{w}|)_k}{k!}\lambda^k$ times the term of order 0; when a coefficient vanishes on a face of a shape simplex the terms need not decay geometrically.

For one table read at $\rho_0 = 1$ the two letters on the totals are linear, and so are their reference letters, one constant for each coefficient. Rescaling the scale by the ratio of the references of the letter K carries its reference letter to a multiple of $x_R + x_B$, and that of Ψ to a multiple of the linear letter whose scale coefficient over its constant coefficient is the reference of the even letter of the level d over that of the level r; the rescaling multiplies the Breta coefficient by a power of the ratio of the references of K. Two letters on the line of the totals have two zeros besides its two endpoints, and four points of a line have one modulus, the ratio of the references of the even letters of the two levels: the Breta coefficients depend on the references only through their scales and this modulus, defined up to its inverse by the naming of the two levels. What one reference for each coefficient adds is that every deviation can vanish at its own peak, at the price of the one modulus in the coefficients. When the modulus is 1 both reference letters are multiples of x_R+x_B, which is 1 on the simplex, and Corollary 19.31 applies. With N tables there are $2N$ letters on the totals and $2N-1$ such moduli.

Corollary 19.32 (The ladder on the totals)**.** *In Theorem 19.28(ii) let every $w_i > 0$, put $w' = \sum_{i\neq i_0} w_i$, and let $\zeta > 0$.*

(i) At shapes where every ratio ξ_{i_0}/ξ_i lies in $(0, 2\zeta)$,

$$\int_{\Delta_{\{B,R\}}} x_B^{\kappa_B-1} x_R^{\kappa_B^{\mathrm{top}}-1} \prod_{i\neq i_0} \Big(x_R + x_B\,\frac{\xi_{i_0}}{\xi_i}\Big)^{-w_i} d\omega(x)$$
$$= \sum_{\mathbf{n}} \prod_{i\neq i_0} \frac{(w_i)_{n_i}}{n_i!} \Big(\zeta - \frac{\xi_{i_0}}{\xi_i}\Big)^{n_i}\ \mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B+|\mathbf{n}|,\kappa_B^{\mathrm{top}}),\, w'+|\mathbf{n}|;\, I_{\{B,R\}}}(\zeta, 1),$$

absolutely: the linear letters are moved to the common ratio ζ, where they merge into one, and the coefficients do not depend on the shapes. At $\zeta = 1$ they are the Beta corners $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\kappa_B+|\mathbf{n}|,\ \kappa_B^{\mathrm{top}}\\ \kappa_B+\kappa_B^{\mathrm{top}}+|\mathbf{n}|\end{smallmatrix}\right]$. For one linear letter this is the ladder of the Breta in its first coefficient (Theorem 3.5(ii)), $\partial_\zeta \mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),w;I_{\{B,R\}}}(\zeta,1) = -w\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B+1,\kappa_B^{\mathrm{top}}),w+1;I_{\{B,R\}}}(\zeta,1)$, summed as a Taylor series.

(ii) Let $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converge and let every ratio ξ_{i_0}/ξ_i lie in $[\zeta_-,\zeta_+] \subset (0,2\zeta)$ on $\Delta_B\times\Delta_R$. Then (i) integrates term by term against ϕ of Theorem 19.28(i) multiplied by $\xi_{i_0}^{\kappa_B}\prod_i (L_i^{(0)})^{-w_i}$ and by the further factor of Theorem 19.28(ii), and the terms of $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ with $|\mathbf{n}| = n$ add up in absolute value to at most $C\frac{(w')_n}{n!}\vartheta^n P_F(\mathbf{m};\mathbf{s},\mathbf{t})$, where $\vartheta = \max(1-\zeta_-/\zeta,\ \zeta_+/\zeta - 1) < 1$ and $C = \mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),w';I_{\{B,R\}}}(\zeta,1)\big/\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),w';I_{\{B,R\}}}(\zeta_+,1)$. If $\zeta \geq \sup \xi_{i_0}/\xi_i$ for every i the terms are nonnegative and no lower bound on the ratios is needed; both sides are then finite or infinite together.

(iii) For one table with i_0 the letter of K, under the hypotheses of (ii) the only other ratio is $\varrho_B(u)/\varrho_R(v)$, and

$$P_F(\mathbf{m};s,t) = \sum_{n\geq 0} \frac{(t)_n}{n!}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B+n,\kappa_B^{\mathrm{top}}),\,t+n;\,I_{\{B,R\}}}(\zeta,1) \int_{\Delta_B\times\Delta_R} \big(\zeta\mathbf{E}^r - \mathbf{E}^d\big)^n\,(\mathbf{E}^r)^{-t-n}\,\phi(du,dv),$$

with $\mathbf{E}^d$, $\mathbf{E}^r$ and ϕ of Corollary 19.29. With i_0 the letter of Ψ the ratio is $\varrho_R(v)/\varrho_B(u)$, and the same holds with s for t, the moments $\int(\zeta\mathbf{E}^d - \mathbf{E}^r)^n(\mathbf{E}^d)^{-s-n}$ and the measure

$\phi = \big(\Psi_{\mathcal{B}/R}\Psi_{\mathcal{B}|R}\big)^{2\delta^d}\big(\Psi_{\mathcal{B}|B}\Psi_{\mathcal{B}/B}\big)^{2\delta^r} u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}d\omega_B\,d\omega_R$. *So one table closes on the totals with nonnegative terms about any* $\zeta \geq \sup\varrho_B/\inf\varrho_R$ *when no slope is positive, and with* i_0 *the letter of* Ψ *about any point at least* $\sup\varrho_R/\inf\varrho_B$ *when no slope is negative (Lemma 19.15). With* $\zeta = \zeta'\zeta''$, $\zeta' \geq \sup_{\Delta_B}\varrho_B$ *and* $\zeta'' \geq 1/\inf_{\Delta_R}\varrho_R$,

$$\zeta\mathbf{E}^r - \mathbf{E}^d = \zeta'\mathbf{E}_{\mathcal{B}|B}(u)\big(\zeta''\mathbf{E}_{\mathcal{B}/B}(v) - \mathbf{E}_{\mathcal{B}|R}(v)\big) + \mathbf{E}_{\mathcal{B}|R}(v)\big(\zeta'\mathbf{E}_{\mathcal{B}|B}(u) - \mathbf{E}_{\mathcal{B}/R}(u)\big),$$

both terms nonnegative and each a function on the bunch times a function on the room.

(iv) Since $x_R + x_B\xi_{i_0}/\xi_i = (\xi_{i_0}/\xi_i)\big(x_B + x_R\xi_i/\xi_{i_0}\big)$, *the integral of (i) is* $\prod_{i\neq i_0}(\xi_i/\xi_{i_0})^{w_i}$ *times the same integral with* κ_B *and* κ_B^{top} *swapped and every ratio replaced by its inverse. So (i) and (ii) hold in the inverse ratios, the factor* $\prod_{i\neq i_0}(\xi_i/\xi_{i_0})^{w_i}$ *joining* ϕ, *with nonnegative terms about any common point at least* $\sup\xi_i/\xi_{i_0}$: *the ladder closes whenever the ratios are bounded on one side.*

The three closings of the even share read the same linear letters. At a wall the even share closes toward the level whose even letter is the smaller, which needs ϱ_B/ϱ_R on one side of 1 with room (Corollary 19.21(i)); the ladder moves the linear letters to a common ratio, which needs the quotient bounded on one side, and its coefficients are constants of the scale; the peak closing moves each linear letter to a constant reference, which needs the coefficients bounded and asks nothing of their ratio. In all three the terms pair moments of the bunch with moments of the room. With $\varrho_B(u) < \varrho_R(v)$ throughout and $\sup\varrho_B = \inf\varrho_R$, the ladder about $\zeta = 1$ is a series of nonnegative terms whose coefficients are Beta corners, where Proposition 19.23 sums the lattices by scaling the level weights. Where the quotient takes every positive value, of the three only the peak closing applies; the gauge closing applies as well, its terms not separating over the bunch and the room (Proposition 19.24).

19.8 Width zero

Proposition 19.33 (A loop or a coloop on the simplex)**.** *Let a mark* e *be in every table a loop* ($\eta_j = -1$) *or a coloop* ($\eta_j = +1$), *let* F' *be the family of Corollary 15.16 on* $\mathcal{E} - e$, *and put* $\nu_e = \mu_e - \sum_j \eta_j\delta_j$, *so that* $\kappa' = \kappa_{\mathcal{E}} - \nu_e$ *is the imbalance of* F' *at the masses* $\mathbf{m}|_{\mathcal{E}-e}$. *Where* $\nu_e > 0$ *and* $\kappa' > 0$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\nu_e,\ \kappa'\\ \kappa_{\mathcal{E}}\end{bmatrix}\ P_{F'}(\mathbf{m}|_{\mathcal{E}-e};\mathbf{s},\mathbf{t}),$$

the mark rescaling every other mark by the common factor $1 - u_e$; *it is the compound formula of Theorem 16.17 at* $S = \{e\}$, *with the walls* $\kappa_{\{e\}} = \nu_e$ *and* $\kappa_{\mathcal{E}\setminus e} = \kappa'$. *Marks* $e_1,\dots,e_k$ *of this kind combine first among themselves and then with the rest,* $\Gamma_{\!\!\begin{smallmatrix}k+1\\1\end{smallmatrix}}\!\begin{bmatrix}(\nu_{e_i})_i,\ \kappa'\\ \kappa_{\mathcal{E}}\end{bmatrix} = \Gamma_{\!\!\begin{smallmatrix}k\\1\end{smallmatrix}}\!\begin{bmatrix}(\nu_{e_i})_i\\ a\end{bmatrix}\ \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}a,\ \kappa'\\ \kappa_{\mathcal{E}}\end{bmatrix}$ *with* $a = \sum_i \nu_{e_i}$ *and* κ' *the imbalance once all of them are removed (Theorem 3.12). On the balance wall* $\kappa' = -\nu_e$ *and the period of the compound diverges (Proposition 8.16): the Foreman tree splits molecules first (Definition 14.4).*

The bunch $\{e\}$ has width zero: the room does not see it at all, and the period integrates it out as the Beta corner $\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\nu_e,\ \kappa'\\ \kappa_{\mathcal{E}}\end{bmatrix}$, whose Gamma $\Gamma(\nu_e)$ is the Gamma $\Gamma(\mu_e - \langle\boldsymbol{\delta}\rangle_e)$ of the port at e (Theorem 15.15).

19.9 Pooling is a move

Theorem 19.34 (Pooling is a move)**.** *Let a bunch B have width one in every table of a family of tables with positive rational weights, the weights of every table factoring across B and R with rational factors, at rational exponents and masses on the balance wall.*

(i) Theorem 19.8 is one change of variables, $(\rho, u, \alpha_R) \mapsto (\rho u, \alpha_R)$, the scale and the shape of B: it is birational over $\mathbb{Q}$ with the inverse $\alpha \mapsto (|\alpha_B|, \alpha_B/|\alpha_B|, \alpha_R)$, bijective on the open domains, and it carries the face of $S \subseteq \mathcal{E}$ to $\{\rho = 0\} \times \{\alpha_{S\cap R} = 0\}$ if $S \supseteq B$ and to $\{u_{S\cap B} = 0\} \times \{\alpha_{S\cap R} = 0\}$ otherwise, with the walls matched by Theorem 19.9: by (i) there when $S \supseteq B$ or $S \subseteq R$, by (ii) there when $S \subseteq B$, and by the gluing of the counts otherwise.

(ii) When the factorizations, with rational factors, give every active letter the one hinge value γ of Theorem 19.10, that theorem is a composite of moves: the change of variables (i); the change $\rho \mapsto \rho\,\gamma(u)$, birational over $\mathbb{Q}$ and positive on the open domain; the product of the datum of the shape with that of the cell family of the room; and the integral of the hinge, the Beta corner of a linear letter (Proposition 15.1 with one coefficient), itself a change of variables and a product with the datum of the list on two marks whose bases are the two singletons, at the exponents $(0, 2\sigma)$. Every integral in the chain converges exactly when $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ does (Theorem 19.10).

So the identities of Theorems 19.8 and 19.10 hold between period data, not only between their values: when every active letter has one hinge value the period datum of the family times that of the list on two marks is the product of the data of its two cell families.

In the interior Theorem 19.8 writes the period as an integral over the shape simplex, whose faces carry the walls of Theorem 19.9(ii), of the periods of the pooled families $F_{B\to g}(u)$, whose walls are those of Theorem 19.9(i).

19.10 Cleaning a molecule

Definition 19.35 (Cleaning)**.** A *cleaning* of a list is a sequence of poolings of parallel and series bunches of width one, each in the list left by the previous ones, until at most two marks or no such class remain.

Proposition 19.36 (Poolings of disjoint classes commute)**.** *Let X and Y be disjoint parallel or series bunches of width one of a list. Pooling X keeps Y a class of the same kind and width, with the same restriction to Y and the same contraction of its room, and the two orders of pooling give the same list on $(\mathcal{E} \setminus (X \cup Y)) \sqcup \{g_X, g_Y\}$, the same cell kernels and the same hinge values. A cleaning ends after at most $|\mathcal{E}| - 2$ poolings.*

Proposition 19.37 (Cleaning commutes with maps)**.** *A bijection that carries one table to another (Definition 11.13) maps the parallel and series bunches of the first onto those of the second, with the same widths and, the weights factoring across a bunch and its room and the factorization carried along, the same hinge values. An exchange maps parallel bunches onto series bunches and swaps the hinge values by K and by Ψ, the weight θ'_p of a pattern p of a bunch B becoming that of the pattern $g(B \setminus p)$ of its image. Between lists it maps the cleaning of the first list at the exponents (s, t) onto the cleaning of the second at (t, s).*

Theorem 19.38 (Cleaning has one result)**.** *Let a list be one molecule. Every cleaning of it ends at the same list up to the names of its marks. If that list has more than two marks, every cleaning*

ends with the same marks, each the same set of marks of the original list; a list that ends with two marks may divide its marks between them in more than one way. Pooling the parallel bunches first, then the series bunches, and repeating, is one cleaning, and every other order ends at the same place.

Pooling keeps a molecule a molecule, and two poolings at one list can be continued to one list; so the end of a cleaning does not depend on the order of the poolings. The cleaned list has no parallel or series bunch of width one left, each of its marks gathers a set of marks of the molecule, and the period of the molecule is the integral, over the shape simplices of the pooled bunches against their cell kernels, of the periods of the pooled families of the cleaned list.

Chapter 20

Words of a tree

A period is read back from what the Foreman procedure leaves: the bunches peeled at the nodes of a tree of bunches and the leaves at its bottom. The period of a family is the word of such a tree, an iterated fibre integral of stacked Gammas and of the periods of the leaves; opened at the atoms along a chain, or wherever no table is opened on two childcules, the word is one Gramma on the marks and the letters. The word closes in three ways: by draws, with nonnegative terms at nonnegative exponents; at the centre, through the half-sum and the difference of the two letters of a table; and node by node in one scale, where each side of a node of width one is an interval from the decay of its transform to its reach. Read on one tree, the period shows what each node adds, what the leaves contain, and how the integrations of the simplex are shared out.

20.1 The two letters of a table at once

Proposition 20.1 (The pair ladder is the expansion about the midpoint)**.** *For $a, b > 0$, $s, t > 0$ and $x = \frac{a-b}{a+b} \in (-1, 1)$,*

$$a^{-s} b^{-t} = \Big(\frac{a+b}{2}\Big)^{-s-t} (1+x)^{-s} (1-x)^{-t},$$

and the coefficients of the power series $(1+x)^{-s}(1-x)^{-t}$ are

$$[x^k](1+x)^{-s}(1-x)^{-t} = \sum_{i+i'=k} (-1)^i \frac{(s)_i\,(t)_{i'}}{i!\,i'!} = \frac{(s+t)_k}{k!} \Gamma_{\begin{smallmatrix}1\\2\end{smallmatrix}}\begin{bmatrix} s+t \\ s,\ t \end{bmatrix} \int_0^1 w^{s-1}(1-w)^{t-1}(1-2w)^k\, dw.$$

So $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} s,\ t \\ s+t+k \end{bmatrix} [x^k](1+x)^{-s}(1-x)^{-t} = \frac{1}{k!}\int_0^1 w^{s-1}(1-w)^{t-1}(1-2w)^k\, dw$; swapping s and t multiplies the k-th coefficient by $(-1)^k$; and at $s = t$ only even k occur, $[x^{2l}](1-x^2)^{-s} = (s)_l/l!$.

The two letters of a table enter a period as $\Lambda(z)^{-s}\Lambda(-z)^{-t}$. The pair ladder expands such a pair about its midpoint; under one power, the shape of a two-element bunch in the tower decides how much of each letter is taken (Proposition 20.3).

Definition 20.2 (Side shape; half-sum and difference; pattern coefficients)**.** For table j of a family and $w \in [0,1]$ the letter of the *side shape* w is $L_{j,w}(z) = w\Lambda_j(z) + (1-w)\Lambda_j(-z)$; $L_{j,1/2}(z) = \frac{1}{2}\big(\Lambda_j(z) + \Lambda_j(-z)\big)$ is the *half-sum* and $\partial_w L_{j,w}(z) = \Lambda_j(z) - \Lambda_j(-z)$ the *difference* of the two letters of table j. For a bunch B with room R and the Taylor cube $\Lambda_j(z) = \sum_{p \subseteq B} e^{\langle z_B, \chi_p\rangle} \Lambda_{j,(p)}(z_R)$

(Theorem 12.7), $\Lambda_{j,(p)} = 0$ when p is not a pattern of table j, the *pattern coefficients* of the side shape are

$$c_{j,p}(w; z_R) = w\,\Lambda_{j,(p)}(z_R) + (1-w)\,\Lambda^*_{j,(p)}(z_R), \qquad \Lambda^*_{j,(p)}(z_R) = \Lambda_{j,(B\setminus p)}(-z_R),$$

so that $L_{j,w}(z) = \sum_p e^{\langle z_B, \chi_p\rangle} c_{j,p}(w; z_R)$. For one table the index j is omitted.

Proposition 20.3 (The side shape, and what an exchange does to it)**.** *Let a bijection g carry or exchange a family F, with the bijection h of its tables (Definition 11.13). If g carries F, then $L_{h(j),w}(z) = L_{j,w}(g^*z)$; if it exchanges F, then $L_{h(j),w}(z) = L_{j,1-w}(g^*z)$ (Definition 20.2). For $s_j, t_j > 0$, by the tower at the unit configuration on a two-element set J (Theorem 3.2),*

$$\begin{aligned}\Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j} &= \Gamma_{\frac12}\begin{bmatrix} s_j+t_j \\ s_j,\ t_j\end{bmatrix}\, \mathsf{B}^{\{J\}}_{(s_j,t_j),\, s_j+t_j;\, I_J}\big(\Lambda_j(z), \Lambda_j(-z)\big)\\ &= \Gamma_{\frac12}\begin{bmatrix} s_j+t_j \\ s_j,\ t_j\end{bmatrix} \int_0^1 w^{s_j-1}(1-w)^{t_j-1}\, \mathsf{Z}_{s_j+t_j;(1)}\big(L_{j,w}(z)\big)\, dw,\end{aligned}$$

and for one table an exchange together with $w \mapsto 1-w$ keeps $\mathsf{Z}_{s+t;(1)}(L_w(z))$, keeping the factor $w^{s-1}(1-w)^{t-1}$ exactly when $s = t$.

The side shape weighs the basis letter against the cobasis letter, and an exchange of the table turns it over. The half-sum is even in z and the difference odd: $\Lambda(z)\Lambda(-z) = L_{1/2}(z)^2 - \frac14\big(\partial_w L_w(z)\big)^2$.

20.2 The junction in pattern form

Theorem 20.4 (The junction in pattern form)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a molecule on n marks with real exponents $s_j, t_j \geq 0$, its masses on the balance wall with every proper wall positive, and let B be a proper bunch with room R and $g \in R$. For a table at a full lean put $w_j = 1$ when $t_j = 0$ and $w_j = 0$ when $s_j = 0$. With the pattern coefficients $c_{j,p}(w; z_R)$ of the side shapes (Definition 20.2),*

$$\begin{aligned}P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \frac{2^{n-1}}{\prod_{j:\, s_j>0}\Gamma(s_j)\prod_{j:\, t_j>0}\Gamma(t_j)} \int \prod_{j:\, s_j, t_j>0} w_j^{s_j-1}(1-w_j)^{t_j-1} \int_{\mathbb{R}^{R-g}} e^{2\langle \boldsymbol{\mu}|_R, z_R\rangle}\\ \times \int \prod_{j\ \mathit{active}} y_j^{s_j+t_j-1}\, \mathsf{G}_{2\boldsymbol{\mu}|_B;(\chi_p)_p}\Big(\Big(\sum_{j\ \mathit{active}} y_j\, c_{j,p}(w_j; z_R)\Big)_p\Big)\, d\mathbf{y}\, dz_{R-g}\Big|_{z_g=0}\, d\mathbf{w},\end{aligned}$$

with $d\mathbf{y} = \prod_j dy_j$ over the active tables, $y_j \in (0,\infty)$, and $d\mathbf{w} = \prod_j dw_j$ over the tables with $s_j, t_j > 0$, $w_j \in [0,1]$; the Gramma on the scales e^{z_B} with one term for each $p \subseteq B$ whose coefficient $\sum_j y_j c_{j,p}(w_j; z_R)$ is not 0. A table at a full lean has no integral over its side shape: at $(s_j, 0)$ its coefficients are $y_j\Lambda_{j,(p)}(z_R)$, at $(0, t_j)$ they are $y_j\Lambda^*_{j,(p)}(z_R)$.

The two letters of every table become one letter, the side shapes are integrated last, and the bunch is integrated on its own orthant against the patterns of all the tables at once: each pattern is one term of one Gramma on the bunch, whose coefficient adds the pattern coefficients of the tables, each at its own scale y_j, and the room sits in the coefficients. A map that keeps the bunch moves the coefficients with the side shapes and the tables.

Proposition 20.5 (What a map keeping the bunch gives the coefficients)**.** *In Theorem 20.4 let a bijection $\tilde{g}$ of $\mathcal{E}$ keep B, with $\tilde{g}_R$ its restriction to the room, and carry or exchange F with the bijection h of its tables. If $\tilde{g}$ carries F, then $c_{h(j),\tilde{g}(p)}(w; z_R) = c_{j,p}(w; \tilde{g}_R^* z_R)$ for every j, p and w; if it exchanges F, then $c_{h(j),\tilde{g}(p)}(1-w; z_R) = c_{j,p}(w; \tilde{g}_R^* z_R)$.*

Proposition 20.6 (Level form of the half-sum)**.** *In Theorem 20.4 the half-sum and the difference of table j have the pattern coefficients $c_{j,p}(\frac{1}{2}; z_R) = \frac{1}{2}\big(\Lambda_{j,(p)} + \Lambda^*_{j,(p)}\big)(z_R)$ and $\partial_w c_{j,p}(w; z_R) = \big(\Lambda_{j,(p)} - \Lambda^*_{j,(p)}\big)(z_R)$.*

(i) For every table, $c_{j,B\setminus p}(\frac{1}{2}; -z_R) = c_{j,p}(\frac{1}{2}; z_R)$ and $\partial_w c_{j,B\setminus p}(w; -z_R) = -\partial_w c_{j,p}(w; z_R)$.

(ii) The half-sum of table j has level form on B (Proposition 14.12), $c_{j,p}(\frac{1}{2}; \cdot)$ depending on $|p|$ alone, exactly when every transposition of two marks of B fixing the rest carries the multiset $\mathcal{B}_j \uplus \{\mathcal{E} \setminus T : T \in \mathcal{B}_j\}$, with its weights, onto itself; if $2r_j \neq n$ this is the condition that every such transposition carries table j (Proposition 14.12).

(iii) If every such transposition carries or exchanges table j, its half-sum has level form and the square of its difference is unchanged by every permutation of B.

20.3 The word of a tree

Definition 20.7 (Tree of bunches)**.** A *tree of bunches* of a family F on $\mathcal{E}$ has *vertices* v, each with marks $M_v \subseteq \mathcal{E}$ and tables on M_v with their exponents, the root having $M_v = \mathcal{E}$ and the active tables of F. A vertex whose tables form one molecule is a *node*. A node v that is not a leaf carries a proper bunch $B_v \subsetneq M_v$; the tables at v have their children at B_v, indexed by the table and the pattern as in the record of the peel (Definition 14.2) and not merged, the *tables made at* v, each carrying the letters its parent carries, its exponent in each letter a positive multiple of the parent's, as at every native split (Definition 12.25). The molecules $R_{v,1}, \dots, R_{v,q_v}$ of the tables made at v on $M_v \setminus B_v$, in a fixed order, are the childcules of v (Definition 14.1), each a node whose tables are the restrictions to it of the tables made at v. The *room* of v is the one childcule when $q_v = 1$; when $q_v \geq 2$ it is a *split vertex*, with the marks $M_v \setminus B_v$ and the tables made at v, whose children are the childcules. The root of a family whose active tables form two or more molecules is likewise a split vertex, whose children are the molecules with the restrictions of its tables. A leaf v has $B_v = M_v$. The *width* of a node is the largest width $c(B_v)$ of its bunch in the tables at v, zero at a leaf. Merging equal children with their exponents added, as the peel does (Definition 14.1), keeps every letter when the exponents are nonnegative, and can cancel a letter when exponents of both signs meet; the tables made at a node are therefore indexed. The tables at a node are thus the children of the tables of F at the patterns shown on the bunches above it, restricted to its marks (Proposition 14.5); with a native gauge they carry as exponents the shares of the native splits at the nodes above (Theorem 12.26). Its descents are the sequences of nodes from the root to a leaf, each a childcule of the one before, as on the Foreman tree (Definition 14.17), which is a tree of bunches (Definition 14.4).

Corollary 20.8 (The tree of a linked molecule is a chain)**.** *Let some active table of a molecule F be connected, as for a molecule of one table. Every tree of bunches of F is then a chain $M_0 \supset M_1 \supset \cdots \supset M_d$ with $M_{k+1} = M_k \setminus B_k$, above one leaf: the tables at M_k are the children of the tables of F at $S_k = B_0 \sqcup \cdots \sqcup B_{k-1}$ (Proposition 14.5), which form one molecule (Lemma 18.14). With a native gauge their exponents are the shares of the native split, and on the Foreman tree the family at M_k is the peel $F_{S_k}(\mathbf{a}^\star, \mathbf{b}^\star)$ (Definition 14.1).*

Corollary 20.9 (A linked molecule)**.** *Let some active table of F be connected, as for a molecule of one table. Every peel then leaves one childcule, so the tree is a chain (Corollary 20.8), there are no twins, and the room of each node is the next node. For every chain that peels a class at each node, every class of a node other than its bunch lies inside one class of the next node, and the leaf is a union of whole classes of every node above it (Theorems 14.20(ii) and 14.22(i)). For the Foreman tree, which is then a chain, at every realization, the centre and every lean, Theorem 14.22(ii) with no twins gives*

$$|B_0| \le |B_1| \le \cdots \le |B_{d-1}| \le |M_d| :$$

the leaf alone has at least as many marks as every bunch.

Corollary 20.10 (The word of a tree)**.** *Let F have real data with $s_j, t_j \ge 0$, a native gauge and its masses on the balance wall, and let the children's weights factor across the childcules at every node of a tree of bunches. Run the port of Theorem 18.13(ii) at every node, on the native splits. A table $\mathcal{B}'$ made at a node, with the exponents $(a_{\mathcal{B}'}, b_{\mathcal{B}'})$, is the direct sum of its restrictions to the childcules there (Proposition 7.9); let $q_{\mathcal{B}'}$ be the number of those childcules, other than leaves, on which its letter is not a monomial. Then*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int_{\mathcal{X}} \prod_{\mathcal{B}'} \big(\Gamma(a_{\mathcal{B}'})\Gamma(b_{\mathcal{B}'})\big)^{1-q_{\mathcal{B}'}} \prod_{leaves\ v} P_{F_v}(\mathbf{m}|_{M_v};\mathbf{a}_v,\mathbf{b}_v),$$

the word of the tree*: an iterated fibre integral through the native splits, over the fibre $\mathcal{X}$ cut out by the share sums and by the $|B_v|+q_v-1$ pins of every node v that is not a leaf (Theorem 18.13(ii)), a monomial restriction entering the pins of its node as a load; F_v is the family of the tables at the leaf v with their exponents $(\mathbf{a}_v,\mathbf{b}_v)$, and a letter with exponent zero carries no shares and no Gamma. A table opened on one childcule carries no factor, and when an active table is connected every $q_{\mathcal{B}'}$ is 0 or 1 (Corollary 20.8).*

Theorem 20.11 (Building back up)**.** *In Corollary 20.10 open every leaf by the atom formula on the balance wall (Corollary 18.16), a unit being a monomial, and write the shares (a,b) of one pattern of one table in pair coordinates, $\sigma = \frac{a+b}{2}$ and $\delta = \frac{a-b}{2}$.*

(i) Polycules multiply. *At every node and at every point of its fibre on the wall contour of the native split (Theorem 18.13(iii)), the integral over the shares below the node is the product over its childcules of their words: the polycule of the childcules is evaluated as the product of its molecules, a unit contributing a monomial in the weights.*

(ii) The pins are odd. *If every table has $s_j, t_j > 0$, every pin other than the share sums involves the odd shares alone: the pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$ of the bunch, the pins of the childcules,*

$$\kappa_{R_i} = \mu_{R_i} - \sum_{j,p} \delta_{j,p}\big(2r_{j,(p)}(R_i) - |R_i|\big) = 0,$$

the sum over the tables made at the node, indexed as in Theorem 18.13 by the table j at the node and the pattern p, and the pins of the atoms. The share sums are $\sum_p \sigma_{j,p} = \sigma_j$ and $\sum_p \delta_{j,p} = \delta_j$, so the fibre is the product of an even fibre, cut by the even sums alone, and an odd fibre.

(iii) One Gramma. *If at every node every table made there is a monomial on all its childcules but at most one, in particular along a chain (Corollary 20.8), the leaf shares are one for each basis $T \in \mathcal{B}_j$ and letter, the pins of the tree are, for a fixed mark g, the pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$,*

$e \in \mathcal{E}-g$, of Corollary 18.16 after a change of determinant ±1, and the word is the atom formula, one Gramma on the marks and the letters:

$$\prod_{j:\,s_j>0}\Gamma(s_j)\prod_{j:\,t_j>0}\Gamma(t_j)\,P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\mathsf{G}_{(\boldsymbol{\mu}|_{\mathcal{E}-g};\mathbf{s};\mathbf{t});\,t_F^{(g)}}(\boldsymbol{\theta}\oplus\boldsymbol{\theta}),$$

with $t_F^{(g)}$ the configuration t_F of Theorem 8.9 without its unit columns and without the row of g: on the scales $(\mathcal{E}-g)\sqcup\{v_j\}\sqcup\{w_j\}$, the columns $(\frac12\chi_T|_{\mathcal{E}-g};\mathbf{e}_j;0)$ and $(-\frac12\chi_T|_{\mathcal{E}-g};0;\mathbf{e}_j)$, $T\in\mathcal{B}_j$, each with the coefficient $\theta_{j,T}$, a letter with exponent zero omitted: one scale for each mark but g, which carries an odd pin, and one for each letter.

(iv) The even shares close jointly. *In (iii) let every table have $s_j,t_j>0$, and write $c_{j,T}=b_{j,T}-a_{j,T}=-2\delta_{j,T}$. At fixed differences the K-shares of table j close into one scale,*

$$\int_{\sum_T a_{j,T}=s_j}\prod_{T\in\mathcal{B}_j}\Gamma(a_{j,T})\,\Gamma(a_{j,T}+c_{j,T})\,\theta_{j,T}^{-2a_{j,T}}=\int_0^\infty y^{s_j-1}\prod_{T\in\mathcal{B}_j}\mathsf{G}_{c_{j,T};(1,-1)}\big(1,\theta_{j,T}^2y\big)\,dy,$$

and $\prod_j\Gamma(s_j)\Gamma(t_j)P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the fibre integral over the differences, with the pins $\sum_T c_{j,T}=t_j-s_j$ and $-\frac12\sum_{j,T}c_{j,T}\chi_T(e)=\mu_e$, $e\in\mathcal{E}-g$, of $\prod_j\prod_T\theta_{j,T}^{-c_{j,T}}$ times these scale integrals; on the wall contour of the native split (Theorem 18.13(iii)) every integral converges absolutely. The fibre of differences has dimension $\sum_j(|\mathcal{B}_j|-1)-(n-1)$. At the centre, every table riding the line, $s_j=t_j$, with the masses $\sigma\mathbf{1}$, the pins are $\sum_{T\in\mathcal{B}_j}c_{j,T}=0$ for every j and $\sum_j M_j^{\mathsf{T}}\mathbf{c}_j=0$, with $\mathbf{c}_j=(c_{j,T})_{T\in\mathcal{B}_j}$ and the incidence matrices $(M_j)_{Te}=\mathbf{1}_T(e)$, and the contour is $\operatorname{Re}\mathbf{c}=0$.

Built back up, the word is the period read from its atoms. At every node the childcules multiply and the pins see only the odd shares; along a chain nothing is opened twice, and the word is the atom formula itself, one Gramma on the marks and the letters whose configuration is that of the interest with the masses' own columns removed. The even shares are free but for their sums, and they close table by table into one scale of Grammas $\mathsf{G}_{\nu;(1,-1)}$, as at the centre of a molecule (Theorem 20.11(iv)) and for the table that is one circuit (Corollary 18.22).

20.4 Closing the tree by draws

Proposition 20.12 (The draw series)**.** *Let $U=(U_Q)_{Q\in\mathcal{P}}$ have independent blocks $U_Q\sim\operatorname{Dir}_{\mathbf{m}|_Q}$ on a partition $\mathcal{P}$ of the marks, and let $P_1,\dots,P_q$ be nonzero polynomials with nonnegative coefficients, P_i homogeneous of degree $g_{i,Q}$ in the variables of each block Q. For an exponent vector $\mathbf{a}$ with block sums $g_{i,Q}$ put $b_{\mathbf{a}}=\prod_Q g_{i,Q}!/\prod_e a_e!$ and $C_i=\max_{\mathbf{a}}\big([u^{\mathbf{a}}]P_i\big)/b_{\mathbf{a}}$. Then $\prod_Q\big(\sum_{e\in Q}u_e\big)^{g_{i,Q}}-P_i/C_i$ has nonnegative coefficients and equals $1-P_i/C_i\in[0,1)$ on the interior of the source, and for $\nu_i\ge0$*

$$\mathbb{E}_{U\sim\bigotimes_Q\operatorname{Dir}_{\mathbf{m}|_Q}}\Big[\prod_i P_i(U)^{-\nu_i}\Big]=\prod_i C_i^{-\nu_i}\sum_{\mathbf{k}\in\mathbb{N}^q}\prod_i\frac{(\nu_i)_{k_i}}{k_i!}\times\mathbb{E}_{U\sim\bigotimes_Q\operatorname{Dir}_{\mathbf{m}|_Q}}\Big[\prod_i\Big(1-\frac{P_i(U)}{C_i}\Big)^{k_i}\Big],$$

every term nonnegative, both sides finite or infinite together. The left side is the Trailmoment of Proposition 5.22,

$$\Big(\mathsf{T}_{\boldsymbol{\nu};I_q}[(P_i)_i]\bigotimes_Q\operatorname{Dir}_{\mathbf{m}|_Q}\Big)\Big(\prod_Q\Delta_Q\Big),$$

and each expectation on the right is a finite nonnegative combination of the moments of Theorem 3.11. For a family $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ *with* $s_j, t_j \geq 0$*, one block,* $P_{2j-1} = K_j$ *and* $P_{2j} = \Psi_j$ *with the indices* (s_j, t_j)*, the constants are* $C_{2j-1} = \theta_{j,\max}/r_j!$ *and* $C_{2j} = \theta_{j,\max}/(n-r_j)!$*,* $\theta_{j,\max} = \max_T \theta_{j,T}$*; the left side times* $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$ *is* $\int_{\Delta_\mathcal{E}} u^{\mathbf{m-1}} \prod_j K_j(u)^{-s_j} \Psi_j(u)^{-t_j}\, d\omega_\mathcal{E}(u)$*, which on the balance wall is* $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$*; and the expectation of the term* $\mathbf{k} = (k_j, l_j)_j$ *is the chance that, in the urn that draws a mark with probability proportional to its current mass and then adds one to that mass, for* $j = 1, \dots, N$ *in turn* k_j *rounds of* r_j *draws and then* l_j *rounds of* $n - r_j$ *draws all fail for table* j*, a round failing when its marks are not a basis, respectively not a cobasis, of table* j*, and, when they are the basis* T*, respectively the cobasis* $\mathcal{E} \setminus T$*, with probability* $1 - \theta_{j,T}/\theta_{j,\max}$*: a finite nonnegative combination of Dirichlet moments, a finite sum of them for lists. This series is Theorem 15.13(i) with the references* $M_{2j-1} = C_{2j-1}$ *and* $M_{2j} = C_{2j}$*.*

Corollary 20.13 (At $\delta = 0$ and at a full lean the period reads the law of one letter for each table)**.** *Let a family* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$*, every table active, carry its masses* $\mathbf{m}$ *on the balance wall with every proper wall positive, every table riding the line,* $(s_j, t_j) = (\sigma_j, \sigma_j)$*, or at a full lean,* $(s_j, 0)$ *or* $(0, t_j)$*. Put* $L_j = \mathbf{E}_j = K_j \Psi_j$ *and* $\nu_j = \sigma_j$ *when table* j *rides the line,* $L_j = K_j$ *and* $\nu_j = s_j$ *at* $(s_j, 0)$*, and* $L_j = \Psi_j$ *and* $\nu_j = t_j$ *at* $(0, t_j)$*, and let* $\phi_\mathbf{m}$ *be the joint law of* $(L_1(U), \dots, L_N(U))$ *for* $U \sim \mathrm{Dir}_\mathbf{m}$ *(Proposition 8.12). The kernel* $\prod_j a_j^{-s_j} b_j^{-t_j}$ *of the letters* $a_j = K_j$ *and* $b_j = \Psi_j$ *is the kernel* $\mathsf{Z}_{\boldsymbol{\nu}; I_N}\big((L_j)_j\big)$*, one scale for each table, so*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \big(\mathsf{T}_{\boldsymbol{\nu}; I_N}[\mathrm{id}]\, \phi_\mathbf{m}\big)\big((0,\infty)^N\big);$$

with one table $\phi_\mathbf{m}$ *is a law on the half-line and the right side is* $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} (\mathsf{T}_{\nu;(1)}[x]\, \phi_\mathbf{m})((0,\infty))$*, at* $\delta = 0$ *the law* $\mathbf{E}_* \mathrm{Dir}_\mathbf{m}$ *of the even letter* $\mathbf{E}(U) = K(U)\Psi(U)$ *(Theorem 5.9). The law* $\phi_\mathbf{m}$ *is carried by* $\prod_j (0, \max_{\Delta_\mathcal{E}} L_j]$*, it is determined by its moments, and its moments are finite sums of Dirichlet moments (Theorem 3.11).*

Theorem 20.14 (The draw series along a chain)**.** *Let a family* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ *have the exponents* $s_j, t_j \geq 0$ *and masses* $\mathbf{m} \in \mathbb{R}^\mathcal{E}_{>0}$*, and let* $M_0 = \mathcal{E} \supset M_1 \supset \cdots \supset M_d$ *be a chain of sets with* $B_i = M_i \setminus M_{i+1}$*,* $i < d$*, and* $S_i = B_0 \sqcup \cdots \sqcup B_{i-1}$*,* $S_0 = \varnothing$*,* $S_{d+1} = \mathcal{E}$*. A round of draws* extends on S *for table* j *when its draws inside* S *are distinct and are* $T \cap S$ *for some basis* T *of table* j*. In this statement write* $\theta_j(S; P)$ *for the largest weight* $\theta_{j,T}$ *over the bases* T *of table* j *with* $T \cap S = P$*, and say that a round that extends on* S_i *and on* S_{i+1}*, with the traces* P_i *and* P_{i+1}*, is* accepted at S_{i+1} *with probability* $\theta_j(S_{i+1}; P_{i+1})/\theta_j(S_i; P_i)$*, independently for each* i*. Let* $\mathfrak{f}_{j,i}(u)$*,* $0 \leq i \leq d$*, be the chance that a round of* r_j *draws from* u *extends on* S_i *and is accepted at* $S_1, \dots, S_i$ *but does not both extend on* S_{i+1} *and get accepted there, and* $\mathfrak{f}^*_{j,i}$ *the same for rounds of* $n - r_j$ *draws and the cobases* $\mathcal{E} \setminus T$ *with the weights* $\theta_{j,T}$*. Then, with* $\theta_{j,\max}$ *the largest weight of table* j*,* $1 - r_j!\, K_j/\theta_{j,\max} = \sum_i \mathfrak{f}_{j,i}$*,* $1 - (n-r_j)!\, \Psi_j/\theta_{j,\max} = \sum_i \mathfrak{f}^*_{j,i}$*, and*

$$\int_{\Delta_\mathcal{E}} u^{\mathbf{m-1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, d\omega_\mathcal{E} = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \prod_j \Big(\frac{r_j!}{\theta_{j,\max}}\Big)^{s_j} \Big(\frac{(n-r_j)!}{\theta_{j,\max}}\Big)^{t_j} \times \sum_{\mathbf{k},\mathbf{l}} \prod_j \frac{(s_j)_{|\mathbf{k}_j|}\, (t_j)_{|\mathbf{l}_j|}}{\mathbf{k}_j!\, \mathbf{l}_j!}\ \mathbb{E}_{U \sim \mathrm{Dir}_\mathbf{m}}\Big[\prod_{j,i} \mathfrak{f}_{j,i}(U)^{k_{j,i}}\, \mathfrak{f}^*_{j,i}(U)^{l_{j,i}}\Big],$$

over $\mathbf{k}_j, \mathbf{l}_j \in \mathbb{N}^{d+1}$*, every term a finite nonnegative combination of Dirichlet moments, a finite sum of them for lists, whose rounds are always accepted. The numbers* $\sum_{i' \geq i} k_{j,i'}$ *of failing basis rounds of table* j *that pass* $B_0, \dots, B_{i-1}$ *decrease with* i*: they are nested, as the coordinates of the region* X_N *of a trail are (Definition 5.23). Let a bijection* g *keep each* B_i *and the masses, let* h *be a bijection of the tables, and put* $(g_* u)_{g(e)} = u_e$*. If* g *maps the bases of every table* j *onto those of table* $h(j)$ *with their weights, then* $\mathfrak{f}_{h(j),i}(g_* u) = \mathfrak{f}_{j,i}(u)$ *and* $\mathfrak{f}^*_{h(j),i}(g_* u) = \mathfrak{f}^*_{j,i}(u)$*; if* $T \mapsto \mathcal{E} \setminus g(T)$ *does, then* $\mathfrak{f}^*_{h(j),i}(g_* u) = \mathfrak{f}_{j,i}(u)$ *and* $\mathfrak{f}_{h(j),i}(g_* u) = \mathfrak{f}^*_{j,i}(u)$*, and the array of expectations is unchanged when every* $(\mathbf{k}_j, \mathbf{l}_j)$ *is replaced by* $(\mathbf{l}_{h^{-1}(j)}, \mathbf{k}_{h^{-1}(j)})$*; for one table, where an exchange forces* $2r = n$*, this is the symmetry under* $\mathbf{k} \leftrightarrow \mathbf{l}$*.*

Theorem 20.15 (The draw series along a tree)**.** *Let a molecule* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_j$ *on* $\mathcal{E}$ *carry* $s_j, t_j \geq 0$ *and masses* $\mathbf{m} \in \mathbb{R}^{\mathcal{E}}_{>0}$*, and take a tree of bunches (Definition 20.7) at every node of which the children's weights factor across the childcules, as they do for lists: for a table* $\mathcal{B}'$ *at a node* v *and a pattern* p *of* $\mathcal{B}'$ *at* B_v*, the restrictions* $\mathcal{B}'_{(p)}|R_{v,i}$ *of the child to the childcules (Proposition 7.9) carry weights whose products are the weights of the child,* $\theta_{(p),X} = \prod_i \theta_{(p),i}(X \cap R_{v,i})$*. Write* $\theta_{\max}(\mathcal{B}')$ *for the largest weight of a table* $\mathcal{B}'$*, and say that a round of draws that is a basis* T *of* $\mathcal{B}'$ *is* accepted *with probability* $\theta_T / \theta_{\max}(\mathcal{B}')$*. A cobasis round of table* j *is a basis round of its dual table, whose children are the duals of the children, with the same weights (Lemma 14.7(ii) with the identity). The* coordinates of the tree *are, at each node* v*, the totals* $x^{(v)} \in \Delta_{\{B_v, R_{v,1}, \dots, R_{v,q_v}\}}$ *of the shape of* M_v *on its blocks and the shape* $u^{(v)} \in \Delta_{B_v}$ *of the bunch; under* $\mathrm{Dir}_{\mathbf{m}}$ *they are independent, with the laws* $\mathrm{Dir}_{(m_{B_v}, m_{R_{v,1}}, \dots, m_{R_{v,q_v}})}$ *and* $\mathrm{Dir}_{\mathbf{m}|_{B_v}}$ *(Theorem 3.12). For a table* $\mathcal{B}'$ *at a node write its chances as functions of the coordinates of the subtree of that node.*

(i) A round passes a node, then each childcule. *For a table* $\mathcal{B}'$ *of rank* r' *at a node* v*, the chance* $\mathfrak{s}_{\mathcal{B}'} = r'! \, K_{\mathcal{B}'} / \theta_{\max}(\mathcal{B}')$ *that* r' *draws from the shape of* M_v *are a basis of* $\mathcal{B}'$ *and are accepted is*

$$\mathfrak{s}_{\mathcal{B}'} = \sum_p \mathfrak{p}_{\mathcal{B}',p} \prod_i \mathfrak{s}_{\mathcal{B}'_{(p)}|R_{v,i}},$$

$$\mathfrak{p}_{\mathcal{B}',p} = \frac{\prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_{v,i}\big)}{\theta_{\max}(\mathcal{B}')} \, \frac{r'!}{\prod_i r'_{p,i}!} \, \big(x^{(v)}_{B_v}\big)^{|p|} \big(u^{(v)}\big)^p \prod_i \big(x^{(v)}_{R_{v,i}}\big)^{r'_{p,i}},$$

over the patterns p *of* $\mathcal{B}'$ *at* B_v*, with* $r'_{p,i}$ *the rank on* $R_{v,i}$ *of the child* $\mathcal{B}'_{(p)}$*:* $\mathfrak{p}_{\mathcal{B}',p}$ *is the chance that the round* passes v *showing* p*, its draws in* B_v *being the marks of* p *and its numbers of draws in the childcules the* $r'_{p,i}$*, and is accepted there, with the probability given by its first factor, which is at most* 1*; and* $\mathfrak{s}_{\mathcal{B}'_{(p)}|R_{v,i}}$ *is the chance that its draws in* $R_{v,i}$ *are a basis of the restriction of the child there and are accepted. At a leaf* $\mathfrak{s}_{\mathcal{B}'} = r'! \, K_{\mathcal{B}'} / \theta_{\max}(\mathcal{B}')$*.*

(ii) A failure is met at one node. *The chance* $1 - \mathfrak{s}_{\mathcal{B}'}$ *that the round fails, by not being a basis or by not being accepted, is* $\sum_w \mathfrak{f}_{\mathcal{B}' \to w}$ *over the nodes and leaves* w *of the subtree of* v*, where* $\mathfrak{f}_{\mathcal{B}' \to v} = 1 - \sum_p \mathfrak{p}_{\mathcal{B}',p}$ *and, for* w *in the subtree of* $R_{v,i}$*,*

$$\mathfrak{f}_{\mathcal{B}' \to w} = \sum_p \mathfrak{p}_{\mathcal{B}',p} \prod_{i' < i} \mathfrak{s}_{\mathcal{B}'_{(p)}|R_{v,i'}} \; \mathfrak{f}_{\mathcal{B}'_{(p)}|R_{v,i} \to w},$$

the chance that the round first fails at w*. Each* $\mathfrak{f}_{\mathcal{B}' \to w}$ *is a finite sum of products, over the blocks of the tree, of polynomials with nonnegative coefficients in one block each.*

(iii) The series. *With* $\mathfrak{f}_{j,w}$ *and* $\mathfrak{f}^*_{j,w}$ *these chances for the basis and the cobasis rounds of table* j *of* F *from the root, and* $\theta_{j,\max}$ *the largest weight of table* j*,*

$$\int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, d\omega_{\mathcal{E}} = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \prod_j \Big(\frac{r_j!}{\theta_{j,\max}}\Big)^{s_j} \Big(\frac{(n-r_j)!}{\theta_{j,\max}}\Big)^{t_j}$$
$$\times \sum_{\mathbf{k},\mathbf{l}} \prod_j \frac{(s_j)_{|\mathbf{k}_j|}\,(t_j)_{|\mathbf{l}_j|}}{\mathbf{k}_j!\,\mathbf{l}_j!}\ \mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}\Big[\prod_{j,w} \mathfrak{f}_{j,w}(U)^{k_{j,w}}\, \mathfrak{f}^*_{j,w}(U)^{l_{j,w}}\Big],$$

every term nonnegative, both sides finite or infinite together; on the balance wall the left side is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$. Each expectation is a finite nonnegative combination of products over the blocks of Dirichlet moments, a finite sum of them for lists: at every node, the node's own moments times the product over its childcules of theirs.

(iv) Nesting along the tree. *The numbers $k_{j,\geq w} = \sum_{w'} k_{j,w'}$, over the w' in the subtree of w, of the basis rounds of table j that first fail in the subtree of w, satisfy $k_{j,\geq w} = k_{j,w} + \sum_{w'} k_{j,\geq w'}$ over the childcules w' of w, and likewise for the cobasis rounds: along every descent they are nested, as the coordinates of the region X_N of a trail are (Definition 5.23). On a chain this is Theorem 20.14.*

At nonnegative exponents every share of the word closes at once. The draw series closes each letter about its largest coefficient on the simplex, and the tree only sorts its failures: a round that fails is charged to the first node at which it stops passing, the chances of passing are monomials in the coordinates of that node, and the coordinates of different nodes are independent. So every term is a nonnegative combination of products, over the blocks of the tree, of Dirichlet moments, and the counts of failures nest along the descents as the coordinates of the region X_N of a trail do.

Corollary 20.16 (What each node of a chain integrates)**.** *Let the tree of a molecule on n marks be a chain $M_0 \supset M_1 \supset \cdots \supset M_d$ with bunches B_k and rooms $R_k = M_{k+1}$. The simplex of the molecule is the product*

$$\Delta_{M_0} \cong \prod_{k=0}^{d-1} \Big(\Delta_{B_k} \times \Delta_{\{B_k,R_k\}}\Big) \times \Delta_{M_d}$$

of the shape of each bunch, the totals of the bunch and its room at each node, and the shape of the leaf, and under $\mathrm{Dir}_{\mathbf{m}}$ these coordinates are independent: they are the coordinates of the tree of Theorem 20.15, which use no weights. The node of B_k has $|B_k|$ coordinates, $|B_k|-1$ of shape and one total, the leaf has $|M_d|-1$, and they add up to $n-1$. When some active table is connected, the Foreman tree, which is then a chain, has, at the centre and at every lean, no node with more coordinates than the leaf has marks (Corollary 20.9).

Proposition 20.17 (Rates at the centre)**.** *Let a family on $n \geq 3$ marks be at the centre, every table with the exponents (σ_j,σ_j), with the masses $\sigma\mathbf{1}$, $\sigma = \sum_j \sigma_j$, and let every active table be connected.*

(i) The walls are $\kappa_S = \sum_j \sigma_j c_j(S)$ (Proposition 8.21(ii)), so $\kappa_S = \kappa_{\mathcal{E}\setminus S}$.

(ii) The rate of the peak closing (Theorem 15.13(ii)) is $\varpi = \sigma/\max_j \max(r_j-1, n-r_j-1)$, the maximum over the active tables, attained at the faces $S = \mathcal{E}-e$, where every active table has $c_j = 1$, $d_j = r_j - 1$ and $d_j^* = n - r_j - 1$.

(iii) Near the vertex of e, with $u_e = 1-x$ and $u_{\mathcal{E}-e} = xv$, $v \in \Delta_{\mathcal{E}-e}$, every active table has

$$K_j(u) = (1-x)\,x^{r_j-1} K_{j,/e}(v) + x^{r_j} K_{j,\setminus e}(v),$$
$$\Psi_j(u) = (1-x)\,x^{n-r_j-1} \Psi_{j,\setminus e}(v) + x^{n-r_j} \Psi_{j,/e}(v);$$

the vertex product *of e in table j is $K_{j,/e}\Psi_{j,\setminus e}$, the product of the letters of the two minors that dominate as $x \to 0$.*

(iv) For two marks e, f and a bijection β with $\beta(e) = f$ that carries or exchanges the family, with the bijection h of its tables, the restriction $\beta : \mathcal{E} - e \to \mathcal{E} - f$ takes the vertex product of e in table j to that of f in table $h(j)$. For one table such a β exists exactly when e and f lie in one class at the centre (Corollary 11.16).

20.5 Devices at the centre

Proposition 20.18 (The half-sum and the difference of a table with $2r = n$)**.** *Let $F = ((\mathcal{B}, \theta); \sigma, \sigma)$, $\sigma > 0$, have $2r = n$ and masses $\mathbf{m}$ on its balance wall. The half-sum $\frac{1}{2}(K + \Psi) = \frac{1}{2}\sum_T \theta_T\big(\alpha^T + \alpha^{\mathcal{E}\setminus T}\big)$ is homogeneous of degree r and unchanged by $T \mapsto \mathcal{E} \setminus T$; the difference $K - \Psi$ vanishes at the table's gauge; in signed coordinates they are $e^{\langle \mathbf{1}, z\rangle} L_{1/2}(z)$ and $e^{\langle \mathbf{1}, z\rangle}\partial_w L_w(z)$ (Definition 20.2). Then $K\Psi = \big(\frac{1}{2}(K + \Psi)\big)^2 - \frac{1}{4}(K - \Psi)^2$ with $|K - \Psi| < K + \Psi$ on the orthant, and by Proposition 20.1*

$$(K\Psi)^{-\sigma} = \sum_{l \geq 0} \frac{(\sigma)_l}{l!\, 4^l}\, (K - \Psi)^{2l} \Big(\frac{K + \Psi}{2}\Big)^{-2\sigma - 2l},$$

a series of nonnegative terms. For a mark e with room $R = \mathcal{E} - e$ write $\frac{1}{2}(K + \Psi) = h_0(\alpha_R) + \alpha_e h_1(\alpha_R)$, so that, by Definition 7.3, $h_0 = \frac{1}{2}(K_{\setminus e} + \Psi_{/e})$, $h_1 = \frac{1}{2}(K_{/e} + \Psi_{\setminus e})$ and $h_1(u) = u^R h_0(u^{-1})$. Expanding $(K - \Psi)^{2l} = \sum_{\mathbf{a}} \big([\alpha^{\mathbf{a}}](K - \Psi)^{2l}\big)\, \alpha^{\mathbf{a}}$ and integrating the mark e of each term as the Beta corner of a linear letter (Proposition 15.1 with one coefficient), with $w_l = 2\sigma + 2l$,

$$P_F(\mathbf{m}; \sigma, \sigma) = \sum_{l \geq 0} \frac{(\sigma)_l}{l!\, 4^l} \sum_{\mathbf{a}} \big([\alpha^{\mathbf{a}}](K - \Psi)^{2l}\big)\, \Gamma_{\substack{2\\1}}\begin{bmatrix} m_e + a_e, & w_l - m_e - a_e \\ & w_l \end{bmatrix} \times \int_{\Delta_R} u^{(\mathbf{m}+\mathbf{a})|_R - \mathbf{1}} h_0(u)^{-(w_l - m_e - a_e)} h_1(u)^{-(m_e + a_e)}\, d\omega_R(u),$$

whenever the terms converge absolutely.

The half-sum is a positive polynomial whose support is closed under complements, but it is not in general the basis polynomial of a table: the bases together with the cobases need not have the exchange property, even for connected tables, so the integrals left over the room are integrals of a positive polynomial against its complement and not periods of a family. When complementation maps the bases onto themselves the difference vanishes and the room is a period.

Corollary 20.19 (Merged Skeleton values)**.** *If $T \mapsto \mathcal{E} \setminus T$ maps the bases of a table $(\mathcal{B}, \theta)$ onto themselves with $\theta_{\mathcal{E}\setminus T} = \theta_T$, then $2r = n$, no mark is a loop or a coloop, $\Psi = K$, $\Psi_{\setminus e} = K_{/e}$ and $\Psi_{/e} = K_{\setminus e}$ for every mark e, and the two Skeleton values of the scale of e coincide at every gauge (Corollary 15.4). For $F = ((\mathcal{B}, \theta); s, t)$ with $s, t \geq 0$, $s + t > 0$, its masses on the balance wall with every proper wall positive, and a mark e, one has $0 < m_e < s + t$ and*

$$P_F(\mathbf{m}; s, t) = \Gamma_{\substack{2\\1}}\begin{bmatrix} m_e, & s + t - m_e \\ & s + t \end{bmatrix}\, P_{\mathcal{B}\setminus e}\big(\mathbf{m}|_{\mathcal{E}-e};\, s + t - m_e,\, m_e\big),$$

the period of the deletion $(\mathcal{B}\setminus e, \theta)$ on $\mathcal{E} - e$ at the exponents $(s + t - m_e, m_e)$, on its balance wall; so $P_F(\mathbf{m}; s, t)$ depends on the exponents only through $s + t$, and at the centre, $s = t = \sigma$ and $\mathbf{m} = \sigma\mathbf{1}$, the identity reads $P_F(\sigma\mathbf{1}; \sigma, \sigma) = \Gamma_{\substack{2\\1}}\begin{bmatrix} \sigma, & \sigma \\ & 2\sigma \end{bmatrix}\, P_{\mathcal{B}\setminus e}(\sigma\mathbf{1}|_{\mathcal{E}-e}; \sigma, \sigma)$.

Theorem 20.20 (The scale integral of one mark)**.** *Let a family $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_j$ carry real exponents $s_j, t_j \geq 0$, its masses on the balance wall with every proper wall positive, and let e be a mark that is neither a loop nor a coloop in any active table, with room $R = \mathcal{E} - e$; write $\Lambda_{j,\backslash e}$ and $\Lambda_{j,/e}$ for the centred polynomials of the deletion and the contraction of table j, so that $\Lambda_j(z) = e^{z_e}\Lambda_{j,/e}(z_R) + e^{-z_e}\Lambda_{j,\backslash e}(z_R)$ (Theorem 12.7).*

(i) At the gauge $(1, e^{2z_R})$ the scale integral of the bunch $\{e\}$ (Theorem 15.3) is

$$\begin{aligned}&\mathrm{Sc}_{\{e\}}\big((1, e^{2z_R}); \mathbf{m}; \mathbf{s}, \mathbf{t}\big)\\ &= e^{-2\sigma\langle \mathbf{1}, z_R\rangle}\int_0^\infty x^{m_e-1}\prod_j \big(\Lambda_{j,\backslash e}(z_R) + x\Lambda_{j,/e}(z_R)\big)^{-s_j}\big(\Lambda_{j,/e}(-z_R) + x\Lambda_{j,\backslash e}(-z_R)\big)^{-t_j}dx,\end{aligned}$$

$\sigma = \sum_j \sigma_j$ the total strength: up to the factor $e^{-2\sigma\langle \mathbf{1}, z_R\rangle}$ it depends only on m_e, the exponents and the functions $\Lambda_{j,\backslash e}$ and $\Lambda_{j,/e}$ on the room, and Corollary 16.6 at the bunch $\{e\}$ reads, for every $g \in R$,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{n-2}\int_{\mathbb{R}^{R-g}} e^{2\langle \mathbf{m}|_R, z_R\rangle}\ \mathrm{Sc}_{\{e\}}\big((1, e^{2z_R}); \mathbf{m}; \mathbf{s}, \mathbf{t}\big)\, dz_{R-g}\Big|_{z_g=0}.$$

(ii) At an atom $\Upsilon = ((s_j, t_j)_j; \alpha^\star)$ with its target $\boldsymbol{\tau}$, under $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ the law of the log-ratios $(y_{g_k} - y_e)_{k=1}^{n-1}$, read in an order $\mathbf{g}_e$ of the room that writes the potential from e as the one pair of functions of Proposition 10.4, does not depend on e.

(iii) For one table at the centre, $s = t = \sigma$ and $\boldsymbol{\mu} = 0$, with the even statistics $E_{\backslash e}$ and $E_{/e}$ of the deletion and the contraction (Theorem 7.16) and a two-element set J, the functions of z_R understood,

$$\begin{aligned}&\int_0^\infty x^{\sigma-1}\big(\Lambda_{\backslash e} + x\Lambda_{/e}\big)^{-\sigma}\big(\Lambda_{/e}(-\cdot) + x\Lambda_{\backslash e}(-\cdot)\big)^{-\sigma}dx\\ &\qquad = e^{-\frac{\sigma}{2}(E_{\backslash e}+E_{/e})}\,\mathsf{B}^{\{J\}}_{(\sigma,\sigma),\,\sigma;\,I_J}\big(e^{\frac{1}{2}(E_{\backslash e}-E_{/e})},\, e^{\frac{1}{2}(E_{/e}-E_{\backslash e})}\big),\end{aligned}$$

and the Breta is $\Gamma_{\!2}\!\left[\begin{matrix}\sigma,\ \sigma\\ 2\sigma\end{matrix}\right]$ where the two even statistics agree.

For one table at the centre the scale integral of a mark is the factor $e^{-\frac{\sigma}{2}(E_{\backslash e}+E_{/e})}$ times a Breta that sees the room through one number, the difference of the even statistics of the deletion and the contraction; at an atom the law of the log-ratios read from a mark does not depend on the mark. Proposition 20.21 reads the scale integral of the mark with the exponents in the shape index.

Proposition 20.21 (A mark is the Breta of its children's even letters)**.** *For one table, a mark e that is neither a loop nor a coloop, exponents $s, t > 0$, $0 < m_e < s+t$ and the top wall $\kappa^{\mathrm{top}}_{\{e\}} = s+t-m_e$, integrating $\alpha_e = x$ with the room fixed,*

$$\int_0^\infty x^{m_e-1}K^{-s}\Psi^{-t}\,dx = K_{/e}^{\,t-m_e}\Psi_{\backslash e}^{\,s-m_e}\,\Gamma_{\!2}\!\left[\begin{matrix}m_e,\ \kappa^{\mathrm{top}}_{\{e\}}\\ s,\ t\end{matrix}\right]\ \mathsf{B}^{\{J\}}_{(s,t),\,\kappa^{\mathrm{top}}_{\{e\}};\,I_J}\big(\mathbf{E}_{\backslash e}, \mathbf{E}_{/e}\big),$$

with $\mathbf{E}_{\backslash e} = K_{\backslash e}\Psi_{\backslash e}$ and $\mathbf{E}_{/e} = K_{/e}\Psi_{/e}$ the even letters of the deletion and the contraction, J a two-element set, and

$$\mathsf{B}^{\{J\}}_{(s,t),\,\kappa;\,I_J}(a, b) = \int_0^1 w^{s-1}(1-w)^{t-1}\,\mathsf{Z}_{\kappa;(1)}\big(wa + (1-w)b\big)\,dw,$$

the shape w of J weighing the deletion against the contraction. At the centre, $s = t = m_e = \sigma$, the prefactor and the stacked Gamma are 1.

The same scale integral is the Breta of Corollary 19.29 for the bunch $\{e\}$, whose levels d and r have the children $\mathcal{B}\setminus e$ and $\mathcal{B}/e$: there the walls $(m_e, \kappa^{\mathrm{top}}_{\{e\}})$ fill the shape index and the exponent t the scale index, here the exponents (s,t) fill the shape index and the top wall the scale index. Both read the scale integral of the mark through the even letters of the two children, the deletion and the contraction.

20.6 The junction under level form

Lemma 20.22 (An even power of $e^{v}-e^{-v}$ on a symmetric two-sided scale)**.** *For $\xi>0$ and $l\in\mathbb{N}$,*

$$\int_{\mathbb{R}} \big(e^{v}-e^{-v}\big)^{2l} e^{-\xi(e^{v}+e^{-v})}\,dv = 4^{l}\big(\tfrac{1}{2}\big)_{l}\,\xi^{-l}\,\mathsf{G}_{l;(1,-1)}(\xi,\xi).$$

Theorem 20.23 (The junction under level form)**.** *Let $F=((\mathcal{B}_j,\theta_j);\sigma_j,\sigma_j)_{j=1}^{N}$, every $\sigma_j>0$, be a molecule on n marks with the masses $\sigma\mathbf{1}$, $\sigma=\sum_j\sigma_j$, let B, $b=|B|$, be a proper bunch with room R such that every transposition of two marks of B fixing the rest carries or exchanges every table, and let $g\in R$. By Proposition 20.6 every half-sum has level form; in this statement write $q_{j,k}(z_R)$ for the common value of $c_{j,p}(\frac{1}{2};z_R)$ over $|p|=k$ and $\mathbf{q}_j=(q_{j,0},\dots,q_{j,b})$, so that $q_{j,b-k}(-z_R)=q_{j,k}(z_R)$ and*

$$L_{j,1/2}(z)=\sum_{k=0}^{b} e^{-\langle\mathbf{1},z_B\rangle}e_k\big(e^{2z_B}\big)\,q_{j,k}(z_R),\qquad e^{-\langle\mathbf{1},z_B\rangle}e_k\big(e^{2z_B}\big)=\sum_{|p|=k}e^{\langle z_B,\chi_p\rangle},$$

the centred polynomial of the list on B whose bases are the k-subsets of B; and for each $\mathbf{l}=(l_j)_j\in\mathbb{N}^N$

$$\prod_j\big(\partial_w L_{j,w}(z)\big)^{2l_j}=e^{-2|\mathbf{l}|\langle\mathbf{1},z_B\rangle}\sum_{|\mathbf{h}|=2|\mathbf{l}|}\prod_{k=0}^{b}e_k\big(e^{2z_B}\big)^{h_k}\,c^{(\mathbf{l})}_{\mathbf{h}}(z_R),$$

with $c^{(\mathbf{l})}_{\mathbf{h}}$ Laurent polynomials in e^{z_R}.

(i) With the Gramma $\mathsf{G}_{0;(\chi_p)_{p\subseteq B}}\big((\upsilon_{|p|})_{p\subseteq B}\big)$, $\boldsymbol{\upsilon}=(\upsilon_0,\dots,\upsilon_b)$, whose coefficients depend on $|p|$ alone and which depends on B through b alone,

$$P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})=2^{n-1}\sum_{\mathbf{l}\in\mathbb{N}^N}\prod_j\frac{(\sigma_j)_{l_j}}{l_j!\,4^{l_j}\,\Gamma(2\sigma_j+2l_j)}\int_{\mathbb{R}^{R-g}}\int_{(0,\infty)^N}\prod_j y_j^{2\sigma_j+2l_j-1}$$
$$\times\sum_{|\mathbf{h}|=2|\mathbf{l}|}c^{(\mathbf{l})}_{\mathbf{h}}(z_R)\Big((-\partial_{\boldsymbol{\upsilon}})^{\mathbf{h}}\,\mathsf{G}_{0;(\chi_p)_p}\big((\upsilon_{|p|})_p\big)\Big)\Big|_{\boldsymbol{\upsilon}=\sum_j y_j\mathbf{q}_j(z_R)}\,d\mathbf{y}\,dz_{R-g}\Big|_{z_g=0},$$

the inner sum being $\int_{\mathbb{R}^B}\prod_j\big(\partial_w L_{j,w}(z)\big)^{2l_j}e^{-\sum_j y_j L_{j,1/2}(z)}\,dz_B\ge 0$, a level with $\sum_j y_j q_{j,k}=0$ absent.

(ii) One mark at a time. *For $e\in B$, $B'=B-e$ and $\boldsymbol{\upsilon}>0$,*

$$\mathsf{G}_{0;(\chi_p)_{p\subseteq B}}\big((\upsilon_{|p|})_p\big)=\int_0^{\infty}\mathsf{G}_{0;(\chi_p)_{p\subseteq B'}}\big((x\,\upsilon_{|p|+1}+x^{-1}\upsilon_{|p|})_p\big)\,\frac{dx}{x},$$

and for $B=\{e\}$ this Gramma is $\mathsf{G}_{0;(1,-1)}(\upsilon_1,\upsilon_0)$.

(iii) The share side. *Let G be the family on B of the lists whose bases are the k-subsets of B, $1 \le k \le b-1$, with the exponents $(a'_k, 0)$, at the masses $\frac{1}{b}\sum_{k=1}^{b-1} k a'_k\, \mathbf{1}_B$ on its balance wall. Where every $\operatorname{Re} a'_k > 0$ its period converges, is unchanged when the exponents of the lists of ranks k and $b-k$ are swapped, and is read on the hyperplane $\langle \mathbf{1}, w\rangle = 0$:*

$$P_G\Big(\tfrac{1}{b}\sum_{k=1}^{b-1} k a'_k\, \mathbf{1}_B; (a'_k)_{k=1}^{b-1}, \mathbf{0}\Big) = 2^{b-1}\sqrt{b}\int_{\langle \mathbf{1}, w\rangle = 0} \prod_{k=1}^{b-1} e_k\big(e^{2w}\big)^{-a'_k}\, d\mathrm{vol}_{b-1}(w).$$

For $\boldsymbol{\upsilon} > 0$ and $\mathbf{h} \in \mathbb{N}^{b+1}$, on a contour with $\operatorname{Re} a_k > h_k$ for every k and with $a'_k = a_k - h_k$,

$$\Big((-\partial_{\boldsymbol{\upsilon}})^{\mathbf{h}}\, \mathsf{G}_{0;(\chi_p)_p}\big((\upsilon_{|p|})_p\big)\Big)(\boldsymbol{\upsilon}) = 2^{1-b}\int_{\sum_{k=0}^{b}(2k-b)(a_k-h_k)=0} \prod_{k=0}^{b} \Gamma(a_k)\, \upsilon_k^{-a_k}\; P_G\Big(\tfrac{1}{b}\sum_{k=1}^{b-1} k a'_k\, \mathbf{1}_B; (a'_k)_{k=1}^{b-1}, \mathbf{0}\Big).$$

(iv) Two marks. *If $B = \{e, f\}$ and the swap of e and f exchanges every table, then with $z_\pm = z_e \pm z_f$, for every j,*

$$L_{j,1/2}(z) = e^{-z_+} c_{j,\varnothing}(\tfrac{1}{2}; z_R) + e^{z_+} c_{j,\varnothing}(\tfrac{1}{2}; -z_R) + \big(e^{z_-} + e^{-z_-}\big) c_{j,\{e\}}(\tfrac{1}{2}; z_R),$$
$$\partial_w L_{j,w}(z) = \big(e^{z_-} - e^{-z_-}\big)\, \partial_w c_{j,\{e\}}(w; z_R),$$

the functions $c_{j,\{e\}}(\frac{1}{2}; \cdot)$ and $\partial_w c_{j,\{e\}}$ are even, and, writing in this statement $C_\varnothing(z_R) = \sum_j y_j c_{j,\varnothing}(\frac{1}{2}; z_R)$ and $C_e(z_R) = \sum_j y_j c_{j,\{e\}}(\frac{1}{2}; z_R)$,

$$P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma}) = 2^{n-2} \sum_{\mathbf{l}\in\mathbb{N}^N} \big(\tfrac{1}{2}\big)_{|\mathbf{l}|} \prod_j \frac{(\sigma_j)_{l_j}}{l_j!\,\Gamma(2\sigma_j + 2l_j)} \int_{\mathbb{R}^{R-g}} \prod_j \big(\partial_w c_{j,\{e\}}\big)^{2l_j}$$
$$\times \int_{(0,\infty)^N} \prod_j y_j^{2\sigma_j + 2l_j - 1}\, C_e^{-|\mathbf{l}|}\, \mathsf{G}_{0;(1,-1)}\big(C_\varnothing(-z_R),\, C_\varnothing(z_R)\big)$$
$$\times\, \mathsf{G}_{|\mathbf{l}|;(1,-1)}\big(C_e(z_R),\, C_e(z_R)\big)\, d\mathbf{y}\, dz_{R-g}\Big|_{z_g = 0},$$

a series of nonnegative terms.

Under level form the bunch enters only through its levels: one Gramma with one coefficient for each level, which depends on the number of its marks alone and is computed one mark at a time, is differentiated at the level coefficients of the half-sums, added over the tables with their scales, and its share side is a period of the lists of k-subsets of the bunch's own marks. For a pair whose swap exchanges every table the bunch splits along $z_e + z_f$ and $z_e - z_f$ into two Grammas $\mathsf{G}_{\nu;(1,-1)}$.

20.7 The node in one scale

At a bunch of width one the integrand of Theorem 19.13 is, in the even share, the product of two Mellin transforms, one carrying the bunch and one the room. Read back on the half-line, the period is one integral over one scale of two functions, the bunch transform and the room transform, each an integral of a Gramma $\mathsf{G}_{\nu;(1,-1)}$ over its own shape. A closing toward a level is the expansion of one of them in powers of the scale, integrated term by term against the other. The expansion

of a bunch is measured by one number, its reach: every bunch has a reach, no reach lies below the least value of the bunch's ratio, and a node closes on its lattice toward every level at which the growths of its two families multiply to less than one. Each side of a node is an interval, from the decay of its transform to its reach, and, against a room and a bunch bounded below, a node closes on a geometric lattice exactly when the two intervals are disjoint.

Definition 20.24 (Node of width one; its two transforms)**.** A *node of width one*, a node of a tree of bunches (Definition 20.7) whose bunch has width one, is a family $F = ((\mathcal{B}, \theta); s, t)$ with real data, $s, t > 0$, its masses on the balance wall with every proper wall positive, and a bunch B of width one with room R, the weights factoring across them: the setting of Theorem 19.13 for one table. Along the level d its level shares are $(a^d, b^d, a^r, b^r) = (s - \kappa_B + b^d,\ b^d,\ \kappa_B - b^d,\ t - b^d)$, the odd share being fixed at $2\delta^d = s - \kappa_B$; its ratios ϱ_B on Δ_B and ϱ_R on Δ_R are those of Definition 19.2; and σ_*^d is a real value of the even share at which both families converge and the four arguments of its stacked Gamma are positive, so that the line $\operatorname{Re}\sigma^d = \sigma_*^d$ is a contour of Theorem 19.13, with $b_* = \sigma_*^d - \delta^d > 0$ the real part of b^d there. The node carries the nonnegative measures

$$\phi_B(du) = u^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}/R}^{-(s-\kappa_B)} K_{\mathcal{B}|B}^{-\kappa_B} \Psi_{\mathcal{B}|B}^{-t}\, d\omega_B(u) \quad \text{on } \Delta_B,$$
$$\phi_R(dv) = v^{\mathbf{m}|_R - \mathbf{1}} K_{\mathcal{B}|R}^{-(s-\kappa_B)} K_{\mathcal{B}/B}^{-\kappa_B} \Psi_{\mathcal{B}/B}^{-t}\, d\omega_R(v) \quad \text{on } \Delta_R,$$

the forms of the bunch family and of the room family at the level share $b^d = 0$; Chapter 23 calls ϕ_B the base form of the cell of the bunch $\mathcal{B}_B^+$ at the exponents (s, t). At the share b^d the periods of the two families are

$$P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b}) = \int_{\Delta_B} \varrho_B^{-b^d}\, d\phi_B, \qquad P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}) = \int_{\Delta_R} \varrho_R^{b^d}\, d\phi_R,$$

continued meromorphically in b^d by their sector forms (Theorem 8.5, Lemma 8.4): the *continued periods along the level d*. For $y > 0$ the *bunch transform* and the *room transform* are

$$f_B(y) = \int_{\Delta_B} \mathsf{G}_{s-\kappa_B;(1,-1)}\big(1, \varrho_B(u)\, y\big)\, \phi_B(du),$$
$$f_R(y) = \int_{\Delta_R} \big(\varrho_R(v)\, y\big)^{\kappa_B}\, \mathsf{G}_{t-\kappa_B;(1,-1)}\big(1, \varrho_R(v)\, y\big)\, \phi_R(dv),$$

with values in $[0, \infty]$. Swapping B and R gives the *swapped node*, with the bunch R, the room B and the two levels swapped (Theorem 19.13).

The bunch family at the share b^d has the form ϕ_B times $(\mathbf{E}_{\mathcal{B}/R}/\mathbf{E}_{\mathcal{B}|B})^{-b^d} = \varrho_B^{-b^d}$, and the room family the form ϕ_R times $(\mathbf{E}_{\mathcal{B}|R}/\mathbf{E}_{\mathcal{B}/B})^{-b^d} = \varrho_R^{b^d}$: the even share enters through the ratio of the hinge values of each side alone (Lemma 19.15).

Lemma 20.25 (Two letters in one scale)**.** *For linear letters $L_i(\rho) = L_i^{(0)} + \rho L_i^{(1)}$ with positive coefficients and the Skeleton values $\xi_i = L_i^{(0)}/L_i^{(1)}$, $i = 1, 2$, exponents $s, t > 0$ and $0 < \kappa < s + t$,*

$$\int_0^\infty \rho^{\kappa-1} L_1(\rho)^{-s} L_2(\rho)^{-t}\, d\rho$$
$$= \frac{(L_1^{(0)})^{\kappa-s} (L_1^{(1)})^{-\kappa} (L_2^{(0)})^{-t}}{\Gamma(s)\Gamma(t)} \int_0^\infty x^{\kappa-1}\, \mathsf{G}_{s-\kappa;(1,-1)}\Big(1, \frac{\xi_1}{\xi_2}\, x\Big)\, \mathsf{G}_{t-\kappa;(1,-1)}(1, x)\, dx.$$

Theorem 20.26 (The node in one scale)**.** *Let F and B be a node of width one (Definition 20.24).*

(i) The period is one integral over one scale of the two transforms,

$$P_F(\mathbf{m}; s, t) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty f_B(y)\, f_R(y)\, \frac{dy}{y},$$

the integrand nonnegative.

(ii) With the hinge values of Definition 19.2,

$$f_B(y) = \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}|B}^{-s} \Psi_{\mathcal{B}|B}^{-t}\, \mathsf{G}_{s-\kappa_B;(1,-1)}\big(\gamma_B^K(u)^{-1},\, y\,\gamma_B^\Psi(u)\big)\, d\omega_B(u) :$$

the bunch meets the room through its two hinge values, as in pooling (Theorem 19.8).

(iii) For real $b^d > 0$ with $a^d > 0$, and for real b^d with $a^r > 0$ and $b^r > 0$, respectively, in $[0, \infty]$,

$$\int_0^\infty y^{b^d - 1} f_B(y)\, dy = \Gamma(b^d)\, \Gamma(a^d) \int_{\Delta_B} \varrho_B^{-b^d}\, d\phi_B,$$
$$\int_0^\infty y^{-b^d - 1} f_R(y)\, dy = \Gamma(a^r)\, \Gamma(b^r) \int_{\Delta_R} \varrho_R^{b^d}\, d\phi_R.$$

Where the integrals over the shapes are finite they are the continued periods of the bunch family and of the room family at the share b^d, with their Gamma pairs, and (i), written through these two integrals in the share b^d, is Theorem 19.13 for one table.

(iv) If no slope on the room is positive, so that $\inf \varrho_R > 0$ (Lemma 19.15), then $\phi_R(\Delta_R) < \infty$, and with $a = \frac{t+\kappa_B}{2} - \frac{1}{4}$

$$f_R(y) \le \sqrt{\pi}\, e^{(t-\kappa_B)^2/4}\, \phi_R(\Delta_R)\, \big(y \inf \varrho_R\big)^a\, e^{-2\sqrt{y \inf \varrho_R}} \qquad \Big(y \ge \frac{\max(1, a^2)}{\inf \varrho_R}\Big) :$$

the room transform decays at the least ratio of the room.

Lemma 20.27 (The bunch transform on the orthant)**.** *For a node of width one and $y > 0$,*

$$f_B(y) = \int_{\mathbb{R}^B_{>0}} x^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}|B}(x)^{-s} \Psi_{\mathcal{B}|B}(x)^{-t} \exp\Big(- \frac{1}{\gamma_B^K(x)} - y\,\gamma_B^\Psi(x)\Big)\, dx.$$

The bunch transform is a Laplace transform in the hinge value γ_B^Ψ, damped by the hinge value γ_B^K: the room meets the bunch through its two hinge values and nothing else. Its expansion in powers of y is measured by its reach.

Definition 20.28 (Lattice expansion; reach)**.** A function f on $(0, \infty)$ has a *lattice expansion of reach* $x \ge 0$ when there are finitely many reals $\eta_1, \dots, \eta_q$, an integer $J \ge 0$ and coefficients $\beta_{\eta,j}$, for η in $A = \bigcup_i (\eta_i + \mathbb{N})$ and $0 \le j \le J$, such that

$$f(y) = \sum_{\eta \in A} \sum_{j=0}^{J} \beta_{\eta,j}\, y^\eta (-\log y)^j \quad (y > 0), \qquad |\beta_{\eta,j}| \le C(1+\eta)^D\, \frac{x^\eta}{\Gamma(1+\eta)^2} \quad (\eta \in A,\ \eta \ge 0),$$

with constants C and D; the series then converges absolutely at every $y > 0$, and $\sum_{\eta,j} |\beta_{\eta,j}| < \infty$. Its *least exponent* is the least η with some $\beta_{\eta,j} \ne 0$. A function has at most one lattice expansion. The *reach of the bunch* of a node of width one, $\mathrm{reach}(B)$, is the infimum of the x for which the bunch transform has a lattice expansion of reach x, and the *reach of the room*, $\mathrm{reach}(R)$, is the reach of R as the bunch of the swapped node.

Lemma 20.29 (The lattice of one mark)**.** *Let ν be real.*

(i) For $x > 0$, $\mathsf{G}_{\nu;(1,-1)}(1,x)$ is the sum of the residues of $\Gamma(z)\Gamma(z+\nu)x^{-z}$ at the poles of $\Gamma(z)\Gamma(z+\nu)$. Off the integers

$$\mathsf{G}_{\nu;(1,-1)}(1,x) = \sum_{k\geq 0} \frac{(-1)^k}{k!}\Big(\Gamma(\nu-k)\,x^k + \Gamma(-\nu-k)\,x^{\nu+k}\Big);$$

at $\nu = l \in \mathbb{N}$

$$\mathsf{G}_{l;(1,-1)}(1,x) = \sum_{k<l} \frac{(-1)^k}{k!}\Gamma(l-k)\,x^k + (-1)^l \sum_{k\geq l} \frac{x^k}{k!\,(k-l)!}\Big((\log\Gamma)'(1+k) + (\log\Gamma)'(1+k-l) - \log x\Big),$$

and $\mathsf{G}_{-l;(1,-1)}(1,x) = x^{-l}\mathsf{G}_{l;(1,-1)}(1,x)$.

(ii) For $x \geq 1$, $0 < \mathsf{G}_{\nu;(1,-1)}(1,x) \leq \sqrt{\pi}\, e^{\nu^2/4}\, x^{\nu/2-1/4} e^{-2\sqrt{x}}$.

(iii) For $w > 0$, $y \mapsto \mathsf{G}_{\nu;(1,-1)}(1,w^2y)$ has a lattice expansion of reach w^2, with exponents in $\mathbb{N} \cup (\nu+\mathbb{N})$, in $-l+\mathbb{N}$ at $\nu = -l$, and at most one power of $-\log y$.

Lemma 20.30 (Products of lattice expansions)**.** *If f_1 and f_2 have lattice expansions of reaches x_1 and x_2, then $f_1 f_2$ has one of reach $(\sqrt{x_1}+\sqrt{x_2})^2$, whose terms are the products of theirs collected by exponent, and for real a, $y^a f_1$ has one of reach x_1.*

Lemma 20.31 (Closing a lattice expansion against a decaying function)**.** *Let f have a lattice expansion $\sum_{\eta,j} \beta_{\eta,j} y^\eta (-\log y)^j$ of reach x with least exponent η_0, and let g be measurable on $(0,\infty)$ and integrable on its compact subsets, with $\int_0^1 y^{\eta_0-1-\epsilon}|g(y)|\,dy < \infty$ for some $\epsilon > 0$ and $|g(y)| \leq C' y^a e^{-2\sqrt{x'y}}$ for $y \geq y_0$, where $x < x'$. Then*

$$\int_0^\infty f(y)\,g(y)\,\frac{dy}{y} = \sum_{\eta,j} \beta_{\eta,j} \int_0^\infty y^{\eta-1}(-\log y)^j\, g(y)\,dy,$$

every integral on the right and the series converging absolutely, and the terms with $\eta \geq N$ add up in absolute value to at most $C''(1+N)^{D''}(x/x')^N$, with C'' and D'' independent of N.

Lemma 20.32 (Continued cube integrals grow at most exponentially)**.** *Let G and h be analytic on a neighbourhood of $[0,1]^m$ in $\mathbb{C}^m$, with $h > 0$ on $[0,1]^m$, and let a_k, b_k be real with $a_k > 0$ wherever $b_k = 0$. The integral*

$$I(z) = \int_{[0,1]^m} \prod_k q_k^{a_k + b_k z - 1}\, G(q)\, h(q)^{-z}\, dq,$$

convergent where every $a_k + b_k \operatorname{Re} z > 0$, continues meromorphically to $z \in \mathbb{C}$, with poles only in the set $\mathcal{W}$ of the real points at which some $a_k + b_k z$ lies in $-\mathbb{N}$ (Lemma 8.4). For every $\vartheta > 0$ there is M, and then for every $\epsilon > 0$ there are C and D, with

$$|I(z)| \leq C(1+|z|)^D\, M^{|\operatorname{Re} z|}\, e^{\vartheta |\operatorname{Im} z|}$$

at every z at distance at least ϵ from $\mathcal{W}$.

Definition 20.33 (Growth along a level; growth at the integers)**.** For a node of width one, the *growth* $\operatorname{growth}(B)$ of the bunch family along the level d is the infimum of the $M \geq 0$ for which, for every $\epsilon > 0$, there are C, D and $\vartheta < \pi$ with

$$\left|P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\right| \leq C(1+|b^d|)^D\, M^{-\operatorname{Re} b^d}\, e^{\vartheta|\operatorname{Im} b^d|}$$

for the continued period at the share b^d, at every b^d with $\operatorname{Re} b^d \leq 0$ at distance at least ϵ from the set $\mathcal{W}_B$ of the real points at which a proper wall of the bunch family lies in $-\mathbb{N}$, which holds its poles; the growth $\operatorname{growth}(R)$ of the room family is the same with $P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$ and the set $\mathcal{W}_R$ of the room family. Along the level r the growths are those of the swapped node. The *growth at the integers* of the bunch is

$$\operatorname{growth}_{\mathbb{Z}}(B) = \limsup_{k\to\infty} \left|P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\right|^{1/k},$$

the continued period of the bunch family taken at $b^d = -k$, $k \in \mathbb{N}$.

Theorem 20.34 (A node closes at the growth of its two families)**.** *Let F and B be a node of width one.*

(i) Every bunch has a reach. *The growths are finite;* $\operatorname{growth}(B) \leq \sup \varrho_B$ *and* $\operatorname{growth}(R) \leq 1/\inf \varrho_R$ *when these are finite; and* $\operatorname{reach}(B) \leq \operatorname{growth}(B)$. *The exponents of the lattice expansion of f_B are the η for which $b^d = -\eta$ is a pole of $\Gamma(b^d)\Gamma(a^d)P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})$ to the left of the line $\operatorname{Re} b^d = b_*$, and $\sum_j j!\,\beta_{\eta,j}(b^d+\eta)^{-j-1}$ is its principal part there: the lattice of the bunch is the lattice of its continued period with its Gamma pair, the wall residues joining.*

(ii) Against a room bounded below. *Let no slope on the room be positive, and let* $\operatorname{reach}(B) < \inf \varrho_R$. *Then*

$$P_F(\mathbf{m};s,t) = \frac{1}{\Gamma(s)\Gamma(t)} \sum_{\eta,j} \beta_{\eta,j}\, \partial_{b^d}^j \Big[\Gamma(a^r)\Gamma(b^r)\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})\Big]_{b^d=-\eta},$$

absolutely, every room period in it convergent, and for every $x > \operatorname{reach}(B)$ the terms with $\eta \geq N$ add up in absolute value to at most $C(1+N)^D(x/\inf\varrho_R)^N$. The terms are the residues of the integrand of Theorem 19.13 at its poles toward the level d: the even share closes on the lattice of continued periods of the bunch family at the root rate $\operatorname{reach}(B)/\inf\varrho_R$.

(iii) Both families continued. *Whatever the slopes, if* $\operatorname{growth}(B)\operatorname{growth}(R) < 1$, *then $P_F(\mathbf{m};s,t)$ is the sum of the residues of the integrand*

$$\Gamma_{\!\!\begin{smallmatrix}4\\2\end{smallmatrix}}\!\begin{bmatrix} a^d,\ b^d,\ a^r,\ b^r \\ s,\ t \end{bmatrix} P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$$

of Theorem 19.13, continued in b^d, at its poles with $\operatorname{Re} b^d < b_*$; for every $M_B > \operatorname{growth}(B)$ and $M_R > \operatorname{growth}(R)$ the residues at the poles with real part in $(-N-1,-N]$ add up in absolute value to at most $C(1+N)^D(M_BM_R)^N$. The even share closes on the lattices of continued periods of both families, the wall residues of both joining.

(iv) Toward the level r. *Swapping B and R leaves the period and Theorem 19.13 unchanged; the swapped node has the bunch R with the ratio ϱ_R and the room B with the ratio ϱ_B. Applied to it, (i) to (iii) close the even share toward the level r: against a bunch with* $\inf\varrho_B > 0$ *at the root rate* $\operatorname{reach}(R)/\inf\varrho_B$, *and in general at a root rate at most the product of the growths of the two families along the level r.*

A node is therefore closed on its lattice toward the level at which the growths of its two families multiply to less than one, each growth read on its family alone; where neither level has a product below one, the peak closing (Theorem 19.30) closes the node with nonnegative terms and needs no root rate. Against a room bounded below the bunch enters through its reach alone, and the domination is a reach against a decay.

Proposition 20.35 (No reach is below the least ratio)**.** *Let F and B be a node of width one.*

(i) If $\inf \varrho_B > 0$, there are C and a with $f_B(y) \le C y^a \, e^{-2\sqrt{y \inf \varrho_B}}$ for $y \ge 1$, and for every $x' > \inf \varrho_B$ there is $C_- > 0$ with $f_B(y) \ge C_- e^{-2\sqrt{x' y}}$ for $y \ge 1$: the bunch transform decays at the least ratio of the bunch.

(ii) $\operatorname{reach}(B) \ge \inf \varrho_B$*: no lattice expansion of the bunch transform has a reach below the least ratio.*

(iii) When ϱ_B is bounded, $\operatorname{reach}(B) = \operatorname{growth}(B) = \sup \varrho_B$*: the reach of a bounded bunch is its largest ratio.*

Against a room bounded below, the root rate $\operatorname{reach}(B)/\inf \varrho_R$ of Theorem 20.34(ii) therefore lies between $\inf \varrho_B / \inf \varrho_R$ and $\sup \varrho_B / \inf \varrho_R$. The series bunches take the lower end, and the bounded bunches the upper, which is $\sup(\varrho_B/\varrho_R)$, the root rate of Corollary 19.21(i).

Proposition 20.36 (A series bunch is its marks)**.** *Let the bunch B of a node of width one be a series bunch, θ'_B the weight of its pattern B and θ'_{B-e} that of its pattern $B - e$ (Definition 19.2, Proposition 19.3), and in this statement write $w_e = \theta'_{B-e}/\theta'_B$.*

(i) The bunch transform is the product over the marks of B:

$$\begin{aligned} f_B(y) &= (\theta'_B)^{-s-t} \prod_{e \in B} \mathsf{G}_{m_e - s;(1,-1)}(w_e y,\, w_e) \\ &= (\theta'_B)^{-s-t} \, y^{s-\kappa_B} \prod_{e \in B} w_e^{s-m_e} \, \mathsf{G}_{m_e-s;(1,-1)}\big(1,\, w_e^2 y\big). \end{aligned}$$

(ii) $\varrho_B(u) = \big(\sum_e w_e/u_e\big)\big(\sum_e w_e u_e\big) \ge \big(\sum_e w_e\big)^2$, with equality exactly at the point of equal coordinates of Δ_B, and ϱ_B is unbounded: whatever the weights, the least ratio of a series bunch is taken at the point of equal coordinates of its shape.

(iii) f_B has the lattice expansion of reach $\big(\sum_e w_e\big)^2 = \inf \varrho_B$ that is the product of the lattices of its marks, so $\operatorname{reach}(B) = \inf \varrho_B$*. With $\boldsymbol{\nu} = \mathbf{m}|_B - s\mathbf{1}_B$ and no ν_e an integer, its exponents lie in the lattices $-\nu_S + \mathbb{N}$, $S \subseteq B$: $S = \varnothing$ and $S = B$ are the lattices of the Gamma pair, and for every other S the lattice is where the wall of the bunch family at $B \setminus S$, continued, meets $-\mathbb{N}$. Finally $f_B(y) \le C \, y^a \, e^{-2(\sum_e w_e)\sqrt{y}}$ for $y \ge \max_e w_e^{-2}$, with constants C and a.*

Proposition 20.37 (A bunch in series is the product of its parts)**.** *Let the bunch of a node of width one be $B = B_1 \sqcup \cdots \sqcup B_q$, and on each B_i let $\mathcal{X}_i$ and $\mathcal{Y}_i$ be tables, $\mathcal{Y}_i$ of rank one less than $\mathcal{X}_i$, with the hinge values $\gamma_i^K = K_{\mathcal{X}_i}/K_{\mathcal{Y}_i}$ and $\gamma_i^\Psi = \Psi_{\mathcal{Y}_i}/\Psi_{\mathcal{X}_i}$ and the ratio $\varrho_i = \mathbf{E}_{\mathcal{Y}_i}/\mathbf{E}_{\mathcal{X}_i}$, such that for constants $C, C' > 0$*

$$K_{\mathcal{B}|B} = C \prod_i K_{\mathcal{X}_i}, \qquad \Psi_{\mathcal{B}|B} = C' \prod_i \Psi_{\mathcal{X}_i}, \qquad \frac{1}{\gamma_B^K} = \sum_i \frac{1}{\gamma_i^K}, \qquad \gamma_B^\Psi = \sum_i \gamma_i^\Psi :$$

B is the series connection *of the parts B_i. A series bunch is the series connection of its marks, with $K_{\mathcal{X}_e} = \alpha_e$, $\Psi_{\mathcal{X}_e} = 1$, $K_{\mathcal{Y}_e} = w_e$, $\Psi_{\mathcal{Y}_e} = w_e \alpha_e$ and $\varrho_e = w_e^2$ (Proposition 20.36).*

(i) $f_B = C^{-s}C'^{-t}\prod_i f_i^\circ$, *with the* orthant transforms *of the parts*

$$f_i^\circ(y) = \int_{\mathbb{R}_{>0}^{B_i}} x^{\mathbf{m}|_{B_i}-\mathbf{1}} K_{\mathcal{X}_i}(x)^{-s}\Psi_{\mathcal{X}_i}(x)^{-t}\exp\Big(-\frac{1}{\gamma_i^K(x)} - y\,\gamma_i^\Psi(x)\Big)\,dx;$$

and $\sqrt{\inf\varrho_B} = \sum_i \sqrt{\inf\varrho_i}$.

(ii) If every f_i° *has a lattice expansion of reach* x_i, *then* f_B *has one of reach* $\big(\sum_i\sqrt{x_i}\big)^2$.

(iii) Let every part be a single mark, or have a bounded ratio and a finite measure $u^{\mathbf{m}|_{B_i}-\mathbf{1}}K_{\mathcal{X}_i}^{-s}\Psi_{\mathcal{X}_i}^{-t}(\gamma_i^\Psi)^{-a_i}\,d\omega_{B_i}$ *on* Δ_{B_i}, *where* $a_i = m_{B_i} - s\,r_i - t\,(|B_i| - r_i)$ *with* r_i *the rank of* $\mathcal{X}_i$; *and let no sum of the* a_i *over a nonempty set of parts be an integer. Then* f_i° *has the reach* $\sup\varrho_i$, *and* $\operatorname{reach}(B) = \big(\sum_i\sqrt{\sup\varrho_i}\big)^2$: *in a series connection the square roots of the reaches add.*

Lemma 20.38 (The point of equal coordinates is stationary)**.** *Let* $\mathcal{X}$ *be a table on a set* S, *with any weights. At every point* u *of* $(0,\infty)^S$ *with equal coordinates,* $u_e\,\partial_{u_e}\log\mathbf{E}_{\mathcal{X}}(u) = 1$ *for every* $e \in S$. *So a ratio of even letters of tables on* S *is stationary at the point of equal coordinates of* Δ_S: ϱ_B *on* Δ_B, ϱ_R *on* Δ_R, *and* ϱ_B/ϱ_R *at the pair of these points.*

Theorem 20.39 (The two intervals of a node)**.** *For a node of width one, the bunch transform of the swapped node is* $y^{-t}f_R(y)$, *so that* $\inf\varrho_B$ *and* $\inf\varrho_R$ *are the decays of the two transforms, and* $\operatorname{reach}(B)$ *and* $\operatorname{reach}(R)$ *their reaches (Proposition 20.35(i), Lemma 20.30).*

(i) No transform reaches less far than it decays: $\inf\varrho_B \le \operatorname{reach}(B)$ *and* $\inf\varrho_R \le \operatorname{reach}(R)$.

(ii) Toward the level d. *Let no slope on the room be positive, so that* $\inf\varrho_R > 0$, *and index the terms of the lattice of Theorem 20.34(ii) by* (η, j). *Their root rate is* $\operatorname{reach}(B)/\inf\varrho_R$: *for every* $x > \operatorname{reach}(B)$ *they are at most* $C(1+\eta)^D(x/\inf\varrho_R)^\eta$ *in modulus, and for every* $\epsilon > 0$ *infinitely many of them exceed* $\big(\operatorname{reach}(B)/\inf\varrho_R - \epsilon\big)^\eta$. *So the lattice converges absolutely when* $\operatorname{reach}(B) < \inf\varrho_R$, *and its terms do not tend to zero when* $\operatorname{reach}(B) > \inf\varrho_R$.

(iii) Toward the level r. *Let no slope on the bunch be negative, so that* $\inf\varrho_B > 0$. *The terms of the lattice of continued periods toward the level* r *have the root rate* $\operatorname{reach}(R)/\inf\varrho_B$ *in the same sense: the lattice converges absolutely when* $\operatorname{reach}(R) < \inf\varrho_B$, *and its terms do not tend to zero when* $\operatorname{reach}(R) > \inf\varrho_B$.

(iv) One level at most. $\operatorname{reach}(B) < \inf\varrho_R$ *and* $\operatorname{reach}(R) < \inf\varrho_B$ *never hold together. Under both hypotheses of (ii) and (iii) the node therefore closes on a geometrically convergent lattice exactly when the intervals* $[\inf\varrho_B, \operatorname{reach}(B)]$ *of the bunch and* $[\inf\varrho_R, \operatorname{reach}(R)]$ *of the room are disjoint, toward the level of the lower one, at the ratio of the upper end of the lower interval to the lower end of the upper. When each interval extends beyond the lower end of the other, neither lattice converges, and the share closes at its peaks (Theorem 19.30); when they touch, the root rate is one and domination decides nothing.*

Each side of a node is read by two numbers of its own transform, its decay and the reach of its expansion, and no expansion reaches less far than its transform decays. A series bunch has both at its least ratio, and its interval is the point $(\sum_e w_e)^2$ (Proposition 20.36); a bounded bunch spreads over its whole range $[\inf\varrho_B, \sup\varrho_B]$ (Proposition 20.35(iii)); in a series connection of marks and bounded parts the square roots of the reaches add (Proposition 20.37(iii)). When the bunch and the room are both series bunches the two intervals are points: the node closes toward the level of the smaller point at the ratio of the two points, its terms toward the other level grow at the inverse ratio, and where the points meet domination decides nothing. Where a far side is unbounded, the

level toward it can still close with both families continued, at a root rate at most the product of their growths (Theorem 20.34(iii)). By Lemma 20.38 the value of ϱ_B/ϱ_R at the points of equal coordinates of Δ_B and Δ_R is a stationary value of the node's ratio, but the root rate is read at the extremes of the two transforms. The reach of a bunch that is neither bounded nor built from parallel and series bunches is read on the law of its ratio (Section 20.8).

Theorem 20.40 (A series connection closes at its reach)**.** *(i) Let the bunch B of a node of width one be the series connection of parts $B_1, \dots, B_q$ (Proposition 20.37), every orthant transform f_i° having a lattice expansion of reach x_i, and let no slope on the room be positive. If $\left(\sum_i \sqrt{x_i}\right)^2 < \inf \varrho_R$, the even share closes toward the level d on the lattice of continued periods of the bunch family (Theorem 20.34(ii)): for every $x > \left(\sum_i \sqrt{x_i}\right)^2$ its terms with $\eta \geq N$ add up in absolute value to at most a constant times $(1+N)^D (x/\inf \varrho_R)^N$, and its coefficients $\beta_{\eta,j}$ are those of $C^{-s}C'^{-t}\prod_i f_i^\circ$, finite sums of products, over the parts, of the coefficients of their lattice expansions. Under the hypotheses of Proposition 20.37(iii) the root rate of the terms is $\left(\sum_i \sqrt{\sup \varrho_i}\right)^2 / \inf \varrho_R$ (Theorem 20.39(ii)). For a series bunch, the series connection of its marks with w_e as in Proposition 20.36 and $x_e = w_e^2$, the root rate is $\inf \varrho_B / \inf \varrho_R = \left(\sum_e w_e\right)^2 / \inf \varrho_R$ and the coefficients are those of $(\theta'_B)^{-s-t} y^{s-\kappa_B} \prod_e w_e^{s-m_e} \mathsf{G}_{m_e-s;(1,-1)}(1, w_e^2 y)$: finite sums of products, over the marks of B, of the coefficients of Lemma 20.29, Laurent coefficients of the Gamma pairs $\Gamma(z)\Gamma(z+m_e-s)$.*

(ii) Let B be the series connection of parts $B_1, \dots, B_q$ in the dual table at the exponents (t, s) (Proposition 8.15), its orthant transforms there having lattice expansions of reaches x_i^*, and let no slope on the room be negative. If $\left(\sum_i \sqrt{x_i^*}\right)^2 \sup \varrho_R < 1$, the even share closes toward the level r on the lattice of continued periods of the bunch family, at a root rate at most $\left(\sum_i \sqrt{x_i^*}\right)^2 \sup \varrho_R$. A parallel bunch B is a series bunch of the dual table (Proposition 19.37), with $w_e = \theta'_{\{e\}}/\theta'_\varnothing$, $\theta'_\varnothing$ and $\theta'_{\{e\}}$ the weights of its patterns $\varnothing$ and $\{e\}$; then $x_e^* = w_e^2$ and $\sup \varrho_B = \left(\sum_e w_e\right)^{-2}$, taken at the point of equal coordinates of Δ_B, so the share closes when $\sup \varrho_R < \sup \varrho_B$, at the root rate $\sup \varrho_R / \sup \varrho_B$.

No order of the values is asked: the room's ratio need only be bounded on the side away from the closing, and the bunch's ratio may be unbounded, its walls falling toward the level closed on. When the bunch and the room are both series bunches, the root rate $\inf \varrho_B / \inf \varrho_R$ is read at the points of equal coordinates of both shapes. When the extreme ratio of the room sits at a vertex of its shape simplex the root rate is not read at the points of equal coordinates, which are only stationary.

Theorem 20.41 (Products of Grammas $\mathsf{G}_{\nu;(1,-1)}$ close on nested lattices)**.** *For a finite set S, numbers $w_e > 0$ and real indices ν_e, put $w_S = \sum_{e\in S} w_e$ and $\nu_S^- = \sum_{e\in S} \min(0, \nu_e)$, and in this statement write, for $\operatorname{Re} z > -\nu_S^-$,*

$$\mathcal{M}_S(z) = \int_0^\infty y^{z-1} \prod_{e\in S} \mathsf{G}_{\nu_e;(1,-1)}\big(1, w_e^2 y\big)\, dy.$$

(i) $\mathcal{M}_{\{e\}}(z) = w_e^{-2z}\, \Gamma(z)\Gamma(z+\nu_e)$, and for two marks with $w_e = w_f = w$

$$\mathcal{M}_{\{e,f\}}(z) = w^{-2z}\, {}_4\Gamma_1\left[\begin{matrix} z,\ z+\nu_e,\ z+\nu_f,\ z+\nu_e+\nu_f \\ 2z+\nu_e+\nu_f \end{matrix}\right].$$

(ii) For $S = S_1 \sqcup S_2$ with $w_{S_1} < w_{S_2}$, let $\prod_{e\in S_1} \mathsf{G}_{\nu_e;(1,-1)}(1, w_e^2 y) = \sum \beta_{\eta,j} y^\eta (-\log y)^j$ be the product of the lattices of the marks of S_1 (Lemmas 20.29 and 20.30). For $\operatorname{Re} z > -\nu_S^-$ and $i \in \mathbb{N}$,

$$(-\partial_z)^i \mathcal{M}_S(z) = \sum_{\eta,j} \beta_{\eta,j}\, (-\partial_z)^{i+j} \mathcal{M}_{S_2}(z+\eta),$$

every $z+\eta$ in the domain of $\mathcal{M}_{S_2}$, absolutely, the terms with $\eta \geq N$ adding up in absolute value to at most $C(1+N)^D (w_{S_1}/w_{S_2})^{2N}$.

(iii) Removing at each step a mark e of least w_e, until one mark is left or two with equal w_e, writes $\mathcal{M}_S(z)$ and its derivatives as nested lattice sums of products of Laurent coefficients of Gamma pairs at the points of the lattices, ending at (i): at most $|S|-1$ nested lattices, summed from the inside out, the one that removes e from S' converging absolutely at every point of the lattices above it, at the root rate $\big(w_e/(w_{S'} - w_e)\big)^2$, its constant depending on that point.

Corollary 20.42 (A table that is one circuit is a nested lattice of Gamma data)**.** *In Corollary 18.22 the integral over y is the integral of Theorem 20.41 with $S = \mathcal{E}$, $w_e = \theta_e$ and the indices $\nu_e = m_e - s$, taken at $z = s$, and $s > -\sum_e \min(0, \nu_e)$. So the period of every table that is one circuit, at all exponents $s, t > 0$, all masses of Corollary 18.22 and all weights, is a nested lattice sum of products of Laurent coefficients of Gamma pairs, with at most $n-1$ nested lattices, each converging absolutely at every point of the lattices above it and summed from the inside out. With unit weights, removing one mark at a time, the root rates are $(n-1)^{-2}, (n-2)^{-2}, \dots, \frac{1}{4}$, and the last two marks give four Gammas over one; splitting the marks into a lighter part S_1 and a heavier part S_2 converges at the root rate $(\theta_{S_1}/\theta_{S_2})^2$. At integer data with unit weights every coefficient is a rational combination of products of values $(\log\Gamma)^{(i)}(k)$, $i, k \geq 1$, which are Trailmoments of the counting input (Propositions 5.13 and 5.14).*

20.8 The reach read on the law

The reach of a bunch is defined by the expansion of its transform (Definition 20.28) and is bounded below by the least ratio and above by the growth of the continued family (Proposition 20.35, Theorem 20.34). Read on one measure on the half-line, the law of the ratio under the base form, the reach is the point beyond which the law has a density equal to its own wall series (Definition 20.43), the series whose terms are the poles of the continued family; every point at which the law fails to be analytic therefore bounds the reach from below, and the least ratio and the largest ratio of a bounded bunch are such points. From above the reach is bounded by the ceiling of a continuation (Definition 20.47): in each sector the falling variables are carried around zero on circles, as far as the sector letters allow, and the largest modulus of the ratio met there bounds the reach. From below it is bounded by the growth of the continued family at the integers; a continuation whose largest values sit at simple tops that do not cancel has that growth as its ceiling, so that the reach is at least the modulus of a stationary value of the ratio on a flat face, and equal to it when the ceilings come down to it.

Definition 20.43 (The law of a bunch; wall series)**.** The *law* of the bunch of a node of width one is the image $(\varrho_B)_*\phi_B$ of its base form under its ratio, a measure on $(0,\infty)$ (Definition 5.4). The continued period of the bunch family at the share b^d is $\int_0^\infty x^{-b^d}\, d\big((\varrho_B)_*\phi_B\big)(x)$ wherever the integral converges, which is for $\operatorname{Re} b^d$ in an open interval containing b_*, the *domain interval* of the law, the walls being affine in b^d and the conditions of Theorem 8.7 strict. A function g on (x_0,∞),

$x_0 \geq 0$, is a *wall series of radius* x_0 when there are finitely many reals η_i, an integer $J \geq 0$ and coefficients $a_{\eta,j}$, for $\eta \in A = \bigcup_i (\eta_i + \mathbb{N})$ and $0 \leq j \leq J$, with $\sum_{\eta,j} |a_{\eta,j}| x'^{-\eta} < \infty$ for every $x' > x_0$, such that

$$g(x) = \sum_{\eta,j} a_{\eta,j}\, x^{-\eta-1} (\log x)^j \qquad (x > x_0).$$

The series then converges absolutely and locally uniformly on $\{x \in \mathbb{C} : \ |x| > x_0, \ |\arg x| < \pi\}$, so a wall series is real-analytic.

Theorem 20.44 (The reach is read on the law)**.** *Let F and B be a node of width one.*

(i) On $(\mathrm{reach}(B), \infty)$ *the law* $(\varrho_B)_* \phi_B$ *has a density, and the density is a wall series of radius* $\mathrm{reach}(B)$*. Its terms are the poles of the continued period of the bunch family to the left of the domain interval: a pole* $b^d = -\eta$ *with the principal part* $\sum_j j!\, a_{\eta,j} (b^d + \eta)^{-j-1}$ *gives the terms* $a_{\eta,j} x^{-\eta-1} (\log x)^j$*, and there are no others.*

(ii) If on some (x_0, ∞) *the law has a density that is a wall series of radius* x_0*, then* $\mathrm{reach}(B) \leq x_0$*.*

(iii) So the reach is the least $x_0 \geq 0$ *for which the law has on* (x_0, ∞) *a density that is a wall series of radius* x_0*. In particular* $\mathrm{reach}(B) \geq x_1$ *at every* $x_1 > 0$ *near which* $(\varrho_B)_* \phi_B$ *has no real-analytic density: at the least ratio when it is positive, and at the largest ratio of a bounded bunch.*

So no reach lies below a point where the law of its ratio is singular, and Proposition 20.35(ii), with the lower half of (iii), is the case of the two ends of the law. The law of a family at a node is likewise its own wall series beyond its growth, so that $\mathrm{reach}(B) \leq \mathrm{growth}(B)$ of Theorem 20.34(i) is an equality (Corollary 32.11); along a direction of the shares the law reads the reach of a piece of any width (Theorem 32.10). From above the reach is bounded by the ceilings of continuations.

Lemma 20.45 (Finite parts on a keyhole)**.** *Let $0 < r \leq 1$, $r' > r$, a real, and let G be holomorphic on $U \times \mathbb{C}$ for an open U containing $\{|q| \leq r'\} \cup [r, 1]$. Then $I(z) = \int_0^1 q^{a+z-1} G(q, z)\, dq$, defined for $\mathrm{Re}\, z > -a$, continues meromorphically to $\mathbb{C}$ with poles only where $a + z \in -\mathbb{N}$, and*

$$I(z) = \sum_{l \geq 0} g_l(z)\, \frac{r^{a+z+l}}{a+z+l} + \int_r^1 q^{a+z-1} G(q, z)\, dq, \qquad g_l(z) = \frac{1}{2\pi i} \oint_{|\zeta| = r'} \zeta^{-l-1} G(\zeta, z)\, d\zeta.$$

Where $|a + z + l| \geq \epsilon$ for every $l \in \mathbb{N}$,

$$|I(z)| \leq \frac{r'}{\epsilon (r' - r)}\, r^{a + \mathrm{Re}\, z} \max_{|\zeta| = r'} |G(\zeta, z)| + \int_r^1 q^{a + \mathrm{Re}\, z - 1} |G(q, z)|\, dq.$$

If $0 < q_0 \leq 1$ and U contains the closed disc D with diameter $[-q_0, 1]$, then off the points with $a + z \in \mathbb{Z}$

$$I(z) = \frac{1}{e^{2\pi i (a+z)} - 1} \oint_{\partial D} q^{a+z-1} G(q, z)\, dq,$$

the circle run once counterclockwise from 1, with q^{a+z-1} continued along it from its positive value at 1.

Definition 20.46 (Falling and base variables; keyholes)**.** For a node of width one and an order π of its bunch B let $q = (q_k)_k$ be its sector coordinates and U_k its tails, and let $k_{\mathcal{B}|B,\pi}$, $\psi_{\mathcal{B}|B,\pi}$, $k_{\mathcal{B}/R,\pi}$, $\psi_{\mathcal{B}/R,\pi}$ be the sector letters of the two tables of the bunch family (Theorem 8.5(ii)). Along the

level d the wall of the bunch family at U_k is $\kappa_k + \mathrm{sl}_k\, b^d$ at the share b^d, with its *slope* $\mathrm{sl}_k \in \{-1, 0, 1\}$ (Lemma 19.15). On the sector of π the base form is

$$\phi_\pi = \prod_k q_k^{\kappa_k - 1}\, k_{\mathcal{B}/R,\pi}^{-(s-\kappa_B)}\, k_{\mathcal{B}|B,\pi}^{-\kappa_B}\, \psi_{\mathcal{B}|B,\pi}^{-t}\, dq,$$

and the ratio is

$$\varrho_\pi = \frac{k_{\mathcal{B}/R,\pi}\, \psi_{\mathcal{B}/R,\pi}}{k_{\mathcal{B}|B,\pi}\, \psi_{\mathcal{B}|B,\pi}} \prod_k q_k^{-\mathrm{sl}_k},$$

continued to the complex points of the coordinates at which the sector letters do not vanish; so the continued period of the bunch family at the share b^d is $\sum_\pi \int_{[0,1]^{|B|-1}} \varrho_\pi^{-b^d}\, \phi_\pi$. The q_k with $\mathrm{sl}_k = 1$ are the *falling variables* of π, whose walls fall as $\mathrm{Re}\, b^d$ decreases, and the others its *base variables*, *flat* when $\mathrm{sl}_k = 0$ and *rising* when $\mathrm{sl}_k = -1$. Where some flat variables vanish the sector reaches a *flat face* of the shape, on which every wall crossed has slope 0 and ϱ_π has the finite limit given by the leading parts of the letters; where some falling variable vanishes it reaches a *falling face*. For $0 < r \le b \le 1$ the *keyhole* and its *outline* are $Y_r[0,b] = \{q \in \mathbb{C} : |q| \le r\} \cup [r,b]$ and $\mathcal{O}_r[0,b] = \{|q| = r\} \cup [r,b]$, with $Y_r = Y_r[0,1]$ and $\mathcal{O}_r = \mathcal{O}_r[0,1]$.

Definition 20.47 (Keyhole continuations; ceiling)**.** A *keyhole continuation* of the bunch family of a node of width one assigns to every order π of B finitely many closed boxes Ω with disjoint interiors covering the cube of its base variables, and to every such box and falling variable q_k a radius $r_{\pi,\Omega,k} \in (0,1]$, such that none of the four sector letters vanishes on the set $Y = \Omega \times \prod_k Y_{r_{\pi,\Omega,k}}$, the product over the falling variables. The boxes may also divide the sides of the falling variables: they then cover the whole cube of the order, a box being the product of its part in the base variables and of its sides in the falling variables, with radii of its own; a falling variable with the side $[0, b_k]$ carries a radius $r \le b_k$ and the sets $Y_r[0,b_k]$ and $\mathcal{O}_r[0,b_k]$, and one with the side $[a_k, b_k]$, $a_k > 0$, is kept real, its sets Y and $\mathcal{O}$ being $[a_k, b_k]$. A box, divided or not, is written Ω with its radii and its sets Y and $\mathcal{O} = \Omega \times \prod_k \mathcal{O}_{r_{\pi,\Omega,k}}$, where Ω stands for its part in the variables without keyholes and the products run over the falling variables with keyholes. The *ceiling* $H_\mathcal{O}$ of the continuation is the largest value of $|\varrho_\pi|$ on the sets $\mathcal{O}$ of its boxes, an order without falling variables contributing the largest value of ϱ_π on its cube.

The sector letters are positive on the closed cube, with positive constant terms (Theorem 8.5(ii)), so small radii give keyhole continuations, and every bunch has one of finite ceiling. Whether a letter vanishes on a keyhole is decided on its outline and at one point.

Lemma 20.48 (Zeros on a keyhole are decided on its outline and at one point)**.** *Let Ω be a connected set, J a finite set, $0 < r_j \le b_j \le 1$ for $j \in J$, and L a continuous function on $\Omega \times \mathbb{C}^J$, a polynomial in $q = (q_j)_{j \in J}$. If L has no zero on $\Omega \times \prod_j \mathcal{O}_{r_j}[0,b_j]$, and for some $v_0 \in \Omega$ and each j the polynomial in q_j obtained from $L(v_0, \cdot)$ by putting $q_k = r_k$ for $k \ne j$ has no zero in the disc $|q_j| < r_j$, then L has no zero on $\Omega \times \prod_j Y_{r_j}[0,b_j]$.*

So a box and its radii keep a letter from vanishing on the set Y of the box when the letter has no zero on its set $\mathcal{O}$, where the ceiling is read, and, at one point of Ω with the other falling coordinates at their radii, none in the disc of each keyhole; the falling variables kept real count among the coordinates of Ω.

Theorem 20.49 (No reach exceeds the ceiling)**.** *For a node of width one, the reach of its bunch is at most the ceiling of every keyhole continuation:* $\inf \varrho_B \le \mathrm{reach}(B) \le H_\mathcal{O}$ *for the ceiling $H_\mathcal{O}$ of each. When ϱ_B is bounded, every keyhole continuation has the ceiling* $\sup \varrho_B$*, and* $\mathrm{reach}(B) = \sup \varrho_B$.

The letters of the contraction enter the keyhole continuations because the family is continued at points off the integers, where $\varrho_\pi^{-b^d}$ branches at their zeros; at the integers $-k$, $k \in \mathbb{N}$, when $s - \kappa_B$ is zero or a negative integer, only the letters of the restriction $\mathcal{B}|B$ matter. When a zero of a letter of the contraction is met by the keyholes that would bring $|\varrho_\pi|$ down, the ceiling can exceed the reach. From below, the reach is read at the integers, where $\psi_{\mathcal{B}/R,\pi}$ does not enter, nor $k_{\mathcal{B}/R,\pi}$ when $s - \kappa_B$ is an integer, and a continuation whose largest values sit at simple tops that do not cancel cannot be lowered.

Definition 20.50 (Integer keyhole continuations; simple tops)**.** For a node of width one let $\nu = s - \kappa_B$. An *integer keyhole continuation* is defined as in Definition 20.47 with only $k_{\mathcal{B}|B,\pi}$ and $\psi_{\mathcal{B}|B,\pi}$, and $k_{\mathcal{B}/R,\pi}$ when ν is not an integer, required not to vanish on the sets Y of its boxes; its ceiling is defined in the same way. For an order π and a box Ω write J for its falling variables with keyholes and r_j for their radii, and let C_j be the circle $|q_j| = r_j$ run once counterclockwise from r_j. The *keyhole contour* of the continuation is the formal sum over the orders and their boxes of

$$\Omega \times \prod_{j \in J} \Big(\frac{C_j}{e^{2\pi i \kappa_j} - 1} + [r_j, 1] \Big),$$

the segment being $[r_j, b_j]$ on a divided side $[0, b_j]$, a falling variable kept real on $[a_j, b_j]$ entering with that segment, and an order without falling variables contributing its cube; on the keyhole contour of π it carries the form ϕ_π, the powers continued along the circles from their positive values at r_j. A *top* is a point of the keyhole contour at which $|\varrho_\pi|$ equals the ceiling. A top p is *simple* when it lies on no face of its box other than faces $q_j = 0$ of flat variables, its falling coordinates lie on the circles away from r_j or inside the segments, and, with $\mathbf{x}$ the coordinates that vanish at p and $\mathbf{y}$ real coordinates of the keyhole contour for the others, vanishing at p (the angle on a circle),

$$\log \frac{\varrho_\pi}{\varrho_\pi(p)} = -\langle \boldsymbol{\lambda}, \mathbf{x} \rangle - \tfrac{1}{2}\, \mathbf{y}^{\mathsf{T}} \mathbf{Q}\, \mathbf{y} + O\big(|\mathbf{x}|^2 + |\mathbf{x}|\,|\mathbf{y}| + |\mathbf{y}|^3\big)$$

on the keyhole contour near p, with $\operatorname{Re} \lambda_j > 0$ for every j and $\operatorname{Re} \mathbf{Q}$ positive definite. The *exponent* of p is $\operatorname{ex}(p) = \sum_j \kappa_j + \frac{1}{2} \dim \mathbf{y}$, the sum over the coordinates $\mathbf{x}$, and its *coefficient* is

$$A(p) = \phi_p \prod_j \Gamma(\kappa_j)\, \lambda_j^{-\kappa_j}\, (2\pi)^{\dim \mathbf{y}/2} (\det \mathbf{Q})^{-1/2},$$

ϕ_p the value at p of the density of ϕ_π with the coefficient $1/(e^{2\pi i \kappa} - 1)$ of each circle of the keyhole contour through p, κ the wall of its falling variable, against $\prod_j x_j^{\kappa_j - 1}\, d\mathbf{x}\, d\mathbf{y}$, the powers and the root continued from real $\boldsymbol{\lambda}$ and real $\mathbf{Q}$ along segments.

The coordinates $\mathbf{x}$ belong to flat variables, whose walls κ_j are positive, so the Gammas are finite.

Theorem 20.51 (A simple top is the growth at the integers)**.** *For a node of width one let the walls κ_j of the falling variables of every order be off the integers, and let* $\operatorname{growth}_{\mathbb{Z}}(B)$ *be the growth at the integers (Definition 20.33).*

(i) For every integer keyhole continuation and every integer $k \geq \max(0, s - \kappa_B)$, the continued period at $b^d = -k$ is the integral of $\varrho_\pi^k \phi_\pi$ over its keyhole contour. So its modulus is at most $C H_{\mathcal{O}}^k$, $H_{\mathcal{O}}$ the ceiling, and $\operatorname{growth}_{\mathbb{Z}}(B) \leq H_{\mathcal{O}}$.

(ii) $\operatorname{reach}(B) \geq \operatorname{growth}_{\mathbb{Z}}(B)$.

(iii) Let an integer keyhole continuation of ceiling $H_{\mathcal{O}}$ have finitely many tops, all simple, let ex_0 *be the least of their exponents, and for $|\zeta| = 1$ let A_ζ be the sum of the coefficients $A(p)$ of the tops with* $\mathrm{ex}(p) = \mathrm{ex}_0$ *and $\varrho_\pi(p) = H_{\mathcal{O}}\zeta$. Each $A(p)$ is nonzero, and each top is a stationary point of the restriction of ϱ_π to the flat face $\mathbf{x} = 0$, the sector itself when there are no coordinates $\mathbf{x}$. If some $A_\zeta \neq 0$, then* $\mathrm{growth}_{\mathbb{Z}}(B) = H_{\mathcal{O}} \leq \mathrm{reach}(B)$*: the reach is at least the modulus of a stationary value of the ratio on a flat face.*

(iv) If moreover there are keyhole continuations with ceilings arbitrarily close to $H_{\mathcal{O}}$, then $\mathrm{reach}(B) = H_{\mathcal{O}}$*. In particular, if a keyhole continuation, with all four letters kept from vanishing, has finitely many tops, all simple, and some $A_\zeta \neq 0$, the reach is its ceiling.*

So $\mathrm{growth}_{\mathbb{Z}}(B) \leq \mathrm{reach}(B)$ always, with equality to the ceiling whenever the tops of an integer keyhole continuation are finitely many, all simple, and do not cancel, and the keyhole continuations come down to it: the reach is then the modulus of a stationary value of the ratio on a flat face, the one at which the best continuation is pinned. The stationary point need not be a positive point of the face.

The upper end needs the letters of the contraction only where they meet the falling faces. Elsewhere a zero of $k_{\mathcal{B}/R,\pi}$ or $\psi_{\mathcal{B}/R,\pi}$ is a zero of ϱ_π, and the law is read near infinity on the fibres $\varrho_\pi = x$ with $|x|$ large, which never pass through it.

Lemma 20.52 (A falling variable takes each large value once)**.** *Let g be meromorphic on a neighbourhood of the closed disc $|q| \leq r$, with a simple pole at 0 as its only pole there, and $|g| \leq M$ on the circle $|q| = r$. Then g takes every value x with $|x| > M$ exactly once in the disc, with $g' \neq 0$ there, and maps the open set $\{|q| < r : \ |g(q)| > M\} \cup \{0\}$ biholomorphically onto $\{|x| > M\} \cup \{\infty\}$.*

Lemma 20.53 (A wall series continued is a wall series)**.** *If the sum of a wall series of radius x_0 continues analytically along every path in $\{|x| > x_1\}$, with $x_1 < x_0$, then the series is a wall series of radius x_1.*

Definition 20.54 (Fibre continuations; stationary bound)**.** A *fibre continuation* is defined as in Definition 20.47, divided sides allowed, with only $k_{\mathcal{B}|B,\pi}$ and $\psi_{\mathcal{B}|B,\pi}$ required not to vanish on the sets Y of its boxes, and $k_{\mathcal{B}/R,\pi}$ and $\psi_{\mathcal{B}/R,\pi}$ only at their points with a falling coordinate 0; its ceiling $H_{\mathcal{O}}$ is defined in the same way. Its *stationary bound* S is the supremum of $|\varrho_\pi|$ over the points of these sets with no falling coordinate 0 at which at least two falling coordinates lie inside their discs, $|q_j| < r_j$, and the derivatives of ϱ_π in all the falling coordinates inside their discs vanish; $S = 0$ when there is no such point.

With one falling coordinate inside its disc and the others on their outlines, ϱ_π is in that coordinate as in Lemma 20.52, with the bound $H_{\mathcal{O}}$ on the circle, or identically 0 when a rising base coordinate is 0; so it has no stationary point there with $|\varrho_\pi| > H_{\mathcal{O}}$, and the stationary bound concerns two or more falling coordinates at once.

Theorem 20.55 (The contraction enters only on the falling faces)**.** *For a node of width one and every fibre continuation, of ceiling $H_{\mathcal{O}}$ and stationary bound S,* $\mathrm{reach}(B) \leq \max(H_{\mathcal{O}}, S)$*. So, under the hypotheses of Theorem 20.51, if a fibre continuation that is also an integer keyhole continuation has $S \leq H_{\mathcal{O}}$ and finitely many tops, all simple, with some $A_\zeta \neq 0$, then* $\mathrm{reach}(B) = \mathrm{growth}_{\mathbb{Z}}(B) = H_{\mathcal{O}}$*: the reach is read at the integers.*

The letters of the contraction, which Theorem 20.49 keeps from vanishing on whole keyholes, matter for the upper bound only on the falling faces, where the coefficient of the pole of the ratio vanishes, and through the stationary points of the ratio in two or more falling variables. The two bounds from above work box by box, so they can share one continuation.

Theorem 20.56 (Keyhole boxes and fibre boxes mix)**.** *Let a continuation assign to every order π of the bunch B of a node of width one boxes and radii as in Definition 20.47, the sides of the falling variables divided or not, each box either a* keyhole box*, on whose sets Y none of the four sector letters vanishes, or a* fibre box*, on whose sets Y the conditions of Definition 20.54 hold. Let $H_{\mathcal{O}}$ be its ceiling and S the stationary bound of Definition 20.54 taken over its fibre boxes alone. Then* $\operatorname{reach}(B) \leq \max(H_{\mathcal{O}}, S)$.

Corollary 20.57 (One falling variable needs only the restriction)**.** *In an order with at most one falling variable, a box on whose sets Y the letters of the restriction $\mathcal{B}|B$ do not vanish is a fibre box, and it adds nothing to the stationary bound. So in a continuation of Theorem 20.56 such boxes need only the letters of the restriction, and the reach is at most* $\max(H_{\mathcal{O}}, S)$ *with S taken over the fibre boxes of the orders with two or more falling variables.*

So the stationary bound enters only where a letter of the contraction vanishes inside the keyholes of a sector with two or more falling variables. The zeros of the contraction on the falling faces can still keep every such bound above the reach.

These theorems bound the right end of each interval $[\inf \varrho_B, \operatorname{reach}(B)]$ of Theorem 20.39 from both sides. The reach need not be read on the positive shape: the ratio is real at the real points of a flat face whatever their signs, and a keyhole of a falling variable crosses the negative axis nearer to 0 than every pole of the ratio on it, so a simple top can sit at a point with a negative coordinate.

20.9 One tree

Read on one tree, the period of a point, built back up, is the periods of the leaves multiplied by the share kernels of the nodes and integrated over the free shares; its $n-1$ integrations are shared out among the bunches and the leaves; at integer data the share kernels bring the Trailmoments of the counting input order by order; and where the free shares close on integer points the period is the sum of the residues along the lattice of the tree.

Definition 20.58 (Share kernels, contents, integration count)**.** Let a molecule of lists with a native gauge be read on a tree of bunches, its Foreman tree or another, with the port of the balance wall run at every node on the native splits (Corollary 20.10). The share kernels of the tree (Corollary 4.4) are what the ports place on the shares: one stacked Gamma $\Gamma_{\substack{k \\ 1}}\begin{bmatrix}\mathbf{a} \\ a\end{bmatrix}$ for each letter opened at a node, k the number of its patterns, $\mathbf{a}$ its shares and a its exponent, which in Corollary 20.10 leave the factors $\big(\Gamma(a_{\mathcal{B}'})\Gamma(b_{\mathcal{B}'})\big)^{1-q_{\mathcal{B}'}}$ and $\prod_j \Gamma(s_j)^{-1}\Gamma(t_j)^{-1}$; each is balanced (Definition 1.2), its shares adding to its exponent on the fibre. The *free shares* are coordinates of the fibre of the word. Let $\mathcal{Z}$ be the $\mathbb{Q}$-algebra generated by the Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $k \geq 2$, of the counting input; it maps into the content $\mathcal{C}^{(1)}$ of Chapter 45, the algebra of trails of lattice inputs, which also holds the sums along the lattices of paired words (Chapters 43 and 45). The *content* $\mathcal{C}_M$ of a childcule M of a node, a leaf in particular, is the $\mathcal{Z}$-module spanned by the values, at the integer exponents and masses at which it converges, of the period of a family on M whose tables are those of M, and of its derivatives in the exponents and masses along the balance wall; the content of the molecule is the same with its own tables. The content $\mathcal{C}_B$ of the bunch B of a node with room R is the same for the families on B whose tables, each with both letters, are the minors $\mathcal{B}'|B$ and $\mathcal{B}'/R$ of the tables $\mathcal{B}'$ at the node: those of the face of B (Theorem 16.25), at width one those of the bunch family (Theorem 19.13). The content $\mathcal{C}_R$ of the room is the same for the families on R whose tables are the $\mathcal{B}'|R$ and $\mathcal{B}'/B$.

The *integration count* of an integral as written is the number of its integrations, assigned to the written expression by these rules. The period of a family on k marks has the integration count $k-1$, its simplex. A trail $(\mathsf{T}_{\mathbf{a};I_d}[y]\,\mathbb{I}_{(b+\mathbb{Z})^d})(\{b<y_1<\cdots<y_d\})$ of a lattice input with integer indices, $b>-1$ (Definition 5.23), has the integration count $a_1+\cdots+a_d$, its total index; the counting trails are the case $b=0$. A Gamma has the integration count one, $\Gamma(a)=\int_0^\infty x^{a-1}e^{-x}\,dx$ being one integration, so a stacked Gamma with k Gammas above and l below has the integration count $k-l$. A derivative in an exponent or a mass adds one, the logarithm of a letter or of a mark. A product has the sum of the integration counts of its factors, and a fibre integral of dimension D (Definition 4.1) the integration count of its integrand less D, each share integrated inverting one Mellin transform. A real number has *integration count at most* w when it is a rational combination of values, at rational data with unit weights, of integrals of integration count at most w.

The openings keep the integration count. The power $(X_1+\cdots+X_k)^{-s}$ is the share integral of $\Gamma_{\substack{k\\1}}\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix}\prod_i X_i^{-a_i}$, of integration count $(k-1)-(k-1)=0$ (Corollary 4.4), and e^{-y} is the integral of $\Gamma(b)y^{-b}$ over one share, of integration count $1-1=0$. A Beta corner $\Gamma_{\substack{2\\1}}\begin{bmatrix}a,\ b\\ a+b\end{bmatrix}$ has the integration count one, as the period of the list on two marks whose bases are the two singletons; the stacked Gamma $\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}$ of the shape integral has the integration count $|B|-1$, its simplex (Theorem 3.9); the interest $J_F=\Gamma(\kappa_{\mathcal{E}})P_F$ has the integration count n, its orthant (Theorem 8.7). The atom formula of a family on n marks (Corollary 18.16) has the integration count $\sum(|\mathcal{B}_j|-1)$ of its stacked Gammas, over the active letters, less its dimension $\sum(|\mathcal{B}_j|-1)-(n-1)$: $n-1$, the integration count of the period. Taking the residue at a pole of one Gamma removes that Gamma and adds one pin (Theorem 4.5), so every term of a closing has the integration count of the integral closed.

Theorem 20.59 (One tree suffices)**.** *Let F be a molecule of lists on n marks with real exponents $s_j,t_j\geq 0$ and masses $\mathbf{m}$ on the balance wall, with $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ convergent, and let its Foreman tree be read at its exponent pairs, at the centre or at a lean (Proposition 16.19, Theorem 14.6). Run the port of the balance wall at every node on the native splits of the point, as in Corollary 20.10. Parts (i), (ii), (iv) and (v) hold as well for every tree of bunches in place of the Foreman tree.*

(i) The word starts at the leaves. *$P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the word of the tree (Corollary 20.10): the periods of the leaves, each read at its shares and each entering once on its descent, multiplied by the share kernels and integrated over the free shares. At a node the words of the childcules multiply, and along a chain, its leaf opened by the atom formula, the word is one Gramma on the marks and the letters (Theorem 20.11).*

(ii) The integrations are counted. *The word has the integration count $n-1$, that of the period, and at rational data $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ has integration count at most $n-1$. At a node with bunch B and q childcules the integrations of the simplex are the $|B|+q-1$ pins of the node, $|B|-1$ for the shape of the bunch and q for the totals of the bunch and of the childcules, $q+1$ numbers that add to one; in a leaf L they are the $|L|-1$ integrations of its period.*

(iii) The leaf bounds the bunches. *Along every descent of the Foreman tree, at the centre and at every lean, the copies of every bunch have at most as many marks as the copies of the leaf (Theorem 14.22). When some active list is connected, as for a molecule of one list, the tree is a chain and $|B_0|\leq\cdots\leq|B_{d-1}|\leq|L|$ (Corollary 20.9): the shape of every bunch has at most as many integrations as the leaf, and its node adds one, the total of the bunch against its room.*

(iv) At integer data the share kernels bring $\mathcal{Z}$. *Let the exponents and masses be integers and let x be a point of the fibre at which every share is an integer. The pins and the share sums have integer coefficients, so the directions of the fibre have a rational basis; write the fibre near x as $x + V\mathbf{v}$ with V rational. Near x the share kernels are*

$$c \prod_i \ell_i(\mathbf{v})^{-\epsilon_i} \exp h(\mathbf{v}),$$

with c rational, $\ell_i(\mathbf{v})$ the deviation from its value at x of each argument of the share kernels that is a nonpositive integer at x, ϵ_i equal to 1 for a Gamma above and to -1 for one below, and h a power series without constant term whose part of degree k has its coefficients in $\mathbb{Q} + \mathbb{Q}\,(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ for $k \geq 2$ and in $\mathbb{Q}$ for $k = 1$: the terms of order k of the share kernels carry total index at most k. Where the periods of the leaves converge at x, every residue of the word at x, taken in turn along rational directions spanning the fibre, is therefore a finite $\mathcal{Z}$-combination of products, over the leaves, of elements of their contents.

(v) Summed along the tree. *A free share closed on the poles of the share kernels, where they alone have poles on the way and the closing converges, writes the word as a sum over a lattice of terms of its own integration count, each a fibre integral with one more pin (Theorem 4.5); where every free share closes so, on points at which every share is an integer and the periods of the leaves converge, the period at integer data is the sum along that lattice of the residues of (iv), finite $\mathcal{Z}$-combinations of products of the contents of the leaves. Across a bunch of width one, grouping its patterns by level puts the period of the bunch family in the word in place of the bunch (Theorem 19.13); for one table the even share then closes on its lattice whenever $\sup \varrho_B < \inf \varrho_R$ or $\inf \varrho_B > \sup \varrho_R$, and at integer data its residues are finite $\mathcal{Z}$-combinations of products of an element of $\mathcal{C}_B$ and one of $\mathcal{C}_R$ (Corollary 19.21).*

Whatever the order of the even letters, with $s, t > 0$, the even share of a node of width one closes on the lattices of continued periods of the bunch family and the room family toward a level at which their growths multiply to less than one (Theorem 20.34); a series bunch closes toward the level d against every room with $\inf \varrho_R > \inf \varrho_B$, and a series connection against every room with $\sqrt{\inf \varrho_R}$ above the sum of the square roots of the reaches of its parts, the room's ratio bounded only on the side away from the closing (Theorem 20.40); and against a room bounded below and a bunch bounded below the node closes on a geometric lattice exactly when the intervals of the bunch and of the room are disjoint, at its peaks otherwise (Theorem 20.39). Off the integers the level shares of several tables close in the same way across bunches of width one whose values stand in one order, node after node down a tree of splits of width one (Theorem 19.20, Corollary 19.22). At nonnegative exponents the period is in every case the sum, along the draws nested on the descents of the tree, of products over its blocks of Dirichlet moments (Theorem 20.15).

The pins need not be unimodular, so a closing can reach points at which some share is not an integer, even at integer data. When the active tables link all the marks the pins reach every integer point of their target (Proposition 18.17), and the fibre has integral coordinates through an integer point; the shares that a closing leaves after pinning the others at the poles of their Gammas are solved from a square block of the pins, whose determinant can exceed 1 in modulus: for the list $\binom{[4]}{2}$ at the data $(\mathbf{1}; 1, 1)$ (Exercise 49.3), the fibre has points at which every share off such a block is an integer and some share is $\frac{1}{2}$. There the share kernels have rational arguments, and their expansion runs in heads and Trailmoments of the lattice inputs of those arguments (Proposition 5.13); that is why (iv) and (v) speak of integer points. Where a node has no free share its sum has one term. For a list whose bunch has width one, at a full lean, $(s, 0)$ or $(0, t)$, the level shares are pinned and the

period is a Beta corner times the periods of the bunch family and the room family (Corollary 19.14); at rational data this is an identity of period data (Theorem 19.34), at integer data the Beta corner is rational, and with the room a leaf the period is a rational multiple of a product of an element of $\mathcal{C}_B$ and one of the content of the leaf. When the two hinge values of a bunch of width one agree the shape and the room separate at every pair of exponents (Proposition 19.12), and at the wall of a set, with the weights factoring on its bottom levels, the residue of the period is the product of the periods of its two faces (Theorem 16.25).

The integration count of a period bounds that of its value, and the theorem is stated for lists because the weights of a table can carry integrations of their own, which the integration count of a period leaves out. With unit weights and rational data nothing is carried in. On an atom the tree is the atom and the theorem says no more than its own content; its atom formula is a word of the same kind whose leaves are single marks, and the argument of (iv) puts each of its residues at integer points in $\mathcal{Z}$. The lattice can make what no bunch or leaf shows: a tree whose bunches and leaves have their contents in $\mathcal{Z}$ can sum, along the lattice of one free share, to a Trailmoment of a lattice input with a frequency (Definition 5.4) that none of them carries. The finite form of the theorem, the period in the $\mathcal{Z}$-algebra generated by the contents of the bunches and the leaves, holds where the shares are pinned at an integer point; where they are free the tree gives the sum along its lattice. For a word of one free share made of Gammas alone, as the atom formula leaves it at integer data, that sum lies in the content of Definition 45.1 when the word is paired (Propositions 43.2 and 45.3), and otherwise it is a combination of Trailmoments of live inputs (Theorem 43.1).

Theorem 20.59 is the Foreman procedure read, step by step, as a statement about numbers. *Split first*: a compound is split into its polycule before any molecule is asked whether it answers yes, and at every node the polycule of the childcules is evaluated as the product of its molecules (Theorem 20.11(i)). *Read at the table's gauge, then ask*: each molecule is read at its table's gauge, native for the masses it uses and inherited by its children (Proposition 12.27), whether it answers yes is read at the centre (Theorem 11.8), and a realization that is not atomic is peeled. *Peel small*: the peel is taken at a class with the fewest marks, it splits a class only across twin childcules, each part inside one class (Theorem 14.20), so the leaf with its copies holds whole classes of every node above it and is at least as large as every bunch with its copies (Theorem 14.22). *The share kernels reapply*: what a node adds is its port, stacked Gammas of the shares of its bunch and the pins of its shape and its totals; near integer data a balanced stacked Gamma brings $\mathcal{Z}$ order by order in total index, and in the draw series each node contributes its own Dirichlet moments to every product over the blocks. *One tree*: every tree of bunches computes the same period and shares out the same $n-1$ integrations, and the Foreman tree is one in which the copies of no bunch have more marks than the copies of the leaf, and when some active list is connected no bunch has more marks than the leaf. *Pinned shares*: where the shares of a node are pinned, as for one list at a full lean across a bunch of width one, the node contributes an identity of period data (Theorem 19.34).

Chapter 34 lifts the statements of this section from periods to motivic periods: at integer data the left factors of the coaction of the motivic period of a family are read along every tree of bunches, the Foreman tree among them, through the steps attached to its vertices (Theorem 34.13); at its nodes, at any width, through the joint spectrum of the bunch (Theorem 34.16); and where the shares of a node are pinned the identities of period data above are identities of motivic periods (Theorem 34.17, Corollary 34.18).

Part V

Construction and composition

Chapter 21

Gluing and what crosses it

Every table read at a bunch is the gluing of what its bunch and its room carry (Proposition 12.5), and at a bunch of width one the gluing is along one mark, the hinge (Theorem 19.4). Gluing is an operation with its own calculus. A table is built from cells, tables with one marked hinge, substituted into the marks of a core; the bases, the weights, the multivectors, the letters and the walls of the result are read off the cells and the core; the blocks of Theorem 21.5 and the series connections of Section 20.7 are the cells and cores of the simplest kinds; two cell families glued along their hinges have hinge walls that add to 2σ on the balance wall; and every connected table whose weights factor at its bunches of width one, every list and every determinantal table among them, is a tree of such substitutions whose nodes are prime tables and series and parallel nodes. Cells with several hinges splice in pairs along any graph of pairs, closed paths included, and stay tables, their channels composing multilinearly. What a room receives from its bunch, at any width, for any family and any multivector, is the joint law of finitely many bunch letters: as many as the ranks of the level weights, which for a multivector are bounded by the squares of its interface ranks and for a determinantal table by $\binom{c(B)}{i}^2$ at the level $d(B)+i$, whatever the numbers of marks; for a determinantal table the bunch crosses its gluing as one positive definite matrix of size $c(B)$, whose determinant at width one is the hinge value γ_B^K, and one scalar.

21.1 Gluing along a mark

Definition 21.1 (Cells, cores, substitution)**.** A *cell* is a table $\mathcal{A}$ on $B \sqcup \{g\}$ in which the mark g, its *hinge*, is neither a loop nor a coloop; it may carry two systems of positive weights on its bases, θ^K for its letter $K_{\mathcal{A}}$ and θ^Ψ for its letter $\Psi_{\mathcal{A}}$ (Definition 8.1), and it is *plain* when they agree, as for one table. Its *hinge minors* are the deletion $\mathcal{A} \setminus g$ and the contraction $\mathcal{A}/g$, tables on B of ranks $r_{\mathcal{A}}$ and $r_{\mathcal{A}} - 1$ with the inherited weights (Definition 7.3), so that

$$K_{\mathcal{A}} = K_{\mathcal{A}\setminus g} + \alpha_g K_{\mathcal{A}/g}, \qquad \Psi_{\mathcal{A}} = \alpha_g \Psi_{\mathcal{A}\setminus g} + \Psi_{\mathcal{A}/g};$$

its hinge values and its ratio, as for a bunch of width one (Definition 19.2), are

$$\gamma_{\mathcal{A}}^K = \frac{K_{\mathcal{A}\setminus g}}{K_{\mathcal{A}/g}}, \qquad \gamma_{\mathcal{A}}^\Psi = \frac{\Psi_{\mathcal{A}/g}}{\Psi_{\mathcal{A}\setminus g}}, \qquad \varrho_{\mathcal{A}} = \frac{\gamma_{\mathcal{A}}^\Psi}{\gamma_{\mathcal{A}}^K} = \frac{\mathbf{E}_{\mathcal{A}/g}}{\mathbf{E}_{\mathcal{A}\setminus g}},$$

functions of α_B homogeneous of the degrees 1, 1 and 0. The cell of a bunch $\mathcal{B}_B^+$ of Theorem 19.4 is a cell, with the hinge minors $\mathcal{B}|B$ and $\mathcal{B}/R$, the hinge values γ_B^K, γ_B^Ψ and the ratio ϱ_B of Definition 19.2. A *core* is a table $\mathcal{H}$ on marks $h_1, \dots, h_q$, none of them a loop or a coloop, plain or

with two systems of weights as a cell. For cells $\mathcal{A}_i$ on $B_i \sqcup \{g_i\}$, the B_i disjoint, the *substitution* $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ is the family of sets $\bigsqcup_i X_i \subseteq \bigsqcup_i B_i$ for which some basis T of $\mathcal{H}$ has $X_i \in \mathcal{A}_i \setminus g_i$ when $h_i \in T$ and $X_i \in \mathcal{A}_i/g_i$ when $h_i \notin T$, with the weight, in each letter,

$$\theta_{\sqcup_i X_i} = \theta^{\mathcal{H}}_T \prod_{h_i \in T} \theta^{\mathcal{A}_i}_{X_i} \prod_{h_i \notin T} \theta^{\mathcal{A}_i}_{X_i + g_i},$$

every θ read as the weight θ^K of the letter K throughout, or as θ^Ψ throughout; T is determined by the sizes, $h_i \in T$ exactly when $|X_i| = r_{\mathcal{A}_i}$. A mark h is *left in place* by the cell $\binom{\{h,g\}}{1}$, the list on $\{h, g\}$ with the bases $\{h\}$ and $\{g\}$, whose hinge minors have the single bases $\{h\}$ and $\varnothing$ and whose hinge values are both α_h. A plain cell is a *dual pair* when its contraction $\mathcal{A}/g$ is the dual table of its deletion $\mathcal{A} \setminus g$, weights included: the bases of $\mathcal{A}/g$ are the $B \setminus X$, X a basis of $\mathcal{A} \setminus g$, with $\theta^{\mathcal{A}}_{(B\setminus X)+g} = \theta^{\mathcal{A}}_X$; a mark left in place is one. One cell $\mathcal{A}$ at one mark h of $\mathcal{H}$, the other marks left in place, is the *gluing of $\mathcal{A}$ and $\mathcal{H}$ along h*: the gluing (Definition 12.4) of the communicatroid of B whose children are $\theta^{\mathcal{A}}_X$ times $\mathcal{H}/h$ at the bases X of $\mathcal{A} \setminus g$ and $\theta^{\mathcal{A}}_{X+g}$ times $\mathcal{H} \setminus h$ at the bases X of $\mathcal{A}/g$. A family is *substituted* table by table, $(\mathcal{B}_j, \theta_j) = \mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$, the cells of all tables at the mark h_i sharing the marks $B_i \sqcup \{g_i\}$; its masses are on $\mathcal{E} = \bigsqcup_i B_i$.

Lemma 21.2 (Gluing along a mark gives a table)**.** *Let $\mathcal{A}$ be a cell on $B \sqcup \{g\}$ and $\mathcal{A}'$ a cell on $R \sqcup \{g'\}$. The sets $X \sqcup Y$ with $X \in \mathcal{A} \setminus g$ and $Y \in \mathcal{A}'/g'$, or $X \in \mathcal{A}/g$ and $Y \in \mathcal{A}' \setminus g'$, are the bases of a table on $B \sqcup R$ of rank $r_{\mathcal{A}} + r_{\mathcal{A}'} - 1$, in which B is a bunch of width one whose cell of the bunch and cell of the room (Theorem 19.4) are $\mathcal{A}$ and $\mathcal{A}'$, the hinges read as the one mark g.*

Proposition 21.3 (Substitution)**.** *(i) $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ is a table of rank $r_{\mathcal{H}} + \sum_i (r_{\mathcal{A}_i} - 1)$; it is the gluing of the cells to $\mathcal{H}$ along its marks one at a time, in any order; each B_i is a bunch of it of width one whose cell of the bunch is $\mathcal{A}_i$, its weights factoring across B_i and its room.*

(ii) Substituting into a mark of a cell is substituting the substituted cell: for a mark x of B_1 and a cell $\mathcal{A}'$ at x,

$$\big(\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle\big)\langle \mathcal{A}' \text{ at } x\rangle = \mathcal{H}\big\langle \mathcal{A}_1\langle \mathcal{A}' \text{ at } x\rangle, \mathcal{A}_2, \dots, \mathcal{A}_q\big\rangle.$$

(iii) Conversely, a table with a bunch B of width one, the weights factoring across B and R, is the gluing of its cell of the bunch $\mathcal{B}^+_B$ and its cell of the room $\mathcal{B}^+_R$ along g (Theorem 19.4): every gluing along a mark is that decomposition read backwards, and every such decomposition is a gluing.

(iv) If $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ is connected, so are $\mathcal{H}$ and every cell. A connected table with a bunch B of width one, $|B|, |R| \geq 2$, the weights factoring across B and R, is the gluing along a mark of two connected tables on fewer marks.

Theorem 21.4 (The letters of a substitution)**.** *Let K and Ψ be the letters of $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$, each with its own weights, and read the hinge minors and the hinge values of each cell on its own marks. Then*

$$K(\alpha) = \prod_i K_{\mathcal{A}_i/g_i}(\alpha_{B_i})\ K_{\mathcal{H}}\big(\gamma^K_{\mathcal{A}_1}(\alpha_{B_1}), \dots, \gamma^K_{\mathcal{A}_q}(\alpha_{B_q})\big),$$
$$\Psi(\alpha) = \prod_i \Psi_{\mathcal{A}_i\setminus g_i}(\alpha_{B_i})\ \Psi_{\mathcal{H}}\big(\gamma^\Psi_{\mathcal{A}_1}(\alpha_{B_1}), \dots, \gamma^\Psi_{\mathcal{A}_q}(\alpha_{B_q})\big).$$

For one cell, the other marks left in place, this is Theorem 19.4.

The core reads each of its cells as one mark, at the hinge value of the cell by the letter: in K the mark h_i stands at $\gamma^K_{\mathcal{A}_i}$ and in Ψ at $\gamma^\Psi_{\mathcal{A}_i}$, and what is left over is a product of one letter of a hinge minor of every cell. Nothing in the core sees more of a cell than its two hinge values.

A family can be generated rather than found: seed tables assembled under a set of bijections give atoms (Proposition 11.6), and a mark that is neither a loop nor a coloop, replaced by a block of new marks, gives a bunch of width one, which the room reads as the mark it replaced.

Theorem 21.5 (Replacement of marks by blocks)**.** *Let $(\mathcal{B},\theta)$ be a table of rank r on $\mathcal{E}=\mathcal{E}_1\sqcup\mathcal{E}_2$, and replace every mark h by a* block *X_h of $q_h\geq 1$ new marks, by the* first rule *at the marks of $\mathcal{E}_1$ and by the* complementary rule*, the first rule read on the dual table, at those of $\mathcal{E}_2$: a basis T gives the sets that contain one mark of X_h for each $h\in T\cap\mathcal{E}_1$, none for $h\in\mathcal{E}_1\setminus T$, all of X_h for $h\in T\cap\mathcal{E}_2$ and all of X_h but one for $h\in\mathcal{E}_2\setminus T$, each with the weight of T. They are the bases of a table on $\bigsqcup_h X_h$ of rank $r+\sum_{h\in\mathcal{E}_2}(q_h-1)$; a block of one mark leaves its mark in place, and when no mark is a loop or a coloop the table is the substitution into $(\mathcal{B},\theta)$ of the* parallel blocks $\binom{X_h\sqcup\{g_h\}}{1}$ *at $\mathcal{E}_1$ and the* series blocks $\binom{X_h\sqcup\{g_h\}}{q_h}$ *at $\mathcal{E}_2$ (Definition 21.1). Its Foreman polynomials are, each e_k read on the marks of its block,*

$$K'=\prod_{h\in\mathcal{E}_2}e_{q_h-1}\cdot K\Big(\big(e_1\big)_{h\in\mathcal{E}_1},\big(\tfrac{e_{q_h}}{e_{q_h-1}}\big)_{h\in\mathcal{E}_2}\Big),\qquad \Psi'=\prod_{h\in\mathcal{E}_1}e_{q_h-1}\cdot\Psi\Big(\big(\tfrac{e_{q_h}}{e_{q_h-1}}\big)_{h\in\mathcal{E}_1},\big(e_1\big)_{h\in\mathcal{E}_2}\Big);$$

for one block $X=\{x_1,\dots,x_q\}$ at the mark h, the other marks left in place, the first rule gives $K'=K_{\setminus h}+e_1(\alpha_X)\,K_{/h}$ and $\Psi'=e_q(\alpha_X)\,\Psi_{\setminus h}+e_{q-1}(\alpha_X)\,\Psi_{/h}$, and the complementary rule gives $K'=e_q(\alpha_X)\,K_{/h}+e_{q-1}(\alpha_X)\,K_{\setminus h}$ and $\Psi'=\Psi_{/h}+e_1(\alpha_X)\,\Psi_{\setminus h}$ (Definition 7.3). In scale and shape, $\alpha_X=\rho u$ with $u\in\Delta_X$: $e_1(\alpha_X)=\rho$, $e_q(\alpha_X)=\rho^q e_q(u)$, $e_{q-1}(\alpha_X)=\rho^{q-1}e_{q-1}(u)$, and $\alpha_X^{\mathbf{m}|_X-\mathbf{1}}d\alpha_X=\rho^{m_X-1}d\rho\;u^{\mathbf{m}|_X-\mathbf{1}}d\omega_X(u)$ (Lemma 2.7).

Theorem 21.6 (Counts, walls and atoms under replacement)**.** *Let $(\mathcal{B},\theta)$ be a table of rank r on $\mathcal{E}$, $n=|\mathcal{E}|$, and replace every mark h by a block X_h of $q_h\geq 1$ new marks as in Theorem 21.5, by the first rule at the marks of $\mathcal{E}_1$ and by the complementary rule at those of $\mathcal{E}_2$, $\mathcal{E}=\mathcal{E}_1\sqcup\mathcal{E}_2$.*

(i) The marks become $\sum_h q_h$; the rank becomes $r+\sum_{h\in\mathcal{E}_2}(q_h-1)$; the total weight becomes

$$\sum_{T\in\mathcal{B}}\theta_T\prod_{h\in T\cap\mathcal{E}_1}q_h\prod_{h\in\mathcal{E}_2\setminus T}q_h=\prod_{h\in\mathcal{E}_2}q_h\cdot K\Big((q_h)_{h\in\mathcal{E}_1},(q_h^{-1})_{h\in\mathcal{E}_2}\Big),$$

which is $K((q_h)_h)$ under the first rule alone and $\Psi((q_h)_h)$ under the complementary rule alone.

(ii) For a set Y of new marks let $\lceil Y\rceil=\{h:\ X_h\cap Y\neq\varnothing\}$ and $\lfloor Y\rfloor=\{h:\ X_h\subseteq Y\}$, and let r', d', $d^{*\prime}$, c' be the counts of the new table (Definition 7.12). Then

$$d'(Y)=\sum_{h\in\mathcal{E}_2}\max\big(|X_h\cap Y|-1,0\big)+d\big((\lfloor Y\rfloor\cap\mathcal{E}_1)\sqcup(\lceil Y\rceil\cap\mathcal{E}_2)\big),$$
$$d^{*\prime}(Y)=\sum_{h\in\mathcal{E}_1}\max\big(|X_h\cap Y|-1,0\big)+d^*\big((\lceil Y\rceil\cap\mathcal{E}_1)\sqcup(\lfloor Y\rfloor\cap\mathcal{E}_2)\big),$$

and $c'(Y)=|Y|-d'(Y)-d^{*\prime}(Y)$. Under the first rule alone $r'(Y)=r(\lceil Y\rceil)$, $d'(Y)=r-r(\mathcal{E}\setminus\lfloor Y\rfloor)$ and $c'(Y)=r(\lceil Y\rceil)+r(\mathcal{E}\setminus\lfloor Y\rfloor)-r$; under the complementary rule alone $c'(Y)=|\lceil Y\rceil|-|\lfloor Y\rfloor|+r(\mathcal{E}\setminus\lceil Y\rceil)+r(\lfloor Y\rfloor)-r$. If $Y=\bigcup_{h\in S}X_h$, then $c'(Y)=c(S)$; when the seed table is connected on at least two marks, a proper nonempty part of one block has width one.

(iii) Let a family of seed tables with positive exponent pairs be connected on at least two marks and carried to itself by a set H of bijections as in Proposition 11.6, and replace every mark by a block of the same q marks, all by the first rule or all by the complementary rule. Then the bijections of H lift to the blocks, the permutations inside a block carry every replaced table to itself, the replaced family is connected, and for every q its realization at the table's gauge is an atom.

The shape of a block is integrated by Theorem 19.8, and its scale is the mark of the seed. A block X_h of Theorem 21.5 with $q_h \geq 2$ whose mark is neither a loop nor a coloop of the seed is a parallel bunch of width one under the first rule and a series bunch of width one under the complementary rule (Definition 19.1).

Corollary 21.7 (Blocks and closures)**.** *(i) The series block on a set B of ℓ marks (Theorem 21.5) is the cell $\binom{B\sqcup\{g\}}{\ell}$, whose hinge minors have $K_{\backslash g} = e_\ell(\alpha_B)$, $K_{/g} = e_{\ell-1}(\alpha_B)$, $\Psi_{\backslash g} = 1$ and $\Psi_{/g} = e_1(\alpha_B)$; the parallel block is the cell $\binom{B\sqcup\{g\}}{1}$, with $K_{\backslash g} = e_1(\alpha_B)$, $K_{/g} = 1$, $\Psi_{\backslash g} = e_{\ell-1}(\alpha_B)$ and $\Psi_{/g} = e_\ell(\alpha_B)$. Series blocks at the marks h_i, $i \in S$, of a core $\mathcal{H}$ and parallel blocks at the others replace the marks of $\mathcal{H}$ by blocks as in Theorem 21.5, the series blocks by its complementary rule and the parallel blocks by its first rule, and give the letters of that theorem, each e_k read on the marks of its block: a series block is seen by Ψ through its total and by K through the hinge value $e_\ell/e_{\ell-1} = \big(\sum_x \alpha_x^{-1}\big)^{-1}$, a parallel block the other way round.*

(ii) The series closure *of cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ is the substitution $\binom{\{h_1,\dots,h_q\}}{q-1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ into the core of unit weights; its letters are*

$$K = \prod_i K_{\mathcal{A}_i\backslash g_i} \cdot \sum_i \frac{K_{\mathcal{A}_i/g_i}}{K_{\mathcal{A}_i\backslash g_i}}, \qquad \Psi = \prod_i \Psi_{\mathcal{A}_i\backslash g_i} \cdot \sum_i \frac{\Psi_{\mathcal{A}_i/g_i}}{\Psi_{\mathcal{A}_i\backslash g_i}},$$

so that the bunch $B_1 \sqcup \cdots \sqcup B_{q-1}$ is the series connection of the parts B_i with the tables $\mathcal{A}_i \setminus g_i$ and $\mathcal{A}_i/g_i$ (Proposition 20.37, with $C = C' = 1$). The parallel closure $\binom{\{h_1,\dots,h_q\}}{1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ *has the letters $K = \prod_i K_{\mathcal{A}_i/g_i} \cdot \sum_i K_{\mathcal{A}_i\backslash g_i}/K_{\mathcal{A}_i/g_i}$ and $\Psi = \prod_i \Psi_{\mathcal{A}_i/g_i} \cdot \sum_i \Psi_{\mathcal{A}_i\backslash g_i}/\Psi_{\mathcal{A}_i/g_i}$; it is the dual of the series closure of the dual cells.*

(iii) The gluing of two cells along their hinges is $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}_1, \mathcal{A}_2\rangle$, with the letters $K = K_{\mathcal{A}_1\backslash g_1} K_{\mathcal{A}_2/g_2} + K_{\mathcal{A}_1/g_1} K_{\mathcal{A}_2\backslash g_2}$ and $\Psi = \Psi_{\mathcal{A}_1/g_1}\Psi_{\mathcal{A}_2\backslash g_2} + \Psi_{\mathcal{A}_1\backslash g_1}\Psi_{\mathcal{A}_2/g_2}$.

Proposition 21.8 (Dual pairs)**.** *Let $\mathcal{A}$ be a plain cell on $B \sqcup \{g\}$ of rank $r_\mathcal{A}$ that is a dual pair (Definition 21.1), and put $k = K_{\mathcal{A}\backslash g}$ and $\psi = \Psi_{\mathcal{A}\backslash g}$.*

(i) A plain cell is a dual pair exactly when its bases are closed under complement with equal weights; then $|B| = 2r_\mathcal{A} - 1$, so that k has the degree $r_\mathcal{A}$ and ψ the degree $r_\mathcal{A} - 1$.

(ii) $K_{\mathcal{A}/g} = \psi$ and $\Psi_{\mathcal{A}/g} = k$; the two hinge values agree, $\gamma_\mathcal{A}^K = \gamma_\mathcal{A}^\Psi = k/\psi$, and $\varrho_\mathcal{A} = 1$.

(iii) Let $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ be a substitution of dual pairs into a plain core $\mathcal{H}$, of rank $r_\mathcal{H}$, with k_i and ψ_i as above and $\pi = \psi_1 \cdots \psi_q$; a mark left in place is a dual pair with $k_i = \alpha_h$ and $\psi_i = 1$. Then, as identities of polynomials, with $\pi k_i/\psi_i = k_i \prod_{l\neq i} \psi_l$ and K, Ψ the letters of the substitution,

$$K_\mathcal{H}\big((\pi k_i/\psi_i)_i\big) = \pi^{\,r_\mathcal{H}-1} K, \qquad \Psi_\mathcal{H}\big((\pi k_i/\psi_i)_i\big) = \pi^{\,q-r_\mathcal{H}-1}\Psi.$$

Proposition 21.9 (The multivector of a substitution)**.** *Let $\mathcal{H}$ and the cells carry multivectors $\boldsymbol{\omega}_{\mathcal{H}}$ and $\boldsymbol{\omega}_i$ (Definition 7.20), the hinge first in the order of the marks of each cell, and order the marks of the substitution by putting B_i, in its order in $\mathcal{A}_i$, in the place of h_i in the order of $\mathcal{H}$. Then*

$$\boldsymbol{\omega}\Big(\bigsqcup_i X_i\Big) = \boldsymbol{\omega}_{\mathcal{H}}(T) \prod_{h_i\in T} \boldsymbol{\omega}_i(X_i) \prod_{h_i\notin T} \boldsymbol{\omega}_i(g_i + X_i)$$

is a multivector of $\mathcal{H}\langle\mathcal{A}_1,\dots,\mathcal{A}_q\rangle$; it is multilinear in $(\boldsymbol{\omega}_{\mathcal{H}},\boldsymbol{\omega}_1,\dots,\boldsymbol{\omega}_q)$, a substitution of decomposable multivectors is decomposable, and

$$\mathrm{rk}(\boldsymbol{\omega}) \le \mathrm{rk}(\boldsymbol{\omega}_{\mathcal{H}}) \prod_i \mathrm{rk}(\boldsymbol{\omega}_i).$$

A substitution of determinantal tables is determinantal.

Theorem 21.10 (The walls of a substitution)**.** *For $S = \bigsqcup_i S_i$, $S_i \subseteq B_i$, let $D(S)$ be the set of the marks h_i of the core with $d_{\mathcal{A}_i}(S_i+g_i) > d_{\mathcal{A}_i}(S_i)$ and $D^*(S)$ that of the h_i with $d^*_{\mathcal{A}_i}(S_i+g_i) > d^*_{\mathcal{A}_i}(S_i)$, the two differences being 0 or 1. The counts of the substitution are*

$$d(S) = \sum_i d_{\mathcal{A}_i}(S_i) + d_{\mathcal{H}}\big(D(S)\big), \qquad d^*(S) = \sum_i d^*_{\mathcal{A}_i}(S_i) + d^*_{\mathcal{H}}\big(D^*(S)\big),$$

and for a substituted family, with $D_j(S)$ and $D^*_j(S)$ read in table j, its walls are

$$\kappa_S = \sum_i \Big(m_{S_i} - \sum_j \big(s_j d_{\mathcal{A}_{ij}}(S_i) + t_j d^*_{\mathcal{A}_{ij}}(S_i)\big)\Big) - \sum_j \Big(s_j d_{\mathcal{H}_j}\big(D_j(S)\big) + t_j d^*_{\mathcal{H}_j}\big(D^*_j(S)\big)\Big).$$

At $S = B_i$, $D(S) = D^*(S) = \{h_i\}$ and the wall is $\kappa_{B_i} = m_{B_i} - \sum_j \big(s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big)$, the mass that pooling gives the hinge of B_i (Theorem 19.8).

Corollary 21.11 (The width of the marks left in place)**.** *Let a substitution have the core $\mathcal{H}$ on $L \sqcup H'$, the marks of L left in place and cells with at least two marks at the marks of H'. Every $S \subseteq L$ has the width it has in the core, and*

$$c(S) = c_{\mathcal{H}}(S) \le |L \setminus S| + |H'|.$$

So a bunch of width c that is all the marks left in place needs at least c other cells, and every width occurs: in $\binom{B\sqcup\{h_1,\dots,h_c\}}{c}$ with $|B| = c$, the marks of B left in place, B has the width c, whatever the cells at $h_1,\dots,h_c$.

Lemma 21.12 (Two-sums of cells; hinge walls)**.** *Let $\mathcal{A}$ be a cell on $B \sqcup \{g\}$ and $\mathcal{A}'$ a cell on $R \sqcup \{g\}$ with the one hinge g, $B \cap R = \varnothing$, and let $\mathcal{A} \oplus_g \mathcal{A}'$ be their gluing along the hinge, $\binom{\{h_1,h_2\}}{1}\langle\mathcal{A},\mathcal{A}'\rangle$ (Corollary 21.7(iii)), each basis carrying in each letter the product of the weights of the two bases of $\mathcal{A}$ and $\mathcal{A}'$ it is made of: a table on $B \sqcup R$ of rank $r_{\mathcal{A}} + r_{\mathcal{A}'} - 1$, plain when $\mathcal{A}$ and $\mathcal{A}'$ are, in which B is a bunch of width one with the cell of the bunch $\mathcal{A}$ and the cell of the room $\mathcal{A}'$ (Lemma 21.2); every table with a bunch of width one, the weights factoring across it, is the two-sum of its two cells (Proposition 21.3(iii)). Let the cell families $F_B = ((\mathcal{A}_j, \theta^{\mathcal{A}_j}); s_j, t_j)_{j=1}^N$ on $B \sqcup \{g\}$ and $F_R = ((\mathcal{A}'_j, \theta^{\mathcal{A}'_j}); s_j, t_j)_{j=1}^N$ on $R \sqcup \{g\}$ be glued table by table into the family $F = ((\mathcal{A}_j \oplus_g \mathcal{A}'_j); s_j, t_j)_j$ on $B \sqcup R$, with masses $\mathbf{m}$. The walls of F at B and at R are the hinge walls of the two cell families, for one table those of the two cells, $\kappa_{\mathcal{A}}$ and $\kappa_{\mathcal{A}'}$,*

$$\kappa_{F_B} = m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big), \qquad \kappa_{F_R} = m_R - \sum_j \big(s_j(r_{\mathcal{A}'_j} - 1) + t_j(|R| - r_{\mathcal{A}'_j})\big),$$

and $\kappa_{F_B} + \kappa_{F_R} = 2\sigma + \kappa_{\mathcal{E}}$, σ the total strength and $\kappa_{\mathcal{E}}$ the imbalance of F (Definition 8.1): on the balance wall the two hinge walls add to $2\sigma = \sum_j (s_j + t_j)$, and the mass $2\sigma - \kappa_{F_B}$ on g puts the cell family F_B itself on its balance wall.

Each letter of a two-sum is a letter of a hinge minor of one cell times the letter of the other cell read at one hinge value of the first (Theorem 21.4): K sees $\mathcal{A}$ through $\gamma_{\mathcal{A}}^{K}$ and Ψ through $\gamma_{\mathcal{A}}^{\Psi}$. The hinge wall of a cell is a number of the cell alone, its masses and its exponents, and balance makes the two hinge walls of a two-sum complementary; the period of a two-sum is read through them in Theorem 23.9.

21.2 Splicing hinges in pairs

A cell has one hinge. Gluing along several marks at once, or around a closed path of cells, needs cells with several hinges and a splicing of hinges in pairs. The splicing stays inside tables because the levels of a bunch are tables (Proposition 12.17); one pair joining two cells is the gluing of Lemma 21.2.

Definition 21.13 (Cells with several hinges; channels)**.** A *cell with the hinges* G is a table $\mathcal{A}$ on $B \sqcup G$ with two systems of positive weights θ^K, θ^Ψ on its bases; $n_{\mathcal{A}} = |B|$ is its number of marks other than its hinges. Its *channels* are the letters of the children of the bunch G (Definition 12.1): for a pattern $p \in \mathrm{pat}(G)$, the basis polynomial $K_{\mathcal{A},(p)}$ of the child at p with the weights θ^K and its cobasis polynomial $\Psi_{\mathcal{A},(p)}$ with the weights θ^Ψ, polynomials on B, so that by the Taylor cube (Theorem 12.7)

$$K_{\mathcal{A}} = \sum_{p \in \mathrm{pat}(G)} \alpha_G^p \, K_{\mathcal{A},(p)}(\alpha_B), \qquad \Psi_{\mathcal{A}} = \sum_{p \in \mathrm{pat}(G)} \alpha_G^{G \setminus p} \, \Psi_{\mathcal{A},(p)}(\alpha_B) :$$

p is the set of hinges a basis occupies, in both letters. A channel has positive coefficients on its support and the degree

$$\deg K_{\mathcal{A},(p)} = r_{\mathcal{A}} - |p|, \qquad \deg \Psi_{\mathcal{A},(p)} = n_{\mathcal{A}} - r_{\mathcal{A}} + |p|.$$

The *channel vector* $\mathrm{ch}(\mathcal{A})$ lists the channels by their *tags*, the pairs of a letter and a set $p \subseteq G$, $2^{|G|+1}$ of them; a tag whose set is not a pattern is an absent channel, counted as zero. A cell of Definition 21.1 is the case of one hinge g that is neither a loop nor a coloop, with $K_{\mathcal{A},(\varnothing)} = K_{\mathcal{A}\setminus g}$, $K_{\mathcal{A},(\{g\})} = K_{\mathcal{A}/g}$, $\Psi_{\mathcal{A},(\varnothing)} = \Psi_{\mathcal{A}\setminus g}$ and $\Psi_{\mathcal{A},(\{g\})} = \Psi_{\mathcal{A}/g}$. *Splicing* a pair $\{a, b\}$ of hinges keeps the bases that occupy exactly one of them and forgets the pair; on either letter it is

$$L \longmapsto (\partial_{\alpha_a} + \partial_{\alpha_b}) L \big|_{\alpha_a = \alpha_b = 0}.$$

Cells are spliced along pairs of their hinges by splicing the pairs in their direct sum (Proposition 7.9).

Theorem 21.14 (Splicing hinges in pairs)**.** *(i) Splicing a pair $\{a, b\}$ of hinges of a cell $\mathcal{A}$ at which some basis occupies exactly one of a, b gives a cell on $B \sqcup (G \setminus \{a, b\})$ of rank $r_{\mathcal{A}} - 1$: its bases are the sets $T \setminus \{a, b\}$ with $|T \cap \{a, b\}| = 1$, each carrying in each letter the sum of the weights of the bases T that give it.*

(ii) Let a direct sum of cells carry a set of disjoint pairs of hinges, forming any graph on the cells: closed paths, a pair inside one cell and several pairs between two cells are allowed. If

some choice of one basis in each cell occupies exactly one hinge of every pair, splicing the pairs one at a time, in any order, gives one cell; its rank is the sum of the ranks less the number of pairs, its bases are the unions of the unpaired parts of such choices, and its weight in each letter at a basis is the sum, over the choices that give it, of the products of the weights chosen. Plain cells give a plain cell.

(iii) Under the hypothesis of (ii), let the cells be $\mathcal{A}_1, \dots, \mathcal{A}_m$ with the hinges $G_1, \dots, G_m$, and in this statement let G' be the set of the hinges that lie in no pair, k the number of pairs and, for $p \subseteq G'$, let J run over the sets of paired hinges that contain exactly one hinge of every pair. The spliced cell $\mathcal{A}$ has the channels

$$K_{\mathcal{A},(p)} = \sum_J \prod_{i=1}^{m} K_{\mathcal{A}_i,((p \sqcup J)\cap G_i)}, \qquad \Psi_{\mathcal{A},(p)} = \sum_J \prod_{i=1}^{m} \Psi_{\mathcal{A}_i,((p \sqcup J)\cap G_i)},$$

absent channels counting as zero; it has $\sum_i n_{\mathcal{A}_i}$ marks besides its hinges G' and the rank $\sum_i r_{\mathcal{A}_i} - k$, and every product on the right has the degree of the channel on the left. The map $\big(\mathrm{ch}(\mathcal{A}_1), \dots, \mathrm{ch}(\mathcal{A}_m)\big) \mapsto \mathrm{ch}(\mathcal{A})$ is multilinear. For two cells $\mathcal{A}_1$ and $\mathcal{A}_2$ spliced along k pairs, each pair named once, so that the paired hinges of either cell form one set I of k names and the other hinges form G'_1 and G'_2, it is the bilinear map

$$K_{\mathcal{A}_{12},(p_1 \sqcup p_2)} = \sum_{I' \subseteq I} K_{\mathcal{A}_1,(p_1 \sqcup I')}\, K_{\mathcal{A}_2,(p_2 \sqcup (I \setminus I'))},$$
$$\Psi_{\mathcal{A}_{12},(p_1 \sqcup p_2)} = \sum_{I' \subseteq I} \Psi_{\mathcal{A}_1,(p_1 \sqcup I')}\, \Psi_{\mathcal{A}_2,(p_2 \sqcup (I \setminus I'))},$$

$p_1 \subseteq G'_1$, $p_2 \subseteq G'_2$; with $k = 1$ and no other hinges it is Corollary 21.7(iii).

Chapter 27 turns the bilinearity of Theorem 21.14(iii) into finite moment vectors and transfer matrices.

21.3 What crosses a gluing

Definition 21.15 (Level weights, level factorizations, interface)**.** Let $(\mathcal{B}, \theta)$ be a table of rank r with a bunch B, of any width, and room R. For a level k, $d(B) \le k \le r(B)$, the *level weights* are the matrix $\Theta_B^{(k)} = (\theta_{p \sqcup X})_{p,X}$ indexed by the k-sets $p \subseteq B$ and the $(r-k)$-sets $X \subseteq R$, with $\theta_{p \sqcup X} = 0$ when $p \sqcup X$ is not a basis. A *level factorization* writes every level letter (Definition 12.16) as a sum of products over a finite set C_k,

$$K_B^{(k)}(\alpha) = \sum_{c \in C_k} V_{k,c}(\alpha_B)\, L_{k,c}(\alpha_R),$$

each $V_{k,c}$ a real linear combination of the monomials α_B^p, $|p| = k$, and each $L_{k,c}$ a real linear combination of the monomials α_R^X, $|X| = r - k$: equivalently, a writing of $\Theta_B^{(k)}$ as a sum of $|C_k|$ products of a column and a row, so that $|C_k| \ge \operatorname{rank} \Theta_B^{(k)}$, with equality for some factorization. The $V_{k,c}$ are the *bunch letters* and the $L_{k,c}$ the *room letters*. With $V^*_{k,c}(\alpha_B) = \alpha_B^B V_{k,c}(\alpha_B^{-1})$ and $L^*_{k,c}(\alpha_R) = \alpha_R^R L_{k,c}(\alpha_R^{-1})$ the same factorization writes the colevel letter of the colevel $|B| - k$,

$$\Psi_B^{(|B|-k)}(\alpha) = \sum_{c \in C_k} V^*_{k,c}(\alpha_B)\, L^*_{k,c}(\alpha_R)\,:$$

each level is one function read twice (Proposition 7.2). The *stations* of B are the maximal sets of patterns whose children are positive multiples of one another, each with a fixed representative, the *child of the station*; every station lies in one level. The *children factorization* lets c run over the stations of level k, with $V_{k,c} = \sum_{p\in c} w_p\, \alpha_B^p$ for the child at p equal to w_p times the child of the station, and $L_{k,c}$ the basis polynomial of the child of the station: all its terms are nonnegative. The *interface dimension* of B is $\sum_k \operatorname{rank} \Theta_B^{(k)}$; for a family it is summed over the tables.

Along the scale of B, $\alpha_B = \rho u$ with $u \in \Delta_B$ (Lemma 2.7), $V_{k,c}(\rho u) = \rho^k V_{k,c}(u)$ and $V^*_{k,c}(\rho u) = \rho^{|B|-k} V^*_{k,c}(u)$; since $|B| - r(B) = d^*(B)$,

$$K(\rho u, \alpha_R) = \rho^{d(B)} \sum_k \rho^{k-d(B)} \sum_c V_{k,c}(u)\, L_{k,c}(\alpha_R),$$
$$\Psi(\rho u, \alpha_R) = \rho^{d^*(B)} \sum_k \rho^{r(B)-k} \sum_c V^*_{k,c}(u)\, L^*_{k,c}(\alpha_R),$$

each letter counted from its own bottom level, both powers running over $\rho^0, \dots, \rho^{c(B)}$.

Theorem 21.16 (What crosses a gluing)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family on $\mathcal{E}$ with real data and masses $\mathbf{m}$ on its balance wall, B a bunch with room R, and fix a level factorization of every table, with the bunch letters $V_{j,k,c}$, the room letters $L_{j,k,c}$ and the starred letters $V^*_{j,k,c}$, $L^*_{j,k,c}$. Let $\mathbf{V}(u) = \big(V_{j,k,c}(u), V^*_{j,k,c}(u)\big)_{j,k,c}$ be the* letter vector *of $u \in \Delta_B$. Then, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_R} v^{\mathbf{m}|_R - \mathbf{1}} \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \int_0^\infty \rho^{\kappa_B - 1} \prod_j \Big(\sum_{k,c} \rho^{k-d_j(B)} V_{j,k,c}(u)\, L_{j,k,c}(v) \Big)^{-s_j} \times \Big(\sum_{k,c} \rho^{r_j(B)-k} V^*_{j,k,c}(u)\, L^*_{j,k,c}(v) \Big)^{-t_j} d\rho\, d\omega_B(u)\, d\omega_R(v),$$

κ_B the wall of B, and the integrand, apart from the measure $u^{\mathbf{m}|_B-\mathbf{1}} d\omega_B(u)$, depends on u only through $\mathbf{V}(u)$. So the period sees the bunch only through its wall κ_B, its widths $c_j(B)$ and the image $\mathbf{V}_*\big(u^{\mathbf{m}|_B-\mathbf{1}} d\omega_B(u)\big)$, the joint law of its bunch letters, unnormalized: two families that agree on the room, with the same room letters, and whose bunches have equal walls, widths and laws of their letter vectors, have equal periods, whatever the numbers of marks of the bunches; the bunches are interchangeable in every room. The same holds for the interest $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ at any real masses, with $\alpha_R \in \mathbb{R}^R_{>0}$ in place of $v \in \Delta_R$ and the factor $e^{-\rho - |\alpha_R|}$ in the integrand.

At width one with factoring weights the children factorization has one set at each level, and the letter vector is $(K_{\mathcal{B}/R}, K_{\mathcal{B}|B}; \Psi_{\mathcal{B}|B}, \Psi_{\mathcal{B}/R})$, each letter from its bottom level: Theorem 19.8, which keeps the scale as the hinge g, keeps of it the cell kernel (Definition 22.2) and the two hinge values γ_B^K, γ_B^Ψ. The theorem holds whatever the weights; what the weights decide is how many bunch letters there are.

Theorem 21.17 (Multivectors bound what crosses)**.** *Let the table carry a multivector $\boldsymbol{\omega}$ of any tensor rank (Definition 7.20), the marks of B ordered before those of R, so that $\boldsymbol{\omega}_{p\sqcup X}$ is the coefficient of $\mathbf{e}_p \wedge \mathbf{e}_X$, $\mathbf{e}_p$ the wedge of the unit vectors of the marks of p in their order. At the level k let the* interface rank $\operatorname{ir}_k(\boldsymbol{\omega})$ *of $\boldsymbol{\omega}$ at B be the rank of the matrix $(\boldsymbol{\omega}_{p\sqcup X})_{p,X}$, and write $\boldsymbol{\omega}_{p\sqcup X} = \sum_{i=1}^{\operatorname{ir}_k(\boldsymbol{\omega})} f_i(p)\, f_i'(X)$.*

(i) $\operatorname{rank}\Theta_B^{(k)} \le \operatorname{ir}_k(\boldsymbol{\omega})^2$, *and* $\operatorname{rank}\Theta_B^{(k)} \le \frac12 \operatorname{ir}_k(\boldsymbol{\omega})\big(\operatorname{ir}_k(\boldsymbol{\omega})+1\big)$ *when* $\boldsymbol{\omega}$ *is real. With the positive semidefinite Hermitian matrices*

$$\mathbf{C}_k(\alpha_B) = \sum_p \alpha_B^p\, f(p) f(p)^*, \qquad \mathbf{C}'_k(\alpha_R) = \sum_X \alpha_R^X\, f'(X) f'(X)^*,$$

with $f(p) = \big(f_i(p)\big)_i$ *and* $f'(X) = \big(f'_i(X)\big)_i$, *the level letter is* $K_B^{(k)}(\alpha) = \operatorname{tr}\big(\mathbf{C}_k(\alpha_B)\,\overline{\mathbf{C}'_k(\alpha_R)}\big)$, *and the real and imaginary parts of the entries of* $\mathbf{C}_k(\alpha_B)$ *are the bunch letters of a real level factorization with at most* $\operatorname{ir}_k(\boldsymbol{\omega})^2$ *terms.*

(ii) If $\boldsymbol{\omega} = \sum_{l=1}^{\operatorname{rk}(\boldsymbol{\omega})} \boldsymbol{\omega}^{(l)}$ *with slices* $A^{(l)}$, *let the* width of the slice c_l *be the dimension of the intersection of the column spaces of the columns* $A_B^{(l)}$ *of* B *and* $A_R^{(l)}$ *of* R, *and its* floor $d_l = r - \dim\big($*column space of* $A_R^{(l)}\big)$, *which at tensor rank one are* $c(B)$ *and* $d(B)$. *Then* $\operatorname{ir}_k(\boldsymbol{\omega}) \le \sum_l \binom{c_l}{k-d_l}$.

(iii) So $\operatorname{rank}\Theta_B^{(k)} \le \big(\sum_l \binom{c_l}{k-d_l}\big)^2$, *and the interface dimension of* B *is at most* $\sum_k \big(\sum_l \binom{c_l}{k-d_l}\big)^2$, *a bound in the widths and floors of the slices alone, whatever the numbers of marks of* B *and* R; *for a real multivector each square may be replaced by* $\frac12 i_k(i_k+1)$, $i_k = \sum_l \binom{c_l}{k-d_l}$. *For a determinantal table, of tensor rank one with* $c_1 = c(B)$ *and* $d_1 = d(B)$, $\operatorname{ir}_k(\boldsymbol{\omega}) \le \binom{c(B)}{k-d(B)}$, $\operatorname{rank}\Theta_B^{(k)} \le \binom{c(B)}{k-d(B)}^2$ *and the interface dimension is at most* $\binom{2c(B)}{c(B)}$; *at width one each level has one bunch letter, as Theorem 12.19 says.*

Theorem 21.18 (The interface of a determinantal table)**.** *Let a table be determinantal over* $\mathbb{F}$ *with slice* A, $\theta_T = |\det A_T|^2$, *and let* B *have width* $c(B) = w$, *the minors* $\mathcal{B}|B$ *and* $\mathcal{B}/R$ *on* B *and* $\mathcal{B}|R$ *and* $\mathcal{B}/B$ *on* R *carrying the factoring weights of Theorem 12.19. There are Hermitian* $w \times w$ *matrices* $\mathbf{S}_B(\alpha_B)$ *and* $\mathbf{S}_R(\alpha_R)$, *positive definite on the open orthant, each entry a rational function homogeneous of degree one, with*

$$K(\alpha) = K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}/B}(\alpha_R)\, \det\big(\mathbf{S}_B(\alpha_B) + \mathbf{S}_R(\alpha_R)\big), \qquad \det \mathbf{S}_B = \frac{K_{\mathcal{B}|B}}{K_{\mathcal{B}/R}}, \qquad \det \mathbf{S}_R = \frac{K_{\mathcal{B}|R}}{K_{\mathcal{B}/B}},$$

and $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1})$. *Along the scale of* B, $K(\rho\alpha_B, \alpha_R) = \rho^{d(B)} K_{\mathcal{B}/R}(\alpha_B) K_{\mathcal{B}/B}(\alpha_R) \det\big(\rho\,\mathbf{S}_B + \mathbf{S}_R\big)$, *so the Skeleton values of* B *(Definition 12.10) are the eigenvalues of* $\mathbf{S}_B^{-1/2}\mathbf{S}_R\mathbf{S}_B^{-1/2}$, *real and positive. At width one* $\mathbf{S}_B = \gamma_B^K$, *the hinge value by* K *(Definition 19.2). So a bunch of width* w *of a determinantal table crosses its gluing as one positive definite* $w \times w$ *matrix and one scalar, and the same read at* α^{-1} *for* Ψ.

For a determinantal table a bunch B is thus read by its room through one positive definite matrix, $\binom{c(B)+1}{2}$ real entries when it is symmetric and $c(B)^2$ real parameters when it is Hermitian, and one scalar for each letter, and the bunch letters of Theorem 21.17 are the compound determinants of that matrix. For a multivector of tensor rank $\operatorname{rk}(\boldsymbol{\omega})$ the level letters carry the entries of the matrices $\mathbf{C}_k$ of Theorem 21.17(i), whose size is fixed by the interface ranks. In every case Theorem 21.16 says what crosses, and Theorems 21.17 and 21.18 say how much.

Chapter 22

Periods of constructions

A construction (Definition 33.1) hands its period a structure: the period of a substitution is read from the core and the cells, for any family. The core reads each cell through the joint law of its hinge values, which is pooling done at every cell at once; opened on the core's own shares, the cells ride on the core's fibre, each entering through the period of its own family at the exponents the shares assign it; and where every active letter reads a cell alike, the cell's share is pinned; where the active letters read every cell by one hinge value, as at a full lean of one table, the period factors through the core. Chains of cells are the powers of one operator, and the letters of any chain of bunches, and their integer joint moments, are products of finite matrices. A bunch of any width opens on one share per term of its level factorization; in level form, one term per level, one table at a full lean keeps $c(B) - 1$ free shares. The same periods are read through the laws of the cell families, one measure on $(0,\infty)^{2N-1}$ for each (Chapter 23), and through the periods at the full leans of the nodes (Chapters 25 and 26).

22.1 The core reads its cells

Definition 22.1 (Twists)**.** A *twist* of a family F on $\mathcal{E}$ assigns to every table j two vectors $\mathbf{x}_j^K, \mathbf{x}_j^\Psi \in (0,\infty)^{\mathcal{E}}$. The *twisted family* $F[\mathbf{x}]$, $\mathbf{x} = (\mathbf{x}_j^K, \mathbf{x}_j^\Psi)_j$, has the bases and the exponents of F, the basis weights $\theta_{j,T}^K(\mathbf{x}_j^K)^T$ and the cobasis weights $\theta_{j,T}^\Psi(\mathbf{x}_j^\Psi)^{\mathcal{E}\setminus T}$: its letters are $K_j(\mathbf{x}_j^K \circ \alpha)$ and $\Psi_j(\mathbf{x}_j^\Psi \circ \alpha)$, with $\mathbf{x} \circ \alpha = (x_e \alpha_e)_e$. A twist *at a mark* g has all its coordinates 1 off g and is written by its values $\mathbf{x} = (x_j^K, x_j^\Psi)_j$ at g; then $F[\mathbf{x}@g]$ multiplies, in table j, the weights in K_j of the bases through g by x_j^K and the weights in Ψ_j of the bases avoiding g by x_j^Ψ. A twist of one table $\mathcal{H}$ in its cobasis letter alone, $\mathbf{x}^K = \mathbf{1}$, is written $\mathcal{H}[\mathbf{x}]$ with $\mathbf{x} = \mathbf{x}^\Psi$: $K_{\mathcal{H}[\mathbf{x}]} = K_{\mathcal{H}}$ and $\Psi_{\mathcal{H}[\mathbf{x}]}(\alpha) = \Psi_{\mathcal{H}}(\mathbf{x} \circ \alpha)$. On the balance wall the period obeys the *torus relation*: for every twist $\mathbf{x}$, every $\boldsymbol{\lambda} \in (0,\infty)^{\mathcal{E}}$ and all constants $c_j, c_j^* > 0$,

$$P_{F[(c_j \boldsymbol{\lambda} \circ \mathbf{x}_j^K,\ c_j^* \boldsymbol{\lambda} \circ \mathbf{x}_j^\Psi)_j]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \boldsymbol{\lambda}^{-\mathbf{m}} \prod_j c_j^{-r_j s_j}\, c_j^{*\,-r_j^* t_j}\, P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}),$$

both sides converging together, in $[0,\infty]$ for real data; at the scalar $\boldsymbol{\lambda} = \mu\mathbf{1}$ with $c_j = c_j^* = \mu^{-1}$ the twist is unchanged and the factor is 1, by balance. So a twist at a mark g, $\boldsymbol{\lambda}$ equal to 1 off g, has $P_{F[\lambda \mathbf{x}@g]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \lambda^{-m_g} P_{F[\mathbf{x}@g]}(\mathbf{m}; \mathbf{s}, \mathbf{t})$ and is read with $x_1^K = 1$, a point of $(0,\infty)^{2N-1}$; for one table it is one number, $x = x_1^\Psi$, and only the cobasis letter moves. A twist of every mark of one letter by one factor multiplies the period by that factor to the power $-r_j s_j$ or $-r_j^* t_j$; for one table in its cobasis letter, $P_{\mathcal{H}[\mu\mathbf{x}]}(\mathbf{m}; s, t) = \mu^{-r^* t} P_{\mathcal{H}[\mathbf{x}]}(\mathbf{m}; s, t)$.

Definition 22.2 (Cell kernels; the core family read at a shape)**.** Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a substituted family (Definition 21.1), table j being $\mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$ with the cells $\mathcal{A}_{ij}$ on $B_i \sqcup \{g_i\}$. The *cell kernel* of B_i is the measure

$$u^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K_{\mathcal{A}_{ij}/g_i}(u)^{-s_j} \Psi_{\mathcal{A}_{ij}\setminus g_i}(u)^{-t_j}\, d\omega_{B_i}(u) \qquad \text{on } \Delta_{B_i}.$$

The *core family* is $F_{\mathcal{H}} = ((\mathcal{H}_j, \theta^{\mathcal{H}_j}); s_j, t_j)_j$ on the marks $h_1, \dots, h_q$. For $u = (u_i)_i \in \prod_i \Delta_{B_i}$ the *core family read at* u is its twist $F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]$ (Definition 22.1) by the hinge values, $\boldsymbol{\gamma}(u)$ putting $\gamma^K_{\mathcal{A}_{ij}}(u_i)$ in K_j and $\gamma^\Psi_{\mathcal{A}_{ij}}(u_i)$ in Ψ_j at the mark h_i: table j has the basis weights $\theta_T^{\mathcal{H}_j} \prod_{h_i \in T} \gamma^K_{\mathcal{A}_{ij}}(u_i)$ and the cobasis weights $\theta_T^{\mathcal{H}_j} \prod_{h_i \notin T} \gamma^\Psi_{\mathcal{A}_{ij}}(u_i)$, and the letters $K_{\mathcal{H}_j}\big((\gamma^K_{\mathcal{A}_{ij}}(u_i)\,\alpha_{h_i})_i\big)$ and $\Psi_{\mathcal{H}_j}\big((\gamma^\Psi_{\mathcal{A}_{ij}}(u_i)\,\alpha_{h_i})_i\big)$.

Theorem 22.3 (The core reads its cells)**.** *For a substituted family F the core family read at u has the imbalance of F, and*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\prod_i \Delta_{B_i}} \prod_i \Big(u_i^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K_{\mathcal{A}_{ij}/g_i}(u_i)^{-s_j} \Psi_{\mathcal{A}_{ij}\setminus g_i}(u_i)^{-t_j}\, d\omega_{B_i}(u_i)\Big) \\ \times P_{F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]}\big((\kappa_{B_1}, \dots, \kappa_{B_q});\mathbf{s},\mathbf{t}\big),$$

both sides converging together; the same holds with J for P, and on the balance wall. So the period of a substitution depends on each cell only through its wall κ_{B_i} and the joint law, under its cell kernel, of its hinge values $\big(\gamma^K_{\mathcal{A}_{ij}}, \gamma^\Psi_{\mathcal{A}_{ij}}\big)_j$: two cells with equal walls and equal laws are interchangeable in every core. For one cell this is Theorem 19.8.

Pooling, done at every cell at once, leaves the core: each cell is integrated over its shape with its kernel and collapses to one mark of the core, which keeps the mass κ_{B_i} and is read through the cell's hinge values as weights. The cells meet only in the core.

Theorem 22.4 (The cells ride on the core's fibre)**.** *Let F be a substituted family with complex data, every exponent* 0 *or of positive real part. Open $K_{\mathcal{H}_j}$ with one share $a_{j,T}$ and $\Psi_{\mathcal{H}_j}$ with one share $b_{j,T}$ per basis T of $\mathcal{H}_j$ (in this statement $|\mathcal{H}_j|$ is the number of bases of $\mathcal{H}_j$), $\sum_T a_{j,T} = s_j$ and $\sum_T b_{j,T} = t_j$, a letter with exponent zero opened by nothing, and let the* loads *of the mark h_i be $a_j(h_i) = \sum_{T \ni h_i} a_{j,T}$ and $b_j(h_i) = \sum_{T \not\ni h_i} b_{j,T}$. The* load family $F_i(\mathbf{a},\mathbf{b})$ *is the family on B_i with, for every j, the table $\mathcal{A}_{ij} \setminus g_i$ at the exponents $\big(a_j(h_i),\, t_j - b_j(h_i)\big)$ and the table $\mathcal{A}_{ij}/g_i$ at $\big(s_j - a_j(h_i),\, b_j(h_i)\big)$; its imbalance is $\kappa_{B_i} - \sum_j \big(a_j(h_i) + b_j(h_i)\big)$. In this statement write $\mathbf{s}^{(i)} = \big((a_j(h_i))_j, (s_j - a_j(h_i))_j\big)$ and $\mathbf{t}^{(i)} = \big((t_j - b_j(h_i))_j, (b_j(h_i))_j\big)$ for its exponents, the tables $\mathcal{A}_{ij} \setminus g_i$ listed first. On every contour of positive real parts of the shares at which all load families converge absolutely,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j} \prod_j \Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{a}_j \\ s_j\end{bmatrix} \Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{b}_j \\ t_j\end{bmatrix} \prod_{j,T} \big(\theta_T^{\mathcal{H}_j}\big)^{-a_{j,T}-b_{j,T}} \\ \times \prod_i J_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big),$$

with $J_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big) = \Gamma\big(\kappa_{B_i} - \sum_j (a_j(h_i) + b_j(h_i))\big) P_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big)$. The integrand without the periods of the load families is that of the interest $J_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)$ of the core family (Definition 22.2) opened with one share per basis (Corollary 4.4).

The cells ride on the core's share space: the core's letters are opened over its bases, every K-share of a basis loads the marks it occupies and every Ψ-share the marks it leaves out, and each cell enters through the period of its own two hinge minors at the exponents that the loads of its mark assign them.

22.2 Pinned cells

Theorem 22.5 (A cell read alike is pinned)**.** *Let F be a substituted family at every cell of which every active letter reads alike: for every i there are a function γ_i on Δ_{B_i} and constants $c_{ij}, c^*_{ij} > 0$ with $\gamma^K_{\mathcal{A}_{ij}} = c_{ij}\gamma_i$ whenever $s_j \neq 0$ and $\gamma^\Psi_{\mathcal{A}_{ij}} = c^*_{ij}\gamma_i$ whenever $t_j \neq 0$, and $c_{ij} = 1$ when $s_j = 0$, $c^*_{ij} = 1$ when $t_j = 0$. Let $F_\mathcal{H}[\mathbf{c}]$ be the core family twisted by these constants at the marks h_i, c_{ij} in K_j and c^*_{ij} in Ψ_j (Definitions 22.1 and 22.2). Then on the balance wall*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{F_\mathcal{H}[\mathbf{c}]}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i\int_{\Delta_{B_i}}\gamma_i^{-\kappa_{B_i}}\,u^{\mathbf{m}|_{B_i}-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i}\Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}\,d\omega_{B_i}(u),$$

the core at the walls of the cells times the pinned shape integrals *of the cells. In general the free shares at the cell i are one fewer than the distinct hinge values of B_i by the active letters, as at a bunch of width one (Theorem 19.13).*

Corollary 22.6 (Read by one hinge value, the period factors through the core)**.** *Let F be a substituted family with real data on its balance wall (Definition 21.1), table j being $\mathcal{H}_j\langle\mathcal{A}_{1j},\dots,\mathcal{A}_{qj}\rangle$, at every cell of which the active letters read the cell by one hinge value: for every i there is a function γ_i on Δ_{B_i} with $\gamma^K_{\mathcal{A}_{ij}} = \gamma_i$ whenever $s_j \neq 0$ and $\gamma^\Psi_{\mathcal{A}_{ij}} = \gamma_i$ whenever $t_j \neq 0$, as at a mark left in place, whose two hinge values are both α_h. Let σ be the total strength (Definition 8.1) and $F_i = ((\mathcal{A}_{ij},\theta^{\mathcal{A}_{ij}});s_j,t_j)_j$ the cell family of the cells at B_i, on $B_i \sqcup \{g_i\}$. Then, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{F_\mathcal{H}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i\int_{\Delta_{B_i}}\gamma_i^{-\kappa_{B_i}}\,u^{\mathbf{m}|_{B_i}-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i}\Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}\,d\omega_{B_i}(u),$$

and at a cell with $0 < \kappa_{B_i} < 2\sigma$ the family F_i at the masses $(\mathbf{m}|_{B_i}, 2\sigma - \kappa_{B_i})$ is on its balance wall and its pinned shape integral is $P_{F_i}\big((\mathbf{m}|_{B_i},2\sigma-\kappa_{B_i});\mathbf{s},\mathbf{t}\big)/\Gamma_1^2\begin{bmatrix}\kappa_{B_i},\ 2\sigma-\kappa_{B_i}\\ 2\sigma\end{bmatrix}$. If F has every proper wall positive, then $0 < \kappa_{B_i} < 2\sigma$ for every cell, and

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})\prod_i\Gamma_1^2\begin{bmatrix}\kappa_{B_i},\ 2\sigma-\kappa_{B_i}\\ 2\sigma\end{bmatrix} = P_{F_\mathcal{H}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i P_{F_i}\big((\mathbf{m}|_{B_i},2\sigma-\kappa_{B_i});\mathbf{s},\mathbf{t}\big),$$

every factor on its balance wall with every proper wall positive, the hinge g_i at the mass $2\sigma - \kappa_{B_i}$ and the mark h_i of the core family at κ_{B_i}. For one cell this is Theorem 19.10, and the first display is the shape-integral form of Proposition 19.12.

Corollary 22.7 (The pinned shape integral is the period of a load family)**.** *In Corollary 22.6, let the loads a_{ij}, b_{ij} of the cell i be real, with $a_{ij} = 0$ when $s_j = 0$, $b_{ij} = 0$ when $t_j = 0$ and $\sum_j(a_{ij} + b_{ij}) = \kappa_{B_i}$. The load family $F_i(\mathbf{a},\mathbf{b})$ of Theorem 22.4 at the loads $a_j(h_i) = a_{ij}$ and $b_j(h_i) = b_{ij}$, with the tables $\mathcal{A}_{ij} \setminus g_i$ at $(a_{ij}, t_j - b_{ij})$ and $\mathcal{A}_{ij}/g_i$ at $(s_j - a_{ij}, b_{ij})$ and the masses $\mathbf{m}|_{B_i}$, has the imbalance zero, and its form does not depend on the loads: in $[0,\infty]$ its period is the pinned shape integral of the cell,*

$$P_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big) = \int_{\Delta_{B_i}}\gamma_i^{-\kappa_{B_i}}\,u^{\mathbf{m}|_{B_i}-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i}\Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}\,d\omega_{B_i}(u),$$

and $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i P_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big)$*. If every proper wall of* F *is positive and* $s_j, t_j \geq 0$*, there are such loads with* $0 \leq a_{ij} \leq s_j$ *and* $0 \leq b_{ij} \leq t_j$*, integers when the data are; the load family then has nonnegative exponents, and every proper wall of it is positive when* P_F *is finite.*

22.3 Chains and transfer

Theorem 22.8 (A chain is the power of one operator)**.** *Let* $F = ((\mathcal{A}_j,\theta_j);s_j,t_j)_{j=1}^N$ *be a family on* $\mathcal{E}$ *with two marks* $g_{\text{in}} \neq g_{\text{out}}$*, each neither a loop nor a coloop in every table, real exponents, masses on the balance wall with* $m_{g_{\text{in}}} + m_{g_{\text{out}}} = \sum_j (s_j + t_j)$*, and let* $F^{\triangleright k}$ *be* k *copies of* F *glued table by table along marks,* $g_{\text{out}}^{(i)}$ *with* $g_{\text{in}}^{(i+1)}$ *(Lemma 21.12). For* $\mathbf{x} = (x_j^K, x_j^\Psi)_j \in (0,\infty)^{2N}$ *let* $F^{\triangleright k}[\mathbf{x}@g_{\text{in}}^{(1)}]$ *be* $F^{\triangleright k}$ *twisted at* $g_{\text{in}}^{(1)}$ *(Definition 22.1), and in this statement write* $P_k(\mathbf{x})$ *for its period at the masses of the copies and* $\mathcal{A}_j^{\mathbf{x}}$ *for table* j *of* $F[\mathbf{x}@g_{\text{in}}]$*, read as a cell with the hinge* g_{out}*. With* $B = \mathcal{E}\setminus\{g_{\text{out}}\}$ *define on nonnegative functions* H *of the twist*

$$(\mathcal{Q}H)(\mathbf{x}) = \int_{\Delta_B} H\Big(\big(\gamma^K_{\mathcal{A}_j^{\mathbf{x}}}(u),\gamma^\Psi_{\mathcal{A}_j^{\mathbf{x}}}(u)\big)_j\Big)\, u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{A}_j^{\mathbf{x}}/g_{\text{out}}}(u)^{-s_j}\,\Psi_{\mathcal{A}_j^{\mathbf{x}}\setminus g_{\text{out}}}(u)^{-t_j}\, d\omega_B(u),$$

$$P_0(\mathbf{x}) = \int_0^1 v^{m_{g_{\text{in}}}-1}(1-v)^{m_{g_{\text{out}}}-1}\prod_j \big(1-v+x_j^K v\big)^{-s_j}\big(1-v+x_j^\Psi v\big)^{-t_j}\, dv.$$

Then $P_k = \mathcal{Q}P_{k-1}$*, and in* $[0,\infty]$

$$P_{F^{\triangleright k}}(\mathbf{m};\mathbf{s},\mathbf{t}) = (\mathcal{Q}^k P_0)(\mathbf{1}).$$

The operator acts on functions of the $2N$ *twists, of which one is removed by the scale of the hinge. Where the twist is pinned, the active letters reading the cell by one hinge value,* $\gamma^K_{\mathcal{A}_j}$ *for* $s_j \neq 0$ *and* $\gamma^\Psi_{\mathcal{A}_j}$ *for* $t_j \neq 0$ *all equal, as at a full lean of one table, every* P_k *is a multiple of one power of the twist along the diagonal* $\mathbf{x} = \lambda\mathbf{1}$*,* $\mathcal{Q}$ *multiplies it by* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} m_{g_{\text{in}}}, & m_{g_{\text{out}}} \\ m_{g_{\text{in}}}+m_{g_{\text{out}}} & \end{bmatrix}$*, and*

$$P_{F^{\triangleright k}}(\mathbf{m};\mathbf{s},\mathbf{t}) = \frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})^k}{\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} m_{g_{\text{in}}}, & m_{g_{\text{out}}} \\ m_{g_{\text{in}}}+m_{g_{\text{out}}} & \end{bmatrix}^{k-1}}.$$

Hinge values that are only proportional, with constants other than one, still pin each cell (Theorem 22.5) but carry the twist off the diagonal.

A chain is pooled one copy at a time: the first copy, read as the cell of its bunch, collapses to its hinge and twists the next copy by its hinge values, so the period of the chain is one operator on functions of the twist applied once per copy, and the two-mark table left at the end supplies the starting function. Where the hinge values pin the twist the operator is a multiplication, and the chain costs a power.

Theorem 22.9 (Letters by transfer)**.** *Let* $(\mathcal{B},\theta)$ *be a table with any weights and* $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$ *a chain of bunches, with the steps* $B_{l-1} = S_l \setminus S_{l-1}$*. Group the patterns of each* S_l *into its stations (Definition 21.15), the children being tables on* $\mathcal{E}\setminus S_l$ *and the child at* p *being* w_p *times the child of its station* c*. For a station* c *of* S_{l-1} *and a station* c' *of* S_l *let*

$$\mathbf{M}_l^K(c,c') = \sum_{p'} w(c,p')\,\alpha^{p'}_{B_{l-1}}, \qquad \mathbf{M}_l^\Psi(c,c') = \sum_{p'} w(c,p')\,\alpha^{B_{l-1}\setminus p'}_{B_{l-1}},$$

the sums over the patterns p' of B_{l-1} in the child of the station c whose child is $w(c,p')$ times the child of the station c'. Then, with $\mathbf{e}$ the unit vector of the station of the empty pattern of S_0 and $\boldsymbol{\theta}_\varnothing$ the vector of the weights of the one basis $\varnothing$ of the children of the stations of S_L,

$$K = \mathbf{e}^{\mathsf{T}}\mathbf{M}_1^K\mathbf{M}_2^K\cdots\mathbf{M}_L^K\,\boldsymbol{\theta}_\varnothing, \qquad \Psi = \mathbf{e}^{\mathsf{T}}\mathbf{M}_1^\Psi\cdots\mathbf{M}_L^\Psi\,\boldsymbol{\theta}_\varnothing.$$

The number of states at S_l is the number of its stations, at least the interface dimension of S_l (Definition 21.15), and the states can be reduced to that many linearly independent ones; at a bunch S_l of width one with factoring weights it is two. For a table carrying a multivector $\boldsymbol{\omega}$ of any tensor rank (Definition 7.20), with the widths c_i and the floors d_i of its slices at S_l (Theorem 21.17(ii)), the interface dimension of S_l is at most $\sum_k\big(\sum_i\binom{c_i}{k-d_i}\big)^2$, each square q^2 replaced by $\frac{1}{2}q(q+1)$ when $\boldsymbol{\omega}$ is real; for a determinantal table it is at most $\binom{2c(S_l)}{c(S_l)}$ (Theorem 21.17).

Theorem 22.10 (Integer joint moments by fixed matrices)**.** *Let the letters of a family $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ be products of matrix functions of the variables of the steps $B_0,\dots,B_{L-1}$ of one chain of bunches, $K_j = \mathbf{e}_j^{\mathsf{T}}\mathbf{M}_{j,0}^K(\alpha_{B_0})\cdots\mathbf{M}_{j,L-1}^K(\alpha_{B_{L-1}})\boldsymbol{\theta}_{j,\varnothing}$ and $\Psi_j = \mathbf{e}_j^{\mathsf{T}}\mathbf{M}_{j,0}^\Psi(\alpha_{B_0})\cdots\mathbf{M}_{j,L-1}^\Psi(\alpha_{B_{L-1}})\boldsymbol{\theta}_{j,\varnothing}$ with constant vectors $\mathbf{e}_j$ and $\boldsymbol{\theta}_{j,\varnothing}$, as Theorem 22.9 gives for every table along any chain of bunches; and let the steps carry independent positive measures $\phi_0,\dots,\phi_{L-1}$ with the required moments finite. For integers $a_j,b_j\ge 0$ put*

$$\mathbf{A}_l = \int\bigotimes_j\Big(\mathbf{M}_{j,l}^K(\alpha)^{\otimes a_j}\otimes\mathbf{M}_{j,l}^\Psi(\alpha)^{\otimes b_j}\Big)\,\phi_l(d\alpha),$$

one α in all the factors. Then

$$\int\prod_j K_j^{a_j}\Psi_j^{b_j}\prod_{l<L}\phi_l(d\alpha_{B_l}) = \Big(\bigotimes_j\mathbf{e}_j^{\otimes(a_j+b_j)}\Big)^{\mathsf{T}}\mathbf{A}_0\mathbf{A}_1\cdots\mathbf{A}_{L-1}\Big(\bigotimes_j\boldsymbol{\theta}_{j,\varnothing}^{\otimes(a_j+b_j)}\Big).$$

For a chain of identical steps, with one pair of matrix functions $\mathbf{M}_j^K,\mathbf{M}_j^\Psi$ for each table, the states of all the S_l indexed by one set, $\boldsymbol{\theta}_{j,\varnothing}$ vanishing at the states that do not occur at S_L, and one measure ϕ, every $\mathbf{A}_l$ is one matrix $\mathbf{A}$: the moment sequence along L is the power of one matrix, of size $\prod_j(\text{number of states of table } j)^{a_j+b_j}$, and satisfies a linear recurrence with constant coefficients of that order. With independent laws Gamma_{m_x} on the marks (Definition 1.1(vii)) the moment $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\big[\prod_j K_j(U)^{a_j}\Psi_j(U)^{b_j}\big]$ is that integral divided by the rising factorial $(m_{\mathcal{E}})_{\sum_j(a_jr_j+b_jr_j^*)}$; with the law $\mathrm{Dir}_{\mathbf{m}|_{B_l}}$ on each step the entries are the Dirichlet moments $\prod_{x\in B_l}(m_x)_{k_x}/(m_{B_l})_{|\mathbf{k}|}$ of the monomials $\alpha_{B_l}^{\mathbf{k}}$ (Theorem 3.11).

22.4 A bunch of any width

Theorem 22.11 (Gluing across a bunch of any width)**.** *Let F be a family with real data on its balance wall with every proper wall positive, B a bunch of any width with room R, and fix level factorizations with nonnegative terms (Definition 21.15), the children factorization for instance, with the bunch letters $V_{j,k,c}$ and the room letters $L_{j,k,c}$; drop the terms whose bunch letter or room letter vanishes identically, and let C_j be the set of the terms (k,c) left for table j. Open every active letter with one share per term, $a_{j,k,c}$ for the term $V_{j,k,c}L_{j,k,c}$ of K_j with $\sum_{k,c}a_{j,k,c}=s_j$ and $b_{j,k,c}$ for the term $V_{j,k,c}^*L_{j,k,c}^*$ of Ψ_j with $\sum_{k,c}b_{j,k,c}=t_j$, pinned by the scale of B,*

$$\sum_{j,k,c}\Big(\big(k-d_j(B)\big)a_{j,k,c}+\big(r_j(B)-k\big)b_{j,k,c}\Big)=\kappa_B.$$

On every contour of positive real parts at which the two integrals below converge,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j,\ pin} \prod_j \Gamma_{|C_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|C_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}$$
$$\times \int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}} \prod_{j,k,c} V_{j,k,c}^{-a_{j,k,c}} {V^*_{j,k,c}}^{-b_{j,k,c}}\, d\omega_B(u) \int_{\Delta_R} v^{\mathbf{m}|_R-\mathbf{1}} \prod_{j,k,c} L_{j,k,c}^{-a_{j,k,c}} {L^*_{j,k,c}}^{-b_{j,k,c}}\, d\omega_R(v),$$

the two inner integrals projective integrals of degree zero, the periods of the bunch word *and the* room word *at the shares. Their letters are the bunch and room letters of the level factorization, which need not be the letters of tables, so their convergence is that of the integrals themselves. The free shares are the terms less the active letters less one. For one table at a full lean $(0,t)$ in level form, one station per level, with the share b_k and the bunch and room letters V_k^*, L_k^* of the station of the level k, $d(B)\le k\le r(B)$, the pins are $\sum_k b_k = t$ and $\sum_k (r(B)-k)\,b_k = \kappa_B$, the free shares are the $c(B)-1$ shares of the levels strictly between $d(B)$ and $r(B)$, and on a contour of such shares*

$$P_F(\mathbf{m};0,t) = \frac{1}{(2\pi i)^{c(B)-1}} \int \frac{\prod_{k=d(B)}^{r(B)} \Gamma(b_k)}{c(B)\,\Gamma(t)} \int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}} \prod_k V_k^{*-b_k}\, d\omega_B(u)$$
$$\times \int_{\Delta_R} v^{\mathbf{m}|_R-\mathbf{1}} \prod_k L_k^{*-b_k}\, d\omega_R(v) \prod_{d(B)<k<r(B)} db_k,$$

with $b_{d(B)} = \frac{1}{c(B)}\big(\kappa_B - \sum_{d(B)<k<r(B)} (r(B)-k)\,b_k\big)$ and $b_{r(B)} = t - b_{d(B)} - \sum_{d(B)<k<r(B)} b_k$. At $c(B)=2$, with $b_{d(B)+1} = z$, the shares are $\frac{\kappa_B - z}{2}$, z and $\frac{2t-\kappa_B-z}{2}$; at $c(B)=1$ no share is free, and the period is $\Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa_B,\ t-\kappa_B\\ t\end{bmatrix}$ times the two inner periods (Corollary 19.14). At width one with factoring weights, every letter active, the free shares are $2N-1$, and this is Theorem 19.13.

For one table in level form at a full lean $(0,t)$ the scale can also be integrated before the shapes. Writing $\Psi(\rho u, v) = \rho^{d^*(B)} \sum_{i=0}^{c(B)} \rho^i V^*_{r(B)-i}(u) L^*_{r(B)-i}(v)$ and rescaling ρ so that the two extreme terms become 1 and $x^{c(B)}$, the period is the integral over $\Delta_B \times \Delta_R$ of a form that factors into a bunch part and a room part, times the scale integral $\int_0^\infty x^{\kappa_B-1}\big(1 + \sum_{i=1}^{c(B)-1} \zeta_i x^i + x^{c(B)}\big)^{-t} dx$, whose coefficients

$$\zeta_i = \frac{V^*_{r(B)-i}(u)}{V^*_{r(B)}(u)^{1-i/c(B)}\, V^*_{d(B)}(u)^{i/c(B)}} \cdot \frac{L^*_{r(B)-i}(v)}{L^*_{r(B)}(v)^{1-i/c(B)}\, L^*_{d(B)}(v)^{i/c(B)}}$$

are each a bunch part times a room part: the room receives $c(B)-1$ ratios. At $c(B)=1$ the scale integral is $\Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa_B,\ t-\kappa_B\\ t\end{bmatrix}$, which is Theorem 19.10; its Mellin transform in $(\zeta_1,\dots,\zeta_{c(B)-1})$ at $(z_1,\dots,z_{c(B)-1})$ is $\prod_{k=d(B)}^{r(B)} \Gamma(b_k)/\big(c(B)\Gamma(t)\big)$ at the shares $b_{r(B)-i} = z_i$ of Theorem 22.11, its share kernel with the factor $1/c(B)$ of the pin, at $c(B)=2$ the kernel $\Gamma(z)\Gamma\big(\frac{\kappa_B-z}{2}\big)\Gamma\big(\frac{2t-\kappa_B-z}{2}\big)/\big(2\Gamma(t)\big)$. With the factorization taken to be the canonical one of the seam of the bunch, the bunch word is written through one law, which with the wall is exactly what the bunch contributes (Chapter 28).

Chapter 23

The law of a cell

A table with a bunch of width one across which its weights factor is two cells glued along a hinge that neither of them shows to the other (Lemma 21.12). What a cell family must carry so that its contribution to every family built on it is fixed is its hinge wall and one measure on $(0,\infty)^{2N-1}$, the law of its ratio vector under its base form, for one cell a measure on the half-line, and nothing less (Theorem 23.11). A cell family enters a period only through that law (Theorem 23.9); the law is read in two ways, by the transform of the cell family and by its dual, and the two transforms expand one function on the two sides of one strip (Proposition 23.8). The step that makes all of this compose is Lemma 23.2: a cell sitting at a mark of any family can be integrated away, leaving the family with that mark twisted by the ratio vector of the cell family; pooling (Theorem 19.8) and Theorem 22.3, in which the core sees each cell as one mark, are its cases. The series closure of any cells, at any positive exponents, is one integral over the shared scales of the product of the transforms of the cells, so identical cells enter as one transform to a power (Theorem 23.3). A core with cells at its markers is read the same way: the law of a substitution is the mixture, over the laws of its cell families, of the laws of the twisted core family (Theorem 23.6); at a series node the transforms multiply and at a parallel node the dual transforms multiply (Theorem 23.14); and the period of a family built along a common hinge tree is read from the leaves up, with one law per edge, computed once for each distinct subtree (Theorem 23.17, Proposition 23.18). The calculus extends to bunches of every width (Chapter 28).

23.1 Cells and their laws

Definition 23.1 (The law of a cell family)**.** A *cell family* is a family $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ on $B \sqcup \{g\}$ whose tables are cells with the one hinge g (Definition 21.1), with real exponents, carrying masses $\mathbf{m}$ on B; one cell $\mathcal{A}$ at the exponents (s,t) is the cell family $((\mathcal{A},\theta);s,t)$, and its objects carry the subscript $\mathcal{A}$. Its *hinge wall* is

$$\kappa_F = m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big),$$

for one cell the hinge wall $\kappa_{\mathcal{A}}$ of Lemma 21.12. Its *base form* and its *ratio vector* are

$$\phi_F(du) = u^{\mathbf{m-1}} \prod_j K_{\mathcal{A}_j/g}(u)^{-s_j}\, \Psi_{\mathcal{A}_j\setminus g}(u)^{-t_j}\, \gamma^K_{\mathcal{A}_1}(u)^{-\kappa_F}\, d\omega_B(u) \quad \text{on } \Delta_B,$$

$$\boldsymbol{\varrho}_F = \Big(\frac{\gamma^K_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}},\ \frac{\gamma^\Psi_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}}\Big)_j :\ \Delta_B \to (0,\infty)^{2N-1},$$

the coordinate $\gamma^K_{\mathcal{A}_1}/\gamma^K_{\mathcal{A}_1} = 1$ omitted, and its *law* is the image $(\boldsymbol{\varrho}_F)_*\phi_F$, a positive measure on $(0,\infty)^{2N-1}$. For $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$, also written as the pair $(\mathbf{z}_K, \mathbf{z}_\Psi)$ of $\mathbf{z}_K = (z_{j,K})_j$ and $\mathbf{z}_\Psi = (z_{j,\Psi})_j$, write

$$\boldsymbol{\varrho}_F^{-\mathbf{z}} = \prod_{j\ge 2}\Big(\frac{\gamma^K_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}}\Big)^{-z_{j,K}} \prod_j \Big(\frac{\gamma^\Psi_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}}\Big)^{-z_{j,\Psi}},$$

the coordinates of the ratio vector to the powers $-z_{j,K}$ $(j \ge 2)$ and $-z_{j,\Psi}$; the *Mellin transform* of the law, $\int_{\Delta_B} \boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F$, lies in $[0,\infty]$ for real $\mathbf{z}$ and is defined for complex $\mathbf{z}$ at whose real part it is finite, with modulus at most that value. For one cell the ratio vector is the ratio $\varrho_{\mathcal{A}}$ (Definition 21.1), the base form is

$$\phi_{\mathcal{A}}(du) = u^{\mathbf{m-1}} K_{\mathcal{A}/g}(u)^{\kappa_{\mathcal{A}}-s} K_{\mathcal{A}\setminus g}(u)^{-\kappa_{\mathcal{A}}} \Psi_{\mathcal{A}\setminus g}(u)^{-t}\, d\omega_B(u),$$

the law $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}$ is a measure on $(0,\infty)$, and, with $z = z_{1,\Psi}$, the Mellin transform is $\int_{\Delta_B} \varrho_{\mathcal{A}}^{-z}\, d\phi_{\mathcal{A}} = \int_0^\infty x^{-z}\, d\big((\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}\big)(x)$. For a cell family with $s_j, t_j > 0$ the *transform*, the *dual transform* and the two *hinge measures* are, for $\mathbf{x}, \mathbf{y} \in (0,\infty)^N$,

$$f_F(\mathbf{x},\mathbf{y}) = \int_{\mathbb{R}^B_{>0}} \alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j\setminus g} \Psi^{-t_j}_{\mathcal{A}_j\setminus g} \exp\Big(-\sum_j \frac{x_j}{\gamma^K_{\mathcal{A}_j}} - \sum_j y_j\, \gamma^\Psi_{\mathcal{A}_j}\Big)\, d\alpha,$$

$$\check f_F(\mathbf{x},\mathbf{y}) = \int_{\mathbb{R}^B_{>0}} \alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j/g} \Psi^{-t_j}_{\mathcal{A}_j/g} \exp\Big(-\sum_j \frac{x_j}{\gamma^\Psi_{\mathcal{A}_j}} - \sum_j y_j\, \gamma^K_{\mathcal{A}_j}\Big)\, d\alpha,$$

with values in $[0,\infty]$, the measure $\mathfrak{r}_F$, the image of $\alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j\setminus g} \Psi^{-t_j}_{\mathcal{A}_j\setminus g}\, d\alpha$ under $\alpha \mapsto \big((1/\gamma^K_{\mathcal{A}_j}(\alpha))_j, (\gamma^\Psi_{\mathcal{A}_j}(\alpha))_j\big)$, and the measure $\check{\mathfrak{r}}_F$, the image of $\alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j/g} \Psi^{-t_j}_{\mathcal{A}_j/g}\, d\alpha$ under $\alpha \mapsto \big((1/\gamma^\Psi_{\mathcal{A}_j}(\alpha))_j, (\gamma^K_{\mathcal{A}_j}(\alpha))_j\big)$, positive measures on $(0,\infty)^{2N}$. For $c > 0$, $f_F(\mathbf{x},\mathbf{y}) = c^{\kappa_F - |\mathbf{s}|} f_F(\mathbf{x}/c, c\mathbf{y})$ and $\check f_F(\mathbf{x},\mathbf{y}) = c^{\kappa_F - |\mathbf{t}|} \check f_F(\mathbf{x}/c, c\mathbf{y})$, so each is fixed by its values at $x_1 = 1$. For one cell they are read there: the transform and the dual transform are $f_{\mathcal{A}}(y) = f_F(1,y)$ and $\check f_{\mathcal{A}}(y) = \check f_F(1,y)$, $y > 0$, and the hinge measures $\mathfrak{r}^\Psi_{\mathcal{A}}$ and $\mathfrak{r}^K_{\mathcal{A}}$ are the images of $e^{-\lambda}\mathfrak{r}_F(d\lambda, d\mu)$ and of $e^{-\lambda}\check{\mathfrak{r}}_F(d\lambda, d\mu)$ under $(\lambda,\mu) \mapsto \mu$, positive measures on $(0,\infty)$.

The base form of a cell is the restriction to Δ_B of the orthant form $x^{\mathbf{m-1}} K^{\kappa_{\mathcal{A}}-s}_{\mathcal{A}/g} K^{-\kappa_{\mathcal{A}}}_{\mathcal{A}\setminus g} \Psi^{-t}_{\mathcal{A}\setminus g} dx$, homogeneous of degree $m_B + (\kappa_{\mathcal{A}} - s)(r_{\mathcal{A}} - 1) - \kappa_{\mathcal{A}} r_{\mathcal{A}} - t(|B| - r_{\mathcal{A}}) = 0$, so it is read on any section of a positive function of degree one. When $\mathcal{A}$ is the cell of a bunch B of width one in a table, its hinge minors are the restriction $\mathcal{B}|B$ and the contraction $\mathcal{B}/R$, $\kappa_{\mathcal{A}}$ is the wall κ_B, $\phi_{\mathcal{A}}$ is the base form ϕ_B, $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}$ is the law of the bunch, $\int \varrho_{\mathcal{A}}^{-b^d} d\phi_{\mathcal{A}}$ is the continued period of the bunch family at the share b^d, and $f_{\mathcal{A}}$ is the bunch transform f_B (Definitions 20.24 and 20.43, Lemma 20.27); the cell kernel of Definition 22.2 is $(\gamma^K_{\mathcal{A}})^{\kappa_{\mathcal{A}}}\phi_{\mathcal{A}}$, and for a cell family $(\gamma^K_{\mathcal{A}_1})^{\kappa_F}\phi_F$. None of these objects mentions a room: a cell carries them by itself, and they are all that a room ever sees of it (Theorem 23.11). The hinge wall is read from the other side of the hinge: with the mass $s + t - \kappa_{\mathcal{A}}$ on g the table $\mathcal{A}$ itself is on its balance wall, and in a two-sum on its balance wall that mass is the hinge wall of the other cell (Lemma 21.12).

23.2 Absorbing a cell

Lemma 23.2 (Absorbing a cell)**.** *Let $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a cell family on $B \sqcup \{g\}$ with masses $\mathbf{m}$ on B, its hinge wall κ_F, base form ϕ_F and ratio vector $\boldsymbol{\varrho}_F$, whose coordinates are*

written $x_{j,K}$ and $x_{j,\Psi}$, $x_{1,K}=1$. Let Z be a finite set disjoint from $B\sqcup\{g\}$, $\mathbf{m}_Z$ masses on Z, and $f_1,\dots,f_p$ positive continuous functions on $(0,\infty)^Z\times(0,\infty)$, each assigned to one letter of F, with real exponents a_i such that the exponents assigned to each letter add to its exponent, s_j for $K_{\mathcal{A}_j}$ and t_j for $\Psi_{\mathcal{A}_j}$. In this statement write L_i, γ_i and x_i for the letter of the hinge minor, the hinge value and the coordinate of the letter of f_i: $L_i=K_{\mathcal{A}_j/g}$, $\gamma_i=\gamma^K_{\mathcal{A}_j}$ and $x_i=x_{j,K}$ when it is $K_{\mathcal{A}_j}=K_{\mathcal{A}_j/g}\,(\gamma^K_{\mathcal{A}_j}+\alpha_g)$, and $L_i=\Psi_{\mathcal{A}_j\setminus g}$, $\gamma_i=\gamma^\Psi_{\mathcal{A}_j}$ and $x_i=x_{j,\Psi}$ when it is $\Psi_{\mathcal{A}_j}=\Psi_{\mathcal{A}_j\setminus g}\,(\gamma^\Psi_{\mathcal{A}_j}+\alpha_g)$. Let $H\ge 0$ be a Borel function on $(0,\infty)^p$. If the form

$$\omega=\alpha_B^{\mathbf{m}-\mathbf{1}}\alpha_Z^{\mathbf{m}_Z-\mathbf{1}}\,H\Big(\big(f_i(\alpha_Z,\gamma_i(\alpha_B))\big)_i\Big)\prod_{i=1}^{p}\Big[L_i(\alpha_B)\,f_i\big(\alpha_Z,\gamma_i(\alpha_B)\big)\Big]^{-a_i}d\alpha_B\,d\alpha_Z$$

is homogeneous of degree zero on $(0,\infty)^{B\sqcup Z}$, then, in $[0,\infty]$,

$$\int_{\{h=1\}}\omega=\int_{(0,\infty)^{2N-1}}\Big(\int_{\{h'=1\}}\omega_{\mathbf{x}}\Big)\,d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)(\mathbf{x}),$$
$$\omega_{\mathbf{x}}=\alpha_Z^{\mathbf{m}_Z-\mathbf{1}}\alpha_g^{\kappa_F-1}\,H\Big(\big(f_i(\alpha_Z,x_i\alpha_g)\big)_i\Big)\prod_i f_i(\alpha_Z,x_i\alpha_g)^{-a_i}\,d\alpha_Z\,d\alpha_g,$$

the integrals taken over any sections of positive functions h, h' of degree one on the two orthants.

For one table the ratio vector is the ratio of the cell. The lemma says that whatever the rest of a family does with a cell, provided it sees the cell through the cell's letters at the hinge, it sees it only through the law: the cell collapses to its hinge, which keeps the mass κ_F, and the letters that see the hinge see it stretched by the coordinates of the ratio vector. Pooling (Theorem 19.8) is the case of one cell family read on the shape before the law is formed; Theorem 22.3, in which the core sees each cell as one mark, is the lemma at every cell of a core, with the hinge values of the cells kept as weights of the core instead of their ratios.

23.3 Closures

Theorem 23.3 (The series closure at positive exponents)**.** *Let every table of F be a series closure, $(\mathcal{B}_j,\theta_j)=\binom{\{h_1,\dots,h_q\}}{q-1}\langle\mathcal{A}_{1j},\dots,\mathcal{A}_{qj}\rangle$ (Corollary 21.7(ii)), with exponents $s_j,t_j>0$, and let f_{F_i} be the transform of the cell family $F_i=((\mathcal{A}_{ij},\theta^{\mathcal{A}_{ij}});s_j,t_j)_j$ of the cells at B_i, on $B_i\sqcup\{g_i\}$ (Definition 23.1). On the balance wall, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\prod_j\frac{1}{\Gamma(s_j)\Gamma(t_j)}\int\prod_j x_j^{s_j-1}\prod_{j\ge2}y_j^{t_j-1}\;\prod_{i=1}^{q}f_{F_i}\big(\mathbf{x},(1,y_2,\dots,y_N)\big)\;d\mathbf{x}\,dy_2\cdots dy_N,$$

the integral over $(0,\infty)^N\times(0,\infty)^{N-1}$: the cells communicate only through $2N-1$ shared scales. A cell entering k times enters as its transform to the power k: one transform per kind of cell, its multiplicity a power, the order of the cells immaterial; for one table the closure is $\frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{t-1}\prod_i f_{\mathcal{A}_i}(y)\,dy$. With $|\mathbf{s}|=\sum_j s_j$ and the wall κ_{B_i} of the cell B_i, its hinge wall κ_{F_i} (Theorem 21.10), each transform is one Gramma integrated over the shape of the cell,

$$f_{F_i}(\mathbf{x},\mathbf{y})=\int_{\Delta_{B_i}}\mathsf{G}_{|\mathbf{s}|-\kappa_{B_i};(1,-1)}\Big(\sum_j\frac{x_j}{\gamma^K_{\mathcal{A}_{ij}}(u)},\;\sum_j y_j\,\gamma^\Psi_{\mathcal{A}_{ij}}(u)\Big)$$
$$\times\,u^{\mathbf{m}|_{B_i}-\mathbf{1}}\prod_j K_{\mathcal{A}_{ij}\setminus g_i}(u)^{-s_j}\,\Psi_{\mathcal{A}_{ij}\setminus g_i}(u)^{-t_j}\,d\omega_{B_i}(u),$$

against the form of the family of the hinge minors $F_i \setminus g_i = ((\mathcal{A}_{ij} \setminus g_i, \theta^{\mathcal{A}_{ij}}); s_j, t_j)_j$ *at the masses* $\mathbf{m}|_{B_i}$*. When that family is on its balance wall,* $\kappa_{B_i} = |\mathbf{s}|$*, as at every cell of one table at the centre,* $s = t = \sigma$ *with every mass* σ*, the index is* 0*, the Gramma depends on its two coefficients through their product,* $\mathsf{G}_{0;(1,-1)}(a,b) = \mathsf{G}_{0;(1,-1)}(1,ab)$ *(Theorem 2.4(iii)), and, when the family of the hinge minors has every proper wall positive,*

$$f_{F_i}(\mathbf{x},\mathbf{y}) = P_{F_i\setminus g_i}(\mathbf{m}|_{B_i};\mathbf{s},\mathbf{t})\ \mathbb{E}_{U\sim \mathrm{Breta}^\circ(F_i\setminus g_i;\mathbf{m}|_{B_i};\mathbf{s},\mathbf{t})}\Big[\mathsf{G}_{0;(1,-1)}\Big(1,\ \sum_j \frac{x_j}{\gamma^K_{\mathcal{A}_{ij}}(U)} \sum_j y_j\, \gamma^{\Psi}_{\mathcal{A}_{ij}}(U)\Big)\Big],$$

for one table $P_{\mathcal{A}_i\setminus g_i}(\mathbf{m}|_{B_i}; s, t)\, \mathbb{E}\big[\mathsf{G}_{0;(1,-1)}\big(1, xy\, \varrho_{\mathcal{A}_i}(U)\big)\big]$*: the image under the ratio of the form of the hinge minor, read through* $\mathsf{G}_{0;(1,-1)}$*. So* $k \geq 2$ *copies of one cell family closed in series have the period* $\prod_j (\Gamma(s_j)\Gamma(t_j))^{-1} \int \prod_j x_j^{s_j-1} \prod_{j\geq 2} y_j^{t_j-1} f_{F_1}\big(\mathbf{x}, (1, y_2, \ldots, y_N)\big)^k\, d\mathbf{x}\, dy_2 \cdots dy_N$*.*

For one table the closure of q cells is a node of width one (Definition 20.24) with the bunch $B_1 \sqcup \cdots \sqcup B_{q-1}$, a series connection (Proposition 20.37), against the room B_q. Theorem 20.26 reads it as $\frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty f_B(y) f_R(y) \frac{dy}{y}$, and Theorem 23.3 is the same integral written symmetrically in all the cells, $\frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} \prod_i f_{\mathcal{A}_i}(y)\, dy$ with the transforms of the cells (Definition 23.1), which are the orthant transforms f_i° of Proposition 20.37; for k equal cells at the centre the formula for k copies is Theorem 24.2(i), with $k-1$ copies at a series node and one copy as the room. The table that is one circuit, $\binom{\mathcal{E}}{n-1}$, is the series closure of n single marks, each with the transform $\mathsf{G}_{s-m_e;(1,-1)}(1,y)$ up to its weight (Corollary 18.22). The parallel closure is the dual statement (Proposition 8.15).

23.4 Cores and their markers

A table is built from cells by two-sums at the markers of cores.

Definition 23.4 (Markers; the twisted core)**.** Let the exponents be $s, t > 0$. A *core with markers* is a plain table $\mathcal{H}$ on $\mathcal{E}_\mathcal{H} \sqcup G_\mathcal{H}$ none of whose marks is a loop or a coloop, a core of Definition 21.1, whose *markers* are the marks of $G_\mathcal{H}$, the marks at which cells may be substituted (the marks h_i of Definition 21.1). *Cells at the markers* are plain cells $\mathcal{A}_1, \ldots, \mathcal{A}_k$ on $B_i \sqcup \{g_i\}$ whose hinges are identified with distinct markers $g_1, \ldots, g_k$, the sets $\mathcal{E}_\mathcal{H}$ and B_i disjoint, and $\mathcal{H}\langle\mathcal{A}_1, \ldots, \mathcal{A}_k\rangle$ is their substitution (Definition 21.1), the marks of $\mathcal{E}_\mathcal{H}$ and the markers without a cell left in place. For $\mathbf{x} \in (0,\infty)^k$ the *twisted core* $\mathcal{H}[\mathbf{x}]$ is $\mathcal{H}$ twisted in its cobasis letter by x_i at every g_i (Definition 22.1), carrying the masses $\kappa_{\mathcal{A}_i}$ on the markers g_i and the given masses $\mathbf{m}|_{\mathcal{E}_\mathcal{H}}$ on $\mathcal{E}_\mathcal{H}$; it has the basis weights θ_T and the cobasis weights $\theta_T \prod_{g_i \notin T} x_i$, and its period is that of a table with two systems of weights (Definition 8.1). With the cells at the shapes $u = (u_i)_i$, the twisted core $\mathcal{H}[(\varrho_{\mathcal{A}_i}(u_i))_i]$ is the core read at u of Definition 22.2 with the scale of every marker g_i divided by $\gamma^K_{\mathcal{A}_i}(u_i)$. A core with one marker g_0 left free is a *hinged core*; then $\mathcal{H}[\mathbf{x}]$ is a cell with the hinge g_0, not plain, and its hinge wall, base form, ratio and law are those of Definition 23.1.

Lemma 23.5 (Substitution at the markers)**.** *The substitution* $\mathcal{H}\langle\mathcal{A}_1, \ldots, \mathcal{A}_k\rangle$ *does not depend on the order of the two-sums (Proposition 21.3(i)). Its bases are the sets* $(T \cap \mathcal{E}_\mathcal{H}) \sqcup \bigsqcup_i C_i$ *with* T *a basis of* $\mathcal{H}$ *and, for each* i*,* C_i *a basis of* $\mathcal{A}_i$ *avoiding* g_i *when* $g_i \in T$ *and* $C_i + g_i$ *a basis of* $\mathcal{A}_i$ *when* $g_i \notin T$*, with the product weights. Its rank is* $r_\mathcal{H} + \sum_i (r_{\mathcal{A}_i} - 1)$*, and with* G' *the markers left free (Theorem 21.4)*

$$K_{\mathcal{H}\langle\mathcal{A}_1,\ldots,\mathcal{A}_k\rangle} = \prod_i K_{\mathcal{A}_i/g_i} \cdot K_\mathcal{H}\big(\alpha_{\mathcal{E}_\mathcal{H}}, (\gamma^K_{\mathcal{A}_i})_i, \alpha_{G'}\big),$$

$$\Psi_{\mathcal{H}\langle\mathcal{A}_1,\dots,\mathcal{A}_k\rangle}=\prod_i \Psi_{\mathcal{A}_i\setminus g_i}\cdot\Psi_{\mathcal{H}}\big(\alpha_{\mathcal{E}_\mathcal{H}},(\gamma^{\Psi}_{\mathcal{A}_i})_i,\alpha_{G'}\big).$$

For a hinged core the letters of the two hinge minors $\mathcal{H}\langle\mathcal{A}_1,\dots,\mathcal{A}_k\rangle\setminus g_0$ and $\mathcal{H}\langle\mathcal{A}_1,\dots,\mathcal{A}_k\rangle/g_0$ are the same products with the letters of $\mathcal{H}\setminus g_0$ and $\mathcal{H}/g_0$ in place of $K_\mathcal{H}$ and $\Psi_\mathcal{H}$.

Theorem 23.6 (The law of a substitution)**.** *Let F be a family with real data whose table j is $\mathcal{H}_j\langle\mathcal{A}_{1j},\dots,\mathcal{A}_{kj}\rangle$, the cores $\mathcal{H}_j$ plain on $\mathcal{E}_\mathcal{H}\sqcup G_\mathcal{H}$ with $\{g_1,\dots,g_k\}\subseteq G_\mathcal{H}$, the cells of all tables at g_i forming the cell family $F_i=((\mathcal{A}_{ij},\theta^{\mathcal{A}_{ij}});s_j,t_j)_j$ on $B_i\sqcup\{g_i\}$ (Definition 23.1), with masses on $\mathcal{E}_\mathcal{H}$ and on the marks of the cells. For $\mathbf{x}=(\mathbf{x}_i)_i$, $\mathbf{x}_i=(x_{i,j,K},x_{i,j,\Psi})_j\in(0,\infty)^{2N-1}$ with $x_{i,1,K}=1$, let $F_\mathcal{H}[\mathbf{x}]$ be the family $((\mathcal{H}_j,\theta^{\mathcal{H}_j});s_j,t_j)_j$ twisted at every g_i by $\mathbf{x}_i$, $x_{i,j,K}$ in K_j and $x_{i,j,\Psi}$ in Ψ_j (Definition 22.1), with the masses κ_{F_i} on the markers g_i and the given masses on $\mathcal{E}_\mathcal{H}$.*

(i) If $G_\mathcal{H}=\{g_1,\dots,g_k\}$ and F is on its balance wall, then, in $[0,\infty]$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int P_{F_\mathcal{H}[\mathbf{x}]}\big((\mathbf{m}|_{\mathcal{E}_\mathcal{H}},\kappa_{F_1},\dots,\kappa_{F_k});\mathbf{s},\mathbf{t}\big)\prod_i d\big((\boldsymbol{\varrho}_{F_i})_*\phi_{F_i}\big)(\mathbf{x}_i).$$

With the core $\binom{\{g_1,g_2\}}{1}$ in every table this is the gluing of two cell families F_1 and F_2, $\kappa_{F_1}+\kappa_{F_2}=2\sigma$, σ the total strength (Definition 8.1): the period of the twisted core family is $\int_0^\infty v^{\kappa_{F_1}-1}\prod_j(x_{1,j,K}v+x_{2,j,K})^{-s_j}(x_{1,j,\Psi}v+x_{2,j,\Psi})^{-t_j}\,dv$, a Breta of one bunch with at most $2N$ coefficients (Proposition 15.1), unchanged when $(\kappa_{F_1},\mathbf{x}_1)$ and $(\kappa_{F_2},\mathbf{x}_2)$ are swapped.

(ii) If $G_\mathcal{H}=\{g_0,g_1,\dots,g_k\}$, F is a cell family with the hinge g_0; its hinge wall is that of $F_\mathcal{H}[\mathbf{x}]$, which does not depend on $\mathbf{x}$,

$$\kappa_F=m_{\mathcal{E}_\mathcal{H}}+\sum_i\kappa_{F_i}-\sum_j\big(s_j(r_{\mathcal{H}_j}-1)+t_j(|\mathcal{E}_\mathcal{H}|+k-r_{\mathcal{H}_j})\big),$$

and its law is the mixture of the twisted laws: for every Borel $H\ge0$ on $(0,\infty)^{2N-1}$,

$$\int H\,d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)=\int\Big(\int H\,d\big((\boldsymbol{\varrho}_{F_\mathcal{H}[\mathbf{x}]})_*\phi_{F_\mathcal{H}[\mathbf{x}]}\big)\Big)\prod_i d\big((\boldsymbol{\varrho}_{F_i})_*\phi_{F_i}\big)(\mathbf{x}_i).$$

In particular $\int\boldsymbol{\varrho}_F^{-\mathbf{z}}d\phi_F=\int\big(\int\boldsymbol{\varrho}_{F_\mathcal{H}[\mathbf{x}]}^{-\mathbf{z}}d\phi_{F_\mathcal{H}[\mathbf{x}]}\big)\prod_i d\big((\boldsymbol{\varrho}_{F_i})_*\phi_{F_i}\big)(\mathbf{x}_i)$ for every real $\mathbf{z}$.

For one table with $s,t>0$ the laws are measures on $(0,\infty)$ and the twisted core family is the twisted core $\mathcal{H}[\mathbf{x}]$ of Definition 23.4.

The core is the only place where cells interact. Below a core each cell family is one measure on $(0,\infty)^{2N-1}$, for one table on a half-line; at the core the cell families are fed into the core family as twists of both letters of every table at the markers, of the cobasis letter alone for one table (Lemma 23.2), and the core family, with the masses κ_{F_i} on its markers, is integrated once for each value of the twists. So the work at a core is governed by the core itself and by the number of its markers, not by the sizes of the cells: a cell with a thousand marks and a cell with one mark cost the same once their laws are known. Theorem 19.13 is the gluing of two cell families read through Mellin transforms (Theorem 23.9(iv)).

23.5 Two cells glued

Lemma 23.7 (Two marks)**.** *For $s,t>0$, $\kappa,\kappa'>0$ with $\kappa+\kappa'=s+t$, and $x,x'>0$, let $\binom{\{g,g'\}}{1}[(x,x')]$ be the list $\binom{\{g,g'\}}{1}$ twisted in its cobasis letter by x at g and by x' at g' (Definition 22.1). With the masses κ on g and κ' on g' its period is*

$$P_{\binom{\{g,g'\}}{1}[(x,x')]}\big((\kappa,\kappa');s,t\big)=\int_0^\infty v^{\kappa-1}(1+v)^{-s}(xv+x')^{-t}\,dv$$

$$= x^{-t}\,\Gamma_{\overline{2}}\!\begin{bmatrix}\kappa,\ \kappa'\\ s,\ t\end{bmatrix}\ \mathsf{B}^{\{\{0,1\}\}}_{(s,t),\,\kappa';\,I_{\{0,1\}}}\big((1,x'/x)\big)$$
$$= \frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{t-1}\,\mathsf{G}_{s-\kappa;(1,-1)}(1,xy)\,\mathsf{G}_{s-\kappa';(1,-1)}(1,x'y)\,dy$$

(Proposition 15.1, the two coefficients merging into one of letter exponent $s+t$ when $x=x'$), unchanged when (κ,x) and (κ',x') are swapped. For a family of tables the kernel is the period of the twisted core family of Theorem 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$ in every table.

Proposition 23.8 (Faces of one law)**.** *Let $F=((\mathcal{A}_j,\theta_j);s_j,t_j)_{j=1}^N$ be a cell family on $B\sqcup\{g\}$ with plain tables, $s_j,t_j>0$, masses $\mathbf{m}$ on B and the hinge wall $\kappa=\kappa_F$; in this statement the points of $(0,\infty)^{2N-1}$ are written $\mathbf{z}=(z_{j,K},z_{j,\Psi})_j$, $z_{1,K}=1$, and $\mathbf{x}'=(x_2,\dots,x_N)$.*

(i) Law forms. *For $\mathbf{x},\mathbf{y}\in(0,\infty)^N$,*

$$f_F(\mathbf{x},\mathbf{y})=\int \mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j\frac{x_j}{z_{j,K}},\ \sum_j y_j z_{j,\Psi}\Big)\prod_j z_{j,K}^{-s_j}\,d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)(\mathbf{z}),$$
$$\check f_F(\mathbf{x},\mathbf{y})=\int \mathsf{G}_{|\mathbf{t}|-\kappa;(1,-1)}\Big(\sum_j x_j\frac{z_{1,\Psi}}{z_{j,\Psi}},\ \sum_j y_j\frac{z_{j,K}}{z_{1,\Psi}}\Big)\prod_j\Big(\frac{z_{j,\Psi}}{z_{1,\Psi}}\Big)^{-t_j} z_{1,\Psi}^{-\kappa}\,d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)(\mathbf{z}):$$

the two transforms depend on F only through its hinge wall and its law. For one cell, for $y>0$,

$$f_{\mathcal{A}}(y)=\int_{\Delta_B}\mathsf{G}_{s-\kappa;(1,-1)}\big(1,\varrho_{\mathcal{A}}y\big)\,d\phi_{\mathcal{A}},\qquad \check f_{\mathcal{A}}(y)=\int_{\Delta_B}\mathsf{G}_{t-\kappa;(1,-1)}\big(1,y/\varrho_{\mathcal{A}}\big)\,\varrho_{\mathcal{A}}^{-\kappa}\,d\phi_{\mathcal{A}}.$$

(ii) Hinge measures.

$$f_F(\mathbf{x},\mathbf{y})=\int e^{-\langle\mathbf{x},\boldsymbol{\lambda}\rangle-\langle\mathbf{y},\boldsymbol{\mu}\rangle}\,d\mathfrak{r}_F(\boldsymbol{\lambda},\boldsymbol{\mu}),\qquad \check f_F(\mathbf{x},\mathbf{y})=\int e^{-\langle\mathbf{x},\boldsymbol{\lambda}\rangle-\langle\mathbf{y},\boldsymbol{\mu}\rangle}\,d\check{\mathfrak{r}}_F(\boldsymbol{\lambda},\boldsymbol{\mu}),$$

and $\mathfrak{r}_F$ is the image of the product of $\prod_j z_{j,K}^{-s_j}\,(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z})$ and $p^{\kappa-|\mathbf{s}|-1}dp$ under $(\mathbf{z},p)\mapsto\big((1/(pz_{j,K}))_j,(pz_{j,\Psi})_j\big)$. So f_F and $\check f_F$ are completely monotone in their $2N$ variables, $(-1)^{|\mathbf{n}|}\partial^{\mathbf{n}}f_F\ge 0$ for every multi-index $\mathbf{n}$ on the interior of the set where f_F is finite and the same for $\check f_F$, and log-convex along every segment on which they are finite. For one cell, $f_{\mathcal{A}}(y)=\int e^{-y\lambda}d\mathfrak{r}^{\Psi}_{\mathcal{A}}(\lambda)$ and $\check f_{\mathcal{A}}(y)=\int e^{-y\lambda}d\mathfrak{r}^{K}_{\mathcal{A}}(\lambda)$; $\mathfrak{r}^{\Psi}_{\mathcal{A}}$ is the image of $\alpha^{\mathbf{m}-\mathbf{1}}K_{\mathcal{A}\setminus g}^{-s}\Psi_{\mathcal{A}\setminus g}^{-t}e^{-1/\gamma^K_{\mathcal{A}}}d\alpha$ under $\gamma^{\Psi}_{\mathcal{A}}$ and of $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dx)\otimes p^{s-\kappa-1}e^{-p}dp$ under $(x,p)\mapsto x/p$, and $\mathfrak{r}^{K}_{\mathcal{A}}$ the image of $\alpha^{\mathbf{m}-\mathbf{1}}K_{\mathcal{A}/g}^{-s}\Psi_{\mathcal{A}/g}^{-t}e^{-1/\gamma^{\Psi}_{\mathcal{A}}}d\alpha$ under $\gamma^{K}_{\mathcal{A}}$ and of $x^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dx)\otimes p^{t-\kappa-1}e^{-p}dp$ under $(x,p)\mapsto 1/(xp)$; so $f_{\mathcal{A}}$ and $\check f_{\mathcal{A}}$ are decreasing and log-convex wherever finite.

(iii) Mellin transforms. *For real $\mathbf{a},\mathbf{b}\in(0,\infty)^N$ with $\sum_j b_j-\sum_j a_j=\kappa-|\mathbf{s}|$, in $[0,\infty]$,*

$$\int_{(0,\infty)^{2N-1}}\prod_{j\ge2}x_j^{a_j-1}\prod_j y_j^{b_j-1}\,f_F\big((1,\mathbf{x}'),\mathbf{y}\big)\,d\mathbf{x}'\,d\mathbf{y}$$
$$=\prod_j\Gamma(a_j)\Gamma(b_j)\int_{\Delta_B}\boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a},\mathbf{b})}\,d\phi_F,$$

and for real $\mathbf{a},\mathbf{b}\in(0,\infty)^N$ *with* $\sum_j b_j-\sum_j a_j=\kappa-|\mathbf{t}|$ *the same holds for* $\check f_F$ *with* $\int_{\Delta_B}\boldsymbol{\varrho}_F^{-(\mathbf{b},\mathbf{t}-\mathbf{a})}\,d\phi_F$ *on the right; both Mellin transforms of the law are taken on the hyperplane* $\sum_j(z_{j,K}+z_{j,\Psi})=\kappa$ *of their arguments. For one cell, with* $z=b$*, these are*

$$\int_0^\infty y^{z-1}f_{\mathcal A}(y)\,dy=\Gamma(z)\Gamma(z+s-\kappa)\int_{\Delta_B}\varrho_{\mathcal A}^{-z}\,d\phi_{\mathcal A}\qquad\big(z>\max(0,\kappa-s)\big),$$
$$\int_0^\infty y^{z-1}\check f_{\mathcal A}(y)\,dy=\Gamma(z)\Gamma(z+t-\kappa)\int_{\Delta_B}\varrho_{\mathcal A}^{z-\kappa}\,d\phi_{\mathcal A}\qquad\big(z>\max(0,\kappa-t)\big).$$

(iv) The dual cell family. *The dual tables* $\mathcal A_j^*$*, with the hinge* g*, at the exponents* (t_j,s_j) *form a cell family* F^* *with the hinge wall* κ*, the hinge values* $\gamma^K_{\mathcal A_j^*}=\gamma^\Psi_{\mathcal A_j}$ *and* $\gamma^\Psi_{\mathcal A_j^*}=\gamma^K_{\mathcal A_j}$*, the base form* $(\gamma^\Psi_{\mathcal A_1}/\gamma^K_{\mathcal A_1})^{-\kappa}\phi_F$*, the law the image of* $z_{1,\Psi}^{-\kappa}\,(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z})$ *under* $\mathbf{z}\mapsto(z_{j,\Psi}/z_{1,\Psi},\,z_{j,K}/z_{1,\Psi})_j$*, the transform* $\check f_F$ *and the dual transform* f_F*. For one cell the ratio is* $1/\varrho_{\mathcal A}$*, the base form* $\varrho_{\mathcal A}^{-\kappa}\phi_{\mathcal A}$*, the law the image of* $x^{-\kappa}(\varrho_{\mathcal A})_*\phi_{\mathcal A}(dx)$ *under* $x\mapsto1/x$*, and* $\int\varrho_{\mathcal A^*}^{-z}d\phi_{\mathcal A^*}=\int\varrho_{\mathcal A}^{z-\kappa}d\phi_{\mathcal A}$*.*

(v) Change of transform. *For real* $\mathbf{a},\mathbf{b}$ *with* $0<a_j<s_j$*,* $0<b_j<t_j$ *and* $\sum_j b_j-\sum_j a_j=\kappa-|\mathbf{s}|$*, the left side of (iii) for* $\check f_F$ *at* $(\mathbf{t}-\mathbf{b},\mathbf{s}-\mathbf{a})$ *times* $\prod_j\Gamma(a_j)\Gamma(b_j)$ *equals that for* f_F *at* $(\mathbf{a},\mathbf{b})$ *times* $\prod_j\Gamma(t_j-b_j)\Gamma(s_j-a_j)$*, in* $(0,\infty]$*. For one cell, with* $z=s-a$*: for real* z *with* $\max(0,\kappa-t)<z<\min(s,\kappa)$*, in* $(0,\infty]$*,*

$$\int_0^\infty y^{z-1}\check f_{\mathcal A}(y)\,dy=\frac{\Gamma(z)\Gamma(z+t-\kappa)}{\Gamma(\kappa-z)\Gamma(s-z)}\int_0^\infty y^{\kappa-z-1}f_{\mathcal A}(y)\,dy.$$

When $\mathcal A$ *is the cell of the bunch of a node of width one (Definition 20.24) and* $\int\varrho_{\mathcal A}^{-b}d\phi_{\mathcal A}<\infty$ *at some* b *with* $\max(0,\kappa_{\mathcal A}-s)<b<\min(t,\kappa_{\mathcal A})$*, the expansion of* $f_{\mathcal A}$ *at* $y\to0$ *is the sum of the residues of* $y^{-z}\Gamma(z)\Gamma(z+s-\kappa_{\mathcal A})\int\varrho_{\mathcal A}^{-z}d\phi_{\mathcal A}$ *to the left of the line* $\operatorname{Re}z=b$ *(Theorem 20.34(i)), and that of* $\check f_{\mathcal A}$*, the transform of* $\mathcal A^*$ *read as the bunch of the dual node of width one at the exponents* (t,s) *(Proposition 8.15), is the sum of the residues of* $y^{-z}\Gamma(z)\Gamma(z+t-\kappa_{\mathcal A})\int\varrho_{\mathcal A}^{z-\kappa_{\mathcal A}}d\phi_{\mathcal A}$ *to the left of the line* $\operatorname{Re}z=\kappa_{\mathcal A}-b$*, at the poles of the Gamma pair and at the points* $\kappa_{\mathcal A}-w$*,* w *a pole of the continued* $\int\varrho_{\mathcal A}^{-w}d\phi_{\mathcal A}$ *to the right of the line* $\operatorname{Re}w=b$*: the two transforms expand one function on the two sides of its strip.*

So one law has four faces. The base form and its image, the law, are where the cell family lives; the transform reads the law through Grammas of index $|\mathbf{s}|-\kappa_F$ and is the Laplace transform of the hinge measure $\mathfrak r_F$, carried by the reciprocals of the hinge values by K and the hinge values by Ψ; the dual transform reads the law turned over through Grammas of index $|\mathbf{t}|-\kappa_F$ and is the Laplace transform of $\check{\mathfrak r}_F$, carried by the reciprocals of the hinge values by Ψ and the hinge values by K; and on the Mellin side both are the one function $\int\boldsymbol{\varrho}_F^{-\mathbf{z}}d\phi_F$ with two Gamma products. Passing from one transform to the other costs nothing on the Mellin side, a reflection and a ratio of Gammas, and costs integral transforms on the side of the transforms; for one cell, on the half-line, the two expansions at $y\to0$ are read on opposite sides of the strip: the transform expands where the law is large, the dual transform where it is small.

Theorem 23.9 (Two cell families glued)**.** *Let* F_1 *and* F_2 *be cell families with plain tables on* $B\sqcup\{g\}$ *and* $R\sqcup\{g\}$*, at the exponents* $(s_j,t_j)_j$ *with* $s_j,t_j>0$*, and let* $F_1\oplus_g F_2$*, whose table* j *is the two-sum of the tables* j *of* F_1 *and* F_2 *(Lemma 21.12), carry masses* $\mathbf{m}$ *on its balance wall, so that* $\kappa_{F_1}+\kappa_{F_2}=2\sigma$*,* σ *the total strength (Definition 8.1); let* $0<\kappa_{F_1}<2\sigma$*. In this statement the points of the laws of* F_1 *and* F_2 *are written* $\mathbf{z}$ *and* $\mathbf{z}'$*,* $\mathcal G(\mathbf{z},\mathbf{z}')=\int_0^\infty v^{\kappa_{F_1}-1}\prod_j(z_{j,K}v+z'_{j,K})^{-s_j}(z_{j,\Psi}v+z'_{j,\Psi})^{-t_j}\,dv$ *is*

the period of the twisted core family of Theorem 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$ *in every table, the transforms are read at* $x_1 = 1$*, with* $\mathbf{x}' = (x_2, \dots, x_N)$*, and the hinge measures of* F_1 *act in the variables* $(\boldsymbol{\lambda}, \boldsymbol{\mu})$*, those of* F_2 *in* $(\boldsymbol{\lambda}', \boldsymbol{\mu}')$*. Then, in* $[0, \infty]$*,*

$$P_{F_1 \oplus_g F_2}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \iint \mathcal{G}(\mathbf{z}, \mathbf{z}')\, d\big((\boldsymbol{\varrho}_{F_1})_* \phi_{F_1}\big)(\mathbf{z})\, d\big((\boldsymbol{\varrho}_{F_2})_* \phi_{F_2}\big)(\mathbf{z}') \tag{i}$$

$$= \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1}\, f_{F_1} f_{F_2}\, d\mathbf{x}'\, d\mathbf{y}$$

$$= \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{t_j - 1} \prod_j y_j^{s_j - 1}\, \check{f}_{F_1} \check{f}_{F_2}\, d\mathbf{x}'\, d\mathbf{y} \tag{ii}$$

$$= \frac{1}{\Gamma(s_1)} \iint e^{-\lambda_1 - \lambda_1'} \prod_{j \geq 2} (\lambda_j + \lambda_j')^{-s_j} \prod_j (\mu_j + \mu_j')^{-t_j}\, d\mathfrak{r}_{F_1}\, d\mathfrak{r}_{F_2}$$

$$= \frac{1}{\Gamma(t_1)} \iint e^{-\lambda_1 - \lambda_1'} \prod_{j \geq 2} (\lambda_j + \lambda_j')^{-t_j} \prod_j (\mu_j + \mu_j')^{-s_j}\, d\check{\mathfrak{r}}_{F_1}\, d\check{\mathfrak{r}}_{F_2}, \tag{iii}$$

and these are finite exactly when every proper wall of $F_1 \oplus_g F_2$ *is positive. On every contour of the level shares of Theorem 19.13 on which its stacked Gammas have arguments of positive real part and the two Mellin transforms below converge absolutely (Definition 23.1),*

$$P_{F_1 \oplus_g F_2}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{N-1} \int_{\sum_j \delta_j^d = \frac{1}{2}(|\mathbf{s}| - \kappa_{F_1})} \prod_j \Gamma_2 \begin{bmatrix} a_j^d, & b_j^d, & a_j^r, & b_j^r \\ & & s_j, & t_j \end{bmatrix}$$
$$\times \int_{\Delta_B} \boldsymbol{\varrho}_{F_1}^{-(\mathbf{a}^r, \mathbf{b}^d)}\, d\phi_{F_1} \int_{\Delta_R} \boldsymbol{\varrho}_{F_2}^{-(\mathbf{a}^d, \mathbf{b}^r)}\, d\phi_{F_2}, \tag{iv}$$

the formula of Theorem 19.13, the periods of its bunch family and room family at the level shares being these two Mellin transforms. For one table the laws are $(\varrho_{\mathcal{A}})_* \phi_{\mathcal{A}}$ *and* $(\varrho_{\mathcal{A}'})_* \phi_{\mathcal{A}'}$ *and the kernel of (i) is the period of Lemma 23.7; (ii) reads* $\frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} f_{\mathcal{A}} f_{\mathcal{A}'}\, dy = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{s-1} \check{f}_{\mathcal{A}} \check{f}_{\mathcal{A}'}\, dy$*, its first form Theorem 20.26(i) with the room transform* $f_R(y) = y^t f_{\mathcal{A}'}(y)$*; (iii) reads* $\frac{1}{\Gamma(s)} \iint (\lambda + \lambda')^{-t}\, d\mathfrak{r}_{\mathcal{A}}^{\Psi}\, d\mathfrak{r}_{\mathcal{A}'}^{\Psi} = \frac{1}{\Gamma(t)} \iint (\lambda + \lambda')^{-s}\, d\mathfrak{r}_{\mathcal{A}}^{K}\, d\mathfrak{r}_{\mathcal{A}'}^{K}$*; and (iv) holds on a line* $\operatorname{Re} z = z_0$ *with* $\max(0, \kappa_{\mathcal{A}} - s) < z_0 < \min(t, \kappa_{\mathcal{A}})$ *on which* $\int \varrho_{\mathcal{A}}^{-z_0} d\phi_{\mathcal{A}}$ *and* $\int \varrho_{\mathcal{A}'}^{z_0 - t} d\phi_{\mathcal{A}'}$ *are finite, as*

$$P_{\mathcal{A} \oplus_g \mathcal{A}'}(\mathbf{m}; s, t) = \frac{1}{\Gamma(s)\Gamma(t)} \frac{1}{2\pi i} \int_{\operatorname{Re} z = z_0} \Gamma(z)\Gamma(z + s - \kappa_{\mathcal{A}})\Gamma(t - z)\Gamma(\kappa_{\mathcal{A}} - z)$$
$$\times \int_{\Delta_B} \varrho_{\mathcal{A}}^{-z}\, d\phi_{\mathcal{A}} \int_{\Delta_R} \varrho_{\mathcal{A}'}^{z - t}\, d\phi_{\mathcal{A}'}\, dz.$$

The period of a two-sum is a pairing of two laws by one kernel, the period of the twisted core family on two marks, for one table the period of two marks. Read through the Grammas the pairing is one integral over the scales of the two transforms, symmetric in the two sides; for one table, read through the hinge measures, it is the moment of order $-t$ of a sum: under $\mathfrak{r}_{\mathcal{A}}^{\Psi}$ the variable is the hinge value $\gamma_{\mathcal{A}}^{\Psi}$ of the cell (Proposition 23.8(ii)), and gluing the two cells into one table adds their hinge values by Ψ, as $\Psi = \Psi_{\mathcal{A} \setminus g} \Psi_{\mathcal{A}' \setminus g} (\gamma_{\mathcal{A}}^{\Psi} + \gamma_{\mathcal{A}'}^{\Psi})$ says.

23.6 What a cell retains

Definition 23.10 (One weighted mark)**.** For $\mathbf{x}_0 = (x_{0,j,K}, x_{0,j,\Psi})_{j=1}^N$ with $x_{0,1,K} = 1$, a point of $(0, \infty)^{2N-1}$, and a mark e with the mass m_e, the *one-mark cell family of ratio vector* $\mathbf{x}_0$ has in

table j the bases $\{e\}$ and $\{g\}$, with the weights $x_{0,j,K}$ and 1 in its letter K_j and the weights 1 and $x_{0,j,\Psi}$ in its letter Ψ_j (Definition 21.1): $K_j = x_{0,j,K}\alpha_e + \alpha_g$ and $\Psi_j = \alpha_g + x_{0,j,\Psi}\alpha_e$. Its hinge values are $x_{0,j,K}\alpha_e$ and $x_{0,j,\Psi}\alpha_e$, its ratio vector is the constant $\mathbf{x}_0$, its hinge wall is m_e, and its law is $\delta_{\mathbf{x}_0}$. For one table and $x_0 > 0$, the *one-mark cell of ratio* x_0 is the plain table $\left(\binom{\{e,g\}}{1}, (1, x_0^{1/2})\right)$ on $\{e,g\}$, the bases $\{e\}$ and $\{g\}$ with the weights 1 and $x_0^{1/2}$. Its hinge minors have $K_{\setminus g} = \alpha_e$, $K_{/g} = x_0^{1/2}$, $\Psi_{\setminus g} = 1$ and $\Psi_{/g} = x_0^{1/2}\alpha_e$, its ratio is the constant x_0, its hinge wall is m_e, and its law is $x_0^{(m_e-s)/2}\,\delta_{x_0}$.

Theorem 23.11 (The law is what a cell family retains)**.** *Let F and F' be cell families with plain tables, at the exponents $(s_j, t_j)_j$ with $s_j, t_j > 0$, with masses on their marks other than the hinge and one hinge wall $\kappa \in (0, 2\sigma)$, σ the total strength (Definition 8.1), and let there be a real $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$ with $0 < z_{j,K} < s_j$, $0 < z_{j,\Psi} < t_j$ and $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa$ at which $\int \boldsymbol{\varrho}_F^{-\mathbf{z}} d\phi_F$ and $\int \boldsymbol{\varrho}_{F'}^{-\mathbf{z}} d\phi_{F'}$ are finite (Definition 23.1). The following are equivalent.*

(a) $(\boldsymbol{\varrho}_F)_* \phi_F = (\boldsymbol{\varrho}_{F'})_* \phi_{F'}$.

(b) $P_{F \oplus_g F''} = P_{F' \oplus_g F''}$, *in* $[0, \infty]$, *for every cell family F'' with the hinge wall $2\sigma - \kappa$, the masses of F, F' and F'' kept.*

(c) (b) holds for the one-mark cell families F'' of ratio vector $\mathbf{x}_0$ and mass $2\sigma - \kappa$ (Definition 23.10), for every $\mathbf{x}_0$ in a nonempty open subset of $(0, \infty)^{2N-1}$.

(d) $\int \boldsymbol{\varrho}_F^{-\mathbf{w}} d\phi_F = \int \boldsymbol{\varrho}_{F'}^{-\mathbf{w}} d\phi_{F'}$ *at every complex $\mathbf{w}$ with real part $\mathbf{z}$.*

So the hinge wall and the law are the whole of what a cell family contributes to the families built on it, and the one-mark cell families, whose laws are single points, already separate any two laws. For one table the plain one-mark cells of ratio x_0 suffice in (c), and (d) is the agreement of $\int \varrho_{\mathcal{A}}^{-w} d\phi_{\mathcal{A}}$ and $\int \varrho_{\mathcal{A}'}^{-w} d\phi_{\mathcal{A}'}$ on the line $\operatorname{Re} w = z_{1,\Psi}$, $\max(0, \kappa - s) < z_{1,\Psi} < \min(t, \kappa)$.

What determines the contribution of a cell family to a larger construction is its hinge wall, which balance hands to the other side of the hinge, and its law, one positive measure on $(0, \infty)^{2N-1}$, on the half-line for one cell. Anything finer about the cells, their marks, their masses one by one, their letters, is invisible from outside, and anything coarser fails already against one weighted mark. A cell family is therefore replaceable, in every family it can sit in, by any other cell family of the same wall and the same law, and a cell family whose law is one point is a mark. Theorem 21.16 says the same at any width for the joint law of the bunch letters; at width one the joint law collapses to the law of the ratio vector.

Corollary 23.12 (Cells that are marks)**.** *Let F be a cell family with plain tables and $s_j, t_j > 0$, and $\mathcal{A}$ a plain cell with exponents $s, t > 0$.*

(i) The law of F is carried by a single point, $(\boldsymbol{\varrho}_F)_* \phi_F = \phi_F(\Delta_B)\,\delta_{\mathbf{x}_0}$, exactly when the ratio vector is constant, $\boldsymbol{\varrho}_F \equiv \mathbf{x}_0$; then $\phi_F(\Delta_B) \in (0, \infty]$, finite whenever F sits in a gluing of finite period, and in every gluing F acts as $\phi_F(\Delta_B)$ times the one-mark cell family of ratio vector $\mathbf{x}_0$ and mass κ_F. For one cell with $\varrho_{\mathcal{A}} \equiv x_0$ this is $\phi_{\mathcal{A}}(\Delta_B)\, x_0^{(s-\kappa_{\mathcal{A}})/2}$ times the one-mark cell of ratio x_0 and mass $\kappa_{\mathcal{A}}$.

(ii) If $\mathcal{A}/g$ is the dual table of $\mathcal{A} \setminus g$ with the identity on B and the same weights, then $\varrho_{\mathcal{A}} \equiv 1$ and $\phi_{\mathcal{A}}(\Delta_B) = P_{\mathcal{A}\setminus g}(\mathbf{m}; \kappa_{\mathcal{A}}, s + t - \kappa_{\mathcal{A}})$.

(iii) If a table $(\mathcal{B}, \theta)$ is identically self-dual*, its bases closed under complement with $\theta_{\mathcal{E}\setminus T} = \theta_T$, then for every mark g and masses on the balance wall with every proper wall positive,*

$$P_F(\mathbf{m}; s, t) = \Gamma_{\binom{2}{1}} \begin{bmatrix} m_g, & s + t - m_g \\ & s + t \end{bmatrix} P_{F\setminus g}\big(\mathbf{m}|_{\mathcal{E}\setminus g}; s + t - m_g, m_g\big),$$

$F = ((\mathcal{B},\theta); s, t)$ and $F \setminus g = ((\mathcal{B} \setminus g, \theta); s + t - m_g, m_g)$ the family of the deletion.

So (iii) splits off any mark of an identically self-dual table, by one Beta corner, at every pair of exponents, and Proposition 19.12, every letter active, is (i) read across a bunch of width one, the ratio vector being constant. The relation is a consequence of structure alone: the mark g and the rest of the table see each other through a ratio that is identically one, so the hinge integral separates. For a table that is its own dual only up to a relabelling of its marks the hypothesis of (iii) fails, and (iii) does not apply.

23.7 Series and parallel nodes

Definition 23.13 (Series and parallel cores)**.** For $q \geq 2$ and the markers $G = \{g_0, g_1, \dots, g_q\}$ a *series core* is a table with the bases of the list $\binom{G}{q}$ and a *parallel core* a table with the bases of the list $\binom{G}{1}$, with any positive weights. With cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ at $g_1, \dots, g_q$ and the hinge g_0, $\binom{G}{q}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ and $\binom{G}{1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$, the cores with unit weights, are the *series connection* of the cells, whose bunch $\bigsqcup_i B_i$ is the series connection of the parts B_i of Proposition 20.37, and their *parallel connection*; at the vertices of a hinge tree series and parallel cores are *series nodes* and *parallel nodes*.

A series bunch of marks is the series connection of its marks, each a one-mark cell, and a parallel bunch the parallel connection (Proposition 19.3); and the series closure $\binom{\{h_1,\dots,h_q\}}{q-1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ of Corollary 21.7(ii) is the series connection of $\mathcal{A}_1, \dots, \mathcal{A}_{q-1}$ glued to $\mathcal{A}_q$ along g_0.

Theorem 23.14 (Series and parallel nodes)**.** *Let $F_1, \dots, F_q$, $q \geq 2$, be cell families with plain tables at the exponents $(s_j, t_j)_j$, $s_j, t_j > 0$, the table j of F_i being $\mathcal{A}_{ij}$, and $G = \{g_0, \dots, g_q\}$.*

(i) Series. *The cell family F whose table j is $\mathcal{A}_j = \binom{G}{q}\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$ has $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{s}|$, its hinge values combine in every table as $1/\gamma^K_{\mathcal{A}_j} = \sum_i 1/\gamma^K_{\mathcal{A}_{ij}}$ and $\gamma^\Psi_{\mathcal{A}_j} = \sum_i \gamma^\Psi_{\mathcal{A}_{ij}}$, and*

$$f_F = \prod_{i=1}^{q} f_{F_i}, \qquad \mathfrak{r}_F = \mathfrak{r}_{F_1} * \cdots * \mathfrak{r}_{F_q},$$

the convolution on $(0,\infty)^{2N}$. For one table, $\mathcal{A} = \binom{G}{q}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ has $f_{\mathcal{A}} = \prod_i f_{\mathcal{A}_i}$ and $\mathfrak{r}^\Psi_{\mathcal{A}} = \mathfrak{r}^\Psi_{\mathcal{A}_1} * \cdots * \mathfrak{r}^\Psi_{\mathcal{A}_q}$, the convolution on $(0,\infty)$, and the series core twisted by $\mathbf{x}$ at $g_1, \dots, g_q$ has, on the simplex of its markers $g_1, \dots, g_q$ with the coordinates v, the ratio and the base form

$$\Big(\sum_i \frac{1}{v_i}\Big)\Big(\sum_i x_i v_i\Big), \qquad \prod_i v_i^{\kappa_{\mathcal{A}_i}-1}\, e_{q-1}(v)^{\kappa_{\mathcal{A}}-s}\Big(\prod_i v_i\Big)^{-\kappa_{\mathcal{A}}} d\omega_{\{g_1,\dots,g_q\}}(v).$$

(ii) Parallel. *The cell family F whose table j is $\mathcal{A}_j = \binom{G}{1}\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$ has $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{t}|$, its hinge values combine as $\gamma^K_{\mathcal{A}_j} = \sum_i \gamma^K_{\mathcal{A}_{ij}}$ and $1/\gamma^\Psi_{\mathcal{A}_j} = \sum_i 1/\gamma^\Psi_{\mathcal{A}_{ij}}$, and*

$$\check{f}_F = \prod_{i=1}^{q} \check{f}_{F_i}, \qquad \check{\mathfrak{r}}_F = \check{\mathfrak{r}}_{F_1} * \cdots * \check{\mathfrak{r}}_{F_q};$$

for one table $\check{f}_{\mathcal{A}} = \prod_i \check{f}_{\mathcal{A}_i}$ and $\mathfrak{r}^K_{\mathcal{A}} = \mathfrak{r}^K_{\mathcal{A}_1} * \cdots * \mathfrak{r}^K_{\mathcal{A}_q}$.

(iii) Extremes of the ratio. *For one table, with plain cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ at the exponents (s,t): in a series connection $\sqrt{\inf \varrho_{\mathcal{A}}} = \sum_i \sqrt{\inf \varrho_{\mathcal{A}_i}}$ (Proposition 20.37(i)) and $\sup \varrho_{\mathcal{A}} = \infty$; in a parallel connection $1/\sqrt{\sup \varrho_{\mathcal{A}}} = \sum_i 1/\sqrt{\sup \varrho_{\mathcal{A}_i}}$ and $\inf \varrho_{\mathcal{A}} = 0$, with $1/\sqrt{\infty} = 0$.*

(iv) Reaches. *For one table: if every $f_{\mathcal{A}_i}$ has a lattice expansion of reach ζ_i (Definition 20.28), the series connection has $f_{\mathcal{A}}$ with a lattice expansion of reach $\big(\sum_i \sqrt{\zeta_i}\big)^2$ (Proposition 20.37(ii)); if every $\check{f}_{\mathcal{A}_i}$ has one of reach ζ_i, the parallel connection has $\check{f}_{\mathcal{A}}$ with one of reach $\big(\sum_i \sqrt{\zeta_i}\big)^2$.*

A series node adds the hinge values by Ψ of its cells and a parallel node their hinge values by K; in the matching transform each is a product, and in the matching hinge measure a convolution. In the other transform neither node is a product: the dual orthant form of a series connection does not factor, since $K_{\mathcal{A}/g_0} = \prod_i K_{\mathcal{A}_i/g_i}\, e_{q-1}\big((\gamma^K_{\mathcal{A}_i})_i\big)$ and $1/\gamma^K_{\mathcal{A}} = \sum_i 1/\gamma^K_{\mathcal{A}_i}$ mix the cells. What changes at the edges where the kind of node changes is the transform, not the cells (Proposition 23.8(v)). The ceilings of Section 20.8 are read on a bunch as a whole, where the letters $K_{\mathcal{A}/g_0}$ and $\Psi_{\mathcal{A}/g_0}$ of the hinge minors of a series connection mix the cells and their zeros can raise a ceiling above the reach; at a node the cells are separate factors and the reach is read cell by cell (Lemma 20.30). So ceilings are to be applied to the prime nodes of a hinge tree, and reaches combined across series and parallel nodes by (iv).

23.8 Hinge trees

Definition 23.15 (Hinge tree)**.** A *hinge tree* of a table on $\mathcal{E}$ is a finite tree $\mathfrak{T}$ with a core $\mathcal{H}_v$ on $\mathcal{E}_v \sqcup G_v$ at each vertex v, its *node*, where the $\mathcal{E}_v$ partition $\mathcal{E}$ and $G_v = \{g_{v,\mathfrak{e}}\}$ has one marker for each edge $\mathfrak{e}$ at v, such that the table is obtained from the $\mathcal{H}_v$ by the two-sums identifying $g_{v,\mathfrak{e}}$ with $g_{w,\mathfrak{e}}$ for every edge $\mathfrak{e} = \{v,w\}$. Rooted at a vertex o, every vertex $v \neq o$ has a parent edge $\mathfrak{e}_v$, and the *subtree cell* $\mathcal{A}_v$ is the substitution of the $\mathcal{H}_w$ over the vertices w below v, v included, with the hinge $g_{v,\mathfrak{e}_v}$; its cells are the subtree cells of the vertices directly below v. The tables of a family $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_j$ have a *common hinge tree* when one tree $\mathfrak{T}$, with the same sets $\mathcal{E}_v$ and markers G_v, is a hinge tree of every table, table j with the nodes $\mathcal{H}_{v,j}$; the nodes then form the *node families* $F_{\mathcal{H}_v} = ((\mathcal{H}_{v,j}, \theta^{\mathcal{H}_{v,j}}); s_j, t_j)_j$ and, rooted at o, the subtree cells $\mathcal{A}_{v,j}$ the cell families $F_v = ((\mathcal{A}_{v,j}, \theta^{\mathcal{A}_{v,j}}); s_j, t_j)_j$ of the vertices $v \neq o$ (Definition 23.1).

Theorem 23.16 (A connected table is a tree of prime tables)**.** *Call a connected table on at least four marks* prime *when it has no bunch of width one with at least two marks on each side. Every connected table on at least three marks whose weights factor across each bunch of width one at which it is decomposed, a list or a determinantal table for instance (Theorem 12.19), is obtained by substitutions along a tree from prime tables and the series and parallel nodes $\binom{S}{|S|-1}$ and $\binom{S}{1}$, $|S| \geq 3$ (Definition 23.13), the weights carried along. With no two series nodes and no two parallel nodes adjacent in the tree, the tree, the* canonical hinge tree *(Definition 23.15), is unique up to the names of the markers. The prime nodes of a list are therefore invariants of the list.*

By Lemma 23.5, applied leaf by leaf, the substitution along a tree does not depend on the order of the two-sums: its bases are the sets $\bigsqcup_v (T_v \cap \mathcal{E}_v)$ with T_v a basis of $\mathcal{H}_v$ and every edge's marker in exactly one of the two bases at its ends. Every connected table has a hinge tree with one vertex, and decomposing at bunches of width one with at least two marks on each side, the weights factoring across each such bunch, markers counted as marks, refines it until no node has such a bunch. A node on at least four marks without one is prime (Theorem 23.16); the nodes on three marks, with the bases of $\binom{G}{2}$ or of $\binom{G}{1}$, $|G| = 3$, are series and parallel nodes, and merging adjacent

series nodes, and adjacent parallel nodes, gives the series and parallel nodes of any size; with no two series and no two parallel nodes adjacent this is the canonical hinge tree of Theorem 23.16.

Theorem 23.17 (The period along a hinge tree)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ have real data on its balance wall and its tables a common hinge tree rooted at o, with the node families $F_{\mathcal{H}_v}$ and the cell families F_v of the vertices $v \neq o$ (Definition 23.15), and let $F_{\mathcal{H}_v}[\mathbf{x}]$ be $F_{\mathcal{H}_v}$ twisted at the markers of the edges to the vertices w directly below v by $\mathbf{x}_w \in (0,\infty)^{2N-1}$, with the masses κ_{F_w} there and $\mathbf{m}|_{\mathcal{E}_v}$ on $\mathcal{E}_v$ (Theorem 23.6).*

(i) For $v \neq o$, κ_{F_v} is the hinge wall of $F_{\mathcal{H}_v}[\mathbf{x}]$, and

$$\int H\, d\big((\boldsymbol{\varrho}_{F_v})_*\phi_{F_v}\big) = \int \Big(\int H\, d\big((\boldsymbol{\varrho}_{F_{\mathcal{H}_v}[\mathbf{x}]})_*\phi_{F_{\mathcal{H}_v}[\mathbf{x}]}\big)\Big) \prod_w d\big((\boldsymbol{\varrho}_{F_w})_*\phi_{F_w}\big)(\mathbf{x}_w)$$

for every Borel $H \geq 0$ on $(0,\infty)^{2N-1}$.

(ii) The period is

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int P_{F_{\mathcal{H}_o}[\mathbf{x}]}\big((\mathbf{m}|_{\mathcal{E}_o}, (\kappa_{F_w})_w);\mathbf{s},\mathbf{t}\big) \prod_w d\big((\boldsymbol{\varrho}_{F_w})_*\phi_{F_w}\big)(\mathbf{x}_w),$$

the product over the vertices w directly below the root.

(iii) The law of F_v is determined by the tables $\mathcal{H}_{w,j}$ and the masses on $\mathcal{E}_w$ for the vertices w below v, and by the exponents; nothing above v enters it.

For one table the cell families are the subtree cells $\mathcal{A}_v$, with laws on $(0,\infty)$.

Where the active letters read every subtree cell by one hinge value, as at a full lean of one table, every cell of the tree is pinned (Theorem 22.5), and Chapter 26 writes the period along the tree as a product and carries it to every load.

Proposition 23.18 (What the work depends on)**.** *In Theorem 23.17:*

(i) Two vertices whose subtree cells are carried onto each other, in every table, by one bijection keeping the tables, the weights, the masses and the hinge have equal hinge walls and equal laws. So the laws to be computed are one for each orbit of subtree cells under such bijections.

(ii) Let $s_j, t_j > 0$. At a series node v whose cell families directly below fall into orbits $c_1, \dots, c_p$ with multiplicities $k_1, \dots, k_p$, $f_{F_v} = \prod_l f_{c_l}^{k_l}$ and $\kappa_{F_v} = \sum_l k_l \kappa_{c_l} - \big(\sum_l k_l - 1\big)|\mathbf{s}|$, f_{c_l} and κ_{c_l} the transform and the hinge wall of a member of c_l; at a parallel node $\check{f}_{F_v} = \prod_l \check{f}_{c_l}^{k_l}$ and $\kappa_{F_v} = \sum_l k_l \kappa_{c_l} - \big(\sum_l k_l - 1\big)|\mathbf{t}|$. The multiplicities enter as exponents.

(iii) A change at a vertex v_0 of the weights, of the masses keeping their total, or of the tables $\mathcal{H}_{v_0,j}$ keeping their ranks and their numbers of marks, changes no hinge wall and changes the laws of the cell families F_v only for the vertices v on the path from v_0 to the root. A change of masses at two vertices v_0, v_1 that keeps the total mass changes the hinge wall κ_{F_v} exactly for the vertices v whose subtree contains one of v_0, v_1 and not the other, the edges on the path between v_0 and v_1, by the change of the total mass below $\mathfrak{e}_v$.

A hinge tree is read from the leaves up by a computation whose reusable data are the laws on the edges. Its cost is governed by the prime nodes, where cells interact through the node's own integral, and by the number of distinct subtrees, since equal subtrees have equal laws; series and parallel nodes cost no integration in the matching transform, and repeated subtrees enter as powers, as the hinge tree procedure makes explicit.

Definition 23.19 (The hinge tree procedure)**.** *Given*: a common hinge tree of the tables of a family (Definition 23.15), the masses on the marks of its nodes, exponents $s_j, t_j > 0$, and a root. *Reusable data*: for every orbit of subtree cells (Proposition 23.18(i)) its hinge wall κ and its law, kept in one of three representations: (a) the transform f and the dual transform $\check{f}$ on a grid of $(\mathbf{x}, \mathbf{y})$ with $x_1 = 1$, of y for one table; (b) the Mellin transform $\int \boldsymbol{\varrho}^{-\mathbf{z}} d\phi$ of the law on a vertical subspace of its domain of convergence, for one table $\int \varrho^{-z} d\phi$ on a vertical line of its strip; (c) for one table, the lattice expansion of f at $y \to 0$, the residues of $y^{-z}\Gamma(z)\Gamma(z+s-\kappa)\int \varrho^{-z} d\phi$ to the left of the line (Theorem 20.34(i)). *Operations*, from the leaves up:

(1) a one-mark leaf, the one-mark cell family of ratio vector $\mathbf{x}_0$ (Definition 23.10), has $\kappa = m_e$, the law $\delta_{\mathbf{x}_0}$ and the transforms

$$f(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,K}^{-s_j}\, \mathsf{G}_{|\mathbf{s}|-m_e;(1,-1)}\Big(\sum_j \frac{x_j}{x_{0,j,K}},\ \sum_j y_j x_{0,j,\Psi}\Big),$$
$$\check{f}(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,\Psi}^{-t_j}\, \mathsf{G}_{|\mathbf{t}|-m_e;(1,-1)}\Big(\sum_j \frac{x_j}{x_{0,j,\Psi}},\ \sum_j y_j x_{0,j,K}\Big);$$

for one table the one-mark cell of ratio x_0 has the law $x_0^{(m_e-s)/2}\,\delta_{x_0}$ and

$$f(y) = x_0^{(m_e-s)/2}\mathsf{G}_{s-m_e;(1,-1)}(1, x_0 y), \qquad \check{f}(y) = x_0^{(m_e-s)/2-m_e}\mathsf{G}_{t-m_e;(1,-1)}(1, y/x_0);$$

(2) at a series node multiply the transforms of the vertices directly below, in (a) pointwise, in (b) by Mellin convolution of the Mellin transforms of Proposition 23.8(iii), for one table $\Gamma(z)\Gamma(z+s-\kappa)\int \varrho^{-z} d\phi$, in (c) by the product of lattice expansions (Lemma 20.30); at a parallel node the same with the dual transforms;

(3) along an edge where the kind of node changes, change the transform by Proposition 23.8(v): in (b) a reflection and a ratio of Gammas, in (a) a Mellin transform followed by an inverse Mellin transform, each one integral on $(0,\infty)^{2N-1}$, on the half-line for one table;

(4) at a prime node evaluate the Mellin transform of the law of the twisted node family, or its transform, as a period of the twisted node family, and integrate it against the laws of the vertices directly below (Theorem 23.6(ii));

(5) at the root evaluate Theorem 23.6(i); for a series root, with the cell families F_i directly below and their transforms read at $x_1 = 1$, $\mathbf{x}' = (x_2, \dots, x_N)$, the period is

$$\prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j\ge 2} x_j^{s_j-1} \prod_j y_j^{t_j-1} \prod_i f_{F_i}\, d\mathbf{x}'\, d\mathbf{y},$$
$$\text{for one table} \quad \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} \prod_i f_{\mathcal{A}_i}(y)\, dy,$$

and for a parallel root the same with s_j and t_j swapped and the dual transforms (Theorem 23.9(ii)).

Output: the period, and with it every law on every edge, ready to be reused when the tree is changed (Chapter 24).

Corollary 23.20 (Series-parallel families)**.** *Let the tables of a family with exponents $s_j, t_j > 0$ have a common hinge tree whose nodes are series and parallel nodes and whose leaves are one-mark cell families, and at every edge joining a series node to a parallel node let the law of the subtree below*

have a Mellin transform finite at a real $\mathbf{z}$ *with* $0 < z_{j,K} < s_j$ *and* $0 < z_{j,\Psi} < t_j$ *for every* j *(Proposition 23.8(v)). Then its period is obtained from the transforms of the leaves, Grammas of the marks (Definition 23.19(1)), by pointwise products, by two integral transforms on* $(0,\infty)^{2N-1}$ *for each edge joining a series node to a parallel node, and by one final integral on* $(0,\infty)^{2N-1}$*; the number of integrations depends on the alternations of the tree and not on the number of marks, and subtrees repeated at a node enter as powers. For one table the integral transforms are on the half-line.*

Chapter 24

Repetition and the change of query

A construction that repeats a cell should not repeat the work, and a query that changes one weight or one mass should not restart the computation. A cell family repeated k times at a series node enters its period as the k-th power of its transform, and repetition without bound forces its hinge wall to be $\sum_j s_j$; for one cell the growth in k is then factorial, with the base $A_1/\min(s,t)$, A_1 the period of the cell with its hinge deleted, and a prefactor read from one further constant of the cell and one leading coefficient of the room (Theorems 24.1 and 24.2); a chain of repeated cell families is a power of one positive operator (Proposition 24.4); the weight of a hinge enters through a Mellin convolution whose two expansions are the two closings of the node of Section 20.7 (Theorem 24.5); and a change of data at one node moves only the laws on one path of the tree (Proposition 24.6).

24.1 Repetition at a node

For a cell family F with plain tables $\mathcal{A}_j$ and $k \geq 1$, F_k^{s} and F_k^{p} are the cell families whose table j is the series and the parallel connection of k copies of $\mathcal{A}_j$ (Definition 23.13), with $F_1^{\mathrm{s}} = F_1^{\mathrm{p}} = F$; for one plain cell $\mathcal{A}$ they are written $\mathcal{A}_k^{\mathrm{s}}$ and $\mathcal{A}_k^{\mathrm{p}}$. The cell family is *series-balanced* when $\kappa_F = |\mathbf{s}|$ and *parallel-balanced* when $\kappa_F = |\mathbf{t}|$; by Definition 23.1 the first says that the deletions $\mathcal{A}_j \setminus g$, at the exponents (s_j, t_j), form a family on its balance wall at the masses $\mathbf{m}$, $m_B = \sum_j \big(s_j r_{\mathcal{A}_j} + t_j(|B| - r_{\mathcal{A}_j})\big)$, and the second that the contractions $\mathcal{A}_j/g$ do; for one cell they say $\kappa_{\mathcal{A}} = s$ and $\kappa_{\mathcal{A}} = t$.

Theorem 24.1 (How often a cell family repeats)**.** *Let F be a cell family with plain tables and real exponents.*

(i) $\kappa_{F_k^{\mathrm{s}}} = |\mathbf{s}| + k(\kappa_F - |\mathbf{s}|)$ *and* $\kappa_{F_k^{\mathrm{p}}} = |\mathbf{t}| + k(\kappa_F - |\mathbf{t}|)$.

(ii) If F' is a cell family with $F_k^{\mathrm{s}} \oplus_g F'$ on its balance wall and of convergent period, then $0 < |\mathbf{s}| + k(\kappa_F - |\mathbf{s}|) < 2\sigma$, σ the total strength (Definition 8.1). So series connections of any number of copies of F occur in convergent periods only if F is series-balanced; otherwise $k < |\mathbf{s}|/(|\mathbf{s}| - \kappa_F)$ when $\kappa_F < |\mathbf{s}|$ and $k < |\mathbf{t}|/(\kappa_F - |\mathbf{s}|)$ when $\kappa_F > |\mathbf{s}|$. For parallel connections the same holds with $|\mathbf{t}|$ in place of $|\mathbf{s}|$ and $|\mathbf{s}|$ in place of $|\mathbf{t}|$.

The hinge wall of a series connection drifts by $\kappa_F - |\mathbf{s}|$ for every copy, and the room must absorb the drift; only a series-balanced cell family repeats without limit. It is the cell family whose deletions at the hinge form a balanced family, and for one cell the period of that deletion is the number that governs the repetition.

Theorem 24.2 (Series repetition)**.** *Let F be a series-balanced cell family on $B \sqcup \{g\}$ and $F' = ((\mathcal{A}'_j, \theta'_j); s_j, t_j)_j$ a cell family on $R \sqcup \{g\}$ with $\kappa_{F'} = |\mathbf{t}|$, plain tables, exponents $s_j, t_j > 0$, so that $F \overset{\mathrm{s}}{_k}\!\oplus_g F'$ is on its balance wall for every k; in this statement write $P(k)$ for its period, at the masses of the copies and of F', and read the transforms at $x_1 = 1$, $\mathbf{x}' = (x_2, \dots, x_N)$.*

(i) In $[0, \infty]$,

$$P(k) = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1} \, f_F^k f_{F'} \, d\mathbf{x}' \, d\mathbf{y},$$

the k-th moment of the image under f_F of the measure

$$\prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1} \, f_{F'}(\mathbf{x}, \mathbf{y}) \, d\mathbf{x}' \, d\mathbf{y}.$$

Where finite, $P(k)^2 \leq P(k-1)P(k+1)$, and for $z \geq 0$, with $P(0)$ the period at $\mathbf{m}|_R$ of the family of the contractions $((\mathcal{A}'_j/g, \theta'_j); s_j, t_j)_j$, the hinge of F' contracted,

$$\sum_{k \geq 0} P(k) \frac{z^k}{k!} = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1} \, e^{z f_F} f_{F'} \, d\mathbf{x}' \, d\mathbf{y}.$$

For one cell, $P(k) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} f_{\mathcal{A}}(y)^k f_{\mathcal{A}'}(y) \, dy$ and $P(0) = \int \varrho_{\mathcal{A}'}^{-t} d\phi_{\mathcal{A}'} = P_{\mathcal{A}'/g}(\mathbf{m}|_R; s, t)$.

(ii) For one cell $\mathcal{A}$ and one cell $\mathcal{A}'$, at the exponents (s, t), let $\mathcal{A} \backslash g$ have every proper wall positive at the masses of $\mathcal{A}$ and the exponents (s, t), and put

$$A_1 = \int_{\Delta_B} d\phi_{\mathcal{A}} = P_{\mathcal{A}\backslash g}(\mathbf{m}|_B; s, t), \qquad A_0 = 2A_1(\log\Gamma)'(1) - \int_{\Delta_B} \log \varrho_{\mathcal{A}} \, d\phi_{\mathcal{A}}.$$

Then $\int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}}$ is finite on a neighbourhood of $z = 0$, and for every $\varepsilon \in (0, 1)$ at which $\int \varrho_{\mathcal{A}}^{\varepsilon} d\phi_{\mathcal{A}} < \infty$ there is C with $\big| f_{\mathcal{A}}(y) - A_1(-\log y) - A_0 \big| \leq C y^{\varepsilon}$ for all $y > 0$.

(iii) Let moreover $\int \varrho_{\mathcal{A}'}^{-z} d\phi_{\mathcal{A}'}$ be finite on a neighbourhood of $z = \max(0, t - s)$, and let the polynomial of the room *be*

$$\mathcal{R}(v) = \begin{cases} \Gamma(s-t) \int d\phi_{\mathcal{A}'} & (s > t), \\ \Gamma(t-s) \int \varrho_{\mathcal{A}'}^{s-t} d\phi_{\mathcal{A}'} & (s < t), \\ v \int d\phi_{\mathcal{A}'} + 2(\log\Gamma)'(1) \int d\phi_{\mathcal{A}'} - \int \log \varrho_{\mathcal{A}'} \, d\phi_{\mathcal{A}'} & (s = t), \end{cases}$$

with the coefficients $\mathcal{R}(v - A_0/A_1) = \sum_i \mathcal{R}_i v^i$; let $P(1) < \infty$. Then there is $\varepsilon > 0$ such that for every $\vartheta \in \big(\min(s,t)/(\min(s,t) + \varepsilon), 1\big)$

$$P(k) = \frac{A_1^k \, e^{\min(s,t) A_0/A_1}}{\Gamma(s)\Gamma(t)} \sum_i \mathcal{R}_i \, \frac{\Gamma(k + i + 1)}{\min(s,t)^{k+i+1}} \big(1 + O(\vartheta^k)\big).$$

In particular $P(k) \sim \frac{c \, e^{\min(s,t) A_0/A_1}}{\Gamma(s)\Gamma(t)} \Gamma(k + 1 + j) A_1^k \min(s,t)^{-k-1-j}$, with $j = 1$ and $c = \int d\phi_{\mathcal{A}'}$ when $s = t$ and $j = 0$ and $c = \mathcal{R}$ otherwise: the growth is factorial, with the base $A_1/\min(s,t)$.

(iv) Under the hypotheses of (iii) the series of (i) converges for $0 \leq z < \min(s,t)/A_1$ and diverges at $z = \min(s,t)/A_1$.

So a repeated cell family enters its construction as a power, and the whole sequence $P(k)$ is the sequence of moments of one positive measure: log-convex, with an exponential generating function that is one integral. For one cell the factorial growth is not a property of the room: the transform of a series-balanced cell grows like $A_1 \log(1/y)$ at small scales, its k-th power concentrates where $\log(1/y)$ is of the order $k/\min(s,t)$, and there only two numbers of the cell survive, the period A_1 of the cell with its hinge deleted and the constant A_0, and only the first lattice term of the room, through $\min(s,t)$ and $\mathcal{R}$. The error is exponentially small, not a power of $1/k$: the next terms of both lattice expansions sit at distance ε to the right and contribute in the ratio $\big(\min(s,t)/(\min(s,t)+\varepsilon)\big)^k$. Theorem 23.3 reads the series closure of any cells, for any family, through their transforms; for k equal cells at the centre it is (i) with $k-1$ copies at the series node and one copy as the room.

Corollary 24.3 (Parallel repetition)**.** *Let F be a parallel-balanced cell family on $B \sqcup \{g\}$ and $F' = ((\mathcal{A}'_j, \theta'_j); s_j, t_j)_j$ a cell family on $R \sqcup \{g\}$ with $\kappa_{F'} = |\mathbf{s}|$, plain tables, exponents $s_j, t_j > 0$, and in this statement write $P(k)$ for the period of $F_k^{\mathrm{p}} \oplus_g F'$, the transforms read at $x_1 = 1$, $\mathbf{x}' = (x_2, \dots, x_N)$. Then*

$$P(k) = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \ge 2} x_j^{t_j - 1} \prod_j y_j^{s_j - 1}\, \check{f}_F^{\,k} \check{f}_{F'}\, d\mathbf{x}'\, d\mathbf{y},$$

with $P(0)$ the period at $\mathbf{m}|_R$ of the family of the deletions $((\mathcal{A}'_j \setminus g, \theta'_j); s_j, t_j)_j$; for one cell, $P(k) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{s-1} \check{f}_{\mathcal{A}}(y)^k \check{f}_{\mathcal{A}'}(y)\, dy$. For one cell $\mathcal{A}$ and one cell $\mathcal{A}'$ let $\mathcal{A}/g$ have every proper wall positive at the masses of $\mathcal{A}$ and the exponents (s,t), and put

$$A_1 = \int \varrho_{\mathcal{A}}^{-t} d\phi_{\mathcal{A}} = P_{\mathcal{A}/g}(\mathbf{m}|_B; s, t), \qquad A_0 = 2A_1 (\log\Gamma)'(1) + \int \varrho_{\mathcal{A}}^{-t} \log \varrho_{\mathcal{A}}\, d\phi_{\mathcal{A}}.$$

Then Theorem 24.2(ii) to (iv) hold with s and t swapped and with these A_1 and A_0, the dual transform $\check{f}_{\mathcal{A}}$ in place of the transform $f_{\mathcal{A}}$, the finiteness of $\int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}}$ near $z = t$ in place of near $z = 0$, the condition $\int \varrho_{\mathcal{A}}^{-t-\varepsilon} d\phi_{\mathcal{A}} < \infty$ in place of $\int \varrho_{\mathcal{A}}^{\varepsilon} d\phi_{\mathcal{A}} < \infty$, the finiteness of $\int \varrho_{\mathcal{A}'}^{-z} d\phi_{\mathcal{A}'}$ near $z = \min(s,t)$ as the hypothesis on $\mathcal{A}'$, and the polynomial of the room

$$\mathcal{R}(v) = \begin{cases} \Gamma(t-s) \int \varrho_{\mathcal{A}'}^{-s} d\phi_{\mathcal{A}'} & (s < t), \\ \Gamma(s-t) \int \varrho_{\mathcal{A}'}^{-t} d\phi_{\mathcal{A}'} & (s > t), \\ v \int \varrho_{\mathcal{A}'}^{-s} d\phi_{\mathcal{A}'} + 2(\log\Gamma)'(1) \int \varrho_{\mathcal{A}'}^{-s} d\phi_{\mathcal{A}'} + \int \varrho_{\mathcal{A}'}^{-s} \log \varrho_{\mathcal{A}'}\, d\phi_{\mathcal{A}'} & (s = t), \end{cases}$$

whose leading coefficient is the constant c of the equivalent in (iii).

24.2 Chains

Proposition 24.4 (A chain is a power of one operator)**.** *Let a* chain cell family *U have in every table a hinged core $\mathcal{U}_j$ (Definition 23.4) with one further marker g_{in}, its hinge g_{out}, own marks $\mathcal{E}_{\mathcal{U}}$ with masses, and real exponents (s_j, t_j); further fixed cell families may be substituted into U beforehand. For a cell family F_0 let $F_{l+1} = U\langle F_l \rangle$, table by table, the cell at g_{in}.*

(i) $\kappa_{F_{l+1}} = \kappa_{F_l} + \kappa_U$, with $\kappa_U = m_{\mathcal{E}_{\mathcal{U}}} - \sum_j \big(s_j(r_{\mathcal{U}_j} - 1) + t_j(|\mathcal{E}_{\mathcal{U}}| + 1 - r_{\mathcal{U}_j})\big)$ the hinge wall of U with the mass 0 on g_{in}. For a fixed cell family F', the families $F_l \oplus_g F'$ are all on their balance walls exactly when $F_0 \oplus_g F'$ is and $\kappa_U = 0$; then $\kappa_{F_l} = \kappa_{F_0}$.

(ii) Let $\kappa_U = 0$, and in this statement write λ_l for the law of F_l and $\lambda^U_{\mathbf{x}}$ for the law of U twisted at g_{in} by $\mathbf{x} \in (0,\infty)^{2N-1}$ (Definition 22.1) with the mass κ_{F_0} there. On positive measures μ and nonnegative functions H on $(0,\infty)^{2N-1}$ put

$$\mathcal{Q}\mu = \int \lambda^U_{\mathbf{x}}\, d\mu(\mathbf{x}), \qquad (\mathcal{Q}^*H)(\mathbf{x}) = \int H\, d\lambda^U_{\mathbf{x}},$$

$$h_{F'}(\mathbf{x}) = \int \Big(\int_0^\infty v^{\kappa_{F_0}-1} \prod_j (x_{j,K}v + x'_{j,K})^{-s_j} (x_{j,\Psi}v + x'_{j,\Psi})^{-t_j}\, dv \Big)\, d\big((\varrho_{F'})_*\phi_{F'}\big)(\mathbf{x}'),$$

the inner integral the period of the twisted core family of Theorem 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$ in every table. Then $\lambda_l = \mathcal{Q}^l\lambda_0$ and $P_{F_l \oplus_g F'} = \int (\mathcal{Q}^{*l} h_{F'})\, d\lambda_0$; for $z \ge 0$, $\sum_l z^l P_{F_l \oplus_g F'} = \int \sum_l z^l (\mathcal{Q}^{*l} h_{F'})\, d\lambda_0$, in $[0,\infty]$.

(iii) If $H > 0$ and $M > 0$ have $\mathcal{Q}^*H \le MH$ and $h_{F'} \le CH$, then $P_{F_l \oplus_g F'} \le CM^l \int H\, d\lambda_0$; if $H' \ge 0$ and $M' > 0$ have $\mathcal{Q}^*H' \ge M'H'$ and $h_{F'} \ge c\,H'$, then $P_{F_l \oplus_g F'} \ge c\,M'^l \int H'\, d\lambda_0$.

(iv) Let the tables be plain and $s_j, t_j > 0$. If U is the series core on $\{g_{\text{out}}, g_{\text{in}}, g'\}$ with a fixed cell family F'' at g', then $f_{F_{l+1}} = f_{F''} f_{F_l}$ and $\kappa_U = \kappa_{F''} - |\mathbf{s}|$; if it is the parallel core, then $\check{f}_{F_{l+1}} = \check{f}_{F''} \check{f}_{F_l}$ and $\kappa_U = \kappa_{F''} - |\mathbf{t}|$.

For one table the laws are measures on $(0,\infty)$, the twist is one number, and the inner integral of $h_{F'}$ is the period of Lemma 23.7.

A chain is one kernel applied l times: the law of the chain cell family twisted at g_{in}, computed once, carries the law of the chain from one length to the next, and the period at length l pairs the l-th iterate with the room. The work for all lengths up to l is l applications of one operator, and for one length $\lceil \log_2 l \rceil$ compositions of kernels. A series chain cell family is diagonal in the transform and a parallel one in the dual transform, so for them the operator is a multiplication and the length is an exponent; a chain cell family that alternates the two kinds mixes the two hinge values, and its operator is not diagonal in either. Theorem 22.8 is the same chain read on functions of the twists, a family glued to copies of itself at two marks, with the pinned case $P_F(\mathbf{m};\mathbf{s},\mathbf{t})^k / \Gamma_2 \binom{m_{g_{\text{in}}},\ m_{g_{\text{out}}}}{m_{g_{\text{in}}} + m_{g_{\text{out}}}}^{k-1}$; Proposition 24.4 reads it on laws, and positive test functions bound it from both sides. Every polynomial Gamma moment of a chain is the power of one finite matrix (Theorem 27.5).

24.3 The weight of a hinge

Theorem 24.5 (The weight of a hinge)**.** *For plain cells $\mathcal{A}$ on $B \sqcup \{g\}$ and $\mathcal{A}'$ on $R \sqcup \{g\}$ with exponents $s, t > 0$, and $w > 0$, let $w\mathcal{A}$ be $\mathcal{A}$ with the weights of its bases through g multiplied by w, so that in $(w\mathcal{A}) \oplus_g \mathcal{A}'$ the bases made of a basis of $\mathcal{A}/g$ and a basis of $\mathcal{A}' \setminus g$, the second kind of Lemma 21.2, carry the factor w; let it be on its balance wall, put $q = w^2$, and in this statement write $\Pi(q) = q^{(s-\kappa_{\mathcal{A}})/2} P_{(w\mathcal{A})\oplus_g \mathcal{A}'}(\mathbf{m}; s, t)$.*

(i) For a cell family F with plain tables $\mathcal{A}_j$ and $\mathbf{w} \in (0,\infty)^N$, let $\mathbf{w}F$ be F with the weights of the bases through g multiplied by w_j in table j. Its hinge wall is κ_F, its ratio vector has the coordinates $(w_1/w_j)\,\gamma^K_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}$ and $w_1 w_j\, \gamma^\Psi_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}$, the hinge values of F being read, $\phi_{\mathbf{w}F} = w_1^{\kappa_F} \prod_j w_j^{-s_j}\, \phi_F$, and, for $s_j, t_j > 0$, $f_{\mathbf{w}F}(\mathbf{x},\mathbf{y}) = f_F(\mathbf{w}\circ\mathbf{x}, \mathbf{w}\circ\mathbf{y})$ and $\check{f}_{\mathbf{w}F}(\mathbf{x},\mathbf{y}) = \prod_j w_j^{-s_j-t_j}\, \check{f}_F(\mathbf{x}/\mathbf{w}, \mathbf{y}/\mathbf{w})$, the products and quotients taken coordinatewise. For one cell, $\varrho_{w\mathcal{A}} = w^2 \varrho_{\mathcal{A}}$, $\phi_{w\mathcal{A}} = w^{\kappa_{\mathcal{A}}-s}\phi_{\mathcal{A}}$, $f_{w\mathcal{A}}(y) = w^{\kappa_{\mathcal{A}}-s} f_{\mathcal{A}}(w^2 y)$ and $\check{f}_{w\mathcal{A}}(y) = w^{-\kappa_{\mathcal{A}}-s}\check{f}_{\mathcal{A}}(y/w^2)$.

(ii) $\Pi(q) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} f_{\mathcal{A}}(qy) f_{\mathcal{A}'}(y)\, dy$, *a decreasing function of* q, *and for real* z *with* $\max(0, \kappa_{\mathcal{A}} - s) < z < \min(t, \kappa_{\mathcal{A}})$, *in* $(0, \infty]$,

$$\int_0^\infty q^{z-1} \Pi(q)\, dq = \frac{\Gamma(z)\Gamma(z+s-\kappa_{\mathcal{A}}) \int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}} \cdot \Gamma(t-z)\Gamma(\kappa_{\mathcal{A}} - z) \int \varrho_{\mathcal{A}'}^{z-t} d\phi_{\mathcal{A}'}}{\Gamma(s)\Gamma(t)}.$$

(iii) If this is finite at a real c *in that interval, then* Π *is finite and continuous on* $(0, \infty)$ *and is the value there of the function* $\frac{1}{2\pi i} \int_{(c)} q^{-z} (\cdots)\, dz$, *with* $(\cdots)$ *the right side of the display in (ii), holomorphic on the sector* $|\arg q| < 2\pi$ *of the logarithmic cover of* $\mathbb{C} \setminus \{0\}$.

(iv) In the setting of Section 20.7, with $\mathcal{A}$ *the cell of the bunch and* $\mathcal{A}'$ *that of the room, let* $f_{\mathcal{A}}$ *have a lattice expansion* $\sum_{\eta,j} \beta_{\eta,j} y^\eta (-\log y)^j$ *of every reach above* $\operatorname{reach}(B)$ *(Definition 20.28), and let* $\inf \varrho_{\mathcal{A}'} > 0$. *Then for* $0 < q < \inf \varrho_{\mathcal{A}'} / \operatorname{reach}(B)$, *absolutely,*

$$\Pi(q) = \frac{1}{\Gamma(s)\Gamma(t)} \sum_{\eta,j} \beta_{\eta,j}\, q^\eta \sum_{i=0}^{j} \binom{j}{i} (-\log q)^{j-i} \times (-\partial_z)^i \Big[\Gamma(z)\Gamma(z+s-\kappa_{\mathcal{A}'}) \int \varrho_{\mathcal{A}'}^{-z} d\phi_{\mathcal{A}'} \Big]_{z=t+\eta};$$

at $q = 1$ *this is the closing of Theorem 20.34(ii) toward the level* d. *Swapping* $\mathcal{A}$ *and* $\mathcal{A}'$, q^{-t} *times the same expansion in* $1/q$ *holds for* $q > \operatorname{reach}(R) / \inf \varrho_{\mathcal{A}}$, *and is the closing toward the level* r. *The two ranges never overlap.*

The closings of a node of Section 20.7 are therefore not two separate devices but the two expansions, at $q \to 0$ and at $q \to \infty$, of one function of the weight of its hinge, holomorphic on two sheets around the origin and given everywhere by the contour integral of (iii), whose integrand is the product of the Mellin transforms of the two sides over $\Gamma(s)\Gamma(t)$ (Proposition 23.8(iii)). The unweighted period is the value at $q = 1$; a closing reaches it when 1 lies inside its range, its terms do not tend to zero when 1 lies beyond it, and at the common end of the two ranges the root rate is one and domination decides nothing (Theorem 20.39(ii) to (iv), the ranges being read at the reaches of the two sides); and the coefficients of each expansion are the reusable lattice data of one side paired with the Mellin data of the other, the same for every weight.

24.4 Changing masses and weights

Proposition 24.6 (Carrying a calculation through a change)**.** *Let the tables of a family have a common hinge tree rooted at* o *(Definition 23.15), with every law computed by Definition 23.19.*

(i) A change at a vertex v' *of the weights, of the masses keeping their total, or of the tables* $\mathcal{H}_{v',j}$ *keeping their ranks and their numbers of marks, is carried through by recomputing the laws at the vertices on the path from* v' *to* o, *each from the new law of the vertex directly below it on the path and the old laws of the other vertices directly below it, and then the root.*

(ii) A change of the weight of the hinge of the edge $\mathfrak{e}_v$ *by the factor* w_j *in table* j *replaces the law of* F_v *by* $w_1^{\kappa_{F_v}} \prod_j w_j^{-s_j}$ *times its image under* $(x_{j,K}, x_{j,\Psi})_j \mapsto \big((w_1/w_j)\, x_{j,K},\, w_1 w_j\, x_{j,\Psi}\big)_j$ *(Theorem 24.5(i)), for one table by* $w^{\kappa_{\mathcal{A}_v} - s}$ *times its image under* $x \mapsto w^2 x$; *nothing below* v *changes, and above* v *(i) applies along the path from* v *to* o.

(iii) A change of masses at two vertices v', v'' *keeping the total mass changes the hinge walls named in Proposition 23.18(iii), those on the edges of the path between* v' *and* v'', *and is carried through by recomputing the laws on the paths from* v' *and from* v'' *to* o.

A query about one part of a construction is answered by the part of the computation that sees it. The laws on the edges off the path are kept, and with them the work that made them; a construction with many equal subtrees, or one asked many queries about one of its nodes, pays for each subtree once and for each query along one path.

Chapter 25

Holds and the lean convolution

A table built along a hinge tree is read from the laws of its subtrees, and at a prime node the twisted core is integrated over its own simplex once for every value of the twists (Definition 23.19(4)). Holding the marks removes that integral, and with it the dependence of the reusable data on the query. Every mark of every letter of a family can be twisted, and a twist can be *held* at a load, by a Mellin transform. Twists separate any word in positive letters into its factors, its period becoming one fibre integral of the periods of the factors (Theorem 25.4), with the convergence read off the Newton sums (Theorem 25.2); the chapter opens with these words. Holding every mark separates the letters of the family completely: the hold is the product of the periods of its letters alone, the periods of its tables at their full leans, each read at its own load (Lemma 25.11). Undoing the hold writes the period at all exponents, every mass and every twist as one fibre integral of dimension $(2N-1)(n-1)$ over the splits of the masses among the letters (Theorem 25.12). The periods at the full leans of a table built along a hinge tree are the products of those of its nodes over Beta corners at its edges, with no integral at all (Theorem 26.5), so the period of a family of such tables is one fibre integral of node data (Theorem 26.13).

25.1 Newton words

Twists separate any product of powers of positive letters into its factors, and nothing in this uses the exchange property. The convergence criterion of Theorem 8.5, whose sectors rest on the greedy bases that need the exchange property (Lemma 8.3), extends to such products through the Gramma and its domain (Theorem 25.2), and on its domain the separation holds (Theorem 25.4). The holds of the next section are the case of the letters of a family (Lemma 25.11).

Definition 25.1 (Positive letters; Newton words and sums)**.** A *positive letter* on $\mathcal{E}$ is a nonzero homogeneous polynomial $L = \sum_k c_k x^{t_k}$ of degree $\deg L \geq 1$ with positive coefficients c_k on distinct exponents $t_k \in \mathbb{N}^{\mathcal{E}}$, its *support*; its Newton polytope is $\mathcal{N}_L = \operatorname{conv}\{t_k\}$, in the hyperplane $|x| = \deg L$. A *product of powers* of positive letters $W = \prod_i L_i^{-w_i}$, with exponents $w_i \in \mathbb{C}$, has the degree $D = \sum_i w_i \deg L_i$, and for masses $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ with $m_{\mathcal{E}} = D$ its period $P_W(\mathbf{m})$ is the integral of the form $\alpha^{\mathbf{m}-\mathbf{1}} W \, d\alpha$ over a section (Lemma 16.2), wherever it converges absolutely; it does not depend on the section. It is a *Newton word* when every $w_i > 0$, with the *Newton sum* $\mathcal{N}(W) = \sum_i w_i \mathcal{N}_{L_i}$, in the hyperplane $|x| = D$. A Newton word is *split into factors* when it is written $W = W_1 \cdots W_p$, each factor a product of some of its letters with their exponents. The family word of a family is a product of powers of its letters, with $P_W(\mathbf{m}) = P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$, and a Newton word in its active letters when its exponents are nonnegative; for a table $\mathcal{N}_K = P(\mathcal{H})$ and $\mathcal{N}_\Psi = P(\mathcal{H}^*)$, and for a family $\mathcal{N}(W)$ is the Newton sum $\mathcal{N}_F$ of Theorem 8.19.

Theorem 25.2 (Convergence of a Newton word)**.** *$P_W(\mathbf{m})$ converges exactly when $\mathcal{N}(W)$ has dimension $n-1$ and $\operatorname{Re}\mathbf{m} \in \operatorname{ri}\mathcal{N}(W)$; it is then holomorphic in $\mathbf{m}$ on the balance hyperplane. For a family of tables this is the condition that every proper wall be positive (Theorems 8.7 and 8.19(iii)).*

Proposition 25.3 (One positive letter)**.** *Let $L = \sum_k c_k x^{t_k}$ be a positive letter with $\dim\mathcal{N}_L = n-1$, $\mathbf{a} \in \mathbb{C}^{\mathcal{E}}$ with $\operatorname{Re}|\mathbf{a}| > 0$, and $w = |\mathbf{a}|/\deg L$.*

(i) $P_{L^{-w}}(\mathbf{a}) = \deg L\, \mathsf{G}_{\mathbf{a};(t_k)_k}\big((c_k)_k\big)/\Gamma(w)$*, both sides converging together.*

(ii) $P_{L^{-w}}(\mathbf{a})$ *converges exactly when* $\operatorname{Re}\mathbf{a} \in (\operatorname{Re} w)\operatorname{ri}\mathcal{N}_L$*.*

(iii) For real $\mathbf{a}$ at which it converges, every $\boldsymbol{\eta} \in \mathbb{R}^{\mathcal{E}}$ with $|\boldsymbol{\eta}| = 0$ and every $\varepsilon \in (0,1)$,

$$\big|P_{L^{-w}}(\mathbf{a}+i\boldsymbol{\eta})\big| \le \min\Big(1,\ \big(\sin\tfrac{\pi\varepsilon}{2}\big)^{-w} e^{-(1-\varepsilon)h_L(\boldsymbol{\eta})}\Big)\, P_{L^{-w}}(\mathbf{a}),$$

with h_L the support function of $\{y :\ |\langle t_k, y\rangle| < \frac{\pi}{2}$ for every $k\}$, a norm on $\mathbb{R}^{\mathcal{E}}$.

Theorem 25.4 (Separation into factors)**.** *Let $W = W_1 \cdots W_p$ be split into factors with $\dim\mathcal{N}(W_i) = n-1$ for every i, of degrees D_i, and let $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ have $m_{\mathcal{E}} = D$. Points $(\mathbf{c}_i)_i$ with $\mathbf{c}_i \in \operatorname{ri}\mathcal{N}(W_i)$ and $\sum_i \mathbf{c}_i = \operatorname{Re}\mathbf{m}$ exist exactly when $\operatorname{Re}\mathbf{m} \in \operatorname{ri}\mathcal{N}(W)$, and on each of them, for a fixed mark e_0 and all twists $\mathbf{x}_i \in (0,\infty)^{\mathcal{E}}$ of the factors,*

$$\int_{\alpha_{e_0}=1} \alpha^{\mathbf{m}-\mathbf{1}} \prod_i W_i(\mathbf{x}_i \circ \alpha)\, d\alpha' = \int_{\mathbf{a}_1+\cdots+\mathbf{a}_p=\mathbf{m},\ |\mathbf{a}_i|=D_i\ (i<p)} \ \prod_i P_{W_i}(\mathbf{a}_i)\, \mathbf{x}_i^{-\mathbf{a}_i},$$

a fibre integral of dimension $(p-1)(n-1)$, absolutely convergent, P_{W_i} the period of the factor W_i at its own masses.

Corollary 25.5 (Exponents add on one letter)**.** *For a positive letter L with $\dim\mathcal{N}_L = n-1$, exponents $w, w' > 0$ and $\operatorname{Re}\mathbf{m} \in (w+w')\operatorname{ri}\mathcal{N}_L$,*

$$P_{L^{-w-w'}}(\mathbf{m}) = \int_{\mathbf{a}+\mathbf{a}'=\mathbf{m},\ |\mathbf{a}|=w\deg L} P_{L^{-w}}(\mathbf{a})\, P_{L^{-w'}}(\mathbf{a}').$$

At $L = e_1(x) = \sum_{e\in\mathcal{E}} x_e$, $P_{L^{-w}}(\mathbf{a}) = \Gamma_1^n \begin{bmatrix}\mathbf{a}\\ w\end{bmatrix}$ and this is Theorem 4.6; for an identically self-dual table $\Psi = K$, and the lean convolution is this identity at $L = K$, $w = s$, $w' = t$, $P_{\mathcal{H}}(\mathbf{m}; s, t) = P_{\mathcal{H}}(\mathbf{m}; s+t, 0)$.

Proposition 25.6 (Simplicial letters)**.** *If the support of a positive letter L consists of n linearly independent exponents $t_1, \dots, t_n$, so that L is* simplicial*, with the configuration $t = (t_1, \dots, t_n)$, then for $\mathbf{a}$ with $|\mathbf{a}| = w\deg L$ and the shares $\mathbf{b} = t^{-1}\mathbf{a}$,*

$$P_{L^{-w}}(\mathbf{a}) = \frac{\deg L}{|\det t|}\, \Gamma_1^n \begin{bmatrix}\mathbf{b}\\ w\end{bmatrix} \prod_k c_k^{-b_k}, \qquad |\mathbf{b}| = w,$$

converging exactly when $\operatorname{Re}\mathbf{b} > 0$: the Gramma of Proposition 25.3(i) at a square configuration (Corollary 2.6). A table that is one molecule has a simplicial basis letter exactly when it has n bases.

Lemma 25.7 (Concatenation of Grammas)**.** *Let $t^{(1)}, \dots, t^{(p)}$, $p \ge 2$, be configurations on $\mathcal{E}$ with $t^{(2)}, \dots, t^{(p)}$ spanning $\mathbb{R}^{\mathcal{E}}$, $\boldsymbol{\xi}^{(1)}, \dots, \boldsymbol{\xi}^{(p)}$ coefficients, and $\mathbf{c}_i \in \operatorname{int}\operatorname{cone}(t^{(i)})$ for $i \ge 2$ with $\operatorname{Re}\nu - \sum_{i\ge2}\mathbf{c}_i \in \operatorname{int}\operatorname{cone}(t^{(1)})$. Then*

$$\mathsf{G}_{\nu;t^{(1)}\oplus\cdots\oplus t^{(p)}}\big(\boldsymbol{\xi}^{(1)}\oplus\cdots\oplus\boldsymbol{\xi}^{(p)}\big)$$
$$=\frac{1}{(2\pi i)^{(p-1)n}}\int_{\mathbf{c}_2+i\mathbb{R}^{\mathcal{E}}}\cdots\int_{\mathbf{c}_p+i\mathbb{R}^{\mathcal{E}}}\mathsf{G}_{\nu-\sum_{i\geq 2}\mathbf{w}_i;t^{(1)}}(\boldsymbol{\xi}^{(1)})\prod_{i\geq 2}\mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})\,d\mathbf{w}_2\cdots d\mathbf{w}_p,$$

absolutely convergent: a Gramma of a concatenated configuration is the Mellin convolution of the Grammas of its parts.

For one table the concatenation $(\mathbf{1}_T)_T\oplus(\mathbf{1}_{\mathcal{E}\setminus T})_T$ with the coefficients $\boldsymbol{\theta}\oplus\lambda\boldsymbol{\theta}$ is the Gramma $\mathsf{G}_{\mathbf{m};(\mathbf{1}_T)_T\oplus(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta}\oplus\lambda\boldsymbol{\theta})=\int x^{\mathbf{m}-\mathbf{1}}e^{-K(x)-\lambda\Psi(x)}dx$, and Lemma 25.7 writes it as $(2\pi i)^{-n}\int\mathsf{G}_{\mathbf{m}-\mathbf{b};(\mathbf{1}_T)_T}(\boldsymbol{\theta})\,\mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta})\,\lambda^{-|\mathbf{b}|/r^*}d\mathbf{b}$, by the torus homogeneity of $\mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_T}$. The Mellin transform in λ at t pins $|\mathbf{b}|=r^*t$, the pin contributing the factor r^*, and through Proposition 25.10(i) returns $\Gamma(s)\Gamma(t)/r$ times the lean convolution of Theorem 25.12 at $\mathbf{x}=\mathbf{1}$; Proposition 26.22 states the resulting identity $\int_0^\infty\lambda^{t-1}\mathsf{G}_{\mathbf{m};(\mathbf{1}_T)_T\oplus(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta}\oplus\lambda\boldsymbol{\theta})\,d\lambda=\Gamma(s)\Gamma(t)P_{\mathcal{H}}(\mathbf{m};s,t)/r$. So at fixed masses the lean is the Mellin variable of the relative scale λ of the two letters, and the two-step integration separates what the masses fix, the Gramma of the concatenated configuration, from what the exponents choose, one Mellin transform in one variable.

25.2 Holds and the periods at the full leans

Definition 25.8 (Holding)**.** Let $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ be a family on $\mathcal{E}$, $n=|\mathcal{E}|\geq 2$, with exponents $s_j,t_j>0$, whose tables are each one molecule, table j of rank r_j and corank $r_j^*=n-r_j$, so that $0<r_j<n$ and the bases of every table span $\mathbb{R}^{\mathcal{E}}$: a vector orthogonal to every $\mathbf{1}_T$, $T\in\mathcal{B}_j$, is constant by Lemma 7.8, and then zero since $r_j>0$. $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the period (Theorem 8.7) on the balance wall $m_{\mathcal{E}}=\sum_j(r_js_j+r_j^*t_j)$, and $F[\mathbf{x}]$ is F twisted by $\mathbf{x}=(\mathbf{x}_j^K,\mathbf{x}_j^\Psi)_j$ (Definition 22.1). For $G\subseteq\mathcal{E}$, masses $\mathbf{m}$ and real loads $\boldsymbol{\beta}=(\boldsymbol{\beta}_j^K,\boldsymbol{\beta}_j^\Psi)_j$, $\boldsymbol{\beta}_j^K,\boldsymbol{\beta}_j^\Psi\in\mathbb{R}^G$, with $\boldsymbol{\beta}_1^K$ omitted, *holding the marks of G at $\boldsymbol{\beta}$* is the Mellin transform in the twists of every letter but K_1 at the marks of G, the twist of K_1 and the twists at the other marks kept at $\mathbf{1}$,

$$\mathrm{Hold}_G(\boldsymbol{\beta})=\int_{((0,\infty)^G)^{2N-1}}\prod_{j\geq 2}(\mathbf{x}_j^K)^{\boldsymbol{\beta}_j^K-\mathbf{1}}\prod_{j=1}^N(\mathbf{x}_j^\Psi)^{\boldsymbol{\beta}_j^\Psi-\mathbf{1}}\,P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})\,d\mathbf{x},$$

in $[0,\infty]$ for real masses and loads; for complex masses or loads it is taken where it converges absolutely, that is where it is finite at the real parts.

The twists of every letter but K_1 are held. By the torus relation of Definition 22.1, a twist of every mark of one letter by one factor multiplies the period by that factor to the power $-r_js_j$, or $-r_j^*t_j$ for a cobasis letter, so that a hold of every mark is taken with one twist of each held letter fixed; and a twist of every letter by one vector $\boldsymbol{\lambda}$ multiplies the period by $\boldsymbol{\lambda}^{-\mathbf{m}}$, so that the twist of K_1 is kept at $\mathbf{1}$. A twist is a change of weights: Proposition 26.20 carries product weights on the bases into twists.

Definition 25.9 (Loads; admissible points)**.** For a split of the masses among the letters of F, $\mathbf{m}=\sum_j(\mathbf{a}_j+\mathbf{b}_j)$, $\mathbf{a}_j$ is the *K_j-load* and $\mathbf{b}_j$ the *Ψ_j-load*, as a K-share of a pattern loads its marks and a Ψ-share the marks it leaves out (Definition 12.24); for one table, $\mathbf{m}=\mathbf{a}+\mathbf{b}$ with the K-load $\mathbf{a}$ and the Ψ-load $\mathbf{b}$. The periods of a table $\mathcal{H}$ of F, with the weights $\boldsymbol{\theta}$, of rank r and corank r^*, at its two full leans are read as functions of their loads, $P_{\mathcal{H}}(\mathbf{a};|\mathbf{a}|/r,0)$ and $P_{\mathcal{H}}(\mathbf{b};0,|\mathbf{b}|/r^*)$, the

exponent of the one active letter being the total load divided by its degree. Its letters give the Grammas (Definition 2.1)

$$\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_{T\in\mathcal{H}}}(\boldsymbol{\theta})=\int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{a}-\mathbf{1}}e^{-K_{\mathcal{H}}(x)}dx,\qquad \mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_{T\in\mathcal{H}}}(\boldsymbol{\theta})=\int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{b}-\mathbf{1}}e^{-\Psi_{\mathcal{H}}(x)}dx.$$

A real load $\mathbf{a}$ with $|\mathbf{a}|=rs$, $s>0$, is *admissible for the full basis lean* when $\mathbf{a}\in s\cdot\operatorname{ri}P(\mathcal{H})$, $P(\mathcal{H})$ the base polytope of $\mathcal{H}$ (Theorem 7.18) and ri the relative interior, and a real $\mathbf{b}$ with $|\mathbf{b}|=r^*t$, $t>0$, is *admissible for the full cobasis lean* when $\mathbf{b}\in t\cdot\operatorname{ri}P(\mathcal{H}^*)$, $P(\mathcal{H}^*)$ the base polytope of the dual table. A real point $\mathbf{c}=(\mathbf{c}_j^K,\mathbf{c}_j^\Psi)_j$ is *admissible* for $(\mathbf{m};\mathbf{s},\mathbf{t})$ when $\sum_j(\mathbf{c}_j^K+\mathbf{c}_j^\Psi)=\operatorname{Re}\mathbf{m}$, every $\mathbf{c}_j^K$ is admissible for the full basis lean of table j at s_j and every $\mathbf{c}_j^\Psi$ for its full cobasis lean at t_j; for one table it is named by its Ψ-load $\mathbf{c}=\mathbf{c}_1^\Psi$, with $\operatorname{Re}\mathbf{m}-\mathbf{c}$ admissible for the full basis lean and $\mathbf{c}$ for the full cobasis lean.

Proposition 25.10 (The periods at the two full leans)**.** *In (i), (ii) and (iv) $\mathcal{H}$ is a table of F, with the weights $\boldsymbol{\theta}$, of rank r and corank r^*.*

(i) For $\mathbf{a}\in\mathbb{C}^{\mathcal{E}}$ with $\operatorname{Re}|\mathbf{a}|>0$ and $s=|\mathbf{a}|/r$, $P_{\mathcal{H}}(\mathbf{a};s,0)=r\,\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta})/\Gamma(s)$, both sides converging together; likewise $P_{\mathcal{H}}(\mathbf{b};0,t)=r^*\mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta})/\Gamma(t)$ with $t=|\mathbf{b}|/r^*$. By Cremona, $P_{\mathcal{H}}(\mathbf{b};0,t)=P_{\mathcal{H}}(t\mathbf{1}-\mathbf{b};t,0)$.

(ii) $P_{\mathcal{H}}(\mathbf{a};s,0)$ converges exactly when $\operatorname{Re}a_S>(\operatorname{Re}s)\,d(S)$ for every nonempty proper S, which for real $s>0$ is $\operatorname{Re}\mathbf{a}\in s\cdot\operatorname{ri}P(\mathcal{H})$; $P_{\mathcal{H}}(\mathbf{b};0,t)$ converges exactly when $\operatorname{Re}b_S>(\operatorname{Re}t)\,d^*(S)$ for every nonempty proper S, which for real $t>0$ is $\operatorname{Re}\mathbf{b}\in t\cdot\operatorname{ri}P(\mathcal{H}^*)$.

(iii) At every gauge α the occupation $R_j(\alpha)$ of table j lies in $\operatorname{ri}P(\mathcal{B}_j)$ and its dual occupation $R_j^*(\alpha)$ in $\operatorname{ri}P(\mathcal{B}_j^*)$ (Theorem 7.16). At a native gauge $\alpha^\star$ of real masses $\mathbf{m}$ on the balance wall (Definition 8.17), which exists when every proper wall is positive, the native loads *$\mathbf{a}_j^\star=s_jR_j(\alpha^\star)$ and $\mathbf{b}_j^\star=t_jR_j^*(\alpha^\star)$ are a split of $\mathbf{m}$ among the letters of F, and $\mathbf{c}=(\mathbf{a}_j^\star,\mathbf{b}_j^\star)_j$ is admissible.*

(iv) On a vertical line the periods at the full leans decay exponentially: for admissible real $\mathbf{a}$, every $\mathbf{y}\in\mathbb{R}^{\mathcal{E}}$ with $|\mathbf{y}|=0$ and every $\varepsilon\in(0,1)$,

$$\big|P_{\mathcal{H}}(\mathbf{a}+i\mathbf{y};s,0)\big|\le\min\Big(1,\ \big(\sin\tfrac{\pi\varepsilon}{2}\big)^{-s}e^{-(1-\varepsilon)h(\mathbf{y})}\Big)\,P_{\mathcal{H}}(\mathbf{a};s,0),$$

with h the support function of the set $\{y:\ |\langle\mathbf{1}_T,y\rangle|<\frac{\pi}{2}$ for every basis $T\}$, a norm on $\mathbb{R}^{\mathcal{E}}$; likewise at the full cobasis lean with the vectors $\mathbf{1}_{\mathcal{E}\setminus T}$.

Items (i), (ii) and (iv) hold for every positive letter with a Newton polytope of dimension $n-1$, in the form of Proposition 25.3.

Lemma 25.11 (Holding every mark separates the letters)**.** *Fix $e_0\in\mathcal{E}$ and let $\mathbf{m}\in\mathbb{C}^{\mathcal{E}}$ have $m_{\mathcal{E}}=\sum_j(r_js_j+r_j^*t_j)$. For real loads $\boldsymbol{\beta}=(\boldsymbol{\beta}_j^K,\boldsymbol{\beta}_j^\Psi)_j$ on $\mathcal{E}\setminus e_0$, $\boldsymbol{\beta}_1^K$ omitted, let $\mathbf{a}_j$ ($j\ge2$) extend $\boldsymbol{\beta}_j^K$ by $a_{j,e_0}=r_js_j-|\boldsymbol{\beta}_j^K|$, let $\mathbf{b}_j$ extend $\boldsymbol{\beta}_j^\Psi$ by $b_{j,e_0}=r_j^*t_j-|\boldsymbol{\beta}_j^\Psi|$, and put $\mathbf{a}_1=\mathbf{m}-\sum_{j\ge2}\mathbf{a}_j-\sum_j\mathbf{b}_j$. Then, in $[0,\infty]$ for real $\mathbf{m}$,*

$$\operatorname{Hold}_{\mathcal{E}\setminus e_0}(\boldsymbol{\beta})=\prod_{j=1}^{N}P_{\mathcal{B}_j}(\mathbf{a}_j;s_j,0)\,P_{\mathcal{B}_j}(\mathbf{b}_j;0,t_j),$$

the twists at e_0 being 1; for complex $\mathbf{m}$ and complex $\boldsymbol{\beta}$ the same holds whenever the integral converges at the real parts.

The lemma is the whole mechanism. Every letter but K_1 sees its own twists and K_1 sees none; a Mellin transform in all of them at once absorbs every mark of each twisted letter into its own load, and what is left of K_1 is its own period at the remaining load. Nothing is integrated against anything: the hold is a product.

25.3 The lean convolution

Theorem 25.12 (The lean convolution)**.** *Let $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ have $m_{\mathcal{E}} = \sum_j (r_j s_j + r_j^* t_j)$.*

(i) Admissible points for $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ exist exactly when every proper wall of $\operatorname{Re}\mathbf{m}$ is positive; they form a convex set, and for real $\mathbf{m}$ the native loads (Proposition 25.10(iii)) are one of them.

(ii) On every admissible point and for every twist $\mathbf{x} = (\mathbf{x}_j^K, \mathbf{x}_j^\Psi)_j$ (Definition 22.1),

$$P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\substack{\sum_j (\mathbf{a}_j + \mathbf{b}_j) = \mathbf{m} \\ |\mathbf{a}_j| = r_j s_j \ (j \geq 2),\ |\mathbf{b}_j| = r_j^* t_j}} \prod_{j=1}^{N} P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{B}_j}(\mathbf{b}_j; 0, t_j)\, (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j},$$

the fibre integral of Definition 4.1, of dimension $(2N-1)(n-1)$, absolutely convergent; in the coordinates of the loads of every letter but K_1 at the marks other than a fixed mark e_0 it is $(2\pi i)^{-(2N-1)(n-1)} \int \cdots$ over the imaginary translate through the admissible point, the loads at e_0 and $\mathbf{a}_1$ being fixed by the pins. At $\mathbf{x} = \mathbf{1}$ it is the period.

(iii) The modulus of the integrand is at most its value at the real point,

$$\prod_j P_{\mathcal{B}_j}(\mathbf{c}_j^K; s_j, 0)\, P_{\mathcal{B}_j}(\mathbf{c}_j^\Psi; 0, t_j)\, (\mathbf{x}_j^K)^{-\mathbf{c}_j^K} (\mathbf{x}_j^\Psi)^{-\mathbf{c}_j^\Psi},$$

and it decays exponentially in the imaginary part of the load of every letter but K_1, at the rate of Proposition 25.10(iv) for its table.

With exponents $s_j, t_j \geq 0$, not all zero, the same holds over the active letters, p of them, any one of them in the place of K_1, in dimension $(p-1)(n-1)$.

For one table, at $\mathcal{H} = \binom{\{1,2\}}{1}$ the lean convolution is Corollary 4.7, and at the circuit $\binom{\mathcal{E}}{n-1}$ it is Corollary 18.22, the periods at the full leans being those of Proposition 26.9. Opening each of them on its share side (Proposition 4.2) gives the atom formula of Corollary 18.16, one share per basis and letter: the atom formula opens every letter over its bases and every mass over the letters at once, while the lean convolution keeps each letter closed, read at its full lean, and opens only the split of each mass among the letters.

Corollary 25.13 (What a family retains)**.** *The Grammas $\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_{T \in \mathcal{B}_j}}(\boldsymbol{\theta}_j)$, $j = 1, \ldots, N$, each one function of the n variables $\mathbf{a}$, determine every period of F and of all its twists: at all exponents $s_j, t_j \geq 0$, not all zero, at all masses on the balance wall with every proper wall positive, and at every twist. At the exponents $(\mathbf{s}, \mathbf{t})$ only the values of the Gramma of table j on the hyperplanes $|\mathbf{a}| = r_j s_j$ and $|t_j \mathbf{1} - \mathbf{b}| = r_j t_j$ enter, through $P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0) = r_j\, \mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(s_j)$ and $P_{\mathcal{B}_j}(\mathbf{b}; 0, t_j) = r_j\, \mathsf{G}_{t_j \mathbf{1} - \mathbf{b};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(t_j)$. Conversely the periods of table j at its full basis leans determine its Gramma, through Proposition 25.10(i).*

A cell of a hinge tree retains, for one pair of exponents, one measure on the half-line (Theorem 23.11); a family retains, for all exponents and all masses at once, one function of the loads for each of its tables. The two retentions serve different queries; Lemma 25.16 relates them, and Section 26.6 compares them.

25.4 Holding some marks

Theorem 25.14 (Pinned holds)**.** *Let every proper wall of* $\operatorname{Re}\mathbf{m}$ *be positive,* e_0 *a fixed mark,* $\varnothing \neq G \subseteq \mathcal{E}\setminus e_0$*, and* $\boldsymbol{\beta}=(\boldsymbol{\beta}_j^K,\boldsymbol{\beta}_j^\Psi)_j$*,* $\boldsymbol{\beta}_1^K$ *omitted, the restriction to* G *of the loads of an admissible point. Then for every real* $\boldsymbol{\eta}=(\boldsymbol{\eta}_j^K,\boldsymbol{\eta}_j^\Psi)_j$ *of the same shape the hold* $\operatorname{Hold}_G(\boldsymbol{\beta}+i\boldsymbol{\eta})$ *converges absolutely and equals the fibre integral of Theorem 25.12(ii) at* $\mathbf{x}=\mathbf{1}$ *with the* $(2N-1)|G|$ *further pins* $\mathbf{a}_j|_G=\boldsymbol{\beta}_j^K+i\boldsymbol{\eta}_j^K$ *(*$j\geq 2$*) and* $\mathbf{b}_j|_G=\boldsymbol{\beta}_j^\Psi+i\boldsymbol{\eta}_j^\Psi$*,*

$$\operatorname{Hold}_G(\boldsymbol{\beta}+i\boldsymbol{\eta})=\int_{\sum_j(\mathbf{a}_j+\mathbf{b}_j)=\mathbf{m},\ |\mathbf{a}_j|=r_js_j\ (j\geq 2),\ |\mathbf{b}_j|=r_j^*t_j,\ \text{the pins on } G}\ \prod_{j=1}^{N}P_{\mathcal{B}_j}(\mathbf{a}_j;s_j,0)\,P_{\mathcal{B}_j}(\mathbf{b}_j;0,t_j),$$

of dimension $(2N-1)(n-1-|G|)$*, in the coordinates of the loads of every letter but* K_1 *at the marks off* $G\cup e_0$ *the integral* $(2\pi i)^{-(2N-1)(n-1-|G|)}\int\cdots$*. At* $G=\mathcal{E}\setminus e_0$ *it is Lemma 25.11.*

For one mark the hold has a closed form in the minors at that mark. For a mark g that is neither a loop nor a coloop of any table, the patterns of the bunch $\{g\}$ in table j are $\varnothing$ and $\{g\}$, with the children the deletion $\mathcal{B}_j\setminus g$, of rank r_j and corank r_j^*-1, and the contraction $\mathcal{B}_j/g$, of rank r_j-1 and corank r_j^* (Definition 12.1). For K-loads a_j and Ψ-loads b_j at g the children family of $\{g\}$ (Definition 12.24) at the K-shares (s_j-a_j,a_j) and the Ψ-shares (b_j,t_j-b_j) of the patterns $\varnothing$ and $\{g\}$ is

$$F_{\{g\}}\big((s_j-a_j,a_j),(b_j,t_j-b_j)\big)_j=\Big(\big((\mathcal{B}_j\setminus g,\theta_j);\ s_j-a_j,\ b_j\big),\ \big((\mathcal{B}_j/g,\theta_j);\ a_j,\ t_j-b_j\big)\Big)_j,$$

a family of $2N$ tables on $\mathcal{E}\setminus g$ whose word is $\prod_j K_{j\setminus g}^{a_j-s_j}\Psi_{j\setminus g}^{-b_j}K_{j/g}^{-a_j}\Psi_{j/g}^{b_j-t_j}$. At masses $\mathbf{m}|_{\mathcal{E}\setminus g}$ with $\sum_j(a_j+b_j)=m_g$ it is on its balance wall: its degree is $\sum_j\big((s_j-a_j)r_j+b_j(r_j^*-1)+a_j(r_j-1)+(t_j-b_j)r_j^*\big)=\sum_j(r_js_j+r_j^*t_j)-m_g$.

Proposition 25.15 (Holding one mark)**.** *Let* g *be a mark that is neither a loop nor a coloop of any table,* $\mathbf{m}$ *real masses on the balance wall,* a_j *(*$j\geq 2$*) and* b_j *(*$j\leq N$*) real, and* $a_1=m_g-\sum_{j\geq 2}a_j-\sum_j b_j$*. The hold*

$$\operatorname{Hold}_{\{g\}}\big((a_j)_{j\geq 2},(b_j)_j\big)=\int_{(0,\infty)^{2N-1}}\prod_{j\geq 2}(x_j^K)^{a_j-1}\prod_{j=1}^{N}(x_j^\Psi)^{b_j-1}\,P_{F[\mathbf{x}@g]}(\mathbf{m};\mathbf{s},\mathbf{t})\,d\mathbf{x},\qquad x_1^K=1,$$

is finite exactly when $0<a_j<s_j$ *and* $0<b_j<t_j$ *for every* j *and every proper wall of* $F_{\{g\}}\big((s_j-a_j,a_j),(b_j,t_j-b_j)\big)_j$ *at* $\mathbf{m}|_{\mathcal{E}\setminus g}$ *is positive, and then*

$$\operatorname{Hold}_{\{g\}}\big((a_j)_{j\geq 2},(b_j)_j\big)=\prod_{j=1}^{N}\Gamma_1^2\begin{bmatrix}a_j,\ s_j-a_j\\ s_j\end{bmatrix}\Gamma_1^2\begin{bmatrix}b_j,\ t_j-b_j\\ t_j\end{bmatrix}\,P_{F_{\{g\}}((s_j-a_j,a_j),(b_j,t_j-b_j))_j}\big(\mathbf{m}|_{\mathcal{E}\setminus g}\big);$$

the identity holds as well for complex loads and complex masses whose real parts are such data. Where the loads are those at g *of an admissible point, Theorem 25.14 writes the same value as the lean convolution with the pins at* g*. If* $m_g>0$*, then for every letter of* F *whose exponent* u *exceeds* m_g*, as the twists at* g *of all the other letters decrease to* 0*, the twist of that letter staying* 1*,* $P_{F[\mathbf{x}@g]}(\mathbf{m};\mathbf{s},\mathbf{t})$ *increases, in* $(0,\infty]$*, to* $\Gamma_1^2\begin{bmatrix}m_g,\ u-m_g\\ u\end{bmatrix}$ *times the period at* $\mathbf{m}|_{\mathcal{E}\setminus g}$ *of the children family of* $\{g\}$ *at the shares that put the whole mass* m_g *on that letter:* (s_j-m_g,m_g) *if it is* K_j*,* (m_g,t_j-m_g) *if it is* Ψ_j*,* $(s_{j'},0)$ *for every other letter* $K_{j'}$ *and* $(0,t_{j'})$ *for every other letter* $\Psi_{j'}$*.*

The ends of the twist are the faces of the children family at which every letter but one has the mark g switched off: the chosen letter keeps both its minors and every other letter keeps one. For one table the two limits are those of $P_{\mathcal{H}[x@g]}$ as $x \downarrow 0$ and of $x^{m_g} P_{\mathcal{H}[x@g]}$ as $x \uparrow \infty$ (Definition 22.1), and a weight on a mark driven to one end of its range leaves the deletion and the contraction of the mark, two tables carrying three active letters between them. With Proposition 26.20 and Definition 22.1, a weight w on the bases through g is the twist w^{-2} up to the factor w^{-m_g-t}, so the period behaves like w^{-m_g-t} times the first limit as $w \to \infty$ and like w^{m_g-t} times the second as $w \to 0$, where these are finite.

Lemma 25.16 (The Mellin transform of a law is the period of a children family)**.** *Let $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a cell family on $B \sqcup \{g\}$ with masses $\mathbf{m}$ on B, hinge wall κ_F, base form ϕ_F and ratio vector $\boldsymbol{\varrho}_F$ (Definition 23.1). In this statement let $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$ have $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa_F$, and write $\boldsymbol{\varrho}_F^{-\mathbf{z}} = \prod_{j \geq 2} \left(\gamma^K_{\mathcal{A}_j} / \gamma^K_{\mathcal{A}_1}\right)^{-z_{j,K}} \prod_j \left(\gamma^\Psi_{\mathcal{A}_j} / \gamma^K_{\mathcal{A}_1}\right)^{-z_{j,\Psi}}$, the coordinates of the ratio vector to the powers $-z_{j,K}$ $(j \geq 2)$ and $-z_{j,\Psi}$. For real $\mathbf{z}$, in $[0, \infty]$,*

$$\int_{\Delta_B} \boldsymbol{\varrho}_F^{-\mathbf{z}} \, d\phi_F = P_{F_{\{g\}}((z_{j,K},\, s_j - z_{j,K}),(t_j - z_{j,\Psi},\, z_{j,\Psi}))_j}(\mathbf{m}),$$

the period of the children family of the bunch $\{g\}$ of F (Definition 12.24), its tables $\mathcal{A}_j \setminus g$ at the exponents $(z_{j,K}, t_j - z_{j,\Psi})$ and $\mathcal{A}_j / g$ at $(s_j - z_{j,K}, z_{j,\Psi})$, on its balance wall at $\mathbf{m}$. Let moreover $s_j, t_j > 0$, and put the mass $\sum_j (s_j + t_j) - \kappa_F$ on g, which puts the family of the tables $\mathcal{A}_j$ on its balance wall. For complex $\mathbf{z}$ whose real part has $0 < \operatorname{Re} z_{j,K} < s_j$ and $0 < \operatorname{Re} z_{j,\Psi} < t_j$ for every j and is a point at which $\int \boldsymbol{\varrho}_F^{-\operatorname{Re}\mathbf{z}} d\phi_F$ is finite,

$$\int_{(0,\infty)^{2N-1}} \prod_{j \geq 2} (x_j^K)^{s_j - z_{j,K} - 1} \prod_{j=1}^N (x_j^\Psi)^{t_j - z_{j,\Psi} - 1} \, P_{F[\mathbf{x}@g]}\Big(\big(\mathbf{m}, \textstyle\sum_j (s_j + t_j) - \kappa_F\big); \mathbf{s}, \mathbf{t}\Big) \, d\mathbf{x}$$
$$= \prod_{j=1}^N \Gamma_{\binom{2}{1}}\begin{bmatrix} s_j - z_{j,K}, & z_{j,K} \\ & s_j \end{bmatrix} \Gamma_{\binom{2}{1}}\begin{bmatrix} t_j - z_{j,\Psi}, & z_{j,\Psi} \\ & t_j \end{bmatrix} \int_{\Delta_B} \boldsymbol{\varrho}_F^{-\mathbf{z}} \, d\phi_F,$$

$F[\mathbf{x}@g]$ the family of the tables $\mathcal{A}_j$ twisted at g, $x_1^K = 1$ (Definition 22.1). So the Mellin transform of the law of a cell family is the Mellin transform of its own period in the twists of its hinge, divided by $2N$ Beta corners.

Corollary 25.17 (Identically self-dual tables twisted at one mark)**.** *Let the family be one table $\mathcal{H}$ at the exponents (s, t), identically self-dual (Corollary 23.12(iii)), g a mark, and $\mathbf{m}$ on the balance wall with every proper wall positive. For every $x > 0$,*

$$P_{\mathcal{H}[x@g]}(\mathbf{m}; s, t) = P_{\mathcal{H}\setminus g}\big(\mathbf{m}|_{\mathcal{E}\setminus g};\ s + t - m_g,\ m_g\big) \, P_{\binom{\{g,g'\}}{1}[(x,1)]}\big((m_g, s + t - m_g); s, t\big),$$

the second factor the period $\int_0^\infty v^{m_g - 1}(1 + v)^{-s}(1 + xv)^{-t} dv$ of the twisted list on two marks (Lemma 23.7); and the hold at g is $\Gamma_{\binom{2}{1}}\begin{bmatrix} m_g - b, & s - m_g + b \\ & s \end{bmatrix} \Gamma_{\binom{2}{1}}\begin{bmatrix} b, & t - b \\ & t \end{bmatrix}$ times the same period of $\mathcal{H} \setminus g$ for every b in its strip: the children family does not depend on the split. At $x = 1$ this is Corollary 23.12(iii), and Corollary 4.7 recovers it from the hold,

$$\frac{1}{2\pi i} \int \Gamma_{\binom{2}{1}}\begin{bmatrix} m_g - b, & s - m_g + b \\ & s \end{bmatrix} \Gamma_{\binom{2}{1}}\begin{bmatrix} b, & t - b \\ & t \end{bmatrix} db = \Gamma_{\binom{2}{1}}\begin{bmatrix} m_g, & s + t - m_g \\ & s + t \end{bmatrix}.$$

Chapter 26

Periods at the full leans and networks along a hinge tree

At a full lean the period of a two-sum is the product of the periods of its two cells over one Beta corner (Theorem 19.10), and along any hinge tree this is Corollary 26.11, which for one table at a full lean is Theorem 26.5 at the loads $\mathbf{m}$. Along a hinge tree this needs only bookkeeping, and the bookkeeping is affine: every marker receives a load that is a fixed affine function of the loads of the marks, the same function at every pair of exponents. So the periods of a substitution at its full leans are products of those of its nodes, read at affine loads, over Beta corners at its edges, with no integral. Where the active letters read every cell by one hinge value, the period along a hinge tree is the product of the periods of the nodes over Beta corners at the edges: at a full lean of a table whose weights factor at its bunches of width one, a product of stacked Gammas, monomials in the weights and the periods of the prime nodes of its canonical hinge tree, and k copies glued along any tree cost one period and a power. Blocks aggregate exactly where every active letter reads them by one hinge value, their total or the inverse of the sum of the reciprocals of their marks (Theorem 26.12). The nodes whose periods at the full leans cost nothing are characterized by counting bases. The period of a family of such tables, at all exponents, every mass and every twist, is then one fibre integral of node data, of dimension $(2N-1)(n-1)$: the lean convolution of Chapter 25 with the periods of every letter at its full lean read along the hinge tree of its table (Theorem 26.13).

26.1 Molecules and marker loads

Lemma 26.1 (Substitutions of molecules)**.** *For a core $\mathcal{H}$ and cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ at its marks (Definition 21.1), the substitution $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q \rangle$ is one molecule exactly when $\mathcal{H}$ and the $\mathcal{A}_i$, as tables on their marks and hinges, are each one molecule. In particular the two-sum $\mathcal{A} \oplus_g \mathcal{A}'$ of two cells is one molecule exactly when $\mathcal{A}$ and $\mathcal{A}'$ are.*

So a substitution of nodes that are molecules is one molecule, and a table with every proper wall positive, which is one molecule by Proposition 8.16, has every cell of its hinge trees one molecule.

Definition 26.2 (Branches of a hinge tree)**.** Let $\mathcal{B}$ be a table on $\mathcal{E}$, $|\mathcal{E}| = n$, of rank r and corank r^*, with a hinge tree $\mathfrak{T}$ (Definition 23.15) whose nodes $\mathcal{H}_v$ on $\mathcal{E}_v \sqcup G_v$ are each one molecule, of rank r_v and corank $r_v^* = |\mathcal{E}_v| + |G_v| - r_v$; the weights of $\mathcal{B}$ are the products of the weights of the nodes (Lemma 21.12). For an edge $\mathfrak{e} = \{v, w\}$ of $\mathfrak{T}$ and its end v, the *branch* $\mathfrak{T}_v^{\mathfrak{e}}$ is the component

of $\mathfrak{T}-\mathfrak{e}$ containing v, $\mathcal{E}_v^{\mathfrak{e}}$ the union of the $\mathcal{E}_u$ over its vertices, and $\mathcal{A}_v^{\mathfrak{e}}$ the substitution of its nodes, a cell on $\mathcal{E}_v^{\mathfrak{e}} \sqcup \{g_{v,\mathfrak{e}}\}$ of rank $r_v^{\mathfrak{e}}$ and corank $r_v^{*\mathfrak{e}} = |\mathcal{E}_v^{\mathfrak{e}}| + 1 - r_v^{\mathfrak{e}}$; $\mathcal{B}$ is the two-sum of $\mathcal{A}_v^{\mathfrak{e}}$ and $\mathcal{A}_w^{\mathfrak{e}}$.

Definition 26.3 (Marker loads; node loads)**.** For a table with a hinge tree as in Definition 26.2 and loads $\mathbf{a}, \mathbf{b} \in \mathbb{C}^{\mathcal{E}}$ with $|\mathbf{a}| = rs$ and $|\mathbf{b}| = r^*t$, the *marker loads* at the end v of the edge $\mathfrak{e}$ are

$$k_{v,\mathfrak{e}} = s\, r_v^{\mathfrak{e}} - a_{\mathcal{E}_v^{\mathfrak{e}}}, \qquad l_{v,\mathfrak{e}} = t\, r_v^{*\mathfrak{e}} - b_{\mathcal{E}_v^{\mathfrak{e}}},$$

and the *node loads* of v are $\mathbf{a}_v = \big(\mathbf{a}|_{\mathcal{E}_v}, (k_{v,\mathfrak{e}})_{\mathfrak{e}\ni v}\big)$ and $\mathbf{b}_v = \big(\mathbf{b}|_{\mathcal{E}_v}, (l_{v,\mathfrak{e}})_{\mathfrak{e}\ni v}\big)$, the load of the marker $g_{v,\mathfrak{e}}$ being $k_{v,\mathfrak{e}}$, respectively $l_{v,\mathfrak{e}}$.

Lemma 26.4 (Node loads)**.** *In Definition 26.3:*

(i) $k_{v,\mathfrak{e}} + k_{w,\mathfrak{e}} = s$ *and* $l_{v,\mathfrak{e}} + l_{w,\mathfrak{e}} = t$ *on every edge.*

(ii) $|\mathbf{a}_v| = r_v s$ *and* $|\mathbf{b}_v| = r_v^* t$ *at every node.*

(iii) If $\mathbf{a}+\mathbf{b} = \mathbf{m}$ *with* $\mathbf{m}$ *on the balance wall of* $\mathcal{B}$*, then* $k_{v,\mathfrak{e}} + l_{v,\mathfrak{e}} = s + t - \kappa_{\mathcal{A}_v^{\mathfrak{e}}} = \kappa_{\mathcal{A}_w^{\mathfrak{e}}}$*: the node masses* $\mathbf{a}_v + \mathbf{b}_v$ *are* $\mathbf{m}|_{\mathcal{E}_v}$ *on the marks and the hinge walls of the branches on the markers, the masses of the twisted core in Theorem 23.6.*

26.2 The periods of a substitution at its full leans

Theorem 26.5 (Periods at the full leans along a hinge tree)**.** *In Definition 26.3, for real loads* $\mathbf{a}$ *with* $|\mathbf{a}| = rs$*, if every marker load* $k_{v,\mathfrak{e}}$ *lies in* $(0, s)$*, then, in* $(0, \infty]$*,*

$$P_{\mathcal{B}}(\mathbf{a}; s, 0) = \frac{\prod_v P_{\mathcal{H}_v}(\mathbf{a}_v; s, 0)}{\prod_{\mathfrak{e}=\{v,w\}} \Gamma_{\!2}\!\left[\begin{matrix} k_{v,\mathfrak{e}}, & k_{w,\mathfrak{e}} \\ & s \end{matrix}\right]_1},$$

and if some marker load lies outside $(0, s)$*, then* $P_{\mathcal{B}}(\mathbf{a}; s, 0) = +\infty$*. So* $P_{\mathcal{B}}(\mathbf{a}; s, 0)$ *is finite exactly when every marker load lies in* $(0, s)$ *and the period of every node at its full basis lean is finite at its node loads. The same holds at the full cobasis leans, with* $\mathbf{b}$*, the loads* $l_{v,\mathfrak{e}}$ *and* t*. For complex loads both identities hold on the tube over the real loads at which they are finite, both sides being holomorphic there.*

In the language of bunches this is Theorem 19.10 at every bunch of width one of the tree at once: the bunch $\mathcal{E}_v^{\mathfrak{e}}$ has, at $(s, 0)$, the wall $k_{w,\mathfrak{e}}$, and its two cells are $\mathcal{A}_v^{\mathfrak{e}}$ and $\mathcal{A}_w^{\mathfrak{e}}$ (Lemma 21.12). What the tree adds is that the loads of all markers are affine in the loads of the marks, fixed once for every pair of exponents, so that the periods of the substitution at its full leans are evaluated at any loads by evaluating each node once.

Corollary 26.6 (Convergence is decided at the nodes)**.** *Let* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ *have exponents* $s_j, t_j > 0$*, and let table* j *carry a hinge tree* $\mathfrak{T}_j$ *as in Definition 26.2, its nodes* $\mathcal{H}_{j,v}$ *on* $\mathcal{E}_{j,v} \sqcup G_{j,v}$ *each one molecule, with the node loads of Definition 26.3 read in table* j*. For masses* $\mathbf{m}$ *on the balance wall of* F *every proper wall of* F *is positive exactly when there is a real split* $\mathbf{m} = \sum_j (\mathbf{a}_j + \mathbf{b}_j)$ *with* $|\mathbf{a}_j| = r_j s_j$ *and* $|\mathbf{b}_j| = r_j^* t_j$ *at which every node of every table has*

$$(\mathbf{a}_{j,v})_S > s_j\, d_{j,v}(S) \quad \text{and} \quad (\mathbf{b}_{j,v})_S > t_j\, d_{j,v}^*(S) \qquad (\varnothing \neq S \subsetneq \mathcal{E}_{j,v} \sqcup G_{j,v}),$$

$d_{j,v}$ and $d_{j,v}^*$ the counts of $\mathcal{H}_{j,v}$. This is a system of $\sum_j \sum_v 2(2^{|\mathcal{E}_{j,v}|+|G_{j,v}|} - 2)$ *strict linear inequalities in* $(2N-1)(n-1)$ *unknowns, a number linear in the number of nodes when the nodes have bounded size, against the* $2^n - 2$ *walls of* F*; the native loads of* F *solve it.*

26.3 What a node costs

Lemma 26.7 (Bases outnumber marks)**.** *A table on $n \geq 2$ marks that is one molecule has at least n bases, and exactly n only when its rank is 1 or $n-1$, that is, as a list, when it is $\binom{\mathcal{E}}{1}$ or $\binom{\mathcal{E}}{n-1}$.*

Corollary 26.8 (The cost of a node)**.** *For a node $\mathcal{H}_v$ that is one molecule on $n_v = |\mathcal{E}_v| + |G_v|$ marks with $|\mathcal{H}_v|$ bases, the share side of its period at each of its two full leans (Proposition 4.2) is a fibre integral of dimension $|\mathcal{H}_v| - n_v \geq 0$. The dimension is zero exactly when $\mathcal{H}_v$ has the bases of $\binom{\mathcal{E}_v \sqcup G_v}{n_v - 1}$ or of $\binom{\mathcal{E}_v \sqcup G_v}{1}$, with any weights: the series and the parallel nodes up to their weights, whose periods at the full leans are stacked Gammas times monomials in the weights (Proposition 25.6; Proposition 26.9, next); a prime node on at least four marks has $|\mathcal{H}_v| > n_v$.*

Proposition 26.9 (Series and parallel nodes at their full leans)**.** *On a set $\mathcal{X}$ of $q+1$ marks let $\binom{\mathcal{X}}{q}$ carry the weight θ_e on its basis $\mathcal{X} \setminus e$ and $\binom{\mathcal{X}}{1}$ the weight θ_e on its basis $\{e\}$, the series and the parallel node with weights (Definition 23.13). Then*

$$P_{\binom{\mathcal{X}}{q}}(\mathbf{a}; s, 0) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(s-a_e)_e\\ s\end{bmatrix}\prod_e \theta_e^{a_e - s}, \qquad P_{\binom{\mathcal{X}}{q}}(\mathbf{b}; 0, t) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(b_e)_e\\ t\end{bmatrix}\prod_e \theta_e^{-b_e},$$

$$P_{\binom{\mathcal{X}}{1}}(\mathbf{a}; s, 0) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(a_e)_e\\ s\end{bmatrix}\prod_e \theta_e^{-a_e}, \qquad P_{\binom{\mathcal{X}}{1}}(\mathbf{b}; 0, t) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(t-b_e)_e\\ t\end{bmatrix}\prod_e \theta_e^{b_e - t},$$

with $|\mathbf{a}| = qs$, $|\mathbf{b}| = t$ at the series node and $|\mathbf{a}| = s$, $|\mathbf{b}| = qt$ at the parallel node; each converges exactly when every Gamma above has an argument of positive real part. These are Proposition 25.6 at the configurations $(\mathbf{e}_x)_{x\in\mathcal{X}}$, of the letters K of $\binom{\mathcal{X}}{1}$ and Ψ of $\binom{\mathcal{X}}{q}$, and $(\mathbf{1} - \mathbf{e}_x)_{x\in\mathcal{X}}$, of the letters K of $\binom{\mathcal{X}}{q}$ and Ψ of $\binom{\mathcal{X}}{1}$.

Corollary 26.10 (Series-parallel tables at their full leans)**.** *In Definition 26.2 let every node of $\mathfrak{T}$ be a series or a parallel node (Definition 23.13), the node v on $\mathcal{E}_v \sqcup G_v$ carrying the weight $\theta_{v,x}$ on its basis that omits x if it is a series node and on its basis $\{x\}$ if it is a parallel node, and write $(\mathbf{a}_v)_x$ and $(\mathbf{b}_v)_x$ for the node loads at $x \in \mathcal{E}_v \sqcup G_v$ (Definition 26.3). Then*

$$P_{\mathcal{B}}(\mathbf{a}; s, 0) = \frac{1}{\Gamma(s)} \prod_{e \text{ at series}} \Gamma(s - a_e) \prod_{e \text{ at parallel}} \Gamma(a_e) \prod_{\mathfrak{e} \text{ series–parallel}} \frac{\Gamma(s - k_{\mathfrak{e}})}{\Gamma(k_{\mathfrak{e}})}$$
$$\times \prod_{v \text{ series}} \prod_x \theta_{v,x}^{(\mathbf{a}_v)_x - s} \prod_{v \text{ parallel}} \prod_x \theta_{v,x}^{-(\mathbf{a}_v)_x},$$

$$P_{\mathcal{B}}(\mathbf{b}; 0, t) = \frac{1}{\Gamma(t)} \prod_{e \text{ at series}} \Gamma(b_e) \prod_{e \text{ at parallel}} \Gamma(t - b_e) \prod_{\mathfrak{e} \text{ series–parallel}} \frac{\Gamma(l_{\mathfrak{e}})}{\Gamma(t - l_{\mathfrak{e}})}$$
$$\times \prod_{v \text{ series}} \prod_x \theta_{v,x}^{-(\mathbf{b}_v)_x} \prod_{v \text{ parallel}} \prod_x \theta_{v,x}^{(\mathbf{b}_v)_x - t},$$

the products over the marks by the kind of their node, over the edges joining a series node to a parallel node, with $k_{\mathfrak{e}}$, $l_{\mathfrak{e}}$ the marker loads at the series end, and over the nodes and their marks and markers x; an edge joining two nodes of one kind contributes no Gamma. Each of these periods is finite exactly when every Gamma above its line has an argument with positive real part.

The period of a series-parallel table at a full lean is written down from its tree in time linear in the size of the tree, one Gamma for each mark, two for each change of kind along an edge and a monomial in the weights of each node, while the table may have a number of bases exponential in n. The atom formula of the same period (Corollary 18.16) has one share for each basis.

Corollary 26.11 (Periods along a hinge tree read by one hinge value)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ have real data on its balance wall with every proper wall positive, let its tables have a common hinge tree $\mathfrak{T}$ rooted at o, with the node families $F_{\mathcal{H}_v}$ (Definition 23.15), and let the active letters read every subtree cell by one hinge value: for every $v \neq o$ there is a function γ_v with $\gamma^{K}_{\mathcal{A}_{v,j}} = \gamma_v$ whenever $s_j \neq 0$ and $\gamma^{\Psi}_{\mathcal{A}_{v,j}} = \gamma_v$ whenever $t_j \neq 0$. With σ the total strength (Definition 8.1),*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \frac{\prod_v P_{F_{\mathcal{H}_v}}(\mathbf{m}_v; \mathbf{s}, \mathbf{t})}{\prod_{edges\ \mathfrak{e}} \Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa_{\mathfrak{e}}, & 2\sigma - \kappa_{\mathfrak{e}}\\ & 2\sigma\end{smallmatrix}\right]},$$

each node family on its balance wall with every proper wall positive, its masses $\mathbf{m}_v$ those of its marks and, on the marker at an end of the edge $\mathfrak{e}$, the wall of the marks across $\mathfrak{e}$: $\kappa_{\mathfrak{e}}$ on one side and $2\sigma - \kappa_{\mathfrak{e}}$ on the other. For one table at a full lean, $(0, t)$ or $(s, 0)$, one letter is active, the hypothesis holds on every hinge tree, and this is Theorem 26.5 at the loads $\mathbf{m}$. With the canonical hinge tree of a connected table whose weights factor across each bunch of width one at which it is decomposed, a list or a determinantal table for instance (Theorem 23.16), a series or a parallel node, with any weights on its bases, contributes at a full lean a stacked Gamma times a monomial in its weights (Proposition 26.9), so the period at a full lean of such a table is a product of stacked Gammas, monomials in the weights and the periods of its prime nodes. In particular connected tables $\mathcal{A}_1, \ldots, \mathcal{A}_k$ at $(0, t)$, each on its balance wall with every proper wall positive, glued along any tree, every gluing joining a mark of mass $\kappa_{\mathfrak{e}}$ of one to a mark of mass $t - \kappa_{\mathfrak{e}}$ of another, have the period

$$\frac{\prod_v P_{\mathcal{A}_v}(\mathbf{m}_v; 0, t)}{\prod_{\mathfrak{e}} \Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa_{\mathfrak{e}}, & t - \kappa_{\mathfrak{e}}\\ & t\end{smallmatrix}\right]},$$

whatever the shape of the tree; for k copies of one table glued at the masses $(\kappa, t - \kappa)$ this is $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa, & t - \kappa\\ & t\end{smallmatrix}\right] \big(P_{\mathcal{A}}(\mathbf{m}; 0, t)/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa, & t - \kappa\\ & t\end{smallmatrix}\right]\big)^{k}$. On the chain $F^{\triangleright k}$ of Theorem 22.8, with every proper wall positive and the active letters reading each copy by one hinge value, the product is $P_F^k/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}m_{g_{\mathrm{in}}}, & m_{g_{\mathrm{out}}}\\ & 2\sigma\end{smallmatrix}\right]^{k-1}$, the display of that theorem.

Where the active letters read every cell through one hinge value, as at a full lean of one table, where only one letter sees a hinge, Ψ at $(0, t)$ and K at $(s, 0)$, every cell is pinned, every hinge integral is the Beta corner of a linear letter (Proposition 15.1 with one coefficient), and the period of the whole construction is the product of the periods of its nodes over the Beta corners of its hinges. Nothing is integrated against anything.

26.4 Blocks aggregate

Theorem 26.12 (Blocks read by one hinge value aggregate exactly)**.** *Let F be a family substituted with blocks (Corollary 21.7(i)) into the tables $\mathcal{H}_j$ of a core family $F_{\mathcal{H}} = ((\mathcal{H}_j, \theta^{\mathcal{H}_j}); s_j, t_j)_j$, with unit weights in the blocks, the block B_i of ℓ_i marks. The active letters read B_i* through its total *when every table with $s_j \neq 0$ has a parallel block and every table with $t_j \neq 0$ a series block at B_i, and through $e_{\ell_i}/e_{\ell_i - 1} = \big(\sum_{x \in B_i} \alpha_x^{-1}\big)^{-1}$ when every table with $s_j \neq 0$ has a series block and every table with $t_j \neq 0$ a parallel block (Corollary 21.7(i)). Let $U \sim \mathrm{Dir}_{\mathbf{m}}$ (Definition 3.10), with the block totals $|U_{B_i}|$.*

(i) If the active letters read every block through its total, the joint law of all the active letters, $\big(K_j(U)\big)_{s_j\neq 0}$ and $\big(\Psi_j(U)\big)_{t_j\neq 0}$, with the totals $(|U_{B_i}|)_i$ themselves, is that of $\big(K_{\mathcal{H}_j}(U')\big)_{s_j\neq 0}$, $\big(\Psi_{\mathcal{H}_j}(U')\big)_{t_j\neq 0}$ and U' for $U'\sim \mathrm{Dir}_{(m_{B_i})_i}$: every nonnegative function of these variables, and every absolutely integrable one, has the same expectation on both sides.

(ii) Let the active letters read B_i through its total for $i\in I$ and through e_{ℓ_i}/e_{ℓ_i-1} for $i\notin I$, and let σ be the total strength (Definition 8.1). The wall of a block is $\kappa_{B_i}=m_{B_i}$ for $i\in I$ and $\kappa_{B_i}=m_{B_i}-(\ell_i-1)\,2\sigma$ for $i\notin I$, and on the balance wall, in $[0,\infty]$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\prod_{i\in I}\Gamma_{\substack{\ell_i\\1}}\begin{bmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{bmatrix}\prod_{i\notin I}\Gamma_{\substack{\ell_i\\1}}\begin{bmatrix}(2\sigma-m_x)_{x\in B_i}\\ 2\sigma-\kappa_{B_i}\end{bmatrix}\;P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big),$$

a stacked Gamma read as $+\infty$ when one of its upper arguments is not positive; both sides are finite together when these are positive.

(iii) If the active letters read every block through its total, then at fixed pooled masses m_{B_i}, dividing each among any number of marks leaves the expectation $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)]$ of the family word unchanged and multiplies the period by the stacked Gammas $\Gamma_{\substack{|B_i|\\1}}\begin{bmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{bmatrix}$ alone: lengthening the blocks creates no new integral.

A block read through its total by some active letters and through $e_\ell/e_{\ell-1}$ by others keeps the block as one mark of Theorem 22.3, with its kernel and its hinge values.

26.5 The network theorem

The lean convolution of a family of substitutions, the periods of its letters at their full leans written along the hinge trees of its tables, is one fibre integral whose integrand is a product of node data at affine loads. Read as one integral over all the loads, it has dimension $(2N-1)(n-1)$ for every family whose tables carry hinge trees, and for series-parallel tables its integrand is a written-down stacked Gamma times an exponential of the loads. Integrated from the leaves up, when the tables share one hinge tree, the same integral is computed node by node with one rule at every node, prime or not (Theorem 26.16): a node enters through the periods of its tables at their full leans and its cell families through the Mellin transforms of their laws; for one table this is the hinge tree procedure of Chapter 23.

Theorem 26.13 (The network)**.** *Let $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ be a family on $\mathcal{E}$ with exponents $s_j,t_j>0$ whose table j carries a hinge tree $\mathfrak{T}_j$ as in Definition 26.2, its nodes $\mathcal{H}_{j,v}$ each one molecule, and let $\mathbf{m}$ be on the balance wall of F with every proper wall positive. Then every table is one molecule, and on every admissible point (Definition 25.9) and for every twist $\mathbf{x}$ (Definition 22.1),*

$$P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})=\int\prod_{j=1}^{N}\frac{\prod_{v\in\mathfrak{T}_j}P_{\mathcal{H}_{j,v}}(\mathbf{a}_{j,v};s_j,0)\,P_{\mathcal{H}_{j,v}}(\mathbf{b}_{j,v};0,t_j)}{\prod_{\mathfrak{e}=\{v,w\}\in\mathfrak{T}_j}\Gamma_{\substack{2\\1}}\begin{bmatrix}k_{j;v,\mathfrak{e}},\;k_{j;w,\mathfrak{e}}\\ s_j\end{bmatrix}\Gamma_{\substack{2\\1}}\begin{bmatrix}l_{j;v,\mathfrak{e}},\;l_{j;w,\mathfrak{e}}\\ t_j\end{bmatrix}}\;(\mathbf{x}_j^K)^{-\mathbf{a}_j}(\mathbf{x}_j^\Psi)^{-\mathbf{b}_j},$$

the fibre integral of Theorem 25.12 over the loads $(\mathbf{a}_j,\mathbf{b}_j)_j$ of the marks, of dimension $(2N-1)(n-1)$, with the node loads $\mathbf{a}_{j,v},\mathbf{b}_{j,v}$ and the marker loads $k_{j;v,\mathfrak{e}},l_{j;v,\mathfrak{e}}$ of Definition 26.3 read in table j at its loads $(\mathbf{a}_j,\mathbf{b}_j)$ and its exponents (s_j,t_j). At every admissible point the periods of every node at both full leans are finite and every marker load of table j lies in $(0,s_j)$, respectively $(0,t_j)$;

the native loads of F are an admissible point. The nodes enter only through their periods at the two full leans, as functions of their loads, the same functions for all exponents, all masses and all twists.

Corollary 26.14 (Series-parallel periods)**.** *In Theorem 26.13 let every node of every table be a series or a parallel node with the weights of Corollary 26.10, $\theta_{j,v,x}$ at the node v of $\mathfrak{T}_j$. Then*

$$\prod_j \Gamma(s_j)\Gamma(t_j)\, P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}) = \int \prod_j \Bigg[\prod_{e \text{ at series}} \Gamma(s_j - a_{j,e})\Gamma(b_{j,e}) \prod_{e \text{ at parallel}} \Gamma(a_{j,e})\Gamma(t_j - b_{j,e}) \times \prod_{\mathfrak{e} \text{ series–parallel}} \frac{\Gamma(s_j - k_{j,\mathfrak{e}})\,\Gamma(l_{j,\mathfrak{e}})}{\Gamma(k_{j,\mathfrak{e}})\,\Gamma(t_j - l_{j,\mathfrak{e}})} \prod_{v \text{ series}} \prod_x \theta_{j,v,x}^{(\mathbf{a}_{j,v})_x - (\mathbf{b}_{j,v})_x - s_j} \times \prod_{v \text{ parallel}} \prod_x \theta_{j,v,x}^{(\mathbf{b}_{j,v})_x - (\mathbf{a}_{j,v})_x - t_j} \, (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j} \Bigg],$$

the kinds of the nodes and the marker loads $k_{j,\mathfrak{e}}$, $l_{j,\mathfrak{e}}$ at the series end read in $\mathfrak{T}_j$, the fibre integral of Theorem 26.13, of dimension $(2N-1)(n-1)$, on any point at which every Gamma above the line has an argument of positive real part. Every period of a family of series-parallel tables with any weights on their nodes, at all exponents, at all masses with positive walls and at every twist, is a fibre integral of dimension $(2N-1)(n-1)$ of an explicit stacked Gamma times an exponential of the loads, whatever the numbers of bases, and at a full lean of one table no integral is left (Corollary 26.10).

Proposition 26.15 (Pooling a pair)**.** *In Corollary 26.14, for one table twisted in its cobasis letter by $\mathbf{x}$ (Definition 22.1), let e, f be two marks of one series node v whose twists times the squares of their weights agree, $x_e\theta_{v,e}^2 = x_f\theta_{v,f}^2$, and $c_e = s - m_e$, $c_f = s - m_f$. Their loads enter the integrand through $\Gamma(c_e + b_e)\Gamma(b_e)\Gamma(c_f + b_f)\Gamma(b_f)$ and otherwise through $b_{\{e,f\}} = b_e + b_f$ alone, and at fixed $b_{\{e,f\}}$*

$$\frac{1}{2\pi i} \int \Gamma(c_e + b_e)\Gamma(b_e)\Gamma\big(c_f + b_{\{e,f\}} - b_e\big)\Gamma\big(b_{\{e,f\}} - b_e\big)\, db_e = \Gamma_{\,1}^{\,4}\begin{bmatrix} c_e + c_f + b_{\{e,f\}},\ c_e + b_{\{e,f\}},\ c_f + b_{\{e,f\}},\ b_{\{e,f\}} \\ c_e + c_f + 2b_{\{e,f\}} \end{bmatrix}.$$

For two marks of one parallel node v with $x_e\theta_{v,e}^{-2} = x_f\theta_{v,f}^{-2}$, at fixed $b_{\{e,f\}}$, the pair gives

$$\Gamma_{\,1}^{\,4}\begin{bmatrix} m_e + m_f - b_{\{e,f\}},\ m_e + t - b_{\{e,f\}},\ m_f + t - b_{\{e,f\}},\ 2t - b_{\{e,f\}} \\ m_e + m_f + 2t - 2b_{\{e,f\}} \end{bmatrix}.$$

Each pooled pair lowers the dimension of the fibre by one, and disjoint pairs can be pooled at once.

The pooled pair is a series pair read as one cell: its factor is $\Gamma_{\,1}^{\,2}\begin{bmatrix} c_e + b_{\{e,f\}},\ c_f + b_{\{e,f\}} \\ c_e + c_f + 2b_{\{e,f\}} \end{bmatrix} \Gamma(c_e + c_f + b_{\{e,f\}})\Gamma(b_{\{e,f\}})$, the Beta corner being the Mellin transform of the law of the pair at $b_{\{e,f\}}$ and the two Gammas those of its hinge in the periods of the node at its full leans (Theorem 26.16). A series bunch of three or more marks at one node can be pooled pair by pair only once: the pooled factor of a pair is no longer of the form $\Gamma(c+b)\Gamma(b)$, and the class is a circuit of its marks (Corollary 18.22).

26.6 Nodes in hold form

Theorem 26.16 (Nodes in hold form)**.** *Let F be a substituted family (Definition 21.1) whose tables are each one molecule, with exponents $s_j, t_j > 0$, table j being $\mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$ with the cores $\mathcal{H}_j$ on $\mathcal{E}_{\mathcal{H}} \sqcup G_{\mathcal{H}}$ and the cells $\mathcal{A}_{ij}$ on $B_i \sqcup \{g_i\}$ at the markers $g_1, \dots, g_k$ (Definition 23.4), and let $F_i = ((\mathcal{A}_{ij}, \theta^{\mathcal{A}_{ij}}); s_j, t_j)_j$ be the cell family at g_i (Definition 23.1). For loads $(\mathbf{a}_j, \mathbf{b}_j)_j$ on the marks and markers of the cores write $\mathbf{z}_i = (a_{j,g_i}, b_{j,g_i})_j$ for their values at g_i, and in this statement write $\int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} d\phi_{F_i}$ for the Mellin transform of the law of F_i of Lemma 25.16 at $\mathbf{z}_i$.*

(i) Root. *If $G_{\mathcal{H}} = \{g_1, \dots, g_k\}$ and F is on its balance wall with every proper wall positive, let $\mathcal{F}$ be the fibre of the loads of the cores with the pins $\sum_j (\mathbf{a}_j + \mathbf{b}_j) = (\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}, \kappa_{F_1}, \dots, \kappa_{F_k})$, $|\mathbf{a}_j| = r_{\mathcal{H}_j} s_j$ $(j \geq 2)$ and $|\mathbf{b}_j| = r^*_{\mathcal{H}_j} t_j$. On the node loads at the cores (Definition 26.3, in every table) of any admissible point of F and for twists $\mathbf{x}$ of F with every coordinate 1 off $\mathcal{E}_{\mathcal{H}}$,*

$$P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\mathcal{F}} \prod_{j=1}^{N} P_{\mathcal{H}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{H}_j}(\mathbf{b}_j; 0, t_j)\, (\mathbf{x}_j^K)^{-\mathbf{a}_j|_{\mathcal{E}_{\mathcal{H}}}} (\mathbf{x}_j^{\Psi})^{-\mathbf{b}_j|_{\mathcal{E}_{\mathcal{H}}}} \prod_{i=1}^{k} \int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i}\, d\phi_{F_i},$$

of dimension $(2N-1)(|\mathcal{E}_{\mathcal{H}}| + k - 1)$, every $\int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} d\phi_{F_i}$ finite at the real point.

(ii) Hinged core. *If $G_{\mathcal{H}} = \{g_0, g_1, \dots, g_k\}$, let F' be the cell family with the hinge g_0 whose table j is $\mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$, with the hinge wall $\kappa_{F'}$, and for $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$ with $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa_{F'}$ let $\mathcal{F}_{\mathbf{z}}$ be the fibre of the loads of the cores with the pins $\sum_j (\mathbf{a}_j + \mathbf{b}_j) = \big(\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}, \kappa_{F_1}, \dots, \kappa_{F_k}, \sum_j (s_j + t_j) - \kappa_{F'}\big)$, $|\mathbf{a}_j| = r_{\mathcal{H}_j} s_j$ $(j \geq 2)$, $|\mathbf{b}_j| = r^*_{\mathcal{H}_j} t_j$, $a_{j,g_0} = s_j - z_{j,K}$ $(j \geq 2)$ and $b_{j,g_0} = t_j - z_{j,\Psi}$. For $\mathbf{z}$ whose real part is $(s_j - c^K_{j,g_0}, t_j - c^{\Psi}_{j,g_0})_j$, $\mathbf{c}$ the node loads at the cores of an admissible point of the family of the tables of F' with the mass $\sum_j (s_j + t_j) - \kappa_{F'}$ on g_0,*

$$\prod_{j=1}^{N} \Gamma_{\!1}^{\!2}\begin{bmatrix} s_j - z_{j,K}, & z_{j,K} \\ & s_j \end{bmatrix} \Gamma_{\!1}^{\!2}\begin{bmatrix} t_j - z_{j,\Psi}, & z_{j,\Psi} \\ & t_j \end{bmatrix} \int \boldsymbol{\varrho}_{F'}^{-\mathbf{z}}\, d\phi_{F'}$$
$$= \int_{\mathcal{F}_{\mathbf{z}}} \prod_{j=1}^{N} P_{\mathcal{H}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{H}_j}(\mathbf{b}_j; 0, t_j) \prod_{i=1}^{k} \int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i}\, d\phi_{F_i},$$

of dimension $(2N-1)(|\mathcal{E}_{\mathcal{H}}| + k - 1)$.

The cores and the cells are molecules by Lemma 26.1.

Corollary 26.17 (The glue and the series and parallel rules in hold form)**.** *On the lines of Theorem 26.16:*

(i) For one table with $\mathcal{H} = \binom{\{g_1, g_2\}}{1}$ and the cells $\mathcal{A}$ and $\mathcal{A}'$, Theorem 26.16(i) is the integral of Theorem 23.9(iv), with the same integrand; for a family with the core $\binom{\{g_1, g_2\}}{1}$ in every table it is the integral of Theorem 19.13 over the scale shares.

(ii) For one table and a series node $\binom{G}{q}$, Theorem 26.16(ii) reads

$$\Gamma(z)\Gamma(z + s - \kappa_{\mathcal{A}}) \int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}} = \int_{z_1 + \dots + z_q = z} \prod_{i=1}^{q} \Gamma(z_i)\Gamma(z_i + s - \kappa_{\mathcal{A}_i}) \int \varrho_{\mathcal{A}_i}^{-z_i} d\phi_{\mathcal{A}_i},$$

the Mellin transform of $f_{\mathcal{A}}$ as the Mellin convolution of those of the $f_{\mathcal{A}_i}$ (Proposition 23.8(iii)): the Mellin transform, on these lines, of the product rule $f_{\mathcal{A}} = \prod_i f_{\mathcal{A}_i}$ of Theorem 23.14(i).

(iii) For one table and a parallel node $\binom{G}{1}$ *it reads, with* $z' = \kappa_{\mathcal{A}} - z$ *and* $z'_i = \kappa_{\mathcal{A}_i} - z_i$,

$$\Gamma(z')\Gamma(z' + t - \kappa_{\mathcal{A}}) \int \varrho_{\mathcal{A}}^{z' - \kappa_{\mathcal{A}}} d\phi_{\mathcal{A}} = \int_{z'_1 + \cdots + z'_q = z'} \prod_{i=1}^{q} \Gamma(z'_i)\Gamma(z'_i + t - \kappa_{\mathcal{A}_i}) \int \varrho_{\mathcal{A}_i}^{z'_i - \kappa_{\mathcal{A}_i}} d\phi_{\mathcal{A}_i},$$

the Mellin convolution of the transforms of the $\check{f}_{\mathcal{A}_i}$*: the Mellin transform, on these lines, of the product rule* $\check{f}_{\mathcal{A}} = \prod_i \check{f}_{\mathcal{A}_i}$ *of Theorem 23.14(ii).*

So the three rules of the hinge tree procedure, the glue and the series and parallel products (Definition 23.19), are one rule read at three nodes, on the lines that the network certifies, the rules themselves holding on their own wider ranges; and the same rule applies at a prime node, where the procedure integrates the twisted core over its simplex for every value of the twists (Definition 23.19(4)). In Theorem 26.16 the prime node enters through the periods of its tables at their full leans, functions of its loads that do not depend on the twists, on the exponents or on the masses, and the cell families enter through the Mellin transforms of their laws on imaginary translates, for one table the representation (b) of Definition 23.19; Theorem 26.16(i) is Theorem 23.6(i) with the twisted core replaced by its lean convolution and the integration against the laws carried out through Mellin transforms.

26.7 The procedure and its cost

Definition 26.18 (The network procedure)**.** *Given*: a hinge tree of every table of a family (Definition 23.15) with the weights and the marks of its nodes; exponents $s_j, t_j > 0$ and masses $\mathbf{m}$ on the balance wall, or several of each; optionally twists $\mathbf{x}$, or product weights (Proposition 26.20 below). *Reusable data*: for every orbit of nodes, over all the tables, under bijections keeping the table and its weights, the Gramma $\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta})$ of its table, a function of the loads alone (Corollary 25.13), kept as (a) the closed form at series and parallel nodes (Proposition 26.9); (b) the share side, a fibre integral of dimension $|\mathcal{H}_v| - n_v$ (Corollary 26.8); (c) for an identically self-dual node, the reduction to the deletion of one mark (Corollaries 23.12(iii) and 25.17); (d) values on the imaginary translates in use. *Operations*:

(1) *Contour.* Find a split satisfying the node inequalities of Corollary 26.6, a linear feasibility problem with $\sum_j \sum_v 2(2^{|\mathcal{E}_{j,v}|+|G_{j,v}|} - 2)$ strict inequalities in $(2N-1)(n-1)$ unknowns; the native loads solve it, and infeasibility proves divergence.

(2) *Assemble.* The node loads of every table by Definition 26.3, the integrand of Theorem 26.13, at series-parallel nodes that of Corollary 26.14, with $\prod_j (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j}$ for twists.

(3) *Pool*, for one table, disjoint pairs of marks of one series or parallel node whose twists and weights satisfy the condition of Proposition 26.15, by that proposition.

(4) *Integrate* the remaining loads on the contour; in the imaginary parts the integrand is analytic on a strip about the contour, the admissible points forming an open set, and decays exponentially there (Theorem 25.12(iii)), so the trapezoidal rule with step ς has an error of the order $e^{-2\pi d/\varsigma}$, d the half-width of the strip; or close contours on lattices (Theorem 4.5).

(5) *Recursive Mellin evaluation*, an alternative to (2) to (4) when the hinge trees of the tables are one rooted tree with the same marks $\mathcal{E}_v$ at every vertex, so that at every vertex v other than the root the subtree cells of the tables (Definition 23.15) form a cell family, the *subtree cell family* of v (Definition 23.1): from the leaves up, compute the Mellin transform of the law of each orbit of subtree cell families on an imaginary translate by Theorem 26.16(ii),

one fibre integral of dimension $(2N-1)(|\mathcal{E}_v|+|G_v|-2)$ per point, and finish at the root by Theorem 26.16(i); for one table, at series and parallel nodes with repeated subtrees, equal and with equal masses, pass to the transforms, where repetition is a power (Proposition 23.18(ii)).

Output: the period, and with the same reusable data the period at other exponents, masses and twists (Proposition 26.21 below).

Proposition 26.19 (What the network costs)**.** *In Definition 26.18:*

(i) The fibre of the loads in (1)–(4) has dimension $(2N-1)(n-1)$, for one table less one for each pooled pair; opening every prime node on the share sides of the periods of its table at the two full leans adds $2(|\mathcal{H}_{j,v}|-n_{j,v})$ for each, and series and parallel nodes add nothing. The dimension does not depend on the numbers $|\mathcal{B}_j|$ of bases of the tables, against $\sum_j 2|\mathcal{B}_j|-2N-n+1$ for the atom formula (Corollary 18.16) and $\sum_j|\mathcal{B}_j|-n+1$ after its even shares close (Theorem 20.11(iv)).

(ii) The reusable data are one function of n_v variables per orbit of nodes, the same for every node of the orbit, in every table, and for every query; a value of the period of a node at a full lean at given loads, once computed, serves every node of the orbit and every query at which those loads occur.

(iii) In the recursive Mellin evaluation the work is one Mellin transform on an imaginary translate for each orbit of subtree cell families (as in Proposition 23.18(i)), each point costing one fibre integral of dimension $(2N-1)(|\mathcal{E}_v|+|G_v|-2)$ at its top node, and at the root one of dimension $(2N-1)(|\mathcal{E}_o|+|G_o|-1)$, whose integrand needs the periods of the tables of the node at their full leans, at the loads of the fibre: closed forms at series and parallel nodes, and at a prime node an integral of its own, of dimension $|\mathcal{H}_{j,v}|-n_v$ on its share side, unless another representation of Definition 26.18 is cheaper.

In the recursive Mellin evaluation equal subtrees at a series node enter as a power of one transform (Theorems 23.14(i) and 23.9(ii)), whose growth in the number of copies Theorem 24.2 reads when the subtree is series-balanced.

26.8 Changing the query in hold form

A query about a period is a choice of exponents, of masses, of weights and of the kind of answer asked for. In the hold form each of them enters at one place: exponents and masses move the pins and the affine marker loads, twists and product weights multiply the integrand by an exponential of the loads, and a replaced node, or a general change of the weights of one node, replaces that node's two functions and nothing else. Product weights are read as twists, and the exponents at fixed masses as the Mellin variables of the relative scales of the letters.

Proposition 26.20 (Product weights are twists)**.** *For a family F on $\mathcal{E}$ and $\boldsymbol{\zeta}=(\boldsymbol{\zeta}_j)_j$, $\boldsymbol{\zeta}_j\in(0,\infty)^{\mathcal{E}}$, let $F^{\boldsymbol{\zeta}}$ be F with the weights $\theta_{j,T}\boldsymbol{\zeta}_j^T$ in table j. Then $K_{F^{\boldsymbol{\zeta}},j}(\alpha)=K_j(\boldsymbol{\zeta}_j\circ\alpha)$ and $\Psi_{F^{\boldsymbol{\zeta}},j}(\alpha)=\boldsymbol{\zeta}_j^{\mathcal{E}}\Psi_j(\alpha/\boldsymbol{\zeta}_j)$: $F^{\boldsymbol{\zeta}}$ has the letters of the twist $F[\mathbf{x}]$ with $\mathbf{x}_j^K=\boldsymbol{\zeta}_j$ and $\mathbf{x}_j^\Psi=\boldsymbol{\zeta}_j^{-1}$ (Definition 22.1), its cobasis letters multiplied by the constants $\boldsymbol{\zeta}_j^{\mathcal{E}}$, and*

$$P_{F^{\boldsymbol{\zeta}}}(\mathbf{m};\mathbf{s},\mathbf{t})=\prod_j\boldsymbol{\zeta}_j^{-t_j\mathbf{1}}\,P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}),$$

$$P_{\mathcal{B}_j^{\boldsymbol{\zeta}_j}}(\mathbf{a};s_j,0)=\boldsymbol{\zeta}_j^{-\mathbf{a}}P_{\mathcal{B}_j}(\mathbf{a};s_j,0),\qquad P_{\mathcal{B}_j^{\boldsymbol{\zeta}_j}}(\mathbf{b};0,t_j)=\boldsymbol{\zeta}_j^{\mathbf{b}-t_j\mathbf{1}}P_{\mathcal{B}_j}(\mathbf{b};0,t_j),$$

so that the lean convolution of $F^{\boldsymbol\zeta}$ is that of F with the exponential $\prod_j \boldsymbol\zeta_j^{-t_j\mathbf 1}\boldsymbol\zeta_j^{\mathbf b_j-\mathbf a_j}$ of the loads. With one $\boldsymbol\zeta$ for every table, $P_{F^{\boldsymbol\zeta}}(\mathbf m;\mathbf s,\mathbf t)=\boldsymbol\zeta^{-\mathbf m-|\mathbf t|\mathbf 1}P_{F[\mathbf x']}(\mathbf m;\mathbf s,\mathbf t)$, $\mathbf x'$ the twist of every cobasis letter by $\boldsymbol\zeta^{-2}$ and $|\mathbf t|=\sum_j t_j$. Weights $\zeta_{j,g}$ on the bases through one mark g, in each table j, are the twist at g with $x_j^K=\zeta_{j,g}$ and $x_j^\Psi=\zeta_{j,g}^{-1}$, up to the factor $\prod_j\zeta_{j,g}^{-t_j}$: the weight of a hinge of Theorem 24.5, at any mark.

Proposition 26.21 (Carrying a calculation through a change, in hold form)**.** *In Theorem 26.13, with its node data computed:*

(i) a change of the exponents (s_j,t_j) changes the pins $|\mathbf a_j|=r_js_j$, $|\mathbf b_j|=r_j^*t_j$ and the marker loads of table j, which are affine in (s_j,t_j), and nothing else; the native loads of the new data are a new contour;

(ii) a change of the masses changes the load $\mathbf a_1=\mathbf m-\sum_{j\ge2}\mathbf a_j-\sum_j\mathbf b_j$ in the coordinates of Theorem 25.12(ii), and through it the marker loads $k_{1;v,\mathfrak e}$ of table 1, and in the integrand nothing else; in (i) and (ii) the contour must move when the old one is no longer admissible;

(iii) a change of twists, or of product weights on the bases of the tables, multiplies the integrand by an exponential of the loads (Proposition 26.20); product weights on the bases of one node multiply its periods at the two full leans by exponentials of its node loads;

(iv) replacing a node of a table by another table on the same marks and markers, of the same rank, one molecule with its markers neither loops nor coloops, or changing the weights of a node in any way, replaces the periods of that node at its two full leans and nothing else;

(v) in the recursive Mellin evaluation (Definition 26.18(5)), a change at one node changes the Mellin transforms of the laws of the subtree cell families on the path from it to the root only, each recomputed by Theorem 26.16(ii) from the new one of the vertex directly below it on the path and the kept ones of the other vertices directly below it, as Proposition 24.6 says of laws.

The reusable data of Chapter 23, the laws of the subtrees, depend on the exponents and on the masses below each edge (Theorem 23.17(iii)); the reusable data of the network do not, and a sweep over exponents, over masses, over twists or over product weights recomputes no node. What does change is the contour, and Corollary 26.6 decides at the nodes whether the new query has an answer.

Proposition 26.22 (The exponents are moment indices)**.** *Let $F=((\mathcal B_j,\theta_j);s_j,t_j)_{j=1}^N$ be a family on $\mathcal E$, fix real masses $\mathbf m$, and let I be the set of exponents $(\mathbf s,\mathbf t)\in(0,\infty)^{2N}$ with $\sum_j(r_js_j+r_j^*t_j)=m_{\mathcal E}$ at which every proper wall of $(\mathbf m;\mathbf s,\mathbf t)$ is positive: a relatively open convex subset of that hyperplane, possibly empty. In this statement write $L_1,\dots,L_{2N}$ for the letters K_j and Ψ_j of F, w_i for their exponents and g_i for their degrees, and for each i and $\boldsymbol\lambda\in(0,\infty)^{2N-1}$ put*

$$\mathsf G_i(\mathbf m;\boldsymbol\lambda)=\int_{\mathbb R^{\mathcal E}_{>0}}x^{\mathbf m-\mathbf 1}\exp\Big(-L_i(x)-\sum_{i'\ne i}\lambda_{i'}L_{i'}(x)\Big)\,dx,$$

the Gramma (Definition 2.1) of the concatenated configuration of all the letters, with the coefficients of L_i and $\lambda_{i'}$ times those of $L_{i'}$. For $(\mathbf s,\mathbf t)\in I$ and every i,

$$\Gamma(w_i)\,P_F(\mathbf m;\mathbf s,\mathbf t)=g_i\int_{\mathbb R^{\mathcal E}_{>0}}x^{\mathbf m-\mathbf 1}e^{-L_i(x)}\prod_{i'\ne i}L_{i'}(x)^{-w_{i'}}\,dx$$

$$= \frac{g_i}{\prod_{i' \neq i} \Gamma(w_{i'})} \int_{(0,\infty)^{2N-1}} \prod_{i' \neq i} \lambda_{i'}^{w_{i'}-1} \, \mathsf{G}_i(\mathbf{m}; \boldsymbol{\lambda}) \, d\boldsymbol{\lambda}.$$

At fixed masses the numbers $\Gamma(w_i) P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})/g_i$ *on* I*, functions of the exponents of the other letters, are the integrals of* $\prod_{i' \neq i} L_{i'}^{-w_{i'}}$ *against the one positive measure* $x^{\mathbf{m}-\mathbf{1}} e^{-L_i(x)} dx$*, and* $\prod_{i'} \Gamma(w_{i'}) \, P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})/g_i$ *is the Mellin transform at* $(w_{i'})_{i' \neq i}$ *of the one function* $\mathsf{G}_i(\mathbf{m}; \cdot)$ *of the relative scales of the other letters to* L_i*, completely monotone in each of them. On* I *the period* $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ *is log-convex, by Proposition 8.12(iii) and Hölder's inequality.*

So the period of a family at fixed masses is one integral over the orthant with any one of its letters closed: the closed letter leaves one positive measure on the orthant, the integrand of its Gramma at the index $\mathbf{m}$, and the period is a joint moment of the other letters under it, their exponents with the sign changed being its orders.

Chapter 27

Finite interfaces and certified enclosures

A cell enters every construction through its hinge wall and its law (Theorem 23.11), and a cell with several hinges through the joint law of its channels (Theorem 21.16, Definition 21.13). A law is not a finite list of numbers. Finitely many numbers are retained for every prescribed polynomial moment, the joint moments of the channels under independent Gamma draws on the marks, up to a degree D: they compose by the bilinear map of Theorem 21.14, a repeated chain cell acts on them by one finite matrix, so that every polynomial Gamma moment of a repetition satisfies a linear recurrence with constant coefficients, and the walls turn a moment of finite degree into a certified enclosure of the period. The period is retained as well over a neighbourhood of the query: the walls give the neighbourhood explicitly, the Taylor coefficients in the weights are periods with walls at least as high, and a finite parameter bank with two explicit errors answers every query inside it. Exact representation size, arithmetic work and accuracy are separate quantities.

27.1 The joint law of the channels and its moments

Let $\mathcal{A}$ be a cell with hinges G on its marks B, or cells with the hinges G on the same marks B kept together (Definition 27.1), with channel vector $\mathrm{ch}(\mathcal{A})$ (Definition 21.13), with positive masses $\mathbf{m}$ on B; the hinges carry no mass. Its total mass m_B and the product $\prod_{e\in B}\Gamma(m_e)$ are kept as data: at a splicing the total masses add and these products multiply. For a multi-index $\mathbf{k}$ on the channels, $\mathrm{ch}(\mathcal{A})^{\mathbf{k}}$ is the product of the channels with these multiplicities, a polynomial of the degree $\deg\mathbf{k}=\sum_c k_c\deg c$, the sum over the channels c.

Definition 27.1 (The channel law; moment vectors)**.** Cells $\mathcal{A}_1,\dots,\mathcal{A}_N$ with the hinges G on the same marks B are *kept together* when their channels are read jointly: $\mathcal{A}=(\mathcal{A}_1,\dots,\mathcal{A}_N)$ has the channel vector $\mathrm{ch}(\mathcal{A})=(\mathrm{ch}(\mathcal{A}_j))_j$, listing the channels of all of them by the tags (j,letter,p), $N2^{|G|+1}$ of them, absent channels counting as zero; one cell is the case $N=1$, and the tables of a family, with $G=\varnothing$, are kept together with their letters as channels. The *channel law* of $\mathcal{A}$ is the image $\mathrm{ch}(\mathcal{A})_*\,\mathrm{Dir}_{\mathbf{m}}$ of the Dirichlet law of the shape (Definition 3.10) under the channel vector. The *Gamma draws* on the marks are $\alpha\sim\mathrm{Gamma}_{\mathbf{m}}$, independent Gamma_{m_e} variables (Proposition 8.12), and the *moments* and the *moment vector of degree* D of $\mathcal{A}$ are

$$\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})=\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}\big[\,\mathrm{ch}(\mathcal{A})(\alpha)^{\mathbf{k}}\big],\qquad \mathrm{Mom}_{\mathcal{A}}^{\le D}=\big(\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})\big)_{|\mathbf{k}|\le D},$$

at most $\binom{q+D}{D}$ numbers, q the number of channels. A *polynomial Gamma moment* of degree at

most D of $\mathcal{A}$ is $\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}\big[p(\mathrm{ch}(\mathcal{A})(\alpha))\big]$ for a polynomial p of degree at most D in the channels, and a *shape moment* is the same with $U\sim\mathrm{Dir}_{\mathbf{m}}$ in place of α. A cell without marks has constant channels, a point as its law and the products of its channels as its moments, with total mass 0 and the empty product of Gammas.

The channels must be kept jointly: the moments of the channels one at a time do not determine a product of different channels exposed to the same room.

Proposition 27.2 (The channel law composes)**.** *Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be cells, or cells kept together, with disjoint marks B_1, B_2 and masses $\mathbf{m}_1$, $\mathbf{m}_2$, spliced table by table along pairs of their hinges to $\mathcal{A}_{12}$ (Theorem 21.14), both with marks. Let $U_1\sim\mathrm{Dir}_{\mathbf{m}_1}$, $U_2\sim\mathrm{Dir}_{\mathbf{m}_2}$ and $(V_1,V_2)\sim\mathrm{Dir}_{(m_{B_1},m_{B_2})}$ be independent; replace every channel c of $\mathcal{A}_1$ by $V_1^{\deg c}c(U_1)$ and every channel c of $\mathcal{A}_2$ by $V_2^{\deg c}c(U_2)$. The law of the image of the two channel vectors so replaced under the bilinear map of Theorem 21.14(iii), applied table by table, is the channel law of $\mathcal{A}_{12}$. For the tables of a family, kept together without hinges, the channels are their letters, and the period of the family is Proposition 8.12(i) read on the channel law: one Trailmoment of the channel law on its source space (Definition 5.1, Theorem 5.21). For a cell family $F=((\mathcal{A}_j,\theta_j);s_j,t_j)_{j=1}^N$ with the one hinge g (Definition 23.1), its cells kept together, with the coordinates $(x_{j,0},x_{j,1},x^*_{j,0},x^*_{j,1})$ of the channels $K_{\mathcal{A}_j\setminus g}$, $K_{\mathcal{A}_j/g}$, $\Psi_{\mathcal{A}_j\setminus g}$, $\Psi_{\mathcal{A}_j/g}$, the law of F is the weighted image*

$$(\boldsymbol{\varrho}_F)_*\phi_F=\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}\ (R_F)_*\Big[\prod_j x_{j,1}^{-s_j}x_{j,0}^{*\,-t_j}\Big(\frac{x_{1,0}}{x_{1,1}}\Big)^{-\kappa_F}\ \mathrm{ch}(\mathcal{A})_*\,\mathrm{Dir}_{\mathbf{m}}\Big],$$

with $R_F=\big((x_{j,0}x_{1,1}/(x_{j,1}x_{1,0}))_{j\ge2},(x^*_{j,1}x_{1,1}/(x^*_{j,0}x_{1,0}))_j\big)$, an equality of positive measures, possibly infinite: the channel law, which knows nothing of the exponents, gives the law of the cell family at all exponents by a weight and an image.

Theorem 27.3 (Gamma draws close the moments)**.** *(i)* Leaves. *If $\mathrm{ch}(\mathcal{A})(x)^{\mathbf{k}}=\sum_{\mathbf{h}}c_{\mathbf{h}}x^{\mathbf{h}}$, then $\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})=\sum_{\mathbf{h}}c_{\mathbf{h}}\prod_e(m_e)_{h_e}$, rational at rational masses and weights.*

(ii) Assembly. *Let $\mathrm{ch}(\mathcal{A}_{12})=\mathrm{sp}\big(\mathrm{ch}(\mathcal{A}_1),\mathrm{ch}(\mathcal{A}_2)\big)$ with the bilinear map* sp *of Theorem 21.14(iii), applied table by table to cells kept together, and write $\mathrm{sp}(c,c')^{\mathbf{k}}=\sum_{\mathbf{k}_1,\mathbf{k}_2}b_{\mathbf{k};\mathbf{k}_1,\mathbf{k}_2}c^{\mathbf{k}_1}c'^{\mathbf{k}_2}$, a sum over $|\mathbf{k}_1|=|\mathbf{k}_2|=|\mathbf{k}|$. Then*

$$\mathrm{Mom}_{\mathcal{A}_{12}}(\mathbf{k})=\sum_{\mathbf{k}_1,\mathbf{k}_2}b_{\mathbf{k};\mathbf{k}_1,\mathbf{k}_2}\,\mathrm{Mom}_{\mathcal{A}_1}(\mathbf{k}_1)\,\mathrm{Mom}_{\mathcal{A}_2}(\mathbf{k}_2),$$

and no moment of degree above $|\mathbf{k}|$ of either cell is needed; a splicing inside one cell is linear and keeps the degree.

(iii) Return to the shape. *For a cell with marks, $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\big[\mathrm{ch}(\mathcal{A})(U)^{\mathbf{k}}\big]=\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})/(m_B)_{\deg\mathbf{k}}$: the joint moments of the channel law through degree D are read off $\mathrm{Mom}_{\mathcal{A}}^{\le D}$, the rank, the sizes and the tags fixing the denominators.*

(iv) Sufficiency. *Two cells with the same hinges, channel tags and degrees and the same $\mathrm{Mom}^{\le D}$ give every final polynomial Gamma moment of degree at most D in the channels, in any assembly with cells on disjoint marks, the same value, even where an intermediate cell exposes more hinges than the replaced one; with the same total mass as well, they give every final shape moment of degree at most D of the channels of the root, under its Dirichlet law, the same value.*

(v) The whole law. *For a cell with marks, the* $\mathrm{Mom}_{\mathcal{A}}^{\le D}$ *for all* D*, with* m_B *and the degrees, determine its channel law, and with it the expectation of every integrable function of the channels (Proposition 27.2).*

The draws are used for nonnegative moments of the channels. A negative moment of a period integrand under the draws diverges on the balance wall: the scale contributes $\int_0^\infty S^{-1}e^{-S}dS$, the pole of $\Gamma(\kappa_{\mathcal{E}})$ in Proposition 8.12. The return to the shape, followed by Proposition 27.2 or by the peak closing (Theorem 15.13(i)) with its stopping rule (Theorem 27.7), is the change of representation that the period needs.

Corollary 27.4 (The work follows the distinct interfaces)**.** *Fix a degree* D*, an assembly that exposes at most* h *hinges at every cell it keeps, and identifications of its repeated subassemblies keeping their data and hinges. Every requested moment of the root is computed exactly by keeping at most* $\binom{2^{h+1}+D}{D}$ *numbers for each distinct cell,* $\binom{N2^{h+1}+D}{D}$ *for* N *tables kept together. The arithmetic is* $O\big(\sum_v E_v(D)\big)$ *operations once the local expansions and the leaf moments are built,* $E_v(D)$ *the number of nonzero coefficients* $b_{\mathbf{k};\mathbf{k}_1,\mathbf{k}_2}$ *at a distinct splicing* v *of two cells, at most the product of the sizes of its three moment vectors, or of two for a splicing inside one cell. Equal copies share one moment vector and remain independent draws: their moments are multiplied as in Theorem 27.3(ii), never identified.*

This counts operations. Building the local expansions, finding an assembly of small width, the bit lengths of exact coefficients and the degree a moment needs are separate parts of the cost; finding the cheapest assembly is a search, as for constructions (Section 33.1).

27.2 Repetition is a finite matrix

Theorem 27.5 (A finite matrix for every polynomial Gamma moment)**.** *Let a chain cell* $\mathcal{U}$ *(Proposition 24.4, here with several hinges) be a cell with two disjoint sets of hinges* G_{in} *and* G_{out}*, each in a fixed bijection with the hinges of a cell* $\mathcal{A}_0$*, and let* $\mathcal{A}_{l+1}$ *be* $\mathcal{U}$*, with fresh marks and the same masses and weights at every use, spliced onto* $\mathcal{A}_l$ *by pairing* G_{in} *with the hinges of* $\mathcal{A}_l$*; so* $\mathcal{A}_{l+1}$ *exposes* G_{out}*, named through the bijection as the hinges of* $\mathcal{A}_l$*, with the same channel tags, and its channel vector is the image of* $\mathrm{ch}(\mathcal{A}_l)$ *and that of the copy of* $\mathcal{U}$ *under the map* sp*. Keep one list of channel tags for every* l*, absent channels being zero. For every* D *there is a matrix* $\mathbf{T}_D$*, independent of* l *and computed from* $\mathcal{U}$ *by finite expansion and Theorem 27.3(i), with*

$$\mathrm{Mom}_{\mathcal{A}_l}^{\le D} = \mathbf{T}_D^{\,l}\ \mathrm{Mom}_{\mathcal{A}_0}^{\le D}\,.$$

Every fixed polynomial Gamma moment of degree at most D *of* $\mathcal{A}_l$*, a fixed terminal cell spliced on it included, is* $\mathbf{e}^{\mathsf{T}}\mathbf{T}_D^{\,l}\,\mathrm{Mom}_{\mathcal{A}_0}^{\le D}$ *for a fixed row* $\mathbf{e}^{\mathsf{T}}$*; so its generating series in* l *is* $\mathbf{e}^{\mathsf{T}}(I-\zeta\mathbf{T}_D)^{-1}\,\mathrm{Mom}_{\mathcal{A}_0}^{\le D}$*, rational in* ζ*, the sequence satisfies the linear recurrence with constant coefficients given by the characteristic polynomial of* $\mathbf{T}_D$*, of order at most* $\binom{q+D}{D}$*, and one length* l *costs* $O\big(\binom{q+D}{D}^3\log(l{+}1)\big)$ *operations by repeated squaring.* $\mathbf{T}_D$ *is block diagonal by the degree, and a moment of one degree uses its block alone; a repeated sequence of different chain cells is the power of the product of their matrices.*

On a cell with one hinge g, a chain cell $\mathcal{U}$ with the hinges i and o, i spliced to g, and the channels $K_{\mathcal{U},(P)}$, $\Psi_{\mathcal{U},(P)}$, $P\subseteq\{i,o\}$, acts by

$$\begin{pmatrix} K_{\mathcal{A}_{l+1},(\varnothing)} \\ K_{\mathcal{A}_{l+1},(\{o\})} \end{pmatrix} = \begin{pmatrix} K_{\mathcal{U},(\{i\})} & K_{\mathcal{U},(\varnothing)} \\ K_{\mathcal{U},(\{i,o\})} & K_{\mathcal{U},(\{o\})} \end{pmatrix} \begin{pmatrix} K_{\mathcal{A}_l,(\varnothing)} \\ K_{\mathcal{A}_l,(\{g\})} \end{pmatrix},$$

and by the same matrix of the $\Psi_{\mathcal{U},(P)}$ on the Ψ channels, which is Theorem 21.14(iii) with one pair. A moment of degree a in the K channels and b in the Ψ channels lives on $(a+1)(b+1)$ monomials, and the block of $\mathbf{T}_D$ on them is the joint expectation over the draws of $\mathcal{U}$ of the expanded actions: the two matrices share the marks of $\mathcal{U}$ and are not independent. For a family at integer exponents, the joint moments of its letters along a chain of bunches with matrix letters are the case of Theorem 22.10. An iterate of an integral operator gives no finite closing relation by itself (Theorem 22.8, Proposition 24.4); every polynomial Gamma moment of a repetition does. The period is recovered from a sequence of such moments with certified tails, in Section 27.3.

Corollary 27.6 (Shape moments have recurrences)**.** *For the repetition of Theorem 27.5 let the total masses be $m_{\mathcal{A}_l} = m_{\mathcal{A}_0} + l\, m_{\mathcal{U}}$, $m_{\mathcal{U}}$ that of $\mathcal{U}$, let p be a polynomial in the channels, the same for every l, homogeneous of degree $d_l = d_0 + lv$ in the marks, v an integer, and let M_l be the polynomial Gamma moment of $\mathcal{A}_l$ for p. The shape moment for p is $M_l/(m_{\mathcal{A}_l})_{d_l}$, and every recurrence $\sum_i c_i M_{l+i} = 0$ gives $\sum_i c_i \frac{(m_{\mathcal{A}_{l+i}})_{d_{l+i}}}{(m_{\mathcal{A}_l})_{d_l}} \frac{M_{l+i}}{(m_{\mathcal{A}_{l+i}})_{d_{l+i}}} = 0$. If the masses are rational and $h \geq 1$ is an integer with $h m_{\mathcal{U}} \in \mathbb{Z}$, then on each residue of l modulo h, with $\mathbf{T}_D^h$ for $\mathbf{T}_D$, the coefficients are rational functions of l over the field of the weights, polynomials after clearing denominators, in $\mathbb{Q}[l]$ at rational weights.*

27.3 A certified enclosure of the period

Theorem 27.7 (Explicit stopping from the walls)**.** *Let a family F on $n \geq 2$ marks have real masses on its balance wall and exponents $s_j, t_j \geq 0$, with every proper wall positive. Write its active letters, the K_j with $s_j > 0$ and the Ψ_j with $t_j > 0$, as L_i with exponents $w_i > 0$, degrees g_i and counts $d_i(S)$, as in Theorem 15.13, so that $m_{\mathcal{E}} = \sum_i g_i w_i$ and $\kappa_S = m_S - \sum_i w_i d_i(S)$. Choose references $M_i \geq \max_{\Delta_{\mathcal{E}}} L_i$, and in this statement put $Y_i = L_i/M_i \in (0,1]$ on the open simplex. For an order π of the marks let $U_{\pi,k}$ be its tails, $L_{i,\pi}$ the sector letter of L_i, the $k_{j,\pi}$ or $\psi_{j,\pi}$ of Theorem 8.5(ii), and $\underline{L}_{i,\pi} > 0$ its constant term. For integers $Q_i \geq 0$ put*

$$P^{\leq \mathbf{Q}} = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \prod_i M_i^{-w_i} \sum_{0 \leq k_i \leq Q_i} \prod_i \frac{(w_i)_{k_i}}{k_i!}\ \mathbb{E}_{U \sim \mathrm{Dir}_{\mathbf{m}}} \Big[\prod_i \big(1 - Y_i(U)\big)^{k_i} \Big],$$

the partial sums of the peak closing (Theorem 15.13(i)), computed exactly from the moments of the letters through the degree $\sum_i Q_i$, each monomial with its own rising factorial (Theorem 27.3(iii)). For every choice of $\epsilon_i > 0$ with

$$\kappa_S - \epsilon_i d_i(S) > 0 \qquad (\varnothing \neq S \subsetneq \mathcal{E}),$$

which is exactly $\epsilon_i < \varpi_i$, ϖ_i the least of the ratios $\kappa_S/d_i(S)$ in Theorem 15.13(ii) (any $\epsilon_i > 0$ when no proper S has $d_i(S) > 0$), put

$$C_i = M_i^{\epsilon_i}\, n^{g_i \epsilon_i} \sum_{\pi} \underline{L}_{i,\pi}^{-\epsilon_i} \prod_{i'} \underline{L}_{i',\pi}^{-w_{i'}} \prod_{k=2}^{n} \frac{1}{\kappa_{U_{\pi,k}} - \epsilon_i d_i(U_{\pi,k})}.$$

Then

$$0 \leq P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) - P^{\leq \mathbf{Q}} \leq \sum_i C_i\, \frac{\Gamma(w_i + \epsilon_i)\, \Gamma(w_i + Q_i + 1)}{\Gamma(w_i)\, \Gamma(w_i + Q_i + 1 + \epsilon_i)},$$

and for $\epsilon_i \leq 1$ the last factor is at most $(w_i/(w_i + Q_i + 1))^{\epsilon_i}$. So finitely many moments and finitely many walls enclose the period to any prescribed accuracy; the same moments answer every other

choice of exponents w_i with $\sum_i g_i w_i = m_{\mathcal{E}}$ at the same masses, only the coefficients of $P^{\leq \mathbf{Q}}$, the walls and the certificate changing.

At rational data the constants of the certificate are positive rationals and their rational powers, and no period other than the one sought enters it; the stacked Gamma $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$ and the Gamma ratios stay exact products and quotients of values of the Gamma function, so the enclosure is exact without being rational. The certificate can itself be expensive, C_i being a sum over all $n!$ orders, and it is crude: it replaces every sector letter by its least value. The true tail is of the order $Q^{-\varpi}(\log Q)^{k-1}$ (Theorem 15.13(ii)); the certified decay $Q_i^{-\epsilon_i}$ tends to it as $\epsilon_i \to \varpi_i$, at the cost of the factor $\prod_k (\kappa_{U_{\pi,k}} - \epsilon_i d_i(U_{\pi,k}))^{-1}$, which grows without bound there. Sharper references and sharper sector bounds lower the degree needed without changing the moments. The degree needed can grow with the size of the assembly, so this is a terminating certified procedure, not a uniform fast approximation.

27.4 Changing the query with finite data

Proposition 27.8 (Parameters and insertions)**.** *Fix the local supports and the assembly.*

(i) For each D the leaf moments and the splicing and transfer coefficients are polynomials in the local weights and in the masses; they can be kept symbolic and specialized at any positive values.

(ii) Carrying the leaf moments, Theorem 27.3(ii) and Theorem 27.5 in the truncated ring $\mathbb{Q}(\boldsymbol{\theta}, \mathbf{m})[\boldsymbol{\zeta}]/(\boldsymbol{\zeta})^{h+1}$ of p local parameters $\boldsymbol{\zeta}$, over the field generated by the weights $\boldsymbol{\theta}$ and the masses, gives every derivative of the moments through the order h, at most $\binom{p+h}{h}$ coefficients per entry.

(iii) A further polynomial observable, a mark total for instance, can be made a channel when every local update is separately affine in each independent cell vector, a constant channel adjoined.

(iv) Raising the masses by nonnegative integers $\boldsymbol{\gamma}$ is a polynomial insertion: for every observable g,

$$\mathbb{E}_{\alpha \sim \mathrm{Gamma}_{\mathbf{m}+\boldsymbol{\gamma}}}[g(\alpha)] = \mathbb{E}_{\alpha \sim \mathrm{Gamma}_{\mathbf{m}}}\left[\alpha^{\boldsymbol{\gamma}} g(\alpha)\right] \Big/ \prod_e (m_e)_{\gamma_e}.$$

The moments do not depend on the exponents asked for: changing the exponents reuses every moment, and changing a local weight or mass changes the local data and their ancestors in the assembly only, as Proposition 24.6 says of laws. Shared parameters may appear in many copies, whose draws stay independent.

27.5 Certified parameter banks

A family F on its balance wall, its letters K_j and Ψ_j written L_i, $i = 1, \ldots, \ell$, with the exponents w_i (the s_j and t_j), the degrees g_i and the counts $d_i(S) = \min\{\gamma_S : [u^\gamma] L_i > 0\}$, so that $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_i L_i(u)^{-w_i} d\omega_{\mathcal{E}}(u)$, with $m_{\mathcal{E}} = \sum_i g_i w_i$ and the walls $\kappa_S = m_S - \sum_i w_i d_i(S) > 0$ at the starting data, is asked a neighbourhood of queries: p complex parameters $\boldsymbol{\zeta}$, and a bank of Taylor coefficients through the total order h, at most $\binom{h+p}{p}$ numbers. The order of a bank is the number of differentiations, not the degree of a polynomial moment of Section 27.1.

Theorem 27.9 (Taylor coefficients in the weights are periods with walls at least as high)**.** *Let the exponents w_i be real and change the coefficients of the letters, $L_i(\boldsymbol{\zeta}) = L_i + \sum_{j=1}^{p} \zeta_j D_{ij}$, each D_{ij} homogeneous of degree g_i with support in that of L_i and coefficients of either sign, and in this statement write $P(\boldsymbol{\zeta}) = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_i L_i(\boldsymbol{\zeta})(u)^{-w_i}\, d\omega_{\mathcal{E}}(u)$, so that $P(\mathbf{0}) = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$. Then, with $l_i = \sum_j l_{ij}$,*

$$[\boldsymbol{\zeta}^{\mathbf{k}}]P = \sum_{\substack{l_{ij}\geq 0 \\ \sum_i l_{ij} = k_j}} \prod_i \frac{(-1)^{l_i}(w_i)_{l_i}}{\prod_j l_{ij}!} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_i \Big(L_i^{-w_i - l_i} \prod_j D_{ij}^{l_{ij}} \Big)\, d\omega_{\mathcal{E}},$$

and after the numerators are expanded every term is a convergent period of the family at masses $\mathbf{m} + \boldsymbol{\gamma}$ and exponents $w_i + l_i$, balanced, with every proper wall $\kappa'_S = \kappa_S + \gamma_S - \sum_i l_i d_i(S) \geq \kappa_S$.

Theorem 27.10 (A uniform certificate for weights)**.** *In the setting of Theorem 27.9, with $P(\boldsymbol{\zeta})$ as there, write $L_i = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} u^{\boldsymbol{\gamma}}$ with $c_{i,\boldsymbol{\gamma}} > 0$ and $D_{ij} = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} v_{i,\boldsymbol{\gamma},j} u^{\boldsymbol{\gamma}}$, $V_{ij} = \max_{\boldsymbol{\gamma}} |v_{i,\boldsymbol{\gamma},j}|$, and choose radii $R_j > 0$ with $r_i = \sum_j R_j V_{ij} < 1$. On $|\zeta_j| \leq R_j$, $|P(\boldsymbol{\zeta})| \leq P(\mathbf{0}) \prod_i \max\big((1-r_i)^{-w_i}, (1+r_i)^{-w_i}\big) =: M$, made explicit by $P(\mathbf{0}) \leq \sum_\pi \prod_i \max\big(L_{i,\pi}(\mathbf{0})^{-w_i}, L_{i,\pi}(\mathbf{1})^{-w_i}\big) \prod_{k=2}^{n} \kappa_{U_{\pi,k}}^{-1}$, with the sector letters $L_{i,\pi}$ of the letters on the sector of the order π, the $k_{j,\pi}$ or $\psi_{j,\pi}$ of Theorem 8.5(ii), or by the upper end of the enclosure of Proposition 15.12; P is holomorphic near the closed polydisc, $|[\boldsymbol{\zeta}^{\mathbf{k}}]P| \leq M\mathbf{R}^{-\mathbf{k}}$, and its Taylor polynomial $P^{[h]}$ through the order h has, on $|\zeta_j| \leq \vartheta R_j$ with $0 \leq \vartheta < 1$,*

$$|P(\boldsymbol{\zeta}) - P^{[h]}(\boldsymbol{\zeta})| \leq M \sum_{k>h} \binom{k+p-1}{p-1} \vartheta^k \leq M\, \frac{\binom{h+p}{p-1}\vartheta^{h+1}}{1 - \vartheta(h+p+1)/(h+2)},$$

the second bound when $\vartheta(h+p+1) < h+2$.

Theorem 27.11 (Wall-certified parameter banks)**.** *Keep the letters and change the masses and exponents affinely, $m_e(\boldsymbol{\zeta}) = m_e^0 + \sum_j a_{ej}\zeta_j$ and $w_i(\boldsymbol{\zeta}) = w_i^0 + \sum_j b_{ij}\zeta_j$ with real slopes and balance in every direction, so that $\kappa_S(\boldsymbol{\zeta}) = \kappa_S^0 + \sum_j c_{Sj}\zeta_j$ with $c_{Sj} = \sum_{e\in S} a_{ej} - \sum_i b_{ij} d_i(S)$, and in this statement write $P(\boldsymbol{\zeta}) = P_F(\mathbf{m}(\boldsymbol{\zeta});\mathbf{s}(\boldsymbol{\zeta}),\mathbf{t}(\boldsymbol{\zeta}))$, the exponents $w_i(\boldsymbol{\zeta})$ on the letters. Choose $R_j > 0$ with*

$$\underline{\kappa}_S := \kappa_S^0 - \sum_j |c_{Sj}| R_j > 0 \qquad (\varnothing \neq S \subsetneq \mathcal{E}).$$

On the sector of an order π every letter is its monomial times its sector letter $L_{i,\pi}$, the $k_{j,\pi}$ or $\psi_{j,\pi}$ of Theorem 8.5(ii), with nonnegative coefficients in the sector coordinates; put $\underline{L}_{i,\pi} = L_{i,\pi}(\mathbf{0})$, $\overline{L}_{i,\pi} = L_{i,\pi}(\mathbf{1})$, $\eta_{i,\pi} = \max(|\log \underline{L}_{i,\pi}|, |\log \overline{L}_{i,\pi}|)$, $\nu_i = \sum_j |b_{ij}| R_j$,

$$B_\pi = \prod_i \max\big(\underline{L}_{i,\pi}^{-w_i^0}, \overline{L}_{i,\pi}^{-w_i^0}\big) e^{\eta_{i,\pi}\nu_i}, \qquad M_{\mathbf{R}} = \sum_\pi B_\pi \prod_{k=2}^{n} \underline{\kappa}_{U_{\pi,k}}^{-1}.$$

Then P is holomorphic near $|\zeta_j| \leq R_j$ with $|P| \leq M_{\mathbf{R}}$, every $|[\boldsymbol{\zeta}^{\mathbf{k}}]P| \leq M_{\mathbf{R}}\mathbf{R}^{-\mathbf{k}}$, the bound of Theorem 27.10 holds with $M_{\mathbf{R}}$ for M, and

$$[\boldsymbol{\zeta}^{\mathbf{k}}]P = \frac{1}{\mathbf{k}!} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}^0-\mathbf{1}} \prod_i L_i^{-w_i^0} \prod_j \Big(\sum_e a_{ej} \log u_e - \sum_i b_{ij} \log L_i \Big)^{k_j} d\omega_{\mathcal{E}},$$

absolutely integrable and of degree zero. The starting exponents may be any reals at which the proper walls are positive.

Theorem 27.12 (A finite compiler with two errors)**.** *In the setting of Theorem 27.11, with $P(\boldsymbol{\zeta})$ as there, put $y_{i,\pi} = 1 - L_{i,\pi}/\overline{L}_{i,\pi} \in [0, \lambda_\pi]$, $\lambda_\pi = \max_i(1 - \underline{L}_{i,\pi}/\overline{L}_{i,\pi}) < 1$, $\beta_i = |w_i^0| + \nu_i$ and $A_\pi = \prod_i \overline{L}_{i,\pi}^{-w_i^0} e^{|\log \overline{L}_{i,\pi}|\nu_i} \prod_{k=2}^n \underline{\kappa}_{U_{\pi,k}}^{-1}$. Expand $\prod_i L_{i,\pi}^{-w_i(\boldsymbol{\zeta})} = \prod_i \overline{L}_{i,\pi}^{-w_i(\boldsymbol{\zeta})} \sum_{\mathbf{l}} \prod_i \frac{(w_i(\boldsymbol{\zeta}))_{l_i}}{l_i!} y_{i,\pi}^{l_i}$, truncate at $|\mathbf{l}| \le Q$, and integrate the finitely many terms exactly, each $\prod_i y_{i,\pi}^{l_i} = \sum_{\boldsymbol{\gamma}} c_{\boldsymbol{\gamma}} q^{\boldsymbol{\gamma}}$ in the sector coordinates q contributing*

$$\prod_i \overline{L}_{i,\pi}^{-w_i(\boldsymbol{\zeta})} \prod_i \frac{(w_i(\boldsymbol{\zeta}))_{l_i}}{l_i!} \sum_{\boldsymbol{\gamma}} c_{\boldsymbol{\gamma}} \prod_{k=2}^n \frac{1}{\kappa_{U_{\pi,k}}(\boldsymbol{\zeta}) + \gamma_k},$$

to obtain $\widetilde{P}_Q(\boldsymbol{\zeta})$. Then on the whole polydisc

$$|P(\boldsymbol{\zeta}) - \widetilde{P}_Q(\boldsymbol{\zeta})| \le \mathrm{err}_Q := \sum_\pi A_\pi \sum_{k>Q} \frac{(|\boldsymbol{\beta}|)_k}{k!} \lambda_\pi^k \le \sum_\pi A_\pi \frac{(|\boldsymbol{\beta}|)_{Q+1} \lambda_\pi^{Q+1}}{(Q+1)!\,(1 - r_{\pi,Q})},$$

the second bound when $r_{\pi,Q} = \lambda_\pi \max\left(1, \frac{|\boldsymbol{\beta}|+Q+1}{Q+2}\right) < 1$ for every π; and the Taylor polynomial $\widetilde{P}_Q^{[h]}$ of $\widetilde{P}_Q$ through the order h, computed by finite arithmetic in a truncated ring, has on $|\zeta_j| \le \vartheta R_j$

$$|P(\boldsymbol{\zeta}) - \widetilde{P}_Q^{[h]}(\boldsymbol{\zeta})| \le \frac{\mathrm{err}_Q}{(1-\vartheta)^p} + M_{\mathbf{R}} \sum_{k>h} \binom{k+p-1}{p-1} \vartheta^k.$$

With no parameters, $p = 0$, so that $\nu_i = 0$ and $\underline{\kappa}_S = \kappa_S$, and with every $w_i^0 \ge 0$, the first bound is that of Proposition 15.12(ii): there $\overline{L}_{i,\pi} = L_i(\mathbf{1})$ for every π, $\widetilde{P}_Q = \prod_i L_i(\mathbf{1})^{-w_i^0} \sum_\pi \sum_{M \le Q} I_{\pi,M}$, and the explicit tail is $\prod_i L_i(\mathbf{1})^{-w_i^0} \sum_\pi \mathrm{err}_{\pi,Q}$, the letters being the K_j and Ψ_j with the exponents s_j and t_j.

For fixed data and $\vartheta < 1$ the orders Q and h grow like the logarithm of the inverse error, up to the polynomial factors of the scalar tails; expanding the sector polynomials still costs work that depends on the dimension and the supports. The walls also cover paths: along a segment on which every proper wall is positive, with κ_- the least proper wall at the two ends and a the largest change of a proper wall along it, every point of the segment, parametrized by $\lambda \in [0,1]$, has the disc of radius $\kappa_-/(2a)$ in the complex parameter on which every proper wall keeps a real part at least $\kappa_-/2$, the walls being affine, and a compact path has a finite cover by such discs. A bank at one point does not give the coefficients at another; what is reused everywhere is the compiled sector data, the coefficient procedure and the wall test.

27.6 Procedures

Definition 27.13 (The interface procedure)**.** *Given*: cells with their weights and masses; the pairs of hinges to splice and an assembly order with its identifications of repeated parts; a polynomial moment, a peak truncation or an accuracy, with, for an accuracy, references and a wall certificate. *Steps*: (1) read the channels of the leaves by coefficient extraction, with their degrees, the sizes, the total masses and the products $\prod_e \Gamma(m_e)$; (2) choose D, the degree of the moment or $\sum_i Q_i$ from Theorem 27.7; (3) compute the leaf moments by Theorem 27.3(i), only at the multi-indices the moment pulls back to; (4) compile every distinct splicing by Theorem 27.3(ii), and every repeated chain cell once into $\mathbf{T}_D$ and its power (Theorem 27.5), reusing a vector only when its tags and data match; (5) splice the pairs at the root, return to the shape by Theorem 27.3(iii),

and evaluate $P^{\leq \mathbf{Q}}$ with its certificate. *Output*: an exact polynomial moment, a recurrence or rational generating series for a repetition, or a certified enclosure of the period, with the vectors, the splicing coefficients, the parameter expressions and the certificate kept for the next query.

Definition 27.14 (The bank procedure)**.** *Given*: a family on its balance wall with positive walls, and affine directions of change of its masses and exponents, or of the weights of its letters. *Steps* along masses and exponents: compute the walls and a box of radii satisfying the strict inequalities of Theorem 27.11; form the sector constants and $M_{\mathbf{R}}$; choose Q and h until the bound of Theorem 27.12 is below the tolerance; compile the retained sector terms in the truncated ring. Along weights, where the sector letters move with $\boldsymbol{\zeta}$ and Theorem 27.12 does not apply: choose radii by Theorem 27.10, with its bound made explicit there; each coefficient through the order h is the finite signed sum of periods with walls at least as high of Theorem 27.9, each enclosed by Proposition 15.12, and the tail beyond h is bounded by Theorem 27.10. *Output*: the parameter bank, its domain and its certified error; every query inside the smaller box is a polynomial evaluation, and the exact coefficients remain available as the periods of Theorem 27.9 or the integrals of Theorem 27.11.

Chapter 28

Bunches of any width

A bunch of width one is one mark to its room (Theorem 19.4): its patterns fill two levels, each level is one child of the bunch against one table of the room, and what the room sees of the bunch is its wall and the law of its ratio vector, one ratio for one table (Lemma 23.2, Theorem 23.11). At every width a room sees its bunch through the wall, the widths and the joint law of the bunch letters of a level factorization (Theorem 21.16); that statement has a canonical form, and the calculus of width one carries to every width. At the width c the patterns fill $c+1$ levels, and a level between the extremes pairs several stations of patterns with several stations of room sides: which pairs meet, and with which constants, is the seam of the bunch (Definition 28.1), graded by the excess, which no description by marks or by level totals contains. The room sees the bunch through the letters of the tables it presents to it, one for each station of room sides, and the law of these letters is what the bunch retains: it suffices against every partner, and the tinkers of one partner already separate any two laws (Theorem 28.19). A bunch of width c is absorbed into its partner as one stand-in mark of degree c (Lemma 28.8), a mark of a table exactly at width one (Theorem 28.24), and a nonempty family of sets of one size is the family of bases of a table exactly when, at one bunch, its children and room children are tables, adjacent ones are quotients, the extreme ones are one, and its levels are lifts (Theorem 28.30). The circuits of the seam are the directions of the shares of the compatible pairs that neither side sees, and the period integrates over them through the Gramma of the compatible pairs (Proposition 28.14). Width one is the case of the hinge throughout: the seam of a bunch of width one is its hinge (Proposition 28.3(ii)), and its law is the law of its ratio vector read through one fixed change of scale (Proposition 28.11). Bunches compose along chains and trees (Chapter 29), the glue closes along the directions of its shares (Chapter 31), and a law has regions of equality and a reach at its ends (Chapter 32).

28.1 The seam of a bunch

Definition 28.1 (Seam)**.** Let a table $(\mathcal{B}, \theta)$ have the bunch B and the room R. A level k of B is named by its excess $i = k - d(B)$ (Definition 12.16), $0 \le i \le c(B)$: its patterns have $d(B)+i$ marks and its room sides $d(R)+c(B)-i$, the room having the width of the bunch (Definition 19.1). Fix an excess i.

(i) Two patterns of excess i lie in one station when their children (Definition 12.1) are positive multiples of one another, weights included, as at the bunches S_l of Theorem 22.9. The *room child* at a room side X is the child of the bunch R at its pattern X: the table on B with the bases $\mathcal{B}_{(X)} = \{p : \ p \sqcup X \in \mathcal{B}\}$ and the weights $\theta_{(X),p} = \theta_{p \sqcup X}$ (Lemma 12.2). Two room sides of excess i lie in one station when their room children are positive multiples of one another.

The stations of patterns of excess i form the set $\mathcal{K}_i$ and those of room sides the set $\mathcal{D}_i$: the stations of the bunch and of the room are $\mathcal{K} = \bigsqcup_i \mathcal{K}_i$ and $\mathcal{D} = \bigsqcup_i \mathcal{D}_i$, each station carrying its excess $i(C)$ or $i(D)$.

(ii) Within a station C the child at $p \in C$ is w_p times one weighted table on R, the *child of the station* C, with the bases $\mathcal{B}_{(C)}$ and the weights $\theta_{(C)}$, the *multipliers* $w_p > 0$ being normalized by $\max_{p\in C} w_p = 1$; likewise the room child at $X \in D$ is w'_X times the *room child of the station* D, a table on B with the bases $\mathcal{B}_{(D)}$ and the weights $\theta_{(D)}$, with $\max_{X\in D} w'_X = 1$. The *pair constant* of $C \in \mathcal{K}_i$ and $D \in \mathcal{D}_i$ is $h_{CD} = \theta_{p\sqcup X}/(w_p w'_X)$ for any $p \in C$ and $X \in D$ with $p \sqcup X \in \mathcal{B}$, which does not depend on the choice, and $h_{CD} = 0$ when there is none; (C, D) is *compatible* when $h_{CD} > 0$. The *seam matrix* of the excess i is $H_i = (h_{CD})_{C\in\mathcal{K}_i,\, D\in\mathcal{D}_i}$, and the *seam* of B is $(H_0, \dots, H_{c(B)})$ with its stations; a family has one seam in each table.

(iii) The *station letters* are $\mathsf{x}_C = \sum_{p\in C} w_p\, \alpha_B^p$ and $\mathsf{x}^*_C = \sum_{p\in C} w_p\, \alpha_B^{B\setminus p}$ on B, and $\mathsf{y}_D = \sum_{X\in D} w'_X\, \alpha_R^X$ and $\mathsf{y}^*_D = \sum_{X\in D} w'_X\, \alpha_R^{R\setminus X}$ on R. The *seam letters* of B are the letters $K_{(D)}$ and $\Psi_{(D)}$ of the room children of the stations $D \in \mathcal{D}$, polynomials on B; those of R are the letters $K_{(C)}$ and $\Psi_{(C)}$ of the children of the stations $C \in \mathcal{K}$, polynomials on R.

For a list every multiplier and every pair constant is 1. The normalization by the largest multiplier chooses nothing and commutes with every bijection of the marks; any other normalization multiplies x_C and y_D by positive constants and h_{CD} by their inverses, and leaves every product $h_{CD}\,\mathsf{x}_C\,\mathsf{y}_D$ unchanged. In table j of a family the stations, the letters and the constants carry the index j: $\mathcal{K}_j$, $\mathcal{D}_j = \bigsqcup_i \mathcal{D}_{j,i}$, x_{jC}, y_{jD}, $h_{j,CD}$, $K_{j,(D)}$ and so on.

Proposition 28.2 (The seam identity)**.** *In Definition 28.1:*

(i) The seam letters are the station letters through the seam:

$$K_{(D)} = \sum_C h_{CD}\,\mathsf{x}_C, \qquad \Psi_{(D)} = \sum_C h_{CD}\,\mathsf{x}^*_C,$$
$$K_{(C)} = \sum_D h_{CD}\,\mathsf{y}_D, \qquad \Psi_{(C)} = \sum_D h_{CD}\,\mathsf{y}^*_D :$$

the seam letters of a side are the letters of the tables that the other side presents to it.

(ii) If (C, D) is compatible, every $p \in C$ and $X \in D$ make a basis, with $\theta_{p\sqcup X} = h_{CD}\, w_p\, w'_X$, and the bases of excess i are the disjoint union of these rectangles $C \times D$ *over the compatible pairs.*

(iii) With x_i, x^*_i, y_i, y^*_i the vectors of the station letters of excess i,

$$K = \sum_{i=0}^{c(B)} \mathsf{x}_i^{\mathsf{T}} H_i\, \mathsf{y}_i = \sum_{D\in\mathcal{D}} K_{(D)}\,\mathsf{y}_D = \sum_{C\in\mathcal{K}} \mathsf{x}_C\, K_{(C)},$$
$$\Psi = \sum_{i=0}^{c(B)} \mathsf{x}_i^{*\mathsf{T}} H_i\, \mathsf{y}^*_i = \sum_{D\in\mathcal{D}} \Psi_{(D)}\,\mathsf{y}^*_D = \sum_{C\in\mathcal{K}} \mathsf{x}^*_C\, \Psi_{(C)},$$

and along the scale of B, $\alpha_B = \rho u$ with $u \in \Delta_B$ (Lemma 2.7),

$$K(\rho u, \alpha_R) = \rho^{d(B)} \sum_{i=0}^{c(B)} \rho^i \sum_{D\in\mathcal{D}_i} K_{(D)}(u)\,\mathsf{y}_D(\alpha_R),$$

$$\Psi(\rho u, \alpha_R) = \rho^{d^*(B)} \sum_{i=0}^{c(B)} \rho^{c(B)-i} \sum_{D \in \mathcal{D}_i} \Psi_{(D)}(u)\, \mathsf{y}^*_D(\alpha_R).$$

The writing by the stations of the room, with the bunch letters $K_{(D)}$ and the room letters y_D, is a level factorization with nonnegative terms (Definition 21.15), the room-station factorization*; the writing by the stations of the bunch is the children factorization.*

(iv) Within one excess no two stations of $\mathcal{K}_i$ have proportional rows of H_i, and no two stations of $\mathcal{D}_i$ proportional columns.

(v) At the excesses 0 and $c(B)$ every pattern meets every room side in a basis (Theorem 12.19). H_0, respectively $H_{c(B)}$, is a single positive entry exactly when the weights of that level factor along these products, as for unit weights, for weights factoring across B and R and for tables determinantal over $\mathbb{R}$ or $\mathbb{C}$; when both do, the room child of the one station of excess 0 is $\mathcal{B}/R$ and that of excess $c(B)$ is $\mathcal{B}|B$ (Proposition 12.17), these tables carrying the weights of the room children of the two stations, and the seam letters at the two extremes are $K_{\mathcal{B}/R}$, $\Psi_{\mathcal{B}/R}$ and $K_{\mathcal{B}|B}$, $\Psi_{\mathcal{B}|B}$.

(vi) Swapping B and R transposes every H_i and reverses the excesses, $i \mapsto c(B) - i$; the seam letters of R are then those of the bunch R.

A compatible pair fills its rectangle: inside it every tuple of the bunch meets every tuple of the room, with weights of rank one. What the marks do not show is the pattern of the rectangles, which stations of the bunch meet which stations of the room. At an extreme excess every station of the bunch meets every station of the room, in one rectangle when the weights of that level factor; between the extremes a station need not meet every station of the other side, and the seam matrix H_i, not the marks, says how they meet.

Proposition 28.3 (The width ladder)**.** *In Definition 28.1:*

(i) $c(B) = 0$ exactly when B is a union of molecules of the table; there is one excess, every pattern meets every room side, and $K = \mathsf{x}_0^{\mathsf{T}} H_0\, \mathsf{y}_0$, $\Psi = \mathsf{x}_0^{*\mathsf{T}} H_0\, \mathsf{y}_0^*$. The seam is one rectangle $C \times D$ exactly when the weights factor across B and R, as for lists; then $K = h_{CD}\, \mathsf{x}_C\, \mathsf{y}_D$ and $\Psi = h_{CD}\, \mathsf{x}_C^*\, \mathsf{y}_D^*$: the letters factor over the bunch and the room.

(ii) $c(B) = 1$ exactly when there are two excesses, both extreme. Each is one rectangle exactly when the weights factor across B and R; then for the stations D_0 and D_1 of the excesses 0 and 1 the seam letters are $K_{(D_0)} = K_{\mathcal{B}/R}$, $K_{(D_1)} = K_{\mathcal{B}|B}$, $\Psi_{(D_0)} = \Psi_{\mathcal{B}/R}$ and $\Psi_{(D_1)} = \Psi_{\mathcal{B}|B}$, the room station letters are $\mathsf{y}_{D_0} = K_{\mathcal{B}|R}$ and $\mathsf{y}_{D_1} = K_{\mathcal{B}/B}$, these two tables carrying the multipliers, the largest 1, and the room-station factorization is the writing

$$K = K_{\mathcal{B}/R}\, K_{\mathcal{B}|R} + K_{\mathcal{B}|B}\, K_{\mathcal{B}/B}$$

of Theorem 19.4: the letter adds over two levels, and the seam is the hinge.

(iii) For $c(B) \geq 2$ the excesses $0 < i < c(B)$ can carry several rectangles.

At every width an excess with one station on one side has one station on the other, and the children at the patterns of each excess are equal, B being in level form, exactly when every transposition of two marks of B, fixing every other mark, carries the table (Proposition 14.12); then every excess is one rectangle.

28.2 Pieces, their laws, and the glue

The seam identity writes a family as a pairing of what the bunch presents to the room and what the room presents to the bunch, station by station. The pairing uses nothing of the exchange property and holds for pieces of any kind.

Definition 28.4 (Pieces with stations; their glue)**.** Let $N \geq 1$ and $c_1, \dots, c_N \geq 0$ be integers. A set of *stations* for N tables of the widths c_j is a family $\mathcal{D} = (\mathcal{D}_j)_{j=1}^{N}$ of finite sets graded by excesses, $\mathcal{D}_j = \bigsqcup_{i=0}^{c_j} \mathcal{D}_{j,i}$, with $\mathcal{D}_{j,0}$ and $\mathcal{D}_{j,c_j}$ nonempty; a station $D \in \mathcal{D}_{j,i}$ has the excess $i(D) = i$. A *piece* with the stations $\mathcal{D}$ on a finite set B, at exponents $(s_j, t_j)_j$ with $s_j, t_j \geq 0$, carries for every j and $D \in \mathcal{D}_j$ two nonzero polynomials V_{jD} and V^*_{jD} on B with nonnegative coefficients, its *seam letters* (positive letters, Definition 25.1, where their degree is positive), homogeneous of the degrees $d_j + i(D)$ and $d^*_j + c_j - i(D)$ for integers $d_j, d^*_j \geq 0$, and masses $\mathbf{m}_B$; its *wall* and its *top wall* are

$$\kappa_B = m_B - \sum_j \big(s_j d_j + t_j d^*_j\big), \qquad \kappa_B^{\mathrm{top}} = \sum_j (s_j + t_j) c_j - \kappa_B.$$

A *partner* is a piece with the same stations on a set R disjoint from B, read with the reversed excesses $c_j - i(D)$: its letters, the *room letters* L_{jD} and L^*_{jD} (as in Definition 21.15), have the degrees $e_j + c_j - i(D)$ and $e^*_j + i(D)$, and its wall is $\kappa_R = m_R - \sum_j (s_j e_j + t_j e^*_j)$. A *family of Newton words* is a sequence $(K_j, \Psi_j; s_j, t_j)_{j=1}^{N}$ of pairs of positive letters on one set with exponents $s_j, t_j \geq 0$, carrying masses; its word is $\prod_j K_j^{-s_j} \Psi_j^{-t_j}$ and its period is the period of that word (Definition 25.1). The *glue* of the piece and the partner is the family of Newton words on $B \sqcup R$ with the letters

$$K_j = \sum_{D \in \mathcal{D}_j} V_{jD}\, L_{jD}, \qquad \Psi_j = \sum_{D \in \mathcal{D}_j} V^*_{jD}\, L^*_{jD},$$

the exponents (s_j, t_j) and the masses $(\mathbf{m}_B, \mathbf{m}_R)$; its period is written $P_{B \sqcup R}(\mathbf{m})$, and its balance wall is $\kappa_R = \kappa_B^{\mathrm{top}}$. A piece is named by its set, or by a calligraphic letter $\mathcal{X}$ when several pieces are compared, and its objects carry that name.

By Proposition 28.2(iii) a bunch of a family of tables, with its seam letters $V_{jD} = K_{j,(D)}$ and $V^*_{jD} = \Psi_{j,(D)}$, is a piece with the stations of its room, $d_j = d_j(B)$ and $d^*_j = d^*_j(B)$; its room, with the room station letters $L_{jD} = \mathsf{y}_{jD}$ and $L^*_{jD} = \mathsf{y}^*_{jD}$, is a partner, $e_j = d_j(R)$ and $e^*_j = d^*_j(R)$; and their glue is the family. By (vi) the same holds with B and R swapped, on the stations $\mathcal{K}$. The walls are the walls κ_B and κ_R of the bunch and of the room, and the top wall is the top wall κ_B^{top} of the bunch (Definition 8.1). A glue need not be a family of tables; its period converges exactly when its Newton sum has the dimension $|B \sqcup R| - 1$ and the masses lie in its relative interior (Theorem 25.2), and every statement about pieces is a statement about the period.

Definition 28.5 (The law of a piece)**.** Let a piece on B with the stations $\mathcal{D}$ have exponents with $\sum_j (s_j + t_j) c_j > 0$. Its active letters (Definition 8.1) are K_j when $s_j > 0$ and Ψ_j when $t_j > 0$; every coordinate, share, pin, product and count indexed by a letter runs over the active letters only, an inactive letter entering no period. The *excess sums* are $V_j^{(i)} = \sum_{D \in \mathcal{D}_{j,i}} V_{jD}$ and $V_j^{*(i)} = \sum_{D \in \mathcal{D}_{j,i}} V^*_{jD}$, the superscript counting excesses. On Δ_B put

$$\gamma_B^{\rho} = \prod_j \Big(\frac{V_j^{(c_j)}}{V_j^{(0)}}\Big)^{s_j/(\kappa_B + \kappa_B^{\mathrm{top}})} \Big(\frac{V_j^{*(0)}}{V_j^{*(c_j)}}\Big)^{t_j/(\kappa_B + \kappa_B^{\mathrm{top}})},$$

$$\phi_B^\rho(du) = u^{\mathbf{m}_B-\mathbf{1}} \prod_j \big(V_j^{(0)}\big)^{-s_j} \big(V_j^{*(c_j)}\big)^{-t_j} \, (\gamma_B^\rho)^{-\kappa_B} \, d\omega_B(u),$$

$$\varrho_{jD}^\rho = \frac{V_{jD}}{V_j^{(0)} \, (\gamma_B^\rho)^{i(D)}}, \qquad \varrho_{jD}^{\rho *} = \frac{V_{jD}^*}{V_j^{*(c_j)} \, (\gamma_B^\rho)^{c_j - i(D)}},$$

the *scale of the piece*, homogeneous of degree one, the *scale form*, and the coordinates of the *seam vector* $\boldsymbol{\varrho}_B^\rho = (\varrho_{jD}^\rho, \varrho_{jD}^{\rho *})_{j,D}$, homogeneous of degree zero, with a coordinate ϱ_{jD}^ρ for each active K_j and $D \in \mathcal{D}_j$ and a coordinate $\varrho_{jD}^{\rho *}$ for each active Ψ_j and $D \in \mathcal{D}_j$. Its values, the *seam points*, lie on the *seam torus*: the set of the points $\mathbf{x} = (x_{jD}, x_{jD}^*)_{j,D}$ with positive coordinates on which $\sum_{D \in \mathcal{D}_{j,0}} x_{jD} = 1$ for every active K_j, $\sum_{D \in \mathcal{D}_{j,c_j}} x_{jD}^* = 1$ for every active Ψ_j, and

$$\prod_j \Big(\sum_{D \in \mathcal{D}_{j,c_j}} x_{jD} \Big)^{s_j} \Big(\sum_{D \in \mathcal{D}_{j,0}} x_{jD}^* \Big)^{t_j} = 1,$$

of dimension $n_{\mathcal{D}}$, the number of the coordinates less the number of the active letters less one, which is $2|\mathcal{D}| - 2N - 1$, $|\mathcal{D}| = \sum_j |\mathcal{D}_j|$, when every letter is active. The *law* of the piece is the image $(\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho$, a positive measure carried by the seam torus. Its Mellin transform, at shares $(\mathbf{a}, \mathbf{b}) = (a_{jD}, b_{jD})_{j,D}$ of the active letters,

$$\int_{\Delta_B} (\boldsymbol{\varrho}_B^\rho)^{-(\mathbf{a},\mathbf{b})} \, d\phi_B^\rho = \int \mathbf{x}^{-(\mathbf{a},\mathbf{b})} \, d\big((\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho\big)(\mathbf{x}), \qquad \mathbf{x}^{-(\mathbf{a},\mathbf{b})} = \prod_{j,D} x_{jD}^{-a_{jD}} \, {x_{jD}^*}^{-b_{jD}},$$

is the *seam function*, in $[0, \infty]$ at real shares and defined at complex shares wherever it converges absolutely. A partner has its scale, scale form, seam vector, law and seam function by the same definition, with its room letters and the reversed excesses.

The scale form is projective: its orthant form has the degree $m_B - \sum_j (s_j d_j + t_j d_j^*) - \kappa_B = 0$, so it is read on any section (Lemma 16.2). The seam points lie on the seam torus: the sums at the excesses 0 and c_j are 1 by definition, and the product is $(\gamma_B^\rho)^{\kappa_B + \kappa_B^{\mathrm{top}}} (\gamma_B^\rho)^{-\sum_j (s_j + t_j) c_j} = 1$. The scale of the piece is the geometric mean of the oriented extreme ratios of the active letters, each taken to the root of its width: K_j measures the scale by $(V_j^{(c_j)} / V_j^{(0)})^{1/c_j}$ and Ψ_j by $(V_j^{*(0)} / V_j^{*(c_j)})^{1/c_j}$, in the proportions $s_j c_j$ and $t_j c_j$; a letter of width zero does not see the scale. Any other positive scale $q \, \gamma_B^\rho$ of degree one, q a positive function of the seam vector, multiplies ϕ_B^ρ by $q^{-\kappa_B}$ and the coordinates of the seam vector by powers of q: a fixed bijection of the tori with a density, so the law is determined up to it. The scale named is canonical: it uses the seam letters of the active letters alone, treats the two letters alike, and is kept at the centre by the bijections exchanging the family (Proposition 28.6(iii)). With weights factoring at the extreme levels, $V_j^{(0)} = K_{\mathcal{B}_j/R}$, $V_j^{(c_j)} = K_{\mathcal{B}_j|B}$, $V_j^{*(0)} = \Psi_{\mathcal{B}_j/R}$ and $V_j^{*(c_j)} = \Psi_{\mathcal{B}_j|B}$ (Proposition 28.2(v)); so

$$\gamma_B^\rho = \prod_j \Big(\frac{K_{\mathcal{B}_j|B}}{K_{\mathcal{B}_j/R}} \Big)^{s_j/(\kappa_B + \kappa_B^{\mathrm{top}})} \Big(\frac{\Psi_{\mathcal{B}_j/R}}{\Psi_{\mathcal{B}_j|B}} \Big)^{t_j/(\kappa_B + \kappa_B^{\mathrm{top}})},$$

$$\phi_B^\rho(du) = u^{\mathbf{m}_B-\mathbf{1}} \prod_j K_{\mathcal{B}_j/R}^{-s_j} \Psi_{\mathcal{B}_j|B}^{-t_j} \, (\gamma_B^\rho)^{-\kappa_B} \, d\omega_B(u).$$

The law is read off the measure of Theorem 21.16. For the room-station factorization the letter vector is $\mathbf{V}(u) = \big(K_{j,(D)}(u), \Psi_{j,(D)}(u)\big)_{j,D}$; the density of ϕ_B^ρ against $u^{\mathbf{m}_B-\mathbf{1}} d\omega_B$ and the seam vector are functions of $\mathbf{V}(u)$, so the law is an image of the measure $\mathbf{V}_*\big(u^{\mathbf{m}_B-\mathbf{1}} d\omega_B(u)\big)$ with

a density. The seam vector forgets the seam letters of the inactive letters and, of the others, one constant for each active letter and the scale, $V_{jD} \mapsto \lambda^{i(D)} V_{jD}$ and $V_{jD}^* \mapsto \lambda^{c_j - i(D)} V_{jD}^*$: the directions that the pins of Lemma 28.7 fix, on which the scale form changes by constant factors.

Proposition 28.6 (Symmetries of a seam)**.** *Let a bijection g of the marks $\mathcal{E}$ of a family map B onto B' and R onto R'.*

(i) If g carries table j to table j' (Definition 11.13), it maps the patterns and the room sides of B of each excess in table j onto those of B' of the same excess in table j', stations onto stations and compatible pairs onto compatible pairs, and the station letters and seam letters of the image stations are those of the stations read at $g^*\alpha$, the normalization of the multipliers commuting with g.

(ii) If g exchanges table j with table j', the windows of B and B' correspond as in Lemma 14.10, and $p \mapsto B' \setminus g(p)$, $X \mapsto R' \setminus g(X)$ map the stations of B of excess i in table j onto those of B' of excess $c(B) - i$ in table j' and compatible pairs onto compatible pairs, the station letter x_C and the seam letter $K_{(D)}$ going to the station letter $\mathsf{x}^*_{C'}$ and the seam letter $\Psi_{(D')}$ of the image stations C', D', and back.

(iii) So the bijections that keep B and carry or exchange the family permute the stations of its seams, with their station letters and seam letters, up to the bijection. Let such a bijection also keep the masses. A carrying one maps the seam torus of B onto itself, permuting the coordinates, and keeps the law $(\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho$. An exchanging one, sending table j to table j', maps the seam torus of B at the lean $\boldsymbol{\delta}$ onto the seam torus of B at the lean $-\boldsymbol{\delta}$, the coordinates ϱ_{jD}^ρ of table j going to the coordinates $\varrho_{j'D'}^{\rho *}$ of table j' and back, and maps the law of B at $\boldsymbol{\delta}$ onto the law of B at $-\boldsymbol{\delta}$; at the centre it keeps the law. At the centre the potential is kept by the carrying and the exchanging maps alike, at a lean by the carrying maps (Proposition 11.14): the centre does not tell apart stations joined by a bijection carrying or exchanging the family, nor a lean stations joined by a carrying one.

At the centre two stations can lie in one orbit through an exchange and in different orbits at every lean: the centre sees them as one, a lean separates them. These are the confusable stations of the seam, the analogue for stations of the halves C_K, C_Ψ of a class of marks (Theorem 11.17); the seam carries them down the tree with the classes of marks (Theorem 14.20).

Lemma 28.7 (The seam function is the period of the seam word)**.** *On the* pins of the seam*, the letter totals $\sum_D a_{jD} = s_j$ for every active K_j and $\sum_D b_{jD} = t_j$ for every active Ψ_j and the* scale pin

$$\sum_{j,D} \Big(i(D)\, a_{jD} + \big(c_j - i(D)\big)\, b_{jD} \Big) = \kappa_B,$$

the seam function is the period at the masses $\mathbf{m}_B$ of the seam word $W_B(\mathbf{a}, \mathbf{b}) = \prod_{j,D} V_{jD}^{-a_{jD}} {V_{jD}^*}^{-b_{jD}}$ *(Definition 25.1):*

$$\int_{\Delta_B} (\boldsymbol{\varrho}_B^\rho)^{-(\mathbf{a},\mathbf{b})}\, d\phi_B^\rho = P_{W_B(\mathbf{a},\mathbf{b})}(\mathbf{m}_B) = \int_{\Delta_B} u^{\mathbf{m}_B - \mathbf{1}} \prod_{j,D} V_{jD}(u)^{-a_{jD}} V_{jD}^*(u)^{-b_{jD}}\, d\omega_B(u).$$

For the piece of a bunch of a family the seam word is the bunch word of the room-station factorization (Theorem 22.11), and its period on the pins is the Mellin transform of one law.

Lemma 28.8 (Absorbing a piece of any width)**.** *Let a piece on B with the stations $\mathcal{D}$ carry masses $\mathbf{m}_B$ and exponents $(s_j, t_j)_j$ with $\sum_j (s_j + t_j) c_j > 0$, its wall κ_B and its seam vector $\boldsymbol{\varrho}_B^\rho$, whose*

values are written $\mathbf{x} = (x_{jD}, x^*_{jD})_{j,D}$. *Let* g *be a mark outside* B, *the* stand-in *of the piece,* Z *a finite set disjoint from* $B \sqcup \{g\}$, $\mathbf{m}_Z$ *masses on* Z, *and* $f_1, \dots, f_p$ *positive continuous functions, each assigned to one active letter of the piece,* f_l *defined on* $(0,\infty)^Z \times (0,\infty)^{\mathcal{D}_j}$ *when its letter is* K_j *or* Ψ_j *and homogeneous of degree one in its second argument,* $f_l(\alpha_Z, \lambda\mathbf{y}) = \lambda\, f_l(\alpha_Z, \mathbf{y})$ *for* $\lambda > 0$, *with real exponents* a_l *such that the exponents assigned to each letter add to its exponent. In this statement write* $\mathbf{V}_l$ *for the vector of the seam letters of the letter of* f_l, $(V_{jD})_{D\in\mathcal{D}_j}$ *for* K_j *and* $(V^*_{jD})_{D\in\mathcal{D}_j}$ *for* Ψ_j; V_l *for its extreme excess sum,* $V^{(0)}_j$ *for* K_j *and* $V^{*(c_j)}_j$ *for* Ψ_j; *and* $\mathbf{x}_l(\alpha_g)$ *for the vector* $(x_{jD}\,\alpha_g^{\,i(D)})_D$ *for* K_j *and* $(x^*_{jD}\,\alpha_g^{\,c_j - i(D)})_D$ *for* Ψ_j. *Let* $H \geq 0$ *be a Borel function on* $(0,\infty)^p$. *If the form*

$$\omega = \alpha_B^{\mathbf{m}_B - \mathbf{1}} \alpha_Z^{\mathbf{m}_Z - \mathbf{1}}\, H\Big(\Big(\frac{f_l\big(\alpha_Z, \mathbf{V}_l(\alpha_B)\big)}{V_l(\alpha_B)}\Big)_l\Big) \prod_{l=1}^{p} f_l\big(\alpha_Z, \mathbf{V}_l(\alpha_B)\big)^{-a_l}\, d\alpha_B\, d\alpha_Z$$

is homogeneous of degree zero on $(0,\infty)^{B\sqcup Z}$, *then, in* $[0,\infty]$,

$$\int_{\{h=1\}} \omega = \int \Big(\int_{\{h'=1\}} \omega_{\mathbf{x}} \Big)\, d\big((\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B\big)(\mathbf{x}),$$
$$\omega_{\mathbf{x}} = \alpha_Z^{\mathbf{m}_Z - \mathbf{1}} \alpha_g^{\kappa_B - 1}\, H\Big(\big(f_l(\alpha_Z, \mathbf{x}_l(\alpha_g))\big)_l\Big) \prod_l f_l\big(\alpha_Z, \mathbf{x}_l(\alpha_g)\big)^{-a_l}\, d\alpha_Z\, d\alpha_g,$$

the integrals taken over any sections of positive functions h, h' *of degree one on the two orthants.*

The piece collapses to its stand-in g, which carries the mass κ_B; a letter of table j that sees the piece sees the stand-in with the degree c_j, at its stations twisted by the seam point. At width one this is Lemma 23.2 read through Proposition 28.11.

Lemma 28.9 (The native split at the stations)**.** *Let a family of tables have real data on its balance wall, with* $s_j, t_j \geq 0$, *not all zero, and* $\mathbf{m} > 0$, *and a native gauge* $\alpha^\star$ *(Definition 8.17),* B *a bunch with* $\sum_j (s_j + t_j) c_j(B) > 0$, *and put, for the active letters,*

$$a_{jD} = s_j\, \mathbb{P}_{j,\alpha^\star}(T \cap R \in D), \qquad b_{jD} = t_j\, \mathbb{P}^*_{j,\alpha^\star}(T \cap R \in D), \qquad D \in \mathcal{D}_j,$$

the native split (Definition 12.25) aggregated to the stations of room sides. Then $(\mathbf{a}, \mathbf{b})$ *is a point of the pins of the seam with positive coordinates, at which* $\mathbf{m}_B$ *and* $\mathbf{m}_R$ *lie in the relative interiors of the Newton sums of the bunch word and of the room word (Theorem 22.11); the two words converge there exactly when these Newton sums have the full dimensions* $|B|-1$ *and* $|R|-1$ *(Theorem 25.2).*

Theorem 28.10 (The glue across a seam)**.** *Let a piece on* B *and a partner on* R *with the stations* $\mathcal{D}$ *be glued, with masses on the balance wall of the glue and* $\sum_j (s_j + t_j) c_j > 0$.

(i) The piece absorbed. *In* $[0,\infty]$,

$$P_{B\sqcup R}(\mathbf{m}) = \int P_{R^{(\mathbf{x})}}(\mathbf{m}_R, \kappa_B)\, d\big((\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B\big)(\mathbf{x}),$$

with $R^{(\mathbf{x})}$ *the* partner with its stand-in at $\mathbf{x}$*: the family of Newton words on* $R \sqcup \{g\}$ *with the letters* $\sum_D x_{jD}\, \alpha_g^{\,i(D)} L_{jD}$ *and* $\sum_D x^*_{jD}\, \alpha_g^{\,c_j - i(D)} L^*_{jD}$, *the exponents* (s_j, t_j) *and the mass* κ_B *on* g.

(ii) Both absorbed. *On the scales $\{g\}$ and one scale for each active letter let $t_{\mathcal{D}}$ be the configuration whose column at the coordinate (j,D) of K_j is $i(D)\,\mathbf{e}_g+\mathbf{e}_{K_j}$ and at that of Ψ_j is $(c_j-i(D))\,\mathbf{e}_g+\mathbf{e}_{\Psi_j}$. In $[0,\infty]$,*

$$P_{B\sqcup R}(\mathbf{m})=\iint \Gamma(\kappa_B)\,\mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x}\circ\mathbf{x}')\,d\big((\boldsymbol{\varrho}_B^{\rho})_*\phi_B^{\rho}\big)(\mathbf{x})\,d\big((\boldsymbol{\varrho}_R^{\rho})_*\phi_R^{\rho}\big)(\mathbf{x}'),$$

$\mathbf{x}\circ\mathbf{x}'$ the coordinatewise product and the index $(\kappa_B,\mathbf{s},\mathbf{t})$ read on the active letters (Definition 2.12). The kernel is the period of the stand-in pair*, the marks g and g' with the masses κ_B and κ_R and the letters $\sum_D x_{jD}x'_{jD}\,\alpha_g^{i(D)}\alpha_{g'}^{c_j-i(D)}$ and $\sum_D x^*_{jD}x'^*_{jD}\,\alpha_g^{c_j-i(D)}\alpha_{g'}^{i(D)}$, read on one scale:*

$$\Gamma(\kappa)\,\mathsf{Z}_{(\kappa,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x}\circ\mathbf{x}')=\int_0^{\infty}v^{\kappa-1}\prod_j\Big(\sum_{D\in\mathcal{D}_j}x_{jD}x'_{jD}\,v^{i(D)}\Big)^{-s_j}\Big(\sum_{D\in\mathcal{D}_j}x^*_{jD}x'^*_{jD}\,v^{c_j-i(D)}\Big)^{-t_j}dv.$$

(iii) One share per station. *On every contour of shares a_{jD}, b_{jD} of the active letters with positive real parts, on the pins of the seam (Lemma 28.7), which on the balance wall are the pins of the partner as well, at which the seam functions of the piece and of the partner converge,*

$$P_{B\sqcup R}(\mathbf{m})=\int_{pins}\prod_{j:\,s_j>0}\Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\prod_{j:\,t_j>0}\Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\int_{\Delta_B}(\boldsymbol{\varrho}_B^{\rho})^{-(\mathbf{a},\mathbf{b})}\,d\phi_B^{\rho}\int_{\Delta_R}(\boldsymbol{\varrho}_R^{\rho})^{-(\mathbf{a},\mathbf{b})}\,d\phi_R^{\rho},$$

a fibre integral of the dimension $n_{\mathcal{D}}$ of the seam torus: the two seam functions are evaluated at the same shares, one per station and letter.

Parts (i) and (ii) hold in every case; (iii) needs a point of the pins at which both seam functions are finite. Full dimension of the Newton sums of the seam word and of the room word is necessary for it (Theorem 25.2), and when the glue is a family of tables with a native gauge, every proper wall positive, it is sufficient, the native split at the stations being such a point (Lemma 28.9); without it the share side carries one pin more for each direction that a side does not see (Theorem 28.12 below).

The glue is one pairing at every width: the bunch is a law on its seam torus, the room another, and the kernel is the period of two stand-in marks whose letters pair the stations. At width one the seam torus has one dimension less than the number of active letters, a half-line for one table with $s,t>0$ and $(0,\infty)^{2N-1}$ when every letter is active (Proposition 28.11), and the kernel is the period of two marks in every table (Lemma 23.7, Theorem 23.6(i)); at the width c the seam torus has the dimension $n_{\mathcal{D}}$ and the kernel is one scale with at most c_j+1 levels in the letters of table j, exactly c_j+1 for a table. Opened on its shares, the glue is the product of the two seam functions against the share kernels of Corollary 4.4, one for each active letter.

Proposition 28.11 (Width one is the hinge)**.** *Let B have width one in every table of a family with exponents $s_j,t_j>0$, the weights factoring across B and R, $\theta_{j,T}=\theta'_{j,T\cap B}\,\theta''_{j,T\cap R}$, with θ''_j of largest value 1 on the room sides of each excess, so that θ''_j gives the multipliers of the room sides (Definition 28.1), with the stations D_0 and D_1 of the excesses 0 and 1 in every table (Proposition 28.3(ii)), the hinge values $\gamma^K_{j,B}$, $\gamma^{\Psi}_{j,B}$ of Definition 19.2, and the cell family $F_B^+=((\mathcal{B}^+_{j,B},\theta'_j);s_j,t_j)_j$ of the bunch (Definition 23.1), with its hinge wall κ_B, base form $\phi_{F_B^+}$ and ratio vector $\boldsymbol{\varrho}_{F_B^+}$, whose values are written $\mathbf{x}=(x_{j,K},x_{j,\Psi})_j$, $x_{1,K}=1$. In this statement write $\psi(\mathbf{x})=\prod_j x_{j,K}^{s_j/(2\sigma)}x_{j,\Psi}^{t_j/(2\sigma)}$, $\sigma=\sum_j\sigma_j$ the total strength (Definition 8.1).*

(i) $\gamma^{\rho}_{B} = \prod_j (\gamma^{K}_{j,B})^{s_j/(2\sigma)} (\gamma^{\Psi}_{j,B})^{t_j/(2\sigma)} = \gamma^{K}_{1,B}\,\psi(\boldsymbol{\varrho}_{F^+_B})$ *and* $\phi^{\rho}_{B} = \psi(\boldsymbol{\varrho}_{F^+_B})^{-\kappa_B}\,\phi_{F^+_B}$.

(ii) The seam vector of the piece has $\varrho^{\rho}_{jD_0} = \varrho^{\rho*}_{jD_1} = 1$, $\varrho^{\rho}_{jD_1} = x_{j,K}/\psi(\mathbf{x})$ *and* $\varrho^{\rho*}_{jD_0} = x_{j,\Psi}/\psi(\mathbf{x})$ *at* $\mathbf{x} = \boldsymbol{\varrho}_{F^+_B}$*; the map* $\mathbf{x} \mapsto \big(x_{j,K}/\psi(\mathbf{x}),\, x_{j,\Psi}/\psi(\mathbf{x})\big)_j$ *is a bijection of* $(0,\infty)^{2N-1}$ *onto the seam torus, and under it the law* $(\boldsymbol{\varrho}^{\rho}_{B})_*\phi^{\rho}_{B}$ *is the image of* $\psi^{-\kappa_B}$ *times the law* $(\boldsymbol{\varrho}_{F^+_B})_*\phi_{F^+_B}$ *of the cell family.*

(iii) The partner with its stand-in at the seam point of $\mathbf{x}$ *is, after* $\alpha_g \mapsto \psi(\mathbf{x})\,\alpha_g$*, the family* $((\mathcal{B}^{+}_{j,R}, \theta''_j); s_j, t_j)_j$ *of the cells of the room twisted at* g *by* $\mathbf{x}$*,* $x_{j,K}$ *in* K_j *and* $x_{j,\Psi}$ *in* Ψ_j *(Definition 22.1); the change of variable multiplies* $\alpha_g^{\kappa_B - 1} d\alpha_g$ *by* $\psi(\mathbf{x})^{\kappa_B}$*, which cancels the density of (ii), and Lemma 28.8 is Lemma 23.2 for the cell family* F^{+}_{B}.

(iv) The room, read as the partner, has the same description through the cell family $F^{+}_{R} = ((\mathcal{B}^{+}_{j,R}, \theta''_j); s_j, t_j)_j$ *of the room, with its hinge wall* κ_R *and ratio vector* $\boldsymbol{\varrho}_{F^+_R}$*, whose values are written* $\mathbf{x}'$*,* $x'_{1,K} = 1$*: its scale is* $\gamma^{K}_{1,R}\,\psi(\boldsymbol{\varrho}_{F^+_R})$*, its scale form* $\psi(\boldsymbol{\varrho}_{F^+_R})^{-\kappa_R}\phi_{F^+_R}$*, its seam vector has* $\varrho^{\rho}_{jD_1} = \varrho^{\rho*}_{jD_0} = 1$*,* $\varrho^{\rho}_{jD_0} = x'_{j,K}/\psi(\mathbf{x}')$ *and* $\varrho^{\rho*}_{jD_1} = x'_{j,\Psi}/\psi(\mathbf{x}')$ *at* $\mathbf{x}' = \boldsymbol{\varrho}_{F^+_R}$*, and its law is the image of* $\psi^{-\kappa_R}$ *times the law* $(\boldsymbol{\varrho}_{F^+_R})_*\phi_{F^+_R}$ *under the bijection of (ii). At the seam points the coordinate ratios* $\varrho^{\rho*}_{jD_0}/\varrho^{\rho}_{jD_1}$ *of the piece and* $\varrho^{\rho*}_{jD_1}/\varrho^{\rho}_{jD_0}$ *of the partner are the ratios* $\varrho_{j,B}$ *and* $\varrho_{j,R}$ *of Definition 19.2. At the seam points* $\mathbf{p}$ *of the piece and* $\mathbf{p}'$ *of the partner that are the images of* $\mathbf{x}$ *and* $\mathbf{x}'$*,*

$$\begin{aligned}\Gamma(\kappa_B)\,\mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{p}\circ\mathbf{p}') &= \int_0^{\infty} v^{\kappa_B-1} \prod_j \Big(\frac{x_{j,K}\,v}{\psi(\mathbf{x})} + \frac{x'_{j,K}}{\psi(\mathbf{x}')}\Big)^{-s_j} \Big(\frac{x_{j,\Psi}\,v}{\psi(\mathbf{x})} + \frac{x'_{j,\Psi}}{\psi(\mathbf{x}')}\Big)^{-t_j} dv \\ &= \psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}\,P_{F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]}\big((\kappa_B,\kappa_R);\mathbf{s},\mathbf{t}\big),\end{aligned}$$

$F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]$ *the core* $\binom{\{g,g'\}}{1}$ *in every table twisted at* g *by* $\mathbf{x}$ *and at* g' *by* $\mathbf{x}'$*, with the masses* κ_B *and* κ_R *(Theorem 23.6(i)). So Theorem 28.10(i) is Lemma 23.2 at the hinge, by part (iii); Theorem 28.10(ii) is Theorem 23.6(i) with that core and the cell families* F^{+}_{B} *at* g *and* F^{+}_{R} *at* g'*, the factor* $\psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}$ *cancelling the densities of the two laws; and Theorem 28.10(iii) is Theorem 19.13.*

At width one the scale of the piece is the geometric mean of the hinge values in the proportions of the exponents, and the general law is the law of the cell family of the bunch with a fixed power of its ratio vector as density. Every statement about pieces therefore specializes to its statement about cells through this one fixed change, and the calculus of cells (Chapter 23) is the case of width one of the calculus of pieces.

Theorem 28.12 (The share side with transverse pins)**.** *In Theorem 28.10(iii), the period of the glue converging, so that its Newton sum has full dimension (Theorem 25.2), let* $\vec{\mathcal{N}}_B \subseteq \mathbb{R}^B$ *and* $\vec{\mathcal{N}}_R \subseteq \mathbb{R}^R$ *be the spaces of the directions of the Newton sums of the seam word of the piece and of the room word of the partner at shares with positive real parts, which do not depend on the shares, and* $M = \vec{\mathcal{N}}_B^{\perp} \oplus \vec{\mathcal{N}}_R^{\perp} \subseteq \mathbb{R}^{B \sqcup R}$*, which contains* $\mathbf{1}$*; let* $\mathbf{v}_1, \ldots, \mathbf{v}_q$ *complete* $\mathbf{1}$ *to a basis of the lattice* $M \cap \mathbb{Z}^{B\sqcup R}$*. For every active letter and station fix the exponents* $\mathbf{n}_{jD}$*,* $\mathbf{n}^{*}_{jD}$*,* $\mathbf{n}'_{jD}$ *and* $\mathbf{n}'^{*}_{jD}$ *of one monomial of each of* V_{jD}*,* V^{*}_{jD}*,* L_{jD} *and* L^{*}_{jD}*, and in this statement write*

$$\mathbf{n}(\mathbf{a},\mathbf{b}) = \sum_{j,D} \Big(a_{jD}\big(\mathbf{n}_{jD} + \mathbf{n}'_{jD}\big) + b_{jD}\big(\mathbf{n}^{*}_{jD} + \mathbf{n}'^{*}_{jD}\big)\Big),$$

which modulo $\vec{\mathcal{N}}_B \oplus \vec{\mathcal{N}}_R$ *does not depend on the monomials fixed. For a word* W *on* B *whose Newton sum has the space of directions* $\vec{\mathcal{N}}_B$*, its* reduced period *is*

$$P^\flat_W(\mathbf{m}_B) = \int_{\mathbb{R}^B/\vec{\mathcal{N}}_B^\perp} e^{\langle \mathbf{m}_B, y\rangle}\, W(e^y)\, dy,$$

dy the quotient of Lebesgue measure by the measure on $\vec{\mathcal{N}}_B^\perp$ *that gives its integer lattice the covolume one; likewise on* R*. On every contour of shares of the active letters with positive real parts, pinned by the letter totals and by* $\langle \mathbf{n}(\mathbf{a},\mathbf{b}), \mathbf{v}_l\rangle = \langle \mathbf{m}, \mathbf{v}_l\rangle$*,* $l \le q$*, pins that are independent by the full dimension, at which the reduced periods of the two words converge,*

$$P_{B\sqcup R}(\mathbf{m}) = \int_{pins} \prod_{j:\, s_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \prod_{j:\, t_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\; P^\flat_{W_B(\mathbf{a},\mathbf{b})}(\mathbf{m}_B)\, P^\flat_{W_R(\mathbf{a},\mathbf{b})}(\mathbf{m}_R),$$

$W_R(\mathbf{a},\mathbf{b}) = \prod_{j,D} L_{jD}^{-a_{jD}} {L^*_{jD}}^{-b_{jD}}$ *the room word of the partner; on the pins the integrands of the reduced periods are invariant under* $\vec{\mathcal{N}}_B^\perp$ *and* $\vec{\mathcal{N}}_R^\perp$*. When* $\vec{\mathcal{N}}_B$ *and* $\vec{\mathcal{N}}_R$ *are the hyperplanes of sum zero,* $q = 1$*, the pin is the scale pin, the reduced periods are the periods, and this is Theorem 28.10(iii). When every room letter of the partner is a monomial, as for the room of a table whose stations of room sides are single room sides, with the coefficients* 1*,* $\vec{\mathcal{N}}_R = 0$*, the pins on* R *say that the exponent of the partner's monomials is* $\mathbf{m}_R$*, and the reduced period of the partner is the product of the coefficients of these monomials raised to minus the shares.*

So a degenerate seam word or room word costs pins, not convergence: the directions that a side does not see are fixed by the masses, one pin each, and the integral over them is gone.

28.3 The Gramma of a set of pairs

The station letters are what the bunch is; the seam letters are what the room sees. The seam joins the two, and its compatible pairs enter the period through one Gramma.

Definition 28.13 (The Gramma of a set of pairs)**.** Let $\mathfrak{R} \subseteq \mathcal{K}\times\mathcal{D}$ be a set of pairs of stations of two finite sets, every station in some pair, with coefficients $h_{CD} > 0$ on its pairs. Its *station map* sends $\mathbf{e} = (e_{CD}) \in \mathbb{C}^{\mathfrak{R}}$ to $\partial\mathbf{e}$, with $(\partial\mathbf{e})_C = \sum_D e_{CD}$ and $(\partial\mathbf{e})_D = \sum_C e_{CD}$. Two pairs lie in one *part* when a chain of pairs, each sharing a station with the next, joins them; a *circuit* is a minimal set of pairs whose columns $\mathbf{e}_C + \mathbf{e}_D$ are linearly dependent, which is the set of pairs of a closed chain $C_1D_1C_2D_2\cdots C_mD_mC_1$, $m \ge 2$, of distinct stations with (C_l, D_l) and (C_{l+1}, D_l) in $\mathfrak{R}$. Choose one station of $\mathcal{D}$ in each part, and call the other stations *kept*. For *loads* $(\mathbf{a},\mathbf{a}') \in \mathbb{C}^{\mathcal{K}}\oplus\mathbb{C}^{\mathcal{D}}$ whose totals on the two sides of every part agree, the *Gramma of* $\mathfrak{R}$ is the Gramma $\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h})$ (Definition 2.1) on the scales of the kept stations whose configuration $\partial\mathfrak{R}$ has at the pair (C,D) the column $\mathbf{e}_C + \mathbf{e}_D$, a station that is not kept read as 0, whose index is the loads of the kept stations, and whose coefficients are the h_{CD}. By Proposition 4.2 it is the fibre integral

$$\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h}) = \int_{\partial\mathbf{e}=(\mathbf{a},\mathbf{a}')} \prod_{(C,D)\in\mathfrak{R}} \Gamma(e_{CD})\, h_{CD}^{-e_{CD}},$$

pinned by the sums at the kept stations, which imply the others. Another choice of the stations not kept changes the pins by an integer matrix of determinant ±1, which leaves the fibre integral unchanged; its dimension is $|\mathfrak{R}| - |\mathcal{K}| - |\mathcal{D}|$ plus the number of parts. The *seam Gramma* of the excess i is the Gramma of the set $\mathfrak{R}_i$ of the compatible pairs of H_i, with the pair constants as its coefficients.

Proposition 28.14 (The seam transfers the station loads of the bunch to those of the room)**.** *For a bunch of a family let the* station word *of the bunch be* $\prod_{j,C} \mathsf{x}_{jC}^{-a_{jC}} {\mathsf{x}^*_{jC}}^{-b_{jC}}$*, with the station loads* a_{jC} *of the active* K_j *and* b_{jC} *of the active* Ψ_j *on the pins* $\sum_C a_{jC} = s_j$*,* $\sum_C b_{jC} = t_j$ *and* $\sum_{j,C} \big(i(C)a_{jC} + (c_j - i(C))b_{jC}\big) = \kappa_B$*, and let the station loads* a_{jD} *and* b_{jD} *of the room be shares on the pins of the seam (Lemma 28.7). On contours of positive real parts at which the station word of the bunch converges,*

$$\prod_{j:\, s_j>0} \prod_{D\in\mathcal{D}_j} \Gamma(a_{jD}) \prod_{j:\, t_j>0} \prod_{D\in\mathcal{D}_j} \Gamma(b_{jD})\; P_{W_B(\mathbf{a},\mathbf{b})}(\mathbf{m}_B)$$
$$= \int \prod_{j:\, s_j>0} \prod_{i=0}^{c_j} \mathsf{G}_{(\mathbf{a}_{j,\mathcal{K}_i},\mathbf{a}_{j,\mathcal{D}_i});\partial\mathfrak{R}_{j,i}}(\mathbf{h}_j) \prod_{j:\, t_j>0} \prod_{i=0}^{c_j} \mathsf{G}_{(\mathbf{b}_{j,\mathcal{K}_i},\mathbf{b}_{j,\mathcal{D}_i});\partial\mathfrak{R}_{j,i}}(\mathbf{h}_j)\; P_{\prod_{j,C} \mathsf{x}_{jC}^{-a_{jC}} {\mathsf{x}^*_{jC}}^{-b_{jC}}}(\mathbf{m}_B),$$

the integral over the station loads of the bunch with equal totals on the two sides of every part of every set $\mathfrak{R}_{j,i}$ *of the compatible pairs of table* j *at the excess* i*,* $\mathbf{h}_j$ *its pair constants.*

Along a circuit the pair terms $h_{CD}\,\mathsf{x}_C\,\mathsf{y}_D$ of a level have the alternating product $\prod_l h_{C_lD_l}/h_{C_{l+1}D_l}$, a constant: a direction $\mathbf{e}$ of the pair shares with $\partial\mathbf{e} = 0$ moves neither the station loads of the bunch, which the bunch sees, nor those of the room, which the room sees, and the period integrates over it through the seam Gramma alone. A part without circuits has a fibre of dimension zero, and its Gramma is the product $\prod \Gamma(e_{CD}) h_{CD}^{-e_{CD}}$ at the one solution of $\partial\mathbf{e} = (\mathbf{a}, \mathbf{a}')$ (Corollary 2.6).

Proposition 28.15 (The Gramma of a set of pairs in the Foreman's order)**.** *Let* $\mathfrak{R}$ *have the parts* $\mathfrak{R}_1, \dots, \mathfrak{R}_\nu$*, the part* $\mathfrak{R}_l$ *on the stations* $\mathcal{K}_l \sqcup \mathcal{D}_l$*, and loads with* $|\mathbf{a}_{\mathcal{K}_l}| = |\mathbf{a}'_{\mathcal{D}_l}|$ *for every* l*; in this statement write* $L_C(v) = \sum_{D:\,(C,D)\in\mathfrak{R}} h_{CD}\, v_D$ *for* $C \in \mathcal{K}_l$ *and* $v \in \Delta_{\mathcal{D}_l}$*. Then*

$$\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h}) = \prod_{C\in\mathcal{K}} \Gamma(a_C) \prod_{l=1}^{\nu} \int_{\Delta_{\mathcal{D}_l}} v^{\mathbf{a}'_{\mathcal{D}_l}-\mathbf{1}} \prod_{C\in\mathcal{K}_l} L_C(v)^{-a_C}\, d\omega_{\mathcal{D}_l}(v)$$
$$= \prod_{C\in\mathcal{K}} \Gamma(a_C) \prod_{l=1}^{\nu} P_{\prod_{C\in\mathcal{K}_l} L_C^{-a_C}}(\mathbf{a}'_{\mathcal{D}_l}),$$

both sides converging together, the period of the part l *being that of the word* $\prod_{C\in\mathcal{K}_l} L_C^{-a_C}$ *on* $\mathcal{D}_l$ *at the masses* $\mathbf{a}'_{\mathcal{D}_l}$*; with* $\operatorname{Re} a_C > 0$ *it converges exactly when, for every* l*,* $\operatorname{Re}\mathbf{a}'_{\mathcal{D}_l}$ *lies in the relative interior of* $\sum_{C\in\mathcal{K}_l} \operatorname{Re} a_C \operatorname{conv}\{\mathbf{e}_D :\ (C,D)\in\mathfrak{R}\}$ *(Theorem 25.2).*

The stations of the bunch are integrated first, each on its half-line, and those of the room last, on the simplex of each part: the Foreman's order of Definition 16.14, the bunch inside the integral over its room. For a seam the inner step returns, at each station C of the bunch, its child letter $K_{(C)} = \sum_D h_{CD}\,\mathsf{y}_D$ with every room station letter y_D replaced by a scale v_D. No station is singled out and no order is chosen.

Proposition 28.16 (Closing the Gramma of a set of pairs)**.** *In Proposition 28.15 let the loads be real and positive in the domain for (i) and (iii); in this statement write, for a part* $\mathfrak{R}_l$*,* $L_C(v) = \sum_{D:\,(C,D)\in\mathfrak{R}} h_{CD}\, v_D$ *for* $C \in \mathcal{K}_l$ *and* $v \in \Delta_{\mathcal{D}_l}$*,* $M_C = \max_D h_{CD}$ *for its largest coefficient, its largest value on the simplex, and* $\mathsf{G}_{(\mathbf{a},\mathbf{a}')}$ *for* $\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h})$*.*

(i) Peak closing. *For every part* $\mathfrak{R}_l$*, with* $\bar{L}_C = 1 - L_C/M_C \geq 0$ *on* $\Delta_{\mathcal{D}_l}$*,*

$$\int_{\Delta_{\mathcal{D}_l}} v^{\mathbf{a}'_{\mathcal{D}_l}-\mathbf{1}} \prod_{C\in\mathcal{K}_l} L_C(v)^{-a_C}\, d\omega_{\mathcal{D}_l}(v)$$
$$= \prod_{C\in\mathcal{K}_l} M_C^{-a_C}\, \Gamma_{\substack{|\mathcal{D}_l|\\1}}\!\begin{bmatrix}\mathbf{a}'_{\mathcal{D}_l}\\ |\mathbf{a}'_{\mathcal{D}_l}|\end{bmatrix} \sum_{\mathbf{n}\in\mathbb{N}^{\mathcal{K}_l}} \prod_{C\in\mathcal{K}_l} \frac{(a_C)_{n_C}}{n_C!}\; \mathbb{E}_{V\sim \mathrm{Dir}_{\mathbf{a}'_{\mathcal{D}_l}}}\Big[\prod_{C\in\mathcal{K}_l} \bar{L}_C(V)^{n_C}\Big],$$

a series of nonnegative terms, each a finite nonnegative combination of Dirichlet moments (Theorem 3.11), rational at rational loads and coefficients; it is Theorem 15.13(i) with the sharp references M_C, and $\mathsf{G}_{(\mathbf{a},\mathbf{a}')}$ is $\prod_C \Gamma(a_C)$ times the product of these right sides over the parts.

(ii) Relations at shifted loads. *At all loads in the domain, real or complex, for every C and D,*

$$a_C\, \mathsf{G}_{(\mathbf{a},\mathbf{a}')} = \sum_{D'\colon (C,D')\in\mathfrak{R}} h_{CD'}\, \mathsf{G}_{(\mathbf{a}+\mathbf{e}_C,\, \mathbf{a}'+\mathbf{e}_{D'})},$$
$$a'_D\, \mathsf{G}_{(\mathbf{a},\mathbf{a}')} = \sum_{C'\colon (C',D)\in\mathfrak{R}} h_{C'D}\, \mathsf{G}_{(\mathbf{a}+\mathbf{e}_{C'},\, \mathbf{a}'+\mathbf{e}_D)}.$$

(iii) Residues. *A part that is one circuit $C_1D_1\cdots C_mD_m$ has one share z moving round the circuit with alternating signs: for a real point $\mathbf{q}$ of the fibre the pairs (C_l, D_l) carry $q_l + z$ and the pairs (C_{l+1}, D_l) carry $q'_l - z$. With $x = \prod_l h_{C_lD_l}/h_{C_{l+1}D_l}$ and $\Sigma = \sum_l (q_l + q'_l)$, the total load of the part, its Gramma is a constant times*

$$\frac{1}{2\pi i}\int_{(0)} x^{-z} \prod_{l=1}^{m} \Gamma(q_l + z)\, \Gamma(q'_l - z)\, dz.$$

If $x < 1$, or $x = 1$ and $\Sigma < m$, the contour moves to the left across the poles of the $\Gamma(q_l + z)$, and if $x > 1$, or $x = 1$ and $\Sigma < m$, to the right across those of the $\Gamma(q'_l - z)$ (Theorem 4.5): the Gramma is the limit of the sums of the residues at the poles crossed, a series over m lattices, absolutely convergent when $x \neq 1$ or $\Sigma < m - 1$. The residue at a simple pole is a product of the other Gammas at the shifted point, and where poles of several factors meet, the q_l being congruent modulo $\mathbb{Z}$, it is a Laurent coefficient. At $x = 1$ with $\Sigma \geq m$ the shifted contours need not tend to zero and the residues need not sum.

Theorem 28.17 (Shares inside the rectangles close)**.** *Fix a table and an excess i, and give every basis T of excess i a share a_T. For pattern loads $\mathbf{a}^B = (a^B_p)_p$ and room-side loads $\mathbf{a}^R = (a^R_X)_X$ whose station loads $a^B_C = \sum_{p\in C} a^B_p$ and $a^R_D = \sum_{X\in D} a^R_X$ have equal totals on every part of $\mathfrak{R}_i$, on contours of positive real parts,*

$$\int_{\sum_X a_{p\sqcup X}=a^B_p,\ \sum_p a_{p\sqcup X}=a^R_X} \prod_T \Gamma(a_T)\, \theta_T^{-a_T}$$
$$= \prod_{C\in\mathcal{K}_i} \Gamma_{\substack{|C|\\1}}\!\begin{bmatrix}\mathbf{a}^B|_C\\ a^B_C\end{bmatrix} \prod_p w_p^{-a^B_p} \prod_{D\in\mathcal{D}_i} \Gamma_{\substack{|D|\\1}}\!\begin{bmatrix}\mathbf{a}^R|_D\\ a^R_D\end{bmatrix} \prod_X {w'_X}^{-a^R_X}\; \mathsf{G}_{(\mathbf{a}^B_{\mathcal{K}_i},\mathbf{a}^R_{\mathcal{D}_i});\partial\mathfrak{R}_i}(\mathbf{h}),$$

$\mathbf{a}^B_{\mathcal{K}_i}$ and $\mathbf{a}^R_{\mathcal{D}_i}$ the vectors of the station loads, the fibre on the left pinned by these sums at all patterns and at all room sides but one in each part of $\mathfrak{R}_i$, which imply the others.

Inside the rectangle of a compatible pair the shares spread freely and close to stacked Gammas, one for each station; what does not close is how the stations share one another, the seam Gramma.

28.4 What a piece retains

Definition 28.18 (Tinkers)**.** A *tinker* of a partner $\mathcal{Y}$ with the stations $\mathcal{D}$ is a vector $\mathbf{v} = (v_{jD}, v^*_{jD})_{j,D}$ of positive numbers, one for each coordinate of the seam vector; the *tinkered partner* $\mathcal{Y}_{\mathbf{v}}$ has the room letters $v_{jD}L_{jD}$ and $v^*_{jD}L^*_{jD}$. A tinker is *plain* when $v_{jD} = v^*_{jD}$ for every coordinate. On the room of a table a plain tinker multiplies the weights of the room sides of each station by one constant, which keeps the bases and gives a table (Proposition 12.9, read for the bunch R); a tinker that is not plain gives the room two systems of weights, one for each letter, as a cell may carry (Definition 21.1).

Theorem 28.19 (The law is what a piece retains, at every width)**.** *Let $\mathcal{X}$ and $\mathcal{X}'$ be pieces with the stations $\mathcal{D}$, on sets B and B' of any sizes, with the exponents $(s_j, t_j)_j$, $\sum_j (s_j + t_j)c_j > 0$, and one wall κ. Let $\mathcal{Y}_0$ be a partner of both with the wall $\sum_j (s_j + t_j)c_j - \kappa$, and let some real point $\mathbf{c}$ of the pins of the seam, with positive coordinates, lie in the interior, relative to the pins, of the domains of the seam functions of $\mathcal{X}$, $\mathcal{X}'$ and $\mathcal{Y}_0$. The following are equivalent.*

(a) $(\boldsymbol{\varrho}^{\rho}_{\mathcal{X}})_*\phi^{\rho}_{\mathcal{X}} = (\boldsymbol{\varrho}^{\rho}_{\mathcal{X}'})_*\phi^{\rho}_{\mathcal{X}'}$.

(b) $P_{\mathcal{X}\sqcup\mathcal{Y}} = P_{\mathcal{X}'\sqcup\mathcal{Y}}$, *in* $[0,\infty]$, *for every partner* $\mathcal{Y}$ *with the wall* $\sum_j (s_j + t_j)c_j - \kappa$.

(c) $P_{\mathcal{X}\sqcup\mathcal{Y}_{0,\mathbf{v}}} = P_{\mathcal{X}'\sqcup\mathcal{Y}_{0,\mathbf{v}}}$ *for every tinker* $\mathbf{v}$ *of* $\mathcal{Y}_0$.

(d) The seam functions of $\mathcal{X}$ and $\mathcal{X}'$ agree on the contour of the pins through $\mathbf{c}$.

When every table j with both letters active has $|\mathcal{D}_j| = 1$, except at most one, which has $c_j \geq 1$ and $|\mathcal{D}_j| = 2$, the plain tinkers suffice in (c). For one table at width one with factoring weights, $|\mathcal{D}| = 2$, and the theorem is Theorem 23.11 read through Proposition 28.11: the one-mark cell of ratio 1 (Definition 23.10), glued at the hinge, is a partner whose plain tinkers $(1, q^{1/2})$ are the one-mark cells of ratio q.

So the wall and the law are the whole of what a piece contributes to the families glued on it, and the tinkers of one partner already separate any two laws. A plain tinker multiplies the two room letters of a station by one constant, so with every letter active, and some table of positive width, the plain tinkers of one partner move only $|\mathcal{D}| - N$ of the $2|\mathcal{D}| - 2N - 1$ directions of the seam torus. The other directions are reached by partners with one system of weights whose weights and masses vary.

Definition 28.20 (Plain partners)**.** A partner on R with the stations $\mathcal{D}$ is *plain* when for every table j with an active letter and every $D \in \mathcal{D}_j$ there are a nonempty family $\mathcal{S}_{jD}$ of subsets of R of $e_j + c_j - i(D)$ marks, its *room sides*, and positive weights $w'_{jD,X}$, $X \in \mathcal{S}_{jD}$, with

$$L_{jD} = \sum_{X \in \mathcal{S}_{jD}} w'_{jD,X}\, \alpha^X_R, \qquad L^*_{jD} = \sum_{X \in \mathcal{S}_{jD}} w'_{jD,X}\, \alpha^{R\setminus X}_R,$$

one system of weights for the two letters of a station; then $e^*_j = |R| - e_j - c_j$. The room of a family of tables is plain (Proposition 28.2(iii)), its stations of room sides being disjoint, with $w'_{jD,X}$ the multiplier of X in table j (Definition 28.1).

Definition 28.21 (The station transform of a piece)**.** For a piece $\mathcal{X}$ on B with the stations $\mathcal{D}$, $\sum_j (s_j + t_j)c_j > 0$, the wall $\kappa_{\mathcal{X}}$ and the masses $\mathbf{m}_B$, and for $\mathbf{p} = (p_{jD}, p^*_{jD})_{j,D}$ with positive coordinates, one for each coordinate of the seam vector, the *station transform* is

$$\mathcal{I}_{\mathcal{X}}(\mathbf{p}) = \int \Gamma(\kappa_{\mathcal{X}})\, \mathsf{Z}_{(\kappa_{\mathcal{X}},\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{p})\, d\big((\boldsymbol{\varrho}^{\rho}_{\mathcal{X}})_*\phi^{\rho}_{\mathcal{X}}\big)(\mathbf{x})$$

$$= \int_{(0,\infty)^B} \alpha^{\mathbf{m}_B - \mathbf{1}} \prod_j \Big(\sum_D p_{jD} V_{jD} \Big)^{-s_j} \Big(\sum_D p^*_{jD} V^*_{jD} \Big)^{-t_j} d\alpha,$$

in $[0,\infty]$: the kernel of Theorem 28.10(ii) read against the law, the period of the piece against the constant room letters $\mathbf{p}$, the second expression by the computation of Lemma 28.8 with $\alpha = \rho u$ and $v = \rho\, \gamma^\rho_{\mathcal{X}}(u)$.

Theorem 28.22 (Plain partners separate laws)**.** *Let $\mathcal{X}$ and $\mathcal{X}'$ be pieces as in Theorem 28.19, with a real point $\mathbf{c}$ of the pins, with positive coordinates, in the interior of the domains of their seam functions on the pins. Fix room sides $\mathcal{S}_{jD}$ on a set R, every family with at least two members, such that the room word of the plain partners with these room sides has at $\mathbf{c}$ a Newton sum of the dimension $|R|-1$, as when one family of an active letter has a Newton polytope of that dimension, and masses $\mathbf{m}^0_R$ in its relative interior, with $\kappa_R = \sum_j (s_j + t_j) c_j - \kappa$. Then $(\boldsymbol{\varrho}^\rho_{\mathcal{X}})_* \phi^\rho_{\mathcal{X}} = (\boldsymbol{\varrho}^\rho_{\mathcal{X}'})_* \phi^\rho_{\mathcal{X}'}$ exactly when $P_{\mathcal{X} \sqcup \mathcal{Y}} = P_{\mathcal{X}' \sqcup \mathcal{Y}}$ for every plain partner $\mathcal{Y}$ with these room sides, at every positive weights and every masses in a neighbourhood of $\mathbf{m}^0_R$ on the hyperplane $\kappa_R = \sum_j (s_j + t_j) c_j - \kappa$. Such room sides exist for all stations and degrees: $e_j = 1$, $|R| \geq \max_j c_j + 2$, and every $\mathcal{S}_{jD}$ all the subsets of $1 + c_j - i(D)$ marks.*

The room of a family of tables is a plain partner, and multiplying the weights of the bases of each table by positive numbers that depend on the room side alone keeps the bases and the stations, and multiplies the members of each station of room sides by these numbers. So when a bunch of a family has at least two room sides in every station of every table with an active letter and the room word of its room has a Newton sum of full dimension, these families, with the masses of the room near the family's and those of the bunch kept, separate its law from every other law whose seam function converges, as its own does, at the native split (Lemma 28.9, which puts the masses of the room in the relative interior there).

Lemma 28.23 (Partners on one mark see a point mass)**.** *Let $\mathcal{X}$ be a piece on B with the stations $\mathcal{D}$ (Definition 28.4), the exponents $(s_j, t_j)_j$ with $w = \sum_j (s_j + t_j) c_j > 0$, the masses $\mathbf{m}_B$ and the wall $\kappa = \kappa_{\mathcal{X}}$ with $0 < \kappa < w$. For a point $\mathbf{x}$ of its seam torus (Definition 28.5), the* one-mark piece at $\mathbf{x}$ *is the piece on one mark g with the seam letters $x_{jD} \alpha_g^{i(D)}$ and $x^*_{jD} \alpha_g^{c_j - i(D)}$ and the mass κ; its scale is α_g, its seam vector is $\mathbf{x}$, its law is $\delta_{\mathbf{x}}$, and its station transform (Definition 28.21) is $\Gamma(\kappa) \mathsf{Z}_{(\kappa,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{p})$. If for some $c > 0$ and $\mathbf{x}_0$ on the seam torus*

$$\mathcal{I}_{\mathcal{X}}(\mathbf{p}) = c\, \Gamma(\kappa)\, \mathsf{Z}_{(\kappa,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x}_0 \circ \mathbf{p}) \qquad \text{for every } \mathbf{p},$$

then the law of $\mathcal{X}$ is $c\, \delta_{\mathbf{x}_0}$. The station transform at $\mathbf{p}$ is the period of the glue of $\mathcal{X}$ with the partner on one mark b with the room letters $p_{jD} \alpha_b^{c_j - i(D)}$ and $p^*_{jD} \alpha_b^{i(D)}$ and the mass $w - \kappa$, the tinker $\mathbf{p}$ of one partner on one mark (Definition 28.18). So a piece that acts on these partners as c times one mark has the law of c times one mark, with no condition on the domains of the seam functions as in Theorem 28.19.

28.5 Storage, and what replaces it

Theorem 28.24 (A bunch is stored on one mark exactly at width one)**.** *In Theorem 28.10(i) let the partner be the room of a family of tables. The partner with its stand-in at $\mathbf{x}$, $R^{(\mathbf{x})}$, is a family of tables on $R \sqcup \{g\}$, with a system of weights for each letter as a cell may carry (Definition 21.1), at every point of the seam torus exactly when every table with an active letter has $c_j \leq 1$, and $c_j = 1$ when both of its letters are active. At width one with factoring weights, normalized as*

in Proposition 28.11, it is the family of the cells $\mathcal{B}^{+}_{j,R}$ of the room with g twisted in both letters, which the change of variable of Proposition 28.11(iii) turns into the twist of Definition 22.1, and Lemma 28.8 is Lemma 23.2 with the scale changed from $\gamma^{K}_{1,B}$ to γ^{ρ}_{B}, a fixed bijection of the tori with a density (Proposition 28.11). A table with $c_j = 0$ does not see the stand-in, in either letter, and at the width $c_j \geq 2$ the stand-in enters the letters of table j with the degree c_j, the coefficient of $\alpha_g^{c_j}$ in K_j being $\sum_{D \in \mathcal{D}_{j,c_j}} x_{jD} L_{jD} \neq 0$: the partner with its stand-in is a family of Newton words, whose period converges exactly when its Newton sum has the dimension $|R|$ and the masses lie in its relative interior (Theorem 25.2).

So the single mark of width one is the case $c = 1$ of a stand-in of degree c. What replaces storage at every width is the stand-in itself, the station-wise pairing of Theorem 28.10: every statement about pieces is made on pieces with stations, whose glue is a family of Newton words, and table structure is never used to compose.

Theorem 28.25 (A point mass is a pinned set)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ be a family on $\mathcal{E}$ with real exponents $s_j, t_j \geq 0$ and masses on its balance wall with every proper wall positive, and let $\varnothing \neq S \subsetneq \mathcal{E}$, with the room $S' = \mathcal{E} \setminus S$, be read as a piece with the stations of its room and S' as a partner (Definition 28.4). Call S* pinned *when every active table has width at most one at S, with its weights factoring across S and S', and the active letters of the tables of width one read S alike (Theorem 22.5): there are a function γ on Δ_S and constants $c_j, c_j^* > 0$ with $\gamma^{K}_{j,S} = c_j \gamma$ whenever $s_j > 0$ and $\gamma^{\Psi}_{j,S} = c_j^* \gamma$ whenever $t_j > 0$ (Definition 19.2). Then $w = \sum_j (s_j + t_j) c_j(S) > 0$ and $0 < \kappa_S < w$, and the following are equivalent.*

- *(i) The law of S is carried by one point: $(\boldsymbol{\varrho}^{\rho}_{S})_* \phi^{\rho}_{S} = c\, \delta_{\mathbf{x}_0}$ (Definition 28.5) with $c > 0$ and $\mathbf{x}_0$ on the seam torus.*
- *(ii) S is pinned.*
- *(iii) S* splits off*: there are $c > 0$ and $\mathbf{x}_0$ on the seam torus such that every partner $\mathcal{Y}$ of S with the wall $w - \kappa_S$ has $P_{S \sqcup \mathcal{Y}} = c\, P_{\mathcal{Y}^{(\mathbf{x}_0)}}$ in $[0, \infty]$, $\mathcal{Y}^{(\mathbf{x}_0)}$ the partner with its stand-in at $\mathbf{x}_0$ carrying the mass κ_S (Theorem 28.10(i)).*
- *(iv) S splits off against the partners on one mark: its station transform is c times that of the one-mark piece at $\mathbf{x}_0$ with the mass κ_S (Lemma 28.23).*

When they hold, the seam vector is the constant $\mathbf{x}_0$, $c = \int_{\Delta_S} \phi^{\rho}_{S}$, and $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = c\, P_{S'^{(\mathbf{x}_0)}}$. For a family of lists, read at the factorization with the weights 1, the active letters of the tables of width one read S alike exactly when their hinge values are equal, $c_j = c_j^* = 1$, and then $\mathbf{x}_0 = \mathbf{1}$. For one list $\mathcal{B}$ with $s, t > 0$ and $|S| \geq 2$, S is pinned exactly when it has width one and $\varrho_S = 1$, that is $\mathbf{E}_{\mathcal{B}|S} = \mathbf{E}_{\mathcal{B}/S'}$.

When $\mathcal{B}|S$ and $\mathcal{B}/S'$ are connected, Chapter 30 reads the last condition as a dual pair (Corollary 30.18).

Corollary 28.26 (Which sets split off)**.** *Let F and S be as in Theorem 28.25.*

- *(i) If some active table has width two or more at S, then S does not split off, and its law is carried by no point.*
- *(ii) If $F = ((\mathcal{B}, \theta); s, t)$ is one prime table (Theorem 23.16) with $s + t > 0$, no set with at least two marks on each side splits off.*

(iii) *If $F = (\mathcal{B}; s, 0)$ is one list with $s > 0$ and e is a mark that is neither a loop nor a coloop, then $\mathcal{E} \setminus \{e\}$ splits off, and*

$$P_F(\mathbf{m}; s, 0) = \Gamma_{\substack{2\\1}}\begin{bmatrix} s - m_e, & m_e \\ & s \end{bmatrix} P_{((\mathcal{B}/e;\, m_e, 0),\, (\mathcal{B}\setminus e;\, s-m_e, 0))}\big(\mathbf{m}|_{\mathcal{E}\setminus\{e\}}\big),$$

the period of the bunch family of Corollary 19.14; at $(0, t)$, $t > 0$, the same holds with $\Gamma_{\substack{2\\1}}\begin{bmatrix} t - m_e, & m_e \\ & t \end{bmatrix}$ and the family $((\mathcal{B}/e;\, 0, t - m_e), (\mathcal{B} \setminus e;\, 0, m_e))$.

(iv) *If $F = (\mathcal{B}; s, t)$ is one list on at least three marks with $s, t > 0$ and e is neither a loop nor a coloop, then $\mathcal{E} \setminus \{e\}$ splits off exactly when $\mathbf{E}_{\mathcal{B}\setminus e} = \mathbf{E}_{\mathcal{B}/e}$.*

Lemma 28.27 (The level cell of a bunch)**.** *Let B have the width $c = c(B) \geq 1$ in a table, and let G be a set of c new marks. The sets $p \sqcup F$, $F \subseteq G$ and p a pattern of excess $c - |F|$, are the bases of a table $\mathcal{B}_B^\natural$ on $B \sqcup G$ of rank $r(B)$, the* level cell *of B, with any positive weights; $\mathcal{B}_B^\natural \setminus G = \mathcal{B}|B$ and $\mathcal{B}_B^\natural/G = \mathcal{B}/R$. At width one $\mathcal{B}_B^\natural$ is the cell of the bunch $\mathcal{B}_B^+$ of Theorem 19.4.*

Theorem 28.28 (The level envelope)**.** *Let the weights of a table factor across B and R, $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$, with $c = c(B) \geq 1$, and let $\mathcal{B}_B^\natural$ on $B \sqcup G$ and $\mathcal{B}_R^\natural$ on $R \sqcup G'$ be the level cells of the bunch and of the room, G' a copy of G, with the weights θ'_p on $p \sqcup F$ and $\theta''_X\binom{c}{|F'|}^{-1}$ on $X \sqcup F'$. Splicing the c pairs of a mark of G and its copy (Theorem 21.14) gives the table $\mathcal{B}^{\mathrm{env}}$ on $\mathcal{E}$, the* level envelope *of the bunch, whose bases are the sets $p \sqcup X$ of a pattern p and a room side X of one excess, with the weights $\theta'_p\,\theta''_X$. It has the patterns and the room sides of the table at every excess, its seam has one rectangle at every excess, and $\mathcal{B} \subseteq \mathcal{B}^{\mathrm{env}}$, with equality exactly when every excess of the seam of B is one rectangle.*

A bunch in level form is therefore stored on $c(B)$ pairs of hinges, the table being the splicing of its two level cells, and every bunch is read off its envelope by its seam: the table keeps, at each excess, the rectangles of the compatible pairs.

Proposition 28.29 (When a glue of seamed pieces is a table: what is necessary)**.** *Let $\mathcal{B}$ be a nonempty family of r-sets of $B \sqcup R$, with children and room children as in Definition 28.1. If $\mathcal{B}$ is the family of bases of a table, then:*

(a) every child and every room child is a table;

(b) the children at adjacent patterns p and $p + b$ are quotients, the child at $p + b$ of the child at p, and likewise the room children at adjacent room sides;

(c) at the least and at the largest number of marks of a pattern all the children are one table, and likewise all the room children;

(d) with d and $d + c$ the least and the largest numbers of marks of a pattern, for every k with $d \leq k \leq d + c$ the patterns of k marks are the k-sets independent in the table of the patterns of $d + c$ marks that span the table of the patterns of d marks, the bases of the $(k - d)$-th lift of the latter toward the former (Proposition 12.17); and likewise for the room sides.

Theorem 28.30 (The gluing criterion)**.** *Conditions (a)–(d) of Proposition 28.29 suffice: a nonempty family of r-sets of $B \sqcup R$ satisfying them is the family of bases of a table. So a nonempty family of sets of one size is the family of bases of a table exactly when, at one bunch, its children and room children are tables, adjacent ones are quotients, the extreme ones are one, and its levels are lifts (Proposition 12.17).*

Each of (c) and (d) is needed, and (d) on each side. The criterion is what storage of a bunch on new marks of a table asks, and nothing in the periods depends on it: the glue of Theorem 28.10 composes pieces whether or not their glue is a table.

Chapter 29

Composing bunches

The pieces of Chapter 28 compose. A family with several bunches is read through a core in which every bunch is replaced by its stand-in at its seam point, and its period is the mixture of the periods of these cores over the laws of the bunches (Theorem 29.1); a bunch inside a bunch makes the law of the larger one a mixture of the laws of its nodes (Theorem 29.2); and along a chain of bunches the letters are products of transfer matrices indexed by the stations of room sides, the laws are carried from bunch to bunch by one transfer each, and the period opens as a flow of shares through the stations (Theorem 29.3, Corollary 29.5). Every operation is a rule on the seam functions on contours or on the station transforms (Theorem 29.6), a closed chain of cells is read the same way (Theorem 29.8), and the Foreman tree with nodes of any width is evaluated by one procedure whose reusable data are the laws on its bunches (Definition 29.9). Read backward, the period as a function of the masses determines the potential, so the classes of marks and the tree are read off the periods, and with retention at every node this is the correspondence between a realization and its word (Theorems 29.11 and 29.12).

29.1 Cores, nested pieces and chains of bunches

Theorem 29.1 (The core reads its pieces, at any width)**.** *Let $B_1, \dots, B_q$ be disjoint bunches of a family F on its balance wall, each with $\sum_j (s_j + t_j) c_j(B_l) > 0$, with their scales $\gamma^\rho_{B_l}$, scale forms $\phi^\rho_{B_l}$ and seam vectors $\boldsymbol{\varrho}^\rho_{B_l}$ (Definition 28.5), and let Z be the marks outside them. For $u = (u_l)_l \in \prod_l \Delta_{B_l}$ read the letters of F at $\alpha_{B_l} = \alpha_{g_l} u_l / \gamma^\rho_{B_l}(u_l)$, for new marks g_l, and divide each letter of table j by the product over l of the extreme excess sums of the bunches B_l in that letter (Definition 28.5), $V_j^{(0)}$ for K_j and $V_j^{*(c_j(B_l))}$ for Ψ_j, read at the same point. The letters obtained depend on u only through the seam points $\mathbf{x}_l = \boldsymbol{\varrho}^\rho_{B_l}(u_l)$; with them, the exponents of F and the masses κ_{B_l} on g_l and $\mathbf{m}_Z$ on Z they form the* stand-in core $F^{(\mathbf{x}_1, \dots, \mathbf{x}_q)}$*, a family of Newton words on $Z \sqcup \{g_1, \dots, g_q\}$ (Definition 28.4) in which the stand-in g_l enters the letters K_j and Ψ_j with the degree $c_j(B_l)$. In $[0, \infty]$,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\prod_l \Delta_{B_l}} \prod_l \phi^\rho_{B_l}(du_l)\ P_{F^{(\boldsymbol{\varrho}^\rho_{B_1}(u_1), \dots, \boldsymbol{\varrho}^\rho_{B_q}(u_q))}}\big(\mathbf{m}_Z, \kappa_{B_1}, \dots, \kappa_{B_q}\big)$$
$$= \int P_{F^{(\mathbf{x}_1, \dots, \mathbf{x}_q)}}\big(\mathbf{m}_Z, \kappa_{B_1}, \dots, \kappa_{B_q}\big) \prod_l d\big((\boldsymbol{\varrho}^\rho_{B_l})_* \phi^\rho_{B_l}\big)(\mathbf{x}_l).$$

At width one with factoring weights the stand-in core is the twisted family $F_{\mathcal{H}}[\mathbf{x}]$ of Theorem 23.6(i), up to the change of scale of Proposition 28.11, and this is Theorem 23.6(i); before the laws are

formed it is Theorem 22.3 with the scale of every marker divided by the scale $\gamma^{\rho}_{B_l}$ of its piece.

The core meets its pieces only through their seam points: every bunch collapses to one stand-in mark carrying the mass of its wall, and enters each letter with the degree of its width, at its stations twisted by its seam point. The cores of Chapter 22 are the case in which every piece is a cell.

Theorem 29.2 (The law of a piece from its sub-pieces)**.** *Let S be a bunch of a family with $\sum_j (s_j + t_j) c_j(S) > 0$, and $S_1, \dots, S_k \subseteq S$ disjoint bunches with $\sum_j (s_j + t_j) c_j(S_l) > 0$.*

(i) The seam letters of S are linear in those of each S_l, with nonnegative coefficients: in each table, for S_1, every station D of room sides of S and any member $X \in D$,

$$K_{(D)} = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{\,q}\, K_{(D_1(q))}, \qquad \Psi_{(D)} = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{(S \setminus S_1) \setminus q}\, \Psi_{(D_1(q))},$$

the sums over the sets $q \subseteq S \setminus S_1$ for which $q \sqcup X$ is a room side of S_1, $D_1(q)$ its station, $w'_{q \sqcup X}$ its multiplier there and w'_X that of X at S; the coefficients and the stations $D_1(q)$ do not depend on the member X.

(ii) Let the node $S^{(\mathbf{x}_1, \dots, \mathbf{x}_k)}$*, the node at S of the tree of the bunches $S \supseteq S_1, \dots, S_k$, be the piece on $(S \setminus \bigcup_l S_l) \sqcup \{g_1, \dots, g_k\}$ whose seam letters are those of S with each S_l replaced by its stand-in at $\mathbf{x}_l$, as in Theorem 29.1, with the masses κ_{S_l} on g_l. The node has the wall of S, and, in this statement writing λ_S, λ_{S_l} and $\lambda_{\mathbf{x}}$ for the laws of S, of S_l and of the node at $\mathbf{x} = (\mathbf{x}_l)_l$, the law of S is the mixture of the laws of its nodes:*

$$\int H \, d\lambda_S = \int \Big(\int H \, d\lambda_{\mathbf{x}} \Big) \prod_l d\lambda_{S_l}(\mathbf{x}_l) \qquad \text{for every Borel } H \geq 0,$$

and so are the seam functions, $\int \mathbf{z}^{-(\mathbf{a},\mathbf{b})}\, d\lambda_S(\mathbf{z}) = \int \big(\int \mathbf{z}^{-(\mathbf{a},\mathbf{b})}\, d\lambda_{\mathbf{x}}(\mathbf{z}) \big) \prod_l d\lambda_{S_l}(\mathbf{x}_l)$ at all shares $(\mathbf{a}, \mathbf{b})$.

At width one, with factoring weights, this is Theorem 23.6(ii), up to the change of scale of Proposition 28.11.

Theorem 29.3 (A chain of bunches)**.** *Let $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$, $L \geq 2$, be a chain of bunches of a family with the steps $B_{l-1} = S_l \setminus S_{l-1}$, and for $0 < l < L$ let $\mathcal{D}(S_l)$ be the stations of room sides of the bunch S_l, in each table, the index j dropped.*

(i) Letters. *For $2 \leq l \leq L-1$, $D' \in \mathcal{D}(S_{l-1})$ and $D \in \mathcal{D}(S_l)$ put, for any $X \in D$,*

$$\mathbf{N}^K_l(D', D) = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{\,q}_{B_{l-1}}, \qquad \mathbf{N}^{\Psi}_l(D', D) = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{B_{l-1} \setminus q}_{B_{l-1}},$$

the sums over the $q \subseteq B_{l-1}$ for which $q \sqcup X$ is a room side of S_{l-1} in the station D'. With $\mathbf{k} = (K_{(D)})_{D \in \mathcal{D}(S_1)}$ and $\mathbf{k}^* = (\Psi_{(D)})_{D \in \mathcal{D}(S_1)}$ the seam letters of S_1 and $\mathsf{y} = (\mathsf{y}_D)_{D \in \mathcal{D}(S_{L-1})}$ and $\mathsf{y}^* = (\mathsf{y}^*_D)_{D \in \mathcal{D}(S_{L-1})}$ the room station letters of S_{L-1},

$$K = \mathbf{k}^{\mathsf{T}} \mathbf{N}^K_2 \cdots \mathbf{N}^K_{L-1}\, \mathsf{y}, \qquad \Psi = \mathbf{k}^{*\mathsf{T}} \mathbf{N}^{\Psi}_2 \cdots \mathbf{N}^{\Psi}_{L-1}\, \mathsf{y}^*,$$

products of matrices of polynomials with nonnegative coefficients, indexed by the stations of room sides of the bunches S_l: Theorem 22.9 with these stations as the states.

(ii) Laws. *Let $\sum_j (s_j + t_j) c_j(S_l) > 0$ for every $0 < l < L$, and in this statement write λ_l for the law of S_l. For $2 \le l \le L-1$ let $S_l^{(\mathbf{x})}$ be the node of S_l with its sub-bunch S_{l-1} at $\mathbf{x}$ (Theorem 29.2(ii)), the piece on $B_{l-1} \sqcup \{g\}$ with the seam letters $\sum_{D'} x_{D'}\, \alpha_g^{i(D')} \mathbf{N}_l^K(D', D)$ and $\sum_{D'} x^*_{D'}\, \alpha_g^{c(S_{l-1}) - i(D')} \mathbf{N}_l^\Psi(D', D)$, $D \in \mathcal{D}(S_l)$, and the mass $\kappa_{S_{l-1}}$ on g, and let $\lambda_{\mathbf{x}}^{(l)}$ be its law. Then λ_l is the image of λ_{l-1} under the* transfer $\mathcal{Q}_l$,

$$\lambda_l = \mathcal{Q}_l \lambda_{l-1} = \int \lambda_{\mathbf{x}}^{(l)}\, d\lambda_{l-1}(\mathbf{x}),$$

and $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int P_{B_{L-1}^{(\mathbf{x})}}(\mathbf{m}_{B_{L-1}}, \kappa_{S_{L-1}})\, d\lambda_{L-1}(\mathbf{x})$, the room B_{L-1} of the last bunch S_{L-1} with its stand-in at $\mathbf{x}$ (Theorem 28.10(i)).

(iii) Repetition. *If the nodes $S_l^{(\cdot)}$, $2 \le l \le L-1$, are one piece up to bijections of marks keeping seam letters and masses, the transfers are one operator $\mathcal{Q}$ and $\lambda_l = \mathcal{Q}^{l-1} \lambda_1$: the chain is the power of one operator, whatever the widths of its bunches. At width one with factoring weights this is Theorem 22.8 and Proposition 24.4, the seam point being the twist up to the change of scale of Proposition 28.11.*

Lemma 29.4 (The native split along a chain)**.** *Let a family of tables have real data on its balance wall, with $s_j, t_j \ge 0$, not all zero, and $\mathbf{m} > 0$, and a native gauge $\alpha^\star$ (Definition 8.17), and let $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$ be a chain of bunches as in Theorem 29.3, with the steps $B_{l-1} = S_l \setminus S_{l-1}$. For the active letters put*

$$e^{(l)}_{jD'D} = s_j\, \mathbb{P}_{j,\alpha^\star}\big(T \setminus S_{l-1} \in D',\ T \setminus S_l \in D\big), \qquad f^{(l)}_{jD'D} = t_j\, \mathbb{P}^*_{j,\alpha^\star}\big(T \setminus S_{l-1} \in D',\ T \setminus S_l \in D\big),$$

and $a^{(l)}_{jD} = s_j\, \mathbb{P}_{j,\alpha^\star}(T \setminus S_l \in D)$, $b^{(l)}_{jD} = t_j\, \mathbb{P}^*_{j,\alpha^\star}(T \setminus S_l \in D)$, the native split at the stations of the bunch S_l (Lemma 28.9). Then the entry shares conserve the loads, $\sum_{D'} e^{(l)}_{jD'D} = a^{(l)}_{jD} = \sum_{D''} e^{(l+1)}_{jDD''}$, and likewise with f and b; and $\mathbf{m}_{B_{l-1}}$ lies in the relative interior of the Newton sum of the word of the node l at its shares, which converges there exactly when that Newton sum has the dimension $|B_{l-1}| - 1$.

Corollary 29.5 (The word of a chain of bunches)**.** *In Theorem 29.3 read the room B_{L-1} of the last bunch S_{L-1} as a last node whose transfer matrix from $\mathcal{D}(S_{L-1})$ to one terminal station is the column y, and y^* for Ψ, and open every active letter with one share per nonzero entry of every transfer matrix: $e^{(l)}_{jD'D}$ for the entry (D', D) of $\mathbf{N}_l^K$ and $f^{(l)}_{jD'D}$ for that of $\mathbf{N}_l^\Psi$, $2 \le l \le L$. A station D of a bunch S_l, $2 \le l \le L-1$, carries the load $a^{(l)}_{jD} = \sum_{D'} e^{(l)}_{jD'D}$ in and $\sum_{D''} e^{(l+1)}_{jDD''}$ out, and a station D of the first bunch S_1 the load $a^{(1)}_{jD} = \sum_{D''} e^{(2)}_{jDD''}$; likewise b with f. In this statement write $W_l(\mathbf{e}, \mathbf{f}) = \prod_{D',D} \mathbf{N}_l^K(D', D)^{-e_{D'D}}\, \mathbf{N}_l^\Psi(D', D)^{-f_{D'D}}$ for the word of the node l on B_{l-1}, the products over the active letters of every table understood. On contours of positive real parts at which the seam function of S_1 and the words of the nodes converge, which needs their Newton sums to have full dimension and, for a family with a native gauge, holds as soon as they do (Lemmas 28.9 and 29.4),*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \prod_{j:\, s_j > 0} \frac{\prod_{l=2}^{L} \prod_{D',D} \Gamma(e^{(l)}_{jD'D})}{\Gamma(s_j) \prod_{l=2}^{L-1} \prod_D \Gamma(a^{(l)}_{jD})} \prod_{j:\, t_j > 0} \frac{\prod_{l=2}^{L} \prod_{D',D} \Gamma(f^{(l)}_{jD'D})}{\Gamma(t_j) \prod_{l=2}^{L-1} \prod_D \Gamma(b^{(l)}_{jD})}$$
$$\times\, P_{W_{S_1}(\mathbf{a}^{(1)}, \mathbf{b}^{(1)})}(\mathbf{m}_{B_0}) \prod_{l=2}^{L} P_{W_l(\mathbf{e}^{(l)}, \mathbf{f}^{(l)})}(\mathbf{m}_{B_{l-1}}),$$

W_{S_1} the seam word of the first bunch S_1 (Lemma 28.7), the fibre pinned by the letter totals, the equality of the loads in and out at every station of the bunches $S_2, \dots, S_{L-1}$, and the homogeneity of the first bunch and of every node but one. At a bunch or a node whose Newton sum is degenerate the pins and the reduced period of Theorem 28.12 take the place of the period. At $L = 2$ this is Theorem 28.10(iii).

The word of a chain of bunches is a flow: one share per transfer entry, conserved at every station, each node contributing the period of its own word at the loads of its entries. Every station through which the flow passes is paid once by a Gamma in the denominator, except at the first bunch, where the seam function of the piece takes the loads. At width one with factoring weights every bunch S_l has two stations, and this is the word of the tree (Corollary 20.10) grouped by the two stations of each bunch.

Theorem 29.6 (Two transforms of a law, and their rules)**.** *Let a piece $\mathcal{X}$ on B with the stations $\mathcal{D}$ have $\sum_j (s_j + t_j)c_j > 0$, and a real point $\mathbf{c}$ of the pins of the seam, with positive coordinates, in the interior of the domain of its seam function on the pins.*

(i) Two transforms. *The law $(\boldsymbol{\varrho}^\rho_{\mathcal{X}})_* \phi^\rho_{\mathcal{X}}$ is determined by its seam function on the contour through $\mathbf{c}$, and by its station transform $\mathcal{I}_{\mathcal{X}}$ (Definition 28.21) on any nonempty open set of points $\mathbf{p}$; the two are related by*

$$\mathcal{I}_{\mathcal{X}}(\mathbf{p}) = \int_{pins} \prod_{j:\, s_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j \\ s_j\end{bmatrix} \prod_{j:\, t_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j \\ t_j\end{bmatrix} \; \mathbf{p}^{-(\mathbf{a},\mathbf{b})} \int_{\Delta_B} (\boldsymbol{\varrho}^\rho_{\mathcal{X}})^{-(\mathbf{a},\mathbf{b})}\, d\phi^\rho_{\mathcal{X}},$$

$\mathbf{p}^{-(\mathbf{a},\mathbf{b})} = \prod_{j,D} p_{jD}^{-a_{jD}} {p^*_{jD}}^{-b_{jD}}$.

(ii) The glue. *For a partner $\mathcal{Y}$ with the wall $\sum_j (s_j+t_j)c_j - \kappa_{\mathcal{X}}$, $P_{\mathcal{X}\sqcup\mathcal{Y}} = \int \mathcal{I}_{\mathcal{X}}(\mathbf{x}')\, d\big((\boldsymbol{\varrho}^\rho_{\mathcal{Y}})_* \phi^\rho_{\mathcal{Y}}\big)(\mathbf{x}')$, the station transform read against the law of the partner; on the pins, at a point where the seam function of the partner converges too, it is the share side of Theorem 28.10(iii). It extends the node of width one in one scale (Theorem 20.26) to any width.*

(iii) Nesting. *In Theorem 29.2, $\mathcal{I}_S(\mathbf{p}) = \int \mathcal{I}_{S(\mathbf{x}_1,\dots,\mathbf{x}_k)}(\mathbf{p}) \prod_l d\big((\boldsymbol{\varrho}^\rho_{S_l})_* \phi^\rho_{S_l}\big)(\mathbf{x}_l)$. With one sub-piece S_1, in this statement write $\mathbf{N}^K_{2,j}(\alpha)$ and $\mathbf{N}^\Psi_{2,j}(\alpha)$, $\alpha = \alpha_{S\setminus S_1}$, for the matrices $\mathbf{N}^K_2$ and $\mathbf{N}^\Psi_2$ in table j of the chain $\varnothing \subset S_1 \subset S \subset \mathcal{E}$ (Theorem 29.3(i)), the transfer matrices from the bunch S_1 to the bunch S. Then*

$$\mathcal{I}_S(\mathbf{p}) = \int_{(0,\infty)^{S\setminus S_1}} \alpha^{\mathbf{m}_{S\setminus S_1} - \mathbf{1}}\, \mathcal{I}_{S_1}\big(\big(\mathbf{N}^K_{2,j}(\alpha)\, \mathbf{p}_j\big)_j, \big(\mathbf{N}^\Psi_{2,j}(\alpha)\, \mathbf{p}^*_j\big)_j\big)\, d\alpha,$$

$\mathbf{p}_j = (p_{jD})_D$ and $\mathbf{p}^*_j = (p^*_{jD})_D$; and at shares $(\mathbf{a}, \mathbf{b})$ on the pins of the seam of S, on contours of positive real parts at which the seam function of S_1 and the word of the node converge,

$$\int_{\Delta_S} (\boldsymbol{\varrho}^\rho_S)^{-(\mathbf{a},\mathbf{b})}\, d\phi^\rho_S = \int \prod_{j:\, s_j>0} \frac{\prod_{D_1,D} \Gamma(e_{jD_1D})}{\prod_D \Gamma(a_{jD})} \prod_{j:\, t_j>0} \frac{\prod_{D_1,D} \Gamma(f_{jD_1D})}{\prod_D \Gamma(b_{jD})} \times \int_{\Delta_{S_1}} (\boldsymbol{\varrho}^\rho_{S_1})^{-(\mathbf{a}^{(1)},\mathbf{b}^{(1)})}\, d\phi^\rho_{S_1}\; P_{W(\mathbf{e},\mathbf{f})}(\mathbf{m}_{S\setminus S_1}),$$

the shares e_{jD_1D} and f_{jD_1D} one per nonzero entry of the transfer matrices, pinned by $\sum_{D_1} e_{jD_1D} = a_{jD}$, $\sum_{D_1} f_{jD_1D} = b_{jD}$ and the homogeneity of the node, with the loads $a^{(1)}_{jD_1} = \sum_D e_{jD_1D}$ and $b^{(1)}_{jD_1} = \sum_D f_{jD_1D}$, and $W(\mathbf{e},\mathbf{f})$ the word of the entries on $S \setminus S_1$ as in Corollary 29.5.

(iv) Repetition. *The rule of (iii) for a node repeated along a chain is one operator on seam functions, and on station transforms one operator on functions of* $\mathbf{p}$*; a chain of repeated nodes is its power applied to the first bunch.*

So every operation along a chain is a rule on the seam functions on contours, or on the station transforms: the representations (b) and (a) of Definition 23.19, at every width.

Definition 29.7 (Cyclic chains)**.** A family on $\mathcal{E} = B_1 \sqcup \cdots \sqcup B_k$, $k \geq 2$, the *segments* B_i indexed by $\mathbb{Z}_k$, is a *cyclic chain* with the finite sets of *states* $\mathcal{J}_0, \ldots, \mathcal{J}_{k-1}$, $\mathcal{J}_k = \mathcal{J}_0$, and the *top layer* $h \in \mathbb{N}$, when every active letter L is

$$L = [\varepsilon^h] \operatorname{tr} \prod_{i=1}^{k} \mathbf{M}_{L,i}(\alpha_{B_i}; \varepsilon),$$

the coefficient of ε^h in the trace of a product of matrices $\mathbf{M}_{L,i}$ indexed by $\mathcal{J}_{i-1} \times \mathcal{J}_i$, the *segment matrices*, whose entries are polynomials in α_{B_i} and ε with nonnegative coefficients.

Theorem 29.8 (Cyclic chains)**.** *(i) Let every table j of a family be a closed chain of cells $\mathcal{A}_{j,1}, \ldots, \mathcal{A}_{j,k}$ with several hinges spliced along hinge pairs (Theorem 21.14(ii)), the cells $\mathcal{A}_{j,i}$ of all tables having the marks B_i besides their hinges, and the pairs I_i joining the cell i to the cell $i+1$ named alike in every table, each pair named once. The family is a cyclic chain with the top layer* 0*: its states at the boundary i are the sets $J \subseteq I_i$ of the pairs whose hinge on the side of the cell i is occupied, and the segment matrices of the letters of table j are the channels of its cells (Definition 21.13),*

$$\mathbf{M}_{K_j,i}[J, J'] = K_{\mathcal{A}_{j,i},((I_{i-1} \setminus J) \sqcup J')}, \qquad \mathbf{M}_{\Psi_j,i}[J, J'] = \Psi_{\mathcal{A}_{j,i},((I_{i-1} \setminus J) \sqcup J')}.$$

(ii) Widen the states by layers, $\mathcal{J}_i \times \{0, \ldots, h\}$, a term in ε^μ of an entry raising the layer by μ, and write $\widetilde{\mathbf{M}}_{L,i}$ for the widened segment matrices. Then $L = \sum_{J \in \mathcal{J}_0} \mathbf{e}_{(J,0)}^{\mathsf{T}} \prod_i \widetilde{\mathbf{M}}_{L,i}\, \mathbf{e}_{(J,h)}$: for every starting state a chain in the sense of Theorem 29.3(i), the segment matrices as transfer matrices. In this statement write w_L for the exponent of the active letter L. Open every active letter first over the starting states J at which its term $\mathbf{e}_{(J,0)}^{\mathsf{T}} \prod_i \widetilde{\mathbf{M}}_{L,i}\, \mathbf{e}_{(J,h)}$ is not zero, with shares $c_{L,J}$, $\sum_J c_{L,J} = w_L$, and then every chain as in Corollary 29.5: one share $e_{L,i}^{(J)}$ per starting state J and per nonzero entry of a widened segment matrix that lies on a path from $(J,0)$ to (J,h), the shares of each starting state a flow from $(J,0)$, with the total $c_{L,J}$, to (J,h), conserved at every boundary, $a_{L,i}^{(J)}$ the loads it leaves at the boundary after the segment i, and $e_{L,1}^{(J)} = a_{L,1}^{(J)}$. On contours of positive real parts at which the words of the segments converge,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \prod_L \frac{1}{\Gamma(w_L)} \prod_{L,J} \frac{\prod_{i=2}^{k} \prod \Gamma(e_{L,i}^{(J)})}{\prod_{i=2}^{k-1} \prod \Gamma(a_{L,i}^{(J)})} \prod_{i=1}^{k} P_{W_i}(\mathbf{m}_{B_i}),$$

the inner products over the entries and over the states, W_i the word on B_i of the entries of the segment i at their total shares $\sum_J e_{L,i}^{(J)}$, and the fibre pinned by the conservation and by the homogeneity of every segment but one.

(iii) For k copies of one segment the integrand of (ii), read at each boundary as a function of the loads of all the starting states, is the kernel of one operator $\mathcal{C}$, restricted at each boundary to

the loads of the states that lie on a path there, which integrates the loads out over the fibre pinned by the homogeneity of the segment. With $\mathcal{C}_\circ$ the same kernel without that pin, which for one segment is implied by those of the others, the letter totals and the balance, the period is

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int \prod_L \frac{\prod_J \Gamma(c_{L,J})}{\Gamma(w_L)} \left(\mathcal{C}^{k-1}\mathcal{C}_\circ\right)(\mathbf{c},\mathbf{c})\, d\mathbf{c},$$

a power of one operator taken from the loads $\mathbf{c}$ at the starting states back to them. Its price is that of one segment and the binary digits of k (Chapter 33); evaluated, the power takes at most $\lfloor \log_2 k \rfloor$ squarings of $\mathcal{C}$ and as many products.

A closed chain is a chain whose last boundary is glued back to its first: the trace sums over the states at one boundary, and each starting state carries its own flow around the closed chain. Cells spliced along any graph of hinge pairs stay tables (Theorem 21.14); around a closed chain their letters are traces of products of channel matrices, and repeated cells give the power of one operator.

29.2 The procedure

The Newton sums of Lemmas 28.9 and 29.4 can be degenerate. When every station of room sides is a single room side, every room station letter is a monomial, the room word is a monomial, and its period diverges at every share; the period is then read by Theorem 28.10(i) or (ii), or on the share side of Theorem 28.12 with $\vec{\mathcal{N}}_R = 0$.

Definition 29.9 (The seam procedure)**.** *Given*: a family F with real data on its balance wall, every proper wall positive. *Classification*, from the tables alone: the molecules, the readings at the centre and the Foreman tree (Definition 16.14, steps (1)–(5)); for a molecule with a connected active table the tree is a chain whose bunches $B_0, \dots, B_{d-1}$ (Corollary 20.8) are the steps of the chain of bunches $S_k = B_0 \sqcup \cdots \sqcup B_{k-1}$, the leaf being the last step. *Seams*, from the tables alone, at every bunch S_k:

(1) group the patterns of S_k of each excess by their weighted children up to positive multiples, and the room sides by their weighted room children, with the multipliers normalized by their largest value (Definition 28.1); the stations, the station letters and the pair constants are determined;

(2) record the seam matrices H_i and the transfer matrices $\mathbf{N}_k^K$, $\mathbf{N}_k^\Psi$ of Theorem 29.3(i);

(3) record the orbits of the stations under the bijections that keep S_k and keep the potential (Proposition 28.6, Lemma 14.18).

Evaluation at a point $(\mathbf{m};\mathbf{s},\mathbf{t})$:

(4) the law of $S_1 = B_0$ (Definition 28.5), with its canonical scale;

(5) the transfers of the laws from bunch to bunch (Theorem 29.3(ii)), one operator for repeated nodes (Theorem 29.3(iii)), the transfer entries of pairs of stations in one orbit being images of one another (Proposition 28.6(i));

(6) the period at the leaf with its stand-in at the seam point of the last bunch S_d, integrated against the law of that bunch (Theorem 28.10(i)); equivalently the word of Corollary 29.5 on the contour through the native shares of Lemma 29.4, with the transverse pins of Theorem 28.12 at every bunch or node whose Newton sum is degenerate, every node a Newton

word with multilinear letters closed by its peak series with the references of Theorem 15.13(i) and every seam Gramma by its peak closing with the sharp references (Proposition 28.16(i)).

Output: the period, with the seams, the transfer matrices and the law on every bunch, ready to be reused when a node is changed. At a general Foreman tree the bunches of a node with several childcules are composed by Theorem 29.1, the childcules of a peel being the pieces of the core.

Nothing in the procedure is chosen. The tree is the Foreman tree (Definition 14.4); the stations are fixed by the children, and their multipliers by the normalization of the largest one; the scale of a piece is the geometric mean of the oriented extreme ratios of the active letters, each taken to the root of its width, kept by the symmetries of the family (Proposition 28.6(iii)); inside a seam Gramma the stations of the bunch are integrated before those of the room, the Foreman's order (Proposition 28.15), and another choice of the stations that are not kept changes the pins by a unimodular matrix; the real point of the share contour, where the word is used, is the native split at the stations and the entries (Lemmas 28.9 and 29.4); the references of the closings are those of Theorem 15.13(i) and the sharp ones, read off the coefficients. Repetition is paid once: stations in one orbit carry data that are images of one another, and repeated nodes one operator.

Proposition 29.10 (What the work depends on)**.** *(i) The law of a piece depends on the piece only through the joint law of the seam letters of its active letters under the measure $u^{\mathbf{m}_B-\mathbf{1}}d\omega_B$, with its exponents and wall: two pieces, on sets of any sizes, whose seam letters have the same joint law have equal laws and are interchangeable in every glue (Theorem 28.19); in particular so are two pieces whose seam letters and masses correspond under a bijection of their marks.*

(ii) Along a chain the letters cost one product of matrices of the sizes $|\mathcal{D}(S_l)|$, and a seam Gramma of a part without circuits costs no integration (Definition 28.13, Corollary 2.6).

(iii) The law of a bunch depends on the room only through the room children of the bunch, the tables on the bunch that the room presents to it: changing the room while keeping its room children and their stations changes no law below the bunch.

29.3 The correspondence

The procedure reads a realization forward, from its tables to its word. The other direction asks what the periods distinguish, and it has two answers: as a function of the masses the period determines the potential, which carries the classes of marks and the tree; at one realization each bunch is seen by its room through its wall and its law, and only through them.

Theorem 29.11 (The period function determines the potential)**.** *Let a family have real exponents $s_j, t_j \geq 0$ with $s_j + t_j > 0$ for every j and a period that converges at some masses on its balance wall, and let U be the set of the complex masses on the balance wall at which it converges.*

(i) With the measure dy on the hyperplane $\mathbf{1}^\perp = \{y \in \mathbb{R}^{\mathcal{E}} : \langle \mathbf{1}, y\rangle = 0\}$ of the section $\prod_e \alpha_e = 1$ (Lemma 16.2),

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\mathbf{1}^\perp} e^{\langle \mathbf{m}, y\rangle - \Phi(y)}\, dy, \qquad \mathbf{m} \in U,$$

Φ the potential at the table's gauge (Definition 8.18). The function $P_F(\cdot; \mathbf{s}, \mathbf{t})$ on U determines Φ, and is determined by it.

(ii) Two families on the same marks with the same exponents have the same period function exactly when they have the same irreducible factors with their exponents $(L, w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))$

(Theorem 10.6) and the same constant $\Phi(0) = \sum_j (s_j + t_j) \log K_j(\mathbf{1})$*; at the centre, every* $\delta_j = 0$*, exactly when they have the same* $(L, w_{\mathbf{E}}(L))$ *and the same constant. In particular a family and the family of its dual tables (Proposition 8.15), the bases replaced by their complements, have the same period function at the centre.*

(iii) A bijection g *of the marks keeps the period function,* $P_F(g^*\mathbf{m}; \mathbf{s}, \mathbf{t}) = P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ *on* U*, exactly when it keeps the potential, both parts. For a molecule, two marks lie in one class exactly when a bijection keeping the period function takes one to the other (Lemma 14.18), and when its children's weights factor across the childcules at every node, every such bijection carries the Foreman tree onto itself, with its classes, units, atoms, bunches and childcules (Proposition 14.21).*

(iv) At the centre the bijections that carry the family and those that exchange it keep the period function; at a lean the carrying ones keep it, and an exchanging one sends it to the period function of the opposite lean (Proposition 11.14). For a molecule of one table two marks lie in one class at the centre exactly when a bijection carrying or exchanging the table takes one to the other, and at a lean exactly when a carrying one does (Corollary 11.16), and a class of the centre that is not a class at a lean is there two classes of equal size, which an exchange swaps (Theorem 11.17).

So two marks of a molecule are told apart by the period function of the centre, or of a lean, exactly when no bijection keeping that potential joins them; for one table these are, at the centre, the bijections carrying or exchanging the table, and at a lean the carrying ones. The confusable classes, the classes of a lean that share a class of the centre, are exactly what the period function of the centre does not separate and that of a lean does. The tree is read off the period function but does not determine it; the irreducible factors with their exponents, and the constant $\Phi(0)$, do.

Theorem 29.12 (The correspondence)**.** *Let a molecule be read at a realization with masses on its balance wall and every proper wall positive, its children's weights factoring across the childcules at every node.*

(i) Forward. *Its period is the reassembly of the laws of the bunches of its Foreman tree with the leaves, by the seam procedure: the law of the first bunch on its seam torus, transferred along the chain by the later bunches as nodes, the childcules of a node composed at its core, and glued to the leaves by the stand-ins; nothing in it is chosen (Definition 29.9, Theorems 29.1 and 29.3).*

(ii) Backward at the realization. *At the node reached after* k *peels the room sees the bunch* S_k*, the union of the bunches peeled down to it, through its wall and its law, and only through them: bunches with the same stations, exponents, walls and laws are interchangeable in every family glued on that room, and under the hypotheses of Theorem 28.19 two different laws are told apart by the tinkers of one room and by plain partners (Theorems 28.19 and 28.22); the laws of nested bunches compose by Theorems 29.2 and 29.3.*

Whether the molecule has an atomic realization is read at a centre: it has one exactly when some centre is atomic, and an atom at a lean is an atom at the centre (Theorems 11.8 and 11.1). The word of the tree is the forward procedure written out, and retention is the statement that it loses nothing that a room could detect.

Chapter 30

Foreman fusion

A table of rank r and corank r^* is the contraction of $c = r^* - r$ new marks in an identically self-dual table, in which the new marks form a bunch in level form and their deletion leaves the dual table, exactly when every cocircuit of the table is a union of circuits and, at $c = 0$, the table has the weights of its dual. The bases of the larger table are then unique, and at $c \leq 1$ the larger table itself: they are those of the level cell of the pair formed by the dual table and the table, the new marks are its fusion bunch, and their children on the marks of the table are the lifts of the table toward its dual, one for each level, whose weights between the two extremes, the middle weights, are a choice paired by complement, all 1 unless others are named. This is Foreman fusion, and the table is Foreman-embedded in its fusion. The fused table is identically self-dual, so each of its marks comes off by one Beta corner at every pair of exponents (Corollary 20.19); its automorphisms contain those of the table that keep the middle weights, all of them when the middle weights are 1, times the permutations of the fusion bunch, and are often more; and every identically self-dual table is the fusion of each of its one-mark contractions. At the fusion width one the fusion bunch is one mark g, the letter of the fused table is $\Psi + \alpha_g K$, and the table is read off it by one Beta corner: the law of the fusion bunch is a point. At the width c the seam of the fusion bunch is one rectangle at every excess, its seam letters are the elementary symmetric polynomials of the new marks up to constants, all 1 for a list, and the fused table is the glue (Theorem 28.10) of a universal piece of width c, which does not depend on the table, with the lift partner, whose room letters are the letters of the lifts. The law of the universal piece is the image of a measure on the torus of the fusion bunch, of dimension $c-1$, and the law of the lift partner is a point only when its lift ratios are constant, which they are not at $c \geq 2$: these two laws are the price of width. Down the partial fusions the first step is a Beta corner and every later step is a cell whose law is not a point. Every one-mark contraction of the fused table has the fusion width one and fuses back into it: these cousins share one period up to Beta corners, and an automorphism of the fused table carries the cousin at a mark onto the cousin at the image of that mark. Across a two-sum of lists the widths add, the condition is read at the hinge by four conditions on each cell, and at the total width one a two-sum fuses exactly when both cells fuse, the cell of the width zero being then identically self-dual. A table of the fusion width $c \geq 0$ that does not fuse has uncovered marks, or has the width zero and the bases but not the weights of its dual; doubling the uncovered marks in parallel, or one mark in the second case, gives a table that fuses. The chapter ends with the branch these statements set at every node of the Foreman tree.

30.1 Quotients, lifts and the level cell of a pair

Definition 30.1 (Quotients; lifts; the level cell of a pair)**.** Let $\mathcal{M}$ and $\mathcal{N}$ be tables on one set of marks $\mathcal{E}$, of the ranks $r_{\mathcal{M}}$ and $r_{\mathcal{N}}$, with the rank functions $r_{\mathcal{M}}(Y)$ and $r_{\mathcal{N}}(Y)$ (Definition 7.4). $\mathcal{N}$ is a *quotient* of $\mathcal{M}$ when every set that is maximal for its rank in $\mathcal{N}$ is maximal for its rank in $\mathcal{M}$, as the child at $p+b$ is of the child at p (Proposition 12.6); its *rank difference* is $c = r_{\mathcal{M}} - r_{\mathcal{N}}$. For $0 \le j \le c$ the *j-th lift of $\mathcal{N}$ toward $\mathcal{M}$* is the table $\mathcal{L}_j(\mathcal{M},\mathcal{N})$ on $\mathcal{E}$ with the rank function

$$r_{\mathcal{L}_j(\mathcal{M},\mathcal{N})}(Y) = \min\big(r_{\mathcal{M}}(Y),\ r_{\mathcal{N}}(Y)+j\big) \qquad (Y \subseteq \mathcal{E}),$$

a table by Proposition 30.2(iii) below; for a bunch B of a table and $(\mathcal{M},\mathcal{N}) = (\mathcal{B}|B, \mathcal{B}/R)$ these are the lifts of Proposition 12.17. For a set G of c new marks the *level cell* of the pair $(\mathcal{M},\mathcal{N})$ is the table on $\mathcal{E} \sqcup G$ whose bases are the sets $X \sqcup Q$ with $Q \subseteq G$ and X a basis of $\mathcal{L}_{c-|Q|}(\mathcal{M},\mathcal{N})$; for $(\mathcal{B}|B, \mathcal{B}/R)$ it is the level cell of the bunch B (Lemma 28.27). It carries any positive weights, and G is in level form in it (Proposition 14.12) when the weight of $X \sqcup Q$ depends on X and $|Q|$ alone.

Proposition 30.2 (The level cell of a pair)**.** *Let $\mathcal{N}$ be a quotient of $\mathcal{M}$ on $\mathcal{E}$ with the rank difference c, let G be a set of c new marks, and in this statement write $\mathcal{L}_j$ for $\mathcal{L}_j(\mathcal{M},\mathcal{N})$.*

(i) For tables $\mathcal{M}$ and $\mathcal{N}$ on $\mathcal{E}$, $\mathcal{N}$ is a quotient of $\mathcal{M}$ exactly when $r_{\mathcal{M}}(Y) - r_{\mathcal{M}}(X) \ge r_{\mathcal{N}}(Y) - r_{\mathcal{N}}(X)$ for all $X \subseteq Y \subseteq \mathcal{E}$. Then $c \ge 0$, every set independent in $\mathcal{N}$ is independent in $\mathcal{M}$, every set spanning $\mathcal{M}$ spans $\mathcal{N}$, and a quotient of $\mathcal{N}$ is a quotient of $\mathcal{M}$.

(ii) The level cell of $(\mathcal{M},\mathcal{N})$ is a table of rank $r_{\mathcal{M}}$ with the rank function

$$r^\natural(Y \sqcup P) = \min\big(r_{\mathcal{M}}(Y) + |P|,\ r_{\mathcal{N}}(Y) + c\big) \qquad (Y \subseteq \mathcal{E},\ P \subseteq G).$$

In it every subset of G is a pattern of the bunch G, $d(G) = 0$ and $c(G) = c$; the child at $Q \subseteq G$ is $\mathcal{L}_{c-|Q|}$; the deletion of G is $\mathcal{M}$ and the contraction of G is $\mathcal{N}$; and every permutation of G maps its bases onto themselves.

(iii) Each $\mathcal{L}_j$ is a table of rank $r_{\mathcal{N}}+j$ whose bases are the sets of $r_{\mathcal{N}}+j$ marks that are independent in $\mathcal{M}$ and span $\mathcal{N}$; $\mathcal{L}_0 = \mathcal{N}$, $\mathcal{L}_c = \mathcal{M}$, and $\mathcal{L}_j$ is a quotient of $\mathcal{L}_{j'}$ for $j \le j'$. Lifts compose: for $0 \le a \le b \le c$ and $0 \le i \le b-a$,

$$\mathcal{L}_i(\mathcal{L}_b, \mathcal{L}_a) = \mathcal{L}_{a+i}.$$

(iv) For disjoint $C, D \subseteq G$ the minor of the level cell contracting C and deleting D is the level cell of the pair $(\mathcal{L}_{c-|C|}, \mathcal{L}_{|D|})$ on $\mathcal{E} \sqcup \big(G \setminus (C \cup D)\big)$.

The level cell stores a whole chain of lifts on c new marks: one level of the bunch G for each lift, the deletion of G at the top and its contraction at the bottom, and every interval of the chain as a minor at the marks of G. Nothing in its bases is chosen except the names of the new marks, which every permutation of G moves without moving the bases; its weights are a choice.

30.2 The fused table

Lemma 30.3 (Complements of the lifts toward the dual)**.** *Let $\mathcal{B}$ be a table of rank r on $\mathcal{E}$, $n = |\mathcal{E}|$, which is a quotient of its dual table $\mathcal{B}^*$, of rank $r^* = n - r$ (Definition 30.1), let $c = r^* - r$, and let $\mathcal{L}_j = \mathcal{L}_j(\mathcal{B}^*, \mathcal{B})$, $0 \le j \le c$. Then X is a basis of $\mathcal{L}_j$ exactly when $\mathcal{E} \setminus X$ is a basis of $\mathcal{L}_{c-j}$: the dual table of $\mathcal{L}_j$ has the bases of $\mathcal{L}_{c-j}$.*

Definition 30.4 (Fusion width; Foreman fusion; majors)**.** Let $(\mathcal{B},\theta)$ be a table of rank r and corank r^* on $\mathcal{E}$, $n=|\mathcal{E}|$, with its dual table $(\mathcal{B}^*,\theta^*)$, $\theta^*_{\mathcal{E}\setminus T}=\theta_T$ (Definition 7.1), and with the rank function $r(Y)$ and the dual rank function $r^*(Y)$ (Definition 7.12). Its *fusion width* is $c=r^*-r=n-2r$, and $(\mathcal{B},\theta)$ is *fusable* when $\mathcal{B}$ is a quotient of $\mathcal{B}^*$ (Definition 30.1) and, at the fusion width zero, $\theta^*=\theta$. A quotient forces $c\ge 0$ (Proposition 30.2(i)), and a quotient of the same rank has the same bases, so a fusable table of the fusion width zero is identically self-dual (Corollary 23.12(iii)), and a list of the fusion width zero is fusable exactly when it is identically self-dual. Let $c\ge 0$, and let G be a set of c new marks, the *fusion bunch*.

(i) A *major* of $(\mathcal{B},\theta)$ is a table on $\mathcal{E}\sqcup G$ whose contraction of G is $(\mathcal{B},\theta)$ and whose deletion of G is $(\mathcal{B}^*,\theta^*)$.

(ii) The *lifts* of a fusable $(\mathcal{B},\theta)$ are $\mathcal{L}_j=\mathcal{L}_j(\mathcal{B}^*,\mathcal{B})$, $0\le j\le c$, with the weights $\theta^{(j)}$: $\theta^{(0)}=\theta$ on $\mathcal{L}_0=\mathcal{B}$, $\theta^{(c)}=\theta^*$ on $\mathcal{L}_c=\mathcal{B}^*$, and for $0<j<c$ the *middle weights*, positive weights with $\theta^{(c-j)}_{\mathcal{E}\setminus X}=\theta^{(j)}_X$, complement mapping the bases of $\mathcal{L}_j$ onto those of $\mathcal{L}_{c-j}$ (Lemma 30.3), a choice, all 1 unless others are named; so the lifts of a list are lists unless other middle weights are named. Its *lift letters* are $L_l=K_{\mathcal{L}_{c-l}}$, $0\le l\le c$, read with these weights, so that $L_0=\Psi$ and $L_c=K$, and its *lift ratios* are $z_j=K^2_{\mathcal{L}_j}/(K_{\mathcal{L}_{j-1}}K_{\mathcal{L}_{j+1}})$, $0<j<c$.

(iii) The *Foreman fusion* of a fusable $(\mathcal{B},\theta)$, with its middle weights, is the level cell of $(\mathcal{B}^*,\mathcal{B})$ on $\mathcal{E}\sqcup G$ with the weight $\theta^{(c-|Q|)}_X$ on $X\sqcup Q$, written $(\mathcal{B}^\sharp,\theta^\sharp)$; $(\mathcal{B},\theta)$, the contraction of the fusion bunch in it, is its *Foreman embedding*. For $G'\subseteq G$ the *partial fusion* at G' is $\mathcal{B}^\sharp/(G\setminus G')$, written $\mathcal{B}^\sharp_i$ for $|G'|=i$ when the marks of G' are not named; $\mathcal{B}^\sharp_0=\mathcal{B}$ and $\mathcal{B}^\sharp_c=\mathcal{B}^\sharp$.

At the fusion width zero G is empty and $\mathcal{B}^\sharp=\mathcal{B}$. A table of negative fusion width c has a dual of the fusion width $-c$, and the dual is the table that is fused.

Theorem 30.5 (Foreman fusion)**.** *Let $(\mathcal{B},\theta)$ be fusable, of the fusion width $c\ge 0$, rank r and corank r^*, with its Foreman fusion $(\mathcal{B}^\sharp,\theta^\sharp)$ on $\mathcal{E}\sqcup G$ and its lifts $\mathcal{L}_j$.*

(i) $\mathcal{B}^\sharp$ is a table of rank r^* on $n+c=2r^*$ marks with the rank function

$$r^\sharp(Y\sqcup P)=\min\big(r^*(Y)+|P|,\ r(Y)+c\big)\qquad (Y\subseteq\mathcal{E},\ P\subseteq G),$$

its contraction of G is $(\mathcal{B},\theta)$ and its deletion of G is $(\mathcal{B}^*,\theta^*)$: it is a major of $(\mathcal{B},\theta)$.

(ii) The dual of each lift is a lift, weights included: $\mathcal{L}^*_j=\mathcal{L}_{c-j}$. So $(\mathcal{B}^\sharp,\theta^\sharp)$ is identically self-dual (Corollary 23.12(iii)), and

$$K^\sharp=\Psi^\sharp=\sum_{l=0}^{c}e_l(\alpha_G)\,L_l(\alpha_{\mathcal{E}}),\qquad L_l=K_{\mathcal{L}_{c-l}}=\Psi_{\mathcal{L}_l}.$$

(iii) Every basis of every major of $(\mathcal{B},\theta)$ is a basis of $\mathcal{B}^\sharp$.

(iv) Every major of $(\mathcal{B},\theta)$ in which G is in level form, equivalently that every permutation of G carries, has the bases of $\mathcal{B}^\sharp$, the weights θ on the bases $X\sqcup G$ and θ^* on the bases $X\sqcup\varnothing$, and on the others positive weights depending on X and $|Q|$ alone; it is identically self-dual exactly when these weights, read as middle weights $\theta^{(c-|Q|)}_X$ on the bases $X\sqcup Q$, are paired as in Definition 30.4(ii), and then it is the Foreman fusion with these middle weights. So the identically self-dual majors in level form are the Foreman fusions with all the choices of middle weights, and for a list $\mathcal{B}^\sharp$ is the only one that is a list. For every bijection β of $\mathcal{E}$ carrying $(\mathcal{B},\theta)$ with $\theta^{(j)}\circ\beta=\theta^{(j)}$ for $0<j<c$, as for the middle weights 1, and every permutation π of G, $\beta\sqcup\pi$ carries $(\mathcal{B}^\sharp,\theta^\sharp)$.

(v) *At the fusion width one, $G = \{g\}$, the bases of $\mathcal{B}^\sharp$ are the sets $T + g$ and $\mathcal{E} \setminus T$, $T \in \mathcal{B}$, both with the weight θ_T, $K^\sharp = \Psi + \alpha_g K$, and $(\mathcal{B}^\sharp, \theta^\sharp)$ is the only major of $(\mathcal{B}, \theta)$.*

The fusion stores the table and its dual as the two extreme children of one bunch, and fills the levels between them with the lifts, which is the only way to fill them in level form that the exchange property allows (Proposition 28.29(d)): the containment (iii) says that no major has a basis the fusion lacks, and level form at G selects the bases of the fusion among the majors, its weights between the extremes being a choice paired by complement. Self-duality comes from the chain of lifts turning over under duality, the j-th lift onto the $(c - j)$-th. At the fusion width one there is no level between the extremes, and the fusion is the only major; at the width c the middle levels admit majors that are not in level form, among them identically self-dual ones, and the bases of the fusion are the ones that every permutation of the new marks keeps.

Theorem 30.6 (The tell)**.** *For a table $(\mathcal{B}, \theta)$ of the fusion width c, conditions (a), (b) and (c) are equivalent and imply $c \geq 0$. The table is fusable exactly when they hold and, at $c = 0$, $\theta^* = \theta$; so a list, or a table of the fusion width $c \geq 1$, is fusable exactly when (a) holds. For $c \geq 0$ the table is fusable exactly when (d) holds.*

(a) *$\mathcal{B}$ is a quotient of $\mathcal{B}^*$.*

(b) *Every cocircuit of $\mathcal{B}$ is a union of circuits of $\mathcal{B}$ (Definition 7.4).*

(c) *For every mark e and every split $\mathcal{E} \setminus \{e\} = A \sqcup A'$, the rank of A or the rank of A' does not grow when e is added to it.*

(d) *$(\mathcal{B}, \theta)$ has a major (Definition 30.4(i)); for $c \geq 1$, equivalently, some table on $\mathcal{E} \sqcup G$, $|G| = c$, has $\mathcal{B}^*$ and $\mathcal{B}$ as the children of the empty pattern and of the full pattern of its bunch G.*

A table that satisfies (b) has no coloop. For $c < 0$ the dual table has the fusion width $-c$, and the theorem read on it has (b) in the form: every circuit of $\mathcal{B}$ is a union of cocircuits.

The tell is read on the Skeleton of the table (Definition 7.12): the cocircuit layer must be covered by the circuit layer, cocircuit by cocircuit, before anything is integrated. In the form (c) it is a statement about one mark at a time: however the other marks are split in two, one half spans the mark. A coloop fails it at once, being a cocircuit of one mark; a loop lies in no cocircuit and never fails it. The weights enter only at the width zero, where the bases of a table that passes are those of its dual and the table fuses when its weights are those of its dual as well.

Corollary 30.7 (The fused tables are the identically self-dual tables)**.** *(i) Let $(\mathcal{H}, \theta)$ be identically self-dual on $\mathcal{E}'$ and $C \subseteq \mathcal{E}'$ independent, every transposition of two marks of C carrying $(\mathcal{H}, \theta)$. Then $\mathcal{H}/C$ is fusable of the fusion width $|C|$, and $(\mathcal{H}, \theta)$ is its Foreman fusion with the fusion bunch C and the middle weights read off θ. In particular every identically self-dual table on at least one mark is the Foreman fusion of each of its one-mark contractions, and the fused tables are exactly the identically self-dual tables.*

(ii) *If $\mathcal{B}$ is connected and fusable, $\mathcal{B}^\sharp$ is connected.*

Proposition 30.8 (Partial fusions and inner lifts)**.** *Let $(\mathcal{B}, \theta)$ be fusable of the fusion width $c \geq 1$, with its lifts $\mathcal{L}_j$.*

(i) *For disjoint $C, D \subseteq G$, $\mathcal{B}^\sharp/C \setminus D$ is the level cell of $(\mathcal{L}_{c-|C|}, \mathcal{L}_{|D|})$ on $\mathcal{E} \sqcup \big(G \setminus (C \cup D)\big)$, with the weights of $\mathcal{B}^\sharp$.*

(ii) *The partial fusion at G', $|G'| = i$, is the level cell of $(\mathcal{L}_i, \mathcal{B})$ on $\mathcal{E} \sqcup G'$. It has the fusion width $c - i$, it is fusable, and its Foreman fusion with the middle weights read off $\theta^\sharp$ is $(\mathcal{B}^\sharp, \theta^\sharp)$ with*

the fusion bunch $G \setminus G'$. So $\mathcal{B} = \mathcal{B}_0^\sharp, \mathcal{B}_1^\sharp, \dots, \mathcal{B}_c^\sharp = \mathcal{B}^\sharp$, each the contraction of the next at one mark, all fuse into one table.

(iii) *$\mathcal{L}_j$ has the fusion width $c - 2j$. For $2j \le c$ it is fusable, with the Foreman fusion $\mathcal{B}^\sharp/C \setminus D$, $|C| = |D| = j$, its middle weights read off $\theta^\sharp$. For even c the middle lift $\mathcal{L}_{c/2}$ is identically self-dual; for odd c the lift $\mathcal{L}_{(c-1)/2}$ has the fusion width one and fuses into $\mathcal{B}^\sharp/C \setminus D$ with $|C| = |D| = (c-1)/2$.*

So the fused table holds, at its minors on the fusion bunch, the fusion of every table in the chain between $\mathcal{B}$ and $\mathcal{B}^*$: each partial fusion fuses into it again, and each inner lift fuses into the minor that deletes as many marks of G as it contracts.

Remark 30.9 (Examples). (i) The uniform list $\binom{\mathcal{E}}{r}$ with $n \ge 2r$ is fusable, its cocircuits having $r^* + 1 \ge r + 1$ marks; its lifts are the uniform lists $\binom{\mathcal{E}}{r+j}$, and its fusion is $\binom{\mathcal{E} \sqcup G}{r^*}$: the fusion of $\binom{[n]}{r}$ is $\binom{[2r^*]}{r^*}$, and the fusion of $\binom{[5]}{2}$ is $\binom{[6]}{3}$; for $\binom{[2r+1]}{r}$, $r \ge 2$, the fusion $\binom{[2r+2]}{r+1}$ is the table one mark larger of Proposition 39.48(ii).

(ii) A list of rank two with no loop has as its bases the pairs of marks in different classes of a partition of its marks, the classes of two or more marks being its parallel classes (Definition 19.1). With $k \ge 3$ classes it is fusable exactly when $k \ge 4$, or $k = 3$ and every class has at least two marks: its cocircuits are the complements of the classes, and a mark of a class P' lies in a circuit inside the complement of another class P exactly when P' has a second mark or two classes other than P and P' exist. So the list $\binom{[3]}{2}$ and every such list with three classes one of which is a single mark fail, and those with four or more classes pass at every fusion width (receipt `t03_rank2_census.py`, every connected list of rank two on 5 to 9 marks).

(iii) The list of rank two on $n \ge 5$ marks with one parallel pair P and every other mark in a class of its own has the fusion width $n - 4$, and its fusion has exactly two sets of $n - 2$ marks that are not bases, the complementary halves $P \sqcup G$ and $\mathcal{E} \setminus P$; its automorphisms are the $2((n-2)!)^2$ bijections keeping this pair of halves, transitive on the marks. At $n = 5$ the fusion is the two-sum of two copies of $\binom{[4]}{2}$.

(iv) Of the 108 lists of rank three on seven marks, up to the names of the marks, 22 are fusable, of the fusion width one, and they fuse into the 17 identically self-dual lists on eight marks; of the 18 that are prime (Theorem 23.16), 10 are fusable, those whose lines all have three marks and meet pairwise (receipts `census7.py`, `atoms7.py`).

(v) The majors of $\binom{[4]}{1}$ that are lists, of the fusion width two, are 79 lists on six marks; 7 of them are identically self-dual and one is in level form at G, and that one is the fusion $\binom{[6]}{3}$ (receipt `t02_majors_U14.py`). At rank three the following are fusable, of the fusion widths two and three: $\binom{[8]}{3}$ and $\binom{[9]}{3}$; the list F_7 on the marks $\mathbb{Z}/7$ whose dependent sets of three marks are the seven sets $\{i, i+1, i+3\}$, with a free mark added; the list A_9 on the marks $\mathbb{F}_3^2$ whose dependent sets of three marks are the twelve sets $\{a, a+d, a+2d\}$, $d \ne 0$, and A_9 less a mark; the list Z_9 on the marks a_i, b_i, c_i, $i = 1, 2, 3$, whose dependent sets of three marks are $\{a_1, a_2, a_3\}$, $\{b_1, b_2, b_3\}$, $\{c_1, c_2, c_3\}$ and the six sets $\{a_i, b_j, c_{i+j-2}\}$ with $i \ne j$; and five further lists of rank three whose dependent sets of three marks form configurations of lines; and their fusions have the seam of Proposition 30.10 (receipt `w01_seam_rank3.py`).

30.3 The seam of the fusion bunch and the glue

Proposition 30.10 (The seam of the fusion bunch)**.** *Let $(\mathcal{B}, \theta)$ be fusable of the fusion width $c \ge 1$, with its fusion $(\mathcal{B}^\sharp, \theta^\sharp)$, its lifts $\mathcal{L}_j$ and its lift letters L_l.*

(i) *In $\mathcal{B}^\sharp$ the bunch G has $\mathrm{pat}(G) = 2^G$, $d(G) = 0$ and $c(G) = c$, and it is in level form. At the excess l (Definition 28.1) there is one station of patterns, all $Q \subseteq G$ with $|Q| = l$, with the multipliers 1 and the station letters $e_l(\alpha_G)$ and $e_{c-l}(\alpha_G)$, and one station D_l of room sides, the bases of $\mathcal{L}_{c-l}$, whose room children are positive multiples of $\binom{G}{l}$. Every excess is one rectangle, $\mathcal{B}^\sharp$ is its own level envelope (Theorem 28.28), and the room-station factorization (Proposition 28.2(iii)) is*

$$K^\sharp = \sum_{l=0}^{c} e_l(\alpha_G)\, L_l(\alpha_\mathcal{E}), \qquad \Psi^\sharp = \sum_{l=0}^{c} e_{c-l}(\alpha_G)\, L_{c-l}(\alpha_\mathcal{E}).$$

The seam letters of G are $K_{(D_l)} = \mu_l e_l(\alpha_G)$ and $\Psi_{(D_l)} = \mu_l e_{c-l}(\alpha_G)$, with μ_l the largest weight of $\mathcal{L}_{c-l}$, the room-station letters are L_l/μ_l and L_{c-l}/μ_l, and the pair constants are μ_l; for a list every μ_l is 1.

(ii) *The interaction of G (Definition 12.18) and its dual interaction are*

$$\iota_G = \iota_G^* = \sum_{j=0}^{c} (-1)^j \binom{c}{j} \log K_{\mathcal{L}_j}(\alpha_\mathcal{E}),$$

and $\iota_G = -\log z_1$ at $c = 2$.

(iii) *At the table's gauge the pattern probabilities of the excess l add up to $\Pi_l = \binom{c}{l} L_l(\mathbf{1})/K^\sharp(\mathbf{1})$, with $\Pi_l = \Pi_{c-l}$, and the peel of $((\mathcal{B}^\sharp, \theta^\sharp); s, t)$ at G (Definition 14.1) is the family of the lifts $((\mathcal{L}_j, \theta^{(j)}); \Pi_j s, \Pi_j t)_{0 \le j \le c}$, whose word is $\prod_j K_{\mathcal{L}_j}^{-2\sigma\Pi_j}$.*

Between the extremes the seam of a bunch can carry several rectangles (Proposition 28.3(iii)); the fusion bunch carries one at every excess, with the multipliers 1 on its patterns and the pair constants μ_l, and what it presents to its room at the excess l is the elementary symmetric polynomial of degree l of its own marks. These letters do not depend on the table. Everything the table contributes sits on the other side of the seam, in the letters of its lifts.

Definition 30.11 (The universal piece; the lift partner)**.** For $c \ge 1$ the *universal piece of width* c is the piece on a set G of c marks for one table (Definition 28.4) with the stations $D_0, \dots, D_c$, D_l of the excess l, the seam letters $V_{D_l} = e_l(\alpha_G)$ and $V_{D_l}^* = e_{c-l}(\alpha_G)$, and $d = d^* = 0$. The *lift partner* of a fusable $(\mathcal{B}, \theta)$ of the fusion width c is the partner on $\mathcal{E}$ with the same stations, the room letters $L_{D_l} = L_l$ and $L_{D_l}^* = L_{c-l}$, and $e = e^* = r$.

By Proposition 30.10(i) and Proposition 28.2(iii) the glue of the universal piece of width c and the lift partner of $(\mathcal{B}, \theta)$ is the fusion $(\mathcal{B}^\sharp, \theta^\sharp)$. Since $\mathcal{B}^\sharp$ is identically self-dual its word at the exponents (s, t) is $K^{\sharp\,-2\sigma}$, $\sigma = \frac{s+t}{2}$ the strength (Definition 8.1), and the fused table is read below as the family $((\mathcal{B}^\sharp, \theta^\sharp); 2\sigma, 0)$, with the one active letter $K^\sharp$; its walls are those at (s, t), d and d^* agreeing on an identically self-dual table. Its balance wall is $m_\mathcal{E} + m_G = 2\sigma r^*$. In this section $H_G = \{w \in \mathbb{R}^G : \sum_g w_g = 0\}$ is the torus of G in logarithmic coordinates, with the measure $d\lambda_G(w) = dw_{g_1} \cdots dw_{g_{c-1}}$ for an order $g_1, \dots, g_c$ of G, and $\mathbf{e}(y) = (e_0(y), \dots, e_c(y))$ for $y \in (0, \infty)^G$.

Theorem 30.12 (The glue of a fused table)**.** *Let $(\mathcal{B}, \theta)$ be fusable of the fusion width $c \ge 1$, with its fusion $(\mathcal{B}^\sharp, \theta^\sharp)$ at a strength $\sigma > 0$ and masses $\mathbf{m}$ on $\mathcal{E} \sqcup G$ on its balance wall. For $\mathbf{x} = (x_0, \dots, x_c) \in (0, \infty)^{c+1}$ let $W[\mathbf{x}]$ be the Newton word on $\mathcal{E} \sqcup \{g\}$, g the stand-in of the universal piece, with the one letter $\sum_{l=0}^{c} x_l\, \alpha_g^l\, L_l(\alpha_\mathcal{E})$ at the exponent 2σ (Definition 25.1).*

(i) The universal law. *The scale of the universal piece is* $\gamma_G^{\rho} = e_c(\alpha_G)^{1/c}$, *its seam vector at the excess* l *is* $e_l(\alpha_G)/\gamma_G^{\rho}(\alpha_G)^l$, *and its law is the image of the measure* $c\,e^{\langle \mathbf{m}|_G, w\rangle}\,d\lambda_G(w)$ *on* H_G *under* $w \mapsto \mathbf{e}(e^w)$.

(ii) The torus form. *In* $[0,\infty]$,

$$P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t) = c \int_{H_G} e^{\langle \mathbf{m}|_G, w\rangle}\, P_{W[\mathbf{e}(e^w)]}\big(\mathbf{m}|_{\mathcal{E}}, m_G\big)\, d\lambda_G(w),$$

the stand-in carrying the mass m_G, *the wall of the universal piece.*

(iii) The scale of the stand-in. *At a fixed* $\alpha_{\mathcal{E}}$ *write* $\sum_l x_l L_l\, y^l = x_c L_c \prod_{i=1}^{c}(y + \xi_i)$; *the* ξ_i, *in* $\mathbb{C} \setminus (-\infty, 0]$, *are the Skeleton values of the stand-in, read on the letter of* $W[\mathbf{x}]$ *as in Definition 12.10. For* $0 < m_G < 2\sigma c$,

$$\int_0^\infty \rho^{m_G - 1}\Big(\sum_l x_l \rho^l L_l\Big)^{-2\sigma} d\rho = (x_c L_c)^{-2\sigma}\, \Gamma_c^2\begin{bmatrix} m_G, & 2\sigma c - m_G \\ 2\sigma, & \dots,\ 2\sigma \end{bmatrix}\, \mathsf{B}^{\{J\}}_{(2\sigma)_{i\in J},\, 2\sigma c - m_G;\, I_J}(\boldsymbol{\xi}),$$

the Breta of one bunch $J = \{1, \dots, c\}$ *of Proposition 15.1, coincident Skeleton values merging.*

(iv) One share per level. *On every contour of shares* $\mathbf{a} = (a_0, \dots, a_c)$ *with positive real parts on the pins* $\sum_l a_l = 2\sigma$ *and* $\sum_l l\, a_l = m_G$ *at which the two periods below converge, as at the native split at the stations when every proper wall of* $\mathcal{B}^\sharp$ *is positive (Theorem 28.10(iii)),*

$$P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t) = \int_{\sum_l a_l = 2\sigma,\ \sum_l l\, a_l = m_G} \Gamma_1^{c+1}\begin{bmatrix} \mathbf{a} \\ 2\sigma \end{bmatrix}\, P_{\prod_{l=1}^{c} e_l^{-a_l}}\big(\mathbf{m}|_G\big)\, P_{\prod_{l=0}^{c} L_l^{-a_l}}\big(\mathbf{m}|_{\mathcal{E}}\big),$$

a fibre integral of the dimension $c - 1$.

(v) The universal factor. *For* $c \geq 2$, $P_{\prod_{l=1}^{c} e_l^{-a_l}}(\mathbf{m}|_G) = P_{\prod_{l=1}^{c-1} e_l^{-a_l}}(\mathbf{m}|_G - a_c\mathbf{1})$, *the period on* G *of the uniform lists* $\binom{G}{l}$, $0 < l < c$, *at the exponents* $(a_l, 0)$ *and the masses* $\mathbf{m}|_G - a_c\mathbf{1}$; *at* $c = 1$ *it is* 1.

The universal piece is absorbed into its stand-in at its own scale, the geometric mean of the scales of G, and what is left of it is a point of the torus of G, entering the letter of the stand-in through the elementary symmetric polynomials of the exponentials of that point. The stand-in enters with the degree c, so $W[\mathbf{x}]$ is not the word of a table for $c \geq 2$ (Theorem 28.24), and integrating its scale is one Breta over the c Skeleton values of the stand-in. Opened on its shares, the fused table carries one share per level and two pins, the strength and the scale, so $c - 1$ shares are free; the universal factor is the period of the uniform lists on G, and everything that depends on the table is the period of the family of its lifts, in which the table itself is the member with $a_1 = \dots = a_{c-1} = 0$, $P_{L_0^{-a_0} L_c^{-a_c}}(\mathbf{m}|_{\mathcal{E}}) = P_{\mathcal{B}}(\mathbf{m}|_{\mathcal{E}}; a_c, a_0)$.

Corollary 30.13 (Width one)**.** *At the fusion width one, on the balance wall of* $\mathcal{B}^\sharp$ *with every proper wall positive,*

$$P_{\mathcal{B}^\sharp}(\mathbf{m}|_{\mathcal{E}}, m_g; s, t) = \Gamma_1^2\begin{bmatrix} m_g, & 2\sigma - m_g \\ & 2\sigma \end{bmatrix}\, P_{\mathcal{B}}\big(\mathbf{m}|_{\mathcal{E}}; m_g, 2\sigma - m_g\big)\,:$$

the exponents of the table are read off the mass of the fusion bunch. The universal piece is the mark g *itself, the laws of the universal piece and of the lift partner are points, and the share form of Theorem 30.12(iv) has no free share.*

Corollary 30.14 (Width two). *Let $c=2$, $G=\{g_1,g_2\}$, with the masses m_1,m_2 on g_1,g_2.*

(i) On a line of one free share a_2, with $a_1=m_1+m_2-2a_2$ and $a_0=2\sigma-m_1-m_2+a_2$,

$$P_{\mathcal{B}^\sharp}(\mathbf{m};s,t)=\frac{1}{2\pi i}\int\frac{\Gamma(a_0)\,\Gamma(a_2)\,\Gamma(m_1-a_2)\,\Gamma(m_2-a_2)}{\Gamma(2\sigma)}\,P_{L_0^{-a_0}L_1^{-a_1}L_2^{-a_2}}\big(\mathbf{m}|\mathcal{E}\big)\,da_2:$$

the universal factor is the Beta corner $\Gamma_{\,1}^{\,2}\Big[\begin{matrix}m_1-a_2,\ m_2-a_2\\ a_1\end{matrix}\Big]$, and the Gamma of the middle level cancels.

(ii) The universal law is the image of $2v^{m_1-m_2-1}dv$ on $(0,\infty)$ under $v\mapsto(1,v+v^{-1},1)$, and

$$P_{\mathcal{B}^\sharp}(\mathbf{m};s,t)=2\int_0^\infty v^{m_1-m_2-1}\,P_{(L_0+(v+v^{-1})\alpha_gL_1+\alpha_g^2L_2)^{-2\sigma}}\big(\mathbf{m}|\mathcal{E},m_1+m_2\big)\,dv.$$

(iii) At a fixed $\alpha_{\mathcal{E}}$, for $0<m_1<2\sigma$ and $0<m_2<2\sigma$,

$$\int_{(0,\infty)^2}\alpha_{g_1}^{m_1-1}\alpha_{g_2}^{m_2-1}K^{\sharp\,-2\sigma}\,d\alpha_G$$
$$=\Gamma_{\,1}^{\,2}\Big[\begin{matrix}m_2,\ 2\sigma-m_2\\ 2\sigma\end{matrix}\Big]\,L_0^{m_1+m_2-2\sigma}L_1^{-m_1-m_2}\int_0^\infty x^{m_1-1}(1+x)^{m_2-2\sigma}\big(1+z_1^{-1}x\big)^{-m_2}dx,$$

a Beta corner times a Breta of one bunch with the two coefficients 1 and z_1 (Proposition 15.1). At $m_1=m_2=1$, $2\sigma=2$ it is $\log z_1/(L_1^2-L_0L_2)$ where $z_1\neq1$ and L_1^{-2} where $z_1=1$, and at integer $m_1,m_2,2\sigma$, where $z_1\neq1$, it is $L_0^{m_1+m_2-2\sigma}L_1^{-m_1-m_2}\big(R_1(z_1)+R_2(z_1)\log z_1\big)$ with rational functions R_1,R_2.

30.4 The price of width

Lemma 30.15 (Products of lifts). *Let $\mathcal{N}$ be a quotient of $\mathcal{M}$ with the rank difference c, its lifts $\mathcal{L}_j=\mathcal{L}_j(\mathcal{M},\mathcal{N})$ carrying any positive weights. For $0\le i'<i\le j<j'\le c$ with $i+j=i'+j'$, $K_{\mathcal{L}_i}K_{\mathcal{L}_j}$ has a monomial in which exactly $r_{\mathcal{N}}+i$ exponents equal 2, while in every monomial of $K_{\mathcal{L}_{i'}}K_{\mathcal{L}_{j'}}$ at most $r_{\mathcal{N}}+i'$ do; so the two products are not proportional. In particular, for $c\ge2$, no ratio $K_{\mathcal{L}_j}^2/(K_{\mathcal{L}_{j-1}}K_{\mathcal{L}_{j+1}})$, $0<j<c$, is constant on the orthant.*

Theorem 30.16 (The price of width). *Let $(\mathcal{B},\theta)$ be fusable of the fusion width $c\ge1$, with its fusion read at a strength $\sigma>0$ as in Theorem 30.12, its lift letters L_l and its lift ratios z_j.*

(i) The law of the universal piece is carried by one point exactly when $c=1$. The law of the lift partner is carried by one point exactly when every lift ratio is constant, that is exactly when $c=1$.

(ii) The Skeleton braid. *Write $\sum_l\binom{c}{l}L_l\,x^l=L_0\prod_{i=1}^c(1+\eta_ix)$; the η_i are the reciprocals of the Skeleton values of G at the gauge $(\mathbf{1}_G,\alpha_{\mathcal{E}})$ (Definition 12.10). Then*

$$K^\sharp=\frac{L_0}{c!}\sum_{\pi\in S_c}\prod_{i=1}^c\big(1+\alpha_{g_i}\eta_{\pi(i)}\big)=\frac{L_0}{c!}\,\mathrm{per}\,\big(\mathbf{1}\mathbf{1}^{\mathsf{T}}+\alpha_G\boldsymbol{\eta}^{\mathsf{T}}\big),$$

$\mathbf{1}\mathbf{1}^{\mathsf{T}}$ the $c\times c$ matrix of ones and per *the permanent: $K^\sharp$ is L_0 times the average, over the bijections from the marks of G to the Skeleton values, of products of c letters of one mark. At a gauge $\alpha_{\mathcal{E}}$ the η_i coincide exactly when every lift ratio is 1 there, and for $c\ge2$ they coincide on no open set of the orthant: $K^\sharp$ is not $L_0\prod_{g\in G}(1+\alpha_g\eta)$ for any one function η.*

(iii) The partial fusions. *For $1 \le i \le c$, $G' \subseteq G$ with $|G'| = i$ and $g \in G'$, read the partial fusion at G' as a cell with the hinge g (Definition 21.1). For $i = c$ its ratio is* 1*, and on the balance wall of $\mathcal{B}^\sharp$ with every proper wall positive*

$$P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t) = \Gamma_{\substack{2\\1}}\begin{bmatrix} m_g, & 2\sigma - m_g \\ & 2\sigma \end{bmatrix} P_{\mathcal{B}^\sharp/g}\big(\mathbf{m}|_{\mathcal{E}\sqcup G-g};\, m_g,\, 2\sigma - m_g\big);$$

for $i < c$ its ratio is not constant, and at any exponents $s, t > 0$ and masses its law is not carried by one point (Corollary 23.12(i)).

At the fusion width one both laws are points and the glue is a product: one Beta corner. At the width $c \ge 2$ neither is: the universal law spreads over the $(c-1)$-dimensional torus of G, and the lift partner's seam vector moves with the room because the lifts are not a geometric chain, the lift ratios being not constant. Read on the Skeleton values of the fusion bunch, the letter of the fused table is an average over the bijections from the marks of G to the Skeleton values, and it is one product of c letters of one mark on no open set, and at no gauge for a list (Remark 30.19(i)); one product is what c independent fusions of width one would give. Read down the partial fusions, only the step at the top is free.

Among the plain extensions of a table by one mark with connected hinge minors, the fusion is the only one that is free in this way, up to one constant on the weights of the bases avoiding the new mark.

Proposition 30.17 (The free extensions by one mark)**.** *Let $\mathcal{A}$ be a plain cell on $\mathcal{E} \sqcup \{g\}$, $|\mathcal{E}| \ge 2$, whose hinge minors $\mathcal{A}\setminus g$ and $\mathcal{A}/g$ are connected. Its ratio $\varrho_{\mathcal{A}}$ (Definition 21.1) is constant exactly when $\mathcal{A}/g$ is the dual of $\mathcal{A}\setminus g$ with its weights multiplied by one constant $\mu > 0$; then $\varrho_{\mathcal{A}} \equiv \mu^2$, $\mathcal{A}/g$ has the fusion width one, and $\mu K_{\mathcal{A}}(\alpha_{\mathcal{E}}, \alpha_g) = K^\sharp(\alpha_{\mathcal{E}}, \mu\alpha_g)$, $K^\sharp$ the letter of the Foreman fusion of $\mathcal{A}/g$: $\mathcal{A}$ is that fusion with the weights of its bases avoiding g divided by μ. For a list, $\varrho_{\mathcal{A}}$ is constant exactly when $\mathcal{A}$ is identically self-dual, that is when $\mathcal{A}$ is the Foreman fusion of $\mathcal{A}/g$. So, at any exponents $s, t > 0$ and masses, the law of $\mathcal{A}$ is a point (Corollary 23.12(i)) exactly when $\mathcal{A}$ is the Foreman fusion of $\mathcal{A}/g$ up to one constant on the weights of the bases avoiding g, and for a list exactly when it is that fusion.*

Corollary 30.18 (A pinned set of one list is a dual pair)**.** *Let $\mathcal{B}$ be a list on $\mathcal{E}$, let $s, t > 0$, and let $S \subsetneq \mathcal{E}$, $|S| \ge 2$, with $S' = \mathcal{E} \setminus S$, be such that $\mathcal{B}|S$ and $\mathcal{B}/S'$ are connected. Then S is pinned (Theorem 28.25) exactly when it has width one and $\mathcal{B}/S' = (\mathcal{B}|S)^*$: the cell of the bunch $\mathcal{B}_S^+$ (Theorem 19.4) is a dual pair.*

Remark 30.19 (The lift ratios at a gauge; width three). (i) Let $\mathcal{B}$ be a list. The room sides of G in $\mathcal{B}^\sharp$ are the sets independent in $\mathcal{B}^*$ that span $\mathcal{B}$: the bases of the identity map from $\mathcal{B}^*$ to its quotient $\mathcal{B}$ read as a morphism of matroids, the arena property that Theorem 30.6(a) supplies. For such a morphism Eur and Huh (*Logarithmic concavity for morphisms of matroids*, Corollary 4.6) show that the basis generating polynomial, each basis counted once and homogenized by one more variable to the power n less the size of the basis, is Lorentzian, and a linear change of the variables with nonnegative coefficients keeps that property (Brändén and Huh), so at every gauge $\alpha_{\mathcal{E}}$ the values $K_{\mathcal{L}_j}(\alpha_{\mathcal{E}})/\binom{n}{r+j}$ form a log-concave sequence:

$$z_j(\alpha_{\mathcal{E}}) \;\ge\; \frac{(k+1)(n-k+1)}{k(n-k)} > 1, \qquad k = r + j,\ 0 < j < c.$$

So for a list at $c \ge 2$ the lift ratios exceed one at every gauge and the η_i of Theorem 30.16(ii) never coincide. The bound holds on the 6720 exact ratios of the receipt `t12_ratios_and_tell.py`,

with equality in 35 of them, and it is attained at the table's gauge whenever $\mathcal{L}_{j-1}$, $\mathcal{L}_j$ and $\mathcal{L}_{j+1}$ are uniform lists, as for every uniform list: there $K_{\mathcal{L}_i}(\mathbf{1}) = \binom{n}{r+i}$. The receipt `t08_logconcave.py` finds every lift ratio above one on 6720 exact ratios of its own. With weights the lift ratios can fall below one: $\binom{[4]}{1}$ with the weights 2 and the middle weights 1 has $K_{\mathcal{L}_0}(\mathbf{1}) = K_{\mathcal{L}_2}(\mathbf{1}) = 8$ and $K_{\mathcal{L}_1}(\mathbf{1}) = 6$, so $z_1 = \frac{9}{16}$ at the table's gauge.

(ii) At $c = 2$ the discriminant of $L_0 + 2L_1x + L_2x^2$ is $4L_0L_2(z_1 - 1)$, so the Skeleton values of G at $(\mathbf{1}_G, \alpha_{\mathcal{E}})$ are real and distinct where $z_1 > 1$, at every gauge for a list, and complex conjugate where $z_1 < 1$.

(iii) At $c = 3$, with the scale a of one mark of G held fixed, $K^\sharp$ is the letter of width two with the lift letters $L'_l = L_l + aL_{l+1}$, $0 \le l \le 2$; so at the masses 1 on G and $2\sigma = 2$ the integral over the scales of G at a fixed room point is, by Corollary 30.14(iii),

$$\int_0^\infty \frac{\log\big(L_1'^2/(L'_0L'_2)\big)}{L_1'^2 - L'_0L'_2}\,da, \qquad L_1'^2 - L'_0L'_2 = -\big((L_1L_3 - L_2^2)a^2 + (L_0L_3 - L_1L_2)a + (L_0L_2 - L_1^2)\big),$$

the coefficients of the quadratic being the 2×2 minors of the matrix with the rows (L_0, L_1, L_2) and (L_1, L_2, L_3), and its discriminant $-\frac{1}{27}$ times that of the diagonal level cubic $L_0+3L_1x+3L_2x^2+L_3x^3$ (receipts `t10_weight.py`, a rendering, and `t13_faces_and_cubic.py`). The cubic is real-rooted at all 560 gauges sampled on 8 lists of the fusion width three (receipts `w02_skeleton_braid.py`, `w03_real_c3.py`).

30.5 Cousins and hidden symmetry

Proposition 30.20 (Cousins)**.** *The* cousins *of an identically self-dual table are its one-mark contractions. Let $(\mathcal{B}, \theta)$ be fusable of the fusion width $c \ge 1$, and x a mark of $\mathcal{E} \sqcup G$.*

(i) The cousin $\mathcal{B}^\sharp/x$ has the fusion width one and is fusable, $\mathcal{B}^\sharp \setminus x$ is its dual, and its Foreman fusion is $(\mathcal{B}^\sharp, \theta^\sharp)$ with the fusion bunch $\{x\}$. For $x \in G$ the cousin is the partial fusion $\mathcal{B}^\sharp_{c-1}$, and at $c = 1$ the table $\mathcal{B}$ is the cousin at g.

(ii) On the balance wall of $\mathcal{B}^\sharp$ with every proper wall positive, for every two marks x, y,

$$\begin{aligned} P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t) &= \Gamma\begin{bmatrix} m_x, & 2\sigma - m_x \\ 1 & 2\sigma \end{bmatrix} P_{\mathcal{B}^\sharp/x}\big(\mathbf{m}_{-x}; m_x, 2\sigma - m_x\big) \\ &= \Gamma\begin{bmatrix} m_y, & 2\sigma - m_y \\ 1 & 2\sigma \end{bmatrix} P_{\mathcal{B}^\sharp/y}\big(\mathbf{m}_{-y}; m_y, 2\sigma - m_y\big), \end{aligned}$$

$\mathbf{m}_{-x}$ the masses of the marks other than x: the cousins share one period up to Beta corners, each reading its exponents off the mass of the mark it lacks.

(iii) For every bijection β of $\mathcal{E} \sqcup G$ carrying $(\mathcal{B}^\sharp, \theta^\sharp)$, $P_{\mathcal{B}^\sharp}(\beta^*\mathbf{m}; s, t) = P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t)$, with $(\beta^*\mathbf{m})_e = m_{\beta(e)}$. At $c = 1$ the function

$$(\mathbf{m}|_{\mathcal{E}}, s') \;\mapsto\; \Gamma\begin{bmatrix} s', & 2\sigma - s' \\ 1 & 2\sigma \end{bmatrix} P_{\mathcal{B}}\big(\mathbf{m}|_{\mathcal{E}}; s', 2\sigma - s'\big)$$

on the part of the hyperplane $m_{\mathcal{E}} + s' = 2\sigma r^*$ where every proper wall of $\mathcal{B}^\sharp$ is positive, which the automorphisms keep, is invariant under the automorphisms of $\mathcal{B}^\sharp$ acting on $\mathcal{E} \sqcup \{g\}$, s' read as the mass of g: an automorphism β with $\beta(g) = e \ne g$ makes the mass m_e the basis exponent and gives s' to the mark $\beta^{-1}(g)$ as its mass, and the transposition of g and e, when it is an automorphism, exchanges s' and m_e.

What fusion buys at every width is one step below the fused table. The fused table is identically self-dual and its automorphisms permute its marks; each cousin is the fused table with one mark integrated by a Beta corner, so the cousins are tables of fusion width one, often not isomorphic to one another, whose periods are one function. At the fusion width one, the table itself is a cousin and the hidden symmetry acts on its exponent and its masses together; at the width $c \geq 2$ the partial fusion $\mathcal{B}^{\sharp}_{c-1}$ is the cousin, and the table lies $c-1$ cells below it.

Remark 30.21 (Hidden symmetry). The index of the automorphisms of $\mathcal{B}$ times the permutations of G in those of $\mathcal{B}^{\sharp}$ measures the symmetry the fusion adds. For $\binom{\mathcal{E}}{r}$ it is $\binom{n+c}{c}$, all bijections of $\mathcal{E} \sqcup G$ carrying $\binom{\mathcal{E} \sqcup G}{r^*}$; for the list of rank two with one parallel pair of Remark 30.9(iii) it is $(n-2)(n-3)$; for the 10 fusable prime lists of rank three on seven marks the fusion is transitive and its group is 8 times that of the list (receipts `census_extra.py`, `atoms7.py`). At the fusion width two the index is 45 for $\binom{[8]}{3}$, 20 for one line of three marks and five free marks, 15 for one line of four marks and four free marks, 6 for two lines of three marks meeting in a mark and three free marks, 2 for two disjoint lines of three marks and two free marks, and 1 for F_7 with a free mark, for A_9 less a mark and for two disjoint lines of four marks; at the width three it is 220 for $\binom{[9]}{3}$ and 1 for A_9 and for Z_9 (receipt `w01_seam_rank3.py`). At $\binom{[2r+1]}{r}$, $r \geq 2$, the symmetric group of $\binom{[2r+2]}{r+1}$ accounts for a nonzero submodule that the closings miss at its centre, with every symmetry of the four kinds of its centre ray in the group (Proposition 39.48); at $\binom{[5]}{2}$ its part in the span of a box is a line (Exercise 52.20). Among 40 identically self-dual lists of rank five on ten marks drawn at random, in each of which the sets of five marks that are not bases form a family closed under complement whose members meet pairwise in at most three marks, 23 have ten cousins no two of which are isomorphic (receipt `random10.py`), and all the cousins of these 23 are prime (receipt `random10b.py`).

30.6 Two-sums and repair

Definition 30.22 (Covered marks; the conditions at a hinge). Let $\mathcal{B}$ be a table on $\mathcal{E}$. A mark e of a cocircuit D is *covered in* D when some circuit C has $e \in C \subseteq D$, and e is *uncovered* when some cocircuit contains it without covering it. For a cell $\mathcal{A}$ on $B \sqcup \{g\}$ with the hinge g (Definition 21.1) the *conditions at the hinge* are:

(a) every cocircuit avoiding g is a union of circuits;

(u) every cocircuit D through g covers every mark of $D \setminus \{g\}$;

(s) every cocircuit D through g covers every mark of $D \setminus \{g\}$ by circuits avoiding g;

(h) every cocircuit through g covers g.

For two cells $\mathcal{A}_1$, $\mathcal{A}_2$ the conditions are written a_1, u_1, s_1, h_1 and a_2, u_2, s_2, h_2.

Theorem 30.23 (Fusion across a two-sum). *Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be cells that are lists, on $\mathcal{E}_1 \sqcup \{g\}$ and $\mathcal{E}_2 \sqcup \{g\}$ with n_1 and n_2 marks, of the ranks r_1, r_2 and the fusion widths c_1, c_2, and $\mathcal{S} = \mathcal{A}_1 \oplus_g \mathcal{A}_2$ their two-sum on $\mathcal{E}_1 \sqcup \mathcal{E}_2$ (Lemma 21.12).*

(i) $\mathcal{S}$ has the fusion width $c_1 + c_2$. Its circuits are the circuits of $\mathcal{A}_1$ and of $\mathcal{A}_2$ that avoid g and the sets $(C_1 \setminus g) \cup (C_2 \setminus g)$ with C_i a circuit of $\mathcal{A}_i$ through g; its cocircuits are the same with cocircuits in place of circuits; and $\mathcal{S}^* = \mathcal{A}_1^* \oplus_g \mathcal{A}_2^*$.

(ii) $\mathcal{S}$ is fusable exactly when

$$a_1 \text{ and } u_1 \text{ and } a_2 \text{ and } u_2 \text{ and } (s_1 \text{ or } h_2) \text{ and } (s_2 \text{ or } h_1),$$

and a cell, read as a table, is fusable exactly when a, u and h hold. So two fusable cells give a fusable two-sum.

(iii) If $\mathcal{S}$ is fusable, then $c_1, c_2 \geq 0$, and in $\mathcal{S}^\sharp$ the sets $\mathcal{E}_1$ and $\mathcal{E}_2$ are bunches of the widths c_1+1 and c_2+1.

(iv) If both cells are fusable, the two-sum $\mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2^\sharp$, with the fusion bunch $G_1 \sqcup G_2$, is an identically self-dual major of $\mathcal{S}$. It is $\mathcal{S}^\sharp$ when $c_1 = 0$ or $c_2 = 0$. When $c_1, c_2 \geq 1$ its children at one mark of G_1 and at one mark of G_2 have the ranks $r_1^* - 1$ and r_1^* on $\mathcal{E}_1$, $r_1^* = n_1 - r_1$ the corank of $\mathcal{A}_1$, so $G_1 \sqcup G_2$ is not in level form in it and it is a major other than $\mathcal{S}^\sharp$.

(v) If $c_1 + c_2 = 1$, $\mathcal{S}$ is fusable exactly when both cells are fusable; then, for $c_2 = 0$, $\mathcal{A}_2$ is identically self-dual, $\mathcal{S}^\sharp = \mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2$, and the hinge of $\mathcal{S}$ is a hinge of $\mathcal{S}^\sharp$.

The condition is local at the hinge: away from it each cell answers for itself, and through it a cocircuit of the two-sum joins a cocircuit of each cell, so a mark is covered either by a circuit of its own cell avoiding the hinge or by a circuit through the hinge continued by a circuit through the hinge on the other side; that continuation exists exactly when the other cell covers its hinge. In the fused table the hinge of a cell of positive width thickens into a bunch of that width plus one. Fusing cell by cell gives a major, identically self-dual, that is not the fusion as soon as two cells have positive width: the canonical fusion of a two-sum is the global level cell. At the total width one the hinge survives the fusion, with the identically self-dual cell on one side and the fusion of the other cell on the other.

Corollary 30.24 (A two-sum with an identically self-dual cell)**.** *Let $\mathcal{A}_1$ be identically self-dual and $s, t > 0$, with the masses $\mathbf{m}$ of $\mathcal{S} = \mathcal{A}_1 \oplus_g \mathcal{A}_2$ on its balance wall with every proper wall positive, and let $\kappa = m_{\mathcal{E}_1} - s(r_1 - 1) - t(|\mathcal{E}_1| - r_1)$ be the hinge wall of $\mathcal{A}_1$ (Definition 23.1). Then*

$$P_{\mathcal{S}}(\mathbf{m}; s, t) = P_{\mathcal{A}_1}\big(\mathbf{m}|_{\mathcal{E}_1}, 2\sigma - \kappa; s, t\big)\, P_{\mathcal{A}_2}\big(\mathbf{m}|_{\mathcal{E}_2}, \kappa; s, t\big) \Big/ \Gamma_1^2 \begin{bmatrix} 2\sigma - \kappa, & \kappa \\ & 2\sigma \end{bmatrix},$$

the hinge carrying $2\sigma - \kappa$ in $\mathcal{A}_1$ and κ in $\mathcal{A}_2$. When the cells are lists and $\mathcal{A}_2$ is fusable of the fusion width one, this is the period of $\mathcal{S}$ read across the hinge that $\mathcal{S}^\sharp = \mathcal{A}_1 \oplus_g \mathcal{A}_2^\sharp$ keeps.

By Corollary 20.19 at the hinge, the first factor divided by the Beta corner is the period $P_{\mathcal{A}_1 \backslash g}(\mathbf{m}|_{\mathcal{E}_1}; \kappa, 2\sigma - \kappa)$ of the deletion at the hinge. So identically self-dual cells attached at their hinges to one cell that is not identically self-dual cost the period of that cell and, for each of them, the period of its deletion at the hinge; two cells that are not identically self-dual on the two sides of a hinge pair two laws, and the hinge is a convolution (Theorem 23.9).

Proposition 30.25 (Uncovered marks and repair)**.** *Let $(\mathcal{B}, \theta)$ be a table on $\mathcal{E}$ of the fusion width c.*

(i) $\mathcal{B}$ has no uncovered mark (Definition 30.22) exactly when every cocircuit of $\mathcal{B}$ is a union of circuits. So a list, or a table of the fusion width $c \neq 0$, is fusable exactly when it has no uncovered mark, and a table of the fusion width zero exactly when it is identically self-dual.

(ii) For a set S of marks, none a loop, let $(\mathcal{B}_S, \theta)$ be the table in which each $h \in S$ is replaced by a block $\{h, h'\}$ of two marks by the first rule of Theorem 21.5, a parallel pair, every new basis keeping the weight of its basis, and $S' = \{h' : h \in S\}$. Then $\mathcal{B}_S$ has the fusion width $c + |S|$, $(\mathcal{B}_S, \theta) \setminus S' = (\mathcal{B}, \theta)$, and $\mathcal{B}_S$ has no uncovered mark exactly when S contains every uncovered mark of $\mathcal{B}$; when $\mathcal{B}$ is a list or $c + |S| \neq 0$, $\mathcal{B}_S$ is fusable exactly then.

(iii) If $\mathcal{B}_S$ is fusable, $(\mathcal{B},\theta) = (\mathcal{B}_S)^\sharp/G \setminus S'$ is a minor of an identically self-dual table. If $c \geq 0$ and $(\mathcal{B},\theta)$ is not fusable, $\mathcal{B}_S$ is fusable for S the set of the uncovered marks when there is one, and for S any one mark when there is none, no mark being then a loop. Doubling every mark of a list that is not a loop gives a fusable list.

(iv) At the full basis lean, where every other proper wall is positive, the residue of $P_{\mathcal{B}_S}(\mathbf{m};s,0)$ at the wall $m_{h'}=0$ of one copy h' is $P_{\mathcal{B}_{S-h}}(\mathbf{m}_{-h'};s,0)$ (Theorem 16.25); so $P_{\mathcal{B}}(\cdot;s,0)$ is the residue of $P_{\mathcal{B}_S}(\cdot;s,0)$ at the walls of the copies, taken one at a time.

A mark is uncovered when a cocircuit through it has no circuit through it inside: the cocircuit is a set the bases must meet, and the mark is not tied by any circuit to the other marks there. A parallel copy is a circuit of two marks through it inside every cocircuit through it, so doubling an uncovered mark covers it, and each doubled mark adds one to the fusion width. The list $\binom{[3]}{2}$, of the fusion width -1, has every mark uncovered: one or two doubled marks do not fuse, all three do, at the width two (receipts `t04_twosums.py`, `w04_repair.py`); its dual $\binom{[3]}{1}$, which the procedure below reads first, fuses as it is, into $\binom{[4]}{2}$.

30.7 The procedure

The statements of this chapter set a branch at every node of the Foreman tree (Definition 14.4), read before anything is integrated.

(1) *Peel.* If the realization of the node is not an atom, peel it at its Foreman bunch and continue on the room (Definition 14.4).

(2) *Go down.* If the table of the node is identically self-dual, every mark comes off by one Beta corner at every pair of exponents (Corollary 20.19); its cousins share its period (Proposition 30.20).

(3) *Go up.* Otherwise read the fusion width $c = r^* - r$, passing to the dual table when it is negative, and the tell on the Skeleton (Theorem 30.6). If the table is fusable, fuse it (Theorem 30.5); at $c = 0$ it is not, a fusable table of the width zero being identically self-dual. At $c = 1$ the fusion is free, one Beta corner (Corollary 30.13): the automorphisms of the identically self-dual table $\mathcal{B}^\sharp$ act on the exponent and the masses of the table, its cousins share its period, and its module carries over (Chapter 39). At $c \geq 2$ the table is the child of the full pattern of the fusion bunch of $\mathcal{B}^\sharp$; the fused table is the universal piece of width c glued to the lift partner (Theorem 30.12), both laws spread (Theorem 30.16), the step from $\mathcal{B}^\sharp$ to $\mathcal{B}^\sharp_{c-1}$ is a Beta corner and the $c-1$ steps below it are cells whose laws are not points.

(4) *Two-sums.* On a two-sum of lists the tell is read at the hinge (Theorem 30.23(ii)); at the total width one the hinge survives the fusion and the period is a product over one Beta corner (Corollary 30.24); when both cells are fusable and one has the width zero the fusion is the two-sum of the fusions of the cells, and when both cells have positive width it is the global level cell, not that two-sum (Theorem 30.23(iv)).

(5) *Repair.* If the table is not fusable and has uncovered marks, they are the obstruction: doubling exactly those in parallel gives a fusable table with the table as its deletion at the copies (Proposition 30.25), at the fusion width c plus the number of uncovered marks; a table of the width zero with no uncovered mark that is not identically self-dual, which needs weights, becomes fusable when any one mark is doubled.

The fusion width is the price, and it is read off the rank and the number of marks before any integral is taken: at the width one the fusion bunch is a single mark with a point for its law; at the width $c \geq 2$ it is the universal piece, whose law spreads over a torus of the dimension $c - 1$ against a lift partner whose law is not a point.

Chapter 31

Closing a glue along a direction

A glue opened on its shares is a fibre integral of Gammas and two seam functions (Theorem 28.10(iii)). It is closed by moving the contour along a direction of the shares, collecting a fibre integral of residues at every polar hyperplane crossed and showing that what is left tends to zero. The glue closes on the lattices of its Gammas along every direction whose rate, read off the two laws through the largest values of the monomials of their letter vectors, is negative (Theorem 31.2), and past the walls of the piece, or of the piece and the partner both, at the growths of the continued seam functions (Theorem 31.4). Moving the contour is exact at every step, and the limit needs only a bound that travels with it (Lemma 31.3): the series depends on the direction only through its chamber (Theorem 31.6), the laws can be closed one source point at a time and averaged, at the rate read on the support of the source measure rather than on its convex hull (Theorem 31.7), and past the walls of a piece the bound is read on the outline of a keyhole continuation, the modulus and the phase of the sector letters taken at one point (Theorem 31.11). At simple nonresonant crossings of a rational ray, a crossing term kept with its falling shares integrated on their strips is a finite sum of joint Trailmoments of fixed functions, a falling Gamma becoming a positive share and one Beta variable and a continued source factor a segment and circle integrals (Theorem 31.18); for a positive kernel its root rate is the exponential of the least Gamma rate of its chamber in every fibre dimension and, averaged, of the essential supremum over the source of that least rate, whose sign, when it is not zero, decides whether the terms tend to zero (Theorem 31.26, Corollary 31.27); and an averaged closing can converge where no direction of its chamber certifies it, at a family of tables whose seam has single stations (Theorem 31.8). Coincident crossings are packets of factors polar on one crossed hyperplane, assembled before a sign or a top is read (Theorem 31.30), and at the root rate one a positive moment sequence is decided by its law near its top (Theorem 31.32).

31.1 Closing along a direction

Definition 31.1 (Letter vectors)**.** Let a piece on B and a partner on R with the stations $\mathcal{D}$ be glued with masses on the balance wall of the glue (Definition 28.4), the glue read on its shares as in Theorem 28.10(iii) or Theorem 28.12. The *letter vectors* of the piece and of the partner are $\mathbf{V}_B = (V_{jD}, V_{jD}^*)_{j,D}$ and $\mathbf{V}_R = (L_{jD}, L_{jD}^*)_{j,D}$, and for a vector $\mathbf{v}$ of the coordinates of the shares $\mathbf{V}_B^{\mathbf{v}} = \prod_{j,D} V_{jD}^{v_{jD}} {V_{jD}^*}^{v_{jD}^*}$, and likewise $\mathbf{V}_R^{\mathbf{v}}$.

Theorem 31.2 (Closing the glue along a direction)**.** *In Theorem 28.12 let V be the space of the free directions of the shares, so that for $\mathbf{v} \in V$ the monomials $\mathbf{V}_B^{\mathbf{v}}$ and $\mathbf{V}_R^{\mathbf{v}}$ (Definition 31.1) are*

homogeneous of degree zero, and their suprema over the orthants do not change under $\vec{\mathcal{N}}_B^{\perp}$ *and* $\vec{\mathcal{N}}_R^{\perp}$. *When the seam word of the piece has a Newton sum of full dimension,* $\sup \mathbf{V}_B^{\mathbf{v}} = \sup_{u\in\Delta_B} \boldsymbol{\varrho}_B^{\rho}(u)^{\mathbf{v}}$, *the largest value of the monomial* $\mathbf{x}^{\mathbf{v}}$ *on the support of the law of the piece, and when the room word of the partner has a Newton sum of full dimension,* $\sup \mathbf{V}_R^{\mathbf{v}} = \sup_{v\in\Delta_R} \boldsymbol{\varrho}_R^{\rho}(v)^{\mathbf{v}}$, *the largest value on the support of the law of the partner; where the room letters of the partner are monomials,* $\sup \mathbf{V}_R^{\mathbf{v}} = \boldsymbol{\xi}^{\mathbf{v}}$, $\boldsymbol{\xi}$ *their coefficients. For* $\boldsymbol{\sigma} \in V$ *put*

$$h(\boldsymbol{\sigma}) = \sum_{j,D} \big(\sigma_{jD} \log|\sigma_{jD}| + \sigma^*_{jD} \log|\sigma^*_{jD}|\big) + \log\sup \mathbf{V}_B^{-\boldsymbol{\sigma}} + \log\sup \mathbf{V}_R^{-\boldsymbol{\sigma}},$$

with $0\log 0 = 0$, *the first sum over the shares of the active letters, the* rate *of the direction; it is homogeneous of degree one. Let* $h(\boldsymbol{\sigma}) < 0$, *and let the real point* $\mathbf{c}$ *of the pins lie off the null set of the points whose ray* $\mathbf{c} + \tau\boldsymbol{\sigma}$, $\tau \geq 0$, *meets two polar hyperplanes of the Gammas that are not parallel on the fibre at one point, or, for arbitrarily large* τ, *at points less than* τ^{-3} *apart. Then the ray meets only the hyperplanes* $a_{jD} = -n$ *with* $\sigma_{jD} < 0$ *and* $b_{jD} = -n$ *with* $\sigma^*_{jD} < 0$, $n \in \mathbb{N}$, *and*

$$P_{B\sqcup R}(\mathbf{m}) = \sum_{crossings} \mathcal{R},$$

the sum, in the order of crossing, of the fibre integrals $\mathcal{R}$ *of the residues of the integrand at the hyperplanes crossed, each pinned also by its hyperplane, of the dimension* $\dim V - 1$ *and of the same kind, at the point where the ray meets it; a hyperplane of the fibre on which several Gammas have poles is crossed once, for every* $\mathbf{c}$, *and its residue is a Laurent coefficient (Theorem 4.5). The sum converges absolutely: the residue at a crossing at* τ, *and the integral left after it, are* $O\big(e^{\tau(h(\boldsymbol{\sigma})+\epsilon)}\big)$ *for every* $\epsilon > 0$.

So the glue closes on the lattices of its Gammas along every direction of the cone $\{h < 0\}$ of V, the terms falling at least by the factor $e^{h(\boldsymbol{\sigma})}$ per unit of $\boldsymbol{\sigma}$. The cone and the rate are read off the laws through the largest values of the monomials of their letter vectors, and the term $\sum \sigma \log|\sigma|$ of the rate is the cost of moving shares between stations: it vanishes at width one with the stations of one excess merged, every letter then moving its two shares oppositely, and along the one direction of a node of width one the condition is that the ranges of the two ratios be separated, as in Section 20.7. A direction along which a law is unbounded, its supremum infinite, crosses the walls of its seam function as well. The seam function of a piece continues past its walls with a growth that its sector integrals bound, and the partner can be read in its monomials, which have no walls, or continued past its own walls in the same way.

31.2 Closing along a path

The closings of Theorems 31.2 and 31.4 move the contour of the share fibre along a ray, collect a fibre integral of residues at every polar hyperplane crossed, and show that what is left tends to zero. The two steps come apart. Moving the contour is exact at every step: the period is the sum of the crossing terms met so far and of the fibre integral at the current point, by Cauchy's theorem on bounded contours. Only the last step is a limit, and it asks for a bound on the remaining fibre integral at the points that the path passes through, not for a domination of the double family of crossings and fibres, which is what interchanging the sum over the crossings with the fibre integrals would ask, and what fails past the walls. The step is Lemma 31.3, with three consequences: a closing depends on its direction only through its chamber (Theorem 31.6); the laws can be closed one source point at a time and averaged (Theorem 31.7); and past the walls of a piece the bound

that travels is read on the outline of a keyhole continuation (Theorem 31.11). The bounds multiply over products, survive averaging and pass to the crossing terms (Proposition 31.13).

Lemma 31.3 (Closing along a path)**.** *Let A be the affine space of the real points of the real pins of a fibre integral and V its space of directions, of dimension $n \geq 1$, with the measure $d\boldsymbol{\vartheta}$, the fibre integral through a real point p being $\int_{p+iV} I$ (Definition 4.1) wherever it converges absolutely; and let $\Omega \subseteq A$ be open. Let I be meromorphic on the tube $\Omega + iV$, with its poles on the complex hyperplanes $\{\ell_H = 0\}$ of a family $\mathfrak{H}$ of real hyperplanes $H = \{\ell_H = 0\}$ of A, locally finite in Ω, each ℓ_H real, affine and not constant. Let $\gamma : [0, \infty) \to \Omega$ be continuous and piecewise linear, with finitely many breaks on every bounded interval and $\gamma(0) = \mathbf{c}$ off $\bigcup \mathfrak{H}$, and assume:*

(H0) Clean crossings. *γ meets $\bigcup \mathfrak{H}$ at a discrete set of times, its* crossings*, none of them a break of γ; at a crossing τ^* the hyperplanes of $\mathfrak{H}$ through $\gamma(\tau^*)$ are one hyperplane H, and $\ell_H \circ \gamma$ changes sign at τ^* with a nonzero slope.*

(H1) Local decay. *For every compact $Q \subset \Omega$ and $\delta > 0$ there are $C, c > 0$ with $|I(x+i\boldsymbol{\vartheta})| \leq Ce^{-c|\boldsymbol{\vartheta}|}$ whenever $x \in Q$ and $x + i\boldsymbol{\vartheta}$ lies at distance at least δ from every hyperplane $\{\ell_H = 0\}$ of the tube.*

(H2) A travelling bound. *There are times $\tau_1 < \tau_2 < \cdots$ tending to infinity, none a crossing, at which*

$$\beta(\tau_k) = \int_V \big|I(\gamma(\tau_k) + i\boldsymbol{\vartheta})\big|\, d\boldsymbol{\vartheta}$$

is finite and tends to zero.

Then for every k

$$\int_{\mathbf{c}+iV} I = \sum_{\tau^* < \tau_k} \mathcal{R}_{\tau^*} + \int_{\gamma(\tau_k)+iV} I, \qquad \Big|\int_{\gamma(\tau_k)+iV} I\Big| \leq (2\pi)^{-n}\beta(\tau_k),$$

so that $\int_{\mathbf{c}+iV} I = \lim_k \sum_{\tau^*<\tau_k} \mathcal{R}_{\tau^*}$. The crossing term *at a crossing τ^* of the hyperplane H is*

$$\mathcal{R}_{\tau^*} = \pm(2\pi)^{1-n} \int_{V_H} \mathrm{Res}_{w=0}\, I\big(\gamma(\tau^*) + w\,\mathbf{e} + i\boldsymbol{\vartheta}'\big)\, d\mu_H(\boldsymbol{\vartheta}'),$$

with $V_H \subset V$ the kernel of the linear part ℓ_H^{lin} of ℓ_H, μ_H the measure on V_H with $d\boldsymbol{\vartheta} = d\mu_H\, d\ell_H^{\mathrm{lin}}(\boldsymbol{\vartheta})$, $\mathbf{e} \in V$ any vector with $\ell_H^{\mathrm{lin}}(\mathbf{e}) = 1$, on which the term does not depend, and the sign $+$ when ℓ_H decreases along γ and $-$ when it increases. The residue is that of the whole integrand, a Laurent coefficient where several of its factors are polar on H, and the crossing term is its fibre integral, pinned also by $\ell_H = 0$. If some τ_k lies between any two consecutive crossings and $\sum_k \beta(\tau_k) < \infty$, the crossing terms are absolutely summable, a crossing between τ_k and τ_{k+1}, the only one there, having $|\mathcal{R}_{\tau^*}| \leq (2\pi)^{-n}\big(\beta(\tau_k) + \beta(\tau_{k+1})\big)$.

Theorem 31.4 (Closing past the walls)**.** *Let the seam function of the piece converge at a real point of the pins of the seam with positive shares, so that its seam word has a Newton sum of full dimension (Theorem 25.2). Read the partner in its monomials, $L_{jD} = \sum_k \xi_{jDk}\, \alpha_R^{\mathbf{n}_{jDk}}$ and $L^*_{jD} = \sum_k \xi^*_{jDk}\, \alpha_R^{\mathbf{n}^*_{jDk}}$, with one share a_{jDk} or b_{jDk} for each monomial, and write $\bar{a}_{jD} = \sum_k a_{jDk}$ and $\bar{b}_{jD} = \sum_k b_{jDk}$; bars mark sums over the monomials.*

(i) On every contour of shares with positive real parts, pinned by the letter totals and by $\sum_{j,D,k}(a_{jDk}\mathbf{n}_{jDk} + b_{jDk}\mathbf{n}^*_{jDk}) = \mathbf{m}_R$, at which the seam function of the piece converges, the

period of the glue converging,

$$P_{B\sqcup R}(\mathbf{m}) = \int_{pins} \prod_{j:\, s_j>0} \frac{\prod_{D,k} \Gamma(a_{jDk})\, \xi_{jDk}^{-a_{jDk}}}{\Gamma(s_j)} \prod_{j:\, t_j>0} \frac{\prod_{D,k} \Gamma(b_{jDk})\, {\xi^*_{jDk}}^{-b_{jDk}}}{\Gamma(t_j)}\ P_{W_B(\bar{\mathbf{a}},\bar{\mathbf{b}})}(\mathbf{m}_B).$$

(ii) Fix $e_0 \in B$ and a fan of the seam word*: finitely many rational simplicial cones of dimension $|B|-1$ with disjoint interiors covering $\mathbb{R}^{B\setminus e_0}$, each contained, for every seam letter V of an active letter, in a set $\{\mathbf{v} :\ \langle \mathbf{t}, \mathbf{v}\rangle \le \langle \mathbf{t}', \mathbf{v}\rangle$ for every exponent $\mathbf{t}'$ of the support of $V\}$, $\mathbf{t}$ one exponent of that support, the pairings taken on $B \setminus e_0$. On a cone π of it, a* sector*, with the primitive integral generators $\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1}$, write the logarithmic coordinates $x = \log \alpha_{B\setminus e_0}$ of the section $\alpha_{e_0} = 1$ as $x = -\sum_k u_k \mathbf{v}_{\pi,k}$, $u_k \ge 0$; the $q_k = e^{-u_k} \in (0,1]$ are the* sector coordinates*. On the section a seam letter V of an active letter is $V = \prod_k q_k^{\langle \mathbf{t}, \mathbf{v}_{\pi,k}\rangle}\, k_{V,\pi}(q)$, $\mathbf{t}$ the exponent of its support at which every generator of π pairs least, and its* sector letter *$k_{V,\pi}$ is a polynomial in q with nonnegative coefficients and a positive constant term. Write $k_{jD,\pi}$ and $\psi_{jD,\pi}$ for the sector letters of V_{jD} and V^*_{jD}, $\mathbf{t}_{jD,\pi}$ and $\mathbf{t}^*_{jD,\pi}$ for their exponents, and for a generator $\mathbf{v}$ of π call*

$$\kappa_{\mathbf{v}}(\mathbf{z}) = \Big\langle \mathbf{m}_B - \sum_{j,D} \big(\bar{a}_{jD}\, \mathbf{t}_{jD,\pi} + \bar{b}_{jD}\, \mathbf{t}^*_{jD,\pi}\big),\ \mathbf{v}\Big\rangle, \qquad \mathbf{z} = (\bar{\mathbf{a}}, \bar{\mathbf{b}}),$$

the pairing on $B \setminus e_0$, the wall *of $\mathbf{v}$: affine in the shares with a rational linear part, and the same for every sector of which $\mathbf{v}$ is a generator. On the pins of the seam, at every real point at which every wall is positive,*

$$P_{W_B(\mathbf{z})}(\mathbf{m}_B) = \sum_{\pi} |\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \int_{[0,1]^{|B|-1}} \prod_k q_k^{\kappa_{\mathbf{v}_{\pi,k}}(\mathbf{z})-1} \prod_{j,D} k_{jD,\pi}(q)^{-\bar{a}_{jD}}\, \psi_{jD,\pi}(q)^{-\bar{b}_{jD}}\, dq,$$

and the seam function converges at a real point exactly when every wall is positive there. It continues meromorphically to all complex shares, with poles, of order at most $|B|-1$, only on the hyperplanes on which a wall lies in $-\mathbb{N}$, and for every $\theta > 0$ there are M, C_0 and n_1 with

$$\big|P_{W_B(\mathbf{z})}(\mathbf{m}_B)\big| \le C_0\, \delta^{-(|B|-1)} (1+|\mathbf{z}|)^{n_1} M^{|\operatorname{Re}\mathbf{z}|_1} e^{\theta |\operatorname{Im}\mathbf{z}|_1},$$

δ the smaller of 1 and the distance of $\mathbf{z}$ from these hyperplanes. The piece is of tables *when at every station the seam letters of the active letters are the basis and the cobasis polynomials of one table on B with the bases $\mathcal{B}_{jD}$ and positive weights, as for a bunch of a family of tables (Proposition 28.2(i)). Its seam word then has a Newton sum of full dimension exactly when the family of these tables is connected; the sectors of the orders π of B (Theorem 8.5(ii)) form a fan of it, every determinant being 1, the sector letters being those of the tables $\mathcal{B}_{jD}$ and the generators $\mathbf{1}_{U_k} - (\mathbf{1}_{U_k})_{e_0}\mathbf{1}$ on $B \setminus e_0$, U_k the tails; the wall of the generator of U is the proper wall $\kappa_U = m_U - \sum_{j,D} \big(\bar{a}_{jD}\, d_{jD}(U) + \bar{b}_{jD}\, d^*_{jD}(U)\big)$, d_{jD} and d^*_{jD} the counts of $\mathcal{B}_{jD}$.*

(iii) For a direction $\boldsymbol{\sigma}$ of the shares of (i) and $\theta \ge 0$ let $g(\boldsymbol{\sigma};\theta)$, at the real point $\bar{\mathbf{c}}$ of the pins of the seam, be the infimum of the M for which there are C, n_1 and n_2 with

$$\big|P_{W_B(\bar{\mathbf{c}}+\tau\bar{\boldsymbol{\sigma}}+i\bar{\boldsymbol{\vartheta}})}(\mathbf{m}_B)\big| \le C\, \delta^{-n_2} (1+\tau+|\boldsymbol{\vartheta}|)^{n_1} M^{\tau} e^{\theta|\boldsymbol{\vartheta}|_1}$$

for $\tau \ge 0$ and $\boldsymbol{\vartheta}$ in the directions of the shares, δ the smaller of 1 and the distance from the polar hyperplanes: the growth *of the continued seam function along $\boldsymbol{\sigma}$ at the angle θ, the*

growth of Definition 20.33 read along any direction with the angle fixed. It is finite for $\theta > 0$, and $g(\boldsymbol{\sigma}; 0) = \sup \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$ when no wall of the seam word falls along $\bar{\boldsymbol{\sigma}}$, which is exactly when that supremum is finite. Put

$$h_\Gamma(\boldsymbol{\sigma}) = \sum_l \sigma_l \log|\sigma_l| - \sum_l \sigma_l \log \xi_l, \qquad h_\theta(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \log g(\boldsymbol{\sigma}; \theta) + \frac{2\theta^2}{\pi} \sum_{l:\, \sigma_l > 0} \sigma_l,$$

l running over the shares and ξ_l being the coefficient of the monomial of the share l; h_Γ is the Gamma rate *of the direction. Let the real point $\mathbf{c}$ of the pins lie off the null set of Theorem 31.2, taken with the polar hyperplanes of the seam function as well, and let g and h_θ be taken at its sums $\bar{\mathbf{c}}$. If $h_\theta(\boldsymbol{\sigma}) < 0$ for some $\theta \in [0, \pi/8)$, then*

$$P_{B \sqcup R}(\mathbf{m}) = \sum_{crossings} \mathcal{R},$$

the sum, in the order of crossing, of the fibre integrals of the residues of the integrand at the polar hyperplanes met by the ray $\mathbf{c} + \tau\boldsymbol{\sigma}$: the hyperplanes $a_l = -n$ of the shares with $\sigma_l < 0$, $n \in \mathbb{N}$, and the hyperplanes on which a wall of (ii) that falls along $\bar{\boldsymbol{\sigma}}$ lies in $-\mathbb{N}$, a hyperplane polar for several factors being crossed once, with its full residue. The sum converges absolutely, the terms at τ and the integrals left after them being $O\big(e^{\tau(h_\theta(\boldsymbol{\sigma}) + \epsilon)}\big)$ for every $\epsilon > 0$.

(iv) Both sides continued. *Let the glue be read on the share side of Theorem 28.10(iii), one share a_{jD} or b_{jD} per station, in place of the reading of (i), and let the seam functions of the piece and of the partner both converge at a real point $\mathbf{c}$ of the pins with positive shares. By (ii), applied to the piece and to the partner, a piece on R read with the reversed excesses (Definition 28.4), both continue meromorphically to all complex shares of the pins, with their poles on the hyperplanes on which a wall of the seam word of the piece or of the room word of the partner lies in $-\mathbb{N}$, each with the bound of (ii). For a direction $\boldsymbol{\sigma}$ of the shares and $\theta \geq 0$ let $g_B(\boldsymbol{\sigma}; \theta)$ and $g_R(\boldsymbol{\sigma}; \theta)$ be the growths of the two continued seam functions along $\boldsymbol{\sigma}$ at $\mathbf{c}$, defined as in (iii) with the shares in place of their sums, and put*

$$h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) = \sum_l \sigma_l \log|\sigma_l| + \log g_B(\boldsymbol{\sigma}; \theta_B) + \log g_R(\boldsymbol{\sigma}; \theta_R) + \frac{2(\theta_B + \theta_R)^2}{\pi} \sum_{l:\, \sigma_l > 0} \sigma_l.$$

Let $\mathbf{c}$ lie off the null set of Theorem 31.2, taken with the polar hyperplanes of both seam functions as well. If $h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) < 0$ for some $\theta_B, \theta_R \geq 0$ with $\theta_B + \theta_R < \pi/8$, then $P_{B \sqcup R}(\mathbf{m}) = \sum_{crossings} \mathcal{R}$, the sum, in the order of crossing, of the fibre integrals of the residues of the integrand at the polar hyperplanes met by the ray $\mathbf{c} + \tau\boldsymbol{\sigma}$: the hyperplanes $a_l = -n$ of the shares with $\sigma_l < 0$, $n \in \mathbb{N}$, and those on which a wall of either word that falls along $\boldsymbol{\sigma}$ lies in $-\mathbb{N}$, a hyperplane polar for several factors being crossed once, with its full residue. The sum converges absolutely, the terms at τ and the integrals left after them being $O\big(e^{\tau(h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) + \epsilon)}\big)$ for every $\epsilon > 0$. A side along whose direction no wall of its word falls has the growth $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$ of its letter vector at every angle, the largest value of the monomial on the support of its law (Theorem 31.2).

Along a direction in which the seam function converges, (iii) at $\theta = 0$ is the bound of Theorem 31.2 for the glue read in the monomials of the partner, and past a wall the residues of the seam function join those of the Gammas at the cost $\log g$ of its continuation, as the walls of the two families join the lattice at width one (Theorem 20.34(iii)). Read in its monomials as well,

any piece contributes only its coefficients: the integrand is then Gammas and coefficients, every supremum of a monomial of the letter vectors is a monomial in the coefficients, and every direction of the shares with $h_\Gamma(\boldsymbol{\sigma}) < 0$ closes, by Theorem 31.2. In which representation to close is a choice of rate: a law bounded along the direction closes at the largest value of its monomial, a side past its walls at the growth of its continuation, a side read in monomials at its coefficients, at the price of one pin for each of its marks. So (iii) and (iv) read each side at its own rate, the coefficients $-\sum_l \sigma_l \log \xi_l$ for a side read in its monomials and the logarithm of the growth of its continued seam function for a side read in its seam letters; at width one, where the share side has one free share, (iv) continues both families of the node, as Theorem 20.34(iii) does.

Expanding each active letter L, with its exponent e_L, in some of its terms, $L = Y_L + X_L$ with $X_L = \sum_k X_{L,k}$ the chosen terms, gives the multiple lattice whose terms are $(-1)^{|\mathbf{n}_L|}(e_L)_{|\mathbf{n}_L|}/\mathbf{n}_L!$ times the periods of $\alpha^{\mathbf{m}-\mathbf{1}} \prod_L Y_L^{-e_L-|\mathbf{n}_L|} \prod_k X_{L,k}^{n_{L,k}}$, positive words up to sign. Summed term by term in absolute value, it converges absolutely exactly when $X_L \le Y_L$ on the orthant for every active letter and the integral of $\alpha^{\mathbf{m}-\mathbf{1}} \prod_L (Y_L - X_L)^{-e_L}$ over a section is finite (Proposition 2.10). This needs the Newton polytope of X_L inside that of Y_L, so it never holds for a letter of a table, all of whose monomials are vertices of its Newton polytope: for families of tables the lattices close only in the order of a direction.

Theorems 31.2 and 31.4 are the lemma along a ray: (H0) holds off a null set of starting points, (H1) from the exponential decay of the Gamma factors in the imaginary directions (Theorem 2.11 at one scale, carried to negative real parts by the shift of Proposition 1.3(iii)), and (H2) from the largest values of the monomials on the supports of the laws or from the growth of the continued seam function. Two closings of one fibre integral along two paths agree at all times up to their remainders,

$$\sum_{\tau^* < \tau_k} \mathcal{R}_{\tau^*} - \sum_{\tau'^* < \tau'_k} \mathcal{R}'_{\tau'^*} = \int_{\gamma'(\tau'_k)+iV} I - \int_{\gamma(\tau_k)+iV} I,$$

and one of them may converge while the other does not. Each crossing term is a fibre integral of one dimension less, of an integrand of the same kind, which the lemma can close again along a path of its own when a bound travels with it there; this is how multiple lattices arise.

Definition 31.5 (Chambers of a closing)**.** Let I be meromorphic on the tube $\Omega + iV$ of Lemma 31.3 over an open convex $\Omega \subseteq A$, with its poles on the complex hyperplanes on which one of finitely many real affine functions ℓ on A, not constant, lies in $-\mathbb{N}$: its *polar functions*. A direction $\boldsymbol{\sigma} \in V$ *falls* a polar function ℓ when $\partial_{\boldsymbol{\sigma}} \ell < 0$, at the *falling rate* $-\partial_{\boldsymbol{\sigma}} \ell$. For a real point $\mathbf{c} \in \Omega$ at which every polar function is positive, the *chamber* $C(\boldsymbol{\sigma})$ of a direction $\boldsymbol{\sigma}$ whose ray $\mathbf{c} + \tau\boldsymbol{\sigma}$, $\tau \ge 0$, lies in Ω is the set of the $\boldsymbol{\sigma}' \in V$ whose ray lies in Ω and which fall exactly the polar functions that $\boldsymbol{\sigma}$ falls, each at the same falling rate; it is convex. In Theorem 31.2 the polar functions are the shares of the active letters, and Ω is the interior of the set of the real points of the pins at which the seam functions of the piece and of the partner converge, their reduced periods in Theorem 28.12, which is convex by Hölder's inequality and contains the ray $\mathbf{c} + \tau\boldsymbol{\sigma}$, $\tau \ge 0$, from a point $\mathbf{c} \in \Omega$ whenever $\sup \mathbf{V}_B^{-\boldsymbol{\sigma}}$ and $\sup \mathbf{V}_R^{-\boldsymbol{\sigma}}$ are finite. In Theorem 31.4(iii) they are the shares and the walls of the seam word, read as affine functions of the shares through their sums (Theorem 31.4(ii)), and $\Omega = A$. In a fibre integral of Gammas and coefficients, as the share side of a Gramma (Proposition 4.2) and the kernel of Theorem 31.7 below, they are the shares and $\Omega = A$, so that

$$C(\boldsymbol{\sigma}) = \{\boldsymbol{\sigma}' \in V : \ \sigma'_l = \sigma_l \text{ where } \sigma_l < 0, \ \ \sigma'_l \ge 0 \text{ elsewhere}\},$$

a compact convex polytope when the pins fix the total of the shares of every letter; the walls of source factors that are cube integrals (Definition 31.16 below) are polar functions as well.

Theorem 31.6 (A closing depends only on its chamber)**.** *Let I, $\mathbf{c}$ and $\boldsymbol{\sigma}$ be as in Definition 31.5, let I have the local decay (H1) of Lemma 31.3 on Ω, and let $\boldsymbol{\sigma}' \in C(\boldsymbol{\sigma})$. The rays $\mathbf{c}+\tau\boldsymbol{\sigma}$ and $\mathbf{c}+\tau\boldsymbol{\sigma}'$ meet the same polar hyperplanes at the same times, and (H0) holds along one of them exactly when it holds along the other; at every other time the fibre integrals through their two points are equal, and at a crossing their crossing terms are equal. So the closings along the two directions are one series, with the same remainders: the closing along $\boldsymbol{\sigma}$ converges, with the bounds of its terms and of its remainders, as soon as the closing along some direction of its chamber does. In Theorem 31.2, or in Theorem 31.4(iii), if $\mathbf{c}$ lies off the null set of Theorem 31.2, taken for Theorem 31.4(iii) with the polar hyperplanes of the seam function as well, the closing along $\boldsymbol{\sigma}$ converges absolutely as soon as*

$$\inf_{\boldsymbol{\sigma}' \in C(\boldsymbol{\sigma})} h(\boldsymbol{\sigma}') < 0,$$

h the rate of Theorem 31.2, the infimum being taken for Theorem 31.4(iii) over the rates h_θ, $\theta \in [0, \pi/8)$, as well; its terms and the integrals left after them are then $O(e^{\tau(h'+\epsilon)})$ for every $\epsilon > 0$, h' that infimum. On the chamber the rate h of Theorem 31.2 is convex.

So the closing along a direction converges as soon as some direction of its chamber is in the cone of Theorem 31.2, and its best certified rate is a convex problem in the free coordinates of the chamber. A chamber has positive dimension when the shares that do not fall have free directions on the pins, which stations of one excess supply. Theorem 31.2 reads both laws along one direction, through the largest values of the monomials of their letter vectors. The glue is also the average over the laws of the stand-in kernel (Theorem 28.10(ii)), whose integrand at one source pair has no walls, and, with the partner read in its monomials, the average over the law of the piece alone of an integrand without walls; each source point can therefore close along its own best direction of the chamber, and the averages of the crossing terms close the glue.

Theorem 31.7 (Averaging the laws)**.** *Take a share side, on pins that fix the total of the shares of every active letter, with V the space of the directions of the shares on the pins, whose integrand is*

$$\prod_{j:\, s_j>0} \frac{\prod \Gamma(a_l)}{\Gamma(s_j)} \prod_{j:\, t_j>0} \frac{\prod \Gamma(a_l)}{\Gamma(t_j)} \int e^{-\langle \mathbf{a}, y(\zeta)\rangle}\, d\lambda(\zeta),$$

the products over the shares l of K_j and of Ψ_j, $\mathbf{a}$ the vector of all the shares, λ a positive measure on a source, the source measure*, whose points are the* source points *ζ, and $y = (y_l)_l$ a continuous map of the source to the space of the shares. In Theorem 28.10 two share sides are of this kind, their shares l those of the active letters:*

- *(A) that of Theorem 28.10(iii), with one share per station; $\lambda = (\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho \otimes (\boldsymbol{\varrho}_R^\rho)_* \phi_R^\rho$, a source point $\zeta = (\mathbf{x}, \mathbf{x}')$ is a seam point of the piece and one of the partner, and $y_l(\zeta) = \log(x_{jD} x'_{jD})$ or $\log(x^*_{jD} x'^*_{jD})$ for the share of the station D of the letter;*
- *(B) that of Theorem 28.12 with the partner read in its monomials, one share per monomial, written as in Theorem 31.4(i), the seam word of the piece having a Newton sum of full dimension; $\lambda = (\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho$ is the law of the piece, a source point is a seam point $\zeta = \mathbf{x}$ of the piece, and $y_l(\zeta) = \log x_{jD} + \log \xi_l$ or $\log x^*_{jD} + \log \xi_l$ for the share of the monomial of the coefficient ξ_l at the station D.*

Let S_λ be the support of the image of λ under y, the log-support *of the source measure, $\mathbf{c}$ a real point of the pins with positive coordinates at which $\int e^{-\langle \mathbf{c}, y\rangle}\, d\lambda$ converges (in (A) the seam functions of the piece and of the partner converge there, in (B) that of the piece), $\boldsymbol{\sigma} \in V$, and $C = C(\boldsymbol{\sigma})$ the*

chamber of Definition 31.5 of the kernel below, whose polar functions are the shares:

$$C = \{\boldsymbol{\sigma}' \in V :\ \sigma'_l = \sigma_l \text{ where } \sigma_l < 0,\quad \sigma'_l \geq 0 \text{ elsewhere}\},$$

a compact convex polytope. Two shares are in one bundle *when they belong to one letter and have one column in the pins: in (A) when their stations have one excess, in (B) when moreover they carry one monomial of the partner. For y in the space of the shares put*

$$h(\boldsymbol{\sigma}'; y) = \sum_l \sigma'_l \log|\sigma'_l| - \langle \boldsymbol{\sigma}', y\rangle, \qquad h_C(y) = \min_{\boldsymbol{\sigma}' \in C} h(\boldsymbol{\sigma}'; y), \qquad \hat{h}_C = \sup_{y \in S_\lambda} h_C(y) \in (-\infty, \infty],$$

the rate *of $\boldsymbol{\sigma}'$ at y, the* chamber rate *of the kernel at y, concave in y, and the* averaged rate.

Let $\mathbf{c}$ be interior, relative to the pins, to the set of the real points at which the Laplace transform $\int e^{-\langle \mathbf{a}, y\rangle}\, d(y_*\lambda)(y)$ of $y_*\lambda$ converges; equivalently, for a basis $\mathbf{v}_1, \dots, \mathbf{v}_d$ of V there is $\eta > 0$ with

$$\sum_{\boldsymbol{\epsilon} \in \{-1,1\}^d} \int \exp\Big(- \Big\langle \mathbf{c} - \eta \sum_i \epsilon_i \mathbf{v}_i,\, y \Big\rangle \Big)\, d(y_*\lambda)(y) < \infty, \tag{31.1}$$

a finite certificate that $\mathbf{c}$ is interior to the set where that Laplace transform converges, that set being convex. Put $|y|_V = \sum_i |\langle \mathbf{v}_i, y\rangle|$, let p be the largest total order of the Gamma factors polar on one hyperplane that the ray of $\boldsymbol{\sigma}$ crosses, at most their number, and in this statement write $w(y) = e^{-\langle \mathbf{c}, y\rangle}(1 + |y|_V)^{p-1}$. Every power of $1 + |y|_V$ is at most a constant times $e^{\eta |y|_V} \leq \sum_{\boldsymbol{\epsilon}} \exp(\eta \sum_i \epsilon_i \langle \mathbf{v}_i, y\rangle)$, so (31.1) gives $\int w\, d(y_*\lambda) < \infty$.

(i) The kernel. *The* kernel *at a source point ζ is the fibre integral on the pins*

$$\int_{pins} \prod_{j:\, s_j > 0} \frac{\prod \Gamma(a_l)}{\Gamma(s_j)} \prod_{j:\, t_j > 0} \frac{\prod \Gamma(a_l)}{\Gamma(t_j)}\ e^{-\langle \mathbf{a}, y(\zeta)\rangle},$$

of an integrand without walls, and the fibre integral of the share side is its integral against λ, which in (A) and (B) is $P_{B \sqcup R}(\mathbf{m})$; in (A) the kernel is $\Gamma(\kappa_B)\, \mathsf{Z}_{(\kappa_B, \mathbf{s}, \mathbf{t}); t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{x}')$ of Theorem 28.10(ii). Let $\mathbf{c}$ lie off the null set of Theorem 31.2 for the direction $\boldsymbol{\sigma}$. For every $\epsilon > 0$ there is C_ϵ, not depending on the source point, such that the crossing terms $\mathcal{R}_k(\zeta)$ of the closing of the kernel along $\boldsymbol{\sigma}$, at their crossing times τ_k, satisfy

$$|\mathcal{R}_k(\zeta)| \leq C_\epsilon\, w(y)\, e^{\tau_k (h_C(y) + \epsilon)}, \qquad y = y(\zeta),$$

and the integrals left at times separating the crossings satisfy the same bound at the time τ with $e^{-\langle \mathbf{c}, y\rangle}$ in place of $w(y)$. When every crossed hyperplane carries a simple pole, $p = 1$ and $w(y) = e^{-\langle \mathbf{c}, y\rangle}$.

(ii) The averaged closing. *If $\hat{h}_C < 0$, the fibre integral of the share side is*

$$\sum_k \int \mathcal{R}_k\, d\lambda,$$

the sum in the order of crossing, absolutely convergent, its terms $O(e^{\tau_k(\hat{h}_C + \epsilon)})$ for every $\epsilon > 0$. At a crossing whose point along some $\boldsymbol{\sigma}' \in C$ lies in the interior of the set where $\int e^{-\langle \mathbf{a}, y\rangle}\, d\lambda$ converges, the averaged term is the crossing term of the share side there; so when it converges along the ray of $\boldsymbol{\sigma}$, the averaged closing is the closing of the share side along $\boldsymbol{\sigma}$, in (A) and (B) that of Theorem 31.2.

(iii) One direction reads the convex hull. $\min_{\boldsymbol{\sigma}'\in C}\sup_{y\in S_\lambda} h(\boldsymbol{\sigma}';y) = \sup_{y\in\operatorname{conv} S_\lambda} h_C(y) \geq \hat{h}_C$, *and* $\sup_{y\in S_\lambda} h(\boldsymbol{\sigma}';y)$ *is the rate of the closing of the share side along* $\boldsymbol{\sigma}'$ *read through the largest value of* $e^{-\langle\boldsymbol{\sigma}',y\rangle}$ *on the log-support, in (A) and (B) the rate* $h(\boldsymbol{\sigma}')$ *of Theorem 31.2: the best single direction of the chamber reads the convex hull of the log-support, the averaged closing the log-support itself. In a bundle* G *of shares that* $\boldsymbol{\sigma}$ *does not fall, the least value of* $\sum_{l\in G}(\sigma'_l\log\sigma'_l - \sigma'_l y_l)$ *over the* $\sigma'_l \geq 0$ *with* $\sum_G \sigma'_l = g$ *is* $g\log g - g\log\sum_{l\in G} e^{y_l}$*, attained at* $\sigma'_l = g\,e^{y_l}/\sum_{l'\in G} e^{y_{l'}}$*: at each source point the best direction merges the bundle into the sum* $e^{y_G} = \sum_{l\in G} e^{y_l}$*, which in (A) is the sum* $\sum_D x_{jD}x'_{jD}$ *that the kernel reads at the excess, splitting* g *in the proportions of its terms.*

At width one the share side of a node merged by its bundles has at most one free share and the seam torus is a half-line, so the log-support is an interval, its own hull, and one direction loses nothing. A gap between the averaged rate and the best single direction needs a chamber of positive dimension and a log-support that is not convex in its directions: the best split at each source point moves with it, and one direction must take one split for all of them. Averaging can reach a rate that no single direction certifies, and an averaged closing is in general no closing of one share side along one ray: its terms are averages of the crossing terms of the kernel, and it can converge past walls that every ray of the chamber meets. Where the averaged rate is infinite, a closing past the walls can still converge, at the rate of the best source point: past its walls the law is read through its expansion at its falling end, not through its moments, as at width one (Theorem 20.44). Past the walls of a piece the bound that travels is read on the sector letters (Theorem 31.11).

Theorem 31.8 (An averaging gap at a family of tables)**.** *There are three tables with the bases* $\binom{\{1,2,3,4\}}{2}$*, at the exponents* $s_j = 1$*,* $t_j = 0$ *and the masses* $\frac{3}{2}$ *on every mark, whose bunch* $\{1,2\}$*, with the room* $\{3,4\}$*, read through its seam with the product of the two laws as the source measure, the kind (A) of Theorem 31.7, has one station at each excess of each letter, so that every bundle of Theorem 31.7 is a single share, and a chamber* C *with*

$$\min_{\boldsymbol{\sigma}\in C}\sup_{y\in S_\lambda} h(\boldsymbol{\sigma};y) = 0, \qquad \hat{h}_C = \sup_{y\in S_\lambda}\min_{\boldsymbol{\sigma}\in C} h(\boldsymbol{\sigma};y) < 0 :$$

no direction of the chamber certifies the closing (Theorem 31.2), and the averaged closing converges geometrically (Theorem 31.7(ii)). Two directions of the chamber, the better one taken at each source point, already give the bound.

Remark 31.9 (What the gap separates). The gap lies between the averaged closing and the directions of one share side, the seam of the bunch with its bundles merged, and no merging of stations removes it, every bundle being a single share. It is not a gap against every closing of the period: closing each source point along its own direction, as the bound does with two directions, is itself a closing of the period, and so is completing another inner read before closing. A gap in every closing is a statement about a specified set of closings; for the closings of one share side by the directions of its chamber, the family of Theorem 31.8 has one, while at a seam whose two stations of one excess multiply the same monomial of the partner, merging the two stations can remove such a gap.

Definition 31.10 (Keyhole continuations along a direction; ceiling and phase)**.** In Theorem 31.4 let $\boldsymbol{\sigma}$ be a direction of the shares and $\bar{\boldsymbol{\sigma}}$ its sums over the monomials of the partner, and fix a fan of the seam word (Theorem 31.4(ii)). For a sector π of it let $q = (q_k)_k$ be its sector coordinates, $\kappa_k = \kappa_{\mathbf{v}_{\pi,k}}$ the walls of its generators and $\mathbf{k}_\pi = (k_{jD,\pi}, \psi_{jD,\pi})_{j,D}$ the vector of its sector letters, so

that the seam function on the pins is the sum over the sectors of

$$|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|\int_{[0,1]^{|B|-1}}\prod_k q_k^{\kappa_k(\bar{\mathbf{a}},\bar{\mathbf{b}})-1}\,\mathbf{k}_\pi(q)^{-(\bar{\mathbf{a}},\bar{\mathbf{b}})}\,dq,$$

the walls affine in the shares; for a piece of tables and the sectors of the orders π of B these are the walls κ_{U_k} of the tails and the sector letters of the tables $\mathcal{B}_{jD}$ (Theorem 8.5(ii)). Put $\dot\kappa_k=\partial_{\bar{\boldsymbol\sigma}}\kappa_k$, the derivative of the wall along $\bar{\boldsymbol\sigma}$, and

$$\varrho_{\pi,\bar{\boldsymbol\sigma}}(q)=\prod_k q_k^{\dot\kappa_k}\;\mathbf{k}_\pi(q)^{-\bar{\boldsymbol\sigma}},$$

the *directional ratio in the sector* of π; its *falling variables* along $\bar{\boldsymbol\sigma}$ are the q_k with $\dot\kappa_k<0$. A *keyhole continuation along* $\bar{\boldsymbol\sigma}$ is defined as in Definition 20.47, with the sectors in place of the orders, with these falling variables, and with the sector letters of $\mathbf{k}_\pi$ required not to vanish on the sets Y of its boxes; its *outline* $\mathcal{O}$ is the union of the sets $\mathcal{O}$ of its boxes, and its *ceiling along* $\bar{\boldsymbol\sigma}$ is the largest value of $|\varrho_{\pi,\bar{\boldsymbol\sigma}}|$ on the outline, π the sector of the box. Each set Y is a product of a box and of keyholes, simply connected, and on a neighbourhood of it the sector letters have logarithms real at the real points; the *phase* of the continuation at a point q of the outline of a box of π is the linear function $\boldsymbol\vartheta\mapsto\langle\bar{\boldsymbol\vartheta},\arg\mathbf{k}_\pi(q)\rangle$ on the directions of the shares, arg the imaginary part of the logarithm and the bars marking sums over the monomials of the partner.

The modulus $|\varrho_{\pi,\bar{\boldsymbol\sigma}}(q)|$ is the factor by which the part of the integrand that grows with τ along $\bar{\boldsymbol\sigma}$ is multiplied per unit of τ at the point; the phase is the linear function by which the imaginary parts of the shares enter it. With no falling variables the outline is the real cube, the letters are positive there, the phase vanishes, and the ceiling is $\sup\mathbf{V}_B^{-\bar{\boldsymbol\sigma}}$, the largest value of the monomial of Theorem 31.2; at width one, along the fall of the share b^d of Section 20.7, the directional ratio in the sector is the ratio ϱ_π of Definition 20.46.

Theorem 31.11 (The travelling bound past the walls)**.** *In Theorem 31.4 let $\boldsymbol\sigma$ be a direction of the shares and take a keyhole continuation along $\bar{\boldsymbol\sigma}$ (Definition 31.10) with the outline $\mathcal{O}$.*

(i) For every $\epsilon>0$ there is C such that for $\tau\ge0$ and $\boldsymbol\vartheta$ in the directions of the shares

$$\big|P_{W_B(\bar{\mathbf{c}}+\tau\bar{\boldsymbol\sigma}+i\bar{\boldsymbol\vartheta})}(\mathbf{m}_B)\big|\le C\,\delta^{-(|B|-1)}\sup_{q\in\mathcal{O}}\Big(|\varrho_{\pi,\bar{\boldsymbol\sigma}}(q)|^\tau\,e^{\tau\epsilon+\langle\bar{\boldsymbol\vartheta},\arg\mathbf{k}_\pi(q)\rangle}\Big),$$

δ the smaller of 1 and the distance of $\bar{\mathbf{c}}+\tau\bar{\boldsymbol\sigma}$ from the hyperplanes on which a wall of the fan of Definition 31.10 lies in $-\mathbb{N}$, which hold the poles of the seam function.

(ii) Let $\boldsymbol\upsilon$ be a bounded function on $\mathcal{O}$ with values in the vectors indexed by the shares, with $\langle\bar{\boldsymbol\vartheta},\arg\mathbf{k}_\pi(q)\rangle=\sum_l\upsilon_l(q)\vartheta_l$ for $\boldsymbol\vartheta$ in the directions of the shares and $|\upsilon_l(q)|\le\frac{\pi}{2}-\epsilon_0$ for some $\epsilon_0>0$; representatives differing by a vector orthogonal to the directions give the same phase on the fibre. Put

$$h_{\boldsymbol\upsilon}(\boldsymbol\sigma)=h_\Gamma(\boldsymbol\sigma)+\sup_{q\in\mathcal{O}}\Big(\log|\varrho_{\pi,\bar{\boldsymbol\sigma}}(q)|-\sum_{l:\,\sigma_l>0}\sigma_l\log\cos\upsilon_l(q)\Big),$$

h_Γ the Gamma rate of Theorem 31.4(iii). If $h_{\boldsymbol\upsilon}(\boldsymbol\sigma)<0$ and the real point $\mathbf{c}$ of the pins lies off the null set of Theorem 31.2, taken with these hyperplanes as well, the conclusion of Theorem 31.4(iii) holds along $\boldsymbol\sigma$, its terms and the integrals left after them being $O(e^{\tau(h_{\boldsymbol\upsilon}(\boldsymbol\sigma)+\epsilon)})$ for every $\epsilon>0$.

Corollary 31.12 (When the phase costs nothing)**.** *In Theorem 31.11(ii) let $H_{\mathcal{O}}$ be the ceiling of the continuation along $\bar{\boldsymbol{\sigma}}$. Then $h_{\boldsymbol{\upsilon}}(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \log H_{\mathcal{O}}$ in each of two cases:*

(i) $\upsilon_l(q) = 0$ for every $q \in \mathcal{O}$ and every share with $\sigma_l > 0$: the phase sits on shares that fall or stay, whose Gammas absorb it;

(ii) $-\sum_{l:\,\sigma_l>0} \sigma_l \log\cos\upsilon_l(q) \le \log H_{\mathcal{O}} - \log|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}(q)|$ for every $q \in \mathcal{O}$: the modulus lost away from the highest points pays for the phase.

With no falling variables (i) holds with $\boldsymbol{\upsilon} = 0$, and the rate is that of Theorem 31.2 for the glue read in the monomials of the partner.

Theorem 31.4(iii) pays for an allowed growth $e^{\theta|\operatorname{Im}\mathbf{z}|_1}$ of the seam function with the term $\frac{2\theta^2}{\pi}\sum_{\sigma_l>0}\sigma_l$; Theorem 31.11 pays only for the phase that the letters have at the points where the ceiling is read, and at the exact price $-\log\cos\upsilon_l$ of the rotation. Enlarging the radii of the keyholes, as far as the sector letters do not vanish, can lower the ceiling; it comes down to the reach at simple tops that do not cancel (Theorem 32.15(iii)). The joint words of Section 31.3 complete the retained fibre on its strips before a modulus is taken, and their bound $r_G c_{\max}$ has no phase term; the ceiling $c_{\max}$ of the joint word need not be the ceiling of this outline.

Proposition 31.13 (How travelling bounds compose)**.** *(i)* Products. *When the share integrand carries several continued factors, each bounded as in Theorem 31.11(i) on its own outline, their product is bounded on the product of the outlines by the products of their moduli and of the exponentials of their phases, and Theorem 31.11(ii) holds with these on the product outline: the phases of different factors at one point of it may cancel.*

(ii) Averaging. *Let integrands I_ζ, for ζ in a source with a measure λ, have fibre integrals that close along one path with their crossings at the same times, with the crossing terms $\mathcal{R}_k(\zeta)$, and with crossing terms and remainders, at times τ_k separating the crossings, at most $M(\zeta)\beta(\tau_k)$, where M is integrable against λ and $\sum_k \beta(\tau_k) < \infty$. Then $\int \big(\int_{\mathbf{c}+iV} I_\zeta\big)\, d\lambda(\zeta) = \sum_k \int \mathcal{R}_k\, d\lambda$, absolutely convergent, the k-th term at most $\|M\|_1\beta(\tau_k)$. When for some $\hat{h} < 0$ the bound holds with $\beta(\tau) = e^{\tau(\hat{h}+\epsilon)}$ for every small $\epsilon > 0$, M depending on ϵ, the terms are $O(e^{\tau(\hat{h}+\epsilon)})$: the rate is the bound $\hat{h}$ shared by the source points, read on the support of λ, the uniformity over the source being the hypothesis on M. Where the fibre integral and the source integral of a crossing term converge absolutely together, $\int \mathcal{R}_k\, d\lambda$ is the crossing term of the averaged integrand $\int I_\zeta\, d\lambda$.*

(iii) Crossing. *The crossing terms of the closings of Theorems 31.4(iii) and 31.11 are fibre integrals of one dimension less of integrands of the same kind. At a share crossing $a_l = -n$ the Gamma becomes $(-1)^n/n!$ and the other factors are read on the hyperplane. At a wall crossing where the pole of the seam function is simple, its residue is the* face coefficient *of the walls through the hyperplane: the sum, over the generators $\mathbf{v}$ of a fan of the seam word (Theorem 31.4(ii)) whose wall $\kappa_{\mathbf{v}}$ takes a value $-n_{\mathbf{v}} \in -\mathbb{N}$ on the hyperplane and over the sectors π of which they are generators, of $|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|$ times the cube integral over the other coordinates of the sector of the $n_{\mathbf{v}}$-th Taylor coefficient at $q_{\mathbf{v}} = 0$ of the rest of the integrand (Lemma 8.4), $q_{\mathbf{v}}$ the sector coordinate of $\mathbf{v}$, each divided by the derivative of $\kappa_{\mathbf{v}}$ along the vector $\mathbf{e}$ of Lemma 31.3; its letters are the sector letters at $q_{\mathbf{v}} = 0$, positive on the face, and its numerators are polynomials in the shares and in the derivatives of the letters. For a piece of tables and the sectors of the orders the generators are those of the tails U, and the sectors the orders of which they are tails. Where the pole is multiple, because several factors, or two walls of one sector, are polar on the hyperplane, the residue is a*

Laurent coefficient, a finite sum of derivatives in the shares of products of such terms, which brings powers of the logarithms of the letters and of the q_k. A keyhole continuation of the seam function restricts to one of each face coefficient, $q_{\mathbf{v}} = 0$ lying in the disc of the keyhole of $q_{\mathbf{v}}$, and Cauchy's estimate on its circle bounds the Taylor coefficients by the values on the outline; so each crossing term can be closed again by Lemma 31.3 whenever a travelling bound of its own tends to zero.

Completing an inner read is not a crossing. Merging the stations of one excess by Corollary 4.4, integrating a source before closing, or finishing a transverse fibre replaces the integrand by a completed one, with a smaller fibre; the identity between the two integrands holds where the completed read converges and persists under continuation, but the travelling bound does not pass from one to the other, and the completed object can grow along the outer path where the uncompleted integrand does not. Completing an inner read is free when the completed object obeys, along the outer path, a bound already in hand. Merging the stations of one excess reads the law through the sum of their seam letters, whose expansion at the falling end of the law can stop at an edge of the support nearer than the far edge that the stations read, and the merged closing can then diverge where the station closing converges. The two closings agree at every step up to their remainders (Lemma 31.3), and only the remainder of the station closing need tend to zero. On the shares, merging the stations is conditioning on their merged seam letter: integration commutes with completing the fibre, and continuation commutes with it only while the level sets of the merged seam letter stay inside the region where the expansion of the law at its falling end represents it. Completing the inner fibre before closing and keeping its inherited cut are two representations, whose finite heads agree together with their boundary words; the regions on which a law equals its expansion, with their cuts and boundary words, are built in Section 32.1, and a convergent germ alone does not equal the law.

31.3 Retained crossings as joint words

31.3.1 Completing the retained fibre before taking its size

Theorem 31.11 bounds the share integrand before its fibre is completed, and pays for the phase of the letters where it reads the ceiling. The procedure for joint words completes the retained shares first. It keeps the sectors and coordinate cuts of every continued source and its finite heads, and its auxiliary variables are part of one joint input of a Trailmoment (Definition 5.1), their number not growing along a retained cohort (Definition 31.16): a falling Gamma kept inside a crossing term becomes a positive share and one Beta variable, and a continued source strip becomes a segment and a few circle integrals whose powers on the circles have integral exponents.

Lemma 31.14 (A Gamma strip as a Beta input)**.** *Let $\delta \geq 1$ be an integer, $b = a + \delta$, and $0 < \operatorname{Re} b < 1$. For a nonzero coefficient ξ, with one fixed logarithm,*

$$\Gamma(a)\,\xi^{-a} = \frac{(-1)^{\delta}\xi^{\delta}}{\Gamma(\delta)} \int_0^1 (1-u)^{\delta-1}\,\Gamma(b)\,(\xi u)^{-b}\,du.$$

The identity is pointwise in every remaining share. So on a fibre $P\mathbf{a} = \boldsymbol{\pi}$ replacing a_l by $b_l = a_l + \delta_l$ changes the pins to $P\mathbf{b} = \boldsymbol{\pi} + P(\delta_l)_l$ and does not move the contour. A share with positive real part is left unchanged, with $\delta_l = 0$ and no auxiliary variable. In a family $\operatorname{Re} a_l(n) = a_l^0 - d_l n$ with $d_l \in \mathbb{N}$ and $a_l^0 \notin \mathbb{Z}$ one may take $\delta_l(n) = d_l n + \delta_l(0)$ with $0 < a_l^0 + \delta_l(0) < 1$, the lift being used at the n with $\delta_l(n) \geq 1$, all but finitely many when $d_l \geq 1$: one Beta input serves at every n.

Lemma 31.15 (A continued source strip in six integrals)**.** *Let g be holomorphic near $\{|q| \le r_2\} \cup [r,1]$, and choose $0 < r < r_1 < rT < r_2$ and $T > 1$. If $\ell \ge 1$ is an integer and $0 < \operatorname{Re}(\kappa + \ell) < 1$, the continued integral $I_\kappa = \operatorname{Reg} \int_0^1 q^{\kappa-1} g(q)\, dq$ (Definition 1.5) is*

$$I_\kappa = \int_r^1 q^{\kappa-1} g(q)\, dq + R_\ell + R'_\ell,$$

all circles run counterclockwise, with

$$R_\ell = \frac{1}{2\pi i} \int_0^r q^{\kappa+\ell-1} \oint_{|\zeta|=r_1} \frac{g(\zeta)\, \zeta^{-\ell}}{\zeta - q}\, d\zeta\, dq,$$
$$R'_\ell = -\frac{r^\kappa}{2\pi i} \Big(\int_1^T u^{\kappa-1} \oint_{|\zeta|=r_2} + \int_T^\infty u^{\kappa-1} \oint_{|\zeta|=r_1} \Big) g(\zeta)\, \frac{1 - (ru/\zeta)^\ell}{\zeta - ru}\, d\zeta\, du.$$

Splitting the numerator of R'_ℓ gives four absolutely convergent integrals, six in all. This remains true with holomorphic parameters and other source variables on compact sets, uniformly when the strip and the radius inequalities have positive margins. For $\kappa(\mathbf{a}, n) = \kappa_0(\mathbf{a}) - dn$ and $\ell(n) = \ell_0 + dn$, $d \in \mathbb{N}$, each of the six has a fixed domain and is a fixed function times an n-th power; the exponent of ζ that contains $\ell(n)$ is integral and independent of the shares that move. At a simple wall $\kappa = -N$, $N \in \mathbb{N}$, the residue is the single coefficient input

$$[q^N] g(q) = \frac{1}{2\pi i} \oint_{|\zeta|=r_1} g(\zeta)\, \zeta^{-N-1}\, d\zeta.$$

A cut upper end $q_0 > 0$ is kept by applying the same formula after $q = q_0 v$, the factor q_0^κ included; a positive lower cut is an ordinary segment throughout.

Two of the six integrals simplify by Cauchy's formula: the integral of R'_ℓ over $[1, T]$ with the numerator 1 is $-\int_r^{rT} q^{\kappa-1} g(q)\, dq$, and the integral over (T, ∞) with the numerator 1 vanishes, its pole ru lying outside the smaller circle. So I_κ is the segment $\int_{rT}^1 q^{\kappa-1} g$, the remainder R_ℓ on the circle of radius r_1, and the two integrals of R'_ℓ with the numerator $(ru/\zeta)^\ell$; along a family $\kappa_0 - dn$ the segment carries the n-th power of q^{-d}, at most $(rT)^{-d}$, and the circle integrals that of ζ^{-d}, of modulus r_1^{-d} or r_2^{-d}. The radius r splits the contour, not the source: the sum does not depend on it.

Definition 31.16 (A rational retained cohort)**.** A *retained cohort* is one sequence of simple crossing terms, or one sequence of travelling fibre integrals, of a fibre integral of Gammas, coefficients and source factors whose pins carry a *grading*, a rational vector $\mathbf{v}$ with $\langle t_l, \mathbf{v} \rangle = 1$ for the column t_l of every share in the pins, such as the share side of Theorem 31.4 and the kernel of Theorem 31.7, graded by their letter totals, and the share side of a Gramma whose configuration has a grading (Proposition 4.2), with the following data retained: the pins; the inherited contour; the crossed hyperplane and its orientation, if any; the source sectors and their coordinate cuts. The contours have the real points $\mathbf{a}^0 + n\mathbf{d}$ with $\mathbf{d}$ rational. Nonresonance is tested against the full displayed arrangement of the polar hyperplanes of the Gammas and of the source walls, including candidate walls which might cancel after the sectors are added: the real points lie on none of these hyperplanes other than the one being crossed, and at a source crossing each contributing sector has just one wall on that hyperplane. The constant terms of the shares and walls and the indices are real. Finitely many initial terms may be removed and kept as a finite head. Passing to finitely many arithmetic subsequences makes the changes of all falling shares and walls integral; this is the *integer step* of

the cohort, and its index n always refers to that step. The same definition permits a finite product of source factors that are cube integrals in their charts (Lemma 8.4)

$$\operatorname{Reg}\int_{(0,1)^k} q^{\boldsymbol{\kappa}(\mathbf{a})-\mathbf{1}}\prod_i P_i(q)^{-w_i(\mathbf{a})}\,dq,$$

the P_i with nonnegative coefficients and positive constant terms and the affine functions κ_i, w_i with rational linear parts. A growing wall is eventually positive; a stationary wall may be negative and off the integers, and is then continued with a constant integer ℓ in Lemma 31.15. Finite coordinate box cuts are part of the retained data. The sector integrals of a fan of a seam word (Theorem 31.4(ii)), those of tables among them (Corollary 8.6), have these properties.

Proposition 31.17 (Computing the cohorts of a rational ray)**.** *Let the pins and the finitely many candidate polar functions ℓ_i be rational, and let a rational open box in the affine space of the pins be given on which the required initial walls and shares are positive. For a prescribed rational direction a rational starting point in that box can be chosen such that distinct polar hyperplanes never meet the ray at the same time; factors on the same hyperplane are kept together. After a finite head, the crossing order and the separating gaps are a finite repeated pattern, and its crossing cohorts, endpoint cohorts, the integers ℓ of Lemma 31.15 and a uniform positive separation are computed by finite rational arithmetic. More precisely, with the direction multiplied by a common denominator so that every falling polar function decreases at an integer falling rate $d_i > 0$, there are at most $\sum_i d_i$ crossings in each interval of length one of this parameter; neither that number nor the stored source sectors depend on how many such intervals are read.*

Theorem 31.18 (A finite procedure for a retained cohort)**.** *For a rational retained cohort there is a finite procedure, using the pins and the monomials of the source factors, with the following output, the* joint word *of the cohort, after a finite head and a possible change to the integer step:*

$$\mathcal{R}_n = G_n\sum_i\int_{Y_i} c_i(x)^n\,\phi_i(dx) = G_n\sum_i\big(\mathsf{T}_{-n;(1)}[c_i]\,\phi_i\big)(Y_i),\qquad n\in\mathbb{N},\tag{31.2}$$

a finite sum over i. Here Y_i is a compact cube times finitely many circles, the coefficient function c_i is continuous on it, and the input ϕ_i is a complex measure of finite total variation; the coefficients, their branches, the orientations and every cut are retained. On every Y_i each coefficient of a letter is a real Laurent monomial times a coefficient factor*, holomorphic there, with its real part bounded below by a positive constant and its modulus bounded above, uniformly on the compact variables; these two bounds are the* margins *of the coefficient factors, and the radii $r < r_1 < rT < r_2$ of Lemma 31.15 are admissible while the margins hold. At the integer index $-n$ the Zamma of the one-term configuration is the polynomial $\mathsf{Z}_{-n;(1)}(c) = c^n$, at $c = 0$ as well, so the second expression is the Trailmoment of Definition 5.1 extended to a continuous complex coefficient function and a complex input of finite variation, as Theorem 5.21(ii) extends its reads to complex data where every integral converges absolutely.*

The scalar is explicit. Let $\mathbf{v}$ be the grading of the pins (Definition 31.16) and $w = \langle\mathbf{v},\boldsymbol{\nu}\rangle$, $\boldsymbol{\nu}$ the values of the pins, the total of the shares on the pins, which on the share sides of Theorems 31.4 and 31.7 is $|\mathbf{s}|+|\mathbf{t}|$, the sum of the exponents of the active letters. Let $\mathcal{S}_-$ be the shares treated by Lemma 31.14, with $\delta_l(n) = \delta_l(0) + d_l n$. At a Gamma crossing let $-N(n)$, with $N(n) = N_0 + d_0 n$, be the pole of its Gamma that is crossed; at a source crossing or for a travelling fibre omit N and the factor $\Gamma(N+1)$ of G_n, and put $d_0 = 0$. Up to fixed nonzero factors of the share side, as $1/\Gamma(s_j)$

and $1/\Gamma(t_j)$, and the inherited crossing sign,

$$G_n = \frac{(-1)^{N(n)+\sum_{l\in\mathcal{S}_-}\delta_l(n)}\,\Gamma\big(w+N(n)+\sum_{l\in\mathcal{S}_-}\delta_l(n)\big)}{\Gamma(N(n)+1)\prod_{l\in\mathcal{S}_-}\Gamma(\delta_l(n))},$$

fixed nonzero constants may equally be put in the inputs. With $\bar d = d_0+\sum_{l\in\mathcal{S}_-} d_l$,

$$\lim_{n\to\infty}|G_n|^{1/n} = r_G = \frac{\bar d^{\,\bar d}}{d_0^{d_0}\prod_{l\in\mathcal{S}_-} d_l^{d_l}}, \qquad 0^0 = 1;$$

when $\bar d = 0$ the scalar is constant and $r_G = 1$, and in all cases $|G_n| \le C(n+1)^p r_G^n$ for computable C and p. The procedure introduces at most six integrals per continued source wall and one Beta variable per negative retained Gamma, whatever n. Its remaining work is a finite common Newton fan and simplicial subdivision; this fan, the inputs and the coefficient functions c_i serve for every n, and the procedure does not expand the Taylor head of degree $\ell(n)-1$ of Lemma 31.15 or a residue coefficient of degree $N(n)$ into its monomials.

Corollary 31.19 (Finite bounds and the exact positive root rate)**.** *For the joint word of Theorem 31.18 put $c_{\max} = \max_i \max_{Y_i}|c_i|$ and $M = \sum_i \|\phi_i\|$. Then*

$$|\mathcal{R}_n| \le M\,|G_n|\,c_{\max}^n \le CM(n+1)^p (r_G c_{\max})^n,$$

and $c_{\max}$ is the ceiling *of the joint word. Every quantity in this bound has a finite certificate: the margins of the coefficient factors and the sector walls bound M, and compact subdivision bounds $c_{\max}$; with computable input coefficients a strict inequality $r_G c_{\max} < r < 1$ is certified by finite subdivision and gives a computable geometric tail. If the original source has a convergent positive representation on every contour of the cohort, no source continuation or coefficient input is needed, and the remaining constant terms are real, then the output of the procedure has, after its common scalar sign is removed, one positive input ϕ and a coefficient function $c \ge 0$, and*

$$\lim_{n\to\infty}|\mathcal{R}_n|^{1/n} = r_G \operatorname*{ess\,sup}_{\phi} c.$$

For a positive read on an unbounded domain, before it is compactified, the same holds with an infinite essential supremum whenever the moments exist. So averaging over a positive source measure takes the essential supremum on the joint input of the source, Beta and shape variables; it does not reverse the order of an infimum over directions and a supremum over source points.

Definition 31.20 (Simple tops of the joint word)**.** For (31.2) with $c_{\max} > 0$, tops are read as in Definition 20.50, with the coefficient function c_i in place of ϱ_π, the oriented Y_i in place of the keyhole contour and the input ϕ_i in place of ϕ_π: a simple top is a point p with $|c_i(p)| = c_{\max}$ for which local real coordinates on the oriented Y_i are $(\mathbf{x},\mathbf{y}) \in [0,\epsilon)^{b_1}\times(-\epsilon,\epsilon)^{b_2}$, and

$$\log\frac{c_i(\mathbf{x},\mathbf{y})}{c_i(p)} = -\langle\boldsymbol{\lambda},\mathbf{x}\rangle - \tfrac12\,\mathbf{y}^{\mathsf T}\mathbf{Q}\,\mathbf{y} + O\big(|\mathbf{x}|^2+|\mathbf{x}|\,|\mathbf{y}|+|\mathbf{y}|^3\big), \qquad \phi_i = \prod_{j=1}^{b_1} x_j^{\kappa_j-1}\,g(\mathbf{x},\mathbf{y})\,d\mathbf{x}\,d\mathbf{y},$$

with $\kappa_j > 0$, $g(0,0)\neq 0$, $\operatorname{Re}\lambda_j > 0$ and $\operatorname{Re}\mathbf{Q}$ positive definite, $\mathbf{Q}$ complex symmetric, and g, continuous and bounded near $(0,0)$, including the orientation and every factor of the circle integrals. Its exponent and its coefficient are

$$\operatorname{ex}(p) = \sum_j \kappa_j + \tfrac{b_2}{2}, \qquad A(p) = g(0,0)\,(2\pi)^{b_2/2}(\det\mathbf{Q})^{-1/2}\prod_j \Gamma(\kappa_j)\,\lambda_j^{-\kappa_j},$$

the square root continued from positive definite real matrices. The subdivisions of the procedure, the splits of the Beta intervals and of the scales and the cones of the fan, divide one domain into parts on which the integrand is the restriction of one function; tops are read with these parts reassembled, a point on a common boundary of the parts being a simple top when the expansion holds on its reassembled neighbourhood, in the coordinates before the subdivision. Parts that are different integrals, as the segment and the circle integrals of Lemma 31.15, keep their own coefficients, and the contributions at one point are added before noncancellation is tested.

Theorem 31.21 (The joint tops give the root rate of a complex word)**.** *Let the largest values of $|c_i|$ on the joint word of Theorem 31.18 sit at finitely many simple tops. Let ex_0 be the least of their exponents, and for $|\zeta| = 1$ let A_ζ be the sum of the coefficients $A(p)$ of the tops with $\mathrm{ex}(p) = \mathrm{ex}_0$ and $c_i(p) = c_{\max}\,\zeta$. If some A_ζ is not zero, then*

$$\limsup_{n\to\infty} |\mathcal{R}_n|^{1/n} = r_G\, c_{\max}.$$

No separate price of the phase is added to $c_{\max}$: the completed fibre and its source phases belong to the one coefficient function and the oriented input. A largest value of $|c_i|$ alone is not the hypothesis; the expansion includes the stationarity of the complex phase in every tangential variable.

This is Theorem 20.51 for joint words: the coordinates $\mathbf{x}$ that vanish at a top carry the Gammas of the boundary, the coordinates $\mathbf{y}$ the factor $(2\pi)^{b_2/2}(\det \mathbf{Q})^{-1/2}$, and the coefficients of the tops of least exponent and one phase add before the root rate is read.

Remark 31.22 (Where the tops of a continued source factor sit)*.* In the integrals of Lemma 31.15 a falling wall carries the n-th power of q^{-d} on the segment and of ζ^{-d} on the circles. On a circle the modulus of ζ^{-d} is constant and its phase turns, so a largest value of $|c_i|$ there is a simple top only where the other factors make the phase stationary, a saddle in the radius. The radii r, r_1, r_2 and T are free within the margins of the coefficient factors, and the sum does not depend on them; Theorem 31.21 is applied after the radii are chosen. When the largest values of $|c_i|$ sit at simple tops of the segment, of the shape or of the Beta inputs, or at saddles of the circles, and do not cancel, the root rate is $r_G c_{\max}$; a largest value spread over a circle, or one at the end $q = rT$ of the segment, which moves with the radii, certifies the upper bound only, and the radii may be moved, within the margins, to lower it. Where the law has log expansions the continuation is made on the law, and its bound has no phase and no margin (Proposition 32.14).

Corollary 31.23 (Closing along a path under an endpoint condition)**.** *In Lemma 31.3 keep (H0) and (H1) and replace (H2) by the weaker condition $\int_{\gamma(\tau_k)+iV} I \to 0$ along times $\tau_k \to \infty$ off the poles. Then the sums of the signed crossing terms before τ_k converge to $\int_{\mathbf{c}+iV} I$. If the crossing terms are absolutely summable, this is their absolutely convergent ordered sum; (H2) remains a sufficient condition for the endpoint condition.*

Corollary 31.24 (Closing with the joint words)**.** *For a rational ray with simple nonresonant crossings, compute the finitely many arithmetic cohorts of crossings and one cohort of endpoints off the poles in each repeating gap. If each crossing cohort and each required endpoint cohort has a certified upper root rate $r_G c_{\max} < 1$, the crossing terms are absolutely summable and their ordered sum is the initial fibre integral. The finite head is retained, and the bounds of Corollary 31.19 give an effective stopping rule. A cohort with the positive hypothesis of Corollary 31.19, or the noncancelling simple tops of Theorem 31.21, has the stated exact root rate and cannot tend to zero when it exceeds one.*

The procedure keeps the falling rates, which fix the crossed heads, and the cut source; within the chamber the growing shares can still be changed by Theorem 31.6, the completed word being the same. The bound $r_G c_{\max}$ has no phase term: $c_{\max}$ is the ceiling of this joint word, which need not equal the ceiling of a keyhole continuation read before the fibre is completed, nor the least ceiling over all joint words.

Lemma 31.25 (Positive capacity and the growing shares)**.** *Let $t_g \in \mathbb{R}^r$, $g \in G$, be a finite nonempty configuration with a grading $\langle t_g, \mathbf{v}\rangle = 1$, let $\xi_g > 0$, $w > 0$ and $\langle \mathbf{v}, \nu\rangle = w$, and in this statement write $L(X) = \sum_{g\in G} \xi_g X^{t_g}$ and $\Pi(\nu, w) = \{\mathbf{b} \in [0,\infty)^G : \ \sum_g b_g = w,\ \sum_g b_g t_g = \nu\}$. With $0^0 = 1$ and the infimum of the empty set equal to $+\infty$,*

$$\sup_{X\in(0,\infty)^r} \frac{X^\nu}{L(X)^w} = \inf_{\mathbf{b}\in\Pi(\nu,w)} \prod_{g\in G} \Big(\frac{b_g}{w\,\xi_g}\Big)^{b_g},$$

the left side being the capacity *of L at (ν, w). The supremum may equally be taken on any logarithmic section transverse to $\mathbf{v}$. When ν/w lies in the relative interior of the convex hull of the t_g, the infimum has a unique positive minimizer and is attained by a gauge X, unique modulo the directions constant on all the t_g; on a proper face the same computation on that face gives the value, approached by suppressing the other monomials.*

The supremum on the left is the largest value of a monomial of the letter vector, and the logarithm of the infimum on the right is the least Gamma rate over the polytope $\Pi(\nu, w)$ of shares, less $w \log w$: the lemma is the duality that turns the chamber rate of a positive kernel into a capacity of its letter.

Theorem 31.26 (The exact positive rate of a retained chamber)**.** *Take a Gramma $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ with rational configuration and real index read on its share side (Proposition 4.2), the pins $t\mathbf{a} = \nu$ with the grading of the letter totals and positive coefficients $\boldsymbol{\xi}$. Let a nonzero rational ray start at a positive real point of the pins, and consider any simple nonresonant Gamma crossing cohort of that ray, or any cohort of travelling fibres off the poles, with its integer step as in Definition 31.16. Write its direction per step as $\boldsymbol{\sigma}$, and put*

$$\mathcal{S}_- = \{l :\ \sigma_l < 0\}, \qquad d_l = -\sigma_l \ \ (l \in \mathcal{S}_-), \qquad \mathcal{S}_+ = \mathcal{S}_-^c,$$

t_l the columns of t, with the chamber of $\boldsymbol{\sigma}$ (Definition 31.5), the polar functions of the share side being the shares,

$$C = \{\boldsymbol{\sigma}' :\ t\boldsymbol{\sigma}' = 0,\ \sigma'_l = -d_l\ (l \in \mathcal{S}_-),\ \sigma'_l \geq 0\ (l \in \mathcal{S}_+)\}.$$

Then the retained terms have a common sign at each index, independent of $\boldsymbol{\xi}$, after their crossing orientation is fixed, and

$$\lim_{n\to\infty} |\mathcal{R}_n(\boldsymbol{\xi})|^{1/n} = \exp \min_{\boldsymbol{\sigma}'\in C} h_\Gamma(\boldsymbol{\sigma}'),$$

h_Γ the Gamma rate of Theorem 31.4(iii), the coefficients being those of the Gramma: the exponential of the least Gamma rate of the chamber. This minimizes over the growing shares, the falling rates d_l being fixed; if the integer step has the length Δ in the parameter of the ray, the logarithm is Δ times the rate per unit of that parameter.

Corollary 31.27 (Exact positive averaging at every retained dimension)**.** *In Theorem 31.26 let the coefficients be measurable positive functions $\xi_l(\zeta)$ of a source with a nonzero positive measure λ, the configuration, the index, the chamber and the integer step fixed across the source, and put*

$$h_C(\zeta) = \min_{\boldsymbol{\sigma}' \in C} \Big(\sum_l \sigma'_l \log|\sigma'_l| - \sum_l \sigma'_l \log \xi_l(\zeta) \Big), \qquad \hat{h}_C = \operatorname*{ess\,sup}_{\lambda} h_C.$$

Let one averaged retained term be absolutely integrable, so that after multiplication by $\prod_l \xi_l(\zeta)^{-a_l^0}$, $\mathbf{a}^0$ the real point at which the cohort starts (Definition 31.16), the joint positive input is finite. If $\hat{h}_C < \infty$, every later averaged term is integrable and

$$\lim_{n\to\infty} \Big| \int \mathcal{R}_n(\zeta)\, d\lambda(\zeta) \Big|^{1/n} = e^{\hat{h}_C};$$

if all these terms are finite and $\hat{h}_C = +\infty$, their root rate is $+\infty$. No estimate uniform over the source and no retained fibre of dimension one is needed. This applies to the kernel of Theorem 31.7 on rational pins along a rational ray whose crossings are simple and nonresonant, under a finite averaged rate and the certificate (31.1)*, whose bound supplies the finite first term. For a continuous coefficient map the essential supremum is the supremum on the support of the image of λ; it is $\operatorname{ess\,sup}_\zeta \min_C$, with no interchange to $\min_C \operatorname{ess\,sup}_\zeta$.*

Theorem 31.28 (Averaging is sharp)**.** *In the setting of Theorem 31.7, with its chamber C, its averaged rate $\hat{h}_C$, its source measure λ and the crossing terms $\mathcal{R}_k$ of the closing of its kernel at the crossing times τ_k, let the pins be rational, as in the kinds (A) and (B), let $\boldsymbol{\sigma}$ have rational coordinates, let the crossings of its ray be simple and nonresonant (Definition 31.16), and let $\hat{h}_C < \infty$. Along each cohort of crossings, with its integer step of length Δ in τ, the averaged terms have one sign at each index and $\lim_{n\to\infty} \big| \int \mathcal{R}_n\, d\lambda \big|^{1/n} = e^{\Delta \hat{h}_C}$ (Theorem 31.26, Corollary 31.27); so for every $\epsilon > 0$, $\big| \int \mathcal{R}_k\, d\lambda \big| \geq e^{\tau_k(\hat{h}_C - \epsilon)}$ for all large k, and the terms of the averaged closing, which are those of the closing of the share side along $\boldsymbol{\sigma}$ when $\int e^{-\langle \mathbf{a}, y\rangle}\, d\lambda$ converges along it, tend to zero when $\hat{h}_C < 0$ and do not when $\hat{h}_C > 0$. When every share that $\boldsymbol{\sigma}$ falls is alone in its bundle and the share side read with one share per bundle (Corollary 4.4 merging each bundle) has a fibre of dimension one, a positive multiple of $\boldsymbol{\sigma}$ has a direction with rational coordinates in its chamber, along which the closing is the same series after a change of scale of τ (Theorem 31.6); the crossings are simple and nonresonant as soon as no two lattices of the falling shares meet; and every crossing term is*

$$\mathcal{R}_k(\zeta) = r_k \prod_G e^{-\bar{a}_{k,G}\, y_G(\zeta)}, \qquad y_G = \log \sum_{l \in G} e^{y_l},$$

with r_k real and not depending on the source point and $\bar{\mathbf{a}}_k$ the crossing point of the merged shares.

So Theorem 31.28 holds in every retained dimension, with no condition on the bundles: the falling shares need not be alone in them, and the share side merged by its bundles may have any number of free shares. The chamber minimum is a finite convex problem, shared by all the cohorts of the ray once their common integer step is chosen, and the completed source average is one joint Trailmoment. The positivity belongs to the completed source average; a continued source factor, with its oriented circle integrals, is read by Theorem 31.21, its phases not being replaced by a supremum over a positive support. A completed crossing and the tail it inherits are different series: closing a retained share only through finitely many of its poles leaves an exact Taylor remainder whose rate depends on how many poles are taken with each crossing, and it can grow while the completed crossings fall.

31.3.2 Coincident crossings and the boundary at root rate one

A residue is taken after the factors on its hyperplane have been assembled. A deformation may separate those factors for computation, but its separate residues are then one packet. The finite operation of Theorem 31.30 retains that packet, including its vanishing factors, and extends Theorem 31.18 to coincident crossings.

Lemma 31.29 (A source pole with a uniform parameter margin)**.** *Use the radii of Lemma 31.15. Near $\kappa = -N$, $N \in \mathbb{N}$, put $z = \kappa + N$ and fix $0 < \eta < 1$. On $|\operatorname{Re} z| \le \eta$ the same continued source factor is exactly*

$$I_\kappa = \int_r^1 q^{\kappa-1} g(q)\, dq + R_{N+1} + R'_N + g_N \frac{r^z}{z}, \qquad g_N = \frac{1}{2\pi i} \oint g(\zeta)\, \zeta^{-N-1}\, d\zeta,$$

R_{N+1} and R'_N given by the formulas of that lemma with these indices, $R'_0 = 0$. The first three terms are uniformly absolutely convergent on the whole strip; splitting R'_N gives at most seven integrals in all, the circle integral g_N carrying the explicit pole r^z/z. If $N = N_0 + dn$, the domains stay fixed and their dependence on n is again in fixed powers; in particular a small complex circle of the parameter about $z = 0$ has a common integrable source representation after its displayed pole is removed.

Theorem 31.30 (A finite jet for a coincident crossing)**.** *Consider a rational affine family of crossing fibres, indexed by n, with n in one fixed arithmetic progression. Fix the inherited contour and the whole packet of factors polar on its crossed hyperplane, let the other polar hyperplanes be separated from this packet on the real contour by a positive constant, and let the total order of the packet be at most p, independent of n. Perturb the constant terms of all shares and walls by one parameter ε, with a fixed rational vector of perturbations, keeping the affine pins and their prescribed continuation; the vector can be chosen outside finitely many rational hyperplanes so that distinct members of the packet separate, factors forced to remain coincident being kept as one member. Let $\mathcal{R}_{n,j}(\varepsilon)$ be the inherited crossing terms of the separated members. Then*

$$\mathcal{R}_n = [\varepsilon^0] \sum_j \mathcal{R}_{n,j}(\varepsilon).$$

Each member has a pole of order at most $p - 1$ in ε if its separated pole is simple, and the same bound suffices after keeping forced multiple members intact; so the constant coefficient uses only the Taylor coefficients through the degree $p - 1$ of the holomorphic parts after their poles are removed. Further meromorphic normalizing factors raise this order by their known pole orders. Every coefficient is a jet of the Gammas, of the source or of the parameter; no limit in n is part of the computation. More generally, whenever a finite exact procedure returns terms $\mathcal{R}_{n,j} = \varepsilon^{-o_j} \widetilde{\mathcal{R}}_{n,j}$ with $\widetilde{\mathcal{R}}_{n,j}$ holomorphic, the required coefficients are exactly $[\varepsilon^{o_j}]\widetilde{\mathcal{R}}_{n,j}$; positive orders of a factor that vanishes at $\varepsilon = 0$ are retained just as negative orders of a polar factor are. The local decay hypothesis of Lemma 31.3 is used on the common enclosing contour, uniformly on a small disc of the parameter. For the Gamma words and the continued sector integrals a uniform procedure is available: factors whose Gamma poles lie on the crossed hyperplane are constant on its residual fibre and are put in the scalar Gamma word, each source wall coincident on that hyperplane is treated by Lemma 31.29, and the other walls and Gamma strips have strict margins.

The auxiliary parameter separates the poles of one assembled crossing; it does not identify the independent regulators of a product of words, which keep their separate constant terms (Proposition 44.2). If the value asked for is a finite part, its prescribed Mellin regulator is retained and its

constant coefficient taken as in Theorem 1.7 and Corollary 1.8; finite parts in different coordinates are not interchangeable (Lemma 1.9: a dilation of the coordinate moves a finite part by a polynomial in the logarithm of the dilation). Several such regulators give finitely many multivariate Laurent coefficients, with the coefficient order fixed by the value asked for, the retention of finitely many numbers at every corner (Theorem 46.10) applied to a whole crossing fibre. The separation hypothesis concerns the actual residual contour: a transverse meeting of different hyperplanes needs the finite ordered packet of the chosen crossing flag, with its iterated transverse contours, and does not allow an arbitrary order of point residues. Nor does a Laurent coefficient keep a sharp root rate by itself: the coefficient can vanish.

Proposition 31.31 (Bounds and top coefficients pass to jets)**.** *In Theorem 31.30 let the procedure of Theorem 31.18 for the separated terms give a uniform bound on a circle of the parameter,* $|\varepsilon^{o_j}\mathcal{R}_{n,j}(\varepsilon)| \le C_\delta(1+n)^{p'}(q+\delta)^n$ *for every* $\delta > 0$*, the circle allowed to shrink with* δ*. Then* $\limsup_n |\mathcal{R}_n|^{1/n} \le q$*. For perturbations of the constant terms only, the slopes in* n*, the linear parts of the pins, the letters and the coefficients fixed, Lemma 31.29 and the extraction of the scalar give more: one common fan serves on a small disc of the parameter, each coefficient function* c_i *is independent of the parameter, only the fixed amplitudes, the explicit polar factors and the constant terms of the Gammas varying, and their endpoint powers have positive margins once the explicit poles of Lemma 31.29 are removed; so the finitely many Laurent coefficients of each order have the same upper root rate as their common collection of coefficient functions and Gamma slopes, with only polynomial and logarithmic losses in* n*. If the joint word has finitely many simple tops (Definition 31.20), uniformly separated from the rest of its domain, its finitely many Laurent coefficients supply every prescribed finite order of the expansion of* $\mathcal{R}_n$ *at the tops: differentiate the exact integrand to the required order in the parameter, expand its phase and amplitude to the required Taylor order at each top, use* $\mathbf{x} = \mathbf{X}/n$ *in its decaying boundary coordinates and* $\mathbf{y} = \mathbf{Y}/\sqrt{n}$ *in its coordinates* $\mathbf{y}$*, integrate the resulting monomials against the Gamma densities in* $\mathbf{X}$ *and the density* $e^{-\frac{1}{2}\mathbf{Y}^{\mathsf{T}}\mathbf{Q}\mathbf{Y}}$ *in* $\mathbf{Y}$*, and expand the Gamma prefactors to the same order. Equal top phases are added before any test of nonvanishing. A surviving coefficient at the root rate* q *proves* $\limsup_n |\mathcal{R}_n|^{1/n} = q$*; if the tested coefficients cancel, the upper bound remains and no sharp lower bound follows.*

At the root rate one the boundary is decided exactly, without simple or isolated tops, by the law of the coefficient function that the procedure retains in the positive case; its geometry is computed once and answers several queries at the boundary.

Theorem 31.32 (The test at root rate one for a positive moment sequence)**.** *Let* ϕ *be a finite positive measure,* $0 \le c \le 1$ *on its support,* $m_n = \int c^n \, d\phi$*, and*

$$b_n = G_n \, m_n, \qquad G_n > 0, \qquad G_n \sim C \, n^\beta, \quad C > 0, \ \beta \in \mathbb{R} :$$

the form of Corollary 31.19 at the root rate one, the coefficient function divided by its positive largest value $c_{\max}$ *and the scalar multiplied by* $c_{\max}^n$*, the exponent* β *read from the explicit Gamma quotient and the scalar sign a fixed number of modulus one to the power* n*. Finitely many initial terms have no effect.*

(i) The series $\sum_n b_n$ *is finite exactly when:* $\beta < -1$*, always;* $\beta = -1$ *and* $\int \log \frac{1}{1-c} \, d\phi < \infty$*;* $\beta > -1$ *and* $\int (1-c)^{-\beta-1} \, d\phi < \infty$*; the integrands of the last two conditions having the value* $+\infty$ *at* $c = 1$*. These are also the exact conditions for absolute convergence after multiplication by any* ζ^n*,* $|\zeta| = 1$*.*

(ii) Let moreover G_n be a balanced Gamma quotient with its exponential factor removed, or more generally let its forward differences satisfy $\Delta^k G_n = O(n^{\beta-k})$ for every fixed k. For $|\zeta| = 1$, $\zeta \neq 1$, the ordered series $\sum_n \zeta^n b_n$ converges exactly when $b_n \to 0$, and with $M(t) = \phi\{c > 1-t\}$ this is: $\beta < 0$, always; $\beta = 0$ and $\phi\{c = 1\} = 0$; $\beta > 0$ and $M(t) = o(t^\beta)$ as $t \downarrow 0$.

So (i) and (ii) tell absolute convergence, conditional convergence and the failure of the term test apart with one retained law.

A certified law near its largest value of the form $M(t) \asymp t^a(\log(1/t))^j$ turns these criteria into comparisons of powers and logarithms; the criteria stay exact for nonisolated or degenerate maxima, Dirac measures and mixtures. They do not change the crossing order: a periodic family must be reassembled in its original order before a conditional verdict is used, since splitting an alternating series into its even and odd subseries loses exactly that information.

Proposition 31.33 (Finite certificates at the boundary from the jets of the tops)**.** *Let the assembled terms at the root rate one have a certified finite expansion*

$$\mathcal{R}_n = \sum_{\zeta \in Z} \zeta^n \sum_{\beta \geq -1} \sum_{j=0}^{J_\beta} c_{\zeta,\beta,j}\, n^\beta (\log n)^j + O(n^{-1-\delta}), \qquad \delta > 0,$$

Z a finite subset of the unit circle, the real exponents β a finite set and equal triples combined; the jets of the tops of Proposition 31.31 produce such a certificate to every order needed when its hypotheses on simple tops and uniformity hold. Then:

(i) the series is absolutely convergent if and only if every displayed coefficient vanishes;

(ii) it converges in its given order if and only if every coefficient with $\zeta = 1$ vanishes and every coefficient with $\beta \geq 0$ vanishes;

(iii) when (ii) holds and (i) does not, the convergence is conditional.

The verdict uses exact certificates of zero or nonzero for the finitely many assembled coefficients; where those are unavailable, the finitely many coefficients are the output, a numerical near cancellation not being a certificate of zero.

Chapter 32

Laws at their ends: regions of equality and reach

A closing past the walls reads a law where its moments diverge, at its falling ends, through its expansion there. The regions on which a law equals its expansion at an end are built from the sector letters, with the map from the motion of the shares to the exponents, the coefficients, the truncation bounds and exact boundary words (Theorems 32.2 and 32.4); the domain where a series converges and the region where it equals the law are kept apart (Proposition 32.6). The reach is read on the law along every rational direction of the shares, as at width one: along a rational direction the seam function is the Mellin transform of one measure on the half-line, the image of the law, multiplied by $\mathbf{x}^{-\mathbf{w}_0}$ for the starting point $\mathbf{w}_0$, under the directional ratio, and the reach is the far boundary of the region on which that measure is its own wall series (Theorem 32.10). Run on the laws of the two families at a node of width one, the same argument reads the growth of each family on its law and makes that of the bunch its reach (Corollary 32.11). For a piece the reach is bounded by the ceilings of the keyhole continuations along the direction, with no phase, and attained at their simple tops that do not cancel (Theorem 32.15). A closing past the walls converges when the exponential of the Gamma rate times the reach of its tube, the imaginary translates of the line through its starting point, is less than one, its terms then having at most that root rate, and a law with log expansions bounds that reach in every direction with nothing paid for the phase; with one free share the tube is the line, and the terms reach that root rate under conditions that keep the poles apart (Theorems 32.16 and 32.17, Corollary 32.18).

32.1 Regions on which a law is represented

A region of equality belongs to an expansion of a law: to the coordinates advanced, the coordinates retained, and the grouping of its terms. A disc in a sector variable is not by itself a region of equality of the law: the map to the seam torus can identify points, have critical values, or have an image of smaller dimension than its torus. The procedure of Theorem 32.4 keeps that map, and gives a law, with its tangential source retained, before any assertion about a density on the whole torus.

Definition 32.1 (A chart at an end of a law)**.** Let a part of the source of a law have the coordinates $(q, z) \in (0, r_1) \times \cdots \times (0, r_k) \times Z$ and the form

$$q^{\boldsymbol{\kappa}-\mathbf{1}} A(q, z)\, dq\, \phi(dz), \qquad \kappa_i > 0,$$

ϕ a finite positive measure retained as an input and A holomorphic in q on a neighbourhood of a closed polydisc, uniformly bounded there for $z \in Z$; complex densities are allowed, and a fixed

monomial in the seam coordinates can be included in the integrand and removed after integration. The selected seam coordinates of the law at the end are $\check{x}_i^{-1}$, with

$$\check{x}_i = \prod_{j=1}^{k} q_j^{N_{ij}}\, \mathfrak{u}_i(q,z), \qquad 1 \le i \le k,$$

$\mathbf{N} = (N_{ij})$ an invertible integer matrix and the *chart factors* $\mathfrak{u}_i$ positive on the real source and holomorphic and nonzero on the complex neighbourhood; the other coordinates of the seam point are a retained map $\mathbf{C}(q,z)$, and branches are continued from the positive source. The *chart coordinates* are

$$\mathfrak{t}_j = \prod_i \check{x}_i^{(\mathbf{N}^{-1})_{ji}} = q_j \prod_i \mathfrak{u}_i(q,z)^{(\mathbf{N}^{-1})_{ji}}.$$

A certificate of the chart supplies positive radii R_i and a uniform analytic inverse $q_j = \mathfrak{t}_j\, g_j(\mathfrak{t},z)$ for $|\mathfrak{t}_i| < R_i$, with bounds for the inverse, its Jacobian and the holomorphic functions used in Theorem 32.2 and Corollary 32.3 below; the cuts on Z are retained. Write $\mathbf{1}_{\mathrm{cut}}(\mathfrak{t},z)$ for the indicator of any further retained cut of the source, $\mathbf{1}_{\mathrm{cut}} = 1$ when there is none; if a cut of the source moves in these coordinates, its indicator is retained in the integrand and is neither differentiated nor included in a holomorphic coefficient.

Theorem 32.2 (Computing and using an end chart)**.** *For a chart of Definition 32.1 the following computations are effective.*

(i) A smaller certified polydisc exists whenever the holomorphy and nonvanishing conditions hold uniformly on a compact tangential source. In its coordinates the source form is

$$\mathfrak{t}^{\boldsymbol{\kappa}-\mathbf{1}}\, \widetilde{A}(\mathfrak{t},z)\, d\mathfrak{t}\, \phi(dz), \qquad \widetilde{A} = A(\mathfrak{t}g,z) \prod_j g_j^{\kappa_j - 1} \det \frac{\partial(\mathfrak{t}g)}{\partial \mathfrak{t}},$$

the determinant positive on a small enough positive chart. The seam map is $(\mathfrak{t},z) \mapsto \big(\mathfrak{t}^{\mathbf{N}}, \mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)\big)$; with $\mathbf{C}_{\mathfrak{t}}(z) = \mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)$, its density in the normal coordinates $\check{x}$, a measure in the tangential ones, is

$$\frac{\check{x}^{-\mathbf{1}}\mathfrak{t}^{\boldsymbol{\kappa}}}{|\det \mathbf{N}|}\, (\mathbf{C}_{\mathfrak{t}})_* \big(\mathbf{1}_{\mathrm{cut}}(\mathfrak{t},\cdot)\widetilde{A}(\mathfrak{t},\cdot)\,\phi\big)\, d\check{x},$$

a density only in the selected normal coordinates, whose tangential value may be a singular measure.

(ii) Write $\widetilde{A} = \sum_{\mathbf{n}\ge 0} \widetilde{A}_{\mathbf{n}}(z)\, \mathfrak{t}^{\mathbf{n}}$. The coefficients are tangential words computed by finite jet arithmetic on the original letters; the inverse certifies that they represent the source and is not evaluated. If $|\widetilde{A}| \le M$ on $|\mathfrak{t}_i| = R_i$, then $|\widetilde{A}_{\mathbf{n}}(z)| \le M \prod_i R_i^{-n_i}$. For every bounded measurable test function H, the law on the chart is, for real $0 < \mathfrak{t}_i < R_i$,

$$\mathfrak{t}^{\boldsymbol{\kappa}-\mathbf{1}} \sum_{\mathbf{n}\ge 0} \mathfrak{t}^{\mathbf{n}} \int_Z \widetilde{A}_{\mathbf{n}}(z)\, \mathbf{1}_{\mathrm{cut}}(\mathfrak{t},z)\, H\big(\mathfrak{t}^{\mathbf{N}}, \mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)\big)\, \phi(dz)\, d\mathfrak{t}. \tag{32.1}$$

The same seam map and the same retained cut occur in every term. With rational data the inverse branch is algebraic, specified by the positive branch and the certificate; alternatively every exact pairing with a test function is kept on the original source (q,z) with that cut pulled back, and an inverse function need not be added to the operations. The series converges in

total variation on compact subcharts, the normal power included. If the tangential map and the cut do not depend on $\mathfrak{t}$*, its coefficients are the fixed face laws* $\mathbf{C}_*(\widetilde{A}_{\mathbf{n}}\phi)$*; otherwise* (32.1) *is a series of pairings on the one tangential source, not an assertion that those images are fixed.*

(iii) With $\vartheta_i = |\mathfrak{t}_i|/R_i < 1$*, truncating to* $0 \le n_i < k_i$ *has, before the normal power and for* $\|H\|_\infty \le 1$*, the error at most*

$$M\,\phi(Z)\prod_i (1-\vartheta_i)^{-1}\sum_i \vartheta_i^{k_i}. \tag{32.2}$$

It also has an exact boundary word: with T_i *the Taylor operator through the degree* $k_i - 1$ *in* $\mathfrak{t}_i$*,*

$$\widetilde{A} = \prod_i T_i\widetilde{A} + \sum_{\varnothing\neq I\subseteq\{1,\dots,k\}}\prod_{i\in I}(1-T_i)\prod_{j\notin I}T_j\widetilde{A}, \tag{32.3}$$

$$(1-T_i)\widetilde{A} = \frac{\mathfrak{t}_i^{k_i}}{2\pi i}\oint_{|\zeta_i|=R_i}\frac{\widetilde{A}(\mathfrak{t}_1,\dots,\zeta_i,\dots,\mathfrak{t}_k,z)}{\zeta_i^{k_i}(\zeta_i-\mathfrak{t}_i)}\,d\zeta_i :$$

there are $2^k - 1$ *boundary words, whatever the truncation orders* k_i*.*

(iv) The chart records the map from the motion of the shares to the exponents. The following statements about walls and continuation concern normal Mellin powers with an analytic tangential composite, or a further word with its own sector certificate; the bounded measurable test functions of (ii) have only the convergent identity and the error bound, and no continuation is asserted for them. For the Mellin power in the seam coordinates $\check{x}_i^{-1}$ *at the shares* $\mathbf{z}$*, the normal exponent of a term is* $\boldsymbol{\kappa} + \mathbf{N}^{\mathsf{T}}\mathbf{z} + \mathbf{n} - \mathbf{1}$*. For a fixed tangential cut and a normal Mellin power, integrating these powers gives the walls* $\kappa_j + (\mathbf{N}^{\mathsf{T}}\mathbf{z})_j + n_j = 0$*; a moving cut keeps its indicator in the integrand, with no such set of poles. A motion* $\boldsymbol{\sigma}$ *of the shares in the coordinates of these Mellin powers has the wall slopes* $\mathbf{N}^{\mathsf{T}}\boldsymbol{\sigma}$*; for the original coordinates of the shares one first applies to* $\boldsymbol{\sigma}$ *the recorded linear change of the exponents of the Mellin powers. Other requested coordinates remain in the tangential word, and are not replaced by their values on the face unless that replacement is part of the stated test function.*

Corollary 32.3 (Face jets for an analytic test function)**.** *Let the tangential cut be fixed, the map* $\mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)$ *holomorphic on a specified polydisc about* $\mathfrak{t} = 0$*, that point included, and* H *a bounded holomorphic tangential test function on a neighbourhood of its image. Expand* $\widetilde{A}(\mathfrak{t},z)\,H\big(\mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)\big)$ *instead of* $\widetilde{A}$ *alone: its coefficient of degree* $\mathbf{n}$ *is a finite sum of tangential source integrals of derivatives of* H *at* $\mathbf{C}(0,z)$*, multiplied by jets of* $\widetilde{A}$*,* $\mathbf{C}$ *and* g *of degrees at most* $\mathbf{n}$*. These are fixed face functionals, and their series and boundary words obey the same coefficient bounds as in Theorem 32.2(ii), with the bound of this composite. A change of analytic tangential test function needs its jets, not a new continuation of the source. The normal monomial* $\mathfrak{t}^{\mathbf{N}}$ *remains unexpanded in the integrand; a normal test function may be expanded as well when its composite is holomorphic and bounded on the polydisc and its jets are supplied by finite operations, a further condition, since negative entries of* $\mathbf{N}$ *can give poles at* $\mathfrak{t} = 0$*.*

Theorem 32.4 (Certified regions of equality and composition)**.** *For rational positive polynomial letters and rational monomial seam coordinates the following is a procedure.*

(i) The finite sector data determine the exponent matrices and the chart factors of Definition 32.1. Every invertible normal minor gives a proposed end chart, with the other variables

and test functions retained; enumerating rational tangential boxes and radii and verifying the inequalities of the certificate and of nonvanishing produces certified charts. Every compact tangential set on the end face $q = 0$ on which the proposed chart has nonvanishing chart factors and an invertible normal matrix has a small enough normal neighbourhood covered by finitely many charts produced in this way. Extending a chart away from that face needs its certificate, or a new certificate of inverse functions for the full Jacobian; invertibility of $\mathbf{N}$ *alone does not supply it.*

(ii) The region of equality supplied by one chart is the image of its certified polydisc in $\mathfrak{t}$ *under* $\check{x} = \mathfrak{t}^{\mathbf{N}}$*, with the specified branches and cuts; its positive part satisfies* $\sum_i (\mathbf{N}^{-1})_{ji} \log \check{x}_i < \log R_j$*,* $1 \le j \le k$*, and need not be a rectangle in the original seam coordinates. A finite atlas is certified for a requested region of equality only after its source patches cover the inverse image of that region; a patch not yet covered is retained as its original source word. So at every step the output is an exact finite head together with the boundary words* (32.3) *and the uncovered source word, with bounds for the parts certified small; an uncovered word is never omitted.*

(iii) Atlases may be restricted to common source patches, added, transported by their seam maps, and composed with a further test function. On compact certified subregions every order of the absolutely convergent reads has the same value. At finite truncations the equality uses the boundary words (32.3)*, and moving a cut retains the source between the old and the new cut. For words whose sector certificate supplies a meromorphic continuation, the same finite identities hold there, the full Laurent coefficient taken before pairing with a test function at coincident poles; an arbitrary bounded test function is not thereby continued. An infinite read in another grouping is asserted only with its own bound of convergence or as the continuation of the retained word.*

The retained data are the sector syntax, the matrix $\mathbf{N}$, the cuts, the recursion for the inverse jets and its certificate, and the tangential coefficient words, computed once per distinct source chart. For a prescribed error on $|\mathfrak{t}_i| \le \vartheta_i R_i$ the orders k_i are read off (32.2); the finite head has $\prod_i k_i$ coefficient words and its exact remainder $2^k - 1$ boundary words. A repeated chart reuses its coefficient words, a new analytic test function supplies its finite face jets, and the certificate and the retained boundary remain valid when only the truncation changes.

Proposition 32.5 (When a retained chart carries a density)**.** *A retained chart is in general not a chart of an ambient density. It becomes one in the selected normal coordinates and further tangential coordinates* $\mathbf{y}$ *when their image has a certified local product parametrization over* $(\mathfrak{t}, \mathbf{y})$*, with compact fibres, fixed cuts, a holomorphic form and an integrable uniform bound; integration over those fibres then gives a holomorphic coefficient function, and the series of Theorem 32.2, or of Corollary 32.3, represents that density with the same bounds. For a source form made of integrable powers of the boundary coordinates of a cube times a holomorphic density, such parametrizations are built near every compact regular fibre of a rational map* $\mathbf{C}$ *on the cube, when the map is defined on a neighbourhood of the closed cube, its restriction to every face meeting the fibre has a surjective differential, and its letters are nonzero. To assert a density for the whole law, the target patch must have no other or uncovered preimages, and in particular no omitted pole face may supply further source patches; otherwise the procedure describes the covered part of the law only, and the further preimages remain uncovered source words under Theorem 32.4. With the cube coordinates* v_i *put*

$$\mathbf{J}_{ji} = v_i(1 - v_i)\, \partial_{v_i} C_j, \qquad X_j = \sum_i v_i(1 - v_i) \big[\mathbf{J}^{\mathsf{T}} (\mathbf{J}\mathbf{J}^{\mathsf{T}})^{-1}\big]_{ij} \partial_{v_i};$$

then $D\mathbf{C}(X_j) = \mathbf{e}_j$. On a small enough neighbourhood the flows of these vector fields carry the reference fibre to the neighbouring fibres, keep all faces and give the product parametrization. Uniform nonvanishing of $\det(\mathbf{J}\mathbf{J}^{\mathsf{T}})$*, of the letters and of the relevant Jacobians, together with a bound for the vector fields on a larger neighbourhood, is a finite certificate; the iteration of the integral equations of the flows, certified to converge geometrically, computes the local flows and their holomorphic extensions.*

The possible failures of this procedure are explicit: zeros of letters, loss of properness at an omitted face of the source, or loss of surjectivity of the differential of the seam map on a face. Their images are candidate boundaries, not automatically singularities of the law, since different contributions can cancel. If the source has smaller dimension than the number of requested seam coordinates, the retained law and its atlas still exist but no ambient density is asserted. A genuine edge of the support is an obstruction: if the law vanishes on an open set on one side of an edge and has positive measure arbitrarily close on the other, it has no analytic density across that edge, by the identity theorem, as in Theorem 20.44(iii).

Proposition 32.6 (Convergence domains and equality regions)**.** *Let the fixed face laws of a certified chart, or the face functionals for a fixed bounded set of analytic test functions, be $\widetilde{A}_{\mathbf{n}}$ in a Banach space. The interior Ω_{abs} of*

$$\Big\{\mathfrak{t} : \sum_{\mathbf{n}\geq 0} \|\widetilde{A}_{\mathbf{n}}\|\,|\mathfrak{t}^{\mathbf{n}}| < \infty\Big\}$$

is the logarithmically convex domain of absolute convergence *of that series, which contains with each of its points $\mathfrak{t}$ every $\mathfrak{t}'$ with $|\mathfrak{t}'_i| \leq |\mathfrak{t}_i|$ for all i, and every certified polydisc lies in it. Its* region of equality *is retained separately: the chart patches covered by the source on which the series is proved to read the law, together with the patches reached by a chain of transports of charts that keep the equality. The series represents the law only on the intersection of the domain Ω_{abs} and the region of equality; analytic continuation of the sum alone is not a transport of the equality across an edge of the support or a singular value of the law. For a positive integer vector $\mathbf{d}$ and a fixed point $\mathfrak{t}$ of the chart retain the grouping*

$$\widetilde{A}_{(N)}(\mathfrak{t}) = \sum_{\langle \mathbf{d},\mathbf{n}\rangle = N} \widetilde{A}_{\mathbf{n}}\,\mathfrak{t}^{\mathbf{n}};$$

the grouped analytic germ in one variable v on the curve $(\mathfrak{t}_i v^{d_i})_i$ has the exact radius R with $R^{-1} = \limsup_N \|\widetilde{A}_{(N)}(\mathfrak{t})\|^{1/N}$: inside it the grouped series converges, and outside it its grouped terms do not tend to zero. These statements concern the germ, and they become statements about the law only where its region of equality is certified as well; on the boundary of convergence the coefficients decide. Completing another read replaces these grouped terms by their images, whose radius can be larger, the equality with the new expansion of the law still using the transported source and cuts. Neither radius is a reach of the law by definition, nor is it replaced by the supremum of an unintegrated modulus of the source.

The comparison of radii for a bounded read coefficient by coefficient concerns the same normal coordinates and the same bank of coefficients. Merging a law and then choosing a new coordinate is another operation: the source is carried by that map and its expansion is regraded in the new coordinate, whose coefficients and grouped terms need not be bounded images of the old ones, and whose boundary of equality can come earlier. The old and the new expansion agree through their common source, with the cuts, the finite heads and the boundary words transported; their

domains of convergence and atlases of equality are recorded in their own coordinates. Maximal analytic continuation, and deciding every cancellation at a singular boundary, are neither used by the procedure nor concluded from a ceiling. In one selected coordinate with a fixed tangential test function the procedure certifies local patches of equality with the wall series, and the global inversion of Theorem 32.10 (Theorem 20.44 at width one) identifies the reach of the law along that coordinate; in several coordinates the certified atlas of equality, the domain of convergence of the series and its retained grouping are stored separately.

32.2 The reach along a direction

At width one the reach of a bunch is read on the law of its ratio, as the point beyond which the law is its own wall series (Theorem 20.44); it is bounded above by the ceilings of the keyhole continuations and attained at their simple tops that do not cancel (Theorems 20.49 and 20.51), and against a room bounded below the terms of the node have the root rate of the reach over the least ratio of the room, which infinitely many of them reach (Theorem 20.39). A closing past the walls at any width advances along one direction of the shares, and along a rational direction the seam function is the Mellin transform of one measure on the half-line, the directional law (Definition 32.9). The theory of width one applies to that measure line by line, and its counterparts are stated along a direction.

Definition 32.7 (Walls and growth bounds)**.** Let a piece have its seam function $P_{W_B(\mathbf{w})}(\mathbf{m}_B)$ on the pins of the seam, an affine space A_B of real shares $\mathbf{w} = (\mathbf{a}, \mathbf{b})$ of the active letters with the space of directions $\vec{A}_B$, of dimension $n_{\mathcal{D}}$; for $\mathbf{w} \in A_B + i\vec{A}_B$ it is $\int \mathbf{x}^{-\mathbf{w}}\, d\big((\boldsymbol{\varrho}_B^{\rho})_*\phi_B^{\rho}\big)(\mathbf{x})$ where this converges, the law being carried by the closure of the image of the open simplex under the seam vector. The seam function has *walls* when

(W1) there are finitely many affine functions κ_U on A_B with rational linear parts, its *walls*, such that it converges at a real point exactly when every wall is positive there, which happens on an open convex set $\Omega_B \subseteq A_B$, and it continues meromorphically to $A_B + i\vec{A}_B$ with its poles on the hyperplanes on which a wall lies in $-\mathbb{N}$;

and it has *growth bounds* when

(W2) for every $\theta > 0$ there are M, C_0, n_1 and n_2 with $|P_{W_B(\mathbf{z})}(\mathbf{m}_B)| \le C_0\, \delta^{-n_2}(1 + |\mathbf{z}|)^{n_1} M^{|\operatorname{Re}\mathbf{z}|_1} e^{\theta|\operatorname{Im}\mathbf{z}|_1}$, δ the smaller of 1 and the distance of $\mathbf{z}$ from these hyperplanes.

The same definitions apply to the period of any word whose exponents are affine functions of the shares.

Every piece whose seam function converges at a real point has walls and growth bounds, and so does the seam function of its seam letters merged by bundles: the seam word of a piece is a word in positive letters, and so is the word of the merged seam letters of Theorem 31.7, sums of seam letters of several stations of one excess, which are no longer letters of tables; such a word continues past its walls as the seam word of every piece does (Theorem 31.4(ii)).

Lemma 32.8 (A word in positive letters continues past its walls)**.** *(i) Let $W = \prod_l L_l^{-w_l}$ be a product of powers of positive letters on a set B (Definition 25.1), whose exponents w_l and masses $\mathbf{m}_B$ are affine functions of $\mathbf{z} \in \mathbb{C}^p$ with rational linear parts, the masses on the balance hyperplane of W for every $\mathbf{z}$, and let its period converge at a real point $\mathbf{z}_0$, the exponents there being of any sign. Then there are finitely many affine functions κ_U on $\mathbb{C}^p$ with rational linear parts, its walls, such that the period converges at a real point exactly when every κ_U is*

positive there, and $P_W(\mathbf{z})$ *continues meromorphically to* $\mathbb{C}^p$*, with its poles, of order at most* $|B|-1$*, on the hyperplanes* $\kappa_U(\mathbf{z}) \in -\mathbb{N}$*; and for every* $\theta > 0$ *there are* M*,* C_0 *and* n_1 *with*

$$|P_W(\mathbf{z})| \le C_0\, \delta^{-(|B|-1)} (1+|\mathbf{z}|)^{n_1} M^{|\operatorname{Re}\mathbf{z}|_1} e^{\theta |\operatorname{Im}\mathbf{z}|_1},$$

δ *the smaller of* 1 *and the distance of* $\mathbf{z}$ *from these hyperplanes.*

(ii) For a piece and a partner read in its monomials, as in Theorem 31.7(B), merge the shares of each bundle G *of Theorem 31.7 into* $\bar a_G$*, and put* $L_G = \sum_{l\in G} \xi_l\, V_{jD(l)}$ *for a bundle of shares of* K_j*, and* L_G *alike with the seam letters* V^*_{jD} *for a bundle of shares of* Ψ_j*. On the pins of the merged shares the* merged seam word $W^{\mathrm{m}}_B(\bar{\mathbf{a}}) = \prod_G L_G^{-\bar a_G}$ *is such a product at the masses* $\mathbf{m}_B$*, and its period, the* merged seam function*,*

$$P_{W^{\mathrm{m}}_B(\bar{\mathbf{a}})}(\mathbf{m}_B) = \int \prod_G e^{-\bar a_G\, y_G}\, d\lambda, \qquad y_G = \log \sum_{l\in G} e^{y_l},$$

reads the merged values y_G *at the source points, against the source measure* λ *(Theorem 31.28); it converges at the sums over the bundles of every real point with positive coordinates at whose sums over the monomials the seam function of the piece converges.*

For the seam word of a piece, at the masses $\mathbf{m}_B$ and with the shares as its exponents, Lemma 32.8(i) gives walls and growth bounds whenever the seam function converges at a real point, the functions κ_U of the lemma serving as the walls, with $n_2 = |B|-1$; when the shares are positive at that point the seam word has a Newton sum of full dimension (Theorem 25.2), and the walls of the generators of a fan of the seam word serve, with the sector integrals of Theorem 31.4(ii), the proper walls for a piece of tables. A piece whose seam function on the pins is a product of Beta corners $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} x,\ y \\ x+y \end{bmatrix}$ of affine functions of the shares with rational linear parts and of real exponentials of the shares, converging exactly where those affine functions are positive, has walls, those functions, and growth bounds, by the reflection formula (Proposition 1.3(v)) and the asymptotic expansion of $\log\Gamma$ away from the negative real axis, and growth bounds even at $\theta = 0$ when for each Beta corner the linear parts of its two arguments on $\vec{A}_B$ are proportional, as when one argument does not depend on the shares.

Definition 32.9 (Directional ratios and laws; the reach along a direction)**.** Let a piece have walls and growth bounds (Definition 32.7), and let $\bar{\boldsymbol\sigma} \in \vec{A}_B \setminus \{0\}$ have rational coordinates. Its *directional ratio* is the positive function $\mathbf{x}^{-\bar{\boldsymbol\sigma}}$ on the seam torus. For $\mathbf{w}_0 \in \Omega_B$ and $\bar{\boldsymbol\vartheta} \in \vec{A}_B$ the *directional law* at $\mathbf{w}_0 + i\bar{\boldsymbol\vartheta}$ is the image

$$\lambda_{\bar{\boldsymbol\sigma},\mathbf{w}_0+i\bar{\boldsymbol\vartheta}} = \big(\mathbf{x}^{-\bar{\boldsymbol\sigma}}\big)_*\Big(\mathbf{x}^{-\mathbf{w}_0 - i\bar{\boldsymbol\vartheta}}\, (\boldsymbol\varrho^\rho_B)_* \phi^\rho_B\Big),$$

a complex measure on $(0,\infty)$ whose variation is at most the positive directional law $\lambda_{\bar{\boldsymbol\sigma},\mathbf{w}_0}$, which integrates to the seam function at $\mathbf{w}_0$; it is *oscillating* when $\bar{\boldsymbol\vartheta} \ne 0$, and the line $\{\mathbf{w}_0 + i\bar{\boldsymbol\vartheta} + z\bar{\boldsymbol\sigma} : z \in \mathbb{R}\}$ is the *imaginary translate* by $\bar{\boldsymbol\vartheta}$ of the line through $\mathbf{w}_0$. Its *line function* is

$$\ell(z) = P_{W_B(\mathbf{w}_0+i\bar{\boldsymbol\vartheta}+z\bar{\boldsymbol\sigma})}(\mathbf{m}_B) = \int_0^\infty X^z\, d\lambda_{\bar{\boldsymbol\sigma},\mathbf{w}_0+i\bar{\boldsymbol\vartheta}}(X),$$

absolutely convergent for $\operatorname{Re} z$ in the open interval of the x with $\mathbf{w}_0 + x\bar{\boldsymbol\sigma} \in \Omega_B$, which contains 0, and continued meromorphically by (W1). A wall falls along $\bar{\boldsymbol\sigma}$ when its falling rate $-\dot\kappa_U$,

$\dot\kappa_U = \partial_{\bar{\boldsymbol\sigma}}\kappa_U$, is positive (Definition 31.5). The poles of ℓ with $\operatorname{Re} z \ge 0$ lie among the *candidate poles*

$$z = \big(\kappa_U(\mathbf{w}_0) + n + i\,\partial_{\bar{\boldsymbol\vartheta}}\kappa_U\big)/(-\dot\kappa_U), \qquad n \in \mathbb{N},$$

of the falling walls: on finitely many horizontal lines, with real parts in finitely many cosets of one lattice $N_0^{-1}\mathbb{Z}$, the falling rates being rational. Let Σ be the half-strip of the z with $\operatorname{Re} z \ge 0$ and $\operatorname{Im} z$ within 1 of the interval spanned by 0 and these lines. The *reach along* $\bar{\boldsymbol\sigma}$ at $\mathbf{w}_0 + i\bar{\boldsymbol\vartheta}$ is

$$r(\bar{\boldsymbol\sigma}; \mathbf{w}_0 + i\bar{\boldsymbol\vartheta}) = \inf\big\{M \ge 0 :\ \text{for every } \delta > 0 \text{ there are } C, n' \text{ with } |\ell(z)| \le C(1+|z|)^{n'} M^{\operatorname{Re} z}\big\},$$

the bound asked at every $z \in \Sigma$ at distance at least δ from the candidate poles; it is finite by (W2), and by the maximum modulus principle on circles about the candidates it does not change when the actual poles are used. A *wall series* is as in Definition 20.43, the η_i allowed to be complex and its condition read as $\sum_{\eta,j} |a_{\eta,j}| x'^{-\operatorname{Re}\eta} < \infty$ for every $x' > x_0$, x_0 its radius. At $\bar{\boldsymbol\vartheta} = 0$ the directional law is the *law along* $\bar{\boldsymbol\sigma}$. The same definitions apply to the merged seam function of Lemma 32.8(ii) on the pins of the merged shares, the merged values e^{y_G} in place of the seam coordinates and the source measure in place of the law.

The falling end of the directional law is $X \to \infty$, where the walls that fall along $\bar{\boldsymbol\sigma}$ make its moments diverge. In an end chart of Theorem 32.2 the Mellin power at the shares $\mathbf{z}$ is $\mathfrak{t}^{\mathbf{N}^{\mathsf T}\mathbf{z}}$, so the directional ratio is $\mathfrak{t}^{-\mathbf{d}}$, $\mathbf{d} = -\mathbf{N}^{\mathsf T}\bar{\boldsymbol\sigma}$ the falling rates of the walls of the chart, times a power of the tangential coordinates: the wall series of the directional law groups the terms of the chart by their exponents $(\kappa_j + n_j)/d_j$ along the direction, and the reach is the far boundary, in X, of the region on which the law equals that expansion. At width one with one table and factoring weights, normalized as in Proposition 28.11, the seam torus is a curve, $\vec{A}_B$ is the line of $\bar{\boldsymbol\sigma} = (-1, 1, -1, 1)$ in the shares $(a_{D_0}, a_{D_1}, b_{D_0}, b_{D_1})$, and the directional ratio is $x^*_{D_0}/x_{D_1}$, which is ϱ_B at the seam points (Proposition 28.11); every coordinate of the torus is a power of ϱ_B, so the directional law is a power of ϱ_B times the law of Definition 20.43, and by Theorems 32.10 and 20.44 the reach along $\bar{\boldsymbol\sigma}$ is the reach $\operatorname{reach}(B)$ of the bunch.

Theorem 32.10 (The reach along a direction is read on the law)**.** *Let $\bar{\boldsymbol\sigma}$, $\mathbf{w}_0$, $\bar{\boldsymbol\vartheta}$, the directional law $\lambda = \lambda_{\bar{\boldsymbol\sigma}, \mathbf{w}_0 + i\bar{\boldsymbol\vartheta}}$, its line function ℓ and its reach r be as in Definition 32.9.*

(i) On (r, ∞) the measure λ has a density which is a wall series of radius r. Its terms are the poles of ℓ with $\operatorname{Re} z > 0$: a pole η with the principal part $\sum_j (-1)^{j+1} j!\, a_{\eta,j}(z-\eta)^{-j-1}$ gives the terms $a_{\eta,j} X^{-\eta-1}(\log X)^j$, and there are no others.

(ii) If on some (x_0, ∞) the measure λ has a density which is a wall series of radius x_0, then $r \le x_0$; and for every $x' > x_0$ there is C with $|\ell(z)| \le C\delta^{-J-1} x'^{\operatorname{Re} z}$ at every z with $\operatorname{Re} z \ge 0$ at distance at least $\delta \le 1$ from the poles, whatever $\operatorname{Im} z$, J the largest power of the logarithm in the series.

(iii) So r is the least $x_0 \ge 0$ for which λ has on (x_0, ∞) a density which is a wall series of radius x_0. It does not change when $\mathbf{w}_0$ moves along $\bar{\boldsymbol\sigma}$, $r(c\bar{\boldsymbol\sigma}; \cdot) = r(\bar{\boldsymbol\sigma}; \cdot)^c$ for rational $c > 0$, and r is at least every point near which λ has no real-analytic density. At $\bar{\boldsymbol\vartheta} = 0$ the law λ is positive and carried by the closure of the values of $\mathbf{x}^{-\bar{\boldsymbol\sigma}}$ on the support of the law of the piece: $r \ge X_{\min}$, the least of these values, when it is positive, the least value of the directional ratio taking the place of the least ratio (Proposition 20.35); and no wall falls along $\bar{\boldsymbol\sigma}$ exactly when $\mathbf{x}^{-\bar{\boldsymbol\sigma}}$ is bounded on that support, and then $r = X_{\max}$, the largest. When the seam word has a Newton sum of full dimension, as for a piece of tables whose tables form a connected family (Theorem 31.4(ii)), $X_{\min} = (\sup \mathbf{V}_B^{\bar{\boldsymbol\sigma}})^{-1}$ and $X_{\max} = \sup \mathbf{V}_B^{-\bar{\boldsymbol\sigma}}$, the suprema of the monomials of the letter vector of Theorem 31.2 (Definition 31.1).

(iv) *At $\bar{\boldsymbol{\vartheta}} = 0$ let $\mathbb{L} = \{z_0 + kz_1 : \ k \in \mathbb{N}\}$, $z_1 > 0$ rational, be a progression at positive distance from the poles of ℓ, and put $r_{\mathrm{res}} = \limsup_\eta \max_j |a_{\eta,j}|^{1/\eta}$ over the poles and $r_{\mathbb{L}} = \limsup_k |\ell(z_0 + kz_1)|^{1/(z_0+kz_1)}$. Then $r = \max(r_{\mathrm{res}}, r_{\mathbb{L}})$: the reach is carried by the residues of the line or by its values on every progression.*

(v) *For every $\theta \geq 0$, r is the infimum of the $M \geq 0$ for which, for every $\delta > 0$, there are C and n' with $|\ell(z)| \leq C(1 + |z|)^{n'} M^{\operatorname{Re} z} e^{\theta |\operatorname{Im} z|}$ at every z with $\operatorname{Re} z \geq 0$ at distance at least δ from the candidate poles: the growth of the line function on its whole half-plane, at any allowance in the imaginary direction, is the reach.*

At width one the same argument, run on the law of a family at a node, reads the growth of the family on its law, and that of the bunch is its reach.

Corollary 32.11 (At width one the growth of a family is read on its law)**.** *In Section 20.7 let λ_B be the image of the base form ϕ_B under ϱ_B and λ_R the image of ϕ_R under $1/\varrho_R$. For $X \in \{B, R\}$ the growth* $\mathrm{growth}(X)$ *along the level d (Definition 20.33) is the least $x_0 \geq 0$ for which λ_X has on (x_0, ∞) a density that is a wall series of radius x_0. In particular* $\mathrm{growth}(B) = \mathrm{reach}(B)$*: the inequality* $\mathrm{reach}(B) \leq \mathrm{growth}(B)$ *of Theorem 20.34(i) is an equality, and Theorem 20.34(iii) closes a node on the lattices of both families when the two points beyond which their laws are their own wall series multiply to less than one.*

A closing of the share side moves the imaginary parts of the shares over a whole fibre, so its bounds read the seam function on every imaginary translate of a line as well as on the line itself.

Corollary 32.12 (The reach of a tube)**.** *Let $\bar{\boldsymbol{\sigma}}$ and $\mathbf{w}_0$ be as in Definition 32.9 and $V' \subseteq \vec{A}_B$ a subspace containing $\bar{\boldsymbol{\sigma}}$. For $\theta \geq 0$ let $g_{V'}(\bar{\boldsymbol{\sigma}}; \theta)$ be the infimum of the M for which there are C, n_1 and n_2 with*

$$\left|P_{W_B(\mathbf{w}_0 + \tau\bar{\boldsymbol{\sigma}} + i\bar{\boldsymbol{\vartheta}})}(\mathbf{m}_B)\right| \leq C\,\delta^{-n_2}(1 + \tau + |\bar{\boldsymbol{\vartheta}}|)^{n_1} M^\tau e^{\theta|\bar{\boldsymbol{\vartheta}}|_1} \qquad (\tau \geq 0,\ \bar{\boldsymbol{\vartheta}} \in V'),$$

δ the smaller of 1 and the distance from the hyperplanes of (W1), and let $r^\sharp_{V'}(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = \lim_{\theta \downarrow 0} g_{V'}(\bar{\boldsymbol{\sigma}}; \theta)$ be the reach of the tube *over V'.*

(i) *$r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}}) \leq r^\sharp_{V'}$ for every $\bar{\boldsymbol{\vartheta}} \in V'$; in particular $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) \leq r^\sharp_{V'}$, with equality when $V' = \mathbb{R}\bar{\boldsymbol{\sigma}}$.*

(ii) A certificate on the real fibres. *Let $\mathbf{v}_1, \ldots, \mathbf{v}_{n_{\mathcal{D}}-1}$ complete $\bar{\boldsymbol{\sigma}}$ to a basis of $\vec{A}_B$, and $y_i = \langle \mathbf{v}_i, \log \mathbf{x} \rangle$ on the torus. If the image of the measure $\mathbf{x}^{-\mathbf{w}_0}(\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B$ under $(\mathbf{x}^{-\bar{\boldsymbol{\sigma}}}, \mathbf{y})$, on the part where $\mathbf{x}^{-\bar{\boldsymbol{\sigma}}} > x_0$, is $\sum_{\eta,j} X^{-\eta-1}(\log X)^j dX \otimes \phi_{\eta,j}(d\mathbf{y})$, $j \leq J$, with finite signed measures $\phi_{\eta,j}$, the η in finitely many sets $\eta_i + \mathbb{N}$, and $\sum_{\eta,j} \|\phi_{\eta,j}\| x'^{-\eta} < \infty$ for every $x' > x_0$, then $g_{\vec{A}_B}(\bar{\boldsymbol{\sigma}}; 0) \leq x_0$, so that $r^\sharp_{\vec{A}_B} \leq x_0$: an expansion at the face over the real fibres certifies the tube, with no allowance for the phase.*

The certificate asks the expansion to hold over whole fibres of the directional ratio, with fixed measures on them, along one direction. Read in the coordinates in which the shares pair with the law linearly, expansions with fixed tangential measures at the falling ends certify the tube in every direction at once: in those coordinates a complex share multiplies the law by a factor of modulus one, and the continuation is made on the law itself.

Definition 32.13 (Log expansions of a law)**.** For a piece, with the space $\vec{A}_B$ of the directions of its shares of Definition 32.7 and its dual $\vec{A}^*_B$, the *log-ratio map* sends a point $\mathbf{x}$ of the seam torus

to the linear function $\mathbf{z} \mapsto \langle \mathbf{z}, \log \mathbf{x} \rangle$ on $\vec{A}_B$, written $\log \mathbf{x}|_{\vec{A}_B}$; for a real point $\bar{\mathbf{c}}$ of the pins at which the seam function converges, $\lambda_{\bar{\mathbf{c}}} = (\log \mathbf{x}|_{\vec{A}_B})_* \big(\mathbf{x}^{-\bar{\mathbf{c}}} (\boldsymbol{\varrho}_B^{\rho})_* \phi_B^{\rho}\big)$ is a finite positive measure on $\vec{A}_B^*$, and the seam function at $\bar{\mathbf{c}} + \mathbf{z}$ is $\int e^{-\langle \mathbf{z}, f \rangle}\, d\lambda_{\bar{\mathbf{c}}}(f)$ for $\mathbf{z} \in \vec{A}_B + i\vec{A}_B$ wherever the integral converges absolutely. A *log expansion* is the image of the complex measure

$$\sum_{\boldsymbol{\eta}, \mathbf{j}} \mathbf{p}^{\mathbf{j}} e^{-\langle \boldsymbol{\eta}, \mathbf{p} \rangle}\, d\mathbf{p} \otimes \phi_{\boldsymbol{\eta}, \mathbf{j}}(df) \qquad \text{on } [0, \infty)^k \times Y_Q$$

under $(\mathbf{p}, f) \mapsto f - \sum_{h \le k} p_h u_h$, where $u_1, \dots, u_k \in \vec{A}_B^*$ are linearly independent, $\boldsymbol{\eta}$ runs over the union of finitely many translates of $\mathbb{N}^k$ by positive vectors, $\mathbf{j} \le \mathbf{J}$, and the $\phi_{\boldsymbol{\eta}, \mathbf{j}}$ are complex measures on a compact set $Y_Q \subset \vec{A}_B^*$, its *inner face*, with $\sum \|\phi_{\boldsymbol{\eta}, \mathbf{j}}\| < \infty$. The law *has log expansions at* $\bar{\mathbf{c}}$ when $\lambda_{\bar{\mathbf{c}}}$ is the sum of finitely many log expansions Q and of a complex measure ϕ_0 carried by a compact set Y_0. Its *inner bound* along a direction $\bar{\boldsymbol{\sigma}} \in \vec{A}_B$ is the largest value of the directional ratio $e^{-\langle \bar{\boldsymbol{\sigma}}, f \rangle}$ on Y_0 and on the inner faces,

$$X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}}) = \max \Big(\max_{f \in Y_0} e^{-\langle \bar{\boldsymbol{\sigma}}, f \rangle},\ \max_{Q} \max_{f \in Y_Q} e^{-\langle \bar{\boldsymbol{\sigma}}, f \rangle} \Big),$$

whose logarithm is convex in $\bar{\boldsymbol{\sigma}}$.

An end chart of Theorem 32.2 on a compact tangential source, whose tangential map and cut do not depend on $\mathfrak{t}$ and whose normal coordinates enter the log-ratio map through k linearly independent elements of $\vec{A}_B^*$, gives finitely many log expansions and a compact part: the log-ratio map is linear in $\log \mathfrak{t}$ and in the logarithms of the tangential coordinates; with $p_h = \log(R_h'/\mathfrak{t}_h)$ on the polydisc of the radii $R_h' < R_h$ the exponents are $\boldsymbol{\kappa} + \mathbf{n}$ and the coefficients $\widetilde{A}_{\mathbf{n}} R'^{\boldsymbol{\kappa} + \mathbf{n}}$, summable by Theorem 32.2(ii); where some $\mathfrak{t}_h$ lie in $[R_h', R_h)$ these join the tangential coordinates, the others keeping their expansions, summable uniformly by Cauchy's estimate on the closed polydisc; and where all of them do, the part is compact. A law with independent coordinates has such log expansions from the expansions of its factors at their ends.

Proposition 32.14 (A law with log expansions continues without phase)**.** *Let the law of a piece have log expansions at* $\bar{\mathbf{c}}$ *(Definition 32.13). Then for* $\mathbf{z} \in \vec{A}_B + i\vec{A}_B$ *off the hyperplanes* $\langle \mathbf{z}, u_h \rangle = \eta_h$ *of the log expansions,*

$$P_{W_B(\bar{\mathbf{c}} + \mathbf{z})}(\mathbf{m}_B) = \int e^{-\langle \mathbf{z}, f \rangle}\, d\phi_0(f) + \sum_{Q} \sum_{\boldsymbol{\eta}, \mathbf{j}} \prod_{h \le k} \frac{j_h!}{(\eta_h - \langle \mathbf{z}, u_h \rangle)^{j_h + 1}} \int_{Y_Q} e^{-\langle \mathbf{z}, f \rangle}\, d\phi_{\boldsymbol{\eta}, \mathbf{j}}(f),$$

continued: the seam function at $\bar{\mathbf{c}} + \cdot$ *is meromorphic on* $\vec{A}_B + i\vec{A}_B$ *with its poles on these hyperplanes, and there are* C *and* n *with*

$$\big| P_{W_B(\bar{\mathbf{c}} + \mathbf{z})}(\mathbf{m}_B) \big| \le C\, \delta^{-n} \max \Big(\max_{f \in Y_0} e^{-\langle \operatorname{Re} \mathbf{z}, f \rangle},\ \max_{Q} \max_{f \in Y_Q} e^{-\langle \operatorname{Re} \mathbf{z}, f \rangle} \Big),$$

δ *the smaller of* 1 *and the distance of* $\mathbf{z}$ *from these hyperplanes; the same holds, with another constant, for* δ *the smaller of* 1 *and the distance from the polar hyperplanes of the seam function alone. On the contour through a point* $\bar{\mathbf{c}} + \tau\bar{\boldsymbol{\sigma}}$, $\tau \ge 0$, *the bound is* $C\delta^{-n} X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}})^{\tau}$, *with no factor in the imaginary parts and no power of* τ. *So, for a piece whose seam word has a Newton sum of full dimension and a rational* $\bar{\boldsymbol{\sigma}}$, $r(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}}) \le r^{\sharp}_{\vec{A}_B}(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}}) \le g_{\vec{A}_B}(\bar{\boldsymbol{\sigma}}; 0) \le X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}})$, *and the growth of Theorem 31.4(iii) is* $g(\boldsymbol{\sigma}; 0) \le X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}})$ *for every direction* $\boldsymbol{\sigma}$ *of the shares of Theorem 31.4(i) whose sums over the monomials are* $\bar{\boldsymbol{\sigma}}$.

The certificate of Corollary 32.12(ii) is one expansion along the direction itself, with fibre measures of any extent and no condition on the rest of the law: on the contours through the points of a ray of $\bar{\boldsymbol{\sigma}}$ the factor $\mathbf{x}^{-i\bar{\boldsymbol{\vartheta}}}$ has modulus one and only the values of the directional ratio enter, not the compactness of the faces. Neither certificate applies as stated where the tangential coordinates of a face move with the normal ones, as when the support of the law curves in its log-ratio coordinates, or where a cut of the fibres moves, as in Theorem 32.2: the measures on the fibres are then not fixed, and the imaginary translates are read one at a time by Theorem 32.10, or bounded by the ceilings of Theorem 32.15, which carry the phase of the letters on these lines through their constants.

Without such a certificate a tube can reach farther than every one of its lines. There are sector integrals in two shares, with a single wall and growth bounds, for which every line of the tube along the fall of the wall has a finite reach while the reach of the tube over all the directions of the shares is infinite: the factor $\mathbf{x}^{-i\bar{\boldsymbol{\vartheta}}}$ enters the Taylor coefficients of a letter at the falling face through the powers of $\bar{\boldsymbol{\vartheta}}$. The seam function of a piece can do the same: when its seam function on the pins is a Beta corner whose two arguments move independently on the pins, it has growth bounds only at positive θ, and on an imaginary translate along which one argument alone moves in the imaginary direction the Beta corner grows like a power of the imaginary part whose degree grows with the real parts, so that its tube reaches infinitely far while each of its lines has a finite reach. The sector letters of a piece of tables are never monomials in the falling variables, through which alone the factor $\mathbf{x}^{-i\bar{\boldsymbol{\vartheta}}}$ would enter without reaching the Taylor coefficients at the falling face. Theorem 32.16(i) is the limit $\theta \to 0$ of Theorem 31.4(iii); when the reach of the tube is infinite, the bound of that theorem at a positive θ, which pays for the phase, is the one that travels.

Theorem 32.15 (Ceilings, simple tops and fibres along a direction)**.** *Let the seam function of the piece converge at a real point of the pins with positive shares, let $\bar{\boldsymbol{\sigma}}$ and $\mathbf{w}_0$ be as in Definition 32.9, the walls being those of a fan of the seam word (Theorem 31.4(ii)), and for a sector π of it let $\varrho_{\pi,\bar{\boldsymbol{\sigma}}} = \prod_k q_k^{\dot{\kappa}_k}\, \mathbf{k}_\pi^{-\bar{\boldsymbol{\sigma}}}$ be the directional ratio in the sector (Definition 31.10), the sector letters L of $\mathbf{k}_\pi$ entering with the coordinates $\bar{\sigma}_L$ of $\bar{\boldsymbol{\sigma}}$ at their stations and $w_{0,L}$ of $\mathbf{w}_0$. The keyhole continuations along a direction $\boldsymbol{\sigma}$ depend on $\boldsymbol{\sigma}$ only through $\bar{\boldsymbol{\sigma}}$, and the falling variables of π are those with $\dot{\kappa}_k < 0$; for a piece of tables the sectors of the orders π of B may be taken.*

(i) No reach exceeds the ceiling. *For every keyhole continuation along $\bar{\boldsymbol{\sigma}}$, of ceiling $H_{\mathcal{O}}$, and every $\bar{\boldsymbol{\vartheta}} \in \vec{A}_B$, $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}}) \le H_{\mathcal{O}}$: the phase of the letters does not enter the reach.*

(ii) The values at the integers. *Let $\bar{\boldsymbol{\sigma}}$ be scaled so that every $\dot{\kappa}_k$ is an integer, and let the walls of the falling variables of every sector be off the integers at $\mathbf{w}_0$. An* integer keyhole continuation along $\bar{\boldsymbol{\sigma}}$ *is defined as in Definition 31.10 with every sector letter kept from vanishing except those L whose exponent $-w_{0,L} - m\bar{\sigma}_L$ at $\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}}$ is a nonnegative integer for every large integer m, its ceiling defined in the same way; its keyhole contour, its tops and its simple tops, with their exponents* $\exp(p)$ *and coefficients $A(p)$, are those of Definition 20.50 with $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in place of ϱ_π, the sector form $\phi_{\pi,\mathbf{w}_0} = |\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \prod_k q_k^{\kappa_k(\mathbf{w}_0)-1} \mathbf{k}_\pi^{-\mathbf{w}_0} dq$ in place of ϕ_π, the walls $\kappa_j(\mathbf{w}_0)$ of the sector in place of κ_j, and the faces $q_j = 0$ with $\dot{\kappa}_j = 0$ in place of the faces of the level variables. Then for every large integer m the seam function at $\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}}$ is the integral of $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}^m\, \phi_{\pi,\mathbf{w}_0}$ over the keyhole contour, and the growth at the integers* $\mathrm{growth}_{\mathbb{Z}}(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = \limsup_m |P_{W_B(\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B)|^{1/m}$ *is at most the ceiling of every integer keyhole continuation along $\bar{\boldsymbol{\sigma}}$, and at most $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0)$.*

(iii) Simple tops. *If an integer keyhole continuation along $\bar{\boldsymbol{\sigma}}$ of ceiling $H_{\mathcal{O}}$ has finitely many tops, all simple, and some $A_\zeta \neq 0$ (Theorem 20.51(iii)), then* $\mathrm{growth}_{\mathbb{Z}}(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = H_{\mathcal{O}} \le r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0)$*; if moreover some keyhole continuation along $\bar{\boldsymbol{\sigma}}$ has the ceiling $H_{\mathcal{O}}$, then $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = H_{\mathcal{O}}$.*

(iv) Fibres. *Let $\bar{\boldsymbol{\sigma}}$ be scaled so that its coordinates and every $\dot{\kappa}_k$ are integers. Call a box of a continuation along $\bar{\boldsymbol{\sigma}}$ a* fibre box *when on its sets Y the letters with $\bar{\sigma}_L \geq 0$ do not vanish, and those with $\bar{\sigma}_L < 0$ do not vanish at the points with a falling coordinate 0; its stationary bound, as in Definition 20.54 with $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in place of ϱ_π, is the supremum of $|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}|$ over the points of these sets with no falling coordinate 0 at which some falling coordinates lie inside their discs and the derivatives of $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in them vanish, the points with only one such coordinate, of falling rate one, left out. For every continuation along $\bar{\boldsymbol{\sigma}}$ whose boxes are keyhole boxes or fibre boxes, of ceiling $H_{\mathcal{O}}$, with the stationary bound S of its fibre boxes, $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) \leq \max(H_{\mathcal{O}}, S)$.*

So the reach along a direction sits between the growth at the integers and the ceiling of every keyhole continuation along it, and the two meet at simple tops that do not cancel when a keyhole continuation along the direction has the same ceiling, as at width one; and, by Theorem 32.10, without any top it is the far boundary of the region on which the directional law is its own wall series. The phase of the letters, which Theorem 31.11 pays for at the points where the ceiling is read, does not enter the reach: on a line the imaginary parts stay bounded. It enters only through the transverse fibre of a closing, which Theorem 32.16 reads.

The integer scaling in (iv) is needed: a sector with one falling variable of falling rate one and a letter entering with a coordinate that is not an integer can have a fibre box of stationary bound 0 and a reach above its ceiling, the directional ratio not being single valued in the disc, so that Lemma 20.52 does not apply; scaled to integers, the stationary point that the bound left out is counted.

Theorem 32.16 (The closing past the walls is read along its direction)**.** *Read the glue as in Theorem 31.4(i), the partner in its monomials, for a piece with walls and growth bounds whose seam word has a Newton sum of full dimension. Let V be the space of the directions of the shares, of dimension n, the number of free shares, $\boldsymbol{\sigma} \in V$ a direction with rational coordinates, $\bar{\boldsymbol{\sigma}}$ its sums, $V' \subseteq \vec{A}_B$ the space of the sums of the directions in V, and $\mathbf{c}$ the real starting point, with $\bar{\mathbf{c}} \in \Omega_B$, off the null set of Theorem 31.2 taken with the polar hyperplanes of the seam function. Let h_Γ be the Gamma rate of Theorem 31.4(iii), so that the rate of that theorem, which holds for such a piece (Theorem 31.4, its seam function converging at the sums $\bar{\mathbf{c}} \in \Omega_B$ of the positive shares of the starting point), is $h_\theta(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \log g(\boldsymbol{\sigma}; \theta) + \frac{2\theta^2}{\pi} \sum_{\sigma_l > 0} \sigma_l$, and let I be the integrand of the share side.*

(i) The tube bounds the closing. *If $h_\Gamma(\boldsymbol{\sigma}) + \log r^\sharp_{V'}(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}}) < 0$, the closing along $\boldsymbol{\sigma}$ converges absolutely, its terms and the integrals left after them being $O\big(e^{\tau(h_\Gamma(\boldsymbol{\sigma}) + \log r^\sharp_{V'} + \epsilon)}\big)$ for every $\epsilon > 0$; the closing along $\boldsymbol{\sigma}$ is the closing along every direction of its chamber (Theorem 31.6), whose rates bound it as well. For the share side of Theorem 31.4(iv), with $\boldsymbol{\sigma}$ a direction of its shares with rational coordinates and $\mathbf{c}$ off the null set of that part, the closing along $\boldsymbol{\sigma}$ converges absolutely when $\sum_l \sigma_l \log|\sigma_l| + \log r^\sharp_B + \log r^\sharp_R < 0$, in this sentence $r^\sharp_B$ and $r^\sharp_R$ being the reaches of the tubes over the space of the directions of its shares (Corollary 32.12) of the seam functions of the piece and of the partner along $\boldsymbol{\sigma}$ at $\mathbf{c}$; its terms and the integrals left after them are then $O\big(e^{\tau(\sum_l \sigma_l \log|\sigma_l| + \log r^\sharp_B + \log r^\sharp_R + \epsilon)}\big)$ for every $\epsilon > 0$.*

(ii) One free share. *Let $n = 1$, and put $r = r(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}})$ and $\ell(z) = P_{W_B(\bar{\mathbf{c}} + z\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B)$. Then $r^\sharp_{V'} = r$, and for every $r' > r$ the terms are $O\big((1+\tau)^{n'} e^{\tau(h_\Gamma(\boldsymbol{\sigma}) + \log r')}\big)$; so the closing converges absolutely when $h_\Gamma(\boldsymbol{\sigma}) + \log r < 0$. Let $r > 0$, let $\mathbb{L}$ be the progression of the crossings of a falling share l_0, and let the data keep the poles apart: from some point on, $\mathbb{L}$ meets no pole of ℓ and no pole of another Gamma, and no pole of ℓ is a pole of a Gamma. If the poles*

of ℓ crossed by the ray are simple from some point on, or the values of ℓ on $\mathbb{L}$ carry the reach (Theorem 32.10(iv)), then for every $\epsilon > 0$ infinitely many terms have modulus at least $e^{\tau(h_\Gamma(\boldsymbol{\sigma})+\log r-\epsilon)}$: the rate $h_\Gamma(\boldsymbol{\sigma}) + \log r$ is exact, and the terms do not tend to zero when it is positive. Without either proviso, if those poles have order at most p, the same holds for all but at most $p-1$ of the partners obtained by multiplying the coefficients by $e^{\tau_0 v_l}$, $\tau_0 \in \mathbb{R}$, for a fixed $\mathbf{v}$ with $\langle \mathbf{v}, \boldsymbol{\sigma}\rangle = 1$, the rate of the partner of τ_0 being $h_\Gamma(\boldsymbol{\sigma}) - \tau_0 + \log r$. The same holds for the share side of Theorem 28.10(iii) with one free share and a partner whose seam function converges along the ray, with $\sum_l \sigma_l \log|\sigma_l| + \log\sup \mathbf{V}_R^{-\bar{\boldsymbol{\sigma}}}$ in place of $h_\Gamma(\boldsymbol{\sigma})$, the partners being the tinkered partners that multiply the room letters of each station D by $e^{\tau_0 v_D}$ (Definition 28.18); at a node of width one, where the share side has one free share, this rate is $\log\big(\operatorname{reach}(B)/\inf \varrho_R\big)$, the logarithm of the root rate of Theorem 20.39(ii).

(iii) Several free shares. *Let $n \geq 2$. Fix a complement $V_\perp$ of $\mathbb{R}\boldsymbol{\sigma}$ in V, a Lebesgue measure $d\boldsymbol{\vartheta}_\perp$ on it and $k_\sigma > 0$ with $d\boldsymbol{\vartheta} = k_\sigma\, dy\, d\boldsymbol{\vartheta}_\perp$ for $\boldsymbol{\vartheta} = y\boldsymbol{\sigma} + \boldsymbol{\vartheta}_\perp$, $d\boldsymbol{\vartheta}$ the measure of Definition 4.1. Let T_σ be the space of the* transverse tinkers, *the tinkers of the partner read in its monomials (Definition 28.18, with one multiplier e^{v_l} for each monomial in place of each coordinate of the seam vector), multiplying the coefficients read on V as $\xi_l \mapsto \xi_l e^{v_l}$, with $\langle \mathbf{v}, \boldsymbol{\sigma}\rangle = 0$, identified with the dual of $V_\perp$ and carrying the dual Lebesgue measure $d\mathbf{v}$. The crossings of the closing along $\boldsymbol{\sigma}$ are the same for every tinker, and at a crossing at the time τ^**

$$\int_{T_\sigma} e^{\langle \mathbf{v},\mathbf{c}\rangle}\, \mathcal{R}_{\tau^*}\big(\boldsymbol{\xi} e^{\mathbf{v}}\big)\, d\mathbf{v} = -k_\sigma \operatorname*{Res}_{z=\tau^*} I(\mathbf{c} + z\boldsymbol{\sigma}),$$

$\mathcal{R}_{\tau^*}(\boldsymbol{\xi} e^{\mathbf{v}})$ the crossing term of the partner with the coefficients $\boldsymbol{\xi} e^{\mathbf{v}}$. Integrated over the transverse tinkers against the factor $e^{\langle \mathbf{v},\mathbf{c}\rangle}$, the terms of the closing are $-k_\sigma$ times the residues of the integrand on the real line through $\mathbf{c}$ in the direction $\boldsymbol{\sigma}$; these are $O\big((1+\tau)^{n'} e^{\tau(h_\Gamma(\boldsymbol{\sigma})+\log r')}\big)$ for every $r' > r = r(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}})$, and under the conditions of (ii) on that line infinitely many of them have modulus at least $e^{\tau(h_\Gamma(\boldsymbol{\sigma})+\log r-\epsilon)}$.

Theorem 32.17 (A closing with one free share is read on the law along it)**.** *Take, in Theorem 31.7, the share side of the kind (B), its bundles merged by Corollary 4.4, the bars marking sums over the bundles, and in this statement write*

$$I(\bar{\mathbf{a}}) = \frac{\prod_G \Gamma(\bar{a}_G)}{\prod_j \Gamma(s_j) \prod_j \Gamma(t_j)}\; P_{W_B^{\mathrm{m}}(\bar{\mathbf{a}})}(\mathbf{m}_B),$$

the products over the bundles and over the active letters, with the merged seam function of Lemma 32.8. Let the pins of the merged shares leave a fibre of dimension one, spanned by a rational direction $\bar{\boldsymbol{\sigma}}$, and let $\bar{\mathbf{c}}$ be a real point of the merged pins with positive coordinates at which the merged seam function converges. Let the merged law along $\bar{\boldsymbol{\sigma}}$, the law along $\bar{\boldsymbol{\sigma}}$ of the merged seam function (Definition 32.9), be the image of $\prod_G e^{-\bar{c}_G y_G}\lambda$ under $X = \prod_G e^{-\bar{\sigma}_G y_G}$, λ the source measure, its line function $\ell(\zeta) = P_{W_B^{\mathrm{m}}(\bar{\mathbf{c}}+\zeta\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B)$ and its reach r, the least $x_0 \geq 0$ for which it has on (x_0, ∞) a density that is a wall series of radius x_0 (Definition 32.9, Theorem 32.10(iii)); and in this statement write $h_\Gamma^{\mathrm{m}}(\bar{\boldsymbol{\sigma}}) = \sum_G \bar{\sigma}_G \log|\bar{\sigma}_G|$, the Gamma rate of the merged share side, whose coefficients are inside the merged values.

(i) There is $k_P > 0$, depending only on the pins, with $P_{B\sqcup R}(\mathbf{m}) = \frac{k_P}{2\pi i}\int_{\operatorname{Re}\zeta=0} I(\bar{\mathbf{c}} + \zeta\bar{\boldsymbol{\sigma}})\, d\zeta$, and the crossing term (Lemma 31.3) at a crossing ζ^* of the ray is $-k_P \operatorname{Res}_{\zeta=\zeta^*} I(\bar{\mathbf{c}} + \zeta\bar{\boldsymbol{\sigma}})$: at the crossing $\bar{a}_f = -n$ of the falling share f, at which no other factor of I is polar, it is

$$k_P\, \frac{(-1)^n}{n!\,|\bar{\sigma}_f|}\, \frac{\prod_{G\neq f} \Gamma(\bar{a}_G)}{\prod_j \Gamma(s_j) \prod_j \Gamma(t_j)}\; P_{W_B^{\mathrm{m}}(\bar{\mathbf{a}})}(\mathbf{m}_B)$$

at $\bar{\mathbf{a}} = \bar{\mathbf{c}} + \zeta^* \bar{\boldsymbol{\sigma}}$*, and at a simple pole of the merged seam function at which no Gamma is polar it is* $-k_P \prod_G \Gamma(\bar{a}_G) / \big(\prod_j \Gamma(s_j) \prod_j \Gamma(t_j) \big)$ *there times* $\operatorname{Res}_{\zeta^*} \ell$*.*

(ii) For every $r' > r$ *there are* C *and* n' *such that the crossing terms at their times* τ*, and the integrals left at the times at a fixed distance from the crossings, are at most* $C(1+\tau)^{n'} e^{\tau h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r'^{\tau}$*.*

(iii) Let the ray $\bar{\mathbf{c}} + \tau \bar{\boldsymbol{\sigma}}$*,* $\tau \geq 0$*, cross its polar points simply: no crossing point of a falling merged share, one with* $\bar{\sigma}_G < 0$*, is a pole of the merged seam function or a crossing point of another falling share, and the poles of the merged seam function on the ray are simple. Then for each falling share* f *the crossing terms of* f *and those at the poles of the merged seam function have together the root rate* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r$*: the upper limit of* $|\mathcal{R}|^{1/\tau}$ *over them is* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r$*. The same holds when the data keep the poles apart only from some point on, as in Theorem 32.16(ii), read on* ℓ *and on the progression of the crossings of* f*, and the poles of* ℓ *crossed by the ray are simple from some point on or the values of* ℓ *on that progression carry the reach; without either proviso, if those poles have order at most* p*, it holds for all but at most* $p-1$ *of the partners obtained by multiplying the coefficients of the shares of each bundle* G *by* $e^{\tau_0 v_G}$*,* $\tau_0 \in \mathbb{R}$*, for a fixed* $\mathbf{v}$ *with* $\sum_G v_G \bar{\sigma}_G = 1$*, whose merged laws are constant multiples of the images of the merged law under* $X \mapsto e^{-\tau_0} X$*, with the root rates* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}}) - \tau_0} r$*.*

(iv) So the closing of the merged share side along $\bar{\boldsymbol{\sigma}}$*,* $P_{B \sqcup R}(\mathbf{m}) = \sum \mathcal{R}$ *over the crossings of the ray in their order, converges absolutely when* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r < 1$*, and under the conditions of (iii) its terms do not tend to zero when* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r > 1$*. No hypothesis on tops or on the phase of the letters enters, only the separations of the poles in (iii): the rate is read on the merged law along the direction. Where* X *is bounded on the support of the law of the piece,* r *is its largest value there, and when moreover the falling shares are alone in their bundles,* $h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}}) + \log r$ *is the averaged rate* $\hat{h}_C$ *of Theorem 31.28; where* X *is unbounded,* $\hat{h}_C$ *is infinite and the averaged terms diverge past the walls, and (iii) carries the sharpness of that theorem past them, with the reach of the merged law in place of the largest value of* X*. When every bundle is a single share,* $\boldsymbol{\sigma} = \bar{\boldsymbol{\sigma}}$*, the merged seam function is the product of* $\boldsymbol{\xi}^{-\mathbf{a}}$ *and of the seam function at the sums over the monomials (Theorem 31.4(i)),* r *is* $e^{-\sum_l \sigma_l \log \xi_l}$ *times the reach of the piece along the sums of* $\boldsymbol{\sigma}$ *(Definition 32.9), and* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r = e^{h_0(\boldsymbol{\sigma})}$*,* h_0 *the rate of Theorem 31.4(iii) at* $\theta = 0$*, whose growth* $g(\boldsymbol{\sigma}; 0)$ *is on a fibre of dimension one that reach: on such a fibre the rate of Theorem 31.4(iii) at* $\theta = 0$ *is exact under the conditions of (iii), and (iii) is Theorem 32.16(ii).*

The merged share side and the share side at the stations, before the stations of one excess are merged (Corollary 4.4), are two representations of one period, with two laws along their directions; completing the splits of a bundle can move the edge of the support at which the law stops being its own wall series. The merged law can stop being its own wall series at the image of a nearer edge of the support, where a level set of the merged seam letter ends the expansion of the law at its falling end, while the station closings, with more free shares, can read a farther edge: the merged closing can diverge where the station closing converges.

Corollary 32.18 (A law with log expansions bounds a closing of any fibre dimension)**.** *In Theorem 31.4(iii), with the starting point* $\mathbf{c}$ *off its null set, let the law of the piece have log expansions at the sums* $\bar{\mathbf{c}}$ *of* $\mathbf{c}$ *(Definition 32.13), with the inner bound* X_{in}*. The function* $h_\Gamma(\boldsymbol{\sigma}') + \log X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}}')$ *is convex on the chamber* $C(\boldsymbol{\sigma})$ *of Definition 31.5. If its infimum* h' *on the chamber is negative, the closing along* $\boldsymbol{\sigma}$ *converges absolutely to* $P_{B \sqcup R}(\mathbf{m})$*, its terms and the integrals left after them being* $O(e^{\tau(h' + \epsilon)})$ *for every* $\epsilon > 0$*. The fibre may have any dimension, and no margin of coefficient*

factors and no phase enter: the rate is read on the compact part of the law and on the inner faces of its log expansions. In Theorem 31.4(iv), with $\mathbf{c}$ *off the null set of that part, if the laws of the piece and of the partner have log expansions at* $\mathbf{c}$*, with the inner bounds written* X^B_{in} *and* X^R_{in} *in this sentence, the same holds with* $\sum_l \sigma'_l \log|\sigma'_l| + \log X^B_{\mathrm{in}}(\boldsymbol{\sigma'}) + \log X^R_{\mathrm{in}}(\boldsymbol{\sigma'})$*, convex on the chamber* $C(\boldsymbol{\sigma})$*, in place of* $h_\Gamma(\boldsymbol{\sigma'}) + \log X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}}')$*.*

So the counterparts of the reach theorems of width one hold along every rational direction. The tinkers of Definition 28.18 play two parts: all the tinkers of one partner separate the laws (Theorem 28.19), and the integral over the transverse tinkers of a closing, the partner read in its monomials, reads the reach of the real line (Theorem 32.16(iii)).

For a positive kernel the root rate of a crossing term is the exponential of the least Gamma rate of its chamber in every fibre dimension (Theorem 31.26), and averaging reads the log-support of the source measure (Theorem 31.7); Corollary 32.18 reads, past the walls, the compact part and the inner faces of the log expansions. It is an upper bound: with two or more free shares a crossing term is a fibre integral of positive dimension, whose size the law along one direction does not fix, and the best direction of a chamber can lose against the averaged closing of the same source (Theorem 31.8); the exact rate is then read on a joint word at a noncancelling simple top (Theorem 31.21). One partner does not read the real line: it averages over its transverse fibre, and its terms can fall below the rate of every real line of its chamber, the integral of the terms over the transverse tinkers then being carried by the other tinkered partners. Whatever the closing, the reach along the direction of the closing is at least the least value of the directional ratio on the support of the law, and at least every point near which the directional law, or after merging the merged law, has no real-analytic density (Theorem 32.10(iii)).

32.3 What the calculus of bunches settles

Remark 32.19 (What the calculus of bunches settles, and what it leaves open)*.* A closing of the glue along a direction of its share fibre is exact at every step and depends only on its chamber (Lemma 31.3, Theorem 31.6); Theorems 31.2, 31.4 and 31.11 give sufficient conditions for it, and the closings can be averaged over the source (Theorem 31.7). The cases that these conditions leave are settled as follows.

(i) *Crossing terms of positive dimension.* Along a rational ray, at simple nonresonant crossings and after a finite head, a crossing term kept with its falling shares integrated on their strips is, along its cohort, a scalar G_n times a finite sum $\sum_i \big(\mathsf{T}_{-n;(1)}[c_i]\,\phi_i\big)(Y_i)$ of joint Trailmoments of fixed functions on fixed compact domains, the joint word (Theorem 31.18): a falling Gamma becomes a positive share and one Beta variable (Lemma 31.14), a continued source strip a segment and circle integrals (Lemma 31.15), and nothing is extracted from the term. For a positive kernel its rate is the least Gamma rate of the chamber, in every fibre dimension and with no hypothesis on the tops (Theorem 31.26); averaged over a positive source it is the averaged rate $\hat{h}_C$ (Corollary 31.27), so a nonzero averaged rate decides by its sign whether the averaged terms tend to zero, with no condition on the bundles or the dimension (Theorem 31.28), and at $\hat{h}_C = 0$ Theorem 31.32 decides. The joint word has the root rate $r_G c_{\max}$ at noncancelling simple tops, r_G the root rate of the scalar and $c_{\max}$ the ceiling of the word (Theorem 31.21). Coincident crossings are packets of factors polar on one crossed hyperplane, assembled before a sign or a top is read (Theorem 31.30, Proposition 31.31); at the root rate one a positive moment sequence is decided by its law near the top (Theorem 31.32), and an expansion by its coefficients (Proposition 31.33).

(ii) *The phase.* A joint word is bounded by $|G_n| \sum_i \|\phi_i\| \, c_{\max}^n$ (Corollary 31.19): the fibre is completed before a modulus is taken, and the phase of the letters enters the coefficient functions, where it can cancel, and costs no term of the bound. It confines the radii of the procedure to the margins of the coefficient factors (Theorem 31.18, Remark 31.22). Read on the law, the phase costs nothing where the law has log expansions: in its log-ratio coordinates the shares pair with the law linearly, and the compact part and the inner faces of its log expansions bound the continued seam function on whole contours (Proposition 32.14). Where the tangential coordinates move with the normal coordinates at a falling end, the phase enters the coefficients of the expansion there, and a tube can reach infinitely far while each of its lines has a finite reach.

(iii) *Regions of equality and bounds at any width.* The regions on which a law equals its expansion at a falling end are certified from the sector letters, with the map from the motion of the shares to the exponents, the coefficients, the truncation bounds and exact boundary words (Theorems 32.2 and 32.4); the domain where a series converges and the region where it equals the law are kept apart (Proposition 32.6), and density charts are built at regular fibres (Proposition 32.5). The terms of a closing have at most the root rate $r_G c_{\max}$ of every joint word whose radii keep within the holomorphy of its source and the margins of its coefficient factors, $c_{\max}$ the ceiling of the word, and have that root rate when the word has a noncancelling simple top.

(iv) *The reach at any width.* Along every rational direction of the shares the reach is read on the law: it is the far boundary of the region on which the directional law, the law multiplied by $\mathbf{x}^{-\mathbf{w}_0}$ for the starting point $\mathbf{w}_0$ and read through the directional ratio, is its own wall series, whose exponents lie among the values $(\kappa_U(\mathbf{w}_0) + n)/(-\dot{\kappa}_U)$ of the walls divided by their falling rates, and it is at least every point at which that law is not analytic, the least value of the directional ratio among them (Theorem 32.10). The walls and the growth bounds that this uses hold for every seam function of a piece that converges at a real point and for its seam letters merged by bundles (Lemma 32.8), and at width one the same argument reads the growth of a family on its law, that of the bunch being its reach (Corollary 32.11). For a piece the reach is at most the ceiling of every keyhole continuation along the direction, with no phase, equal to it at simple tops that do not cancel when a keyhole continuation has the same ceiling, and bounded by fibres where letters vanish inside the keyholes, the direction scaled to integers (Theorem 32.15). A closing past the walls converges when the exponential of the Gamma rate times the reach of its tube is less than one, its terms then having at most that root rate (Theorem 32.16(i)); an expansion at the face over the real fibres bounds the reach of the tube along one direction and log expansions in every direction, with no allowance for the phase (Corollary 32.12, Proposition 32.14), and otherwise the tube can reach farther than every one of its lines; a law with log expansions bounds a closing of any fibre dimension by the rate read on the compact part and the inner faces of its log expansions (Corollary 32.18). With one free share, as at a node of width one, the tube is the line, and the terms attain the rate that the reach along the direction gives when the poles of the line and of the Gammas keep apart and the poles of the line are eventually simple or its values on a progression of crossings carry the reach (Theorem 32.16); the closing of a share side merged by its bundles to one free share, a series other than the station closings, has under the same conditions the root rate of Theorem 32.17(iii), read on the merged law along its direction, its terms not tending to zero beyond it, which carries the sharpness of Theorem 31.28 past the walls (Theorem 32.17). In every number of free shares the terms integrated over the tinkers of the partner transverse to the direction are the residues of the

real line, which attain its rate under the same conditions (Theorem 32.16), while the terms of one partner average over its transverse fibre and can fall below every real line of its chamber.

(v) *A gap at a family of tables.* A family of three tables whose seam has single stations has a chamber in which no direction certifies the closing while the averaged closing converges geometrically (Theorem 31.8); the gap is relative to the closings of one share side by the directions of its chamber (Remark 31.9).

At width one the reach is read on the law, as the point beyond which the law is its own wall series, and it is the ceiling of a keyhole continuation whose tops are simple and do not cancel (Theorems 20.44 and 20.51). At any width the first holds along every rational direction (Theorem 32.10), and the second along every rational direction for every piece, under the conditions of Theorem 32.15(ii) and (iii), and for joint words (Theorem 31.21). What a node of width one does not have is a transverse fibre. With two or more free shares the crossing terms of one partner integrate over the transverse fibre the imaginary translates of the line through its starting point. Theorem 32.16(i) bounds them by the reach of the tube when the exponential of the Gamma rate times that reach is less than one; a law with log expansions bounds the tube by the directional ratio on the compact part and the inner faces of its log expansions, and a closing of any fibre dimension with nothing paid for the phase (Corollary 32.18), while without log expansions the reach of the tube can be infinite though every line of the tube has a finite reach, and the bound that travels pays for the phase at a positive θ (Theorem 31.4(iii)). Theorem 32.16(iii) gives the integral of the terms over the transverse tinkers the rate of the real line, under the conditions of its part (ii) on that line; Theorem 31.21 gives the rate of one partner at the noncancelling simple tops of its joint words. The real line does not give the rate of one partner, whose terms can fall below every real line of its chamber, and the best direction of a chamber can lose against averaging (Theorem 31.8). Before the walls, under the hypotheses of Theorem 31.28, the rate of one partner is averaged over the source, and past them the closing of a share side merged by its bundles to one free share has, under the conditions of Theorem 32.17(iii), the rate read on the merged law along its direction. Two problems are open. The first is the rate past the walls of one partner whose joint words have no noncancelling simple top: at two or more free shares, where it asks for an averaged closing of the continued law, and at one free share when the data do not keep the poles apart or no proviso of Theorems 32.16(ii) and 32.17(iii) holds for that partner. The second is to read the growth $g(\boldsymbol{\sigma};\theta)$ on the law, as Theorem 32.10 reads the reach of a line. At $\theta = 0$ the reach of the line bounds it below and log expansions bound it above by their inner bound (Corollary 32.12, Proposition 32.14), and the two meet when, for every $\epsilon > 0$, the law has log expansions whose compact part and inner faces lie where the directional ratio is at most the reach plus ϵ; in general, and without log expansions, where it can be infinite at $\theta = 0$, it is known only through these bounds and those of Theorems 31.4(iii) and 31.11 and of Corollary 31.19, within the margins of the coefficient factors (Theorem 31.18). Degenerate and cancelling tops are open at width one as well.

Chapter 33

Price and complexity

A construction is a description of a family, and the exact theorems of Part V turn it into a description of the period; both have a price. The price of a construction is the length of its description, a repeated subtree described once and its multiplicity written in binary; the work of a word is the dimension of the integrals it asks for, an integral that recurs counted once. Repetition then costs what it should: M copies of a cell glued along a tree at a full lean, or closed in series at any exponents, or chained, have periods whose words cost the word of the cell and the binary length of M, and a departure from the repetition costs the cells that depart. The laws of cells (Chapter 23), the networks of holds (Chapters 25 and 26) and the finite interfaces (Chapter 27) are priced the same way: a law is computed once for each distinct subtree, a node of a network costs its periods at the two full leans whatever the number of bases of the whole table, and a polynomial Gamma moment of a repetition is the power of one finite matrix. What a value can contain is bounded the same way along a ray (Theorem 38.24).

33.1 Constructions and their price

Definition 33.1 (Construction)**.** A *construction* $\mathfrak{C}$ is a finite rooted tree with a family at each leaf, its marks, bases, weights, masses and exponents given, and at each inner node one of the operations: the direct sum of its subtrees; the substitution of its subtrees, each with a named hinge, into the marks of a core family named at the node, a mark left without a subtree being left in place (Definition 21.1); and the *powers* of a subtree $\mathfrak{C}_1$ with an integer M: $\mathfrak{C}_1^{\oplus M}$, M copies in direct sum; $\circ(\mathfrak{C}_1^{M})$, the series closure of M copies of the cell $\mathfrak{C}_1$ (Corollary 21.7(ii)); and $\mathfrak{C}_1^{\triangleright M}$, the chain of M copies glued along two named marks (Theorem 22.8). A subtree may be given by a reference to one already described. A construction builds one family, by Proposition 21.3 at each substitution.

Definition 33.2 (Price; work)**.** The *price* $\mathrm{price}(\mathfrak{C})$ of a construction is the number of bits of its description as written: for every leaf its marks, bases, weights, masses and exponents, the numbers written as rationals in binary; for every node its operation, its core and its attachments; for every power the $\lceil \log_2(M+1) \rceil$ bits of M; a subtree may be written once and referred to by an index wherever it recurs. The price of a word (Definition 1.1(vi)) is defined in the same way, and its *work* $\mathrm{work}(w)$ is the sum of the dimensions of its distinct integrals, an integral that recurs, under a power or an iterate in particular, counted once. The integration count of Definition 20.58 counts every occurrence: the work of w^M is that of w, and its integration count is M times that of w.

The work counts the distinct integrals a word is written with, and the price the length of what

has to be written down; neither counts the digits of an evaluated value, nor the cost of evaluating: a scalar power costs one exponentiation, a power of a $k \times k$ matrix costs $O(k^3 \log M)$ operations by repeated squaring, and an iterate $\mathcal{Q}^M$ of an operator costs M applications of $\mathcal{Q}$, nested integrals of M times its dimension, or $O(\log M)$ matrix products when $\mathcal{Q}$ leaves a finite-dimensional space of functions invariant. Every polynomial Gamma moment of degree D of a repetition (Chapter 27) is such a matrix power, a fixed row times a power of the transfer matrix $\mathbf{T}_D$ applied to the moment vector of the start (Theorem 27.5), and the period itself is reached through such moments with the certified stopping rule of Theorem 27.7, at a degree that depends on the accuracy asked.

Theorem 33.3 (Pricing by repetition)**.** *Let $\mathcal{A}$ be a family, a cell where a hinge is needed, described by a construction of price* $\mathrm{price}(\mathcal{A})$*, with the period $P_{\mathcal{A}}$ at its data, and let $M \geq 1$.*

(i) Direct sum. *The molecules of $\mathcal{A}^{\oplus M}$, $\mathcal{A}$ connected, are its M copies, and the product of their periods is $P_{\mathcal{A}}^{M}$ (Proposition 7.9; the period of the direct sum itself diverges on the balance wall, Proposition 8.16): price at most* $\mathrm{price}(P_{\mathcal{A}}) + \lceil \log_2(M+1) \rceil + O(1)$*, work that of $P_{\mathcal{A}}$.*

(ii) A tree of copies at a full lean. *For a table $\mathcal{A}$ at the exponents $(0,t)$, M copies glued along any tree at the masses $(\kappa, t-\kappa)$ on the two marks of every gluing have the word* $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa, & t-\kappa\\ & t\end{smallmatrix}\right] \big(P_{\mathcal{A}}(\mathbf{m};0,t)/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa, & t-\kappa\\ & t\end{smallmatrix}\right]\big)^{M}$ *(Corollary 26.11): price at most* $\mathrm{price}(P_{\mathcal{A}}) + \lceil \log_2(M+1) \rceil + O(1)$*, work that of $P_{\mathcal{A}}$, whatever the shape of the tree.*

(iii) A series closure. *The series closure of n_k copies of cells $\mathcal{A}_k$, $k \leq d$, at positive exponents of N tables on the balance wall has the word of Theorem 23.3,* $\prod_j (\Gamma(s_j)\Gamma(t_j))^{-1} \int \prod_j x_j^{s_j-1} \prod_{j \geq 2} y_j^{t_j-1} \prod_k \mathfrak{L}_k(x,y)^{n_k}\, dx\, dy$ *with the responses $\mathfrak{L}_k$ of the cells: price $O\big(\sum_k \mathrm{price}(\mathcal{A}_k)\big) + \sum_k \lceil \log_2(n_k+1) \rceil + O(N)$, the letters of a constructed cell being written through Theorem 21.4, and work at most $\sum_k |B_k| + 2N - 1$, B_k the marks of $\mathcal{A}_k$ besides its hinge, independent of the multiplicities and of the order of the cells.*

(iv) A chain. *$\mathcal{A}^{\triangleright M}$ has the word $(\mathcal{Q}^M P_0)(\mathbf{1})$ of Theorem 22.8, with the operator $\mathcal{Q}$ and the function P_0 of that theorem: price at most* $\mathrm{price}(\mathcal{A}) + \lceil \log_2(M+1) \rceil + O(1)$*, work at most $|\mathcal{E}_{\mathcal{A}}| - 1$; evaluated, the iterate unrolls to nested integrals of total dimension $M(|\mathcal{E}_{\mathcal{A}}| - 2) + 1$, that of the expanded period, unless $\mathcal{Q}$ leaves a finite-dimensional space of functions invariant; where the twist is pinned the word is* $P_{\mathcal{A}}^{M} / \Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}m_{g_{\mathrm{in}}}, & m_{g_{\mathrm{out}}}\\ & m_{g_{\mathrm{in}}} + m_{g_{\mathrm{out}}}\end{smallmatrix}\right]^{M-1}$*, $m_{g_{\mathrm{in}}}, m_{g_{\mathrm{out}}}$ the masses of the two named marks $g_{\mathrm{in}}, g_{\mathrm{out}}$, of work $|\mathcal{E}_{\mathcal{A}}| - 1$ evaluated as well.*

(v) Integer joint moments *of a repeated transfer matrix, with Gamma laws on the marks or Dirichlet laws on the steps, have as word a fixed row times the power $\mathbf{A}^M$ of the one matrix of Theorem 22.10 times a fixed column, with rational entries at rational masses (Theorem 22.10): price at most that of the matrices of the cell plus $\lceil \log_2(M+1) \rceil + O(1)$, work zero.*

(vi) Blocks *read through their totals cost one stacked Gamma each,* $\Gamma_{\begin{smallmatrix}|B_i|\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{smallmatrix}\right]$ *at the block B_i (Theorem 26.12): the work of the period of the family with blocks is that of the period of its core family, whatever the lengths of the blocks.*

(vii) A series repetition. *M copies of a series-balanced cell $\mathcal{A}$ at a series node against a cell $\mathcal{A}'$, at the exponents (s,t), have the word* $(\Gamma(s)\Gamma(t))^{-1} \int_0^\infty y^{t-1} f_{\mathcal{A}}(y)^M f_{\mathcal{A}'}(y)\, dy$*, with the transforms $f_{\mathcal{A}}, f_{\mathcal{A}'}$ of Theorem 24.2: price at most* $\mathrm{price}(\mathcal{A}) + \mathrm{price}(\mathcal{A}') + \lceil \log_2(M+1) \rceil + O(1)$*, work $|B| + |R| - 1$ for the marks B of $\mathcal{A}$ and R of $\mathcal{A}'$ besides the hinge, whatever M; at a*

series or parallel node of a hinge tree the multiplicities of equal subtrees enter as exponents (Proposition 23.18).

(viii) Series-parallel tables. *A table with a hinge tree of series and parallel nodes over marks has periods at the full leans that are stacked Gammas times monomials in the weights of its nodes, with one Gamma for each mark and two for each change of kind along an edge, and every period, at every lean and every twist, a fibre integral of dimension $n-1$ of an explicit stacked Gamma times an exponential of the loads, and a family of N such tables one of dimension $(2N-1)(n-1)$ (Corollaries 26.10 and 26.14): price linear in the size of the tree, work at most $n-1$, less one for each pooled pair (Proposition 26.15), and at most $(2N-1)(n-1)$ for the family, whatever the numbers of bases, which can be exponential in n.*

(ix) Polynomial Gamma moments of a repetition *have as word a fixed row times a power of the transfer matrix $\mathbf{T}_D$ applied to a fixed vector, with rational entries at rational data (Theorem 27.5): price that of the matrix of the chain cell plus $\lceil \log_2(M+1) \rceil + O(1)$, work zero.*

In the cases with a multiplicity M the expanded family has of the order of M times as many marks, and its period written as one integral over its simplex has the work of one less than its number of marks.

Theorem 33.4 (Repeated subtrees and departures)**.** *(i) In a construction, replacing identical subtrees, with matching hinges and data, by one subtree and references leaves the family built, the period and every exact word of Part V unchanged, and lowers the price by the price of the copies removed, less that of the references.*

(ii) In (ii), (iii) and (iv) of Theorem 33.3, replacing q of the M copies by cells $\mathcal{A}'_1, \dots, \mathcal{A}'_q$ changes the word to $\Gamma_1^2 \begin{bmatrix} \kappa, & t-\kappa \\ & t \end{bmatrix} \left(P_{\mathcal{A}} / \Gamma_1^2 \begin{bmatrix} \kappa, & t-\kappa \\ & t \end{bmatrix} \right)^{M-q} \prod_i \left(P_{\mathcal{A}'_i} / \Gamma_1^2 \begin{bmatrix} \kappa_i, & t-\kappa_i \\ & t \end{bmatrix} \right)$ at a full lean, to the product of $\mathfrak{L}_{\mathcal{A}}^{M-q}$ and of the responses of the new cells in a series closure, and to the composite of $\mathcal{Q}^{M-q}$ with the q operators of the new cells in a chain: the price grows by $\sum_i \operatorname{price}(\mathcal{A}'_i) + O(q \log M)$ and the work by at most the dimensions of the integrals of the new cells. A change of q leaves in a tree with at most h nodes on every path from its root to a leaf touches at most qh of its nodes.

Repetition is thus priced at the binary length of its count, as 2^M is written with M rather than with its digits, and the work follows the distinct subtrees. The construction is the certificate of the compression: a family given expanded, without its construction, can still be read for its bunches of width one, its prime nodes (Theorem 23.16) and its twins (Theorem 14.22), but finding its cheapest construction is a search. The prime nodes of the canonical hinge tree of a list are invariants of the list (Theorem 23.16); at a full lean its period is a product of stacked Gammas and the periods at the full leans of those nodes (Corollary 26.11, Theorem 26.5), and at every lean a fibre integral of their periods at the full leans, of dimension $n-1$ (Theorem 26.13); a prime node $\mathcal{H}_v$ on n_v marks with $|\mathcal{H}_v|$ bases has periods at the full leans of dimension $|\mathcal{H}_v| - n_v > 0$ on their share sides (Corollary 26.8). Symmetry is a second source of cheapness, distinct from repetition: a table assembled from a seed by a set of bijections (Proposition 11.6) is described by the seed and the bijections, and the table that is one circuit on n marks, prime for no n, is a series closure of single marks whose period is one integral over one scale of a product of n Grammas (Corollary 18.22), of work one whatever n; the list $\binom{\mathcal{E}}{r}$ with $2 \le r \le |\mathcal{E}| - 2$ is prime and cheap to describe, and its period at each full lean is one fibre integral of dimension $\binom{|\mathcal{E}|}{r} - |\mathcal{E}|$ on its share side (Corollary 26.8).

Part VI

The arithmetic of the output

Chapter 34

The motive of a family and the coaction along a tree

A family of lists at integer data whose proper walls are all at least one has as its period the period of a mixed motive: the relative cohomology of the toric variety of the braid fan, less the closures of the zero loci of its letters and of the linear letters, relative to its toric boundary. The walls are the orders of the form along the boundary divisors, and every boundary stratum is the product of the models of the steps of its flag. Brown's coaction acts on the motivic period, and birational changes of variables, Cremona among them, are identities of motivic periods. Along any tree of bunches the left factors of the coaction of weight at most k are sums of products of open periods of steps of at most $k+1$ marks; each step is attached to one vertex of the tree, its letters are letters of children of the tables there, and its open periods are rational combinations of constant terms of continued periods of families of lists on the step, hence of convergent periods of such families and of their derivatives in the masses. At a node of width c the bunch and its room meet only through the Skeleton values of the letters, at most c for each letter: every open period of a node step is a shape integral of rational functions against finite parts of scale integrals, constant terms of stacked Gammas times Bretas of one bunch at the joint spectrum, and at the root so is every left factor; at width one each Skeleton value is a quotient of hinge values and the share side separates the bunch from the room. Cells read through one hinge value split off as identities of motivic periods, whatever the widths of the bunches of the core; so do the cells of dual pairs at every pair of exponents, and at a full lean the motivic period of a list is a rational multiple of the product of those of its prime nodes. A pinned set of at least two marks at which every active table has width one splits off motivically.

34.1 The model

The motive of a family lives on the toric variety of the braid fan. Its charts are the sector cubes of Theorem 8.5(ii), and its boundary strata are indexed by chains of sets of marks.

Definition 34.1 (The model of a family)**.** Let $F = ((\mathcal{B}_j; s_j, t_j))_{j=1}^{N}$ be a family of lists on $\mathcal{E}$, $n = |\mathcal{E}| \geq 2$ (Definition 8.1), with integer exponents and integer masses $\mathbf{m}$ on its balance wall; it is *convergent* when $\kappa_S \geq 1$ for every $\varnothing \neq S \subsetneq \mathcal{E}$. Its *letters* are $\mathrm{Let}(F) = \{K_j : \ s_j \neq 0\} \cup \{\Psi_j : t_j \neq 0\}$; the *linear letters* are the $\alpha_J = \sum_{e \in J} \alpha_e$, $J \subseteq \mathcal{E}$, the basis polynomials of the lists $\binom{J}{1}$; and the *augmented letters* are $\mathrm{Let}^{+}(F) = \mathrm{Let}(F) \cup \{\alpha_J : \ J \subseteq \mathcal{E}, \ |J| \geq 2\}$. For an order $e_1, \ldots, e_n$ of

$\mathcal{E}$ put

$$\Omega_{\mathcal{E}} = \sum_{i=1}^{n} (-1)^{i-1} \alpha_{e_i}\, d\alpha_{e_1} \wedge \cdots \wedge \widehat{d\alpha_{e_i}} \wedge \cdots \wedge d\alpha_{e_n},$$

which another order changes by its sign; the *form* of F is

$$\omega_F = \alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, \frac{\Omega_{\mathcal{E}}}{\alpha^{\mathcal{E}}},$$

a rational form of degree $n-1$ on $\mathbb{P}(\mathbb{Q}^{\mathcal{E}})$. Let $X_{\mathcal{E}}$ be the toric variety over $\mathbb{Q}$ of the *braid fan* in $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$, whose rays are the images ρ_S of the $\mathbf{1}_S$, $\varnothing \neq S \subsetneq \mathcal{E}$, and whose cones are spanned by the rays of the chains $S_1 \subsetneq \cdots \subsetneq S_\ell$; D_S is the divisor of ρ_S, $D = \bigcup_S D_S$ the *boundary*, and the *positive cell* $\widetilde{\sigma}_{\mathcal{E}}$ is the closure in $X_{\mathcal{E}}(\mathbb{R})$ of the set $\widetilde{\sigma}^\circ_{\mathcal{E}}$ of the points of the torus with every $\alpha_e > 0$. An order $e_1, \dots, e_n$ of $\mathcal{E}$ gives the *chart* $U \cong \mathbb{A}^{n-1}$ of the cone of the chain $S_i = \{e_1, \dots, e_i\}$, $i < n$, with the coordinates $\varepsilon_i = \alpha_{e_i}/\alpha_{e_{i+1}}$, so that $\alpha_{e_j}/\alpha_{e_n} = \prod_{i \geq j} \varepsilon_i$. For a polynomial f on $\mathcal{E}$, $\widetilde{Y}_f$ is the closure in $X_{\mathcal{E}}$ of its zero locus in the torus. The *model* of F is the pair

$$V_F = X_{\mathcal{E}} \setminus \bigcup_{f \in \mathrm{Let}^+(F)} \widetilde{Y}_f, \qquad D_F = D \cap V_F,$$

with the relative cohomology $H_F = H^{n-1}(V_F, D_F)$.

Definition 34.2 (Flags and steps)**.** A *flag* of $\mathcal{E}$ is a chain $\mathfrak{f}: \varnothing = S_0 \subsetneq S_1 \subsetneq \cdots \subsetneq S_\ell \subsetneq S_{\ell+1} = \mathcal{E}$, $\ell \geq 0$. Its *steps* are the sets $Z_i = S_i \setminus S_{i-1}$, $1 \leq i \leq \ell+1$, as in Theorem 22.9, and its *dimension* is $\sum_i (|Z_i| - 1) = n - 1 - \ell$. In the model of a family F (Definition 34.1) its *stratum* is $D_{\mathfrak{f}} = D_{S_1} \cap \cdots \cap D_{S_\ell} \cap V_F$, the model V_F itself at $\ell = 0$. The *letters of the step* $(Z_i; \mathfrak{f})$ are the step letters at Z_i (Proposition 12.20) of the augmented letters of F that are not monomials: K of the step list of $\mathcal{B}_j$ at Z_i at the bottom of the chain for every $K_j \in \mathrm{Let}(F)$, Ψ of its step list at the top for every $\Psi_j \in \mathrm{Let}(F)$, and the α_J with $J \subseteq Z_i$, $|J| \geq 2$; for $\ell = 0$ the step lists at the one step $\mathcal{E}$ are the lists of F themselves. They form a set $\mathrm{Let}(Z_i; \mathfrak{f})$ of polynomials on Z_i, which depends on S_{i-1} and S_i alone. The *model of the step* is

$$V_{Z_i;\mathfrak{f}} = X_{Z_i} \setminus \bigcup_{g \in \mathrm{Let}(Z_i;\mathfrak{f})} \widetilde{Y}_g,$$

a point when $|Z_i| = 1$, with its toric boundary $\partial V_{Z_i;\mathfrak{f}}$.

Lemma 34.3 (Charts, strata and the letters in a chart)**.** *In Definition 34.1:*

(i) $X_{\mathcal{E}}$ is smooth and projective of dimension $n-1$, and the charts of the orders cover it. In the chart of the order $e_1, \dots, e_n$, $D_{S_i} \cap U = \{\varepsilon_i = 0\}$ for $i < n$ and $D_S \cap U = \varnothing$ for every other S; D is a divisor with simple normal crossings and smooth components, and $D_{S_1} \cap \cdots \cap D_{S_\ell}$ is not empty exactly when the S_i form a chain. In the chart $\widetilde{\sigma}_{\mathcal{E}} \cap U = [0, \infty)^{n-1}$, and the closure of the set of the points of $\widetilde{\sigma}^\circ_{\mathcal{E}}$ with $\alpha_{e_1} < \cdots < \alpha_{e_n}$, the sector *of the order, is the cube $[0,1]^{n-1}$; the sectors of the orders cover $\widetilde{\sigma}_{\mathcal{E}}$, two of them meeting in a set of measure zero.*

(ii) Let f be a letter K or Ψ of a list $\mathcal{B}$ on $\mathcal{E}$, or a linear letter α_J, of the degree δ, with the floor *$d_f(S)$: $d(S)$ for K, $d^*(S)$ for Ψ (Theorem 7.13), and 1 or 0 for α_J as $J \subseteq S$ or not. In the chart of an order,*

$$f(\alpha)/\alpha_{e_n}^{\delta} = \prod_{i<n} \varepsilon_i^{d_f(S_i)}\, \widetilde{f}(\varepsilon),$$

with $\widetilde{f}$ a polynomial with nonnegative integer coefficients and constant term 1, the sector letter of Theorem 8.5(ii); so $\widetilde{f} \geq 1$ where $\varepsilon \geq 0$. For $I \subseteq \{1, \dots, n-1\}$, the restriction of $\widetilde{f}$ to $\{\varepsilon_i = 0 : \ i \in I\}$ is the product, over the steps Z of the chain $(S_i)_{i \in I}$, of the polynomials that this rule attaches to the step letters of f at Z (Proposition 12.20) in the chart of X_Z of the order induced on Z, whose coordinates are the ε_k with $e_k, e_{k+1} \in Z$. Moreover $\widetilde{Y}_f \cap U = \{\widetilde{f} = 0\}$.

(iii) In the chart, with the walls κ_S of F,

$$\omega_F = \pm \prod_{i<n} \varepsilon_i^{\kappa_{S_i}-1} \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j} \, d\varepsilon_1 \wedge \cdots \wedge d\varepsilon_{n-1}.$$

Proposition 34.4 (The model)**.** *Let F be a family as in Definition 34.1.*

(i) $\widetilde{\sigma}_{\mathcal{E}}$ is compact, $\widetilde{\sigma}_{\mathcal{E}} \setminus \widetilde{\sigma}_{\mathcal{E}}^{\circ} \subseteq D$, and $\widetilde{\sigma}_{\mathcal{E}}$ misses $\widetilde{Y}_f$ for every letter f of a list on $\mathcal{E}$ and every linear letter; so $\widetilde{\sigma}_{\mathcal{E}} \subseteq V_F(\mathbb{R})$.

(ii) ω_F has the order $\kappa_S - 1$ along D_S for every $\varnothing \neq S \subsetneq \mathcal{E}$ and neither zeros nor poles on the points of the torus in V_F. So ω_F is a regular form on V_F exactly when F is convergent, and then its integral over $\widetilde{\sigma}_{\mathcal{E}}$, oriented so that it is positive, converges and is the period $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ (Theorem 8.7).

(iii) V_F is a smooth affine variety, D_F is a divisor in it with simple normal crossings and smooth components, and every stratum is smooth and affine. More generally $X_{\mathcal{E}} \setminus \bigcup_{g \in \mathcal{G}} \widetilde{Y}_g$ is affine for every finite set $\mathcal{G}$ of polynomials on $\mathcal{E}$ that contains the linear letters α_J, $|J| \geq 2$.

(iv) The stratum of a flag $\mathfrak{f}$ with the steps $Z_1, \dots, Z_{\ell+1}$ is isomorphic to $\prod_i V_{Z_i;\mathfrak{f}}$ (Definition 34.2), compatibly with the boundaries, and $\widetilde{\sigma}_{\mathcal{E}}$ meets it in $\prod_i \widetilde{\sigma}_{Z_i}$.

(v) For the tables of F and complex masses and exponents on the balance wall, wherever every proper wall has positive real part,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \sum_{\pi} \operatorname{cube}\big(\boldsymbol{\kappa}_\pi(\mathbf{m}); h_\pi\big), \qquad \boldsymbol{\kappa}_\pi(\mathbf{m}) = \big(\kappa_{S_i^\pi}(\mathbf{m})\big)_{i<n}, \quad h_\pi = \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j},$$

the sum over the orders π of $\mathcal{E}$, with the chains S_i^π and the sector letters of their charts, and cube *the cube integral of Lemma 8.4. The right side continues meromorphically along the balance wall, jointly in the exponents, with poles only on the hyperplanes $\kappa_S = -j$, $\varnothing \neq S \subsetneq \mathcal{E}$, $j \in \mathbb{N}$: it is the* continued period *of Theorems 8.5(iii) and 8.7.*

The walls are thus the orders of the form along the boundary, and convergence is regularity. The linear letters are removed so that the model is affine; they are the letters of the lists $\binom{J}{1}$, and the motivic period does not see them (Lemma 34.7).

34.2 Motivic periods

Definition 34.5 (Motivic periods; Galois conjugates; open spaces)**.** An object M of Brown's category of realizations over $\mathbb{Q}$ (F. Brown, *Notes on motivic periods*, Commun. Number Theory Phys. 11 (2017), 557–655, §3.1) has a Betti realization M_{B} and a de Rham realization M_{dR}; its *motivic periods* $[M, \gamma, \omega]^{\mathrm{mot}}$, $\gamma \in M_{\mathrm{B}}^\vee$ and $\omega \in M_{\mathrm{dR}}$, span the ring $\mathcal{P}^{\mathrm{mot}}$, bilinear in (γ, ω),

with $[M,\phi^{\mathrm{t}}\gamma',\omega]^{\mathrm{mot}} = [M',\gamma',\phi(\omega)]^{\mathrm{mot}}$ for every morphism $\phi : M \to M'$; its de Rham periods $[M,f,\omega]^{\mathrm{dR}}$, $f \in M_{\mathrm{dR}}^{\vee}$, span the ring $\mathcal{P}^{\mathrm{dR}}$; and the *coaction* $\Delta : \mathcal{P}^{\mathrm{mot}} \to \mathcal{P}^{\mathrm{mot}} \otimes \mathcal{P}^{\mathrm{dR}}$ is

$$\Delta[M,\gamma,\omega]^{\mathrm{mot}} = \sum_i [M,\gamma,e_i]^{\mathrm{mot}} \otimes [M,e_i^{\vee},\omega]^{\mathrm{dR}}$$

for a basis (e_i) of M_{dR} with the dual basis $(e_i^{\vee})$; the $[M,\gamma,e_i]^{\mathrm{mot}}$ are its *left factors*. For $\xi \in \mathcal{P}^{\mathrm{mot}}$ the space of its *Galois conjugates* is

$$\mathrm{C}(\xi) = \big\{(\mathrm{id}\otimes\varphi)\Delta\xi :\ \varphi \in \mathrm{Hom}_{\mathbb{Q}}(\mathcal{P}^{\mathrm{dR}},\mathbb{Q})\big\};$$

for an object M and $\gamma \in M_{\mathrm{B}}^{\vee}$, $L(M,\gamma) = \{[M,\gamma,\omega]^{\mathrm{mot}} :\ \omega \in M_{\mathrm{dR}}\}$; and $\mathcal{A}\cdot\mathcal{A}'$ is the span of the products of the elements of two subspaces. For a convergent family F (Definition 34.1) the *motivic period* is

$$P_F^{\mathrm{mot}} = [H_F, \widetilde{\sigma}_{\mathcal{E}}, \omega_F]^{\mathrm{mot}},$$

the positive cell, oriented so that $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}}$ is positive on $\widetilde{\sigma}_{\mathcal{E}}^{\circ}$, read as a class of relative homology, and the form as a class of relative de Rham cohomology; its period is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Proposition 34.4(ii)). For a step $(Z;\mathfrak{f})$ of a flag of $\mathcal{E}$ (Definition 34.2) with $|Z| \geq 2$, its *open space* is the set of motivic periods

$$\mathrm{Open}(Z;\mathfrak{f}) = \Big\{ \big[H^{|Z|-1}(V_{Z;\mathfrak{f}}, \partial V_{Z;\mathfrak{f}}), \widetilde{\sigma}_Z, \eta\big]^{\mathrm{mot}} :\ \eta \text{ a regular form of degree } |Z|-1 \text{ on } V_{Z;\mathfrak{f}} \Big\},$$

$\widetilde{\sigma}_Z$ oriented as $\widetilde{\sigma}_{\mathcal{E}}$ is, and $\mathrm{Open}(Z;\mathfrak{f}) = \mathbb{Q}$ when $|Z| = 1$.

Lemma 34.6 (Galois conjugates)**.** *For $\xi,\xi' \in \mathcal{P}^{\mathrm{mot}}$, an object M and $\gamma \in M_{\mathrm{B}}^{\vee}$:*

(i) $\xi \in \mathrm{C}(\xi)$*;*

(ii) $\mathrm{C}([M,\gamma,\omega]^{\mathrm{mot}}) \subseteq L(M,\gamma)$*, and* $\Delta L(M,\gamma) \subseteq L(M,\gamma)\otimes\mathcal{P}^{\mathrm{dR}}$*;*

(iii) $\mathrm{C}(\xi+\xi') \subseteq \mathrm{C}(\xi)+\mathrm{C}(\xi')$*,* $\mathrm{C}(\xi\xi') \subseteq \mathrm{C}(\xi)\cdot\mathrm{C}(\xi')$*, and* $\mathrm{C}(q\,\xi) = \mathrm{C}(\xi)$ *for* $q \in \mathbb{Q}$*,* $q \neq 0$*.*

Lemma 34.7 (Birational invariance)**.** *For $i = 1,2$ let X_i be a smooth projective variety over $\mathbb{Q}$ of dimension m, $D_i \subset X_i$ a divisor with simple normal crossings and smooth components, $Y_i \subset X_i$ a closed set, $X_i^{\circ} = X_i \setminus Y_i$ and $D_i^{\circ} = D_i \setminus Y_i$. Let $\sigma_i \subset X_i(\mathbb{R})$ be compact, the closure of an oriented open semialgebraic set $\sigma_i^{\circ} \subseteq (X_i \setminus (D_i \cup Y_i))(\mathbb{R})$ of dimension m, with $\sigma_i \cap Y_i = \varnothing$ and $\sigma_i \setminus \sigma_i^{\circ} \subseteq D_i$, and let ω_i be a regular form of degree m on X_i°. If an isomorphism $\Phi : U_1 \to U_2$ of dense open sets $U_i \subseteq X_i \setminus (D_i \cup Y_i)$ with $\sigma_i^{\circ} \subseteq U_i(\mathbb{R})$ maps σ_1° onto σ_2° preserving the orientations and has $\Phi^*\omega_2 = \omega_1$, then*

$$\big[H^m(X_1^{\circ}, D_1^{\circ}), \sigma_1, \omega_1\big]^{\mathrm{mot}} = \big[H^m(X_2^{\circ}, D_2^{\circ}), \sigma_2, \omega_2\big]^{\mathrm{mot}}.$$

In particular, with Φ the identity, removing a larger closed set that misses σ_1 leaves the motivic period unchanged; the motivic period of a convergent family is the same in the model without the linear letters, $X_{\mathcal{E}} \setminus \bigcup_{f\in\mathrm{Let}(F)} \widetilde{Y}_f$; and two convergent families on $\mathcal{E}$ with one form have one motivic period.

Corollary 34.8 (Cremona between motivic periods)**.** *Let $F = ((\mathcal{B}_j; s_j, t_j))_j$ be a convergent family with the masses $\mathbf{m}$ and the total strength σ, and F^{c} the family of the same tables with the exponents (t_j, s_j) and the masses $2\sigma\mathbf{1} - \mathbf{m}$. Then F^{c} is convergent, its wall at S is the wall of F at $\mathcal{E} \setminus S$, and $P_{F^{\mathrm{c}}}^{\mathrm{mot}} = P_F^{\mathrm{mot}}$: Theorem 8.14 is an identity of motivic periods.*

34.3 Steps along a tree of bunches

The boundary strata are indexed by flags, and a stratum is a product over the steps of its flag (Proposition 34.4(iv)). A tree of bunches (Definition 20.7) assigns every step to one of its vertices.

Definition 34.9 (Attachment)**.** Let $\mathcal{T}$ be a tree of bunches of a family F (Definition 20.7) and $(Z;\mathfrak{f})$ a step of a flag of $\mathcal{E}$. Starting at the root, Z descends through vertices whose marks contain it:

(i) at a split vertex, if Z lies in one of its children it descends to that child; otherwise Z is *attached* there as a *node step*;

(ii) at a node v that is not a leaf, with its bunch B_v: if $Z \subseteq B_v$, Z is attached to v as a *bunch step*; if Z meets B_v and $M_v \setminus B_v$, as a *node step*; and if $Z \subseteq M_v \setminus B_v$, it descends to the room of v;

(iii) at a leaf, Z is attached as a *leaf step*.

Every step is attached to exactly one vertex.

Lemma 34.10 (The letters of an attached step)**.** *Let $\mathcal{T}$ be a tree of bunches of a family F as in Definition 34.1, and let the step $(Z;\mathfrak{f})$ be attached to the vertex v, with the marks M_v. Every letter of the step other than the linear letters is K or Ψ of the child, at a pattern of $M_v \setminus Z$ (Definition 12.1), of a table at v that carries that letter.*

Every family has trees of bunches: at every node that is not a leaf choose a bunch and pass to its room, which has fewer marks. For nonnegative exponents the Foreman tree is one of them (Definition 14.4); with nonnegative exponents merging equal children keeps every letter, so the record and the peel give the same tree, and it is fixed by the tables and the rays of the exponent pairs (Theorem 14.6). The statements below hold along every tree of bunches, and along the Foreman tree the vertex at which each step is attached, and the node form there, are fixed by the family.

34.4 The weights of the coaction

Lemma 34.11 (One form carries every period)**.** *Let V be a smooth affine variety over $\mathbb{Q}$ of dimension m and $D \subset V$ a divisor with simple normal crossings and smooth components. Every class of the relative de Rham cohomology $H^m_{\mathrm{dR}}(V,D)$ is the class of a regular form of degree m on V. So, for a convergent family F (Definition 34.1), $L(H_F,\widetilde{\sigma}_{\mathcal{E}})$ is the open space of the one step $\mathcal{E}$ of the flag $\varnothing \subsetneq \mathcal{E}$, and for every step, $L\big(H^{|Z|-1}(V_{Z;\mathfrak{f}},\partial V_{Z;\mathfrak{f}}),\widetilde{\sigma}_Z\big) = \mathrm{Open}(Z;\mathfrak{f})$.*

So the open space of the whole set of marks already holds every period of H_F on its cell, and a decomposition of the left factors into products over steps carries information only when it is graded. The steps enter through the weight filtration W of the de Rham realization, by Brown's weight-graded face maps.

Proposition 34.12 (Weight-graded faces)**.** *Let F be convergent (Definition 34.1), $k < n-1$ and $e \in W_k(H_F)_{\mathrm{dR}}$. Then*

$$[H_F,\widetilde{\sigma}_{\mathcal{E}},e]^{\mathrm{mot}} \in \sum_{\mathfrak{f}\ \textit{of dimension } k}\ \prod_{\textit{steps } Z\ \textit{of } \mathfrak{f}} \mathrm{Open}(Z;\mathfrak{f}).$$

Theorem 34.13 (Small steps along the tree)**.** *Let F be convergent, $\mathcal{T}$ a tree of bunches of F, and (e_i) a basis of $(H_F)_{\mathrm{dR}}$ adapted to the weight filtration W. For $k < n-1$ every left factor $[H_F, \widetilde{\sigma}_{\mathcal{E}}, e_i]^{\mathrm{mot}}$ of the coaction of P_F^{mot} with $e_i \in W_k$ satisfies*

$$[H_F, \widetilde{\sigma}_{\mathcal{E}}, e_i]^{\mathrm{mot}} \in \sum_{\mathfrak{f} \text{ of dimension } k} \ \prod_{\text{steps } Z \text{ of } \mathfrak{f}} \mathrm{Open}(Z; \mathfrak{f}).$$

Only steps of at most $k+1$ marks occur. Each is attached to exactly one vertex v of $\mathcal{T}$, and every letter of it other than the linear letters is K or Ψ of the child, at a pattern of $M_v \setminus Z$, of a table at v that carries that letter.

The theorem is read on Brown's affine model, which removes the linear letters α_J to make the complement affine. They are the letters of the lists $\binom{J}{1}$, and they enter the families of the steps below as tables. Each open period is a constant term of continued periods, and those constant terms are read on convergent data.

Proposition 34.14 (Laurent coefficients are read on convergent data)**.** *Let $F = ((\mathcal{B}_j; s_j, t_j))_{j=1}^N$ be a family of lists on $\mathcal{E}$ with integer exponents and integer masses $\mathbf{m}$ on its balance wall, and $\mathbf{v} \in \mathbb{Q}^{\mathcal{E}}$ with $v_{\mathcal{E}} = 0$ and $v_S \neq 0$ for every $\varnothing \neq S \subsetneq \mathcal{E}$. Then $\varepsilon \mapsto P_F(\mathbf{m} + \varepsilon\mathbf{v}; \mathbf{s}, \mathbf{t})$, the continued period (Proposition 34.4(v)), is meromorphic near $\varepsilon = 0$, and there are finitely many convergent families $F_k = ((\mathcal{B}_j; s_j^{(k)}, t_j^{(k)}))_j$ on the same tables, with $\mathbf{s}^{(k)} - \mathbf{s},\ \mathbf{t}^{(k)} - \mathbf{t} \in \mathbb{N}^N$ and integer masses $\mathbf{m}^{(k)}$ on their balance walls, and rational functions $R_k \in \mathbb{Q}(\varepsilon)$, with*

$$P_F(\mathbf{m} + \varepsilon\mathbf{v}; \mathbf{s}, \mathbf{t}) = \sum_k R_k(\varepsilon)\, P_{F_k}\big(\mathbf{m}^{(k)} + \varepsilon\mathbf{v}; \mathbf{s}^{(k)}, \mathbf{t}^{(k)}\big)$$

near $\varepsilon = 0$. So every coefficient of the Laurent expansion at $\varepsilon = 0$ is a rational combination of the absolutely convergent integrals

$$\partial_\varepsilon^l P_{F_k}\big(\mathbf{m}^{(k)} + \varepsilon\mathbf{v}; \mathbf{s}^{(k)}, \mathbf{t}^{(k)}\big)\Big|_{\varepsilon=0} = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}^{(k)} - \mathbf{1}} \langle \mathbf{v}, \log u \rangle^l \prod_j K_j(u)^{-s_j^{(k)}} \Psi_j(u)^{-t_j^{(k)}}\, d\omega_{\mathcal{E}}(u),$$

$l \in \mathbb{N}$. If $\mathbf{s}, \mathbf{t} \geq 0$, every F_k has nonnegative exponents and positive masses.

Corollary 34.15 (Open periods from families of lists)**.** *Let $(Z; \mathfrak{f})$ be a step of a flag of $\mathcal{E}$ with $|Z| \geq 2$ and η a regular form of degree $|Z| - 1$ on $V_{Z;\mathfrak{f}}$. On the torus,*

$$\eta = \sum_t c_t\, \omega_{F_t}, \qquad c_t \in \mathbb{Q},$$

a finite sum over families F_t of lists on Z at integer data on their balance walls: the tables of F_t are the lists of the letters of the step, the step list for its K or its Ψ and $\binom{J}{1}$ for α_J, each carrying its letter with an exponent $k_{t,g} \in \mathbb{N}$, and the masses are some $\mathbf{a}_t \in \mathbb{Z}^Z$. For every $\mathbf{v} \in \mathbb{Q}^Z$ with $v_Z = 0$ and $v_S \neq 0$ for every $\varnothing \neq S \subsetneq Z$, each continued period $P_{F_t}(\mathbf{a}_t + \varepsilon\mathbf{v})$ (Proposition 34.4(v)) is meromorphic in ε near 0, and

$$\int_{\widetilde{\sigma}_Z} \eta = \sum_t c_t \operatorname*{CT}_{\varepsilon=0} P_{F_t}(\mathbf{a}_t + \varepsilon\mathbf{v}).$$

By Proposition 34.14 each constant term is a rational combination of convergent periods of families of lists on Z with nonnegative integer exponents and positive integer masses, values of the serif family (Proposition 8.12), and of their derivatives along $\mathbf{v}$ in the masses. So every element of $\mathrm{Open}(Z; \mathfrak{f})$ has such a number as its period, and by Theorem 34.13 every left factor of weight at most k has as its period a sum of products of such numbers over the steps of the flags of dimension k, whose letters are letters of children of the tables at the vertices where the steps are attached.

34.5 The node

Theorem 34.16 (The node form)**.** *Let F be convergent (Definition 34.1), $\mathcal{T}$ a tree of bunches of F, v a node of $\mathcal{T}$ that is not a leaf, with its bunch B_v of width c (Definition 20.7), and $(Z;\mathfrak{f})$ a node step of v (Definition 34.9), with $Z_B = Z \cap B_v$ and $Z_R = Z \cap (M_v \setminus B_v)$. Read the bunch side in scale and shape and the room side on its simplex, $\alpha_{Z_B} = \rho u$ with $\rho > 0$ and $u \in \Delta_{Z_B}$, and $\alpha_{Z_R} = u' \in \Delta_{Z_R}$ (Lemma 2.7).*

(i) Every letter g of the step is the basis polynomial of a list $\mathcal{N}_g$ on Z: the step list at the bottom for a letter K, the dual of the step list at the top for a letter Ψ (Proposition 7.2), and $\binom{J}{1}$, its marks outside J loops, for α_J. The width c_g of Z_B in $\mathcal{N}_g$ is at most c, and $c \geq 1$.

(ii) With d_g the floor of Z_B in $\mathcal{N}_g$ and the Skeleton values $\xi_{g,1}, \dots, \xi_{g,c_g}$ of $K_{\mathcal{N}_g}$ along Z_B at (u, u') (Definition 12.10),

$$g(\rho u, u') = \rho^{d_g}\, K_{\mathcal{N}_g|Z_B}(u)\, K_{\mathcal{N}_g/Z_B}(u') \prod_{i=1}^{c_g} \big(\rho + \xi_{g,i}(u,u')\big),$$

the leading coefficient being the top level of Z_B (Theorem 12.19); these Skeleton values, each with its letter, form the joint spectrum of Z_B at (u, u') (Definition 12.14).

(iii) For every regular form η of degree $|Z|-1$ on $V_{Z;\mathfrak{f}}$ there are finitely many terms t, with rational functions A_t on Δ_{Z_B} and B_t on Δ_{Z_R}, integers ν_t and letter exponents $k_{t,g} \in \mathbb{N}$, such that

$$\int_{\widetilde{\sigma}_Z} \eta = \int_{\Delta_{Z_B} \times \Delta_{Z_R}} \sum_t A_t(u)\, B_t(u')\, \mathrm{Sc}_t(u,u')\, d\omega_{Z_B}(u)\, d\omega_{Z_R}(u'),$$

the integrand absolutely integrable. Here $\mathrm{Sc}_t = 0$ when no letter with $k_{t,g} > 0$ has a Skeleton value, and otherwise

$$\begin{aligned} \mathrm{Sc}_t &= \mathrm{Reg} \int_0^\infty \rho^{\nu_t - 1} \prod_{g,i} \big(\rho + \xi_{g,i}\big)^{-k_{t,g}}\, d\rho \\ &= \operatorname*{CT}_{\kappa=\nu_t} \Gamma_{|J_t|}^{\,2} \begin{bmatrix} \kappa, & |\mathbf{w}_t| - \kappa \\ & \mathbf{w}_t \end{bmatrix} \mathsf{B}^{\{J_t\}}_{\mathbf{w}_t, |\mathbf{w}_t|-\kappa;\, I_{J_t}}\big(\boldsymbol{\xi}(u,u')\big), \end{aligned}$$

J_t indexing the Skeleton values $\xi_{g,i}$ of the letters with $k_{t,g} > 0$, each with the letter exponent $w_{t,(g,i)} = k_{t,g}$, so that $|\mathbf{w}_t| = \sum_g k_{t,g} c_g$: the finite part in ρ (Definition 1.5) is the constant term at ν_t of the stacked Gamma times the Breta of one bunch of Proposition 15.1, continued from the strip $0 < \mathrm{Re}\,\kappa < |\mathbf{w}_t|$. This is the node form *of η at v: the bunch and the room meet only through the joint spectrum, as the coefficients of Bretas of one bunch at the unit configuration.*

*(iv) (*Width one.*) If $c = 1$, every letter has at most one Skeleton value, the room's hinge value over the bunch's, $\xi_g = \gamma_R^g(u')/\gamma_B^g(u)$ with $\gamma_B^g = K_{\mathcal{N}_g|Z_B}/K_{\mathcal{N}_g/Z_R}$ and $\gamma_R^g = K_{\mathcal{N}_g|Z_R}/K_{\mathcal{N}_g/Z_B}$ (Corollary 19.5), positive. For κ in the strip, the share side (Corollary 15.5) writes the stacked Gamma times the Breta of the term t as an integral over $|J_t| - 1$ shares of stacked Gammas times powers $\xi_g^{-b_g} = \gamma_R^g(u')^{-b_g} \gamma_B^g(u)^{\,b_g}$, the b_g affine in the shares and in κ, with real parts constant along the contour. On every such contour on which these stacked Gammas have arguments with positive real parts and*

$$\int_{\Delta_{Z_B}} |A_t| \prod_g (\gamma_B^g)^{\mathrm{Re}\, b_g}\, d\omega_{Z_B} < \infty, \qquad \int_{\Delta_{Z_R}} |B_t| \prod_g (\gamma_R^g)^{-\mathrm{Re}\, b_g}\, d\omega_{Z_R} < \infty,$$

the integral over $\Delta_{Z_B} \times \Delta_{Z_R}$ of $A_t B_t$ times the stacked Gamma times the Breta is the share integral of the stacked Gammas times $\int_{\Delta_{Z_B}} A_t \prod_g (\gamma_B^g)^{b_g} d\omega_{Z_B}$ times $\int_{\Delta_{Z_R}} B_t \prod_g (\gamma_R^g)^{-b_g} d\omega_{Z_R}$: the bunch and the room separate. When $0 < \nu_t < |\mathbf{w}_t|$, Sc_t is the convergent scale integral at $\kappa = \nu_t$, and the term itself separates.

(v) If the root of $\mathcal{T}$ is a node that is not a leaf, the period of every element of $L(H_F, \widetilde{\sigma}_{\mathcal{E}})$, every left factor of the coaction of P_F^{mot} among them, is given by the node form at the root.

(vi) At a split vertex and a node step Z of it that meets the child M_1, the same holds with $Z_B = Z \cap M_1$, $Z_R = Z \setminus M_1$ and $c = 1$: the letters other than the linear letters have width zero at Z_B, and only the α_J with J meeting Z_B and Z_R have a Skeleton value.

At width two or more the Skeleton values are the negatives of the roots of Skeleton polynomials of degree c_g, whose extreme coefficients are products of a bunch letter and a room letter and whose middle coefficients are sums over the patterns of their levels (Theorem 12.19); the node form reads them as the coefficients of a Breta of one bunch at any width, while hinge values are defined at width one only (Definition 19.2). For the period itself, $\eta = \omega_F$ at the root, the glue reads the bunch and the room through their two laws (Theorem 28.10(ii)), opened with one share per station (Theorem 28.10(iii)) or with one more pin for every direction a side does not see (Theorem 28.12); at width one in one table this is Theorem 19.13, and with one Skeleton value Corollary 19.14. Theorem 34.16 extends the node form from the period to every form on every node step, and so to every left factor; width one enters it only in (iv).

34.6 Pinned cells

Theorem 34.17 (Pinned cells split off motivically)**.** *Let $F = ((\mathcal{B}_j; s_j, t_j))_{j=1}^N$ be a convergent family (Definition 34.1) on $\mathcal{E} = L \sqcup R_1 \sqcup \cdots \sqcup R_q$, $q \geq 1$, $|R_i| \geq 2$, whose active tables are substituted table by table (Definition 21.1): $\mathcal{B}_j = \mathcal{H}_j\langle \mathcal{A}_{1j}, \ldots, \mathcal{A}_{qj} \rangle$, the marks of L left in place, with the cores $\mathcal{H}_j$, lists on $L \sqcup \{h_1, \ldots, h_q\}$, and the cells $\mathcal{A}_{ij}$, lists on $R_i \sqcup \{g_i\}$; here and below j runs over the active tables, the others carrying no letter. Let the active letters read every cell through one hinge value: for every i there is a function γ_i on Δ_{R_i} with $\gamma^K_{\mathcal{A}_{ij}} = \gamma_i$ whenever $s_j \neq 0$ and $\gamma^\Psi_{\mathcal{A}_{ij}} = \gamma_i$ whenever $t_j \neq 0$. Let $\kappa_i = \kappa_{R_i}$, the wall of the cell (Theorem 21.10), put $2\sigma = \sum_j (s_j + t_j)$, and fix integer loads a_{ij}, b_{ij} with $a_{ij} = 0$ when $s_j = 0$, $b_{ij} = 0$ when $t_j = 0$ and $\sum_j (a_{ij} + b_{ij}) = \kappa_i$.*

(i) $1 \leq \kappa_i \leq 2\sigma - 1$. The core family *$F_{\mathcal{H}} = ((\mathcal{H}_j; s_j, t_j))_j$ on $L \sqcup \{h_1, \ldots, h_q\}$, with the masses $\mathbf{m}|_L$ and κ_i at h_i, is convergent, and so is, for every i, the* pinned family *G_i on R_i, the load family at zero imbalance of Corollary 22.7: the tables $\mathcal{A}_{ij} \setminus g_i$ at $(a_{ij}, t_j - b_{ij})$ and $\mathcal{A}_{ij}/g_i$ at $(s_j - a_{ij}, b_{ij})$, with the masses $\mathbf{m}|_{R_i}$.*

(ii) The motivic period factors through the core:

$$P_F^{\mathrm{mot}} = P_{F_{\mathcal{H}}}^{\mathrm{mot}} \prod_{i=1}^{q} P_{G_i}^{\mathrm{mot}}, \qquad \mathrm{C}(P_F^{\mathrm{mot}}) \subseteq \mathrm{C}(P_{F_{\mathcal{H}}}^{\mathrm{mot}}) \cdot \prod_{i=1}^{q} \mathrm{C}(P_{G_i}^{\mathrm{mot}}).$$

(iii) The cell family *$F_i = ((\mathcal{A}_{ij}; s_j, t_j))_j$ on $R_i \sqcup \{g_i\}$, with the masses $\mathbf{m}|_{R_i}$ and $2\sigma - \kappa_i$ at g_i, is convergent, and*

$$P_{F_i}^{\mathrm{mot}} = \Gamma_{\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}} \begin{bmatrix} \kappa_i, & 2\sigma - \kappa_i \\ & 2\sigma \end{bmatrix} P_{G_i}^{\mathrm{mot}},$$

the Beta corner a positive rational number.

(iv) The form, the period and the motivic period of G_i do not depend on the loads: its form is $\gamma_i^{-\kappa_i}\alpha^{\mathbf{m}|_{R_i}}\prod_j K_{\mathcal{A}_{ij}/g_i}^{-s_j}\Psi_{\mathcal{A}_{ij}\setminus g_i}^{-t_j}\,\Omega_{R_i}/\alpha^{R_i}$, and its period is the pinned shape integral of the cell (Theorem 22.5).

(v) Every $S \subseteq L$ has in $\mathcal{B}_j$ the width it has in $\mathcal{H}_j$ (Corollary 21.11): a bunch among the marks left in place keeps the width it has in the core, and no condition is placed on it.

One hinge value confines the set it reads to width at most one (Lemma 19.6), and the marks of a cell always form a set of width one (Proposition 21.3(i)); so it is the cells that have width one, and a bunch among the marks left in place has any width up to the number of the other marks and cells (Corollary 21.11). With q cells the letters read the room through q hinge values, one for each cell.

Corollary 34.18 (Dual pairs and full leans)**.** *Let $F = (\mathcal{B}; s, t)$ be a convergent family of one list (Definition 34.1).*

*(a) (*Dual pairs.*) Let $\mathcal{B} = \mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ as in Theorem 34.17, with every cell a dual pair (Definition 21.1). Then every cell is read through one hinge value at all integers s, t; the pinned family of the cell i has the form of the family $(\mathcal{B}|R_i;\ \kappa_i,\ s+t-\kappa_i)$ at the masses $\mathbf{m}|_{R_i}$, with $\kappa_i = m_{R_i} - (s+t)(r(R_i)-1)$; and*

$$P_F^{\mathrm{mot}} = P_{(\mathcal{H};s,t)}^{\mathrm{mot}} \prod_{i=1}^{q} P_{(\mathcal{B}|R_i;\,\kappa_i,\,s+t-\kappa_i)}^{\mathrm{mot}},$$

the core at the masses $\mathbf{m}|_L$ and κ_i at h_i.

*(b) (*Full leans.*) Let $t = 0$ or $s = 0$. Split $\mathcal{B}$ at a set of width one with at least two marks on each side into the substitution of its two cells (Proposition 21.3(iii)), and repeat at the nodes so obtained until no node has such a set. The nodes on at least four marks are then prime (Theorem 23.16), and those on two or three marks are the lists $\binom{S}{1}$ and $\binom{S}{|S|-1}$. Give each prime node $\mathcal{H}_v$ the family $F_v = (\mathcal{H}_v; s, t)$ with the masses of its marks and, at each marker, the wall of F at the marks on the other side of the split that made it. Every F_v is convergent, and*

$$P_F^{\mathrm{mot}} = c\prod_v P_{F_v}^{\mathrm{mot}}, \qquad c \in \mathbb{Q}_{>0};$$

so $\mathrm{C}(P_F^{\mathrm{mot}}) \subseteq \prod_v \mathrm{C}(P_{F_v}^{\mathrm{mot}})$, and a list without prime nodes has a rational motivic period at its full leans.

Part (b) is the motivic form of Corollary 26.11 and Theorem 26.5: the series and parallel nodes contribute stacked Gammas at integer arguments (Proposition 26.9), which are rational, and their motivic periods are these rationals. At a full lean the width of a bunch plays no part, and the two full leans of one list are exchanged by Cremona, as motivic periods (Corollary 34.8). In a prime list no set with two marks on each side splits off (Corollary 28.26(ii)), so the splitting of (b) ends at the prime nodes; their only cells are the complements of single marks, which at a full lean split off by one Beta corner and, with both letters active, exactly where the deletion and the contraction have one even letter (Corollary 28.26(iii),(iv)).

Theorem 34.19 (A pinned set splits off motivically)**.** *Let $F = ((\mathcal{B}_j; s_j, t_j))_j$ be a convergent family (Definition 34.1) with $s_j, t_j \geq 0$, and let S, $|S| \geq 2$, be pinned (Theorem 28.25) with width one in every active table, $S' = \mathcal{E} \setminus S$. Every active table is then the substitution of its cell of the bunch $\mathcal{B}_{j,S}^{+}$ into its cell of the room $\mathcal{B}_{j,S'}^{+}$ (Theorem 19.4, Proposition 21.3(iii)), the marks of S' left in*

place, and the active letters read the cell through the one hinge value $\gamma = \gamma^{K}_{j,S} = \gamma^{\Psi}_{j,S}$. *The core family of Theorem 34.17 is the room with its stand-in at* $\mathbf{1}$, $S'^{(\mathbf{1})}$ *(Theorem 28.10(i)), the family* $((\mathcal{B}^{+}_{j,S'}; s_j, t_j))_j$ *of the active tables with the masses* $(\mathbf{m}|_{S'}, \kappa_S)$*; the period of its pinned family* G *is* $c = \int_{\Delta_S} \phi^{\rho}_{S}$*, the total mass of the law of* S*; and*

$$P^{\mathrm{mot}}_{F} = P^{\mathrm{mot}}_{S'^{(\mathbf{1})}}\, P^{\mathrm{mot}}_{G}, \qquad \mathrm{C}(P^{\mathrm{mot}}_{F}) \subseteq \mathrm{C}(P^{\mathrm{mot}}_{S'^{(\mathbf{1})}}) \cdot \mathrm{C}(P^{\mathrm{mot}}_{G}) :$$

the identity $P_F = c\, P_{S'^{(\mathbf{1})}}$ *of Theorem 28.25 is an identity of motivic periods.*

Every period of a family glued on a set is the integral, against the law of the set, of the period of the partner with a stand-in for it (Theorem 28.10(i)). Where the law is a point the integral is one evaluation (Theorem 28.25); where the set is moreover a cell, the change of variables behind the evaluation is birational and the evaluation is an identity of motivic periods. No other set splits off, and a set of width two or more enters every period glued on it through a law that is not a point (Corollary 28.26(i)); at every node, of every width, the open periods of node steps have the node form of Theorem 34.16.

Chapter 35

Positive words

The output of the theory is a value: a period, a Breta, a closing of fibre integrals, and such values are read arithmetically. A closing at rational data writes a value as a rational combination of Trailmoments of lattice inputs and rational heads (Proposition 5.13, Lemma 38.25); a sequence of such values that do not vanish, cleared by integers that grow more slowly than the values decay, shows that some Trailmoment it carries is irrational (Theorem 37.2, Chapter 44). What keeps a value from vanishing is positivity, and positive words are made in four ways: second jets of a period in its weights, even powers of the contrast of two bases, Beta ports on the pairs of a manufactured family, and Beta ports read against one mark with the room integrated. The jets and the contrasts are combinations of periods of the same family at shifted data on its balance wall, bounded by the period times a factor that their statements control; the port words of a manufactured family are combinations of its interests at shifted masses over one normalization, and the reads of the Beta ports are bounded by the read of the constant insertion.

35.1 Coefficient jets and basis contrasts

Theorem 35.1 (Positive coefficient jets)**.** *Let F have real exponents $s_j, t_j \geq 0$ and real masses on its balance wall, every proper wall positive. For an integer $n \geq 1$, the* depth*, the data $(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$ are again on the balance wall, with walls $n\kappa_S$; in this statement $\mathbb{E}$ is the expectation of $U \sim \mathrm{Breta}^{\circ}(F; n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$. Choose* weight directions $\mathbf{b}_{j,T} \in \mathbb{Q}^k$*, one for each table j and basis $T \in \mathcal{B}_j$, and put*

$$\partial_{\mathbf{b}} K_j = \sum_{T \in \mathcal{B}_j} \mathbf{b}_{j,T}\, \alpha^{T}, \qquad \partial_{\mathbf{b}} \Psi_j = \sum_{T \in \mathcal{B}_j} \mathbf{b}_{j,T}\, \alpha^{\mathcal{E} \setminus T},$$

the derivatives of the letters of table j along the weights in these directions.

(i) For $v \in \mathbb{R}^k$ and small real ε, replace every table j by the two tables $(\mathcal{B}_j, \theta_j \pm \varepsilon\, v^{\mathsf{T}} \mathbf{b}_j)$, each with the exponents $(ns_j/2, nt_j/2)$. The period of this doubled family *at the masses $n\mathbf{m}$ is*

$$P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})\ \mathbb{E}\Big[\Big(\prod_j \Big(1 - \varepsilon^2 \Big(\frac{v^{\mathsf{T}} \partial_{\mathbf{b}} K_j}{K_j}\Big)^2\Big)^{-\frac{ns_j}{2}} \Big(1 - \varepsilon^2 \Big(\frac{v^{\mathsf{T}} \partial_{\mathbf{b}} \Psi_j}{\Psi_j}\Big)^2\Big)^{-\frac{nt_j}{2}}\Big)(U)\Big],$$

whose coefficient of ε^2 is $v^{\mathsf{T}} \mathbf{H}_n v$, with the jet matrix

$$\mathbf{H}_n = \frac{n}{2}\, P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})\ \mathbb{E}\Big[\sum_j \Big(s_j\, \frac{\partial_{\mathbf{b}} K_j\, \partial_{\mathbf{b}} K_j^{\mathsf{T}}}{K_j^2} + t_j\, \frac{\partial_{\mathbf{b}} \Psi_j\, \partial_{\mathbf{b}} \Psi_j^{\mathsf{T}}}{\Psi_j^2}\Big)(U)\Big].$$

$\mathbf{H}_n$ *is positive semidefinite, and positive definite when the directions* $\mathbf{b}_{j,T}$ *of the active tables span* $\mathbb{R}^k$.

(ii) For a rational positive semidefinite $k\times k$ *matrix* Q, $\operatorname{tr}(Q\mathbf{H}_n) > 0$ *exactly when* $\mathbf{b}_{j,T}^{\mathsf{T}} Q\, \mathbf{b}_{j,T} > 0$ *for some active table* j *and basis* T. *Completing squares writes* $Q = \sum_l c_l v_l v_l^{\mathsf{T}}$ *with rational* $c_l > 0$ *and rational* v_l, *and so writes* $\operatorname{tr}(Q\mathbf{H}_n) = \sum_l c_l\, v_l^{\mathsf{T}} \mathbf{H}_n v_l$ *as a sum of directional second jets with positive rational coefficients.*

(iii) In periods of F *(Theorem 27.9 at order two, the two copies of each table merged),*

$$\operatorname{tr}(Q\mathbf{H}_n) = \frac{n}{2} \sum_j \sum_{T,T'\in\mathcal{B}_j} \mathbf{b}_{j,T}^{\mathsf{T}} Q\, \mathbf{b}_{j,T'} \Big[s_j\, P_F\big(n\mathbf{m} + \mathbf{1}_T + \mathbf{1}_{T'};\, n\mathbf{s} + 2\mathbf{e}_j,\, n\mathbf{t}\big) + t_j\, P_F\big(n\mathbf{m} + \mathbf{1}_{\mathcal{E}\setminus T} + \mathbf{1}_{\mathcal{E}\setminus T'};\, n\mathbf{s},\, n\mathbf{t} + 2\mathbf{e}_j\big)\Big] :$$

every term stays on its balance wall, a basis insertion changes the wall of S *by* $|T\cap S| + |T'\cap S| - 2d_j(S) \geq 0$, *and a cobasis insertion by* $|S\setminus T| + |S\setminus T'| - 2d_j^*(S) \geq 0$.

(iv) Whenever (ii) gives positivity, $0 < \operatorname{tr}(Q\mathbf{H}_n) \leq n\, C(Q)\, P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, *with* $C(Q) = \frac{1}{2}\sum_j (s_j + t_j) \max_T \mathbf{b}_{j,T}^{\mathsf{T}} Q\, \mathbf{b}_{j,T}/\theta_{j,T}^2$.

(v) With the basis weights and the cobasis weights of every table taken as independent variables, the part of $\mathbf{H}_n$ *carried by* K_j *is the Hessian of* $P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$ *in the basis weights of table* j, *paired with the directions* $\mathbf{b}_{j,T}$, *divided by* $2(ns_j + 1)$, *and the part carried by* Ψ_j *is the same in the cobasis weights divided by* $2(nt_j + 1)$. *On the share side of the period (Corollary 18.16) a weight derivative multiplies the integrand by minus the share divided by its weight, so every pin and every candidate pole hyperplane stays fixed.*

(vi) At the centre, $s_j = t_j$ *for every* j *and* $\mathbf{m} = \sigma\mathbf{1}$, *the doubled family is again at its centre, and the table's gauge* $\mathbf{1}$ *stays native (Proposition 8.21).*

A weight direction moves the two copies of a table apart symmetrically, so the period moves at second order only, by the expectation of a square under the Breta shape law: a jet of the period in its weights is a positive word. Under the basis law $\partial_{\mathbf{b}} K_j/K_j$ is the mean of the directions $\mathbf{b}_{j,T}/\theta_{j,T}$, $\mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[\mathbf{b}_{j,T}/\theta_{j,T}]$, so the jet is the second moment, under the shape law, of the mean weight direction that the bases drawn at the shape carry. Table by table it is the period's own Hessian in the weights divided by $2(ns_j + 1)$ or $2(nt_j + 1)$, it is a finite combination of periods of the same family on the balance wall at data whose walls are no smaller, and it costs at most a factor linear in the depth against the period.

Lemma 35.2 (Rounding inside a rational subspace)**.** *Let* V *be the subspace of the real symmetric* $k\times k$ *matrices on which finitely many linear functionals* $Q \mapsto \operatorname{tr}(MQ)$, M *rational, vanish, spanned over* $\mathbb{R}$ *by integer symmetric matrices* $G_1, \dots, G_r$, *and let* $Q_* = \sum_j u_j G_j \in V$ *with* $Q_* \succeq \varepsilon I$, $\varepsilon > 0$. *For an integer* $\ell \geq 1$ *put* $Q = \sum_j \operatorname{round}(\ell u_j)\, G_j$ *and* $\beta = \frac{1}{2}\sum_j \big(\operatorname{tr} G_j^2\big)^{1/2}$. *Then* Q *is an integer matrix in* V *with* $Q \succeq (\ell\varepsilon - \beta)I$ *and* $\operatorname{tr} Q \leq \ell \operatorname{tr} Q_* + k\beta$; *for* $\ell > \beta/\varepsilon$ *it is positive definite.*

The jet word $\operatorname{tr}(Q\mathbf{H}_n)$ is linear in Q, so conditions on its coefficients of Trailmoments, such as the vanishing of chosen ones, define a rational subspace of matrices Q. A positive definite matrix in that subspace rounds to an integer one in the same subspace, and the jet word keeps its positivity and the vanishing of the chosen coefficients together at integer data.

Theorem 35.3 (Two bases generate positive contrast words of any depth)**.** *Let F have real data $(\mathbf{m};\mathbf{s},\mathbf{t})$ on its balance wall with every proper wall positive, with the pair coordinates $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ (Definition 8.1) and the centred potential Φ° (Theorem 8.19). Let $T \neq T'$ be two bases of a table j, and for integers $n, h \geq 1$ let the* contrast word *be*

$$\frac{1}{2}\,P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})\;\mathbb{E}_{U\sim\mathrm{Breta}^\circ(F;n\mathbf{m};n\mathbf{s},n\mathbf{t})}\Big[\big(\mathbb{P}_{j,U}(T)-\mathbb{P}_{j,U}(T')\big)^{2h}+\big(\mathbb{P}^*_{j,U}(T)-\mathbb{P}^*_{j,U}(T')\big)^{2h}\Big]>0;$$

the two contrasts have modulus below one on the open orthant.

(i) The first half of the contrast word is

$$\frac{1}{2}\sum_{l=0}^{2h}(-1)^l\binom{2h}{l}\theta_{j,T}^{2h-l}\theta_{j,T'}^{l}\,P_F\big(n\mathbf{m}+(2h-l)\mathbf{1}_T+l\mathbf{1}_{T'};\,n\mathbf{s}+2h\mathbf{e}_j,\,n\mathbf{t}\big),$$

and the second is the same with $\mathbf{1}_{\mathcal{E}\setminus T}$, $\mathbf{1}_{\mathcal{E}\setminus T'}$ in the masses and $2h\mathbf{e}_j$ added to the cobasis exponents; every term is balanced with its walls weakly raised. The inversion $\alpha\mapsto\alpha^{-1}$ carries the second half at $(\mathbf{m};\mathbf{s},\mathbf{t})$ to the first half at $(2\sigma\mathbf{1}-\mathbf{m};\mathbf{t},\mathbf{s})$, the data of Theorem 8.14; at the centre the two halves are equal.

(ii) For rational $\eta>0$ and $h=\eta n$, along the integers n with $\eta n\in\mathbb{N}$, $\frac{1}{n}\log$ of the contrast word tends to

$$-\inf_{z\in\mathbf{1}^\perp}\Big(\Phi^\circ(z)-2\langle\boldsymbol{\mu}_\perp,z\rangle-2\eta\log\max\Big(\big|\mathbb{P}_{j,e^{2z}}(T)-\mathbb{P}_{j,e^{2z}}(T')\big|,\;\big|\mathbb{P}^*_{j,e^{2z}}(T)-\mathbb{P}^*_{j,e^{2z}}(T')\big|\Big)\Big);$$

the infimum exceeds the minimum of $\Phi^\circ-2\langle\boldsymbol{\mu}_\perp,\cdot\rangle$ on $\mathbf{1}^\perp$ and tends to infinity with η; that minimum is $-\lim_n\frac{1}{n}\log P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$.

(iii) Let the exponents be nonnegative, and write $q=2h$ and $H_q(x,y)=\sum_{l=0}^{q}(-1)^l\binom{q}{l}(x)_{q-l}(y)_l$. When $s_j>0$, the first half is the share side of $P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$ (Corollary 18.16) with its integrand multiplied by $H_q(a_{j,T},a_{j,T'})/\big(2(ns_j)_q\big)$, a the basis shares; when $t_j>0$, the second half is the same with $H_q(b_{j,T},b_{j,T'})/\big(2(nt_j)_q\big)$, b the cobasis shares: polynomial insertions that keep every pin and every pole hyperplane.

Two bases of one table give a contrast of modulus below one, which stays away from one on the compact sets where the Breta shape law concentrates; so its even powers are positive words that become exponentially small against the period, at a rate that grows without bound with the ratio of the power to the depth. At the centre with unit weights the basis law gives every basis the same chance at the table's gauge, where the law concentrates, and the contrast vanishes there. On the share side the contrast is a polynomial in the shares, so the contrast words of every depth keep the fibre, the pins and the poles of the period.

35.2 Manufactured tables and their port words

Definition 35.4 (Manufactured tables)**.** Let *seed letters* on $\mathcal{E}$ be polynomials $L_i(x)=\sum_{P\in\operatorname{supp}L_i}\theta_{i,P}\,x^P$, $i\in I$, with supports $\operatorname{supp}L_i\subseteq 2^{\mathcal{E}}$, weights $\theta_{i,P}>0$ and exponents $w_i\geq 0$; a table $(\mathcal{B}_j,\theta_j)$ of a family with exponents (s_j,t_j) gives the two seed letters K_j, with exponent s_j, and Ψ_j, with exponent t_j. Choose $\beta_i\in(0,1)$ and replace every mark e by a pair $\{e^+,e^-\}$ of new marks, on $\mathcal{E}^\pm=\{e^+,e^-:e\in\mathcal{E}\}$. The polynomials

$$\widetilde{K}_i(\alpha)=L_i(\mathbf{1})\prod_{e\in\mathcal{E}}(\alpha_{e^+}+\alpha_{e^-})-\beta_i\sum_P\theta_{i,P}\prod_{e\in P}\alpha_{e^+}\prod_{e\notin P}(\alpha_{e^+}+\alpha_{e^-}),$$

$$\widetilde{\Psi}_i(\alpha) = L_i(\mathbf{1}) \prod_{e\in\mathcal{E}} (\alpha_{e^+} + \alpha_{e^-}) - \beta_i \sum_P \theta_{i,P} \prod_{e\in P} \alpha_{e^-} \prod_{e\notin P} (\alpha_{e^+} + \alpha_{e^-})$$

are the basis and cobasis polynomials of a table on $\mathcal{E}^{\pm}$ of rank $|\mathcal{E}|$, whose bases $U^+ \cup (\mathcal{E}\backslash U)^-$, $U \subseteq \mathcal{E}$, take one mark of each pair, with the weights $\widetilde{\theta}_{i,U} = L_i(\mathbf{1}) - \beta_i \sum_{P\subseteq U} \theta_{i,P} \geq L_i(\mathbf{1})(1-\beta_i) > 0$; the seed weights are recovered from $\big(L_i(\mathbf{1}) - \widetilde{\theta}_{i,U}\big)/\beta_i = \sum_{P\subseteq U} \theta_{i,P}$ by inversion over the subsets. The *manufactured family* $\widetilde{F}$ has these tables with the exponents $(w_i, 0)$; for masses $\mathbf{m}$ on $\mathcal{E}^{\pm}$ its *pair walls* are $\kappa_{\{e^+,e^-\}} = m_{e^+} + m_{e^-} - |\mathbf{w}|$, $|\mathbf{w}| = \sum_i w_i$, and its *pair shapes* are $x_e = \alpha_{e^+}/(\alpha_{e^+} + \alpha_{e^-})$. For orders $\mathbf{n} \in \mathbb{N}^{\mathcal{E}}$, positive masses and positive pair walls, its *port word* is

$$M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) = \frac{J_{\widetilde{F}}(\mathbf{m};\mathbf{w},\mathbf{0})}{\prod_e \Gamma\big(\kappa_{\{e^+,e^-\}}\big)}\ \mathbb{E}_{\alpha\sim \mathrm{Breta}(\widetilde{F};\mathbf{m};\mathbf{w},\mathbf{0})}\Big[\prod_{e\in\mathcal{E}} \Pi_{n_e}^{m_{e^+},m_{e^-}}(x_e)\Big],$$

and $M^*_{\widetilde{F},\mathbf{n}}(\mathbf{m})$ is the same at the exponents $(0, w_i)$, with $J_{\widetilde{F}}(\mathbf{m};\mathbf{0},\mathbf{w})$ and $\mathrm{Breta}(\widetilde{F};\mathbf{m};\mathbf{0},\mathbf{w})$. An *occupancy* is a family $\mathbf{o} = (o_{i,P})$ of integers $o_{i,P} \geq 0$, zero when $w_i = 0$, with the totals $|o_i| = \sum_P o_{i,P}$ and $|\mathbf{o}| = \sum_i |o_i|$, the *loads* $\mathrm{load}_e(\mathbf{o}) = \sum_i \sum_{P\ni e} o_{i,P}$, and the *coefficient*

$$\mathrm{cf}(\mathbf{o}) = \prod_i \Big[\frac{(w_i)_{|o_i|}}{L_i(\mathbf{1})^{w_i} \prod_P o_{i,P}!} \prod_P \Big(\frac{\beta_i\, \theta_{i,P}}{L_i(\mathbf{1})}\Big)^{o_{i,P}}\Big].$$

Theorem 35.5 (Interest of a manufactured family; the occupancy sum)**.** *In the pair scales and shapes,* $(\alpha_{e^+}, \alpha_{e^-}) = \rho_e(x_e, 1-x_e)$,

$$\widetilde{K}_i = \prod_e \rho_e\big(L_i(\mathbf{1}) - \beta_i L_i(x)\big), \qquad \widetilde{\Psi}_i = \prod_e \rho_e\big(L_i(\mathbf{1}) - \beta_i L_i(\mathbf{1}-x)\big).$$

Let the masses be positive and the pair walls positive.

(i) Every wall of $\widetilde{F}$ *at the exponents* $(w_i, 0)$ *is positive, and*

$$J_{\widetilde{F}}(\mathbf{m};\mathbf{w},\mathbf{0}) = \prod_e \Gamma\big(\kappa_{\{e^+,e^-\}}\big) \int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{m_{e^+}-1}(1-x_e)^{m_{e^-}-1} \prod_i \big(L_i(\mathbf{1}) - \beta_i L_i(x)\big)^{-w_i}\, dx,$$

the integral convergent at all positive masses; the simplex integral $P_{\widetilde{F}}(\mathbf{m};\mathbf{w},\mathbf{0})$ *is* $\Gamma_{1\,|\mathcal{E}|}\begin{bmatrix}(\kappa_{\{e^+,e^-\}})_e \\ \sum_e \kappa_{\{e^+,e^-\}}\end{bmatrix}$ *times the same integral. Under* $\mathrm{Breta}(\widetilde{F};\mathbf{m};\mathbf{w},\mathbf{0})$ *the pair scales are independent,* $\rho_e \sim \mathrm{Gamma}_{\kappa_{\{e^+,e^-\}}}$*, and independent of the pair shapes, whose density on the cube is proportional to the integrand. So* $M_{\widetilde{F},\mathbf{0}}(\mathbf{m})$ *is the integral displayed, and* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m})$ *is that integral with* $\prod_e \Pi_{n_e}^{m_{e^+},m_{e^-}}(x_e)$ *inserted; with* $L_i(\mathbf{1}-x)$ *in place of* $L_i(x)$ *the same holds for* $M^*_{\widetilde{F},\mathbf{n}}(\mathbf{m})$.

(ii) The integral is the series of nonnegative terms

$$M_{\widetilde{F},\mathbf{0}}(\mathbf{m}) = \sum_{\mathbf{o}} \mathrm{cf}(\mathbf{o}) \prod_e \Gamma_{2\,1}\begin{bmatrix} m_{e^+} + \mathrm{load}_e(\mathbf{o}),\ m_{e^-} \\ m_{e^+} + m_{e^-} + \mathrm{load}_e(\mathbf{o})\end{bmatrix}$$

over the occupancies.

Theorem 35.6 (The port word of a manufactured family)**.** *Let the masses be positive, the pair walls positive and* $\mathbf{n} \in \mathbb{N}^{\mathcal{E}}$*. Then*

$$M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) = \sum_{\mathbf{o}} \mathrm{cf}(\mathbf{o}) \prod_{e} \binom{\mathrm{load}_e(\mathbf{o})}{n_e} \Gamma_{1}^{2}\begin{bmatrix} m_{e^+} + \mathrm{load}_e(\mathbf{o}), & m_{e^-} + n_e \\ m_{e^+} + m_{e^-} + \mathrm{load}_e(\mathbf{o}) + n_e \end{bmatrix}$$
$$= \frac{1}{\prod_e n_e!} \int_{[0,1]^{\mathcal{E}}} \prod_{e} x_e^{m_{e^+}+n_e-1}(1-x_e)^{m_{e^-}+n_e-1}\, \partial_x^{\mathbf{n}} \prod_{i} \big(L_i(\mathbf{1}) - \beta_i L_i(x)\big)^{-w_i}\, dx.$$

Every occupancy with $\mathrm{load}_e(\mathbf{o}) < n_e$ *for some* e *is annihilated, every surviving term is positive, every mixed derivative of* $\prod_i (L_i(\mathbf{1}) - \beta_i L_i(x))^{-w_i}$ *is nonnegative on the cube, and* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) > 0$ *exactly when every mark* e *with* $n_e > 0$ *lies in a member of the support of a seed letter with* $w_i > 0$*. At integer masses the Beta ports have integer coefficients, and* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m})$ *is an integer combination of the* $M_{\widetilde{F},\mathbf{0}}$ *at the masses shifted by* $(k_e, n_e - k_e)$ *on the pair of* e*,* $0 \le k_e \le n_e$*: of the interests of* $\widetilde{F}$ *at these masses, all of pair walls* $\kappa_{\{e^+,e^-\}} + n_e$*, over the one normalization* $\prod_e \Gamma(\kappa_{\{e^+,e^-\}} + n_e)$*.*

Theorem 35.7 (Suppression selected by incidence)**.** *In Theorem 35.6 write* $\mathrm{occ}(\mathbf{n}) = \min\{|\mathbf{o}| : \mathrm{load}_e(\mathbf{o}) \ge n_e$ *for every* $e\}$ *over the occupancies, and* $\beta = \max_i \beta_i < 1$ *over the seed letters with* $w_i > 0$*. Then*

$$0 \le M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) \le \prod_{e} \Big[\Gamma_{1}^{2}\begin{bmatrix} m_{e^+}, & m_{e^-} \\ m_{e^+} + m_{e^-} \end{bmatrix} \frac{(m_{e^-})_{n_e}}{n_e!}\Big] \prod_{i} L_i(\mathbf{1})^{-w_i} \sum_{l \ge \mathrm{occ}(\mathbf{n})} \frac{(|\mathbf{w}|)_l}{l!}\, \beta^l.$$

For an integer demand $\mathbf{d} \in \mathbb{N}^{\mathcal{E}}$ *let* $\mathrm{fc}(\mathbf{d}) = \min\big\{ \sum_{i,P} \lambda_{i,P} : \ \lambda \ge 0,\ \sum_{i,P} \lambda_{i,P} \mathbf{1}_P \ge \mathbf{d}\big\}$*, the* fractional cover *of the demand by the members of the supports of the seed letters with* $w_i > 0$*. Then* $\mathrm{occ}(q\mathbf{d}) = q\,\mathrm{fc}(\mathbf{d}) + O(1)$ *as* $q \to \infty$*, and* $\limsup_q \frac{1}{q} \log M_{\widetilde{F},q\mathbf{d}}(\mathbf{m}) \le \mathrm{fc}(\mathbf{d}) \log \beta < 0$ *for a nonzero demand that the supports cover. When the seed letters are the letters* K_j, Ψ_j *of the tables of a family, every surviving occupancy satisfies* $\sum_j \big(r_j(S)\,|o_{K_j}| + (|S| - d_j(S))\,|o_{\Psi_j}|\big) \ge \sum_{e \in S} n_e$ *for every* $S \subseteq \mathcal{E}$*,* $|o_{K_j}|$ *and* $|o_{\Psi_j}|$ *the total occupancies of the two letters of table* j*.*

Theorem 35.8 (Reflection of the port words of a manufactured family)**.** *The swap* $\alpha_{e^+} \leftrightarrow \alpha_{e^-}$ *on every pair,* $x_e \leftrightarrow 1 - x_e$*, swaps* $\widetilde{K}_i$ *with* $\widetilde{\Psi}_i$ *and* m_{e^+} *with* m_{e^-}*, and keeps every seed weight and every* β_i*: with* $\mathbf{m}'$ *the masses swapped on every pair,* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) = (-1)^{|\mathbf{n}|} M^{*}_{\widetilde{F},\mathbf{n}}(\mathbf{m}')$*.*

A manufactured table turns any positive polynomials, the two letters of every table of a family among them, into a table whose bases are the transversals of the pairs. On the pair shapes its letter is the constant $L_i(\mathbf{1})$ less a copy of the seed scaled by β_i, so the port word is a series of nonnegative terms indexed by occupancies of the members of the seed, and a Beta port of order n_e on a pair keeps only the occupancies that load the mark at least n_e times. The port words are therefore positive exactly when the incidence of the supports can meet the demand, and they are suppressed at the exponential rate of the cheapest fractional cover of the demand: the incidence structure of the seed decides which port words survive and how small they are.

35.3 Mark relations with the room integrated

Definition 35.9 (Mark reads)**.** Select a mark with coordinate x, and let the coordinates α_R of its room range over a measure space with a density $\Omega(\alpha_R) \ge 0$. The letters (Section 3.2) in the selected mark are polynomials $L_i(x, \alpha_R)$ in x, $i = 1, \dots, h$, positive where x ranges, with exponents w_i; they are *linear* when $L_i = c_i(\alpha_R) + d_i(\alpha_R)\, x$ with $c_i > 0$ and $d_i \ge 0$, and a linear letter *sees*

the mark when $w_i d_i > 0$ on a set of positive measure. For $a, b > 0$ the *mark read* of an insertion $g(x, \alpha_R)$ on the box is

$$\langle g \rangle_{a,b;\mathbf{w}} = \int\!\!\int_0^1 g\, x^{a-1}(1-x)^{b-1} \prod_i L_i^{-w_i}\, dx\; \Omega(\alpha_R)\, d\alpha_R,$$

and for $a, \lambda > 0$ its read on the half-line is

$$\langle g \rangle^{\lambda}_{a;\mathbf{w}} = \int\!\!\int_0^\infty g\, x^{a-1} e^{-\lambda x} \prod_i L_i^{-w_i}\, dx\; \Omega(\alpha_R)\, d\alpha_R,$$

wherever these converge; at a fixed point of the room, $\Omega(\alpha_R)\, d\alpha_R$ is the Dirac measure there. A sequence of reads at the data $a_n + j$, b_n, $\mathbf{w}(n)$ and Ω_n keeps the depth n (Theorem 35.1) apart from the *moment index* j of the selected mark.

Theorem 35.10 (The mark relation)**.** *Let the letters be polynomial in the selected mark, with exponents w_i, and let $A(x, \alpha_R)$ be a polynomial in x with coefficients depending on the room.*

(a) On the box, for $a, b > 0$:

$$\Big\langle x(1-x)\,\partial_x A + \big(a - (a+b)x\big) A - x(1-x)\, A \sum_i w_i\, \partial_x \log L_i \Big\rangle_{a,b;\mathbf{w}} = 0.$$

(b) On the half-line, for $a, \lambda > 0$, whenever the reads converge:

$$\Big\langle x\,\partial_x A + (a - \lambda x) A - x\, A \sum_i w_i\, \partial_x \log L_i \Big\rangle^{\lambda}_{a;\mathbf{w}} = 0.$$

(c) On a restricted interval the difference of the flux at its two ends is retained, and a kernel coupled to x contributes its product-rule term.

Multiplying by $\prod_i L_i$ gives a polynomial numerator; on a box of several marks the relation holds in each coordinate with the others as room.

Theorem 35.11 (The moments of a selected mark close in a finite span)**.** *On a family select a mark e, $x = \alpha_e$, with room $R = \mathcal{E} - e$; splitting the bases and the cobases by e gives the two coefficients of every letter in x. Pull the letters free of x into the room, absorb pure powers of x into its mass, and write the remaining letters $L_i = c_i(\alpha_R) + d_i(\alpha_R)\, x$, $c_i, d_i > 0$, $i = 1, \dots, h$, with exponents w_i; read them on the half-line at a fixed point of the room (Definition 35.9), with $m, \lambda > 0$. With $\prod_i L_i = \sum_{k=0}^{h} p_k x^k$, $p_{-1} = 0$, $q_0 = 0$ and $q_k = \sum_i w_i d_i\, [x^{k-1}] \prod_{i' \neq i} L_{i'}$,*

$$\lambda p_h\, \langle x^{n+h+1} \rangle^{\lambda}_{m;\mathbf{w}} = \sum_{k=0}^{h} \big[(m+n+k)\, p_k - q_k - \lambda p_{k-1}\big]\, \langle x^{n+k} \rangle^{\lambda}_{m;\mathbf{w}},$$

a recurrence whose coefficients are polynomials in the room coordinates, with leading coefficient λp_h, $p_h = \prod_i d_i > 0$. For a mark f of the room,

$$\partial_{\alpha_f} \langle x^n \rangle^{\lambda}_{m;\mathbf{w}} = -\sum_i w_i \Big[(\partial_{\alpha_f} d_i)\, \langle x^{n+1} \rangle^{\lambda}_{m;\mathbf{w}+\mathbf{e}_i} + (\partial_{\alpha_f} c_i)\, \langle x^n \rangle^{\lambda}_{m;\mathbf{w}+\mathbf{e}_i} \Big],$$

and the moments with at most ℓ such derivatives lie in the span, over rational functions of the room, of $\{\langle x^k \rangle^{\lambda}_{m;\mathbf{w}+v} : 0 \le k \le h,\ v \in \mathbb{N}^h,\ |v| \le \ell\}$: at most $(h+1)\binom{h+\ell}{\ell}$ states.

Theorem 35.12 (Positive reads of the Beta ports)**.** *Let the letters be linear, $L_i = c_i + d_i x$, with exponents $w_i \geq 0$, let $a, b > 0$ and $q \in \mathbb{N}$, and let J be a set of letters, with $w_J = \sum_{i\in J} w_i$.*

(i) The Beta port of order q reads (Theorem 3.15(iii), integrated over the room)

$$\big\langle (-1)^q \Pi_q^{a,b} \big\rangle_{a,b;\mathbf{w}} = \sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!} \Big\langle \big(x(1-x)\big)^q \prod_i \Big(\frac{d_i}{L_i}\Big)^{v_i} \Big\rangle_{a,b;\mathbf{w}} \in \Big[0, \frac{(|\mathbf{w}|)_q}{q!} \langle 1\rangle_{a,b;\mathbf{w}}\Big].$$

(ii) The part of this sum over the v supported in J lies in $\big[0, \frac{(w_J)_q}{q!}\langle 1\rangle_{a,b;\mathbf{w}}\big]$, and for $q > 0$ it is positive when some letter of J sees the mark.

(iii) At $q = 1$,

$$a\,\langle 1\rangle_{a,b;\mathbf{w}} - (a+b)\,\langle x\rangle_{a,b;\mathbf{w}} - \sum_{i\notin J} w_i \Big\langle \frac{x(1-x)\,d_i}{L_i} \Big\rangle_{a,b;\mathbf{w}} = \sum_{i\in J} w_i \Big\langle \frac{x(1-x)\,d_i}{L_i} \Big\rangle_{a,b;\mathbf{w}} \in \big[0,\, w_J \langle 1\rangle_{a,b;\mathbf{w}}\big].$$

Along a sequence at the data $a_n + j$, b_n, $\mathbf{w}(n)$ the bounds hold with $\langle 1\rangle_{a_n,b_n;\mathbf{w}(n)} \geq \langle 1\rangle_{a_n+j,b_n;\mathbf{w}(n)}$ in place of the read at the moment index j, uniformly in j.

Proposition 35.13 (The finite-band polynomial)**.** *In Theorem 35.12 write $L_{J^c} = \prod_{i\notin J} L_i$. The polynomial*

$$A = \big[a - (a+b)x\big] L_{J^c} + x(1-x)\Big(\partial_x L_{J^c} - L_{J^c} \sum_{i\notin J} \frac{w_i d_i}{L_i}\Big),$$

of degree at most $h - |J| + 1$ in x, has the read

$$\langle A\rangle_{a,b;\mathbf{w}} = \sum_{i\in J} w_i \Big\langle \frac{x(1-x)\,d_i\,L_{J^c}}{L_i} \Big\rangle_{a,b;\mathbf{w}},$$

positive when a letter of J sees the mark and the reads are finite. On the half-line, $A = a\,L_{J^c} + x\,\partial_x L_{J^c} - \lambda x L_{J^c} - x L_{J^c} \sum_{i\notin J} w_i d_i/L_i$ has $\langle A\rangle^{\lambda}_{a;\mathbf{w}} = \sum_{i\in J} w_i \big\langle x\,d_i\,L_{J^c}/L_i\big\rangle^{\lambda}_{a;\mathbf{w}}$.

Theorem 35.14 (A positive sequence of Beta ports on an interval of a selected mark)**.** *In a projective chart take a selected mark at a mass $m_e \in (0,1]$, restricted to an interval $0 < \lambda_0 < \alpha_e < \lambda_1$, and write $\alpha_e = \lambda_0 + (\lambda_1 - \lambda_0)x$, $x \in [0,1]$; its letters are linear in α_e with nonnegative coefficients, $L_i = c_i + d_i x$ with $d_i \geq 0$ and $c_i \geq \lambda_0 d_i/(\lambda_1 - \lambda_0)$, and exponents $w_i \geq 0$, the letters constant in x going into the room density, and the factor $\alpha_e^{m_e - 1}$ of the mass is counted among them as the letter $\lambda_0 + (\lambda_1 - \lambda_0)x$ at the exponent $1 - m_e$. So $x(1-x)\,d_i/L_i \leq (\sqrt{\lambda_1} - \sqrt{\lambda_0})/(\sqrt{\lambda_1} + \sqrt{\lambda_0})$ on $[0,1]$ for every letter. Let $\langle 1\rangle_{1,1;\mathbf{w}}$ be finite. By Theorem 35.12(i) at $a = b = 1$ and the bound of Theorem 3.15(iii), integrated over the room, for $n \in \mathbb{N}$,*

$$0 < \big\langle (-1)^n \Pi_n^{1,1} \big\rangle_{1,1;\mathbf{w}} \leq \frac{(|\mathbf{w}|)_n}{n!} \Big(\frac{\sqrt{\lambda_1} - \sqrt{\lambda_0}}{\sqrt{\lambda_1} + \sqrt{\lambda_0}}\Big)^n \langle 1\rangle_{1,1;\mathbf{w}},$$

the first inequality when a letter sees the mark, as the letter of the mass does when $m_e < 1$.

Theorem 35.15 (The finite-band recurrence in the order of the Beta ports)**.** *At a fixed point of the room let the letters be linear, $L_i = c_i + d_i x$ with $c_i > 0$ and $d_i \geq 0$, $i = 1, \dots, h$, with exponents $w_i(n)$ depending on $n \in \mathbb{N}$, let $a_n, b_n > 0$, and in this statement write $L = \prod_i L_i = \sum_k p_k x^k$ and*

$$A_n = \big(a_n - (a_n + b_n)x\big)\, L + x(1-x)\, \partial_x L - x(1-x)\, L \sum_i \frac{w_i(n)\, d_i}{L_i}.$$

For every polynomial g, $\big\langle L\, x(1-x)\, \partial_x g + A_n g \big\rangle_{a_n, b_n; \mathbf{w}(n)} = 0$ (Theorem 35.10(a) with the polynomial Lg). Expanding $L\, x(1-x)\, \partial_x g + A_n g$ at $g = (-1)^r \Pi_r^{a_n, b_n}$ in the Beta ports $(-1)^{r+s} \Pi_{r+s}^{a_n, b_n}$ by the finite rules of Theorem 3.15(v) gives the relation

$$\sum_{s=-h-1}^{h+1} \mathfrak{b}_{n,r,s} \left\langle (-1)^{r+s} \Pi_{r+s}^{a_n, b_n} \right\rangle_{a_n, b_n; \mathbf{w}(n)} = 0$$

of width at most $2h + 3$, whose coefficients, written $\mathfrak{b}_{n,r,s}$ in this statement, are rational in n and r when a_n, b_n and the $w_i(n)$ are, and polynomial in the room data. With $|\mathbf{w}(n)| = \sum_i w_i(n)$ and $p_h = \prod_i d_i$, the furthest forward coefficient is

$$\mathfrak{b}_{n,r,h+1} = \big[|\mathbf{w}(n)| - a_n - b_n - h - r\big]\, p_h\, (-1)^{h+1} \binom{a_n + b_n + 2r - 2}{r} \Big/ \binom{a_n + b_n + 2r + 2h}{r + h + 1}.$$

Lemma 35.16 (A rational eventual sign certificate)**.** *Let $Q_n(X) = n^d q_d(X) + \sum_{k<d} n^k q_k(X)$ with polynomials q_k, and let rationals $X_1 \geq X_{\max} > 0$ and $\delta > 0$ satisfy $q_d \geq \delta$ on $[0, X_1]$. With $B_k = \sum_i \big|[X^i] q_k\big|\, X_1^i$, $Q_n \geq \delta n^d / 2$ on $[0, X_{\max}]$ for every $n \geq \max\big(1, 2 \sum_{k<d} B_k / \delta\big)$. Composition of recurrences with variable coefficients keeps its shifts: with* Sh *the shift $n \mapsto n + 1$, $\big(\sum_j b_j(n)\, \mathrm{Sh}^j\big)\big(\sum_i a_i(n)\, \mathrm{Sh}^i\big) = \sum_{i,j} b_j(n)\, a_i(n + j)\, \mathrm{Sh}^{i+j}$.*

With the room held fixed, a mark is integrated alone against its letters, and the mark relation is the integration by parts that this allows: the flux of the density $x^{a-1}(1-x)^{b-1}$ times a polynomial vanishes at both ends of the box, and that of the Gamma density at both ends of the half-line. Two uses follow. Against the Gamma density a mark of a family closes its moments into a recurrence of bounded order whose coefficients are polynomials in the room, and the derivatives in the room stay in a finite span: one mark closes all its moments. Against a Beta port the same relation gives positive reads, sums of the ratios $x(1-x)d_i/L_i$ that a letter seeing the mark keeps positive and the total exponent bounds, and at every pair $a_n, b_n > 0$ the Beta ports $\Pi_r^{a_n, b_n}$ obey a recurrence of bounded width in their order r whose furthest coefficient is explicit: when every d_i is positive it vanishes at the one order $r = |\mathbf{w}(n)| - a_n - b_n - h$ at most.

Chapter 36

The critical-line law

Let $\mathcal{E}$ have two marks, and let F be the family of the one list $\binom{\mathcal{E}}{1}$, whose bases are its two singletons, at the exponents $(1,1)$ and the masses $\mathbf{1}$ (Theorem 36.2). Its period is one, $P_F(\mathbf{1};1,1)=1$. On the fibre of the atom formula on the balance wall (Corollary 18.16), one share per basis and letter, the pins leave one free share: the basis share and the cobasis share of one basis are equal, $a=b=\frac{1}{2}+iy$, and the fibre is the line $\operatorname{Re} a=\frac{1}{2}$, the critical line. The law of y, the imaginary part of that share, is the critical-line law $\mathbb{P}_{\mathrm{crit}}$ (Definition 36.1). Its characteristic function is a stacked Gamma, its moments are rational, and its negative moments, read as finite parts, are rational multiples of the odd Trailmoments of the lattice input on the coset $\frac{1}{2}+\mathbb{Z}$ (Theorem 36.2). A function R of the share, read against an odd derivative of $\mathbb{P}_{\mathrm{crit}}$ along the critical line, gives the critical-line word $\mathrm{crit}_k[R]$ (Definition 36.3): it carries one Trailmoment of odd index with a rational part, and it is positive whenever R is the Fourier transform of a positive measure (Theorem 36.8).

Definition 36.1 (The critical-line law)**.** The *critical-line law* is the probability law $\mathbb{P}_{\mathrm{crit}}(dy)=p_{\mathrm{crit}}(y)\,dy$ on $\mathbb{R}$ with

$$p_{\mathrm{crit}}(y)=\frac{1}{2\pi}\,\Gamma_4{}_0\big[\tfrac{1}{2}+iy,\ \tfrac{1}{2}+iy,\ \tfrac{1}{2}-iy,\ \tfrac{1}{2}-iy\big]=\frac{2\pi}{(e^{\pi y}+e^{-\pi y})^2},$$

the stacked Gamma of Lemma 5.16 on the line $\operatorname{Re} a=\frac{1}{2}$ divided by 2π.

Theorem 36.2 (The critical-line law and its moments)**.** *Let $\mathcal{E}$ have two marks and let F be the family of the one list $\binom{\mathcal{E}}{1}$, whose bases are its two singletons, at the exponents $(1,1)$ and the masses $\mathbf{1}$. On the fibre of Corollary 18.16 the basis and cobasis shares of one basis are equal, $a=b=\frac{1}{2}+iy$, the fibre integral is $P_F(\mathbf{1};1,1)=1$, and the imaginary part Y of the share has the law $\mathbb{P}_{\mathrm{crit}}$. In this statement $\mathbb{E}$ is the expectation of $Y\sim\mathbb{P}_{\mathrm{crit}}$.*

(i) $\mathbb{E}[e^{i\xi Y}]$ *is the Fourier read of Lemma 5.16(iii) at $t=\xi$; every odd moment of Y vanishes and every even moment is rational.*

(ii) For $j\geq 1$ the finite part at $y=0$ (Definition 1.5) of $\mathbb{E}[Y^{-2j}]$ is

$$\begin{aligned}\operatorname{Reg}\mathbb{E}\big[Y^{-2j}\big]&=(-1)^j\,2j\,(\mathsf{T}_{2j+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty))\\&=(-1)^j\,2j\,(2^{2j+1}-1)\,(\mathsf{T}_{2j+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),\end{aligned}$$

the second line by the lattice relation (Theorem 5.12(i)) at the step $q=2$, the lattice $\frac{1}{2}\mathbb{Z}$ splitting into $\mathbb{Z}$ and $\frac{1}{2}+\mathbb{Z}$.

(iii) With $X=\frac{1}{2}+iY$, $\mathbb{E}[e^{tX}]=\dfrac{te^t}{e^t-1}=t+\dfrac{t}{e^t-1}$, and $d_{m+1}\,\mathbb{E}[X^m]\in\mathbb{Z}$ for every $m\in\mathbb{N}$.

The list $\binom{\mathcal{E}}{1}$ on two marks is the smallest table with a share that moves: at the centre the pins hold its basis and cobasis shares equal, on the critical line, and the period, one, is the integral of the law $\mathbb{P}_{\mathrm{crit}}$ of their imaginary part. Everything about that law is read off the stacked Gamma $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]$ of Lemma 5.16. Its characteristic function is the stacked Gamma $\Gamma_{\substack{2\\0}}\big[1+\frac{i\xi}{2\pi},\ 1-\frac{i\xi}{2\pi}\big]$, its moments are rational with denominators bounded by the clearing integers, and its negative moments, which diverge at the origin, have finite parts that are rational multiples of the odd Trailmoments of the lattice input on the coset $\frac{1}{2}+\mathbb{Z}$, and so of the counting input.

36.1 Critical-line words

Definition 36.3 (Critical-line words)**.** For an integer $k \geq 1$ and a function R on the line $\operatorname{Re} u = \frac{1}{2}$, the *critical-line word* of order k is

$$\mathrm{crit}_k[R] = -\frac{2}{(2k)!}\,\frac{1}{2\pi i}\int_{(1/2)} \frac{R(u)}{1-2u}\,\partial_u^{2k-1}\Gamma_{\substack{4\\0}}\big[u,\ u,\ 1-u,\ 1-u\big]\,du,$$

wherever it converges; the integrand is regular at $u=\frac{1}{2}$, where the odd derivative of the stacked Gamma, which is even about $\frac{1}{2}$, vanishes. On the line, $u=\frac{1}{2}+iy$,

$$\mathrm{crit}_k[R] = \frac{(-1)^k}{(2k)!}\int_{\mathbb{R}} R(\tfrac{1}{2}+iy)\,\frac{p_{\mathrm{crit}}^{(2k-1)}(y)}{y}\,dy.$$

Proposition 36.4 (Critical-line words of polynomials)**.** *For an integer $k \geq 1$ and a polynomial $R(u) = (1-2u)\,Q(u) + R(\frac{1}{2})$, with $\mathbb{E}$ the expectation of $Y \sim \mathbb{P}_{\mathrm{crit}}$,*

$$\begin{aligned}\mathrm{crit}_k[R] &= R(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) + \frac{2}{(2k)!}\,\mathbb{E}\big[Q^{(2k-1)}(\tfrac{1}{2}+iY)\big]\\ &= (2^{2k+1}-1)\,R(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + \frac{2}{(2k)!}\,\mathbb{E}\big[Q^{(2k-1)}(\tfrac{1}{2}+iY)\big],\end{aligned}$$

the second line by Theorem 5.12(i), as in Theorem 36.2(ii).

Definition 36.5 (Rows; clearing)**.** A value written as $x = \sum_{i=1}^{m} A_i v_i - B$, a rational combination of fixed real numbers v_i, its *coordinates*, such as Trailmoments of the counting input, less a rational number B, its *rational part*, has the *row* $(A_1,\dots,A_m;B)$; a positive integer *clears* the row when it makes all these coefficients integers.

Corollary 36.6 (The rational part of an integer polynomial argument)**.** *For $P \in \mathbb{Z}[x]$ of degree D and $R(u) = P(2u)$, the rational part of $\mathrm{crit}_k[R]$ (Definition 36.5) is cleared by $\frac{(2k)!}{2}\,d_{\max(1,D-2k+1)}$.*

Proposition 36.7 (Binomial moments of the critical-line law)**.** *Let $X = \frac{1}{2}+iY$ with $Y \sim \mathbb{P}_{\mathrm{crit}}$, $\mathbb{E}$ its expectation, and $H_m = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,m])$.*

(i) For complex z with $|z|<1$, $\mathbb{E}\big[(1+z)^X\big] = (1+z)\log(1+z)/z$.

(ii) $\mathbb{E}\big[\binom{X}{0}\big] = 1$ and $\mathbb{E}\big[\binom{X}{i}\big] = \dfrac{(-1)^{i+1}}{i(i+1)}$ for $i \geq 1$.

(iii) For $i \geq 1$, $\mathbb{E}\Big[\dfrac{d}{du}\dbinom{u}{i}\Big|_{u=X}\Big] = 2(-1)^i\,\dfrac{H_{i-1}-1}{i(i+1)}$. So for every polynomial $Q = \sum_i q_i\binom{u}{i}$,

$$\mathrm{crit}_1\left[(1-2u)\,Q\right] = 2\sum_{i\geq 1} q_i\,(-1)^i\,\frac{H_{i-1}-1}{i(i+1)}.$$

In the binomial basis the critical-line word of a polynomial has explicit coefficients, harmonic numbers over $i(i+1)$, whose denominators a prime q divides only to bounded powers; this is the basis in which the clearings of the critical-line words of polynomials are counted, prime by prime.

Theorem 36.8 (Positive arguments for the critical-line word)**.** *For an integer $k \geq 1$ and real t, with $\mathbb{E}$ the expectation of $Y \sim \mathbb{P}_{\mathrm{crit}}$,*

$$\mathrm{crit}_k\left[e^{t(u-\frac{1}{2})}\right] = \frac{1}{(2k)!}\int_{|t|}^{\infty} x^{2k-1}\,\mathbb{E}\big[e^{ixY}\big]\,dx > 0,$$

$$\mathrm{crit}_k[1] = (\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = (2^{2k+1}-1)\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \quad \textit{(Theorem 5.12(i)).}$$

If $R(\frac{1}{2}+iy) = \int e^{ity}\,\phi(dt)$ for a finite nonzero positive measure ϕ on $\mathbb{R}$, then $\mathrm{crit}_k[R] = \int \mathrm{crit}_k\left[e^{t(u-\frac{1}{2})}\right]\phi(dt) > 0$.

On the critical line the word reads its argument against an odd derivative of the density of the critical-line law divided by the imaginary part of the share, a kernel without a fixed sign. Its Fourier transform, the word of an exponential argument, is the integral beyond $|t|$ of the characteristic function times a power, a positive function, so an argument that is the Fourier transform of a positive measure has a positive word. This is how positive words carrying one odd Trailmoment are made.

Theorem 36.9 (A positive rational argument with one odd index)**.** *Let $M \in \mathbb{N}$ and $p(v) = \sum_{j=0}^{D} b_j v^j \in \mathbb{Q}[v]$, nonnegative on $[0,1]$ and not identically zero, and let*

$$R_{M,p}(u) = \sum_{j=0}^{D} b_j\Big(\frac{1}{u+M+j} + \frac{1}{1-u+M+j}\Big), \qquad R_{M,p}(\tfrac{1}{2}) = 4\sum_j \frac{b_j}{2M+2j+1} > 0.$$

Then for every $k \geq 1$

$$\begin{aligned}
\mathrm{crit}_k[R_{M,p}] &= \frac{2}{(2k)!}\int_0^1\!\!\int_0^1 \frac{v^M x^{-1/2}\,p(v)\,[-\log(vx)]^{2k}}{1-vx}\,dx\,dv \\
&= R_{M,p}(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) \\
&\qquad - 4\sum_j \frac{b_j}{2M+2j+1}\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((M+j,\infty)) \\
&= (2^{2k+1}-2)\,R_{M,p}(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \\
&\qquad + 4\sum_j \frac{b_j}{2M+2j+1}\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,M+j]) > 0.
\end{aligned}$$

If $Cb_j \in \mathbb{Z}$ for every j, then $C\,d_{2M+2D+1}\,d_{M+D}^{2k+1}$ clears both rational coefficients; the Trailmoments of $\mathbb{I}_{\frac{1}{2}+\mathbb{Z}}$ are read in those of $\mathbb{I}_{\mathbb{Z}}$ by Theorem 5.12(i).

Corollary 36.10 (Arguments of two Beta corners)**.** *For $p(v) = (1-v)^q$, $q \in \mathbb{N}$, the argument of Theorem 36.9 is*

$$R_{M,(1-v)^q}(u) = \Gamma\begin{matrix}2\\1\end{matrix}\!\left[\begin{matrix}u+M,\ q+1\\ u+M+q+1\end{matrix}\right] + \Gamma\begin{matrix}2\\1\end{matrix}\!\left[\begin{matrix}1-u+M,\ q+1\\ 2-u+M+q\end{matrix}\right].$$

Its words are positive and carry one Trailmoment of the counting input, of the odd index $2k+1$, at every k. For fixed M, $\mathrm{crit}_k[R_{M,(1-v)^q}] \sim \frac{4\,\Gamma(M+1)}{(2k)!}\,q^{-M-1}(\log q)^{2k}$ as $q \to \infty$.

Proposition 36.11 (Poles at the half-integers)**.** *For integers $k \geq 1$ and $l \geq 0$,*

$$\frac{1}{2\pi i}\int_{(1/2)} \frac{1}{u-\frac{1}{2}-l}\sum_{m\in\mathbb{Z}}\frac{du}{(u+m)^{2k+1}} = -(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((l,\infty)),$$

the integrand being regular on the line at $l = 0$; and for $l \geq 1$, with the critical-line word of Definition 36.3,

$$\operatorname{crit}_k\Big[\frac{1}{u-\frac{1}{2}-l}\Big] = -\operatorname{crit}_k\Big[\frac{1}{u-\frac{1}{2}+l}\Big] = -\frac{1}{l}\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,l]).$$

So the critical-line word of a simple pole at a half-integer other than $\frac{1}{2}$ is a head of the lattice input on the coset $\frac{1}{2}+\mathbb{Z}$, a rational number, and with Proposition 36.4 the word is read on every polynomial plus simple poles at these points. The rational arguments of Theorem 36.9 have their simple poles at the integers; the half-integers give heads and tails on the coset $\frac{1}{2}$.

Proposition 36.12 (The decay of the word of an exponential argument)**.** *For an integer $k \geq 1$ and $t > 0$, with the polynomial $c_k(t) = \sum_{l=0}^{2k} \frac{(2k)!}{l!}\,2^{2k-l+1}\,t^l$,*

$$\frac{e^{-t/2}}{(2k)!}\,c_k(t) \leq \operatorname{crit}_k\left[e^{t(u-\frac{1}{2})}\right] \leq \frac{e^{-t/2}}{(2k)!\,(1-e^{-t})}\,c_k(t).$$

A rational argument whose positive measure is the exponential profile of a polynomial nonnegative on $[0,1]$ gives a positive word with one odd Trailmoment of the counting input and a rational part cleared by $d_{2M+2D+1}d_{M+D}^{2k+1}$. The words of exponential arguments decay like $e^{-t/2}$ times a polynomial of degree $2k$, so the words of the arguments of two Beta corners decay like a power of their exponent q: they are positive, carry one Trailmoment and are controlled, the raw material of the relation blocks of Chapter 44.

Chapter 37

The gap and the shape words

A value read at rational data has a row, its coefficients on Trailmoments of the counting input and a rational part (Definition 36.5). The gap theorem turns a sequence of nonzero values whose rows are cleared by integers that grow more slowly than the values decay into irrationality, and the shape words supply such sequences. They are integrals over a cube against one letter; opened by one share they close on the stacked Gamma $\Gamma_{0}^{4}\big[a,\ a,\ 1-a,\ 1-a\big]$ of Lemma 5.16, so their rows are explicit, cleared by powers of the clearing integers d_n and carried by the Trailmoments $(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of index at most one more than the number of shapes. Beta ports inserted on one shape keep the words positive while they move the row, filters built from recurrences remove unwanted coordinates, and Beta ports whose order grows with the depth trade a slower decay for a selected coordinate.

37.1 Clearing and the gap

Lemma 37.1 (The growth of the clearing integers)**.** *For integers $n \geq 0$ and $s \geq 1$, d_n^s is the least positive integer D with $D\,(\mathsf{T}_{s;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,n]) \in \mathbb{Z}$ for every integer density f, and*

$$\limsup_{n\to\infty} \frac{\log d_n}{n} \leq c_{\mathrm{lcm}} = \frac{14}{25}\log 2 + \frac{9}{25}\log 3 + \frac{1}{5}\log 5 = 1.10555\ldots < 1.106.$$

Theorem 37.2 (The gap)**.** *Let $v_1, \ldots, v_m$ be real, and let real numbers $x_n = \sum_i A_{n,i} v_i - B_n$, with $A_{n,i}, B_n \in \mathbb{Q}$, have positive integers D_n clearing all these coefficients, with $0 < |D_n x_n| \to 0$.*

(i) Some v_i is irrational.

(ii) For $m = 1$, $A_{n,1} \neq 0$ for all large n.

The hypothesis $x_n \neq 0$ holds in particular when x_n is the value of a word with a strictly positive read.

An integer of modulus below one is zero; nothing else is used. What the theory contributes is the two other hypotheses: positivity, which keeps the cleared values from vanishing, and rows whose clearing integers grow more slowly than the values decay.

37.2 Shape words with polynomial insertions

Definition 37.3 (Shape words)**.** For integers $n \geq 0$ and $k \geq 2$ let $L = \bar{x}_0 + x_0 x_1 \cdots x_k$ on the cube $[0,1]^{k+1}$ with coordinates $x_0, \ldots, x_k$. The *shape word* of depth n with k shapes of a polynomial

insertion $P \in \mathbb{Q}[x_0,\dots,x_k]$ is

$$I_n(k;P)=\int_{[0,1]^{k+1}} P(x)\,\frac{\prod_{i=0}^{k}(x_i\bar{x}_i)^n}{L^{n+1}}\,dx=\int_{[0,1]^{k+1}} P\,f_k^n\,\frac{dx}{L},\qquad f_k=\frac{\prod_{i=0}^{k}x_i\bar{x}_i}{L},$$

with the *base* f_k. For a monomial x^p, $p=(p_0,\dots,p_k)$, the *share function* is

$$g_{n,p}(a)=\frac{(n!)^{k-2}\,(1-a)_n^2\,(a)_{p_0}}{(n+1)_{p_0}\prod_{i=1}^{k}(a+p_i)_{n+1}},$$

and for $P=\sum_p\big([x^p]P\big)x^p$ it is $g_{n,P}=\sum_p\big([x^p]P\big)\,g_{n,p}$, with the coefficients $c_{n,h,s}(P)$ of its principal parts $c_{n,h,s}(P)(a+h)^{-s}$ at the poles $a=-h$ and the *column sums* $C_{n,s}(P)=\sum_h c_{n,h,s}(P)$. The *port factor* is $\mathfrak{a}_k=x_0\,x_1\bar{x}_1\,x_2\cdots x_k/L$, and $M_k(\eta)=\sup_{(0,1)^{k+1}} f_k\mathfrak{a}_k^{\eta}$ for $\eta\ge 0$.

Theorem 37.4 (Shape words: the one-share closing)**.** *Let P have degree at most D, with $p_0\le(k-2)(n+1)$ on every monomial x^p of P.*

(i) Share side. *For a monomial, on $0<\operatorname{Re}a=c<n+1$,*

$$\begin{aligned}I_n(k;x^p)&=\frac{1}{2\pi i}\int_{(c)}\Gamma\,{}_2^2\!\left[\begin{matrix}a,\ n+1-a\\ n+1\end{matrix}\right]\Gamma\,{}_2^2\!\left[\begin{matrix}a+p_0,\ n+1-a\\ n+1+p_0\end{matrix}\right]\prod_{i=1}^{k}\Gamma\,{}_2^1\!\left[\begin{matrix}a+p_i,\ n+1\\ a+p_i+n+1\end{matrix}\right]da\\&=\frac{1}{2\pi i}\int_{(c)}\Gamma\,{}_4^0\big[a,\ a,\ 1-a,\ 1-a\big]\ g_{n,p}(a)\,da.\end{aligned}$$

(ii) Closing. $I_n(k;P)=-\sum_{m\ge1}g'_{n,P}(m)$ *(Lemma 5.16(i)). Every pole of $g_{n,P}$ lies among $0,-1,\dots,-n-D$ with order at most k; the integers $1,\dots,n$ are zeros of order at least two of each $g_{n,p}$; $g_{n,p}(a)=O(|a|^{-2})$.*

(iii) The row. $C_{n,1}(P)=0$ *and*

$$\begin{aligned}I_n(k;P)&=\sum_{h=0}^{n+D}\sum_{s=1}^{k}s\,c_{n,h,s}(P)\,(\mathsf{T}_{s+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((h,\infty))\\&=\sum_{s=2}^{k}s\,C_{n,s}(P)\,(\mathsf{T}_{s+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))-\sum_{h,s}s\,c_{n,h,s}(P)\,(\mathsf{T}_{s+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,h]).\end{aligned}$$

Proposition 37.5 (An integer certificate for inserted words)**.** *In Theorem 37.4 let P have integer coefficients. Then $(n+1)_D\,d_{n+D}^{k-s}\,c_{n,h,s}(P)\in\mathbb{Z}$, and $(n+1)_D\,d_{n+D}^{k+1}$ clears every coefficient of the row, its rational part included.*

Theorem 37.6 (Integer-valued insertion clearing)**.** *In Theorem 37.4 let P have integer coefficients, let $p_{\max}$ be its largest exponent of a shape x_i, $i\ge1$, and $p_{0,\max}$ its largest exponent of x_0, and let $b_P=\operatorname{lcm}\big\{\binom{n+j}{j}:\ j$ an exponent of x_0 in $P\big\}$. Then $b_P\,d^{k-s}_{\max(n+p_{\max},\,p_{0,\max})}\,c_{n,h,s}(P)\in\mathbb{Z}$ and $b_P\,d^{k+1}_{\max(n+p_{\max},\,p_{0,\max})}$ clears the whole row, and b_P divides $d_{n+p_{0,\max}}$; insertions free of x_0 are cleared by $d^{k+1}_{n+p_{\max}}$ whatever their degrees. For $k\ge3$ the insertion x_0^n retains the top coordinate:*

$$C_{n,k}(x_0^n)=(-1)^{n(k+1)}\binom{2n}{n},\qquad\big[(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\big]\,I_n(k;x_0^n)=k\,(-1)^{n(k+1)}\binom{2n}{n}.$$

Theorem 37.7 (The exact maximum under a growing Beta port)**.** *Fix $k \ge 2$, with the base f_k and the port factor $\mathfrak{a}_k$ of Definition 37.3, so that $I_n(k;P) = \int P f_k^n \, dx/L$. For $q \in \mathbb{N}$,*

$$I_n\big(k; (-1)^q \Pi_q^{n+1,n+1}(x_1)\big) = \binom{n+q}{q} \int_{[0,1]^{k+1}} f_k^n \, \mathfrak{a}_k^q \, \frac{dx}{L} > 0.$$

For every real $\eta \ge 0$, $f_k \mathfrak{a}_k^\eta$ has a unique maximum on the open cube: with a the unique root in $0 < a < 1/(2+\eta)$ of

$$(1-2a)\big(1-(2+\eta)a\big)(1+\eta a)^{k-1} = (1+\eta)^k a^{k+2},$$

it is at $x_1 = a$, $x_2 = \cdots = x_k = \frac{(1+\eta)a}{1+\eta a}$, $x_0 = \frac{(1+\eta)a}{1-a}$, with value

$$M_k(\eta) = \frac{(1-a)^{k-2}}{(1+\eta a)^{k-1}} \, (1-2a)^{1+\eta} \, \big(1-(2+\eta)a\big),$$

and $0 < I_n\big(k; (-1)^q \Pi_q^{n+1,n+1}(x_1)\big) \le \binom{n+q}{q} M_k(q/n)^n \, I_0(k;1)$.

Proposition 37.8 (The root rate of the shape words)**.** *For $k \ge 2$:*

(i) $0 < I_n(k;1) \le M_k(0)^n \, I_0(k;1)$ (Theorem 37.7 at $q = 0$);

(ii) $\lim_n I_n(k;1)^{1/n} = M_k(0)$, where

$$M_k(0) = \max_{(0,1)^{k+1}} f_k = \max_{0<a<1} \frac{a^k (1-a)^k}{(1+a^{k/2})^2} \in \Big[\frac{4^{-k}}{(1+2^{-k/2})^2}, \; 4^{-k}\Big].$$

The maximum over a is at the root $a \in (0, \frac{1}{2})$ of $a^{k/2+1} + 2a = 1$, the root of Theorem 37.7 at $\eta = 0$. For every $k \ge 2$, $-\log M_k(0) > (k+1)\, c_{\mathrm{lcm}}$, so that $d_n^{k+1} I_n(k;1) \to 0$.

Theorem 37.9 (An irrational Trailmoment of the counting input)**.** *The shape word $I_n(2;1)$ is positive, and with $a_l = (-1)^l \binom{n}{l}\binom{n+l}{l}$, the coefficient of x^l in $(-1)^n \Pi_n^{1,1}(x)$ (Theorem 3.15(v)), and $b_l = 2a_l \sum_{j \ne l} \frac{a_j}{j-l}$,*

$$\begin{aligned} I_n(2;1) &= \sum_{l=0}^{n} \Big(2a_l^2 \, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((l,\infty)) + b_l \, (\mathsf{T}_{2;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((l,\infty)) \Big) \\ &= 2\big(A_n \, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - B_n \big), \\ A_n &= \sum_l a_l^2 \in \mathbb{Z}, \qquad d_n^3 B_n \in \mathbb{Z}, \end{aligned}$$

$$B_n = \sum_l a_l^2 \, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,l]) + \sum_l a_l \, (\mathsf{T}_{2;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,l]) \sum_{j \ne l} \frac{a_j}{j-l},$$

$I_n(2;1) \le 2\,(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))\,(\sqrt{2}-1)^{4n}$, and $(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N,\infty)) \notin \mathbb{Q}$ for every integer cut $N \ge 0$.

A shape word is the cube against one letter, and one share opens the letter: the Beta corners of the shapes and of x_0 leave a single line integral of stacked Gammas, the stacked Gamma $\Gamma_0^4\big[a,\ a,\ 1-a,\ 1-a\big]$ times a rational share function, which closes on the positive integers. The share function has poles of order at most k at the nonpositive integers, so the row carries the Trailmoments of the counting input of index at most $k+1$ with rational parts cleared by powers

of d_{n+D}, and the coefficients of the simple poles sum to zero because the share function decays quadratically. At two shapes the row has one coordinate besides the rationals, the decay $(\sqrt{2}-1)^{4n}$ beats the clearing d_n^3, and the gap gives the irrationality of $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$ at every integer cut N. With more shapes the decay still beats the clearing, but the row carries several coordinates, and the filters and the growing Beta ports select one (Proposition 37.18, Theorem 37.19).

37.3 Rows

Proposition 37.10 (The row of a closing in rational terms)**.** *Let a closing give $x_n = \sum_{t\ge t_0} r_n(t)$, the sum over the integers $t \ge t_0$, with $r_n(t) = \sum_{k,s} \tilde{c}_{n,k,s}(t+k)^{-s} = O(t^{-2})$, $\tilde{c}_{n,k,s} \in \mathbb{Q}$, integer pole positions and $h_k = t_0 + k - 1 \ge 0$. With $\tilde{C}_{n,s} = \sum_k \tilde{c}_{n,k,s}$,*

$$x_n = \sum_{k,s} \tilde{c}_{n,k,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((h_k,\infty)) = \sum_{s\ge 2} \tilde{C}_{n,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - B_n,$$

$$B_n = \sum_{k,s} \tilde{c}_{n,k,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,h_k]),$$

the index-one Trailmoments regularized, and $D_n = \operatorname{lcm}_{k,s}\left(\operatorname{den}(\tilde{c}_{n,k,s})\,d_{h_k}^s\right)$ clears the row.

Proposition 37.11 (The primitive row)**.** *For a nonzero rational pair (A,B), a common clearing integer D and $g = \gcd(DA,DB)$, the pair $g^{-1}(DA,DB) \in \mathbb{Z}^2$ is, up to sign, independent of D and of every nonzero rational rescaling of (A,B), and with $(a,b) = g^{-1}(DA,DB)$, $av - b = (D/g)(Av - B)$ for every real v.*

Corollary 37.12 (The top-index criterion)**.** *Let $k \ge 3$, and suppose there are integer polynomials P_n of degree at most a fixed D with $C_{n,s}(P_n) = 0$ for $2 \le s < k$ and $I_n(k;P_n) > 0$ for all large n, P_n possibly changing sign on the cube. If the sums $\|P_n\|_1$ of the absolute values of the coefficients of P_n satisfy*

$$\limsup_n \frac{1}{n}\log\|P_n\|_1 < -\log M_k(0) - (k+1)\limsup_n \frac{1}{n}\log d_n,$$

then $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) \notin \mathbb{Q}$ for every integer cut $N \ge 0$.

A closing in rational terms has a row whose clearing is read off its pole positions and orders, and a row is a primitive integer pair up to the clearing. With k shapes the row of $I_n(k;P)$ carries $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),\dots,(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; an insertion that kills the lower column sums and keeps the word positive at a cost below the margin between the decay and the clearing separates $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. The insertion may change sign: positivity is asked of the word, not of the insertion.

37.4 The Beta port ladder and the filters

Theorem 37.13 (The Beta port ladder of the shape words)**.** *For the shape word with $k \ge 2$ shapes at depth $n \ge 1$ write its letter as $L = \bar{x}_0 + x_0x_2\cdots x_k\,x_1$, and in this statement write $P_r = (-1)^r\Pi_r^{n+1,n+1}(x_1)$, $g_{n,r} = g_{n,P_r}$ and $g_n = g_{n,0}$, the share function of the insertion 1 (Definition 37.3).*

(i) $0 < I_n(k;P_r) \le \binom{n+r}{r} I_n(k;1)$ *(Theorem 37.7, the port factor being at most one).*

(ii) $g_{n,r}(a) = \binom{n+r}{r}\dfrac{(n+1-a)_r}{(n+1+a)_r}\,g_n(a)$ *with* $g_n(a) = \dfrac{(n!)^{k-2}(1-a)_n^2}{(a)_{n+1}^k}$, *and with* $b = n+r+1$, $(r+1)(a+b)\,g_{n,r+1}(a) = b\,(b-a)\,g_{n,r}(a)$.

(iii) For a function φ *on the integers and a finite set* $\mathcal{M}$ *of positive integers write* $S_{n,r}[\varphi] = \sum_{m\in\mathcal{M}} g_{n,r}(m)\varphi(m)$ *and* $S'_{n,r}[\varphi] = -\sum_{m\in\mathcal{M}} g'_{n,r}(m)\varphi(m)$, *the sums over the common finite set* $\mathcal{M}$, *with functions of the summation variable* m *inside the brackets. Then, with* $b = n+r+1$,

$$(r+1)\,S'_{n,r+1}\big[(m+b)\varphi\big] = b\,S'_{n,r}\big[(b-m)\varphi\big] + (r+1)\,S_{n,r+1}[\varphi] + b\,S_{n,r}[\varphi].$$

(iv) $\dfrac{g_{n+1,r}(a)}{g_{n,r}(a)} = \dfrac{(n+1)^{k-3}(n+r+1)(n+1-a)(n+r+1-a)}{(a+n+1)^{k-1}(a+n+r+1)}$, *and with* $U(m) = (m+n+1)^{k-1}(m+n+r+1)$ *and* $V(m) = (n+1)^{k-3}(n+r+1)(n+1-m)(n+r+1-m)$,

$$S_{n+1,r}[U\varphi] = S_{n,r}[V\varphi], \qquad S'_{n+1,r}[U\varphi] - S_{n+1,r}[U'\varphi] = S'_{n,r}[V\varphi] - S_{n,r}[V'\varphi].$$

(v) For $0 \le r \le h$ *the insertions* P_r *have degree at most* h, *their rows are cleared by* d_{n+h}^{k+1} *(Theorem 37.6), and their positive reads cost at most* $\binom{n+h}{h}$ *times the base word* $I_n(k;1)$.

(vi) In the selected mark x_1 *the furthest forward coefficient of Theorem 35.15, for the one letter* L *of exponent* $n+1$, *is nonzero on the open cube.*

Proposition 37.14 (The insertion index supplies auxiliary states)**.** *Fix* $k \ge 3$ *and* $n \ge 1$ *and, in this statement, write* $P_r = (-1)^r \Pi_r^{n+1,n+1}(x_1)$, *an insertion of the shape word with* k *shapes. As functions of* r, $C_{n,k}(P_r) \in \mathbb{Q}[r]$ *and* $C_{n,k-1}(P_r) = -2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r])\,C_{n,k}(P_r) + q_n(r)$ *with* $q_n \in \mathbb{Q}(r)$. *Consequently a scalar finite shift recurrence over* $\mathbb{Q}(r)$ *annihilating* $C_{n,k-1}(P_r)$ *for all large* r *also annihilates* $C_{n,k}(P_r)$.

Proposition 37.15 (The shape step is a two-state operator relation)**.** *For* $k \ge 2$ *and* $n \ge 0$ *let* g_n *be the share function of the insertion* 1, $g_n(a) = (n!)^{k-2}(1-a)_n^2/(a)_{n+1}^k$, *and for integer cuts* $0 \le N_0 < N_1$ *write* $S_n[\varphi] = \sum_{N_0<m\le N_1} g_n(m)\varphi(m)$ *and* $S'_n[\varphi] = -\sum_{N_0<m\le N_1} g'_n(m)\varphi(m)$. *Then*

$$\begin{aligned} S_{n+1}\big[(m+n+1)^k\varphi\big] &= (n+1)^{k-2}\,S_n\big[(m-n-1)^2\varphi\big], \\ S'_{n+1}\big[(m+n+1)^k\varphi\big] &= k\,S_{n+1}\big[(m+n+1)^{k-1}\varphi\big] + (n+1)^{k-2}\,S'_n\big[(m-n-1)^2\varphi\big] \\ &\quad - 2(n+1)^{k-2}\,S_n\big[(m-n-1)\varphi\big], \end{aligned}$$

and the same holds over all $m > N_0$ *whenever the sums converge one by one: for bounded* φ, *when* $(k-2)(n+1) > 1$ *in the first identity and* $(k-2)(n+1) > 0$ *in the second.*

Proposition 37.16 (A polynomial filter from a bounded local recurrence)**.** *Let unwanted closing coefficients satisfy* $c_{n,j} = \ell(n)\,y_{n,j}$ *and* $y_{n+1,j} = \mathbf{A}(n)\,y_{n,j}$, $1 \le j \le r$, *with state vectors* $y_{n,j}$ *of a fixed length* h, *a* $1\times h$ *matrix* $\ell(n)$ *and an* $h\times h$ *matrix* $\mathbf{A}(n)$ *over* $\mathbb{Q}(n)$. *Write* $\mathbf{A} = \widehat{\mathbf{A}}/q$ *and* $\ell = \hat{\ell}/s$ *with integer polynomial matrices* $\widehat{\mathbf{A}}$, $\hat{\ell}$ *and integer polynomials* q, s, *put* $\mathbf{A}^{[0]} = I$, $\mathbf{A}^{[i]}(n) = \mathbf{A}(n+i-1)\cdots\mathbf{A}(n)$, $Q(n) = \prod_{t=0}^{h-1} q(n+t)\prod_{t=0}^{h} s(n+t)$ *and the* $1\times h$ *matrices* $\mathbf{r}_i(n) = Q(n)\,\ell(n+i)\,\mathbf{A}^{[i]}(n)$, $0 \le i \le h$. *If the* $\mathbf{r}_i$ *span* $\mathbb{Q}(n)^{1\times h}$ *and* $a_i(n)$ *is* $(-1)^i$ *times the determinant of the* $h\times h$ *matrix stacking the* $\mathbf{r}_{i'}$, $i' \ne i$, *in order, then the* a_i *are integer polynomials, not all zero, and*

$$\sum_{i=0}^{h} a_i(n)\,c_{n+i,j} = 0 \qquad (1 \le j \le r)$$

away from the integer zeros of the denominators; a target sequence x_n is retained at n exactly when $\sum_i a_i(n)\,x_{n+i} \neq 0$. With $D_{\mathbf{A}} = \max(\deg \widehat{\mathbf{A}}, \deg q)$, $D_s = \deg s$ and $D_\ell = \deg \hat{\ell}$: $\deg a_i \leq h(hD_{\mathbf{A}} + hD_s + D_\ell)$. For two states, with $\ell = (1,0)$ and $\mathbf{A} = (\beta_{il})$, the filter is $\beta_{12}c_{n+2} - (\beta_{11}^{+}\beta_{12} + \beta_{12}^{+}\beta_{22})c_{n+1} + \beta_{12}^{+}\det \mathbf{A}(n)\,c_n = 0$, $\beta_{il}^{+} = \beta_{il}(n+1)$; for one state, $q(n)c_{n+1} = p(n)c_n$ gives the filter $q(n)\,\mathrm{Sh} - p(n)$, Sh the shift $n \mapsto n+1$.

Proposition 37.17 (The finite jet recurrence)**.** *If a state has the recurrence $U_{n+1}(\varepsilon) = \mathbf{A}(n,\varepsilon)\,U_n(\varepsilon)$, regular at $\varepsilon = 0$, then with $U_n(\varepsilon) = \sum_j U_{n,j}\varepsilon^j$ and $\mathbf{A}(n,\varepsilon) = \sum_j \mathbf{A}_j(n)\varepsilon^j$, $U_{n+1,j} = \sum_{i=0}^{j} \mathbf{A}_i(n)\,U_{n,j-i}$ for $0 \leq j \leq J$: a lower block-triangular recurrence for the jets of order at most J, with diagonal blocks $\mathbf{A}_0(n)$.*

Proposition 37.18 (A recurrence filters the positive word)**.** *Let $x_n = \int X^n\,d\mu$ with $0 < X \leq X_{\max} < 1$ on the support of a finite positive measure μ, and $x_n = \sum_{j=1}^{r} \tilde{C}_{n,j}v_j - B_n$ with $\tilde{C}_{n,j}, B_n \in \mathbb{Q}$. Let $a_0(n), \dots, a_h(n) \in \mathbb{Z}[n]$, $Q_n(X) = \sum_i a_i(n)X^i$ and $\tilde{x}_n = \sum_i a_i(n)\,x_{n+i}$, and suppose that for all large n, $\sum_i a_i(n)\tilde{C}_{n+i,j} = 0$ for $j \neq j_*$ and Q_n has one sign on the support of μ, strictly on a set of positive measure. Then $\tilde{x}_n = \int X^n Q_n\,d\mu = A_n v_{j_*} - \sum_i a_i(n)B_{n+i}$ with $A_n = \sum_i a_i(n)\tilde{C}_{n+i,j_*}$, and $0 < |\tilde{x}_n| \leq O(n^{\max_i \deg a_i})\,X_{\max}^n \mu(\mathbb{R})$. If positive integers $D_{n,h}$ clear the rows of $x_n, \dots, x_{n+h}$, and $\limsup_n \frac{1}{n}\log D_{n,h} < -\log X_{\max}$, then $v_{j_*} \notin \mathbb{Q}$ and $A_n \neq 0$ for all large n. The sign of Q_n on $(0, X_{\max}]$ is certified when $Q_n(X) = \sum_{l=0}^{h} \beta_l(n)X^l(X_1 - X)^{h-l}$ for a rational $X_1 > X_{\max}$ with all $\beta_l(n) \geq 0$ and one of them positive.*

The Beta ports of one shape move the row of a shape word along a ladder: the share function of the Beta port of order $r+1$ is that of order r times a rational factor, so the sums over a common finite set of integers satisfy finite operator relations in the order of the Beta ports and in the depth. Unwanted coordinates whose coefficients follow a bounded local recurrence in the depth are annihilated by a filter with integer polynomial coefficients, read off the determinants of that recurrence, and jets in a regulator follow a block-triangular recurrence with the same diagonal. A filter applied to a positive word stays positive as long as its polynomial keeps one sign on the support of the word's measure, which a nonnegative expansion in the products $X^l(X_1 - X)^{h-l}$ certifies, and then the filtered word separates the coordinate it keeps, at the cost of clearing the rows of $h+1$ consecutive values.

37.5 Growing Beta ports

Theorem 37.19 (An arithmetic budget for growing Beta ports)**.** *For $k \geq 2$, $\eta > 0$ and $q_n = \lfloor \eta n \rfloor$, in this statement write $\ell_k(\eta) = \log M_k(\eta) + (1+\eta)\log(1+\eta) - \eta\log\eta$. Then*

$$\lim_n \frac{1}{n}\log I_n\big(k; (-1)^{q_n}\Pi_{q_n}^{n+1,n+1}(x_1)\big) = \ell_k(\eta),$$

$$\limsup_n \frac{1}{n}\log\Big(d_{n+q_n}^{k+1}\, I_n\big(k; (-1)^{q_n}\Pi_{q_n}^{n+1,n+1}(x_1)\big)\Big) \leq \ell_k(\eta) + (k+1)(1+\eta)\,c_{\mathrm{lcm}}.$$

The integer $d_{n+q_n}^{k+1}$ clears the whole row, which stays on the coordinates $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, …, $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and a rational part, and the coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ is

$$k\binom{n+q_n}{q_n}\sum_{j=0}^{n}(-1)^{jk}\binom{n}{j}^k\binom{n+j}{n}^2\frac{(n+1+j)_{q_n}}{(n+1-j)_{q_n}},$$

positive for even k.

Corollary 37.20 (A growing Beta port with a certified gap)**.** *For every $k \geq 2$ there is a rational $\eta_0 > 0$ such that $\ell_k(\eta) + (k+1)(1+\eta)\, c_{\mathrm{lcm}} < 0$ for every rational $\eta \in (0, \eta_0]$ (Theorem 37.19), the root of Theorem 37.7 at each such η being certified as a root of an integer polynomial in a rational interval. So with $q_n = \lfloor \eta n \rfloor$ the positive words $I_n\big(k; (-1)^{q_n} \Pi_{q_n}^{n+1,n+1}(x_1)\big)$ satisfy*

$$\limsup_n \frac{1}{n} \log \Big(d_{n+q_n}^{k+1}\, I_n\big(k; (-1)^{q_n} \Pi_{q_n}^{n+1,n+1}(x_1)\big)\Big) < 0,$$

on the fixed coordinates $(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$, …, $(\mathsf{T}_{k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$ and a rational part, with a positive coefficient of $(\mathsf{T}_{k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$ when k is even.

A Beta port whose order grows in proportion to the depth pulls the word toward the port factor: the word decays at the maximum value $M_k(\eta)$ of $f_k \mathfrak{a}_k^{\eta}$, the binomial of the Beta port adds its entropy $(1+\eta)\log(1+\eta) - \eta \log \eta$, and the clearing grows to $d_{n+q_n}^{k+1}$. Where the sum is negative the cleared words tend to zero on a fixed set of coordinates, and for even k the top coordinate keeps a positive coefficient. For every number of shapes the margin is positive at small rational ratios, by continuity from the base word (Proposition 37.8), each certified by an explicit algebraic root, and the cleared positive words decay exponentially on the coordinates $(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)), \ldots, (\mathsf{T}_{k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$ with a rational part.

Part VII

Rays

Chapter 38

Rays, native gauges and tiers

A ray moves the data of a family along an integral direction. Along it the values of every insertion obey two relations, integration by parts in every mark and the shift by the stride, wherever the integrals converge and, when the walls of the direction are positive, at every strength as identities of meromorphic functions; these relations make the values a holonomic sequence: they satisfy a recurrence of order at most the dimension of the ray. The complex points at which the direction is native, the native gauges of the ray, carry formal solutions of the relations and bound the dimension from below, and the frame of any recurrence is a power times a stacked Gamma. Every ray with real origin is one function, carried by a pencil, with a branch at the native gauge of its direction, and on the centre ray of a family every sign is a candidate native gauge; a ray along a wall carries the product of the rays of its two faces. What a value can contain is bounded: finitely many values span every value along a ray, and a closing in rational terms carries Trailmoments of index at most the largest order of its poles (Theorem 38.24, Lemma 38.25).

38.1 The relations along a ray

Definition 38.1 (Ray, stride, insertions, values)**.** Let F be a family on $\mathcal{E}$. A *ray* of F is the sequence of data $(\mathbf{m}_0 + k\mathbf{m}_1;\ \mathbf{s}_0 + k\mathbf{s}_1;\ \mathbf{t}_0 + k\mathbf{t}_1)$, $k \in \mathbb{N}$, the points at the integer times of the half-line from its *origin* $(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0)$, which is complex, along its *direction* $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$, which is integral, $\mathbf{m}_1 \in \mathbb{Z}^{\mathcal{E}}$ and $s_{1j}, t_{1j} \in \mathbb{N}$, both on the balance wall; the walls along it are $\kappa_{0,S} + k\kappa_{1,S}$, with $\kappa_{0,S}$ and $\kappa_{1,S}$ the walls of the origin and of the direction. On the chart $y_{e_0} = 0$ of a fixed mark e_0, with $dy = \prod_{e \neq e_0} d\alpha_e/\alpha_e$, the *stride* and the *origin form* are

$$f = \alpha^{\mathbf{m}_1} \prod_j K_j^{-s_{1j}} \Psi_j^{-t_{1j}}, \qquad \omega_0 = \alpha^{\mathbf{m}_0} \prod_j K_j^{-s_{0j}} \Psi_j^{-t_{0j}}\, dy,$$

f a rational function of degree 0. The *letter complement* of F is $X_F = \{\alpha \in (\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times : K_j(\alpha)\Psi_j(\alpha) \neq 0 \text{ for every } j\}$, and the insertions along the ray are its regular functions, the algebra $\mathbb{C}[X_F]$, which is generated by the monomials α^v, $v \in \mathbb{Z}^{\mathcal{E}}$ with $v_{\mathcal{E}} = 0$, and the chances of the bases under the basis and cobasis laws, $\alpha \mapsto \mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha)$ and $\alpha \mapsto \mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$, $T \in \mathcal{B}_j$. The *values* along the ray are $\langle g\rangle_k = \int_{y_{e_0}=0} g\, f^k \omega_0$ for $g \in \mathbb{C}[X_F]$, and over a compact real $(n-1)$-dimensional chain D in X_F, $\langle g\rangle_{k;D} = \int_D g\, f^k \omega_0$. The *defects* of the origin and of the direction are the masses less the usages,

$$c_e^{(i)} = m_{i,e} - \sum_j \big(s_{ij} R_{j,e} + t_{ij} R^*_{j,e}\big), \qquad i = 0, 1,$$

with the occupation $R_{j,e}$ and the dual occupation $R^*_{j,e}$ of table j (Theorem 7.16).

Lemma 38.2 (Insertions). *(i) $\mathbb{C}[X_F]$ is closed under every ∂_{y_e}: $\partial_{y_e}\alpha^v = v_e\alpha^v$, $\partial_{y_e}\mathbb{P}_{j,\alpha}(T) = (\mathbf{1}_T(e) - R_{j,e})\,\mathbb{P}_{j,\alpha}(T)$ and $\partial_{y_e}\mathbb{P}^*_{j,\alpha}(T) = (\mathbf{1}_{\mathcal{E}\setminus T}(e) - R^*_{j,e})\,\mathbb{P}^*_{j,\alpha}(T)$; the occupations and the defects lie in $\mathbb{C}[X_F]$.*

(ii) The stride lies in $\mathbb{C}[X_F]$, and $c^{(1)}_e = \partial_{y_e}\log f$, $c^{(0)}_e + k\,c^{(1)}_e = \partial_{y_e}\log(f^k\omega_0/dy)$.

(iii) Every value $\langle g\rangle_k$ is a finite combination, with products of weights as coefficients, of periods P_F at data $(\mathbf{m}_0 + k\mathbf{m}_1 + v;\ \mathbf{s}_0 + k\mathbf{s}_1 + \mathbf{a};\ \mathbf{t}_0 + k\mathbf{t}_1 + \mathbf{b})$ on the balance wall, $v \in \mathbb{Z}^{\mathcal{E}}$ and $\mathbf{a}, \mathbf{b} \in \mathbb{N}^N$.

Theorem 38.3 (The relations along a ray). *If $\kappa_{1,S} > 0$ for every nonempty proper S, then for every insertion g the value $\langle g\rangle_k$ converges for all large k, and for every mark e*

$$\big\langle \big(c^{(0)}_e + k\,c^{(1)}_e + \partial_{y_e}\big)g\big\rangle_k = 0, \qquad \langle fg\rangle_k = \langle g\rangle_{k+1}.$$

In Theorem 38.3 the first relation is integration by parts in the mark e against the form $f^k\omega_0$, whose logarithmic derivative is the defect; the second is the definition of the stride. Proposition 38.5 carries both to every strength: wherever the integrals converge, whatever the walls of the direction, and, when these walls are positive, as identities of meromorphic functions of k. Both act on the insertions, which are closed under the derivatives, so the values of all insertions along a ray form one module over the operators in k, and every relation among insertions modulo these two is a recurrence among values.

38.2 Native gauges carry solutions

Definition 38.4 (Native gauges of a ray; the dimension). A *native gauge* of a ray is a point α_c of X_F at which the direction is native, $c^{(1)}(\alpha_c) = 0$: a native gauge of the direction at a complex point (Definition 8.17); its *native value* is $f(\alpha_c)$, and it is *nondegenerate* when the Hessian of $\log f$ in y modulo $\mathbb{C}\mathbf{1}$ is nonsingular there. Near a nondegenerate native gauge there are holomorphic coordinates $v \in \mathbb{C}^{n-1}$ with $\log f = \log f(\alpha_c) - \frac{1}{2}\sum_i v_i^2$; the image D_c of a real ball $\{v \in \mathbb{R}^{n-1} : |v| \le \varepsilon\}$ is a *steepest-descent disc*. The *relation space* $\mathcal{L} \subseteq \mathbb{C}[X_F] \otimes \mathbb{C}(k)$ is the $\mathbb{C}(k)$-span of the elements $(c^{(0)}_e + k\,c^{(1)}_e + \partial_{y_e})g$, and the *dimension* of the ray is $\dim_{\mathbb{C}(k)}\big(\mathbb{C}[X_F] \otimes \mathbb{C}(k)\big)/\mathcal{L}$.

Proposition 38.5 (The relations at every strength). *Let a ray of a family F be given. For an insertion $g = \alpha^u \prod_j K_j^{-a_j}\Psi_j^{-b_j}$ with $u \in \mathbb{Z}^{\mathcal{E}}$, $\mathbf{a}, \mathbf{b} \in \mathbb{N}^N$ and $u_{\mathcal{E}} = \sum_j (a_j r_j + b_j r_j^*)$, such insertions spanning $\mathbb{C}[X_F]$, write $\kappa_U(g) = u_U - \sum_j \big(a_j d_j(U) + b_j d_j^*(U)\big)$ for the walls of its data $(u; \mathbf{a}, \mathbf{b})$ (Definition 8.1).*

(i) Whatever the walls of the direction, at every complex k with $\operatorname{Re}\kappa_{0,U} + \kappa_U(g) + \operatorname{Re}(k)\,\kappa_{1,U} > 0$ for every nonempty proper U the values $\langle g\rangle_k$ and $\big\langle (c^{(0)}_e + k\,c^{(1)}_e + \partial_{y_e})g\big\rangle_k$ converge absolutely, and the second is 0 for every mark e. So if polynomials $p_0, \dots, p_h \in \mathbb{C}[k]$, an insertion g' and finite combinations G_e of insertions of the above kind, with coefficients in $\mathbb{C}(k)$, satisfy

$$\sum_{j=0}^{h} p_j(k)\, f^j g' = \sum_{e\in\mathcal{E}} \big(c^{(0)}_e + k\,c^{(1)}_e + \partial_{y_e}\big)G_e,$$

an explicit element of the relation space $\mathcal{L}$ (Definition 38.4), a certificate *of the recurrence, then $\sum_j p_j(k)\langle g'\rangle_{k+j} = 0$ at every k off the poles of those coefficients at which the values $\langle g'\rangle_{k+j}$ converge absolutely and the condition above holds for every insertion in the G_e.*

(ii) If $\kappa_{1,S} > 0$ for every nonempty proper S, every value $\langle g\rangle_k$ continues to a meromorphic function of $k \in \mathbb{C}$, whose poles have order at most $n-1$ and lie among the k with $\kappa_{0,U} + \kappa_U(g) + k\,\kappa_{1,U} \in \{0, -1, -2, \dots\}$ for some nonempty proper U. For every element $\sum_l r_l(k)\, g_l$ of $\mathcal{L}$ the continuation of $\sum_l r_l(k)\langle g_l\rangle_k$ vanishes identically, and $\langle fg\rangle_k = \langle g\rangle_{k+1}$ identically. So a recurrence $\sum_j p_j(k)\langle g\rangle_{k+j} = 0$ that follows from the relations, its element $\sum_j p_j(k) f^j g$ lying in $\mathcal{L}$, holds as an identity of meromorphic functions of k, and at every k at which its terms converge.

Theorem 38.6 (A native gauge carries a solution)**.** *Let α_c be a nondegenerate native gauge of a ray and D_c a steepest-descent disc through it.*

(i) As $k \to \infty$,

$$\langle g\rangle_{k;D_c} \sim f(\alpha_c)^k\, k^{-(n-1)/2} \sum_{i\geq 0} a_i(g)\, k^{-i},$$

where $a_0(g) = (2\pi)^{(n-1)/2}(\omega_0/dv)(\alpha_c)\, g(\alpha_c)$ with $(\omega_0/dv)(\alpha_c) \neq 0$, and each $a_i(g)$ is a finite combination of Taylor coefficients of $g\,\omega_0/dv$ and of $\log f$ at α_c. Further $\langle fg\rangle_{k;D_c} = \langle g\rangle_{k+1;D_c}$, and $\big\langle (c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})g\big\rangle_{k;D_c}$ is at most a constant times $|f(\alpha_c)|^k e^{-k\varepsilon^2/2}$; so the expansions satisfy both relations to all orders.

(ii) For a finite set C of distinct nondegenerate native gauges α_c, the maps $A^{(c)} : \mathbb{C}[X_F] \otimes \mathbb{C}(k) \to \mathbb{C}((1/k))$, $g \mapsto \sum_l a_l^{(c)}(g)\, k^{-l}$, extended $\mathbb{C}(k)$-linearly, vanish on $\mathcal{L}$ and are linearly independent over $\mathbb{C}((1/k))$.

(iii) The dimension of the ray is at least the number of its nondegenerate native gauges.

Corollary 38.7 (The symbol)**.** *Let g be an insertion and $p_0, \dots, p_h \in \mathbb{C}[k]$, not all zero, with $\sum_j p_j(k)\, f^j g \in \mathcal{L}$. Under the hypothesis of Theorem 38.3, $\sum_j p_j(k)\, \langle g\rangle_{k+j} = 0$ for all large k. With d the largest degree of the p_j and c_j the coefficient of k^d in p_j, the* symbol *$\sum_j c_j \lambda^j$ vanishes at the native value of every nondegenerate native gauge α_c with $g(\alpha_c) \neq 0$. So h is at least the number of distinct native values of such native gauges.*

Proposition 38.8 (The top and the bottom of a certificate)**.** *Let g be an insertion and $\sum_{j=0}^h p_j(k)\, f^j g = \sum_e \big(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}\big) G_e$ a certificate (Proposition 38.5(i)) in which the $p_j \in \mathbb{C}[k]$ have the largest degree $d \geq 1$ and every $G_e = \sum_{i<d} k^i\, G_e^{[i]}$ is a polynomial in k of degree less than d with coefficients $G_e^{[i]} \in \mathbb{C}[X_F]$. With c_j the coefficient of k^d in p_j and the symbol $\chi(\lambda) = \sum_j c_j \lambda^j$ (Corollary 38.7),*

$$\chi(f)\, g = \sum_e c_e^{(1)}\, G_e^{[d-1]}, \qquad \sum_{j=0}^h p_j(0)\, f^j g = \sum_e \big(c_e^{(0)} + \partial_{y_e}\big) G_e^{[0]}.$$

So χ vanishes at the native value of every native gauge α_c of the ray with $g(\alpha_c) \neq 0$, nondegenerate or not, and for the origin 0 the bottom piece $\sum_j p_j(0) f^j g$ is the divergence $\sum_e \partial_{y_e} G_e^{[0]}$.

Proposition 38.9 (The native value bounds the order)**.** *Let a ray of a family have $\kappa_{1,S} > 0$ for every nonempty proper S. Then $\langle 1\rangle_k \neq 0$ for all large k, positive when the origin is real, and $\langle 1\rangle_{k+1}/\langle 1\rangle_k \to f(\alpha^\star)$, the native value of the native gauge $\alpha^\star$ of the direction on $\mathbb{R}^{\mathcal{E}}_{>0}$, which is the largest value of f there. If $\sum_{j=0}^h p_j(k)\, \langle 1\rangle_{k+j} = 0$ for all large k, with $p_j \in \mathbb{Q}[k]$ not all zero, then the symbol vanishes at $f(\alpha^\star)$: $f(\alpha^\star)$ is algebraic, $h \geq [\mathbb{Q}(f(\alpha^\star)) : \mathbb{Q}]$, and every conjugate of $f(\alpha^\star)$ is a root of the symbol. In particular, when $f(\alpha^\star)$ is irrational the values $\langle 1\rangle_k$ satisfy no recurrence of order one over $\mathbb{Q}(k)$, and when it is transcendental none at all.*

Proposition 38.10 (The conjugates of the native value of the direction)**.** *Let a ray of a family whose weights lie in a subfield $\Bbbk$ of $\mathbb{C}$ have $\kappa_{1,S} > 0$ for every nonempty proper S, and let $\alpha^\star$ be the native gauge of its direction on the positive orthant, $\alpha^\star_{e_0} = 1$. Then the coordinates of $\alpha^\star$ are algebraic over $\Bbbk$, and for every embedding τ of $\Bbbk(\alpha^\star)$ in $\mathbb{C}$ over $\Bbbk$ the point $\tau(\alpha^\star)$ is a nondegenerate native gauge of the ray, of native value $\tau\big(f(\alpha^\star)\big)$. So every conjugate of $f(\alpha^\star)$ over $\Bbbk$ is the native value of a nondegenerate native gauge, the dimension of the ray is at least $[\Bbbk(\alpha^\star) : \Bbbk]$, and the symbol of every recurrence for $\langle 1\rangle_k$ that follows from the relations vanishes at every conjugate of $f(\alpha^\star)$.*

Proposition 38.11 (The frame of a recurrence is a power times a stacked Gamma)**.** *Let $\sum_{j=0}^{h} p_j(k)\, x_{k+j} = 0$ for $k \geq k_0$, with $p_j \in \mathbb{C}[k]$ and p_0, p_h without zeros at the integers $k \geq k_0$, and let $x^{(1)}, \dots, x^{(h)}$ be solutions. Their* frame $C_k = \det\big[x^{(i)}_{k+l}\big]_{1\leq i\leq h,\ 0\leq l<h}$ *satisfies $C_{k+1} = (-1)^h \frac{p_0(k)}{p_h(k)}\, C_k$, and writing $\frac{p_0(k)}{p_h(k)} = c\, \frac{\prod_{i=1}^{\deg p_0}(k-a_i)}{\prod_{i=1}^{\deg p_h}(k-b_i)}$,*

$$C_k = C_{k_0}\big((-1)^h c\big)^{k-k_0}\, \Gamma_{\substack{\deg p_0\\ \deg p_h}}\begin{bmatrix} k-a_1,\ \dots \\ k-b_1,\ \dots \end{bmatrix} \Big/ \Gamma_{\substack{\deg p_0\\ \deg p_h}}\begin{bmatrix} k_0-a_1,\ \dots \\ k_0-b_1,\ \dots \end{bmatrix}.$$

The tier *of a sequence over a field $\Bbbk \subseteq \mathbb{C}$ is the least order of a recurrence with coefficients in $\Bbbk[k]$ and $p_0 p_h \neq 0$ that it satisfies for large k; a sequence of tier one is a power times a stacked Gamma, and so is every frame.*

Proposition 38.12 (The growth of the frame)**.** *In Proposition 38.11 let $C_{k_0} \neq 0$, let d_0 and d_h be the degrees of p_0 and p_h, and let c be the quotient of their leading coefficients. As $k \to \infty$,*

$$\log|C_k| = k\log|c| + (d_0 - d_h)(k\log k - k) + O(\log k).$$

When $d_0 = d_h$ is the largest degree of the p_j, the symbol $\sum_j c_j \lambda^j$ (Corollary 38.7) has the degree h and $c_0 \neq 0$, the product of its roots is $(-1)^h c$, and $|C_k|^{1/k} \to |c|$.

A native gauge of a ray is a complex point at which the direction uses every mark exactly as its masses ask, the complex counterpart of the native gauge of a family. Through it a steepest-descent disc carries a value that satisfies both relations up to exponentially small terms, and its asymptotic expansion is a formal solution; distinct native gauges give independent solutions, so their number bounds the dimension, and their native values are roots of the symbol of every recurrence that follows from the relations. On the real orthant the native gauge of the direction governs the growth of the values, and an irrational native value forbids a recurrence of order one. Whatever the order, the frame of a recurrence is a power times a stacked Gamma: the sequences of tier one are the bottom of the hierarchy of sequences that the relations produce.

38.3 The centre ray of a family

Definition 38.13 (The centre ray)**.** Let F be a family of N tables on $n \geq 2$ marks, and let strengths $\sigma_j \in \mathbb{N}$, not all zero, make it connected: every nonempty proper S has $c_j(S) > 0$ for some j with $\sigma_j > 0$ (Definition 8.1); put $\boldsymbol{\sigma} = (\sigma_j)_j$ and $\sigma = \sum_j \sigma_j$. The *centre ray* of F at the strengths $\boldsymbol{\sigma}$ is its ray with origin 0 and direction $(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$. Its stride is

$$f = \alpha^{\sigma\mathbf{1}} \prod_j \big(K_j \Psi_j\big)^{-\sigma_j} = \prod_j \big(K_j(e^y) K_j(e^{-y})\big)^{-\sigma_j},$$

its values are $\langle 1\rangle_k = \int_{y_{e_0}=0} f^k\,dy = P_F(k\sigma\mathbf{1}; k\boldsymbol{\sigma}, k\boldsymbol{\sigma})$, $k \geq 1$, and its direction walls are $\kappa_{1,S} = \sum_j \sigma_j c_j(S) > 0$ (Proposition 8.21).

Theorem 38.14 (A ray is one function)**.** *Let a ray of a family on $n \geq 2$ marks have real origin and $\kappa_{1,S} > 0$ for every nonempty proper S, let $\alpha^\star$ be the native gauge of its direction on $\mathbb{R}^{\mathcal{E}}_{>0}$, with $\alpha^\star_{e_0} = 1$ (Theorem 8.19), and let $k_0 \geq 0$ be an integer with $\kappa_{0,S} + k_0\kappa_{1,S} > 0$ for every nonempty proper S.*

(i) $0 < f \leq f(\alpha^\star)$ on $\mathbb{R}^{\mathcal{E}}_{>0}$, with equality only at $\alpha \propto \alpha^\star$, and $f \leq C'e^{-c'\|y\|}$ on the chart for some $c', C' > 0$; every value $\langle 1\rangle_k$, $k \geq k_0$, converges and is positive.

(ii) For $|q| < f(\alpha^\star)^{-1}$ the generating function $G(q) = \sum_{k\geq k_0}\langle 1\rangle_k q^{k-k_0}$ is

$$G(q) = \int_{y_{e_0}=0} \frac{f^{k_0}\omega_0}{1-qf} = \int_{y_{e_0}=0} \frac{\alpha^{\mathbf{m}_0+k_0\mathbf{m}_1}\prod_j K_j^{-s_{0j}-(k_0-1)s_{1j}}\Psi_j^{-t_{0j}-(k_0-1)t_{1j}}}{\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}} - q\,\alpha^{\mathbf{m}_1}}\,dy,$$

$$G^{(l)}(q) = l!\int_{y_{e_0}=0} \frac{f^{k_0+l}\omega_0}{(1-qf)^{l+1}},$$

and the positive integral converges for every real $q < f(\alpha^\star)^{-1}$: the whole ray is carried by the pencil $\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}} - q\,\alpha^{\mathbf{m}_1}$.

(iii) For $k \geq k_0$ and $h \geq 1$,

$$\det\big[\langle 1\rangle_{k+i+i'}\big]_{i,i'=0}^{h-1} = \frac{1}{h!}\int \prod_{a=1}^{h} f(y_a)^k\,\frac{\omega_0}{dy}(y_a)\prod_{a<b}\big(f(y_a)-f(y_b)\big)^2\prod_a dy_a > 0,$$

and G is not a rational function of q.

On the centre ray of a family (Definition 38.13) one may take $k_0 = 1$, and $\alpha^\star = \mathbf{1}$.

Theorem 38.15 (The branch at the native gauge of the direction)**.** *For a ray as in Theorem 38.14, with its generating function G and the native gauge $\alpha^\star$ of its direction, put $A = f(\alpha^\star)^{-1}$, $\varepsilon = 1 - q/A$ and $a = \frac{n-1}{2}$, and let H be the Hessian of $-\log f$ at $\log\alpha^\star$ on the chart,*

$$H = \Big(\sum_j \big(s_{1j}\,\kappa_{j,\alpha^\star}(X_e, X_{e'}) + t_{1j}\,\kappa_{j,(\alpha^\star)^{-1}}(X_e, X_{e'})\big)\Big)_{e,e'\neq e_0},$$

the covariances of the indicators under the basis laws and the cobasis laws at $\alpha^\star$ (Definition 7.17, Theorem 7.16); it is positive definite. As $q \uparrow A$ there are functions U, V holomorphic at $\varepsilon = 0$ with

$$G(q) = U(\varepsilon) + \varepsilon^{a-1}V(\varepsilon) \quad (n \text{ even}), \qquad G(q) = U(\varepsilon) + \varepsilon^{a-1}\log\varepsilon\,V(\varepsilon) \quad (n \text{ odd}),$$

$$V(0) = \frac{\Gamma(1-a)\,(2\pi)^a\,(f^{k_0-1}\omega_0/dy)(\alpha^\star)}{A\sqrt{\det H}} \quad (n \text{ even}),$$

$$V(0) = \frac{(-1)^a(2\pi)^a\,(f^{k_0-1}\omega_0/dy)(\alpha^\star)}{\Gamma(a)\,A\sqrt{\det H}} \quad (n \text{ odd}).$$

The derivatives $G^{(l)}(A)$ are finite exactly for $l < a-1$, and

$$G(q) = \sum_{l<a-1}\frac{G^{(l)}(A)}{l!}\big(q-A\big)^l + V(0)\,\varepsilon^{a-1}\big(1 \text{ or } \log\varepsilon\big) + O\big(\varepsilon^{\lceil a-1\rceil}\big).$$

On the centre ray of a family, with $k_0 = 1$, $A = \prod_j K_j(\mathbf{1})^{2\sigma_j}$, the density $f^{k_0-1}\omega_0/dy$ is 1, and $H = 2\sum_j \sigma_j\big(\kappa_{j,\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\neq e_0}$, the covariances under the weight laws.

Proposition 38.16 (The native gauges of a centre ray)**.** *On the centre ray of a family at the strengths $\boldsymbol{\sigma}$ (Definition 38.13) the defects of the direction are*

$$c_e^{(1)}(\alpha) = \sum_j \sigma_j \big(R_{j,e}(\alpha^{-1}) - R_{j,e}(\alpha)\big),$$

so a point α of X_F is a native gauge of the ray exactly when the occupations weighted by the strengths agree at α and at α^{-1}, $\sum_j \sigma_j R_j(\alpha) = \sum_j \sigma_j R_j(\alpha^{-1})$. Its native value is $\prod_j \big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$.

Theorem 38.17 (Native gauges at the signs)**.** *For the centre ray of a family at the strengths $\boldsymbol{\sigma}$ (Definition 38.13), on the chart $\alpha_{e_0} = 1$, with the family word $W = \prod_j K_j^{-\sigma_j}\Psi_j^{-\sigma_j}$ at the exponents $(\boldsymbol{\sigma}, \boldsymbol{\sigma})$, so that $f = \alpha^{\sigma\mathbf{1}}W$:*

(i) $f(\alpha^{-1}) = f(\alpha)$; the points fixed by $\alpha \mapsto \alpha^{-1}$ are the 2^{n-1} signs $\eta \in \{\pm 1\}^{\mathcal{E}}$, $\eta_{e_0} = 1$; every native gauge of the centre ray is a sign or is paired with its image, a native gauge of the same native value.

(ii) A sign with $K_j(\eta) = \sum_T \theta_{j,T}\eta^T \neq 0$ for every j is a native gauge, of native value $\prod_j K_j(\eta)^{-2\sigma_j}$. In the chart $\alpha_e = \eta_e e^{y_e}$, $y_{e_0} = 0$, the Hessian of $-\log f$ at $y = 0$ is $2\sum_j \sigma_j \big(\kappa_{j,\eta}(X_e, X_{e'})\big)_{e,e' \neq e_0}$, with $\kappa_{j,\eta}$ the covariances of the indicators under the signed measure $\mathbb{P}_{j,\eta}(T) = \theta_{j,T}\eta^T/K_j(\eta)$ of total mass one at the sign.

(iii) The critical polynomials $\sigma W^{-1} - \alpha_e \partial_e W^{-1}$, $e \neq e_0$, vanish identically modulo $\alpha_e^2 = 1$, and $W^{-1} \equiv \alpha^{\sigma\mathbf{1}} \prod_j K_j^{2\sigma_j}$ there. In $\mathbb{Q}(\boldsymbol{\theta})[\alpha_e : e \neq e_0]/(\alpha_e^2 - 1)$, $\mathbb{Q}(\boldsymbol{\theta})$ the field of the weights, localized at $\prod_j K_j$, the idempotents $2^{1-n}\prod_{e \neq e_0}(1+\eta_e\alpha_e)$ of the signs with $K_j(\eta) \neq 0$ for every j form a basis, and multiplication by $W^{-1}/\alpha^{\sigma\mathbf{1}}$ acts on the idempotent of η by $\prod_j K_j(\eta)^{2\sigma_j}$.

Theorem 38.18 (Weights that separate the signs)**.** *Let a family of N tables on $n \geq 2$ marks be connected at the strengths $\sigma_j \in \mathbb{N}$, of total σ, with its centre ray (Definition 38.13). There are weights $\theta_{j,T} \in \big\{1, \ldots, 2^{n-1}\big((2n-1)N + (2^{n-1}-1)\sigma\big) + 1\big\}$ for which every sign has $K_j(\eta) \neq 0$ for every j and is a nondegenerate native gauge of the centre ray, and the 2^{n-1} native values $\prod_j K_j(\eta)^{-2\sigma_j}$ are distinct. For such weights the dimension of the centre ray is at least 2^{n-1}, and every recurrence for $\langle 1\rangle_k$ that follows from the relations (Corollary 38.7) has order at least 2^{n-1}.*

Along a ray with real origin the whole sequence of values is one function of the pencil parameter, holomorphic up to the reciprocal of the native value of the direction. The determinants $\det\big[\langle 1\rangle_{k+i+i'}\big]$ of the values are positive integrals of squared differences, so the generating function is not rational; at the native gauge of the direction it branches, with the exponent set by the number of marks and the coefficient by the value of $f^{k_0-1}\omega_0/dy$ there and the determinant of the covariances of the basis and cobasis laws. The centre ray of a family is its period at the centre read at every multiple of its strengths; its native gauge is the table's gauge, where it branches at $\prod_j K_j(\mathbf{1})^{2\sigma_j}$ with the covariances of the weight laws. Every sign is a candidate native gauge, the signed measures $\mathbb{P}_{j,\eta}$ playing the part of the weight laws, and with weights chosen to separate the signs every recurrence that the relations produce for the centre ray of a family on n marks has order at least 2^{n-1}.

38.4 What a wall carries

Lemma 38.19 (Where native gauges can reach a wall)**.** *Let B be a proper bunch with room $R \ni e_0$, $\alpha_B = \rho u$, and in this statement write $K_j^{[l]} = K_{j,B}^{(d_j(B)+l)}(u, \alpha_R)$ and $\Psi_j^{[l]} = \Psi_{j,B}^{(d_j^*(B)+l)}(u, \alpha_R)$, so*

that $K_j = \rho^{d_j(B)} \sum_l \rho^l K_j^{[l]}$ *and* $\Psi_j = \rho^{d_j^*(B)} \sum_l \rho^l \Psi_j^{[l]}$ *(Theorem 16.25), with the bottom letters* $K_j^{[0]}$ *and* $\Psi_j^{[0]}$.

(i) At every native gauge α *of a direction* $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$ *with* $s_{1j}, t_{1j} \geq 0$, $m_{1,B} = \sum_j \big(s_{1j} \sum_T |T \cap B|\, \mathbb{P}_{j,\alpha}(T) + t_{1j} \sum_T |B \setminus T|\, \mathbb{P}^*_{j,\alpha}(T) \big)$*: the direction's mass on* B *is its usage of* B.

(ii) Let native gauges of such directions $\mathbf{v}_i \to \mathbf{v}$ *have bunch scales* $\rho_i \to 0$ *and* $(u_i, \alpha_{R,i}) \to (u^*, \alpha_R^*)$ *in the torus, with* $\kappa_{1,B}(\mathbf{v}) > 0$. *Then some bottom letter vanishes at* (u^*, α_R^*), *and if only one does, its gradient in* (u, α_R) *vanishes there as well: the product* $\prod_j K_j^{[0]} \Psi_j^{[0]}$ *of the bottom letters vanishes at* (u^*, α_R^*) *together with its gradient in* (u, α_R).

So the native gauges of a family leave for the wall of a bunch only at singular points of the zero set of the product of its bottom letters.

Theorem 38.20 (A ray along a wall carries its faces)**.** *Let* B *be a proper bunch with room* $R \ni e_0$ *on whose bottom levels the weights factor, and let a ray have its direction on the wall,* $\kappa_{1,B} = 0$, *and* $\operatorname{Re} \kappa_{0,B} > 0$. *With* $\alpha_B = \rho u$, *and writing in this statement* $K_j^{[l]} = K_{j,B}^{(d_j(B)+l)}(u, \alpha_R)$ *and* $\Psi_j^{[l]} = \Psi_{j,B}^{(d_j^*(B)+l)}(u, \alpha_R)$ *as in Theorem 16.25,*

$$f = f_B(u)\, f_R(\alpha_R) \big(1 + \rho\, h_1 + O(\rho^2)\big), \qquad h_1 = -\sum_j \Big(s_{1j} \frac{K_j^{[1]}}{K_j^{[0]}} + t_{1j} \frac{\Psi_j^{[1]}}{\Psi_j^{[0]}} \Big),$$

where f_B *and* f_R *are the strides along the ray of the faces* F^B *and* F^R *of Theorem 16.25, and* $\omega_0 = \rho^{\kappa_{0,B}-1} d\rho \wedge \omega_0^B \wedge \omega_0^R\, (1 + O(\rho))$ *with the origin forms of the faces. Let* α_c^B *and* α_c^R *be nondegenerate native gauges of the rays of the faces with* $h_1(\alpha_c^B, \alpha_c^R) \neq 0$, *let* $\operatorname{Re}\big(e^{i\vartheta} h_1\big) < 0$ *near* (α_c^B, α_c^R), *and let* $D = e^{i\vartheta}[0, \rho_1] \times D_B \times D_R$ *with steepest-descent discs* D_B, D_R *of the faces. Then the values* $\langle g \rangle_{k;D}$ *satisfy both relations up to terms exponentially smaller than* $|f_B(\alpha_c^B) f_R(\alpha_c^R)|^k$, *and*

$$\langle 1 \rangle_{k;D} = \Gamma(\kappa_{0,B}) \big(-k\, h_1(\alpha_c^B, \alpha_c^R) \big)^{-\kappa_{0,B}} \langle 1 \rangle_{k;D_B}^{F^B} \langle 1 \rangle_{k;D_R}^{F^R} \big(1 + O(1/k)\big),$$

with $\langle 1 \rangle_{k;D_B}^{F^B}$ *and* $\langle 1 \rangle_{k;D_R}^{F^R}$ *the values of the rays of the faces over their discs (Theorem 38.6), and the power taken continuously along* $e^{i\vartheta}(0, \infty)$.

Definition 38.21 (The whole cluster port)**.** On a bunch scale ρ take letters with *lower coefficients* $\boldsymbol{\lambda}_i \in \mathbb{C}^{r_i}$,

$$L_i(\rho, \boldsymbol{\lambda}_i) = \sum_{l < r_i} \lambda_{i,l}\, \rho^l + \rho^{r_i} C_i(\rho), \qquad C_i(0) \neq 0,$$

with exponents w_i, and a word $\rho^{\kappa-1} G(\rho) \prod_i L_i^{-w_i}$, with G and the C_i holomorphic on a neighbourhood of $[0, \infty)$, each $|C_i|$ bounded below by a positive constant there, and the word integrable at ∞; put $\kappa^\flat = \kappa - \sum_i r_i w_i$. Fix $b > 0$ with every C_i zero-free on $|\rho| \leq b$; for $\boldsymbol{\lambda}$ in a small polydisc each L_i has exactly r_i roots in $|\rho| < b$, the *cluster*, none on $|\rho| = b$ and none on $[b, \infty)$. The *cluster port* of the word is its integral over the *cluster contour*, which comes in along the half-line $[b, \infty)$ of the scale, goes once around $|\rho| = b$ and goes out on the new sheet:

$$\operatorname{Port}_\kappa \Big[G \prod_i L_i^{-w_i} \Big](\boldsymbol{\lambda}) = \frac{1}{2\pi i} \Big[(e^{2\pi i \kappa^\flat} - 1) \int_b^\infty \rho^{\kappa-1} G \prod_i L_i^{-w_i}\, d\rho + \oint_{|\rho|=b} \rho^{\kappa-1} G \prod_i L_i^{-w_i}\, d\rho \Big].$$

Theorem 38.22 (Relations of the whole cluster port)**.** *(i) One turn around* $|\rho| = b$ *multiplies the word by* $e^{2\pi i \kappa^\flat}$; *the cluster port does not depend on* b, *is holomorphic in all the lower*

coefficients jointly on the polydisc, and the contour has no end at which the word is nonzero, so every integral of a ρ-derivative over it vanishes.

(ii) On the stratum $\boldsymbol{\lambda}=0$ the word is $\rho^{\kappa^\flat-1}h$, with the stratum word $h=G\prod_i C_i^{-w_i}$, *and*

$$\operatorname{Port}_\kappa\Big[G\prod_i L_i^{-w_i}\Big](0)=\frac{e^{2\pi i\kappa^\flat}-1}{2\pi i}\operatorname{Reg}\int_0^\infty \rho^{\kappa^\flat-1}h\,d\rho\quad(\kappa^\flat\notin\mathbb{Z}),$$

$$\operatorname{Port}_\kappa\Big[G\prod_i L_i^{-w_i}\Big](0)=[\rho^{-\kappa^\flat}]\,h\quad(\kappa^\flat\in\mathbb{Z}_{\le 0}),\qquad \operatorname{Port}_\kappa\Big[G\prod_i L_i^{-w_i}\Big](0)=0\quad(\kappa^\flat\in\mathbb{Z}_{>0}).$$

(iii) For multi-indices $\mathbf{a}_i$, with $A_i=|\mathbf{a}_i|$ and $D=\sum_i\sum_l(r_i-l)\,a_{i,l}$, the derivative $\partial^{\mathbf{a}}_{\boldsymbol{\lambda}}$ of the cluster port at $\boldsymbol{\lambda}=0$ is $\prod_i(-1)^{A_i}(w_i)_{A_i}$ times the value of (ii) for the stratum word $G\prod_i C_i^{-w_i-A_i}$ at the index $\kappa^\flat-D$; at $\kappa^\flat=0$ it is $\prod_i(-1)^{A_i}(w_i)_{A_i}\cdot\frac{1}{D!}\,\partial_\rho^D\big[G\prod_i C_i^{-w_i-A_i}\big]_{\rho=0}$.

Proposition 38.23 (Small lower coefficients on one parameter)**.** *Let $C_1,\dots,C_p$ be bounded below by a positive constant on $[0,\infty)$, G and the C_i holomorphic near 0 and continuous on $[0,\infty)$, $\xi_1,\dots,\xi_p\in\mathbb{C}\setminus(-\infty,0]$, $\operatorname{Re}\kappa>0$ and $\kappa^\flat=\kappa-|\mathbf{w}|\notin\mathbb{Z}$, with $\rho^{\kappa-1}G\prod_i(\rho C_i)^{-w_i}$ integrable at ∞ together with every $\rho^{\kappa^\flat-|\mathbf{a}|-1}G\prod_i C_i^{-w_i-a_i}$, $\mathbf{a}\in\mathbb{N}^p$.*

(i) For small $\lambda>0$,

$$\int_0^\infty \rho^{\kappa-1}G(\rho)\prod_i\big(\lambda\xi_i+\rho\,C_i(\rho)\big)^{-w_i}d\rho=A(\lambda)+\lambda^{\kappa^\flat}B(\lambda)$$

with A,B holomorphic at 0,

$$A(\lambda)=\sum_{\mathbf{a}\in\mathbb{N}^p}\prod_i\binom{-w_i}{a_i}(\lambda\xi_i)^{a_i}\operatorname{Reg}\int_0^\infty\rho^{\kappa^\flat-|\mathbf{a}|-1}G\prod_i C_i^{-w_i-a_i}\,d\rho,$$

$$B(0)=G(0)\int_0^\infty u^{\kappa-1}\prod_i\big(\xi_i+u\,C_i(0)\big)^{-w_i}\,du,$$

the last integral continued in κ and $\mathbf{w}$; where every $\operatorname{Re}w_i>0$ it is

$$\prod_i C_i(0)^{-w_i}\,\Gamma_{\!p}^{2}\begin{bmatrix}\kappa,\ |\mathbf{w}|-\kappa\\ \mathbf{w}\end{bmatrix}\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}\big((\xi_i/C_i(0))_i\big),\qquad J=\{1,\dots,p\}$$

(Proposition 15.1). The analytic part opens with the finite part of the stratum word along the whole scale, and the branch part with $G(0)$ times the scale integral of the letters linearized at the wall.

(ii) When G and the C_i are holomorphic on a neighbourhood of $[0,\infty)$, the cluster port of Definition 38.21 for these letters, $r_i=1$ with the lower coefficients $\lambda\xi_i$, is $\frac{e^{2\pi i\kappa^\flat}-1}{2\pi i}A(\lambda)$.

At a wall the ray splits: the stride is the product of the strides of the two faces up to a first-order correction in the scale of the bunch, and integrating that scale against the correction contributes one Gamma of the origin wall, so a ray along a wall carries the product of the native values of its faces. The native gauges of a family reach a wall only where the product of its bottom letters is singular. When the bottom letters of a bunch are perturbed by lower coefficients, the scale integral is read on the cluster contour, which surrounds the whole cluster of roots at once: the cluster port is holomorphic in the lower coefficients, its value on the stratum where they vanish is the finite

part of the stratum word, and its derivatives there are the finite parts of the stratum words with raised exponents. With small lower coefficients on one parameter the scale integral splits into an analytic part, the cluster port up to a constant, and a branch part led by the scale integral of the letters linearized at the wall, a stacked Gamma times a Breta of one bunch.

38.5 What a value can contain

Theorem 38.24 (The values along a ray)**.** *Let a ray of a family have weights and origin in a subfield $\Bbbk$ of $\mathbb{C}$, with $\kappa_{1,S} > 0$ for every nonempty proper S.*

(i) The dimension of the ray is finite: it is the dimension over $\mathbb{C}(k)$ of the top cohomology of the twisted algebraic de Rham complex of the letter complement X_F, which is finite. It is at least the number of nondegenerate native gauges of the ray (Theorem 38.6). The native gauges are the critical points of the stride f, a product of powers of the letters with the exponents of the direction; for a direction outside a countable union of proper subvarieties they are $(-1)^{n-1}\operatorname{eu}(X_F)$ in number, eu *the Euler characteristic, and all nondegenerate, so that the dimension is then at least $(-1)^{n-1}\operatorname{eu}(X_F)$. An integral direction can lie in that exceptional set.*

(ii) For every insertion g with coefficients in $\Bbbk$ there are $p_0, \dots, p_h \in \Bbbk[k]$, $p_h \neq 0$, h at most the dimension, with $\sum_j p_j(k)\,\langle g\rangle_{k+j} = 0$ for all large k.

(iii) So there is k_1 such that every value $\langle g\rangle_k$, $k \geq k_1$, is a $\Bbbk$-linear combination of the h values $\langle g\rangle_{k_1}, \dots, \langle g\rangle_{k_1+h-1}$: the values along a ray span a $\Bbbk$-space of dimension at most that of the ray, and whatever Trailmoments those h values contain, no later value contains others.

Lemma 38.25 (Index by pole count)**.** *Let $\Bbbk$ be a subfield of $\mathbb{C}$, and let $r \in \Bbbk(k)$ have all its poles at negative integers, with largest order p, and $r(k) = O(k^{-2})$. Then $\sum_{k\geq 0} r(k) \in \Bbbk + \sum_{j=2}^{p} \Bbbk\,(\mathsf{T}_{j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. If moreover $r(-M-k) = -r(k)$ for an integer M, only odd j occur. If the poles lie in one coset $-b+\mathbb{Z}$, $b \in \mathbb{Q}$, the same holds with the Trailmoments of the lattice input $\mathbb{I}_{b+\mathbb{Z}}$ at its cuts; poles in several cosets leave, besides, differences of the index-one values of the cosets.*

A closing whose terms are rational functions of the lattice index brings Trailmoments of the counting input of index at most the largest order of its poles, and a reflection of the terms keeps only the odd indices. With Theorem 38.24 this is what bounds the complexity of a family along a ray: the space of values is finite-dimensional, spanned by finitely many of them, and where a closing writes the values as sums of rational terms the index is read off the orders of the poles, which along the ray can be bounded once for all its points.

Chapter 39

The two ends of a ray

The values of the insertions along a ray form one module over the rational functions of the strength k, and masters write the step from one strength to the next as one rational matrix, the one-step matrix. Finitely many relations of the period write the step on the Laurent part of the module, which is all of it when the direction uses every letter: on a box of insertions bounded by the powers of the letters in their denominators and by their growth against the reciprocal stride, one elimination over the rational functions of the strength gives a spanning set of that part, the matrix of the step on it and a certificate, and a search through the boxes stops. A family of functionals that respects the two relations of the ray is a solution of the module, and the one-step matrix is fixed by its solutions: by a full set of them outright, and by fewer exactly when they annihilate no submodule. The solutions come from the two ends of strength. At the high end the sources of the stride, components of its critical locus, carry formal solutions: one for a nondegenerate native gauge, one for each class of paths at a native gauge of corank one, one for every cycle on a clean family of native gauges. At the low end the values continue to the nonpositive integers, where their top principal parts are the constant terms of the powers of the reciprocal stride: the closings, one exact solution read on the compact torus. The symmetries of the ray, built from the automorphisms and the exchanges of the family and from the Cremona, act on the module; the values live on its invariant part, every orbit of nondegenerate native gauges contributes to it, and a degenerate native gauge only through the classes on which its stabilizer acts trivially. At a lean of a family with an exchange the exchange survives only composed with the Cremona: the integral keeps a symmetry that carries the basis half of a class of the centre onto its cobasis half, while the reading, which is oriented, separates the halves. Over the formal series at the high end the module splits by native values into pieces, each mapped to itself by the symmetries, and the closings glue the pieces back into the rational matrix. The closings, the values and the expansion at the native gauge of the direction vanish on the same elements, the positive orthant and the torus through that gauge meeting there and nowhere else: the closings determine the one-step matrix exactly where the values do, which is where that gauge sees every submodule, and a submodule that it does not see is invisible to the integral itself. Such a submodule can come from a table one mark larger: when the list is the contraction of an identically self-dual table, the integral over the contracted scale makes the automorphisms of the larger table act on the module, and the closings see only what they fix; each orbit of nondegenerate native gauges of the larger table that the symmetries of the ray split into s orbits adds $s-1$ to the dimension of what the closings miss, and at the centre of every $\binom{[2r+1]}{r}$, $r \geq 2$, the signs of $\binom{[2r+2]}{r+1}$ with two scales -1 split in two, with every symmetry of the ray of the four kinds in the group. On a wheel the letters are traces of one holonomy in SL_2, and the sources of its centre ray are read off the critical locus of the pair of traces and the

stratum where both holonomies are $-I$.

39.1 The module of a ray

Definition 39.1 (The module of a ray; masters; the one-step matrix)**.** Let a ray of a family F have the stride f, the origin form ω_0 and the relation space $\mathcal{L}$ (Definitions 38.1 and 38.4). The *module* of the ray is the $\mathbb{C}(k)$-vector space

$$\mathfrak{M} = \big(\mathbb{C}[X_F] \otimes \mathbb{C}(k)\big)/\mathcal{L},$$

whose dimension is the dimension of the ray, finite when $\kappa_{1,S} > 0$ for every nonempty proper S (Theorem 38.24(i)); $[g]$ is the class of $g \otimes 1$. The *shift* is $\mathrm{Sh}[g \otimes r(k)] = [fg \otimes r(k+1)]$; it is well defined and bijective, and $\mathrm{Sh}(r(k)\,m) = r(k+1)\,\mathrm{Sh}(m)$. A *submodule* is a $\mathbb{C}(k)$-subspace $\mathfrak{N}$ with $\mathrm{Sh}(\mathfrak{N}) = \mathfrak{N}$, and $\mathfrak{N} \neq 0$ is *irreducible* when its only submodules are 0 and $\mathfrak{N}$. *Masters* of a finite-dimensional submodule $\mathfrak{N}$ are insertions $b_1, \dots, b_{\mathfrak{m}}$ whose classes form a basis of $\mathfrak{N}$, $\mathfrak{m} = \dim_{\mathbb{C}(k)} \mathfrak{N}$, and their *one-step matrix* $\mathbf{B_b} \in GL_{\mathfrak{m}}(\mathbb{C}(k))$ is the matrix of Sh^{-1} on them,

$$[f^{-1} b_i] = \sum_{l=1}^{\mathfrak{m}} \mathbf{B}_{\mathbf{b},il}(k)\,[b_l], \qquad 1 \le i \le \mathfrak{m}.$$

The *cyclic submodule* of $[1]$ is the span of the $\mathrm{Sh}^i[1]$, $i \in \mathbb{Z}$. When it is finite-dimensional, of dimension h, $[1], \mathrm{Sh}[1], \dots, \mathrm{Sh}^{h-1}[1]$ is a basis of it, and the *law* of $[1]$ is the relation

$$\sum_{i=0}^{h} p_i(k)\,\mathrm{Sh}^i[1] = 0, \qquad p_i \in \mathbb{C}[k] \text{ without a common factor,}$$

unique up to a constant factor; $p_0 p_h \neq 0$.

The law of $[1]$ is the shortest relation among the strides of 1 that the relations of the ray produce, so by Corollary 38.7 the values obey it, $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ for all large k; its order h is at most the dimension of the ray.

Definition 39.2 (Solutions)**.** The *spaces of germs* are: the sequences on the integers $k \ge k_0$, for some k_0, two of them identified when they agree for all large k, the *germs at* $+\infty$; the same on the integers $k \le k_0$, the *germs at* $-\infty$; and, for $\lambda \in \mathbb{C}^\times$ and $\rho \in \mathbb{C}$, the *formal series* $\lambda^k k^\rho\,\mathbb{C}((1/k))$. Each carries the multiplication by $\mathbb{C}(k)$ and the shift ς: $(\varsigma x)(k) = x(k+1)$ on germs, and $\varsigma\big(\lambda^k k^\rho \varphi(k)\big) = \lambda^k k^\rho \cdot \lambda\big(1+\frac{1}{k}\big)^\rho \varphi(k+1)$ on formal series, re-expanded in powers of $1/k$; in each case $\varsigma(rx) = r(k+1)\,\varsigma(x)$ for $r \in \mathbb{C}(k)$. A *solution* of the ray with values in a space of germs $\mathcal{G}$, or in a finite direct sum of spaces of formal series, is a $\mathbb{C}(k)$-linear map $V : \mathfrak{M} \to \mathcal{G}$ with

$$V \circ \mathrm{Sh} = \varsigma \circ V.$$

A solution with values in $\lambda^k k^\rho\,\mathbb{C}((1/k))$ has the *type* $(\lambda, \rho + \mathbb{Z})$ and the *normalization* $\hat{V} = \lambda^{-k} k^{-\rho} V : \mathfrak{M} \to \mathbb{C}((1/k))$. For masters $\mathbf{b}$ the *solution vector* of V is $v_V = (V[b_1], \dots, V[b_{\mathfrak{m}}])$. The *annihilator* of a set $\mathcal{S}$ of solutions is $\mathrm{Ann}(\mathcal{S}) = \{m \in \mathfrak{M} :\ V(m) = 0 \text{ for every } V \in \mathcal{S}\}$.

When $\kappa_{1,S} > 0$ for every nonempty proper S, the values along the ray are a solution with values in germs at $+\infty$, $V[g] = \langle g\rangle_k$ (Theorem 38.3). At a nondegenerate native gauge α_c the expansion $f(\alpha_c)^k k^{-(n-1)/2} \sum_i a_i(g) k^{-i}$ of Theorem 38.6 is a solution of type $\big(f(\alpha_c), -\frac{n-1}{2} + \mathbb{Z}\big)$, written A_c; its normalization is the map $A^{(c)}$ there.

Lemma 39.3 (Solutions of distinct types)**.** *Let $A_1, \dots, A_r$ be solutions with values in formal series such that, for every type, those of that type among them are linearly independent over $\mathbb{C}$. Then their normalizations are linearly independent over $\mathbb{C}((1/k))$, and $\dim_{\mathbb{C}(k)} \mathfrak{N} \geq r$ for every submodule $\mathfrak{N}$ on which their restrictions are, type by type, linearly independent over $\mathbb{C}$.*

Theorem 39.4 (What solutions determine)**.** *Let $\mathbf{b}$ be masters of a submodule $\mathfrak{N}$, $\mathbf{B} = \mathbf{B}_\mathbf{b}$, and let $\mathcal{S}$ be a set of solutions.*

(i) Every solution V has $\varsigma^{-1} v_V = \mathbf{B}\, v_V$, that is $v_V(k-1) = \mathbf{B}(k)\, v_V(k)$. When the values along the ray are a solution, $\langle b_i \rangle_{k-1} = \sum_l \mathbf{B}_{il}(k)\, \langle b_l \rangle_k$ for all large k.

(ii) $\mathrm{Ann}(\mathcal{S})$ *is a submodule.*

(iii) A matrix $\mathbf{B}' \in \mathrm{Mat}_\mathfrak{m}(\mathbb{C}(k))$ has $\varsigma^{-1} v_V = \mathbf{B}' v_V$ for every $V \in \mathcal{S}$ if and only if every row of $\mathbf{B}' - \mathbf{B}$, read as the element $\sum_l (\mathbf{B}' - \mathbf{B})_{il} [b_l]$ of $\mathfrak{N}$, lies in $\mathrm{Ann}(\mathcal{S})$*. So $\mathbf{B}$ is the only such matrix exactly when* $\mathrm{Ann}(\mathcal{S}) \cap \mathfrak{N} = 0$*.*

(iv) Let $\mathrm{Ann}(\mathcal{S}) \cap \mathfrak{N} = 0$*, write $\mathbf{B} = P_0/Q_0$ with Q_0 the monic least common denominator of its entries, and put $D_0 = \max(\deg P_0, \deg Q_0)$. For $D \geq 0$ the equations $Q(k)\, \varsigma^{-1} v_V = P(k)\, v_V$, $V \in \mathcal{S}$, in pairs (P, Q) of a matrix and a scalar of polynomials of degree at most D, read at every integer beyond a bound past which (i) holds for a germ and at every power of k for a formal series, are linear and homogeneous in the coefficients of (P, Q). Their solutions are the pairs (cP_0, cQ_0) with $c \in \mathbb{C}[k]$ of degree at most $D - D_0$: only 0 for $D < D_0$, one line at $D = D_0$. For each D finitely many of the equations have the same solutions as all of them.*

Corollary 39.5 (One solution of an irreducible module)**.** *If $\mathfrak{N}$ is irreducible, every solution that does not vanish on $\mathfrak{N}$ determines $\mathbf{B}_\mathbf{b}$ in the sense of Theorem 39.4(iii). If the cyclic submodule of $[1]$ is finite-dimensional and irreducible, then for every solution V with $V[1] \neq 0$, $V[1]$ satisfies the law of $[1]$, $\sum_i p_i(k)\, \varsigma^i V[1] = 0$, and satisfies no relation $\sum_{i<h} q_i(k)\, \varsigma^i V[1] = 0$ with $q_i \in \mathbb{C}(k)$ not all zero.*

Masters are a basis of a submodule, the module itself or its invariant part (Theorem 39.8), while a box of insertions reduces to a spanning set of a submodule that contains [1], which is a set of masters exactly when it is independent (Section 39.3); the one-step matrix is the matrix of the inverse shift, so that the values of the masters step down in strength by $\mathbf{B}_\mathbf{b}$. A solution is anything that respects the two relations: the values over the orthant, the expansion at a native gauge, the closings on the torus (Theorem 39.29). Each solution gives $\mathfrak{m}$ linear equations on the numerator and the denominator of $\mathbf{B}_\mathbf{b}$ for every integer or every power of k, and the equations of a set of solutions determine $\mathbf{B}_\mathbf{b}$ exactly when the solutions annihilate no submodule: a submodule annihilated by all of them is a direction in which a row of the matrix can move without any of them noticing. An irreducible module is determined by any one of its nonzero solutions, and the law of [1] by any solution that does not vanish at [1] when the cyclic submodule of [1] is irreducible.

Proposition 39.6 (The law of [1] read from a one-step matrix)**.** *Let insertions $b_1 = 1, b_2, \dots, b_\mathfrak{m}$ have classes that span a submodule $\mathfrak{N}$, and let $\mathbf{B} \in \mathrm{Mat}_\mathfrak{m}(\mathbb{C}(k))$ have $[f^{-1} b_i] = \sum_l \mathbf{B}_{il}(k) [b_l]$ for every i; for masters, $\mathbf{B} = \mathbf{B}_\mathbf{b}$. Put $w_0 = (1, 0, \dots, 0)$ and $w_i(k) = w_{i-1}(k-1)\, \mathbf{B}(k)$, so that $w_i(k) = w_0\, \mathbf{B}(k-i+1) \cdots \mathbf{B}(k)$.*

(i) $\mathrm{Sh}^{-i}[1] = \sum_l w_{i,l}(k) [b_l]$*. So every solution V has $\varsigma^{-i} V[1] = w_i\, v_V$, with $v_V = (V[b_1], \dots, V[b_\mathfrak{m}])$ as in Definition 39.2, and when $\kappa_{1,S} > 0$ for every nonempty proper S, $\langle 1 \rangle_{k-i} = \sum_l w_{i,l}(k) \langle b_l \rangle_k$ as meromorphic functions of k.*

(ii) *Let h be the least index at which $w_0, \dots, w_h$ are linearly dependent over $\mathbb{C}(k)$, so that $h \le \mathfrak{m}$, let $\sum_{i=0}^{h} q_i(k)\, w_i(k) = 0$ with $q_i \in \mathbb{C}[k]$ without a common factor, and put $p_j(k) = q_{h-j}(k+h)$. Then $\sum_{j=0}^{h} p_j(k)\, \mathrm{Sh}^j[1] = 0$, that is $\sum_j p_j(k) f^j \in \mathcal{L}$; $p_0 \neq 0$, and $p_h \neq 0$ when $\mathbf{B}$ is invertible. The order h is at least the dimension of the cyclic submodule of $[1]$, with equality when $\mathbf{b}$ are masters, and then the relation is the law of $[1]$.*

(iii) *Let $\mathbf{W} = [w_0; \dots; w_{h-1}]$ be the matrix of the first h rows and J a set of h columns with $\det \mathbf{W}_J \neq 0$. Then $(q_i)_i$ is proportional over $\mathbb{C}(k)$ to the signed maximal minors $\big((-1)^i \det[w_0; \dots; \widehat{w_i}; \dots; w_h]_J\big)_i$. With Q the monic least common denominator of the entries of $\mathbf{B}$, q_h divides the polynomial $\det \mathbf{W}(k)_J \prod_{j=0}^{h-1} Q(k-j)^{h-j}$. So at a zero k_0 of q_h with $Q(k_0 - j) \neq 0$ for $0 \le j < h$ the rows $w_0(k_0), \dots, w_{h-1}(k_0)$ are linearly dependent: there $\langle 1 \rangle_{k_0}, \dots, \langle 1 \rangle_{k_0-h+1}$, written by (i) as linear forms in the values of $\mathbf{b}$ at k_0, are dependent forms.*

(iv) *If $h = \mathfrak{m}$, then $q_0(k) \det \mathbf{W}(k) = (-1)^h q_h(k) \det \mathbf{W}(k-1) \det \mathbf{B}(k)$.*

(v) *The same holds for every element $m = \sum_l c_l(k)[b_l]$ of $\mathfrak{N}$, with the row c in place of w_0, m in place of $[1]$, and the cyclic submodule of m, the span of the $\mathrm{Sh}^i m$, in place of that of $[1]$.*

The first dependence among the rows w_i, which carry $\langle 1 \rangle$ to the lower strengths in the coordinates of a spanning set, is a recurrence of the values of 1, and on masters it is the law of $[1]$. Its end coefficient $p_0(k) = q_h(k+h)$ vanishes only at shifted zeros of the denominator of $\mathbf{B}$ and at strengths where the rows line up, and when the rows reach the whole span, $h = \mathfrak{m}$, the end coefficients are tied to the determinant of $\mathbf{B}$. A spanning set that is not independent gives a relation whose order may exceed that of the law of $[1]$.

39.2 Symmetries of a ray

Proposition 39.7 (The symmetries of a ray)**.** *Let a ray of F have the origin $(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0)$ and the direction $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$, and put $\sigma_i = \frac{1}{2} \sum_j (s_{ij} + t_{ij})$, $i = 0, 1$. On the torus $(\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times$ put, for a bijection β of the marks, $\phi_\beta(\alpha) = \alpha \circ \beta$ with $(\alpha \circ \beta)_e = \alpha_{\beta(e)}$; the* Cremona $\mathrm{cr}(\alpha) = \alpha^{-1}$*; and the* reversed *map $\bar{\phi}_\beta = \mathrm{cr} \circ \phi_\beta$, $\bar{\phi}_\beta(\alpha) = (\alpha \circ \beta)^{-1}$. A* symmetry *of the ray is a map ϕ among the ϕ_β and the $\bar{\phi}_\beta$ that keeps the stride and the density of the origin form, $f \circ \phi = f$ and $(\omega_0/dy) \circ \phi = \omega_0/dy$, the latter on the positive orthant with the principal powers when the origin is not integral; the symmetries form a group. Let h be a bijection of the tables, write $u \circ \beta^{-1}$ for $e \mapsto u_{\beta^{-1}(e)}$ and $\mathbf{a} \circ h^{-1}$ for $j \mapsto a_{h^{-1}(j)}$, and say that β* carries, *respectively* exchanges, *the tables with h when it satisfies the conditions of Definition 11.13 on the bases and the weights, without its condition on the pairs.*

(i) *If β carries the tables, $K_j \circ \phi_\beta = K_{h(j)}$ and $\Psi_j \circ \phi_\beta = \Psi_{h(j)}$. If β exchanges them, $K_j \circ \phi_\beta = \Psi_{h(j)}$, $\Psi_j \circ \phi_\beta = K_{h(j)}$, $K_j \circ \bar{\phi}_\beta = \alpha^{-\mathcal{E}} K_{h(j)}$ and $\Psi_j \circ \bar{\phi}_\beta = \alpha^{-\mathcal{E}} \Psi_{h(j)}$. Always $K_j \circ \mathrm{cr} = \alpha^{-\mathcal{E}} \Psi_j$ and $\Psi_j \circ \mathrm{cr} = \alpha^{-\mathcal{E}} K_j$.*

(ii) *The following maps are symmetries when, for $i = 0, 1$, every mark e and every table j:*

map	*masses*	*exponents*
ϕ_β, *β carrying*	$m_{i,\beta(e)} = m_{i,e}$	$(s_{i,h(j)}, t_{i,h(j)}) = (s_{ij}, t_{ij})$
ϕ_β, *β exchanging*	$m_{i,\beta(e)} = m_{i,e}$	$(s_{i,h(j)}, t_{i,h(j)}) = (t_{ij}, s_{ij})$
cr	$\mathbf{m}_i = \sigma_i \mathbf{1}$	$s_{ij} = t_{ij}$
$\bar{\phi}_\beta$, *β exchanging*	$m_{i,\beta(e)} + m_{i,e} = 2\sigma_i$	$(s_{i,h(j)}, t_{i,h(j)}) = (s_{ij}, t_{ij})$

A symmetry maps X_F and the positive orthant onto themselves, has $\phi^*\omega_0 = \pm\omega_0$ when the origin is integral, and keeps the values, $\langle g' \circ \phi\rangle_k = \langle g'\rangle_k$ for every insertion g'.

(iii) On the insertions $\alpha^u \prod_j K_j^{-a_j}\Psi_j^{-b_j}$, written $(u;\mathbf{a},\mathbf{b})$, the four kinds act by $(u;\mathbf{a},\mathbf{b}) \mapsto$

$$(u\circ\beta^{-1};\mathbf{a}\circ h^{-1},\mathbf{b}\circ h^{-1}) \ \ (\phi_\beta,\ \beta \text{ carrying}), \qquad (-u+|\mathbf{a}+\mathbf{b}|\mathbf{1};\mathbf{b},\mathbf{a}) \ \ (\mathrm{cr}),$$
$$(u\circ\beta^{-1};\mathbf{b}\circ h^{-1},\mathbf{a}\circ h^{-1}) \ \ (\phi_\beta,\ \beta \text{ exchanging}),$$
$$(-u\circ\beta^{-1}+|\mathbf{a}+\mathbf{b}|\mathbf{1};\mathbf{a}\circ h^{-1},\mathbf{b}\circ h^{-1}) \ \ (\bar\phi_\beta,\ \beta \text{ exchanging}),$$

with $|\mathbf{a}+\mathbf{b}| = \sum_j (a_j+b_j)$.

The four kinds are the monomial maps of the torus that a list can certify: a permutation of the marks that carries the bases, one that carries them onto the cobases, the inversion of every scale, and the composite of the last two. The first keeps each letter, the second and the third swap the two letters of every table, and the fourth keeps each letter up to the monomial $\alpha^{-\mathcal{E}}$, which the masses absorb when the masses of a mark and of its image add up to $2\sigma_i$, twice the total strength. Composites of symmetries are symmetries, so a group generated by maps of the four kinds can be used whole.

Theorem 39.8 (The invariant module)**.** *Let G be a finite group of symmetries of a ray.*

(i) The maps $\phi^*[g'] = [g'\circ\phi]$, $\phi \in G$, are well defined on $\mathfrak{M}$ and commute with the shift. With $e_G = |G|^{-1}\sum_{\phi\in G}\phi^*$, $\mathfrak{M} = \mathfrak{M}^G \oplus (1-e_G)\mathfrak{M}$ as submodules, and $\mathfrak{M}^G = e_G\mathfrak{M}$ consists of the classes of the invariant insertions.

(ii) A solution V with $V\circ\phi^* = V$ for every $\phi \in G$, the values along the ray among them, vanishes on $(1-e_G)\mathfrak{M}$: $\langle g'\rangle_k = \langle e_G g'\rangle_k$. The cyclic submodule of $[1]$ lies in $\mathfrak{M}^G$, so the law of $[1]$ is read in $\mathfrak{M}^G$, on masters among the invariant insertions.

(iii) If $\phi \in G$ fixes a nondegenerate native gauge α_c, then $A_c \circ \phi^* = A_c$. For the G-orbit O of α_c, with stabilizer G_c, the sum $A_O = \sum_{\phi G_c \in G/G_c} A_c\circ\phi^*$ is invariant and does not vanish on $\mathfrak{M}^G$, and $\dim_{\mathbb{C}(k)}\mathfrak{M}^G$ is at least the number of G-orbits of nondegenerate native gauges.

Theorem 39.9 (A reversed exchange at a lean)**.** *Let β exchange a family with itself (Definition 11.13), with the bijection h of its tables, and let $\Upsilon = \Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ be a realization at the table's gauge with masses $\boldsymbol{\tau}$ (Definition 9.1).*

(i) $R_{h(j),\beta(e)}(\mathbf{1}) = 1 - R_{j,e}(\mathbf{1})$ for every table and mark, and $\tau_{\beta(e)} + \tau_e = 2\sigma$. So $\bar\phi_\beta$ keeps the integrand $\alpha^{\boldsymbol{\tau}}\prod_j K_j^{-s_j}\Psi_j^{-t_j}$ of the period at the data of Υ, and it is a symmetry of the ray with origin 0 and direction $(c\boldsymbol{\tau};c\mathbf{s},c\mathbf{t})$ whenever that direction is integral. The exchange ϕ_β and the Cremona each take that integrand to the integrand at the data of $\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}$.

(ii) For every mark e and partners $\mathbf{g} = (g_1,\dots,g_q)$, with $\beta(\mathbf{g}) = (\beta(g_1),\dots,\beta(g_q))$, the views (Definition 10.1) satisfy

$$\Phi^{\mathbf{E}}_{\beta(e),\beta(\mathbf{g})} = \Phi^{\mathbf{E}}_{e,\mathbf{g}}, \qquad \Phi^{\mathbf{O}}_{\beta(e),\beta(\mathbf{g})} = -\Phi^{\mathbf{O}}_{e,\mathbf{g}}:$$

the reading of $\beta(e)$ at $\boldsymbol{\delta}$ is the reading of e at $-\boldsymbol{\delta}$. At the centre β maps every class onto itself; at a lean it permutes the classes of Υ.

(iii) In a one-table molecule, β maps every class C of the centre onto itself, and when C splits at the lean into C_K and C_Ψ (Theorem 11.17), $\beta(C_K) = C_\Psi$ and $\beta(C_\Psi) = C_K$.

An exchange carries the bases of a family onto the cobases, and the integrand at a lean weighs the two unequally: the exchange alone takes the lean to its opposite, and so does the Cremona, which inverts every scale. Their composite keeps the integrand at every lean at the table's gauge, because there the masses of a mark and of its image add up to 2σ. So the integral at a lean keeps a symmetry carrying each mark to its image under the exchange, and insertions related by it have equal values; but the symmetry reverses every scale, $y \mapsto -\beta^* y$, and the reading, whose odd part changes sign under a reversal, sees $\beta(e)$ as e read at the opposite lean. A bijection that keeps the potential fixes every class (Lemma 14.18); the reversed exchange permutes the classes, and on a class of the centre that the lean splits it swaps the basis half and the cobasis half. At the centre the odd part is absent and the exchange, the Cremona and their composite all keep the integrand: the reversal is invisible to the reading there, and the classes it joins at the centre are those the lean separates.

Remark 39.10. The condition $m_{\beta(e)} + m_e = 2\sigma$ of Proposition 39.7 is met off the table's gauge as well, for instance by the uniform masses $\sigma\mathbf{1}$ at every lean of a family with $\sum_j \delta_j(2r_j - n) = 0$, as for one table with $2r = n$. The native gauge of such a realization on the positive orthant is fixed by $\bar{\phi}_\beta$ up to scale, and the views of $\beta(e)$ are those of e at the reflected arguments $v \mapsto -v$ up to affine terms in v, by the identities $\Phi_{\mathbf{O}}(-\beta^* y) = \Phi_{\mathbf{O}}(y)$ and $\Phi_{\mathbf{E}}(-\beta^* y) = \Phi_{\mathbf{E}}(y) - 2\sigma\langle \mathbf{1}, y\rangle$ of Section 48.38. When that gauge is not proportional to $\mathbf{1}$ the realization is not an atom (Theorem 11.4), though the integral keeps the reversed exchange.

39.3 The box of the walls

The module of a ray is the quotient by the relations of the period, and finitely many of them give a spanning set of its Laurent part and the step on it. An insertion is measured by its level, the powers of the letters in its denominator, and by its wall degree, the power of the reciprocal stride that dominates it, read on the walls of the direction. A box is the finite set of symbols of insertions within given levels and wall degree; its relations, the letter identities among them and the mark relations of its stems, are relations of the module, so they hold as identities of meromorphic functions of the strength. One elimination of a box leaves a spanning set of a submodule that contains [1] and the matrix of the inverse shift on it, with a certificate supported on the stems, and a search through the boxes stops by the finite dimension of the ray. When the origin carries no letter exponents, the relations $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})(\alpha^w f^{-1})$ involve Laurent insertions alone, and they do not suffice: the evaluation at a singular point of the zero set of f^{-1} on the torus, when there is one, vanishes on all of them and not on the law of [1]. Insertions are the regular functions on the letter complement X_F of Definition 38.1, on which every letter is inverted whatever its exponent along the ray; a box need carry no unit of a letter whose exponent along the ray vanishes identically, and its relations are relations of the module all the same.

Definition 39.11 (Levels, wall degree, boxes)**.** Let a ray of a family F of N tables on $n \geq 2$ marks have $\kappa_{1,S} > 0$ for every nonempty proper S, and let $\Bbbk \subseteq \mathbb{C}$ be a field that contains the weights and the origin.

(i) The *symbols* are the data $(u; \mathbf{a}, \mathbf{b})$ with $u \in \mathbb{Z}^{\mathcal{E}}$, $\mathbf{a}, \mathbf{b} \in \mathbb{N}^N$ and $u_{\mathcal{E}} = \sum_j (a_j r_j + b_j r_j^*)$. The *evaluation* of $(u; \mathbf{a}, \mathbf{b})$ is the insertion $\alpha^u \prod_j K_j^{-a_j} \Psi_j^{-b_j}$ (Proposition 39.7(iii)); its *level* is $(\mathbf{a}, \mathbf{b})$, and the symbols at the level $(\mathbf{0}, \mathbf{0})$ are the *Laurent* symbols α^u, $u_{\mathcal{E}} = 0$. A *formal combination* is a finite combination of symbols with coefficients in $\Bbbk(k)$, and its evaluation is the same combination of the insertions, an element of $\mathbb{C}[X_F] \otimes \mathbb{C}(k)$. The evaluations of the symbols span $\mathbb{C}[X_F]$.

(ii) The *walls* of a symbol $g = (u; \mathbf{a}, \mathbf{b})$ are those of its data, $\kappa_U(g) = u_U - \sum_j \big(a_j d_j(U) + b_j d_j^*(U)\big)$, and its *wall degree* is

$$\deg_\kappa g = \max_{\varnothing \neq U \subsetneq \mathcal{E}} \frac{-\kappa_U(g)}{\kappa_{1,U}},$$

the least real x with $\kappa_U(g) + x\,\kappa_{1,U} \geq 0$ for every nonempty proper U.

(iii) The *mark relation* of a symbol $g = (u; \mathbf{a}, \mathbf{b})$ at a mark e is the formal combination

$$\begin{aligned}\mathrm{MR}_e(g) = \big(m_{0,e} + k\,m_{1,e} + u_e\big)\,g - \sum_j \big(s_{0j} + k\,s_{1j} + a_j\big) \sum_{T \ni e} \theta_{j,T}\,(u + \mathbf{1}_T; \mathbf{a} + \mathbf{e}_j, \mathbf{b}) \\ - \sum_j \big(t_{0j} + k\,t_{1j} + b_j\big) \sum_{T \not\ni e} \theta_{j,T}\,(u + \mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{a}, \mathbf{b} + \mathbf{e}_j),\end{aligned}$$

the sums over the bases T of table j, the terms whose coefficient vanishes identically in k omitted. Its evaluation is $(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})$ applied to that of g. The *letter identities* of g are $g - \sum_{T \in \mathcal{B}_j} \theta_{j,T}(u + \mathbf{1}_T; \mathbf{a} + \mathbf{e}_j, \mathbf{b})$, for K_j, and $g - \sum_{T \in \mathcal{B}_j} \theta_{j,T}(u + \mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{a}, \mathbf{b} + \mathbf{e}_j)$, for Ψ_j; they evaluate to 0.

(iv) A letter K_j, or Ψ_j, is *active along the ray* when its exponent $s_{0j} + k\,s_{1j}$, or $t_{0j} + k\,t_{1j}$, does not vanish identically. *Level bounds* are $\bar{\mathbf{a}}, \bar{\mathbf{b}} \in \mathbb{N}^N$ with $\bar{a}_j \geq 1$ when K_j is active and $\bar{b}_j \geq 1$ when Ψ_j is active; put $\bar{a}_j^- = \max(\bar{a}_j - 1, 0)$ and $\bar{b}_j^- = \max(\bar{b}_j - 1, 0)$. For $\mathfrak{t} \in \mathbb{N}$ the *box* $\mathfrak{B} = \mathfrak{B}_\mathfrak{t}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ is the set of the symbols of wall degree at most $\mathfrak{t}$ at the levels $(\mathbf{a}, \mathbf{b})$ with $\mathbf{a} \leq \bar{\mathbf{a}}$ and $\mathbf{b} \leq \bar{\mathbf{b}}$, and its *stems* are its symbols at the level $(\bar{\mathbf{a}}^-, \bar{\mathbf{b}}^-)$. Its formal combinations form the space $\mathcal{F}_\mathfrak{B}$, and its *relations* $\mathcal{R}_\mathfrak{B} \subseteq \mathcal{F}_\mathfrak{B}$ are the $\Bbbk(k)$-span of the letter identities of its symbols $(u; \mathbf{a}, \mathbf{b})$, for K_j when $a_j < \bar{a}_j$ and for Ψ_j when $b_j < \bar{b}_j$, and of the mark relations of its stems at every mark. The box with *one unit of each active letter* has $\bar{a}_j = 1$ exactly when K_j is active and $\bar{b}_j = 1$ exactly when Ψ_j is; its stems are the Laurent symbols of wall degree at most $\mathfrak{t}$.

(v) The *Laurent part* $\mathfrak{M}_\mathrm{L}$ of the module $\mathfrak{M}$ (Definition 39.1) is the span of the classes of the Laurent insertions.

(vi) A group of symmetries generated by maps of the four kinds of Proposition 39.7(ii) carries every letter to a letter times a monomial (Proposition 39.7(i)): each of its elements ϕ permutes the letters, and the symbols, $g \mapsto \phi^* g$ (Proposition 39.7(iii)), the evaluation of $\phi^* g$ being that of g composed with ϕ. Such a finite group G *keeps* the box when its permutations of the letters keep the level bounds, $\bar{a}_j$ being the bound of K_j and $\bar{b}_j$ that of Ψ_j.

Proposition 39.12 (The wall degree)**.** *For a ray as in Definition 39.11 and its symbols g:*

(i) For real x, $|g|\,f^x$ is bounded on the positive orthant exactly when $x \geq \deg_\kappa g$: the symbols of wall degree at most x are those whose evaluation is dominated by f^{-x}. For the origin 0 the value $\langle g \rangle_k$ converges absolutely exactly when $\operatorname{Re} k > \deg_\kappa g$, and for any origin it converges absolutely at every k with $\operatorname{Re} \kappa_{0,U} + (\operatorname{Re} k - \deg_\kappa g)\,\kappa_{1,U} > 0$ for every nonempty proper U.

(ii) A Laurent symbol α^u has the wall degree $\max_U(-u_U)/\kappa_{1,U} \geq 0$, zero exactly for $u = 0$. Every monomial of the Laurent polynomial f^{-1} has wall degree at most 1, and every monomial of $f^{-1}\alpha^u$ wall degree at most $\deg_\kappa \alpha^u + 1$.

(iii) The terms of the mark relations and of the letter identities of g have wall degree at most $\deg_\kappa g$.

(iv) At every level and for every real x there are finitely many symbols of wall degree at most x.

(v) For every element ϕ of a group as in Definition 39.11(vi), $\deg_\kappa \phi^* g = \deg_\kappa g$, and ϕ^* maps Laurent symbols to Laurent symbols.

Lemma 39.13 (The Laurent part)**.** *For a ray as in Definition 39.11, the Laurent part $\mathfrak{M}_{\mathrm{L}}$ is a submodule. It contains* [1]*, and the class of every insertion g with $f^{-l}g$ a Laurent polynomial for some $l \in \mathbb{N}$; when every letter exponent of the direction is positive, $s_{1j}, t_{1j} \geq 1$, it is $\mathfrak{M}$. For a finite group G of symmetries, $e_G\mathfrak{M}_{\mathrm{L}} = \mathfrak{M}_{\mathrm{L}} \cap \mathfrak{M}^G$ is a submodule, spanned by the classes of the invariant Laurent insertions.*

Proposition 39.14 (The relations of a box)**.** *Let $\mathfrak{B} = \mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ be a box of a ray as in Definition 39.11.*

(i) Every term of every relation of the box lies in the box. The letter identities evaluate to 0 *and the mark relation $\mathrm{MR}_e(g)$ of a stem to the element $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})g$ of $\mathcal{L}$: every relation evaluates into $\mathcal{L}$, and its value vanishes identically in k (Proposition 38.5(ii)).*

(ii) A formal combination of symbols of the box whose evaluation is 0 *is a combination of the letter identities of the box.*

(iii) The mark relation at every mark of every symbol of wall degree at most $\mathfrak{t}$ at a level at most $(\bar{\mathbf{a}}^-, \bar{\mathbf{b}}^-)$ lies in $\mathcal{R}_{\mathfrak{B}}$.

(iv) If a finite group G of symmetries keeps the box, every $\phi \in G$ maps the box onto itself, its stems onto its stems and $\mathcal{R}_{\mathfrak{B}}$ onto itself; so, with $e_G = |G|^{-1} \sum_{\phi \in G} \phi^*$ and $\mathcal{F}_{\mathfrak{B}}^G = e_G \mathcal{F}_{\mathfrak{B}}$, $e_G \mathcal{R}_{\mathfrak{B}} = \mathcal{R}_{\mathfrak{B}} \cap \mathcal{F}_{\mathfrak{B}}^G$.

Proposition 39.15 (The closing relations have more solutions than the ray)**.** *Let a ray as in Definition 39.11 have an origin with $\mathbf{s}_0 = \mathbf{t}_0 = \mathbf{0}$, and write $f^{-1} = \sum_v \lambda_v \alpha^v$. For a Laurent insertion α^w and a mark e the* closing relation

$$\big(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}\big)\big(\alpha^w f^{-1}\big) = \sum_v \lambda_v \big(m_{0,e} + w_e + (1-k)\, v_e\big)\, \alpha^{w+v}$$

is an element of $\mathcal{L}$ whose terms are Laurent insertions. Let $z \in (\mathbb{C}^\times)^{\mathcal{E}}$ be a point at which f^{-1} and every $\partial_{y_e} f^{-1}$ vanish, a singular point of the zero set of f^{-1} on the torus.

(i) The evaluation at z, $\alpha^w \mapsto z^w$ on the Laurent insertions, extended $\mathbb{C}(k)$-linearly, vanishes on every closing relation.

(ii) It takes the value $q_0(k)$ on every element $\sum_{i=0}^h q_i(k) f^{-i}$, $q_i \in \mathbb{C}(k)$. So no element of $\mathcal{L}$ of this form with $q_0 \neq 0$ is a combination of closing relations, while $\mathcal{L}$ contains such elements: the law $\sum_j p_j(k)\, \mathrm{Sh}^j[1] = 0$ of [1]*, with $p_0 p_h \neq 0$ (Definition 39.1), gives $\sum_{i=0}^h p_{h-i}(k-h) f^{-i} \in \mathcal{L}$. No $\mathbb{C}(k)$-linear form that vanishes on $\mathcal{L}$ agrees with the evaluation at z on the Laurent insertions.*

The point z lies off X_F, a letter that the direction uses vanishing there. The relations of $\mathcal{L}$ that the closing relations lack involve the insertions with inverse letters, through the letter identities and the mark relations of the occupations, and a box supplies them through its units of the letters.

Definition 39.16 (The reduction of a box)**.** Let a finite group G of symmetries keep a box $\mathfrak{B} = \mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ (Definition 39.11); G may be trivial.

(i) The *orbit average* of a symbol g of the box is $e_G g$, with e_G as in Proposition 39.14(iv); the orbit averages of the orbits of symbols form a basis of $\mathcal{F}_{\mathfrak{B}}^G$, and $\mathcal{R}_{\mathfrak{B}}^G = e_G \mathcal{R}_{\mathfrak{B}}$.

(ii) A *ranking* is a total order of these orbits in which every orbit at a positive level ranks above every Laurent orbit and, of two Laurent orbits, the one of larger wall degree ranks higher; the level of the symbols of an orbit is $(\mathbf{0},\mathbf{0})$ for all of them or for none, and they all have the same wall degree (Proposition 39.12(v)). The *leading* orbit of a nonzero element of $\mathcal{F}^G_{\mathfrak{B}}$ is the highest-ranked orbit in it; an orbit is a *pivot* when it leads an element of $\mathcal{R}^G_{\mathfrak{B}}$, and *free* otherwise. The *reduction* of $x \in \mathcal{F}^G_{\mathfrak{B}}$ is the unique combination of the averages of the free orbits that is congruent to x modulo $\mathcal{R}^G_{\mathfrak{B}}$; if x involves no orbit ranked above a given orbit, neither does its reduction.

(iii) For $\mathfrak{s} \in \mathbb{N}$ with $\mathfrak{s} < \mathfrak{t}$, the *spanning set* of the box at $\mathfrak{s}$ is $\mathbf{b} = (b_1, \dots, b_{\mathfrak{m}})$, the averages of the free Laurent orbits of wall degree at most $\mathfrak{s}$, listed with $b_1 = 1$, the orbit of 1 being free. Its *targets* are the invariant Laurent polynomials $f^{-1}b_i$, elements of $\mathcal{F}^G_{\mathfrak{B}}$. The box is *accepted* at $\mathfrak{s}$ when the reduction of every target is a combination of the spanning set, $f^{-1}b_i \equiv \sum_l \mathbf{B}_{il}\, b_l$ modulo $\mathcal{R}^G_{\mathfrak{B}}$ with $\mathbf{B}_{il} \in \Bbbk(k)$; $\mathbf{B}$ is then the *matrix* of the box.

Theorem 39.17 (The box gives a spanning set, the one-step matrix and a certificate)**.** *Let a finite group G of symmetries keep a box $\mathfrak{B} = \mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ of a ray as in Definition 39.11, and let the box be accepted at $\mathfrak{s}$ (Definition 39.16), with the spanning set $\mathbf{b}$ and the matrix $\mathbf{B}$.*

(i) In the module, $[f^{-1}b_i] = \sum_l \mathbf{B}_{il}(k)[b_l]$ for every i. The classes of $\mathbf{b}$ span a submodule $\mathfrak{N}$ of $\mathfrak{M}^G$ that contains $[1]$: the span of the classes of the invariant combinations of Laurent symbols of wall degree at most $\mathfrak{s}$, which lies in $e_G\mathfrak{M}_{\mathrm{L}}$ (Lemma 39.13). The spanning set is a set of masters of $\mathfrak{N}$ exactly when its classes are linearly independent over $\mathbb{C}(k)$, and then $\mathbf{B} = \mathbf{B}_{\mathbf{b}}$ (Definition 39.1).

(ii) The certificate. *There are formal combinations $G_{i,e}$ of the stems, with coefficients in $\Bbbk(k)$, such that*

$$f^{-1}b_i - \sum_l \mathbf{B}_{il}(k)\, b_l = \sum_{e\in\mathcal{E}} \big(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}\big) G_{i,e}$$

as functions on X_F, the $G_{i,e}$ evaluated: a certificate of the one-step relation (Proposition 38.5(i)).

(iii) Every solution V has $\varsigma^{-1}v_V = \mathbf{B}\, v_V$, with $v_V = (V[b_1], \dots, V[b_{\mathfrak{m}}])$, and $\langle b_i\rangle_{k-1} = \sum_l \mathbf{B}_{il}(k)\langle b_l\rangle_k$ as an identity of meromorphic functions of k. At every real k off the poles of the coefficients with $\operatorname{Re}\kappa_{0,U} + (k-\mathfrak{t})\,\kappa_{1,U} > 0$ for every nonempty proper U, all the values in (ii) converge absolutely and every term $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})s$, s a stem, has the value 0, so the certificate integrates over the chart term by term.

(iv) Proposition 39.6 applies to $\mathbf{b}$ and $\mathbf{B}$: the first dependence of the rows gives a relation $\sum_j p_j(k) f^j \in \mathcal{L}$, with a certificate assembled from those of (ii), whose order is at least that of the law of $[1]$, with equality when the spanning set is a set of masters, and which then is the law of $[1]$.

Theorem 39.18 (The search stops)**.** *Let G be a finite group of symmetries of a ray as in Definition 39.11, generated by maps of the four kinds of Proposition 39.7(ii) and possibly trivial, and fix a ranking for every box that G keeps. There are $\mathfrak{t} \geq 1$, level bounds kept by G and $\mathfrak{s} < \mathfrak{t}$ at which the box $\mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ modulo G is accepted, with a spanning set whose classes span $e_G\mathfrak{M}_{\mathrm{L}}$, which is $\mathfrak{M}^G$ when every letter exponent of the direction is positive. So a search that runs through all the $(\mathfrak{t}, \bar{\mathbf{a}}, \bar{\mathbf{b}}, \mathfrak{s})$ with level bounds kept by G, in any order that reaches each, and tests the acceptance by one elimination over $\Bbbk(k)$ at each, stops.*

So the construction is finite and every step is an identity: a box is finite (Proposition 39.12(iv)), its relations are relations of the period read at the data of the ray (Proposition 39.14), so that whatever the box, the one-step relation and the recurrence it produces hold at every strength, and a search through the boxes stops (Theorem 39.18). The spanning set spans a submodule that contains [1], which is all a recurrence of the values of 1 needs; it is a set of masters exactly when it is independent, and then the recurrence that its rows give is the law of [1] and its matrix is $\mathbf{B_b}$.

39.4 Sources at the high end

Definition 39.19 (Sources; regular points; the normal density)**.** The *critical locus* of a ray is the set of its native gauges, $\{\alpha \in X_F : \ c^{(1)}(\alpha) = 0\}$, the critical points of $\log f$. A *source* is an irreducible component Z of it; f is constant on Z, with value f_Z, its *native value*. A point $z \in Z$ is *regular* when it lies on no other source, Z is smooth at z of dimension $d = \dim Z$, and the Hessian of $\log f$ in y at z has rank $n-1-d$ on the tangent space of the torus; its kernel is then the tangent space of Z, and it induces a nondegenerate quadratic form on the normal space. Z is *clean* when it has a regular point; the regular points then form a dense open subset Z°. In local holomorphic coordinates (u, w) at a regular point, with $Z = \{w = 0\}$ and $-\log f = -\log f_Z + \frac{1}{2} w^{\mathsf{T}} A(u) w + O(|w|^3)$, the *normal density* is the d-form

$$\nu = \det A(u)^{-1/2} \, \frac{\omega_0}{du \wedge dw}\Big|_{w=0} du,$$

which does not depend on the coordinates up to its sign. For an integral origin ω_0 is single-valued on X_F, and the *normal cover* $\tilde{Z}^\circ \to Z^\circ$ is the double cover on which ν is single-valued, its deck involution changing the sign of ν. A source of dimension 0 is an isolated native gauge, clean exactly when it is nondegenerate.

Theorem 39.20 (A closed cycle on a source carries a solution)**.** *Let a ray have an integral origin, let Z be a clean source of dimension d, $m = n-1-d$, and let γ be a compact oriented cycle of real dimension d in $\tilde{Z}^\circ$.*

(i) Over γ there is a chain $N(\gamma)$ of real dimension $n-1$ in X_F, a union of discs, one through each point z of γ, transverse to Z, on which $|f| \le |f_Z| e^{-c|v|^2}$ at the displacement v from z for some $c > 0$, oriented by the sheet of ν; its boundary lies where $|f| \le |f_Z| e^{-\varepsilon}$ for some $\varepsilon > 0$. Two such chains over homologous cycles differ by a boundary and by pieces on which $|f| \le |f_Z| e^{-\varepsilon}$.

(ii) As $k \to \infty$,

$$\int_{N(\gamma)} g \, f^k \omega_0 \sim f_Z^k \, k^{-m/2} \sum_{i \ge 0} c_i(\gamma; g) \, k^{-i}, \qquad c_0(\gamma; g) = (2\pi)^{m/2} \int_\gamma g \, \nu.$$

Where one family of holomorphic normal slices $\{u = \text{const}\}$, coordinates (u, w) as in Definition 39.19, covers Z°, $c_i(\gamma; g) = \int_\gamma a_i(\cdot\,; g)$, with $f_Z^k k^{-m/2} \sum_i k^{-i} a_i(u; g)$ the Laplace expansion of the integral over the disc in the slice at u: $a_0(\cdot\,; g) = (2\pi)^{m/2} g\,\nu$, and each $a_i(\cdot\,; g)$ is a holomorphic d-form given by finitely many w-derivatives of g, of $\omega_0/(du \wedge dw)$ and of $\log f$ at $w = 0$.

(iii) The map $A_\gamma : g \mapsto f_Z^k k^{-m/2} \sum_i c_i(\gamma; g) \, k^{-i}$ is a solution of type $\big(f_Z, -\frac{m}{2} + \mathbb{Z}\big)$. It depends only on the homology class of γ in $\tilde{Z}^\circ$, and the image of γ under the deck involution gives $-A_\gamma$.

(iv) If $\gamma_1,\dots,\gamma_r$ are such cycles and $g_1,\dots,g_r$ insertions with $\det\big[\int_{\gamma_a} g_b\,\nu\big]_{a,b}\neq 0$, the A_{γ_a} are linearly independent over $\mathbb{C}$, and their normalizations over $\mathbb{C}((1/k))$.

For $d=0$, γ is the point with a sheet and this is Theorem 38.6. For an origin that is not integral, ν is multivalued along Z°, and the same holds for cycles with coefficients in the local system of its branches.

Corollary 39.21 (A curve of native gauges: residues and periods)**.** *Let a ray have an integral origin, let Z be a clean source of dimension one, and let $\bar Z$ be the smooth compact curve that contains $\tilde Z^\circ$ as the complement of finitely many* punctures*: the points over the points of Z that are not regular, among them its crossings with other sources and the points where the normal form degenerates, and the points where Z leaves X_F. The solutions of Z are spanned by those of a small loop around each puncture and those of a basis of the cycles of $\bar Z$. When Z is the closure of $\{\alpha_0 b^{\mathbf{v}} : b\in\mathbb{C}^\times\}$, $\mathbf{v}\in\mathbb{Z}^{\mathcal{E}}$, with $b\mapsto\alpha_0 b^{\mathbf{v}}$ injective, and the normal coordinates are those of one fixed complement N of $\mathbb{C}\mathbf{v}$ in $\mathbb{C}^{\mathcal{E}}/\mathbb{C}\mathbf{1}$, $\alpha=\alpha_0 b^{\mathbf{v}}e^{w}$ with $w\in N$, the determinant $\det A(b)$ is rational in b, every $a_i(\cdot\,;g)$ is $r_i(b;g)\det A(b)^{-1/2}\,db/b$ with r_i rational in b, $\tilde Z^\circ$ is an open part of the curve $Y^2=\det A(b)$, and the loop around a puncture q gives $2\pi i$ times the residue at q of $f_Z^k k^{-m/2}\sum_i k^{-i}a_i(\cdot\,;g)$.*

Theorem 39.22 (A native gauge of corank at most one)**.** *Let α_c be an isolated native gauge of a ray at which the Hessian of $\log f$ in y has corank at most one. There are holomorphic coordinates $(X,w)\in\mathbb{C}\times\mathbb{C}^{n-2}$ at α_c with*

$$\log f=\log f(\alpha_c)-\tfrac12\sum_i w_i^2+c\,X^{\mu+1},\qquad c\neq 0,$$

for an integer $\mu\geq 1$, the multiplicity*; $\mu=1$ exactly when α_c is nondegenerate, and otherwise ∂_X spans the kernel of the Hessian at α_c. Then α_c is an A_μ* point*, and:*

(i) There are μ solutions $A_{c,i}$, $0\leq i<\mu$, $A_{c,i}$ of type $\big(f(\alpha_c),-\frac{n-2}{2}-\frac{i+1}{\mu+1}+\mathbb{Z}\big)$: the expansions of the integrals of $g\,f^k\omega_0$ over the chains (real normal disc)$\times\Gamma_i$, where Γ_i is the combination of paths between the $\mu+1$ sectors on which $\operatorname{Re}(cX^{\mu+1})<0$ with $\int_{\Gamma_i}X^j e^{cX^{\mu+1}}dX=\delta_{ij}$ for $0\leq j<\mu$. The leading coefficient of $A_{c,i}(g)$ is $(2\pi)^{(n-2)/2}$ times the coefficient of X^i in the Taylor expansion of $g\,\omega_0/(dX\wedge dw)$ at α_c. They are linearly independent, their exponents lying in distinct classes modulo $\mathbb{Z}$, and $A_{c,0}=(2\pi)^{-1/2}A_c$ for $\mu=1$.

(ii) If a symmetry ϕ of the ray fixes α_c and $\mu\geq 2$, its differential multiplies the kernel by a root of unity ζ_ϕ with $\zeta_\phi^{\mu+1}=1$, and $A_{c,i}\circ\phi^*=\zeta_\phi^i A_{c,i}$.

Corollary 39.23 (Signs at a centre)**.** *On the centre ray of a family (Definition 38.13) let a sign η lie in X_F (Theorem 38.17). The Cremona is a symmetry of the ray that fixes η and acts on the tangent space there by -1.*

(i) If η is nondegenerate, $A_\eta\circ\mathrm{cr}^*=A_\eta$.

(ii) If η is isolated and the Hessian at η has corank one, η is an A_μ point with $\mu\geq 3$ odd, and $A_{\eta,i}\circ\mathrm{cr}^*=(-1)^iA_{\eta,i}$: the $\frac{\mu+1}{2}$ solutions of even i do not vanish on the Cremona-invariant insertions, and the $\frac{\mu-1}{2}$ of odd i vanish on them. For $\mu=3$ the three exponents are $-\frac{n-2}{2}-\frac14$, $-\frac{n-1}{2}$ and $-\frac{n-2}{2}-\frac34$, the middle one odd.

Proposition 39.24 (Symmetries on the cycles of a source)**.** *Let a ray have an integral origin, and let a symmetry ϕ, with $\phi^*\omega_0=\varepsilon_\phi\,\omega_0$, map a clean source Z onto a source. It maps Z° onto $\phi(Z)^\circ$ and lifts to $\tilde\phi:\tilde Z^\circ\to\widetilde{\phi(Z)}^\circ$, carrying each oriented normal disc to its image, and*

$$A_\gamma\circ\phi^*=\varepsilon_\phi\,A_{\tilde\phi_*\gamma}.$$

For a finite group G of symmetries, on $\mathfrak{M}^G$ the solution of γ is that of the averaged cycle $|G|^{-1}\sum_{\phi\in G}\varepsilon_\phi\,\tilde{\phi}_*\gamma$; a cycle class with $\sum_{\phi\in G_Z}\varepsilon_\phi\,\tilde{\phi}_*[\gamma]=0$, G_Z the stabilizer of Z, gives a solution that vanishes on $\mathfrak{M}^G$.

Theorem 39.25 (What the invariant module sees at the high end)**.** *Let G be a finite group of symmetries of a ray, and let α_c be an A_μ point with stabilizer G_c. The solutions $A_{c,i}$ for which $\phi\mapsto\zeta_\phi^i$ is trivial on G_c (all of them when $\mu=1$, with $\zeta_\phi^0=1$) average over G to invariant solutions that do not vanish on $\mathfrak{M}^G$; the others vanish on $\mathfrak{M}^G$. So*

$$\dim_{\mathbb{C}(k)}\mathfrak{M}^G\ \geq\ \sum_{O}\#\big\{0\leq i<\mu_O:\ \zeta_\phi^i=1\ \textit{for every}\ \phi\in G_{c_O}\big\},$$

the sum over the G-orbits O of the isolated native gauges of corank at most one, α_{c_O} a point of O and μ_O its multiplicity.

Where the critical locus of the stride is not a set of nondegenerate points, its components still carry solutions. Over a clean family of native gauges the integral over a normal disc is a Gaussian, whose expansion on holomorphic normal slices is a holomorphic form of top degree on the family, so a closed one, with values in the orientation of the normal discs; every closed cycle carries a solution, because integration by parts is exact under the normal Gaussian and integrates to zero over a cycle along the family. On a curve the cycles are the loops around its punctures, among them its crossings with other sources and its degenerate points, and the cycles of its compactification: residues and periods, on the double cover where the determinant of the normal form has a square root. A native gauge of corank one carries one solution for each class of paths between the sectors where its reduced function decays, with exponents spaced by $\frac{1}{\mu+1}$. A symmetry that fixes a native gauge acts on its solutions: trivially at a nondegenerate one, and through the powers of its eigenvalue on the kernel at a degenerate one, so that at a sign of a centre, fixed by the Cremona, a point of multiplicity three carries an odd solution that every Cremona-invariant insertion misses. The invariant module counts each orbit of sources by the solutions its stabilizer leaves alone.

39.5 Closings at the low end

Definition 39.26 (Closings)**.** Let a ray have an integral origin, $\mathbf{m}_0\in\mathbb{Z}^{\mathcal{E}}$ and $s_{0j},t_{0j}\in\mathbb{Z}$, and a direction in which every letter has a positive exponent, $s_{1j},t_{1j}\geq 1$. For an insertion g let $j(g)$ be the least integer such that $g\,f^{-j}\omega_0/dy$ is a Laurent polynomial for every $j\geq j(g)$. The *closings* of g are the constant terms

$$\mathsf{C}_j(g)=\mathrm{CT}\left(g\,f^{-j}\,\omega_0/dy\right)=\frac{1}{(2\pi i)^{n-1}}\oint_{|\alpha_e|=1,\ e\neq e_0}g\,f^{-j}\,\omega_0,\qquad j\geq j(g),$$

taken on the chart $\alpha_{e_0}=1$, where they do not depend on e_0.

Proposition 39.27 (One form on the orthant and on the torus)**.** *Let a ray have origin 0 and $\kappa_{1,S}>0$ for every nonempty proper S, let $\alpha^\star$ be the native gauge of its direction on the positive orthant, $\alpha^\star_{e_0}=1$, and put $A=f(\alpha^\star)^{-1}$. On the chart $\alpha_{e_0}=1$, with $dy=\prod_{e\neq e_0}d\alpha_e/\alpha_e$, the rational form $\varpi_q=dy/(f^{-1}-q)$ has*

$$\int_{y_{e_0}=0}\varpi_q=\sum_{k\geq 1}\langle 1\rangle_k\,q^{k-1}\qquad(|q|<A),$$

$$\frac{1}{(2\pi i)^{n-1}}\oint_{|\alpha_e|=\alpha_e^\star,\ e\neq e_0}\varpi_q=-\sum_{j\geq 0}\mathrm{CT}\left(f^{-j}\right)q^{-j-1}\qquad(|q|>A),$$

where $\mathrm{CT}(f^{-j})$ *is the constant term of the Laurent polynomial* f^{-j}*, the closing* $\mathsf{C}_j(1)$ *when every letter exponent of the direction is positive (Definition 39.26). So the generating function of the values near* $q=0$ *and that of the constant terms near* $q=\infty$ *are the integrals of one form, whose denominator is the pencil* $\alpha^{-\mathbf{m}_1}\big(\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}-q\,\alpha^{\mathbf{m}_1}\big)$ *of Theorem 38.14(ii), over the orthant and over the torus through* $\alpha^\star$*.*

Proposition 39.28 (The constant terms of a centre ray and its signs)**.** *On the centre ray of a family at the strengths* $\boldsymbol{\sigma}$ *(Definition 38.13), with the tables indexed by* l*, for every* $j\in\mathbb{N}$

$$\frac{1}{2^{n-1}}\sum_\eta\prod_l K_l(\eta)^{2j\sigma_l}=\sum_{v\in 2\mathbb{Z}^{\mathcal{E}}}[\alpha^v]\,f^{-j}\ \geq\ \mathrm{CT}\left(f^{-j}\right),$$

the sum over the 2^{n-1} *signs* η *of Theorem 38.17: the mean of the* j*-th powers of the values* $f^{-1}(\eta)=\prod_l K_l(\eta)^{2\sigma_l}$ *at the signs is the sum of the coefficients of* f^{-j} *at the monomials with even exponents.*

Theorem 39.29 (The closings are a solution)**.** *For a ray as in Definition 39.26,* $\mathsf{C}_j\big((c_e^{(0)}-j\,c_e^{(1)}+\partial_{y_e})g\big)=0$ *and* $\mathsf{C}_j(fg)=\mathsf{C}_{j-1}(g)$ *for* $j>j(g)$*, and*

$$V_{\mathsf{C}}\Big(\sum_i g_i\otimes r_i\Big)(k)=\sum_i r_i(k)\,\mathsf{C}_{-k}(g_i),$$

at the integers $k\leq-\max_i j(g_i)$ *off the poles of the* r_i*, is a well defined solution with values in germs at* $-\infty$*. So for masters* $\mathbf{b}$*,* $\mathsf{C}_{j+1}(\mathbf{b})=\mathbf{B}_{\mathbf{b}}(-j)\,\mathsf{C}_j(\mathbf{b})$ *for all large* j*, with* $\mathsf{C}_j(\mathbf{b})=(\mathsf{C}_j(b_1),\dots,\mathsf{C}_j(b_{\mathtt{m}}))$*, and the law of* [1] *gives* $\sum_i p_i(-j)\,\mathsf{C}_{j-i}(1)=0$ *for all large* j*.*

Proposition 39.30 (The closings are the top principal parts)**.** *Let a ray as in Definition 39.26 have origin* 0*, so that* $\omega_0=dy$ *and* $\langle g\rangle_k=\int_{y_{e_0}=0}g\,f^k\,dy$*, and* $\kappa_{1,S}>0$ *for every nonempty proper* S*, and put*

$$V_1=\sum_\pi\prod_{l=2}^n\kappa_{1,U_l}^{-1}>0,$$

the sum over the orders π *of the marks with tails* U_l *(Theorem 17.5(i)). For every insertion* g *the value* $\langle g\rangle_k$ *continues to a meromorphic function of* k *whose poles have order at most* $n-1$*, and for every integer* $j\geq j(g)$

$$\lim_{k\to -j}(k+j)^{n-1}\langle g\rangle_k=V_1\,\mathsf{C}_j(g).$$

Proposition 39.31 (The low end, for every direction)**.** *Let a ray have* $\kappa_{1,S}>0$ *for every nonempty proper* S*; its letter exponents* $s_{1j},t_{1j}\in\mathbb{N}$ *may vanish.*

(i) For an integral origin, every pole of the value of an insertion $g=\alpha^u\prod_j K_j^{-a_j}\Psi_j^{-b_j}$ *is a* k *with* $\kappa_{0,U}+\kappa_U(g)+k\,\kappa_{1,U}\in\{0,-1,-2,\dots\}$ *for some nonempty proper* U *(Proposition 38.5(ii)), and so lies on one of the lines* $\kappa_{1,U}\,k\in\mathbb{Z}$ *set by the walls of the direction; its order is at most* $n-1$*.*

(ii) For the origin 0*, for every insertion* g *and every* $j\in\mathbb{N}$ *with* $f^{-j}g$ *a Laurent polynomial, with* CT *its constant term as in Definition 39.26,*

$$\lim_{k\to -j}(k+j)^{n-1}\langle g\rangle_k=V_1\ \mathrm{CT}\left(f^{-j}g\right),\qquad V_1=\sum_\pi\prod_{l=2}^n\kappa_{1,U_l}^{-1}$$

as in Proposition 39.30; in particular $\lim_{k\to -j}(k+j)^{n-1}\langle 1\rangle_k = V_1\,\mathrm{CT}(f^{-j})$ *for every* $j\in\mathbb{N}$.

Corollary 39.32 (The law read at the low end)**.** *For a ray as in Proposition 39.30, with the law* $\sum_{i=0}^{h} p_i(k)\,\mathrm{Sh}^i[1]=0$ *of* $[1]$*, the closings of* 1*,* $\mathsf{C}_j(1)=\mathrm{CT}(f^{-j})$*, satisfy*

$$\sum_{i=0}^{\min(h,j)} p_i(-j)\,\mathsf{C}_{j-i}(1)=0 \qquad \text{for every } j\ge 0.$$

On the centre ray of a family with every $\sigma_j\ge 1$*,* $f^{-1}=\prod_j\big(K_j(\alpha)K_j(\alpha^{-1})\big)^{\sigma_j}$*, and* $\mathsf{C}_j(1)$ *is the constant term of its* j*-th power.*

Definition 39.33 (The low-end coefficients)**.** For a ray with origin 0 and $\kappa_{1,S}>0$ for every nonempty proper S, the *low-end coefficients* L_i, $i\in\mathbb{N}$, are the Laurent coefficients of the values of 1 at $k=0$,

$$\langle 1\rangle_k=\sum_{i\ge 0}\mathsf{L}_i\,k^{i-n+1} \qquad \text{near } k=0;$$

$\mathsf{L}_0=V_1>0$.

Corollary 39.34 (The low-end relation)**.** *Let a ray have origin* 0 *and* $\kappa_{1,S}>0$ *for every nonempty proper* S*, whatever its letter exponents, and let* $p_0,\dots,p_h\in\mathbb{C}[k]$ *have* $\sum_i p_i(k)f^i\in\mathcal{L}$*: the law of* $[1]$*, or a relation given by Proposition 39.6, by a box among them (Theorem 39.17(iv)).*

(i) $\sum_i p_i(k)\langle 1\rangle_{k+i}=0$ *as meromorphic functions of* k*, and at every* k *with* $\mathrm{Re}\,k>0$*; and* $\sum_{i=0}^{\min(h,j)} p_i(-j)\ \mathrm{CT}\left(f^{i-j}\right)=0$ *for every* $j\in\mathbb{N}$.

(ii) p_0 *vanishes to order at least* $n-1$ *at* $k=0$*, and the* low-end relation

$$[k^{n-1}]p_0\cdot\mathsf{L}_0+\sum_{i=1}^{h}p_i(0)\,\langle 1\rangle_i=0$$

holds.

(iii) If $\sum_i p_i(k)f^i=\sum_e\big(k\,c_e^{(1)}+\partial_{y_e}\big)G_e$ *is a certificate whose* G_e *have no pole at* $k=0$*, then* $\sum_{i\ge 1}p_i(0)f^i=\sum_e\partial_{y_e}G_e(0)$*, and on the chart the flux of the field* $\big(G_e(0)-G_{e_0}(0)\big)_{e\ne e_0}$ *out of the cube* $\{|y_e|\le R,\ e\ne e_0\}$ *tends to* $-[k^{n-1}]p_0\cdot\mathsf{L}_0$ *as* $R\to\infty$*: the low-end relation is the flux at infinity of the bottom piece of the certificate (Proposition 38.8).*

Proposition 39.35 (The low-end coefficients are integrals over the faces of the sectors)**.** *Let a ray have origin* 0 *and* $\kappa_{1,S}>0$ *for every nonempty proper* S*. For an order* π *of* $\mathcal{E}$ *with tails* U_l *write* $\kappa_l=\kappa_{1,U_l}$ *and, in the sector coordinates* $q\in[0,1]^{n-1}$ *(Theorem 8.5(ii)),* $f=\prod_{l=2}^{n}q_l^{\kappa_l}\,\varphi_\pi(q)$ *with* $\varphi_\pi=\prod_j k_{j,\pi}^{-s_{1j}}\psi_{j,\pi}^{-t_{1j}}$*, positive and smooth on a neighbourhood of the cube. For* $A\subseteq\{2,\dots,n\}$ *and a function* h *on the cube put*

$$h_{[A]}=\sum_{B\subseteq A}(-1)^{|A\setminus B|}\,h\big|_{q_l=0,\ l\notin B},$$

a function of q_A *that vanishes where some* q_l*,* $l\in A$*, vanishes. Then*

$$\mathsf{L}_i=\sum_\pi\sum_A\prod_{l\notin A}\kappa_l^{-1}\sum_{m+|\boldsymbol{\gamma}|=i-|A|}\frac{1}{m!}\int_{[0,1]^A}\prod_{l\in A}\frac{(\kappa_l\log q_l)^{\gamma_l}}{\gamma_l!\,q_l}\left(\log^m\varphi_\pi\right)_{[A]}dq_A,$$

the second sum over $A \subseteq \{2,\dots,n\}$ and the third over $m \in \mathbb{N}$ and $\boldsymbol{\gamma} \in \mathbb{N}^A$, every integral converging. So $\mathsf{L}_0 = V_1$ and

$$\mathsf{L}_1 = \sum_{\pi} \prod_{l} \kappa_l^{-1} \log \varphi_\pi(0) = -\sum_{\pi} \prod_{l} \kappa_l^{-1} \sum_{j} \left(s_{1j} \log \theta_{j,T^+_{j,\pi}} + t_{1j} \log \theta_{j,T^-_{j,\pi}} \right),$$

with the greedy bases of Lemma 8.3; up to $i = n-1$ only the faces $A \neq \{2,\dots,n\}$ of the cube enter, $1_{[A]}$ being 0 for $A \neq \varnothing$, and the whole cube enters from $i = n$ on.

At the low end of strength the values of a ray from the origin have poles at the nonpositive integers, and the leading coefficient of each pole is one constant, the volume read by the walls of the direction, times the constant term of a Laurent polynomial: the closings. Integration by parts lowers the pole of every monomial but the constant one, so only the constant term survives at the top order. The closings are the values over the compact torus, a cycle on which every negative power of the stride is single-valued, and they obey every relation of the ray; read at $k = -j$, the law of [1] is a recurrence for the constant terms of the powers of f^{-1}, from $j = 0$ on.

Remark 39.36. The constant term of a Laurent polynomial is its mean over every torus $|\alpha_e| = \rho_e$. When the walls of the direction are positive, on the torus through the native gauge $\alpha^\star$ of the direction $|f^{-1}|$ is largest at $\alpha^\star$ alone, and as $j \to \infty$ the closings localize there, with the expansion of $A_\star$ read at $k = -j$ (Theorem 39.41). Where another torus can be deformed into descent cycles of $|f^{-1}|$, the closings decompose over the sources met on the way, with integer multiplicities, a family of native gauges among them. The decomposition does not enlarge what they see: they annihilate exactly what $A_\star$ annihilates (Section 39.7), and what they supply is a cheap stream of equations with the strength of that one solution.

39.6 Separating the sources and gluing them

Proposition 39.37 (The high end reads the formal type)**.** *Let $\mathbf{B} = \mathbf{B_b}$ be regular at $k = \infty$, $\mathbf{B}(k) = \mathbf{B}_0 + \mathbf{B}_1 k^{-1} + O(k^{-2})$, and let V be a solution of type $(\lambda, \rho + \mathbb{Z})$ whose solution vector is $v_V = \lambda^k k^\rho (a_0 + a_1 k^{-1} + \cdots)$ with $a_0 \neq 0$. Then $\mathbf{B}_0 a_0 = \lambda^{-1} a_0$, and for every row vector w with $w\mathbf{B}_0 = \lambda^{-1} w$ and $w a_0 \neq 0$, $\rho = -\lambda\, w\mathbf{B}_1 a_0/(w a_0)$. So the native value of every source with a solution that does not vanish on $\mathfrak{N}$ is the reciprocal of an eigenvalue of $\mathbf{B}(\infty)$; its exponent, congruent modulo $\mathbb{Z}$ to $-\frac{n-1-d}{2}$ for a clean source of dimension d and to $-\frac{n-2}{2} - \frac{i+1}{\mu+1}$ for the solution $A_{c,i}$ of an A_μ point, is read from $\mathbf{B}_1$ when a left eigenvector of $\mathbf{B}_0$ for λ^{-1} does not annihilate a_0, as when λ^{-1} is a simple eigenvalue.*

Proposition 39.38 (The spectrum at infinite strength; changes of masters)**.** *Let insertions $b_1 = 1, b_2, \dots, b_{\mathfrak{m}}$ have classes that span a submodule $\mathfrak{N}$, and let $\mathbf{B} \in \mathrm{Mat}_{\mathfrak{m}}(\mathbb{C}(k))$ have $[f^{-1} b_i] = \sum_l \mathbf{B}_{il} [b_l]$: masters with $\mathbf{B_b}$, or the spanning set and the matrix of an accepted box (Theorem 39.17). For a nondegenerate native gauge α_c put $\beta_c = \big(b_l(\alpha_c)\big)_l$, nonzero as $b_1 = 1$, and let $\hat{A}_c$ be the normalization of the expansion A_c (Definition 39.2).*

(i) If the vectors β_c of a set C of nondegenerate native gauges span $\mathbb{C}^{\mathfrak{m}}$, the classes of the b_l are linearly independent: they are masters of $\mathfrak{N}$, $\dim_{\mathbb{C}(k)} \mathfrak{N} = \mathfrak{m}$, and $\mathbf{B} = \mathbf{B_b}$.

(ii) If $\mathbf{B}$ is regular at $k = \infty$, then $\mathbf{B}(\infty)\, \beta_c = f(\alpha_c)^{-1} \beta_c$ for every nondegenerate native gauge α_c: the reciprocal of every nondegenerate native value is an eigenvalue of $\mathbf{B}(\infty)$, with the values of the b_l at the gauge as an eigenvector. So gauges of distinct native values have independent vectors β_c, there are at most $\mathfrak{m}$ distinct nondegenerate native values, and under

(i) $\mathbf{B}(\infty)$ *is diagonalizable with the eigenvalues* $f(\alpha_c)^{-1}$, $\alpha_c \in C$. *In particular, when two nondegenerate native gauges of distinct native values have proportional vectors* β_c, $\mathbf{B}$ *is not regular at* $k = \infty$.

(iii) *Let* $T \in GL_{\mathfrak{m}}(\mathbb{C}(k))$ *make* $\tilde{\mathbf{B}}(k) = T(k-1)\,\mathbf{B}(k)\,T(k)^{-1}$, *the matrix of* Sh^{-1} *on the elements* $\tilde{b}_i = \sum_l T_{il}(k)[b_l]$, *regular at* $k = \infty$. *Then* $f(\alpha_c)^{-1}$ *is an eigenvalue of* $\tilde{\mathbf{B}}(\infty)$ *for every nondegenerate native gauge* α_c, *with the leading coefficient of the series vector* $T(k)\big(\hat{A}_c(b_l)\big)_l$ *as an eigenvector; and if* $T(k-1)T(k)^{-1} \to I$ *as* $k \to \infty$, *the eigenvalues of* $\mathbf{B}(k)$ *tend to those of* $\tilde{\mathbf{B}}(\infty)$, *with their multiplicities.*

(iv) *The condition* $T(k-1)T(k)^{-1} \to I$ *cannot be dropped. For a constant* $\mathbf{B}_\circ = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \in GL_2(\mathbb{C})$ *with* $x_{12} \neq 0$ *and* $S(k) = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$, *the matrix* $\mathbf{B}(k) = S(k-1)\,\mathbf{B}_\circ\,S(k)^{-1}$ *is not regular at* $k = \infty$, *the change* $T = S^{-1}$ *makes it* $\mathbf{B}_\circ$, *with* $T(k-1)T(k)^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, *and at every* k *the eigenvalues of* $\mathbf{B}(k)$ *are the roots of* $\lambda^2 - (x_{11} + x_{22} - x_{12})\lambda + x_{11}x_{22} - x_{12}x_{21}$, *which are not those of* $\mathbf{B}_\circ$.

Theorem 39.39 (Separating the sources)**.** *Let* $\mathbf{B} = \mathbf{B}_{\mathfrak{b}}$ *be regular at* $k = \infty$, *and let the spectrum of* $\mathbf{B}(\infty)$ *be the disjoint union of nonempty sets* $\Lambda_1, \dots, \Lambda_s$. *Over* $\mathbb{C}((1/k))$, *with the shift* ς,

$$\mathfrak{N} \otimes \mathbb{C}((1/k)) = \bigoplus_{a=1}^{s} \hat{\mathfrak{N}}_a,$$

uniquely, with every $\hat{\mathfrak{N}}_a$ *stable under the shift and the matrix of* Sh^{-1} *on it regular at* ∞ *in a suitable basis, with value at* ∞ *of spectrum* Λ_a; *equivalently there is* $T \in GL_{\mathfrak{m}}(\mathbb{C}[[1/k]])$ *with* $T(k-1)\,\mathbf{B}(k)\,T(k)^{-1}$ *block diagonal. A solution of type* $(\lambda, \rho + \mathbb{Z})$ *with* $\lambda^{-1} \in \Lambda_a$ *vanishes on every* $\hat{\mathfrak{N}}_b$ *with* $b \neq a$, *and every group of symmetries that maps* $\mathfrak{N}$ *onto itself maps each* $\hat{\mathfrak{N}}_a$ *onto itself.*

Corollary 39.40 (Gluing at the two ends)**.** *Let* $\mathcal{S}$ *consist of formal solutions and the closings* V_{C}, *and let* $\mathfrak{N}_{\mathsf{C}} = \mathrm{Ann}(V_{\mathsf{C}}) \cap \mathfrak{N}$ *be the submodule the closings miss. Then* $\mathcal{S}$ *determines* $\mathbf{B}_{\mathfrak{b}}$ *exactly when no nonzero submodule of* $\mathfrak{N}_{\mathsf{C}}$ *is annihilated by all the formal solutions in* $\mathcal{S}$. *The closings alone determine it exactly when* $\mathfrak{N}_{\mathsf{C}} = 0$, *as they do when* $\mathfrak{N}$ *is irreducible and* V_{C} *does not vanish on it; formal solutions alone determine it when* $\mathfrak{m}$ *of them are linearly independent over* $\mathbb{C}$ *on* $\mathfrak{N}$, *type by type.*

At the high end the module comes apart: over the formal series in $1/k$ it is the direct sum of one piece for each cluster of native values, the pieces are canonical, and every symmetry of the ray maps each piece to itself. This is where distinguishable sources are separated before the symmetric procedure runs: a piece is read with its own solutions, each class at a degenerate point and each cycle on a family counted by its character under the stabilizer, and the formal type of the one-step matrix at infinity, its eigenvalues and its exponents, contains the native values and the exponents of the sources that the module sees. The pieces need not be rational. The rational one-step matrix glues them, and the closings, an exact solution at the other end of strength, supply the glue: they determine the matrix on every submodule that meets their annihilator only in 0, every irreducible submodule on which they do not vanish among them, and formal solutions are needed only on the submodule that the closings miss.

39.7 What the closings see

Theorem 39.41 (The closings, the values and the native gauge)**.** *Let a ray have an integral origin and an integral direction in which every letter has a positive exponent, $s_{1j}, t_{1j} \geq 1$, with $\kappa_{1,S} > 0$ for every nonempty proper S, and let $\alpha^\star$ be the native gauge of the direction on the positive orthant, where f is largest (Theorem 38.14(i)). Then $\alpha^\star$ is a nondegenerate native gauge of the ray; write $A_\star$ for the expansion over the orthant near it. For every $m \in \mathfrak{M}$ the following are equivalent:*

(i) the closings vanish on m: $V_{\mathsf{C}}(m) = 0$;

(ii) the values vanish on m: $\langle m \rangle_k = 0$ for all large integers k;

(iii) the expansion at $\alpha^\star$ vanishes on m: $A_\star(m) = 0$;

and when they hold, the continuation of $\langle m \rangle_k$ in k vanishes identically. So $\operatorname{Ann}(V_{\mathsf{C}}) = \operatorname{Ann}(\langle\cdot\rangle) = \operatorname{Ann}(A_\star)$, and the closings, the values and $A_\star$ each determine $\mathbf{B}_{\mathbf{b}}$ on exactly the same submodules (Theorem 39.4(iii)). Moreover, for every insertion g,

$$j^{(n-1)/2} f(\alpha^\star)^j \, \mathsf{C}_j(g) \;\sim\; (2\pi)^{-(n-1)} \, \hat{A}_\star(g)\big|_{k=-j} \qquad (j \to \infty).$$

The positive orthant and the torus through $\alpha^\star$ meet only at $\alpha^\star$. The values are read on the orthant, where f is largest at $\alpha^\star$. The closings are means of Laurent polynomials over any torus, and on the torus through $\alpha^\star$ the reciprocal stride has its largest modulus at $\alpha^\star$ alone, because the letters have positive coefficients and the walls of the direction are positive. So the two ends of strength expand at one point, and their expansions are one formal solution, $A_\star$, read as $k \to \infty$ and at $k = -j$. An element on which $A_\star$ vanishes has a density of values that vanishes near the top of its range, and analyticity carries the vanishing to the whole range; and the values reach the closings through their principal parts (Proposition 39.30). The closings see exactly what the values see, and both see exactly what the native gauge of the direction sees.

Definition 39.42 (The value module; what a solution sees)**.** For a ray with $\kappa_{1,S} > 0$ for every nonempty proper S, the *kernel of the values* is $\operatorname{Ann}(\langle\cdot\rangle)$, the submodule of the elements whose values vanish for all large k (Theorem 39.4(ii)), and the *value module* is $\mathfrak{M}/\operatorname{Ann}(\langle\cdot\rangle)$. A solution *sees* a submodule when it does not vanish on it, and a nondegenerate native gauge α_c sees it when A_c does.

Corollary 39.43 (When the closings miss nothing)**.** *Under the hypotheses of Theorem 39.41, let $\mathfrak{N}$ be a submodule. The closings annihilate no nonzero submodule of $\mathfrak{N}$ exactly when the values vanish on no nonzero element of $\mathfrak{N}$, and exactly when $\alpha^\star$ sees every nonzero submodule of $\mathfrak{N}$. Every symmetry of the ray fixes $\alpha^\star$, so for every finite group G of symmetries the closings, the values and $A_\star$ vanish on $(1 - e_G)\mathfrak{M}$: the condition is to be read on $\mathfrak{M}^G$. Let $\mathbf{B} = \mathbf{B}_{\mathbf{b}}$, for masters of $\mathfrak{N}$, be regular at $k = \infty$ with $1/f(\alpha^\star)$ a simple eigenvalue of $\mathbf{B}(\infty)$, let $\hat{\mathfrak{N}}_\star$ be the line of Theorem 39.39 for that eigenvalue, and let $\alpha^\star$ see $\mathfrak{N}$. Then $\alpha^\star$ sees a submodule $\mathfrak{N}'$ exactly when $\hat{\mathfrak{N}}_\star \subseteq \mathfrak{N}' \otimes \mathbb{C}((1/k))$; when it sees every nonzero submodule, $\mathfrak{N}$ has exactly one minimal nonzero submodule; and when $\mathfrak{N} = \mathfrak{N}_1 \oplus \mathfrak{N}_2$ with $\mathfrak{N}_1, \mathfrak{N}_2$ nonzero submodules, the closings annihilate one of the two.*

Corollary 39.44 (What the closings determine)**.** *Under the hypotheses of Theorem 39.41, the closings determine the one-step matrix of the value module: for insertions whose classes modulo $\operatorname{Ann}(\langle\cdot\rangle)$ form a basis of the image of a submodule, Theorem 39.4(iii) and (iv) hold on the value module with the closings alone. For polynomials $p_0, \dots, p_h$, $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ for all large k exactly when $\sum_i p_i(-j)\, \mathsf{C}_{j-i}(1) = 0$ for all large j: the values of 1 and the closings of 1 satisfy the same recurrences, read at k and at $-j$, and have the same tier (Proposition 38.11).*

Proposition 39.45 (The closings measure what they miss)**.** *Let* $\mathbf{b}$ *be masters of a submodule* $\mathfrak{N}$*,* $\mathfrak{m}$ *of them, under the hypotheses of Theorem 39.29; let* $d = \dim_{\mathbb{C}(k)}\big(\operatorname{Ann}(V_{\mathsf{C}}) \cap \mathfrak{N}\big)$*, the dimension of the largest submodule of* $\mathfrak{N}$ *that the closings annihilate, and for* $D \geq 0$ *let* $\mathcal{S}_D$ *be the space of pairs* (P, Q) *of a matrix and a scalar of polynomials of degree at most* D *with* $Q(-j)\,\mathsf{C}_{j+1}(\mathbf{b}) = P(-j)\,\mathsf{C}_j(\mathbf{b})$ *for all large* j*. Then* $\dim \mathcal{S}_D = (\mathfrak{m}d + 1)D + O(1)$ *as* $D \to \infty$*. When* $d = 0$*,* $\dim \mathcal{S}_D = \max(0, D - D_0 + 1)$ *with* D_0 *as in Theorem 39.4(iv); so a single step* $\dim \mathcal{S}_{D+1} - \dim \mathcal{S}_D > 1$ *proves* $d > 0$*.*

The closings therefore certify a one-step matrix by themselves exactly when the native gauge of the direction sees every nonzero submodule. When it does not, the closings show it, by a step of more than one in $\dim \mathcal{S}_D$, and the slope of $\dim \mathcal{S}_D$ is $\mathfrak{m}d + 1$; a slope of one over the degrees computed is evidence that nothing is missed, not a certificate, the degrees of the missed part not being bounded here. What the closings miss is missed by the values as well. It is no defect of the torus: a submodule that the native gauge of the direction does not see is invisible to the integral itself, and an element of it is a relation among values that the relations of the ray do not produce. The symmetries of the ray account for some of these, on $(1 - e_G)\mathfrak{M}$, and so can symmetries that are not maps of its torus at all: when the list is the contraction of an identically self-dual table, the integral over the contracted scale carries the module of the larger table onto that of the ray, and the automorphisms of the larger table act through it.

Proposition 39.46 (A table one mark larger)**.** *Let* $(\mathcal{B}', \theta')$ *be a connected identically self-dual table on* $\mathcal{E}' = \mathcal{E} \sqcup \{g\}$*,* $|\mathcal{E}| \geq 2$ *(Corollary 23.12(iii)), and* $(\mathcal{B}, \theta)$ *its contraction at* g *(Definition 7.3), with letters* K, Ψ*; then* $(\mathcal{B}, \theta)$ *is connected, and the letter of* $\mathcal{B}'$ *is* $K' = \Psi + \alpha_g K$*. Let* F' *and* F *be the two tables at the strength* 1*, read on their centre rays (Definition 38.13),* F' *on a chart* $y_{e_0} = 0$ *with* $e_0 \in \mathcal{E}$*; their strides are* $f' = \alpha^{\mathbf{1}}\alpha_g/K'^2$ *and* $f = \alpha^{\mathbf{1}}/(K\Psi)$*, their modules* $\mathfrak{M}'$ *and* $\mathfrak{M}$*. For a Laurent polynomial* h *on* $\mathcal{E}$*,* $a \in \mathbb{Z}$ *and* $j \in \mathbb{N}$ *with* $\alpha_g^a K'^{-j} h$ *of degree* 0*, put*

$$\pi_*\big(\alpha_g^a K'^{-j} h\big) = r_{a,j}(k)\Big(\frac{\Psi}{K}\Big)^a \Psi^{-j} h, \qquad r_{a,j}(k) = \frac{\Gamma(k+a)\,\Gamma(k+j-a)\,\Gamma(2k)}{\Gamma(k)^2\,\Gamma(2k+j)} \in \mathbb{Q}(k).$$

(i) For every insertion g' *of* F'*, at the points of the positive orthant and for all large* k*,* $\int_0^\infty g' f'^k \, d\alpha_g/\alpha_g = \Gamma\!\left[\begin{matrix} k,\ k \\ 2k \end{matrix}\right] f^k\, \pi_* g'$*. So* $\langle g' \rangle'_k = \Gamma\!\left[\begin{matrix} k,\ k \\ 2k \end{matrix}\right] \langle \pi_* g' \rangle_k$ *for all large* k*, and* $\mathsf{C}'_j(g') = \binom{2j}{j}\,\mathsf{C}_j\big(\pi_* g'|_{k=-j}\big)$ *for all large* j*. The native gauges of* F' *are the points* $\alpha_{c'} = \big(\alpha_c, \Psi(\alpha_c)/K(\alpha_c)\big)$ *over the native gauges* α_c *of* F*, with* $f'(\alpha_{c'}) = f(\alpha_c)/4$*, and* $\alpha_{c'}$ *is non-degenerate exactly when* α_c *is; then* $A'_{c'}(g') = \Gamma\!\left[\begin{matrix} k,\ k \\ 2k \end{matrix}\right] A_c(\pi_* g')$*, the stacked Gamma read as its expansion at* $k = \infty$ *and the disc at* $\alpha_{c'}$ *taken over the disc at* α_c*.*

(ii) π_* *induces a bijection* $\mathfrak{M}' \to \mathfrak{M}$*, with* $\pi_* \operatorname{Sh}' = \frac{k}{2(2k+1)} \operatorname{Sh} \pi_*$*.*

(iii) Every automorphism β *of* $(\mathcal{B}', \theta')$ *acts on* $\mathfrak{M}$ *by* $T_\beta = \pi_* \phi_\beta^* \pi_*^{-1}$*, which commutes with* Sh*;* $T_\beta = \phi_\beta^*$ *when* $\beta(g) = g$*, and when the transposition* β *of a mark* $e \in \mathcal{E}$ *and* g *is an automorphism,*

$$T_\beta[\alpha^u] = \frac{\Gamma(k+u_e)\,\Gamma(k-u_e)}{\Gamma(k)^2}\Big[\alpha^{u - u_e \mathbf{1}_e}\Big(\frac{\Psi}{K}\Big)^{u_e}\Big].$$

With the Cremona, which π_* *carries to the Cremona, the* T_β *generate a finite group* G'*.*

(iv) The values, the closings and $A_\star$*, at the native gauge* $\alpha^\star = \mathbf{1}$*, are invariant under* G' *and vanish on* $(1 - e_{G'})\mathfrak{M}$*. For a group* $G \subseteq G'$ *of symmetries of* F*,* $\mathfrak{M}^G = \mathfrak{M}^{G'} \oplus (1 - e_{G'})\mathfrak{M}^G$

as submodules, and π_* maps the invariant module of F' under its automorphisms and the Cremona onto $\mathfrak{M}^{G'}$.

Corollary 39.47 (The cousins of a fused table)**.** *Let $(\mathcal{B},\theta)$ be a connected fusable table on n marks, of the fusion width $c\geq 1$, with $n+c\geq 3$, and $(\mathcal{B}^\sharp,\theta^\sharp)$ its Foreman fusion (Definition 30.4). Then $\mathcal{B}^\sharp$ is connected, and Proposition 39.46 applies to $\mathcal{B}^\sharp$ and each of its cousins (Proposition 30.20): the automorphisms of $\mathcal{B}^\sharp$ act on the module of the centre ray of every cousin.*

The scale of g enters the letter of the larger table linearly, so the integral over it is a Beta corner, and it carries the centre of F' onto the centre of F insertion by insertion: a power of α_g becomes the same power of Ψ/K, the ratio at which the scale of g is native, times a rational function of k. The relations at the scale of g are exactly what that integral forgets, so the two modules are one, with the shift twisted by the Beta corner. An automorphism of $\mathcal{B}'$ that moves g is no symmetry of F, nor a map of its torus: it moves insertions, not points, and on native gauges it is the lift $\alpha_c\mapsto\alpha_{c'}$ followed by the permutation and the projection. The values, the closings and the expansion at $\alpha^\star$ are read on the orthant, the torus and the point $\mathbf{1}$ of F', which every automorphism keeps; so they see only the part of $\mathfrak{M}$ that G' fixes, and when $\mathfrak{M}^G$ is larger than $\mathfrak{M}^{G'}$ the closings miss the difference, with every symmetry of F in G. A sign η of F lifts to the sign $(\eta,\eta^{\mathcal{E}})$ of F', since $\Psi(\eta)=\eta^{\mathcal{E}}K(\eta)$, with $f(\eta)=4/K'(\eta,\eta^{\mathcal{E}})^2$; signs that the automorphisms of $\mathcal{B}'$ carry to one another have one native value. Proposition 39.48 measures what is missed in this way: an orbit of nondegenerate native gauges of F' that the symmetries of F split into several orbits gives, orbit against orbit, solutions that vanish on $\mathfrak{M}^{G'}$ and not on $\mathfrak{M}^G$.

Proposition 39.48 (What a table one mark larger hides from the closings)**.** *Let $(\mathcal{B}',\theta')$, g, F', F, π_* and G' be as in Proposition 39.46. Let Γ' be the group of symmetries of the centre ray of F' generated by the maps ϕ_β, β an automorphism of $(\mathcal{B}',\theta')$, and the Cremona, and G the group of symmetries of F generated by the maps ϕ_β, β an automorphism of $(\mathcal{B}',\theta')$ with $\beta(g)=g$ read on $\mathcal{E}$, and the Cremona, so that $G\subseteq G'$ (Proposition 39.46(iii)). A native gauge α_c of F lies* below *the native gauge $\alpha_{c'}=\big(\alpha_c,\Psi(\alpha_c)/K(\alpha_c)\big)$ of F' (Proposition 39.46(i)).*

(i) Let $O'_1,\dots,O'_q$ be distinct orbits of Γ' of nondegenerate native gauges of F', and let the native gauges of F below O'_a form s_a orbits of G. Then

$$\dim_{\mathbb{C}(k)}(1-e_{G'})\mathfrak{M}^G\ \geq\ \sum_{a=1}^{q}(s_a-1),$$

and the closings, the values and $A_\star$ vanish on the submodule $(1-e_{G'})\mathfrak{M}^G$. For an orbit O' fix an expansion $A'_{c'_0}$ at one of its points, put $A'_{\gamma(c'_0)}=A'_{c'_0}\circ\gamma^*$ for $\gamma\in\Gamma'$, and for α_c below O' put $A_c=\Gamma{\scriptstyle\begin{bmatrix}k,\ k\\ 2k\end{bmatrix}}^{-1}_{2\ 1}A'_{c'}\circ\pi_*^{-1}$, the stacked Gamma read as its expansion at $k=\infty$. If O_1 and O_2 are two orbits of G below O', the solution $|O_1|^{-1}\sum_{c\in O_1}A_c-|O_2|^{-1}\sum_{c\in O_2}A_c$ is invariant under G, vanishes on $\mathfrak{M}^{G'}$ and does not vanish on $(1-e_{G'})\mathfrak{M}^G$.

(ii) For $r\geq 2$ let $\mathcal{B}'=\binom{[2r+2]}{r+1}$ with unit weights and $g=2r+2$, so that F is $\binom{[2r+1]}{r}$ read on its centre ray at the strength 1; G is then the group generated by the symmetries of the four kinds of that ray (Proposition 39.7): the maps ϕ_β, β a bijection of $[2r+1]$, and the Cremona. With $C_r=\frac{1}{r+1}\binom{2r}{r}$, the signs of F with one and with two scales -1, $2r+1$ and $\binom{2r+1}{2}$ of them, are nondegenerate native gauges of the native value C_r^{-2} and form two orbits of G, and the native gauges above them are the $\binom{2r+2}{2}$ signs of F' with two scales -1, one orbit of Γ'. At the sign of F' with the scales -1 on $D=\{a_1,a_2\}$, with $C=[2r+2]\setminus D$, the Hessian

H' of $\log f'$ *is such that* $-H'$ *has, modulo* $\mathbf{1}$*, the eigenvalue* $r+1$ *on* $\mathbf{1}_{a_1}-\mathbf{1}_{a_2}$*, the eigenvalue* $\frac{r+1}{2r-1}$ *on every* $\mathbf{1}_c-\mathbf{1}_{c'}$ *with* $c,c'\in C$*, and the eigenvalue* $-(r+1)$ *on* $r\mathbf{1}_D-\mathbf{1}_C$*; at the signs of* F *with one and with two scales* -1*, on a chart,* $\det(-H)=-(r+1)^{2r}/(2r-1)^{2r-1}$*. So* $(1-e_{G'})\mathfrak{M}^G\neq 0$*: at the centre of every* $\binom{[2r+1]}{r}$*,* $r\geq 2$*, the closings miss a nonzero submodule of* $\mathfrak{M}^G$*, with every symmetry of the four kinds in* G*.*

The two kinds of signs of $\binom{[2r+1]}{r}$ share their native value because the signs above them are one orbit of the symmetric group of $\binom{[2r+2]}{r+1}$, while the symmetries of the ray keep the two kinds apart; the mean of the solutions over the one kind less the mean over the other is a solution that sees what the larger table hides. Exercise 52.20 reads the small centres with these statements; at $r=2$ the part of the span of a box that the larger table hides from the closings is a line.

39.8 Wheels: the letters as a holonomy

Proposition 39.49 (The holonomy of a wheel)**.** *For* $p\geq 3$ *let* W_p *be the list of the wheel graph: a hub and rim vertices* $1,\dots,p$ *in cyclic order, the marks the spokes* s_i*, joining the hub to* i*, and the rims* r_i*, joining* i *to* $i+1$*, indices modulo* p*; its bases are the spanning trees, of rank* p *on* $n=2p$ *marks. With*

$$\mathbf{U}(a)=\begin{bmatrix}1 & a\\ 0 & 1\end{bmatrix},\qquad \mathbf{L}(b)=\begin{bmatrix}1 & 0\\ b & 1\end{bmatrix},\qquad \mathbf{h}(\alpha)=\prod_{i=1}^{p}\mathbf{U}(\alpha_{s_i})\,\mathbf{L}(\alpha_{r_i}^{-1})\in SL_2,$$

the product taken in the order $i=1,\dots,p$*,*

$$K(\alpha)=\prod_i \alpha_{r_i}\big(\operatorname{tr}\mathbf{h}(\alpha)-2\big),\qquad \Psi(\alpha)=\prod_i \alpha_{s_i}\big(\operatorname{tr}\mathbf{h}(\alpha^{-1})-2\big).$$

The bijection x *with* $x(s_i)=r_{p+1-i}$ *and* $x(r_i)=s_{p+1-i}$ *has* $\operatorname{tr}\mathbf{h}(\alpha\circ x)=\operatorname{tr}\mathbf{h}(\alpha^{-1})$*, so* $K\circ\phi_x=\Psi$ *and* x *exchanges* W_p*. At the table's gauge* $\mathbf{h}(\mathbf{1})=\begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}^p$*, whose eigenvalues are* $\lambda_\pm^p$ *with* $\lambda_\pm=\frac{3\pm\sqrt{5}}{2}$*, the squares of the golden ratio and of its inverse; so* $K(\mathbf{1})=\lambda_+^p+\lambda_-^p-2$*, the Lucas number of index* $2p$ *less* 2*.*

Proposition 39.50 (The centre ray of a wheel)**.** *Put* $\mathfrak{d}(\alpha)=\operatorname{tr}\mathbf{h}(\alpha)-2$ *and* $\mathfrak{d}^*(\alpha)=\operatorname{tr}\mathbf{h}(\alpha^{-1})-2=\mathfrak{d}(\alpha\circ x)$*. The centre ray of* W_p *at the strength* σ *has the stride* $f=(\mathfrak{d}\,\mathfrak{d}^*)^{-\sigma}$*, and* $X_F=\{\mathfrak{d}\,\mathfrak{d}^*\neq 0\}$*.*

(i) Its native gauges are the points of X_F *with* $\mathfrak{d}^*\,d\mathfrak{d}+\mathfrak{d}\,d\mathfrak{d}^*=0$*. At every one* $d\mathfrak{d}$ *and* $d\mathfrak{d}^*$ *are linearly dependent: the native gauges lie on the critical locus of the pair of traces* $(\mathfrak{d},\mathfrak{d}^*)$*, where its differential has rank at most one. The native value is* $(\mathfrak{d}\,\mathfrak{d}^*)^{-\sigma}$*.*

(ii) At a sign η*,* $\mathfrak{d}^*(\eta)=\mathfrak{d}(\eta)$ *and the native value is* $\mathfrak{d}(\eta)^{-2\sigma}$*; at the table's gauge it is* $(\lambda_+^p+\lambda_-^p-2)^{-2\sigma}$*.*

(iii) Every point with $\mathbf{h}(\alpha)=-I$ *has* $d\mathfrak{d}=0$*. The* torsion stratum $\{\mathbf{h}(\alpha)=\mathbf{h}(\alpha^{-1})=-I\}$ *lies in* X_F *and consists of native gauges of native value* $16^{-\sigma}$ *at which* $d\mathfrak{d}=d\mathfrak{d}^*=0$*; it is cut out of the torus of dimension* $2p-1$ *by six equations, so that each of its components has dimension at least* $2p-7$*.*

Proposition 39.51 (The classes of a wheel)**.** *At the table's gauge the occupation of a spoke is*

$$R_{\mathrm{s}}=\frac{\lambda_+^p-\lambda_-^p}{\sqrt{5}\,\big(\lambda_+^p+\lambda_-^p-2\big)},$$

equal to $\frac{1}{2}$ *exactly for* $p = 3$*; the occupation of a rim is* $R_{\mathrm{r}} = 1 - R_{\mathrm{s}}$*, and the masses of* $\Upsilon_{\sigma,\delta;\mathbf{1}}$ *are* $\sigma + \delta(2R_{\mathrm{s}} - 1)$ *on the spokes and* $\sigma - \delta(2R_{\mathrm{s}} - 1)$ *on the rims.* W_p *is an atom at the centre. For* $p \geq 4$ *its classes at every lean are the spokes and the rims, which the reversed exchange* $\bar{\phi}_x$*, a symmetry of the integrand at every lean (Theorem 39.9), interchanges;* $W_3 = M(K_4)$ *is an atom at every lean.*

Remark 39.52. The uniform masses $\sigma\mathbf{1}$ lie on the balance wall of W_p at every lean, since $2r = n$, and they meet the condition of Proposition 39.7 for $\bar{\phi}_x$, which therefore keeps the integrand of every such realization; its native gauge is constant on the spokes and on the rims. For $p = 3$ the uniform masses are the masses of the table's gauge at every lean, R_{s} being $\frac{1}{2}$, and the native gauge stays at $\mathbf{1}$; for $p \geq 4$ they are not, and the native gauge leaves $\mathbf{1}$ as soon as the realization leans. For W_4, at the gauge c on the spokes and 1 on the rims, $K = c(c+2)^2(c+4)$ and $\Psi = (2c+1)^2(4c+1)$. At the full basis lean nativity asks every occupation to be $\frac{1}{2}$, which on a spoke reads $c^2 + 2c - 4 = 0$: the native gauge on the orthant has spokes $\sqrt{5} - 1$, twice the inverse of the golden ratio, against rims 1, and at the full cobasis lean, where the condition reads $4c^2 - 2c - 1 = 0$, spokes $\frac{\sqrt{5}+1}{4}$, half the golden ratio, its reciprocal. The other roots, $-\sqrt{5} - 1$ and $\frac{1-\sqrt{5}}{4}$, are native gauges as well, with negative spokes and in X_F: sources off the orthant. At the basis lean the integrand is $c^{4\sigma}K^{-2\sigma} = \big(c/((c+2)^2(c+4))\big)^{2\sigma}$, whose base is $\frac{5\sqrt{5}-11}{8}$ at the positive root and $-\frac{5\sqrt{5}+11}{8}$ at the negative one, a quarter of the inverse fifth power of the golden ratio and minus a quarter of its fifth power; the cobasis lean gives the same two numbers. Off the centre the gauge on the orthant is not proportional to $\mathbf{1}$, so the realization is not an atom (Theorem 11.4); its classes are the spokes and the rims, and $\bar{\phi}_x$ interchanges them. The torsion stratum of W_4 is a union of curves.

The letters of a wheel are traces of one product of unipotent matrices, the holonomy of a transfer around the rim, and the planar self-duality of the wheel is the transpose of that product. On the centre ray the stride is the inverse product of the two traces, at α and at α^{-1}, so a native gauge lies on the critical locus of the pair of traces, and wherever both holonomies are $-I$ both traces are critical at once: a stratum of native gauges of native value $16^{-\sigma}$, of dimension at least $2p - 7$ wherever it is not empty. At a sign η the two traces agree, in an integer $\mathfrak{d}(\eta)$, and the native value is $\mathfrak{d}(\eta)^{-2\sigma}$; at the table's gauge the holonomy is the p-th power of the step $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, whose eigenvalues are the squares of the golden ratio and of its inverse. The wheel is an atom at the centre and splits into spokes and rims at every lean once $p \geq 4$, the reversed exchange interchanging them; with uniform masses its native gauge then leaves the table's gauge as soon as the realization leans, and for W_4 at a full lean the ratio of the scales of spokes and rims is twice the inverse of the golden ratio or half the golden ratio.

39.9 The procedure

The statements of this chapter assemble into a procedure that reads the one-step matrix and the law of [1] of a ray from its two ends, with no box (Definition 39.11) reduced.

(1) *Symmetries.* From the list read the automorphisms and the exchanges, and keep the maps of Proposition 39.7 that are symmetries of the ray; at a lean of a self-dual family at the table's gauge these include the reversed exchanges (Theorem 39.9). Work in $\mathfrak{M}^G$ with invariant insertions (Theorem 39.8). When the list is the contraction of an identically self-dual table at a mark, the automorphisms of that table act as well, through the integral over the scale of the mark, and the values live on the part they fix, which the centre of the larger table reads directly (Proposition 39.46).

(2) *Sources.* Read the critical locus of the stride modulo G: orbits of nondegenerate native gauges, one solution each; A_μ points, with the classes of trivial character (Theorems 39.22 and 39.25); clean families, with their averaged cycles (Theorem 39.20, Corollary 39.21, Proposition 39.24). Their native values and exponents enter the formal type of the one-step matrix at infinity (Proposition 39.37).

(3) *Closings.* Compute the closings of the candidate masters, constant terms of Laurent polynomials (Definition 39.26).

(4) *Masters.* Choose invariant insertions whose solution vectors over the supplied sources and the closings are linearly independent; there are at most $\dim \mathfrak{M}^G$ of them, and the count of Theorem 39.25 bounds $\dim \mathfrak{M}^G$ below.

(5) *The one-step matrix.* Solve the homogeneous linear equations of Theorem 39.4(iv) at increasing degree D. When the supplied solutions annihilate no submodule, the first degree with a nonzero solution is D_0, the solution is unique up to a constant factor, and it is $\mathbf{B}_\mathbf{b}$.

(6) *The law.* The law of [1] is the least relation among $[1], \mathrm{Sh}[1], \dots$, computed with $\mathbf{B}_\mathbf{b}$; on a ray from the origin it is read again on the closings of 1 at $-j$ (Corollary 39.32).

A case is certified when the supplied solutions annihilate no submodule: by Corollary 39.40 this holds when $\mathfrak{m}$ supplied formal solutions are linearly independent over $\mathbb{C}$ on $\mathfrak{M}^G$ type by type, a full set from the high end; and when the walls of the direction are positive the closings alone certify exactly when the native gauge of the direction sees every nonzero submodule of $\mathfrak{M}^G$ (Corollary 39.43), as when $\mathfrak{M}^G$ is irreducible and they do not vanish on it. The closings show it when they miss something: the dimension of the solutions of their equations then has slope $\mathfrak{m}d + 1$, with d the dimension of the largest missed submodule $\mathrm{Ann}(V_\mathsf{C}) \cap \mathfrak{M}^G$, and a single step of more than one proves $d > 0$ (Proposition 39.45); a slope of one over the degrees computed is evidence that nothing is missed, not a certificate. When they miss something they still give the one-step matrix of the value module and the law of the values (Corollary 39.44), and the missing part is read from sources whose solutions see it: when it comes from a table one mark larger, the centre of that table reads $\mathfrak{M}^{G'}$ (Proposition 39.46), and the differences of the solutions over the orbits of native gauges that the symmetries of the ray split do not vanish on the rest (Proposition 39.48). With fewer formal solutions and no such knowledge of $\mathfrak{M}^G$, a unique solution at the least degree that also meets equations held out is evidence, not a certificate: a submodule annihilated by every supplied solution would leave a different matrix that meets all of them. Masters that span $\mathfrak{M}^G$ come from a box when every letter exponent of the direction is positive: the spanning set of a box accepted at $\mathfrak{s}$ spans the classes of the invariant Laurent combinations of wall degree at most $\mathfrak{s}$ (Theorem 39.17(i)), which increase with $\mathfrak{s}$, and some box is accepted with a spanning set whose classes span $\mathfrak{M}^G$ (Theorem 39.18). That spanning set is a set of masters of $\mathfrak{M}^G$ when it is independent: when as many solutions as it has elements are linearly independent on its span, type by type (Lemma 39.3), or when its values at nondegenerate native gauges span $\mathbb{C}^\mathfrak{m}$ (Proposition 39.38(i)).

Chapter 40

The stride's law and the gauge end

The values of a ray with real origin and positive walls are the moments of one measure, the stride's law: the image of the origin form on the positive part of the chart under the stride. It is carried by $(0, f(\alpha^\star)]$, and the values are read from it three ways, as a Trailmoment on a line, through one Gramma on the Laplace side, and as the log-partition of the exponential family that the Breta shape laws form along the ray. Its two ends are the two ends of strength: near 0 its mass grows like a power of the logarithm, with the volume of the direction as coefficient, and at the native value $f(\alpha^\star)$ it ends in a power $(f(\alpha^\star) - t)^{\frac{n-3}{2}}$ times a function holomorphic there, whose value is read off the Hessian at the native gauge. Its density is real-analytic in between. Every recurrence of the values is a differential equation for the density, the recurrence read backwards, and holds at every strength once it holds at the large integers. At the native value of the direction, when it is a simple root of the symbol, the germs of solutions that the equation lets end there form a line; the density spans it, its exponent is forced to $\frac{n-3}{2}$, and its continuation along the open support is the density: so the recurrence, the native gauge of the direction and the Hessian there give the values and their derivatives in the strength exactly, with no datum from the low end and no inequality. At a centre the same native gauge lies on the unit torus, where the stride is real and at least $f(\alpha^\star)$; the torus value continues the closings in the strength, and the densities of the chart and of the torus at the native value are the two cut solutions of one function, on its two sides.

40.1 The stride's law

Definition 40.1 (The stride's law)**.** Let a ray of a family on $n \geq 2$ marks have a real origin and $\kappa_{1,S} > 0$ for every nonempty proper S (Definition 38.1). On the positive part of the chart, $\alpha_{e_0} = 1$ and $y = \log \alpha \in \mathbb{R}^{\mathcal{E} \setminus e_0}$, the origin form $\omega_0 = (\omega_0 / dy)\, dy$ is a positive measure, its density $\alpha^{\mathbf{m}_0} \prod_j K_j^{-s_{0j}} \Psi_j^{-t_{0j}}$ taken with the principal powers. The *stride's law* of the ray is its image under the stride,

$$\mu_f(A) = \int_{\{y:\ f(e^y) \in A\}} \omega_0 \qquad (A \subseteq (0, \infty) \text{ a Borel set}),$$

a measure on $(0, \infty)$, and ϱ_f is its density where it has one. For the origin 0, $\omega_0 = dy$, and μ_f is the image under f of the Lebesgue input of the chart (Definition 5.4).

Proposition 40.2 (A ray is one Trailmoment on a line and one Gramma)**.** *Let a ray be as in Definition 40.1, let $\alpha^\star$ be the native gauge of its direction on the positive orthant, $\alpha^\star_{e_0} = 1$ (Theorem 38.14), and write in this statement $f^\star = f(\alpha^\star)$.*

(i) *μ_f is carried by $(0, f^\star]$, it is finite on $[t, f^\star]$ for every $t > 0$, and at every complex k at which $\langle 1\rangle_k$ converges*

$$\langle 1\rangle_k = \int_{(0,f^\star]} t^k \, \mu_f(dt) = \big(\mathsf{T}_{k;(1)}[t^{-1}]\,\mu_f\big)\big((0,\infty)\big) = (\mathsf{T}_{k;(1)}[x]\,\iota_*\mu_f)((0,\infty)), \qquad \iota(t) = t^{-1}:$$

the values are the Trailmoments on a line of index k, one unit exponent and the coefficient function x of one input, the image of μ_f under the reciprocal.

(ii) *Let the origin carry no letter exponents, $\mathbf{s}_0 = \mathbf{t}_0 = \mathbf{0}$, so that $\omega_0 = \alpha^{\mathbf{m}_0} dy$, and write $f^{-1} = \sum_i w_i\, \alpha^{\mathbf{v}_i}$ on the chart, a Laurent polynomial in the scales α_e, $e \neq e_0$, with positive coefficients w_i and exponents $\mathbf{v}_i \in \mathbb{Z}^{\mathcal{E}\setminus e_0}$; put $\mathbf{v} = (\mathbf{v}_i)_i$ and $\mathbf{w} = (w_i)_i$. The cone of $\mathbf{v}$ is $\mathbb{R}^{\mathcal{E}\setminus e_0}$, so the Gramma $\mathsf{G}_{\nu;\mathbf{v}}(u\mathbf{w})$ is entire in ν (Theorem 2.2), and at every k with $\operatorname{Re} k > 0$ at which $\langle 1\rangle_k$ converges*

$$\Gamma(k)\,\langle 1\rangle_k = \int_0^\infty u^{k-1}\, \mathsf{G}_{\mathbf{m}_0|_{\mathcal{E}\setminus e_0};\,\mathbf{v}}(u\,\mathbf{w})\, du,$$

the Laplace side (Theorem 5.11) of (i), whose Laplace transform beyond the cut 0 is the Gramma of the reciprocal stride.

(iii) *For real k at which $\langle 1\rangle_k$ converges, let $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}_0 + k\mathbf{m}_1; \mathbf{s}_0 + k\mathbf{s}_1, \mathbf{t}_0 + k\mathbf{t}_1)$ (Definition 8.10). The law of $f(U)$ is $t^k \mu_f(dt)/\langle 1\rangle_k$: along the ray the shape laws are the exponential family with the sufficient statistic $\log f$, the base measure μ_f and the log-partition $\log\langle 1\rangle_k$, so that*

$$\frac{d}{dk}\log\langle 1\rangle_k = \mathbb{E}\big[\log f(U)\big], \qquad \frac{d^2}{dk^2}\log\langle 1\rangle_k = \mathrm{Var}\,\big[\log f(U)\big] > 0.$$

Proposition 40.3 (The two ends of the stride's law)**.** *Let a ray be as in Definition 40.1, with $\alpha^\star$ as in Proposition 40.2 and H the Hessian of $-\log f$ at $\log\alpha^\star$ on the chart (Theorem 38.15); write in this statement $f^\star = f(\alpha^\star)$ and $w^\star = (\omega_0/dy)(\alpha^\star)$.*

(i) The open support. *μ_f has a density ϱ_f, real-analytic and positive on $(0, f^\star)$.*

(ii) The native value. *Near $f^\star$, $\varrho_f(t) = (f^\star - t)^{\frac{n-3}{2}} g(f^\star - t)$ for $t < f^\star$, with g holomorphic at 0 and*

$$g(0) = \frac{w^\star}{\Gamma(\frac{n-1}{2})\sqrt{\det H}}\Big(\frac{2\pi}{f^\star}\Big)^{\frac{n-1}{2}}.$$

(iii) The low end. *Let the origin be 0, let κ be the wall at a tilt of the question of the direction (Definitions 17.1 and 17.3), read at y, and put $C_- = \prod_j K_j(\mathbf{1})^{-s_{1j}-t_{1j}}$, $C_+ = \prod_j \theta_{j,\min}^{-s_{1j}-t_{1j}}$, $\theta_{j,\min}$ the least weight of table j. Then $C_- e^{-\kappa(y)} \le f(e^y) \le C_+ e^{-\kappa(y)}$ on the chart, $\int e^{-\kappa(y)} dy = V_1$ with the volume $V_1 = \sum_\pi \prod_{l=2}^n \kappa_{1,U_l}^{-1}$ of the direction (Proposition 39.30, Theorem 17.5(i)), and for $0 < t \le C_-$*

$$\frac{V_1}{(n-1)!}\Big(\log\frac{C_-}{t}\Big)^{n-1} \le \mu_f\big([t, f^\star]\big) \le \frac{V_1}{(n-1)!}\Big(\log\frac{C_+}{t}\Big)^{n-1}.$$

So $\mu_f([t, f^\star]) \sim \frac{V_1}{(n-1)!}(\log\frac{1}{t})^{n-1}$ as $t \downarrow 0$, and $C_-^k V_1 \le k^{n-1}\langle 1\rangle_k \le C_+^k V_1$ for every real $k > 0$.

The two ends of the stride's law are the two ends of strength. At the low end the mass of μ_f near 0 is set by the walls of the direction alone, through the volume of the direction, and the values blow up at $k = 0$ with that volume as coefficient (Proposition 39.31). At the high end μ_f ends at the native value of the direction in a power fixed by the number of marks, with a coefficient read off the Hessian of the stride at its native gauge; the branch of the generating function in Theorem 38.15 is this end read through the pencil.

40.2 The gauge end

Definition 40.4 (The density operator of a recurrence)**.** Let $\mathbf{p} = (p_0, \dots, p_h)$ be polynomials in $\mathbb{C}[k]$, not all zero, of largest degree D, with the symbol $\chi(\lambda) = \sum_i c_i \lambda^i$, $c_i = [k^D]p_i$ (Corollary 38.7), and the *second symbol* $\chi_1(\lambda) = \sum_i [k^{D-1}]p_i\, \lambda^i$. Its *density operator* is

$$\mathfrak{D}_{\mathbf{p}} = \sum_{i=0}^{h} t^i\, p_i(-1-i-\theta), \qquad \theta = t\frac{d}{dt},$$

a differential operator with polynomial coefficients acting on functions and distributions on $(0,\infty)$; written $\sum_{j=0}^{D} a_j(t)\, \partial_t^j$, its leading coefficient is $a_D = (-1)^D t^D \chi(t)$. At a root $\lambda \neq 0$ of χ, in the variable $s = \lambda - t$, write $\mathfrak{D}_{\mathbf{p}} = \sum_j b_j(s)\, \partial_s^j$ and $b_{j,l} = [s^l]\, b_j$. A function u on an interval $(\lambda-\delta, \lambda)$, or $(\lambda, \lambda+\delta)$, *extends by zero* at λ when it is integrable there, $\mathfrak{D}_{\mathbf{p}} u = 0$ there, and the distribution $u\, \mathbf{1}_{(\lambda-\delta,\lambda)}$, or $u\, \mathbf{1}_{(\lambda,\lambda+\delta)}$, is annihilated by $\mathfrak{D}_{\mathbf{p}}$ on $(\lambda-\delta, \lambda+\delta)$; this is a property of the germ of u at λ.

The density operator is the recurrence read on the other side of the moments: $\mathfrak{D}_{\mathbf{p}}$ is the operator whose formal adjoint sends t^k to $\sum_i p_i(k)\, t^{k+i}$, so a measure on $(0,\infty)$ annihilated by it has moments that satisfy the recurrence, and its singular points in $\mathbb{C}^\times$ are the roots of the symbol.

Proposition 40.5 (A recurrence of the values is an equation for the density)**.** *Let a ray be as in Definition 40.1, let $f^\star$ be as in Proposition 40.2 and $\mathbf{p} = (p_0, \dots, p_h)$ as in Definition 40.4. The following are equivalent:*

(a) $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ *for all large integers k;*

(b) $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ *at every complex k at which every $\langle 1\rangle_{k+i}$ converges, and as an identity of meromorphic functions of k (Proposition 38.5(ii));*

(c) the distribution $\mathfrak{D}_{\mathbf{p}}\, \mu_f$ on $(0,\infty)$ vanishes.

When they hold, $\mathfrak{D}_{\mathbf{p}} \varrho_f = 0$ on $(0, f^\star)$ and the germ of ϱ_f at $f^\star$ extends by zero; and for the origin 0, the conclusions of Corollary 39.34(i) and (ii) hold for $p_0, \dots, p_h$: $\sum_{i \le \min(h,j)} p_i(-j)\, \mathrm{CT}(f^{i-j}) = 0$ for every $j \in \mathbb{N}$, and p_0 vanishes to order at least $n-1$ at 0 with $[k^{n-1}]p_0 \cdot \mathsf{L}_0 + \sum_{i\ge 1} p_i(0)\langle 1\rangle_i = 0$.

So a recurrence that the values satisfy at the large integers, whether or not it follows from the relations of the ray, holds at every strength: no bound on the pieces of a certificate is needed down to $k \to 0$, and the low-end relation follows for it as for the law of [1].

Theorem 40.6 (The gauge end)**.** *Let a ray be as in Definition 40.1, with $\alpha^\star$, $f^\star = f(\alpha^\star)$, the density ϱ_f and the function g of Proposition 40.3, and let $\mathbf{p} = (p_0, \dots, p_h)$ satisfy the equivalent conditions of Proposition 40.5; let $f^\star$ be a simple root of the symbol χ, and use the notation of Definition 40.4 at $\lambda = f^\star$. Put $[x]_j = x(x-1)\cdots(x-j+1)$ and $R_m(x) = \sum_j b_{j,m-D+1+j}\, [x]_j$, with $b_{j,l} = 0$ for $l < 0$.*

(i) $D \ge 1$, $b_{D,0} = 0$, $b_{D,1} = -(f^\star)^D \chi'(f^\star) \neq 0$, and $s^{D-1}\mathfrak{D}_{\mathbf{p}} = \sum_{m\ge 0} s^m R_m(s\partial_s)$ with

$$R_0(x) = b_{D,1}\, [x]_{D-1}\, (x-\gamma), \qquad \gamma = D - 1 - \frac{b_{D-1,0}}{b_{D,1}} = -1 + \frac{\chi_1(f^\star)}{f^\star \chi'(f^\star)}\,:$$

$s = 0$ is a regular singular point of $\mathfrak{D}_{\mathbf{p}}$ with the exponents $0, 1, \dots, D-2$ and γ.

(ii) The line. *The germs at $f^\star$ of functions on the chart's side, $t < f^\star$, that extend by zero form a space of dimension at most one.*

(iii) *The germ of ϱ_f extends by zero, so it spans that space, and $\gamma = \frac{n-3}{2}$.*

(iv) The germ. *If $\gamma \notin \{0, 1, \dots, D-2\}$, then $g(s) = g(0)\,\Phi(s)$, where $\Phi = \sum_N \Phi_N s^N$, $\Phi_0 = 1$, is fixed by $R_0(\gamma+N)\Phi_N = -\sum_{m=1}^{N} R_m(\gamma+N-m)\Phi_{N-m}$, $N \geq 1$. If $\gamma \in \{0, 1, \dots, D-2\}$, then $\varrho_f = \sum_r u_r s^r$ near $f^\star$ with $u_r = 0$ for $r < \gamma$, $u_\gamma = g(0)$, the u_r with $\gamma < r \leq D-2$ fixed one at a time by $\mathsf{p}_r(u) = 0$, where*

$$\mathsf{p}_r(u) = \sum_{i \leq r} \sum_{i < j \leq D} (-1)^l \binom{j-1-i}{l}\, i!\, l!\, b_{j,l}\, u_i, \qquad l = j + r + 1 - D - i \geq 0,$$

whose coefficient of u_r is $r!\, b_{D,1}(r-\gamma)$, and the u_N with $N \geq D-1$ by $R_0(N)u_N = -\sum_{m=1}^{N} R_m(N-m)u_{N-m}$.

(v) The values. *ϱ_f extends holomorphically to a neighbourhood of $(0, f^\star)$ in $\mathbb{C}$ on which $\mathfrak{D}_{\mathbf{p}}\varrho_f = 0$: it is the continuation of the germ of (iv) along $(0, f^\star)$, around the roots of χ on the segment by small arcs on either side. For every $j \in \mathbb{N}$ and every k at which $\langle 1 \rangle_k$ converges,*

$$\frac{\partial^j}{\partial k^j} \langle 1 \rangle_k = \int_0^{f^\star} t^k\, (\log t)^j\, \varrho_f(t)\, dt.$$

The theorem reads a ray from its gauge end. The recurrence fixes the operator, and the native gauge of the direction fixes the rest: $f^\star$, the exponent, which the recurrence must reproduce, and the scale $g(0)$, through the Hessian. For odd n the exponent is an integer and the line lies among the holomorphic solutions; when the exponent is at most $D-2$, the equations $\mathsf{p}_r(u) = 0$, the absence of point masses at $f^\star$, single it out. For even n it is the half-integer class. Nothing from the low end enters, nor any inequality, nor the native values on the segment $(0, f^\star)$, which the density passes analytically.

Proposition 40.7 (The law knows the dimension of its chart)**.** *Let a ray have $\kappa_{1,S} > 0$ for every nonempty proper S, let $\mathbf{p} = (p_0, \dots, p_h)$ have $\sum_i p_i(k) f^i \in \mathcal{L}$ (Definition 38.4), as the law of* [1] *has, and let λ be a simple root of its symbol that is the native value of a nondegenerate native gauge. Then the formal solutions of $\sum_i p_i(k) x(k+i) = 0$ of type $(\lambda, \rho + \mathbb{Z})$ (Definition 39.2) form a line, spanned by $\lambda^k k^{\rho_\lambda}(1 + \sum_{m \geq 1} e_m k^{-m})$ with $\rho_\lambda = -\chi_1(\lambda)/(\lambda\chi'(\lambda)) = -\frac{n-1}{2}$, and λ is a regular singular point of $\mathfrak{D}_{\mathbf{p}}$ with the exponents $0, 1, \dots, D-2$ and $-1-\rho_\lambda = \frac{n-3}{2}$.*

So a recurrence read off closings, integers indexed by one strength, has to carry the dimension $n-1$ of the chart at every simple root of its symbol that is a native value: a test that a candidate law must pass. A root of the symbol need not be a native value, and at such a root the exponent is whatever the recurrence makes it.

Proposition 40.8 (The germ at the gauge end is local data at the native gauge)**.** *Let a ray have the origin 0 and $\kappa_{1,S} > 0$ for every nonempty proper S, with $\alpha^\star$, $f^\star = f(\alpha^\star)$, H and g as in Proposition 40.3; in the coordinates y of the chart, centred at $\log \alpha^\star$, write $\log f = \log f^\star - \frac{1}{2}H_{ab}y^a y^b + \frac{1}{6}h_{abc}y^a y^b y^c + \frac{1}{24}h_{abcd}y^a y^b y^c y^d + \cdots$, with the summation over repeated indices in $\mathcal{E} \setminus e_0$ and $G = H^{-1}$.*

(i) *For $m \geq 2$ the Taylor tensor of order m of $\log f$ at $\log \alpha^\star$ is -2^{-m} times rung m of the still of the realization $\big((s_{1j}, t_{1j})_j; \alpha^\star\big)$ (Definition 9.4), that is $-\sum_j \big(s_{1j}\kappa_{j,\alpha^\star} + (-1)^m t_{1j}\kappa_{j,(\alpha^\star)^{-1}}\big)$ in the cumulants of the basis laws (Proposition 9.6).*

(ii) As $k \to \infty$,

$$\langle 1\rangle_k = (f^\star)^k \Big(\frac{2\pi}{k}\Big)^{\frac{n-1}{2}} \frac{1}{\sqrt{\det H}} \Big(1 + \frac{c_1}{k} + O(k^{-2})\Big),$$

with

$$c_1 = \tfrac{1}{8} h_{abcd} G^{ab} G^{cd} + \tfrac{1}{8} h_{abc} h_{def} G^{ab} G^{cd} G^{ef} + \tfrac{1}{12} h_{abc} h_{def} G^{ad} G^{be} G^{cf}.$$

(iii) With $g(s) = \sum_m g_m s^m$, the asymptotic expansion of $\langle 1\rangle_k$ is $\sum_m g_m (f^\star)^{k+\frac{n-1}{2}+m} \mathrm{B}\big(k + 1, \frac{n-1}{2} + m\big)$, read term by term, and the term of g_m begins at $(f^\star)^k k^{-\frac{n-1}{2}-m}$ with the coefficient $(f^\star)^{\frac{n-1}{2}+m}\Gamma(\frac{n-1}{2} + m) g_m$. So the coefficients of the expansion of $(f^\star)^{-k} k^{\frac{n-1}{2}} \langle 1\rangle_k$ in powers of k^{-1} and the Taylor coefficients of g determine each other, the m-th by the first $m + 1$.

Proposition 40.9 (Slopes from the gauge end)**.** *Let a ray and $\mathbf{p}$ be as in Theorem 40.6, with $\varrho_f(t) = (f^\star - t)^{\frac{n-3}{2}} \sum_m g_m (f^\star - t)^m$ near $f^\star$.*

(i) For every $j \in \mathbb{N}$, $\sum_{l=0}^{j} \binom{j}{l} \sum_i p_i^{(j-l)}(k)\, \partial_k^l \langle 1\rangle_{k+i} = 0$ identically in k: the j-th derivative of the values satisfies the recurrence, with an inhomogeneity made of the lower derivatives.

(ii) If the series $\sum_m g_m s^m$ converges on $[0, f^\star)$ and $\sum_m |g_m| (f^\star)^m \mathrm{B}\big(K' + 1, \frac{n-1}{2} + m\big) < \infty$ for a real K', then for every real $K > K'$ and every $j \in \mathbb{N}$

$$\partial_k^j \langle 1\rangle_k \big|_{k=K} = \sum_{m \geq 0} g_m \frac{\partial^j}{\partial a^j} \Big[(f^\star)^{a+\frac{n-1}{2}+m}\, \mathrm{B}\big(a + 1, \tfrac{n-1}{2} + m\big)\Big]_{a=K}.$$

The derivatives of the values in the strength, the slopes, need no datum from the low end either: they are the logarithmic moments of the same density, and the recurrence differentiated carries them down from large strengths once they are read there.

Definition 40.10 (The procedure that reads a ray from its gauge end)**.** Input: a ray as in Definition 40.1 with its data in a field $\Bbbk \subseteq \mathbb{R}$, and polynomials $\mathbf{p} = (p_0, \dots, p_h)$ over $\Bbbk$ with $\sum_i p_i(k) \langle 1\rangle_{k+i} = 0$ for all large integers k, $f^\star = f(\alpha^\star)$ a simple root of their symbol. Output: the germ of the density at $f^\star$, and the values and slopes at the positive integers.

(1) *The native gauge.* $\alpha^\star$, the unique critical point of $\log f$ on the positive part of the chart, where $-\log f$ is strictly convex (Theorem 17.5(ii)); $f^\star$, $\det H$ and $(\omega_0/dy)(\alpha^\star)$; the constant $g(0)$ (Proposition 40.3(ii)).

(2) *The operator.* $\mathfrak{D}_{\mathbf{p}}$ and its coefficients $b_{j,l}$ at $f^\star$ (Definition 40.4), and γ (Theorem 40.6(i)); a recurrence with $\gamma \neq \frac{n-3}{2}$ is not a recurrence of the values (Theorem 40.6(iii)).

(3) *The germ.* The Taylor coefficients of g, by Theorem 40.6(iv), over $\Bbbk(f^\star)$ up to the factor $g(0)$.

(4) *Large strengths.* For K large and $0 < \delta$ less than the radius of convergence of g, $\langle 1\rangle_{K+i} = \int_{f^\star - \delta}^{f^\star} t^{K+i} \varrho_f(t)\, dt + \epsilon_{K+i}$, $0 \leq i < h$, the integral read from the germ, and $0 \leq \epsilon_k \leq (f^\star - \delta)^{k-k'} \langle 1\rangle_{k'}$ for every $k' \leq k$ at which $\langle 1\rangle_{k'}$ converges; for the origin 0, $\langle 1\rangle_{k'} \leq C_+^{k'} V_1 k'^{1-n}$ (Proposition 40.3(iii)).

(5) *The run back.* The recurrence, which holds at every positive integer (Proposition 40.5), solved for $\langle 1\rangle_k$ at the $k < K$ with $p_0(k) \neq 0$, from $k = K - 1$ down; at a positive integer zero of p_0, and below it, the values are read as moments of ϱ_f continued along $(0, f^\star)$ (Theorem 40.6(v)).

(6) *Slopes.* The same with $(\log t)^j$ in the integrals and the differentiated recurrence (Proposition 40.9).

(7) *Checks.* For the origin 0, the low-end relation (Corollary 39.34(ii)) read on the output returns V_1; two depths K give the same values.

Every step but the truncations in step (4) is an identity: the germ is exact over $\Bbbk(f^\star)$ up to $g(0)$, the error ϵ_k of the cut at $f^\star-\delta$ is bounded by step (4), and the run back is exact arithmetic on exact coefficients. The truncation of the series of g on $[f^\star-\delta, f^\star]$ is estimated, by the size of the last terms kept and by the agreement of two depths. The run back multiplies an error at K by the growth of the other solutions of the recurrence from K down to 1, which is why the depth K grows with the precision asked. The exercises read the centres of three and five marks this way (Exercises 52.3 and 52.15), and a ray of a list on four marks whose symbol has a root on the segment $(0, f^\star)$ (Exercise 53.2).

40.3 The torus value

Definition 40.11 (The torus value)**.** Let the centre ray of a family at the strengths $\boldsymbol{\sigma}$ (Definition 38.13) be given, with its stride $f=\prod_j \big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$ and $f^\star=f(\mathbf{1})$. On the torus $\mathbb{T}=\{\alpha:\ |\alpha_e|=1\ (e\in\mathcal{E}),\ \alpha_{e_0}=1\}$, $\alpha_e=e^{i\vartheta_e}$, with the normalized measure $d\vartheta/(2\pi)^{n-1}$, the stride is $f=\prod_j |K_j(\alpha)|^{-2\sigma_j}$, a function with values in $[f^\star,\infty]$, equal to ∞ exactly where an active letter vanishes. The *torus law* $\nu_f^{\mathbb{T}}$ is the image of $d\vartheta/(2\pi)^{n-1}$ under f, a probability measure on $[f^\star,\infty)$, $\varrho_f^{\mathbb{T}}$ its density where it has one, and the *torus value* is

$$\langle 1\rangle_k^{\mathbb{T}}=\int t^k\,\nu_f^{\mathbb{T}}(dt)=\frac{1}{(2\pi)^{n-1}}\int_{\mathbb{T}} f^k\,d\vartheta \qquad (\operatorname{Re}k\le 0).$$

When no letter K_j of the family vanishes on $\mathbb{T}$, so that $\mathbb{T}$ lies in X_F, it is $(2\pi i)^{1-n}\langle 1\rangle_{k;\mathbb{T}}$ in the notation of Definition 38.1, $\mathbb{T}$ oriented by $d\vartheta$.

Proposition 40.12 (The torus is the second chain)**.** *For the centre ray of a family at the strengths $\boldsymbol{\sigma}$, with the notation of Definition 40.11, ϱ_f and g as in Proposition 40.3:*

(i) On $\mathbb{T}$, $f>f^\star$ off $\mathbf{1}$; the zero set of every active letter on $\mathbb{T}$ has measure zero. $\langle 1\rangle_k^{\mathbb{T}}$ is continuous on $\operatorname{Re}k\le 0$ and holomorphic on $\operatorname{Re}k<0$, and $\langle 1\rangle_{-j}^{\mathbb{T}}=\mathrm{CT}(f^{-j})$ for $j\in\mathbb{N}$, the closing $\mathsf{C}_j(1)$ when every $\sigma_j\ge 1$ (Definition 39.26).

(ii) Let every $\sigma_j\ge 1$, and let $\mathbf{p}=(p_0,\dots,p_h)$ satisfy $\sum_i p_i(k)\langle 1\rangle_{k+i}=0$ for all large integers k, as the law of [1] does. Then $\sum_i p_i(k)\langle 1\rangle_{k+i}^{\mathbb{T}}=0$ for every k with $\operatorname{Re}k\le -h$, and the distribution $\mathfrak{D}_{\mathbf{p}}\,\nu_f^{\mathbb{T}}$ on $(0,\infty)$ vanishes.

(iii) Near $f^\star$, $\nu_f^{\mathbb{T}}$ has a density, and with the one function g

$$\varrho_f(t)=(f^\star-t)^{\frac{n-3}{2}}g(f^\star-t)\quad(t<f^\star),\qquad \varrho_f^{\mathbb{T}}(t)=(2\pi)^{1-n}(t-f^\star)^{\frac{n-3}{2}}g(f^\star-t)\quad(t>f^\star).$$

(iv) For $\mathbf{p}$ as in (ii) with $f^\star$ a simple root of its symbol, the germ of ϱ_f on the chart's side spans the line of Theorem 40.6(ii), the germ of $\varrho_f^{\mathbb{T}}$ extends by zero on the torus's side, and the two are the cut solutions $|f^\star-t|^{\frac{n-3}{2}}g(f^\star-t)$ of one germ on its two sides, the second with the factor $(2\pi)^{1-n}$.

At a centre the native gauge of the direction lies on the positive orthant and on the unit torus, and the two meet only there (Theorem 39.41). The torus value continues the closings in the strength, as the values continue them at the poles of the low end (Proposition 39.30), and it satisfies every recurrence of the values. Its density is the density of the chart seen from the other side of the native value: the Morse function of the stride at **1** changes sign between the orthant and the torus, so the one function g enters both, and the asymptotic statement at the end of Theorem 39.41 is the shadow of this exact one. Exercise 52.4 works the three-mark centre, where the torus value, the values and the solution at the sign are tied by an explicit connection.

Chapter 41

Reading a ray

A ray is read from its list and its direction. A box of its relations gives a spanning set of a submodule that contains [1], the one-step matrix on it with a certificate, and a recurrence of the values that holds at every strength (Section 39.3); the gauge end then gives the values themselves (Chapter 40). This chapter assembles the reading and says what each step certifies. At the low end a recurrence of the values meets the values at the positive integers through the poles at the nonpositive integers: the low-end relation at 0 and, where the coefficients of the recurrence vanish to the order of the poles, second low-end relations at the other poles, with coefficients in the field of the recurrence, the weights and the volume of the direction. At the high end the generating function of the values is holomorphic off the reciprocals of $(0, f(\alpha^\star)]$, and its jump across them is the density of the stride's law. When every other root of the symbol lies outside the closed disc of radius $f(\alpha^\star)$, the values are, up to scale, the only solution of the recurrence whose exponential rate of growth is at most $f(\alpha^\star)$; the low-end relation fixes the scale, and solutions over the field of the recurrence approximate the values at the rate $f(\alpha^\star)/\lambda'$, λ' the least modulus of the other roots. Positivity is the moment problem of the stride's law on $[0, f(\alpha^\star)]$. Against a solution carried by a segment that ends at a real point λ off $[0, f(\alpha^\star)]$ its brackets shrink at the rate of the Green function of the plane cut along $[0, f(\alpha^\star)]$, read at λ; it cannot see a solution carried by $[0, f(\alpha^\star)]$ and dominated by the stride's law, which the analyticity of the density excludes; and for a recurrence of order two it gives the value at $k = 1$ as the volume of the direction times a continued fraction in the coefficients. The procedure that reads a ray from its list runs the box by evaluation at strengths drawn at random, modulo primes for a list, and bounds the chance of a wrong output by degrees that the box fixes, or removes it with enough strengths. Its cost is set by the stencils of the relations on a lattice, by the degrees of the one-step matrix, which the number of pivots bounds, and by the order and the degree of the recurrence. The nondegenerate native gauges bound the sizes of a reading from below; over all exponents at once they form the graph of one map, through which they are found by continuation and certified one at a time.

41.1 The low end and the slit

The low-end relation of a recurrence of the values is Corollary 39.34(ii), with the low-end coefficients of Definition 39.33; by Proposition 40.5 it holds for every recurrence that the values satisfy at the large integers, whether or not it follows from the relations of the ray. The poles at the negative integers (Proposition 39.31) give more relations.

Proposition 41.1 (The low-end relations at every pole)**.** *Let a ray have the origin* 0 *and* $\kappa_{1,S} > 0$ *for every nonempty proper* S*, with the volume* V_1 *of its direction (Proposition 39.30), and let*

$p_0, \dots, p_h \in \mathbb{C}[k]$ *satisfy* $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ *for all large integers* k. *Let* $j \in \mathbb{N}$, *and let* $m_0 \in \{0, \dots, n-1\}$ *be such that every* p_i *with* $i \le \min(h, j)$ *vanishes to order at least* m_0 *at* $k = -j$.

(i) The function $\sum_{i \le \min(h,j)} p_i(k)\langle 1\rangle_{k+i}$ *is holomorphic at* $k = -j$, *with the value* $-\sum_{i=j+1}^{h} p_i(-j)\langle 1\rangle_{i-j}$ *there.*

(ii) If $m_0 < n-1$, *then* $\sum_{i \le \min(h,j)} [(k+j)^{m_0}] p_i \cdot \mathrm{CT}\left(f^{i-j}\right) = 0$.

(iii) If $m_0 = n-1$, *then*

$$V_1 \sum_{i \le \min(h,j)} [(k+j)^{n-1}] p_i \cdot \mathrm{CT}\left(f^{i-j}\right) + \sum_{i=j+1}^{h} p_i(-j)\,\langle 1\rangle_{i-j} = 0.$$

At $j = 0$ *this is the low-end relation; at* $j \ge 1$ *it is the* second low-end relation *at* $-j$, *a linear relation among the values at the positive integers whose coefficients lie in the field generated by those of the* p_i, *the weights and* V_1.

At $m_0 = 0$, (ii) is the recurrence of the constant terms, $\sum_i p_i(-j)\,\mathrm{CT}(f^{i-j}) = 0$ (Corollary 39.34(i)). At larger m_0 it ties the derivatives of order m_0 of $p_0, \dots, p_j$ at $-j$, and (i) ties the values at the positive integers to the Laurent coefficients of the values at the poles $-j, \dots, 0$, those at 0 being integrals over the faces of the sector cubes (Proposition 39.35).

The generating function of the values is read on the plane cut along the reciprocals of $(0, f(\alpha^\star)]$. The segment $(0, f(\alpha^\star))$ of the plane of native values is the *slit*.

Proposition 41.2 (The values live on the slit plane)**.** *Let a ray have a real origin and* $\kappa_{1,S} > 0$ *for every nonempty proper* S, *with* $\alpha^\star$, $f^\star = f(\alpha^\star)$, μ_f *and* ϱ_f *as in Proposition 40.3, and let* k_0 *be an integer at which* $\langle 1\rangle_{k_0}$ *converges.*

(i) For $|q| < 1/f^\star$, $G(q) = \sum_{k \ge k_0} \langle 1\rangle_k q^{k-k_0} = \int t^{k_0} \mu_f(dt)/(1-qt)$, *and the right side is holomorphic on* $\mathbb{C} \setminus [1/f^\star, \infty)$: G *continues to the plane cut along the reciprocals of* $(0, f^\star]$.

(ii) For $x > 1/f^\star$, G *continues holomorphically across a neighbourhood of* x *in* $(1/f^\star, \infty)$ *from either side, with the limits* $G(x \pm i0)$, *and* $G(x+i0) - G(x-i0) = 2\pi i\, x^{-1-k_0}\, \varrho_f(1/x)$: *the jump of* G *across the cut is the density of the stride's law.*

(iii) On the centre ray of a family, every native gauge on the torus $\mathbb{T}$ *of Definition 40.11 has its native value in* $[f^\star, \infty)$, *off the slit.*

(iv) Every conjugate of $f^\star$ *over the field of the weights is the native value of a nondegenerate native gauge (Proposition 38.10), and a root of the symbol of every recurrence that the values satisfy at the large integers with coefficients in that field; the real conjugates in* $(0, f^\star)$ *lie on the slit.*

A root λ of a symbol on the slit sits on the cut, at $1/\lambda \in (1/f^\star, \infty)$, where the generating function of the values continues from either side and jumps by the density of the stride's law; a root off $[0, f^\star]$ sits where it is holomorphic. Not every root of a symbol is a native value: whenever the values satisfy $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$, they satisfy the recurrence with the coefficients $p_{i-1}(k+1) - \lambda_0\, p_i(k)$, $0 \le i \le h+1$, $p_{-1} = p_{h+1} = 0$, which follows from the relations when the first does and has the symbol $(\lambda - \lambda_0)\chi(\lambda)$, for every λ_0. In the other direction, the native value of every nondegenerate native gauge is a root of the symbol of every recurrence that follows from the relations (Corollary 38.7), and that of every native gauge when the recurrence has a certificate as in Proposition 38.8. Exercises 53.1 and 53.2 work second low-end relations on a list of four marks.

41.2 Positive selection

Theorem 41.3 (Positive selection)**.** *Let a ray have a real origin and $\kappa_{1,S} > 0$ for every nonempty proper S, with $f^\star = f(\alpha^\star)$, and let $p_0, \dots, p_h \in \mathbb{C}[k]$, $h \geq 1$, satisfy $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ for all large integers k, so that the recurrence holds at every integer at which the values converge (Proposition 40.5). Let p_0 and p_h have the largest degree D, so that the symbol χ has degree h and $\chi(0) \neq 0$; then $f^\star$ is a root of χ, and let the roots of χ be distinct, $\lambda_1 = f^\star, \lambda_2, \dots, \lambda_h$. Let k_1 be an integer at which the values converge, with $p_0(k)p_h(k) \neq 0$ for every integer $k \geq k_1$. A* solution *is a sequence $(x(k))_{k\geq k_1}$ with $\sum_i p_i(k)x(k+i) = 0$ for every $k \geq k_1$; the solutions form a space $\mathcal{V}$ of dimension h, and the values are one of them.*

(i) For every solution, $X(q) = \sum_{k\geq k_1} x(k)q^{k-k_1}$ satisfies $L_q X = R_x$ as a formal power series, with

$$L_q = \sum_{i=0}^{h} q^{h-i}\, p_i\Big(q\frac{d}{dq} + k_1 - i\Big)$$

and R_x a polynomial of degree less than h that depends linearly on $x(k_1), \dots, x(k_1+h-1)$. The coefficient of $(d/dq)^D$ in L_q is $q^{D+h}\chi(1/q)$, so the singular points of L_q in $\mathbb{C}^\times$ are the points $1/\lambda_c$. For the values, X continues holomorphically to $\mathbb{C}\setminus[1/f^\star, \infty)$ (Proposition 41.2).

(ii) The solutions with $x(k) = O(r^k)$ for every $r > f^\star$ form a subspace of $\mathcal{V}$ whose dimension is the number of roots in the closed disc $|\lambda| \leq f^\star$; the values lie in it, $\langle 1\rangle_k = C\,(f^\star)^k k^{-\frac{n-1}{2}}\big(1+o(1)\big)$ with $C > 0$, and the exponent $-\chi_1(f^\star)/(f^\star\chi'(f^\star))$ of the root $f^\star$ is $-\frac{n-1}{2}$.

(iii) Let every root other than $f^\star$ have modulus greater than $f^\star$, and put $\lambda' = \min_{c\geq 2}|\lambda_c|$. Then the solutions with $x(k) = O(r^k)$ for some $r < \lambda'$ are the multiples of the values, and so are those whose generating function converges on $|q| < 1/f^\star$: the values are, up to a constant factor, the solution of least growth*. Let moreover the origin be 0, $k_1 = 1$ and $a = [k^{n-1}]p_0 \neq 0$, and put $\ell(x) = -a^{-1}\sum_{i=1}^{h} p_i(0)\,x(i)$. Then $\ell(\langle 1\rangle) = V_1$, and $\langle 1\rangle_k = V_1\,x(k)/\ell(x)$ for every nonzero solution x of least growth: the values divided by the volume of the direction are a function of the recurrence alone.*

(iv) Under the hypotheses of (iii), let $x^{(0)} \in \mathcal{V}$ have $\ell(x^{(0)}) = V_1$ and let $x^{(1)}, \dots, x^{(h-1)}$ be a basis of $\{\ell = 0\}$, so that $\langle 1\rangle = x^{(0)} + \sum_{i\geq 1} M_i x^{(i)}$ for one $M \in \mathbb{C}^{h-1}$. For all large K the equations $x^{(0)}(K+l) + \sum_{i\geq 1} M_i^{(K)} x^{(i)}(K+l) = 0$, $0 \leq l \leq h-2$, have exactly one solution $M^{(K)}$, and $|M^{(K)} - M| = O\big((f^\star/\lambda')^K K^{C'}\big)$ for some C'. When the p_i have their coefficients in a field $\Bbbk$ that contains V_1, the $x^{(i)}$ can be taken with values in $\Bbbk$, and then every $M^{(K)}$ lies in $\Bbbk^{h-1}$.

The selection runs on growth alone: among the solutions, the values are the one that grows like the native value of the direction, and the other roots of the symbol, outside the closed disc of radius $f^\star$, give solutions that grow faster and are cut away. The hypothesis of (iii) is what makes the values the solution of least growth: a root in the closed disc, in particular one on the slit, gives a solution whose exponential rate of growth is at most that of the values, and growth cannot remove it (Section 41.3). The low-end relation then fixes the scale through the volume of the direction, so the values over that volume are a function of the recurrence alone.

41.3 Positivity as a moment problem

Proposition 41.4 (Positivity is the moment problem of the stride's law)**.** *Let a ray have a real origin and $\kappa_{1,S} > 0$ for every nonempty proper S, with $f^\star = f(\alpha^\star)$ and μ_f as in Proposition 40.3.*

For a real polynomial $p = \sum_m p_m t^m$, a real K and a sequence x defined at $K + \mathbb{N}$, put $L_{K,p}(x) = \sum_m p_m\, x(K+m)$.

(i) If $\langle 1\rangle_K$ converges, $p \ge 0$ on $[0, f^\star]$ and $p \ne 0$, then $L_{K,p}(\langle 1\rangle) = \int t^K p(t)\, \mu_f(dt) > 0$.

(ii) Every real polynomial of degree at most N that is nonnegative on $[0, f^\star]$ is a sum of polynomials q^2, $t\,q^2$, $(f^\star - t)\,q^2$ and $t(f^\star - t)\,q^2$ of degree at most N, with real q. So for a real sequence x, $L_{K,p}(x) \ge 0$ for every such p exactly when the Hankel matrices

$$\begin{matrix} \big[x(K+i+i')\big], & \big[x(K+1+i+i')\big], \\ \big[f^\star x(K+i+i') - x(K+1+i+i')\big], & \big[f^\star x(K+1+i+i') - x(K+2+i+i')\big], \end{matrix}$$

with $0 \le i, i' < \lfloor \frac{N}{2} \rfloor + 1$, $\lfloor \frac{N+1}{2} \rfloor$, $\lfloor \frac{N+1}{2} \rfloor$ and $\lfloor \frac{N}{2} \rfloor$ respectively, are positive semidefinite.

(iii) A real sequence $(x(K+m))_{m\in\mathbb{N}}$ has $L_{K,p}(x) \ge 0$ for every real $p \ge 0$ on $[0, f^\star]$ exactly when $x(K+m) = \int t^m \nu(dt)$ for every m, for a positive measure ν on $[0, f^\star]$, which is then unique; for the values, $\nu = t^K \mu_f$.

So the positivity of the values is the statement that they are the moments of a positive measure on $[0, f^\star]$, and Theorem 38.14(iii) is its first family, the Hankel determinants of the values. For a family $x^{(0)} + \sum_i M_i x^{(i)}$ of solutions of a recurrence, as in Theorem 41.3(iv), each condition $L_{K,p} \ge 0$ is one linear inequality in the coordinates M, and the conditions of degree at most N at the strength K cut out a closed convex set, the *bracket* of degree N, which contains the coordinates of the values.

Theorem 41.5 (The rate of positivity off the segment)**.** *Let a ray be as in Proposition 41.4, K an integer at which $\langle 1\rangle_K$ converges, $\lambda \in \mathbb{R} \setminus [0, f^\star]$ and I_λ the closed segment between 0 and λ. Let $u(k) = \int_{I_\lambda} t^k\, \nu(t)\, dt$ for the integers $k \ge K$, where ν is a real function on I_λ with $\int_{I_\lambda} |t|^K |\nu(t)|\, dt < \infty$ and $\nu(t) = c\,|\lambda - t|^a \big(1 + o(1)\big)$ as $t \to \lambda$ in I_λ, with $c \ne 0$ and $a > -1$. Let E_N be the set of the real ε with $L_{K,p}(\langle 1\rangle + \varepsilon u) \ge 0$ for every real $p \ge 0$ on $[0, f^\star]$ of degree at most N. Then E_N is a closed interval that contains 0, and there is a constant C with its length at most $C\, N^{a+1} e^{-N g(\lambda)}$ for all large N, where*

$$g(\lambda) = \log \big| w + \sqrt{w^2 - 1} \big| > 0, \qquad w = \frac{2\lambda}{f^\star} - 1,$$

the branch of the square root with $|w + \sqrt{w^2-1}| > 1$. For $\lambda > f^\star$, with $\rho = f^\star/\lambda$,

$$e^{-g(\lambda)} = \frac{1 - \sqrt{1-\rho}}{1 + \sqrt{1-\rho}} = \frac{\rho}{4} + O(\rho^2).$$

The rate is read off the Green function g of the plane cut along $[0, f^\star]$, with its pole at infinity, at the end λ of the segment that carries u: the farther λ lies from $[0, f^\star]$, the faster positivity separates the values from the values plus a multiple of u. For a recurrence of order two, with the coordinate M of Theorem 41.3(iv) and $u = x^{(1)}$ of this form, E_N is the bracket of degree N of M less the value of M.

Proposition 41.6 (What positivity cannot see)**.** *Let a ray be as in Proposition 41.4, K a real at which $\langle 1\rangle_K$ converges, and ν a real signed measure on $[0, f^\star]$ with $|\nu| \le C\mu_f$ for a constant C; put $u(k) = \int t^k \nu(dt)$ for $k \ge K$.*

(i) For $|\varepsilon| \le 1/C$, $\langle 1\rangle + \varepsilon u$ is from K on the moment sequence of the positive measure $\mu_f + \varepsilon\nu$, and $L_{K',p}(\langle 1\rangle + \varepsilon u) \ge 0$ for every real $K' \ge K$ and every real $p \ge 0$ on $[0, f^\star]$.

(ii) *If on an open interval $J \subseteq (0, f^\star)$ the measure ν has a density that agrees almost everywhere with no real-analytic function on J, then for $\varepsilon \neq 0$ the measure $t^K(\mu_f + \varepsilon\nu)$, the only finite measure on $[0, f^\star]$ whose moments are the $\langle 1\rangle_{K+m} + \varepsilon u(K+m)$, $m \in \mathbb{N}$, has no density on $(0, f^\star)$ that is real-analytic, while $t^K\mu_f$ has one (Proposition 40.3(i)).*

(iii) *Let the origin be 0, let $\mathbf{p} = (p_0, \dots, p_h)$ satisfy the conditions of Proposition 40.5, and let ν have a density that is bounded near 0, with $\mathfrak{D}_{\mathbf{p}}\nu = 0$ as distributions on $(0, \infty)$. Then $u(k) = \int t^k \nu(dt)$ converges at every k with $\operatorname{Re} k > -1$, $\sum_i p_i(k)u(k+i) = 0$ there, and $\sum_{i=1}^{h} p_i(0)\, u(i) = 0$: u satisfies the recurrence and contributes nothing to the low-end relation.*

So a sequence u as in (i) and (iii), a solution of the recurrence carried by $[0, f^\star]$ and dominated by the stride's law, is invisible to the recurrence, to the low-end relation and to positivity together: read as conditions on candidates, they leave the direction of u free. For a root λ of the symbol on the slit, a density ν on $(0, \lambda)$ with $\mathfrak{D}_{\mathbf{p}}\big(\nu\, \mathbf{1}_{(0,\lambda)}\big) = 0$ as distributions on $(0, \infty)$, a solution of the density equation on $(0, \lambda)$ that extends by zero at λ (Definition 40.4), bounded near 0 and with $|\nu| \leq C\varrho_f$ on $(0, \lambda)$, gives such a u. The gauge end does not choose among candidates: it constructs the density, and by (ii) the analyticity of the density on the open support is as much a property of the stride's law as its positivity.

Proposition 41.7 (A recurrence of order two is the volume of the direction times a continued fraction)**.** *Let a ray have the origin 0 and $\kappa_{1,S} > 0$ for every nonempty proper S, with $f^\star = f(\alpha^\star)$, and let $p_0, p_1, p_2 \in \mathbb{R}[k]$ satisfy $p_0(k)\langle 1\rangle_k + p_1(k)\langle 1\rangle_{k+1} + p_2(k)\langle 1\rangle_{k+2} = 0$ for all large integers k; put $a = [k^{n-1}]p_0$ and $\rho_k = \langle 1\rangle_{k+1}/\langle 1\rangle_k$ for the integers $k \geq 1$.*

(i) *$0 < \rho_k < f^\star$, and $\rho_k\big(p_1(k) + p_2(k)\rho_{k+1}\big) = -p_0(k)$ for every integer $k \geq 1$.*

(ii) *$\langle 1\rangle_1\big(p_1(0) + p_2(0)\rho_1\big) = -a\, V_1$. When $-a/p_1(0)$ and $-a/(p_1(0) + f^\star p_2(0))$ are finite and positive, $\langle 1\rangle_1/V_1$ lies between them, strictly when they differ.*

(iii) *Let the hypotheses of Theorem 41.3(iii) hold with $h = 2$ and $k_1 = 1$, so that $\lambda' = |\lambda_2|$, and let $a \neq 0$; put $\mathsf{M}_k(r) = -p_0(k)/\big(p_1(k) + p_2(k)\, r\big)$. There are K_0, C and C' such that for every $K \geq K_0$ and every $r \in [0, f^\star]$ the composite $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r)$ is defined and lies within $C(f^\star/\lambda')^K K^{C'}$ of ρ_1. So ρ_1 is the value of a continued fraction in the coefficients of the recurrence, and $\langle 1\rangle_1 = -aV_1/(p_1(0) + p_2(0)\rho_1)$. The sets of the $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r)$ over the $r \in (0, f^\star)$ whose intermediate values $\mathsf{M}_k \circ \cdots \circ \mathsf{M}_K(r)$, $2 \leq k \leq K$, lie in $(0, f^\star)$ are nested in K and contain ρ_1.*

The low-end relation fixes the value at $k = 1$ through the volume of the direction once the ratio ρ_1 is known, and positivity alone places every ρ_k in $(0, f^\star)$. The bracket of (ii) is the image of $\rho_1 \in (0, f^\star)$ under $\rho \mapsto -a/(p_1(0) + p_2(0)\rho)$; the nested sets of (iii) sharpen that condition with the conditions $\langle 1\rangle_k > 0$ and $f^\star\langle 1\rangle_k > \langle 1\rangle_{k+1}$ at every $k \leq K+1$, and they shrink to ρ_1 at the rate $(f^\star/\lambda')^K$, up to a power of K. The exercises work the brackets and the rate of Theorem 41.5 on the centre of three marks (Exercises 52.1 and 52.19), and the continued fraction on W_3 at $(\mathbf{1}; 0, 2)$ (Exercise 52.18).

41.4 The procedure

Definition 41.8 (The procedure that reads a ray from its list)**.** Input: a family whose weights lie in a field $\Bbbk \subseteq \mathbb{R}$, $\mathbb{Q}$ for a list; a ray of it with $\kappa_{1,S} > 0$ for every nonempty proper S and its origin in $\Bbbk$, the origin 0 unless stated; the precision wanted. Output: a spanning set and its one-step matrix with a certificate, a recurrence of the values that holds at every strength, the values at the positive integers, and brackets for them.

(1) *Walls and symmetries.* The walls $\kappa_{1,S}$ and the volume $V_1 = \sum_\pi \prod_{l=2}^{n} \kappa_{1,U_l}^{-1}$ of the direction (Proposition 39.30); a finite group G generated by maps of the four kinds of Proposition 39.7(ii) that are symmetries of the ray.

(2) *The box.* Through the boxes $\mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ with level bounds kept by G, each with a ranking, and the $\mathfrak{s} < \mathfrak{t}$, in an order that reaches each, the first box accepted at some $\mathfrak{s}$, with its spanning set and its stems (Definition 39.16); the search stops (Theorem 39.18). Acceptance is decided by eliminations of the relations at strengths drawn at random, for a list modulo a prime (Proposition 41.10(iii) and (v), below).

(3) *The one-step matrix.* At strengths drawn at random, for a list modulo primes, the reductions of the targets, from the relations they need (Proposition 41.10(vi)); the entries of $\mathbf{B}$ over $\Bbbk(k)$ by rational reconstruction from these values (Proposition 41.10(ii)), for a list with Chinese remaindering and rational reconstruction of the coefficients over the primes; a check against eliminations of the whole box at strengths and primes drawn afresh (Proposition 41.10(iv) and (v)). A certificate supported on the stems exists (Theorem 39.17(ii)), and every solution, the values among them, satisfies $\varsigma^{-1} v_V = \mathbf{B} v_V$ (Theorem 39.17(iii)).

(4) *The recurrence.* The first dependence of the rows $w_i(k) = w_{i-1}(k-1)\,\mathbf{B}(k)$ (Proposition 39.6(ii)), read at points and reconstructed as in step (3), its coefficients being proportional to signed maximal minors (Proposition 39.6(iii)): $\sum_j p_j(k) f^j \in \mathcal{L}$, the law of [1] when the spanning set is independent (Theorem 39.17(iv)), as it is when the values of the spanning set at the nondegenerate native gauges span (Proposition 39.38(i)). Its end coefficient p_0 vanishes only at the shifted zeros of the denominator of $\mathbf{B}$ and at the strengths where the rows line up (Proposition 39.6(iii)).

(5) *The values.* Either the gauge end, from the recurrence, $\alpha^\star$ and $\det H$, when $f(\alpha^\star)$ is a simple root of its symbol (Definition 40.10); or, when the hypotheses of Theorem 41.3(iii) hold, positive selection with the low-end relation (Theorem 41.3(iv)), with brackets from positivity (Proposition 41.4).

(6) *Checks, outside the construction.* When the origin is integral and every letter exponent of the direction is at least 1, the closings of the spanning set satisfy $\mathsf{C}_{j+1}(\mathbf{b}) = \mathbf{B}(-j)\,\mathsf{C}_j(\mathbf{b})$ for all large j (Theorem 39.29, Theorem 39.17(iii)); for the origin 0, the constant terms of the f^{-j} satisfy the reflected recurrence (Corollary 39.34(i)); when $\mathbf{B}$ is regular at ∞, the native values of nondegenerate native gauges found by other means are reciprocal eigenvalues of $\mathbf{B}(\infty)$ (Proposition 39.38(ii)); the exponent at $f(\alpha^\star)$ is $\frac{n-3}{2}$ (Theorem 40.6(iii)); the low-end relation returns V_1 from the values read; two readings from different depths agree.

Every relation of a box is a relation of the module, so every identity that the elimination gives over $\Bbbk(k)$ holds at every strength (Proposition 39.14, Theorem 39.17), and the recurrence of step (4) holds for the values at every strength at which they converge (Proposition 40.5). Chance enters only through the strengths and primes at which steps (2) to (4) evaluate: Proposition 41.10 bounds it by degrees that the box fixes, and enough strengths remove it. The procedure needs no bound on the box: the search stops (Theorem 39.18).

The procedure of Section 39.9 reads the same module from its solutions, with no box reduced. Its output is certified when the solutions supplied annihilate no submodule (Corollary 39.40), and the closings alone certify it exactly when the native gauge of the direction sees every nonzero submodule (Corollary 39.43); otherwise a unique solution of its equations is evidence. The procedure above certifies the matrix of its spanning set by construction: the certificate of the box is an identity of the relations, whatever any solution sees, and the matrix is the one-step matrix of the

module exactly when the spanning set is independent. The two check each other: every solution that Section 39.9 uses satisfies $\varsigma^{-1}v_V = \mathbf{B}v_V$ on the spanning set of the box (Theorem 39.17(iii)), which is step (6) for the closings, and where the supplied solutions annihilate a submodule, the matrix of the box is still certified by its certificate. Section 39.9 ends with the law of [1]; the procedure above goes on to the values, at the gauge end or by positive selection. A recurrence read off the closings is a recurrence of the values when it holds for all large j and every letter exponent of the direction is positive (Corollary 39.44); finitely many closings do not show that, while the recurrence of step (4) carries a certificate.

Proposition 41.9 (The relations of a box are difference equations)**.** *Let a box $\mathfrak{B} = \mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ of a ray be given (Definition 39.11).*

(i) The symbols at a level $(\mathbf{a}, \mathbf{b})$ are the points of the affine lattice $\{u \in \mathbb{Z}^{\mathcal{E}} : u_{\mathcal{E}} = \sum_j (a_j r_j + b_j r_j^*)\}$, of rank $n-1$. There is a constant C, depending on the ray and the level, such that the coordinates u_e, $e \neq e_0$, of the symbols of wall degree at most $\mathfrak{t}$ at that level lie in intervals of length at most $C(1+\mathfrak{t})$; so they number at most $\big(C(1+\mathfrak{t})+1\big)^{n-1}$.

(ii) A letter identity of table j ties a symbol to the $|\mathcal{B}_j|$ symbols one unit of that letter up, with weights as coefficients; the mark relation of a symbol at a mark ties it to at most $\sum_j |\mathcal{B}_j|$ symbols one unit of a letter up, with coefficients affine in k. The shifts of $(u; \mathbf{a}, \mathbf{b})$ that occur lie in one finite set that does not depend on the symbol.

(iii) In the basis of the symbols the relations of the box form a matrix $A(k) = A_0 + kA_1$ with entries in the field of the weights and the origin; A_1 vanishes outside the rows of the mark relations, where its entries are $m_{1,e}$, $-s_{1j}\theta_{j,T}$ and $-t_{1j}\theta_{j,T}$. The same holds for the orbit averages of the relations modulo a finite group of symmetries that keeps the box, in the basis of the orbit averages of the symbols.

Proposition 41.10 (What a reading by evaluation certifies)**.** *Let a finite group G of symmetries keep a box $\mathfrak{B}$ of a ray, with the field $\Bbbk$ of Definition 39.11 and a ranking (Definition 39.16). Let $A(k)$ be the matrix whose rows are the orbit averages of the letter identities and of the mark relations that span $\mathcal{R}_{\mathfrak{B}}$, in the basis of the orbit averages ordered by the ranking, so that its rows span $\mathcal{R}_{\mathfrak{B}}^G$ and its entries are affine in k (Proposition 41.9(iii)). Let P be the pivot orbits and F the free orbits, $r = |P|$, and let $A_P(k)$ and $A_F(k)$ be r rows of A that are independent over $\Bbbk(k)$, read on the columns P and F.*

(i) r is the rank of A, $A_P(k)$ is invertible over $\Bbbk(k)$, and the reduction of $x \in \mathcal{F}_{\mathfrak{B}}^G$ with coefficients in $\Bbbk$ is $x_F - x_P A_P(k)^{-1} A_F(k)$; $\det A_P$ is a nonzero polynomial of degree at most r. At $k_0 \in \Bbbk$ the elimination of all the rows of $A(k_0)$ in the order of the ranking has the pivots P exactly when some minor of order r of $A(k_0)$ on the columns P is nonzero, as when $\det A_P(k_0) \neq 0$; then the reduction it gives of every such x is the value at k_0 of the reduction over $\Bbbk(k)$.

(ii) When the box is accepted, every entry of its matrix $\mathbf{B}$ is a polynomial of degree at most r divided by $\det A_P$. A rational function whose numerator and denominator have degrees at most d_1 and d_2 is fixed, among such functions, by its values at any $d_1 + d_2 + 1$ points at which the denominators do not vanish; with a common denominator known, a numerator of degree at most d_1 is fixed by $d_1 + 1$ values.

(iii) The verdict of the elimination at k_0 on acceptance at $\mathfrak{s}$, whether the reductions of the targets are combinations of the free Laurent orbits of wall degree at most $\mathfrak{s}$, is the verdict over $\Bbbk(k)$ except at no more than $2r$ strengths k_0.

(iv) Let the box be accepted, and let $\mathbf{B}'$ *be a matrix over* $\Bbbk(k)$ *whose entries have a common denominator* Q'*, with* Q' *and the numerators of degree at most* d*. If* $\mathbf{B}' \neq \mathbf{B}$*, then* $\mathbf{B}'(k_0)$ *agrees with the matrix that the elimination at* k_0 *gives at no more than* $r+d$ *strengths* k_0 *with* $Q'(k_0) \neq 0$ *at which the elimination has the pivots* P*. So agreement at* $r+d+1$ *such strengths proves* $\mathbf{B}' = \mathbf{B}$*, and a wrong* $\mathbf{B}'$ *passes the check at a strength drawn uniformly from a finite set* $S \subseteq \Bbbk$ *with probability at most* $(r+d)/|S|$*.*

(v) Let $\Bbbk = \mathbb{Q}$*, and let* q *be a prime that divides no denominator of an entry of* A*, with* $\det A_P$ *not identically* 0 *modulo* q*. Then* A *modulo* q *has the rank* r *and the pivots* P *over* $\mathbb{F}_q(k)$*, its reductions are those over* $\mathbb{Q}(k)$ *taken modulo* q*, and (i) to (iv) hold over* $\mathbb{F}_q$ *with strengths in* $\mathbb{F}_q$*. A* $\mathbf{B}'$ *over* $\mathbb{Q}(k)$ *with* $\mathbf{B}' \neq \mathbf{B}$ *passes the check modulo* q *at a strength drawn uniformly from* $\mathbb{F}_q$ *with probability at most* $(r+d)/q$*, for every prime* q *outside a finite set fixed by* A *and* $\mathbf{B}'$*.*

(vi) If a subspace $\mathcal{R}' \subseteq \mathcal{R}^G_{\mathfrak{B}}$ *contains* $x - \bar{x}$ *for every target* x*,* $\bar{x}$ *its reduction, then the reduction of every target modulo* $\mathcal{R}'$*, defined as in Definition 39.16(ii) with* $\mathcal{R}'$ *in place of* $\mathcal{R}^G_{\mathfrak{B}}$*, is* $\bar{x}$*.*

An elimination over $\Bbbk(k)$ carries polynomials whose degrees are bounded only by the number r of pivots, which grows with the box. By (ii) a reconstruction from evaluations needs only as many strengths as the degrees of the entries of $\mathbf{B}$ ask, and by (iv) a check at one fresh strength drawn from a set S bounds the chance of a wrong result by $(r+d)/|S|$, while $r+d+1$ checks remove it.

A ray with a real origin is read in the same way: the box carries the origin in its mark relations (Definition 39.11(iii)), and the stride's law, its density and the gauge end are those of Chapter 40, which allow a real origin, the constant $g(0)$ carrying the factor $(\omega_0/dy)(\alpha^\star)$ (Proposition 40.3(ii)); the low-end relation and Theorem 41.3(iii) and (iv) ask for the origin 0. Data with rational masses are read through an origin.

Proposition 41.11 (Data with rational masses)**.** *Let data* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *of a family on the balance wall have* $\mathbf{m} \in \mathbb{Q}^{\mathcal{E}}$*,* $\mathbf{s},\mathbf{t} \in \mathbb{N}^N$ *and every wall* κ_U*,* U *nonempty and proper, positive, and put* $f_{\mathbf{m}} = \alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ *on the positive orthant.*

(i) Let $\mathbf{m}' \in \mathbb{Z}^{\mathcal{E}}$ *put* $(\mathbf{m}';\mathbf{s},\mathbf{t})$ *on the balance wall with every wall positive. Then the ray with the origin* $(\mathbf{m}-\mathbf{m}';\mathbf{0},\mathbf{0})$ *and the direction* $(\mathbf{m}';\mathbf{s},\mathbf{t})$ *has positive walls at every real* $k \geq 1$*, its values converge there, and* $\langle 1\rangle_1 = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*.*

(ii) Let $N \geq 1$ *be an integer with* $N\mathbf{m} \in \mathbb{Z}^{\mathcal{E}}$*. Then the ray with the origin* 0 *and the direction* $N(\mathbf{m};\mathbf{s},\mathbf{t})$ *has positive walls, its stride is* $f_{\mathbf{m}}^N$*,* $\langle 1\rangle_{1/N} = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*, and the wall degree of a symbol* g *for it is* $\frac{1}{N}\max_U\big(-\kappa_U(g)/\kappa_U\big)$*, the maximum over the nonempty proper* U*.*

So data with rational masses are read at $k=1$ on a ray with an origin, by the gauge end, or at $k=\frac{1}{N}$ on a ray from the origin 0, whose box of wall degree $\mathfrak{t}$ holds the symbols of wall degree $N\mathfrak{t}$ read on the walls of the data.

41.5 What a reading costs

A box has at most $\big(C(1+\mathfrak{t})+1\big)^{n-1}$ symbols at each level (Proposition 41.9(i)), and modulo a group G of symmetries its orbits number at least its symbols divided by $|G|$; one evaluation in step (3) of Definition 41.8 is one elimination of its relations, and the number of evaluations is set by the degrees of $\mathbf{B}$ (Proposition 41.10(ii)). The relations are affine in the strength, so one factorization at one strength gives every Taylor coefficient there.

Proposition 41.12 (One factorization serves every coefficient)**.** *In the setting of Proposition 41.10, let $k_0 \in \Bbbk$ have $\det A_P(k_0) \neq 0$, and put $M = A_P(k_0)$ and $E = A_P(k_0+1) - A_P(k_0)$, the part of A_1 on the rows and columns of A_P, so that $A_P(k_0+\epsilon) = M + \epsilon E$. For every row x_P over $\Bbbk$, as power series in ϵ convergent for small $|\epsilon|$,*

$$x_P\, A_P(k_0+\epsilon)^{-1} = \sum_{j\geq 0} \lambda_j \epsilon^j, \qquad \lambda_0 = x_P M^{-1}, \quad \lambda_j = -\lambda_{j-1} E M^{-1}.$$

So the Taylor coefficients at k_0 of the reductions, and of the entries of $\mathbf{B}$, up to any order J come from one factorization of M and J products by EM^{-1}; and a rational function whose numerator and denominator have degrees at most d_1 and d_2, the denominator not vanishing at k_0, is fixed, among such functions, by its Taylor coefficients of order at most $d_1 + d_2$ at k_0.

So one factorization at one strength can replace many evaluations; each step of the series is one solve with the factorization of M, and what it saves depends on the fill of that factorization.

Proposition 41.13 (The densities of a spanning set)**.** *Let a ray have a real origin and $\kappa_{1,S} > 0$ for every nonempty proper S, with $\alpha^\star$, $f^\star = f(\alpha^\star)$ and the function g of Proposition 40.3, and let $b_1, \dots, b_{\mathfrak{m}}$ be Laurent insertions with real coefficients and Q, $\tilde{P}_{il}$ polynomials with $Q(k)\langle b_i\rangle_{k-1} = \sum_l \tilde{P}_{il}(k)\langle b_l\rangle_k$ as meromorphic functions, as a box gives with $\mathbf{B} = \tilde{P}/Q$ (Theorem 39.17(iii)). For a Laurent insertion b with real coefficients let $\mu_{f,b}$ be the image under f of the signed measure $b\,\omega_0$ on the positive part of the chart, a signed measure on $(0, f^\star]$, finite on $[t, f^\star]$ for $t > 0$.*

- *(i) $\langle b\rangle_k = \int t^k \mu_{f,b}(dt)$ wherever $\langle b\rangle_k$ converges absolutely; $\mu_{f,b}$ has a density $\varrho_{f,b}$, real-analytic on $(0, f^\star)$, and near $f^\star$, $\varrho_{f,b}(t) = (f^\star - t)^{\frac{n-3}{2}} g_b(f^\star - t)$ with g_b holomorphic at 0 and $g_b(0) = b(\alpha^\star)\, g(0)$.*
- *(ii) As distributions on $(0,\infty)$, $Q(-1-\theta)\big(t^{-1}\mu_{f,b_i}\big) = \sum_l \tilde{P}_{il}(-1-\theta)\,\mu_{f,b_l}$ for every i, $\theta = t\frac{d}{dt}$: a system of order at most $\max(\deg Q, \deg \tilde{P}_{il})$ for the $\mathfrak{m}$ densities.*
- *(iii) With $v(k) = (\langle b_l\rangle_k)_l$ and every $K \geq 2$ at which $\mathbf{B}(2), \dots, \mathbf{B}(K)$ are finite and $v(K)$ converges, the continuation of v is finite at 1 and $v(1) = \mathbf{B}(2)\mathbf{B}(3)\cdots\mathbf{B}(K)\, v(K)$: the values of the spanning set run back from K with no scalar recurrence.*

The density operator of a scalar recurrence has the order of the degree of its coefficients, which carry the strengths where the rows line up (Proposition 39.6(iii)); the system of (ii) has the order of the degrees of the entries of $\mathbf{B}$, for $\mathfrak{m}$ densities whose germs at $f^\star$ start from the values $b_l(\alpha^\star)$ times the one constant $g(0)$.

Proposition 41.14 (The native gauges of all exponents)**.** *Let a family of N tables on $n \geq 2$ marks have the letter complement X_F (Definition 38.1), and for $\alpha \in X_F$ and complex exponents $(\mathbf{s}, \mathbf{t}) \in \mathbb{C}^{2N}$ write $\mathbf{u}(\alpha; \mathbf{s}, \mathbf{t}) = \sum_j \big(s_j R_j(\alpha) + t_j R_j^*(\alpha)\big) \in \mathbb{C}^{\mathcal{E}}$ for the usage (Definition 8.17).*

- *(i) The data $\big(\mathbf{u}(\alpha; \mathbf{s}, \mathbf{t}); \mathbf{s}, \mathbf{t}\big)$ lie on the balance wall. Call $\alpha \in X_F$* native for *data $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ on the balance wall when $\mathbf{m} = \mathbf{u}(\alpha; \mathbf{s}, \mathbf{t})$: the native gauges of a ray are the points native for its direction (Definition 38.4), and for $m_{\mathcal{E}} \neq 0$ the points native for data are its native points (Definition 8.17). So the set of the $(\alpha, \mathbf{m}, \mathbf{s}, \mathbf{t})$ with α native for $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ is the graph of the map $(\alpha, \mathbf{s}, \mathbf{t}) \mapsto \mathbf{u}(\alpha; \mathbf{s}, \mathbf{t})$ on $X_F \times \mathbb{C}^{2N}$, a smooth irreducible variety of dimension $n-1+2N$, and every $\alpha \in X_F$ is native for the data $\big(\mathbf{u}(\alpha; \mathbf{s}, \mathbf{t}); \mathbf{s}, \mathbf{t}\big)$, whatever $(\mathbf{s}, \mathbf{t})$.*
- *(ii) A ray of F with $\kappa_{1,S} > 0$ for every nonempty proper S has at most $(-1)^{n-1}\,\mathrm{eu}(X_F)$ nondegenerate native gauges, the number of native gauges of the directions outside the exceptional set of Theorem 38.24(i).*

The nondegenerate native gauges bound the sizes of a reading from below: the dimension of the invariant module is at least the number of their orbits (Theorem 39.8(iii)), and the order of the law of [1] at least the number of their distinct native values (Corollary 38.7 at $g = 1$); Proposition 41.14(ii) bounds their number from above. These are inequalities, with X_F inverting every letter whatever its exponent along the ray (Definition 38.1).

Definition 41.15 (The procedure that finds native gauges by continuation)**.** Input: a family and a ray of it with $\kappa_{1,S} > 0$ for every nonempty proper S. Output: nondegenerate native gauges of the ray, each with a certificate, and their native values.

(1) *A start.* Draw $(\alpha_0, \mathbf{s}_0, \mathbf{t}_0) \in X_F \times \mathbb{C}^{2N}$ at random; α_0 is native for the data $w_0 = \big(\mathbf{u}(\alpha_0; \mathbf{s}_0, \mathbf{t}_0); \mathbf{s}_0, \mathbf{t}_0\big)$ (Proposition 41.14(i)).

(2) *More points native for w_0.* Follow the points native for w_0 found so far along closed loops $w_0 \to w_1 \to w_2 \to w_0$ of random data on the balance wall, by Newton's method along the segments, and add every new endpoint; stop when several loops in a row add nothing.

(3) *To the direction.* Follow each point native for w_0 by Newton's method along a path of data from w_0 to the direction of the ray, and keep the endpoints that lie in X_F.

(4) *Certificates.* Around each endpoint, a box in the coordinates α_e, $e \neq e_0$, of the chart on which interval enclosures of the letters exclude 0; Krawczyk's interval Newton test on the equations $c_e^{(1)} = 0$, $e \neq e_0$, in the box, and an interval enclosure of the determinant of the Hessian of $\log f$ on the box that excludes 0. When both succeed, the box holds a native gauge of the ray, and every native gauge in it is nondegenerate. Disjoint boxes hold distinct native gauges, and disjoint enclosures of f on them prove distinct native values.

The certified native gauges of step (4) are distinct nondegenerate native gauges of the ray, so their number bounds the dimension of the ray from below (Theorem 38.6(iii)), the number of their orbits under a finite group of symmetries bounds $\dim \mathfrak{M}^G$ from below (Theorem 39.8(iii)), and the number of their distinct native values bounds the order of every recurrence that follows from the relations (Corollary 38.7); by Proposition 41.14(ii) their number is at most $(-1)^{n-1}\,\mathrm{eu}(X_F)$. The stopping rule of step (2) is not a certificate that every point native for w_0 was found, so the procedure gives lower bounds. Nothing in it writes a relation of a box. Exercise 53.10 reads the dimensions and the orders of the laws by the box, ray by ray.

Chapter 42

Inputs along a ray

A function that satisfies a recurrence with polynomial coefficients in its argument, above all the values of 1 along a ray continued in the strength, gives two live inputs on a lattice: its values and its slopes. The slopes satisfy the same recurrence, inhomogeneous by the values read through the derivatives of its coefficients. A Trailmoment at index zero of a polynomial word against either input is closed by summation by parts in the adjoint form of the recurrence: what crosses the cut is read off the Laurent data of the function at the low end, for a ray the low-end relation and its derivative, and when the symbol does not vanish at 1 everything else lands on a fixed block of $2D$ Trailmoments, D the degree of the recurrence. A mark of a list opened on its share side and closed at an integer strength is such a pairing: a polynomial word against the slope input of the room, less its derivative against the values input. The recurrence read at the poles of an input ties its initial slopes to residues and finite parts, and minimality, the root rate of the input, removes the components along the faster solutions. A recurrence of the values that is self-adjoint under the reflection of the strength pairs the values with the closings in one constant, the low-end relation, whose limit at high strength ties the volume of the direction to the Hessian at its native gauge through the native values. The last section reads all of this on the centre ray of the list $\binom{\mathcal{E}}{1}$, for every number n of marks: its native gauges are the signs, and at a lean the points whose occupations take two values; a parallel class enters a list through the closings and the signs of $\binom{P}{1}$; a mark is added by closing the critical-line kernel against the slopes of n marks; the law of the values is the n-th symmetric power of the second-order relation that the Gramma of one mark satisfies in its coefficient, read in the index, of order $\lceil n/2 \rceil$, with the native values $(n-2j)^{-2}$ as the roots of its symbol, and one identity carries it from n marks to $n+1$; the lowered settings are its solutions without low-end coefficient; and the exact pairing holds at every strength. The n-th power of the Gramma of one mark is a mixture of its dilations by the stride's law, and the balanced directions with one total and one multiset of offsets have one value up to Beta corners.

42.1 Inputs read from a ray

Definition 42.1 (Recurrences; the values input and the slope input)**.** A *recurrence* of order h is an operator $(L\varphi)(x) = \sum_{i=0}^{h} c_i(x)\,\varphi(x+i)$ with polynomials $c_i \in \mathbb{C}[x]$ and $c_0 c_h \neq 0$. Its *adjoint* and its *derivative* are

$$(L^{*}\varphi)(x) = \sum_{i=0}^{h} c_i(x-i)\,\varphi(x-i), \qquad (L'\varphi)(x) = \sum_{i=0}^{h} c_i'(x)\,\varphi(x+i),$$

L'^* is the adjoint of L', its *degree* is $D = \max_i \deg c_i$, and its *symbol* is $\chi(\lambda) = \sum_i [x^D] c_i\, \lambda^i$, as in Corollary 38.7. Let p be meromorphic on $\mathbb{C}$ with $Lp = 0$ identically, holomorphic at the points of a coset $b + \mathbb{Z}$ beyond a cut N, and put $\tilde{p} = -p'$. The *values input* of p is $p\,\mathbb{I}_{b+\mathbb{Z}}$ and its *slope input* is $\tilde{p}\,\mathbb{I}_{b+\mathbb{Z}}$ (Definition 5.4), both read beyond the cut N. When p continues the values $\langle 1\rangle_k$ of a ray and L is a recurrence of these values, they are the *inputs read from the ray.*

Proposition 42.2 (The values of a ray as an input)**.** *Let a ray of a family on $n \geq 2$ marks have origin 0, every letter exponent of its direction positive and $\kappa_{1,S} > 0$ for every nonempty proper S, and let L be a recurrence of order h with $\sum_{i=0}^{h} c_i(k)\langle 1\rangle_{k+i} = 0$ for all large integers k.*

(i) $\langle 1\rangle_k$ is holomorphic in k on $\operatorname{Re} k > 0$ and continues to a meromorphic function p on $\mathbb{C}$ with $Lp = 0$ identically. At every $j \in \mathbb{N}$, $\lim_{k\to -j}(k+j)^{n-1}p(k) = V_1\, \mathsf{C}_j(1)$, with V_1 of Proposition 39.30, and $\sum_{i=0}^{\min(h,j)} c_i(-j)\, \mathsf{C}_{j-i}(1) = 0$.

(ii) The slope input satisfies $L\tilde{p} = L'p$ identically: a recurrence of the same order, inhomogeneous by the values read through the derivative of L.

(iii) Along the positive integers p has the root rate $f(\alpha^\star)$, the native value of the native gauge $\alpha^\star$ of the direction on the positive orthant, and $\tilde{p}$ a root rate at most $f(\alpha^\star)$.

The values of a ray are therefore an input of the kind of Chapter 6: their density is read from the serif family, and the recurrence gives the values input and the slope input recurrences of one finite order. The slope input carries what the derivative in the strength adds, and the values input enters its recurrence only through L'.

42.2 Closing an input

Proposition 42.3 (Summation by parts at the cut)**.** *Let p and L be as in Definition 42.1, p holomorphic at the integers $x \geq 1$, with p and $\tilde{p}$ of root rate below one along them, and for $0 \leq j < h$ put $\beta_j(x) = \sum_{i=0}^{j} c_i(x)\, p(x+i)$, which is holomorphic at $x = -j$ because $\beta_j = -\sum_{i>j} c_i(x)\, p(x+i)$. For every polynomial R,*

$$(\mathsf{T}_{0;(1)}[x]\,(L^*R)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\sum_{j=0}^{h-1} R(-j)\,\beta_j(-j),$$

$$(\mathsf{T}_{0;(1)}[x]\,(L^*R)\,\tilde{p}\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,(L'^*R)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + \sum_{j=0}^{h-1} R(-j)\,\beta'_j(-j).$$

If every c_i vanishes to order two at the integers $-j$ with $1 \leq j < i$, only $j = 0$ remains, with $\beta_0 = c_0\, p$: the boundary is the value and the derivative of $c_0 p$ at the cut.

What a summation by parts puts on the inputs beyond the cut is the adjoint word; what it leaves at the cut is read off p at $0, -1, \ldots, -(h-1)$, where for the values of a ray the poles sit. The same move closes the slope input, once through L and once through L'.

Proposition 42.4 (The fixed block)**.** *Let L be a recurrence of degree D whose symbol has $\chi(1) \neq 0$, and let $\Bbbk$ be a field containing the coefficients of the c_i.*

(i) L^* raises the degree of every nonzero polynomial by exactly D, and $\Bbbk[x] = L^*(\Bbbk[x]) \oplus \{r \in \Bbbk[x] :\ \deg r < D\}$.

(ii) In the setting of Proposition 42.3, for every polynomial $W \in \Bbbk[x]$ the two Trailmoments

$$(\mathsf{T}_{0;(1)}[x]\, W\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad (\mathsf{T}_{0;(1)}[x]\, W\, \tilde{p}\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$$

lie in the $\Bbbk$-span of the $2D$ block values

$$(\mathsf{T}_{0;(1)}[x]\, (x-1)^a\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad (\mathsf{T}_{0;(1)}[x]\, (x-1)^a\, \tilde{p}\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) \qquad (0 \le a < D)$$

and of the boundary numbers $\beta_j(-j)$ and $\beta'_j(-j)$, $0 \le j < h$.

This is the counterpart at the index zero of Theorem 6.9: the words of a closing are polynomials, the shifts of the input are moved onto the word, and the input is read on one block of evaluations fixed once for the recurrence. When the boundary numbers are known, as they are at the cut of a ray from the low-end relation, a closing lands on the block plus a number of $\Bbbk$.

42.3 A mark opened at an integer strength

A mark that is neither a loop nor a coloop is a bunch of width one, and Theorem 19.13 opens the centre on it: at the strength s the period is the integral over one even share d of $\Gamma_{\substack{4\\2}}\begin{bmatrix} d,\ d,\ s-d,\ s-d \\ s,\ s \end{bmatrix}$ against the period of the room family, the deletion at the share d and the contraction at $s-d$. Closing the share (Corollary 19.21) at an integer strength turns its heads into one polynomial word.

Corollary 42.5 (A mark opened at an integer strength)**.** *Let $\mathcal{B}$ be a connected list on $\mathcal{E}$, $|\mathcal{E}| \ge 3$, and e a mark that is neither a loop nor a coloop; on the chart $y_{e_0} = 0$ of $R = \mathcal{E} - e$ let $X_\backslash = K_{\mathcal{B}\backslash e}(\alpha) K_{\mathcal{B}\backslash e}(\alpha^{-1})$ and $X_/ = K_{\mathcal{B}/e}(\alpha) K_{\mathcal{B}/e}(\alpha^{-1})$, the even letters of the deletion and of the contraction over α^R. Suppose that one lies below the other: $\{X_\flat, X_\sharp\} = \{X_\backslash, X_/\}$ with $\rho = \sup X_\flat / X_\sharp < 1$ on the chart. For $s \in \{1, 2, \dots\}$ and $x \ge 0$ put*

$$q_s(x) = \int_{y_{e_0}=0} \Big(\frac{X_\flat}{X_\sharp}\Big)^x X_\sharp^{-s}\, dy, \qquad \tilde{q}_s = -\partial_x q_s, \qquad V_s(x) = \big((x+1)(x+2)\cdots(x+s-1)\big)^2.$$

Then q_s converges and is holomorphic on a neighbourhood of $[0,\infty)$, and

$$\Gamma(s)^2\, P_{\mathcal{B}}(s\mathbf{1}; s, s) = \sum_{x \in \mathbb{N}} \big(V_s(x)\, \tilde{q}_s(x) - V_s'(x)\, q_s(x)\big),$$

the terms of root rate at most ρ: the centre at the strength s is a polynomial word against the slope input of q_s less its derivative against its values input, read on $\mathbb{N}$.

The function q_s is read along the direction of the ratio $X_\flat / X_\sharp$ of the two even letters, with the origin form $X_\sharp^{-s} dy$, which decays where the ratio does not; where it satisfies a recurrence in x, Sections 42.2 and 42.4 close it, and its values half $\sum_x V_s(x)\, q_s(x)$ with it.

42.4 Boundaries and minimality

Proposition 42.6 (Boundary relations)**.** *Let p and L be as in Definition 42.1, let $x_0 \in \mathbb{C}$ and $0 \le j < h$, and let p be holomorphic at $x_0 + i$ for $j < i \le h$. Then $\beta_{x_0,j}(x) = \sum_{i=0}^{j} c_i(x)\, p(x+i)$*

is holomorphic at x_0, its value and its derivative there are read off the Laurent expansions of p at $x_0, \dots, x_0 + j$ and the Taylor expansions of $c_0, \dots, c_j$ at x_0, and

$$\sum_{i=j+1}^{h} c_i(x_0)\, p(x_0+i) = -\beta_{x_0,j}(x_0),$$

$$\sum_{i=j+1}^{h} \big(c_i(x_0)\, \tilde{p}(x_0+i) - c_i'(x_0)\, p(x_0+i)\big) = \beta'_{x_0,j}(x_0).$$

Where c_h has no zero at the integers $x \geq 1$, the slope input on these integers is fixed by its initial slopes $\tilde{p}(1), \dots, \tilde{p}(h)$ through $L\tilde{p} = L'p$, and every second relation at an integer x_0 with $x_0 + j \geq 0$ is an affine relation among them; at $x_0 = -j$ its right side is read off the Laurent data of p at the nonpositive integers $-j, \dots, 0$ alone. At the cut of the values of a ray (Proposition 42.2), with $c_0(x) = \sum_m \gamma_m x^m$ and $p(x) = \sum_{i \geq 0} \mathsf{L}_i x^{i-n+1}$ near $x = 0$, the low-end coefficients of Definition 39.33, c_0 vanishes to order at least $n-1$ at 0, $\mathsf{L}_0 = V_1$, and at $x_0 = 0$, $j = 0$ the two relations are

$$\sum_{i=1}^{h} c_i(0)\, p(i) = -\gamma_{n-1} V_1, \qquad \sum_{i=1}^{h} \big(c_i(0)\, \tilde{p}(i) - c_i'(0)\, p(i)\big) = \gamma_{n-1}\mathsf{L}_1 + \gamma_n V_1,$$

the low-end relation (Corollary 39.34(ii), which holds for every recurrence of the values at the large integers by Proposition 40.5) and its derivative; when every table of the family has unit weights, $\mathsf{L}_1 = 0$.

At a pole of the input that a wall or a zero share puts on the lattice, the right sides are its principal part and its finite part: at a simple pole x_0 where c_0 has a simple zero, $\beta_{x_0,0}(x_0) = c_0'(x_0) \operatorname{Res}_{x_0} p$ and $\beta'_{x_0,0}(x_0) = c_0'(x_0)\, a + \frac{1}{2} c_0''(x_0) \operatorname{Res}_{x_0} p$, a the constant term of p at x_0. Each independent relation removes one direction from the initial slopes; what the relations leave free, the root rate of the input fixes in the directions that grow faster than the input.

Proposition 42.7 (Minimality)**.** *Let p and L be as in Definition 42.1, p holomorphic at the integers $x \geq 1$, with $c_0(x)c_h(x) \neq 0$ at these integers, every c_i of degree at most $\deg c_0 = \deg c_h$, and the roots $\lambda_1, \dots, \lambda_h$ of the symbol of distinct moduli. Let $\lambda > 0$ be at least the root rates of p and of $\tilde{p}$ along the positive integers (Definition 1.1(vii)).*

(i) The solutions u of $Lu = 0$ on the integers $x \geq 1$ of root rate at most λ form a space U_λ of dimension $\#\{c : \ |\lambda_c| \leq \lambda\}$, and every nonzero $u \in U_\lambda$ has $u(x+1)/u(x) \to \lambda_c$ for one of these roots, so the root rate $|\lambda_c|$.

(ii) The functions φ on the integers $x \geq 1$ with $L\varphi = L'p$ and root rate at most λ are the $\tilde{p} + u$, $u \in U_\lambda$, and those with $L\varphi = 0$ are the $p + u$. So the slope input is the only one of them that satisfies a set of second relations of Proposition 42.6 whose left sides, read on u, vanish together on no nonzero $u \in U_\lambda$; the values input is singled out by the first relations in the same way.

So the initial slopes are fixed by as many boundary relations as there are roots of the symbol of modulus at most the root rate of the input; the roots beyond it are removed by the root rate alone.

42.5 The two ends paired

Proposition 42.8 (The two ends paired)**.** *Let a ray be as in Proposition 42.2, with a recurrence L of its values whose coefficients satisfy $c_i(-x-i) = \epsilon\, c_i(x)$ for every i, $\epsilon \in \{1,-1\}$, and put $\mathsf{C}_j = \mathsf{C}_j(1)$ for $j \geq 0$ and $\mathsf{C}_j = 0$ for $j < 0$.*

(i) $\sum_{i=0}^{h} c_i(\sigma - i)\, \mathsf{C}_{\sigma-i} = 0$ *for every integer σ: the closings, read at the strength σ, solve the adjoint recurrence.*

(ii) The concomitant

$$B(\sigma) = \sum_{i=1}^{h} \sum_{j=1}^{i} c_i(\sigma - j)\, \mathsf{C}_{\sigma-j}\, \langle 1 \rangle_{\sigma-j+i}$$

takes one value at every integer $\sigma \geq 1$, $B(\sigma) = \sum_{i=1}^{h} c_i(0)\langle 1\rangle_i = -\gamma_{n-1} V_1$ with $\gamma_{n-1} = [x^{n-1}]c_0$: the low-end relation, carried to every strength.

(iii) On the centre ray of a family (Definition 38.13), if every c_i has degree at most $n-1$, then $c_0 = \chi(0)\, x^{n-1}$, $\lambda_\star = f(\alpha^\star)$ is a simple root of the symbol, and

$$V_1 \det H = -\frac{\lambda_\star\, \chi'(\lambda_\star)}{\chi(0)} = \prod_{c \neq \star} \Big(1 - \frac{\lambda_\star}{\lambda_c}\Big),$$

with H the Hessian of $-\log f$ at $\mathbf{1}$ on the chart (Theorem 38.15) and λ_c the roots of the symbol, counted with multiplicity.

The values at the high end and the closings at the low end are two solutions of one recurrence and of its adjoint, and their concomitant is the constant that the low end fixes. At high strength the two decay and grow at the reciprocal rates $f(\alpha^\star)^{\pm k}$ with the Gaussian constants of the native gauge of the direction, and the constant becomes an identity between the volume V_1 of the direction, read by its walls, and the Hessian at its native gauge, through the other native values.

42.6 The centre ray of $\binom{\mathcal{E}}{1}$

For the list $\binom{\mathcal{E}}{1}$ on n marks, whose bases are the singletons, $K(\alpha) = \sum_e \alpha_e$ and $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1})$, and the stride of the centre ray at the strength σ is $\big(K(\alpha)K(\alpha^{-1})\big)^{-\sigma} = \big(\sum_{e,e'} \alpha_e/\alpha_{e'}\big)^{-\sigma}$. Its widths are all one, so $V_1 = n!$ at the strength one. Its share side is one Gramma read n times: by Corollary 18.22 for the dual table, whose bases are the sets $\mathcal{E} - e$, and Proposition 8.15, for masses $\mathbf{m}$ on the balance wall, $m_{\mathcal{E}} = s + (n-1)t$, with $m_S > t(|S|-1)$ for every nonempty proper S,

$$\Gamma(s)\,\Gamma(t)\, P_{\binom{\mathcal{E}}{1}}(\mathbf{m}; s, t) = \int_0^\infty y^{t-1} \prod_{e \in \mathcal{E}} \mathsf{G}_{m_e - t;(1,-1)}(1, y)\, dy,$$

and at the centre $\Gamma(s)^2 P_{\binom{\mathcal{E}}{1}}(s\mathbf{1}; s, s) = \int_0^\infty y^{s-1} \mathsf{G}_{0;(1,-1)}(1,y)^n\, dy$. With $\vartheta = y\, d/dy$, the ladder and the exchange relation of the Gramma (Theorem 2.4) give $\vartheta^2 \mathsf{g} = y\, \mathsf{g}$ for $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y)$, and the partner $\mathsf{h}(y) = \sum_{m \geq 0} y^m/m!^2$ of its lattice expansion (Lemma 20.29(i), $\mathsf{g} = -\mathsf{h}\log y + \sum_m 2(\log\Gamma)'(1+m)\, y^m/m!^2$) satisfies the same relation. The statements below read the centre ray from these two functions.

Proposition 42.9 (The native gauges of $\binom{\mathcal{E}}{1}$)**.** *Let $n \geq 2$, let $s, t \geq 1$ and m be integers with $nm = s + (n-1)t$, and read a ray of $\binom{\mathcal{E}}{1}$ in the direction $(m\mathbf{1}; s, t)$; in this statement $u_e(\alpha) = \alpha_e/K(\alpha)$ is the occupation of the mark e and $\lambda(\alpha) = \big(K(\alpha)K(\alpha^{-1})\big)^{-1}$.*

(i) A point α of X_F is a native gauge exactly when its occupations take at most two values, u_+ on $n-j$ marks and u_- on j marks, $0 \le j \le n$, $2j \ne n$, with

$$(n-2j)\,u_+ = 1 - j\,\frac{s-t}{ns}, \qquad u_+ + u_- = \frac{s-t}{ns}, \qquad u_+u_- \ne 0;$$

then $\lambda(\alpha) = -s\,u_+u_-/t$ and the native value is $f(\alpha) = \big(\prod_e u_e\big)^{m-t}\lambda(\alpha)^t$. Up to scale these points are $\alpha = (u_e)_e$, one for each set of j marks, a set and its complement giving the same point.

(ii) At the centre, $s = t = \sigma$, $m = \sigma$, the native gauges are exactly the signs η with $K(\eta) = \sum_e \eta_e \ne 0$, of the native values $K(\eta)^{-2\sigma} = (n-2j)^{-2\sigma}$, j the number of marks at -1: there are $\lceil n/2 \rceil$ native values. Each of these signs is nondegenerate: on the chart $\alpha_e = \eta_e e^{y_e}$, $y_{e_0} = 0$, the Hessian of $-\log f$ at $y = 0$ has the determinant $(2\sigma)^{n-1}\prod_e \eta_e/K(\eta)^n$.

A lean moves every sign to a real point with the same pattern: the occupations keep two values, on the same marks, and only the two values change. Every list has the signs with $K(\eta) \ne 0$ among the native gauges of its centre ray (Theorem 38.17); the list $\binom{\mathcal{E}}{1}$ has nothing else.

Proposition 42.10 (A parallel class enters through its list)**.** *Let $\mathcal{H}$ be a connected list on the marks $1, \dots, q$, $q \ge 2$, and replace every mark i by a parallel class P_i of $s_i \ge 1$ marks by the first rule of Theorem 21.5: the list $\mathcal{B}$ on $\mathcal{E} = \bigsqcup_i P_i$ whose bases are the sets with one mark in P_i for every $i \in T$ and none in the other classes, T a basis of $\mathcal{H}$. Write $A_i = \sum_{e \in P_i} \alpha_e$.*

(i) $K_{\mathcal{B}}(\alpha) = K_{\mathcal{H}}(A_1, \dots, A_q)$, and for $v \in \mathbb{N}^{\mathcal{E}}$, with $w_i = v_{P_i}$, $[\alpha^v]K^k_{\mathcal{B}} = \big([A^w]K^k_{\mathcal{H}}\big)\prod_i \binom{w_i}{v|_{P_i}}$, with the multinomial coefficients $\binom{w_i}{v|_{P_i}}$.

(ii) The closings of the centre ray of $\mathcal{B}$ at the strength one (Definition 39.26) are

$$\mathsf{C}^{\mathcal{B}}_k(1) = \sum_{w \in \mathbb{N}^q} \big([A^w]K^k_{\mathcal{H}}\big)^2 \prod_{i=1}^{q} \mathsf{C}^{\binom{P_i}{1}}_{w_i}(1), \qquad \mathsf{C}^{\binom{P}{1}}_{w}(1) = \sum_{v \in \mathbb{N}^P,\ |v| = w} \binom{w}{v}^2,$$

the second the closings of the centre ray of $\binom{P}{1}$ at the strength one, read as 1 when $|P| = 1$.

(iii) At a sign η of $\mathcal{B}$, $K_{\mathcal{B}}(\eta) = K_{\mathcal{H}}(a_1, \dots, a_q)$ with $a_i = \sum_{e \in P_i}\eta_e \in \{s_i, s_i - 2, \dots, -s_i\}$, the sign values of $\binom{P_i}{1}$: on the centre ray of $\mathcal{B}$ at the strength σ the native values of the signs with $K_{\mathcal{B}}(\eta) \ne 0$ (Theorem 38.17(ii)) are the $K_{\mathcal{H}}(\mathbf{a})^{-2\sigma}$ at these points $\mathbf{a}$.

A parallel class enters the closings of a list only through the closings of its own list $\binom{P}{1}$, and the signs only through the sign values of that list, as a parallel block is read by K through its total and by Ψ through its hinge value (Corollary 21.7(i)).

Proposition 42.11 (A mark added is a closing of the critical-line kernel)**.** *Let $n \ge 2$; in this statement $p_n(s) = P_{\binom{\mathcal{E}}{1}}(s\mathbf{1}; s, s) = \int_{y_{e_0}=0} \big(K(\alpha)K(\alpha^{-1})\big)^{-s} dy$ for $|\mathcal{E}| = n$ and $\operatorname{Re} s > 0$, $\tilde{p}_n = -p_n'$, and $W_s(x) = \prod_{j=1}^{s-1}(x-j)^2$.*

(i) For $\operatorname{Re} s > 0$ and $0 < c < \operatorname{Re} s$, $\Gamma(s)^2 p_{n+1}(s) = \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2\,\Gamma(s-d)^2\,p_n(d)\,dd$. At $s = 1$ the kernel on the line $\operatorname{Re} d = \frac{1}{2}$ is 2π times the critical-line density (Definition 36.1): $p_{n+1}(1) = \mathbb{E}\big[p_n(\frac{1}{2} + iY)\big]$, $Y \sim \mathbb{P}_{\mathrm{crit}}$.

(ii) For $s \in \{1, 2, \dots\}$,

$$\Gamma(s)^2 p_{n+1}(s) = \big(\mathsf{T}_{0;(1)}[x]\,(W_s\tilde{p}_n - W_s' p_n)\,\mathbb{I}_{\mathbb{Z}}\big)((0,\infty)),$$

$$\int_0^\infty y^{s-1}\mathsf{G}_{0;(1,-1)}(1,y)^n\sum_{m\ge0}\frac{y^m}{m!^2}\,dy=(\mathsf{T}_{0;(1)}[x]\,W_s\,p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty));$$

at $s=1$, $p_{n+1}(1)=(\mathsf{T}_{0;(1)}[x]\,\tilde p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$: the centre of $n+1$ marks at the strength one is the slope input of n marks summed over the positive integers.

(iii) On the share side every mark carries on average the share $\frac{1}{n+1}$ of the strength:

$$(n+1)\Big((\mathsf{T}_{0;(1)}[x]\,p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty))-(\mathsf{T}_{0;(1)}[x]\,(x-1)\,\tilde p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big)=(\mathsf{T}_{0;(1)}[x]\,\tilde p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Theorem 42.12 (The law of the values of $\binom{\mathcal{E}}{1}$)**.** *Let $n\ge1$ and $h=\lceil n/2\rceil$; in this statement $\vartheta=y\,d/dy$, and the operators $\mathbf{R}_i$ are $\mathbf{R}_0=1$, $\mathbf{R}_1=\vartheta$ and $\mathbf{R}_{i+1}=\vartheta\mathbf{R}_i-i(n+1-i)\,y\,\mathbf{R}_{i-1}$.*

(i) $\mathbf{Q}_n=\mathbf{R}_{n+1}$ annihilates every product of n solutions of $\vartheta^2u=yu$. Written $\mathbf{Q}_n=\sum_{k\ge0}y^kq_{n,k}(\vartheta)$ with polynomials $q_{n,k}$, $q_{n,0}=\vartheta^{n+1}$ and $\deg q_{n,k}\le n+1-2k$, so that $q_{n,k}=0$ for $k>h$; with $a_{n,k}$ the coefficient of ϑ^{n+1-2k} in $q_{n,k}$,

$$\sum_k a_{n,k}\,\eta^k\xi^{n+1-2k}=\prod_{j=0}^{n}\big(\xi-(n-2j)\sqrt{\eta}\big)=\xi^{n+1-2h}\prod_{0\le j<n/2}\big(\xi^2-(n-2j)^2\eta\big),$$

so $q_{n,h}\ne0$. The formal adjoint for the measure dy/y is $\mathbf{Q}_n^*=(-1)^{n+1}\mathbf{Q}_n$: $q_{n,k}(-\vartheta-k)=(-1)^{n+1}q_{n,k}(\vartheta)$.

(ii) Put $c_{n,0}(s)=(-1)^{n+1}s^{n-1}$ and $c_{n,k}(s)=q_{n,k}(-s-k)\big((s+1)(s+2)\cdots(s+k-1)\big)^2$ for $1\le k\le h$. Every $c_{n,k}$ has degree $n-1$, $c_{n,k}(-s-k)=(-1)^{n+1}c_{n,k}(s)$, $c_{n,k}$ vanishes to order two at $s=-1,\dots,-(k-1)$, $c_{n,h}$ has no zero at the integers $s\ge0$, and the symbol is

$$\chi_n(\lambda)=(-1)^{n+1}\prod_{0\le j<n/2}\big(1-(n-2j)^2\lambda\big),$$

whose roots are the native values $(n-2j)^{-2}$ of the signs (Proposition 42.9). For $n\ge2$ the values satisfy

$$\sum_{k=0}^{h}c_{n,k}(s)\,P_{\binom{\mathcal{E}}{1}}\big((s+k)\mathbf{1};s+k,s+k\big)=0$$

identically in s.

(iii) For $n\ge2$ the values $P_{\binom{\mathcal{E}}{1}}(k\mathbf{1};k,k)$ satisfy no recurrence of order less than h: their tier (Proposition 38.11) is h, and the recurrence of (ii) is their least recurrence. The closings $\mathsf{C}_j(1)=\sum_{|v|=j}\binom{j}{v}^2$ satisfy $\sum_{k=0}^{\min(h,j)}c_{n,k}(-j)\,\mathsf{C}_{j-k}(1)=0$ for every $j\in\mathbb{N}$ and no recurrence of order less than h, and the cyclic submodule of $[1]$ in the value module (Definition 39.42) has dimension h.

The law of the values of $\binom{\mathcal{E}}{1}$ is the relation of one mark raised to the n-th symmetric power and read in the index, where ϑ becomes $-s$ and y the shift. Its leading coefficients are those of the product of the $n+1$ factors $\xi-(n-2j)\sqrt{\eta}$, which pair into the native values of the signs, the tie $n=2j$ at even n leaving the factor ξ; every coefficient $c_{n,k}$ carries the square of a rising factorial, which is why the boundaries below the cut vanish. One identity carries it from n marks to $n+1$.

Proposition 42.13 (A mark added carries the law of the values)**.** *In the notation of Theorem 42.12, for every $n\ge1$:*

(i) For every solution u of $\vartheta^2 u = yu$ and every smooth function F on $(0,\infty)$,

$$\mathbf{Q}_{n+1}(uF) = u\,\vartheta\,\mathbf{Q}_n(F) + (n+2)\,(\vartheta u)\,\mathbf{Q}_n(F).$$

(ii) With $\binom{a}{b} = \Gamma(a+1)/\big(\Gamma(b+1)\Gamma(a-b+1)\big)$ and $A_s(x) = \big((n+1)x-(n+2)s\big)\binom{x}{s}^2$,

$$\sum_{k=0}^{\lceil (n+1)/2\rceil} c_{n+1,k}(s)\binom{x-1}{s+k-1}^2 = \sum_{j=0}^{\lceil n/2\rceil} c_{n,j}(x-j)\,A_s(x-j),$$

an identity of rational functions of (x,s) after division by $\binom{x}{s}^2$. At integers $s, x \geq 1$ the words $\binom{x-1}{s+k-1}^2$ are the words $W_{s+k}(x)/\Gamma(s+k)^2$ of the closing of Proposition 42.11(ii): weighted by the coefficients $c_{n+1,k}(s)$, they are the adjoint of the recurrence of n marks applied to the one word A_s, which vanishes at $x = 0, 1, \ldots, s-1$.

Proposition 42.14 (The lowered settings)**.** *Let $n \geq 2$ and $h = \lceil n/2\rceil$, with $c_{n,k}$ of Theorem 42.12; in this statement $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y)$ and $\mathsf{h}(y) = \sum_{m\geq 0} y^m/m!^2$, and for $0 \leq j < n/2$ the* lowered setting *with j marks lowered is*

$$\xi_{n,j}(s) = \Gamma(s)^{-2}\int_0^\infty y^{s-1}\,\mathsf{g}(y)^{n-j}\,\mathsf{h}(y)^j\,dy \qquad (\operatorname{Re} s > 0),$$

so that $\xi_{n,0}(s) = P_{\binom{\mathcal{E}}{1}}(s\mathbf{1}; s, s)$, $|\mathcal{E}| = n$.

(i) Every $\xi_{n,j}$ satisfies the recurrence of Theorem 42.12(ii) on $\operatorname{Re} s > 0$.

(ii) For $j \geq 1$, $s^{n-1}\xi_{n,j}(s) \to 0$ as $s \downarrow 0$, and $\sum_{k=1}^h c_{n,k}(0)\,\xi_{n,j}(k) = 0$, against $\sum_{k=1}^h c_{n,k}(0)\,\xi_{n,0}(k) = (-1)^n n!$: the lowered settings have no low-end coefficient.

(iii) $\xi_{n,j}(k+1)/\xi_{n,j}(k) \to (n-2j)^{-2}$ along the integers, so $\xi_{n,j}$ has the root rate $(n-2j)^{-2}$: the setting with j marks lowered carries the native value of the signs with j marks at -1.

(iv) The h sequences $(\xi_{n,j}(k))_{k\geq 1}$ are a basis of the solutions of the recurrence on the integers $k \geq 1$, and its solutions x with $\sum_{k=1}^h c_{n,k}(0)\,x(k) = (-1)^n n!$ are the $\xi_{n,0} + \sum_{j=1}^{h-1} b_j\,\xi_{n,j}$, $b_j \in \mathbb{C}$.

So the recurrence and the low-end relation fix the values only up to the $h-1$ lowered settings with $j \geq 1$, which satisfy both with the right side zero; $h-1$ further numbers fix them, for instance the first $h-1$ values (Corollary 42.15). The integrand $\mathsf{g}^{n-j}\mathsf{h}^j$ of the setting with j marks lowered decays like $e^{-2(n-2j)\sqrt{y}}$ up to powers of y, and its integral converges on all of $\operatorname{Re} s > 0$ exactly when $j < n/2$: at even n the tie $j = n/2$, the sign with $K(\eta) = 0$, is no native gauge and gives no setting.

Corollary 42.15 (The value span of $\binom{\mathcal{E}}{1}$)**.** *Let $n \geq 2$, $h = \lceil n/2\rceil$ and $p_n(s) = P_{\binom{\mathcal{E}}{1}}(s\mathbf{1}; s, s)$, $|\mathcal{E}| = n$, with $c_{n,k}$ of Theorem 42.12.*

(i) $\sum_{k=1}^h c_{n,k}(0)\,p_n(k) = (-1)^n n!$, and every value $p_n(k)$, $k \geq 1$, lies in $\mathbb{Q} + \sum_{i=1}^{h-1}\mathbb{Q}\,p_n(i)$; so do the values $p_n(k\sigma)$ of the centre ray at every strength σ.

(ii) For even n the symbol has $\chi_n(1) \neq 0$, and for every polynomial W with rational coefficients, $(\mathsf{T}_{0;(1)}[x]\,W\,p_n\,\mathbb{I}_\mathbb{Z})((0,\infty))$ and $(\mathsf{T}_{0;(1)}[x]\,W\,\tilde{p}_n\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $\tilde{p}_n = -p_n'$, lie in $\mathbb{Q}$ plus the $\mathbb{Q}$-span of the $2(n-1)$ block values $(\mathsf{T}_{0;(1)}[x]\,(x-1)^a\,p_n\,\mathbb{I}_\mathbb{Z})((0,\infty))$ and $(\mathsf{T}_{0;(1)}[x]\,(x-1)^a\,\tilde{p}_n\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $a < n-1$. In particular so do $\Gamma(s)^2 P_{\binom{[n+1]}{1}}(s\mathbf{1}; s, s)$ and $\Gamma(s)^2\xi_{n+1,1}(s)$ at every integer $s \geq 1$.

Corollary 42.16 (The exact pairing of $\binom{\mathcal{E}}{1}$)**.** *Let $n \geq 2$, $h = \lceil n/2 \rceil$, and p_n and $c_{n,k}$ as in Corollary 42.15; put $\mathsf{C}_l = \sum_{v \in \mathbb{N}^{\mathcal{E}},\ |v|=l} \binom{l}{v}^2$ for $l \geq 0$ and $\mathsf{C}_l = 0$ for $l < 0$. For every integer $\sigma \geq 1$,*

$$\sum_{i=1}^{h} \sum_{j=1}^{i} c_{n,i}(\sigma - j)\, \mathsf{C}_{\sigma-j}\, p_n(\sigma - j + i) = (-1)^n n!,$$

and its limit at high strength is the identity

$$n! \cdot \frac{2^{n-1}}{n^n} = \prod_{j=1}^{h-1} \Big(1 - \frac{(n-2j)^2}{n^2}\Big) = \prod_{j=1}^{h-1} \frac{4j(n-j)}{n^2}$$

of Proposition 42.8(iii), with $V_1 = n!$, $\det H = 2^{n-1}/n^n$, $\lambda_\star = n^{-2}$ and the other roots $(n-2j)^{-2}$.

Proposition 42.17 (The power of one mark as a mixture)**.** *Let $n \geq 2$; in this statement $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1, y)$, and f_n, f_{n+1} are the strides $\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$ of the centre rays of $\binom{\mathcal{E}}{1}$ at the strength one for $|\mathcal{E}| = n$ and for $|\mathcal{E}| = n+1$, with the stride's laws μ_{f_n} and $\mu_{f_{n+1}}$ (Definition 40.1): the images of the Lebesgue measure dy of the chart $y_{e_0} = 0$ under the strides, carried by $(0, n^{-2}]$ and by $(0, (n+1)^{-2}]$.*

(i) $\mathsf{g}(y)^n = \int \mathsf{g}(y/t)\, \mu_{f_n}(dt)$ *for every $y > 0$: the n-th power of the Gramma of one mark is a mixture of its dilations by the stride's law.*

(ii) For $t > 0$ let $\phi_t(x) = x/\big((x + t^{-1/2})(1 + t^{-1/2}x)\big)$ on $(0, \infty)$, the stride of the centre ray at the strength one of the table $\binom{\{1,2\}}{1}$ with the weights $t^{-1/2}$ on its basis $\{1\}$ and 1 on $\{2\}$, on the chart $\alpha_1 = 1$, $\alpha_2 = x$; its stride's law μ_{ϕ_t} is the image of dx/x under ϕ_t. Then $\mu_{f_{n+1}} = \int \mu_{\phi_t}\, \mu_{f_n}(dt)$: adding a mark mixes the stride's law of two marks over the stride's law of n.

Proposition 42.18 (Balanced directions of $\binom{\mathcal{E}}{1}$)**.** *Let $|\mathcal{E}| = n \geq 2$, let $s, t \geq 1$ be integers and $\mathbf{m} \in \mathbb{Z}^{\mathcal{E}}$ with $m_{\mathcal{E}} = s + (n-1)t$ and $m_S > t(|S| - 1)$ for every nonempty proper S, and put $c_e = m_e - t$, so that $\sum_e c_e = s - t$. For real $k > 0$, with $P(k) = P_{\binom{\mathcal{E}}{1}}(k\mathbf{m}; ks, kt)$,*

$$\Gamma(sk)\, \Gamma(tk)\, P(k) = \int_0^\infty y^{\frac{(s+t)k}{2} - 1} \prod_{e \in \mathcal{E}} \mathsf{G}_{c_e k;(1,-1)}\big(\sqrt{y}, \sqrt{y}\big)\, dy,$$

which depends on the direction only through $s + t$ and the multiset $\{|c_e|\}_e$. Two such directions $(\mathbf{m}; s, t)$ and $(\mathbf{m}'; s', t')$ with $s + t = s' + t'$ and the same multiset have, with $P'(k) = P_{\binom{\mathcal{E}}{1}}(k\mathbf{m}'; ks', kt')$,

$$\frac{P'(k)}{P(k)} = \Gamma_1^2 \begin{bmatrix} sk,\ tk \\ (s+t)k \end{bmatrix} \Big/ \Gamma_1^2 \begin{bmatrix} s'k,\ t'k \\ (s+t)k \end{bmatrix},$$

a quotient of Beta corners at one total, rational at the integers k, so that their values span one $\mathbb{Q}$-space; the inversion $\alpha \mapsto \alpha^{-1}$ carries the integrand at $(\mathbf{m}; s, t)$ to the integrand at $((s+t)\mathbf{1} - \mathbf{m}; t, s)$, which is $c \mapsto -c$.

The exercises work these statements on the lists of a few marks: the closings and the law of the values of $\binom{[n]}{1}$ for $n = 3, 4, 5$ (Chapter 52), the value at the centre of five marks as a Trailmoment of a ray input (Exercise 52.16), and the lean $(2\mathbf{1}; 4, 1)$ of three marks (Exercise 52.5).

Chapter 43

Arithmetic along a ray

A free share closed along a ray brings at every lattice point a rational number of tier one in the point times Trailmoments of the counting input at cuts that move with the point; summed, these are Trailmoments of live inputs, and along the ray the same finitely many live inputs carry every value, the step from one point of the ray to the next entering as rational words. The terms are rational in the lattice point, up to a sign, exactly when the lower Gammas of the word whose slope is at least two in modulus pair off; the word is then a product of sines times a rational function, and its closings are combinations of Trailmoments of lattice inputs with coefficients polynomial in $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. The frame of a law of the values bounds every clearing of their rows from below, by the product of the roots of the symbol, and the merit of the rows, the decay at the native gauge of the direction against the growth of the clearing, decides the gap along the ray. On a ray from the origin the law of [1] vanishes to the order $n-1$ at its low end, which gives one linear relation among the first values; at order two every value is then a row in the one coordinate $\langle 1\rangle_1$, and when the law is reflexive the coefficient of that coordinate is a closing of the ray, the constant term of a power of the reciprocal stride, read with no lattice sum. A hinge closing is a row over finitely many periods of the room as soon as the room's periods along the lattice of the closing satisfy a recurrence of finite order. The critical primes of a native gauge, the primes at which its native value is not integral, are primes at which the letters degenerate, and when every root of the symbol of a law is such a native value they are the primes of its leading coefficient.

43.1 Closing a free share along a ray

Theorem 43.1 (Closing one free share along a ray)**.** *Let A and B be finite sets with integers b_γ and u_γ for $\gamma \in A \sqcup B$,* slopes *$\epsilon_\alpha \in \{1,-1\}$ for $\alpha \in A$ and $\epsilon_\beta \in \mathbb{Z}\setminus\{0\}$ for $\beta \in B$, balanced in the slopes, $\sum_{\alpha\in A}\epsilon_\alpha = \sum_{\beta\in B}\epsilon_\beta$. In this statement write, for $k \in \mathbb{N}$ and a share a,*

$$W_k(a) = \frac{\prod_{\alpha\in A}\Gamma(\epsilon_\alpha a + b_\alpha + k u_\alpha)}{\prod_{\beta\in B}\Gamma(\epsilon_\beta a + b_\beta + k u_\beta)},$$

the stacked Gamma (Definition 1.2) of one free share at the k-th point of the ray along the integral direction $(u_\gamma)_\gamma$; write $A_\pm$ for the $\alpha \in A$ with $\epsilon_\alpha = \pm 1$, $B_\pm$ for the $\beta \in B$ with $\pm\epsilon_\beta > 0$, $w = |A_+| - |B_+|$, and

$$r_W = \prod_{\beta\in B_+} \epsilon_\beta^{\epsilon_\beta} \Big/ \prod_{\beta\in B_-} |\epsilon_\beta|^{|\epsilon_\beta|}.$$

(i) The step factor. $W_k = W_0\, g_k$ *with*

$$g_k(a) = \frac{\prod_{\alpha\in A}(\epsilon_\alpha a + b_\alpha)_{k u_\alpha}}{\prod_{\beta\in B}(\epsilon_\beta a + b_\beta)_{k u_\beta}}, \qquad (x)_{-r} = \frac{1}{(x-r)_r} \quad (r\in\mathbb{N}),$$

a rational function of the share with rational coefficients.

(ii) The terms. *The poles of W_k lie at integers. Let n_1 be an integer beyond which, at $a=-n$, the Gammas of A_+ and of B_+ have nonpositive integer arguments and those of A_- and of B_- positive ones, and for $\gamma\in A\sqcup B$ let $N_\gamma(n)$ be minus the argument of its Gamma at $a=-n$ in the first case and that argument less one in the second; $N_\gamma(n)$ is affine in n with the slope $|\epsilon_\gamma|$. For every integer $n>n_1$ and small ε,*

$$W_k(-n+\varepsilon) = C_n\,\varepsilon^{-w}\exp\Big(\sum_{i\ge1}\Lambda_{n,i}\,\varepsilon^i\Big),$$

where C_n is a nonzero rational number, a signed product of the factorials $N_\gamma(n)!$, to the power 1 for $\gamma\in A_-\cup B_+$ and -1 for $\gamma\in A_+\cup B_-$, and of the slopes of B_+; $C_{n+1}=\rho(n)\,C_n$ for a rational function ρ with rational coefficients, so that $(C_n)_n$ has tier at most one (Proposition 38.11), and $|C_n|^{1/n}\to r_W$, so that r_W is the root rate (Definition 1.1(viii)) of the C_n; $\Lambda_{n,1}$ is a rational combination of the heads $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$; and for $i\ge2$, $\Lambda_{n,i}$ is a rational combination of $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ and the heads $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$. The coefficients of these combinations do not depend on n or k. The residue of W_k at $-n$ is C_n times the coefficient of ε^{w-1} in the exponential, a polynomial with rational coefficients, not depending on n, in the Trailmoments $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $2\le i\le w-1$, and the heads $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$, $1\le i\le w-1$, each of its monomials of total index $w-1$; it is zero when $w\le0$.

(iii) The closing. *Let W_k close to the left on a line $\operatorname{Re}a=c$ that separates the poles of the Gammas of A_+ from those of A_- (Theorem 4.5). Then $\frac{1}{2\pi i}\int_{(c)}W_k(a)\,da$ is the sum of the residues of W_k at the integers $-n<c$. When $r_W<1$ it is a finite rational combination of products of Trailmoments $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $i\ge2$, with 1 and with the absolutely convergent sums*

$$(\mathsf{T}_{0;(1)}[x]\,\psi\,\mathbb{I}_\mathbb{Z})((n_1,\infty)) = \sum_{n>n_1}\psi(n), \qquad \psi(n) = C_n\prod_{l=1}^{q}(\mathsf{T}_{i_l;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_{\gamma_l}(n)]),$$

over the finitely many products with $\gamma_l\in A\sqcup B$, $i_l\ge1$ and $i_1+\cdots+i_q\le w-1$; each such ψ is a live input, its density read from the serif family at the moving cut, and it satisfies a recurrence of order at most 2^q with polynomial coefficients and nonzero extreme coefficients.

(iv) Along the ray. *For two steps k and k' and every n beyond both bounds n_1, C_n at k is C_n at k' times a rational function of n with rational coefficients, and every $N_\gamma(n)$ at k is $N_\gamma(n)$ at k' plus an integer that does not depend on n. So every input ψ of (iii) at the step k is a combination, with coefficients rational functions of n with rational coefficients, of the inputs ψ of (iii) at the step k': when $r_W<1$ every value along the ray is a finite rational combination of products of Trailmoments $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ with 1 and with Trailmoments at index zero of the finitely many live inputs of one step against rational functions of the point with rational coefficients, whose rows Theorem 6.10 reads.*

Closing to the right is the same statement after the share a is replaced by $-a$.

A word of one free share arises from the atom formula (Corollary 18.16) when its other shares are closed or merged, for instance by the four Gammas on a line (Corollary 4.7): integrating $\Gamma(a+z)\Gamma(b+z)\Gamma(c-z)\Gamma(d-z)$ over z leaves four upper Gammas and the lower Gamma $\Gamma(a+b+c+d)$, whose slope in a remaining share is the sum of the slopes of a, b, c and d. A lower Gamma of slope v contributes v^v to the root rate of the terms and one of slope $-v$ contributes v^{-v}, so a lone lower Gamma of slope -2 makes the terms of a left closing fall like 4^{-n}, while lower Gammas that pair off leave terms that do not fall geometrically and are rational in n up to a sign.

Proposition 43.2 (Paired words)**.** *In Theorem 43.1 the following are equivalent.*

(a) The lower Gammas of slope at least two in modulus pair off: for every integer $v \geq 2$ as many $\beta \in B$ have $\epsilon_\beta = v$ as have $\epsilon_\beta = -v$.

(b) For some k, and then for every k, there are $\varsigma \in \{0,1\}$ and a rational function R with $C_n = (-1)^{\varsigma n} R(n)$ for every $n > n_1$.

When they hold, put $p = |A_+| - \#\{\beta \in B : \epsilon_\beta = 1\}$ and let $B^{\geq 2}$ be the set of the β with $\epsilon_\beta \geq 2$, so that $w = p - |B^{\geq 2}|$. Then

$$W_k(a) = \pm\, \pi^w \, \frac{\prod_{\beta \in B^{\geq 2}} \sin(\pi \epsilon_\beta a)}{\sin(\pi a)^p} \, R_k(a),$$

with R_k a rational function with rational coefficients, and for $n > n_1$ the residue of W_k at $-n$ is

$$\pm(-1)^{\varsigma n} \sum_{0 \leq l \leq (w-1)/2} \tau_l \, \pi^{2l} \, \frac{R_k^{(w-1-2l)}(-n)}{(w-1-2l)!},$$

with $\varsigma \equiv p + \sum_{\beta \in B^{\geq 2}} \epsilon_\beta \pmod 2$, rational τ_l that depend only on the slopes, and $\tau_0 = \prod_{\beta \in B^{\geq 2}} \epsilon_\beta$. So a closing of Theorem 43.1(iii) is, up to finitely many terms in $\mathbb{Q}[\pi^2]$, a sum over the lattice of $(-1)^{\varsigma n} r(n)$ with r a rational function of n whose coefficients lie in $\mathbb{Q}[\pi^2]$, and Lemma 38.25 over the field $\mathbb{Q}(\pi^2)$, applied to $m \mapsto r(n_1+1+m)$ when $\varsigma = 0$ and to $m \mapsto r(n_1+1+2m) - r(n_1+2+2m)$ when $\varsigma = 1$, writes it over 1, the Trailmoments of the lattice inputs on the cosets of the poles at their cuts and differences of the index-one values of these inputs, with coefficients in $\mathbb{Q}[\pi^2]$; here $\pi^2 = 6\,(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ (Theorem 5.17(iv)).

So the pairing of the lower Gammas, not their number, keeps the terms rational. The two-mark word $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]$ of Lemma 5.16 has no lower Gamma; the word of two merged pairs on four marks, with one lower Gamma of slope 2 and one of slope -2, pairs off, and the poles of its rational factor lie on the coset $\frac{1}{2} + \mathbb{Z}$; the word of one merged pair on three marks, with one lower Gamma of slope -2, does not pair off, and its terms carry a live input of root rate $\frac{1}{4}$. Exercise 52.7 works the word on four marks.

43.2 The clearing along a ray

Theorem 43.3 (The frame bounds the clearing from below)**.** *Let $p_0, \dots, p_h \in \mathbb{Q}[k]$, $h \geq 1$, with p_0 and p_h without zeros at the integers $k \geq k_0$; let d_0, d_h and c be the degrees of p_0 and p_h and the quotient of their leading coefficients.*

(i) Let rational sequences $x^{(1)}, \dots, x^{(h)}$ on the integers $k \geq k_0$ satisfy $\sum_{j=0}^{h} p_j(k)\, x^{(i)}_{k+j} = 0$ and be linearly independent, with frame C_k (Proposition 38.11). If a positive integer D_k makes

every $D_k\, x^{(i)}_{k+l}$, $1 \le i \le h$, $0 \le l < h$, an integer, then $D_k^h\, |C_k| \ge 1$, and as $k \to \infty$

$$\frac{\log D_k}{k} \;\ge\; \frac{d_h - d_0}{h}\,(\log k - 1) + \frac{1}{h}\log\frac{1}{|c|} + O\Big(\frac{\log k}{k}\Big).$$

When $d_0 = d_h$ is the largest degree of the p_j, $\liminf_k \frac{1}{k}\log D_k \ge \frac{1}{h}\log\frac{1}{|c|}$, and $|c|$ is the modulus of the product of the roots of the symbol (Proposition 38.12).

(ii) Let $x_k = \sum_{l=1}^m A_{k,l} v_l - B_k$, $k \ge k_0$, with rational $A_{k,l}$ and B_k and real v_l such that $1, v_1, \dots, v_m$ are linearly independent over $\mathbb{Q}$, satisfy $\sum_j p_j(k)\, x_{k+j} = 0$. Then each of the $m+1$ sequences $(A_{k,l})_k$ and $(B_k)_k$ satisfies the same recurrence, and if h of them are linearly independent, every D_k clearing the rows of $x_k, \dots, x_{k+h-1}$ (Definition 36.5) satisfies (i).

The bound reads the clearing on h consecutive rows together, through one determinant: the frame grows like the k-th power of the product of the roots of the symbol, and the clearing raised to the power h must undo it. For the values $\langle 1\rangle_k$ along a ray with $\kappa_{1,S} > 0$ for every nonempty proper S, the roots of the symbol of every law of the values with rational coefficients include the native value $f(\alpha^\star)$ of the native gauge of the direction and its conjugates (Proposition 38.9). The frame does not say how the clearing is shared among the coordinates; on the rays of Section 43.4 the denominator of the ratio of a reflexive law falls on the coefficient of the one coordinate, at its full power up to what the closings absorb.

43.3 The merit of a ray

Definition 43.4 (The merit of rows along a ray)**.** Let a ray of a family have real origin and $\kappa_{1,S} > 0$ for every nonempty proper S, with the native gauge $\alpha^\star$ of its direction on $\mathbb{R}^{\mathcal{E}}_{>0}$ (Theorem 38.14), and let its values have rows $\langle 1\rangle_k = \sum_{l=1}^m A_{k,l} v_l - B_k$, $k \ge k_0$, over fixed real numbers $v_1, \dots, v_m$ (Definition 36.5), cleared by positive integers D_k. The *merit* of these rows is

$$-\log f(\alpha^\star) - \limsup_{k\to\infty} \frac{1}{k}\log D_k \;\in [-\infty, \infty).$$

Theorem 43.5 (The merit decides the gap along a ray)**.** *In Definition 43.4 let $k_1 \ge k_0$ be an integer with $\kappa_{0,S} + k_1\kappa_{1,S} > 0$ for every nonempty proper S.*

(i) For every $k \ge k_1$, $0 < D_k\langle 1\rangle_k \le D_k\, f(\alpha^\star)^{k-k_1}\, \langle 1\rangle_{k_1}$.

(ii) If the merit is positive, then $D_k\langle 1\rangle_k \to 0$ and some v_l is irrational; for $m = 1$, $v_1 \notin \mathbb{Q}$ and $A_{k,1} \ne 0$ for all large k.

(iii) If $D_k \le C\gamma^k\, d^w_{\lfloor ak+b\rfloor}$ for all large k, with constants $C, \gamma > 0$, $a, w \ge 0$ and b, and $\lfloor ak+b\rfloor \ge 0$, the merit is at least $-\log f(\alpha^\star) - \log\gamma - a\, w\, c_{\mathrm{lcm}}$ (Lemma 37.1), and $\gamma\, e^{awc_{\mathrm{lcm}}} f(\alpha^\star) < 1$ gives (ii).

(iv) If each D_k also clears the rows of $\langle 1\rangle_{k+1}, \dots, \langle 1\rangle_{k+h-1}$, $1, v_1, \dots, v_m$ are linearly independent over $\mathbb{Q}$, the values satisfy a law $\sum_{j=0}^h p_j(k)\langle 1\rangle_{k+j} = 0$, $k \ge k_0$, as in Theorem 43.3 with p_0 and p_h of the largest degree, and h of the sequences $(A_{k,l})_k$, $(B_k)_k$ are linearly independent, then the merit is at most $-\log f(\alpha^\star) + \frac{1}{h}\log|c|$, with $|c|$ the modulus of the product of the roots of the symbol.

On the centre ray of a family, $f(\alpha^\star) = \prod_j K_j(\mathbf{1})^{-2\sigma_j}$ (Theorem 38.15), and a clearing $D_k = \gamma^{k-1} d^w_{k-1}$ passes the gap when $\gamma\, e^{wc_{\mathrm{lcm}}} < \prod_j K_j(\mathbf{1})^{2\sigma_j}$, the minimum of the reciprocal stride on the

orthant. Only the upper bound of the values is used: positivity and the largest value of the stride on the orthant, both read before any closing. By (iv), with the native value among the roots of the symbol, a law of order h leaves positive merit only to a ray whose symbol has roots of modulus larger than $f(\alpha^\star)$.

43.4 Laws at the low end and their closings

Proposition 43.6 (The low end of the law)**.** *Let a ray be as in Proposition 39.30, on n marks, with origin 0, every letter of positive exponent in the direction, $\kappa_{1,S} > 0$ for every nonempty proper S, and the volume V_1 of the direction; let the law of $[1]$ (Definition 39.1) be $\sum_{i=0}^{h} p_i(k)\,\mathrm{Sh}^i[1] = 0$ with $p_0, \dots, p_h \in \mathbb{Q}[k]$, and write $x_k = \langle 1\rangle_k$.*

(i) The values satisfy the law at every integer $k \geq 1$. Further k^{n-1} divides p_0, and with ϖ the coefficient of k^{n-1} in p_0 the values satisfy the low-end relation

$$\varpi\, V_1 + \sum_{i=1}^{h} p_i(0)\, x_i = 0;$$

V_1 is rational. For a solution y of the law on the integers $k \geq 1$, $\sum_i p_i(k) y_{k+i} = 0$ for every $k \geq 1$, call $-\sum_{i=1}^{h} p_i(0)\, y_i$ its low-end coefficient*; that of the values is ϖV_1, the value at $k = 0$ of the continuation of $p_0(k)\langle 1\rangle_k$.*

(ii) Let $h = 2$ and $p_2(k) \neq 0$ at every integer $k \geq 0$, and let c and r be the solutions of the law on $k \geq 1$ with $(c_1, c_2) = \big(1, -p_1(0)/p_2(0)\big)$ and $(r_1, r_2) = \big(0, -\varpi V_1/p_2(0)\big)$. They are rational, the low-end coefficient of c vanishes, that of r is ϖV_1, and

$$x_k = x_1\, c_k + r_k \qquad (k \geq 1):$$

every value has a row in the one coordinate x_1, its coefficient the solution whose low-end coefficient vanishes and whose first term is 1.

Definition 43.7 (Reflexive laws)**.** A recurrence $\sum_{i=0}^{h} p_i(k)\, x_{k+i} = 0$ with $p_i \in \mathbb{Q}[k]$ is *reflexive* with the *ratio* $\lambda \in \mathbb{Q}^\times$ when there is a constant $\gamma \neq 0$ with

$$p_{h-i}(-k-h+1) = \gamma\, \lambda^i\, p_i(k) \qquad (0 \leq i \leq h)$$

as polynomials in k. When p_0 and p_h have the largest degree d of the p_i, the roots of the symbol of a reflexive recurrence are exchanged by $\varrho \mapsto \lambda/\varrho$, and at $h = 2$ their product is $\pm\lambda$.

Theorem 43.8 (The closings read the coordinate of a reflexive law)**.** *Let a ray be as in Proposition 43.6, with the law of $[1]$ reflexive with the ratio λ, and let $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j})$ be the closings of 1 (Definition 39.26).*

(i) $y_k = \lambda^{k-1}\, \mathsf{C}_{k-1}(1)$, $k \geq 1$, is a solution of the law on $k \geq 1$, and its low-end coefficient vanishes.

(ii) If $h = 2$ and $p_2(k) \neq 0$ at every integer $k \geq 0$, then y is the solution c of Proposition 43.6(ii):

$$\langle 1\rangle_k = \lambda^{k-1}\, \mathsf{C}_{k-1}(1)\, \langle 1\rangle_1 + r_k \qquad (k \geq 1),$$

the coefficient of the coordinate read from the closings of the ray, with no lattice sum. If moreover $\langle 1\rangle_1 = A_1 v - B_1$ is a row in one coordinate v, the coefficient of v in $\langle 1\rangle_k$ is $A_1 \lambda^{k-1} \mathsf{C}_{k-1}(1)$.

(iii) If moreover the weights are integers, $\mathsf{C}_j(1) \in \mathbb{Z}$, and with $\lambda = a/b$ in lowest terms, $b > 0$, the coefficient of $\langle 1\rangle_1$ in $\langle 1\rangle_k$ has the denominator $b^{k-1}/\gcd\big(b^{k-1}, \mathsf{C}_{k-1}(1)\big)$.

Corollary 43.9 (The merit of a reflexive ray of order two)**.** *In Theorem 43.8(iii), every clearing D_k of the rows $\langle 1\rangle_k = \lambda^{k-1}\mathsf{C}_{k-1}(1)\,\langle 1\rangle_1 + r_k$ in the coordinate $\langle 1\rangle_1$ has merit (Definition 43.4) at most*

$$-\log f(\alpha^\star) - \log b + \liminf_{k\to\infty} \frac{1}{k} \log\gcd\big(b^{k-1}, \mathsf{C}_{k-1}(1)\big).$$

The closings are the constant terms of the powers of the reciprocal stride, finite sums of products of weights; on a reflexive ray of order two they are the coefficient of the coordinate at every step, up to the power λ^{k-1}, and the clearing pays the full power b^{k-1} of the denominator of the ratio except for what the closings absorb. The ray is then judged by the divisibility of its closings. The centre rays of the lists $\binom{[3]}{1}$ and $\binom{[4]}{1}$ have reflexive laws of order two with the ratios $\frac{1}{9}$ and $\frac{1}{64}$; the exercises derive both laws (Exercises 52.2 and 52.8), and on four marks the rows, the coefficient of the coordinate and the clearing (Exercises 52.9, 52.10 and 52.11).

43.5 The hinge closing as a row over the room

Proposition 43.10 (The hinge closing as a row over the room)**.** *For one table in Theorem 19.13 let $\sup_{\Delta_B} \varrho_B < \inf_{\Delta_R} \varrho_R$, so that the even share closes on the lattices of the level d (Corollary 19.21(i)). Group the residues by the index n of their points and write the closed value as $P_F(\mathbf{m}; s, t) = \sum_{n\ge 0}\langle \Phi_n, \mathcal{X}_n\rangle$, where $\mathcal{X}_n$ lists the data of the room family that the residues at the n-th points contain, its periods at the level shares of those points and, where the two lattices meet, their derivatives in the even share (Corollary 19.21(ii)), and Φ_n lists the coefficients that Corollary 19.21(ii) puts in front of them.*

(i) Let there be numbers $\chi_1, \dots, \chi_h$ and lists $\mathbf{c}_1(n), \dots, \mathbf{c}_h(n)$ with $\mathcal{X}_n = \sum_{i=1}^h \chi_i\, \mathbf{c}_i(n)$ for every n and $\sum_n |\langle \Phi_n, \mathbf{c}_i(n)\rangle| < \infty$ for every i. Then

$$P_F(\mathbf{m}; s, t) = \sum_{i=1}^{h} \chi_i\, \Xi_i, \qquad \Xi_i = \sum_{n\ge 0} \langle \Phi_n, \mathbf{c}_i(n)\rangle :$$

a row over h data of the room, with coefficients that are sums along the lattice of the bunch's data against the sequences $\mathbf{c}_i$.

(ii) When $2\delta^d \notin \mathbb{Z}$ every pole is simple and $\mathcal{X}_n$ lists the two room periods $x_n^\pm$ at the points $\sigma^d = \pm\delta^d - n$. If each sequence $x^\pm$ satisfies a recurrence $\sum_{i=0}^{h_\pm} p_i^\pm(n)\, x_{n+i}^\pm = 0$, $n \ge 0$, with $p_{h_\pm}^\pm(n) \ne 0$ for $n \ge 0$, let $c_i^\pm$, $0 \le i < h_\pm$, be its solutions with the initial values $c_i^\pm(l) = \delta_{il}$, $0 \le l < h_\pm$, rational when the $p_i^\pm$ have rational coefficients. Then $x_n^\pm = \sum_i x_i^\pm\, c_i^\pm(n)$ for every n, and whenever the sums $\sum_n \Phi_n^\pm\, c_i^\pm(n)$ converge absolutely, (i) holds with the $h_+ + h_-$ initial periods $x_0^\pm, \dots, x_{h_\pm - 1}^\pm$.

(iii) When $2\delta^d \in \mathbb{Z}$ the poles lie on the one lattice $\sigma^d = |\delta^d| - n$, $n \in \mathbb{N}$. Let the room period at $\sigma^d = \sigma - n$, a function $x_n(\sigma)$ of σ near $\sigma_0 = |\delta^d|$, satisfy a recurrence $\sum_{i=0}^h p_i(n, \sigma)\, x_{n+i}(\sigma) = 0$ whose coefficients are polynomials in n and σ, with $p_h(n, \sigma_0) \ne 0$ for $n \ge 0$. Then the values $x_n(\sigma_0)$ and the derivatives $x_n'(\sigma_0)$ satisfy the block-triangular recurrence of Proposition 37.17, they are combinations of the $2h$ initial data $x_i(\sigma_0)$, $x_i'(\sigma_0)$, $i < h$, with coefficients that do not depend on the room's periods, and whenever the resulting sums converge absolutely (i) holds with these $2h$ data.

The same holds with the bunch and the room exchanged when $\inf_{\Delta_B} \varrho_B > \sup_{\Delta_R} \varrho_R$*, the share closing on the level* r*.*

Read down a tree of bunches, a hinge closing writes a period as a row over finitely many periods of its room, the coefficients being sums along the lattice of the hinge of the bunch's data against the fundamental solutions of the room's recurrence. The room's periods along the lattice are the moments of one ratio against one form (Corollary 19.21(ii)); when they have tier one, each lattice contributes one room period times one sum along the lattice. Proposition 45.12 gives a condition on the roots of the symbols under which the sums converge.

43.6 Critical primes

Definition 43.11 (Critical primes)**.** Let a ray of a family have an integral direction with the stride $f = \alpha^{\mathbf{m}_1} \prod_L L^{-w_L}$, the product over the distinct letters L of positive exponent w_L, and let its weights lie in a number field $\Bbbk$ that contains the coordinates of a native gauge α_c (Definition 38.4) on the chart $\alpha_{e_0} = 1$. A prime $\mathfrak{p}$ of $\Bbbk$ at which the weights are integral and the coordinates of α_c are units is a *critical prime* of α_c when the native value is not integral there, $v_{\mathfrak{p}}\big(f(\alpha_c)\big) < 0$.

Theorem 43.12 (The letters degenerate at a critical prime)**.** *Let* $\mathfrak{p}$ *be a critical prime of a native gauge* α_c*, with the residue field* κ *and the reduction* $\bar{\alpha}_c \in (\kappa^\times)^{\mathcal{E}}$*, and let* S *be the set of the letters of positive exponent whose valuation* $v_{\mathfrak{p}}(L(\alpha_c))$ *is the largest, a positive number.*

(i) Every letter of S *vanishes at* $\bar{\alpha}_c$ *after reduction, and there are units* u_L *of the local ring at* $\mathfrak{p}$ *with*

$$\sum_{L \in S} \overline{w_L u_L}\, \nabla \bar{L}(\bar{\alpha}_c) = 0,$$

∇ *the gradient in the coordinates* α_e*,* $e \in \mathcal{E}$*, and the bar the reduction.*

(ii) If the characteristic of κ *divides none of the exponents* w_L*,* $L \in S$*, the zero sets of the reduced letters of* S *are not smooth hypersurfaces meeting transversally at* $\bar{\alpha}_c$*: when* S *has one letter, that letter is singular at* $\bar{\alpha}_c$*, and when it has several, their gradients there are linearly dependent.*

Proposition 43.13 (Critical primes and the symbol)**.** *Let* $\sigma(\lambda) = \sum_{j=0}^{h} c_j \lambda^j \in \mathbb{Z}[\lambda]$*,* $c_h \neq 0$*,* $h \geq 1$*, be a symbol (Corollary 38.7) with coprime coefficients, and* K *a number field containing its roots.*

(i) For every root λ *of* σ*,* $c_h\lambda$ *is an algebraic integer.*

(ii) A prime p *divides* c_h *exactly when some root of* σ *has negative valuation at some prime of* K *above* p*; and then at every prime of* K *above* p *some root has negative valuation.*

(iii) Let every root of σ *be the native value of a native gauge whose coordinates lie in* K*, at which the weights are integral and the coordinates are units at the primes above* p*. Then* p *divides* c_h *exactly when some prime of* K *above* p *is a critical prime of one of these native gauges. When* σ *is the symbol of a recurrence whose* p_0 *and* p_h *have the largest degree,* c_0/c_h *is the quotient of their leading coefficients, whose* k*-th power enters the frame (Proposition 38.11).*

At a critical prime the native gauge reduces to a point at which the letters of largest valuation meet badly, so the critical primes of a ray are read from the stride and its critical points before any value is computed, and under (iii) they are the primes of the leading coefficient of the symbol, which the frame carries. The value span of a ray can see fewer primes: on the centre ray of $\binom{[4]}{1}$ the

critical prime 2 sits in the frame and in the denominators of the coefficients, and the coordinate is the Trailmoment $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of the counting input. On the centre ray of $\binom{[3]}{1}$ the native value $\frac{1}{9}$ reduces modulo 3 to a tangency of the letters, and on that of $\binom{[4]}{1}$ the native value $\frac{1}{16}$ reduces modulo 2 to the collapse of three intersection points of the letters onto the native gauge.

Part VIII

Relations and the content

Chapter 44

Relations and separation

A value is the evaluation of a word, and two words of equal value either are related by the proved relations of the theory or are not, a problem that relation blocks make finite. The admitted words form an algebra with an evaluation, the proved relations generate an ideal inside its kernel, and completeness asks for equality. Relation blocks reduce a finite space of words to representatives, each word with a certificate; when the representatives have values independent over the rationals, the block is complete. Words of polynomials on a box are complete outright, a finite census of insertions separates numerators, and the critical-line words of order k of polynomials form a block whose values are the rational combinations of 1 and $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; its two representatives have values independent over the rationals exactly when $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \notin \mathbb{Q}$, and the block is then complete. Independence of values is proved by separators: families of rational matrices whose rows are small on the vector of values.

44.1 Words and relations

Definition 44.1 (Words, evaluation, relations, completeness)**.** An *admitted word* is a word (Definition 1.1(vi)) given by finite rational data (tables and weights, coefficients, masses, exponents, inputs with a finite presentation, cuts) together with the rule that assigns its value: a convergent integral, a finite part or a continuation, as proved for it. The admitted words span a rational algebra $\mathfrak{W}$, whose product tensors the data, with independent integration variables and separate regulators, and *evaluation* is the map $\mathrm{ev} : \mathfrak{W} \to \mathbb{C}$. The *relation ideal* $\mathfrak{I} \subseteq \mathfrak{W}$ is generated by the proved relations, each on its stated domain: linearity; changes of variables and of order; scale and shape; split, port and gluing; the Gramma relations; the letter relations and the mark relations (Theorems 8.9 and 35.10); the subtraction of a common boundary term from the two sides of a relation; the cut relation of the Trailmoment (Proposition 5.3); the closings. A set of words is *complete* when each of its rational combinations of value zero lies in $\mathfrak{I}$.

Proposition 44.2 (Products keep separate regulators)**.** *If $A(\varepsilon)$ and $A'(\varepsilon')$ have Laurent expansions in disjoint sets of regulators, then $\mathrm{CT}_{\varepsilon}\,\mathrm{CT}_{\varepsilon'}\,\big(A(\varepsilon)A'(\varepsilon')\big) = \big(\mathrm{CT}_{\varepsilon}\,A\big)\big(\mathrm{CT}_{\varepsilon'}\,A'\big)$, and evaluation is multiplicative on $\mathfrak{W}$. Identifying regulators is a further operation: $\mathrm{CT}(\varepsilon^{-1}) = \mathrm{CT}(\varepsilon) = 0$, while $\mathrm{CT}(\varepsilon^{-1}\varepsilon) = 1$.*

Proposition 44.3 (Soundness)**.** $\mathfrak{I} \subseteq \ker(\mathrm{ev})$.

Proposition 44.4 (A finite matrix of mark relations)**.** *On $[0,1]^d$ let $a_l, b_l > 0$, let $D_1, \dots, D_h$ be rational polynomials positive on the closed box with exponents $p_1, \dots, p_h \in \mathbb{Q}$, and in this statement*

write $D = \prod_k D_k = \sum_\gamma D_\gamma x^\gamma$ *and* $x_i \sum_k p_k\, \partial_i D_k \prod_{k' \neq k} D_{k'} = \sum_\gamma D_\gamma^{(i)} x^\gamma$. *For every polynomial* A *and every coordinate* i,

$$\mathrm{mr}_i(A) = D\big[x_i(1-x_i)\,\partial_i A + \big(a_i - (a_i+b_i)x_i\big)A\big] - x_i(1-x_i)\,A \sum_k p_k\, \partial_i D_k \prod_{k' \neq k} D_{k'}$$

has $\int_{[0,1]^d} \mathrm{mr}_i(A) \prod_l x_l^{a_l - 1}(1-x_l)^{b_l - 1} \prod_k D_k^{-p_k - 1}\, dx = 0$ *(Theorem 35.10(a) with the letters* D_k *at the exponents* $p_k + 1$ *and the polynomial* AD*), and*

$$\mathrm{mr}_i(x^\beta) = \sum_\gamma \Big[\big((\beta_i + a_i)D_\gamma - D_\gamma^{(i)}\big)x^{\beta+\gamma} - \big((\beta_i + a_i + b_i)D_\gamma - D_\gamma^{(i)}\big)x^{\beta+\gamma+\mathbf{e}_i}\Big].$$

For a degree bound these vectors form a finite rational relation matrix; a rational solution of its column equation for a numerator writes the numerator as a sum of polynomials $\mathrm{mr}_i(A_i)$*, so that its word vanishes.*

Theorem 44.5 (Words of polynomials on a box are complete)**.** *For* $P \in \mathbb{Q}[x]$ *and rational* $a_i, b_i > 0$ *the words* $\int_{[0,1]^d} P(x) \prod_i x_i^{a_i - 1}(1-x_i)^{b_i - 1}\, dx$ *have the mark relations* $x^{\beta + \mathbf{e}_i} \equiv \frac{\beta_i + a_i}{\beta_i + a_i + b_i}\, x^\beta$*, and every* P *reduces to*

$$\sum_\beta \big([x^\beta]P\big) \prod_i \frac{(a_i)_{\beta_i}}{(a_i + b_i)_{\beta_i}},$$

the expectation of P *under the Dirichlet laws* $\mathrm{Dir}_{(a_i, b_i)}$ *of the pairs* $(x_i, 1-x_i)$*, times the base word, of value* $\prod_i \Gamma_2 \left[\begin{matrix} a_i, & b_i \\ & a_i + b_i \end{matrix}\right] > 0$*. A word vanishes exactly when its reduction is zero. The same holds for every finite tensor product of such words.*

Theorem 44.6 (A finite insertion census separates numerators)**.** *Let* ϕ *be a measure on* $[0,1]^d$ *with an integrable density positive on the interior, and* D *a rational polynomial positive on the closed box. For* $|\beta|, |\gamma| \leq h$ *the matrix* $\mathbf{G}_{\beta\gamma} = \int x^{\beta+\gamma} D^{-1}\, d\phi$ *is positive definite. For* $A = \sum_\beta a_\beta x^\beta$ *of degree at most* h *the insertion values* $v_\beta = \int x^\beta A D^{-1}\, d\phi$ *satisfy* $\mathbf{v} = \mathbf{G}\mathbf{a}$ *and* $\mathrm{adj}(\mathbf{G})\,\mathbf{v} = \det(\mathbf{G})\,\mathbf{a}$*: these* $\binom{d+h}{d}$ *values determine* A*. Equivalently, the Laplace transform* $t \mapsto \int A D^{-1} e^{-\langle t, x\rangle}\, d\phi$ *has all its Taylor coefficients of degree at most* h *zero at* $t = 0$ *only for* $A = 0$*.*

Evaluation is multiplicative because a product of words keeps the regulators of its factors apart; identifying two regulators is a new operation, not a relation. Every relation the theory proves lies in the kernel of evaluation, and completeness asks whether the kernel holds anything more. On a box against positive letters the mark relations form a finite rational matrix, so that vanishing is certified by linear algebra; for polynomials on a box without a letter the relations reduce every insertion to one rational multiple of the base word, and with a letter the values of finitely many monomial insertions, through the positive definite matrix $\mathbf{G}$ of Theorem 44.6, recover the insertion itself.

44.2 Relation blocks

Definition 44.7 (Relation block)**.** A *relation block* is a finite-dimensional rational space V of admitted words with a matrix C whose columns are proved relations in V, each with its derivation. Rational row reduction gives representatives $\mathfrak{w}_0, \ldots, \mathfrak{w}_s$ of $V/\operatorname{im} C$ and for every $\mathfrak{w} \in V$ a certificate $\mathfrak{w} = C\mathbf{v} + \sum_j c_j \mathfrak{w}_j$.

Proposition 44.8 (Separation completes a block)**.** *If* $\mathrm{ev}(\mathfrak{w}_0),\dots,\mathrm{ev}(\mathfrak{w}_s)$ *are linearly independent over* $\mathbb{Q}$*, then* $\ker(\mathrm{ev}\,|_V) = \operatorname{im} C$*.*

Theorem 44.9 (Completeness from separated blocks)**.** *If every finite set of words of a subalgebra* $\mathfrak{W}' \subseteq \mathfrak{W}$ *lies in a relation block whose representatives have values linearly independent over* $\mathbb{Q}$*, then* $\mathfrak{W}'$ *is complete, and each of its words of value zero has a finite derivation, its certificate.*

Proposition 44.10 (Tensoring relation blocks)**.** *For blocks* (V_1, C_1) *and* (V_2, C_2) *the induced relations on* $V_1 \otimes V_2$ *have the matrix* $\left[\, C_1 \otimes I \;\middle|\; I \otimes C_2 \,\right]$*, with quotient* $(V_1/\operatorname{im} C_1) \otimes (V_2/\operatorname{im} C_2)$*, and the values of the representatives are the products of the values (Proposition 44.2).*

Proposition 44.11 (The critical polynomial block)**.** *For an integer* $k \geq 1$ *and* $P \in \mathbb{Q}[x]$ *put* $Q_P(u) = \frac{P(2u)-P(1)}{1-2u}$ *and* $h(P) = \frac{2}{(2k)!}\,\mathbb{E}_{Y\sim\mathbb{P}_{\mathrm{crit}}}\big[Q_P^{(2k-1)}(\frac{1}{2}+iY)\big]$*. Then* $h(P) \in \mathbb{Q}$ *and*

$$\mathrm{crit}_k[P(2u)] = h(P) + (2^{2k+1}-1)\,P(1)\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

So the values of the rational constants and of the critical-line words of order k *of polynomials factor through the surjective coordinate map* $(c,P) \mapsto \big(c+h(P), (2^{2k+1}-1)P(1)\big) \in \mathbb{Q}^2$ *followed by* $(a,b) \mapsto a+b\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$*, which is injective exactly when* $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \notin \mathbb{Q}$*; moving the cut of the Trailmoment subtracts a rational head.*

A relation block is the finite form of completeness: inside a finite space of words, rational row reduction by the proved relations leaves representatives and certificates, and independence of the values of the representatives over the rationals makes the block complete (Proposition 44.8). Blocks tensor, their quotients tensoring and the values of their representatives multiplying, so a product of blocks is complete as soon as the products of those values are independent over the rationals. The critical-line words of order k of polynomials have a block whose quotient has dimension at most two: every such word is a rational number plus a rational multiple of $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, read off two coordinates of the polynomial, and the values of the two representatives are independent over the rationals exactly when that Trailmoment is irrational, the block being then complete.

44.3 Separators

Theorem 44.12 (A primitive exterior separator)**.** *Let* $v = (v_1,\dots,v_s,1) \in \mathbb{R}^{s+1}$*. For a rational* $s \times (s+1)$ *matrix* A *of rank* s *with rows* r_i*, let* $w \in \mathbb{Q}^{s+1}$ *be its cofactor vector,* $\det[x;A] = \langle x, w\rangle$ *for all* x*, and* $z = Mw$ *its primitive integral multiple,* $M > 0$*. If* D_i *clears* r_i*, then* $M = \prod_i D_i/c$ *with* c *the greatest common divisor of the maximal minors of* $\mathrm{diag}(D_i)A$*. Put* $y_i = \langle r_i, v\rangle$*, let* $\hat{A}_i$ *be the other rows restricted to the first* s *columns, and*

$$\beta(A) = M \sum_i |y_i| \sum_{J\subseteq\{1,\dots,s\},\ |J|=s-1} \big|\det \hat{A}_i[:,J]\big|.$$

If for all large n *finitely many such matrices* $A_1^{(n)},\dots,A_{k_n}^{(n)}$ *have primitive cofactor vectors spanning* $\mathbb{Q}^{s+1}$ *and* $\max_l \beta(A_l^{(n)}) \to 0$*, then* $1, v_1, \dots, v_s$ *are linearly independent over* $\mathbb{Q}$*.*

Corollary 44.13 (Separation with bounded integer entries)**.** *If integer matrices* $A_l^{(n)}$ *with spanning cofactor vectors have entries at most* $H_n \geq 1$ *and* $|\langle r_i, v\rangle| \leq \epsilon_n$ *for every row, with* $H_n^{s-1}\epsilon_n \to 0$*, then* $1, v_1, \dots, v_s$ *are linearly independent over* $\mathbb{Q}$*. In rates: if rational matrices with spanning cofactor vectors have entries at most* $e^{an+o(n)}$*, a common clearing integer at most* $e^{bn+o(n)}$ *and row values at most* $e^{-cn+o(n)}$*, with* $c > sb + (s-1)a$*, the same conclusion holds.*

Corollary 44.14 (Consecutive rows)**.** *Let rows $r_k \in \mathbb{Q}^{s+1}$ be such that the square matrix of the rows $r_k, \dots, r_{k+s}$ is invertible for large k. Then for each such k the $s+1$ matrices omitting one of these rows have spanning primitive cofactor vectors, and Theorem 44.12 applies with these matrices. If the coordinate sequences of the rows satisfy one recurrence of order $s+1$ whose extreme coefficients have no integer zeros from k_0 on, invertibility of that matrix at k_0 gives it for every $k \geq k_0$.*

Theorem 44.15 (Function level: pole pinning)**.** *Let f and g be meromorphic on a connected open subset of $\mathbb{C}$. If f has infinitely many poles at which g is holomorphic, and the values of g at those poles include infinitely many distinct numbers, then no nonzero polynomial $Q \in \mathbb{C}[X, Y]$ has $Q(f, g) \equiv 0$.*

A separator is a family of rational matrices whose rows are nearly orthogonal to the vector of values: a primitive integral multiple of the cofactor vector of each matrix pairs with any integer relation to an integer, and if the rows are small enough against the clearing and the bounds on the entries, no rational relation among the values survives. Consecutive rows of a recurrence give such matrices as soon as one square matrix of $s+1$ consecutive rows is invertible. At the level of functions, a function with infinitely many poles at which a second function takes infinitely many values satisfies no polynomial relation with it.

Chapter 45

The content and its coaction

The content is the algebra of the trails of lattice inputs at rational frequencies, on rational cosets and at rational cuts, read modulo the proved relations. At the level M it is spanned by the roots of unity of order M times the trails of the counting lattice at the cut 0 whose frequencies have the denominator M; it is closed under products, it holds every trail with rational data at the level that its denominators fix, and it holds the differences of the index-one values of lattice inputs on rational cosets, so the sums along the lattices of paired words lie in it. Its values are the chains of level M: the iterated integrals from 0 to 1 whose letters are 0 and the roots of unity of order M, each a trail whose frequencies are the ratios of its consecutive nonzero letters; read between 0 and a root of unity in either order a chain is again a value of the content. The cutting rule keeps a subsequence of the letters of a chain and reads every run of the letters it drops between the two letters that bound it; on the symbols of chains it is a coassociative coaction, graded by the weight and compatible with the rotations and the reversal of chains. A live input has a letter too, its generating function, which satisfies a linear differential equation singular at the reciprocals of the roots of the symbol of its recurrence; the letter of a product of two live inputs continues across every point that is not a product of singular points of the letters of the factors, and the row of a hinge closing is the room's periods against the letters of the live inputs built on the room's fundamental solutions.

45.1 The content

Definition 45.1 (The content)**.** Let $M \geq 1$ be an integer and μ_M the group of the roots of unity of order dividing M. A *trail of level M* is the trail (Definition 5.23) at the cut 0 of a lattice input on $\mathbb{Z}^d$ at frequencies in $\frac{1}{M}\mathbb{Z}$, with positive integer indices and no constant columns,

$$T = \big(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\; e^{2\pi i\langle \mathbf{a},y\rangle}\,\mathbb{I}_{\mathbb{Z}^d}\big)\big(\{0<y_1<\cdots<y_d\}\big), \qquad d\geq 1,\ \boldsymbol{\nu}\in\{1,2,\dots\}^d,\ \mathbf{a}\in\big(\tfrac{1}{M}\mathbb{Z}\big)^d,$$

with $e^{2\pi i a_d}\neq 1$ when $\nu_d=1$, read as the limit of its heads over $\{0<y_1<\cdots<y_d\leq N\}$ as $N\to\infty$, which exists and is the Trailmoment itself when $\nu_d\geq 2$ (Theorem 5.27); $|\boldsymbol{\nu}|$ is its *weight* and d its *depth*. The *content of level M*, $\mathcal{C}^{(M)}$, is the $\mathbb{Q}$-span in $\mathfrak{W}/\mathfrak{I}$ (Definition 44.1) of the classes of the words ζ and ζT, for $\zeta\in\mu_M$, read as $e^{2\pi i r/M}$, and T a trail of level M; $\mathcal{C}^{(M)}_w$ is the span of those with T of weight at most w, and the *content* is $\mathcal{C}=\bigcup_M \mathcal{C}^{(M)}$.

The content is not the content $\mathcal{C}_M$ of a childcule M in Definition 20.58, a $\mathcal{Z}$-module of values of periods; the level is written as an upper index to keep the two apart. The algebra $\mathcal{Z}$ of Defini-

tion 20.58 maps onto the subalgebra of $\mathcal{C}^{(1)}$ generated by its trails of depth one, the Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $k \geq 2$ (Proposition 45.2(i)).

Proposition 45.2 (The content is an algebra)**.** *(i) $\mathcal{C}^{(M)}$ contains the classes of 1 and of every product of Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $k \geq 2$, so it contains the image of $\mathcal{Z}$ (Definition 20.58); and $\mathcal{C}^{(M)} \subseteq \mathcal{C}^{(M')}$ when $M \mid M'$.*

(ii) Let T and T' be trails of level M with the data $(\nu_i, a_i)_{i\leq d}$ and $(\nu'_j, a'_j)_{j\leq d'}$. For a surjection ϕ from the disjoint union $\{1,\dots,d\} \sqcup \{1',\dots,d'\}$ onto $\{1,\dots,e\}$, strictly increasing on each of the two parts, let T_ϕ be the trail of level M of depth e whose r-th point has the index $\sum_{\phi(x)=r} \nu_x$ and the frequency $\sum_{\phi(x)=r} a_x$. Then

$$TT' = \sum_{\phi} T_\phi,$$

the sum over these surjections, of depths between $\max(d,d')$ and $d+d'$, every T_ϕ of the weight $|\boldsymbol{\nu}| + |\boldsymbol{\nu'}|$.

(iii) So $\mathcal{C}^{(M)}$ is a $\mathbb{Q}$-algebra with $\mathcal{C}^{(M)}_w\,\mathcal{C}^{(M)}_{w'} \subseteq \mathcal{C}^{(M)}_{w+w'}$, and $\mathcal{C}$ is a $\mathbb{Q}$-algebra.

Proposition 45.3 (Cosets, cuts and index one)**.** *Let $M, M' \geq 1$ be integers.*

(i) Let a trail have a lattice input on $\prod_{i=1}^{d}(b_i + \mathbb{Z})$ with $b_i \in \frac{1}{M}\mathbb{Z}$, at frequencies $a_i \in \frac{1}{M'}\mathbb{Z}$, with positive integer indices, no constant columns and a cut $N \in \frac{1}{M}\mathbb{Z}$, $N \geq 0$, and with $e^{2\pi i a_d} \neq 1$ when $\nu_d = 1$, read as the limit of its heads. Then its class lies in $\mathcal{C}^{(MM')}$, and the class of every head of it lies in the span of the classes of $\mu_{MM'}$; its value is a combination, with coefficients in $\mathbb{Q}(\mu_{MM'})$, of 1 and the trails of level MM' of depths at most d.

(ii) For $b, b' \in \frac{1}{M}\mathbb{Z}$ and cuts $N, N' \in \frac{1}{M}\mathbb{Z}$, $N, N' \geq 0$, the difference $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b'+\mathbb{Z}})((N',\infty))$ of index-one values (Definition 5.2) has its class in $\mathcal{C}^{(M)}$.

So the sum along a lattice of a paired word at integer data (Proposition 43.2), a combination of Trailmoments of lattice inputs on rational cosets at their cuts and differences of their index-one values with coefficients in $\mathbb{Q}[(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))]$, lies in the content. The values met on the centre rays of the lists of three and four marks are trails of depth one: at the frequency $\frac{1}{3}$ and the index 2, and on the coset $\frac{1}{2}$ at the index 3, which the coset relation returns to the counting input; Exercise 54.1 reads them.

45.2 Chains

Definition 45.4 (Chains)**.** For distinct points $a_0, a_{w+1} \in \mathbb{C}$ and points $a_1,\dots,a_w$, $w \geq 0$, none on the open segment from a_0 to a_{w+1}, the *chain* is the iterated integral along that segment, $\gamma(t) = a_0 + t(a_{w+1} - a_0)$,

$$\mathrm{I}(a_0; a_1, \dots, a_w; a_{w+1}) = \int_{0<t_1<\cdots<t_w<1} \prod_{i=1}^{w} \frac{\gamma'(t_i)\,dt_i}{\gamma(t_i) - a_i},$$

with $\mathrm{I}(a_0;\varnothing;a_{w+1}) = 1$, wherever it converges. The $a_1,\dots,a_w$ are its *letters*, a_0 and a_{w+1} its *ends* and w its *weight*. It converges when its first letter differs from its first end and its last letter from its last end (Proposition 45.6(i), below). A chain is *of level M* when its ends and letters lie in $\{0\} \cup \mu_M$.

Proposition 45.5 (The content as chains). *(i) A trail T of level M with the data $(\nu_i, a_i)_{i\le d}$ is*

$$T = (-1)^d \,\mathrm{I}\big(0;\ c_1^{-1}, 0^{\nu_1-1},\ \dots,\ c_d^{-1}, 0^{\nu_d-1};\ 1\big), \qquad c_i = e^{2\pi i(a_i+\cdots+a_d)} \in \mu_M,$$

a chain of level M from 0 to 1 whose first letter is not 0 and whose last letter is not 1 (Theorem 5.27).

(ii) Conversely, let $\mathrm{I}(0; a_1, \dots, a_w; 1)$ be a chain of level M with $a_1 \neq 0$ and $a_w \neq 1$, let $j_1 < \cdots < j_d$ be the positions of its nonzero letters, $j_{d+1} = w+1$, and put $c_i = a_{j_i}^{-1}$, $\nu_i = j_{i+1} - j_i$, $c_{d+1} = 1$. With frequencies $a'_i \in \frac{1}{M}\mathbb{Z}$ such that $e^{2\pi i a'_i} = c_i/c_{i+1}$, the chain is $(-1)^d$ times the trail of level M with the data $(\nu_i, a'_i)_{i\le d}$.

(iii) So the values of $\mathcal{C}^{(M)}$ form the $\mathbb{Q}(\mu_M)$-span of 1 and of the chains of level M from 0 to 1 with first letter not 0 and last letter not 1, and those of $\mathcal{C}_w^{(M)}$ the span of 1 and of such chains of weight at most w.

Proposition 45.6 (Chains with an end at zero). *(i) A chain whose first letter differs from its first end and whose last letter differs from its last end converges absolutely.*

(ii) For $\lambda \in \mathbb{C}^\times$ and $\eta \in \mathbb{C}$, $\mathrm{I}(\lambda a_0 + \eta; \lambda a_1 + \eta, \dots, \lambda a_w + \eta; \lambda a_{w+1} + \eta) = \mathrm{I}(a_0; a_1, \dots, a_w; a_{w+1})$.

(iii) $\mathrm{I}(a_{w+1}; a_w, \dots, a_1; a_0) = (-1)^w \,\mathrm{I}(a_0; a_1, \dots, a_w; a_{w+1})$.

(iv) Every chain of level M with one end 0 and the other in μ_M, whose first letter differs from its first end and whose last letter differs from its last end, is ± 1 times a chain of level M from 0 to 1 with first letter not 0 and last letter not 1, so its value lies in the values of $\mathcal{C}^{(M)}$.

A chain between two singular points, two points of $\{0\} \cup \mu_M$, is the verb that the cutting rule uses on its right factors; with one end at 0 it is a value of the content, and on the symbols below it is read as it stands.

45.3 The cutting rule

Definition 45.7 (Symbols of chains). For an integer $M \ge 1$ the *symbols of level M* form the commutative $\mathbb{Q}(\mu_M)$-algebra $\mathfrak{S}_M$ of polynomials in the *formal chains* $\mathrm{I}^{\mathfrak{s}}(a_0; a_1, \dots, a_w; a_{w+1})$, $w \ge 1$, with $a_0, \dots, a_{w+1} \in \{0\} \cup \mu_M$ and $a_0 \neq a_{w+1}$, as independent variables. In $\mathfrak{S}_M$ one writes $\mathrm{I}^{\mathfrak{s}}(a; \varnothing; b) = 1$ for all a, b, and $\mathrm{I}^{\mathfrak{s}}(a; a_1, \dots, a_w; a) = 0$ for $w \ge 1$. The *weight* of $\mathrm{I}^{\mathfrak{s}}(a_0; a_1, \dots, a_w; a_{w+1})$ is w, and $\mathfrak{S}_M$ is graded by the weight, the weight of a product being the sum. The *symbol* of a trail of level M is $(-1)^d \,\mathrm{I}^{\mathfrak{s}}\big(0; c_1^{-1}, 0^{\nu_1-1}, \dots, c_d^{-1}, 0^{\nu_d-1}; 1\big)$ (Proposition 45.5(i)), and the *realization* of a formal chain with one end 0, the other in μ_M, first letter not its first end and last letter not its last end, is the chain with the same entries (Proposition 45.6(i)).

Definition 45.8 (The cutting rule). The *cutting rule* is the homomorphism of $\mathbb{Q}(\mu_M)$-algebras $\Delta : \mathfrak{S}_M \to \mathfrak{S}_M \otimes_{\mathbb{Q}(\mu_M)} \mathfrak{S}_M$ given on the formal chains by

$$\Delta\, \mathrm{I}^{\mathfrak{s}}(a_0; a_1, \dots, a_w; a_{w+1}) = \sum_{S \subseteq \{1,\dots,w\}} \mathrm{I}^{\mathfrak{s}}(a_0; a_S; a_{w+1}) \otimes \prod_{p=0}^{|S|} \mathrm{I}^{\mathfrak{s}}\big(a_{s_p}; a_{s_p+1}, \dots, a_{s_{p+1}-1}; a_{s_{p+1}}\big),$$

where $s_1 < \cdots < s_{|S|}$ lists S, $s_0 = 0$, $s_{|S|+1} = w+1$ and $a_S = (a_{s_1}, \dots, a_{s_{|S|}})$: the left factor keeps the letters of S between the same ends, and each right factor is a run of dropped letters read between the two kept letters, or ends, that bound it. The *counit* $\varepsilon : \mathfrak{S}_M \to \mathbb{Q}(\mu_M)$ is the homomorphism

that vanishes on the formal chains. For $\lambda \in \mu_M$ the *rotation* ρ_λ multiplies every entry of every formal chain by λ, and the *reversal* ϱ maps $\mathrm{I}^{\mathfrak{s}}(a_0; a_1, \dots, a_w; a_{w+1})$ to $(-1)^w\, \mathrm{I}^{\mathfrak{s}}(a_{w+1}; a_w, \dots, a_1; a_0)$; both extend to automorphisms of the algebra.

Theorem 45.9 (The cutting rule is a coaction)**.** *(i) Δ is coassociative and counital: $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$ and $(\varepsilon \otimes \mathrm{id}) \circ \Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon) \circ \Delta$.*

(ii) Δ preserves the weight: it maps a symbol of weight w to a sum of tensors of weights w' and w'' with $w' + w'' = w$; and for every formal chain x, $\Delta x - x \otimes 1 - 1 \otimes x$ is a sum of tensors whose two factors have positive weights.

(iii) $\Delta \circ \rho_\lambda = (\rho_\lambda \otimes \rho_\lambda) \circ \Delta$ for $\lambda \in \mu_M$, and $\Delta \circ \varrho = (\varrho \otimes \varrho) \circ \Delta$.

On the symbol of a trail of level M the left factors are the symbols of the chains from 0 to 1 of the letters kept, up to sign the symbol of a trail of level M whenever the kept letters begin with a letter other than 0 and end with one other than 1 (Proposition 45.5(ii)), and the right factors are runs of the trail's letters read between singular points, which at tier one are 0 and roots of unity of order M. Exercise 54.2 computes the rule on the symbols of the trails $(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0, \infty))$ and of index 2 at the frequency $\frac{1}{3}$.

45.4 The letter of a live input

Definition 45.10 (The letter of an input)**.** The *letter* of the input with density ψ on the integers $n \geq n_0$ of the counting input, $n_0 \in \mathbb{N}$, is the power series $L_\psi(z) = \sum_{n \geq n_0} \psi(n)\, z^n$. At $z = e^{-u}$ it is the Laplace transform of the input beyond the cut $n_0 - 1$ (Definition 5.4), and where the series converges at $z = 1$ its value there is the Trailmoment at index zero $(\mathsf{T}_{0;(1)}[x]\, \psi\, \mathbb{I}_{\mathbb{Z}})((n_0 - 1, \infty)) = \sum_{n \geq n_0} \psi(n)$. The letter of the lattice input at the frequency a on $\mathbb{Z}$ beyond the cut 0 is $e^{2\pi i a} z/(1 - e^{2\pi i a} z)$, singular at the letter $e^{-2\pi i a}$ of its chains (Proposition 45.5).

Proposition 45.11 (The letter of a live input)**.** *Let ψ satisfy $\sum_{i=0}^{h} p_i(n)\, \psi(n + i) = 0$ for the integers $n \geq n_0$, with $p_i \in \mathbb{C}[n]$, let d be the largest degree of the p_i and $\sigma(\lambda) = \sum_i c_i \lambda^i$ the symbol, c_i the coefficient of n^d in p_i; write $\theta = z \frac{d}{dz}$.*

(i) As formal power series, $\sum_{i=0}^{h} z^{h-i} p_i(\theta - i)\, L_\psi = Q$ for a polynomial Q of degree less than $n_0 + h$: a linear differential equation of order d whose coefficient of $\frac{d^d}{dz^d}$ is $z^d \sum_i c_i z^{h-i} = z^{d+h} \sigma(1/z)$. Its singular points in $\mathbb{C}^\times$ are the reciprocals of the nonzero roots of σ.

(ii) Let $c_h \neq 0$, that is, p_h of the largest degree, and let ϱ be the largest modulus of a root of σ. Then for every $\eta > 0$, $|\psi(n)| \leq C_\eta (\varrho + \eta)^n$ for $n \geq n_0$; L_ψ converges on $|z| < 1/\varrho$ and continues analytically along every path in $\mathbb{C}^\times$ that starts in that disc and avoids the reciprocals of the nonzero roots of σ, and to the plane cut along the rays $\{t/\lambda : t \geq 1\}$, λ a nonzero root of σ.

(iii) The termwise product of two sequences that satisfy, for large n, recurrences of orders h and h' with polynomial coefficients and nonzero extreme coefficients satisfies, for large n, one of order at most hh' with nonzero extreme coefficients.

(iv) Let ψ and ψ' be as in (ii), with symbols σ and σ'. Then the letter of $\psi\psi'$ converges on $|z| < 1/(\varrho\varrho')$ and continues analytically to the plane cut along the rays $\{t/(\lambda\lambda') : t \geq 1\}$, λ and λ' nonzero roots of σ and σ'.

Proposition 45.12 (The row of a hinge closing read by letters)**.** *In Proposition 43.10(ii) let the recurrences of the room have nonzero extreme coefficients and $p^{\pm}_{h_\pm}$ of the largest degree, with $\varrho^{\pm}$ the largest modulus of a root of their symbols, and let the coefficients $\Phi^{\pm}_n$ of the room periods $x^{\pm}_n$ satisfy, for large n, recurrences of the same kind, with $\varrho^{\pm}_\Phi$ the largest modulus of a root of their symbols. If $\varrho^{\pm}_\Phi \varrho^{\pm} < 1$, then:*

(i) every sum $\Xi^{\pm}_i = \sum_n \Phi^{\pm}_n \, c^{\pm}_i(n)$ converges absolutely, it is the value $L_{\psi^{\pm}_i}(1)$ of the letter of the live input $\psi^{\pm}_i(n) = \Phi^{\pm}_n \, c^{\pm}_i(n)$, and

$$P_F(\mathbf{m}; s, t) = \sum_{\pm} \sum_{i<h_\pm} x^{\pm}_i \, L_{\psi^{\pm}_i}(1) :$$

the room's $h_+ + h_-$ initial periods against the letters of the live inputs built on its fundamental solutions;

(ii) each $L_{\psi^{\pm}_i}$ satisfies a linear differential equation with polynomial coefficients and continues analytically to the plane cut along the rays $\{t/(\lambda\lambda') : t \geq 1\}$, λ a nonzero root of the symbol of the room's recurrence and λ' one of the symbol of the coefficients' recurrence.

At tier one the room's recurrence has order one, its fundamental solution is a power times a stacked Gamma (Proposition 38.11), and each lattice carries one letter; at higher tier the room's periods open into its h fundamental solutions, and the letters of the row are singular only at products of the reciprocals of the roots of the two symbols.

Chapter 46

Finite algebra for scale reads

A bunch read on its scale against linear letters is a Breta of one bunch at the unit configuration (Proposition 15.1), and a Breta of unit columns on one bunch is, up to a stacked Gamma, such a scale integral. Its values at integer shifts of its index and its letter exponents satisfy two relations, the letter split and the scale exchange, and these relations are complete: every linear relation between the shifted values, with coefficients rational in the index and the letter exponents, is a combination of shifted letter splits and scale exchanges, and the values modulo the relations form a space whose dimension is the number of letters. The same holds at every corner, with the coefficients ξ_ℓ of the letters $x+\xi_\ell$ as the variables. For polynomial letters on one scale the same two relations reduce every insertion to one read per distinct root of the product of the letters, counted without finding a root; at a numerical corner three Laurent coefficients per read suffice, every integer ray has a recurrence with a certificate, and the jets in the coefficients of the letters are reads of rational insertions.

46.1 Complete relations for linear letters and unit columns

Definition 46.1 (Shifts, relations between shifts, the relation space)**.** Let $J=\{0,\dots,L\}$ carry distinct coefficients $\xi_\ell \in \mathbb{C}\setminus(-\infty,0]$, with the unit configuration I_J, and let $\mathbb{K}=\mathbb{C}(\kappa,w_0,\dots,w_L)$ be the field of rational functions of an index κ and letter exponents $\mathbf{w}$. Let $\mathcal{M}$ be the vector space over $\mathbb{K}$ with one basis element $[a,\mathbf{b}]$ for each *shift* $(a,\mathbf{b})\in\mathbb{Z}\times\mathbb{Z}^J$. The *value* of $[a,\mathbf{b}]$ is the scale integral at the shifted index and letter exponents,

$$\int_0^\infty x^{\kappa+a-1}\prod_{\ell\in J}(x+\xi_\ell)^{-w_\ell-b_\ell}\,dx \\ = \Gamma_{\,2}^{\,|J|}\begin{bmatrix}\kappa+a,\ |\mathbf{w}|+|\mathbf{b}|-\kappa-a\\ \mathbf{w}+\mathbf{b}\end{bmatrix}\mathsf{B}^{\{J\}}_{\mathbf{w}+\mathbf{b},\,|\mathbf{w}|+|\mathbf{b}|-\kappa-a;\,I_J}(\boldsymbol{\xi})$$

which converges on $0<\operatorname{Re}\kappa+a<\operatorname{Re}|\mathbf{w}|+|\mathbf{b}|$, the second form holding where moreover $\operatorname{Re}(w_\ell+b_\ell)>0$ for every ℓ (Proposition 15.1); finitely many shifts converge together on a nonempty convex open set. An element $\sum_i c_i[a_i,\mathbf{b}_i]$ of $\mathcal{M}$ is a *relation between shifts* when the combination of the values of the $[a_i,\mathbf{b}_i]$ with the coefficients $c_i(\kappa,\mathbf{w})$ vanishes at every point where all its terms converge and every c_i is defined. At the shift $(a,\mathbf{b})$ the letter split of the letter $x+\xi_\ell$ and the scale exchange are

$$\operatorname{split}_\ell(a,\mathbf{b})=[a,\mathbf{b}]-[a+1,\mathbf{b}+\mathbf{e}_\ell]-\xi_\ell\,[a,\mathbf{b}+\mathbf{e}_\ell],$$

$$\operatorname{exch}(a,\mathbf{b}) = (\kappa + a)\,[a,\mathbf{b}] - \sum_{\ell\in J}(w_\ell + b_\ell)\,[a+1,\mathbf{b}+\mathbf{e}_\ell],$$

the letter split and the scale exchange of Proposition 3.6 for the letters $x+\xi_\ell$. The *relation space* $\mathcal{L}_J \subseteq \mathcal{M}$ is the span over $\mathbb{K}$ of all letter splits and scale exchanges, and the *dimension* of the shifts is $\dim_{\mathbb{K}} \mathcal{M}/\mathcal{L}_J$; on one scale these play the part of the relation space and the dimension of a ray (Definition 38.4).

Theorem 46.2 (The relations of a scale integral are complete)**.** *Let the coefficients range over distinct points of* $\mathbb{C}\setminus(-\infty,0]$.

(i) Every element of $\mathcal{L}_J$ *is a relation between shifts.*

(ii) $\mathcal{M} = \mathcal{L}_J + \sum_{\ell\in J}\mathbb{K}\,[0,\mathbf{e}_\ell]$*: the dimension is at most* $|J|$.

(iii) The values of the $|J|$ *shifts* $[0,\mathbf{e}_\ell]$,

$$\Gamma_{\,2}{}_{|J|}\begin{bmatrix}\kappa,\ |\mathbf{w}|+1-\kappa\\ \mathbf{w}+\mathbf{e}_\ell\end{bmatrix}\,\mathsf{B}^{\{J\}}_{\mathbf{w}+\mathbf{e}_\ell,\,|\mathbf{w}|+1-\kappa;\,I_J}(\boldsymbol{\xi}),\qquad \ell\in J,$$

are linearly independent over $\mathbb{K}$*: a combination of them with coefficients in* $\mathbb{K}$ *that vanishes on a nonempty open subset of* $\{0<\operatorname{Re}\kappa<\operatorname{Re}|\mathbf{w}|+1\}$ *has every coefficient* 0.

(iv) The dimension is $|J|$*, and every relation between shifts lies in* $\mathcal{L}_J$*: it is a combination over* $\mathbb{K}$ *of shifted letter splits and scale exchanges.*

(v) Coefficients as variables. *Let the coefficients be rational functions of parameters* $\mathbf{u}=(u_1,\dots,u_d)$*, distinct and not* 0 *as rational functions, defined with distinct values in* $\mathbb{C}\setminus(-\infty,0]$ *on a nonempty open subset of* $\mathbb{R}^d$ *or of* $\mathbb{C}^d$*. Then (i), (ii) and (iv) hold with the field* $\mathbb{C}(\mathbf{u},\kappa,\mathbf{w})$ *in the place of* $\mathbb{K}$*, a relation between shifts being an element whose combination of values vanishes at every* $\mathbf{u}$ *of that set and every* $(\kappa,\mathbf{w})$ *where all its terms converge and every coefficient is defined.*

The letter splits are the partial fractions of the integrand, and the scale exchange is one integration by parts along the scale. Part (ii) reduces every shift to the $|J|$ values with one letter exponent raised by one, by these two relations alone, and part (iii) says that those values are linearly independent over $\mathbb{K}$. So a relation between shifts that is not a combination of shifted letter splits and scale exchanges does not exist, and the number of independent values is the number of letters; for one letter the dimension is one: every shift of the Beta corner of a linear letter is a rational multiple of one value. By Theorem 15.3 the scale integral of a family at a bunch is such an integral, with the points of the joint spectrum as its coefficients, so its relations in its index and its letter exponents are complete; relations between scale integrals of a family at shifted masses and exponents, where the letter exponents are tied to the exponents of the family, are a further problem.

What decides the relations at a corner, where the index and the letter exponents are numbers and the coefficients are the variables, is the behaviour of the values as one coefficient tends to 0: its letter becomes a power of the scale and joins the index.

Lemma 46.3 (A coefficient tends to 0)**.** *Let* $L\geq 1$*, let the coefficients be positive, and let the index and the letter exponents be numbers with* $\operatorname{Re} w_\ell > 0$ *for every* ℓ *and* $0\leq\operatorname{Re}\kappa'<1$*, where* $\kappa' = \kappa - w_0$*. In this statement* $J' = J\setminus\{0\}$*,* $\mathbf{w}' = \mathbf{w}|_{J'}$*,* $\boldsymbol{\xi}' = \boldsymbol{\xi}|_{J'}$*,* f_ℓ *is the value of* $[0,\mathbf{e}_\ell]$ *(Definition 46.1), and for letter exponents* $\mathbf{v}$ *on* J' *with* $\operatorname{Re}|\mathbf{v}|>0$ *and* $h_{\mathbf{v}}(x) = \prod_{j\in J'}(x+\xi_j)^{-v_j}$*,*

$$S'(z;\mathbf{v}) = \frac{h_{\mathbf{v}}(0)}{z} + \int_0^1 x^{z-1}\big(h_{\mathbf{v}}(x) - h_{\mathbf{v}}(0)\big)\,dx + \int_1^\infty x^{z-1}h_{\mathbf{v}}(x)\,dx$$

for $-1 < \operatorname{Re} z < \operatorname{Re}|\mathbf{v}|$, $z \neq 0$. *For* $\operatorname{Re} z > 0$ *and* $\operatorname{Re} v_j > 0$ *for every* j, $S'(z;\mathbf{v}) = \Gamma_{|J'|}^{\,2}\begin{bmatrix} z,\ |\mathbf{v}|-z \\ \mathbf{v} \end{bmatrix} \mathsf{B}^{\{J'\}}_{\mathbf{v},\,|\mathbf{v}|-z;\,I_{J'}}(\boldsymbol{\xi}')$, *the scale integral with the coefficients* $\boldsymbol{\xi}'$, *so* S' *continues it in its index across the wall* $\operatorname{Re} z = 0$, *with a simple pole at* $z = 0$ *of residue* $h_{\mathbf{v}}(0)$*; its constant term* $\operatorname{CT}_{z=0} S'(z;\mathbf{v})$ *is the sum of the two integrals at* $z = 0$, *and its values off the pole, at complex* z *and complex letter exponents, are the finite parts* $\operatorname{Reg}\int_0^\infty x^{z-1} h_{\mathbf{v}}\,dx$, *its constant term at* $z = 0$ *being* $\operatorname{Reg}\int_0^\infty x^{-1} h_{\mathbf{v}}\,dx$ *(Theorem 1.7). Fix* $\boldsymbol{\xi}'$ *and let* $\xi_0 \to 0^+$. *Then, with* $j \in J'$ *in (ii)–(v):*

(i) $\xi_0^{1-\kappa'} f_0 \to \Gamma_1^{\,2}\begin{bmatrix} \kappa,\ 1-\kappa' \\ w_0+1 \end{bmatrix} h_{\mathbf{w}'}(0)$, *which is not* 0.

(ii) If $\operatorname{Re}\kappa' > 0$, *then* $f_j \to S'(\kappa';\mathbf{w}'+\mathbf{e}_j)$.

(iii) If $\operatorname{Re}\kappa' = 0$ *and* $\kappa' \neq 0$, *then*

$$f_j = \Gamma_1^{\,2}\begin{bmatrix} \kappa,\ -\kappa' \\ w_0 \end{bmatrix} \frac{h_{\mathbf{w}'}(0)}{\xi_j}\,\xi_0^{\kappa'} + S'(\kappa';\mathbf{w}'+\mathbf{e}_j) + o(1).$$

(iv) If $\kappa' = 0$, *then*

$$f_j = \frac{h_{\mathbf{w}'}(0)}{\xi_j}\Big(\log\frac{1}{\xi_0} + (\log\Gamma)'(1) - (\log\Gamma)'(\kappa)\Big) + \operatorname*{CT}_{z=0} S'(z;\mathbf{w}'+\mathbf{e}_j) + o(1).$$

(v) The relations across the wall. *For* $-1 < \operatorname{Re} z \le 0$, $z \neq 0$,

$$S'(z;\mathbf{v}) = S'(z+1;\mathbf{v}+\mathbf{e}_j) + \xi_j\, S'(z;\mathbf{v}+\mathbf{e}_j), \qquad z\,S'(z;\mathbf{v}) = \sum_{i\in J'} v_i\, S'(z+1;\mathbf{v}+\mathbf{e}_i),$$

and $\operatorname{CT}_{z=0} S'(z;\mathbf{v}) = S'(1;\mathbf{v}+\mathbf{e}_j) + \xi_j\ \operatorname{CT}_{z=0} S'(z;\mathbf{v}+\mathbf{e}_j)$.

As the coefficient ξ_0 tends to 0 its letter $x+\xi_0$ becomes the power x. The value with the exponent of that letter raised grows like $\xi_0^{\kappa'-1}$, and when $\operatorname{Re}\kappa' > 0$ the others tend to the scale integrals of the remaining coefficients at the index $\kappa' = \kappa - w_0$, the index left when the letter exponent w_0 has joined it. When $\operatorname{Re}\kappa' = 0$ that index lies on the wall of the remaining integral, and a second term of the same size stands next to the integral continued across its wall: the oscillating power $\xi_0^{\kappa'}$, and at $\kappa' = 0$ a logarithm, which grows. The letter split and the scale exchange hold across the wall, and at the pole $z = 0$ the scale exchange keeps only its residue, the first identity of Theorem 46.4(ii) for the remaining coefficients. These are the first terms of the closing between two separated groups (Corollary 19.19), the letter of ξ_0 being the lower group: one lattice carries the powers $\xi_0^{\kappa'+n}$, with the wall residues of the other letters as coefficients, and the other carries the integer powers ξ_0^n, with the scale integral of the other coefficients continued in its index; at $\kappa' = 0$ the two lattices meet, and what their poles leave is the logarithm of (iv).

Theorem 46.4 (The relations at a corner)**.** *Let the index* κ *and the letter exponents* $\mathbf{w}$ *be numbers with* $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|+1$, *so that the values of the* $|J|$ *shifts* $[0,\mathbf{e}_\ell]$ *converge, and let the coefficients be distinct points of* $\mathbb{C}\setminus(-\infty,0]$. *A shift* $[a,\mathbf{b}]$ converges at the corner *when* $0 < \operatorname{Re}\kappa + a < \operatorname{Re}|\mathbf{w}| + |\mathbf{b}|$, *and a letter split or a scale exchange at the shift* $(a,\mathbf{b})$, *with the index and the letter exponents at their numbers, converges at the corner when* $[a,\mathbf{b}]$ *does; all its terms then converge. A* linear relation *between shifts that converge at the corner is a combination* $\sum_i r_i[a_i,\mathbf{b}_i]$ *of such shifts, with coefficients* r_i *rational in* $\boldsymbol{\xi}$, *whose value vanishes for* $\boldsymbol{\xi}$ *in a nonempty open set; it is a combination of letter splits and scale exchanges when it is one as a formal combination of shifts. The values are holomorphic in* $\boldsymbol{\xi}$ *on* $(\mathbb{C}\setminus(-\infty,0])^J$, *coincident coefficients included, and functions of* $\boldsymbol{\xi}$ *are* independent over $\boldsymbol{\xi}$ *when no combination of them with coefficients polynomial in* $\boldsymbol{\xi}$, *not all* 0, *vanishes at all* $\boldsymbol{\xi} > 0$, *equivalently for* $\boldsymbol{\xi}$ *in a nonempty open set.*

(i) Reduction. *If no letter exponent lies in* $\{-1,-2,\dots\}$*, the value of every shift that converges at the corner is a combination of the values of the* $[0,\mathbf{e}_\ell]$ *with coefficients rational in* $\boldsymbol{\xi}$*; the combination follows from letter splits and scale exchanges that converge at the corner, taken at shifts* $(a',\mathbf{b}')$ *with* $b'_\ell \geq \min(b_\ell,0)$ *for every* ℓ*.*

(ii) The two walls. *If* $\operatorname{Re}|\mathbf{w}|>0$*, the values of the* $[0,\mathbf{e}_\ell]$ *at the index* 1 *and at the index* $|\mathbf{w}|$ *satisfy*

$$\sum_{\ell\in J} w_\ell \int_0^\infty \prod_{\ell'\in J} (x+\xi_{\ell'})^{-w_{\ell'}-\mathbf{1}[\ell'=\ell]}\,dx = \prod_{\ell\in J}\xi_\ell^{-w_\ell},$$
$$\sum_{\ell\in J} w_\ell\,\xi_\ell \int_0^\infty x^{|\mathbf{w}|-1}\prod_{\ell'\in J} (x+\xi_{\ell'})^{-w_{\ell'}-\mathbf{1}[\ell'=\ell]}\,dx = 1,$$

the integrals being, where every $\operatorname{Re} w_\ell>0$*, the stacked Gammas times the Bretas of Definition 46.1.*

(iii) Independence. *The values of the* $|J|$ *shifts* $[0,\mathbf{e}_\ell]$ *are independent over* $\boldsymbol{\xi}$ *if and only if no letter exponent lies in* $\{-1,-2,\dots\}$*.*

(iv) Completeness. *Whatever the letter exponents, every linear relation between shifts that converge at the corner is a combination, with coefficients rational in* $\boldsymbol{\xi}$*, of letter splits and scale exchanges that converge at the corner, and over the rational functions of* $\boldsymbol{\xi}$ *the values of these shifts span a space of dimension* $|J|$*: at every corner the dimension is* $|J|$*.*

(v) A coefficient equal to 1. *Parts (i), (iii) and (iv) hold as well when one coefficient is the constant* 1 *and the others are the variables.*

At a corner the relations are the same, and when no letter exponent lies in $\{-1,-2,\dots\}$ they reduce every shift that converges to the $|J|$ values with one letter exponent raised without leaving the corner: every relation of the reduction has all its terms convergent, and the leading coefficients $\kappa-r$, $r\geq 1$, and $\kappa+j-|\mathbf{w}|$, $j\geq 0$, by which the descents at the origin and at infinity divide vanish only at shifts that have left the domain. The two identities of (ii) are the relations at the two walls of the domain: the first is the scale exchange at the shift $(-1,\mathbf{0})$ at the index 1, whose member on the wall is replaced by its limit $\prod_\ell \xi_\ell^{-w_\ell}$, and the second is the scale exchange less the letter splits multiplied by their letter exponents at the shift $(0,\mathbf{0})$, at the index $|\mathbf{w}|$, where $|\mathbf{w}|-\kappa$ times the value of $[0,\mathbf{0}]$ tends to 1; at a wall one combination of the $|J|$ values is a monomial in $\boldsymbol{\xi}$. A letter exponent in $\{-1,-2,\dots\}$ is not on a wall: its letter has become a polynomial factor of the integrand, the $|J|$ values with one letter exponent raised are dependent and no longer span the shifts that converge; the corner with that letter exponent moved up to 0 has the same shifts, and its $|J|$ values with one letter exponent raised span them. Independence is read from the coefficients one at a time: as ξ_0 tends to 0 the value with the letter exponent w_0 raised grows faster than the others (Lemma 46.3), so in a relation the polynomial multiplying it vanishes at $\xi_0=0$, and what is left is a relation between scale integrals of the remaining coefficients, at the index $\kappa-w_0$, or on their wall, where the relations across the wall carry it to the next index.

Corollary 46.5 (Unit columns on one bunch)**.** *Let* B *be a bunch with the unit configuration* I_B *and distinct coefficients* $\xi_e>0$*. By Proposition 15.1 with* $J=B$*,* $\mathbf{w}=\mathbf{m}$ *and* $\kappa=m_B-\nu$*, the value* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{a},\,\nu+n;\,I_B}(\boldsymbol{\xi})$*,* $\mathbf{a}\in\mathbb{Z}^B$*,* $n\in\mathbb{Z}$*, is the value of the shift* $[|\mathbf{a}|-n,\mathbf{a}]$ *of Definition 46.1 divided by* $\Gamma_{\substack{2\\|B|}}\begin{bmatrix} m_B-\nu+|\mathbf{a}|-n,\ \nu+n\\ \mathbf{m}+\mathbf{a}\end{bmatrix}$*, wherever that proposition applies; this stacked Gamma is the*

one at $(\mathbf{m},\nu)$ *times a rational function of* $(\mathbf{m},\nu)$*. Each relation below is read where all its Bretas converge.*

(i) The mark relation. *For each* $e \in B$*,*

$$m_e\, \mathsf{B}^{\{B\}}_{\mathbf{m},\,\nu;\,I_B}(\boldsymbol{\xi}) = (m_B - \nu)\, \mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{e}_e,\,\nu;\,I_B}(\boldsymbol{\xi}) + \nu\, \xi_e\, \mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{e}_e,\,\nu+1;\,I_B}(\boldsymbol{\xi}) :$$

the sum over $f \in B$ *of the shape exchanges of Theorem 3.5(iv) for the pairs* (e,f)*, reduced by the shape split and the scale exchange; it is the letter split of the letter* $x + \xi_e$*.*

(ii) The shape split, the scale exchange and the shape exchanges of Theorem 3.5, at all integer shifts of $(\mathbf{m},\nu)$*, span over* $\mathbb{C}(\mathbf{m},\nu)$ *every linear relation with coefficients in* $\mathbb{C}(\mathbf{m},\nu)$ *between the values* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{a},\,\nu+n;\,I_B}(\boldsymbol{\xi})$*, and the quotient of the space of shifts by these relations has dimension* $|B|$*. The same holds over* $\mathbb{C}(\boldsymbol{\xi},\mathbf{m},\nu)$ *for relations whose coefficients are rational in* $\boldsymbol{\xi}$ *as well and which hold at all distinct* $\boldsymbol{\xi} > 0$ *at which their coefficients are defined.*

(iii) At a corner. *Let* $\mathbf{m}$ *and* ν *be numbers with* $\operatorname{Re} m_e > 0$ *for every* e *and* $\operatorname{Re}\nu > 0$*, and call a value* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{a},\,\nu+n;\,I_B}(\boldsymbol{\xi})$ interior *when* $\operatorname{Re}(m_e + a_e) > 0$ *for every* e *and* $\operatorname{Re}\nu + n > 0$*; the Breta converges there. Over the rational functions of* $\boldsymbol{\xi}$ *the interior values span a space of dimension* $|B|$*, and every linear relation between interior values with coefficients rational in* $\boldsymbol{\xi}$ *is a combination, with such coefficients, of shape splits, scale exchanges and shape exchanges all of whose terms are interior. If* $\operatorname{Re}\nu < \operatorname{Re} m_B$*, the* $|B|$ *values* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{e}_e,\,\nu+1;\,I_B}(\boldsymbol{\xi})$*,* $e \in B$*, are a basis of that space.*

Completeness holds for the scale integral of one bunch against linear letters and for unit columns on one bunch, which are the same integrals: the coefficients are the same, the shape index is the letter exponents, and the index of the scale integral is the amount $m_B - \nu$ by which the scale index falls short of the mass m_B. It holds at a free index and free letter exponents, and at every corner with the coefficients as the variables, one of them possibly the constant 1. For a Breta with a kernel, with letters of more than two patterns or of levels other than 0 and 1, or with several bunches, the relations of Theorem 3.5 hold, and their completeness is a further problem.

46.2 Polynomial letters on one scale

A word on one scale with polynomial letters, such as the scale integral of a bunch at fixed shapes (Theorems 15.3 and 19.28), has the same two relations, the letter splits becoming partial fractions over the distinct factors of the letters, and the same finiteness: the retained data are one read per distinct root of the product of the letters, counted without finding a root, and at a numerical corner three Laurent coefficients per read. Every rational insertion, every integer shift of the index and the letter exponents, every derivative in the coefficients of the letters and every integer ray is then finite algebra, with a certificate.

Definition 46.6 (Insertions and reads on one scale)**.** Let $\mathbb{F} \subseteq \mathbb{C}$ be a field, the field of the letters, and let $P_1,\dots,P_N \in \mathbb{F}[x]$ be nonconstant letters without roots on $[0,\infty)$, a constant letter being moved into a scalar; put $\mathbf{P} = (P_1,\dots,P_N)$. For an index κ and letter exponents $w_1,\dots,w_N$ in a field $\mathbb{F}' \supseteq \mathbb{F}(\kappa,\mathbf{w})$ embedded in $\mathbb{C}$, the insertions, the regular functions of the letter complement as in Definition 38.1, are the elements of the ring $\mathfrak{A} = \mathbb{F}'[x, x^{-1}, P_1^{-1},\dots,P_N^{-1}]$, and the *read* of an insertion R is

$$\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}} = \int_0^\infty x^{\kappa-1} \prod_j P_j(x)^{-w_j}\, R(x)\, dx$$

where it converges absolutely, each power $P_j^{-w_j}$ being taken on $[0,\infty)$ by continuity from its principal value at 0, and continued in the index and the letter exponents elsewhere. The shift of the index by a and of the letter exponents by b_j, integers, is the insertion $x^a\prod_j P_j^{-b_j}$, and a derivative in the coefficients of a letter is an insertion. A squarefree decomposition with gcd splitting gives pairwise coprime squarefree $q_1,\dots,q_M\in\mathbb{F}[x]$, none vanishing at 0, with $P_j=c_j\prod_k q_k^{m_{jk}}$, without roots and without a field extension; $Q=q_1\cdots q_M$ is the monic radical of $P_1\cdots P_N$, of degree $D>0$. The *scale exchange* on insertions is

$$\eth g = xg' + \Big(\kappa - \sum_j w_j \frac{xP_j'}{P_j}\Big)g, \qquad \text{so that} \qquad \Big(x^\kappa \prod_j P_j^{-w_j}\, g\Big)' = x^{\kappa-1}\prod_j P_j^{-w_j}\,\eth g.$$

For the linear letters $x+\xi_\ell$ with distinct $\xi_\ell\in\mathbb{C}\setminus(-\infty,0]$, the reads of the shifts are the values of Definition 46.1, $D=|J|$, and $\eth$ applied to $x^a\prod_\ell(x+\xi_\ell)^{-b_\ell}$ is the scale exchange $\mathrm{exch}(a,\mathbf{b})$ read under the integral.

Theorem 46.7 (Reduction by the scale exchange)**.** *Let the numbers*

$$\kappa - r \quad (r\ge 1), \qquad \kappa + a - \sum_j w_j \deg P_j \quad (a\ge 0), \qquad \sum_j m_{jk}w_j + h \quad (h\ge 1,\ 1\le k\le M)$$

be nonzero in $\mathbb{F}'$. Every $R\in\mathfrak{A}$ has a unique decomposition

$$R = \eth g + \sum_{i=0}^{D-1} c_i \frac{x^i}{Q}, \qquad g\in\mathfrak{A},\quad c_i\in\mathbb{F}',$$

obtained by polynomial arithmetic, partial fractions and the extended Euclidean algorithm; so $\dim_{\mathbb{F}'}\mathfrak{A}/\eth\mathfrak{A}=D$, and on the domain where the reads converge $\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}}=\sum_i c_i\,\langle x^i/Q\rangle_{\kappa,\mathbf{w};\mathbf{P}}$. For the linear letters $x+\xi_\ell$ with distinct $\xi_\ell\in\mathbb{C}\setminus(-\infty,0]$, $D=|J|$ and this is Theorem 46.2(ii) over $\mathbb{K}$, the x^i/Q spanning the same space as the $(x+\xi_\ell)^{-1}$.

The dimension is that of the quotient by the scale exchange, the dimension of the ray on one scale (Theorem 38.24); it does not assert that the reads of the D insertions x^i/Q are independent once parameters are tied, and symmetries of the letters can relate them further. For linear letters Theorem 46.2(iii) gives the independence. The three conditions are the leading coefficients of the reduction at the origin, at infinity and at a repeated factor. Exact computation needs exact arithmetic with a zero test in $\mathbb{F}'$, numerical exponents being taken in an effective extension; the analytic statements hold at all complex exponents.

Definition 46.8 (Deformation; masters)**.** At numerical exponents $(\kappa,\mathbf{w})$ fix a direction $(\lambda_0;\boldsymbol{\lambda})\in\mathbb{R}\times\mathbb{R}^N$ of the index and the letter exponents that is *admissible*: $\lambda_0>0$, $\sum_j\lambda_j\deg P_j>\lambda_0$ and $\sum_j m_{jk}\lambda_j\neq 0$ for every k. The *deformation* moves the data along it, to $(\kappa+\lambda_0\varepsilon,\ \mathbf{w}+\varepsilon\boldsymbol{\lambda})$; the three conditions of Theorem 46.7 hold over $\mathbb{F}'(\varepsilon)$, and for an insertion R the coefficients $c_i(\varepsilon)$ are those of its decomposition there. The *masters* are the deformed reads

$$H_i(\varepsilon) = \big\langle x^i/Q\big\rangle_{\kappa+\lambda_0\varepsilon,\ \mathbf{w}+\varepsilon\boldsymbol{\lambda};\ \mathbf{P}}, \qquad 0\le i<D.$$

For large real ε every finite set of deformed reads converges, the exponent at the origin growing with the slope λ_0 and the decay at infinity with the slope $\sum_j\lambda_j\deg P_j-\lambda_0>0$, so the reductions hold there and then as identities of meromorphic functions. The masters are finite combinations of reads of shifts of the original letters: $1/Q=\big(P_1\cdots P_N/Q\big)\prod_j P_j^{-1}$ with $P_1\cdots P_N/Q$ a polynomial, so each term of $x^iP_1\cdots P_N/Q$ shifts the index by an integer and raises every letter exponent by one. The admissible directions form a nonempty open cone; the direction $(\frac12;\mathbf{1})$ is admissible.

Lemma 46.9 (Simple poles at a corner)**.** *For every insertion R independent of ε, or holomorphic in ε, the coefficients $c_i(\varepsilon)$ have at most a simple pole at $\varepsilon = 0$, and every master $H_i(\varepsilon)$ is meromorphic near 0 with at most a simple pole.*

Theorem 46.10 (Finitely many numbers at every corner)**.** *Along an admissible direction (Definition 46.8), for an insertion R write $H_i(\varepsilon) = h_{i,-1}\varepsilon^{-1} + h_{i,0} + h_{i,1}\varepsilon + O(\varepsilon^2)$ and $c_i(\varepsilon) = c_{i,-1}\varepsilon^{-1} + c_{i,0} + c_{i,1}\varepsilon + O(\varepsilon^2)$. The $3D$* corner numbers *$(h_{i,-1}, h_{i,0}, h_{i,1})_{i<D}$ determine every absolutely convergent read:*

$$\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}} = \sum_{i=0}^{D-1} \big(c_{i,-1}h_{i,1} + c_{i,0}h_{i,0} + c_{i,1}h_{i,-1}\big).$$

They depend on the letters, the exponents and the direction only, not on the insertion or the size of the shift. The reads of $R\left(\lambda_0 \log x - \sum_j \lambda_j \log P_j\right)^b$, the derivatives ∂_ε^b at $\varepsilon = 0$ of the deformed reads $\langle R\rangle_{\kappa+\lambda_0\varepsilon,\,\mathbf{w}+\varepsilon\boldsymbol{\lambda};\,\mathbf{P}}$, need the masters through the order $b+1$: $(b+3)D$ numbers. When $\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}}$ converges absolutely, a read of $R\,(\log x)^{k_0}\prod_j(\log P_j)^{k_j}$, $b = k_0 + \sum_j k_j$, is a rational combination of the reads of $R\left(\lambda_0 \log x - \sum_j \lambda_j \log P_j\right)^b$ along finitely many admissible directions, each with its own corner numbers.

The residues are finite extractions of endpoint coefficients, the other numbers finite parts and derivatives in the deformation. The D values of the masters suffice when every pivot of the reduction is regular and the masters are holomorphic at the corner, as for linear letters at a corner with no letter exponent in $\{-1, -2, \dots\}$ and convergent masters (Theorem 46.4(i)); regular pivots alone do not suffice, and at an arbitrary corner the corner numbers are what is retained.

Theorem 46.11 (Integer rays have certified recurrences)**.** *Fix integers $a, b_1, \dots, b_N$, and let $\eth_n$ be the scale exchange at the deformed data $\left(\kappa + an + \frac{\varepsilon}{2},\ \mathbf{w} + n\mathbf{b} + \varepsilon\mathbf{1}\right)$. A finite algorithm gives polynomials $p_0(n,\varepsilon), \dots, p_D(n,\varepsilon)$ and a certificate g, rational in x, n and ε, with*

$$\eth_n g = \sum_{k=0}^{D} p_k(n,\varepsilon)\, x^{ak} \prod_j P_j^{-b_j k},$$

the polynomials $p_k(n) = p_k(n,0)$ not all zero; then

$$\sum_{k=0}^{D} p_k(n)\, \langle 1\rangle_{\kappa+a(n+k),\ \mathbf{w}+(n+k)\mathbf{b};\ \mathbf{P}} = 0$$

at every integer n at which the displayed reads converge.

The recurrence need not have the least order, and its leading coefficient may vanish at finitely many integers, read off that polynomial; there further values are given or another dependence is formed. On one scale this supplies, for every ray, the certificate that a recurrence asks for: an insertion g whose scale exchange is the recurrence.

Theorem 46.12 (Coefficient jets on one scale)**.** *Let $P_j(x;u) = P_j(x) + \sum_{a=1}^{p} u_a P_{ja}(x)$ with $\deg P_{ja} \le \deg P_j$, choose bounds $M_{ja} \ge \sup_{x>0} |P_{ja}/P_j|$, finite since the quotient is continuous on $[0,\infty]$, and let $\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}}$ converge absolutely. On the polydisc where $\sum_a |u_a| M_{ja} < 1$ for every*

j the read $\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}(\,\cdot\,;u)}$, the powers of the changed letters being continued in u from $u=0$, is holomorphic and

$$\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}(\,\cdot\,;u)} = \sum_{(n_{ja})\geq 0} \prod_{j} \frac{(-1)^{|n_j|}(w_j)_{|n_j|}}{\prod_a n_{ja}!} \prod_a u_a^{\sum_j n_{ja}} \Big\langle \prod_{j,a} \Big(\frac{P_{ja}}{P_j}\Big)^{n_{ja}} \Big\rangle_{\kappa,\mathbf{w};\mathbf{P}},$$

$|n_j| = \sum_a n_{ja}$, absolutely; every coefficient is the read of a rational insertion, given by the corner numbers of Theorem 46.10.

Splitting of repeated roots under the change costs nothing, the expansion being made against the original letters; a change that raises a degree is outside the statement. For coefficients that are symbolic parameters with fixed gcds and multiplicities, differentiating the masters gives rational insertions, so Theorem 46.7 gives matrices $\partial_{u_a} H = \mathbf{A}_a H$, with certificates, for the vector H of the masters, whose mixed derivatives agree because they agree before the reduction; at a corner where a leading coefficient of the reduction vanishes the corner numbers replace them. The structural data reused are the squarefree base, Q and the inverses of xq_k' modulo q_k; the analytic data are the D masters or the $3D$ corner numbers. A rational insertion with polynomial part of degree A_+, a pole of order A_- at the origin and of order h_k at q_k takes at most $(A_+ + 1) + A_- + \sum_k \max(h_k - 1, 0)$ elimination rounds, counted in operations, the growth of exact coefficients charged separately; a repeated letter adds m_{jk} times its letter exponent to the exponent of each squarefree factor q_k and leaves Q and D unchanged. A recurrence in a count of repetitions needs the integer shifts of Theorem 46.11, and a multiplicity written in binary, which is what the price of Chapter 33 counts, does not by itself make the evaluation logarithmic in it.

Part IX

The family as inference

Chapter 47

The family as inference

A family is a model of data in two ways. A sample of bases, drawn from the basis and cobasis laws of the tables of a family at an unknown gauge, is itself a question: its counts are the masses, the sizes of its parts are the exponents, and its chance as a function of the tilt is the integrand of the period. The law of a question is then the conjugate prior of the tilt, and every posterior, every predictive chance and every evidence is a ratio of periods; read at a bunch, a sample is a sample of the children at the shares it realizes, and each of its walls is a sum of independent excesses of the levels and colevels of its draws over their floors, whose law the Skeleton polynomials give. A sample of tilts drawn from the law of a question makes the period the cumulant function of an exponential family. Between the two sits the family word read at Dirichlet draws: the negative logarithms of its letters have the Grey law, whose Laplace transform in the exponents is the period divided by the stacked Gamma of the masses over their total; the pseudo-MGF and the pseudoexpectation are that Laplace transform fixed at the Foreman's question and at the centre. A draw of the shape law, read as a vector of numbers, has an order that is a sector of the period, best and worst bases that are functions of the order, and entropies and moments that are steps of the data. What a bunch all of whose subsets are patterns reads of its room is its interaction (Definition 12.18), the total over the scales of the bunch of the joint cumulants of the use of its marks under the basis and cobasis laws, weighted by the exponents; for a regression design the interaction of a split of the prior marks into two blocks is minus the logarithm of the generalized variance inflation factor.

47.1 Samples of bases

Proposition 47.1 (A sample of bases is a question)**.** *For a family at a gauge $\alpha = e^{2z}$ draw, for every table j, s_j bases $T_{j,1}, \dots, T_{j,s_j}$ from its basis law $\mathbb{P}_{j,\alpha}$ and t_j bases $T'_{j,1}, \dots, T'_{j,t_j}$ from its cobasis law $\mathbb{P}^*_{j,\alpha}$, all independent, let m_e count the $T_{j,i}$ that hold e and the $T'_{j,i}$ that miss e, and let $\theta_{\mathrm{draws}} = \prod_{j,i} \theta_{j,T_{j,i}} \prod_{j,i} \theta_{j,T'_{j,i}}$ be the product of the weights of the drawn bases.*

(i) As a function of the tilt z, the chance of the sample is

$$\theta_{\mathrm{draws}} \cdot \exp\Big(2\langle \boldsymbol{\mu}_\perp, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)\Big),$$

$2^{1-n}\theta_{\mathrm{draws}}$ times the integrand of the period of the question $(\mathbf{m}; \mathbf{s}, \mathbf{t})$, of pair coordinates $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ (Definition 17.1): the counts are the masses, the sizes of the parts are the exponents, and $m_{\mathcal{E}} = \sum_j (s_j r_j + t_j r^*_j)$ puts the question on the balance wall.

(ii) Its logarithmic gradient in z is $2\big(\mathbf{m}-\sum_j(s_jR_j(\alpha)+t_jR_j^*(\alpha))\big)$, twice the excess of the masses over the usage at α. For a connected family the maximum likelihood tilt is the native gauge of the question, and it exists, unique on $\mathcal{T}$, exactly when every proper wall of the question is positive.

(iii) $\kappa_S=\sum_j\Big(\sum_{i=1}^{s_j}\big(|T_{j,i}\cap S|-d_j(S)\big)+\sum_{i=1}^{t_j}\big(|S\setminus T'_{j,i}|-d_j^*(S)\big)\Big)\geq 0$: a wall is the excess of the sample's use of S over its least use.

Proposition 47.2 (The conjugate posterior)**.** *Let F be a connected family. Take as prior on the tilt of the gauge the law of a question $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ with every proper wall positive, and observe a sample as in Proposition 47.1, with counts $\mathbf{m}$, sizes $\mathbf{s},\mathbf{t}$ and the product θ_{draws} of the weights of the drawn bases.*

(i) The posterior is the law of the question $(\mathbf{m}_0+\mathbf{m};\mathbf{s}_0+\mathbf{s},\mathbf{t}_0+\mathbf{t})$: the masses add and the exponents add.

(ii) The chance of the sample with the tilt integrated out is

$$\theta_{\text{draws}}\cdot\frac{P_F(\mathbf{m}_0+\mathbf{m};\mathbf{s}_0+\mathbf{s},\mathbf{t}_0+\mathbf{t})}{P_F(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)}.$$

(iii) At the posterior data $(\mathbf{m}';\mathbf{s}',\mathbf{t}')$ the chance that one more draw from the basis law of table j is T is $\theta_{j,T}\,P_F(\mathbf{m}'+\mathbf{1}_T;\mathbf{s}'+\mathbf{e}_j,\mathbf{t}')/P_F(\mathbf{m}';\mathbf{s}',\mathbf{t}')$, and from its cobasis law $\theta_{j,T}\,P_F(\mathbf{m}'+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{s}',\mathbf{t}'+\mathbf{e}_j)/P_F(\mathbf{m}';\mathbf{s}',\mathbf{t}')$.

(iv) The posterior mean of the usage $\sum_j(s'_jR_j+t'_jR_j^*)$ at the posterior exponents is the posterior masses $\mathbf{m}_0+\mathbf{m}$.

(v) Every wall of the posterior is the wall of the prior plus the excess of Proposition 47.1(iii): a prior inside its walls gives a posterior inside its walls for every sample, the empty one included.

Proposition 47.3 (What the data see of the bases)**.** *(i) For a circuit C of $k\geq 2$ marks of a table and each $e\in C$ some basis has $T\cap C=C\setminus e$: $r(C)=k-1$, every $(k-1)$-subset of C is a pattern of C and C is not; for $k\geq 3$ every two marks of C lie in a common basis, and C in none.*

(ii) For a sample of a family at a gauge α as in Proposition 47.1 and $S\subseteq\mathcal{E}$, the level $|T_{j,i}\cap S|$ of every basis draw of table j lies in $[d_j(S),r_j(S)]$ and the colevel $|S\setminus T'_{j,i}|$ of every cobasis draw in $[d_j^*(S),|S|-d_j(S)]$; the chance that no basis draw of table j reaches the largest level $r_j(S)$ of its window is $(1-p_j)^{s_j}$ with $p_j=\mathbb{P}_{j,\alpha}(|T\cap S|=r_j(S))>0$, and likewise at the floor $d_j(S)$ and for the cobasis draws at the ends of their window, with the cobasis law and t_j.

Proposition 47.4 (The exact laws of a sample of bases)**.** *Let a sample of a family be drawn at a gauge α as in Proposition 47.1, s_j basis and t_j cobasis draws of table j, with counts $\mathbf{m}$, and let B be a bunch with room R.*

(i) For a basis draw of table j, $\mathbb{P}_{j,\alpha}(T)=\pi_{j,p}(\alpha)\,\mathbb{P}_{j,(p),\alpha_R}(T\setminus p)$ with $p=T\cap B$, and for a cobasis draw $\mathbb{P}^*_{j,\alpha}(T)=\pi^*_{j,p}(\alpha)\,\mathbb{P}^*_{j,(p),\alpha_R}(T\setminus p)$, with the pattern probabilities $\pi_{j,p}(\alpha)=\alpha_B^pK_{j,(p)}(\alpha_R)/K_j(\alpha)$ and $\pi^*_{j,p}(\alpha)=\alpha_B^{B\setminus p}\Psi_{j,(p)}(\alpha_R)/\Psi_j(\alpha)$ (Theorem 12.8) and the basis and cobasis laws of the child (j,p) at α_R. So the numbers $a_{j,p}$ of the basis draws and $b_{j,p}$ of the cobasis draws of table j that meet B in p are independent multinomial vectors with the chances $\pi_{j,p}(\alpha)$ and $\pi^*_{j,p}(\alpha)$; they are shares of B, $|\mathbf{a}_j|=s_j$ and $|\mathbf{b}_j|=t_j$ (Definition 12.24);

the counts on the bunch are their loads, $m_e = \mathrm{load}_e$ for $e \in B$; and given them the parts in the room are a sample of the children $F_B(\mathbf{a}, \mathbf{b})$ at α_R, $a_{j,p}$ basis and $b_{j,p}$ cobasis draws of the child (j,p), with the counts $\mathbf{m}|_R$. The mean shares are $s_j \pi_{j,p}(\alpha)$ and $t_j \pi^*_{j,p}(\alpha)$, at a native gauge the native split (Definition 12.25).

(ii) Given $\mathbf{m}$, each sample with these counts has the probability θ_{draws} divided by the coefficient of $\alpha^{\mathbf{m}}$ in $\prod_j K_j(\alpha)^{s_j} \Psi_j(\alpha)^{t_j}$, free of the gauge.

(iii) For every nonempty proper S the wall κ_S of the sample (Proposition 47.1(iii)) is the sum of the independent excesses of the levels of its basis draws and of the colevels of its cobasis draws over their floors, with the generating function

$$\mathbb{E}\big[x^{\kappa_S}\big] = \prod_j \mathsf{Q}^K_{j,S}(x;\alpha)^{s_j}\, \mathsf{Q}^{\Psi}_{j,S}(x;\alpha)^{t_j}$$

of the Skeleton polynomials (Definition 12.10). Its values fill $\{0, 1, \dots, \sum_j (s_j + t_j) c_j(S)\}$; its mean is the excess $\sum_j \big(s_j R_j(\alpha) + t_j R^*_j(\alpha)\big)_S - \sum_j \big(s_j d_j(S) + t_j d^*_j(S)\big)$ of the usage at α over the least use, and its cumulants of order $k \geq 2$ are $\partial^k_S \Phi(\log \alpha)$, $\partial_S = \sum_{e \in S} \partial_{y_e}$ (Proposition 12.12); it vanishes, every draw sitting at the floor of its window on S, with the chance $\prod_j \mathsf{Q}^K_{j,S}(0;\alpha)^{s_j} \mathsf{Q}^{\Psi}_{j,S}(0;\alpha)^{t_j} > 0$; and when the Skeleton spectra of S are positive it is distributed as a sum of independent $\{0,1\}$-valued variables whose means are the coins, s_j copies of the coins of $\mathsf{Q}^K_{j,S}$ and t_j of those of $\mathsf{Q}^{\Psi}_{j,S}$ (Proposition 12.11(iv)).

Proposition 47.5 (Tests that do not see the gauge)**.** *For a family F draw a sample at a gauge α as in Proposition 47.1, with counts $\mathbf{m}$, and let $\Theta_F(\mathbf{m})$ be the coefficient of $\alpha^{\mathbf{m}}$ in $\prod_j K_j(\alpha)^{s_j} \Psi_j(\alpha)^{t_j}$, the total of θ_{draws} over the set $\mathfrak{F}_F(\mathbf{m})$ of the samples with these counts.*

(i) Given $\mathbf{m}$, the sample has the law $\theta_{\mathrm{draws}}/\Theta_F(\mathbf{m})$ on $\mathfrak{F}_F(\mathbf{m})$, free of the gauge; for a family of lists it is uniform, and $|\mathfrak{F}_F(\mathbf{m})| = \Theta_F(\mathbf{m})$.

(ii) A test of F that rejects on an event A whose samples with counts $\mathbf{m}$ carry at most $\varepsilon\, \Theta_F(\mathbf{m})$ of the weight θ_{draws}, for every $\mathbf{m}$, has size at most ε at every gauge.

(iii) If every basis of every table of a family F' on $\mathcal{E}$ is a basis of the same table of F, with the same weight, then given $\mathbf{m}$ the chance under F that every draw is a basis of its table in F' is $p_{F'|F}(\mathbf{m}) = \Theta_{F'}(\mathbf{m})/\Theta_F(\mathbf{m})$, and rejecting F when this happens and $p_{F'|F}(\mathbf{m}) \leq \varepsilon$ has size at most ε at every gauge; one draw that is a basis of its table in F and not in F' rejects F' without error.

(iv) If each family of a collection has a test as in (ii), the families not rejected contain the true family with probability at least $1 - \varepsilon$, at every gauge.

Corollary 47.6 (The evidence of a family)**.** *Let every family F of a collection on $\mathcal{E}$, each with N tables, carry as prior the law of its centre of strengths $\boldsymbol{\sigma}_0 = (\sigma_{0,j})_j$, the question $(\sigma_0 \mathbf{1}; \boldsymbol{\sigma}_0, \boldsymbol{\sigma}_0)$ with $\sigma_0 = \sum_j \sigma_{0,j}$, inside its walls. For a sample as in Proposition 47.5, with the sizes $(s_j, t_j)_j$ and counts $\mathbf{m}$, the probability of the sample under F with the gauge integrated out is*

$$\mathrm{evidence}(F) = \theta^F_{\mathrm{draws}}\, \mathbf{1}[\textit{every draw is a basis of its table of } F]\, \frac{P_F(\sigma_0 \mathbf{1} + \mathbf{m}; \boldsymbol{\sigma}_0 + \mathbf{s}, \boldsymbol{\sigma}_0 + \mathbf{t})}{P_F(\sigma_0 \mathbf{1}; \boldsymbol{\sigma}_0, \boldsymbol{\sigma}_0)},$$

$\theta^F_{\mathrm{draws}}$ the product of the weights in F of the drawn bases, 1 for lists; and if the family is drawn from the collection with the chances $p(F)$, the posterior chance of F is proportional to $p(F)\, \mathrm{evidence}(F)$.

A sample of bases asks the family a question: the counts are its masses and the sample sizes its exponents, and a wall of the question is how far the sample's use of a set exceeds the least use any basis makes of it, a sum of the independent excesses of the levels and colevels of the draws over their floors, whose generating function is the product of the Skeleton polynomials of the set raised to the sizes. Read at a bunch, a sample is a sample of the children at the shares it realizes: the loads of these shares are the counts of the bunch, and their means at a native gauge are the native split. The law of a question is the conjugate prior of the tilt, so the posterior, the predictive chances and the evidence of a family are ratios of periods, and a prior inside its walls stays inside them. Given the counts, a sample has a law that does not see the gauge, uniform for a family of lists, so a test built on the samples with those counts has a size that no gauge can change.

47.2 Samples of tilts

Proposition 47.7 (A sample of tilts: the period is the cumulant function)**.** *Let a question* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *of pair coordinates* $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ *have every proper wall positive, let* $Z_1,\dots,Z_q$ *be the tilts of independent draws from* $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, and let* $\bar Z$*,* $\bar E_j$*,* $\bar O_j$ *be the sample means of* Z_i*,* $E_j(Z_i)$*,* $O_j(Z_i)$*. In this statement* $\mathbb{E}$ *is the expectation of* $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ *and* Z *is its tilt.*

(i) The sample has the density

$$2^{q(n-1)}\exp\Big(q\Big[2\langle\boldsymbol{\mu}_\perp,\bar Z\rangle-\sum_j\big(\sigma_j\bar E_j+\delta_j\bar O_j\big)\Big]\Big)\Big/P_F(\mathbf{m};\mathbf{s},\mathbf{t})^q$$

on $\mathcal{T}^q$*, so* $(\bar Z;\bar E_j,\bar O_j)_j$ *is sufficient for the question.*

(ii) The means are $\mathbb{E}[Z]=\frac12\nabla_{\boldsymbol{\mu}_\perp}\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*,* $\mathbb{E}[E_j(Z)]=-\partial_{\sigma_j}\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $\mathbb{E}[O_j(Z)]=-\partial_{\delta_j}\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*, the derivatives of Definition 17.1, and the likelihood equations equate them to the sample means.*

(iii) Against a second question $(\mathbf{m}';\mathbf{s}',\mathbf{t}')$ *of pair coordinates* $(\boldsymbol{\mu}'_\perp;\boldsymbol{\sigma}',\boldsymbol{\delta}')$ *the logarithm of the likelihood ratio is*

$$q\Big[2\langle\boldsymbol{\mu}'_\perp-\boldsymbol{\mu}_\perp,\bar Z\rangle-\sum_j\big((\sigma'_j-\sigma_j)\bar E_j+(\delta'_j-\delta_j)\bar O_j\big)\Big]-q\log\frac{P_F(\mathbf{m}';\mathbf{s}',\mathbf{t}')}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}.$$

(iv) The sums $q\bar Z$*,* $q\bar E_j$*,* $q\bar O_j$ *have as Laplace transform that of Corollary 17.2 raised to the power* q*.*

Definition 47.8 (The Foreman's question; the pseudo-MGF)**.** Let $\omega=W\,d\alpha_{\mathbf{m}}$ have real data on a connected family, $s_j,t_j\ge0$ not all zero and $\mathbf{m}>0$, with target $\boldsymbol{\tau}$. The *Foreman's question* of ω is the question $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ at its target, of pair coordinates $\big(\boldsymbol{\tau}-\frac{\tau\varepsilon}{n}\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\delta}\big)$; off the pole hyperplanes its period $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ is the pseudointegral $\wr W\,d\alpha_{\mathbf{m}}$ (Definition 16.10(iii)). Where the target lies on no pole hyperplane and $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})\neq0$, the *pseudo-MGF* of ω is

$$\mathsf{M}_\omega(c;\mathbf{a},\mathbf{b})=\frac{P_F\big(\boldsymbol{\tau}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)}{P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})}\qquad(c\in\mathbf{1}^\perp\otimes\mathbb{C},\ \mathbf{a},\mathbf{b}\in\mathbb{C}^N),$$

continued meromorphically (Lemma 16.9), and the *pseudo-cumulants* of ω are the derivatives of $\log\mathsf{M}_\omega$ at 0: the derivatives of $\log P_F$ at the Foreman's question in its pair coordinates.

Proposition 47.9 (The pseudo-MGF inside the walls)**.** *In Definition 47.8 let ω have a native gauge, and let Z be the tilt of $U \sim \mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})$, the Foreman's law of Proposition 16.34(i); the law of Z is the law of the Foreman's question.*

(i) Wherever the real parts of the shifted data $\big(\boldsymbol{\tau}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)$ have every proper wall positive, a region containing a neighbourhood of 0, $\mathsf{M}_{\omega}(c;\mathbf{a},\mathbf{b})$ is the moment generating function of the statistics $(2Z;-E_j(Z),-O_j(Z))_j$ under the Foreman's law (Corollary 17.2 at the Foreman's question); the pseudo-cumulants are their joint cumulants.

(ii) For a balanced shift $(\dot{\mathbf{m}};\dot{\mathbf{s}},\dot{\mathbf{t}})$ the function $U^{\dot{\mathbf{m}}}\prod_j K_j(U)^{-\dot{s}_j}\Psi_j(U)^{-\dot{t}_j}$ has degree 0, a function of the ray of U, and where the shifted data have every proper wall positive its expectation under the Foreman's law is $P_F(\boldsymbol{\tau}+\dot{\mathbf{m}};\mathbf{s}+\dot{\mathbf{s}},\mathbf{t}+\dot{\mathbf{t}})/P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ (the case of Proposition 47.22(i) below at the Foreman's question); $\mathsf{M}_{\omega}(c;\mathbf{a},\mathbf{b})$ is this ratio at the shift of pair coordinates $(c;\mathbf{a},\mathbf{b})$, so M_{ω} is the moment generating function of $(\log U;-\log K_j(U),-\log\Psi_j(U))_j$ along the balanced shifts.

(iii) The normaliser is $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t}) = P_{\mathrm{c}}\,\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$ (Theorem 18.1(i)). The centre ω_{c} has the target $\sigma\mathbf{1}$ and the Foreman's question $(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, and the pseudoexpectation is the pseudo-MGF of the centre at the departure:

$$\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big] = \mathsf{M}_{\omega_{\mathrm{c}}}\Big(\boldsymbol{\tau}-\frac{\tau_{\mathcal{E}}}{n}\mathbf{1};\ \mathbf{0},\ \boldsymbol{\delta}\Big) = \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[e^{2\langle\boldsymbol{\tau}-\frac{\tau_{\mathcal{E}}}{n}\mathbf{1},Z\rangle-\sum_j\delta_jO_j(Z)}\Big],$$

Z the tilt of U: at fixed strengths the pseudoexpectations of all the questions are values of one moment generating function, that of $(2Z;-O_j(Z))_j$ under the law of the centre.

Past the walls M_{ω} is the continuation, as the moments of the distribution of Theorem 16.17(i) are, and $\mathsf{M}_{\omega}(0)=1$ throughout.

Proposition 47.10 (Two layers: the word evaluated at the draws)**.** *Let a family have real data with $\mathbf{m}>0$, and let $U_1,\dots,U_q$ be independent draws from $\mathrm{Dir}_{\mathbf{m}}$, the shapes $\alpha/|\alpha|$ of $\alpha\sim\mathrm{Gamma}_{\mathbf{m}}$ (Proposition 8.12(ii)).*

(i) By Proposition 8.12(i), $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)] = P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$, and it is finite exactly when every proper wall at $(\mathbf{m};\mathbf{s},\mathbf{t})$ is positive.

(ii) $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)^k] = P_F(\mathbf{m};k\mathbf{s},k\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$, and the sample mean $\overline{W}_q=\frac{1}{q}\sum_i W(U_i)$ has the mean $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ and the variance

$$\frac{1}{q}\left[\frac{P_F(\mathbf{m};2\mathbf{s},2\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}-\left(\frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}\right)^2\right],$$

finite exactly when every proper wall at $(\mathbf{m};2\mathbf{s},2\mathbf{t})$ is positive.

(iii) Where every proper wall at $(\mathbf{m};\mathbf{s},\mathbf{t})$ is positive, $\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ has the density $W\,\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ with respect to $\mathrm{Dir}_{\mathbf{m}}$, and under it a second word $W'=\prod_j K_j^{-s'_j}\Psi_j^{-t'_j}$ of the same tables has $\mathbb{E}_{U\sim\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})}[W'(U)^k] = P_F(\mathbf{m};\mathbf{s}+k\mathbf{s}',\mathbf{t}+k\mathbf{t}')/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$.

Definition 47.11 (The Grey law)**.** For a family F and masses $\mathbf{m} > 0$ the *Grey law* $\mathrm{Grey}(F;\mathbf{m})$ is the joint law of

$$(G_j, G_j^*)_j = \big(-\log K_j(U),\, -\log \Psi_j(U)\big)_j \qquad \text{for } U \sim \mathrm{Dir}_{\mathbf{m}};$$

it depends on the tables of F and on $\mathbf{m}$, and not on the exponents.

Proposition 47.12 (The Grey law and its exponential family)**.** *Let $\mathbf{m} > 0$ and let $(G_j, G_j^*)_j \sim \mathrm{Grey}(F;\mathbf{m})$.*

(i) For complex exponents $(\mathbf{s},\mathbf{t})$ whose real parts keep every proper wall at $\mathbf{m}$ positive,

$$\mathbb{E}_{(G_j,G_j^*)_j \sim \mathrm{Grey}(F;\mathbf{m})}\Big[e^{\sum_j (s_j G_j + t_j G_j^*)}\Big] = \frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}} \quad \textit{(Proposition 8.12(i),(iii))}:$$

on real exponents the moment generating function, on imaginary exponents the characteristic function; the joint cumulants are the derivatives of $\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ in the exponents at $(\mathbf{0},\mathbf{0})$.

(ii) For real exponents with every proper wall positive, under $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ the vector $\big(-\log K_j(U), -\log \Psi_j(U)\big)_j$ has the law

$$e^{\sum_j (s_j g_j + t_j g_j^*)}\, \frac{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}\, \mathrm{Grey}(F;\mathbf{m})(dg\, dg^*):$$

over the exponents these laws form the exponential family with base $\mathrm{Grey}(F;\mathbf{m})$ and cumulant function $\log\big(P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\big)$.

(iii) For a real direction $(\mathbf{a},\mathbf{b}) \neq 0$ of the exponents let $g(S) = \sum_j \big(a_j d_j(S) + b_j d_j^*(S)\big)$ and $k^* = \min\{m_S/g(S) :\ S \text{ nonempty proper},\ g(S) > 0\}$, the value of λ at which the ray $\lambda(\mathbf{a},\mathbf{b})$, $\lambda > 0$, first meets a wall at $\mathbf{m}$, with $k^* = \infty$ if no $g(S)$ is positive. If $k^* < \infty$,

$$\mathbb{P}\Big(\sum_j \big(a_j G_j + b_j G_j^*\big) > x\Big) = e^{-k^* x + o(x)} \qquad (x \to \infty),$$

and if $k^* = \infty$ the sum is bounded above.

(iv) For q independent copies the sum has the Laplace transform $\big(P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\big)^q$, and the sample mean $\overline{W}_q$ of Proposition 47.10(ii) is the empirical counterpart of the transform of (i) at the exponents $(\mathbf{s},\mathbf{t})$.

Corollary 47.13 (The Grey law at the target)**.** *Let a connected family carry real exponents $s_j, t_j \geq 0$, not all zero, and masses $\mathbf{m} > 0$ with a native gauge and the target $\boldsymbol{\tau}$. Then*

$$\mathbb{E}_{U\sim \mathrm{Dir}_{\boldsymbol{\tau}}}[W(U)] = \mathbb{E}_{(G_j,G_j^*)_j \sim \mathrm{Grey}(F;\boldsymbol{\tau})}\Big[e^{\sum_j (s_j G_j + t_j G_j^*)}\Big] = \frac{P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\boldsymbol{\tau}\\ \tau_{\mathcal{E}}\end{bmatrix}}$$

$$= \frac{P_{\mathrm{c}}}{\Gamma_1^n\begin{bmatrix}\boldsymbol{\tau}\\ \tau_{\mathcal{E}}\end{bmatrix}}\, \mathsf{E}_{U\sim \mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big]:$$

the moment generating function of the Grey law at the target, read at the exponents, is the pseudoexpectation times the period of the centre divided by $\Gamma_1^n\begin{bmatrix}\boldsymbol{\tau}\\ \tau_{\mathcal{E}}\end{bmatrix}$. *With* $\boldsymbol{\tau}$ *held at the target, the logarithmic derivatives of* $(\mathbf{s}',\mathbf{t}')\mapsto P_F(\boldsymbol{\tau};\mathbf{s}',\mathbf{t}')$ *at* $(\mathbf{s},\mathbf{t})$ *are the joint cumulants of* $\big(-\log K_j(U),-\log\Psi_j(U)\big)_j$ *under the Foreman's law* $\mathrm{Breta}^\circ(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})$.

Proposition 47.14 (A Laplace transform with a moving target)**.** *Let a connected family carry real exponents* $s_j,t_j\geq 0$, *not all zero, and masses* $\mathbf{m}>0$ *whose target* $\boldsymbol{\tau}$ *has every proper wall positive, and let* $g(S)=\sum_j\big(s_jd_j(S)+t_jd_j^*(S)\big)$ *be the set function of Theorem 8.19. With the tails* U_k *of the orders* π *of* $\mathcal{E}$ *(Lemma 8.3),*

$$\lim_{\lambda\to 0^+}\mathbb{E}_{V\sim\mathrm{Dir}_{\lambda\boldsymbol{\tau}}}\Big[\prod_j K_j(V)^{-\lambda s_j}\Psi_j(V)^{-\lambda t_j}\Big]=\frac{\sum_\pi\prod_{k=2}^n\big(\tau_{U_k}-g(U_k)\big)^{-1}}{\sum_\pi\prod_{k=2}^n\tau_{U_k}^{-1}}\ \geq\ 1,$$

with equality exactly when $g(S)=0$ *for every nonempty proper* S. *The limit is the ratio of the limits of Theorem 17.5(i) for the question of data* $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ *and for the list* $\binom{\mathcal{E}}{1}$ *at a full basis lean at the masses* $\boldsymbol{\tau}$, *whose walls are the* τ_S *(Proposition 16.3).*

The word read at Dirichlet draws is a random variable whose expectation, as a function of the exponents, is the period divided by the stacked Gamma of the masses over their total: the Laplace transform of the Grey law of the family. At fixed masses the Grey law is the base of an exponential family over the exponents, and its tail along a direction of exponents is set by the first wall met along that direction. Read at the target of each exponent, as the normaliser $\binom{\mathcal{E}}{1}$ of Proposition 16.35(ii) reads it, the expectation of the word is not a moment generating function in the exponents: its limit at the origin exceeds 1 as soon as the word has a floor on a proper set, since at each exponent it reads the Grey law at another target. The pseudo-MGF fixes the target once, at the Foreman's question, and the pseudoexpectation fixes the law once for given strengths, at the centre.

Proposition 47.15 (The exponent is a dial between the mean and the maximum)**.** *In Proposition 47.10 let the exponents be real, not all zero, and put* $Y_i=\log W(U_i)$ *and* $Y=\log W(U)$ *for* $U\sim\mathrm{Dir}_{\mathbf{m}}$; *in this statement* $\mathbb{E}$ *is the expectation of independent* $U,U_1,\dots,U_q\sim\mathrm{Dir}_{\mathbf{m}}$. *Let* $g(S)=\sum_j\big(s_jd_j(S)+t_jd_j^*(S)\big)$, $k^*=\min\{m_S/g(S):\ S$ *nonempty proper,* $g(S)>0\}$, *with* $k^*=\infty$ *if no* $g(S)$ *is positive, and for* $k>0$ *let* $A_k=\frac{1}{k}\log\big(\frac{1}{q}\sum_i e^{kY_i}\big)$.

(i) A_k *is nondecreasing in* k, *tends to the sample mean of the* Y_i *as* $k\to 0$ *and to* $\max_i Y_i$ *as* $k\to\infty$, *and* $\max_i Y_i-\frac{\log q}{k}\leq A_k\leq\max_i Y_i$.

(ii) For $k<k^*$, $\mathbb{E}[e^{kA_k}]=P_F(\mathbf{m};k\mathbf{s},k\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$, *and* e^{kA_k} *has a finite variance exactly when* $2k<k^*$.

(iii) For $k\geq k^*$, $\mathbb{E}[e^{kY}]=\infty$; *and* $\mathbb{P}(\max_i Y_i\leq x)=\mathbb{P}(Y\leq x)^q$.

(iv) If $k^*<\infty$, $\mathbb{P}(Y>x)=e^{-k^*x+o(x)}$ *as* $x\to\infty$, *and if* $k^*=\infty$, Y *is bounded above (Proposition 47.12(iii)). For* $s_j,t_j\geq 0$ *and a set* S *attaining* k^* *the scale* $\rho=|U_S|$ *has* $(\rho,1-\rho)\sim\mathrm{Dir}_{(m_S,m_{\mathcal{E}\setminus S})}$, *so* ρ *has density proportional to* $\rho^{m_S-1}(1-\rho)^{m_{\mathcal{E}\setminus S}-1}$, *and* $W(U)\geq\prod_j K_j(\mathbf{1})^{-s_j-t_j}\rho^{-g(S)}$.

Proposition 47.16 (The even and odd statistics from a centre)**.** *Let a family carry the centre of strengths* $\boldsymbol{\sigma}=(\sigma_j)_j$, $\sigma_j\geq 0$, *inside its walls, let* Z *be the tilt of* $U\sim\mathrm{Breta}^\circ(F;\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, *with density proportional to* $e^{-\sum_j\sigma_jE_j(z)}$ *on* $\mathcal{T}$, *and let* $\ell^*_{\boldsymbol{\sigma}}$ *be the slope of Proposition 17.7. In this statement* $\mathbb{E}$ *is the expectation of* $U\sim\mathrm{Breta}^\circ(F;\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$.

(i) $\sum_j \sigma_j E_j(Z) \geq 2\sum_j \sigma_j \log K_j(\mathbf{1})$, *with equality only at* $Z = 0$ *(Proposition 17.4), where* $\sum_j \sigma_j E_j$ *has the Hessian* $8\sum_j \sigma_j\big(\kappa_{j,\mathbf{1}}(X_e, X_f)\big)_{e,f}$, *rung two of the still of the realization* $\big((\sigma_j,\sigma_j)_j;\mathbf{1}\big)$ *of the active tables, eight times the covariance of the indicators under the weight laws; and each* $E_j(Z) \geq 2\log K_j(\mathbf{1})$, *with equality exactly where* Z *is constant on the molecules of table* j. *For real* $\mathbf{k} = (k_j)_j$,

$$\mathbb{E}\Big[e^{-\sum_j k_j E_j(Z)}\Big] = \frac{P_F\big((\sigma + |\mathbf{k}|)\mathbf{1}; \boldsymbol{\sigma} + \mathbf{k}, \boldsymbol{\sigma} + \mathbf{k}\big)}{P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})},$$

the ratio of the periods of the centres of strengths $\boldsymbol{\sigma} + \mathbf{k}$ *and* $\boldsymbol{\sigma}$, *finite exactly when* $\sum_j (\sigma_j + k_j)c_j(S) > 0$ *for every nonempty proper* S: *along the even direction* $\mathbf{k} = \lambda\boldsymbol{\sigma}$ *every wall closes at once, at* $\lambda = -1$, *and along any other even direction* $\lambda\mathbf{k}$, $\lambda > 0$, *the first wall closes at the least of the ratios* $\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})/(-\kappa_S^{\mathbf{E}}(\mathbf{k}))$ *with* $\kappa_S^{\mathbf{E}}(\mathbf{k}) < 0$.

(ii) $\big(Z, (E_j(Z))_j, (O_j(Z))_j\big)$ *and* $\big(-Z, (E_j(Z))_j, (-O_j(Z))_j\big)$ *have one law, so every joint moment of odd total degree in* Z *and the* $O_j(Z)$ *vanishes. For real* $c \in \mathbf{1}^\perp$ *and* $\mathbf{b} \in \mathbb{R}^N$, $\mathbb{E}\big[e^{2\langle c, Z\rangle - \sum_j b_j O_j(Z)}\big]$ *is finite exactly when* $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) > 1$ *(Proposition 17.7(i)), where it is*

$$P_F\big(\mathbf{m}_F(c; \boldsymbol{\sigma}, \mathbf{b});\, \boldsymbol{\sigma} + \mathbf{b},\, \boldsymbol{\sigma} - \mathbf{b}\big)/P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma}),$$

for $|b_j| \leq \sigma_j$ *the pseudoexpectation of the question of pair coordinates* $(c; \boldsymbol{\sigma}, \mathbf{b})$. *If* $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) < \infty$, *both tails of* $Y = 2\langle c, Z\rangle - \sum_j b_j O_j(Z)$ *have the rate* $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b})$: $\mathbb{P}(\pm Y > x) = e^{-\ell^*_{\boldsymbol{\sigma}}(c,\mathbf{b})x + o(x)}$ *as* $x \to \infty$; *if it is infinite,* Y *is bounded (Proposition 17.8).*

(iii) *At* $\alpha = e^{2z}$, $K_j(\alpha)^{-s_j} = e^{-s_j\langle \mathbf{1}, z\rangle} e^{-\frac{s_j}{2}(E_j(z) + O_j(z))}$ *and* $\Psi_j(\alpha)^{-t_j} = e^{-t_j\langle \mathbf{1}, z\rangle} e^{-\frac{t_j}{2}(E_j(z) - O_j(z))}$: *every letter is its table at a full lean,* $K_j^{-s_j}$ *at the pair coordinates* $(\sigma, \delta) = (\frac{s_j}{2}, \frac{s_j}{2})$ *and* $\Psi_j^{-t_j}$ *at* $(\frac{t_j}{2}, -\frac{t_j}{2})$, *with the masses shifted by* $-\frac{s_j}{2}\mathbf{1}$, *resp.* $-\frac{t_j}{2}\mathbf{1}$; *a family is the family of its letters, and its pair coordinates are the sums of theirs.*

Proposition 47.17 (What the two statistics are)**.** *For table* j *of a family and a tilt* z, $\log \Lambda_j(z) = \log \sum_T \theta_{j,T} e^{\langle \chi_T, z\rangle}$ *is a soft maximum and* $-\log \Lambda_j(-z)$ *a soft minimum of the scores* $\langle \chi_T, z\rangle$ *of its bases, weighted by* $\theta_{j,T}$.

(i) $E_j(z)$ *is the soft maximum less the soft minimum, and* $O_j(z)$ *is their sum; the centred potential* $\Phi^\circ = \sum_j (\sigma_j E_j + \delta_j O_j)$ *is the sum over the tables of* s_j *times the soft maximum less* t_j *times the soft minimum.*

(ii) *With* $Y_j = \langle \chi_T, z\rangle$ *for* T *under the weight law of table* j, *near* $z = 0$

$$E_j(z) = 2\log K_j(\mathbf{1}) + \sum_{k\ even} \frac{2}{k!}\,\kappa_{j,\mathbf{1}}(Y_j^{\times k}),$$
$$O_j(z) = \sum_{k\ odd} \frac{2}{k!}\,\kappa_{j,\mathbf{1}}(Y_j^{\times k}),$$

so $E_j(z) = 2\log K_j(\mathbf{1}) + \kappa_{j,\mathbf{1}}(Y_j, Y_j) + \frac{1}{12}\kappa_{j,\mathbf{1}}(Y_j^{\times 4}) + \cdots$ *and* $O_j(z) = 2\kappa_{j,\mathbf{1}}(Y_j) + \frac{1}{3}\kappa_{j,\mathbf{1}}(Y_j^{\times 3}) + \cdots$, *with* $\kappa_{j,\mathbf{1}}(Y_j) = \sum_e (2R_{j,e}(\mathbf{1}) - 1)z_e$ *and, for* $k \geq 2$, $\kappa_{j,\mathbf{1}}(Y_j^{\times k}) = 2^k \sum_{w \in \mathcal{E}^k} \kappa_{j,\mathbf{1}}(w)\, z^w$, $z^w = z_{w_1} \cdots z_{w_k}$. *Summed with the strengths and the leans, the terms of order* $k \geq 2$ *of* Φ° *are* $\frac{1}{k!}\sum_{w \in \mathcal{E}^k} 2^k \sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w)\, z^w$, *which for the exponents of a realization* $\Upsilon = ((s_j, t_j)_j; \mathbf{1})$ *is* $\frac{1}{k!}\sum_w \kappa_\Upsilon(w)\, z^w$, *rung* k *of its still (Proposition 9.7): the even statistics carry the even rungs and the odd statistics the odd rungs.*

(iii) *As* $\lambda \to \infty$, $E_j(\lambda z)/\lambda \to \mathrm{wd}_j(z)$, *the width of Proposition 17.4, and* $O_j(\lambda z)/\lambda \to h_j(z) - h_j(-z)$, *the maximum of the scores being at the greedy basis* $T^+_{j,\pi}$ *and the minimum at* $T^-_{j,\pi}$ *for an order* π *along which the coordinates of* z *decrease (Lemma 8.3). So* $\Phi^\circ(\lambda z)/\lambda \to \langle 2v_\pi - 2\sigma\mathbf{1}, z\rangle$ *with* $v_\pi = \sum_j \big(s_j \mathbf{1}_{T^+_{j,\pi}} + t_j \mathbf{1}_{\mathcal{E}\setminus T^-_{j,\pi}}\big)$, *for* $s_j, t_j \geq 0$ *a vertex of the Newton sum at which* $\langle \cdot, z\rangle$ *is largest, the* greedy vertex *of* π*; and for a question the exponent of Proposition 17.4 has* $X(\lambda z)/\lambda \to -2\kappa(z) = 2\langle \mathbf{m} - v_\pi, z\rangle$ *on* $\mathbf{1}^\perp$*: the walls of the tails of* π *are the tail sums of the masses less the greedy vertex,* $\kappa_{U_k} = m_{U_k} - (v_\pi)_{U_k}$.

Proposition 47.18 (The laws of the questions are an exponential family)**.** *Let a connected family be such that* 1, *the coordinates* z_e, $e \neq g$, *of a chart* $z_g = 0$ *of* $\mathcal{T}$ *and the statistics* E_j, O_j *are linearly independent functions on* $\mathcal{T}$. *Then the laws of its questions with every proper wall positive form a regular exponential family of dimension* $n-1+2N$, *with natural parameter* $(2\boldsymbol{\mu}_\perp; -\boldsymbol{\sigma}, -\boldsymbol{\delta})$, *statistic* $(z; E_j(z), O_j(z))_j$ *and cumulant function* $\log P_F\big(\mathbf{m}_F(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta}); \boldsymbol{\sigma} + \boldsymbol{\delta}, \boldsymbol{\sigma} - \boldsymbol{\delta}\big)$. *For the sample means* $(\bar{Z}; \bar{E}_j, \bar{O}_j)_j$ *of* q *independent tilts, as in Proposition 47.7:*

(i) $(\bar{Z}; \bar{E}_j, \bar{O}_j)_j$ *is sufficient, and a function of it with expectation zero under every law of the family vanishes almost surely;*

(ii) *it is the unbiased estimator of least variance of the mean parameters*

$$\Big(\frac{1}{2}\nabla_{\boldsymbol{\mu}_\perp} \log P_F;\ -\partial_{\sigma_j} \log P_F,\ -\partial_{\delta_j} \log P_F\Big)_j,$$

with covariance $\frac{1}{q}$ *times the Hessian of* $\log P_F$ *in the natural parameter;*

(iii) *the likelihood equations have a solution, unique, exactly when* $(\bar{Z}; \bar{E}_j, \bar{O}_j)_j$ *is interior to the convex hull of the image of* $z \mapsto (z; E_j(z), O_j(z))_j$*;*

(iv) *for* $\delta_j \leq \delta_0$ *against* $\delta_j > \delta_0$, *with the other parameters unknown, rejecting for small* $\bar{O}_j$ *given the other coordinates of the sample means is uniformly most powerful among unbiased tests, with a conditional null law free of the other parameters; likewise for* σ_j *and* $\bar{E}_j$*;*

(v) *the law of the sums* $q\bar{Z}$, $q\bar{E}_j$, $q\bar{O}_j$ *has the Laplace transform of Proposition 47.7(iv), so the power of every test is a computation in the family.*

When the complement of every basis of table j *is a basis with the same weight,* $O_j \equiv 0$ *and the statements hold with* δ_j *removed.*

A word read at Dirichlet draws is a dial: its power means run from the mean of the logarithms to their maximum as the exponent grows, and they keep a finite expectation up to the first wall met along the ray of exponents, where the tail of the logarithm takes over with the rate of that wall. At a centre the centred potential $\sum_j \sigma_j E_j$ is least at the table's gauge and its exponential moments along the strengths close every wall at once, while the odd statistics are jointly symmetric and every odd direction has both tails at the rate of its slope. The two statistics of a table are the soft maximum and the soft minimum of the scores of its bases, read as their difference and their sum: at the table's gauge their Taylor terms are the rungs of the still, the even ones carried by the strengths and the odd ones by the leans, and at infinity they read the greedy vertex of the Newton sum, whose tail sums against the masses are the walls. The laws of the questions are a regular exponential family in the tilt and these statistics, so the classical optimal estimators and tests are computations with periods.

47.3 Draws, orders and races

Proposition 47.19 (A sorted draw lives on the sector of its order)**.** *Let a family have real data on its balance wall with every proper wall positive, and let* $U \sim \mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$*. For an order* π *of* $\mathcal{E}$ *let* κ_k *be the walls of its tails and* I_π *its sector integral (Corollary 8.6), so that* $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_\pi I_\pi$ *(Theorem 8.5(ii)).*

(i) $\mathbb{P}\big(U_{\pi(1)} > \cdots > U_{\pi(n)}\big) = I_\pi / P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*.*

(ii) Given that order, the gaps $u_k = \log\big(U_{\pi(k-1)}/U_{\pi(k)}\big)$*,* $2 \le k \le n$*, have the density*

$$I_\pi^{-1} \exp\Big(-\sum_{k=2}^{n} \kappa_k u_k\Big) \prod_j k_{j,\pi}(e^{-u})^{-s_j}\, \psi_{j,\pi}(e^{-u})^{-t_j}$$

on $[0,\infty)^{n-1}$*, with the sector letters* $k_{j,\pi}, \psi_{j,\pi}$ *of Theorem 8.5(ii).*

(iii) $\theta_{j,T^+_{j,\pi}} \le k_{j,\pi} \le K_j(\mathbf{1})$ *and* $\theta_{j,T^-_{j,\pi}} \le \psi_{j,\pi} \le K_j(\mathbf{1})$ *on the cube, so the density of (ii) lies between constant multiples of the density of independent exponential gaps of the rates* κ_k*; for* $s_j, t_j \ge 0$*,* I_π *lies between* $\prod_j K_j(\mathbf{1})^{-s_j-t_j} \prod_k \kappa_k^{-1}$ *and* $\prod_j \theta_{j,T^+_{j,\pi}}^{-s_j}\, \theta_{j,T^-_{j,\pi}}^{-t_j} \prod_k \kappa_k^{-1}$*.*

(iv) On the sector of π *the largest monomial* U^T *over the bases of table* j *is at its greedy basis* $T^+_{j,\pi}$ *and the smallest at* $T^-_{j,\pi}$ *(Lemma 8.3): the best and the worst basis of a draw are functions of its order, and the best basis of table* j *at* U *is* T *with probability* $\sum_{\pi:\, T^+_{j,\pi}=T} I_\pi / P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*.*

Proposition 47.20 (An ordered sample and a cut)**.** *Let* $Y_1, \dots, Y_q$ *be independent with a probability law* ϕ *on* $\mathbb{R}$*, an input (Definition 5.4) that charges no point.*

(i) The ordered sample has the law $q!\,\phi^{\otimes q}$ *on the region* $\{y_1 < \cdots < y_q\}$*.*

(ii) The number of observations above a cut N *is binomial with* q *trials and the chance* $\phi((N,\infty))$*, which for* ϕ *living on* $(0,\infty)$ *and* $N \ge 0$ *is the Trailmoment* $(\mathsf{T}_{0;(1)}[x]\,\phi)((N,\infty))$ *at index* 0*; given that it is* d*, the ordered observations above the cut have the law* $d!\,\phi^{\otimes d}/\phi((N,\infty))^d$ *on the trail region* $X_N = \{N < y_1 < \cdots < y_d\}$*.*

(iii) If ϕ *lives on* $(0,\infty)$ *and* $N \ge 0$*, the empirical input* $\hat\phi = \frac{1}{q}\sum_i \delta_{Y_i}$ *has* $(\mathsf{T}_{\nu;(1)}[x]\,\hat\phi)((N,\infty)) = \frac{1}{q}\sum_{Y_i > N} Y_i^{-\nu}$*, with the mean* $(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ *and the variance*

$$\frac{1}{q}\Big[(\mathsf{T}_{2\nu;(1)}[x]\,\phi)((N,\infty)) - (\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))^2\Big].$$

Proposition 47.21 (The races and the dilation relation)**.** *For table* j *of a family and a tilt* z*, with* $\alpha = e^{2z}$*, the laws of the winners of the basis race and the cobasis race (Theorem 18.5) are* $\mathbb{P}_{j,\alpha}$ *and* $\mathbb{P}^*_{j,\alpha}$*; let*

$$\mathrm{Ent}_j(z) = -\sum_T \mathbb{P}_{j,\alpha}(T) \log \frac{\mathbb{P}_{j,\alpha}(T)}{\theta_{j,T}}, \qquad \mathrm{Ent}^*_j(z) = -\sum_T \mathbb{P}^*_{j,\alpha}(T) \log \frac{\mathbb{P}^*_{j,\alpha}(T)}{\theta_{j,T}}$$

be their entropies relative to the weights.

(i) $E_j(z) = \langle z, \nabla E_j(z)\rangle + \mathrm{Ent}_j(z) + \mathrm{Ent}^*_j(z)$ *and* $O_j(z) = \langle z, \nabla O_j(z)\rangle + \mathrm{Ent}_j(z) - \mathrm{Ent}^*_j(z)$*, with*

$$\langle z, \nabla E_j(z)\rangle = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}\big[\langle \chi_T, z\rangle\big] - \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}\big[\langle \chi_T, z\rangle\big],$$
$$\langle z, \nabla O_j(z)\rangle = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}\big[\langle \chi_T, z\rangle\big] + \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}\big[\langle \chi_T, z\rangle\big].$$

Let the family be connected and let a question $(\mathbf{m};\mathbf{s},\mathbf{t})$ *of pair coordinates* $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ *have every proper wall positive. In the rest of this statement* $\mathbb{E}$ *is the expectation of* $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, Z is its tilt and* $\mathbf{u}(\alpha)=\sum_j\big(s_jR_j(\alpha)+t_jR_j^*(\alpha)\big)$ *the usage.*

(ii) Dilation relation.

$$2\langle\boldsymbol{\mu}_\perp,\mathbb{E}[Z]\rangle-\sum_j\big(\sigma_j\mathbb{E}[E_j(Z)]+\delta_j\mathbb{E}[O_j(Z)]\big)=-(n-1)-\sum_j\big(s_j\mathbb{E}[\mathrm{Ent}_j(Z)]+t_j\mathbb{E}[\mathrm{Ent}_j^*(Z)]\big);$$

at a centre, $\sum_j\sigma_j\mathbb{E}[E_j(Z)]=n-1+\sum_j\sigma_j\mathbb{E}[\mathrm{Ent}_j(Z)+\mathrm{Ent}_j^*(Z)]$.

(iii) $\mathrm{Cov}\big(\mathbf{u}_e(U),Z_f\big)=\frac12\big(\mathbf{1}[e=f]-\frac1n\big)$*, Z read on* $\mathbf{1}^\perp$.

(iv) $\log P_F$ *is convex in the pair coordinates of the question, so for a balanced direction* $v=(\dot{\mathbf{m}};\dot{\mathbf{s}},\dot{\mathbf{t}})$ *with the questions at the data* $(\mathbf{m}\pm\dot{\mathbf{m}};\mathbf{s}\pm\dot{\mathbf{s}},\mathbf{t}\pm\dot{\mathbf{t}})$ *inside their walls,*

$$\log\frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}{P_F(\mathbf{m}-\dot{\mathbf{m}};\mathbf{s}-\dot{\mathbf{s}},\mathbf{t}-\dot{\mathbf{t}})}\le D_v\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})\le\log\frac{P_F(\mathbf{m}+\dot{\mathbf{m}};\mathbf{s}+\dot{\mathbf{s}},\mathbf{t}+\dot{\mathbf{t}})}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$

bracketing each mean of Proposition 47.7(ii) by neighbouring values of the period.

Proposition 47.22 (Moments are steps; entropies are derivatives one column up)**.** *Let a connected family carry a question* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *with every proper wall positive, and in this statement let* $\mathbb{E}$ *be the expectation of* $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$.

(i) For a real balanced shift $(\dot{\mathbf{m}};\dot{\mathbf{s}},\dot{\mathbf{t}})$ *with every proper wall of the shifted question positive,*

$$\mathbb{E}\Big[U^{\dot{\mathbf{m}}}\prod_j K_j(U)^{-\dot{s}_j}\Psi_j(U)^{-\dot{t}_j}\Big]=\frac{P_F(\mathbf{m}+\dot{\mathbf{m}};\mathbf{s}+\dot{\mathbf{s}},\mathbf{t}+\dot{\mathbf{t}})}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}.$$

(ii) For a basis T of table j the columns $v_{j,T}=(\mathbf{1}_T;\mathbf{e}_j,\mathbf{0})$ *and* $v_{j,T}^*=(\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{0},\mathbf{e}_j)$ *are balanced, and for* $a\ge0$

$$\mathbb{E}\Big[\Big(\frac{\mathbb{P}_{j,U}(T)}{\theta_{j,T}}\Big)^a\Big]=\frac{P_F(\mathbf{m}+a\mathbf{1}_T;\,\mathbf{s}+a\mathbf{e}_j,\,\mathbf{t})}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$
$$\mathbb{E}\Big[\Big(\frac{\mathbb{P}^*_{j,U}(T)}{\theta_{j,T}}\Big)^a\Big]=\frac{P_F(\mathbf{m}+a\mathbf{1}_{\mathcal{E}\setminus T};\,\mathbf{s},\,\mathbf{t}+a\mathbf{e}_j)}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$

the walls only growing along a column.

(iii) With the entropies of Proposition 47.21,

$$\mathbb{E}[\mathrm{Ent}_j(Z)]=-\sum_T\theta_{j,T}\,\frac{(D_{v_{j,T}}P_F)(\mathbf{m}+\mathbf{1}_T;\,\mathbf{s}+\mathbf{e}_j,\,\mathbf{t})}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$
$$\mathbb{E}[\mathrm{Ent}_j^*(Z)]=-\sum_T\theta_{j,T}\,\frac{(D_{v^*_{j,T}}P_F)(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T};\,\mathbf{s},\,\mathbf{t}+\mathbf{e}_j)}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$

Z the tilt of U: an entropy is a derivative of the period one column up.

Corollary 47.23 (A derivative adds one finite sum)**.** *The series of Corollary 15.11 may be differentiated term by term along a balanced direction* $v=(\dot{\mathbf{m}};\dot{\mathbf{s}},\dot{\mathbf{t}})$ *at a point inside the walls, with* $\dot\kappa_S$ *the change of the walls; each factor of a term has an elementary logarithmic derivative,*

$$D_v\log\frac{(s_j)_{n_j}}{n_j!}=\dot s_j\sum_{i<n_j}\frac{1}{s_j+i},\qquad D_v\log\frac{(t_j)_{n'_j}}{n'_j!}=\dot t_j\sum_{i<n'_j}\frac{1}{t_j+i},$$

$$D_v \log K_j(\mathbf{1})^{-s_j - t_j} = -(\dot{s}_j + \dot{t}_j) \log K_j(\mathbf{1}),$$
$$D_v \log {}_1^2\Gamma \begin{bmatrix} \kappa_{U_k} + \beta_k,\ \gamma_k + 1 \\ \kappa_{U_k} + \beta_k + \gamma_k + 1 \end{bmatrix} = -\dot{\kappa}_{U_k} \sum_{i=0}^{\gamma_k} \frac{1}{\kappa_{U_k} + \beta_k + i},$$

so every quantity of Proposition 47.22 is the same positive series, at the question or one column up, with each term multiplied by a finite sum of reciprocals and logarithms of total weights.

Sorting a draw of the shape law lands it in one sector of the period: the chance of an order is the fraction of the period carried by its sector, the gaps of the sorted draw are exponential at the rates of the walls of the tails up to bounded factors, and the best and the worst basis of the draw are the greedy bases of its order. The races read the statistics as entropies, and the dilation relation is the balance between the strength of a question and the entropies of its winners. Every moment of a letter, of a winner's chance or of the shape is a step of the data, and every entropy is a derivative one column up, so on the closing series of the period each is a positive series whose terms carry one extra finite sum.

47.4 The interaction of a bunch

Proposition 47.24 (The interaction belongs to the room)**.** *Let a family have a bunch B, $|B| \geq 2$, with $\mathrm{pat}_j(B) = 2^B$ in every active table.*

(i) $\iota_{j,B}$ and $\iota^*_{j,B}$, and so $\iota^{\mathbf{E}}_B$ and $\iota^{\mathbf{O}}_B$, are homogeneous of degree 0 in α_R.

(ii) At a gauge α put, for $P \subseteq B$, $w(P) = \prod_j \pi_{j,P}(\alpha)^{s_j} \pi^*_{j,B\setminus P}(\alpha)^{t_j}$, with the pattern probabilities $\pi_{j,p}(\alpha) = \alpha_B^p K_{j,(p)}(\alpha_R)/K_j(\alpha)$ and $\pi^*_{j,p}(\alpha) = \alpha_B^{B\setminus p} \Psi_{j,(p)}(\alpha_R)/\Psi_j(\alpha)$ of Definition 12.25; for a sample as in Proposition 47.1 it is the chance that every draw uses exactly P on B, its basis draws meeting B in P and its cobasis draws missing exactly P. In the expansion $\log w(P) = \sum_{A \subseteq P} \lambda_A$ over the subsets of B, the top term is $\lambda_B = \iota^{\mathbf{E}}_B + \iota^{\mathbf{O}}_B$, and for $A \subsetneq B$ with $|A| \geq 2$, λ_A is the sum $\iota^{\mathbf{E}}_A + \iota^{\mathbf{O}}_A$ of the interaction of A in the face $F^{\mathcal{E}\setminus(B\setminus A)}$ of the wall at $B \setminus A$ (Theorem 16.25), whose tables are the restrictions $\mathcal{B}_j|\big(\mathcal{E} \setminus (B \setminus A)\big)$, the deletions of $B \setminus A$, at the exponents $(s_j, 0)$ and the contractions $\mathcal{B}_j/(B \setminus A)$ at $(0, t_j)$; none of them depends on α_B. For one table at the exponents $(1, 0)$, $w(P) = \pi_P(\alpha)$ and $\lambda_B = \iota_B$, the top term of the log-linear expansion of the pattern probabilities.

Theorem 47.25 (The interaction totals the joint cumulants over the bunch's scales)**.** *Let a family with real exponents have a bunch $B = \{e_1, \ldots, e_k\}$, $k \geq 2$, with $\mathrm{pat}_j(B) = 2^B$ in every active table. Then for both parts $\mathbf{X} \in \{\mathbf{E}, \mathbf{O}\}$*

$$\iota^{\mathbf{X}}_B(\alpha_R) = \int_{\mathbb{R}^B} \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \Phi_{\mathbf{X}}(y_B, \log \alpha_R)\, dy_B,$$

converging absolutely; at a realization the integrand of the sum of the parts is $2^{-k} \kappa_\Upsilon(e_1, \ldots, e_k) = \sum_j \big(s_j \kappa_{j,\alpha}(e_1, \ldots, e_k) + (-1)^k t_j \kappa_{j,\alpha^{-1}}(e_1, \ldots, e_k)\big)$, with κ_Υ the still at the gauge $\alpha = (e^{y_B}, \alpha_R)$ (Proposition 9.6). For one table at the exponents $(1, 0)$ the integrand is the joint cumulant of the indicators of the marks of B under the basis law at (e^{y_B}, α_R) (Definition 7.17), and the total is $\iota_B(\alpha_R)$.

Proposition 47.26 (Both sides, and parity)**.** *For a table with $\mathrm{pat}(B) = 2^B$, with $\Psi = \sum_p \alpha_B^{B\setminus p} \Psi_{(p)}(\alpha_R)$, as functions of the tilt z_R of the room $\iota^*_B(z_R) = (-1)^{|B|} \iota_B(-z_R)$. So for a*

family

$$\iota_B^{\mathbf{E}}(z_R) = \sum_j \sigma_j \big(\iota_{j,B}(z_R) + (-1)^{|B|}\iota_{j,B}(-z_R)\big), \qquad \iota_B^{\mathbf{O}}(z_R) = \sum_j \delta_j \big(\iota_{j,B}(z_R) - (-1)^{|B|}\iota_{j,B}(-z_R)\big) :$$

for $|B|$ even the even part of the interaction is even in z_R and the odd part is odd, for $|B|$ odd the reverse, and at the table's gauge of the room the interaction is $\big(2\sum_j \sigma_j\iota_{j,B}, 0\big)$ for $|B|$ even and $\big(0, 2\sum_j \delta_j\iota_{j,B}\big)$ for $|B|$ odd: the strengths read the even bunches and the leans the odd ones, as they read the even and the odd rungs of the still at the table's gauge (Proposition 9.7). The total of Theorem 47.25 for the logarithm $-\Phi$ of the family word is $-\sum_j \big(s_j\iota_{j,B}(z_R) + (-1)^{|B|}t_j\iota_{j,B}(-z_R)\big)$.

Proposition 47.27 (A design)**.** *Let $\mathbf{X} \in \mathbb{R}^{\mathcal{E}\times C}$ have full column rank, with rows x_e, and for $J \subseteq C$ let $K_J(\alpha) = \det\big(\mathbf{X}_J^{\mathsf{T}} \operatorname{diag}(\alpha)\mathbf{X}_J\big)$, $\mathbf{X}_J$ the columns in J, with $K_\varnothing = 1$.*

(i) The marks are the observations of the determinantal table with the slice $\mathbf{X}^{\mathsf{T}}$ (Definition 7.20) and the weights $\theta_T = \det(\mathbf{X}_{T,C})^2$, $|T| = |C|$: $K = K_C$, and the occupation of a mark is its leverage, $R_e(\alpha) = \alpha_e\, x_e^{\mathsf{T}}\big(\mathbf{X}^{\mathsf{T}} \operatorname{diag}(\alpha)\mathbf{X}\big)^{-1}x_e$.

(ii) Add one prior mark *γ^{p} for each column $\gamma \in C$, with the rows of the identity and the scales β: $K(\alpha,\beta) = \det\big(\mathbf{X}^{\mathsf{T}} \operatorname{diag}(\alpha)\mathbf{X} + \operatorname{diag}(\beta)\big) = \sum_{J\subseteq C} \beta^{C\setminus J} K_J(\alpha)$. The prior marks form a bunch C^{p} with $\operatorname{pat}(C^{\mathrm{p}}) = 2^{C^{\mathrm{p}}}$, and the child of the pattern q^{p}, $q \subseteq C$, is $K_{C\setminus q}$.*

(iii) For a split $C = J_1 \sqcup J_2$ the generalized variance inflation factor is $\frac{K_{J_1}K_{J_2}}{K_C K_\varnothing} = \exp\big(-\iota_{(J_1^{\mathrm{p}}, J_2^{\mathrm{p}})}\big)$, and for two columns γ, γ', one minus the squared partial correlation given the rest is $\exp(\iota_{\{\gamma^{\mathrm{p}},\gamma'^{\mathrm{p}}\}})$ in the table with the other prior marks deleted.

(iv) The dual multivector of the table of (ii) has the slice $[\, I_{\mathcal{E}} \mid -\mathbf{X}\,]$, and the basis polynomial of its dual table, $\det\big(\operatorname{diag}(\alpha') + \mathbf{X}\operatorname{diag}(\beta')\mathbf{X}^{\mathsf{T}}\big)$, is the polynomial of (ii) for the matrix $\mathbf{X}^{\mathsf{T}}$: the columns of $\mathbf{X}$ are its observations, at the scales β', and the marks of $\mathcal{E}$ its prior marks, at the scales α'. Its child with every prior mark deleted, $\det\big(\mathbf{X}\operatorname{diag}(\beta')\mathbf{X}^{\mathsf{T}}\big)$, is nonzero exactly when $\mathbf{X}$ has rank $|\mathcal{E}|$.

A bunch all of whose subsets are patterns sees the room through its children, one for each subset of the bunch, and the alternating sum of their logarithms is the interaction: it is homogeneous of degree zero in the room, it is the top term of the log-linear expansion of the chance that a sample uses one pattern of the bunch throughout, and it is the total, over the bunch's scales, of the joint cumulants of the use of the bunch's marks under the basis and cobasis laws, weighted by the exponents. The dual table reads the same interaction at the opposite tilt with the sign of the parity of the bunch, so a family reads the interaction of an even bunch through its strengths and that of an odd bunch through its leans, as it reads the rungs of the same parities. In a regression design the prior marks of the columns are such a bunch: the interactions are minus the logarithms of the generalized variance inflation factors and the logarithms of one minus the squared partial correlations, and the occupations are the leverages.

Part X

Proofs of the general forms

Chapter 48

Proofs of the general forms

This part proves the statements that are more general than their counterparts in the reference *The Foreman and the serif family* (16 October 2026), and those with no counterpart there; in the source file the comment before each such statement says so. Statements are cited by their numbers, and every symbol and term is that of the chapter the statement belongs to. A statement of the reference that is a special case of a statement proved here is not proved again; where reading it off needs a word, the word follows the proof. Lemmas that belong to this part alone are numbered S1, S2, ... The chapters are taken in order, one section each, and the last section records the checks run on small families.

48.1 Chapter 1: conventions, the stacked Gamma and the finite part

Lemma S1. *Let $\Theta \subset \mathbb{R}$ be finite and $\phi(y) = \sum_{\theta\in\Theta} a_\theta e^{i\theta y}$ with $a_\theta \in \mathbb{C}$. If $\phi(y) \to 0$ as $y \to -\infty$, then every a_θ is 0.*

Proof. Fix $\theta_0 \in \Theta$. For $Y > 0$,

$$\frac{1}{Y}\int_{-2Y}^{-Y} \phi(y)\, e^{-i\theta_0 y}\, dy = a_{\theta_0} + \sum_{\theta\neq\theta_0} a_\theta\, \frac{1}{Y}\int_{-2Y}^{-Y} e^{i(\theta-\theta_0)y}\, dy,$$

and each integral on the right has modulus at most $2/|\theta-\theta_0|$, so the sum is $O(Y^{-1})$. The left side has modulus at most $\sup_{y\le -Y}|\phi(y)|$, which tends to 0. Letting $Y \to \infty$ gives $a_{\theta_0} = 0$. □

Lemma S2. *Let $g(x) = \sum_{(k,j)\in F} c_{k,j}\, x^k \log^j x$ for $x \in (0,1)$, where $F \subset \mathbb{C}\times\mathbb{N}$ is finite and nonempty and every $c_{k,j}$ is nonzero. Let κ be the least real part of the exponents k, $(k,j) \in F$, and J the largest j such that $(k,j) \in F$ for some k with $\operatorname{Re} k = \kappa$. Then $g(x)/(x^\kappa \log^J x)$ does not tend to 0 as $x \to 0$.*

Proof. Put $y = \log x$, so that $y \to -\infty$ as $x \to 0$ and $x^k = e^{ky}$. Then

$$\frac{g(x)}{x^\kappa \log^J x} = \sum_\theta c_{\kappa+i\theta,J}\, e^{i\theta y} + \sum_{(k,j)} c_{k,j}\, e^{(k-\kappa)y}\, y^{j-J},$$

the first sum over the real θ with $(\kappa + i\theta, J) \in F$ and the second over the other elements of F. Every term of the second sum tends to 0: if $\operatorname{Re} k = \kappa$, then $j < J$, $|e^{(k-\kappa)y}| = 1$ and $|y|^{j-J} \to 0$;

if $\operatorname{Re} k > \kappa$, then $|e^{(k-\kappa)y} y^{j-J}| = e^{(\operatorname{Re} k-\kappa)y} |y|^{j-J} \to 0$. The first sum is a function $\phi(y)$ as in Lemma S1 with a nonzero coefficient, so it does not tend to 0 as $y \to -\infty$, and neither does the left side. □

Definition 1.5 (The finite part). As stated: the exponents of a log-power expansion are complex numbers, and its coefficients are determined by the function.

Proof. What is to be shown is that the coefficients of a log-power expansion are determined by h, so that the finite parts are well defined; the rest of the definition is that of the reference, with complex exponents.

(1) One remainder. Let two finite sums S_1 and S_2 of terms $c\,x^k \log^j x$, every exponent of real part below N, both differ from h by $O(x^N)$ as $x \to 0$. Their difference $g = S_1 - S_2$ is a finite sum of the same kind with $g(x) = O(x^N)$. If g had a nonzero coefficient, then, with $\kappa < N$ and J as in Lemma S2, $|g(x)/(x^\kappa \log^J x)| \le C x^{N-\kappa} |\log x|^{-J} \to 0$, which that lemma excludes. So S_1 and S_2 have the same coefficients.

(2) Two remainders. Let $N < N'$, and let S and S' be finite sums for N and for N'. A term $x^k \log^j x$ of S' with $\operatorname{Re} k \ge N$ is $O(x^{N-\delta})$ for every $\delta > 0$, so the part $S'_{<N}$ of S' with exponents of real part below N has $h - S'_{<N} = O(x^{N-\delta})$. Choose $\delta > 0$ smaller than $N - \operatorname{Re} k$ for every exponent k of S and of $S'_{<N}$. Then S and $S'_{<N}$ both differ from h by $O(x^{N-\delta})$, with all exponents of real part below $N - \delta$, so they coincide by (1). Hence the coefficient of a term $x^k \log^j x$ is the same for every $N > \operatorname{Re} k$, and the coefficient of $x^0 \log^0 x$, read with any $N > 0$, is determined by h. At ∞ the same holds for $h(1/x)$ at 0: the term $x^{-k} \log^j x$ is $(-1)^j x'^k \log^j x'$ in $x' = 1/x$, and $x^0 \log^0 x$ is $x'^0 \log^0 x'$.

(3) The primitive. Two primitives of f differ by a constant C, whose expansion at either end is the single term $C\,x^0 \log^0 x$; by (2) the finite parts of the second primitive at 0 and at ∞ are those of the first plus C. So $\operatorname{Reg}_{x\to\infty} h - \operatorname{Reg}_{x\to 0} h$ and $h(b) - \operatorname{Reg}_{x\to 0} h$ do not depend on the primitive. □

Proposition 1.6 (A convergent integral is its own finite part). *As stated, for log-power expansions with complex exponents.*

Proof. Let h be a primitive of f with log-power expansions at both ends. Since $\int_0^\infty f\,dx$ converges, $h(x) = h(1) - \int_x^1 f$ has a limit L_0 as $x \to 0$, h has a limit L_∞ as $x \to \infty$, and $\int_0^\infty f\,dx = L_\infty - L_0$. Take the expansion at 0 for some $N > 0$, and let G be its part with exponents of real part at most 0. The other terms and the remainder tend to 0, so $G(x) \to L_0$. Write $G - L_0$ as a finite sum of terms $c\,x^k \log^j x$, the constant $-L_0$ joining the coefficient of $x^0 \log^0 x$, and suppose it has a nonzero coefficient; let $\kappa \le 0$ and J be as in Lemma S2. If $(\kappa, J) \ne (0,0)$, then $|x^\kappa \log^J x| \to \infty$, so $(G - L_0)/(x^\kappa \log^J x) \to 0$. If $(\kappa, J) = (0,0)$, every exponent of $G - L_0$ has real part 0 and every power of the logarithm is 0, and $(G - L_0)/(x^0 \log^0 x) = G - L_0 \to 0$. Both are excluded by Lemma S2. So G is the constant L_0: the expansion at 0 has no term with an exponent of real part at most 0 other than $L_0\,x^0 \log^0 x$, and $\operatorname{Reg}_{x\to 0} h = L_0$. In the same way, for $h(1/x)$ at 0, $\operatorname{Reg}_{x\to\infty} h = L_\infty$. Hence $\operatorname{Reg} \int_0^\infty f\,dx = L_\infty - L_0 = \int_0^\infty f\,dx$. □

Theorem 1.7 (A finite part is a constant term). *As stated, for log-power expansions with complex exponents.*

Proof. The steps are those of the reference, proof of 1.7; they hold at complex exponents as follows.

The end at 0. On $(0,1]$ write $f = \sum c_{b,j} x^b \log^j x + r_N$, the sum over the finitely many terms of the expansion at 0 with $\operatorname{Re} b < N$, where $N > 0$ and $r_N = O(x^N)$ as $x \to 0$; r_N is integrable on $(0,1]$. For $b \neq -1$ put

$$Q_{b,j}(y) = \sum_{i=0}^{j} (-1)^{j-i} \frac{j!}{i!\,(b+1)^{j-i+1}}\, y^i;$$

then $(b+1)Q_{b,j} + Q'_{b,j} = y^j$, so $\frac{d}{dx}\big(x^{b+1} Q_{b,j}(\log x)\big) = x^b \log^j x$, and $Q_{b,j}(0) = (-1)^j j!/(b+1)^{j+1}$. Hence

$$\int_x^1 t^b \log^j t\, dt = Q_{b,j}(0) - x^{b+1} Q_{b,j}(\log x) \quad (b \neq -1), \qquad \int_x^1 t^{-1} \log^j t\, dt = -\frac{\log^{j+1} x}{j+1},$$

and $\int_x^1 r_N = \int_0^1 r_N - \int_0^x r_N$ with $\int_0^x r_N = O(x^{N+1})$. So $\int_x^1 f$ has a log-power expansion at 0, and its coefficient of $x^0 \log^0 x$ is $\sum_{b \neq -1} c_{b,j} Q_{b,j}(0) + \int_0^1 r_N$: a term $x^{b+1} \log^i x$ has the exponent $b+1 \neq 0$, and a term $\log^{j+1} x$ has a positive power of the logarithm. For a primitive h of f, $h(1) - \operatorname{Reg}_{x\to 0} h = \operatorname{Reg}_{x\to 0} \int_x^1 f$, so by 1.5

$$\operatorname{Reg} \int_0^1 f\, dx = \sum_{b \neq -1} c_{b,j} \frac{(-1)^j j!}{(b+1)^{j+1}} + \int_0^1 r_N\, dx.$$

On the other side, by $\int_0^1 x^{s-1} \log^j x\, dx = (-1)^j j!/s^{j+1}$ for $\operatorname{Re} s > 0$,

$$\int_0^1 x^\varepsilon f\, dx = \sum c_{b,j} \frac{(-1)^j j!}{(b+1+\varepsilon)^{j+1}} + \int_0^1 x^\varepsilon r_N\, dx$$

for $\operatorname{Re}\varepsilon$ large, the last term holomorphic on $\operatorname{Re}\varepsilon > -N-1$; this is the meromorphic continuation there. Its constant term at $\varepsilon = 0$ is the sum displayed above: a term with $b \neq -1$ gives its value, and a term with $b = -1$ is the pure pole $(-1)^j j!\,\varepsilon^{-j-1}$, of constant term 0.

The end at ∞. By $x \mapsto 1/x$, $\int_1^\infty x^\varepsilon f\, dx = \int_0^1 x^{-\varepsilon} x^{-2} f(1/x)\, dx$ and $\int_1^x f = \int_{1/x}^1 t^{-2} f(1/t)\, dt$; the function $x^{-2} f(1/x)$ has a log-power expansion at 0, the coefficient of $x^0 \log^0 x$ at ∞ is that of $x'^0 \log^0 x'$ at 0 in $x' = 1/x$, and a constant term at $\varepsilon = 0$ does not change under $\varepsilon \mapsto -\varepsilon$. So $\operatorname{Reg}_{x\to\infty} \int_1^x f = \operatorname{CT}_{\varepsilon=0} \int_1^\infty x^\varepsilon f\, dx$ by the end at 0.

Adding, $\operatorname{Reg} \int_0^\infty f\, dx = \operatorname{Reg}_{x\to\infty} h - \operatorname{Reg}_{x\to 0} h = \operatorname{Reg}_{x\to\infty} \int_1^x f + \operatorname{Reg}_{x\to 0} \int_x^1 f$ is the constant term of the sum of the two continuations. □

Corollary 1.8 (Finite parts at a pole of a Mellin integral). *As stated: for every f with a log-power expansion at* 0*, at every $a \in \mathbb{C}$.*

Proof. Fix a and $N > 1 - \operatorname{Re} a$, and write $f = \sum c_{k,j} x^k \log^j x + r_N$ on $(0,1]$, the finitely many terms of the expansion with $\operatorname{Re} k < N$, $r_N = O(x^N)$ at 0 and integrable on $[\delta, 1]$ for every $\delta > 0$. For $\operatorname{Re} a'$ large,

$$I(a') = \sum c_{k,j} \frac{(-1)^j j!}{(a'+k)^{j+1}} + \int_0^1 x^{a'-1} r_N\, dx,$$

the last term holomorphic on $\operatorname{Re} a' > -N$. This is the meromorphic continuation of I on $\operatorname{Re} a' > -N$; its poles lie at the points $-k$, a term $x^k \log^j x$ giving a pole of order at most $j+1$, and since N is arbitrary this describes the continuation on $\mathbb{C}$. Its constant term at $a' = a$ is

$$\operatorname*{CT}_{a'=a} I(a') = \sum_{a+k \neq 0} c_{k,j} \frac{(-1)^j j!}{(a+k)^{j+1}} + \int_0^1 x^{a-1} r_N\, dx,$$

a term with $a+k=0$ being the pure pole $(-1)^j j!\,(a'-a)^{-j-1}$; at an a that is not a pole it is $I(a)$. On the other side, $x^{a-1}f$ has the log-power expansion at 0 with the terms $c_{k,j}x^{a+k-1}\log^j x$ and the remainder $x^{a-1}r_N = O(x^{N+\operatorname{Re} a-1})$, $N+\operatorname{Re} a-1>0$. The end at 0 in the proof of 1.7 above, with $b=a+k-1$ (so $b\neq -1$ exactly when $a+k\neq 0$, and $b+1=a+k$), gives for $\operatorname{Reg}\int_0^1 x^{a-1}f\,dx$ the same sum. □

Lemma 1.9 (A finite part depends on the coordinate). *As stated: the dilation rule for every function with log-power expansions.*

Proof. (a) The finite part of $h(cx^p)$. Let $h(y)=\sum c_{k,j}\,y^{-k}\log^j y+O(y^{-N})$ as $y\to\infty$, finitely many terms with $\operatorname{Re} k<N$, so that $a_j=c_{0,j}$. For $y=cx^p$, $y^{-k}=c^{-k}x^{-pk}$ and $\log y=\log c+p\log x$, so

$$h(cx^p)=\sum c_{k,j}\,c^{-k}x^{-pk}\sum_{i=0}^{j}\binom{j}{i}(\log c)^{j-i}p^i\log^i x+O(x^{-pN}),$$

a log-power expansion at ∞ with the exponents pk, of real parts below pN; as N is arbitrary, so is pN. The term $x^0\log^0 x$ comes from $pk=0$, that is $k=0$, and $i=0$; by 1.5 its coefficient $\sum_j c_{0,j}(\log c)^j=\sum_j a_j(\log c)^j$ is $\operatorname{Reg}_{x\to\infty}h(cx^p)$. At 0 the computation is the same.

(b) The finite part of the integral. Let $F(x)=\int_1^x f$. At ∞, $f(t)=\sum f_{k,j}t^{-k}\log^j t+O(t^{-N})$ with $N>1$; integrating from 1 to x, a term with $k\neq 1$ gives $x^{1-k}Q_{-k,j}(\log x)$ up to a constant (with $Q_{b,j}$ as in the proof of 1.7 above), a term with $k=1$ gives $\log^{j+1}x/(j+1)$, and the remainder gives a constant plus $O(x^{1-N})$. So F has a log-power expansion at ∞, and for $i\geq 1$ its coefficient of $x^0\log^i x$ comes from the term $k=1$, $j=i-1$ alone: it is f^∞_{i-1}/i. In the same way, at 0 the coefficient of $x^0\log^i x$ in F is f^0_{i-1}/i for $i\geq 1$. Write A^∞_i and A^0_i for the coefficients of $x^0\log^i x$ of F at ∞ and at 0; $A^\infty_0-A^0_0=\operatorname{Reg}\int_0^\infty f\,dx$. The function $G(x)=F(cx^p)$ is a primitive of $f(cx^p)\,c\,p\,x^{p-1}$, and by (a) at both ends

$$\operatorname{Reg}\int_0^\infty f(cx^p)\,c\,p\,x^{p-1}dx=\sum_{i\geq 0}\big(A^\infty_i-A^0_i\big)(\log c)^i=\operatorname{Reg}\int_0^\infty f\,dx+\sum_{i\geq 1}\frac{f^\infty_{i-1}-f^0_{i-1}}{i}(\log c)^i,$$

which is the display with $i=j+1$. At $c=1$ the correction vanishes: a power of the coordinate leaves the finite part unchanged. □

Remark. The reference states the instance $f(x)=1/(1+cx)$, $c>0$. This f has the coefficient $f^\infty_0=1/c$ of x^{-1} at ∞ and no term $x^{-1}\log^j x$ at 0, and with the dilation c^{-1} and $p=1$ the lemma gives $\frac{1}{c}\operatorname{Reg}\int_0^\infty\frac{dx'}{1+x'}=\operatorname{Reg}\int_0^\infty\frac{dx}{1+cx}-\frac{\log c}{c}$. The left side is 0, the primitive $\log(1+x')=\log x'+\log(1+1/x')$ having no constant term at either end; so $\operatorname{Reg}\int_0^\infty\frac{dx}{1+cx}=\frac{\log c}{c}$, the value of the reference. A second check, with a logarithm: $f(x)=\log(1+x)/(1+x)$ has the primitive $\frac12\log^2(1+x)$, with no constant term at either end, so $\operatorname{Reg}\int_0^\infty f\,dx=0$; at ∞, $f=x^{-1}\log x+O(x^{-2}\log x)$, so $f^\infty_1=1$ and $f^\infty_0=0$, and at 0 it has no term $x^{-1}\log^j x$. The lemma gives $\operatorname{Reg}\int_0^\infty f(cx)\,c\,dx=\frac12\log^2 c$; directly, the primitive $\frac12\log^2(1+cx)=\frac12\big(\log c+\log x+\log(1+\frac{1}{cx})\big)^2$ has the constant term $\frac12\log^2 c$ at ∞ and none at 0.

48.2 Chapter 2: the Gramma and the Zamma

Proposition 2.3 (Affine changes of the logarithmic coordinates). *As stated: one statement for the reflection, the linear changes of the coordinates and the torus homogeneity.*

Proof. Let $A \in GL_{\mathcal{E}}(\mathbb{R})$ and $\mathbf{h} \in \mathbb{R}^{\mathcal{E}}$. The map $y' \mapsto y = Ay' + \mathbf{h}$ is a bijection of $\mathbb{R}^{\mathcal{E}}$ with $dy = |\det A|\, dy'$, and $\langle \nu, y\rangle = \langle A^{\mathsf{T}}\nu, y'\rangle + \langle \nu, \mathbf{h}\rangle$, $\langle t_k, y\rangle = \langle A^{\mathsf{T}}t_k, y'\rangle + \langle t_k, \mathbf{h}\rangle$. In the logarithmic form of 2.1,

$$e^{\langle \nu, y\rangle} \exp\Big(-\sum_k \xi_k e^{\langle t_k, y\rangle}\Big)\, dy = |\det A|\, e^{\langle \nu, \mathbf{h}\rangle}\, e^{\langle A^{\mathsf{T}}\nu, y'\rangle} \exp\Big(-\sum_k \big(e^{\langle t_k, \mathbf{h}\rangle}\xi_k\big)\, e^{\langle A^{\mathsf{T}}t_k, y'\rangle}\Big)\, dy'.$$

The moduli of the two sides are the same expressions with ν replaced by $\operatorname{Re}\nu$, so the integral of the left side converges exactly when that of the right side does. The left side is the integrand of $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ and the right side, up to the constant factor, that of $\mathsf{G}_{A^{\mathsf{T}}\nu;A^{\mathsf{T}}t}\big((e^{\langle t_k, \mathbf{h}\rangle}\xi_k)_k\big)$, with positive coefficients; by 2.2, $\nu \in \mathcal{T}_t$ exactly when $A^{\mathsf{T}}\nu \in \mathcal{T}_{A^{\mathsf{T}}t}$, that is $\mathcal{T}_{A^{\mathsf{T}}t} = A^{\mathsf{T}}\mathcal{T}_t$, and integrating gives the display. At $A = -I_{\mathcal{E}}$ and $\mathbf{h} = 0$, $|\det A| = 1$ and $A^{\mathsf{T}}t = -t$: the reflection. At $A = I_{\mathcal{E}}$ and $\mathbf{h} = \log\lambda$, $\lambda \in \mathbb{R}^{\mathcal{E}}_{>0}$, $e^{\langle t_k, \mathbf{h}\rangle} = \lambda^{t_k}$ and $e^{\langle \nu, \mathbf{h}\rangle} = \lambda^{\nu}$: the torus homogeneity. □

Theorem 2.5 (Product). *As stated, for any number of blocks.*

Proof. Write $x = (x^{(1)}, \dots, x^{(q)})$ with $x^{(i)} \in \mathbb{R}^{\mathcal{E}_i}_{>0}$. Each t_k is supported in one $\mathcal{E}_i$, so $x^{t_k} = (x^{(i)})^{t_k|\mathcal{E}_i}$ for that i, and

$$x^{\nu-\mathbf{1}} \exp\Big(-\sum_k \xi_k x^{t_k}\Big) = \prod_{i=1}^{q} (x^{(i)})^{\nu|\mathcal{E}_i - \mathbf{1}} \exp\Big(-\sum_{k\in K_i} \xi_k\, (x^{(i)})^{t_k|\mathcal{E}_i}\Big),$$

K_i the set of the terms whose exponents are supported in $\mathcal{E}_i$, the zero vectors counted in K_1. The moduli of the factors are positive functions of separate variables, so by Tonelli the modulus of the product is integrable exactly when every factor is integrable over $\mathbb{R}^{\mathcal{E}_i}_{>0}$; by 2.2 this says $\mathcal{T}_t = \mathcal{T}_{t^{(1)}} \times \cdots \times \mathcal{T}_{t^{(q)}}$, and Fubini gives the product. □

Theorem 2.8 (Scale and shape along sets). *As stated, for several disjoint sets at once.*

Proof. Apply 2.7 on each B_i: $x_{B_i} = \rho_i u_i$ with $\rho_i > 0$ and $u_i \in \Delta_{B_i}$, and $x_{B_i}^{\nu|_{B_i}-\mathbf{1}} dx_{B_i} = \rho_i^{\nu_{B_i}-1} u_i^{\nu|_{B_i}-\mathbf{1}}\, d\rho_i\, d\omega_{B_i}(u_i)$. With x_R the coordinates on R, the map $(x_R, (\rho_i)_i, (u_i)_i) \mapsto x$ is a bijection onto $\mathbb{R}^{\mathcal{E}}_{>0}$, and since $(\rho_i u_i)^{t_k|_{B_i}} = \rho_i^{\langle t_k, \mathbf{1}_{B_i}\rangle} u_i^{t_k|_{B_i}}$,

$$x^{\nu-\mathbf{1}}dx = x_R^{\nu|_R-\mathbf{1}} \prod_i \rho_i^{\nu_{B_i}-1} u_i^{\nu|_{B_i}-\mathbf{1}}\, d\rho\, d\omega(u)\, dx_R, \qquad \xi_k x^{t_k} = b_k(u, x_R) \prod_i \rho_i^{\ell_{k,i}}.$$

At $\nu \in \mathcal{T}_t$ the Gramma converges (2.2), that is, its integrand is absolutely integrable. By Tonelli and Fubini the integral may be taken with the integral over $\rho = (\rho_i)_i \in \mathbb{R}^q_{>0}$ innermost, that integral converging absolutely for almost every (u, x_R), and it is

$$\int_{\mathbb{R}^q_{>0}} \rho^{(\nu_{B_i})_i - \mathbf{1}} \exp\Big(-\sum_k b_k(u, x_R)\, \rho^{\ell_k}\Big)\, d\rho = \mathsf{G}_{(\nu_{B_i})_i;\ell}\big(\mathbf{b}(u, x_R)\big),$$

the Gramma of 2.1 on q scales with the configuration ℓ. When R is empty there are no coordinates x_R. □

Proposition 2.13 (First facts about the Zamma). *As stated: (i) at every invertible square configuration and (iv) on any number of scales; (ii) and (iii) are those of the reference.*

Proof. (i) By 2.6, $\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det t|^{-1}\prod_k \Gamma(b_k)\,\xi_k^{-b_k}$ on $\mathcal{T}_t = \{\nu : \operatorname{Re}\mathbf{b} \in (0,\infty)^N\}$; dividing by $\prod_e \Gamma(\nu_e)$ (2.12) gives the display, the quotient of Gammas being the stacked Gamma of 1.2, and the display carries the continuation. At the unit configuration $\mathbf{b} = \nu$ and the display is the monomial; on one scale with $t_1 > 0$, $|\det t| = t_1$ and $b_1 = \nu/t_1$, which is the formula of the reference.

(iv) Write $t'' = I_{\mathcal{E}} \oplus t'$ and $\boldsymbol{\xi}'' = \mathbf{1} \oplus \boldsymbol{\xi}'$. Since $\operatorname{cone}(t'') \supseteq \operatorname{cone}(I_{\mathcal{E}}) = [0,\infty)^{\mathcal{E}}$, every $\nu \in \mathbb{R}^{\mathcal{E}}_{>0}$ lies in $\operatorname{int}\operatorname{cone}(t'')$, hence in $\mathcal{T}_{t''}$. By 2.1,

$$\mathsf{G}_{\nu;t''}(\boldsymbol{\xi}'') = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\nu-\mathbf{1}} e^{-\sum_e x_e} \exp\Big(-\sum_k \xi'_k x^{t'_k}\Big)\,dx = \prod_e \Gamma(\nu_e)\,\mathbb{E}_{U\sim\mathrm{Gamma}_\nu}\Big[\exp\Big(-\sum_k \xi'_k U^{t'_k}\Big)\Big],$$

Gamma_ν having the density $\prod_e x_e^{\nu_e-1}e^{-x_e}/\Gamma(\nu_e)$ (1.1(vii)); dividing by $\prod_e \Gamma(\nu_e)$ gives the display. The variable $\exp(-\sum_k \xi'_k U^{t'_k})$ lies in $(0,1]$ and decreases in each ξ'_k, and so does its expectation. Put $X = \sum_k \xi'_k U^{t'_k} \geq 0$. Where every $\nu + t'_k$ lies in $\mathbb{R}^{\mathcal{E}}_{>0}$, the coordinates of U being independent, $\mathbb{E}[U^{t'_k}] = \prod_e \mathbb{E}[U_e^{t'_{k,e}}] = \prod_e \Gamma(\nu_e + t'_{k,e})/\Gamma(\nu_e)$, so $\mathbb{E}X = \sum_k \xi'_k \prod_e \Gamma_1\begin{bmatrix}\nu_e + t'_{k,e}\\ \nu_e\end{bmatrix}$ is finite. The function $s \mapsto e^{-s}$ is convex, so $e^{-s} \geq e^{-s_0} - e^{-s_0}(s - s_0)$ for all real s, s_0; at $s = X$ and $s_0 = \mathbb{E}X$, taking expectations, $\mathbb{E}[e^{-X}] \geq e^{-\mathbb{E}X}$, which is the bound. □

Proposition 2.14 (Powers of a polynomial in the scales). *As stated: (i) and (iii) for every Zamma; (ii) and (iv) are those of the reference.*

Proof. Write $\boldsymbol{\xi}(\rho) = \boldsymbol{\xi}^{(0)} + \boldsymbol{\Delta}(\rho)$ with $\Delta_k(\rho) = \sum_{l\neq 0} \rho^l \xi_k^{(l)}$, a polynomial vanishing at $\rho = 0$.

(1) Derivatives. Away from the unit configuration, $\mathsf{Z}_{\nu;t} = \mathsf{G}_{\nu;t}/\prod_e \Gamma(\nu_e)$ on $(0,\infty)^N$. By 2.4(i), $\partial^{\mathbf{w}}\mathsf{G}_{\nu;t} = (-1)^{|\mathbf{w}|}\mathsf{G}_{\nu+t\mathbf{w};t}$, every $\nu + t\mathbf{w}$ lying in $\mathcal{T}_t$ because $\operatorname{int}\operatorname{cone}(t) + \operatorname{cone}(t) \subseteq \operatorname{int}\operatorname{cone}(t)$; so $\mathsf{Z}_{\nu;t}$ is infinitely differentiable on $(0,\infty)^N$ and

$$\partial^{\mathbf{w}}\mathsf{Z}_{\nu;t} = (-1)^{|\mathbf{w}|}\frac{\mathsf{G}_{\nu+t\mathbf{w};t}}{\prod_e \Gamma(\nu_e)} = (-1)^{|\mathbf{w}|}\prod_e \Gamma_1\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}\,\mathsf{Z}_{\nu+t\mathbf{w};t},$$

the second form wherever no entry of $\nu + t\mathbf{w}$ lies in $-\mathbb{N}$, the product being read as the first form otherwise, as in the statement. At the unit configuration $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) = \boldsymbol{\xi}^{-\nu}$ (2.12), holomorphic near $\boldsymbol{\xi}^{(0)}$ with the principal branch, and $\partial^{\mathbf{w}}\boldsymbol{\xi}^{-\nu} = (-1)^{|\mathbf{w}|}\prod_e (\nu_e)_{w_e}\,\boldsymbol{\xi}^{-\nu-\mathbf{w}}$ with $(\nu_e)_{w_e} = \nu_e(\nu_e + 1)\cdots(\nu_e + w_e - 1)$.

(2) Taylor coefficients. Each monomial of $\boldsymbol{\Delta}(\rho)^{\mathbf{w}}$ has total degree at least $|\mathbf{w}|$ in ρ, every $l \neq 0$ having $|l| \geq 1$. So Taylor's formula for $\mathsf{Z}_{\nu;t}$ at $\boldsymbol{\xi}^{(0)}$ to the order $|q|$, whose remainder is $O(|\boldsymbol{\Delta}|^{|q|+1}) = O(|\rho|^{|q|+1})$, gives

$$c_q^\nu = \sum_{|\mathbf{w}|\leq|q|} \partial^{\mathbf{w}}\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}^{(0)})\,[\rho^q]\,\frac{\boldsymbol{\Delta}(\rho)^{\mathbf{w}}}{\mathbf{w}!}.$$

By the multinomial theorem, $\Delta_k^{w_k}/w_k! = \sum \prod_{l\neq 0} (\rho^l \xi_k^{(l)})^{n_{l,k}}/n_{l,k}!$, the sum over the $(n_{l,k})_l$ in $\mathbb{N}$ with $\sum_l n_{l,k} = w_k$; multiplying over k and taking the coefficient of ρ^q leaves the arrays $\mathbf{n}$ with $\sum_{l,k} l\,n_{l,k} = q$ and $\mathbf{w} = (\sum_l n_{l,k})_k$, $|\mathbf{w}| = |\mathbf{n}|$. With (1) this is (i). At one unit exponent, $t = (1)$, $\boldsymbol{\xi} = L$ and the index λ, the product in (i) is $\Gamma_1\begin{bmatrix}\lambda + |\mathbf{n}|\\ \lambda\end{bmatrix}(L^{(0)})^{-\lambda-|\mathbf{n}|}\prod_l (L^{(l)})^{n_l}/n_l!$, which is the formula of the reference.

(3) The coefficients as variables. The function $(\rho, (\xi_k^{(l)})) \mapsto \mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho))$ is infinitely differentiable, so derivatives in the coefficients commute with taking Taylor coefficients in ρ. Since $\partial \xi_k(\rho)/\partial \xi_k^{(l)} = \rho^l$, (1) with $\mathbf{w} = \mathbf{e}_k$ gives

$$\partial_{\xi_k^{(l)}} \mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho)) = -\rho^l \prod_e \Gamma_1 \begin{bmatrix} \nu_e + t_{k,e} \\ \nu_e \end{bmatrix} \mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}(\rho)),$$

and the coefficient of ρ^q is $-\prod_e \Gamma_1 \begin{bmatrix} \nu_e + t_{k,e} \\ \nu_e \end{bmatrix} c_{q-l}^{\nu+t_k}$, which is (iii). At $t = (1)$ the factor is $\Gamma_1 \begin{bmatrix} \lambda+1 \\ \lambda \end{bmatrix} = \lambda$, the formula of the reference. □

48.3 Chapter 3: the Breta

Proposition 3.4 (The Breta on the orthant). *As stated, for any data.*

Proof. By 3.1 and 2.12, with the dressed coefficients $\eta_k = u^{\pi_k}\xi_k$ and the bunch scales $\rho = (\rho_Q)_{Q\in\mathcal{P}}$,

$$\mathsf{Z}_{\boldsymbol{\nu};\ell}(\boldsymbol{\eta}) = \frac{1}{\prod_Q \Gamma(\nu_Q)} \int_{\mathbb{R}^{\mathcal{P}}_{>0}} \rho^{\boldsymbol{\nu}-\mathbf{1}} \exp\Big(-\sum_k \xi_k\, u^{\pi_k} \rho^{\ell_k} \Big)\, d\rho.$$

On each bunch, 2.7 with $x_Q = \rho_Q u_Q$ gives $x_Q^{\mathbf{m}|_Q - \mathbf{1}} dx_Q = \rho_Q^{m_Q-1} u_Q^{\mathbf{m}|_Q-\mathbf{1}} d\rho_Q\, d\omega_Q(u_Q)$, hence

$$u_Q^{\mathbf{m}|_Q-\mathbf{1}} \rho_Q^{\nu_Q-1}\, d\rho_Q\, d\omega_Q(u_Q) = x_Q^{\mathbf{m}|_Q-\mathbf{1}} |x_Q|^{\nu_Q-m_Q}\, dx_Q,$$

and each dressed term is

$$\xi_k\, u^{\pi_k}\rho^{\ell_k} = \xi_k \prod_Q \Big(\frac{x_Q}{|x_Q|}\Big)^{\pi_k|_Q} |x_Q|^{\ell_{k,Q}} = \xi_k\, x^{\pi_k} \prod_Q |x_Q|^{\ell_{k,Q}-\langle \pi_k, \mathbf{1}_Q\rangle}.$$

So, under the bijection $x \leftrightarrow (\rho_Q, u_Q)_Q$ of 2.7, the Breta with its Zamma written as the scale integral is the integral of the right side over $\mathbb{R}^{\mathcal{E}}_{>0}$. The moduli of the two integrands are the same functions with $\mathbf{m}$ and $\boldsymbol{\nu}$ replaced by their real parts, so, as in the reference, proof of 3.4, Tonelli gives that either side converges when the other does, and Fubini gives the equality. For homogeneous data the inner powers of the bunch totals are absent, and at $\nu_Q = m_Q$ so are the outer ones: the right side is then $\mathsf{G}_{\mathbf{m};t}(\boldsymbol{\xi})/\prod_Q \Gamma(m_Q)$, which is 3.2. □

Theorem 3.5 (Relations of the Breta). *As stated: (v) for any terms; (i) to (iv) are those of the reference.*

Proof. The proofs of (i) to (iv) in the reference, proof of 3.5, use no homogeneity. For (v), let $e \neq f$ be scales of the bunch Q, and in this proof write, with the products over the bunches Q' of $\mathcal{P}$,

$$F(x) = x^{\mathbf{m}-\mathbf{1}} \prod_{Q'} |x_{Q'}|^{\nu_{Q'}-m_{Q'}} \exp\Big(-\sum_k \xi_k \varphi_k(x) \Big), \qquad \varphi_k(x) = x^{\pi_k} \prod_{Q'} |x_{Q'}|^{\ell_{k,Q'}-\langle \pi_k, \mathbf{1}_{Q'}\rangle},$$

the integrand of 3.4 at $(\mathbf{m}, \boldsymbol{\nu})$ without the factor $1/\prod_{Q'} \Gamma(\nu_{Q'})$.

(1) Derivatives. For $e \in Q$, $x_e \partial_{x_e} |x_Q| = x_e$ and $x_e \partial_{x_e} |x_{Q'}| = 0$ for $Q' \neq Q$, so

$$x_e \partial_{x_e} F = F\Big[(m_e - 1) + (\nu_Q - m_Q)\frac{x_e}{|x_Q|} - \sum_k \xi_k \varphi_k \Big(\pi_{k,e} + \big(\ell_{k,Q} - \langle \pi_k, \mathbf{1}_Q \rangle\big)\frac{x_e}{|x_Q|}\Big)\Big],$$

and the same holds with f in place of e.

(2) The integrands. By 3.4: F is the integrand of $\prod_{Q'} \Gamma(\nu_{Q'})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi})$; $F\, x_e/|x_Q|$ is that of $\prod_{Q'} \Gamma(\nu_{Q'})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi})$, the power of $|x_Q|$ being $\nu_Q - (m_Q + 1)$; $F\varphi_k$ is that of $\prod_{Q'} \Gamma(\nu_{Q'} + \ell_{k,Q'})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})$, the power of each $|x_{Q'}|$ being $(\nu_{Q'} + \ell_{k,Q'}) - (m_{Q'} + \langle \pi_k, \mathbf{1}_{Q'} \rangle)$; and $F\varphi_k\, x_e/|x_Q|$ is that of $\prod_{Q'} \Gamma(\nu_{Q'} + \ell_{k,Q'})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_e,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})$. Here $\prod_{Q'} \Gamma(\nu_{Q'} + \ell_{k,Q'}) = g_k \prod_{Q'} \Gamma(\nu_{Q'})$. These Bretas, and those with f in place of e, are the Bretas of (v), which converge by hypothesis; so all these functions are integrable on $\mathbb{R}^{\mathcal{E}}_{>0}$.

(3) No boundary terms. By (1) and (2), $\partial_{x_e}(x_e F) = F + x_e \partial_{x_e} F$ is integrable. By Fubini, for almost every value x' of the coordinates other than x_e the function $G(s) = s\, F(x', s)$ of $s = x_e > 0$ has G' and $G(s)/s$ integrable on $(0, \infty)$. Then $G(s) = G(1) + \int_1^s G'$ has limits at 0 and at ∞, and both are 0: a nonzero limit L at one end would make $|G(s)/s| \geq |L|/(2s)$ near that end, which is not integrable there. So $\int_0^\infty G'(s)\, ds = 0$, and by Fubini $\int \partial_{x_e}(x_e F)\, dx = 0$; likewise for f. Hence

$$\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \big(x_e \partial_{x_e} - x_f \partial_{x_f}\big) F\, dx = \int \big[\partial_{x_e}(x_e F) - F\big] dx - \int \big[\partial_{x_f}(x_f F) - F\big] dx = 0.$$

(4) Collecting. By (1), $(x_e \partial_{x_e} - x_f \partial_{x_f})F$ is F times

$$(m_e - m_f) + (\nu_Q - m_Q)\frac{x_e - x_f}{|x_Q|} - \sum_k \xi_k \varphi_k \Big((\pi_{k,e} - \pi_{k,f}) + \big(\ell_{k,Q} - \langle \pi_k, \mathbf{1}_Q \rangle\big)\frac{x_e - x_f}{|x_Q|}\Big).$$

Integrating, using (2) and dividing by $\prod_{Q'} \Gamma(\nu_{Q'})$ gives (v). For homogeneous data $\ell_{k,Q} = \langle \pi_k, \mathbf{1}_Q \rangle$ and the second line of the right side of (v) is absent.

(5) The closing sentences. A term k raises the shape total of a bunch Q' by $\langle \pi_k, \mathbf{1}_{Q'} \rangle$ and its scale index by $\ell_{k,Q'}$, which are equal for homogeneous data; so the ladder, the scale exchange and the shifts $(\mathbf{m} + \pi_k, \boldsymbol{\nu} + \ell_k)$ of (v) keep the condition $\nu_Q = m_Q$, and in (v) the shifts $\mathbf{m} + \mathbf{e}_e$, $\mathbf{m} + \mathbf{e}_f$ of the shape index alone carry the factor $\nu_Q - m_Q$. The shape split and (iv) shift the shape index alone. □

Remark. Relation (v) was checked in exact arithmetic on one bunch of two scales with one term whose level is not the degree of its shape exponent, where the Breta has a closed form, for shape exponents $(1, 0)$, $(\frac{3}{2}, -\frac{1}{3})$ and symbolic (`sweep/s0107/breta_v.py`).

Proposition 3.6 (Letter presentation). *As stated, for a partition of the scales B into any number of bunches.*

Proof. (1) The presentation. The bunches of $\mathcal{P}$ are those of $\mathcal{Q}$, with the scales ρ_Q, and the singletons $\{v_i\}$, whose scales are written v_i and whose shapes are the point $u_{v_i} = 1$ of $\Delta_{\{v_i\}}$, with $\omega_{\{v_i\}}$ the Dirac measure (1.1(iii)). The dressed coefficients of the columns are $g_j u^{\pi_j}$ and $C_{i,p} u^{\pi_{i,p}}$, and $u^{(\mathbf{m},\mathbf{s})-\mathbf{1}} = u^{\mathbf{m}-\mathbf{1}}$. By 2.1, the Gramma on the bunch scales with the configuration $\widetilde{\ell}$ and these coefficients is

$$\int \rho^{\boldsymbol{\nu}-\mathbf{1}} \prod_i v_i^{s_i - 1} \exp\Big(-\sum_j g_j u^{\pi_j} \rho^{\lambda_j} - \sum_{i,p} C_{i,p} u^{\pi_{i,p}} \rho^{\ell_{i,p}} v_i\Big)\, d\rho\, dv,$$

the level of the pattern (i,p) on $\{v_i\}$ being 1. For $\operatorname{Re} s_i > 0$, integrating v_i gives $\Gamma(s_i)\,P_i(\rho,u)^{-s_i}$ (2.6 on one scale, $P_i > 0$). Dividing by $\prod_Q \Gamma(\nu_Q)\prod_i \Gamma(s_i)$ gives the Zamma (2.12), and integrating against $u^{\mathbf{m}-\mathbf{1}}d\omega(u)$ over $\prod_Q \Delta_Q$ gives $1/\prod_Q \Gamma(\nu_Q)$ times the left side of the display; Tonelli on the moduli and Fubini justify the order of the integrals, either side converging when the other does.

(2) The relations. They are 3.5(ii),(iii) for the terms $(\widetilde{\pi},\widetilde{\ell})$. For the scale exchange at $\{v_i\}$ the columns of nonzero level there are the patterns (i,p), of level 1, with the factor $\Gamma_1{}_1\begin{bmatrix} s_i+1 \\ s_i+1\end{bmatrix} = 1$ at $\{v_i\}$; the other singletons give $\Gamma_1{}_1\begin{bmatrix} s_{i'} \\ s_{i'}\end{bmatrix} = 1$ and each $Q \in \mathcal{Q}$ gives $\Gamma_1{}_1\begin{bmatrix} \nu_Q+\ell_{i,p,Q} \\ \nu_Q\end{bmatrix}$: this is the letter split. In the ladders the factor g_k of 3.5 is $\prod_Q \Gamma_1{}_1\begin{bmatrix} \nu_Q+\lambda_{j,Q} \\ \nu_Q\end{bmatrix}$ for a kernel column and $s_i\prod_Q \Gamma_1{}_1\begin{bmatrix} \nu_Q+\ell_{i,p,Q} \\ \nu_Q\end{bmatrix}$ for a pattern, by $\Gamma_1{}_1\begin{bmatrix} s_i+1 \\ s_i\end{bmatrix} = s_i$. The scale exchange on $Q \in \mathcal{Q}$ is 3.5(iii) multiplied by ν_Q, with $\nu_Q\Gamma_1{}_1\begin{bmatrix} \nu_Q+\lambda \\ \nu_Q+1\end{bmatrix} = \Gamma_1{}_1\begin{bmatrix} \nu_Q+\lambda \\ \nu_Q\end{bmatrix}$ (1.3(iii)), the singletons contributing the factors 1 for a kernel column and s_i for a pattern. □

Proposition 3.7 (Letter data on the orthant). *As stated, for any levels and any number of bunches.*

Proof. In the left side of the display of 3.6 apply 2.7 on each bunch $Q \in \mathcal{Q}$, $\alpha_Q = \rho_Q u_Q$: $u_Q^{\mathbf{m}|_Q-\mathbf{1}}\rho_Q^{\nu_Q-1}d\rho_Q\,d\omega_Q = \alpha_Q^{\mathbf{m}|_Q-\mathbf{1}}|\alpha_Q|^{\nu_Q-m_Q}d\alpha_Q$, and for $\lambda \in \mathbb{R}^{\mathcal{Q}}$, $\pi \in \mathbb{R}^B$, $\rho^\lambda u^\pi = \alpha^\pi\prod_Q|\alpha_Q|^{\lambda_Q-\langle\pi,\mathbf{1}_Q\rangle}$. So each kernel term becomes $g_j\alpha^{\pi_j}\prod_Q|\alpha_Q|^{\lambda_{j,Q}-\langle\pi_j,\mathbf{1}_Q\rangle}$ and each letter $P_i(\alpha)$. Tonelli on the moduli gives the equivalence of convergence and Fubini the equality, and 3.6 identifies the left side with $\prod_Q\Gamma(\nu_Q)$ times the Breta. For homogeneous letter data every exponent $\lambda_Q-\langle\pi,\mathbf{1}_Q\rangle$ is 0, and at $\nu_Q = m_Q$ so is $\nu_Q - m_Q$. □

Theorem 3.13 (The Shruffle). *As stated, for any number of bunches.*

Proof. Apply 3.12 to $A = A_1\sqcup\cdots\sqcup A_q$ with $J = \{1,\dots,q\}$: $u|_{A_i} = w_i v^{(i)}$ with $w \in \Delta_J$, $v^{(i)} \in \Delta_{A_i}$, and

$$u^{\mathbf{m}-\mathbf{1}}d\omega_A = \prod_i w_i^{m_{A_i}-1}\,d\omega_J(w)\prod_i (v^{(i)})^{\mathbf{m}|_{A_i}-\mathbf{1}}d\omega_{A_i}(v^{(i)}), \qquad \bigotimes_i f_i(u) = \prod_i f_i(v^{(i)}).$$

Scaling by $w_i > 0$ keeps the order of the coordinates of $u|_{A_i}$, which is that of $v^{(i)}$. Up to the null set where two coordinates of u are equal, the points u whose restrictions follow every ς_i form the disjoint union of the sets Δ_A^ς over the orders ς of A that restrict to every ς_i, and they are the images of $\Delta_J\times\prod_i\Delta_{A_i}^{\varsigma_i}$. The integrals converge absolutely, so by Fubini

$$\sum_\varsigma I_A^\varsigma\Big(\bigotimes_i f_i;\mathbf{m}\Big) = \int_{\Delta_J}\prod_i w_i^{m_{A_i}-1}\,d\omega_J(w)\;\prod_i I_{A_i}^{\varsigma_i}(f_i;\mathbf{m}|_{A_i})$$
$$= \Gamma_q{}_1\begin{bmatrix} m_{A_1},\ \dots,\ m_{A_q} \\ m_A\end{bmatrix}\prod_i I_{A_i}^{\varsigma_i}(f_i;\mathbf{m}|_{A_i}),$$

by 3.9 on J, where $\operatorname{Re} m_{A_i} > 0$. Dividing by the stacked Gamma, whose reciprocal is $\Gamma_1{}_q\begin{bmatrix} m_A \\ m_{A_1},\ \dots,\ m_{A_q}\end{bmatrix}$, gives the statement; at $q = 2$ it is the factor $\Gamma_1{}_2\begin{bmatrix} m_A+m_B \\ m_A,\ m_B\end{bmatrix}$ of the reference. □

Theorem 3.15 (Relations of the Beta port). *As stated: (iii) for linear letters of either slope and complex exponents, and (v) at every pair of exponents; (i), (ii) and (iv) are those of the reference.*

Proof. (iii) On $[0,1]$ every letter $L_i = c_i + d_i x$ is positive, since it is affine with the positive values c_i and $c_i + d_i$ at the ends, so $L_i^{-w_i}$ is infinitely differentiable there for every $w_i \in \mathbb{C}$, with $\partial_x^v L_i^{-w_i} = (-1)^v (w_i)_v d_i^v L_i^{-w_i - v}$, d_i real. Leibniz's rule gives

$$\partial_x^q \prod_i L_i^{-w_i} = \sum_{|v|=q} \frac{q!}{\prod_i v_i!} \prod_i (-1)^{v_i} (w_i)_{v_i} d_i^{v_i} L_i^{-w_i - v_i},$$

which is the first display. Then (ii), with $g = \prod_i L_i^{-w_i}$ and $\operatorname{Re} a, \operatorname{Re} b > 0$, gives the second display, by $x^{a+q-1}(1-x)^{b+q-1} = x^{a-1}(1-x)^{b-1} \prod_i (x(1-x))^{v_i}$ for $|v| = q$. Let $a, b > 0$ and $w_i \geq 0$. Each integrand on the right is $x^{a-1}(1-x)^{b-1} \prod_i L_i^{-w_i} > 0$ times $\prod_i (x(1-x) d_i / L_i)^{v_i}$, which is nonnegative if every $d_i \geq 0$, and is $(-1)^{|v|} = (-1)^q$ times a nonnegative function if every $d_i \leq 0$; the coefficients $\prod_i (w_i)_{v_i} / v_i!$ are nonnegative. This gives the two signs. If $x(1-x)|d_i|/L_i \leq \varepsilon$ on $[0,1]$ for every i, each integrand has modulus at most $\varepsilon^q x^{a-1}(1-x)^{b-1} \prod_i L_i^{-w_i}$, and

$$\sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!} = \frac{(|\mathbf{w}|)_q}{q!},$$

the coefficient of y^q on both sides of $\prod_i (1-y)^{-w_i} = (1-y)^{-|\mathbf{w}|}$, $|y| < 1$, by $(1-y)^{-w} = \sum_v \frac{(w)_v}{v!} y^v$. This gives the bound.

(v) Fix $q \geq 1$, write $s = a + b$, and first let $\operatorname{Re} a, \operatorname{Re} b > 0$.

(1) Coefficients. Expanding $x^k (1-x)^{q-k}$ in the second form of 3.14,

$$[x^j]\, \Pi_q^{a,b} = (-1)^{q+j} \sum_k \binom{a+q-1}{q-k} \binom{q-k}{q-j} \binom{b+q-1}{k}.$$

By $\binom{N}{M}\binom{M}{L} = \binom{N}{L}\binom{N-L}{M-L}$, a polynomial identity in N, at $N = a+q-1$, $M = q-k$, $L = q-j$, this is $(-1)^{q+j}\binom{a+q-1}{q-j} \sum_k \binom{a+j-1}{j-k}\binom{b+q-1}{k}$, and $\sum_k \binom{A}{j-k}\binom{B}{k} = \binom{A+B}{j}$, a polynomial identity in A and B that holds at nonnegative integers by counting the j-subsets of a disjoint union, gives $\binom{s+q+j-2}{j}$: the coefficient formula. At $j = q$, $\lambda_q := [x^q]\Pi_q^{a,b} = \binom{s+2q-2}{q} = (s+q-1)_q / q!$, which is nonzero since $\operatorname{Re} s > 0$. Also

$$\frac{[x^{q-1}]\Pi_q^{a,b}}{\lambda_q} = -\frac{q(a+q-1)}{s+2q-2}, \qquad \frac{[x^q]\Pi_{q+1}^{a,b}}{\lambda_{q+1}} = -\frac{(q+1)(a+q)}{s+2q}.$$

(2) Pairing. Put $\langle g, h \rangle_\circ = \int_0^1 x^{a-1}(1-x)^{b-1} g\, h\, dx$ for polynomials g, h. By (i) at $c = a + j$, $\langle \Pi_k^{a,b}, x^j \rangle_\circ = \binom{j}{k} \Gamma \begin{bmatrix} a+j,\ b+k \\ s+j+k \end{bmatrix}$, which is 0 for $j < k$ and $\beta_k := \Gamma \begin{bmatrix} a+k,\ b+k \\ s+2k \end{bmatrix} \neq 0$ for $j = k$.

(3) Span. Let g be a polynomial of degree at most $q+1$ with $\langle g, x^j \rangle_\circ = 0$ for $j \leq q-2$. Since $\deg \Pi_k^{a,b} = k$ by (1), $g = \sum_{k \leq q+1} \gamma_k \Pi_k^{a,b}$; pairing with $x^0, x^1, \ldots, x^{q-2}$ in turn, by (2) the pairing with x^j involves only $\gamma_0, \ldots, \gamma_j$, with γ_j multiplied by $\beta_j \neq 0$, so $\gamma_0 = \cdots = \gamma_{q-2} = 0$ and $g \in \operatorname{span}(\Pi_{q-1}^{a,b}, \Pi_q^{a,b}, \Pi_{q+1}^{a,b})$.

(4) Multiplication by x. $\langle x\Pi_q^{a,b}, x^j \rangle_\circ = \langle \Pi_q^{a,b}, x^{j+1} \rangle_\circ = 0$ for $j \leq q-2$, so by (3) $x\Pi_q^{a,b} = A\Pi_{q+1}^{a,b} + B\Pi_q^{a,b} + C\Pi_{q-1}^{a,b}$. The coefficients of x^{q+1} give $A = \lambda_q / \lambda_{q+1} = \frac{(q+1)(s+q-1)}{(s+2q)(s+2q-1)}$. Pairing

with x^{q-1} gives $\beta_q = C\beta_{q-1}$, so $C = \frac{(a+q-1)(b+q-1)}{(s+2q-1)(s+2q-2)}$. The coefficients of x^q give $[x^{q-1}]\Pi_q^{a,b} = A\,[x^q]\Pi_{q+1}^{a,b} + B\lambda_q$, so by (1) $B = \frac{(q+1)(a+q)}{s+2q} - \frac{q(a+q-1)}{s+2q-2}$.

(5) The operator $D = x(1-x)\partial_x$. For a polynomial g, integrating by parts, the boundary terms $x^{a+j}(1-x)^b g$ vanishing at both ends,

$$\langle Dg, x^j\rangle_\circ = \int_0^1 x^{a+j}(1-x)^b g'\,dx = -(a+j)\langle g, x^j\rangle_\circ + (s+j)\langle g, x^{j+1}\rangle_\circ,$$

which is 0 at $g = \Pi_q^{a,b}$ for $j \le q-2$. By (3), $D\Pi_q^{a,b} \in \operatorname{span}(\Pi_{q-1}^{a,b}, \Pi_q^{a,b}, \Pi_{q+1}^{a,b})$, and eliminating $\Pi_{q+1}^{a,b}$ by (4), $A \neq 0$, $D\Pi_q^{a,b} = (e\,x + e_0)\Pi_q^{a,b} + e_1\Pi_{q-1}^{a,b}$. Since $Dx^k = kx^k - kx^{k+1}$, the coefficients of x^{q+1} give $e = -q$. Pairing with x^{q-1}: the left side is $(s+q-1)\beta_q$ and the right side $-q\beta_q + e_1\beta_{q-1}$, so $e_1 = (s+2q-1)\beta_q/\beta_{q-1} = \frac{(a+q-1)(b+q-1)}{s+2q-2}$. The coefficients of x^q give $q\lambda_q - (q-1)[x^{q-1}]\Pi_q^{a,b} = -q[x^{q-1}]\Pi_q^{a,b} + e_0\lambda_q$, so $e_0 = q + [x^{q-1}]\Pi_q^{a,b}/\lambda_q = \frac{q(b+q-1)}{s+2q-2}$.

(6) Continuation. The three displays compare polynomials in x whose coefficients are rational functions of (a,b), the coefficients of the Beta ports being polynomials in (a,b). They agree on the open set $\operatorname{Re} a, \operatorname{Re} b > 0$, hence wherever the coefficients are finite, which is the condition $a+b+2q-i \neq 0$, $i = 0,1,2$; the coefficient formula is a polynomial identity and holds for all a, b and every $q \in \mathbb{N}$. □

Remark. At $a = b = n+1$ the middle coefficient of the three-term relation is $\frac{1}{2}$, and the relations become those of the reference: $(1-2x)\Pi_r^{n+1,n+1} = -2A\,\Pi_{r+1}^{n+1,n+1} - 2C\,\Pi_{r-1}^{n+1,n+1}$ and $x(1-x)\partial_x\Pi_r^{n+1,n+1} = \frac{r}{2}(1-2x)\Pi_r^{n+1,n+1} + \frac{r+n}{2}\Pi_{r-1}^{n+1,n+1}$, with the leading coefficient $\binom{2n+2r}{r}$, and at $a = b = 1$ the coefficient formula gives $(-1)^r\Pi_r^{1,1} = \sum_k (-1)^k\binom{r}{k}\binom{r+k}{k}x^k$. The three displays and the coefficient formula were checked in exact arithmetic with a and b symbolic for $q \le 9$, together with these diagonal forms (`sweep/s0107/betaport.py`).

48.4 Chapter 4: fibre integrals and closing

Theorem 4.3 (The native gauge of a configuration). *As stated: (iii) with a fixed offset ν' of the index; (i) and (ii) are those of the reference.*

Proof. By (i), $\Phi_{\boldsymbol{\xi}}$ is smooth and strictly convex with its minimum at $y^\star$, where its Hessian $H^\star = \sum_k w_k^\star t_k t_k^{\mathsf{T}}$ is positive definite.

(1) The integral. In the logarithmic form of 2.1,

$$\mathsf{G}_{\lambda\nu+\nu';t}(\lambda\boldsymbol{\xi}) = \int_{\mathbb{R}^{\mathcal{E}}} e^{\langle\nu',y\rangle}\,e^{-\lambda\Phi_{\boldsymbol{\xi}}(y)}\,dy,$$

which converges when $\lambda\nu+\nu' \in \mathcal{T}_t$ (2.2). Since $\operatorname{int}\operatorname{cone}(t)$ is an open cone containing ν, it contains $\nu + \lambda^{-1}\operatorname{Re}\nu'$ for λ large, so the integral converges for $\lambda \ge \lambda_0$, some $\lambda_0 > 0$.

(2) Away from $y^\star$. Fix $\delta > 0$ and put $\mu_\delta = \min_{|y-y^\star|=\delta}\big(\Phi_{\boldsymbol{\xi}}(y) - \Phi_{\boldsymbol{\xi}}(y^\star)\big) > 0$. For a unit vector d the function $\sigma \mapsto \Phi_{\boldsymbol{\xi}}(y^\star + \sigma d) - \Phi_{\boldsymbol{\xi}}(y^\star)$ is convex and vanishes at $\sigma = 0$, so for $\sigma \ge \delta$ it is at least $\sigma\mu_\delta/\delta$. With $|e^{\langle\nu',y\rangle}| \le e^{|\operatorname{Re}\nu'|\,|y^\star|}e^{|\operatorname{Re}\nu'|\,|y-y^\star|}$,

$$\Big|\int_{|y-y^\star|\ge\delta} e^{\langle\nu',y\rangle}e^{-\lambda(\Phi_{\boldsymbol{\xi}}(y)-\Phi_{\boldsymbol{\xi}}(y^\star))}\,dy\Big| \le e^{|\operatorname{Re}\nu'|\,|y^\star|}\int_{|u|\ge\delta} e^{-(\lambda\mu_\delta/\delta - |\operatorname{Re}\nu'|)|u|}\,du,$$

which, once $\lambda\mu_\delta/\delta \ge 2|\operatorname{Re}\nu'|$ and $\lambda\mu_\delta \ge 2\delta$, is at most a constant times $e^{-\lambda\mu_\delta/4}$: on $|u| \ge \delta$, $e^{-\Lambda|u|} \le e^{-\Lambda\delta/2}e^{-\Lambda|u|/2}$ for $\Lambda = \lambda\mu_\delta/\delta - |\operatorname{Re}\nu'| \ge \lambda\mu_\delta/(2\delta) \ge 1$. So it is $O(\lambda^{-|\mathcal{E}|/2-1})$.

(3) Near $y^\star$. Laplace's method at the nondegenerate minimum $y^\star$, with the smooth amplitude $e^{\langle \nu', y\rangle}$, gives

$$\int_{|y-y^\star|<\delta} e^{\langle \nu', y\rangle} e^{-\lambda(\Phi_{\boldsymbol{\xi}}(y)-\Phi_{\boldsymbol{\xi}}(y^\star))}\, dy = \Big(\frac{2\pi}{\lambda}\Big)^{|\mathcal{E}|/2} \det(H^\star)^{-1/2}\Big(e^{\langle \nu', y^\star\rangle} + O(\lambda^{-1})\Big),$$

as in the reference, proof of 4.3, where the amplitude is 1: the amplitude enters the leading term through its value at $y^\star$, and its derivatives only the term $O(\lambda^{-1})$.

Adding (2) and (3), multiplying by $e^{-\lambda\Phi_{\boldsymbol{\xi}}(y^\star)}$ and factoring out $e^{\langle \nu', y^\star\rangle} \neq 0$ gives (iii). □

Corollary 4.7 (Four Gammas on a line). *As stated: the display is 4.6 on two coefficients.*

Proof. (1) A straight contour. Let first $\max(-\operatorname{Re} a, -\operatorname{Re} b) < \min(\operatorname{Re} c, \operatorname{Re} d)$, and take the line $\operatorname{Re} z = \gamma$ with $\max(-\operatorname{Re} a, -\operatorname{Re} b) < \gamma < \min(\operatorname{Re} c, \operatorname{Re} d)$; it separates the two sets of poles. Apply 4.6 with $k = 2$, the exponents $a + c$ and $b + d$, and $(a + d,\, b + c)$, of total $a + b + c + d$, in the place of the merged shares. In the shares $(\alpha_1, \alpha_2, \beta_1, \beta_2)$ the fibre has the pins $\alpha_1 + \alpha_2 = a + c$, $\alpha_1 + \beta_1 = a + d$, $\alpha_2 + \beta_2 = b + c$, and it is parametrized by $\alpha = (a + z,\, c - z)$, $\beta = (d - z,\, b + z)$. The matrix P of the pins has the rows $(1, 1, 0, 0)$, $(1, 0, 1, 0)$, $(0, 1, 0, 1)$, so $\det PP^{\mathsf{T}} = 4$, and its kernel is spanned by $(1, -1, -1, 1)$, of length 2. By 4.1, with the admissible real point on the line $\operatorname{Re} z = \gamma$, where all four shares have positive real parts, the fibre integral of a function G is $(2\pi)^{-1} \cdot 4^{-1/2} \cdot 2 \int G\, d\tau = \frac{1}{2\pi i} \int_{\gamma + i\mathbb{R}} G\, dz$, $z = \gamma + i\tau$. The integrand of 4.6 is

$$\Gamma_{\,1}^{\,2}\begin{bmatrix} a+z,\ c-z \\ a+c \end{bmatrix} \Gamma_{\,1}^{\,2}\begin{bmatrix} d-z,\ b+z \\ b+d \end{bmatrix} = \frac{\Gamma(a+z)\Gamma(b+z)\Gamma(c-z)\Gamma(d-z)}{\Gamma(a+c)\Gamma(b+d)},$$

and its right side is $\Gamma_{\,1}^{\,2}\begin{bmatrix} a+d,\ b+c \\ a+b+c+d \end{bmatrix}$. Multiplying by $\Gamma(a + c)\Gamma(b + d)$ (1.3(i)) gives the display.

(2) Continuation. A contour separating the poles exists exactly when no pole $-a-n$ or $-b-n$ of the first two factors coincides with a pole $c + n'$ or $d + n'$ of the last two, $n, n' \in \mathbb{N}$, that is on the set U of $(a, b, c, d) \in \mathbb{C}^4$ with $a+c, a+d, b+c, b+d \notin -\mathbb{N}$. Near a point of U one fixed contour separates the poles, and by 1.3(iii) and 2.11 on one scale the four Gammas decay exponentially along it, locally uniformly in (a, b, c, d); so the left side is holomorphic on U, and by Cauchy's theorem it does not depend on the separating contour. The right side is holomorphic on U. The set U is connected, being the complement in $\mathbb{C}^4$ of countably many complex hyperplanes, and contains the open set of (1), so the identity holds on U. □

48.5 Chapter 5: the Trailmoment

Theorem 5.7 (The gauge of the coefficient function). *As stated: the last sentence at the unit configuration on any number of scales.*

Proof. The first two sentences are those of the reference (the reference, proof of 5.7). At the unit configuration $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\mathbf{c}) = \prod_e c_e^{-\nu_e} = \mathbf{c}^{-\nu}$ for every $\nu \in \mathbb{C}^{\mathcal{E}}$ (2.12), so by 5.1 $(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = \int_{Y'} \mathbf{c}^{-\nu}\, d\phi$, either side converging when the other does. □

Theorem 5.8 (Exchange and ladder of the Trailmoment). *As stated: (iii) at the unit configuration on any number of scales; (i) and (ii) are those of the reference.*

Proof. At $t = I_{\mathcal{E}}$ the column t_k is $\mathbf{e}_k$, $k \in \mathcal{E}$, so in (i) at the scale e only $k = e$ has $t_{k,e} \neq 0$, with $t_{e,e} = 1$ and the factor $\Gamma_1\begin{bmatrix}\nu_e+1\\1\end{bmatrix}\prod_{f\neq e}\Gamma_1\begin{bmatrix}\nu_f\\1\end{bmatrix} = 1$. Directly, for every ν, $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\mathbf{c}) = \mathbf{c}^{-\nu} = c_e\,\mathbf{c}^{-\nu-\mathbf{e}_e} = c_e\,\mathsf{Z}_{\nu+\mathbf{e}_e;I_{\mathcal{E}}}(\mathbf{c})$ (2.12); integrating against ϕ over Y' gives the first identity.

For the derivative, let $D \subseteq \mathbb{C}^{\mathcal{E}}$ be the set of indices where the Trailmoment converges, $\int_{Y'}\mathbf{c}^{-\operatorname{Re}\nu}\,d|\phi| < \infty$, and ν_0 an interior point of D. Take $\delta > 0$ such that $\nu_0 \pm \delta\mathbf{e}_e \in D$. For $\nu = \nu_0 + \zeta\mathbf{e}_e$ with $|\zeta| < \delta/2$, $|\mathbf{c}^{-\nu}\log c_e| = \mathbf{c}^{-\operatorname{Re}\nu_0}c_e^{-\operatorname{Re}\zeta}|\log c_e|$. For $\eta > 0$ and $y > 0$, $|\log y| \le \eta^{-1}(y^{\eta} + y^{-\eta})$; with $\eta = \delta/2$, and $y^{\theta} \le y^{\delta} + y^{-\delta}$ for $|\theta| \le \delta$,

$$\left|\mathbf{c}^{-\nu}\log c_e\right| \le \frac{4}{\delta}\Big(\mathbf{c}^{-(\operatorname{Re}\nu_0-\delta\mathbf{e}_e)} + \mathbf{c}^{-(\operatorname{Re}\nu_0+\delta\mathbf{e}_e)}\Big),$$

an integrable function independent of ζ. So $\zeta \mapsto \int_{Y'}\mathbf{c}^{-\nu_0-\zeta\mathbf{e}_e}d\phi$ may be differentiated under the integral sign, and $\partial_\zeta\mathbf{c}^{-\nu} = -(\log c_e)\,\mathbf{c}^{-\nu}$ gives the second identity. □

Proposition 5.14 (Laurent data of the Gamma at its poles). *As stated: the residue and the Laurent coefficients of a product of any number of Gammas with a common pole.*

Proof. The quotient form of $(-1)^j\varepsilon\Gamma(-j+\varepsilon)$, the expansion of its logarithm and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))$ are those of the reference (the reference, proof of 5.14). On $|\varepsilon| < 1$ each $\phi_i(\varepsilon) := (-1)^{j_i}\varepsilon\,\Gamma(-j_i+\varepsilon) = \Gamma(1+\varepsilon)\Gamma(1-\varepsilon)/\Gamma(1+j_i-\varepsilon)$ is holomorphic and nonzero, and it is the exponential of the expansion at j_i. Put $b = -j+\varepsilon$; then $b + k_i = -j_i + \varepsilon$ and $\Gamma(b+k_i) = (-1)^{j_i}\varepsilon^{-1}\phi_i(\varepsilon)$, so

$$h(b)\prod_{i=1}^{p}\Gamma(b+k_i) = (-1)^{j_1+\cdots+j_p}\,\varepsilon^{-p}\,h(-j+\varepsilon)\prod_{i=1}^{p}\phi_i(\varepsilon).$$

The residue at $\varepsilon = 0$ is $(-1)^{\sum_i j_i}$ times the coefficient of ε^{p-1} in $h(-j+\varepsilon)\prod_i\phi_i(\varepsilon)$, and with $h(-j+\varepsilon) = \sum_l h^{(l)}(-j)\varepsilon^l/l!$ the Cauchy product gives the displayed sum. At $p = 2$ and $k_1 = k_2 = 0$, $\phi_1\phi_2 = \exp(2\log\phi_1) = (j!)^{-2}\big(1 - 2(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))\varepsilon + O(\varepsilon^2)\big)$, so the residue is $(j!)^{-2}\big(h'(-j) - 2(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))h(-j)\big)$, which is the formula of the reference. The Laurent coefficients of $\prod_i\Gamma(b+k_i)$ at $-j$ are $(-1)^{\sum_i j_i}$ times the Taylor coefficients of $\prod_i\phi_i = \prod_i(j_i!)^{-1}\exp\big(\sum_i(\log\phi_i + \log j_i!)\big)$; the series $\sum_i(\log\phi_i + \log j_i!)$ has no constant term and its coefficients are rational multiples of the Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j_i,\infty))$ and $(\mathsf{T}_{2k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, so the Taylor coefficients of its exponential are polynomials in these Trailmoments with rational coefficients. □

Remark. The residue formula was checked in exact arithmetic for $p = 2, 3, 4$ at several j and (k_i) (`sweep/s0107/resonance.py`).

Proposition 5.22 (The joint Laplace transform of Dirichlet inputs). *As stated, for finite sums of monomials with nonnegative real exponents.*

Proof. (1) The series. Every exponent $\mathbf{a}$ lies in $[0,\infty)^{\mathcal{E}}$, so $0 < u^{\mathbf{a}} \le 1$ on the source $\prod_Q\Delta_Q$ and $|P_i(u)| \le \|P_i\| := \sum_{\mathbf{a}}\big|[u^{\mathbf{a}}]P_i\big|$. Hence $\sum_{\mathbf{k}}\frac{|v|^{\mathbf{k}}}{\mathbf{k}!}\prod_i|P_i(U)|^{k_i} \le \exp\big(\sum_i|v_i|\,\|P_i\|\big)$, the exponential series of $\exp(-\sum_i v_iP_i(U))$ converges uniformly on the source for each $v \in \mathbb{C}^q$ and integrates term by term, and $\mathcal{F}(v) = \sum_{\mathbf{k}}\frac{(-v)^{\mathbf{k}}}{\mathbf{k}!}\mathbb{E}[\prod_i P_i(U)^{k_i}]$, a power series converging on $\mathbb{C}^q$, hence entire.

(2) The moments. Multiplying out, $\prod_i P_i^{k_i} = \sum_{\mathbf{a}}[u^{\mathbf{a}}]\big(\prod_i P_i^{k_i}\big)u^{\mathbf{a}}$, a finite sum with exponents in $[0,\infty)^{\mathcal{E}}$. The blocks of U are independent, so $\mathbb{E}[U^{\mathbf{a}}] = \prod_Q\mathbb{E}_{U_Q\sim\mathrm{Dir}_{\mathbf{m}|_Q}}[U_Q^{\mathbf{a}|_Q}]$, and by 3.11 at the

exponents $\mathbf{a}|_Q$, whose real parts satisfy $m_e + a_e > 0$, together with 1.3(i),(ii),

$$\mathbb{E}_{U_Q \sim \mathrm{Dir}_{\mathbf{m}|_Q}}\big[U_Q^{\mathbf{a}|_Q}\big] = \frac{\Gamma_{1}^{|Q|}\begin{bmatrix}\mathbf{m}|_Q + \mathbf{a}|_Q \\ m_Q + a_Q\end{bmatrix}}{\Gamma_{1}^{|Q|}\begin{bmatrix}\mathbf{m}|_Q \\ m_Q\end{bmatrix}} = \prod_{e\in Q} \Gamma_1\begin{bmatrix}m_e + a_e \\ m_e\end{bmatrix}\ \Gamma_1\begin{bmatrix}m_Q \\ m_Q + a_Q\end{bmatrix}.$$

Summing gives $M(\mathbf{k})$; for integer exponents 3.11 gives the rising factorials.

(3) The read. Where every P_i is positive on the interior of the source, which carries the law, 2.12 and 2.1 give $\mathsf{Z}_{\nu;t}(\mathbf{P}(U)) = \frac{1}{\prod_j \Gamma(\nu_j)} \int_{(0,\infty)^d} x^{\nu-\mathbf{1}} \exp\big(-\sum_i P_i(U) x^{t_i}\big) dx$ where this converges. By Tonelli on the moduli and Fubini, wherever the double integral converges, the expectation of the left side is the integral of $\mathbb{E}[\exp(-\sum_i P_i(U)x^{t_i})] = \mathcal{F}(x^{t_1}, \dots, x^{t_q})$ against $x^{\nu-\mathbf{1}}dx/\prod_j \Gamma(\nu_j)$. The left side is the Trailmoment $(\mathsf{T}_{\nu;t}[(P_i)_i] \otimes_Q \mathrm{Dir}_{\mathbf{m}|_Q})(\prod_Q \Delta_Q)$ by 5.1.

(4) Uniqueness. With real coefficients the vector $(P_i(U))_i$ takes its values in the compact box $\prod_i [-\|P_i\|, \|P_i\|]$, and its moments $\mathbb{E}[\prod_i P_i(U)^{k_i}] = M(\mathbf{k})$ are the integrals of the monomials against its law. Polynomials are dense in the continuous functions on the box, so a law on the box is determined by these integrals; and the read $\mathbb{E}[\mathsf{Z}_{\nu;t}(\mathbf{P}(U))]$ depends on the law of $\mathbf{P}(U)$ alone. □

Theorem 5.26 (The Shruffle on a line). *As stated: (i) and (iii) for any number of strings and of trails; (ii) is that of the reference.*

Proof. (i) Let $w^{(i)}$ have length n_i, $n = \sum_i n_i$, and $S_i = \{x_1^{(i)} > \cdots > x_{n_i}^{(i)} > 0\}$. By Tonelli the function $\prod_i \prod_j w_j^{(i)}(x_j^{(i)})$ is integrable on $S_1 \times \cdots \times S_r$, with integral $\prod_i \mathrm{chain}(w^{(i)})$ by Fubini. The points with two equal coordinates form a null set. At every other point the n coordinates are distinct, and their decreasing order is a total order of the n positions (i,j) that restricts to the order of each string: an interleaving ς, the string of the n functions in that order. The points of one interleaving are carried by a permutation of the coordinates, which keeps Lebesgue measure, onto $\{y_1 > \cdots > y_n > 0\}$, the integrand becoming $\prod_m \varsigma_m(y_m)$; so their integral is $\mathrm{chain}(\varsigma)$. The interleavings are the ways of distributing the n places among the strings, $n!/(n_1! \cdots n_r!)$ of them.

(iii) Let r trails be as in (ii), with strings $\sigma^{(1)}, \dots, \sigma^{(r)}$ and cuts $N_1, \dots, N_r$. By (ii) the trail i is $\mathrm{chain}(\sigma^{(i)})$, and by (i) the product is $\sum_\varsigma \mathrm{chain}(\varsigma)$ over the interleavings. An interleaving begins with the first function of one of the strings, a Laplace transform $e^{-N_i \cdot} L_1^{(i)}$, which is $\int e^{-(s+N_i)\cdot} \psi(ds)$ for the first increment ψ of trail i: the Laplace transform of its image under $s \mapsto s + N_i$, carried by $(0,\infty)$ by the hypothesis of (ii). The other Laplace transforms of ς are those of nonnegative increments or of such images. Reading ς as in (ii), each Laplace transform followed by its run of constant functions 1 gives an increment with the index one more than the length of the run; the trail at the cut 0 with these increments has no constant columns and positive integer indices, and by (ii) it is $\mathrm{chain}(\varsigma)$. □

Theorem 5.27 (Trails at frequencies are iterated integrals). *As stated.*

Proof. For a word $\ell = (\ell_1, \dots, \ell_w)$ with $\ell_1 \neq 0$ and letters in $\{0\} \cup \{|z| = 1\}$ put $F_\varnothing = 1$ and $F_\ell(z) = \int_0^z F_{\ell'}(t)\, dt/(t - \ell_w)$ on the disc $|z| < 1$, ℓ' the word without its last letter, along the segment. By induction F_ℓ is holomorphic on the disc with $F_\ell(0) = 0$ for $w \geq 1$: for $\ell_w \neq 0$ the factor $1/(t - \ell_w)$ is holomorphic there, and for $\ell_w = 0$, $F_{\ell'}(t)/t$ is, as $F_{\ell'}(0) = 0$, the word ℓ' being nonempty since $\ell_1 \neq 0$. Unwinding the recursion, $F_\ell(z)$ is the iterated integral over $0 < t_1 < \cdots < t_w < z$ along the segment.

For $|z| < 1$ let $S(z) = \sum_{0<n_1<\cdots<n_d} \prod_i \zeta_i^{n_i} n_i^{-\nu_i} z^{n_d}$, absolutely convergent, the trail with z^{y_d} in its density, $S(0) = 0$. If $\nu_d \geq 2$, $zS'(z)$ is the same series with $\nu_d - 1$ in place of ν_d, whose word

is ℓ without its last letter 0, and $S(z) = \int_0^z S_{\nu_d-1}(t)\,dt/t$. If $\nu_d = 1$, summing the geometric series over $n_d > n_{d-1}$,

$$S'(z) = \sum_{0<n_1<\cdots<n_{d-1}} \prod_{i<d} \zeta_i^{n_i} n_i^{-\nu_i}\ \frac{\zeta_d\,(\zeta_d z)^{n_{d-1}}}{1-\zeta_d z} = \frac{\zeta_d}{1-\zeta_d z}\,S^\flat(z),$$

where $S^\flat$ is the series of depth $d-1$ with the last frequency $a_{d-1}+a_d$, so $\zeta_{d-1}\zeta_d$, and $S^\flat = 1$ when $d = 1$. Its c_i are those of S for $i < d$, its word is ℓ without the last letter $c_d^{-1} = \zeta_d^{-1}$, and $\zeta_d/(1-\zeta_d z) = -1/(z-\zeta_d^{-1})$, so $S(z) = -\int_0^z S^\flat(t)\,dt/(t-\zeta_d^{-1})$. By induction on w, $S = (-1)^d F_\ell$ on the disc, which is the last sentence.

Write $S(z) = \sum_n b_n z^n$ with $b_n = A(n)\zeta_d^n n^{-\nu_d}$, $A(n) = \sum_{0<n_1<\cdots<n_{d-1}<n} \prod_{i<d} \zeta_i^{n_i} n_i^{-\nu_i}$, $A = 1$ when $d = 1$; then $|A(n)| \le (1+\log n)^{d-1}$, and the head over $y_d \le N$ is $\sum_{n\le N} b_n$. For $\nu_d \ge 2$ the series $\sum b_n$ converges absolutely, so the heads converge to the trail. For $\nu_d = 1$ the partial sums of ζ_d^n are bounded by $2/|1-\zeta_d|$, and $A(n)/n \to 0$ with $\sum_n |A(n+1)/(n+1) - A(n)/n| \le \sum_n C(1+\log n)^{d-1} n^{-2} < \infty$, since $A(n+1) - A(n)$ is the sum of the terms with $n_{d-1} = n$; summation by parts makes $\sum b_n$ converge. By Abel's theorem on power series, $S(z) \to \sum_n b_n$ as $z \to 1$ along $[0,1)$.

On the other side, the integrand $\prod_j 1/|t_j - \ell_j|$ on the simplex of $[0,1]$ is at most $C\prod_{\ell_j=0} t_j^{-1} \prod_{\ell_j=1}(1-t_j)^{-1}$, the other letters lying off $[0,1]$. Split the simplex by the number of the t_j below $\frac12$; on the part below, with $\ell_1 \ne 0$, the iterated integrals $G_j(t)$ of the first j factors over $0 < t_1 < \cdots < t_j < t$ satisfy $G_1(t) \le Ct$ and $G_j(t) \le \int_0^t G_{j-1}(s)\,C(1+1/s)\,ds \le C_j t$; on the part above the same holds after $t \mapsto 1-t$, as $\ell_w \ne 1$ ($\ell_w = 0$ when $\nu_d \ge 2$, $\ell_w = \zeta_d^{-1} \ne 1$ when $\nu_d = 1$). So the integral over $0 < t_1 < \cdots < t_w < 1$ converges absolutely, and by dominated convergence $F_\ell(z)$ tends to it as $z \to 1$ along $[0,1)$. Hence the limit of the heads is $(-1)^d$ times that integral. □

Theorem 5.31 (Two convolutions). *As stated: (ii) and (iii) at the unit configuration on any number of scales; (i) and the displays of (ii) for inputs carried by $(0,\infty)$ are those of the reference.*

Proof. (ii) By 2.12, $\mathsf{Z}_{\nu;I_\mathcal{E}}(\mathbf{c}\,\mathbf{c}') = \prod_e (c_e c'_e)^{-\nu_e} = \mathsf{Z}_{\nu;I_\mathcal{E}}(\mathbf{c})\,\mathsf{Z}_{\nu;I_\mathcal{E}}(\mathbf{c}')$ at every point of $Y_1 \times Y_2$. The two Trailmoments on the right converge, so by Tonelli the kernel is integrable for $|\phi| \otimes |\psi|$ on $Y_1' \times Y_2'$, and Fubini gives the product (5.1).

(iii) *The kernel on sums.* First, g has no zero: if $g(s_0) = 0$, then $g(s_0/2)^2 = g(s_0) = 0$, so $g(s_0/2^n) = 0$ for every n; every $s \in (0,\infty)^\mathcal{E}$ has $s - s_0/2^n \in (0,\infty)^\mathcal{E}$ for n large, and $g(s) = g(s_0/2^n)\,g(s - s_0/2^n) = 0$, against the hypothesis. Fix $e \in \mathcal{E}$. For $s, s'' \in (0,\infty)^\mathcal{E}$ and $\tau > 0$, $g(s+\tau\mathbf{e}_e)\,g(s'') = g(s+s''+\tau\mathbf{e}_e) = g(s''+\tau\mathbf{e}_e)\,g(s)$, so $\varphi_e(\tau) = g(s+\tau\mathbf{e}_e)/g(s)$ does not depend on s. It is continuous on $(0,\infty)$, has no zero, and $\varphi_e(\tau+\tau') = \varphi_e(\tau)\,\varphi_e(\tau')$, by the independence of the base point applied at s and at $s+\tau\mathbf{e}_e$. By the converse on one scale, the case $|\mathcal{E}| = 1$ (the reference, proof of 5.31), $\varphi_e(\tau) = e^{-u_e\tau}$ for one $u_e \in \mathbb{C}$, and then $g(s+\tau\mathbf{e}_e) = e^{-u_e\tau} g(s)$ also for $\tau < 0$ with $s + \tau\mathbf{e}_e \in (0,\infty)^\mathcal{E}$, applying it at the base point $s+\tau\mathbf{e}_e$. Moving one coordinate at a time from a fixed point s_1, $g(s) = C\,e^{-\langle u,s\rangle}$ on $(0,\infty)^\mathcal{E}$ with $C = g(s_1)e^{\langle u,s_1\rangle} \ne 0$, and $g(s+s') = g(s)\,g(s')$ gives $C = C^2$, so $C = 1$.

The kernel on products. If $h(x) \ne 0$, then $h(x) = h(x)\,h(\mathbf{1})$ gives $h(\mathbf{1}) = 1$, and $h(x)\,h(x^{-1}) = h(\mathbf{1}) = 1$ shows that h has no zero. The function $s \mapsto h(e^s)$ on $(0,\infty)^\mathcal{E}$ is continuous, not identically zero and satisfies the hypothesis on sums, so $h(e^s) = e^{-\langle u,s\rangle}$, that is $h(x) = x^{-u}$ for every x with all coordinates above 1. Any $x \in (0,\infty)^\mathcal{E}$ is x'/x'' with $x''_e > \max(1, 1/x_e)$ and $x' = x\,x''$, both with all coordinates above 1, and $h(x) = h(x')/h(x'') = x^{-u}$. So $h = \mathsf{Z}_{u;I_\mathcal{E}}$. □

48.6 Chapter 6: rational words and live inputs

Theorem 6.4 (Pole columns are a rational difference quotient). *As stated: coefficients in any subfield of $\mathbb{C}$, and no decay condition.*

Proof. (1) The decomposition. For $k \geq 0$ and $s \geq 1$, telescoping gives $\Delta \sum_{j=0}^{k-1} (t+j)^{-s} = \sum_{j=0}^{k-1} \big((t+j+1)^{-s} - (t+j)^{-s}\big) = (t+k)^{-s} - t^{-s}$. So $\Delta H(t) = \sum_{k,s} c_{k,s}\big((t+k)^{-s} - t^{-s}\big) = R(t) - \sum_s C_s t^{-s}$, and H has its coefficients in $\mathbb{K}$.

(2) The decay. As $t \to \infty$, $(t+k)^{-s} = t^{-s}(1+k/t)^{-s} = t^{-s} + O(t^{-s-1})$, so $R(t) = C_1 t^{-1} + O(t^{-2})$, which is $O(t^{-2})$ exactly when $C_1 = 0$.

(3) The converse. A proper rational function H with poles on the nonpositive integer lattice has one partial-fraction presentation $H(t) = \sum_{k,s} d_{k,s}(t+k)^{-s}$, finitely many terms with $k \in \mathbb{N}$, $s \geq 1$. Then $\Delta H(t) = \sum_{k,s} d_{k,s}\big((t+k+1)^{-s} - (t+k)^{-s}\big)$, whose coefficient of $(t+k)^{-s}$ is $d_{k-1,s} - d_{k,s}$ (with $d_{-1,s} = 0$), and the column sum over k telescopes to 0. The column sums $R \mapsto (C_s)_s$ are linear on rational words, presentations being unique, and vanish on such differences. A combination $\sum_s a_s t^{-s}$ has the column sums $(a_s)_s$, so it is a difference only if it is 0: the functions $t^{-1}, \ldots, t^{-w}$ are independent modulo differences, and by (1) they span the rational words of maximum order w modulo differences. A difference is a rational word with column sums zero, hence $O(t^{-2})$ by (2); and a word that is $O(t^{-2})$ has $C_1 = 0$, so it is congruent to $\sum_{s \geq 2} C_s t^{-s}$: modulo differences these words have the basis $t^{-2}, \ldots, t^{-w}$.

(4) Cosets. Translation by one keeps each coset $b + \mathbb{Z}$, so the steps above apply coset by coset, with $(t+b)^{-s}$ in the place of t^{-s}. □

Theorem 6.5 (Summation by parts with a live input). *As stated: (iv) without the condition $C_1 = 0$; (i) to (iii) are those of the reference.*

Proof. (iv) For an integer $M \geq 1$ the head is $(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) = \sum_{N < m \leq N+M} R(m)$, the kernel at index 0 being 1. By 6.4, $R(m) = \sum_s C_s m^{-s} + H(m+1) - H(m)$, so

$$(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) = \sum_{s=1}^{w} C_s\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) + H(N+M+1) - H(N+1).$$

As $M \to \infty$: $H(N+M+1) \to 0$, H being proper; for $s \geq 2$, $(\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) \to (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, \infty))$; and $(\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) - \log M \to (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, \infty))$, the regularized value of 5.2. This gives the first display. When $C_1 = 0$, $R(m) = O(m^{-2})$ by 6.4, the series $\sum_{m>N} R(m)$ converges absolutely, and the left side is $(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((N, \infty))$, the term $s = 1$ being absent. For a density f, when the terms converge, $\sum_{m>N} R(m) f(m) = \sum_{s=1}^{w} C_s \sum_{m>N} m^{-s} f(m) + \sum_{m>N} \Delta H(m) f(m)$, and (ii) gives the last two terms of the second display when the upper boundary vanishes. □

Theorem 6.8 (A Breta input with a recurrence of finite order). *As stated: complex a, b with positive real parts, real $q_i < 1$ of either sign, and complex s_i.*

Proof. (1) The letters. For real $q_i < 1$ and $x \in [0,1]$, $1 - q_i x$ lies between 1 and $1 - q_i$, so in $[\min(1, 1-q_i), \max(1, 1-q_i)] \subset (0, \infty)$. So $\varrho(x) = \prod_i (1 - q_i x)^{-s_i}$ is infinitely differentiable on $[0,1]$, and $|\varrho| = \prod_i (1 - q_i x)^{-\operatorname{Re} s_i}$ lies between two positive constants.

(2) The size. For $m \geq 1$, by $x \leq e^{-(1-x)}$ on $[0,1]$ and $y = 1 - x$,

$$|f(m)| \leq \max |\varrho| \int_0^1 x^{m + \operatorname{Re} a - 1} (1-x)^{\operatorname{Re} b - 1} dx$$

$$\le \max|\varrho| \int_0^\infty e^{-(m+\operatorname{Re} a-1)y} y^{\operatorname{Re} b-1} dy = \max|\varrho|\,\Gamma(\operatorname{Re} b)\,(m+\operatorname{Re} a-1)^{-\operatorname{Re} b},$$

so $f(m) = O(m^{-\operatorname{Re} b})$.

(3) The recurrence. Differentiating the logarithm, $\frac{d}{dx}\big[x^{m+a}(1-x)^b \prod_i (1-q_i x)^{1-s_i}\big]$ is $x^{m+a-1}(1-x)^{b-1}\varrho(x)$ times

$$(m+a)(1-x)Q(x) - bxQ(x) + x(1-x)\sum_i (s_i-1) q_i Q_i(x) = B_m(x).$$

The bracket tends to 0 at both ends, since $\operatorname{Re}(m+a) > 0$ and $\operatorname{Re} b > 0$, and $x^{m+a-1}(1-x)^{b-1}\varrho B_m$ is integrable; so integrating over $[0,1]$ gives $\sum_\ell B_\ell(m) f(m+\ell) = 0$.

(4) The extreme coefficients. $B_0(m) = B_m(0) = (m+a)Q(0) = m+a$. Every $q_i \neq 0$, so Q has degree r and leading coefficient $(-1)^r \prod_i q_i$, and Q_i has degree $r-1$ and leading coefficient $(-1)^{r-1}\prod_{i'\neq i} q_{i'}$. The coefficient of x^{r+1} in B_m is therefore

$$(-1)^{r+1}\prod_i q_i \Big((m+a)+b+\sum_i (1-s_i)\Big) = (-1)^{r+1}\Big(\prod_i q_i\Big)\Big(m+a+b+r-\sum_i s_i\Big).$$

$B_0(m) \neq 0$ since $\operatorname{Re} a > 0$, and $B_{r+1}(m) \neq 0$ for every integer $m \ge 0$ exactly when $a+b+r-\sum_i s_i \notin -\mathbb{N}$. Then the recurrence solved for $f(m+r+1)$, and for $f(m)$, expresses every shift through $(f(m),\dots,f(m+r))$ with coefficients rational in m.

(5) The Breta. Let $\operatorname{Re} s_i > 0$. On the bunch $\{1,2\}$ with the scale ρ and the shape $u = (x, 1-x)$, $\omega_{\{1,2\}}$ being dx in the coordinate $x = u_1$, the kernel terms ρu_1 and ρu_2 with unit coefficients sum to ρ, and the letter $P_i = \rho\big((1-q_i)u_1 + u_2\big) = \rho(1-q_i x)$ has the positive pattern coefficients $1-q_i$ and 1. At the shape index $(m+a, b)$ and the scale index $1+|\mathbf{s}|$ the left side of 3.6 is

$$\int_0^1 x^{m+a-1}(1-x)^{b-1} \int_0^\infty \rho^{|\mathbf{s}|} e^{-\rho} \rho^{-|\mathbf{s}|} \prod_i (1-q_i x)^{-s_i}\, d\rho\, dx = f(m),$$

the inner integral being $\Gamma(1) = 1$; 3.6 identifies it with $\Gamma(1+|\mathbf{s}|)$ times the Breta. □

Theorem 6.9 (A fixed set of input-dependent evaluations). *As stated, for the data of 6.8.*

Proof. The steps of the reference, proof of 6.9, hold for these data; they are as follows. Write $F_{p,k}(N) = (\mathsf{T}_{p;(1)}[x+k]\, f\, \mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} f(m)(m+k)^{-p}$, and, for $0 \le x < 1$, $S_p(x) = \sum_{m>N} x^m (m+k)^{-p}$, with an integer cut $N \ge 0$ and $k \ge 0$; $S_0(x) = x^{N+1}/(1-x)$.

(1) The identity behind (ii). As in the proof of 6.8, $(xC\eta)' = B\eta$, with $xC\eta = x^a(1-x)^b \prod_i (1-q_i x)^{1-s_i}$. Since $xS_p'(x) = \sum_{m>N} m\, x^m (m+k)^{-p} = S_{p-1}(x) - kS_p(x)$,

$$\frac{d}{dx}\big[xC\eta\, S_p\big] = \eta\big[(B-kC)S_p + C\, S_{p-1}\big] \qquad (p \ge 1).$$

At $x = 0$ the bracket is $O(|x^{a+N+1}|) \to 0$, since $\operatorname{Re} a > 0$. At $x = 1$ it is bounded by a constant times $|(1-x)^b|$ for $p \ge 2$, where $S_p(1)$ is finite, and times $|(1-x)^b|\,|\log(1-x)|$ for $p = 1$, where $S_1(x) = O(|\log(1-x)|)$; it tends to 0 since $\operatorname{Re} b > 0$. The functions $|\eta|\, S_p$ and $|\eta|\,|\log(1-x)|$ are integrable on $[0,1)$, so integrating over $[0,1)$ and summing term by term (Tonelli on the moduli) gives, for $p \ge 2$,

$$\sum_{m>N} (m+k)^{-p} \int_0^1 x^m (B-kC)\eta\, dx + \sum_{m>N} (m+k)^{1-p} \int_0^1 x^m C\eta\, dx = 0.$$

With $\int_0^1 x^{m+\ell}\eta\,dx = f(m+\ell)$ and, for $k \ge r+1 \ge \ell$, $\sum_{m>N} f(m+\ell)(m+k)^{-p} = F_{p,k-\ell}(N) - \sum_{v=1}^{\ell} f(N+v)(N+v+k-\ell)^{-p}$, this is (ii) for $p \ge 2$. At $p=1$ the term with S_0 is $\int_0^1 \eta\, C\, S_0\, dx = \int_0^1 \eta\, x^{N+1} Q\, dx = \sum_{\ell=0}^{r} Q_\ell f(N+1+\ell)$, by $C\,S_0 = x^{N+1}Q$, which gives (ii) at $p=1$.

(2) The reduction (i). In (ii) the coefficient of $F_{p,k}(N)$, the term $\ell = 0$, is $B_0 - kC_0 = a - k$, since $B(0) = aQ(0) = a$ and $C(0) = 1$. For $k \ge r+1$ with $k \ne a$, (ii) expresses $F_{p,k}(N)$ through the $F_{p,k-\ell}(N)$, $1 \le \ell \le r+1$, the $F_{p-1,k-\ell}(N)$ (none at $p=1$), and the values $f(N+v)$, $1 \le v \le r+1$. Taking the indices p in increasing order and, for each, the shifts k in increasing order, every $F_{p,k}(N)$ with $p \le A$ reduces to the $F_{p',k'}(N)$ with $p' \le p$ and $k' \le r$, the $F_{p',a}(N)$ when a is an integer at least $r+1$, and $f(N+1), \dots, f(N+r+1)$.

(3) The cuts (iii) are 5.3 for the counting input.

(4) Denominators (iv). For rational data, let c_* clear the coefficients of B, C and Q, so that $c_*(a-k)$, $c_*(B_\ell - kC_\ell)$, c_*C_ℓ and c_*Q_ℓ are integers for integer k. By induction along the order of (2), the coefficients of $F_{p,k}(N)$, $k \ge r+1$, in the reduced form are cleared by $D_{p,k} = d_{N+k}^{p} \prod_{r+1 \le k' \le k,\ k' \ne a} |c_*(a-k')|^p$. Multiply (ii) by c_* and divide by $c_*(a-k)$: the terms $F_{p,k-\ell}(N)$ and $F_{p-1,k-\ell}(N)$ carry integer factors, and their reduced forms are cleared by $D_{p,k-\ell}$ and $D_{p-1,k-\ell}$, which divide $D_{p,k}/|c_*(a-k)|$; the terms in $f(N+v)$ carry the denominators $(N+v+k-\ell)^p$ and $(N+v+k-\ell)^{p-1}$, which divide d_{N+k}^{p}, since $1 \le N+v+k-\ell \le N+k$. So $D_{p,k}$ clears $F_{p,k}(N)$, and $d_{N+J}^{A} \prod_{r+1 \le k \le J,\ k \ne a} |c_*(a-k)|^A$, a multiple of every $D_{p,k}$ with $p \le A$ and $k \le J$, clears all shifts through J and indices through A. □

Theorem 6.10 (The row of a rational function against an input with a recurrence). *As stated.*

Proof. All the sums converge absolutely: $m + h \ge N + 1 + h_* > 0$ for $m > N$ and $h \ge h_*$, and $\sum_{m>N} |f(m)|(1+|m|)^d < \infty$. (i) is linearity, $P(m) = \sum_r ([m^r]P) m^r$.

(ii) Expand p_i about $-h$: $p_i(m) = \sum_e \pi_{i,e}(h)(m+h)^e$ with $\pi_{i,e}(h) = p_i^{(e)}(-h)/e!$. Multiply the recurrence at m by $(m+h)^{-s}$, sum over $m > N$ and put $m' = m + i$:

$$\sum_{i=0}^{\tau} \sum_{e} \pi_{i,e}(h) \sum_{m' > N+i} f(m')\,(m'+h-i)^{e-s} = 0.$$

Moving the cut of each inner sum to N (5.3) and writing, for $e \ge s$, $(m'+h-i)^{e-s} = \sum_{r \le e-s} \binom{e-s}{r} (h-i)^{e-s-r} m'^r$,

$$\begin{aligned} p_0(-h)\,T_s(h) = &\sum_{i=1}^{\tau} \sum_{v=1}^{i} \sum_{e} \pi_{i,e}(h)\, f(N+v)\,(N+v+h-i)^{e-s} \\ &- \sum_{(i,e) \ne (0,0),\ e<s} \pi_{i,e}(h)\, T_{s-e}(h-i) - \sum_{e \ge s} \sum_{i} \pi_{i,e}(h) \sum_{r \le e-s} \binom{e-s}{r} (h-i)^{e-s-r} M_r, \end{aligned}$$

with $\pi_{0,0}(h) = p_0(-h)$. At $h = h_* + j$, $\tau \le j \le J$, divide by $p_0(-h) \ne 0$: the right side holds $T_{s'}$ with $s' < s$ at the same shift, or with $s' \le s$ at the smaller shifts $h - i \ge h_*$, moments M_r with $r \le d - s \le d-1$, and the values $f(N+v)$, $v \le \tau$. Induction on j, and for fixed j on s, gives (ii); all coefficients are rational.

(iii) Put $Q(s,j) = d_{u(N+h_*+j)}^{s} \prod_{j'=\tau}^{j} |u^d p_0(-h_* - j')|^s$; we show by the same induction that $Q(s,j)$ clears the coefficients of $T_s(h_*+j)$, and $Q(s,j)$ divides the integer of (iii). Since $h \in \frac{1}{u}\mathbb{Z}$ and $p_i \in \mathbb{Z}[m]$, $\pi_{i,e}(h)$ has a denominator dividing $u^{\deg p_i - e}$, and $u^d p_0(-h)$ is an integer. Write the relation as $T_s(h) = \big(u^d p_0(-h)\big)^{-1} u^d\,[\cdots]$ and take its terms in turn.

- $T_{s-e}(h-i)$ with $i \geq 1$ or $e \geq 1$, $s-e \geq 1$: the factor $u^d \pi_{i,e}(h)$ is an integer, and the coefficients of $T_{s-e}(h-i)$ are 1 if it is a coordinate and are cleared by $Q(s-e, j-i)$ otherwise, by induction; $|u^d p_0(-h)|\, Q(s-e, j-i)$ divides $Q(s,j)$, as $s-e \leq s$, $j-i \leq j$, and when $i = 0$, $e \geq 1$ leaves the factor $|u^d p_0(-h)|^e$ free in $Q(s,j)/Q(s-e,j)$.
- The moments: $u^d \pi_{i,e}(h)(h-i)^{e-s-r}$ has a denominator dividing $u^{-d+\deg p_i - e + e - s - r}$, an integer power of u with exponent at most 0, so it is an integer.
- The values $f(N+v)$: for $e < s$, $(N+v+h-i)^{e-s} = u^{s-e}\big(u(N+v+h-i)\big)^{e-s}$, where $u(N+v+h-i)$ is an integer with $1 \leq u(N+v+h-i) \leq u(N+h)$, positive since $h-i \geq h_*$ and $N+1+h_* > 0$; so its power $s-e \leq s$ divides $d^s_{u(N+h_*+j)}$, and $u^d \pi_{i,e}(h) u^{s-e}$ is an integer. For $e \geq s$ the term $u^d \pi_{i,e}(h)(N+v+h-i)^{e-s}$ has a denominator dividing $u^{\deg p_i - e + (e-s) - d}$, a power of u with exponent at most 0, so it is an integer.

Every coefficient of $T_s(h_* + j)$ is a sum of such products, so $Q(s,j)$ clears it; and for $s \leq w$, $j \leq J$, $Q(s,j)$ divides $d^w_{u(N+h_*+J)} \prod_{j'=\tau}^{J} |u^d p_0(-h_* - j')|^w$, as $d_n \mid d_{n'}$ for $n \leq n'$. Poles in several cosets are reduced coset by coset, each with its own u and h_*. □

48.7 Chapter 7: tables

Proposition 7.6 (Duality of minors). *As stated.*

Proof. Every basis has r marks, so $|T \cap X|$ is largest exactly when $|T \cap X'|$ is least, that is exactly when the cobasis $\mathcal{E} \setminus T$ has $|(\mathcal{E} \setminus T) \cap X'| = |X'| - |T \cap X'|$ largest: T is at the top of X exactly when $\mathcal{E} \setminus T$ is at the top of X' in $\mathcal{B}^*$ (7.5). By 7.5 the bases of $\mathcal{B}^*/X'$ are the sets $(\mathcal{E} \setminus T) \setminus X' = X \setminus (T \cap X)$ for these T, the complements in X of the bases $T \cap X$ of $\mathcal{B}|X$, and the bases of $\mathcal{B}^*|X'$ are the sets $(\mathcal{E} \setminus T) \cap X' = X' \setminus (T \setminus X)$, the complements in X' of the bases $T \setminus X$ of $\mathcal{B}/X$. The weight of the cobasis $\mathcal{E} \setminus T$ is $\theta_T = \theta'_{T \cap X} \theta''_{T \setminus X}$, the factor $\theta''_{T \setminus X}$ belonging to its part $X' \setminus (T \setminus X)$ and the factor $\theta'_{T \cap X}$ to its part $X \setminus (T \cap X)$; these are the weights of the dual tables of 7.1. The letters follow from 7.2. □

Theorem 7.15 (One function, two Foreman polynomials). *As stated: (iii) along every direction constant on the molecules of the table, with the converse; (i) and (ii) are those of the reference.*

Proof. The directions. By 7.9 for the family whose one table is this table, the table is the direct sum of its restrictions to its molecules M: every basis T is the union of one basis of each restriction $\mathcal{B}|M$, and a basis of $\mathcal{B}|M$ has $r(M)$ marks (7.5). So $|T \cap M| = r(M)$, and for v constant on each M, with value v_M,

$$\langle v, \chi_T \rangle = \sum_M v_M \big(|T \cap M| - |M \setminus T| \big) = \sum_M v_M \big(2r(M) - |M| \big)$$

for every basis. By 7.14, $\Lambda(z+v) = \sum_T \theta_T e^{\langle z, \chi_T \rangle} e^{\langle v, \chi_T \rangle} = \exp\big(\sum_M v_M (2r(M) - |M|) \big) \Lambda(z)$. At $v = \lambda \mathbf{1}$ the exponent is $\lambda(2r - n)$, the $r(M)$ adding up to r, since a basis meets each molecule in $r(M)$ marks, and the $|M|$ to n.

The converse. Let $v \in \mathbb{R}^{\mathcal{E}}$ and real $\lambda \neq 0$ be such that $\Lambda(z + \lambda v) = c\, \Lambda(z)$ for every $z \in \mathbb{R}^{\mathcal{E}}$, with c independent of z. Then $\sum_T \theta_T \big(e^{\lambda \langle v, \chi_T \rangle} - c \big) e^{\langle z, \chi_T \rangle} = 0$ for every z. Distinct bases have distinct signed indicators, and exponentials $e^{\langle z, \chi \rangle}$ with finitely many distinct vectors χ are linearly independent functions of z: on a line $z = \sigma y$ with the $\langle y, \chi \rangle$ distinct, as $\sigma \to \infty$ the term

with the largest $\langle y, \chi \rangle$ dominates, so its coefficient vanishes, and the others follow in turn. So $\theta_T(e^{\lambda\langle v,\chi_T\rangle} - c) = 0$, and since $\theta_T > 0$, $e^{\lambda\langle v,\chi_T\rangle} = c$ for every basis. Hence $\langle v, \chi_T\rangle = 2\langle v, \mathbf{1}_T\rangle - \langle v, \mathbf{1}\rangle$ is constant over the bases, and so is $\langle v, \mathbf{1}_T\rangle$. By 7.8, $v_a = v_b$ for any two marks linked by a circuit of the table; the molecules of the table alone are the smallest nonempty sets of marks containing, with each mark, every mark so linked to it (7.4), so v is constant on each of them. □

Theorem 7.16 (The inner moment generating function). *As stated: the even statistic at every gauge, with the basis law and the cobasis law there.*

Proof. For real z and w, $e^{E(z+w)} = \Lambda(z+w)\,\Lambda(-z-w)$. By the first display of the theorem, $\Lambda(z+w)/\Lambda(z) = \mathbb{E}_{T\sim\mathbb{P}_\alpha}[e^{\langle w,\chi_T\rangle}]$, and the same display at the gauge $\alpha^{-1} = e^{-2z}$, whose basis law is $\mathbb{P}_{\alpha^{-1}} = \mathbb{P}^*_\alpha$, gives $\Lambda(-z-w)/\Lambda(-z) = \mathbb{E}_{T'\sim\mathbb{P}^*_\alpha}[e^{-\langle w,\chi_{T'}\rangle}]$. Multiplying, and using the independence of T and T' under $\mathbb{P}_\alpha \otimes \mathbb{P}^*_\alpha$,

$$\frac{e^{E(z+w)}}{e^{E(z)}} = \mathbb{E}_{(T,T')\sim\mathbb{P}_\alpha\otimes\mathbb{P}^*_\alpha}\big[e^{\langle w,\chi_T-\chi_{T'}\rangle}\big], \qquad \chi_T - \chi_{T'} = 2(\mathbf{1}_T - \mathbf{1}_{T'}),$$

which is the display; its logarithm says that $w \mapsto E(z+w) - E(z)$ is the cumulant generating function of $\chi_T - \chi_{T'}$. At $z = 0$, $e^{E(0)} = \Lambda(0)^2 = K(\mathbf{1})^2$ by (i) of 7.15, and $\mathbb{P}^*_{\mathbf{1}} = \mathbb{P}_{\mathbf{1}}$; with w written z this is the display of the reference. □

Remark. The display at a general gauge, and (iii) of 7.15, were checked in exact arithmetic on the table whose bases are the unions of a 2-subset of $\{0,1,2\}$ and a 1-subset of $\{3,4\}$, with arbitrary positive weights (`sweep/s0107/tables.py`).

Corollary 7.24 (Moments and cumulants). *As stated: the joint cumulants of the table at every order; (i) and the cumulants of the components are those of the reference.*

Proof. For a function Y of the index (k,l) write $\mathbb{E}_w[Y] = \sum_{k,l} w_{kl} Y(k,l)$; $\mathbb{E}_w[1] = 1$ by 7.23.

(1) Generating functions. By 7.23 at $x = e^v$, $v \in \mathbb{C}^B$,

$$\mathbb{E}_{T\sim\mathbb{P}_\alpha}\Big[\exp\Big(\sum_{e\in B} v_e X_e\Big)\Big] = \sum_{k,l} w_{kl}\, M_{kl}(v), \qquad M_{kl}(v) = \det\big(I + (D_{e^v} - I)\Pi^{(kl)}_{BB}\big),$$

with $M_{kl}(0) = 1$. So $C_{kl}(v) = \log M_{kl}(v)$ is a power series in v without constant term, and for distinct marks the cumulant $\kappa^{(kl)}(e_1,\dots,e_p)$ of the component is the coefficient of $v_{e_1}\cdots v_{e_p}$ in C_{kl}; the reference computes it by the cyclic formula. The joint cumulant $\kappa_\alpha(e_1,\dots,e_p)$ is the coefficient of $v_{e_1}\cdots v_{e_p}$ in $\log\sum_{k,l} w_{kl} e^{C_{kl}(v)}$, a power series since $\sum_{k,l} w_{kl} = 1$.

(2) Cumulants over the index. Let $(Y_i)_{i\in I}$ be finitely many functions of the index, $\lambda = (\lambda_i)_{i\in I}$ formal variables, and $\mu(\lambda) = \mathbb{E}_w[\exp(\sum_i \lambda_i Y_i)]$, a power series with constant term 1. For $D \subseteq I$ write $\lambda^D = \prod_{i\in D}\lambda_i$; the coefficient of λ^D in μ is $\mathbb{E}_w[\prod_{i\in D} Y_i]$. In $\log\mu = \sum_{j\geq 1} \frac{(-1)^{j-1}}{j}(\mu-1)^j$ the coefficient of λ^I comes from the products of j monomials of $\mu - 1$ without repeated variables and with disjoint supports covering I; each partition $\mathfrak{q}$ of I into j blocks arises from $j!$ such ordered products. So

$$[\lambda^I]\log\mu = \sum_{j\geq 1} \frac{(-1)^{j-1}}{j}\, j! \sum_{|\mathfrak{q}|=j} \prod_{D\in\mathfrak{q}} \mathbb{E}_w\Big[\prod_{i\in D} Y_i\Big] = \kappa_w\big((Y_i)_{i\in I}\big),$$

and, setting the variables outside a subset D to 0, $[\lambda^D]\log\mu = \kappa_w((Y_i)_{i\in D})$.

(3) Substitution. Fix distinct marks $e_1,\dots,e_p$ of B; for nonempty $C \subseteq \{1,\dots,p\}$ put $v^C = \prod_{c\in C} v_{e_c}$ and $Y_C(k,l) = \kappa^{(kl)}(e_C)$, the coefficient of v^C in C_{kl}. Setting the variables v_e with

$e \notin \{e_1, \dots, e_p\}$ to 0 and dropping every monomial divisible by some $v_{e_c}^2$ is a ring homomorphism onto the ring $\mathcal{R}$ of polynomials in $v_{e_1}, \dots, v_{e_p}$ with $v_{e_c}^2 = 0$; it keeps the coefficient of $v_{e_1} \cdots v_{e_p}$ and commutes with exp and log of series without constant term. In $\mathcal{R}$, C_{kl} becomes $\sum_{C \neq \varnothing} Y_C(k,l)\, v^C$, the image of $\sum_C \lambda_C Y_C(k,l)$ under $\lambda_C \mapsto v^C$. So the coefficient of $v_{e_1} \cdots v_{e_p}$ in $\log \sum_{k,l} w_{kl} e^{C_{kl}}$ is that of the image of $\log \mu(\lambda)$, with I the set of nonempty C. A monomial $\prod_C \lambda_C^{n_C}$ maps to $\prod_C (v^C)^{n_C}$, which is $v_{e_1} \cdots v_{e_p}$ exactly when the C with $n_C > 0$ have $n_C = 1$ and form a partition $\mathfrak{p}$ of $\{1, \dots, p\}$, and is 0 or another monomial otherwise. By (2),

$$\kappa_\alpha(e_1, \dots, e_p) = \sum_{\mathfrak{p}} [\lambda^{\mathfrak{p}}] \log \mu = \sum_{\mathfrak{p}} \kappa_w\Big(\big(\kappa^{(kl)}(e_C)\big)_{C \in \mathfrak{p}} \Big).$$

At $p = 1$ the single partition gives $\kappa_\alpha(e) = \mathbb{E}_w[\Pi_{ee}^{(kl)}]$, and $\kappa_\alpha(e) = \mathbb{P}_\alpha(e \in T) = R_e(\alpha)$. Every step is an identity of power series in which the w_{kl} enter linearly inside μ, with $\sum_{k,l} w_{kl} = 1$, so it holds for complex mixing coefficients. □

Remark. At $p = 2$ and $p = 3$ the formula gives $\kappa_\alpha(e,f) = \mathbb{E}_w[\kappa^{(kl)}(e,f)] + \kappa_w(\Pi_{ee}^{(kl)}, \Pi_{ff}^{(kl)})$ and $\kappa_\alpha(e,f,g) = \mathbb{E}_w[\kappa^{(kl)}(e,f,g)] + \sum_3 \kappa_w(\Pi_{ee}^{(kl)}, \kappa^{(kl)}(f,g)) + \kappa_w(\Pi_{ee}^{(kl)}, \Pi_{ff}^{(kl)}, \Pi_{gg}^{(kl)})$, the sum over the three choices of the mark that stands alone: the covariances and the third cumulant over the index of the reference. The formula was checked in exact arithmetic against the generating function of a mixture of three components with complex mixing coefficients for $p = 2, 3, 4, 5$ (`sweep/s0107/cumulants2.py`).

48.8 Chapter 8: families

Proposition 8.11 (The interest over latent vectors). *As stated: every table determinantal over $\mathbb{R}$ or over $\mathbb{C}$, each letter opened with as many latent vectors as its own exponent asks.*

Proof. The masses are real and positive. For a mass $m_e \leq 0$ the interest diverges by 8.5(i), the wall $\kappa_{\{e\}} = m_e - \sum_j \big(s_j d_j(\{e\}) + t_j d_j^*(\{e\})\big)$ being at most m_e, and so does the integral over α_e in step (4).

Fix a table j, write $\mathbb{F} = \mathbb{F}_j$, $\beta = \beta_j$, $r = r_j$, and let $A \in \mathbb{F}^{r \times n}$ be a slice of it: $\det A_T = \boldsymbol{\omega}_T$, so that $|\det A_T|^2 = \theta_{j,T}$ for $T \in \mathcal{B}_j$ and $\det A_T = 0$ for every other set T of r marks (7.20). For a matrix X with n columns write X^{H} for its conjugate transpose, and $D_\alpha = \operatorname{diag}(\alpha)$. Expanding by the maximal minors of X,

$$\det\big(X D_\alpha X^{\mathsf{H}}\big) = \sum_W |\det X_W|^2\, \alpha^W,$$

the sum over the sets W of as many marks as X has rows; this is the expansion behind 7.21, and with $X = A$ it gives $K_j(\alpha) = \det(A D_\alpha A^{\mathsf{H}})$. So $K_j(\mathbf{1}) = \det(AA^{\mathsf{H}}) > 0$, A has rank r, and its row space $V_j = \{A^{\mathsf{H}} y :\ y \in \mathbb{F}^r\}$, the space onto which $\Pi^{(11)}$ of 7.23 projects, has dimension r over $\mathbb{F}$.

(1) The basis letter. The matrix $\hat{A} = (AA^{\mathsf{H}})^{-1/2} A$ has orthonormal rows and $\{\hat{A}^{\mathsf{H}} y\} = V_j$, and $\det \hat{A}_T = \det(AA^{\mathsf{H}})^{-1/2} \det A_T$. So $\theta_{j,T} = K_j(\mathbf{1})\, |\det \hat{A}_T|^2$ and $K_j(\alpha) = K_j(\mathbf{1}) \det(\hat{A} D_\alpha \hat{A}^{\mathsf{H}})$.

(2) The cobasis letter. Choose $\hat{C} \in \mathbb{F}^{(n-r) \times n}$ with orthonormal rows and $\{\hat{C}^{\mathsf{H}} y\} = V_j^{\perp}$, the orthogonal complement for the form $\sum_e u_e \overline{v_e}$. Then $\hat{A}\hat{C}^{\mathsf{H}} = 0$, and the square matrix U whose rows are those of $\hat{A}$ followed by those of $\hat{C}$ is unitary. For an invertible matrix, $\det U[I,J] = \pm \det U \cdot \det U^{-1}[J^c, I^c]$ for every set I of rows and set J of columns of equal size; with $U^{-1} = U^{\mathsf{H}}$ and $|\det U| = 1$ this gives $|\det U[I,J]| = |\det U[I^c, J^c]|$. With I the rows of $\hat{A}$ and $J = T$,

$|\det \hat{C}_{\mathcal{E}\setminus T}| = |\det \hat{A}_T|$ for every set T of r marks. The expansion above, for $X = \hat{C}$ and the sets $W = \mathcal{E} \setminus T$, gives

$$\det\big(\hat{C} D_\alpha \hat{C}^{\mathsf{H}}\big) = \sum_T |\det \hat{A}_T|^2 \alpha^{\mathcal{E}\setminus T} = \frac{1}{K_j(\mathbf{1})} \sum_{T \in \mathcal{B}_j} \theta_{j,T}\, \alpha^{\mathcal{E}\setminus T} = \frac{\Psi_j(\alpha)}{K_j(\mathbf{1})}.$$

(3) The Gaussian opening. For a Hermitian positive definite matrix G of size d over $\mathbb{F}$, a unitary change of variables, which keeps the Lebesgue measure of $\mathbb{F}^d \cong \mathbb{R}^{\beta d}$, makes G diagonal, and $\int_{\mathbb{F}} e^{-\lambda|z|^2} dz = (\pi/\lambda)^{\beta/2}$ for $\lambda > 0$; so $\int_{\mathbb{F}^d} e^{-z^{\mathsf{H}} G z} dz = \pi^{\beta d/2} \det(G)^{-\beta/2}$. For an integer $k \geq 0$ and $s = \beta k/2$ it follows that

$$\det(G)^{-s} = \pi^{-ds} \int_{(\mathbb{F}^d)^k} \exp\Big(-\sum_{i=1}^{k} z_i^{\mathsf{H}} G z_i \Big)\, dz_1 \cdots dz_k.$$

Take $G = \hat{A} D_\alpha \hat{A}^{\mathsf{H}}$, $d = r$ and $k = 2s_j/\beta$. The map $z \mapsto x = \hat{A}^{\mathsf{H}} z$ is an $\mathbb{F}$-linear isometry of $\mathbb{F}^r$ onto V_j, so it carries the Lebesgue measure of $\mathbb{F}^r$ to that of V_j, and $z^{\mathsf{H}} \hat{A} D_\alpha \hat{A}^{\mathsf{H}} z = \sum_e \alpha_e |x_e|^2$. With step (1),

$$K_j(\alpha)^{-s_j} = K_j(\mathbf{1})^{-s_j} \pi^{-r_j s_j} \int_{V_j^{2s_j/\beta}} \exp\Big(-\sum_{i,e} \alpha_e |(x_{j,i})_e|^2 \Big) \prod_i dx_{j,i},$$

and in the same way, with step (2), $G = \hat{C} D_\alpha \hat{C}^{\mathsf{H}}$, $d = n - r_j = r_j^*$ and $k = 2t_j/\beta$,

$$\Psi_j(\alpha)^{-t_j} = K_j(\mathbf{1})^{-t_j} \pi^{-r_j^* t_j} \int_{(V_j^\perp)^{2t_j/\beta}} \exp\Big(-\sum_{i,e} \alpha_e |(x^*_{j,i})_e|^2 \Big) \prod_i dx^*_{j,i}.$$

An exponent 0 asks for no vectors, and the formula then reads $1 = 1$.

(4) The scales. Multiplying over j, with $c_F = \prod_j K_j(\mathbf{1})^{s_j + t_j} \pi^{r_j s_j + r_j^* t_j}$,

$$W(\alpha) = \frac{1}{c_F} \int \exp\Big(-\sum_e \alpha_e \mathfrak{q}_e \Big) \prod_{j,i} dx_{j,i}\, dx^*_{j,i}.$$

Insert this into $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int \alpha^{\mathbf{m-1}} e^{-|\alpha|} W(\alpha)\, d\alpha$. Every integrand is nonnegative, so the integrals may be taken in either order, and

$$\int_0^\infty \alpha_e^{m_e - 1} e^{-\alpha_e(1+\mathfrak{q}_e)}\, d\alpha_e = \Gamma(m_e)\,(1+\mathfrak{q}_e)^{-m_e}.$$

Integrating the scales first gives the right side of the statement, and integrating the latent vectors first gives $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$: both are the integral of one nonnegative function, finite or infinite together.

(5) The Breta law. Where the two sides are finite, the function

$$f(\alpha, x, x^*) = \alpha^{\mathbf{m-1}} \exp\Big(-\sum_e \alpha_e (1 + \mathfrak{q}_e) \Big)$$

is integrable, with integral $c_F J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ by step (4). Integrating out the latent vectors, its marginal in α is $c_F\, \alpha^{\mathbf{m-1}} e^{-|\alpha|} W(\alpha)$, proportional to the density $\alpha^{\mathbf{m-1}} e^{-|\alpha|} W(\alpha)/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ of $\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ (8.10; the interest is finite exactly when every wall is positive, 8.5(i)). Given the latent vectors, f is the product over the marks of $\alpha_e^{m_e-1} e^{-\alpha_e(1+\mathfrak{q}_e)}$: the scales are independent, and $\alpha_e(1+\mathfrak{q}_e)$ has the density $u^{m_e-1}e^{-u}/\Gamma(m_e)$ on $(0,\infty)$, the law Gamma_{m_e}.

(6) Equal exponents over $\mathbb{R}$. If $\mathbb{F}_j = \mathbb{R}$ and $s_j = t_j$, the vectors of the two letters come in pairs $(x_{j,i}, x^*_{j,i})$, $i \le 2s_j$. The map $V_j \times V_j^\perp \to \mathbb{R}^{\mathcal{E}}$, $(x, x^*) \mapsto x + x^*$, is a linear isometry, so it carries the product of the Lebesgue measures to that of $\mathbb{R}^{\mathcal{E}}$, and for $u = x + x^*$, $x = \mathrm{pr}_j u$ and $x^* = (I - \mathrm{pr}_j)u$. The constant of table j is then $K_j(\mathbf{1})^{2s_j}\pi^{ns_j}$. □

Remark. The reference, proof of 8.11, treats $s_j = t_j = k_j/2$ over $\mathbb{R}$ and opens the two letters of table j with one set of k_j vectors $u = \hat{A}^{\mathsf{T}} y + \hat{C}^{\mathsf{T}} y^*$; that is step (6). In the proof above each letter is opened on its own space, with as many vectors as its exponent asks, and over $\mathbb{C}$ by the complex Gaussian integral, $\beta = 2$.

Lemma S3. *For a family on $\mathcal{E}$ and a nonempty set $S \subseteq \mathcal{E}$, $c_j(S) = 0$ for every table j of the family exactly when S is a union of molecules.*

Proof. The molecules are the smallest nonempty sets of marks that contain, with each mark, every mark linked to it (7.4); so a set of marks with that property is a union of molecules, and a union of molecules has it.

Let S be a union of molecules. By 7.9 every table is the direct sum of its restrictions to the molecules: its bases are the unions of one basis of each restriction, and a basis of the restriction to a molecule M has $r_j(M)$ marks (7.5). So $|T \cap S| = \sum_{M \subseteq S} r_j(M)$ on every basis T of table j, and by 7.13, $d_j(S) = r_j(S)$, that is $c_j(S) = 0$.

Let S not be a union of molecules. Then some $e \in S$ is linked to some $f \notin S$: e and f lie in a common circuit of some table j. By 7.7 some basis T of table j has $e \in T$, $f \notin T$ and $T' = (T \setminus \{e\}) \cup \{f\}$ a basis. Then $|T' \cap S| = |T \cap S| - 1$, so $|T \cap S|$ takes two values on the bases of table j, and $c_j(S) \ge 1$ by 7.13. □

Remark. Since the weight law charges every basis, Lemma S3 is the first sentence of 13.9 read on the bases: $\mathrm{Var}_{T \sim \mathbb{P}_{j,\mathbf{1}}} |T \cap U| = 0$ exactly when $|T \cap U|$ is constant on $\mathcal{B}_j$, that is when $c_j(U) = 0$.

Proposition 8.16 (The wall of a compound). *As stated: the identity holds for every nonempty proper S, and only the tables with $\sigma_j \neq 0$ enter it.*

Proof. 8.13 gives, as identities of functions of the data, $\kappa_S + \kappa_S^{\mathrm{top}} = 2\sum_j \sigma_j c_j(S)$ and $\kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}} + \kappa_S^{\mathrm{top}}$. Adding the two,

$$\kappa_S + \kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}} + 2\sum_j \sigma_j c_j(S).$$

If $c_j(S) = 0$ in every table with $\sigma_j \neq 0$, the sum vanishes and $\kappa_S + \kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}}$; this holds in particular when S is a union of molecules, where $c_j(S) = 0$ in every table (Lemma S3). On the balance wall $\kappa_{\mathcal{E}} = 0$, so $\mathrm{Re}\,\kappa_S + \mathrm{Re}\,\kappa_{\mathcal{E}\setminus S} = 0$ and one of the two walls has real part at most zero. By 8.7 the period converges only where every nonempty proper wall has positive real part, so it diverges. □

Remark. The reference, 8.16, takes $c_j(S) = 0$ in every table. With complex exponents a table with $s_j = -t_j \neq 0$ is active and has $\sigma_j = 0$; it does not enter the identity.

Definition 8.17 (Target; native gauge). As stated: the target for all data with $m_{\mathcal{E}} \neq 0$ and on the balance wall, and native points of $(\mathbb{C}^\times)^{\mathcal{E}}$ off the zeros of the letters, the native gauges being the native points in $\mathbb{R}^{\mathcal{E}}_{>0}$.

Proof. The definition is consistent.

(1) Scale. K_j and Ψ_j are homogeneous polynomials of degrees r_j and $r_j^* = n - r_j$, every basis having r_j marks and every complement $n - r_j$ (7.1). So for $c \in \mathbb{C}^\times$, $K_j(c\alpha) = c^{r_j} K_j(\alpha)$ and $(\partial_e K_j)(c\alpha) = c^{r_j - 1}(\partial_e K_j)(\alpha)$, whence $R_{j,e}(c\alpha) = R_{j,e}(\alpha)$ wherever $K_j(\alpha) \neq 0$, and likewise $R^*_{j,e}(c\alpha) = R^*_{j,e}(\alpha)$ wherever $\Psi_j(\alpha) \neq 0$. The condition $K_j(\alpha)\Psi_j(\alpha) \neq 0$ is unchanged by scaling and the target does not involve the point, so the usage, and with it nativity, is a property of the point up to scale.

(2) The two clauses of the target. On the balance wall $m_{\mathcal{E}} = \sum_j (s_j r_j + t_j r_j^*)$ (8.1); where this is not zero the formula gives $\boldsymbol{\tau} = \mathbf{m}$, the value assigned on the balance wall. So the two clauses agree where both apply, and the second also covers the balance wall with $m_{\mathcal{E}} = 0$.

(3) Positive points. At $\alpha \in \mathbb{R}^{\mathcal{E}}_{>0}$ the letters are positive, the weights being positive, so every positive point lies off their zeros, and $R_{j,e}(\alpha) = \mathbb{P}_{j,\alpha}(e \in T)$ and $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T)$ (7.16). □

Remark. The direction of a ray lies on the balance wall (38.1), so its target is its masses $\mathbf{m}_1$, and the native gauges at complex points of the direction are the points α_c of the letter complement with $c^{(1)}(\alpha_c) = 0$: the native gauges of the ray (38.4).

Theorem 8.19 (The native gauge is the saddle). *As stated, with the native gauges of Definition 8.17.*

Proof. At a gauge $\alpha = e^y$ the usage of Definition 8.17 is $\nabla\Phi(y)$: by the chain rule $\partial_{y_e} \log K_j(e^y) = \alpha_e \partial_e K_j(\alpha)/K_j(\alpha) = R_{j,e}(\alpha)$ and $\partial_{y_e} \log \Psi_j(e^y) = R^*_{j,e}(\alpha)$, and Φ is $\sum_j \left(s_j \log K_j + t_j \log \Psi_j\right)$ read at e^y (8.18). For real data with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, $m_{\mathcal{E}} > 0$, so the target of that definition is the one given by its formula, and its native gauges are the gauges of (i). With this reading the statement is the reference, proof of 8.19. □

Proposition 8.20 (The native gauges of a compound). *As stated: the weights need not factor except in the sentence on products, and the molecules are those of the active family.*

Proof. Write $g = \sum_j (s_j d_j + t_j d_j^*)$. An inactive table enters neither the potential, nor the target, nor g, nor the Newton sum $\mathcal{N}_F$, its terms carrying the factor zero; every statement below concerns the active family, whose molecules are $\mathcal{E}_1, \ldots, \mathcal{E}_q$.

(1) The criterion of the saddle. By 8.19(ii) a native gauge exists exactly when $\boldsymbol{\tau}$ lies in the relative interior of $\mathcal{N}_F = \sum_j (s_j P_j + t_j P_j^*)$, and it is then unique modulo the space V of directions along which Φ is affine. By 8.19(iii), whose first sentence uses no connectedness, $\mathcal{N}_F = \{x : \ x_{\mathcal{E}} = g(\mathcal{E}),\ x_S \geq g(S) \text{ for all } S\}$.

(2) The extent of x_S on $\mathcal{N}_F$. The least value of a linear function on a Minkowski sum with nonnegative coefficients is the same sum of its least values on the summands. The least value of x_S on $P_j = \operatorname{conv}\{\mathbf{1}_T\}$ is $\min_T |T \cap S| = d_j(S)$, and on $P_j^* = \operatorname{conv}\{\mathbf{1}_{\mathcal{E}\setminus T}\}$ it is $\min_T |S \setminus T| = d_j^*(S)$ (7.13). So the least value of x_S on $\mathcal{N}_F$ is $g(S)$, and that of $x_{\mathcal{E}\setminus S}$ is $g(\mathcal{E} \setminus S)$; since $x_{\mathcal{E}} = g(\mathcal{E})$ on $\mathcal{N}_F$ and $\mathcal{N}_F$ is convex, x_S takes on $\mathcal{N}_F$ exactly the values of $[g(S),\ g(\mathcal{E}) - g(\mathcal{E} \setminus S)]$. At any masses $\kappa_S + \kappa_{\mathcal{E}\setminus S} - \kappa_{\mathcal{E}} = g(\mathcal{E}) - g(S) - g(\mathcal{E} \setminus S)$, the masses cancelling, so by 8.16 the length of this interval is $2\sum_j \sigma_j c_j(S) = \sum_j (s_j + t_j) c_j(S)$.

(3) The relative interior. An active table has $s_j + t_j > 0$. So the interval of step (2) is a point exactly when $c_j(S) = 0$ for every active table, that is, by Lemma S3 applied to the active family, when S is a union of the $\mathcal{E}_i$. The relative interior of a polyhedron given by linear equalities and inequalities consists of its points at which every inequality that is not an equality on the whole polyhedron holds strictly. So $\boldsymbol{\tau}$ lies in the relative interior of $\mathcal{N}_F$ exactly when $\tau_S = g(S)$ for every union S of the $\mathcal{E}_i$ and $\tau_S > g(S)$ for every other nonempty S.

(4) Molecule by molecule. By 7.9 for the active family, every active table is the direct sum of its restrictions to the $\mathcal{E}_i$, its bases being the unions of one basis of each restriction. Taking the least of $|T\cap S|$, and of $|S\setminus T|$, restriction by restriction, $d_j(S)=\sum_i d_j(S\cap\mathcal{E}_i)$ and $d_j^*(S)=\sum_i d_j^*(S\cap\mathcal{E}_i)$ for every active j; so $g(S)=\sum_i g(S\cap\mathcal{E}_i)$ and $\tau_S-g(S)=\sum_i\big(\tau_{S\cap\mathcal{E}_i}-g(S\cap\mathcal{E}_i)\big)$. Hence the conditions of step (3) hold exactly when $\tau_{\mathcal{E}_i}=g(\mathcal{E}_i)$ for every i and $\tau_S>g(S)$ for every S with $\varnothing\neq S\subsetneq\mathcal{E}_i$ for some i: given these, a union of the $\mathcal{E}_i$ has $\tau_S-g(S)=0$, and any other nonempty S meets some $\mathcal{E}_i$ in a nonempty proper subset, whose term is positive while the others are nonnegative; conversely each $\mathcal{E}_i$ is a union of the $\mathcal{E}_i$, and a nonempty proper subset of one $\mathcal{E}_i$ is not. Since $c_j(\mathcal{E}_i)=0$ for every active j, $d_j(\mathcal{E}_i)=r_j(\mathcal{E}_i)$, so $r_j(\mathcal{E}\setminus\mathcal{E}_i)=r_j-r_j(\mathcal{E}_i)$ (7.12, $d_j(S)=r_j-r_j(\mathcal{E}\setminus S)$) and $d_j^*(\mathcal{E}_i)=|\mathcal{E}_i|-r_j(\mathcal{E}_i)=r_j^*(\mathcal{E}_i)$. So $g(\mathcal{E}_i)=\sum_j\big(s_jr_j(\mathcal{E}_i)+t_jr_j^*(\mathcal{E}_i)\big)$, and the conditions are the balance of every $\mathcal{E}_i$ at the target and the positivity of its proper walls there. This proves the criterion.

(5) The free scales. By 7.17 the second derivative of $\log K_j(e^y)$ along a direction v is the variance of $\langle v,\mathbf{1}_T\rangle$ under $\mathbb{P}_{j,e^y}$, and that of $\log\Psi_j(e^y)=\langle\mathbf{1},y\rangle+\log K_j(e^{-y})$ (7.2) is the variance of $\langle v,\mathbf{1}_T\rangle$ under $\mathbb{P}_{j,e^{-y}}$. Both laws charge every basis, so each variance vanishes exactly when $\langle v,\mathbf{1}_T\rangle$ is constant over $\mathcal{B}_j$. The second derivative of Φ along v is the sum of these variances with the coefficients $s_j,t_j\geq 0$, and every active table has a positive coefficient; so Φ is affine along v exactly when $\langle v,\mathbf{1}_T\rangle$ is constant over the bases of every active table. By 7.8 for the active family such a v is constant on every $\mathcal{E}_i$. Conversely, if v is constant on every $\mathcal{E}_i$, with value v_i, then $\langle v,\mathbf{1}_T\rangle=\sum_i v_i|T\cap\mathcal{E}_i|=\sum_i v_ir_j(\mathcal{E}_i)$ on every basis of an active table j, by step (4). So $V=\operatorname{span}\{\mathbf{1}_{\mathcal{E}_1},\dots,\mathbf{1}_{\mathcal{E}_q}\}$, and the native gauges are the gauges $e^v\alpha^\star$, $v\in V$: the gauges β with $\beta/\alpha^\star$ constant on every $\mathcal{E}_i$.

(6) Factoring weights. When the weights factor across the $\mathcal{E}_i$, the letters of every active table factor,

$$K_j(\alpha)=\prod_i K_{j,i}(\alpha_{\mathcal{E}_i}),\qquad \Psi_j(\alpha)=\prod_i\Psi_{j,i}(\alpha_{\mathcal{E}_i}),$$

with the Foreman polynomials of the restrictions (7.9). So $\Phi(y)=\sum_i\Phi_i(y_{\mathcal{E}_i})$, with Φ_i the potential of the restriction of the family to $\mathcal{E}_i$, and the condition $\nabla\Phi=\boldsymbol{\tau}$ splits into the conditions $\nabla\Phi_i=\boldsymbol{\tau}|_{\mathcal{E}_i}$, each the native condition of the restriction to $\mathcal{E}_i$ at the masses $\boldsymbol{\tau}|_{\mathcal{E}_i}$, which lie on its balance wall by step (4), so that its target is $\boldsymbol{\tau}|_{\mathcal{E}_i}$ (8.17). The native gauges are the gauges whose restriction to every $\mathcal{E}_i$ is native there: the products of native gauges of the molecules.

(7) Units. A loop is a circuit by itself, so it lies in no other circuit; and a coloop e lies in no circuit, since a circuit C through e would lie in every basis containing the independent set $C\setminus\{e\}$, such a basis containing e, while a circuit lies in no basis. So a mark e that is a loop or a coloop in every table is linked to no other mark, and $\{e\}$ is a unit. Every basis of table j contains e, or none does, so $K_j(\alpha)=\alpha_e^{\epsilon_j}\tilde K_j(\alpha_{\mathcal{E}\setminus e})$ and $\Psi_j(\alpha)=\alpha_e^{1-\epsilon_j}\tilde\Psi_j(\alpha_{\mathcal{E}\setminus e})$ with $\epsilon_j\in\{0,1\}$: the letters of $\{e\}$ are monomials. □

Remark. The criterion is stated for the molecules of the active family because linking (7.4) runs through the circuits of every table, while an inactive table enters neither the potential, nor the target, nor the walls. So a molecule of the family may join several molecules of the active family, and then a proper wall of that molecule vanishes at the target although a native gauge exists. For instance, on $\mathcal{E}=\{1,2,3,4\}$ let the list with the bases $\{1,3\},\{1,4\},\{2,3\},\{2,4\}$ carry $(s,t)=(1,0)$ and the list $\binom{\mathcal{E}}{1}$ carry $(0,0)$, with the masses $\frac12\mathbf{1}$ on the balance wall. Through the second table $\mathcal{E}$ is one molecule of the family, while the active family has the molecules $\{1,2\}$ and $\{3,4\}$. Every mark lies in half of the bases of the first table, so the gauge $\mathbf{1}$ is native, and the wall at $\{1,2\}$ is $\tau_{\{1,2\}}-d_1(\{1,2\})=1-1=0$.

Remark. At the centre of 8.21 the gauge **1** is native, and step (5) shows that the native gauges are the gauges constant on every molecule of the active family: this is the uniqueness clause of 8.21(i).

48.9 Chapter 9: realizations and the still

Proposition 9.2 (A realization sits on a wall only along its molecules). *As stated, for every family.*

Proof. (1) The walls as expectations. By 7.16, $R_{j,e}(\alpha) = \mathbb{P}_{j,\alpha}(e \in T)$ and $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T)$. Summing the usage over $e \in S$,

$$m_S = \sum_j \Big(s_j \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[N_S] + t_j \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}\big[|S| - N_S\big] \Big),$$

and with $d^*_j(S) = |S| - r_j(S)$ (7.12) the wall $\kappa_S = m_S - \sum_j \big(s_j d_j(S) + t_j d^*_j(S)\big)$ is the displayed sum.

(2) The sign. By 7.13, $d_j(S) \le N_S \le r_j(S)$ on every basis of table j. So both brackets are nonnegative, and so is κ_S, the exponents being nonnegative.

(3) Equality. The laws $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha)$ and $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$ charge every basis. So the first bracket of table j vanishes exactly when $N_S = d_j(S)$ on every basis, and the second exactly when $N_S = r_j(S)$ on every basis; since the values of N_S fill $\{d_j(S), \dots, r_j(S)\}$ (7.13), each happens exactly when $c_j(S) = 0$. Every table has $s_j > 0$ or $t_j > 0$, so $\kappa_S = 0$ exactly when $c_j(S) = 0$ for every table, that is, by Lemma S3, when S is a union of molecules.

(4) The last sentence. $\mathcal{E}$ is a union of molecules, so $\kappa_{\mathcal{E}} = 0$. At a realization every table is active, so the family is connected (8.1) exactly when no nonempty proper S has $c_j(S) = 0$ for every table, that is, by Lemma S3, when no nonempty proper set is a union of molecules; then every proper wall is positive by step (3). □

Remark. The reference, proof of 9.2, has steps (1) and (2) and the positivity of the bracket of an active table with $c_j(S) \ge 1$, which connectedness supplies for every nonempty proper S. Step (3) reads off the equality case, which holds for every family.

Proposition 9.3 (Masses at a gauge). *As stated, at every gauge e^{y_0}.*

Proof. By 9.1 the masses are $\tau_e = \sum_j \big(s_j R_{j,e}(\alpha) + t_j R^*_{j,e}(\alpha)\big)$, with $R_{j,e} = \alpha_e \partial_e K_j/K_j$ the occupation and $R^*_{j,e} = \alpha_e \partial_e \Psi_j/\Psi_j$ the dual occupation, and $R^*_{j,e}(\alpha) = 1 - R_{j,e}(\alpha^{-1})$ (7.16). At $\alpha = e^{y_0}$ this is the first expression. Put $a_j = R_{j,e}(e^{y_0})$ and $b_j = R_{j,e}(e^{-y_0})$, so that $R^{\pm}_{j,e} = \frac{1}{2}(a_j \pm b_j)$. With $s_j = \sigma_j + \delta_j$ and $t_j = \sigma_j - \delta_j$,

$$s_j a_j + t_j(1 - b_j) = \sigma_j(1 + a_j - b_j) + \delta_j(a_j + b_j - 1) = \sigma_j(1 + 2R^-_{j,e}) + \delta_j(2R^+_{j,e} - 1).$$

At $y_0 = 0$, $a_j = b_j = R_{j,e}(\mathbf{1})$: $R^-_{j,e} = 0$ and $R^+_{j,e} = R_{j,e}(\mathbf{1})$. □

Remark. By 7.17, $R_{j,e}(e^y) = \partial_{y_e} \log K_j(e^y)$, whose Taylor series at 0 is

$$\sum_{i\ge 0} \frac{1}{i!} \sum_{u\in\mathcal{E}^i} \kappa_{j,\mathbf{1}}(e,u) \prod_{l=1}^{i} y_{u_l}$$

in its domain of convergence, as in 9.6. At $y = \pm y_0$ the term of order i carries $(\pm 1)^i$, so $R^+_{j,e}$ collects the terms of even order and $R^-_{j,e}$ those of odd order. Rung one at a deviation has the

parity of the odd rungs of 9.6, σ_j with the odd part of the occupations and δ_j with the even part, besides the constant $\sum_j(\sigma_j-\delta_j)=\sum_j t_j$.

Proposition 9.5 (The potential of a compound splits by molecules). *As stated; the last sentence is new.*

Proof. The first two sentences are the reference, proof of 9.5. Let e be a loop or a coloop in every table. By 8.20 $\{e\}$ is a unit, and every basis of table j contains e, or none does, so $K_j(\alpha)=\alpha_e^{\epsilon_j}\tilde K_j(\alpha_{\mathcal{E}\setminus e})$ and $\Psi_j(\alpha)=\alpha_e^{1-\epsilon_j}\tilde\Psi_j(\alpha_{\mathcal{E}\setminus e})$ with $\epsilon_j\in\{0,1\}$. So $\Phi(y)=\sum_j\big(s_j\epsilon_j+t_j(1-\epsilon_j)\big)y_e$ plus a function of $y_{\mathcal{E}\setminus e}$ (8.18), every derivative of Φ of order at least two that involves y_e vanishes, every entry of the still on a string through e is zero, and rung one at e is the constant $\sum_j\big(s_j\epsilon_j+t_j(1-\epsilon_j)\big)$, the mass of e (9.4). The same holds for the two parts $\Phi_{\mathbf{E}}$ and $\Phi_{\mathbf{O}}$, which are $\sum_j\sigma_j\log(K_j\Psi_j)$ and $\sum_j\delta_j\log(K_j/\Psi_j)$ read at e^y. □

Proposition 9.10 (Occupations at a gauge). *As stated, at every gauge e^{y_0}.*

Proof. By 9.4, $\varphi_e^{\mathbf{X}}(0)=\partial_{y_e}\Phi_{\mathbf{X}}(\log\alpha)$, and by 8.18, $\Phi_{\mathbf{E}}(y)=\sum_j\sigma_j\big(\log K_j(e^y)+\log\Psi_j(e^y)\big)$ and $\Phi_{\mathbf{O}}(y)=\sum_j\delta_j\big(\log K_j(e^y)-\log\Psi_j(e^y)\big)$. By 8.19, $\partial_{y_e}\log K_j(e^y)=R_{j,e}$ and $\partial_{y_e}\log\Psi_j(e^y)=R^*_{j,e}$, and $R^*_{j,e}(\alpha)=1-R_{j,e}(\alpha^{-1})$ (7.16). At $\alpha=e^{y_0}$, with $a_j=R_{j,e}(e^{y_0})$ and $b_j=R_{j,e}(e^{-y_0})$,

$$\varphi_e^{\mathbf{E}}(0)=\sum_j\sigma_j(a_j+1-b_j)=\sum_j\sigma_j(1+2R^-_{j,e}),$$
$$\varphi_e^{\mathbf{O}}(0)=\sum_j\delta_j(a_j-1+b_j)=\sum_j\delta_j(2R^+_{j,e}-1).$$

At $\alpha=\mathbf{1}$, $R^-_{j,e}=0$ and $R^+_{j,e}=R_{j,e}(\mathbf{1})$. The two parts add to the mass of 9.3. □

48.10 Chapter 10: the reading, classes, atoms

Proposition 10.2 (A view through the patterns). *As stated; the inclusion laws are read at every gauge.*

Proof. The expression of the views through $K_{j;e,\mathbf{g}}$ and $\Psi_{j;e,\mathbf{g}}$ and the closing sentence on one table are proved in the reference, proof of 10.2. What follows is the expansion at a gauge α and its reading at the table's gauge. Since $\prod_{i\in p}(1+x_i)=\sum_{A\subseteq p}x^A$,

$$K_{j;e,\mathbf{g}}(\mathbf{1}+x)=\sum_p\vartheta^K_{j,p}\sum_{A\subseteq p}x^A=\sum_A x^A\sum_{p\supseteq A}\vartheta^K_{j,p},$$
$$\sum_{p\supseteq A}\vartheta^K_{j,p}=\sum_{T:\ p(T)\supseteq A}\theta_{j,T}\alpha^T=K_j(\alpha)\,\mathbb{P}_{j,\alpha}\big(p(T)\supseteq A\big),$$

the basis law at α being $\mathbb{P}_{j,\alpha}(T)=\theta_{j,T}\alpha^T/K_j(\alpha)$ (7.16). Likewise $(\mathbf{1}+x)^{\bar p}=\prod_{i\notin p}(1+x_i)=\sum_{A\cap p=\varnothing}x^A$, so

$$\Psi_{j;e,\mathbf{g}}(\mathbf{1}+x)=\sum_A x^A\sum_{p\cap A=\varnothing}\vartheta^\Psi_{j,p},$$
$$\sum_{p\cap A=\varnothing}\vartheta^\Psi_{j,p}=\sum_{T:\ p(T)\cap A=\varnothing}\theta_{j,T}\alpha^{\mathcal{E}\setminus T}=\Psi_j(\alpha)\,\mathbb{P}^*_{j,\alpha}\big(p(T)\cap A=\varnothing\big),$$

with $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$. So the polynomials $K_{j;e,\mathbf{g}}$ and $\Psi_{j;e,\mathbf{g}}$, and with them the view, are determined by $K_j(\alpha)$, $\Psi_j(\alpha)$ and the joint inclusion laws of $e, g_1, \dots, g_k$ under $\mathbb{P}_{j,\alpha}$ and $\mathbb{P}^*_{j,\alpha}$. At $\alpha = \mathbf{1}$, $\alpha^T = \alpha^{\mathcal{E}\setminus T} = 1$: $\vartheta^K_{j,p} = \vartheta^\Psi_{j,p} = \vartheta_{j,p}$, both laws are the weight law (7.16), and $x^{\bar{p}} = x^{\{0,\dots,k\}}(x^{-1})^p$ gives $\Psi_{j;e,\mathbf{g}}(x) = x^{\{0,\dots,k\}}K_{j;e,\mathbf{g}}(x^{-1})$. □

Proposition 10.8 (Computing the classes). *As stated; at a gauge α the identities have their coefficients in the field generated by the weights and the coordinates of α.*

Proof. The steps of the computation compare views and writings, as in the reference, proof of 10.8; only the field is new. Let k be the field generated by the weights and the coordinates of α. By 10.2, a view is the pair $\sum_j \sigma_j \log(K_{j;e,\mathbf{g}}\Psi_{j;e,\mathbf{g}})$, $\sum_j \delta_j \log(K_{j;e,\mathbf{g}}/\Psi_{j;e,\mathbf{g}})$ in $x_i = e^{v_i}$, and the coefficients $\vartheta^K_{j,p} = \sum_{T:\,p} \theta_{j,T}\alpha^T$ and $\vartheta^\Psi_{j,p} = \sum_{T:\,p} \theta_{j,T}\alpha^{\mathcal{E}\setminus T}$ of these polynomials are polynomials with integer coefficients in the weights and the coordinates of α, so they lie in k. Factor every $K_{j;e,\mathbf{g}}$ and $\Psi_{j;e,\mathbf{g}}$ into irreducible polynomials over k. Each factor of a polynomial positive on the orthant vanishes nowhere on it and so has constant sign there, and it is normalized to the value 1 at $x = \mathbf{1}$ by a constant of k. Two different normalized polynomials irreducible over k have no common factor over $\mathbb{C}$, the greatest common divisor of two polynomials over k being their greatest common divisor over $\mathbb{C}$. So the factors over $\mathbb{C}$ of 10.6 are the factors over $\mathbb{C}$ of the factors over k, and a factor over $\mathbb{C}$ divides exactly one normalized factor over k. If two views carry over $\mathbb{C}$ the same factors with the same pairs of exponents, a factor over k of one and a factor over k of the other that share a factor over $\mathbb{C}$ are equal, and comparing the exponents of that shared factor gives equal pairs of exponents. So by 10.6 two views, or two writings, are equal exactly when they carry the same normalized polynomials irreducible over k with the same pairs of exponents: every step is an identity in $k[x]$. At $\alpha = \mathbf{1}$, k is the field generated by the weights, $\mathbb{Q}$ for rational weights, and the coefficients are integers for lists. □

Proposition 10.9 (Masses and occupations on a class). *As stated; the first sentence holds for each of the two parts of the mass.*

Proof. By 9.4, $\varphi^{\mathbf{X}}_e(0) = \partial_{y_e}\Phi_{\mathbf{X}}(\log\alpha)$ is the part $\mathbf{X}$ of rung one, and $\tau_e(\Upsilon) = \partial_{y_e}\Phi(\log\alpha) = \varphi^{\mathbf{E}}_e(0) + \varphi^{\mathbf{O}}_e(0)$. By 10.3(iii) the reading of e determines its self-line functions $\varphi^{\mathbf{E}}_e$ and $\varphi^{\mathbf{O}}_e$, hence their values at 0, and the classes are the level sets of the reading (10.1). So the two parts, and their sum, are constant on every class. For a family whose weights factor across its molecules, $\Phi_{\mathbf{X}}$ is the sum of the potentials of the restrictions to the molecules (9.5), so $\varphi^{\mathbf{X}}_e(0)$ is the same in the family and in the realization of the molecule of e, on which its classes are read. Items (i) and (ii) are proved in the reference, proof of 10.9. □

48.11 Chapter 11: lean, gauge and label

Lemma 11.11 (No degenerate pair). *As stated, at every realization of every family, with the equality case.*

Proof. (1) A sum of covariances. By 9.4 and 8.18, $\kappa^{\mathbf{E}}_\Upsilon(e,f) = 4\sum_j \sigma_j\big(\partial_{y_e}\partial_{y_f}\log K_j(e^y) + \partial_{y_e}\partial_{y_f}\log\Psi_j(e^y)\big)$ at $y = \log\alpha$. By 7.17 the first derivative is $\kappa_{j,\alpha}(e,f)$, and since $\log\Psi_j(e^y) = \langle\mathbf{1}, y\rangle + \log K_j(e^{-y})$ (7.2) the second is $\kappa_{j,\alpha^{-1}}(e,f)$. So $\kappa^{\mathbf{E}}_\Upsilon = 4\sum_j \sigma_j\big(C_{j,\alpha} + C_{j,\alpha^{-1}}\big)$, $C_{j,\beta}$ the covariance matrix of the indicators $(X_e)_e$ under $\mathbb{P}_{j,\beta}$, and for $w \in \mathbb{R}^{\mathcal{E}}$

$$w^{\mathsf{T}}\kappa^{\mathbf{E}}_\Upsilon w = 4\sum_j \sigma_j\Big(\operatorname{Var}_{T\sim\mathbb{P}_{j,\alpha}}\langle w, \mathbf{1}_T\rangle + \operatorname{Var}_{T\sim\mathbb{P}_{j,\alpha^{-1}}}\langle w, \mathbf{1}_T\rangle\Big) \;\geq\; 0,$$

since $\sigma_j > 0$ at a realization.

(2) The inequality. The principal 2×2 submatrix of $\kappa^{\mathbf{E}}_{\Upsilon}$ on $\{e, f\}$ is positive semidefinite, so its determinant $\kappa^{\mathbf{E}}_{\Upsilon}(e,e)\kappa^{\mathbf{E}}_{\Upsilon}(f,f) - \kappa^{\mathbf{E}}_{\Upsilon}(e,f)^2$ is nonnegative. It vanishes exactly when that submatrix is singular, that is when some $w = \lambda \mathbf{1}_e + \mu \mathbf{1}_f \neq 0$ has $w^{\mathsf{T}} \kappa^{\mathbf{E}}_{\Upsilon} w = 0$, a positive semidefinite matrix annihilating exactly the vectors at which its form vanishes.

(3) The vectors of form zero. Both laws in step (1) charge every basis (7.16), so $w^{\mathsf{T}} \kappa^{\mathbf{E}}_{\Upsilon} w = 0$ exactly when $\langle w, \mathbf{1}_T \rangle$ is constant over the bases of every table. By 7.8 such a w is constant on every molecule. Conversely, if w is constant on every molecule, with value w_M on the molecule M, then $\langle w, \mathbf{1}_T \rangle = \sum_M w_M r_j(M)$ on every basis of table j, each table being the direct sum of its restrictions to the molecules (7.9).

(4) The cases. If e is a unit, $w = \mathbf{1}_e$ is constant on every molecule, and equality holds; likewise if f is a unit. If neither is a unit and they lie in different molecules, the molecule of e has a mark other than e, not f, at which $w = \lambda \mathbf{1}_e + \mu \mathbf{1}_f$ vanishes, so $\lambda = 0$; likewise $\mu = 0$, and the inequality is strict. If e and f lie in one molecule M: when M has a third mark, w vanishes there, so $\lambda = \mu = 0$ and the inequality is strict; when $M = \{e, f\}$, $w = \mathbf{1}_e + \mathbf{1}_f$ is constant on every molecule, and equality holds. A molecule with at least three marks contains neither a unit nor a molecule of two marks, so the inequality is strict for any two of its marks. □

Remark. The reference, proof of 11.11, has steps (1) to (3) and concludes for a molecule with at least three marks; the case analysis of step (4) gives the equality case for every family.

Lemma 11.12 (Equal diagonals on a class). *As stated: on every class, for both parts and every rung.*

Proof. By 9.4, $\varphi^{\mathbf{X}}_e(v) = \partial_{y_e} \Phi_{\mathbf{X}}(\log \alpha + v \mathbf{1}_e)$, so $(\varphi^{\mathbf{X}}_e)^{(k-1)}(0) = \partial^k_{y_e} \Phi_{\mathbf{X}}(\log \alpha)$, and the still of each part gives

$$\kappa^{\mathbf{X}}_{\Upsilon}(e, \dots, e) = 2^k \, \partial^k_{y_e} \Phi_{\mathbf{X}}(\log \alpha) = 2^k (\varphi^{\mathbf{X}}_e)^{(k-1)}(0).$$

By 10.3(iii) the reading of e determines $\varphi^{\mathbf{E}}_e$ and $\varphi^{\mathbf{O}}_e$; the marks of a class have one reading (10.1), so they have the same self-line functions, and the same derivatives of every order at 0. At an atom there is one class. □

Remark. The reference states 11.12 at an atom of a molecule with at least three marks, for $\kappa^{\mathbf{E}}(e,e)$, and proves it through the rung-two tallies and 11.11. The self-line functions carry every order and both parts, and make the statement one about classes.

Corollary 11.15 (An exchange negates odd cumulants). *As stated, for an exchange of any family.*

Proof. (1) The weight laws. By 11.13, $T \mapsto T' = \mathcal{E} \setminus g(T)$ maps $\mathcal{B}_j$ onto $\mathcal{B}_{h(j)}$ with $\theta_{h(j),T'} = \theta_{j,T}$, and it is injective because g is. So $K_{h(j)}(\mathbf{1}) = K_j(\mathbf{1})$, and for $T \sim \mathbb{P}_{j,\mathbf{1}}$ the basis T' has the law $\mathbb{P}_{h(j),\mathbf{1}}$.

(2) The cumulants. $X_{g(e)}(T') = \mathbf{1}[g(e) \notin g(T)] = 1 - X_e(T)$. Joint cumulants of order at least two are multilinear and unchanged when constants are added to the variables, so for a string $w = (e_1, \dots, e_k)$, $k \geq 2$,

$$\begin{aligned} \kappa_{h(j),\mathbf{1}}(g(w)) &= \kappa\big(X_{g(e_1)}(T'), \dots, X_{g(e_k)}(T')\big) \\ &= \kappa\big(1 - X_{e_1}(T), \dots, 1 - X_{e_k}(T)\big) = (-1)^k \kappa_{j,\mathbf{1}}(w). \end{aligned}$$

(3) The stills. By 9.7, $\kappa_{\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}}(w) = 2^k \sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w)$. The condition on the pairs in 11.13 gives $(s_{h(j)}, t_{h(j)}) = (s_j, t_j)$, and h is a bijection of the tables; reindexing the sum by h and using step (2),

$$\begin{aligned}\kappa_{\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}}(g(w)) &= 2^k \sum_j \big(s_{h(j)} + (-1)^k t_{h(j)}\big)\kappa_{h(j),\mathbf{1}}(g(w)) \\ &= 2^k \sum_j \big(t_j + (-1)^k s_j\big)\kappa_{j,\mathbf{1}}(w),\end{aligned}$$

the still at $\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}$, whose exponents are (t_j, s_j).

(4) The classes of the centre. By 11.14 with $F' = F$ at the table's gauge, the reading of $g(e)$ at $-\boldsymbol{\delta}$ equals that of e at $\boldsymbol{\delta}$; at the centre $-\boldsymbol{\delta} = \boldsymbol{\delta} = 0$, so $g(e)$ lies in the class of e. Hence g maps every class of the centre into itself, and onto itself, g being injective and the class finite. □

Remark. The reference, proof of 11.15, is the case of one table, where h is the identity.

48.12 Chapter 12: restriction to a bunch

Lemma 12.3 (Children do not widen). *As stated.*

Proof. The child at p is a table on R (12.2), and a set $X \subseteq R$ is one of its bases exactly when $p \sqcup X$ is a basis of the table. Since $p, W \subseteq B$ and $X, A \subseteq R$,

$$|X \cap A| = |(p \sqcup X) \cap (A \cup W)| - |p \cap W|.$$

So the traces of A on the bases of the child are traces of $A \cup W$ on bases of the table, all shifted by the one constant $|p \cap W|$. The width is the largest trace less the least (7.13), so $c_{(p)}(A) \le c(A \cup W)$. □

Proposition 12.13 (Coins of a tensorminental table). *As stated: tensor rank one over* $\mathbb{R}$ *or over* $\mathbb{C}$.

Proof. The first display is proved in the reference, proof of 12.13. At tensor rank one there is one slice $A \in \mathbb{F}^{r\times n}$, of rank r since $K(\mathbf{1}) = \det(AA^{\mathsf{H}}) > 0$ (7.21); the only pair is $(1,1)$, with $w_{11} = 1$, and $\Pi^{(11)} = A^{\mathsf{H}} M^{-1} A D_\alpha$, $M = A D_\alpha A^{\mathsf{H}}$ (7.23). So $x^{d(B)} \mathsf{Q}_B^K(x;\alpha) = \prod_{i=1}^{|B|}(1 - p_i + p_i x)$ with p_i the eigenvalues of $\Pi^{(11)}_{BB}$.

(1) The projection. Put $P = D_\alpha^{1/2} \Pi^{(11)} D_\alpha^{-1/2} = D_\alpha^{1/2} A^{\mathsf{H}} M^{-1} A D_\alpha^{1/2}$. Then $P^{\mathsf{H}} = P$ and $P^2 = D_\alpha^{1/2} A^{\mathsf{H}} M^{-1} (A D_\alpha A^{\mathsf{H}}) M^{-1} A D_\alpha^{1/2} = P$, and $P\, D_\alpha^{1/2} A^{\mathsf{H}} = D_\alpha^{1/2} A^{\mathsf{H}}$; so P is the orthogonal projection onto $\{D_\alpha^{1/2} A^{\mathsf{H}} y\}$, the row space of $A D_\alpha^{1/2}$ in the sense of 7.23.

(2) Similarity. D_α is diagonal, so $P_{BB} = D_B^{1/2} \Pi^{(11)}_{BB} D_B^{-1/2}$, with D_B the diagonal block on B, and P_{BB} has the eigenvalues p_i.

(3) Principal angles. P_{BB} is the compression of P to $\mathbb{F}^B$, the vectors supported on B: $v^{\mathsf{H}} P_{BB} v = \|Pv\|^2$ for $v \in \mathbb{F}^B$. With E_B the inclusion of $\mathbb{F}^B$ and Q a matrix whose columns are an orthonormal basis of the range of P, $P_{BB} = (E_B^{\mathsf{H}} Q)(E_B^{\mathsf{H}} Q)^{\mathsf{H}}$. Its eigenvalues are the squares of the singular values of $E_B^{\mathsf{H}} Q$, completed by zeros, which are the cosines of the principal angles between $\mathbb{F}^B$ and the range of P; they lie in $[0,1]$.

(4) The eigenvalue 1. For $v \in \mathbb{F}^B$, $P_{BB} v = v$ exactly when $\|Pv\| = \|v\|$, that is when v lies in the range of P. So the eigenspace is $\{D_\alpha^{1/2} A^{\mathsf{H}} y :\ (D_\alpha^{1/2} A^{\mathsf{H}} y)_R = 0\} = \{D_\alpha^{1/2} A^{\mathsf{H}} y :\ A_R^{\mathsf{H}} y = 0\}$, of dimension $r - \operatorname{rank} A_R$, the map $y \mapsto D_\alpha^{1/2} A^{\mathsf{H}} y$ being injective. The bases of the table are the

sets T with $\det A_T \neq 0$ (7.20), so $r(R) = \operatorname{rank} A_R$, and $r - r(R) = d(B)$ (7.12, $d(B) = |B| - r^*(B)$ with $r^*(B) = |B| + r(R) - r$).

(5) The eigenvalue 0. For $v \in \mathbb{F}^B$, $P_{BB}v = 0$ exactly when $Pv = 0$, that is $AD_\alpha^{1/2}v = A_B D_B^{1/2} v_B = 0$. So the eigenspace has dimension $|B| - \operatorname{rank} A_B = |B| - r(B) = d^*(B)$.

(6) The coins. P_{BB} is Hermitian, so of its $|B|$ eigenvalues $d(B)$ are 1, $d^*(B)$ are 0, and the other $|B| - d(B) - d^*(B) = c(B)$ lie in $(0,1)$. In $x^{d(B)}\mathsf{Q}_B^K(x;\alpha) = \prod_i (1 - p_i + p_i x)$ the eigenvalues 1 give the factor $x^{d(B)}$ and the eigenvalues 0 the factor 1; so the coins of Q_B^K are the eigenvalues $p_i = \cos^2 \vartheta_i \in (0,1)$, and the Skeleton values are $\xi_i = (1 - p_i)/p_i = \tan^2 \vartheta_i > 0$ (12.10). □

Remark. Over $\mathbb{R}$ this is the reference, proof of 12.13. Over $\mathbb{C}$ the projection is Hermitian, and each step reads the same with conjugate transposes.

Definition 12.18: the claims. $\operatorname{pat}(B) = 2^B$ holds exactly when B is independent (B is a pattern) and some basis avoids B ($\varnothing$ is a pattern): if both hold, $d(B) = 0$, so the contraction $\mathcal{B}/R$ has rank 0, and every subset of B is independent with $r_{\mathcal{B}/R} = 0 = d(B)$, a pattern by Proposition 12.17(i). Equivalently $r(B) = |B|$ and $d(B) = 0$, that is $c(B) = |B|$. The dual table has the bases $\mathcal{E} \setminus T$ and the patterns $B \setminus p$, with the children of cobasis letters $\Psi_{(p)}$; so every subset of B is a pattern of the dual table when every subset is a pattern of the table, and its interaction is $\sum_p (-1)^{|B \setminus (B \setminus p)|} \log \Psi_{(p)} = \iota_B^*$. The parts: $\Phi_{\mathbf{E}} = \sum_j \sigma_j (\log K_j + \log \Psi_j)$ and $\Phi_{\mathbf{O}} = \sum_j \delta_j (\log K_j - \log \Psi_j)$ (Definition 8.18), and $\sigma_j \pm \delta_j$ are s_j and t_j.

Proposition 12.20 (The bottom of a chain factors over its steps). *As stated.*

Proof. Existence. Order the marks of each step arbitrarily and the steps as they come; the sets S_i are initial segments of this order of $\mathcal{E}$, which are the tails of the reversed order, and 8.3 gives a basis T^+ with $|T^+ \cap U| = d(U)$ for every tail U, in particular at the bottom of every S_i.

One step is exchanged at a time. Let T and T' be at the bottom of every S_k, and $1 \leq i \leq \ell + 1$. For $i \geq 2$, by 12.19 at the bunch S_{i-1} the bases at its bottom are all the unions of a pattern of that level and a room side of that level, so $T'' = (T' \cap S_{i-1}) \cup (T \setminus S_{i-1})$ is a basis; for $i = 1$ put $T'' = T$. Its traces are $|T'' \cap S_k| = |T' \cap S_k| = d(S_k)$ for $k < i$ and $d(S_{i-1}) + |T \cap S_k| - |T \cap S_{i-1}| = d(S_k)$ for $k \geq i$. For $i \leq \ell$, 12.19 at the bunch S_i, applied to T'' and T', both at its bottom, gives the basis $(T'' \cap S_i) \cup (T' \setminus S_i) = (T' \setminus Z_i) \cup (T \cap Z_i)$; for $i = \ell + 1$ this set is T'' itself. Its trace on S_k is that of T' for $k < i$, and for $k \geq i$ it is

$$d(S_{i-1}) + \big(d(S_i) - d(S_{i-1})\big) + \big(d(S_k) - d(S_i)\big) = d(S_k),$$

since T and T' meet Z_i in $d(S_i) - d(S_{i-1})$ marks and $S_k \setminus S_i$ in $d(S_k) - d(S_i)$. So the part on one step of a basis at the bottom of every S_k can be replaced by that of any other, and these bases are exactly the unions of their traces on the steps, chosen independently.

The step tables. If $(T_0 \setminus Z_i) \sqcup X$ is a basis with $X \subseteq Z_i$, then $|X| = |T_0 \cap Z_i|$, its traces on the S_k are those of T_0, and it is at the bottom of every S_k. Conversely, for T at the bottom of every S_k the exchange gives the basis $(T_0 \setminus Z_i) \cup (T \cap Z_i)$. So the traces on Z_i are the bases of the child at the pattern $T_0 \setminus Z_i$, a table (12.2), the same set for every T_0. A basis T at the bottom of S_{i-1} and of S_i alone has the same trace on Z_i as one at the bottom of every S_k: the two exchanges above with $T' = T_0$ use only that T is at the bottom of S_{i-1} and of S_i, and give the basis $(T_0 \setminus Z_i) \cup (T \cap Z_i)$, whose traces were computed above. So the step table depends on S_{i-1} and S_i alone.

The top. $|T \cap S_i|$ is largest exactly when the cobasis $\mathcal{E} \setminus T$ meets S_i least, so the bases at the top of every S_i are the complements of the bases at the bottom of every S_i in the dual table (7.1), and the above applies there. For q a pattern of $\mathcal{E} \setminus Z_i$ in the dual table, $X \subseteq Z_i$ is a basis of its

child exactly when $\big((\mathcal{E}\setminus Z_i)\setminus q\big)\sqcup(Z_i\setminus X)$ is a basis of $\mathcal{B}$: the child of the dual table at q is the dual of the child of $\mathcal{B}$ at $(\mathcal{E}\setminus Z_i)\setminus q$. At $q=(\mathcal{E}\setminus T_1)\setminus Z_i$ this is the dual of $\mathcal{B}_{(T_1\setminus Z_i)}$.

The letters. The degree of α^T in α_{S_i} is $|T\cap S_i|$, whose least value is $d(S_i)$, attained for every i at once by the bases at the bottom of every S_i; for a list their sum is $\sum_{X_1,\dots,X_{\ell+1}}\prod_i \alpha^{X_i}=\prod_i K_{\mathcal{B}_{(T_0\setminus Z_i)}}(\alpha_{Z_i})$, and with weights $\theta_T=\prod_i\theta_i(T\cap Z_i)$ the same with the weights θ_i. The degree of $\alpha^{\mathcal{E}\setminus T}$ in α_{S_i} is $|S_i\setminus T|$, least exactly when T is at the top of S_i, and by the dual statement the terms of Ψ of the least degree for every i at once are $\prod_i K_{(\mathcal{B}_{(T_1\setminus Z_i)})^*}(\alpha_{Z_i})=\prod_i\Psi_{\mathcal{B}_{(T_1\setminus Z_i)}}(\alpha_{Z_i})$ (7.2). The bases of $\binom{J}{1}$ on $\mathcal{E}$ are the $\{j\}$, $j\in J$, and $d(S)$ is 1 or 0 as $J\subseteq S$ or not; so $\{j\}$ is at the bottom of every S_k exactly when $j\notin S_k$ for every k with $J\not\subseteq S_k$, that is when $j\in J\cap Z_{i_0}$, Z_{i_0} the first step with $J\subseteq S_{i_0}$. Its step table at Z_{i_0} has the bases $\{j\}$, $j\in J\cap Z_{i_0}$, and its step tables at the other steps the one basis $\varnothing$.

A maximal chain. When every step is one mark, a basis at the bottom of every S_i meets the step $\{e_i\}$ in $d(S_i)-d(S_{i-1})$ marks, so it is determined by the chain, and it exists; its term in K is one monomial, with the coefficient 1 for a list. □

Corollary 12.23 (The native level split is interior). *As stated: the directions come from chains of circuits joining the marks of each side, which a connected table with a positive exponent supplies.*

Proof. The first two sentences, the relative interior and the walls with their exceptions, are proved in the reference, proof of 12.23, without any connectedness. Write $\mathcal{L}$ for the set of level and colevel letters with positive exponent, $c_L>0$ for the exponent of $L\in\mathcal{L}$, N_L for the convex hull of its exponent vectors, and $Q=\sum_{L\in\mathcal{L}}c_LN_L$ for the Minkowski sum of the statement.

(1) A circuit with two marks on one side. Let $a\neq a'$ lie both in B or both in R, and in a common circuit of a table j with a positive exponent. By 7.7 some basis T of table j has $a\in T$, $a'\notin T$ and $T'=(T\setminus\{a\})\cup\{a'\}$ a basis, and $|T\cap B|=|T'\cap B|=:k$, the two marks lying on one side. At the native gauge every basis has positive probability under both laws (7.16). If $s_j>0$, the level letter $K^{(k)}_{j,B}$ has the positive exponent $s_j\mathbb{P}_{j,\alpha^\star}(N_B=k)$, and $N_{K^{(k)}_{j,B}}$ contains $\mathbf{1}_T$ and $\mathbf{1}_{T'}$, whose difference is $\mathbf{1}_a-\mathbf{1}_{a'}$. If $t_j>0$, the colevel letter $\Psi^{(|B|-k)}_{j,B}$ has the positive exponent $t_j\mathbb{P}^*_{j,\alpha^\star}(N^*_B=|B|-k)$, and its polytope contains $\mathbf{1}_{\mathcal{E}\setminus T}$ and $\mathbf{1}_{\mathcal{E}\setminus T'}$, whose difference is $\mathbf{1}_{a'}-\mathbf{1}_a$. Fixing points of the other summands, $c_L(\mathbf{1}_a-\mathbf{1}_{a'})$ is the difference of two points of Q for one $L\in\mathcal{L}$, so every v with $\langle v,x\rangle$ constant on Q has $v_a=v_{a'}$.

(2) The directions. Under the hypothesis the chains join all the marks of B and all the marks of R, so by step (1) such a v is constant on B and on R. Conversely, a monomial α^T of a level letter of level k has $\langle\mathbf{1}_B,\mathbf{1}_T\rangle=k$, and a monomial $\alpha^{\mathcal{E}\setminus T}$ of a colevel letter of colevel k has $\langle\mathbf{1}_B,\mathbf{1}_{\mathcal{E}\setminus T}\rangle=|B\setminus T|=k$; so $\langle\mathbf{1}_B,x\rangle$ is constant on every N_L, hence on Q, and so is $\langle\mathbf{1}_R,x\rangle=\langle\mathbf{1},x\rangle-\langle\mathbf{1}_B,x\rangle$, each letter being homogeneous. So the directions v with $\langle v,x\rangle$ constant on Q are $\operatorname{span}\{\mathbf{1}_B,\mathbf{1}_R\}$.

(3) The exceptions. A linear function is constant on a Minkowski sum with positive coefficients exactly when it is constant on every summand. So S is an exception exactly when $\mathbf{1}_S\in\operatorname{span}\{\mathbf{1}_B,\mathbf{1}_R\}$, that is $S\in\{\varnothing,B,R,\mathcal{E}\}$.

(4) Convergence. Write $\alpha_B=\rho_Bu$ with $\rho_B=\langle\mathbf{1}_B,\alpha_B\rangle$ and $\alpha_B=e^{y_B}$; then $d\alpha_B=\rho_B^{|B|-1}d\rho_B\,d\omega_B(u)=\prod_{e\in B}\alpha_e\,dy_B$, and splitting y_B into its component t along the unit vector $|B|^{-1/2}\mathbf{1}_B$, which changes ρ_B by the factor $e^{t|B|^{-1/2}}$ and leaves u unchanged, and its component orthogonal to $\mathbf{1}_B$, on which u depends, gives $d\rho_B/\rho_B=|B|^{-1/2}dt$ and so $d\omega_B(u)=|B|^{1/2}\prod_{e\in B}u_e\,dy_B$ on the hyperplane $\langle y_B,\mathbf{1}_B\rangle=0$; likewise on R. So with $Y=\{y:\ \langle y_B,\mathbf{1}_B\rangle=$

$\langle y_R, \mathbf{1}_R\rangle = 0\}$ the integral of the statement is $(|B||R|)^{1/2}\int_Y u^{\boldsymbol\tau|_B} v^{\boldsymbol\tau|_R}\prod_{L\in\mathcal{L}} L(u,v)^{-c_L}dy$, with $u = e^{y_B}/\rho_B$ and $v = e^{y_R}/\rho_R$. Each L is homogeneous in α_B and in α_R, and the degrees weighted by the exponents add to τ_B and to τ_R: for the level letters of table j they add to $s_j\mathbb{E}_{T\sim\mathbb{P}_{j,\alpha^\star}}[N_B] = s_j\sum_{e\in B} R_{j,e}(\alpha^\star)$ in α_B, for its colevel letters to $t_j\mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha^\star}}[N^*_B] = t_j\sum_{e\in B} R^*_{j,e}(\alpha^\star)$, and the usage at the native gauge is $\boldsymbol\tau$ (8.17); likewise on R. So the powers of ρ_B and ρ_R cancel, and the integrand is $e^{\langle\boldsymbol\tau,y\rangle}\prod_L L(e^y)^{-c_L}$. Each $L(e^y)$ is at least its least coefficient times $e^{\max_{x\in N_L}\langle x,y\rangle}$, the maximum being attained at an exponent vector; so the integrand is at most $C\exp\big(-(h(y)-\langle\boldsymbol\tau,y\rangle)\big)$, with $h(y) = \max_{x\in Q}\langle x,y\rangle = \sum_L c_L\max_{x\in N_L}\langle x,y\rangle$. By step (2) the affine hull of Q has the direction space $\operatorname{span}\{\mathbf{1}_B,\mathbf{1}_R\}^\perp = Y$, and $\boldsymbol\tau$ lies in the relative interior of Q; so for some $\varepsilon>0$, $\boldsymbol\tau + w\in Q$ for all $w\in Y$ with $\|w\|\le\varepsilon$, and $h(y)\ge\langle\boldsymbol\tau,y\rangle+\varepsilon\|y\|$ for $y\in Y$. The integrand is at most $Ce^{-\varepsilon\|y\|}$, integrable over Y.

(5) A connected table. In a connected table any two marks lie in a common circuit, the fact used in the reference, proof of 12.23; so a connected table with a positive exponent joins any two marks of B, and any two marks of R, by a chain of length one. □

Remark. The hypothesis cannot be replaced by connectedness of the family. On $\mathcal{E} = \{a,a',b,b'\}$ with $B = \{a,a'\}$, let the three tables of rank one with the bases $\{a\},\{b\}$, then $\{a'\},\{b'\}$, then $\{a\},\{b'\}$ carry positive exponents, read at a native gauge. The family is connected, but every level of every table holds one basis, so every level and colevel polytope is a point, every set is an exception, and the wall at $\{a\}$ vanishes. Each circuit $\{a,b\}$, $\{a',b'\}$, $\{a,b'\}$ meets both sides, and no chain on one side joins a to a'.

48.13 Chapter 13: the order of classes and the Foreman bunch

Proposition 13.10 (Mirror classes). *As stated: levels mirrored up to a bijection of the tables, and the exchange of any family.*

Proof. (i) Fix j. Under $\mathbb{P}_{h(j),\mathbf{1}}$ the level N_D takes the value a with probability $\mathbb{P}_{j,\mathbf{1}}(N_C = k_j - a)$, so it has the law of $k_j - N_C$ under $\mathbb{P}_{j,\mathbf{1}}$. A cumulant of order $a\ge 2$ is unchanged by adding a constant and is homogeneous of degree a, so $\kappa_{h(j),\mathbf{1}}(N_D^{\times a}) = (-1)^a\kappa_{j,\mathbf{1}}(N_C^{\times a})$. Since h is a bijection of the tables with $\sigma_{h(j)} = \sigma_j$,

$$\sum_j \sigma_j\kappa_{j,\mathbf{1}}(N_D^{\times a}) = \sum_j \sigma_{h(j)}\kappa_{h(j),\mathbf{1}}(N_D^{\times a}) = (-1)^a\sum_j \sigma_j\kappa_{j,\mathbf{1}}(N_C^{\times a}),$$

and likewise with the δ_j. By 13.4(i) the coefficients of v_0^a, $a\ge 2$, in the **E** part of stage 0 are $\frac{2}{a!}\sum_j\sigma_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ at even a and zero at odd a, and in its **O** part $\frac{2}{a!}\sum_j\delta_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ at odd a and zero at even a; so the **E** parts of $\mathcal{V}_0(C)$ and $\mathcal{V}_0(D)$ are equal and their **O** parts are negatives.

(ii) Stage 0 holds one pair. By (i) the **E** parts agree, so the pairs are compared by their **O** parts (13.5(iii)). Their coefficients vanish at every even a and at every odd a below the least odd a with $c = \frac{2}{a!}\sum_j\delta_j\kappa_{j,\mathbf{1}}(N_C^{\times a})\neq 0$, where C has c and D has $-c$. So stage 0 separates C and D, and $C\prec D$ exactly when $c<-c$, that is $c<0$. At the opposite lean c changes sign, being linear in $\boldsymbol\delta$, and the order is the other one.

(iii) At the centre the **O** parts vanish and the **E** parts agree by (i), so stage 0 does not separate C and D. They are different classes with equally many marks and $\prec$ is a strict total order on them (13.7(i)), so a later stage does.

The two instances. If C and D together make up the molecule, $|T\cap D| = r_j - |T\cap C|$ on every basis of table j: h is the identity and $k_j = r_j$. If g exchanges the family, with h the bijection of its

tables, then for $T \sim \mathbb{P}_{j,\mathbf{1}}$ the basis $T' = \mathcal{E} \setminus g(T)$ has the law $\mathbb{P}_{h(j),\mathbf{1}}$ (proof of 11.15, step (1)) and $|T' \cap g(C)| = |g(C) \setminus g(T)| = |C \setminus T| = |C| - |T \cap C|$, so $\mathbb{P}_{h(j),\mathbf{1}}(|T \cap g(C)| = a) = \mathbb{P}_{j,\mathbf{1}}(|T \cap C| = |C| - a)$ and $k_j = |C|$. Moreover $g(C)$ is a class at the lean with $|C|$ marks: g maps the classes at $\boldsymbol{\delta}$ onto those at $-\boldsymbol{\delta}$ (11.14), which are the classes at $\boldsymbol{\delta}$ (11.2). □

Remark. The reference, proof of 13.10, takes the mirror table by table, h the identity, and the exchange of a one-table molecule. Its conclusions use only the sums over the tables weighted by the σ_j and the δ_j, which a bijection of the tables preserving the pairs leaves unchanged.

48.14 Chapter 14: peel, children and the Foreman tree

Corollary 14.9 (A molecule with an exchange). *As stated: a molecule with any number of tables; the classes of the centre are those of 11.15.*

Proof. Read M as the family $F|M$ of the restrictions of the tables to M (14.1), with the pair coordinates (σ_j, δ_j) of its tables. By 11.13 an exchange g of M comes with a bijection h of the tables such that $T \mapsto M \setminus g(T)$ maps the bases of table j onto those of table $h(j)$, with their weights, and $(\sigma_{h(j)}, \delta_{h(j)}) = (\sigma_j, \delta_j)$. The first two conditions concern the tables alone, and the third holds with $\boldsymbol{\delta}$ replaced by $-\boldsymbol{\delta}$ or by 0 when it holds at $\boldsymbol{\delta}$. So g exchanges M at each of the leans $\boldsymbol{\delta}$, $-\boldsymbol{\delta}$ and 0; likewise a bijection carrying M at $\boldsymbol{\delta}$ carries it at these three leans.

Trees and bunches. 14.8 with $F = F' = M$ gives $g\big(\operatorname{Tree}(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big) = \operatorname{Tree}(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$, each recorded bunch going onto the bunch recorded at the same place. At the root the recorded bunches are $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ and $B(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$ (14.4), so $g\big(B(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big) = B(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$. At the lean 0 the two trees are one tree, which gives $g\big(\operatorname{Tree}(M;\boldsymbol{\sigma},0)\big) = \operatorname{Tree}(M;\boldsymbol{\sigma},0)$ and $g\big(B(M;\boldsymbol{\sigma},0)\big) = B(M;\boldsymbol{\sigma},0)$.

Classes of the centre. 11.15, applied to the family $F|M$.

A carrying map. Let g carry M. 11.14 at the tables' gauges, where $g^* \log \mathbf{1} = \log \mathbf{1}$, says that the reading of $g(e)$ equals that of e at $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$, for every mark e. So $g(e)$ lies in the class of e, that is $g(C) \subseteq C$ for every class C, and $g(C) = C$ since g is injective and C is finite. The last sentence of 14.8, at each of the three leans at which g carries M, maps each of the trees at 0, $\boldsymbol{\delta}$ and $-\boldsymbol{\delta}$ onto itself. □

Lemma 14.10 (An exchange turns a window over). *As stated: an exchange of one family with another, table by table through the bijection h of the tables.*

Proof. Let F be on $\mathcal{E}$ and F' on $\mathcal{E}'$. By 11.13 the map $T \mapsto \mathcal{E}' \setminus g(T)$ is a bijection from the bases of table j of F onto those of table $h(j)$ of F'. For a basis T, g being a bijection,

$$\big|(\mathcal{E}' \setminus g(T)) \cap g(S)\big| = |g(S)| - |g(T) \cap g(S)| = |S| - |T \cap S|.$$

By 7.13 the numbers $|T \cap S|$, T a basis of table j, fill $\{d_j(S), \dots, r_j(S)\}$. So the numbers $|T' \cap S'|$, T' a basis of table $h(j)$, fill $\{|S| - r_j(S), \dots, |S| - d_j(S)\}$, a window of the same width $c_j(S) = r_j(S) - d_j(S)$, and the bijection takes the bases of table j at level k on S to those of table $h(j)$ at level $|S| - k$ on S'. This is the reference, proof of 14.10, with h. □

Proposition 14.12 (Level form). *As stated: table by table in a family, and for the family through the bijections that carry it with the identity on its tables.*

Proof. Fix a table j and a transposition τ of two marks of B fixing every other mark. A basis T with $T \cap B = p$ is $p \sqcup X$ with X a basis of the child at p, of weight $\theta_{j,(p),X} = \theta_{j,T}$, and

$K_{j,(p)}(\alpha_R) = \sum_X \theta_{j,(p),X}\alpha_R^X$ (12.1, 12.7). Distinct bases of one child give distinct monomials, so $K_{j,(p)}$ determines the child at p with its weights, and $K_{j,(p)} = 0$ when p is not a pattern.

Let $K_{j,(p)} = K_{j,(p')}$ whenever $|p| = |p'|$. For a basis $T = p \sqcup X$, $\tau(T) = \tau(p) \sqcup X$, since τ fixes R, and $|\tau(p)| = |p|$. So X is a basis of the child at $\tau(p)$ with the weight $\theta_{j,(p),X}$: $\tau(T)$ is a basis of table j and $\theta_{j,\tau(T)} = \theta_{j,T}$. Hence τ maps the bases of table j into themselves with their weights, and onto them, being injective on a finite set.

Conversely, let every such transposition map the bases of table j onto themselves with their weights, and let $p, p' \subseteq B$ have $|p| = |p'|$. The transpositions of marks of B generate the permutations of B, so a product ς of them has $\varsigma(p) = p'$; it maps the bases onto themselves with their weights and fixes R. So $p \sqcup X \mapsto p' \sqcup X$ is a bijection from the bases with trace p onto those with trace p' keeping the weights, and $K_{j,(p')} = K_{j,(p)}$. This is the reference, proof of 14.12, for table j, where 14.7(i) is used with the identity on R.

B is in level form in the family when it is in level form in every table, that is, by the above, when every such transposition maps the bases of every table onto themselves with their weights. That is the condition of 11.13 with $F' = F$ and h the identity, whose condition on the pairs, $(\sigma_{h(j)}, \delta_{h(j)}) = (\sigma_j, \delta_j)$, then holds by itself. □

Definition 14.17 (Keeping the potential; twins; descents; copies). As stated: keeping and taking the potential at every realization; twins, descents and copies at the table's gauge.

Lemma 14.18 (The writings certify the potential). *As stated, at every realization* $\Upsilon = ((s_j, t_j)_j; \alpha)$.

Proof. Let M have n marks, and let $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ be its potential. For a mark x and an order $w = (w_1, \ldots, w_{n-1})$ of its room, the potential written from x in the order w is the pair of functions

$$\Phi^{\mathbf{X}}_{x,w}(v_0, \ldots, v_{n-1}) = \Phi_{\mathbf{X}}\Big(\log\alpha + v_0\mathbf{1}_x + \sum_{i=1}^{n-1} v_i\mathbf{1}_{w_i}\Big)$$

(10.1). For a bijection g and a mark a, $g^*\mathbf{1}_{g(a)} = \mathbf{1}_a$, since $(g^*\mathbf{1}_{g(a)})_e = (\mathbf{1}_{g(a)})_{g(e)} = \mathbf{1}[e = a]$.

(i) By 10.4 there are orders $(a_1, \ldots, a_{n-1})$ of the room of e and $(b_1, \ldots, b_{n-1})$ of the room of f with $\Phi^{\mathbf{X}}_{e,(a_i)_i} = \Phi^{\mathbf{X}}_{f,(b_i)_i}$ for both parts. Put $a_0 = e$ and $b_0 = f$, and let β be the bijection with $\beta(a_i) = b_i$ for $0 \le i \le n-1$; so $\beta(e) = f$. For $y \in \mathbb{R}^M$ put $v_i = y_{b_i}$. Then $\sum_i v_i\mathbf{1}_{b_i} = y$ and $\sum_i v_i\mathbf{1}_{a_i} = \sum_i y_{\beta(a_i)}\mathbf{1}_{a_i} = \beta^*y$, and the equality of the two writings at $(v_0, \ldots, v_{n-1})$ reads

$$\Phi_{\mathbf{X}}(\log\alpha + \beta^*y) = \Phi_{\mathbf{X}}(\log\alpha + y)$$

for every y and both parts: β keeps the potential (14.17).

(ii) Let g keep the potential. For a mark x, an order w of its room and $v \in \mathbb{R}^n$, put $y = v_0\mathbf{1}_{g(x)} + \sum_i v_i\mathbf{1}_{g(w_i)}$; then $g^*y = v_0\mathbf{1}_x + \sum_i v_i\mathbf{1}_{w_i}$, and

$$\Phi^{\mathbf{X}}_{x,w}(v) = \Phi_{\mathbf{X}}(\log\alpha + g^*y) = \Phi_{\mathbf{X}}(\log\alpha + y) = \Phi^{\mathbf{X}}_{g(x),g(w)}(v), \qquad g(w) = (g(w_1), \ldots, g(w_{n-1})).$$

As w runs over the orders of the room of x, $g(w)$ runs over those of the room of $g(x)$, once each. So the readings of x and of $g(x)$, the multisets of these pairs of functions, are equal, and $g(x)$ lies in the class of x. Thus $g(C) \subseteq C$ for every class C, and $g(C) = C$, g being injective and C finite.

Let $g : M \to M'$ take the potential of M, read at Υ, to that of M', read at a realization with the gauge α'. The same computation gives $\Phi^{\mathbf{X}}_{x,w}(v) = \Phi'^{\mathbf{X}}_{g(x),g(w)}(v) + c_{\mathbf{X}}$: the reading of x in M is the reading of $g(x)$ in M' with every function shifted by the constants $c_{\mathbf{X}}$. So two marks of M

have equal readings exactly when their images have equal readings in M', and g maps the classes of M onto those of M'.

The centre of $\Upsilon = \Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha}$ is $\Upsilon_{\boldsymbol{\sigma},0;\alpha}$, read at the same gauge (10.1). Its potential is $(\Phi_{\mathbf{E}}, 0)$, $\Phi_{\mathbf{E}} = \sum_j \sigma_j \log \mathbf{E}_j(e^y)$ being the same function at every lean (8.18). A bijection keeping the potential at Υ keeps it at the centre, and one taking the potential of M to that of M' at their realizations takes it at their centres, with the constants $(c_{\mathbf{E}}, 0)$; so the two preceding paragraphs hold at the centre as well.

For the last sentence: if e and f read alike, (i) gives a bijection keeping the potential that takes e to f; conversely, a bijection keeping the potential maps the class of e onto itself by (ii), so if it takes e to f, then f lies in that class. □

Remark. At the table's gauge $\log \alpha = 0$, the two conditions of 14.17 read $\Phi_{\mathbf{X}} \circ g^* = \Phi_{\mathbf{X}}$ and $\Phi_{\mathbf{X}} \circ g^* = \Phi'_{\mathbf{X}} + c_{\mathbf{X}}$, and the proof above is the reference, proof of 14.18. The statements that keep or take the potential at the table's gauge, 14.20, 14.21, 14.22, 29.9 and 29.11, and the twins, descents and copies of 14.17, read these conditions as in the reference.

48.15 Chapter 15: the integral at a bunch

Proposition 15.2 (Where the Breta of one bunch converges). *As stated.*

Proof. The set. If a convex combination of the coordinates of $t\boldsymbol{\xi} + (1-t)\mathbf{1}$, $0 < t \le 1$, were a point $p \in (-\infty, 0]$, the same combination of the ξ_ℓ would be $(p-1+t)/t \in (-\infty, 0]$; at $t = 0$ the combination is 1. So Ξ_J is star-shaped about $\mathbf{1}$, hence connected. The convex hull of the ξ_ℓ is compact and avoids the closed set $(-\infty, 0]$, so it lies at a distance $\delta > 0$ from it, and moving every ξ_ℓ by less than δ moves every convex combination by less than δ: Ξ_J is open. Convex combinations of points with positive real parts have positive real parts.

The integral. Put $\nu = |\mathbf{w}| - \kappa$, so $\operatorname{Re}\nu > 0$, and $z(u) = \sum_\ell \xi_\ell u_\ell$, which for u in the closed simplex lies in the convex hull of the ξ_ℓ. On a compact neighbourhood of a point of Ξ_J inside Ξ_J these hulls stay at a distance at least some $\delta' > 0$ from $(-\infty, 0]$, so the principal branch $z^{-\nu} = \exp(-\nu \log z)$ is holomorphic in $\boldsymbol{\xi}$ and $|z^{-\nu}| = |z|^{-\operatorname{Re}\nu} e^{\operatorname{Im}\nu \arg z} \le \delta'^{-\operatorname{Re}\nu} e^{\pi|\operatorname{Im}\nu|}$ there. Since $u^{\mathbf{w}-\mathbf{1}}$ is integrable on Δ_J for $\operatorname{Re} w_\ell > 0$ (3.9), the integral converges absolutely, locally uniformly, and is holomorphic in $\boldsymbol{\xi}$ on Ξ_J. At positive coefficients it is the Breta: with the unit configuration on one bunch the Zamma is $\Gamma(\nu)^{-1} \int_0^\infty x^{\nu-1} e^{-x\sum_\ell \xi_\ell u_\ell} dx = z(u)^{-\nu}$ (2.1, 2.12).

The identity. The left side of 15.1 is holomorphic on $(\mathbb{C} \setminus (-\infty, 0])^J \supseteq \Xi_J$, and the stacked Gamma does not depend on $\boldsymbol{\xi}$; the two sides agree on $(0, \infty)^J$. A holomorphic function on the connected open set Ξ_J that vanishes on the nonempty open set $(0, \infty)^J$ of its real points vanishes identically, its Taylor coefficients at a real point being its real derivatives there; so they agree on Ξ_J. □

Corollary 15.4 (The scale of a bunch of width one). *As stated: every bunch, table by table, with one Skeleton value in each table of width one.*

Proof. Fix a table j and write $d = d_j(B)$, $d^* = d_j^*(B)$ and $c = c_j(B)$. By 7.13 the levels $|T \cap B|$ fill $\{d, \dots, d+c\}$, and the colevels $|B \setminus T|$ fill $\{|B| - d - c, \dots, |B| - d\} = \{d^*, \dots, d^* + c\}$, since $d^* = |B| - r_j(B) = |B| - d - c$ (7.12). Each level letter $K_{j,B}^{(k)}$ and each colevel letter $\Psi_{j,B}^{(k)}$ in these windows is a sum of positive terms over a nonempty set of bases, positive on the orthant. By

12.16,

$$x^d \mathsf{Q}^K_{j,B}(x;\alpha) = \sum_{k=d}^{d+c} x^k \frac{K^{(k)}_{j,B}(\alpha)}{K_j(\alpha)}, \qquad x^{d^*} \mathsf{Q}^\Psi_{j,B}(x;\alpha) = \sum_{k=d^*}^{d^*+c} x^k \frac{\Psi^{(k)}_{j,B}(\alpha)}{\Psi_j(\alpha)}.$$

If $c = 0$ both Skeleton polynomials are the constant 1, of degree 0, and the spectra are empty (12.10). If $c = 1$, then $K_j = K^{(d)}_{j,B} + K^{(d+1)}_{j,B}$, and with $\xi = K^{(d)}_{j,B}(\alpha)/K^{(d+1)}_{j,B}(\alpha) > 0$,

$$\mathsf{Q}^K_{j,B}(x;\alpha) = \frac{K^{(d)}_{j,B}(\alpha) + xK^{(d+1)}_{j,B}(\alpha)}{K^{(d)}_{j,B}(\alpha) + K^{(d+1)}_{j,B}(\alpha)} = \frac{x+\xi}{1+\xi}.$$

So the Skeleton value is ξ, the level letter at the floor over the level letter at the largest level, and the coin is $1/(1+\xi) = K^{(d+1)}_{j,B}(\alpha)/K_j(\alpha)$, the probability under $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha)$ of the bases at level $d+1$ (7.16), that is $\mathbb{P}_{j,\alpha}(N_B = d+1)$. The same computation with the colevel letters gives the Skeleton value $\Psi^{(d^*)}_{j,B}(\alpha)/\Psi^{(d^*+1)}_{j,B}(\alpha)$ and the coin $\Psi^{(d^*+1)}_{j,B}(\alpha)/\Psi_j(\alpha) = \mathbb{P}^*_{j,\alpha}(N^*_B = d^*+1)$, since $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$.

Let $B = \{e\}$ with e neither a loop nor a coloop of table j. Then $d = 0$ and $r_j(\{e\}) = 1$, so $c = 1$ and $d^* = 0$. The bases at level 0 avoid e and those at level 1 contain it, so $K^{(0)}_{j,\{e\}} = K_{j,\setminus e}(\alpha_R)$ and $K^{(1)}_{j,\{e\}} = \alpha_e K_{j,/e}(\alpha_R)$; the bases at colevel 0 contain e and those at colevel 1 avoid it, so $\Psi^{(0)}_{j,\{e\}} = \Psi_{j,/e}(\alpha_R)$ and $\Psi^{(1)}_{j,\{e\}} = \alpha_e \Psi_{j,\setminus e}(\alpha_R)$ (7.3). The coins are $\alpha_e K_{j,/e}/K_j = \mathbb{P}_{j,\alpha}(e \in T) = R_{j,e}(\alpha)$ and $\alpha_e \Psi_{j,\setminus e}/\Psi_j = \mathbb{P}^*_{j,\alpha}(e \notin T) = R^*_{j,e}(\alpha)$ (7.16). This is the reference, proof of 15.4. □

Theorem 15.6 (The scale in its spectral gaps). *As stated: complex coefficients and letter exponents, expanded about any point r of the axis with $|\xi_\ell - r| < r$ for every ℓ.*

Proof. The left side is the value of the scale integral $\int_0^\infty x^{\kappa-1} \prod_{\ell\in J}(x+\xi_\ell)^{-w_\ell} dx$, which converges since $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$ (15.1). For $x \geq 0$ put $q_\ell(x) = (r-\xi_\ell)/(x+r)$, so that $x + \xi_\ell = (x+r)(1-q_\ell(x))$ and $|q_\ell(x)| \leq |r-\xi_\ell|/r = |g_\ell| < 1$.

Expansion. Since $x + r > 0$ and $\operatorname{Re}(1-q_\ell(x)) > 0$, the principal argument of $x+\xi_\ell$ is that of $1-q_\ell(x)$, which lies in $(-\pi/2, \pi/2)$; so $(x+\xi_\ell)^{-w_\ell} = (x+r)^{-w_\ell}(1-q_\ell(x))^{-w_\ell}$ with principal powers, and the binomial series $(1-q)^{-w} = \sum_{n\geq 0} \frac{(w)_n}{n!} q^n$ converges absolutely for $|q| < 1$. Hence

$$x^{\kappa-1} \prod_\ell (x+\xi_\ell)^{-w_\ell} = \sum_{\mathbf{n}\in\mathbb{N}^J} \prod_\ell \frac{(w_\ell)_{n_\ell}}{n_\ell!} (r-\xi_\ell)^{n_\ell}\, x^{\kappa-1}(x+r)^{-|\mathbf{w}|-|\mathbf{n}|}.$$

Term by term. $|(w)_n| = \prod_{i<n}|w+i| \leq \prod_{i<n}(|w|+i) = (|w|)_n$, and $(x+r)^{-|\mathbf{n}|} \leq r^{-|\mathbf{n}|}$. So the moduli of the terms sum to at most

$$x^{\operatorname{Re}\kappa-1}(x+r)^{-\operatorname{Re}|\mathbf{w}|} \prod_\ell \sum_{n\geq 0} \frac{(|w_\ell|)_n}{n!} |g_\ell|^n = x^{\operatorname{Re}\kappa-1}(x+r)^{-\operatorname{Re}|\mathbf{w}|} \prod_\ell (1-|g_\ell|)^{-|w_\ell|},$$

which is integrable on $(0,\infty)$ since $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$; so the series may be integrated term by term. For $0 < \operatorname{Re}\kappa < \operatorname{Re} a$ the integral of a linear letter, in the paragraph after 15.1 with $c_0 = r$ and $c_1 = 1$, is

$$\int_0^\infty x^{\kappa-1}(x+r)^{-a} dx = \Gamma \begin{bmatrix} \kappa, & a-\kappa \\ & a \end{bmatrix} r^{\kappa-a},$$

and at $a = |\mathbf{w}| + |\mathbf{n}|$, with $(r-\xi_\ell)^{n_\ell} r^{-n_\ell} = g_\ell^{n_\ell}$, the term of $\mathbf{n}$ is the term of the statement.

Root rate. Let A_M be the sum of the terms with $|\mathbf{n}| = M$, $v = \sum_\ell |w_\ell|$ and $\bar g = \max_\ell |g_\ell|$. Comparing the coefficients of z^M in $\prod_\ell (1-z)^{-|w_\ell|} = (1-z)^{-v}$ gives $\sum_{|\mathbf{n}|=M} \prod_\ell \frac{(|w_\ell|)_{n_\ell}}{n_\ell!} = \frac{(v)_M}{M!}$. The Beta corner of order M is the integral $\int_0^1 x^{\kappa-1}(1-x)^{|\mathbf{w}|+M-\kappa-1}dx$ (3.9), of modulus at most $b_0 = \int_0^1 x^{\operatorname{Re}\kappa-1}(1-x)^{\operatorname{Re}|\mathbf{w}|-\operatorname{Re}\kappa-1}dx < \infty$. So $|A_M| \le |r^{\kappa-|\mathbf{w}|}|\, b_0 \frac{(v)_M}{M!}\bar g^M$, and $\frac{(v)_M}{M!} = \prod_{i=1}^M \left(1+\frac{v-1}{i}\right) \le e^{|v-1|(1+\log M)}$ grows at most polynomially in M. Hence $\limsup_M |A_M|^{1/M} \le \bar g$ (1.1(viii)).

The remaining sentences. A coefficient $\xi_\ell = r$ has $g_\ell = 0$, and every term with $n_\ell > 0$ vanishes. For $0 < \xi_\ell \le r$, $w_\ell > 0$ and $0 < \kappa < |\mathbf{w}|$ every factor of every term is positive, a factor g_ℓ^0 being 1. For positive coefficients write $\xi_{\min}$ and $\xi_{\max}$ for the least and the largest; $|1-\xi/r|$ is largest over $\xi \in [\xi_{\min}, \xi_{\max}]$ at an end, so $\max_\ell |g_\ell| = \max\left(|1-\xi_{\min}/r|, |1-\xi_{\max}/r|\right)$. For $r \le \xi_{\min}$ this is $\xi_{\max}/r - 1$, decreasing in r; for $r \ge \xi_{\max}$ it is $1-\xi_{\min}/r$, increasing in r; for $\xi_{\min} \le r \le \xi_{\max}$ it is the larger of $1-\xi_{\min}/r$, increasing, and $\xi_{\max}/r - 1$, decreasing, which are equal at $r = \frac12(\xi_{\min}+\xi_{\max})$, with the value $\frac{\xi_{\max}-\xi_{\min}}{\xi_{\max}+\xi_{\min}}$. At that r, $|\xi_\ell - r| \le \frac12(\xi_{\max}-\xi_{\min}) < r$ for every ℓ. At $r = \xi_{\max}$ the bound is $1-\xi_{\min}/\xi_{\max}$, the largest coefficient drops out, and the statement is the reference, proof of 15.6. □

Corollary 15.11 (The period closes on the ordering sectors). *As stated: complex data with $\kappa_{\mathcal{E}} = 0$ and every proper wall of positive real part.*

Proof. Sector form. Let Ω be the set of data at which every proper wall has positive real part. It is open and convex, the walls being affine in the data with real coefficients, and it contains points at which $\operatorname{Re}\kappa_{\mathcal{E}} > 0$ as well (raise every mass). By 8.5(ii), valid for all complex exponents, and 8.7, wherever every wall has positive real part

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_\pi I_\pi, \qquad I_\pi = \int_{[0,1]^{n-1}} \prod_{k=2}^n q_k^{\kappa_{U_k}-1} h_\pi(q)\, dq, \quad h_\pi = L_\pi^{-\kappa_{\mathcal{E}}} \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}.$$

The sector letters are polynomials with nonnegative coefficients and positive constant terms, and $L_\pi \in [1,n]$ on the cube, so h_π is smooth on a neighbourhood of the cube, with every q-derivative continuous in q and the data and entire in the data, and $|h_\pi|$ is bounded on the cube locally uniformly in the data. The tails U_k, $k \ge 2$, of the orders are the nonempty proper sets of marks. So every I_π converges and is holomorphic on Ω, and so is P_F (8.7); Ω being connected, $P_F = \sum_\pi I_\pi$ on Ω. At $\kappa_{\mathcal{E}} = 0$, $h_\pi = \prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$.

The identity and the range. In the sector coordinates $\alpha^T = \rho^{|T|}\prod_k q_k^{N_{U_k}(T)}$, so $k_{j,\pi}(q) = \sum_T \theta_{j,T}\prod_k q_k^{N_{U_k}(T)-d_j(U_k)}$ (8.5(ii)), with exponents in $\mathbb{N}$, all zero only at $T = T^+_{j,\pi}$ (8.3). So $k_{j,\pi}/K_j(\mathbf{1})$ increases in each q_k on the cube, from $\theta_{j,T^+_{j,\pi}}/K_j(\mathbf{1})$ at $q = 0$ to 1 at $q = \mathbf{1}$, and $0 \le 1 - k_{j,\pi}/K_j(\mathbf{1}) \le \lambda$. The identity is the expectation under $\mathbb{P}_{j,\mathbf{1}}$ of $1-\prod_k q_k^{e_k} = \sum_k \prod_{k'<k} q_{k'}^{e_{k'}}(1-q_k^{e_k})$, $e_k = N_{U_k} - d_j(U_k)$, with $1-q^e = (1-q)\sum_{0\le i<e} q^i$. The same holds for $\psi_{j,\pi}$ with N^*, d_j^* and $T^-_{j,\pi}$, and $\Psi_j(\mathbf{1}) = K_j(\mathbf{1})$. These are statements about the tables alone, as in the reference, proof of 15.11.

Expansion. With $a_j = 1 - k_{j,\pi}/K_j(\mathbf{1})$ and $b_j = 1 - \psi_{j,\pi}/K_j(\mathbf{1})$ in $[0,\lambda]$, $k_{j,\pi}^{-s_j} = K_j(\mathbf{1})^{-s_j}\sum_{n\ge0}\frac{(s_j)_n}{n!}a_j^n$ and $\psi_{j,\pi}^{-t_j} = K_j(\mathbf{1})^{-t_j}\sum_{n\ge0}\frac{(t_j)_n}{n!}b_j^n$, with principal powers of positive numbers. By $|(s)_n| \le (|s|)_n$ the moduli of the terms of the product of these series sum to at most $\prod_j K_j(\mathbf{1})^{-\operatorname{Re}(s_j+t_j)}(1-\lambda)^{-|s_j|-|t_j|}$ on the cube, and $\int \prod_k q_k^{\operatorname{Re}\kappa_{U_k}-1}dq = \prod_k(\operatorname{Re}\kappa_{U_k})^{-1} < \infty$. So the series may be integrated term by term over the cube, which gives the displayed series, convergent.

Terms. By the identity, a_j and b_j are sums, with nonnegative coefficients, of products $(1-q_k)q^{\boldsymbol{\beta}}$, $\boldsymbol{\beta}\in\mathbb{N}^{n-1}$; so $\prod_j a_j^{n_j}b_j^{n'_j}$ is a finite sum, with nonnegative coefficients, of products $\prod_k(1-q_k)^{\gamma_k}q_k^{\beta_k}$, and for $\operatorname{Re}\kappa>0$, $\int_0^1 q^{\kappa+\beta-1}(1-q)^{\gamma}dq=\Gamma\,{}_2_1\!\left[\begin{matrix}\kappa+\beta,\ \gamma+1\\ \kappa+\beta+\gamma+1\end{matrix}\right]$ (3.9). For real data with $s_j,t_j\ge 0$ the coefficients $(s_j)_n/n!$ and $(t_j)_n/n!$ and the integrands are nonnegative.

Bound. The terms with $|\mathbf{n}|+|\mathbf{n}'|=M$ have moduli summing to at most

$$\prod_j K_j(\mathbf{1})^{-\operatorname{Re}(s_j+t_j)}\sum_{|\mathbf{n}|+|\mathbf{n}'|=M}\prod_j\frac{(|s_j|)_{n_j}(|t_j|)_{n'_j}}{n_j!\,n'_j!}\,\lambda^M\sum_{\pi}\prod_k(\operatorname{Re}\kappa_{U_k})^{-1},$$

and comparing coefficients of z^M in $\prod_j(1-z)^{-|s_j|}(1-z)^{-|t_j|}=(1-z)^{-w}$ gives $\frac{(w)_M}{M!}$ for the inner sum, $w=\sum_j(|s_j|+|t_j|)$. □

Proposition 15.12 (An enclosure of the period in the field of its data). *As stated: weights, masses and exponents in a subfield k of $\mathbb{R}$, the exponents of any sign.*

Proof. The data are real, so by the hypotheses of 15.11 every proper wall is positive, and every $\kappa_{U_k}>0$.

(i) The weights lie in k, so the coefficients of the sector letters, sums of weights (8.5(ii)), and $K_j(\mathbf{1})$ lie in k, and $a_{j,\pi},b_{j,\pi}\in k[q]$. For $s\in k$ and $n\in\mathbb{N}$, $(s)_n/n!\in k$. So $f_{\pi,M}=\sum_{|\mathbf{n}|+|\mathbf{n}'|=M}\prod_j\frac{(s_j)_{n_j}(t_j)_{n'_j}}{n_j!\,n'_j!}a_{j,\pi}^{n_j}b_{j,\pi}^{n'_j}$, the coefficient of ζ^M in the product of the binomial series, lies in $k[q]$. Each κ_{U_k} is an integer combination of masses and exponents and lies in k, and $\int_{[0,1]^{n-1}}\prod_k q_k^{\kappa_{U_k}-1+\beta_k}dq=\prod_k(\kappa_{U_k}+\beta_k)^{-1}\in k$ for $\boldsymbol{\beta}\in\mathbb{N}^{n-1}$; so $I_{\pi,M}\in k$. Grouping the absolutely convergent series of 15.11 by $M=|\mathbf{n}|+|\mathbf{n}'|$ gives the identity. When $s_j,t_j\ge 0$ the coefficients are nonnegative, and $a_{j,\pi},b_{j,\pi}\ge 0$ on the cube, so $I_{\pi,M}\ge 0$.

(ii) On the cube $0\le a_{j,\pi},b_{j,\pi}\le\lambda_\pi$ (15.11), so by $|(s)_n|\le(|s|)_n$ and the coefficient identity of the proof of 15.11, $|f_{\pi,M}|\le\frac{(w)_M}{M!}\lambda_\pi^M$ and $|I_{\pi,M}|\le\prod_k\kappa_{U_k}^{-1}\frac{(w)_M}{M!}\lambda_\pi^M$. The ratio of consecutive bounds is $\frac{w+M}{M+1}\lambda_\pi$, and $M\mapsto\frac{w+M}{M+1}$ is nonincreasing for $w\ge 1$ and below 1 for $w<1$; so for $M\ge Q+1$ the ratio is at most $r_{\pi,Q}<1$, and summing the geometric majorant gives $\big|\sum_{M>Q}I_{\pi,M}\big|\le\operatorname{err}_{\pi,Q}$. The numbers w, λ_π, $r_{\pi,Q}$ and $\operatorname{err}_{\pi,Q}$ lie in k, an ordered subfield of $\mathbb{R}$ being closed under $|\cdot|$ and under maxima of finitely many elements. As $Q\to\infty$, $r_{\pi,Q}\to\lambda_\pi<1$, and $\frac{(w)_{Q+1}}{(Q+1)!}\lambda_\pi^{Q+1}\to 0$, the factor $\frac{(w)_{Q+1}}{(Q+1)!}$ growing at most polynomially; so $\operatorname{err}_{\pi,Q}\to 0$. With $P^{\le Q}=\sum_\pi\sum_{M\le Q}I_{\pi,M}\in k$, the left side of (i) lies between $P^{\le Q}-\sum_\pi\operatorname{err}_{\pi,Q}$ and $P^{\le Q}+\sum_\pi\operatorname{err}_{\pi,Q}$, and for $s_j,t_j\ge 0$, the remainders being nonnegative, between $P^{\le Q}$ and $P^{\le Q}+\sum_\pi\operatorname{err}_{\pi,Q}$.

(iii) $\partial_\zeta\log f=\sum_i w_iA_i/(1-\zeta A_i)$; multiply by $f\prod_i(1-\zeta A_i)$. Comparing the coefficients of ζ^M on the two sides gives a linear recurrence for the coefficients of f, with coefficients polynomial in M and in the A_i. With rational data this is the reference, proof of 15.12. □

Theorem 15.14 (Continued Beta corners). *As stated: complex data, every index off the nonpositive integers, several indices of nonpositive real part at once.*

Proof. A continuation as a linear functional. For $\kappa\in\mathbb{C}$ and $h\in\mathbb{N}$ with $\operatorname{Re}\kappa>-h-1$ and $\kappa\notin\{0,-1,\dots,-h\}$, and f smooth on $[0,1]$, 8.4 with $m=1$ and $M=h+1$ gives the continuation in κ of $\int_0^1 x^{\kappa-1}f\,dx$:

$$\mathcal{C}(\kappa;f)=\int_0^1 x^{\kappa-1}\big(f-\operatorname{Tay}_h f\big)dx+\sum_{l\le h}\frac{f^{(l)}(0)}{l!\,(\kappa+l)},$$

Tay_h taking the Taylor polynomial of degree h at 0; it is the finite part $\mathrm{Reg}\int_0^1 x^{\kappa-1}f\,dx$ (1.8). By Taylor's formula $|f-\mathrm{Tay}_h f|\le \frac{x^{h+1}}{(h+1)!}\sup|f^{(h+1)}|$, so

$$|\mathcal{C}(\kappa;f)|\le \frac{\sup|f^{(h+1)}|}{(h+1)!\,(\mathrm{Re}\,\kappa+h+1)}+\sum_{l\le h}\frac{|f^{(l)}(0)|}{l!\,|\kappa+l|}:$$

$\mathcal{C}(\kappa;\cdot)$ is linear and bounded by a constant times the largest of the sup-norms of $f,\dots,f^{(h+1)}$. On a cube $[0,1]^m$, with a number $L_k\in\mathbb{N}$ for each coordinate, $\mathrm{Re}\,x_k+L_k>0$ and $x_k\notin\{0,\dots,1-L_k\}$, 8.4 repeated in each coordinate gives the continuation of $\int\prod_k q_k^{x_k-1}f\,dq$ as a finite sum, over the sets A of coordinates and the $\mathbf{l}$ with $l_k<L_k$ for $k\in A$, of $\prod_{k\in A}(x_k+l_k)^{-1}$ times the integral over the other coordinates of $\prod_{k\notin A}q_k^{x_k-1}$ against $\prod_{k\notin A}(1-\mathrm{Tay}_{L_k-1})$ applied to $\partial^{\mathbf{l}}_{q_A}f/\mathbf{l}!$ at $q_A=0$ (as in 8.6), with $\mathrm{Tay}_{-1}=0$. Taylor's formula in each coordinate outside A bounds each term by a constant, depending only on the x_k, times the largest sup-norm of the derivatives of f of total order at most $L=\sum_k L_k$; so this continuation is again linear and bounded on $C^L([0,1]^m)$, and it is the iterated finite part, by 1.8 in one coordinate at a time.

Leibniz bounds. Let finitely many functions f_i be smooth on a cube, with $0\le f_i\le\lambda$, $0<\lambda<1$, and let $D\ge 1$ bound their derivatives of order at most L. For a multi-index $\mathbf{n}$ over the i and $f_{\mathbf{n}}=\prod_i f_i^{n_i}$, a derivative of total order $l\le L$ distributes over the $|\mathbf{n}|$ factors in at most $|\mathbf{n}|^l$ ways, each product having at most l differentiated factors, each bounded by D, and the others bounded by λ; so $|\partial^{\boldsymbol\beta}f_{\mathbf{n}}|\le D^l|\mathbf{n}|^l\lambda^{|\mathbf{n}|-l}$ for $|\boldsymbol\beta|=l$. Hence the sum, over the $\mathbf{n}$ with $|\mathbf{n}|=M$, of $\prod_i\frac{|(w_i)_{n_i}|}{n_i!}$ times the C^L-norm of $f_{\mathbf{n}}$ is at most $\frac{(w)_M}{M!}D^L(1+M)^L\lambda^{M-L}$, $w=\sum_i|w_i|$, by the coefficient identity of the proof of 15.6. These bounds are summable over M, so $\sum_{\mathbf{n}}\prod_i\frac{(w_i)_{n_i}}{n_i!}f_{\mathbf{n}}$ converges in C^L, to $\prod_i(1-f_i)^{-w_i}$; a bounded linear functional on C^L commutes with the sum, and applied to $f_{\mathbf{n}}$ it is at most a constant times $D^L(1+|\mathbf{n}|)^L\lambda^{|\mathbf{n}|-L}$. If $\lambda=0$ every f_i vanishes and only $\mathbf{n}=0$ contributes.

The two sectors of a scale. In 15.9 take $f_i=1-\mathsf{Q}_{S,i}(\cdot;\alpha)$ on $[0,1]$: each $\mathsf{Q}_{S,i}$ has nonnegative coefficients, positive constant term and value 1 at $x=1$ (12.10), so $0\le f_i\le 1-\mathsf{Q}_{S,i}(0;\alpha)\le\lambda_S$ there, and $\prod_i\mathsf{Q}_{S,i}^{-w_i}=\sum_{\mathbf{n}}\prod_i\frac{(w_i)_{n_i}}{n_i!}f_{\mathbf{n}}$. Let κ be the index of the sector, not a nonpositive integer. If $\mathrm{Re}\,\kappa>0$, take $\mathcal{C}(\kappa;f)=\int_0^1 x^{\kappa-1}f\,dx$, bounded by $\sup|f|/\mathrm{Re}\,\kappa$, with $L=0$; otherwise take h as in the statement and $L=h+1$. By the preceding paragraph,

$$\mathcal{C}\Big(\kappa;\prod_i\mathsf{Q}_{S,i}^{-w_i}\Big)=\sum_{\mathbf{n}}\prod_i\frac{(w_i)_{n_i}}{n_i!}\,\mathcal{C}(\kappa;f_{\mathbf{n}}),$$

the terms of total order M having moduli summing to at most $\frac{(w)_M}{M!}\lambda_S^M$ times a polynomial in M of degree L, $w=\sum_j(|s_j|+|t_j|)$. The left side is the finite part of the sector integral and its continuation in κ. By 15.8, $f_{\mathbf{n}}=(1-x)^{|\mathbf{n}|}\prod_i\mathsf{Q}_{S,i,>}^{n_i}$, a polynomial, and $\mathcal{C}\big(\kappa;x^k(1-x)^{|\mathbf{n}|}\big)$ is the continuation of $\int_0^1 x^{\kappa+k-1}(1-x)^{|\mathbf{n}|}dx=\Gamma\begin{bmatrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{bmatrix}$ (3.9), the meromorphic stacked Gamma, finite since $\kappa+k\notin-\mathbb{N}$. So the series of 15.9(ii), with these Beta corners, converges to the finite part, with the stated bound. By 15.9(i) the scale integral is the product of the letters at α and the sum of the two sectors at the indices κ_B and κ_B^{top}, and its continuation in the data is the same expression with the sectors continued.

The ordering sectors. By the first step of the proof of 15.11 and 8.4, $P_F=\sum_\pi I_\pi$ as meromorphic functions of the data, each I_π continued by 8.4, whose formula expresses the continuation at a point through the values of h_π at that point; on the balance wall $h_\pi=\prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$. At a

point of the balance wall at which no proper wall is a nonpositive integer, take for an order π the numbers $L_k = h_k + 1$ when $\operatorname{Re}\kappa_{U_k} \le 0$, with $-h_k - 1 < \operatorname{Re}\kappa_{U_k} \le -h_k$, and $L_k = 0$ otherwise. With f running over the a_j, b_j of the proof of 15.11, in $[0,\lambda]$ on the cube, the preceding paragraphs give the continuation of I_π as the series of 15.11 for π, each integral replaced by its continuation; and the continuation of $\int \prod_k q_k^{\kappa_{U_k}+\beta_k-1}(1-q_k)^{\gamma_k}dq$ is the product over k of the continued Beta corners, each finite since $\kappa_{U_k} + \beta_k \notin -\mathbb{N}$. The terms of total order M have moduli summing to at most $\frac{(w)_M}{M!}\lambda^M$ times a polynomial in M of degree $\sum_k L_k$, and the tails of one order being distinct proper sets, $\sum_k L_k$ is at most the sum of the integers $h+1$ over the proper walls of real part at most 0. Every proper set of marks is a tail of some order, so the sector series computes P_F at every point of the balance wall off the pole hyperplanes, with no positivity of the walls. For one real index in $(-h-1, -h)$ this is the reference, proof of 15.14. □

Corollary 15.16 (A loop or coloop in every table). *As stated: a set of marks, each a loop or a coloop in every table.*

Proof. For a table j let p_j be the set of the marks of A that are coloops of table j. Every basis contains every coloop and avoids every loop, so $T \cap A = p_j$ for every basis T of table j. The sets $T \setminus A$ are then the bases of the table on $\mathcal{E} \setminus A$ obtained from table j by contracting the marks of p_j and deleting those of $A \setminus p_j$, one mark at a time (7.3; a loop is never a coloop, so each step is a table), with the weights $\theta_{j,T}$; with K'_j, Ψ'_j its Foreman polynomials and $\alpha' = \alpha_{\mathcal{E}\setminus A}$,

$$K_j(\alpha) = \alpha_A^{p_j} K'_j(\alpha'), \qquad \Psi_j(\alpha) = \alpha_A^{A\setminus p_j}\Psi'_j(\alpha'),$$

since $\mathcal{E} \setminus T = (A \setminus p_j) \sqcup \big((\mathcal{E} \setminus A) \setminus (T \setminus A)\big)$. So the integrand of $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the product of

$$\prod_{e\in A} \alpha_e^{m_e - 1 - \sum_j (s_j \mathbf{1}[e\in p_j] + t_j \mathbf{1}[e\notin p_j])} e^{-\alpha_e}$$

and of the integrand of $J_{F'}(\mathbf{m}|_{\mathcal{E}\setminus A};\mathbf{s},\mathbf{t})$. With $s_j = \sigma_j + \delta_j$, $t_j = \sigma_j - \delta_j$ and $m_e = \mu_e + \sigma$, the exponent of α_e is $\mu_e - \sum_j \eta_{j,e}\delta_j - 1$. The integrand is a product of functions of disjoint sets of variables, so by Tonelli on the modulus its integral converges exactly when every factor's does, and then, by Fubini, it is the product of the factors' integrals, with $\int_0^\infty \alpha_e^{\nu-1}e^{-\alpha_e}d\alpha_e = \Gamma(\nu)$ for $\operatorname{Re}\nu > 0$. As meromorphic functions the identity persists (8.5(iii)). For one mark this is the reference, proof of 15.16, there read through 15.15 at $B = \{e\}$. □

48.16 Chapter 16: integrals, expectations and the Foreman procedure

Proposition 16.5 (Snips are completions). *As stated: (iii) at every imbalance, through the appended list $\binom{B}{1}$ of the singletons of the bunch.*

Proof. Items (i) and (ii) are the reference, proof of 16.5. For (iii) let $R = \mathcal{E} \setminus B$.

Projectivity. The family word W is homogeneous of degree $-\sum_j(s_j r_j + t_j r_j^*) = \kappa_{\mathcal{E}} - m_{\mathcal{E}}$ (8.1), so $\omega = \alpha^{\mathbf{m}-\mathbf{1}} W\, d\alpha$ is multiplied by $\lambda^{\kappa\varepsilon}$ under $\alpha \mapsto \lambda\alpha$, and $|\alpha_B|^{-\kappa\varepsilon}\omega$ is unchanged: it is projective (1.1(vii)). On the section $|\alpha_B| = 1$ the multiplier is 1, so the two forms induce the same form there.

The appended table. The sets $\{e\}$, $e \in B$, have one element each and the exchange property, two of them differing in one mark; so they are the bases of a table on $\mathcal{E}$ of rank 1 (7.1), in which

the marks of R lie in no basis, and with unit weights its basis polynomial is $\sum_{e\in B}\alpha_e = |\alpha_B|$. So $|\alpha_B|^{-\kappa_\varepsilon}W$ is the family word of F' at the exponents $\big((\mathbf{s},\kappa_\mathcal{E}),(\mathbf{t},0)\big)$, the appended table carrying $(\kappa_\mathcal{E},0)$, and $|\alpha_B|^{-\kappa_\varepsilon}\omega$ is the form of F' at the masses $\mathbf{m}$.

Walls. For the appended table $d_0(S) = \min_{e\in B}|\{e\}\cap S| = \mathbf{1}[B\subseteq S]$ (7.13), and its exponent on Ψ is 0; so by 8.1 the walls of F' at $\mathbf{m}$ are $\kappa_S - \kappa_\mathcal{E}\mathbf{1}[B\subseteq S]$, and its imbalance is $\kappa_\mathcal{E} - \kappa_\mathcal{E}\cdot 1 = 0$, the appended table having rank 1.

Value. $|\alpha_B|$ is positive and homogeneous of degree one on the orthant. Where every proper wall of F' has positive real part, the period of F' converges, and the integral of its form over the section $\{|\alpha_B|=1\}$ is that period (8.7); this is the display.

The cases. For $B=\{g\}$ the appended table has the one basis $\{g\}$ and the letter α_g, and $\alpha_g^{-\kappa_\varepsilon}W\,d\alpha_\mathbf{m} = W\,d\alpha_{\mathbf{m}-\kappa_\mathcal{E}\mathbf{e}_g} = \omega_{\mathbf{m}^{(g)}}$ (16.4): this is (i). For $B=\mathcal{E}$ the appended table is the list $\binom{\mathcal{E}}{1}$, and the statement is 16.3. For $k=-\kappa_\mathcal{E}\in\mathbb{N}$, $k\geq 1$: on the section $\omega = |\alpha_B|^k\omega$, $|\alpha_B|^k = \sum_{|\mathbf{n}|=k}\binom{k}{\mathbf{n}}\alpha_B^\mathbf{n}$, and $\alpha_B^\mathbf{n}\omega = W\,d\alpha_{\mathbf{m}+\mathbf{n}}$ is the completion along $\mathbf{n}/k$ (16.4: $\mathbf{m}-\kappa_\mathcal{E}\mathbf{n}/k = \mathbf{m}+\mathbf{n}$), projective, whose integral over the section is $P_F(\mathbf{m}+\mathbf{n};\mathbf{s},\mathbf{t})$ wherever that period converges (8.7). This last case is the reference, proof of 16.5(iii). □

Corollary 16.6 (The period fibres over the scale of a bunch). *As stated: every section $\{h(\alpha_R)=1\}$ of the room, and off the balance wall.*

Proof. For a positive function h on an orthant $\mathbb{R}^X_{>0}$, homogeneous of degree one, write λ_h for the measure on $\Sigma_h=\{\beta:\ h(\beta)=1\}$ with $d\alpha = \rho^{|X|-1}d\rho\,d\lambda_h(\beta)$ under the bijection $(\rho,\beta)\mapsto\rho\beta$ of $(0,\infty)\times\Sigma_h$ onto $\mathbb{R}^X_{>0}$: the measure of 16.2 (the reference, proof of 16.2). For $h=\alpha_g$ it is the Lebesgue measure in the coordinates α_e, $e\neq g$, at $\alpha_g=1$, and for $h=|\alpha|$ it is $d\omega$ (2.7).

Let h be a function of α_R as in the statement, and put $\tilde h(\alpha)=h(\alpha_R)$ on $\mathbb{R}^\mathcal{E}_{>0}$, positive and homogeneous of degree one. The map $(\rho,\beta_B,\beta_R)\mapsto(\rho\beta_B,\rho\beta_R)$, $\beta_B\in\mathbb{R}^B_{>0}$, $\beta_R\in\Sigma_h$, is a bijection onto $\mathbb{R}^\mathcal{E}_{>0}$, with $d\alpha_R = \rho^{|R|-1}d\rho\,d\lambda_h(\beta_R)$ and $d\alpha_B = \rho^{|B|}d\beta_B$ at fixed ρ. So $d\alpha = \rho^{n-1}d\rho\,d\beta_B\,d\lambda_h(\beta_R)$, and the measure of the section $\{\tilde h=1\}$ is $d\beta_B\,d\lambda_h(\beta_R)$. The integral of $W\,d\alpha_\mathbf{m}$ over it is

$$\int_{\Sigma_h}\beta_R^{\mathbf{m}|_R-\mathbf{1}}\int_{\mathbb{R}^B_{>0}}\beta_B^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_j(\beta_B,\beta_R)^{-s_j}\Psi_j(\beta_B,\beta_R)^{-t_j}\,d\beta_B\,d\lambda_h(\beta_R),$$

by Tonelli on the modulus, which is the same integrand at the real parts of the data, and then Fubini. With $\beta_B = xu$, $x>0$, $u\in\Delta_B$, $\beta_B^{\mathbf{m}|_B-\mathbf{1}}d\beta_B = x^{m_B-1}u^{\mathbf{m}|_B-\mathbf{1}}dx\,d\omega_B(u)$ (2.7), so the inner integral is $\int_{\Delta_B}u^{\mathbf{m}|_B-\mathbf{1}}\,\mathrm{Sc}_B(u,\beta_R;\mathbf{m};\mathbf{s},\mathbf{t})\,d\omega_B(u)$, with the scale integral of 15.3 at the gauge (u,β_R). This is the display.

The scale integral converges exactly when $\operatorname{Re}\kappa_B>0$ and $\operatorname{Re}\kappa_B^{\mathrm{top}}>0$ (15.3), conditions on the data alone; by Tonelli it converges at almost every (u,β_R), hence at every point, and there it is the product of 15.3. On the balance wall the form is projective and its integral over every section is the period (8.7). At $h=\alpha_g$ the integral over $\{\alpha_g=1\}$ is $P_F(\mathbf{m}^{(g)};\mathbf{s},\mathbf{t})$ wherever it converges (16.5(i)), and λ_h is the Lebesgue measure in α_{R-g} at $\alpha_g=1$; at $h=|\alpha_R|$, $\Sigma_h=\Delta_R$ and $\lambda_h=d\omega_R$. At $h=\alpha_g$ on the balance wall this is the reference, proof of 16.6. □

Lemma 16.9 (The period is meromorphic in the data). *As stated, for every family and all complex data.*

Proof. Meromorphy. By the first step of the proof of 15.11 above, which uses neither connectedness nor real data, $P_F=\sum_\pi I_\pi$ on the open convex set Ω of the data at which every proper wall has positive real part, with $I_\pi = \int_{[0,1]^{n-1}}\prod_{k=2}^n q_k^{\kappa_{U_k}-1}h_\pi\,dq$ and $h_\pi = L_\pi^{-\kappa_\varepsilon}\prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$ smooth near

the closed cube, its derivatives continuous in the variables and the data and entire in the data. By 8.4 each I_π extends to a meromorphic function of the data, a quotient of a holomorphic function by products of affine functions $\kappa_{U_k}+i$, $i\in\mathbb{N}$, with poles only on the hyperplanes $\kappa_{U_k}=-i$. Every nonempty proper set of marks is a tail of some order, and no tail is $\mathcal{E}$. So P_F is meromorphic in the masses and the exponents, with poles only where a proper wall is a nonpositive integer.

The restriction to the balance wall. For a nonempty proper U the affine function κ_U+i of the data has the part m_U in the masses, which is not proportional to the part $m_{\mathcal{E}}$ of $\kappa_{\mathcal{E}}$; so the hyperplane $\kappa_U=-i$ is not the balance wall $\{\kappa_{\mathcal{E}}=0\}$, and it meets the balance wall in a hyperplane of it. Near every point of the balance wall P_F is a quotient of a holomorphic function by a product of such affine functions, none vanishing identically on the balance wall; so the restriction of P_F to the balance wall is meromorphic, with poles only where a proper wall is a nonpositive integer. □

Proposition 16.12 (The form of a compound at one point). *As stated, for any number $q\ge 2$ of parts.*

Proof. The form. Write $r_{j,i}$ for the rank of the restriction of table j to $\mathcal{E}_i$, so that $K_{j,i}$ is homogeneous of degree $r_{j,i}$, $\Psi_{j,i}$ of degree $|\mathcal{E}_i|-r_{j,i}$, and $\kappa_i=m_{\mathcal{E}_i}-\sum_j\big(s_jr_{j,i}+t_j(|\mathcal{E}_i|-r_{j,i})\big)$; summing over i, $\sum_i\kappa_i=\kappa_{\mathcal{E}}=0$. The section $\{|\alpha_q|=1\}$ carries the measure $\prod_{i<q}d\alpha_i\,d\omega_{\mathcal{E}_q}(u_q)$, by the computation in the proof of 16.6 above with the room there replaced by $\mathcal{E}_q$. For $i<q$, $\alpha_i=\rho_iu_i$ gives $\alpha_i^{\mathbf{m}_i-\mathbf{1}}d\alpha_i=\rho_i^{m_{\mathcal{E}_i}-1}u_i^{\mathbf{m}_i-\mathbf{1}}d\rho_i\,d\omega_{\mathcal{E}_i}(u_i)$ (2.7), $K_{j,i}(\rho_iu_i)=\rho_i^{r_{j,i}}K_{j,i}(u_i)$ and $\Psi_{j,i}(\rho_iu_i)=\rho_i^{|\mathcal{E}_i|-r_{j,i}}\Psi_{j,i}(u_i)$. The powers of ρ_i collect to $\rho_i^{\kappa_i-1}$, which gives the displayed form.

(i) For real κ, $\int_0^\infty\rho^{\kappa-1}d\rho$ diverges, at ∞ if $\kappa\ge0$ and at 0 if $\kappa\le0$. The form is positive, and by Tonelli its integral is the product of the $q-1$ integrals $\int_0^\infty\rho_i^{\kappa_i-1}d\rho_i$, each infinite, and of the q shape integrals, each positive; so it is infinite.

(ii) A primitive of $\rho^{\kappa-1}$ is ρ^κ/κ, or $\log\rho$ at $\kappa=0$. It is its own log-power expansion at 0 and at ∞, with no term $\rho^0\log^0\rho$, so both regularized limits are 0 and the finite part is 0 (1.5).

(iii) At $\mathbf{m}(\boldsymbol{\eta})$, $\kappa_i=\kappa_{F_i}(\mathbf{m}_i^0)+i\eta_i|w_i|=i\eta_i$, and $\sum_i\kappa_i=0$ since $\sum_i\eta_i=0$. The walls of F_i at $\mathbf{m}_i^0+i\eta_iw_i$ have the real parts of those at $\mathbf{m}_i^0$, positive, so $f(\boldsymbol{\eta}')=\prod_{i=1}^qP_{F_i}(\mathbf{m}_i^0+i\eta_iw_i;\mathbf{s},\mathbf{t})$, with $\boldsymbol{\eta}'=(\eta_1,\dots,\eta_{q-1})$ and $\eta_q=-\sum_{i<q}\eta_i$, is a smooth function of $\boldsymbol{\eta}'$, each period being holomorphic in the masses there (8.7). In $y_i=\log\rho_i$, $\rho_i^{\kappa_i-1}d\rho_i=e^{i\eta_iy_i}dy_i$. For a test function H of $\boldsymbol{\eta}'$ of compact support, the integral of the form against H, taken over the shapes, then over $\boldsymbol{\eta}'$, then over $\mathbf{y}=(y_1,\dots,y_{q-1})$, is

$$\int_{\mathbb{R}^{q-1}}\int_{\mathbb{R}^{q-1}}H(\boldsymbol{\eta}')f(\boldsymbol{\eta}')\,e^{i\langle\boldsymbol{\eta}',\mathbf{y}\rangle}\,d\boldsymbol{\eta}'\,d\mathbf{y}=(2\pi)^{q-1}H(0)f(0),$$

by Fourier inversion for the smooth function Hf of compact support, whose Fourier transform is integrable; and $f(0)=\prod_iP_{F_i}(\mathbf{m}_i^0;\mathbf{s},\mathbf{t})$. For $q=2$ this is the reference, proof of 16.12. □

Lemma S4. *Let S be a nonempty proper set of marks of a family F on $\mathcal{E}$ with $c_j(S)=0$ in every active table, and put $S'=\mathcal{E}\setminus S$.*

(i) In every active table j, every basis meets S in $r_j(S)$ marks and S' in $r_j(S')$ marks; the bases of $\mathcal{B}_j|S$ and of $\mathcal{B}_j|S'$ are the sets $T\cap S$ and $T\cap S'$, $T\in\mathcal{B}_j$; and $X\sqcup Y\in\mathcal{B}_j$ for every basis X of $\mathcal{B}_j|S$ and every basis Y of $\mathcal{B}_j|S'$.

(ii) In this lemma write

$$I_S(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_S\times\Delta_{S'}}\alpha^{\mathbf{m}-\mathbf{1}}\prod_jK_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}\,d\omega_S\,d\omega_{S'}.$$

It converges, and is holomorphic in the data, wherever $\operatorname{Re}\kappa_U > 0$ *for every nonempty proper subset* U *of* S *and every nonempty proper subset* U *of* S'*; and it extends to a meromorphic function of the data whose poles lie on the hyperplanes* $\kappa_U = -i$, $i \in \mathbb{N}$, U *a nonempty proper subset of* S *or of* S'.

(iii) Wherever every proper wall has positive real part, $\kappa_{\mathcal{E}} = \kappa_S + \kappa_{S'}$ *has positive real part and* $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_S)\Gamma(\kappa_{S'})\, I_S(\mathbf{m};\mathbf{s},\mathbf{t})$.

Proof. (i) Fix an active table j. By 7.13 the numbers $|T\cap S|$ fill $\{d_j(S),\dots,r_j(S)\}$, a single value since $c_j(S)=0$; so every basis meets S in $r_j(S)$ marks and S' in $r_j - r_j(S)$ marks, a constant, which is then $r_j(S')$. By 7.5 the bases of $\mathcal{B}_j|S$ are the sets $T\cap S$ with $|T\cap S| = r_j(S)$, that is all of them, and likewise for S'. Let $X = T_1\cap S$ and $Y = T_2\cap S'$ for bases T_1, T_2; induct on $k = |(T_2\cap S)\setminus X|$. If $k=0$, then $T_2\cap S\subseteq X$, both of size $r_j(S)$, so $T_2 = X\sqcup Y$. If $k>0$, take $a\in (T_2\cap S)\setminus X$; then $a\notin T_1$. The exchange property (7.1) gives $b\in T_1\setminus T_2$ with $T_3 = (T_2\setminus\{a\})\cup\{b\}$ a basis. As $|T_3\cap S| = r_j(S) = |T_2\cap S|$ and $a\in S$, $b\in S$, so $b\in X\setminus T_2$. Then $T_3\cap S' = Y$ and $|(T_3\cap S)\setminus X| = k-1$, and by induction $X\sqcup Y$ is a basis.

Homogeneity in each block. By (i), for an active table j, $K_j(\rho\alpha_S,\rho'\alpha_{S'}) = \rho^{r_j(S)}\rho'^{r_j(S')}K_j(\alpha)$ and $\Psi_j(\rho\alpha_S,\rho'\alpha_{S'}) = \rho^{|S|-r_j(S)}\rho'^{|S'|-r_j(S')}\Psi_j(\alpha)$. For $U\in\{S,S'\}$, $c_j(U) = 0$ (7.13: $c_j(S') = c_j(S)$), so $d_j(U) = r_j(U)$ and $d_j^*(U) = |U| - r_j(U)$ (7.12). An inactive table enters with the exponents 0. So with $g(U) = \sum_j \big(s_j d_j(U) + t_j d_j^*(U)\big) = m_U - \kappa_U$, the function $\phi(\alpha) = \alpha^{\mathbf{m}-\mathbf{1}}\prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}$ has

$$\phi(\rho\alpha_S,\rho'\alpha_{S'}) = \rho^{\kappa_S-|S|}\rho'^{\kappa_{S'}-|S'|}\phi(\alpha),$$

and $|\phi|$ is the same function at the real parts of the data.

(iii) With $\alpha_S = \rho u$ and $\alpha_{S'} = \rho' u'$ (2.7 in each block), $d\alpha = \rho^{|S|-1}\rho'^{|S'|-1}d\rho\, d\rho'\, d\omega_S(u)\, d\omega_{S'}(u')$, and the integrand of J_F becomes $\rho^{\kappa_S-1}e^{-\rho}\rho'^{\kappa_{S'}-1}e^{-\rho'}\phi(u,u')$. Wherever every proper wall has positive real part, $\operatorname{Re}\kappa_{\mathcal{E}} = \operatorname{Re}\kappa_S + \operatorname{Re}\kappa_{S'} > 0$, by 8.16, $c_j(S) = 0$ holding in every table with $\sigma_j\neq 0$, these being active. Then every wall has positive real part, and J_F at the real parts of the data is finite (8.5(i)); by Tonelli it is $\Gamma(\operatorname{Re}\kappa_S)\Gamma(\operatorname{Re}\kappa_{S'})$ times the integral of $|\phi|$ over $\Delta_S\times\Delta_{S'}$, which is therefore finite, and Fubini gives $J_F = \Gamma(\kappa_S)\Gamma(\kappa_{S'})I_S$.

(ii) *Charts.* For a finite set X, a mark $x\in X$ and a function f on $\mathbb{R}^X_{>0}$, homogeneous of degree $-|X|$ and nonnegative or integrable on Δ_X, $\int_{\Delta_X} f\, d\omega_X = \int f|_{\beta_x=1}\, d\beta_{X-x}$, and the same holds on the cones over subsets of Δ_X and their traces on the chart: for a nonnegative function υ on $(0,\infty)$ with $\int_0^\infty \upsilon(\rho)\,d\rho/\rho = 1$, both sides equal $\int_{\mathbb{R}^X_{>0}} f(\alpha)\upsilon(|\alpha|)\,d\alpha$, by 2.7 on the left, and on the right by the change of variables $\alpha = \rho(1,\beta)$, $d\alpha = \rho^{|X|-1}d\rho\, d\beta$, with $\int_0^\infty \upsilon(\rho|(1,\beta)|)\,d\rho/\rho = 1$. The function $\psi(\alpha) = |\alpha_S|^{-\kappa_S}|\alpha_{S'}|^{-\kappa_{S'}}\phi(\alpha)$ is homogeneous of degree $-|S|$ in α_S and $-|S'|$ in $\alpha_{S'}$, and equal to ϕ on $\Delta_S\times\Delta_{S'}$.

For an order π of S, with tails $U_k = \{\pi(k),\dots,\pi(|S|)\}$, $2\le k\le |S|$, use on the chart $\beta_{\pi(1)} = 1$ the coordinates $\beta_{\pi(k)} = q_2\cdots q_k$, $q\in(0,1]^{|S|-1}$, which cover the trace of the sector $\{\beta_{\pi(1)}\ge\cdots\ge\beta_{\pi(|S|)}\}$; the exponent of q_k in β^X is $|X\cap U_k|$, $d\beta_{S-\pi(1)} = \prod_k q_k^{|S|-k}dq$, $\beta^{\mathbf{m}|_S-\mathbf{1}}d\beta_{S-\pi(1)} = \prod_k q_k^{m_{U_k}-1}dq$, and $|\beta| = L_\pi(q) = \sum_{i=1}^{|S|}\prod_{k=2}^{i} q_k \in [1,|S|]$. Likewise π', U'_l, q' and $L_{\pi'}$ on S'. Decomposing each simplex into its sectors, and applying the identity above in each block, by Tonelli on the modulus and then Fubini,

$$I_S = \sum_{\pi,\pi'}\int \prod_k q_k^{m_{U_k}-1}\prod_l q_l'^{m_{U'_l}-1}\, L_\pi(q)^{-\kappa_S}L_{\pi'}(q')^{-\kappa_{S'}}\prod_j K_j(\beta,\beta')^{-s_j}\Psi_j(\beta,\beta')^{-t_j}\,dq\,dq',$$

the integral over $(0,1]^{|S|-1}\times(0,1]^{|S'|-1}$.

The letters on a product of sectors. Fix an active table j. By (i) and 7.5, $\mathcal{B}_j|S$ is a table on S, and by 8.3 for π it has bases $X^\pm$ with $|X^+\cap U_k|=\min_X|X\cap U_k|$ and $|U_k\setminus X^-|=\min_X|U_k\setminus X|$, the minima over its bases; by (i) these minima are $\min_T|T\cap U_k|=d_j(U_k)$ and $\min_T|U_k\setminus T|=d_j^*(U_k)$ (7.13). Likewise $\mathcal{B}_j|S'$ has bases $Y^\pm$ for π'. For a basis T, $\beta^{T\cap S}/\beta^{X^+}=\prod_k q_k^{|T\cap U_k|-d_j(U_k)}$ with exponents in $\mathbb{N}$, and likewise on S'. So

$$K_j(\beta,\beta')=\prod_k q_k^{d_j(U_k)}\prod_l {q_l'}^{d_j(U_l')}\,k_{j,\pi,\pi'}(q,q'),$$
$$k_{j,\pi,\pi'}=\sum_T\theta_{j,T}\prod_k q_k^{|T\cap U_k|-d_j(U_k)}\prod_l {q_l'}^{|T\cap U_l'|-d_j(U_l')},$$

a polynomial with nonnegative coefficients. Its constant term is the sum of $\theta_{j,T}$ over the bases with $|T\cap U_k|=d_j(U_k)$ for $k\ge2$ and $|T\cap U_l'|=d_j(U_l')$ for $l\ge2$. As $|T\cap S|=|X^+|$, these counts determine $T\cap S=X^+$, the mark $\pi(k)$ lying in T exactly when $|T\cap U_k|-|T\cap U_{k+1}|=1$, with $U_1=S$ and $U_{|S|+1}=\varnothing$; likewise $T\cap S'=Y^+$. By (i) $X^+\sqcup Y^+$ is a basis, so the constant term is $\theta_{j,X^+\sqcup Y^+}>0$. In the same way $\Psi_j(\beta,\beta')=\prod_k q_k^{d_j^*(U_k)}\prod_l {q_l'}^{d_j^*(U_l')}\,\psi_{j,\pi,\pi'}(q,q')$, with nonnegative coefficients and the constant term $\theta_{j,X^-\sqcup Y^-}>0$. As $m_U-g(U)=\kappa_U$,

$$I_S=\sum_{\pi,\pi'}\int\prod_{k=2}^{|S|}q_k^{\kappa_{U_k}-1}\prod_{l=2}^{|S'|}{q_l'}^{\kappa_{U_l'}-1}\,h_{\pi,\pi'}(q,q')\,dq\,dq',\qquad h_{\pi,\pi'}=L_\pi^{-\kappa_S}L_{\pi'}^{-\kappa_{S'}}\prod_j k_{j,\pi,\pi'}^{-s_j}\psi_{j,\pi,\pi'}^{-t_j},$$

the product over the active tables.

Continuation. The polynomials $k_{j,\pi,\pi'}$, $\psi_{j,\pi,\pi'}$, L_π and $L_{\pi'}$ are at least their positive constant terms on the closed cube, hence positive on a neighbourhood of it; so $h_{\pi,\pi'}$ is smooth there, with every derivative continuous in the variables and the data and entire in the data, and $|h_{\pi,\pi'}|$, which is $h_{\pi,\pi'}$ at the real parts of the data, lies between positive constants on the cube, locally uniformly in the data. So the term of (π,π') converges, and is holomorphic, exactly where $\operatorname{Re}\kappa_{U_k}>0$ and $\operatorname{Re}\kappa_{U_l'}>0$ for all k,l; the tails U_k, $k\ge2$, of the orders of S are the nonempty proper subsets of S, and likewise for S'. This gives the convergence and the holomorphy of I_S. By 8.4, each term is a quotient of a holomorphic function of the data by products of the affine functions $\kappa_{U_k}+i$ and $\kappa_{U_l'}+i$, $i\in\mathbb{N}$, each a nonconstant function of the data (the coefficient of m_e, $e\in U$, being 1); so it extends meromorphically with poles only on these hyperplanes, and so does I_S. □

Theorem 16.17 (The pseudointegral off the domain). *As stated: the compound formula, with the value* 0 *on the balance wall, for every family and all complex data; the clause on targets, (i) and (ii) under the hypotheses of the reference.*

Proof. The compound formula. Let S be nonempty and proper with $c_j(S)=0$ in every active table, $S'=\mathcal{E}\setminus S$, and I_S the integral of Lemma S4, the integral of the statement. On the open convex set Ω of the data at which every proper wall has positive real part, $\operatorname{Re}\kappa_{\mathcal{E}}>0$ and $J_F=\Gamma(\kappa_S)\Gamma(\kappa_{S'})I_S$ (Lemma S4(iii)), and $J_F=\Gamma(\kappa_{\mathcal{E}})P_F$ (8.7); so $P_F={}_2\Gamma_1\left[\begin{matrix}\kappa_S,\ \kappa_{S'}\\ \kappa_{\mathcal{E}}\end{matrix}\right]I_S$ on Ω, with $\kappa_S+\kappa_{S'}=\kappa_{\mathcal{E}}$ (8.16). Both sides are meromorphic in the data, by 16.9, by Lemma S4(ii) and by 1.2, and they agree on the nonempty open set Ω; so they agree as meromorphic functions.

The value 0 *on the balance wall.* Put $G=\Gamma(\kappa_S)\Gamma(\kappa_{S'})I_S$, meromorphic in the data. The poles of $\Gamma(\kappa_S)\Gamma(\kappa_{S'})$ lie on the hyperplanes $\kappa_S=-i$ and $\kappa_{S'}=-i$, $i\in\mathbb{N}$, and those of I_S on the hyperplanes $\kappa_U=-i$ with U a nonempty proper subset of S or of S' (Lemma S4(ii)). In each case

U is a nonempty proper set of marks, so, as in the proof of 16.9, none of these hyperplanes contains the balance wall, and G restricts to a meromorphic function on the balance wall, holomorphic off a proper analytic subset of it. The function $1/\Gamma$ is entire and vanishes at 0, so $1/\Gamma(\kappa_{\mathcal{E}})$ is entire in the data and vanishes on the balance wall. Hence $P_F = G \cdot \Gamma(\kappa_{\mathcal{E}})^{-1}$ vanishes at every point of the balance wall at which G is holomorphic, and the meromorphic restriction of P_F to the balance wall is 0.

The clause on targets, (i) and (ii) are under the hypotheses of the reference, connected with real data and $s_j, t_j \geq 0$ not all zero: the reference, proof of 16.17. □

Definition 16.28 (Period datum; moves). As stated, for tables with positive rational weights.

Proof. The definition asks of the family only that its form be algebraic over $\mathbb{Q}$ and homogeneous of degree zero. The letters K_j and Ψ_j are polynomials whose coefficients are the weights, here rational; with rational exponents and masses $W\, d\alpha_{\mathbf{m}}$ is a product of rational powers of polynomials over $\mathbb{Q}$ and of the coordinates, times $d\alpha$, hence algebraic over $\mathbb{Q}$, and on the balance wall its degree is $\kappa_{\mathcal{E}} = 0$ (8.1). The walls of the faces are those of 8.5, for any weights. □

Proposition 16.32 (Sections off the balance wall). *As stated: a bunch section at every imbalance.*

Proof. The clause on a bunch section is 16.5(iii), proved above; the rest is the reference, proof of 16.32. In (ii) the walls of the chart completion at g are those of 16.5(iii) at $B = \{g\}$, $\kappa_S - \kappa_{\mathcal{E}} \mathbf{1}_S(g)$. □

Definition 16.29 (The pseudointegral over a region). As stated; the last sentence of (i) is stated for families of tables with positive rational weights, where the reference takes a list.

Proof. Let the tables have positive rational weights and the data be rational, on the balance wall. The letters K_j and Ψ_j are polynomials whose coefficients are the weights, so the form $W\, d\alpha_{\mathbf{m}}$ is algebraic over $\mathbb{Q}$ and the family has a period datum (16.28), whose moves are defined over $\mathbb{Q}$ by that definition. The three regions named in (i) are cut out by polynomial inequalities over $\mathbb{Q}$: the chart $\alpha_g = 1$ by $\alpha_e > 0$ for $e \neq g$; the simplex, in the coordinates $(u_e)_{e \neq e_0}$, by $u_e > 0$ and $1 - \sum_{e \neq e_0} u_e > 0$; a bunch section $|\alpha_B| = 1$ by these inequalities on B together with $\alpha_e > 0$ on the room.

A change of variables of 16.28 keeps this property. Let $\mathcal{R}$ lie in the open domain and be cut out there by $p_1 > 0, \ldots, p_k > 0$ with $p_i \in \mathbb{Q}[x]$, and let ϕ be birational over $\mathbb{Q}$ and carry the open domain bijectively onto the open domain. Each coordinate of ϕ^{-1} is a quotient of polynomials over $\mathbb{Q}$ whose denominator does not vanish on the open domain, where ϕ^{-1} is defined; so $p_i \circ \phi^{-1} = A_i / D_i$ with $A_i, D_i \in \mathbb{Q}[y]$, D_i a product of these denominators, and

$$\phi(\mathcal{R}) = \{y \text{ in the open domain} : \ A_i(y)\, D_i(y) > 0 \text{ for } 1 \leq i \leq k\},$$

the open domain being an orthant, cut out by the positivity of its coordinates. The region of a product of data is the product of the regions, cut out by the inequalities of the two factors in disjoint coordinates; linear combinations and boundary terms are formed on the regions already reached, as for a list. Of the family the argument of the reference uses only that its period datum and its moves are defined over $\mathbb{Q}$, which holds for positive rational weights. □

48.17 Chapter 17: the family as one Laplace transform

48.17.1 What the one-table proofs read, and what a family reads

The one-table statements read the exponent of the law of a question, $\sigma E + \delta O$ at offsets zero, through the width wd of the Newton polytope of Λ and through the convexity of E; the walls of the wedge, its slope and the volume at low strength are all values of the width. For a family the exponent is

$$X(z) = 2\langle \boldsymbol{\mu}_\perp, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big),$$

and what replaces the width is the wall at a tilt (Definition 17.3): on the sector of every order the exponent is minus twice the wall at the tilt less the logarithms of the sector letters (Proposition 17.4(i)), which is Theorem 8.5(ii) read in the signed coordinates. Every statement below is read off that identity, the convexity of the centred potential, and, for samples of bases, the factorization of the chance of a sample over its draws. Throughout, $n = |\mathcal{E}| \geq 2$, the data are real and on the balance wall unless said otherwise, a question is written $q = (\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$, its data are $\big(\mathbf{m}_F(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta}); \boldsymbol{\sigma} + \boldsymbol{\delta}, \boldsymbol{\sigma} - \boldsymbol{\delta}\big)$, and $P_F(q)$ denotes its period. On the sector H_π of an order π, read in the tilts (Definition 17.3), the differences of a tilt along π are written $v_k = z_{\pi(k-1)} - z_{\pi(k)} \geq 0$, $2 \leq k \leq n$, so that $\kappa(z) = \sum_k \kappa_{U_k} v_k$; at $\alpha = e^{2z}$ the gaps of Corollary 8.6 are $2v_k$.

Lemma S5 (The walls on the balance wall)**.** *On the balance wall, for every nonempty proper S,*

$$\kappa_S = \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}) + \kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_\perp, \boldsymbol{\delta}), \qquad \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}) = \sum_j \sigma_j c_j(S), \qquad \kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_\perp, \boldsymbol{\delta}) = \mu_{\perp,S} - \sum_j \delta_j \varepsilon_j^\perp(S),$$

with $\varepsilon_j^\perp(S) = \varepsilon_j(S) - |S|\frac{2r_j - n}{n}$; and $\kappa_{\mathcal{E}\setminus S}^{\mathbf{E}} = \kappa_S^{\mathbf{E}}$, $\kappa_{\mathcal{E}\setminus S}^{\mathbf{O}} = -\kappa_S^{\mathbf{O}}$, so $\kappa_{\mathcal{E}\setminus S} = \kappa_S^{\mathbf{E}} - \kappa_S^{\mathbf{O}}$. In particular every wall is a linear function of the pair coordinates $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ of the question.

Proof. By Theorem 8.2(iii), $\kappa_S = \mu_S + \sum_j \big(\sigma_j c_j(S) - \delta_j \varepsilon_j(S)\big)$ and $\kappa_{\mathcal{E}} = \mu_{\mathcal{E}} - \sum_j \delta_j(2r_j - n)$. On the balance wall $\mu_{\mathcal{E}} = \sum_j \delta_j(2r_j - n)$, and $\boldsymbol{\mu} = \boldsymbol{\mu}_\perp + \frac{\mu_{\mathcal{E}}}{n}\mathbf{1}$ gives $\mu_S = \mu_{\perp,S} + \frac{|S|}{n}\sum_j \delta_j(2r_j - n)$; substituting gives the decomposition. By Theorem 7.13, $c_j(\mathcal{E} \setminus S) = c_j(S)$ and $\varepsilon_j(\mathcal{E} \setminus S) = 2r_j - n - \varepsilon_j(S)$, so $\varepsilon_j^\perp(\mathcal{E} \setminus S) = 2r_j - n - \varepsilon_j(S) - \frac{n-|S|}{n}(2r_j - n) = -\varepsilon_j^\perp(S)$, and $\mu_{\perp,\mathcal{E}\setminus S} = -\mu_{\perp,S}$ as $\boldsymbol{\mu}_\perp \in \mathbf{1}^\perp$. □

48.17.2 The exponent of a question

Definition 17.3: the claims. Let $z \in H_\pi$. A difference v_k is positive exactly when $z_{\pi(k-1)} > z_{\pi(k)}$, and then $U_k = \{e : \ z_e < z_{\pi(k-1)}\}$, a set $\{e : \ z_e < z_f\}$ with $z_f > \min_e z_e$; so the tails with a positive difference, and their differences, are read off z alone, and $\kappa(z)$ does not depend on the order π whose sector holds z. On H_π the differences are linear in z, so κ is linear there, and $\kappa(cz) = c\kappa(z)$ for $c \geq 0$. The tilt $-\mathbf{1}_S$ lies in the sectors of the orders that list $\mathcal{E} \setminus S$ first; there its only positive difference is 1, at the tail $U_k = S$, so $\kappa(-\mathbf{1}_S) = \kappa_S$. In the chart $z_{\pi(1)} = 0$ the coordinates are $z_{\pi(i)} = -(v_2 + \cdots + v_i)$, a triangular change of coordinates with determinant ± 1, so $dz = dv_2 \cdots dv_n$ on H_π; the chart does not matter (Definition 17.1). Finally $\kappa(z)$ is linear in the walls, which are linear in the question by Lemma S5.

Proposition 17.4 (The exponent of a question reads its walls; the even statistic is least at the table's gauge). *As stated, for a question with real exponents of any signs.*

Proof. (i) Write $\boldsymbol{\mu} = \mathbf{m} - \sigma\mathbf{1}$ for the offsets, so that $\boldsymbol{\mu} - \boldsymbol{\mu}_\perp \in \mathbb{R}\mathbf{1}$ and $\mu_{\mathcal{E}} = \sum_j \delta_j(2r_j - n)$ on the balance wall. The function $\tilde{X}(z) = 2\langle\boldsymbol{\mu}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)$ on $\mathbb{R}^{\mathcal{E}}$ is unchanged by $z \mapsto z + c\mathbf{1}$, since E_j is unchanged and O_j moves by $2c(2r_j - n)$ (Theorem 7.15(iii)), the move of the whole being $2c\kappa_{\mathcal{E}} = 0$ (Proposition 16.7); on $\mathbf{1}^\perp$ it is X. So X may be read at any representative of a tilt. At $\alpha = e^{2z}$, Theorem 7.15(i) gives $\sigma_j E_j + \delta_j O_j = s_j \log \Lambda_j(z) + t_j \log \Lambda_j(-z) = s_j \log K_j(\alpha) + t_j \log \Psi_j(\alpha) - (s_j + t_j)\langle\mathbf{1}, z\rangle$, and $2\langle\boldsymbol{\mu}, z\rangle + 2\sigma\langle\mathbf{1}, z\rangle = \langle\mathbf{m}, \log\alpha\rangle$, so

$$X(z) = \log\Big(\alpha^{\mathbf{m}} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j}\Big),$$

the logarithm of the integrand of the period against the measure $d\alpha/\alpha^{\mathbf{1}}$. Take the sector coordinates of Theorem 8.5(ii) for π: $\rho = \alpha_{\pi(1)} = e^{2z_{\pi(1)}}$ and $q_k = \alpha_{\pi(k)}/\alpha_{\pi(k-1)} = e^{-2v_k}$. Since $\alpha_{\pi(i)} = \rho q_2 \cdots q_i$ and $\pi(i) \in U_k$ exactly when $k \le i$, $\alpha^{\mathbf{m}} = \rho^{m_{\mathcal{E}}} \prod_k q_k^{m_{U_k}}$; with $K_j = \rho^{r_j} \prod_k q_k^{d_j(U_k)} k_{j,\pi}(q)$ and $\Psi_j = \rho^{n-r_j} \prod_k q_k^{d_j^*(U_k)} \psi_{j,\pi}(q)$,

$$\alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} = \rho^{\kappa_{\mathcal{E}}} \prod_{k=2}^{n} q_k^{\kappa_{U_k}} \prod_j k_{j,\pi}(q)^{-s_j} \psi_{j,\pi}(q)^{-t_j},$$

and $\rho^{\kappa_{\mathcal{E}}} = 1$ on the balance wall. The logarithm, with $\log q_k = -2v_k$, is the identity of (i).

On H_π the greedy basis $T^+_{j,\pi}$ has the largest monomial α^T among the bases of table j, and $T^-_{j,\pi}$ the largest $\alpha^{\mathcal{E}\setminus T}$ (Lemma 8.3); as $-\chi_T = \chi_{\mathcal{E}\setminus T}$, $h_j(z) = \langle\chi_{T^+_{j,\pi}}, z\rangle$ and $h_j(-z) = \langle\chi_{\mathcal{E}\setminus T^-_{j,\pi}}, z\rangle$, whatever the signs of the exponents. Write $z = z_{\pi(1)}\mathbf{1} - \sum_k v_k \mathbf{1}_{U_k}$; for $A \subseteq \mathcal{E}$, $\langle\chi_A, z\rangle = z_{\pi(1)}(2|A| - n) - \sum_k v_k(2|A \cap U_k| - |U_k|)$, and $|T^+_{j,\pi} \cap U_k| = d_j(U_k)$, $|U_k \setminus T^-_{j,\pi}| = d_j^*(U_k)$. So

$$\sum_j \big(s_j h_j(z) + t_j h_j(-z)\big) = 2z_{\pi(1)} \sum_j \delta_j(2r_j - n) + 2\sum_k v_k\Big(\sigma|U_k| - \sum_j \big(s_j d_j(U_k) + t_j d_j^*(U_k)\big)\Big),$$

while $2\langle\boldsymbol{\mu}, z\rangle = 2z_{\pi(1)}\mu_{\mathcal{E}} - 2\sum_k v_k \mu_{U_k}$. The difference is $2\sum_k v_k\big(m_{U_k} - \sum_j (s_j d_j(U_k) + t_j d_j^*(U_k))\big) = 2\kappa(z)$, the term in $z_{\pi(1)}$ vanishing on the balance wall; on $\mathbf{1}^\perp$, $\langle\boldsymbol{\mu}, z\rangle = \langle\boldsymbol{\mu}_\perp, z\rangle$. For $s_j, t_j \ge 0$: the support functions of P_j and P_j^* are $\frac{1}{2}(h_j(z) + \langle\mathbf{1}, z\rangle)$ and $\frac{1}{2}(h_j(-z) + \langle\mathbf{1}, z\rangle)$, support functions add over a Minkowski sum with nonnegative coefficients, and so $2\kappa(z) = 2h_{\mathcal{N}_F}(z) - 2\sigma\langle\mathbf{1}, z\rangle - 2\langle\boldsymbol{\mu}, z\rangle = 2h_{\mathcal{N}_F}(z) - 2\langle\mathbf{m}, z\rangle$, twice the support function of $\mathcal{N}_F - \mathbf{m}$, which lies in $\mathbf{1}^\perp$ and so reads tilts.

The sector letters are polynomials in q with nonnegative coefficients and the constant terms $\theta_{j,T^+_{j,\pi}}$, $\theta_{j,T^-_{j,\pi}}$, and their value at $q = \mathbf{1}$ is $K_j(\mathbf{1}) = \Psi_j(\mathbf{1})$; on $[0,1]^{n-1}$ they increase in every q_k and so lie between these (Theorem 8.5(ii)), whatever the signs of the exponents. Taking logarithms and weighting by s_j and t_j gives both bounds.

(ii) If every proper wall is positive, on H_π, $\kappa(z) = \sum_k \kappa_{U_k} v_k \ge \kappa_{\min} \sum_k v_k = \kappa_{\min}(z_{\pi(1)} - z_{\pi(n)})$, and $\max_e z_e - \min_e z_e$ vanishes on $\mathcal{T}$ only at 0. Conversely $\kappa(-\mathbf{1}_S) = \kappa_S$.

(iii) At the centre $\boldsymbol{\mu}_\perp = 0$ and $\boldsymbol{\delta} = 0$, so $X = -\sum_j \sigma_j E_j$, and the walls are $\kappa_S = \sum_j \sigma_j c_j(S)$ (Proposition 8.21(ii)). By (i), $2\kappa = \sum_j \sigma_j (h_j(z) + h_j(-z)) = \sum_j \sigma_j \operatorname{wd}_j(z)$, and $\operatorname{wd}_j(z) = \max_T\langle\chi_T, z\rangle - \min_T\langle\chi_T, z\rangle = 2\big(\max_T\langle\mathbf{1}_T, z\rangle - \min_T\langle\mathbf{1}_T, z\rangle\big)$ is twice the width of P_j in the direction z, equal to $2(r_j(S) - d_j(S)) = 2c_j(S)$ at $z = \mathbf{1}_S$ (Theorem 7.13). The bounds of (i) at $s_j = t_j = \sigma_j$ are the displayed ones. The Hessian of $\log\Lambda_j$ at z is the covariance of χ_T under $\mathbb{P}_{j,e^{2z}}$ (Theorem 7.16), so $\log\Lambda_j$ is convex, and so is $E_j(z) = \log\Lambda_j(z) + \log\Lambda_j(-z)$, which is even; hence $E_j(z) = \frac{1}{2}(E_j(z) + E_j(-z)) \ge E_j(0) = 2\log\Lambda_j(0) = 2\log K_j(\mathbf{1})$.

For $v \in \mathbf{1}^{\perp}$,

$$v^{\mathsf{T}} \nabla^2 \Big(\sum_j \sigma_j E_j \Big)(z)\, v = \sum_j \sigma_j \Big(\operatorname{Var}_{T \sim \mathbb{P}_{j,e^{2z}}} \langle \chi_T, v \rangle + \operatorname{Var}_{T \sim \mathbb{P}_{j,e^{-2z}}} \langle \chi_T, v \rangle \Big).$$

Every basis has positive chance under both laws, so the form vanishes only if $\langle \chi_T, v \rangle = 2\langle \mathbf{1}_T, v \rangle - \langle \mathbf{1}, v \rangle$ is constant over the bases of every table with $\sigma_j > 0$. Then v is constant on every molecule of the family of these tables (Lemma 7.8), and every such molecule M has $c_j(M) = 0$ in each of them, the table being the direct sum of its restrictions to the molecules (Proposition 7.9), so that $|T \cap M| = r_j(M)$ for every basis. If $v \notin \mathbb{R}\mathbf{1}$, the union S of the molecules on which v is largest is nonempty and proper with $c_j(S) = 0$ for every table with $\sigma_j > 0$: the family is not connected. So on a connected family the form is positive on $\mathcal{T} \setminus 0$ and $\sum_j \sigma_j E_j$ is strictly convex on $\mathcal{T}$. Finally $\sum_j \sigma_j \operatorname{wd}_j = 2\kappa$ is positive on $\mathcal{T} \setminus 0$ by (ii), the walls $\sum_j \sigma_j c_j(S)$ of the centre being positive exactly when the family is connected (Proposition 8.21(ii)). □

Remark. For one table at the centre of strength σ, (iii) is the statement of the reference, with $2\kappa = \sigma \operatorname{wd}$; the family statement adds (i), in which the width of one table is replaced by the wall at the tilt of any question, and (ii). The identity of (i) is the sector form of the period read pointwise: integrated against $dz = dv$ on each sector it is $P_F(q) = 2^{n-1} \sum_\pi \int_{[0,\infty)^{n-1}} e^{-2\sum_k \kappa_{U_k} v_k} h_\pi(e^{-2v})\, dv$, with $h_\pi = \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}$, which is $\sum_\pi I_\pi$ of Corollary 8.6 in the gaps $2v_k$.

48.17.3 The two ends of strength

The high end needs one fact about integrals of $e^{-\lambda f}$ with a convex exponent, for complex λ in a half-plane; it is proved here once, in the form in which the period uses it.

Lemma S6 (A convex exponent at high strength)**.** *Let f be a smooth function on a real vector space V of dimension $d \geq 1$ with Lebesgue measure dz and a norm $|\cdot|$, such that*

(a) every derivative of f of order at least one is bounded;

(b) f is convex, with a critical point $z^\star$ at which $A_0 = \nabla^2 f(z^\star)$ is positive definite;

(c) $f(z) - f(z^\star) \geq c|z| - C$ for some $c > 0$ and C.

Then $z^\star$ is the only critical point, and as $|\lambda| \to \infty$ with $\operatorname{Re}\lambda \geq \lambda_0$, for a fixed $\lambda_0 > 0$,

$$\int_V e^{-\lambda f(z)}\, dz = e^{-\lambda f(z^\star)} \Big(\frac{2\pi}{\lambda} \Big)^{\frac{d}{2}} \det(A_0)^{-\frac{1}{2}} \big(1 + O(|\lambda|^{-1}) \big),$$

the determinant in the coordinates of dz and the power principal.

Proof. Use a Euclidean norm, equivalent to the given one. *The gradient away from $z^\star$.* By convexity $\langle \nabla f(z), z - z^\star \rangle \geq f(z) - f(z^\star) \geq c|z| - C$, so $|\nabla f(z)| \geq (c|z| - C)/|z - z^\star|$, which tends to c as $|z| \to \infty$. A convex function with a positive definite Hessian at a critical point has its strict minimum there, and its critical points form the convex set of its minima, which is $\{z^\star\}$. So for every neighbourhood V_0 of $z^\star$, $|\nabla f| \geq c_0 > 0$ on $V \setminus V_0$: on a large ball by compactness, outside it by the bound.

A chart at $z^\star$. Put $A(z) = 2\int_0^1 (1-t)\, \nabla^2 f\big(z^\star + t(z - z^\star)\big)\, dt$, smooth and symmetric with $A(z^\star) = A_0$. Since $\nabla f(z^\star) = 0$, $f(z) = f(z^\star) + \frac{1}{2}\langle A(z)(z - z^\star), z - z^\star \rangle$. On a ball V_1 about $z^\star$ on which $A(z)$ is positive definite put $w(z) = A(z)^{1/2}(z - z^\star)$, the positive square root being smooth there; then $f = f(z^\star) + \frac{1}{2}|w|^2$, and $Dw(z^\star) = A_0^{1/2}$ is invertible, so w is a diffeomorphism of a smaller ball V_2 onto a neighbourhood of 0.

Near $z^\star$. Take a smooth χ with compact support in V_2 and $\chi = 1$ near $z^\star$. In the coordinates w, $\int \chi e^{-\lambda f} dz = e^{-\lambda f(z^\star)} \int_{\mathbb{R}^d} e^{-\lambda |w|^2/2} g(w)\, dw$, with g smooth of compact support and $g(0) = |\det Dw(z^\star)|^{-1} = \det(A_0)^{-1/2}$. For $\operatorname{Re}\lambda > 0$ and $\xi \in \mathbb{R}^d$, $\int e^{-\lambda|w|^2/2 + i\langle \xi, w\rangle} dw = (2\pi/\lambda)^{d/2} e^{-|\xi|^2/(2\lambda)}$: both sides are holomorphic in λ on the half-plane and agree for $\lambda > 0$, completing the square in each coordinate. Writing $g(w) = (2\pi)^{-d} \int \hat g(\xi) e^{i\langle \xi, w\rangle} d\xi$ with $\hat g$ rapidly decreasing,

$$\int e^{-\lambda|w|^2/2} g(w)\, dw = \Big(\frac{2\pi}{\lambda}\Big)^{\frac{d}{2}} (2\pi)^{-d} \int \hat g(\xi)\, e^{-|\xi|^2/(2\lambda)}\, d\xi = \Big(\frac{2\pi}{\lambda}\Big)^{\frac{d}{2}} \big(g(0) + \varrho(\lambda)\big),$$

and since $\operatorname{Re}(1/\lambda) \geq 0$ and $|e^{-x} - 1| \leq |x|$ for $\operatorname{Re} x \geq 0$, $|\varrho(\lambda)| \leq (2|\lambda|)^{-1} (2\pi)^{-d} \int |\xi|^2 |\hat g(\xi)|\, d\xi$.

Away from $z^\star$. On the support of $1 - \chi$, $|\nabla f| \geq c_0$. The operator $L = -\lambda^{-1} |\nabla f|^{-2} \langle \nabla f, \nabla \rangle$ has $L e^{-\lambda f} = e^{-\lambda f}$, and its transpose is $L^{\mathsf t} \varphi = \lambda^{-1} \operatorname{div}\big(\varphi\, |\nabla f|^{-2} \nabla f\big)$. Integrating by parts N times, $\int (1-\chi) e^{-\lambda f} dz = \int e^{-\lambda f} (L^{\mathsf t})^N (1-\chi)\, dz$, with no boundary terms: $1 - \chi$ vanishes near $z^\star$, and on large spheres $|e^{-\lambda f}| = e^{-\operatorname{Re}\lambda f}$ decays exponentially by (c) while the coefficients stay bounded. Now $(L^{\mathsf t})^N (1-\chi) = \lambda^{-N} G_N$ with G_N a polynomial in the derivatives of χ, in those of f of orders 1 to $N+1$ and in $|\nabla f|^{-2}$, bounded by (a) and $|\nabla f| \geq c_0$. Since $f \geq f(z^\star)$ and $\operatorname{Re}\lambda \geq \lambda_0$,

$$\Big| \int (1-\chi) e^{-\lambda f} dz \Big| \leq |\lambda|^{-N} \sup |G_N|\; e^{-\operatorname{Re}\lambda\, f(z^\star)} \int_V e^{-\lambda_0 (f - f(z^\star))} dz,$$

the last integral finite by (c). With $N > \frac{d}{2} + 1$ this is $O(|\lambda|^{-1})$ times the modulus $(2\pi/|\lambda|)^{d/2} e^{-\operatorname{Re}\lambda f(z^\star)} \det(A_0)^{-1/2}$ of the leading term, and the two parts add to the statement. (On the real half-line this is Laplace's method, along the imaginary direction stationary phase, and w is the Morse chart.) □

Theorem 17.5 (The two ends of strength). *As stated.*

Proof. The walls and the exponents of λq are λ times those of q (Lemma S5), so its walls $\lambda \kappa_S$ have positive real parts for $\operatorname{Re}\lambda > 0$. At such data the period is the convergent integral of Theorem 8.7, the powers of the positive letters principal, and the change of variables of Definition 17.1, which uses only $K_j(e^{2z}) = e^{\langle \mathbf{1}, z\rangle} \Lambda_j(z)$ and $\Psi_j(e^{2z}) = e^{\langle \mathbf{1}, z\rangle} \Lambda_j(-z)$, gives $P_F(\lambda q) = 2^{n-1} \int_{\mathcal T} e^{\lambda X(z)} dz$, absolutely convergent as $|e^{\lambda X}| = e^{\operatorname{Re}\lambda\, X}$; here X is the exponent of q, and the exponent of λq is λX, the exponent being linear in the question.

(i) By Proposition 17.4(i) and $dz = dv$ on each sector, with $h_\pi = \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}$,

$$P_F(\lambda q) = 2^{n-1} \sum_\pi \int_{[0,\infty)^{n-1}} e^{-2\lambda \sum_k \kappa_{U_k} v_k}\, h_\pi(e^{-2v})^\lambda\, dv,$$

and with $v = w/\lambda$,

$$\lambda^{n-1} P_F(\lambda q) = 2^{n-1} \sum_\pi \int_{[0,\infty)^{n-1}} e^{-2\sum_k \kappa_{U_k} w_k}\, h_\pi(e^{-2w/\lambda})^\lambda\, dw.$$

On the cube $|\log h_\pi| \leq C$ (Proposition 17.4(i)), so for $0 < \lambda \leq 1$ the integrand lies between $e^{\mp C} e^{-2\sum_k \kappa_{U_k} w_k}$, whose integral is $\prod_k (2\kappa_{U_k})^{-1}$; it tends to $e^{-2\sum_k \kappa_{U_k} w_k}$, and by domination $\lambda^{n-1} P_F(\lambda q) \to 2^{n-1} \sum_\pi \prod_k (2\kappa_{U_k})^{-1} = \sum_\pi \prod_k \kappa_{U_k}^{-1}$. For the next order, $|h^\lambda - 1 - \lambda \log h| \leq \frac{1}{2} \lambda^2 C^2 e^{\lambda C}$, and for w in the open orthant $e^{-2w/\lambda} \to 0$, so $\log h_\pi(e^{-2w/\lambda}) \to \log h_\pi(0)$, boundedly; hence

$$\lambda^{-1} \Big(\lambda^{n-1} P_F(\lambda q) - \sum_\pi \prod_k \kappa_{U_k}^{-1} \Big) \to \sum_\pi \prod_k \kappa_{U_k}^{-1} \log h_\pi(0),$$

and $h_\pi(0) = \prod_j \theta_{j,T^+_{j,\pi}}^{-s_j}\theta_{j,T^-_{j,\pi}}^{-t_j}$ is the sector integrand at $q = 0$, so that $\log h_\pi(0) = -\sum_j \big(s_j \log\theta_{j,T^+_{j,\pi}} + t_j \log\theta_{j,T^-_{j,\pi}}\big)$, the coefficient stated. Finally κ is positively homogeneous of degree one and positive on $\mathcal{T}\setminus 0$ (Proposition 17.4(ii)), so $\operatorname{vol}\{\kappa\le r\} = r^{n-1}\operatorname{vol}\{\kappa\le 1\}$ and

$$2^{n-1}\int_{\mathcal{T}} e^{-2\kappa(z)}dz = 2^{n-1}\operatorname{vol}\{\kappa\le 1\}\,(n-1)\int_0^\infty e^{-2r}r^{n-2}dr = (n-1)!\operatorname{vol}\{\kappa\le 1\},$$

while on the sectors the same integral is $2^{n-1}\sum_\pi\prod_k(2\kappa_{U_k})^{-1}$. (For $s_j,t_j\ge 0$, $\{\kappa\le 1\}$ is the polar of $\mathcal{N}_F-\mathbf{m}$ in $\mathbf{1}^\perp$.)

(ii) Let $s_j,t_j\ge 0$ and $f=-X$ on $\mathcal{T}$, of dimension $n-1$ with the measure dz. The hypotheses of Lemma S6 hold. (a) Every derivative of $\log\Lambda_j$ of order $k\ge 1$ is a joint cumulant of order k of $\chi_T\in\{\pm1\}^{\mathcal{E}}$ under $\mathbb{P}_{j,e^{2z}}$ (Theorem 7.16), bounded by a constant depending on k alone, and $f=\sum_j\big(s_j\log\Lambda_j(z)+t_j\log\Lambda_j(-z)\big)-2\langle\boldsymbol{\mu}_\perp,z\rangle$. (b) $v^{\mathsf T}\nabla^2 f(z)v=\sum_j\big(s_j\operatorname{Var}_{\mathbb{P}_{j,e^{2z}}}\langle\chi_T,v\rangle+t_j\operatorname{Var}_{\mathbb{P}_{j,e^{-2z}}}\langle\chi_T,v\rangle\big)$; a question with $s_j,t_j\ge 0$ inside its walls is on a connected family, as $\kappa_S+\kappa_{\mathcal{E}\setminus S}=2\sum_j\sigma_j c_j(S)>0$ (Lemma S5) puts some active table of positive width on every S, and the argument of Proposition 17.4(iii) makes the form positive definite on $\mathcal{T}$ at every z; so f is strictly convex. (c) $f\ge 2\kappa-C\ge 2\kappa_{\min}(\max_e z_e-\min_e z_e)-C$ by Proposition 17.4(i)–(ii), and $\max_e z_e-\min_e z_e$ is a norm on $\mathcal{T}$. A strictly convex f with (c) attains its minimum at one point $z^\star$, its only critical point. By Theorem 7.16, $\nabla\log\Lambda_j(z)=2R_j(\alpha)-\mathbf{1}$ and $\nabla[\log\Lambda_j(-z)]=\mathbf{1}-2R_j(\alpha^{-1})=2R_j^*(\alpha)-\mathbf{1}$ at $\alpha=e^{2z}$, so the derivative of f along $v\in\mathbf{1}^\perp$ is $2\langle\mathbf{u}(\alpha)-\mathbf{m},v\rangle$, $\mathbf{u}=\sum_j(s_jR_j+t_jR_j^*)$ the usage, $\mathbf{m}-\sigma\mathbf{1}-\boldsymbol{\mu}_\perp$ being a multiple of $\mathbf{1}$. It vanishes for every v exactly when $\mathbf{u}(\alpha)-\mathbf{m}\in\mathbb{R}\mathbf{1}$, that is $\mathbf{u}(\alpha)=\mathbf{m}$, the two having the total $\sum_j(s_jr_j+t_jr_j^*)$ on the balance wall; and $\mathbf{m}$ is the target of q there, so $z^\star$ is its native gauge (Definition 8.17). The usage and the masses of λq are λ times those of q, so $z^\star$ is the native gauge of every λq, $\lambda>0$. Lemma S6 with $d=n-1$ gives the expansion, the determinant taken on a chart $z_g=0$, that is of the Hessian of the invariant form $\tilde X$ of the proof of Proposition 17.4 with the row and the column of g deleted. With $y=2z$, $\Phi(y)=\sigma\langle\mathbf{1},y\rangle+\Phi^\circ(z)$ (Theorem 8.19), so $\partial_{z_e}\partial_{z_f}(-\tilde X)=\partial_{z_e}\partial_{z_f}\Phi^\circ=4\partial_{y_e}\partial_{y_f}\Phi$, rung two of the still of the realization $\big((s_j,t_j)_j;e^{2z^\star}\big)$ of the active tables (Definition 9.4). At the centre $f=\sum_j\sigma_jE_j$ is even and strictly convex, so $z^\star=0$, $f(0)=2\sum_j\sigma_j\log K_j(\mathbf{1})$, and $\nabla^2 f(0)=\sum_j\sigma_j\cdot 2\operatorname{Cov}_{\mathbb{P}_{j,\mathbf{1}}}(\chi_T)=8\sum_j\sigma_j\big(\kappa_{j,\mathbf{1}}(X_e,X_f)\big)_{e,f}$, as $\chi_T=2X-\mathbf{1}$, read on the chart $z_g=0$ with the row and the column of g deleted. □

Remark. At one table and the centre of strength one, $\kappa=\frac12\,\mathrm{wd}$ and $(n-1)!\operatorname{vol}\{\kappa\le 1\}=2^{n-1}(n-1)!\operatorname{vol}\{\mathrm{wd}\le 1\}$, the reference's low end; its high end is (ii) at offsets zero. Divided by the same limit for the list $\binom{\mathcal{E}}{1}$ at a full basis lean, whose walls are the masses, (i) is the limit of Proposition 47.14: there $\Gamma_{\binom{n}{1}}\begin{bmatrix}\lambda\boldsymbol{\tau}\\ \lambda\tau_{\mathcal{E}}\end{bmatrix}$ has $\lambda^{n-1}\Gamma_{\binom{n}{1}}\begin{bmatrix}\lambda\boldsymbol{\tau}\\ \lambda\tau_{\mathcal{E}}\end{bmatrix}\to\tau_{\mathcal{E}}/\prod_e\tau_e=\sum_\pi\prod_k\tau_{U_k}^{-1}$. In (ii) the real ray is Proposition 18.9 written in the signed coordinates; what the lemma adds is the whole half-plane, uniformly.

48.17.4 How far a question can lean, and the tails

Proposition 17.7 (How far a question can lean; the values of the statistics). *As stated.*

Proof. The decomposition of the walls is Lemma S5. Every proper wall is positive exactly when, for every S, $\kappa_S^{\mathbf{E}}+\kappa_S^{\mathbf{O}}>0$ and $\kappa_S^{\mathbf{E}}-\kappa_S^{\mathbf{O}}>0$, that is $|\kappa_S^{\mathbf{O}}|<\kappa_S^{\mathbf{E}}$. The fold of Theorem 16.18 sends the pair coordinates $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ to $(-\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},-\boldsymbol{\delta})$, negating the offsets and the leans and keeping

the strengths (Definition 9.1); so it negates $\kappa_S^{\mathbf{O}}$ and swaps κ_S with $\kappa_{\mathcal{E}\setminus S}$. When every $\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})$ is positive the conditions read $\kappa_S^{\mathbf{E}}/|\kappa_S^{\mathbf{O}}| > 1$ wherever $\kappa_S^{\mathbf{O}} \neq 0$, that is $\ell^*_{\boldsymbol{\sigma}}(\boldsymbol{\mu}_\perp, \boldsymbol{\delta}) > 1$: finitely many strict linear inequalities in the odd part, each with its negative, an open convex polyhedron symmetric under the fold. It is bounded exactly when the linear map $(c, \mathbf{b}) \mapsto (\kappa_S^{\mathbf{O}}(c, \mathbf{b}))_S$ is injective: then it lies in the preimage of a bounded box, and otherwise a line through 0 in the kernel lies in it. The homogeneities and the evenness in $(c, \mathbf{b})$ are read off the definition. For one connected table at offsets zero, $\kappa_S^{\mathbf{E}} = \sigma c(S)$ and $\kappa_S^{\mathbf{O}} = -\delta\varepsilon^\perp(S)$, so the conditions are $\sigma > 0$ and $|\delta| < \sigma \min c(S)/|\varepsilon^\perp(S)| = \ell_1^*(0,1)\,\sigma$.

(i) The exponent is linear in the question, so with $q' = (c; \boldsymbol{\sigma} + \mathbf{a}, \mathbf{b})$ and the exponent $X_{q'}$ of q', the expectation is $2^{n-1} P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})^{-1} \int_{\mathcal{T}} e^{X_{q'}(z)} dz$, an integral of a positive function. By Proposition 17.4(i), whose identity holds at every real exponents, $e^{X_{q'}}$ lies between constant multiples of $e^{-2\kappa'(z)}$, κ' the wall at the tilt of q', and $\int_{H_\pi} e^{-2\kappa'(z)} dz = \int_{[0,\infty)^{n-1}} e^{-2\sum_k \kappa_{U_k}(q')v_k} dv$ is finite exactly when every $\kappa_{U_k}(q')$ is positive. Every nonempty proper S is a tail of some order, so the expectation is finite exactly when every proper wall of q' is positive, and it is then $P_F(q')/P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$ by Definition 17.1. The walls of q' are $\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma} + \mathbf{a}) \pm \kappa_S^{\mathbf{O}}(c, \mathbf{b})$; at $\mathbf{a} = 0$ their positivity is $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) > 1$.

(ii) With $|\delta_j| \le \sigma_j$ the exponents are nonnegative, and by the proof of Theorem 17.5(ii) $-X$ is strictly convex on $\mathcal{T}$ with its least value at the native gauge $z^\star$, at which alone it is attained. $-X(z)$ is the value of the linear form $-2\langle \boldsymbol{\mu}_\perp, x\rangle + \sum_j (\sigma_j y_j + \delta_j w_j)$ at the point $(z; E_j(z), O_j(z))_j$, and z is a coordinate of that point, so the least value of the form on the set of values of the statistics is taken at $\big(z^\star; E_j(z^\star), O_j(z^\star)\big)_j$ and only there. At the centre $z^\star = 0$ (Proposition 17.4(iii)), $E_j(0) = 2\log K_j(\mathbf{1})$ and $O_j(0) = 0$. □

Proposition 17.8 (The tails of the statistics are read at the walls). *As stated.*

Proof. By Lemma S5 the walls are linear in the pair coordinates, and for the direction v of pair coordinates $(c; \mathbf{a}, \mathbf{b})$, $\kappa_S(v) = \kappa_S^{\mathbf{E}}(\mathbf{a}) + \kappa_S^{\mathbf{O}}(c, \mathbf{b})$, the offset direction c entering through $c_S = \sum_{e\in S} c_e$. With $X_v(z) = 2\langle c, z\rangle - \sum_j (a_j E_j(z) + b_j O_j(z))$, $Y = X_v(Z)$ and $X_q + \lambda X_v = X_{q+\lambda v}$, so for real $\lambda \ge 0$, $\mathbb{E}[e^{\lambda Y}] = 2^{n-1} P_F(q)^{-1} \int_{\mathcal{T}} e^{X_{q+\lambda v}} dz$, finite, by the argument of Proposition 17.7(i), exactly when every $\kappa_S(q) + \lambda\kappa_S(v)$ is positive, that is for $\lambda < \lambda^*$, and then the ratio of the periods.

Upper bound. For $0 < \lambda < \lambda^*$, $\mathbb{P}(Y > x) \le e^{-\lambda x}\mathbb{E}[e^{\lambda Y}]$, so $\limsup_x x^{-1}\log \mathbb{P}(Y > x) \le -\lambda$ for every $\lambda < \lambda^*$.

Lower bound. Let $\lambda^* < \infty$, let S^* attain it, and let π list $\mathcal{E} \setminus S^*$ first, so that $S^* = U_{k^*}$ is a tail of π. Write $\kappa_q(z)$ and $\kappa_v(z) = \sum_k \kappa_{U_k}(v)\big(z_{\pi(k-1)} - z_{\pi(k)}\big)$ for the walls at the tilt of q and of v, the latter read as balanced data with the exponents $(\mathbf{a} + \mathbf{b}, \mathbf{a} - \mathbf{b})$. By Proposition 17.4(i), whose identity holds at any balanced real data, $|X_q + 2\kappa_q| \le C_q$ and $|X_v + 2\kappa_v| \le C_v$ on $\mathcal{T}$. For $\varepsilon > 0$ put $a_x = (1+\varepsilon)x/(-2\kappa_{S^*}(v))$ and let $A_x \subset H_\pi$ be the set of the tilts whose difference along π at the tail S^* lies in $[a_x, a_x + 1]$ and whose other differences lie in $[0, 1]$, of dz-volume 1. On A_x, as $\kappa_{S^*}(v) < 0$,

$$Y \ge -2\kappa_v(z) - C_v \ge (1+\varepsilon)x - 2\sum_k |\kappa_{U_k}(v)| - C_v > x \quad \text{for large } x,$$

$$X_q \ge -2\kappa_q(z) - C_q \ge -2\kappa_{S^*}(q)(a_x + 1) - 2\sum_k |\kappa_{U_k}(q)| - C_q = -(1+\varepsilon)\lambda^* x - C',$$

since $2\kappa_{S^*}(q)a_x = (1+\varepsilon)\lambda^* x$. So $\mathbb{P}(Y > x) \ge 2^{n-1} P_F(q)^{-1} e^{-(1+\varepsilon)\lambda^* x - C'}$ for large x, $\liminf_x x^{-1}\log \mathbb{P}(Y > x) \ge -(1+\varepsilon)\lambda^*$ for every $\varepsilon > 0$, and with the upper bound the limit is $-\lambda^*$.

If $\lambda^* = \infty$, every $\kappa_S(v) \ge 0$, so $\kappa_v \ge 0$ on every sector and $Y \le C_v$.

At the centre of strengths $\boldsymbol{\sigma}$ and the direction of pair coordinates $(c;\mathbf{0},\mathbf{b})$, $\kappa_S(q)=\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})$ and $\kappa_S(v)=\kappa_S^{\mathbf{O}}(c,\mathbf{b})$. The first is unchanged and the second negated by $S\mapsto\mathcal{E}\setminus S$, so the sets on which $\kappa_S(v)<0$ are the complements of those on which it is positive, and $\lambda^*=\min_S \kappa_S^{\mathbf{E}}/|\kappa_S^{\mathbf{O}}|=\ell^*_{\boldsymbol{\sigma}}(c,\mathbf{b})$, for Y and for $-Y$ alike. □

Remark. The one-table rate of both tails of $O(Z)$ at the centre of strength σ_0 is $\ell^*_{\sigma_0}(0,1)=\sigma_0\,\ell_1^*(0,1)$. The same proposition, applied to the family with the list $\binom{\mathcal{E}}{1}$ added, gives the tails of the Grey law (Proposition 47.12(iii), proved in Section 48.46.3).

48.18 Chapter 18: the Breta law and the port on the balance wall

Proposition 18.3 (Scale and shapes along a bunch). *As stated: off the balance wall and at complex exponents, the scale integral then carrying the appended list* $\binom{\mathcal{E}}{1}$.

Proof. The display. The masses are positive and every proper wall has positive real part, so $P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}}W(u)\,d\omega_\mathcal{E}(u)$ converges (8.7). By 3.12 with the blocks B and R, $u=(w_Bv_B,w_Rv_R)$ with $(w_B,w_R)\in\Delta_{\{B,R\}}$, $v_B\in\Delta_B$ and $v_R\in\Delta_R$, and

$$u^{\mathbf{m}-\mathbf{1}}d\omega_\mathcal{E}=w_B^{m_B-1}w_R^{m_R-1}dw_B\cdot v_B^{\mathbf{m}|_B-\mathbf{1}}d\omega_B\cdot v_R^{\mathbf{m}|_R-\mathbf{1}}d\omega_R.$$

With $x=w_B/w_R\in(0,\infty)$, $w_B=x/(1+x)$, $w_R=1/(1+x)$ and $dw_B=(1+x)^{-2}dx$, so $w_B^{m_B-1}w_R^{m_R-1}dw_B=x^{m_B-1}(1+x)^{-m_\mathcal{E}}dx$. The word W is homogeneous of degree $\kappa_\mathcal{E}-m_\mathcal{E}$ (8.1), so $W(w_Bv_B,w_Rv_R)=w_R^{\kappa_\mathcal{E}-m_\mathcal{E}}W(xv_B,v_R)=(1+x)^{m_\mathcal{E}-\kappa_\mathcal{E}}W(xv_B,v_R)$, and the powers of $1+x$ collect to $(1+x)^{-\kappa_\mathcal{E}}$. By Tonelli on the modulus, which is the same integrand at the real parts of the exponents, and Fubini,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_B}\int_{\Delta_R} v_B^{\mathbf{m}|_B-\mathbf{1}}v_R^{\mathbf{m}|_R-\mathbf{1}}\int_0^\infty x^{m_B-1}(1+x)^{-\kappa_\mathcal{E}}W(xv_B,v_R)\,dx\,d\omega_R(v_R)\,d\omega_B(v_B).$$

The densities of $\mathrm{Dir}_{\mathbf{m}|_B}$ and $\mathrm{Dir}_{\mathbf{m}|_R}$ are $v_B^{\mathbf{m}|_B-\mathbf{1}}/\Gamma_{\binom{|B|}{1}}\begin{bmatrix}\mathbf{m}|_B\\ m_B\end{bmatrix}$ and $v_R^{\mathbf{m}|_R-\mathbf{1}}/\Gamma_{\binom{|R|}{1}}\begin{bmatrix}\mathbf{m}|_R\\ m_R\end{bmatrix}$ (3.10), which gives the display. At (xU_B',U_R') the letter $|\alpha|$ of the list $\binom{\mathcal{E}}{1}$ is $x|U_B'|+|U_R'|=1+x$, so the inner integral is the scale integral (15.3) of the family with that table appended at the exponents $(\kappa_\mathcal{E},0)$ (16.3). On the balance wall the factor $(1+x)^{-\kappa_\mathcal{E}}$ is 1, the inner integral is $\mathrm{Sc}_B(U_B',U_R';\mathbf{m};\mathbf{s},\mathbf{t})$, and the double integral above is 16.6 at $h=|\alpha_R|$.

The conditional density. For real data with every proper wall positive the same computation, which needs no sign of the masses, shows that under $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ the variables $\rho=|U_B|/|U_R|$, V_B and V_R have the joint density

$$\rho^{m_B-1}(1+\rho)^{-\kappa_\mathcal{E}}W(\rho V_B,V_R)\,V_B^{\mathbf{m}|_B-\mathbf{1}}V_R^{\mathbf{m}|_R-\mathbf{1}}/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$$

against $d\rho\,d\omega_B\,d\omega_R$. Given (V_B,V_R), the density of ρ is proportional to $\rho^{m_B-1}(1+\rho)^{-\kappa_\mathcal{E}}W(\rho V_B,V_R)$. On the balance wall this is the reference, proof of 18.3. □

Corollary 18.7 (Conjugacy for the continued period). *As stated.*

Proof. Let Ω be the set of the data at which every nonempty proper wall has positive real part, open and convex, and not empty: at $\mathbf{s}=\mathbf{t}=\mathbf{0}$ the walls are $\kappa_S=m_S$. Adding $\mathbf{1}_T$ to the masses and $\mathbf{e}_j$ to $\mathbf{s}$ adds r_j to $m_\mathcal{E}$ and to $\sum_i(s_ir_i+t_ir_i^*)$, so it keeps the imbalance and maps the balance

wall into itself, and it adds $|T\cap S|-d_j(S)\ge 0$ to the wall κ_S (8.1), so it maps Ω into Ω; adding $\mathbf{1}_{\mathcal{E}\setminus T'}$ to the masses and $\mathbf{e}_j$ to $\mathbf{t}$ does the same, adding $|S\setminus T'|-d_j^*(S)\ge 0$ to κ_S. At data in Ω the three periods converge absolutely as integrals over $\Delta_{\mathcal{E}}$ (8.7). On $\Delta_{\mathcal{E}}$, $K_j(u)=\sum_T \theta_{j,T}u^{\mathbf{1}_T}$ and $\Psi_j(u)=\sum_{T'}\theta_{j,T'}u^{\mathbf{1}_{\mathcal{E}\setminus T'}}$ (7.1), so

$$u^{\mathbf{m}-\mathbf{1}}\prod_i K_i(u)^{-s_i}\Psi_i(u)^{-t_i}=\sum_T \theta_{j,T}\,u^{\mathbf{m}+\mathbf{1}_T-\mathbf{1}}K_j(u)^{-s_j-1}\prod_{i\ne j}K_i(u)^{-s_i}\prod_i\Psi_i(u)^{-t_i},$$

the powers of the positive letters being principal, and integrating term by term gives the first identity on Ω; the second is the same with Ψ_j. By 16.9 the three sides are meromorphic in the masses and the exponents, the shifted ones being P_F composed with translations, and they agree on the nonempty open set Ω, so they agree as meromorphic functions. Their restrictions to the balance wall, meromorphic by 16.9 and mapped into themselves by the translations, then agree as well. □

Lemma S7. *Let G be a convex function of class C^2 on $\mathbb{R}^d$ with $G(0)=0$, $\nabla G(0)=0$ and $\nabla^2G(0)$ positive definite, and let b be a measurable function on $\mathbb{R}^d$, continuous at 0, with $|b(y)|\le e^{c_0(1+|y|)}$ for a constant $c_0\ge 0$. Then $\int_{\mathbb{R}^d}|b|e^{-\lambda G}dy<\infty$ for every large λ, and for every bounded continuous function H on $\mathbb{R}^d$*

$$\lim_{\lambda\to+\infty}\lambda^{d/2}\int_{\mathbb{R}^d}b(y)H(\sqrt{\lambda}\,y)e^{-\lambda G(y)}dy=b(0)\int_{\mathbb{R}^d}H(x)e^{-\frac12\langle x,\nabla^2G(0)x\rangle}dx.$$

At $H=1$ the right side is $b(0)(2\pi)^{d/2}\det\big(\nabla^2G(0)\big)^{-1/2}$.

Proof. Let $\epsilon>0$ be the least eigenvalue of $\nabla^2G(0)$. By Taylor's formula $G(y)=\frac12\langle y,\nabla^2G(0)y\rangle+o(|y|^2)$; choose $\delta>0$ with $G(y)\ge\frac14\langle y,\nabla^2G(0)y\rangle\ge\frac{\epsilon}{4}|y|^2$ and $|b(y)|\le|b(0)|+1$ for $|y|\le\delta$.

Linear growth. For $|y|\ge\delta$ put $t=\delta/|y|\in(0,1]$. Convexity and $G(0)=0$ give $G(ty)\le tG(y)$, so $G(y)\ge t^{-1}G(ty)\ge\frac{|y|}{\delta}\cdot\frac{\epsilon}{4}\delta^2=\gamma|y|$, with $\gamma=\frac{\epsilon\delta}{4}$.

The outer part. For $\lambda\gamma\ge 2c_0$ and $|y|\ge\delta$, $|b(y)|e^{-\lambda G(y)}\le e^{c_0-\frac14\lambda\gamma|y|}\le e^{c_0-\frac14\lambda\gamma\delta}e^{-\frac14\lambda\gamma|y|}$. So $\int_{|y|\ge\delta}|b|e^{-\lambda G}dy\le e^{c_0-\frac14\lambda\gamma\delta}\big(\frac{4}{\lambda\gamma}\big)^d\int_{\mathbb{R}^d}e^{-|x|}dx$, and $\lambda^{d/2}$ times it tends to 0. With the bound on $|y|\le\delta$, the integral of $|b|e^{-\lambda G}$ is finite.

The inner part. With $y=x/\sqrt{\lambda}$,

$$\lambda^{d/2}\int_{|y|<\delta}b(y)H(\sqrt{\lambda}\,y)e^{-\lambda G(y)}dy=\int_{\mathbb{R}^d}\mathbf{1}\big[|x|<\delta\sqrt{\lambda}\big]\,b(x/\sqrt{\lambda})H(x)e^{-\lambda G(x/\sqrt{\lambda})}dx.$$

As $\lambda\to\infty$ the integrand tends to $b(0)H(x)e^{-\frac12\langle x,\nabla^2G(0)x\rangle}$ at every x, by the continuity of b at 0 and $\lambda G(x/\sqrt{\lambda})\to\frac12\langle x,\nabla^2G(0)x\rangle$, and it is bounded by $(|b(0)|+1)\sup|H|\,e^{-\epsilon|x|^2/4}$, which is integrable. Dominated convergence gives the limit. At $H=1$, $\int e^{-\frac12\langle x,Ax\rangle}dx=(2\pi)^{d/2}(\det A)^{-1/2}$ for positive definite A. □

Proposition 18.9 (The period along a ray of data). *As stated: a base point of complex data on the balance wall added to the scaled data.*

Proof. Write $\mathbf{d}_\lambda=(\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$, and in this proof $G(y)=\Phi(y)-\langle\boldsymbol{\tau},y\rangle$, Φ the potential at $(\mathbf{s},\mathbf{t})$.

The data. The walls and the imbalance are affine in the data with real coefficients. At $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$, real data with $\boldsymbol{\tau}>0$, $s_j,t_j\ge 0$ not all zero and a native gauge on a connected family, every proper

wall is positive (8.19(iii)), and both $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ and $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ lie on the balance wall. So $\kappa_{\mathcal{E}}(\mathbf{d}_\lambda)=0$, and $\operatorname{Re}\kappa_S(\mathbf{d}_\lambda)=\operatorname{Re}\kappa_S(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)+\lambda\kappa_S(\boldsymbol{\tau};\mathbf{s},\mathbf{t})\to+\infty$ for every nonempty proper S: for λ large every proper wall of $\mathbf{d}_\lambda$ has positive real part.

The chart integral. For such λ, $P_F(\mathbf{d}_\lambda)$ is the integral over the chart $\alpha_g=1$ (16.2). In $y=\log\alpha$, $y_g=0$, the measure is $\alpha^{\mathbf{m}-\mathbf{1}}d\alpha_{\mathcal{E}-g}=e^{\langle\mathbf{m},y\rangle}dy_{\mathcal{E}-g}$, and the family word at the exponents $(\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$ is $W_0(e^y)e^{-\lambda\Phi(y)}$ (8.18). So

$$P_F(\mathbf{d}_\lambda)=\int_{\{y_g=0\}}b(y)\,e^{-\lambda G(y)}\,dy_{\mathcal{E}-g},\qquad b(y)=e^{\langle\mathbf{m}_0,y\rangle}W_0(e^y).$$

The minimum. Φ is convex (8.19(ii)), and $\nabla G(y)=\nabla\Phi(y)-\boldsymbol{\tau}$ is the usage at e^y less the target, which vanishes at $y^\star=\log\alpha^\star$ (8.19(i)); normalize $\alpha^\star_g=1$, so that $y^\star$ lies on the chart. For $v\in\mathbb{R}^{\mathcal{E}}$ and $\alpha=e^y$,

$$\langle v,\nabla^2\Phi(y)v\rangle=\sum_j\Big(s_j\operatorname{Var}_{T\sim\mathbb{P}_{j,\alpha}}\langle v,\mathbf{1}_T\rangle+t_j\operatorname{Var}_{T\sim\mathbb{P}^*_{j,\alpha}}\langle v,\mathbf{1}_T\rangle\Big),$$

since $\partial_c^2\log K_j(e^{y+cv})$ at $c=0$ is the variance of $\langle v,\mathbf{1}_T\rangle$ under $\mathbb{P}_{j,\alpha}(T)=\theta_{j,T}\alpha^T/K_j(\alpha)$ (7.16), and likewise for Ψ_j under $\mathbb{P}^*_{j,\alpha}$, with $\langle v,\mathbf{1}_{\mathcal{E}\setminus T}\rangle=\langle v,\mathbf{1}\rangle-\langle v,\mathbf{1}_T\rangle$. Both laws charge every basis. So if $\langle v,\nabla^2\Phi(y)v\rangle=0$, then $\langle v,\mathbf{1}_T\rangle$ is constant on the bases of every active table, v is constant on every molecule of the active family (7.8), and the active family of a connected family is one molecule (Lemma S3); so $v\in\mathbb{R}\mathbf{1}$. Conversely $\nabla^2\Phi\,\mathbf{1}=0$, $\Phi(y+c\mathbf{1})-\Phi(y)$ being linear in c. So the restriction of $\nabla^2\Phi(y^\star)$ to the chart $\{v_g=0\}$, the matrix $\big(\nabla^2\Phi(y^\star)\big)_{(g)}$, is positive definite; it is the Hessian at $y^\star$ of G on the chart.

The amplitude. For every basis T of table j, $|\langle\mathbf{1}_T,y\rangle|\le r_j|y|$, so $\log\min_T\theta_{j,T}-r_j|y|\le\log K_j(e^y)\le\log K_j(\mathbf{1})+r_j|y|$, and likewise for Ψ_j with $n-r_j$. Hence $|b(y)|=e^{\langle\operatorname{Re}\mathbf{m}_0,y\rangle}\prod_j K_j(e^y)^{-\operatorname{Re}s_{0,j}}\Psi_j(e^y)^{-\operatorname{Re}t_{0,j}}\le e^{c_0(1+|y|)}$ for a constant c_0, and b is continuous and nowhere 0.

Laplace's method. Lemma S7 on the chart, with $d=n-1$, the function $G(y^\star+x)-G(y^\star)$ and the amplitude $b(y^\star+x)$, bounded by $e^{c_0(1+|y^\star|+|x|)}$, gives

$$P_F(\mathbf{d}_\lambda)=e^{-\lambda G(y^\star)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(\nabla^2\Phi(y^\star)\big)_{(g)}^{-1/2}\big(b(y^\star)+o(1)\big),$$

with $b(y^\star)=(\alpha^\star)^{\mathbf{m}_0}W_0(\alpha^\star)\ne0$ and $G(y^\star)=\Phi(\log\alpha^\star)-\langle\boldsymbol{\tau},\log\alpha^\star\rangle$: the display. The function $\alpha\mapsto\alpha^{\mathbf{m}_0}W_0(\alpha)$ is homogeneous of degree $\kappa_{\mathcal{E}}(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$, so $b(y^\star)$ does not depend on the scale of $\alpha^\star$; likewise $G(y+c\mathbf{1})=G(y)$, the data $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ being balanced, and $\nabla^2\Phi$ does not change along $\mathbf{1}$. By 9.4, rung two of the still of $((s_j,t_j)_j;\alpha^\star)$ is $4\nabla^2\Phi(\log\alpha^\star)$.

The normal limit. For real $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ and λ large, $\operatorname{Breta}^\circ(F;\mathbf{d}_\lambda)$ is defined (8.10). For a bounded continuous function H of y modulo $\mathbb{R}\mathbf{1}$, the function $u\mapsto H(\log u)u^{\mathbf{m}-\mathbf{1}}W(u)$ is homogeneous of degree $-n$, so its integral over $\Delta_{\mathcal{E}}$ is its integral over the chart, as in the proof of Lemma S4(ii). So the law of $Y=\log U$ modulo $\mathbb{R}\mathbf{1}$, read on the chart, has the density $b(y)e^{-\lambda G(y)}/P_F(\mathbf{d}_\lambda)$, with $b>0$ continuous. Applying Lemma S7 as above to $H(\sqrt{\lambda}(y-y^\star))$ and to 1, and dividing,

$$\mathbb{E}\big[H\big(\sqrt{\lambda}(Y-y^\star)\big)\big]\to\int H(x)e^{-\frac12\langle x,\nabla^2\Phi(y^\star)_{(g)}x\rangle}dx\Big/\int e^{-\frac12\langle x,\nabla^2\Phi(y^\star)_{(g)}x\rangle}dx,$$

the expectation of H under the centred normal law on the chart with covariance $\big(\nabla^2\Phi(y^\star)_{(g)}\big)^{-1}$: the centred normal law on $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$ with covariance $(\nabla^2\Phi)^{-1}$, the quadratic form of $\nabla^2\Phi(y^\star)$ on $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$ being read on the chart. At $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$, $b=1$ and this is the reference, proof of 18.9. □

Remark. At $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$ let q be the question of the data $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ (17.1), with the exponent $X(z)=2\langle\boldsymbol{\mu}_\perp,z\rangle-\sum_j\big(\sigma_jE_j(z)+\delta_jO_j(z)\big)$, read at the representative of z in $\mathbf{1}^\perp$. By 8.19, $\Phi(y)=\sigma\langle\mathbf{1},y\rangle+\sum_j\big(\sigma_jE_j(z)+\delta_jO_j(z)\big)$ in $y=2z$, and $\langle\boldsymbol{\tau},y\rangle=2\langle\boldsymbol{\mu},z\rangle+\sigma\langle\mathbf{1},y\rangle$, $\boldsymbol{\tau}$ being the masses of the question; so $\langle\boldsymbol{\tau},y\rangle-\Phi(y)=X(z)$ for $z\in\mathbf{1}^\perp$, and both sides are unchanged along $\mathbf{1}$. Hence $e^{-\lambda(\Phi(y^\star)-\langle\boldsymbol{\tau},y^\star\rangle)}=e^{\lambda X(z^\star)}$, $z^\star$ the tilt of $\frac12y^\star$. The Hessian in z is four times the Hessian in y, so on a chart $\det\big(-\nabla^2X(z^\star)\big)=4^{n-1}\det\big(\nabla^2\Phi(y^\star)\big)_{(g)}$, and

$$2^{n-1}e^{\lambda X(z^\star)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(-\nabla^2X(z^\star)\big)^{-\frac12}=e^{-\lambda(\Phi(y^\star)-\langle\boldsymbol{\tau},y^\star\rangle)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(\nabla^2\Phi(y^\star)\big)_{(g)}^{-1/2}:$$

the leading term of 17.5(ii) at real λ is the display of 18.9.

Definition 18.10 (Skeleton law). As stated, at real nonzero letter exponents.

Proof of the last clause. For a real coefficient $\xi_\ell>0$ and $y>0$, $(y+\xi_\ell)^{-w_\ell}>0$. A coefficient that is not real comes with its conjugate and the same real letter exponent w_ℓ; $y+\xi_\ell$ and $y+\bar\xi_\ell$ are conjugate and off $(-\infty,0]$, so their principal powers are conjugate and $(y+\xi_\ell)^{-w_\ell}(y+\bar\xi_\ell)^{-w_\ell}=|y+\xi_\ell|^{-2w_\ell}>0$. So $y^{\kappa-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}$ is positive and continuous on $(0,\infty)$. As $y\to0$ it is asymptotic to $y^{\kappa-1}\prod_\ell\xi_\ell^{-w_\ell}$, the product being positive by the same pairing, and as $y\to\infty$ to $y^{\kappa-1-|\mathbf{w}|}$; so it is integrable on $(0,\infty)$ exactly when $0<\kappa<|\mathbf{w}|$. Its integral is the value of 15.1, which asks no sign of the letter exponents; so the normalizer is that value, positive, and the density is that of a law. □

Theorem 18.11 (The Skeleton law). *As stated: (i) off the balance wall and at exponents of any sign; (ii) at real nonzero letter exponents; (iii) about every point $r\ge\max_\ell\xi_\ell$ of the axis.*

Proof. (i) By 18.3, given the internal shapes (V_B,V_R) the scale ρ of B has a density proportional to $\rho^{m_B-1}(1+\rho)^{-\kappa_\mathcal{E}}\prod_jK_j(\rho V_B,V_R)^{-s_j}\Psi_j(\rho V_B,V_R)^{-t_j}$. At the gauge $\alpha=(V_B,V_R)$, 12.10 gives

$$K_j(\rho\alpha_B,\alpha_R)=\rho^{d_j(B)}K_j(\alpha)\prod_i\frac{\rho+\xi_{j,i}}{1+\xi_{j,i}},\qquad\Psi_j(\rho\alpha_B,\alpha_R)=\rho^{d_j^*(B)}\Psi_j(\alpha)\prod_i\frac{\rho+\xi^*_{j,i}}{1+\xi^*_{j,i}},$$

over the Skeleton values $\xi_{j,i}$ and $\xi^*_{j,i}$ of the two letters. The Skeleton polynomials are real, so their values not real come in conjugate pairs of equal multiplicity; the exponents are real, so the principal powers of conjugate factors are conjugate, and $K_j(\rho\alpha_B,\alpha_R)^{-s_j}=\rho^{-s_jd_j(B)}K_j(\alpha)^{-s_j}\prod_i(\rho+\xi_{j,i})^{-s_j}(1+\xi_{j,i})^{s_j}$, likewise for Ψ_j. The power of ρ collects to $\rho^{m_B-1-\sum_j(s_jd_j(B)+t_jd_j^*(B))}=\rho^{\kappa_B-1}$ (8.1), and the linear factors collect to $\prod_\ell(\rho+\xi_\ell)^{-w_\ell}$ over the distinct points of the joint spectrum of the letters with nonzero exponent, with the letter exponents w_ℓ of 15.3. The factor $(1+\rho)^{-\kappa_\mathcal{E}}$ is the coefficient 1 with the letter exponent $\kappa_\mathcal{E}$, added to the letter exponent of 1 when 1 is a point of the joint spectrum; discarding the coefficients of letter exponent 0 changes nothing. So the conditional density is proportional to $\rho^{\kappa_B-1}\prod_\ell(\rho+\xi'_\ell)^{-w'_\ell}$, with coefficients in $\mathbb{C}\setminus(-\infty,0]$ closed under conjugation and real nonzero letter exponents, equal on conjugate coefficients. Their total is $|\mathbf{w}|+\kappa_\mathcal{E}=\kappa_B+\kappa_B^{\text{top}}+\kappa_\mathcal{E}$ (15.3), which is $\kappa_B+\kappa_{\mathcal{E}\setminus B}$ (8.13); B and $\mathcal{E}\setminus B$ are nonempty proper sets and every proper wall is positive, so $0<\kappa_B<|\mathbf{w}'|$, and the conditional law is $\mathrm{Skel}_{\mathbf{w}',\kappa_B}(\boldsymbol{\xi}')$ (18.10). On the balance wall this is the reference, proof of 18.11(i).

(ii) For $-\kappa<\operatorname{Re}k<|\mathbf{w}|-\kappa$, $\mathbb{E}[Y^k]$ is $\int_0^\infty y^{\kappa+k-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}dy$ over $\int_0^\infty y^{\kappa-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}dy$, both convergent, the values of 15.1 at the indices $\kappa+k$ and κ; for positive letter exponents the factors $\Gamma(w_\ell)$ of the two stacked Gammas cancel, which is the display. For the inversion put $y=1/z$: $dy=z^{-2}dz$, and $y+\xi=\xi z^{-1}(z+\xi^{-1})$ with $\operatorname{Log}(y+\xi)=\operatorname{Log}\xi-\log z+\operatorname{Log}(z+\xi^{-1})$, the

principal arguments of ξ and of $z+\xi^{-1}$ having opposite signs and sum in $(-\pi,\pi)$. So the density of $1/Y$ is proportional to $z^{1-\kappa}z^{-2}z^{|\mathbf{w}|}\prod_\ell(z+\xi_\ell^{-1})^{-w_\ell} = z^{|\mathbf{w}|-\kappa-1}\prod_\ell(z+\xi_\ell^{-1})^{-w_\ell}$, the constant $\prod_\ell \xi_\ell^{-w_\ell}$ being positive; with $0<|\mathbf{w}|-\kappa<|\mathbf{w}|$ this is $\mathrm{Skel}_{\mathbf{w},|\mathbf{w}|-\kappa}(\boldsymbol{\xi}^{-1})$. This is the reference, proof of 18.11(ii), with 15.7, which uses no sign of the letter exponents.

(iii) The coefficients are real, hence positive, and $r\ge\max_\ell\xi_\ell$, so $|\xi_\ell-r|<r$ and $0\le g_\ell<1$. By the expansion in the proof of 15.6,

$$y^{\kappa-1}\prod_\ell(y+\xi_\ell)^{-w_\ell} = \sum_{\mathbf{n}\in\mathbb{N}^J}\prod_\ell \frac{(w_\ell)_{n_\ell}}{n_\ell!}(r-\xi_\ell)^{n_\ell}\,y^{\kappa-1}(y+r)^{-|\mathbf{w}|-|\mathbf{n}|},$$

with nonnegative terms, the letter exponents being positive. For $a>\kappa>0$, the law $\mathrm{Skel}_{a,\kappa}(1)$ has the density $y^{\kappa-1}(1+y)^{-a}/\Gamma_1^2\left[\begin{matrix}\kappa,\ a-\kappa\\ a\end{matrix}\right]$, the Breta of one coefficient 1 being 1 (the paragraph after 15.1); so rY_a, $Y_a\sim\mathrm{Skel}_{a,\kappa}(1)$, has the density $r^{a-\kappa}y^{\kappa-1}(y+r)^{-a}/\Gamma_1^2\left[\begin{matrix}\kappa,\ a-\kappa\\ a\end{matrix}\right]$. With $a=|\mathbf{w}|+|\mathbf{n}|$ and $(r-\xi_\ell)^{n_\ell}=r^{n_\ell}g_\ell^{n_\ell}$, the density of Y is the sum over $\mathbf{n}$ of $r^{\kappa-|\mathbf{w}|}\prod_\ell\frac{(w_\ell)_{n_\ell}}{n_\ell!}g_\ell^{n_\ell}\Gamma_1^2\left[\begin{matrix}\kappa,\ |\mathbf{w}|+|\mathbf{n}|-\kappa\\ |\mathbf{w}|+|\mathbf{n}|\end{matrix}\right]$ times the density of $rY_{\mathbf{n}}$, divided by the normalizer; the coefficients are nonnegative and, by Tonelli, sum to 1. So Y is the stated mixture. At $r=\max_\ell\xi_\ell$ the largest coefficient has $g_\ell=0$, and only $n_\ell=0$ contributes.

(iv) On the balance wall with $s_j,t_j\ge0$, given the shapes the density of the scale ρ is proportional to the integrand of the scale integral at $\alpha=(V_B,V_R)$, by (i). By the proof of 15.9(i) it is, on $(0,1)$, proportional to $\rho^{\kappa_B-1}\prod_i\mathsf{Q}_{B,i}(\rho;\alpha)^{-w_i}$, and the density of $1/\rho$ on $(0,1)$ is proportional to $y^{\kappa_B^{\mathrm{top}}-1}\prod_i\mathsf{Q}_{R,i}(y;\alpha)^{-w_i}$. The expansions of 15.9(ii) have nonnegative terms, each the term of $\mathbf{n}$ and k times $y^{\kappa+k-1}(1-y)^{|\mathbf{n}|}/\Gamma_1^2\left[\begin{matrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{matrix}\right]$, the density of the first coordinate of $\mathrm{Dir}_{(\kappa+k,|\mathbf{n}|+1)}$ (3.9); by Tonelli the conditional laws are the stated mixtures, and the two sectors carry probabilities proportional to their Trailmoments. This is the reference, proof of 18.11(v). □

Definition 18.12 (Wall contour; joint period). As stated, at complex data on the balance wall with $\mathrm{Re}\,s_j>0$ or $s_j=0$, and $\mathrm{Re}\,t_j>0$ or $t_j=0$.

Remark. A wall contour is fixed by real parts alone, and the fibre integral at B has complex pins, whose admissible real points (4.1) are the wall contours. The two descriptions agree: $\mathrm{load}_e=\sigma+\langle\boldsymbol{\delta}\rangle_e$ on the share sums and $m_e=\sigma+\mu_e$, so $\mathrm{Re}\,\mathrm{load}_e=\mathrm{Re}\,m_e$ exactly when $\mathrm{Re}\langle\boldsymbol{\delta}\rangle_e=\mathrm{Re}\,\mu_e$. For real data with $s_j,t_j\ge0$ these are the conditions of the reference.

Theorem 18.13 (The port on the balance wall). *As stated, at complex data on the balance wall with* $\mathrm{Re}\,s_j>0$ *or* $s_j=0$, *and* $\mathrm{Re}\,t_j>0$ *or* $t_j=0$.

Proof. A letter with exponent 0 is 1 and is not opened; the shares, the stacked Gammas and the products below run over the letters with nonzero exponent, whose exponents have positive real part.

The real parts. The walls and the imbalance are affine in the data with real coefficients. So the data $(\mathrm{Re}\,\mathbf{m};\mathrm{Re}\,\mathbf{s},\mathrm{Re}\,\mathbf{t})$ lie on the balance wall, and their proper walls are the real parts of those of $(\mathbf{m};\mathbf{s},\mathbf{t})$, positive since $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converges (8.7). Their exponents are nonnegative and not all zero: otherwise $\kappa_S=\mathrm{Re}\,m_S$ for every S, and $0=\kappa_{\mathcal{E}}=\kappa_B+\kappa_R>0$. Their masses are positive: $\mathrm{Re}\,m_e=\kappa_{\{e\}}+\sum_j\big(\mathrm{Re}\,s_j\,d_j(\{e\})+\mathrm{Re}\,t_j\,d_j^*(\{e\})\big)>0$, $\{e\}$ being a nonempty proper set ($n\ge2$, as B is proper). A nonempty proper S with $c_j(S)=0$ in every active table would have

$\kappa_S+\kappa_{\mathcal{E}\setminus S}=\kappa_{\mathcal{E}}=0$ at the real parts (8.16), against the positivity of both walls; so F is connected, and its active family, whose tables are those of the real parts, is one molecule (Lemma S3). By 8.19(iii) the real parts have a native gauge.

(i), convergence and the room integral. By 7.15(i), $K_j=e^{\langle\mathbf{1},z\rangle}\Lambda_j(z)$ and $\Psi_j=e^{\langle\mathbf{1},z\rangle}\Lambda_j(-z)$ at $\alpha=e^{2z}$, so with $\sigma=\sum_j\sigma_j$ the integrand of P_F is $f(\alpha)=\alpha^{\boldsymbol{\mu}-\mathbf{1}}\prod_j\Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j}$, homogeneous of degree $\kappa_{\mathcal{E}}-n=-n$ (7.15(iii), 8.2(iii)), and $|f|$ is the same function at the real parts of the data. At the real parts, 8.7 on the section $|\alpha_R|=1$, whose measure is $d\alpha_B\,d\omega_R$ (proof of 16.6 above), gives $\int_{\Delta_R}\int_{\mathbb{R}^B_{>0}}|f(\alpha_B,v)|\,d\alpha_B\,d\omega_R(v)<\infty$; and $f(\alpha_B,v)$ is $v^{\boldsymbol{\mu}|_R-\mathbf{1}}$ times the integrand of $\Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})$. By Tonelli that integral converges absolutely at almost every $v\in\Delta_R$. For two room points α_R,α'_R there is $C\geq1$ with $\Lambda_{j,(p)}(\pm z_R)\leq C\Lambda_{j,(p)}(\pm z'_R)$ for all j,p, these being finite sums of positive terms; by 12.7, $\Lambda_j(z)=\sum_p e^{\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(z_R)$, so $\Lambda_j(z_B,z_R)\geq C^{-1}\Lambda_j(z_B,z'_R)$, likewise at $-z$, and the modulus of the integrand of Ξ_B at α_R is at most $C^{\sum_j(\operatorname{Re}s_j+\operatorname{Re}t_j)}$ times that at α'_R. So Ξ_B converges at every α_R, and Fubini gives $P_F=\int_{\Delta_R}v^{\boldsymbol{\mu}|_R-\mathbf{1}}\Xi_B(v)\,d\omega_R(v)$.

Independence of the pins. Let $R_1,\dots,R_q$ be the molecules of the children (for $q=1$ there are no pins κ_{R_i} below), and write $\kappa_{R_i}=m_{R_i}-\ell_i$ with $\ell_i(\mathbf{a},\mathbf{b})=\sum_{j,p}\big(a_{j,p}r_{j,(p)}(R_i)+b_{j,p}(|R_i|-r_{j,(p)}(R_i))\big)$. The linear forms $|\mathbf{a}_j|$ and $|\mathbf{b}_j|$ of the letters, $\langle\boldsymbol{\delta}\rangle_e$ for $e\in B$, and ℓ_i for $i<q$, are linearly independent. Indeed, let $\sum_{e\in B}\gamma_e\langle\boldsymbol{\delta}\rangle_e+\sum_{i<q}w_i\ell_i=\sum_j(u_j|\mathbf{a}_j|+u'_j|\mathbf{b}_j|)$, and put $V\in\mathbb{R}^{\mathcal{E}}$ equal to γ on B, to w_i on R_i for $i<q$ and to 0 on R_q. For a basis T of table j with $T\cap B=p$, $T\setminus B$ is a basis of the child (j,p), which meets each R_i in $r_{j,(p)}(R_i)$ marks, R_i being a union of molecules of that child (7.9); and $\frac12\langle\gamma,\chi_p\rangle=\langle\gamma,\mathbf{1}_p\rangle-\frac12\langle\gamma,\mathbf{1}_B\rangle$. Comparing the coefficients of $a_{j,p}$ gives $\langle V,\mathbf{1}_T\rangle=u_j+\frac12\langle\gamma,\mathbf{1}_B\rangle$, and comparing those of $b_{j,p}$ gives $\langle V,\mathbf{1}_T\rangle=\langle V,\mathbf{1}_R\rangle+\frac12\langle\gamma,\mathbf{1}_B\rangle-u'_j$, for every basis T of table j whenever that letter is opened. So $\langle V,\mathbf{1}_T\rangle$ is constant on the bases of every active table; by 7.8 V is constant on the one molecule $\mathcal{E}$ of the active family, and as it vanishes on $R_q\neq\varnothing$, $V=0$: $\gamma=0$ and every $w_i=0$.

(i), the fibre formula. Fix α_R and a wall contour. Put $X_{j,p}=e^{\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(z_R)$ and $Y_{j,p}=e^{-\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(-z_R)$, positive, so that $\Lambda_j(z)=\sum_p X_{j,p}$ and $\Lambda_j(-z)=\sum_p Y_{j,p}$. For a letter with more than one pattern, 4.4 on the contour $\operatorname{Re}a_{j,p}=c_{j,p}$, admissible since every $c_{j,p}>0$ and $\sum_p c_{j,p}=\operatorname{Re}s_j$, gives $\Lambda_j(z)^{-s_j}=\int_{|\mathbf{a}_j|=s_j}\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\prod_p X_{j,p}^{-a_{j,p}}$; for one pattern this is the evaluation $a_{j,p}=s_j$. Likewise for $\Lambda_j(-z)^{-t_j}$ with the shares $\mathbf{b}_j$. Since $\prod_{j,p}e^{-\langle z_B,\chi_p\rangle(a_{j,p}-b_{j,p})}=e^{-2\langle z_B,\langle\boldsymbol{\delta}\rangle\rangle}=\alpha_B^{-\langle\boldsymbol{\delta}\rangle}$ (12.24), in $y=\log\alpha_B$, where $\alpha_B^{\boldsymbol{\mu}|_B-\mathbf{1}}d\alpha_B=e^{\langle\boldsymbol{\mu}|_B,y\rangle}dy$,

$$\Xi_B(\alpha_R)=\int_{\mathbb{R}^B}\Big(\int_{\mathcal{X}_0}\mathcal{G}(\mathbf{a},\mathbf{b})\,e^{\langle\boldsymbol{\mu}|_B-\langle\boldsymbol{\delta}\rangle,y\rangle}\Big)dy,$$

$$\mathcal{G}=\prod_j\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\prod_{j,p}\Lambda_{j,(p)}(z_R)^{-a_{j,p}}\Lambda_{j,(p)}(-z_R)^{-b_{j,p}},$$

the inner integral the product of the share integrals, over the set $\mathcal{X}_0$ of shares on the contour with the share sums pinned. On a wall contour $\operatorname{Re}\langle\boldsymbol{\delta}\rangle_e=\operatorname{Re}\mu_e$ for $e\in B$, so $e^{\langle\boldsymbol{\mu}|_B-\langle\boldsymbol{\delta}\rangle,y\rangle}=e^{i\langle\operatorname{Im}\boldsymbol{\mu}|_B-\operatorname{Im}\langle\boldsymbol{\delta}\rangle,y\rangle}$. By Stirling's formula $|\Gamma(c+i\eta)|\leq C_c(1+|\eta|)^{c-1/2}e^{-\pi|\eta|/2}$, so $|\mathcal{G}|$, whose factors $\Lambda_{j,(p)}^{-a_{j,p}}$ have moduli constant on the contour, decays exponentially along every imaginary direction of $\mathcal{X}_0$, uniformly on parallel affine subspaces through a compact set. The map $(\mathbf{a},\mathbf{b})\mapsto\operatorname{Im}\langle\boldsymbol{\delta}\rangle\in\mathbb{R}^B$ is affine on $\mathcal{X}_0$, with linear part onto $\mathbb{R}^B$ by the independence of the pins; so the image of $\mathcal{G}$ times the measure of the share integrals has a density g on $\mathbb{R}^B$, continuous and integrable, and the inner integral is $e^{i\langle\operatorname{Im}\boldsymbol{\mu}|_B,y\rangle}\hat g(y)$, $\hat g(y)=\int g(\nu)e^{-i\langle\nu,y\rangle}d\nu$. The modulus $|\hat g(y)|$ is the modulus of the integrand of Ξ_B in y, integrable by the first step. By Fourier inversion, $\Xi_B(\alpha_R)=\int\hat g(y)e^{i\langle\operatorname{Im}\boldsymbol{\mu}|_B,y\rangle}dy=$

$(2\pi)^{|B|}g(\operatorname{Im}\boldsymbol{\mu}|_B)$, which is the fibre integral of $\mathcal{G}$ over the fibre $\mathcal{X}_1$ cut by the pins $|\mathbf{a}_j| = s_j$, $|\mathbf{b}_j| = t_j$ and $\langle\boldsymbol{\delta}\rangle = \boldsymbol{\mu}|_B$ (4.1: the $|B|$ further pins raise $(2\pi)^{q-N}$ by $(2\pi)^{|B|}$ and add the factor $\delta^{|B|}(\operatorname{Im}\langle\boldsymbol{\delta}\rangle - \operatorname{Im}\boldsymbol{\mu}|_B)$), with the wall contour as its admissible real point. This is the display of (i). For real data this is the reference, proof of 18.13(i), where $\operatorname{Im}\boldsymbol{\mu}|_B = 0$.

(ii). On the pins of (i), $\operatorname{load}_e = m_e$ for $e \in B$, and by 12.2 the child (j,p) has rank $r_j - |p|$; so

$$\sum_{i=1}^{q}\kappa_{R_i} = m_R - \sum_{j,p}\big(a_{j,p}(r_j-|p|)+b_{j,p}(|R|-r_j+|p|)\big) = m_R - \sum_j\big(s_jr_j+t_jr_j^*\big)+\sum_{e\in B}\operatorname{load}_e = \kappa_{\mathcal{E}} = 0,$$

using $\sum_{e\in B}\operatorname{load}_e = \sum_{j,p}\big(a_{j,p}|p|+b_{j,p}(|B|-|p|)\big)$; so $\kappa_{R_q} = 0$ holds on $\mathcal{X}$. Take the section $\{|\alpha_{R_q}| = 1\}$, with the measure $d\alpha_B\prod_{i<q}d\alpha_{R_i}\,d\omega_{R_q}$, and as in (i) $P_F = \int \alpha_R^{\boldsymbol{\mu}|_R-\mathbf{1}}\,\Xi_B(\alpha_R)\prod_{i<q}d\alpha_{R_i}\,d\omega_{R_q}$, absolutely convergent. Write $\alpha_{R_i} = e^{y_i}u_i$, $u_i \in \Delta_{R_i}$, for $i<q$, and $u_q = \alpha_{R_q}$; then $\alpha_{R_i}^{\boldsymbol{\mu}|_{R_i}-\mathbf{1}}d\alpha_{R_i} = e^{y_i\mu_{R_i}}u_i^{\boldsymbol{\mu}|_{R_i}-\mathbf{1}}dy_i\,d\omega_{R_i}$. Each R_i is a union of molecules of every child, so every basis X of the child (j,p) has $\langle\chi_X,\mathbf{1}_{R_i}\rangle = 2r_{j,(p)}(R_i) - |R_i|$, and $\Lambda_{j,(p)}(z_R + \frac12 y_i\mathbf{1}_{R_i}) = e^{\frac12 y_i(2r_{j,(p)}(R_i)-|R_i|)}\Lambda_{j,(p)}(z_R)$. Inserting the fibre formula of (i) and using $\sum_{j,p}\sigma_{j,p} = \sigma$, the exponent of e^{y_i} collects to $\mu_{R_i} - \sum_{j,p}\delta_{j,p}\big(2r_{j,(p)}(R_i) - |R_i|\big) = \kappa_{R_i}(\mathbf{a},\mathbf{b})$, and the rest is $\mathcal{G}$ at u times $u^{\boldsymbol{\mu}|_R-\mathbf{1}}$, that is, by 7.15 and $\sum_{j,p}\sigma_{j,p} = \sigma$, the stacked Gammas times the integrand of the joint period of the children at the masses $\mathbf{m}|_R$. For fixed $\mathbf{y} = (y_i)_{i<q}$, on a wall contour with $\operatorname{Re}\kappa_{R_i} = 0$ for $i<q$, the modulus of this integrand over $u \in \prod_i\Delta_{R_i}$ and the shares is the modulus of the stacked Gammas times the joint integrand at the real parts $(\operatorname{Re}\mathbf{m}|_R;\mathbf{c},\mathbf{c}')$, integrable by the hypothesis on $P^{\otimes}$ and by Stirling; so by Fubini the integral over u and the shares is

$$\hat g_1(\mathbf{y}) = \int_{\mathcal{X}_1}\prod_j \Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\;P^{\otimes}_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})\,e^{i\sum_{i<q}y_i\operatorname{Im}\kappa_{R_i}(\mathbf{a},\mathbf{b})},$$

$\mathcal{X}_1$ the fibre of (i), and $\int|\hat g_1(\mathbf{y})|\,d\mathbf{y}$ is at most P_F at the real parts, finite. The joint period is continuous in the shares and bounded by its value at the real parts, by dominated convergence. The map $(\mathbf{a},\mathbf{b}) \mapsto (\operatorname{Im}\kappa_{R_i})_{i<q}$ is affine on $\mathcal{X}_1$ with linear part onto $\mathbb{R}^{q-1}$, by the independence of the pins. So, as in (i), Fourier inversion in $\mathbf{y}$ gives $P_F = \int\hat g_1(\mathbf{y})\,d\mathbf{y}$ as the fibre integral over $\mathcal{X}$, the pins $\kappa_{R_i} = 0$, $i<q$, holding in real part on the contour and in imaginary part by inversion. For $q=1$ there is no $\mathbf{y}$, and $P_F = \hat g_1$ by Fubini.

(iii). The real parts have a native gauge (first step). For real data the statement is the reference, proof of 18.13(iii): the native split is a wall contour, lies on the real fibre, and $P^{\otimes}$ converges there. A wall contour, and the real parts of the pins of $\mathcal{X}$, $\operatorname{Re}|\mathbf{a}_j| = \operatorname{Re}s_j$, $\operatorname{Re}|\mathbf{b}_j| = \operatorname{Re}t_j$, $\operatorname{Re}\langle\boldsymbol{\delta}\rangle = \operatorname{Re}\boldsymbol{\mu}|_B$ and $\operatorname{Re}\kappa_{R_i} = \kappa_{R_i}(\operatorname{Re}\mathbf{m}|_R;\mathbf{c},\mathbf{c}')$, involve the data only through their real parts. So the native split of the real parts is a wall contour for the data, an admissible real point of $\mathcal{X}$, and $P^{\otimes}_{F_B(\mathbf{c},\mathbf{c}')}(\operatorname{Re}\mathbf{m}|_R;\mathbf{c},\mathbf{c}')$ converges there, as (ii) asks. □

Proposition 18.17 (The pins reach every integer point). *As stated.*

Proof. The pins are a linear map $\mathbb{Z}^{\text{shares}} \to \mathbb{Z}^{\text{rows}}$, the rows being the share sums of the active letters and the loads at the marks $e \neq g$; a share $a_{j,T}$ has 1 in the row of its sum and in the loads of the marks of $T-g$, and a share $b_{j,T}$ has 1 in the row of its sum and in the loads of the marks of $\mathcal{E}\setminus T$ other than g. Write $\mathbf{e}_x$ for the unit vector of the load of $x \neq g$ and $\mathbf{e}_g = 0$. If $x \neq y$ lie in a common circuit of table j, 7.7 gives a basis $T \ni x$, $T \not\ni y$ with $T' = (T\setminus\{x\})\cup\{y\}$ a basis; if K_j is active the difference of the columns of $a_{j,T}$ and $a_{j,T'}$ is $\mathbf{e}_x - \mathbf{e}_y$, and if Ψ_j is active that of $b_{j,T}$ and $b_{j,T'}$ is $\mathbf{e}_y - \mathbf{e}_x$. Along the sequences of the hypothesis every $\mathbf{e}_x - \mathbf{e}_y$ lies in the image,

and with $y = g$ every $\mathbf{e}_x$, $x \neq g$. Subtracting from the column of a share the loads it carries leaves the unit vector of its share sum. So the image is all of $\mathbb{Z}^{\text{rows}}$.

At integer data the target point is an integer point, so it has an integer preimage x_0, and the integer points of the fibre are $x_0 + \Lambda$ with Λ the kernel of the map on $\mathbb{Z}^{\text{shares}}$. The map is onto a free module of the rank of the number of rows, so the rows are independent and Λ, a subgroup of $\mathbb{Z}^{\text{shares}}$, is free of rank the number of shares less the number of rows, the dimension of the fibre. With a basis $v_1, \dots, v_D$ of Λ the point $x_0 + \sum_i z_i v_i$ is an integer point exactly when $z \in \mathbb{Z}^D$. □

48.19 Chapter 19: bunches of width one: pooling and cleaning

Throughout this section, for a bunch B of width one of table j whose weights factor across B and R, $\theta_{j,T} = \theta'_{j,T\cap B}\,\theta''_{j,T\cap R}$, the minors $\mathcal{B}_j|B$ and $\mathcal{B}_j/R$ on B carry θ'_j and the minors $\mathcal{B}_j|R$ and $\mathcal{B}_j/B$ on R carry θ''_j (19.2), and every integral of a positive function is read in $[0,\infty]$ (Tonelli).

Proposition 19.3 (Hinge values of parallel and series bunches). *As stated, for classes of any table whose weights factor across them.*

Proof. That a parallel or series bunch has width at most one, and width zero exactly when it is a union of molecules, is the reference, proof of prop:classreadings. Let B be a parallel bunch of width one: $r(B) = 1$ and $c(B) = r(B) - d(B) = 1$, so $d(B) = 0$; $\mathcal{B}/R$ has rank 0, with the single basis $\varnothing$ and the weight $\theta'_\varnothing$, and $\mathcal{B}|B$ has rank one, with the singletons of B as bases, none of its marks being a loop, and the weights $\theta'_{\{e\}} = \theta'_\varnothing w_e$. Then $K_{\mathcal{B}/R} = \theta'_\varnothing$, $K_{\mathcal{B}|B} = \sum_e \theta'_{\{e\}}\alpha_e = \theta'_\varnothing \sum_e w_e\alpha_e$, $\Psi_{\mathcal{B}/R} = \theta'_\varnothing \alpha^B = \theta'_\varnothing e_{|B|}(\alpha_B)$ and $\Psi_{\mathcal{B}|B} = \sum_e \theta'_{\{e\}}\alpha^{B-e} = \theta'_\varnothing e_{|B|}(\alpha_B)\sum_e w_e\alpha_e^{-1}$, and the hinge values are the quotients of 19.2. Let B be a series bunch of width one: no mark of B is a coloop, so $d(B) \geq |B|-1$, and width one gives $r(B) = |B|$ and $d(B) = |B|-1$; $\mathcal{B}|B$ has the single basis B, with the weight θ'_B, and $\mathcal{B}/R$ the sets $B-e$, every one of them occurring since a basis avoids e, with the weights $\theta'_{B-e} = \theta'_B w_e$. Then $K_{\mathcal{B}|B} = \theta'_B e_{|B|}(\alpha_B)$, $K_{\mathcal{B}/R} = \theta'_B \sum_e w_e\alpha^{B-e} = \theta'_B e_{|B|}(\alpha_B)\sum_e w_e\alpha_e^{-1}$, $\Psi_{\mathcal{B}|B} = \theta'_B$ and $\Psi_{\mathcal{B}/R} = \theta'_B \sum_e w_e\alpha_e$, which give γ_B^Ψ and γ_B^K. In a list with $\theta' = \theta'' = 1$ every w_e is 1, and for a pair with the shape $(x, 1-x)$: parallel, $\gamma_B^K = 1$ and $\gamma_B^\Psi = (x^{-1}+(1-x)^{-1})^{-1} = x(1-x)$; series, $\gamma_B^\Psi = 1$ and $\gamma_B^K = x(1-x)$. □

Remark. A change of the factorization (19.2) multiplies the factors w_e of a parallel bunch by c, the weights of its level $r(B) = 1$ over those of its level $d(B) = 0$, and those of a series bunch by c^{-1}, its patterns $B - e$ lying on the level $d(B)$; either way γ_B^K is multiplied by c and γ_B^Ψ by c^{-1}. In exact arithmetic (receipt `check_classes_realization.py`): a parallel bunch of three marks whose cell of the bunch is $\binom{\{1,2,3,g\}}{1}$ with the weights $2, \frac{3}{7}, 5$ on the singletons of the bunch and $\frac{4}{3}$ on $\{g\}$, and a series bunch of three marks whose cell is $\binom{\{1,2,3,g\}}{3}$ with the weights $\frac{3}{2}$ on $\{1,2,3\}$ and $\frac{1}{5}, 7, \frac{2}{3}$ on the sets through g, each glued to a weighted room cell, have the hinge values of the statement as rational functions, and $\varrho_B = (\sum_e w_e)^{-2}$ at the centre of the shape simplex of the parallel bunch.

Theorem 19.4 (A bunch of width one is one mark to its room). *As stated; the converse holds with any number of products.*

Proof. All but the converse is the reference, proof of thm:widthone. On the open orthant let $K(\alpha) = \sum_{l=0}^{k} A_l(\alpha_B)C_l(\alpha_R)$, C_l homogeneous of degree D_l. Grouping the bases by their level $i = |T \cap B|$, which takes every value $d(B) \leq i \leq r(B)$ (7.13), $K = \sum_i K^{(i)}$ with $K^{(i)}(\alpha_B, \lambda\alpha_R) =$

$\lambda^{r-i}K^{(i)}(\alpha)$ and $K^{(i)} > 0$ on the open orthant. At a fixed α, with $\lambda\alpha_R$ for α_R,

$$\sum_{i=d(B)}^{r(B)} \lambda^{r-i}K^{(i)}(\alpha) = \sum_D \lambda^D \sum_{l:\ D_l=D} A_l(\alpha_B)\, C_l(\alpha_R) \qquad (\lambda > 0),$$

the sum on the right over the distinct values D among the D_l. The functions $\lambda \mapsto \lambda^D$ with distinct real D are linearly independent on $(0,\infty)$, so each of the $c(B)+1$ exponents $r-i$ on the left, whose coefficient is positive, is one of the D_l: $c(B)+1 \le k+1$. With $k=1$ this is the converse of the reference. □

Lemma 19.6 (One hinge value of any degree is width at most one). *As stated.*

Proof. Write $d = d(B)$, $c = c(B)$ and $K^{(k)} = \sum_{p\in \mathrm{pat}^{(k)}(B)} \alpha_B^p K_{(p)}(\alpha_R)$ for the level letters (12.16), nonzero exactly for $d \le k \le d+c$ (7.13); each child $K_{(p)}$ has positive coefficients, the weights. Let a be the degree of A and $P(\alpha_R, y) = \sum_{i=0}^D P_i(\alpha_R) y^i$. For $\tau > 0$,

$$\sum_k \tau^k K^{(k)}(\alpha) = K(\tau\alpha_B, \alpha_R) = \sum_{i=0}^D \tau^{a+i} A(\alpha_B)\gamma(\alpha_B)^i P_i(\alpha_R),$$

and at a fixed α the functions $\tau \mapsto \tau^x$ with distinct real x are linearly independent on $(0,\infty)$. So a is an integer, $K^{(d)}$ being nonzero, and $K^{(k)} = A\gamma^{k-a}P_{k-a}$ for every k, the right side read as 0 when $k-a \notin \{0,\dots,D\}$.

Let $d \le k \le d+c$. Then $P_{k-a} \not\equiv 0$; fixing α_R where $P_{k-a}(\alpha_R) \ne 0$ shows that $A\gamma^{k-a}$ is a polynomial $\widehat{f_k}$ in α_B, so $K^{(k)}(\alpha) = \widehat{f_k}(\alpha_B)P_{k-a}(\alpha_R)$ as polynomials. The coefficient of $\alpha_B^p\alpha_R^X$ in $K^{(k)}$ is the weight of $p \sqcup X$, distinct patterns giving distinct monomials α_B^p, so the matrix of these coefficients, rows indexed by the patterns of the level and columns by the monomials of the room, has rank one: the children at the patterns of the level are positive multiples of one polynomial C_k, $K_{(p)} = \lambda_p C_k$ with $\lambda_p > 0$, and $K^{(k)} = f_k(\alpha_B)C_k(\alpha_R)$ with $f_k = \sum_{p\in\mathrm{pat}^{(k)}(B)} \lambda_p \alpha_B^p$. Comparing with $A\gamma^{k-a}P_{k-a}$ at a fixed α_R where C_k and P_{k-a} do not vanish, $f_k = \mu_k A\gamma^{k-a}$ with a constant $\mu_k > 0$.

Suppose $c \ge 2$. Then $f_{d+1}^2 = \mu\, f_d f_{d+2}$ on the open orthant with $\mu = \mu_{d+1}^2/(\mu_d\mu_{d+2}) > 0$, an identity of polynomials. For a pattern p of $d+1$ marks the monomial $\alpha_B^{\mathbf{2}1_p}$ has the coefficient $\lambda_p^2 > 0$ in f_{d+1}^2, since a product $\alpha_B^{p'}\alpha_B^{p''}$ of two monomials of sets equals it only for $p' = p'' = p$; in $f_d f_{d+2}$ it would need two sets q, q' with $q = q' = p$, against $|q| = d \ne d+2 = |q'|$. So $c \le 1$.

Conversely, with the weights factoring across B and R: at $c = 0$ the one level consists of the unions of a pattern and a room side, all combinations occurring (12.19), so $K = K_{\mathcal{B}|B}(\alpha_B)K_{\mathcal{B}|R}(\alpha_R)$ with the factored weights; at $c = 1$ this is 19.4. For Ψ, apply the statement to the dual table (7.2), in which the window of B is read on the cobases and $c(B)$ is unchanged (7.12, 7.13). □

Proposition 19.7 (The freedom of the factorization). *As stated.*

Proof. By 19.4 the bases of the level $d(B)$ are the sets $X \sqcup Y$ with $X \in \mathcal{B}/R$ and $Y \in \mathcal{B}|R$, and those of the level $r(B)$ the sets $X \sqcup Y$ with $X \in \mathcal{B}|B$ and $Y \in \mathcal{B}/B$, every such set being a basis. Let $\theta_{X\sqcup Y} = \theta'_X\theta''_Y = \phi'_X\phi''_Y$ be two factorizations and fix a basis $X_d \sqcup Y_d$ of the level $d(B)$. For every pattern X of this level $X \sqcup Y_d$ is a basis, so $\phi'_X = \lambda_d\theta'_X$ with $\lambda_d = \theta''_{Y_d}/\phi''_{Y_d}$; for every room side Y of this level $X_d \sqcup Y$ is a basis, so $\phi''_Y = \theta'_{X_d}\theta''_Y/\phi'_{X_d} = \lambda_d^{-1}\theta''_Y$. Likewise at the level $r(B)$ with a constant λ_r. Conversely any $\lambda_d, \lambda_r > 0$ give a factorization, the patterns of the two levels

having different sizes, and so have their room sides. The new factors multiply $K_{\mathcal{B}/R}$ and $\Psi_{\mathcal{B}/R}$ by λ_d, $K_{\mathcal{B}|B}$ and $\Psi_{\mathcal{B}|B}$ by λ_r, $K_{\mathcal{B}|R}$ and $\Psi_{\mathcal{B}|R}$ by λ_d^{-1}, and $K_{\mathcal{B}/B}$ and $\Psi_{\mathcal{B}/B}$ by λ_r^{-1}. With $c = \lambda_r/\lambda_d$,

$$\gamma_B^K = \frac{K_{\mathcal{B}|B}}{K_{\mathcal{B}/R}} \mapsto c\,\gamma_B^K, \quad \gamma_B^\Psi = \frac{\Psi_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|B}} \mapsto c^{-1}\gamma_B^\Psi, \quad \gamma_R^K = \frac{K_{\mathcal{B}|R}}{K_{\mathcal{B}/B}} \mapsto c\,\gamma_R^K, \quad \gamma_R^\Psi = \frac{\Psi_{\mathcal{B}/B}}{\Psi_{\mathcal{B}|R}} \mapsto c^{-1}\gamma_R^\Psi,$$

so $\varrho_B = \gamma_B^\Psi/\gamma_B^K$ and $\varrho_R = \mathbf{E}_{\mathcal{B}/B}/\mathbf{E}_{\mathcal{B}|R} = \gamma_R^\Psi/\gamma_R^K$ are multiplied by c^{-2}, and γ_R^K/γ_B^K, $\gamma_R^\Psi/\gamma_B^\Psi$ and ϱ_B/ϱ_R are unchanged. □

Remark. The quotients γ_R^K/γ_B^K and $\gamma_R^\Psi/\gamma_B^\Psi$ are the Skeleton values of 19.5, which belong to the table and its gauge alone. What depends on the factorization is the split of a Skeleton value into a hinge value of the bunch and one of the room, and with it the cells, the cell families and the pooled families; the identities of the chapter hold at every factorization, each side changing with it.

Two lemmas are used below.

Lemma S8. *Let a bunch B of a table $(\mathcal{B}, \theta)$ of rank r on $\mathcal{E}$, $n = |\mathcal{E}|$, have width one, the weights factoring across B and R. With $\alpha_B = \rho u$,*

$$\rho^{-d(B)} K(\rho u, \alpha_R) = K_{\mathcal{B}/R}(u)\, K_{\mathcal{B}|R}(\alpha_R) + \rho\, K_{\mathcal{B}|B}(u)\, K_{\mathcal{B}/B}(\alpha_R),$$
$$\rho^{-d^*(B)} \Psi(\rho u, \alpha_R) = \Psi_{\mathcal{B}|B}(u)\, \Psi_{\mathcal{B}/B}(\alpha_R) + \rho\, \Psi_{\mathcal{B}/R}(u)\, \Psi_{\mathcal{B}|R}(\alpha_R);$$

the cells have the ranks $r(B) = d(B) + 1$ and $r - d(B)$ and the letters

$$K_{\mathcal{B}_B^+} = K_{\mathcal{B}/R}\,(\alpha_g + \gamma_B^K), \quad \Psi_{\mathcal{B}_B^+} = \Psi_{\mathcal{B}|B}\,(\alpha_g + \gamma_B^\Psi),$$
$$K_{\mathcal{B}_R^+} = K_{\mathcal{B}|R} + \alpha_g K_{\mathcal{B}/B}, \quad \Psi_{\mathcal{B}_R^+} = \alpha_g \Psi_{\mathcal{B}|R} + \Psi_{\mathcal{B}/B};$$

and $d(B) + d(R) = r - 1$, $d^*(B) + d^*(R) = n - r - 1$. For a family with masses $\mathbf{m}$ in which B has width one in every table, $\kappa_B + \kappa_R = \kappa_\mathcal{E} + 2\sigma$, σ the total strength.

Proof. By 19.4 the bases of the level $d(B)$ are the $X \sqcup Y$ with $X \in \mathcal{B}/R$, $Y \in \mathcal{B}|R$, and those of the level $r(B) = d(B) + 1$ the $X \sqcup Y$ with $X \in \mathcal{B}|B$, $Y \in \mathcal{B}/B$, with $\theta_{X\sqcup Y} = \theta'_X \theta''_Y$ and the complement $(B \setminus X) \sqcup (R \setminus Y)$. Summing over the two levels, $K = K_{\mathcal{B}/R}K_{\mathcal{B}|R} + K_{\mathcal{B}|B}K_{\mathcal{B}/B}$ and $\Psi = \Psi_{\mathcal{B}/R}\Psi_{\mathcal{B}|R} + \Psi_{\mathcal{B}|B}\Psi_{\mathcal{B}/B}$, and on B the polynomials $K_{\mathcal{B}/R}$, $K_{\mathcal{B}|B}$, $\Psi_{\mathcal{B}|B}$, $\Psi_{\mathcal{B}/R}$ are homogeneous of the degrees $d(B)$, $d(B)+1$, $|B| - r(B) = d^*(B)$ and $|B| - d(B) = d^*(B) + 1$. The deletions and the contractions of the hinge are $\mathcal{B}_B^+ \setminus g = \mathcal{B}|B$, $\mathcal{B}_B^+/g = \mathcal{B}/R$, $\mathcal{B}_R^+ \setminus g = \mathcal{B}|R$ and $\mathcal{B}_R^+/g = \mathcal{B}/B$ (19.4), which gives the ranks and, by 7.3 and 19.2, the letters. By 7.12, $d(S) = r - r(\mathcal{E} \setminus S)$, so $d(B) + d(R) = 2r - r(B) - r(R) = r - c(B) = r - 1$, and $d^*(B) + d^*(R) = n - r(B) - r(R) = n - r - 1$. Table by table, $\kappa_B + \kappa_R = m_\mathcal{E} - \sum_j \big(s_j(r_j - 1) + t_j(n - r_j - 1)\big) = \kappa_\mathcal{E} + \sum_j (s_j + t_j)$. □

Lemma S9. *Let a family F on $\mathcal{E} = B \sqcup R$, B and R nonempty, have real data and be on its balance wall, and let $f(\alpha) = \alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j}$. Then*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_R} \int_{\mathbb{R}^B_{>0}} f(x, v)\, dx\, d\omega_R(v) = \int_{\Delta_R} \int_{\Delta_B} \int_0^\infty \rho^{|B|-1} f(\rho u, v)\, d\rho\, d\omega_B(u)\, d\omega_R(v).$$

Proof. On the balance wall f is homogeneous of degree $\kappa_\mathcal{E} - n = -n$, $n = |\mathcal{E}|$. Let φ be the indicator of $[1, e]$, so that $\int_0^\infty \varphi(\lambda)\, d\lambda/\lambda = 1$, and compute $I = \int_{\mathbb{R}^\mathcal{E}_{>0}} f(\alpha)\varphi(|\alpha_R|)\, d\alpha$ in two ways. With $\alpha = \lambda w$, $w \in \Delta_\mathcal{E}$, $d\alpha = \lambda^{n-1} d\lambda\, d\omega_\mathcal{E}(w)$ (2.7) and $f(\lambda w) = \lambda^{-n} f(w)$, so

$I = \int_{\Delta_\mathcal{E}} f(w) \int_0^\infty \varphi(\lambda|w_R|) \frac{d\lambda}{\lambda}\, d\omega_\mathcal{E}(w) = \int_{\Delta_\mathcal{E}} f\, d\omega_\mathcal{E} = P_F$ (8.7). With $\alpha_R = \lambda v$, $v \in \Delta_R$, and $\alpha_B = \lambda x$, $x \in \mathbb{R}^B_{>0}$, $d\alpha = \lambda^{|R|-1} d\lambda\, d\omega_R(v)\, \lambda^{|B|} dx$ and $f(\lambda x, \lambda v) = \lambda^{-n} f(x,v)$, so $I = \int_0^\infty \varphi(\lambda) \frac{d\lambda}{\lambda} \int_{\Delta_R} \int_{\mathbb{R}^B_{>0}} f(x,v)\, dx\, d\omega_R(v)$, the first form. The second is 2.7 for $x = \rho u$. For $|B| = 1$, Δ_B is a point with the Dirac measure (1.1(iii)). □

Theorem 19.9 (Walls glue). *As stated: (i) and (ii) for every family; in (iii) the identities for every family and the gluing when every active letter reads B alike.*

Proof. The gluing of the counts of one table, the counts of its cells,

$$\begin{aligned} d_{\mathcal{B}^+_B}(S_B) &= d_{\mathcal{B}/R}(S_B), \quad d_{\mathcal{B}^+_B}(S_B + g) = d_{\mathcal{B}|B}(S_B), \\ d^*_{\mathcal{B}^+_B}(S_B) &= d^*_{\mathcal{B}|B}(S_B), \quad d^*_{\mathcal{B}^+_B}(S_B + g) = d^*_{\mathcal{B}/R}(S_B), \\ d_{\mathcal{B}^+_R}(S_R) &= d_{\mathcal{B}/B}(S_R), \quad d_{\mathcal{B}^+_R}(S_R + g) = d_{\mathcal{B}|R}(S_R), \\ d^*_{\mathcal{B}^+_R}(S_R) &= d^*_{\mathcal{B}|R}(S_R), \quad d^*_{\mathcal{B}^+_R}(S_R + g) = d^*_{\mathcal{B}/B}(S_R), \end{aligned}$$

and the inequalities $d_{\mathcal{B}/R} \le d_{\mathcal{B}|B} \le d_{\mathcal{B}/R} + 1$ and $d^*_{\mathcal{B}|B} \le d^*_{\mathcal{B}/R} \le d^*_{\mathcal{B}|B} + 1$ on the subsets of B, $d_{\mathcal{B}/B} \le d_{\mathcal{B}|R} \le d_{\mathcal{B}/B} + 1$ and $d^*_{\mathcal{B}|R} \le d^*_{\mathcal{B}/B} \le d^*_{\mathcal{B}|R} + 1$ on those of R, are the reference, proof of thm:wallsglue. For table j and $S = S_B \sqcup S_R$ call

$$\begin{aligned} a^K_j &= d_{\mathcal{B}_j|B}(S_B) - d_{\mathcal{B}_j/R}(S_B), \quad b^K_j = d_{\mathcal{B}_j|R}(S_R) - d_{\mathcal{B}_j/B}(S_R), \\ a^\Psi_j &= d^*_{\mathcal{B}_j/R}(S_B) - d^*_{\mathcal{B}_j|B}(S_B), \quad b^\Psi_j = d^*_{\mathcal{B}_j/B}(S_R) - d^*_{\mathcal{B}_j|R}(S_R), \end{aligned}$$

all in $\{0,1\}$, in this proof the *jumps* of the letters K_j and Ψ_j on the side of the bunch and on the side of the room. The gluing of the counts reads

$$d_j(S) = d_{\mathcal{B}_j/R}(S_B) + d_{\mathcal{B}_j/B}(S_R) + \min(a^K_j, b^K_j), \qquad d^*_j(S) = d^*_{\mathcal{B}_j|B}(S_B) + d^*_{\mathcal{B}_j|R}(S_R) + \min(a^\Psi_j, b^\Psi_j).$$

At $S_B = \varnothing$ the jumps on the side of the bunch are 0, and at $S_R = \varnothing$ those on the side of the room. At $S_B = B$ those on the side of the bunch are 1: the ranks give $d_{\mathcal{B}_j|B}(B) = d_j(B) + 1$, $d_{\mathcal{B}_j/R}(B) = d_j(B)$, $d^*_{\mathcal{B}_j/R}(B) = |B| - d_j(B) = d^*_j(B) + 1$ and $d^*_{\mathcal{B}_j|B}(B) = d^*_j(B)$. At $S_R = R$ those on the side of the room are 1: $d_{\mathcal{B}_j|R}(R) = r_j - d_j(B) = d_{\mathcal{B}_j/B}(R) + 1$ and $d^*_{\mathcal{B}_j/B}(R) = |R| - r_j + d_j(B) + 1 = d^*_{\mathcal{B}_j|R}(R) + 1$.

(i) The tables of $F_{B\to g}(u)$ of index j have the bases of $\mathcal{B}^+_{j,R}$, at $(s_j, 0)$ and at $(0, t_j)$, so its wall at $S \subseteq R + g$ is $m'_S - \sum_j \big(s_j d_{\mathcal{B}^+_{j,R}}(S) + t_j d^*_{\mathcal{B}^+_{j,R}}(S)\big)$, $\mathbf{m}' = (\kappa_B, \mathbf{m}|_R)$. At $S = S_R$ the counts of the cell are $d_{\mathcal{B}_j/B}(S_R)$ and $d^*_{\mathcal{B}_j|R}(S_R)$, which are $d_j(S_R)$ and $d^*_j(S_R)$ by the gluing at $S_B = \varnothing$: the wall is κ_{S_R}. At $S = S_R + g$ they are $d_{\mathcal{B}_j|R}(S_R)$ and $d^*_{\mathcal{B}_j/B}(S_R)$, and the gluing at $S_B = B$ gives $d_j(B \sqcup S_R) = d_j(B) + d_{\mathcal{B}_j|R}(S_R)$ and $d^*_j(B \sqcup S_R) = d^*_j(B) + d^*_{\mathcal{B}_j/B}(S_R)$; with $\kappa_B = m_B - \sum_j (s_j d_j(B) + t_j d^*_j(B))$ the wall is $\kappa_{B \sqcup S_R}$.

(ii) With α_{S_B} replaced by $\varepsilon \alpha_{S_B}$, $K_{\mathcal{B}_j/R}$ is $\varepsilon^{d_{\mathcal{B}_j/R}(S_B)}$ times a polynomial positive at $\varepsilon = 0$ on the open orthant, its monomials having at least $d_{\mathcal{B}_j/R}(S_B)$ marks in S_B and some exactly that many (7.13); likewise $\Psi_{\mathcal{B}_j|B}$ with $d^*_{\mathcal{B}_j|B}(S_B)$. By the gluing at $S_R = \varnothing$ these are $d_j(S_B)$ and $d^*_j(S_B)$, so the exponent of the density at the face is κ_{S_B}.

(iii) The walls of G_B and G_R are read with the counts of the cells and the masses $(\mathbf{m}|_B, 2\sigma - \kappa_B)$ and $(\kappa_B, \mathbf{m}|_R)$. With

$$X = m_{S_B} + m_{S_R} - \sum_j \Big(s_j \big(d_{\mathcal{B}_j/R}(S_B) + d_{\mathcal{B}_j/B}(S_R) \big) + t_j \big(d^*_{\mathcal{B}_j|B}(S_B) + d^*_{\mathcal{B}_j|R}(S_R) \big) \Big),$$

the counts of the cells give $\kappa^{G_B}_{S_B}+\kappa^{G_R}_{S_R}=X$ and, the two hinge masses adding up to $2\sigma=\sum_j(s_j+t_j)$, $\kappa^{G_B}_{S_B+g}+\kappa^{G_R}_{S_R+g}=X+\sum_j\big(s_j(1-a_j^K-b_j^K)+t_j(1-a_j^\Psi-b_j^\Psi)\big)$, while $\kappa_{S_B\sqcup S_R}=X-\sum_j\big(s_j\min(a_j^K,b_j^K)+t_j\min(a_j^\Psi,b_j^\Psi)\big)$. Since $1-a-b=(1-\max(a,b))-\min(a,b)$ for $a,b\in\{0,1\}$,

$$\begin{aligned}\kappa^{G_B}_{S_B}+\kappa^{G_R}_{S_R}-\kappa_{S_B\sqcup S_R}&=\sum_j\big(s_j\min(a_j^K,b_j^K)+t_j\min(a_j^\Psi,b_j^\Psi)\big),\\ \kappa^{G_B}_{S_B+g}+\kappa^{G_R}_{S_R+g}-\kappa_{S_B\sqcup S_R}&=\sum_j\big(s_j(1-\max(a_j^K,b_j^K))+t_j(1-\max(a_j^\Psi,b_j^\Psi))\big),\end{aligned}\tag{48.1}$$

at all masses and exponents. At $S_R=\varnothing$ the first line gives $\kappa^{G_B}_{S_B}=\kappa_{S_B}$, the wall at the empty set being 0; at $S_B=\varnothing$ it gives $\kappa^{G_R}_{S_R}=\kappa_{S_R}$. The imbalance of G_B is

$$\kappa^{G_B}_{B+g}=m_B+2\sigma-\kappa_B-\sum_j\big(s_j(d_j(B)+1)+t_j(d_j^*(B)+1)\big)=2\sigma-\sum_j(s_j+t_j)=0,$$

the cell $\mathcal{B}^+_{j,B}$ having rank $d_j(B)+1$ on $|B|+1=d_j(B)+d_j^*(B)+2$ marks: G_B is on its balance wall at all data. At $S_B=B$ the second line then gives $\kappa^{G_R}_{S_R+g}=\kappa_{B\sqcup S_R}$; at $S_R=R$ in particular $\kappa^{G_R}_{R+g}=\kappa_{\mathcal{E}}$, so G_R is on its balance wall exactly when F is, and then the second line at $S_R=R$ gives $\kappa^{G_B}_{S_B+g}=\kappa_{S_B\sqcup R}$. These are the four identities.

Let now $s_j,t_j\geq 0$ and let every active letter read B alike. The jump of a letter on the side of the bunch at S_B is the order to which its hinge value vanishes at the face $u_{S_B}=0$: by the expansion in (ii) applied to $\mathcal{B}_j|B$ and $\mathcal{B}_j/R$, with $\varepsilon\alpha_{S_B}$ for α_{S_B}, $\gamma^K_{j,B}=K_{\mathcal{B}_j|B}/K_{\mathcal{B}_j/R}$ is $\varepsilon^{a_j^K}$ times a function positive at $\varepsilon=0$, and $\gamma^\Psi_{j,B}=\Psi_{\mathcal{B}_j/R}/\Psi_{\mathcal{B}_j|B}$ is $\varepsilon^{a_j^\Psi}$ times one. Constant multiples of one function vanish to one order at each face, so at S_B the active letters have one jump $a\in\{0,1\}$ on the side of the bunch. If $a=0$ every term of the first line of (48.1) vanishes, a letter of exponent zero contributing nothing; if $a=1$ every term of the second does. Both lines are nonnegative, the exponents being nonnegative, so $\kappa_{S_B\sqcup S_R}$ is the smaller of the two sums. One active letter reads B alike by itself.

Positivity. If both cell families have every proper wall positive, let $S=S_B\sqcup S_R$ be nonempty and proper. In the first sum a nonempty part is a proper subset of the marks of its cell family, $S_B\subseteq B\subsetneq B+g$ and $S_R\subseteq R\subsetneq R+g$, and an empty part has the wall 0: the first sum is positive. In the second both S_B+g and S_R+g are nonempty, and at most one of them is all the marks of its cell family, S being proper, its wall then being the imbalance 0: the second sum is positive. So $\kappa_S>0$. Conversely, by the identities, $\kappa^{G_B}_{S_B}=\kappa_{S_B}$ for $\varnothing\neq S_B\subseteq B$ and $\kappa^{G_B}_{S_B+g}=\kappa_{S_B\sqcup R}$ for $S_B\subsetneq B$ are proper walls of F, and likewise for G_R; this direction does not use the hypothesis. □

Remark. Without the reading alike, (iii) keeps the identities and loses the gluing. On $\mathcal{E}=\{1,2,3,4\}$ with $B=\{1,2\}$, the family of the lists $\binom{\mathcal{E}}{3}$ and $\binom{\mathcal{E}}{1}$, each at $(0,1)$, has B of width one in both tables, with the cells $\binom{\{1,2,g\}}{2}$, $\binom{\{3,4,g\}}{2}$ and $\binom{\{1,2,g\}}{1}$, $\binom{\{3,4,g\}}{1}$, and the hinge values $\gamma^\Psi_{1,B}=\alpha_1+\alpha_2$ and $\gamma^\Psi_{2,B}=\alpha_1\alpha_2/(\alpha_1+\alpha_2)$, which are not constant multiples of one function. At the masses $(\frac{2}{5},\frac{8}{5},\frac{2}{5},\frac{8}{5})$, on the balance wall, $\kappa_B=1$ and both cell families have every proper wall positive, the least being $\frac{2}{5}$, while $\kappa_{\{1,3\}}=\frac{2}{5}+\frac{2}{5}-1=-\frac{1}{5}$, against $\frac{4}{5}$ from the gluing; the walls at $\{1,4\}$, $\{2,3\}$ and $\{2,4\}$ do not glue either. In exact rational arithmetic (receipt `check_walls.py`): this family; the four identities at every subset at two partial leans of the same tables, where four walls do not glue; and the gluing at every subset for two tables sharing one weighted cell of the

bunch at cobasis full leans, three pairs of exponents and three sets of masses, for a table at $(s,0)$ with the cell $\binom{\{1,2,g\}}{1}$ together with a table at $(0,t)$ with the cell $\binom{\{1,2,g\}}{2}$, whose hinge values $\gamma^K_{1,B}$ and $\gamma^\Psi_{2,B}$ are both $\alpha_1+\alpha_2$, and for $\binom{\mathcal{E}}{2}$ with B three of its four marks at two partial leans, where $\varrho_B=1$.

Theorem 19.10 (Pooling factors at one hinge value). *As stated: at real exponents of any sign, every active letter having the one hinge value γ.*

Proof. The walls $\kappa_{\{e\}}$ at the singletons are positive and the exponents real, so the masses are real, and the integrals below are of positive functions.

Bounds. κ_B and κ_R are proper walls of F, hence positive, and $\kappa_B+\kappa_R=2\sigma$ on the balance wall (Lemma S8): $0<\kappa_B<2\sigma$.

The cell families. By the identities of 19.9(iii), which hold for every family on the balance wall, G_B and G_R are on their balance walls and each of their proper walls is a proper wall of F, hence positive; their periods converge (8.7).

The family. Put $\phi(u)=u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{B}_j/R}(u)^{-s_j}\Psi_{\mathcal{B}_j|B}(u)^{-t_j}$ on Δ_B. By 19.4, with $K_{\mathcal{B}_j/R}(\rho u)=\rho^{d_j(B)}K_{\mathcal{B}_j/R}(u)$, $\Psi_{\mathcal{B}_j|B}(\rho u)=\rho^{d^*_j(B)}\Psi_{\mathcal{B}_j|B}(u)$ and the hinge values homogeneous of degree one, Lemma S9 gives

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_B}\phi(u)\int_{\Delta_R}\int_0^\infty \rho^{\kappa_B-1}v^{\mathbf{m}|_R-\mathbf{1}}\\ \times\prod_j K_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^K_{j,B}(u),v\big)^{-s_j}\Psi_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^\Psi_{j,B}(u),v\big)^{-t_j}d\rho\,d\omega_R(v)\,d\omega_B(u),$$

the powers of ρ collecting to $\rho^{m_B-1-\sum_j(s_jd_j(B)+t_jd^*_j(B))}=\rho^{\kappa_B-1}$. Every active letter has the hinge value $\gamma(u)$, and a letter of exponent zero is the factor 1. The substitution $x=\rho\gamma(u)$ at fixed u, $\rho^{\kappa_B-1}d\rho=\gamma(u)^{-\kappa_B}x^{\kappa_B-1}dx$, and Lemma S9 for G_R on $\{g\}\sqcup R$ at the masses $(\kappa_B,\mathbf{m}|_R)$ give

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=P_{G_R}\big((\kappa_B,\mathbf{m}|_R);\mathbf{s},\mathbf{t}\big)\int_{\Delta_B}\phi(u)\,\gamma(u)^{-\kappa_B}\,d\omega_B(u).$$

The cell family of the bunch. By Lemma S8, with y the variable of g, $K_{\mathcal{B}^+_{j,B}}(u,y)=K_{\mathcal{B}_j/R}(u)\big(y+\gamma^K_{j,B}(u)\big)$ and $\Psi_{\mathcal{B}^+_{j,B}}(u,y)=\Psi_{\mathcal{B}_j|B}(u)\big(y+\gamma^\Psi_{j,B}(u)\big)$; every active letter has the factor $y+\gamma(u)$, and the exponents of the active letters add up to $\sum_j(s_j+t_j)=2\sigma$. Lemma S9 for G_B on $\{g\}\sqcup B$ at the masses $(\mathbf{m}|_B,2\sigma-\kappa_B)$, and the integral of a linear letter (the paragraph after 15.1, with $c_0=\gamma(u)$, $c_1=1$ and $0<2\sigma-\kappa_B<2\sigma$), give

$$P_{G_B}\big((\mathbf{m}|_B,2\sigma-\kappa_B);\mathbf{s},\mathbf{t}\big)=\int_{\Delta_B}\phi(u)\int_0^\infty y^{2\sigma-\kappa_B-1}\big(y+\gamma(u)\big)^{-2\sigma}dy\,d\omega_B(u)\\ =\Gamma_{\binom{2}{1}}\begin{bmatrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{bmatrix}\int_{\Delta_B}\phi(u)\,\gamma(u)^{-\kappa_B}\,d\omega_B(u).$$

Multiplying the display for P_F by $\Gamma_{\binom{2}{1}}\begin{bmatrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{bmatrix}$ gives the statement, every factor being finite and positive. The parenthesis of the statement is 19.2: the constants λ_d,λ_r of table j multiply $\gamma^K_{j,B}$ by c and $\gamma^\Psi_{j,B}$ by c^{-1}, so a table with one active letter is normalized by its own constant; at full leans of all tables only the hinge values of different tables are compared; one table with both letters active and ϱ_B constant is normalized by $c=\varrho_B^{1/2}$, and $\gamma^K_B=\gamma^\Psi_B$ forces $\varrho_B=1$. One table at $(0,t)$ has $s=0$ and $2\sigma=t$, and at $(s,0)$, $2\sigma=s$. □

Remark. One table at $(0,t)$ has the one active letter Ψ, whose hinge value γ_B^{Ψ} serves as γ at any factorization; the cell families are the cells $(\mathcal{B}_B^+,\theta')$ and $(\mathcal{B}_R^+,\theta'')$ at $(0,t)$, and the display is that of the reference, $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\kappa_B,\ t-\kappa_B\\ t\end{smallmatrix}\right] P_F(\mathbf{m};0,t) = P_{\mathcal{B}_B^+}\big((\mathbf{m}|_B, t-\kappa_B);0,t\big)P_{\mathcal{B}_R^+}\big((\kappa_B,\mathbf{m}|_R);0,t\big)$; at $(s,0)$ the same holds with $\gamma=\gamma_B^K$ and s for t. The proof above is the proof of the reference with γ_B^{Ψ} replaced by the common hinge value of the active letters. A family at full leans, every table at $(s_j,0)$ or $(0,t_j)$, has one active letter in each table, normalized by the factorization of its table; the hypothesis then asks only that the hinge values of the active letters of different tables be constant multiples of one another, a condition across the tables, and 22.6 is the same condition at every cell of a core. One table with $s,t>0$ has both letters active; the hypothesis is ϱ_B constant, reached at the factorization with $\varrho_B=1$, and the display then holds at every pair (s,t) with $2\sigma=s+t$, the separation of 19.12 completed by the Beta corner of the hinge.

Proposition 19.11 (Pooling a realization). *As stated, for every family and every realization.*

Proof. The masses of Υ are $\tau_e=\sum_j\big(s_j\alpha_e\partial_e\log K_j(\alpha)+t_j\alpha_e\partial_e\log\Psi_j(\alpha)\big)$ (9.1). With $u=\alpha_B/|\alpha_B|$ fixed, the tables of index j of $F_{B\to g}(u)$, read as one table with two systems of weights, have the letters $K_j^u(\alpha_g,\alpha_R)=K_{\mathcal{B}_{j,R}^+}(\gamma_{j,B}^K(u)\alpha_g,\alpha_R)$ and $\Psi_j^u(\alpha_g,\alpha_R)=\Psi_{\mathcal{B}_{j,R}^+}(\gamma_{j,B}^{\Psi}(u)\alpha_g,\alpha_R)$ (19.8). By 19.4 and the homogeneity of the hinge values, $\gamma_{j,B}^K(\alpha_B)=\rho\gamma_{j,B}^K(u)$ with $\rho=|\alpha_B|$, and

$$K_j(\alpha)=K_{\mathcal{B}_j/R}(\alpha_B)\,K_j^u(\rho,\alpha_R),\qquad \Psi_j(\alpha)=\Psi_{\mathcal{B}_j|B}(\alpha_B)\,\Psi_j^u(\rho,\alpha_R).$$

For $e\in R$ the first factors do not depend on α_e, so τ_e is the usage of e in $F_{B\to g}(u)$ at the gauge (ρ,α_R) with the exponents (s_j,t_j). Summing over $e\in B$, Euler's identity gives $\sum_{e\in B}\alpha_e\partial_e\log K_{\mathcal{B}_j/R}(\alpha_B)=d_j(B)$ and $\sum_{e\in B}\alpha_e\partial_e\log\Psi_{\mathcal{B}_j|B}(\alpha_B)=d_j^*(B)$, and, $\gamma_{j,B}^K$ being homogeneous of degree one,

$$\sum_{e\in B}\alpha_e\partial_e\log K_{\mathcal{B}_{j,R}^+}\big(\gamma_{j,B}^K(\alpha_B),\alpha_R\big)=\big[y\partial_y\log K_{\mathcal{B}_{j,R}^+}(y,\alpha_R)\big]_{y=\gamma_{j,B}^K(\alpha_B)}$$
$$=\big[\alpha_g\partial_{\alpha_g}\log K_j^u(\alpha_g,\alpha_R)\big]_{\alpha_g=\rho},$$

and likewise for Ψ. So $\tau_B=\sum_j\big(s_jd_j(B)+t_jd_j^*(B)\big)+\tau_g^u$, with τ_g^u the usage of g in $F_{B\to g}(u)$ at (ρ,α_R), and τ_g^u is the wall κ_B at the masses $\boldsymbol{\tau}$. The exponents are those of a realization and (ρ,α_R) is a gauge, so $((s_j,t_j)_j;(\rho,\alpha_R))$ is a realization of $F_{B\to g}(u)$ with these masses, its usage. For one table at $(0,t)$, $\Psi^u(\rho,\alpha_R)=\Psi_{\mathcal{B}_R^+}(\gamma_B^{\Psi}(\alpha_B),\alpha_R)$, so the usage of $(\mathcal{B}_R^+,\theta'')$ at $(0,t)$ and at the gauge $(\gamma_B^{\Psi}(\alpha_B),\alpha_R)$ is $\boldsymbol{\tau}|_R$ on R and, by the chain rule above, $\tau_B-t\,d^*(B)$ at g. □

Remark. The reference is one list at $(0,t)$, its last sentence. In exact rational arithmetic (receipt `check_classes_realization.py`): two tables on five marks, each the two-sum of a weighted cell of the bunch on $\{1,2,g\}$ and a weighted cell of the room on $\{3,4,5,g\}$, at the partial leans $(\frac{2}{3},\frac{5}{4})$ and $(3,\frac{1}{2})$ and a rational gauge; the usage of $F_{B\to g}(u)$ at $(|\alpha_B|,\alpha_R)$ is $(\kappa_B(\boldsymbol{\tau}),\boldsymbol{\tau}|_R)$.

Proposition 19.12 (Hinge values that agree). *As stated, for families.*

Proof. As in the proof of 19.10, with $\phi(u)=u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{B}_j/R}(u)^{-s_j}\Psi_{\mathcal{B}_j|B}(u)^{-t_j}$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_B}\phi(u)\int_{\Delta_R}\int_0^{\infty}\rho^{\kappa_B-1}v^{\mathbf{m}|_R-\mathbf{1}}$$
$$\times\prod_j K_{\mathcal{B}_{j,R}^+}\big(\rho\gamma_{j,B}^K(u),v\big)^{-s_j}\Psi_{\mathcal{B}_{j,R}^+}\big(\rho\gamma_{j,B}^{\Psi}(u),v\big)^{-t_j}d\rho\,d\omega_R(v)\,d\omega_B(u).$$

For $s_j \neq 0$, $K_{\mathcal{B}^+_{j,R}}(\rho c_j\gamma(u), v) = K_{\mathcal{B}|R}(v) + \rho c_j\gamma(u) K_{\mathcal{B}/B}(v)$ is the letter K of $\mathcal{B}^+_{j,R}$ with its weights multiplied by c_j on its bases through g, read at $(\rho\gamma(u), v)$; for $t_j \neq 0$, $\Psi_{\mathcal{B}^+_{j,R}}(\rho c_j^*\gamma(u), v)$ is the letter Ψ of $\mathcal{B}^+_{j,R}$ with its weights multiplied by c_j^* on its bases avoiding g, read at $(\rho\gamma(u), v)$ (Lemma S8). These are the letters of $G^{\mathbf{c}}_R$, and the letters of exponent zero are the factor 1. The substitution $x = \rho\gamma(u)$ and Lemma S9 for $G^{\mathbf{c}}_R$, which has the bases and so the walls of G_R and is on its balance wall by 19.9(iii), give the display. Its period converges, each proper wall of G_R being a proper wall of F, so the shape integral is finite.

For one table with ϱ_B constant, 19.7 reaches $\varrho_B = 1$ by $c = \varrho_B^{1/2}$; then $\gamma^K_B = \gamma^\Psi_B$, $c = c^* = 1$ with $\gamma = \gamma^K_B$, and $G^{\mathbf{c}}_R$ is the cell $(\mathcal{B}^+_R, \theta'')$ at (s, t), which is the reference. □

Theorem 19.13 (Gluing across a bunch of width one). *As stated: at nonnegative exponents, a letter of exponent zero being opened by nothing.*

Proof. The letters on the scale. For table j and $(u, v) \in \Delta_B \times \Delta_R$ put

$$A^K_j = K_{\mathcal{B}_j/R}(u) K_{\mathcal{B}_j|R}(v), \quad C^K_j = K_{\mathcal{B}_j|B}(u) K_{\mathcal{B}_j/B}(v),$$
$$A^\Psi_j = \Psi_{\mathcal{B}_j|B}(u) \Psi_{\mathcal{B}_j/R}(v), \quad C^\Psi_j = \Psi_{\mathcal{B}_j/R}(u) \Psi_{\mathcal{B}_j|R}(v),$$

all positive, so that $\rho^{-d_j(B)} K_j(\rho u, v) = A^K_j + \rho C^K_j$ and $\rho^{-d^*_j(B)} \Psi_j(\rho u, v) = A^\Psi_j + \rho C^\Psi_j$ (Lemma S8). The minors of the level d sit in A^K_j and in C^Ψ_j, those of the level r in C^K_j and in A^Ψ_j.

The scale. By Lemma S9, the powers of ρ collecting to $\rho^{\kappa_B - 1}$,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_R} \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} v^{\mathbf{m}|_R - \mathbf{1}} \, I(u, v) \, d\omega_B(u) \, d\omega_R(v),$$
$$I(u, v) = \int_0^\infty \rho^{\kappa_B - 1} \prod_{\ell=1}^{p} (A_\ell + \rho C_\ell)^{-w_\ell} d\rho,$$

where ℓ runs over the p active letters, with their exponents $w_\ell > 0$ and their coefficients (A_ℓ, C_ℓ), a letter of exponent zero being the factor 1. Here $\sum_\ell w_\ell = \sum_j (s_j + t_j) = 2\sigma$, and κ_B and $\kappa_R = 2\sigma - \kappa_B$ (Lemma S8, $\kappa_\mathcal{E} = 0$) are proper walls: $0 < \kappa_B < 2\sigma$.

Opening the scale. If $p = 1$, the integral of a linear letter (the paragraph after 15.1) gives $I = \Gamma^{\,2}_{\,1}\!\begin{bmatrix} w_1 - \kappa_B, & \kappa_B \\ & w_1 \end{bmatrix} A_1^{\kappa_B - w_1} C_1^{-\kappa_B}$. If $p \geq 2$, choose $c_\ell \in (0, w_\ell)$ for $\ell < p$ with $0 < \kappa_B - \sum_{\ell<p} c_\ell < w_p$, possible as $0 < \kappa_B < \sum_\ell w_\ell$, and open each letter $\ell < p$ by 4.4 with $k = 2$ on the line $\operatorname{Re} x_\ell = c_\ell$:

$$(A_\ell + \rho C_\ell)^{-w_\ell} = \frac{1}{2\pi i} \int_{c_\ell + i\mathbb{R}} \Gamma^{\,2}_{\,1}\!\begin{bmatrix} w_\ell - x_\ell, & x_\ell \\ & w_\ell \end{bmatrix} A_\ell^{x_\ell - w_\ell} (\rho C_\ell)^{-x_\ell} \, dx_\ell.$$

The modulus of the integrand is its value at the real parts times the modulus of the stacked Gamma, which decays exponentially along the line (Stirling), and $\int_0^\infty \rho^{\kappa_B - \sum_{\ell<p} c_\ell - 1} (A_p + \rho C_p)^{-w_p} d\rho$ is finite; so Fubini puts the integral over ρ inside, where the integral of a linear letter at the complex index $x_p = \kappa_B - \sum_{\ell<p} x_\ell$, $0 < \operatorname{Re} x_p < w_p$, gives $\Gamma^{\,2}_{\,1}\!\begin{bmatrix} w_p - x_p, & x_p \\ & w_p \end{bmatrix} A_p^{x_p - w_p} C_p^{-x_p}$. Hence

$$I(u, v) = \frac{1}{(2\pi i)^{p-1}} \int_{c_1 + i\mathbb{R}} \cdots \int_{c_{p-1} + i\mathbb{R}} \prod_{\ell=1}^{p} \Gamma^{\,2}_{\,1}\!\begin{bmatrix} w_\ell - x_\ell, & x_\ell \\ & w_\ell \end{bmatrix} A_\ell^{x_\ell - w_\ell} C_\ell^{-x_\ell} \, dx_1 \cdots dx_{p-1},$$
$$x_p = \kappa_B - \sum_{\ell<p} x_\ell,$$

the share integral $\int_{x_1+\cdots+x_p=\kappa_B}$ of 4.1: the factor $(2\pi i)^{-1}$ of each of the $p-1$ openings is its normalization $(2\pi i)^{1-p}$, and its dimension is $p-1$. At $p=1$ the share integral has dimension zero, one share and the pin $P=(1)$, and 4.1 gives the value of the integrand at the one admissible point $x_1=\kappa_B$: the formula for $p=1$ above.

The level shares. For the letter K_j write $x_\ell=a_j^r$ and $w_\ell-x_\ell=a_j^d$, for Ψ_j write $x_\ell=b_j^d$ and $w_\ell-x_\ell=b_j^r$, and give the letters of exponent zero the shares 0. The pin $\sum_\ell x_\ell=\kappa_B$ is $\sum_j(a_j^r+b_j^d)=\kappa_B$; the factor of K_j is $\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}a_j^d,\ a_j^r\\ s_j\end{smallmatrix}\right](A_j^K)^{-a_j^d}(C_j^K)^{-a_j^r}$ and that of Ψ_j is $\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}b_j^d,\ b_j^r\\ t_j\end{smallmatrix}\right](C_j^\Psi)^{-b_j^d}(A_j^\Psi)^{-b_j^r}$. The product of the powers over all letters, times $u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}$, is $f(u)\,g(v)$, with

$$f(u)=u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{B}_j/R}^{-a_j^d}\Psi_{\mathcal{B}_j/R}^{-b_j^d}K_{\mathcal{B}_j|B}^{-a_j^r}\Psi_{\mathcal{B}_j|B}^{-b_j^r},\qquad g(v)=v^{\mathbf{m}|_R-\mathbf{1}}\prod_j K_{\mathcal{B}_j|R}^{-a_j^d}\Psi_{\mathcal{B}_j|R}^{-b_j^d}K_{\mathcal{B}_j/B}^{-a_j^r}\Psi_{\mathcal{B}_j/B}^{-b_j^r},$$

the integrands of the periods of $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ on Δ_B and of $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$ on Δ_R (8.7).

Balance on the pin. The tables $\mathcal{B}_j/R$ and $\mathcal{B}_j|B$ on B have the ranks $d_j(B)$ and $d_j(B)+1$, and $\mathcal{B}_j|R$ and $\mathcal{B}_j/B$ on R the ranks $r_j-d_j(B)$ and $r_j-d_j(B)-1$. With $a_j^d+a_j^r=s_j$, $b_j^d+b_j^r=t_j$ and $|B|-d_j(B)-1=d_j^*(B)$, the imbalance of $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ at $\mathbf{m}|_B$ is

$$m_B-\sum_j\big(a_j^d d_j(B)+b_j^d(|B|-d_j(B))+a_j^r(d_j(B)+1)+b_j^r d_j^*(B)\big)=\kappa_B-\sum_j(a_j^r+b_j^d);$$

with $|R|+1-r_j+d_j(B)=n-r_j-d_j^*(B)$, that of $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$ at $\mathbf{m}|_R$ is

$$m_R-\sum_j\big(s_j(r_j-d_j(B))+t_j(|R|+1-r_j+d_j(B))\big)+\sum_j(a_j^r+b_j^d)=\kappa_{\mathcal{E}}-\kappa_B+\sum_j(a_j^r+b_j^d).$$

Both vanish on the pin, F being on its balance wall.

Fubini. On a contour of the statement the modulus of $f(u)g(v)$ is $f(u)g(v)$ at the real parts of the shares, which do not move along the contour, and its integral over $\Delta_B\times\Delta_R$ is the product of the two periods at the real parts, finite by hypothesis; the stacked Gammas decay exponentially along the contour. So Fubini exchanges the share integral with the integral over $\Delta_B\times\Delta_R$, and the inner integrals are $P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})$ and $P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$: the general display.

Pair coordinates. When every s_j and t_j is positive, $p=2N$ and the shares are $(a_j^r,b_j^d)_j$. With $a_j^d=\sigma_j^d+\delta_j^d$, $b_j^d=\sigma_j^d-\delta_j^d$ and $a_j^r=s_j-a_j^d$, the affine map $(\sigma_j^d,\delta_j^d)_j\mapsto(a_j^r,b_j^d)_j$ has in each table the linear part $(y_\sigma,y_\delta)\mapsto(-y_\sigma-y_\delta,\ y_\sigma-y_\delta)$, of determinant 2. So the imaginary parts $\mathbf{y}$ of the shares and $\mathbf{y}'$ of the pair coordinates have $d\mathbf{y}=2^N d\mathbf{y}'$ and $\sum_\ell y_\ell=-2\sum_j y'_{\delta,j}$, and with $\operatorname{Im}\kappa_B=0$ and $\delta^1(-2z)=\frac12\delta^1(z)$, 4.1 gives

$$\int_{\sum_\ell x_\ell=\kappa_B}h=(2\pi)^{1-2N}\int_{\mathbb{R}^{2N}}h\,\delta^1\Big(\sum_\ell y_\ell\Big)d\mathbf{y}=2^{N-1}(2\pi)^{1-2N}\int_{\mathbb{R}^{2N}}h\,\delta^1\Big(\sum_j y'_{\delta,j}\Big)d\mathbf{y}',$$

2^{N-1} times the fibre integral in the pair coordinates pinned by $\sum_j\delta_j^d=\frac12(\sum_j s_j-\kappa_B)$, the image of the pin $\sum_j(a_j^r+b_j^d)=\sum_j(s_j-2\delta_j^d)=\kappa_B$, the admissible real points corresponding under the map. With $\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}a_j^d,\ a_j^r\\ s_j\end{smallmatrix}\right]\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}b_j^d,\ b_j^r\\ t_j\end{smallmatrix}\right]=\Gamma_{\!\!\begin{smallmatrix}4\\2\end{smallmatrix}}\!\left[\begin{smallmatrix}a_j^d,\ b_j^d,\ a_j^r,\ b_j^r\\ s_j,\ t_j\end{smallmatrix}\right]$ this is the first display, whose N coordinates σ_j^d are not pinned. For one table the pin fixes δ^d, $2^{N-1}=1$, and the fibre integral is $(2\pi)^{-1}\int dy=(2\pi i)^{-1}\int d\sigma^d$ along the line in σ^d. The other forms of the pin, the value $2\delta^d=s-\kappa_B=\delta(1+\varepsilon(B))-\mu_B$ (8.2(iii) with $c(B)=1$) and the swap of B and R are the reference, proof of thm:twosides. □

Remark. The first display needs both letters of every table active: a letter of exponent zero would put $\Gamma(0)$ into the stacked Gamma of its table. The general display carries one Beta corner ${}_1^2\Gamma\left[\begin{smallmatrix}\cdot, & \cdot \\ & w\end{smallmatrix}\right]$ for each active letter, and at a full lean of one table it has no free share, which is 19.14.

Corollary 19.14 (One Skeleton value pins the share). *As stated: real exponents of any sign, one Skeleton value for all active letters.*

Proof. $0 < \kappa_B < 2\sigma$ as in the proof of 19.10. With the coefficients of the proof of 19.13, Lemma S9 gives, all functions being positive,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_R}\int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}} \times \int_0^\infty \rho^{\kappa_B-1}\prod_j (A_j^K+\rho C_j^K)^{-s_j}(A_j^\Psi+\rho C_j^\Psi)^{-t_j}\,d\rho\,d\omega_B(u)\,d\omega_R(v).$$

The Skeleton values are $A_j^K/C_j^K = \gamma_{j,R}^K(v)/\gamma_{j,B}^K(u)$ and $A_j^\Psi/C_j^\Psi = \gamma_{j,R}^\Psi(v)/\gamma_{j,B}^\Psi(u)$ (19.5), and by the hypothesis, the constants cancelling, every active letter has $A_\ell = \xi C_\ell$ with $\xi = \gamma_R(v)/\gamma(u)$. The exponents of the active letters add up to 2σ, so the integral of a linear letter gives $\int_0^\infty \rho^{\kappa_B-1}(\xi+\rho)^{-2\sigma}d\rho = {}_1^2\Gamma\left[\begin{smallmatrix}\kappa_B, & 2\sigma-\kappa_B \\ & 2\sigma\end{smallmatrix}\right]\xi^{\kappa_B-2\sigma}$, and

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = {}_1^2\Gamma\left[\begin{smallmatrix}\kappa_B, & 2\sigma-\kappa_B \\ & 2\sigma\end{smallmatrix}\right]\int_{\Delta_R}\int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}} \times \prod_{s_j\neq 0}(C_j^K)^{-s_j}\prod_{t_j\neq 0}(C_j^\Psi)^{-t_j}\,\xi^{\kappa_B-2\sigma}\,d\omega_B(u)\,d\omega_R(v).$$

For level shares as in the statement, $(A_j^K)^{-a_j^d}(C_j^K)^{-a_j^r} = (C_j^K)^{-s_j}\xi^{-a_j^d}$ for an active K_j, $(A_j^\Psi)^{-b_j^r}(C_j^\Psi)^{-b_j^d} = (C_j^\Psi)^{-t_j}\xi^{-b_j^r}$ for an active Ψ_j, and an inactive letter contributes 1; since $\sum_j(a_j^d+b_j^r) = 2\sigma - \sum_j(a_j^r+b_j^d) = 2\sigma-\kappa_B$, the integrand above is $f(u)g(v)$ with f and g the integrands of the periods of $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ and $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$ (proof of 19.13), whatever the shares. At complex shares $|f(u)g(v)|$ is $f(u)g(v)$ at the real parts, which satisfy the same constraints; its integral is finite by the display at the real parts, and both factors are positive, so $|f|$ and $|g|$ are integrable and Fubini gives the statement, the product of the two periods not depending on the shares. With one active letter the hypothesis holds with γ and γ_R its hinge values of B and of R. One table at $(0,t)$: $\mathbf{a}=0$, $b^d+b^r=t$ and $b^d=\kappa_B$, so the level shares are $(0,\kappa_B,0,t-\kappa_B)$ and the two families have the forms of the statement; at $(s,0)$, $a^r=\kappa_B$ and $a^d=s-\kappa_B$. □

Remark. The reference, rem:pinnedproduct, is one table at a full lean, the case of one active letter. The hypothesis does not depend on the factorizations: a change of the factorization of table j multiplies the hinge values of a letter of table j at B and at R by one constant (19.2), so the Skeleton values are unchanged.

Lemma 19.15 (Slopes and the range of the ratios). *As stated: for every table of a family, both letters of every table active.*

Proof. With $a_j^\ell = \sigma_j^\ell+\delta_j^\ell$ and $b_j^\ell = \sigma_j^\ell-\delta_j^\ell$, $K^{-a}\Psi^{-b} = \mathbf{E}^{-\sigma}(K/\Psi)^{-\delta}$ for every minor, $\mathbf{E}=K\Psi$. So the factors of table j in the word of the bunch family (proof of 19.13) are

$$\mathbf{E}_{\mathcal{B}_j/R}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j|B}^{-\sigma_j^r}\Big(\frac{K_{\mathcal{B}_j/R}}{\Psi_{\mathcal{B}_j/R}}\Big)^{-\delta_j^d}\Big(\frac{K_{\mathcal{B}_j|B}}{\Psi_{\mathcal{B}_j|B}}\Big)^{-\delta_j^r}, \qquad \mathbf{E}_{\mathcal{B}_j/R}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j|B}^{-\sigma_j+\sigma_j^d} = \varrho_{j,B}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j|B}^{-\sigma_j},$$

and in the word of the room family $\mathbf{E}_{\mathcal{B}_j|R}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j/B}^{-\sigma_j+\sigma_j^d} = \varrho_{j,R}^{\sigma_j^d}\mathbf{E}_{\mathcal{B}_j/B}^{-\sigma_j}$ (19.2), the other factors not involving σ_j^d at fixed δ_j^d: σ_j^d enters through $\varrho_{j,B}(u)^{-\sigma_j^d}\varrho_{j,R}(v)^{\sigma_j^d}$.

Slopes. The wall at $S \subseteq B$ of the bunch family is $m_S - \sum_j \big(a_j^d d_{\mathcal{B}_j/R}(S) + b_j^d d^*_{\mathcal{B}_j/R}(S) + a_j^r d_{\mathcal{B}_j|B}(S) + b_j^r d^*_{\mathcal{B}_j|B}(S)\big)$, affine in the shares; raising σ_j^d by one at fixed δ_j^d raises a_j^d, b_j^d and lowers a_j^r, b_j^r by one, so the slope is $c_{\mathcal{B}_j/R}(S) - c_{\mathcal{B}_j|B}(S)$, $c = |S| - d - d^*$. With r^+ the rank function of $\mathcal{B}_{j,B}^+$, $\mathcal{B}_j|B = \mathcal{B}_{j,B}^+ \setminus g$ has the rank function r^+ on B and $\mathcal{B}_j/R = \mathcal{B}_{j,B}^+/g$ the rank function $Z \mapsto r^+(Z+g) - 1$, and $c_Z(S) = r_Z(S) + r_Z(B \setminus S) - r_Z(B)$ for a table Z on B (7.12, 7.13); the difference is the first displayed slope. On R the table of the level d is $\mathcal{B}_j|R = \mathcal{B}_{j,R}^+ \setminus g$ and that of the level r is $\mathcal{B}_j/B = \mathcal{B}_{j,R}^+/g$, which gives the second.

Range. On the sector of an order of B, $\alpha_{\pi(1)} = \rho_0$ and $\alpha_{\pi(k)} = \rho_0 q_2 \cdots q_k$ with $q_k \in (0,1]$ and tails U_k, the letters of a table Z on B are $K_Z = \rho_0^{r_Z}\prod_k q_k^{d_Z(U_k)} k_{Z,\pi}$ and $\Psi_Z = \rho_0^{|B|-r_Z}\prod_k q_k^{d_Z^*(U_k)}\psi_{Z,\pi}$, with sector letters positive on the closed cube (8.5(ii)), hence between positive constants. So

$$\varrho_{j,B} = \frac{\mathbf{E}_{\mathcal{B}_j/R}}{\mathbf{E}_{\mathcal{B}_j|B}} = \prod_k q_k^{c_{\mathcal{B}_j|B}(U_k) - c_{\mathcal{B}_j/R}(U_k)} \times (\text{a function between positive constants}),$$

the exponent of q_k being minus the slope of U_k. Every nonempty proper subset of B is a tail of some order, the q_k range over $(0,1]$ independently, and finitely many sectors cover Δ_B: $\varrho_{j,B}$ is bounded exactly when no slope on B is positive (a positive slope at a tail U_k sends it to ∞ as $q_k \to 0$), and bounded below by a positive constant exactly when none is negative. On R, $\varrho_{j,R} = \mathbf{E}_{\mathcal{B}_j/B}/\mathbf{E}_{\mathcal{B}_j|R}$ carries q_k to the power of the slope of U_k: $\inf \varrho_{j,R} > 0$ exactly when no slope on R is positive and $\sup \varrho_{j,R} < \infty$ exactly when none is negative. A wall falls as σ_j^d decreases, toward the level d, exactly when its slope is positive, which forces $\sup \varrho_{j,B} = \infty$ or $\inf \varrho_{j,R} = 0$, so $\sup \varrho_{j,B}/\inf \varrho_{j,R} = \infty$; toward the level r, a negative slope forces $\inf \varrho_{j,B}/\sup \varrho_{j,R} = 0$. A change of the factorization of table j multiplies $\varrho_{j,B}$ and $\varrho_{j,R}$ by one constant c^{-2} (19.7), which keeps the quotient, the finiteness of the suprema, the positivity of the infima and the two ratios; the slopes are counts. □

Proposition 19.24 (The same shares closed at the gauge). *As stated: at complex exponents and masses, with the bound in absolute value.*

Proof. At width one the letters on the scale are linear in ρ (Lemma S8), and divided by their values at $\rho = 1$ they are the Skeleton polynomials of the statement (19.16): with the coefficients of the proof of 19.13, $\pi_{j,d} = A_j^K/(A_j^K + C_j^K)$, $\pi_{j,r} = 1 - \pi_{j,d}$, $\pi^*_{j,r} = A_j^\Psi/(A_j^\Psi + C_j^\Psi)$ and $\pi^*_{j,d} = 1 - \pi^*_{j,r}$, the chances of the levels under the basis law and the cobasis law at (u,v), all in $(0,1)$. Each polynomial is $1 - (1-x)\epsilon$ with a constant exceedance polynomial ϵ: $\pi_{j,r}$ for K_j and $\pi^*_{j,d}$ for Ψ_j on the bunch, and, read from the top of the scale, $\pi_{j,d}$ and $\pi^*_{j,r}$ on the room. In 15.9(ii) only $k = 0$ remains, $[x^k]\prod_i \epsilon_i^{n_i} = 0$ for $k > 0$, and the term of $\mathbf{n}$ is $\prod_i \frac{(w_i)_{n_i}}{n_i!}\epsilon_i^{n_i}\Gamma_2\left[\begin{matrix}\kappa,\ |\mathbf{n}|+1\\ \kappa+|\mathbf{n}|+1\end{matrix}\right]$; the two sectors of 15.9(i) give the display, of nonnegative terms for real data with $s_j, t_j \geq 0$. For complex exponents and masses with $\operatorname{Re}\kappa_B > 0$ and $\operatorname{Re}\kappa_B^{\text{top}} > 0$ the series converges absolutely to the same value (15.9, last sentence; 15.3 for the convergence of the scale integral). The bound: $|(w)_n| \leq (|w|)_n$; $\prod_j \pi_{j,r}^{n_j}\pi_{j,d}^{*n'_j} \leq \lambda_B^M$; $|\Gamma_2\left[\begin{matrix}\kappa_B,\ M+1\\ \kappa_B+M+1\end{matrix}\right]| = \big|\int_0^1 x^{\kappa_B-1}(1-x)^M dx\big| \leq \int_0^1 x^{\operatorname{Re}\kappa_B - 1}dx = 1/\operatorname{Re}\kappa_B$; and $\sum_{|\mathbf{n}|+|\mathbf{n}'|=M}\prod_j \frac{(|s_j|)_{n_j}(|t_j|)_{n'_j}}{n_j!\,n'_j!} = \frac{(\sum_j(|s_j|+|t_j|))_M}{M!}$, the coefficient of z^M in $\prod_j(1-z)^{-|s_j|-|t_j|}$. Likewise

on the room with λ_R and $\operatorname{Re}\kappa_B^{\text{top}}$. The bounds are summable over M, $\sum_M \frac{(W)_M}{M!}\lambda^M = (1-\lambda)^{-W}$ for $\lambda < 1$. □

Proposition 19.33 (A loop or a coloop on the simplex). *As stated: a mark that is a loop or a coloop in every table of a family.*

Proof. In a table where e is a loop no basis contains e, so $K_j = K_{j,\setminus e}$ and $\Psi_j = \alpha_e \Psi_{j,\setminus e}$; where e is a coloop every basis contains e, so $K_j = \alpha_e K_{j,/e}$ and $\Psi_j = \Psi_{j,/e}$ (7.3). The letters of F' (15.16) are these minors, of the ranks $r'_j = r_j$ at a loop and $r_j - 1$ at a coloop on $n-1$ marks, so the family word is $\alpha_e^{-\sum_{\eta_j=+1} s_j - \sum_{\eta_j=-1} t_j} W'(\alpha_{\mathcal{E}-e})$, with W' the word of F', homogeneous of degree $\kappa' - m_{\mathcal{E}-e}$, κ' the imbalance of F' at $\mathbf{m}|_{\mathcal{E}-e}$. On $\Delta_{\mathcal{E}}$ put $u = (u_e, (1-u_e)v)$, $u_e \in (0,1)$, $v \in \Delta_{\mathcal{E}-e}$, so that $u^{\mathbf{m}-\mathbf{1}} d\omega_{\mathcal{E}}(u) = u_e^{m_e-1}(1-u_e)^{m_{\mathcal{E}-e}-1} du_e\, v^{\mathbf{m}|_{\mathcal{E}-e}-\mathbf{1}} d\omega_{\mathcal{E}-e}(v)$ (3.12). Then

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_0^1 u_e^{\nu_e-1}(1-u_e)^{\kappa'-1} du_e\; P_{F'}(\mathbf{m}|_{\mathcal{E}-e};\mathbf{s},\mathbf{t}), \qquad \nu_e = m_e - \sum_{\eta_j=+1} s_j - \sum_{\eta_j=-1} t_j,$$

for real data in $[0,\infty]$, and the integral over u_e is $\Gamma_1^2 \begin{bmatrix} \nu_e,\ \kappa' \\ \nu_e + \kappa' \end{bmatrix}$ where $\nu_e, \kappa' > 0$. This ν_e is the wall $\kappa_{\{e\}}$, since $d_j(\{e\}) = 1$ and $d_j^*(\{e\}) = 0$ at a coloop and the reverse at a loop, and by 8.2(iii), with $c_j(\{e\}) = 0$ and $\varepsilon_j(\{e\}) = \eta_j$, it is $\mu_e - \sum_j \eta_j \delta_j$. The imbalance of F' is $m_{\mathcal{E}} - m_e - \sum_{\eta_j=+1} \big(s_j(r_j - 1) + t_j(n - r_j)\big) - \sum_{\eta_j=-1} \big(s_j r_j + t_j(n-1-r_j)\big) = \kappa_{\mathcal{E}} - \nu_e$, so $\nu_e + \kappa' = \kappa_{\mathcal{E}}$. Several such marks are removed one at a time, each keeping its kind in the minors, and the product of the Beta corners is $\Gamma_1^{k+1} \begin{bmatrix} (\nu_{e_i})_i,\ \kappa' \\ \kappa_{\mathcal{E}} \end{bmatrix}$, split by 3.12 as stated. On the balance wall $\kappa' = -\nu_e$, and ν_e and κ' are not both positive.

The set $S = \{e\}$ has $c_j(S) = 0$ in every table, as 16.17 asks; there $\kappa_{\{e\}} = \nu_e$ and, by 8.2(iii) and 7.13 ($c_j(\mathcal{E}-e) = c_j(\{e\}) = 0$, $\varepsilon_j(\mathcal{E}-e) = 2r_j - n - \eta_j$), $\kappa_{\mathcal{E}\setminus e} = \kappa_{\mathcal{E}} - \nu_e = \kappa'$. The integral of 16.17 over $\Delta_{\{e\}} \times \Delta_{\mathcal{E}-e}$, $\Delta_{\{e\}}$ the point $u_e = 1$ with the Dirac measure, is the integral over $\Delta_{\mathcal{E}-e}$ of $v^{\mathbf{m}|_{\mathcal{E}-e}-\mathbf{1}}$ times the family word at $(1,v)$, which is $W'(v)$: it is $P_{F'}(\mathbf{m}|_{\mathcal{E}-e};\mathbf{s},\mathbf{t})$, and the compound formula at $S = \{e\}$ is the display. □

Theorem 19.34 (Pooling is a move). *As stated: for tables with positive rational weights, and in (ii) whenever the active letters have one hinge value at rational factorizations.*

Proof. (i) is the reference, proof of thm:cleaningmove(i): the map $(\rho, u, \alpha_R) \mapsto (\rho u, \alpha_R)$ and its inverse are defined over $\mathbb{Q}$ whatever the weights, and the faces correspond as stated. After the change of variables the form is $\rho^{\kappa_B - 1}\phi(u)$ times the form of $F_{B\to g}(u)$ at the gauge (ρ, α_R) (proof of 19.8), with ϕ as in the proof of 19.10; its coefficients lie in the field of the weights and of the factors, which is $\mathbb{Q}$. The walls are matched by 19.9(i), (ii) and the gluing of the counts, for families.

(ii) The common hinge value γ is the hinge value of an active letter, a quotient of polynomials with rational coefficients, positive on the open simplex. The change $(\rho, u, \alpha_R) \mapsto (\rho\gamma(u), u, \alpha_R)$ is birational over $\mathbb{Q}$, with the inverse $(x, u, \alpha_R) \mapsto (x/\gamma(u), u, \alpha_R)$, and bijective on the open domain; by the proof of 19.10 it carries the form of (i) to the product of the form $\phi(u)\gamma(u)^{-\kappa_B} d\omega_B(u)$ of the shape and the form of G_R at the masses $(\kappa_B, \mathbf{m}|_R)$. In the datum of G_B, read on the section $|\alpha_B| = 1$ (Lemma S9), the change $y = \gamma(u)z$, birational over $\mathbb{Q}$ and bijective on the open domain, carries the form to $\phi(u)\gamma(u)^{-\kappa_B} d\omega_B(u)$ times $z^{2\sigma-\kappa_B-1}(1+z)^{-2\sigma} dz$, the form of the list on two marks whose bases are the two singletons, at the exponents $(0, 2\sigma)$ and the masses $(\kappa_B, 2\sigma - \kappa_B)$, read in its chart at the first mark, where its letter is $1 + z$. So the data of F and of that list,

and the data of G_B and G_R, are related by these moves to the same product of three data, with rational masses κ_B and $2\sigma - \kappa_B$. If P_F converges, every proper wall of F is positive (8.7) and 19.10 gives the convergence of the periods of G_B and G_R, the Beta corner, and the shape integral, which is P_F over P_{G_R}; and P_F is the first integral of the chain. □

Remark. Positive rational weights that factor across B and R factor with rational factors: fix a basis $X_\ell \sqcup Y_\ell$ at each level $\ell \in \{d(B), r(B)\}$ and put $\theta'_X = \theta_{X \sqcup Y_\ell}$ and $\theta''_Y = \theta_{X_\ell \sqcup Y}/\theta_{X_\ell \sqcup Y_\ell}$ for the patterns X and the room sides Y of the level ℓ. For any factorization $\theta = \vartheta'\vartheta''$, $\theta_{X \sqcup Y_\ell}\theta_{X_\ell \sqcup Y}/\theta_{X_\ell \sqcup Y_\ell} = \vartheta'_X \vartheta''_{Y_\ell} \vartheta'_{X_\ell} \vartheta''_Y / (\vartheta'_{X_\ell} \vartheta''_{Y_\ell}) = \theta_{X \sqcup Y}$. The hypothesis of (ii) asks more, rational factors at which the active letters share one hinge value; for one table with both letters active it asks a rational factorization with $\varrho_B = 1$.

Proposition 19.37 (Cleaning commutes with maps). *As stated, for maps between any tables.*

Proof. A bijection g that carries $(\mathcal{B}, \theta)$ to $(\mathcal{B}_1, \theta_1)$ maps bases to bases with their weights. Parallel and series bunches, the counts and the widths are defined by the bases, so they are carried to those of the image; a factorization θ', θ'' across a class B is carried to $\theta'_{1,g(p)} = \theta'_p$, $\theta''_{1,g(Y)} = \theta''_Y$, a factorization across $g(B)$, and the minors of $g(B)$ are the images of those of B with their weights, so $\gamma^K_{g(B)}(\alpha \circ g^{-1}) = \gamma^K_B(\alpha)$ and likewise for Ψ. An exchange maps T to $\mathcal{E}_1 \setminus g(T)$, with the weight θ_T. A parallel bunch B has $|T \cap B| \leq 1$ for every basis and no loop, so $g(B)$ has $|g(B) \cap T_1| \geq |B| - 1$ for every basis T_1 of the image and no coloop: a series bunch, and conversely. The patterns of $g(B)$ are the sets $g(B \setminus p)$, p a pattern of B, and putting $\theta'_{1,g(B\setminus p)} = \theta'_p$ and $\theta''_{1,g(R\setminus Y)} = \theta''_Y$ factors the weights of the image. The level of $g(B \setminus p)$ is $|B| - |p|$, so the restriction of the image to $g(B)$ has the patterns $g(B \setminus p)$, $p \in \mathcal{B}/R$, and its contraction of the room the patterns $g(B \setminus p)$, $p \in \mathcal{B}|B$; hence $K_{\mathcal{B}_1|g(B)}(\alpha \circ g^{-1}) = \sum_{p \in \mathcal{B}/R} \theta'_p \alpha^{B\setminus p} = \Psi_{\mathcal{B}/R}(\alpha)$ and $K_{\mathcal{B}_1/g(R)}(\alpha \circ g^{-1}) = \Psi_{\mathcal{B}|B}(\alpha)$, so the hinge value by K of $g(B)$ is γ^Ψ_B; likewise its hinge value by Ψ is γ^K_B. The statement on the cleanings of lists is the reference, proof of prop:cleanmaps. □

48.20 Chapter 20: words of a tree

Definition 20.2 (Side shape; half-sum and difference; pattern coefficients). As stated, for every table of a family.

Proof. The one claim is the identity $L_{j,w}(z) = \sum_p e^{\langle z_B, \chi_p \rangle} c_{j,p}(w; z_R)$. Theorem 12.7 for table j gives $\Lambda_j(z) = \sum_{p \subseteq B} e^{\langle z_B, \chi_p \rangle} \Lambda_{j,(p)}(z_R)$, with $\Lambda_{j,(p)} = 0$ off the patterns of table j. Replacing z by $-z$ gives

$$\Lambda_j(-z) = \sum_p e^{-\langle z_B, \chi_p \rangle} \Lambda_{j,(p)}(-z_R).$$

Since $\chi_{B\setminus p} = -\chi_p$ on B, the substitution $p \mapsto B \setminus p$ turns this into

$$\Lambda_j(-z) = \sum_p e^{\langle z_B, \chi_p \rangle} \Lambda_{j,(B\setminus p)}(-z_R) = \sum_p e^{\langle z_B, \chi_p \rangle} \Lambda^*_{j,(p)}(z_R).$$

So

$$w\Lambda_j(z) + (1-w)\Lambda_j(-z) = \sum_p e^{\langle z_B, \chi_p \rangle} \big(w\Lambda_{j,(p)}(z_R) + (1-w)\Lambda^*_{j,(p)}(z_R) \big).$$

□

Proposition 20.3 (The side shape, and what an exchange does to it). *As stated, for every table of a family and every bijection that carries or exchanges the family.*

Proof. For $T \subseteq \mathcal{E}$ and $z \in \mathbb{R}^{\mathcal{E}}$, with $(g^*z)_e = z_{g(e)}$,

$$\langle z, \chi_{g(T)} \rangle = \sum_{e \in \mathcal{E}} z_{g(e)}\, \chi_{g(T)}(g(e)) = \sum_{e \in \mathcal{E}} (g^*z)_e\, \chi_T(e) = \langle g^*z, \chi_T \rangle.$$

If g carries F, then $g(\mathcal{B}_j) = \mathcal{B}_{h(j)}$ with $\theta_{h(j),g(T)} = \theta_{j,T}$ (Definition 11.13), so

$$\Lambda_{h(j)}(z) = \sum_{T \in \mathcal{B}_j} \theta_{j,T}\, e^{\langle z, \chi_{g(T)} \rangle} = \Lambda_j(g^*z),$$

and $\Lambda_{h(j)}(-z) = \Lambda_j(-g^*z)$, as g^* is linear; hence $L_{h(j),w}(z) = L_{j,w}(g^*z)$. If g exchanges F, then $T \mapsto \mathcal{E} \setminus g(T)$ maps $\mathcal{B}_j$ onto $\mathcal{B}_{h(j)}$ with $\theta_{h(j),\mathcal{E}\setminus g(T)} = \theta_{j,T}$, and $\chi_{\mathcal{E}\setminus g(T)} = -\chi_{g(T)}$, so $\Lambda_{h(j)}(z) = \Lambda_j(-g^*z)$ and $\Lambda_{h(j)}(-z) = \Lambda_j(g^*z)$; hence

$$L_{h(j),w}(z) = w\, \Lambda_j(-g^*z) + (1-w)\, \Lambda_j(g^*z) = L_{j,1-w}(g^*z).$$

The display holds for any two positive numbers a, b in place of $\Lambda_j(z), \Lambda_j(-z)$ and any $s_j, t_j > 0$: it is Theorem 3.2 at the unit configuration on a two-element set with the coarsest partition, whose Zamma is $(wa + (1-w)b)^{-(s_j+t_j)} = \mathsf{Z}_{s_j+t_j;(1)}(wa + (1-w)b)$ at the shape $(w, 1-w)$, that is, the two-factor form $a^{-s_j} b^{-t_j} = \Gamma_{\frac{1}{2}}\begin{bmatrix} s_j + t_j \\ s_j,\ t_j \end{bmatrix} \int_0^1 w^{s_j - 1}(1-w)^{t_j - 1}(wa + (1-w)b)^{-s_j - t_j}\, dw$ (the reference, proof of prop:sideexchange). Taking $a = \Lambda_j(z)$ and $b = \Lambda_j(-z)$, $wa + (1-w)b = L_{j,w}(z)$.

For one table an exchange has h the identity, so that $L_w(z) = L_{1-w}(g^*z)$ for every w. Hence

$$\mathsf{Z}_{s+t;(1)}\big(L_w(g^*z)\big) = \mathsf{Z}_{s+t;(1)}\big(L_{1-w}(z)\big)\,:$$

the exchange together with $w \mapsto 1-w$ keeps the Zamma. The substitution $w \mapsto 1-w$ takes $w^{s-1}(1-w)^{t-1}$ to $w^{t-1}(1-w)^{s-1}$, which is the same function on $(0,1)$ exactly when $s = t$. □

Theorem 20.4 (The junction in pattern form). *As stated, for every family.*

Proof. The proof is that of the reference (proof of thm:junctionpattern), with a product over the tables in each step.

Signed coordinates. By Theorem 8.7 the period is the integral of $\alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, d\alpha$ over the section $\alpha_g = 1$. Put $\alpha = e^{2z}$, so that $d\alpha_e = 2\alpha_e\, dz_e$ for each of the $n-1$ marks $e \neq g$. Theorem 7.15(i) for each table gives $K_j(\alpha) = e^{\langle \mathbf{1}, z \rangle} \Lambda_j(z)$ and $\Psi_j(\alpha) = e^{\langle \mathbf{1}, z \rangle} \Lambda_j(-z)$, so, with $\sum_j (s_j + t_j) = 2\sigma$ and $\boldsymbol{\mu} = \mathbf{m} - \sigma\mathbf{1}$ (Definition 8.1),

$$\begin{aligned} \alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} &= e^{2\langle \mathbf{m}, z \rangle - 2\sigma \langle \mathbf{1}, z \rangle} \prod_j \Lambda_j(z)^{-s_j} \Lambda_j(-z)^{-t_j} \\ &= e^{2\langle \boldsymbol{\mu}, z \rangle} \prod_j \Lambda_j(z)^{-s_j} \Lambda_j(-z)^{-t_j}, \end{aligned}$$

and, with $e^{2\langle \boldsymbol{\mu}, z \rangle} = e^{2\langle \boldsymbol{\mu}|_B, z_B \rangle} e^{2\langle \boldsymbol{\mu}|_R, z_R \rangle}$,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{n-1} \int_{\mathbb{R}^{\mathcal{E}-g}} e^{2\langle \boldsymbol{\mu}, z \rangle} \prod_j \Lambda_j(z)^{-s_j} \Lambda_j(-z)^{-t_j}\, dz_{\mathcal{E}-g} \Big|_{z_g = 0}.$$

Side shapes. For j with $s_j, t_j > 0$, Proposition 20.3 writes the factor of table j as

$$\Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j} = \Gamma_{\!2}\!\begin{bmatrix} s_j+t_j \\ s_j,\ t_j \end{bmatrix} \int_0^1 w_j^{s_j-1}(1-w_j)^{t_j-1} L_{j,w_j}(z)^{-(s_j+t_j)}\, dw_j.$$

For $t_j = 0 < s_j$ the factor is $\Lambda_j(z)^{-s_j} = L_{j,1}(z)^{-s_j}$, for $s_j = 0 < t_j$ it is $\Lambda_j(-z)^{-t_j} = L_{j,0}(z)^{-t_j}$, and an inactive table contributes 1. So every active table contributes $L_{j,w_j}(z)^{-(s_j+t_j)}$, integrated over w_j when $s_j, t_j > 0$ and with the value of w_j fixed in the statement otherwise.

Scales. For each active j, $L_{j,w_j}(z) > 0$ and $L^{-a} = \Gamma(a)^{-1}\int_0^\infty y^{a-1}e^{-yL}dy$ with $a = s_j + t_j > 0$. The constants combine to $\Gamma_{\!2}\!\begin{bmatrix} s_j+t_j \\ s_j,\ t_j \end{bmatrix}\Gamma(s_j+t_j)^{-1} = \Gamma(s_j)^{-1}\Gamma(t_j)^{-1}$ when $s_j, t_j > 0$, and to $\Gamma(s_j)^{-1}$ or $\Gamma(t_j)^{-1}$ for a table at a full lean: the prefactor of the statement.

Patterns. By Definition 20.2,

$$\sum_{j \text{ active}} y_j L_{j,w_j}(z) = \sum_{p\subseteq B} e^{\langle z_B, \chi_p\rangle} C_p, \qquad C_p = \sum_{j \text{ active}} y_j\, c_{j,p}(w_j; z_R) \geq 0,$$

each $c_{j,p}(w; z_R)$ being a combination with the nonnegative coefficients w and $1-w$ of the nonnegative functions $\Lambda_{j,(p)}(z_R)$ and $\Lambda^*_{j,(p)}(z_R)$.

The Gramma. With $x = e^{z_B}$, Definition 2.1 in its logarithmic form gives

$$\int_{\mathbb{R}^B} e^{2\langle \boldsymbol{\mu}|_B, z_B\rangle} \exp\Big(-\sum_p C_p\, e^{\langle \chi_p, z_B\rangle}\Big) dz_B = \mathsf{G}_{2\boldsymbol{\mu}|_B;(\chi_p)_p}\big((C_p)_p\big),$$

the configuration having one column χ_p for each p with $C_p \neq 0$.

Order of integration. Every integrand is nonnegative, so Tonelli's theorem allows the integrals to be taken in the order of the statement: the side shapes outermost, then z_{R-g} at $z_g = 0$, then the scales y_j, then z_B. The two sides agree in $[0,\infty]$ and are finite, every proper wall being positive (Theorem 8.7). □

Remark. With one table the formula is that of the reference, and at a full lean of that table the integral over the side shape is absent, which is the reference's last sentence.

Proposition 20.5 (What a map keeping the bunch gives the coefficients). *As stated, for every family.*

Proof. Since $\tilde g$ keeps B, it maps R onto R, so $(\tilde g^* z)_R = \tilde g_R^* z_R$, and for $p \subseteq B$, substituting $e' = \tilde g(e)$,

$$\langle (\tilde g^* z)_B, \chi_p\rangle = \sum_{e\in B} z_{\tilde g(e)}\chi_p(e) = \sum_{e'\in B} z_{e'}\chi_{\tilde g(p)}(e') = \langle z_B, \chi_{\tilde g(p)}\rangle.$$

If $\tilde g$ carries F, Proposition 20.3 and Definition 20.2 give

$$\begin{aligned}\sum_{q\subseteq B} e^{\langle z_B,\chi_q\rangle} c_{h(j),q}(w; z_R) &= L_{h(j),w}(z) = L_{j,w}(\tilde g^* z) \\ &= \sum_{p\subseteq B} e^{\langle z_B, \chi_{\tilde g(p)}\rangle} c_{j,p}(w; \tilde g_R^* z_R).\end{aligned}$$

The vectors χ_q, $q \subseteq B$, are distinct, and at fixed z_R the exponentials $e^{\langle z_B,\chi_q\rangle}$ with distinct exponents are linearly independent functions of z_B: along a generic direction the largest exponent dominates (the reference, proof of prop:mapcoeff). Comparing the coefficients at $q = \tilde g(p)$ gives $c_{h(j),\tilde g(p)}(w; z_R) = c_{j,p}(w; \tilde g_R^* z_R)$. If $\tilde g$ exchanges F, the same comparison with $L_{h(j),w}(z) = L_{j,1-w}(\tilde g^* z)$ gives $c_{h(j),\tilde g(p)}(w; z_R) = c_{j,p}(1-w; \tilde g_R^* z_R)$ for every w, and $w \mapsto 1-w$ gives the statement. □

Proposition 20.6 (Level form of the half-sum). *As stated, for every table of the family of Theorem 20.4.*

Proof. Every claim concerns one table j, and the reference's proof (proof of prop:levelformQ) applies to it. The two displayed coefficients are $c_{j,p}(w;z_R)$ at $w=\frac12$ and its derivative in w (Definition 20.2).

(i) $c_{j,B\setminus p}(\frac12;-z_R) = \frac12\big(\Lambda_{j,(B\setminus p)}(-z_R)+\Lambda_{j,(p)}(z_R)\big) = c_{j,p}(\frac12;z_R)$, and $\partial_w c_{j,B\setminus p}(w;-z_R) = \Lambda_{j,(B\setminus p)}(-z_R)-\Lambda_{j,(p)}(z_R) = -\partial_w c_{j,p}(w;z_R)$.

(ii) Let τ be a transposition of two marks of B fixing the rest. As in the proof of Proposition 20.5, $L_{j,1/2}(\tau^*z)=\sum_p e^{\langle z_B,\chi_{\tau(p)}\rangle}c_{j,p}(\frac12;z_R)$. If $c_{j,p}(\frac12;\cdot)$ depends on $|p|$ alone, this is $L_{j,1/2}(z)$, since $|\tau(p)|=|p|$. Conversely, if $L_{j,1/2}(\tau^*z)=L_{j,1/2}(z)$ for every such τ, the independence of the exponentials gives $c_{j,\tau(p)}(\frac12;\cdot)=c_{j,p}(\frac12;\cdot)$, and these transpositions act transitively on the subsets of B of each size. So level form is the invariance of $L_{j,1/2}$ under these transpositions. Further,

$$L_{j,1/2}(z)=\tfrac12\sum_{T\in\mathcal{B}_j}\theta_{j,T}\big(e^{\langle z,\chi_T\rangle}+e^{\langle z,\chi_{\mathcal{E}\setminus T}\rangle}\big),$$
$$L_{j,1/2}(\tau^*z)=\tfrac12\sum_{T\in\mathcal{B}_j}\theta_{j,T}\big(e^{\langle z,\chi_{\tau(T)}\rangle}+e^{\langle z,\chi_{\mathcal{E}\setminus\tau(T)}\rangle}\big),$$

by the computation of the proof of Proposition 20.3. The exponentials $e^{\langle z,\chi_S\rangle}$, $S\subseteq\mathcal{E}$, are linearly independent, so the invariance under τ says that τ maps the multiset $\mathcal{B}_j\uplus\{\mathcal{E}\setminus T:\ T\in\mathcal{B}_j\}$, with its weights, onto itself. When $2r_j\neq n$ a basis and a complement of a basis have different sizes, τ keeps sizes, and the condition says that τ maps $\mathcal{B}_j$ onto itself with its weights, that is, carries table j (Definition 11.13), as in Proposition 14.12.

(iii) If τ carries table j, it keeps $\Lambda_j(z)$ and $\Lambda_j(-z)$ under $z\mapsto\tau^*z$; if it exchanges table j, it swaps them (Proposition 20.3, the bijection of the tables being the identity). Either way $L_{j,1/2}(\tau^*z)=L_{j,1/2}(z)$, so the half-sum has level form by (ii), and $\partial_w L_{j,w}(\tau^*z)=\pm\partial_w L_{j,w}(z)$, so the square of the difference is unchanged by τ. The transpositions of B generate the permutations of B. □

Theorem 20.11 (Building back up). *As stated; the last sentence of (iv) is stated for every molecule.*

Proof. Parts (i) to (iii), and (iv) up to its last sentence, are proved in the reference (proof of thm:buildup). The last sentence of (iv) makes three claims.

Dimension. In (iv) every table has $s_j,t_j>0$ and is active. The unknowns are the $c_{j,T}$, $T\in\mathcal{B}_j$, $\sum_j|\mathcal{B}_j|$ of them, subject to the N share sums $\sum_T c_{j,T}=t_j-s_j$ and the $n-1$ pins $-\frac12\sum_{j,T}c_{j,T}\chi_T(e)=\mu_e$, $e\in\mathcal{E}-g$. Their linear parts are independent. Indeed, let $\sum_j\lambda_j\sum_T c_{j,T}+\sum_{e\neq g}d_e\sum_{j,T}c_{j,T}\chi_T(e)$ vanish identically, and put $d_g=0$. The coefficient of $c_{j,T}$ is $\lambda_j+\langle d,\chi_T\rangle=\lambda_j+2\langle d,\mathbf{1}_T\rangle-\langle d,\mathbf{1}\rangle$, so $\langle d,\mathbf{1}_T\rangle$ is constant over the bases of every table. By Lemma 7.8, d is constant on every molecule; the family is a molecule, so d is constant, and $d_g=0$ makes it 0; then every $\lambda_j=0$. So the fibre of differences has dimension $\sum_j|\mathcal{B}_j|-N-(n-1)=\sum_j(|\mathcal{B}_j|-1)-(n-1)$.

The pins at the centre. At the centre $t_j=s_j$ and $\boldsymbol{\mu}=0$, so the share sums read $\sum_T c_{j,T}=0$. With $\chi_T=2\mathbf{1}_T-\mathbf{1}$,

$$\sum_T c_{j,T}\chi_T(e)=2\big(M_j^{\mathsf T}\mathbf{c}_j\big)_e-\sum_T c_{j,T},$$

so on the share sums the pin of $e \neq g$ reads $(\sum_j M_j^{\mathsf{T}} \mathbf{c}_j)_e = 0$. The equation at g follows, since $\sum_{e \in \mathcal{E}} (M_j^{\mathsf{T}} \mathbf{c}_j)_e = \sum_T c_{j,T} |T| = r_j \sum_T c_{j,T} = 0$. So the pins and the share sums together are equivalent to $\sum_T c_{j,T} = 0$ for every j and $\sum_j M_j^{\mathsf{T}} \mathbf{c}_j = 0$.

The contour. At the centre the gauge $\mathbf{1}$ is native (Proposition 8.21(i)). The leaves are opened by the atom formula with one share per basis and letter (Corollary 18.16), and the native split there (Definition 12.25) gives the basis T of table j the shares $a_{j,T} = s_j \theta_{j,T} / K_j(\mathbf{1})$ and $b_{j,T} = t_j \theta_{j,T} / \Psi_j(\mathbf{1})$, the probabilities of T under the basis law and the cobasis law at $\mathbf{1}$ times the exponents. As $\Psi_j(\mathbf{1}) = \sum_T \theta_{j,T} = K_j(\mathbf{1})$ and $s_j = t_j$, $c_{j,T} = b_{j,T} - a_{j,T} = 0$ there, so the wall contour of the native split (Theorem 18.13(iii)) is $\operatorname{Re} \mathbf{c} = 0$. □

Remark. With one table $\sum_T c_T = \frac{1}{r} \sum_e (M^{\mathsf{T}} \mathbf{c})_e$ follows from $M^{\mathsf{T}} \mathbf{c} = 0$, and the dimension is $|\mathcal{B}_1| - n$: the reference's case of one table at the centre.

Proposition 20.12 (The draw series). *As stated; the reading of its last part is stated for every family.*

Proof. The first part, through the Trailmoment and the Dirichlet moments, is the reference's proposition (proof of prop:jointdraw). For the last part take one block, $q = 2N$, $P_{2j-1} = K_j$ with the index s_j and $P_{2j} = \Psi_j$ with the index t_j; an index 0 contributes the factor 1.

Constants. K_j is homogeneous of degree r_j, with the squarefree monomials u^T, $T \in \mathcal{B}_j$, and the coefficients $\theta_{j,T}$. For $\mathbf{a} = \mathbf{1}_T$, $b_{\mathbf{a}} = r_j! / \prod_e a_e! = r_j!$, and every other $\mathbf{a}$ has $[u^{\mathbf{a}}] K_j = 0$; so $C_{2j-1} = \theta_{j,\max} / r_j!$. Likewise Ψ_j has degree $n - r_j$ and the monomials $u^{\mathcal{E} \setminus T}$ with the coefficients $\theta_{j,T}$, so $C_{2j} = \theta_{j,\max} / (n - r_j)!$.

The period. $\mathbb{E}_{U \sim \mathrm{Dir}_{\mathbf{m}}}[f(U)] = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}^{-1} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} f \, d\omega_{\mathcal{E}}$ for $f \geq 0$ (Theorem 3.9), so the left side times $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$ is the stated integral, which on the balance wall is $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ (Theorem 8.7).

The urn. For u in the simplex, $(\sum_e u_e)^g = \sum_{|\mathbf{a}|=g} \frac{g!}{\mathbf{a}!} u^{\mathbf{a}} = 1$ is the total chance of g independent draws from u, grouped by their counts $\mathbf{a}$. For $P_{2j-1} = K_j$,

$$\frac{P_{2j-1}}{C_{2j-1}} = \sum_{T \in \mathcal{B}_j} \frac{\theta_{j,T}}{\theta_{j,\max}} \, r_j! \, u^T$$

is the chance that a round of r_j draws is a basis T of table j and does not fail, a round that is the basis T failing with probability $1 - \theta_{j,T} / \theta_{j,\max}$; so $Q_{2j-1} = 1 - P_{2j-1} / C_{2j-1}$ is the chance that the round fails, and likewise Q_{2j} for a round of $n - r_j$ draws and the cobases. Given u, independent rounds all fail with the chance $\prod_i Q_i(u)^{k_i}$. The urn that draws a mark with probability proportional to its current mass and then adds one to that mass gives a sequence of draws with the counts $\mathbf{b}$ the chance $\prod_e (m_e)_{b_e} / (m_{\mathcal{E}})_{|\mathbf{b}|} = \mathbb{E}_{U \sim \mathrm{Dir}_{\mathbf{m}}}[U^{\mathbf{b}}]$ (Theorem 3.11), the chance of the same sequence when $U \sim \mathrm{Dir}_{\mathbf{m}}$ is drawn first and the marks are then drawn independently from U. Summing over the sequences whose rounds all fail, $\mathbb{E}_{U \sim \mathrm{Dir}_{\mathbf{m}}}[\prod_i Q_i(U)^{k_i}]$ is the urn's chance of the event of the statement; the order of the rounds does not change it, the product being symmetric in its factors. For lists every ratio $\theta_{j,T} / \theta_{j,\max}$ is 1, the Q_i have integer coefficients, and the expectation is a finite sum of Dirichlet moments.

Theorem 15.13(i). With the references $M_i = C_i$, on the simplex

$$M_i^{-w_i - n_i} (M_i - L_i)^{n_i} = C_i^{-w_i} (1 - L_i / C_i)^{n_i} = C_i^{-w_i} Q_i^{n_i},$$

so the series of Theorem 15.13(i), divided by $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$, is the present series term by term. □

Corollary 20.13 (At $\delta = 0$ and at a full lean the period reads the law of one letter for each table). *As stated, for every family.*

Proof. Letters. For a table riding the line, $K_j^{-\sigma_j}\Psi_j^{-\sigma_j} = \mathbf{E}_j^{-\sigma_j}$; at $(s_j, 0)$ the factor is $K_j^{-s_j}$ and at $(0, t_j)$ it is $\Psi_j^{-t_j}$. So on the open simplex

$$\prod_j K_j^{-s_j}\Psi_j^{-t_j} = \prod_j L_j^{-\nu_j} = \mathsf{Z}_{\boldsymbol{\nu};I_N}\big((L_j)_j\big),$$

the Zamma at the unit configuration being $\prod_j \xi_j^{-\nu_j}$ (Definition 2.12).

The period. By Theorem 8.7 and the Dirichlet density (Theorem 3.9),

$$\begin{aligned} P_F(\mathbf{m};\mathbf{s},\mathbf{t}) &= \int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j}\Psi_j^{-t_j}\, d\omega_\mathcal{E} \\ &= \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_\mathcal{E} \end{bmatrix} \mathbb{E}_{U\sim \mathrm{Dir}_\mathbf{m}}\Big[\mathsf{Z}_{\boldsymbol{\nu};I_N}\big((L_j(U))_j\big)\Big]. \end{aligned}$$

The expectation is the integral of the kernel against the image $\phi_\mathbf{m}$ of $\mathrm{Dir}_\mathbf{m}$ under $(L_j)_j$, the Trailmoment $(\mathsf{T}_{\boldsymbol{\nu};I_N}[\mathrm{id}]\,\phi_\mathbf{m})((0,\infty)^N)$ with the identity as coefficient function (Definition 5.1), as in the display of Theorem 18.1(i). With one table it is the Trailmoment $(\mathsf{T}_{\nu;(1)}[x]\,\phi_\mathbf{m})((0,\infty))$ of a law on the half-line, at $\delta = 0$ the image of $\mathrm{Dir}_\mathbf{m}$ under the even letter (Theorem 5.9).

The law. Each L_j is a polynomial, positive on the open simplex and continuous on the closed one, and $\mathrm{Dir}_\mathbf{m}$ charges only the open simplex; so $\phi_\mathbf{m}$ is carried by $\prod_j(0, \max_{\Delta_\mathcal{E}} L_j]$. A law carried by a compact set is determined by its moments, polynomials being dense in the continuous functions on that set. Every moment $\mathbb{E}[\prod_j L_j(U)^{k_j}]$ is the expectation of a polynomial in U, a finite sum of Dirichlet moments (Theorem 3.11). □

Theorem 20.14 (The draw series along a chain). *As stated, for every family of tables.*

Proof. The proof is that of the reference (proof of thm:drawchain), with the acceptances and a product over the tables.

The failures split. Fix table j and a round of r_j draws. A round that extends on S_{i+1} extends on S_i, with the trace $P_i = P_{i+1} \cap S_i$, and every round extends on $S_0 = \varnothing$ with the trace $\varnothing$. So the events A_i, that the round extends on S_i and is accepted at $S_1, \dots, S_i$, decrease with i, A_0 is certain, and $\mathfrak{f}_{j,i}$ is the chance of $A_i \setminus A_{i+1}$; hence $\sum_{i=0}^d \mathfrak{f}_{j,i} = 1 - \Pr(A_{d+1})$.

The acceptances telescope. A round extends on $S_{d+1} = \mathcal{E}$ exactly when it is a basis T of table j with distinct draws, which has the chance $r_j!\,u^T$; its traces are then $P_i = T \cap S_i$, and its acceptances multiply to

$$\prod_{i=0}^{d} \frac{\theta_j(S_{i+1}; T\cap S_{i+1})}{\theta_j(S_i; T\cap S_i)} = \frac{\theta_j(\mathcal{E};T)}{\theta_j(\varnothing;\varnothing)} = \frac{\theta_{j,T}}{\theta_{j,\max}}.$$

So $\Pr(A_{d+1}) = \sum_T r_j!\,u^T\theta_{j,T}/\theta_{j,\max} = r_j!\,K_j(u)/\theta_{j,\max}$. The cobasis rounds are the basis rounds of the dual table, whose bases $\mathcal{E}\setminus T$ carry the weights $\theta_{j,T}$, whose largest weight is $\theta_{j,\max}$ and whose basis polynomial is Ψ_j (Proposition 7.2); so $\sum_i \mathfrak{f}^*_{j,i} = 1 - (n-r_j)!\,\Psi_j/\theta_{j,\max}$.

Polynomials. Each $\mathfrak{f}_{j,i}(u)$ is a finite sum, over the sequences of r_j draws, of the monomial of the sequence times a product of acceptance and rejection probabilities in $[0,1]$: a polynomial with nonnegative coefficients, with integer coefficients for lists, whose rounds are always accepted.

The series. Proposition 20.12, with one block and the constants $C_{2j-1} = \theta_{j,\max}/r_j!$, $C_{2j} = \theta_{j,\max}/(n-r_j)!$ of its last part, gives

$$\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\Big[\prod_j K_j^{-s_j}\Psi_j^{-t_j}\Big] = \prod_j C_{2j-1}^{-s_j}C_{2j}^{-t_j}\sum_{\mathbf{k}\in\mathbb{N}^{2N}}\prod_j \frac{(s_j)_{k_{2j-1}}(t_j)_{k_{2j}}}{k_{2j-1}!\,k_{2j}!} \times \mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\Big[\prod_j\Big(\sum_i \mathfrak{f}_{j,i}\Big)^{k_{2j-1}}\Big(\sum_i \mathfrak{f}^*_{j,i}\Big)^{k_{2j}}\Big],$$

both sides finite or infinite together. The multinomial theorem expands

$$\frac{(s_j)_k}{k!}\Big(\sum_i \mathfrak{f}_{j,i}\Big)^k = \sum_{|\mathbf{k}_j|=k}\frac{(s_j)_{|\mathbf{k}_j|}}{\mathbf{k}_j!}\prod_i \mathfrak{f}_{j,i}^{k_{j,i}},$$

with nonnegative terms, and likewise for the cobasis rounds; Tonelli's theorem exchanges the sums and the expectation, and multiplying by $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ (Theorem 3.9) gives the display. Each expectation is that of a polynomial with nonnegative coefficients, a finite nonnegative combination of Dirichlet moments (Theorem 3.11).

Nesting. $\sum_{i'\ge i} k_{j,i'}$ is a tail sum of nonnegative integers, so it decreases with i.

Bijections. Let g keep each B_i, hence each S_i, and the masses. A sequence of draws D from u has the chance of the sequence $g(D)$ from g_*u. If g maps the bases of table j onto those of table $h(j)$ with their weights, then $g(T\cap S_i) = g(T)\cap S_i$, so D extends on S_i for table j exactly when $g(D)$ extends on S_i for table $h(j)$, the traces corresponding under g, and $\theta_j(S_i;P) = \theta_{h(j)}(S_i;g(P))$, so the acceptances agree: $\mathfrak{f}_{h(j),i}(g_*u) = \mathfrak{f}_{j,i}(u)$, and likewise for the cobasis rounds, g carrying the dual tables as well. If $T\mapsto \mathcal{E}\setminus g(T)$ maps the bases of table j onto those of table $h(j)$ with their weights, then $g(T)$ is a cobasis of table $h(j)$ for every basis T of table j, the round lengths agree, $r_j = n - r_{h(j)}$, the largest weights over the traces agree in the same way, and $\mathfrak{f}^*_{h(j),i}(g_*u) = \mathfrak{f}_{j,i}(u)$, $\mathfrak{f}_{h(j),i}(g_*u) = \mathfrak{f}^*_{j,i}(u)$. Since the masses are kept, $g_*U\sim\mathrm{Dir}_{\mathbf{m}}$ when $U\sim\mathrm{Dir}_{\mathbf{m}}$, so in the second case

$$\mathbb{E}\Big[\prod_{j,i}\mathfrak{f}_{j,i}(U)^{k_{j,i}}\mathfrak{f}^*_{j,i}(U)^{l_{j,i}}\Big] = \mathbb{E}\Big[\prod_{j,i}\mathfrak{f}^*_{h(j),i}(U)^{k_{j,i}}\mathfrak{f}_{h(j),i}(U)^{l_{j,i}}\Big],$$

the expectation at the indices $(\mathbf{k}'_{j'},\mathbf{l}'_{j'}) = (\mathbf{l}_{h^{-1}(j')},\mathbf{k}_{h^{-1}(j')})$. With one table h is the identity, an exchange maps the bases onto complements of bases, so $r = n-r$, and the array is symmetric under $\mathbf{k}\leftrightarrow\mathbf{l}$. □

Theorem 20.15 (The draw series along a tree). *As stated, for every molecule whose children's weights factor across the childcules.*

Proof. The proof is that of the reference (proof of thm:drawtree), the weights entering through the acceptances; the coordinates of the tree and their independence (Theorem 3.12 at every node) are unchanged.

(i) Let $\mathcal{B}'$ have rank r' at the node v, and write $R_i = R_{v,i}$, $x = x^{(v)}$, $u = u^{(v)}$ and $v^{(i)}$ for the shape of R_i. By Proposition 7.9 the bases of $\mathcal{B}'$ are the sets $T = p\sqcup\bigsqcup_i X_i$, p a pattern at B_v and X_i a basis of $\mathcal{B}'_{(p)}|R_i$ of size $r'_{p,i}$, and by the hypothesis $\theta_T = \prod_i \theta_{(p),i}(X_i)$. The shape of M_v is $x_{B_v}u$ on B_v and $x_{R_i}v^{(i)}$ on R_i, so by homogeneity

$$K_{\mathcal{B}'} = \sum_p x_{B_v}^{|p|}u^p\prod_i x_{R_i}^{r'_{p,i}}K_{\mathcal{B}'_{(p)}|R_i}\big(v^{(i)}\big),$$

the restrictions carrying the weights $\theta_{(p),i}$. Multiply by $r'!/\theta_{\max}(\mathcal{B}')$, write $r'! = \frac{r'!}{\prod_i r'_{p,i}!}\prod_i r'_{p,i}!$, and multiply and divide the term of p by

$$\prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big);$$

this gives $\mathfrak{s}_{\mathcal{B}'} = \sum_p \mathfrak{p}_{\mathcal{B}',p}\prod_i \mathfrak{s}_{\mathcal{B}'_{(p)}|R_i}$. The hypothesis fixes the weights of the restrictions up to constant factors whose product is 1, which change neither the $\mathfrak{s}$ of the restrictions nor the first factor of $\mathfrak{p}_{\mathcal{B}',p}$. That factor is at most 1, since

$$\theta_{\max}(\mathcal{B}') = \max_p \max_{(X_i)_i}\prod_i \theta_{(p),i}(X_i) = \max_p \prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big).$$

The chance that r' independent draws from the shape of M_v are the basis T with distinct draws is $r'!$ times its monomial, and the acceptance with probability $\theta_T/\theta_{\max}(\mathcal{B}')$ gives the chance $r'!K_{\mathcal{B}'}/\theta_{\max}(\mathcal{B}')$ of an accepted basis. The round passes v showing p, its draws in B_v being the marks of p and its numbers of draws in the R_i the $r'_{p,i}$, with the chance $\frac{r'!}{\prod_i r'_{p,i}!}x_{B_v}^{|p|}u^p\prod_i x_{R_i}^{r'_{p,i}}$, and given this its draws in R_i are $r'_{p,i}$ independent draws from $v^{(i)}$. The acceptance splits as

$$\frac{\theta_T}{\theta_{\max}(\mathcal{B}')} = \frac{\prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big)}{\theta_{\max}(\mathcal{B}')}\ \prod_i \frac{\theta_{(p),i}(X_i)}{\theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big)},$$

an acceptance at v followed by independent acceptances in the childcules. So $\mathfrak{p}_{\mathcal{B}',p}$ is the chance of passing v showing p and being accepted there, and $\mathfrak{s}_{\mathcal{B}'_{(p)}|R_i}$ the chance of an accepted basis of the restriction in R_i.

(ii) By (i) the round fails at v with the chance $1-\sum_p \mathfrak{p}_{\mathcal{B}',p}$, and otherwise passes v showing one pattern p and is accepted there; it then first fails at w in the subtree of R_i exactly when it succeeds in $R_1,\dots,R_{i-1}$ and first fails at w inside R_i, which is the recursion. Write $\mathfrak{p}_{\mathcal{B}',p} = a_p\,\mathfrak{p}^\circ_{\mathcal{B}',p}$ with $a_p \le 1$ the first factor. Then

$$1-\sum_p \mathfrak{p}_{\mathcal{B}',p} = \Big(1-\sum_p \mathfrak{p}^\circ_{\mathcal{B}',p}\Big) + \sum_p (1-a_p)\,\mathfrak{p}^\circ_{\mathcal{B}',p}.$$

The bracket is $1 = (x_{B_v}\sum_{e\in B_v}u_e + \sum_i x_{R_i})^{r'}$, expanded by the multinomial theorem, less some of its terms, distinct patterns giving distinct terms (the reference, proof of thm:drawtree, part (ii)); it has nonnegative coefficients in the coordinates of v, and so has the sum. With $1-\prod_i S_i = \sum_i \prod_{i'<i} S_{i'}(1-S_i)$ and induction on the subtree, every $\mathfrak{f}_{\mathcal{B}'\to w}$ is a finite sum of products, over the blocks of the tree, of polynomials with nonnegative coefficients in one block each.

(iii) On the simplex $r_j!K_j/\theta_{j,\max} = \mathfrak{s}_{\mathcal{B}_j} = 1-\sum_w \mathfrak{f}_{j,w}$, with the sum in $[0,1)$ on the interior. The letter Ψ_j is the basis polynomial of the dual table (Proposition 7.2), whose bases $\mathcal{E}\setminus T$ carry the weights $\theta_{j,T}$, so that its largest weight is $\theta_{j,\max}$; its children are the duals of the children with the same weights (Lemma 14.7(ii) with the identity), and the dual of a direct sum is the direct sum of the duals of its parts with the product weights, so the factoring passes to the dual and (i), (ii) apply to its rounds. Then $K_j^{-s_j} = (r_j!/\theta_{j,\max})^{s_j}(1-\sum_w \mathfrak{f}_{j,w})^{-s_j}$, expanded as in Proposition 20.12 and by the multinomial theorem as in Theorem 20.14, with nonnegative terms; Tonelli's theorem, the independence of the blocks under $\mathrm{Dir}_{\mathbf{m}}$ (Theorem 3.12) and Theorem 3.11 give the series and the form of its expectations. On the balance wall the left side is the period (Theorem 8.7).

(iv) A round that first fails in the subtree of w passes every node above w, which gives the nesting. On a chain every node has one childcule. The table at the node M_k that a round of table j reaches is the child of table j at its trace P_k on S_k, whose largest weight is $\theta_j(S_k;P_k)$ in the notation of Theorem 20.14; so the first factor of $\mathfrak{p}$ at that node is $\theta_j(S_{k+1};P_{k+1})/\theta_j(S_k;P_k)$, the acceptance at S_{k+1}, and passing the node is extending on S_{k+1}. The chances $\mathfrak{f}$ of the two theorems agree, and the series is that of Theorem 20.14. □

Proposition 20.17 (Rates at the centre). *As stated, for every family whose active tables are connected.*

Proof. (i) is Proposition 8.21(ii), with $c_j(\mathcal{E}\setminus S)=c_j(S)$ (Theorem 7.13).

(ii) The rate is $\varpi=\min_i\varpi_i$ over the letters, ϖ_i the least of $\kappa_S/d_i(S)$ over the nonempty proper S with $d_i(S)>0$ (Theorem 15.13(ii)); an inactive table has $\sigma_j=0$ and no letters. A connected table has $c_j(S)\geq 1$ on every nonempty proper S (Definition 8.1), and so no loops and no coloops, a loop or a coloop e having $c_j(\{e\})=r_j(\{e\})-d_j(\{e\})=0$ (Theorem 7.13) with $\{e\}$ proper as $n\geq 3$. So $\kappa_S=\sum_j\sigma_j c_j(S)\geq\sum_j\sigma_j=\sigma$.

For the letter K_j, $d_j(S)=|S|-r_j^*(S)=r_j-r_j(\mathcal{E}\setminus S)\leq r_j-1$ (Definition 7.12), as $\mathcal{E}\setminus S$ is nonempty and contains no loop. For Ψ_j, $d_j^*(S)=|S|-r_j(S)$ is the nullity of S, which increases with the set, and $S\subseteq\mathcal{E}-e$ for some e, of nullity $n-1-r_j(\mathcal{E}-e)=n-1-r_j$, as e is not a coloop. So $\kappa_S/d_j(S)\geq\sigma/(r_j-1)$ wherever $d_j(S)>0$, and $\kappa_S/d_j^*(S)\geq\sigma/(n-r_j-1)$ wherever $d_j^*(S)>0$.

At $S=\mathcal{E}-e$, every table without loops or coloops has $c_j(\mathcal{E}-e)=c_j(\{e\})=r_j(\{e\})-d_j(\{e\})=1$ (Theorem 7.13), so $\kappa_{\mathcal{E}-e}=\sigma$; and $d_j(\mathcal{E}-e)=r_j-r_j(\{e\})=r_j-1$, $d_j^*(\mathcal{E}-e)=n-1-r_j$. Hence $\varpi_{K_j}=\sigma/(r_j-1)$ when $r_j\geq 2$ and $\varpi_{\Psi_j}=\sigma/(n-r_j-1)$ when $n-r_j\geq 2$, attained at the faces $\mathcal{E}-e$; a letter with no S of positive count does not enter the minimum. As $(r_j-1)+(n-r_j-1)=n-2\geq 1$, at least one letter of every active table enters, and $\varpi=\sigma/\max_j\max(r_j-1,n-r_j-1)$.

(iii) By Definition 7.3, $K_j=K_{j,\setminus e}+u_eK_{j,/e}$ and $\Psi_j=u_e\Psi_{j,\setminus e}+\Psi_{j,/e}$, the deletion and the contraction being tables since e is neither a loop nor a coloop of the active table j. They are homogeneous in $u_{\mathcal{E}-e}$ of the degrees r_j, r_j-1, $n-1-r_j$ and $n-r_j$; substituting $u_e=1-x$ and $u_{\mathcal{E}-e}=xv$ gives the display, and as $x\to 0$ the terms of $x^{r_j-1}K_{j,/e}(v)$ and $x^{n-r_j-1}\Psi_{j,\setminus e}(v)$ dominate.

(iv) Let β carry the family, with $v'_{\beta(e')}=v_{e'}$ on $\mathcal{E}-e$. It maps the bases of table j onto those of table $h(j)$ with their weights, the bases through e onto those through f, so

$$K_{h(j),/f}(v')=\sum_{T\ni e}\theta_{j,T}\,{v'}^{\beta(T)-f}=\sum_{T\ni e}\theta_{j,T}\,v^{T-e}=K_{j,/e}(v),$$

and in the same way $\Psi_{h(j),\setminus f}(v')=\Psi_{j,\setminus e}(v)$, the bases avoiding e going to those avoiding f. Let β exchange the family: $T\mapsto\mathcal{E}\setminus\beta(T)$ maps the bases of table j through e onto the bases of table $h(j)$ avoiding f, with their weights, so

$$\Psi_{h(j),\setminus f}(v')=\sum_{T\ni e}\theta_{j,T}\,{v'}^{(\mathcal{E}-f)\setminus(\mathcal{E}\setminus\beta(T))}=\sum_{T\ni e}\theta_{j,T}\,{v'}^{\beta(T)-f}=K_{j,/e}(v),$$

and in the same way $K_{h(j),/f}(v')=\Psi_{j,\setminus e}(v)$. Either way the vertex product $K_{j,/e}\Psi_{j,\setminus e}$ goes to $K_{h(j),/f}\Psi_{h(j),\setminus f}$. With one table, a bijection taking e to f that carries or exchanges it exists exactly when e and f lie in one class at the centre, by Corollary 11.16 at the table's gauge, the gauge of the centre. □

Remark. Connectedness of every active table is needed for (ii). On the marks $\{0,1,2,3\}$, for the family of the list whose bases are all the two-element sets and the disconnected list with the bases $\{a,b\}$, $a\in\{0,2\}$, $b\in\{1,3\}$, at the strengths $(1,3)$, the face $S=\{0,2\}$ has $\kappa_S=1\cdot 2+3\cdot 0=2$ and $d_2(S)=1$, so $\varpi\le 2$, while $\sigma/\max_j\max(r_j-1,n-r_j-1)=4$.

Corollary 20.19 (Merged Skeleton values). *As stated, at every pair of exponents and every mass on the balance wall.*

Proof. The letters. With $\theta_{\mathcal{E}\setminus T}=\theta_T$, $\Psi(\alpha)=\sum_T\theta_T\alpha^{\mathcal{E}\setminus T}=\sum_T\theta_{\mathcal{E}\setminus T}\alpha^{\mathcal{E}\setminus T}=K(\alpha)$, the last sum running over the same bases. For a mark e the bases $T\not\ni e$ correspond to the bases $T'=\mathcal{E}\setminus T\ni e$, with $(\mathcal{E}-e)\setminus T=T'-e$ and $\theta_T=\theta_{T'}$, so

$$\Psi_{\setminus e}=\sum_{T\not\ni e}\theta_T\,\alpha^{(\mathcal{E}-e)\setminus T}=\sum_{T'\ni e}\theta_{T'}\,\alpha^{T'-e}=K_{/e}$$

(Definition 7.3), and likewise $\Psi_{/e}=K_{\setminus e}$. The Skeleton values of the scale of e are $K_{\setminus e}/(\alpha_e K_{/e})$ and $\Psi_{/e}/(\alpha_e\Psi_{\setminus e})$ (Corollary 15.4), which coincide.

The table. Bases and complements of bases are both bases, so $r=n-r$. A coloop lies in every basis, hence in no complement of one, and a loop lies in no basis, hence in every complement; both contradict the complementation, so no mark is a loop or a coloop, and the deletion $(\mathcal{B}\setminus e,\theta)$ is a table of rank r on $\mathcal{E}-e$.

The walls at $\{e\}$ and at $\mathcal{E}-e$. $d(\{e\})=d^*(\{e\})=0$, so $\kappa_{\{e\}}=m_e$. Also $d(\mathcal{E}-e)=r-r(\{e\})=r-1$ and $d^*(\mathcal{E}-e)=n-1-r(\mathcal{E}-e)=n-1-r=r-1$. On the balance wall $m_{\mathcal{E}}=sr+t(n-r)=(s+t)r$, so $\kappa_{\mathcal{E}-e}=m_{\mathcal{E}}-m_e-(s+t)(r-1)=s+t-m_e$. The proper walls being positive, $0<m_e<s+t$.

Integrating the mark. Let $g\neq e$ and $R=\mathcal{E}-e$. By Theorem 8.7 the period is the integral of $\alpha^{\mathbf{m}-\mathbf{1}}K^{-s}\Psi^{-t}=\alpha^{\mathbf{m}-\mathbf{1}}K^{-(s+t)}$ over the section $\alpha_g=1$. By Tonelli's theorem the integral over $\alpha_e=x$ may be taken first, at fixed α_R: with $K=K_{\setminus e}+xK_{/e}$, the Beta corner of a linear letter (Proposition 15.1 with one coefficient) gives

$$\int_0^\infty x^{m_e-1}\big(K_{\setminus e}+xK_{/e}\big)^{-(s+t)}dx=\Gamma\begin{bmatrix}m_e,\ s+t-m_e\\ s+t\end{bmatrix}K_{\setminus e}^{-(s+t-m_e)}K_{/e}^{-m_e},$$

convergent as $0<m_e<s+t$, and $K_{/e}=\Psi_{\setminus e}$.

The deletion. What remains is $\Gamma\begin{bmatrix}m_e,\ s+t-m_e\\ s+t\end{bmatrix}$ times the integral, over the section $\alpha_g=1$ of the orthant of R, of

$$\alpha_R^{\mathbf{m}|_R-\mathbf{1}}K_{\setminus e}^{-(s+t-m_e)}\Psi_{\setminus e}^{-m_e}\,d\alpha_{R-g},$$

the family word of the deletion at the exponents $(s+t-m_e,m_e)$. Its balance wall holds: $m_{\mathcal{E}-e}=(s+t)r-m_e$, which is $(s+t-m_e)r+m_e(n-1-r)$ as $n-1-r=r-1$. By Theorem 8.7 the section integral is $P_{\mathcal{B}\setminus e}(\mathbf{m}|_{\mathcal{E}-e};s+t-m_e,m_e)$; it is finite because P_F is, so the walls of the deletion are positive there. Since $K=\Psi$, $P_F(\mathbf{m};s,t)$ depends on the exponents only through $s+t$. At $s=t=\sigma$ and $\mathbf{m}=\sigma\mathbf{1}$, $m_e=\sigma$, which is the last identity. □

Theorem 20.20 (The scale integral of one mark). *As stated; parts (i) and (ii) are stated for every family.*

Proof. (i) By Theorem 15.3 with $B=\{e\}$ and $m_B=m_e$, at the gauge (α_e,α_R),

$$\mathrm{Sc}_{\{e\}}(\alpha;\mathbf{m};\mathbf{s},\mathbf{t})=\int_0^\infty x^{m_e-1}\prod_j K_j(x\alpha_e,\alpha_R)^{-s_j}\Psi_j(x\alpha_e,\alpha_R)^{-t_j}\,dx,$$

convergent since the walls $\kappa_{\{e\}}$ and $\kappa^{\mathrm{top}}_{\{e\}}$ are positive. At $\alpha_e = 1$, Definition 7.3 gives $K_j = K_{j,\backslash e}(\alpha_R) + xK_{j,/e}(\alpha_R)$ and $\Psi_j = x\Psi_{j,\backslash e}(\alpha_R) + \Psi_{j,/e}(\alpha_R)$ for every active table, the deletion and the contraction being tables on R as e is neither a loop nor a coloop there. At $\alpha_R = e^{2z_R}$, Theorem 7.15(i) for these tables gives $K_{j,\star}(\alpha_R) = e^{\langle \mathbf{1}, z_R\rangle}\Lambda_{j,\star}(z_R)$ and $\Psi_{j,\star}(\alpha_R) = e^{\langle \mathbf{1}, z_R\rangle}\Lambda_{j,\star}(-z_R)$ for $\star \in \{\backslash e, /e\}$. So

$$K_j = e^{\langle \mathbf{1}, z_R\rangle}\big(\Lambda_{j,\backslash e}(z_R) + x\Lambda_{j,/e}(z_R)\big),$$
$$\Psi_j = e^{\langle \mathbf{1}, z_R\rangle}\big(\Lambda_{j,/e}(-z_R) + x\Lambda_{j,\backslash e}(-z_R)\big),$$

and the powers carry the factor $e^{-\sum_j (s_j+t_j)\langle \mathbf{1}, z_R\rangle} = e^{-2\sigma\langle \mathbf{1}, z_R\rangle}$, an inactive table contributing 1. This is the display, and the rest of the integrand depends on the room only through the functions $\Lambda_{j,\backslash e}$ and $\Lambda_{j,/e}$. Corollary 16.6 at the bunch $\{e\}$, whose shape simplex is the point $u_e = 1$, gives $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \alpha_R^{\mathbf{m}|_R - \mathbf{1}} \operatorname{Sc}_{\{e\}}((1, \alpha_R); \mathbf{m}; \mathbf{s}, \mathbf{t})\, d\alpha_{R-g}$ on the section $\alpha_g = 1$, $g \in R$. With $\alpha_R = e^{2z_R}$, $\alpha_R^{\mathbf{m}|_R - \mathbf{1}} d\alpha_{R-g} = 2^{n-2} e^{2\langle \mathbf{m}|_R, z_R\rangle} dz_{R-g}$ there, $R - g$ having $n - 2$ marks.

(ii) A realization that is an atom is a realization of a molecule (Definition 10.1), so F is connected, and Theorem 11.4 gives $\alpha^\star \propto \mathbf{1}$. By Theorem 18.1(ii), in $y = \log u$ modulo $\mathbb{R}\mathbf{1}$ the law $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ has a density proportional to $e^{\langle \boldsymbol{\tau}, y\rangle - \Phi(y)}$. The target is the usage at $\alpha^\star$ (Definition 8.17),

$$\tau_e = \sum_j \big(s_j R_{j,e}(\alpha^\star) + t_j R^*_{j,e}(\alpha^\star)\big) = \partial_{y_e}\Phi(y^\star), \qquad y^\star = \log \alpha^\star,$$

with the potential Φ of Definition 8.18. With $y = y^\star + v$ the density is proportional to

$$\exp\big(\langle \nabla\Phi(y^\star), v\rangle - \Phi(y^\star + v)\big).$$

By Proposition 10.4 there are orders $\mathbf{g}_e$ of the rooms in which the potential written from e at the gauge $\alpha^\star$ (Definition 10.1),

$$(v_0, v_1, \ldots, v_{n-1}) \longmapsto \Phi_{\mathbf{X}}\Big(y^\star + v_0\mathbf{1}_e + \sum_k v_k \mathbf{1}_{g_k}\Big) \qquad (\mathbf{X} \in \{\mathbf{E}, \mathbf{O}\}),$$

is one pair of functions for every e. Hence $v \mapsto \Phi(y^\star + v)$ read in the coordinates $(v_e, v_{g_1}, \ldots, v_{g_{n-1}})$ is one function for every e, and so is its gradient at 0; the density read in these coordinates is one function for every e, and dv is invariant under the renaming. The density is invariant along $\mathbf{1}$, so it is a function of the differences $v_{g_k} - v_e$, and the law of these differences does not depend on e. As $y^\star$ is a multiple of $\mathbf{1}$, $y_{g_k} - y_e = v_{g_k} - v_e$.

(iii) The reference (proof of thm:atomforward) proves this part, a statement for one table. □

Theorem 20.23 (The junction under level form). *As stated, for every family at the centre.*

Proof. Parts (ii) and (iii) concern the Gramma $\mathsf{G}_{0;(\chi_p)_p}((v_{|p|})_p)$ alone and are proved in the reference (proof of thm:uniformjunction, parts (ii) and (iii)).

The preamble. By Proposition 20.6(iii) for every table, every half-sum has level form and the square of every difference is unchanged by the permutations of B; $q_{j,b-k}(-z_R) = q_{j,k}(z_R)$ is Proposition 20.6(i). On z_B each $(\partial_w L_{j,w})^2$ has exponents in $\{-2, 0, 2\}^B$, being a sum of products of two terms $e^{\langle z_B, \chi_p\rangle}$. A function of z_B of this kind that is unchanged by the permutations of B is a combination of the products $e^{-2\langle \mathbf{1}, z_B\rangle} e_k(e^{2z_B}) e_{k'}(e^{2z_B})$, $k \le k'$: such a product is the symmetric sum of the monomials with k exponents 2 and $b - k'$ exponents -2, plus terms with fewer of

both, and induction removes the terms (the reference, proof of thm:uniformjunction, part (i)). The coefficients are Laurent polynomials in e^{z_R}, the $\Lambda_{j,(p)}$ being such. The product over j of $(\partial_w L_{j,w})^{2l_j}$ therefore expands in products of $2|\mathbf{l}|$ of the e_k, with Laurent polynomial coefficients.

(i) At the centre $\boldsymbol{\mu} = 0$, and the first step of the proof of Theorem 20.4 gives

$$P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma}) = 2^{n-1}\int_{\mathbb{R}^{\mathcal{E}-g}} \prod_j \big(\Lambda_j(z)\Lambda_j(-z)\big)^{-\sigma_j}\, dz_{\mathcal{E}-g}\Big|_{z_g=0}.$$

For each table $\Lambda_j(z)\Lambda_j(-z) = L_{j,1/2}(z)^2 - \frac14(\partial_w L_{j,w}(z))^2$, and Proposition 20.1 with $a = \Lambda_j(z)$, $b = \Lambda_j(-z)$, $s = t = \sigma_j$, the midpoint $L_{j,1/2}(z)$ and $x = \partial_w L_{j,w}(z)/(2L_{j,1/2}(z)) \in (-1,1)$ gives

$$\big(\Lambda_j(z)\Lambda_j(-z)\big)^{-\sigma_j} = \sum_{l_j\ge 0} \frac{(\sigma_j)_{l_j}}{l_j!\,4^{l_j}} \big(\partial_w L_{j,w}(z)\big)^{2l_j} L_{j,1/2}(z)^{-2\sigma_j-2l_j},$$

a series of nonnegative terms. Open each power of the half-sum,

$$L_{j,1/2}^{-2\sigma_j-2l_j} = \Gamma(2\sigma_j+2l_j)^{-1}\int_0^\infty y_j^{2\sigma_j+2l_j-1} e^{-y_j L_{j,1/2}}\, dy_j;$$

by Tonelli's theorem the sums and integrals may be taken in the order of the statement. With $\upsilon_k = \sum_j y_j q_{j,k}(z_R)$,

$$\sum_j y_j L_{j,1/2}(z) = \sum_k \upsilon_k\, e^{-\langle \mathbf{1}, z_B\rangle} e_k\big(e^{2z_B}\big),$$

and inserting $e^{-\langle \mathbf{1}, z_B\rangle} e_k(e^{2z_B})$ under the exponential of minus this sum is $-\partial_{\upsilon_k}$. With the expansion of the preamble and Definition 2.1 (index 0, the integral over z_B),

$$\int_{\mathbb{R}^B} \prod_j \big(\partial_w L_{j,w}(z)\big)^{2l_j} e^{-\sum_j y_j L_{j,1/2}(z)}\, dz_B$$
$$= \sum_{|\mathbf{h}|=2|\mathbf{l}|} c_{\mathbf{h}}^{(\mathbf{l})}(z_R)\Big((-\partial_{\boldsymbol{\upsilon}})^{\mathbf{h}}\, \mathsf{G}_{0;(\chi_p)_p}\big((\upsilon_{|p|})_p\big)\Big)\Big|_{\boldsymbol{\upsilon}=\sum_j y_j \mathbf{q}_j(z_R)}.$$

Each term converges. The family being a molecule, some active table j has $c_j(B) \ge 1$ (Definition 8.1), so its patterns at B fill at least two consecutive levels $k, k+1$ (Theorem 12.7), and $q_{j,k}, q_{j,k+1} > 0$, each being a sum of nonnegative terms one of which is positive; by $q_{j,b-k}(-z_R) = q_{j,k}(z_R)$ also $q_{j,b-k}, q_{j,b-k-1} > 0$. The vectors χ_p of these four levels are closed under negation, $\chi_{B\setminus p} = -\chi_p$, and span $\mathbb{R}^B$, since $\chi_{p\cup\{e\}} - \chi_p = 2\mathbf{e}_e$ for $|p| = k$, $e \notin p$; so they positively span $\mathbb{R}^B$, the exponential of $-\sum_j y_j L_{j,1/2}(z)$ decays in z_B faster than every exponential, and the inserted exponentials are integrable against it.

(iv) For each table the swap τ of e and f keeps B, fixes R and exchanges table j, the bijection of the tables being the identity. Proposition 20.5 gives $c_{j,\tau(p)}(1-w;z_R) = c_{j,p}(w;z_R)$. At $w = \frac12$, $c_{j,\{f\}}(\frac12;\cdot) = c_{j,\{e\}}(\frac12;\cdot)$; in the derivative, $\partial_w c_{j,\varnothing} = \partial_w c_{j,B} = 0$ and $\partial_w c_{j,\{f\}} = -\partial_w c_{j,\{e\}}$. With $c_{j,B}(\frac12;z_R) = c_{j,\varnothing}(\frac12;-z_R)$ (Proposition 20.6(i)) this gives the two displayed forms. Evenness: $c_{j,\{e\}}(\frac12;-z_R) = c_{j,\{f\}}(\frac12;z_R) = c_{j,\{e\}}(\frac12;z_R)$ and $\partial_w c_{j,\{e\}}(w;-z_R) = -\partial_w c_{j,\{f\}}(w;z_R) = \partial_w c_{j,\{e\}}(w;z_R)$, by Proposition 20.6(i). In (i) change (z_e, z_f) to (z_+, z_-), with $dz_e\, dz_f = \frac12 dz_+ dz_-$. Then $\prod_j(\partial_w L_{j,w})^{2l_j} = (e^{z_-} - e^{-z_-})^{2|\mathbf{l}|}\prod_j(\partial_w c_{j,\{e\}})^{2l_j}$ and

$$\sum_j y_j L_{j,1/2}(z) = e^{-z_+}C_\varnothing(z_R) + e^{z_+}C_\varnothing(-z_R) + \big(e^{z_-} + e^{-z_-}\big)C_e(z_R).$$

The integral over z_+ is $\mathsf{G}_{0;(1,-1)}\big(C_\varnothing(-z_R), C_\varnothing(z_R)\big)$ (Definition 2.1 on one scale e^{z_+}), and the integral over z_- is Lemma 20.22 with $\xi = C_e(z_R)$ and the power $|\mathbf{l}|$, $4^{|\mathbf{l}|}(\frac{1}{2})_{|\mathbf{l}|}C_e^{-|\mathbf{l}|}\mathsf{G}_{|\mathbf{l}|;(1,-1)}(C_e, C_e)$. The constants collect to $2^{n-1}\cdot\frac{1}{2}\cdot 4^{|\mathbf{l}|}\prod_j 4^{-l_j} = 2^{n-2}$, which is the display; every term is nonnegative. □

Remark. With one table, $(\frac{1}{2})_l(\sigma)_l/(l!\,\Gamma(2\sigma+2l)) = \frac{1}{l!}\Gamma\,{}_2^{}{}_3\!\left[\begin{matrix}\sigma+l,\ \frac{1}{2}+l\\ 2\sigma+2l,\ \sigma,\ \frac{1}{2}\end{matrix}\right]$ and $y^{2\sigma+2l-1}(yc_{\{e\}})^{-l} = y^{2\sigma+l-1}c_{\{e\}}^{-l}$, which is the reference's form of (iv).

Theorem 20.40 (A series connection closes at its reach). *As stated, for series connections and for series connections of the dual table.*

Proof. (i) By Proposition 20.37(i), $f_B = C^{-s}C'^{-t}\prod_i f_i^\circ$, and by Proposition 20.37(ii), which is Lemma 20.30 applied $q-1$ times, f_B has a lattice expansion of reach $(\sum_i\sqrt{x_i})^2$ whose terms are the products of the terms of the parts collected by exponent. A function has at most one lattice expansion (Definition 20.28), so these are the coefficients $\beta_{\eta,j}$ of Theorem 20.34, and $\operatorname{reach}(B) \le (\sum_i\sqrt{x_i})^2 < \inf\varrho_R$, the reach being an infimum. Theorem 20.34(ii) applies: the even share closes toward the level d on the lattice of continued periods of the bunch family, with the bound for every $x > \operatorname{reach}(B)$, in particular for every $x > (\sum_i\sqrt{x_i})^2$. Under the hypotheses of Proposition 20.37(iii), $\operatorname{reach}(B) = (\sum_i\sqrt{\sup\varrho_i})^2$, and Theorem 20.39(ii) gives the root rate $\operatorname{reach}(B)/\inf\varrho_R$. For a series bunch the parts are the marks (Proposition 20.37), with

$$\begin{aligned} f_e^\circ(y) &= \int_0^\infty x^{m_e-s-1}e^{-w_e/x-yw_ex}\,dx = \mathsf{G}_{m_e-s;(1,-1)}(yw_e, w_e)\\ &= (yw_e)^{s-m_e}\mathsf{G}_{m_e-s;(1,-1)}\big(1, w_e^2y\big)\end{aligned}$$

by the torus homogeneity of the Gramma; by Lemma 20.29(iii) it has a lattice expansion of reach w_e^2, a power of y keeping the reach (Lemma 20.30). Proposition 20.36(iii) gives $\operatorname{reach}(B) = \inf\varrho_B = (\sum_e w_e)^2$ exactly, and Proposition 20.36(i) gives the coefficients.

(ii) The reference (proof of thm:classcloses, part (ii)) gives the argument for a parallel bunch, and it applies to a connection. The dual table at the exponents (t, s) has the same period (Proposition 8.15). Since $K^* = \Psi$ and $\Psi^* = K$ (Proposition 7.2), the dual node has the bunch B, of width one in the dual table, the room R, the ratios $1/\varrho_B$ and $1/\varrho_R$, and its levels swapped: its level d carries the even letter of the level r of the given node. No slope on the room being negative, $\sup\varrho_R < \infty$ (Lemma 19.15), so $\inf(1/\varrho_R) > 0$ and no slope on the room of the dual node is positive (the same lemma for the dual node). Part (i) for the dual node, whose bunch is a series connection with the reaches x_i^*, closes the even share toward its level d, the level r of the given node, when $(\sum_i\sqrt{x_i^*})^2 < 1/\sup\varrho_R$, at a root rate at most $(\sum_i\sqrt{x_i^*})^2\sup\varrho_R$. For a parallel bunch, the dual pattern weights are $\theta'^*_B = \theta'_\varnothing$ and $\theta'^*_{B-e} = \theta'_{\{e\}}$, so the w_e of the dual series bunch are $\theta'_{\{e\}}/\theta'_\varnothing$, $x_e^* = w_e^2$, and the root rate is exact by (i). The ratio of the bunch of the dual node is $1/\varrho_B = (\sum_e w_e/u_e)(\sum_e w_eu_e)$, whose least value $(\sum_e w_e)^2$ is taken at the point of equal coordinates of Δ_B (Proposition 20.36(ii)); so $\sup\varrho_B = (\sum_e w_e)^{-2}$, the condition reads $\sup\varrho_R < \sup\varrho_B$, and the root rate is $(\sum_e w_e)^2\sup\varrho_R = \sup\varrho_R/\sup\varrho_B$. □

Theorem 20.59 (One tree suffices). *As stated; the chain clause of (iii) is stated whenever some active list is connected.*

Proof. All of the theorem but the chain clause of (iii) is the reference's (proof of thm:onetree). For the chain clause, let some active list of the molecule F be connected. The weights of lists factor

across every split, so Corollary 20.9 applies: every peel leaves one childcule, the Foreman tree is a chain (Corollary 20.8), and at every realization, the centre and every lean, $|B_0| \le |B_1| \le \cdots \le |B_{d-1}| \le |L|$, L the leaf. The shape of the bunch B_k then has $|B_k| - 1 \le |L| - 1$ integrations, and its node adds one, the total of the bunch against its room, by (ii). A molecule of one list is such an F, its list being connected. □

48.21 Chapter 21: gluing and what crosses it

Theorem 21.5 (Replacement of marks by blocks). *As stated: every mark is replaced, by a block of any size and by either rule.*

Proof. One mark, first rule. Let h be replaced by a block X of q marks by the first rule, the other marks left in place: the new sets are the bases $T \not\ni h$ and the sets $(T - h) \sqcup \{x\}$, $T \ni h$, $x \in X$, each with the weight θ_T. If h is neither a loop nor a coloop, $(\mathcal{B}, \theta)$ is a cell on $(\mathcal{E} - h) \sqcup \{h\}$ with the hinge h, and the parallel block $\binom{X \sqcup \{g\}}{1}$ is a cell on $X \sqcup \{g\}$ whose deletion of g has the singletons of X as bases and whose contraction of g has the single basis $\varnothing$. By 21.2 for these two cells, the sets $\{x\} \sqcup (T - h)$, $T \ni h$, and $\varnothing \sqcup T$, $T \not\ni h$, are the bases of a table of rank $1 + r - 1 = r$: they are the new sets. If h is a loop, no basis contains h, and the new sets are the bases of $\mathcal{B}$ read on $\mathcal{E} - h$, the marks of X being loops: a table of rank r. If h is a coloop, every basis contains h, and the new sets are the sets $(T - h) \sqcup \{x\}$: the direct sum of the contraction of h and the table $\binom{X}{1}$, a table of rank r, the exchange property of a direct sum holding in each summand.

One mark, complementary rule. The dual table has the bases $\mathcal{E} \setminus T$ (7.2). The first rule at h on the dual table gives the sets $\mathcal{E} \setminus T$ for $T \ni h$ and $(\mathcal{E} \setminus T) - h + x$ for $T \not\ni h$, $x \in X$, the bases of a table on $(\mathcal{E} - h) \sqcup X$ by the first case. Their complements in $(\mathcal{E} - h) \sqcup X$ are $(T - h) \sqcup X$ for $T \ni h$ and $T \sqcup (X - x)$ for $T \not\ni h$, the sets of the complementary rule, and the complements of the bases of a table are the bases of its dual. Its rank is $(|\mathcal{E}| - 1 + q) - (|\mathcal{E}| - r) = r + q - 1$.

All marks. The rule at a mark h decides the part of a new set in X_h from whether $h \in T$ alone, so replacing the marks one at a time, in any order, gives the sets of the statement; by the two cases each step gives a table, and the step at $h \in \mathcal{E}_2$ raises the rank by $q_h - 1$. A new set T' determines $T = \{h \in \mathcal{E}_1 : X_h \cap T' \ne \varnothing\} \sqcup \{h \in \mathcal{E}_2 : X_h \subseteq T'\}$ and the choice made in every block, so it carries the one weight θ_T. A block of one mark gives back its mark under either rule.

Substitution. When no mark is a loop or a coloop, $(\mathcal{B}, \theta)$ is a core (21.1), and the parallel block $\binom{X_h \sqcup \{g_h\}}{1}$ and the series block $\binom{X_h \sqcup \{g_h\}}{q_h}$ are cells, g_h being neither a loop nor a coloop of either. In the parallel block the deletion of g_h has the singletons of X_h as bases and the contraction the single basis $\varnothing$; in the series block the deletion has the single basis X_h and the contraction the sets $X_h - x$. So the sets of the substitution contain, for a basis T of the core, one mark of X_h for $h \in T \cap \mathcal{E}_1$, none for $h \in \mathcal{E}_1 \setminus T$, all of X_h for $h \in T \cap \mathcal{E}_2$ and all but one for $h \in \mathcal{E}_2 \setminus T$, each with the weight θ_T, every weight of a block being 1: the table of the statement.

Letters. By the correspondence between the new sets and the pairs of a basis T and a choice in every block, $K'(\alpha)$ is the sum over T of θ_T times the product over the blocks of the sum, over the choices, of the monomials of the chosen marks: $e_1(\alpha_{X_h})$ for $h \in T \cap \mathcal{E}_1$, 1 for $h \in \mathcal{E}_1 \setminus T$, $e_{q_h}(\alpha_{X_h})$ for $h \in T \cap \mathcal{E}_2$ and $e_{q_h - 1}(\alpha_{X_h})$ for $h \in \mathcal{E}_2 \setminus T$. Taking out $\prod_{h \in \mathcal{E}_2} e_{q_h - 1}$,

$$K' = \prod_{h \in \mathcal{E}_2} e_{q_h - 1} \sum_T \theta_T \prod_{h \in T \cap \mathcal{E}_1} e_1 \prod_{h \in T \cap \mathcal{E}_2} \frac{e_{q_h}}{e_{q_h - 1}},$$

the first display. The complement of a new set meets X_h in all of X_h but the chosen mark for $h \in T \cap \mathcal{E}_1$, in all of X_h for $h \in \mathcal{E}_1 \setminus T$, in nothing for $h \in T \cap \mathcal{E}_2$ and in the missing mark for

$h \in \mathcal{E}_2 \setminus T$, so

$$\Psi' = \sum_T \theta_T \prod_{h \in T \cap \mathcal{E}_1} e_{q_h - 1} \prod_{h \in \mathcal{E}_1 \setminus T} e_{q_h} \prod_{h \in \mathcal{E}_2 \setminus T} e_1 = \prod_{h \in \mathcal{E}_1} e_{q_h - 1} \sum_T \theta_T \prod_{h \in \mathcal{E}_1 \setminus T} \frac{e_{q_h}}{e_{q_h - 1}} \prod_{h \in \mathcal{E}_2 \setminus T} e_1,$$

which is $\prod_{h \in \mathcal{E}_1} e_{q_h - 1}\, \Psi(\beta^*)$ with $\beta^*_h = e_{q_h}/e_{q_h - 1}$ on $\mathcal{E}_1$ and $\beta^*_h = e_1$ on $\mathcal{E}_2$, since $\Psi(\beta^*) = \sum_T \theta_T(\beta^*)^{\mathcal{E} \setminus T}$. For one block at h, every other block of one mark, K and Ψ are affine in α_h, $K = K_{\setminus h} + \alpha_h K_{/h}$ and $\Psi = \alpha_h \Psi_{\setminus h} + \Psi_{/h}$ (7.3), and the two displays become the four formulas of the statement. In scale and shape, e_k is homogeneous of degree k, $e_1(u) = |u| = 1$ on Δ_X, and the measure is 2.7. □

Theorem 21.6 (Counts, walls and atoms under replacement). *As stated: (i) and (ii) for blocks of any sizes and either rule at each mark; (iii) at every positive exponent pairs.*

Proof. (i) The marks and the rank are 21.5. At $\alpha = \mathbf{1}$, $e_1 = q_h$, $e_{q_h} = 1$ and $e_{q_h - 1} = q_h$ on a block of q_h marks, so the first display of 21.5 gives the total weight $K'(\mathbf{1}) = \prod_{h \in \mathcal{E}_2} q_h \cdot K\big((q_h)_{h \in \mathcal{E}_1}, (q_h^{-1})_{h \in \mathcal{E}_2}\big)$, and expanding K over the bases gives the sum displayed; it counts q_h choices at $h \in T \cap \mathcal{E}_1$ and q_h at $h \in \mathcal{E}_2 \setminus T$. With $\mathcal{E}_2 = \varnothing$ it is $K((q_h)_h)$; with $\mathcal{E}_1 = \varnothing$ it is $\prod_h q_h\, K((q_h^{-1})_h) = \Psi((q_h)_h)$ (7.2).

(ii) By 7.13, $d'(Y) = \min_{T'} |T' \cap Y|$ and $d^{*\prime}(Y) = \min_{T'} |Y \setminus T'|$ over the new bases T', which are the pairs of a basis T and a choice in every block (proof of 21.5). Fix T and minimize over the choices, block by block. In $|T' \cap Y|$ a block contributes, at its least: for $h \in T \cap \mathcal{E}_1$, 1 if $X_h \subseteq Y$ and 0 otherwise, the chosen mark avoiding Y when it can; for $h \in \mathcal{E}_1 \setminus T$, 0; for $h \in T \cap \mathcal{E}_2$, $|X_h \cap Y|$; for $h \in \mathcal{E}_2 \setminus T$, $\max(|X_h \cap Y| - 1, 0)$, the missing mark taken in Y when it can. Writing $|X_h \cap Y| = \max(|X_h \cap Y| - 1, 0) + [h \in \lceil Y \rceil]$ for $h \in \mathcal{E}_2$, the least value at T is

$$\sum_{h \in \mathcal{E}_2} \max\big(|X_h \cap Y| - 1, 0\big) + \big|T \cap \big((\lfloor Y \rfloor \cap \mathcal{E}_1) \sqcup (\lceil Y \rceil \cap \mathcal{E}_2)\big)\big|,$$

and the least over T is $d'(Y)$ as stated (7.13). In $|Y \setminus T'|$ a block contributes, at its least: for $h \in T \cap \mathcal{E}_1$, $\max(|X_h \cap Y| - 1, 0)$; for $h \in \mathcal{E}_1 \setminus T$, $|X_h \cap Y|$; for $h \in T \cap \mathcal{E}_2$, 0; for $h \in \mathcal{E}_2 \setminus T$, 1 if $X_h \subseteq Y$ and 0 otherwise. In the same way $d^{*\prime}(Y)$ is as stated, and $c'(Y) = |Y| - d'(Y) - d^{*\prime}(Y)$ is 7.12. Under the first rule alone, $r'(Y) = \max_{T'} |T' \cap Y| = \max_T |T \cap \lceil Y \rceil| = r(\lceil Y \rceil)$, $d'(Y) = d(\lfloor Y \rfloor) = r - r(\mathcal{E} \setminus \lfloor Y \rfloor)$ (7.12) and $c'(Y) = r'(Y) - d'(Y)$ (7.13). Under the complementary rule alone, $|Y| = \sum_h \max(|X_h \cap Y| - 1, 0) + |\lceil Y \rceil|$, so

$$c'(Y) = |\lceil Y \rceil| - d(\lceil Y \rceil) - d^*(\lfloor Y \rfloor) = |\lceil Y \rceil| - |\lfloor Y \rfloor| + r(\mathcal{E} \setminus \lceil Y \rceil) + r(\lfloor Y \rfloor) - r.$$

If $Y = \bigcup_{h \in S} X_h$, then $\lceil Y \rceil = \lfloor Y \rfloor = S$, $d'(Y) = \sum_{h \in S \cap \mathcal{E}_2} (q_h - 1) + d(S)$, $d^{*\prime}(Y) = \sum_{h \in S \cap \mathcal{E}_1} (q_h - 1) + d^*(S)$ and $|Y| = \sum_{h \in S} q_h$, so $c'(Y) = |S| - d(S) - d^*(S) = c(S)$. If $\varnothing \neq Y \subsetneq X_h$, then $\lceil Y \rceil = \{h\}$ and $\lfloor Y \rfloor = \varnothing$: for $h \in \mathcal{E}_1$, $d'(Y) = 0$ and $d^{*\prime}(Y) = |Y| - 1 + d^*(\{h\})$, and for $h \in \mathcal{E}_2$, $d'(Y) = |Y| - 1 + d(\{h\})$ and $d^{*\prime}(Y) = 0$; a seed table connected on at least two marks has $c(\{h\}) = 1 - d(\{h\}) - d^*(\{h\}) > 0$, so $d(\{h\}) = d^*(\{h\}) = 0$ and $c'(Y) = 1$.

(iii) Write $X_e = \{e\} \times \{1, \dots, q\}$. For $h \in H$, carrying table j to table $\tau(j)$ (11.13), put $\hat{h}(e, i) = (h(e), i)$; the rule and q are the same at every mark, so $\hat{h}$ maps the replaced table j onto the replaced table $\tau(j)$ with the same weights, and the two tables have the same pairs: $\hat{h}$ carries the replaced family to itself. A permutation of X_e fixing the other new marks maps every replaced table onto itself with its weights, the sets of the rule being symmetric in the marks of X_e. The group generated by these bijections is transitive on the new marks, H being transitive on $\mathcal{E}$.

The replaced family is connected. The seed family is one molecule: a molecule $M \neq \mathcal{E}$ would be a union of molecules of every table (7.9), with $|T \cap M|$ constant on the bases of every table and so $c_j(M) = 0$ for every j (7.13), all tables being active at positive pairs. Let $e \neq f$ be linked in table j: by 7.7 some basis T of table j has $e \in T$, $f \notin T$ and $T - e + f$ a basis. Under the first rule, a new basis T' from T choosing x in X_e gives $T' - x + x'$, the new basis from T choosing x', and $T' - x + y$, $y \in X_f$, the new basis from $T - e + f$ choosing y with the same choices elsewhere; under the complementary rule, the new basis T' from $T - e + f$ missing x' in X_e gives $T' - x + x'$, the one missing x, and the new basis T'' from T missing y in X_f gives $T'' - x + y$, the new basis from $T - e + f$ missing x. By 7.7 any two marks of X_e are linked, and every mark of X_e is linked to every mark of X_f. Every mark is linked to another mark, the seed family being one molecule on at least two marks, so the new marks form one molecule. Then no nonempty proper set Y of new marks has $c'_j(Y) = 0$ in every table: by 7.13 $|T' \cap Y|$ would be constant on the bases of every table, and by 7.8 no circuit would join Y to its complement.

Finally $((s_j, t_j)_j; \mathbf{1})$ is a realization of the replaced family, and every bijection above fixes the table's gauge, $\hat{h}^* \log \mathbf{1} = \log \mathbf{1}$. By 11.14 the reading of the image of a mark equals the reading of the mark, so the reading is constant on the new marks, which form one molecule: the realization is an atom (10.1). □

Remark. In exact rational arithmetic (receipt `check_blocks_splice.py`): on the seed of rank two on $\{a, b, c, d\}$ whose bases are the pairs other than $\{c, d\}$, with weights $1, 2, \frac{3}{2}, 5, \frac{1}{3}$, blocks of $2, 3, 1, 2$ marks, the first rule at a, c and the complementary rule at b, d, the new sets satisfy the exchange property, the rank is $r + \sum_{h \in \mathcal{E}_2}(q_h - 1)$, the total weight is that of (i), $d'(Y)$ and $d^{*\prime}(Y)$ are those of (ii) at every one of the 2^8 sets Y, and unions of full blocks keep the width; with the first rule at every mark and the same sizes, $r'(Y) = r(\lceil Y \rceil)$, $d'(Y) = r - r(\mathcal{E} \setminus \lfloor Y \rfloor)$ and the total weight is $K((q_h)_h)$.

Proposition 21.8 (Dual pairs). *As stated.*

Proof. Write θ for the one system of weights of $\mathcal{A}$. (i) The bases of $\mathcal{A}$ are the $X \in \mathcal{A} \setminus g$ and the $Y + g$ with $Y \in \mathcal{A}/g$ (7.3). If $\mathcal{A}$ is a dual pair, the complement $(B \setminus X) + g$ of X is a basis with the weight θ_X, and the complement $B \setminus Y$ of $Y + g$ is a basis of $\mathcal{A} \setminus g$ with the weight θ_{Y+g}: the bases are closed under complement with equal weights. Conversely, if they are, $X \in \mathcal{A} \setminus g$ gives the basis $(B \setminus X) + g$ with the weight θ_X, so $B \setminus X \in \mathcal{A}/g$, and $Y \in \mathcal{A}/g$ gives $B \setminus Y \in \mathcal{A} \setminus g$: $\mathcal{A}/g$ is the dual of $\mathcal{A} \setminus g$ with its weights. Since g is neither a loop nor a coloop, $\mathcal{A} \setminus g$ has the rank $r_{\mathcal{A}}$ and $\mathcal{A}/g$ the rank $r_{\mathcal{A}} - 1$, while the dual of $\mathcal{A} \setminus g$ has the rank $|B| - r_{\mathcal{A}}$; so $|B| = 2r_{\mathcal{A}} - 1$.

(ii) $K_{\mathcal{A}/g} = \sum_Y \theta_{Y+g} \alpha^Y = \sum_X \theta_X \alpha^{B \setminus X} = \psi$ and $\Psi_{\mathcal{A}/g} = \sum_Y \theta_{Y+g} \alpha^{B \setminus Y} = \sum_X \theta_X \alpha^X = k$. So $\gamma^K_{\mathcal{A}} = K_{\mathcal{A} \setminus g}/K_{\mathcal{A}/g} = k/\psi$ and $\gamma^{\Psi}_{\mathcal{A}} = \Psi_{\mathcal{A}/g}/\Psi_{\mathcal{A} \setminus g} = k/\psi$ (21.1).

(iii) The cell $\binom{\{h,g\}}{1}$ has the deletion with the basis $\{h\}$ and the contraction with the basis $\varnothing$, the dual of $\{h\}$ on $\{h\}$, so it is a dual pair with $k = \alpha_h$ and $\psi = 1$. By 21.4 and (ii), $K = \prod_i K_{\mathcal{A}_i/g_i} K_{\mathcal{H}}\big((k_i/\psi_i)_i\big) = \pi K_{\mathcal{H}}\big((k_i/\psi_i)_i\big)$ and $\Psi = \prod_i \Psi_{\mathcal{A}_i \setminus g_i} \Psi_{\mathcal{H}}\big((k_i/\psi_i)_i\big) = \pi \Psi_{\mathcal{H}}\big((k_i/\psi_i)_i\big)$. $K_{\mathcal{H}}$ and $\Psi_{\mathcal{H}}$ are homogeneous of the degrees $r_{\mathcal{H}}$ and $q - r_{\mathcal{H}}$, so multiplying the arguments by π gives the two identities; both sides are polynomials, $r_{\mathcal{H}} \geq 1$ and $q - r_{\mathcal{H}} \geq 1$ since no mark of the core is a loop or a coloop. □

Corollary 21.11 (The width of the marks left in place). *As stated.*

Proof. In 21.10 take $S \subseteq L$: the part of S in the cell $\binom{\{l,g\}}{1}$ of a mark $l \in S$ is $\{l\}$, with $d(\{l\}) = d^*(\{l\}) = 0$ and $d(\{l, g\}) = d^*(\{l, g\}) = 1$, the bases being $\{l\}$ and $\{g\}$ (7.13); every other cell meets S in $\varnothing$, and its hinge, neither a loop nor a coloop, has $d(\{g\}) = d^*(\{g\}) = 0$. So

$D(S) = D^*(S) = S$, read on the marks of the core, and $d(S) = d_{\mathcal{H}}(S)$, $d^*(S) = d^*_{\mathcal{H}}(S)$, whence $c(S) = |S| - d(S) - d^*(S) = c_{\mathcal{H}}(S)$ (7.12). By 7.12, $d_{\mathcal{H}}(S) = r_{\mathcal{H}} - r_{\mathcal{H}}(L \sqcup H' \setminus S)$, so

$$c_{\mathcal{H}}(S) = r_{\mathcal{H}}(S) + r_{\mathcal{H}}(L \sqcup H' \setminus S) - r_{\mathcal{H}} \le r_{\mathcal{H}}(L \sqcup H' \setminus S) \le |L \setminus S| + |H'|,$$

the rank of a set being at most its size and $r_{\mathcal{H}}(S) \le r_{\mathcal{H}}$. In $\binom{B \sqcup \{h_1,\dots,h_c\}}{c}$ the bases are the sets of c marks, which meet B in every number from 0 to c, so B has the width c (7.13); no h_i is a loop or a coloop, and by the first part the width stays c after any cells are substituted at the h_i. □

Lemma 21.12 (Two-sums of cells; hinge walls). *As stated, for cell families glued table by table.*

Proof. That $\mathcal{A} \oplus_g \mathcal{A}'$ is a table of rank $r_{\mathcal{A}} + r_{\mathcal{A}'} - 1$, plain when the cells are, in which B is a bunch of width one with the cells $\mathcal{A}$ and $\mathcal{A}'$, and that a table with a bunch of width one and factoring weights is the two-sum of its two cells, is the reference, proof of lem:twosumpieces(i), the weights multiplied letter by letter. The two-sum is $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}, \mathcal{A}' \rangle$: the bases $\{h_1\}$ and $\{h_2\}$ of the core give the two kinds of sets of 21.2 (21.1). So F is substituted table by table, with the core $\binom{\{h_1,h_2\}}{1}$ in every table and the cells $\mathcal{A}_j$ at h_1 and $\mathcal{A}'_j$ at h_2, and 21.10 at $S = B$ gives the wall $m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big) = \kappa_{F_B}$, and at $S = R$ the wall κ_{F_R}. Table j of F has the rank $r_j = r_{\mathcal{A}_j} + r_{\mathcal{A}'_j} - 1$ on $n = |B| + |R|$ marks, so

$$\kappa_{F_B} + \kappa_{F_R} = m_{\mathcal{E}} - \sum_j \big(s_j(r_j - 1) + t_j(n - r_j - 1)\big) = \kappa_{\mathcal{E}} + \sum_j (s_j + t_j) = \kappa_{\mathcal{E}} + 2\sigma.$$

With the mass $2\sigma - \kappa_{F_B}$ on g, the imbalance of F_B on $B \sqcup \{g\}$ is

$$m_B + 2\sigma - \kappa_{F_B} - \sum_j \big(s_j r_{\mathcal{A}_j} + t_j(|B| + 1 - r_{\mathcal{A}_j})\big) = 2\sigma - \sum_j (s_j + t_j) = 0,$$

on the balance wall of F or off it. □

Theorem 21.14 (Splicing hinges in pairs). *As stated; (iii) holds for any number of cells and any graph of pairs.*

Proof. (i) and (ii) are the reference, proof of thm:hingepairs(i),(ii). (iii) In the direct sum of the cells the letters are $K = \prod_i K_{\mathcal{A}_i}$ and $\Psi = \prod_i \Psi_{\mathcal{A}_i}$ (7.9), and by 21.13, with $G = \bigsqcup_i G_i$,

$$K = \sum_{(p_i)_i} \alpha_G^{\sqcup_i p_i} \prod_i K_{\mathcal{A}_i,(p_i)}, \qquad \Psi = \sum_{(p_i)_i} \alpha_G^{G \setminus \sqcup_i p_i} \prod_i \Psi_{\mathcal{A}_i,(p_i)},$$

over $p_i \subseteq G_i$, absent channels counting as zero. Splicing a pair $\{a, b\}$ acts on a monomial α_G^P by $(\partial_{\alpha_a} + \partial_{\alpha_b})\alpha_G^P|_{\alpha_a = \alpha_b = 0}$, which is α_G^{P-a} when $a \in P \not\ni b$, α_G^{P-b} when $b \in P \not\ni a$, and 0 otherwise; on $\alpha_G^{G \setminus P}$ it is likewise nonzero exactly when P contains one of a, b. The splicings of disjoint pairs commute and act on different variables, so splicing all pairs keeps the terms in which $P = \sqcup_i p_i$ is $p \sqcup J$ with $p \subseteq G'$ and J containing exactly one hinge of every pair, and leaves $\alpha_{G'}^p$ in K and $\alpha_{G'}^{G' \setminus p}$ in Ψ. By (ii) the result is the letters of the spliced cell $\mathcal{A}$, and by the Taylor cube (12.7) its channels are the coefficients of these monomials:

$$K_{\mathcal{A},(p)} = \sum_J \prod_i K_{\mathcal{A}_i,((p \sqcup J) \cap G_i)}, \qquad \Psi_{\mathcal{A},(p)} = \sum_J \prod_i \Psi_{\mathcal{A}_i,((p \sqcup J) \cap G_i)}.$$

The marks other than hinges are $\bigsqcup_i B_i$, and the rank is $\sum_i r_{\mathcal{A}_i} - k$ by (ii). The degrees add up: $\sum_i \big(r_{\mathcal{A}_i} - |(p \sqcup J) \cap G_i|\big) = \sum_i r_{\mathcal{A}_i} - k - |p|$, since $|J| = k$, and $\sum_i \big(n_{\mathcal{A}_i} - r_{\mathcal{A}_i} + |(p \sqcup J) \cap G_i|\big) = n_{\mathcal{A}} - r_{\mathcal{A}} + |p|$, since $n_{\mathcal{A}} - r_{\mathcal{A}} = \sum_i (n_{\mathcal{A}_i} - r_{\mathcal{A}_i}) + k$. Each channel of $\mathcal{A}$ is a sum of products of one channel of each cell, so the map is multilinear. For two cells with the pairs named by I, a set J is given by the set $I' \subseteq I$ of the pairs whose hinge on the side of $\mathcal{A}_1$ it contains, $(p \sqcup J) \cap G_1 = p_1 \sqcup I'$ and $(p \sqcup J) \cap G_2 = p_2 \sqcup (I \setminus I')$: the bilinear map displayed. With $k = 1$ and no other hinges, $K_{\mathcal{A},(\varnothing)} = K_{\mathcal{A}_1 \setminus g_1} K_{\mathcal{A}_2/g_2} + K_{\mathcal{A}_1/g_1} K_{\mathcal{A}_2 \setminus g_2}$ and $\Psi_{\mathcal{A},(\varnothing)} = \Psi_{\mathcal{A}_1 \setminus g_1} \Psi_{\mathcal{A}_2/g_2} + \Psi_{\mathcal{A}_1/g_1} \Psi_{\mathcal{A}_2 \setminus g_2}$, the letters of 21.7(iii). □

Remark. In exact arithmetic (receipt `check_blocks_splice.py`): three cells in a closed path, a weighted $\binom{\{x_1,x_2\} \sqcup G_1}{3}$ with three hinges, one of them unpaired, a weighted $\binom{\{y_1,y_2\} \sqcup G_2}{2}$ and a weighted $\binom{\{z_1\} \sqcup G_3}{1}$ with two hinges each, spliced along three pairs, one between each two cells: both channels of the spliced cell at both values of the unpaired hinge are the sums over J of the products of the channels of the three cells.

Theorem 21.17 (Multivectors bound what crosses). *As stated; (iii) holds at any tensor rank.*

Proof. (i) and (ii) are the reference, proof of thm:cutrank(i),(ii), with the interface rank for the cut rank; (ii) is proved there at any tensor rank, the interface rank being subadditive. (iii) By (i) and (ii), $\operatorname{rank} \Theta_B^{(k)} \le \operatorname{ir}_k(\boldsymbol{\omega})^2 \le i_k^2$ with $i_k = \sum_l \binom{c_l}{k - d_l}$, and summing over the levels bounds the interface dimension $\sum_k \operatorname{rank} \Theta_B^{(k)}$ (21.15); for a real multivector (i) gives $\operatorname{rank} \Theta_B^{(k)} \le \frac{1}{2} \operatorname{ir}_k(\boldsymbol{\omega})(\operatorname{ir}_k(\boldsymbol{\omega}) + 1)$, and $x \mapsto \frac{1}{2} x(x+1)$ increases on $[0, \infty)$. The bounds involve only the widths and the floors of the slices. A determinantal table has a multivector of tensor rank one, $\boldsymbol{\omega}_T = \det A_T$ (7.20); a set T is a basis exactly when $\det A_T \neq 0$, so the column space of A_S has the dimension $r(S)$, the column spaces of A_B and A_R span $\mathbb{F}^r$ and meet in dimension $r(B) + r(R) - r = c(B)$, and $d_1 = r - r(R) = d(B)$ (7.12). So $\operatorname{ir}_k(\boldsymbol{\omega}) \le \binom{c(B)}{k - d(B)}$, and $\sum_k \binom{c(B)}{k - d(B)}^2 = \sum_i \binom{c(B)}{i}^2 = \binom{2c(B)}{c(B)}$. At width one $\binom{1}{k - d(B)} = 1$ at $k = d(B)$ and $k = d(B) + 1$: one bunch letter at each level. □

48.22 Chapter 22: periods of constructions

Definition 22.1 (Twists). As stated: the torus relation for every twist, every vector $\boldsymbol{\lambda}$ on the marks and all constants c_j, c_j^*.

Proof. Put $\mathbf{x}' = (c_j \boldsymbol{\lambda} \circ \mathbf{x}_j^K,\ c_j^* \boldsymbol{\lambda} \circ \mathbf{x}_j^\Psi)_j$. Every basis of table j has r_j marks and its complement r_j^*, so K_j is homogeneous of degree r_j and Ψ_j of degree r_j^*, and the letters of $F[\mathbf{x}']$ at α are

$$K_j(c_j \boldsymbol{\lambda} \circ \mathbf{x}_j^K \circ \alpha) = c_j^{r_j} K_j\big(\mathbf{x}_j^K \circ (\boldsymbol{\lambda} \circ \alpha)\big), \qquad \Psi_j(c_j^* \boldsymbol{\lambda} \circ \mathbf{x}_j^\Psi \circ \alpha) = {c_j^*}^{r_j^*} \Psi_j\big(\mathbf{x}_j^\Psi \circ (\boldsymbol{\lambda} \circ \alpha)\big).$$

Write $\omega_{\mathbf{x}} = \beta^{\mathbf{m-1}} \prod_j K_j(\mathbf{x}_j^K \circ \beta)^{-s_j} \Psi_j(\mathbf{x}_j^\Psi \circ \beta)^{-t_j}\, d\beta$ for the form of $F[\mathbf{x}]$. The map $\Phi : \alpha \mapsto \boldsymbol{\lambda} \circ \alpha$ is a linear bijection of the open orthant onto itself that commutes with positive scalings, and with $\beta_e = \lambda_e \alpha_e$ one has $\beta^{\mathbf{m-1}} d\beta = \boldsymbol{\lambda}^{\mathbf{m}} \alpha^{\mathbf{m-1}} d\alpha$. So the form of $F[\mathbf{x}']$ is

$$\omega_{\mathbf{x}'} = \boldsymbol{\lambda}^{-\mathbf{m}} \prod_j c_j^{-r_j s_j} {c_j^*}^{-r_j^* t_j}\ \Phi^* \omega_{\mathbf{x}}.$$

A twist does not change degrees, so on the balance wall both forms are homogeneous of degree zero, and each period is the integral of its form over the section $\{h = 1\}$ of any positive function h

homogeneous of degree one (8.7). Since $\int_{\{h=1\}} \Phi^* \omega_{\mathbf{x}} = \int_{\{h\circ\Phi^{-1}=1\}} \omega_{\mathbf{x}}$ and $h \circ \Phi^{-1}$ is again positive and homogeneous of degree one, the two periods differ by the constant factor of the display. The moduli of the two integrands are related by the same change of variables and a positive constant, so the two sides converge together; for real data the integrands are nonnegative and the identity holds in $[0, \infty]$.

At $\boldsymbol{\lambda} = \mu \mathbf{1}$ and $c_j = c_j^* = \mu^{-1}$ every coordinate of $\mathbf{x}'$ equals that of $\mathbf{x}$, and the factor is $\mu^{-m_{\mathcal{E}}} \mu^{\sum_j (r_j s_j + r_j^* t_j)} = 1$, since $m_{\mathcal{E}} = \sum_j (r_j s_j + r_j^* t_j)$ on the balance wall. With $\boldsymbol{\lambda}$ equal to 1 off g and to λ at g, and $c_j = c_j^* = 1$, the factor is λ^{-m_g}; with $\lambda = 1/x_1^K$ this reads every twist at g with $x_1^K = 1$, up to the factor $(x_1^K)^{-m_g}$, and for one table it leaves the one number x_1^Ψ / x_1^K, carried by the cobasis letter. With $\boldsymbol{\lambda} = \mathbf{1}$, one c_j or one c_j^* different from 1 and the other constants 1, the factor is $c_j^{-r_j s_j}$ or $c_j^{*\,-r_j^* t_j}$; for one table twisted in its cobasis letter by $\mu \mathbf{x}$ it is $\mu^{-r^* t}$. □

Remark. The reference, 22.1, rescales a twist at one mark, normalized by the scale of the mark, and the relation for one table twisted in its cobasis letter is used with 25.8. The exact check of the scalar case is check 11 of S21-23_work/checks.py.

Theorem 22.4 (The cells ride on the core's fibre). *As stated: at complex data, every exponent* 0 *or of positive real part, on contours on which the load families converge absolutely.*

Proof. The steps are those of the reference, proof of 22.4, with the moduli carried along. By 21.4, table j has the letters

$$K_j = \prod_i K_{\mathcal{A}_{ij}/g_i}(\alpha_{B_i})\, K_{\mathcal{H}_j}\big((\gamma^K_{\mathcal{A}_{ij}}(\alpha_{B_i}))_i\big), \qquad \Psi_j = \prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i}(\alpha_{B_i})\, \Psi_{\mathcal{H}_j}\big((\gamma^\Psi_{\mathcal{A}_{ij}}(\alpha_{B_i}))_i\big).$$

For $\operatorname{Re} s_j > 0$, 4.4 opens $K_{\mathcal{H}_j}(X)^{-s_j} = \big(\sum_T \theta_T^{\mathcal{H}_j} X^T\big)^{-s_j}$ with one share $a_{j,T}$ per basis on a contour of positive real parts, and at $X_i = \gamma^K_{\mathcal{A}_{ij}}$ the monomials give $\prod_T (X^T)^{-a_{j,T}} = \prod_i (\gamma^K_{\mathcal{A}_{ij}})^{-a_j(h_i)}$; in the same way $\Psi_{\mathcal{H}_j}(Y)^{-t_j} = \big(\sum_T \theta_T^{\mathcal{H}_j} Y^{\{h\}\setminus T}\big)^{-t_j}$ is opened with the shares $b_{j,T}$, and $\prod_T (Y^{\{h\}\setminus T})^{-b_{j,T}} = \prod_i (\gamma^\Psi_{\mathcal{A}_{ij}})^{-b_j(h_i)}$. A letter with exponent 0 is the constant 1 and is opened by nothing. Since $\gamma^K_{\mathcal{A}} = K_{\mathcal{A}\setminus g}/K_{\mathcal{A}/g}$ and $\gamma^\Psi_{\mathcal{A}} = \Psi_{\mathcal{A}/g}/\Psi_{\mathcal{A}\setminus g}$ (21.1),

$$K^{-s_j}_{\mathcal{A}_{ij}/g_i}(\gamma^K_{\mathcal{A}_{ij}})^{-a_j(h_i)} = K^{-(s_j - a_j(h_i))}_{\mathcal{A}_{ij}/g_i} K^{-a_j(h_i)}_{\mathcal{A}_{ij}\setminus g_i}, \qquad \Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}(\gamma^\Psi_{\mathcal{A}_{ij}})^{-b_j(h_i)} = \Psi^{-(t_j - b_j(h_i))}_{\mathcal{A}_{ij}\setminus g_i} \Psi^{-b_j(h_i)}_{\mathcal{A}_{ij}/g_i},$$

and $e^{-|\alpha|} = \prod_i e^{-|\alpha_{B_i}|}$, so for fixed shares the integrand of J_F is the product over the cells of the integrands of the interests of the load families $F_i(\mathbf{a}, \mathbf{b})$, times the share kernels and the powers of the weights of the cores.

It remains to exchange the share integral and the integral over the orthant. For $X > 0$, $|X^{-a}| = X^{-\operatorname{Re} a}$, and $|\alpha^{\mathbf{m}-\mathbf{1}}| = \alpha^{\operatorname{Re}\mathbf{m}-\mathbf{1}}$; so the modulus of the joint integrand is

$$\prod_j \big|\Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{a}_j \\ s_j\end{bmatrix}\big|\, \big|\Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{b}_j \\ t_j\end{bmatrix}\big| \prod_{j,T} \big(\theta_T^{\mathcal{H}_j}\big)^{-\operatorname{Re}(a_{j,T}+b_{j,T})}$$

times the product of the integrands of the interests of the load families at the real parts of their data and exponents. Integrated over the orthant, the second factor gives the product of these interests at the real parts, which are finite by the hypothesis and constant along the contour, since they depend on the shares only through their real parts. On the contour, $a_{j,T} = c_{j,T} + i\tau_{j,T}$ with $c_{j,T} > 0$ fixed and $\sum_T \tau_{j,T} = \operatorname{Im} s_j$, and the classical bound $|\Gamma(c + i\tau)| \le C_c\, e^{-\pi|\tau|/2}(1 + |\tau|)^{c-1/2}$ makes $\prod_T |\Gamma(a_{j,T})|$ integrable over the imaginary parts of all shares but one; the same holds for

the shares $b_{j,T}$. So the joint integrand is absolutely integrable, the integrals may be taken in either order, and the display follows. The identity $J_{F_i(\mathbf{a},\mathbf{b})} = \Gamma(\kappa_{B_i} - \sum_j (a_j(h_i) + b_j(h_i)))\, P_{F_i(\mathbf{a},\mathbf{b})}$ is 8.7 at the imbalance of the load family: its tables $\mathcal{A}_{ij} \setminus g_i$ and $\mathcal{A}_{ij}/g_i$ have the degrees $r_{\mathcal{A}_{ij}}$, $|B_i| - r_{\mathcal{A}_{ij}}$, $r_{\mathcal{A}_{ij}} - 1$ and $|B_i| - r_{\mathcal{A}_{ij}} + 1$, and the masses $\mathbf{m}|_{B_i}$ exceed their degree by $\kappa_{B_i} - \sum_j (a_j(h_i) + b_j(h_i))$, with κ_{B_i} of 21.10. □

Theorem 22.5 (A cell read alike is pinned). *As stated: the hypothesis on every active letter, whatever the sign of its exponent.*

Proof. By 22.3, which holds for every substituted family at all data, both sides converging together,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \prod_i \beta_i(du_i)\; P_{F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]}\big((\kappa_{B_i})_i; \mathbf{s}, \mathbf{t}\big),$$

β_i the cell kernels and $F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]$ the core family read at u. A letter with exponent 0 is the constant 1 in every period, so the twists of the inactive letters do not enter the period of the core family read at u; give them the values $c_{ij} = 1$, $c^*_{ij} = 1$. Under the hypothesis the twists of the active letters at h_i are $c_{ij}\gamma_i(u_i)$ in K_j and $c^*_{ij}\gamma_i(u_i)$ in Ψ_j, so $F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]$ has the period of $F_{\mathcal{H}}[\mathbf{c}]$ twisted further by the vector $\boldsymbol{\lambda} = (\gamma_i(u_i))_i$ on the marks of the core, in both letters of every table. The core family read at u has the imbalance of F (22.3), so it is on its balance wall, and the torus relation of 22.1 with this $\boldsymbol{\lambda}$ and constants 1 gives

$$P_{F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]}\big((\kappa_{B_i})_i; \mathbf{s}, \mathbf{t}\big) = \prod_i \gamma_i(u_i)^{-\kappa_{B_i}}\, P_{F_{\mathcal{H}}[\mathbf{c}]}\big((\kappa_{B_i})_i; \mathbf{s}, \mathbf{t}\big).$$

Inserting this and integrating cell by cell gives the display, by Tonelli for real data and by Fubini where the integrals converge absolutely. The step is the reference proof of 22.5, which nowhere uses the sign of an exponent; the hypothesis there is written for the letters with positive exponent. □

Corollary 22.6 (Read by one hinge value, the period factors through the core). *As stated: for every substituted family with real data at every cell of which the active letters read the cell by one hinge value.*

Proof. (1) The first display. The hypothesis is that of 22.5 with all constants $c_{ij} = c^*_{ij} = 1$, and $F_{\mathcal{H}}[\mathbf{c}] = F_{\mathcal{H}}$; 22.5 gives the first display, in $[0, \infty]$ since the data are real. At a mark h left in place, the cell $\binom{\{h,g\}}{1}$ has both hinge values equal to α_h (21.1), so the hypothesis holds there with $\gamma = \alpha_h$.

(2) The cells. The cell family F_i has the tables $\mathcal{A}_{ij}$, cells on $|B_i| + 1$ marks of rank $r_{\mathcal{A}_{ij}}$, and by 21.10

$$\kappa_{B_i} = m_{B_i} - \sum_j \big(s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big),$$

so $m_{B_i} + 2\sigma - \kappa_{B_i} = \sum_j \big(s_j r_{\mathcal{A}_{ij}} + t_j(|B_i| + 1 - r_{\mathcal{A}_{ij}})\big)$, the degree of the family word of F_i: at the masses $(\mathbf{m}|_{B_i}, 2\sigma - \kappa_{B_i})$ it is on its balance wall. Read its period on the section $\{|\alpha_{B_i}| = 1\}$ (8.7), with $\alpha_{B_i} = u \in \Delta_{B_i}$ and $\alpha_{g_i} = v > 0$. By 21.1, $K_{\mathcal{A}_{ij}} = K_{\mathcal{A}_{ij}/g_i}(u)\big(\gamma^K_{\mathcal{A}_{ij}}(u) + v\big)$ and $\Psi_{\mathcal{A}_{ij}} = \Psi_{\mathcal{A}_{ij}\setminus g_i}(u)\big(v + \gamma^{\Psi}_{\mathcal{A}_{ij}}(u)\big)$. Under the hypothesis every active letter carries the factor $(v + \gamma_i(u))$ to the power of its exponent, and an inactive letter has exponent 0; so the word of F_i is the density of the cell kernel without $u^{\mathbf{m}|_{B_i}-\mathbf{1}}$ times $(v + \gamma_i(u))^{-2\sigma}$, $2\sigma = \sum_j (s_j + t_j)$. For $0 < \kappa < 2\sigma$ and $\gamma > 0$, the substitution $v = \gamma w$ and the Beta integral give

$$\int_0^\infty v^{2\sigma-\kappa-1}(v+\gamma)^{-2\sigma}\, dv = \gamma^{-\kappa} \int_0^\infty w^{2\sigma-\kappa-1}(1+w)^{-2\sigma}\, dw = \Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}\kappa, & 2\sigma-\kappa\\ & 2\sigma\end{bmatrix}\gamma^{-\kappa},$$

the Beta corner of a linear letter (15.1 with one coefficient). Integrating v first (Tonelli),

$$P_{F_i}\big((\mathbf{m}|_{B_i}, 2\sigma-\kappa_{B_i}); \mathbf{s}, \mathbf{t}\big) = \Gamma_1^2\begin{bmatrix}\kappa_{B_i}, & 2\sigma-\kappa_{B_i}\\ & 2\sigma\end{bmatrix}\int_{\Delta_{B_i}} \gamma_i^{-\kappa_{B_i}}\, d\beta_i,$$

which identifies the pinned shape integral.

(3) The walls. The marks of a core are neither loops nor coloops (21.1), so a core has at least two marks, $q \geq 2$, and B_i is a nonempty proper set of marks; it has width one in every table (21.3(i)). Width one in table j means $d_j(B_i)+d_j(\mathcal{E}\setminus B_i) = r_j - 1$ and $d_j^*(B_i)+d_j^*(\mathcal{E}\setminus B_i) = r_j^* - 1$, so

$$\kappa_{B_i} + \kappa_{\mathcal{E}\setminus B_i} = m_{\mathcal{E}} - \sum_j \big(s_j(r_j-1) + t_j(r_j^*-1)\big) = \kappa_{\mathcal{E}} + 2\sigma,$$

which is 2σ on the balance wall (21.12; check 2 of S21-23_work/checks.py). If every proper wall is positive, $\kappa_{B_i} > 0$ and $\kappa_{\mathcal{E}\setminus B_i} > 0$, so $0 < \kappa_{B_i} < 2\sigma$.

(4) The second display. Multiply the first display by $\prod_i \Gamma_1^2\begin{bmatrix}\kappa_{B_i}, & 2\sigma-\kappa_{B_i}\\ & 2\sigma\end{bmatrix} \in (0,\infty)$ and use step (2). The left side is finite (8.7), every factor on the right lies in $(0,\infty]$, so every factor is finite. The core family at $(\kappa_{B_i})_i$ has the imbalance of F (22.3), the cell families are balanced by step (2), and a family on its balance wall with a finite period has every proper wall positive (8.7).

(5) One cell. With one cell at h_1 and the other marks of the core left in place, the left-in-place cell family at h, the list $\binom{\{h,g\}}{1}$ in every table with the masses $(m_h, 2\sigma-m_h)$, has the word $(\alpha_h+\alpha_g)^{-2\sigma}$ and the period $\Gamma_1^2\begin{bmatrix}m_h, & 2\sigma-m_h\\ & 2\sigma\end{bmatrix}$, while $\kappa_{\{h\}} = m_h$; these factors cancel the Beta corners of the marks left in place. The core family is then the family of the cells of the room at the masses $(\kappa_B, \mathbf{m}|_R)$ and F_1 that of the cells of the bunch (21.3(iii)), and the second display is the identity of 19.10. In the first display the marks left in place contribute 1, their simplex being a point and their kernel the constant 1, and it is the identity of 19.12 with all constants 1. □

Remark. The reference proof (22.6, one table at $(0,t)$) uses of the full lean only that the letter which sees the cells sees each through one function. A family at mixed full leans is covered: on $B=\{1,2\}$ let table 1 at $(s_1,0)$ carry the series block $\binom{\{1,2,g\}}{2}$ and table 2 at $(0,t_2)$ the parallel block $\binom{\{1,2,g\}}{1}$. Both active letters read B through e_2/e_1 (21.7(i)), $\kappa_B = m_B - 2\sigma$, the cell family has the word $e_2(\alpha_1,\alpha_2,\alpha_g)^{-2\sigma}$ and the period $\Gamma(2\sigma-m_1)\Gamma(2\sigma-m_2)\Gamma(\kappa_B)/\Gamma(2\sigma)$ (26.9), and the pinned shape integral is $\int_0^1 u^{m_1-1}(1-u)^{m_2-1}(u(1-u))^{-\kappa_B}du = \Gamma(2\sigma-m_1)\Gamma(2\sigma-m_2)/\Gamma(4\sigma-m_B)$, whose quotient is $\Gamma_1^2\begin{bmatrix}\kappa_B, & 2\sigma-\kappa_B\\ & 2\sigma\end{bmatrix}$, as step (2) says. At $(2\sigma, m_1, m_2) = (3,2,2)$ the period is $\frac{1}{2}$, and the identity holds exactly there and at $(\frac{7}{2},\frac{3}{2},\frac{5}{2})$ (checks 7a to 7d of S21-23_work/checks.py).

Corollary 22.7 (The pinned shape integral is the period of a load family). *As stated.*

Proof. A load $a_{ij} \neq 0$ occurs only where $s_j \neq 0$, where $\gamma^K_{\mathcal{A}_{ij}} = K_{\mathcal{A}_{ij}\setminus g_i}/K_{\mathcal{A}_{ij}/g_i} = \gamma_i$, and a load $b_{ij} \neq 0$ only where $t_j \neq 0$, where $\gamma^\Psi_{\mathcal{A}_{ij}} = \Psi_{\mathcal{A}_{ij}/g_i}/\Psi_{\mathcal{A}_{ij}\setminus g_i} = \gamma_i$ (21.1). Since $\sum_j(a_{ij}+b_{ij}) = \kappa_{B_i}$,

$$\gamma_i^{-\kappa_{B_i}}\prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i}\Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i} = \prod_j K^{-a_{ij}}_{\mathcal{A}_{ij}\setminus g_i}\Psi^{-(t_j-b_{ij})}_{\mathcal{A}_{ij}\setminus g_i}K^{-(s_j-a_{ij})}_{\mathcal{A}_{ij}/g_i}\Psi^{-b_{ij}}_{\mathcal{A}_{ij}/g_i},$$

the word of the load family, whose imbalance $\kappa_{B_i} - \sum_j(a_{ij}+b_{ij})$ is zero (22.4). Its period is the integral of $u^{\mathbf{m}|_{B_i}-\mathbf{1}}$ times its word over the shape simplex (8.7), the pinned shape integral, and the

product formula is 22.6. If every proper wall of F is positive, then $0 < \kappa_{B_i} < 2\sigma = \sum_j (s_j + t_j)$ (22.6); with $s_j, t_j \geq 0$, raising the loads one after another from 0, each up to its bound s_j or t_j and only on the active letters, reaches the total κ_{B_i}, and at integer data κ_{B_i} is an integer (21.10) and the loads can be integers. The exponents a_{ij}, $t_j - b_{ij}$, $s_j - a_{ij}$ and b_{ij} are then nonnegative. When P_F is finite every factor of the product formula is finite, the factors being positive, and the load family, on its balance wall, has every proper wall positive (8.7). □

Theorem 22.8 (A chain is the power of one operator). *As stated: at real exponents, the pinned case for the active letters.*

Proof. The reference proof of 22.8 applies verbatim at real exponents. It uses pooling (19.8, for every family at all data, both sides converging together), the nonnegativity of the integrands (at real exponents the letters are positive, and so is every power of them), the scalar twist of a mark (22.1) and the Beta integral $\int_0^1 v^{m_{g_{\mathrm{in}}}-1}(1-v)^{m_{g_{\mathrm{out}}}-1}(1-v+\lambda v)^{-2\sigma}dv = \lambda^{-m_{g_{\mathrm{in}}}}\Gamma\,{}_2{}_1\!\left[\begin{matrix} m_{g_{\mathrm{in}}},\ m_{g_{\mathrm{out}}} \\ 2\sigma \end{matrix}\right]$ at $m_{g_{\mathrm{in}}} + m_{g_{\mathrm{out}}} = 2\sigma$; none of these uses the sign of an exponent. In the pinned case the reference assumes that all the active letters read the cell through one function; a letter of exponent 0 is absent from every period, which is the hypothesis as stated. □

Theorem 22.9 (Letters by transfer). *As stated: the reduction of the states to linearly independent ones, and the bound on the interface dimension for a multivector of any tensor rank.*

Proof. The products of transfer matrices are the reference's. *The stations.* The stations of level k of S_l give a level factorization with as many terms (the children factorization, 21.15), so their number is at least $\operatorname{rank}\Theta^{(k)}_{S_l}$, and the number of states is at least the interface dimension.

The reduction. The child of a pattern p of S_l at level k is $\sum_X \theta_{p\sqcup X}\alpha^X$, the polynomial of the row p of $\Theta^{(k)}_{S_l}$, so the children of level k span a space of dimension $\operatorname{rank}\Theta^{(k)}_{S_l}$. Splitting X along the step B_{l-1}, the child at p of S_{l-1} is

$$\sum_X \theta_{p\sqcup X}\alpha^X = \sum_{p'\subseteq B_{l-1}} \alpha^{p'} \sum_{X'} \theta_{(p\sqcup p')\sqcup X'}\alpha^{X'},$$

the sum over p' of $\alpha^{p'}$ times the child at $p\sqcup p'$ of S_l. Choose at every S_l and every level a basis of the span of the children and write every child in it; the display expresses each basis element at S_{l-1}, a combination of children of S_{l-1}, through the basis elements at S_l, with coefficients polynomial in $\alpha_{B_{l-1}}$. These are transfer matrices on $\sum_k \operatorname{rank}\Theta^{(k)}_{S_l}$ states at S_l, and the products give K; the same coefficients, read on the complements, give Ψ (21.15).

The bound. By 21.17(i), $\operatorname{rank}\Theta^{(k)}_{S_l} \leq \operatorname{ir}_k(\boldsymbol{\omega})^2$, and $\leq \frac{1}{2}\operatorname{ir}_k(\boldsymbol{\omega})(\operatorname{ir}_k(\boldsymbol{\omega})+1)$ when $\boldsymbol{\omega}$ is real; by (ii), $\operatorname{ir}_k(\boldsymbol{\omega}) \leq \sum_i \binom{c_i}{k-d_i}$. The functions $q \mapsto q^2$ and $q \mapsto \frac{1}{2}q(q+1)$ increase on $[0,\infty)$, and summing over k gives the bound. For a determinantal table, of tensor rank one with $c_1 = c(S_l)$ and $d_1 = d(S_l)$ (21.17(ii)), the bound is $\sum_m \binom{c}{m}^2 = \binom{2c}{c}$, $c = c(S_l)$, counting the c-subsets of a set of $2c$ elements by their intersection with a fixed half. □

Theorem 22.10 (Integer joint moments by fixed matrices). *As stated: all the letters of a family at once, along a chain of bunches with any steps and any independent measures on them.*

Proof. This is the reference proof of 22.10 with the tensor product taken over all the tables and with one matrix per step. A product of scalars $\mathbf{u}_i^{\mathsf{T}} A_{i,0}\cdots A_{i,L-1}\mathbf{v}_i$, $i = 1,\dots,n$, equals $(\bigotimes_i \mathbf{u}_i)^{\mathsf{T}}\big(\bigotimes_i A_{i,0}\big)\cdots\big(\bigotimes_i A_{i,L-1}\big)(\bigotimes_i \mathbf{v}_i)$ by the mixed-product rule $(\bigotimes_i A_i)(\bigotimes_i B_i) =$

$\bigotimes_i (A_i B_i)$. Apply it to the a_j copies of K_j and the b_j copies of Ψ_j for every j: the factor of the step l is $\bigotimes_j \big(\mathbf{M}^K_{j,l}(\alpha_{B_l})^{\otimes a_j} \otimes \mathbf{M}^\Psi_{j,l}(\alpha_{B_l})^{\otimes b_j}\big)$, a matrix function of α_{B_l} alone. Every entry of the product is a finite sum of products of one entry of each step factor; the measures of the steps are independent, so the integral of such a product is the product of the integrals, by Tonelli for nonnegative entries and by Fubini under the stated finiteness, and integrating entrywise replaces the factor of the step l by $\mathbf{A}_l$.

For identical steps every $\mathbf{A}_l$ is one matrix $\mathbf{A}$ of the stated size, and the scalars $\mathbf{u}^\top \mathbf{A}^L \mathbf{v}$ satisfy the linear recurrence whose coefficients are those of the characteristic polynomial of $\mathbf{A}$, which annihilates $\mathbf{A}$. With independent laws Gamma_{m_x} on the marks, $\prod_j K_j^{a_j} \Psi_j^{b_j}$ is homogeneous of degree $d = \sum_j (a_j r_j + b_j r_j^*)$, and the draw splits into an independent total of law $\mathrm{Gamma}_{m_\mathcal{E}}$ and a shape of law $\mathrm{Dir}_\mathbf{m}$ (8.12(ii)), the total having the moment $(m_\mathcal{E})_d$ of order d. With $\mathrm{Dir}_{\mathbf{m}|_{B_l}}$ on a step, the entries of $\mathbf{A}_l$ are the Dirichlet moments of the monomials (3.11). □

Theorem 22.11 (Gluing across a bunch of any width). *As stated: the share display for one table at a full lean at every width, and the count $2N-1$ of the free shares at width one.*

Proof. The main display is the reference's. For one table at the full lean $(0,t)$ in level form, one station per level, the active letter Ψ has the $c(B)+1$ terms of the levels $k = d(B), \dots, r(B)$, so the stacked Gamma of the display is $\prod_k \Gamma(b_k)/\Gamma(t)$ and the pins are $\sum_k b_k = t$ and $\sum_k (r(B)-k) b_k = \kappa_B$, the main pin without the shares of K. The fibre integral (4.1) has $c(B)+1$ shares and two pins. Integrating the delta of the pins against the imaginary parts of $b_{d(B)}$ and $b_{r(B)}$ divides by $|\det|$ of the matrix of the two pins in these shares,

$$\begin{pmatrix} 1 & 1 \\ c(B) & 0 \end{pmatrix}, \qquad |\det| = c(B)$$

(check 8a of S21-23_work/checks.py), and leaves $(2\pi)^{2-(c(B)+1)}$ times the integral over the imaginary parts of the $c(B)-1$ free shares, which is $\frac{1}{(2\pi i)^{c(B)-1}} \int \prod db_k$. The pins solved for $b_{d(B)}$ and $b_{r(B)}$ give the stated values; at $c(B)=2$ they are $\frac{\kappa_B - z}{2}$ and $\frac{2t-\kappa_B-z}{2}$ with $b_{d(B)+1} = z$ (check 8b). At $c(B)=1$ the fibre has dimension 0, the matrix of the pins has $|\det| = 1$, and the integral is the value at $(b_{d(B)}, b_{r(B)}) = (\kappa_B, t-\kappa_B)$, the Beta corner times the two inner periods, which is 19.14: the bunch and room letters of the two levels are the minors $\Psi_{\mathcal{B}/R}, \Psi_{\mathcal{B}|B}$ and $\Psi_{\mathcal{B}|R}, \Psi_{\mathcal{B}/B}$ up to constant factors of the factorization that cancel in the product. At width one with factoring weights each level has one station (12.19), so each of the $2N$ active letters has two terms, and the free shares are $4N - 2N - 1 = 2N-1$, the dimension of the fibre integral of 19.13, of which the main display is then the formula. □

Remark. The paragraph after 22.11 gives the Mellin transform of the scale integral $I(\boldsymbol{\zeta}) = \int_0^\infty x^{\kappa_B - 1} \big(1 + \sum_{i=1}^{c-1} \zeta_i x^i + x^c\big)^{-t} dx$, $c = c(B)$. By Tonelli and the substitutions $\zeta_i = \eta_i x^{-i}$, then $x^c = w$,

$$\int \prod_i \zeta_i^{z_i - 1} I(\boldsymbol{\zeta})\, d\boldsymbol{\zeta} = \frac{\prod_i \Gamma(z_i)\, \Gamma(t - \sum_i z_i)}{\Gamma(t)} \int_0^\infty x^{\kappa_B - \sum_i i z_i - 1} (1+x^c)^{\sum_i z_i - t} dx = \frac{\prod_{k=d(B)}^{r(B)} \Gamma(b_k)}{c\, \Gamma(t)}$$

at $b_{r(B)-i} = z_i$, $b_{d(B)} = (\kappa_B - \sum_i i z_i)/c$ and $b_{r(B)} = t - b_{d(B)} - \sum_i z_i$, wherever these have positive real parts: the share kernel of the theorem with the factor $1/c$ of the pin.

48.23 Chapter 23: the law of a cell

Definition 23.1 (The law of a cell family). As stated: the Mellin transform of the law of a cell family, and the transforms and the hinge measures of a cell family, those of one cell being their readings.

Proof. The definition asserts three things. *The Mellin transform.* For complex $\mathbf{z}$ and real positive coordinates, $|\boldsymbol{\varrho}_F^{-\mathbf{z}}| = \boldsymbol{\varrho}_F^{-\operatorname{Re}\mathbf{z}}$, so the integral converges absolutely where it is finite at the real part, with modulus at most that value; for one cell $\boldsymbol{\varrho}_F^{-\mathbf{z}} = (\gamma_{\mathcal{A}}^{\Psi}/\gamma_{\mathcal{A}}^{K})^{-z_{1,\Psi}} = \varrho_{\mathcal{A}}^{-z}$, and the second form is the image of the measure. The base form of one cell is ϕ_F with $(\gamma_{\mathcal{A}}^{K})^{-\kappa} = K_{\mathcal{A}/g}^{\kappa} K_{\mathcal{A}\setminus g}^{-\kappa}$.

Homogeneity. Substitute $\alpha = c\beta$ in the orthant forms: $\alpha^{\mathbf{m}-\mathbf{1}} d\alpha = c^{m_B} \beta^{\mathbf{m}-\mathbf{1}} d\beta$, the letters of the deletions have the degrees $r_{\mathcal{A}_j}$ and $|B| - r_{\mathcal{A}_j}$, those of the contractions $r_{\mathcal{A}_j} - 1$ and $|B| - r_{\mathcal{A}_j} + 1$, and the hinge values have degree one. So

$$f_F(\mathbf{x}, \mathbf{y}) = c^{m_B - \sum_j (s_j r_{\mathcal{A}_j} + t_j(|B| - r_{\mathcal{A}_j}))} f_F(\mathbf{x}/c, c\mathbf{y}) = c^{\kappa_F - |\mathbf{s}|} f_F(\mathbf{x}/c, c\mathbf{y}),$$

and in the same way $\check{f}_F(\mathbf{x}, \mathbf{y}) = c^{\kappa_F - |\mathbf{t}|} \check{f}_F(\mathbf{x}/c, c\mathbf{y})$. With $c = x_1$ each transform is fixed by its values at $x_1 = 1$.

One cell. At $N = 1$, $f_F(1, y)$ and $\check{f}_F(1, y)$ are the integrals called the orthant forms of the transforms of one cell, and the images of $e^{-\lambda} \mathfrak{r}_F$ and $e^{-\lambda} \check{\mathfrak{r}}_F$ under $(\lambda, \mu) \mapsto \mu$ are the images of $\alpha^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}\setminus g}^{-s} \Psi_{\mathcal{A}\setminus g}^{-t} e^{-1/\gamma_{\mathcal{A}}^{K}} d\alpha$ under $\gamma_{\mathcal{A}}^{\Psi}$ and of $\alpha^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}/g}^{-s} \Psi_{\mathcal{A}/g}^{-t} e^{-1/\gamma_{\mathcal{A}}^{\Psi}} d\alpha$ under $\gamma_{\mathcal{A}}^{K}$. □

Theorem 23.3 (The series closure at positive exponents). *As stated: the closure through the transforms of the cell families, each transform one Gramma of index $|\mathbf{s}| - \kappa_{B_i}$ at every data, and its Breta shape reading wherever the family of the hinge minors is balanced.*

Proof. The closure. The responses of the reference, proof and statement of 23.3, are the integrals over $\mathbb{R}_{>0}^{B_i}$ of $\alpha^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K_{\mathcal{A}_{ij}\setminus g_i}^{-s_j} \Psi_{\mathcal{A}_{ij}\setminus g_i}^{-t_j}$ against $\exp\big(- \sum_j x_j K_{\mathcal{A}_{ij}/g_i}/K_{\mathcal{A}_{ij}\setminus g_i} - \sum_j y_j \Psi_{\mathcal{A}_{ij}/g_i}/\Psi_{\mathcal{A}_{ij}\setminus g_i}\big)$ with $y_1 = 1$; since $K_{\mathcal{A}/g}/K_{\mathcal{A}\setminus g} = 1/\gamma_{\mathcal{A}}^{K}$ and $\Psi_{\mathcal{A}/g}/\Psi_{\mathcal{A}\setminus g} = \gamma_{\mathcal{A}}^{\Psi}$, they are the transforms $f_{F_i}(\mathbf{x}, (1, y_2, \dots, y_N))$ of 23.1, and the first display is the reference's. The statement on repeated cells is read off the product.

One table. By the homogeneity of 23.1 with $c = x$, $f_{F_i}(x, 1) = x^{\kappa_{B_i} - s} f_{\mathcal{A}_i}(x)$. The core $\binom{\{h_1,\dots,h_q\}}{q-1}$ has rank $q - 1$ and corank 1, and the core family read at u is balanced (22.3), so $\sum_i \kappa_{B_i} = (q-1)s + t$. The power of x is then $x^{s-1+(q-1)s+t-qs} = x^{t-1}$, and the closure is $\frac{1}{\Gamma(s)\Gamma(t)} \int_0^{\infty} x^{t-1} \prod_i f_{\mathcal{A}_i}(x)\, dx$.

The Gramma. Fix i and write $\alpha = \rho u$, $u \in \Delta_{B_i}$, $\rho > 0$, so that $\alpha^{\mathbf{m}|_{B_i}-\mathbf{1}} d\alpha = \rho^{m_{B_i}-1} u^{\mathbf{m}|_{B_i}-\mathbf{1}} d\rho\, d\omega$ (2.7). The marks g_i being neither loops nor coloops, $K_{\mathcal{A}_{ij}\setminus g_i}$ is homogeneous of degree $r_{\mathcal{A}_{ij}}$ and $\Psi_{\mathcal{A}_{ij}\setminus g_i}$ of degree $|B_i| - r_{\mathcal{A}_{ij}}$, and the hinge values of degree one. So the power of ρ is $\rho^{\nu-1}$ with

$$\nu = m_{B_i} - \sum_j \big(s_j r_{\mathcal{A}_{ij}} + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big) = \kappa_{B_i} - |\mathbf{s}|,$$

and the exponential is $\exp(-a(u)/\rho - b(u)\rho)$ with $a = \sum_j x_j/\gamma_{\mathcal{A}_{ij}}^{K}$ and $b = \sum_j y_j \gamma_{\mathcal{A}_{ij}}^{\Psi}$. By 2.1 and the reflection (2.3),

$$\int_0^{\infty} \rho^{\nu-1} e^{-b\rho - a/\rho}\, d\rho = \mathsf{G}_{\nu;(1,-1)}(b, a) = \mathsf{G}_{-\nu;(1,-1)}(a, b),$$

and Tonelli gives the display of the transform.

Index 0. The family of the hinge minors has the degree $\sum_j \big(s_j r_{\mathcal{A}_{ij}} + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big)$ at the masses $\mathbf{m}|_{B_i}$, so it is on its balance wall exactly when $\kappa_{B_i} = |\mathbf{s}|$; at the centre of one table, every mass σ and $s = t = \sigma$, $\kappa_{B_i} = \sigma|B_i| - \sigma(r-1) - \sigma(|B_i| - r) = s$. Torus homogeneity (2.4(iii)) with $\lambda = a$ gives $\mathsf{G}_{0;(1,-1)}(a,b) = \mathsf{G}_{0;(1,-1)}(1,ab)$. With every proper wall positive, the form of the family of the hinge minors divided by its period is its Breta shape law (8.10), which gives the expectation; for one table, $ab = xy\,\gamma^{\Psi}_{\mathcal{A}_i}/\gamma^{K}_{\mathcal{A}_i} = xy\varrho_{\mathcal{A}_i}$. The period of k copies is the first display with all F_i equal. □

Theorem 23.6 (The law of a substitution). *As stated: (i) and (ii) for families at real data; the present (i) was the gluing of a family, and (ii) for one table at $s, t > 0$.*

Proof. (ii) This is the reference proof of 23.6(ii) with the index j, the steps being those of 23.2 applied to one cell family after another. Write $B = \mathcal{E}_{\mathcal{H}} \sqcup \bigsqcup_i B_i$ and $\mathcal{A}_j = \mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$, a cell with the hinge g_0 (23.5). For a Borel $H \geq 0$, $\int H\, d((\boldsymbol{\varrho}_F)_*\phi_F)$ is the integral over the section $\{|\alpha| = 1\}$ of the orthant form

$$\omega = \alpha^{\mathbf{m-1}} H(\boldsymbol{\varrho}_F)\, K_{\mathcal{A}_1/g_0}^{\kappa_F - s_1} K_{\mathcal{A}_1\setminus g_0}^{-\kappa_F} \prod_{j\geq 2} K_{\mathcal{A}_j/g_0}^{-s_j} \prod_j \Psi_{\mathcal{A}_j\setminus g_0}^{-t_j}\, d\alpha,$$

homogeneous of degree $m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big) - \kappa_F = 0$, the base form being $u^{\mathbf{m-1}} \prod_j K_{\mathcal{A}_j/g_0}^{-s_j} \Psi_{\mathcal{A}_j\setminus g_0}^{-t_j} (K_{\mathcal{A}_1\setminus g_0}/K_{\mathcal{A}_1/g_0})^{-\kappa_F} d\omega$. By 23.5, with the hinge values of the cells at their own marks,

$$K_{\mathcal{A}_j\setminus g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}\, K_{\mathcal{H}_j\setminus g_0}\big(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^K_{\mathcal{A}_{ij}})_i\big), \qquad K_{\mathcal{A}_j/g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}\, K_{\mathcal{H}_j/g_0}\big(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^K_{\mathcal{A}_{ij}})_i\big),$$

and the same for Ψ with $\prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i}$ and the hinge values $\gamma^{\Psi}_{\mathcal{A}_{ij}}$. The products over the cells cancel in the quotients, so the hinge values of $\mathcal{A}_j$ at g_0 are those of $\mathcal{H}_j$ read at $(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^K_{\mathcal{A}_{ij}})_i)$ and $(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^{\Psi}_{\mathcal{A}_{ij}})_i)$.

Apply 23.2 to F_1 with $Z = \mathcal{E}_{\mathcal{H}} \sqcup \bigsqcup_{i\geq 2} B_i$. Assign to the letter $K_{\mathcal{A}_{1j}}$ the functions $\prod_{i\geq 2} K_{\mathcal{A}_{ij}/g_i} \cdot K_{\mathcal{H}_j/g_0}$ and $\prod_{i\geq 2} K_{\mathcal{A}_{ij}/g_i} \cdot K_{\mathcal{H}_j\setminus g_0}$, read at the hinge value of the cell at g_1, with the exponents s_j and 0 for $j \geq 2$ and $s_1 - \kappa_F$ and κ_F for $j = 1$; and to the letter $\Psi_{\mathcal{A}_{1j}}$ the functions $\prod_{i\geq 2} \Psi_{\mathcal{A}_{ij}\setminus g_i} \cdot \Psi_{\mathcal{H}_j\setminus g_0}$ and $\prod_{i\geq 2} \Psi_{\mathcal{A}_{ij}\setminus g_i} \cdot \Psi_{\mathcal{H}_j/g_0}$ with the exponents t_j and 0. The exponents assigned to each letter add to its exponent, the letters of the hinge minors of F_1 collect to $K_{\mathcal{A}_{1j}/g_1}^{-s_j} \Psi_{\mathcal{A}_{1j}\setminus g_1}^{-t_j}$, the product of the assigned powers is the rest of ω, and $H(\boldsymbol{\varrho}_F)$ is a Borel function of the values of the assigned functions, which give the hinge values of every $\mathcal{A}_j$ as quotients. The lemma collapses B_1 to g_1 with the mass κ_{F_1} and replaces the hinge values of the cells at g_1 by $x_{1,j,K}\alpha_{g_1}$ and $x_{1,j,\Psi}\alpha_{g_1}$, the twist at g_1 by the point $\mathbf{x}_1$ of the law of F_1. Repeat for $F_2, \dots, F_k$, moving the integrals over the laws outside (Tonelli). What remains is the integral over a section of the form of the cell family $F_{\mathcal{H}}[\mathbf{x}]$, its hinge minors at g_0 read with the twists, its masses $\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}$ and κ_{F_i}, the power κ_F in place of its hinge wall, and H of its ratio vector inserted. Its hinge wall is

$$m_{\mathcal{E}_{\mathcal{H}}} + \sum_i \kappa_{F_i} - \sum_j \big(s_j(r_{\mathcal{H}_j} - 1) + t_j(|\mathcal{E}_{\mathcal{H}}| + k - r_{\mathcal{H}_j})\big),$$

and it equals κ_F: insert $\kappa_{F_i} = m_{B_i} - \sum_j (s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(|B_i| - r_{\mathcal{A}_{ij}}))$, $r_{\mathcal{A}_j} = r_{\mathcal{H}_j} + \sum_i (r_{\mathcal{A}_{ij}} - 1)$ and $|B| = |\mathcal{E}_{\mathcal{H}}| + \sum_i |B_i|$ (23.5); the coefficients of s_j and of t_j agree term by term (check 1 of S21-23_work/checks.py). So the remaining integral is $\int H\, d((\boldsymbol{\varrho}_{F_{\mathcal{H}}[\mathbf{x}]})_*\phi_{F_{\mathcal{H}}[\mathbf{x}]})$, and the mixture follows;

the hinge wall does not depend on $\mathbf{x}$. With $H = \boldsymbol{\varrho}^{-\mathbf{z}}$, a nonnegative Borel function for real $\mathbf{z}$, the Mellin transforms mix in the same way.

(i) The same steps with the family word of F in place of the base form: on the balance wall the form of F is homogeneous of degree zero, its letters are $\prod_i K_{\mathcal{A}_{ij}/g_i} \cdot K_{\mathcal{H}_j}$ and $\prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i} \cdot \Psi_{\mathcal{H}_j}$ read at the hinge values (23.5), the factors $K_{\mathcal{H}_j}$ and $\Psi_{\mathcal{H}_j}$ are assigned to the letters of each cell family in turn with the exponents s_j and t_j, and 23.2 collapses every cell to its marker with the mass κ_{F_i} and the twist by the point of its law. This is the reference's remark on families (rem:familiesnodes): 23.2 holds at real exponents, its proof collecting degrees and using Tonelli. For the core $\binom{\{g_1,g_2\}}{1}$, twisted at g_1 by $\mathbf{x}_1$ and at g_2 by $\mathbf{x}_2$, table j has $K_j = x_{1,j,K}\alpha_{g_1} + x_{2,j,K}\alpha_{g_2}$ and $\Psi_j = x_{1,j,\Psi}\alpha_{g_1} + x_{2,j,\Psi}\alpha_{g_2}$, and on the section $\alpha_{g_2} = 1$ its period is the displayed integral, a Breta of one bunch (15.1); the substitution $v \mapsto 1/v$ multiplies the integrand by $v^{2\sigma}$ and turns $v^{\kappa_{F_1}-1}dv$ into $v^{-\kappa_{F_1}-1}dv$, which gives the swap since $\kappa_{F_1} + \kappa_{F_2} = 2\sigma$. □

Proposition 23.8 (Faces of one law). *As stated, for every cell family with plain tables and $s_j, t_j > 0$.*

Proof. Write $\kappa = \kappa_F$ and γ_j^K, γ_j^Ψ for the hinge values of $\mathcal{A}_j$. The reference proof of 23.8 is the case $N = 1$; the steps below are that proof with the index j, the changes being the bookkeeping of the powers of γ_1^K and γ_1^Ψ, which pass the integrals from the simplex to the law.

(i) In the orthant form of f_F write $\alpha = \rho u$ (2.7). As in the proof of 23.3 in this part, the power of ρ is $\rho^{\kappa-|\mathbf{s}|-1}$ and

$$f_F(\mathbf{x},\mathbf{y}) = \int_{\Delta_B} \mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j \frac{x_j}{\gamma_j^K(u)},\ \sum_j y_j\gamma_j^\Psi(u)\Big)\, u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j\setminus g}^{-s_j}\Psi_{\mathcal{A}_j\setminus g}^{-t_j}\, d\omega(u).$$

Since $K_{\mathcal{A}_j\setminus g} = \gamma_j^K K_{\mathcal{A}_j/g}$, the measure is $(\gamma_1^K)^\kappa \prod_j (\gamma_j^K)^{-s_j}\phi_F = (\gamma_1^K)^{\kappa-|\mathbf{s}|}\prod_j(\gamma_j^K/\gamma_1^K)^{-s_j}\phi_F$. Torus homogeneity (2.4(iii)) with $\lambda = 1/\gamma_1^K$, $\mathsf{G}_{\nu;(1,-1)}(\lambda\xi_1,\xi_2/\lambda) = \lambda^{-\nu}\mathsf{G}_{\nu;(1,-1)}(\xi_1,\xi_2)$, gives, at $\mathbf{z} = \boldsymbol{\varrho}_F(u)$,

$$\mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j \frac{x_j}{\gamma_j^K},\ \sum_j y_j\gamma_j^\Psi\Big) = (\gamma_1^K)^{|\mathbf{s}|-\kappa}\,\mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j \frac{x_j}{z_{j,K}},\ \sum_j y_j z_{j,\Psi}\Big).$$

The powers of γ_1^K cancel, and the image under $\boldsymbol{\varrho}_F$ gives the first law form. For $\check{f}_F$ the contracted letters have the degrees $r_{\mathcal{A}_j} - 1$ and $|B| - r_{\mathcal{A}_j} + 1$, so the power of ρ is $\rho^{\kappa-|\mathbf{t}|-1}$, the index is $|\mathbf{t}| - \kappa$, and the coefficients are $\sum_j x_j/\gamma_j^\Psi$ and $\sum_j y_j\gamma_j^K$; the measure is $u^{\mathbf{m}-\mathbf{1}}\prod_j K_{\mathcal{A}_j/g}^{-s_j}\Psi_{\mathcal{A}_j/g}^{-t_j}d\omega = (\gamma_1^K)^\kappa\prod_j(\gamma_j^\Psi)^{-t_j}\phi_F$, since $\Psi_{\mathcal{A}_j/g} = \gamma_j^\Psi\Psi_{\mathcal{A}_j\setminus g}$. Torus homogeneity with $\lambda = 1/\gamma_1^\Psi$ extracts $(\gamma_1^\Psi)^{|\mathbf{t}|-\kappa}$, and $(\gamma_1^K)^\kappa\prod_j(\gamma_j^\Psi)^{-t_j}(\gamma_1^\Psi)^{|\mathbf{t}|-\kappa} = z_{1,\Psi}^{-\kappa}\prod_j(z_{j,\Psi}/z_{1,\Psi})^{-t_j}$, with $z_{1,\Psi}/z_{j,\Psi} = \gamma_1^\Psi/\gamma_j^\Psi$ and $z_{j,K}/z_{1,\Psi} = \gamma_j^K/\gamma_1^\Psi$: the second law form. At $N = 1$, $z_{1,K} = 1$, $z_{1,\Psi} = \varrho_{\mathcal{A}}$ and $\mathbf{x} = 1$ give $\mathsf{G}_{s-\kappa;(1,-1)}(1,\varrho_{\mathcal{A}}y)$ and $\mathsf{G}_{t-\kappa;(1,-1)}(1,y/\varrho_{\mathcal{A}})\varrho_{\mathcal{A}}^{-\kappa}$.

(ii) The Laplace forms are the definitions of the images. For the law form of $\mathfrak{r}_F$ write $\alpha = \rho u$ and $p = \rho\gamma_1^K(u)$: then $1/\gamma_j^K(\alpha) = 1/(pz_{j,K})$ and $\gamma_j^\Psi(\alpha) = pz_{j,\Psi}$ at $\mathbf{z} = \boldsymbol{\varrho}_F(u)$, and $\rho^{\kappa-|\mathbf{s}|-1}d\rho = (\gamma_1^K)^{|\mathbf{s}|-\kappa}p^{\kappa-|\mathbf{s}|-1}dp$, which with the measure of (i) leaves $\prod_j z_{j,K}^{-s_j}\phi_F(du)\otimes p^{\kappa-|\mathbf{s}|-1}dp$. A Laplace transform of a positive measure is completely monotone on the interior of the set where it is finite, by differentiation under the integral, the derivatives being dominated near each such point by the integrand at a point with smaller coordinates, and it is log-convex along segments by Hölder's inequality. For one cell, with $\lambda = 1/p$ and $p = 1/q$, the measure $e^{-\lambda}\mathfrak{r}_F$ becomes $e^{-q}q^{s-\kappa-1}dq$ against the law, and the coordinate $\mu = pz$ becomes z/q; for $\check{\mathfrak{r}}_F$, with $p = \rho\gamma^\Psi(u)$, the same

computation gives the image of $z^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dz)\otimes p^{\kappa-t-1}dp$ under $(z,p)\mapsto(1/p,p/z)$, and $e^{-\lambda}$ and $p=1/q$ give the image of $z^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dz)\otimes q^{t-\kappa-1}e^{-q}dq$ under $(z,q)\mapsto 1/(zq)$.

(iii) Insert the Laplace form of (ii) and integrate first in x_j, $j\geq 2$, and y_j (Tonelli): $\int_0^\infty x^{a-1}e^{-x\lambda}dx=\Gamma(a)\lambda^{-a}$. With $\lambda_j=1/(pz_{j,K})$, $\mu_j=pz_{j,\Psi}$ and $x_1=1$,

$$\prod_{j\geq 2}\Gamma(a_j)\prod_j\Gamma(b_j)\iint e^{-1/p}\prod_{j\geq 2}(pz_{j,K})^{a_j}\prod_j(pz_{j,\Psi})^{-b_j}\prod_j z_{j,K}^{-s_j}\,p^{\kappa-|\mathbf{s}|-1}dp\,(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z}).$$

The power of p is p^{-a_1-1} with $a_1=\sum_j b_j-\sum_{j\geq 2}a_j-\kappa+|\mathbf{s}|$, the given a_1 by the hyperplane, and $\int_0^\infty e^{-1/p}p^{-a_1-1}dp=\Gamma(a_1)$ for $a_1>0$; the power of $\mathbf{z}$ is $\prod_j z_{j,K}^{-(s_j-a_j)}z_{j,\Psi}^{-b_j}$, $z_{1,K}=1$ (check 5). For $\check{f}_F$ the law form of $\check{\mathfrak{r}}_F$, obtained as in (ii) with $p=\rho\gamma_1^{\Psi}$, is the image of $z_{1,\Psi}^{-\kappa}\prod_j(z_{j,\Psi}/z_{1,\Psi})^{-t_j}(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z})\otimes p^{\kappa-|\mathbf{t}|-1}dp$ under $(\mathbf{z},p)\mapsto\big((z_{1,\Psi}/(pz_{j,\Psi}))_j,(pz_{j,K}/z_{1,\Psi})_j\big)$; the same integration leaves p^{-a_1-1} with $a_1=\sum_j b_j-\sum_{j\geq 2}a_j-\kappa+|\mathbf{t}|$, and the total power of $z_{1,\Psi}$ is $-\sum_{j\geq 2}a_j+\sum_j b_j-\kappa+\sum_{j\geq 2}t_j=a_1-t_1$, so the power of $\mathbf{z}$ is $\prod_j z_{j,K}^{-b_j}z_{j,\Psi}^{-(t_j-a_j)}$. Both points satisfy $\sum_j(z_{j,K}+z_{j,\Psi})=\kappa$. At $N=1$, $b=z$ and $a=z+s-\kappa$, the first gives $\Gamma(z)\Gamma(z+s-\kappa)\int\varrho_{\mathcal{A}}^{-z}d\phi_{\mathcal{A}}$, and the second, with $a=z+t-\kappa$, $\Gamma(z)\Gamma(z+t-\kappa)\int\varrho_{\mathcal{A}}^{-(\kappa-z)}d\phi_{\mathcal{A}}$.

(iv) Table by table, the dual table $\mathcal{A}^*$ has the bases $(B\sqcup\{g\})\setminus T$ with the weights θ_T, and splitting by whether $g\in T$,

$$K_{\mathcal{A}^*/g}=\Psi_{\mathcal{A}\setminus g},\quad K_{\mathcal{A}^*\setminus g}=\Psi_{\mathcal{A}/g},\quad \Psi_{\mathcal{A}^*\setminus g}=K_{\mathcal{A}/g},\quad \Psi_{\mathcal{A}^*/g}=K_{\mathcal{A}\setminus g},$$

its rank is $|B|+1-r_{\mathcal{A}}$, and g is neither a loop nor a coloop of it. So $\gamma^K_{\mathcal{A}^*}=\gamma^{\Psi}_{\mathcal{A}}$ and $\gamma^{\Psi}_{\mathcal{A}^*}=\gamma^K_{\mathcal{A}}$, and the hinge wall at (t_j,s_j) is $m_B-\sum_j\big(t_j(|B|-r_{\mathcal{A}_j})+s_j(r_{\mathcal{A}_j}-1)\big)=\kappa$. The base form of F^* is $u^{\mathbf{m}-\mathbf{1}}\prod_j\Psi_{\mathcal{A}_j\setminus g}^{-t_j}K_{\mathcal{A}_j/g}^{-s_j}(\gamma_1^{\Psi})^{-\kappa}=(\gamma_1^{\Psi}/\gamma_1^K)^{-\kappa}\phi_F$, its ratio vector is $(\gamma_j^{\Psi}/\gamma_1^{\Psi},\gamma_j^K/\gamma_1^{\Psi})_j=(z_{j,\Psi}/z_{1,\Psi},z_{j,K}/z_{1,\Psi})_j$ at $\mathbf{z}=\boldsymbol{\varrho}_F$, which gives the law, and the orthant forms of f_{F^*} and $\check{f}_{F^*}$ are those of $\check{f}_F$ and f_F. At $N=1$ the ratio is $1/\varrho_{\mathcal{A}}$, and $\int\varrho_{\mathcal{A}^*}^{-z}d\phi_{\mathcal{A}^*}=\int\varrho_{\mathcal{A}}^{z}\varrho_{\mathcal{A}}^{-\kappa}d\phi_{\mathcal{A}}$.

(v) By (iii) for f_F at $(\mathbf{a},\mathbf{b})$ and for $\check{f}_F$ at $(\mathbf{t}-\mathbf{b},\mathbf{s}-\mathbf{a})$, whose hyperplane condition $\sum_j(s_j-a_j)-\sum_j(t_j-b_j)=\kappa-|\mathbf{t}|$ is that of $(\mathbf{a},\mathbf{b})$, the Mellin transform of the law appears at the same point $(\mathbf{s}-\mathbf{a},\mathbf{b})$ on both sides, with the Gamma products $\prod_j\Gamma(a_j)\Gamma(b_j)$ and $\prod_j\Gamma(t_j-b_j)\Gamma(s_j-a_j)$; all these Gammas are finite and positive, which gives the relation in $(0,\infty]$. At $N=1$, with $z=s-a$ and $b=\kappa-z$, the ratio of the Gammas is that of the displayed one-cell relation (check 6). The expansion clause concerns the node of width one of one table and is the reference's. □

Lemma S10. *Let F be a cell family with plain tables, $s_j,t_j>0$, and $\mathbf{a},\mathbf{b}\in(0,\infty)^N$ real with $\sum_j b_j-\sum_j a_j=\kappa_F-|\mathbf{s}|$ and $\int\boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a},\mathbf{b})}d\phi_F<\infty$. Write $\mathbf{u}=(x_2,\dots,x_N,y_1,\dots,y_N)\in(0,\infty)^{2N-1}$, $g(\mathbf{u})=f_F((1,x_2,\dots,x_N),\mathbf{y})$, $\mathbf{c}=(a_2,\dots,a_N,b_1,\dots,b_N)$, and for $\boldsymbol{\tau}=(\tau_2,\dots,\tau_N,\tau'_1,\dots,\tau'_N)\in\mathbb{R}^{2N-1}$ put $\tau_1=\sum_j\tau'_j-\sum_{j\geq 2}\tau_j$ and*

$$G(\boldsymbol{\tau})=\prod_j\Gamma(a_j+i\tau_j)\Gamma(b_j+i\tau'_j)\int\boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a}-i\boldsymbol{\tau}_K,\ \mathbf{b}+i\boldsymbol{\tau}_\Psi)}d\phi_F,$$

$\boldsymbol{\tau}_K=(\tau_j)_j$ and $\boldsymbol{\tau}_\Psi=(\tau'_j)_j$. Then g is finite and continuous on $(0,\infty)^{2N-1}$, G is the Mellin transform of g on the vertical subspace through $\mathbf{c}$ and is integrable, and

$$g(\mathbf{u})=(2\pi)^{1-2N}\int_{\mathbb{R}^{2N-1}}G(\boldsymbol{\tau})\,\mathbf{u}^{-\mathbf{c}-i\boldsymbol{\tau}}\,d\boldsymbol{\tau}\qquad(\mathbf{u}\in(0,\infty)^{2N-1}).$$

Proof. The computation of 23.8(iii) at the complex exponents $\mathbf{c}+i\boldsymbol{\tau}$ converges absolutely, its integrand having at every point the modulus of the integrand at $\mathbf{c}$, so it computes the Mellin transform of g at $\mathbf{c}+i\boldsymbol{\tau}$ as $G(\boldsymbol{\tau})$; the point $(\mathbf{s}-\mathbf{a}-i\boldsymbol{\tau}_K, \mathbf{b}+i\boldsymbol{\tau}_\Psi)$ stays on the hyperplane, which fixes τ_1. By 23.1, $|G(\boldsymbol{\tau})| \le \prod_j |\Gamma(a_j+i\tau_j)\Gamma(b_j+i\tau'_j)| \int \boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a},\mathbf{b})} d\phi_F$, and the bound $|\Gamma(c+i\tau)| \le C_c\, e^{-\pi|\tau|/2}(1+|\tau|)^{c-1/2}$ makes G integrable on $\mathbb{R}^{2N-1}$. The Mellin transform of g at the real point $\mathbf{c}$ is finite, so $g<\infty$ almost everywhere. By 23.8(ii), g is the Laplace transform of a positive measure in the variables $\mathbf{u}$, hence nonincreasing in each of them: if g were infinite at a point it would be infinite on the set of points below it, which has positive measure. So g is finite everywhere, and continuous by dominated convergence, the integrand near a point being dominated by the integrand at a point with smaller coordinates. In the coordinates $\mathbf{v}=\log\mathbf{u}$ the function $g(e^{\mathbf{v}})e^{\langle \mathbf{c},\mathbf{v}\rangle}$ is integrable with the integrable Fourier transform G, so the Fourier inversion theorem gives the display almost everywhere, and everywhere since both sides are continuous. □

Theorem 23.9 (Two cell families glued). *As stated, for two cell families with plain tables and $s_j, t_j > 0$.*

Proof. (i) 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$ in every table, F_1 at g_1 and F_2 at g_2, the period of the twisted core family being $\mathcal{G}$.

(ii), first form. The gluing of two cells along the hinge is the series closure $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}_1, \mathcal{A}_2\rangle$ (21.7(iii)), so 23.3 at $q=2$ gives

$$P = C\int \prod_j x_j^{s_j-1} \prod_{j\ge 2} y_j^{t_j-1}\, f_{F_1} f_{F_2}\big(\mathbf{x},(1,\mathbf{y}')\big)\, d\mathbf{x}\, d\mathbf{y}', \qquad C=\prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)}.$$

Substitute $\mathbf{x}=x_1(1,\boldsymbol{\xi})$, $d\mathbf{x}=x_1^{N-1}dx_1\, d\boldsymbol{\xi}$, and use the homogeneity of 23.1 with $c=x_1$, $f_{F_i}(\mathbf{x},(1,\mathbf{y}')) = x_1^{\kappa_{F_i}-|\mathbf{s}|} f_{F_i}((1,\boldsymbol{\xi}), x_1(1,\mathbf{y}'))$; then substitute $\boldsymbol{\eta}=x_1(1,\mathbf{y}')$, $dx_1\, d\mathbf{y}' = x_1^{1-N} d\boldsymbol{\eta}$. The powers of $x_1=\eta_1$ collect to

$$(|\mathbf{s}|-N)+(N-1)+(\kappa_{F_1}+\kappa_{F_2}-2|\mathbf{s}|)+(1-N)-(|\mathbf{t}|-t_1)+(N-1) = t_1-1,$$

since $\kappa_{F_1}+\kappa_{F_2}=|\mathbf{s}|+|\mathbf{t}|$ (check 10 of S21-23_work/checks.py): the first form.

(ii), second form. The complements of the bases of a two-sum of cells are the bases of the two-sum of the dual cells: for X a basis of $\mathcal{A}\setminus g$ and $Y\sqcup\{g\}$ one of $\mathcal{A}'$, $B\setminus X$ is a basis of $\mathcal{A}^*/g$ and $R\setminus Y$ one of $\mathcal{A}'^*\setminus g$, and symmetrically (21.2), with the product weights. By 8.15 the period of $F_1\oplus_g F_2$ at $(\mathbf{s},\mathbf{t})$ is that of $F_1^*\oplus_g F_2^*$ at $(\mathbf{t},\mathbf{s})$, and by 23.8(iv) the dual cell families have the hinge walls $\kappa_{F_1},\kappa_{F_2}$ and the transforms $\check{f}_{F_1}, \check{f}_{F_2}$. The first form for them is the second form.

(iii) Insert the Laplace forms of 23.8(ii) into (ii) and integrate first in x_j, $j\ge 2$, and y_j (Tonelli): $\int_0^\infty x^{s-1}e^{-x(\lambda+\lambda')}dx = \Gamma(s)(\lambda+\lambda')^{-s}$, and $x_1=1$ leaves $e^{-\lambda_1-\lambda'_1}$; the factors $\Gamma(s_j)$, $j\ge 2$, and $\Gamma(t_j)$ cancel against C. The period converges exactly when every proper wall is positive (8.7).

(iv) Let the contour have the real point $(\mathbf{a}^d,\mathbf{b}^d,\mathbf{a}^r,\mathbf{b}^r)$, all coordinates positive, with $\mathbf{a}^d+\mathbf{a}^r=\mathbf{s}$, $\mathbf{b}^d+\mathbf{b}^r=\mathbf{t}$ and $\sum_j(a_j^r+b_j^d)=\kappa_{F_1}$, and put $\mathbf{a}=\mathbf{a}^d$, $\mathbf{b}=\mathbf{b}^d$; then $\sum_j b_j-\sum_j a_j = \kappa_{F_1}-|\mathbf{s}|$, and the Mellin transform of the law of F_1 at $(\mathbf{s}-\mathbf{a},\mathbf{b})=(\mathbf{a}^r,\mathbf{b}^d)$ is finite. By Lemma S10, f_{F_1} is the inverse Mellin integral of G_1 through $\mathbf{c}$. Insert it into the first form of (ii). The double integral converges absolutely: integrating first in $\mathbf{u}$ gives $\int |G_1|\, d\boldsymbol{\tau}$ times $\int \prod_{j\ge2} x_j^{s_j-a_j-1}\prod_j y_j^{t_j-b_j-1} f_{F_2}\, d\mathbf{x}'\, d\mathbf{y}$, which is 23.8(iii) for F_2 at $(\mathbf{s}-\mathbf{a},\mathbf{t}-\mathbf{b})$: its first coordinate is $|\mathbf{t}|-|\mathbf{b}|-(|\mathbf{s}|-s_1-\sum_{j\ge2}a_j)-\kappa_{F_2}+|\mathbf{s}| = s_1-a_1$, by $\kappa_{F_2}=2\sigma-\kappa_{F_1}$, so it equals $\prod_j\Gamma(s_j-a_j)\Gamma(t_j-b_j)\int \boldsymbol{\varrho}_{F_2}^{-(\mathbf{a}^d,\mathbf{b}^r)}d\phi_{F_2}<\infty$. By Fubini,

$$P = C\,(2\pi)^{1-2N}\int_{\mathbb{R}^{2N-1}} \prod_j \Gamma(a_j^d)\Gamma(b_j^d)\Gamma(a_j^r)\Gamma(b_j^r) \int \boldsymbol{\varrho}_{F_1}^{-(\mathbf{a}^r,\mathbf{b}^d)}d\phi_{F_1}\int \boldsymbol{\varrho}_{F_2}^{-(\mathbf{a}^d,\mathbf{b}^r)}d\phi_{F_2}\, d\boldsymbol{\tau},$$

the shares moving with the imaginary parts $\boldsymbol{\tau}$ of $(a_2^d,\dots,a_N^d,b_1^d,\dots,b_N^d)$, the others fixed by the pins, and $C\prod_j\Gamma(\cdot)$ is $\prod_j\Gamma_{\,2}^{\,4}\left[\begin{matrix}a_j^d,\ b_j^d,\ a_j^r,\ b_j^r\\ s_j,\ t_j\end{matrix}\right]$. This is the fibre integral of 4.1 over the level shares with their $2N+1$ pins: solved for a_j^r, b_j^r and a_1^d the matrix of the pins has $|\det|=1$, so the fibre integral is $(2\pi)^{1-2N}$ times the integral over $\boldsymbol{\tau}$. In the pair coordinates of 19.13, $\sigma_j^d=\frac12(a_j^d+b_j^d)$ and $\delta_j^d=\frac12(a_j^d-b_j^d)$ with the one pin $\sum_j\delta_j^d=\frac12(|\mathbf{s}|-\kappa_{F_1})$, the linear map from $\boldsymbol{\tau}$ to the imaginary parts of $(\sigma_1^d,\dots,\sigma_N^d,\delta_2^d,\dots,\delta_N^d)$ has $|\det|=2^{1-N}$: the map $(a,b)\mapsto(\frac{a+b}{2},\frac{a-b}{2})$ of $\mathbb{R}^{2N}$ has $|\det|=2^{-N}$, it sends the normal coordinate $\sum_j(a_j-b_j)$ of the pin to twice the normal coordinate $\sum_j\delta_j$, and the quotient is 2^{1-N}. So the integral is 2^{N-1} times the fibre integral in the pair coordinates, the formula of 19.13. That the two Mellin transforms are the periods of the bunch family and of the room family at the level shares is the identity $u^{\mathbf{m}-\mathbf{1}}\prod_j K_{\mathcal{A}_j/g}^{-a_j^d}\Psi_{\mathcal{A}_j/g}^{-b_j^d}K_{\mathcal{A}_j\setminus g}^{-a_j^r}\Psi_{\mathcal{A}_j\setminus g}^{-b_j^r}=\boldsymbol{\varrho}_{F_1}^{-(\mathbf{a}^r,\mathbf{b}^d)}(\gamma_{\mathcal{A}_1}^K)^{\kappa_{F_1}-\sum_j(a_j^r+b_j^d)}$ times the density of ϕ_{F_1}, the last power being 1 on the pin, with $\mathcal{B}_j|B=\mathcal{A}_j\setminus g$ and $\mathcal{B}_j/R=\mathcal{A}_j/g$ (21.12), and its analogue for the room, whose pin $\sum_j(a_j^d+b_j^r)=2\sigma-\kappa_{F_1}=\kappa_{F_2}$ follows from the letter totals.

One table. At $N=1$ the laws are measures on $(0,\infty)$, the kernel is that of 23.7, (ii) and (iii) have no variables x_j, the image of $e^{-\lambda}\mathfrak{r}_{F_i}$ under μ being $\mathfrak{r}^\Psi$, and in (iv) the fibre is the line of $b^d=z$, with $a^d=s-\kappa_{\mathcal{A}}+z$, $a^r=\kappa_{\mathcal{A}}-z$, $b^r=t-z$; the conditions on the contour are $\max(0,\kappa_{\mathcal{A}}-s)<z_0<\min(t,\kappa_{\mathcal{A}})$ and the finiteness of $\int\varrho_{\mathcal{A}}^{-z_0}d\phi_{\mathcal{A}}$ and $\int\varrho_{\mathcal{A}'}^{z_0-t}d\phi_{\mathcal{A}'}$, and $d\sigma^d=dz$. The first form of (ii) is 20.26(i) with the room transform $y^tf_{\mathcal{A}'}(y)$. □

Definition 23.10 (One weighted mark). As stated: the one-mark cell family of a ratio vector, with two systems of weights in every table.

Proof. Table j has the bases $\{e\}$ and $\{g\}$, with $\theta^K=(x_{0,j,K},1)$ and $\theta^\Psi=(1,x_{0,j,\Psi})$, so $K_j=x_{0,j,K}\alpha_e+\alpha_g$ and $\Psi_j=\theta^\Psi_{\{e\}}\alpha_g+\theta^\Psi_{\{g\}}\alpha_e=\alpha_g+x_{0,j,\Psi}\alpha_e$, two systems of weights being allowed (21.1, 8.1). The hinge minors are $K_{\setminus g}=x_{0,j,K}\alpha_e$, $K_{/g}=1$, $\Psi_{\setminus g}=1$ and $\Psi_{/g}=x_{0,j,\Psi}\alpha_e$, so the hinge values are $x_{0,j,K}\alpha_e$ and $x_{0,j,\Psi}\alpha_e$ and the ratio vector is the constant $\mathbf{x}_0$, $x_{0,1,K}=1$. The rank is 1 and $|B|=1$, so the hinge wall is m_e. The simplex $\Delta_{\{e\}}$ is the point $u_e=1$, carrying the unit mass, and there the density of the base form is $1\cdot1\cdot(x_{0,1,K})^{-m_e}=1$: the law is $\delta_{\mathbf{x}_0}$. □

Theorem 23.11 (The law is what a cell family retains). *As stated, for every pair of cell families with plain tables and $s_j,t_j>0$.*

Proof. (a)⇒(b). By 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$, the period of $F\oplus_gF''$ depends on F only through κ and its law; this needs no plainness of F''. (b)⇒(c) is a special case.

(c)⇒(d). By 23.6(i) and the law $\delta_{\mathbf{x}_0}$ of the one-mark cell family (23.10), the period against it is $\Pi_F(\mathbf{x}_0)=\int\mathcal{G}(\mathbf{z},\mathbf{x}_0)\,d((\boldsymbol{\varrho}_F)_*\phi_F)(\mathbf{z})$, $\mathcal{G}(\mathbf{z},\mathbf{x}_0)=\int_0^\infty v^{\kappa-1}\prod_j(z_{j,K}v+x_{0,j,K})^{-s_j}(z_{j,\Psi}v+x_{0,j,\Psi})^{-t_j}dv$. Since $(zv+x)/(zv+x')$ lies between $\min(1,x/x')$ and $\max(1,x/x')$, $\mathcal{G}(\cdot,\mathbf{x}_0)$ and $\mathcal{G}(\cdot,\mathbf{x}_0')$ are comparable within constants that depend on $\mathbf{x}_0,\mathbf{x}_0'$ only, so Π_F is finite everywhere on $(0,\infty)^{2N-1}$ or nowhere.

Take the Mellin transform of Π_F in the coordinates of $\mathbf{x}_0$ other than $x_{0,1,K}=1$, with the exponents $s_j-z_{j,K}$ at $x_{0,j,K}$, $j\ge2$, and $t_j-z_{j,\Psi}$ at $x_{0,j,\Psi}$. By Tonelli and $\int_0^\infty q^{w-1}(c+q)^{-s}dq=c^{w-s}\Gamma_{\,2}^{\,2}\left[\begin{matrix}w,\ s-w\\ s\end{matrix}\right]$, $0<w<s$, each coordinate gives a Beta corner and the power $(z_{j,K}v)^{-z_{j,K}}$ or $(z_{j,\Psi}v)^{-z_{j,\Psi}}$, and the remaining integral is $\int_0^\infty v^{z_{1,K}-1}(1+v)^{-s_1}dv=\Gamma_{\,2}^{\,2}\left[\begin{matrix}z_{1,K},\ s_1-z_{1,K}\\ s_1\end{matrix}\right]$, the

power of v collecting to $v^{z_{1,K}-1}$ by the hyperplane. So the Mellin transform is

$$\prod_{j\ge 2}\Gamma_2^1\begin{bmatrix}s_j-z_{j,K},\ z_{j,K}\\ s_j\end{bmatrix}\,\Gamma_2^1\begin{bmatrix}z_{1,K},\ s_1-z_{1,K}\\ s_1\end{bmatrix}\prod_j\Gamma_2^1\begin{bmatrix}t_j-z_{j,\Psi},\ z_{j,\Psi}\\ t_j\end{bmatrix}\int \boldsymbol{\varrho}_F^{-\mathbf{z}}\,d\phi_F<\infty.$$

In particular Π_F is finite almost everywhere, hence everywhere. For complex exponents with these real parts the same computation converges absolutely, and the Beta corners at complex arguments of positive real part do not vanish. Where finite, Π_F extends holomorphically to the product of right half-planes: $|(zv+\zeta)^{-s}|\le(zv+\operatorname{Re}\zeta)^{-s}$ for $\operatorname{Re}\zeta>0$ and $s>0$, so dominated convergence and Morera's theorem give holomorphy in each coordinate, the extension being locally bounded, and hence jointly. So Π_F and $\Pi_{F'}$ are real-analytic on $(0,\infty)^{2N-1}$, equal on a nonempty open set, and equal everywhere; their Mellin transforms on the vertical subspace through $\mathbf{z}$ agree, which is (d).

(d)⇒(a). The finite measures $\boldsymbol{\varrho}^{-\mathbf{z}}(\boldsymbol{\varrho}_F)_*\phi_F$ and $\boldsymbol{\varrho}^{-\mathbf{z}}(\boldsymbol{\varrho}_{F'})_*\phi_{F'}$ on $(0,\infty)^{2N-1}$, read in the logarithms of the coordinates, have the Fourier transforms $\boldsymbol{\tau}\mapsto\int\boldsymbol{\varrho}^{-(\mathbf{z}+i\boldsymbol{\tau})}d\phi$, equal by (d); a finite measure on $\mathbb{R}^{2N-1}$ is determined by its Fourier transform, so the two measures agree, and multiplying by the positive continuous function $\boldsymbol{\varrho}^{\mathbf{z}}$ the laws agree. This replaces the Gaussian smoothing of the reference proof of 23.11, which is the same argument on the line.

One table. The plain one-mark cell of ratio x_0 has the law $x_0^{(m_e-s)/2}\delta_{x_0}$, a positive multiple of the point law, and its ratios fill $(0,\infty)$, so (c) with the plain cells gives the equality of Π_F on an open interval; (d) is the case $N=1$, $\mathbf{w}=(w_{1,K},w)$ with $w_{1,K}$ not entering. □

Corollary 23.12 (Cells that are marks). *As stated: (i) for every cell family with plain tables and $s_j,t_j>0$.*

Proof. (i) This is the reference proof of 23.12(i) with the ratio vector in place of the ratio. If $\boldsymbol{\varrho}_F\equiv\mathbf{x}_0$, the image of ϕ_F is $\phi_F(\Delta_B)\delta_{\mathbf{x}_0}$. Conversely, ϕ_F has a positive continuous density on the open simplex and $\boldsymbol{\varrho}_F$ is continuous, so if the law is carried by $\mathbf{x}_0$ the open set $\{\boldsymbol{\varrho}_F\ne\mathbf{x}_0\}$ has ϕ_F-measure 0 and is empty. The mass $\phi_F(\Delta_B)$ is positive. By 23.6(i), the period of a gluing $F\oplus_g F''$ is $\phi_F(\Delta_B)\int\mathcal{G}(\mathbf{x}_0,\mathbf{z}')\,d((\boldsymbol{\varrho}_{F''})_*\phi_{F''})(\mathbf{z}')$, a positive multiple of $\phi_F(\Delta_B)$; so a finite period forces $\phi_F(\Delta_B)<\infty$, and the integral is the period of the gluing of the one-mark cell family of ratio vector $\mathbf{x}_0$ and mass κ_F with F'', whose law is $\delta_{\mathbf{x}_0}$ and whose hinge wall is κ_F. For one plain cell, the one-mark cell of ratio x_0 and mass $\kappa_{\mathcal{A}}$ has the law $x_0^{(\kappa_{\mathcal{A}}-s)/2}\delta_{x_0}$, which gives the factor. (ii) and (iii) are the reference's. □

Theorem 23.14 (Series and parallel nodes). *As stated: (i) and (ii) for cell families with plain tables and $s_j,t_j>0$.*

Proof. This is the reference proof of 23.14(i),(ii), table by table. *(i)* The series core $\binom{G}{q}$, $G=\{g_0,\dots,g_q\}$, has rank q; its deletion at g_0 has the one basis $\{g_1,\dots,g_q\}$, so $K_{\binom{G}{q}\setminus g_0}(\mathbf{v})=\prod_i v_i$ and $\Psi_{\binom{G}{q}\setminus g_0}=1$, and its contraction has the bases of size $q-1$ of $\{g_1,\dots,g_q\}$, so $K_{\binom{G}{q}/g_0}=e_{q-1}(\mathbf{v})$ and $\Psi_{\binom{G}{q}/g_0}(\mathbf{v})=\sum_i v_i$. By 23.5,

$$K_{\mathcal{A}_j\setminus g_0}=\prod_i K_{\mathcal{A}_{ij}/g_i}\gamma^K_{\mathcal{A}_{ij}}=\prod_i K_{\mathcal{A}_{ij}\setminus g_i},\qquad \Psi_{\mathcal{A}_j\setminus g_0}=\prod_i\Psi_{\mathcal{A}_{ij}\setminus g_i},$$

and the hinge values of $\mathcal{A}_j$ are $\prod_i\gamma^K_{\mathcal{A}_{ij}}/e_{q-1}((\gamma^K_{\mathcal{A}_{ij}})_i)$, whose reciprocal is $\sum_i 1/\gamma^K_{\mathcal{A}_{ij}}$, and $\sum_i\gamma^\Psi_{\mathcal{A}_{ij}}$. In the orthant form of f_F the measure $\alpha^{\mathbf{m-1}}\prod_j K_{\mathcal{A}_j\setminus g_0}^{-s_j}\Psi_{\mathcal{A}_j\setminus g_0}^{-t_j}d\alpha$ is the product over i of the

measures of the cell families, and the exponential is the product of their exponentials, so Tonelli gives $f_F = \prod_i f_{F_i}$, and the image of the product measure under the sum of the maps of the factors is the convolution: $\mathfrak{r}_F = *_i \mathfrak{r}_{F_i}$. The hinge-wall formula of 23.6(ii), with $\mathcal{E}_{\mathcal{H}} = \varnothing$, $k = q$ and $r_{\mathcal{H}_j} = q$, gives $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{s}|$ (check 3a of S21-23_work/checks.py). For one table, $f_{\mathcal{A}}(y) = f_F(1, y) = \prod_i f_{F_i}(1, y)$, and since $e^{-\lambda} = \prod_i e^{-\lambda_i}$ for $\lambda = \sum_i \lambda_i$, the measure $e^{-\lambda}\mathfrak{r}_F$ is the convolution of the $e^{-\lambda_i}\mathfrak{r}_{F_i}$, whose images under μ convolve: $\mathfrak{r}^{\Psi}_{\mathcal{A}} = *_i \mathfrak{r}^{\Psi}_{\mathcal{A}_i}$. The twisted series core is the reference's.

(ii) The parallel core $\binom{G}{1}$ has rank 1: $K_{\binom{G}{1}\backslash g_0} = \sum_i v_i$, $K_{\binom{G}{1}/g_0} = 1$, $\Psi_{\binom{G}{1}\backslash g_0} = e_{q-1}(\mathbf{v})$ and $\Psi_{\binom{G}{1}/g_0} = \prod_i v_i$. So $K_{\mathcal{A}_j/g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}$, $\Psi_{\mathcal{A}_j/g_0} = \prod_i \Psi_{\mathcal{A}_{ij}\backslash g_i} \gamma^{\Psi}_{\mathcal{A}_{ij}} = \prod_i \Psi_{\mathcal{A}_{ij}/g_i}$, the hinge values combine as stated, the orthant form of $\check{f}_F$ factors, and $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{t}|$ (check 3b). Equally, 23.8(iv) exchanges (i) and (ii) through the dual cell families, the dual of a parallel core being a series core. (iii) and (iv) are the reference's. □

Definition 23.15 (Hinge tree). As stated: the common hinge tree of the tables of a family, with its node families and its cell families.

Remark. The added objects are definitions. Each $\mathcal{A}_{v,j}$ is a cell with the hinge $g_{v,\mathfrak{e}_v}$ on the marks below v (23.15), the same for every j, so the $\mathcal{A}_{v,j}$ form a cell family in the sense of 23.1; and the substitution along the tree does not depend on the order of the two-sums, table by table (23.5).

Theorem 23.17 (The period along a hinge tree). *As stated: for every family whose tables have a common hinge tree, at real data.*

Proof. Induction from the leaves. At a leaf v the cell family F_v is the node family with its one marker as hinge. At an inner vertex $v \neq o$, table j of F_v is the substitution of the node $\mathcal{H}_{v,j}$, a hinged core with the hinge $g_{v,\mathfrak{e}_v}$, with the subtree cells $\mathcal{A}_{w,j}$ of the vertices w directly below it at the markers of their edges (23.15), so 23.6(ii) gives (i), the masses on $\mathcal{E}_v$ being those of F. At the root, the node has no hinge, and 23.6(i) gives (ii). (iii) is the recursion of (i): the law at v is computed from the nodes and masses at v and the laws of the vertices directly below, and by induction from the data below v. This is the reference proof of 23.17 (thm:contraction), with 23.6 for families, which the reference's remark on families (rem:familiesnodes) gives verbatim. □

Proposition 23.18 (What the work depends on). *As stated, for a family whose tables have a common hinge tree.*

Proof. (i) A bijection of marks carrying the subtree cells onto each other in every table, with their weights, masses and hinge, carries the letters onto the letters, hence the base forms and the ratio vectors onto each other; the ranks, sizes and masses are kept, and so is the hinge wall.

(ii) At a series node of unit weights, F_v is the series connection of the cell families directly below (23.13), so 23.14(i) gives $f_{F_v} = \prod_w f_{F_w}$ and $\kappa_{F_v} = \sum_w \kappa_{F_w} - (q-1)|\mathbf{s}|$, $q = \sum_l k_l$; members of one orbit have equal transforms and hinge walls by (i), and collecting them gives the powers. A weight θ_{g_i} on the basis of a series node that omits the marker g_i is carried, by the weight formula of 21.1, by the cell at g_i as the factor $\theta_{g_i}/\theta_{g_0}$ on its bases through its hinge, the weight θ_{g_0} of the basis omitting the hinge remaining as a constant factor of the table; so a series node with weights is a series connection of cells so weighted, and the orbits are taken of these. The parallel node is the same with 23.14(ii).

(iii) By 23.1 the hinge wall of F_v is $m_{B_v} - \sum_j \big(s_j(r_{\mathcal{A}_{v,j}} - 1) + t_j(|B_v| - r_{\mathcal{A}_{v,j}})\big)$, B_v the marks below v, and by 23.5 the ranks $r_{\mathcal{A}_{v,j}}$ are sums of the ranks of the nodes below v less one per edge. A change at v_0 keeping the total mass at v_0, the ranks and the numbers of marks of the nodes,

keeps every m_{B_v}, rank and size, since B_v contains all of $\mathcal{E}_{v_0}$ or none of it; so no hinge wall changes, and by 23.17(iii) the law of F_v changes only if v_0 lies below v, that is, for v on the path from v_0 to the root. A change of masses at v_0 and v_1 keeping the total changes m_{B_v} exactly when B_v contains one of $\mathcal{E}_{v_0}, \mathcal{E}_{v_1}$ and not the other, by the change of the total mass below $\mathfrak{e}_v$. □

Definition 23.19 (The hinge tree procedure). As stated: for a family whose tables have a common hinge tree, the representations (a) and (b) in $2N-1$ variables, (c) for one table.

Remark. Each operation rests on a statement for families. The laws and hinge walls of the orbits are those of 23.17 and 23.18(i). (1) The one-mark leaf: by 23.8(i) at the point law $\delta_{\mathbf{x}_0}$ with $\kappa = m_e$, $f(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,K}^{-s_j} \mathsf{G}_{|\mathbf{s}|-m_e;(1,-1)}(\sum_j x_j/x_{0,j,K}, \sum_j y_j x_{0,j,\Psi})$, and the second law form with torus homogeneity at $\lambda = x_{0,1,\Psi}$ gives $\check{f}(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,\Psi}^{-t_j} \mathsf{G}_{|\mathbf{t}|-m_e;(1,-1)}(\sum_j x_j/x_{0,j,\Psi}, \sum_j y_j x_{0,j,K})$, the powers of $x_{0,1,\Psi}$ cancelling; the orthant forms give the same directly. For one table, $\mathbf{x}_0 = (1, x_0)$ and the plain cell's law $x_0^{(m_e-s)/2}\delta_{x_0}$ give the displayed one-table transforms, with $\mathsf{G}_{t-m_e;(1,-1)}(x_0^{-1}, y) = x_0^{t-m_e}\mathsf{G}_{t-m_e;(1,-1)}(1, y/x_0)$. (2) 23.14(i),(ii), with 23.18(ii) for the weights; a pointwise product of transforms is a Mellin convolution of their Mellin transforms of 23.8(iii). (3) 23.8(v) on the Mellin side, and Lemma S10 for the inverse Mellin transform, applied to the dual cell family for $\check{f}$ (23.8(iv)). (4) 23.6(ii) for families. (5) 23.6(i), and for a series or parallel root 23.9(ii) with the series or parallel connection of all but one cell family on one side.

Corollary 23.20 (Series-parallel families). *As stated, for a family whose tables have a common hinge tree of series and parallel nodes over one-mark cell families.*

Proof. This is the reference proof of 23.20 with the family statements. The leaves have their transforms in closed form, Grammas of the marks (the remark after 23.19 in this part). At a series node the transform of the subtree is the product of those below, and at a parallel node the dual transform (23.14), so no integral is taken there; repeated subtrees enter as powers (23.18(ii)). At an edge joining a series node to a parallel node the transform of the subtree below is changed into its dual transform by 23.8(v): the Mellin transform of f on a vertical subspace through a real point $(\mathbf{a},\mathbf{b})$, $0 < a_j < s_j$, $0 < b_j < t_j$, of its domain, an integral on $(0,\infty)^{2N-1}$; multiplication by the ratio of Gammas; and the inverse Mellin transform of the result, which is the Mellin transform of $\check{f}$ (23.8(iii) at $(\mathbf{t}-\mathbf{b}, \mathbf{s}-\mathbf{a})$), an integral on $(0,\infty)^{2N-1}$ that converges absolutely and returns $\check{f}$ by Lemma S10 applied to the dual cell family (23.8(iv)). The hypothesis is the finiteness of the Mellin transform of the law at $\mathbf{z} = (\mathbf{s}-\mathbf{a}, \mathbf{b})$, which these points require. At the root one integral on $(0,\infty)^{2N-1}$ remains, 23.9(ii). The count of the integrations depends only on the edges where the kind of node changes and on the root. For one table the variables are one. □

48.24 Chapter 24: repetition and the change of query

Theorem 24.1 (How often a cell family repeats). *As stated: for every cell family with plain tables, at real exponents.*

Proof. (i) The series connection of k copies of $\mathcal{A}_j$ has $k|B|$ marks besides the hinge, the masses repeated, and the rank $k + k(r_{\mathcal{A}_j} - 1) = kr_{\mathcal{A}_j}$ (23.5, the series core of rank k). By 23.1, since $s_j(kr_{\mathcal{A}_j} - 1) = ks_j(r_{\mathcal{A}_j} - 1) + (k-1)s_j$ and $t_j(k|B| - kr_{\mathcal{A}_j}) = kt_j(|B| - r_{\mathcal{A}_j})$,

$$\kappa_{F_k^{\mathrm{s}}} = km_B - \sum_j \big(s_j(kr_{\mathcal{A}_j} - 1) + t_j(k|B| - kr_{\mathcal{A}_j})\big) = k\kappa_F - (k-1)|\mathbf{s}| = |\mathbf{s}| + k(\kappa_F - |\mathbf{s}|).$$

The parallel connection has the rank $1 + k(r_{\mathcal{A}_j} - 1)$, and the same count gives $k\kappa_F - (k-1)|\mathbf{t}|$ (checks 3a and 3b of S21-23_work/checks.py, which are this count at the exponents as symbols). No sign of an exponent enters.

(ii) The walls of $F_k^{\mathrm{s}} \oplus_g F'$ at the marks of the copies and at those of F' are the hinge walls $\kappa_{F_k^{\mathrm{s}}}$ and $\kappa_{F'}$, and on the balance wall they add to 2σ (21.12). A convergent period has every proper wall positive (8.7, at any complex exponents), so $0 < \kappa_{F_k^{\mathrm{s}}} < 2\sigma$. If $\kappa_F \neq |\mathbf{s}|$ the middle term is unbounded in k, which bounds k: $|\mathbf{s}| - k(|\mathbf{s}| - \kappa_F) > 0$ gives $k < |\mathbf{s}|/(|\mathbf{s}| - \kappa_F)$ when $\kappa_F < |\mathbf{s}|$, and $|\mathbf{s}| + k(\kappa_F - |\mathbf{s}|) < |\mathbf{s}| + |\mathbf{t}|$ gives $k < |\mathbf{t}|/(\kappa_F - |\mathbf{s}|)$ when $\kappa_F > |\mathbf{s}|$. The parallel case is the same with the roles of $|\mathbf{s}|$ and $|\mathbf{t}|$ exchanged. □

Theorem 24.2 (Series repetition). *As stated: (i) for cell families with plain tables and $s_j, t_j > 0$; (ii) to (iv) are the reference's, for one cell.*

Proof. (i) By 23.14(i) with k copies of F, $f_{F_k^{\mathrm{s}}} = f_F^k$; by 24.1(i), $\kappa_{F_k^{\mathrm{s}}} = |\mathbf{s}|$, which lies in $(0, 2\sigma)$ since $|\mathbf{s}|, |\mathbf{t}| > 0$. So the first form of 23.9(ii) applies to F_k^{s} and F' and gives the display. With ν the measure of the statement, $P(k) = \int f_F^k \, d\nu = \int w^k \, d((f_F)_*\nu)(w)$. Where finite, $P(k)^2 = \big(\int f_F^{(k-1)/2} f_F^{(k+1)/2} d\nu\big)^2 \leq P(k-1)P(k+1)$ by the Cauchy–Schwarz inequality, and for $z \geq 0$ Tonelli gives $\sum_k P(k) z^k/k! = \int e^{z f_F} d\nu$.

The value $P(0)$. Insert the orthant form of $f_{F'}$ and integrate first in x_j, $j \geq 2$, and y_j (Tonelli): $\int_0^\infty x^{s_j - 1} e^{-x/\gamma^K} dx = \Gamma(s_j)(\gamma^K)^{s_j}$ and $\int_0^\infty y^{t_j - 1} e^{-y\gamma^\Psi} dy = \Gamma(t_j)(\gamma^\Psi)^{-t_j}$. Since $K_{\mathcal{A}'_j \backslash g}(\gamma^K_{\mathcal{A}'_j})^{-1} = K_{\mathcal{A}'_j/g}$ and $\Psi_{\mathcal{A}'_j \backslash g} \gamma^\Psi_{\mathcal{A}'_j} = \Psi_{\mathcal{A}'_j/g}$, what remains is

$$\prod_{j \geq 2} \Gamma(s_j) \prod_j \Gamma(t_j) \int \alpha^{\mathbf{m}|_R - \mathbf{1}} e^{-1/\gamma^K_{\mathcal{A}'_1}} K_{\mathcal{A}'_1 \backslash g}^{-s_1} \prod_{j \geq 2} K_{\mathcal{A}'_j/g}^{-s_j} \prod_j \Psi_{\mathcal{A}'_j/g}^{-t_j} \, d\alpha.$$

In $\alpha = \rho u$ the power of ρ is $\rho^{\kappa_{F'} - |\mathbf{t}| - s_1 - 1} = \rho^{-s_1 - 1}$, by $\kappa_{F'} = |\mathbf{t}|$ (check 9 of S21-23_work/checks.py), and $\int_0^\infty \rho^{-s_1 - 1} e^{-1/(\rho\gamma)} d\rho = \Gamma(s_1)\gamma^{s_1}$ turns $K_{\mathcal{A}'_1 \backslash g}^{-s_1}$ into $K_{\mathcal{A}'_1/g}^{-s_1}$. So $P(0)$ is the period at $\mathbf{m}|_R$ of the family of the contractions, whose degree is $\sum_j (s_j(r_{\mathcal{A}'_j} - 1) + t_j(|R| - r_{\mathcal{A}'_j} + 1))$, so that it is balanced exactly when $\kappa_{F'} = |\mathbf{t}|$. For one cell, $\int \varrho_{\mathcal{A}'}^{-t} d\phi_{\mathcal{A}'}$ with $\kappa_{\mathcal{A}'} = t$ has the density $u^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}'/g}^{t-s} K_{\mathcal{A}' \backslash g}^{-t} \Psi_{\mathcal{A}' \backslash g}^{-t} (\gamma^\Psi/\gamma^K)^{-t} = u^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}'/g}^{-s} \Psi_{\mathcal{A}'/g}^{-t}$, the same period.

(ii) to (iv) are the reference's, for one cell. □

Remark. For a family the asymptotics of $P(k)$ is a Laplace problem in $2N - 1$ variables: a series-balanced cell family has $f_F = \int \mathsf{G}_{0;(1,-1)}(1, ab) \, d\phi_{F \backslash g}$ by 23.3, of the order of $P_{F \backslash g}$ times the logarithm of the scale of $\mathbf{y}$ plus a term that depends on the direction of $\mathbf{y}$ and on $\mathbf{x}'$, and the uniformity of the error in these directions does not follow from the one-variable argument. So (ii) to (iv) are stated for one cell.

Corollary 24.3 (Parallel repetition). *As stated: the first display for cell families with plain tables and $s_j, t_j > 0$; the rest is the reference's, for one cell.*

Proof. By 23.14(ii), $\check{f}_{F_k^{\mathrm{p}}} = \check{f}_F^k$, and by 24.1(i), $\kappa_{F_k^{\mathrm{p}}} = |\mathbf{t}| \in (0, 2\sigma)$; the second form of 23.9(ii) gives the display. For $P(0)$, integrate the orthant form of $\check{f}_{F'}$ in x_j, $j \geq 2$, and y_j: $\int x^{t_j - 1} e^{-x/\gamma^\Psi} dx = \Gamma(t_j)(\gamma^\Psi)^{t_j}$ and $\int y^{s_j - 1} e^{-y\gamma^K} dy = \Gamma(s_j)(\gamma^K)^{-s_j}$ turn $\Psi_{\mathcal{A}'_j/g}^{-t_j}$ into $\Psi_{\mathcal{A}'_j \backslash g}^{-t_j}$, $j \geq 2$, and every $K_{\mathcal{A}'_j/g}^{-s_j}$ into $K_{\mathcal{A}'_j \backslash g}^{-s_j}$; in $\alpha = \rho u$ the power of ρ is $\rho^{\kappa_{F'} - |\mathbf{s}| - t_1 - 1} = \rho^{-t_1 - 1}$, by $\kappa_{F'} = |\mathbf{s}|$, and $\int \rho^{-t_1 - 1} e^{-1/(\rho\gamma_1^\Psi)} d\rho = \Gamma(t_1)(\gamma_1^\Psi)^{t_1}$ turns $\Psi_{\mathcal{A}'_1/g}^{-t_1}$ into $\Psi_{\mathcal{A}'_1 \backslash g}^{-t_1}$. What remains is the period of the family of the deletions at $\mathbf{m}|_R$, balanced exactly when $\kappa_{F'} = |\mathbf{s}|$. Equally, the dual cell families at

$(\mathbf{t},\mathbf{s})$ turn the statement into (i) of 24.2 (23.8(iv), 8.15), the dual of the contractions being the deletions of the duals. The rest is the reference's, for one cell. □

Proposition 24.4 (A chain is a power of one operator). *As stated, for chain cell families at real exponents, (iv) with plain tables and $s_j, t_j > 0$.*

Proof. This is the reference proof of 24.4 with the laws on $(0,\infty)^{2N-1}$. *(i)* 23.6(ii) with the one cell family F_l at g_{in}: its hinge-wall formula, $\mathcal{U}_j$ having the marks $\mathcal{E}_{\mathcal{U}} \sqcup \{g_{\text{in}}\}$ besides its hinge, is $\kappa_{F_{l+1}} = \kappa_{F_l} + \kappa_U$. By 21.12, $F_l \oplus_g F'$ is on its balance wall exactly when $\kappa_{F_l} + \kappa_{F'} = 2\sigma$; with $\kappa_{F_l} = \kappa_{F_0} + l\kappa_U$ this holds for every l exactly when it holds at $l = 0$ and $\kappa_U = 0$.

(ii) 23.6(ii) with the one cell family F_l at g_{in} gives $\lambda_{l+1} = \int \lambda_{\mathbf{x}}^U \, d\lambda_l(\mathbf{x}) = \mathcal{Q}\lambda_l$, the twisted family carrying the mass $\kappa_{F_l} = \kappa_{F_0}$ at g_{in}; by induction $\lambda_l = \mathcal{Q}^l\lambda_0$. 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$ gives $P_{F_l \oplus_g F'} = \int h_{F'} \, d\lambda_l$, and $\int H \, d(\mathcal{Q}\mu) = \int (\mathcal{Q}^* H) \, d\mu$ for $H \geq 0$ by Tonelli, so $\int h_{F'} \, d\lambda_l = \int (\mathcal{Q}^{*l} h_{F'}) \, d\lambda_0$. The series in $z \geq 0$ is summed under the integral by Tonelli.

(iii) $\mathcal{Q}^*$ preserves inequalities between nonnegative functions, so $\mathcal{Q}^* H \leq MH$ gives $\mathcal{Q}^{*l} H \leq M^l H$, and $h_{F'} \leq CH$ gives $\mathcal{Q}^{*l} h_{F'} \leq CM^l H$; integrate against λ_0. The lower bound is the same.

(iv) For the series core $\binom{\{g_{\text{out}},g_{\text{in}},g'\}}{2}$ with F'' at g', F_{l+1} is the series connection of F_l and F'', so 23.14(i) gives $f_{F_{l+1}} = f_{F''} f_{F_l}$, and the hinge-wall formula with the mass 0 at g_{in}, the marks $\{g_{\text{in}}\} \sqcup B''$ and the ranks $2 + r_{\mathcal{A}_j''} - 1$, gives $\kappa_U = m_{B''} - \sum_j \left(s_j r_{\mathcal{A}_j''} + t_j(|B''| - r_{\mathcal{A}_j''}) \right) = \kappa_{F''} - |\mathbf{s}|$. For the parallel core the ranks are $r_{\mathcal{A}_j''}$, the formula gives $\kappa_{F''} - |\mathbf{t}|$, and 23.14(ii) gives the product of the dual transforms. □

Theorem 24.5 (The weight of a hinge). *As stated: (i) for cell families with plain tables, one weight per table; (ii) to (iv) are the reference's, for one cell.*

Proof. (i) The bases through g carry the monomials of $K_{\mathcal{A}_j/g}$ and of $\Psi_{\mathcal{A}_j/g}$, which are multiplied by w_j; the deletions keep their weights (the reference proof of 24.5(i), table by table). So the hinge wall is kept, $\gamma^K_{\mathcal{A}_j} = K_{\mathcal{A}_j \setminus g}/K_{\mathcal{A}_j/g}$ becomes $\gamma^K_{\mathcal{A}_j}/w_j$ and $\gamma^\Psi_{\mathcal{A}_j} = \Psi_{\mathcal{A}_j/g}/\Psi_{\mathcal{A}_j \setminus g}$ becomes $w_j \gamma^\Psi_{\mathcal{A}_j}$; the coordinates of the ratio vector become $(w_1/w_j)\gamma^K_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}$ and $w_1 w_j \gamma^\Psi_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}$, and the base form $u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j/g}^{-s_j} \Psi_{\mathcal{A}_j \setminus g}^{-t_j} (\gamma^K_{\mathcal{A}_1})^{-\kappa_F} d\omega$ is multiplied by $\prod_j w_j^{-s_j} \cdot w_1^{\kappa_F}$. In the orthant forms, $x_j/\gamma^K_{\mathcal{A}_j}$ becomes $w_j x_j/\gamma^K_{\mathcal{A}_j}$ and $y_j \gamma^\Psi_{\mathcal{A}_j}$ becomes $w_j y_j \gamma^\Psi_{\mathcal{A}_j}$ with the measure of f_F unchanged, and for $\check{f}_F$ the measure is multiplied by $\prod_j w_j^{-s_j - t_j}$ while $x_j/\gamma^\Psi_{\mathcal{A}_j}$ and $y_j \gamma^K_{\mathcal{A}_j}$ are divided by w_j. For one cell, $w_1 = w$: the ratio becomes $w^2 \varrho_{\mathcal{A}}$, the base form $w^{\kappa_{\mathcal{A}} - s} \phi_{\mathcal{A}}$, and the homogeneity of 23.1 with $c = w$ gives $f_{w\mathcal{A}}(y) = f_F(w, wy) = w^{\kappa_{\mathcal{A}} - s} f_{\mathcal{A}}(w^2 y)$, and with $c = 1/w$, $\check{f}_{w\mathcal{A}}(y) = w^{-s-t} \check{f}_F(1/w, y/w) = w^{-s-\kappa_{\mathcal{A}}} \check{f}_{\mathcal{A}}(y/w^2)$. (ii) to (iv) are the reference's. □

Remark. For a family the weights act on the law through an N-dimensional subtorus of $(0,\infty)^{2N-1}$, and the Mellin transform in $\mathbf{w}$ is a restricted Mellin transform of the law against the transform of the room, without the product formula of (ii); (iii) and (iv) concern the one-variable expansions of the node of width one. So (ii) to (iv) are stated for one cell.

Proposition 24.6 (Carrying a calculation through a change). *As stated, for a family whose tables have a common hinge tree.*

Proof. (i) and (iii) are 23.18(iii) with the recursion of 23.17(i): the law at a vertex is recomputed from the new law of the vertex directly below it on the path and the kept laws of the others, and the root by 23.17(ii). (ii) is 24.5(i) for the cell family F_v: the weight of the hinge multiplies the base form by $w_1^{\kappa_{F_v}} \prod_j w_j^{-s_j}$ and moves the ratio vector by the stated map, so the law is multiplied

by that factor and carried by that map; the cell families below v are unchanged, and above v the change of the law of F_v is a change at a vertex directly below, carried by (i). $\square$

48.25 Chapter 25: holds and the lean convolution

In this section $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ is a family as in Definition 25.8: its tables are each one molecule on $\mathcal{E}$, $n = |\mathcal{E}| \geq 2$, and its exponents are positive, unless a statement says otherwise. Its $2N$ letters are written X, running over $K_1, \dots, K_N, \Psi_1, \dots, \Psi_N$. The letter X of table j has the exponent w_X, which is s_j or t_j, the degree $\deg X$, which is r_j or r_j^*, the twist $\mathbf{x}^X$, which is $\mathbf{x}_j^K$ or $\mathbf{x}_j^\Psi$, and the Newton polytope $\mathcal{N}_X$, which is $P(\mathcal{B}_j)$ or $P(\mathcal{B}_j^*)$; for masses $\mathbf{a}$ with $|\mathbf{a}| = w_X \deg X$ the period of its one-letter word X^{-w_X} is written $P_X(\mathbf{a})$, which is $P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0)$ or $P_{\mathcal{B}_j}(\mathbf{a}; 0, t_j)$. $\mathcal{X}'$ is the set of the $2N-1$ letters other than K_1, and in a split $\mathbf{m} = \sum_X \mathbf{a}^X$ the load of X is $\mathbf{a}^X$, so that $\mathbf{a}^{K_j} = \mathbf{a}_j$ and $\mathbf{a}^{\Psi_j} = \mathbf{b}_j$.

Two facts are used throughout. First, the base polytope of a table that is one molecule, and that of its dual table, have dimension $n-1$: if $w \in \mathbb{R}^{\mathcal{E}}$ is orthogonal to every difference $\mathbf{1}_T - \mathbf{1}_{T'}$ of bases, then $\langle w, \mathbf{1}_T \rangle$ is constant over the bases and w is constant by Lemma 7.8; so these differences span $\mathbf{1}^\perp$, and so do the differences $\mathbf{1}_{\mathcal{E}\setminus T'} - \mathbf{1}_{\mathcal{E}\setminus T}$. Second, a point $\sum_k p_k v_k$ with every $p_k > 0$ and $\sum_k p_k = 1$ lies in the relative interior of the convex hull of the finitely many points v_k: for $x = \sum_k q_k v_k$ in the hull and $0 < \varepsilon \leq \min_k p_k$, the point $\sum_k p_k v_k + \varepsilon\big(\sum_k p_k v_k - x\big) = \sum_k \big((1+\varepsilon)p_k - \varepsilon q_k\big) v_k$ has nonnegative coefficients of sum 1; so every segment from a point of the hull to the given point extends beyond it within the hull, which characterizes the relative interior of a convex set.

Definition 25.1 (Positive letters; Newton words and sums). As stated: the period of a product of powers of positive letters at complex exponents, wherever it converges absolutely, a Newton word being the case of positive exponents.

Remark. The independence of the section is the proof of Lemma 16.2, which uses of the integrand $f = \alpha^{\mathbf{m}-\mathbf{1}} W$ only that it is measurable, homogeneous, $f(\lambda\alpha) = \lambda^{-n} f(\alpha)$ for $\lambda > 0$ when $m_{\mathcal{E}} = D$, and absolutely integrable over one section; here $|f| = \alpha^{\operatorname{Re}\mathbf{m}-\mathbf{1}} \prod_i L_i^{-\operatorname{Re} w_i}$. The family word at any exponents is a product of powers of its letters with $D = \sum_j (s_j r_j + t_j r_j^*)$, and on the balance wall its period is that of Theorem 8.7.

Proposition 25.3 (One positive letter). *As stated.*

Proof. Put $D = \deg L$ and $t = (t_k)_k$. Every t_k has $|t_k| = D$, and $\mathcal{N}_L$ has dimension $n-1$ in the hyperplane $|x| = D$, which does not contain 0; so the t_k span $\mathbb{R}^{\mathcal{E}}$.

(i) By Lemma 2.7, $x = \rho u$ with $\rho > 0$ and $u \in \Delta_{\mathcal{E}}$ gives $x^{\mathbf{a}-\mathbf{1}} dx = \rho^{|\mathbf{a}|-1} u^{\mathbf{a}-\mathbf{1}} d\rho\, d\omega_{\mathcal{E}}(u)$, and $L(\rho u) = \rho^D L(u)$. For $\operatorname{Re}|\mathbf{a}| > 0$ the substitution $\sigma = \rho^D L(u)$ gives

$$\int_0^\infty \rho^{|\mathbf{a}|-1} e^{-\rho^D L(u)}\, d\rho = \frac{\Gamma(w)}{D}\, L(u)^{-w}, \qquad w = |\mathbf{a}|/D.$$

So $\mathsf{G}_{\mathbf{a};t}((c_k)_k) = \frac{\Gamma(w)}{D} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{a}-\mathbf{1}} L(u)^{-w} d\omega_{\mathcal{E}}(u) = \frac{\Gamma(w)}{D}\, P_{L^{-w}}(\mathbf{a})$, the last integral being the period over the section $\Delta_{\mathcal{E}}$ (Definition 25.1). The same computation at $\operatorname{Re}\mathbf{a}$, by Tonelli's theorem, shows that the two integrals converge absolutely together.

(ii) $|u^{\mathbf{a}-\mathbf{1}} L(u)^{-w}| = u^{\operatorname{Re}\mathbf{a}-\mathbf{1}} L(u)^{-\operatorname{Re} w}$, so $P_{L^{-w}}(\mathbf{a})$ converges absolutely exactly when the period of the Newton word $L^{-\operatorname{Re} w}$, $\operatorname{Re} w > 0$, converges at $\operatorname{Re}\mathbf{a}$. Its Newton sum is $(\operatorname{Re} w)\mathcal{N}_L$, of dimension $n-1$, and Theorem 25.2 gives the condition $\operatorname{Re}\mathbf{a} \in \operatorname{ri}\big((\operatorname{Re} w)\mathcal{N}_L\big) = (\operatorname{Re} w)\operatorname{ri}\mathcal{N}_L$.

(iii) For real $\mathbf{a}$ and $|\boldsymbol{\eta}| = 0$ the exponent w is the same real number at $\mathbf{a}$ and at $\mathbf{a} + i\boldsymbol{\eta}$, so by (i) $P_{L^{-w}}(\mathbf{a} + i\boldsymbol{\eta}) = D\,\mathsf{G}_{\mathbf{a}+i\boldsymbol{\eta};t}((c_k)_k)/\Gamma(w)$. If $P_{L^{-w}}(\mathbf{a})$ converges, so does $\mathsf{G}_{\mathbf{a};t}$, and $\mathbf{a} \in \operatorname{int}\operatorname{cone}(t)$ (Theorem 2.2). Theorem 2.11 gives $|\mathsf{G}_{\mathbf{a}+i\boldsymbol{\eta};t}((c_k)_k)| \le \mathsf{G}_{\mathbf{a};t}(\lambda(c_k)_k)\, e^{-(1-\varepsilon)h_t(\boldsymbol{\eta})}$ with $\lambda = \sin\frac{\pi\varepsilon}{2}$, and $h_t = h_L$ is a norm since the t_k span. Every t_k having total D, the substitution $x = \lambda^{-1/D}y$ gives $\mathsf{G}_{\mathbf{a};t}(\lambda(c_k)_k) = \lambda^{-w}\mathsf{G}_{\mathbf{a};t}((c_k)_k)$; multiplying by $D/\Gamma(w)$ gives the second bound. The bound by $P_{L^{-w}}(\mathbf{a})$ itself is that of the modulus of the integrand by its value at the real parts. □

Remark. At the letter $K_{\mathcal{H}}$ of a table that is one molecule, with the support $(\mathbf{1}_T)_T$, the coefficients $\boldsymbol{\theta}$ and $\mathcal{N}_{K_{\mathcal{H}}} = P(\mathcal{H})$, items (i), (ii) and (iii) are Proposition 25.10(i), (ii) and (iv) at the full basis lean, and at $\Psi_{\mathcal{H}}$ at the full cobasis lean. At a simplicial letter, (i) with Corollary 2.6 at the square configuration t is Proposition 25.6: $\mathsf{G}_{\mathbf{a};t}((c_k)_k) = |\det t|^{-1}\prod_k \Gamma(b_k)c_k^{-b_k}$ with $\mathbf{b} = t^{-1}\mathbf{a}$, and $|\mathbf{b}| = w$ since $\mathbf{1}^{\mathsf{T}}t = D\mathbf{1}^{\mathsf{T}}$.

Lemma 25.7 (Concatenation of Grammas). *As stated, for any number of parts.*

Proof. For $i \ge 2$ the configuration $t^{(i)}$ spans $\mathbb{R}^{\mathcal{E}}$, so by Theorem 2.11 the Gramma $\mathsf{G}_{\mathbf{w};t^{(i)}}(\boldsymbol{\xi}^{(i)})$ on $\mathbf{c}_i + i\mathbb{R}^{\mathcal{E}}$ is at most a constant times $e^{-(1-\varepsilon)h_{t^{(i)}}(\operatorname{Im}\mathbf{w})}$, $h_{t^{(i)}}$ a norm, and is integrable there. In $\mathbf{y} = \log x$ it is the Fourier transform of the continuous integrable function $e^{\langle \mathbf{c}_i,\mathbf{y}\rangle}\exp\big(-\sum_k \xi_k^{(i)} e^{\langle t_k^{(i)},\mathbf{y}\rangle}\big)$ (Definition 2.1), so Fourier inversion gives, for every $x \in \mathbb{R}^{\mathcal{E}}_{>0}$,

$$\exp\Big(-\sum_k \xi_k^{(i)} x^{t_k^{(i)}}\Big) = \frac{1}{(2\pi i)^n}\int_{\mathbf{c}_i + i\mathbb{R}^{\mathcal{E}}} \mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})\, x^{-\mathbf{w}_i}\, d\mathbf{w}_i.$$

Multiply these for $i = 2,\dots,p$ by $x^{\nu-\mathbf{1}}\exp\big(-\sum_k \xi_k^{(1)}x^{t_k^{(1)}}\big)$ and integrate over x. The modulus of the integrand is $x^{\operatorname{Re}\nu-\sum_{i\ge2}\mathbf{c}_i-\mathbf{1}}\exp\big(-\sum_k \xi_k^{(1)}x^{t_k^{(1)}}\big)\prod_{i\ge2}|\mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})|$, whose integral is $\mathsf{G}_{\operatorname{Re}\nu-\sum_{i\ge2}\mathbf{c}_i;t^{(1)}}(\boldsymbol{\xi}^{(1)})\prod_{i\ge2}\int|\mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})|\,|d\mathbf{w}_i|$, finite by Theorem 2.2 since $\operatorname{Re}\nu - \sum_{i\ge2}\mathbf{c}_i \in \operatorname{int}\operatorname{cone}(t^{(1)})$. So Fubini's theorem exchanges the integrals; the integral over x is $\mathsf{G}_{\nu-\sum_{i\ge2}\mathbf{w}_i;t^{(1)}}(\boldsymbol{\xi}^{(1)})$, and the left side is the Gramma of the concatenation (Definition 2.1). □

Definition 25.8 (Holding). As stated: the hold of a family, the Mellin transform in the twists of its letters other than K_1.

Remark. That the bases of every table span $\mathbb{R}^{\mathcal{E}}$ is the first fact above. By the torus relation of Definition 22.1, a factor $\mu > 0$ on every coordinate of one twist $\mathbf{x}^X$ multiplies the period by $\mu^{-w_X \deg X}$, and a twist of every letter by one vector $\boldsymbol{\lambda}$ multiplies it by $\boldsymbol{\lambda}^{-\mathbf{m}}$; so the Mellin transform in all the twists of all the letters at all the marks diverges, and the hold keeps the twist of K_1 at $\mathbf{1}$ and, in Lemma 25.11, the twists at one mark.

Definition 25.9 (Loads; admissible points). As stated: the loads and the admissible points of a family, an admissible point of one table being named by its Ψ-load.

Proposition 25.10 (The periods at the two full leans). *As stated; (iii) is stated for every family, and the closing sentence is new.*

Proof. Items (i), (ii) and (iv) concern one table $\mathcal{H}$ of F, one molecule, and are the reference, proof of prop:fullleans. The closing sentence is Proposition 25.3, proved above; at the letter $K_{\mathcal{H}}$, with the support $(\mathbf{1}_T)_T$, the coefficients $\boldsymbol{\theta}$ and the Newton polytope $P(\mathcal{H})$ of dimension $n-1$, it gives back (i), (ii) and (iv) at the full basis lean, and at $\Psi_{\mathcal{H}}$ at the full cobasis lean.

(iii) Let α be a gauge. By Theorem 7.16, $R_j(\alpha) = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[\mathbf{1}_T] = \sum_{T\in\mathcal{B}_j}\mathbb{P}_{j,\alpha}(T)\,\mathbf{1}_T$ with $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha) > 0$ for every basis, and $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T)$, so $R^*_j(\alpha) = \sum_T \mathbb{P}^*_{j,\alpha}(T)\,\mathbf{1}_{\mathcal{E}\setminus T}$ with $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha) > 0$. By the second fact above $R_j(\alpha) \in \operatorname{ri} P(\mathcal{B}_j)$ and $R^*_j(\alpha) \in \operatorname{ri} P(\mathcal{B}^*_j)$, the two polytopes being the convex hulls of the $\mathbf{1}_T$ and of the $\mathbf{1}_{\mathcal{E}\setminus T}$.

Let $\mathbf{m}$ be real on the balance wall with every proper wall positive. A mark of a table that is one molecule on at least two marks is neither a loop nor a coloop, so $d_j(\{e\}) = d^*_j(\{e\}) = 0$ and $m_e = \kappa_{\{e\}} > 0$ for every e (Definition 8.1). A nonempty proper S with $c_j(S) = 0$ would have $|T \cap S|$ constant over the bases of table j, and so would be a union of molecules of it by Lemma 7.10, which is impossible, table j being one molecule; so $c_j(S) > 0$ in every table, every table being active, and F is connected. By Theorem 8.19(ii) and (iii) a native gauge $\alpha^\star$ exists, the target being $\mathbf{m}$ on the balance wall (Definition 8.17) and the walls at the target positive. Its usage is the target, $\sum_j \big(s_j R_j(\alpha^\star) + t_j R^*_j(\alpha^\star)\big) = \mathbf{m}$: the native loads are a split of $\mathbf{m}$ among the letters, with $|\mathbf{a}^\star_j| = s_j r_j$ and $|\mathbf{b}^\star_j| = t_j r^*_j$, every $\mathbf{1}_T$ having r_j marks and every $\mathbf{1}_{\mathcal{E}\setminus T}$ having r^*_j. By the preceding paragraph $\mathbf{a}^\star_j \in s_j \operatorname{ri} P(\mathcal{B}_j)$ and $\mathbf{b}^\star_j \in t_j \operatorname{ri} P(\mathcal{B}^*_j)$, so the native loads are admissible (Definition 25.9). □

Lemma 25.11 (Holding every mark separates the letters). *As stated, for every family.*

Proof. Let the data be real. By Theorem 8.7 the period of $F[\mathbf{x}]$ is the integral over the section $\{\alpha_{e_0} = 1\}$,

$$P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{(0,\infty)^{\mathcal{E}\setminus e_0}} \alpha^{\mathbf{m}-\mathbf{1}} K_1(\alpha)^{-s_1} \prod_{X\in\mathcal{X}'} X(\mathbf{x}^X\circ\alpha)^{-w_X}\, d\alpha', \qquad \alpha = (\alpha', 1),$$

the twists at e_0 being 1 and the twist of K_1 being $\mathbf{1}$. Insert this into the hold and integrate first over the twists, by Tonelli's theorem, every function being positive. For fixed α' substitute $\mathbf{v}^X = \mathbf{x}^X\circ\alpha'$ on $\mathcal{E}\setminus e_0$ for every $X \in \mathcal{X}'$: then $(\mathbf{x}^X)^{\boldsymbol{\beta}^X-\mathbf{1}} d\mathbf{x}^X = \alpha'^{-\boldsymbol{\beta}^X}(\mathbf{v}^X)^{\boldsymbol{\beta}^X-\mathbf{1}} d\mathbf{v}^X$ and $X(\mathbf{x}^X\circ\alpha) = X(\mathbf{v}^X, 1)$, so the integral over $\mathbf{x}^X$ is $\alpha'^{-\boldsymbol{\beta}^X}$ times

$$\int_{(0,\infty)^{\mathcal{E}\setminus e_0}} (\mathbf{v}^X)^{\boldsymbol{\beta}^X-\mathbf{1}} X(\mathbf{v}^X, 1)^{-w_X}\, d\mathbf{v}^X = P_X(\mathbf{a}^X),$$

the integral over the same section of the form of the one-letter word X^{-w_X} at the extension $\mathbf{a}^X$ of $\boldsymbol{\beta}^X$ of total $w_X \deg X$ (Theorem 8.7); here $\boldsymbol{\beta}^{K_j} = \boldsymbol{\beta}^K_j$ and $\boldsymbol{\beta}^{\Psi_j} = \boldsymbol{\beta}^\Psi_j$, and $\mathbf{a}^X$ is the $\mathbf{a}_j$ or $\mathbf{b}_j$ of the statement. What is left is $\int \alpha'^{\mathbf{m}'-\mathbf{1}-\sum_X \boldsymbol{\beta}^X} K_1(\alpha', 1)^{-s_1} d\alpha'$, $\mathbf{m}' = \mathbf{m}|_{\mathcal{E}\setminus e_0}$, which is $P_{K_1}(\mathbf{a}_1)$ for $\mathbf{a}_1 = \mathbf{m} - \sum_{X\in\mathcal{X}'} \mathbf{a}^X$: its coordinates off e_0 are $\mathbf{m}' - \sum_X \boldsymbol{\beta}^X$, and its total is $m_{\mathcal{E}} - \sum_{X\in\mathcal{X}'} w_X \deg X = r_1 s_1$ on the balance wall. This is the identity in $[0,\infty]$. For complex $\mathbf{m}$ and $\boldsymbol{\beta}$ the moduli of all the integrands are their values at the real parts, where the integrals converge by assumption, and Fubini's theorem replaces Tonelli's. □

Lemma S11. *Let shares $\mathbf{v} \in \mathbb{C}^M$ be pinned by $P\mathbf{v} = \boldsymbol{\pi}$, with q pins, as in Definition 4.1.*

(i) The fibre integral does not change when the shares are replaced by $A^{-1}\mathbf{v}$, the integrand by $f\circ A$ and the pins by PA, for a real $M\times M$ matrix A with $|\det A| = 1$, nor when the pins are replaced by $CP\mathbf{v} = C\boldsymbol{\pi}$, for a real $q\times q$ matrix C with $|\det C| = 1$.

(ii) Let $\mathbf{v} = (\mathbf{u}, \mathbf{w})$ with $\mathbf{w} \in \mathbb{C}^q$, so that $P = (P_{\mathbf{u}}, P_{\mathbf{w}})$, and let $|\det P_{\mathbf{w}}| = 1$. Then

$$\int_{P\mathbf{v}=\boldsymbol{\pi}} f = \frac{1}{(2\pi i)^{M-q}} \int_{\mathbf{c}_{\mathbf{u}}+i\mathbb{R}^{M-q}} f\big(\mathbf{u}, P_{\mathbf{w}}^{-1}(\boldsymbol{\pi} - P_{\mathbf{u}}\mathbf{u})\big)\, d\mathbf{u},$$

$\mathbf{c}_{\mathbf{u}}$ the $\mathbf{u}$-part of the admissible real point, the two sides converging absolutely together.

(iii) Let $\mathbf{v} = (\mathbf{v}_0, \mathbf{v}_1, \dots, \mathbf{v}_k)$ and let the pins be the union of pins $P_l\mathbf{v}_l = \boldsymbol{\pi}_l$, each on one block. If the fibre integral of f converges absolutely, it is the integral over the fibre $P_0\mathbf{v}_0 = \boldsymbol{\pi}_0$ of the fibre integral over $P_1\mathbf{v}_1 = \boldsymbol{\pi}_1, \dots, P_k\mathbf{v}_k = \boldsymbol{\pi}_k$ at fixed $\mathbf{v}_0$, the inner one converging absolutely for almost every point of the outer imaginary translate.

Proof. For $\mathbf{y}_0 \in \mathbb{R}^q$ let $\mu_{P,\mathbf{y}_0} = (\det PP^{\mathsf{T}})^{-1/2}\mathrm{vol}_{M-q}$ on $\{\mathbf{y} \in \mathbb{R}^M : P\mathbf{y} = \mathbf{y}_0\}$, so that the fibre integral is $(2\pi)^{q-M}\int f(\mathbf{c} + i\mathbf{y})\, d\mu_{P,\mathrm{Im}\,\boldsymbol{\pi}}(\mathbf{y})$. By the coarea formula for the linear map P, whose Jacobian is $(\det PP^{\mathsf{T}})^{1/2}$, these measures disintegrate Lebesgue measure, $\int_{\mathbb{R}^M}\varphi\, d\mathbf{y} = \int_{\mathbb{R}^q}\big(\int \varphi\, d\mu_{P,\mathbf{y}_0}\big)d\mathbf{y}_0$ for integrable φ, and $\mathbf{y}_0 \mapsto \int \varphi\, d\mu_{P,\mathbf{y}_0}$ is continuous for continuous φ of compact support; these two properties determine the family $(\mu_{P,\mathbf{y}_0})_{\mathbf{y}_0}$.

(i) The substitution $\mathbf{y} = A\mathbf{y}'$ preserves Lebesgue measure and carries the disintegration along PA to that along P; the substitution $\mathbf{y}_0 = C^{-1}\mathbf{y}_0'$ preserves Lebesgue measure on $\mathbb{R}^q$ and carries the disintegration along CP to that along P, with $\mu_{CP,C\mathbf{y}_0} = \mu_{P,\mathbf{y}_0}$. The admissible real point and the factor $(2\pi)^{q-M}$ do not change.

(ii) The map $\mathbf{y} \mapsto (\mathbf{y}_\mathbf{u}, P\mathbf{y})$ has the block triangular matrix $\begin{pmatrix} I & 0 \\ P_\mathbf{u} & P_\mathbf{w} \end{pmatrix}$, of determinant of modulus 1; so $d\mathbf{y} = d\mathbf{y}_\mathbf{u}\, d\mathbf{y}_0$, and $\mu_{P,\mathbf{y}_0}$ is the image of Lebesgue measure on $\mathbb{R}^{M-q}$ under $\mathbf{y}_\mathbf{u} \mapsto \big(\mathbf{y}_\mathbf{u}, P_\mathbf{w}^{-1}(\mathbf{y}_0 - P_\mathbf{u}\mathbf{y}_\mathbf{u})\big)$. With $\mathbf{u} = \mathbf{c}_\mathbf{u} + i\mathbf{y}_\mathbf{u}$ and $d\mathbf{u} = i^{M-q}d\mathbf{y}_\mathbf{u}$ this is the display; both sides are one integral against one measure.

(iii) The pin matrix is block diagonal, so the disintegration along it is the product of the disintegrations along the blocks, $\mu_{P,(\mathbf{y}_{0,l})_l} = \bigotimes_l \mu_{P_l,\mathbf{y}_{0,l}}$, and the factor $(2\pi)^{q-M}$ is the product of the factors of the blocks. Fubini's theorem gives the iterated integral. □

Remark. In Theorem 25.12(ii) the shares are the loads of the $2N$ letters, $M = 2Nn$, with the $q = n+2N-1$ pins $\sum_X \mathbf{a}^X = \mathbf{m}$ and $|\mathbf{a}^X| = w_X \deg X$ for $X \in \mathcal{X}'$, and the eliminated coordinates are $\mathbf{a}_1$ and the loads $a^X_{e_0}$, $X \in \mathcal{X}'$. With the rows ordered as the mass pins at $e \neq e_0$, the mass pin at e_0 and the totals, and the columns as $a_{1,e}$ for $e \neq e_0$, a_{1,e_0} and the $a^X_{e_0}$, the block $P_\mathbf{w}$ is

$$\begin{pmatrix} I_{n-1} & 0 & 0 \\ 0 & 1 & \mathbf{1}^{\mathsf{T}} \\ 0 & 0 & I_{2N-1} \end{pmatrix},$$

of determinant 1, and Lemma S11(ii) gives the coordinate form stated there, with the factor $(2\pi i)^{-(2N-1)(n-1)}$. With the further pins on G of Theorem 25.14 the loads a^X_e, $X \in \mathcal{X}'$, $e \in G$, are eliminated as well, each by its own pin with the coefficient 1, and the block stays unit upper triangular. The function `eliminated_block_det` of `checks/bookkeeping.py` computes this determinant exactly for $n \leq 4$ and up to six letters.

Lemma S12. *Let $A, B > 0$, $w > 0$ and $c \in \mathbb{C}$. If $0 < \mathrm{Re}\, c < w$,*

$$\int_0^\infty x^{c-1}(A + xB)^{-w}\, dx = \Gamma\,{}_2^{}{}_1\!\left[\begin{matrix} c, & w-c \\ & w \end{matrix}\right] A^{c-w}B^{-c},$$

absolutely convergent; for real $c \notin (0, w)$ the integral is $+\infty$.

Proof. The substitution $x = (A/B)y$ gives $A^{c-w}B^{-c}\int_0^\infty y^{c-1}(1+y)^{-w}dy$. The last integral is the integral over the section $\{u_2 = 1\}$ of the form of the word $(u_1+u_2)^{-w}$ on two marks at the masses $(c, w-c)$: the family of the list $\binom{\{1,2\}}{1}$ with unit weights at the exponents $(w, 0)$, on its balance wall, whose walls are c and $w-c$. For $0 < \mathrm{Re}\, c < w$ its period converges absolutely (Theorem 8.7), and by Lemma 16.2 the integral equals that over $\Delta_{\{1,2\}}$, where $u_1 + u_2 = 1$; by Theorem 3.9 it is

$\int_\Delta u^{(c,w-c)-\mathbf{1}}d\omega = \Gamma_{\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}c,\ w-c\\ w\end{smallmatrix}\right]$. For real $c \le 0$ the integrand is at least $2^{-w}y^{c-1}$ on $(0,1)$, and for real $c \ge w$ at least $2^{-w}y^{c-w-1}$ on $(1,\infty)$; both integrals diverge. □

Theorem 25.12 (The lean convolution). *As stated, for every family; the closing sentence takes the active letters of a family with nonnegative exponents.*

Proof. (i) Let $\mathbf{c}$ be admissible. By Proposition 25.10(ii), $\mathbf{c}_j^K \in s_j \operatorname{ri} P(\mathcal{B}_j)$ is the condition $(\mathbf{c}_j^K)_S > s_j d_j(S)$ for every nonempty proper S, and likewise $(\mathbf{c}_j^\Psi)_S > t_j d_j^*(S)$; summing over the letters, $\operatorname{Re}\kappa_S = \operatorname{Re} m_S - \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big) > 0$ (Definition 8.1). Conversely, if every proper wall of $\operatorname{Re}\mathbf{m}$ is positive, the native loads of $\operatorname{Re}\mathbf{m}$ are admissible for $(\mathbf{m};\mathbf{s},\mathbf{t})$ by Proposition 25.10(iii), admissibility involving $\operatorname{Re}\mathbf{m}$ only. The admissible points are cut out by linear equations and the convex conditions $\mathbf{c}^X \in w_X \operatorname{ri}\mathcal{N}_X$, so they form a convex set.

(ii) Apply Theorem 25.4 to the family word $W = \prod_X X^{-w_X}$ split into the $2N$ factors $W_X = X^{-w_X}$, of degrees $w_X \deg X$. The Newton polytope of W_X is $w_X\mathcal{N}_X$, of dimension $n-1$ by the first fact above, so its relative interior is $w_X \operatorname{ri}\mathcal{N}_X$, and the points $(\mathbf{c}_i)_i$ of Theorem 25.4 are exactly the admissible points. With the twists $\mathbf{x}^X$ of the factors, the left side of Theorem 25.4 is the integral over the section $\{\alpha_{e_0} = 1\}$ of the word of $F[\mathbf{x}]$, whose letters are $X(\mathbf{x}^X \circ \alpha)$ (Definition 22.1), that is $P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 8.7); the periods of the factors are the $P_X(\mathbf{a}^X)$. This is the display, absolutely convergent, of dimension $(2N-1)(n-1)$; K_1 is taken as the last factor of Theorem 25.4, whose total pin is omitted, and that pin holds by the others and the balance wall. The coordinate form is the remark after Lemma S11.

(iii) For a real point $\mathbf{a}$ at which $P_X(\mathbf{a})$ converges, $|P_X(\mathbf{a}+i\mathbf{y})| \le P_X(\mathbf{a})$, the modulus of the integrand of a period being its value at the real parts, and $|(\mathbf{x}^X)^{-\mathbf{a}^X}| = (\mathbf{x}^X)^{-\operatorname{Re}\mathbf{a}^X}$. On the imaginary translate through $\mathbf{c}$ the imaginary part $\mathbf{y}^X$ of every load has $|\mathbf{y}^X| = 0$, by the pins, and Proposition 25.10(iv) bounds $|P_X(\mathbf{c}^X + i\mathbf{y}^X)|$ by $\big(\sin\frac{\pi\varepsilon}{2}\big)^{-w_X} e^{-(1-\varepsilon)h_X(\mathbf{y}^X)} P_X(\mathbf{c}^X)$, h_X the norm of that proposition for the letter X.

The closing sentence is Theorem 25.4 with one factor for each active letter, the inactive letters being the constant 1; any factor may be taken last, and the walls and the admissible points are read with the active letters alone. □

Corollary 25.13 (What a family retains). *As stated, for every family.*

Proof. By Theorem 25.12(ii) and its closing sentence every period of F and of its twists, at exponents $s_j, t_j \ge 0$ not all zero and at masses on the balance wall with every proper wall positive, is a fibre integral of products of the periods P_X of its active letters at their loads, times the twists. By Proposition 25.10(i), $P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0) = r_j\, \mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(s_j)$ for $s_j > 0$, and by the same item and Cremona $P_{\mathcal{B}_j}(\mathbf{b}; 0, t_j) = P_{\mathcal{B}_j}(t_j\mathbf{1} - \mathbf{b}; t_j, 0) = r_j\, \mathsf{G}_{t_j\mathbf{1}-\mathbf{b};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(t_j)$ for $t_j > 0$. The loads run over $|\mathbf{a}| = r_j s_j$ and $|\mathbf{b}| = r_j^* t_j$, that is $|t_j\mathbf{1} - \mathbf{b}| = nt_j - r_j^* t_j = r_j t_j$. The converse is Proposition 25.10(i) for table j. □

Theorem 25.14 (Pinned holds). *As stated, for every family.*

Proof. Put $d = (2N-1)|G|$. In the coordinates of Theorem 25.12(ii), the loads of the letters of $\mathcal{X}'$ at the marks other than e_0, split the coordinates into their values $\mathbf{z}_G$ on G and the rest $\mathbf{z}'$, and let $\mathcal{I}(\mathbf{z}_G, \mathbf{z}')$ be the integrand at $\mathbf{x} = \mathbf{1}$, the product of the periods P_X at the loads that $(\mathbf{z}_G, \mathbf{z}')$ and the pins determine. Let $\mathbf{c}$ be the admissible point of the statement, $\boldsymbol{\beta} = \mathbf{c}_G$, and for $\mathbf{y} \in \mathbb{R}^d$ put

$$\Phi(\mathbf{y}) = \frac{1}{(2\pi i)^{(2N-1)(n-1-|G|)}} \int \mathcal{I}(\mathbf{c}_G + i\mathbf{y}, \mathbf{z}')\, d\mathbf{z}',$$

over the imaginary translate through $\mathbf{c}$ in $\mathbf{z}'$. By Lemma S11(ii) and the remark after it, $\Phi(\boldsymbol{\eta})$ is the fibre integral of the statement at $\boldsymbol{\beta}+i\boldsymbol{\eta}$.

(a) By Theorem 25.12(iii), $|\mathcal{I}|$ on the imaginary translate is at most a constant times $\prod_{X\in\mathcal{X}'} e^{-(1-\varepsilon)h_X(\mathbf{y}^X)}$, $\mathbf{y}^X$ the imaginary part of the load of X, a vector of total 0 determined by its coordinates off e_0; since h_X is a norm on the vectors of total 0, this bound is integrable in all the imaginary parts. So Φ is defined everywhere, continuous by dominated convergence, and integrable, with $\int|\Phi|\le(2\pi)^{(2N-1)(n-1-|G|)}\int|\mathcal{I}|$.

(b) For a twist at the marks of G alone, $\mathbf{x}^X=e^{\mathbf{u}^X}$ on G with $\mathbf{u}^X\in\mathbb{R}^G$ for $X\in\mathcal{X}'$ and $\mathbf{x}^{K_1}=\mathbf{1}$, the factor $\prod_X(\mathbf{x}^X)^{-\mathbf{a}^X}$ of Theorem 25.12(ii) is $e^{-\langle\mathbf{c}_G+i\mathbf{y},\mathbf{u}\rangle}$ on the translate, $\mathbf{u}=(\mathbf{u}^X)_X$. By Fubini's theorem in $(\mathbf{y},\mathbf{z}')$, $d\mathbf{z}_G=i^d\,d\mathbf{y}$,

$$e^{\langle\mathbf{c}_G,\mathbf{u}\rangle}P_{F[e^{\mathbf{u}}]}(\mathbf{m};\mathbf{s},\mathbf{t})=\frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\Phi(\mathbf{y})\,e^{-i\langle\mathbf{y},\mathbf{u}\rangle}\,d\mathbf{y}=:\widehat{\Phi}(\mathbf{u}).$$

(c) Let first $\mathbf{m}$ be real. Then $\widehat{\Phi}\ge0$, and for $\varepsilon>0$, by Fubini's theorem and the Gaussian integral,

$$\int_{\mathbb{R}^d}\widehat{\Phi}(\mathbf{u})\,e^{-\varepsilon|\mathbf{u}|^2/2}\,d\mathbf{u}=\int_{\mathbb{R}^d}\Phi(\mathbf{y})\,\frac{e^{-|\mathbf{y}|^2/(2\varepsilon)}}{(2\pi\varepsilon)^{d/2}}\,d\mathbf{y}\longrightarrow\Phi(\mathbf{0})\qquad(\varepsilon\downarrow0),$$

Φ being integrable and continuous at $\mathbf{0}$. By monotone convergence $\int\widehat{\Phi}\,d\mathbf{u}=\Phi(\mathbf{0})<\infty$, and with $d\mathbf{x}/\mathbf{x}=d\mathbf{u}$ this integral is $\mathrm{Hold}_G(\mathbf{c}_G)$: the statement at $\boldsymbol{\eta}=\mathbf{0}$.

(d) For complex $\mathbf{m}$, $|P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})|\le P_{F[\mathbf{x}]}(\mathrm{Re}\,\mathbf{m};\mathbf{s},\mathbf{t})$, and $\mathbf{c}$ is admissible for $\mathrm{Re}\,\mathbf{m}$ as well; by (c) at $\mathrm{Re}\,\mathbf{m}$ the hold converges absolutely at every $\mathbf{c}_G+i\boldsymbol{\eta}$. Then $\widehat{\Phi}$ of (b) is integrable, Φ is integrable and continuous, and Fourier inversion gives, at every $\boldsymbol{\eta}$,

$$\mathrm{Hold}_G(\mathbf{c}_G+i\boldsymbol{\eta})=\int_{\mathbb{R}^d}e^{\langle\mathbf{c}_G+i\boldsymbol{\eta},\mathbf{u}\rangle}P_{F[e^{\mathbf{u}}]}(\mathbf{m};\mathbf{s},\mathbf{t})\,d\mathbf{u}=\int_{\mathbb{R}^d}\widehat{\Phi}(\mathbf{u})\,e^{i\langle\boldsymbol{\eta},\mathbf{u}\rangle}\,d\mathbf{u}=\Phi(\boldsymbol{\eta}).$$

At $G=\mathcal{E}\setminus e_0$ the fibre has dimension 0, and its value is the product of Lemma 25.11. □

Proposition 25.15 (Holding one mark). *As stated, for every family.*

Proof. Let the data be real. Take the section $\{h=1\}$ of Theorem 8.7 with $h(\alpha)=\sum_{e\ne g}\alpha_e$: on it $\alpha=(\alpha',\alpha_g)$ with $\alpha'\in\Delta_{\mathcal{E}\setminus g}$ and $\alpha_g\in(0,\infty)$, and the measure is $d\omega_{\mathcal{E}\setminus g}(\alpha')\,d\alpha_g$. By Definitions 22.1 and 21.1 the twist at g acts on the letters as

$$K_j(\mathbf{x}@g\circ\alpha)=K_{\mathcal{B}_j\setminus g}(\alpha')+x_j^K\alpha_gK_{\mathcal{B}_j/g}(\alpha'),\qquad\Psi_j(\mathbf{x}@g\circ\alpha)=x_j^\Psi\alpha_g\Psi_{\mathcal{B}_j\setminus g}(\alpha')+\Psi_{\mathcal{B}_j/g}(\alpha'),$$

the minors carrying the weights of table j, all four nonzero since g is neither a loop nor a coloop. Insert the period into the hold and integrate first over the twists of the letters of $\mathcal{X}'$, by Tonelli's theorem. By Lemma S12, for $j\ge2$, respectively for every j,

$$\int_0^\infty x^{a_j-1}\big(K_{\mathcal{B}_j\setminus g}+x\alpha_gK_{\mathcal{B}_j/g}\big)^{-s_j}dx=\Gamma{}_1^2\begin{bmatrix}a_j,\ s_j-a_j\\ s_j\end{bmatrix}K_{\mathcal{B}_j\setminus g}^{a_j-s_j}\big(\alpha_gK_{\mathcal{B}_j/g}\big)^{-a_j},$$
$$\int_0^\infty x^{b_j-1}\big(\Psi_{\mathcal{B}_j/g}+x\alpha_g\Psi_{\mathcal{B}_j\setminus g}\big)^{-t_j}dx=\Gamma{}_1^2\begin{bmatrix}b_j,\ t_j-b_j\\ t_j\end{bmatrix}\Psi_{\mathcal{B}_j/g}^{b_j-t_j}\big(\alpha_g\Psi_{\mathcal{B}_j\setminus g}\big)^{-b_j},$$

finite exactly when $0 < a_j < s_j$, respectively $0 < b_j < t_j$, and $+\infty$ otherwise. The power of α_g left is $\alpha_g^{m_g-1-\sum_{j\geq 2} a_j - \sum_j b_j} = \alpha_g^{a_1-1}$, and by the same lemma

$$\int_0^\infty \alpha_g^{a_1-1}\big(K_{\mathcal{B}_1\setminus g} + \alpha_g K_{\mathcal{B}_1/g}\big)^{-s_1} d\alpha_g = \Gamma_1^2\begin{bmatrix} a_1, & s_1 - a_1 \\ & s_1 \end{bmatrix} K_{\mathcal{B}_1\setminus g}^{a_1-s_1} K_{\mathcal{B}_1/g}^{-a_1},$$

finite exactly when $0 < a_1 < s_1$. What remains is

$$\int_{\Delta_{\mathcal{E}\setminus g}} \alpha'^{\mathbf{m}|_{\mathcal{E}\setminus g}-\mathbf{1}} \prod_j K_{\mathcal{B}_j\setminus g}^{-(s_j-a_j)}\, \Psi_{\mathcal{B}_j\setminus g}^{-b_j}\, K_{\mathcal{B}_j/g}^{-a_j}\, \Psi_{\mathcal{B}_j/g}^{-(t_j-b_j)}\, d\omega_{\mathcal{E}\setminus g},$$

the period of the children family $F_{\{g\}}\big((s_j-a_j,a_j),(b_j,t_j-b_j)\big)_j$ at $\mathbf{m}|_{\mathcal{E}\setminus g}$, which is on its balance wall (the paragraph before the proposition); it is finite exactly when every proper wall of it is positive (Theorem 8.7). The integrands being positive, the hold is the product of these factors in $[0,\infty]$ and is finite exactly when all of them are. For complex loads and masses whose real parts are such data the moduli are dominated by the real computation, Lemma S12 holds in the strip, and Fubini's theorem gives the same identity. The sentence on admissible points is Theorem 25.14 with $G = \{g\}$.

For the limits, let the letter Y have the exponent $u > m_g > 0$ and let the twists at g of the other letters decrease to 0. Each other letter decreases pointwise to one of its minors, $K_{\mathcal{B}_j\setminus g}$ or $\Psi_{\mathcal{B}_j/g}$, and its power with a negative exponent increases; by monotone convergence the period increases to the integral of the limit. In the limit only Y contains α_g, as $A+\alpha_g B$ with A and B its two minors, and by Lemma S12 the integral over α_g of $\alpha_g^{m_g-1}(A+\alpha_g B)^{-u}$ is $\Gamma_1^2\begin{bmatrix} m_g, & u-m_g \\ & u \end{bmatrix} A^{m_g-u}B^{-m_g}$. What is left is the period of the children family at the shares of the statement: for $Y = K_j$, $A = K_{\mathcal{B}_j\setminus g}$ carries the exponent $s_j - m_g$ and $B = K_{\mathcal{B}_j/g}$ the exponent m_g; for $Y = \Psi_j$, $A = \Psi_{\mathcal{B}_j/g}$ carries $t_j - m_g$ and $B = \Psi_{\mathcal{B}_j\setminus g}$ carries m_g; every other $K_{j'}$ is $K_{\mathcal{B}_{j'}\setminus g}$ at $s_{j'}$ and every other $\Psi_{j'}$ is $\Psi_{\mathcal{B}_{j'}/g}$ at $t_{j'}$. □

Remark. For one table a twist at g in the cobasis letter is $x = x_1^\Psi$ with $x_1^K = 1$ (Definition 22.1), and $x \downarrow 0$ is the limit with the letter K chosen. With the letter Ψ chosen the twist y of K decreases to 0 with that of Ψ at 1; by the torus relation of Definition 22.1 at the mark g, $P_{\mathcal{H}[(y,1)@g]} = y^{-m_g} P_{\mathcal{H}[(1,1/y)@g]} = x^{m_g} P_{\mathcal{H}[x@g]}$ with $x = 1/y$. So the two limits are those of $P_{\mathcal{H}[x@g]}$ as $x \downarrow 0$ and of $x^{m_g} P_{\mathcal{H}[x@g]}$ as $x \uparrow \infty$, as the paragraph after the proposition reads them.

Lemma 25.16 (The Mellin transform of a law is the period of a children family). *As stated, for every cell family.*

Proof. By Definitions 23.1 and 21.1, with $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa_F$ the power of $\gamma_{\mathcal{A}_1}^K$ in $\boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F$ is $-\kappa_F + \sum_{j\geq 2} z_{j,K} + \sum_j z_{j,\Psi} = -z_{1,K}$. With $\gamma_{\mathcal{A}_j}^K = K_{\mathcal{A}_j\setminus g}/K_{\mathcal{A}_j/g}$ and $\gamma_{\mathcal{A}_j}^\Psi = \Psi_{\mathcal{A}_j/g}/\Psi_{\mathcal{A}_j\setminus g}$,

$$\begin{aligned} \boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F &= u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j/g}^{-s_j}\, \Psi_{\mathcal{A}_j\setminus g}^{-t_j} \big(\gamma_{\mathcal{A}_j}^K\big)^{-z_{j,K}} \big(\gamma_{\mathcal{A}_j}^\Psi\big)^{-z_{j,\Psi}}\, d\omega_B \\ &= u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j\setminus g}^{-z_{j,K}}\, \Psi_{\mathcal{A}_j\setminus g}^{-(t_j-z_{j,\Psi})}\, K_{\mathcal{A}_j/g}^{-(s_j-z_{j,K})}\, \Psi_{\mathcal{A}_j/g}^{-z_{j,\Psi}}\, d\omega_B. \end{aligned}$$

This is the form of the children family of $\{g\}$ at the shares of the statement, its tables $\mathcal{A}_j \setminus g$, of rank $r_{\mathcal{A}_j}$ and corank $|B| - r_{\mathcal{A}_j}$, at $(z_{j,K}, t_j - z_{j,\Psi})$ and $\mathcal{A}_j/g$, of rank $r_{\mathcal{A}_j} - 1$ and corank $|B| - r_{\mathcal{A}_j} + 1$, at $(s_j - z_{j,K}, z_{j,\Psi})$. Its degree is

$$\sum_j \Big(r_{\mathcal{A}_j} z_{j,K} + (|B| - r_{\mathcal{A}_j})(t_j - z_{j,\Psi}) + (r_{\mathcal{A}_j} - 1)(s_j - z_{j,K}) + (|B| - r_{\mathcal{A}_j} + 1) z_{j,\Psi} \Big)$$
$$= \kappa_F + \sum_j \big(s_j (r_{\mathcal{A}_j} - 1) + t_j (|B| - r_{\mathcal{A}_j}) \big) = m_B,$$

so the children family is on its balance wall, and the integral over Δ_B is its period (Theorem 8.7), in $[0, \infty]$ for real $\mathbf{z}$. This is the first identity.

With the mass $\sum_j (s_j + t_j) - \kappa_F$ on g the family of the tables $\mathcal{A}_j$ is on its balance wall (Lemma 21.12). Apply Proposition 25.15 to it at the mark g, a hinge and so neither a loop nor a coloop, with $a_j = s_j - z_{j,K}$ for $j \geq 2$ and $b_j = t_j - z_{j,\Psi}$. Then

$$a_1 = \sum_j (s_j + t_j) - \kappa_F - \sum_{j \geq 2} (s_j - z_{j,K}) - \sum_j (t_j - z_{j,\Psi}) = s_1 - z_{1,K},$$

the children family of Proposition 25.15 at the shares $(s_j - a_j, a_j) = (z_{j,K}, s_j - z_{j,K})$ and $(b_j, t_j - b_j) = (t_j - z_{j,\Psi}, z_{j,\Psi})$ is that of the first identity, its Beta corners are $\Gamma \begin{matrix} 2 \\ 1 \end{matrix} \begin{bmatrix} s_j - z_{j,K}, & z_{j,K} \\ & s_j \end{bmatrix}$ and $\Gamma \begin{matrix} 2 \\ 1 \end{matrix} \begin{bmatrix} t_j - z_{j,\Psi}, & z_{j,\Psi} \\ & t_j \end{bmatrix}$, and its hold is the left side of the second identity. At the real part of $\mathbf{z}$ the conditions of Proposition 25.15 hold: $0 < a_j < s_j$ and $0 < b_j < t_j$ are $0 < \operatorname{Re} z_{j,K} < s_j$ and $0 < \operatorname{Re} z_{j,\Psi} < t_j$, and the walls of the children family are positive since its period, $\int \boldsymbol{\varrho}_F^{-\operatorname{Re}\mathbf{z}} d\phi_F$ by the first identity, is finite. So Proposition 25.15 gives the second identity for complex $\mathbf{z}$ with this real part. □

Remark. The balance of the children family, the balance of the family of the tables $\mathcal{A}_j$ with the mass $\sum_j (s_j + t_j) - \kappa_F$ on g, and the identity $a_1 = s_1 - z_{1,K}$ are computed exactly by `check_hingesplit_balance` in `checks/bookkeeping.py`, and the balance of the children family of Proposition 25.15 by `check_holdone_balance`.

48.26 Chapter 26: periods at the full leans and networks along a hinge tree

The notation of Section 48.25 is kept. When table j of a family carries a hinge tree $\mathfrak{T}_j$ as in Definition 26.2, its marker loads and node loads are read in table j (Definition 26.3), at its loads $(\mathbf{a}_j, \mathbf{b}_j)$ and its exponents (s_j, t_j).

Lemma 26.1 (Substitutions of molecules). *As stated, for any number of cells.*

Proof. The case of two cells is the reference, proof of lem:twosummol. By Proposition 21.3(i) the substitution is the gluing of the cells to $\mathcal{H}$ along its marks one at a time: $\mathcal{B}_0 = \mathcal{H}$, and $\mathcal{B}_i$ is the gluing of $\mathcal{A}_i$ and $\mathcal{B}_{i-1}$ along h_i, the marks $h_{i+1}, \ldots, h_q$ left in place. By Definition 21.1 and Proposition 21.3(iii) this gluing is the two-sum of $\mathcal{A}_i$ and $\mathcal{B}_{i-1}$, the latter read as a cell with the hinge h_i. This h_i is neither a loop nor a coloop of $\mathcal{B}_{i-1}$: $\mathcal{H}$ has bases through h_i and bases avoiding it, and each extends to a basis of $\mathcal{B}_{i-1}$ by bases of the hinge minors of the cells glued before, which exist since their hinges are neither loops nor coloops. By the case of two cells $\mathcal{B}_i$ is one molecule exactly when $\mathcal{A}_i$ and $\mathcal{B}_{i-1}$ are, and by induction $\mathcal{B}_q$ is one molecule exactly when $\mathcal{H}$ and all the $\mathcal{A}_i$ are. □

Corollary 26.6 (Convergence is decided at the nodes). *As stated, for every family.*

Proof. Every table is one molecule by Lemma 26.1 applied along its hinge tree. By Theorem 26.5 in table j, at real loads, $P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0)$ is finite exactly when every marker load $k_{j;v,\mathfrak{e}}$ lies in $(0, s_j)$ and the period of every node at its full basis lean is finite at its node loads. By Proposition 25.10(ii) at the node, one molecule, the latter is $(\mathbf{a}_{j,v})_S > s_j d_{j,v}(S)$ for every nonempty proper S of its marks and markers; at $S = \{g\}$ and at the complement of a marker g these inequalities are $k > 0$ and $k < s_j$ for the marker load k at g, since g is neither a loop nor a coloop of the node and $|\mathbf{a}_{j,v}| = r_{j,v} s_j$ (Lemma 26.4(ii)). Likewise at the full cobasis lean. So the node inequalities hold at a split exactly when every load is admissible for its letter (Proposition 25.10(ii) in table j), that is when the split is an admissible point, and by Theorem 25.12(i) such a point exists exactly when every proper wall is positive; the native loads are one. The unknowns are the loads of the $2N$ letters at n marks less $n + 2N - 1$ independent pins, $(2N-1)(n-1)$ of them, and the node v of table j contributes $2(2^{|\mathcal{E}_{j,v}|+|G_{j,v}|} - 2)$ inequalities. □

Proposition 26.9 (Series and parallel nodes at their full leans). *As stated; the last sentence, which reads the four closed forms off Proposition 25.6, is new.*

Proof. The closed forms are the reference, proof of prop:spleans. The letter K of $\binom{\mathcal{X}}{1}$ is $\sum_x \theta_x \alpha_x$, and so is the letter Ψ of $\binom{\mathcal{X}}{q}$, whose basis $\mathcal{X} \setminus x$ has the cobasis $\{x\}$: both are simplicial with the configuration $I = (\mathbf{e}_x)_x$, of degree 1, $|\det I| = 1$ and the shares $\mathbf{b} = \mathbf{a}$, and Proposition 25.6 gives $\Gamma_{\substack{q+1\\1}}\begin{bmatrix}\mathbf{a}\\|\mathbf{a}|\end{bmatrix} \prod_x \theta_x^{-a_x}$, the forms of $P_{\binom{\mathcal{X}}{1}}(\mathbf{a}; s, 0)$ and $P_{\binom{\mathcal{X}}{q}}(\mathbf{b}; 0, t)$. The letter K of $\binom{\mathcal{X}}{q}$ and the letter Ψ of $\binom{\mathcal{X}}{1}$ are $\sum_x \theta_x \alpha^{\mathcal{X}\setminus x}$, simplicial with the configuration $J - I = (\mathbf{1} - \mathbf{e}_x)_x$ of degree q, J the matrix of ones. $J - I$ has the eigenvalues q, once, and -1, q times, so $|\det(J - I)| = q$, and $(J - I)^{-1} = \frac{1}{q}J - I$, so the shares of $\mathbf{a}$ with $|\mathbf{a}| = qs$ are $\frac{|\mathbf{a}|}{q}\mathbf{1} - \mathbf{a} = (s - a_x)_x$, of sum s. Proposition 25.6 gives $\frac{q}{q}\Gamma_{\substack{q+1\\1}}\begin{bmatrix}(s-a_x)_x\\ s\end{bmatrix} \prod_x \theta_x^{a_x - s}$, the form of $P_{\binom{\mathcal{X}}{q}}(\mathbf{a}; s, 0)$, and in the same way that of $P_{\binom{\mathcal{X}}{1}}(\mathbf{b}; 0, t)$ with $|\mathbf{b}| = qt$. The criteria of convergence agree, the positivity of the real parts of the shares being that of the arguments of the Gammas. □

Remark. The function `check_spleans_is_simplicial` of `checks/bookkeeping.py` verifies exactly, at $q = 4$, the shares $s - a_x$ and the prefactor $q/|\det(J - I)| = 1$.

Corollary 26.10 (Series-parallel tables at their full leans). *As stated, with any weights on the bases of the series and parallel nodes.*

Proof. By Theorem 26.5 the period of $\mathcal{B}$ at its full basis lean is the product over the nodes of their periods at their node loads over the Beta corners $\Gamma_{\substack{2\\1}}\begin{bmatrix}k_{v,\mathfrak{e}}, & k_{w,\mathfrak{e}}\\ & s\end{bmatrix}$ of the edges, the weights of $\mathcal{B}$ being the products of those of the nodes (Definition 26.2). Proposition 26.9 gives each node: $\Gamma_{\substack{n_v\\1}}\begin{bmatrix}(s-(\mathbf{a}_v)_x)_x\\ s\end{bmatrix} \prod_x \theta_{v,x}^{(\mathbf{a}_v)_x - s}$ at a series node and $\Gamma_{\substack{n_v\\1}}\begin{bmatrix}((\mathbf{a}_v)_x)_x\\ s\end{bmatrix} \prod_x \theta_{v,x}^{-(\mathbf{a}_v)_x}$ at a parallel node, $n_v = |\mathcal{E}_v| + |G_v|$. The Gammas, the factors $\Gamma(s)$ and the corners combine as in the reference, proof of cor:spfulllean: a tree has one node more than edges, which leaves $1/\Gamma(s)$; a mark e contributes $\Gamma(s - a_e)$ at a series node and $\Gamma(a_e)$ at a parallel node; at an edge $\mathfrak{e} = \{v, w\}$, where $k_{v,\mathfrak{e}} + k_{w,\mathfrak{e}} = s$ (Lemma 26.4(i)), the two markers contribute $\Gamma(s - k_{v,\mathfrak{e}})\Gamma(s - k_{w,\mathfrak{e}}) = \Gamma(k_{w,\mathfrak{e}})\Gamma(k_{v,\mathfrak{e}})$ between two series nodes and $\Gamma(k_{v,\mathfrak{e}})\Gamma(k_{w,\mathfrak{e}})$ between two parallel nodes, which the corner cancels, and $\Gamma(s - k_{v,\mathfrak{e}})\Gamma(k_{w,\mathfrak{e}})$ when v is a series and w a parallel node, which the corner leaves as $\Gamma(s - k_{\mathfrak{e}})/\Gamma(k_{\mathfrak{e}})$, $k_{\mathfrak{e}} = k_{v,\mathfrak{e}}$. The monomials of the nodes stand as they are. The full cobasis lean is the same with the second pair of forms of Proposition 26.9: $\Gamma(b_e)$ at series and $\Gamma(t - b_e)$ at parallel nodes, and at

an edge with the series end v the quotient $\Gamma(l_{v,\mathfrak{e}})\Gamma(t-l_{w,\mathfrak{e}})/\big(\Gamma(l_{v,\mathfrak{e}})\Gamma(l_{w,\mathfrak{e}})\big)=\Gamma(l_{\mathfrak{e}})/\Gamma(t-l_{\mathfrak{e}})$. The criterion of finiteness does not see the weights, which enter through monomials, positive and finite at real loads; it is the reference's. □

Corollary 26.11 (Periods along a hinge tree read by one hinge value). *As stated: for a family whose tables have a common hinge tree, at real data, wherever the active letters read every subtree cell by one hinge value.*

Proof. Induction on the number of edges of $\mathfrak{T}$. With one vertex there is nothing to prove.

Removing a leaf. A tree with an edge has two leaves, so some leaf v differs from o; let p be its parent and $\mathfrak{e}=\mathfrak{e}_v$. The subtree cell $\mathcal{A}_{v,j}$ is the node $\mathcal{H}_{v,j}$ with the hinge $g_{v,\mathfrak{e}}$ (23.15). The two-sums along a tree may be taken in any order (23.5, 21.3(i),(ii)), so table j of F is the substitution $\mathcal{H}'_j\langle\mathcal{A}_{v,j}\rangle$ of the cell $\mathcal{A}_{v,j}$ at the mark $g_{p,\mathfrak{e}}$ of the table $\mathcal{H}'_j$ obtained from the other nodes along $\mathfrak{T}-v$, every other mark of $\mathcal{H}'_j$ left in place. The cell on $\mathcal{E}_v$ is read by the active letters through γ_v, and a mark left in place through α_h, so 22.6 applies; the factors of the marks left in place cancel (its step (5)), and with $\kappa_{\mathfrak{e}}=\kappa_{\mathcal{E}_v}$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})\,\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}\kappa_{\mathfrak{e}}, & 2\sigma-\kappa_{\mathfrak{e}}\\ & 2\sigma\end{bmatrix}=P_{F'}(\mathbf{m}';\mathbf{s},\mathbf{t})\,P_{F_{\mathcal{H}_v}}(\mathbf{m}_v;\mathbf{s},\mathbf{t}),$$

F' the family of the tables $\mathcal{H}'_j$ at the masses $\mathbf{m}'$ equal to $\mathbf{m}$ on its marks and to $\kappa_{\mathcal{E}_v}$ at $g_{p,\mathfrak{e}}$, and $\mathbf{m}_v$ equal to $\mathbf{m}|_{\mathcal{E}_v}$ and to $2\sigma-\kappa_{\mathcal{E}_v}=\kappa_{\mathcal{E}\setminus\mathcal{E}_v}$ at $g_{v,\mathfrak{e}}$; every factor is on its balance wall with every proper wall positive.

The walls of F'. By 21.10, a set S' of marks of F' containing $g_{p,\mathfrak{e}}$ has the wall of $(S'\setminus\{g_{p,\mathfrak{e}}\})\sqcup\mathcal{E}_v$ in F, the cell entering with its whole set $\mathcal{E}_v$, of wall $\kappa_{\mathcal{E}_v}$, and the marker in D_j and D_j^*; a set avoiding $g_{p,\mathfrak{e}}$ has the same wall in both. So the walls across the edges of $\mathfrak{T}-v$ are those of F.

The hypothesis passes to F'. F' has the common hinge tree $\mathfrak{T}-v$, with the same nodes and $g_{p,\mathfrak{e}}$ a mark of the node p. Its subtree cells are those of F except at the vertices w on the path from p to o, where $\mathcal{A}_{w,j}=\mathcal{A}'_{w,j}\langle\mathcal{A}_{v,j}\rangle$, the cell at the mark $g_{p,\mathfrak{e}}$ (21.3(ii)). By 21.4 the hinge minors of $\mathcal{A}_{w,j}$ are those of $\mathcal{A}'_{w,j}$ with $\alpha_{g_{p,\mathfrak{e}}}$ replaced by $\gamma^K_{\mathcal{A}_{v,j}}$ in the letters K and by $\gamma^\Psi_{\mathcal{A}_{v,j}}$ in the letters Ψ, times factors of $\mathcal{A}_{v,j}$ that cancel in the quotients, so

$$\gamma^K_{\mathcal{A}_{w,j}}(\alpha)=\gamma^K_{\mathcal{A}'_{w,j}}\big(\alpha',\gamma^K_{\mathcal{A}_{v,j}}(\alpha_{\mathcal{E}_v})\big),\qquad \gamma^\Psi_{\mathcal{A}_{w,j}}(\alpha)=\gamma^\Psi_{\mathcal{A}'_{w,j}}\big(\alpha',\gamma^\Psi_{\mathcal{A}_{v,j}}(\alpha_{\mathcal{E}_v})\big),$$

α' the variables of the other marks. For the active letters both inner hinge values are $\gamma_v(\alpha_{\mathcal{E}_v})$, and γ_v, positive and homogeneous of degree one, takes every value in $(0,\infty)$. So for every value $c>0$ at $g_{p,\mathfrak{e}}$ the hinge values of $\mathcal{A}'_{w,j}$ by the active letters at (α',c) all equal $\gamma_w(\alpha)$ for any α with $\gamma_v(\alpha_{\mathcal{E}_v})=c$: the active letters read $\mathcal{A}'_{w,j}$ by one hinge value, and the induction hypothesis applies to F'. Its product formula and the display give that of F, with the masses on the markers the walls across the edges.

One table at a full lean. One letter is active, and the hypothesis is empty. At $(s,0)$ on the balance wall $|\mathbf{m}|=rs$, so the loads $\mathbf{a}=\mathbf{m}$ are admissible in 26.5; the marker load $k_{v,\mathfrak{e}}=sr_v^{\mathfrak{e}}-m_{\mathcal{E}_v^{\mathfrak{e}}}$ is $s-\kappa_{\mathcal{E}_v^{\mathfrak{e}}}$, the wall of the branch at v being $m_{\mathcal{E}_v^{\mathfrak{e}}}-s(r_v^{\mathfrak{e}}-1)$, and the two walls across an edge add to s; so the node loads are the masses $\mathbf{m}_v$ and the two identities are the same (26.4). At $(0,t)$ the same holds with $\mathbf{b}$ and t. The series and parallel nodes are 26.9.

Tables glued along a tree. The glued table has the hinge tree whose nodes are the tables $\mathcal{A}_v$ and whose markers are the glued marks; the glued marks of connected tables are neither loops nor coloops, and a two-sum has the product weights (21.12). The masses of the glued marks are positive, every singleton being a proper wall of its table, so every $\kappa_{\mathfrak{e}}$ lies in $(0,t)$. Each table

$\mathcal{A}_v$ is balanced, $m_{\mathcal{E}_{\mathcal{A}_v}} = tr^*_{\mathcal{A}_v}$, and the glued table, of corank $\sum_v r^*_{\mathcal{A}_v} - (k-1)$ and total mass $\sum_v tr^*_{\mathcal{A}_v} - (k-1)t$, is on its balance wall. Removing a leaf as above, with the first display of 22.6 in $[0,\infty]$ and its step (2), the cell at a leaf v glued along its mark g has the wall $t - m_g$ and its cell family is $\mathcal{A}_v$ at its own masses, so the period of the glued table is $\prod_v P_{\mathcal{A}_v} / \prod_{\mathfrak{e}} \Gamma_{\substack{2\\1}}\!\left[\begin{smallmatrix}\kappa_{\mathfrak{e}}, & t-\kappa_{\mathfrak{e}}\\ & t\end{smallmatrix}\right]$ in $[0,\infty]$; it is finite, and the glued table, balanced, has every proper wall positive (8.7).

A chain. By 21.12 the chain $F^{\triangleright k}$ of 22.8 has the hinge tree whose nodes are the copies, a path, the glued marks its markers; by the propagation of the hypothesis above, its subtree cells are read by the active letters through one hinge value when the copy is, and the walls across its edges are $m_{g_{\mathrm{in}}}$ and $m_{g_{\mathrm{out}}}$, so with every proper wall positive the display gives the product $P_F^k / \Gamma_{\substack{2\\1}}\!\left[\begin{smallmatrix}m_{g_{\mathrm{in}}}, & m_{g_{\mathrm{out}}}\\ & 2\sigma\end{smallmatrix}\right]^{k-1}$. □

Theorem 26.12 (Blocks read by one hinge value aggregate exactly). *As stated: in (ii) blocks read through $e_\ell/e_{\ell-1}$ as well as blocks read through their totals, and the active letters at nonzero exponents of either sign.*

Proof. (i) and (iii). These are the reference proof of 26.12(i),(iii), the letters with positive exponent replaced by the active letters: at a parallel block $K_{\mathcal{A}/g} = 1$ and $\gamma^K_{\mathcal{A}} = e_1$, at a series block $\Psi_{\mathcal{A}\setminus g} = 1$ and $\gamma^\Psi_{\mathcal{A}} = e_1$ (21.7(i)), so by 21.4 every active letter is the letter of its core read at the block totals, and the totals of $U \sim \mathrm{Dir}_{\mathbf{m}}$ have the law $\mathrm{Dir}_{(m_{B_i})_i}$ (3.10). The sign of an exponent does not enter.

(ii). When a block has $\ell_i \geq 2$ marks a table with both letters active would need a block that is both parallel and series; so at such a block each table has at most one active letter, and the inactive letters, of exponent 0, enter neither the walls nor the kernels. By 21.7(i), for $i \in I$ the active hinge values are $\gamma^K = e_1$ at the parallel blocks and $\gamma^\Psi = e_1$ at the series blocks, and for $i \notin I$ they are $\gamma^K = e_{\ell_i}/e_{\ell_i-1}$ at the series blocks and $\gamma^\Psi = e_{\ell_i}/e_{\ell_i-1}$ at the parallel blocks. So 22.5 applies with all constants 1 and $\gamma_i = e_1$ for $i \in I$, $\gamma_i = e_{\ell_i}/e_{\ell_i-1}$ for $i \notin I$:

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i S_i, \qquad S_i = \int_{\Delta_{B_i}} \gamma_i^{-\kappa_{B_i}}\, d\beta_i,$$

in $[0,\infty]$. By 21.10, $\kappa_{B_i} = m_{B_i} - \sum_j \big(s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(\ell_i - r_{\mathcal{A}_{ij}})\big)$ with the ranks 1 of a parallel block and ℓ_i of a series block.

For $i \in I$ the active letters see ranks that make every term vanish, so $\kappa_{B_i} = m_{B_i}$; the density of the cell kernel is $u^{\mathbf{m}|_{B_i}-\mathbf{1}}$, the active minors $K_{\mathcal{A}/g}$ and $\Psi_{\mathcal{A}\setminus g}$ being 1; and $e_1 = 1$ on the simplex, so $S_i = \int_{\Delta_{B_i}} u^{\mathbf{m}|_{B_i}-\mathbf{1}} d\omega$, which is $\Gamma_{\substack{\ell_i\\1}}\!\left[\begin{smallmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{smallmatrix}\right]$ when every m_x is positive and $+\infty$ otherwise, by the shape integral (3.11).

For $i \notin I$ each active letter contributes its exponent times $\ell_i - 1$, so $\kappa_{B_i} = m_{B_i} - (\ell_i - 1)2\sigma$. The active minors are $K_{\mathcal{A}/g} = e_{\ell_i-1}$ at a series block and $\Psi_{\mathcal{A}\setminus g} = e_{\ell_i-1}$ at a parallel block, so the density of the cell kernel is $u^{\mathbf{m}|_{B_i}-\mathbf{1}} e_{\ell_i-1}(u)^{-2\sigma}$, and with $e_{\ell_i} = \prod_{x\in B_i} u_x$,

$$S_i = \int_{\Delta_{B_i}} \Big(\frac{e_{\ell_i}}{e_{\ell_i-1}}\Big)^{-\kappa_{B_i}} u^{\mathbf{m}|_{B_i}-\mathbf{1}} e_{\ell_i-1}^{-2\sigma}\, d\omega = \int_{\Delta_{B_i}} u^{\mathbf{m}|_{B_i}-\kappa_{B_i}\mathbf{1}-\mathbf{1}}\, e_{\ell_i-1}(u)^{-(2\sigma-\kappa_{B_i})}\, d\omega.$$

This is the period of the series node $\binom{B_i}{\ell_i-1}$ with unit weights, whose letter K is e_{ℓ_i-1}, at the full basis lean $(2\sigma - \kappa_{B_i}, 0)$ and the masses $m_x - \kappa_{B_i}$; they add to $m_{B_i} - \ell_i\kappa_{B_i} = (\ell_i - 1)(2\sigma - \kappa_{B_i})$, its balance wall. By 26.9 it is $\Gamma_{\substack{\ell_i\\1}}\!\left[\begin{smallmatrix}(2\sigma - \kappa_{B_i} - m_x + \kappa_{B_i})_x\\ 2\sigma - \kappa_{B_i}\end{smallmatrix}\right] = \Gamma_{\substack{\ell_i\\1}}\!\left[\begin{smallmatrix}(2\sigma - m_x)_x\\ 2\sigma - \kappa_{B_i}\end{smallmatrix}\right]$, convergent exactly

when every $2\sigma - m_x$ is positive, and $+\infty$ otherwise, the integrand being positive. At $\ell_i = 1$ both kinds give $\kappa_{B_i} = m_x$ and $S_i = 1$.

So the display holds in $[0, \infty]$ with the stated reading of the stacked Gammas, and when these are finite the two sides are finite together.

The last sentence. At $\ell \geq 2$ the functions e_1 and $e_\ell / e_{\ell-1}$ are not proportional on the simplex, their quotient $e_1 e_{\ell-1}/e_\ell = (\sum_x u_x)(\sum_x u_x^{-1})$ being unbounded; so 22.5 does not apply to a block read both ways, and 22.3 keeps it as one mark with its kernel and both hinge values. □

Remark. Checks 7a to 7d of S21-23_work/checks.py are the case $\ell = 2$ of the blocks with $i \notin I$, with one table at $(s_1, 0)$ and one at $(0, t_2)$: $S = \Gamma(2\sigma - m_1)\Gamma(2\sigma - m_2)/\Gamma(4\sigma - m_B)$, and the period of the cell family is $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}\kappa, & 2\sigma - \kappa \\ & 2\sigma\end{bmatrix} S$.

Theorem 26.13 (The network). *As stated, for every family.*

Proof. Every table is one molecule by Lemma 26.1 along its hinge tree, and every proper wall is positive, so Theorem 25.12(ii) writes $P_{F[\mathbf{x}]}$ as the fibre integral of $\prod_j P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0) P_{\mathcal{B}_j}(\mathbf{b}_j; 0, t_j)$ times the twists, on the imaginary translate through an admissible point $\mathbf{c}$. At $\mathbf{c}$ every $P_{\mathcal{B}_j}(\mathbf{c}_j^K; s_j, 0)$ and $P_{\mathcal{B}_j}(\mathbf{c}_j^\Psi; 0, t_j)$ is finite (Proposition 25.10(ii)), so by Theorem 26.5 in table j every marker load lies in $(0, s_j)$, respectively $(0, t_j)$, every node period is finite at its node loads, and both factorizations of Theorem 26.5 hold on the tube over these real loads, which contains the translate. Inserting them gives the integrand of the statement. The native loads are admissible by Proposition 25.10(iii). The node periods are functions of the node loads alone, the exponent of the one active letter of a node being its total load divided by its degree (Definition 25.9). □

Corollary 26.14 (Series-parallel periods). *As stated, for every family and any weights on the bases of the nodes.*

Proof. Insert Corollary 26.10 for both full leans of every table into Theorem 26.13 and multiply by $\prod_j \Gamma(s_j)\Gamma(t_j)$. At a series node the monomials of the two leans multiply to $\theta_{j,v,x}^{(\mathbf{a}_{j,v})_x - s_j} \theta_{j,v,x}^{-(\mathbf{b}_{j,v})_x}$, at a parallel node to $\theta_{j,v,x}^{-(\mathbf{a}_{j,v})_x} \theta_{j,v,x}^{(\mathbf{b}_{j,v})_x - t_j}$. At a real point at which every Gamma above the line has an argument of positive real part all the periods of the tables at their full leans are finite (Corollary 26.10), so the point is admissible (Proposition 25.10(ii)) and Theorem 26.13 applies on it. The dimension is that of Theorem 26.13, and the integrand does not involve the bases of the tables. □

Proposition 26.15 (Pooling a pair). *As stated, with any weights on the bases of the node.*

Proof. For one table the pins are $\mathbf{a} + \mathbf{b} = \mathbf{m}$ and $|\mathbf{b}| = r^* t$, the twist $\mathbf{x}$ is on the cobasis letter, and the integrand of Corollary 26.14 is a function of $\mathbf{b}$, with $\Gamma(s - a_e) = \Gamma(c_e + b_e)$ at a mark e of a series node. The marks e and f of one node v lie on one side of every edge, so every marker load, an affine function of sums of loads over branches (Definition 26.3), depends on b_e and b_f through $b_{\{e,f\}}$ alone; so do the Gammas of the other marks, the monomials of the other nodes and those of the markers of v. What is left at a series node is $\Gamma(c_e + b_e)\Gamma(b_e)\Gamma(c_f + b_f)\Gamma(b_f)$ times

$$\theta_{v,e}^{a_e - b_e - s} \theta_{v,f}^{a_f - b_f - s} x_e^{-b_e} x_f^{-b_f} = \theta_{v,e}^{m_e - s} \theta_{v,f}^{m_f - s} \left(x_e \theta_{v,e}^2\right)^{-b_e} \left(x_f \theta_{v,f}^2\right)^{-b_f},$$

a function of $b_{\{e,f\}}$ when $x_e \theta_{v,e}^2 = x_f \theta_{v,f}^2$; at a parallel node it is $\Gamma(m_e - b_e)\Gamma(t - b_e)\Gamma(m_f - b_f)\Gamma(t - b_f)$ times $\theta_{v,e}^{-m_e - t} \theta_{v,f}^{-m_f - t} \left(x_e \theta_{v,e}^{-2}\right)^{-b_e} \left(x_f \theta_{v,f}^{-2}\right)^{-b_f}$, from the monomials $\theta_{v,x}^{b_x - a_x - t}$, a function of $b_{\{e,f\}}$ when

$x_e\theta_{v,e}^{-2} = x_f\theta_{v,f}^{-2}$. The rest is the reference, proof of prop:poolpair, with $x_e\theta_{v,e}^{\pm 2}$ in place of x_e: at fixed $b_{\{e,f\}}$ the integral over b_e is the integral of four Gammas displayed in the statement, and disjoint pairs occupy disjoint coordinates. □

Theorem 26.16 (Nodes in hold form). *As stated, for every substituted family.*

Proof. The cores $\mathcal{H}_j$ and the cells $\mathcal{A}_{ij}$ are molecules by Lemma 26.1, and the markers are neither loops nor coloops of their cores and cells. Table j carries the hinge tree whose nodes are $\mathcal{H}_j$ and the $\mathcal{A}_{ij}$, each cell read as a core on $B_i \sqcup \{g_i\}$ and joined to $\mathcal{H}_j$ by an edge $\mathfrak{e}_i$ (Definition 23.15), the table being obtained by these two-sums (Proposition 21.3(i)). Each B_i is nonempty, a hinge being neither a loop nor a coloop; fix $e_i \in B_i$.

(i) *Step 1: the network.* Let $\mathbf{c}$ be an admissible point of F. By Theorem 26.13 with these trees, $P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})$ is the fibre integral, over the loads of F with the pins $\sum_j(\mathbf{a}_j+\mathbf{b}_j)=\mathbf{m}$, $|\mathbf{a}_j| = r_j s_j$ ($j \geq 2$) and $|\mathbf{b}_j| = r_j^* t_j$, on the translate through $\mathbf{c}$, of

$$\prod_j \frac{P_{\mathcal{H}_j}(\mathbf{a}_{j,\mathcal{H}};s_j,0)P_{\mathcal{H}_j}(\mathbf{b}_{j,\mathcal{H}};0,t_j)\prod_i P_{\mathcal{A}_{ij}}(\mathbf{a}_{j,i};s_j,0)P_{\mathcal{A}_{ij}}(\mathbf{b}_{j,i};0,t_j)}{\prod_i \Gamma_1^2\begin{bmatrix} a_{j,g_i},\ s_j-a_{j,g_i} \\ s_j\end{bmatrix}\Gamma_1^2\begin{bmatrix} b_{j,g_i},\ t_j-b_{j,g_i} \\ t_j\end{bmatrix}} (\mathbf{x}_j^K)^{-\mathbf{a}_j|\varepsilon_{\mathcal{H}}}(\mathbf{x}_j^\Psi)^{-\mathbf{b}_j|\varepsilon_{\mathcal{H}}}.$$

Here, by Definition 26.3 with the branch B_i on the side of the cell, of rank $r_{\mathcal{A}_{ij}}$ and corank $|B_i|+1-r_{\mathcal{A}_{ij}}$, and by Lemma 26.4(i), the marker loads of the core at g_i are

$$a_{j,g_i} = a_{j,B_i} - s_j\big(r_{\mathcal{A}_{ij}}-1\big), \qquad b_{j,g_i} = b_{j,B_i} - t_j\big(|B_i|-r_{\mathcal{A}_{ij}}\big),$$

those of the cell at g_i are $s_j - a_{j,g_i}$ and $t_j - b_{j,g_i}$, the node loads of the core are $\mathbf{a}_{j,\mathcal{H}} = (\mathbf{a}_j|_{\mathcal{E}_{\mathcal{H}}}, (a_{j,g_i})_i)$ and $\mathbf{b}_{j,\mathcal{H}} = (\mathbf{b}_j|_{\mathcal{E}_{\mathcal{H}}}, (b_{j,g_i})_i)$, and those of the cell $\mathcal{A}_{ij}$ are $\mathbf{a}_{j,i} = (\mathbf{a}_j|_{B_i}, s_j - a_{j,g_i})$ and $\mathbf{b}_{j,i} = (\mathbf{b}_j|_{B_i}, t_j - b_{j,g_i})$. The twists are 1 off $\mathcal{E}_{\mathcal{H}}$.

Step 2: the marker loads as coordinates. In every table j and for every i replace the coordinate a_{j,e_i} by a_{j,g_i} and b_{j,e_i} by b_{j,g_i}. Each new coordinate is the old one plus other old coordinates plus a constant, so the change of coordinates is unit triangular, of determinant 1, and by Lemma S11(i) the fibre integral does not change. The new coordinates are the *core coordinates*, the loads $(\mathbf{a}_{j,\mathcal{H}}, \mathbf{b}_{j,\mathcal{H}})_j$ of the cores on $\mathcal{E}_{\mathcal{H}} \sqcup \{g_1,\dots,g_k\}$, and for each i the *coordinates of* B_i, the loads $a_{j,e}$ and $b_{j,e}$ at the marks $e \in B_i \setminus e_i$, $j = 1,\dots,N$.

Step 3: the core masses. Replace the mass pin at e_i by the sum C_i of the mass pins over B_i, a row operation with the coefficient 1 on the replaced pin, of determinant 1 (Lemma S11(i)). In the new coordinates

$$C_i: \qquad \sum_j \big(a_{j,g_i}+b_{j,g_i}\big) = m_{B_i} - \sum_j \Big(s_j\big(r_{\mathcal{A}_{ij}}-1\big) + t_j\big(|B_i|-r_{\mathcal{A}_{ij}}\big)\Big) = \kappa_{F_i}$$

by Definition 23.1: the mass of the cores at g_i is the hinge wall of the cell family at g_i.

Step 4: the totals, and the split of the pins. By Proposition 21.3(i), $r_j = r_{\mathcal{H}_j} + \sum_i (r_{\mathcal{A}_{ij}}-1)$, and so $r_j^* = n - r_j = r_{\mathcal{H}_j}^* + \sum_i(|B_i| - r_{\mathcal{A}_{ij}})$ with $r_{\mathcal{H}_j}^* = |\mathcal{E}_{\mathcal{H}}| + k - r_{\mathcal{H}_j}$ and $n = |\mathcal{E}_{\mathcal{H}}| + \sum_i |B_i|$. In the new coordinates therefore

$$|\mathbf{a}_j| = \big|\mathbf{a}_j|_{\mathcal{E}_{\mathcal{H}}}\big| + \sum_i \Big(a_{j,g_i} + s_j\big(r_{\mathcal{A}_{ij}}-1\big)\Big), \qquad \text{so} \qquad |\mathbf{a}_j| = r_j s_j \iff |\mathbf{a}_{j,\mathcal{H}}| = r_{\mathcal{H}_j} s_j,$$

and likewise $|\mathbf{b}_j| = r_j^* t_j \iff |\mathbf{b}_{j,\mathcal{H}}| = r_{\mathcal{H}_j}^* t_j$. The pins are now the *core pins*, namely the mass pins at $\mathcal{E}_{\mathcal{H}}$, the C_i and the totals of the cores, on the core coordinates alone, which are the pins of

$\mathcal{F}$; and for each i the *pins of* B_i, the mass pins at $e \in B_i \setminus e_i$, on the coordinates of B_i alone. The pin matrix is block diagonal, and since the fibre integral converges absolutely (Theorem 25.12(ii)), Lemma S11(iii) writes it as the integral over $\mathcal{F}$, on the translate through the core node loads of $\mathbf{c}$, of the fibre integrals over the coordinates of the B_i at fixed core coordinates. The corners and the twists depend on the core coordinates alone and leave the inner integrals, and the inner integral for B_i is

$$\Phi_i = \int \prod_j P_{\mathcal{A}_{ij}}(\mathbf{a}_{j,i}; s_j, 0)\, P_{\mathcal{A}_{ij}}(\mathbf{b}_{j,i}; 0, t_j)$$

over the coordinates of B_i with the pins of B_i.

Step 5: the inner integrals are holds. Let $F_i^\natural$ be the family of the tables $\mathcal{A}_{ij}$ on $B_i \sqcup \{g_i\}$ with the masses $\mathbf{m}|_{B_i}$ and $\sum_j (s_j + t_j) - \kappa_{F_i}$ on g_i, on its balance wall (Lemma 21.12). The node loads $(\mathbf{a}_{j,i}, \mathbf{b}_{j,i})_j$ of the cells are loads of its letters: for every j, $|\mathbf{a}_{j,i}| = a_{j,B_i} + s_j - a_{j,g_i} = r_{\mathcal{A}_{ij}} s_j$ and $|\mathbf{b}_{j,i}| = t_j(|B_i| + 1 - r_{\mathcal{A}_{ij}})$, their masses at $e \in B_i$ are the m_e, and at g_i they add to $\sum_j (s_j + t_j - a_{j,g_i} - b_{j,g_i}) = \sum_j (s_j + t_j) - \kappa_{F_i}$ by C_i. At $\mathbf{c}$ every period of a cell at its node loads is finite (Theorem 26.5), so the node loads of the cells at $\mathbf{c}$ are an admissible point of $F_i^\natural$ (Proposition 25.10(ii)), and every proper wall of $F_i^\natural$ is positive (Theorem 25.12(i)). In Theorem 25.14 for $F_i^\natural$ take e_i as the fixed mark and $G = \{g_i\}$: the pinned fibre integral at the pins $s_j - a_{j,g_i}$ $(j \geq 2)$ and $t_j - b_{j,g_i}$ at g_i is, in the coordinates of the loads of the letters other than K_1 at the marks of $B_i \setminus e_i$, the integral with the factor $(2\pi i)^{-(2N-1)(|B_i|-1)}$, the other loads solved from its pins. The integral Φ_i, with $a_{1,e}$ for $e \in B_i \setminus e_i$ solved from the mass pin at e, has the same coordinates, the same factor by Lemma S11(ii), and the same integrand, since the loads $(\mathbf{a}_{j,i}, \mathbf{b}_{j,i})_j$ satisfy all the pins of $F_i^\natural$. The real parts of the core coordinates being those of $\mathbf{c}$, Theorem 25.14 gives

$$\Phi_i = \mathrm{Hold}_{\{g_i\}}\big((s_j - a_{j,g_i})_{j\geq 2}, (t_j - b_{j,g_i})_j\big) \qquad \text{for } F_i^\natural.$$

Step 6: the corners cancel. Apply Lemma 25.16 to the cell family F_i at $\mathbf{z}_i = (a_{j,g_i}, b_{j,g_i})_j$, of sum κ_{F_i} by C_i. Its conditions hold: $0 < \mathrm{Re}\, a_{j,g_i} < s_j$ and $0 < \mathrm{Re}\, b_{j,g_i} < t_j$, the marker loads at $\mathbf{c}$ lying in these intervals (Theorem 26.5); and the hold of Step 5 at the real parts is finite, so by Proposition 25.15 and the first identity of Lemma 25.16 $\int \boldsymbol{\varrho}_{F_i}^{-\mathrm{Re}\,\mathbf{z}_i} d\phi_{F_i}$ is finite. So

$$\Phi_i = \prod_j \Gamma_{\!1}^{\,2}\begin{bmatrix} s_j - a_{j,g_i}, & a_{j,g_i} \\ & s_j \end{bmatrix} \Gamma_{\!1}^{\,2}\begin{bmatrix} t_j - b_{j,g_i}, & b_{j,g_i} \\ & t_j \end{bmatrix} \int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} d\phi_{F_i},$$

and these are the corners of the edge $\mathfrak{e}_i$ in Step 1, $\Gamma_{\!1}^{\,2}\begin{bmatrix} a, & b \\ & a+b \end{bmatrix}$ being symmetric in a and b. They cancel, and what is left is the integral of (i) over $\mathcal{F}$. Its dimension is the number $2N(|\mathcal{E}_\mathcal{H}| + k)$ of core coordinates less the $|\mathcal{E}_\mathcal{H}| + k + 2N - 1$ core pins, $(2N-1)(|\mathcal{E}_\mathcal{H}| + k - 1)$.

(ii) Let $F'^\natural$ be the family of the tables of F' with the mass $\sum_j (s_j + t_j) - \kappa_{F'}$ on g_0, on its balance wall (Lemma 21.12), and $\mathbf{c}$ an admissible point of it as in the statement. Its loads at g_0 lie in $(0, s_j)$ and $(0, t_j)$, g_0 being neither a loop nor a coloop, so $\mathrm{Re}\,\mathbf{z}$ meets the conditions of Lemma 25.16 for F'; the finiteness of $\int \boldsymbol{\varrho}_{F'}^{-\mathrm{Re}\,\mathbf{z}} d\phi_{F'}$ follows as in Step 6 from that of the hold of $F'^\natural$ at g_0 at the loads of $\mathbf{c}$ (Theorem 25.14). By the second identity of Lemma 25.16 the left side of (ii) is the hold of $F'^\natural$ at g_0 at the loads $(s_j - z_{j,K})_{j\geq 2}$ and $(t_j - z_{j,\Psi})_j$, and by Theorem 25.14 with $G = \{g_0\}$ it is the fibre integral of Theorem 25.12(ii) for $F'^\natural$ at $\mathbf{x} = \mathbf{1}$ with the pins at g_0. Steps 1 to 6 apply to it with g_0 a mark of the cores, its pins being core pins, and give the integral over $\mathcal{F}_\mathbf{z}$, of dimension $(2N-1)(|\mathcal{E}_\mathcal{H}| + k) - (2N-1) = (2N-1)(|\mathcal{E}_\mathcal{H}| + k - 1)$. □

Remark. Let the core of every table be $\binom{\{g_1,g_2\}}{1}$ with unit weights, the weights of the tables being those of the cells (Lemma 21.12), so that $\mathcal{E}_{\mathcal{H}} = \varnothing$ and $B = B_1$ is a bunch of width one in every table. By Proposition 26.9 at $q = 1$,

$$P_{\mathcal{H}_j}(\mathbf{a}_{j,\mathcal{H}}; s_j, 0)\, P_{\mathcal{H}_j}(\mathbf{b}_{j,\mathcal{H}}; 0, t_j) = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} a_{j,g_1},\ a_{j,g_2} \\ s_j \end{bmatrix} \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} b_{j,g_2},\ b_{j,g_1} \\ t_j \end{bmatrix} = \Gamma_{\!\!\begin{smallmatrix}4\\2\end{smallmatrix}}\!\begin{bmatrix} a_j^d,\ b_j^d,\ a_j^r,\ b_j^r \\ s_j,\ t_j \end{bmatrix}$$

of Theorem 19.13, with the level shares $a_j^r = a_{j,g_1}$, $a_j^d = a_{j,g_2}$, $b_j^d = b_{j,g_1}$ and $b_j^r = b_{j,g_2}$. With $\mathbf{z}_1 = (a_j^r, b_j^d)_j$ the children family of F_1 in Lemma 25.16 has the tables $\mathcal{A}_{1j} \setminus g_1 = \mathcal{B}_j | B$ at $(a_j^r, t_j - b_j^d) = (a_j^r, b_j^r)$ and $\mathcal{A}_{1j}/g_1 = \mathcal{B}_j/R$ at $(s_j - a_j^r, b_j^d) = (a_j^d, b_j^d)$: it is the bunch family of Theorem 19.13; with $\mathbf{z}_2 = (a_j^d, b_j^r)_j$ the children family of F_2 is the room family. The core pins are $|\mathbf{a}_{j,\mathcal{H}}| = s_j$ $(j \geq 2)$, $|\mathbf{b}_{j,\mathcal{H}}| = t_j$ and the masses κ_{F_1} and κ_{F_2}; solving the totals for a_j^d and b_j^r by Lemma S11(ii), with the block of the totals and of the mass κ_{F_2} unit triangular after the row operations of Lemma S11(i), leaves the share integral over $(a_j^r, b_j^d)_j$ pinned by $\sum_j (a_j^r + b_j^d) = \kappa_{F_1} = \kappa_B$. This is the integral of Theorem 19.13 over the scale shares, with the same integrand. In the pair coordinates $\sigma_j^d = \frac{1}{2}(a_j^d + b_j^d)$ and $\delta_j^d = \frac{1}{2}(a_j^d - b_j^d)$, the map $(a_j^r, b_j^d) \mapsto (\sigma_j^d, \delta_j^d)$, with $a_j^d = s_j - a_j^r$, has the determinant $\frac{1}{2}$ in each table, so $d(\mathbf{a}^r, \mathbf{b}^d) = 2^N\, d(\boldsymbol{\sigma}^d, \boldsymbol{\delta}^d)$; and the pin $\sum_j (a_j^r + b_j^d) = \kappa_B$ reads $-2\sum_j \delta_j^d = \kappa_B - \sum_j s_j$, its row being -2 times that of $\sum_j \delta_j^d = \frac{1}{2}(\sum_j s_j - \kappa_B)$, which by Definition 4.1 multiplies the fibre measure by $\frac{1}{2}$. The two factors give the 2^{N-1} of the first display of Theorem 19.13, a consistency check of the bookkeeping above. In `checks/bookkeeping.py`, `check_twosides_pairs` computes $2^N \cdot \frac{1}{2} - 2^{N-1} = 0$ exactly; `check_core_family_balance` computes the balance of the core family at the masses $(\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}, \kappa_{F_1}, \ldots, \kappa_{F_k})$ and the corank identity of Step 4; and `check_holdnode_pins` computes, for small families, the unit Jacobian of Step 2, the split of the pins of Steps 3 and 4 with the core masses κ_{F_i} and the totals of the cores, and the totals and the hinge mass of the node loads of the cells in Step 5.

Corollary 26.17 (The glue and the series and parallel rules in hold form). *As stated; in (i) the reading for a family is new.*

Proof. For one table the point $\mathbf{z}_i$ of Theorem 26.16 is $(\kappa_{\mathcal{A}_i} - b_{g_i}, b_{g_i})$, by the core mass at g_i, and $\boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} = \varrho_{\mathcal{A}_i}^{-b_{g_i}}$, the ratio vector of one cell being its ratio (Definition 23.1); so (i) for one table, (ii) and (iii) are the reference, proof of cor:holdrules. The reading of (i) for a family is the remark after the proof of Theorem 26.16. □

Definition 26.18 (The network procedure). As stated, for a family whose tables carry hinge trees; pooling and the passage to the transforms are for one table, and the recursive Mellin evaluation asks the tables to share one rooted tree.

Remark. The operations rest on the statements they cite, proved above for families: (1) is Corollary 26.6, infeasibility meaning that some proper wall is not positive, so that the period diverges (Theorem 8.7); (2) is Theorem 26.13 and Corollary 26.14; (4) uses that the admissible points form an open subset of the real points of the pins, the conditions $\mathbf{c}^X \in w_X \operatorname{ri} \mathcal{N}_X$ being open there, and the bounds of Theorem 25.12(iii), uniform on compact sets of admissible points; (5) is Theorem 26.16, the substituted family at a vertex v having the cores $\mathcal{H}_{j,v}$ and, as its cell families, the subtree cell families of the vertices directly below v. The reusable data are those of Corollary 25.13, one Gramma for each orbit of nodes.

Proposition 26.19 (What the network costs). *As stated, for a family.*

Proof. (i) The fibre of Theorem 26.13 has dimension $(2N-1)(n-1)$, and for one table each pooled pair removes one coordinate (Proposition 26.15). Opening the period of a prime node $\mathcal{H}_{j,v}$ at one full lean on its share side adds $|\mathcal{H}_{j,v}|-n_{j,v}$ coordinates (Corollary 26.8), so the two full leans add $2(|\mathcal{H}_{j,v}|-n_{j,v})$; series and parallel nodes have closed forms (Proposition 26.9). The atom formula (Corollary 18.16) has the shares $\mathbf{a}_j, \mathbf{b}_j \in \mathbb{C}^{\mathcal{B}_j}$ against the $2N$ total pins and the $n-1$ pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$, $e \neq g$: dimension $\sum_j 2|\mathcal{B}_j| - 2N - n + 1$. After its even shares close (Theorem 20.11(iv)) the fibre of differences has dimension $\sum_j (|\mathcal{B}_j|-1) - (n-1)$, and each table keeps one scale integral: $\sum_j |\mathcal{B}_j| - n + 1$ in all.

(ii) is Corollary 25.13: one Gramma of n_v variables for each orbit of nodes, the orbits taken over all the tables.

(iii) By Theorem 26.16(ii) a point of the Mellin transform of the law of a subtree cell family with the top node v costs a fibre integral of dimension $(2N-1)(|\mathcal{E}_v| + k_v - 1)$, $k_v = |G_v| - 1$ the number of vertices directly below v, that is $(2N-1)(|\mathcal{E}_v| + |G_v| - 2)$; at the root Theorem 26.16(i) costs $(2N-1)(|\mathcal{E}_o| + |G_o| - 1)$. Their integrands need the periods of the tables of the node at their full leans at the loads of the fibre, closed at series and parallel nodes and on the share side at a prime node (Corollary 26.8). □

Proposition 26.20 (Product weights are twists). *As stated, for every family.*

Proof. $K_{F^{\boldsymbol{\zeta}},j}(\alpha) = \sum_T \theta_{j,T} \boldsymbol{\zeta}_j^T \alpha^T = K_j(\boldsymbol{\zeta}_j \circ \alpha)$, and $\Psi_{F^{\boldsymbol{\zeta}},j}(\alpha) = \sum_T \theta_{j,T} \boldsymbol{\zeta}_j^T \alpha^{\mathcal{E}\setminus T} = \boldsymbol{\zeta}_j^{\mathcal{E}} \sum_T \theta_{j,T} (\alpha/\boldsymbol{\zeta}_j)^{\mathcal{E}\setminus T} = \boldsymbol{\zeta}_j^{\mathcal{E}} \Psi_j(\alpha/\boldsymbol{\zeta}_j)$. These are the letters of $F[\mathbf{x}]$ with $\mathbf{x}_j^K = \boldsymbol{\zeta}_j$ and $\mathbf{x}_j^\Psi = \boldsymbol{\zeta}_j^{-1}$ (Definition 22.1), the cobasis letters multiplied by $\boldsymbol{\zeta}_j^{\mathcal{E}}$, and $(\boldsymbol{\zeta}_j^{\mathcal{E}})^{-t_j} = \boldsymbol{\zeta}_j^{-t_j \mathbf{1}}$ gives the first identity. By the torus relation of Definition 22.1 for the family of the one letter K_j, the period of $K_j(\boldsymbol{\zeta}_j \circ \alpha)^{-s_j}$ at $\mathbf{a}$ is $\boldsymbol{\zeta}_j^{-\mathbf{a}} P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0)$, and for the one letter Ψ_j the period of $\big(\boldsymbol{\zeta}_j^{\mathcal{E}} \Psi_j(\alpha/\boldsymbol{\zeta}_j)\big)^{-t_j}$ at $\mathbf{b}$ is $\boldsymbol{\zeta}_j^{-t_j \mathbf{1}} \boldsymbol{\zeta}_j^{\mathbf{b}} P_{\mathcal{B}_j}(\mathbf{b}; 0, t_j)$. In the lean convolution of $F^{\boldsymbol{\zeta}}$ the integrand is therefore that of F times $\prod_j \boldsymbol{\zeta}_j^{-t_j \mathbf{1}} \boldsymbol{\zeta}_j^{\mathbf{b}_j - \mathbf{a}_j}$. With one $\boldsymbol{\zeta}$ for every table, the torus relation with the vector $\boldsymbol{\zeta}^{-1}$ on every letter gives $P_{F[\mathbf{x}]} = \boldsymbol{\zeta}^{-\mathbf{m}} P_{F[\mathbf{x}']}$ for $\mathbf{x}_j^K = \boldsymbol{\zeta}$, $\mathbf{x}_j^\Psi = \boldsymbol{\zeta}^{-1}$ and $\mathbf{x}_j'^K = \mathbf{1}$, $\mathbf{x}_j'^\Psi = \boldsymbol{\zeta}^{-2}$, and $\prod_j \boldsymbol{\zeta}^{-t_j \mathbf{1}} = \boldsymbol{\zeta}^{-|\mathbf{t}|\mathbf{1}}$. Weights at one mark g are the $\boldsymbol{\zeta}_j$ equal to 1 off g. □

Proposition 26.21 (Carrying a calculation through a change, in hold form). *As stated, for every family.*

Proof. In Theorem 26.13 the integrand is the product of the node periods at the node loads over the Beta corners at the marker loads, times the twists. (i) The exponents of table j enter the pins $|\mathbf{a}_j| = r_j s_j$ and $|\mathbf{b}_j| = r_j^* t_j$ and the marker loads $k_{j;v,\mathfrak{e}} = s_j r_{j,v}^{\mathfrak{e}} - a_{j,\mathcal{E}_v^{\mathfrak{e}}}$ and $l_{j;v,\mathfrak{e}} = t_j r_{j,v}^{*\mathfrak{e}} - b_{j,\mathcal{E}_v^{\mathfrak{e}}}$, affine in (s_j, t_j), and the corners through them; the node periods are functions of the node loads alone (Definition 25.9), and the native loads of the new data are admissible (Proposition 25.10(iii)). (ii) In the coordinates of Theorem 25.12(ii) the masses enter $\mathbf{a}_1 = \mathbf{m} - \sum_{j \geq 2} \mathbf{a}_j - \sum_j \mathbf{b}_j$ alone, the loads at e_0 of the other letters being fixed by their totals. (iii) is Proposition 26.20, for one node read for its table. (iv) The node periods are the only data of a node in the integrand; a replaced node of the same rank keeps the ranks of the branches and so the marker loads, and Theorem 26.13 applies to the new trees, whose nodes are molecules with markers neither loops nor coloops. (v) By Theorem 26.16(ii) the transform at a vertex is computed from the node there and the transforms of the vertices directly below it, so a change at one node changes the transforms on its path to the root only. □

Proposition 26.22 (The exponents are moment indices). *As stated, for every family, any of its letters being closed.*

Proof. The walls are affine in $(\mathbf{s},\mathbf{t})$ (Definition 8.1), so I, the intersection of the hyperplane with the open orthant and the open half-spaces $\kappa_S>0$, is relatively open and convex. On I the period converges (Theorem 8.7), and $\mathbf{m}>0$, every singleton wall $m_e-\sum_j(s_j d_j(\{e\})+t_j d_j^*(\{e\}))$ being positive and the counts nonnegative. Fix i and apply Lemma 16.2 to the form of F with $h=L_i^{1/g_i}$, positive and homogeneous of degree one, and $\varphi(\rho)=\rho^{w_i g_i}e^{-\rho^{g_i}}$, for which $\int_0^\infty\varphi(\rho)\,d\rho/\rho=\Gamma(w_i)/g_i$: the integrand $x^{\mathbf{m}-\mathbf{1}}\prod_{i'}L_{i'}^{-w_{i'}}$ times $\varphi(h)=L_i^{w_i}e^{-L_i}$ is $x^{\mathbf{m}-\mathbf{1}}e^{-L_i}\prod_{i'\neq i}L_{i'}^{-w_{i'}}$, which gives the first identity. Every letter being positive on the orthant and every $w_{i'}>0$, $L_{i'}(x)^{-w_{i'}}=\Gamma(w_{i'})^{-1}\int_0^\infty\lambda^{w_{i'}-1}e^{-\lambda L_{i'}(x)}d\lambda$, and Tonelli's theorem gives the second identity, the integral over x being $\mathsf{G}_i(\mathbf{m};\boldsymbol{\lambda})$, the Gramma of the concatenated configuration (Definition 2.1). The derivative $\partial^k_{\lambda_{i'}}\mathsf{G}_i$ is $(-1)^k$ times the integral with the further factor $L_{i'}^k$, which gives the complete monotonicity. By Proposition 8.12(iii), $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ is the moment generating function, in the exponents, of $\big(-\log K_j(U),-\log\Psi_j(U)\big)_j$ for $U\sim\mathrm{Dir}_{\mathbf{m}}$, and by Hölder's inequality a moment generating function is log-convex on the convex set where it is finite; $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ does not depend on the exponents. □

48.27 Chapter 27: finite interfaces and certified enclosures

Definition 27.1 (The channel law; moment vectors). As stated; cells with the same hinges on the same marks are kept together by reading their channels jointly.

Proposition 27.2 (The channel law composes). *As stated, for cells kept together and for every cell family.*

Proof. By Theorem 3.12 with the blocks B_1 and B_2, $U\sim\mathrm{Dir}_{(\mathbf{m}_1,\mathbf{m}_2)}$ on $\Delta_{B_1\sqcup B_2}$ is $U|_{B_l}=V_lU_l$ with $(V_1,V_2)\sim\mathrm{Dir}_{(m_{B_1},m_{B_2})}$ and $U_l\sim\mathrm{Dir}_{\mathbf{m}_l}$, independent. In every table j the channels of the spliced cell are the bilinear map of Theorem 21.14(iii) applied to the channels of the two cells of table j, and a channel c has $c(V_lU_l)=V_l^{\deg c}c(U_l)$. So $\mathrm{ch}(\mathcal{A}_{12})(U)$ is the image of the replaced channel vectors under the bilinear map applied table by table, and its law is the channel law of $\mathcal{A}_{12}$. For the tables of a family kept together without hinges the channels are the letters K_j and Ψ_j, and Proposition 8.12(i) is the stated reading.

For the cell family F, by Definitions 23.1 and 21.1 at the point $\mathrm{ch}(\mathcal{A})(u)$, $\gamma^K_{\mathcal{A}_j}=x_{j,0}/x_{j,1}$ and $\gamma^\Psi_{\mathcal{A}_j}=x^*_{j,1}/x^*_{j,0}$, so $\gamma^K_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}=x_{j,0}x_{1,1}/(x_{j,1}x_{1,0})$ and $\gamma^\Psi_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}=x^*_{j,1}x_{1,1}/(x^*_{j,0}x_{1,0})$: $\boldsymbol{\varrho}_F=R_F\circ\mathrm{ch}(\mathcal{A})$. The base form is $\phi_F(du)=\varpi(\mathrm{ch}(\mathcal{A})(u))\,u^{\mathbf{m}-\mathbf{1}}d\omega_B(u)$ with $\varpi=\prod_j x_{j,1}^{-s_j}x_{j,0}^{*-t_j}(x_{1,0}/x_{1,1})^{-\kappa_F}$, the factor $\gamma^K_{\mathcal{A}_1}(u)^{-\kappa_F}$ being $(x_{1,0}/x_{1,1})^{-\kappa_F}$, and $u^{\mathbf{m}-\mathbf{1}}d\omega_B=\Gamma_1^{|B|}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}\mathrm{Dir}_{\mathbf{m}}(du)$ (Definition 3.10). For a positive measure μ and positive functions, $(R\circ c)_*\big((\varpi\circ c)\,\mu\big)=R_*\big(\varpi\,c_*\mu\big)$, both sides giving $\int H(R(c(u)))\,\varpi(c(u))\,d\mu(u)$ against every Borel $H\geq0$. With $c=\mathrm{ch}(\mathcal{A})$ and $\mu=\Gamma_1^{|B|}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}\mathrm{Dir}_{\mathbf{m}}$ this is the display. □

Theorem 27.3 (Gamma draws close the moments). *As stated, for cells kept together.*

Proof. The reference, proof of thm:gammalift, uses of the channel vector only that it is a vector of polynomials on the marks, homogeneous of known degrees, that the Gamma draws on disjoint sets of marks are independent, and that a splicing acts on the channel vectors by a bilinear map

whose every product has the degree of the channel it gives. For cells kept together the channel vector is the joint vector $(\mathrm{ch}(\mathcal{A}_j))_j$, and a splicing acts table by table, so on the joint vectors by the direct sum over the tables of the maps of Theorem 21.14(iii), again bilinear with that property. So (i), (iii) and (iv) hold word for word. In (ii) the joint channels of $\mathcal{A}_{12}$ at α are $\mathrm{sp}\big(\mathrm{ch}(\mathcal{A}_1)(\alpha|_{B_1}), \mathrm{ch}(\mathcal{A}_2)(\alpha|_{B_2})\big)$, each a sum of products of one joint channel of each cell, so a monomial of degree $|\mathbf{k}|$ in them expands into monomials of degree $|\mathbf{k}|$ in the joint channels of each cell, and the independence of $\alpha|_{B_1}$ and $\alpha|_{B_2}$ gives the formula. In (v) the joint channels are bounded on Δ_B, so by (iii) their moments determine their law, and Proposition 27.2 composes it. □

Theorem 27.9 (Taylor coefficients in the weights are periods with walls at least as high). *As stated, at real exponents of any sign.*

Proof. For $u \in \Delta_{\mathcal{E}}$ put $y_i(u) = \sum_j \zeta_j D_{ij}(u)/L_i(u)$. Write $L_i = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} u^{\boldsymbol{\gamma}}$ with $c_{i,\boldsymbol{\gamma}} > 0$; D_{ij} has its support in that of L_i, so $D_{ij} = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} v_{i,\boldsymbol{\gamma},j} u^{\boldsymbol{\gamma}}$ with real $v_{i,\boldsymbol{\gamma},j}$, and with $V_{ij} = \max_{\boldsymbol{\gamma}} |v_{i,\boldsymbol{\gamma},j}|$, $|D_{ij}| \le V_{ij} L_i$ on the orthant and $|y_i| \le r_i(\boldsymbol{\zeta}) := \sum_j |\zeta_j| V_{ij}$; let every $r_i(\boldsymbol{\zeta}) < 1$. For real w and $|y| < 1$, $(1+y)^{-w} = \sum_{l \ge 0} (-1)^l \frac{(w)_l}{l!} y^l$, and $|(w)_l| \le (|w|)_l$ since $|w+k| \le |w| + k$, so the series of the moduli is at most $\sum_l \frac{(|w|)_l}{l!} |y|^l = (1-|y|)^{-|w|}$. With the multinomial expansion of $y_i^{l_i}$,

$$L_i(\boldsymbol{\zeta})^{-w_i} = L_i^{-w_i}(1+y_i)^{-w_i} = \sum_{(l_{ij})_j} \frac{(-1)^{l_i}(w_i)_{l_i}}{\prod_j l_{ij}!}\, L_i^{-w_i - l_i} \prod_j \big(\zeta_j D_{ij}\big)^{l_{ij}}, \qquad l_i = \sum_j l_{ij},$$

and the product over i of the series of the moduli is at most $\prod_i L_i^{-w_i}(1 - r_i(\boldsymbol{\zeta}))^{-|w_i|}$. Against $u^{\mathbf{m}-\mathbf{1}} d\omega_{\mathcal{E}}$ this bound is integrable, $P(\mathbf{0})$ converging at the real data; so termwise integration is justified, P is a convergent power series on $\{r_i(\boldsymbol{\zeta}) < 1 \ \forall i\}$, and its coefficients are the stated ones. After the numerators are expanded every term is a coefficient times $u^{\mathbf{m}+\boldsymbol{\gamma}-\mathbf{1}} \prod_i L_i^{-w_i - l_i}$, $\boldsymbol{\gamma}$ a sum, over i, of l_i exponents of the support of L_i. So $|\boldsymbol{\gamma}| = \sum_i l_i g_i$ and the term is balanced, and $\gamma_S \ge \sum_i l_i d_i(S)$ with $d_i(S)$ the least value of $\langle \mathbf{1}_S, \cdot \rangle$ on the support of L_i, which is the count $d_j(S)$ or $d_j^*(S)$ of the letter in the walls (Definition 8.1). So $\kappa'_S = \kappa_S + \gamma_S - \sum_i l_i d_i(S) \ge \kappa_S > 0$, and the term converges by Theorem 8.7, whose criterion holds for exponents of any sign. □

Theorem 27.10 (A uniform certificate for weights). *As stated, at real exponents of any sign.*

Proof. Let $|\zeta_j| \le R_j$, so that $|y_i| \le r_i < 1$ in the notation of the proof of Theorem 27.9. For real w_i, $|(1+y_i)^{-w_i}| = |1+y_i|^{-w_i}$, the real part of $1+y_i$ being positive, and $1 - r_i \le |1+y_i| \le 1 + r_i$, so $|L_i(\boldsymbol{\zeta})^{-w_i}| \le L_i^{-w_i} \max\big((1-r_i)^{-w_i}, (1+r_i)^{-w_i}\big)$; integrating, $|P(\boldsymbol{\zeta})| \le M$. The radii can be enlarged a little with every $r_i < 1$ kept, so P is holomorphic near the closed polydisc by Theorem 27.9, and the Cauchy estimates give $|[\boldsymbol{\zeta}^{\mathbf{k}}]P| \le M \mathbf{R}^{-\mathbf{k}}$; the bounds on the Taylor remainder follow from these as in the reference, proof of thm:weightbank, there being $\binom{k+p-1}{p-1}$ multi-indices of degree k. For the explicit bound on $P(\mathbf{0})$: by Theorem 8.5(ii) and Theorem 8.7, off the balance wall and then on it by continuity, $P(\mathbf{0}) = \sum_\pi \int_{[0,1]^{n-1}} \prod_k q_k^{\kappa_{U_{\pi,k}}^{-1}} \prod_i L_{i,\pi}(q)^{-w_i}\, dq$. Each sector letter has nonnegative coefficients and is increasing in every q_k, so $L_{i,\pi}(\mathbf{0}) \le L_{i,\pi}(q) \le L_{i,\pi}(\mathbf{1})$ and $L_{i,\pi}(q)^{-w_i} \le \max\big(L_{i,\pi}(\mathbf{0})^{-w_i}, L_{i,\pi}(\mathbf{1})^{-w_i}\big)$ for every real w_i, and $\int_{[0,1]^{n-1}} \prod_k q_k^{\kappa_{U_{\pi,k}}^{-1}} dq = \prod_k \kappa_{U_{\pi,k}}^{-1}$. The enclosure of Proposition 15.12 holds for real data of any sign, its error bound being written with $\sum_j (|s_j| + |t_j|)$, so its upper end bounds $\prod_j K_j(\mathbf{1})^{s_j+t_j} P(\mathbf{0})$ from above. □

48.28 Chapter 28: bunches of any width

Definition 28.1 (Seam). As stated: the pair constant does not depend on the choice.

Proof. Let $p \in C$ and $X \in D$ with $p \sqcup X \in \mathcal{B}$. The child at p carries the weight $\theta_{p\sqcup X}$ on its basis X (12.2) and is w_p times the child of the station C, so $\theta_{p\sqcup X} = w_p\,\theta_{(C),X}$; the room child at X carries $\theta_{p\sqcup X}$ on its basis p and is w'_X times the room child of the station D, so $\theta_{p\sqcup X} = w'_X\,\theta_{(D),p}$. Hence $\theta_{p\sqcup X}/(w_p w'_X)$ is $\theta_{(C),X}/w'_X$, which does not depend on p, and $\theta_{(D),p}/w_p$, which does not depend on X. The children in a station are positive multiples of one another and so have the same bases: $p' \sqcup X \in \mathcal{B}$ for every $p' \in C$ and $p \sqcup X' \in \mathcal{B}$ for every $X' \in D$. Changing p and then X one at a time, the quotient takes one value at all pairs of $C \times D$. □

Proposition 28.2 (The seam identity). *As stated; in (v) the seam matrix of an extreme excess is a single entry exactly when the weights of that level factor.*

Proof. Parts (i)–(iv) and (vi), and in (v) the room children and the seam letters at the two extremes, are the reference, proof of prop:seamidentity. Let $i \in \{0, c(B)\}$. By 12.19 every pattern of excess i and every room side of excess i make a basis, so every station of patterns of excess i is compatible with every station of room sides of excess i.

Let the weights of the level factor along these products, $\theta_{p\sqcup X} = \theta'_p\,\theta''_X$ for the patterns p and the room sides X of excess i, with positive θ' and θ''. The child at a pattern p of excess i is the table on R whose bases are all the room sides of excess i, with the weights $\theta'_p\,\theta''_X$: it is θ'_p times one table. So the patterns of excess i form one station and H_i has one row. The columns of a matrix with one row are proportional, and by (iv) no two stations of $\mathcal{D}_i$ have proportional columns; so H_i has one column, and its entry is positive, the pair being compatible.

Conversely let H_i be a single entry h_{CD}. The patterns of excess i then form the one station C and its room sides the one station D, and (ii) gives $\theta_{p\sqcup X} = h_{CD}\,w_p\,w'_X$ for every such p and X: the weights of the level factor along these products, with the factors $h_{CD}\,w_p$ and w'_X. □

Proposition 28.3 (The width ladder). *As stated; in (i) and (ii) the seam is one rectangle at each excess exactly when the weights factor across B and R, and the closing sentence holds at every width.*

Proof. (i) That $c(B) = 0$ exactly when B is a union of molecules, with one excess, every pattern meeting every room side, and the letters $K = \mathsf{x}_0^{\mathsf{T}} H_0\,\mathsf{y}_0$ and $\Psi = \mathsf{x}_0^{*\mathsf{T}} H_0\,\mathsf{y}_0^*$, is the reference, proof of prop:ladder(i). The one excess is extreme, so by 28.2(v) the seam is one rectangle exactly when the weights of its level factor along the products $p \sqcup X$. Every basis lies in this level, so this is a factoring $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$ across B and R, and conversely. With one station C and one station D, 28.2(iii) reads $K = h_{CD}\,\mathsf{x}_C\,\mathsf{y}_D$ and $\Psi = h_{CD}\,\mathsf{x}_C^*\,\mathsf{y}_D^*$.

(ii) That $c(B) = 1$ exactly when there are two excesses, both extreme, is the reference, proof of prop:ladder(ii). By 28.2(v) the excess i is one rectangle exactly when the weights of its level factor, $\theta_{p\sqcup X} = \theta'_{i,p}\,\theta''_{i,X}$ for the patterns p and the room sides X of excess i. If both levels factor, put $\theta'_p = \theta'_{i(p),p}$ and $\theta''_X = \theta''_{i(X),X}$, each pattern and each room side having one excess; the pattern and the room side of a basis have one excess, so $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$ for every basis T. Conversely a factoring across B and R restricts to a factoring of each level. The seam letters, the room station letters and the room-station factorization are the reference's (proof of prop:ladder(ii), and exercise ex:widthone).

(iii) and the closing sentence. A matrix with one row has proportional columns, and a matrix with one column proportional rows; so by 28.2(iv) an excess with one station on one side has one

station on the other, at every width. The criterion for level form is 14.12, which does not restrict the width. In level form the children at the patterns of one excess are equal, so these patterns form one station; the excess then has one station on each side and is one rectangle. That an excess $0 < i < c(B)$ can carry several rectangles is the reference's (prop:ladder(iii) and exercise ex:prismseam). □

Proposition 28.6 (Symmetries of a seam). *As stated; in (iii) an exchanging bijection is read at every lean, and it carries the law of B at the lean $\boldsymbol{\delta}$ to its law at $-\boldsymbol{\delta}$.*

Proof. Parts (i) and (ii), the first sentence of (iii), the carrying bijections and the sentence on the potential are the reference, proof of prop:seamsym.

Let g keep B and the masses and exchange the family, and let $j \mapsto j'$ be the bijection of its tables of 11.13. The pairs are kept, $(\sigma_{j'}, \delta_{j'}) = (\sigma_j, \delta_j)$, so at the lean $-\boldsymbol{\delta}$ table j' carries the exponents $(t_{j'}, s_{j'}) = (t_j, s_j)$. Read every function of α_B at $g^*\alpha$. This permutes the coordinates of Δ_B and keeps $u^{\mathbf{m}_B - \mathbf{1}} d\omega_B$, since g keeps B and the masses. By 14.10, $c_{j'}(B) = c_j(B)$, $d_{j'}(B) = |B| - r_j(B) = d_j^*(B)$ and $d_{j'}^*(B) = d_j(B)$.

(a) By (ii) the seam letter V_{jD} of K_j at the excess $i(D)$ is, at $g^*\alpha$, the seam letter $V_{j'D'}^*$ of $\Psi_{j'}$ at the excess $c_j - i(D)$, and V_{jD}^* is $V_{j'D'}$. Summing over the stations of one excess, $V_j^{(i)} = V_{j'}^{*(c_j - i)}$ and $V_j^{*(i)} = V_{j'}^{(c_j - i)}$.

(b) K_j is active at $\boldsymbol{\delta}$ exactly when $s_j > 0$, that is exactly when $\Psi_{j'}$ is active at $-\boldsymbol{\delta}$; likewise Ψ_j and $K_{j'}$. The sum $\kappa_B + \kappa_B^{\mathrm{top}} = \sum_j (s_j + t_j) c_j$ is the same at the two leans.

(c) By (a) and (b) the scale of the piece at $\boldsymbol{\delta}$ (28.5) is

$$\prod_j \Big(\frac{V_{j'}^{*(0)}}{V_{j'}^{*(c_j)}} \Big)^{s_j/(\kappa_B + \kappa_B^{\mathrm{top}})} \Big(\frac{V_{j'}^{(c_j)}}{V_{j'}^{(0)}} \Big)^{t_j/(\kappa_B + \kappa_B^{\mathrm{top}})},$$

which is the scale at $-\boldsymbol{\delta}$, where $\Psi_{j'}$ carries the exponent s_j and $K_{j'}$ the exponent t_j.

(d) The wall at $-\boldsymbol{\delta}$ is $m_B - \sum_j \big(t_j d_{j'}(B) + s_j d_{j'}^*(B) \big) = m_B - \sum_j \big(s_j d_j(B) + t_j d_j^*(B) \big) = \kappa_B$.

(e) By (a), (c) and (d) the scale form at $\boldsymbol{\delta}$,

$$u^{\mathbf{m}_B - \mathbf{1}} \prod_j (V_j^{(0)})^{-s_j} (V_j^{*(c_j)})^{-t_j} (\gamma_B^\rho)^{-\kappa_B} d\omega_B,$$

is the scale form at $-\boldsymbol{\delta}$.

(f) For a station D of excess $i(D)$ and its image D' of excess $c_j - i(D)$,

$$\varrho_{jD}^\rho = \frac{V_{jD}}{V_j^{(0)} \, (\gamma_B^\rho)^{i(D)}} = \frac{V_{j'D'}^*}{V_{j'}^{*(c_j)} \, (\gamma_B^\rho)^{c_j - i(D')}},$$

the coordinate $\varrho_{j'D'}^{\rho *}$ at $-\boldsymbol{\delta}$; likewise $\varrho_{jD}^{\rho *}$ is the coordinate $\varrho_{j'D'}^\rho$ at $-\boldsymbol{\delta}$.

So the seam vector at $\boldsymbol{\delta}$, followed by the map of the coordinates, is the seam vector at $-\boldsymbol{\delta}$ read at $g^*\alpha$, and by (e) the scale forms correspond. The image of the law at $\boldsymbol{\delta}$ is therefore the law at $-\boldsymbol{\delta}$. By (a) and (b) the relations defining the seam torus at $\boldsymbol{\delta}$ go over into those at $-\boldsymbol{\delta}$. At the centre $\boldsymbol{\delta} = -\boldsymbol{\delta}$, and the law is kept. □

Lemma 28.8 (Absorbing a piece of any width). *As stated, with functions f_l homogeneous of degree one in their second argument.*

Proof. Take the section $h = \gamma_B^\rho(\alpha_B) + |\alpha_Z|$, a positive function of degree one on $(0,\infty)^{B\sqcup Z}$, and write $\alpha_B = \lambda u$ with $\lambda > 0$ and $u \in \Delta_B$ (2.7).

- *The seam letters along the scale.* Let f_l be assigned to K_j. By 28.5, $V_{jD} = V_j^{(0)}\,(\gamma_B^\rho)^{i(D)}\varrho_{jD}^\rho$ on Δ_B, and V_{jD} is homogeneous of degree $d_j + i(D)$. Hence

$$V_{jD}(\lambda u) = \lambda^{d_j} V_j^{(0)}(u)\,\big(\lambda\gamma_B^\rho(u)\big)^{i(D)}\varrho_{jD}^\rho(u),$$
$$\text{that is}\quad \mathbf{V}_l(\lambda u) = V_l(\lambda u)\,\mathbf{x}_l(\alpha_g),$$

  at $\alpha_g = \lambda\gamma_B^\rho(u)$ and $\mathbf{x} = \boldsymbol{\varrho}_B^\rho(u)$, with $V_l(\lambda u) = \lambda^{d_j} V_j^{(0)}(u) > 0$. For Ψ_j the same holds with $V_{jD}^* = V_j^{*(c_j)}(\gamma_B^\rho)^{c_j - i(D)}\varrho_{jD}^{\rho *}$, of degree $d_j^* + c_j - i(D)$, and $V_l(\lambda u) = \lambda^{d_j^*} V_j^{*(c_j)}(u)$.

- *Homogeneity.* Since f_l is homogeneous of degree one in its second argument,

$$f_l\big(\alpha_Z, \mathbf{V}_l(\lambda u)\big) = V_l(\lambda u)\, f_l\big(\alpha_Z, \mathbf{x}_l(\alpha_g)\big).$$

  So the arguments of H are the $f_l(\alpha_Z, \mathbf{x}_l(\alpha_g))$. The exponents assigned to each active letter add to its exponent, so

$$\prod_l f_l\big(\alpha_Z, \mathbf{V}_l(\lambda u)\big)^{-a_l} = \lambda^{-\sum_j (s_j d_j + t_j d_j^*)} \prod_j V_j^{(0)}(u)^{-s_j} V_j^{*(c_j)}(u)^{-t_j} \prod_l f_l\big(\alpha_Z, \mathbf{x}_l(\alpha_g)\big)^{-a_l}.$$

- *The scale and the stand-in.* With $\alpha_B^{\mathbf{m}_B - \mathbf{1}} d\alpha_B = \lambda^{m_B - 1} u^{\mathbf{m}_B - \mathbf{1}}\, d\lambda\, d\omega_B(u)$, the power of λ is $\lambda^{\kappa_B - 1}$. At fixed u substitute $\alpha_g = \lambda\gamma_B^\rho(u)$, so that $\lambda^{\kappa_B - 1} d\lambda = \gamma_B^\rho(u)^{-\kappa_B}\alpha_g^{\kappa_B - 1} d\alpha_g$. In the coordinates (u, α_g, α_Z) the form ω is then $\phi_B^\rho(du)$ times $\omega_{\boldsymbol{\varrho}_B^\rho(u)}$ (28.5), and $h = \alpha_g + |\alpha_Z|$.

- *The integral.* At fixed u the map $(\lambda, \alpha_Z) \mapsto (\alpha_g, \alpha_Z)$ intertwines the scalings of the two orthants, so $\omega_{\boldsymbol{\varrho}_B^\rho(u)}$ is homogeneous of degree zero together with ω. By 16.2, the integral of a form homogeneous of degree zero over a section does not depend on the section. Tonelli's theorem, applied to the nonnegative integrand, then gives

$$\int_{\{h=1\}} \omega = \int_{\Delta_B} \phi_B^\rho(du) \int_{\{\alpha_g + |\alpha_Z| = 1\}} \omega_{\boldsymbol{\varrho}_B^\rho(u)} = \int \Big(\int_{\{h'=1\}} \omega_{\mathbf{x}} \Big)\, d\big((\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho\big)(\mathbf{x}).$$

This is the reference's proof (proof of lem:absorbwidth). There, linearity of the f_l enters only through the identity $f_l(\alpha_Z, \mathbf{V}_l(\lambda u)) = V_l(\lambda u)\, f_l(\alpha_Z, \mathbf{x}_l(\alpha_g))$, which homogeneity of degree one gives. □

Remark. At width one with factoring weights, normalized as in 28.11, the seam letters of K_j are $(K_{\mathcal{B}_j/R}, K_{\mathcal{B}_j|B})$ and those of Ψ_j are $(\Psi_{\mathcal{B}_j/R}, \Psi_{\mathcal{B}_j|B})$ (Lemma S13). The positive continuous functions homogeneous of degree one in the second argument are, for K_j, $f_l(\alpha_Z, (y_0, y_1)) = y_0\, f(\alpha_Z, y_1/y_0)$, and for Ψ_j, $f_l(\alpha_Z, (y_0, y_1)) = y_1\, f(\alpha_Z, y_0/y_1)$, with f positive and continuous on $(0,\infty)^Z \times (0,\infty)$. At the seam letters they are $K_{\mathcal{B}_j/R}\, f(\alpha_Z, \gamma_{j,B}^K)$ and $\Psi_{\mathcal{B}_j|B}\, f(\alpha_Z, \gamma_{j,B}^\Psi)$. These are the factors $L_i\, f_i(\alpha_Z, \gamma_i)$ of 23.2 for the cell family F_B^+, with arbitrary f_i. Linear functions f_l give only the f_i that are affine in the hinge value; homogeneity of degree one gives all of them.

Lemma S13. *Let a bunch B have width one in table j of a family, the weights factoring across B and R, $\theta_{j,T} = \theta'_{j,T\cap B}\,\theta''_{j,T\cap R}$, with the stations D_0 and D_1 of the excesses 0 and 1 (28.3(ii)). In this statement write $\hat\theta_i$ for the largest value of θ''_j on the room sides of excess i, and read $\mathcal{B}_j/R$ and $\mathcal{B}_j|B$ with the weights θ'_j and $\mathcal{B}_j|R$ and $\mathcal{B}_j/B$ with the weights θ''_j.*

(i) $K_{j,(D_0)} = \hat\theta_0 K_{\mathcal{B}_j/R}$, $\Psi_{j,(D_0)} = \hat\theta_0\Psi_{\mathcal{B}_j/R}$, $K_{j,(D_1)} = \hat\theta_1 K_{\mathcal{B}_j|B}$ *and* $\Psi_{j,(D_1)} = \hat\theta_1\Psi_{\mathcal{B}_j|B}$; *and* $\mathsf{y}_{jD_0} = K_{\mathcal{B}_j|R}/\hat\theta_0$, $\mathsf{y}^*_{jD_0} = \Psi_{\mathcal{B}_j|R}/\hat\theta_0$, $\mathsf{y}_{jD_1} = K_{\mathcal{B}_j/B}/\hat\theta_1$ *and* $\mathsf{y}^*_{jD_1} = \Psi_{\mathcal{B}_j/B}/\hat\theta_1$.

(ii) For $i = 0, 1$, divide θ''_j by $\hat\theta_i$ on the room sides of excess i and multiply θ'_j by $\hat\theta_i$ on the patterns of excess i. This gives a factorization of the same weights with $\hat\theta_0 = \hat\theta_1 = 1$, in which θ''_j is the system of the multipliers of the room sides (28.1). The seam letters and the room station letters do not depend on the factorization. In the normalized factorization

$$\gamma^K_{j,B} = \frac{K_{j,(D_1)}}{K_{j,(D_0)}}, \qquad \gamma^\Psi_{j,B} = \frac{\Psi_{j,(D_0)}}{\Psi_{j,(D_1)}}, \qquad \gamma^K_{j,R} = \frac{\mathsf{y}_{jD_0}}{\mathsf{y}_{jD_1}}, \qquad \gamma^\Psi_{j,R} = \frac{\mathsf{y}^*_{jD_1}}{\mathsf{y}^*_{jD_0}}.$$

In a factorization with the constants $\hat\theta_0$ and $\hat\theta_1$, the four ratios on the right are $(\hat\theta_1/\hat\theta_0)\gamma^K_{j,B}$, $(\hat\theta_0/\hat\theta_1)\gamma^\Psi_{j,B}$, $(\hat\theta_1/\hat\theta_0)\gamma^K_{j,R}$ and $(\hat\theta_0/\hat\theta_1)\gamma^\Psi_{j,R}$.

Proof. (i) By 19.4 the bases of table j are the $X \sqcup Y$ with $X \in \mathcal{B}_j|B$ and $Y \in \mathcal{B}_j/B$, at the level $r_j(B)$, and with $X \in \mathcal{B}_j/R$ and $Y \in \mathcal{B}_j|R$, at the level $d_j(B)$.

- So the patterns and the room sides of excess 0 are the bases of $\mathcal{B}_j/R$ and of $\mathcal{B}_j|R$, those of excess 1 the bases of $\mathcal{B}_j|B$ and of $\mathcal{B}_j/B$, and every pattern meets every room side of its excess.
- The room child at a room side X of excess 0 (28.1) is the table on B with the bases of $\mathcal{B}_j/R$ and the weights $\theta'_{j,p}\,\theta''_{j,X}$, that is $\theta''_{j,X}$ times $\mathcal{B}_j/R$ with the weights θ'_j.
- These room children are positive multiples of one another, so the room sides of excess 0 form the one station D_0. Its multipliers are $w'_X = \theta''_{j,X}/\hat\theta_0$, the largest equal to 1, and the room child of the station is $\hat\theta_0$ times $\mathcal{B}_j/R$ with θ'_j.
- Hence $K_{j,(D_0)} = \hat\theta_0 K_{\mathcal{B}_j/R}$ and $\Psi_{j,(D_0)} = \hat\theta_0\Psi_{\mathcal{B}_j/R}$, and the station letters are $\mathsf{y}_{jD_0} = \sum_X w'_X\alpha^X_R = K_{\mathcal{B}_j|R}/\hat\theta_0$ and $\mathsf{y}^*_{jD_0} = \sum_X w'_X\alpha^{R\setminus X}_R = \Psi_{\mathcal{B}_j|R}/\hat\theta_0$.
- The excess 1 is the same with $\mathcal{B}_j|B$, $\mathcal{B}_j/B$ and $\hat\theta_1$.

(ii) A basis has its pattern and its room side at one excess i, so the new weights multiply to $(\hat\theta_i\,\theta'_{j,p})(\theta''_{j,X}/\hat\theta_i) = \theta_{j,p\sqcup X}$. The new θ''_j has the largest value 1 on each excess, so it is the system of the multipliers w'_X. The stations, the multipliers, the seam letters and the station letters are defined from the weights θ_j alone (28.1), so they do not change. In the normalized factorization, (i) holds with $\hat\theta_0 = \hat\theta_1 = 1$, and the four ratios are the hinge values of 19.2, read with the minors carrying the normalized weights. In a factorization with the constants $\hat\theta_0$ and $\hat\theta_1$, (i) gives the four ratios as stated. □

Proposition 28.11 (Width one is the hinge). *As stated: for a family, with θ''_j normalized on each excess, and with part (iv) for the family.*

Proof. The letters. With θ''_j of largest value 1 on the room sides of each excess, Lemma S13 gives the seam letters of the piece of the bunch and the room letters of its room, read as its partner (the paragraph after 28.4):

$$V_{jD_0} = K_{\mathcal{B}_j/R}, \quad V_{jD_1} = K_{\mathcal{B}_j|B}, \quad V^*_{jD_0} = \Psi_{\mathcal{B}_j/R}, \quad V^*_{jD_1} = \Psi_{\mathcal{B}_j|B},$$

$$L_{jD_0} = K_{\mathcal{B}_j|R}, \quad L_{jD_1} = K_{\mathcal{B}_j/B}, \quad L^*_{jD_0} = \Psi_{\mathcal{B}_j|R}, \quad L^*_{jD_1} = \Psi_{\mathcal{B}_j/B},$$

the minors on B carrying the weights θ'_j and those on R the weights θ''_j. The hinge values of 19.2 are $\gamma^K_{j,B} = V_{jD_1}/V_{jD_0}$, $\gamma^\Psi_{j,B} = V^*_{jD_0}/V^*_{jD_1}$, $\gamma^K_{j,R} = L_{jD_0}/L_{jD_1}$ and $\gamma^\Psi_{j,R} = L^*_{jD_1}/L^*_{jD_0}$. Every letter is active and $c_j = 1$, so $\kappa_B + \kappa_B^{\rm top} = \sum_j (s_j + t_j) = 2\sigma$, and on the balance wall of the glue $\kappa_B + \kappa_R = 2\sigma$.

The cells.

- The cell of the bunch $\mathcal{B}^+_{j,B}$ has the bases of $\mathcal{B}_j|B$ avoiding g and the $X + g$, $X \in \mathcal{B}_j/R$, through g, with the weights θ'_j (19.4). So $\mathcal{B}^+_{j,B} \setminus g = \mathcal{B}_j|B$ and $\mathcal{B}^+_{j,B}/g = \mathcal{B}_j/R$, its hinge values are $\gamma^K_{j,B}$ and $\gamma^\Psi_{j,B}$ (21.1), and its rank is $d_j(B) + 1$.
- Since $d^*_j(B) = |B| - d_j(B) - 1$, the hinge wall of F^+_B (23.1) is $m_B - \sum_j \big(s_j d_j(B) + t_j(|B| - d_j(B) - 1)\big) = \kappa_B$.
- Likewise $\mathcal{B}^+_{j,R} \setminus g = \mathcal{B}_j|R$ and $\mathcal{B}^+_{j,R}/g = \mathcal{B}_j/B$, with the weights θ''_j and the hinge values $\gamma^K_{j,R}$ and $\gamma^\Psi_{j,R}$.
- The hinge wall of F^+_R is $m_R - \sum_j \big(s_j(r(\mathcal{B}_j|R) - 1) + t_j(|R| - r(\mathcal{B}_j|R))\big)$. This is the wall κ_R of the partner, whose letters at D_0 have the degrees $r(\mathcal{B}_j|R) = e_j + 1$ and $|R| - r(\mathcal{B}_j|R) = e^*_j$ (28.4).

(i) By 28.5,

$$\gamma^\rho_B = \prod_j \Big(\frac{V_{jD_1}}{V_{jD_0}}\Big)^{s_j/(2\sigma)} \Big(\frac{V^*_{jD_0}}{V^*_{jD_1}}\Big)^{t_j/(2\sigma)} = \prod_j (\gamma^K_{j,B})^{s_j/(2\sigma)} (\gamma^\Psi_{j,B})^{t_j/(2\sigma)}.$$

As $\sum_j (s_j + t_j)/(2\sigma) = 1$, this is $\gamma^K_{1,B} \prod_j (\gamma^K_{j,B}/\gamma^K_{1,B})^{s_j/(2\sigma)} (\gamma^\Psi_{j,B}/\gamma^K_{1,B})^{t_j/(2\sigma)} = \gamma^K_{1,B}\, \psi(\boldsymbol{\varrho}_{F^+_B})$, the ratio vector of F^+_B being $(\gamma^K_{j,B}/\gamma^K_{1,B}, \gamma^\Psi_{j,B}/\gamma^K_{1,B})_j$ (23.1). The scale form is

$$\phi^\rho_B(du) = u^{\mathbf{m}_B - \mathbf{1}} \prod_j K^{-s_j}_{\mathcal{B}_j/R} \Psi^{-t_j}_{\mathcal{B}_j|B} \big(\gamma^K_{1,B}\psi\big)^{-\kappa_B} d\omega_B(u) = \psi^{-\kappa_B} \phi_{F^+_B}(du),$$

by the base form of 23.1, whose letters are $K_{\mathcal{B}^+_{j,B}/g} = K_{\mathcal{B}_j/R}$ and $\Psi_{\mathcal{B}^+_{j,B}\setminus g} = \Psi_{\mathcal{B}_j|B}$, with the hinge wall κ_B.

(ii) By 28.5, $\varrho^\rho_{jD_0} = V_{jD_0}/V^{(0)}_j = 1$ and $\varrho^{\rho*}_{jD_1} = V^*_{jD_1}/V^{*(1)}_j = 1$. Further $\varrho^\rho_{jD_1} = V_{jD_1}/(V_{jD_0}\gamma^\rho_B) = \gamma^K_{j,B}/(\gamma^K_{1,B}\psi) = x_{j,K}/\psi(\mathbf{x})$ and $\varrho^{\rho*}_{jD_0} = V^*_{jD_0}/(V^*_{jD_1}\gamma^\rho_B) = x_{j,\Psi}/\psi(\mathbf{x})$, at $\mathbf{x} = \boldsymbol{\varrho}_{F^+_B}$. Write $\Phi(\mathbf{x}) = \big(x_{j,K}/\psi(\mathbf{x}),\, x_{j,\Psi}/\psi(\mathbf{x})\big)_j$.

- The seam torus is the set of the points with $x_{jD_0} = x^*_{jD_1} = 1$ and $\prod_j x^{s_j}_{jD_1} x^{*t_j}_{jD_0} = 1$. Since $\prod_j (x_{j,K}/\psi)^{s_j} (x_{j,\Psi}/\psi)^{t_j} = \psi^{2\sigma}\psi^{-2\sigma} = 1$, Φ maps $(0,\infty)^{2N-1}$ into it.
- Let a point of the torus have $x_{jD_1} = a_j$ and $x^*_{jD_0} = b_j$. The vector $\mathbf{x}$ with $x_{j,K} = a_j/a_1$ and $x_{j,\Psi} = b_j/a_1$ has $\psi(\mathbf{x}) = \big(\prod_j a^{s_j}_j b^{t_j}_j\big)^{1/(2\sigma)}/a_1 = 1/a_1$, so $\Phi(\mathbf{x})$ is the point.
- Conversely $\Phi(\mathbf{x})$ determines $\psi(\mathbf{x})$ as the inverse of its coordinate $x_{1,K}/\psi(\mathbf{x})$, and then $\mathbf{x}$. So Φ is a bijection.

As $\boldsymbol{\varrho}^\rho_B = \Phi \circ \boldsymbol{\varrho}_{F^+_B}$ and $\phi^\rho_B = \psi(\boldsymbol{\varrho}_{F^+_B})^{-\kappa_B} \phi_{F^+_B}$, the law $(\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B$ is the image under Φ of $\psi^{-\kappa_B}$ times $(\boldsymbol{\varrho}_{F^+_B})_* \phi_{F^+_B}$.

(iii) The partner with its stand-in at $\Phi(\mathbf{x})$ (28.10(i)) has the letters

$$\sum_D x_{jD}\, \alpha_g^{i(D)} L_{jD} = K_{\mathcal{B}_j|R} + \frac{x_{j,K}}{\psi(\mathbf{x})}\, \alpha_g\, K_{\mathcal{B}_j/B},$$

$$\sum_D x^*_{jD}\,\alpha_g^{1-i(D)}L^*_{jD} = \frac{x_{j,\Psi}}{\psi(\mathbf{x})}\,\alpha_g\,\Psi_{\mathcal{B}_j|R} + \Psi_{\mathcal{B}_j/B}.$$

After $\alpha_g \mapsto \psi(\mathbf{x})\,\alpha_g$ these are $K_{\mathcal{B}_j|R} + x_{j,K}\,\alpha_g K_{\mathcal{B}_j/B}$ and $x_{j,\Psi}\,\alpha_g\Psi_{\mathcal{B}_j|R} + \Psi_{\mathcal{B}_j/B}$. These are the letters of $\mathcal{B}^+_{j,R}$ twisted at g by $x_{j,K}$ in K_j and by $x_{j,\Psi}$ in Ψ_j: by 22.1, the twist multiplies the weights in K_j of the bases through g and the weights in Ψ_j of the bases avoiding g. The mass κ_B on g turns $\alpha_g^{\kappa_B-1}d\alpha_g$ into $\psi(\mathbf{x})^{\kappa_B}\alpha_g^{\kappa_B-1}d\alpha_g$, which cancels the density of (ii). The functions of 28.8, homogeneous of degree one in the vector (y_0, y_1) of the seam letters of a letter, are $y_0\,f(\alpha_Z, y_1/y_0)$ for K_j and $y_1\,f(\alpha_Z, y_0/y_1)$ for Ψ_j, with f positive and continuous. At the seam letters they are $K_{\mathcal{B}_j/R}\,f(\alpha_Z, \gamma^K_{j,B})$ and $\Psi_{\mathcal{B}_j|B}\,f(\alpha_Z, \gamma^\Psi_{j,B})$, the factors $L_i\,f_i(\alpha_Z, \gamma_i)$ of 23.2 for F^+_B. At $\mathbf{x}_l(\alpha_g)$, after the change of variable, they are $f(\alpha_Z, x_{j,K}\alpha_g)$ and $f(\alpha_Z, x_{j,\Psi}\alpha_g)$, the factors of $\omega_{\mathbf{x}}$ in 23.2. So the two lemmas are one, read through Φ.

(iv) *The room.* Read with the reversed excesses, the station D_1 has the partner excess 0 and D_0 the partner excess 1. By 28.5 for the partner:

- its scale is
$$\prod_j \Big(\frac{L_{jD_0}}{L_{jD_1}}\Big)^{s_j/(2\sigma)}\Big(\frac{L^*_{jD_1}}{L^*_{jD_0}}\Big)^{t_j/(2\sigma)} = \prod_j (\gamma^K_{j,R})^{s_j/(2\sigma)}(\gamma^\Psi_{j,R})^{t_j/(2\sigma)} = \gamma^K_{1,R}\,\psi(\boldsymbol{\varrho}_{F^+_R}),$$
as in (i);
- its scale form is $v^{\mathbf{m}_R-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{B}_j/B}\Psi^{-t_j}_{\mathcal{B}_j|R}(\gamma^K_{1,R}\psi)^{-\kappa_R}d\omega_R(v) = \psi^{-\kappa_R}\phi_{F^+_R}$, since the base form of F^+_R has the letters $K_{\mathcal{B}^+_{j,R}/g} = K_{\mathcal{B}_j/B}$ and $\Psi_{\mathcal{B}^+_{j,R}\setminus g} = \Psi_{\mathcal{B}_j|R}$ and the hinge wall κ_R;
- with γ_R its scale, its seam vector has $\varrho^\rho_{jD_1} = \varrho^{\rho*}_{jD_0} = 1$, $\varrho^\rho_{jD_0} = L_{jD_0}/(L_{jD_1}\gamma_R) = x'_{j,K}/\psi(\mathbf{x}')$ and $\varrho^{\rho*}_{jD_1} = L^*_{jD_1}/(L^*_{jD_0}\gamma_R) = x'_{j,\Psi}/\psi(\mathbf{x}')$ at $\mathbf{x}' = \boldsymbol{\varrho}_{F^+_R}$;
- its law follows as in (ii).

The coordinate ratios are $x_{j,\Psi}/x_{j,K} = \gamma^\Psi_{j,B}/\gamma^K_{j,B} = \varrho_{j,B}$ and $x'_{j,\Psi}/x'_{j,K} = \varrho_{j,R}$ (19.2).

The kernel. At $c_j = 1$ the configuration $t_{\mathcal{D}}$ has the column $\mathbf{e}_{K_j}$ at (j, D_0) and $\mathbf{e}_g + \mathbf{e}_{K_j}$ at (j, D_1) for K_j, and $\mathbf{e}_g + \mathbf{e}_{\Psi_j}$ at D_0 and $\mathbf{e}_{\Psi_j}$ at D_1 for Ψ_j. The display of 28.10(ii) gives

$$\Gamma(\kappa_B)\,\mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{p}\circ\mathbf{p}') = \int_0^\infty v^{\kappa_B-1}\prod_j \big(p_{jD_0}p'_{jD_0} + p_{jD_1}p'_{jD_1}v\big)^{-s_j}\big(p^*_{jD_0}p'^*_{jD_0}v + p^*_{jD_1}p'^*_{jD_1}\big)^{-t_j}dv,$$

and the coordinates of (ii) and of the partner give the first line of (iv). Substitute $v = (\psi(\mathbf{x})/\psi(\mathbf{x}'))\,w$. Each factor $x_{j,K}v/\psi(\mathbf{x}) + x'_{j,K}/\psi(\mathbf{x}')$ becomes $(x_{j,K}w + x'_{j,K})/\psi(\mathbf{x}')$, and likewise for Ψ_j. Since $\sum_j(s_j+t_j) - \kappa_B = \kappa_R$, the integral is

$$\Big(\frac{\psi(\mathbf{x})}{\psi(\mathbf{x}')}\Big)^{\kappa_B}\psi(\mathbf{x}')^{\sum_j(s_j+t_j)}\int_0^\infty w^{\kappa_B-1}\prod_j(x_{j,K}w + x'_{j,K})^{-s_j}(x_{j,\Psi}w + x'_{j,\Psi})^{-t_j}dw = \psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}\,P_{F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]}\big((\kappa_B,\kappa_R);\mathbf{s},\mathbf{t}\big).$$

The last step is the period of the twisted core of 23.6(iii) with the core $\binom{\{g,g'\}}{1}$ in every table, read on the section $\alpha_{g'} = 1$ (16.2); its letters are $x_{j,K}\alpha_g + x'_{j,K}\alpha_{g'}$ and $x_{j,\Psi}\alpha_g + x'_{j,\Psi}\alpha_{g'}$ (22.1).

The reductions.

- 28.10(i) is part (iii).

- 28.10(ii): insert the laws of (ii) and of the partner. The densities $\psi(\mathbf{x})^{-\kappa_B}$ and $\psi(\mathbf{x}')^{-\kappa_R}$ cancel the factor of the kernel, and the period becomes

$$\iint P_{F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]}\big((\kappa_B,\kappa_R);\mathbf{s},\mathbf{t}\big)\,d\big((\boldsymbol{\varrho}_{F_B^+})_*\phi_{F_B^+}\big)(\mathbf{x})\,d\big((\boldsymbol{\varrho}_{F_R^+})_*\phi_{F_R^+}\big)(\mathbf{x}').$$

  This is 23.6(iii) for the family whose table j is $\binom{\{g,g'\}}{1}\langle\mathcal{B}_{j,B}^+,\mathcal{B}_{j,R}^+\rangle$, which is the table $(\mathcal{B}_j,\theta_j)$ by 21.3(iii).

- 28.10(iii): on the pins the seam word of the piece is $\prod_j K_{\mathcal{B}_j/R}^{-a_{jD_0}}K_{\mathcal{B}_j|B}^{-a_{jD_1}}\Psi_{\mathcal{B}_j/R}^{-b_{jD_0}}\Psi_{\mathcal{B}_j|B}^{-b_{jD_1}}$. This is the word of the bunch family $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ of 19.13 at the level shares $(a_j^d,a_j^r,b_j^d,b_j^r)=(a_{jD_0},a_{jD_1},b_{jD_0},b_{jD_1})$, and the room word is that of $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$.
  - The seam functions on the pins are the periods of these words (28.7), and

$$\Gamma\begin{smallmatrix}2\\1\end{smallmatrix}\begin{bmatrix}a_{jD_0},\ a_{jD_1}\\ s_j\end{bmatrix}\Gamma\begin{smallmatrix}2\\1\end{smallmatrix}\begin{bmatrix}b_{jD_0},\ b_{jD_1}\\ t_j\end{bmatrix}=\Gamma\begin{smallmatrix}4\\2\end{smallmatrix}\begin{bmatrix}a_j^d,\ b_j^d,\ a_j^r,\ b_j^r\\ s_j,\ t_j\end{bmatrix}.$$

  - The scale pin $\sum_j(a_{jD_1}+b_{jD_0})=\kappa_B$ is the pin of 19.13.
  - In the notation of 4.1, eliminating the letter totals leaves the shares $(a_j^d,b_j^d)_j$ with the one pin $\sum_j(b_j^d-a_j^d)=\kappa_B-\sum_j s_j$, with the same power of 2π.
  - In the pair coordinates $\sigma_j^d=\frac12(a_j^d+b_j^d)$ and $\delta_j^d=\frac12(a_j^d-b_j^d)$, Lebesgue measure gains the factor 2^N, and the pin becomes $-2\sum_j\delta_j^d$, whose delta is $\frac12$ that of $\sum_j\delta_j^d$.

  So the fibre integral of 28.10(iii) is 2^{N-1} times the fibre integral in the pair coordinates, which is 19.13.

□

Remark. The constants match exactly under the normalization. In a factorization with the constants $\hat\theta_{0j},\hat\theta_{1j}$ of Lemma S13, the scale of the piece, its seam vector and its law are unchanged, since they are defined from the seam letters, while the hinge values move. Then (i) holds up to the constant factor $\prod_j(\hat\theta_{1j}/\hat\theta_{0j})^{\delta_j/\sigma}$, and the coordinate $\varrho^{\rho}_{jD_1}$ of (ii) is $(\hat\theta_{1j}/\hat\theta_{0j})\prod_l(\hat\theta_{1l}/\hat\theta_{0l})^{-\delta_l/\sigma}$ times $x_{j,K}/\psi(\mathbf{x})$. Normalizing θ_j'' on each excess removes these constants and costs no generality, by Lemma S13(ii). For one table, $\mathbf{x}=(1,\varrho_B)$ and $\mathbf{x}'=(1,\varrho_R)$. The family $F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]$ is then the list $\binom{\{g,g'\}}{1}$ twisted in its cobasis letter by ϱ_B at g and by ϱ_R at g', and $\psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}=\varrho_B^{\kappa_B t/(s+t)}\varrho_R^{\kappa_R t/(s+t)}$. So (iv) is the identity with 23.7, and the reductions are 23.9(i) and (iv) together with 19.13.

Proposition 28.15 (The Gramma of a set of pairs in the Foreman's order). *As stated, for a set of pairs with any number of parts.*

Proof. The configuration $\partial\mathfrak{R}$ of 28.13 has at the pair (C,D) the column $\mathbf{e}_C+\mathbf{e}_D$ on the kept stations, the station not kept in each part read as 0. A pair of $\mathfrak{R}_l$ has both its stations in $\mathcal{K}_l\sqcup\mathcal{D}_l$, so its column is supported on the kept stations of $\mathfrak{R}_l$, and different parts have different kept stations. By 2.5, applied to one part after another, the Gramma of $\mathfrak{R}$ is the product of the Grammas of its parts,

$$\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h})=\prod_{l=1}^{\nu}\mathsf{G}_{(\mathbf{a}_{\mathcal{K}_l},\mathbf{a}'_{\mathcal{D}_l});\partial\mathfrak{R}_l}(\mathbf{h}),$$

and it converges exactly when every factor does, its domain being the product of their domains. For one part the statement is the reference, proof of prop:relgrammaforeman. There the scales of $\mathcal{K}_l$ are integrated first, $\int_0^\infty x^{a_C-1}e^{-xL_C(v)}dx = \Gamma(a_C)L_C(v)^{-a_C}$. The form left on the stations of $\mathcal{D}_l$ is homogeneous of degree $|\mathbf{a}'_{\mathcal{D}_l}| - |\mathcal{D}_l| - |\mathbf{a}_{\mathcal{K}_l}| = -|\mathcal{D}_l|$, so its integral over a chart is its integral over $\Delta_{\mathcal{D}_l}$ (16.2), and the convergence criterion is 25.2. Multiplying over the parts gives the statement. □

Proposition 28.16 (Closing the Gramma of a set of pairs). *As stated: any number of parts, and (ii) at real and complex loads.*

Proof. (i) On each part the reference's proof (proof of prop:relgrammaclose(i)) applies on the simplex $\Delta_{\mathcal{D}_l}$. On its interior $0 \le \bar{L}_C < 1$ almost everywhere, and $L_C^{-a_C} = M_C^{-a_C}\sum_n \frac{(a_C)_n}{n!}\bar{L}_C^{\,n}$ with nonnegative terms. On the simplex $\bar{L}_C$ is a linear function with nonnegative coefficients, and Tonelli's theorem with 3.11 gives the series. By 28.15 the Gramma is $\prod_C \Gamma(a_C)$ times the product of these integrals over the parts.

(ii) The exchange relation of 2.4(ii) holds at every index of the domain, real or complex.

- At a kept station of $\mathcal{K}$ or of $\mathcal{D}$ it is the stated relation, the column of (C, D) being $\mathbf{e}_C + \mathbf{e}_D$ with the coefficient h_{CD}.
- At the station not kept of a part it is the sum of the relations at the stations of $\mathcal{K}$ of that part less those at its other stations of $\mathcal{D}$, the totals of a part being equal (the reference, proof of prop:relgrammaclose(ii)).
- A shifted load $(\mathbf{a} + \mathbf{e}_C, \mathbf{a}' + \mathbf{e}_{D'})$ with $(C, D') \in \mathfrak{R}$ keeps the totals of every part equal, since C and D' lie in one part, and it lies in the domain with $(\mathbf{a}, \mathbf{a}')$ (2.4).

Neither (i) nor (ii) uses one part, and (ii) does not use real loads.

(iii) concerns one part and is the reference, proof of prop:relgrammaclose(iii). □

Theorem 28.19 (The law is what a piece retains, at every width). *As stated; the plain tinkers suffice under the stated condition on the tables with both letters active.*

Proof. The equivalence of (a)–(d) is the reference, proof of thm:retainwidth. Its step (c)⇒(d) reads as follows in the notation of 4.1.

The step (c)⇒(d). Let V be the real space of the directions of the pins of the seam (28.7): the vectors $\boldsymbol{\vartheta} = (\vartheta_{jD}, \vartheta^*_{jD})_{j,D}$, one coordinate for each coordinate of the seam vector, with

$$\sum_D \vartheta_{jD} = 0 \text{ for every active } K_j, \qquad \sum_D \vartheta^*_{jD} = 0 \text{ for every active } \Psi_j,$$
$$\sum_{j,D} \big(i(D)\vartheta_{jD} + (c_j - i(D))\vartheta^*_{jD}\big) = 0.$$

For a tinker $\mathbf{v}$ of $\mathcal{Y}_0$ with the logarithms $\boldsymbol{\ell} = \log \mathbf{v}$, the seam word of $\mathcal{Y}_{0,\mathbf{v}}$ is $\mathbf{v}^{-(\mathbf{a},\mathbf{b})}$ times that of $\mathcal{Y}_0$. So, by 28.7 and by 28.10(iii) on the contour through $\mathbf{c}$,

$$e^{\langle \mathbf{c}, \boldsymbol{\ell}\rangle}\, P_{\mathcal{X}\sqcup\mathcal{Y}_{0,\mathbf{v}}} = (2\pi)^{-n_{\mathcal{D}}} \int_V e^{-i\langle \boldsymbol{\ell}, \boldsymbol{\vartheta}\rangle}\, f(\boldsymbol{\vartheta})\, d\boldsymbol{\vartheta},$$

where $f(\boldsymbol{\vartheta})$ is the product of the stacked Gammas of 28.10(iii) and of the seam functions of $\mathcal{X}$ and of $\mathcal{Y}_0$ at $\mathbf{c} + i\boldsymbol{\vartheta}$. The stacked Gammas have no zero, and f is continuous and integrable on V (the reference, proof of thm:retainwidth). The proof uses (c) only to know the Fourier transform

of f at every point of the dual of V; then f is determined. The rest is unchanged: the stacked Gammas have no zero, the seam function of $\mathcal{Y}_0$ is positive at $\mathbf{c}$, the identity theorem applies, and the projection is injective on the seam torus.

The plain tinkers. Their logarithms form the space Λ of the $\boldsymbol{\ell}$ with $\ell_{jD} = \ell^*_{jD}$ for every table j with both letters active (28.18). A table with one active letter has one coordinate for each station, and on it a plain tinker is a general one. The restrictions to V of the functionals $\boldsymbol{\vartheta} \mapsto \langle \boldsymbol{\ell}, \boldsymbol{\vartheta} \rangle$, $\boldsymbol{\ell} \in \Lambda$, exhaust the dual of V exactly when no nonzero $\boldsymbol{\vartheta} \in V$ is orthogonal to Λ.

No orthogonal vector. Let $\boldsymbol{\vartheta} \in V$ be orthogonal to Λ.

- Then $\vartheta_{jD} = 0$ in a table with only K_j active, $\vartheta^*_{jD} = 0$ in a table with only Ψ_j active, and $\vartheta^*_{jD} = -\vartheta_{jD}$ in a table with both letters active.
- The letter totals give $\sum_D \vartheta_{jD} = 0$ in every table with both letters active. The scale pin becomes $\sum_{j,D} \big(2i(D) - c_j\big)\vartheta_{jD} = 0$, the sum running over the tables with both letters active.
- A table with $|\mathcal{D}_j| = 1$ has $\vartheta_{jD} = 0$ by its total.
- The exceptional table has $c_j \geq 1$, and $\mathcal{D}_{j,0}$ and $\mathcal{D}_{j,c_j}$ are nonempty (28.4), so its two stations D' and D'' have the excesses 0 and c_j. Its total gives $\vartheta_{jD'} = -\vartheta_{jD''} = \tau$, and the scale pin reads $-c_j\tau - c_j\tau = 0$, so $\tau = 0$.

Hence $\boldsymbol{\vartheta} = 0$, and the plain tinkers suffice in (c). □

Remark. In general, let J_2 be the tables with both letters active. The orthogonal of Λ in V has the dimension $\sum_{j \in J_2}(|\mathcal{D}_j| - 1)$, less one when some table of J_2 has positive width. Indeed, the vectors with $\sum_D \vartheta_{jD} = 0$ on the tables of J_2 and zero elsewhere form a space of the first dimension, and on it the scale pin $\sum_{j,D}(2i(D) - c_j)\vartheta_{jD}$ vanishes identically exactly when every table of J_2 has width zero. With every letter active and some table of positive width, $n_{\mathcal{D}} = 2|\mathcal{D}| - 2N - 1$ and the orthogonal has the dimension $|\mathcal{D}| - N - 1$, so the plain tinkers of one partner move $|\mathcal{D}| - N$ of the directions of the seam torus, as the paragraph after the theorem says.

Lemma 28.23 (Partners on one mark see a point mass). *As stated.*

Proof. The one-mark piece. On one mark the seam letters have $d_j = d^*_j = 0$, the wall is the mass κ, and by the relations of the seam torus the excess sums are $V_j^{(0)} = \sum_{D \in \mathcal{D}_{j,0}} x_{jD} = 1$, $V_j^{*(c_j)} = 1$, $V_j^{(c_j)} = \alpha_g^{c_j} \sum_{D \in \mathcal{D}_{j,c_j}} x_{jD}$ and $V_j^{*(0)} = \alpha_g^{c_j} \sum_{D \in \mathcal{D}_{j,0}} x^*_{jD}$. So the scale and the seam vector of 28.5 are

$$\alpha_g \Big(\prod_j \Big(\sum_{D \in \mathcal{D}_{j,c_j}} x_{jD} \Big)^{s_j} \Big(\sum_{D \in \mathcal{D}_{j,0}} x^*_{jD} \Big)^{t_j} \Big)^{1/w} = \alpha_g,$$

$$\Big(\frac{x_{jD}\, \alpha_g^{i(D)}}{\alpha_g^{i(D)}},\ \frac{x^*_{jD}\, \alpha_g^{c_j - i(D)}}{\alpha_g^{c_j - i(D)}} \Big)_{j,D} = \mathbf{x},$$

and the scale form on the one point of $\Delta_{\{g\}}$ has the mass 1: the law is $\delta_{\mathbf{x}}$, and the station transform is the kernel at $\mathbf{x}$ (28.21). In the chart $\alpha_b = 1$ the period of the glue of $\mathcal{X}$ with the partner on b is $\int_{(0,\infty)^B} \alpha^{\mathbf{m}_B - \mathbf{1}} \prod_j (\sum_D p_{jD} V_{jD})^{-s_j} (\sum_D p^*_{jD} V^*_{jD})^{-t_j} d\alpha$, the second expression of 28.21; the partner has the degrees $e_j = e^*_j = 0$, the wall $w - \kappa$, and its room letters are those of the partner with the coefficients 1 tinkered by $\mathbf{p}$ (28.18).

Coordinates. Index the coordinates of the seam vector by the pairs of an active letter l and a station of its table, $n_{\mathcal{X}}$ of them; write $\mathbf{p} = e^{\mathbf{P}}$ with $\mathbf{P} \in \mathbb{R}^{n_{\mathcal{X}}}$, and pair shares $(\mathbf{a}, \mathbf{b}) \in \mathbb{R}^{n_{\mathcal{X}}}$, one for each coordinate, with $\mathbf{P}$ by $\langle (\mathbf{a}, \mathbf{b}), \mathbf{P} \rangle$. Let $\mathbf{u}_l$ be the indicator of the coordinates of the letter

l, U' the span of the $\mathbf{u}_l$, $\mathbf{u}_0$ the vector with the entries $i(D)$ at the coordinates of the letters K_j and $c_j - i(D)$ at those of the letters Ψ_j, and $U = U' + \mathbb{R}\mathbf{u}_0$. Since $w > 0$ some active letter has $c_j > 0$, and on its coordinates $\mathbf{u}_0$ takes the values 0 and c_j at the excesses 0 and c_j, which have stations; so $\mathbf{u}_0 \notin U'$. The pins of the seam (28.7) are $\langle(\mathbf{a},\mathbf{b}),\mathbf{u}_l\rangle$ equal to the exponent of l and $\langle(\mathbf{a},\mathbf{b}),\mathbf{u}_0\rangle = \kappa$; let $\mathcal{A}$ be the set of the shares with positive coordinates on the pins. For one letter of width $c_j > 0$ and exponent s, the sums $\sum_D i(D)a_{jD}$ over positive shares with $\sum_D a_{jD} = s$ fill $(0, sc_j)$, and they are 0 at width zero; so on the pins the scale pin takes every value in $(0, w)$, and $\mathcal{A}$ is not empty, open in the affine space of the pins, whose direction space is $V = U^\perp$.

The station transform opens the seam function. Fix $(\mathbf{a},\mathbf{b}) \in \mathcal{A}$ and put $f(\mathbf{P}) = e^{\langle(\mathbf{a},\mathbf{b}),\mathbf{P}\rangle}\,\Gamma(\kappa)\mathsf{Z}_{(\kappa,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(e^{\mathbf{P}})$. By the kernel of 28.10(ii) with $v = e^\tau$,

$$f(\mathbf{P}) = \int_{\mathbb{R}} g(\mathbf{P} + \tau\mathbf{u}_0)\,d\tau, \qquad g(\mathbf{Q}) = e^{\langle(\mathbf{a},\mathbf{b}),\mathbf{Q}\rangle} \prod_{j:\,s_j>0} \Big(\sum_D e^{Q_{jD}}\Big)^{-s_j} \prod_{j:\,t_j>0} \Big(\sum_D e^{Q^*_{jD}}\Big)^{-t_j},$$

using $\langle(\mathbf{a},\mathbf{b}),\mathbf{u}_0\rangle = \kappa$. Translation by $\mathbf{u}_l$ multiplies the letter l by e^λ and the exponential by $e^{\lambda s}$, s its exponent: g is invariant under U', and f under U, a translation by $\mathbf{u}_0$ being absorbed by τ. Let W be a linear complement of U and W' the subspace on which the coordinate of one chosen station D_l of every active letter vanishes, a complement of U'. Since $\mathbf{u}_0 \notin U'$, $(\mathbf{P},\tau) \mapsto \mathbf{P} + \tau\mathbf{u}_0$ is a linear isomorphism of $W \times \mathbb{R}$ onto $\mathbb{R}^{n_{\mathcal{X}}}/U' \cong W'$, and by Tonelli and the invariance of g

$$\int_W f\,d\mathbf{P} = c_W \int_{W'} g = c_W \prod_{j:\,s_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \prod_{j:\,t_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix} =: \beta(\mathbf{a},\mathbf{b}) \in (0,\infty),$$

with $c_W > 0$ fixed by W, W' and their Lebesgue measures. On W' the integral factors over the letters; for a letter K_j, with $y_D = e^{Q_{jD}}$ and $a_{jD_l} = s_j - \sum_{D\neq D_l} a_{jD} > 0$, it is $\int_{(0,\infty)^{\mathcal{D}_j\setminus D_l}} \prod_{D\neq D_l} y_D^{a_{jD}-1}\big(1 + \sum_{D\neq D_l} y_D\big)^{-s_j} d\mathbf{y}$, which is the stacked Gamma, as $(1+|\mathbf{y}|)^{-s_j} = \Gamma(s_j)^{-1}\int_0^\infty x^{s_j-1}e^{-x(1+|\mathbf{y}|)}dx$ and Tonelli integrate it to $\prod_D \Gamma(a_{jD})/\Gamma(s_j)$; likewise for a letter Ψ_j. For $\mathbf{x}$ on the seam torus and $\mathbf{X} = \log\mathbf{x}$, $e^{\langle(\mathbf{a},\mathbf{b}),\mathbf{P}\rangle}\Gamma(\kappa)\mathsf{Z}(\mathbf{x}\circ e^{\mathbf{P}}) = e^{-\langle(\mathbf{a},\mathbf{b}),\mathbf{X}\rangle} f(\mathbf{P}+\mathbf{X})$, and $\int_W f(\mathbf{P}+\mathbf{X})\,d\mathbf{P} = \int_W f\,d\mathbf{P}$ by the invariance of f under U, Lebesgue measure on W being invariant under translation. By the first expression of 28.21 and Tonelli,

$$\int_W e^{\langle(\mathbf{a},\mathbf{b}),\mathbf{P}\rangle}\,\mathcal{I}_{\mathcal{X}}(e^{\mathbf{P}})\,d\mathbf{P} = \beta(\mathbf{a},\mathbf{b}) \int \mathbf{x}^{-(\mathbf{a},\mathbf{b})}\,d\mu(\mathbf{x}),$$

μ the law of $\mathcal{X}$. The hypothesis makes the left side $c\,\beta(\mathbf{a},\mathbf{b})\,\mathbf{x}_0^{-(\mathbf{a},\mathbf{b})}$, so $\int \mathbf{x}^{-(\mathbf{a},\mathbf{b})}d\mu = c\,\mathbf{x}_0^{-(\mathbf{a},\mathbf{b})}$ for every $(\mathbf{a},\mathbf{b}) \in \mathcal{A}$: the seam function is finite on all of $\mathcal{A}$.

A point mass. Fix $(\mathbf{a}_0,\mathbf{b}_0) \in \mathcal{A}$ and let $\nu = \mathbf{x}^{-(\mathbf{a}_0,\mathbf{b}_0)}\mu$, a finite measure of the total mass $C = c\,\mathbf{x}_0^{-(\mathbf{a}_0,\mathbf{b}_0)}$. For $\boldsymbol{\delta} \in V$ near 0, $(\mathbf{a}_0,\mathbf{b}_0) \pm \boldsymbol{\delta} \in \mathcal{A}$, and so $\int(\mathbf{x}/\mathbf{x}_0)^{\mp\boldsymbol{\delta}}d\nu = C$ and

$$\int \Big((\mathbf{x}/\mathbf{x}_0)^{\boldsymbol{\delta}/2} - (\mathbf{x}/\mathbf{x}_0)^{-\boldsymbol{\delta}/2}\Big)^2 d\nu(\mathbf{x}) = C - 2C + C = 0.$$

So $\langle\log\mathbf{x} - \log\mathbf{x}_0, \boldsymbol{\delta}\rangle = 0$ for ν-almost every $\mathbf{x}$, for every $\boldsymbol{\delta}$ of a basis of V, and $\log\mathbf{x} - \log\mathbf{x}_0 \in V^\perp = U$ for μ-almost every $\mathbf{x}$, μ and ν having the same null sets. Two points $\mathbf{x},\mathbf{x}'$ of the seam torus with $\log\mathbf{x} - \log\mathbf{x}' \in U$ are equal: $\log x_{jD} - \log x'_{jD} = \lambda_j + \lambda i(D)$ and $\log x^*_{jD} - \log x'^*_{jD} = \lambda^*_j + \lambda(c_j - i(D))$; the relations at the excess 0 of the letters K_j and at the excess c_j of the letters Ψ_j give $\lambda_j = \lambda^*_j = 0$, and then the product relation gives $e^{\lambda w} = 1$, so $\lambda = 0$. Hence μ is carried by $\mathbf{x}_0$, and $\mu = c\,\delta_{\mathbf{x}_0}$. □

Theorem 28.24 (A bunch is stored on one mark exactly at width one). *As stated, the clause at width one read with the normalization of 28.11.*

Proof. Everything except the clause at width one is the reference, proof of thm:storeone. At width one with factoring weights, Lemma S13(i) writes the letters of the partner with its stand-in at a seam point $\mathbf{x}$, where $x_{jD_0} = x^*_{jD_1} = 1$, as

$$\frac{1}{\hat\theta_0}\Big(K_{\mathcal{B}_j|R} + \frac{\hat\theta_0}{\hat\theta_1}\, x_{jD_1}\, \alpha_g\, K_{\mathcal{B}_j/B}\Big), \qquad \frac{1}{\hat\theta_1}\Big(\frac{\hat\theta_1}{\hat\theta_0}\, x^*_{jD_0}\, \alpha_g\, \Psi_{\mathcal{B}_j|R} + \Psi_{\mathcal{B}_j/B}\Big),$$

the constants $\hat\theta_0, \hat\theta_1$ being those of table j. Under the normalization of 28.11 the constants are 1. The letters are then those of $\mathcal{B}^+_{j,R}$ twisted at g by x_{jD_1} in K_j and by $x^*_{jD_0}$ in Ψ_j (19.4, 22.1). At the seam point of a value $\mathbf{y}$ of the ratio vector of F^+_B these twists are $y_{j,K}/\psi(\mathbf{y})$ and $y_{j,\Psi}/\psi(\mathbf{y})$ (28.11(ii)), and the change of variable $\alpha_g \mapsto \psi(\mathbf{y})\alpha_g$ of 28.11(iii) turns them into the twist by $\mathbf{y}$, as stated. Without the normalization the two letters carry the constant factors $1/\hat\theta_0$ and $1/\hat\theta_1$ and the twists $(\hat\theta_0/\hat\theta_1)x_{jD_1}$ and $(\hat\theta_1/\hat\theta_0)x^*_{jD_0}$. The last sentence of the clause is 28.11(i)–(iii). □

Theorem 28.25 (A point mass is a pinned set). *As stated.*

Proof. The walls. On the balance wall $\kappa_S + \kappa_{S'} = 2\sum_j \sigma_j c_j(S) = w$ (8.16), and both walls are positive; so $w > 0$ and $0 < \kappa_S < w$. By the paragraph after 28.4, S is a piece with the seam letters $V_{jD} = K_{j,(D)}$ and $V^*_{jD} = \Psi_{j,(D)}$ of the stations D of its room, S' is a partner, and their glue is F (28.2(iii)); the wall of the piece is κ_S and its top wall $w - \kappa_S = \kappa_{S'}$.

(i) and the seam vector. The scale form has a positive continuous density on the open simplex Δ_S, the excess sums and the scale being positive there, and the seam vector is continuous. So the law is carried by one point exactly when the seam vector is constant, $\mathbf{x}_0$ say, and then it is $c\,\delta_{\mathbf{x}_0}$ with $c = \int_{\Delta_S} \phi^\rho_S > 0$. By 28.10(i) with the partner S', $P_F = c\, P_{S'(\mathbf{x}_0)}$ in $[0,\infty]$; P_F is finite and $P_{S'(\mathbf{x}_0)} > 0$, so c and $P_{S'(\mathbf{x}_0)}$ are finite.

(ii)⇒(i). An active table of width zero at S with its weights factoring across S and S' has one station of room sides (28.3(i)), whose coordinates of the seam vector are $V_{jD}/V^{(0)}_j = 1$ and $V^*_{jD}/V^{*(c_j)}_j = 1$. An active table of width one with factoring weights has the stations D_0 and D_1 and the seam letters $K_{\mathcal{B}_j/S'}, K_{\mathcal{B}_j|S}$ and $\Psi_{\mathcal{B}_j/S'}, \Psi_{\mathcal{B}_j|S}$ (28.3(ii), the paragraph after 28.5), so its coordinates are 1, $\gamma^K_{j,S}/\gamma^\rho_S$, $\gamma^\Psi_{j,S}/\gamma^\rho_S$ and 1, the hinge values read at the factorization that normalizes the multipliers. Under (ii) the scale is $\gamma^\rho_S = \prod_j (c_j\gamma)^{s_j/w}(c^*_j\gamma)^{t_j/w} = C\gamma$, the product over the tables of width one with $C = \prod_j c_j^{s_j/w} c_j^{*\,t_j/w}$, and every coordinate is a constant: c_j/C, c^*_j/C or 1.

(i)⇒(ii). Let the seam vector be the constant $\mathbf{x}_0$. For every active letter K_j, $V_{jD} = x_{0,jD} V^{(0)}_j (\gamma^\rho_S)^{i(D)}$, and by the room-station factorization (28.2(iii))

$$K_j = \sum_D V_{jD}\, \mathsf{y}_{jD} = V^{(0)}_j(\alpha_S) \sum_D x_{0,jD}\, \gamma^\rho_S(\alpha_S)^{i(D)}\, \mathsf{y}_{jD}(\alpha_{S'}),$$

so K_j reads S through the one function γ^ρ_S, positive and homogeneous of degree one, with $V^{(0)}_j$ positive and homogeneous: by 19.6, $c_j(S) \le 1$. Likewise for every active letter Ψ_j, with $V^{*(c_j)}_j$ and $(\gamma^\rho_S)^{c_j - i(D)}$. Two stations $D \ne D'$ of one excess would have proportional seam letters, $V_{jD}/V_{jD'} = x_{0,jD}/x_{0,jD'}$ (or the same with V^*), so proportional room children, against 28.1(i); so every excess of every active table has one station of room sides, hence one station of patterns (28.3), and is one rectangle, which at the extreme excesses, and so at widths zero and one, means that the weights

factor across S and S' (28.3(i),(ii)). At width one the coordinates above are constant, so the active hinge values are positive multiples of γ_S^ρ: they read S alike, with $\gamma = \gamma_S^\rho$.

(i)⇒(iii)⇒(iv)⇒(i). The first is 28.10(i) with the law $c\,\delta_{\mathbf{x}_0}$. For the second take the partner on one mark b with the room letters $p_{jD}\alpha_b^{c_j - i(D)}$ and $p_{jD}^*\alpha_b^{i(D)}$ and the mass $w - \kappa_S$: the period of its glue with S is the station transform at $\mathbf{p}$ (28.23), and the partner with its stand-in at $\mathbf{x}_0$ is the glue of the one-mark piece at $\mathbf{x}_0$ with it, whose period is the kernel $\Gamma(\kappa_S)\mathsf{Z}_{(\kappa_S,\mathbf{s},\mathbf{t});t_\mathcal{D}}(\mathbf{x}_0 \circ \mathbf{p})$ (28.10(ii)). The third is 28.23.

Lists. With the weights 1 the seam letters are the letters themselves and the multipliers are 1 (28.1). Let two active hinge values of tables of width one be proportional, $\gamma_{j,S}^K = \mu\,\gamma_{j',S}^\Psi$ say, that is $K_{\mathcal{B}_j|S}\Psi_{\mathcal{B}_{j'}|S} = \mu\,K_{\mathcal{B}_j/S'}\Psi_{\mathcal{B}_{j'}/S'}$ (19.2). Along a maximal chain of subsets of S the terms of least degree of each of these four letters, at every set of the chain at once, form one monomial with the coefficient 1 (12.20); for a product of two polynomials with nonnegative coefficients the least degrees add and the terms of least degree are the products of those of the factors, so the two sides have such terms $1 \cdot \alpha^{\mathbf{a}}$ and $\mu\,\alpha^{\mathbf{a}'}$, and $\mu = 1$. The other pairs of letters are alike. So reading alike is equality, every c_j and c_j^* is 1, $C = 1$, and $\mathbf{x}_0 = \mathbf{1}$.

One list. With both letters active, $w = (s+t)c(S) > 0$, so pinned means $c(S) = 1$ and γ_S^K and γ_S^Ψ equal, by the above, that is $\varrho_S = 1$, which is $\mathbf{E}_{\mathcal{B}|S} = \mathbf{E}_{\mathcal{B}/S'}$ (19.2). The cell of the bunch $\mathcal{B}_S^+$ is a list with the hinge minors $\mathcal{B}|S$ and $\mathcal{B}/S'$ (19.4) and the ratio ϱ_S (21.1). □

Corollary 28.26 (Which sets split off). *As stated.*

Proof. (i) A pinned set has width at most one in every active table; apply 28.25. (ii) A prime table is connected, so a set with at least two marks on each side has positive width (7.10), and it has no bunch of width one with two marks on each side (23.16); so such a set has width at least two, and (i) applies. (iii) The set $S = \mathcal{E} \setminus \{e\}$ has the width $c(\{e\}) = r(\{e\}) - d(\{e\}) = 1$ (7.13), and one letter is active, so S is pinned with $\gamma = \gamma_S^K$, and it splits off. At $(s,0)$, $d(\{e\}) = 0$ gives $\kappa_{\{e\}} = m_e$ and $\kappa_S = s - m_e$, as $\kappa_S + \kappa_{\{e\}} = s$ (8.16). 19.14 with the bunch S and the room $\{e\}$, at the level shares $(s - \kappa_S, 0, \kappa_S, 0)$, gives the Beta corner $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} s - m_e, & m_e \\ & s \end{bmatrix}$ times the period of the bunch family, the tables $\mathcal{B}/\{e\} = \mathcal{B}/e$ at $(m_e, 0)$ and $\mathcal{B}|S = \mathcal{B} \setminus e$ at $(s - m_e, 0)$ (7.5), times the period of the room family on one mark, which is 1. At $(0,t)$, $d^*(\{e\}) = 0$ gives $\kappa_{\{e\}} = m_e$, $\kappa_S = t - m_e$, and the level shares $(0, \kappa_S, 0, t - \kappa_S)$ give the stated family. (iv) $S = \mathcal{E} \setminus \{e\}$ has width one and $|S| \geq 2$, and its hinge minors are $\mathcal{B}|S = \mathcal{B} \setminus e$ and $\mathcal{B}/\{e\} = \mathcal{B}/e$; apply the last part of 28.25. □

48.29 Chapter 29: composing bunches

Theorem 29.6 (Two transforms of a law, and their rules). *As stated; the first rule of (iii) holds at any number of sub-pieces.*

Proof. Parts (i), (ii) and (iv), and the second and third rules of (iii), are the reference, proof of thm:lawreadings. For the first rule:

- Let $S_1, \ldots, S_k$ be as in 29.2, and $\mathbf{p}$ a point with positive coordinates, one for each coordinate of the seam vector of S.
- By 28.21, $\mathcal{I}_S(\mathbf{p}) = \int H\,d\lambda_S$, with the law $\lambda_S = (\boldsymbol{\varrho}_S^\rho)_*\phi_S^\rho$ and the Borel function $H(\mathbf{z}) = \Gamma(\kappa_S)\,\mathsf{Z}_{(\kappa_S,\mathbf{s},\mathbf{t});t_\mathcal{D}}(\mathbf{z} \circ \mathbf{p}) \geq 0$, $\mathcal{D}$ being the stations of room sides of S.
- 29.2(ii) gives $\int H\,d\lambda_S = \int \big(\int H\,d\lambda_{\mathbf{x}}\big) \prod_l d\lambda_{S_l}(\mathbf{x}_l)$, with $\lambda_{\mathbf{x}}$ the law of the node $S^{(\mathbf{x}_1,\ldots,\mathbf{x}_k)}$.

- The node is a piece with the stations $\mathcal{D}$, the exponents of the family and the wall κ_S (29.2(ii)), so by the same definition $\int H\, d\lambda_{\mathbf{x}} = \mathcal{I}_{S^{(\mathbf{x}_1,\dots,\mathbf{x}_k)}}(\mathbf{p})$.

This is the reference's argument for one sub-piece, which does not use $k = 1$. The second and third rules use the transfer matrices from one sub-piece: 29.2(i) gives linearity in the seam letters of each S_l separately, with coefficients that depend on the marks of the other sub-pieces. □

Theorem 29.8 (Cyclic chains). *As stated: (i) holds for a family, and in (ii) each letter is opened over the starting states at which its term is not zero.*

Proof. (i) Fix a table j.

- By 21.14(ii), the bases of table j are the unions of the unpaired parts of the choices of one basis in every cell $\mathcal{A}_{j,i}$ that occupy exactly one hinge of every pair. Its weight in each letter at a basis is the sum, over the choices that give it, of the products of the weights chosen.
- A pair of I_{i-1} occupied on the side of the cell $i-1$ is free on the side of the cell i, and conversely. So, with $J_i \subseteq I_i$ the pairs occupied on the side of the cell i, the basis chosen in $\mathcal{A}_{j,i}$ occupies the hinges $(I_{i-1} \setminus J_{i-1}) \sqcup J_i$.
- Its part $X_i \subseteq B_i$ is a basis of the child of the bunch of its hinges at that pattern. With their weights, these choices sum to the channel $K_{\mathcal{A}_{j,i},((I_{i-1}\setminus J_{i-1})\sqcup J_i)}(\alpha_{B_i})$ (21.13).
- A basis $\bigsqcup_i X_i$ leaves the cobasis $\bigsqcup_i (B_i \setminus X_i)$, so Ψ_j is summed the same way with the channels of Ψ.
- Summing over the sequences $(J_0, \dots, J_{k-1})$, $J_k = J_0$, gives $K_j = \operatorname{tr} \prod_i \mathbf{M}_{K_j,i}$ and $\Psi_j = \operatorname{tr} \prod_i \mathbf{M}_{\Psi_j,i}$. The entries are polynomials in α_{B_i} with nonnegative coefficients and no ε, which is the top layer 0.
- The pairs are named alike in every table, so the sets of states $\mathcal{J}_i = \{J :\ J \subseteq I_i\}$ serve every letter, as 29.7 asks.

This is the reference's proof (proof of thm:cyclicchain(i)) applied table by table.

(ii) and (iii) are the reference, proof of thm:cyclicchain. In (ii), L is the sum over the starting states J of the terms $\mathbf{e}_{(J,0)}^{\mathsf{T}} \prod_i \widetilde{\mathbf{M}}_{L,i}\, \mathbf{e}_{(J,h)}$, each a sum over the paths from $(J,0)$ to (J,h) of products of entries with nonnegative coefficients. 4.4 opens a sum of positive terms, one share for each. A starting state whose term vanishes, the state of no path, therefore carries no share and no flow. □

48.30 Chapter 30: Foreman fusion

The proofs use the rank function of a table, $r(Y) = \max_T |T \cap Y|$ (Definition 7.4), through its standard properties: $r(\varnothing) = 0$, $r(Y) \le r(Y+e) \le r(Y)+1$ and $r(Y \cup Y') + r(Y \cap Y') \le r(Y) + r(Y')$; conversely a function on the subsets of a finite set with these three properties is the rank function of the table whose bases are the sets Y with $|Y| = r(Y)$ equal to the value at the whole set (the rank axioms, which hold for the bases of Definition 7.1 by the arena property, the exchange property of the bases). A set is *closed* when it is maximal for its rank, that is when adding any mark raises its rank; the *closure* of Y is the set of the marks whose addition does not raise the rank of Y, it is closed, and it lies in every closed set that contains Y. A mark $e \notin Y$ lies in the closure of Y exactly when some circuit C has $e \in C \subseteq Y + e$: a basis I of Y with $I + e$ dependent contains, with e, a circuit through e, and a circuit C through e inside $Y + e$ puts e in the closure of $C - e$. For a minor contracting an independent set C and deleting D, the rank of a set Y of the remaining

marks is $r(Y \cup C) - r(C)$ (Definition 7.5). The dual rank function is $r^*(Y) = |Y| + r(\mathcal{E} \setminus Y) - r$ (Definition 7.12), and $d(Y) = |Y| - r^*(Y)$.

Definition 30.1 (Quotients; lifts; the level cell of a pair). As stated.

Proof. That the lifts and the level cell are tables is Proposition 30.2(ii) and (iii). For a bunch B with the room R, $\mathcal{B}/R$ is a quotient of $\mathcal{B}|B$: for $X \subseteq Y \subseteq B$, submodularity gives $r(Y) - r(X) \geq r(Y \cup R) - r(X \cup R)$, which is the inequality of Proposition 30.2(i) for the rank functions $r(\cdot)$ of $\mathcal{B}|B$ and $r(\cdot \cup R) - r(R)$ of $\mathcal{B}/R$. The independent sets of $\mathcal{L}_j(\mathcal{B}|B, \mathcal{B}/R)$ are the X with $|X| \leq r(X)$ and $|X| \leq r_{\mathcal{B}/R}(X) + j$, the independent sets of Proposition 12.17(ii) for $i = j$, so the two lifts are one table, and the level cells of Definition 30.1 and Lemma 28.27 have the same bases. Level form at G is Proposition 14.12: a transposition of two marks of G fixes X and $|Q|$. □

Proposition 30.2 (The level cell of a pair). *As stated.*

Proof. (i) Let the inequality hold, A be closed in $\mathcal{N}$ and $e \notin A$. Then $r_{\mathcal{N}}(A + e) = r_{\mathcal{N}}(A) + 1$, the inequality at $X = A$, $Y = A + e$ gives $r_{\mathcal{M}}(A + e) = r_{\mathcal{M}}(A) + 1$, and A is closed in $\mathcal{M}$. Conversely, let every closed set of $\mathcal{N}$ be closed in $\mathcal{M}$. Along a chain from X to Y adding one mark at a time it suffices to take $Y = X + e$. If $r_{\mathcal{N}}(X + e) = r_{\mathcal{N}}(X)$ the inequality holds, $r_{\mathcal{M}}$ being monotone. Otherwise e is not in the closure A of X in $\mathcal{N}$; A is closed in $\mathcal{M}$ and contains X, so the closure of X in $\mathcal{M}$ lies in A and avoids e, and $r_{\mathcal{M}}(X + e) = r_{\mathcal{M}}(X) + 1$. At $X = \varnothing$, $Y = \mathcal{E}$ the inequality gives $c \geq 0$; at $X = \varnothing$ it gives $r_{\mathcal{M}}(Y) \geq r_{\mathcal{N}}(Y)$, so a set independent in $\mathcal{N}$ is independent in $\mathcal{M}$; at $Y = \mathcal{E}$ it gives $r_{\mathcal{N}}(\mathcal{E}) - r_{\mathcal{N}}(X) \leq r_{\mathcal{M}}(\mathcal{E}) - r_{\mathcal{M}}(X)$, so a set spanning $\mathcal{M}$ spans $\mathcal{N}$; and two such inequalities add.

(ii) Put $f_1(Y \sqcup P) = r_{\mathcal{M}}(Y) + |P|$ and $f_2(Y \sqcup P) = r_{\mathcal{N}}(Y) + c$. Each is monotone, rises by at most one when a mark is added, and is submodular, being a rank function plus a function that is modular; $f_1(\varnothing) = 0 \leq c = f_2(\varnothing)$; and $f_1 - f_2 = r_{\mathcal{M}}(Y) - r_{\mathcal{N}}(Y) + |P| - c$ is nondecreasing under inclusion by (i). The minimum of two submodular functions with a nondecreasing difference is submodular: for sets Z, Z' the cases where one function attains the minimum at both are its own submodularity, and if $\min(Z) = f_1(Z)$ and $\min(Z') = f_2(Z')$, then

$$f_1(Z) + f_2(Z') - f_1(Z \cap Z') - f_2(Z \cup Z') \geq \big(f_2(Z) - f_2(Z \cap Z')\big) - \big(f_2(Z \cup Z') - f_2(Z')\big) \geq 0,$$

the first step because $(f_1 - f_2)(Z) \geq (f_1 - f_2)(Z \cap Z')$ and the second by the submodularity of f_2, so that $\min(Z) + \min(Z') \geq f_1(Z \cap Z') + f_2(Z \cup Z') \geq \min(Z \cap Z') + \min(Z \cup Z')$; the case with f_1 and f_2 exchanged is the same with the roles of Z and Z' exchanged. So $r^\natural = \min(f_1, f_2)$ satisfies the rank axioms. Its value at $\mathcal{E} \sqcup G$ is $\min(r_{\mathcal{M}} + c, r_{\mathcal{N}} + c) = r_{\mathcal{M}}$, and $Y \sqcup P$ with $|Y| + |P| = r_{\mathcal{M}}$ is a basis exactly when $r_{\mathcal{M}}(Y) \geq |Y|$ and $r_{\mathcal{N}}(Y) \geq |Y| + |P| - c = r_{\mathcal{N}}$: Y independent in $\mathcal{M}$ and spanning $\mathcal{N}$, with $|Y| = r_{\mathcal{N}} + c - |P|$. For $Q \subseteq G$, $r^\natural(Q) = |Q|$, so the minor contracting Q and deleting $G \setminus Q$ has the rank function

$$Y \mapsto r^\natural(Y \sqcup Q) - |Q| = \min\big(r_{\mathcal{M}}(Y),\ r_{\mathcal{N}}(Y) + c - |Q|\big),$$

that of $\mathcal{L}_{c-|Q|}$, provided some basis contains Q and avoids $G \setminus Q$. Such bases exist for every Q: a basis Y_0 of $\mathcal{N}$ is independent in $\mathcal{M}$ by (i), it extends to a basis of $\mathcal{M}$ of $r_{\mathcal{N}} + c$ marks, and every set between the two is independent in $\mathcal{M}$ and spans $\mathcal{N}$. So every Q is a pattern, $d(G) = 0$, $r(G) = c$ and $c(G) = c$, and the child at Q is $\mathcal{L}_{c-|Q|}$ (Lemma 12.2). At $Q = \varnothing$ the rank function is $r_{\mathcal{M}}$, since $r_{\mathcal{M}}(Y) \leq r_{\mathcal{N}}(Y) + c$ by (i) at $X = Y$ and $\mathcal{E}$; at $Q = G$ it is $r_{\mathcal{N}}$, since $r_{\mathcal{N}} \leq r_{\mathcal{M}}$. The rank function depends on P through $|P|$ alone, so every permutation of G maps the bases onto themselves.

(iii) By (ii) $\mathcal{L}_j$ is the child at any Q of $c-j$ marks, a table, with the stated rank and bases, and $\mathcal{L}_0 = \mathcal{N}$, $\mathcal{L}_c = \mathcal{M}$. For $j \le j'$ take $Q' \subseteq Q$ with $|Q| = c-j$ and $|Q'| = c-j'$ and a chain of patterns from Q' to Q adding one mark at a time; each child is a quotient of the one before (Proposition 12.6), and by (i) $\mathcal{L}_j$ is a quotient of $\mathcal{L}_{j'}$. With the rank difference $b-a$ of $(\mathcal{L}_b, \mathcal{L}_a)$,

$$\min\big(r_{\mathcal{L}_b},\ r_{\mathcal{L}_a}+i\big) = \min\big(r_{\mathcal{M}},\ r_{\mathcal{N}}+b,\ r_{\mathcal{M}}+i,\ r_{\mathcal{N}}+a+i\big) = \min\big(r_{\mathcal{M}},\ r_{\mathcal{N}}+a+i\big),$$

since $i \ge 0$ and $a+i \le b$.

(iv) C is a pattern, so the minor is defined, and its bases are the sets $Y \sqcup (Q \setminus C)$ with $C \subseteq Q$, $Q \cap D = \varnothing$ and Y a basis of $\mathcal{L}_{c-|Q|}$. With $Q'' = Q \setminus C \subseteq G'' = G \setminus (C \cup D)$ and $c'' = c - |C| - |D| = |G''|$, $c - |Q| = |D| + (c'' - |Q''|)$, and by (iii) $\mathcal{L}_{|D|+i} = \mathcal{L}_i(\mathcal{L}_{c-|C|}, \mathcal{L}_{|D|})$ for $0 \le i \le c''$, the rank difference of that pair being c''. □

Lemma 30.3 (Complements of the lifts toward the dual). *As stated.*

Proof. Write $r(Y)$ and $r^*(Y)$ for the rank functions of $\mathcal{B}$ and $\mathcal{B}^*$, so that $r(\mathcal{E} \setminus X) = |\mathcal{E} \setminus X| + r^*(X) - r^*$ and $r^*(\mathcal{E} \setminus X) = |\mathcal{E} \setminus X| + r(X) - r$. By Proposition 30.2(iii) $\mathcal{L}_j$ is a table of rank $r+j$, with the rank function $r_{\mathcal{L}_j}(Y) = \min\big(r^*(Y),\ r(Y)+j\big)$ (30.1), and the rank function of its dual table is

$$\begin{aligned} |X| + r_{\mathcal{L}_j}(\mathcal{E} \setminus X) - r - j &= |X| + \min\big(n - |X| + r(X) - r,\ n - |X| + r^*(X) - r^* + j\big) - r - j \\ &= \min\big(r(X) + c - j,\ r^*(X)\big) = r_{\mathcal{L}_{c-j}}(X), \end{aligned}$$

using $n - 2r = c$ and $n - r^* - r = 0$. Two tables with one rank function have the same bases, the sets Y with $|Y| = r_{\mathcal{L}}(Y)$ equal to the rank; so the dual table of $\mathcal{L}_j$, whose bases are the complements of those of $\mathcal{L}_j$, has the bases of $\mathcal{L}_{c-j}$. □

Definition 30.4 (Fusion width; Foreman fusion; majors). As stated.

Proof. $c \ge 0$ for a quotient is Proposition 30.2(i). A quotient $\mathcal{N}$ of $\mathcal{M}$ of the same rank has $r_{\mathcal{N}} = r_{\mathcal{M}}$: (i) of that proposition gives $r_{\mathcal{M}}(Y) \ge r_{\mathcal{N}}(Y)$ at $X = \varnothing$, and $r_{\mathcal{N}}(X) \ge r_{\mathcal{M}}(X)$ at $Y = \mathcal{E}$; so the two have the same bases, the sets Y with $|Y| = r_{\mathcal{N}}(Y) = r_{\mathcal{N}}(\mathcal{E})$. At $c = 0$ a fusable table therefore has $\mathcal{B} = \mathcal{B}^*$ and $\theta^* = \theta$, its bases closed under complement with $\theta_{\mathcal{E}\setminus T} = \theta_T$; a list has $\theta^* = \theta$, so at $c = 0$ it is fusable exactly when $\mathcal{B} = \mathcal{B}^*$. The lifts are then the one table $\mathcal{L}_0 = \mathcal{L}_c$, with $\theta^{(0)} = \theta = \theta^* = \theta^{(c)}$. The condition on the weights pairs the bases of $\mathcal{L}_j$ with those of $\mathcal{L}_{c-j}$ by complement, which is a bijection by Lemma 30.3; at $j = 0$ it is the definition of θ^*. $L_0 = K_{\mathcal{B}^*} = \Psi$ (Proposition 7.2) and $L_c = K_{\mathcal{B}} = K$. Every permutation of G maps the bases of $\mathcal{B}^\sharp$ onto themselves (Proposition 30.2(ii)) and keeps their weights, which depend on X and $|Q|$ alone, so the partial fusion depends on G' only through $|G'|$, up to the names of its marks. □

Theorem 30.5 (Foreman fusion). *As stated.*

Proof. (i) Proposition 30.2(ii) with $(\mathcal{M}, \mathcal{N}) = (\mathcal{B}^*, \mathcal{B})$, $r_{\mathcal{M}} = r^*$, $r_{\mathcal{N}} = r$, gives the table, its rank, its rank function and the two minors; $n + c = 2n - 2r = 2r^*$. The bases $X \sqcup G$ carry $\theta^{(0)}_X = \theta_X$ and the bases $X \sqcup \varnothing$ carry $\theta^{(c)}_X = \theta^*_X$, so the contraction of G is $(\mathcal{B}, \theta)$ and the deletion is $(\mathcal{B}^*, \theta^*)$ (Definition 7.5).

(ii) By Lemma 30.3, X is a basis of $\mathcal{L}_j$ exactly when $\mathcal{E} \setminus X$ is a basis of $\mathcal{L}_{c-j}$, and with the weights $\theta^{(c-j)}_{\mathcal{E}\setminus X} = \theta^{(j)}_X$ the dual table of $(\mathcal{L}_j, \theta^{(j)})$ is $(\mathcal{L}_{c-j}, \theta^{(c-j)})$. A basis $X \sqcup Q$ of $\mathcal{B}^\sharp$, X a basis of $\mathcal{L}_{c-|Q|}$, has the complement $(\mathcal{E} \setminus X) \sqcup (G \setminus Q)$ with $\mathcal{E} \setminus X$ a basis of $\mathcal{L}_{|Q|} = \mathcal{L}_{c-|G\setminus Q|}$ and the

weight $\theta^{(|Q|)}_{\mathcal{E}\setminus X} = \theta^{(c-|Q|)}_X$: complementation maps the bases onto themselves with their weights. Summing $\theta^\sharp\alpha^{X\sqcup Q}$ over Q by its size gives $K^\sharp = \sum_l e_l(\alpha_G)K_{\mathcal{L}_{c-l}}$; $\Psi^\sharp = K^\sharp$ by Corollary 20.19; and $\Psi_{\mathcal{L}_l} = K_{\mathcal{L}^*_l} = K_{\mathcal{L}_{c-l}}$ by Proposition 7.2.

(iii) Let L be a major and $X\sqcup Q$ one of its bases; at $c=0$, $L=(\mathcal{B},\theta)=\mathcal{B}^\sharp$. In L the children of the bunch G at $\varnothing$ and at G are $\mathcal{B}^*$ and $\mathcal{B}$, so $\varnothing$ and G are patterns, $d(G)=0$ and G is independent; the contraction of the room onto G has the rank $d(G)=0$, so by Proposition 12.17(i) every subset of G is a pattern of G in L, and along chains of patterns from $\varnothing$ to Q and from Q to G the child at Q, of which X is a basis, is a quotient of $\mathcal{B}^*$ and has $\mathcal{B}$ as a quotient (Propositions 12.6 and 30.2(i)). So X is independent in $\mathcal{B}^*$ and spans $\mathcal{B}$ (Proposition 30.2(i)); it has $r^*-|Q| = r+c-|Q|$ marks, so it is a basis of $\mathcal{L}_{c-|Q|}$ (Proposition 30.2(iii)) and $X\sqcup Q$ is a basis of $\mathcal{B}^\sharp$.

(iv) The weights of $\mathcal{B}^\sharp$ depend on X and $|Q|$, so G is in level form in it (Definition 30.1). Let L be a major in which G is in level form. By Proposition 28.29(d) for L and its bunch G, the room sides of $r+j$ marks are the bases of the j-th lift of the table of the room sides of r marks, those of $L/G=\mathcal{B}$, toward the table of the room sides of $r+c$ marks, those of $L\setminus G=\mathcal{B}^*$: the bases of $\mathcal{L}_j$. A room side X of $r+j$ marks makes a basis with some pattern of $c-j$ marks, and in level form the children at all patterns of $c-j$ marks are one table, so $X\sqcup Q$ is a basis of L for every Q of $c-j$ marks: every basis of $\mathcal{B}^\sharp$ is a basis of L, and by (iii) the two have the same bases. The weights of L at $X\sqcup G$ and $X\sqcup\varnothing$ are those of its minors, θ and θ^*, and level form makes the others depend on X and $|Q|$ alone (Proposition 14.12); L is identically self-dual exactly when the weight of $X\sqcup Q$ equals that of $(\mathcal{E}\setminus X)\sqcup(G\setminus Q)$, that is when the middle weights are paired as in Definition 30.4(ii), and L is then the Foreman fusion whose middle weight $\theta^{(c-|Q|)}_X$ is the weight of $X\sqcup Q$. When L and $\mathcal{B}$ are lists all these weights are 1, and $L=\mathcal{B}^\sharp$. The transpositions generate the permutations of G, which gives the equivalent form (Proposition 14.12). A bijection β carrying $(\mathcal{B},\theta)$ carries $(\mathcal{B}^*,\theta^*)$, keeps r and r^*, and so keeps the rank function of every $\mathcal{L}_j$; with $\theta^{(j)}\circ\beta=\theta^{(j)}$, $\beta\sqcup\pi$ maps each basis $X\sqcup Q$ to the basis $\beta(X)\sqcup\pi(Q)$ of the same weight.

(v) At $c=1$, $\mathcal{L}_0=\mathcal{B}$ and $\mathcal{L}_1=\mathcal{B}^*$: the bases are $X+g$, $X\in\mathcal{B}$, with θ_X, and $X\in\mathcal{B}^*$ with θ^*_X, that is $\mathcal{E}\setminus T$ with θ_T; $K^\sharp = e_0L_0+e_1L_1 = \Psi+\alpha_gK$. A major L has $L/g=(\mathcal{B},\theta)$ and $L\setminus g=(\mathcal{B}^*,\theta^*)$, which fixes its bases through g and avoiding g with their weights: $L=\mathcal{B}^\sharp$. □

Theorem 30.6 (The tell). *As stated.*

Proof. Three facts about a table $\mathcal{B}$ of rank r. (α) A set A is closed in $\mathcal{B}^*$ exactly when $\mathcal{E}\setminus A$ is a union of circuits of $\mathcal{B}$: for $e\notin A$, $r^*(A+e)-r^*(A) = 1+r(\mathcal{E}\setminus A-e)-r(\mathcal{E}\setminus A)$, which is 1 exactly when e lies in the closure of $\mathcal{E}\setminus A-e$, that is in a circuit inside $\mathcal{E}\setminus A$. (β) The cocircuits are the complements of the closed sets of rank $r-1$: D meets every basis exactly when $\mathcal{E}\setminus D$ contains none, that is $r(\mathcal{E}\setminus D)\le r-1$, and D is minimal with this property exactly when $\mathcal{E}\setminus D$ is maximal with it, a closed set of rank $r-1$. (γ) The complement of a closed set A is a union of cocircuits: for $e\notin A$, a basis I of A has $I+e$ independent, $I+e$ extends to a basis T, and the closure H of $T-e$ has rank $r-1$, contains A, which lies in the closure of I, and avoids e, T being independent; so $\mathcal{E}\setminus H$ is a cocircuit through e inside $\mathcal{E}\setminus A$, by (β).

(a)⇒(b): the complement of a cocircuit is closed in $\mathcal{B}$ by (β), so in $\mathcal{B}^*$, and the cocircuit is a union of circuits by (α). (b)⇒(a): the complement of a closed set A of $\mathcal{B}$ is a union of cocircuits by (γ), each a union of circuits by (b), so A is closed in $\mathcal{B}^*$ by (α). (b)⇒(c): if e is not in the closure of A, (γ) applied to that closure gives a cocircuit D with $e\in D$ and $D\cap A=\varnothing$, so $D-e\subseteq A'$; by (b) a circuit C has $e\in C\subseteq D$, and e lies in the closure of $C-e\subseteq A'$. (c)⇒(b): for a cocircuit D and $e\in D$ split $\mathcal{E}\setminus\{e\}$ into $A=\mathcal{E}\setminus D$ and $A'=D-e$; A is closed by (β) and avoids e, so by (c) e lies in the closure of $D-e$, in a circuit inside D. By Proposition 30.2(i) (a) implies $c\ge 0$.

By Definition 30.4 the table is fusable exactly when (a) holds and, at $c = 0$, $\theta^* = \theta$, which a list always has. At $c = 0$, G is empty, a major is a table equal to $(\mathcal{B}, \theta)$ and to $(\mathcal{B}^*, \theta^*)$, and one exists exactly when $\mathcal{B} = \mathcal{B}^*$ and $\theta^* = \theta$, which is fusability, a quotient of the same rank having the same bases (proof of Definition 30.4). Let $c \geq 1$. A fusable table has a major, its Foreman fusion (Theorem 30.5(i), whose proof uses Proposition 30.2 alone). Conversely, in a major the children of the bunch G at $\varnothing$ and at G are $\mathcal{B}^*$ and $\mathcal{B}$, so $\varnothing$ and G are patterns, $d(G) = 0$ and G is independent; the contraction of the room onto G has rank $d(G) = 0$, and by Proposition 12.17(i) every subset of G is a pattern. Along a chain from $\varnothing$ to G adding one mark at a time each child is a quotient of the one before (Proposition 12.6), and $\mathcal{B}$ is a quotient of $\mathcal{B}^*$ by Proposition 30.2(i): the table is fusable. A table on $\mathcal{E} \sqcup G$ whose children at $\varnothing$ and at G have the bases of $\mathcal{B}^*$ and of $\mathcal{B}$ is a major once its bases $X \sqcup G$ carry θ_X and its bases $X \sqcup \varnothing$ carry θ^*_X, the two sets of bases being disjoint as $G \neq \varnothing$; this is the equivalent form of (d).

A coloop e makes $\{e\}$ a cocircuit, and a circuit inside $\{e\}$ would make e a loop; so (b) fails. The circuits of $\mathcal{B}^*$ are the cocircuits of $\mathcal{B}$ and its cocircuits the circuits of $\mathcal{B}$, which gives (b) for the dual. □

Corollary 30.7 (The fused tables are the identically self-dual tables). *As stated.*

Proof. (i) By Corollary 20.19 $\mathcal{H}$ has rank $|\mathcal{E}'|/2$ and no loop or coloop; C is independent in $\mathcal{H}$ and so in $\mathcal{H}^* = \mathcal{H}$. $\mathcal{H}/C$ has rank $|\mathcal{E}'|/2 - |C|$ on $|\mathcal{E}'| - |C|$ marks, hence the fusion width $|C|$. Its dual has the bases $(\mathcal{E}' \setminus C) \setminus (T \setminus C) = \mathcal{E}' \setminus T$, $T \supseteq C$, with the weights $\theta_T = \theta_{\mathcal{E}' \setminus T}$: the bases of $\mathcal{H}$ avoiding C with their weights, so $(\mathcal{H}/C)^* = \mathcal{H} \setminus C$ and $\mathcal{H}$ is a major of $\mathcal{H}/C$. By Theorem 30.6 $\mathcal{H}/C$ is fusable; C is in level form in $\mathcal{H}$ (Proposition 14.12), and $\mathcal{H}$ is identically self-dual, so by Theorem 30.5(iv) it is the Foreman fusion of $\mathcal{H}/C$ with the middle weights $\theta^{(j)}_X = \theta_{X \sqcup Q}$, $|Q| = c - j$. A one-mark set is independent, $\mathcal{H}$ having no loop, and it has no transposition. With Theorem 30.5(ii) the fused tables are exactly the identically self-dual ones.

(ii) $c(S) = |S| - d(S) - d^*(S)$ does not change under duality, which exchanges d and d^*, so $\mathcal{B}^*$ is connected and its circuits link all of $\mathcal{E}$ (Proposition 28.3(i)). The circuits of $\mathcal{B}^* = \mathcal{B}^\sharp \setminus G$ are circuits of $\mathcal{B}^\sharp$, so $\mathcal{E}$ lies in one molecule of $\mathcal{B}^\sharp$. For $g \in G$ take a basis $X \sqcup Q$ with $g \notin Q$; the circuit through g inside $X \sqcup Q + g$ does not lie in the independent set G, so it meets $\mathcal{E}$, and g lies in the molecule of $\mathcal{E}$. So $\mathcal{B}^\sharp$ is one molecule, connected by Proposition 28.3(i). □

Proposition 30.8 (Partial fusions and inner lifts). *As stated.*

Proof. (i) is Proposition 30.2(iv) for $(\mathcal{B}^*, \mathcal{B})$, the minor carrying the weights of $\mathcal{B}^\sharp$ (Definition 7.5). (ii) With $C = G \setminus G'$ and $D = \varnothing$, (i) gives the level cell of $(\mathcal{L}_i, \mathcal{B})$, of rank $r + i$ on $n + i$ marks, so of the fusion width $c - i$. Being identically self-dual, $\mathcal{B}^\sharp$ has $(\mathcal{B}^\sharp/(G \setminus G'))^* = \mathcal{B}^\sharp \setminus (G \setminus G')$ with the weights, as in the proof of Corollary 30.7(i); $G \setminus G'$ is independent and in level form in $\mathcal{B}^\sharp$ (Proposition 30.2(ii), Definition 30.1); so Corollary 30.7(i) makes $\mathcal{B}^\sharp_i$ fusable with the Foreman fusion $\mathcal{B}^\sharp$, its middle weights read off $\theta^\sharp$. (iii) $\mathcal{L}_j$ has the rank $r+j$ and the corank $n-r-j = r+c-j$, so the fusion width $c - 2j$. For $2j \leq c$, $\mathcal{L}_j$ is a quotient of $\mathcal{L}_{c-j} = \mathcal{L}^*_j$ (Proposition 30.2(iii) and Theorem 30.5(ii)), and at $2j = c$, where $\mathcal{L}^*_j = \mathcal{L}_j$, its weights are those of its dual, $\theta^{(j)}_{\mathcal{E} \setminus X} = \theta^{(j)}_X$; so it is fusable, and its lifts toward its dual are the $\mathcal{L}_{j+i}$, $0 \leq i \leq c - 2j$ (Proposition 30.2(iii)). Its level cell is $\mathcal{B}^\sharp/C \setminus D$ with $|C| = |D| = j$ by (i), whose weights are paired by complement as those of $\mathcal{B}^\sharp$ are, so it is the Foreman fusion of $(\mathcal{L}_j, \theta^{(j)})$ with the middle weights read off $\theta^\sharp$. At $j = c/2$ the lift is identically self-dual; at $j = (c-1)/2$ the fusion width is one. □

Proposition 30.10 (The seam of the fusion bunch). *As stated.*

Proof. (i) $\mathrm{pat}(G) = 2^G$, $d(G) = 0$, $c(G) = c$ and level form are Proposition 30.2(ii) and Definition 30.1. A pattern Q of l marks has the child $(\mathcal{L}_{c-l}, \theta^{(c-l)})$, the same for all of them, so they form one station with the multipliers 1 (Definition 28.1); its station letters are $\sum_{|Q|=l} \alpha_G^Q = e_l(\alpha_G)$ and $\sum_{|Q|=l} \alpha_G^{G\setminus Q} = e_{c-l}(\alpha_G)$. A room side X of excess l is a basis of $\mathcal{L}_{c-l}$, its size fixing l, and its room child has the bases of $\binom{G}{l}$, each with the weight $\theta_X^{(c-l)}$: all the room children of the excess l are positive multiples of $\binom{G}{l}$, one station D_l, with the multipliers $w'_X = \theta_X^{(c-l)}/\mu_l$ and the room child of the station $\mu_l\binom{G}{l}$, of the letters $\mu_l e_l$ and $\mu_l e_{c-l}$. The room-station letters are $\sum_X w'_X \alpha^X = L_l/\mu_l$ and $\sum_X w'_X \alpha^{\mathcal{E}\setminus X} = \Psi_{\mathcal{L}_{c-l}}/\mu_l = L_{c-l}/\mu_l$ (Theorem 30.5(ii)), and the pair constant is $\theta_X^{(c-l)}/(1 \cdot w'_X) = \mu_l$. One station on each side at each excess is one rectangle, and the room-station factorization of Proposition 28.2(iii) is the stated one. The weight of $X \sqcup Q$ depends on X alone, $|X|$ fixing $|Q|$, so the weights factor across G and $\mathcal{E}$ and Theorem 28.28 applies: with one rectangle at every excess the table is its level envelope.

(ii) By Definition 12.18, with $K_{(Q)} = K_{\mathcal{L}_{c-|Q|}}$ and $\Psi_{(Q)} = \Psi_{\mathcal{L}_{c-|Q|}} = K_{\mathcal{L}_{|Q|}}$,

$$\iota_G = \sum_{l=0}^{c} \binom{c}{l} (-1)^{c-l} \log K_{\mathcal{L}_{c-l}}, \qquad \iota_G^* = \sum_{l=0}^{c} \binom{c}{l} (-1)^{l} \log K_{\mathcal{L}_l},$$

both equal to $\sum_j (-1)^j \binom{c}{j} \log K_{\mathcal{L}_j}$; at $c = 2$ this is $\log K_{\mathcal{L}_0} - 2\log K_{\mathcal{L}_1} + \log K_{\mathcal{L}_2} = -\log z_1$.

(iii) At the table's gauge the pattern probability of Q is $K_{(Q)}(\mathbf{1})/K^\sharp(\mathbf{1}) = L_{|Q|}(\mathbf{1})/K^\sharp(\mathbf{1})$ (Definition 12.25); the $\binom{c}{l}$ patterns of l marks give Π_l. $L_l(\mathbf{1})$ is the total weight of $(\mathcal{L}_{c-l}, \theta^{(c-l)})$, which equals that of its dual $(\mathcal{L}_l, \theta^{(l)})$, $L_{c-l}(\mathbf{1})$; so $\Pi_l = \Pi_{c-l}$. The peel (Definition 14.1) gives the child at Q the exponents $\pi_Q(\mathbf{1})(s,t)$, and merging the equal children of each excess gives $(\mathcal{L}_{c-l}; \Pi_l s, \Pi_l t)$, that is $(\mathcal{L}_j; \Pi_j s, \Pi_j t)$. Its word is $\prod_j K_{\mathcal{L}_j}^{-\Pi_j s} \Psi_{\mathcal{L}_j}^{-\Pi_j t} = \prod_j K_{\mathcal{L}_j}^{-\Pi_j s} \prod_j K_{\mathcal{L}_{c-j}}^{-\Pi_j t} = \prod_j K_{\mathcal{L}_j}^{-\Pi_j (s+t)}$, reindexing the second product by $j \mapsto c - j$. □

Definition 30.11 (The universal piece; the lift partner). As stated.

Proof. The seam letters e_l and e_{c-l} of the station D_l are nonzero, with nonnegative coefficients, homogeneous of the degrees $l = d + l$ and $c - l = d^* + c - l$ with $d = d^* = 0$, and positive letters where the degree is positive; the room letters $L_l = K_{\mathcal{L}_{c-l}}$ and L_{c-l} have the degrees $r + c - l = e + c - l$ and $r + l = e^* + l$ with $e = e^* = r$, as Definition 28.4 asks of a partner read with the reversed excesses. □

Theorem 30.12 (The glue of a fused table). *As stated.*

Proof. The family $((\mathcal{B}^\sharp, \theta^\sharp); 2\sigma, 0)$ has the word $K^{\sharp\,-2\sigma} = K^{\sharp\,-s} \Psi^{\sharp\,-t}$ (Theorem 30.5(ii)), and it is the glue of the universal piece and the lift partner with the one active letter K (Definition 28.4, Proposition 30.10(i)).

(i) In Definition 28.5 with the one active letter at the exponent 2σ: $\kappa_G = m_G$, $\kappa_G + \kappa_G^{\mathrm{top}} = 2\sigma c$, $V^{(0)} = e_0 = 1$ and $V^{(c)} = e_c$, so $\gamma_G^\rho = e_c^{1/c}$, the scale form is $\phi_G^\rho(du) = u^{\mathbf{m}|_G - \mathbf{1}} e_c(u)^{-m_G/c}\, d\omega_G(u)$, and the seam vector at D_l is $e_l/(\gamma_G^\rho)^l$. The form $\nu_G = \alpha^{\mathbf{m}|_G - \mathbf{1}} e_c(\alpha)^{-m_G/c}\, d\alpha$ on $(0,\infty)^G$ and the seam vector are homogeneous of degree zero. For a Borel $\psi \geq 0$ on the seam torus and $\mathbf{1}_{[1,e]}$ the indicator of $[1,e]$, with $\int_0^\infty \mathbf{1}_{[1,e]}(\rho)\, d\rho/\rho = 1$, the integral of $\psi(\boldsymbol{\varrho}_G^\rho(\alpha))\, \mathbf{1}_{[1,e]}(h(\alpha))\, \nu_G$ over the orthant is, for every positive h of degree one, the integral of $\psi(\boldsymbol{\varrho}_G^\rho)$ against the law: in the coordinates $\alpha = \rho u$ of scale and shape (Lemma 2.7) the form is $\mathbf{1}_{[1,e]}(\rho)\, d\rho/\rho$ times ϕ_G^ρ, the integrand not depending on ρ, and a change of the section h is a change of ρ along each ray, which leaves $\int \mathbf{1}_{[1,e]}(\rho)\, d\rho/\rho$ unchanged (the computation of Lemma 16.2, in $[0,\infty]$ by Tonelli). Take

$h = e_c^{1/c}$ and write $\alpha = \rho\, e^w$ with $\rho > 0$, $w \in H_G$. In the coordinates $y = \log\alpha = \lambda\mathbf{1} + w$, $\lambda = \log\rho$, the map $(\lambda, w_{g_1}, \dots, w_{g_{c-1}}) \mapsto y$, $w_{g_c} = -\sum_{i<c} w_{g_i}$, is linear with determinant $\pm c$: adding all rows to the last turns it into $(c, 0, \dots, 0)$. So $d\alpha = \alpha^{\mathbf{1}} c\, d\lambda\, d\lambda_G(w)$, and since $e_c(\rho e^w) = \rho^c$,

$$\nu_G = e^{m_G\lambda + \langle \mathbf{m}|_G, w\rangle}\, e^{-m_G\lambda}\, c\, d\lambda\, d\lambda_G(w) = c\, e^{\langle \mathbf{m}|_G, w\rangle}\, \frac{d\rho}{\rho}\, d\lambda_G(w),$$

while $\boldsymbol{\varrho}_G^{\rho}(\rho e^w) = \mathbf{e}(e^w)$. So the integral of ψ against the law is $c\int_{H_G} \psi(\mathbf{e}(e^w))\, e^{\langle \mathbf{m}|_G, w\rangle}\, d\lambda_G(w)$ for every ψ, which is the statement.

(ii) By Lemma 16.2, in $[0,\infty]$ by Tonelli, $P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t)$ is the integral over the orthant of $\alpha^{\mathbf{m}-\mathbf{1}} K^{\sharp\,-2\sigma} \mathbf{1}_{[1,e]}(h(\alpha))\, d\alpha$ with $h = e_c(\alpha_G)^{1/c}$. In the coordinates $(\alpha_{\mathcal{E}}, \rho, w)$, $\alpha_G = \rho e^w$, as in (i), $\alpha_G^{\mathbf{m}|_G - \mathbf{1}} d\alpha_G = c\, e^{\langle \mathbf{m}|_G, w\rangle} \rho^{m_G - 1} d\rho\, d\lambda_G(w)$ and $K^\sharp = \sum_l \rho^l e_l(e^w) L_l(\alpha_{\mathcal{E}})$ (Theorem 30.5(ii)). By Tonelli the integral is $c\int_{H_G} e^{\langle \mathbf{m}|_G, w\rangle} I(w)\, d\lambda_G(w)$ with

$$I(w) = \int_{(0,\infty)^{\mathcal{E}} \times (0,\infty)} \alpha_{\mathcal{E}}^{\mathbf{m}|_{\mathcal{E}} - \mathbf{1}} \rho^{m_G - 1} \Big(\sum_l e_l(e^w) \rho^l L_l(\alpha_{\mathcal{E}}) \Big)^{-2\sigma} \mathbf{1}_{[1,e]}(\rho)\, d\alpha_{\mathcal{E}}\, d\rho.$$

The letter $\sum_l x_l \alpha_g^l L_l$ is homogeneous of degree $r + c = r^*$ on $\mathcal{E} \sqcup \{g\}$, so the form of $W[\mathbf{x}]$ at the masses $(\mathbf{m}|_{\mathcal{E}}, m_G)$ has degree $m_{\mathcal{E}} + m_G - 2\sigma r^* = 0$ on the balance wall, and $I(w)$ is its integral against $\mathbf{1}_{[1,e]}$ of the section function α_g: by Lemma 16.2 it is $P_{W[\mathbf{e}(e^w)]}(\mathbf{m}|_{\mathcal{E}}, m_G)$.

(iii) $\sum_l x_l L_l y^l$ is a polynomial of degree c in y with positive coefficients, $x_c L_c \neq 0$, so its roots $-\xi_i$ avoid $[0,\infty)$; read as in Definition 12.10 on the bunch $\{g\}$, whose floor is 0, its Skeleton polynomial at $\alpha_g = 1$ is $\prod_i (x + \xi_i)/(1 + \xi_i)$. Non-real roots come in conjugate pairs, and for $\rho > 0$ the product of the principal powers $(\rho + \xi_i)^{-2\sigma}$ is the positive power of the product. Proposition 15.1 with $J = \{1, \dots, c\}$, the letter exponents $w_i = 2\sigma$, $|\mathbf{w}| = 2\sigma c$ and $\kappa = m_G$ gives the formula for $0 < m_G < 2\sigma c$, coincident values merging.

(iv) This is Theorem 28.10(iii) for the universal piece and the lift partner with the one active letter: one share $a_l = a_{D_l}$ per station, the letter total $\sum_l a_l = 2\sigma$ and the scale pin $\sum_l i(D_l) a_l = \kappa_G = m_G$, the share kernel $\Gamma_{\substack{c+1 \\ 1}} \begin{bmatrix} \mathbf{a} \\ 2\sigma \end{bmatrix}$, and the two seam functions, which on the pins are the periods of the seam word $\prod_l e_l^{-a_l}$ at $\mathbf{m}|_G$ and of the room word $\prod_l L_l^{-a_l}$ at $\mathbf{m}|_{\mathcal{E}}$ (Lemma 28.7), $e_0 = 1$ dropping out. Two pins on $c+1$ shares leave a fibre of dimension $c - 1$.

Let the masses be real and every proper wall of $\mathcal{B}^\sharp$ positive; we show that both periods converge at the native split at the stations. The walls of $((\mathcal{B}^\sharp, \theta^\sharp); 2\sigma, 0)$ are $\kappa_U = m_U - 2\sigma d^\sharp(U)$, with $d^\sharp(U) = r^* - r^\sharp(\mathcal{E}^\sharp \setminus U)$ on $\mathcal{E}^\sharp = \mathcal{E} \sqcup G$; at one mark, no mark being a coloop (Corollary 20.19), $d^\sharp$ is 0, so the masses are positive. A set U of the width zero in $\mathcal{B}^\sharp$ has $r^\sharp(U) + r^\sharp(\mathcal{E}^\sharp \setminus U) = r^*$, so $d^\sharp(U) = r^\sharp(U)$ and $d^\sharp(\mathcal{E}^\sharp \setminus U) = r^\sharp(\mathcal{E}^\sharp \setminus U)$, and $\kappa_U + \kappa_{\mathcal{E}^\sharp \setminus U} = m_{\mathcal{E}^\sharp} - 2\sigma r^* = 0$ on the balance wall; so no nonempty proper U has the width zero, $\mathcal{B}^\sharp$ is connected (Proposition 28.3(i)), and it has a native gauge (Theorem 8.19(iii)). By Lemma 28.9, $c(G) = c \geq 1$, the native split at the stations is a point of the pins with positive shares a_l at which $\mathbf{m}|_G$ and $\mathbf{m}|_{\mathcal{E}}$ lie in the relative interiors of the Newton sums of the seam word and of the room word, and the two words converge there once these Newton sums have the full dimensions $c - 1$ and $n - 1$; the seam letters $\mu_l e_l$ and the room-station letters L_l/μ_l of the bunch G (Proposition 30.10(i)) differ from e_l and L_l by constant factors, which change neither the Newton sums nor the convergence. The Newton sum $\sum_{l \geq 1} a_l \mathcal{N}_{e_l}$ of the seam word contains a translate of $a_1 \operatorname{conv}\{\mathbf{1}_{\{g\}} : g \in G\}$, of the dimension $c - 1$. The Newton sum $\sum_l a_l \mathcal{N}_{L_l}$ of the room word, $\mathcal{N}_{L_l}$ the base polytope of $\mathcal{L}_{c-l}$, has as its directions the sums of the directions of its terms, so a vector $w \in \mathbb{R}^{\mathcal{E}}$ orthogonal to them has $\langle w, \mathbf{1}_X \rangle$ constant over the bases X of every lift, and w is constant on every molecule of the family

of the lifts (Lemma 7.8). That family is connected. Otherwise a proper nonempty $S \subset \mathcal{E}$ is a union of its molecules, hence of molecules of every lift (Proposition 7.9), and has the width zero in $\mathcal{L}_0 = \mathcal{B}$ and in $\mathcal{L}_1$, whose rank function is $r(Y) + \min(-\varepsilon(Y), 1)$, $\varepsilon(Y) = r(Y) - r^*(Y) \le 0$ the slant of Y in $\mathcal{B}$ (Definition 7.12, Proposition 30.2(i)). The two widths give $r(S) + r(\mathcal{E} \setminus S) = r$ and $\min(-\varepsilon(S), 1) + \min(-\varepsilon(\mathcal{E} \setminus S), 1) = 1$, so $\varepsilon(S) = 0$ or $\varepsilon(\mathcal{E} \setminus S) = 0$, say $\varepsilon(S) = 0$, exchanging S and $\mathcal{E} \setminus S$ if needed. Then by Theorem 30.5(i) $r^\sharp(S) = \min(r^*(S), r(S) + c) = r(S)$ and $r^\sharp((\mathcal{E} \setminus S) \sqcup G) = \min(r^*(\mathcal{E} \setminus S) + c,\ r(\mathcal{E} \setminus S) + c) = r(\mathcal{E} \setminus S) + c$, whose sum is $r + c = r^*$: S has the width zero in $\mathcal{B}^\sharp$, which is excluded. So w is constant, the Newton sum of the room word has the dimension $n - 1$, and both words converge at the native split (Lemma 28.9, Theorem 25.2).

(v) $e_c(\alpha_G) = \alpha_G^{\mathbf{1}}$, so $\alpha_G^{\mathbf{m}|_G - \mathbf{1}} \prod_{l=1}^{c} e_l^{-a_l} = \alpha_G^{(\mathbf{m}|_G - a_c\mathbf{1}) - \mathbf{1}} \prod_{l=1}^{c-1} e_l^{-a_l}$: the two forms are one, and so are their periods; e_l is the letter of $\binom{G}{l}$. At $c = 1$ the pins give $a_1 = m_g$, and the form $\alpha_g^{m_g - 1} \alpha_g^{-a_1} d\alpha_g$ on one mark has the period 1. At $c = 2$, $u_1 + u_2 = 1$ on Δ_G, so the period of $e_1^{-a_1}$ at $(m_1 - a_2, m_2 - a_2)$ is $\int_{\Delta_G} u^{(\mathbf{m}|_G - a_2\mathbf{1}) - \mathbf{1}} d\omega_G = \Gamma_{\substack{2\\1}}\!\left[\begin{matrix} m_1 - a_2,\ m_2 - a_2 \\ a_1 \end{matrix}\right]$ (Theorem 3.9), $m_1 + m_2 - 2a_2 = a_1$ on the pins. □

Corollary 30.13 (Width one). *As stated.*

Proof. $\mathcal{B}^\sharp$ is identically self-dual (Theorem 30.5(ii)), and Corollary 20.19 at the mark g, which allows the exponents $(2\sigma, 0)$, gives $P_{\mathcal{B}^\sharp} = \Gamma_{\substack{2\\1}}\!\left[\begin{matrix} m_g,\ 2\sigma - m_g \\ 2\sigma \end{matrix}\right] P_{\mathcal{B}^\sharp \setminus g}(\mathbf{m}|_{\mathcal{E}}; 2\sigma - m_g, m_g)$. $\mathcal{B}^\sharp \setminus g = (\mathcal{B}^*, \theta^*)$, whose letters are Ψ and K (Proposition 7.2), so its period at $(2\sigma - m_g, m_g)$ is $P_{\mathcal{B}}(\mathbf{m}|_{\mathcal{E}}; m_g, 2\sigma - m_g)$. At $c = 1$, $H_G = \{0\}$ and the seam vector of the universal piece is $(1, 1)$; that of the lift partner, $\big(L_0/(L_1\gamma_{\mathcal{E}}),\ L_1/L_1\big)$ with $\gamma_{\mathcal{E}} = L_0/L_1$ (Definition 28.5, read with the reversed excesses), is $(1, 1)$; and the two pins fix $a_0 = 2\sigma - m_g$ and $a_1 = m_g$. □

Corollary 30.14 (Width two). *As stated.*

Proof. (i) Theorem 30.12(iv) and (v) at $c = 2$: the pins give $a_1 = m_1 + m_2 - 2a_2$ and $a_0 = 2\sigma - m_1 - m_2 + a_2$, solved for (a_0, a_1) by a matrix of determinant one, and the kernel times the universal factor is $\frac{\Gamma(a_0)\Gamma(a_1)\Gamma(a_2)}{\Gamma(2\sigma)} \cdot \frac{\Gamma(m_1 - a_2)\Gamma(m_2 - a_2)}{\Gamma(a_1)}$. (ii) Theorem 30.12(i) and (ii) at $c = 2$: $H_G = \{(w, -w)\}$ with $d\lambda_G = dw$, $v = e^w$, $\mathbf{e}(e^w) = (1, v + v^{-1}, 1)$ and $2e^{(m_1 - m_2)w} dw = 2v^{m_1 - m_2 - 1} dv$. (iii) $K^\sharp = A + \alpha_{g_2} B$ with $A = L_0 + \alpha_{g_1} L_1$ and $B = L_1 + \alpha_{g_1} L_2$, and $\int_0^\infty \rho^{\nu - 1}(A + B\rho)^{-2\sigma} d\rho = \Gamma_{\substack{2\\1}}\!\left[\begin{matrix} \nu,\ 2\sigma - \nu \\ 2\sigma \end{matrix}\right] A^{\nu - 2\sigma} B^{-\nu}$ for $0 < \nu < 2\sigma$ (Proposition 15.1) at $\nu = m_2$. With $\alpha_{g_1} = (L_0/L_1)x$, $A = L_0(1 + x)$ and $B = L_1(1 + z_1^{-1}x)$, which gives the display by Tonelli; the x-integral is $z_1^{m_2} \int_0^\infty x^{m_1 - 1}(x + 1)^{-(2\sigma - m_2)}(x + z_1)^{-m_2} dx$, the scale integral of Proposition 15.1 with the coefficients $1, z_1$, the letter exponents $2\sigma - m_2, m_2$ and $\kappa = m_1$, finite exactly for $0 < m_1 < 2\sigma$. At $m_1 = m_2 = 1$, $2\sigma = 2$, $\int_0^\infty \frac{dx}{(1+x)(1+x/z_1)} = \frac{z_1 \log z_1}{z_1 - 1}$ for $z_1 \ne 1$ and 1 for $z_1 = 1$, and $L_1^{-2} z_1 \log z_1/(z_1 - 1) = \log z_1/(L_1^2 - L_0 L_2)$. At integer data and $z_1 \ne 1$ the integrand is a rational function of x with poles at -1 and $-z_1$ only and of order at most x^{-2} at infinity, $m_1 < 2\sigma$; its partial fractions give an antiderivative $R(x) + A' \log(1 + x) + B' \log(1 + x/z_1)$ with R rational and $A' + B' = 0$, all coefficients rational in z_1, whose increment over $(0, \infty)$ is $R(\infty) - R(0) + A' \log z_1$ (receipt `t10b_weight_exact.py` for $m_1 = m_2 = k$, $2\sigma = 2k$, $k \le 4$). □

Lemma 30.15 (Products of lifts). *As stated.*

Proof. Let A be a basis of $\mathcal{L}_i$. Since $\mathcal{L}_i$ is a quotient of $\mathcal{L}_j$ (Proposition 30.2(iii)), A is independent in $\mathcal{L}_j$ (Proposition 30.2(i)) and lies in a basis C of $\mathcal{L}_j$. The monomial $\alpha^{\mathbf{1}_A+\mathbf{1}_C} = \alpha^{2\mathbf{1}_A+\mathbf{1}_{C\setminus A}}$ occurs in $K_{\mathcal{L}_i}K_{\mathcal{L}_j}$ with a positive coefficient, every coefficient being nonnegative, and exactly $|A| = r_{\mathcal{N}}+i$ of its exponents equal 2. A monomial $\alpha^{\mathbf{1}_{A'}+\mathbf{1}_{C'}}$ of $K_{\mathcal{L}_{i'}}K_{\mathcal{L}_{j'}}$ has the exponent 2 exactly on $A'\cap C'$, at most $|A'| = r_{\mathcal{N}}+i' < r_{\mathcal{N}}+i$ marks. Both products are nonzero with nonnegative coefficients, so a positive constant multiple of one has the support of the one, and they are not proportional. The ratio is the case $(i',i,j,j') = (j-1,j,j,j+1)$, $0<j<c$; a constant ratio would make $K_{\mathcal{L}_j}^2 - \lambda K_{\mathcal{L}_{j-1}}K_{\mathcal{L}_{j+1}}$ vanish on the orthant, hence identically. □

Theorem 30.16 (The price of width). *As stated.*

Proof. (i) By Theorem 30.12(i) the universal law is the image of a measure with a positive continuous density on H_G, of dimension $c-1$, under the continuous map $w \mapsto \mathbf{e}(e^w)$. At $c=1$ it is the unit point mass at $(1,1)$. At $c\geq 2$, $e_1(e^w) = \sum_g e^{w_g}$ is c at $w=0$ and larger at every other point of H_G by the inequality of the arithmetic and geometric means, $\prod_g e^{w_g} = 1$; a law carried by one point would make it constant almost everywhere, hence everywhere by continuity. The law of the lift partner is the image of its scale form, which has a positive continuous density on the open simplex $\Delta_{\mathcal{E}}$, under its continuous seam vector; it is carried by one point exactly when the seam vector is constant. Read with the reversed excesses, $V^{(0)} = L_c$, $V^{(c)} = L_0$, the scale is $\gamma_{\mathcal{E}} = (L_0/L_c)^{1/c}$ and the seam vector at D_l is $L_l/(L_c\gamma_{\mathcal{E}}^{c-l}) = L_l L_0^{-(c-l)/c} L_c^{-l/c}$ (Definition 28.5). Put $f_l = \log L_l - \log L_0 - \frac{l}{c}(\log L_c - \log L_0)$, so that $f_0 = f_c = 0$ and the second differences of $(f_l)_l$ are those of $(\log L_l)_l$, which are $-\log z_{c-l}$, since $L_l = K_{\mathcal{L}_{c-l}}$. The seam vector is constant exactly when every f_l is constant; then every lift ratio is constant, and conversely constant second differences with $f_0 = f_c = 0$ determine f_l as constants. At $c=1$ there is no lift ratio and the seam vector is $(1,1)$; at $c\geq 2$ Lemma 30.15 excludes constant lift ratios.

(ii) For $a\in\mathbb{C}^c$ and a set S of l indices, $\pi(S)$ runs over the sets of l indices as π runs over S_c, each reached $l!\,(c-l)!$ times, so

$$\sum_{\pi\in S_c}\prod_{i=1}^{c}\bigl(1+a_i\eta_{\pi(i)}\bigr) = \sum_S a^S \sum_\pi \prod_{i\in S}\eta_{\pi(i)} = \sum_{l=0}^{c} l!\,(c-l)!\,e_l(a)\,e_l(\boldsymbol{\eta}),$$

and the left side is $\operatorname{per}(\mathbf{1}\mathbf{1}^{\mathsf{T}}+a\boldsymbol{\eta}^{\mathsf{T}})$ by the definition of the permanent. With $L_l = L_0 e_l(\boldsymbol{\eta})/\binom{c}{l}$, read off $\sum_l \binom{c}{l}L_l x^l = L_0\prod_i(1+\eta_i x)$, division by $c!$ gives $\sum_l e_l(\alpha_G)L_l = K^\sharp$. The Skeleton polynomial of G at $(\mathbf{1}_G,\alpha_{\mathcal{E}})$ is $K^\sharp(x\mathbf{1}_G,\alpha_{\mathcal{E}})/K^\sharp(\mathbf{1}_G,\alpha_{\mathcal{E}}) = \sum_l\binom{c}{l}L_l x^l/K^\sharp$, $d(G)=0$, and with $L_0>0$ its roots $-\xi_i$ give $\sum_l\binom{c}{l}L_l x^l = L_0\prod_i(1+x/\xi_i)$: $\eta_i = 1/\xi_i$. At a gauge all η_i equal η exactly when $\sum_l\binom{c}{l}L_l x^l = L_0(1+\eta x)^c$, that is $L_l = L_0\eta^l$ for every l, which holds exactly when $L_l^2 = L_{l-1}L_{l+1}$ for $0<l<c$ (then $L_l = L_0(L_1/L_0)^l$ by induction), that is every lift ratio is 1. Coincidence on an open set would make $K_{\mathcal{L}_j}^2 - K_{\mathcal{L}_{j-1}}K_{\mathcal{L}_{j+1}}$ vanish there and so identically, against Lemma 30.15; and $K^\sharp = L_0\prod_g(1+\alpha_g\eta) = L_0\sum_l e_l(\alpha_G)\eta^l$ would give $L_l = L_0\eta^l$ everywhere.

(iii) By Proposition 30.8(ii) the partial fusion $\mathcal{A}$ at G' is the level cell of $(\mathcal{L}_i,\mathcal{B})$ on $\mathcal{E}\sqcup G'$, in which g lies in some bases and avoids others, so it is a cell with the hinge g; by Proposition 30.2(iv) its hinge minors are $\mathcal{A}/g$, the level cell of $(\mathcal{L}_{i-1},\mathcal{B})$, and $\mathcal{A}\setminus g$, the level cell of $(\mathcal{L}_i,\mathcal{L}_1)$, both on $\mathcal{E}\sqcup(G'-g)$. At $i=c$, $\mathcal{A} = \mathcal{B}^\sharp$ is identically self-dual, $\mathcal{A}/g$ is the dual of $\mathcal{A}\setminus g$ with the same weights (as in the proof of Corollary 30.7(i)), so $K_{\mathcal{A}/g} = \Psi_{\mathcal{A}\setminus g}$ and $\Psi_{\mathcal{A}/g} = K_{\mathcal{A}\setminus g}$ (Proposition 7.2) and the ratio is 1; and Corollary 20.19 at g, with $P_{\mathcal{A}\setminus g}$ at $(2\sigma-m_g,m_g)$ equal to $P_{\mathcal{A}/g}$ at $(m_g,2\sigma-m_g)$, gives the display. Let $i<c$. The letters of a level cell of $(\mathcal{M}',\mathcal{N}')$ on $\mathcal{E}\sqcup G''$ are $K = \sum_Q \alpha^Q K_{\mathcal{L}'_{c''-|Q|}}$ and $\Psi = \sum_Q \alpha^{G''\setminus Q}\Psi_{\mathcal{L}'_{c''-|Q|}}$, which at $\alpha_{G''}=0$ are $K_{\mathcal{M}'}$ and $\Psi_{\mathcal{N}'}$. So at $\alpha_{G'-g}=0$ the even

letters of the hinge minors are

$$\mathbf{E}_{\mathcal{A}/g} = K_{\mathcal{L}_{i-1}}\Psi_{\mathcal{B}} = K_{\mathcal{L}_{i-1}}K_{\mathcal{L}_c}, \qquad \mathbf{E}_{\mathcal{A}\setminus g} = K_{\mathcal{L}_i}\Psi_{\mathcal{L}_1} = K_{\mathcal{L}_i}K_{\mathcal{L}_{c-1}}.$$

A constant ratio $\varrho_{\mathcal{A}} = \mathbf{E}_{\mathcal{A}/g}/\mathbf{E}_{\mathcal{A}\setminus g}$ on the orthant would make $\mathbf{E}_{\mathcal{A}/g} - \lambda\mathbf{E}_{\mathcal{A}\setminus g}$ vanish identically, also at $\alpha_{G'-g} = 0$, against Lemma 30.15 with $(i', i, j, j') = (i-1, i, c-1, c)$. The law of the cell is the image of its base form, with a positive continuous density on the open simplex, under the continuous ratio, and it is carried by one point only if the ratio is constant (Corollary 23.12(i)). □

Proposition 30.17 (The free extensions by one mark). *As stated.*

Proof. Write $K_\backslash, \Psi_\backslash$ and $K_/, \Psi_/$ for the letters of $\mathcal{A}\setminus g$ and $\mathcal{A}/g$, of the degrees $r_{\mathcal{A}}$, $|\mathcal{E}| - r_{\mathcal{A}}$, $r_{\mathcal{A}} - 1$ and $|\mathcal{E}| - r_{\mathcal{A}} + 1$; $\mathcal{A}$ being plain, each hinge minor carries one system of weights in its two letters. If $\mathcal{A}/g$ is the dual of $\mathcal{A}\setminus g$ with its weights multiplied by μ, then $K_/ = \mu\Psi_\backslash$ and $\Psi_/ = \mu K_\backslash$ (Proposition 7.2), and $\varrho_{\mathcal{A}} = \mathbf{E}_{\mathcal{A}/g}/\mathbf{E}_{\mathcal{A}\setminus g} = \mu^2$. Conversely let $\varrho_{\mathcal{A}}$ be a constant λ. Connected tables on at least two marks are single molecules (Proposition 28.3(i)), so their letters are irreducible (Theorem 7.11), and $K_/\Psi_/ = \lambda K_\backslash\Psi_\backslash$ matches the irreducible factors on the two sides up to constant factors. $K_/$ and $K_\backslash$ have different degrees, so $K_/ = \mu\Psi_\backslash$ with a constant μ, positive since both have positive coefficients; comparing monomials, the bases of $\mathcal{A}/g$ are the complements in $\mathcal{E}$ of the bases T of $\mathcal{A}\setminus g$, with $\theta_{(\mathcal{E}\setminus T)+g} = \mu\theta_T$. The degrees give $r_{\mathcal{A}} - 1 = |\mathcal{E}| - r_{\mathcal{A}}$, so $\mathcal{A}/g$ has rank $r_{\mathcal{A}} - 1$ on $2r_{\mathcal{A}} - 1$ marks, the fusion width one; it is fusable by Theorem 30.6(d), $\mathcal{A}$ having $\mathcal{A}/g$ and $\mathcal{A}\setminus g$, with the bases of $(\mathcal{A}/g)^*$, as its children at the full and at the empty pattern of $\{g\}$. By Theorem 30.5(v) its fusion has the letter $K^\sharp = \Psi_/ + \alpha_g K_/ = \mu K_\backslash + \alpha_g K_/$, and $K^\sharp(\alpha_{\mathcal{E}}, \mu\alpha_g) = \mu K_\backslash + \mu\alpha_g K_/ = \mu K_{\mathcal{A}}(\alpha_{\mathcal{E}}, \alpha_g)$: $\mathcal{A}$ has the bases of that fusion, the weights $\theta_{T'+g}$ on the bases $T' + g$ and $\theta_{T'+g}/\mu$ on the bases $\mathcal{E}\setminus T'$, T' a basis of $\mathcal{A}/g$. For a list every coefficient is 1, so $\mu = 1$: the complement in $\mathcal{E}\sqcup\{g\}$ of a basis T avoiding g is the basis $(\mathcal{E}\setminus T) + g$ and conversely, $\mathcal{A}$ is identically self-dual, $\mathcal{A}\setminus g = (\mathcal{A}/g)^*$, and $\mathcal{A}$ is the only major of $\mathcal{A}/g$, its fusion; an identically self-dual $\mathcal{A}$ has the ratio 1, its hinge minors being dual with the same weights. At exponents $s, t > 0$ a ratio is constant exactly when the law is carried by one point, the base form having a positive continuous density on the open simplex (Corollary 23.12(i)). □

Corollary 30.18 (A pinned set of one list is a dual pair). *As stated.*

Proof. By Theorem 28.25, S is pinned exactly when it has width one and $\varrho_S = 1$. The cell of the bunch $\mathcal{B}_S^+$ is a list with the hinge minors $\mathcal{B}|S$ and $\mathcal{B}/S'$ (19.4) and the ratio ϱ_S (21.1). Its hinge minors being connected, its ratio is constant exactly when it is identically self-dual (Proposition 30.17), that is a dual pair (Proposition 21.8(i)), and then the ratio is 1 (Proposition 21.8(ii)). So $\varrho_S = 1$ exactly when $\mathcal{B}_S^+$ is a dual pair, that is $\mathcal{B}/S' = (\mathcal{B}|S)^*$. □

Proposition 30.20 (Cousins). *As stated.*

Proof. (i) is Corollary 30.7(i) with $C = \{x\}$, and its proof gives $\mathcal{B}^\sharp\setminus x = (\mathcal{B}^\sharp/x)^*$; for $x \in G$, $\mathcal{B}^\sharp/x$ is the partial fusion at $G - x$, and at $c = 1$, $\mathcal{B}^\sharp/g = \mathcal{B}$. (ii) Corollary 20.19 at x, which allows every pair of exponents with $s + t = 2\sigma$, gives $P_{\mathcal{B}^\sharp} = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} m_x,\ 2\sigma - m_x \\ 2\sigma \end{bmatrix} P_{\mathcal{B}^\sharp\setminus x}(\mathbf{m}_{-x}; 2\sigma - m_x, m_x)$, and the dual $\mathcal{B}^\sharp\setminus x$ of $\mathcal{B}^\sharp/x$ has the letters exchanged (Proposition 7.2); the same holds at y. (iii) $K^\sharp(\beta^*\alpha) = K^\sharp(\alpha)$, and the change of coordinates $u \mapsto \beta^* u$ permutes the coordinates of $\Delta_{\mathcal{E}\sqcup G}$ and keeps $d\omega$, so the period at $\beta^*\mathbf{m}$ is the period at $\mathbf{m}$. At $c = 1$ Corollary 30.13 writes $P_{\mathcal{B}^\sharp}$ on its balance wall, where every proper wall is positive, as the stated function of $(\mathbf{m}|_{\mathcal{E}}, s')$, $s' = m_g$; an automorphism permutes the proper walls, $\kappa_U(\beta^*\mathbf{m}) = \kappa_{\beta(U)}(\mathbf{m})$, so it keeps that part of the

hyperplane, and the function inherits the invariance; at $\beta^*\mathbf{m}$ the mass of g is $m_{\beta(g)}$ and the mass of $\beta^{-1}(g)$ is $m_g = s'$. □

Theorem 30.23 (Fusion across a two-sum). *As stated.*

Proof. (i) $\mathcal{S}$ has the rank $r_1 + r_2 - 1$ on $n_1 + n_2 - 2$ marks (Lemma 21.12), so the fusion width $c_1 + c_2$. By Lemma 21.2 its bases are the $X \sqcup Y$ with X a basis of $\mathcal{A}_1 \setminus g$ and Y of $\mathcal{A}_2/g$, or X of $\mathcal{A}_1/g$ and Y of $\mathcal{A}_2 \setminus g$; since independent sets of a table extend to bases inside any set that contains them and has full rank, a set $I_1 \sqcup I_2$ is independent exactly when I_1 and $I_2 + g$ are independent in $\mathcal{A}_1$ and $\mathcal{A}_2$, or $I_1 + g$ and I_2 are. A dependent set inside $\mathcal{E}_1$ is dependent in $\mathcal{A}_1$, $\{g\}$ being independent in $\mathcal{A}_2$, so the circuits inside $\mathcal{E}_1$ are the circuits of $\mathcal{A}_1$ avoiding g, and likewise for $\mathcal{E}_2$. A circuit $C_1' \sqcup C_2'$ meeting both sides contains no such circuit, so C_1' and C_2' are independent; it is dependent exactly when $C_1' + g$ and $C_2' + g$ are dependent, that is when g lies in the closures of C_1' and of C_2'; and removing $e \in C_1'$ leaves an independent set exactly when $C_1' - e + g$ is independent, $C_2' + g$ being dependent. So the minimal ones are those with $C_1' + g$ and $C_2' + g$ circuits, and every such set is a circuit. The complement of $X \sqcup Y$ is $(\mathcal{E}_1 \setminus X) \sqcup (\mathcal{E}_2 \setminus Y)$, and $(\mathcal{A}_1 \setminus g)^* = \mathcal{A}_1^*/g$, $(\mathcal{A}_2/g)^* = \mathcal{A}_2^* \setminus g$, so $\mathcal{S}^* = \mathcal{A}_1^* \oplus_g \mathcal{A}_2^*$, g being neither a loop nor a coloop of $\mathcal{A}_i^*$; its circuits, the cocircuits of $\mathcal{S}$, are read off the same description.

(ii) A cocircuit D of $\mathcal{A}_1$ avoiding g contains no circuit of $\mathcal{A}_2$ and no set $(C_1 \setminus g) \cup (C_2 \setminus g)$, $C_2 \setminus g$ being a nonempty part of $\mathcal{E}_2$; so it is a union of circuits of $\mathcal{S}$ exactly when it is one of $\mathcal{A}_1$, and these cocircuits answer a_1; likewise a_2. For $D = (D_1 \setminus g) \cup (D_2 \setminus g)$ and $e \in D_1 \setminus g$, the circuits of $\mathcal{S}$ inside D through e are the circuits of $\mathcal{A}_1$ inside D_1 avoiding g through e and the sets $(C_1 \setminus g) \cup (C_2 \setminus g)$ with $e \in C_1 \subseteq D_1$ and $C_2 \subseteq D_2$ circuits through g. So e is covered in D exactly when D_1 covers it by a circuit avoiding g, or by one through g while D_2 covers g. D_1 and D_2 run independently over the cocircuits through g. If h_2 holds, the marks of the side $\mathcal{E}_1$ are covered for every D_2 exactly when u_1; if h_2 fails, some D_2 does not cover g, and they are covered exactly when s_1, which implies u_1. So the side $\mathcal{E}_1$ asks u_1 and (s_1 or h_2), the side $\mathcal{E}_2$ asks u_2 and (s_2 or h_1), and Theorem 30.6(b) gives the rule. For one cell, its cocircuits avoiding g answer a, and those through g are unions of circuits exactly when their marks other than g are covered, u, and g is, h. Two fusable cells have a_i, u_i, h_i, and the rule holds.

(iii) By the bases of (i), $r_{\mathcal{S}}(\mathcal{E}_2) = r_2$ and $d_{\mathcal{S}}(\mathcal{E}_2) = r_2 - 1$, so $r^*_{\mathcal{S}}(\mathcal{E}_2) = |\mathcal{E}_2| - d_{\mathcal{S}}(\mathcal{E}_2) = n_2 - r_2$. A fusable $\mathcal{S}$ has $r_{\mathcal{S}} \le r^*_{\mathcal{S}}$ pointwise, $\mathcal{S}$ being a quotient of $\mathcal{S}^*$ (Proposition 30.2(i)): $r_2 \le n_2 - r_2$, $c_2 \ge 0$, and likewise $c_1 \ge 0$. In $\mathcal{S}^\sharp$, $r^\sharp(\mathcal{E}_2) = r^*_{\mathcal{S}}(\mathcal{E}_2) = n_2 - r_2$, the rank of a set avoiding G being its rank in $\mathcal{S}^\sharp \setminus G = \mathcal{S}^*$. An identically self-dual table has $r^*(A) = r(A)$, so $d(A) = |A| - r(A)$ and $c(A) = 2r(A) - |A|$, which for $A = \mathcal{E}_2$ is $2(n_2 - r_2) - (n_2 - 1) = c_2 + 1$; likewise for $\mathcal{E}_1$.

(iv) In $\mathcal{A}_i^\sharp$ the mark g lies in a basis $X \sqcup G_i$, X a basis of $\mathcal{A}_i$ through g, and avoids a basis X of $\mathcal{A}_i^*$, so $\mathcal{A}_i^\sharp$ is a cell with the hinge g and $W = \mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2^\sharp$ is a table on $\mathcal{E}_1 \sqcup \mathcal{E}_2 \sqcup G_1 \sqcup G_2$. By (i) $W^* = \mathcal{A}_1^{\sharp *} \oplus_g \mathcal{A}_2^{\sharp *} = W$. Contracting $G = G_1 \sqcup G_2$ commutes with the minors at g, marks of different sets (Definition 7.5), so the bases of W containing G, less G, are those of $(\mathcal{A}_1^\sharp/G_1) \oplus_g (\mathcal{A}_2^\sharp/G_2) = \mathcal{S}$, and likewise $W \setminus G = \mathcal{A}_1^* \oplus_g \mathcal{A}_2^* = \mathcal{S}^*$: W is an identically self-dual major. If $c_2 = 0$, $\mathcal{A}_2$ is a quotient of $\mathcal{A}_2^*$ of the same rank, so the two rank functions agree (Proposition 30.2(i) at $X \subseteq Y$ and at $Y \subseteq \mathcal{E}$) and $\mathcal{A}_2 = \mathcal{A}_2^* = \mathcal{A}_2^\sharp$; every transposition of two marks of G_1 carries $\mathcal{A}_1^\sharp$, hence W, and $W = \mathcal{S}^\sharp$ by Theorem 30.5(iv); likewise for $c_1 = 0$. Let $c_1, c_2 \ge 1$, $x_1 \in G_1$, $x_2 \in G_2$, and let $r_1^* = n_1 - r_1$, the rank of $\mathcal{A}_1^\sharp$. A basis $Z_1 \sqcup Z_2$ of W has $|Z_1| \in \{r_1^*, r_1^* - 1\}$. At the pattern $\{x_1\}$ of G, $|Z_1 \cap \mathcal{E}_1| = |Z_1| - 1 \le r_1^* - 1$, with equality for $Z_1 = X_1 \sqcup \{x_1\}$, X_1 a basis of $\mathcal{L}_{c_1-1}(\mathcal{A}_1^*, \mathcal{A}_1)$ avoiding g, and $Z_2 = X_2 - g$, X_2 a basis of $\mathcal{A}_2^*$ through g; such bases exist, g being independent in $\mathcal{A}_1$ and so in its lift $\mathcal{L}_1 = \mathcal{L}^*_{c_1-1}$, and not a loop of $\mathcal{A}_2^*$. At the pattern $\{x_2\}$, Z_1 a basis of $\mathcal{A}_1^*$

avoiding g and $Z_2 = (X_2 \sqcup \{x_2\}) - g$, X_2 a basis of $\mathcal{L}_{c_2-1}(\mathcal{A}_2^*, \mathcal{A}_2)$ through g, give $|Z_1 \cap \mathcal{E}_1| = r_1^*$. So the two children have the ranks $r_1^* - 1$ and r_1^* on $\mathcal{E}_1$, they differ, and G is not in level form in W, while it is in $\mathcal{S}^\sharp$.

(v) If both cells are fusable, $\mathcal{S}$ is fusable by (ii), and for $c_2 = 0$ the fusable list $\mathcal{A}_2$ of the width zero equals its dual, as in (iv), so it is identically self-dual, and $\mathcal{S}^\sharp = \mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2$ by (iv). Conversely let $\mathcal{S}$ be fusable; by (iii) $\{c_1, c_2\} = \{0, 1\}$; let $c_2 = 0$. By (iii) $\mathcal{E}_2$ is a bunch of width one of $\mathcal{S}^\sharp$, a list, and by Theorem 19.4 and Proposition 21.3(iii) $\mathcal{S}^\sharp$ is the two-sum of the cell $\mathcal{N}_2$ of the bunch $\mathcal{E}_2$ and the cell $\mathcal{N}_1$ of its room $\mathcal{E}_1 \sqcup G$, along a hinge h. The cell of $\mathcal{E}_2$ in the dual table has the bases $\mathcal{S}^{\sharp *}|\mathcal{E}_2 = (\mathcal{S}^\sharp/(\mathcal{E}_1 \sqcup G))^*$ and $\mathcal{S}^{\sharp *}/(\mathcal{E}_1 \sqcup G) + h = (\mathcal{S}^\sharp|\mathcal{E}_2)^* + h$, the complements in $\mathcal{E}_2 + h$ of the bases of $\mathcal{N}_2$; $\mathcal{S}^\sharp$ being identically self-dual, $\mathcal{N}_2$ is. In $\mathcal{S} = \mathcal{S}^\sharp/G$ the cell of $\mathcal{E}_2$ is $\mathcal{A}_2$ (Lemma 21.2); its bases through the hinge, $\mathcal{S}/\mathcal{E}_1 = \mathcal{S}^\sharp/(\mathcal{E}_1 \sqcup G)$, are those of $\mathcal{N}_2$, and its bases avoiding the hinge are those of $(\mathcal{S}^\sharp/G)|\mathcal{E}_2$. Here $r^\sharp(\mathcal{E}_2) = n_2 - r_2 = r_2$ by (iii) and $c_2 = 0$, while $r^\sharp(\mathcal{E}_2 \sqcup G) = r_{\mathcal{S}}(\mathcal{E}_2) + |G| = r_2 + |G|$; by submodularity every independent $I \subseteq \mathcal{E}_2$ of $\mathcal{S}^\sharp$ has $r^\sharp(I \sqcup G) \geq |I| + r^\sharp(\mathcal{E}_2 \sqcup G) - r^\sharp(\mathcal{E}_2) = |I| + |G|$, so $(\mathcal{S}^\sharp/G)|\mathcal{E}_2 = \mathcal{S}^\sharp|\mathcal{E}_2$ and $\mathcal{A}_2 = \mathcal{N}_2$, identically self-dual. The cell of the room $\mathcal{E}_1$ in $\mathcal{S}$ has the bases of $\mathcal{S}|\mathcal{E}_1 = (\mathcal{S}^\sharp|(\mathcal{E}_1 \sqcup G))/G$ and of $\mathcal{S}/\mathcal{E}_2 + h = (\mathcal{S}^\sharp/\mathcal{E}_2)/G + h$, so $\mathcal{A}_1 = \mathcal{N}_1/G$; $\mathcal{N}_1$ is identically self-dual by the same argument as $\mathcal{N}_2$, so $\mathcal{N}_1 \setminus G = (\mathcal{N}_1/G)^*$ and $\mathcal{N}_1$ is a major of $\mathcal{A}_1$: $\mathcal{A}_1$ is fusable (Theorem 30.6), of the width $|G| = 1$, and $\mathcal{N}_1 = \mathcal{A}_1^\sharp$ (Theorem 30.5(v)). So $\mathcal{S}^\sharp = \mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2$, as (iv) also gives: $\mathcal{S}^\sharp$ is the two-sum along the hinge g of $\mathcal{S}$, with $\mathcal{E}_2$ a bunch of width one. □

Corollary 30.24 (A two-sum with an identically self-dual cell). *As stated.*

Proof. Every proper wall of $\mathcal{S}$ positive makes $P_{\mathcal{S}}$ finite (Theorem 8.7); κ and $2\sigma - \kappa$ are the walls of $\mathcal{S}$ at $\mathcal{E}_1$ and $\mathcal{E}_2$ (Lemma 21.12), so $0 < \kappa < 2\sigma$. $\mathcal{A}_1$ is identically self-dual, so $\mathcal{A}_1/g$ is the dual of $\mathcal{A}_1 \setminus g$ with the same weights, its ratio is 1, and $\phi_{\mathcal{A}_1}(\Delta_{\mathcal{E}_1}) = P_{\mathcal{A}_1 \setminus g}(\mathbf{m}|_{\mathcal{E}_1}; \kappa, 2\sigma - \kappa)$ (Corollary 23.12(ii)), finite since $\mathcal{A}_1$ sits in a gluing of finite period. By Corollary 23.12(i) $\mathcal{A}_1$ acts in $\mathcal{S}$ as this number times the one-mark cell of ratio 1 and mass κ, whose two-sum with $\mathcal{A}_2$ is $\mathcal{A}_2$ with its hinge carrying κ: $P_{\mathcal{S}} = \phi_{\mathcal{A}_1}(\Delta_{\mathcal{E}_1})\, P_{\mathcal{A}_2}(\mathbf{m}|_{\mathcal{E}_2}, \kappa; s, t)$. The Beta corner of Corollary 23.12(iii) at g with the mass $2\sigma - \kappa$, which puts $\mathcal{A}_1$ on its balance wall (Lemma 21.12), is the integral over the scale of g, valid in $[0, \infty]$ for $0 < \kappa < 2\sigma$ by Tonelli: $P_{\mathcal{A}_1}(\mathbf{m}|_{\mathcal{E}_1}, 2\sigma - \kappa; s, t) = \Gamma_{\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}} \begin{bmatrix} 2\sigma - \kappa, & \kappa \\ & 2\sigma \end{bmatrix} P_{\mathcal{A}_1 \setminus g}(\mathbf{m}|_{\mathcal{E}_1}; \kappa, 2\sigma - \kappa)$. When the cells are lists and $\mathcal{A}_2$ is fusable of the fusion width one, $\mathcal{A}_1$, identically self-dual, is fusable of the width zero, and Theorem 30.23(v), with the roles of the two cells exchanged, gives $\mathcal{S}^\sharp = \mathcal{A}_1 \oplus_g \mathcal{A}_2^\sharp$. □

Proposition 30.25 (Uncovered marks and repair). *As stated.*

Proof. (i) A cocircuit is a union of circuits exactly when it covers each of its marks, so $\mathcal{B}$ has no uncovered mark exactly when (b) of Theorem 30.6 holds, equivalently (a). By that theorem a list, or a table of the width $c \geq 1$, is fusable exactly when (a) holds; at $c < 0$ neither (a) holds nor is the table fusable; and at $c = 0$ the table is fusable exactly when it is identically self-dual (Definition 30.4). (ii) By Theorem 21.6(i) and (ii), under the first rule alone $\mathcal{B}_S$ has $n + |S|$ marks, the rank r and the rank function $Y \mapsto r(\lceil Y \rceil)$, $\lceil Y \rceil$ the marks of $\mathcal{B}$ whose blocks meet Y. So Y is independent exactly when it meets each block in at most one mark and $\lceil Y \rceil$ is independent: the circuits of $\mathcal{B}_S$ are the pairs $\{h, h'\}$, $h \in S$, and the sets meeting each block in at most one mark whose shadow $\lceil \cdot \rceil$ is a circuit of $\mathcal{B}$. A closed set contains with a mark its whole block, adding a mark of a block already met not raising the rank; so the closed sets are the unions of the blocks over the closed sets of $\mathcal{B}$, and since the cocircuits are the complements of the closed sets of rank $r - 1$ (fact (β) in the proof of Theorem 30.6) the cocircuits of $\mathcal{B}_S$ are $D^+ = D \cup \{h' : \ h \in D \cap S\}$,

D a cocircuit of $\mathcal{B}$. In D^+ the marks h, h' of a block are covered by $\{h, h'\}$, and a mark $e \in D \setminus S$ is covered exactly when a circuit of $\mathcal{B}_S$ through e lies in D^+, that is when a circuit of $\mathcal{B}$ through e, its shadow, lies in D. So $\mathcal{B}_S$ has no uncovered mark exactly when every mark of $\mathcal{E} \setminus S$ is covered in every cocircuit of $\mathcal{B}$ through it: when S contains the uncovered marks; and by (i) for $\mathcal{B}_S$, a list or a table of the width $c + |S| \neq 0$, it is then fusable exactly when it has no uncovered mark. Its fusion width is $n + |S| - 2r$. The bases of $\mathcal{B}_S$ avoiding S' are the bases T of $\mathcal{B}$, each with its weight θ_T (Theorem 21.5), so $(\mathcal{B}_S, \theta) \setminus S' = (\mathcal{B}, \theta)$. (iii) $(\mathcal{B}_S)^\sharp/G = (\mathcal{B}_S, \theta)$ and $(\mathcal{B}_S, \theta) \setminus S' = (\mathcal{B}, \theta)$, the two minors being taken at disjoint sets. Let $c \geq 0$, $(\mathcal{B}, \theta)$ not fusable, and U its set of uncovered marks, none a loop, since a loop lies in no cocircuit. If $U \neq \varnothing$, $S = U$ gives the width $c + |U| \geq 1$ and no uncovered mark, and $\mathcal{B}_U$ is fusable by (ii). If $U = \varnothing$, (i) leaves $c = 0$ and $(\mathcal{B}, \theta)$ not identically self-dual, while $\mathcal{B} = \mathcal{B}^*$ by (a) and the proof of Definition 30.4; a loop of $\mathcal{B}$ would be a coloop of $\mathcal{B}^* = \mathcal{B}$, so there is none, and $\mathcal{E} \neq \varnothing$, since on no marks the one basis $\varnothing$ is its own complement with its own weight. For any mark h, $S = \{h\}$ gives the width one and no uncovered mark, and $\mathcal{B}_S$ is fusable by (ii). For a list $\mathcal{B}$, the set of the marks that are not loops contains U, and the list $\mathcal{B}_S$ it doubles has no uncovered mark and is fusable by (i). (iv) The copy h' is neither a loop nor a coloop, so its wall at $t = 0$ is $\kappa_{\{h'\}} = m_{h'}$. Theorem 16.25 at the bunch $\{h'\}$, whose bottom level holds the bases avoiding h', with the empty pattern alone, so that the weights factor on it, gives the residue as the product of the periods of the faces: on $\{h'\}$ the contraction of the room, of rank $d(\{h'\}) = 0$ and letter 1, whose period on one mark is 1; on the room, $(\mathcal{B}_S|(\mathcal{E}_S \setminus h'); s, 0) = (\mathcal{B}_{S-h}; s, 0)$. □

48.31 Chapter 31: closing a glue along a direction

Theorem 31.2 (Closing the glue along a direction). *As stated, the largest values of the monomials of the letter vector of the partner being read on its law when its room word has a Newton sum of full dimension.*

Proof. Apart from the sentence on the partner, the statement is the reference's, with its proof (the reference, proof of thm:closecone). Let the room word of the partner have a Newton sum of full dimension. The partner is a piece on R with the stations $\mathcal{D}$, read with the reversed excesses $c_j - i(D)$ (28.4); its scale γ_R^ρ and its seam vector $\boldsymbol{\varrho}_R^\rho = (\varrho_{R,jD}^\rho, \varrho_{R,jD}^{\rho *})_{j,D}$ are those of 28.5 with its room letters and these excesses. In this proof write $L_{j,R}^{(0)}$ for the sum of the L_{jD} over the stations of excess c_j, which have the reversed excess 0, and $L_{j,R}^{*(c)}$ for the sum of the L_{jD}^* over the stations of excess 0, which have the reversed excess c_j. By 28.5,

$$L_{jD} = L_{j,R}^{(0)} \, (\gamma_R^\rho)^{c_j - i(D)} \, \varrho_{R,jD}^\rho, \qquad L_{jD}^* = L_{j,R}^{*(c)} \, (\gamma_R^\rho)^{i(D)} \, \varrho_{R,jD}^{\rho *}.$$

A direction $\mathbf{v} \in V$ keeps the letter totals, $\sum_D v_{jD} = 0$ and $\sum_D v_{jD}^* = 0$ for every active letter. It also keeps the pin of 28.12 for every vector of M, the pin of $\mathbf{1}$ holding by the letter totals on the balance wall, and $\mathbf{1}_R \in \vec{\mathcal{N}}_R^\perp \subseteq M$, the room letters being homogeneous. The exponents in $\mathbf{n}(\mathbf{a}, \mathbf{b})$ of the monomials of V_{jD} and V_{jD}^* vanish on R, and those of the monomials of L_{jD} and L_{jD}^* have the totals $\deg L_{jD} = e_j + c_j - i(D)$ and $\deg L_{jD}^* = e_j^* + i(D)$ on R (28.4); so, by the letter totals,

$$0 = \langle \mathbf{n}(\mathbf{v}), \mathbf{1}_R \rangle = \sum_{j,D} \big(v_{jD} \deg L_{jD} + v_{jD}^* \deg L_{jD}^* \big) = \sum_{j,D} \big((c_j - i(D)) v_{jD} + i(D) v_{jD}^* \big).$$

So in $\mathbf{V}_R^\mathbf{v} = \prod_{j,D} L_{jD}^{v_{jD}} {L_{jD}^*}^{v_{jD}^*}$ the sums $L_{j,R}^{(0)}$ and $L_{j,R}^{*(c)}$ and the scale γ_R^ρ enter with the exponent 0, and $\mathbf{V}_R^\mathbf{v} = (\boldsymbol{\varrho}_R^\rho)^\mathbf{v}$ on the open orthant of R, where these sums are positive, both sides being homogeneous of degree zero; hence $\sup \mathbf{V}_R^\mathbf{v} = \sup_{v \in \Delta_R} \boldsymbol{\varrho}_R^\rho(v)^\mathbf{v}$, the supremum over the open simplex.

The scale form ϕ_R^ρ has a positive continuous density on the open simplex and the seam vector is continuous there, so the law $(\boldsymbol{\varrho}_R^\rho)_*\phi_R^\rho$ is carried by the closure of the image of the open simplex, and every point of that image lies in its support: the preimage of a neighbourhood of an image point is a nonempty open set, of positive mass. The supremum of the continuous function $\mathbf{x}^{\mathbf{v}}$ over the image is therefore its largest value on the support of the law, in $(0,\infty]$. □

Lemma S14. *Let $L_1,\dots,L_p$ be positive letters on a finite set B (25.1), $e_0\in B$, $e_1,\dots,e_p$ real exponents and $\mathbf{m}$ real masses on B with $m_B=\sum_\ell e_\ell \deg L_\ell$; the pairings below are taken on $B\setminus e_0$.*

(i) There are finitely many rational simplicial cones of dimension $|B|-1$ with disjoint interiors covering $\mathbb{R}^{B\setminus e_0}$, each contained, for every ℓ, in a set $\{\mathbf{v}:\ \langle\mathbf{t},\mathbf{v}\rangle\le\langle\mathbf{t}',\mathbf{v}\rangle$ for every exponent $\mathbf{t}'$ of the support of $L_\ell\}$ for one exponent $\mathbf{t}$ of that support.

(ii) On such a cone π, with the primitive integral generators $\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1}$, write $x=\log\alpha_{B\setminus e_0}=-\sum_k u_k\mathbf{v}_{\pi,k}$ on the section $\alpha_{e_0}=1$, $u_k\ge0$, and $q_k=e^{-u_k}$. The exponent $\mathbf{t}_{\ell,\pi}$ of (i) is unique, and $L_\ell=\prod_k q_k^{\langle\mathbf{t}_{\ell,\pi},\mathbf{v}_{\pi,k}\rangle}k_{\ell,\pi}(q)$ on the section, with $k_{\ell,\pi}$ a polynomial in q with nonnegative coefficients whose constant term is the coefficient of $\alpha^{\mathbf{t}_{\ell,\pi}}$ in L_ℓ.

(iii) For a generator $\mathbf{v}$ of π the number $\kappa_{\mathbf{v}}=\langle\mathbf{m}-\sum_\ell e_\ell\mathbf{t}_{\ell,\pi},\mathbf{v}\rangle$ is the same for every cone π of which $\mathbf{v}$ is a generator, and, in $[0,\infty]$,

$$\int_{\alpha_{e_0}=1}\alpha^{\mathbf{m}-\mathbf{1}}\prod_\ell L_\ell^{-e_\ell}\,d\alpha_{B\setminus e_0}$$
$$=\sum_\pi|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|\int_{[0,1]^{|B|-1}}\prod_k q_k^{\kappa_{\mathbf{v}_{\pi,k}}-1}\prod_\ell k_{\ell,\pi}(q)^{-e_\ell}\,dq;$$

both sides are finite exactly when every $\kappa_{\mathbf{v}}$ is positive, and they are then the period of the word $\prod_\ell L_\ell^{-e_\ell}$ at $\mathbf{m}$, the integral over a section of 25.1.

Proof. (i) Identify $\mathbb{R}^{B\setminus e_0}$ with the hyperplane $v_{e_0}=0$ of $\mathbb{R}^B$. For each ℓ the sets $C_{\ell,\mathbf{t}}=\{\mathbf{v}:\ \langle\mathbf{t},\mathbf{v}\rangle\le\langle\mathbf{t}',\mathbf{v}\rangle$ for every $\mathbf{t}'\}$, $\mathbf{t}$ in the support of L_ℓ, are rational polyhedral cones, defined by the inequalities $\langle\mathbf{t}-\mathbf{t}',\mathbf{v}\rangle\le0$: the cones of the inner normal fan of the Newton polytope of L_ℓ cut by that hyperplane, which form a fan covering $\mathbb{R}^{B\setminus e_0}$, every $\mathbf{v}$ having an exponent of least pairing. The cones $\bigcap_\ell C_{\ell,\mathbf{t}_\ell}\cap O$, O running over the closed coordinate orthants of $\mathbb{R}^{B\setminus e_0}$, are pointed rational polyhedral cones which with their faces form a fan covering $\mathbb{R}^{B\setminus e_0}$, the common refinement of these fans and of the fan of the orthants. Order its rays and triangulate each cone by pulling: the cone is the union of the cones spanned by its first ray and by the triangulated faces that do not contain that ray. A face F of a pointed cone C has $\operatorname{span}F\cap C=F$, so the first ray of C lies off $\operatorname{span}F$ for such a face, and the cones produced are simplicial; the pulling triangulation of a cone restricts on each face to the pulling triangulation of that face, which depends on the face alone, so the triangulations of two cones agree on their common face. This gives finitely many rational simplicial cones of dimension $|B|-1$ with disjoint interiors covering $\mathbb{R}^{B\setminus e_0}$, each in some $\bigcap_\ell C_{\ell,\mathbf{t}_\ell}$.

(ii) If $\mathbf{t}$ and $\mathbf{t}''$ both pair least with every $\mathbf{v}_{\pi,k}$, then $\langle\mathbf{t}-\mathbf{t}'',\mathbf{v}_{\pi,k}\rangle=0$ for every k; the generators span $\mathbb{R}^{B\setminus e_0}$, so $\mathbf{t}$ and $\mathbf{t}''$ agree on $B\setminus e_0$, and as $|\mathbf{t}|=|\mathbf{t}''|=\deg L_\ell$ they are equal. On the section $\alpha^{\mathbf{t}'}=e^{\langle\mathbf{t}',x\rangle}=\prod_k q_k^{\langle\mathbf{t}',\mathbf{v}_{\pi,k}\rangle}$, so with $L_\ell=\sum_{\mathbf{t}'}c_{\mathbf{t}'}\alpha^{\mathbf{t}'}$ and $\mathbf{t}=\mathbf{t}_{\ell,\pi}$,

$$L_\ell=\prod_k q_k^{\langle\mathbf{t},\mathbf{v}_{\pi,k}\rangle}\sum_{\mathbf{t}'}c_{\mathbf{t}'}\prod_k q_k^{\langle\mathbf{t}'-\mathbf{t},\mathbf{v}_{\pi,k}\rangle}.$$

Every exponent $\langle \mathbf{t}' - \mathbf{t}, \mathbf{v}_{\pi,k} \rangle$ is an integer, the exponents and the generators being integral, and it is nonnegative by the choice of $\mathbf{t}$; by the uniqueness just proved, all of them vanish only at $\mathbf{t}' = \mathbf{t}$. So the sum is a polynomial $k_{\ell,\pi}$ with nonnegative coefficients and the constant term $c_{\mathbf{t}} > 0$.

(iii) A cone π of which $\mathbf{v}$ is a generator has $\langle \mathbf{t}_{\ell,\pi}, \mathbf{v} \rangle = \min_{\mathbf{t}'} \langle \mathbf{t}', \mathbf{v} \rangle$, the minimum over the support of L_ℓ, which does not depend on π; so neither does $\kappa_{\mathbf{v}}$. On π the map $(u_k)_k \mapsto x = -\sum_k u_k \mathbf{v}_{\pi,k}$ is a linear bijection of $[0,\infty)^{|B|-1}$ onto $-\pi$ with the Jacobian $|\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})|$; on the section $d\alpha_e = \alpha_e \, dx_e$ for $e \neq e_0$, $\alpha^{\mathbf{m}} = e^{\langle \mathbf{m}, x \rangle}$, and $du_k = -dq_k/q_k$. By (ii), on the image of $-\pi$, the form $\alpha^{\mathbf{m}-\mathbf{1}} \prod_\ell L_\ell^{-e_\ell} \, d\alpha_{B\setminus e_0} = e^{\langle \mathbf{m}, x \rangle} \prod_\ell L_\ell^{-e_\ell} \, dx$ is

$$|\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \prod_k q_k^{\langle \mathbf{m} - \sum_\ell e_\ell \mathbf{t}_{\ell,\pi}, \mathbf{v}_{\pi,k} \rangle - 1} \prod_\ell k_{\ell,\pi}(q)^{-e_\ell} \, dq.$$

The cones $-\pi$, like the cones π, cover $\mathbb{R}^{B\setminus e_0}$ with disjoint interiors, their boundaries are null and the integrands are positive, so the two sides agree in $[0,\infty]$ (Tonelli's theorem). On $[0,1]^{|B|-1}$ each $k_{\ell,\pi}$ lies between its constant term and the sum of its coefficients, so $\prod_\ell k_{\ell,\pi}^{-e_\ell}$ is bounded above and below by positive constants, and the integral over the cube is finite exactly when every exponent $\kappa_{\mathbf{v}_{\pi,k}}$ is positive; every generator is a generator of some cone. The form $\alpha^{\mathbf{m}-\mathbf{1}} \prod_\ell L_\ell^{-e_\ell} d\alpha$ is projective, the masses lying on the balance hyperplane, and α_{e_0} is positive and homogeneous of degree one, so the integral over the section is the period (25.1, 16.2). □

Theorem 31.4 (Closing past the walls). *As stated, for every piece whose seam function converges at a real point of the pins of the seam with positive shares, with the sector form of (ii) for every fan of the seam word and with part (iv).*

Proof. The hypothesis. At a real point of the pins with positive shares the seam word $W_B(\mathbf{a}, \mathbf{b})$ is a Newton word (25.1), and the masses $\mathbf{m}_B$ lie on its balance hyperplane: by the letter totals and the scale pin,

$$\sum_{j,D} \Big(a_{jD}\big(d_j + i(D)\big) + b_{jD}\big(d_j^* + c_j - i(D)\big) \Big) = \sum_j \big(s_j d_j + t_j d_j^* \big) + \kappa_B = m_B$$

(28.4, 28.7). Its period, the seam function, converges there, so its Newton sum has the dimension $|B| - 1$ (25.2).

(i). The reference proves (i) for a piece of tables (the reference, proof of thm:pastwalls(i)). Of the piece it uses only that its seam word has a Newton sum of full dimension, so that its reduced period in 28.12 is its seam function at the sums over the monomials, the seam letters repeating over the monomials of the partner; the reference obtains that fact from the connectedness of the family of tables, and here it is the hypothesis. The rest of that proof concerns the partner, whose room letters are read in their monomials, so that $\vec{\mathcal{N}}_R = 0$ and its reduced period is $\prod \xi^{-\mathbf{a}} \prod \xi^{*-\mathbf{b}}$ (28.12); with $M = \mathbb{R}\mathbf{1}_B \oplus \mathbb{R}^R$ and the lattice basis $\mathbf{1}$, $\mathbf{e}_r$ ($r \in R$), the pins of 28.12 are the displayed ones, the scale pin following by the balance.

(ii), the sector form. On the pins of the seam the exponents $\bar{a}_{jD}$, $\bar{b}_{jD}$ of the seam word are affine in the shares with rational linear parts, and the masses $\mathbf{m}_B$ lie on its balance hyperplane at every share, by the display above with the sums in place of the shares. Lemma S14, applied to the seam letters of the active letters with the exponents $\bar{a}_{jD}$, $\bar{b}_{jD}$, gives the fans of the seam word, the sector coordinates, the sector letters with their exponents, the walls and the identity at every real point of the pins, and shows that the seam function converges at a real point exactly when every wall is positive there. The coefficients $\langle \mathbf{t}_{jD,\pi}, \mathbf{v} \rangle$ and $\langle \mathbf{t}^*_{jD,\pi}, \mathbf{v} \rangle$ of the shares in a wall

are integers, so the wall is affine with a rational linear part, and it does not depend on the sector (Lemma S14(iii)).

(ii), continuation and bound. On a sector the integrand is $\prod_k q_k^{\kappa_{\mathbf{v}_{\pi,k}}(\mathbf{z})-1} h(q,\mathbf{z})$ with $h = \prod_{j,D} k_{jD,\pi}^{-\bar a_{jD}} \psi_{jD,\pi}^{-\bar b_{jD}}$. The sector letters are at least their positive constant terms on the closed cube, so their real parts are positive on a neighbourhood of it, where their principal logarithms are holomorphic; so h is holomorphic in q near $[0,1]^{|B|-1}$ and entire in the shares. 8.4, with $x_k = \kappa_{\mathbf{v}_{\pi,k}}(\mathbf{z})$ and the shares as its parameter, continues each sector integral meromorphically to all complex shares, with its poles on the hyperplanes on which a wall of a generator of the sector lies in $-\mathbb{N}$; a wall constant on the pins is positive there, the seam function converging, and gives no hyperplane. The formula of 8.4, repeated in the $|B|-1$ coordinates, writes each term with at most $|B|-1$ factors $(\kappa_{\mathbf{v}_{\pi,k}}+l)^{-1}$, which bounds the orders of the poles by $|B|-1$. For the bound, the proof of 20.32 is run on each sector integral with the several parameters entering through the powers of the sector letters and through the walls, $|\operatorname{Re}\mathbf{z}|_1$ and $|\operatorname{Im}\mathbf{z}|_1$ in place of $|\operatorname{Re} z|$ and $|\operatorname{Im} z|$ (the reference, proof of thm:pastwalls(ii)): each of the at most $|B|-1$ factors $(\kappa+l)^{-1}$ of a term is at most a constant times δ^{-1}, and nothing else depends on δ.

(ii), a piece of tables. Let the seam letters at the station D of the active letters be the basis and the cobasis polynomials of a table $\mathcal{B}_{jD}$ on B with positive weights, and consider the family of these tables at the exponents $\bar a_{jD}, \bar b_{jD}$ at a point with positive shares. Its potential $\sum_{j,D} \left(\bar a_{jD} \log K_{\mathcal{B}_{jD}}(e^y) + \bar b_{jD} \log \Psi_{\mathcal{B}_{jD}}(e^y)\right)$ (8.18) is affine along a direction $\mathbf{w}$ exactly when, in every letter, all the exponents of the support have one pairing with $\mathbf{w}$, a logarithm of a sum of exponentials with positive coefficients being strictly convex along every direction that separates two of its exponents: exactly when $\mathbf{w}$ is orthogonal to the directions of the Newton sum. If the family is connected, that space is $\mathbb{R}\mathbf{1}$ (8.19(iii), a space that does not involve the masses), and the Newton sum has the dimension $|B|-1$. If it is not, some nonempty proper $S \subset B$ has the width $c_{jD}(S) = 0$ in every table (8.1), so that $|T \cap S| = d_{jD}(S)$ for every basis T of $\mathcal{B}_{jD}$ (7.13). The base polytope of $\mathcal{B}_{jD}$ then lies in $\{x_S = d_{jD}(S)\}$ and that of its dual in $\{x_S = |S| - d_{jD}(S)\}$, so the Newton sum lies in a hyperplane $\{x_S = \text{const}\}$, and as $\mathbf{1}_S \notin \mathbb{R}\mathbf{1}$ its dimension is less than $|B|-1$.

In the coordinates of 8.5(ii) on the sector H_π of an order π of B, $\alpha_{\pi(1)} = \rho$ and $\alpha_{\pi(k)} = \rho q_2 \cdots q_k$, so that, with $u_k = -\log q_k$,

$$\log \alpha_{\pi(m)} - \log \alpha_{\pi(1)} = \sum_{k=2}^{m} \log q_k = -\sum_{k=2}^{|B|} u_k \, (\mathbf{1}_{U_k})_{\pi(m)},$$

because $\pi(m) \in U_k$ exactly when $k \le m$. So, modulo $\mathbf{1}$, H_π is the cone with the generators $\mathbf{1}_{U_k}$, $2 \le k \le |B|$, read on the section as $\mathbf{1}_{U_k} - (\mathbf{1}_{U_k})_{e_0}\mathbf{1}$. These $|B|-1$ vectors form a basis of $\mathbb{Z}^B/\mathbb{Z}\mathbf{1}$, since the differences $\mathbf{1}_{U_k} - \mathbf{1}_{U_{k+1}} = \mathbf{e}_{\pi(k)}$ and $\mathbf{1}_{U_1} = \mathbf{1}$ span $\mathbb{Z}^B$; so the cone is simplicial of dimension $|B|-1$ with the determinant ± 1, and the sectors of the orders cover $\mathbb{R}^B/\mathbb{R}\mathbf{1}$ with disjoint interiors. For a basis T, $\alpha^T/\alpha^{T_\pi^+} = \prod_k q_k^{|T\cap U_k| - |T_\pi^+ \cap U_k|}$, and 8.3, $\alpha^T \le \alpha^{T_\pi^+}$ on H_π for every basis, gives, with one q_k small and the others 1, $\langle \mathbf{1}_{T_\pi^+}, \mathbf{1}_{U_k}\rangle \le \langle \mathbf{1}_T, \mathbf{1}_{U_k}\rangle$ for every k. As $\mathbf{1}_T$ and $\mathbf{1}_{T_\pi^+}$ have one total, their pairings on $B \setminus e_0$ with the generator $\mathbf{1}_{U_k} - (\mathbf{1}_{U_k})_{e_0}\mathbf{1}$ differ by the difference of their pairings with $\mathbf{1}_{U_k}$. So the sector lies in the set of (ii) for $K_{\mathcal{B}_{jD}}$ with $\mathbf{t} = \mathbf{1}_{T_\pi^+}$, and likewise for $\Psi_{\mathcal{B}_{jD}}$ with $\mathbf{t} = \mathbf{1}_{B\setminus T_\pi^-}$, the greedy bases being those of $\mathcal{B}_{jD}$; the sectors of the orders form a fan of the seam word. Its sector letters are $\sum_T \theta_{jD,T} \prod_k q_k^{|T\cap U_k| - d_{jD}(U_k)}$ and their cobasis counterparts, the sector letters of 8.5(ii). The vector $\mathbf{m}_B - \sum_{j,D}(\bar a_{jD}\mathbf{1}_{T^+} + \bar b_{jD}\mathbf{1}_{B\setminus T^-})$ has the total 0 on the pins, so its pairing with the generator of a tail U is its pairing with $\mathbf{1}_U$; by 8.3, $|T^+ \cap U| = d_{jD}(U)$ and $|U \setminus T^-| = d^*_{jD}(U)$, and the wall of that generator is $m_U - \sum_{j,D}(\bar a_{jD} d_{jD}(U) + \bar b_{jD} d^*_{jD}(U))$, the proper wall κ_U. Every nonempty proper U is a tail of some order.

(iii). The reference proves (iii) for a piece of tables (the reference, proof of thm:pastwalls(iii)), using three facts about the piece. The first is the continuation with the bound of (ii). The second is that the seam function converges at a real point exactly when its walls are positive there, so that it converges along a ray along which no wall falls, the walls staying above their positive values at $\bar{\mathbf{c}}$, and diverges past a wall that falls. The third is that at real points it is $\int \mathbf{x}^{-\mathbf{w}}\, d\big((\boldsymbol{\varrho}_B^{\rho})_* \phi_B^{\rho}\big)$, with the scale form positive and continuous on the open simplex and the seam vector continuous there, which with 31.2 gives $g(\boldsymbol{\sigma}; 0) = \sup \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$. The first two are (ii), and the third holds for every piece (28.5, 28.7). The rest of that proof (the Gamma bounds, Stirling's formula with $\sum_l \sigma_l = 0$, the crossings and the null set) does not use the tables.

(iv). The partner is a piece on R read with the reversed excesses. The pins of its seam, the letter totals and its scale pin $\sum_{j,D} \big((c_j - i(D))a_{jD} + i(D)b_{jD}\big) = \kappa_R$, are on the balance wall $\kappa_R = \kappa_B^{\mathrm{top}}$ those of the piece (28.10(iii)), and its seam function converges at $\mathbf{c}$; so (ii) applies to the piece and to the partner. On the share side of 28.10(iii) the integrand at $\mathbf{c} + \tau\boldsymbol{\sigma} + i\boldsymbol{\vartheta}$ is $\prod_{j:\, s_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j \\ s_j\end{bmatrix} \prod_{j:\, t_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j \\ t_j\end{bmatrix}$ times the two seam functions. For $M_B > g_B(\boldsymbol{\sigma}; \theta_B)$ and $M_R > g_R(\boldsymbol{\sigma}; \theta_R)$ the definitions of the growths give C, n_1 and n_2 with

$$\big|P_{W_B}\, P_{W_R}\big| \le C^2 \delta^{-2n_2} (1 + \tau + |\boldsymbol{\vartheta}|)^{2n_1} (M_B M_R)^{\tau} e^{(\theta_B + \theta_R)|\boldsymbol{\vartheta}|_1}$$

for $\tau \ge 0$, δ the smaller of 1 and the distance from the polar hyperplanes of both seam functions. From here the proof of (iii) applies with $\theta = \theta_B + \theta_R < \pi/8$ and $M_B M_R$ in place of M, the coefficients sitting inside the seam functions:

- the contour moves along times at distance at least τ^{-3} from the crossings, where δ^{-2n_2} is a power of τ, and the power of $1 + |\boldsymbol{\vartheta}|$ is absorbed into $e^{\eta|\boldsymbol{\vartheta}|_1}$ with η small;
- a share whose real part x grows has $|\Gamma(x + iy)| e^{\theta|y|} \le \Gamma(x) e^{2\theta^2 x/\pi}$, and a share that falls or stays absorbs $e^{\theta|y|}$ against $e^{-\pi|y|/2}$;
- integrating over the directions of the shares, Stirling's formula and $\sum_l \sigma_l = 0$, by the letter totals, give the factor $e^{\tau(h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) + \epsilon)}$ as $M_B \downarrow g_B$ and $M_R \downarrow g_R$.

The ray crosses the hyperplanes of the falling shares and the hyperplanes on which a falling wall of either word lies in $-\mathbb{N}$, and each crossing adds the fibre integral of the residue of the whole integrand (31.3), a Laurent coefficient where several factors are polar. The pins and the walls of both words have rational linear parts, so crossings on parallel hyperplanes coincide or keep a fixed positive distance. Crossings on hyperplanes that are not parallel, closer than T^{-3} near $\tau = T$, occur only for $\mathbf{c}$ in a set of measure $O(T^{-3})$, summable over T (Borel and Cantelli). The local decay (H1) of 31.3 holds because both seam functions grow at most like $e^{\theta' |\operatorname{Im} \mathbf{z}|_1}$ for every $\theta' > 0$, by (ii), against the decay of the Gammas at bounded real parts. (H2) holds at the times above when $h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) < 0$, and 31.3 gives the closing with the stated bounds.

For the last sentence of (iv), let no wall of the word of a side fall along $\boldsymbol{\sigma}$. Both words have Newton sums of full dimension, their seam functions converging at $\mathbf{c}$ (25.2), so the largest value of $\mathbf{x}^{-\boldsymbol{\sigma}}$ on the support of the law of the side is $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$ of its letter vector (31.2, for the partner by its sentence on the partner). At $\theta = 0$ the proof of (iii), run on that side, gives the growth $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$, finite, and the growth does not increase with the angle, a bound with a smaller angle being a bound with a larger one. On the real ray, where $\boldsymbol{\vartheta} = 0$, the walls stay above their positive values at $\mathbf{c}$, so δ is bounded below there, and the seam function at $\mathbf{c} + \tau\boldsymbol{\sigma}$ is at least a positive constant times $e^{\tau h'}$ for every $h' < \log \sup \mathbf{V}^{-\boldsymbol{\sigma}}$ (the proof of (iii)); so every admissible M, at every angle, is at least $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$. □

Definition 31.5 (Chambers of a closing). As stated, for every fibre integral with finitely many affine polar functions.

Proof. The chamber is convex. If $\boldsymbol{\sigma}', \boldsymbol{\sigma}'' \in C(\boldsymbol{\sigma})$ and $0 \le s \le 1$, then $\partial_{s\boldsymbol{\sigma}'+(1-s)\boldsymbol{\sigma}''}\ell = s\,\partial_{\boldsymbol{\sigma}'}\ell + (1-s)\,\partial_{\boldsymbol{\sigma}''}\ell$ equals $\partial_{\boldsymbol{\sigma}}\ell$ for every polar function that $\boldsymbol{\sigma}$ falls and is nonnegative for every other. The ray of $s\boldsymbol{\sigma}' + (1-s)\boldsymbol{\sigma}''$ consists of the points $s(\mathbf{c}+\tau\boldsymbol{\sigma}') + (1-s)(\mathbf{c}+\tau\boldsymbol{\sigma}'')$ of the convex Ω.

The instances:

- In 31.2 the integrand is the Gammas times the reduced periods of the two words. These are holomorphic where they converge absolutely, on a set convex by Hölder's inequality (the reference, def:chamber). On the pins the integrand of a reduced period at $\mathbf{c}' + \tau\boldsymbol{\sigma}$ is its integrand at $\mathbf{c}'$ times $\mathbf{V}^{-\tau\boldsymbol{\sigma}}$ read at the point of the orthant (the reference, proof of thm:closecone), at most $(\sup \mathbf{V}^{-\boldsymbol{\sigma}})^{\tau}$; so when both suprema are finite a ball about $\mathbf{c}$ inside the set of convergence, translated by $\tau\boldsymbol{\sigma}$, stays inside it, and the ray lies in Ω.
- In 31.4(iii) the seam function continues to all complex shares, with its poles where a wall lies in $-\mathbb{N}$ (31.4(ii)), the walls read through the sums of the shares being affine in the shares.
- For a fibre integral of Gammas and coefficients the integrand $\prod_l \Gamma(a_l)\xi_l^{-a_l}$, up to a constant, is meromorphic on all of $A+iV$ with its poles where a share lies in $-\mathbb{N}$. So the polar functions are the shares, $\Omega = A$, and $\partial_{\boldsymbol{\sigma}'} a_l = \sigma_l'$, which gives the chamber displayed. When the pins fix the total of the shares of every letter, every $\boldsymbol{\sigma}' \in V$ has $\sum_{l\in K_j} \sigma_l' = 0$ and $\sum_{l\in\Psi_j} \sigma_l' = 0$, so on the chamber $0 \le \sigma_l' \le \sum |\sigma_{l'}|$, the sum over the falling shares l' of the letter of l: the chamber is a compact convex polytope.
- A source factor that is a cube integral continued by Reg is meromorphic on all of $A + iV$, with its poles where one of its walls lies in $-\mathbb{N}$ (8.4).

□

Theorem 31.6 (A closing depends only on its chamber). *As stated, for every fibre integral of 31.5 with the local decay (H1) of 31.3.*

Proof. A polar function ℓ that $\boldsymbol{\sigma}$ falls has $\ell(\mathbf{c}+\tau\boldsymbol{\sigma}) = \ell(\mathbf{c}) + \tau\partial_{\boldsymbol{\sigma}}\ell = \ell(\mathbf{c}+\tau\boldsymbol{\sigma}')$. So its hyperplanes $\ell = -n$ are met by the two rays at the same times, and ℓ is constant on the segment joining their points at a time τ. A polar function that $\boldsymbol{\sigma}$ does not fall is not fallen by $\boldsymbol{\sigma}'$ either; it is at least $\ell(\mathbf{c}) > 0$ at both points, hence on the segment, and does not lie in $-\mathbb{N}$ there.

So the two rays meet the same polar hyperplanes at the same times. At a crossing time τ^* the polar hyperplanes through $\mathbf{c} + \tau^*\boldsymbol{\sigma}$ and through $\mathbf{c} + \tau^*\boldsymbol{\sigma}'$ are the same, with $\partial_{\boldsymbol{\sigma}}\ell_H = \partial_{\boldsymbol{\sigma}'}\ell_H$, so (H0) holds along one ray exactly when it holds along the other.

At a time τ that is no crossing, the segment avoids every polar hyperplane at a positive distance and lies in the convex Ω. By (H1) on a compact neighbourhood of the segment in Ω, step (1) of the proof of 31.3 (Cauchy's theorem on rectangles whose horizontal sides vanish by (H1), then Fubini's theorem) equates the fibre integrals through its two ends. A crossing term is the difference of the fibre integrals just before and just after the crossing (step (2) of that proof), so the crossing terms of the two rays at a common crossing are equal. Equal terms and equal remainders carry convergence and bounds from one closing to the other.

The rest is the reference's (the reference, proof of thm:chamber). The integrands of 31.2 and of 31.4(iii) have the local decay (H1), by Stirling's formula and the bounds of their proofs, so the first part applies to them. For every $\epsilon > 0$ some $\boldsymbol{\sigma}' \in C(\boldsymbol{\sigma})$, with some θ in the case of 31.4(iii), has a rate below $h' + \epsilon/2$. The null set of 31.2 is the same for every direction of the chamber, its crossings being those of the falling polar functions, and the bounds of 31.2 or of 31.4(iii) along $\boldsymbol{\sigma}'$

bound the terms and the remainders along $\boldsymbol{\sigma}$. On the chamber the falling shares are fixed, $x \log x$ is convex on $[0,\infty)$, and $\log \sup \mathbf{V}_B^{-\boldsymbol{\sigma}'}$ and $\log \sup \mathbf{V}_R^{-\boldsymbol{\sigma}'}$ are suprema of linear functions of $\boldsymbol{\sigma}'$, so the rate h is convex there. □

Theorem 31.7 (Averaging the laws). *As stated, for every share side whose integrand is the Gammas of the shares times a Laplace transform $\int e^{-\langle \mathbf{a},y\rangle}\, d\lambda$ of a source measure, the kinds (A) and (B) among them.*

Proof. (i). At a source point ζ the kernel is a fibre integral of Gammas and the positive coefficients $e^{y_l(\zeta)}$: the share side of a Gramma (4.2) divided by the constant $\prod_j \Gamma(s_j) \prod_j \Gamma(t_j)$. At the real point $\mathbf{c}$ the Gammas are integrable over the fibre, $|e^{-\langle \mathbf{c}+i\boldsymbol{\vartheta},y\rangle}| = e^{-\langle \mathbf{c},y\rangle}$, and $\int e^{-\langle \mathbf{c},y\rangle}\, d\lambda < \infty$. So the fibre and the source integrals converge absolutely together, and by Fubini's theorem the fibre integral of the share side is the integral of the kernel against λ. In (A) and (B) that fibre integral is $P_{B\sqcup R}(\mathbf{m})$ (28.10(iii), 31.4(i)). In (A) the kernel is $\Gamma(\kappa_B)\mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x}\circ\mathbf{x}')$ (the reference, proof of thm:pool(i)): each active letter of the stand-in pair of 28.10(ii) is opened by 4.4 with one share per station, and the integral over the scale is the pin of the scale.

The bound on the crossing terms is the reference's (the reference, proof of thm:pool(i)). It uses three things:

- 31.6 for the kernel, whose polar functions are the shares, with $\Omega = A$ and (H1) holding by Stirling's formula, so that the closings along $\boldsymbol{\sigma}$ and along every $\boldsymbol{\sigma}' \in C$ are one series;
- Stirling's formula for the Gammas, uniformly on the compact C, the terms $\tau x \log \tau$ cancelling over each letter by its total;
- at a crossing, the expansion of $e^{-\langle \mathbf{a},y\rangle} = e^{-\langle \mathbf{c}+\tau\boldsymbol{\sigma}',y\rangle} e^{-i\langle \boldsymbol{\vartheta}',y\rangle} e^{-w\langle \mathbf{e},y\rangle}$ in the transverse variable w to the order $p-1$, with $|\langle \mathbf{e},y\rangle| \le C|y|_V$, against the Laurent coefficients of the Gammas, which Cauchy's estimate bounds off the null set.

None of these steps uses the kind of the share side.

(ii). The source points with $y(\zeta) \in S_\lambda$ carry all of λ, and there $h_C(y) \le \hat{h}_C$. By (i) the averaged terms are absolutely convergent integrals, at most $C_\epsilon e^{\tau_k(\hat{h}_C+\epsilon)} \int w\, d(y_*\lambda)$, which is finite by the certificate of 31.7. At each source point the kernel is the sum of its crossing terms before a time T separating the crossings and of the integral left at T (31.3, whose identity at a finite time needs no limit), and that integral is at most $C_\epsilon e^{-\langle \mathbf{c},y\rangle} e^{T(\hat{h}_C+\epsilon)}$. Integrating against λ, the fibre integral of the share side is the sum of the averaged terms before T up to $C_\epsilon e^{T(\hat{h}_C+\epsilon)} \int e^{-\langle \mathbf{c},y\rangle}\, d\lambda$; the crossings up to T number $O(T)$, and with $\hat{h}_C + \epsilon < 0$ both the sum and the bound follow. At a crossing point, along some $\boldsymbol{\sigma}' \in C$, inside the set where $\int e^{-\langle \mathbf{a},y\rangle}\, d\lambda$ converges absolutely, with its derivatives in the shares, Fubini's theorem (the Gammas decaying by Stirling's formula) exchanges the fibre and the source integrals in the crossing term of the share side. The inner integral is then the crossing term of the kernel at that point, which is $\mathcal{R}_k(\zeta)$ by 31.6.

(iii). The function $h(\boldsymbol{\sigma}';y)$ is continuous, convex in $\boldsymbol{\sigma}'$ on C (the falling coordinates being fixed and $x \log x$ convex on $[0,\infty)$), and affine in y; C is compact and convex. Sion's minimax theorem on $C \times \operatorname{conv} S_\lambda$ gives the identity, the supremum of an affine function over $\operatorname{conv} S_\lambda$ being its supremum over S_λ.

For the reading through the log-support,

$$\Big| \int e^{-\langle \mathbf{c}+\tau\boldsymbol{\sigma}'+i\boldsymbol{\vartheta},y\rangle}\, d\lambda \Big| \le e^{\tau \sup_{y\in S_\lambda}(-\langle \boldsymbol{\sigma}',y\rangle)} \int e^{-\langle \mathbf{c},y\rangle}\, d\lambda.$$

With the Gamma bounds of the proof of 31.2, this bounds the closing of the share side along $\boldsymbol{\sigma}'$ at the rate $\sum_l \sigma'_l \log|\sigma'_l| + \sup_{S_\lambda}(-\langle \boldsymbol{\sigma}', y\rangle) = \sup_{y\in S_\lambda} h(\boldsymbol{\sigma}'; y)$. In (A) and (B) this is the rate $h(\boldsymbol{\sigma}')$ of 31.2 (the reference, proof of thm:pool(iii)):

- the seam words of the sides that are laws have Newton sums of full dimension, their seam functions converging at $\mathbf{c}$ (25.2);
- so the largest values of the monomials of their letter vectors are read on the supports of their laws (31.2; for the partner in (A), its sentence on the partner);
- in (B), $\sup \mathbf{V}_R^{-\boldsymbol{\sigma}'} = \boldsymbol{\xi}^{-\boldsymbol{\sigma}'}$;
- the largest value of $-\langle \boldsymbol{\sigma}', y\rangle$ on the log-support of a product of laws is the sum of its largest values on the log-supports of the factors.

In a bundle G, Jensen's inequality for the logarithm, in the proportions σ'_l/g, gives $\sum_G \sigma'_l \log(\sigma'_l e^{-y_l}) \geq -g\log\left(g^{-1}\sum_G e^{y_l}\right)$, with equality exactly when e^{y_l}/σ'_l is constant on G. □

Definition 31.10 (Keyhole continuations along a direction; ceiling and phase). As stated, for every piece of 31.4 and the sectors of any fan of its seam word.

Proof. The displayed sum is 31.4(ii) at the sums of the shares over the monomials of the partner. A set Y of a box is compact and is the product of a box, of intervals, and of keyholes, a closed disc about 0 joined to a segment of the positive axis (20.47); each factor is star-shaped, about the centre of its disc for a keyhole, so Y is star-shaped, and so is Y plus a small open polydisc about 0. The sector letters do not vanish on Y, by the definition of a keyhole continuation, hence on such a neighbourhood of Y, which is simply connected, so they have logarithms there. The real points of Y form a product of intervals, connected, containing points with nonnegative coordinates, where the letters, with their nonnegative coefficients and positive constant terms, are positive; real and without zeros on that set, they are positive on it, and the logarithms can be taken real at the real points. For a piece of tables and the sectors of the orders, the walls of the generators are the proper walls of the tails and the sector letters are those of 8.5(ii) (31.4(ii)). □

Theorem 31.11 (The travelling bound past the walls). *As stated, for every piece of 31.4.*

Proof. This is the reference's proof (the reference, proof of thm:keyholebound), with the sectors of the fan in place of the orders of B and the generators in place of the tails.

(i) Fix a sector π, a box Ξ of it, the set J of its falling variables with keyholes and their radii r_k, and choose $r_k^- \in (0, r_k)$ with $\sum_{k\in J}(-\dot\kappa_k)\log(r_k/r_k^-) < \epsilon/2$. The term of π is $|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})| \int \prod_{k\in J} q_k^{\kappa_k(\bar{\mathbf{z}})-1}\Phi\, dq$, with $\Phi = \prod_{k\notin J} q_k^{\kappa_k(\bar{\mathbf{z}})-1}\mathbf{k}_\pi(q)^{-\bar{\mathbf{z}}}$ holomorphic in the variables of J on a neighbourhood of their keyholes, because the sector letters do not vanish on the sets Y. The following steps apply as in the reference:

- 20.45, applied in the variables of J one at a time with r_k^- and r_k;
- the maximum modulus principle in the variables of $J \setminus A$ that lie in their discs;
- the exact integration of the falling monomials, which carries no phase;
- the identity $|k^{-z}| = |k|^{-\operatorname{Re} z} e^{\operatorname{Im} z \arg k}$ at the points of the outline;
- the bounds of the letters above and below on the compact outline;
- the integrability of $q_k^{\kappa_k(\bar{\mathbf{c}})-1}$ over the base variables, the walls being positive at $\bar{\mathbf{c}}$;

- $|\kappa_k(\bar{\mathbf{z}}) + l_k| \geq c\,\delta$ for at most $|B| - 1$ factors.

Each of these uses only the sector form of 31.4(ii): walls affine in the shares and positive at $\bar{\mathbf{c}}$, and sector letters polynomial, positive on the closed cube and nonvanishing on the sets Y. The determinant is a constant, and the boxes, the sets A and the sectors are finitely many.

(ii) The integrand of the share side of 31.4(i) at $\mathbf{c} + \tau\boldsymbol{\sigma} + i\boldsymbol{\vartheta}$ is the Gammas and the coefficients times the seam function, which (i) bounds. The steps are those of the reference:

- for a growing share, the exact bound $|\Gamma(x + iy)|e^{\varphi y} \leq \Gamma(x)(\cos\varphi)^{-x}$ for $|\varphi| < \frac{\pi}{2}$, from $\Gamma(z) = e^{i\varphi z}\int_0^\infty u^{z-1}e^{-ue^{i\varphi}}du$, applied with $\varphi = \upsilon_l(q) + \eta\,\operatorname{sign} y$;
- falling and staying shares absorb $e^{\upsilon_l(q)y}$ against $e^{-\pi|y|/2}$, as $|\upsilon_l| \leq \frac{\pi}{2} - \epsilon_0$;
- Stirling's formula with $\sum_l \sigma_l = 0$, the supremum over q being taken after the integration over the directions of the shares.

Together these give the travelling bound at the rate $h_{\boldsymbol{\upsilon}}(\boldsymbol{\sigma})$. Then 31.3 applies: (H0) off the null set; (H1) by 31.4(ii) with any $\theta < \pi/2$ and Stirling's formula; (H2) at times at distance at least τ^{-3} from the crossings. □

Corollary 31.12 (When the phase costs nothing). *As stated, for every piece of 31.4.*

Proof. In (i) the price $-\sum_{\sigma_l>0}\sigma_l \log\cos\upsilon_l$ vanishes. In (ii), $\log|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}| - \sum_{\sigma_l>0}\sigma_l\log\cos\upsilon_l \leq \log H_{\mathcal{O}}$ on the outline, with equality approached where $|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}|$ approaches $H_{\mathcal{O}}$, the price being nonnegative.

With no falling variables, the outline is the union of the real cubes of the sectors, where the sector letters are positive and the phase vanishes. Differentiating the wall of 31.4(ii) gives $\dot{\kappa}_k = -\langle\sum_{j,D}(\bar{\sigma}_{jD}\mathbf{t}_{jD,\pi} + \bar{\sigma}^*_{jD}\mathbf{t}^*_{jD,\pi}), \mathbf{v}_{\pi,k}\rangle$. So on the section $\varrho_{\pi,\bar{\boldsymbol{\sigma}}} = \prod_{j,D}\big(\prod_k q_k^{\langle\mathbf{t}_{jD,\pi},\mathbf{v}_{\pi,k}\rangle}k_{jD,\pi}\big)^{-\bar{\sigma}_{jD}}\big(\prod_k q_k^{\langle\mathbf{t}^*_{jD,\pi},\mathbf{v}_{\pi,k}\rangle}\psi_{jD,\pi}\big)^{-\bar{\sigma}^*_{jD}} = \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$, read in the sector coordinates. The cubes of the sectors cover the section, so the ceiling is $\sup\mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$, the monomial being homogeneous of degree zero, and the rate is that of 31.2 for the glue read in the monomials of the partner. □

Proposition 31.13 (How travelling bounds compose). *As stated, the face coefficient of (iii) being read on the sectors of any fan of the seam word.*

Proof. (i) and (ii) are the reference's (the reference, proof of prop:pathcompose). For (iii), the reference's proof applies with one sentence changed: a wall $\kappa_{\mathbf{v}}$ is the exponent of q_k in each sector of which $\mathbf{v} = \mathbf{v}_{\pi,k}$ is a generator (31.4(ii)).

- The pole is simple, as opposed to the multiple poles of the statement, when no other factor and no second wall of a sector is polar on the hyperplane. Then, by 8.4, the residue of the sector integral in the direction $\mathbf{e}$ at $\kappa_{\mathbf{v}} = -n$ is $|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|$ times the integral over the other coordinates of the n-th Taylor coefficient in q_k of the rest of the integrand, divided by $\partial_{\mathbf{e}}\kappa_{\mathbf{v}}$. Summing over the sectors gives the face coefficient.
- The other powers $q_{k'}^{\kappa_{k'}-1}$ do not depend on q_k, so the Taylor coefficient is that of the product of the powers of the sector letters: a finite sum of products of derivatives of the letters and powers of the letters at $q_k = 0$, with coefficients polynomial in the exponents.
- Where two walls of one sector lie in $-\mathbb{N}$ on the hyperplane, the expansion of 8.4 in both variables gives a pole of the second order. A Laurent coefficient of a product is a finite sum of products of Laurent coefficients, and the derivatives in the shares of $q^{\kappa(\mathbf{z})}$ and of k^{-z} bring $\log q$ and $-\log k$.

- The wall crossed falls along the direction, so q_k is a falling variable and $q_k = 0$ lies in its keyhole, where the letters do not vanish; Cauchy's estimate on $|q_k| = r_k$ bounds the n-th Taylor coefficient by r_k^{-n} times the largest modulus on the circle.

□

Definition 31.16 (A rational retained cohort). As stated: the fibre integrals of Gammas, coefficients and source factors whose pins carry a grading.

Proof. The share side of 31.4(i) and the kernel of 31.7 are graded by their letter totals: the sum of the rows of the pins that fix the totals of the letters takes the value 1 on the column of every share, each share belonging to one letter. By 31.4(ii), the sector integrals of a fan of a seam word are cube integrals $\int_{[0,1]^k} q^{\boldsymbol{\kappa}(\mathbf{a})-\mathbf{1}} \prod_i P_i(q)^{-w_i(\mathbf{a})} dq$. Here the P_i are the sector letters, with nonnegative coefficients and positive constant terms, and the κ_i and w_i are affine with rational linear parts; for tables these are the sector integrals of 8.6. □

Remark. The grading is part of the definition because 31.18 extracts the scalar of a joint word through the change of logarithmic scales $z = \rho\mathbf{v} + E\log X$, which needs a vector $\mathbf{v}$ taking the value 1 on every column. Without one the form of that theorem can fail, as follows.

Take the Gramma $\mathsf{G}_{\nu;(1,2)}(\xi_1,\xi_2) = \int_0^\infty x^{\nu-1} e^{-\xi_1 x - \xi_2 x^2} dx$, $\nu, \xi_1, \xi_2 > 0$. Its configuration has no grading, since $\langle 1, v\rangle = \langle 2, v\rangle = 1$ is impossible. Its share side is pinned by $b_1 + 2b_2 = \nu$, and its closing along $(-2, 1)$ crosses $b_1 = -N$, $N \in \mathbb{N}$, with the crossing terms

$$\frac{(-\xi_1)^N}{N!} \cdot \frac{1}{2}\Gamma\Big(\frac{\nu+N}{2}\Big)\xi_2^{-(\nu+N)/2},$$

the terms of the expansion of the integral in powers of ξ_1, using $\int_0^\infty x^{\nu+N-1} e^{-\xi_2 x^2} dx = \frac{1}{2}\Gamma(\frac{\nu+N}{2})\xi_2^{-(\nu+N)/2}$.

The scalar $\Gamma(\frac{\nu+N}{2})/\Gamma(N+1)$ has an N-th root tending to 0: its logarithm divided by N is $-\frac{1}{2}\log N + O(1)$. A joint word with the scalar of 31.18, here with $r_G = 1$, would in this positive case have the root rate $\operatorname{ess\,sup} c > 0$, or vanishing terms (31.19).

Theorem 31.18 (A finite procedure for a retained cohort). *As stated, for every retained cohort, the scalar being written through the grading of its pins.*

Proof. This is the reference's proof (the reference, proof of thm:retainedchain); only the radial step involves the totals of the shares.

- At a Gamma crossing, the crossed share is fixed at $-N(n)$, and its Gamma is replaced by the residue $(-1)^{N(n)}/\Gamma(N(n)+1)$.
- The crossed share and its added pin are eliminated together. This leaves the columns t_l, $l \neq f$, of the original pins, with the values $\boldsymbol{\nu} + N(n)t_f$, t_f the column of the crossed share.
- By 31.14 each share of $\mathcal{S}_-$ is translated, $b_l = a_l + \delta_l(n)$, with the factor $(-1)^{\delta_l(n)}\xi_l^{\delta_l(n)}/\Gamma(\delta_l(n))$, the power of ξ_l staying in the joint input.

The completed retained fibre is then the Gramma $\mathsf{G}_{\boldsymbol{\mu}_n;t}$ (4.2), with the remaining columns and the index $\boldsymbol{\mu}_n = \boldsymbol{\nu} + N(n)t_f + \sum_{l\in\mathcal{S}_-} \delta_l(n) t_l$. Every remaining column is a column of the original pins, so $\langle t_l, \mathbf{v}\rangle = 1$ for the grading $\mathbf{v}$, and

$$S_n = \langle \mathbf{v}, \boldsymbol{\mu}_n\rangle = w + N(n) + \sum_{l\in\mathcal{S}_-} \delta_l(n).$$

The change of logarithmic scales $z = \rho\mathbf{v} + E\log X$, with E a rational matrix whose columns complement $\mathbf{v}$, gives $\mathsf{G}_{\boldsymbol{\mu}_n;t} = J_{\mathbf{v}}\Gamma(S_n)\int X^{E^{\mathsf{T}}\boldsymbol{\mu}_n{}^{-\mathbf{1}}}\mathcal{L}(X)^{-S_n}dX$: the integral over ρ is $\Gamma(S_n)\mathcal{L}^{-S_n}$. Collecting the residue, the factors of the translations and $\Gamma(S_n)$ gives the stated G_n. Its slopes in n agree in the numerator and the denominator, $d_0 + \sum_{\mathcal{S}_-} d_l = \bar{d}$, so the terms $n\log n$ of Stirling's formula cancel, and the rest of Stirling's formula gives r_G and the polynomial bound. On the share sides of 31.4 and 31.7, $\mathbf{v}$ is the sum of the letter-total rows and $w = |\mathbf{s}| + |\mathbf{t}|$. The other steps of the procedure (the source strips, the circles, the joint read, the resolution of the endpoints and the fan) do not involve the grading. □

Theorem 31.26 (The exact positive rate of a retained chamber). *As stated, with a real index.*

Proof. The reference's proof (the reference, proof of thm:retainedchamberrate) uses rationality only of the configuration and of the direction, through the cohort and its integer step.

- 31.16 allows real constant terms and indices. The crossing times of a falling share are $(a_l^0 + N)/d_l$, $N \in \mathbb{N}$, and passing to the integer step makes the changes of all falling shares integral, whatever the constant terms.
- The positive construction uses 31.14 with $\delta_l(n) = \delta_l(0) + d_l n$ and real starting values.
- The rate is computed by 31.25, at the slope $\sum_f d_f t_f$ of the index, which does not involve its constant term.
- The root limit is that of 31.19, whose positive case allows real constant terms.

The chamber polytope of the statement is the chamber of 31.5 of the share side, whose polar functions are the shares (31.5), for the direction per step. □

Theorem 31.28 (Averaging is sharp). *As stated, for every share side of Theorem 31.7 and every direction with rational coordinates whose crossings are simple and nonresonant.*

Proof. At each source point the kernel of 31.7(i) is, up to the constant factor $\prod_j \Gamma(s_j)^{-1}\prod_j \Gamma(t_j)^{-1}$, the share side of the Gramma whose configuration is the matrix of the pins, rational by hypothesis, with the grading of the letter totals; whose index is the vector of the values of the pins, real; and whose coefficients are $e^{y_l(\zeta)}$, positive and continuous in ζ (4.2). Let $\boldsymbol{\sigma}$ be rational and the crossings of its ray simple and nonresonant. A falling share l crosses its poles at the times $(c_l + N)/|\sigma_l|$, $N \in \mathbb{N}$; for a rational $\Delta > 0$ with every $\Delta\sigma_l$ an integer these times fall into finitely many progressions of step Δ, so the crossing terms form finitely many cohorts with the integer step of length Δ in τ (31.16), the direction per step being $\Delta\boldsymbol{\sigma}$, from the positive point $\mathbf{c}$.

- 31.26, with a real index, gives every retained term a sign at each index that does not depend on the coefficients, hence not on ζ, so the averaged term has that sign.
- 31.27 applies to each cohort: one averaged term is absolutely integrable by 31.7(i) and its certificate, $\hat{h}_C < \infty$, and the coefficient map is continuous, so the essential supremum of the chamber rate is its supremum on S_λ.
- The chamber of the step direction is ΔC, and since $\sum_l \sigma_l' = 0$ on V, $h(\Delta\boldsymbol{\sigma}'; y) = \Delta h(\boldsymbol{\sigma}'; y)$. So the root rate per step is the limit $e^{\Delta\hat{h}_C}$.

Hence for every $\epsilon > 0$ and every cohort, $|\int \mathcal{R}_n\, d\lambda| \geq e^{n\Delta(\hat{h}_C - \epsilon/2)}$ for all large n. The crossing time being $\tau_n = \tau_0 + n\Delta$, this exceeds $e^{\tau_n(\hat{h}_C - \epsilon)}$ for all large n, the quotient being $e^{n\Delta\epsilon/2 - \tau_0(\hat{h}_C - \epsilon)}$; the cohorts being finitely many, the bound holds for all large k. With 31.7(ii), the terms tend to zero when $\hat{h}_C < 0$ and do not when $\hat{h}_C > 0$.

Now let every falling share be alone in its bundle and the merged fibre have dimension one.

- The merged pins are rational, so the merged directions form a line spanned by a rational vector $\mathbf{r}$. The merged direction of $\boldsymbol{\sigma}$, nonzero since $\boldsymbol{\sigma}$ falls a share, which is alone in its bundle, is $\mu\mathbf{r}$ with $\mu \neq 0$, and $\mu > 0$ after the choice of the sign of $\mathbf{r}$. The falling coordinates of the positive multiple $\mu^{-1}\boldsymbol{\sigma}$ of $\boldsymbol{\sigma}$ are coordinates of $\mathbf{r}$, rational.
- The chamber of $\mu^{-1}\boldsymbol{\sigma}$ is a nonempty polytope defined by rational equations and inequalities, so its vertices have rational coordinates. By 31.6 the closing along $\mu^{-1}\boldsymbol{\sigma}$, which is the closing along $\boldsymbol{\sigma}$ after a change of scale of τ, is the closing along that rational direction.
- At a crossing time the growing shares are positive at the real point of the ray, so the crossing is simple and nonresonant exactly when no other falling share lies in $-\mathbb{N}$ there, that is, when no two lattices of the falling shares meet.

The form of the crossing terms is the reference's (the reference, proof of thm:pool(iv)). The fibre is a bundle over the merged line whose fibres are the splits of the bundles at fixed sums, and 4.4 completes each split, $\int_{\sum_{l\in G} a_l=\bar{a}} \prod_{l\in G} \Gamma(a_l) e^{-a_l y_l} = \Gamma(\bar{a}) e^{-\bar{a} y_G}$. A crossing term on the merged line is a fibre integral of dimension zero: the residue of the Gamma crossed, times the other Gammas at the crossing point and $\prod_G e^{-\bar{a}_{k,G} y_G}$, divided by the Gammas of the exponents and by the determinant of the pins. □

48.32 Chapter 32: laws at their ends: regions of equality and reach

Theorem 32.10 (The reach along a direction is read on the law). *Parts (i) to (iv) as stated, with part (v).*

Proof. Parts (i) to (iv) are the reference's (the reference, proof of thm:dirreach); the piece of tables named in (iii) has a seam word with a Newton sum of full dimension when its tables form a connected family, by 31.4(ii). For (v) fix $\theta \geq 0$ and let g_θ be the infimum of (v).

First, $r \leq g_\theta$. On the half-strip Σ of 32.9 the imaginary parts are bounded, say by Y, so $e^{\theta|\operatorname{Im} z|} \leq e^{\theta Y}$ there. A bound $|\ell(z)| \leq C(1+|z|)^{n'} M^{\operatorname{Re} z} e^{\theta|\operatorname{Im} z|}$ on the half-plane, off the δ-neighbourhoods of the candidate poles, therefore gives on Σ the bound of 32.9 with $Ce^{\theta Y}$, and $r \leq M$.

Second, $g_\theta \leq r$. By (i), on (r,∞) the measure λ has a density that is a wall series of radius r. By (ii) with $x_0 = r$, for every $x' > r$ there is C with $|\ell(z)| \leq C\delta^{-J-1} x'^{\operatorname{Re} z}$ at every z with $\operatorname{Re} z \geq 0$ at distance at least $\delta \leq 1$ from the poles, whatever $\operatorname{Im} z$. The poles with $\operatorname{Re} z \geq 0$ lie among the candidate poles, and those with $\operatorname{Re} z < 0$ keep a distance at least some $\epsilon_0 > 0$ from the half-plane, ℓ converging absolutely on a strip about the imaginary axis (32.9). So at every z with $\operatorname{Re} z \geq 0$ at distance at least δ from the candidate poles, with $\delta_1 = \min(\delta, 1, \epsilon_0)$, $|\ell(z)| \leq C\delta_1^{-J-1} x'^{\operatorname{Re} z}$: the bound of (v) with the constant $C\delta_1^{-J-1}$ and $n' = 0$. So $M = x'$ is admissible in (v) at $\theta = 0$, hence at every $\theta \geq 0$, the factor $e^{\theta|\operatorname{Im} z|}$ being at least 1, and $g_\theta \leq x'$ for every $x' > r$. □

Theorem 32.15 (Ceilings, simple tops and fibres along a direction). *As stated, for every piece whose seam function converges at a real point of the pins with positive shares, with the sectors of any fan of its seam word.*

Proof. By 31.4(ii) the piece has walls and growth bounds (32.7), its walls being those of the generators of the fan, so 32.9 applies with these walls. The proof is the reference's (the reference,

proof of thm:dirheights), with the sectors of the fan in place of the orders of B, the generators in place of the tails, and the constant $|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|$ in the sector forms. At the point $\mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}} + z\bar{\boldsymbol{\sigma}}$ the integrand of the sector integral of π is

$$|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})| \prod_k q_k^{\kappa_k(\mathbf{w}_0)+i\partial_{\bar{\boldsymbol{\vartheta}}}\kappa_k-1}\, \mathbf{k}_\pi^{-\mathbf{w}_0-i\bar{\boldsymbol{\vartheta}}}\, \varrho_{\pi,\bar{\boldsymbol{\sigma}}}^{z}.$$

(i) The proof of 31.11(i) concerns the seam function alone. It holds at the points $\mathbf{w}_0 + \operatorname{Re} z\,\bar{\boldsymbol{\sigma}} + i(\bar{\boldsymbol{\vartheta}} + \operatorname{Im} z\,\bar{\boldsymbol{\sigma}})$, with δ the distance of the complex point from the hyperplanes on which a wall lies in $-\mathbb{N}$, which is at least a constant times the distance of z from the candidate poles (the reference, proof of cor:dirtube). On Σ the phase is bounded, the imaginary parts being bounded there. So $|\ell(z)| \le C\delta^{-(|B|-1)} e^{\operatorname{Re} z(\log H_{\mathcal{O}}+\epsilon)}$, and $r \le H_{\mathcal{O}} e^{\epsilon}$ for every $\epsilon > 0$.

(ii) and (iii) follow from the proof of 20.51(i) and (iii), with the point $\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}}$ in place of $b^d = -k$, $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in place of ϱ_π and $\phi_{\pi,\mathbf{w}_0}$ in place of ϕ_π.

- The falling variables carry $q_k^{\kappa_k(\mathbf{w}_0)-m|\dot{\kappa}_k|-1}$ with $|\dot{\kappa}_k|$ an integer, so the factor $(e^{2\pi i\kappa_k(\mathbf{w}_0)}-1)^{-1}$ of their circles is the same at every m. The letters whose exponents are nonnegative integers at every large m carry polynomials and need not be kept from vanishing.
- Every wall is the wall of a generator of some sector, so its falling rate is one of the integers $|\dot{\kappa}_k|$. The points m lie at a distance bounded below from the candidate poles, the falling walls being off the integers at $\mathbf{w}_0$, so $|\ell(m)| \le C(1+m)^D r'^m$ for every $r' > r$ by 32.9, and $\operatorname{growth}_{\mathbb{Z}} \le r$.
- Under (iii), $\operatorname{growth}_{\mathbb{Z}} = H_{\mathcal{O}}$ as in 20.51(iii), and a keyhole continuation of ceiling $H_{\mathcal{O}}$ gives $r \le H_{\mathcal{O}}$ by (i).

(iv) follows from the proofs of 20.55 and 20.56, applied to the positive directional law, with 32.10(i) and (ii) in place of 20.44.

- A keyhole box is handled as in (i).
- In a fibre box, $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ is meromorphic in each falling variable on its disc, the exponents $\bar{\sigma}_L$ and $\dot{\kappa}_j$ being integers. At a point above the ceiling and the stationary bound, the map is a submersion on its stratum: by 20.52 when one coordinate of falling rate one lies inside its disc, and by the definition of the stationary bound otherwise.
- The Mellin transform of a fibre law is a cube integral as in 20.32, with its poles in finitely many cosets of $(\operatorname{lcm}_j |\dot{\kappa}_j|)^{-1}\mathbb{Z}$, so 20.53 applies.

None of these steps uses more of a sector integral than the form displayed above, which 31.4(ii) supplies for every fan of the seam word of every such piece. □

Theorem 32.16 (The closing past the walls is read along its direction). *As stated, with the last sentence of (i), for the share side of 31.4(iv).*

Proof. Apart from the last sentence of (i), the statement is the reference's (the reference, proof of thm:dirclosing). For that sentence, the piece and the partner have walls and growth bounds (31.4(ii), applied to each), and $\boldsymbol{\sigma}$ has rational coordinates, so 32.12 applies to each of the two seam functions at $\mathbf{c}$, over the whole space V of the directions of the shares. There the growth $g_X(\boldsymbol{\sigma};\theta)$ of 31.4(iv), $X \in \{B,R\}$, is the function $g_V(\boldsymbol{\sigma};\theta)$ of 32.12 for that seam function: the two definitions coincide, the shares carrying no sums and the walls of 31.4(ii) serving as the walls of (W1), so that δ is measured from the same hyperplanes, those of the polar functions of the seam function (31.5).

It does not increase with θ and tends to $r_X^\sharp$ as $\theta \downarrow 0$, so $g_X(\boldsymbol{\sigma};\theta) \le r_X^\sharp$ for every $\theta > 0$. Then

$$h_{\theta/2,\theta/2}(\boldsymbol{\sigma}) \le \sum_l \sigma_l \log|\sigma_l| + \log r_B^\sharp + \log r_R^\sharp + \frac{2\theta^2}{\pi} \sum_{l:\,\sigma_l>0} \sigma_l,$$

which is negative for θ small when $\sum_l \sigma_l \log|\sigma_l| + \log r_B^\sharp + \log r_R^\sharp < 0$. 31.4(iv) gives the closing, its terms and remainders being $O(e^{\tau(h_{\theta/2,\theta/2}(\boldsymbol{\sigma})+\epsilon)})$, and letting $\theta \to 0$ gives the stated bound. □

Corollary 32.18 (A law with log expansions bounds a closing of any fibre dimension). *As stated, with its last sentence, for the share side of 31.4(iv).*

Proof. Apart from the last sentence, the statement is the reference's (the reference, proof of cor:anyfibre). For that sentence, apply 32.14 to the law of the piece and to that of the partner, a piece with the reversed excesses. On the contour through $\mathbf{c} + \tau\boldsymbol{\sigma}'$ each seam function is at most $C\delta^{-n} X_{\text{in}}^X(\boldsymbol{\sigma}')^\tau$, with no factor in the imaginary parts and no power of τ, so $g_X(\boldsymbol{\sigma}';0) \le X_{\text{in}}^X(\boldsymbol{\sigma}')$ for every direction $\boldsymbol{\sigma}'$. Hence

$$h_{0,0}(\boldsymbol{\sigma}') \le \sum_l \sigma_l' \log|\sigma_l'| + \log X_{\text{in}}^B(\boldsymbol{\sigma}') + \log X_{\text{in}}^R(\boldsymbol{\sigma}'), \qquad \boldsymbol{\sigma}' \in C(\boldsymbol{\sigma}).$$

The polar functions of the share side of 31.4(iv) are the shares and the walls of both words, with $\Omega = A$, and (H1) holds by 31.4(ii) and Stirling's formula. So 31.6 makes the closings along the directions of the chamber one series; their crossings and the near coincidences of these are those of the falling polar functions, the same for every direction of the chamber, so $\mathbf{c}$ lies off the null set of 31.4(iv) for each of them; and 31.4(iv) at $\theta_B = \theta_R = 0$, along a direction of the chamber whose bound is within ϵ of the infimum, gives the closing along $\boldsymbol{\sigma}$ with the stated bound. On the chamber the falling coordinates are fixed and the others are nonnegative, where $x \log x$ is convex, and the logarithms of the inner bounds are convex (32.13). □

Remark (The directional ratio at width one, in the paragraph after 32.9). At width one with one table, factoring weights normalized as in 28.11, and $s, t > 0$, the directional ratio $x_{D_0}^*/x_{D_1}$ along $\bar{\boldsymbol{\sigma}} = (-1, 1, -1, 1)$ is the ratio ϱ_B at the seam points, and every coordinate of the seam torus there is a power of ϱ_B.

Proof. On the seam torus $x_{D_0} = x_{D_1}^* = 1$, so $\mathbf{x}^{-\bar{\boldsymbol{\sigma}}} = x_{D_0}\, x_{D_1}^{-1}\, x_{D_0}^*\, x_{D_1}^{*-1} = x_{D_0}^*/x_{D_1}$. At the seam point of u, 28.5 gives $\varrho_{D_0}^{\rho*}/\varrho_{D_1}^{\rho} = V_{D_0}^* V_{D_0}/(V_{D_1}^* V_{D_1})$, the scale cancelling. By Lemma S13(i) this is

$$\Big(\frac{\hat\theta_0}{\hat\theta_1}\Big)^2 \frac{\Psi_{\mathcal{B}/R}\, K_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|B}\, K_{\mathcal{B}|B}} = \Big(\frac{\hat\theta_0}{\hat\theta_1}\Big)^2 \varrho_B,$$

with $\varrho_B = \gamma_B^\Psi/\gamma_B^K$ of 19.2. Under the normalization $\hat\theta_0 = \hat\theta_1 = 1$, so it is ϱ_B. By 28.11(ii) with $N = 1$, $\mathbf{x} = (1, \varrho_B)$ and $\psi = \varrho_B^{t/(s+t)}$, the coordinates of the seam point are 1, $\varrho_B^{-t/(s+t)}$, $\varrho_B^{s/(s+t)}$ and 1, all powers of ϱ_B. □

48.33 Chapter 34: the motive of a family and the coaction along a tree

Five outside works enter these proofs, each named where it is used together with the property of the setting it needs: the theory of toric varieties (W. Fulton, *Introduction to toric varieties*, Princeton

University Press, 1993, Chapters 1 to 3), for the braid fan; F. Brown, *Notes on motivic periods*, Commun. Number Theory Phys. 11 (2017), 557–655 (*Brown's notes*), for motivic periods, the coaction and the weight-graded face maps; F. Brown, *Feynman amplitudes, coaction principle, and cosmic Galois group*, Commun. Number Theory Phys. 11 (2017), 453–556, §5, for the affine model; A. Huber and S. Müller-Stach, *Periods and Nori motives*, Springer, 2017, for the functoriality of relative cohomology in Nori's category, whose objects and morphisms realize in Brown's category; and H. Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero*, Ann. of Math. 79 (1964), 109–326. In this section a *pair* (V, D) is a smooth variety V over $\mathbb{Q}$ of dimension m with a divisor $D \subset V$ with simple normal crossings and smooth components. Its relative cohomology $H^m(V, D)$ is an object of Brown's category, a morphism of pairs giving a morphism of Nori's category and so of Brown's (Huber and Müller-Stach). A regular form η of degree m on V is closed and restricts to zero on the components of D, which have smaller dimension; $[\eta]$ is its class in $H^m_{\mathrm{dR}}(V, D)$. A compact semialgebraic $\sigma \subset V(\mathbb{R})$ that is the closure of an oriented open semialgebraic set $\sigma^\circ \subseteq (V \setminus D)(\mathbb{R})$ of dimension m, with $\sigma \setminus \sigma^\circ \subseteq D$, is a *relative cycle*: a semialgebraic triangulation makes the sum of its oriented top simplices a chain whose boundary lies in $D(\mathbb{R})$, and its class in $H_m(V(\mathbb{C}), D(\mathbb{C}); \mathbb{Q})$ is written σ again. The comparison of the two realizations is integration, so the period of $[H^m(V, D), \sigma, [\eta]]^{\mathrm{mot}}$ is $\int_\sigma \eta$. The positive cells $\widetilde{\sigma}_Z$ are oriented as in Definition 34.5.

Definition 34.1 (The model of a family). As stated.

Proof. $\Omega_{\mathcal{E}}$ is the contraction of $d\alpha_{e_1} \wedge \cdots \wedge d\alpha_{e_n}$ with the vector field $\sum_e \alpha_e \, \partial/\partial\alpha_e$, which does not depend on the order, while the volume form changes by the sign of the permutation. The coefficient $f = \alpha^{\mathbf{m-1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ of $\omega_F = f\,\Omega_{\mathcal{E}}$ is homogeneous of the degree $m_{\mathcal{E}} - \sum_j (s_j r_j + t_j r_j^*) - n = \kappa_{\mathcal{E}} - n = -n$ (Definition 8.1), the masses being on the balance wall. With the affine coordinates $x_i = \alpha_{e_i}/\alpha_{e_n}$, $i < n$, of the chart $\alpha_{e_n} \neq 0$ of $\mathbb{P}(\mathbb{Q}^{\mathcal{E}})$, a direct computation gives $dx_1 \wedge \cdots \wedge dx_{n-1} = (-1)^{n-1} \alpha_{e_n}^{-n} \Omega_{\mathcal{E}}$, so that $f\,\Omega_{\mathcal{E}} = (-1)^{n-1} f(x_1, \ldots, x_{n-1}, 1)\, dx_1 \wedge \cdots \wedge dx_{n-1}$ is the pull-back of a rational form on that chart. These forms agree on the overlaps of the charts, all pulling back to $f\,\Omega_{\mathcal{E}}$ on $\mathbb{Q}^{\mathcal{E}} \setminus \{0\}$; so ω_F is a rational form of degree $n-1$ on $\mathbb{P}(\mathbb{Q}^{\mathcal{E}})$, and on $X_{\mathcal{E}}$, which contains the torus of $\mathbb{P}(\mathbb{Q}^{\mathcal{E}})$ as a dense open set. □

Definition 34.2 (Flags and steps). As stated.

Proof. The steps are $\ell + 1$ nonempty sets that partition $\mathcal{E}$, so $\sum_i (|Z_i| - 1) = n - 1 - \ell$. For $\ell \geq 1$ the step lists at Z_i depend on S_{i-1} and S_i alone (Proposition 12.20), and the α_J with $J \subseteq Z_i$ on Z_i alone; so does $\mathrm{Let}(Z_i; \mathfrak{f})$. When $|Z_i| = 1$, X_{Z_i} is the toric variety of the fan $\{0\}$ in $\mathbb{R}^{Z_i}/\mathbb{R}\mathbf{1} = 0$, a point, and the step has no letter: the basis polynomials of the lists on one mark are 1 and α_e, monomials, and no $J \subseteq Z_i$ has two marks. □

Lemma 34.3 (Charts, strata and the letters in a chart). *As stated.*

Proof. (i) The fan lies in $N_{\mathbb{R}}$, $N = \mathbb{Z}^{\mathcal{E}}/\mathbb{Z}\mathbf{1}$, with the dual lattice $M = \{\mathbf{a} \in \mathbb{Z}^{\mathcal{E}} : \; a_{\mathcal{E}} = 0\}$, whose elements are the characters $\alpha^{\mathbf{a}}$ of the torus. For an order $e_1, \ldots, e_n$ the vectors $\mathbf{1}_{S_1}, \ldots, \mathbf{1}_{S_{n-1}}, \mathbf{1}$ form a basis of $\mathbb{Z}^{\mathcal{E}}$, their matrix in the basis $(\mathbf{e}_{e_k})_k$ being triangular with ones on the diagonal; so the images of $\mathbf{1}_{S_1}, \ldots, \mathbf{1}_{S_{n-1}}$ form a basis of N, with the dual basis $\mathbf{e}_{e_i} - \mathbf{e}_{e_{i+1}}$, $i < n$, of M, since $\langle \mathbf{1}_{S_k}, \mathbf{e}_{e_i} - \mathbf{e}_{e_{i+1}} \rangle$ is 1 for $k = i$ and 0 otherwise. Every $\mathbf{x} \in \mathbb{R}^{\mathcal{E}}$ lies in such a cone modulo $\mathbf{1}$: ordering $\mathcal{E}$ with $x_{e_1} \geq \cdots \geq x_{e_n}$, $\mathbf{x} = \sum_{i<n} (x_{e_i} - x_{e_{i+1}}) \mathbf{1}_{S_i} + x_{e_n} \mathbf{1}$. The cones of two chains meet in the cone of their common sets, a face of both. So the braid fan is a complete fan whose cones are spanned by parts of bases of N; it is the normal fan of the polytope $\sum_{\{a,b\}} \mathrm{conv}\{\mathbf{e}_a, \mathbf{e}_b\}$, the sum over the pairs of marks, since on the interior of the cone of an order a linear form $\mathbf{x}$ has

$x_{e_1} > \cdots > x_{e_n}$ and attains its least value on that polytope at the one vertex $\sum_{\{a,b\}} \mathbf{e}_{\max(a,b)}$, the maximum taken in the order, which is the same on the whole cone and differs for different orders. By the dictionary between fans and toric varieties (Fulton, Chapters 1 to 3), which applies to such a fan, $X_{\mathcal{E}}$ is smooth and complete, and projective, the fan being a normal fan; the chart of the cone of the chain is the affine space with the coordinates $\alpha^{\mathbf{e}_{e_i}-\mathbf{e}_{e_{i+1}}} = \varepsilon_i$; the charts of the maximal cones cover $X_{\mathcal{E}}$; and the closure D_S of the orbit of the ray ρ_S meets the chart exactly when ρ_S is a ray of its cone, in the coordinate hyperplane of the dual coordinate, here $\{\varepsilon_i = 0\}$ for $S = S_i$, several of these closures meeting exactly when their rays span a cone, here when their sets form a chain (the orbit–cone correspondence). In every chart D is a union of coordinate hyperplanes, so it has simple normal crossings and smooth components. In every chart the torus is $\{\varepsilon_i \neq 0 \text{ for all } i\}$ and $\widetilde{\sigma}^\circ_{\mathcal{E}} = (0,\infty)^{n-1}$, since $\alpha_{e_j}/\alpha_{e_n} = \prod_{i\geq j} \varepsilon_i$; its closure in the open set $U(\mathbb{R})$ is $[0,\infty)^{n-1} = \widetilde{\sigma}_{\mathcal{E}} \cap U$. The points of $\widetilde{\sigma}^\circ_{\mathcal{E}}$ with $\alpha_{e_1} < \cdots < \alpha_{e_n}$ are $(0,1)^{n-1}$ in the chart, with the closure $[0,1]^{n-1}$, compact and so closed in $X_{\mathcal{E}}(\mathbb{R})$. Every point of $\widetilde{\sigma}^\circ_{\mathcal{E}}$ is a limit of points with distinct coordinates, so $\widetilde{\sigma}^\circ_{\mathcal{E}}$ lies in the union of the sectors, a finite union of compact sets, and so does its closure $\widetilde{\sigma}_{\mathcal{E}}$, which contains every sector. The open cube $(0,1)^{n-1}$ of one order is open in $X_{\mathcal{E}}(\mathbb{R})$ and misses the open sector of every other order, hence its closure; so two sectors meet in the boundary of a cube, of measure zero.

(ii) For a basis T of a table of rank δ, $\alpha^T/\alpha^\delta_{e_n} = \alpha^{\mathbf{1}_T - \delta\mathbf{e}_{e_n}} = \prod_{i<n} \varepsilon_i^{|T\cap S_i|}$, by the dual basis of (i) and $e_n \notin S_i$. So $K_{\mathcal{B}}/\alpha^\delta_{e_n}$ is the sum of these monomials over the bases. Every exponent $|T \cap S_i|$ is at least $d(S_i)$ (Theorem 7.13), and along the maximal chain $(S_i)_{i<n}$ exactly one basis has $|T \cap S_i| = d(S_i)$ for every i, its term having the coefficient 1 (Proposition 12.20); dividing by its monomial leaves $\widetilde{f}$, a polynomial with nonnegative integer coefficients, the numbers of bases with given traces, and with the constant term 1. For $\Psi_{\mathcal{B}} = K_{\mathcal{B}^*}$ the same holds with the dual table, whose floors are the $d^*(S)$ (Proposition 7.2, Theorem 7.13), and for α_J with the list $\binom{J}{1}$ on $\mathcal{E}$, whose floor at S is 1 or 0 as $J \subseteq S$ or not. In the sector coordinates of Theorem 8.5(ii), with $\pi(k) = e_{n+1-k}$, the coordinates are $q_k = \varepsilon_{n+1-k}$ and the tails are $U_k = S_{n+1-k}$, and $\widetilde{f}$ is the sector letter there. On $\varepsilon \geq 0$, $\widetilde{f} \geq 1$.

On $\{\varepsilon_i = 0 :\ i \in I\}$ only the bases T with $|T \cap S_i| = d(S_i)$ for every $i \in I$ leave a term. By Proposition 12.20 for the chain $(S_i)_{i\in I}$ they are the unions $\bigsqcup_Z X_Z$ of one basis X_Z of the step table at the bottom at each step Z of that chain. For $i \notin I$ let $Z = S_b \setminus S_a$ be the step with $S_a \subsetneq S_i \subsetneq S_b$, where S_a may be $\varnothing$ and S_b may be $\mathcal{E}$; then $|T \cap S_i| = d(S_a) + |X_Z \cap (S_i \setminus S_a)|$. The basis at the bottom of the maximal chain is one of these unions and has the least traces, so $d(S_i) - d(S_a)$ is the floor $d_Z(S_i \setminus S_a)$ of $S_i \setminus S_a$ in the step table at Z, and

$$\widetilde{f}\big|_{\{\varepsilon_i = 0:\, i\in I\}} = \prod_Z \sum_{X_Z} \prod_{a<k<b} \varepsilon_k^{|X_Z \cap (S_k \setminus S_a)| - d_Z(S_k \setminus S_a)}.$$

The factor of Z is the polynomial that the rule above attaches to K of the step table in the chart of X_Z of the order $e_{a+1}, \ldots, e_b$, whose initial segments are the $S_k \setminus S_a$ and whose coordinates are the ε_k, $a < k < b$. For Ψ the same holds with the tops and the dual tables, and for α_J with its step letters (Proposition 12.20). Finally, on the torus part of U, $f = 0$ exactly where $\widetilde{f} = 0$. No irreducible component of $\{\widetilde{f} = 0\}$ lies in a coordinate hyperplane $\{\varepsilon_i = 0\}$, since ε_i does not divide $\widetilde{f}$, whose constant term is 1; so each component is the closure of its part in the torus, and $\{\widetilde{f} = 0\}$ is the closure in U of the zero set of f in the torus, that is $\widetilde{Y}_f \cap U$.

(iii) In the affine coordinates $x_j = \alpha_{e_j}/\alpha_{e_n} = \prod_{i\geq j} \varepsilon_i$, $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}} = (-1)^{n-1} \bigwedge_{j<n} dx_j/x_j$ (proof of Definition 34.1), and $\bigwedge_j d\log x_j = \bigwedge_i d\log \varepsilon_i$, the matrix of the $\log x_j$ in the $\log \varepsilon_i$ being triangular with ones on the diagonal; so $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}} = (-1)^{n-1} \bigwedge_{i<n} d\varepsilon_i/\varepsilon_i$. By (ii), $\alpha^{\mathbf{m}}/\alpha^{m_{\mathcal{E}}}_{e_n} = \prod_i \varepsilon_i^{m_{S_i}}$, $K_j/\alpha^{r_j}_{e_n} = \prod_i \varepsilon_i^{d_j(S_i)} \widetilde{K}_j$ and $\Psi_j/\alpha^{r^*_j}_{e_n} = \prod_i \varepsilon_i^{d^*_j(S_i)} \widetilde{\Psi}_j$. On the balance wall the powers of α_{e_n} cancel,

and the exponent of ε_i is $m_{S_i} - \sum_j \big(s_j d_j(S_i) + t_j d_j^*(S_i)\big) - 1 = \kappa_{S_i} - 1$. □

Lemma S15 (Integrals over the positive cell)**.** *Let ϕ be a measurable function on $\mathbb{R}^{\mathcal{E}}_{>0}$, homogeneous of degree zero, $n = |\mathcal{E}| \geq 2$. Then, in $[0,\infty]$ when $\phi \geq 0$, and whenever one side converges absolutely,*

$$\int_{\widetilde{\sigma}^\circ_{\mathcal{E}}} \phi\, \frac{\Omega_{\mathcal{E}}}{\alpha^{\mathcal{E}}} = \int_{\Delta_{\mathcal{E}}} \phi(u)\, u^{-\mathbf{1}}\, d\omega_{\mathcal{E}}(u) = \int_{\Delta_B \times \Delta_R} \int_0^\infty \phi(\rho u, u')\, \frac{d\rho}{\rho}\; u^{-\mathbf{1}} u'^{-\mathbf{1}}\, d\omega_B(u)\, d\omega_R(u')$$

for every partition $\mathcal{E} = B \sqcup R$ into nonempty sets; in the chart of an order the left side is $\int_{(0,\infty)^{n-1}} \phi \prod_i d\varepsilon_i/\varepsilon_i$. If $\phi\, \Omega_{\mathcal{E}}/\alpha^{\mathcal{E}}$ is the restriction of a form that is regular on an open set containing $\widetilde{\sigma}_{\mathcal{E}}$, the integrals converge absolutely and are its integral over $\widetilde{\sigma}_{\mathcal{E}}$.

Proof. In the logarithmic coordinates $y = \log \alpha$, $\alpha^{-\mathbf{1}} d\alpha = dy$. For an order of $\mathcal{E}$ the linear map $y \mapsto \big((y_{e_i} - y_{e_{i+1}})_{i<n}, y_{e_n}\big)$ has the determinant ± 1, and its first $n-1$ coordinates are the $\log \varepsilon_i$, on which ϕ depends alone. For a positive h homogeneous of degree one and a measurable $\varphi \geq 0$ on $(0,\infty)$, at fixed ε the substitution $\rho = e^{y_{e_n}} h(\alpha/\alpha_{e_n})$ has $d\rho/\rho = dy_{e_n}$; by Tonelli

$$\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \phi(\alpha)\, \varphi\big(h(\alpha)\big)\, \alpha^{-\mathbf{1}} d\alpha = \int_0^\infty \varphi(\rho) \frac{d\rho}{\rho} \int_{(0,\infty)^{n-1}} \phi \prod_i \frac{d\varepsilon_i}{\varepsilon_i},$$

the same holds with $|\phi|$, and with ϕ whenever the integrals of $|\phi|$ are finite. With $h = |\alpha|$, Lemma 2.7 on $\mathcal{E}$ writes the left side as $\int_0^\infty \varphi(\rho) \frac{d\rho}{\rho} \int_{\Delta_{\mathcal{E}}} \phi(u) u^{-\mathbf{1}} d\omega_{\mathcal{E}}(u)$. With $h = |\alpha_R|$, Lemma 2.7 on B and on R, $\alpha_B = \rho' u$ and $\alpha_R = \lambda u'$, gives $\alpha^{-\mathbf{1}} d\alpha = u^{-\mathbf{1}} u'^{-\mathbf{1}} \frac{d\rho'}{\rho'} \frac{d\lambda}{\lambda} d\omega_B(u) d\omega_R(u')$ and $\phi(\rho' u, \lambda u') = \phi(\rho u, u')$ with $\rho' = \lambda\rho$, so the left side is the third expression times $\int \varphi(\lambda) d\lambda/\lambda$. A φ with $\int_0^\infty \varphi\, d\rho/\rho = 1$ gives the identities. In the chart $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}} = (-1)^{n-1} \bigwedge_i d\varepsilon_i/\varepsilon_i$ (proof of Lemma 34.3(iii)), which with the orientation of $\widetilde{\sigma}^\circ_{\mathcal{E}} = (0,\infty)^{n-1}$ is the measure $\prod_i d\varepsilon_i/\varepsilon_i$. If the form is regular near the compact $\widetilde{\sigma}_{\mathcal{E}}$, it is $g\, d\varepsilon$ on each sector cube with g continuous on the cube; so its integral over each sector converges absolutely, the sectors cover $\widetilde{\sigma}_{\mathcal{E}}$ and meet in sets of measure zero, and $\widetilde{\sigma}_{\mathcal{E}} \setminus \widetilde{\sigma}^\circ_{\mathcal{E}}$ lies in the coordinate hyperplanes of the charts (Lemma 34.3(i)). □

Lemma S16 (The continued period off the balance wall)**.** *Let F be a family of lists on $\mathcal{E}$, $n \geq 2$. For an order π with the chain $(S_i^\pi)_{i<n}$ let $\widetilde{K}_j, \widetilde{\Psi}_j$ and $\widetilde{L}_\pi$ be the sector letters of K_j, Ψ_j and of the linear letter $\alpha_{\mathcal{E}} = |\alpha|$ in its chart (Lemma 34.3(ii)), and $h_\pi(\varepsilon; \kappa, \mathbf{s}, \mathbf{t}) = \widetilde{L}_\pi^{-\kappa} \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j}$. As meromorphic functions of $(\mathbf{m}, \mathbf{s}, \mathbf{t}) \in \mathbb{C}^{\mathcal{E}} \times \mathbb{C}^N \times \mathbb{C}^N$,*

$$\frac{J_F(\mathbf{m}; \mathbf{s}, \mathbf{t})}{\Gamma(\kappa_{\mathcal{E}})} = \sum_\pi \operatorname{cube}\Big(\big(\kappa_{S_i^\pi}\big)_{i<n},\ (\kappa_{\mathcal{E}}, \mathbf{s}, \mathbf{t});\ h_\pi\Big),$$

with cube *the cube integral of Lemma 8.4; the right side is holomorphic off the hyperplanes $\kappa_S = -j$, $\varnothing \neq S \subsetneq \mathcal{E}$, $j \in \mathbb{N}$. So the period $P_F = J_F/\Gamma(\kappa_{\mathcal{E}})$ (Theorem 8.7) has its poles on these hyperplanes only, and on the balance wall it is the right side at $\kappa_{\mathcal{E}} = 0$, where $h_\pi = \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j}$.*

Proof. On the region where $\operatorname{Re} \kappa_S > 0$ for every nonempty $S \subseteq \mathcal{E}$, which contains the points with $\mathbf{s} = \mathbf{t} = 0$ and $\operatorname{Re} \mathbf{m} > 0$, Theorem 8.5(ii) is the display, read in the chart coordinates as in the proof of Lemma 34.3(ii): its sector letters $k_{j,\pi}, \psi_{j,\pi}, L_\pi$ are $\widetilde{K}_j, \widetilde{\Psi}_j, \widetilde{L}_\pi$. The sector letters are at least 1 on the cube, so h_π is smooth in ε near $[0,1]^{n-1}$, with every ε-derivative continuous in all the variables and holomorphic in $(\kappa, \mathbf{s}, \mathbf{t}) \in \mathbb{C}^{1+2N}$. The map $(\mathbf{m}, \mathbf{s}, \mathbf{t}) \mapsto \big((\kappa_{S_i^\pi})_{i<n}, \kappa_{\mathcal{E}}, \mathbf{s}, \mathbf{t}\big)$ is a linear isomorphism of $\mathbb{C}^{\mathcal{E}} \times \mathbb{C}^{2N}$, since $\mathbf{m} \mapsto (m_{S_1^\pi}, \ldots, m_{S_{n-1}^\pi}, m_{\mathcal{E}})$ is triangular. Lemma 8.4 continues

each term to a meromorphic function on the whole space with poles on the hyperplanes $\kappa_{S_i^\pi} = -j$ only. The sum agrees with the meromorphic $J_F/\Gamma(\kappa_\mathcal{E})$ (Theorems 8.5(iii) and 8.7) on a nonempty open set, hence everywhere. At $\kappa_\mathcal{E} = 0$, $\widetilde{L}_\pi^{-\kappa_\mathcal{E}} = 1$. □

Proposition 34.4 (The model). *As stated.*

Proof. (i) $\widetilde{\sigma}_\mathcal{E}$ is the union of the finitely many sectors, cubes (Lemma 34.3(i)), so it is compact. In a chart $\widetilde{\sigma}_\mathcal{E} \cap U = [0,\infty)^{n-1}$ and $\widetilde{\sigma}_\mathcal{E}^\circ = (0,\infty)^{n-1}$, whose difference lies in the coordinate hyperplanes, that is in D; and the charts cover $X_\mathcal{E}$. For a letter f of a list or a linear letter, $\widetilde{Y}_f \cap U = \{\widetilde{f} = 0\}$ and $\widetilde{f} \geq 1$ on $[0,\infty)^{n-1}$ (Lemma 34.3(ii)).

(ii) Let $\varnothing \neq S \subsetneq \mathcal{E}$ and take an order whose chain contains $S = S_i$. D_S meets V_F: the face $\{\varepsilon_i = 0\} \cap [0,1]^{n-1}$ of the sector cube lies in $D_S \cap \widetilde{\sigma}_\mathcal{E}$, and $\widetilde{\sigma}_\mathcal{E} \subseteq V_F(\mathbb{R})$ by (i). On $V_F \cap U$ every $\widetilde{K}_j$ with $s_j \neq 0$ and every $\widetilde{\Psi}_j$ with $t_j \neq 0$ is a unit, V_F omitting their zero sets, so by Lemma 34.3(iii) ω_F is a unit times $\prod_k \varepsilon_k^{\kappa_{S_k}-1} d\varepsilon_1 \wedge \cdots \wedge d\varepsilon_{n-1}$ there. Its order along D_S, read at a point of $D_S \cap V_F \cap U$, is $\kappa_S - 1$, and on the torus, where every ε_k is a unit, it has neither zeros nor poles. A prime divisor of V_F either meets the torus or is one of the $D_S \cap V_F$, each irreducible as a nonempty open subset of the irreducible D_S. So the divisor of ω_F on the smooth V_F is $\sum_S (\kappa_S - 1)(D_S \cap V_F)$, and ω_F is regular exactly when this divisor is effective, that is when F is convergent. On the cube of a sector, $\omega_F = \pm \prod_k \varepsilon_k^{\kappa_{S_k^\pi}-1} h_\pi \, d\varepsilon$ with $h_\pi = \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j}$ positive and continuous on the compact cube, so bounded between two positive constants; as $\int_0^1 \varepsilon^{\kappa-1} d\varepsilon$ converges for $\kappa \geq 1$, the integral over $\widetilde{\sigma}_\mathcal{E}$ converges when F is convergent. By Lemma S15 it is the integral over $\Delta_\mathcal{E}$ of $u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}$, the period (Theorem 8.7), positive with the orientation in which $\Omega_\mathcal{E}/\alpha^\mathcal{E}$ is positive on $\widetilde{\sigma}_\mathcal{E}^\circ$.

(iii) V_F is open in the smooth $X_\mathcal{E}$. The complement in $X_\mathcal{E}$ of the $\widetilde{Y}_{\alpha_J}$, $|J| \geq 2$, is Brown's affine model (Brown, *Feynman amplitudes*, §5), which applies because $X_\mathcal{E}$ is the iterated blow-up of $\mathbb{P}(\mathbb{Q}^\mathcal{E})$ along its coordinate linear subspaces, the toric variety of the subdivision of its fan by all chains, and the removed hypersurfaces are the strict transforms of the hyperplanes $\alpha_J = 0$ of all the sets J of at least two marks. In toric terms the divisor $A = \sum_{|J|\geq 2} \widetilde{Y}_{\alpha_J}$ is ample. In the chart of an order, $\alpha_J/\alpha_y = \widetilde{\alpha}_J$ for the last mark y of J in the order (Lemma 34.3(ii)), so $\widetilde{Y}_{\alpha_J}$ is the divisor of the section $\sum_{e\in J} \alpha^{\mathbf{e}_e}$ of the line bundle of the simplex $\Delta_J = \operatorname{conv}\{\mathbf{e}_e : \ e \in J\}$, whose vertex of least value on the cone of the order is $\mathbf{e}_y$; A is then the divisor of a section of the line bundle of the polytope $\sum_J \Delta_J$, whose vertex of least value on the interior of the cone of the order $e_1, \ldots, e_n$ is $\sum_J \mathbf{e}_{y_J} = \sum_k (2^{k-1} - 1)\, \mathbf{e}_{e_k}$, the mark e_k being the last mark of $2^{k-1} - 1$ sets J; it is the only point of least value there and differs for different orders. So the normal fan of $\sum_J \Delta_J$ is the braid fan, the divisor A is ample (Fulton, Chapter 3), a multiple of it is a hyperplane section in a projective embedding, and its complement is affine. A finite set $\mathcal{G}$ of polynomials that contains the α_J removes in addition the $\widetilde{Y}_g$, $g \in \mathcal{G}$, divisors of the smooth $X_\mathcal{E}$, hence Cartier divisors; the complement of a Cartier divisor in an affine variety is affine, the inclusion being locally the complement of the zero set of one function. D_F is the restriction of D to the open V_F, with simple normal crossings and smooth components (Lemma 34.3(i)). A stratum is an intersection of components meeting transversally, so smooth, and closed in the affine V_F, so affine.

(iv) Let τ be the cone of the flag. By the orbit–cone correspondence (Fulton, Chapter 3), $D_{S_1} \cap \cdots \cap D_{S_\ell}$ is the toric variety of the star of τ, the fan in $N/(N \cap \mathbb{R}\tau)$ of the images of the cones that contain τ. Restriction to the steps, $\mathbf{x} \mapsto (\mathbf{x}|_{Z_i} \bmod \mathbb{Z}\mathbf{1}_{Z_i})_i$, maps $\mathbb{Z}^\mathcal{E}$ onto $\bigoplus_i \mathbb{Z}^{Z_i}/\mathbb{Z}\mathbf{1}_{Z_i}$ with the kernel spanned by $\mathbf{1}_{Z_1}, \ldots, \mathbf{1}_{Z_{\ell+1}}$, that is by $\mathbf{1}_{S_1}, \ldots, \mathbf{1}_{S_\ell}$ and $\mathbf{1}$; a set S with $S_{i-1} \subsetneq S \subsetneq S_i$ goes to $\mathbf{1}_{S\setminus S_{i-1}}$ in the summand of Z_i and to 0 in the others. So the cones of the chains that refine the flag go to the products of cones of chains of the steps, the star of τ is the product of the braid

fans of the steps, and $D_{S_1} \cap \cdots \cap D_{S_\ell} \cong \prod_i X_{Z_i}$. The charts that meet the stratum are those of the orders whose chains contain the flag; in such a chart the stratum is $\{\varepsilon_i = 0 :\ i \in I\}$, I the positions of the sets of the flag, and the remaining coordinates are the chart coordinates of the steps. By Lemma 34.3(ii) $\widetilde{Y}_f$ meets it in the union over the steps Z of the zero sets of the polynomials attached to the step letters of f at Z, and by the same lemma on Z these are the $\widetilde{Y}_g \cap U_Z$ for the step letters g, empty for a monomial. The step letters at Z of the augmented letters of F that are not monomials form $\mathrm{Let}(Z;\mathfrak{f})$: the step letter of α_J is $\alpha_{J\cap Z}$ at the first step Z with $J \subseteq S_b$ and 1 elsewhere, and every $\alpha_{J'}$ with $J' \subseteq Z$, $|J'| \geq 2$, arises from $J = J'$. So the stratum of $\mathfrak{f}$ is $\prod_i V_{Z_i;\mathfrak{f}}$. Its boundary, its meeting with the D_S of the sets S outside the flag, is $\{\varepsilon_k = 0\}$ for the other positions k, the union of the pull-backs of the toric boundaries of the factors; and $\widetilde{\sigma}_{\mathcal{E}}$ meets it in $[0,\infty)^{n-1} \cap \{\varepsilon_i = 0 :\ i \in I\} = \prod_Z [0,\infty)^{|Z|-1}$, chart by chart, that is in $\prod_i \widetilde{\sigma}_{Z_i}$.

(v) Where every proper wall has positive real part the period converges (Theorem 8.7), and by Lemma S15, the sectors and Lemma 34.3(iii) it is $\sum_\pi \int_{[0,1]^{n-1}} \prod_i \varepsilon_i^{\kappa_{S_i^\pi}-1} h_\pi \, d\varepsilon = \sum_\pi \mathrm{cube}(\boldsymbol{\kappa}_\pi(\mathbf{m}); h_\pi)$, each term converging absolutely. By Lemma S16 the right side is, on the whole balance wall, the restriction of $J_F/\Gamma(\kappa_{\mathcal{E}})$, meromorphic there jointly in the masses and the exponents, with poles on the hyperplanes $\kappa_S = -j$ only. □

Definition 34.5 (Motivic periods; Galois conjugates; open spaces). As stated.

Proof. The motivic periods, the de Rham periods, their relations and the coaction are those of Brown's notes (§2 and §3.1), where the coaction is shown to be well defined. For a convergent family, (V_F, D_F) is a pair with V_F affine (Proposition 34.4(iii)), so H_F is an object of Brown's category; $\widetilde{\sigma}_{\mathcal{E}}$ is a relative cycle of it, compact, the closure of the open cell, with $\widetilde{\sigma}_{\mathcal{E}} \setminus \widetilde{\sigma}^\circ_{\mathcal{E}} \subseteq D_F$ and $\widetilde{\sigma}_{\mathcal{E}} \subseteq V_F(\mathbb{R})$ (Proposition 34.4(i)); and ω_F is a regular form of degree $n-1$ on V_F (Proposition 34.4(ii)), with its class $[\omega_F]$. The period of P_F^{mot} is $\int_{\widetilde{\sigma}_{\mathcal{E}}} \omega_F = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Proposition 34.4(ii)). For a step with $|Z| \geq 2$ the letters of the step are letters of lists on Z and the α_J, $J \subseteq Z$, $|J| \geq 2$; so by Proposition 34.4(i) and (iii) on Z, $(V_{Z;\mathfrak{f}}, \partial V_{Z;\mathfrak{f}})$ is a pair with $V_{Z;\mathfrak{f}}$ affine and $\widetilde{\sigma}_Z$ is a relative cycle of it. □

Lemma 34.6 (Galois conjugates). *As stated.*

Proof. $\mathcal{P}^{\mathrm{dR}}$ is the ring of functions on the de Rham Galois group, $[M,f,\omega]^{\mathrm{dR}}$ being the function $g \mapsto f(g\,\omega)$ (Brown's notes, §2); evaluation at the identity is a $\mathbb{Q}$-linear map $\varphi_1 : \mathcal{P}^{\mathrm{dR}} \to \mathbb{Q}$ with $\varphi_1([M,f,\omega]^{\mathrm{dR}}) = f(\omega)$.

(i) For $\xi = [M,\gamma,\omega]^{\mathrm{mot}}$, $(\mathrm{id}\otimes\varphi_1)\Delta\xi = \sum_i [M,\gamma,e_i]^{\mathrm{mot}}\, e_i^\vee(\omega) = [M,\gamma,\omega]^{\mathrm{mot}}$ by bilinearity; by linearity the same holds for every ξ.

(ii) $(\mathrm{id}\otimes\varphi)\Delta[M,\gamma,\omega]^{\mathrm{mot}} = \big[M,\gamma,\sum_i \varphi([M,e_i^\vee,\omega]^{\mathrm{dR}})\, e_i\big]^{\mathrm{mot}} \in L(M,\gamma)$, and the left factors of the coaction of an element of $L(M,\gamma)$ lie in $L(M,\gamma)$.

(iii) $\mathrm{C}(\xi)$ is the image of the linear map $\varphi \mapsto (\mathrm{id}\otimes\varphi)\Delta\xi$, a space; so $\mathrm{C}(\xi+\xi') \subseteq \mathrm{C}(\xi)+\mathrm{C}(\xi')$ and $\mathrm{C}(q\,\xi) = q\,\mathrm{C}(\xi) = \mathrm{C}(\xi)$ for $q \neq 0$. Write $\Delta\xi = \sum_i \xi_i \otimes \xi_i'$ with the ξ_i' linearly independent; a φ with $\varphi(\xi_k')$ equal to 1 for $k = i$ and to 0 otherwise shows $\xi_i \in \mathrm{C}(\xi)$. Likewise $\Delta\xi' = \sum_k \eta_k \otimes \eta_k'$ with $\eta_k \in \mathrm{C}(\xi')$. The coaction is a homomorphism of rings (Brown's notes, §2), so $(\mathrm{id}\otimes\varphi)\Delta(\xi\xi') = \sum_{i,k} \xi_i\eta_k\, \varphi(\xi_i'\eta_k') \in \mathrm{C}(\xi)\cdot\mathrm{C}(\xi')$. □

Lemma 34.7 (Birational invariance). *As stated.*

Proof. The resolution. Let $\Gamma \subseteq X_1 \times X_2$ be the closure of the graph of Φ, a projective variety. The graph is closed in $U_1 \times X_2$, so $\Gamma \cap (U_1 \times X_2)$ is the graph; it is also the graph of Φ^{-1}, closed in $X_1 \times U_2$, so $\Gamma \cap (X_1 \times U_2)$ is the graph too; and the graph, isomorphic to U_1, is a smooth dense

open subset of Γ. By resolution of singularities in characteristic zero (Hironaka), in the form that gives for a projective variety and a smooth dense open subset of it a smooth projective variety with a proper birational morphism onto it that is an isomorphism over that subset and under which the complement pulls back to a divisor with simple normal crossings and smooth components, there is such a $\varpi : \widetilde{X} \to \Gamma$ for the graph; the property used is that the graph of Φ is smooth and the ground field $\mathbb{Q}$ has characteristic zero. Let $q_i : \widetilde{X} \to X_i$ be the two projections; by the above $q_1^{-1}(U_1) = q_2^{-1}(U_2) = \varpi^{-1}(\text{graph})$, on which q_1 and $q_2 = \Phi \circ q_1$ are isomorphisms onto U_1 and U_2. Put $\widetilde{D}_i = q_i^{-1}(D_i)$ and $\widetilde{Y} = q_1^{-1}(Y_1) \cup q_2^{-1}(Y_2)$, both closed. A point of $\widetilde{D}_1$ is not over the graph, D_1 missing U_1; so $\widetilde{D}_1$, the support of the pull-back of a Cartier divisor under the dominant q_1, is a divisor contained in the divisor $\varpi^{-1}(\Gamma \setminus \text{graph})$ and so a union of its components; likewise $\widetilde{D}_2$ and $\widetilde{D}_1 \cup \widetilde{D}_2$. All three have simple normal crossings and smooth components.

The cycle. Let $\widetilde{\sigma}$ be the closure in $\widetilde{X}(\mathbb{R})$ of $\widetilde{\sigma}^\circ = q_1^{-1}(\sigma_1^\circ)$, which q_1 maps isomorphically onto σ_1° and q_2 onto σ_2°; orient it by q_1. It is compact, and $q_i(\widetilde{\sigma}) = \sigma_i$, which misses Y_i; so $\widetilde{\sigma}$ misses $\widetilde{Y}$. If $x \in \widetilde{\sigma} \setminus \widetilde{\sigma}^\circ$ had $q_1(x) \in \sigma_1^\circ \subseteq U_1$, x would lie over the graph, where q_1 is injective, and so in $\widetilde{\sigma}^\circ$; hence $q_1(x) \in \sigma_1 \setminus \sigma_1^\circ \subseteq D_1$ and $x \in \widetilde{D}_1$. The same argument with q_2 and $\Phi(\sigma_1^\circ) = \sigma_2^\circ$ gives $x \in \widetilde{D}_2$. So $\widetilde{\sigma}$ is a relative cycle of $(\widetilde{X} \setminus \widetilde{Y}, \widetilde{D}_i \setminus \widetilde{Y})$ for $i = 1, 2$ and of $(\widetilde{X} \setminus \widetilde{Y}, (\widetilde{D}_1 \cup \widetilde{D}_2) \setminus \widetilde{Y})$, and q_i carries its class to that of σ_i, mapping $\widetilde{\sigma}^\circ$ onto σ_i° with the orientation, Φ preserving it, and $\widetilde{\sigma} \setminus \widetilde{\sigma}^\circ$ into D_i.

The form. $q_1^*\omega_1$ and $q_2^*\omega_2$ are rational forms on $\widetilde{X}$ that agree on the dense open $q_1^{-1}(U_1)$, where $q_2 = \Phi \circ q_1$ and $\Phi^*\omega_2 = \omega_1$; so they are one form $\widetilde{\omega}$, regular off $q_1^{-1}(Y_1)$ and off $q_2^{-1}(Y_2)$, hence on $\widetilde{X} \setminus \widetilde{Y}$.

The chain of pairs. The maps

$$(X_1^\circ, D_1^\circ) \xleftarrow{q_1} (\widetilde{X} \setminus \widetilde{Y}, \widetilde{D}_1 \setminus \widetilde{Y}) \xrightarrow{\mathrm{id}} \big(\widetilde{X} \setminus \widetilde{Y}, (\widetilde{D}_1 \cup \widetilde{D}_2) \setminus \widetilde{Y}\big) \xleftarrow{\mathrm{id}} (\widetilde{X} \setminus \widetilde{Y}, \widetilde{D}_2 \setminus \widetilde{Y}) \xrightarrow{q_2} (X_2^\circ, D_2^\circ)$$

are morphisms of pairs, so their maps on relative cohomology are morphisms of Brown's category (Huber and Müller-Stach). Each pulls back the class of the form of degree m to the class of the pulled-back form, and its transpose carries the class of $\widetilde{\sigma}$, or of σ_1, σ_2 at the ends, to the class of the cycle at its target. The relation $[M, \phi^{\mathrm{t}}\gamma', \omega]^{\mathrm{mot}} = [M', \gamma', \phi(\omega)]^{\mathrm{mot}}$ along the four morphisms gives

$$\big[H^m(X_1^\circ, D_1^\circ), \sigma_1, \omega_1\big]^{\mathrm{mot}} = \big[H^m(\widetilde{X} \setminus \widetilde{Y}, \widetilde{D}_1 \setminus \widetilde{Y}), \widetilde{\sigma}, \widetilde{\omega}\big]^{\mathrm{mot}} = \cdots = \big[H^m(X_2^\circ, D_2^\circ), \sigma_2, \omega_2\big]^{\mathrm{mot}}.$$

The particular cases. With $X_1 = X_2$, $D_1 = D_2$, $Y_2 \supseteq Y_1$ closed and missing σ_1, $U_1 = U_2 = X_1 \setminus (D_1 \cup Y_2)$ and Φ the identity, the hypotheses hold, σ_1° lying in $U_1(\mathbb{R})$. For a convergent family, ω_F is regular on $X_{\mathcal{E}} \setminus \bigcup_{f \in \mathrm{Let}(F)} \widetilde{Y}_f$, by the proof of Proposition 34.4(ii), which uses only that the letters of F are units there; and V_F removes in addition the $\widetilde{Y}_{\alpha_J}$, which miss $\widetilde{\sigma}_{\mathcal{E}}$ (Proposition 34.4(i)). Two convergent families F, F' on $\mathcal{E}$ with $\omega_F = \omega_{F'}$ give $X_1 = X_2 = X_{\mathcal{E}}$, $D_1 = D_2 = D$, Y_1 and Y_2 the unions of the $\widetilde{Y}_f$ over the augmented letters of F and of F', both missing $\widetilde{\sigma}_{\mathcal{E}}$, $U_1 = U_2$ the torus less $Y_1 \cup Y_2$ and Φ the identity. □

Lemma S17 (Products of pairs)**.** *For $i = 1, \dots, p$ let (V_i, D_i) be a pair of dimension m_i with a relative cycle σ_i and a regular form η_i of degree m_i on V_i. Let $V = \prod_i V_i$, $D = \bigcup_i \mathrm{pr}_i^{-1}(D_i)$, $\sigma = \prod_i \sigma_i$ with the product orientation, and $\eta = \mathrm{pr}_1^*\eta_1 \wedge \cdots \wedge \mathrm{pr}_p^*\eta_p$. Then (V, D) is a pair, σ is a relative cycle of it, and, with $m = \sum_i m_i$,*

$$\big[H^m(V, D), \sigma, [\eta]\big]^{\mathrm{mot}} = \prod_{i=1}^{p} \big[H^{m_i}(V_i, D_i), \sigma_i, [\eta_i]\big]^{\mathrm{mot}}.$$

Proof. In products of local coordinates D is a union of coordinate hyperplanes of the factors, so (V, D) is a pair; σ is the closure of the oriented $\prod_i \sigma_i^\circ$, of dimension m, and $\sigma \setminus \prod_i \sigma_i^\circ \subseteq D$. The cross product $\mu : \bigotimes_i H^{m_i}(V_i, D_i) \to H^m(V, D)$ is a morphism of Nori's category (Huber and Müller-Stach), hence of Brown's; on de Rham realizations it carries $\bigotimes_i [\eta_i]$ to $[\eta]$, and its transpose carries σ to $\bigotimes_i \sigma_i$, the class of a product of relative cycles being the cross product of their classes; on forms this is Fubini, $\int_\sigma \mathrm{pr}_1^* \theta_1 \wedge \cdots \wedge \mathrm{pr}_p^* \theta_p = \prod_i \int_{\sigma_i} \theta_i$. The relation of Definition 34.5 for μ, with the product of motivic periods defined by the tensor product of objects (Brown's notes, §2), gives the identity. □

Corollary 34.8 (Cremona between motivic periods). *As stated.*

Proof. The walls. With $d(S) = r - r(\mathcal{E} \setminus S)$, $d^*(S) = |S| - r(S)$, $d(\mathcal{E} \setminus S) = r - r(S)$ and $d^*(\mathcal{E} \setminus S) = |\mathcal{E} \setminus S| - r(\mathcal{E} \setminus S)$ in every table (Definition 7.12), and $m_\mathcal{E} = \sum_j (s_j r_j + t_j(n - r_j))$ on the balance wall,

$$\kappa_S(F^\mathrm{c}) = 2\sigma|S| - m_S - \sum_j t_j\big(r_j - r_j(\mathcal{E} \setminus S)\big) - \sum_j s_j\big(|S| - r_j(S)\big),$$
$$\kappa_{\mathcal{E}\setminus S}(F) = m_\mathcal{E} - m_S - \sum_j s_j\big(r_j - r_j(S)\big) - \sum_j t_j\big(n - |S| - r_j(\mathcal{E} \setminus S)\big),$$

and both equal $-m_S + \sum_j s_j r_j(S) + \sum_j t_j\big(|S| - r_j + r_j(\mathcal{E} \setminus S)\big)$, using $2\sigma = \sum_j (s_j + t_j)$. At $S = \mathcal{E}$ the same computation gives $\kappa_\mathcal{E}(F^\mathrm{c}) = -\kappa_\mathcal{E}(F) = 0$. So F^c is on its balance wall, at integer data, and convergent with F.

The forms. Let Φ be the inversion $\alpha \mapsto \alpha^{-1}$ of the torus, an automorphism that maps $\widetilde{\sigma}_\mathcal{E}^\circ$ onto itself. By Proposition 7.2, $K_j(\alpha^{-1}) = \alpha^{-\mathcal{E}} \Psi_j(\alpha)$ and $\Psi_j(\alpha^{-1}) = \alpha^{-\mathcal{E}} K_j(\alpha)$, and $\Phi^*(\Omega_\mathcal{E}/\alpha^\mathcal{E}) = (-1)^{n-1} \Omega_\mathcal{E}/\alpha^\mathcal{E}$, every $d \log \alpha_e$ changing sign. So

$$\Phi^* \omega_{F^\mathrm{c}} = \alpha^{\mathbf{m} - 2\sigma\mathbf{1}}\, \alpha^{2\sigma\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, (-1)^{n-1} \frac{\Omega_\mathcal{E}}{\alpha^\mathcal{E}} = (-1)^{n-1} \omega_F,$$

the factor $\alpha^{2\sigma\mathbf{1}}$ collecting the powers $\alpha^\mathcal{E}$ of the letters, $\sum_j (s_j + t_j) = 2\sigma$. Let U_1 be the torus less the zero sets of the augmented letters of F and the preimages under Φ of those of F^c, and $U_2 = \Phi(U_1)$; these polynomials are positive at the positive points, so $\widetilde{\sigma}_\mathcal{E}^\circ \subseteq U_1(\mathbb{R})$. The periods P_F and $P_{F^\mathrm{c}} = \int \omega_{F^\mathrm{c}}$ over the positively oriented cell are both positive, so Φ maps the oriented $\widetilde{\sigma}_\mathcal{E}^\circ$ onto itself with the sign $(-1)^{n-1}$ of the forms; reversing the orientation of the target cell and the sign of ω_{F^c} together when that sign is -1, Lemma 34.7 applies to $(X_\mathcal{E}, D, Y_F)$ and $(X_\mathcal{E}, D, Y_{F^\mathrm{c}})$, Y the unions of the $\widetilde{Y}_f$ over the augmented letters, and gives $P_{F^\mathrm{c}}^\mathrm{mot} = P_F^\mathrm{mot}$ by bilinearity. □

Definition 34.9 (Attachment). As stated.

Proof. Every vertex of a tree of bunches is a split vertex, a node that is not a leaf, or a leaf (Definition 20.7), and the first two have children: the molecules of a split vertex, and the room of a node. Along the descent $Z \subseteq M_w$ at every vertex w reached: this holds at the root, a split vertex passes Z to the child that contains it, and a node passes Z to its room only when $Z \subseteq M_w \setminus B_w$, the marks of the room. At each vertex exactly one rule applies, Z being nonempty, and the vertex reached next has fewer marks: a child of a split vertex is one of at least two molecules, and $B_w \neq \varnothing$. So the descent ends, and it ends exactly where Z is attached. □

Lemma 34.10 (The letters of an attached step). *As stated.*

Proof. For a vertex w on the descent of Z say that Z *is read at* w when every letter of the step other than the linear letters is K, respectively Ψ, of the child at a pattern of $M_w \setminus Z$ of a table at w that carries K, respectively Ψ. At the root it holds: the step lists of $\mathcal{B}_j$ at Z are the children $\mathcal{B}_{j,(T_0\setminus Z)}$ and $\mathcal{B}_{j,(T_1\setminus Z)}$ at patterns of $\mathcal{E}\setminus Z$ (Proposition 12.20), or $\mathcal{B}_j$ itself when $Z=\mathcal{E}$, and the tables at the root are the active tables of F, $\mathcal{B}_j$ carrying K exactly when $s_j\neq 0$ and Ψ exactly when $t_j\neq 0$. Let it hold at w, and let the descent pass to a child w'.

At a split vertex, $Z\subseteq M'$ for a molecule M', and every table N at w is the direct sum of its restrictions to the molecules (Proposition 7.9). So the bases of N with a given trace p on $M_w\setminus Z$ have their other parts fixed, and the child of N at p is the child of $N|M'$ at the pattern $p\cap M'$ of $M'\setminus Z$; $N|M'$ is a table at w' with the exponents of N.

At a node that is not a leaf, $Z\subseteq M_w\setminus B_w$. A pattern p of $M_w\setminus Z$ in a table N at w is $p_B\sqcup p_R$ with $p_B=p\cap B_w$ a pattern of B_w and $p_R\subseteq (M_w\setminus B_w)\setminus Z$, and the bases T of N with $T\setminus Z=p$ are those with $T\cap B_w=p_B$ whose parts $T\setminus B_w$, bases of the child $N_{(p_B)}$, have the trace p_R on $(M_w\setminus B_w)\setminus Z$. So the child of N at p is the child of $N_{(p_B)}$ at p_R (Proposition 14.5). $N_{(p_B)}$ is a table made at w, carrying the letters N carries with exponents that are positive multiples of those of N (Definition 20.7); it is a table of the room, of the one childcule when the room is a node and of the split vertex otherwise.

So Z is read at the vertex where it is attached, which is the statement. □

Lemma 34.11 (One form carries every period). *As stated.*

Proof. Number the components $D_1,\dots,D_K$ of D and put $D_I=\bigcap_{k\in I}D_k$, $D_\varnothing=V$; each D_I is smooth of dimension $m-|I|$ or empty, and closed in the affine V, so affine. The relative de Rham cohomology $H^\bullet_{\mathrm{dR}}(V,D)$ is the hypercohomology of the double complex $\Omega^\bullet_V\to\bigoplus_{|I|=1}\Omega^\bullet_{D_I}\to\bigoplus_{|I|=2}\Omega^\bullet_{D_I}\to\cdots$ with the alternating restrictions (Brown's notes, §10; Huber and Müller-Stach). Its terms are coherent sheaves on the affine V, with no higher cohomology, so it is the cohomology of the total complex of global sections, $\mathrm{Tot}^q=\bigoplus_I\Omega^{q-|I|}(D_I)$. Since $\Omega^q(D_I)=0$ for $q>m-|I|$, $\mathrm{Tot}^{m+1}=0$ and every $(\eta_I)_I\in\mathrm{Tot}^m$ is a cocycle, η_I a form of top degree on D_I. Let ℓ be the largest $|I|$ with $\eta_I\neq 0$, and suppose $\ell\geq 1$.

Restriction is onto. For $|I'|=\ell-1$, restriction maps $\Omega^{m-\ell}(D_{I'})$ onto $\bigoplus_{k\notin I'}\Omega^{m-\ell}(D_{I'}\cap D_k)$, the top forms on the boundary components of $D_{I'}$. In local coordinates of $D_{I'}$ in which the components $D_{I'}\cap D_k$ through a point are $\{z_1=0\},\dots,\{z_r=0\}$ and dw is the product of the differentials of the other coordinates, the form $\sum_t \widetilde{g}_t\,dz_1\wedge\cdots\widehat{dz_t}\cdots\wedge dz_r\wedge dw$ restricts to $\pm g_t\,dw$ on $\{z_t=0\}$, $\widetilde{g}_t$ any local extension of g_t, and to 0 on the others; so the map of coherent sheaves is onto, and so is its map on global sections, $D_{I'}$ being affine.

The deepest level removed. For $|I|=\ell$ let $I'=I\setminus\{\max I\}$. Choose $\gamma_{I'}\in\Omega^{m-\ell}(D_{I'})$ for every $|I'|=\ell-1$ restricting on $D_{I'\cup\{k\}}$, $k>\max I'$, to the form that cancels $\eta_{I'\cup\{k\}}$ in the total differential, and to 0 on the $D_{I'\cup\{k\}}$ with $k<\max I'$. In $(\eta)-d\gamma$ the entry at $|I|=\ell$ is η_I less the restrictions of the $\gamma_{I\setminus\{k\}}$, $k\in I$, of which only $k=\max I$ contributes, and it is 0; the entries above ℓ stay 0, and those at $\ell-1$ change by $d\gamma_{I'}$. By induction every class is the class of $(\eta_\varnothing,0,\dots,0)$, a regular m-form on V.

For a convergent family, (V_F,D_F) is such a pair (Proposition 34.4(iii)), and $V_{\mathcal{E};\varnothing\subsetneq\mathcal{E}}=V_F$, the letters of the one step $\mathcal{E}$ being the augmented letters of F that are not monomials, whose $\widetilde{Y}_f$ are empty; so $L(H_F,\widetilde{\sigma}_\mathcal{E})=\mathrm{Open}(\mathcal{E};\varnothing\subsetneq\mathcal{E})$. For a step with $|Z|\geq 2$ the same holds on Z (proof of Definition 34.5); for $|Z|=1$ both sides are $\mathbb{Q}$. □

Proposition 34.12 (Weight-graded faces). *As stated.*

Proof. For a chain $\mathfrak{c}$ of nonempty proper sets let $D_{\mathfrak{c}}$ be its stratum in V_F, a smooth affine variety of dimension $n-1-|\mathfrak{c}|$, nonempty since it meets $\widetilde{\sigma}_{\mathcal{E}}$ (Proposition 34.4(iii),(iv)), with its boundary $\partial D_{\mathfrak{c}}$, its meeting with the other components of D_F. The summands of the double complex of the proof of Lemma 34.11 over the chains refining $\mathfrak{c}$ form the double complex of $(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}})$, shifted by $|\mathfrak{c}|$, and their inclusion is the *face map* $\phi_{\mathfrak{c}} : H^{n-1-|\mathfrak{c}|}(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}}) \to H^{n-1}(V_F, D_F)$, a morphism of Brown's category whose transpose takes a relative cycle that is a manifold with corners, with its faces on the components of D_F, to its face on $D_{\mathfrak{c}}$, with a sign (Brown's notes, §10). Here $\widetilde{\sigma}_{\mathcal{E}}$ is such a cycle, $[0,\infty)^{n-1}$ in every chart, and $\phi_{\mathfrak{c}}^{\mathrm{t}}\widetilde{\sigma}_{\mathcal{E}} = \pm\widetilde{\sigma}_{\mathcal{E}} \cap D_{\mathfrak{c}}$. By Brown's weight-graded face maps (Brown's notes, §10), for $k < n-1$ the sum of the face maps of the chains with $n-1-|\mathfrak{c}| = k$ maps $\bigoplus_{\mathfrak{c}} W_k H^k(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}})$ onto $W_k H^{n-1}(V_F, D_F)$; the property it needs is that every stratum is affine, so that its cohomology vanishes above its dimension. Write $e = \sum_{\mathfrak{c}} \phi_{\mathfrak{c}}(e_{\mathfrak{c}})$. By Lemma 34.11 on the pair $(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}})$, $e_{\mathfrak{c}} = [\eta_{\mathfrak{c}}]$ for a regular k-form on $D_{\mathfrak{c}}$, and by the relation for $\phi_{\mathfrak{c}}$

$$[H_F, \widetilde{\sigma}_{\mathcal{E}}, e]^{\mathrm{mot}} = \sum_{\mathfrak{c}} \pm \big[H^k(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}}), \widetilde{\sigma}_{\mathcal{E}} \cap D_{\mathfrak{c}}, [\eta_{\mathfrak{c}}]\big]^{\mathrm{mot}}.$$

The chain $\mathfrak{c}$ is a flag $\mathfrak{f}$ of dimension k, and by Proposition 34.4(iv) its stratum with its boundary is the product of the pairs $(V_{Z_i;\mathfrak{f}}, \partial V_{Z_i;\mathfrak{f}})$ over its steps, with $\widetilde{\sigma}_{\mathcal{E}} \cap D_{\mathfrak{c}} = \prod_i \widetilde{\sigma}_{Z_i}$. The factors are affine, so the regular functions on the product are the tensor product of those on the factors, the top forms likewise, and $\eta_{\mathfrak{c}}$ is a finite sum of products of top forms on the factors. By Lemma S17 each term is a product over the steps of elements of $\mathrm{Open}(Z_i;\mathfrak{f})$, a step of one mark giving a rational number; the signs are absorbed, $\mathrm{Open}(Z;\mathfrak{f})$ being closed under negation. □

Theorem 34.13 (Small steps along the tree). *As stated.*

Proof. Proposition 34.12 at $e = e_i$ is the inclusion. A flag of dimension k has $\sum_i (|Z_i| - 1) = k$, so each of its steps has at most $k+1$ marks. Each step is attached to exactly one vertex (Definition 34.9), and its letters are those of Lemma 34.10. □

Proposition 34.14 (Laurent coefficients are read on convergent data). *As stated.*

Proof. Meromorphy. The line $\mathbf{m} + \varepsilon\mathbf{v}$ lies in the balance wall, $v_{\mathcal{E}} = 0$, and $\kappa_S(\mathbf{m} + \varepsilon\mathbf{v}) = \kappa_S + \varepsilon v_S$ with $v_S \neq 0$, so the line meets every hyperplane $\kappa_S = -j$ in at most one point, and the continued period is meromorphic on it (Proposition 34.4(v)).

A relation at each wall. For lists every weight is 1. Let $\varnothing \neq S \subsetneq \mathcal{E}$. Summing the mark relations of Theorem 8.9(iii) over $e \in S$, and subtracting $s_j d_j(S)$ times the first letter relation of table j and $t_j d_j^*(S)$ times the second, for every j, gives, as meromorphic functions of the data,

$$\kappa_S\, P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \kappa_{\mathcal{E}} \sum_{e\in S} P_F(\mathbf{m}+\mathbf{e}_e;\mathbf{s},\mathbf{t}) + \sum_j s_j \sum_{T\in\mathcal{B}_j} \big(|T\cap S| - d_j(S)\big) P_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t}) + \sum_j t_j \sum_{T\in\mathcal{B}_j} \big(|S\setminus T| - d_j^*(S)\big) P_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{s},\mathbf{t}+\mathbf{e}_j).$$

Each term has its poles on the hyperplanes $\kappa_{S'} = -j$, S' proper, of its own data (Lemma S16), none of which contains the balance wall; so on the balance wall the first term vanishes, and the data of the other terms stay on it, $|T| = r_j$ and $|\mathcal{E}\setminus T| = r_j^*$. On the line $\mathbf{m} + \varepsilon\mathbf{v}$ every remaining term is meromorphic, as above, and

$$P_F(\mathbf{m}+\varepsilon\mathbf{v};\mathbf{s},\mathbf{t}) = \frac{1}{\kappa_S + \varepsilon v_S}\Big(\sum_j s_j \sum_T \big(|T\cap S| - d_j(S)\big) P_F(\mathbf{m}+\mathbf{1}_T+\varepsilon\mathbf{v};\mathbf{s}+\mathbf{e}_j,\mathbf{t})$$

$$+\sum_j t_j \sum_T \big(|S\setminus T| - d_j^*(S)\big) P_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T}+\varepsilon\mathbf{v};\mathbf{s},\mathbf{t}+\mathbf{e}_j)\Big),$$

with $1/(\kappa_S+\varepsilon v_S)\in\mathbb{Q}(\varepsilon)$ and integer coefficients.

The descent. Put $\delta(F)=\sum_{\varnothing\neq S\subsetneq\mathcal{E}}\max(0,1-\kappa_S)$, a nonnegative integer, zero exactly when F is convergent. If $\delta(F)>0$, choose S with $\kappa_S\leq 0$ and apply the relation at S, keeping the terms with nonzero coefficients. Their families are on the same tables, at integer data on their balance walls, with the exponents larger by one unit vector; their walls are $\kappa_{S'}+|T\cap S'|-d_j(S')\geq\kappa_{S'}$, respectively $\kappa_{S'}+|S'\setminus T|-d_j^*(S')\geq\kappa_{S'}$, for every S' (Theorem 7.13), and at S itself, the coefficient being nonzero, at least κ_S+1. As $1-\kappa_S\geq 1$, the term of S in δ drops by one, and δ drops by at least one. Repeating on every family with $\delta>0$ ends after finitely many steps, with convergent families F_k and coefficients $R_k\in\mathbb{Q}(\varepsilon)$, products of the factors above. The exponents only grow, so with $\mathbf{s},\mathbf{t}\geq 0$ every F_k has nonnegative exponents, and then positive masses, $m_e^{(k)}=\kappa_{\{e\}}(F_k)+\sum_j\big(s_j^{(k)}d_j(\{e\})+t_j^{(k)}d_j^*(\{e\})\big)\geq 1$, the singletons being proper as $n\geq 2$.

The coefficients. At the masses $\mathbf{m}^{(k)}+\varepsilon\mathbf{v}$ the walls of F_k have positive real parts for $|\varepsilon|$ small, so its period is the integral over $\Delta_{\mathcal{E}}$, converging absolutely and locally uniformly, holomorphic in ε, with the derivatives under the integral sign, $u^{\varepsilon\mathbf{v}}=\exp(\varepsilon\langle\mathbf{v},\log u\rangle)$. The displayed integrals converge absolutely: on $\Delta_{\mathcal{E}}$, $|\langle\mathbf{v},\log u\rangle|^l\leq C\prod_e u_e^{-\eta}$ for every $\eta>0$, and the walls at $\mathbf{m}^{(k)}-\eta\mathbf{1}$ are positive for small η, so that the integral of $u^{\mathbf{m}^{(k)}-\eta\mathbf{1}-\mathbf{1}}\prod_j K_j^{-s_j^{(k)}}\Psi_j^{-t_j^{(k)}}$ converges (Theorem 8.7, which reads the integral over $\Delta_{\mathcal{E}}$ off the balance wall as well). The Laurent coefficients of $\sum_k R_kP_{F_k}$ at 0 are rational combinations of the Taylor coefficients of the P_{F_k}, those of the R_k being rational. □

Corollary 34.15 (Open periods from families of lists). *As stated.*

Proof. The forms. The torus part of $V_{Z;\mathfrak{f}}$ is the torus less the zero sets of the letters of the step, a principal open subset of the torus, so a regular function on $V_{Z;\mathfrak{f}}$ is there a homogeneous Laurent polynomial divided by a product of powers of the letters, of total degree zero; and Ω_Z/α^Z is a nowhere vanishing form of top degree on the torus. Splitting the Laurent polynomial into its monomials writes η as a finite sum of terms $c_t\,\alpha^{\mathbf{a}_t}\prod_g g^{-k_{t,g}}\,\Omega_Z/\alpha^Z$, $k_{t,g}\in\mathbb{N}$, each the form ω_{F_t} of the family of the statement, a letter K of a step list being K of the table $(N;k_{t,g},0)$, a letter Ψ being Ψ of the table $(N;0,k_{t,g})$, and α_J being K of $(\binom{J}{1};k_{t,g},0)$ (Definition 34.1); $\alpha^{\mathbf{a}_t}\prod_g g^{-k_{t,g}}$ has degree zero, so F_t is on its balance wall.

The period. For $\mathbf{x}\in\mathbb{C}^Z$ with $x_Z=0$ near 0 put $G(\mathbf{x})=\int_{\widetilde{\sigma}_Z}\alpha^{\mathbf{x}}\eta$. In the chart of an order π, $\alpha^{\mathbf{x}}=\prod_i\varepsilon_i^{x_{S_i}}$ (proof of Lemma 34.3(ii)), and η, regular near $\widetilde{\sigma}_Z\subseteq V_{Z;\mathfrak{f}}(\mathbb{R})$ (proof of Definition 34.5), integrates over the sector as $g_\pi\,d\varepsilon$ on $[0,1]^{|Z|-1}$ with g_π smooth near the cube. So $G(\mathbf{x})=\sum_\pi\operatorname{cube}(\mathbf{x}_\pi+\mathbf{1};g_\pi)$, $\mathbf{x}_\pi=(x_{S_i^\pi})_i$, holomorphic near $\mathbf{x}=0$, and $G(0)=\int_{\widetilde{\sigma}_Z}\eta$ (Lemma S15). On the open cube Lemma 34.3(iii) gives, with the same sign for every t,

$$g_\pi=\sum_t c_t\,\varepsilon^{\boldsymbol{\kappa}_\pi(F_t)-\mathbf{1}}h_{t,\pi},\qquad h_{t,\pi}=\prod_g\widetilde{g}^{-k_{t,g}},$$

$\boldsymbol{\kappa}_\pi(F_t)$ the walls of F_t at $\mathbf{a}_t$ along the chain of π. Choose $A\in\mathbb{N}$ with $\boldsymbol{\kappa}_\pi(F_t)+A\mathbf{1}\geq\mathbf{1}$ for every t and π. The cube integrals have $\operatorname{cube}(\mathbf{y}+\mathbf{b};h)=\operatorname{cube}(\mathbf{y};\varepsilon^{\mathbf{b}}h)$ for $\mathbf{b}\in\mathbb{N}^{|Z|-1}$, where both converge and so, both sides being meromorphic (Lemma 8.4), everywhere; they are linear in h. So, as meromorphic functions of $\mathbf{x}$,

$$\sum_t c_t\operatorname{cube}\big(\mathbf{x}_\pi+\boldsymbol{\kappa}_\pi(F_t);h_{t,\pi}\big)=\sum_t c_t\operatorname{cube}\big(\mathbf{x}_\pi+(1-A)\mathbf{1};\varepsilon^{\boldsymbol{\kappa}_\pi(F_t)+(A-1)\mathbf{1}}h_{t,\pi}\big)$$

$$= \operatorname{cube}\big(\mathbf{x}_\pi + (1-A)\mathbf{1}; \varepsilon^{A\mathbf{1}} g_\pi\big) = \operatorname{cube}(\mathbf{x}_\pi + \mathbf{1}; g_\pi),$$

the functions $\varepsilon^{\boldsymbol{\kappa}_\pi(F_t)+(A-1)\mathbf{1}} h_{t,\pi}$ being smooth near the cube. Summing over π and using Proposition 34.4(v) with $\kappa_S(\mathbf{a}_t + \mathbf{x}) = \kappa_S(\mathbf{a}_t) + x_S$, $G(\mathbf{x}) = \sum_t c_t P_{F_t}(\mathbf{a}_t + \mathbf{x})$ as meromorphic functions near $\mathbf{x} = 0$ on $\{x_Z = 0\}$. On the line $\mathbf{x} = \varepsilon\mathbf{v}$, every v_S being nonzero, each term is meromorphic in ε, as in the proof of Proposition 34.14, the sum is $G(\varepsilon\mathbf{v})$, holomorphic at 0 with the value $\int_{\widetilde{\sigma}_Z} \eta$, and the constant term is linear.

The rest. Every F_t has nonnegative exponents, so Proposition 34.14 writes its constant term through convergent families of lists on Z with nonnegative integer exponents and positive integer masses, whose periods are values of the serif family (Proposition 8.12(i)), and their derivatives along $\mathbf{v}$. The period of $[H^{|Z|-1}(V_{Z;\mathfrak{f}}, \partial V_{Z;\mathfrak{f}}), \widetilde{\sigma}_Z, \eta]^{\mathrm{mot}}$ is $\int_{\widetilde{\sigma}_Z} \eta$, and taking periods is a homomorphism of rings; Theorem 34.13 gives the last sentence. □

Theorem 34.16 (The node form). *As stated.*

Proof. (i) A letter K of the step is K of a step list, which by Lemma 34.10 is the child N' of a table N at v at a pattern p of $M_v \setminus Z$, a list being determined by its basis polynomial; a letter Ψ is Ψ of such a child N', that is K of N'^* (Proposition 7.2); a letter α_J is K of $\binom{J}{1}$ read on Z, its marks outside J loops. Lemma 12.3, for the table N on M_v with the bunch $M_v \setminus Z$, its room Z, the pattern p, $A = Z_B$ and $W = B_v \setminus Z$, so that $A \cup W = B_v$, gives $c_{N'}(Z_B) \le c_N(B_v) \le c$; the dual has the same widths (Theorem 7.13). $\binom{J}{1}$ has rank at most one, so the width of Z_B in it is at most one. Finally $c \ge 1$: if every table at v had $c(B_v) = 0$, B_v would be a union of molecules of every table (Lemma 7.10), no circuit of a table at v would meet both B_v and $M_v \setminus B_v$, and B_v would be a union of molecules of the tables at v (Definition 7.4), against $\varnothing \ne B_v \subsetneq M_v$ and the tables at the node v forming one molecule.

(ii) For $g = K_{N_g}$ and the gauge (u, u'), $g(\rho u, u') = \sum_k \rho^k K^{(k)}_{Z_B}(u, u')$ over the levels $d_g \le k \le d_g + c_g$ (Definition 12.16, Theorem 7.13). The bases of the top level are the unions of a basis of $N_g|Z_B$ and one of N_g/Z_B, each of weight 1 (Theorem 12.19), so the top level letter is $K_{N_g|Z_B}(u)\, K_{N_g/Z_B}(u')$. The Skeleton polynomial $\mathsf{Q}^K_{Z_B}(x; (u, u')) = x^{-d_g} g(xu, u')/g(u, u') = \prod_i (x + \xi_{g,i})/(1 + \xi_{g,i})$ (Definition 12.10) shows that $\rho^{-d_g} g(\rho u, u')$ is a polynomial of degree c_g in ρ with the roots $-\xi_{g,i}$, and comparing leading coefficients gives the display. The Skeleton values of $K_{N''^*}$ along Z_B are those of $\Psi_{N''}$ (Definition 12.10), so the union over the letters is the joint spectrum of Z_B in the sense of Definition 12.14 and Theorem 15.3.

(iii) By Corollary 34.15, on the torus $\eta = \phi\, \Omega_Z/\alpha^Z$ with $\phi = \sum_t c_t \alpha^{\mathbf{a}_t} \prod_g g^{-k_{t,g}}$, homogeneous of degree zero. By Lemma S15 for the partition $Z = Z_B \sqcup Z_R$, $\int_{\widetilde{\sigma}_Z} \eta$ is the absolutely convergent integral over $\Delta_{Z_B} \times \Delta_{Z_R} \times (0, \infty)$ of $\phi(\rho u, u')\rho^{-1} u^{-\mathbf{1}} u'^{-\mathbf{1}}$. By (ii) the term t of this integrand is

$$A_t(u)\, B_t(u')\, \rho^{\nu_t - 1} \prod_{g,i} \big(\rho + \xi_{g,i}\big)^{-k_{t,g}},$$

with $A_t = c_t\, u^{\mathbf{a}_t|_{Z_B} - \mathbf{1}} \prod_g K_{N_g|Z_B}^{-k_{t,g}}$, $B_t = u'^{\mathbf{a}_t|_{Z_R} - \mathbf{1}} \prod_g K_{N_g/Z_B}^{-k_{t,g}}$ and $\nu_t = \sum_{e \in Z_B} a_{t,e} - \sum_g k_{t,g} d_g$. By Tonelli, for almost every (u, u') the integral over ρ of the sum converges absolutely, and the integral over $\Delta_{Z_B} \times \Delta_{Z_R}$ of it is the period, with an integrable modulus. At such a point the integral over ρ is its own finite part (Proposition 1.6): each term is $\rho^{\nu_t - 1}$ times a rational function of ρ without poles on $(0, \infty)$, the $\xi_{g,i}$ lying off $(-\infty, 0]$, whose primitive, a sum of rational functions and of logarithms $\log \rho$ and $\log(\rho + \xi_{g,i})$, has log-power expansions at both ends. Finite parts are linear, so the integral over ρ is $\sum_t A_t B_t \operatorname{Sc}_t$. A term in which no letter with $k_{t,g} > 0$ has a Skeleton value integrates $\rho^{\nu_t - 1}$, whose primitive ρ^{ν_t}/ν_t or $\log \rho$ has no constant term at either

end, so $\mathrm{Sc}_t = 0$. Otherwise, by Theorem 1.7, Sc_t is the constant term at $\kappa = \nu_t$ of the integral $\int_0^\infty \rho^{\kappa-1} \prod_{g,i} (\rho + \xi_{g,i})^{-k_{t,g}} d\rho$, continued from the strip $0 < \operatorname{Re}\kappa < |\mathbf{w}_t|$, where it converges, the product having degree $|\mathbf{w}_t| = \sum_g k_{t,g} c_g$ in ρ; on the strip it is the stacked Gamma times the Breta of the statement (Proposition 15.1, coincident Skeleton values merging), and both sides are meromorphic in κ, so their constant terms at ν_t agree.

(iv) At $c = 1$ every c_g is at most one by (i). A letter with $c_g = 1$ has the one Skeleton value $\gamma_R^g(u')/\gamma_B^g(u)$, positive (Corollary 19.5, the weights 1 factoring). Fix κ in the strip. Corollary 15.5, one coefficient ξ_g for each letter of the term with a Skeleton value, writes the stacked Gamma times the Breta as an integral over $|J_t| - 1$ shares a_ℓ on the lines $\operatorname{Re} a_\ell = c_\ell$ of a product of Beta corners in the shares and of the powers $\xi_g^{-b_g}$, $b_\ell = w_{t,\ell} - a_\ell$ for the shared coefficients and $b_0 = w_{t,0} - \kappa + |\mathbf{a}|$ for the last, affine in the shares and in κ, with $\operatorname{Re} b_g$ constant on the contour. The Beta corners are absolutely integrable over the contour: those in one share are bounded by a constant times $e^{-\pi|\operatorname{Im} a_\ell|}(1 + |\operatorname{Im} a_\ell|)^{w_{t,\ell}-1}$ by the classical bound $|\Gamma(c + i\tau)| \le C_c\, e^{-\pi|\tau|/2}(1 + |\tau|)^{c-1/2}$, and the last one by its value at the real parts of its entries, since $|\Gamma(z)| \le \Gamma(\operatorname{Re} z)$ for $\operatorname{Re} z > 0$. As $|\xi_g^{-b_g}| = \gamma_R^g(u')^{-\operatorname{Re} b_g} \gamma_B^g(u)^{\operatorname{Re} b_g}$, the integral of the modulus of $A_t B_t$ times the share integrand over the contour, Δ_{Z_B} and Δ_{Z_R} is the integral of the modulus of the share kernel times the two integrals of the statement, finite; by Fubini the integrals over Δ_{Z_B} and Δ_{Z_R} move inside the share integral, where at fixed shares the integrand is a function of u times a function of u', and they separate. If $0 < \nu_t < |\mathbf{w}_t|$, the integral over ρ of the term converges at $\kappa = \nu_t$ and is Sc_t (Proposition 1.6), and the argument runs at $\kappa = \nu_t$.

(v) By Lemma 34.11, every element of $L(H_F, \widetilde{\sigma}_\mathcal{E})$ is $[H_F, \widetilde{\sigma}_\mathcal{E}, [\eta]]^{\mathrm{mot}}$ for a regular form η on $V_F = V_{\mathcal{E};\varnothing\subsetneq\mathcal{E}}$. The step $\mathcal{E}$ meets the bunch of the root and its complement, so it is attached to the root as a node step, and (iii) applies to η. The left factors are elements of $L(H_F, \widetilde{\sigma}_\mathcal{E})$ by Definition 34.5.

(vi) At a split vertex the tables are direct sums over the molecules (Proposition 7.9), so the children at patterns of $M_v \setminus Z$ are direct sums over the parts of Z in the molecules (proof of Lemma 34.10), and $Z_B = Z \cap M_1$ is a union of molecules of each of them, of width zero (Lemma 7.10); the same holds for the duals. The list $\binom{J}{1}$ has width one at Z_B exactly when it has bases on both sides, that is when J meets Z_B and Z_R, and width zero otherwise. The proofs of (ii) to (iv) use only the widths of the letters at Z_B, here at most one. □

Lemma S18 (Families on two marks)**.** *Every convergent family F on a set $\mathcal{E}$ of two marks (Definition 34.1) has $H_F \cong \mathbb{Q}(0)$, and $P_F^{\mathrm{mot}} = P_F$, the Beta corner $\Gamma_{\binom{2}{1}}\begin{bmatrix}\kappa_{\{1\}}, & \kappa_{\{2\}}\\ \kappa_{\{1\}} + \kappa_{\{2\}}\end{bmatrix}$ of its walls, a positive rational number.*

Proof. The lists on $\mathcal{E} = \{1, 2\}$ have the basis polynomials 1, α_1, α_2, $\alpha_1\alpha_2$ and $\alpha_1 + \alpha_2$, and their cobasis polynomials are among these. $X_\mathcal{E}$ is the projective line, its boundary the two points $\alpha_1 = 0$ and $\alpha_2 = 0$, the monomials have no zeros on the torus, and the linear letter $\alpha_1 + \alpha_2$ is always removed; so V_F is the projective line less one point, an affine line, and D_F two points of it. In the long exact sequence of the pair, a sequence of morphisms of Brown's category (Huber and Müller-Stach), $H^0(V_F) \to H^0(D_F) \to H^1(V_F, D_F) \to H^1(V_F)$, the first map is the diagonal $\mathbb{Q}(0) \to \mathbb{Q}(0)^2$ and H^1 of the affine line is 0, every form $f(x)\,dx$ being exact; so $H_F \cong \mathbb{Q}(0)$. By the relation of Definition 34.5 for this isomorphism every motivic period of H_F is a rational multiple of the unit $[\mathbb{Q}(0), 1, 1]^{\mathrm{mot}} = 1$, the multiple being its period. On $\Delta_\mathcal{E}$ the letter $\alpha_1 + \alpha_2$ is 1 and the monomials shift the masses to the walls, so $P_F = \int_0^1 u^{\kappa_{\{1\}}-1}(1-u)^{\kappa_{\{2\}}-1} du$, with $\kappa_{\{1\}}, \kappa_{\{2\}} \ge 1$ integers. □

Theorem 34.17 (Pinned cells split off motivically). *As stated.*

Proof. Write $\mathbf{h} = \{h_1, \dots, h_q\}$ and $r_{ij} = r_{\mathcal{A}_{ij}}$, so that $K_{\mathcal{A}_{ij}/g_i}$ and $\Psi_{\mathcal{A}_{ij}\setminus g_i}$ have the degrees $r_{ij}-1$ and $|R_i| - r_{ij}$; the core $\mathcal{H}_j$ has at least two marks, as no mark of a table on one mark is neither a loop nor a coloop, so $\mathcal{E} \setminus R_i \neq \varnothing$.

(v) is Corollary 21.11.

The walls of the cells. R_i has width one in every active table (Proposition 21.3(i)), so on the balance wall $\kappa_{R_i} + \kappa_{\mathcal{E}\setminus R_i} = \sum_j (s_j + t_j) = 2\sigma$ (Proposition 8.16); both are at least 1, F being convergent, so $1 \leq \kappa_i \leq 2\sigma - 1$.

The letters. By Theorem 21.4 and the hypothesis every active letter reads the core at one point:

$$K_j = \prod_i K_{\mathcal{A}_{ij}/g_i}(\alpha_{R_i})\, K_{\mathcal{H}_j}(\mathbf{x}), \qquad \Psi_j = \prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i}(\alpha_{R_i})\, \Psi_{\mathcal{H}_j}(\mathbf{x}),$$
$$\mathbf{x} = \big(\alpha_L, \gamma_1(\alpha_{R_1}), \dots, \gamma_q(\alpha_{R_q})\big),$$

a mark of L standing at its own variable, both hinge values of $\binom{\{l,g\}}{1}$ being α_l.

The map. Each γ_i is a quotient of polynomials, homogeneous of degree one, so

$$\Phi : \ \mathbb{P}(\mathbb{Q}^{\mathcal{E}}) \dashrightarrow \mathbb{P}(\mathbb{Q}^{L\sqcup\mathbf{h}}) \times \prod_i \mathbb{P}(\mathbb{Q}^{R_i}), \qquad [\alpha] \mapsto \big([\mathbf{x}], [\alpha_{R_1}], \dots, [\alpha_{R_q}]\big),$$

is well defined, between varieties of the same dimension $|\mathcal{E}| - 1$; off the zero sets of the numerators and denominators of the γ_i it is an isomorphism between open subsets of the tori, with the inverse $\alpha_L = x_L$, $\alpha_{R_i} = \big(x_{h_i}/\gamma_i(v_i)\big)v_i$ at $([\mathbf{x}], [v_1], \dots, [v_q])$, well defined by the same homogeneity. Let U_1 be the torus less these zero sets, those of the augmented letters of F, and the preimages under Φ of those of the augmented letters of the core family and of the pinned families, and $U_2 = \Phi(U_1)$. All these polynomials are positive at the positive points, so Φ maps $\widetilde{\sigma}^\circ_{\mathcal{E}}$ onto $\widetilde{\sigma}^\circ_{L\sqcup\mathbf{h}} \times \prod_i \widetilde{\sigma}^\circ_{R_i}$.

The forms. On the positive points write $\alpha_{R_i} = \lambda_i v_i$ with $\lambda_i > 0$ and v_i on a section of the rays of $\mathbb{R}^{R_i}_{>0}$, so that $x_{h_i} = \lambda_i \gamma_i(v_i)$. In logarithmic coordinates Φ keeps $\log \alpha_L$ and the v_i and adds $\log \gamma_i(v_i)$ to $\log \lambda_i$, a map with Jacobian determinant 1; with $\nu_X = \Omega_X/\alpha^X$, the form Ω_X/α^X being the invariant volume of the logarithmic coordinates modulo $\mathbf{1}$ (proof of Lemma S15), $\Phi^*\big(\nu_{L\sqcup\mathbf{h}} \wedge \bigwedge_i \nu_{R_i}\big) = \pm\nu_{\mathcal{E}}$, an identity of rational forms since it holds at the Zariski dense positive points. The core family $F_{\mathcal{H}}$ is on its balance wall: by Proposition 21.3(i), $r_j = r_{\mathcal{H}_j} + \sum_i (r_{ij} - 1)$, and

$$m_L + \sum_i \kappa_i = m_{\mathcal{E}} - \sum_{i,j} \big(s_j(r_{ij}-1) + t_j(|R_i| - r_{ij})\big) = \sum_j \Big(s_j r_{\mathcal{H}_j} + t_j\big(|L| + q - r_{\mathcal{H}_j}\big)\Big),$$

with $\kappa_i = m_{R_i} - \sum_j \big(s_j(r_{ij}-1) + t_j(|R_i| - r_{ij})\big)$ (Theorem 21.10). The word of G_i is that of Corollary 22.7, $\gamma_i^{-\kappa_i} \prod_j K_{\mathcal{A}_{ij}/g_i}^{-s_j} \Psi_{\mathcal{A}_{ij}\setminus g_i}^{-t_j}$, homogeneous of the degree $-m_{R_i}$, so G_i is on its balance wall. Pulling back $\omega_{F_{\mathcal{H}}}$ and ω_{G_i} by Φ substitutes $\mathbf{x}$ and $v_i = \alpha_{R_i}$ in their coefficients, and the powers $\gamma_i(\alpha_{R_i})^{\pm\kappa_i}$ cancel:

$$\Phi^*\big(\omega_{F_{\mathcal{H}}} \boxtimes \omega_{G_1} \boxtimes \cdots \boxtimes \omega_{G_q}\big) = \pm\, \alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, \nu_{\mathcal{E}} = \pm\, \omega_F.$$

(i) By the change of variables the period of F, of the positive integrand, is the product of the periods of $F_{\mathcal{H}}$ and of the G_i in $(0, \infty]$; this is Corollary 22.6 with Corollary 22.7. P_F is finite, so every factor is finite, and at integer data on its balance wall every proper wall of each factor is then at least 1 (Theorem 8.7): $F_{\mathcal{H}}$ and the G_i are convergent, their tables being lists.

(ii) Apply Lemma 34.7 to $(X_{\mathcal{E}}, D, Y_F)$ and to the product of the toric varieties of $F_{\mathcal{H}}$ and of the G_i with the union of the pull-backs of their boundaries and of their removed sets, a smooth projective variety with a divisor with simple normal crossings and smooth components, with the cells $\widetilde{\sigma}_{\mathcal{E}}$ and $\widetilde{\sigma}_{L\sqcup\mathbf{h}} \times \prod_i \widetilde{\sigma}_{R_i}$ and the forms ω_F and $\omega_{F_{\mathcal{H}}} \boxtimes \omega_{G_1} \boxtimes \cdots \boxtimes \omega_{G_q}$, regular on $V_{F_{\mathcal{H}}} \times \prod_i V_{G_i}$ by (i); the two periods are positive, so the signs of the orientations and of the forms match as in the proof of Corollary 34.8. With Lemma S17,

$$P_F^{\mathrm{mot}} = \Big[H^{n-1}\Big(V_{F_{\mathcal{H}}} \times \prod_i V_{G_i}, \partial\Big), \widetilde{\sigma}_{L\sqcup\mathbf{h}} \times \prod_i \widetilde{\sigma}_{R_i}, \omega_{F_{\mathcal{H}}} \boxtimes \cdots \boxtimes \omega_{G_q}\Big]^{\mathrm{mot}} = P_{F_{\mathcal{H}}}^{\mathrm{mot}} \prod_i P_{G_i}^{\mathrm{mot}},$$

and Lemma 34.6(iii) gives the conjugates.

(iii) Each cell is itself a substitution: $\mathcal{A}_{ij} = \binom{\{x,h\}}{1}\langle\binom{\{g_i,y\}}{1}, \mathcal{A}'_{ij}\rangle$, the mark g_i left in place at x and $\mathcal{A}'_{ij}$, a copy of $\mathcal{A}_{ij}$ with its hinge renamed, at h: the basis $\{h\}$ of the core contributes the bases of $\mathcal{A}_{ij} \setminus g_i$ and the basis $\{x\}$ the sets $Y + g_i$, Y a basis of $\mathcal{A}_{ij}/g_i$ (Definition 21.1). The hinge values of $\mathcal{A}'_{ij}$ are those of $\mathcal{A}_{ij}$, so the cell family F_i is read through the one hinge value γ_i, its wall at R_i is κ_i (Theorem 21.10, with the masses $\mathbf{m}|_{R_i}$), and it is on its balance wall with the mass $2\sigma - \kappa_i$ on g_i (Lemma 21.12). Its core family F'_i has the tables $\binom{\{g_i,h\}}{1}$, whose two letters are both $\alpha_{g_i} + \alpha_h$, with the masses $2\sigma - \kappa_i$ on g_i and κ_i on h, and its pinned family at the same loads is G_i. By the change of variables as in (i), in $(0,\infty]$, $P_{F_i} = P_{F'_i} P_{G_i}$ with $P_{F'_i} = \int_0^1 u^{2\sigma-\kappa_i-1}(1-u)^{\kappa_i-1}du = \Gamma\begin{bmatrix}\kappa_i, & 2\sigma-\kappa_i \\ & 2\sigma\end{bmatrix}_2^1$, finite since $1 \le \kappa_i \le 2\sigma - 1$. So P_{F_i} is finite, F_i is convergent (Theorem 8.7), and (ii) for F_i gives $P_{F_i}^{\mathrm{mot}} = P_{F'_i}^{\mathrm{mot}} P_{G_i}^{\mathrm{mot}}$, with $P_{F'_i}^{\mathrm{mot}}$ the Beta corner by Lemma S18.

(iv) The form of G_i was computed above without reference to the loads; its period is the integral over Δ_{R_i} of its coefficient, the pinned shape integral of Theorem 22.5 with all its constants 1; and two choices of loads give two convergent families on R_i with one form, hence one motivic period (Lemma 34.7). □

Corollary 34.18 (Dual pairs and full leans). *As stated.*

Proof. (a) By Proposition 21.8(ii) each cell has $\gamma^K_{\mathcal{A}_i} = \gamma^{\Psi}_{\mathcal{A}_i} = k_i/\psi_i$, so the one active table reads it through one hinge value whatever s and t, and Theorem 34.17 applies. With $K_{\mathcal{A}_i/g_i} = \Psi_{\mathcal{A}_i \setminus g_i} = \psi_i$, the form of the pinned family (Theorem 34.17(iv)) is $(k_i/\psi_i)^{-\kappa_i}\psi_i^{-s}\psi_i^{-t}\alpha^{\mathbf{m}|_{R_i}}\nu_{R_i} = k_i^{-\kappa_i}\psi_i^{-(s+t-\kappa_i)}\alpha^{\mathbf{m}|_{R_i}}\nu_{R_i}$, the form of $(\mathcal{A}_i \setminus g_i; \kappa_i, s+t-\kappa_i)$ at the masses $\mathbf{m}|_{R_i}$, and $\mathcal{A}_i \setminus g_i = \mathcal{B}|R_i$, the cell of the bunch R_i (Proposition 21.3(i), Theorem 19.4). By Proposition 21.8(i), $|R_i| - r_{\mathcal{A}_i} = r_{\mathcal{A}_i} - 1$ with $r_{\mathcal{A}_i} = r(R_i)$, so Theorem 21.10 gives $\kappa_i = m_{R_i} - (s+t)(r(R_i)-1)$. The family $(\mathcal{B}|R_i; \kappa_i, s+t-\kappa_i)$ at the masses $\mathbf{m}|_{R_i}$ has the form of the convergent pinned family, which is homogeneous of degree zero, so it is on its balance wall; its walls are the orders of the form along the boundary divisors plus one (Proposition 34.4(ii)), those of the pinned family, so it is convergent, and by Lemma 34.7 the two have one motivic period. Theorem 34.17(ii) is the display.

(b) Let $t = 0$; the case $s = 0$ is the same with Ψ. If $s = 0$ too, $\kappa_S + \kappa_{\mathcal{E}\setminus S} = 0$ for every S on the balance wall (Proposition 8.16), against convergence; so $s \neq 0$, and only K is active: every cell of every substitution of $\mathcal{B}$ is read through one hinge value, its γ^K. By Proposition 8.16, $\kappa_S + \kappa_{\mathcal{E}\setminus S} = s\,c(S) \ge 2$ for every nonempty proper S, so $\mathcal{B}$ is connected. We induct on $|\mathcal{E}|$.

If $\mathcal{B}$ has a set R of width one with $|R|, |\mathcal{E} \setminus R| \ge 2$, then $\mathcal{B} = \mathcal{H}\langle\mathcal{A}\rangle$, the marks of $L = \mathcal{E} \setminus R$ left in place, $\mathcal{H}$ the cell of the room and $\mathcal{A}$ the cell of the bunch R (Proposition 21.3(iii)), connected lists on fewer marks (Proposition 21.3(iv)). Theorem 34.17(ii),(iii) with $q = 1$ give

$$P_F^{\mathrm{mot}} = \Gamma\begin{bmatrix}\kappa_R, & s-\kappa_R \\ & s\end{bmatrix}_2^1{}^{-1} P_{(\mathcal{H};s,0)}^{\mathrm{mot}}\, P_{(\mathcal{A};s,0)}^{\mathrm{mot}},$$

the two families convergent at the masses $(\mathbf{m}|_L, \kappa_R)$ and $(\mathbf{m}|_R, s-\kappa_R)$, where $s-\kappa_R=\kappa_L$ (Proposition 8.16); the marker of each carries the wall of F at the marks on the other side of the split. A wall of $(\mathcal{H}; s, 0)$ at a set containing the marker is the wall of F at that set with the marker replaced by R, and at a set of marks of L it is the wall of F there (Theorem 21.10, the hinge g of $\mathcal{A}$ having $d_{\mathcal{A}}(R+g)=d_{\mathcal{A}}(R)+1$ and $d^*_{\mathcal{A}}(R+g)=d^*_{\mathcal{A}}(R)+1$); likewise for $(\mathcal{A}; s, 0)$. So the induction, applied to the two families, writes P_F^{mot} as a positive rational number times the product over the prime nodes of $P_{F_v}^{\mathrm{mot}}$, with the masses of the statement.

Otherwise $\mathcal{B}$ is prime, its own one node, or it has two or three marks. A connected list on two marks is $\binom{\mathcal{E}}{1}$, and one on three marks is $\binom{\mathcal{E}}{1}$ or $\binom{\mathcal{E}}{2}$, having no loop and no coloop. On two marks P_F^{mot} is a positive rational number by Lemma S18. On three marks, for a mark g, neither a loop nor a coloop, $\mathcal{B}$ is the cell $\binom{\{g,h\}}{1}\langle g, \mathcal{A}'\rangle$ of the proof of Theorem 34.17(iii) with the two other marks as R, and Theorem 34.17(ii) writes P_F^{mot} as the product of the motivic periods of a convergent family on the two marks g, h and of a convergent pinned family on the two marks of R, each a positive rational number by Lemma S18. Lemma 34.6(iii) gives the conjugates, a nonzero rational factor changing nothing. □

Theorem 34.19 (A pinned set splits off motivically). *As stated.*

Proof. F is a family of lists, so by Theorem 28.25 the active hinge values at S are equal, $\gamma=\gamma^K_{j,S}=\gamma^\Psi_{j',S}$ for all active K_j and $\Psi_{j'}$. Every active table $\mathcal{B}_j$ has width one at S with factoring weights, so it is the gluing of its cell of the bunch $\mathcal{B}^+_{j,S}$ and its cell of the room $\mathcal{B}^+_{j,S'}$ along their hinges (Proposition 21.3(iii)), that is the substitution $\mathcal{B}^+_{j,S'}\langle\mathcal{B}^+_{j,S}\rangle$ with the marks of S' left in place (Definition 21.1); the hinge values of $\mathcal{B}^+_{j,S}$ are $\gamma^K_{j,S}$ and $\gamma^\Psi_{j,S}$. So the hypothesis of Theorem 34.17 holds with $q=1$, $L=S'$, $R_1=S$ and $\gamma_1=\gamma$, $|S|\geq 2$. Its core family is $((\mathcal{B}^+_{j,S'}; s_j, t_j))_j$ at the masses $(\mathbf{m}|_{S'}, \kappa_S)$. By Proposition 28.3(ii) the room station letters of S' are $\mathsf{y}_{jD_0}=K_{\mathcal{B}_j|S'}$, $\mathsf{y}_{jD_1}=K_{\mathcal{B}_j/S}$, $\mathsf{y}^*_{jD_0}=\Psi_{\mathcal{B}_j|S'}$ and $\mathsf{y}^*_{jD_1}=\Psi_{\mathcal{B}_j/S}$, so the partner with its stand-in at $\mathbf{1}$ has the letters $K_{\mathcal{B}_j|S'}+\alpha_g K_{\mathcal{B}_j/S}$ and $\alpha_g\Psi_{\mathcal{B}_j|S'}+\Psi_{\mathcal{B}_j/S}$ and the mass κ_S on g (Theorem 28.10(i)), the letters of the cell of the room (Theorem 19.4): it is the core family. The period of the pinned family is $\int_{\Delta_S}\gamma^{-\kappa_S}u^{\mathbf{m}|_S-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{B}_j/S'}\Psi^{-t_j}_{\mathcal{B}_j|S}\,d\omega_S$ (Theorem 34.17(iv)), and by the display after Definition 28.5, with the weights factoring, this is $\int_{\Delta_S}\phi^\rho_S$: there $\gamma^\rho_S=\prod_j(\gamma^K_{j,S})^{s_j/w}(\gamma^\Psi_{j,S})^{t_j/w}=\gamma$, $w=\sum_j(s_j+t_j)$ being the sum of the exponents of the active tables, all of width one at S. Theorem 34.17(ii) is the display, and its period is $P_F=c\,P_{S'(\mathbf{1})}$. □

48.34 Chapter 35: positive words

Theorem 35.1 (Positive coefficient jets). *As stated, at positive weights that need not be rational.*

Proof. Write $P=P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, finite and positive because its walls $n\kappa_S$ are positive (8.7), and $\mathbb{E}$ for the expectation of $U\sim\mathrm{Breta}^\circ(F; n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, whose density on the simplex is $P^{-1}u^{n\mathbf{m}-\mathbf{1}}\prod_j K_j(u)^{-ns_j}\Psi_j(u)^{-nt_j}$, positive on the open simplex. For $v\in\mathbb{R}^k$ put $X_j=v^\mathsf{T}\partial_{\mathbf{b}}K_j/K_j$ and $X^*_j=v^\mathsf{T}\partial_{\mathbf{b}}\Psi_j/\Psi_j$. With $u_{j,T}=\mathbf{b}_{j,T}/\theta_{j,T}$,

$$\frac{\partial_{\mathbf{b}}K_j}{K_j}=\mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[u_{j,T}],\qquad \frac{\partial_{\mathbf{b}}\Psi_j}{\Psi_j}=\mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}[u_{j,T}],$$

so $|X_j|$ and $|X^*_j|$ are at most $\beta_j=\max_T|v^\mathsf{T}u_{j,T}|$ on the whole orthant.

(i) For $|\varepsilon|\max_j \beta_j < \frac12$ the weights $\theta_{j,T} \pm \varepsilon v^{\mathsf T}\mathbf{b}_{j,T} = \theta_{j,T}(1 \pm \varepsilon v^{\mathsf T} u_{j,T})$ are positive, and the letters of the two copies of table j are $K_j(1 \pm \varepsilon X_j)$ and $\Psi_j(1 \pm \varepsilon X_j^*)$. The copies have the levels of table j, each counted twice at half the exponents, so the doubled family at the masses $n\mathbf{m}$ has the imbalance 0 and the walls $n\kappa_S$, and its word is

$$\prod_j K_j^{-ns_j}\Psi_j^{-nt_j}\cdot B_\varepsilon, \qquad B_\varepsilon = \prod_j \big(1-\varepsilon^2 X_j^2\big)^{-\frac{ns_j}{2}}\big(1-\varepsilon^2 X_j^{*2}\big)^{-\frac{nt_j}{2}}.$$

Its period is therefore $P\,\mathbb{E}[B_\varepsilon(U)]$, the display of (i). For $|\varepsilon| \le \varepsilon_0 < (2\max_j\beta_j)^{-1}$, B_ε is a power series in ε^2 whose coefficients are polynomials in the bounded functions X_j^2, X_j^{*2}, converging uniformly on the orthant; so $\mathbb{E}[B_\varepsilon(U)]$ is analytic in ε and expands term by term, with the coefficient $\sum_j\big(\frac{ns_j}{2}\mathbb{E}[X_j^2] + \frac{nt_j}{2}\mathbb{E}[X_j^{*2}]\big)$ of ε^2. Since $X_j^2 = v^{\mathsf T}\big(\partial_{\mathbf b}K_j\,\partial_{\mathbf b}K_j^{\mathsf T}/K_j^2\big)v$, P times this coefficient is $v^{\mathsf T}\mathbf{H}_n v$. It is a sum of nonnegative terms. If it vanishes, then for every j with $s_j>0$, $X_j(U)=0$ almost surely; the law of U has a positive density on the open simplex, so the polynomial $\sum_T (v^{\mathsf T}\mathbf{b}_{j,T})\alpha^T$ vanishes on an open set, and every $v^{\mathsf T}\mathbf{b}_{j,T}$ is 0, the monomials of distinct bases being distinct. For $t_j>0$ the same holds with $\alpha^{\mathcal E\setminus T}$. So v is orthogonal to every direction of an active table, and $v=0$ when these span $\mathbb{R}^k$.

(ii) Write $Q = R^{\mathsf T}R$ with R real. Then

$$\operatorname{tr}(Q\mathbf{H}_n) = \frac{n}{2}\,P\,\mathbb{E}\Big[\sum_j\Big(s_j\frac{|R\,\partial_{\mathbf b}K_j|^2}{K_j^2} + t_j\frac{|R\,\partial_{\mathbf b}\Psi_j|^2}{\Psi_j^2}\Big)(U)\Big],$$

a sum of nonnegative terms, which vanishes exactly when, as in (i), $R\mathbf{b}_{j,T}=0$, that is $\mathbf{b}_{j,T}^{\mathsf T}Q\mathbf{b}_{j,T}=0$, for every basis of every active table. If $Q_{11}>0$, then $Q - Q_{11}^{-1}Q_{\cdot1}Q_{\cdot1}^{\mathsf T}$ is rational, positive semidefinite, with first row and column zero; if $Q_{11}=0$, the first row and column of Q vanish, Q being positive semidefinite. Induction on k gives $Q=\sum_l c_l v_l v_l^{\mathsf T}$ with rational $c_l>0$ and rational v_l, and then $\operatorname{tr}(Q\mathbf{H}_n)=\sum_l c_l\, v_l^{\mathsf T}\mathbf{H}_n v_l$.

(iii) $\partial_{\mathbf b}K_j^{\mathsf T}Q\,\partial_{\mathbf b}K_j/K_j^2 = \sum_{T,T'}\mathbf{b}_{j,T}^{\mathsf T}Q\,\mathbf{b}_{j,T'}\,\alpha^{\mathbf 1_T+\mathbf 1_{T'}}K_j^{-2}$, and $P\,\mathbb{E}\big[(\alpha^{\mathbf 1_T+\mathbf 1_{T'}}K_j^{-2})(U)\big]$ is the period whose integrand is that of P times $u^{\mathbf 1_T+\mathbf 1_{T'}}K_j(u)^{-2}$, the period at $\big(n\mathbf m+\mathbf 1_T+\mathbf 1_{T'};n\mathbf s+2\mathbf e_j,n\mathbf t\big)$. Its imbalance is that of $(n\mathbf m;n\mathbf s,n\mathbf t)$ plus $2r_j-2r_j=0$, and its wall of S is $n\kappa_S+|T\cap S|+|T'\cap S|-2d_j(S)\ge n\kappa_S$, by 7.13. The cobasis part is the same with $u^{\mathbf 1_{\mathcal E\setminus T}+\mathbf 1_{\mathcal E\setminus T'}}\Psi_j^{-2}$ and the change $|S\setminus T|+|S\setminus T'|-2d_j^*(S)\ge 0$. This is the case of order two of 27.9 for the doubled family, the two copies of each table merged.

(iv) $x\mapsto x^{\mathsf T}Qx$ is convex, so $\big(\mathbb{E}_T[u_{j,T}]\big)^{\mathsf T}Q\,\mathbb{E}_T[u_{j,T}] \le \mathbb{E}_T\big[u_{j,T}^{\mathsf T}Qu_{j,T}\big] \le \max_T u_{j,T}^{\mathsf T}Qu_{j,T}$ under either law, and $\operatorname{tr}(Q\mathbf{H}_n)\le \frac n2 P\sum_j(s_j+t_j)\max_T \mathbf{b}_{j,T}^{\mathsf T}Q\,\mathbf{b}_{j,T}/\theta_{j,T}^2 = n\,C(Q)\,P$; positivity is (ii).

(v) With the basis weights $\theta^K_{j,T}$ and the cobasis weights $\theta^\Psi_{j,T}$ of every table independent (8.1), $\partial^2 K_j^{-ns_j}/\partial\theta^K_{j,T}\partial\theta^K_{j,T'} = ns_j(ns_j+1)K_j^{-ns_j-2}\alpha^T\alpha^{T'}$. The differentiated integrals are the periods of (iii), convergent locally uniformly in the weights, so differentiation under the integral gives

$$\sum_{T,T'}\mathbf{b}_{j,T}\mathbf{b}_{j,T'}^{\mathsf T}\,\frac{\partial^2 P}{\partial\theta^K_{j,T}\partial\theta^K_{j,T'}} = ns_j(ns_j+1)\,P\,\mathbb{E}\Big[\frac{\partial_{\mathbf b}K_j\,\partial_{\mathbf b}K_j^{\mathsf T}}{K_j^2}(U)\Big],$$

which is $2(ns_j+1)$ times the part $\frac n2 s_j P\,\mathbb{E}[\cdots]$ of $\mathbf{H}_n$ carried by K_j; the cobasis part is the same with t_j. On the share side of 18.16 the weights enter only through $\prod_{j,T}(\theta^K_{j,T})^{-a_{j,T}}(\theta^\Psi_{j,T})^{-b_{j,T}}$: a derivative in $\theta^K_{j,T}$ multiplies the integrand by $-a_{j,T}/\theta^K_{j,T}$, one in $\theta^\Psi_{j,T}$ by $-b_{j,T}/\theta^\Psi_{j,T}$, and neither the contour, nor the pins, nor the share kernels involve the weights.

(vi) At the centre the two copies of table j carry the equal exponents $(\frac{ns_j}{2}, \frac{ns_j}{2})$, and the masses $n\sigma\mathbf{1}$ are the total strength of the doubled family times $\mathbf{1}$. 8.21(i) holds at every family at its centre, whatever its weights: at $\mathbf{1}$ the usage is $\sum_j \sigma_j(R_j + R_j^*)(\mathbf{1}) = \sigma\mathbf{1}$, since $R_{j,e}(\mathbf{1}) + R^*_{j,e}(\mathbf{1})$ is the chance that e lies in the basis plus the chance that it does not. □

Remark. No step uses the rationality of the weights; rational weights are what the arithmetic of the jets asks for, through the rows of the periods of (iii).

Theorem 35.3 (Two bases generate positive contrast words of any depth). *As stated, at every point of the balance wall with real data and positive proper walls, and at positive weights.*

Proof. Write $P_n = P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, finite and positive (8.7), $\mathbb{E}$ for the expectation of $U \sim \mathrm{Breta}^\circ(F; n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, and $c = \mathbb{P}_{j,\alpha}(T) - \mathbb{P}_{j,\alpha}(T')$, $c^* = \mathbb{P}^*_{j,\alpha}(T) - \mathbb{P}^*_{j,\alpha}(T')$. Each chance lies in $(0,1)$ and two chances of distinct bases sum to at most one, so $|c|, |c^*| < 1$ on the open orthant. c vanishes only where $\theta_{j,T}\alpha^{\mathbf{1}_T} = \theta_{j,T'}\alpha^{\mathbf{1}_{T'}}$, a hypersurface, since $\mathbf{1}_T \neq \mathbf{1}_{T'}$; so $c^{2h} > 0$ almost everywhere, and the contrast word, $\frac{1}{2}P_n$ times the expectation of a function that is positive almost everywhere under a law with a positive density, is positive.

(i) $c^{2h} = \sum_{l=0}^{2h}(-1)^l\binom{2h}{l}\theta_{j,T}^{2h-l}\theta_{j,T'}^{l}\,\alpha^{v_l}K_j^{-2h}$ with $v_l = (2h-l)\mathbf{1}_T + l\mathbf{1}_{T'}$, each term homogeneous of degree 0, and $P_n\,\mathbb{E}\big[(\alpha^{v_l}K_j^{-2h})(U)\big]$ is the period whose integrand is that of P_n times $u^{v_l}K_j(u)^{-2h}$, the period at $\big(n\mathbf{m} + v_l; n\mathbf{s} + 2h\mathbf{e}_j, n\mathbf{t}\big)$. Its imbalance changes by $|v_l| - 2hr_j = 0$, and its wall of S by $(2h-l)|T \cap S| + l|T' \cap S| - 2hd_j(S) \geq 0$ (7.13). The second half is the same with $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j$, the masses raised by $(2h-l)\mathbf{1}_{\mathcal{E}\setminus T} + l\mathbf{1}_{\mathcal{E}\setminus T'}$, the cobasis exponent by $2h$, and the wall of S by $(2h-l)|S \setminus T| + l|S \setminus T'| - 2hd^*_j(S) \geq 0$.

From $\Psi_j(\alpha) = \alpha^{\mathbf{1}}K_j(\alpha^{-1})$, $K_j(\alpha^{-1}) = \alpha^{-\mathbf{1}}\Psi_j(\alpha)$ and $\Psi_j(\alpha^{-1}) = \alpha^{-\mathbf{1}}K_j(\alpha)$, so $\mathbb{P}^*_{j,\alpha^{-1}}(T) = \mathbb{P}_{j,\alpha}(T)$. On the chart $y_g = 0$ of 8.7 the integrand of a period at data $(\mathbf{m}'; \mathbf{s}', \mathbf{t}')$ is $e^{\langle \mathbf{m}', y\rangle}\prod_j K_j(e^y)^{-s'_j}\Psi_j(e^y)^{-t'_j}$, and $y \mapsto -y$, which keeps dy, carries it to $e^{\langle 2\sigma'\mathbf{1} - \mathbf{m}', y\rangle}\prod_j K_j(e^y)^{-t'_j}\Psi_j(e^y)^{-s'_j}$, σ' the total strength of $(\mathbf{s}', \mathbf{t}')$. Applied to the second half at $(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$ it gives the first half at $\big(n(2\sigma\mathbf{1} - \mathbf{m}); n\mathbf{t}, n\mathbf{s}\big)$, the depth n of the data $(2\sigma\mathbf{1} - \mathbf{m}; \mathbf{t}, \mathbf{s})$ of 8.14, whose wall of S is the wall of $\mathcal{E} \setminus S$ at $(\mathbf{m}; \mathbf{s}, \mathbf{t})$, positive. At the centre these data are $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ themselves, and the two halves are equal.

(ii) By 17.1, with $X(z) = 2\langle \boldsymbol{\mu}_\perp, z\rangle - \Phi^\circ(z)$ read on $\mathbf{1}^\perp$ and the number of marks written $|\mathcal{E}|$, $P_n = 2^{|\mathcal{E}|-1}\int_{\mathcal{T}} e^{nX}dz$ and the contrast word is $2^{|\mathcal{E}|-2}\int_{\mathcal{T}} e^{nX}\big(c^{2h} + c^{*2h}\big)dz$, with c, c^* at $\alpha = e^{2z}$. For $h = \eta n$ put $\psi = -X - 2\eta\log\max(|c|, |c^*|)$, with values in $(-\infty, \infty]$, continuous where finite and lower semicontinuous; then $e^{-n\psi} \leq e^{nX}(c^{2h} + c^{*2h}) \leq 2e^{-n\psi}$. By 17.4(i) and (ii), $-X(z) \geq 2\kappa_{\min}(\max_e z_e - \min_e z_e) - C$, and the logarithmic term is nonnegative, so ψ grows at least linearly on $\mathbf{1}^\perp$. Laplace's principle follows: outside a large ball the integral of $e^{-n\psi}$ is at most $e^{-n(\inf\psi+1)}$ for large n, inside it is at most $e^{-n\inf\psi}$ times the volume, and near a point z_0 with $\psi(z_0) < \inf\psi + \epsilon$ the integrand is at least $e^{-n(\inf\psi+2\epsilon)}$ on a ball. So $\frac{1}{n}\log$ of the contrast word tends to $-\inf\psi$, and the same with $\eta = 0$ and without the contrasts gives $\frac{1}{n}\log P_n \to \max X = -\min(\Phi^\circ - 2\langle\boldsymbol{\mu}_\perp, \cdot\rangle)$, the maximum attained, X being continuous and tending to $-\infty$. The set $K = \{z \in \mathbf{1}^\perp :\ 2\kappa_{\min}(\max_e z_e - \min_e z_e) - C \leq \inf\psi + 1\}$ is compact and contains every z with $\psi(z) \leq \inf\psi + 1$; on it $\max(|c|, |c^*|) \leq 1 - \delta_K < 1$, so $\inf\psi \geq \min(-X) + 2\eta|\log(1-\delta_K)| > \min(-X)$. For $A > 0$ let B_A be a ball of $\mathbf{1}^\perp$ outside which $2\kappa_{\min}(\max_e z_e - \min_e z_e) - C \geq A$; on B_A, $\psi \geq \min(-X) + 2\eta|\log(1 - \delta_{B_A})|$, which exceeds A for large η, so $\inf\psi \geq A$ for large η.

(iii) Let $s_j > 0$ and $q = 2h$. In the share form of P_F used by 18.16, the letter $K_j^{-ns_j-q}$ of the term l of (i) is opened by one share a'_U per basis, $\sum_U a'_U = ns_j + q$, with the share kernel $\prod_U \Gamma(a'_U)/\Gamma(ns_j + q)$ and the factors $(\theta_{j,U}\alpha^{\mathbf{1}_U})^{-a'_U}$ (4.4). The insertion $\theta_{j,T}^{q-l}\theta_{j,T'}^{l}\alpha^{v_l}$ turns the

factors of T and T' into $(\theta_{j,T}\alpha^{\mathbf{1}_T})^{-a_T}$ and $(\theta_{j,T'}\alpha^{\mathbf{1}_{T'}})^{-a_{T'}}$ with $a_T = a'_T - q + l$, $a_{T'} = a'_{T'} - l$, the other shares unchanged as $a_U = a'_U$. In the shares a the total is ns_j and the factors in α and in the weights are those of $P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, so the integral over α, which produces the pins, gives the pins of $(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, and the kernel is

$$\frac{\Gamma(a_T + q - l)\,\Gamma(a_{T'} + l)\,\prod_{U \neq T,T'} \Gamma(a_U)}{\Gamma(ns_j + q)} = \frac{\prod_U \Gamma(a_U)}{\Gamma(ns_j)} \cdot \frac{(a_T)_{q-l}\,(a_{T'})_l}{(ns_j)_q}.$$

The factor $\Gamma(a_T + q - l)\Gamma(a_{T'} + l)$ is holomorphic for $\operatorname{Re} a_T > -(q - l)$ and $\operatorname{Re} a_{T'} > -l$, so the contour returns to a wall contour of $(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$ without crossing a pole, the share kernels decaying along the imaginary directions. Summing over l with the weights $\frac{1}{2}(-1)^l\binom{q}{l}$ gives the share side of $P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$ with the insertion $H_q(a_{j,T}, a_{j,T'})/\big(2(ns_j)_q\big)$. For $t_j > 0$ the second half is the same with Ψ_j, the cobasis shares b and nt_j. The insertions are polynomials in the shares and change neither the pins nor the share kernels, hence no pole hyperplane. At the centre the two halves are equal by (i), and their insertions add to the one of the reference. □

Theorem 35.14 (A positive sequence of Beta ports on an interval of a selected mark). *As stated, at every mass $m_e \in (0, 1]$.*

Proof. In x, $d\alpha_e = (\lambda_1 - \lambda_0)\,dx$, and the constant goes into the room density. The factor of the mass is $\alpha_e^{m_e - 1} = L_0^{-(1-m_e)}$ with $L_0 = \lambda_0 + (\lambda_1 - \lambda_0)x$, a linear letter with $c_0 = \lambda_0$, $d_0 = \lambda_1 - \lambda_0 > 0$ and the exponent $w_0 = 1 - m_e \geq 0$; it satisfies $c_0 \geq \lambda_0 d_0/(\lambda_1 - \lambda_0)$ with equality, so the read is $\langle\cdot\rangle_{1,1;\mathbf{w}}$ over the letters of the chart and L_0. For a letter with $d_i > 0$ put $\mu = \lambda_0/(\lambda_1 - \lambda_0)$; then $x(1-x)d_i/L_i \leq x(1-x)/(\mu + x)$, whose derivative vanishes on $[0, 1]$ only where $x^2 + 2\mu x - \mu = 0$, at $x = \sqrt{\mu(1+\mu)} - \mu$, with the value $\big(\sqrt{1+\mu} - \sqrt{\mu}\big)^2 = (\sqrt{\lambda_1} - \sqrt{\lambda_0})^2/(\lambda_1 - \lambda_0) = \varepsilon_{01}$, the ratio of the statement; for $d_i = 0$ the quotient is 0. By 35.12(i) at $a = b = 1$,

$$\big\langle (-1)^n \Pi_n^{1,1} \big\rangle_{1,1;\mathbf{w}} = \sum_{|v|=n} \prod_i \frac{(w_i)_{v_i}}{v_i!} \Big\langle \prod_i \Big(\frac{x(1-x)d_i}{L_i} \Big)^{v_i} \Big\rangle_{1,1;\mathbf{w}},$$

every term between 0 and $\prod_i \frac{(w_i)_{v_i}}{v_i!} \varepsilon_{01}^n \langle 1\rangle_{1,1;\mathbf{w}}$, and $\sum_{|v|=n} \prod_i (w_i)_{v_i}/v_i! = (|\mathbf{w}|)_n/n!$ by the Vandermonde identity for rising factorials. A letter L_i that sees the mark makes the term $v = n\mathbf{e}_i$ positive (35.12(ii) with $J = \{i\}$), and for $n = 0$ the read is $\langle 1\rangle_{1,1;\mathbf{w}} > 0$. When $m_e < 1$, $w_0 d_0 = (1 - m_e)(\lambda_1 - \lambda_0) > 0$: the letter of the mass sees the mark. □

Theorem 35.15 (The finite-band recurrence in the order of the Beta ports). *As stated, at every pair (a_n, b_n) with $a_n, b_n > 0$.*

Proof. Write $a = a_n$, $b = b_n$, $w_i = w_i(n)$. The letters are at least $\min_i c_i > 0$ on $[0, 1]$, so every read of a polynomial converges and the flux $x^a(1-x)^b A \prod_i L_i^{-w_i}$ of a polynomial A vanishes at both ends. 35.10(a) with the polynomial Lg reads, by $x(1-x)\partial_x(Lg) = x(1-x)(L\partial_x g + g\partial_x L)$ and $\partial_x \log L_i = d_i/L_i$,

$$x(1-x)\partial_x(Lg) + \big(a - (a+b)x\big)Lg - x(1-x)Lg \sum_i \frac{w_i d_i}{L_i} = L\,x(1-x)\,\partial_x g + A_n g.$$

The coefficient of x^q in $\Pi_q^{a,b}$ is $\binom{a+b+2q-2}{q}$, a product of the q positive numbers $a + b + q - 1, \ldots, a + b + 2q - 2$ over $q!$, so $\Pi_0^{a,b}, \ldots, \Pi_N^{a,b}$ are a basis of the polynomials of degree at most N, and the conditions $a + b + 2q - i \neq 0$ of 3.15(v) hold. By its finite rules, multiplication by x maps

$\Pi^{a,b}_q$ into the span of $\Pi^{a,b}_{q-1}, \Pi^{a,b}_q, \Pi^{a,b}_{q+1}$, and $x(1-x)\partial_x$ maps it into the span of $\Pi^{a,b}_{q-1}, \Pi^{a,b}_q$ and $x\Pi^{a,b}_q$, with coefficients rational in (a,b,q); for $q=0$, $x\Pi^{a,b}_0 = x$ and $\partial_x \Pi^{a,b}_0 = 0$. So multiplication by a polynomial of degree d moves $\Pi^{a,b}_q$ into the span of $\Pi^{a,b}_{q-d}, \dots, \Pi^{a,b}_{q+d}$. L has degree h and A_n degree at most $h+1$, so at $g = (-1)^r \Pi^{a,b}_r$,

$$L\,x(1-x)\,\partial_x g + A_n g = \sum_{s=-h-1}^{h+1} e_s\,\Pi^{a,b}_{r+s},$$

$e_s = 0$ when $r+s<0$, with e_s rational in (a,b,r) and polynomial in the coefficients of L and of A_n, hence in the room data and the w_i; the mark relation is the displayed relation with $\mathfrak{b}_{n,r,s} = (-1)^{r+s} e_s$.

$\Pi^{a,b}_{r+h+1}$ is the only member of the expansion of degree $r+h+1$, so e_{h+1} is the coefficient of x^{r+h+1} divided by $\binom{a+b+2r+2h}{r+h+1}$. The coefficient of x^r in g is $(-1)^r\binom{a+b+2r-2}{r}$. The coefficient of x^{h+1} in A_n is $-(a+b)p_h - hp_h + |\mathbf{w}(n)|\,p_h$, the last from $-x(1-x)\sum_i w_i d_i \prod_{i'\neq i} L_{i'}$, whose leading coefficient is $\sum_i w_i d_i \prod_{i'\neq i} d_{i'} = |\mathbf{w}(n)|\,p_h$; and the coefficient of x^{r+h+1} in $L\,x(1-x)\,\partial_x x^r$ is $-rp_h$. So the coefficient of x^{r+h+1} is $(-1)^r\binom{a+b+2r-2}{r}\,p_h\big(|\mathbf{w}(n)| - a - b - h - r\big)$, and $\mathfrak{b}_{n,r,h+1} = (-1)^{r+h+1} e_{h+1}$ is the stated value. $\square$

At $a_n = b_n = n+1$ the furthest coefficient is

$$\big[|\mathbf{w}(n)| - 2n - 2 - h - r\big]\,p_h\,(-1)^{h+1}\binom{2n+2r}{r}\Big/\binom{2n+2r+2h+2}{r+h+1},$$

that of the reference.

48.35 Chapter 36: the critical-line law

Proposition 36.7 (Binomial moments of the critical-line law). *As stated.*

Proof. (i) The density $p_{\mathrm{crit}}(y) = 2\pi/(e^{\pi y} + e^{-\pi y})^2$ is at most $2\pi e^{-2\pi|y|}$, so $\mathbb{E}[e^{tX}] = e^{t/2}\mathbb{E}[e^{itY}]$ is holomorphic in t on the strip $|\operatorname{Im} t| < 2\pi$, as is $te^t/(e^t-1)$, whose poles $2\pi i k$, $k \neq 0$, lie outside it; they agree for real t (36.2(iii)), hence on the strip. For $|z|<1$, $t = \log(1+z)$ has $|\operatorname{Im} t| < \frac{\pi}{2}$, and $te^t/(e^t-1) = (1+z)\log(1+z)/z$.

(ii) $|\binom{x}{i}| \le \binom{|x|+i-1}{i}$ termwise, and $\sum_i \binom{|x|+i-1}{i}|z|^i = (1-|z|)^{-|x|}$. For $|z| < 1 - e^{-2\pi}$, $(1-|z|)^{-|X|} \le e^{c(\frac12 + |Y|)}$ with $c = -\log(1-|z|) < 2\pi$, which has a finite expectation, so $\mathbb{E}[(1+z)^X] = \sum_i \mathbb{E}[\binom{X}{i}]z^i$ there. The expansion $\log(1+z)/z = \sum_{i\ge0}(-1)^i z^i/(i+1)$ times $1+z$ has the coefficient 1 at $i=0$ and $(-1)^i\big(\frac{1}{i+1} - \frac{1}{i}\big) = \frac{(-1)^{i+1}}{i(i+1)}$ at $i \ge 1$.

(iii) Differentiating $(1+z)^u = \sum_i \binom{u}{i} z^i$ in u gives $\log(1+z)(1+z)^u = \sum_i \frac{d}{du}\binom{u}{i} z^i$, so $\frac{d}{du}\binom{u}{i} = \sum_{m=1}^{i} \frac{(-1)^{m-1}}{m}\binom{u}{i-m}$, an identity of polynomials. By (ii), $\mathbb{E}[\frac{d}{du}\binom{u}{i}|_X]$ is the coefficient of z^i in $\log(1+z)\cdot(1+z)\log(1+z)/z$. From $\frac{d}{dz}\log^2(1+z) = 2\log(1+z)/(1+z) = 2\sum_{n\ge1}(-1)^{n-1}H_n z^n$, $\log^2(1+z) = 2\sum_{n\ge2}(-1)^n H_{n-1} z^n/n$, and the coefficient of z^i in $(1+z)\log^2(1+z)/z$ is

$$2(-1)^{i+1}\frac{H_i}{i+1} + 2(-1)^i\frac{H_{i-1}}{i} = 2(-1)^i\Big(\frac{H_{i-1}}{i} - \frac{H_{i-1} + \frac{1}{i}}{i+1}\Big) = 2(-1)^i\frac{H_{i-1} - 1}{i(i+1)}.$$

By 36.4 at $k=1$, for $R = (1-2u)Q$, $R(\frac12) = 0$ and $\mathrm{crit}_1[R] = \mathbb{E}[Q'(X)]$, which is the sum. $\square$

Proposition 36.11 (Poles at the half-integers). *As stated.*

Proof. Put $S(u)=\sum_m (u+m)^{-(2k+1)}$. From $\Gamma_{\substack{4\\0}}\big[u,\ u,\ 1-u,\ 1-u\big]=\sum_m (u+m)^{-2}$ (5.16), $\partial_u^{2k-1}\Gamma_{\substack{4\\0}}\big[u,\ u,\ 1-u,\ 1-u\big]=-(2k)!\,S(u)$, so $\operatorname{crit}_k[R]=\frac{2}{2\pi i}\int_{(1/2)} R(u)S(u)\,du/(1-2u)$. S has the period 1, poles of order $2k+1$ at the integers, it decays exponentially as $|\operatorname{Im} u|\to\infty$ uniformly in $\operatorname{Re} u$, and $S(1-u)=-S(u)$, so $S(\frac12)=0$.

For $p\notin\mathbb{Z}$ with $\operatorname{Re} p>\frac12$ let $I(p)=\frac{1}{2\pi i}\int_{(1/2)} S(u)\,du/(u-p)$. Move the line to $\operatorname{Re} u=N+\frac12$: the horizontal segments vanish by the decay of S, and on the new line $|1/(u-p)|\le 1/(N+\frac12-\operatorname{Re} p)\to 0$ while $\int|S|$ there is that on $\operatorname{Re} u=\frac12$. So $I(p)$ is minus the sum of the residues to the right of the line: at $u=p$ the residue $S(p)$, and at an integer $i\ge 1$ the residue of $(u-i)^{-2k-1}/(u-p)$, which is $(i-p)^{-2k-1}$. At $p=\frac12+l$, $l\ge 1$, $S(p)=\sum_{x\in\frac12+\mathbb{Z}} x^{-2k-1}=0$, and

$$I(p)=-\sum_{i\ge1}\big(i-\tfrac12-l\big)^{-2k-1}=-\sum_{x\in\frac12+\mathbb{Z},\ x>l} x^{-2k-1}=-(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty)),$$

the points x with $|x|<l$ cancelling in pairs. At $l=0$ the integrand $S(u)/(u-\frac12)$ is regular on the line; for real $p=\frac12+\varepsilon$, $\varepsilon\downarrow 0$, $|S(u)/(u-p)|\le|S(u)/(u-\frac12)|$ on the line, an integrable function, so the integral is $\lim I(p)=-S(\frac12)-\sum_{i\ge1}(i-\frac12)^{-2k-1}=-(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))$.

For the word, $\frac{1}{(u-p)(1-2u)}=\frac{1}{1-2p}\big(\frac{1}{u-p}-\frac{1}{u-\frac12}\big)$, so at $p=\frac12+l$

$$\begin{aligned}\operatorname{crit}_k\Big[\frac{1}{u-p}\Big]&=\frac{2}{1-2p}\Big((\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))-(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty))\Big)\\&=-\frac{1}{l}\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,l])\end{aligned}$$

by the cut relation (5.3). Finally $u\mapsto 1-u$ maps the line onto itself, reversing it, and with $S(1-u)=-S(u)$ it gives $\operatorname{crit}_k[R(1-u)]=\operatorname{crit}_k[R(u)]$; for $R(u)=1/(u-\frac12+l)$, $R(1-u)=-1/(u-\frac12-l)$. □

48.36 Chapter 37: the gap and the shape words

Proposition 37.8 (The root rate of the shape words). *As stated, with the maximizer at every $k\ge 2$.*

Proof. (i), the limit in (ii) and the last sentence are the reference's (the reference, proof of prop:kdepth). For the maximum, fix $x_1,\dots,x_k$ and put $y=x_1\cdots x_k$. The factor of f_k in x_0 is $x_0(1-x_0)/\big(1-(1-y)x_0\big)$; its derivative vanishes where $1-2x_0+(1-y)x_0^2=0$, at $x_0=1/(1+\sqrt{y})$, where $1-x_0=\sqrt{y}/(1+\sqrt{y})$ and $1-(1-y)x_0=\sqrt{y}$, with the value $(1+\sqrt{y})^{-2}$. So $M_k(0)=\sup\prod_{i\ge1}x_i(1-x_i)/\big(1+\sqrt{x_1\cdots x_k}\big)^2$. At a fixed product the numerator is largest when the x_i are equal, $t\mapsto\log(1-e^t)$ being concave, so $M_k(0)=\max_{0<a<1}\phi(a)$ with $\phi(a)=a^k(1-a)^k/(1+a^{k/2})^2$, and

$$\frac{\phi'(a)}{\phi(a)}=k\Big(\frac{1}{a}-\frac{1}{1-a}-\frac{a^{k/2-1}}{1+a^{k/2}}\Big)=\frac{k\left(1-2a-a^{k/2+1}\right)}{a(1-a)(1+a^{k/2})}.$$

The numerator decreases strictly from 1 at $a=0$ to $-2^{-k/2-1}$ at $a=\frac12$, so ϕ increases up to the unique root $a\in(0,\frac12)$ of $a^{k/2+1}+2a=1$ and decreases after it. At $\eta=0$ the equation

of 37.7 is $(1-2a)^2 = a^{k+2}$, whose root in $(0,\frac12)$ solves $1-2a = a^{k/2+1}$: the same a, with $x_2 = \cdots = x_k = a$ and $x_0 = a/(1-a) = 1/(1+a^{k/2})$, since $a^{k/2} = (1-2a)/a$. The interval follows from $\phi(\frac12) \le M_k(0) \le 4^{-k}$. At $k=2$ the root is $\sqrt2 - 1$ and at $k=4$ the root of a^3+2a-1, the maximizers of the reference. □

Proposition 37.14 (The insertion index supplies auxiliary states). *As stated, for every $k \ge 3$.*

Proof. By 37.13(ii), which holds for every $k \ge 2$, the share function of P_r is

$$g_{n,r}(a) = \binom{n+r}{r} \frac{(n!)^{k-2}\,(1-a)_n^2\,(n+1-a)_r}{(a)_{n+1}^k\,(n+1+a)_r}.$$

Its poles in $a \le 0$ are at $a = -h$, $0 \le h \le n$, of order exactly k, the numerator being nonzero there, and at $a = -h$, $n < h \le n+r$, simple. For $k \ge 3$ the coefficients of order $k-1 \ge 2$ come from the first poles only, so $C_{n,k}(P_r) = \sum_{h=0}^{n} R_h(-h)$ and $C_{n,k-1}(P_r) = \sum_{h=0}^{n} R_h'(-h)$, with $R_h(a) = (a+h)^k g_{n,r}(a)$, holomorphic near $-h$. First,

$$R_h(-h) = \binom{n+r}{r} \frac{(n!)^{k-2}(1+h)_n^2}{\prod_{j\ne h}(j-h)^k} \cdot \frac{(n+1+h)_r}{(n+1-h)_r}, \qquad \frac{(n+1+h)_r}{(n+1-h)_r} = \frac{(n+1-h+r)_{2h}}{(n+1-h)_{2h}},$$

the product over $0 \le j \le n$; $\binom{n+r}{r} = (r+1)_n/n!$ and $(n+1-h+r)_{2h}$ are polynomials in r, $n+1-h \ge 1$, so $R_h(-h) \in \mathbb{Q}[r]$ and $C_{n,k}(P_r) \in \mathbb{Q}[r]$. Second, $R_h'(-h) = R_h(-h)\,\rho_h$ with

$$\rho_h = -2\sum_{i=1}^{n} \frac{1}{i+h} - \sum_{i=n+1}^{n+r} \Big(\frac{1}{i+h} + \frac{1}{i-h}\Big) - k\sum_{j\ne h} \frac{1}{j-h},$$

the logarithmic derivatives at $-h$ of $(1-a)_n^2$, $(n+1-a)_r$, $(n+1+a)_r^{-1}$ and $\prod_{j\ne h}(a+j)^{-k}$. The finite Trailmoment $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,N])$ is the sum of m^{-1} over $0 < m \le N$, so $\sum_{i=n+1}^{n+r} \frac{1}{i+h} = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+h]) + \sum_{i=1}^{h} \frac{1}{n+r+i}$ and $\sum_{i=n+1}^{n+r} \frac{1}{i-h} = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n-h]) - \sum_{i=0}^{h-1} \frac{1}{n+r-i}$, and $\rho_h = -2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r])\,C_{n,k}(P_r) + q_n(r)$ with $q_n = \sum_h R_h(-h)\tilde\rho_h \in \mathbb{Q}(r)$.

Let $\sum_{i=0}^{d} b_i(r)\,C_{n,k-1}(P_{r+i}) = 0$ for all large r, $b_i \in \mathbb{Q}(r)$. Since $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r+i]) = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) + \sum_{l=1}^{i} \frac{1}{n+r+l}$, this reads $A(r)\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) + B(r) = 0$ for all large r with $A = -2\sum_i b_i(r) C_{n,k}(P_{r+i})$ and B in $\mathbb{Q}(r)$. If $A \ne 0$, then $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) = -B(r)/A(r)$ for all large r; a rational function is asymptotic to cr^e with $e \in \mathbb{Z}$, while $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) - \log r$ has a limit, which is impossible. So $A = 0$: the recurrence annihilates $C_{n,k}(P_r)$. □

Theorem 37.19 (An arithmetic budget for growing Beta ports). *As stated, with the coefficient of the top coordinate at every $k \ge 2$.*

Proof. The two limits and the clearing are the reference's (the reference, proof of thm:portbudget). The coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ in $I_n(k;P)$ is $k\,C_{n,k}(P)$ (37.4(iii)), and $P = (-1)^q \Pi_q^{n+1,n+1}(x_1)$ is free of x_0, so the theorem applies. As in the proof of 37.14, at $r = q$, $C_{n,k}(P) = \sum_{h=0}^{n} R_h(-h)$ with $\prod_{j\ne h}(j-h) = (-1)^h h!\,(n-h)!$ and $(1+h)_n = (n+h)!/h!$:

$$R_h(-h) = \binom{n+q}{q} \frac{(n!)^{k-2}}{(h!(n-h)!)^k} \frac{((n+h)!)^2}{(h!)^2} (-1)^{hk} \frac{(n+1+h)_q}{(n+1-h)_q}$$

$$= \binom{n+q}{q}(-1)^{hk}\binom{n}{h}^{k}\binom{n+h}{n}^{2}\frac{(n+1+h)_q}{(n+1-h)_q}.$$

For even k every term is positive. □

Corollary 37.20 (A growing Beta port with a certified gap). *As stated, for every $k \geq 2$.*

Proof. The port factor is at most one, since $x_0x_1\bar{x}_1x_2\cdots x_k \leq x_0x_1\cdots x_k \leq L$, so $M_k(\eta) \leq M_k(0)$ for $\eta \geq 0$ and

$$\ell_k(\eta) + (k+1)(1+\eta)\,c_{\mathrm{lcm}} \leq u(\eta) = \log M_k(0) + (1+\eta)\log(1+\eta) - \eta\log\eta + (k+1)(1+\eta)\,c_{\mathrm{lcm}}.$$

u is continuous on $[0,1]$ with $u(0) = \log M_k(0) + (k+1)c_{\mathrm{lcm}} < 0$ (37.8), so $u < 0$ on $[0,\eta_0]$ for a rational $\eta_0 > 0$. At a rational η the equation of 37.7, cleared of denominators, is an integer polynomial equation in a with the unique root in $(0, \frac{1}{2+\eta})$; the difference of its two sides is 1 at $a = 0$ and $-(1+\eta)^k(2+\eta)^{-k-2} < 0$ at $a = \frac{1}{2+\eta}$, so bisection at rational points certifies the root in rational intervals as small as asked. The two displays of 37.19 give the bound, and its last part the sign of the coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ for even k. □

48.37 Chapter 38: rays, native gauges and tiers

The proofs of this chapter rest on two readings on the sectors of the orthant: of an insertion along a ray (Lemma S19) and of the direction of a ray (Lemma S20).

Lemma S19. *Let a ray of a family on $n \geq 2$ marks be given, g an insertion as in Proposition 38.5, and π an order of $\mathcal{E}$ with the tails $U_l = \{\pi(l),\dots,\pi(n)\}$ and the sector coordinates $q = (q_2,\dots,q_n) \in (0,1]^{n-1}$ of Theorem 8.5(ii) on its sector H_π. On H_π*

$$g = \prod_{l=2}^{n} q_l^{\kappa_{U_l}(g)}\, g_\pi(q), \qquad f = \prod_{l=2}^{n} q_l^{\kappa_{1,U_l}}\, \varphi_\pi(q), \qquad \frac{\omega_0}{dy} = \prod_{l=2}^{n} q_l^{\kappa_{0,U_l}}\, \omega_\pi(q),$$

with $g_\pi = \prod_j k_{j,\pi}^{-a_j}\psi_{j,\pi}^{-b_j}$, $\varphi_\pi = \prod_j k_{j,\pi}^{-s_{1j}}\psi_{j,\pi}^{-t_{1j}}$ and $\omega_\pi = \prod_j k_{j,\pi}^{-s_{0j}}\psi_{j,\pi}^{-t_{0j}}$, the powers principal. These are smooth on a neighbourhood of $[0,1]^{n-1}$, and g_π, φ_π and $|\omega_\pi|$ lie between positive constants on the cube. For a function F on the orthant, homogeneous of degree 0 and integrable on the chart, $\int_{y_{e_0}=0} F\,dy = \sum_\pi \int_{(0,1]^{n-1}} F \prod_l \frac{dq_l}{q_l}$.

Proof. In the sector coordinates $\alpha_{\pi(1)} = \rho$, $\alpha_{\pi(i)} = \rho\,q_2\cdots q_i$, one has $\alpha^u = \rho^{u\varepsilon}\prod_l q_l^{u_{U_l}}$, and by Theorem 8.5(ii) $K_j = \rho^{r_j}\prod_l q_l^{d_j(U_l)}k_{j,\pi}$ and $\Psi_j = \rho^{n-r_j}\prod_l q_l^{d_j^*(U_l)}\psi_{j,\pi}$. The three functions are homogeneous of degree 0, the data of g, of the direction and of the origin lying on the balance wall, so ρ cancels, and the exponents of the q_l are the walls at U_l of the data $(u;\mathbf{a},\mathbf{b})$, $(\mathbf{m}_1;\mathbf{s}_1,\mathbf{t}_1)$ and $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$. The sector letters are polynomials with nonnegative coefficients and positive constant terms, so on the cube they are at least their constant terms, and they are positive on a neighbourhood of it; their principal powers are smooth there, the modulus of a principal power x^{-w}, $x > 0$, is $x^{-\operatorname{Re} w}$, and compactness gives the bounds. The sectors cover the orthant and meet in null sets. On the chart the map $y \mapsto (y_{\pi(l-1)} - y_{\pi(l)})_{l=2}^{n}$ is linear with an integral inverse, y being recovered from the differences and from $y_{e_0} = 0$, so it keeps the Lebesgue measure, and $\log q_l = y_{\pi(l)} - y_{\pi(l-1)}$ on the sector gives $dy = \prod_l dq_l/q_l$. □

Proposition 38.5 (The relations at every strength). *As stated.*

Proof. (i) Put $Y = g\,f^k\omega_0/dy$, homogeneous of degree 0. By Lemma S19, on the sector of π, $|Y| = \prod_l q_l^{c_l}\,|g_\pi\varphi_\pi^k\omega_\pi|$ with $c_l = \operatorname{Re}\kappa_{0,U_l} + \kappa_{U_l}(g) + \operatorname{Re}(k)\kappa_{1,U_l} > 0$ and the last factor bounded, every nonempty proper U being a tail of some order. So Y is integrable on the chart, and it tends to 0 at infinity: a sequence leaving every compact set of the chart has, along its terms in any one sector, $\min_l q_l \to 0$, and $\prod_{l'} q_{l'}^{c_{l'}} \le q_l^{c_l}$ for every l. By 38.2(ii), $\partial_{y_e}Y = (c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g\cdot f^k\omega_0/dy$, and by 38.2(i) $(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g$ is $(m_{0,e} + k\,m_{1,e} + u_e)\,g$ less the insertions $(s_{0j} + k\,s_{1j} + a_j)\,\theta_{j,T}\,\alpha^{\mathbf{1}_T}K_j^{-1}g$, $T \ni e$, and $(t_{0j} + k\,t_{1j} + b_j)\,\theta_{j,T}\,\alpha^{\mathbf{1}_{\varepsilon\setminus T}}\Psi_j^{-1}g$, $T \not\ni e$, whose walls are $\kappa_U(g) + |T\cap U| - d_j(U) \ge \kappa_U(g)$ and $\kappa_U(g) + |U\setminus T| - d_j^*(U) \ge \kappa_U(g)$ (7.13). So what precedes applies to them: $\partial_{y_e}Y$ is integrable, and its value converges absolutely. On the chart $\alpha_{e_0} = 1$, for $e \ne e_0$, Fubini's theorem and $\int_{-R}^{R}\partial_{y_e}Y\,dy_e = Y|_{y_e=R} - Y|_{y_e=-R} \to 0$ give $\int \partial_{y_e}Y\,dy = 0$; Y being homogeneous of degree 0, $\sum_e \partial_{y_e}Y = 0$ (Euler's identity), and $\partial_{y_{e_0}}Y = -\sum_{e\ne e_0}\partial_{y_e}Y$ integrates to 0 as well. For the certificate, at such a k every value involved converges absolutely and is linear in the insertion, $\langle f^j g'\rangle_k = \langle g'\rangle_{k+j}$, and the right side has the value 0.

(ii) By Lemma S19, for $\operatorname{Re}k$ large,

$$\langle g\rangle_k = \sum_\pi \int_{[0,1]^{n-1}} \prod_l q_l^{x_l(k)-1} h_\pi(q,k)\,dq, \qquad x_l(k) = \kappa_{0,U_l} + \kappa_{U_l}(g) + k\,\kappa_{1,U_l}, \quad h_\pi = g_\pi\varphi_\pi^k\omega_\pi.$$

h_π is smooth in q on a neighbourhood of the cube and entire in k, with q-derivatives continuous in (q,k) and holomorphic in k. By 8.4 each term extends meromorphically in (x,k), and its recursion, applied in every coordinate, writes it on $\{\operatorname{Re}x_l > -M\ \forall l\}$ as a finite sum of products of factors $(x_l+i)^{-1}$, at most one for each l, with functions holomorphic there. Along $x = x(k)$ each $x_l(k)+i$ is affine in k with the slope $\kappa_{1,U_l} > 0$, so the composite is meromorphic on the half-plane where every $\operatorname{Re}x_l(k) > -M$, with poles only where some $x_l(k)$ is a nonpositive integer, of order at most $n-1$; M being arbitrary, this continues $\langle g\rangle_k$ to $\mathbb{C}$. On the half-plane where the integral converges absolutely it is holomorphic, and it agrees there with the continuation. An element of $\mathcal{L}$ is a finite sum $\sum_{e,i} r_{e,i}(k)(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g_{e,i}$ with insertions $g_{e,i}$ of the above kind; its value is the $\mathbb{C}(k)$-linear extension of $g \mapsto \langle g\rangle_k$, which does not depend on the way the element is written, and by (i), the walls of the direction being positive, it vanishes for $\operatorname{Re}k$ large off the poles of the $r_{e,i}$, hence identically. $\langle fg\rangle_k$ and $\langle g\rangle_{k+1}$ are one integral for $\operatorname{Re}k$ large. A recurrence that follows from the relations is the value of the element $\sum_j p_j(k)f^j g$ of $\mathcal{L}$. □

Proposition 38.8 (The top and the bottom of a certificate). *As stated.*

Proof. For a polynomial $G = \sum_i k^i G^{[i]}$, with $G^{[-1]} = 0$, the coefficient of k^m in $(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})G$ is $(c_e^{(0)} + \partial_{y_e})G^{[m]} + c_e^{(1)}G^{[m-1]}$. At $m = d$, where every $G_e^{[d]}$ vanishes, this gives the first identity, the left side having the coefficient $\sum_j c_j f^j g = \chi(f)\,g$; at $m = 0$ it gives the second. At a native gauge every $c_e^{(1)}$ vanishes (Definition 38.4), so $\chi\big(f(\alpha_c)\big)g(\alpha_c) = 0$. For the origin 0, $c^{(0)} = 0$. □

Lemma S20. *Let a ray of a family on $n \ge 2$ marks have $\kappa_{1,S} > 0$ for every nonempty proper S, with its stride f on the chart $y_{e_0} = 0$.*

(i) $\mathbf{m}_1 > 0$*, the exponents of the direction are not all zero, and the tables with $s_{1j} + t_{1j} > 0$ form a connected family.*

(ii) The direction has a native gauge $\alpha^\star \in \mathbb{R}^{\mathcal{E}}_{>0}$, unique up to scale; with $\alpha^\star_{e_0} = 1$, $0 < f \le f(\alpha^\star)$ on the chart with equality only at $\log\alpha^\star$, and $f \le C'e^{-c'\|y\|}$ there for some $c', C' > 0$.

(iii) The Hessian of $-\log f$ *at* $\log\alpha^\star$ *on the chart is*

$$H = \Big(\sum_j \big(s_{1j}\, \kappa_{j,\alpha^\star}(X_e, X_{e'}) + t_{1j}\, \kappa_{j,(\alpha^\star)^{-1}}(X_e, X_{e'}) \big) \Big)_{e,e' \neq e_0},$$

and it is positive definite.

(iv) $\langle 1 \rangle_k = P_F(\mathbf{m}_0 + k\mathbf{m}_1; \mathbf{s}_0 + k\mathbf{s}_1, \mathbf{t}_0 + k\mathbf{t}_1)$*, and it converges absolutely exactly when* $\operatorname{Re}\kappa_{0,S} + k\kappa_{1,S} > 0$ *for every nonempty proper* S*.*

Proof. (i) Every $\{e\}$ is proper, and $\kappa_{1,\{e\}} = m_{1,e} - \sum_j \big(s_{1j} d_j(\{e\}) + t_{1j} d_j^*(\{e\}) \big)$ with $s_{1j}, t_{1j} \in \mathbb{N}$ and $d_j, d_j^* \geq 0$, so $m_{1,e} \geq \kappa_{1,\{e\}} > 0$. If every exponent of the direction were 0, the balance wall would give $m_{1,\mathcal{E}} = 0$. On the balance wall 8.16 gives $\kappa_{1,S} + \kappa_{1,\mathcal{E}\setminus S} = \sum_j (s_{1j} + t_{1j}) c_j(S)$, so every nonempty proper S has $c_j(S) > 0$ for a table with $s_{1j} + t_{1j} > 0$.

(ii) The direction is real data on its balance wall, where the target is the masses, with nonnegative exponents not all zero and positive masses. By (i) and 8.19(iii) it has a native gauge $\alpha^\star \in \mathbb{R}^{\mathcal{E}}_{>0}$, unique up to scale, the unique point up to scale at which $\alpha^{\mathbf{m}_1} \prod_j K_j^{-s_{1j}} \Psi_j^{-t_{1j}} = f$ is largest. At $y = 2z$ the stride is $e^{X_1(z)}$, X_1 the exponent of the question of the direction (17.1), and 17.4(i) and (ii) give $X_1(z) \leq -2\kappa_{1,\min}(\max_e z_e - \min_e z_e) + C$; on the chart $|y_e| \leq \max y - \min y$, since $y_{e_0} = 0$.

(iii) The Hessian of $\log K_j(e^y)$ in y is the covariance of the indicators of the basis under the basis law at $\alpha = e^y$ (7.16), and $\Psi_j(e^y) = e^{\langle \mathbf{1}, y \rangle} K_j(e^{-y})$, so the Hessian of $\log \Psi_j(e^y)$ is the covariance under the law at α^{-1}; at $\alpha^\star$ this gives H. For v with $v_{e_0} = 0$,

$$v^{\mathsf{T}} H v = \sum_j \Big(s_{1j} \operatorname{Var}_{\mathbb{P}_{j,\alpha^\star}} \langle \mathbf{1}_T, v \rangle + t_{1j} \operatorname{Var}_{\mathbb{P}_{j,(\alpha^\star)^{-1}}} \langle \mathbf{1}_T, v \rangle \Big) \geq 0,$$

both laws charging every basis. Let $v^{\mathsf{T}} H v = 0$ with $v \neq 0$. Then v is not constant, and $\langle \mathbf{1}_T, v \rangle$ is the same at every basis of every table with $s_{1j} + t_{1j} > 0$. Let $S = \{e : v_e = \max v\}$, nonempty and proper, and take by (i) such a table with $c_j(S) > 0$ and a basis T with $|T \cap S| < r_j(S)$. $T \cap S$ is independent in the restriction to S and not a basis of it, so some $f \in S \setminus T$ keeps it independent; the circuit of $T + f$ through f is not contained in S, and with e a mark of it outside S, $T - e + f$ is a basis with $\langle \mathbf{1}_{T-e+f}, v \rangle - \langle \mathbf{1}_T, v \rangle = v_f - v_e > 0$, a contradiction.

(iv) On the chart $\alpha_{e_0} = 1$, $\alpha^{\mathbf{m}-\mathbf{1}} \prod_{e \neq e_0} d\alpha_e = \alpha^{\mathbf{m}} dy$, so the value $\langle 1 \rangle_k = \int f^k \omega_0$ of 38.1 is the period at the point k of the ray in the form of 8.7, which converges exactly when the real parts of its proper walls $\kappa_{0,S} + k\kappa_{1,S}$ are positive; the modulus of its integrand is the integrand at the real parts of the data. □

Proposition 38.9 (The native value bounds the order). *As stated: every ray with positive direction walls, its origin complex.*

Proof. By Lemma S20(iv), $\langle 1 \rangle_k$ converges absolutely for $k \geq k_1$. The amplitude $g_0 = \omega_0 / dy = \alpha^{\mathbf{m}_0} \prod_j K_j^{-s_{0j}} \Psi_j^{-t_{0j}}$, the powers of the positive letters principal, is continuous and nowhere zero on the chart. Let $y^\star = \log \alpha^\star$ with $\alpha^\star_{e_0} = 1$. By Lemma S20(ii) and (iii), $f / f(\alpha^\star)$ is real-analytic, has its unique maximum 1 at $y^\star$, nondegenerate with the Hessian H, and tends to 0 at infinity; so off a ball B about $y^\star$, $f \leq e^{-\delta} f(\alpha^\star)$ for some $\delta > 0$, and for $k \geq k_1$

$$\Big| \int_{y_{e_0} = 0,\ y \notin B} f^k g_0 \, dy \Big| \leq e^{-\delta(k-k_1)} f(\alpha^\star)^{k-k_1} \int f^{k_1} |g_0| \, dy.$$

On B, Laplace's method with the continuous amplitude g_0 gives

$$\int_B f^k g_0\,dy = f(\alpha^\star)^k\Big(\frac{2\pi}{k}\Big)^{\frac{n-1}{2}}(\det H)^{-\frac12}\big(g_0(y^\star)+o(1)\big).$$

Hence $\langle 1\rangle_k = f(\alpha^\star)^k\big(\frac{2\pi}{k}\big)^{\frac{n-1}{2}}(\det H)^{-\frac12}g_0(y^\star)(1+o(1))$ with $g_0(y^\star)\neq 0$: $\langle 1\rangle_k\neq 0$ for all large k and $\langle 1\rangle_{k+1}/\langle 1\rangle_k \to f(\alpha^\star)$. For a real origin $g_0>0$, and $\langle 1\rangle_k>0$ wherever it converges.

Let $\sum_{j=0}^{h} p_j(k)\langle 1\rangle_{k+j}=0$ for all large k, the $p_j\in\mathbb{Q}[k]$ not all zero, d their largest degree and c_j the coefficient of k^d in p_j. Dividing by $k^d\langle 1\rangle_k$ and letting $k\to\infty$ gives $\sum_j c_j f(\alpha^\star)^j=0$, and the symbol $\sum_j c_j\lambda^j$ is a nonzero rational polynomial of degree at most h. So $f(\alpha^\star)$ is algebraic, its minimal polynomial divides the symbol, every conjugate is a root of it, and $h\geq[\mathbb{Q}(f(\alpha^\star)):\mathbb{Q}]$. For $h=1$ this degree is one, and a transcendental native value admits no h. □

Proposition 38.10 (The conjugates of the native value of the direction). *As stated.*

Proof. On the chart $\alpha_{e_0}=1$ the letter complement X_F is the affine variety $\{\alpha\in(\mathbb{C}^\times)^{\mathcal{E}-e_0}:\prod_j K_j(\alpha)\Psi_j(\alpha)\neq 0\}$, and the defects $c_e^{(1)}=m_{1,e}-\sum_j\big(s_{1j}R_{j,e}+t_{1j}R^*_{j,e}\big)$ are regular functions on it with coefficients in $\Bbbk$, the direction being integral. Their sum over e is $\sum_e \partial_{y_e}\log f=0$, f being homogeneous of degree 0, so the native gauges are the common zeros in X_F of the $c_e^{(1)}$, $e\neq e_0$: an algebraic set defined over $\Bbbk$. Its Jacobian matrix in the coordinates y_e, $e\neq e_0$, is the Hessian of $\log f$, nonsingular at $\alpha^\star$ (Lemma S20(iii)); by the holomorphic inverse function theorem $\alpha^\star$ is an isolated point of it, and an isolated point of an algebraic set defined over $\Bbbk$ has coordinates algebraic over $\Bbbk$, the irreducible components of the set being defined over the algebraic closure of $\Bbbk$. An embedding τ over $\Bbbk$ keeps every polynomial identity with coefficients in $\Bbbk$: $\tau(\alpha^\star)$ satisfies the equations of nativity, no letter vanishes at it, the Hessian determinant, a rational function over $\Bbbk$, takes there the value τ of a nonzero number, and $f\big(\tau(\alpha^\star)\big)=\tau\big(f(\alpha^\star)\big)$. Every conjugate of $f(\alpha^\star)$ over $\Bbbk$ is $\sigma\big(f(\alpha^\star)\big)$ for an embedding σ of $\Bbbk\big(f(\alpha^\star)\big)$, which extends to an embedding of the finite extension $\Bbbk(\alpha^\star)$ in the algebraically closed $\mathbb{C}$. The $[\Bbbk(\alpha^\star):\Bbbk]$ embeddings of $\Bbbk(\alpha^\star)$ differ on a coordinate of $\alpha^\star$, so they give as many distinct nondegenerate native gauges, and Theorem 38.6(iii) bounds the dimension. Corollary 38.7 at $g=1$ gives the last statement. □

Proposition 38.12 (The growth of the frame). *As stated.*

Proof. By 38.11, C_k is $C_{k_0}((-1)^h c)^{k-k_0}$ times $\prod_{i\leq d_0}\Gamma(k-a_i)/\prod_{i\leq d_h}\Gamma(k-b_i)$ times a nonzero constant. Stirling's formula, which applies to $\Gamma(k-a)$ for fixed complex a as $k\to+\infty$ along the reals, the argument staying in a sector about the positive axis, gives $\log|\Gamma(k-a)|=(k-\operatorname{Re}a-\frac12)\log k-k+O(1)$; summing, the Gammas contribute $(d_0-d_h)(k\log k-k)+O(\log k)$. When $d_0=d_h=d$ is the largest degree, c_0 and c_h are the leading coefficients of p_0 and p_h, both nonzero, the symbol has degree h, and the product of its roots is $(-1)^h c_0/c_h=(-1)^h c$; the term in $k\log k$ is absent and $\frac1k\log|C_k|\to\log|c|$. □

Definition 38.13 (The centre ray). As stated, for a family connected at strengths $\sigma_j\in\mathbb{N}$.

Proof. The direction $(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$ is integral, and it is on the balance wall: $m_{1,\mathcal{E}}=n\sigma=\sum_j\sigma_j(r_j+r_j^*)$. The origin 0 is on it too. From $\Psi_j(\alpha)=\alpha^{\mathbf{1}}K_j(\alpha^{-1})$, $K_j\Psi_j=\alpha^{\mathbf{1}}K_j(\alpha)K_j(\alpha^{-1})$, and with $\sum_j\sigma_j=\sigma$ the stride is $\alpha^{\sigma\mathbf{1}}\prod_j(K_j\Psi_j)^{-\sigma_j}=\prod_j\big(K_j(e^y)K_j(e^{-y})\big)^{-\sigma_j}$ at $\alpha=e^y$. The origin form is dy, so $\langle 1\rangle_k=\int_{y_{e_0}=0}f^k dy=P_F(k\sigma\mathbf{1};k\boldsymbol{\sigma},k\boldsymbol{\sigma})$ by Lemma S20(iv). The direction walls are $\sigma|S|-\sum_j\sigma_j\big(d_j(S)+d_j^*(S)\big)=\sum_j\sigma_j c_j(S)$, positive on every nonempty proper S exactly when the family is connected at the strengths (8.21(ii)). □

Theorem 38.14 (A ray is one function). *As stated, for every ray with real origin and positive direction walls.*

Proof. (i) The bounds on f are Lemma S20(ii). For $k \geq k_0$ the walls $\kappa_{0,S} + k\kappa_{1,S} \geq \kappa_{0,S} + k_0\kappa_{1,S}$ are real and positive, so $\langle 1 \rangle_k$ converges (Lemma S20(iv)), and its integrand $f^k \omega_0/dy$ is positive.

(ii) For $|q| < f(\alpha^\star)^{-1}$, $|qf| \leq |q| f(\alpha^\star) < 1$ on the chart, so $\sum_{k \geq k_0} q^{k-k_0} f^k = f^{k_0}/(1-qf)$, the partial sums dominated by $f^{k_0}/(1-|q|f(\alpha^\star))$ times the positive density ω_0/dy, which is integrable at $k = k_0$. Summing under the integral gives the first form of G, and the second is the first with numerator and denominator multiplied by $\prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}}$. On smaller discs the same domination allows differentiation under the integral, $\partial_q^l (1-qf)^{-1} = l!\, f^l (1-qf)^{-l-1}$. For real $q < f(\alpha^\star)^{-1}$, $1 - qf \geq \min\big(1, 1 - qf(\alpha^\star)\big) > 0$, so the integrand is positive and at most a multiple of $f^{k_0}\omega_0/dy$.

(iii) With the positive measure $\mu = f^k \omega_0$ on the chart, $\langle 1 \rangle_{k+i+i'} = \int f^{i+i'} d\mu$, and the identity of Andréief gives

$$\det\Big[\int f^{i+i'} d\mu\Big]_{i,i'<h} = \frac{1}{h!} \int \det\big[f(y_a)^i\big]_{i<h,\, a\leq h}^2 \prod_a d\mu(y_a) = \frac{1}{h!} \int \prod_{a<b} \big(f(y_a) - f(y_b)\big)^2 \prod_a d\mu(y_a),$$

positive because f is real-analytic and not constant, so that the set where two values $f(y_a)$ agree is null. If $G = N_1/N_2$ were rational, $N_2(0) \neq 0$, $d = \deg N_2$, then $\sum_{i=0}^{d} [q^i] N_2 \, \langle 1 \rangle_{k_0+m-i} = 0$ for $m > \deg N_1$, a relation with $[q^0]N_2 \neq 0$ among the rows of the Hankel matrix $\big[\langle 1 \rangle_{k+l+i'}\big]_{l,i' \leq d}$ for all large k, whose determinant would vanish, against (iii) at $h = d+1$.

On the centre ray $\kappa_{0,S} = 0 < \kappa_{1,S}$, so $k_0 = 1$, and $\alpha^\star = \mathbf{1}$ by 8.21(i). □

Theorem 38.15 (The branch at the native gauge of the direction). *As stated, for every ray with real origin and positive direction walls.*

Proof. Put $y^\star = \log \alpha^\star$, $\varphi = -\log\big(f/f(\alpha^\star)\big) \geq 0$ on the chart, zero only at $y^\star$, with the Hessian H there, positive definite (Lemma S20(iii)), and $g = f^{k_0}\omega_0/dy$, positive, real-analytic and integrable. At $q = A(1-\varepsilon)$, $1 - qf = 1 - (1-\varepsilon)e^{-\varphi}$, so for $0 < \varepsilon \leq 1$

$$G(q) = \int \frac{g\, dy}{1 - (1-\varepsilon)e^{-\varphi}}.$$

Fix a small $t_1 > 0$. On $\{\varphi \geq t_1\}$, $|1 - (1-\varepsilon)e^{-\varphi}| \geq 1 - |1-\varepsilon| e^{-t_1} > 0$ for $|\varepsilon|$ small, so that part of the integral is holomorphic at $\varepsilon = 0$. Near $y^\star$ the real-analytic Morse lemma gives coordinates $v \in \mathbb{R}^{2a}$, $2a = n-1$, with $\varphi = \frac{1}{2}|v|^2$ and $|\det \partial y/\partial v| = (\det H)^{-1/2}$ at $v = 0$; for t_1 small, $\{\varphi < t_1\}$ lies in their domain. With $\tilde{g} = g\,|\det \partial y/\partial v|$ and

$$\beta(t) = 2^{a-1} \int_{S^{2a-1}} \tilde{g}\big(\sqrt{2t}\,\omega\big)\, d\omega,$$

analytic near $[0, t_1]$, the spherical mean being an even analytic function of the radius, $\int_{\varphi<t} g\, dy$ has the derivative $t^{a-1}\beta(t)$. In $w = e^t - 1$, $1 - (1-\varepsilon)e^{-t} = e^{-t}(w + \varepsilon)$ and $e^t dt = dw$, so the part over $\{\varphi < t_1\}$ is

$$\int_0^{w_1} \frac{w^{a-1}\gamma(w)}{w+\varepsilon}\, dw, \qquad \gamma(w) = \Big(\frac{\log(1+w)}{w}\Big)^{a-1} \beta\big(\log(1+w)\big), \quad \gamma(0) = \beta(0),$$

with $\gamma=\sum_m \gamma_m w^m$ on a disc of radius $R'>w_1$, t_1 being small, so that $|\gamma_m|\le CR'^{-m}$. For $c>0$ not an integer and $|\varepsilon|<w_1$, $\varepsilon\notin(-\infty,0]$,

$$\int_0^{w_1}\frac{w^{c-1}\,dw}{w+\varepsilon}=\frac{\pi}{\sin\pi c}\,\varepsilon^{c-1}-\sum_{l\ge0}\frac{(-\varepsilon)^l w_1^{c-1-l}}{l+1-c},$$

from $\int_0^\infty w^{c-1}dw/(w+\varepsilon)=\frac{\pi}{\sin\pi c}\varepsilon^{c-1}$ at $0<\operatorname{Re}c<1$ and continuation in c; for an integer $c=N\ge1$, $w^{N-1}=\big((w+\varepsilon)-\varepsilon\big)^{N-1}$ gives the integral as $(-1)^N\varepsilon^{N-1}\log\varepsilon$ plus $(-\varepsilon)^{N-1}\log(w_1+\varepsilon)$ plus a polynomial in ε, the two last of modulus $O\big((w_1+2\rho)^{N-1}\big)$ on $|\varepsilon|\le\rho$. Summed with the weights γ_m at $c=a+m$, the singular terms give $\varepsilon^{a-1}V(\varepsilon)$ for n even and $\varepsilon^{a-1}\log\varepsilon\,V(\varepsilon)$ for n odd, with

$$V(\varepsilon)=\frac{\pi}{\sin\pi a}\,\gamma(-\varepsilon)\quad(n\text{ even}),\qquad V(\varepsilon)=(-1)^a\gamma(-\varepsilon)\quad(n\text{ odd}),$$

and the remainders converge to a function holomorphic near 0, since $|l+1-a-m|\ge\frac12$ for n even and $w_1+2\rho<R'$ for small ρ. This is the form of G, with U the sum of the remainders and the part over $\{\varphi\ge t_1\}$. Further $\beta(0)=2^{a-1}|S^{2a-1}|\,g(\alpha^\star)(\det H)^{-1/2}=(2\pi)^a g(\alpha^\star)/\big(\Gamma(a)\sqrt{\det H}\big)$, $\frac{\pi}{\sin\pi a}=\Gamma(a)\Gamma(1-a)$, and $g(\alpha^\star)=f(\alpha^\star)^{k_0}(\omega_0/dy)(\alpha^\star)=(f^{k_0-1}\omega_0/dy)(\alpha^\star)/A$, which gives $V(0)$.

$G^{(l)}(A)=l!\int f^{k_0+l}\omega_0/(1-Af)^{l+1}$ is a positive integral, and near $y^\star$, $1-Af=1-e^{-\varphi}$ is comparable with $|v|^2$, so it is finite exactly when $2(l+1)<2a$. For $l<a-1$, $G^{(l)}(q)\to G^{(l)}(A)$ as $q\uparrow A$ by monotone convergence, while the l-th derivative of $\varepsilon^{a-1}V$, resp. of $\varepsilon^{a-1}\log\varepsilon\,V$, tends to 0; so the Taylor coefficients of U of order $l<a-1$ are those of G at A, with $q-A=-A\varepsilon$, and the rest of U and $\varepsilon^{a-1}(V-V(0))$ are $O(\varepsilon^{\lceil a-1\rceil})$. On the centre ray $k_0=1$, $\alpha^\star=\mathbf{1}=(\alpha^\star)^{-1}$, $A=f(\mathbf{1})^{-1}=\prod_j K_j(\mathbf{1})^{2\sigma_j}$, $f^0\omega_0/dy=1$ and $H=2\sum_j\sigma_j\big(\kappa_{j,\mathbf{1}}(X_e,X_{e'})\big)_{e,e'\ne e_0}$. □

Proposition 38.16 (The native gauges of a centre ray). *As stated.*

Proof. The direction is $(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, so $c_e^{(1)}=\sigma-\sum_j\sigma_j\big(R_{j,e}+R_{j,e}^*\big)$ (Definition 38.1). By 7.2, $\Psi_j(\alpha)=\alpha^{\mathcal{E}}K_j(\alpha^{-1})$, so $R_{j,e}^*(\alpha)=\alpha_e\partial_{\alpha_e}\log\Psi_j(\alpha)=1-R_{j,e}(\alpha^{-1})$ as rational functions, the identity of 7.16 at real gauges; with $\sum_j\sigma_j=\sigma$ this gives the defects. The stride is $\prod_j\big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$, a rational function homogeneous of degree 0 (Definition 38.13). □

Theorem 38.17 (Native gauges at the signs). *As stated, for the centre ray of a family.*

Proof. (i) On the chart $\alpha_{e_0}=1$, $f(\alpha)=\prod_j\big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$ (38.13), so $f(\alpha^{-1})=f(\alpha)$; $\alpha\mapsto\alpha^{-1}$ keeps the letter complement, since $K_j(\alpha^{-1})\Psi_j(\alpha^{-1})=\alpha^{-\mathbf{21}}\Psi_j(\alpha)K_j(\alpha)$. A point of $(\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times$ fixed by the inversion has α_e^2 independent of e, hence $\alpha_e=\pm1$ on the chart: the 2^{n-1} signs. By 38.2(ii) the native gauges of the ray are the critical points of $\log f$ on X_F, and the inversion, which keeps f, permutes them and keeps their native values.

(ii) At a sign η, $\Psi_j(\eta)=\eta^{\mathbf{1}}K_j(\eta)$, so $K_j(\eta)\ne0$ for every j puts η in X_F. In the chart $\alpha=\eta e^y$, $y_{e_0}=0$, $\alpha^{-1}=\eta e^{-y}$, so $y\mapsto\log f(\eta e^y)$ is even and its gradient vanishes at $y=0$: η is a native gauge, of native value $\prod_j\big(K_j(\eta)K_j(\eta)\big)^{-\sigma_j}$. With $K_j(\eta e^y)=\sum_T\theta_{j,T}\eta^T e^{\langle\mathbf{1}_T,y\rangle}=K_j(\eta)\sum_T\mathbb{P}_{j,\eta}(T)e^{\langle\mathbf{1}_T,y\rangle}$, the Hessian of $\log K_j(\eta e^y)$ at 0 is the covariance of the indicators under the signed measure $\mathbb{P}_{j,\eta}$ of total mass one, and $\log K_j(\eta e^{-y})$ has the same Hessian; so the Hessian of $-\log f$ is $2\sum_j\sigma_j\kappa_{j,\eta}$.

(iii) $W^{-1}=\prod_j(K_j\Psi_j)^{\sigma_j}$ is a polynomial, and $\partial_{y_e}\log f=\big(\sigma W^{-1}-\alpha_e\partial_e W^{-1}\big)/W^{-1}$. For a sign η, $h(y)=W^{-1}(\eta e^y)\,e^{-\sigma\langle\mathbf{1},y\rangle}=\prod_j\big(\eta^{\mathbf{1}}K_j(\eta e^y)K_j(\eta e^{-y})\big)^{\sigma_j}$ is even in y, so at $y=0$

$$\partial_{y_e}h=\big(\alpha_e\partial_e W^{-1}-\sigma W^{-1}\big)(\eta)=0$$

at every sign, whether or not the $K_j(\eta)$ vanish, and $W^{-1}(\eta) = \eta^{\boldsymbol{\sigma 1}} \prod_j K_j(\eta)^{2\sigma_j}$. The ring $\mathbb{Q}(\boldsymbol{\theta})[\alpha_e : e \neq e_0]/(\alpha_e^2 - 1)$ is the product of copies of $\mathbb{Q}(\boldsymbol{\theta})$ indexed by the signs, a class being its values at the signs, with the idempotents $2^{1-n}\prod_{e\neq e_0}(1+\eta_e\alpha_e)$, which take the value 1 at η and 0 at the other signs. So the critical polynomials vanish in it, and $W^{-1} \equiv \alpha^{\boldsymbol{\sigma 1}} \prod_j K_j^{2\sigma_j}$. Localizing at $\prod_j K_j$ keeps exactly the factors of the signs with $\prod_j K_j(\eta) \neq 0$, whose idempotents form a basis, and multiplication by $W^{-1}/\alpha^{\boldsymbol{\sigma 1}}$, $\alpha^{\boldsymbol{\sigma 1}}$ being a unit, acts on the idempotent of η by its value $\prod_j K_j(\eta)^{2\sigma_j}$. □

Theorem 38.18 (Weights that separate the signs). *As stated, for a family of N tables connected at its strengths.*

Proof. Treat the weights of all tables as variables. For a sign η, $K_j(\eta) = \sum_T \theta_{j,T}\eta^T$ is a nonzero linear form. Write $M_j = K_j(\eta)^2\big(\kappa_{j,\eta}(X_e, X_{e'})\big)_{e,e'\neq e_0}$, whose entries $K_j(\eta)\sum_{T\ni e,e'}\theta_{j,T}\eta^T - \sum_{T\ni e}\theta_{j,T}\eta^T\sum_{T\ni e'}\theta_{j,T}\eta^T$ are quadratic, and

$$D_\eta = \det\Big[\sum_j \sigma_j M_j \prod_{i\neq j} K_i(\eta)^2\Big] = \prod_i K_i(\eta)^{2(n-1)} \det\Big[\sum_j \sigma_j \kappa_{j,\eta}\Big],$$

of degree $2N(n-1)$. At $\eta = \mathbf{1}$ and unit weights $D_{\mathbf{1}} \neq 0$: the Hessian $2\sum_j \sigma_j\kappa_{j,\mathbf{1}}$ is positive definite, by the argument of Lemma S20(iii) at the strengths. The substitution $\theta_{j,T} \mapsto \theta_{j,T}\eta^T$ carries $K_j(\mathbf{1})$ to $K_j(\eta)$ and $\mathbb{P}_{j,\mathbf{1}}$ to $\mathbb{P}_{j,\eta}$, so D_η is $D_{\mathbf{1}}$ after an invertible linear substitution, and $D_\eta \neq 0$. For signs $\eta \neq \eta'$, $\prod_j K_j(\eta)^{2\sigma_j} - \prod_j K_j(\eta')^{2\sigma_j}$, of degree 2σ, is not zero: otherwise unique factorization, the linear forms $K_j(\eta)$ being irreducible and in disjoint variables for distinct j, makes $K_j(\eta)$ a multiple of $K_j(\eta')$ for every j with $\sigma_j > 0$, that is $\eta^T\eta'^T$ constant on the bases of such a table; with $S = \{e : \eta_e \neq \eta'_e\}$, nonempty and proper since $\eta_{e_0} = \eta'_{e_0}$, the parity of $|T\cap S|$ is then constant, so the window $\{d_j(S), \dots, r_j(S)\}$ of 7.13 is one value and $c_j(S) = 0$ for every j with $\sigma_j > 0$, against connectedness. The product

$$\prod_\eta \Big(D_\eta \prod_j K_j(\eta)\Big) \prod_{\{\eta,\eta'\}} \Big(\prod_j K_j(\eta)^{2\sigma_j} - \prod_j K_j(\eta')^{2\sigma_j}\Big)$$

is a nonzero polynomial of degree $2^{n-1}\big(N+2N(n-1)\big)+2\sigma\binom{2^{n-1}}{2} = 2^{n-1}\big((2n-1)N+(2^{n-1}-1)\sigma\big)$, so by the Schwartz–Zippel bound it does not vanish at some point of $\{1,\dots,D+1\}$ in every variable, D that degree. At such weights every $K_j(\eta) \neq 0$, every sign is a native gauge by 38.17(ii), nondegenerate since $D_\eta \neq 0$, and the native values are distinct. 38.6(iii) bounds the dimension below by 2^{n-1}, and 38.7 at $g = 1$ the order of every recurrence that follows from the relations. □

Lemma 38.19 (Where native gauges can reach a wall). *As stated, for a family.*

Proof. (i) On the balance wall the target of the direction is $\mathbf{m}_1$, so nativity at α reads $\mathbf{m}_1 = \sum_j \big(s_{1j}R_j(\alpha) + t_{1j}R_j^*(\alpha)\big)$ (8.17). $R_{j,e} = \alpha_e\partial_e K_j/K_j = \sum_T \mathbf{1}_T(e)\,\mathbb{P}_{j,\alpha}(T)$ and $R^*_{j,e} = \sum_T \mathbf{1}_{\mathcal{E}\setminus T}(e)\,\mathbb{P}^*_{j,\alpha}(T)$, and summing over $e \in B$ gives (i).

(ii) In the coordinates (ρ, u, α_R), $\alpha^T = \rho^{|T\cap B|}u^{T\cap B}\alpha_R^{T\cap R}$, so $\sum_T |T\cap B|\,\theta_{j,T}\alpha^T = \rho\,\partial_\rho K_j$, and with $\widetilde{K}_j = \sum_l \rho^l K_j^{[l]}$, $\widetilde{\Psi}_j = \sum_l \rho^l \Psi_j^{[l]}$,

$$\kappa_{1,B} = \sum_j \Big(s_{1j}\,\frac{\rho\,\partial_\rho \widetilde{K}_j}{\widetilde{K}_j} + t_{1j}\,\frac{\rho\,\partial_\rho \widetilde{\Psi}_j}{\widetilde{\Psi}_j}\Big)$$

at every native gauge of the direction, by (i) and $\kappa_{1,B} = m_{1,B} - \sum_j \big(s_{1j} d_j(B) + t_{1j} d_j^*(B)\big)$. Along the sequence the exponents stay bounded, and $\rho_i \partial_\rho \widetilde{K}_j = O(\rho_i)$. If no bottom letter vanished at (u^*, α_R^*), every quotient would be $O(\rho_i)$ and the right side would tend to 0, while the left side tends to $\kappa_{1,B}(\mathbf{v}) > 0$. So some bottom letter vanishes there. Let it be the only one, say $K_{j_0}^{[0]}$. Then the terms of the other letters tend to 0, so $s_{1j_0}(\mathbf{v}_i)\, \rho_i \partial_\rho \widetilde{K}_{j_0} / \widetilde{K}_{j_0} \to \kappa_{1,B}(\mathbf{v})$, and since $\rho_i \partial_\rho \widetilde{K}_{j_0} \to 0$, $|s_{1j_0}(\mathbf{v}_i)/\widetilde{K}_{j_0}| \to \infty$. Nativity in a logarithmic coordinate D of u or of α_R reads

$$D \log \big(u^{\mathbf{m}_1|_B} \alpha_R^{\mathbf{m}_1|_R}\big) = \sum_j \Big(s_{1j} \frac{D\widetilde{K}_j}{\widetilde{K}_j} + t_{1j} \frac{D\widetilde{\Psi}_j}{\widetilde{\Psi}_j}\Big);$$

the left side and the terms of the other letters stay bounded, so $s_{1j_0} D\widetilde{K}_{j_0}/\widetilde{K}_{j_0}$ does, and $D\widetilde{K}_{j_0}(\rho_i, u_i, \alpha_{R,i}) \to DK_{j_0}^{[0]}(u^*, \alpha_R^*)$ must vanish. The bottom letter is homogeneous in u and in α_R, so with $K_{j_0}^{[0]}(u^*, \alpha_R^*) = 0$ Euler's identity gives the derivatives along the scalings of u and of α_R as well: the whole gradient vanishes. The same holds for a cobasis letter. If two or more bottom letters vanish, every term of the gradient of the product contains a vanishing factor; if one does, its gradient vanishes. In both cases the product vanishes at (u^*, α_R^*) with its gradient. □

Proposition 38.23 (Small lower coefficients on one parameter). *As stated, with p letters, for real C_i bounded below by a positive constant on $[0, \infty)$.*

Proof. For $\rho \geq 0$ and $\lambda > 0$, $\lambda\xi_i + \rho C_i(\rho)$ is $\lambda\xi_i$ plus a nonnegative number, so it stays in $\mathbb{C}\backslash(-\infty, 0]$, the powers are principal, and the integrand is continuous on $(0, \infty)$. Put $F_{\mathbf{a}} = G \prod_i C_i^{-w_i - a_i}$ and $c_{\mathbf{a}}(\lambda) = \prod_i \binom{-w_i}{a_i} (\lambda\xi_i)^{a_i}$, so that, wherever $|\lambda\xi_i| < |\rho C_i(\rho)|$ for every i,

$$\rho^{\kappa - 1} G \prod_i \big(\lambda\xi_i + \rho C_i\big)^{-w_i} = \rho^{\kappa^\flat - 1} F_{\mathbf{0}} \prod_i \Big(1 + \frac{\lambda\xi_i}{\rho C_i}\Big)^{-w_i} = \sum_{\mathbf{a}} c_{\mathbf{a}}(\lambda)\, \rho^{\kappa^\flat - |\mathbf{a}| - 1} F_{\mathbf{a}},$$

absolutely and uniformly where $|\lambda\xi_i| \leq \frac{1}{2}|\rho C_i(\rho)|$. Choose $b > 0$ with G and the C_i holomorphic and the C_i zero-free on a neighbourhood of $|\rho| \leq b$, and let $|C_i| \geq c > 0$ on $[0, \infty)$ and on $|\rho| \leq b$. For $|\lambda| \leq \lambda_0 = bc/(2 \max_i |\xi_i|)$ each letter $\lambda\xi_i + \rho C_i(\rho)$ has exactly one root in $|\rho| < b$ (Rouché, against $\rho C_i(\rho)$) and none on $|\rho| = b$ or on $[b, \infty)$.

The analytic part. Split the integral at b. On $[b, \infty)$ the expansion above holds uniformly and its terms are dominated by $\sum_{\mathbf{a}} \prod_i |\binom{-w_i}{a_i}| 2^{-|\mathbf{a}|}$ times $|\rho^{\kappa^\flat - 1} F_{\mathbf{0}}|$, integrable at ∞, so $\int_b^\infty = \sum_{\mathbf{a}} c_{\mathbf{a}}(\lambda) \int_b^\infty \rho^{\kappa^\flat - |\mathbf{a}| - 1} F_{\mathbf{a}}\, d\rho$, holomorphic for $|\lambda| < \lambda_0$. Let $\oint$ start at $\rho = b$ on the principal branches and run once around $|\rho| = b$; the integrand continues along it, and the expansion holds on the circle, each $\rho^{\kappa^\flat - |\mathbf{a}| - 1}$ continued along it and $F_{\mathbf{a}}$ single-valued on the disc. For $\operatorname{Re} c' > 0$ and F holomorphic near the closed disc,

$$\oint_{|\rho| = b} \rho^{c' - 1} F\, d\rho = \big(e^{2\pi i c'} - 1\big) \int_0^b \rho^{c' - 1} F\, d\rho,$$

by Cauchy's theorem on the disc cut along $[0, b]$; both sides are holomorphic in c' off $-\mathbb{N}$, the right side read with the finite part $\operatorname{Reg} \int_0^b$ (1.5, 1.7), so the identity holds at every $c' \notin \mathbb{Z}$. At $c' = \kappa^\flat - |\mathbf{a}|$, $e^{2\pi i c'} = e^{2\pi i \kappa^\flat} \neq 1$. Hence

$$A_b(\lambda) = \frac{1}{e^{2\pi i \kappa^\flat} - 1} \oint_{|\rho| = b} \rho^{\kappa - 1} G \prod_i \big(\lambda\xi_i + \rho C_i\big)^{-w_i} d\rho = \sum_{\mathbf{a}} c_{\mathbf{a}}(\lambda)\, \operatorname{Reg} \int_0^b \rho^{\kappa^\flat - |\mathbf{a}| - 1} F_{\mathbf{a}}\, d\rho$$

is holomorphic for $|\lambda| < \lambda_0$, the circle integral of an integrand holomorphic in λ along a contour that no root reaches, and the series converges there. Since $\operatorname{Reg}\int_0^\infty = \operatorname{Reg}\int_0^b + \int_b^\infty$, $A(\lambda) = A_b(\lambda) + \int_b^\infty \cdots$ is the series of the statement, holomorphic at 0.

The branch part. For $\lambda > 0$ put $\rho = \lambda u$ in $\int_0^b$ and in the circle integral; $(\lambda X)^{-w} = \lambda^{-w} X^{-w}$ on the principal branches at $\lambda > 0$, so both carry the factor $\lambda^{\kappa^\flat}$:

$$\int_0^b \cdots d\rho - A_b(\lambda) = \lambda^{\kappa^\flat} D_\lambda(b/\lambda), \qquad D_\lambda(R) = \int_0^R \psi_\lambda \, du - \frac{1}{e^{2\pi i \kappa^\flat} - 1} \oint_{|u|=R} \psi_\lambda \, du,$$

with $\psi_\lambda(u) = u^{\kappa-1} G(\lambda u) \prod_i \big(\xi_i + u C_i(\lambda u)\big)^{-w_i}$. The roots of the letters in u are $\rho_i(\lambda)/\lambda$, where $\rho_i(\lambda)$ is the root in $|\rho| < b$; from $|\rho_i C_i(\rho_i)| = |\lambda \xi_i|$ they have modulus at most $|\xi_i|/c$. Take $R_0 > 2 \max_i |\xi_i|/c$, so that $|u C_i(\lambda u)| > 2|\xi_i|$ on $|u| = R$ for $R \geq R_0$. For $R_0 \leq R \leq b/|\lambda|$, one turn around $|u| = R$ multiplies ψ_λ by $e^{2\pi i \kappa^\flat}$, the factor $u^{\kappa-1}$ giving $e^{2\pi i \kappa}$ and each letter, with one root inside, $e^{-2\pi i w_i}$; so $\frac{d}{dR} \oint_{|u|=R} \psi_\lambda \, du = \big[\psi_\lambda(R e^{i\vartheta}) e^{i\vartheta}\big]_{\vartheta=0}^{2\pi} = (e^{2\pi i \kappa^\flat} - 1)\psi_\lambda(R)$, and $\frac{d}{dR} D_\lambda(R) = 0$. Thus $B(\lambda) = D_\lambda(R_0)$, with R_0 fixed, and $\int_0^\infty = A(\lambda) + \lambda^{\kappa^\flat} B(\lambda)$ for small $\lambda > 0$. For complex λ near 0, ψ_λ is holomorphic in λ on $[0, R_0]$ and on $|u| = R_0$: there $|\lambda u| \leq |\lambda| R_0 < b$, the letters do not vanish, $\xi_i + u C_i(\lambda u)$ stays off $(-\infty, 0]$ on $[0, R_0]$ for λ small by compactness, and $u^{\kappa-1}$ is integrable at 0 since $\operatorname{Re}\kappa > 0$. So B is holomorphic at 0, and

$$B(0) = G(0)\Big[\int_0^{R_0} u^{\kappa-1} \prod_i \big(\xi_i + u C_i(0)\big)^{-w_i} du - \frac{1}{e^{2\pi i \kappa^\flat} - 1} \oint_{|u|=R_0} u^{\kappa-1} \prod_i \big(\xi_i + u C_i(0)\big)^{-w_i} du \Big],$$

independent of $R_0 \geq 2 \max_i |\xi_i|/c$ by the same derivative. When $\operatorname{Re}\kappa^\flat < 0$ the circle integral is $O(R_0^{\operatorname{Re}\kappa^\flat})$ and the bracket tends to the convergent integral $\int_0^\infty$; the bracket is holomorphic in $(\kappa, \mathbf{w})$ off $\kappa^\flat \in \mathbb{Z}$, so it is the continuation of that integral in κ and $\mathbf{w}$. Where every $\operatorname{Re} w_i > 0$, $C_i(0) > 0$ gives $\prod_i (\xi_i + u C_i(0))^{-w_i} = \prod_i C_i(0)^{-w_i} \prod_i (u + \xi_i/C_i(0))^{-w_i}$, and 15.1 with $J = \{1, \ldots, p\}$ and the coefficients $\xi_i/C_i(0)$ gives the stacked Gamma times the Breta.

(ii) When G and the C_i are holomorphic on a neighbourhood of $[0, \infty)$, the cluster port of 38.21 with $r_i = 1$ and the lower coefficients $\lambda \xi_i$, at the b chosen above, is

$$\frac{1}{2\pi i}\Big[\big(e^{2\pi i \kappa^\flat} - 1\big) \int_b^\infty \cdots d\rho + \oint_{|\rho|=b} \cdots d\rho \Big] = \frac{e^{2\pi i \kappa^\flat} - 1}{2\pi i} \Big(\int_b^\infty \cdots d\rho + A_b(\lambda) \Big) = \frac{e^{2\pi i \kappa^\flat} - 1}{2\pi i} A(\lambda),$$

and it does not depend on b (38.22(i)). □

Remark. With $p = 1$, $\xi_1 = 1$ and $C = C_1$, $\int_0^\infty u^{\kappa-1}(1 + uC(0))^{-w} du = C(0)^{-\kappa} \Gamma \, {}_2_1\!\left[\begin{matrix} \kappa, & -\kappa^\flat \\ & w \end{matrix}\right]$, the branch part of the reference.

Theorem 38.24 (The values along a ray). *As stated, over a subfield $\Bbbk$ of $\mathbb{C}$ containing the weights and the origin.*

Proof. (i) is the reference's (the reference, proof of thm:rayspace). The letters have their coefficients in $\Bbbk$, so X_F, the occupations and the stride are defined over $\Bbbk$, and with the origin in $\Bbbk$ and the direction integral the operators $c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}$ map $\Bbbk[X_F] \otimes \Bbbk(k)$ to itself. Let $\mathcal{L}_\Bbbk$ be the $\Bbbk(k)$-span of their images and $M_\Bbbk = \big(\Bbbk[X_F] \otimes \Bbbk(k)\big)/\mathcal{L}_\Bbbk$. Since $\mathbb{C}[X_F] = \Bbbk[X_F] \otimes_\Bbbk \mathbb{C}$, $M_\Bbbk \otimes_{\Bbbk(k)} \mathbb{C}(k)$ is the quotient of 38.4, a cokernel commuting with the extension of scalars; so $\dim_{\Bbbk(k)} M_\Bbbk$ is the dimension d of the ray, finite by (i).

(ii) The classes of $g, fg, \dots, f^d g$ in $M_{\Bbbk}$ are $d+1$ vectors of a space of dimension d, so there are $p_0, \dots, p_d \in \Bbbk(k)$, not all zero, with $\sum_j p_j(k)\, f^j g \in \mathcal{L}_{\Bbbk}$; clearing denominators puts them in $\Bbbk[k]$. With h the largest j with $p_j \neq 0$, $h \leq d$, and 38.7 gives $\sum_{j \leq h} p_j(k) \langle g \rangle_{k+j} = 0$ for all large k.

(iii) Let k_1 exceed the integer zeros of p_h and the bound of (ii). For $k \geq k_1$, $\langle g \rangle_{k+h} = -\sum_{j<h} \frac{p_j(k)}{p_h(k)} \langle g \rangle_{k+j}$ with $p_j(k)/p_h(k) \in \Bbbk$, the values of polynomials over $\Bbbk$ at integers lying in $\Bbbk$; induction on k writes every $\langle g \rangle_k$, $k \geq k_1$, as a $\Bbbk$-combination of $\langle g \rangle_{k_1}, \dots, \langle g \rangle_{k_1+h-1}$. □

Lemma 38.25 (Index by pole count). *As stated, over a subfield $\Bbbk$ of $\mathbb{C}$.*

Proof. The poles of r are rational, so its partial fractions have coefficients in $\Bbbk$: $r(k) = \sum_{h \geq 1} \sum_{s=1}^{p} c_{h,s} (k+h)^{-s}$, finitely many h, $c_{h,s} \in \Bbbk$, with no polynomial part since $r \to 0$, and $\sum_h c_{h,1} = 0$ since $r(k) = O(k^{-2})$. 37.10 at $t_0 = 0$, whose identities are linear in the coefficients and hold at coefficients in any field, gives

$$\sum_{k \geq 0} r(k) = \sum_{s \geq 2} C_s \, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - \sum_{h,s} c_{h,s} \, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0, h-1]), \qquad C_s = \sum_h c_{h,s} \in \Bbbk,$$

the index-one terms regularized and their coefficient $C_1 = 0$, and every $(\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0, h-1])$, a finite sum of m^{-s}, rational. If $r(-M-k) = -r(k)$, uniqueness of the partial fractions gives $(-1)^s c_{M-h,s} = -c_{h,s}$, so $C_s = (-1)^{s+1} C_s$ and $C_s = 0$ for even s. For poles in one coset $-b + \mathbb{Z}$ the same computation runs in the lattice input $\mathbb{I}_{b+\mathbb{Z}}$, the sums $\sum_{k \geq 0} (k+b+h)^{-s}$ being its Trailmoments at its cuts; with several cosets the index-one sums of the cosets are regularized one by one, and only their total coefficient vanishes, which leaves the differences of the index-one values of the cosets. The coefficients are in $\Bbbk$ throughout. □

48.38 Chapter 39: the two ends of a ray

The proofs of this chapter use three facts about the symmetries of Proposition 39.7: in the coordinates $y = \log \alpha$ they are linear, $y \mapsto \pm \beta^* y$; they keep the stride; and an expansion at a native gauge depends only on the Taylor series of the insertion there, which insertions realize to every finite order, $\mathbb{C}[X_F]$ containing the Laurent polynomials of degree 0.

Definition 39.1 (The module of a ray; masters; the one-step matrix). As stated.

Proof. $\mathcal{L}$ is spanned over $\mathbb{C}(k)$ by $\rho_e(g) = (c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}) g$. On $\mathbb{C}[X_F] \otimes \mathbb{C}(k)$ put $\mathrm{Sh}(g \otimes r) = fg \otimes \varsigma r$, $(\varsigma r)(k) = r(k+1)$; then $\mathrm{Sh}(r'x) = \varsigma(r')\, \mathrm{Sh}(x)$. By 38.2(ii), $\partial_{y_e}(fg) = c_e^{(1)} fg + f\, \partial_{y_e} g$, so

$$f \left(c_e^{(0)} + (k+1)\, c_e^{(1)} + \partial_{y_e}\right) g = \left(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}\right)(fg),$$

and $\mathrm{Sh}(r(k) \rho_e(g)) = r(k+1)\, \rho_e(fg) \in \mathcal{L}$. The stride is a unit of $\mathbb{C}[X_F]$, its exponents being integers and the letters units there, so Sh has the inverse $x \mapsto f^{-1} \varsigma^{-1} x$, which maps $\mathcal{L}$ into itself by the same identity read backwards: Sh descends to a bijection of $\mathfrak{M}$ with $\mathrm{Sh}(rm) = \varsigma(r)\, \mathrm{Sh}(m)$. A semilinear bijection maps bases of subspaces to bases of their images, so the classes $\mathrm{Sh}^{-1}[b_i] = [f^{-1} b_i]$ of masters of a submodule form a basis of it, and $\mathbf{B_b}$ is invertible.

Let the cyclic submodule of $[1]$ be finite-dimensional, and let h' be the least index with $\mathrm{Sh}^{h'}[1]$ in the span W of $[1], \dots, \mathrm{Sh}^{h'-1}[1]$. Then $\mathrm{Sh}(W) \subseteq W$, and $\mathrm{Sh}(W)$ has the dimension of W, so $\mathrm{Sh}(W) = W$ and $\mathrm{Sh}^{-1}(W) = W$: W is the cyclic submodule of $[1]$, and $h' = h$. Writing $\mathrm{Sh}^h[1]$ in the basis and clearing denominators gives a relation with $p_h \neq 0$. If $p_0 = 0$, applying Sh^{-1} to $\sum_{i=1}^{h} p_i(k)\, \mathrm{Sh}^i[1] = 0$ gives $\sum_{i=1}^{h} p_i(k-1)\, \mathrm{Sh}^{i-1}[1] = 0$, a relation among $[1], \dots, \mathrm{Sh}^{h-1}[1]$, which

is impossible. Two relations of length h are proportional over $\mathbb{C}(k)$, since otherwise a combination of them eliminating $\mathrm{Sh}^h[1]$ would be a nonzero relation among the basis; with coefficients in $\mathbb{C}[k]$ without a common factor the proportionality factor is a constant. □

Definition 39.2 (Solutions). As stated.

Proof. On germs ς is bijective and $\varsigma(rx) = r(k+1)\varsigma(x)$. On $\lambda^k k^\rho\,\mathbb{C}((1/k))$, a module of rank one over $\mathbb{C}((1/k))$ with the generator $\lambda^k k^\rho$, ς is the composite of the substitution $k \mapsto k+1$, the automorphism ς_0 of the field $\mathbb{C}((1/k))$ with $\frac{1}{k} \mapsto \frac{1}{k}\big(1+\frac{1}{k}\big)^{-1}$, with the multiplication by the unit $\lambda(1+\frac{1}{k})^\rho = \lambda\sum_i \binom{\rho}{i} k^{-i}$; so it is bijective and $\varsigma(rx) = r(k+1)\varsigma(x)$ for $r \in \mathbb{C}(k) \subset \mathbb{C}((1/k))$. □

Lemma 39.3 (Solutions of distinct types). *As stated.*

Proof. With ς_0 the substitution $k \mapsto k+1$ on $\mathbb{C}((1/k))$, $\hat{A}_a(\mathrm{Sh}\,m) = \lambda_a(1+\frac{1}{k})^{\rho_a}\varsigma_0\big(\hat{A}_a(m)\big)$. Suppose $\sum_{a\in S} c_a\hat{A}_a = 0$ on $\mathfrak{M}$ with $c_a \in \mathbb{C}((1/k))$, every $c_a \neq 0$, and S nonempty and minimal. Applying the relation at $\mathrm{Sh}\,m$, and applying ς_0 to it at m, gives

$$\sum_{a\in S} c_a\lambda_a\big(1+\tfrac{1}{k}\big)^{\rho_a}\,\varsigma_0\big(\hat{A}_a(m)\big) = 0, \qquad \sum_{a\in S} \varsigma_0(c_a)\,\varsigma_0\big(\hat{A}_a(m)\big) = 0$$

for every m. The coefficient vectors of relations among the maps $\varsigma_0 \circ \hat{A}_a$, $a \in S$, form a space over $\mathbb{C}((1/k))$ of dimension one: two independent vectors in it would combine to a nonzero one with a zero coordinate, and its image under ς_0^{-1} would be a relation among fewer $\hat{A}_a$. So $c_a\lambda_a(1+\frac{1}{k})^{\rho_a} = \mu\,\varsigma_0(c_a)$ for one μ and every $a \in S$. With $c_a = c_a^0 k^{v_a}\big(1 + c_a' k^{-1} + O(k^{-2})\big)$, $c_a^0 \neq 0$, $v_a \in \mathbb{Z}$, the two sides are $c_a^0 k^{v_a}\lambda_a\big(1 + (\rho_a + c_a')k^{-1} + \cdots\big)$ and $\mu\, c_a^0 k^{v_a}\big(1 + (v_a + c_a')k^{-1} + \cdots\big)$, so $\mu = \lambda_a\big(1 + (\rho_a - v_a)k^{-1} + O(k^{-2})\big)$ for every $a \in S$: one λ, and $\rho_a - v_a$ independent of a, so all $a \in S$ have one type. Fix ρ of that type and write every A_a, $a \in S$, in $\lambda^k k^\rho\,\mathbb{C}((1/k))$, which changes the c_a by integral powers of k; then $c_a\lambda(1+\frac{1}{k})^\rho = \mu\,\varsigma_0(c_a)$ makes $c_a/\varsigma_0(c_a)$ independent of a, so every quotient c_a/c_b is fixed by ς_0. A series $r = \sum_{i\geq i_0} r_i k^{-i}$ with $\varsigma_0 r = r$ is constant: if $r_i \neq 0$ for some $i \neq 0$, the least such i gives in $\varsigma_0 r - r$ the term $-i\,r_i k^{-i-1}$ and no other of that order. So the relation is $c(k)\sum_{a\in S}\gamma_a\hat{A}_a = 0$ with constants $\gamma_a \neq 0$, and $\sum_a \gamma_a A_a = 0$ is a relation over $\mathbb{C}$ among solutions of one type: excluded. Extended $\mathbb{C}((1/k))$-linearly to $\mathfrak{N}\otimes\mathbb{C}((1/k))$, the normalizations are r independent linear forms on a space of dimension $\dim_{\mathbb{C}(k)}\mathfrak{N}$; the argument runs on $\mathfrak{N}$ in place of $\mathfrak{M}$, Sh mapping $\mathfrak{N}$ onto itself. □

Theorem 39.4 (What solutions determine). *As stated.*

Proof. (i) $V[f^{-1}b_i] = V(\mathrm{Sh}^{-1}[b_i]) = \varsigma^{-1}V[b_i]$, and $V[f^{-1}b_i] = \sum_l \mathbf{B}_{il}V[b_l]$ by $\mathbb{C}(k)$-linearity. For the values, $\langle f^{-1}b_i\rangle_k = \langle b_i\rangle_{k-1}$ (38.3).

(ii) $\mathrm{Ann}(\mathcal{S})$ is a $\mathbb{C}(k)$-subspace, and $V(\mathrm{Sh}^{\pm1}\,m) = \varsigma^{\pm1}V(m)$ vanishes with $V(m)$.

(iii) By (i), $\mathbf{B}'v_V - \varsigma^{-1}v_V = (\mathbf{B}'-\mathbf{B})v_V$, whose i-th entry is $V(m_i)$, $m_i = \sum_l(\mathbf{B}'-\mathbf{B})_{il}[b_l] \in \mathfrak{N}$. If a nonzero $m = \sum_l w_l[b_l]$ lies in $\mathrm{Ann}(\mathcal{S}) \cap \mathfrak{N}$, adding w to the first row of $\mathbf{B}$ gives another such matrix.

(iv) For a germ the relation of (i) holds at every integer beyond some bound, and so do the equations of a pair (cP_0, cQ_0); for a formal series they hold as identities. Conversely, let (P, Q) satisfy all the equations. By (i), $Q\,\varsigma^{-1}v_V - Pv_V = (Q\mathbf{B} - P)v_V$, so every row of $Q\mathbf{B} - P$, read in $\mathfrak{N}$, lies in $\mathrm{Ann}(\mathcal{S}) \cap \mathfrak{N} = 0$, and $P = Q\mathbf{B} = QP_0/Q_0$. If $Q = 0$ then $P = 0$. Otherwise Q_0 divides every entry of QP_0; with $d = Q_0/\gcd(Q_0, Q)$, d divides every entry of P_0 and Q_0, whose greatest common divisor is 1, Q_0 being the least common denominator, so $d = 1$, $Q = cQ_0$ and $P = cP_0$, and the degree bounds give $\deg c \leq D - D_0$. The equations are linear and homogeneous in the

coefficients; for each D the solution spaces of the first N of them, in any enumeration, decrease with N in a space of finite dimension, so they are constant from some N on. □

Corollary 39.5 (One solution of an irreducible module). *As stated.*

Proof. $\mathrm{Ann}(\{V\}) \cap \mathfrak{N}$ is a submodule of $\mathfrak{N}$ (Theorem 39.4(ii)) other than $\mathfrak{N}$, hence 0. For the cyclic submodule of $[1]$, $\sum_i p_i(k)\varsigma^i V[1] = V\big(\sum_i p_i(k)\,\mathrm{Sh}^i[1]\big) = 0$. A relation $\sum_{i<h} q_i(k)\varsigma^i V[1] = 0$ is $V(m) = 0$ with $m = \sum_{i<h} q_i(k)\,\mathrm{Sh}^i[1]$, nonzero as the $\mathrm{Sh}^i[1]$, $i < h$, are a basis; then $\mathrm{Ann}(\{V\})$ meets the cyclic submodule in a nonzero submodule, all of it, and $V[1] = 0$. □

Proposition 39.6 (The law of $[1]$ read from a one-step matrix). *As stated.*

Proof. (i) Sh^{-1} is a bijection of $\mathfrak{M}$ with $\mathrm{Sh}^{-1}(r(k)\,m) = r(k-1)\,\mathrm{Sh}^{-1}(m)$ (Definition 39.1). For $i = 0$, $[1] = [b_1]$. If $\mathrm{Sh}^{-(i-1)}[1] = \sum_l w_{i-1,l}(k)[b_l]$, then

$$\mathrm{Sh}^{-i}[1] = \sum_l w_{i-1,l}(k-1)\,[f^{-1}b_l] = \sum_{l,l'} w_{i-1,l}(k-1)\,\mathbf{B}_{ll'}(k)\,[b_{l'}],$$

and induction gives the product form. A solution has $V \circ \mathrm{Sh}^{-i} = \varsigma^{-i} \circ V$ and is $\mathbb{C}(k)$-linear. For the values, $f^{-i} - \sum_l w_{i,l}(k)\,b_l$ lies in $\mathcal{L}$, so its value vanishes identically, and $\langle f^{-i}\rangle_k = \langle 1\rangle_{k-i}$ (Proposition 38.5(ii)).

(ii) The w_i lie in $\mathbb{C}(k)^{\mathfrak{m}}$, so $h \le \mathfrak{m}$. By (i), $\sum_i q_i(k)\,\mathrm{Sh}^{-i}[1] = 0$, and Sh^h, with $\mathrm{Sh}^h(r(k)\,m) = r(k+h)\,\mathrm{Sh}^h(m)$, turns it into $\sum_i q_i(k+h)\,\mathrm{Sh}^{h-i}[1] = 0$, which is the stated relation, $\mathrm{Sh}^j[1] = [f^j]$; the shift of the variable keeps the coefficients without a common factor. $p_0(k) = q_h(k+h) \neq 0$, since $w_0, \dots, w_{h-1}$ are independent. If $\mathbf{B}$ is invertible and $q_0 = 0$, then $\sum_{i\ge 1} q_i(k)\,w_{i-1}(k-1)\,\mathbf{B}(k) = 0$ gives $\sum_{i\ge 1} q_i(k)\,w_{i-1}(k-1) = 0$, and at $k+1$ a dependence among $w_0, \dots, w_{h-1}$ with coefficients not all zero: so $q_0 \neq 0$ and $p_h \neq 0$. Let h' be the dimension of the cyclic submodule of $[1]$; its basis $[1], \mathrm{Sh}[1], \dots, \mathrm{Sh}^{h'-1}[1]$ (Definition 39.1) is carried by the semilinear bijection $\mathrm{Sh}^{-(h'-1)}$ to the basis $\mathrm{Sh}^{-i}[1]$, $0 \le i < h'$. A dependence among $w_0, \dots, w_h$ gives by (i) a dependence among $[1], \dots, \mathrm{Sh}^{-h}[1]$ with the same coefficients, so $h \ge h'$. When the $[b_l]$ are a basis, a dependence among the $\mathrm{Sh}^{-i}[1]$ is one among their coordinate rows w_i, so $h = h'$, and the relation, of length h' with polynomial coefficients without a common factor, is the law of $[1]$ up to a constant factor (Definition 39.1).

(iii) $M = [w_0; \dots; w_h]_J$ is an $(h+1) \times h$ matrix of rank h, so its left kernel is a line. It contains $(q_i)_i$, and also the vector of the signed maximal minors $\mu_i = (-1)^i \det M_{\hat{i}}$, $M_{\hat{i}}$ being M without its row i: for every column l, $\sum_i (-1)^i M_{il} \det M_{\hat{i}}$ is the expansion along its first column of the square matrix $[M_{\cdot l}\,|\,M]$, which has two equal columns; and $\mu_h = (-1)^h \det \mathbf{W}_J \neq 0$. For the divisibility, put $D_i(k) = \prod_{j=0}^{i-1} Q(k-j)$, so that $D_i w_i$ is a row of polynomials, w_i being w_0 times $\mathbf{B}(k-i+1)\cdots\mathbf{B}(k)$. The polynomial matrix $\tilde{M} = \mathrm{diag}(D_0, \dots, D_h)\,M$ has the left kernel spanned by $(q_i/D_i)_i$, and also by its signed maximal minors $\tilde{\mu}_i$, polynomials with $\tilde{\mu}_h = (-1)^h N$, $N = \det \mathbf{W}_J \prod_{i<h} D_i$. Let $\tilde{q}$ be a polynomial vector without a common factor spanning that kernel; then $\tilde{\mu} = \Lambda\tilde{q}$ with Λ a polynomial, since a denominator of Λ would divide every $\tilde{q}_i$. With $D = D_h$, the polynomial vector $(q_i D/D_i)_i$ spans the kernel, so $\tilde{q} = (q_i D/D_i)_i/g$ with g the greatest common divisor of its entries, and $\tilde{q}_h = q_h/g$ divides N. Since g divides every $q_i D$ and the q_i have no common factor, g divides D. So $q_h = g\,\tilde{q}_h$ divides $DN = \det \mathbf{W}_J \prod_{i=0}^{h} D_i = \det \mathbf{W}_J \prod_{j=0}^{h-1} Q(k-j)^{h-j}$. At k_0 with $Q(k_0 - j) \neq 0$ for $j < h$, every $D_i(k_0) \neq 0$ and the rows w_i, $i < h$, are regular, so the order of q_h at k_0 is at most that of $\det \mathbf{W}_J$, for every J with $\det \mathbf{W}_J \neq 0$: if $q_h(k_0) = 0$, every maximal minor of $\mathbf{W}(k_0)$ vanishes.

(iv) Take J all the columns. Since $w_i(k) = w_{i-1}(k-1)\mathbf{B}(k)$, $[w_1;\dots;w_h](k) = \mathbf{W}(k-1)\,\mathbf{B}(k)$, so $\mu_0 = \det\mathbf{W}(k-1)\det\mathbf{B}(k)$ and $\mu_h = (-1)^h \det\mathbf{W}(k)$, and $q_0\mu_h = q_h\mu_0$ by (iii).

(v) $\mathrm{Sh}^{-1}\big(\sum_l c_l(k)[b_l]\big) = \sum_{l,l'} c_l(k-1)\,\mathbf{B}_{ll'}(k)\,[b_{l'}]$, so the rows obey the same recursion, and the proofs of (i) to (iv) run with m in place of [1]. □

Proposition 39.7 (The symmetries of a ray). *As stated.*

Proof. $\phi_\beta \circ \phi_{\beta'} = \phi_{\beta'\circ\beta}$, the Cremona commutes with every ϕ_β and its square is the identity, so the maps ϕ_β and $\bar\phi_\beta$ form a group, and the two conditions defining a symmetry pass to composites and inverses.

(i) If β carries the tables, $K_j(\alpha\circ\beta) = \sum_{T\in\mathcal{B}_j}\theta_{j,T}\alpha^{\beta(T)} = \sum_{T'\in\mathcal{B}_{h(j)}}\theta_{h(j),T'}\alpha^{T'} = K_{h(j)}(\alpha)$, and the same with the complements gives Ψ. If β exchanges them, the sets $\beta(T)$, $T\in\mathcal{B}_j$, are the complements of the bases $\mathcal{E}\setminus\beta(T)$ of table $h(j)$, with the same weights, so $K_j(\alpha\circ\beta) = \Psi_{h(j)}(\alpha)$ and $\Psi_j(\alpha\circ\beta) = K_{h(j)}(\alpha)$. By 7.2, $K_j(\alpha^{-1}) = \alpha^{-\mathcal{E}}\Psi_j(\alpha)$ and $\Psi_j(\alpha^{-1}) = \alpha^{-\mathcal{E}}K_j(\alpha)$; with $(\alpha\circ\beta)^{\mathcal{E}} = \alpha^{\mathcal{E}}$, $K_j\big((\alpha\circ\beta)^{-1}\big) = \alpha^{-\mathcal{E}}\Psi_j(\alpha\circ\beta) = \alpha^{-\mathcal{E}}K_{h(j)}(\alpha)$, and the same for Ψ_j.

(ii) With $(\alpha\circ\beta)^u = \alpha^{u\circ\beta^{-1}}$ and $(\alpha^{-1})^u = \alpha^{-u}$, (i) gives in the four cases

$$f\circ\phi_\beta = \alpha^{\mathbf{m}_1\circ\beta^{-1}}\prod_j K_{h(j)}^{-s_{1j}}\Psi_{h(j)}^{-t_{1j}}, \qquad f\circ\phi_\beta = \alpha^{\mathbf{m}_1\circ\beta^{-1}}\prod_j \Psi_{h(j)}^{-s_{1j}}K_{h(j)}^{-t_{1j}},$$

$$f\circ\mathrm{cr} = \alpha^{-\mathbf{m}_1+2\sigma_1\mathbf{1}}\prod_j \Psi_j^{-s_{1j}}K_j^{-t_{1j}}, \qquad f\circ\bar\phi_\beta = \alpha^{-\mathbf{m}_1\circ\beta^{-1}+2\sigma_1\mathbf{1}}\prod_j K_{h(j)}^{-s_{1j}}\Psi_{h(j)}^{-t_{1j}},$$

each equal to f under the conditions with $i=1$, after reindexing the tables by h; the same computation with the data of the origin gives $(\omega_0/dy)\circ\phi = \omega_0/dy$ under the conditions with $i=0$, on X_F for an integral origin and on the positive orthant with the principal powers otherwise. In the coordinates y the maps are linear, $y\mapsto\pm\beta^*y$, and they keep the form dy on $\mathbb{C}^{\mathcal{E}}/\mathbb{C}\mathbf{1}$ up to the sign of the permutation and the factor $(-1)^{n-1}$ of the inversion; so $\phi^*\omega_0 = \pm\omega_0$ for an integral origin. The letters go to letters times monomials, so X_F is mapped onto itself, and positive gauges go to positive gauges. The value $\langle g'\circ\phi\rangle_k$ is the integral of $(g'\circ\phi)f^k(\omega_0/dy)$ against the Lebesgue measure of the quotient, which does not depend on the chart (17.1); the linear change of variables $y\mapsto\pm\beta^*y$ keeps that measure, and with $f\circ\phi = f$ and $(\omega_0/dy)\circ\phi = \omega_0/dy$ it gives $\langle g'\rangle_k$.

(iii) Substitute (i) in $\alpha^u\prod_j K_j^{-a_j}\Psi_j^{-b_j}$ and reindex the tables by h. □

Theorem 39.8 (The invariant module). *As stated.*

Proof. (i) In the coordinates y, ϕ is $y\mapsto M_\phi y$ with M_ϕ linear. For a constant vector field ξ and a function u, $(\xi u)\circ\phi = (M_\phi^{-1}\xi)(u\circ\phi)$. The relation space is spanned by $\rho_\xi(g') = \xi\big(\log(f^k\omega_0/dy)\big)g' + \xi g' = \sum_e \xi_e\big(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e}\big)g'$, and since $\log(f^k\omega_0/dy)\circ\phi$ differs from $\log(f^k\omega_0/dy)$ by a constant (Proposition 39.7), $\rho_\xi(g')\circ\phi = \rho_{M_\phi^{-1}\xi}(g'\circ\phi)\in\mathcal{L}$. So ϕ^* is well defined on $\mathfrak{M}$, and with $(fg')\circ\phi = f\,(g'\circ\phi)$ it commutes with Sh. Then e_G is an idempotent commuting with Sh, its image and kernel are submodules with $\mathfrak{M} = e_G\mathfrak{M}\oplus(1-e_G)\mathfrak{M}$, $e_G[g'] = [e_Gg']$ with e_Gg' invariant, and the class of an invariant insertion is fixed by e_G.

(ii) $V = V\circ e_G$ vanishes on the kernel of e_G. The values are invariant by Proposition 39.7(ii). 1 is invariant, so $[1]\in\mathfrak{M}^G$, a submodule, which then contains the cyclic submodule of [1].

(iii) ϕ fixes α_c and keeps f, so it maps a steepest-descent disc D_c through α_c to a disc through α_c on which $|f| < |f(\alpha_c)|$ off α_c, with $|f|\le|f(\alpha_c)|e^{-\varepsilon}$ on its boundary. In Morse coordinates v at α_c, $\log f = \log f(\alpha_c) - \frac12\sum_i v_i^2$, the homology of a small ball relative to the part where $|f|\le|f(\alpha_c)|e^{-\varepsilon}$ is $\mathbb{Z}$ in the dimension $n-1$, generated by the real ball; so $\phi(D_c)$ is $\pm D_c$ up to a

boundary and pieces where $|f| \le |f(\alpha_c)|e^{-\varepsilon}$. Fix a branch of ω_0 near α_c; $(\omega_0/dy)\circ\phi$ and ω_0/dy are branches of one function there and $\phi^*dy = \pm dy$, so $\phi^*\omega_0 = c\,\omega_0$ near α_c for a constant $c \neq 0$. Then $\int_{D_c}(g\circ\phi)f^k\omega_0 = c^{-1}\int_{\phi(D_c)} gf^k\omega_0$, and since $gf^k\omega_0$ is a closed form of top degree, $A_c\circ\phi^* = c'A_c$ for a constant c'. The leading coefficients of $A_c(g\circ\phi)$ and of $A_c(g)$ are $(2\pi)^{(n-1)/2}(\omega_0/dv)(\alpha_c)$ times $(g\circ\phi)(\alpha_c) = g(\alpha_c)$ (38.6(i)), and an insertion with $g(\alpha_c) \neq 0$ gives $c' = 1$.

For $\phi \in G$ and $\psi \in G_c$, $A_c\circ(\phi\psi)^* = A_c\circ\psi^*\circ\phi^* = A_c\circ\phi^*$, so A_O is well defined; for $\chi \in G$, $A_O\circ\chi^* = \sum A_c\circ(\chi\phi)^*$ runs over the same cosets, so A_O is invariant, and $A_O\circ e_G = A_O$. By the argument above, $A_c\circ\phi^*$ is a constant multiple of the expansion over the disc $\phi(D_c)$, a steepest-descent disc through the native gauge $\phi(\alpha_c)$. The nondegenerate native gauges are isolated points of the critical locus, a variety, so finitely many; take an insertion g with $g(\alpha_c) = 1$ that vanishes to order N at every other one. Its expansion there vanishes to the relative order $k^{-\lceil N/2\rceil}$, since $a_i(g)$ involves derivatives of g of order at most $2i$, so $A_O(e_Gg) = A_O(g)$ has the nonzero leading coefficient of $A_c(g)$, and A_O does not vanish on $\mathfrak{M}^G$. The same insertions, applied to a relation $\sum_O \gamma_O A_O = 0$ on $\mathfrak{M}^G$ among orbits of one native value, give $\gamma_O = 0$ orbit by orbit. Lemma 39.3 gives the bound. □

Theorem 39.9 (A reversed exchange at a lean). *As stated.*

Proof. (i) $T \mapsto \mathcal{E}\setminus\beta(T)$ is a bijection of $\mathcal{B}_j$ onto $\mathcal{B}_{h(j)}$ keeping the weights, so $K_j(\mathbf{1}) = K_{h(j)}(\mathbf{1})$ and, under the weight laws, $R_{h(j),\beta(e)}(\mathbf{1}) = \mathbb{P}_{j,\mathbf{1}}\big(\beta(e)\notin\beta(T)\big) = \mathbb{P}_{j,\mathbf{1}}(e\notin T) = 1 - R_{j,e}(\mathbf{1})$. By 9.3, $\tau_e = \sum_j\big(\sigma_j + \delta_j(2R_{j,e}(\mathbf{1}) - 1)\big)$; reindexing by h, with $(\sigma_{h(j)},\delta_{h(j)}) = (\sigma_j,\delta_j)$ (11.13), $\tau_{\beta(e)} = \sum_j\big(\sigma_j + \delta_j(1 - 2R_{j,e}(\mathbf{1}))\big) = 2\sigma - \tau_e$. The exponents (s_j,t_j) are carried by h, so Proposition 39.7(ii) applies to $\bar\phi_\beta$ with the masses $\boldsymbol{\tau}$, and to the ray with origin 0 and direction $(c\boldsymbol{\tau}; c\mathbf{s}, c\mathbf{t})$. For ϕ_β and cr, Proposition 39.7(i) turns the integrand into $\alpha^{\boldsymbol{\tau}\circ\beta^{-1}}\prod_j \Psi_{h(j)}^{-s_j}K_{h(j)}^{-t_j}$ and $\alpha^{-\boldsymbol{\tau}+2\sigma\mathbf{1}}\prod_j \Psi_j^{-s_j}K_j^{-t_j}$. Since $\tau_{\beta^{-1}(e)} = 2\sigma - \tau_e = \sum_j\big(\sigma_j - \delta_j(2R_{j,e}(\mathbf{1}) - 1)\big)$, the masses of $\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}$, whose exponents are (t_j,s_j), both are its integrand.

(ii) This is the exchange clause of 11.14 at the table's gauge, read on the views. In the coordinates y, Proposition 39.7(i) gives $(K_j/\Psi_j)\circ\bar\phi_\beta = K_{h(j)}/\Psi_{h(j)}$ and $(K_j\Psi_j)\circ\bar\phi_\beta = \alpha^{-2\mathcal{E}}K_{h(j)}\Psi_{h(j)}$, so, with $(\sigma_{h(j)},\delta_{h(j)}) = (\sigma_j,\delta_j)$,

$$\Phi_{\mathbf{O}}(-\beta^*y) = \Phi_{\mathbf{O}}(y), \qquad \Phi_{\mathbf{E}}(-\beta^*y) = \Phi_{\mathbf{E}}(y) - 2\sigma\langle\mathbf{1},y\rangle.$$

At the table's gauge $\Psi_j(e^y) = e^{\langle\mathbf{1},y\rangle}K_j(e^{-y})$ (7.2), so $\Phi_{\mathbf{O}}(-y) = -\Phi_{\mathbf{O}}(y)$ and $\Phi_{\mathbf{E}}(-y) = \Phi_{\mathbf{E}}(y) - 2\sigma\langle\mathbf{1},y\rangle$. For $w = v_0\mathbf{1}_e + \sum_i v_i\mathbf{1}_{g_i}$ and $w' = v_0\mathbf{1}_{\beta(e)} + \sum_i v_i\mathbf{1}_{\beta(g_i)}$, $\beta^*w' = w$ and $\langle\mathbf{1},w'\rangle = \langle\mathbf{1},w\rangle$, so

$$\Phi_{\mathbf{O}}(w') = \Phi_{\mathbf{O}}(-w) = -\Phi_{\mathbf{O}}(w), \qquad \Phi_{\mathbf{E}}(w') = \Phi_{\mathbf{E}}(-w) + 2\sigma\langle\mathbf{1},w\rangle = \Phi_{\mathbf{E}}(w),$$

the identities of the views at the table's gauge. $\Phi_{\mathbf{O}}$ is linear in $\boldsymbol{\delta}$, so $-\Phi_{\mathbf{O}}^{\boldsymbol{\delta}} = \Phi_{\mathbf{O}}^{-\boldsymbol{\delta}}$, and the views, hence the readings, of $\beta(e)$ at $\boldsymbol{\delta}$ are those of e at $-\boldsymbol{\delta}$, the partners matched by β. At the centre $\Phi_{\mathbf{O}} = 0$ and e and $\beta(e)$ read alike. At a lean the partition at $-\boldsymbol{\delta}$ is that at $\boldsymbol{\delta}$ (11.2): if e and e' lie in one class at $\boldsymbol{\delta}$, they lie in one class at $-\boldsymbol{\delta}$, so $\beta(e)$ and $\beta(e')$ read alike at $\boldsymbol{\delta}$. So β maps classes into classes, and so does β^{-1}, which exchanges the family as well: β permutes the classes.

(iii) By 11.15, β maps every class of the centre onto itself. Let C split at the lean into C_K and C_Ψ, with Q the polynomial written as K from every mark of C_K and as Ψ from every mark of C_Ψ (11.17). With one table, h is the identity and Proposition 39.7(i) gives $K(\alpha\circ\beta) = \Psi(\alpha)$ and $\Psi(\alpha\circ\beta) = K(\alpha)$. For $e \in C_K$ and an order $\mathbf{g}$ of its room with $K_{e,\mathbf{g}} = Q$, put the position variables x_0 at $\beta(e)$ and x_i at $\beta(g_i)$; then $\alpha\circ\beta$ carries x_0 at e and x_i at g_i, so $\Psi_{\beta(e),\beta(\mathbf{g})} = K_{e,\mathbf{g}} = Q$ and $\beta(e) \in C_\Psi$. For $e \in C_\Psi$ with $\Psi_{e,\mathbf{g}} = Q$, $K(\alpha) = \Psi(\alpha\circ\beta)$ gives $K_{\beta(e),\beta(\mathbf{g})} = \Psi_{e,\mathbf{g}} = Q$ and $\beta(e) \in C_K$. C_K and C_Ψ have equally many marks, so $\beta(C_K) = C_\Psi$ and $\beta(C_\Psi) = C_K$. □

Definition 39.11 (Levels, wall degree, boxes). As stated.

Proof. The products of the monomials α^v, $v_{\mathcal{E}} = 0$, and of the chances $\theta_{j,T}\alpha^T/K_j$ and $\theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j$ that generate $\mathbb{C}[X_F]$ (Definition 38.1) are constant multiples of evaluations of symbols, which therefore span it. The wall degree is the largest of finitely many numbers $-\kappa_U(g)/\kappa_{1,U}$, and $\kappa_U(g) + x\,\kappa_{1,U} \geq 0$ for every U exactly when x is at least each of them. By 38.2(i) and Definition 38.1, with $R_{j,e} = \sum_{T\ni e}\theta_{j,T}\alpha^T/K_j$ and $R^*_{j,e} = \sum_{T\not\ni e}\theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j$, the operator $c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e}$ multiplies $\alpha^u \prod_j K_j^{-a_j}\Psi_j^{-b_j}$ by

$$m_{0,e} + k\,m_{1,e} + u_e - \sum_j \big((s_{0j} + k\,s_{1j} + a_j)R_{j,e} + (t_{0j} + k\,t_{1j} + b_j)R^*_{j,e}\big),$$

and $R_{j,e}$ times the evaluation of $(u;\mathbf{a},\mathbf{b})$ is the evaluation of $\sum_{T\ni e}\theta_{j,T}(u + \mathbf{1}_T;\mathbf{a}+\mathbf{e}_j,\mathbf{b})$, and likewise for $R^*_{j,e}$; this is the evaluation of the mark relation. The letter identities evaluate to the evaluation of g times $1 - K_j/K_j$, or $1 - \Psi_j/\Psi_j$. The maps of the four kinds carry each letter to a letter times 1 or $\alpha^{-\mathcal{E}}$ (Proposition 39.7(i)), so do their composites, and the action of Proposition 39.7(iii) is the substitution into the evaluation. □

Proposition 39.12 (The wall degree). *As stated.*

Proof. (i) By Lemma S19, on the sector of an order π, $|g|\,f^x = \prod_l q_l^{\kappa_{U_l}(g) + x\kappa_{1,U_l}}\,|g_\pi|\varphi_\pi^x$ with the last factor between positive constants, so $|g|f^x$ is bounded on the sector exactly when every exponent is nonnegative; every nonempty proper U is a tail of some order, which gives the first statement. Likewise $|g\,f^k\omega_0/dy|$ is, on the sector of π, $\prod_l q_l^{c_l}$ times a factor between positive constants, $c_l = \operatorname{Re}\kappa_{0,U_l} + \kappa_{U_l}(g) + \operatorname{Re}(k)\kappa_{1,U_l}$, and $\int_{(0,1]^{n-1}}\prod_l q_l^{c_l-1}\,dq$ is finite exactly when every $c_l > 0$; so the value converges absolutely exactly when $\operatorname{Re}\kappa_{0,U} + \kappa_U(g) + \operatorname{Re}(k)\kappa_{1,U} > 0$ for every U; for the origin 0 this reads $\operatorname{Re}k > -\kappa_U(g)/\kappa_{1,U}$ for every U, and in general $\kappa_U(g) \geq -\deg_\kappa g\,\kappa_{1,U}$ gives the sufficient condition.

(ii) $\kappa_U(\alpha^u) = u_U$. For $u \neq 0$ with $u_{\mathcal{E}} = 0$ some $u_e < 0$, and $U = \{e\}$ is proper, $n \geq 2$, so the wall degree is positive; for $u = 0$ it is 0. A monomial of $f^{-1} = \alpha^{-\mathbf{m}_1}\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}$ is α^w with $w = -\mathbf{m}_1 + \sum_j\big(\sum_{i\leq s_{1j}}\mathbf{1}_{T_{j,i}} + \sum_{i\leq t_{1j}}\mathbf{1}_{\mathcal{E}\setminus T'_{j,i}}\big)$ for bases $T_{j,i}, T'_{j,i}$ of table j, and by 7.13 $w_U \geq -m_{1,U} + \sum_j\big(s_{1j}d_j(U) + t_{1j}d_j^*(U)\big) = -\kappa_{1,U}$. The walls of Laurent symbols add, so $\kappa_U(\alpha^{u+w}) + (\deg_\kappa\alpha^u + 1)\kappa_{1,U} \geq 0$.

(iii) The terms are g and the symbols $(u + \mathbf{1}_T;\mathbf{a}+\mathbf{e}_j,\mathbf{b})$ and $(u + \mathbf{1}_{\mathcal{E}\setminus T};\mathbf{a},\mathbf{b}+\mathbf{e}_j)$, whose walls are $\kappa_U(g) + |T\cap U| - d_j(U)$ and $\kappa_U(g) + |U\setminus T| - d_j^*(U)$, at least $\kappa_U(g)$ (7.13).

(iv) For a mark e, $\{e\}$ is proper, and $\kappa_{\{e\}}(g) \geq -x\,\kappa_{1,\{e\}}$ bounds u_e from below at a fixed level; with $u_{\mathcal{E}}$ fixed by the level, u is bounded from above as well.

(v) A symmetry maps the positive orthant onto itself and keeps f (Proposition 39.7(ii)), and the evaluation of ϕ^*g is that of g composed with ϕ, so $|\phi^*g|f^x$ and $|g|f^x$ have the same bound, and (i) gives the same wall degree. The action of Proposition 39.7(iii) maps the level $(\mathbf{0},\mathbf{0})$ to itself. □

Lemma 39.13 (The Laurent part). *As stated.*

Proof. $\mathfrak{M}$ is finite-dimensional (Theorem 38.24(i)). The direction has $s_{1j}, t_{1j} \in \mathbb{N}$, so f^{-1} is a Laurent polynomial and $\operatorname{Sh}^{-1}[\alpha^v] = [f^{-1}\alpha^v]$ lies in $\mathfrak{M}_{\mathrm{L}}$: Sh^{-1} maps $\mathfrak{M}_{\mathrm{L}}$ into itself, injectively and semilinearly, hence onto itself, and $\mathfrak{M}_{\mathrm{L}}$ is a submodule. It contains [1], and $[g] = \operatorname{Sh}^l[f^{-l}g]$. When $s_{1j}, t_{1j} \geq 1$ for every j, $f^{-l}\alpha^u\prod_j K_j^{-a_j}\Psi_j^{-b_j}$ is a Laurent polynomial as soon as l is at least every a_j

and every b_j. The symmetries map Laurent insertions to Laurent insertions (Proposition 39.12(v)), so $e_G\mathfrak{M}_{\rm L} \subseteq \mathfrak{M}_{\rm L} \cap \mathfrak{M}^G$, and an element x of the intersection is $e_G x$; the intersection of two submodules is one (Theorem 39.8(i)), and e_G of a Laurent insertion is invariant. □

Proposition 39.14 (The relations of a box). *As stated.*

Proof. (i) The terms of a letter identity of a symbol $(u;\mathbf{a},\mathbf{b})$ of the box for K_j, $a_j < \bar a_j$, are at levels at most the bounds and of wall degree at most that of the symbol (Proposition 39.12(iii)); likewise for Ψ_j. A term of the mark relation of a stem at the level $(\bar{\mathbf{a}}^- + \mathbf{e}_j, \bar{\mathbf{b}}^-)$ has the coefficient $s_{0j} + k\, s_{1j} + \bar a_j^-$; if $\bar a_j \geq 1$ its level is at most $(\bar{\mathbf{a}}, \bar{\mathbf{b}})$, and if $\bar a_j = 0$ then K_j is not active and $\bar a_j^- = 0$, so the coefficient vanishes identically and the term is omitted; likewise for Ψ_j, and the wall degrees do not grow. The evaluations are those of Definition 39.11(iii), and Proposition 38.5(ii) gives the identity in k.

(ii) A letter identity of a symbol $x = (u;\mathbf{a},\mathbf{b})$ of the box for K_j, $a_j < \bar a_j$, makes x congruent to a combination of symbols of the box at the level $(\mathbf{a}+\mathbf{e}_j, \mathbf{b})$, by (i). Repeating, every formal combination y is congruent modulo the letter identities to a combination $y' = \sum_u c_u(u;\bar{\mathbf{a}},\bar{\mathbf{b}})$ of symbols at the top level, with distinct u. The letter identities evaluate to 0, so if y evaluates to 0, so does y', and $\prod_j K_j^{\bar a_j}\Psi_j^{\bar b_j}$ times its evaluation is the Laurent polynomial $\sum_u c_u\alpha^u$, which is 0 only when every c_u is: $y' = 0$, and y is a combination of letter identities.

(iii) Let $x = (u;\mathbf{a},\mathbf{b})$ be such a symbol and $\prod_j K_j^{\bar a_j^- - a_j}\Psi_j^{\bar b_j^- - b_j} = \sum_v \lambda_v\alpha^v$, $\lambda_v \in \Bbbk$. Then x and $\sum_v \lambda_v\, s_v$, $s_v = (u+v;\bar{\mathbf{a}}^-,\bar{\mathbf{b}}^-)$, have one evaluation. Each v is a sum of $\bar a_j^- - a_j$ indicators of bases and $\bar b_j^- - b_j$ of complements of bases of every table j, so $\kappa_U(s_v) \geq \kappa_U(x)$ (7.13) and the s_v are stems. The formal combination $\mathrm{MR}_e(x) - \sum_v \lambda_v \mathrm{MR}_e(s_v)$ evaluates to $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})$ of 0, and its terms lie in the box: those of $\mathrm{MR}_e(s_v)$ by (i), and those of $\mathrm{MR}_e(x)$ because their wall degree is at most $\deg_\kappa x \leq \mathbf{t}$ and a term at the level $(\mathbf{a}+\mathbf{e}_j,\mathbf{b})$, with $a_j \leq \bar a_j^-$, either has a level within the bounds, when $\bar a_j \geq 1$, or the coefficient $s_{0j} + k\, s_{1j}$, identically 0, when $\bar a_j = 0$; likewise for Ψ_j. By (ii) it is a combination of letter identities, and $\mathrm{MR}_e(x) \in \mathcal{R}_{\mathfrak{B}}$.

(iv) ϕ permutes the letters keeping the bounds, so it maps the levels within the bounds onto themselves and the level of the stems onto itself, and it keeps the wall degree (Proposition 39.12(v)); so it permutes the symbols of the box and the stems. The image of a letter identity of the box evaluates to 0 and has its terms in the box, so it lies in the span of the letter identities by (ii). The image of the mark relation of a stem s at e evaluates to $\big((c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})s\big)\circ\phi$, which is $\sum_{e'} \mu_{e'}(c_{e'}^{(0)} + k\, c_{e'}^{(1)} + \partial_{y_{e'}})(s\circ\phi)$ for constants $\mu_{e'}$, ϕ being linear in the coordinates y and keeping $\log(f^k\omega_0/dy)$ up to a constant (proof of Theorem 39.8(i)); so the image less $\sum_{e'}\mu_{e'}\mathrm{MR}_{e'}(\phi^* s)$, with $\phi^* s$ a stem, evaluates to 0 and lies in the span of the letter identities by (ii). So ϕ maps $\mathcal{R}_{\mathfrak{B}}$ into itself, and onto itself, G being finite. Then $e_G\mathcal{R}_{\mathfrak{B}} \subseteq \mathcal{R}_{\mathfrak{B}} \cap \mathcal{F}_{\mathfrak{B}}^G$, and an element x of the intersection is $e_G x$. □

Proposition 39.15 (The closing relations have more solutions than the ray). *As stated.*

Proof. With $\mathbf{s}_0 = \mathbf{t}_0 = \mathbf{0}$, $\omega_0/dy = \alpha^{\mathbf{m}_0}$ and $c_e^{(0)} = m_{0,e}$, and $c_e^{(1)} = \partial_{y_e}\log f = -\partial_{y_e} f^{-1}/f^{-1}$ (38.2(ii)), so

$$\big(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}\big)\big(\alpha^w f^{-1}\big) = \alpha^w\big((m_{0,e}+w_e)\, f^{-1} + (1-k)\,\partial_{y_e} f^{-1}\big),$$

which is the stated Laurent polynomial, $\partial_{y_e}\alpha^v = v_e\alpha^v$; it lies in $\mathcal{L}$, $\alpha^w f^{-1}$ being an insertion. (i) The evaluation at z is multiplicative on Laurent polynomials and gives $z^w\big((m_{0,e}+w_e)f^{-1}(z) + (1-k)\,(\partial_{y_e} f^{-1})(z)\big) = 0$. (ii) It gives $q_0 + \sum_{i\geq 1} q_i\, f^{-1}(z)^i = q_0$. Applying Sh^{-h}, with $\mathrm{Sh}^{-h}(r(k)\, m) =$

$r(k-h)\,\mathrm{Sh}^{-h}(m)$, to the law of [1] gives $\sum_j p_j(k-h)[f^{j-h}]=0$, the element of the statement, whose coefficient at f^0 is $p_h(k-h)\neq 0$. A linear form that vanishes on $\mathcal{L}$ and agrees with the evaluation at z on the Laurent insertions would vanish on that element and take there the value $p_h(k-h)$. □

Definition 39.16 (The reduction of a box). As stated.

Proof. By Proposition 39.14(iv) each $\phi\in G$ permutes the symbols of the box, so $e_G g$ is the mean of the symbols of the orbit of g; orbit averages of distinct orbits have disjoint supports, and they span $\mathcal{F}^G_{\mathfrak{B}}$. The wall degree is constant on an orbit and ϕ^* keeps the Laurent symbols, as do the inverses (Proposition 39.12(v)). Gaussian elimination in the order of the ranking gives a basis (r_p) of $\mathcal{R}^G_{\mathfrak{B}}$, indexed by the pivots, in which r_p is led by p with the coefficient 1 and involves no other pivot. For $x\in\mathcal{F}^G_{\mathfrak{B}}$, $x-\sum_p x_p\,r_p$, x_p the coefficient of the average of p in x, involves no pivot and is congruent to x; two such combinations differ by an element of $\mathcal{R}^G_{\mathfrak{B}}$ that involves no pivot, which is 0, since a nonzero element is led by a pivot. Every r_p involves only orbits ranked at most p, which gives the last statement of (ii). No orbit ranks below that of 1: the other Laurent orbits have positive wall degree (Proposition 39.12(ii)). If it were a pivot, $r_1=1$ would lie in $\mathcal{R}^G_{\mathfrak{B}}\subseteq\mathcal{R}_{\mathfrak{B}}$ (Proposition 39.14(iv)), and 1 would evaluate into $\mathcal{L}$ (Proposition 39.14(i)), $[1]=0$; but the values vanish on $\mathcal{L}$ (Proposition 38.5(ii)), and $\langle 1\rangle_k\neq 0$ for all large k (Proposition 38.9). The monomials of $f^{-1}b_i$ have wall degree at most $\mathfrak{s}+1\leq\mathfrak{t}$ (Proposition 39.12(ii)), and f^{-1} has its coefficients in $\Bbbk$, so $f^{-1}b_i\in\mathcal{F}_{\mathfrak{B}}$; on Laurent combinations the formal and the evaluated products agree, distinct Laurent monomials being independent functions, and $f\circ\phi=f$, so $f^{-1}b_i$ is invariant. □

Theorem 39.17 (The box gives a spanning set, the one-step matrix and a certificate). *As stated.*

Proof. (i) $f^{-1}b_i-\sum_l \mathbf{B}_{il}b_l$ lies in $\mathcal{R}^G_{\mathfrak{B}}\subseteq\mathcal{R}_{\mathfrak{B}}$, so its evaluation lies in $\mathcal{L}$ (Proposition 39.14(i), (iv)), and $[f^{-1}b_i]=\sum_l \mathbf{B}_{il}[b_l]$. With $\mathfrak{N}$ the span of the $[b_l]$, $\mathrm{Sh}^{-1}\mathfrak{N}\subseteq\mathfrak{N}$; Sh^{-1} is injective and semilinear, so it keeps dimensions, $\mathrm{Sh}^{-1}\mathfrak{N}=\mathfrak{N}$, and $\mathfrak{N}$ is a submodule; it contains $[b_1]=[1]$. The b_l are invariant insertions, so $\mathfrak{N}\subseteq\mathfrak{M}^G$ (Theorem 39.8(i)). An invariant combination of Laurent symbols of wall degree at most $\mathfrak{s}$ is a combination of averages of Laurent orbits of wall degree at most $\mathfrak{s}$, and by the ranking every orbit ranked at most the highest of these is a Laurent orbit of wall degree at most $\mathfrak{s}$; so its reduction is a combination of the b_l (Definition 39.16(ii)), congruent to it modulo $\mathcal{R}^G_{\mathfrak{B}}$, and it has a class in $\mathfrak{N}$. When the $[b_l]$ are independent they are a basis of $\mathfrak{N}$, and the matrix of Sh^{-1} on a basis is unique.

(ii) As an element of $\mathcal{R}_{\mathfrak{B}}$, $f^{-1}b_i-\sum_l \mathbf{B}_{il}b_l$ is a combination of letter identities and of mark relations $\mathrm{MR}_e(s)$ of stems s, with coefficients $\nu_{i,e,s}\in\Bbbk(k)$. Evaluated, the letter identities give 0 and $\mathrm{MR}_e(s)$ gives $(c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e})s$; put $G_{i,e}=\sum_s \nu_{i,e,s}\,s$.

(iii) $V[f^{-1}b_i]=V(\mathrm{Sh}^{-1}[b_i])=\varsigma^{-1}V[b_i]$, and by (i) and the $\mathbb{C}(k)$-linearity of V it is $\sum_l \mathbf{B}_{il}V[b_l]$. The certificate is an element of $\mathcal{L}$, so its value vanishes identically, and $\langle f^{-1}b_i\rangle_k=\langle b_i\rangle_{k-1}$ (Proposition 38.5(ii)). At a real k as stated, the stems, the b_l and the monomials of $f^{-1}b_i$ have wall degree at most $\mathfrak{t}$, so their values converge absolutely (Proposition 39.12(i)), and Proposition 38.5(i) gives the value 0 for every term $(c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e})s$.

(iv) Write $\rho_e^{(k)}=c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e}$; by the identity in the proof of Definition 39.1, $f^{-1}\rho_e^{(k-1)}(x)=\rho_e^{(k)}(f^{-1}x)$ and $f^h\rho_e^{(k)}(x)=\rho_e^{(k-h)}(f^h x)$. With the $G_{l,e}$ of (ii), put $G_e^{(0)}=0$ and $G_e^{(i)}(k)=\sum_l w_{i-1,l}(k-1)\,G_{l,e}(k)+f^{-1}G_e^{(i-1)}(k-1)$ for $i\geq 1$, and $R_i(k)=\sum_e \rho_e^{(k)}\big(G_e^{(i)}(k)\big)\in\mathcal{L}$. Then $f^{-i}=\sum_l w_{i,l}(k)\,b_l+R_i(k)$: for $i=0$ this is $1=b_1$, and multiplying the identity for $i-1$ at the

strength $k-1$ by f^{-1} and inserting $f^{-1}b_l = \sum_{l'} \mathbf{B}_{ll'}(k) b_{l'} + \sum_e \rho_e^{(k)}(G_{l,e}(k))$ gives it for i. So $\sum_i q_i(k) f^{-i} = \sum_i q_i(k) R_i(k)$, and multiplying by f^h and replacing k by $k+h$ gives

$$\sum_{j=0}^{h} p_j(k)\, f^j = \sum_e \rho_e^{(k)} \Big(\sum_{i=0}^{h} q_i(k+h)\, f^h G_e^{(i)}(k+h) \Big),$$

a certificate. Proposition 39.6(ii) gives the rest. □

Theorem 39.18 (The search stops). *As stated.*

Proof. Let $\mathcal{L}_\Bbbk$ be the $\Bbbk(k)$-span of the elements $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})x$, $x \in \Bbbk[X_F]$, and $M_\Bbbk = (\Bbbk[X_F] \otimes \Bbbk(k))/\mathcal{L}_\Bbbk$. As in the proof of Theorem 38.24, $M_\Bbbk \otimes_{\Bbbk(k)} \mathbb{C}(k) = \mathfrak{M}$, so $M_\Bbbk$ has the finite dimension of the ray and embeds in $\mathfrak{M}$, and a $\Bbbk(k)$-subspace S of $M_\Bbbk$ has $(S \otimes \mathbb{C}(k)) \cap M_\Bbbk = S$, a basis of S extending to one of $M_\Bbbk$. The symmetries act on $M_\Bbbk$ as on $\mathfrak{M}$, the weights being in $\Bbbk$. Let $\mathfrak{N}_\mathfrak{s} \subseteq M_\Bbbk$ be the span of the classes of the averages of the Laurent orbits of wall degree at most $\mathfrak{s}$. The $\mathfrak{N}_\mathfrak{s}$ increase with $\mathfrak{s}$ in a space of finite dimension, so they are constant from some $\mathfrak{s}_0$ on, and their union is the span of the classes of the invariant Laurent insertions, whose $\mathbb{C}(k)$-span in $\mathfrak{M}$ is $e_G \mathfrak{M}_\mathrm{L}$ (Lemma 39.13).

The averages $\bar g$ of the Laurent orbits of wall degree at most $\mathfrak{s}_0 + 1$ are finitely many (Proposition 39.12(iv)), and each has its class in $\mathfrak{N}_{\mathfrak{s}_0}$:

$$\bar g - \sum_{\bar h} \lambda_{\bar g, \bar h}\, \bar h = \sum_{e,i} \nu_{\bar g, e, i}\, (c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}) G_{\bar g, i},$$

with $\bar h$ running over the averages of the Laurent orbits of wall degree at most $\mathfrak{s}_0$, $\lambda, \nu \in \Bbbk(k)$, and finitely many symbols $G_{\bar g, i}$, the evaluations of the symbols spanning $\Bbbk[X_F]$ over $\Bbbk$. Take an integer $\mathfrak{t} \geq \mathfrak{s}_0 + 1$ at least the wall degree of every $G_{\bar g, i}$, and the level bounds $\bar\ell(L) = 1 + \max \ell(L)$ for every letter L, the maximum over the levels ℓ of the symbols $\phi^* G_{\bar g, i}$, $\phi \in G$, at L. These bounds are at least 1, and kept by G, the set of the $\phi^* G_{\bar g, i}$ being kept by G; every $G_{\bar g, i}$ is at a level at most $(\bar{\mathbf{a}}^-, \bar{\mathbf{b}}^-)$, here $\bar a_j^- = \bar a_j - 1$ and $\bar b_j^- = \bar b_j - 1$. By Proposition 39.14(iii) every $\mathrm{MR}_e(G_{\bar g, i})$ lies in $\mathcal{R}_\mathfrak{B}$, and the formal combination $\bar g - \sum \lambda_{\bar g, \bar h} \bar h - \sum \nu_{\bar g, e, i} \mathrm{MR}_e(G_{\bar g, i})$ has its terms in the box (Proposition 39.12(iii)) and evaluates to 0, so it lies in $\mathcal{R}_\mathfrak{B}$ (Proposition 39.14(ii)). Hence $\bar g - \sum \lambda_{\bar g, \bar h} \bar h$, which is invariant, lies in $\mathcal{R}_\mathfrak{B} \cap \mathcal{F}_\mathfrak{B}^G = \mathcal{R}_\mathfrak{B}^G$ (Proposition 39.14(iv)).

Take $\mathfrak{s} = \mathfrak{s}_0 < \mathfrak{t}$. A target $f^{-1} b_i$ is an invariant combination of Laurent symbols of wall degree at most $\mathfrak{s}_0 + 1$ (Proposition 39.12(ii)), so a combination of the $\bar g$, and it is congruent modulo $\mathcal{R}_\mathfrak{B}^G$ to a combination of averages of Laurent orbits of wall degree at most $\mathfrak{s}_0$. The reduction of that combination, which is the reduction of the target, involves only free Laurent orbits of wall degree at most $\mathfrak{s}_0$ (Definition 39.16(ii)): the spanning set. So the box is accepted at $\mathfrak{s}_0$, and its spanning set spans $\mathfrak{N}_{\mathfrak{s}_0} \otimes \mathbb{C}(k) = e_G \mathfrak{M}_\mathrm{L}$, by Theorem 39.17(i); this is $\mathfrak{M}^G$ when every letter exponent of the direction is positive (Lemma 39.13). A search through all the quadruples reaches this one, and acceptance is decided by finitely many operations in $\Bbbk(k)$. □

Definition 39.19 (Sources; regular points; the normal density). As stated.

Proof. Z is irreducible, so its smooth locus is connected; there $d \log f$ vanishes along Z, so f is constant on it, and on Z by continuity. Differentiating $d \log f = 0$ along Z shows that the tangent space of Z at a smooth point lies in the kernel of the Hessian, whose rank is then at most $n - 1 - d$; the conditions defining a regular point are open, so the regular points form an open set, dense when it is nonempty, Z being irreducible. For another choice of coordinates (u', w')

with $Z=\{w'=0\}$, at $w=0$ one has $u'=U(u)$ and $w'=J(u)w+O(|w|^2)$, the Jacobian of $(u,w)\mapsto(u',w')$ being block triangular there, so $du'\wedge dw'=\det(\partial U/\partial u)\det J\,du\wedge dw$ at $w=0$, $A=J^{\mathsf T}A'J$, and $du'=\det(\partial U/\partial u)\,du$ on Z. So $\det A'^{-1/2}=\pm\det J\,\det A^{-1/2}$ and $\frac{\omega_0}{du'\wedge dw'}=\big(\det(\partial U/\partial u)\det J\big)^{-1}\frac{\omega_0}{du\wedge dw}$, and the product of these with du' is $\pm\det A^{-1/2}\frac{\omega_0}{du\wedge dw}\,du$: ν is defined up to sign. For $d=0$ a point is regular exactly when the Hessian has rank $n-1$, and an isolated point of the critical locus lies on no other component. □

Theorem 39.20 (A closed cycle on a source carries a solution). *As stated.*

Proof. (i) In the coordinates y the torus is, near each point, an open set of the vector space $\mathbb{C}^{\mathcal E}/\mathbb{C}\mathbf{1}$, in which $z+v$ has a meaning for small v. At $z\in Z^\circ$ call a real subspace V of dimension m of the tangent space a *descent subspace* when the real part of the Hessian of $-\log f$ is positive definite on it; then V meets the tangent space of Z only at 0, the Hessian vanishing there. The projection to the normal space $\mathbb{C}^m$, in coordinates in which the normal form is $\sum_i w_i^2$, maps V onto a real subspace V' with $V'\cap i\mathbb{R}^m=0$, the form being negative on $i\mathbb{R}^m$; so $V'=\{x+iBx : x\in\mathbb{R}^m\}$ for a real matrix B, and the form has the real part $|x|^2-|Bx|^2$ on it, positive definite exactly when $\|B\|<1$. So the descent subspaces at z are the graphs over such V' of real linear maps into the tangent space of Z: a contractible set, whose oriented members form two contractible components, one for each sheet of ν. Over the compact $\gamma\subset\tilde Z^\circ$ there is therefore a continuous oriented choice $z\mapsto V_z$, and by Taylor's formula and compactness there are $r,c>0$ with $|f(z+v)|\le|f_Z|e^{-c|v|^2}$ for $v\in V_z$, $|v|\le r$. Put $D_z=z+\{v\in V_z:|v|\le r\}$ and $N(\gamma)=\bigcup_{z\in\gamma}D_z$; as $\partial\gamma=0$, $\partial N(\gamma)$ is the union of the boundaries of the discs, on which $|f|\le|f_Z|e^{-cr^2}$. For two choices over γ, or for $\gamma-\gamma'=\partial\Gamma$, the same construction over $\gamma\times[0,1]$ or over Γ, with discs of the same kind, gives a chain of dimension n whose boundary is the difference of the two chains together with a union of disc boundaries; the discs need not be affine, any smooth family of embedded discs tangent at z to a descent subspace serving as well for r small.

(ii) The integral over $N(\gamma)$ is the integral over $\gamma\times\{|v|\le r\}$ of the pull-back by $(z,v)\mapsto z+v$, $v\in V_z$. At each z, Laplace's method in v, uniform in z on the compact γ, expands the integral over D_z in powers of $1/k$ after the factor $f_Z^k k^{-m/2}$, with coefficients that are continuous d-forms along γ; integrating them over γ gives the $c_i(\gamma;g)$. The leading coefficient is $(2\pi)^{m/2}\det\big[H_z(\mathsf f_a,\mathsf f_b)\big]^{-1/2}g\,\omega_0(\,\cdot\,,\mathsf f_1,\dots,\mathsf f_m)$, H_z the Hessian of $-\log f$ and $\mathsf f_1,\dots,\mathsf f_m$ a basis of V_z, the square root fixed by the orientation; it does not change under a complex linear change of $\mathsf f$, and V_z spans over $\mathbb{C}$ a complement of the tangent space of Z, which projects onto the normal space as the w-space does in coordinates (u,w), so it is $(2\pi)^{m/2}g\nu$ on the sheet. Where holomorphic normal slices $\{u=\text{const}\}$ cover Z°, take every D_z in the slice through z, a real descent disc there, which (i) allows. The integral over the slice at u has the phase $-\log f=-\log f_Z+\frac12 w^{\mathsf T}A(u)w+O(|w|^3)$ and the amplitude $g\,\omega_0/(du\wedge dw)$, and Laplace's method with the holomorphic parameter u gives $f_Z^k k^{-m/2}\sum_i k^{-i}a_i(u;g)$, where $a_i(u;g)$ is du times $(2\pi)^{m/2}\det A(u)^{-1/2}$ times a polynomial in finitely many w-derivatives at $w=0$ of the amplitude and of the phase and in the entries of $A(u)^{-1}$. These coefficients are holomorphic in u, so each $a_i(\cdot\,;g)$ is a holomorphic d-form, $a_0(\cdot\,;g)=(2\pi)^{m/2}g\,\nu$, and $c_i(\gamma;g)=\int_\gamma a_i(\cdot\,;g)$.

(iii) By 38.2(ii), $(c_e^{(0)}+kc_e^{(1)}+\partial_{y_e})g\cdot f^k\omega_0/dy=\partial_{y_e}\big(gf^k\omega_0/dy\big)$, so $(c_e^{(0)}+kc_e^{(1)}+\partial_{y_e})g\cdot f^k\omega_0=d\big(\iota_{\partial_{y_e}}(gf^k\omega_0)\big)$, the form being of top degree and ∂_{y_e} a constant field. By Stokes its integral over $N(\gamma)$ is an integral over $\partial N(\gamma)$, of modulus at most a constant times $|f_Z|^k e^{-kcr^2}$; so the expansion of every relation vanishes to all orders, and A_γ, extended $\mathbb{C}(k)$-linearly, vanishes on $\mathcal L$. The integral of $fg\,f^k\omega_0$ is that of $g\,f^{k+1}\omega_0$, so $A_\gamma(fg)(k)=A_\gamma(g)(k+1)$ after re-expansion, and A_γ is a solution of type $(f_Z,-\frac m2+\mathbb{Z})$. The forms $gf^k\omega_0$ are closed, so by Stokes on the chains of (i) homologous

cycles give integrals differing by terms of modulus at most a constant times $|f_Z|^k e^{-kcr^2}$, hence the same expansion. The deck involution reverses the orientation of every disc. For an origin that is not integral the same construction runs with a branch of ω_0 carried along the chains, that is, for cycles with coefficients in the local system of the branches of ν.

(iv) A relation $\sum_a c_a \hat{A}_{\gamma_a} = 0$ with $c_a \in \mathbb{C}((1/k))$ not all zero, scaled so that the least order of the c_a is 0 with constant terms c_a^0 not all zero, gives at the leading order, applied to g_b, $(2\pi)^{m/2} \sum_a c_a^0 \int_{\gamma_a} g_b \nu = 0$ for every b, so $c^0 = 0$ by the determinant: a contradiction. The relations over $\mathbb{C}$ are among them. □

Corollary 39.21 (A curve of native gauges: residues and periods). *As stated.*

Proof. The first homology of $\tilde{Z}^\circ$, a compact curve less finitely many points, is generated by small loops around the punctures and by the cycles of $\bar{Z}$, and A_γ depends only on the class of γ (Theorem 39.20(iii)). For the family $\alpha = \alpha_0 b^{\mathsf{v}} e^w$ the coordinates $(\log b, w)$ are linear in y, so dy is a constant multiple of $d\log b \wedge dw$. With an integral origin f, g and ω_0/dy are rational functions of α, and so are their derivatives in y, $\partial_{y_e} = \alpha_e \partial_{\alpha_e}$; their w-derivatives at $w = 0$ are constant combinations of these at $\alpha = \alpha_0 b^{\mathsf{v}}$, so rational in b, and so are the entries of $A(b)$ and $A(b)^{-1}$; the slices $\{b = \text{const}\}$ cover Z°, and the Laplace coefficients of Theorem 39.20(ii) are polynomials in these times $\det A(b)^{-1/2}$, so $a_i = r_i(b; g) \det A(b)^{-1/2}\, db/b$ with r_i rational. On the curve $Y^2 = \det A(b)$ these are meromorphic forms, and the integral over a small loop around a puncture is $2\pi i$ times the residue there. □

Theorem 39.22 (A native gauge of corank at most one). *As stated.*

Proof. The Hessian of $\log f$ at α_c has rank at least $n-2$. By the splitting lemma, the holomorphic Morse lemma in the directions where the Hessian is nondegenerate with the remaining coordinate as a parameter, there are coordinates (X, w) with $\log f = \log f(\alpha_c) - \frac{1}{2}\sum_i w_i^2 + R(X)$, $R(0) = R'(0) = 0$; when the Hessian is nondegenerate, X is one of the Morse coordinates and $R = -\frac{1}{2}X^2$. R does not vanish identically, as otherwise $\{w = 0\}$ would be a curve of native gauges through α_c. With $\mu + 1$ its order at 0, $R = cX^{\mu+1}(1 + O(X))$ with $c \neq 0$, and $X(1 + O(X))^{1/(\mu+1)}$ is a coordinate in which $R = cX^{\mu+1}$; $\mu = 1$ exactly when the Hessian is nondegenerate, and otherwise $R''(0) = 0$ and ∂_X spans the kernel.

(i) Let $\varrho_0, \dots, \varrho_\mu$ be the directions with $c\,\varrho^{\mu+1} < 0$, $\varrho_b = \varrho_0 \zeta^b$, $\zeta = e^{2\pi i/(\mu+1)}$; around them lie the sectors on which $\mathrm{Re}(cX^{\mu+1}) < 0$. For the path coming in along $\varrho_0\,\mathbb{R}_{>0}$ and going out along $\varrho_b\,\mathbb{R}_{>0}$,

$$\int X^j e^{cX^{\mu+1}} dX = \big(\varrho_b^{\,j+1} - \varrho_0^{\,j+1}\big) \frac{\Gamma\big(\frac{j+1}{\mu+1}\big)}{(\mu+1)|c|^{(j+1)/(\mu+1)}},$$

and these μ paths generate the homology of the plane relative to the part where $\mathrm{Re}(cX^{\mu+1}) < -1$. The $\mu \times \mu$ matrix of the numbers $\zeta^{b(j+1)} - 1$, $1 \le b \le \mu$, $0 \le j < \mu$, is invertible: its columns, extended by the entry 0 at $b = 0$, are the differences of the columns $l = j+1$ and $l = 0$ of the invertible matrix $(\zeta^{bl})_{0 \le b,l \le \mu}$. So there are complex combinations Γ_i of the paths with $\int_{\Gamma_i} X^j e^{cX^{\mu+1}} dX = \delta_{ij}$, $0 \le j < \mu$. Integration by parts gives $\int_\Gamma X^{j+\mu+1} e^{cX^{\mu+1}} dX = -\frac{j+1}{(\mu+1)c} \int_\Gamma X^j e^{cX^{\mu+1}} dX$, and $\int_\Gamma X^\mu e^{cX^{\mu+1}} dX = 0$, the integrand being a derivative; so $\int_{\Gamma_i} X^j e^{cX^{\mu+1}} dX = 0$ unless $j \equiv i \bmod \mu+1$. On the chain $C_i = D \times \Gamma_i$, D the real ball $|w| \le \varepsilon$ and Γ_i cut off at $|X| = \varepsilon$, the boundary lies where $|f| \le |f(\alpha_c)| e^{-\varepsilon'}$, and the relations and the shift hold up to exponentially smaller terms as in Theorem 39.20(iii). Expand the amplitude $\mathsf{a} = g\,\omega_0/(dX \wedge dw) = \sum \mathsf{a}_{j,\beta} X^j w^\beta$; after $X \mapsto k^{-1/(\mu+1)} X$ and $w \mapsto k^{-1/2} w$ the term (j, β) contributes $k^{-(n-2)/2 - |\beta|/2 - (j+1)/(\mu+1)}$ times a Gaussian moment, zero for odd $|\beta|$, times $\int_{\Gamma_i} X^j e^{cX^{\mu+1}} dX$, and the usual bounds of Laplace's

method make the sum an asymptotic expansion. Only $j \equiv i \bmod \mu+1$ survive, the exponents are $-\frac{n-2}{2} - \frac{i+1}{\mu+1}$ less integers, and the leading one, $j = i$ and $\beta = 0$, has the coefficient $(2\pi)^{(n-2)/2}\mathsf{a}_{i,0}$. Since $0 < \frac{i+1}{\mu+1} < 1$, distinct i give distinct types; each $A_{c,i}$ is nonzero, an insertion with $\mathsf{a}_{i,0} \neq 0$ existing, and Lemma 39.3 gives the independence. For $\mu = 1$, $R = -\frac{1}{2}X^2$, the normalization $\int_{\Gamma_0} e^{-X^2/2}dX = 1$ makes Γ_0 the real line divided by $(2\pi)^{1/2}$, and C_0 is a steepest-descent disc divided by $(2\pi)^{1/2}$: $A_{c,0} = (2\pi)^{-1/2}A_c$.

(ii) In the coordinates y, ϕ is $y \mapsto M_\phi y$, linear, of finite order, fixing $\log \alpha_c$; it keeps $\log f$, so its Hessian and the kernel: $M_\phi \xi = \zeta_\phi \xi$ for ξ spanning the kernel, with ζ_ϕ a root of unity, and M_ϕ maps an invariant complement N of the kernel onto itself. In linear coordinates (X', w') along ξ and N the curve $\{\partial_{w'} \log f = 0\}$ through α_c is smooth and parametrized by X', the Hessian being nondegenerate on N; ϕ maps it onto itself by $X' \mapsto \zeta_\phi X'$, and $\log f$ on it is $\log f(\alpha_c)$ plus a function of order $\mu+1$ in X' with nonzero leading coefficient, the reduced function of the splitting lemma. Its invariance gives $\zeta_\phi^{\mu+1} = 1$.

As in Theorem 39.8(iii), $\phi^*\omega_0 = c_\phi\, \omega_0$ near α_c for a constant $c_\phi \neq 0$, so $A_{c,i}(g\circ\phi)$, the expansion over C_i of $(g\circ\phi)f^k\omega_0 = c_\phi^{-1}\phi^*(gf^k\omega_0)$, is c_ϕ^{-1} times the expansion over $\phi(C_i)$, a chain near α_c with its boundary where $|f|$ is smaller. The relative homology near α_c has rank μ over $\mathbb{C}$ with the basis $C_0, \dots, C_{\mu-1}$, so $A_{c,i} \circ \phi^* = \sum_{i'} \kappa_{i'} A_{c,i'}$. By (i) applied to the insertion $g \circ \phi$, $A_{c,i} \circ \phi^*$ has the type of $A_{c,i}$, and the $A_{c,i'}$ have distinct types, so $A_{c,i} \circ \phi^* = \kappa_i A_{c,i}$. The expansions depend only on the Taylor series of the insertion at α_c, and insertions realize every finite jet there, so the identity holds on formal power series at α_c. Let ℓ be the linear function of $y - \log \alpha_c$ that vanishes on N with $\ell(\xi) = 1$; then $\ell \circ \phi = \zeta_\phi \ell$. In the coordinates of (i), $\ell(X, 0) = \ell_1 X + O(X^2)$ with $\ell_1 \neq 0$, since ∂_X is along ξ at α_c, so the coefficient of X^i in $\ell^i\, \omega_0/(dX \wedge dw)$ at $w = 0$ is $\ell_1^i\, (\omega_0/(dX \wedge dw))(\alpha_c) \neq 0$, and $A_{c,i}(\ell^i) \neq 0$. Then $\kappa_i A_{c,i}(\ell^i) = A_{c,i}(\ell^i \circ \phi) = \zeta_\phi^i A_{c,i}(\ell^i)$ gives $\kappa_i = \zeta_\phi^i$. □

Corollary 39.23 (Signs at a centre). *As stated.*

Proof. The centre ray has origin 0 and direction $(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$, so the Cremona is a symmetry by Proposition 39.7(ii). It fixes $\eta = \eta^{-1}$, and in the chart $\alpha = \eta e^y$ it is $y \mapsto -y$. (i) is Theorem 39.8(iii). In (ii), $\zeta = -1$, so $\mu + 1$ is even, and corank one makes $\mu \geq 2$, so $\mu \geq 3$; Theorem 39.22(ii) gives $A_{\eta,i} \circ \mathrm{cr}^* = (-1)^i A_{\eta,i}$. For a Cremona-invariant insertion g, $A_{\eta,i}(g) = A_{\eta,i}(g \circ \mathrm{cr}) = (-1)^i A_{\eta,i}(g)$ vanishes for odd i; for even i and an insertion g' with $A_{\eta,i}(g') \neq 0$, $\frac{1}{2}(g' + g' \circ \mathrm{cr})$ is invariant with the same value. The exponents are those of Theorem 39.22(i) at $\mu = 3$. □

Proposition 39.24 (Symmetries on the cycles of a source). *As stated.*

Proof. ϕ is an automorphism of X_F keeping f, so it maps the critical locus onto itself, sources onto sources, regular points onto regular points, and descent discs at z onto descent discs at $\phi(z)$, with their orientations; the sheet of ν at $\phi(z)$ that matches the image orientation defines $\tilde{\phi}$, and $\phi(N(\gamma)) = N(\tilde{\phi}_*\gamma)$ as oriented chains. With $f \circ \phi = f$ and $\varepsilon_\phi^2 = 1$, $(g \circ \phi)f^k\omega_0 = \varepsilon_\phi\, \phi^*(gf^k\omega_0)$, so $\int_{N(\gamma)}(g \circ \phi)f^k\omega_0 = \varepsilon_\phi \int_{N(\tilde{\phi}_*\gamma)} gf^k\omega_0$. On $\mathfrak{M}^G$, $A_\gamma = A_\gamma \circ e_G = |G|^{-1} \sum_\phi \varepsilon_\phi A_{\tilde{\phi}_*\gamma}$, which is linear in the cycle. Grouping $\phi \in G$ by the cosets ψG_Z, with $\varepsilon_{\psi\chi} = \varepsilon_\psi \varepsilon_\chi$ and $\widetilde{\psi\chi}_* = \tilde{\psi}_*\tilde{\chi}_*$, each coset contributes $\varepsilon_\psi A_{\tilde{\psi}_* z}$ with $z = \sum_{\chi \in G_Z} \varepsilon_\chi \tilde{\chi}_*\gamma$, zero when $[z] = 0$. □

Theorem 39.25 (What the invariant module sees at the high end). *As stated.*

Proof. For $\phi \in G$ and $\psi \in G_c$, $A_{c,i} \circ (\phi\psi)^* = A_{c,i} \circ \psi^* \circ \phi^* = \zeta_\psi^i A_{c,i} \circ \phi^*$ (Theorem 39.22(ii), and Theorem 39.8(iii) for $\mu = 1$), so

$$A_{c,i} \circ e_G = |G|^{-1} \sum_{\phi G_c} \Big(\sum_{\psi \in G_c} \zeta_\psi^i \Big) A_{c,i} \circ \phi^*.$$

When $\psi \mapsto \zeta_\psi^i$ is a nontrivial character of G_c its sum over G_c vanishes and $A_{c,i}$ vanishes on $\mathfrak{M}^G = e_G \mathfrak{M}$. Otherwise $A_{c,i} \circ e_G$ is $|G_c|/|G|$ times $A_{O,i} = \sum_{\phi G_c} A_{c,i} \circ \phi^*$, which is invariant, and each $A_{c,i} \circ \phi^*$ is a constant multiple of an expansion over the chain $\phi(C_i)$ near $\phi(\alpha_c)$. The isolated native gauges are finitely many. Take an insertion with a Taylor jet at α_c that makes the leading coefficient of $A_{c,i}$ nonzero (Theorem 39.22(i)) and that vanishes to a high order at every other isolated native gauge; on it $A_{O,i}$ has that leading coefficient, so $A_{O,i}$ does not vanish on $\mathfrak{M}^G$, and the same insertions show that the $A_{O,i}$ of one type, over the orbits and the classes, are linearly independent over $\mathbb{C}$ on $\mathfrak{M}^G$, the solutions of one type at one point being a single $A_{c,i}$. Lemma 39.3 gives the bound. □

Definition 39.26 (Closings). As stated.

Proof. $\mathbb{C}[X_F]$ is spanned by the insertions $g = \alpha^u \prod_j K_j^{-a_j} \Psi_j^{-b_j}$ with $a_j, b_j \in \mathbb{N}$, products of the generators of 38.1, and for them $g\, f^{-j} \omega_0/dy = \alpha^{u - j\mathbf{m}_1 + \mathbf{m}_0} \prod_l K_l^{j s_{1l} - s_{0l} - a_l} \Psi_l^{j t_{1l} - t_{0l} - b_l}$, a Laurent polynomial when every exponent of a letter is nonnegative, which holds for large j as $s_{1l}, t_{1l} \geq 1$; a finite sum takes the largest of the bounds. The function is homogeneous of degree 0, so its monomials α^u have $u_\mathcal{E} = 0$, and on the chart $\alpha_{e_0} = 1$ its constant term, the coefficient of the monomials with $u_e = 0$ for $e \neq e_0$, is the coefficient of α^0, whatever e_0. On the chart $dy = \prod_{e \neq e_0} d\alpha_e/\alpha_e$, which gives the integral over the torus. □

Proposition 39.27 (One form on the orthant and on the torus). *As stated.*

Proof. On the orthant $f > 0$ and $1/(f^{-1} - q) = f/(1 - qf)$, so the first integral is the generating function of Theorem 38.14(ii) with $k_0 = 1$, which the origin 0 allows. The Laurent polynomial $f^{-1} = \alpha^{-\mathbf{m}_1} \prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}}$ has nonnegative coefficients, so on the torus through $\alpha^\star$, $|f^{-1}(\alpha)| \leq f^{-1}(\alpha^\star) = A$. For $|q| > A$ the series $1/(f^{-1} - q) = -\sum_{j \geq 0} f^{-j} q^{-j-1}$ converges uniformly on that torus, and term by term $(2\pi i)^{-(n-1)} \oint f^{-j}\, dy$ is the constant term, $\oint \alpha^v\, dy = (2\pi i)^{n-1}$ for $v = 0$ and 0 for the other monomials of degree 0. For every letter exponent positive this is $\mathsf{C}_j(1)$ (Definition 39.26, the origin form being dy). □

Proposition 39.28 (The constant terms of a centre ray and its signs). *As stated.*

Proof. By Definition 38.13, $f^{-1} = \prod_l \big(K_l(\alpha) K_l(\alpha^{-1})\big)^{\sigma_l}$, a Laurent polynomial with nonnegative coefficients, homogeneous of degree 0; at a sign $\eta^{-1} = \eta$, so $f^{-1}(\eta) = \prod_l K_l(\eta)^{2\sigma_l}$. On the chart $\alpha_{e_0} = 1$ a monomial α^v, $v_\mathcal{E} = 0$, takes at η the value $\prod_{e \neq e_0} \eta_e^{v_e}$, whose mean over the signs is 1 when every v_e, $e \neq e_0$, is even, and then $v_{e_0} = -\sum_{e \neq e_0} v_e$ is even too, and 0 otherwise. The constant term is the coefficient at $v = 0$, one of the nonnegative coefficients summed. □

Theorem 39.29 (The closings are a solution). *As stated.*

Proof. By 38.2(ii), $c_e^{(0)} - j\, c_e^{(1)} = \partial_{y_e} \log\big(f^{-j} \omega_0/dy\big)$, so $\big(c_e^{(0)} - j c_e^{(1)} + \partial_{y_e}\big) g \cdot f^{-j} \omega_0/dy = \partial_{y_e}\big(g f^{-j} \omega_0/dy\big)$. For $j \geq j(g)$ the right side is ∂_{y_e} of a Laurent polynomial, which multiplies each monomial α^u by u_e and so has no constant term. $\mathsf{C}_j(fg) = \mathrm{CT}\big(g f^{-(j-1)} \omega_0/dy\big) = \mathsf{C}_{j-1}(g)$ for

$j-1 \ge j(g)$. An element of $\mathcal{L}$ is a finite sum $\sum r_{e,l}(k)\big(c_e^{(0)} + k c_e^{(1)} + \partial_{y_e}\big)h_{e,l}$, and V_{C} of it at an integer $k=-j$ off the poles of the $r_{e,l}$ with $j > j(h_{e,l})$ for all e,l, where the closings of $c_e^{(1)}h_{e,l}$ and of $(c_e^{(0)}+\partial_{y_e})h_{e,l}$ are defined, $c_e^{(0)}$, $c_e^{(1)}$ and ∂_{y_e} costing at most one power of each letter and f^{-1} supplying at least one, is $\sum r_{e,l}(-j)\,\mathsf{C}_j\big((c_e^{(0)} - j c_e^{(1)} + \partial_{y_e})h_{e,l}\big) = 0$; so V_{C} is well defined on $\mathfrak{M}$ as a germ at $-\infty$. It is $\mathbb{C}(k)$-linear, and $V_{\mathsf{C}}(\mathrm{Sh}(g\otimes r))(k) = r(k+1)\,\mathsf{C}_{-k}(fg) = r(k+1)\,\mathsf{C}_{-k-1}(g) = \big(\varsigma V_{\mathsf{C}}(g\otimes r)\big)(k)$. Theorem 39.4(i) with $v(k) = \mathsf{C}_{-k}(\mathbf{b})$ reads $\mathsf{C}_{1-k}(\mathbf{b}) = \mathbf{B}_{\mathbf{b}}(k)\,\mathsf{C}_{-k}(\mathbf{b})$, which at $k=-j$ is the stated identity, and the law of [1] applied to V_{C} gives $\sum_i p_i(k)\,\mathsf{C}_{-k-i}(1) = 0$, at $k=-j$ the last statement. □

Proposition 39.30 (The closings are the top principal parts). *As stated.*

Proof. By 38.2(iii), $\langle g\rangle_k$ is a finite combination of periods P_F at data $(k\mathbf{m}_1 + v; k\mathbf{s}_1 + \mathbf{a}, k\mathbf{t}_1 + \mathbf{b})$ on the balance wall. On the balance wall, where every proper wall has positive real part, 8.5(ii) and 8.7, $J_F = \Gamma(\kappa_{\mathcal{E}})P_F$, give

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_{\pi} \int_{[0,1]^{n-1}} \prod_{l=2}^{n} q_l^{\kappa_{U_l}-1} \prod_j k_{j,\pi}(q)^{-s_j}\psi_{j,\pi}(q)^{-t_j}\,dq,$$

the limit $\kappa_{\mathcal{E}} \downarrow 0$ of the same identity with the factor $L_\pi^{-\kappa\varepsilon}$, $L_\pi \in [1,n]$. The sector letters are positive on the closed cube, so the integrand is smooth in q and holomorphic in the exponents, and 8.4 continues each term meromorphically in the walls with the poles on $\kappa_{U_l} = -i$, $i \in \mathbb{N}$, as a finite sum of products of at most $n-1$ factors $(\kappa_{U_l} + i_l)^{-1}$ with functions holomorphic near the point. Along the ray every wall is $k\kappa_{1,U_l}$ plus a constant, with $\kappa_{1,U_l} > 0$, so $\langle g\rangle_k$ is meromorphic in k with poles of order at most $n-1$.

For an integer $j \ge j(g)$ put $\epsilon = k+j$; then $\langle g\rangle_k = \langle P\rangle_\epsilon$ with the Laurent polynomial $P = gf^{-j}$, an identity for large $\operatorname{Re} k$ that continues. For a monomial α^u with $u \ne 0$ choose e with $u_e \ne 0$. The relation of 38.3 at the strength ϵ, with $c^{(0)} = 0$ for the origin 0, holds for complex ϵ of large real part by the same integration by parts, the integrals converging absolutely there, and gives $u_e\langle\alpha^u\rangle_\epsilon = -\epsilon\,\langle c_e^{(1)}\alpha^u\rangle_\epsilon$, then as meromorphic functions; the right side has a pole of order at most $n-2$ at $\epsilon = 0$. So $\epsilon^{n-1}\langle P\rangle_\epsilon$ and $\mathrm{CT}(P)\,\epsilon^{n-1}\langle 1\rangle_\epsilon$ have the same value at $\epsilon = 0$. Finally $\langle 1\rangle_\epsilon = P_F(\epsilon\mathbf{m}_1;\epsilon\mathbf{s}_1,\epsilon\mathbf{t}_1)$ is the period of the question of the direction at the strength ϵ, whose walls are positive, and 17.5(i) gives $\epsilon^{n-1}\langle 1\rangle_\epsilon \to V_1$ as $\epsilon \downarrow 0$; the function being holomorphic at 0, that is its value. □

Proposition 39.31 (The low end, for every direction). *As stated.*

Proof. (i) For an integral origin $\kappa_{0,U}$ and $\kappa_U(g)$ are integers, so $\kappa_{1,U}k$ is one at such a pole; Proposition 38.5(ii) gives the rest.

(ii) The argument is that of the proof of Proposition 39.30, which uses of the direction only that $P = f^{-j}g$ is a Laurent polynomial. By Proposition 38.5(ii), $\langle g\rangle_k = \langle f^j P\rangle_k = \langle P\rangle_{k+j}$ as meromorphic functions; put $\epsilon = k+j$. For a monomial α^u, $u \ne 0$, take e with $u_e \ne 0$; with the origin 0, $(k\,c_e^{(1)} + \partial_{y_e})\alpha^u$ lies in $\mathcal{L}$ and its value vanishes identically, so $u_e\langle\alpha^u\rangle_\epsilon = -\epsilon\langle c_e^{(1)}\alpha^u\rangle_\epsilon$, whose pole at $\epsilon = 0$ has order at most $n-2$. So $\epsilon^{n-1}\langle P\rangle_\epsilon$ and $\mathrm{CT}(P)\,\epsilon^{n-1}\langle 1\rangle_\epsilon$ have the same value at $\epsilon = 0$. $\langle 1\rangle_\epsilon = P_F(\epsilon\mathbf{m}_1;\epsilon\mathbf{s}_1,\epsilon\mathbf{t}_1)$ (Lemma S20(iv)) is the period of the question of the direction (Definition 17.1) at the strength ϵ, whose walls are positive, and Theorem 17.5(i) gives $\epsilon^{n-1}\langle 1\rangle_\epsilon \to V_1$ as $\epsilon \downarrow 0$; $\epsilon^{n-1}\langle 1\rangle_\epsilon$ is holomorphic at 0, so that is its value. □

Corollary 39.32 (The law read at the low end). *As stated.*

Proof. The law says that $\sum_i p_i(k) f^i$ lies in $\mathcal{L}$, a finite sum $\sum r_{e,l}(k)\big(c_e^{(0)} + k c_e^{(1)} + \partial_{y_e}\big) h_{e,l}$. Integrating against $f^k dy$, by integration by parts for complex k of large real part, where the integrals converge absolutely, $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ there off the poles of the $r_{e,l}$, and so identically as meromorphic functions. At $k = -j$, $j \geq 0$, the term $i \leq j$ has $\langle 1\rangle_{k+i}$ with a pole of order at most $n-1$ and the coefficient $V_1 \mathsf{C}_{j-i}(1)$ of $(k+j)^{-(n-1)}$ (Proposition 39.30, $j(1) = 0$), and $p_i(k) = p_i(-j) + O(k+j)$; the terms $i > j$ are holomorphic there, $k + i$ being near the positive integer $i - j$, where the walls are positive. So the coefficient of $(k+j)^{-(n-1)}$ is $V_1 \sum_{i \leq j} p_i(-j) \mathsf{C}_{j-i}(1) = 0$, with $V_1 > 0$. On the centre ray $f = \prod_j \big(K_j(\alpha) K_j(\alpha^{-1})\big)^{-\sigma_j}$ (38.13). □

Definition 39.33 (The low-end coefficients). As stated.

Proof. By Proposition 38.5(ii) the value of 1 is meromorphic with a pole of order at most $n-1$ at 0, and by Proposition 39.31(ii) at $j = 0$ the coefficient of k^{1-n} is V_1, a sum of products of positive walls. □

Corollary 39.34 (The low-end relation). *As stated.*

Proof. (i) By Proposition 38.5(ii) the value of $\sum_i p_i(k) f^i$, which is $\sum_i p_i(k)\langle 1\rangle_{k+i}$, vanishes identically, and every $\langle 1\rangle_{k+i}$ converges for $\operatorname{Re} k > 0$, the walls $(k+i)\kappa_{1,U}$ having positive real parts. Near $k = -j$, for $i \leq j$, $\lim (k+j)^{n-1}\langle 1\rangle_{k+i} = V_1 \operatorname{CT}(f^{i-j})$ by Proposition 39.31(ii), while for $i > j$ the value $\langle 1\rangle_{k+i}$ is holomorphic at $k = -j$, $k + i$ being near a positive integer. Multiplying by $(k+j)^{n-1}$ and letting $k \to -j$ gives $V_1 \sum_{i \leq \min(h,j)} p_i(-j) \operatorname{CT}(f^{i-j}) = 0$, and $V_1 > 0$.

(ii) Near $k = 0$ the terms $i \geq 1$ are holomorphic, so $p_0(k)\langle 1\rangle_k = p_0(k)\, k^{1-n}\big(\mathsf{L}_0 + O(k)\big)$ is holomorphic at 0; as $\mathsf{L}_0 \neq 0$, p_0 vanishes to order at least $n-1$ there, and the value at 0 of $p_0(k)\langle 1\rangle_k$ is $[k^{n-1}]p_0 \cdot \mathsf{L}_0$.

(iii) With the origin 0, $c^{(0)} = 0$; at $k = 0$ the certificate gives $\sum_i p_i(0) f^i = \sum_e \partial_{y_e} G_e(0)$, and $p_0(0) = 0$ by (ii), $n \geq 2$. The $G_e(0)$ are homogeneous of degree 0, so $\partial_{y_{e_0}} G_{e_0}(0) = -\sum_{e \neq e_0} \partial_{y_e} G_{e_0}(0)$ on the chart, and $\sum_e \partial_{y_e} G_e(0)$ is the divergence of the stated field in the coordinates y_e, $e \neq e_0$. By the divergence theorem its integral over the cube is the flux out of the cube, and as $R \to \infty$ the integral of $\sum_{i \geq 1} p_i(0) f^i$ over the cube tends to $\sum_{i \geq 1} p_i(0)\langle 1\rangle_i$, each f^i, $i \geq 1$, being integrable; by (ii) this is $-[k^{n-1}]p_0 \cdot \mathsf{L}_0$. □

Proposition 39.35 (The low-end coefficients are integrals over the faces of the sectors). *As stated.*

Proof. By Lemma S19 with $g = 1$ and the origin 0, $\langle 1\rangle_k = \sum_\pi \int_{[0,1]^{n-1}} \prod_l q_l^{k\kappa_l - 1} \varphi_\pi^k \, dq$ for $\operatorname{Re} k > 0$. For h smooth on a neighbourhood of the cube, $\sum_A h_{[A]} = h$: the restriction of h to $\{q_l = 0,\ l \notin B\}$ enters with the coefficient $\sum_{A \supseteq B} (-1)^{|A \setminus B|}$, which is 1 for $B = \{2, \dots, n\}$ and 0 otherwise. $h_{[A]}$ depends on q_A only, and where $q_l = 0$, $l \in A$, the terms of B and $B \cup \{l\}$, $l \notin B$, cancel; so, one coordinate at a time (Hadamard's lemma), $h_{[A]} = \prod_{l \in A} q_l \cdot \tilde{h}_A$ with $\tilde{h}_A(q_A) = \int_{[0,1]^A} \big(\prod_{l \in A} \partial_{q_l}\big) h_{[A]}(t\, q_A)\, dt$ smooth, $t\, q_A = (t q_l)_{l \in A}$. With $h = \varphi_\pi^k$, holomorphic in k together with its q-derivatives,

$$\int_{[0,1]^{n-1}} \prod_l q_l^{k\kappa_l - 1} \big(\varphi_\pi^k\big)_{[A]}\, dq = \prod_{l \notin A} \frac{1}{k\kappa_l}\, J_{\pi,A}(k), \qquad J_{\pi,A}(k) = \int_{[0,1]^A} \prod_{l \in A} q_l^{k\kappa_l}\, \tilde{h}_A(q_A, k)\, dq_A,$$

and $J_{\pi,A}$ is holomorphic for $\operatorname{Re} k > -1/\max_l \kappa_l$. So $k^{n-1}\langle 1\rangle_k = \sum_\pi \sum_A k^{|A|} \prod_{l \notin A} \kappa_l^{-1} J_{\pi,A}(k)$ near 0, and $\mathsf{L}_i = \sum_\pi \sum_A \prod_{l \notin A} \kappa_l^{-1} [k^{i-|A|}] J_{\pi,A}$. The Taylor coefficients of $J_{\pi,A}$ at 0 are obtained by differentiating under the integral, the k-derivatives of the integrand being dominated on the cube

by constants times $\prod_{l\in A} q_l^{-\varepsilon}|\log q_l|^N$ for $|k|$ small; by Leibniz's rule, with $\partial_k^m \varphi_\pi^k|_{k=0} = \log^m \varphi_\pi$, $\partial_k^\gamma q^{k\kappa}|_{k=0} = (\kappa \log q)^\gamma$ and $h \mapsto h_{[A]}$, $h \mapsto \tilde{h}_A$ linear and independent of k, the coefficient of $k^{i'}$ is the sum over $m + |\boldsymbol{\gamma}| = i'$ of the integrals of the statement, $\big(\log^m \varphi_\pi\big)_{[A]} / \prod_{l\in A} q_l$ being bounded. For $i = 0$ only $A = \varnothing$ and $m = 0$ enter, giving V_1; for $i = 1$, $A = \varnothing$ with $m = 1$ gives $\sum_\pi \prod_l \kappa_l^{-1} \log \varphi_\pi(0)$, the terms with $|A| = 1$ and $m = 0$ vanishing, $1_{[A]} = 0$; the constant terms of the sector letters are $\theta_{j,T_{j,\pi}^\pm}$ (Theorem 8.5(ii)). For $A = \{2, \dots, n\}$ the exponent m must be at least 1, so the whole cube enters only from $i = n$ on. □

Proposition 39.37 (The high end reads the formal type). *As stated.*

Proof. By Theorem 39.4(i), $\varsigma^{-1} v_V = \mathbf{B} v_V$. Divide by $\lambda^k k^\rho$: $\lambda^{-1}(1 - \frac{1}{k})^\rho \big(a_0 + a_1(k-1)^{-1} + \cdots\big) = \big(\mathbf{B}_0 + \mathbf{B}_1 k^{-1} + \cdots\big)\big(a_0 + a_1 k^{-1} + \cdots\big)$. The order 0 gives $\lambda^{-1} a_0 = \mathbf{B}_0 a_0$, and the order 1 gives $\lambda^{-1}(a_1 - \rho a_0) = \mathbf{B}_0 a_1 + \mathbf{B}_1 a_0$; multiplied by w on the left, with $w\mathbf{B}_0 = \lambda^{-1} w$, it gives $-\lambda^{-1}\rho\, w a_0 = w\mathbf{B}_1 a_0$. For a simple eigenvalue the left and right eigenvectors pair to a nonzero number. A source whose solution does not vanish on $\mathfrak{N}$ has a solution vector of the form of the statement after an integral shift of ρ, the exponents of Theorems 39.20 and 39.22 being defined modulo $\mathbb{Z}$. □

Proposition 39.38 (The spectrum at infinite strength; changes of masters). *As stated.*

Proof. By Theorem 38.6(i) and (ii), $\hat{A}_c$ is $\mathbb{C}(k)$-linear, vanishes on $\mathcal{L}$, and $\hat{A}_c(g) = C_c\, g(\alpha_c) + O(1/k)$ with $C_c = (2\pi)^{(n-1)/2}(\omega_0/dv)(\alpha_c) \neq 0$, the same for every insertion g.

(i) Let $\sum_l \lambda_l(k)[b_l] = 0$ with $\lambda \neq 0$, and multiply λ by a power of k so that every λ_l is regular at ∞ and $\lambda(\infty) \neq 0$. Applying $\hat{A}_c$ and reading the coefficient of k^0 gives $C_c \sum_l \lambda_l(\infty)\, b_l(\alpha_c) = 0$ for every $\alpha_c \in C$; as the β_c span, $\lambda(\infty) = 0$, a contradiction. So the $[b_l]$ are a basis of $\mathfrak{N}$, and the matrix of Sh^{-1} on a basis is unique.

(ii) As in Theorem 39.4(i), $V[f^{-1} b_i] = \varsigma^{-1} V[b_i]$ and $V[f^{-1} b_i] = \sum_l \mathbf{B}_{il} V[b_l]$ for every solution V, whatever the independence of the b_l: $\varsigma^{-1} v_V = \mathbf{B} v_V$ with $v_V = (V[b_l])_l$. For $V = A_c$, of type $\big(f(\alpha_c), -\frac{n-1}{2} + \mathbb{Z}\big)$, the solution vector is $\lambda^k k^\rho \big(a_0 + O(1/k)\big)$ with $\lambda = f(\alpha_c)$, $\rho = -\frac{n-1}{2}$ and $a_0 = C_c \beta_c$, and ς^{-1} multiplies $\lambda^k k^\rho$ by $\lambda^{-1}\big(1 + O(1/k)\big)$ (Definition 39.2); the coefficient of $\lambda^k k^\rho$ in $\varsigma^{-1} v_V = \mathbf{B} v_V$ gives $\lambda^{-1} a_0 = \mathbf{B}(\infty)\, a_0$. Eigenvectors of distinct eigenvalues are independent, and a matrix with an eigenbasis taken among the β_c, $\alpha_c \in C$, is diagonalizable with the eigenvalues of those vectors, every β_c being an eigenvector.

(iii) $\mathrm{Sh}^{-1}\, \tilde{b}_i = \sum_l T_{il}(k-1)[f^{-1} b_l] = \sum_{l,l'} T_{il}(k-1)\mathbf{B}_{ll'}(k)[b_{l'}]$, which is $\sum_{i'} \tilde{\mathbf{B}}_{ii'}(k)\, \tilde{b}_{i'}$, and the solution vector of A_c on the $\tilde{b}_i$ is $T(k)\, v_{A_c}$, which satisfies $\varsigma^{-1}\tilde{v} = \tilde{\mathbf{B}}\tilde{v}$ by the same computation. $T(k)$ is invertible and $\hat{A}_c(b_1)$ has the nonzero constant term C_c, so $T(k)\big(\hat{A}_c(b_l)\big)_l = k^\nu\big(\tilde{\beta}_c + O(1/k)\big)$ with $\nu \in \mathbb{Z}$ and $\tilde{\beta}_c \neq 0$, and the computation of (ii) with $\rho = -\frac{n-1}{2} + \nu$ gives $\tilde{\mathbf{B}}(\infty)\tilde{\beta}_c = f(\alpha_c)^{-1}\tilde{\beta}_c$. Finally $T(k)\,\mathbf{B}(k)\,T(k)^{-1} = \big(T(k-1)T(k)^{-1}\big)^{-1}\tilde{\mathbf{B}}(k)$, which tends to $\tilde{\mathbf{B}}(\infty)$ when $T(k-1)T(k)^{-1} \to I$; it has the characteristic polynomial of $\mathbf{B}(k)$, whose coefficients therefore tend to those of $\tilde{\mathbf{B}}(\infty)$, and the roots of a monic polynomial depend continuously on its coefficients.

(iv) $T(k-1)\,\mathbf{B}(k)\,T(k)^{-1} = S(k-1)^{-1}S(k-1)\,\mathbf{B}_\circ\, S(k)^{-1}S(k) = \mathbf{B}_\circ$, and $T(k-1)T(k)^{-1} = S(k-1)^{-1}S(k) = \begin{bmatrix} 1 & 0 \\ -(k-1) & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. With $S(k)^{-1} = \begin{bmatrix} 1 & 0 \\ -k & 1 \end{bmatrix}$,

$$\mathbf{B}(k) = \begin{bmatrix} x_{11} - x_{12}k & x_{12} \\ (k-1)(x_{11} - x_{12}k) + x_{21} - x_{22}k & (k-1)x_{12} + x_{22} \end{bmatrix},$$

whose entry $(2,1)$ is $-x_{12}k^2 + O(k)$, whose trace is $x_{11} + x_{22} - x_{12}$ and whose determinant is $\det \mathbf{B}_\circ$; the trace of $\mathbf{B}_\circ$ is $x_{11} + x_{22}$. □

Theorem 39.39 (Separating the sources). *As stated.*

Proof. For new masters $\mathbf{b}' = T\mathbf{b}$, T invertible over $\mathbb{C}((1/k))$, $\mathrm{Sh}^{-1} b'_i = \sum_l T_{il}(k-1)\,\mathrm{Sh}^{-1} b_l$, so the matrix of Sh^{-1} becomes $T(k-1)\mathbf{B}(k)T(k)^{-1}$. After a constant change of basis, $\mathbf{B}_0 = \mathrm{diag}(\mathbf{B}_0^{(1)}, \dots, \mathbf{B}_0^{(s)})$ with $\operatorname{spec} \mathbf{B}_0^{(a)} = \Lambda_a$. Seek $T = I + \sum_{i\ge 1} T_i k^{-i}$ and a block diagonal $C = \sum_{i\ge 0} C_i k^{-i}$, $C_0 = \mathbf{B}_0$, with $T(k-1)\mathbf{B}(k) = C(k)T(k)$. Write $T(k-1) = \sum_i \tilde{T}_i k^{-i}$, where $\tilde{T}_i - T_i$ depends on $T_1, \dots, T_{i-1}$ only. The coefficient of k^{-i} is

$$T_i\mathbf{B}_0 - \mathbf{B}_0 T_i = C_i - \mathbf{B}_i + E_i, \qquad E_i = \sum_{0<a<i} \big(C_a T_{i-a} - \tilde{T}_a \mathbf{B}_{i-a}\big) - (\tilde{T}_i - T_i)\mathbf{B}_0,$$

with E_i known from the lower orders. Take C_i the block diagonal part of $\mathbf{B}_i - E_i$, the diagonal blocks of T_i zero, and the block (a,b), $a \ne b$, of T_i the solution X of $X\mathbf{B}_0^{(b)} - \mathbf{B}_0^{(a)} X = (E_i - \mathbf{B}_i)^{(ab)}$, unique since the spectra are disjoint. The blocks of the new masters span the $\hat{\mathfrak{N}}_a$.

If $\bigoplus_a \hat{\mathfrak{N}}'_a$ is another such decomposition, the inclusion of $\hat{\mathfrak{N}}'_a$ followed by the projection onto $\hat{\mathfrak{N}}_b$, $b \ne a$, commutes with the shift; in bases it is a matrix X over $\mathbb{C}((1/k))$ with $C'^{(a)}(k)X(k) = X(k-1)C^{(b)}(k)$. If $X \ne 0$, $X = k^v(X_0 + O(k^{-1}))$ with $X_0 \ne 0$, and the leading order $C_0'^{(a)} X_0 = X_0 C_0^{(b)}$ forces $X_0 = 0$, the spectra being disjoint. So $\hat{\mathfrak{N}}'_a \subseteq \hat{\mathfrak{N}}_a$ for every a, and the sums being direct and equal to the whole space, $\hat{\mathfrak{N}}'_a = \hat{\mathfrak{N}}_a$. A solution of type $(\lambda, \rho + \mathbb{Z})$, extended $\mathbb{C}((1/k))$-linearly and restricted to $\hat{\mathfrak{N}}_b$, has in the basis of $\hat{\mathfrak{N}}_b$ the solution vector $\lambda^k k^\rho w$; if it is not zero, $w = k^{-v}(w_0 + \cdots)$ with $w_0 \ne 0$, and the first statement of Proposition 39.37, whose proof uses only the expansion of the matrix at ∞, gives $C^{(b)}(\infty)w_0 = \lambda^{-1} w_0$, so $\lambda^{-1} \in \Lambda_b$. A symmetry commutes with the shift, so it carries the decomposition to one with the same property, the matrix of Sh^{-1} on the image basis being the same; by uniqueness it maps each $\hat{\mathfrak{N}}_a$ onto itself. □

Corollary 39.40 (Gluing at the two ends). *As stated.*

Proof. With $\mathcal{S}_\mathrm{f}$ the formal solutions in $\mathcal{S}$, $\mathrm{Ann}(\mathcal{S}) \cap \mathfrak{N} = \mathrm{Ann}(\mathcal{S}_\mathrm{f}) \cap \mathfrak{N}_\mathsf{C}$, a submodule of $\mathfrak{N}_\mathsf{C}$ (Theorem 39.4(ii)), and Theorem 39.4(iii) applies. If $\mathfrak{N}$ is irreducible and V_C does not vanish on it, $\mathfrak{N}_\mathsf{C}$ is a submodule other than $\mathfrak{N}$, hence 0. If $\mathfrak{m}$ formal solutions are linearly independent over $\mathbb{C}$ on $\mathfrak{N}$, type by type, Lemma 39.3 makes their normalizations $\mathfrak{m}$ independent linear forms on $\mathfrak{N} \otimes \mathbb{C}((1/k))$, of dimension $\mathfrak{m}$, so no nonzero element of $\mathfrak{N}$ is annihilated by all of them. □

Theorem 39.41 (The closings, the values and the native gauge). *As stated.*

Proof. The origin form is $u\,dy$ with $u = \omega_0/dy$ a unit of $\mathbb{C}[X_F]$, and $\partial_{y_e} \log u = c_e^{(0)}$ (38.2(ii)), so $(k\,c_e^{(1)} + \partial_{y_e})(ug) = u\,(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g$. Multiplication by u therefore carries $\mathcal{L}$ onto the relation space of the ray from the origin with the same direction, commutes with Sh, and carries the values, the closings ($\mathsf{C}_j(g) = \mathrm{CT}(ug\,f^{-j})$) and the expansions at $\alpha^\star$ of the one ray to those of the other; so we may take the origin to be 0, $\omega_0 = dy$.

By Theorem 38.14(i), $0 < f \le f(\alpha^\star)$ on the positive orthant with equality only at $\alpha \propto \alpha^\star$, and $f \le C' e^{-c'\|y\|}$ on the chart. The family is one molecule: a union S of molecules has $c_j(S) = 0$ for every j (7.10), and then $\kappa_{1,S} + \kappa_{1,\mathcal{E}\setminus S} = \kappa_{1,\mathcal{E}} = 0$ (8.16), against the positive walls. The Hessian of $\log K_j(e^y)$ in y is the covariance of the basis indicator $\mathbf{1}_T$ under the basis law at e^y (7.16), every basis having positive weight at a real point, and that of $\log \Psi_j(e^y)$ the covariance of $\mathbf{1}_{\mathcal{E}\setminus T}$; so a real direction u' in the kernel of the Hessian of $\log f$ at a real point has $\langle u', \mathbf{1}_T\rangle$ constant over the bases of every table, and u' is constant on the one molecule (7.8), $u' \in \mathbb{R}\mathbf{1}$. The Hessian H of $\log f$ at $\alpha^\star$ is negative definite on $\mathbb{R}^\mathcal{E}/\mathbb{R}\mathbf{1}$, and $\alpha^\star$ is a nondegenerate native gauge. Write $f^\star = f(\alpha^\star)$, $\hat{A}_\star = (f^\star)^{-k} k^{(n-1)/2} A_\star$, and, for $m = \sum_l r_l(k)[g_l]$, $\langle m\rangle_k = \sum_l r_l(k)\langle g_l\rangle_k$,

$V_{\mathsf C}(m)(-j) = \sum_l r_l(-j)\,\mathsf C_j(g_l)$ for all large j, and $Q(k)$ for a common denominator of the r_l. By Theorem 38.6(i), the orthant near $\alpha^\star$ being a steepest-descent disc and the rest of it contributing $O((f^\star-\delta)^k)$ for some $\delta>0$, $\langle g\rangle_k$ has the asymptotic expansion $A_\star(g)$, and $\langle m\rangle_k$ the expansion $A_\star(m)$.

(ii)⇒(iii). If $\langle m\rangle_k = 0$ at all large integers, $A_\star(m)$ is the asymptotic expansion of 0 along the integers, so $A_\star(m)=0$.

(iii)⇒(ii), and the continuation. For $0<t<f^\star$ the level sets $\Sigma_t=\{f=t\}$ of the real quotient are compact real-analytic hypersurfaces without critical points. By the coarea formula, $\langle g\rangle_k = \int_0^{f^\star} t^k \rho_g(t)\,dt$ for $\operatorname{Re} k$ large, with $\rho_g(t)=\int_{\Sigma_t} g\,dy/df$ real-analytic on $(0,f^\star)$, and by the Morse lemma at $\log\alpha^\star$, $\rho_g(t)=(f^\star-t)^{(n-3)/2}h_g(t)$ with h_g real-analytic at $f^\star$. With $t=f^\star e^{-s}$, $\langle g\rangle_k = (f^\star)^{k+1}\mathcal L[p_g](k+1)$, $\mathcal L$ the Laplace transform and $p_g(s)=\rho_g(f^\star e^{-s})$, real-analytic for $s>0$ and $s^{(n-3)/2}$ times a function analytic at 0. Write $Q(k)r_l(k)=R_l(k+1)$ with polynomials R_l. On distributions supported in $[0,\infty)$, $\mathcal L[Dv](\kappa)=\kappa\,\mathcal L[v](\kappa)$ with $D=d/ds$, so $Q(k)\langle m\rangle_k = (f^\star)^{k+1}\mathcal L[E](k+1)$ for $\operatorname{Re} k$ large, with $E=\sum_l R_l(D)\big(p_{g_l}\mathbf 1_{s>0}\big)$. On $(0,\infty)$, E is the real-analytic function $\sum_l R_l(d/ds)\,p_{g_l}$; near 0 it is a convergent sum $\sum_\lambda c_\lambda s^\lambda$ times $\mathbf 1_{s>0}$, a finite part where needed, with $\lambda\in\frac{n-3}{2}+\mathbb Z$ and never a negative integer (for odd n the function part is analytic at 0), plus finitely many derivatives $\delta^{(b)}$ of the Dirac measure at 0. By Watson's lemma

$$\mathcal L[E](\kappa)\ \sim\ \sum_\lambda c_\lambda\,\Gamma(\lambda+1)\,\kappa^{-\lambda-1}+\sum_b d_b\,\kappa^b,$$

in distinct powers, with $\Gamma(\lambda+1)\neq 0$, and this is the expansion of $(f^\star)^{-k-1}Q(k)\langle m\rangle_k$, which is $(f^\star)^{-1}Q(k)k^{-(n-1)/2}\hat A_\star(m)$. So if $A_\star(m)=0$, every c_λ and every d_b vanishes, E vanishes near 0, hence on $(0,\infty)$ by analyticity, hence $E=0$: $Q(k)\langle m\rangle_k=0$ for $\operatorname{Re} k$ large, and the continuation of $\langle m\rangle_k$ vanishes identically.

(iii)⇒(i). By what precedes the continuation of $\langle m\rangle_k$ vanishes, and by Proposition 39.30, $V_1V_{\mathsf C}(m)(-j)=\lim_{k\to -j}(k+j)^{n-1}\langle m\rangle_k=0$ for all large j.

(i)⇒(iii), and the expansion of the closings. The constant term of a Laurent polynomial is its mean over every torus $|\alpha_e|=\rho_e$. On the one through $\alpha^\star$, $\alpha=\alpha^\star e^{i\theta}$ with $\theta\in\mathbb R^{\mathcal E}/(2\pi\mathbb Z^{\mathcal E}+\mathbb R\mathbf 1)$ and $d\theta$ the measure of the chart $\theta_{e_0}=0$,

$$\mathsf C_j(g)=(2\pi)^{-(n-1)}\int g(\alpha^\star e^{i\theta})\,f(\alpha^\star e^{i\theta})^{-j}\,d\theta,\qquad j\geq j(g).$$

The letters have nonnegative coefficients, so $|K_j(\alpha^\star e^{i\theta})|\leq K_j(\alpha^\star)$, $|\Psi_j(\alpha^\star e^{i\theta})|\leq\Psi_j(\alpha^\star)$ and $|f^{-1}(\alpha^\star e^{i\theta})|\leq 1/f^\star$, with equality only when $\langle\theta,\mathbf 1_T\rangle$ is the same modulo 2π for all bases T of every table. For linked marks e,e' some basis T has $e\in T$, $e'\notin T$ and $(T\setminus\{e\})\cup\{e'\}$ a basis (7.7), so equality gives $\theta_e\equiv\theta_{e'}$; the family being one molecule, θ is constant modulo 2π, and $\alpha=\alpha^\star$. So the torus outside a neighbourhood of $\alpha^\star$ contributes $O\big((1-\delta)^j(f^\star)^{-j}\big)$ for some $\delta>0$, $g\,f^{-j(g)}$ being bounded on it. Near $\theta=0$, $\log f(\alpha^\star e^{i\theta})-\log f^\star=-\frac12\theta^{\mathsf T}H\theta+O(|\theta|^3)$, whose real part has a nondegenerate minimum at 0, and Laplace's method there is the formal expansion at $\alpha^\star$ under the substitution $y=i\theta$, $k=-j$. The coefficient of k^{-i} in $\hat A_\star(g)$ is a sum of Gaussian moments of covariance $(-H)^{-1}$ and of degree $2i+2L$, each times one Taylor coefficient of g and L of $\log f$; the substitution multiplies such a term by $i^{2i+2L}(-1)^L=(-1)^i$ and keeps the covariance, while $(2\pi i)^{-(n-1)}\prod_{e\neq e_0}d\alpha_e/\alpha_e=(2\pi)^{-(n-1)}d\theta$. This gives the expansion of $j^{(n-1)/2}(f^\star)^j\,\mathsf C_j(g)$ in the statement, and the same holds for m, with $\hat A_\star(m)=\sum_l r_l(k)\hat A_\star(g_l)$ read at $k=-j$ as a series in $1/j$. If $V_{\mathsf C}(m)=0$, that series is the expansion of 0, so $\hat A_\star(m)=0$. □

Corollary 39.43 (When the closings miss nothing). *As stated.*

Proof. By Theorem 39.41, $\mathrm{Ann}(V_{\mathsf{C}}) \cap \mathfrak{N}$, $\mathrm{Ann}(\langle\cdot\rangle) \cap \mathfrak{N}$ and $\mathrm{Ann}(A_\star) \cap \mathfrak{N}$ are one submodule (Theorem 39.4(ii)), and it is 0 exactly when $\alpha^\star$ sees every nonzero submodule, being itself a submodule on which $A_\star$ vanishes. A symmetry maps the positive orthant onto itself and keeps f (Proposition 39.7(ii)), so it fixes $\alpha^\star$, the one point there of largest value, and $A_\star \circ \phi^* = A_\star$ (Theorem 39.8(iii)); so $A_\star$, and with it the values and the closings, vanish on $(1 - e_G)\mathfrak{M}$.

For the rest, $\mathbf{B}$ regular at ∞ makes the $\mathbb{C}[[1/k]]$-lattice Λ spanned by the masters stable under Sh^{-1}. For a submodule $\mathfrak{N}'$, $\Lambda' = \Lambda \cap (\mathfrak{N}' \otimes \mathbb{C}((1/k)))$ is a lattice of $\mathfrak{N}' \otimes \mathbb{C}((1/k))$, stable as well and saturated in Λ, so in a basis of Λ' the matrix of Sh^{-1} is regular at ∞, with value there the restriction of $\mathbf{B}(\infty)$ to $\Lambda'/k^{-1}\Lambda' \subseteq \Lambda/k^{-1}\Lambda$. The proof of Theorem 39.39 runs for any basis over $\mathbb{C}((1/k))$ in which the matrix is regular at ∞, so it applies to $\mathfrak{N}'$, and each of its pieces maps by zero to the pieces of $\mathfrak{N}$ of other spectra, as in the uniqueness there, so $\mathfrak{N}' \otimes \mathbb{C}((1/k))$ is the direct sum of its intersections with the $\hat{\mathfrak{N}}_a$. $A_\star$ vanishes on the pieces other than $\hat{\mathfrak{N}}_\star$ (Theorem 39.39), and not on $\hat{\mathfrak{N}}_\star$, since it does not vanish on $\mathfrak{N}$; $\hat{\mathfrak{N}}_\star$ being a line, $A_\star$ vanishes on $\mathfrak{N}'$ exactly when $\hat{\mathfrak{N}}_\star \not\subseteq \mathfrak{N}' \otimes \mathbb{C}((1/k))$. If every nonzero submodule contains $\hat{\mathfrak{N}}_\star$, any two of them meet in a nonzero submodule, completion commuting with intersection, so the intersection of all of them is nonzero and is the only minimal one. Of two nonzero submodules with intersection 0 at most one contains $\hat{\mathfrak{N}}_\star$, and the closings annihilate the other. □

Corollary 39.44 (What the closings determine). *As stated.*

Proof. The kernel of the values is a submodule, so Sh induces on the value module a bijection with $\mathrm{Sh}(r\,m) = \varsigma(r)\,\mathrm{Sh}(m)$, and by Theorem 39.41, V_{C} induces on it a solution that annihilates nothing; the proofs of Theorem 39.4(i), (iii) and (iv) use no more than this. For the recurrences, $\sum_i p_i(k)\langle 1\rangle_{k+i}$ is the value of $m = \sum_i p_i(k)\,\mathrm{Sh}^i[1]$, and $V_{\mathsf{C}}(m)(-j) = \sum_i p_i(-j)\,\mathsf{C}_{j-i}(1)$; by Theorem 39.41 the one vanishes for all large arguments exactly when the other does, so the least orders agree. □

Proposition 39.45 (The closings measure what they miss). *As stated.*

Proof. The closings are a solution (Theorem 39.29), and as in the proof of Theorem 39.4(iii), $(P, Q) \in \mathcal{S}_D$ exactly when every row of $Q\mathbf{B}_{\mathbf{b}} - P$, read in $\mathfrak{N}$, lies in $\mathfrak{A} = \mathrm{Ann}(V_{\mathsf{C}}) \cap \mathfrak{N}$. In the coordinates of the masters $\mathfrak{A}$ is a d-dimensional subspace of $\mathbb{C}(k)^{1\times\mathfrak{m}}$; let $\mathfrak{A}_D$ be its rows of polynomials of degree at most D. Projecting to d coordinates on which $\mathfrak{A}$ projects isomorphically gives $\dim \mathfrak{A}_D \le d(D+1)$, and polynomial rows $a_1, \ldots, a_d$ spanning $\mathfrak{A}$, of degrees ν_i, give the independent rows $\sum_i c_i a_i$, $\deg c_i \le D - \nu_i$, so $\dim \mathfrak{A}_D = dD + O(1)$. On $\mathcal{S}_D$ the map $(P, Q) \mapsto Q$ has fibres that are empty or translates of $\mathfrak{A}_D^{\mathfrak{m}}$, each row of P being fixed modulo $\mathfrak{A}$ by Q; its image lies in the polynomials of degree at most D and contains cQ_0 for $\deg c \le D - D_0$. So $\mathfrak{m}\dim \mathfrak{A}_D + D - D_0 + 1 \le \dim \mathcal{S}_D \le \mathfrak{m}\dim \mathfrak{A}_D + D + 1$ for $D \ge D_0$, which is $(\mathfrak{m}d + 1)D + O(1)$; for $d = 0$, Theorem 39.4(iv) gives $\dim \mathcal{S}_D = \max(0, D - D_0 + 1)$, whose steps are 0 and 1. □

Proposition 39.46 (A table one mark larger). *As stated.*

Proof. By Corollary 20.19, $\mathcal{B}'$ has no loop or coloop and its letters agree, $\Psi' = K'$. The bases of $\mathcal{B}'$ through g, less g, are the bases of $\mathcal{B}$, and complementation in $\mathcal{E}'$ carries the bases through g onto the bases avoiding g, with the same weights; so $K' = \sum_{T \not\ni g} \theta'_T \alpha^T + \alpha_g \sum_{T \ni g} \theta'_T \alpha^{T\setminus g} = \Psi + \alpha_g K$, and $K(\mathbf{1}) = \Psi(\mathbf{1})$. The deletion $\mathcal{B}' \setminus g$ is the dual of the contraction, and a connected table has a connected deletion or a connected contraction at each mark (Tutte), duality keeping connectedness; so $\mathcal{B}$ is connected, and both centre rays have positive walls (Definition 38.13). Write $x_{a,j}(h) = \alpha_g^a K'^{-j} h$; the monomials and the chances of $\mathcal{B}'$ are of this form, so these span $\mathbb{C}[X_{F'}]$.

(i) For α in the positive orthant, $\int_0^\infty \rho^{\nu-1}(c_0+c_1\rho)^{-s}\,d\rho = \Gamma_{\,1}^{\,2}\!\left[\begin{matrix}\nu,\ s-\nu\\ s\end{matrix}\right] c_0^{\nu-s}c_1^{-\nu}$ gives

$$\int_0^\infty \alpha_g^{k+a}(\Psi+\alpha_g K)^{-2k-j}\,\frac{d\alpha_g}{\alpha_g} = \Gamma_{\,1}^{\,2}\!\left[\begin{matrix}k+a,\ k+j-a\\ 2k+j\end{matrix}\right]\Psi^{a-j-k}K^{-a-k}, \qquad k>\max(-a,a-j),$$

and with $f'^{k}=\alpha^{k\mathbf{1}}\alpha_g^k K'^{-2k}$, $f^k=\alpha^{k\mathbf{1}}(K\Psi)^{-k}$ and $\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}k+a,\ k+j-a\\ 2k+j\end{matrix}\right] = r_{a,j}(k)\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}k,\ k\\ 2k\end{matrix}\right]$ this is the identity for $g'=x_{a,j}(h)$. The integral is linear in g', so on a combination of the $x_{a,j}(h)$ the defining formula depends only on the function they sum to, both sides being rational in k and equal for all large k; so π_* is well defined and $\mathbb{C}(k)$-linear on $\mathbb{C}[X_{F'}]\otimes\mathbb{C}(k)$, and the identity holds for every insertion. On the chart $y_{e_0}=0$ the form of F' is $dy\,d\alpha_g/\alpha_g$, and its positive orthant is that of F times $(0,\infty)$; Fubini gives the values. The constant term in α_g of $x_{a,j}(h)f'^{-J}=\alpha_g^{a-J}\alpha^{-J\mathbf{1}}h\,(\Psi+\alpha_g K)^{2J-j}$ is $\binom{2J-j}{J-a}\Psi^{J-j+a}K^{J-a}\alpha^{-J\mathbf{1}}h$, and for $J\ge|a|+j$, $\Gamma(x-J)/\Gamma(-J)=(-1)^x J!/(J-x)!$ gives $r_{a,j}(-J)=\binom{2J-j}{J-a}\big/\binom{2J}{J}$; so that constant term is $\binom{2J}{J}\,\pi_*\big(x_{a,j}(h)\big)\big|_{k=-J}f^{-J}$, and the constant term in the remaining scales gives the closings.

On $X_{F'}$, $c_g'^{(1)}=\partial_{y_g}\log f'=1-2\alpha_g K/K'$, which is -1 where $\Psi=0$ and 1 where $K=0$; where $K\Psi\neq0$ it vanishes exactly at $\alpha_g=\Psi/K$, where $K'=2\Psi$, $f'=f/4$ and $\partial_{y_g}^2\log f'=-2\,\frac{\alpha_g K}{K'}\,\frac{\Psi}{K'}=-\frac12$. So $\log f(\alpha)=\log 4+\log f'(\alpha,\Psi/K)$ on X_F, the defects of F' at $(\alpha,\Psi/K)$ in the marks of $\mathcal{E}$ are those of F at α, the native gauges of F' are the points $\alpha_{c'}$, and the Hessian of $\log f$ at α_c is the Schur complement of the entry $-\frac12$ in the Hessian of $\log f'$ at $\alpha_{c'}$, whose determinant is $-\frac12$ times that of the complement. Over a steepest-descent disc D_c, with Morse coordinates v, put $\alpha_g=e^s\,\Psi/K$; then $d\alpha_g/\alpha_g=ds$ on the fibre and

$$\log f' = \log f(\alpha_c) - \log 4 - \tfrac12\textstyle\sum_i v_i^2 - 2\log\cosh\frac{s}{2},$$

so with the real-analytic coordinate w of the sign of s and $\frac12 w^2=2\log\cosh\frac{s}{2}$, the image of $D_c\times\{|s|\le\varepsilon\}$ is a steepest-descent disc at $\alpha_{c'}$. The integral over all s is the Beta corner, the part with $|s|>\varepsilon$ is exponentially smaller, and the expansion at $\alpha_{c'}$ is that of the Beta corner times $A_c(\pi_*g')$.

(ii) Let $\rho_e'(g')=(k\,c_e'^{(1)}+\partial_{y_e})g'$, so that $f'^{k}\rho_e'(g')=\partial_{y_e}(f'^{k}g')$, and likewise ρ_e for F. Differentiating the identity of (i) under the integral, $\pi_*\rho_e'=\rho_e\pi_*$ for $e\in\mathcal{E}$; and $\pi_*\rho_g'=0$, the integral of $\partial_{y_g}(f'^{k}g')$ over $(0,\infty)$ vanishing for large k. Both are identities between elements of $\mathbb{C}[X_F]\otimes\mathbb{C}(k)$ that hold for all large k, so they hold. Next,

$$\rho_g'\big(x_{a,j}(h)\big) = (k+a)\,x_{a,j}(h) - (2k+j)\,x_{a+1,j+1}(Kh), \qquad x_{a,j}(h)=x_{a,j+1}(\Psi h)+x_{a+1,j+1}(Kh),$$

so modulo the span $\mathcal{L}_g'$ of the $\rho_g'(g')$

$$x_{a,j}(h)\equiv\frac{2k+j}{k+a}\,x_{a+1,j+1}(Kh)\equiv\frac{2k+j}{k+j-a}\,x_{a,j+1}(\Psi h).$$

A finite sum $z=\sum_i c_i x_{a_i,j_i}(h_i)$ is therefore congruent modulo $\mathcal{L}_g'$ to one term $x_{A,A+B}(H)$, for any $A\ge\max_i a_i$ and $B\ge\max_i(j_i-a_i)$, with H a Laurent polynomial, and $\pi_*z=r_{A,A+B}K^{-A}\Psi^{-B}H$; if $\pi_*z=0$ then $H=0$ and $z\in\mathcal{L}_g'$. And π_* is onto: $\pi_*x_{a,a+b}(\alpha^u)=r_{a,a+b}\,\alpha^u K^{-a}\Psi^{-b}$ for $a,b\ge0$, with $r_{a,a+b}\neq0$, and these span $\mathbb{C}[X_F]$. So π_* is a bijection of $(\mathbb{C}[X_{F'}]\otimes\mathbb{C}(k))/\mathcal{L}_g'$ onto $\mathbb{C}[X_F]\otimes\mathbb{C}(k)$ that carries $\mathcal{L}'=\mathcal{L}_g'+\sum_{e\in\mathcal{E}}\rho_e'(\cdot)$ onto $\sum_{e\in\mathcal{E}}\rho_e(\cdot)=\mathcal{L}$, and it induces a bijection $\mathfrak{M}'\to\mathfrak{M}$. Finally $\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}k+1,\ k+1\\ 2k+2\end{matrix}\right]=\frac{k}{2(2k+1)}\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}k,\ k\\ 2k\end{matrix}\right]$, so $\pi_*(f'g')=\frac{k}{2(2k+1)}\,f\,\varsigma(\pi_*g')$.

(iii) By (ii) T_β is a $\mathbb{C}(k)$-linear bijection of $\mathfrak{M}$. ϕ_β is a symmetry of F' (Proposition 39.7(ii)), so ϕ_β^* commutes with Sh' (Theorem 39.8(i)); with $c = \frac{k}{2(2k+1)}$, $\mathrm{Sh} = c^{-1}\pi_* \mathrm{Sh}'\, \pi_*^{-1}$, and then $T_\beta \mathrm{Sh} = c^{-1}\pi_*\phi_\beta^* \mathrm{Sh}'\, \pi_*^{-1} = c^{-1}\pi_* \mathrm{Sh}'\, \phi_\beta^* \pi_*^{-1} = \mathrm{Sh}\, T_\beta$. If $\beta(g) = g$, comparing the coefficients of α_g^0 and α_g^1 in $K' \circ \phi_\beta = K'$ gives $K \circ \phi_\beta = K$ and $\Psi \circ \phi_\beta = \Psi$, so ϕ_β restricted to $\mathcal{E}$ is a symmetry of F and $\pi_* \phi_\beta^* = \phi_\beta^* \pi_*$ on the $x_{a,j}(h)$. For the transposition of e and g, $\phi_\beta^* \alpha^u = x_{u_e,0}(\alpha^{u - u_e \mathbf{1}_e})$, and π_* of it is the stated class, with $r_{u_e,0}(k) = \Gamma(k+u_e)\Gamma(k-u_e)/\Gamma(k)^2$. For the Cremona, $K' \circ \mathrm{cr} = \alpha^{-\mathbf{1}} \alpha_g^{-1} K'$, $K \circ \mathrm{cr} = \alpha^{-\mathbf{1}} \Psi$ and $\Psi \circ \mathrm{cr} = \alpha^{-\mathbf{1}} K$ (Proposition 39.7(i)), so $\mathrm{cr}^*\, x_{a,j}(h) = x_{j-a,j}\big(\alpha^{j\mathbf{1}}\, h \circ \mathrm{cr}\big)$, and both $\pi_* \,\mathrm{cr}^*\, x_{a,j}(h)$ and $\mathrm{cr}^*\, \pi_* x_{a,j}(h)$ are $r_{a,j}\, \Psi^{-a} K^{a-j} \alpha^{j\mathbf{1}}\, h \circ \mathrm{cr}$, as $r_{j-a,j} = r_{a,j}$. The group generated is the image of the finite group generated by the ϕ_β and the Cremona of F'.

(iv) The values and the closings of F' are kept by every ϕ_β and by its Cremona (Proposition 39.7(ii); the torus and the constant term are kept by every permutation of the scales and by their inversion). By (i) the values and the closings of F are those of F' composed with π_*^{-1}, divided by $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}k,\ k\\ 2k\end{bmatrix}$ and by $\binom{2j}{j}$, which do not depend on the insertion; so they are invariant under G'. The native gauge of the direction of F' on the positive orthant is unique up to scale (Theorem 38.14(i)), so it is fixed by every ϕ_β and by the Cremona, and it is $\mathbf{1}$; on the orthant $f' \le f/4$, with equality exactly at $\alpha_g = \Psi/K$, so it lies over $\alpha^\star$, and $\alpha^\star = \mathbf{1}$. Then $A'_\star \circ \phi_\beta^* = A'_\star$ (Theorem 39.8(iii)), and $A_\star \circ T_\beta = A_\star$ by (i). A solution V invariant under G' has $V = V \circ e_{G'}$ and vanishes on $(1 - e_{G'})\mathfrak{M}$. If $G \subseteq G'$, then $e_{G'} e_G = e_{G'}$, so $e_{G'}$ maps $\mathfrak{M}^G$ onto $\mathfrak{M}^{G'} \subseteq \mathfrak{M}^G$, and $\mathfrak{M}^G = e_{G'}\mathfrak{M}^G \oplus (1 - e_{G'})\mathfrak{M}^G$, both submodules since $e_{G'}$ commutes with Sh. Last, π_* intertwines the action on $\mathfrak{M}'$ with G', so it carries the invariant module of F' onto $\mathfrak{M}^{G'}$. □

Corollary 39.47 (The cousins of a fused table). *As stated.*

Proof. $\mathcal{B}^\sharp$ is connected by Corollary 30.7(ii), and identically self-dual. For a mark x of $\mathcal{E} \sqcup G$ the cousin at x is the contraction $\mathcal{B}^\sharp/x$, and Proposition 39.46 applies with $(\mathcal{B}', \theta') = (\mathcal{B}^\sharp, \theta^\sharp)$, its $n + c - 1 \ge 2$ marks other than x, and the mark x in the place of g. □

Proposition 39.48 (What a table one mark larger hides from the closings). *As stated.*

Proof. (i) For a nondegenerate native gauge $\alpha_{c'}$ of F' and $\gamma \in \Gamma'$, $A'_{c'} \circ \gamma^*$ is a nonzero constant multiple of the expansion over the steepest-descent disc $\gamma(D_{c'})$ at $\gamma(\alpha_{c'})$, and $A'_{c'} \circ \psi^* = A'_{c'}$ when $\psi \in \Gamma'$ fixes $\alpha_{c'}$ (proof of Theorem 39.8(iii)). Two γ that carry $\alpha_{c'_0}$ to one point differ by an element of its stabilizer, and $(\gamma\psi)^* = \psi^*\gamma^*$, so the transported $A'_{c'}$ is well defined, and $A'_{c'} \circ \chi^* = A'_{\chi(c')}$ for every $\chi \in \Gamma'$. By Proposition 39.46(i), A_c is a nonzero constant multiple of the expansion of F at α_c: a solution of the type $\big(f(\alpha_c), -\frac{n-1}{2} + \mathbb{Z}\big)$, $n = |\mathcal{E}|$, whose leading coefficient on an insertion h is $C_c\, h(\alpha_c)$ with $C_c \ne 0$ (Theorem 38.6(i)). The native gauges below one orbit O'_a have one native value, $4f'$ at the points of O'_a, which Γ' keeps, and so one type.

Every element of G' is $T_\gamma = \pi_*\gamma^*\pi_*^{-1}$ for some $\gamma \in \Gamma'$ (Proposition 39.46(iii)), and $A_c \circ T_\gamma = \Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}k,\ k\\ 2k\end{bmatrix}^{-1} A'_{\gamma(c')} \circ \pi_*^{-1} = A_{\gamma\cdot c}$, with $\gamma \cdot c$ the native gauge below $\gamma(\alpha_{c'})$. For $\phi \in G$, $\phi^* = T_\gamma$ with γ the map of the same kind on the torus of F', and the lift commutes with ϕ: ϕ_β, $\beta(g) = g$, keeps K and Ψ, and $\Psi(\alpha^{-1})/K(\alpha^{-1}) = K(\alpha)/\Psi(\alpha)$ (Proposition 7.2). So $\gamma \cdot c = \phi(\alpha_c)$ and $A_c \circ \phi^* = A_{\phi(\alpha_c)}$. Let $O_{a,1}, \dots, O_{a,s_a}$ be the orbits of G below O'_a and $V_{a,i} = \sum_{c \in O_{a,i}} A_c$; each $\phi \in G$ permutes $O_{a,i}$, so $V_{a,i}$ is invariant under G. For $x \in \mathfrak{M}^{G'}$ and native gauges α_c, $\alpha_{\tilde c}$ below O'_a, with $\gamma(\alpha_{c'}) = \alpha_{\tilde c'}$, $A_{\tilde c}(x) = A_c(T_\gamma x) = A_c(x)$; so $V_{a,i}(x) = |O_{a,i}|\, A_c(x)$ for any α_c below O'_a, and

$$W_{a,i} = |O_{a,i}|^{-1} V_{a,i} - |O_{a,1}|^{-1} V_{a,1}, \qquad 2 \le i \le s_a,$$

vanish on $\mathfrak{M}^{G'}$.

The $V_{a,i}$ of one type are linearly independent over $\mathbb{C}$ on $\mathfrak{M}^G$. Let $\sum \gamma_{a,i} V_{a,i}$ vanish on $\mathfrak{M}^G$, the sum over those of one type, and fix (a,i) and $\alpha_c \in O_{a,i}$. As in the proof of Theorem 39.8(iii), take an insertion h with $h(\alpha_c) = 1$ that vanishes to the order $N \geq 2$ at the other native gauges below $O'_1, \dots, O'_q$, which are finitely many; the expansion at such a point is smaller than the leading power of the type by the factor $k^{-\lceil N/2 \rceil}$. The class of $e_G h$ lies in $\mathfrak{M}^G$, and every $V_{a',i'}$, being invariant, takes the same value at $e_G h$ and at h; so the leading coefficient of the relation at $e_G h$ is $\gamma_{a,i} C_c$, and $\gamma_{a,i} = 0$. Each $W_{a,i}$ involves its own $V_{a,i}$, so the $W_{a,i}$ of one type are linearly independent over $\mathbb{C}$ on $\mathfrak{M}^G$. They vanish on $\mathfrak{M}^{G'}$, and $\mathfrak{M}^G = \mathfrak{M}^{G'} \oplus (1 - e_{G'})\mathfrak{M}^G$ (Proposition 39.46(iv)), so a combination of them that vanishes on $(1 - e_{G'})\mathfrak{M}^G$ vanishes on $\mathfrak{M}^G$: they are linearly independent over $\mathbb{C}$ on $(1 - e_{G'})\mathfrak{M}^G$, type by type. Lemma 39.3 gives the bound; $(1 - e_{G'})\mathfrak{M}^G$ is a submodule, $e_{G'}$ commuting with Sh, and the closings, the values and $A_\star$ vanish on it by Proposition 39.46(iv). The solution of the last statement is $-W_{a,2}$ for an indexing with $O_{a,1} = O_1$ and $O_{a,2} = O_2$.

(ii) A nonempty proper $S \subseteq [2r+2]$ meets the bases of $\binom{[2r+2]}{r+1}$ in every number of marks from $\max(0, |S| - r - 1)$ to $\min(|S|, r+1)$, so $c(S) > 0$ and the table is connected (Definition 8.1); the complement of a set of $r+1$ of its marks has $r+1$ marks, so it is identically self-dual; and its contraction at g (Definition 7.3) has the bases $T \setminus \{g\}$, $T \ni g$, the sets of r marks of $[2r+1]$, with $K = e_r$ and $\Psi = e_{r+1}$, the elementary symmetric polynomials of $\alpha_1, \dots, \alpha_{2r+1}$. So Proposition 39.46 applies, $|\mathcal{E}| = 2r + 1 \geq 5$. Every bijection of $[2r+2]$ is an automorphism, so G consists of the ϕ_β, β a bijection of $[2r+1]$, and their composites with the Cremona. These are the symmetries of the four kinds of the centre ray (Proposition 39.7(ii), the masses being $\mathbf{1}$ and the exponents $s = t = 1$): a bijection that exchanged the list would carry its bases, of r marks, onto complements of bases, of $r+1$ marks, so there are no exchanges and no reversed exchanges.

At a sign η with i coordinates -1 among $l+i$, $e_m(\eta)$ is the coefficient of z^m in $(1-z)^i(1+z)^l$. With $\binom{2r}{r-1} = \binom{2r}{r+1} = rC_r$, $\binom{2r-1}{r-1} = \binom{2r-1}{r} = \frac{r+1}{2} C_r$ and $\binom{2r-1}{r-2} = \frac{r-1}{2} C_r$,

$$[z^r](1-z)(1+z)^{2r} = C_r, \qquad [z^r](1-z)^2(1+z)^{2r-1} = -C_r,$$

so $K(\eta) = e_r(\eta) = \pm C_r \neq 0$ at the signs of F with one and with two scales -1, which are native gauges of the native value $K(\eta)^{-2} = C_r^{-2}$ (Theorem 38.17(ii)). The bijections of $[2r+1]$ act transitively on the marks and on the pairs of marks, the Cremona fixes every sign, and a sign with one coordinate -1 is not a multiple of one with two, $2r + 1 \geq 5$: the signs form two orbits of G. At a sign $\eta^{-1} = \eta$, so $\Psi(\eta) = \eta^{\mathcal{E}} K(\eta)$ (Proposition 7.2), and the native gauge above η is $(\eta, \eta^{\mathcal{E}})$, with $\eta^{\mathcal{E}} = -1$ for one scale -1 and $\eta^{\mathcal{E}} = 1$ for two: the native gauges above the two kinds are the $\binom{2r+2}{2}$ signs of F' with two scales -1, each once, and the bijections of $[2r+2]$ act transitively on them.

At such a sign η', with $D = \{a_1, a_2\}$ and C as in the statement, $a \in D$ and $c, c' \in C$, the Hessian of $-\log f'$ is 2κ, κ the covariances of the indicators X_e of the marks under the signed measure $\mathbb{P}(T) = \eta'^T / K'(\eta')$ on the sets T of $r+1$ marks (Theorem 38.17(ii), $K' = \Psi'$); as a matrix on $\mathbb{C}^{\mathcal{E}'}$ it has zero row sums, $\sum_e X_e = r + 1$. From $[z^{r+1}](1-z)^2(1+z)^{2r} = 2rC_r - 2(r+1)C_r$, $[z^{r-1}](1-z)(1+z)^{2r-1} = C_r$ and, with $\binom{2r-2}{r-1} = \frac{r(r+1)}{2(2r-1)} C_r$,

$$[z^{r-1}](1-z)^2(1+z)^{2r-2} = \binom{2r-2}{r-1}\Big(1 - \frac{2(r-1)}{r} + \frac{(r-1)(r-2)}{r(r+1)}\Big) = -\frac{r-2}{2r-1}\, C_r,$$

$$K'(\eta') = -2C_r, \qquad \mathbb{P}(e \in T) = \tfrac{1}{2}, \qquad \mathbb{P}(a_1, a_2 \in T) = \frac{rC_r}{-2C_r} = -\frac{r}{2},$$

$$\mathbb{P}(a,c\in T)=\frac12,\qquad \mathbb{P}(c,c'\in T)=\frac{r-2}{2(2r-1)},$$

the first for every mark, as $\eta'_e e_r(\eta'_{\hat e})=-C_r$ for $e\in D$ and for $e\in C$. So $\kappa_{ee}=\frac14$, $\kappa_{a_1a_2}=-\frac{2r+1}{4}$, $\kappa_{ac}=\frac14$ and $\kappa_{cc'}=-\frac{3}{4(2r-1)}$, and

$$\kappa(\mathbf{1}_{a_1}-\mathbf{1}_{a_2})=\tfrac{r+1}{2}(\mathbf{1}_{a_1}-\mathbf{1}_{a_2}),\qquad \kappa(\mathbf{1}_c-\mathbf{1}_{c'})=\tfrac{r+1}{2(2r-1)}(\mathbf{1}_c-\mathbf{1}_{c'}),$$
$$\kappa(r\mathbf{1}_D-\mathbf{1}_C)=-\tfrac{r+1}{2}(r\mathbf{1}_D-\mathbf{1}_C),$$

the last from its coordinates $r(\frac14-\frac{2r+1}{4})-\frac{2r}{4}=-\frac{r(r+1)}{2}$ at a and $\frac{2r}{4}-\frac14+\frac34=\frac{r+1}{2}$ at c, where $r\mathbf{1}_D-\mathbf{1}_C$ has the coordinates r and -1. With $c_1\in C$, the vectors $\mathbf{1}_{a_1}-\mathbf{1}_{a_2}$, $r\mathbf{1}_D-\mathbf{1}_C$ and $\mathbf{1}_{c_1}-\mathbf{1}_c$, $c\in C\setminus\{c_1\}$, are $2r+1$ independent vectors orthogonal to $\mathbf{1}$, which gives the eigenvalues of $-H'=2\kappa$ modulo $\mathbf{1}$; they do not vanish, so η' is nondegenerate, and so is the sign below it (Proposition 39.46(i)). A symmetric matrix A of size N' with $A\mathbf{1}=0$ and rank $N'-1$ has an adjugate of rank one killed by A, so equal to $\lambda\,\mathbf{1}\mathbf{1}^{\mathsf T}/N'$ with λ the sum of its principal minors of size $N'-1$, the product of the eigenvalues of A modulo $\mathbf{1}$; so every such minor is λ/N', and on a chart

$$\det(-H')=\frac{(r+1)\cdot(-(r+1))\cdot\left(\frac{r+1}{2r-1}\right)^{2r-1}}{2r+2}=-\frac{(r+1)^{2r}}{2(2r-1)^{2r-1}}.$$

The Hessian of $\log f$ at η is the Schur complement of the entry $-\frac12$ in that of $\log f'$ at η' (proof of Proposition 39.46(i)), so on charts with $e_0\in\mathcal{E}$, $\det H=-2\det H'$ and $\det(-H)=2\det(-H')$, which is the stated value. Part (i) with the one orbit of the signs of F' with two scales -1, below which lie two orbits of G, gives $(1-e_{G'})\mathfrak{M}^G\neq0$, and the closings vanish on it. □

Proposition 39.49 (The holonomy of a wheel). *As stated.*

Proof. Put $S(a)=\mathbf{U}(a)$ and $R(b)=\begin{bmatrix}b&0\\1&b\end{bmatrix}=b\,\mathbf{L}(b^{-1})$, and $M(\alpha)=\prod_i S(\alpha_{s_i})R(\alpha_{r_i})=\prod_i\alpha_{r_i}\cdot\mathbf{h}(\alpha)$. Index rows and columns by the states $0,1$ and read the factors from left to right as steps around the rim, $S(\alpha_{s_i})$ at the vertex i and $R(\alpha_{r_i})$ at the rim r_i: S allows $0\to0$, $0\to1$ with the weight α_{s_i}, and $1\to1$; R allows $0\to0$ and $1\to1$ with the weight α_{r_i}, and $1\to0$ with the weight 1. Then $\operatorname{tr}M$ is the sum over the closed walks of states of the products of the weights. A spanning tree of the wheel omits at least one rim; the rims it omits cut the rim cycle into arcs of consecutive vertices, joined by the rims it contains, and it contains exactly one spoke at a vertex of each arc, at least one for connection and at most one, two closing a cycle through the hub; every such choice is a spanning tree. To a tree corresponds the closed walk in state 0 at the start of each arc and in state 1 from the vertex of its spoke on, leaving 1 for 0 exactly at the omitted rims; its weight is α^T. The closed walks that never go from 1 to 0 stay in one state all around, taking every rim and no spoke: two walks of weight $\prod_i\alpha_{r_i}$. So $\operatorname{tr}M=K+2\prod_i\alpha_{r_i}$, which is the formula for K. $\det S=1$ and $\det R(b)=b^2$, so $\det\mathbf{h}=1$. By 7.2, $\Psi(\alpha)=\alpha^{\mathcal{E}}K(\alpha^{-1})=\prod_i\alpha_{s_i}\alpha_{r_i}\cdot\prod_i\alpha_{r_i}^{-1}\big(\operatorname{tr}\mathbf{h}(\alpha^{-1})-2\big)$.

The transpose of a product is the product of the transposes in the reverse order, $\mathbf{U}(a)^{\mathsf T}=\mathbf{L}(a)$ and $\mathbf{L}(b)^{\mathsf T}=\mathbf{U}(b)$, so

$$\operatorname{tr}\mathbf{h}(\alpha^{-1})=\operatorname{tr}\Big(\prod_{i=1}^{p}\mathbf{U}(\alpha_{s_i}^{-1})\mathbf{L}(\alpha_{r_i})\Big)^{\mathsf T}=\operatorname{tr}\big(\mathbf{U}(\alpha_{r_p})\mathbf{L}(\alpha_{s_p}^{-1})\cdots\mathbf{U}(\alpha_{r_1})\mathbf{L}(\alpha_{s_1}^{-1})\big)=\operatorname{tr}\mathbf{h}(\alpha\circ x),$$

the factor i of the last product being $\mathbf{U}(\alpha_{r_{p+1-i}})\mathbf{L}(\alpha_{s_{p+1-i}}^{-1})=\mathbf{U}\big((\alpha\circ x)_{s_i}\big)\mathbf{L}\big((\alpha\circ x)_{r_i}^{-1}\big)$. Then $K(\alpha\circ x)=\prod_i\alpha_{x(r_i)}\big(\operatorname{tr}\mathbf{h}(\alpha\circ x)-2\big)=\prod_i\alpha_{s_i}\big(\operatorname{tr}\mathbf{h}(\alpha^{-1})-2\big)=\Psi(\alpha)$. Comparing the monomials,

all with coefficient 1, the sets $x(T)$ of the bases T are the complements of the bases, so $T \mapsto \mathcal{E} \setminus x(T)$ maps the bases onto the bases: x exchanges the list. At $\alpha = \mathbf{1}$, $\mathbf{U}(1)\mathbf{L}(1) = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, of trace 3 and determinant 1, with the eigenvalues $\lambda_\pm$, the squares of the golden ratio and of its inverse, and $\mathbf{h}(\mathbf{1})$ is its p-th power, of trace $\lambda_+^p + \lambda_-^p$, the Lucas number of index $2p$, the Lucas numbers being the sums of the powers of the golden ratio and of minus its inverse. □

Proposition 39.50 (The centre ray of a wheel). *As stated.*

Proof. By 38.13, $f = \alpha^{\sigma\mathcal{E}}(K\Psi)^{-\sigma}$, and by Proposition 39.49, $K\Psi = \alpha^{\mathcal{E}}\, \mathfrak{d}\, \mathfrak{d}^*$; so $f = (\mathfrak{d}\, \mathfrak{d}^*)^{-\sigma}$, and $X_F = \{K\Psi \neq 0\} = \{\mathfrak{d}\, \mathfrak{d}^* \neq 0\}$ on the torus. (i) The native gauges are the critical points of $\log f = -\sigma(\log \mathfrak{d} + \log \mathfrak{d}^*)$ on X_F (38.2(ii), 38.4), where $\mathfrak{d}^* d\mathfrak{d} + \mathfrak{d}\, d\mathfrak{d}^* = 0$, and then $d\mathfrak{d}^* = -(\mathfrak{d}^*/\mathfrak{d})\, d\mathfrak{d}$. (ii) $\eta^{-1} = \eta$, and $\mathfrak{d}(\mathbf{1}) = K(\mathbf{1})$. (iii) Since $\det \mathbf{h} \equiv 1$, $\operatorname{tr}(\mathbf{h}^{-1} d\mathbf{h}) = d \log \det \mathbf{h} = 0$; where $\mathbf{h} = -I$ this is $-\operatorname{tr} d\mathbf{h} = -d\mathfrak{d} = 0$, and the same holds for $\mathfrak{d}^*$ where $\mathbf{h}(\alpha^{-1}) = -I$. On the torsion stratum $\mathfrak{d} = \mathfrak{d}^* = -4$, so it lies in X_F, $d \log f = 0$ there and $f = 16^{-\sigma}$. Each of $\mathbf{h}(\alpha) = -I$ and $\mathbf{h}(\alpha^{-1}) = -I$ is three equations, $\mathbf{h}_{12} = \mathbf{h}_{21} = 0$ and $\mathbf{h}_{11} = -1$, the fourth following from $\det \mathbf{h} = 1$. They do not change under scaling, $\mathbf{h}(c\alpha)$ being $\mathbf{h}(\alpha)$ conjugated by $\operatorname{diag}(c^{1/2}, c^{-1/2})$; on the chart $\alpha_{e_0} = 1$, a torus of dimension $2p - 1$, every component of the zero set of six functions has dimension at least $2p - 7$. □

Proposition 39.51 (The classes of a wheel). *As stated.*

Proof. The number of spanning trees that contain s_1 is the derivative of K in α_{s_1} at $\mathbf{1}$. With S, R and M as in the proof of Proposition 39.49, $K = \operatorname{tr} M - 2\prod_i \alpha_{r_i}$, and the derivative of S is $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, so that number is the entry $(2,1)$ of $R(1)M_1^{p-1}$, $M_1 = S(1)R(1) = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$. With the Fibonacci numbers Fib_m, $M_1 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^2$ and $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^m = \begin{bmatrix} \mathrm{Fib}_{m+1} & \mathrm{Fib}_m \\ \mathrm{Fib}_m & \mathrm{Fib}_{m-1} \end{bmatrix}$; the second row of $R(1)$ is $(1,1)$, so the entry is $\mathrm{Fib}_{2p-1} + \mathrm{Fib}_{2p-2} = \mathrm{Fib}_{2p} = (\lambda_+^p - \lambda_-^p)/\sqrt{5}$. Dividing by $K(\mathbf{1})$ gives R_s. With the Lucas numbers $\mathrm{Luc}_m = \mathrm{Fib}_{m-1} + \mathrm{Fib}_{m+1}$, $\mathrm{Luc}_m - 2\mathrm{Fib}_m = \mathrm{Fib}_{m-3}$, so $2R_\mathrm{s} = 1$ reads $\mathrm{Fib}_{2p-3} = 2$, that is $2p - 3 = 3$ and $p = 3$. The rotation $i \mapsto i + 1$ carries the list, so all spokes have one occupation and all rims one, and Theorem 39.9(i) with x gives $R_\mathrm{r} = 1 - R_\mathrm{s}$; 9.3 gives the masses. By 11.16 the classes at the centre are the orbits of the bijections that carry or exchange the list, and the rotations with x make one class. At a lean they are the orbits of the bijections that carry the list. For $p \geq 4$ the circuits of three marks are the triangles $\{s_i, s_{i+1}, r_i\}$, so a spoke lies in two of them and a rim in one, and a bijection that carries the list, carrying circuits to circuits, maps spokes to spokes: the classes at a lean are the spokes and the rims, the rotations acting transitively on each. By Theorem 39.9(i), $\bar{\phi}_x$ keeps the integrand at every lean, and x maps the spokes onto the rims. For $p = 3$ the permutations of the four vertices of K_4 carry the list and act transitively on its six edges. □

48.39 Chapter 40: the stride's law and the gauge end

In this section the chart is $\alpha_{e_0} = 1$ with the coordinates $y = \log \alpha \in \mathbb{R}^{\mathcal{E} \setminus e_0}$, $f^\star = f(\alpha^\star)$, and a differential operator A with polynomial coefficients on $(0, \infty)$ has the formal adjoint $A^\dagger$, with $\langle AT, \phi \rangle = \langle T, A^\dagger \phi \rangle$ for distributions T and test functions ϕ; for $\theta = t\frac{d}{dt}$, $\theta^\dagger = -1 - \theta$, so that

$p(-1-i-\theta)^\dagger = p(\theta - i)$ and

$$\mathfrak{D}_{\mathbf{p}}^\dagger = \sum_i p_i(\theta - i)\, t^i, \qquad \mathfrak{D}_{\mathbf{p}}^\dagger\, t^k = \sum_i p_i(k)\, t^{k+i}.$$

Five lemmas carry the proofs: a moment lemma, which turns recurrences of moments into equations for measures; the volumes of the sublevel sets of a Morse function; the uniqueness of expansions in powers and logarithms; the formal solutions of a recurrence at a simple root of its symbol; and the point masses that a cut germ produces.

Lemma S21. *Let $\nu_1, \dots, \nu_r$ be complex measures on $(0, R]$ with $\int t^{K_0}\, d|\nu_l| < \infty$ for a real K_0, let $A_1, \dots, A_r$ be differential operators $A_l = \sum_i t^{a_{l,i}} q_{l,i}(\theta)$ with $a_{l,i} \in \mathbb{N}$ and polynomials $q_{l,i}$, and let $E = \sum_l A_l \nu_l$, a distribution on $(0, \infty)$. For $\operatorname{Re} k \geq K_0$ the numbers $M_E(k) = \sum_l \int A_l^\dagger(t^k)\, d\nu_l = \sum_{l,i} q_{l,i}(-1-k-a_{l,i}) \int t^{k+a_{l,i}}\, d\nu_l$ are defined, holomorphic for $\operatorname{Re} k > K_0$.*

(i) If $M_E(k) = 0$ for all large integers k, then $E = 0$.

(ii) If $E = 0$, then $M_E(k) = 0$ for every k with $\operatorname{Re} k \geq K_0$.

Proof. $A_l^\dagger = \sum_i q_{l,i}(-1-\theta)\, t^{a_{l,i}}$, and $\theta\, t^m = m\, t^m$ gives the formula for M_E; the integrals converge absolutely for $\operatorname{Re} k \geq K_0$, t being at most R, and are holomorphic in k for $\operatorname{Re} k > K_0$ by dominated convergence on compact sets.

(i) For $\psi \in C^\infty(\mathbb{R})$ and an integer $K \geq K_0$ put $\Lambda_K(\psi) = \sum_l \int A_l^\dagger(t^K \psi)\, d\nu_l$. Since $\theta(t^m \psi) = t^m(m+\theta)\psi$ and $\theta^c = \sum_{b \leq c} S(c,b)\, t^b \partial^b$ with integers $S(c,b)$, $A_l^\dagger(t^K\psi)$ is a finite combination of the functions $t^{K+a+b}\psi^{(b)}$, b at most the largest order d of the A_l, with coefficients that depend on K; so $|\Lambda_K(\psi)| \leq C_K \max_{b \leq d} \sup_{[0,R]} |\psi^{(b)}|$. For $\psi = t^m$, $\Lambda_K(t^m) = M_E(K+m)$, which vanishes when K is large. Polynomials are dense in $C^d([0,R])$ (the Weierstrass approximation theorem applied to $\psi^{(d)}$, followed by d integrations from 0 with the initial values $\psi^{(b)}(0)$), so $\Lambda_K(\psi) = 0$ for every $\psi \in C^\infty(\mathbb{R})$ and every large K. A test function ϕ on $(0,\infty)$ is $t^K\psi$ with $\psi = t^{-K}\phi$, smooth with compact support in $(0,\infty)$, and $\langle E, \phi\rangle = \sum_l \int A_l^\dagger \phi\, d\nu_l = \Lambda_K(\psi) = 0$.

(ii) Let $\chi \in C^\infty(\mathbb{R})$ vanish on $(-\infty, 1]$ and equal 1 on $[2, \infty)$, put $\chi_\epsilon(t) = \chi(t/\epsilon)$, and let $\tilde\chi$ be a test function equal to 1 on $[\epsilon, R]$. Then $0 = \langle E, t^k \chi_\epsilon \tilde\chi\rangle = \sum_l \int A_l^\dagger(t^k\chi_\epsilon)\, d\nu_l$, the measures being carried by $(0, R]$. The difference $A_l^\dagger(t^k\chi_\epsilon) - \chi_\epsilon A_l^\dagger(t^k)$ involves only the functions $\theta^c\chi_\epsilon(t) = (\theta^c\chi)(t/\epsilon)$ with $c \geq 1$, which are bounded uniformly in ϵ and vanish off $[\epsilon, 2\epsilon]$; so it is at most $C\, t^{\operatorname{Re} k}$ on $[\epsilon, 2\epsilon]$ and 0 elsewhere, and its integral tends to 0. By dominated convergence $\int \chi_\epsilon A_l^\dagger(t^k)\, d\nu_l \to \int A_l^\dagger(t^k)\, d\nu_l$, so $M_E(k) = 0$. □

Lemma S22. *Let F be real-analytic near $0 \in \mathbb{R}^d$, $d \geq 1$, with $F(0) = 0$, $\nabla F(0) = 0$ and $\nabla^2 F(0) = H$ positive definite, and let w be real-analytic near 0.*

(i) There are $\epsilon > 0$ and a function A holomorphic at 0 with $\int_{\{F \leq \sigma\} \cap B_\epsilon} w\, dy = \sigma^{\frac{d}{2}} A(\sigma)$ for small $\sigma > 0$, B_ϵ the ball of radius ϵ, and $A(0) = w(0)\, (2\pi)^{\frac{d}{2}} \big/ \big(\Gamma(\frac{d}{2}+1)\sqrt{\det H}\big)$.

(ii) Let F be even and $w = 1$. Then F extends holomorphically to a neighbourhood of 0 in $\mathbb{C}^d$, $F_{\mathbb{T}}(\vartheta) = -F(i\vartheta)$ is real-analytic and even for real ϑ near 0, with the Hessian H at 0, and with the same A, $\int_{\{F_{\mathbb{T}} \leq \sigma\} \cap B_\epsilon} d\vartheta = \sigma^{\frac{d}{2}} A(-\sigma)$ for small $\sigma > 0$.

Proof. (i) Write $F(y) = \frac{1}{2} y^{\mathsf{T}} Q(y) y$ with $Q(y) = 2\int_0^1 (1-\tau)\nabla^2 F(\tau y)\, d\tau$, symmetric, real-analytic, $Q(0) = H$; near 0, $Q(y)$ has an analytic positive square root, and $v(y) = Q(y)^{1/2} y$ is an analytic change of coordinates with $F = \frac{1}{2}|v|^2$ and $\partial v/\partial y(0) = H^{1/2}$ (the Morse lemma). With

$J(v) = w(y(v))\,|\det \partial y/\partial v| = \sum_\beta J_\beta v^\beta$, convergent on a ball, $J(0) = w(0)(\det H)^{-1/2}$, and for ϵ small the set $\{F \le \sigma\} \cap B_\epsilon$ is the preimage of the ball $|v|^2 \le 2\sigma$ for small σ. The monomials of odd degree integrate to 0 over the ball, and $\int_{|v|^2\le 2\sigma} v^\beta dv = c_\beta (2\sigma)^{\frac{d+|\beta|}{2}}$, so the integral is $\sigma^{\frac d2}\sum_m \big(\sum_{|\beta|=2m} J_\beta c_\beta 2^{\frac d2+m}\big)\sigma^m$, with the series convergent for small σ; at $m=0$, c_0 is the volume $\pi^{\frac d2}/\Gamma(\frac d2+1)$ of the unit ball.

(ii) F is even, so $\nabla^2 F$ and Q are even and v is odd, with real Taylor coefficients in the monomials of odd degree; its inverse $y(v)$ is odd too, and J is even. A function analytic near 0 extends holomorphically to a polydisc, and $F(i\vartheta)$ is real for real ϑ since only even monomials occur, $i^{2m} = (-1)^m$. Put $v_{\mathbb T}(\vartheta) = -i\,v(i\vartheta)$, real for real ϑ since only odd monomials occur, $-i\cdot i^{2m+1} = (-1)^m$. Then $\frac12|v_{\mathbb T}|^2 = \frac12(-i)^2\,v(i\vartheta)\cdot v(i\vartheta) = -F(i\vartheta) = F_{\mathbb T}(\vartheta)$, and $v_{\mathbb T}$ is an analytic change of coordinates with the inverse $\vartheta = -i\,y(i\,v_{\mathbb T})$, whose Jacobian matrix is $(\partial y/\partial v)(i\,v_{\mathbb T})$; so $d\vartheta = J(i\,v_{\mathbb T})\,dv_{\mathbb T}$ with $J(iv) = \sum_\beta J_\beta i^{|\beta|}v^\beta = \sum_\beta (-1)^{|\beta|/2}J_\beta v^\beta$, $|\beta|$ even. The computation of (i) gives $\sigma^{\frac d2}\sum_m(-1)^m\big(\sum_{|\beta|=2m} J_\beta c_\beta 2^{\frac d2+m}\big)\sigma^m = \sigma^{\frac d2}A(-\sigma)$. The Hessian of $F_{\mathbb T}$ at 0 is $-i^2H = H$. □

Lemma S23. *(i) Let (z_c,τ_c), $c=1,\dots,m$, be distinct pairs with $|z_c|=1$ and $\tau_c\in\mathbb R$, and $a_c\in\mathbb C$. If $\sum_c a_c z_c^k\,k^{i\tau_c}\to 0$ as $k\to\infty$ through the integers, every a_c is 0.*

(ii) Let $F(k) = \sum a_{z,\mu,j}\,z^k k^{-\mu}(\log k)^j$ be a finite sum over distinct triples with $|z|=1$, $\mu\in\mathbb C$ and $j\in\mathbb N$. If $F(k) = O(k^{-N})$ along the integers for some $N > \max \operatorname{Re}\mu$, every $a_{z,\mu,j}$ is 0.

Proof. (i) Let $x_k = \sum_c a_c z_c^k k^{i\tau_c}$ and take the logarithmic means $W_M = \frac{1}{\log M}\sum_{k=1}^M |x_k|^2/k$, which tend to 0 when $x_k\to 0$. In $|x_k|^2$ the term of (c,c') is $a_c\bar a_{c'} w^k k^{i\Delta}$ with $w = z_c\bar z_{c'}$ and $\Delta = \tau_c - \tau_{c'}$. For $c = c'$ it contributes $|a_c|^2\frac{1}{\log M}\sum_{k\le M}\frac1k \to |a_c|^2$. If $w\ne 1$, the partial sums of w^k are bounded by $2/|1-w|$ and $k\mapsto k^{i\Delta-1}$ has bounded total variation, $|(k+1)^{i\Delta-1} - k^{i\Delta-1}| \le (1+|\Delta|)k^{-2}$; summation by parts bounds $\sum_{k\le M} w^k k^{i\Delta-1}$ uniformly in M. If $w=1$, then $\Delta\ne 0$ and $\sum_{k\le M} k^{i\Delta-1} = \int_1^M x^{i\Delta-1}dx + O(1) = (M^{i\Delta}-1)/(i\Delta) + O(1)$ is bounded. Divided by $\log M$ these tend to 0, so $W_M\to\sum_c |a_c|^2$, and every a_c vanishes.

(ii) Let σ_0 be the least $\operatorname{Re}\mu$ among the triples with $a\ne 0$ and j_0 the largest j among those with $\operatorname{Re}\mu = \sigma_0$. Then $k^{\sigma_0}(\log k)^{-j_0}F(k)$ is $\sum a_{z,\mu,j_0} z^k k^{-i\operatorname{Im}\mu}$, over $\operatorname{Re}\mu=\sigma_0$, plus terms that tend to 0, and it tends to 0 itself since $N>\sigma_0$; the pairs $(z,-\operatorname{Im}\mu)$ are distinct, so (i) makes these a vanish, against the choice of σ_0 and j_0. □

Lemma S24. *Let $\mathbf p = (p_0,\dots,p_h)$ be as in Definition 40.4, $D\ge 1$, and $\lambda\ne 0$ a simple root of χ. Consider formal sums $\mathcal S = \sum a_{\mu,j}k^\mu(\log k)^j$ over $j\le J$ and μ in finitely many classes modulo $\mathbb Z$ with $\operatorname{Re}\mu$ bounded above, with the operator $L = \sum_i p_i(k)\lambda^i\varsigma^i$, where $\varsigma^i\big(k^\mu(\log k)^j\big) = k^\mu(1+\frac ik)^\mu\big(\log k+\log(1+\frac ik)\big)^j$ is expanded in powers of k^{-1}. If $L\mathcal S = 0$, then $\mathcal S = a\,k^\rho\big(1+\sum_{m\ge1}e_m k^{-m}\big)$ with $\rho = -\chi_1(\lambda)/(\lambda\chi'(\lambda))$, $a\in\mathbb C$, and e_m fixed by $\mathbf p$ and λ; and this series is a formal solution.*

Proof. Each coefficient of $L\mathcal S$ is a finite sum, since L raises the exponent by at most D and ς does not raise it. With $p_i = c_i k^D + c_i' k^{D-1} + O(k^{D-2})$ and $\varsigma^i(k^\mu(\log k)^j) = k^\mu(\log k)^j + i\mu\,k^{\mu-1}(\log k)^j + ij\,k^{\mu-1}(\log k)^{j-1} + \cdots$, the dots of lower exponent,

$$L\big(k^\mu(\log k)^j\big) = \chi(\lambda)k^{\mu+D}(\log k)^j + \big(\mu\lambda\chi'(\lambda)+\chi_1(\lambda)\big)k^{\mu+D-1}(\log k)^j + j\lambda\chi'(\lambda)k^{\mu+D-1}(\log k)^{j-1} + \cdots,$$

using $\sum_i c_i i\lambda^i = \lambda\chi'(\lambda)$, and $\chi(\lambda)=0$. Let σ be the largest $\operatorname{Re}\mu$ in $\mathcal S$ and, for μ with $\operatorname{Re}\mu=\sigma$, j_μ the largest j with $a_{\mu,j}\ne 0$. No term of $\mathcal S$ has an exponent $\mu+r$, $r\ge1$, so the coefficient of

$k^{\mu+D-1}(\log k)^{j_\mu}$ in $L\mathcal{S}$ is $a_{\mu,j_\mu}(\mu\lambda\chi'(\lambda)+\chi_1(\lambda))$, and $\mu=\rho$. The coefficient of $k^{\rho+D-1}(\log k)^{j_\rho-1}$ is then $j_\rho\lambda\chi'(\lambda)a_{\rho,j_\rho}$, so $j_\rho=0$: the terms of $\mathcal{S}$ of largest real exponent are $a\,k^\rho$. For the series $k^\rho\sum_m e_m k^{-m}$, $e_0=1$, the coefficient of $k^{\rho-M+D-1}$ in L of it is $-M\lambda\chi'(\lambda)e_M$ plus a combination of $e_0,\dots,e_{M-1}$, which fixes e_M; so it is a formal solution, and $\mathcal{S}-a\,k^\rho\sum_m e_m k^{-m}$ is a formal solution whose terms all have $\operatorname{Re}\mu<\operatorname{Re}\rho$, hence 0 by what precedes. □

Lemma S25. *Let $\mathcal{D}=\sum_{j=0}^D b_j(s)\,\partial_s^j$ have polynomial coefficients with $b_{D,0}=0$, and let $u=\sum_r u_r s^r$ be holomorphic near $s=0$. Then, with $\mathbf{1}_+=\mathbf{1}_{s>0}$ and p_r as in Theorem 40.6(iv),*

$$\mathcal{D}(u\,\mathbf{1}_+)=(\mathcal{D}u)\,\mathbf{1}_+ + \sum_{r=0}^{D-2}\mathsf{p}_r(u)\,\delta^{(D-2-r)},$$

and the coefficient of u_r in $\mathsf{p}_r(u)$ is $r!\,\big(b_{D-1,0}-(D-1-r)\,b_{D,1}\big)$.

Proof. For $i<j$, $\partial^j(s^i\mathbf{1}_+)=i!\,\delta^{(j-1-i)}$, and the Taylor remainder of order j of u times $\mathbf{1}_+$ has the classical j-th derivative; so $\partial^j(u\mathbf{1}_+)=(\partial^j u)\mathbf{1}_+ + \sum_{i<j} i!\,u_i\,\delta^{(j-1-i)}$. Further $s^l\delta^{(m)}=(-1)^l[m]_l\,\delta^{(m-l)}$ for $l\le m$ and 0 for $l>m$, with $[m]_l=\binom{m}{l}\,l!$. So the term $b_{j,l}s^l\partial^j$ sends $u_i\,s^i$, $i<j$, to $(-1)^l\binom{j-1-i}{l}\,i!\,l!\,b_{j,l}\,u_i\,\delta^{(j-1-i-l)}$. The level $r=D-2-(j-1-i-l)$ has $l=j+r+1-D-i$, and $l\le j-1-i$ exactly when $r\le D-2$; $l\ge 0$ and $j\le D$ give $i\le r+1$, with $i=r+1$ only for $j=D$, $l=0$, excluded by $b_{D,0}=0$. At $i=r$ the terms are $j=D-1$, $l=0$, giving $r!\,b_{D-1,0}$, and $j=D$, $l=1$, giving $-(D-1-r)\,r!\,b_{D,1}$. □

Definition 40.1 (The stride's law). As stated.

Proof. The density of the origin form, $\alpha^{\mathbf{m}_0}\prod_j K_j^{-s_{0j}}\Psi_j^{-t_{0j}}$ with real data, is positive on the positive orthant, the letters being positive there; so ω_0 is a positive measure on the chart and its image under the measurable map $y\mapsto f(e^y)$ is a measure on $(0,\infty)$. □

Proposition 40.2 (A ray is one Trailmoment on a line and one Gramma). *As stated.*

Proof. (i) By Theorem 38.14(i), $0<f\le f^\star$ on the chart and $f\le C'e^{-c'\|y\|}$, so $\{f\ge t\}$ is compact for $t>0$ and carries a finite mass of the continuous density ω_0/dy; μ_f is carried by $(0,f^\star]$. For the image measure $\int t^k\mu_f(dt)=\int f^k\omega_0=\langle 1\rangle_k$, the integrals converging absolutely together. With the kernel $\mathsf{Z}_{k;(1)}(x)=x^{-k}$ (Definition 5.2) the Trailmoment of the input μ_f (Definition 5.4) with the coefficient function t^{-1} over $(0,\infty)$ is $\int t^k\mu_f(dt)$, and Theorem 5.9 with $\tau=\iota$ writes it as $(\mathsf{T}_{k;(1)}[x]\,\iota_*\mu_f)((0,\infty))$.

(ii) If 0 were not an interior point of the cone of $\mathbf{v}$, a nonzero y would have $\langle\mathbf{v}_i,y\rangle\le 0$ for every i (the separation of a point from a closed convex cone), and then $f^{-1}(e^{\lambda y})\le\sum_i w_i$ for $\lambda>0$, against $f\le C'e^{-c'\|y\|}$; so the cone is $\mathbb{R}^{\mathcal{E}\setminus e_0}$, and Theorem 2.2 makes the Gramma entire. For $X>0$ and $\operatorname{Re}k>0$, $\Gamma(k)X^{-k}=\int_0^\infty u^{k-1}e^{-uX}du$. At $X=f^{-1}$, integrated against $\omega_0=\alpha^{\mathbf{m}_0}dy$, Fubini's theorem applies, the double integral of the modulus being $\Gamma(\operatorname{Re}k)\langle 1\rangle_{\operatorname{Re}k}$ at real data, and the inner integral is $\int_{\mathbb{R}^{\mathcal{E}\setminus e_0}_{>0}} x^{\mathbf{m}_0|_{\mathcal{E}\setminus e_0}-\mathbf{1}}\exp\big(-\sum_i uw_i x^{\mathbf{v}_i}\big)dx=\mathsf{G}_{\mathbf{m}_0|_{\mathcal{E}\setminus e_0};\mathbf{v}}(u\mathbf{w})$ (Definition 2.1), $\alpha^{\mathbf{m}_0}$ being $x^{\mathbf{m}_0|_{\mathcal{E}\setminus e_0}}$ on the chart. This is Theorem 5.11 for the Trailmoment of (i), whose Laplace transform beyond the cut 0 is $\int e^{-ux}\,\iota_*\mu_f(dx)=\int e^{-uf^{-1}}\omega_0$.

(iii) The shape law has the density $u^{\mathbf{m}-\mathbf{1}}\prod_j K_j(u)^{-s_j}\Psi_j(u)^{-t_j}$ against $d\omega_{\mathcal{E}}$, divided by the period (Definition 8.10), and the change of variables of Theorem 8.7 that carries the period to the chart, $\langle 1\rangle_k=\int f^k\omega_0$ (Lemma S20(iv)), carries the expectation of every bounded measurable function φ of degree 0 to $\int\varphi\,f^k\omega_0/\langle 1\rangle_k$. At $\varphi=\psi\circ f$ this is $\int\psi(t)\,t^k\mu_f(dt)/\langle 1\rangle_k$. So

$\log\langle 1\rangle_k = \log\int t^k\mu_f(dt)$ is the log-partition of the family $t^k\mu_f(dt)/\langle 1\rangle_k$; it is real-analytic in k, and differentiation under the integral gives the first derivative $\mathbb{E}[\log f(U)]$ and the second $\mathbb{E}[\log^2 f(U)] - \mathbb{E}[\log f(U)]^2$, which is positive because $\log f(U)$ is not almost surely constant: U has a positive continuous density on the chart, and f is continuous there and not constant, $f(\alpha^\star) = f^\star > 0$ while $f \le C'e^{-c'\|y\|}$ (Theorem 38.14(i)). □

Proposition 40.3 (The two ends of the stride's law). *As stated.*

Proof. (i) At $y = 2z$ the stride is $e^{X_1(z)}$, X_1 the exponent of the question of the direction (proof of Lemma S20(ii)), and $-X_1$ is strictly convex (Theorem 17.5(ii)); so $\log f$ is strictly concave on the chart, its only critical point is its maximum $\log\alpha^\star$, and $\nabla f \neq 0$ elsewhere. For $0 < t < f^\star$ the level set $\Sigma_t = \{f = t\}$ is then a compact real-analytic hypersurface, compact because $f \to 0$ at infinity of the chart, and not empty, the chart being connected. Near $t_1 \in (0, f^\star)$ the flow Φ of the real-analytic field $\nabla f/|\nabla f|^2$, along which f grows at the unit rate, maps $\Sigma_{t_1}\times(t_1-\epsilon, t_1+\epsilon)$ real-analytically onto a neighbourhood of Σ_{t_1}, with $f(\Phi(x,\tau)) = \tau$, and $dy = J(x,\tau)\,d\sigma(x)\,d\tau$ with J real-analytic and positive. So μ_f has the density $\varrho_f(\tau) = \int_{\Sigma_{t_1}}(\omega_0/dy)(\Phi(x,\tau))\,J(x,\tau)\,d\sigma(x)$ there, real-analytic and positive; this is the coarea formula.

(ii) Apply Lemma S22(i) to $F(y) = \log f^\star - \log f(e^y)$ at $\log\alpha^\star$, with the Hessian H, positive definite (Lemma S20(iii)), and $w = \omega_0/dy$. Since $f < f^\star - \delta'$ off a neighbourhood of $\log\alpha^\star$, $\mu_f([t, f^\star]) = \int_{\{F\le L\}\cap B_\epsilon} w\,dy = L^{\frac{n-1}{2}}A(L)$ for t near $f^\star$, with $L = \log(f^\star/t)$. Differentiating, with $dL/dt = -1/t$,

$$\varrho_f(t) = -\frac{d}{dt}\mu_f([t,f^\star]) = \frac{1}{t}\,L^{\frac{n-3}{2}}B(L), \qquad B(L) = \tfrac{n-1}{2}A(L) + LA'(L).$$

Put $\lambda(t) = L/(f^\star - t)$, holomorphic and positive near $f^\star$ with $\lambda(f^\star) = 1/f^\star$. Then $\varrho_f(t) = (f^\star - t)^{\frac{n-3}{2}}g(f^\star - t)$ with $g(f^\star - t) = t^{-1}\lambda(t)^{\frac{n-3}{2}}B(L)$, holomorphic at $t = f^\star$, and $g(0) = (f^\star)^{-\frac{n-1}{2}}\frac{n-1}{2}A(0)$; with $A(0)$ from Lemma S22(i) at $d = n-1$ and $\frac{n-1}{2}/\Gamma(\frac{n+1}{2}) = 1/\Gamma(\frac{n-1}{2})$ this is the stated value.

(iii) By Proposition 17.4(i) for the question of the direction, $2\kappa(z) + \sum_j(s_{1j}+t_{1j})\log\theta_{j,\min} \le -X_1(z) \le 2\kappa(z) + \sum_j(s_{1j}+t_{1j})\log K_j(\mathbf{1})$, and $2\kappa(y/2) = \kappa(y)$, κ being positively homogeneous of degree one; with $\log f(e^y) = X_1(y/2)$ this is the squeeze. On the sector of an order π (Definition 17.3) the differences $u_l = y_{\pi(l-1)} - y_{\pi(l)} \ge 0$, $2 \le l \le n$, are coordinates in which dy is the Lebesgue measure and $\kappa(y) = \sum_l \kappa_{1,U_l}u_l$, so $\int_{\text{sector}} e^{-\kappa}dy = \prod_l \kappa_{1,U_l}^{-1}$; the sectors cover the chart and meet in null sets, so $\int e^{-\kappa}dy = V_1$. By homogeneity $\operatorname{vol}\{\kappa \le \sigma\} = \sigma^{n-1}\operatorname{vol}\{\kappa \le 1\}$ and $V_1 = \int_0^\infty e^{-\sigma}\,d\operatorname{vol}\{\kappa\le\sigma\} = (n-1)!\operatorname{vol}\{\kappa\le 1\}$, as in Theorem 17.5(i). For $t \le C_-$ the squeeze gives $\{\kappa \le \log(C_-/t)\} \subseteq \{f \ge t\} \subseteq \{\kappa \le \log(C_+/t)\}$, and $\mu_f([t,f^\star]) = \operatorname{vol}\{f \ge t\}$ for the origin 0. Last, $\int e^{-k\kappa}dy = k^{1-n}V_1$ for $k > 0$, by homogeneity, and $C_-^k e^{-k\kappa} \le f^k \le C_+^k e^{-k\kappa}$. □

Definition 40.4 (The density operator of a recurrence). As stated.

Proof. For a polynomial p of degree at most D, $p(-1-i-\theta) = c(-1)^D\theta^D + \cdots$ with $c = [k^D]p$, and $\theta^D = t^D\partial^D + \cdots$; so the coefficient of ∂^D in $t^i p_i(-1-i-\theta)$ is $(-1)^D c_i t^{D+i}$, and $a_D = (-1)^D t^D\chi(t)$. Extension by zero involves only u near λ: two intervals give the same condition on their common part, $\mathfrak{D}_\mathbf{p}$ being a differential operator. □

Proposition 40.5 (A recurrence of the values is an equation for the density). *As stated.*

Proof. Let k_* be the bound beyond which $\langle 1\rangle_k$ converges absolutely, $\operatorname{Re} k > k_*$ (Lemma S20(iv)), and $K_0 > k_*$. Apply Lemma S21 to $\nu_1 = \mu_f$, carried by $(0, f^\star]$ with $\int t^{K_0} d\mu_f = \langle 1\rangle_{K_0} < \infty$, and $A_1 = \mathfrak{D}_{\mathbf{p}}$, for which $A_1^\dagger t^k = \sum_i p_i(k) t^{k+i}$ and $M_E(k) = \sum_i p_i(k)\langle 1\rangle_{k+i}$, $E = \mathfrak{D}_{\mathbf{p}}\mu_f$. (a) gives (c) by Lemma S21(i), and (c) gives the vanishing for $\operatorname{Re} k \ge K_0$ by (ii), for every $K_0 > k_*$. By Proposition 38.5(ii) the values continue meromorphically to $\mathbb{C}$, so the identity holds as one of meromorphic functions, and at every k where its terms converge; (b) gives (a).

On $(0, f^\star)$, $\mu_f = \varrho_f\, dt$ with ϱ_f real-analytic (Proposition 40.3(i)), so E restricted there is $(\mathfrak{D}_{\mathbf{p}}\varrho_f)\, dt$, and (c) gives $\mathfrak{D}_{\mathbf{p}}\varrho_f = 0$. On $(f^\star - \delta, f^\star + \delta)$, $\mu_f = \varrho_f \mathbf{1}_{(f^\star-\delta, f^\star)} dt$, the point $f^\star$ having no mass, and ϱ_f is integrable there by Proposition 40.3(ii), $\frac{n-3}{2} > -1$; so the germ extends by zero. For the origin 0, the proof of Corollary 39.34(i) and (ii) uses of $p_0, \dots, p_h$ only that the identity holds as one of meromorphic functions, with Proposition 39.31(ii) at $k = -j$ and at $k = 0$, $\mathsf{L}_0 = V_1 \ne 0$ (Definition 39.33). □

Theorem 40.6 (The gauge end). *As stated.*

Proof. (i) If $D = 0$, every p_i is a constant c_i, $\mathfrak{D}_{\mathbf{p}}$ is the multiplication by $\chi(t)$, not identically zero, and $\chi\varrho_f = 0$ on $(0, f^\star)$ (Proposition 40.5) contradicts $\varrho_f > 0$. So $D \ge 1$. With $b_j(s) = (-1)^j a_j(f^\star - s)$, $b_D(s) = (f^\star - s)^D \chi(f^\star - s)$, so $b_{D,0} = 0$ and $b_{D,1} = -(f^\star)^D \chi'(f^\star)$. Write $p_i = c_i k^D + c_i' k^{D-1} + \cdots$; $p_i(-1-i-\theta) = (-1)^D c_i\big(\theta^D + D(1+i)\theta^{D-1}\big) + (-1)^{D-1} c_i' \theta^{D-1} + \cdots$, and $\theta^m = t^m \partial^m + \binom{m}{2} t^{m-1}\partial^{m-1} + \cdots$; so

$$a_{D-1} = (-1)^D t^{D-1}\Big(\big(\tbinom{D}{2} + D\big)\chi(t) + D\, t\, \chi'(t) - \chi_1(t)\Big),$$
$$b_{D-1,0} = -(f^\star)^{D-1}\big(D f^\star \chi'(f^\star) - \chi_1(f^\star)\big),$$

which gives the two forms of γ. Since $s^j\partial_s^j = [s\partial_s]_j$, $s^{D-1}\mathfrak{D}_{\mathbf{p}} = \sum_{j,l} b_{j,l} s^{l+D-1-j}[s\partial_s]_j$; the exponent $m = l + D - 1 - j$ is nonnegative, as $l \ge 1$ for $j = D$, and the terms with $m = 0$ are $(j,l) = (D-1, 0)$ and $(D, 1)$, which give R_0. So $s = 0$ is a regular singular point, R_0 having the degree D of the operator, and its exponents are the roots of R_0.

(ii) Let u extend by zero on $(f^\star - \delta, f^\star)$. By Fuchs's theorem on regular singular points, which applies by (i), $u = \sum_{\lambda,j} s^\lambda (\log s)^j h_{\lambda,j}(s)$ near $s = 0^+$, a finite sum with λ among the exponents, $j \le D$ and $h_{\lambda,j}$ holomorphic at 0; so u has a convergent expansion $\sum c_{\mu,j} s^\mu (\log s)^j$, $\mu \in \{0, \dots, D-2, \gamma\} + \mathbb{N}$, and integrability forces $\operatorname{Re}\mu > -1$ for its terms: the terms of least real part $\operatorname{Re}\mu$ and largest j are $s^{\operatorname{Re}\mu}(\log s)^j$ times a trigonometric polynomial in $\log s$, not identically 0, whose modulus has a positive mean, and this is not integrable at 0 when $\operatorname{Re}\mu \le -1$. Fix ψ smooth, 0 on $(-\infty, f^\star - \frac{2}{3}\delta]$ and 1 on $[f^\star - \frac{1}{3}\delta, \infty)$, and put $T = \psi\, u\, \mathbf{1}_{(f^\star-\delta, f^\star)}$. On $(f^\star - \delta, f^\star + \delta)$, $\mathfrak{D}_{\mathbf{p}} T = \psi\, \mathfrak{D}_{\mathbf{p}}(u\mathbf{1}) + \sum_{a\ge 1}(\dots)\psi^{(a)}$, where the first term vanishes and the others are smooth functions carried by $[f^\star - \frac{2}{3}\delta, f^\star - \frac{1}{3}\delta]$, where u is real-analytic; so $\mathfrak{D}_{\mathbf{p}} T = \mathcal{E}$ is a smooth function with that support, and, T having compact support in $(0, \infty)$,

$$\sum_i p_i(k)\, M(k+i) = \int t^k\, \mathcal{E}(t)\, dt = O\big((f^\star - \tfrac{1}{3}\delta)^k\big), \qquad M(k) = \int t^k T(t)\, dt,$$

for real $k \to \infty$. In $t = f^\star e^{-\sigma}$, $M(k) = f^\star (f^\star)^k \int_0^\infty e^{-(k+1)\sigma}\, T(f^\star e^{-\sigma})\, d\sigma$, and $s = f^\star(1 - e^{-\sigma}) = f^\star\sigma(1 + O(\sigma))$; so near $\sigma = 0$, $T(f^\star e^{-\sigma})$ is a convergent sum of $\sigma^\mu(\log\sigma)^j$ times holomorphic functions, with the same leading terms as u up to the factors $(f^\star)^\mu$, and the part away from 0 contributes $O\big((f^\star - \frac{1}{3}\delta)^k\big)$. Watson's lemma with logarithms, $\int_0^c e^{-\kappa\sigma}\sigma^\mu(\log\sigma)^j d\sigma = \partial_\mu^j\big[\Gamma(\mu+1)\kappa^{-\mu-1}\big] + O(e^{-c\kappa/2})$ for $\operatorname{Re}\mu > -1$, gives $A(k) = (f^\star)^{-k} M(k)$ an asymptotic expansion $\mathcal{S}_u =$

$\sum a_{\mu,j}k^{-\mu-1}(\log k)^j$ along the integers, in finitely many classes of exponents and with $\operatorname{Re}\mu$ bounded below. Its leading terms are $f^\star(f^\star)^\mu c_{\mu,j_0}\Gamma(\mu+1)(-1)^{j_0}k^{-\mu-1}(\log k)^{j_0}$, over the μ of least real part and the largest j_0 among them, with $\Gamma(\mu+1)\neq 0$; by Lemma S23(ii) they do not cancel, so $\mathcal{S}_u\neq 0$ when $u\neq 0$. Applying the recurrence to truncations of the expansion, $\sum_i p_i(k)(f^\star)^i A(k+i)=O(r^k)$, $r<1$, and Lemma S23(ii), order by order, give $L\mathcal{S}_u=0$ with $L=\sum_i p_i(k)(f^\star)^i\varsigma^i$, so by Lemma S24, $\mathcal{S}_u$ is a multiple of one fixed series. The map $u\mapsto\mathcal{S}_u$ is linear and injective, so the germs that extend by zero form a space of dimension at most one.

(iii) By Proposition 40.5 the germ of ϱ_f extends by zero, and it is not zero; so it spans the space of (ii). Put $a=\frac{n-3}{2}$; by Proposition 40.3(ii), $\varrho_f=s^a g(s)$ with $g(0)\neq 0$, and it solves $\mathfrak{D}_{\mathbf{p}}$ on $s>0$. If $a\notin\mathbb{Z}$, the leading coefficient of $s^{D-1}\mathfrak{D}_{\mathbf{p}}(s^a g)=\sum_m s^m R_m(s\partial_s)(s^a g)$ is $R_0(a)g(0)$, so a is a root of R_0 other than $0,\dots,D-2$, and $a=\gamma$. If $a\in\mathbb{N}$, ϱ_f is holomorphic at $s=0$, with $u_r=0$ for $r<a$ and $u_a=g(0)$. When $a\leq D-2$, the germ extends by zero, and by Lemma S25 $\mathsf{p}_a(\varrho_f)=a!\,b_{D,1}(a-\gamma)\,g(0)=0$, the lower coefficients vanishing; so $\gamma=a$. When $a\geq D-1$, the coefficient of s^a in $s^{D-1}\mathfrak{D}_{\mathbf{p}}\varrho_f=0$ is $R_0(a)g(0)$, so a is a root of R_0 and again $\gamma=a$.

(iv) If $\gamma\notin\{0,\dots,D-2\}$, then $\gamma=a$ is not an integer or is an integer at least $D-1$, and in both cases $R_0(\gamma+N)\neq 0$ for $N\geq 1$, the roots of R_0 being $0,\dots,D-2$ and γ. Inserting $s^\gamma\sum_N g_N s^N$ into $\sum_m s^m R_m(s\partial_s)$ and using $s\partial_s(s^{\gamma+N})=(\gamma+N)s^{\gamma+N}$, the coefficient of $s^{\gamma+N}$ is $\sum_{m\leq N}R_m(\gamma+N-m)g_{N-m}$; its vanishing is the stated recurrence, so $g=g(0)\Phi$. If $\gamma\in\{0,\dots,D-2\}$, then by (iii) ϱ_f is holomorphic at $s=0$ with $u_r=0$ for $r<\gamma$ and $u_\gamma=g(0)$; it extends by zero, so $\mathsf{p}_r(\varrho_f)=0$ for $r\leq D-2$ (Lemma S25, the distributions $\delta^{(q)}$ being linearly independent), and for $\gamma<r\leq D-2$ the coefficient $r!\,b_{D,1}(r-\gamma)$ of u_r is not 0, so these equations fix u_r from the lower coefficients. For $N\geq D-1$, $R_0(N)\neq 0$, and the coefficient of s^N in $s^{D-1}\mathfrak{D}_{\mathbf{p}}\varrho_f$ gives the last recurrence.

(v) Each point of $(0,f^\star)$ has a disc in $\mathbb{C}$ to which ϱ_f extends holomorphically (Proposition 40.3(i)); two such discs meet in a convex set that meets the real line, so the extensions agree there, and they define a holomorphic function on the union U, on which $\mathfrak{D}_{\mathbf{p}}\varrho_f=0$ by the identity theorem. The germ at $f^\star$ is that of (iv), and its continuation along a path in U that follows $(0,f^\star)$ is ϱ_f; near a root of χ on the segment, ϱ_f is holomorphic, so the continuations around it on either side agree. For $\operatorname{Re}k>k_*$, $t^k(\log t)^j$ is dominated on $(0,f^\star]$ by $C\,t^K$ for every K with $k_*<K<\operatorname{Re}k$, locally uniformly in k, so the derivatives pass under $\int t^k\mu_f(dt)$. □

Proposition 40.7 (The law knows the dimension of its chart). *As stated.*

Proof. Let α_c be the native gauge with $f(\alpha_c)=\lambda$ and A_c its solution (Theorem 38.6, Definition 39.2). The class of $\sum_i p_i(k)f^i$ in $\mathfrak{M}$ is $\sum_i p_i(k)\operatorname{Sh}^i[1]=0$, so $\sum_i p_i(k)\varsigma^i A_c[1]=0$, and $\lambda^{-k}A_c[1]=k^{-\frac{n-1}{2}}\big(a_0(1)+\cdots\big)$ with $a_0(1)=(2\pi)^{\frac{n-1}{2}}(\omega_0/dv)(\alpha_c)\neq 0$ is a formal solution of $L=\sum_i p_i(k)\lambda^i\varsigma^i$. By Lemma S24 every formal solution is a multiple of $k^{\rho_\lambda}\big(1+\sum_m e_m k^{-m}\big)$, and comparing leading exponents, $\rho_\lambda=-\frac{n-1}{2}$. The computation of Theorem 40.6(i) uses of $f^\star$ only that it is a simple nonzero root of χ, so at λ the point $s=0$ is regular singular with the exponents $0,\dots,D-2$ and $-1+\chi_1(\lambda)/(\lambda\chi'(\lambda))=-1-\rho_\lambda=\frac{n-3}{2}$. □

Proposition 40.8 (The germ at the gauge end is local data at the native gauge). *As stated.*

Proof. (i) The potential of the realization is $\Phi(y)=\sum_j\big(s_{1j}\log K_j(e^y)+t_{1j}\log\Psi_j(e^y)\big)$ (Definition 8.18) and $\log f(e^y)=\langle\mathbf{m}_1,y\rangle-\Phi(y)$, so for $m\geq 2$ the Taylor tensor of $\log f$ is $-\partial^m\Phi=-2^{-m}\kappa_\Upsilon$ (Definition 9.4), and Proposition 9.6 at $y_0=\log\alpha^\star$ gives the cumulants.

(ii) Put $\eta(y)=\log f-\log f^\star+\frac{1}{2}H_{ab}y^a y^b$, of order three. Outside a ball $|y|<\epsilon$ the stride is at most $f^\star e^{-c}$, so that part of $\langle 1\rangle_k$ is $O\big((f^\star e^{-c})^{k-1}\langle 1\rangle_1\big)$. Inside, with $y=v/\sqrt{k}$, $\langle 1\rangle_k=$

$(f^\star)^k k^{-\frac{n-1}{2}} \int_{|v|<\epsilon\sqrt{k}} e^{-\frac12 H_{ab}v^a v^b} e^{k\eta(v/\sqrt{k})} dv$, and $k\eta(v/\sqrt{k}) = k^{-1/2}h_3(v) + k^{-1}h_4(v) + k^{-3/2}h_5(v) + O(k^{-2}|v|^6)$ with $h_3 = \frac16 h_{abc}v^a v^b v^c$, $h_4 = \frac{1}{24}h_{abcd}v^a v^b v^c v^d$ and h_5 the quintic term; for ϵ small $|k\eta(v/\sqrt{k})| \le \frac14 H_{ab}v^a v^b$ on the ball. Expanding the exponential to the order $k^{-3/2}$, with the remainder bounded by $Ck^{-2}(1+|v|)^{12}e^{\frac14 Hvv}$, the terms of order $k^{-1/2}$ and $k^{-3/2}$ are odd in v and integrate to 0 over the ball, and extending the Gaussian integrals to $\mathbb{R}^{n-1}$ costs an exponentially small amount. So $\langle 1\rangle_k = (f^\star)^k (2\pi/k)^{\frac{n-1}{2}} (\det H)^{-1/2}\big(1 + k^{-1}\mathbb{E}[h_4 + \frac12 h_3^2] + O(k^{-2})\big)$ for the centred Gaussian with covariance G. Its moments are $\mathbb{E}[v^a v^b v^c v^d] = G^{ab}G^{cd} + G^{ac}G^{bd} + G^{ad}G^{bc}$ and, for six factors, the sum over the 15 pairings; by the symmetry of h_{abc}, 9 of the pairings of $h_{abc}h_{def}$ join one pair inside each factor and 6 join every index across. So $\mathbb{E}[h_4] = \frac18 h_{abcd}G^{ab}G^{cd}$ and $\frac12\mathbb{E}[h_3^2] = \frac{1}{72}\big(9h_{abc}h_{def}G^{ab}G^{cd}G^{ef} + 6h_{abc}h_{def}G^{ad}G^{be}G^{cf}\big)$.

(iii) On $(f^\star - \delta, f^\star)$, $\varrho_f = \sum_m g_m (f^\star - t)^{\frac{n-3}{2}+m}$, and $\int_0^{f^\star-\delta} t^k \mu_f(dt) = O\big((f^\star-\delta)^{k-1}\langle 1\rangle_1\big)$. For each M, $\int_{f^\star-\delta}^{f^\star} t^k \varrho_f\, dt$ is the sum over $m < M$ of $g_m \int_{f^\star-\delta}^{f^\star} t^k (f^\star - t)^{\frac{n-3}{2}+m} dt$ plus $O\big(\int_0^{f^\star} t^k (f^\star - t)^{\frac{n-3}{2}+M} dt\big)$, and $\int_0^{f^\star} t^k (f^\star - t)^{b-1} dt = (f^\star)^{k+b}\mathrm{B}(k+1, b)$, the part over $(0, f^\star - \delta)$ being exponentially smaller. With $\mathrm{B}(k+1,b) = \Gamma(b)\Gamma(k+1)/\Gamma(k+1+b) = \Gamma(b)k^{-b}(1+O(k^{-1}))$ the term of g_m, $b = \frac{n-1}{2} + m$, begins at $(f^\star)^{k+b}\Gamma(b) g_m k^{-b}$, and the remainder is $O\big((f^\star)^k k^{-\frac{n-1}{2}-M}\big)$. So the expansion of $(f^\star)^{-k} k^{\frac{n-1}{2}}\langle 1\rangle_k$ is triangular in the g_m, with the nonzero diagonal $(f^\star)^{\frac{n-1}{2}+m}\Gamma(\frac{n-1}{2}+m)$. □

Proposition 40.9 (Slopes from the gauge end). *As stated.*

Proof. (i) Differentiate j times the identity of meromorphic functions $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ (Proposition 40.5) by the Leibniz rule.

(ii) By Theorem 40.6(v), $\partial_k^j \langle 1\rangle_k|_{k=K} = \int_0^{f^\star} t^K (\log t)^j \varrho_f(t)\, dt$, and on $(0, f^\star)$ the series gives $\varrho_f = \sum_m g_m (f^\star - t)^{\frac{n-3}{2}+m}$, convergent there. Let $K' < K'' < K$. On $(0, f^\star]$, $t^K |\log t|^j \le C\, t^{K''}$, so $\sum_m |g_m| \int_0^{f^\star} t^K |\log t|^j (f^\star - t)^{\frac{n-3}{2}+m} dt \le C \sum_m |g_m| (f^\star)^{K'' + \frac{n-1}{2} + m} \mathrm{B}\big(K''+1, \frac{n-1}{2}+m\big)$, which is finite because $\mathrm{B}(K''+1, b) \le \mathrm{B}(K'+1, b)$, the Beta function being decreasing in its first argument. So the integral of the sum is the sum of the integrals (Fubini's theorem), and $\int_0^{f^\star} t^a (\log t)^j (f^\star - t)^{b-1} dt = \partial_a^j\big[(f^\star)^{a+b}\mathrm{B}(a+1,b)\big]$, differentiation under the integral being allowed for a near K. □

Definition 40.10 (The procedure that reads a ray from its gauge end). As stated.

Proof. (1) $\log f$ is strictly concave on the positive part of the chart (proof of Proposition 40.3(i)), so its critical point is unique, the native gauge of the direction there (Theorem 38.14(i)). (2) and (3) are Definition 40.4 and Theorem 40.6(i), (iii) and (iv); the coefficients $b_{j,l}$ lie in $\Bbbk(f^\star)$, and so do the ratios u_r/u_γ and Φ_N. (4) $\epsilon_k = \int_{(0, f^\star-\delta)} t^k \mu_f(dt)$ is nonnegative and, for real $k' \le k$, at most $(f^\star - \delta)^{k-k'} \int t^{k'} \mu_f(dt)$; the bound on $\langle 1\rangle_{k'}$ is Proposition 40.3(iii). On $[f^\star - \delta, f^\star]$, ϱ_f is its germ, by Theorem 40.6(iv). (5) The recurrence holds at every positive integer (Proposition 40.5), and $\langle 1\rangle_k = -\sum_{i\ge1} p_i(k)\langle 1\rangle_{k+i}/p_0(k)$ where $p_0(k) \ne 0$; the values are moments of ϱ_f by Theorem 40.6(v). (6) is Proposition 40.9, and (7) is Corollary 39.34(ii), which holds for $\mathbf{p}$ by Proposition 40.5. □

Definition 40.11 (The torus value). As stated.

Proof. On $\mathbb{T}$, $\alpha^{-1} = \bar\alpha$, and the letters have real coefficients, so $K_j(\alpha^{-1}) = \overline{K_j(\alpha)}$ and $f = \prod_j |K_j(\alpha)|^{-2\sigma_j}$; the coefficients being nonnegative, $|K_j(\alpha)| \le K_j(\mathbf{1})$, and $f \ge \prod_j K_j(\mathbf{1})^{-2\sigma_j} = f(\mathbf{1}) = f^\star$, the native gauge of the direction being $\mathbf{1}$ (Theorem 38.14). The zero set of an active letter has measure zero, the letter being a Laurent polynomial that is not zero, so a real-analytic function on the connected torus that is not identically zero, so f is finite almost everywhere and

$\nu_f^{\mathbb{T}}$ is a probability measure on $[f^\star,\infty)$. For $\operatorname{Re} k \le 0$, $|f^k| = f^{\operatorname{Re} k} \le (f^\star)^{\operatorname{Re} k}$, read as 0 where $f=\infty$ and $\operatorname{Re} k<0$, so the integral converges. When no K_j vanishes on $\mathbb{T}$, $\mathbb{T}$ is a compact cycle in X_F, $\Psi_j(\alpha)=\alpha^{\mathcal{E}}K_j(\alpha^{-1})$ (Proposition 7.2) does not vanish either, and on it $dy = i^{n-1}d\vartheta$, the coordinates being $y_e = i\vartheta_e$. □

Proposition 40.12 (The torus is the second chain). *As stated.*

Proof. (i) Equality $|K_j(e^{i\vartheta})| = K_j(\mathbf{1})$ holds exactly when the numbers $\langle \mathbf{1}_T,\vartheta\rangle$, T the bases of table j, agree modulo 2π, all the coefficients being positive. The active tables form a connected family (Definition 38.13), so one molecule, and for linked marks e,e' some basis T has $e\in T$, $e'\notin T$ and $(T\setminus\{e\})\cup\{e'\}$ a basis (Lemma 7.7); equality for every active table therefore gives $\vartheta_e \equiv \vartheta_{e'}$ for all marks, and $\alpha=\mathbf{1}$ on the chart. So $f>f^\star$ off $\mathbf{1}$, as in the proof of Theorem 39.41. The function $|K_j(e^{i\vartheta})|^2$ is a trigonometric polynomial, real-analytic and not identically 0 since its value at $\vartheta=0$ is $K_j(\mathbf{1})^2$; the zero set of such a function on the connected torus has measure zero. For $\operatorname{Re} k\le 0$ the integrand is bounded by $(f^\star)^{\operatorname{Re} k}$; dominated convergence gives the continuity, and Morera's theorem the holomorphy on $\operatorname{Re} k<0$. At $k=-j$, $f^{-j}=\prod_j |K_j|^{2j\sigma_j}$ is the Laurent polynomial $\prod_j (K_j(\alpha)K_j(\alpha^{-1}))^{j\sigma_j}$ restricted to $\mathbb{T}$, whose mean is its constant term; it is $\mathsf{C}_j(1)$ when every letter exponent of the direction is at least 1 (Definition 39.26).

(ii) The hypotheses of Theorem 39.41 hold for the centre ray with every $\sigma_j\ge 1$, so by Corollary 39.44 $\sum_i p_i(-j)\mathsf{C}_{j-i}(1)=0$ for all large j, that is, by (i), $\sum_i p_i(k)\langle 1\rangle^{\mathbb{T}}_{k+i}=0$ at all large negative integers k. With $x=1/t$, $\nu_f^{\mathbb{T}}$ becomes a probability measure $\tilde\nu$ on $(0,1/f^\star]$ with no mass at 0, $\theta = -x\frac{d}{dx}$, and $\mathfrak{D}_{\mathbf{p}}^\dagger = \sum_i p_i(\theta-i)\,t^i$ becomes $\tilde A = \sum_i p_i(-x\frac{d}{dx}-i)\,x^{-i}$. For a test function $\phi(t)=x^K\psi(x)$, $\tilde A(x^K\psi)=\sum_i x^{K-i}p_i(-x\frac{d}{dx}-K)\psi$ is bounded on $(0,1/f^\star]$ by $C_K\max_{b\le D}\sup|\psi^{(b)}|$ for $K\ge h$, and at $\psi=x^m$ its integral against $\tilde\nu$ is $\sum_i p_i(-K-m)\langle 1\rangle^{\mathbb{T}}_{-K-m+i}$, which vanishes for K large. As in the proof of Lemma S21(i), on $[0,1/f^\star]$, the Weierstrass approximation theorem gives $\langle \mathfrak{D}_{\mathbf{p}}\nu_f^{\mathbb{T}},\phi\rangle = \int \tilde A(x^K\psi)\,d\tilde\nu = 0$ for every test function. Conversely, for $\operatorname{Re} k\le -h$ the functions t^{k+i} are bounded on $[f^\star,\infty)$, and the cutoff argument of Lemma S21(ii), at $x\to 0$ in place of $t\to 0$, gives $\sum_i p_i(k)\langle 1\rangle^{\mathbb{T}}_{k+i} = \int \mathfrak{D}_{\mathbf{p}}^\dagger(t^k)\,d\nu_f^{\mathbb{T}} = 0$, the cutoff terms being bounded and carried by $x\in[\epsilon,2\epsilon]$, whose mass under $\tilde\nu$ tends to 0.

(iii) The stride is holomorphic and nonzero near $y=0$ in $\mathbb{C}^{\mathcal{E}\setminus e_0}$, and $F(y)=\log f^\star - \log f(e^y)$ is even, $f(\alpha^{-1})=f(\alpha)$ on the centre ray (Theorem 38.17(i)), with the Hessian H at 0 positive definite (Lemma S20(iii)). On $\mathbb{T}$, $\log f(e^{i\vartheta})-\log f^\star = -F(i\vartheta)=F_{\mathbb{T}}(\vartheta)$, real, and by (i) and compactness $f\ge f^\star+\delta'$ on $\mathbb{T}$ off a neighbourhood of $\mathbf{1}$; so for t near $f^\star$, $\nu_f^{\mathbb{T}}([f^\star,t]) = (2\pi)^{1-n}\operatorname{vol}\{F_{\mathbb{T}}\le \log(t/f^\star)\}\cap B_\epsilon$. By Lemma S22, with $L=\log(f^\star/t)$, $\mu_f([t,f^\star]) = L^{\frac{n-1}{2}}A(L)$ for $t<f^\star$ and $\nu_f^{\mathbb{T}}([f^\star,t]) = (2\pi)^{1-n}(-L)^{\frac{n-1}{2}}A(L)$ for $t>f^\star$, with one A, $w=1$. Differentiating, with $B=\frac{n-1}{2}A+LA'$, $\varrho_f(t)=t^{-1}L^{\frac{n-3}{2}}B(L)$ and

$$\varrho_f^{\mathbb{T}}(t) = \frac{d}{dt}\Big[(2\pi)^{1-n}(-L)^{\frac{n-1}{2}}A(L)\Big] = (2\pi)^{1-n}\,t^{-1}(-L)^{\frac{n-3}{2}}B(L).$$

With $\lambda(t)=L/(f^\star-t)$, positive near $f^\star$, $L=(f^\star-t)\lambda$ and $-L=(t-f^\star)\lambda$, so both densities carry the factor $t^{-1}\lambda(t)^{\frac{n-3}{2}}B(L)$, which is $g(f^\star-t)$ by the proof of Proposition 40.3(ii).

(iv) By Proposition 40.5 and Theorem 40.6(iii) the germ of ϱ_f spans the line. By (ii), $\mathfrak{D}_{\mathbf{p}}\nu_f^{\mathbb{T}}=0$; near $f^\star$, $\nu_f^{\mathbb{T}} = \varrho_f^{\mathbb{T}}\mathbf{1}_{(f^\star,f^\star+\delta)}dt$, integrable by (iii), so its germ extends by zero on the torus's side. (iii) gives the common germ. □

48.40 Chapter 41: reading a ray

In this section $f^\star = f(\alpha^\star)$, and the symbol χ and the second symbol χ_1 of polynomials $p_0, \dots, p_h$ are those of Definition 40.4. One lemma carries the proofs of the selection: the solutions of a recurrence whose symbol has distinct roots, sorted by their growth.

Lemma S26. *Let $p_0, \dots, p_h \in \mathbb{C}[k]$, $h \ge 1$, have $\deg p_0 = \deg p_h = D \ge \deg p_i$ for every i, let χ have distinct roots $\lambda_1, \dots, \lambda_h$, and let k_1 be an integer with $p_0(k)p_h(k) \ne 0$ for every integer $k \ge k_1$. Let $\mathcal{V}$ be the space of the sequences $(x(k))_{k \ge k_1}$ with $\sum_i p_i(k)x(k+i) = 0$ for every integer $k \ge k_1$.*

(i) $\dim \mathcal{V} = h$, and $\mathcal{V}$ has a basis $x_1, \dots, x_h$ with $x_c(k) = \lambda_c^k k^{\rho_c}\big(1 + O(1/k)\big)$ as $k \to \infty$, $\rho_c = -\chi_1(\lambda_c)/(\lambda_c \chi'(\lambda_c))$.

(ii) For $R \ge 0$, the $x \in \mathcal{V}$ with $x(k) = O(r^k)$ for every $r > R$ are the combinations of the x_c with $|\lambda_c| \le R$, and those with $x(k) = O(r^k)$ for some $r < R$ are the combinations of the x_c with $|\lambda_c| < R$.

Proof. (i) The map $x \mapsto (x(k_1), \dots, x(k_1 + h - 1))$ is a bijection of $\mathcal{V}$ onto $\mathbb{C}^h$, the recurrence at k giving $x(k+h)$ from the h values before it, $p_h(k) \ne 0$; so $\dim \mathcal{V} = h$. The leading coefficients $c_i = [k^D]p_i$ have $c_0 c_h \ne 0$, so $\chi(0) \ne 0$ and the roots are nonzero. If $D = 0$ the recurrence has constant coefficients, $\rho_c = 0$, and $x_c(k) = \lambda_c^k$. Let $D \ge 1$. Divided by k^D the recurrence has the coefficients $p_i(k)/k^D$, holomorphic at $k = \infty$ with the values c_i there; written as a system for $(x(k), \dots, x(k+h-1))$, its matrix at $k = \infty$ is the companion matrix of χ/c_h, whose eigenvalues are the λ_c. This is the normal case of Birkhoff's theorem on linear difference equations, which applies because the coefficients are holomorphic at infinity and the eigenvalues of the matrix there are distinct and nonzero: for every c there is a solution, defined at all large integers, with an asymptotic expansion $\lambda_c^k k^\rho\big(1 + \sum_{m \ge 1} e_m k^{-m}\big)$ as $k \to \infty$ that is a formal solution of the recurrence. Divided by λ_c^k it is a formal solution of $\sum_i p_i(k)\lambda_c^i s^i$, so $\rho = \rho_c$ by Lemma S24, λ_c being a simple root. Solving the recurrence for $x(k)$ downward, $p_0(k) \ne 0$, extends each of these solutions to $k \ge k_1$; call them x_c.

(ii) and the independence. Let $x = \sum_c a_c x_c$ with not all a_c zero, $R_x = \max\{|\lambda_c| : a_c \ne 0\}$, $\sigma = \max\{\operatorname{Re}\rho_c : a_c \ne 0,\ |\lambda_c| = R_x\}$, and C_0 the set of the c with $a_c \ne 0$, $|\lambda_c| = R_x$ and $\operatorname{Re}\rho_c = \sigma$. Then

$$\frac{x(k)}{R_x^k k^\sigma} = \sum_{c \in C_0} a_c \Big(\frac{\lambda_c}{R_x}\Big)^k k^{i \operatorname{Im}\rho_c} + o(1),$$

the other terms being $O\big((|\lambda_c|/R_x)^k k^{\operatorname{Re}\rho_c - \sigma}\big)$ with $|\lambda_c| < R_x$, or $O(k^{\operatorname{Re}\rho_c - \sigma})$ with $\operatorname{Re}\rho_c < \sigma$, and the remainders $O(1/k)$. The pairs $(\lambda_c/R_x, \operatorname{Im}\rho_c)$, $c \in C_0$, are distinct, the λ_c being distinct, so by Lemma S23(i) the right side does not tend to 0, and $|x(k)| \ge \eta R_x^k k^\sigma$ for some $\eta > 0$ and infinitely many k. So $x \ne 0$, and the x_c are independent, a basis by (i). And $x(k) = O(r^k)$ fails for every $r < R_x$. If $x(k) = O(r^k)$ for every $r > R$, then $R_x \le R$; if for some $r < R$, then $R_x < R$. Conversely $x_c(k) = O(r^k)$ for every $r > |\lambda_c|$. □

Proposition 41.1 (The low-end relations at every pole). *As stated.*

Proof. By Proposition 40.5 the identity $\sum_{i=0}^h p_i(k)\langle 1\rangle_{k+i} = 0$ holds as one of meromorphic functions of k. For $i > j$ the value $\langle 1\rangle_{k+i}$ is holomorphic at $k = -j$, $k + i$ being near the positive integer $i - j$, where it converges absolutely (Lemma S20(iv), the origin being 0). So $S(k) = \sum_{i \le \min(h,j)} p_i(k)\langle 1\rangle_{k+i} = -\sum_{i=j+1}^h p_i(k)\langle 1\rangle_{k+i}$ is holomorphic at $-j$ with the value of (i). For

$i \le j$, Proposition 39.31(ii) at the integer $j-i$ gives $(k+j)^{n-1}\langle 1\rangle_{k+i} \to V_1\,\mathrm{CT}(f^{i-j})$ as $k \to -j$, and $p_i(k) = (k+j)^{m_0}\big([(k+j)^{m_0}]p_i + O(k+j)\big)$. So

$$(k+j)^{n-1-m_0}S(k) = \sum_{i\le\min(h,j)} \frac{p_i(k)}{(k+j)^{m_0}}\,(k+j)^{n-1}\langle 1\rangle_{k+i} \;\longrightarrow\; V_1 \sum_{i\le\min(h,j)} [(k+j)^{m_0}]p_i\cdot \mathrm{CT}\left(f^{i-j}\right)$$

as $k \to -j$. For $m_0 < n-1$ the left side tends to 0, S being holomorphic at $-j$, and $V_1 > 0$ gives (ii). For $m_0 = n-1$ the left side tends to $S(-j)$, which (i) gives, and this is (iii). The reciprocal stride $f^{-1} = \alpha^{-\mathbf{m}_1}\prod_l K_l^{s_{1l}}\Psi_l^{t_{1l}}$ is a Laurent polynomial whose coefficients are polynomials with integer coefficients in the weights, so every $\mathrm{CT}(f^{i-j})$ lies in the field of the weights. □

Proposition 41.2 (The values live on the slit plane). *As stated.*

Proof. (i) By Proposition 40.2(i), μ_f is carried by $(0, f^\star]$ and $\langle 1\rangle_k = \int t^k\mu_f(dt)$. For $|q| < 1/f^\star$, $\sum_{k\ge k_0}|q|^{k-k_0}\int t^k\mu_f(dt) = \int t^{k_0}\mu_f(dt)/(1-|q|t) \le \langle 1\rangle_{k_0}/(1-|q|f^\star)$, so Fubini's theorem exchanges the sum and the integral and gives the formula. For q in a compact set $Q \subseteq \mathbb{C}\setminus[1/f^\star,\infty)$ the function $(q,t)\mapsto|1-qt|$ is continuous and positive on $Q\times[0,f^\star]$, since $qt = 1$ with $t\in(0,f^\star]$ means $q = 1/t \in [1/f^\star,\infty)$; so it is at least some $\delta_Q > 0$ there, the integrand is holomorphic in q and at most t^{k_0}/δ_Q, which is μ_f-integrable, and the integral is holomorphic on $\mathbb{C}\setminus[1/f^\star,\infty)$ (dominated convergence and Morera's theorem).

(ii) Put $t_0 = 1/x \in (0,f^\star)$. By Proposition 40.3(i), ϱ_f is real-analytic on $(0,f^\star)$, so there is $\delta > 0$ with $[t_0-2\delta, t_0+2\delta] \subseteq (0,f^\star)$ such that $\phi(t) = t^{k_0}\varrho_f(t)$ extends holomorphically to the disc $|t-t_0| \le 2\delta$. Write $G = G_1 + G_2$, with $G_1(q) = \int_{|t-t_0|\ge\delta} t^{k_0}\mu_f(dt)/(1-qt)$ and $G_2(q) = \int_{t_0-\delta}^{t_0+\delta}\phi(t)\,dt/(1-qt)$. For q close enough to x, $1/q$ lies within $\delta/2$ of t_0, so $1-qt$ does not vanish on the support of the first integral, and G_1 is holomorphic near x as in (i). Let Γ_+ and Γ_- be the half circles $t_0+\delta e^{i\vartheta}$ from $t_0-\delta$ to $t_0+\delta$ through the upper and through the lower half plane. For q near x with $\operatorname{Im} q > 0$ the pole $t = 1/q$ of the integrand lies in the disc and below the real axis, $\operatorname{Im}(1/q) = -\operatorname{Im} q/|q|^2$, so Cauchy's theorem moves the segment of G_2 onto Γ_+: $G_2(q) = \int_{\Gamma_+}\phi(t)\,dt/(1-qt)$, and the right side is holomorphic in q on the whole neighbourhood of x where $1/q$ stays within $\delta/2$ of t_0. So G continues from the upper half plane across a neighbourhood of x, with $G(x+i0) = G_1(x) + \int_{\Gamma_+}\phi(t)\,dt/(1-xt)$; likewise from below with Γ_-. The difference is the integral of $\phi(t)/(1-xt)$ along Γ_+ followed by Γ_- reversed, a clockwise circle around t_0, inside which the integrand has one simple pole, at $t_0 = 1/x$, with the residue $-\phi(1/x)/x$, as $1-xt = -x(t-1/x)$. So $G(x+i0) - G(x-i0) = -2\pi i\cdot\big(-\phi(1/x)/x\big) = 2\pi i\,x^{-1-k_0}\varrho_f(1/x)$.

(iii) On $\mathbb{T}$, $f = \prod_j|K_j(\alpha)|^{-2\sigma_j} \ge f^\star$ (proof of Definition 40.11), finite at the points of X_F; so the native value of a native gauge on $\mathbb{T}$ lies in $[f^\star,\infty)$.

(iv) By Proposition 38.10 every conjugate of $f^\star$ over the field $\Bbbk$ of the weights is the native value of a nondegenerate native gauge. Let $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ for all large integers k, with $p_i \in \Bbbk[k]$ not all zero, of largest degree D. By Proposition 38.9, $\langle 1\rangle_k > 0$ for large k and $\langle 1\rangle_{k+i}/\langle 1\rangle_k \to (f^\star)^i$; dividing the recurrence by $k^D\langle 1\rangle_k$ and letting $k\to\infty$ gives $\sum_i[k^D]p_i\cdot(f^\star)^i = \chi(f^\star) = 0$. The symbol is a nonzero polynomial over $\Bbbk$, so the minimal polynomial of $f^\star$ over $\Bbbk$ divides it, and every conjugate of $f^\star$ is a root. A real conjugate in $(0,f^\star)$ lies on the slit by its definition. □

Theorem 41.3 (Positive selection). *As stated.*

Proof. As in the proof of Proposition 41.2(iv), $\chi(f^\star) = 0$; the hypotheses on the degrees make χ of degree h with $\chi(0) = [k^D]p_0 \ne 0$. The walls $\kappa_{0,S} + k\kappa_{1,S}$ increase with k, so the values converge at every integer $k \ge k_1$, and by Proposition 40.5 the recurrence holds there: the values are a solution. Lemma S26(i) gives $\dim\mathcal{V} = h$ and a basis $x_1,\dots,x_h$.

(i) For each i, with $m = k + i$,

$$\sum_{k\ge k_1} p_i(k)x(k+i)\,q^{k-k_1+h} = q^{h-i}\sum_{m\ge k_1+i} p_i(m-i)x(m)\,q^{m-k_1}$$
$$= q^{h-i}p_i\Big(q\frac{d}{dq}+k_1-i\Big)\Big(X-\sum_{m=k_1}^{k_1+i-1}x(m)\,q^{m-k_1}\Big),$$

since $q\frac{d}{dq}$ multiplies q^{m-k_1} by $m-k_1$. Summed over i the left sides vanish, so $L_qX = R_x$ with

$$R_x = \sum_{i=0}^{h}\sum_{m=k_1}^{k_1+i-1} p_i(m-i)\,x(m)\,q^{h-i+m-k_1},$$

a polynomial of degree at most $h-1$, linear in $x(k_1),\dots,x(k_1+h-1)$. With $c_i = [k^D]p_i$, $p_i(q\frac{d}{dq}+c) = c_i(q\frac{d}{dq})^D+\cdots$ and $(q\frac{d}{dq})^D = q^D\frac{d^D}{dq^D}+\cdots$, the dots of lower order in $\frac{d}{dq}$; so the coefficient of $\frac{d^D}{dq^D}$ in L_q is $\sum_i q^{h-i}c_iq^D = q^{D+h}\chi(1/q)$, which vanishes in $\mathbb{C}^\times$ exactly at the $1/\lambda_c$. For the values, X is the generating function of Proposition 41.2(i) with $k_0 = k_1$.

(ii) For $k\ge k_1$, $\langle1\rangle_k = \int t^k\mu_f(dt) \le (f^\star)^{k-k_1}\langle1\rangle_{k_1}$, μ_f being carried by $(0,f^\star]$ (Proposition 40.2(i)); so the values lie in the subspace of the statement, which Lemma S26(ii) at $R = f^\star$ identifies. By Proposition 40.3(ii), with $a = \frac{n-3}{2}$ and a small $\delta>0$,

$$\langle1\rangle_k = \int_0^\delta (f^\star-s)^k\,s^a g(s)\,ds + O\big((f^\star-\delta)^{k-k_1}\langle1\rangle_{k_1}\big).$$

In $\sigma = -\log(1-s/f^\star)$ the integral is $(f^\star)^k\int_0^{\sigma_\delta} e^{-k\sigma}h(\sigma)\,d\sigma$ with $h(\sigma) = s^ag(s)\,f^\star e^{-\sigma} = g(0)(f^\star)^{a+1}\sigma^a\big(1+O(\sigma)\big)$, and Watson's lemma, which applies because $h(\sigma)\sigma^{-a}$ is continuous at 0 and $a>-1$, gives $\langle1\rangle_k = C(f^\star)^k k^{-\frac{n-1}{2}}(1+o(1))$ with $C = g(0)\Gamma(\frac{n-1}{2})(f^\star)^{\frac{n-1}{2}}>0$. Last, $f^\star$ is a simple root of χ and the recurrence meets the conditions of Proposition 40.5, so Theorem 40.6(iii) gives $\frac{n-3}{2} = -1+\chi_1(f^\star)/(f^\star\chi'(f^\star))$.

(iii) By Lemma S26(ii) at $R=\lambda'$, the solutions with $x(k) = O(r^k)$ for some $r<\lambda'$ are the multiples of x_1, $\lambda_1 = f^\star$ being the only root of modulus less than λ'; the values are such a solution, $f^\star<\lambda'$, and they are not 0, so they span that line. A generating function converges on $|q|<1/f^\star$ exactly when $\limsup|x(k)|^{1/k}\le f^\star$, that is when $x(k) = O(r^k)$ for every $r>f^\star$, and Lemma S26(ii) at $R=f^\star$ gives the same line. For the origin 0, Proposition 40.5 gives the low-end relation $a\mathsf{L}_0+\sum_{i\ge1}p_i(0)\langle1\rangle_i = 0$ with $\mathsf{L}_0 = V_1$ (Definition 39.33), that is $\ell(\langle1\rangle) = V_1$; a nonzero solution of least growth is $x = c\langle1\rangle$ with $c\ne0$, and $\ell(x) = cV_1$.

(iv) ℓ is a linear form on $\mathcal{V}$ with $\ell(\langle1\rangle) = V_1\ne0$, so $\{\ell=0\}$ is a hyperplane that does not contain $\langle1\rangle$, and $\langle1\rangle - x^{(0)}\in\{\ell=0\}$ gives M. Take $x_1 = \langle1\rangle$, which (iii) allows. The $z_c = x_c - \frac{\ell(x_c)}{V_1}x_1$, $2\le c\le h$, form a basis of $\{\ell=0\}$, and $x^{(i)} = \sum_c S_{ic}z_c$ with S invertible. The equations of the statement read $\mathbf{X}_KM^{(K)} = -\big(x^{(0)}(K+l)\big)_l$ with $\mathbf{X}_K = \big[x^{(i)}(K+l)\big]_{l,i}$, while $\mathbf{X}_KM = \big(\langle1\rangle_{K+l} - x^{(0)}(K+l)\big)_l$; so, when $\mathbf{X}_K$ is invertible, $M^{(K)}-M = -\mathbf{X}_K^{-1}v_K$ with $v_K = (\langle1\rangle_{K+l})_l$. Now $\mathbf{X}_K = \mathbf{Z}_KS^{\mathsf T}$ with $\mathbf{Z}_K = [z_c(K+l)]_{l,c}$, and $z_c(K+l) = \lambda_c^KK^{\rho_c}\big(\lambda_c^l+o(1)\big)$, since $x_c(K+l) = \lambda_c^{K+l}(K+l)^{\rho_c}\big(1+O(1/K)\big)$ and $x_1(K+l) = O\big((f^\star)^K\big)$ with $f^\star<|\lambda_c|$. So $\mathbf{Z}_K = \mathbf{N}_K\Lambda_K$ with $\Lambda_K = \mathrm{diag}(\lambda_c^KK^{\rho_c})_{c\ge2}$ and $\mathbf{N}_K\to[\lambda_c^l]_{0\le l\le h-2,\ 2\le c\le h}$, the Vandermonde matrix of the distinct $\lambda_2,\dots,\lambda_h$, which is invertible. For large K, $\mathbf{X}_K$ is invertible, $M^{(K)}$ is unique, and with $|v_K| = O\big((f^\star)^KK^{-\frac{n-1}{2}}\big)$ by (ii),

$$|M^{(K)}-M|\le\big\|(S^{\mathsf T})^{-1}\big\|\,\big\|\Lambda_K^{-1}\big\|\,\big\|\mathbf{N}_K^{-1}\big\|\,|v_K|$$

$$= O\Big(\max_{c\ge 2} |\lambda_c|^{-K} K^{-\operatorname{Re}\rho_c} (f^\star)^K K^{-\frac{n-1}{2}} \Big) = O\big((f^\star/\lambda')^K K^{C'} \big).$$

Over $\Bbbk$: the recurrence solved forward from initial values in $\Bbbk$, $p_h(k) \ne 0$, gives solutions with values in $\Bbbk$, and $\mathcal{V}$ has a basis of them; ℓ has its coefficients in $\Bbbk$ and $V_1 \in \Bbbk$, so $x^{(0)}$ and a basis of $\{\ell = 0\}$ can be taken with values in $\Bbbk$, and the equations for $M^{(K)}$ then have their coefficients in $\Bbbk$. □

Proposition 41.4 (Positivity is the moment problem of the stride's law). *As stated.*

Proof. (i) Every $\langle 1\rangle_{K+m}$, $m \in \mathbb{N}$, converges with $\langle 1\rangle_K$, μ_f being carried by $(0, f^\star]$, and $L_{K,p}(\langle 1\rangle) = \sum_m p_m \int t^{K+m} \mu_f(dt) = \int t^K p \, d\mu_f$ (Proposition 40.2(i)). The integrand is nonnegative, and positive off the finitely many zeros of p on $(0, f^\star)$, where μ_f has a positive density (Proposition 40.3(i)); so the integral is positive.

(ii) By the Markov–Lukács theorem on polynomials nonnegative on an interval, which applies to $[0, f^\star]$, a real polynomial of degree $2m$ nonnegative there is $A^2 + t(f^\star - t)B^2$ with real A and B of degrees at most m and $m-1$, and one of degree $2m+1$ is $tA^2 + (f^\star - t)B^2$ with real A and B of degrees at most m; every term has degree at most that of the polynomial. For $q = \sum_i q_i t^i$, $L_{K,q^2}(x) = \sum_{i,i'} q_i q_{i'} \, x(K+i+i')$, and likewise L_{K,tq^2}, $L_{K,(f^\star-t)q^2}$ and $L_{K,t(f^\star-t)q^2}$ are the quadratic forms of the other three Hankel matrices. The four kinds have degree at most N exactly when $\deg q$ is at most $\lfloor \frac{N}{2} \rfloor$, $\lfloor \frac{N-1}{2} \rfloor$, $\lfloor \frac{N-1}{2} \rfloor$ and $\lfloor \frac{N-2}{2} \rfloor$, which gives the sizes. If $L_{K,p}(x) \ge 0$ for every $p \ge 0$ on $[0, f^\star]$ of degree at most N, the four quadratic forms are nonnegative, these p being among them; conversely, if the forms are nonnegative, $L_{K,p}(x) \ge 0$ for every p of the four kinds, and by linearity for every sum of them.

(iii) If $x(K+m) = \int t^m \nu(dt)$ for a positive ν on $[0, f^\star]$, then $L_{K,p}(x) = \int p \, d\nu \ge 0$. Conversely, the linear functional $\Lambda(p) = L_{K,p}(x)$ on real polynomials is nonnegative on those nonnegative on $[0, f^\star]$, so $|\Lambda(p)| \le \Lambda(1) \max_{[0,f^\star]} |p|$, the polynomials $\max|p| \pm p$ being nonnegative there. Polynomials are dense in $C([0, f^\star])$ (the Weierstrass approximation theorem), so Λ extends continuously to $C([0, f^\star])$, and the extension is positive: a continuous $\varphi \ge 0$ is the uniform limit of polynomials p_l, and $p_l + \max|\varphi - p_l| \ge 0$. The Riesz representation theorem gives a positive measure ν on $[0, f^\star]$ with $\Lambda(\varphi) = \int \varphi \, d\nu$, and at $\varphi = t^m$ this is $x(K+m) = \int t^m \nu(dt)$. Two finite measures on $[0, f^\star]$ with the same moments agree on polynomials, hence on $C([0, f^\star])$, hence are equal. For the values, $t^K \mu_f$ is a finite positive measure on $(0, f^\star]$ with the moments $\langle 1\rangle_{K+m}$. □

Theorem 41.5 (The rate of positivity off the segment). *As stated.*

Proof. Put $A(p) = L_{K,p}(\langle 1\rangle) = \int t^K p \, d\mu_f$ and $B(p) = L_{K,p}(u) = \int_{I_\lambda} t^K p(t) \nu(t) \, dt$, both convergent, $|t|^{K+m} \le C_m |t|^K$ on the bounded I_λ. For $p \ge 0$ on $[0, f^\star]$, $A(p) \ge 0$ (Proposition 41.4(i)), and $A(p) + \varepsilon B(p) \ge 0$ holds for every ε when $B(p) = 0$ and otherwise on a closed half line that contains 0. So E_N, the intersection over the p of degree at most N, is a closed interval that contains 0, and if p_1, p_2 of degree at most N have $B(p_1)$ and $B(p_2)$ of opposite signs, its length is at most $A(p_1)/|B(p_1)| + A(p_2)/|B(p_2)|$.

Let $w(t) = 2t/f^\star - 1$, which maps $[0, f^\star]$ onto $[-1, 1]$, and T_m the polynomial of degree m with $T_m(\cos\vartheta) = \cos m\vartheta$; then $|T_m(w)| \le 1$ for $|w| \le 1$ and $|T_m(w)| = \cosh\big(m \operatorname{arccosh}|w|\big)$ for $|w| \ge 1$. On the part of I_λ off $[0, f^\star]$ put $\Gamma(t) = \operatorname{arccosh}|w(t)|$: it is C^1, it grows as t moves from the segment toward λ, $|w(t)|$ doing so, and $\Gamma(\lambda) = \log\big(|w| + \sqrt{w^2-1}\big) = g(\lambda)$ at $w = w(\lambda)$, the branch of the statement; $g(\lambda) > 0$ as $|w(\lambda)| > 1$. Let $\eta(t) = f^\star - t$ if $\lambda > f^\star$ and $\eta(t) = t$ if $\lambda < 0$, so that $\eta \ge 0$ on $[0, f^\star]$ and $\eta(\lambda) < 0$, and take $p_1 = T_m(w)^2$ and $p_2 = \eta \, T_m(w)^2$, nonnegative on $[0, f^\star]$, of degrees $2m$ and $2m+1$. Then $A(p_1) \le \langle 1\rangle_K$ and $A(p_2) \le f^\star \langle 1\rangle_K$.

Fix $\delta > 0$ so small that $I' = \{t \in I_\lambda : \ |t-\lambda| \le \delta\}$ lies off $[0, f^\star]$ and on I': t^K and η have the signs of their values at λ, $|t^K| \ge \frac12 |\lambda|^K$, $|\eta(t)| \ge \frac12 |\eta(\lambda)|$, $\nu(t) = c|\lambda - t|^a(1+\epsilon(t))$ with $|\epsilon| \le \frac12$, and $\Gamma(t) \ge g(\lambda) - \Gamma_1|\lambda - t|$ for a constant Γ_1. On I' the integrand of $B(p_i)$ has the constant sign of $c\lambda^K \eta(\lambda)^{i-1}$, and by $\cosh^2 \ge \frac14 e^{2m\Gamma}$ its modulus is at least $c_1 e^{2mg(\lambda)} e^{-2m\Gamma_1|\lambda-t|}|\lambda - t|^a$ for a constant $c_1 > 0$; so its integral over I' has modulus at least

$$c_1 e^{2mg(\lambda)} \int_0^\delta e^{-2m\Gamma_1 s} s^a \, ds = c_1 e^{2mg(\lambda)} (2m\Gamma_1)^{-a-1} \big(\Gamma(a+1) + o(1)\big) \ge c_2 \, m^{-a-1} e^{2mg(\lambda)}$$

for large m. Off I', $T_m(w)^2 \le \cosh^2\big(m\Gamma(\lambda_\delta)\big) \le e^{2m(g(\lambda)-\eta')}$, with λ_δ the end of I' other than λ and $\eta' = g(\lambda) - \Gamma(\lambda_\delta) > 0$, by the monotonicity of Γ and $|T_m| \le 1$ on $[0, f^\star]$; $\int_{I_\lambda} |t|^K |\nu| \, dt$ being finite and η bounded on I_λ, that part of $B(p_i)$ is at most $C_3 e^{2m(g(\lambda)-\eta')}$ in modulus. So for large m, $B(p_1)$ and $B(p_2)$ have the opposite signs of $c\lambda^K$ and $-c\lambda^K$, and moduli at least $\frac{c_2}{2} m^{-a-1} e^{2mg(\lambda)}$. With $m = \lfloor N/2 \rfloor$ for p_1 and $m = \lfloor (N-1)/2 \rfloor$ for p_2, so that $2m \ge N-2$, the length of E_N is at most $C\, N^{a+1} e^{-Ng(\lambda)}$ for large N.

For $\lambda > f^\star$, $\rho = f^\star/\lambda \in (0,1)$, $w = \frac{2}{\rho} - 1 > 1$, $w^2 - 1 = 4(1-\rho)/\rho^2$ and $w + \sqrt{w^2-1} = \big(2 - \rho + 2\sqrt{1-\rho}\big)/\rho = \big(1 + \sqrt{1-\rho}\big)^2/\rho$. So $e^{-g(\lambda)} = \rho/(1+\sqrt{1-\rho})^2 = (1-\sqrt{1-\rho})/(1+\sqrt{1-\rho})$, using $\rho = (1-\sqrt{1-\rho})(1+\sqrt{1-\rho})$, and $\sqrt{1-\rho} = 1 - \frac{\rho}{2} + O(\rho^2)$ gives $\frac{\rho}{4} + O(\rho^2)$. □

Proposition 41.6 (What positivity cannot see). *As stated.*

Proof. (i) $\mu_f + \varepsilon\nu \ge \mu_f - |\varepsilon|\,|\nu| \ge (1 - |\varepsilon|C)\mu_f \ge 0$. For real $k \ge K$, $t^k \le (f^\star)^{k-K} t^K$ on $(0, f^\star]$, so $\int t^k \, d|\nu| \le C (f^\star)^{k-K} \langle 1 \rangle_K$, and $\langle 1 \rangle_k + \varepsilon u(k) = \int t^k \, d(\mu_f + \varepsilon\nu)$ (Proposition 40.2(i)). For $K' \ge K$ and $p \ge 0$ on $[0, f^\star]$, $L_{K',p}(\langle 1 \rangle + \varepsilon u) = \int t^{K'} p \, d(\mu_f + \varepsilon\nu) \ge 0$.

(ii) $t^K(\mu_f + \varepsilon\nu)$ is a finite measure on $[0, f^\star]$, of total variation at most $(1 + |\varepsilon|C)\langle 1 \rangle_K$, with the stated moments; two finite signed measures on $[0, f^\star]$ with the same moments agree on polynomials, hence on $C([0, f^\star])$ by the Weierstrass approximation theorem, hence are equal. On J its density is $t^K(\varrho_f + \varepsilon\nu')$, ν' the density of ν. If it agreed almost everywhere on $(0, f^\star)$ with a real-analytic function φ, then on J, $\nu' = (t^{-K}\varphi - \varrho_f)/\varepsilon$ almost everywhere, a real-analytic function on J, ϱ_f being real-analytic on $(0, f^\star)$ (Proposition 40.3(i)); this is excluded. $t^K \mu_f$ has the real-analytic density $t^K \varrho_f$.

(iii) Let $|\nu'| \le B$ on $(0, \delta]$. For $K_0 > -1$, $\int t^{K_0} d|\nu| \le B \int_0^\delta t^{K_0} dt + \max_{[\delta, f^\star]} t^{K_0} \cdot C\mu_f([\delta, f^\star]) < \infty$ (Proposition 40.2(i)), so $u(k)$ converges for $\operatorname{Re} k \ge K_0$. Lemma S21(ii), with the measure ν on $(0, f^\star]$, the operator $\mathfrak{D}_{\mathbf{p}}$, for which $\mathfrak{D}_{\mathbf{p}}^\dagger t^k = \sum_i p_i(k) t^{k+i}$, and $E = \mathfrak{D}_{\mathbf{p}}\nu = 0$, gives $\sum_i p_i(k) u(k+i) = 0$ for $\operatorname{Re} k \ge K_0$, for every $K_0 > -1$. At $k = 0$, $p_0(0) = 0$, p_0 vanishing to order at least $n - 1 \ge 1$ at 0 (Proposition 40.5), and what remains is $\sum_{i \ge 1} p_i(0) u(i) = 0$. □

Proposition 41.7 (A recurrence of order two is the volume of the direction times a continued fraction). *As stated.*

Proof. (i) $\langle 1 \rangle_{k+1} = \int t^{k+1} \mu_f(dt) < f^\star \int t^k \mu_f(dt)$, μ_f having a positive density on $(0, f^\star)$ and no mass at $f^\star$, whose preimage is the point $\log \alpha^\star$ (Proposition 40.3(i)); and $\langle 1 \rangle_k > 0$. By Proposition 40.5 the recurrence holds at every integer $k \ge 1$, the origin being 0; divided by $\langle 1 \rangle_{k+1}$ it reads $p_0(k)/\rho_k + p_1(k) + p_2(k)\rho_{k+1} = 0$.

(ii) The low-end relation (Proposition 40.5, Definition 39.33) is $aV_1 + p_1(0)\langle 1 \rangle_1 + p_2(0)\langle 1 \rangle_2 = 0$, and $\langle 1 \rangle_2 = \rho_1 \langle 1 \rangle_1$. Let $\varphi(\rho) = -a/(p_1(0) + p_2(0)\rho)$. If $\varphi(0)$ and $\varphi(f^\star)$ are finite and positive, the affine denominator has the sign of $-a$ at both ends of $[0, f^\star]$, so it does not vanish there, $a \ne 0$, and φ is continuous and monotone on $[0, f^\star]$, strictly unless $p_2(0) = 0$, when the two values agree. As $\rho_1 \in (0, f^\star)$, $\langle 1 \rangle_1 / V_1 = \varphi(\rho_1)$ lies between $\varphi(0)$ and $\varphi(f^\star)$, strictly when they differ.

(iii) The symbol $\chi(\lambda) = c_0 + c_1\lambda + c_2\lambda^2$ has real coefficients and the real root $f^\star$, so λ_2 is real, with $|\lambda_2| = \lambda' > f^\star$. By Lemma S26(i) and the proof of Theorem 41.3(iii), $\mathcal{V}$ has the basis $\langle 1\rangle$, x_2, with $x_2(k) = \lambda_2^k k^{\rho_2}(1 + O(1/k))$. For $r \in [0, f^\star]$ and $K \geq 1$ put

$$\psi = \langle 1\rangle_{K+2} - r\langle 1\rangle_{K+1}, \qquad \phi = x_2(K+2) - r\,x_2(K+1), \qquad y = \phi\,\langle 1\rangle - \psi\,x_2,$$

a solution with $y(K+2) = r\,y(K+1)$. Let $\tau_k = x_2(k+1)/x_2(k)$, defined for $k \geq J$ and tending to λ_2, and $\rho_k \to f^\star$ (Proposition 38.9). Then $\phi = x_2(K+1)(\tau_{K+1} - r)$ with $|\tau_{K+1} - r| \geq \frac{1}{2}(\lambda' - f^\star)$ for large K, so $|\phi| \geq c\,(\lambda')^K K^{\rho_2}$, and $|\psi| \leq 2f^\star\langle 1\rangle_{K+1} = O\big((f^\star)^K K^{-\frac{n-1}{2}}\big)$ by Theorem 41.3(ii), uniformly in r. So $|\psi/\phi| \leq C_1(f^\star/\lambda')^K K^{C'}$. Write $y(k) = \phi\,\langle 1\rangle_k\big(1 - \epsilon_k\big)$ with $\epsilon_k = \frac{\psi}{\phi}\frac{x_2(k)}{\langle 1\rangle_k}$. For $k < J$, $|\epsilon_k| \leq C_J|\psi/\phi| \to 0$. For $J \leq k \leq K+1$,

$$\epsilon_k = \frac{\rho_{K+1} - r}{\tau_{K+1} - r}\prod_{l=k}^{K}\frac{\rho_l}{\tau_l},$$

and choosing J larger if needed, $|\rho_l/\tau_l| \leq \bar{q} = \frac{1}{2}(1 + f^\star/\lambda') < 1$ for $l \geq J$, while $|(\rho_{K+1} - r)/(\tau_{K+1} - r)| \to (f^\star - r)/|\lambda_2 - r| \leq (f^\star - r)/(\lambda' - r) \leq f^\star/\lambda' < \bar{q}$ uniformly in $r \in [0, f^\star]$. So there is K_0 such that for $K \geq K_0$ and every r, $|\epsilon_k| < 1$ for $1 \leq k \leq K+1$, hence $y(k) \neq 0$ there. For such K put $\sigma_k = y(k+1)/y(k)$, $1 \leq k \leq K+1$; $\sigma_{K+1} = r$, and the recurrence at k, divided by $y(k+1)$, gives $p_1(k) + p_2(k)\sigma_{k+1} = -p_0(k)/\sigma_k \neq 0$, so $\sigma_k = \mathsf{M}_k(\sigma_{k+1})$ for $1 \leq k \leq K$. So $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r) = \sigma_1$ is defined, and

$$\sigma_1 = \frac{\langle 1\rangle_2 - (\psi/\phi)x_2(2)}{\langle 1\rangle_1 - (\psi/\phi)x_2(1)} = \rho_1 + O\big((f^\star/\lambda')^K K^{C'}\big),$$

uniformly in r. By (ii) and $a \neq 0$, $p_1(0) + p_2(0)\rho_1 \neq 0$ and $\langle 1\rangle_1 = -aV_1/(p_1(0) + p_2(0)\rho_1)$. Let Σ_K be the set of the $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r)$ over the $r \in (0, f^\star)$ whose intermediate values lie in $(0, f^\star)$. An element of Σ_{K+1}, built from r, is $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r')$ with $r' = \mathsf{M}_{K+1}(r) \in (0, f^\star)$ and the same intermediate values, so it lies in Σ_K; and $\rho_1 \in \Sigma_K$, built from $r = \rho_{K+1}$, its intermediate values being the $\rho_k \in (0, f^\star)$ by (i). □

Definition 41.8 (The procedure that reads a ray from its list). As stated.

Proof. (1) V_1 is the coefficient of Proposition 39.30; the maps of Proposition 39.7(ii) that are symmetries generate a group of symmetries, and a finite one is chosen. (2) The search stops by Theorem 39.18, every box of it being finite (Proposition 39.12(iv)); Proposition 41.10(iii) and (v) bound the strengths and primes at which an elimination misjudges acceptance. (3) The reductions from the relations the targets need are those of the box by Proposition 41.10(vi); the evaluations give the values of the entries of $\mathbf{B}$ at the strengths where the elimination has the generic pivots (Proposition 41.10(i)), the reconstruction is Proposition 41.10(ii), the check is Proposition 41.10(iv) and (v), and the certificate and the one-step relation are Theorem 39.17(ii) and (iii). (4) By Theorem 39.17(iv) Proposition 39.6 applies to the spanning set and its matrix: the first dependence $\sum_i q_i w_i = 0$ gives $\sum_j p_j(k) f^j \in \mathcal{L}$ (Proposition 39.6(ii)), and the ratios q_i/q_h are ratios of signed maximal minors (Proposition 39.6(iii)), rational functions whose values at a point k_0 where $\mathbf{B}(k_0 - h + 1), \ldots, \mathbf{B}(k_0)$ are finite and $w_0(k_0), \ldots, w_{h-1}(k_0)$ independent are the coefficients of the dependence of the rows at k_0, so Proposition 41.10(ii) reconstructs them. The relation is the law of [1] when the spanning set is independent (Theorem 39.17(iv)), as under Proposition 39.38(i). The zeros of $p_0(k) = q_h(k+h)$ are those described, by the divisibility of Proposition 39.6(iii). (5) is Definition 40.10, which applies by Proposition 40.5, or Theorem 41.3(iv) with Proposition 41.4. (6) The closings are a solution with values in germs at $-\infty$ (Theorem 39.29), so Theorem 39.17(iii)

gives $\varsigma^{-1}v = \mathbf{B}v$ for their vector $v(k) = \mathsf{C}_{-k}(\mathbf{b})$, that is $\mathsf{C}_{j+1}(\mathbf{b}) = \mathbf{B}(-j)\mathsf{C}_j(\mathbf{b})$ for all large j; the reflected recurrence and the low-end relation hold by Proposition 40.5 with Corollary 39.34(i) and (ii); the eigenvalues are Proposition 39.38(ii); the exponent is Theorem 40.6(iii). □

Proposition 41.9 (The relations of a box are difference equations). *As stated.*

Proof. (i) By Definition 39.11(i) the symbols at the level $(\mathbf{a}, \mathbf{b})$ are the $(u; \mathbf{a}, \mathbf{b})$ with $u \in \mathbb{Z}^{\mathcal{E}}$ and $u_{\mathcal{E}} = c$, $c = \sum_j (a_j r_j + b_j r_j^*)$: a translate of the lattice $\{u_{\mathcal{E}} = 0\}$, of rank $n-1$. As $\kappa_{1,U} > 0$, the wall degree of g is at most $\mathfrak{t}$ exactly when $\kappa_U(g) + \mathfrak{t}\kappa_{1,U} \geq 0$ for every nonempty proper U (Definition 39.11(ii)). With $L(U) = \sum_j \big(a_j d_j(U) + b_j d_j^*(U)\big)$ and $\kappa_U(g) = u_U - L(U)$, the sets $U = \{e\}$ and $U = \mathcal{E} \setminus \{e\}$, nonempty and proper as $n \geq 2$, give

$$L(\{e\}) - \mathfrak{t}\kappa_{1,\{e\}} \leq u_e \leq c - L(\mathcal{E} \setminus \{e\}) + \mathfrak{t}\kappa_{1,\mathcal{E}\setminus\{e\}},$$

an interval of length at most $C(1+\mathfrak{t})$ with $C = \max_e \max\big(|c - L(\mathcal{E}\setminus\{e\}) - L(\{e\})|,\ \kappa_{1,\{e\}} + \kappa_{1,\mathcal{E}\setminus\{e\}}\big)$, which holds at most $C(1+\mathfrak{t})+1$ integers; and $u_{e_0} = c - \sum_{e \neq e_0} u_e$.

(ii) By Definition 39.11(iii) the letter identity of $g = (u; \mathbf{a}, \mathbf{b})$ for K_j has the terms g and $\theta_{j,T}(u + \mathbf{1}_T; \mathbf{a} + \mathbf{e}_j, \mathbf{b})$, $T \in \mathcal{B}_j$, and that for Ψ_j the terms g and $\theta_{j,T}(u + \mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{a}, \mathbf{b} + \mathbf{e}_j)$; the mark relation at e has g and, for every j, the terms of the $T \in \mathcal{B}_j$ with $T \ni e$ at the level $(\mathbf{a} + \mathbf{e}_j, \mathbf{b})$ and of the $T \not\ni e$ at the level $(\mathbf{a}, \mathbf{b} + \mathbf{e}_j)$, at most $|\mathcal{B}_j|$ for each j, with coefficients affine in k. The shifts are the $(\mathbf{1}_T; \mathbf{e}_j, \mathbf{0})$ and the $(\mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{0}, \mathbf{e}_j)$, $T \in \mathcal{B}_j$.

(iii) The letter identities have the coefficients 1 and $-\theta_{j,T}$, and the mark relations $m_{0,e} + k\, m_{1,e} + u_e$, $-(s_{0j} + k\, s_{1j} + a_j)\theta_{j,T}$ and $-(t_{0j} + k\, t_{1j} + b_j)\theta_{j,T}$, affine in k, with the stated parts in k. For a relation $\sum_g c_g g$, $e_G \sum_g c_g g = \sum_g c_g\, e_G g$, and $e_G g$ is the orbit average of the orbit of g (Definition 39.16(i)); so in the basis of the orbit averages the entry of an orbit is the sum of the c_g over its symbols, again affine in k, the part in k vanishing for the letter identities. □

Proposition 41.10 (What a reading by evaluation certifies). *As stated.*

Proof. (i) Gaussian elimination of the rows of A over $\Bbbk(k)$, the columns taken in the order of the ranking, highest first, gives a basis of the row space $\mathcal{R}_{\mathfrak{B}}^G$ in echelon form, with distinct leading orbits, and the leading orbit of every nonzero element of the row space is one of them; so these are the pivots P of Definition 39.16(ii), and $r = |P|$ is the rank. An element of the row space with zero coordinates on P would have a leading orbit outside P, so it is 0: the projection of $\mathcal{R}_{\mathfrak{B}}^G$ onto the coordinates P is injective, hence bijective. The chosen rows are a basis of $\mathcal{R}_{\mathfrak{B}}^G$, so $A_P(k)$ is invertible, and $E = A_P^{-1}[A_P \,|\, A_F]$, written on the columns P and F, is the basis of $\mathcal{R}_{\mathfrak{B}}^G$ with the identity on P, the unique one by the bijectivity; it is therefore the reduced echelon basis, whose row of a pivot p vanishes on the orbits ranked above p. For x with coefficients in $\Bbbk$, $x - x_P E$ is congruent to x and vanishes on P, so it is the reduction, $x_F - x_P A_P^{-1} A_F$ on F. The entries of A_P are affine in k, so $\det A_P$, not 0, has degree at most r. At k_0: the minors of order $r+1$ of A vanish identically, so $A(k_0)$ has rank at most r. If the minor of order r of the rows J' on the columns P does not vanish at k_0, then $E = A_{J',P}^{-1} A_{J'}$ as well, by the uniqueness, so E is regular at k_0, and $E(k_0)$ has the identity on P and the zeros of an echelon form, identities in k; its r independent rows lie in the row space of $A(k_0)$ and span it, so the elimination at k_0 has the pivots P and gives the reduction $x_F - x_P E_F(k_0)$, the value at k_0. Conversely, if the elimination at k_0 has the pivots P, the projection of the row space of $A(k_0)$ onto the coordinates P is injective, as above, so the columns P of $A(k_0)$ have rank r, and a minor of order r on them does not vanish.

(ii) The targets $f^{-1}b_i$ have their coefficients in $\Bbbk$. By (i) their reductions are $\big(x_F \det A_P - x_P \operatorname{adj}(A_P) A_F\big)/\det A_P$, the adjugate having entries of degree at most $r-1$ and A_F entries of

degree at most 1; the entries of $\mathbf{B}$ are coefficients of these reductions (Definition 39.16(iii)). If N/D and N'/D' have numerators of degree at most d_1 and denominators of degree at most d_2 and agree at d_1+d_2+1 points where D and D' do not vanish, then $ND'-N'D$, of degree at most d_1+d_2, vanishes at these points and is 0. With a common denominator, two numerators of degree at most d_1 that agree at d_1+1 points are equal.

(iii) Write the coefficient of a free orbit c in the reduction of a target as $N_c/\det A_P$, $\deg N_c \le r$, by (ii). If the box is accepted at $\mathfrak{s}$ over $\Bbbk(k)$, every N_c with c outside the free Laurent orbits of wall degree at most $\mathfrak{s}$ vanishes, and at every k_0 with $\det A_P(k_0)\neq 0$ the elimination has the pivots P, so the same free orbits, and gives the values of the reductions: it accepts. If not, some such N_c is a nonzero polynomial, and at every k_0 with $\det A_P(k_0)N_c(k_0)\neq 0$ the elimination has the pivots P and its reduction has a nonzero coefficient on c: it rejects. The exceptions lie among the at most $2r$ zeros of $\det A_P$ and N_c.

(iv) Let $\mathbf{B}_{il}\neq\mathbf{B}'_{il}$, write $\mathbf{B}_{il}=N_{il}/\det A_P$ by (ii) and $\mathbf{B}'_{il}=N'_{il}/Q'$, and put $\Delta=N_{il}Q'-N'_{il}\det A_P$, a nonzero polynomial of degree at most $r+d$. At a strength k_0 with $Q'(k_0)\neq 0$ at which the elimination has the pivots P, it gives $\mathbf{B}(k_0)$, $\mathbf{B}$ being regular at k_0 by (i). If $\det A_P(k_0)\neq 0$, agreement there gives $\Delta(k_0)=0$; if $\det A_P(k_0)=0$, then $N_{il}=\mathbf{B}_{il}\det A_P$ vanishes at k_0, $\mathbf{B}_{il}$ being regular there, and $\Delta(k_0)=0$ again. So the strengths of agreement are among the at most $r+d$ zeros of Δ, and a uniformly drawn strength of S is one of them with probability at most $(r+d)/|S|$.

(v) The entries of A are q-integral, and reduction modulo q is a ring map, so the minors of order $r+1$ of A vanish modulo q and $\det A_P$ modulo q is the determinant of A_P modulo q, which is not 0; the entries of $E=\operatorname{adj}(A_P)A_J/\det A_P$, A_J the chosen rows, reduce to those of $\bar E=\operatorname{adj}(\bar A_P)\bar A_J/\det \bar A_P$, which has the identity on P and the zeros of an echelon form. As in (i), $\bar A$ has the rank r and the pivots P over $\mathbb{F}_q(k)$, and its reductions $x_F-x_P\bar E_F$ are those over $\mathbb{Q}(k)$ taken modulo q; the proofs of (i) to (iv) use only linear algebra over a field and hold over $\mathbb{F}_q$. For the last claim, take an entry with $\mathbf{B}_{il}\neq\mathbf{B}'_{il}$ and write $\mathbf{B}_{il}=\tilde N/\tilde D$ and $\mathbf{B}'_{il}=N'/Q'$ with $\tilde N,\tilde D,N',Q'\in\mathbb{Z}[k]$ and $\tilde D=c_A\det A_P$ for a positive integer c_A; then $\Delta=\tilde NQ'-N'\tilde D\in\mathbb{Z}[k]$ is not 0. The finite set consists of the primes excluded above and those that divide c_A, every coefficient of Q', or every coefficient of Δ. For a prime outside it, $\mathbf{B}'$ modulo q is N'/Q' taken modulo q, Δ modulo q is a nonzero polynomial of degree at most $r+d$, and the argument of (iv) over $\mathbb{F}_q$ bounds the probability by $(r+d)/q$.

(vi) An element of $\mathcal{R}'$ lies in $\mathcal{R}^G_{\mathfrak{B}}$ with the same leading orbit, so the pivots of $\mathcal{R}'$ are pivots of $\mathcal{R}^G_{\mathfrak{B}}$, and the free orbits of $\mathcal{R}^G_{\mathfrak{B}}$ are free for $\mathcal{R}'$. So $\bar x$ is a combination of free orbits of $\mathcal{R}'$ with $x-\bar x\in\mathcal{R}'$. The free orbits of $\mathcal{R}'$ are independent modulo $\mathcal{R}'$, an element of $\mathcal{R}'$ that combines them having a free leading orbit and so being 0; so $\bar x$ is the reduction modulo $\mathcal{R}'$. □

Proposition 41.11 (Data with rational masses). *As stated.*

Proof. (i) Both data lie on the balance wall with the same exponents, so the origin $(\mathbf{m}-\mathbf{m}';\mathbf{0},\mathbf{0})$ has $(\mathbf{m}-\mathbf{m}')_{\mathcal{E}}=0$ and lies on it too. The walls are linear in the data, so the walls along the ray at k are $\kappa_U(\mathbf{m};\mathbf{s},\mathbf{t})+(k-1)\kappa_U(\mathbf{m}';\mathbf{s},\mathbf{t})$, positive for real $k\ge 1$; by Lemma S20(iv) the values converge there and $\langle 1\rangle_1=P_F(\mathbf{m};\mathbf{s},\mathbf{t})$, the data at $k=1$ being $(\mathbf{m};\mathbf{s},\mathbf{t})$.

(ii) The direction $N(\mathbf{m};\mathbf{s},\mathbf{t})$ is integral, on the balance wall, with the walls $N\kappa_U>0$; its stride is $\alpha^{N\mathbf{m}}\prod_j K_j^{-Ns_j}\Psi_j^{-Nt_j}=f_{\mathbf{m}}^N$ on the positive orthant, and by Lemma S20(iv) at $k=\frac{1}{N}$, where the walls are the $\kappa_U>0$, $\langle 1\rangle_{1/N}=P_F(\mathbf{m};\mathbf{s},\mathbf{t})$. The wall degree is that of Definition 39.11(ii) with $\kappa_{1,U}=N\kappa_U$. □

Proposition 41.12 (One factorization serves every coefficient). *As stated.*

Proof. A_P is affine in k, so $A_P(k_0+\epsilon) = M + \epsilon E$. For $|\epsilon|\,\|EM^{-1}\| < 1$, $M+\epsilon E = (I+\epsilon EM^{-1})M$ and $(M+\epsilon E)^{-1} = M^{-1}\sum_{j\ge 0}(-\epsilon EM^{-1})^j$, so $x_P(M+\epsilon E)^{-1} = \sum_j x_P M^{-1}(-EM^{-1})^j\epsilon^j$, which is the recursion. With $A_F(k_0+\epsilon) = A_F(k_0) + \epsilon E_F$, $E_F = A_F(k_0+1) - A_F(k_0)$, the coefficient of ϵ^j, $j \ge 1$, in the reduction $x_F - x_P A_P^{-1} A_F$ is $-\lambda_j A_F(k_0) - \lambda_{j-1}E_F$. Each λ_j is a product of a row by E and one solve with the factorization of M. If N/D and N'/D', with the degree bounds and $D(k_0)D'(k_0) \ne 0$, have the same Taylor coefficients up to the order d_1+d_2 at k_0, then $ND' - N'D = DD'(N/D - N'/D')$ vanishes to the order d_1+d_2+1 at k_0 and has degree at most d_1+d_2, so it is 0. □

Proposition 41.13 (The densities of a spanning set). *As stated.*

Proof. (i) A Laurent insertion b with real coefficients is real and continuous on the positive part of the chart, and $b\,\omega_0$ has the density $b\,\omega_0/dy$; the sets $\{f \ge t\}$, $t > 0$, are compact (proof of Proposition 40.2(i)), so $\mu_{f,b}$ is finite on $[t, f^\star]$, with $|\mu_{f,b}|$ at most the image of $|b|\,\omega_0$. Where $\int |b| f^{\operatorname{Re} k}\omega_0 < \infty$, $\langle b\rangle_k = \int b f^k \omega_0 = \int t^k \mu_{f,b}(dt)$ by the change of variables of an image measure. The proof of Proposition 40.3(i), with the weight $b\,\omega_0/dy$ in place of ω_0/dy, gives a real-analytic density on $(0, f^\star)$; its positivity is not used there. The proof of Proposition 40.3(ii) uses Lemma S22(i), which allows every real-analytic weight w, here $w = b\,\omega_0/dy$, with $w(\log\alpha^\star) = b(\alpha^\star)\,(\omega_0/dy)(\alpha^\star)$; so the germ has the stated form with $g_b(0) = b(\alpha^\star)g(0)$.

(ii) Every monomial of b_l has an absolutely convergent value for large real k (Proposition 38.5(i)), so $\int t^{K_0} d|\mu_{f,b_l}| < \infty$ and $\int t^{K_0} d|t^{-1}\mu_{f,b_i}| < \infty$ for some K_0. Apply Lemma S21 to the measures $t^{-1}\mu_{f,b_i}$ and μ_{f,b_l} on $(0, f^\star]$ and the operators $Q(-1-\theta)$ and $-\tilde P_{il}(-1-\theta)$, whose adjoints are $Q(\theta)$ and $-\tilde P_{il}(\theta)$, $\theta^\dagger = -1-\theta$, with $Q(\theta)t^k = Q(k)t^k$. Its numbers are $M_E(k) = Q(k)\langle b_i\rangle_{k-1} - \sum_l \tilde P_{il}(k)\langle b_l\rangle_k$, which vanish for large k; by Lemma S21(i) the distribution E vanishes, and that is the system.

(iii) The identity $v(k-1) = \mathbf{B}(k)v(k)$ of meromorphic functions, iterated, gives $v(k-K+1) = \mathbf{B}(k-K+2)\cdots\mathbf{B}(k)\,v(k)$. Near $k = K$ the right side is holomorphic, the factors $\mathbf{B}(m)$, $2 \le m \le K$, being finite and v converging in a neighbourhood of K, the walls at K being positive; so the left side, v near 1, is holomorphic at 1 with the stated value. □

Proposition 41.14 (The native gauges of all exponents). *As stated.*

Proof. (i) K_j and Ψ_j are homogeneous of the degrees r_j and r_j^*, so by Euler's identity $\sum_e R_{j,e} = r_j$ and $\sum_e R^*_{j,e} = r_j^*$, and $\mathbf{u}(\alpha;\mathbf{s},\mathbf{t})_{\mathcal{E}} = \sum_j (s_j r_j + t_j r_j^*)$: the balance wall. A native gauge of the ray is a point of X_F with $c_e^{(1)} = m_{1,e} - \sum_j \big(s_{1j}R_{j,e} + t_{1j}R^*_{j,e}\big) = 0$ for every e (Definitions 38.1 and 38.4), that is $\mathbf{m}_1 = \mathbf{u}(\alpha;\mathbf{s}_1,\mathbf{t}_1)$; for data on the balance wall with $m_{\mathcal{E}} \ne 0$ the target of Definition 8.17 is $\mathbf{m}$. The set is the graph of the regular map $(\alpha,\mathbf{s},\mathbf{t}) \mapsto \mathbf{u}(\alpha;\mathbf{s},\mathbf{t})$, isomorphic by the projection to $X_F \times \mathbb{C}^{2N}$; X_F is the complement in the torus $(\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times$, of dimension $n-1$, of the hypersurface $\prod_j K_j\Psi_j = 0$, which does not contain $\mathbf{1}$, so it is a nonempty open subset of an irreducible smooth variety, irreducible and smooth itself.

(ii) Let $\alpha_1,\dots,\alpha_\nu$ be the nondegenerate native gauges of the ray, w_1 its direction, and W the data on the balance wall, a vector space of dimension $n-1+2N$ with the coordinates $(m_e)_{e\ne e_0}$, $\mathbf{s}$, $\mathbf{t}$. In the coordinates $y_e = \log\alpha_e$, $e \ne e_0$, of the chart near each α_c, the map $(y,\mathbf{s},\mathbf{t}) \mapsto \big(\mathbf{u}(e^y;\mathbf{s},\mathbf{t});\mathbf{s},\mathbf{t}\big)$ into W is holomorphic, and its derivative is block triangular, with the identity on $(\mathbf{s},\mathbf{t})$ and the matrix $\big(\partial_{y_{e'}}\mathbf{u}_e\big)_{e,e'\ne e_0} = -\big(\partial_{y_{e'}}c_e^{(1)}\big)$, minus the Hessian of $\log f$ on the chart, which is nonsingular at α_c (Definition 38.4). By the holomorphic inverse function theorem there are disjoint neighbourhoods U_c of the α_c in X_F and a neighbourhood $\mathcal{W}$ of w_1 in W such that every $w \in \mathcal{W}$ has exactly one point native for it in each U_c. The exceptional set of Theorem 38.24(i),

whose count concerns the critical points of the product of powers of the letters with the exponents of the direction, is a countable union of proper subvarieties; a proper subvariety is closed with empty interior in W, and its real points have empty interior in the real data, a polynomial that vanishes on an open set of real points being 0. By the Baire category theorem $\mathcal{W}$ contains data, real if need be, outside the exceptional set; at such data the native points are $(-1)^{n-1}\operatorname{eu}(X_F)$ in number, and they include the ν points in the U_c. □

Definition 41.15 (The procedure that finds native gauges by continuation). As stated.

Proof. (1) is Proposition 41.14(i). Steps (2) and (3) only propose points; nothing in the output rests on them. (4) In the coordinates α_e, $e \neq e_0$, of the chart the map $F = (c_e^{(1)})_{e\neq e_0}$ is rational, with no pole on a box Y on which no letter vanishes; read as a map of a real space of dimension $2(n-1)$, it is continuously differentiable there. Krawczyk's test asks for a point $\tilde{y} \in Y$, a nonsingular real matrix $\mathcal{Y}$ and an interval enclosure $F'(Y)$ of the derivative on Y with $\tilde{y}-\mathcal{Y}F(\tilde{y})+(I-\mathcal{Y}F'(Y))(Y-\tilde{y}) \subseteq Y$. By the mean value theorem in integral form, $y - \mathcal{Y}F(y) = \tilde{y} - \mathcal{Y}F(\tilde{y}) + \int_0^1 \big(I - \mathcal{Y}F'(\tilde{y} + \tau(y - \tilde{y}))\big)d\tau\,(y - \tilde{y})$, and the integral of matrices in the convex set $F'(Y)$ lies in it; so $y \mapsto y - \mathcal{Y}F(y)$ maps the compact convex set Y into itself and has a fixed point (Brouwer's fixed point theorem), a zero of F since $\mathcal{Y}$ is nonsingular. A zero of F in Y is a native gauge of the ray (Definition 38.4, the defects summing to 0 as f has degree 0, so that the defect at e_0 vanishes with the others). The Jacobian of F in the α_e is the Hessian of $\log f$ in the y_e times $\operatorname{diag}(\alpha_e^{-1})$, so an enclosure of the Hessian determinant that excludes 0 makes every native gauge in Y nondegenerate. Disjoint boxes hold distinct points, and distinct values of f at native gauges are distinct native values. □

48.41 Chapter 42: inputs along a ray

In this section $\vartheta = y\,d/dy$ on $(0,\infty)$, and for a function F on $(0,\infty)$, $\hat{F}(s) = \int_0^\infty y^{s-1}F(y)\,dy$ wherever it converges. Two facts about this Mellin read are used throughout.

Lemma S27. *Let F be smooth on $(0,\infty)$ with every $\vartheta^a F$ of size $O\big((1+|\log y|)^N\big)$ as $y \to 0$ and $O\big(y^C e^{-b\sqrt{y}}\big)$ as $y \to \infty$, for some N, C and $b > 0$. Then $\hat{F}$ is holomorphic on $\operatorname{Re} s > 0$, and:*

(i) for every polynomial r and $k \in \mathbb{N}$, $\int_0^\infty y^{s-1}\,y^k r(\vartheta)F\,dy = r(-s-k)\,\hat{F}(s+k)$ on $\operatorname{Re} s > 0$;

(ii) if $\hat{F}$ vanishes on a vertical line in $\operatorname{Re} s > 0$, then $F = 0$.

Proof. The bounds give absolute convergence, locally uniform on $\operatorname{Re} s > 0$, of every integral involved. (i) For $G = \vartheta^a F$ and $\operatorname{Re} a' > 0$, $\int_0^\infty y^{a'-1}\vartheta G\,dy = \int_0^\infty y^{a'}G'\,dy = \big[y^{a'}G\big]_0^\infty - a'\int_0^\infty y^{a'-1}G\,dy$, and the bracket vanishes at both ends by the bounds. So ϑ acts as $-a'$ under $\int_0^\infty y^{a'-1}(\cdot)\,dy$, and with $a' = s+k$ this gives the claim for every monomial $r(\vartheta) = \vartheta^a$, hence for every polynomial. (ii) With $y = e^v$ and $s = c + i\tau$, $\hat{F}(c+i\tau) = \int_{\mathbb{R}} e^{i\tau v}\,e^{cv}F(e^v)\,dv$ is the Fourier transform of the integrable continuous function $e^{cv}F(e^v)$; it vanishes for every τ, so that function vanishes (the uniqueness of the Fourier transform on $L^1(\mathbb{R})$), and $F = 0$. □

Definition 42.1 (Recurrences; the values input and the slope input). As stated.

Proof. The objects are well defined: L, L^* and L' act on functions on a coset of $\mathbb{Z}$, or on meromorphic functions, the coefficients being polynomials; the symbol is that of Corollary 38.7 for the coefficients c_i; the two inputs are inputs in the sense of Definition 5.4, a lattice input with the densities p and $\tilde{p}$ at the points of the coset beyond the cut, where both are finite because p is holomorphic there. □

Proposition 42.2 (The values of a ray as an input). *As stated.*

Proof. (i) By Theorem 38.14(i), $0 < f \le C'e^{-c'\|y\|}$ on the chart, so $\int_{y_{e_0}=0} |f^k|\,dy = \int f^{\operatorname{Re} k}\,dy$ converges for $\operatorname{Re} k > 0$, locally uniformly, and $\langle 1\rangle_k$ is holomorphic there. By Proposition 39.30 it continues to a meromorphic function p on $\mathbb{C}$, with $\lim_{k\to -j}(k+j)^{n-1}p(k) = V_1\,\mathsf{C}_j(1)$ for every $j \ge j(1) = 0$. The class $m = \sum_i c_i(k)\operatorname{Sh}^i[1]$ of the module (Definition 39.1) has the values $\langle m\rangle_k = \sum_i c_i(k)\langle f^i\rangle_k = \sum_i c_i(k)\langle 1\rangle_{k+i}$, which vanish for all large integers k; by Theorem 39.41, whose hypotheses are those of this statement, the continuation of $\langle m\rangle_k$ vanishes identically, and that continuation is $(Lp)(k)$. Multiply $(Lp)(k) = 0$ by $(k+j)^{n-1}$ and let $k \to -j$: for $i \le j$, $(k+j)^{n-1}p(k+i) \to V_1\,\mathsf{C}_{j-i}(1)$, and for $i > j$, $p(k+i)$ is holomorphic at $k = -j$, since $-j+i \ge 1$, so its product with $(k+j)^{n-1}$ tends to 0, $n \ge 2$. So $V_1 \sum_{i\le \min(h,j)} c_i(-j)\,\mathsf{C}_{j-i}(1) = 0$, and $V_1 > 0$.

(ii) Differentiating $\sum_i c_i(x)p(x+i) = 0$ gives $\sum_i c_i'(x)p(x+i) + \sum_i c_i(x)p'(x+i) = 0$, that is $L\tilde{p} = L'p$.

(iii) By Proposition 38.9, $\langle 1\rangle_k > 0$ for large k and $\langle 1\rangle_{k+1}/\langle 1\rangle_k \to f(\alpha^\star)$, so the root rate of p along the integers is $f(\alpha^\star)$. For real $k \ge 1$, $\tilde{p}(k) = \int f^k \log(1/f)\,dy$ by differentiation under the integral, which the bound of (i) allows, and $|\tilde{p}(k)| \le f(\alpha^\star)^{k-1}\int f\,|\log f|\,dy$ because $0 < f \le f(\alpha^\star)$ on the chart (Theorem 38.14(i)). The last integral converges: $\log f$ is a combination of the coordinates y_e and of logarithms of the letters, polynomials with nonnegative coefficients in e^y, so $|\log f| \le C(1+\|y\|)$, while $f \le C'e^{-c'\|y\|}$. So the root rate of $\tilde{p}$ is at most $f(\alpha^\star)$. □

Proposition 42.3 (Summation by parts at the cut). *As stated.*

Proof. For a function φ on the integers $x > -h$ with $\sum_x |R(x)|\,|\varphi(x)|$ and $\sum_x |R(x)\,c_i(x)\,\varphi(x+i)|$ finite over $x \ge 1$, which holds for $\varphi \in \{p, \tilde{p}\}$ since R and the c_i are polynomials and p, $\tilde{p}$ have root rate below one,

$$\sum_{x\ge 1}(L^*R)(x)\,\varphi(x) = \sum_{i=0}^{h}\sum_{x\ge 1} c_i(x-i)R(x-i)\,\varphi(x) = \sum_{i=0}^{h}\sum_{x'\ge 1-i} c_i(x')R(x')\,\varphi(x'+i),$$

with $x' = x - i$. The terms with $x' \ge 1$ add up to $\sum_{x'\ge 1} R(x')\,(L\varphi)(x')$, and those with $1 - i \le x' \le 0$, written $x' = -j$ with $0 \le j < i$, to $\sum_{j=0}^{h-1} R(-j)\,B_j(\varphi)$, $B_j(\varphi) = \sum_{i=j+1}^{h} c_i(-j)\,\varphi(i-j)$. So

$$(\mathsf{T}_{0;(1)}[x]\,(L^*R)\,\varphi\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \sum_{x\ge 1} R(x)\,(L\varphi)(x) + \sum_{j=0}^{h-1} R(-j)\,B_j(\varphi),$$

the Trailmoment at index zero on the counting input being the sum over the integers beyond the cut (Definition 6.1). For $\varphi = p$, $Lp = 0$ and $B_j(p) = \sum_{i>j} c_i(x)p(x+i)\big|_{x=-j} = -\beta_j(-j)$, which is the first identity. For $\varphi = \tilde{p}$, $L\tilde{p} = L'p$ (Proposition 42.2(ii), whose proof uses only $Lp = 0$), and the same rearrangement with L' in place of L gives $\sum_{x\ge 1} R(x)(L'p)(x) = (\mathsf{T}_{0;(1)}[x]\,(L'^*R)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - \sum_j R(-j)B_j'(p)$ with $B_j'(p) = \sum_{i>j} c_i'(-j)\,p(i-j)$. Hence the second identity holds with the boundary $B_j(\tilde{p}) - B_j'(p) = \sum_{i>j}\big(c_i(x)\tilde{p}(x+i) - c_i'(x)p(x+i)\big)\big|_{x=-j} = -\frac{d}{dx}\sum_{i>j} c_i(x)p(x+i)\big|_{x=-j} = \beta_j'(-j)$.

If every c_i with $i > j \ge 1$ vanishes to order two at $-j$, then near $x = -j$ each term of $\beta_j = -\sum_{i>j} c_i(x)p(x+i)$ is a polynomial with a double zero at $-j$ times a function holomorphic there, so $\beta_j(-j) = \beta_j'(-j) = 0$ for $j \ge 1$. □

Proposition 42.4 (The fixed block). *As stated.*

Proof. (i) For R of degree d with leading coefficient $\rho \neq 0$, $c_i(x-i)R(x-i)$ has degree at most $D+d$ and coefficient $[x^D]c_i\,\rho$ at x^{D+d}, so L^*R has degree at most $D+d$ and coefficient $\chi(1)\rho \neq 0$ there. So L^* is injective on $\Bbbk[x]$, and for every d it maps the polynomials of degree at most d, a space of dimension $d+1$, into those of degree at most $D+d$, meeting the polynomials of degree below D only in 0. The dimensions add up, $(d+1)+D = D+d+1$, so the polynomials of degree at most $D+d$ are the direct sum of the two, for every d; this is the decomposition of $\Bbbk[x]$.

(ii) Write $W = L^*B + r$ with $\deg r < D$ by (i), and $L'^*B = L^*B_2 + r_2$ in the same way; B, r, B_2, r_2 have coefficients in $\Bbbk$. By Proposition 42.3,

$$(\mathsf{T}_{0;(1)}[x]\,W\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,r\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - \sum_j B(-j)\,\beta_j(-j),$$

$$\begin{aligned}(\mathsf{T}_{0;(1)}[x]\,W\,\tilde{p}\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) &= (\mathsf{T}_{0;(1)}[x]\,r\,\tilde{p}\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + (\mathsf{T}_{0;(1)}[x]\,r_2\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\\ &\quad - \sum_j B_2(-j)\,\beta_j(-j) + \sum_j B(-j)\,\beta'_j(-j).\end{aligned}$$

A polynomial of degree below D is a $\Bbbk$-combination of the $(x-1)^a$, $a<D$, and the Trailmoment at index zero is linear in the density, which gives the claim. □

Corollary 42.5 (A mark opened at an integer strength). *As stated.*

Proof. The bunch $B=\{e\}$ has width one, $c(\{e\}) = r(\{e\}) + r(R) - r(\mathcal{E}) = 1$ since e is neither a loop nor a coloop, and the unit weights factor across B and R. Its minors on B are $\mathcal{B}|B$, with the one basis $\{e\}$, and $\mathcal{B}/R$, with the one basis $\varnothing$; so the bunch family of Theorem 19.13 has the period 1, and the hinge ratio of B is $\varrho_B = \mathbf{E}_{\mathcal{B}/R}/\mathbf{E}_{\mathcal{B}|B} = \alpha_e/\alpha_e = 1$ (Definition 19.2). The room family is the deletion $\mathcal{B}\setminus e = \mathcal{B}|R$ at the level d and the contraction $\mathcal{B}/e = \mathcal{B}/B$ at the level r, and $\varrho_R = \mathbf{E}_{\mathcal{B}/e}/\mathbf{E}_{\mathcal{B}\setminus e} = X_{/}/X_{\setminus}$, since $\mathbf{E} = K\Psi = \alpha^R K(\alpha)K(\alpha^{-1})$ for each of them. At the centre $(s\mathbf{1}; s, s)$ the wall of B is $\kappa_B = s\,c(B) = s$ (Proposition 8.21(ii)), so the odd share is fixed at $2\delta^d = s - \kappa_B = 0$ and the level shares are $(a^d, b^d, a^r, b^r) = (d, d, s-d, s-d)$, d the even share. The one-table display of Theorem 19.13 reads

$$\Gamma(s)^2 P_{\mathcal{B}}(s\mathbf{1}; s, s) = \frac{1}{2\pi i}\int \Gamma(d)^2\,\Gamma(s-d)^2\,\Phi_s(d)\,dd, \qquad \Phi_s(d) = \int_{y_{e_0}=0} X_{\setminus}^{-d} X_{/}^{-(s-d)}\,dy,$$

the room period at the masses $s\mathbf{1}_R$, whose integrand is $\alpha^{s\mathbf{1}_R} K_{\setminus}^{-d}\Psi_{\setminus}^{-d} K_{/}^{-(s-d)}\Psi_{/}^{-(s-d)} = X_{\setminus}^{-d}X_{/}^{-(s-d)}$.

When $X_\flat = X_{\setminus}$, $\sup\varrho_B = 1 < \inf\varrho_R$, and Corollary 19.21(i) gives $P_{\mathcal{B}}(s\mathbf{1}; s, s) = \sum \text{Res}$ over the poles of $\Gamma(d)^2$ at $d = -x$, $x\in\mathbb{N}$, where the two lattices of the level d coincide, every room period there converging. Then $\Phi_s(-x) = q_s(x)$. When $X_\flat = X_{/}$, $\inf\varrho_B > \sup\varrho_R$, and the same corollary closes on the level r with the opposite sign; the substitution $d = s-\tau$ exchanges the two levels and the two minors and turns that closing into the closing at $\tau = -x$, with $\Phi_s(s-\tau) = \int X_{/}^{-\tau} X_{\setminus}^{-(s-\tau)}\,dy$ and $q_s(x) = \Phi_s(s+x)$. In both cases, writing $\Phi^\flat(\tau) = \int X_\flat^{-\tau} X_\sharp^{-(s-\tau)}\,dy$, so that $q_s(x) = \Phi^\flat(-x)$, the room periods $\Phi^\flat(-x)$, $x\in\mathbb{N}$, converge. The walls of the room family with the minor of $X_\flat$ at the share $-x$ and that of $X_\sharp$ at $s+x$, at the masses $s\mathbf{1}_R$, are $\kappa_S = -x\,c_\flat(S) + (s+x)\,c_\sharp(S)$ with the widths $c_\flat$, $c_\sharp$ of the two minors, by the computation of Proposition 8.21(ii), which is linear in the strengths; they are affine in x and positive at every point of $\mathbb{N}$ (Theorem 8.7), hence on $[0,\infty)$ and on a neighbourhood of it, so q_s converges and is holomorphic there. The closing reads

$$\Gamma(s)^2 P_{\mathcal{B}}(s\mathbf{1}; s, s) = \sum_{x\in\mathbb{N}} \operatorname*{Res}_{\tau=-x}\Big(\Gamma(\tau)^2\,\Gamma(s-\tau)^2\,\Phi^\flat(\tau)\Big).$$

At the integer s, $\Gamma(s-\tau) = (s-1-\tau)\cdots(1-\tau)\,\Gamma(1-\tau)$, so $\Gamma(\tau)^2\Gamma(s-\tau)^2 = V_s(-\tau)\,\Gamma_{4\,0}\big[\tau,\ \tau,\ 1-\tau,\ 1-\tau\big]$, and by Lemma 5.16 the last factor has at $\tau=-x$ the principal part $(\tau+x)^{-2}$ and no simple part. The residue is therefore the derivative of $V_s(-\tau)\Phi^\flat(\tau)$ at $\tau=-x$,

$$-V_s'(x)\,q_s(x) + V_s(x)\,\Phi^{\flat\prime}(-x) = V_s(x)\,\tilde q_s(x) - V_s'(x)\,q_s(x),$$

since $\Phi^{\flat\prime}(-x) = -q_s'(x)$. The ratio $X_\flat/X_\sharp$ is at most ρ, so $0 < q_s(x) \le \rho^x q_s(0)$ and, by differentiation under the integral, $|\tilde q_s(x)| \le \rho^{x-1}\int \log(X_\sharp/X_\flat)\,(X_\flat/X_\sharp)\,X_\sharp^{-s}\,dy$ for $x \ge 1$, the last integral being $\tilde q_s(1)$, finite by the holomorphy at 1; with V_s a polynomial the terms have root rate at most ρ. □

Proposition 42.6 (Boundary relations). *As stated.*

Proof. The identity $Lp = 0$ reads $\beta_{x_0,j}(x) = -\sum_{i>j} c_i(x)\,p(x+i)$ near x_0, and the right side is holomorphic at x_0 by the hypothesis; so is $\beta_{x_0,j}$, and the first relation is this identity at x_0. Differentiating it at x_0 gives $\beta'_{x_0,j}(x_0) = -\sum_{i>j}\big(c_i'(x_0)p(x_0+i) + c_i(x_0)p'(x_0+i)\big)$, which is the second relation since $\tilde p = -p'$. $\beta_{x_0,j} = \sum_{i\le j} c_i\,p(\cdot+i)$ is a finite sum of products of the Taylor expansions of the c_i at x_0 with the Laurent expansions of p at x_0+i; its value and its derivative at x_0 are the coefficients of $(x-x_0)^0$ and $(x-x_0)^1$ of that sum, finitely many products of these coefficients.

When c_h has no zero at the integers $x \ge 1$, $L\tilde p = L'p$ at x expresses $\tilde p(x+h)$ through $\tilde p(x),\dots,\tilde p(x+h-1)$ and values of p, so by induction every $\tilde p(x)$, $x \ge 1$, is an affine function of $\tilde p(1),\dots,\tilde p(h)$; for an integer x_0 with $x_0 + j \ge 0$ the slopes in the second relation are at the integers $x_0 + i \ge 1$, $i > j$, and the relation is affine in the initial slopes.

At the cut of the values of a ray, Proposition 42.2(i) at $j=0$ gives $p(x) = V_1 x^{-(n-1)}(1+O(x))$, $\mathsf{C}_0(1) = 1$, so $\mathsf{L}_0 = V_1$, and p has a pole of order exactly $n-1$ at 0. $\beta_{0,0} = c_0\,p$ is holomorphic at 0, so c_0 vanishes to order at least $n-1$ there, and $c_0(x)p(x) = \gamma_{n-1}\mathsf{L}_0 + (\gamma_{n-1}\mathsf{L}_1 + \gamma_n\mathsf{L}_0)\,x + O(x^2)$. The two relations at $x_0 = 0$, $j = 0$ are therefore the displayed ones; the first is the low-end relation of Corollary 39.34. When every table has unit weights, the values are $\langle 1\rangle_k = P_F(kq)$ for the question q of the direction (Definition 17.1), the direction being a point of the balance wall with real data, and Theorem 17.5(i) gives $k^{n-1}P_F(kq) = V_1 + o(k)$ as $k \downarrow 0$, every $\log\theta_{j,T}$ being 0; since $k^{n-1}p(k) = \mathsf{L}_0 + \mathsf{L}_1 k + O(k^2)$ is holomorphic at 0, $\mathsf{L}_1 = 0$. □

Proposition 42.7 (Minimality). *As stated.*

Proof. (i) Let $D = \deg c_h$. Divided by $c_h(x)$, which has no zero at the integers $x \ge 1$, the recurrence reads $u(x+h) + \sum_{i<h} a_i(x)\,u(x+i) = 0$ with $a_i = c_i/c_h \to [x^D]c_i/[x^D]c_h$ as $x \to \infty$, and $a_0(x) \ne 0$ at these integers. The limit characteristic polynomial is $\chi(\lambda)/[x^D]c_h$, of degree h, with $\chi(0) = [x^D]c_0 \ne 0$ and roots λ_c of distinct moduli. Perron's theorem on such recurrences (coefficients with limits, a_0 without zeros on the range, roots of the limit polynomial of distinct moduli) gives a basis $u_1,\dots,u_h$ of the solutions on the integers $x \ge 1$ with $u_c(x+1)/u_c(x) \to \lambda_c$, so $|u_c(x)|^{1/x} \to |\lambda_c|$. Let $u = \sum_c b_c u_c \ne 0$, and let d be the index with the largest $|\lambda_d|$ among the c with $b_c \ne 0$. For the other such c, $u_c(x)/u_d(x) \to 0$, the root rates being smaller; so $u(x) = b_d u_d(x)\,(1+o(1))$, $u(x+1)/u(x) \to \lambda_d$, and u has the root rate $|\lambda_d|$. Hence u has root rate at most λ exactly when $b_c = 0$ for every c with $|\lambda_c| > \lambda$, and U_λ is spanned by the u_c with $|\lambda_c| \le \lambda$, which are independent.

(ii) $\tilde p$ satisfies $L\tilde p = L'p$ (Proposition 42.2(ii), whose proof uses only $Lp = 0$) and has root rate at most λ; a φ with the same two properties differs from it by $u = \varphi - \tilde p$ with $Lu = 0$ and root rate at most λ, the root rate of a difference being at most the larger of the two; conversely every

$\tilde p + u$, $u \in U_\lambda$, has both properties. The same holds for $L\varphi = 0$ with p. A second relation of Proposition 42.6 at an integer x_0 with $x_0 + j \le 0$ holds for $\tilde p + u$ exactly when its left side vanishes on u, since its right side is read off the Laurent data of p at $x_0, \dots, x_0 + j \le 0$ and involves no slope; relations at other points are consequences of $L\tilde p = L'p$ and single out nothing; so if these left sides vanish together on no nonzero $u \in U_\lambda$, $\tilde p$ is the only solution of root rate at most λ that satisfies the relations. The first relations single out p in the same way. □

Proposition 42.8 (The two ends paired). *As stated.*

Proof. (i) For $\sigma < 0$ every term vanishes. For $\sigma \ge 0$, $c_i(\sigma - i) = \epsilon\, c_i(-\sigma)$ by the hypothesis read at $x = -\sigma$, so the sum is $\epsilon \sum_{i=0}^{\min(h,\sigma)} c_i(-\sigma)\, \mathsf{C}_{\sigma-i}$, which vanishes by Proposition 42.2(i) at $j = \sigma$.

(ii) Write $p(x) = \langle 1 \rangle_x$, and $(L^*\mathsf{C})(\sigma) = \sum_i c_i(\sigma - i)\mathsf{C}_{\sigma-i}$. Shifting j by one,

$$B(\sigma + 1) = \sum_{i=1}^{h} \sum_{j=0}^{i-1} c_i(\sigma - j)\, \mathsf{C}_{\sigma-j}\, p(\sigma - j + i),$$

and the difference with $B(\sigma)$ keeps the term $j = 0$ of the first sum and the term $j = i$ of the second:

$$B(\sigma + 1) - B(\sigma) = \sum_{i=1}^{h} \big(c_i(\sigma)\, \mathsf{C}_\sigma\, p(\sigma + i) - c_i(\sigma - i)\, \mathsf{C}_{\sigma-i}\, p(\sigma) \big) = \mathsf{C}_\sigma\, (Lp)(\sigma) - p(\sigma)\, (L^*\mathsf{C})(\sigma),$$

the terms $i = 0$ cancelling. For integers $\sigma \ge 1$ both vanish, by Proposition 42.2(i) and by (i). At $\sigma = 1$ only $j = 1$ has $\mathsf{C}_{1-j} \ne 0$, with $\mathsf{C}_0 = 1$, so $B(1) = \sum_{i \ge 1} c_i(0)\, p(i)$, which is $-\gamma_{n-1} V_1$ by Proposition 42.6.

(iii) By Proposition 42.6, c_0 vanishes to order at least $n - 1$ at 0; with $\deg c_0 \le n - 1$ and $c_0 \ne 0$, $c_0 = \gamma_{n-1} x^{n-1}$, $\gamma_{n-1} \ne 0$, and the coefficients of x^{n-1} give $\chi(\lambda) = \sum_i [x^{n-1}] c_i \lambda^i$ with $\chi(0) = \gamma_{n-1}$. On the centre ray of a family, $\alpha^\star = \mathbf{1}$ and $\langle 1 \rangle_k = P_F(kq)$ for the centre question q; Theorem 17.5(ii) at the centre, with $-\nabla^2 X(0) = 4H$ on the chart in the coordinates $z = y/2$ (H as in Theorem 38.15) and $e^{X(0)} = f(\mathbf{1})$, gives

$$\langle 1 \rangle_k = f(\mathbf{1})^k \Big(\frac{2\pi}{k} \Big)^{\frac{n-1}{2}} (\det H)^{-1/2} \big(1 + O(k^{-1}) \big),$$

the factor 2^{n-1} of that theorem cancelling against $\det(4H)^{1/2} = 2^{n-1} (\det H)^{1/2}$. Theorem 39.41 at $g = 1$ gives $j^{(n-1)/2} f(\mathbf{1})^j \mathsf{C}_j(1) \to (2\pi)^{-(n-1)} a_0(1)$, where $a_0(1) = (2\pi)^{(n-1)/2} (dy/dv)(\mathbf{1}) = (2\pi)^{(n-1)/2} (\det H)^{-1/2}$ by Theorem 38.6(i), the coordinates v having $\frac{1}{2} \sum v_i^2 = \frac{1}{2} y^{\mathsf{T}} H y + O(|y|^3)$. So for fixed i, as $a \to \infty$, $\mathsf{C}_a \langle 1 \rangle_{a+i} = \lambda_\star^i (\det H)^{-1} a^{-(n-1)} (1 + o(1))$, with $\lambda_\star = f(\mathbf{1})$, while $c_i(a) = [x^{n-1}] c_i\, a^{n-1} + O(a^{n-2})$. Every term of $B(\sigma)$ with $a = \sigma - j$ therefore tends to $[x^{n-1}] c_i\, \lambda_\star^i (\det H)^{-1}$, and

$$\lim_{\sigma \to \infty} B(\sigma) = (\det H)^{-1} \sum_{i=1}^{h} i\, [x^{n-1}] c_i\, \lambda_\star^i = (\det H)^{-1} \lambda_\star \chi'(\lambda_\star).$$

By (ii) the limit is $-\chi(0) V_1$, so $V_1 \det H = -\lambda_\star \chi'(\lambda_\star)/\chi(0)$, which is positive. The symbol vanishes at $\lambda_\star$: $\langle 1 \rangle_{k+1} / \langle 1 \rangle_k \to \lambda_\star$ (Proposition 38.9, a limit that needs no hypothesis on the coefficients of L), so dividing $\sum_i c_i(k) \langle 1 \rangle_{k+i} = 0$ by $k^{n-1} \langle 1 \rangle_k$ and letting $k \to \infty$ gives $\chi(\lambda_\star) = 0$; writing $\chi(\lambda) = \chi(0) \prod_c (1 - \lambda/\lambda_c)$ over its roots with multiplicity, a multiple root at $\lambda_\star$ would give $\chi'(\lambda_\star) = 0$, so $\lambda_\star$ is simple, and $\lambda_\star \chi'(\lambda_\star) = -\chi(0) \prod_{c \ne \star} (1 - \lambda_\star/\lambda_c)$. □

The rest of this section reads the list $\binom{\mathcal{E}}{1}$. There $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y) = \int_0^\infty x^{-1}e^{-x-y/x}\,dx$ and $\mathsf{h}(y) = \sum_{m\ge 0} y^m/m!^2$. The configuration $(1,-1)$ on one scale has $\mathrm{cone}(t) = \mathbb{R}$, so $\mathsf{G}_{\nu;(1,-1)}(1,y)$ is entire in ν (Theorem 2.2); its ladder and its exchange relation (Theorem 2.4(i),(ii)) read $\partial_y \mathsf{G}_{\nu;(1,-1)}(1,y) = -\mathsf{G}_{\nu-1;(1,-1)}(1,y)$ and $\nu\,\mathsf{G}_{\nu;(1,-1)}(1,y) = \mathsf{G}_{\nu+1;(1,-1)}(1,y) - y\,\mathsf{G}_{\nu-1;(1,-1)}(1,y)$. So $\vartheta\mathsf{g} = -y\,\mathsf{G}_{-1;(1,-1)}(1,y)$, the exchange relation at $\nu = -1$ gives $y\,\mathsf{G}_{-2;(1,-1)}(1,y) = \mathsf{g} + \mathsf{G}_{-1;(1,-1)}(1,y)$, and

$$\vartheta^2\mathsf{g} = -y\,\mathsf{G}_{-1;(1,-1)}(1,y) + y^2\mathsf{G}_{-2;(1,-1)}(1,y) = y\,\mathsf{g};$$

and $\vartheta^2\mathsf{h} = \sum_{m\ge 1} y^m/(m-1)!^2 = y\,\mathsf{h}$. By Lemma 20.29(i) at $l = 0$, $\mathsf{g} = -\mathsf{h}\log y + \tilde{\mathsf{g}}$ with the entire function $\tilde{\mathsf{g}}(y) = \sum_m 2(\log\Gamma)'(1+m)\,y^m/m!^2$. The solutions of $\vartheta^2 u = yu$ on $(0,\infty)$ form a space $\mathcal{V}$ of dimension two with the basis g,h, the two being independent since g/h is not constant; the $n+1$ products $\mathsf{g}^{n-j}\mathsf{h}^j$ are independent as well, h^n times the powers of the nonconstant function g/h.

Proposition 42.9 (The native gauges of $\binom{\mathcal{E}}{1}$). *As stated.*

Proof. (i) The occupation of e is $R_e = \alpha_e\partial_{\alpha_e}\log K = u_e$, and since $\Psi(\alpha) = \alpha^{\mathcal{E}}K(\alpha^{-1})$ the dual occupation is $R_e^* = \alpha_e\partial_{\alpha_e}\log\Psi = 1 - v_e$, $v_e = \alpha_e^{-1}/K(\alpha^{-1})$. The defect of the direction (Definition 38.1) is $c_e^{(1)} = m - s\,u_e - t\,(1-v_e) = \frac{s-t}{n} - s\,u_e + t\,v_e$, by the balance $m - t = \frac{s-t}{n}$. On X_F the α_e, $K(\alpha)$ and $K(\alpha^{-1})$ are nonzero, so $u_e, v_e \ne 0$, $u_e v_e = \lambda(\alpha) \ne 0$ for every e, and $\sum_e u_e = \sum_e v_e = 1$.

Let α be a native gauge, $c_e^{(1)} = 0$ for every e. With $v_e = \lambda/u_e$, $s\,u_e^2 - \frac{s-t}{n}u_e - t\lambda = 0$ for every e: the occupations are roots of one quadratic, whose roots u_+, u_- have $u_+ + u_- = \frac{s-t}{ns}$ and $u_+u_- = -t\lambda/s \ne 0$. Let u_+ be taken on $n-j$ marks and u_- on j marks, u_- on none when the occupations are all equal; $\sum_e u_e = 1$ gives $(n-j)u_+ + ju_- = 1$, and with $u_- = \frac{s-t}{ns} - u_+$, $(n-2j)u_+ = 1 - j\frac{s-t}{ns}$. At $2j = n$ this would read $1 = \frac{s-t}{2s}$, that is $s = -t$, which is excluded. Conversely let the occupations of $\alpha \in X_F$ take the value u_+ on $n-j$ marks and u_- on j marks with the three relations. Adding $(n-j)u_+ + ju_- = 1$, which follows from them, to $(n-j)u_- + ju_+$ gives $n(u_+ + u_-) = \frac{s-t}{s}$, so $(n-j)u_- + ju_+ = -\frac{t}{s}$ and

$$\lambda^{-1} = \sum_e \frac{1}{u_e} = \frac{n-j}{u_+} + \frac{j}{u_-} = \frac{(n-j)u_- + ju_+}{u_+u_-} = -\frac{t}{s\,u_+u_-},$$

using $\sum_e v_e = 1$, $v_e = \lambda/u_e$. Then $s\,u_e - t\,v_e = s\,u_e + s\,u_+u_-/u_e = s(u_+ + u_-) = \frac{s-t}{n}$ at both values, so every defect vanishes. Since $\alpha_e = u_eK(\alpha)$, such a point is $(u_e)_e$ up to scale; conversely, for given $u_\pm$ with the three relations and a set of j marks, the point $u = (u_e)_e$ has $K(u) = 1$ and $\Psi(u) = \prod_e u_e \sum_e u_e^{-1} = -\prod_e u_e\,t/(s\,u_+u_-) \ne 0$, so it lies in X_F. The relations for the complement of the set, its $n-j$ marks at the value u_+ in the role of u_- and the other j at u_- in the role of u_+, hold as well: the second is symmetric, and $(n-j)u_+ + ju_- = 1$ with $u_+ = \frac{s-t}{ns} - u_-$ gives $(2j-n)u_- = 1 - (n-j)\frac{s-t}{ns}$; so a set and its complement give the same point. The stride $f = \alpha^{m\mathbf{1}}K^{-s}\Psi^{-t} = \alpha^{(m-t)\mathbf{1}}K(\alpha)^{-s}K(\alpha^{-1})^{-t}$ is homogeneous of degree 0, and at u, where $K(u) = 1$ and $K(u^{-1}) = \lambda^{-1}$, it is $\big(\prod_e u_e\big)^{m-t}\lambda^t$.

(ii) At $s = t = \sigma$ the balance gives $m = \sigma$, the relations give $u_- = -u_+$ and $u_+ = 1/(n-2j)$, so u is $1/(n-2j)$ times the sign η with $\eta_e = -1$ exactly on the j marks; on the chart $\alpha_{e_0} = 1$ the native gauge is the sign $\eta_{e_0}\eta$, with $K \ne 0$. Conversely every sign with $K(\eta) \ne 0$ is a native gauge (Theorem 38.17(ii)). The native value is $\lambda^\sigma = u_+^{2\sigma} = (n-2j)^{-2\sigma} = K(\eta)^{-2\sigma}$, and $|n-2j|$ for $0 \le j \le n$, $2j \ne n$, takes the $\lceil n/2\rceil$ values $n, n-2, \dots$ down to 1 or 2. By Theorem 38.17(ii) the Hessian is $2\sigma\big(\kappa_\eta(X_e, X_{e'})\big)_{e,e'\ne e_0}$, the covariances of the indicators under

$\mathbb{P}_\eta(\{e\}) = \pi_e = \eta_e/K(\eta)$. The indicator X_e takes the value 1 at the basis $\{e\}$ and 0 at the others, so $\kappa_\eta(X_e, X_{e'}) = \delta_{ee'}\pi_e - \pi_e\pi_{e'}$. The determinant of a diagonal matrix less a matrix of rank one, $\det\big(\operatorname{diag}(\pi_e) - \pi\pi^{\mathsf{T}}\big)_{e,e'\neq e_0} = \prod_{e\neq e_0}\pi_e\big(1 - \sum_{e\neq e_0}\pi_e\big)$, is $\prod_e \pi_e$ since $\sum_e \pi_e = 1$, and the Hessian has the determinant $(2\sigma)^{n-1}\prod_e \eta_e/K(\eta)^n \neq 0$. □

Proposition 42.10 (A parallel class enters through its list). *As stated.*

Proof. (i) Theorem 21.5 with every mark replaced by the first rule gives $K_{\mathcal{B}} = K_{\mathcal{H}}\big((e_1(\alpha_{P_i}))_i\big) = K_{\mathcal{H}}(A)$, the weights of a list being one. Write $K_{\mathcal{H}}(A)^k = \sum_w \big([A^w]K_{\mathcal{H}}^k\big)A^w$ and $A^w = \prod_i \sum_{v_i\in\mathbb{N}^{P_i},\ |v_i|=w_i}\binom{w_i}{v_i}\alpha^{v_i}$. The classes are disjoint, so a monomial α^v determines $w_i = v_{P_i}$, and it occurs only in the term of that w, with the coefficient $\big([A^w]K_{\mathcal{H}}^k\big)\prod_i\binom{w_i}{v|_{P_i}}$.

(ii) On the centre ray at the strength one, $f^{-1} = K_{\mathcal{B}}(\alpha)K_{\mathcal{B}}(\alpha^{-1})$ (Corollary 39.32), and $\mathsf{C}_k^{\mathcal{B}}(1)$ is the constant term of $K_{\mathcal{B}}(\alpha)^k K_{\mathcal{B}}(\alpha^{-1})^k$ (Definition 39.26, the origin form being dy), which is $\sum_v \big([\alpha^v]K_{\mathcal{B}}^k\big)^2$, the coefficients being real. By (i) this is $\sum_w \big([A^w]K_{\mathcal{H}}^k\big)^2\prod_i\sum_{|v_i|=w_i}\binom{w_i}{v_i}^2$, and $\sum_{|v_i|=w_i}\binom{w_i}{v_i}^2 = \sum_{v_i}\big([\alpha^{v_i}]K_{P_i}^{w_i}\big)^2$ is the constant term of $\big(K_{P_i}(\alpha)K_{P_i}(\alpha^{-1})\big)^{w_i}$, $K_{P_i} = \sum_{e\in P_i}\alpha_e$, the closing of $\binom{P_i}{1}$; for $|P_i| = 1$ it is 1.

(iii) At a sign, (i) gives $K_{\mathcal{B}}(\eta) = K_{\mathcal{H}}\big((\sum_{e\in P_i}\eta_e)_i\big)$, and the sum of s_i signs takes the values $s_i, s_i - 2, \dots, -s_i$; the native values are those of Theorem 38.17(ii) for one table at the strength σ. □

Proposition 42.11 (A mark added is a closing of the critical-line kernel). *As stated.*

Proof. (i) By the display of Section 42.6 at the centre, from Corollary 18.22 and Proposition 8.15, $\Gamma(s)^2 p_{n+1}(s)$ is, for real $s > 0$, the share integral $\int_{a_1+\cdots+a_{n+1}=s}\prod_e\Gamma(a_e)^2$ of Corollary 18.22, which converges absolutely, $|\Gamma(c+iy)|$ decaying like $e^{-\pi|y|/2}$. In the coordinates $(a_1, \dots, a_{n-1}, d)$ with $d = a_1 + \cdots + a_n$, of Jacobian one, Fubini's theorem splits it into the integral over $\operatorname{Re} d = c$ of $\Gamma(s-d)^2$ times the share integral of n marks at the total d. That inner integral is holomorphic in d on $\operatorname{Re} d > 0$, by absolute and locally uniform convergence, and equals $\Gamma(d)^2 p_n(d)$ at real d by the same display at n marks; so it equals $\Gamma(d)^2 p_n(d)$ on $\operatorname{Re} d > 0$, p_n being holomorphic there by the bound of Theorem 38.14(i), as in the proof of Proposition 42.2(i). This gives the formula at real s; both sides are holomorphic in s on $\operatorname{Re} s > c$, so it holds on $\operatorname{Re} s > 0$. At $s = 1$ and $c = \frac12$, with $d = \frac12 + iy$, the kernel $\Gamma(d)^2\Gamma(1-d)^2$ is $2\pi\, p_{\mathrm{crit}}(y)$ (Definition 36.1) and $dd/(2\pi i) = dy/(2\pi)$.

(ii) At the integer s, $\Gamma(s-d) = (s-1-d)\cdots(1-d)\,\Gamma(1-d)$, so the integrand of (i) is $\Gamma_0^4\big[d,\ d,\ 1-d,\ 1-d\big]\ G(d)$ with $G = W_s\, p_n$, which has double zeros at $1, \dots, s-1$. On $\operatorname{Re} d \geq 1$, $|p_n(d)| \leq p_n(\operatorname{Re} d) \leq 4^{1-\operatorname{Re} d}p_n(1)$, since $0 < (K(\alpha)K(\alpha^{-1}))^{-1} \leq n^{-2} \leq \frac14$ on the chart (Theorem 38.14(i)); and $\big|\Gamma_0^4\big[d,\ d,\ 1-d,\ 1-d\big]\,\big| = \pi^2/|\sin\pi d|^2$ is at most $\pi^2/\cosh^2(\pi\operatorname{Im} d)$ on the lines $\operatorname{Re} d = N + \frac12$ and at most $Ce^{-2\pi|\operatorname{Im} d|}$ for $|\operatorname{Im} d| \geq 1$. So the integrals over the lines $\operatorname{Re} d = N + \frac12$ and over the horizontal segments $|\operatorname{Im} d| = Y$ tend to 0, and moving the line to the right (Theorem 4.5) gives $-\sum_{m\geq1}\operatorname{Res}_{d=m}$. By Lemma 5.16 the kernel has at $d = m$ the principal part $(d-m)^{-2}$ and no simple part, so the residue is $G'(m)$, zero for $m < s$, and

$$\Gamma(s)^2 p_{n+1}(s) = -\sum_{m\geq1}\big(W_s\, p_n\big)'(m) = \sum_{m\geq1}\big(W_s(m)\,\tilde p_n(m) - W_s'(m)\,p_n(m)\big),$$

the series converging geometrically; it is the index-zero Trailmoment cut at 0 (Definition 6.1). For the second identity, $\int_0^\infty y^{s-1}\mathsf{g}^n\,dy = \Gamma(s)^2 p_n(s)$ at every real $s > 0$, and the terms of h are positive,

so by monotone convergence

$$\int_0^\infty y^{s-1}\mathsf{g}^n\mathsf{h}\,dy=\sum_{k\ge0}\frac{\Gamma(s+k)^2}{k!^2}\,p_n(s+k)=\sum_{k\ge0}W_s(s+k)\,p_n(s+k)=\sum_{m\ge1}W_s(m)\,p_n(m),$$

since $\Gamma(s+k)/k!=(k+1)\cdots(k+s-1)$ and W_s vanishes at $1,\dots,s-1$.

(iii) The share integral of (i) is symmetric in the $n+1$ shares, the fibre integral not depending on the coordinates in which it is written (Definition 4.1), and $\sum_e a_e=s$ on the fibre; so for each e, $\int a_e\prod_{e'}\Gamma(a_{e'})^2=\frac{s}{n+1}\Gamma(s)^2p_{n+1}(s)$. For e the last mark, $a_e=s-d$ in the coordinates of (i), and at $s=1$ the closing of (ii), with $G(d)=(1-d)p_n(d)$, gives

$$\frac{1}{n+1}\,p_{n+1}(1)=-\sum_{m\ge1}\big((1-d)p_n(d)\big)'\big|_{d=m}=\sum_{m\ge1}\big(p_n(m)-(m-1)\,\tilde p_n(m)\big).$$

With $p_{n+1}(1)=(\mathsf{T}_{0;(1)}[x]\,\tilde p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ from (ii) this is the stated relation. □

The following bounds give the Mellin reads of the products of g and h the hypotheses of Lemma S27 and fix their growth along the integers.

Lemma S28. *(i)* $\mathsf{g}>0$ *and* $\mathsf{h}\ge1$ *on* $(0,\infty)$*, and for* $y\ge1$

$$\frac{1}{2e}\,y^{-1/2}e^{-2\sqrt y}\le\mathsf{g}(y)\le\sqrt\pi\,y^{-1/4}e^{-2\sqrt y},\qquad e^{-4}y^{-1}e^{2\sqrt y}\le\mathsf{h}(y)\le e^{2\sqrt y}.$$

(ii) For integers $0\le j<n/2$*, every* $\vartheta^a\big(\mathsf{g}^{n-j}\mathsf{h}^j\big)$ *is* $O\big((1+|\log y|)^n\big)$ *as* $y\to0$ *and* $O\big(y^Ce^{-2(n-2j)\sqrt y}\big)$ *as* $y\to\infty$*. For integers* $n/2\le j\le n$*,* $\int_1^\infty y^{s-1}\mathsf{g}^{n-j}\mathsf{h}^j\,dy=\infty$ *for real* $s\ge n$*.*

(iii) For these j*,* $\lim_{k\to\infty}\Big(\Gamma(k)^{-2}\int_0^\infty y^{k-1}\mathsf{g}^{n-j}\mathsf{h}^j\,dy\Big)^{1/k}=(n-2j)^{-2}$ *along the integers.*

Proof. (i) g is the integral of a positive function and $\mathsf{h}\ge1$ term by term. The upper bound on g is Lemma 20.29(ii) at $\nu=0$. For the lower bound restrict the integral to $x=\sqrt y+\tau$, $0\le\tau\le1$, where $x+y/x=2\sqrt y+\tau^2/(\sqrt y+\tau)\le2\sqrt y+1$ and $x^{-1}\ge(2\sqrt y)^{-1}$. For h, $\sum_m y^m/m!^2\le\big(\sum_m y^{m/2}/m!\big)^2=e^{2\sqrt y}$; and with $m=\lfloor\sqrt y\rfloor\ge1$ and $m!\le e\,m^{m+1}e^{-m}$ (from $\log m!\le\int_1^m\log x\,dx+\log m$),

$$\mathsf{h}(y)\ge\frac{y^m}{m!^2}\ge\frac{e^{2m}}{e^2m^2}\Big(\frac{\sqrt y}{m}\Big)^{2m}\ge\frac{e^{2\sqrt y-2}}{e^2\,y}.$$

(ii) By the ladder, $\vartheta\mathsf{g}=-y\,\mathsf{G}_{-1;(1,-1)}(1,y)$, at most $\sqrt\pi e^{1/4}y^{1/4}e^{-2\sqrt y}$ for $y\ge1$ by Lemma 20.29(ii) at $\nu=-1$; and $0\le\vartheta\mathsf{h}=\sum_m my^m/m!^2\le y\,\mathsf{h}$. Since $\vartheta^2=y$ on both, every $\vartheta^a\mathsf{g}$ and $\vartheta^a\mathsf{h}$ is a combination of the function and its ϑ with polynomial coefficients, of size $O(y^Ce^{\mp2\sqrt y})$ at infinity; near 0, $\mathsf{g}=-\mathsf{h}\log y+\tilde{\mathsf{g}}$ with h and $\tilde{\mathsf{g}}$ entire, so every $\vartheta^a\mathsf{g}$ is $O(1+|\log y|)$ and every $\vartheta^a\mathsf{h}$ bounded. By the Leibniz rule $\vartheta^a(\mathsf{g}^{n-j}\mathsf{h}^j)$ is a sum of products of $n-j$ derivatives of g and j of h, which gives the bounds, $e^{-2(n-j)\sqrt y}e^{2j\sqrt y}=e^{-2(n-2j)\sqrt y}$. For $n/2\le j\le n$, (i) gives on $[1,\infty)$

$$\mathsf{g}^{n-j}\mathsf{h}^j\ge(2e)^{-(n-j)}e^{-4j}\,y^{-(n+j)/2}\,e^{2(2j-n)\sqrt y}\ge(2e)^{-n}e^{-4n}\,y^{-n},$$

and $\int_1^\infty y^{s-1-n}\,dy=\infty$ for $s\ge n$.

(iii) Put $b = n - 2j \geq 1$ and $I(k) = \int_0^\infty y^{k-1}\mathsf{g}^{n-j}\mathsf{h}^j\,dy$. The part over $(0,1]$ is at most $\int_0^1 C(1+|\log y|)^n\,dy$, bounded in $k \geq 1$. By (i), on $[1,\infty)$ the integrand lies between $c_1 y^{k-1-n}e^{-2b\sqrt{y}}$ and $c_2 y^{k-1}e^{-2b\sqrt{y}}$ with $c_1, c_2 > 0$, and for a fixed real a, with $y = \tau^2$,

$$\int_1^\infty y^{k-1+a}e^{-2b\sqrt{y}}\,dy = 2\int_1^\infty \tau^{2k+2a-1}e^{-2b\tau}\,d\tau = 2\,\Gamma(2k+2a)\,(2b)^{-2k-2a} + O(1).$$

$\Gamma(2k+2a)/\Gamma(2k)$ has root rate one and $\Gamma(2k)(2b)^{-2k}$ grows faster than every exponential, so $\big(I(k)/\Gamma(2k)\big)^{1/k} \to (2b)^{-2}$; and $\Gamma(2k)/\Gamma(k)^2 = \frac{k}{2}\binom{2k}{k}$ has the root rate 4, so $\big(I(k)/\Gamma(k)^2\big)^{1/k} \to 4(2b)^{-2} = b^{-2}$. □

Theorem 42.12 (The law of the values of $\binom{\mathcal{E}}{1}$). *As stated.*

Proof. (i) *Annihilation.* Let $u \in \mathcal{V}$ and $v = \vartheta u$, so that $\vartheta v = yu$. For $0 \leq i \leq n$, $\mathbf{R}_i(u^n) = \frac{n!}{(n-i)!}u^{n-i}v^i$: this holds at $i = 0, 1$, and $\vartheta(u^{n-i}v^i) = (n-i)u^{n-i-1}v^{i+1} + i\,y\,u^{n-i+1}v^{i-1}$ gives

$$\vartheta\mathbf{R}_i(u^n) = \frac{n!}{(n-i-1)!}u^{n-i-1}v^{i+1} + i(n+1-i)\,y\,\frac{n!}{(n-i+1)!}u^{n-i+1}v^{i-1},$$

whose last term is $i(n+1-i)\,y\,\mathbf{R}_{i-1}(u^n)$; so the recursion gives the claim at $i+1 \leq n$, and at $i = n$ the first term carries the factor $n - i = 0$ and $\mathbf{R}_{n+1}(u^n) = 0$. A product of n solutions is a combination of n-th powers of solutions, $u_1 \cdots u_n = \frac{1}{2^n n!}\sum_{\epsilon\in\{\pm1\}^n}\epsilon_1\cdots\epsilon_n\,(\epsilon_1 u_1 + \cdots + \epsilon_n u_n)^n$, so $\mathbf{Q}_n$ annihilates every such product.

Normal form and leading terms. Since $\vartheta y = y(\vartheta+1)$, every operator built from y and ϑ is $\sum_k y^k r_k(\vartheta)$ with polynomials r_k, uniquely: applied to y^z it gives $\sum_k y^{k+z}r_k(z)$. Write $\mathbf{R}_i = \sum_k y^k \ell_{i,k}(\vartheta)$. By induction $\ell_{i,0} = \vartheta^i$ and $\deg \ell_{i,k} \leq i - 2k$: $\vartheta\, y^k\ell_{i,k}(\vartheta) = y^k(\vartheta+k)\ell_{i,k}(\vartheta)$ has degree at most $i+1-2k$, and so has $y\cdot y^{k-1}\ell_{i-1,k-1}(\vartheta)$. The coefficients $\alpha_{i,k}$ of ϑ^{i-2k} in $\ell_{i,k}$ therefore satisfy $\alpha_{i+1,k} = \alpha_{i,k} - i(n+1-i)\alpha_{i-1,k-1}$, the shift by k not changing a leading coefficient, and $\sigma_i(\xi,\eta) = \sum_k \alpha_{i,k}\eta^k\xi^{i-2k}$ satisfies $\sigma_{i+1} = \xi\sigma_i - i(n+1-i)\,\eta\,\sigma_{i-1}$, $\sigma_0 = 1$, $\sigma_1 = \xi$. Let J be the matrix, indexed by $0,\dots,n$, of the derivation Δ of the forms of degree n in U, V with $\Delta U = V$ and $\Delta V = \eta U$, in the basis $w_i = U^{n-i}V^i$: $\Delta w_i = (n-i)w_{i+1} + i\eta\,w_{i-1}$. The leading principal minors of $\xi - J$, expanded along their last row, satisfy the same recursion, $\det_{i+1} = \xi\det_i - (n+1-i)\,i\eta\,\det_{i-1}$, so $\sigma_{n+1} = \det(\xi - J)$. Over $\mathbb{C}(\eta^{1/2})$ the forms $E_\pm = V \pm \eta^{1/2}U$ have $\Delta E_\pm = \pm\eta^{1/2}E_\pm$, so the basis $E_+^{n-j}E_-^j$ diagonalizes Δ with the eigenvalues $(n-2j)\eta^{1/2}$, and $\det(\xi - J) = \prod_{j=0}^n\big(\xi - (n-2j)\eta^{1/2}\big)$; the factors j and $n-j$ pair into $\xi^2 - (n-2j)^2\eta$, and at even n the factor $j = n/2$ is ξ. With $q_{n,k} = \ell_{n+1,k}$ and $a_{n,k} = \alpha_{n+1,k}$ this is the display; $a_{n,h}$ is the coefficient of the top power η^h, $\prod_{0\leq j<n/2}\big(-(n-2j)^2\big) \neq 0$.

The adjoint. Expanding the recursion, $\mathbf{R}_m$ is the sum over the tilings of $\{1,\dots,m\}$ by monomers $\{a\}$ and dimers $\{a, a+1\}$ of the products of ϑ for each monomer and $-a(n+1-a)\,y$ for each dimer, taken from left to right in decreasing order of position: the tilings of $\{1,\dots,m+1\}$ end with the monomer $\{m+1\}$ or the dimer $\{m, m+1\}$, which are the two terms of the recursion. The formal adjoint for dy/y reverses products, takes ϑ to $-\vartheta$ and keeps y; a tiling of $\{1,\dots,m\}$ with d dimers has $m-2d$ monomers, so $\mathbf{R}_m^*$ is $(-1)^m$ times the same sum with the products taken in increasing order of position. The reflection $a \mapsto m+1-a$ of the positions maps tilings to tilings, reverses the order, and takes the dimer $\{a, a+1\}$ to $\{a', a'+1\}$ with $a' = m-a$, of weight $-(m-a')(n+1-m+a')\,y$. At $m = n+1$ this weight is $-a'(n+1-a')\,y$, so $\mathbf{Q}_n^* = \mathbf{R}_{n+1}^* = (-1)^{n+1}\mathbf{R}_{n+1}$. Since $q(-\vartheta)\,y^k = y^k q(-\vartheta - k)$, $\mathbf{Q}_n^* = \sum_k q_{n,k}(-\vartheta)y^k = \sum_k y^k q_{n,k}(-\vartheta - k)$, and the uniqueness of the normal form gives $q_{n,k}(-\vartheta - k) = (-1)^{n+1}q_{n,k}(\vartheta)$.

(ii) For $k \ge 1$, $c_{n,k}$ has degree at most $(n+1-2k)+2(k-1) = n-1$ with the coefficient $(-1)^{n+1}a_{n,k}$ of s^{n-1}, as $c_{n,0} = (-1)^{n+1}s^{n-1}$ has with $a_{n,0} = 1$. Setting $\xi = 1$ in (i), $\sum_k a_{n,k}\lambda^k = \prod_{0\le j<n/2}\big(1-(n-2j)^2\lambda\big)$, whose coefficients $(-1)^k e_k\big((n-2j)^2\big)$ are nonzero for $0 \le k \le h$; so every $c_{n,k}$ has degree $n-1$ and the symbol is χ_n. By (i), $c_{n,k}(-s-k) = q_{n,k}(s)\big((-s-k+1)\cdots(-s-1)\big)^2 = (-1)^{n+1}q_{n,k}(-s-k)\big((s+1)\cdots(s+k-1)\big)^2 = (-1)^{n+1}c_{n,k}(s)$ for $k \ge 1$, and $c_{n,0}(-s) = s^{n-1} = (-1)^{n+1}c_{n,0}(s)$. The double zeros are those of the squared factor. For odd n, $q_{n,h}$ is the nonzero constant $a_{n,h}$; for even n it has degree one, $q_{n,h}(\vartheta) = a_{n,h}(\vartheta - \vartheta_0)$, and $q_{n,h}(-\vartheta - h) = -q_{n,h}(\vartheta)$ forces $\vartheta_0 = -h/2$, so $q_{n,h}(-s-h) = -a_{n,h}(s+h/2)$; in both cases $c_{n,h}(s) \ne 0$ at the integers $s \ge 0$. For $n \ge 2$ let $M(s) = \Gamma(s)^2 P_{\binom{\varepsilon}{1}}(s\mathbf{1}; s, s) = \int_0^\infty y^{s-1}\mathsf{g}^n\,dy$ (Section 42.6). By Lemma S28(ii), Lemma S27(i) applies to g^n, and $\mathbf{Q}_n(\mathsf{g}^n) = 0$ gives $\sum_k q_{n,k}(-s-k)\,M(s+k) = 0$ on $\operatorname{Re} s > 0$. Dividing by $\Gamma(s+1)^2$, with $\Gamma(s)^2/\Gamma(s+1)^2 = s^{-2}$ and $\Gamma(s+k)^2/\Gamma(s+1)^2 = \big((s+1)\cdots(s+k-1)\big)^2$, and $q_{n,0}(-s) = (-s)^{n+1}$, gives the identity on $\operatorname{Re} s > 0$, and on $\mathbb{C}$ by the meromorphic continuation (Proposition 39.30).

(iii) Write $P(k) = P_{\binom{\varepsilon}{1}}(k\mathbf{1}; k, k)$ and suppose $\sum_{i=0}^m r_i(k)P(k+i) = 0$ for all large integers k, with $r_i \in \mathbb{C}[k]$, $r_0 r_m \ne 0$ and $m < h$. The centre ray at the strength one has origin 0, letter exponents one and walls $\kappa_{1,S} = 1$, so by Proposition 42.2(i) the relation holds identically. On $\operatorname{Re} s > 0$ multiply it by $\Gamma(s+m)^2$: $\sum_i \tilde r_i(s)M(s+i) = 0$ with the polynomials $\tilde r_i(s) = r_i(s)\big((s+i)\cdots(s+m-1)\big)^2$, $\tilde r_m = r_m$. Put $T = \sum_i y^i\,\tilde r_i(-\vartheta - i)$. By Lemma S27(i), with Lemma S28(ii), the Mellin read of $T(\mathsf{g}^n)$ is $\sum_i \tilde r_i(s)M(s+i) = 0$, and by Lemma S27(ii) $T(\mathsf{g}^n) = 0$.

On the ring of polynomials in an indeterminate L with entire coefficients let ϑ act by $\vartheta\big(\sum_l E_l L^l\big) = \sum_l (\vartheta E_l)L^l + \sum_l l E_l L^{l-1}$, and put $\Phi(y, L) = T\big[(\tilde{\mathsf{g}} - \mathsf{h}\,L)^n\big] = \sum_{l\le n} E_l(y)L^l$, E_l entire. Substituting $L = \log y - c$, for a constant c, commutes with this ϑ, so $T\big[(\mathsf{g} + c\,\mathsf{h})^n\big] = \Phi(y, \log y - c)$ on $(0,\infty)$, and $\Phi(y, \log y) = T(\mathsf{g}^n) = 0$. The function $y \mapsto \Phi(y, \log y)$ continues along every path in $\mathbb{C}^\times$, and once around 0 it becomes $\Phi(y, \log y + 2\pi i)$; the continuation of 0 is 0, so $\Phi(y, \log y + 2\pi i\ell) = 0$ for every $\ell \in \mathbb{Z}$ and $y > 0$. For fixed y the polynomial $L \mapsto \Phi(y, L)$ then has infinitely many zeros, so $\Phi = 0$, and $T\big[(\mathsf{g} + c\mathsf{h})^n\big] = 0$ for every $c \in \mathbb{C}$; the coefficients of the powers of c give $T\big(\mathsf{g}^{n-j}\mathsf{h}^j\big) = 0$ for $0 \le j \le n$.

For $0 \le j < n/2$ let $I_j(k) = \int_0^\infty y^{k-1}\mathsf{g}^{n-j}\mathsf{h}^j\,dy$. Lemma S27(i) for $\mathsf{g}^{n-j}\mathsf{h}^j$ (Lemma S28(ii)) gives $\sum_i \tilde r_i(k)I_j(k+i) = 0$ for the integers $k \ge 1$. Take k_1 beyond the zeros of r_m; on the integers $k \ge k_1$ a solution of this recurrence of order m is fixed by its first m values, so the $h > m$ sequences $(I_j(k))_{k\ge k_1}$ are linearly dependent. They are not: if $\sum_j b_j I_j(k) = 0$ for $k \ge k_1$ with some $b_j \ne 0$, let j_0 be the largest such j; by Lemma S28(iii), $I_j(k)/\Gamma(k)^2$ has the exact root rate $(n-2j)^{-2}$, largest at j_0, so $|b_{j_0}|I_{j_0}(k) > \sum_{j<j_0}|b_j|I_j(k)$ for large k, a contradiction. So the tier is at least h, and the recurrence of (ii) has the order h and $c_{n,0}c_{n,h} \ne 0$.

By Proposition 42.2(i) the closings, here the constant terms of $(\sum_e \alpha_e)^j(\sum_e \alpha_e^{-1})^j$ (Corollary 39.32), that is $\sum_{|v|=j}\binom{j}{v}^2$, satisfy the reflected recurrence for every $j \in \mathbb{N}$, and by Corollary 39.44 they satisfy no recurrence that the values do not satisfy, read at k and at $-j$; so none of order less than h. In the value module, a relation $\sum_{i<h} r_i(k)\,\mathrm{Sh}^i[1] \in \mathrm{Ann}(\langle\cdot\rangle)$ with rational r_i not all zero would be a recurrence of order less than h for the values, so the classes of $\mathrm{Sh}^i[1]$, $0 \le i < h$, are independent there; the element $\sum_{i=0}^h c_{n,i}(k)\,\mathrm{Sh}^i[1]$, k the strength, has vanishing values and lies in $\mathrm{Ann}(\langle\cdot\rangle)$, a submodule (Theorem 39.4(ii)), and with $c_{n,0}c_{n,h} \ne 0$ it expresses $\mathrm{Sh}^h[1]$ and $\mathrm{Sh}^{-1}[1]$ through these classes; so their span is the cyclic submodule of [1] in the value module, of dimension h. □

Proposition 42.13 (A mark added carries the law of the values). *As stated.*

Proof. (i) For a fixed solution u both sides are linear differential operators in F of order $n+2$,

with coefficients analytic on $(0,\infty)$. By Theorem 42.12(i), $\mathbf{Q}_{n+1} = \vartheta^{n+2} + \sum_{k\ge1} y^k q_{n+1,k}(\vartheta)$ with $\deg q_{n+1,k} \le n$, so $\mathbf{Q}_{n+1}(uF) = u\,\vartheta^{n+2}F + (n+2)(\vartheta u)\,\vartheta^{n+1}F + (\text{order} \le n)$; and $\mathbf{Q}_n = \vartheta^{n+1} + \sum_{k\ge1} y^k q_{n,k}(\vartheta)$ with $\deg q_{n,k} \le n-1$, so the right side is $u\,\vartheta^{n+2}F + (n+2)(\vartheta u)\,\vartheta^{n+1}F + (\text{order} \le n)$. Their difference is an operator of order at most n that annihilates every F in the space spanned by the products of n solutions, of dimension $n+1$: the right side vanishes there by Theorem 42.12(i), and the left side because uF is a product of $n+1$ solutions. An operator of order at most n whose highest nonzero coefficient does not vanish on an interval has there a space of solutions of dimension at most n, and the $n+1$ analytic functions stay independent on every interval; so the difference is 0.

(ii) Apply (i) with $u = \mathsf{h}$ to $F = y^z$, $z \in \mathbb{C}$, using $\vartheta^a y^w = w^a y^w$ and $\vartheta\mathsf{h} = \sum_m m\,y^m/m!^2$. Both sides are y^z times entire functions of y, and comparing the coefficients of y^{z+N},

$$\sum_{k+m=N} \frac{q_{n+1,k}(z+m)}{m!^2} = \sum_{j+m=N} \frac{\big(z+j+(n+2)m\big)\,q_{n,j}(z)}{m!^2} \qquad (N\in\mathbb{N},\ z\in\mathbb{C}).$$

By Theorem 42.12(i), $q_{n+1,k}(z+m) = (-1)^n q_{n+1,k}(-z-N)$ when $k+m=N$ and $q_{n,j}(z) = (-1)^{n+1} q_{n,j}(w+m)$ with $w = -z-N$ when $j+m=N$; and $z+j+(n+2)m = (n+1)m - w$. So

$$\sum_{k+m=N} \frac{q_{n+1,k}(w)}{m!^2} = \sum_{j+m=N} \frac{\big(w-(n+1)m\big)\,q_{n,j}(w+m)}{m!^2},$$

and applying the same symmetry once more, $q_{n+1,k}(w) = (-1)^n q_{n+1,k}(-w-k)$ and $q_{n,j}(w+m) = (-1)^{n+1} q_{n,j}(-w-N)$,

$$\sum_{k+m=N} \frac{q_{n+1,k}(-s-k)}{m!^2} = \sum_{j+m=N} \frac{\big((n+1)m-s\big)\,q_{n,j}(-s-N)}{m!^2} \qquad (s=w).$$

Multiply by $\Gamma(s+N)^2/\Gamma(s+1)^2$ and put $x = s+N$. By the definitions, $c_{n+1,k}(s) = q_{n+1,k}(-s-k)\,\Gamma(s+k)^2/\Gamma(s+1)^2$ for every $k \ge 0$, so the left side becomes $\sum_k c_{n+1,k}(s)\,\Gamma(x)^2/\big(\Gamma(s+k)^2\Gamma(x-s-k+1)^2\big) = \sum_k c_{n+1,k}(s)\binom{x-1}{s+k-1}^2$. With $c_{n,j}(x-j) = q_{n,j}(-x)\,\Gamma(x)^2/\Gamma(x-j+1)^2$, $(n+1)m-s = (n+1)(x-j)-(n+2)s$ and $A_s(x-j) = \big((n+1)(x-j)-(n+2)s\big)\Gamma(x-j+1)^2/\big(\Gamma(s+1)^2\Gamma(x-j-s+1)^2\big)$, the right side becomes $\sum_j c_{n,j}(x-j)A_s(x-j)$. After division by $\binom{x}{s}^2$ both sides are rational functions of (x,s), each binomial quotient being a quotient of rising factorials, and they agree on the lines $x-s=N$, $N\in\mathbb{N}$; a polynomial in (x,s) vanishing on infinitely many of these lines is divisible by infinitely many distinct factors $x-s-N$, so it is zero, and the identity holds. At integers $s,x \ge 1$, $W_{s+k}(x)/\Gamma(s+k)^2 = \prod_{l=1}^{s+k-1}(x-l)^2/(s+k-1)!^2 = \binom{x-1}{s+k-1}^2$, both sides vanishing for $x < s+k$; the right side is $(L_n^* A_s)(x)$ for the recurrence L_n with the coefficients $c_{n,j}$ (Definition 42.1), and $\binom{x}{s} = x(x-1)\cdots(x-s+1)/s!$ vanishes at $x = 0,1,\dots,s-1$. □

Proposition 42.14 (The lowered settings). *As stated.*

Proof. (i) By Theorem 42.12(i), $\mathbf{Q}_n(\mathsf{g}^{n-j}\mathsf{h}^j) = 0$; Lemma S27(i), which applies by Lemma S28(ii), gives $\sum_k q_{n,k}(-s-k)\,\Gamma(s+k)^2\xi_{n,j}(s+k) = 0$ on $\operatorname{Re} s > 0$, and the division of the proof of Theorem 42.12(ii) gives the recurrence. For $j=0$ the display of Section 42.6 gives $\xi_{n,0}(s) = P_{\binom{\varepsilon}{1}}(s\mathbf{1}; s, s)$.

(ii) On $(0,1]$, $0 < \mathsf{g}^{n-j}\mathsf{h}^j \le C(1+|\log y|)^{n-j}$ by Lemma S28(ii) and $\mathsf{g} = -\mathsf{h}\log y + \tilde{\mathsf{g}}$, so for real $s \in (0,1]$, $\int_0^\infty y^{s-1}\mathsf{g}^{n-j}\mathsf{h}^j\,dy \le C\int_0^1 y^{s-1}(1+|\log y|)^{n-j}dy + C' = O(s^{-(n-j+1)})$, and

with $\Gamma(s)^{-2} = O(s^2)$, $s^{n-1}\xi_{n,j}(s) = O(s^j) \to 0$ for $j \geq 1$. In the recurrence on $(0,\infty)$ the term $c_{n,0}(s)\xi_{n,j}(s) = (-1)^{n+1}s^{n-1}\xi_{n,j}(s)$ then tends to 0 as $s \downarrow 0$, while the others tend to $c_{n,k}(0)\xi_{n,j}(k)$, $\xi_{n,j}$ being continuous on $\operatorname{Re} s > 0$; so $\sum_{k\geq 1} c_{n,k}(0)\xi_{n,j}(k) = 0$. For $j = 0$, $s^{n-1}\xi_{n,0}(s) \to V_1 = n!$ by Proposition 39.30, the walls of the direction being $\kappa_{1,S} = c(S) = 1$ for every nonempty proper S, so that $V_1 = \sum_\pi 1 = n!$; and the same limit gives $\sum_{k\geq 1} c_{n,k}(0)\xi_{n,0}(k) = -(-1)^{n+1}n!$.

(iii) By Lemma S28(iii), $\xi_{n,j}$ has the root rate $\lambda = (n-2j)^{-2}$ along the integers. The recurrence has $c_{n,0}(k)c_{n,h}(k) \neq 0$ at the integers $k \geq 1$, every $c_{n,k}$ of degree $n-1$, and the roots $(n-2j')^{-2}$, $0 \leq j' < n/2$, of its symbol are positive and distinct (Theorem 42.12(ii)); so the proof of Proposition 42.7(i), which uses only these properties, gives $\xi_{n,j}(k+1)/\xi_{n,j}(k) \to \lambda_c$ for a root λ_c of modulus λ, that is $\lambda_c = (n-2j)^{-2}$.

(iv) $c_{n,h}(k) \neq 0$ at the integers $k \geq 1$ (Theorem 42.12(ii)), so a solution on these integers is fixed by its first h values, and the solutions form a space of dimension h. The h sequences are solutions by (i) and are independent by (iii), as in the proof of Theorem 42.12(iii): in a vanishing combination the term with the largest j would dominate. So they are a basis. The linear form $x \mapsto \sum_{k=1}^{h} c_{n,k}(0)x(k)$ is $(-1)^n n!$ on $\xi_{n,0}$ and 0 on the others by (ii), which gives the last assertion. □

Corollary 42.15 (The value span of $\binom{\mathcal{E}}{1}$). *As stated.*

Proof. (i) The first relation is Proposition 42.14(ii) at $j = 0$. The recursion of Theorem 42.12 has integer coefficients, so every $q_{n,k}$ and every $c_{n,k}$ has integer coefficients. $c_{n,h}(0) \neq 0$, so the first relation gives $p_n(h) \in \mathbb{Q} + \sum_{i<h} \mathbb{Q}\, p_n(i)$, and for $k > h$ the recurrence of Theorem 42.12(ii) at $s = k - h \geq 1$, with $c_{n,h}(k-h) \neq 0$, writes $p_n(k)$ through $p_n(k-h), \dots, p_n(k-1)$; by induction every $p_n(k)$ lies in $\mathbb{Q} + \sum_{i=1}^{h-1} \mathbb{Q}\, p_n(i)$. The values of the centre ray at the strength σ are $p_n(k\sigma)$ (Definition 38.13).

(ii) For even n every $n-2j$, $0 \leq j < n/2$, is at least 2, so $\chi_n(1) = (-1)^{n+1}\prod_j \big(1-(n-2j)^2\big) \neq 0$. The values of the centre ray have the root rate $n^{-2} < 1$ and the slopes at most n^{-2} (Proposition 42.2(iii), $f(\alpha^\star) = f(\mathbf{1}) = n^{-2}$). Every $c_{n,k}$ with $k > j \geq 1$ has a double zero at $-j$ (Theorem 42.12(ii)), so by Proposition 42.3 the only boundary numbers are $\beta_0(0)$ and $\beta_0'(0)$ of $\beta_0 = c_{n,0}\, p_n = (-1)^{n+1}s^{n-1}p_n$: by Proposition 42.6, $\beta_0(0) = (-1)^{n+1}V_1 = (-1)^{n+1}n!$, and $\beta_0'(0) = (-1)^{n+1}\mathsf{L}_1 = 0$, the list having unit weights. Proposition 42.4(ii) with $\Bbbk = \mathbb{Q}$ and $D = n-1$ gives the claim. The two last numbers are such Trailmoments by Proposition 42.11(ii), with the rational polynomials W_s and W_s'. □

Corollary 42.16 (The exact pairing of $\binom{\mathcal{E}}{1}$). *As stated.*

Proof. The centre ray of $\binom{\mathcal{E}}{1}$ at the strength one satisfies the hypotheses of Proposition 42.2, and the recurrence of Theorem 42.12(ii) holds for its values with $c_{n,k}(-s-k) = (-1)^{n+1}c_{n,k}(s)$; its closings are the C_l of the statement (proof of Theorem 42.12(iii)). Proposition 42.8(ii), with $\gamma_{n-1} = (-1)^{n+1}$ and $V_1 = n!$, gives the first display. Every $c_{n,k}$ has degree $n-1$, so Proposition 42.8(iii) applies: the roots of χ_n are the simple roots $(n-2j)^{-2}$, $0 \leq j < n/2$, with $\lambda_\star = f(\mathbf{1}) = n^{-2}$ at $j = 0$, and $1 - (n-2j)^2/n^2 = 4j(n-j)/n^2$. The Hessian H of Theorem 38.15 at the centre is $2\big(\kappa_{\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\neq e_0}$, which is the Hessian of Proposition 42.9(ii) at the sign $\mathbf{1}$ and $\sigma = 1$, of determinant $2^{n-1}/n^n$. As a check, the two sides agree directly: for $n = 2l+1$, $h-1 = l$ and $\prod_{j=1}^{l} j(n-j) = l!\,(2l)!/l! = (n-1)!$, and for $n = 2l+2$, $h-1 = l$ and $\prod_{j=1}^{l} j(n-j) = l!\,(2l+1)!/(l+1)! = 2(n-1)!/n$; in both cases $\prod_{j=1}^{h-1} 4j(n-j)/n^2 = 2^{n-1}(n-1)!/n^{n-1} = n!\,2^{n-1}/n^n$. □

Proposition 42.17 (The power of one mark as a mixture). *As stated.*

Proof. Write $p_n(s) = P_{\binom{\mathcal{E}}{1}}(s\mathbf{1}; s, s)$ for $|\mathcal{E}| = n$. By Theorem 38.14(i), $0 < f_n \le f_n(\mathbf{1}) = n^{-2}$ on the chart, so μ_{f_n} is carried by $(0, n^{-2}]$, and $\int t^s \mu_{f_n}(dt) = \int_{y_{e_0}=0} f_n^s \, dy = p_n(s)$ for $\operatorname{Re} s > 0$, the image of a measure under a map (Definition 1.1(vi)); the same holds at $n+1$ marks. The measure μ_{f_n} has infinite mass near 0, and every moment of positive order is finite. By Fubini's theorem and $y = ty'$, $\int_0^\infty y^{s-1}\mathsf{g}(y/t)\, dy = t^s \int_0^\infty y'^{s-1}\mathsf{g}(y')\, dy' = t^s\Gamma(s)^2$, the last because $\int_0^\infty y^{s-1}e^{-y/x}dy = \Gamma(s)x^s$ and $\int_0^\infty x^{s-1}e^{-x}dx = \Gamma(s)$.

(i) Put $G(y) = \int \mathsf{g}(y/t)\, \mu_{f_n}(dt)$. For real $s > 0$, by Tonelli's theorem $\int_0^\infty y^{s-1}G(y)\, dy = \Gamma(s)^2 \int t^s \mu_{f_n}(dt) = \Gamma(s)^2 p_n(s)$, finite, which is $\int_0^\infty y^{s-1}\mathsf{g}^n\, dy$ by the display of Section 42.6. So G is finite almost everywhere; since g is decreasing ($\partial_y \mathsf{g} = -\mathsf{G}_{-1;(1,-1)}(1, y) < 0$), $\mathsf{g}(y/t) \le \mathsf{g}(y_0/t)$ for $y \ge y_0$, and dominated convergence makes G finite and continuous on $(0, \infty)$. For $s = c + i\tau$ the same Fubini argument applies, so the integrable functions $e^{cv}G(e^v)$ and $e^{cv}\mathsf{g}(e^v)^n$ have the same Fourier transform; they are equal, by the uniqueness of the Fourier transform on $L^1(\mathbb{R})$ as in Lemma S27(ii), and both being continuous, $G = \mathsf{g}^n$.

(ii) The table on $\{1, 2\}$ with the bases $\{1\}$ and $\{2\}$ is the table of Corollary 18.22 on two marks, the basis $\{2\} = \{1, 2\} - 1$ carrying $\theta_1 = 1$ and $\{1\}$ carrying $\theta_2 = t^{-1/2}$; its letters are $K = \alpha_2 + t^{-1/2}\alpha_1$ and $\Psi = \alpha_1 + t^{-1/2}\alpha_2$, and its centre stride on the chart $\alpha_1 = 1$, $\alpha_2 = x$, is $\alpha_1\alpha_2/(K\Psi) = x/\big((x+t^{-1/2})(1+t^{-1/2}x)\big) = \phi_t(x)$, and $dy = dx/x$ on that chart. So $\int u^s \mu_{\phi_t}(du) = P(s\mathbf{1}; s, s)$ for that table, and Corollary 18.22 at $\mathbf{m} = s\mathbf{1}$, $\nu_e = 0$, gives $\Gamma(s)^2 \int u^s \mu_{\phi_t}(du) = \int_0^\infty y^{s-1}\mathsf{g}(y)\, \mathsf{g}(\theta_2^2 y)\, dy = \int_0^\infty y^{s-1}\mathsf{g}(y)\, \mathsf{g}(y/t)\, dy$. By (i) and Tonelli's theorem, for real $s > 0$,

$$\begin{aligned}\Gamma(s)^2 \int u^s \mu_{f_{n+1}}(du) &= \int_0^\infty y^{s-1}\mathsf{g}(y)\, \mathsf{g}(y)^n\, dy = \int \mu_{f_n}(dt) \int_0^\infty y^{s-1}\mathsf{g}(y)\, \mathsf{g}(y/t)\, dy \\ &= \Gamma(s)^2 \int \mu_{f_n}(dt) \int u^s \mu_{\phi_t}(du),\end{aligned}$$

and by Fubini's theorem the same holds on $\operatorname{Re} s > 0$. The measures $\mu_{f_{n+1}}$ and $\int \mu_{\phi_t}\, \mu_{f_n}(dt)$ therefore have the same moments $\int u^s(\cdot)$ on the line $\operatorname{Re} s = c > 0$; their images under $u \mapsto v = \log u$, weighted by e^{cv}, are finite measures on $\mathbb{R}$ with the same Fourier transform, hence equal, and so are the two measures. □

Proposition 42.18 (Balanced directions of $\binom{\mathcal{E}}{1}$). *As stated.*

Proof. The walls of $\binom{\mathcal{E}}{1}$ at $(\mathbf{m}; s, t)$ are $\kappa_S = m_S - t(|S| - 1)$, every proper S meeting a basis $\{e\}$ in at most one mark and the cobases $\mathcal{E} - e$ in at least $|S| - 1$, so the hypothesis says that every proper wall is positive, at $(k\mathbf{m}; ks, kt)$ as well. The display of Section 42.6 at these data, from Corollary 18.22 for the dual table at the exponents (kt, ks) and Proposition 8.15, gives $\Gamma(sk)\Gamma(tk)P(k) = \int_0^\infty y^{tk-1} \prod_e \mathsf{G}_{c_e k;(1,-1)}(1, y)\, dy$. By the torus homogeneity (Theorem 2.4(iii)) at $\lambda = y^{1/2}$, $\mathsf{G}_{\nu;(1,-1)}(\lambda, \lambda^{-1}y) = \lambda^{-\nu}\mathsf{G}_{\nu;(1,-1)}(1, y)$, that is $\mathsf{G}_{\nu;(1,-1)}(1, y) = y^{\nu/2}\, \mathsf{G}_{\nu;(1,-1)}(\sqrt{y}, \sqrt{y})$; the product of the powers is $y^{k\sum_e c_e/2} = y^{(s-t)k/2}$, and $tk - 1 + \frac{(s-t)k}{2} = \frac{(s+t)k}{2} - 1$. The reflection of Proposition 2.3 gives $\mathsf{G}_{\nu;(1,-1)}(a, b) = \mathsf{G}_{-\nu;(-1,1)}(a, b) = \mathsf{G}_{-\nu;(1,-1)}(b, a)$, so at equal coefficients the Gramma of one mark is even in its index, and the integral depends only on $s + t$ and the multiset $\{|c_e|\}$. Two directions with the same $s + t$ and multiset therefore have $\Gamma(sk)\Gamma(tk)P(k) = \Gamma(s'k)\Gamma(t'k)P'(k)$, which is the stated quotient of Beta corners, a quotient of factorials at integer k. The image of the integrand $\alpha^{k\mathbf{m}}K^{-ks}\Psi^{-kt}$ under $\alpha \mapsto \alpha^{-1}$, with $K(\alpha^{-1}) = \alpha^{-\mathcal{E}}\Psi(\alpha)$ and $\Psi(\alpha^{-1}) = \alpha^{-\mathcal{E}}K(\alpha)$, is $\alpha^{k((s+t)\mathbf{1}-\mathbf{m})}K^{-kt}\Psi^{-ks}$, the integrand at $((s + t)\mathbf{1} - \mathbf{m}; t, s)$, whose offsets are $(s + t) - m_e - s = -c_e$. □

48.42 Chapter 43: arithmetic along a ray

Throughout, Stirling's formula is $\log\Gamma(z) = (z-\frac12)\log z - z + \frac12\log 2\pi + O(|z|^{-1})$ as $|z|\to\infty$ with $|\arg z| \le \pi - \delta$; it applies below to $\Gamma(k-a)$ and to $N! = \Gamma(N+1)$ with a fixed and $k, N \to +\infty$ along the reals, so that the arguments stay in a sector around the positive axis.

Lemma S29. *Let $\Bbbk$ be a subfield of $\mathbb{C}$. Let $X(n+1) = M(n)\,X(n)$ for the integers $n \ge n_2$, with $X(n) \in \mathbb{C}^m$ and $M(n)$ an $m \times m$ matrix whose entries are rational functions of n with coefficients in $\Bbbk$ and whose determinant is not the zero function, and let $\psi(n) = \ell\,X(n)$ for a fixed row vector $\ell \in \Bbbk^{1\times m}$. Then ψ satisfies, for all large n, a recurrence $\sum_{i=0}^{h} a_i(n)\,\psi(n+i) = 0$ with $a_i \in \Bbbk[n]$, $h \le m$ and $a_0 a_h \ne 0$.*

Proof. If $\ell = 0$, $\psi = 0$ satisfies $\psi(n) = 0$. Otherwise put $\mathbf{r}_0 = \ell$ and $\mathbf{r}_i(n) = \ell\,M(n+i-1)\cdots M(n)$, row vectors over the field $\Bbbk(n)$; then $\psi(n+i) = \mathbf{r}_i(n)X(n)$ wherever the entries are defined, and $\mathbf{r}_i(n) = \mathbf{r}_{i-1}(n+1)M(n)$. The $m+1$ vectors $\mathbf{r}_0, \dots, \mathbf{r}_m$ are dependent over $\Bbbk(n)$; let $h \ge 1$ be the least index with $\mathbf{r}_0, \dots, \mathbf{r}_h$ dependent, and $\sum_{i=0}^{h} a_i\mathbf{r}_i = 0$ with $a_i \in \Bbbk[n]$ after clearing denominators. By minimality $a_h \ne 0$. If $a_0 = 0$, then $\sum_{i=1}^{h} a_i(n)\mathbf{r}_{i-1}(n+1)M(n) = 0$, and $M(n)$ is invertible over $\Bbbk(n)$, so $\sum_{i=1}^{h} a_i(n)\mathbf{r}_{i-1}(n+1) = 0$; replacing n by $n-1$, an automorphism of $\Bbbk(n)$, gives a dependence among $\mathbf{r}_0, \dots, \mathbf{r}_{h-1}$ with the coefficient $a_h(n-1) \ne 0$, against minimality. So $a_0 \ne 0$, and $\sum_i a_i(n)\psi(n+i) = \big(\sum_i a_i(n)\mathbf{r}_i(n)\big)X(n) = 0$ for all n beyond the poles of the entries. □

Theorem 43.1 (Closing one free share along a ray). *As stated.*

Proof. (i) For an integer $r \ge 0$ and x off the poles, $\Gamma(x+r) = (x)_r\Gamma(x)$ (1.1(v)) and $\Gamma(x-r) = \Gamma(x)/(x-r)_r = (x)_{-r}\Gamma(x)$ (1.3(iii)). Applied to every Gamma of W_k with $x = \epsilon_\gamma a + b_\gamma$ and $r = ku_\gamma$ of either sign this is $W_k = W_0\,g_k$; the factors of g_k have integer coefficients.

(ii) The upper Gammas have poles where $\epsilon_\alpha a + b_\alpha + ku_\alpha \in -\mathbb{N}$, at integers since $\epsilon_\alpha = \pm 1$; the lower ones contribute the entire factors $1/\Gamma$. For $n > n_1$ the arguments at $a = -n+\varepsilon$ are $-N_\gamma(n)+\varepsilon$ on A_+, $N_\gamma(n)+1-\varepsilon$ on A_-, $-N_\gamma(n)+\epsilon_\gamma\varepsilon$ on B_+ and $N_\gamma(n)+1-|\epsilon_\gamma|\varepsilon$ on B_-, with $N_\gamma(n) \in \mathbb{N}$, and $N_\gamma(n)$ is $|\epsilon_\gamma|n$ plus an integer. By 5.14, $\Gamma(-N+\delta) = (-1)^N\delta^{-1}\Gamma(1+\delta)\Gamma(1-\delta)/\Gamma(1+N-\delta)$ for $N \in \mathbb{N}$ and $|\delta| < 1$. Put $s_\gamma = 1$ on $A_- \cup B_+$ and $s_\gamma = -1$ on $A_+ \cup B_-$. Then

$$W_k(-n+\varepsilon) = (-1)^{\sum_{A_+\cup B_+} N_\gamma} \prod_{B_+} \epsilon_\gamma \cdot \varepsilon^{-w} \prod_\gamma \Gamma\big(1 + N_\gamma - |\epsilon_\gamma|\varepsilon\big)^{s_\gamma} \frac{\prod_{A_+}\Gamma(1+\varepsilon)\Gamma(1-\varepsilon)}{\prod_{B_+}\Gamma(1+\epsilon_\gamma\varepsilon)\Gamma(1-\epsilon_\gamma\varepsilon)}.$$

At $\varepsilon = 0$ the factor after ε^{-w} is $\prod_\gamma (N_\gamma!)^{s_\gamma}$, which gives C_n as stated, nonzero and rational. By 5.13 at the integer $b = N \ge 0$, where the lattice input $\mathbb{I}_{N+\mathbb{Z}}$ is $\mathbb{I}_{\mathbb{Z}}$, $\log\Gamma(1+N-\delta) = \log N! + \sum_{i\ge1}\frac{\delta^i}{i}(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$, the index-one value regularized, and at $N = 0$ with $\pm\delta$, $\log\Gamma(1+\delta)+\log\Gamma(1-\delta) = \sum_{j\ge1}\frac{\delta^{2j}}{j}(\mathsf{T}_{2j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. So $W_k(-n+\varepsilon) = C_n\varepsilon^{-w}\exp\big(\sum_i \Lambda_{n,i}\varepsilon^i\big)$ with

$$\Lambda_{n,i} = \frac{1}{i}\sum_\gamma s_\gamma|\epsilon_\gamma|^i\,(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N_\gamma(n),\infty)) + [i \text{ even}]\,\frac{2}{i}\Big(|A_+| - \sum_{B_+}\epsilon_\gamma^i\Big)(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

The cut relation (5.3, with the regularized values at index one) gives $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = (\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,N])$. The balance of the slopes gives

$$\sum_\gamma s_\gamma|\epsilon_\gamma| = \#A_- - \#A_+ + \sum_{B_+}\epsilon_\gamma - \sum_{B_-}|\epsilon_\gamma| = -\sum_A \epsilon_\alpha + \sum_B \epsilon_\beta = 0,$$

so at $i = 1$ the regularized value drops out and $\Lambda_{n,1} = -\sum_\gamma s_\gamma|\epsilon_\gamma|\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$, while for $i \ge 2$, $\Lambda_{n,i}$ is $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ times $\frac{1}{i}\sum_\gamma s_\gamma|\epsilon_\gamma|^i + [i \text{ even}]\frac{2}{i}(|A_+| - \sum_{B_+}\epsilon_\gamma^i)$, less $\frac{1}{i}\sum_\gamma s_\gamma|\epsilon_\gamma|^i(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$. These coefficients depend on the slopes alone.

For $n, n+1 > n_1$, $N_\gamma(n+1) = N_\gamma(n) + |\epsilon_\gamma|$, so $N_\gamma(n+1)!/N_\gamma(n)! = (N_\gamma(n)+1)_{|\epsilon_\gamma|}$ and

$$\rho(n) = \frac{C_{n+1}}{C_n} = (-1)^{|A_+|+\sum_{B_+}\epsilon_\gamma}\prod_\gamma \big(N_\gamma(n)+1\big)_{|\epsilon_\gamma|}^{s_\gamma},$$

a rational function with integer coefficients; with $\rho = \rho_1/\rho_2$ in polynomials, $\rho_2(n)C_{n+1} - \rho_1(n)C_n = 0$ is a recurrence of order one with both coefficients nonzero, so the tier is at most one (38.11). By Stirling's formula $\log N! = N\log N - N + O(\log(N+2))$, and with $N_\gamma(n) = |\epsilon_\gamma|n + O(1)$,

$$\log|C_n| = \sum_\gamma s_\gamma\big(|\epsilon_\gamma|n\log n + |\epsilon_\gamma|n\log|\epsilon_\gamma| - |\epsilon_\gamma|n\big) + O(\log n) = n\log r_W + O(\log n),$$

the terms in $n\log n$ and in n cancelling by $\sum_\gamma s_\gamma|\epsilon_\gamma| = 0$, and $|\epsilon_\gamma|\log|\epsilon_\gamma| = 0$ on A. So $|C_n|^{1/n} \to r_W$.

The residue at $-n$ is the coefficient of ε^{-1}, $C_n[\varepsilon^{w-1}]\exp(\sum_i \Lambda_{n,i}\varepsilon^i)$, which is 0 when $w \le 0$, the exponential having no negative powers. For $w \ge 1$ it is $C_n\sum\prod_i \Lambda_{n,i}^{m_i}/m_i!$ over $\sum_i im_i = w-1$, a polynomial with rational coefficients in $\Lambda_{n,1},\dots,\Lambda_{n,w-1}$; inserting the combinations above gives the stated polynomial, each monomial of total index $w-1$.

(iii) Moving the line to the left, the arguments of the Gammas of A_+ fall and cross their poles, those of A_- rise and cross none, and the lower Gammas contribute entire factors; under the hypothesis of the closing, 4.5, with the Laurent coefficients at coincident poles, gives the integral as the sum of the residues at the integers $-n < c$. At each of the finitely many integers $-n < c$ with $n \le n_1$ the residue is a Laurent coefficient of finitely many Gammas at integer arguments: by 5.14 and 5.13 each is a nonzero rational number times the exponential of a series whose coefficient of ε^i is a rational combination of $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ and of heads at integer cuts, which are rational; the coefficient of $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ is $-\epsilon_\alpha\varepsilon$ for an upper Gamma and $\epsilon_\beta\varepsilon$ for a lower one, whatever the sign of the argument, and these cancel by the balance. So these residues are rational combinations of products of $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $i \ge 2$. For $n > n_1$, expand the polynomial of (ii) into monomials: each is a rational number times a product of Trailmoments $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ times $\psi(n)$ for one of the finitely many products ψ of the statement, the heads of the monomial having total index at most $w-1$. Since $|(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N])| \le 1 + \log(N+1)$ for $i \ge 1$ and $N_\gamma(n)$ grows linearly, $|\psi(n)| \le |C_n|\,(c'\log n)^w$ for large n, of root rate $r_W < 1$, so $\sum_{n>n_1}|\psi(n)| < \infty$ (1.1(viii)); the sum of the residues, a finite sum of absolutely convergent series, is the stated combination.

Each ψ is live: C_n is a product of factorials of the moving data, a stacked Gamma, and the heads are Trailmoments at the moving cut. For the recurrence put $H_l(n) = (\mathsf{T}_{i_l;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_{\gamma_l}(n)])$; then $H_l(n+1) = H_l(n) + \eta_l(n)$ with $\eta_l(n) = \sum_{N_{\gamma_l}(n)<j\le N_{\gamma_l}(n)+|\epsilon_{\gamma_l}|} j^{-i_l}$, a rational function of n with rational coefficients. The vector $X(n) = \big(C_n\prod_{l\in T}H_l(n)\big)_{T\subseteq\{1,\dots,q\}}$ satisfies

$$X(n+1)_T = \rho(n)\sum_{U\subseteq T}\Big(\prod_{l\in T\setminus U}\eta_l(n)\Big)X(n)_U,$$

a matrix $\rho(n)$ times a unitriangular one in the order of inclusion, of determinant $\rho(n)^{2^q} \ne 0$. Lemma S29 with ℓ the coordinate of $T = \{1,\dots,q\}$ gives the recurrence of order at most 2^q with nonzero extreme coefficients.

(iv) At the steps k and k', $N_\gamma(n)$ differs by $-(k-k')u_\gamma$ on $A_+ \cup B_+$ and by $(k-k')u_\gamma$ on $A_- \cup B_-$, a constant t_γ. So the sign of C_n changes by a constant, and $N_\gamma(n)!$ at k is $N_\gamma(n)!$ at k' times $(N_\gamma(n)+1)_{t_\gamma}$ or divided by $(N_\gamma(n)+t_\gamma+1)_{-t_\gamma}$, the data at k'; the quotient of the two C_n is a rational function $R_0(n)$ with rational coefficients. Likewise $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N+t])$ is $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N])$ plus or minus the finite sum of j^{-i} between N and $N+t$, a rational function $\eta'(n)$. Hence, with H'_l the heads at k',

$$\psi^{(k)}(n) = R_0(n)\, C_n^{(k')} \prod_l \big(H'_l(n)+\eta'_l(n)\big) = \sum_{T\subseteq\{1,\dots,q\}} R_0(n) \prod_{l\notin T} \eta'_l(n)\; C_n^{(k')} \prod_{l\in T} H'_l(n),$$

and each $C_n^{(k')}\prod_{l\in T} H'_l(n)$ is an input of (iii) at the step k', its heads a sub-multiset of those of ψ. Inserting this into (iii) at the step k, and adding the finitely many terms between the two bounds n_1, which are rational combinations of products of $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $i\ge 2$, by the argument for $n\le n_1$ in (iii), gives the last statement; the sums $\sum_n R(n)\psi^{(k')}(n)$ converge absolutely, $\psi^{(k')}$ having root rate $r_W<1$.

Closing to the right of $W_k(a)$ is closing to the left of $a\mapsto W_k(-a)$, the word with every slope negated, which is again balanced. □

Proposition 43.2 (Paired words). *As stated.*

Proof. Write $N_\gamma(n)=|\epsilon_\gamma|n+e_\gamma$ with $e_\gamma\in\mathbb{Z}$. By the proof of 43.1(ii),

$$\rho(n)=\frac{C_{n+1}}{C_n}=(-1)^{|A_+|+\sum_{B_+}\epsilon_\gamma}\, r_W \prod_\gamma \prod_{t=1}^{|\epsilon_\gamma|} \big(n+\theta_{\gamma,t}\big)^{s_\gamma}, \qquad \theta_{\gamma,t}=\frac{e_\gamma+t}{|\epsilon_\gamma|},$$

using $(N+1)_v = v^v\prod_{t=1}^v (n+(e+t)/v)$ for $N=vn+e$ and $\prod_\gamma |\epsilon_\gamma|^{|\epsilon_\gamma|s_\gamma}=r_W$. For a fixed γ with $|\epsilon_\gamma|=v$ the $\theta_{\gamma,t}$ are v consecutive multiples of $\frac{1}{v}$, so modulo $\mathbb{Z}$ they meet each class j/v once. For $v\ge 1$ put $m_v=\sum_{|\epsilon_\gamma|=v} s_\gamma$, so that $m_v=\#\{\beta:\epsilon_\beta=v\}-\#\{\beta:\epsilon_\beta=-v\}$ for $v\ge 2$, and $\sum_v v\,m_v=\sum_\gamma s_\gamma|\epsilon_\gamma|=0$ by the balance. The signed number of the $-\theta_{\gamma,t}$ in a class of $\mathbb{Q}/\mathbb{Z}$ of exact order v' is $\sum_{v'|v} m_v$.

(b)⇒(a). If $C_n=(-1)^{\varsigma n}R(n)$, then $\rho(n)=(-1)^\varsigma R(n+1)/R(n)$ as rational functions. Writing $R=c\prod_x (n-x)^{\mu_x}$, $R(n+1)/R(n)=\prod_x\big((n-x+1)/(n-x)\big)^{\mu_x}$, whose zeros less poles, counted with multiplicity, add to zero in every class of $\mathbb{C}/\mathbb{Z}$. Comparing with the zeros and poles $-\theta_{\gamma,t}$ of ρ, of multiplicities s_γ, $\sum_{v'|v} m_v=0$ for every $v'\ge 2$, and descending from the largest slope, $m_v=0$ for every $v\ge 2$: that is (a). It does not involve k.

(a)⇒(b). Under (a), $m_v=0$ for $v\ge 2$, hence $m_1=0$ by $\sum_v v m_v=0$, and $r_W=\prod_{v\ge2} v^{vm_v}=1$. Every class of $\mathbb{Q}/\mathbb{Z}$ then holds as many $\theta_{\gamma,t}$ with $s_\gamma=1$ as with $s_\gamma=-1$; pair them within classes, $\theta=\theta'+j$ with $j\in\mathbb{Z}$, and put $R_{\theta,\theta'}(n)=\Gamma(n+\theta)/\Gamma(n+\theta')$, which is $(n+\theta')_j$ or its reciprocal by 1.3(iii), a rational function with $R_{\theta,\theta'}(n+1)/R_{\theta,\theta'}(n)=(n+\theta)/(n+\theta')$. With $\tilde R$ the product of the $R_{\theta,\theta'}$, $\rho(n)=(-1)^\varsigma\tilde R(n+1)/\tilde R(n)$, $\varsigma\equiv|A_+|+\sum_{B_+}\epsilon_\gamma$. For $n>n_1$ the numbers $n+\theta_{\gamma,t}$ are positive, so $\tilde R$ has no zero or pole there, and $C_n=(-1)^{\varsigma n}R(n)$ with $R(n)=(-1)^{\varsigma(n_1+1)}C_{n_1+1}\tilde R(n)/\tilde R(n_1+1)$. As $|A_+|+\sum_{B_+}\epsilon_\gamma=p+2\#\{\epsilon_\beta=1\}+\sum_{B^{\ge2}}\epsilon_\beta$, $\varsigma\equiv p+\sum_{B^{\ge2}}\epsilon_\beta$.

The factorization. By (i) of 43.1, $W_k=W_0 g_k$, so it suffices to factor W_0. For integers b,b', $\Gamma(\pm a+b)/\Gamma(\pm a+b')$ and $\Gamma(\pm va+b)/\Gamma(\pm va)$ are rational functions of a with rational coefficients (1.3(iii)). The balance and (a) give $|A_+|-|A_-|=\#\{\epsilon_\beta=1\}-\#\{\epsilon_\beta=-1\}$, so $p=|A_-|-\#\{\epsilon_\beta=-1\}$ as well. Matching the Gammas of A_+ with those of slope 1 below, and the Gammas of A_-

with those of slope -1, leaves a rational function times $\Gamma(a)^p\Gamma(1-a)^p=\pi^p/\sin(\pi a)^p$ (1.3(v)), read as $\sin(\pi a)^{|p|}/\pi^{|p|}$ when $p<0$; each pair β,β' with $\epsilon_\beta=v=-\epsilon_{\beta'}$, $v\ge 2$, leaves a rational function times $1/(\Gamma(va)\Gamma(1-va))=\sin(\pi va)/\pi$. There are $|B^{\ge 2}|$ pairs, and $w=|A_+|-|B_+|=p-|B^{\ge 2}|$, which gives the display, the sign and R_k collecting the rational factors.

The residue. At $a=-n+\varepsilon$, $\sin(\pi a)=(-1)^n\sin(\pi\varepsilon)$ and $\sin(\pi va)=(-1)^{vn}\sin(\pi v\varepsilon)$, so the trigonometric factor is $(-1)^{n(\sum_{B\ge 2}\epsilon_\beta-p)}\pi^w\prod_\beta\sin(\pi\epsilon_\beta\varepsilon)/\sin(\pi\varepsilon)^p$. Since $\sin(\pi x)/(\pi x)$ is an even power series in πx with rational coefficients and constant term 1, this is $(-1)^{\varsigma n}\varepsilon^{-w}\sum_{l\ge 0}\tau_l\pi^{2l}\varepsilon^{2l}$ with rational τ_l depending on the slopes and $\tau_0=\prod_{B^{\ge 2}}\epsilon_\beta$. For $n>n_1$ the pole of W_k at $-n$ has the order w exactly, $C_n\ne 0$, and the trigonometric factor has a pole of order w there, so R_k is holomorphic at $-n$; the coefficient of ε^{-1} is the displayed sum.

The closing. The finitely many residues at $-n<c$ with $n\le n_1$ are, by the same factorization, combinations of the $\tau_l\pi^{2l}$ with Laurent coefficients of R_k, so they lie in $\mathbb{Q}[\pi^2]$. For $n>n_1$ the residue is $\pm(-1)^{\varsigma n}r(n)$ with $r(n)=\sum_l\tau_l\pi^{2l}R_k^{(w-1-2l)}(-n)/(w-1-2l)!$, a rational function of n whose partial fractions have coefficients in $\mathbb{Q}[\pi^2]$, the poles of R_k being rational. The closing converges; when $\varsigma=0$ its terms are $r(n)$, so $r(n)=O(n^{-2})$, and when $\varsigma=1$, $r(n)\to 0$, so $r(n_1+1+2m)-r(n_1+2+2m)=O(m^{-2})$, and summing consecutive pairs does not change the sum. 38.25 over $\Bbbk=\mathbb{Q}(\pi^2)$, whose identities are linear in the coefficients of the partial fractions (its proof applies 37.10 to them), writes the sum over 1, the Trailmoments of the lattice inputs on the cosets of the poles at their cuts, and the differences of their index-one values, with coefficients in $\mathbb{Q}[\pi^2]$. Finally $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))=\pi^2/6$ by 5.17(iv) at $j=1$. □

Theorem 43.3 (The frame bounds the clearing from below). *As stated.*

Proof. (i) As $p_h(k)\ne 0$ for $k\ge k_0$, a solution is fixed by its values at $k_0,\dots,k_0+h-1$, so independent solutions have independent initial vectors and $C_{k_0}\ne 0$; by 38.11, $C_{k+1}=(-1)^h\frac{p_0(k)}{p_h(k)}C_k$ with $p_0(k)\ne 0$, so $C_k\ne 0$ for every $k\ge k_0$. Then $D_k^hC_k=\det\left[D_kx_{k+l}^{(i)}\right]$ is a nonzero integer and $D_k^h|C_k|\ge 1$. By 38.12, $\log|C_k|=k\log|c|+(d_0-d_h)(k\log k-k)+O(\log k)$, and $h\log D_k\ge-\log|C_k|$ gives the inequality. When $d_0=d_h$ the term in $k\log k$ is absent, and the same proposition identifies $|c|$.

(ii) The coefficients of v_l and of 1 in $\sum_j p_j(k)x_{k+j}=0$ are $\sum_j p_j(k)A_{k+j,l}$ and $-\sum_j p_j(k)B_{k+j}$, rational since $p_j\in\mathbb{Q}[k]$ and k is an integer; as $1,v_1,\dots,v_m$ are independent over $\mathbb{Q}$, all of them vanish. A D_k that clears the rows of $x_k,\dots,x_{k+h-1}$ makes the values at $k,\dots,k+h-1$ of all $m+1$ sequences integers, in particular those of h independent ones, and (i) applies to them. □

Definition 43.4 (The merit of rows along a ray). As stated.

Proof. $f(\alpha^\star)$ is positive and finite (38.14(i)), and $D_k\ge 1$ gives $\limsup_k\frac{1}{k}\log D_k\ge 0$, so the merit lies in $[-\infty,-\log f(\alpha^\star)]$. □

Theorem 43.5 (The merit decides the gap along a ray). *As stated.*

Proof. (i) By 38.14(i), $0<f\le f(\alpha^\star)$ on the orthant and $\langle 1\rangle_k=\int_{y_{e_0}=0}f^k\omega_0$ converges and is positive for $k\ge k_1$; the origin is real, so $\omega_0/dy=\alpha^{\mathbf{m}_0}\prod_j K_j^{-s_{0j}}\Psi_j^{-t_{0j}}$ is positive on the orthant. Hence $\langle 1\rangle_k=\int f^{k-k_1}f^{k_1}\omega_0\le f(\alpha^\star)^{k-k_1}\langle 1\rangle_{k_1}$, and $D_k\ge 1$.

(ii) If the merit is positive, choose $\eta>0$ below it; then $D_k\le f(\alpha^\star)^{-k}e^{-k\eta}$ for all large k, and by (i) $0<D_k\langle 1\rangle_k\le e^{-k\eta}f(\alpha^\star)^{-k_1}\langle 1\rangle_{k_1}\to 0$. With $x_k=\langle 1\rangle_k$, 37.2 gives an irrational v_l, and for $m=1$, $A_{k,1}\ne 0$ for all large k; then v_1 is the irrational one.

(iii) $\frac{1}{k}\log D_k \le \frac{\log C}{k} + \log\gamma + \frac{w}{k}\log d_{\lfloor ak+b\rfloor}$. For $a>0$, $N=\lfloor ak+b\rfloor\to\infty$ with $N/k\to a$, and $\limsup_N \frac{1}{N}\log d_N \le c_{\mathrm{lcm}}$ (37.1), so $\limsup_k \frac{1}{k}\log D_k \le \log\gamma + awc_{\mathrm{lcm}}$; for $a=0$ the last term tends to 0. The merit is then at least $-\log f(\alpha^\star)-\log\gamma-awc_{\mathrm{lcm}}$, positive when $\gamma e^{awc_{\mathrm{lcm}}} f(\alpha^\star)<1$.

(iv) By 43.3(ii) and (i), $\liminf_k \frac{1}{k}\log D_k \ge \frac{1}{h}\log\frac{1}{|c|}$, so the lim sup is at least as large, and the merit is at most $-\log f(\alpha^\star)+\frac{1}{h}\log|c|$. □

Proposition 43.6 (The low end of the law). *As stated.*

Proof. (i) The law says that $\sum_i p_i(k)f^i$ lies in the relation space $\mathcal{L}$ (38.4), a finite sum $\sum r_{e,l}(k)(c_e^{(0)}+kc_e^{(1)}+\partial_{y_e})g_{e,l}$ with $r_{e,l}\in\mathbb{C}(k)$. Integration by parts against $f^k dy$, for complex k of large real part where the integrals converge absolutely (38.3), gives $\sum_i p_i(k)\langle 1\rangle_{k+i}=0$ there off the poles of the $r_{e,l}$, hence identically for the meromorphic continuation of 39.30. At an integer $k\ge 1$ every $\langle 1\rangle_{k+i}$ is a convergent integral, the walls $(k+i)\kappa_{1,S}$ being positive, and the law holds there. Near $k=0$ the terms $i\ge 1$ are holomorphic, and by 39.30 at $j=0$, with $j(1)=0$ and $\mathsf{C}_0(1)=\mathrm{CT}(1)=1$, $\langle 1\rangle_k = V_1 k^{-(n-1)}+O(k^{-(n-2)})$ with $V_1>0$. So $p_0(k)\langle 1\rangle_k$ is holomorphic at 0, which forces p_0 to vanish there to the order $n-1$ at least, and its value at 0 is ϖV_1. The law at $k=0$ is the low-end relation. The walls of the integral direction are integers, $\kappa_{1,S}=m_{1,S}-\sum_j(s_{1j}d_j(S)+t_{1j}d_j^*(S))$ (8.1), so V_1 is rational.

(ii) c and r exist and are rational because $p_2(k)\neq 0$ for $k\ge 1$ and $p_0,p_1,p_2,\varpi V_1$ are rational. The low-end coefficient of c is $-p_1(0)-p_2(0)\big(-p_1(0)/p_2(0)\big)=0$ and that of r is $-p_2(0)\big(-\varpi V_1/p_2(0)\big)=\varpi V_1$. The sequence $x_k-x_1c_k-r_k$ solves the law on $k\ge 1$; it vanishes at $k=1$, and at $k=2$ it is $\big(p_2(0)x_2+p_1(0)x_1+\varpi V_1\big)/p_2(0)=0$ by the low-end relation; so it vanishes identically. □

Definition 43.7 (Reflexive laws). As stated.

Proof. Let c_i be the coefficient of k^d in p_i, so the symbol is $\sigma(\varrho)=\sum_i c_i\varrho^i$ with $c_0c_h\neq 0$ (38.7). The coefficient of k^d in $p_{h-i}(-k-h+1)$ is $(-1)^d c_{h-i}$, so $(-1)^d c_{h-i}=\gamma\lambda^i c_i$, and

$$\varrho^h\sigma(\lambda/\varrho)=\sum_i c_i\lambda^i\varrho^{h-i}=(-1)^d\gamma^{-1}\sum_i c_{h-i}\varrho^{h-i}=(-1)^d\gamma^{-1}\sigma(\varrho).$$

As $\sigma(0)=c_0\neq 0$, $\varrho\mapsto\lambda/\varrho$ maps the roots of σ to roots, with multiplicities. At $h=2$ the identities at $i=0$ and $i=2$ give $c_0/c_2=(-1)^d\gamma^{-1}$ and $c_0/c_2=(-1)^d\gamma\lambda^2$, so $\gamma^2\lambda^2=1$, $\gamma=\pm\lambda^{-1}$, and the product of the roots, c_0/c_2, is $\pm\lambda$. □

Theorem 43.8 (The closings read the coordinate of a reflexive law). *As stated.*

Proof. (i) By 39.32, $\sum_{i=0}^{\min(h,j)} p_i(-j)\mathsf{C}_{j-i}(1)=0$ for every $j\ge 0$. For $k\ge 1$ put $j=k+h-1\ge h$. Reflexivity, $p_i(k)=\gamma^{-1}\lambda^{-i}p_{h-i}(-k-h+1)$, gives

$$\sum_{i=0}^{h} p_i(k)\,\lambda^{k+i-1}\mathsf{C}_{k+i-1}(1)=\gamma^{-1}\lambda^{k-1}\sum_{i=0}^{h}p_{h-i}(-j)\,\mathsf{C}_{j-(h-i)}(1)=0.$$

For the low-end coefficient, reflexivity at $k=0$ gives $p_i(0)=\gamma^{-1}\lambda^{-i}p_{h-i}(-h+1)$, so

$$\sum_{i=1}^{h} p_i(0)\,\lambda^{i-1}\mathsf{C}_{i-1}(1)=\gamma^{-1}\lambda^{-1}\sum_{i'=0}^{h-1}p_{i'}(-(h-1))\,\mathsf{C}_{h-1-i'}(1)=0,$$

by 39.32 at $j=h-1$.

(ii) At $h=2$, $y_1=\mathsf{C}_0(1)=1=c_1$, and the vanishing of the low-end coefficient, $p_1(0)+p_2(0)y_2=0$, gives $y_2=-p_1(0)/p_2(0)=c_2$; both solve the law and $p_2(k)\neq 0$ for $k\geq 1$, so $y=c$, and 43.6(ii) gives the display. With $\langle 1\rangle_1=A_1v-B_1$ and r_k rational, $\langle 1\rangle_k=A_1\lambda^{k-1}\mathsf{C}_{k-1}(1)\,v-\big(B_1\lambda^{k-1}\mathsf{C}_{k-1}(1)-r_k\big)$.

(iii) With integer weights, $f^{-1}=\alpha^{-\mathbf{m}_1}\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}$ is a Laurent polynomial with integer coefficients, and so is each power; at origin 0, $\omega_0/dy=1$, and $\mathsf{C}_j(1)=\mathrm{CT}(f^{-j})\in\mathbb{Z}$. The coefficient is $a^{k-1}\mathsf{C}_{k-1}(1)/b^{k-1}$; as $\gcd(a,b)=1$, its reduced denominator is $b^{k-1}/\gcd\big(b^{k-1},a^{k-1}\mathsf{C}_{k-1}(1)\big)=b^{k-1}/\gcd\big(b^{k-1},\mathsf{C}_{k-1}(1)\big)$. □

Corollary 43.9 (The merit of a reflexive ray of order two). *As stated.*

Proof. The ray has origin 0, real, and $\kappa_{1,S}>0$, so Definition 43.4 applies with $m=1$, $v_1=\langle 1\rangle_1$, $A_{k,1}=\lambda^{k-1}\mathsf{C}_{k-1}(1)$ and $B_k=-r_k$. A clearing D_k makes $D_kA_{k,1}$ an integer, so by 43.8(iii) the integer b^{k-1}/g_k, $g_k=\gcd(b^{k-1},\mathsf{C}_{k-1}(1))$, divides D_k, and $\frac{1}{k}\log D_k\geq\frac{k-1}{k}\log b-\frac{1}{k}\log g_k$. Taking the upper limit, $\limsup_k\frac{1}{k}\log D_k\geq\log b-\liminf_k\frac{1}{k}\log g_k$, which is the bound. □

Proposition 43.10 (The hinge closing as a row over the room). *As stated.*

Proof. (i) By 19.21(i) the closed value is the absolutely convergent sum of the residues, and grouping them by n gives $\sum_n\langle\Phi_n,\mathcal{X}_n\rangle$. Then $\langle\Phi_n,\mathcal{X}_n\rangle=\sum_i\chi_i\langle\Phi_n,\mathbf{c}_i(n)\rangle$, and a finite sum of absolutely convergent series may be summed term by term.

(ii) By 19.21(ii) the two lattices of the level d meet exactly when $2\delta^d\in\mathbb{Z}$, so otherwise every pole is simple and the residue at $\sigma^d=\pm\delta^d-n$ is a coefficient times the room period there. The sequence $\sum_i x_i^{\pm}c_i^{\pm}(n)$ solves the recurrence and agrees with $x^{\pm}$ at $0,\dots,h_{\pm}-1$; as $p_{h_{\pm}}^{\pm}(n)\neq 0$ for $n\geq 0$, the two agree for every n. The $c_i^{\pm}$ are computed from rational initial values by the recurrence, so they are rational when the coefficients are. So $\mathcal{X}_n=\sum_i x_i^{+}\big(c_i^{+}(n),0\big)+\sum_i x_i^{-}\big(0,c_i^{-}(n)\big)$, and (i) applies.

(iii) For $2\delta^d\in\mathbb{Z}$ the lattices $\delta^d-\mathbb{N}$ and $-\delta^d-\mathbb{N}$ are both contained in $|\delta^d|-\mathbb{N}$, and by 19.21(ii) the residue at a common point carries the room period and its derivative in σ^d, at a simple point the room period alone. The room period converges at the points of the lattice (19.21(i)), and near them it is an absolutely convergent integral over the simplex of a function holomorphic in the level shares (8.7), so it is holomorphic in the even share there. The state $U_n(\sigma)=(x_n(\sigma),\dots,x_{n+h-1}(\sigma))$ satisfies $U_{n+1}(\sigma)=\mathbf{A}(n,\sigma)U_n(\sigma)$ with the companion matrix of the recurrence, whose entries $-p_i(n,\sigma)/p_h(n,\sigma)$ are regular at $\sigma=\sigma_0$ for $n\geq 0$. By 37.17 with $\varepsilon=\sigma-\sigma_0$ and $J=1$, $U_{n+1,0}=\mathbf{A}_0(n)U_{n,0}$ and $U_{n+1,1}=\mathbf{A}_0(n)U_{n,1}+\mathbf{A}_1(n)U_{n,0}$, so $(U_{n,0},U_{n,1})$ is a product of block-triangular matrices, which depend on the recurrence alone, applied to $(U_{0,0},U_{0,1})$, the $2h$ initial data. This writes $\mathcal{X}_n$ as in (i), which applies when the resulting sums converge absolutely.

With the bunch and the room exchanged, 19.21(i) closes on the lattices of the level r with the opposite sign, and the same argument applies. □

Definition 43.11 (Critical primes). As stated.

Proof. A native gauge lies in X_F, so every letter is nonzero there, and $f(\alpha_c)\in\Bbbk^{\times}$ since the weights and the coordinates lie in $\Bbbk$; its valuation at $\mathfrak{p}$ is defined. □

Theorem 43.12 (The letters degenerate at a critical prime). *As stated.*

Proof. Let $\mathcal{O}$ be the local ring of $\Bbbk$ at $\mathfrak{p}$, a discrete valuation ring, with a uniformizer $\varpi_{\mathfrak{p}}$, and $v = v_{\mathfrak{p}}$. The letters have coefficients in $\mathcal{O}$ and the coordinates of α_c are units, so every $L(\alpha_c)$ and every $\partial_e L(\alpha_c)$ lies in $\mathcal{O}$, and $L(\alpha_c) \neq 0$. Since $\alpha_c^{\mathbf{m}_1}$ is a unit, $v(f(\alpha_c)) = -\sum_L w_L v(L(\alpha_c)) < 0$, so the largest valuation $v_{\max}$ is positive, and the letters of S vanish at $\bar{\alpha}_c$. The native gauge has $c_e^{(1)} = \partial_{y_e} \log f = 0$ for $e \neq e_0$ (38.4, 38.2(ii)), that is $m_{1,e} = \sum_L w_L\, \alpha_e \partial_e L/L$ at α_c; by Euler's identity for the homogeneous letters, $\sum_e \alpha_e \partial_e L = (\deg L)L$, and the balance $m_{1,\mathcal{E}} = \sum_L w_L \deg L$ of a direction (f has degree 0), the same holds at $e = e_0$. Write $L(\alpha_c) = \varpi_{\mathfrak{p}}^{v_L} u'_L$ with $u'_L \in \mathcal{O}^\times$ and multiply the equation of the mark e by $\varpi_{\mathfrak{p}}^{v_{\max}}$:

$$\sum_L w_L\, \alpha_{c,e}\, \partial_e L(\alpha_c)\, \varpi_{\mathfrak{p}}^{v_{\max} - v_L} u_L'^{-1} = m_{1,e}\, \varpi_{\mathfrak{p}}^{v_{\max}}.$$

Every term lies in $\mathcal{O}$; modulo $\mathfrak{p}$ the right side and the terms with $v_L < v_{\max}$ vanish, and reduction commutes with ∂_e. Dividing by the unit $\bar{\alpha}_{c,e}$ gives (i) with $u_L = u_L'^{-1}$. Under the hypothesis of (ii) every coefficient $\overline{w_L u_L}$ is nonzero, so the gradients of the reduced letters of S at $\bar{\alpha}_c$ satisfy a linear relation with all coefficients nonzero: for one letter its gradient vanishes at a point of its zero set, which is singular there, and for several the gradients are linearly dependent, whereas smooth hypersurfaces meeting transversally have independent gradients. □

Proposition 43.13 (Critical primes and the symbol). *As stated.*

Proof. (i) $c_h^{h-1}\sigma(\lambda) = \sum_j c_j c_h^{h-1-j}(c_h\lambda)^j$ is a monic polynomial with integer coefficients in $c_h\lambda$, which vanishes at a root.

(ii) If $p \nmid c_h$, c_h is a unit at every prime $\mathfrak{P}$ of K above p and $\lambda = c_h\lambda/c_h$ has $v_{\mathfrak{P}}(\lambda) \geq 0$ by (i). Conversely let $p \mid c_h$ and let $\mathfrak{P}$ be a prime above p. If every root had $v_{\mathfrak{P}} \geq 0$, the elementary symmetric functions of the roots, $\pm c_{h-j}/c_h$, would be $\mathfrak{P}$-integral, so $v_{\mathfrak{P}}(c_{h-j}) \geq v_{\mathfrak{P}}(c_h) > 0$ for every j, and p would divide every coefficient, against their coprimality. So some root has negative valuation at $\mathfrak{P}$.

(iii) Under the hypothesis, a root $\lambda = f(\alpha_c)$ with $v_{\mathfrak{P}}(\lambda) < 0$ at a prime $\mathfrak{P}$ above p makes $\mathfrak{P}$ a critical prime of α_c (43.11 with $\Bbbk = K$), and conversely; (ii) gives the equivalence. When p_0 and p_h have the largest degree d, c_0 and c_h are their leading coefficients (38.7), and c_0/c_h is the quotient c of 38.11. □

48.43 Chapter 44: relations and separation

Proposition 44.4 (A finite matrix of mark relations). *As stated, for several letters* $D_1, \dots, D_h$.

Proof. In the coordinate x_i, with the other coordinates as room and the room density $\prod_{l \neq i} x_l^{a_l - 1}(1 - x_l)^{b_l - 1}$, 35.10(a) applies to the letters D_k, positive on the closed box, at the exponents $p_k + 1$, and to the polynomial AD. With $\partial_i D = \sum_k \partial_i D_k \prod_{k' \neq k} D_{k'}$ its expression is

$$\begin{aligned} &x_i(1-x_i)(D\,\partial_i A + A\,\partial_i D) + \big(a_i - (a_i+b_i)x_i\big)AD - x_i(1-x_i)A\sum_k (p_k+1)\,\partial_i D_k \prod_{k' \neq k} D_{k'} \\ &\quad = D\big[x_i(1-x_i)\,\partial_i A + \big(a_i - (a_i+b_i)x_i\big)A\big] - x_i(1-x_i)\,A\sum_k p_k\,\partial_i D_k \prod_{k' \neq k} D_{k'} = \mathrm{mr}_i(A), \end{aligned}$$

and the read is the stated integral. At $A = x^\beta$, $x_i(1-x_i)\partial_i x^\beta + (a_i - (a_i+b_i)x_i)x^\beta = \big((\beta_i + a_i) - (\beta_i + a_i + b_i)x_i\big)x^\beta$, and $x_i(1-x_i)x^\beta \sum_k p_k \partial_i D_k \prod_{k' \neq k} D_{k'} = (1-x_i)x^\beta \sum_\gamma D_\gamma^{(i)} x^\gamma$; expanding

$D = \sum_\gamma D_\gamma x^\gamma$ gives the formula. Its coefficients are rational in (a_i, b_i) over the rational coefficients of the D_k and the p_k, so for a degree bound the vectors $\mathrm{mr}_i(x^\beta)$ form a finite matrix, rational at rational a_i, b_i; by linearity a numerator $\sum_i \mathrm{mr}_i(A_i)$ has the read 0. With one letter, $D_\gamma^{(i)} = p\gamma_i D_\gamma$, the formula of the reference. □

Proposition 44.11 (The critical polynomial block). *As stated, at every order $k \geq 1$.*

Proof. For $R(u) = P(2u)$, $R(\frac{1}{2}) = P(1)$ and Q_P is the polynomial $(R(u) - R(\frac{1}{2}))/(1-2u)$, so 36.4 at the order k gives the display. $Q_P^{(2k-1)}$ has rational coefficients and the moments of $X = \frac{1}{2} + iY$ are rational (36.2(iii)), so $h(P) \in \mathbb{Q}$. The value of $c + \mathrm{crit}_k[P(2u)]$ is $\big(c + h(P)\big) + (2^{2k+1} - 1)P(1)\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$; the coordinate map is onto $\mathbb{Q}^2$, the constant c giving the first coordinate and a constant P, for which $Q_P = 0$, the second; and $a + b\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty)) = 0$ with $(a,b) \in \mathbb{Q}^2$ not zero forces $b \neq 0$ and $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty)) = -a/b$. At the cut N, $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((N,\infty)) = (\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty)) - (\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N])$, a rational head. □

48.44 Chapter 45: the content and its coaction

Every identity proved below between admitted words is a proved relation, so it lies in the relation ideal $\mathfrak{I}$ (44.1) and holds between the classes in $\mathfrak{W}/\mathfrak{I}$; in particular the words $\zeta = e^{2\pi i r/M}$ multiply as the roots of unity do, by the functional equation of the exponential, and a constant factor ζ in front of a trail is a factor of its density, by linearity.

Definition 45.1 (The content). As stated.

Proof. For $\nu_d \geq 2$ the trail converges absolutely: its terms are bounded by $\prod_i n_i^{-\nu_i}$, and $\sum_{0<n_1<\cdots<n_d} \prod_i n_i^{-\nu_i} \leq \sum_{n_d} (1+\log n_d)^{d-1} n_d^{-2} < \infty$; so the limit of the heads is the Trailmoment. For $\nu_d = 1$ and $e^{2\pi i a_d} \neq 1$ the limit exists by 5.27. The data $(\boldsymbol{\nu}, \mathbf{a})$ are finite and rational, so the trails and the words ζ are admitted words, and the spans are defined. □

Proposition 45.2 (The content is an algebra). *As stated.*

Proof. (ii) For an integer $N \geq 1$ let $H_N(T)$ be the head of T over $\{0 < y_1 < \cdots < y_d \leq N\}$. The product $H_N(T)H_N(T')$ is the finite sum, over pairs of increasing tuples $0 < n_1 < \cdots < n_d \leq N$ and $0 < n'_1 < \cdots < n'_{d'} \leq N$, of $\prod_i e^{2\pi i a_i n_i} n_i^{-\nu_i} \prod_j e^{2\pi i a'_j n'_j} {n'_j}^{-\nu'_j}$. The distinct values among the n_i and n'_j, in increasing order $m_1 < \cdots < m_e$, and the positions of the points of each tuple among them give a surjection ϕ, increasing on each part; conversely ϕ and $m_1 < \cdots < m_e$ give back the pair. The term is $\prod_r e^{2\pi i (\sum_{\phi(x)=r} a_x) m_r} m_r^{-\sum_{\phi(x)=r} \nu_x}$, the term of T_ϕ at $(m_r)_r$. So $H_N(T)H_N(T') = \sum_\phi H_N(T_\phi)$. Every T_ϕ is a trail of level M: its frequencies lie in $\frac{1}{M}\mathbb{Z}$, and since ϕ is onto and increasing on each part, its last point holds the last point of T or of T'; if it holds only one of them it has that trail's last index and frequency, and if both, its index is at least 2. As $N \to \infty$ every head converges (45.1), and $TT' = \sum_\phi T_\phi$. The depth e lies between $\max(d, d')$ and $d + d'$ and the indices add to $|\boldsymbol{\nu}| + |\boldsymbol{\nu}'|$.

(i) and (iii). 1 is the word $\zeta = 1$. At level 1 the frequencies are integers, so the trails of depth one of level 1 are the $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ with $k \geq 2$, the index one being excluded by $e^{2\pi i a_1} = 1$. For $\zeta, \zeta' \in \mu_M$, $(\zeta T)(\zeta' T') = \zeta\zeta' \sum_\phi T_\phi$ by (ii), with $\zeta\zeta' \in \mu_M$, so $\mathcal{C}^{(M)}$ is closed under products, with weights adding, and it contains every product of the $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$. When $M \mid M'$, $\frac{1}{M}\mathbb{Z} \subseteq \frac{1}{M'}\mathbb{Z}$ and $\mu_M \subseteq \mu_{M'}$, so $\mathcal{C}^{(M)} \subseteq \mathcal{C}^{(M')}$; as $\mathcal{C}^{(M)}$ and $\mathcal{C}^{(M')}$ both lie in $\mathcal{C}^{(MM')}$, the union $\mathcal{C}$ is an algebra. □

Proposition 45.3 (Cosets, cuts and index one). *As stated.*

Proof. (i) *Cosets.* Put $n_i = My_i$, so that $y_i \in b_i + \mathbb{Z}$ exactly when the integer n_i is congruent to Mb_i modulo M, and the indicator of that congruence is $\frac{1}{M}\sum_{s=0}^{M-1} e^{2\pi i s(n_i - Mb_i)/M}$ (the lattice relation of 5.12, read on one coset). Then $y_i^{-\nu_i} = M^{\nu_i} n_i^{-\nu_i}$ and $e^{2\pi i a_i y_i} = e^{2\pi i (a_i/M) n_i}$, and the heads of the trail up to N' are, term by term,

$$\sum_{\mathbf{s}\in\{0,\dots,M-1\}^d} M^{|\boldsymbol{\nu}|-d} \prod_i e^{-2\pi i s_i b_i} \sum_{MN<n_1<\cdots<n_d\le MN'} \prod_i e^{2\pi i \frac{a_i+s_i}{M} n_i}\, n_i^{-\nu_i}.$$

Here $e^{-2\pi i s_i b_i} = e^{-2\pi i s_i (Mb_i)/M} \in \mu_M$, and the frequencies $(a_i+s_i)/M$ lie in $\frac{1}{MM'}\mathbb{Z}$. If $\nu_d = 1$, then $a_d \notin \mathbb{Z}$, so $(a_d+s_d)/M \notin \mathbb{Z}$: every inner sum is the head of a trail on $\mathbb{Z}^d$ at the cut $c = MN \in \mathbb{N}$ whose last index and frequency satisfy the condition of 45.1, and the trail is the finite combination of their limits.

Cuts. For data $D = (\nu_i, a_i'')_{i\le d}$ on $\mathbb{Z}^d$ let $T_c(D)$ be the trail at the cut $c \in \mathbb{N}$, read as the limit of its heads, $T_c(\varnothing) = 1$, and $D_{>k}$ the last $d-k$ data. Splitting the points $0 < n_1 < \cdots < n_d$ by the number k of them at most c,

$$T_0(D) = \sum_{k=0}^{d} h_k(D)\, T_c(D_{>k}), \qquad h_k(D) = \sum_{0<n_1<\cdots<n_k\le c} \prod_{i\le k} e^{2\pi i a_i'' n_i} n_i^{-\nu_i},$$

first for the heads and then in the limit. So $T_c(D) = T_0(D) - \sum_{k\ge 1} h_k(D) T_c(D_{>k})$, and by induction on the depth $T_c(D)$ is a polynomial in the finite sums $h_k(D_{>l})$ times 1 or a trail $T_0(D_{>l})$, which keeps the last data of D and so is a trail of level MM'. Each h_k is a finite sum of rational multiples of roots of unity in $\mu_{MM'}$, a combination of the words ζ by linearity; with 45.2 the class lies in $\mathcal{C}^{(MM')}$, and the value is as stated. A head of the trail is a finite sum of terms $e^{2\pi i \sum_i a_i y_i} \prod_i y_i^{-\nu_i}$ with $a_i y_i \in \frac{1}{MM'}\mathbb{Z}$, a rational multiple of an element of $\mu_{MM'}$.

(ii) By 5.2, $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,X]) - \log(X-N) \to (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty))$ as $X\to\infty$, and $\log(X-N) - \log(X-N') \to 0$, so the difference D of the two index-one values is the limit of $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,X]) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b'+\mathbb{Z}})((N',X])$. With $n = My$ and the indicator above,

$$(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,X]) = \sum_{s=0}^{M-1} e^{-2\pi i s b} \sum_{MN<n\le MX} \frac{e^{2\pi i s n/M}}{n}.$$

In the difference the terms $s=0$ leave the finite rational sum of $1/n$ between MN and MN', with its sign. For $s\neq 0$, $\sum_{MN<n\le MX} e^{2\pi i s n/M}/n$ is the head up to MX of the trail T_s of level M of depth one, index 1 and frequency s/M, with $e^{2\pi i s/M} \neq 1$, less the finite sum over $0<n\le MN$. So $D = \sum_{s=1}^{M-1} \big(e^{-2\pi i s b} - e^{-2\pi i s b'}\big) T_s$ plus finite sums of rational multiples of roots of unity in μ_M, and $e^{-2\pi i s b} \in \mu_M$ as $Mb \in \mathbb{Z}$. □

Definition 45.4 (Chains). As stated.

Proof. No letter lies on the open segment, so the integrand is continuous on the open simplex, and its only possible singularities on the closure are at $t_i \in \{0,1\}$; the convergence is 45.6(i). □

Proposition 45.5 (The content as chains). *As stated.*

Proof. (i) is 5.27 with $e^{2\pi i a_i} \in \mu_M$: the letters c_i^{-1} lie in μ_M, the first letter is $c_1^{-1} \neq 0$, the last is 0 when $\nu_d \geq 2$ and $c_d^{-1} = e^{-2\pi i a_d} \neq 1$ when $\nu_d = 1$, and no point of $\{0\} \cup \mu_M$ lies on the open segment from 0 to 1.

(ii) The ratios c_i/c_{i+1} lie in μ_M, so frequencies $a_i' \in \frac{1}{M}\mathbb{Z}$ with $e^{2\pi i a_i'} = c_i/c_{i+1}$ exist, and $e^{2\pi i(a_i'+\cdots+a_d')} = c_i/c_{d+1} = c_i$. As $a_1 \neq 0$, $j_1 = 1$, and the word of (i) for the data (ν_i, a_i'), the blocks $(c_i^{-1}, 0^{\nu_i - 1})$, is $a_1, \ldots, a_w$. If $\nu_d = 1$ then $j_d = w$ and $e^{2\pi i a_d'} = c_d = a_w^{-1} \neq 1$, so the trail is of level M, and (i) gives the identity.

(iii) The values of $\mathcal{C}^{(M)}$ are the $\mathbb{Q}$-span of the ζ and the ζT; the $\mathbb{Q}$-span of μ_M is $\mathbb{Q}(\mu_M)$, and by (i) and (ii) the values of the trails of level M are, up to sign, exactly the chains described, the weight $|\boldsymbol{\nu}|$ of a trail being the number w of letters of its chain. □

Proposition 45.6 (Chains with an end at zero). *As stated.*

Proof. (i) Write $\phi_i(t) = \gamma'(t)/(\gamma(t) - a_i)$. As no letter lies on the open segment, $|\phi_i(t)| \leq C/t$ if $a_i = a_0$, $|\phi_i(t)| \leq C/(1-t)$ if $a_i = a_{w+1}$, and $|\phi_i| \leq C$ otherwise. Split the simplex by the number k of the t_i below $\frac{1}{2}$: the part is a product of $\{0 < t_1 < \cdots < t_k < \frac{1}{2}\}$ and $\{\frac{1}{2} \leq t_{k+1} < \cdots < t_w < 1\}$, on which the factors singular at the other end are bounded. On the first, put $G_j(t) = \int_{0<t_1<\cdots<t_j<t} \prod_{i\leq j} |\phi_i(t_i)|\, dt$ for $t \leq \frac{1}{2}$, where $|\phi_i(s)| \leq C(1 + 1/s)$; since $a_1 \neq a_0$, $G_1(t) \leq Ct$, and if $G_{j-1}(s) \leq C_{j-1}s$ then $G_j(t) \leq \int_0^t C_{j-1}s\, C(1+1/s)\, ds \leq C_j t$; so the first factor is finite. The second is the same after $t \mapsto 1 - t$, with $a_w \neq a_{w+1}$. So the integral of the modulus is finite.

(ii) For $\tilde{\gamma} = \lambda\gamma + \eta$, $\tilde{\gamma}'(t)/(\tilde{\gamma}(t) - \lambda a_i - \eta) = \gamma'(t)/(\gamma(t) - a_i)$, and the open segment is mapped onto the open segment of the new ends.

(iii) The segment from a_{w+1} to a_0 is $\tilde{\gamma}(t) = \gamma(1-t)$, with $\tilde{\gamma}' = -\gamma'$; the substitution $t_i = 1 - t'_{w+1-i}$ maps the simplex onto itself, and each of the w factors changes sign.

(iv) For $s \in \mu_M$, (ii) with $\lambda = s^{-1}$ gives $\mathrm{I}(0; a_1, \ldots, a_w; s) = \mathrm{I}(0; a_1/s, \ldots, a_w/s; 1)$, with letters in $\{0\} \cup \mu_M$, first letter $a_1/s \neq 0$ and last letter $a_w/s \neq 1$. For a chain from s to 0, (iii) gives $(-1)^w$ times a chain from 0 to s whose first letter is $a_w \neq 0$ and whose last letter is $a_1 \neq s$, and the first case applies. 45.5(ii) and (iii) place the value in the values of $\mathcal{C}^{(M)}$. □

Definition 45.7 (Symbols of chains). As stated.

Proof. The variables are the formal chains of weight at least one with distinct ends; the expressions with no letter or with equal ends are not variables, so assigning them 1 and 0 involves no relation among the variables. The symbol of a trail depends on the frequencies only through the c_i, which do not change when a frequency moves by an integer, as the trail does not; and the realization converges by 45.6(i), no point of $\{0\} \cup \mu_M$ lying on the open segment between 0 and a root of unity. □

Definition 45.8 (The cutting rule). As stated.

Proof. $\mathfrak{S}_M$ is a polynomial algebra on the formal chains, so a homomorphism is fixed by arbitrary values on them; the right side of the display is an element of $\mathfrak{S}_M \otimes \mathfrak{S}_M$, its degenerate entries read by the conventions of 45.7. The rotation ρ_λ and the reversal map formal chains to formal chains, up to sign, keeping the ends distinct, and the conventions to the conventions; $\rho_{\lambda^{-1}}$ and the reversal itself are their inverses. □

Theorem 45.9 (The cutting rule is a coaction). *As stated.*

Proof. All maps in (i) and (iii) are homomorphisms of algebras, so it suffices to compare them on a formal chain $x = \mathrm{I}^{\mathfrak{s}}(a_0; a_1, \dots, a_w; a_{w+1})$. For a set $U \subseteq \{1, \dots, w\}$ the *gaps* of U are the pairs $t < t'$ of consecutive elements of $U \cup \{0, w+1\}$; for $T \subseteq S$ put

$$\tau(T,S) = \mathrm{I}^{\mathfrak{s}}(a_0; a_T; a_{w+1}) \otimes \prod_{(t,t') \text{ gap of } T} \mathrm{I}^{\mathfrak{s}}\big(a_t; a_{S\cap(t,t')}; a_{t'}\big) \otimes \prod_{(s,s') \text{ gap of } S} \mathrm{I}^{\mathfrak{s}}\big(a_s; a_{(s,s')}; a_{s'}\big),$$

with $a_{(s,s')}$ the letters strictly between the positions s and s', the degenerate entries read by the conventions.

First, for any pair of ends and letters, the display of 45.8 gives Δ of the corresponding expression even when it is not a formal chain. With no letter the expression is 1, and the display has the single term $1 \otimes 1 = \Delta 1$. With equal ends and at least one letter the expression is 0; in the display every term with a kept letter has a left factor with equal ends, which is 0, and the term with no kept letter has the right factor equal to the expression itself, which is 0; so the display gives $0 = \Delta 0$.

(i) $(\Delta \otimes \mathrm{id})\Delta x = \sum_S \Delta\, \mathrm{I}^{\mathfrak{s}}(a_0; a_S; a_{w+1}) \otimes \prod_{\text{gaps of } S}(\cdots)$, and the display for the chain of the letters a_S keeps the letters of $T \subseteq S$ and reads the letters of S between consecutive kept ones: the sum is $\sum_{T\subseteq S} \tau(T,S)$. Next, $(\mathrm{id} \otimes \Delta)\Delta x = \sum_T \mathrm{I}^{\mathfrak{s}}(a_0; a_T; a_{w+1}) \otimes \prod_{(t,t')} \Delta\, \mathrm{I}^{\mathfrak{s}}(a_t; a_{(t,t')}; a_{t'})$, and by the remark above each factor is the display, a sum over the sets $S_{(t,t')}$ of letters kept inside the gap (t,t'). A family of such sets is the same as one set $S \supseteq T$, the middle factors are the chains of the letters of S inside the gaps of T, and the right factors are the gaps of the $S_{(t,t')}$ inside the gaps of T, which are the gaps of S: the sum is again $\sum_{T\subseteq S} \tau(T,S)$. For the counit, ε kills every formal chain, so in $(\varepsilon \otimes \mathrm{id})\Delta x$ only $S = \varnothing$ survives, with the right factor x; in $(\mathrm{id} \otimes \varepsilon)\Delta x$ a right factor contributes 1 only when it has no letter, so only $S = \{1, \dots, w\}$ survives, with the left factor x.

(ii) The term of S has the left weight $|S|$ and the right weight $w - |S|$, the runs between kept letters together holding the letters not kept. The terms $S = \{1, \dots, w\}$ and $S = \varnothing$ are $x \otimes 1$ and $1 \otimes x$, and every other term has $0 < |S| < w$.

(iii) Multiplying every entry by λ commutes with keeping a subsequence and with cutting out runs, so $\Delta\rho_\lambda x = (\rho_\lambda \otimes \rho_\lambda)\Delta x$. The reversal maps the term of S for x to the term of the mirror set $S^* = \{w + 1 - s : s \in S\}$ for the reversed chain: its left factor is $(-1)^{|S|}$ times the reversal of the kept chain read backwards, each run is reversed with the sign $(-1)^{\text{its length}}$, and the runs appear in the opposite order, which does not matter in the commutative algebra. The signs multiply to $(-1)^w$, the sign of ϱx, so $\Delta\varrho x = (\varrho \otimes \varrho)\Delta x$. □

Definition 45.10 (The letter of an input). As stated.

Proof. The Laplace transform beyond the cut $n_0 - 1$ is $\mathcal{F}_{n_0-1}(u) = \sum_{n\geq n_0} \psi(n) e^{-un} = L_\psi(e^{-u})$ (5.4), and $L_\psi(1) = \sum_n \psi(n)$ where the series converges at 1, which is the Trailmoment at index zero (6.1). For $\psi(n) = e^{2\pi i a n}$, $n \geq 1$, $L_\psi(z) = \sum_{n\geq 1}(e^{2\pi i a} z)^n = e^{2\pi i a} z/(1 - e^{2\pi i a} z)$, with its pole at $e^{-2\pi i a}$, which is the letter c_1^{-1} of the depth-one trail at that frequency in 45.5(i). □

Proposition 45.11 (The letter of a live input). *As stated.*

Proof. (i) $p_i(\theta - i) z^n = p_i(n - i) z^n$, so

$$\sum_i z^{h-i} p_i(\theta - i) L_\psi = \sum_i \sum_{n\geq n_0} p_i(n-i)\psi(n) z^{n+h-i} = \sum_m z^{m+h} \sum_{i:\, m+i\geq n_0} p_i(m)\psi(m+i).$$

For $m \geq n_0$ the inner sum is the recurrence, zero; the finitely many m with $n_0 - h \leq m < n_0$ leave a polynomial Q of degree less than $n_0 + h$. Since θ^j is $z^j \frac{d^j}{dz^j}$ plus terms of lower order, the coefficient of $\frac{d^d}{dz^d}$ is $\sum_i z^{h-i} c_i z^d$, nonzero as some $c_i \neq 0$; for $z \neq 0$ it vanishes exactly when $\sigma(1/z) = 0$.

(ii) For large n, $p_h(n) \neq 0$, and the state $U(n) = (\psi(n), \dots, \psi(n+h-1))$ satisfies $U(n+1) = \mathbf{A}(n)U(n)$ with the companion matrix of the recurrence, whose entries $-p_i(n)/p_h(n)$ tend to $-c_i/c_h$, as p_h has the largest degree: $\mathbf{A}(n) \to \mathbf{A}_\infty$, the companion matrix of σ/c_h, of spectral radius ϱ. For $\eta > 0$ there is a norm on $\mathbb{C}^h$ whose operator norm of $\mathbf{A}_\infty$ is at most $\varrho + \eta/2$ (a vector norm adapted to the Jordan form of $\mathbf{A}_\infty$, which exists for every square matrix); then $\|\mathbf{A}(n)\| \le \varrho + \eta$ for $n \ge n_3$, and $\|U(n)\| \le \|U(n_3)\|(\varrho+\eta)^{n-n_3}$, which with the finitely many earlier terms gives the bound. So L_ψ converges on $|z| < 1/(\varrho+\eta)$ for every η. By (i) it satisfies, near any point of that disc other than 0, a linear differential equation with polynomial coefficients and polynomial right side, whose leading coefficient vanishes only at 0 and at the reciprocals of the nonzero roots of σ; the solutions of such an equation continue analytically along every path on which the leading coefficient does not vanish (the existence theorem for linear differential equations with holomorphic coefficients, which applies because the coefficients are polynomials and the leading one is nonzero along the path). The cut plane Ω is simply connected and star-shaped about 0, and contains the disc; a path in Ω meets the singular points only inside the disc, at 0, where L_ψ is holomorphic, so L_ψ continues along every path in Ω, and by the monodromy theorem it defines a holomorphic function on Ω.

(iii) Let the two recurrences have orders h and h', with companion matrices $\mathbf{A}(n)$ and $\mathbf{A}'(n)$, invertible as rational matrices since their determinants are $\pm p_0(n)/p_h(n)$ and $\pm p'_0(n)/p'_{h'}(n)$. The vector $X(n) = U(n) \otimes U'(n)$ satisfies $X(n+1) = (\mathbf{A}(n) \otimes \mathbf{A}'(n))X(n)$ for large n, with an invertible Kronecker product, and the product sequence is the coordinate of $X(n)$ that pairs the first coordinates. Lemma S29 over the field $\Bbbk$ generated by the coefficients gives the recurrence of order at most hh' with nonzero extreme coefficients.

(iv) By (ii), $|\psi(n)\psi'(n)| \le C(\varrho+\eta)^n(\varrho'+\eta)^n$, so the letter of $\psi\psi'$, the termwise product of the coefficients of L_ψ and $L_{\psi'}$, converges on $|z| < 1/(\varrho\varrho')$. By (ii), L_ψ and $L_{\psi'}$ are holomorphic on the planes cut along the rays from the reciprocals of the nonzero roots of σ and of σ'. Hadamard's multiplication theorem, which applies to two power series holomorphic on planes cut along the outward rays from finitely many nonzero points, makes the series of the termwise products holomorphic on the plane cut along the outward rays from the products of those points, the points $1/(\lambda\lambda')$. □

Proposition 45.12 (The row of a hinge closing read by letters). *As stated.*

Proof. (i) The $c_i^\pm$ solve the room's recurrence, whose last coefficient has the largest degree, so by 45.11(ii) $|c_i^\pm(n)| \le C(\varrho^\pm + \eta)^n$; likewise $|\Phi_n^\pm| \le C(\varrho_\Phi^\pm + \eta)^n$ for large n. With η small, $(\varrho^\pm + \eta)(\varrho_\Phi^\pm + \eta) < 1$, and the sums converge absolutely; the radius of convergence of $L_{\psi_i^\pm}$ exceeds 1, so $\Xi_i^\pm = L_{\psi_i^\pm}(1)$. 43.10(ii) gives the row.

(ii) By 45.11(iii) the product $\psi_i^\pm$ satisfies a recurrence with polynomial coefficients for large n, so by 45.11(i) its letter, up to a polynomial from the finitely many first terms, satisfies a linear differential equation with polynomial coefficients; and 45.11(iv), applied to the tails of the two factors, gives the continuation, the finitely many first terms adding a polynomial. □

48.45 Chapter 46: finite algebra for scale reads

The letters of this chapter have complex coefficients and no roots on $[0,\infty)$. Two facts carry the proofs. First, along $[0,\infty)$ a letter P_j has a continuous argument with limits at 0 and at ∞, so for $x \to \infty$, $P_j(x)^{-w_j} = C_j x^{-w_j \deg P_j}(1 + O(x^{-1}))$ with $C_j \neq 0$, and $|P_j^{-w_j}|$ is $|P_j|^{-\operatorname{Re} w_j}$ up to factors bounded above and below on $[0,\infty)$: a read converges absolutely exactly where it does for positive

letters of the same degrees. Second, the reduction of 46.7 is algebraic in $\mathbb{F}'$ and does not use the values of the letters.

Definition 46.6 (Insertions and reads on one scale). As stated, for letters without roots on $[0,\infty)$.

Proof. $P_j(0)\neq 0$ and P_j does not vanish on $[0,\infty)$, so a continuous logarithm of P_j along $[0,\infty)$ starting at the principal logarithm of $P_j(0)$ exists and is unique, and it defines $P_j^{-w_j}$; on $(0,\infty)$ it is locally a holomorphic branch, so $\big(P_j^{-w_j}\big)' = -w_j P_j' P_j^{-w_j-1}$, and the product rule gives $\big(x^\kappa \prod_j P_j^{-w_j} g\big)' = x^{\kappa-1}\prod_j P_j^{-w_j}\,\partial g$. The squarefree decomposition and the gcd splitting are computed in $\mathbb{F}[x]$ by gcds and exact divisions, so the q_k lie in $\mathbb{F}[x]$, and they divide $P_1\cdots P_N$, so none vanishes at 0 or on $[0,\infty)$. For $\xi_\ell\in\mathbb{C}\setminus(-\infty,0]$, $x+\xi_\ell$ stays in $\mathbb{C}\setminus(-\infty,0]$ for $x\geq 0$, so the continuous branch is the principal one, that of 15.1 and 46.1. □

Theorem 46.7 (Reduction by the scale exchange). *As stated, for letters without roots on* $[0,\infty)$.

Proof. Write $\mu_k=\sum_j m_{jk}w_j$, so that $\partial = x\frac{d}{dx}+\kappa-\sum_k \mu_k \frac{xq_k'}{q_k}$ and $\mathfrak{A}=\mathbb{F}'[x,x^{-1},Q^{-1}]$. Every $R\in\mathfrak{A}$ is a Laurent polynomial in x plus principal parts at the q_k.

Three reductions. (a) If R has a pole of order $h+1\geq 2$ at q_k with leading numerator s, $\deg s<\deg q_k$, then

$$\partial\big(r\,q_k^{-h}\big) = -(h+\mu_k)\,\frac{xq_k'\,r}{q_k^{h+1}} + O\big(q_k^{-h}\big),$$

plus simple poles at the other $q_{k'}$ and no polynomial part and no pole at 0, $r\,q_k^{-h}$ vanishing at ∞ and being regular at 0; xq_k' is prime to q_k, q_k being squarefree and $q_k(0)\neq 0$, so the extended Euclidean algorithm gives r with $-(h+\mu_k)xq_k'r\equiv s \bmod q_k$, one division by $h+\mu_k$. (b) If R has a pole x^{-r}, $r\geq 1$, at 0, $\partial(x^{-r}) = (\kappa-r)x^{-r} - \sum_k \mu_k x^{1-r}q_k'/q_k$, whose other terms have poles of order below r at 0, simple poles at the q_k and no polynomial part. (c) If the polynomial part of R has degree $a\geq 0$, $\partial(x^a) = \big(\kappa+a-\sum_j w_j\deg P_j\big)x^a$ plus a polynomial of degree below a plus simple poles at the q_k, since $xq_k'/q_k = \deg q_k + O(x^{-1})$ and $\sum_k \mu_k \deg q_k = \sum_j w_j \deg P_j$. Subtracting ∂ of a multiple at each step lowers the order of the pole, resp. the degree, and leaves $\sum_{i<D} c_i x^i/Q$ once the poles at the q_k are simple and there is no pole at 0 and no polynomial part; each step divides by one of the three numbers of the hypothesis.

Uniqueness. Let $\sum_i c_i x^i/Q = \partial g$ with $g\in\mathfrak{A}$, $g\neq 0$. A pole of order $h\geq 1$ of g at q_k gives ∂g a pole of order $h+1$ by (a), but the left side has simple poles; so g is a Laurent polynomial. A pole x^{-r} of g at 0 gives ∂g the pole $(\kappa-r)x^{-r}$, but $Q(0)\neq 0$; so g is a polynomial, of degree $a\geq 0$, and by (c) ∂g is of order x^a at ∞, while the left side is $O(x^{-1})$. So $g=0$ and every $c_i=0$: $\dim_{\mathbb{F}'}\mathfrak{A}/\partial\mathfrak{A}=D$.

Reads. Let the reads of R and of the x^i/Q converge absolutely, so that $\Phi = x^\kappa\prod_j P_j^{-w_j} g$ has the absolutely integrable derivative $x^{\kappa-1}\prod_j P_j^{-w_j}\big(R-\sum_i c_i x^i/Q\big)$ and limits at 0 and at ∞. With ν_0 the order of g at 0 and a its degree at ∞, $\Phi = c_0 x^{\kappa+\nu_0}(1+o(1))$ at 0 and $\Phi = c_\infty x^{\kappa+a-\sum_j w_j\deg P_j}(1+o(1))$ at ∞, with $c_0,c_\infty\neq 0$ by the first fact above; a limit exists only if the exponent has negative real part at ∞, positive at 0, or is 0. At 0: if $\nu_0\leq -1$ the exponent $\kappa-|\nu_0|$ is not 0 by hypothesis, and if $\nu_0\geq 0$ its real part is positive since the read of $1/Q$ converges, $\operatorname{Re}\kappa>0$. At ∞: if $a\geq 0$ the exponent is not 0 by hypothesis, and if $a\leq -1$ its real part is negative since the read of x^{D-1}/Q converges, $\operatorname{Re}(\kappa-\sum_j w_j\deg P_j)<1$. So both limits are 0, $\langle\partial g\rangle=0$, and $\langle R\rangle = \sum_i c_i\langle x^i/Q\rangle$; at symbolic exponents the identity holds on the open set of

absolute convergence and continues. For the linear letters, $Q = \prod_\ell (x+\xi_\ell)$ is squarefree, the x^i/Q, $i < |J|$, and the $(x+\xi_\ell)^{-1}$ span the same space by partial fractions, and the three numbers are $\kappa - r$, $\kappa + a - |\mathbf{w}|$ and $w_\ell + h$, nonzero in $\mathbb{K}$. □

Definition 46.8 (Deformation; masters). As stated, along every admissible direction.

Proof. Along the deformation the three numbers of 46.7 are $\kappa - r + \lambda_0\varepsilon$, $\big(\kappa + a - \sum_j w_j \deg P_j\big) + \varepsilon\big(\lambda_0 - \sum_j \lambda_j \deg P_j\big)$ and $(\mu_k + h) + \varepsilon \sum_j m_{jk}\lambda_j$, polynomials in ε with nonzero slopes, hence nonzero in $\mathbb{F}'(\lambda_0, \boldsymbol{\lambda})(\varepsilon)$. At large real ε the exponent of a read at 0 has real part growing like $\lambda_0\varepsilon$ and at ∞ falling like $(\sum_j \lambda_j \deg P_j - \lambda_0)\varepsilon$, the moduli of the powers being those of positive letters up to bounded factors; so finitely many reads converge absolutely together, the coefficients of the reductions have no pole there, and 46.7 gives the identities there, hence as meromorphic functions of ε, the reads continuing meromorphically by the expansions of the integrands at 0 and at ∞ (8.4). Q divides $P_1 \cdots P_N$. The conditions are strict inequalities and finitely many inequations, invariant under positive scaling, so the admissible directions form an open cone; $(\frac{1}{2}; \mathbf{1})$ is in it, since $\sum_j \deg P_j \geq 1 > \frac{1}{2}$ and $\sum_j m_{jk} \geq 1$ for every k, q_k dividing some P_j. □

Lemma 46.9 (Simple poles at a corner). *As stated, along every admissible direction and for letters without roots on* $[0, \infty)$.

Proof. The three reductions of the proof of 46.7 act on separate parts of an insertion: (a) on the poles of order at least two at one q_k, (b) on the pole at 0, (c) on the polynomial part, and each step feeds only the lower orders of its own part and the simple poles, which form the remainder $\sum_i c_i x^i/Q$. In (c) the step at the degree a divides by $\big(\kappa + a - \sum_j w_j \deg P_j\big) + \varepsilon\big(\lambda_0 - \sum_j \lambda_j \deg P_j\big)$, which vanishes at $\varepsilon = 0$ for at most one a; in (b) the step at r divides by $\kappa - r + \lambda_0\varepsilon$, vanishing at 0 for at most one r; in (a), for each k, the step at h divides by $\mu_k + h + \varepsilon \sum_j m_{jk}\lambda_j$, vanishing at 0 for at most one h. Along each chain every other division is by a number holomorphic and nonzero near $\varepsilon = 0$. So for R holomorphic in ε, whose principal parts are then holomorphic in ε, every coefficient is a sum of terms with at most one division vanishing at $\varepsilon = 0$: at most a simple pole.

For a master write $H_i = \int_0^1 + \int_1^\infty$. At 0 the integrand is $x^{\kappa + \lambda_0\varepsilon - 1}\phi_\varepsilon(x)$ with ϕ_ε analytic at 0 and holomorphic in ε, and 8.4 gives poles at $\kappa + \lambda_0\varepsilon + l = 0$, $l \in \mathbb{N}$, all simple and at distinct points, at most one of them at $\varepsilon = 0$. At ∞, by the first fact above, the integrand is $x^{e(\varepsilon)-1}$ times a power series in x^{-1} holomorphic in ε, $e(\varepsilon) = \kappa - \sum_j w_j \deg P_j + i - D + \varepsilon\big(\lambda_0 - \sum_j \lambda_j \deg P_j\big)$, and $x \mapsto 1/x$ gives simple poles at $e(\varepsilon) = l$, $l \in \mathbb{N}$, at most one at $\varepsilon = 0$. So H_i has at most a simple pole there. □

Theorem 46.10 (Finitely many numbers at every corner). *As stated, along every admissible direction and for letters without roots on* $[0, \infty)$.

Proof. Absolute convergence of $\langle R \rangle_{\kappa, \mathbf{w}; \mathbf{P}}$ is a strict inequality at 0 and one at ∞, so the deformed read converges absolutely for ε near 0 and is holomorphic there, with the value $\langle R \rangle_{\kappa, \mathbf{w}; \mathbf{P}}$ at 0. By 46.8 it equals $\sum_i c_i(\varepsilon) H_i(\varepsilon)$ as a meromorphic function, and by 46.9 each product has at most a double pole, with the constant term $c_{i,-1}h_{i,1} + c_{i,0}h_{i,0} + c_{i,1}h_{i,-1}$. The constant term of a function holomorphic at 0 is its value. The masters do not involve R. The derivative ∂_ε^b of $x^{\lambda_0\varepsilon} \prod_j P_j^{-\varepsilon\lambda_j}$ is $\big(\lambda_0 \log x - \sum_j \lambda_j \log P_j\big)^b$ times it, the logarithms continuous along $[0, \infty)$, so the read of $R\big(\lambda_0 \log x - \sum_j \lambda_j \log P_j\big)^b$ is $b!$ times the coefficient of ε^b of $\sum_i c_i H_i$, which uses $h_{i,-1}, \ldots, h_{i,b+1}$: $(b+3)D$ numbers. For the last sentence write $\ell_0 = \log x$ and $\ell_j = \log P_j$. The b-th powers of the linear forms $\lambda_0\ell_0 - \sum_j \lambda_j \ell_j$, for $(\lambda_0; \boldsymbol{\lambda})$ in a dense subset of a nonempty open set,

span the forms of degree b in $(\ell_0,\dots,\ell_N)$: a linear functional vanishing on them is a polynomial in $(\lambda_0;\boldsymbol{\lambda})$ vanishing on a dense subset of an open set, hence zero, and its coefficients are, up to multinomial factors, its values on the monomials. The rational admissible directions are such a set, and a linear system with rational coefficients that has a solution has a rational one; so $\ell_0^{k_0}\prod_j \ell_j^{k_j}$ is a rational combination of finitely many such powers, and its read, absolutely convergent with that of R, is the same combination of reads along these directions. □

Theorem 46.11 (Integer rays have certified recurrences). *As stated, for letters without roots on* $[0,\infty)$.

Proof. Over $\mathbb{F}'(n,\varepsilon)$ the three numbers of 46.7 at the data $\big(\kappa+an+\frac{\varepsilon}{2},\mathbf{w}+n\mathbf{b}+\varepsilon\mathbf{1}\big)$ have the nonzero slopes $\frac12$, $\frac12-\sum_j \deg P_j$ and $\sum_j m_{jk}$ in ε. Reduce the $D+1$ insertions $y_k=x^{ak}\prod_j P_j^{-b_jk}$: $y_k=\eth_n g_k+\sum_i c_{ki}x^i/Q$, where the denominators of g_k and of c_{ki} are products of the three numbers. The $(D+1)\times D$ matrix (c_{ki}) has a nonzero vector in its left kernel, found by elimination; clearing denominators gives polynomials $p_k(n,\varepsilon)$, and dividing by the largest power of ε that divides all of them makes the $p_k(n,0)$ not all zero. Then $g=\sum_k p_k g_k$, divided by that power of ε, has $\eth_n g=\sum_k p_k y_k$, and its denominators are products of the three numbers and a power of ε, none vanishing identically in ε at an integer n. Fix an integer n at which the $D+1$ displayed reads converge, absolutely since the reads of a ray are read where they converge absolutely; they converge absolutely for small ε too. For all but finitely many small $\varepsilon\neq0$, g is defined and the exponents of $x^{\kappa+an+\varepsilon/2}\prod_j P_j^{-w_j-nb_j-\varepsilon}g$ at 0 and at ∞, which move with ε, are not 0; the argument of 46.7 then gives $\langle\eth_n g\rangle=0$, so $\sum_k p_k(n,\varepsilon)\langle1\rangle_{\kappa+a(n+k)+\varepsilon/2,\,\mathbf{w}+(n+k)\mathbf{b}+\varepsilon\mathbf{1}}=0$. Letting $\varepsilon\to0$, the reads being continuous there, gives the recurrence at n. □

Theorem 46.12 (Coefficient jets on one scale). *As stated, for letters without roots on* $[0,\infty)$*, with the powers of the changed letters continued in* u.

Proof. P_{ja}/P_j is continuous on $[0,\infty]$, P_j having no roots on $[0,\infty)$ and $\deg P_{ja}\le\deg P_j$, so M_{ja} is finite. On the polydisc $z_j=\sum_a u_aP_{ja}/P_j$ has $|z_j|\le\sum_a|u_a|M_{ja}<1$, so $P_j(x;u)=P_j(x)(1+z_j)$ has no roots on $[0,\infty)$, and the power continued in u from $u=0$, the polydisc being convex, is $P_j(x)^{-w_j}(1+z_j)^{-w_j}$ with the principal branch of the second factor. Then

$$(1+z_j)^{-w_j}=\sum_{m\ge0}\frac{(-1)^m(w_j)_m}{m!}z_j^m,\qquad z_j^m=\sum_{|n_j|=m}\frac{m!}{\prod_a n_{ja}!}\prod_a\Big(u_a\frac{P_{ja}}{P_j}\Big)^{n_{ja}},$$

and the product over j is the displayed series inside the read. It is dominated by $\prod_j\sum_m\frac{|(w_j)_m|}{m!}\rho_j^m$, $\rho_j=\sum_a|u_a|M_{ja}<1$, finite by the ratio test, times the absolutely integrable integrand of $\langle1\rangle_{\kappa,\mathbf{w};\mathbf{P}}$; so the read of the sum is the sum of the reads, absolutely, uniformly on smaller polydiscs, and holomorphic in u. Each coefficient is the read of the rational insertion $\prod_{j,a}(P_{ja}/P_j)^{n_{ja}}$, bounded on $[0,\infty)$, absolutely convergent with $\langle1\rangle_{\kappa,\mathbf{w};\mathbf{P}}$, and 46.10 gives it. The continuation in u fixes the branch: the continuous branch of $P_j(x;u)^{-w_j}$ along $[0,\infty)$ from its principal value at 0 can differ from it by a factor $e^{2\pi i\ell w_j}$, $\ell\in\mathbb{Z}$. □

48.46 Chapter 47: the family as inference

48.46.1 Samples of bases

Every statement of this section rests on one factorization: the chance of a sample of a family is the product over its draws of the chances of single bases, each of which factors over the patterns

of a bunch, and the product over all the draws is $\theta_{\text{draws}}\alpha^{\mathbf{m}}/\prod_j K_j(\alpha)^{s_j}\Psi_j(\alpha)^{t_j}$.

Proposition 47.3 (What the data see of the bases). *As stated.*

Proof. (i) is a property of one table. For $e \in C$, $C \setminus e$ is independent by the minimality of C and lies in a basis T, which does not contain C; so $T \cap C = C \setminus e$. Hence $r(C) = k-1$, every $C \setminus e$ is a pattern of C, and C, not independent, is not. For $k \geq 3$ two marks e, f of C lie in $C \setminus g$ for a third $g \in C$, hence in a common basis.

(ii) The level of a basis draw of table j lies in the window $[d_j(S), r_j(S)]$ of S (Theorem 7.13), and the colevel of a cobasis draw is $|S| - |T' \cap S| \in [|S| - r_j(S), |S| - d_j(S)] = [d_j^*(S), |S| - d_j(S)]$. Some basis reaches $r_j(S)$ and every basis has positive chance, so $p_j > 0$; the s_j basis draws of table j are independent, and none reaches $r_j(S)$ with the chance $(1-p_j)^{s_j}$. The same holds at $d_j(S)$, and for the cobasis draws with $\mathbb{P}^*_{j,\alpha}$ and t_j. □

Proposition 47.4 (The exact laws of a sample of bases). *As stated.*

Proof. (i) With $p = T \cap B$ and the weights $\theta_{j,(p),T\setminus p} = \theta_{j,T}$ of the child (Definition 12.1),

$$\mathbb{P}_{j,\alpha}(T) = \frac{\theta_{j,T}\alpha^T}{K_j(\alpha)} = \frac{\alpha_B^p K_{j,(p)}(\alpha_R)}{K_j(\alpha)} \cdot \frac{\theta_{j,(p),T\setminus p}\,\alpha_R^{T\setminus p}}{K_{j,(p)}(\alpha_R)},$$

and, since $\alpha^{\mathcal{E}\setminus T} = \alpha_B^{B\setminus p}\alpha_R^{R\setminus(T\setminus p)}$,

$$\mathbb{P}^*_{j,\alpha}(T) = \frac{\alpha_B^{B\setminus p}\Psi_{j,(p)}(\alpha_R)}{\Psi_j(\alpha)} \cdot \frac{\theta_{j,(p),T\setminus p}\,\alpha_R^{R\setminus(T\setminus p)}}{\Psi_{j,(p)}(\alpha_R)},$$

the first factors the pattern probabilities (Theorem 12.8) and the second the basis and cobasis laws of the child at α_R. The draws are independent, so the patterns of the basis draws of table j are independent with the chances $\pi_{j,p}(\alpha)$ and their numbers $a_{j,p}$ form a multinomial vector of size s_j, likewise the $b_{j,p}$ of size t_j; and given all the patterns the room parts are independent with the laws of the children. A basis draw with pattern p uses $e \in B$ exactly when $e \in p$, and a cobasis draw with pattern p uses e, by missing it, exactly when $e \notin p$; so $m_e = \sum_j \left(\sum_{p\ni e} a_{j,p} + \sum_{p\not\ni e} b_{j,p}\right) = \text{load}_e$ (Definition 12.24). In the room a basis draw uses the marks of $T \setminus p$, and a cobasis draw the marks of $R \setminus (T' \setminus p)$, which it misses: what the draws of the children use. The mean of a multinomial vector is its size times its chances, which at a native gauge is the native split of Definition 12.25.

(ii) The chance of a sample is

$$\prod_{j,i} \frac{\theta_{j,T_{j,i}}\alpha^{T_{j,i}}}{K_j(\alpha)} \prod_{j,i} \frac{\theta_{j,T'_{j,i}}\alpha^{\mathcal{E}\setminus T'_{j,i}}}{\Psi_j(\alpha)} = \frac{\theta_{\text{draws}}\,\alpha^{\mathbf{m}}}{\prod_j K_j(\alpha)^{s_j}\Psi_j(\alpha)^{t_j}}.$$

Expanding $\prod_j K_j^{s_j}\Psi_j^{t_j}$ draw by draw, every sample contributes the monomial $\theta_{\text{draws}}\alpha^{\mathbf{m}}$ of its counts, so the samples with the counts $\mathbf{m}$ have the total chance $\Theta_F(\mathbf{m})\alpha^{\mathbf{m}}/\prod_j K_j^{s_j}\Psi_j^{t_j}$, $\Theta_F(\mathbf{m})$ the coefficient of $\alpha^{\mathbf{m}}$; the ratio is $\theta_{\text{draws}}/\Theta_F(\mathbf{m})$.

(iii) κ_S is the sum over the draws of $|T_{j,i} \cap S| - d_j(S)$ and $|S \setminus T'_{j,i}| - d_j^*(S)$ (Proposition 47.1(iii)), independent, with the generating functions $\mathsf{Q}^K_{j,S}(x;\alpha) = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[x^{N_S - d_j(S)}]$ and $\mathsf{Q}^\Psi_{j,S}(x;\alpha)$ (Definition 12.10); so the generating function of κ_S is the product. Each factor has degree $c_j(S)$ and every coefficient from degree 0 to $c_j(S)$ positive, every level of the window being reached by some basis (Theorem 7.13); so the values of κ_S fill the stated range and the product at $x = 0$

is positive. The mean of a level excess is $\sum_{e\in S} R_{j,e}(\alpha) - d_j(S)$ and that of a colevel excess $\sum_{e\in S} \mathbb{P}^*_{j,\alpha}(e \notin T) - d^*_j(S) = \sum_{e\in S} R^*_{j,e}(\alpha) - d^*_j(S)$; the cumulants of order $k \geq 2$ of an excess are those of its level, $\sum_i \varkappa_k$ of the coins (Proposition 12.11(iv)), and summed with the sizes they are $\partial^k_S \Phi(\log \alpha)$ (Proposition 12.12). With a positive spectrum each excess is a sum of independent coins (Proposition 12.11(iv)), and κ_S is the sum of all of them. □

Proposition 47.5 (Tests that do not see the gauge). *As stated.*

Proof. (i) is Proposition 47.4(ii); for lists $\theta_{\text{draws}} = 1$. (ii) $\mathbb{P}(A) = \sum_{\mathbf{m}} \mathbb{P}(\mathbf{m})\, \mathbb{P}(A \mid \mathbf{m}) \leq \sum_{\mathbf{m}} \mathbb{P}(\mathbf{m})\, \varepsilon = \varepsilon$, the conditional chances being free of the gauge by (i). (iii) The samples of F' with the counts $\mathbf{m}$ are those of F whose every draw is a basis of its table in F', with the same θ_{draws}; by (i) their conditional chance under F is $\Theta_{F'}(\mathbf{m})/\Theta_F(\mathbf{m})$. The rejection region, the union over the $\mathbf{m}$ with $p_{F'|F}(\mathbf{m}) \leq \varepsilon$ of these samples, carries at most $\varepsilon\Theta_F(\mathbf{m})$ of the weight at every $\mathbf{m}$, and (ii) applies. A draw that is not a basis of its table in F' has chance zero under F' at every gauge. (iv) The true family is rejected with chance at most ε. □

Corollary 47.6 (The evidence of a family). *As stated.*

Proof. The centre of strengths $\boldsymbol{\sigma}_0$ is a question with every proper wall positive on a connected family. If every draw is a basis of its table of F, Proposition 47.2(ii) with the prior question $(\sigma_0 \mathbf{1}; \boldsymbol{\sigma}_0, \boldsymbol{\sigma}_0)$ gives the chance of the sample with the tilt integrated out, which on the balance wall is the gauge integrated out up to its scale, on which the chance of the sample does not depend (Proposition 47.1(i)). Otherwise the sample has chance zero under F at every gauge. The posterior over the collection is the prior times this chance, normalized. □

Remark. At one table, (i) and (ii) of Proposition 47.4 are the reference's statement, with $s = q$ and $t = 0$, and Proposition 47.5 and Corollary 47.6 at one list are the reference's. Part (iii) of Proposition 47.4 is the family reading of Proposition 47.3(ii): the floors of the windows of every table are seen together through the one wall κ_S of the sample, whose chance of vanishing is $\prod_j \mathsf{Q}^K_{j,S}(0;\alpha)^{s_j} \mathsf{Q}^\Psi_{j,S}(0;\alpha)^{t_j}$.

48.46.2 Statistics from a centre

Proposition 47.16 (The even and odd statistics from a centre). *As stated.*

Proof. (i) The law of Z is that of the question $(0; \boldsymbol{\sigma}, \mathbf{0})$, whose exponent is $-\sum_j \sigma_j E_j$ (Definition 17.1). The centre is inside its walls, so the family is connected, and by Proposition 17.4(iii) $\sum_j \sigma_j E_j$ is strictly convex and even, least only at 0, with $\sum_j \sigma_j E_j(0) = 2\sum_j \sigma_j \log K_j(\mathbf{1})$. Its Hessian at 0 is $\sum_j \sigma_j \cdot 2\,\mathrm{Cov}_{\mathbb{P}_{j,\mathbf{1}}}(\chi_T) = 8\sum_j \sigma_j \kappa_{j,\mathbf{1}}(X_e, X_f)$, which is rung two of the still of the realization $\big((\sigma_j, \sigma_j)_j; \mathbf{1}\big)$ by Proposition 9.7. For one table, E_j is convex and even with its least value $2\log K_j(\mathbf{1})$ at 0. If v is constant on the molecules of table j, $\langle \chi_T, v\rangle = c$ for every basis (Proposition 7.9), so $\Lambda_j(z+v) = e^c \Lambda_j(z)$, $\Lambda_j(-z-v) = e^{-c}\Lambda_j(-z)$, and $E_j(z+v) = E_j(z)$; in particular $E_j(v) = E_j(0)$. Conversely if $E_j(z) = E_j(0)$, the convex function $t \mapsto E_j(tz)$ has its least value at $t = 0$ and at $t = 1$ and so is constant on $[0,1]$; its second derivative at 0, $2\,\mathrm{Var}_{\mathbb{P}_{j,\mathbf{1}}}\langle \chi_T, z\rangle$, vanishes, and z is constant on the molecules of table j (Lemma 7.8). The Laplace transform is Proposition 17.7(i) at $c = 0$, $\mathbf{a} = \mathbf{k}$, $\mathbf{b} = 0$: the question $(0; \boldsymbol{\sigma} + \mathbf{k}, \mathbf{0})$ is the centre of strengths $\boldsymbol{\sigma} + \mathbf{k}$, with the data $\big((\sigma + |\mathbf{k}|)\mathbf{1}; \boldsymbol{\sigma} + \mathbf{k}, \boldsymbol{\sigma} + \mathbf{k}\big)$, and its walls are $\kappa^{\mathbf{E}}_S(\boldsymbol{\sigma} + \mathbf{k}) = \kappa^{\mathbf{E}}_S(\boldsymbol{\sigma}) + \kappa^{\mathbf{E}}_S(\mathbf{k})$. Along $\mathbf{k} = \lambda\boldsymbol{\sigma}$ they are $(1+\lambda)\kappa^{\mathbf{E}}_S(\boldsymbol{\sigma})$, all zero at $\lambda = -1$; along $\lambda\mathbf{k}$ the first to vanish is at the least ratio stated.

(ii) The density $e^{-\sum_j \sigma_j E_j(z)}$ is even, and $z \mapsto -z$ keeps every E_j and negates every O_j; so the two triples have one law, and a joint moment of odd total degree in Z and the $O_j(Z)$, finite as the density decays exponentially (Proposition 17.4) while the statistics grow linearly, equals its own negative. The Laplace transform and its domain are Proposition 17.7(i) at $\mathbf{a} = 0$. When $|b_j| \le \sigma_j$ and $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) > 1$, the question $(c; \boldsymbol{\sigma}, \mathbf{b})$ has nonnegative exponents and every proper wall positive, so it has a native gauge (Theorem 8.19(iii)), its target is its masses, and its period over P_{c} is its pseudoexpectation (Definition 16.10(iii)). The tails are Proposition 17.8 at the centre; when $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) = \infty$ both Y and $-Y$ are bounded above, and Y is bounded.

(iii) By Theorem 7.15(i), $K_j(\alpha) = e^{\langle \mathbf{1}, z\rangle}\Lambda_j(z)$ and $\Psi_j(\alpha) = e^{\langle \mathbf{1}, z\rangle}\Lambda_j(-z)$, and $\log \Lambda_j(\pm z) = \frac{1}{2}\big(E_j(z) \pm O_j(z)\big)$. So $K_j^{-s_j} = e^{-s_j\langle \mathbf{1}, z\rangle} e^{-\frac{s_j}{2}(E_j + O_j)}$ is the factor of table j at $(\sigma_j, \delta_j) = (\frac{s_j}{2}, \frac{s_j}{2})$, its factor $e^{-s_j\langle \mathbf{1}, z\rangle} = \alpha^{-\frac{s_j}{2}\mathbf{1}}$ the shift of the masses, and likewise $\Psi_j^{-t_j}$. Splitting each table into its two letters, at $(s_j, 0)$ and $(0, t_j)$, keeps the family word, and the pair coordinates add: $(\frac{s_j}{2} + \frac{t_j}{2}, \frac{s_j}{2} - \frac{t_j}{2}) = (\sigma_j, \delta_j)$. □

Proposition 47.17 (What the two statistics are). *As stated.*

Proof. For scores a_T with weights $\theta_T > 0$, $\max_T(a_T + \log \theta_T) \le \log \sum_T \theta_T e^{a_T} \le \max_T a_T + \log \sum_T \theta_T$: a soft maximum; and $-\log \sum_T \theta_T e^{-a_T}$ is a soft minimum. (i) $E_j = \log \Lambda_j(z) + \log \Lambda_j(-z)$ and $O_j = \log \Lambda_j(z) - \log \Lambda_j(-z)$ are the soft maximum less, and plus, the soft minimum, and $\Phi^\circ = \sum_j \big(s_j \log \Lambda_j(z) + t_j \log \Lambda_j(-z)\big)$ (Theorem 8.19). (ii) By Theorem 7.16 at $\alpha = \mathbf{1}$, $\Lambda_j(z)/\Lambda_j(0) = \mathbb{E}_{T\sim\mathbb{P}_{j,\mathbf{1}}}[e^{Y_j}]$, so near 0, $\log \Lambda_j(z) = \log K_j(\mathbf{1}) + \sum_{k\ge 1} \frac{1}{k!}\kappa_{j,\mathbf{1}}(Y_j^{\times k})$ and $\log \Lambda_j(-z)$ is the same with $(-1)^k$; their sum and difference are the two expansions. $\kappa_{j,\mathbf{1}}(Y_j) = \sum_e z_e \mathbb{E}[\chi_{T,e}] = \sum_e (2R_{j,e}(\mathbf{1}) - 1)z_e$, and for $k \ge 2$ multilinearity and $\chi_T = 2X - \mathbf{1}$ give $\kappa_{j,\mathbf{1}}(Y_j^{\times k}) = \sum_w z^w \kappa_{j,\mathbf{1}}(\chi_{T,w_1}, \dots, \chi_{T,w_k}) = 2^k \sum_w z^w \kappa_{j,\mathbf{1}}(w)$, a constant shift not entering a cumulant of order $k \ge 2$. The terms of order $k \ge 2$ of Φ° are then $\frac{1}{k!}\sum_w 2^k \sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w) z^w$, which at the exponents of a realization is $\frac{1}{k!}\sum_w \kappa_\Upsilon(w) z^w$ (Proposition 9.7). (iii) $|\log \Lambda_j(z) - h_j(z)| \le \max(|\log \theta_{j,\min}|, |\log K_j(\mathbf{1})|)$, so $E_j(\lambda z) - \lambda \operatorname{wd}_j(z)$ and $O_j(\lambda z) - \lambda(h_j(z) - h_j(-z))$ are bounded; divide by λ. On the sector of π the largest score is at $T^+_{j,\pi}$ and the least at $T^-_{j,\pi}$ (Lemma 8.3), and $\sum_j \big(s_j h_j(z) + t_j h_j(-z)\big) = \langle \sum_j (s_j \chi_{T^+_{j,\pi}} + t_j \chi_{\mathcal{E}\setminus T^-_{j,\pi}}), z\rangle = \langle 2v_\pi - 2\sigma\mathbf{1}, z\rangle$; for $s_j, t_j \ge 0$, in the interior of the sector $\mathbf{1}_{T^+_{j,\pi}}$ and $\mathbf{1}_{\mathcal{E}\setminus T^-_{j,\pi}}$ are the only points of P_j and P_j^* at which $\langle \cdot, z\rangle$ is largest, so v_π is the vertex of $\mathcal{N}_F$ at which it is largest. By Proposition 17.4(i), $|X + 2\kappa|$ is bounded, so $X(\lambda z)/\lambda \to -2\kappa(z)$, and on $\mathbf{1}^\perp$, where $\langle \mathbf{1}, z\rangle = 0$ and $\langle \boldsymbol{\mu}_\perp, z\rangle = \langle \mathbf{m}, z\rangle$, $2\kappa(z) = \langle 2v_\pi - 2\sigma\mathbf{1}, z\rangle - 2\langle \boldsymbol{\mu}_\perp, z\rangle = 2\langle v_\pi - \mathbf{m}, z\rangle$. At the tilt $-\mathbf{1}_{U_k} + \frac{|U_k|}{n}\mathbf{1} \in H_\pi \cap \mathbf{1}^\perp$ this reads $\kappa_{U_k} = m_{U_k} - (v_\pi)_{U_k}$, the term in $\mathbf{1}$ dropping because $(v_\pi)_{\mathcal{E}} = \sum_j (s_j r_j + t_j r_j^*) = m_{\mathcal{E}}$ on the balance wall. □

Remark. At one table and the centre of strength σ_0, Proposition 47.16 is the reference's, with $\ell^*_{\sigma_0}(0, b) = \sigma_0\, \ell^*_1(0, 1)/|b|$; Proposition 47.17(i)–(ii) for one table is the reference's statement, and the family adds the still and, in (iii), the greedy vertex of the Newton sum, whose tail sums against the masses are the walls along the order: the reading of the walls of Theorem 8.19(iii) at the vertex that the order selects.

48.46.3 The Grey law, the dial and the sorted draw

Proposition 47.12(iii) (The tails of the Grey law). *As stated, for every real direction of the exponents.*

Proof. Let F_0 be the family of the list $\binom{\mathcal{E}}{1}$, whose bases are the singletons and whose letter is $K_0 = |\alpha|$, and let F' be the family of the tables of F at the exponents $(0,0)$ and the list $\binom{\mathcal{E}}{1}$ at $(m_{\mathcal{E}}, 0)$. These data are on the balance wall of F', the list $\binom{\mathcal{E}}{1}$ having rank one; that table has least level 0 on every proper set, so the walls are $\kappa_S = m_S > 0$; and the law of the question is $\mathrm{Dir}_{\mathbf{m}}$ (Proposition 16.3). Let v be the balanced direction of the data that keeps the masses, moves the exponents of table j of F by (a_j, b_j) and those of the list $\binom{\mathcal{E}}{1}$ by $(-d, 0)$, $d = \sum_j (a_j r_j + b_j r_j^*)$. Along $q + \lambda v$ the walls are $m_S - \lambda g(S)$, so the λ^* of Proposition 17.8 is k^*. The statistic Y of that proposition is $X_v(Z)$, the logarithm of the ratio of the integrands of the periods at $q + v$ and at q (Definition 17.1, and the proof of Proposition 17.4(i)), which at the tilt Z of U is $\log\big(\prod_j K_j(U)^{-a_j} \Psi_j(U)^{-b_j} K_0(U)^d\big) = \sum_j (a_j G_j + b_j G_j^*)$, as $K_0(U) = |U| = 1$ on the simplex. Proposition 17.8 gives the tail when $k^* < \infty$, and the bound above when $k^* = \infty$. □

Remark. The reference reads (iii), for $a_j, b_j \geq 0$, off the abscissa of convergence of the Laplace transform of (i), which gives the limit superior; the argument above gives the limit, at every real direction. In the direction $(\mathbf{s}, \mathbf{t})$ of a word it is the tail of Proposition 47.15(iv), which the reference proves from the collapse of a set attaining k^*.

Proposition 47.15 (The exponent is a dial between the mean and the maximum). *As stated, for real exponents of any signs, not all zero.*

Proof. (i) is a property of power means and does not involve the exponents. (ii) By Proposition 47.10(ii), $\mathbb{E}[e^{kA_k}] = \frac{1}{q}\sum_i \mathbb{E}[W(U_i)^k] = P_F(\mathbf{m}; k\mathbf{s}, k\mathbf{t}) / \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$ wherever every proper wall at $(\mathbf{m}; k\mathbf{s}, k\mathbf{t})$, which is $m_S - k g(S)$, is positive, that is for $0 < k < k^*$; and $e^{kA_k} = \frac{1}{q}\sum_i W(U_i)^k$ has a finite variance exactly when $\mathbb{E}[W(U)^{2k}] < \infty$, that is $2k < k^*$ (Proposition 47.10(i) at the exponents $(2k\mathbf{s}, 2k\mathbf{t})$). (iii) For $k \geq k^*$ some proper wall at $(\mathbf{m}; k\mathbf{s}, k\mathbf{t})$ is not positive, and $\mathbb{E}[W(U)^k] = \infty$ by Proposition 47.10(i); the law of the maximum of independent draws is the power of the law of one. (iv) $Y = \sum_j (s_j G_j + t_j G_j^*)$ with $(G_j, G_j^*)_j \sim \mathrm{Grey}(F; \mathbf{m})$, and Proposition 47.12(iii) in the direction $(\mathbf{s}, \mathbf{t})$ gives the tail and the bound. For $s_j, t_j \geq 0$: $(|U_S|, |U_{\mathcal{E}\setminus S}|) \sim \mathrm{Dir}_{(m_S, m_{\mathcal{E}\setminus S})}$ by the aggregation of the coordinates of a Dirichlet draw; on the simplex every coordinate in S is at most ρ and every coordinate at most 1, so $K_j(U) \leq K_j(\mathbf{1}) \rho^{d_j(S)}$ and $\Psi_j(U) \leq \Psi_j(\mathbf{1}) \rho^{d_j^*(S)} = K_j(\mathbf{1}) \rho^{d_j^*(S)}$, and raising these to the powers $-s_j \leq 0$, $-t_j \leq 0$ gives the bound on $W(U)$. □

Proposition 47.19 (A sorted draw lives on the sector of its order). *As stated, for real exponents of any signs.*

Proof. The family is connected: on the balance wall $\kappa_S + \kappa_{\mathcal{E}\setminus S} = 2\sum_j \sigma_j c_j(S)$ (Lemma S5), so positive walls put on every nonempty proper S a table with $\sigma_j > 0$, hence active, and $c_j(S) > 0$. The tilt Z of U has the density $2^{n-1} e^{X(z)} / P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ with respect to dz (Definition 17.1), and on the sector H_π, where dz is the Lebesgue measure in the differences v_k, Proposition 17.4(i) gives $e^{X(z)} = e^{-2\sum_k \kappa_k v_k} \prod_j k_{j,\pi}(q)^{-s_j} \psi_{j,\pi}(q)^{-t_j}$ with $q_k = e^{-2v_k}$, at any signs of the exponents. In the gaps $u_k = 2v_k = \log(U_{\pi(k-1)}/U_{\pi(k)})$, $du = 2^{n-1} dv$, so on H_π the gaps have the density $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})^{-1} e^{-\sum_k \kappa_k u_k} \prod_j k_{j,\pi}(e^{-u})^{-s_j} \psi_{j,\pi}(e^{-u})^{-t_j}$ on $[0,\infty)^{n-1}$, whose total is $I_\pi / P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ (Corollary 8.6); the boundaries of the sectors are null, and summing over π gives $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \sum_\pi I_\pi$. This is (i) and (ii). (iii) The bounds on the sector letters are those of Proposition 17.4(i); raised to the powers $-s_j$ and $-t_j$, of any signs, they bound $\prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}$ on the cube between two positive constants, and for $s_j, t_j \geq 0$ between $\prod_j K_j(\mathbf{1})^{-s_j - t_j}$ and $\prod_j \theta_{j,T_{j,\pi}^+}^{-s_j} \theta_{j,T_{j,\pi}^-}^{-t_j}$; and $\int_{[0,\infty)^{n-1}} e^{-\sum_k \kappa_k u_k} du = \prod_k \kappa_k^{-1}$. (iv) is Lemma 8.3, which does not involve the exponents. □

Remark. Proposition 47.14 is the ratio of two limits of Theorem 17.5(i): by Proposition 47.10(i), $\mathbb{E}_{V\sim\mathrm{Dir}_{\lambda\boldsymbol{\tau}}}\big[\prod_j K_j(V)^{-\lambda s_j}\Psi_j(V)^{-\lambda t_j}\big] = P_F(\lambda\boldsymbol{\tau};\lambda\mathbf{s},\lambda\mathbf{t})/\Gamma_1^n\begin{bmatrix}\lambda\boldsymbol{\tau}\\ \lambda\tau_{\mathcal{E}}\end{bmatrix}$, the numerator is the period of λ times the question of data $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$, with the walls $\tau_S - g(S)$, and the denominator is the period of the list $\binom{\mathcal{E}}{1}$ at a full basis lean at the masses $\lambda\boldsymbol{\tau}$, with the walls $\lambda\tau_S$ (Proposition 16.3); multiplied by λ^{n-1} they tend to $\sum_\pi\prod_k(\tau_{U_k}-g(U_k))^{-1}$ and $\sum_\pi\prod_k\tau_{U_k}^{-1}$.

48.46.4 The interaction of a bunch

The one-table proofs integrate the joint cumulant of the indicators of a bunch, one logarithm of a Taylor cube; a family integrates a sum of such logarithms with its exponents, one for each letter, so one lemma on one logarithm serves both.

Lemma S30 (An alternating sum over the subsets of a bunch)**.** *Let B be a finite set with $k = |B| \geq 2$, let $(c_p)_{p\subseteq B}$ be positive, and $f(y) = \log\sum_{p\subseteq B} c_p\, e^{\langle \mathbf{1}_p, y\rangle}$ on $\mathbb{R}^B$, and write $\partial^B = \prod_{e\in B}\partial_{y_e}$. Then $|\partial^B f(y)| \leq C\, e^{-\max_{e\in B}|y_e|}$, so $\partial^B f$ is integrable on $\mathbb{R}^B$, and*

$$\int_{\mathbb{R}^B} \partial^B f(y)\, dy = \sum_{p\subseteq B}(-1)^{|B\setminus p|}\log c_p.$$

Proof. The bound. $f(y+w)-f(y)$ is the logarithm of the moment generating function at w of the indicators $(X_e)_{e\in B}$ of a random set $P\subseteq B$ with the law $\pi_y(p)\propto c_p e^{\langle\mathbf{1}_p,y\rangle}$, so $\partial^B f(y)$ is their joint cumulant. Fix $e\in B$. A joint cumulant of $k\geq 2$ variables does not change when one of them is shifted by a constant, and it is a sum, with coefficients depending on k alone, of products of joint moments, one for each block of a partition of the k places, exactly one block holding the place of e. With $X_e - \mathbb{E}X_e$ in that place and $|X_f|\leq 1$, $|\partial^B f(y)| \leq C_k\,\mathbb{E}|X_e - \mathbb{E}X_e| = 2C_k p_e(1-p_e)$, $p_e = \pi_y(e\in P)$. Pairing each $p\not\ni e$ with $p+e$, the odds $p_e/(1-p_e) = \sum_{p\not\ni e} c_{p+e}e^{y_e}w_p \big/ \sum_{p\not\ni e} c_p w_p$, $w_p = e^{\langle\mathbf{1}_p,y\rangle} > 0$, lie between $\rho_- e^{y_e}$ and $\rho_+ e^{y_e}$, $\rho_\pm$ the largest and the least of the ratios c_{p+e}/c_p; so $p_e(1-p_e)\leq \min(\mathrm{odds},\mathrm{odds}^{-1}) \leq \max(\rho_+,\rho_-^{-1})\,e^{-|y_e|}$. The best e gives the bound.

The value, by induction on k. Fix $e\in B$, put $B' = B-e$, $y' = y_{B'}$, and let $f^{/e}$ and $f^{\setminus e}$ be the functions of the same form on B' with the coefficients c_{p+e} and c_p, $p\subseteq B'$: the contraction and the deletion of e in the cube. With $\ell(x) = \log(1+e^x)$,

$$f(y) = f^{/e}(y') + y_e + \ell\big(f^{\setminus e}(y') - f^{/e}(y') - y_e\big) = f^{\setminus e}(y') + \ell\big(y_e + f^{/e}(y') - f^{\setminus e}(y')\big).$$

Every derivative of ℓ is $O(e^x)$ as $x\to-\infty$, and every derivative of order at least one of $f^{/e}$ and $f^{\setminus e}$ is bounded; since $B'\neq\varnothing$, $\partial^{B'}y_e = 0$, so $\partial^{B'}f(y)\to\partial^{B'}f^{/e}(y')$ as $y_e\to+\infty$ and $\partial^{B'}f(y)\to\partial^{B'}f^{\setminus e}(y')$ as $y_e\to-\infty$. Hence $\int_{\mathbb{R}}\partial^B f\,dy_e = \partial^{B'}f^{/e}(y') - \partial^{B'}f^{\setminus e}(y')$ for every y', and by the bound and Fubini

$$\int_{\mathbb{R}^B}\partial^B f\,dy = \int_{\mathbb{R}^{B'}}\big(\partial^{B'}f^{/e} - \partial^{B'}f^{\setminus e}\big)\,dy'.$$

If $k\geq 3$, each term is integrable by the bound, and the induction gives $\sum_{p\subseteq B'}(-1)^{|B'\setminus p|}(\log c_{p+e} - \log c_p) = \sum_{p\subseteq B}(-1)^{|B\setminus p|}\log c_p$. If $k=2$, $B'=\{e'\}$, the integrand is the derivative of $f^{/e}-f^{\setminus e} = \log\big((c_e+c_{ee'}e^{y'})/(c_\varnothing + c_{e'}e^{y'})\big)$, whose limits at $\pm\infty$ are $\log(c_{ee'}/c_{e'})$ and $\log(c_e/c_\varnothing)$; their difference is the alternating sum. □

Theorem 47.25 (The interaction totals the still over the bunch's scales). *As stated.*

Proof. Fix α_R. By the Taylor cube (Theorem 12.7)

$$\log K_j(e^{y_B},\alpha_R)=\log\sum_{p\subseteq B}K_{j,(p)}(\alpha_R)\,e^{\langle\mathbf{1}_p,y_B\rangle},$$

with every $K_{j,(p)}(\alpha_R)>0$ as $\mathrm{pat}_j(B)=2^B$; Lemma S30 gives $\int\partial^B\log K_j\,dy_B=\iota_{j,B}(\alpha_R)$, absolutely. Likewise $\log\Psi_j(e^{y_B},\alpha_R)=\log\sum_{P\subseteq B}\Psi_{j,(B\setminus P)}(\alpha_R)\,e^{\langle\mathbf{1}_P,y_B\rangle}$, and the lemma gives $\sum_P(-1)^{|B\setminus P|}\log\Psi_{j,(B\setminus P)}=\sum_p(-1)^{|p|}\log\Psi_{j,(p)}=\iota^*_{j,B}(\alpha_R)$. The parts of the potential are these logarithms with the coefficients $\sigma_j(1,1)$ and $\delta_j(1,-1)$, and a table with both exponents zero does not enter; integrating term by term gives $\iota^{\mathbf{E}}_B$ and $\iota^{\mathbf{O}}_B$. At a realization $\kappa_\Upsilon(e_1,\dots,e_k)=2^k\partial_{y_{e_1}}\cdots\partial_{y_{e_k}}\Phi$ (Definition 9.4), and Proposition 9.6 writes it through the cumulants at α and at α^{-1}. For one table at $(1,0)$, $\partial^B\log K(e^y)=\kappa_\alpha(e_1,\dots,e_k)$, the joint cumulant of the indicators under $\mathbb{P}_\alpha$ (Definition 7.17). □

Proposition 47.24 (The interaction belongs to the room). *As stated.*

Proof. (i) The child at p has rank $r_j-|p|$ on R (Lemma 12.2), so $K_{j,(p)}$ is homogeneous of degree $r_j-|p|$ and $\Psi_{j,(p)}$ of degree $|R|-r_j+|p|$. Under $\alpha_R\mapsto c\alpha_R$, $\iota_{j,B}$ moves by $\log c\sum_p(-1)^{|B\setminus p|}(r_j-|p|)$ and $\iota^*_{j,B}$ by $\log c\sum_p(-1)^{|p|}(|R|-r_j+|p|)$; both vanish, as $\sum_{p\subseteq B}(-1)^{|p|}=0$ for $|B|\geq 1$ and $\sum_{p\subseteq B}(-1)^{|p|}|p|=0$ for $|B|\geq 2$.

(ii) A basis draw of table j meets B in P with the chance $\pi_{j,P}(\alpha)$, and a cobasis draw misses exactly P on B when it meets B in $B\setminus P$, with the chance $\pi^*_{j,B\setminus P}(\alpha)=\alpha_B^P\Psi_{j,(B\setminus P)}(\alpha_R)/\Psi_j(\alpha)$ (Theorem 12.8); independence gives $w(P)$. So

$$\log w(P)=\sum_j(s_j+t_j)\langle y_B,\mathbf{1}_P\rangle+\sum_j\big(s_j\log K_{j,(P)}(\alpha_R)+t_j\log\Psi_{j,(B\setminus P)}(\alpha_R)\big)-\Phi(\log\alpha).$$

The coefficients of the expansion are $\lambda_A=\sum_{P\subseteq A}(-1)^{|A\setminus P|}\log w(P)$, and for $|A|\geq 2$ the terms constant in P and linear in $\mathbf{1}_P$ drop out:

$$\lambda_A=\sum_j\Big(s_j\sum_{P\subseteq A}(-1)^{|A\setminus P|}\log K_{j,(P)}+t_j\sum_{P\subseteq A}(-1)^{|A\setminus P|}\log\Psi_{j,(B\setminus P)}\Big),$$

free of α_B. At $A=B$ the second inner sum is $\iota^*_{j,B}$, and $\lambda_B=\iota^{\mathbf{E}}_B+\iota^{\mathbf{O}}_B$. For $A\subsetneq B$: the bases of table j that avoid $B\setminus A$ form its deletion of $B\setminus A$, a table as some basis avoids B, in which every subset of A is a pattern, with the children $K_{j,(P)}$, $P\subseteq A$; the bases that contain $B\setminus A$, less $B\setminus A$, form its contraction, a table as B is independent, in which every subset of A is a pattern and the pattern $P'\subseteq A$ has the child of the pattern $P'\cup(B\setminus A)$ of B. With $P'=A\setminus P$, $B\setminus P=P'\cup(B\setminus A)$, so the second inner sum is $\sum_{P'\subseteq A}(-1)^{|P'|}\log\Psi_{j,(P'\cup(B\setminus A))}$, the dual interaction of A in the contraction. The deletion is the restriction to $\mathcal{E}\setminus(B\setminus A)$, some basis avoiding $B\setminus A$, and these are the tables of the face of the wall at $B\setminus A$ on its room (Theorem 16.25), at the exponents $(s_j,0)$ and $(0,t_j)$: every basis of the restriction and of the contraction comes from one basis of the table, whose weight it keeps, so the weights factor on the bottom levels as that theorem asks. So λ_A is $\iota^{\mathbf{E}}_A+\iota^{\mathbf{O}}_A$ computed in that face. At one table and $(1,0)$, $w(P)=\pi_P(\alpha)$. □

Proposition 47.26 (Both sides, and parity). *As stated.*

Proof. The child at p is a table on R, so $\Psi_{(p)}(\alpha_R)=\alpha_R^RK_{(p)}(\alpha_R^{-1})$ (Proposition 7.2); in z_R, $\log\Psi_{(p)}=2\langle\mathbf{1},z_R\rangle+\log K_{(p)}(e^{-2z_R})$, and as $\sum_p(-1)^{|p|}=0$, $\iota^*_B(z_R)=\sum_p(-1)^{|p|}\log K_{(p)}(e^{-2z_R})=(-1)^{|B|}\iota_B(-z_R)$. Substituting in Definition 12.18 gives $\iota^{\mathbf{E}}_B$ and $\iota^{\mathbf{O}}_B$; the parities in z_R and the values at $z_R=0$ are read off. The total for $-\Phi$ is $-(\iota^{\mathbf{E}}_B+\iota^{\mathbf{O}}_B)=-\sum_j(s_j\iota_{j,B}+t_j\iota^*_{j,B})$. □

Remark. At one table and $(1,0)$ these are the reference's statements: the interaction, its homogeneity and log-linear reading, its total as the integral of the joint cumulant, and the reading of the dual table. In a family every subset of the bunch must be a pattern in every active table, and the family reads the alternating sum over the patterns of its whole potential: the term of the pattern P is the family of the children $K_{j,(P)}$ at s_j and $\Psi_{j,(B\setminus P)}$ at t_j, which is where the potential goes as the scales of P grow and those of $B \setminus P$ fall. Proposition 47.27 is one table and is unchanged.

48.47 Receipts

The identities below were checked on small families before or after they were written; the checks are receipts, not part of any proof.

48.47.1 Chapters 35 to 46

In exact rational arithmetic at $(a,b) = (\frac{7}{3}, \frac{5}{4})$, with two linear letters of rational coefficients and the exponents $\frac{9}{7}$ and $-\frac{2}{3}$, and at $r = 3$: the expansion of $L\,x(1-x)\,\partial_x g + A_n g$ at $g = (-1)^r \Pi_r^{a,b}$ in the Beta ports had no component below the order $r-h-1$, and its coefficient of the order $r+h+1$ matched Theorem 35.15. At $(n,k,r) \in \{(2,3,1),(2,3,2),(1,4,3),(2,5,1)\}$ the column sums $C_{n,k}(P_r)$, read off the Laurent expansions of the share functions, matched the sums of the proof of Theorem 37.19, the signs $(-1)^{jk}$ included, and $C_{n,k-1}(P_r) + 2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r])\,C_{n,k}(P_r)$ was rational. For two letters at $\kappa = 0.7$, $\mathbf{w} = (0.4, 0.5)$, $\boldsymbol{\xi} = (1, 1+i)$, $G = e^{-\rho}$, $C_1 = 1+\rho$ and $C_2 = 2+\rho^2$, the quantity $\big(\int_0^\infty \cdots d\rho - A(0)\big)\lambda^{-\kappa^\flat}$ of Proposition 38.23 was $4.10646 - 0.33755i$, $4.10314 - 0.33828i$ and $4.10276 - 0.33837i$ at $\lambda = 10^{-3}, 10^{-4}, 10^{-5}$, against $B(0) = 4.10272 - 0.33838i$.

48.47.2 Chapters 17 and 47

On four marks, with the tables $\binom{\mathcal{E}}{2}$, the rank-two table with one parallel pair, the direct sum of two rank-one tables and a rank-three table with a coloop, all with random weights: the identity of Proposition 17.4(i) held to 10^{-13} at random tilts and random questions with exponents of both signs, and the two forms of the wall at a tilt agreed to 10^{-13}; the period computed on the sectors agreed with a direct cubature of the tilt integral to 10^{-8}; $\lambda^3 P_F(\lambda q)$ approached $\sum_\pi \prod_k \kappa_{U_k}^{-1} = 1.87097$, and $(\sum_\pi \prod_k \kappa_{U_k}^{-1} - \lambda^3 P_F(\lambda q))/\lambda$ approached $\sum_\pi \prod_k \kappa_{U_k}^{-1} \sum_j (s_j \log \theta_{j,T_{j,\pi}^+} + t_j \log \theta_{j,T_{j,\pi}^-}) = 2.75547$ (2.7548 at $\lambda = 3 \cdot 10^{-4}$). On three marks with three tables: the relative error of Theorem 17.5(ii) times $|\lambda|$ was 0.172–0.184 along $\lambda \in \{2, \dots, 40\}$ and along $\lambda = 2 + i\eta$, $\eta \in \{5, 10\}$, the same constant in both directions; the usage at the computed native gauge equalled the masses; and the local slopes of $\log \mathbb{P}(\pm Y > x)$ for an odd statistic at the centre were between -1.4985 and -1.5014 against $-\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) = -1.5$. In exact rational arithmetic, on four marks with sample sizes $\mathbf{s} = (1,0,1,0)$, $\mathbf{t} = (0,1,1,1)$: the conditional law of a sample given its counts was $\theta_{\text{draws}}/\Theta_F(\mathbf{m})$, and the law of the wall κ_S was $\prod_j \mathsf{Q}_{j,S}^{K\,s_j} \mathsf{Q}_{j,S}^{\Psi\,t_j}$ for every S. The total of Theorem 47.25 matched $\iota_B^{\mathbf{E}} + \iota_B^{\mathbf{O}}$ to 10^{-12} for $|B| = 2$ in three tables, and to 10^{-10} for $|B| = 3$, a Hamiltonian path of K_4 in the graphic table of K_4 together with $\binom{\mathcal{E}}{3}$ on six marks; the relation $\iota_B^*(z_R) = (-1)^{|B|} \iota_B(-z_R)$ held to 10^{-12} in each table. All of these were rerun on 20 October 2026 and reproduced (the cubature for $|B| = 2$ on a grid of 1601 points a side, matching to $6 \cdot 10^{-13}$).

In exact rational arithmetic, on six marks with two weighted graphic tables of K_4 and a weighted $\binom{\mathcal{E}}{3}$, at the exponents $(2,1)$, $(1,3)$, $(0,2)$, a rational gauge and the bunch B of a Hamiltonian path:

for every $A \subseteq B$ with $|A| \geq 2$ the coefficient λ_A of Proposition 47.24(ii) satisfied $e^{\lambda_A} = e^{\iota_A^{\mathbf{E}} + \iota_A^{\mathbf{O}}}$ exactly, the right side computed from the deletions and contractions of $B \setminus A$ themselves.

Part XI

Exercises

Each exercise is a computation on a particular list, table or ray, set as a guided example. Its solution follows the procedures and theorems of the chapters step by step: every step names the statement it uses, and every quantity that the next step needs is displayed, in exact arithmetic. A step that is a machine computation too large to display is given by its exact input, its method, the output that the next step uses, the check that was run on it, and the receipt that reproduces it. Decimal values appear only as renderings of exact ones.

Lists are named by an upright capital with the number of marks as a subscript, and each is defined by its marks and bases where it first appears.

Chapter 49

Exercises on tables, families and classification

The exercises of this chapter work the statements of Parts II and III, and the first statements of the reading of rays, on particular lists. The list on two marks is opened on its share side and read along its centre ray, where its values, its closings and the matrix of its branch fit together exactly. The pins of the atom formula of six uniform lists have the elementary divisors 1, and two of them still pin some shares at fractions. Four Foreman trees are drawn: one that is an atom at every lean, one that peels at a bunch of width one into a cell whose contraction is the dual of its deletion, and one whose Foreman bunch at a lean has width two and whose sets do not split off at the centre. Along a set of width two the Skeleton values are shown not to be rational functions of the point, and a family with exponents of both signs is peeled with and without its record. Every number displayed is exact; the receipts that recompute them are the scripts named in each solution.

49.1 The list on two marks

Exercise 49.1 (The list $\binom{[2]}{1}$ on its share side)**.** Let $\mathcal{E} = \{1, 2\}$ and $F = \big(\binom{\mathcal{E}}{1}; s, t\big)$, the list whose bases are the two singletons, with $\operatorname{Re} s, \operatorname{Re} t > 0$ and masses $\mathbf{m} = (m_1, m_2)$ on the balance wall with $\operatorname{Re} m_1, \operatorname{Re} m_2 > 0$.

(i) Write the atom formula of $P_F(\mathbf{m}; s, t)$ (Corollary 18.16): its shares, its pins, its fibre and its measure, and close it.

(ii) At $(\mathbf{1}; 1, 1)$ find the integrand on the line $\operatorname{Re} a = \frac{1}{2}$ and the period.

Solution. Receipt: `ex01_share_side.py`.

(i) *The letters.* The bases are $T_1 = \{1\}$ and $T_2 = \{2\}$, so (Definition 7.1)

$$K(\alpha) = \alpha^{T_1} + \alpha^{T_2} = \alpha_1 + \alpha_2, \qquad \Psi(\alpha) = \alpha^{\mathcal{E}\setminus T_1} + \alpha^{\mathcal{E}\setminus T_2} = \alpha_2 + \alpha_1,$$

the rank and the corank are 1, and the balance wall is $\kappa_{\mathcal{E}} = m_1 + m_2 - (s + t) = 0$ (Definition 8.1), with the strength $\sigma = \frac{s+t}{2}$ and the offset $\mu_1 = m_1 - \sigma$.

The shares and the pins. Corollary 18.16, with $g = 2$, opens each letter with one share per basis: a_1, a_2 for K, a_i the share of α^{T_i}, and b_1, b_2 for Ψ, b_i the share of $\alpha^{\mathcal{E}\setminus T_i}$. Its pins are the two letter totals and the odd load at the mark 1 (Definition 12.24), with $\chi_{T_1}(1) = 1$ and $\chi_{T_2}(1) = -1$:

$$a_1 + a_2 = s, \qquad b_1 + b_2 = t, \qquad \tfrac{1}{2}(a_1 - b_1) - \tfrac{1}{2}(a_2 - b_2) = m_1 - \tfrac{s+t}{2}.$$

With $a = a_1$ they give

$$a_2 = s - a, \qquad b_1 = a + t - m_1, \qquad b_2 = m_1 - a,$$

and the load at the mark 1, $a_1 + b_2$, is m_1, as $\mathrm{load}_1 = \sigma + \langle \boldsymbol{\delta} \rangle_1$ requires (Definition 12.24). The weights are 1, so the factor $\prod_T \theta_T^{-a_T - b_T}$ is 1, and the fibre has the dimension $4 - 3 = 1$.

The measure. The matrix of the pins, the shares in the order (a_1, a_2, b_1, b_2), and its Gram matrix are

$$P = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}, \qquad PP^{\mathsf{T}} = \mathrm{diag}(2, 2, 1), \qquad \sqrt{\det PP^{\mathsf{T}}} = 2,$$

and the fibre runs along $(1, -1, 1, -1)$, of length 2: the imaginary part y of a moves the point by the volume $2\,dy$. So the fibre integral of Definition 4.1 is $(2\pi)^{3-4} \cdot \frac{1}{2} \cdot 2 \int dy = \frac{1}{2\pi i} \int da$:

$$P_F(\mathbf{m}; s, t) = \frac{1}{2\pi i} \int_{(c)} \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} a, & s - a \\ & s \end{bmatrix} \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} a + t - m_1, & m_1 - a \\ & t \end{bmatrix} da,$$

on a line $\mathrm{Re}\, a = c$ on which all four shares have positive real parts, $\max(0, \mathrm{Re}(m_1 - t)) < c < \min(\mathrm{Re}\, s, \mathrm{Re}\, m_1)$; such lines exist, since $\mathrm{Re}\, s, \mathrm{Re}\, t, \mathrm{Re}\, m_1 > 0$ and $\mathrm{Re}(m_1 - t) < \mathrm{Re}\, s$ is $\mathrm{Re}\, m_2 > 0$.

The closing. The integrand is $\Gamma(a)\,\Gamma(t - m_1 + a)\,\Gamma(s - a)\,\Gamma(m_1 - a)$ over $\Gamma(s)\Gamma(t)$: four Gammas on a line that separates the poles of the first two from those of the last two. Corollary 4.7, with its four entries 0, $t - m_1$, s, m_1 and the variable a, adds them in pairs, $0 + s$, $(t - m_1) + m_1$, $0 + m_1$ and $(t - m_1) + s = m_2$:

$$\frac{1}{2\pi i} \int_{(c)} \Gamma(a)\,\Gamma(t - m_1 + a)\,\Gamma(s - a)\,\Gamma(m_1 - a)\,da = \Gamma_{\substack{4 \\ 1}} \begin{bmatrix} s, & t, & m_1, & m_2 \\ & & s + t & \end{bmatrix}.$$

Dividing by $\Gamma(s)\Gamma(t)$ (Proposition 1.3(i),(ii)),

$$P_F(\mathbf{m}; s, t) = \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} m_1, & m_2 \\ m_1 + m_2 & \end{bmatrix}.$$

This agrees with the period read on the simplex (Theorem 8.7): there $K = \Psi = u_1 + u_2 = 1$, and the period is the shape integral $\int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} d\omega_\mathcal{E} = \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} m_1, & m_2 \\ m_1 + m_2 & \end{bmatrix}$ (Theorem 3.9).

(ii) At $(\mathbf{1}; 1, 1)$, $m_1 = s = t = 1$, and the shares are $a_1 = b_1 = a$ and $a_2 = b_2 = 1 - a$: the basis share and the cobasis share of each basis are equal. By Proposition 1.3(i),(v) the integrand is

$$\Gamma_{\substack{2 \\ 1}} \begin{bmatrix} a, & 1 - a \\ & 1 \end{bmatrix} \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} a, & 1 - a \\ & 1 \end{bmatrix} = \Gamma_{\substack{4 \\ 0}} \big[a,\ a,\ 1 - a,\ 1 - a\big] = \Big(\frac{\pi}{\sin \pi a}\Big)^2,$$

regular on $0 < \mathrm{Re}\, a < 1$. On $a = \frac{1}{2} + iy$, where $\sin \pi a = \cosh \pi y$,

$$\frac{1}{2\pi i} \Gamma_{\substack{4 \\ 0}} \big[a,\ a,\ 1 - a,\ 1 - a\big]\ da = \frac{\pi}{2 \cosh^2 \pi y}\, dy = \frac{2\pi}{(e^{\pi y} + e^{-\pi y})^2}\, dy = p_{\mathrm{crit}}(y)\, dy,$$

the density of the critical-line law (Definition 36.1). Its primitive is $\frac{1}{2} \tanh \pi y$, which runs from $-\frac{1}{2}$ to $\frac{1}{2}$, so its integral is 1; and (i) gives the same, $P_F(\mathbf{1}; 1, 1) = \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} 1, & 1 \\ & 2 \end{bmatrix} = 1$ (Theorem 36.2).

Answer. The atom formula is the integral over one share

$$P_F(\mathbf{m};s,t)=\frac{1}{2\pi i}\int_{(c)}\Gamma_{\,1}^{2}\!\begin{bmatrix}a,\ s-a\\ s\end{bmatrix}\Gamma_{\,1}^{2}\!\begin{bmatrix}a+t-m_1,\ m_1-a\\ t\end{bmatrix}\,da=\Gamma_{\,1}^{2}\!\begin{bmatrix}m_1,\ m_2\\ m_1+m_2\end{bmatrix}.$$

At $(\mathbf{1};1,1)$ its integrand on the line $\operatorname{Re}a=\frac12$ is $\Gamma_{\,0}^{4}\big[a,\ a,\ 1-a,\ 1-a\big]=\pi^2/\cosh^2\pi y$, the density of the critical-line law times 2π, and the period is 1. □

Exercise 49.2 (The centre ray of $\binom{[2]}{1}$)**.** For the centre ray of the list $\binom{\mathcal{E}}{1}$, $\mathcal{E}=\{1,2\}$, at the strength 1 (Definition 38.13):

(i) find its stride, its native gauge and native value, its values $\langle1\rangle_k$ and a recurrence of order one that they satisfy, and the integrand of the atom formula at step k against the one at step 1;

(ii) find its closings $\mathsf{C}_j(1)$ (Definition 39.26) and the volume V_1 of its direction, and check Proposition 39.30 and the reflected recurrence at every j;

(iii) find the matrix H of Theorem 38.15 and show that $k\,\langle1\rangle_k\,\mathsf{C}_k(1)=2=(\det H)^{-1}$ for every $k\ge1$; compare with the leading terms of Theorems 38.6 and 39.41.

Solution. Receipt: `ex01_share_side.py`.

(i) *Stride and native gauge.* Definition 38.13 with one table and $\sigma_1=1$ gives the origin 0 and the direction $(\mathbf{1};1,1)$; on the chart $\alpha_2=1$, with $x=\alpha_1$,

$$f=\frac{\alpha^{\mathbf{1}}}{K\Psi}=\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2}=\frac{x}{(1+x)^2}.$$

The gauge $\mathbf{1}$ is native (Proposition 8.21(i)), and $f\le f(\mathbf{1})=\frac14$ on the orthant with equality only at $x=1$ (Theorem 38.14(i)): the native value of the direction is $\frac14$.

Values. By Definition 38.13, $\langle1\rangle_k=P_F(k\mathbf{1};k,k)$, which Exercise 49.1(i) at $m_1=m_2=s=t=k$ gives:

$$\langle1\rangle_k=\Gamma_{\,1}^{2}\!\begin{bmatrix}k,\ k\\ 2k\end{bmatrix}=\frac{((k-1)!)^2}{(2k-1)!}:\qquad 1,\ \tfrac16,\ \tfrac{1}{30},\ \tfrac{1}{140},\ \tfrac{1}{630},\ \tfrac{1}{2772}\quad(k=1,\dots,6).$$

By the shift (Proposition 1.3(iii)) $\langle1\rangle_{k+1}/\langle1\rangle_k=k^2/\big(2k(2k+1)\big)=k/\big(2(2k+1)\big)$, the ratios $\frac16,\frac15,\frac{3}{14},\frac29,\frac{5}{22}$, so

$$2(2k+1)\,\langle1\rangle_{k+1}-k\,\langle1\rangle_k=0,$$

a recurrence of order one. Its symbol, read off the coefficients of k, is $4\lambda-1$, which vanishes at the native value $\frac14$, as Proposition 38.9 requires; and the ratios tend to $\frac14$.

The integrand at step k. At $m_1=s=t=k$ the shares of Exercise 49.1(i) are $a_1=b_1=a$ and $a_2=b_2=k-a$, and the integrand is $\Gamma(a)^2\Gamma(k-a)^2/\Gamma(k)^2$. The shift applied $k-1$ times to $\Gamma(k-a)$ and to $\Gamma(k)$ gives

$$\Gamma_{\,2}^{4}\!\begin{bmatrix}a,\ a,\ k-a,\ k-a\\ k,\ k\end{bmatrix}=\Gamma_{\,0}^{4}\big[a,\ a,\ 1-a,\ 1-a\big]\left(\frac{(1-a)(2-a)\cdots(k-1-a)}{(k-1)!}\right)^2:$$

at $k=2,3,4$ the factor is $(a-1)^2$, $\frac14(a-1)^2(a-2)^2$ and $\frac{1}{36}(a-1)^2(a-2)^2(a-3)^2$. The step multiplies the kernel of step 1 by the square of a polynomial in the share, which adds zeros at $a=1,\dots,k-1$ and no poles; the values are rational numbers.

(ii) *Closings.* The origin is integral and both letters have the exponent 1 in the direction, so Definition 39.26 applies, with $\omega_0/dy = 1$ and $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j})$ (Corollary 39.32). On the chart $f^{-j} = x^{-j}(1+x)^{2j}$, whose constant term is the coefficient of x^j in $(1+x)^{2j}$:

$$\mathsf{C}_j(1) = \binom{2j}{j}: \qquad 1,\ 2,\ 6,\ 20,\ 70,\ 252,\ 924,\ 3432 \quad (j = 0, \dots, 7).$$

The volume. The two orders of $\mathcal{E}$ have the tails $U_2 = \{2\}$ and $U_2 = \{1\}$, and the direction walls are $\kappa_{1,\{e\}} = \sigma_1 c(\{e\}) = 1$ (Proposition 8.21(ii); $c(\{e\}) = r(\{e\}) + r(\mathcal{E} \setminus \{e\}) - r = 1$ by Definition 7.12 and Theorem 7.13). So $V_1 = 1 + 1 = 2$ (Proposition 39.30).

The principal parts. $\langle 1 \rangle_k = \Gamma(k)^2/\Gamma(2k)$ continues meromorphically in k. At $k = -j$, $j \in \mathbb{N}$, the residues of Proposition 1.3(iv) give $\Gamma(k) = \frac{(-1)^j}{j!}(k+j)^{-1} + O(1)$ and, the pole of Γ at $2k = -2j$ having the residue $\frac{1}{(2j)!}$, $\Gamma(2k) = \frac{1}{2(2j)!}(k+j)^{-1} + O(1)$. So the pole has order $1 = n - 1$ and

$$\lim_{k \to -j} (k+j)\, \langle 1 \rangle_k = \frac{(j!)^{-2}}{(2(2j)!)^{-1}} = 2\binom{2j}{j} = V_1\, \mathsf{C}_j(1): \qquad 2,\ 4,\ 12,\ 40,\ 140,\ 504 \quad (j = 0, \dots, 5),$$

Proposition 39.30 at every j.

The reflected recurrence. Corollary 39.44 reads the recurrence of (i), $p_0(k) = -k$ and $p_1(k) = 2(2k+1)$, at $k = -j$ on the closings: $j\, \mathsf{C}_j(1) + 2(1-2j)\, \mathsf{C}_{j-1}(1) = 0$, that is $\mathsf{C}_j(1) = \frac{2(2j-1)}{j}\, \mathsf{C}_{j-1}(1)$ for $j \geq 1$, which $\binom{2j}{j}$ satisfies; at $j = 0$ the sum is $p_0(0)\, \mathsf{C}_0(1) = 0$, as in Corollary 39.32.

(iii) *The matrix H.* In Theorem 38.15, $\alpha^\star = \mathbf{1}$, $e_0 = 2$, and $s_{11} = t_{11} = 1$, so H is the 1×1 matrix $\kappa_{\mathbf{1}}(X_1, X_1) + \kappa_{\mathbf{1}}(X_1, X_1)$, the cobasis law at $(\alpha^\star)^{-1} = \mathbf{1}$ being the weight law (Theorem 7.16). Under the weight law, uniform on $\{1\}, \{2\}$ (Definition 7.17), X_1 is 1 with the probability $\frac{1}{2}$ and has the variance $\frac{1}{4}$; so $H = \frac{1}{2}$. Directly: $-\log f = 2\log(1 + e^y) - y$ on the chart $x = e^y$, with the second derivative $2e^y/(1+e^y)^2 = \frac{1}{2}$ at $y = 0$.

The product. For every $k \geq 1$,

$$k\, \langle 1 \rangle_k\, \mathsf{C}_k(1) = k\, \frac{((k-1)!)^2}{(2k-1)!} \cdot \frac{(2k)!}{(k!)^2} = k \cdot \frac{2k}{k^2} = 2 = (\det H)^{-1}.$$

The leading terms. The native gauge $\mathbf{1}$ is nondegenerate (Theorem 39.41), and near it $\log f = \log \frac{1}{4} - \frac{1}{2}v^2$ with $v = H^{1/2}y + O(y^2)$, so $(\omega_0/dv)(\mathbf{1}) = (dy/dv)(0) = H^{-1/2} = \sqrt{2}$. Theorem 38.6(i) gives $\langle 1 \rangle_k \sim 4^{-k} k^{-1/2} a_0(1)$ with $a_0(1) = (2\pi)^{1/2}\sqrt{2} = 2\sqrt{\pi}$, and Theorem 39.41, in which $\hat{A}_\star(1)$ is that expansion divided by $f(\mathbf{1})^k k^{-1/2} = 4^{-k} k^{-1/2}$, with the leading term $a_0(1)$, gives $\mathsf{C}_j(1) \sim 4^j j^{-1/2} (2\pi)^{-1} 2\sqrt{\pi} = 4^j (\pi j)^{-1/2}$. Their product at $j = k$ is $k^{-1}\, a_0(1)^2/(2\pi) = 2k^{-1}$ to the leading order, $(\det H)^{-1} k^{-(n-1)}$; here the product is $2k^{-1}$ exactly, at every k.

Answer. The stride is $\alpha_1\alpha_2/(\alpha_1 + \alpha_2)^2$, with the native gauge $\mathbf{1}$ and the native value $\frac{1}{4}$; the values are $\langle 1 \rangle_k = \Gamma\, {}_2{}_1 \begin{bmatrix} k,\ k \\ 2k \end{bmatrix}$, with $2(2k+1)\langle 1 \rangle_{k+1} = k \langle 1 \rangle_k$, and the integrand at step k is the one at step 1 times $\big((1-a)_{k-1}/(k-1)!\big)^2$. The closings are $\mathsf{C}_j(1) = \binom{2j}{j}$, $V_1 = 2$, and $\lim_{k \to -j}(k+j)\langle 1 \rangle_k = 2\binom{2j}{j}$ at every j. $H = \frac{1}{2}$, and $k\, \langle 1 \rangle_k\, \mathsf{C}_k(1) = 2 = (\det H)^{-1}$ at every $k \geq 1$. □

49.2 The pins of the atom formula

Exercise 49.3 (The pins of the atom formula of six lists)**.** For a list $(\mathcal{E}, \mathcal{B})$ on n marks at data on the balance wall, the atom formula (Corollary 18.16) is a fibre integral over the shares a_T, b_T,

$T \in \mathcal{B}$, pinned by $|\mathbf{a}| = s$, $|\mathbf{b}| = t$ and $\langle \boldsymbol{\delta} \rangle_e = \mu_e$ for $e \neq g$. For the six lists $\binom{[2]}{1}$, $\binom{[3]}{1}$, $\binom{[3]}{2}$, $\binom{[4]}{2}$, $\binom{[4]}{3}$ and $\binom{[5]}{2}$, with $g = n$:

(i) write the pins with integer coefficients;

(ii) find the elementary divisors of their matrix;

(iii) decide for which of the six lists, at integer data, every point of the fibre at which the shares outside $n+1$ shares with independent columns are integers has all its shares integers, and for the others give such a point with a share that is not an integer.

Solution. Receipt: `ex02_pins.py`.

(i) By Definition 12.24 the load at a mark is $\text{load}_e = \sum_{T \ni e} a_T + \sum_{T \not\ni e} b_T = \sigma + \langle \boldsymbol{\delta} \rangle_e$, with $\sigma = \frac{1}{2}(|\mathbf{a}| + |\mathbf{b}|)$. So, with the two letter totals, the pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$ are the pins $\text{load}_e = m_e$, $e \neq g$, which have the coefficients 0 and 1. Passing from the one set of rows to the other adds to the row of e half the sum of the two rows of the totals: a triangular change with determinant 1, which leaves the delta function of Definition 4.1, and so the fibre integral, unchanged. The matrix of the pins, M, has $n+1$ rows, the two totals and the loads at $e \neq g$, and $2|\mathcal{B}|$ columns:

$$\text{column of } a_T: \ \big(1,\ 0,\ (\mathbf{1}_T(e))_{e \neq g}\big), \qquad \text{column of } b_T: \ \big(0,\ 1,\ (\mathbf{1}_{\mathcal{E} \setminus T}(e))_{e \neq g}\big).$$

(ii) An integer matrix of rank ρ has integer matrices U, V of determinant ± 1 with UMV diagonal, with the entries $d_1 \mid d_2 \mid \cdots \mid d_\rho$ and zeros, and $d_1 \cdots d_k$ is the greatest common divisor of the $k \times k$ minors of M (the Smith normal form of an integer matrix, which applies here because the pins have integer coefficients). So the elementary divisors are all 1 as soon as one maximal minor is ± 1, which also gives the rank $n+1$. With the rows in the order (total of K, total of Ψ, loads at $1, \ldots, n-1$), these minors are ± 1:

$\binom{[2]}{1}$, (a_1, a_2, b_1)

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$\det = 1$

$\binom{[3]}{1}$, (a_1, a_2, a_3, b_1)

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$\det = 1$

$\binom{[3]}{2}$, $(a_{12}, a_{13}, a_{23}, b_{12})$

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

$\det = -1$

$\binom{[4]}{3}$, $(a_{123}, a_{124}, a_{134}, a_{234}, b_{123})$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

$\det = -1$

$\binom{[4]}{2}$, $(a_{12}, a_{13}, a_{14}, a_{23}, b_{12})$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

$\det = 1$

$\binom{[5]}{2}$, $(a_{12}, a_{13}, a_{14}, a_{15}, a_{23}, b_{12})$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$\det = -1$

So every elementary divisor is 1 for the six lists; the receipt computes the Smith normal forms, $\text{diag}(1, \ldots, 1)$ in each case. Then $UMV = (I \mid 0)$: for every integer right side $(s, t, (m_e)_{e \neq g})$ the pins have integer solutions, and the integer points of the fibre form a translate of the integer vectors of the kernel of M, a lattice of rank $2|\mathcal{B}| - n - 1$, the dimension of the fibre.

(iii) Fix a set J of $n+1$ shares whose columns have the minor $\det M_J \neq 0$, and give the other shares integer values. The pins then fix the shares of J by Cramer's rule, with the denominator $\det M_J$. The absolute values of all maximal minors, with their numbers, are

	0	1	2	3
$\binom{[2]}{1}$	0	4	0	0
$\binom{[3]}{1}$, $\binom{[3]}{2}$	3	12	0	0
$\binom{[4]}{3}$	24	32	0	0
$\binom{[4]}{2}$	288	480	24	0
$\binom{[5]}{2}$	17940	19170	1620	30

so for $\binom{[2]}{1}$, $\binom{[3]}{1}$, $\binom{[3]}{2}$ and $\binom{[4]}{3}$ every such point at integer data has integer shares. For $\binom{[4]}{2}$ the minor on $(a_{12}, a_{34}, b_{13}, b_{14}, b_{24})$ is

$$\det \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = 2,$$

and at the centre $(\mathbf{1}; 1, 1)$ with the other seven shares 0 the pins $a_{12} + a_{34} = 1$, $b_{13} + b_{14} + b_{24} = 1$, $a_{12} + b_{24} = 1$ (the load at 1), $a_{12} + b_{13} + b_{14} = 1$ (at 2) and $a_{34} + b_{14} + b_{24} = 1$ (at 3) give

$$a_{12} = a_{34} = b_{13} = b_{24} = \tfrac{1}{2}, \qquad b_{14} = 0.$$

For $\binom{[5]}{2}$ the minor on $(a_{12}, a_{13}, a_{45}, b_{14}, b_{15}, b_{23})$ is

$$\det \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = -3,$$

and at the centre $(\mathbf{1}; 1, 1)$ with the other fourteen shares 0 the pins give $a_{12} = a_{13} = a_{45} = b_{14} = b_{15} = b_{23} = \frac{1}{3}$: every total and every load is $3 \cdot \frac{1}{3} = 1$.

Answer. The pins are the two letter totals and the loads $\mathrm{load}_e = m_e$, $e \neq g$, a 0/1 matrix with $n+1$ rows and $2|\mathcal{B}|$ columns. For all six lists its elementary divisors are 1: at integer data the integer points of the fibre form a lattice of full rank in it. For $\binom{[2]}{1}$, $\binom{[3]}{1}$, $\binom{[3]}{2}$ and $\binom{[4]}{3}$ every maximal minor is 0 or ± 1, and every point with integer shares outside $n+1$ independent ones has integer shares. For $\binom{[4]}{2}$ and $\binom{[5]}{2}$ some maximal minors are ± 2, and ± 3 for $\binom{[5]}{2}$, and at the integer data $(\mathbf{1}; 1, 1)$ the points displayed have the seven, respectively fourteen, shares outside such a minor at the integer 0 and shares $\frac{1}{2}$, respectively $\frac{1}{3}$, on it. So the pins of these two atom formulas are not unimodular in the sense of the paragraph after Theorem 20.59: a closing (Theorem 4.5) that pins the shares outside such a minor at poles of their Gammas can reach shares that are not integers, even at integer data, although the elementary divisors are 1. □

49.3 Foreman trees

Exercise 49.4 (The Foreman trees of $\binom{[4]}{1}$)**.** Let $F = \big(\binom{[4]}{1}; s, t\big)$, the list on $\mathcal{E} = \{1,2,3,4\}$ whose bases are its singletons, read at its realizations $\Upsilon_{\sigma,\delta;\mathbf{1}}$ at the table's gauge, $s = \sigma + \delta$, $t = \sigma - \delta$ (Definition 9.1). Find its classes at the centre and at every lean, and its three Foreman trees (Theorem 14.6).

Solution. Receipt: `ex03_tree_u14.py`.

The letters and the molecule. $K = \alpha_1+\alpha_2+\alpha_3+\alpha_4$ and $\Psi = \alpha_1\alpha_2\alpha_3+\alpha_1\alpha_2\alpha_4+\alpha_1\alpha_3\alpha_4+\alpha_2\alpha_3\alpha_4$ (Definition 7.1). Every nonempty S has $r(S) = 1$, so every proper nonempty S has $c(S) = r(S)+r(\mathcal{E}\setminus S)-1 = 1 > 0$ (Definition 7.12, Theorem 7.13): the family is connected (Definition 8.1) and one molecule.

The masses. Each mark lies in one of the four bases, so the occupation at the table's gauge is $R_e(\mathbf{1}) = \frac{1}{4}$ (Theorem 7.16), and Proposition 9.3 gives the realization $\Upsilon_{\sigma,\delta;\mathbf{1}}$ the masses

$$\tau_e = \sigma + \delta\big(2 \cdot \tfrac{1}{4} - 1\big) = \sigma - \tfrac{\delta}{2} \qquad \text{at every mark,}$$

with the total $4\sigma - 2\delta = s \cdot 1 + t \cdot 3$, on the balance wall.

The writings. By Proposition 10.2, for a mark e and an order $\mathbf{g} = (g_1, g_2, g_3)$ of its room, the pattern of the basis $\{h\}$ is the position of h, and at the table's gauge each pattern of one position has the weight 1:

$$K_{e,\mathbf{g}}(x) = x_0 + x_1 + x_2 + x_3, \qquad \Psi_{e,\mathbf{g}}(x) = x_1x_2x_3 + x_0x_2x_3 + x_0x_1x_3 + x_0x_1x_2,$$

the same two polynomials for every mark and every order (the receipt finds one pair among the $4 \cdot 6$ writings). So one pair of functions, $\Phi^{\mathbf{E}}_{e,\mathbf{g}} = \sigma \log(K_{e,\mathbf{g}}\Psi_{e,\mathbf{g}})$ and $\Phi^{\mathbf{O}}_{e,\mathbf{g}} = \delta \log(K_{e,\mathbf{g}}/\Psi_{e,\mathbf{g}})$ with $x_i = e^{v_i}$, is written from every mark. With no partner, the reading of e (Definition 10.1 at $k = 0$) is the pair of $K = e^v + 3$ and $\Psi = 3e^v + 1$ at every mark.

The classes. By Proposition 10.4 the reading is constant on $\mathcal{E}$ at every lean δ, the centre included: $\mathrm{Cl}(\Upsilon_{\sigma,\delta;\mathbf{1}}) = \{\mathcal{E}\}$. Corollary 11.16 says the same: all 24 permutations of $\mathcal{E}$ carry the table, mapping singletons to singletons, and they take every mark to every other; none exchanges it, a basis having one mark and a cobasis three.

The trees. Every realization is atomic (Definition 10.1), so by Definition 14.4 the molecule is an atom and its Foreman tree is that one leaf, with no bunch and no peel. Theorem 14.6 gives the list at its one pair the three trees $\mathrm{Tree}(F;\sigma,0)$ and $\mathrm{Tree}(F;\sigma,\pm\delta)$, and each is this atom. The molecule answers yes and the label is $(1,1)$ (Definition 11.7).

Answer. At the centre and at every lean the one class is $\mathcal{E}$; the three Foreman trees of $\binom{[4]}{1}$ are one atom, with no bunch. □

Exercise 49.5 (A tree that peels at a bunch of width one)**.** Let P_5 be the list on $\{1,\dots,5\}$ whose bases are the nine pairs other than $\{1,2\}$, read at its realizations $\Upsilon_{\sigma,\delta;\mathbf{1}}$. Find its classes and its Foreman bunch at the centre and at a lean, the width of the bunch, its patterns with their probabilities and children, its record and its peel, and its three Foreman trees. Write P_5 as a substitution through the two cells of Theorem 19.4, and show that the contraction of the cell of the room is the dual of its deletion.

Solution. Receipt: `ex04_five_marks.py`.

The letters. With e_k the elementary symmetric polynomials of α, the rank is 2, the corank 3, and

$$K = e_2(\alpha) - \alpha_1\alpha_2, \qquad \Psi = e_3(\alpha) - \alpha_3\alpha_4\alpha_5.$$

The classes. A bijection of the marks carries P_5 exactly when it maps the one pair that is not a basis, $\{1,2\}$, to itself: these are the 12 products of a permutation of $\{1,2\}$ and one of $\{3,4,5\}$, with the orbits $\{1,2\}$ and $\{3,4,5\}$. No bijection exchanges P_5, a basis having two marks and a cobasis three. By Corollary 11.16 the classes at the centre and at every lean are

$$\mathrm{Cl} = \big\{\{1,2\},\ \{3,4,5\}\big\}.$$

The occupations confirm the split at the centre: $R_1(\mathbf{1}) = R_2(\mathbf{1}) = \frac{3}{9} = \frac{1}{3}$ and $R_3(\mathbf{1}) = R_4(\mathbf{1}) = R_5(\mathbf{1}) = \frac{4}{9}$, and two marks of one class at the centre have equal or complementary occupations (Proposition 10.9(ii)), while $\frac{4}{9} \notin \{\frac{1}{3}, \frac{2}{3}\}$.

The Foreman bunch. By Definition 13.6 the Foreman bunch at the centre is the class with the fewest marks, $B = \{1,2\}$; at a lean the key $(|C|, |C^\circ|, C^\circ, C)$ is least at $C = \{1,2\}$, of two marks, so $B(\mathrm{P}_5; \sigma, \pm\delta) = \{1,2\}$ as well. Its window is $\{0,1\}$ (a basis meets B in at most one mark, and $\{3,4\}$ avoids it), so $d(B) = 0$, $r(B) = 1$ and

$$c(B) = r(B) + r(R) - r = 1 + 2 - 2 = 1$$

(Definition 7.12, Theorem 7.13), $R = \{3,4,5\}$; no basis contains both of its marks, so B is a parallel bunch (Definition 19.1).

Patterns, probabilities, children. The patterns of B are $\varnothing$, $\{1\}$, $\{2\}$ (Definition 12.1), each the trace of three of the nine bases, so each has the probability $\pi_p(\mathbf{1}) = \frac{1}{3}$ at the table's gauge (Definition 12.25). The children on R are

$$\mathcal{B}_{(\varnothing)} = \{34, 35, 45\} = \binom{R}{2}, \qquad \mathcal{B}_{(\{1\})} = \mathcal{B}_{(\{2\})} = \{3,4,5\} = \binom{R}{1},$$

with $K_{\binom{R}{2}} = e_2(\alpha_R)$, $\Psi_{\binom{R}{2}} = e_1(\alpha_R)$, $K_{\binom{R}{1}} = e_1(\alpha_R)$ and $\Psi_{\binom{R}{1}} = e_2(\alpha_R)$.

Record and peel. The record (Definition 14.2) is the three indexed children $\big(p; \mathcal{B}_{(p)}; \frac{s}{3}, \frac{t}{3}\big)$, $p = \varnothing, \{1\}, \{2\}$. The peel (Definition 14.1) merges the two equal children:

$$\Big(\binom{R}{2}; \tfrac{s}{3}, \tfrac{t}{3}\Big), \quad \Big(\binom{R}{1}; \tfrac{2s}{3}, \tfrac{2t}{3}\Big), \qquad \text{word} \quad e_1(\alpha_R)^{-\frac{2s+t}{3}}\, e_2(\alpha_R)^{-\frac{s+2t}{3}}.$$

At the centre, $s = t = \sigma$, the word is $\big(e_1(\alpha_R)e_2(\alpha_R)\big)^{-\sigma}$, the word of the list $\binom{R}{1}$ at its own centre, and the gauge $\mathbf{1}_R$ is native for the masses $\sigma\mathbf{1}_R$ (Proposition 12.27); the balance is $\frac{\sigma}{3}(2+1) + \frac{2\sigma}{3}(1+2) = 3\sigma$.

The trees. The peel is one molecule, the circuit R of $\binom{R}{2}$ linking its marks (Definition 7.4), and every permutation of R carries both of its tables, so it reads every mark alike (Proposition 11.14) and is an atom at every lean (Definition 10.1). So the three Foreman trees (Theorem 14.6) have one shape: the root bunch $\{1,2\}$ of width one and the room $\{3,4,5\}$, an atom, at the exponents of the peel.

The cells. The weights are 1 and factor across B and R. The minors (Definition 7.5) are

$$\mathrm{P}_5|B = \binom{B}{1}, \qquad \mathrm{P}_5/R = \{\varnothing\}, \qquad \mathrm{P}_5|R = \binom{R}{2}, \qquad \mathrm{P}_5/B = \binom{R}{1},$$

P_5/R taking the bases inside R and P_5/B those meeting B. The cell of the bunch (Theorem 19.4) has the bases of $\mathrm{P}_5|B$ and $\varnothing + g$: it is $\binom{\{1,2,g\}}{1}$. The cell of the room has the bases of $\binom{R}{2}$ and $\{x\} + g$, $x \in R$: it is $\mathcal{A} = \binom{\{3,4,5,g\}}{2}$. The hinge values and their ratio (Definition 19.2) are

$$\gamma_B^K = \frac{K_{\mathrm{P}_5|B}}{K_{\mathrm{P}_5/R}} = \alpha_1 + \alpha_2, \qquad \gamma_B^\Psi = \frac{\Psi_{\mathrm{P}_5/R}}{\Psi_{\mathrm{P}_5|B}} = \frac{\alpha_1\alpha_2}{\alpha_1 + \alpha_2}, \qquad \varrho_B = \frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2} = u(1-u)$$

on the shape $(u, 1-u)$, as for every parallel pair (Proposition 19.3).

The substitution. By Proposition 21.3(iii) P_5 is the gluing of the two cells along g: in Definition 21.1,

$$\mathrm{P}_5 = \binom{\{1,2,h\}}{1}\langle 1,2,\mathcal{A}\rangle, \qquad \mathcal{A} = \binom{\{3,4,5,g\}}{2} \text{ at } h,$$

the marks $1, 2$ left in place. Checked on the bases: the core bases $\{1\}$ and $\{2\}$ avoid h and take $X \in \mathcal{A}/g = \binom{R}{1}$, giving the six pairs $\{1,x\}$, $\{2,x\}$; the core basis $\{h\}$ takes $X \in \mathcal{A} \setminus g = \binom{R}{2}$, giving the three pairs inside R. These are the nine bases.

The dual. $\mathcal{A} \setminus g = \binom{R}{2}$ and $\mathcal{A}/g = \binom{R}{1}$, whose bases are the complements in R of the pairs: $\mathcal{A}/g$ is the dual table of $\mathcal{A} \setminus g$, with the weights 1. So $\varrho_{\mathcal{A}} \equiv 1$ (Corollary 23.12(ii)): $\gamma_{\mathcal{A}}^{K} = e_2(\alpha_R)/e_1(\alpha_R) = \gamma_{\mathcal{A}}^{\Psi}$.

Answer. The classes are $\{1,2\}$ and $\{3,4,5\}$ at the centre and at every lean, and the Foreman bunch is $\{1,2\}$, a parallel bunch of width one. Its patterns $\varnothing, \{1\}, \{2\}$ have the probability $\frac{1}{3}$ each and the children $\binom{R}{2}, \binom{R}{1}, \binom{R}{1}$; the peel is $(\binom{R}{2}; \frac{s}{3}, \frac{t}{3})$ with $(\binom{R}{1}; \frac{2s}{3}, \frac{2t}{3})$, at the centre the word $(e_1e_2)^{-\sigma}$ on R, an atom. The three Foreman trees are the bunch $\{1,2\}$ over that atom. And $\mathrm{P}_5 = \binom{\{1,2,h\}}{1}\langle 1,2,\binom{\{3,4,5,g\}}{2}\rangle$, the cell of the room having its contraction $\binom{R}{1}$ dual to its deletion $\binom{R}{2}$, with the ratio 1. □

Exercise 49.6 (A Foreman bunch of width two)**.** Let T_6 be the list of rank three on $\{1, \dots, 6\}$ whose bases are the nineteen sets of three marks other than $\{1,2,3\}$. Show that it is prime (Theorem 23.16); find its classes at the centre and at a lean, its Foreman bunches at a basis lean and at a cobasis lean, and their widths; and draw its three Foreman trees.

Solution. Receipt: `ex05_six_marks.py`.

Ranks and widths. Every set of at most three marks other than $\{1,2,3\}$ is independent, $r(\{1,2,3\}) = 2$, and every set of at least four marks contains a basis and has rank 3. With $c(S) = r(S) + r(\mathcal{E} \setminus S) - 3$ (Definition 7.12, Theorem 7.13):

S	$c(S)$
one mark	$1+3-3=1$
two marks	$2+3-3=2$
three marks, $S \neq \{1,2,3\}, \{4,5,6\}$	$3+3-3=3$
$\{1,2,3\}$ or $\{4,5,6\}$	$2+3-3=2$
four marks	$3+2-3=2$

Every proper nonempty set has positive width, so the list is connected (Definition 8.1); the 50 sets with at least two marks on each side have the widths 2 (32 of them) and 3 (18), so none has width one: T_6 is prime (Theorem 23.16).

Classes. A bijection carries T_6 exactly when it maps the one triple that is not a basis, $\{1,2,3\}$, to itself: the 36 products of a permutation of $\{1,2,3\}$ and one of $\{4,5,6\}$, with the orbits $C = \{1,2,3\}$ and $D = \{4,5,6\}$. The cobases are the triples other than $\{4,5,6\}$, so a bijection exchanges T_6 exactly when it maps $\{1,2,3\}$ onto $\{4,5,6\}$, as $i \mapsto i+3$ modulo 6 does: 36 more bijections, and the 72 together take every mark to every other. By Corollary 11.16, at the centre the one class is $\mathcal{E}$, an atom, and at every lean the classes are C and D, with the occupations $R_e(\mathbf{1}) = \frac{9}{19}$ on C and $\frac{10}{19}$ on D (Proposition 10.9(ii)).

The Foreman bunch at a lean. Both classes have three marks and lie in the one class $\mathcal{E}$ of the centre, so the key of Definition 13.6 is decided by the order $\prec$ at the lean. C and D make up the molecule, so their levels are mirrored (Proposition 13.10, with $k_j = r_j = 3$). Under the weight law,

uniform on the 19 bases (Definition 7.17), the level $N_C = |T \cap C|$ is 0 at $\{4,5,6\}$, 1 at the 9 bases with one mark of C, and 2 at the 9 with two:

$$\mathbb{P}(N_C = 0,1,2) = \tfrac{1}{19},\ \tfrac{9}{19},\ \tfrac{9}{19}; \qquad \mathbb{E}N_C = \tfrac{27}{19}.$$

The deviations $-\frac{27}{19}, -\frac{8}{19}, \frac{11}{19}$ give the cumulants

$$\kappa_{\mathbf{1}}(N_C, N_C) = \frac{9\cdot 64 + 9\cdot 121 + 729}{19^3} = \frac{126}{361},$$
$$\kappa_{\mathbf{1}}(N_C^{\times 3}) = \frac{-19683 - 9\cdot 512 + 9\cdot 1331}{19^4} = -\frac{648}{6859}.$$

The least odd order $a \geq 3$ with $\delta\,\kappa_{\mathbf{1}}(N_C^{\times a}) \neq 0$ is $a = 3$, so by Proposition 13.10(ii) $C \prec D$ exactly when $\delta\,\kappa_{\mathbf{1}}(N_C^{\times 3}) < 0$, that is at a basis lean, $\delta > 0$, and $D \prec C$ at a cobasis lean. (In Corollary 13.4(i) the coefficient of v_0^3 in the odd part of stage 0 is $\frac{2}{3!}\delta\,\kappa_{\mathbf{1}}(N_C^{\times 3}) = -\frac{216}{6859}\delta$ for C and $+\frac{216}{6859}\delta$ for D, the even parts being equal.) So

$$B(\mathrm{T}_6;\sigma,\delta) = \{1,2,3\} \quad (\delta > 0), \qquad B(\mathrm{T}_6;\sigma,\delta) = \{4,5,6\} \quad (\delta < 0),$$

each of width 2, and the exchange $i \mapsto i+3$ maps the one onto the other (Theorem 13.7(iii)).

The trees. At the centre the molecule is an atom (Definition 14.4): one leaf. At a basis lean the root bunch is C, with the patterns (Definition 12.1) and probabilities (Definition 12.25)

p	$\pi_p(\mathbf{1})$	child on $R = \{4,5,6\}$
$\varnothing$	$\frac{1}{19}$	the one basis R
$\{b\},\ b \in C$	$\frac{3}{19}$ each	$\binom{R}{2}$
$\{b,b'\} \subset C$	$\frac{3}{19}$ each	$\binom{R}{1}$

C itself not being a pattern. The peel (Definition 14.1) is the table with the one basis R at $(\frac{s}{19}, \frac{t}{19})$, $\binom{R}{2}$ at $(\frac{9s}{19}, \frac{9t}{19})$ and $\binom{R}{1}$ at $(\frac{9s}{19}, \frac{9t}{19})$; the circuit R of $\binom{R}{2}$ makes R one molecule, and every permutation of R carries the three tables, so the room is an atom (Proposition 11.14). At a cobasis lean the root bunch is D, with the patterns $\{x\}$ (child $\binom{C}{2}$), $\{x,y\}$ (child $\binom{C}{1}$), each of probability $\frac{3}{19}$, and D (child: the one basis $\varnothing$ on C) of probability $\frac{1}{19}$, $\varnothing$ not being a pattern; the room C is again an atom. The exchange $i \mapsto i+3$ maps the tree at δ onto the tree at $-\delta$ (Theorem 14.8).

Answer. T_6 is connected and prime. At the centre it is an atom. At a lean its classes are $\{1,2,3\}$ and $\{4,5,6\}$, and its Foreman bunch is $\{1,2,3\}$ at a basis lean and $\{4,5,6\}$ at a cobasis lean, both of width two. Its three Foreman trees are one atom and two trees whose root bunch, of width two, stands over a room of three marks that is an atom. □

Exercise 49.7 (What a bunch of width two shows, and what splits off)**.** For the list T_6 of Exercise 49.6 at its centre:

(i) find the levels of K along the scale of $C = \{1,2,3\}$, the Skeleton polynomial of C at the table's gauge, its discriminant and the Skeleton values there, and show that the Skeleton values along C are not rational functions of the point;

(ii) show that no set with at least two marks on each side and no complement of a single mark splits off (Theorem 28.25).

Solution. Receipt: `ex05_six_marks.py`.

(i) *The levels.* $d(C) = 0$, the basis $\{4,5,6\}$ avoiding C, and $r(C) = 2$. By the Taylor cube (Theorem 12.7) and Definition 12.16, the level 0 is the pattern $\varnothing$ with the child of the one basis R,

the level 1 the patterns $\{b\}$ with the child $\binom{R}{2}$, and the level 2 the patterns $\{b, b'\}$ with the child $\binom{R}{1}$:

$$K(x\alpha_C, \alpha_R) = \alpha_4\alpha_5\alpha_6 + x\, e_1(\alpha_C)\, e_2(\alpha_R) + x^2\, e_2(\alpha_C)\, e_1(\alpha_R),$$

each level a product of a bunch letter and a room letter.

At the table's gauge. $K(\mathbf{1}) = 19$, and the Skeleton polynomial (Definition 12.10) is

$$\mathsf{Q}^K_C(x; \mathbf{1}) = \frac{K(x\mathbf{1}_C, \mathbf{1}_R)}{K(\mathbf{1})} = \frac{1 + 9x + 9x^2}{19},$$

of degree $c(C) = 2$. The discriminant of $9x^2 + 9x + 1$ is $81 - 36 = 45$, its roots are $x = \frac{-9 \pm 3\sqrt{5}}{18}$, and the Skeleton values are their negatives,

$$\xi_{1,2} = \frac{3 \mp \sqrt{5}}{6},$$

positive and irrational.

Not rational functions. The Skeleton values at α are the negatives of the roots of $K(x\alpha_C, \alpha_R)$ in x, so

$$(\xi_1 - \xi_2)^2\, e_2(\alpha_C)^2\, e_1(\alpha_R)^2 = \Delta(\alpha) := e_1(\alpha_C)^2 e_2(\alpha_R)^2 - 4\, e_2(\alpha_C)\, e_1(\alpha_R)\, \alpha_4\alpha_5\alpha_6.$$

If ξ_1, ξ_2 were rational functions of α, Δ would be the square of a rational function, hence, the polynomial ring being a unique factorization domain, the square of a polynomial, and so would its restriction to every line. On $\alpha = (1, 1, 1, 1, 1, \lambda)$, $e_1(\alpha_C) = e_2(\alpha_C) = 3$, $e_1(\alpha_R) = 2 + \lambda$, $e_2(\alpha_R) = 1 + 2\lambda$ and $\alpha_4\alpha_5\alpha_6 = \lambda$:

$$\Delta = 9(1 + 2\lambda)^2 - 12\lambda(2 + \lambda) = 3\,(8\lambda^2 + 4\lambda + 3),$$

and $8\lambda^2 + 4\lambda + 3$ has the discriminant $16 - 96 = -80$: two simple roots, so Δ is not a square on this line. The value 45 at the gauge shows only that the Skeleton values are irrational numbers there; the line shows that they are not rational functions of the point.

(ii) *Sets with two marks on each side.* T_6 is prime (Exercise 49.6), and at its centre, masses $\sigma\mathbf{1}$, every proper wall is positive (Proposition 8.21(ii)); so by Corollary 28.26(ii) no set with at least two marks on each side splits off. Each such set has width two or three and enters every period of the list through a law carried by no point (Corollary 28.26(i)).

Complements of single marks. $\mathcal{E} \setminus \{e\}$ has the width $c(\{e\}) = 1$ (Theorem 7.13); with both letters active, Corollary 28.26(iv) says it splits off exactly when the deletion and the contraction at e have one even letter, $\mathbf{E}_{\mathrm{T}_6\setminus e} = \mathbf{E}_{\mathrm{T}_6/e}$ (Definition 7.3). For $e \in C$ the deletion has as its bases all ten triples of $\mathcal{E} \setminus \{e\}$ and the contraction the nine pairs $\{x, y\}$ of $\mathcal{E} \setminus \{e\}$ with $\{e, x, y\} \neq \{1, 2, 3\}$; for $e \in D$ the deletion has nine bases and the contraction ten. At the table's gauge $\mathbf{E} = K\Psi$ is the square of the number of bases, so

$$\mathbf{E}_{\mathrm{T}_6\setminus e}(\mathbf{1}) = 100, \quad \mathbf{E}_{\mathrm{T}_6/e}(\mathbf{1}) = 81 \quad (e \in C); \qquad \mathbf{E}_{\mathrm{T}_6\setminus e}(\mathbf{1}) = 81, \quad \mathbf{E}_{\mathrm{T}_6/e}(\mathbf{1}) = 100 \quad (e \in D):$$

the two even letters differ, and no complement of a mark splits off at the centre.

Answer. Along C the levels are $\alpha_4\alpha_5\alpha_6$, $e_1(\alpha_C)e_2(\alpha_R)$ and $e_2(\alpha_C)e_1(\alpha_R)$; at the table's gauge $K(x\mathbf{1}_C, \mathbf{1}_R) = 1 + 9x + 9x^2$, with the discriminant 45 and the Skeleton values $\frac{3 \mp \sqrt{5}}{6}$; on the line $(1, 1, 1, 1, 1, \lambda)$ the discriminant is $3(8\lambda^2 + 4\lambda + 3)$, not a square, so the Skeleton values are not rational functions of the point. At the centre no set with at least two marks on each side and no complement of a mark splits off: T_6 is its own and only factor. (At a full lean each complement of a mark does split off, Corollary 28.26(iii).) □

49.4 Skeleton values at width two

Exercise 49.8 (Skeleton values of $\binom{[4]}{2}$ along a pair)**.** For the list $\binom{[4]}{2}$ and its bunch $B = \{1, 2\}$, find the width of B, the levels of K along the scale of B, the Skeleton polynomial and the Skeleton values at the table's gauge, and show that the Skeleton values are not rational functions of the point. Why does the value of the discriminant at the table's gauge not settle this?

Solution. Receipt: `ex06_skeleton_u24.py`.

The width. The basis $\{3, 4\}$ avoids B and the basis $\{1, 2\}$ is B, so the window of B is $\{0, 1, 2\}$: $d(B) = 0$, $r(B) = 2$ and $c(B) = 2$ (Theorem 7.13).

The levels. The patterns are $\varnothing$ (child: the one basis $\{3, 4\}$), $\{1\}$ and $\{2\}$ (child $\binom{\{3,4\}}{1}$, with $K = \alpha_3 + \alpha_4$) and $\{1, 2\}$ (child: the one basis $\varnothing$). By Theorem 12.7 and Definition 12.16,

$$K(\rho\alpha_1, \rho\alpha_2, \alpha_3, \alpha_4) = \alpha_3\alpha_4 + \rho\,(\alpha_1 + \alpha_2)(\alpha_3 + \alpha_4) + \rho^2\alpha_1\alpha_2.$$

The Skeleton polynomial is $\mathsf{Q}_B^K(x; \alpha) = K(x\alpha_B, \alpha_R)/K(\alpha)$ (Definition 12.10), of degree 2, whose roots $-\xi_1, -\xi_2$ have

$$\xi_1 + \xi_2 = \frac{(\alpha_1 + \alpha_2)(\alpha_3 + \alpha_4)}{\alpha_1\alpha_2}, \qquad \xi_1\xi_2 = \frac{\alpha_3\alpha_4}{\alpha_1\alpha_2}.$$

The list is identically self-dual and $d^*(B) = 0$, so Ψ has the same levels along B and the same Skeleton values.

At the table's gauge. $K(x\mathbf{1}_B, \mathbf{1}_R) = 1 + 4x + x^2$ and $K(\mathbf{1}) = 6$; the discriminant is $16 - 4 = 12$, and the Skeleton values are $\xi = 2 \pm \sqrt{3}$, irrational.

What the value at the gauge decides. $(\xi_1 - \xi_2)^2(\alpha_1\alpha_2)^2 = \Delta(\alpha) := (\alpha_1 + \alpha_2)^2(\alpha_3 + \alpha_4)^2 - 4\alpha_1\alpha_2\alpha_3\alpha_4$, and $\Delta(\mathbf{1}) = 12$ is not the square of a rational number. That shows that the Skeleton values are irrational numbers at the gauge, and no more: if Δ were $3q(\alpha)^2$ for a polynomial q with rational coefficients, $\xi_{1,2} = \frac{(\alpha_1+\alpha_2)(\alpha_3+\alpha_4) \mp \sqrt{3}\,q}{2\alpha_1\alpha_2}$ would be rational functions of the point, with the value 12 at the gauge.

Not rational functions. If ξ_1, ξ_2 are rational functions of α, Δ is the square of a rational function, so of a polynomial (unique factorization in the polynomial ring), and so is its restriction to every line: Δ would be a constant times the square of a polynomial with rational coefficients. On $\alpha = (1, 1, 1, \lambda)$,

$$\Delta = 4(1 + \lambda)^2 - 4\lambda = 4(\lambda^2 + \lambda + 1),$$

whose roots $\frac{-1 \pm \sqrt{-3}}{2}$ are simple. So Δ is not a constant times a square, and the Skeleton values along B are not rational functions of the point.

Answer. B has width two, with the levels $\alpha_3\alpha_4$, $(\alpha_1+\alpha_2)(\alpha_3+\alpha_4)$ and $\alpha_1\alpha_2$; at the table's gauge the Skeleton polynomial is $(1+4x+x^2)/6$ and the Skeleton values are $2 \pm \sqrt{3}$. The discriminant is $4(\lambda^2 + \lambda + 1)$ on the line $(1, 1, 1, \lambda)$, not a constant times a square, so the Skeleton values are not rational functions of the point. The value 12 at the gauge only excludes a rational square, not a constant times a square. At width one, by contrast, the one Skeleton value is the quotient of two hinge values (Corollary 19.5). □

49.5 The record of a peel

Exercise 49.9 (Peeling with and without the record)**.** Let

$$F = \left(\left(\binom{[5]}{2}; 1, 0\right), \left(\binom{[5]}{3}; -1, 0\right), \left(\binom{[5]}{4}; 4, 0\right)\right)$$

on $\mathcal{E} = \{1, \dots, 5\}$, with the masses $3\mathbf{1}$. Show that it is on its balance wall with every proper wall positive and that the table's gauge is native; peel it at $\{1\}$ with the native split of the table's gauge, write its record and its peel, and compare the letters they carry. What changes when every exponent is nonnegative?

Solution. Receipt: `ex07_record.py`.

Walls. The exponents are $s_j = 1, -1, 4$ and $t_j = 0$; the ranks are $2, 3, 4$. The balance is $\sum_j s_j r_j = 2 - 3 + 16 = 15 = m_{\mathcal{E}}$. In $\binom{\mathcal{E}}{r}$ a set S meets every basis in at least $d(S) = \max\big(0, |S| - (5 - r)\big)$ marks (Theorem 7.13), so the walls $\kappa_S = m_S - \sum_j s_j d_j(S)$ (Definition 8.1) depend on $|S|$ alone:

$\lvert S\rvert$	1	2	3	4	5
(d_1, d_2, d_3)	$(0,0,0)$	$(0,0,1)$	$(0,1,2)$	$(1,2,3)$	$(2,3,4)$
κ_S	3	2	2	1	0

every proper wall is positive and $\kappa_{\mathcal{E}} = 0$.

The native gauge. On the balance wall the target is $\mathbf{m}$ (Definition 8.17). At the table's gauge the occupation in $\binom{\mathcal{E}}{r}$ is $\frac{r}{5}$ at every mark, so the usage is $1 \cdot \frac{2}{5} - 1 \cdot \frac{3}{5} + 4 \cdot \frac{4}{5} = 3 = m_e$: the gauge $\mathbf{1}$ is native. An exponent is negative, so F has no realization (Definition 9.1) and no Foreman tree (Definition 14.4); its native split (Definition 12.25), its record and its peel are defined at the native gauge $\mathbf{1}$.

The record. The patterns of $\{1\}$ in $\binom{\mathcal{E}}{r}$ are $\varnothing$, with the child $\binom{R}{r}$ on $R = \{2, 3, 4, 5\}$ and the probability $\binom{4}{r}/\binom{5}{r} = \frac{5-r}{5}$, and $\{1\}$, with the child $\binom{R}{r-1}$ and the probability $\frac{r}{5}$ (Definitions 12.1 and 12.25). The record (Definition 14.2) is the six indexed children, each with the exponents $\pi_{j,p}(\mathbf{1})(s_j, 0)$:

table j	p	child	exponents
1	$\varnothing$	$\binom{R}{2}$	$(\frac{3}{5}, 0)$
1	$\{1\}$	$\binom{R}{1}$	$(\frac{2}{5}, 0)$
2	$\varnothing$	$\binom{R}{3}$	$(-\frac{2}{5}, 0)$
2	$\{1\}$	$\binom{R}{2}$	$(-\frac{3}{5}, 0)$
3	$\varnothing$	$\binom{R}{4}$	$(\frac{4}{5}, 0)$
3	$\{1\}$	$\binom{R}{3}$	$(\frac{16}{5}, 0)$

On R the children lie on their balance wall at the masses $3\mathbf{1}_R$, $\frac{3}{5} \cdot 2 + \frac{2}{5} \cdot 1 - \frac{2}{5} \cdot 3 - \frac{3}{5} \cdot 2 + \frac{4}{5} \cdot 4 + \frac{16}{5} \cdot 3 = 12$, and their usage at $\mathbf{1}_R$ is 3 at every mark (Theorem 12.26). The record determines F (Proposition 14.3): table j is the gluing of its two children, and their exponents add to 1, -1 and 4.

The peel. Definition 14.1 merges the children with the same bases, adding their exponents:

$$\Big(\binom{R}{1}; \tfrac{2}{5}, 0\Big), \qquad \Big(\binom{R}{2}; \tfrac{3}{5} - \tfrac{3}{5}, 0\Big) = \Big(\binom{R}{2}; 0, 0\Big), \qquad \Big(\binom{R}{3}; \tfrac{14}{5}, 0\Big), \qquad \Big(\binom{R}{4}; \tfrac{4}{5}, 0\Big).$$

The table $\binom{R}{2}$ has both exponents zero: it is not active, and the peel carries no letter $e_2(\alpha_R)$ (Definition 8.1). The record carries it twice, in the child $(1, \varnothing)$ at the exponent $\frac{3}{5}$ and in the child $(2, \{1\})$ at $-\frac{3}{5}$. The two words agree,

$$e_1(\alpha_R)^{-2/5}\, e_3(\alpha_R)^{-14/5}\, e_4(\alpha_R)^{-4/5},$$

since $e_2^{-3/5} e_2^{3/5} = 1$; what the peel loses is which pattern of which table each share came from, which the record keeps (the paragraph after Proposition 14.3), and with it the letter $e_2(\alpha_R)$ that the tables 1 and 2 hand to the room.

Nonnegative exponents. When every $s_j, t_j \geq 0$, every child carries $\pi_{j,p}(\mathbf{1})\,(s_j, t_j)$ with $\pi_{j,p}(\mathbf{1}) > 0$ (Proposition 12.27); a merged exponent is a sum of nonnegative terms, zero only when each term is, so merging loses no letter, and a full lean is handed down full, table by table. This is the scope of the realizations, where the Foreman tree is read.

Answer. F is on its balance wall with the walls $3, 2, 2, 1$ at the sets of $1, 2, 3, 4$ marks, and the table's gauge is native. Its record at $\{1\}$ is the six children of the table displayed; its peel is $\binom{R}{1}$, $\binom{R}{3}$, $\binom{R}{4}$ at $\frac{2}{5}$, $\frac{14}{5}$, $\frac{4}{5}$, the child $\binom{R}{2}$ cancelling between the tables 1 and 2: merging loses the letter e_2 on the room, which the record keeps. With nonnegative exponents no letter is lost. $\square$

Chapter 50

Exercises on evaluation and construction

The exercises of this chapter work the statements of Parts IV and V on particular lists. A set of width two whose stations of room sides are single room sides cannot be opened with one share per station, the room's seam function being infinite at every share, and it is opened with the transverse pins instead. The width of the marks left in place by a substitution is read off the core, bounded by the number of the other cells, and every width occurs. The uniform list of rank two on five marks is fused by the procedure of Foreman fusion: its fusion, its one major, its cousins and the symmetry the fusion adds. Every number displayed is exact; the receipts that recompute them are the scripts named in each solution.

50.1 One share per station

Exercise 50.1 (One share per station at a set of width two)**.** Let Q_5 be the list on $\{1,\dots,5\}$ whose bases are the eight sets of three marks other than $\{1,2,5\}$ and $\{3,4,5\}$, at its centre $(\mathbf{1};1,1)$, and let $S=\{1,2,3\}$, with the room $R=\{4,5\}$.

(i) Find the width of S, its patterns and room sides at every excess, its stations, its seam letters and the room station letters (Definition 28.1), and check the room-station factorization.

(ii) Read the period with one share per station (Theorem 28.10(iii)): write the pins of the seam and the room word, and show that the seam function of the room is $+\infty$ at every share.

(iii) Read it with the transverse pins of Theorem 28.12: find the lattice, the pins, the reduced periods and the share side, and check that the native split at the stations lies on the pins, with the masses of S inside the Newton sum of the seam word there.

Solution. Receipt: `ex08_transverse.py`.

(i) *The list.* Its circuits (Definition 7.4) are the two triples $\{1,2,5\}$ and $\{3,4,5\}$ that are not bases and the set $\{1,2,3,4\}$, all of whose triples are bases; the rank is 3 and the corank 2. In S every basis has one to three marks: $r(S)=3$, S being a basis, $r(R)=2$, and by Definition 7.12 and Theorem 7.13

$$d(S)=r-r(R)=1,\qquad d^*(S)=|S|-r(S)=0,\qquad c(S)=r(S)-d(S)=2,$$
$$d(R)=d^*(R)=0,\qquad c(R)=2.$$

Patterns and room sides. An excess i is the level $d(S)+i = 1+i$ (Definition 28.1); its patterns have $1+i$ marks and its room sides $2-i$. With the children (Definition 12.1) and the room children:

i	patterns : children on R	room sides : room children on S
0	$\{1\},\{2\}$: $\{45\}$	$\{4,5\}$: $\{1\},\{2\}$
1	$\{1,2\}$: $\{4\}$; $\{1,3\},\{2,3\}$: $\{4\},\{5\}$	$\{4\}$: $\{12\},\{13\},\{23\}$; $\{5\}$: $\{13\},\{23\}$
2	$\{1,2,3\}$: $\{\varnothing\}$	$\varnothing$: $\{123\}$

The two room sides of the excess 1 have room children with three and with two bases, not multiples of one another; so every room side is its own station: $D_0 = \{\{4,5\}\}$, $D_1 = \{\{4\}\}$, $D_1' = \{\{5\}\}$ and $D_2 = \{\varnothing\}$. The stations of patterns are $\{\{1\},\{2\}\}$, $\{\{1,2\}\}$, $\{\{1,3\},\{2,3\}\}$ and $\{\{1,2,3\}\}$. For a list every multiplier and pair constant is 1.

Seam letters and room station letters. The seam letters are the letters of the room children, on S, and the room station letters are $\mathsf{y}_D = \alpha_R^X$ and $\mathsf{y}_D^* = \alpha_R^{R\setminus X}$ for the one room side X of D (Definition 28.1(iii)):

D	$V_D = K_{(D)}$	$V_D^* = \Psi_{(D)}$	y_D	y_D^*
D_0	$\alpha_1+\alpha_2$	$\alpha_3(\alpha_1+\alpha_2)$	$\alpha_4\alpha_5$	1
D_1	$e_2(\alpha_S)$	$e_1(\alpha_S)$	α_4	α_5
D_1'	$\alpha_3(\alpha_1+\alpha_2)$	$\alpha_1+\alpha_2$	α_5	α_4
D_2	$\alpha_1\alpha_2\alpha_3$	1	1	$\alpha_4\alpha_5$

The room-station factorization (Proposition 28.2(iii)):

$$K = (\alpha_1+\alpha_2)\,\alpha_4\alpha_5 + e_2(\alpha_S)\,\alpha_4 + \alpha_3(\alpha_1+\alpha_2)\,\alpha_5 + \alpha_1\alpha_2\alpha_3,$$
$$\Psi = \alpha_3(\alpha_1+\alpha_2) + e_1(\alpha_S)\,\alpha_5 + (\alpha_1+\alpha_2)\,\alpha_4 + \alpha_4\alpha_5,$$

the eight bases and the eight cobases, each once.

The walls. At $(\mathbf{1};1,1)$, $\kappa_S = m_S - d(S) - d^*(S) = 2$, $\kappa_R = m_R - d(R) - d^*(R) = 2$, and the top wall of S is $(s+t)\,c(S) - \kappa_S = 2 = \kappa_R$ (Definition 28.4): the glue is on its balance wall.

(ii) *The pins of the seam.* One share per station and letter: a_D for K and b_D for Ψ, $D \in \{D_0, D_1, D_1', D_2\}$. By Lemma 28.7 the pins are the two letter totals and the scale pin, whose coefficients are the excesses $i(D)$ for K and $c(S) - i(D)$ for Ψ:

$$a_{D_0} + a_{D_1} + a_{D_1'} + a_{D_2} = 1, \qquad b_{D_0} + b_{D_1} + b_{D_1'} + b_{D_2} = 1,$$
$$a_{D_1} + a_{D_1'} + 2a_{D_2} + 2b_{D_0} + b_{D_1} + b_{D_1'} = \kappa_S = 2.$$

The room word. The room is the partner with the room letters $L_D = \mathsf{y}_D$ and $L_D^* = \mathsf{y}_D^*$ (the paragraph after Definition 28.4), so its word is the monomial

$$W_R(\mathbf{a},\mathbf{b}) = \prod_D \mathsf{y}_D^{-a_D}\,{\mathsf{y}_D^*}^{-b_D} = \alpha_4^{-p_4}\alpha_5^{-p_5},$$
$$p_4 = a_{D_0} + a_{D_1} + b_{D_1'} + b_{D_2}, \qquad p_5 = a_{D_0} + a_{D_1'} + b_{D_1} + b_{D_2},$$

and $p_4 + p_5 = 2\big(|\mathbf{a}| + |\mathbf{b}|\big) - \big(a_{D_1} + a_{D_1'} + 2a_{D_2} + 2b_{D_0} + b_{D_1} + b_{D_1'}\big) = 4 - 2 = 2 = m_4 + m_5$ on the pins.

Its seam function. The seam function of the room is the period of its word at $\mathbf{m}_R = (1,1)$ (Lemma 28.7, read on the partner, whose pins on the balance wall are those of the seam; Theorem 28.10(iii)). On the section $\alpha_5 = 1$ (Definition 25.1) it is

$$\int_0^\infty \alpha_4^{1-p_4}\,\frac{d\alpha_4}{\alpha_4} = +\infty$$

at every real value of p_4: the integrand is a power, which is not integrable at 0 when its exponent is at most -1 and not at ∞ when it is at least -1. Every room letter is a monomial because every room side is its own station, so the Newton sum of the room word is one point, of dimension $0 < |R| - 1$, and Theorem 25.2 excludes convergence at every share. No point of the pins has both seam functions finite: the display of Theorem 28.10(iii), a fibre integral of the dimension $2 \cdot 4 - 2 - 1 = 5$, does not hold here. Its parts (i) and (ii) hold in every case.

(iii) *The Newton sums.* At shares with positive real parts the seam word of S,

$$W_S(\mathbf{a},\mathbf{b}) = \prod_D V_D^{-a_D} V_D^{*-b_D}$$
$$= (\alpha_1+\alpha_2)^{-a_{D_0}-b_{D_1'}} \big(\alpha_3(\alpha_1+\alpha_2)\big)^{-b_{D_0}-a_{D_1'}} e_2(\alpha_S)^{-a_{D_1}} e_1(\alpha_S)^{-b_{D_1}} (\alpha_1\alpha_2\alpha_3)^{-a_{D_2}},$$

has a Newton sum that contains b_{D_1} times the triangle $\operatorname{conv}\{\mathbf{1}_{\{1\}}, \mathbf{1}_{\{2\}}, \mathbf{1}_{\{3\}}\}$ of $e_1(\alpha_S)$ (Definition 25.1): its space of directions is $\vec{\mathcal{N}}_S = \{x \in \mathbb{R}^S : \ x_1 + x_2 + x_3 = 0\}$. The room word is a monomial, $\vec{\mathcal{N}}_R = 0$.

The lattice and the pins. In Theorem 28.12,

$$M = \vec{\mathcal{N}}_S^{\perp} \oplus \vec{\mathcal{N}}_R^{\perp} = \mathbb{R}\mathbf{1}_S \oplus \mathbb{R}^{\{4,5\}},$$

of dimension 3, containing $\mathbf{1}$; $M \cap \mathbb{Z}^5 = \{(a,a,a,b,c)\}$ has the basis $\mathbf{1}, \mathbf{e}_4, \mathbf{e}_5$, since $\mathbf{1}_S = \mathbf{1} - \mathbf{e}_4 - \mathbf{e}_5$. So $q = 2$, $\mathbf{v}_1 = \mathbf{e}_4$ and $\mathbf{v}_2 = \mathbf{e}_5$. The fixed monomials of the seam letters live on S, and those of the room letters are the room letters themselves, so $\langle \mathbf{n}(\mathbf{a},\mathbf{b}), \mathbf{e}_4 \rangle = p_4$ and $\langle \mathbf{n}(\mathbf{a},\mathbf{b}), \mathbf{e}_5 \rangle = p_5$. The pins are the letter totals and

$$a_{D_0} + a_{D_1} + b_{D_1'} + b_{D_2} = m_4 = 1, \qquad a_{D_0} + a_{D_1'} + b_{D_1} + b_{D_2} = m_5 = 1 :$$

four pins on eight shares, a fibre of the dimension 4. With the totals they imply the scale pin, by the identity for $p_4 + p_5$ in (ii).

The reduced periods. $\vec{\mathcal{N}}_S^{\perp} = \mathbb{R}\mathbf{1}_S$, so the reduced period of the seam word is its integral over $\mathbb{R}^S/\mathbb{R}\mathbf{1}_S$, that is over a section: the period $P_{W_S(\mathbf{a},\mathbf{b})}(\mathbf{1}_S)$ (Definition 25.1). $\vec{\mathcal{N}}_R^{\perp} = \mathbb{R}^R$, so the quotient is a point, and the reduced period of the room word is the value of $\alpha_4^{m_4-p_4}\alpha_5^{m_5-p_5}$, which is 1 on the pins, the room letters having the coefficient 1.

The share side. The share kernels are $\Gamma_{\begin{smallmatrix}4\\1\end{smallmatrix}}\begin{bmatrix}\mathbf{a}\\1\end{bmatrix} = \prod_D \Gamma(a_D)$ and $\Gamma_{\begin{smallmatrix}4\\1\end{smallmatrix}}\begin{bmatrix}\mathbf{b}\\1\end{bmatrix} = \prod_D \Gamma(b_D)$, $\Gamma(1) = 1$ (Definition 1.2). So, on every contour of the pins with positive real parts at which P_{W_S} converges,

$$P_{\mathrm{Q}_5}(\mathbf{1}; 1, 1) = \int_{\text{pins}} \prod_D \Gamma(a_D)\,\Gamma(b_D)\; P_{W_S(\mathbf{a},\mathbf{b})}(1,1,1),$$

the pins being $|\mathbf{a}| = |\mathbf{b}| = 1$ and $p_4 = p_5 = 1$.

The native split. At the table's gauge the basis law is uniform on the eight bases and the cobasis law is the same law (Theorem 7.16). $T \cap R$ is $\{4,5\}$ at $145, 245$, $\{4\}$ at $124, 134, 234$, $\{5\}$ at $135, 235$ and $\varnothing$ at 123, so Lemma 28.9 gives

$$a_D^{\star} = b_D^{\star} = \tfrac{1}{4},\ \tfrac{3}{8},\ \tfrac{1}{4},\ \tfrac{1}{8} \qquad (D = D_0, D_1, D_1', D_2),$$

with the totals 1 and 1, $p_4 = \frac{1}{4} + \frac{3}{8} + \frac{1}{4} + \frac{1}{8} = 1$ and $p_5 = \frac{1}{4} + \frac{1}{4} + \frac{3}{8} + \frac{1}{8} = 1$: the native split lies on the transverse pins, with positive coordinates. There $\mathbf{1}_S$ lies in the relative interior of the Newton sum of the seam word (Lemma 28.9); the receipt checks it exactly, the support function

of the Newton sum exceeding $\langle u, \mathbf{1}_S\rangle$ by at least $\frac{9}{8}$ at every edge normal u of its summands. So $P_{W_S(\mathbf{a}^\star,\mathbf{b}^\star)}(\mathbf{1}_S)$ converges (Theorem 25.2), and the contour through the native split carries the display.

Answer. S has width two; its four stations of room sides are the single room sides $\{4,5\}$, $\{4\}$, $\{5\}$, $\varnothing$, the room letters are monomials, and the seam function of the room is $+\infty$ at every share, so the period does not open with one share per station. With the transverse pins $p_4 = p_5 = 1$ in place of the scale pin it opens as the fibre integral of dimension four displayed, the seam word of S against the share kernels, the room contributing 1. □

50.2 The width of the marks left in place

Exercise 50.2 (The width of a bunch of a substitution)**.** Let $\mathrm{M}_8 = \binom{\{b_1,b_2,h_1,h_2\}}{2}\langle b_1, b_2, \mathcal{A}_1, \mathcal{A}_2\rangle$, the marks b_1, b_2 left in place and the cells $\mathcal{A}_i = \binom{\{x_i,y_i,z_i,g_i\}}{2}$ at h_i, with $R_i = \{x_i, y_i, z_i\}$ (Definition 21.1).

(i) Find the rank and the bases of M_8, and the widths of $\{b_1\}$, $\{b_2\}$ and $B = \{b_1, b_2\}$, from the bases and from the core; compare with the bound of Corollary 21.11.

(ii) Write K along the scale of B.

(iii) For $c = 1, 2, 3$ let $B = \{b_1, \dots, b_c\}$ be left in place in the core $\binom{B\sqcup\{h_1,\dots,h_c\}}{c}$, with the cells $\binom{\{x_i,y_i,z_i,g_i\}}{2}$ at the h_i. Find the width of B.

Solution. Receipt: `ex09_width_subst.py`.

(i) *Rank and bases.* The core has the rank 2, each cell $\mathcal{A}_i$ the rank 2, and a mark left in place the cell $\binom{\{b,g\}}{1}$ of rank 1; so M_8 is a table of rank $2 + (2-1) + (2-1) = 4$ on eight marks (Proposition 21.3(i)). Its bases (Definition 21.1) are read off the six bases T of the core: $b \in T \cap B$ gives the mark b, and at h_i the cell gives a basis of $\mathcal{A}_i \setminus g_i = \binom{R_i}{2}$ when $h_i \in T$ and one of $\mathcal{A}_i/g_i = \binom{R_i}{1}$ when $h_i \notin T$, three choices each: $6 \cdot 9 = 54$ bases.

Widths. From the bases, the numbers of marks of $\{b_1\}$, $\{b_2\}$ and B in a basis fill the windows $\{0,1\}$, $\{0,1\}$ and $\{0,1,2\}$ (Theorem 7.13): the widths are 1, 1 and 2. In the core $\binom{\{b_1,b_2,h_1,h_2\}}{2}$ a pair meets one mark in 0 or 1 marks and a pair of marks in 0, 1 or 2: the same widths. That is Corollary 21.11: a set of marks left in place has the width it has in the core. In Theorem 21.10 the cell $\binom{\{b,g\}}{1}$ of a mark b of $S \subseteq \{b_1, b_2\}$ has $d(\{b\}) = d^*(\{b\}) = 0$ and $d(\{b,g\}) = d^*(\{b,g\}) = 1$, the cells $\mathcal{A}_i$ meet S in $\varnothing$ with $d(\{g_i\}) = d^*(\{g_i\}) = 0$, so $D(S) = D^*(S) = S$ and $d(S) = d_{\mathcal{H}}(S) = 0$, $d^*(S) = d^*_{\mathcal{H}}(S) = 0$.

The bound. With $L = \{b_1, b_2\}$ and the two cells, $c(S) \le |L \setminus S| + 2$: $1 \le 3$ for $\{b_1\}$ and $\{b_2\}$, and $2 \le 2$ for B, where it is attained. For the list $\mathrm{P}_5 = \binom{\{1,2,h\}}{1}\langle 1, 2, \binom{\{3,4,5,g\}}{2}\rangle$ of Exercise 49.5 the bound at its Foreman bunch $\{1,2\}$ is $0 + 1 = 1$, attained as well.

(ii) *The letter along B.* The cells have $K_{\mathcal{A}_i/g_i} = e_1(\alpha_{R_i})$ and the hinge values $\gamma_i := \gamma^K_{\mathcal{A}_i} = e_2(\alpha_{R_i})/e_1(\alpha_{R_i})$ (Definition 21.1); their contractions $\binom{R_i}{1}$ are the duals of their deletions $\binom{R_i}{2}$, so also $\gamma^\Psi_{\mathcal{A}_i} = \gamma_i$ (Corollary 23.12(ii)). The core letter is e_2 on four marks, so by Theorem 21.4

$$K(\tau\alpha_B, \alpha_R) = e_1(\alpha_{R_1})\, e_1(\alpha_{R_2})\Big(\gamma_1\gamma_2 + \tau\,(\alpha_{b_1} + \alpha_{b_2})(\gamma_1 + \gamma_2) + \tau^2\alpha_{b_1}\alpha_{b_2}\Big)\,:$$

three levels, the width two, and the room sees B only through the two hinge values of the cells and the marks of B.

(iii) In the core $\binom{B\sqcup\{h_1,\dots,h_c\}}{c}$, $|B| = c$, the sets of c marks meet B in every number of marks from 0 to c, so B has the width c (Theorem 7.13); no h_i is a loop or a coloop, and by Corollary 21.11

the width stays c after the cells are substituted. From the bases:

c	marks	rank	bases	window of B	$c(B)$
1	4	2	6	$\{0,1\}$	1
2	8	4	54	$\{0,1,2\}$	2
3	12	6	540	$\{0,1,2,3\}$	3

At $c=1$ the substitution $\binom{\{b_1,h_1\}}{1}\langle b_1,\mathcal{A}_1\rangle$ is the list $\binom{\{b_1,x_1,y_1,z_1\}}{2}$ itself, and at $c=2$ it is M_8. In each case the bound $|L\setminus B|+c=c$ is attained.

Answer. M_8 is a list of rank four on eight marks with 54 bases; $\{b_1\}$ and $\{b_2\}$ have width one and $B=\{b_1,b_2\}$ width two, as in the core, against the bounds 3, 3 and 2. Along B, $K=e_1(\alpha_{R_1})e_1(\alpha_{R_2})(\gamma_1\gamma_2+\tau(\alpha_{b_1}+\alpha_{b_2})(\gamma_1+\gamma_2)+\tau^2\alpha_{b_1}\alpha_{b_2})$. In $\binom{B\sqcup\{h_1,\dots,h_c\}}{c}$ with cells at the h_i the bunch B has the width c for $c=1,2,3$: a bunch that is all the marks left in place has at most as much width as there are cells, and every width occurs. □

50.3 Fusing a uniform list

Exercise 50.3 (Fusing $\binom{[5]}{2}$)**.** Run the procedure of Section 30.7 on the list $\binom{[5]}{2}$, read at its centre: decide each step, find its fusion width, its tell, its Foreman fusion with its letter, its lifts and its majors, and its cousins; read the period of the fusion through two of its cousins; and find the index of the bijections carrying $\binom{[5]}{2}$, times the permutations of the fusion bunch, among those carrying its fusion.

Solution. Receipt: `ex10_fusion_u25.py`.

(1) Peel. Every permutation of [5] maps pairs to pairs and carries the list, so at the centre, and at every lean, all marks lie in one class (Corollary 11.16): the realization is an atom (Definition 10.1) and nothing is peeled (Definition 14.4).

(2) Go down. The complement of a basis is a triple, not a basis: the list is not identically self-dual (Corollary 23.12(iii)).

(3) Go up. The rank is $r=2$ and the corank $r^*=3$, so the fusion width is $c=r^*-r=1$ (Definition 30.4). The circuits (Definition 7.4) are the ten triples, every pair being independent; the cocircuits, the least sets meeting every pair, are the five sets of four marks. Each cocircuit is the union of the four triples inside it, so the tell (Theorem 30.6(b)) holds. In its form (c): for a mark e and a split of the other four marks into two halves, one half has at least two marks, the rank 2, which e does not raise. The form (a), $r^*(Y)-r^*(X)\geq r(Y)-r(X)$ for all $X\subseteq Y$, the receipt checks on the rank functions. At $c=1$ the weights play no part: $\binom{[5]}{2}$ is fusable.

The fusion. With the fusion bunch $G=\{g\}$, Theorem 30.5(v) gives the bases $T+g$, T a pair of [5], and $[5]\setminus T$, the triples of [5]: ten and ten, all twenty triples of $[5]\sqcup\{g\}$. So

$$\mathcal{B}^\sharp=\binom{[5]\sqcup\{g\}}{3},\qquad K^\sharp=\Psi+\alpha_g K=e_3(\alpha_{[5]})+\alpha_g\,e_2(\alpha_{[5]})=e_3(\alpha_{[5]\sqcup\{g\}})=\Psi^\sharp,$$

identically self-dual, the complement of a triple of six marks being a triple. The lifts (Definition 30.4(ii)) are $\mathcal{L}_0=\binom{[5]}{2}$ and $\mathcal{L}_1=\binom{[5]}{3}$, its dual, with the lift letters $L_0=\Psi=e_3$ and $L_1=K=e_2$; at $c=1$ there are no middle weights and no lift ratios, and $K^\sharp=L_0+\alpha_g L_1$ is Theorem 30.5(ii).

The majors. A table on $[5]\sqcup\{g\}$ whose contraction of g is $\binom{[5]}{2}$ has as its bases through g the sets $T+g$, $T\in\binom{[5]}{2}$, and one whose deletion of g is $\binom{[5]}{3}$ has as its bases avoiding g the triples of

[5] (Definition 7.3). A major (Definition 30.4(i)) has both, so its bases are the twenty triples, and $\mathcal{B}^\sharp$ is the only major, as Theorem 30.5(v) says.

One Beta corner. On the balance wall of the fusion, $m_{[5]}+m_g = 2\sigma r^* = 6\sigma$, with every proper wall positive, Corollary 30.13 gives

$$P_{\binom{[6]}{3}}\big(\mathbf{m}|_{[5]}, m_g; s, t\big) = \Gamma^{\,2}_{\,1}\begin{bmatrix} m_g, & 2\sigma - m_g \\ & 2\sigma \end{bmatrix} P_{\binom{[5]}{2}}\big(\mathbf{m}|_{[5]}; m_g, 2\sigma - m_g\big),$$

[6] written for $[5]\sqcup\{g\}$. It can be read off the letter: the fused table is the family $\big(\binom{[6]}{3}; 2\sigma, 0\big)$ (the paragraph after Definition 30.11), and on the section $|\alpha_{[5]}| = 1$ the scale of g integrates out, with $\alpha_g = (e_3/e_2)v$,

$$\begin{aligned}\int_0^\infty \alpha_g^{\,m_g-1}\big(e_3 + \alpha_g\, e_2\big)^{-2\sigma} d\alpha_g &= e_3^{\,m_g-2\sigma} e_2^{\,-m_g} \int_0^\infty v^{m_g-1}(1+v)^{-2\sigma} dv \\ &= \Gamma^{\,2}_{\,1}\begin{bmatrix} m_g, & 2\sigma - m_g \\ & 2\sigma \end{bmatrix} e_3^{\,m_g-2\sigma} e_2^{\,-m_g},\end{aligned}$$

$e_k = e_k(\alpha_{[5]})$, the last integral being the shape integral of two marks (Theorem 3.9) after $v = u/(1-u)$; what is left is the form of $\binom{[5]}{2}$ with $K = e_2$ at the exponent m_g and $\Psi = e_3$ at $2\sigma - m_g$. The masses of $\binom{[5]}{2}$ are then on its own balance wall, $m_{[5]} = 2m_g + 3(2\sigma - m_g)$.

The cousins. The cousins of $\binom{[6]}{3}$ (Proposition 30.20) are its six one-mark contractions $\binom{[6]\setminus\{x\}}{2}$, each a copy of $\binom{[5]}{2}$, the cousin at g being $\binom{[5]}{2}$ itself; the deletion $\binom{[6]\setminus\{x\}}{3}$ is the dual of each, so each has the fusion width one and fuses back into $\binom{[6]}{3}$ with the fusion bunch $\{x\}$ (Proposition 30.20(i)). For any two marks the period is read through either cousin (Proposition 30.20(ii)); at $x = g$ and $y = 5$, writing $s' = m_g$ and renaming g as 5 in the second cousin,

$$\begin{aligned}&\Gamma^{\,2}_{\,1}\begin{bmatrix} s', & 2\sigma - s' \\ & 2\sigma \end{bmatrix} P_{\binom{[5]}{2}}\big(m_1, \dots, m_5; s', 2\sigma - s'\big) \\ &\qquad = \Gamma^{\,2}_{\,1}\begin{bmatrix} m_5, & 2\sigma - m_5 \\ & 2\sigma \end{bmatrix} P_{\binom{[5]}{2}}\big(m_1, m_2, m_3, m_4, s'; m_5, 2\sigma - m_5\big)\end{aligned}$$

on $m_{[5]} + s' = 6\sigma$: the basis exponent s' of $\binom{[5]}{2}$ and the mass of one mark trade places (Proposition 30.20(iii)). At the centre, $\sigma = 1$ and every mass 1, both sides are $\Gamma^{\,2}_{\,1}\begin{bmatrix} 1, & 1 \\ & 2 \end{bmatrix} P_{\binom{[5]}{2}}(\mathbf{1}; 1, 1) = P_{\binom{[5]}{2}}(\mathbf{1}; 1, 1)$, and $P_{\binom{[6]}{3}}(\mathbf{1}; 1, 1) = P_{\binom{[5]}{2}}(\mathbf{1}; 1, 1)$.

The symmetry the fusion adds. All 720 permutations of $[5]\sqcup\{g\}$ carry $\binom{[6]}{3}$; those carrying $\binom{[5]}{2}$, times the permutations of $G = \{g\}$, are the $120\cdot 1$ permutations fixing g. The index is $720/120 = 6 = \binom{n+c}{c} = \binom{6}{1}$ (Remark 30.21).

(4) Two-sums and (5) repair. Every set of two or three marks has the width $2+2-2 = 2$ (Definition 7.12), so $\binom{[5]}{2}$ is prime (Theorem 23.16) and no two-sum, and being fusable it needs no repair.

Answer. The procedure stops at step (3): $\binom{[5]}{2}$ has the fusion width one, passes the tell, and fuses into $\binom{[5]\sqcup\{g\}}{3}$, with the letter $e_3 = \Psi + \alpha_g K$, its only major. Its cousins are the six copies $\binom{[6]\setminus\{x\}}{2}$ of $\binom{[5]}{2}$, and the period of the fusion is one Beta corner times the period of any cousin, the basis exponent of $\binom{[5]}{2}$ and the masses of its marks trading places under the permutations of six marks; the fusion adds the index 6 to the symmetry. □

Chapter 51

Exercises on the motive and the arithmetic of values

The exercises of this chapter work the statements of Chapter 34 on small lists. The list $\binom{[3]}{2}$ at its centre has a pair of width one that does not split off; read at that pair as its root bunch, its period is a node form with two Skeleton values, and it is three times the Trailmoment at the index two of the lattice input that counts $1+3\mathbb{Z}$ and subtracts $2+3\mathbb{Z}$. The list $\binom{[4]}{2}$ is, at the complement of each mark, a core on two marks with a dual pair substituted, and it has the motivic period of $\binom{[3]}{2}$; its pairs have width two and do not split off. The list $\binom{[4]}{1}$ has rational periods and rational motivic periods at its full leans; at its centre a pair of width one has a law with a density, and the period is the critical-line word $\mathrm{crit}_1[1]$, seven times the Trailmoment at the index three of the counting input. Substitutions of dual pairs on five and on eight marks have the motivic periods $(P^{\mathrm{mot}}_{\binom{[3]}{2}})^2$ and $(P^{\mathrm{mot}}_{\binom{[3]}{2}})^3$. At the exponents $(0,2)$ with unit masses every hinge wall and every Beta corner is 1, and the motivic period of a list is the product of the motivic periods of its prime nodes. Every number displayed is exact, and the receipts that recompute them are the scripts named in each solution.

Throughout, a list $\mathcal{B}$ *at its centre* is the family $(\mathcal{B};1,1)$ at the masses $\mathbf{1}$ (Proposition 8.21), and when it is convergent $P^{\mathrm{mot}}_{\mathcal{B}}$ denotes its motivic period (Definition 34.5); a family at a lean is written out with its exponents and its masses.

51.1 Uniform lists at their centres

Exercise 51.1 (The list $\binom{[3]}{2}$ at its centre)**.** Let $\mathcal{E}=\{1,2,3\}$ and $F=\big(\binom{\mathcal{E}}{2};1,1\big)$ at the masses $\mathbf{1}$.

(i) Show that F is convergent (Definition 34.1), that the pair $S=\{1,2\}$ has width one with the ratio $\varrho_S=1/(u(1-u))$ on its shape $(u,1-u)$, and that S does not split off (Theorem 28.25).

(ii) Take a tree of bunches whose root carries the bunch S. Find the Skeleton values of the letters along S, the node form of ω_F at the root (Theorem 34.16) and its share side.

(iii) Evaluate the period $P_{\binom{\mathcal{E}}{2}}(\mathbf{1};1,1)$ as a Trailmoment of a lattice input.

Solution. Receipt: `ex1_series_u23.py`.

(i) *Letters and walls.* The bases are the three pairs, so $K=e_2(\alpha)=\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3$ and $\Psi=e_1(\alpha)$ (Definition 7.1), with the rank 2 and the corank 1. The imbalance is $\kappa_{\mathcal{E}}=3-(1\cdot 2+1\cdot 1)=0$ (Definition 8.1). By Theorem 7.13 and Definition 7.12, $d(S)=\min_T|T\cap S|$

and $d^*(S) = |S| - r(S)$: a mark has $d = d^* = 0$, and a pair has $d = 1$ and $d^* = 0$. So the walls $\kappa_S = m_S - d(S) - d^*(S)$ are

$$\kappa_{\{e\}} = 1 - 0 - 0 = 1, \qquad \kappa_{\{e,f\}} = 2 - 1 - 0 = 1,$$

the widths of these sets, as Proposition 8.21(ii) says at a centre. Every proper wall is at least 1, so F is convergent, its motivic period $P^{\mathrm{mot}}_{\binom{[3]}{2}}$ is defined (Definition 34.5), and its period is

$$P_{\binom{\mathcal{E}}{2}}(\mathbf{1}; 1, 1) = \int_{\Delta_{\mathcal{E}}} \frac{d\omega_{\mathcal{E}}(u)}{e_1(u)\, e_2(u)}$$

(Proposition 34.4(ii), Theorem 8.7).

The pair. The bases meet S in one or two marks, so $d(S) = 1$, $r(S) = 2$ and $c(S) = 1$ (Theorem 7.13); every basis contains all but at most one mark of S, so S is a series bunch (Definition 19.1). With the room $R = \{3\}$, the restriction $\mathcal{B}|S$ has the one basis S and the contraction $\mathcal{B}/R$ has the bases $\{1\}$ and $\{2\}$ (Definition 7.5). The hinge values and their ratio (Definition 19.2) are

$$\gamma^K_S = \frac{K_{\mathcal{B}|S}}{K_{\mathcal{B}/R}} = \frac{\alpha_1\alpha_2}{\alpha_1+\alpha_2}, \qquad \gamma^\Psi_S = \frac{\Psi_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|S}} = \frac{\alpha_1+\alpha_2}{1}, \qquad \varrho_S = \frac{\gamma^\Psi_S}{\gamma^K_S} = \frac{(\alpha_1+\alpha_2)^2}{\alpha_1\alpha_2} = \frac{1}{u(1-u)}$$

on the shape $(u, 1-u)$, as Proposition 19.3 gives for a series pair.

The pair does not split off. For one list with $s, t > 0$ and $|S| \geq 2$, Theorem 28.25 says that S is pinned exactly when it has width one and $\mathbf{E}_{\mathcal{B}|S} = \mathbf{E}_{\mathcal{B}/R}$. Here

$$\mathbf{E}_{\mathcal{B}|S} = \alpha_1\alpha_2 \cdot 1, \qquad \mathbf{E}_{\mathcal{B}/R} = (\alpha_1+\alpha_2)(\alpha_1+\alpha_2),$$

which differ: their ratio ϱ_S is at least 4, with equality only at $u = \frac{1}{2}$. So S is not pinned; by Theorem 28.25, (ii), (iii) and (i), it does not split off and its law is carried by no point. This is Corollary 28.26(iv) at the mark 3, whose deletion and contraction are $\mathcal{B} \setminus 3 = \mathcal{B}|S$ and $\mathcal{B}/3 = \mathcal{B}/R$ (Definition 7.3).

(ii) *The step at the root.* The list is one molecule, its circuit $\{1, 2, 3\}$ linking its marks (Definition 7.4), so the root of a tree of bunches is a node, and it may carry the bunch S, of width one, its room being the one mark 3 (Definition 20.7). The flag $\varnothing \subsetneq \mathcal{E}$ has the one step $\mathcal{E}$, which meets S and R: it is a node step of the root (Definition 34.9(ii)), with $Z_B = S$ and $Z_R = R$. The form ω_F is regular on V_F (Proposition 34.4(ii)), and its period is read by its node form at the root (Theorem 34.16(v)). Read $\alpha_S = \rho\,(u, 1-u)$ with $\rho > 0$ and $\alpha_3 = 1$, the one point of Δ_R, and put $Q = u(1-u)$.

Letters and Skeleton values. By Theorem 34.16(i) the letter K is the basis polynomial of $\binom{\mathcal{E}}{2}$, in which S has the floor 1, and Ψ is that of the dual list $\binom{\mathcal{E}}{1}$ (Proposition 7.2), in which S has the floor 0; S has the width 1 in both. Along the scale (Theorem 34.16(ii)),

$$K(\rho u, 1) = \rho^2 u(1-u) + \rho\,\big(u + (1-u)\big) = \rho\, Q\,\big(\rho + Q^{-1}\big), \qquad \Psi(\rho u, 1) = \rho + 1,$$

so the joint spectrum of S is $\{Q^{-1}, 1\}$: the Skeleton value of K is $\xi_K = Q^{-1} = \varrho_S$ and that of Ψ is $\xi_\Psi = 1$. At width one each is the hinge value of the room over that of the bunch (Theorem 34.16(iv), Corollary 19.5): with $\gamma^K_R = K_{\mathcal{B}|R}/K_{\mathcal{B}/S} = \alpha_3/1$ and $\gamma^\Psi_R = \Psi_{\mathcal{B}/S}/\Psi_{\mathcal{B}|R} = \alpha_3/1$ (Definition 19.2), on $\Delta_S \times \Delta_R$

$$\xi_K = \frac{\gamma^K_R}{\gamma^K_S} = \frac{1}{Q}, \qquad \xi_\Psi = \frac{\gamma^\Psi_R}{\gamma^\Psi_S} = \frac{1}{1}.$$

The node form. On the section $\alpha_3 = 1$ the masses are $\mathbf{1}$ and $d\alpha_1\, d\alpha_2 = \rho\, d\rho\, du$ (Lemma 2.7), so (Theorem 8.7)

$$P_{\binom{\varepsilon}{2}}(\mathbf{1}; 1, 1) = \int_0^1 \int_0^\infty \frac{\rho\, d\rho}{K(\rho u, 1)\, \Psi(\rho u, 1)}\, du = \int_0^1 Q^{-1} \int_0^\infty \left(\rho + Q^{-1}\right)^{-1} (\rho + 1)^{-1}\, d\rho\, du.$$

This is the node form of Theorem 34.16(iii) with one term: $A_t = Q^{-1}$, $B_t = 1$, $\nu_t = 1$, the letter exponents $k_{t,K} = k_{t,\Psi} = 1$, and $\mathbf{w}_t = (1, 1)$ on the joint spectrum, $|\mathbf{w}_t| = 2$. Since $0 < \nu_t < |\mathbf{w}_t|$, Sc_t is the convergent scale integral, the stacked Gamma times the Breta of one bunch (Proposition 15.1), and at positive coefficients the Breta is the integral over the shape simplex of J (Definition 3.1, Proposition 15.2):

$$\mathrm{Sc}_t = \Gamma_{\binom{2}{2}}\begin{bmatrix} 1, & 1 \\ 1, & 1 \end{bmatrix} \mathsf{B}^{\{J\}}_{(1,1),\,1;\,I_J}(Q^{-1}, 1) = \int_0^1 \frac{dv}{Q^{-1}v + (1 - v)} = \frac{\log Q^{-1}}{Q^{-1} - 1} = \frac{Q \log Q}{Q - 1}.$$

So

$$P_{\binom{\varepsilon}{2}}(\mathbf{1}; 1, 1) = \int_0^1 A_t\, \mathrm{Sc}_t\, du = \int_0^1 \frac{\log Q}{Q - 1}\, du, \qquad Q = u(1 - u).$$

The share side. Corollary 15.5, with $\xi_0 = 1$, $\xi_1 = Q^{-1}$, $w_0 = w_1 = \kappa = 1$ and one share a on a line $\mathrm{Re}\, a = c$, $0 < c < 1$, writes the term as in Theorem 34.16(iv):

$$\begin{aligned} A_t\, \mathrm{Sc}_t &= Q^{-1}\, \frac{1}{2\pi i} \int_{(c)} \Gamma_{\binom{2}{1}}\begin{bmatrix} a, & 1 - a \\ & 1 \end{bmatrix} \Gamma_{\binom{2}{1}}\begin{bmatrix} 1 - a, & a \\ & 1 \end{bmatrix} \left(Q^{-1}\right)^{a-1} da \\ &= \frac{1}{2\pi i} \int_{(c)} \Gamma_{\binom{4}{0}}\big[a,\ a,\ 1 - a,\ 1 - a\big]\ Q^{-a}\, da. \end{aligned}$$

Here $\xi_K^{-b} = (\gamma_S^K)^b$ with $b = 1 - a$, and the condition of Theorem 34.16(iv) on the bunch, $\int_0^1 Q^{-1} Q^{1-c}\, du < \infty$, holds at $c = \frac{1}{2}$, where it is $\int_0^1 (u(1 - u))^{-1/2} du = \Gamma_{\binom{2}{1}}\begin{bmatrix} \frac{1}{2}, & \frac{1}{2} \\ & 1 \end{bmatrix}$ (Theorem 3.9); the room is one point. Closing to the left (Lemma 5.16(ii), with $X = Q \le \frac{1}{4}$), the stacked Gamma has at $a = -j$, $j \in \mathbb{N}$, the principal part $(a + j)^{-2}$ and no simple part, so the residue there is the derivative of Q^{-a} at $a = -j$, which is $-Q^j \log Q$, and

$$\frac{1}{2\pi i} \int_{(1/2)} \Gamma_{\binom{4}{0}}\big[a,\ a,\ 1 - a,\ 1 - a\big]\ Q^{-a}\, da = -\sum_{j \ge 0} Q^j \log Q = \frac{\log Q}{Q - 1},$$

the value of the node form above.

(iii) *The reflection of the shape.* $\log Q = \log u + \log(1 - u)$, and $1 - Q = 1 - u + u^2$ is unchanged by $u \mapsto 1 - u$; so the two logarithms give equal integrals and

$$P_{\binom{\varepsilon}{2}}(\mathbf{1}; 1, 1) = -2 \int_0^1 \frac{\log u}{1 - u + u^2}\, du.$$

The logarithmic coordinate. With $u = e^{-y}$, $du = -e^{-y} dy$, and multiplying numerator and denominator by $1 + e^{-y}$, since $(1 - z + z^2)(1 + z) = 1 + z^3$,

$$P_{\binom{\varepsilon}{2}}(\mathbf{1}; 1, 1) = 2 \int_0^\infty y\, \frac{e^{-y}}{1 - e^{-y} + e^{-2y}}\, dy = 2 \int_0^\infty y\, \frac{e^{-y} + e^{-2y}}{1 + e^{-3y}}\, dy.$$

The lattice input. Let $\phi = \mathbb{I}_{1+6\mathbb{Z}} + \mathbb{I}_{2+6\mathbb{Z}} - \mathbb{I}_{4+6\mathbb{Z}} - \mathbb{I}_{5+6\mathbb{Z}}$, an input with the density ± 1 on four lattice inputs (Definitions 5.2 and 5.4). It is tame beyond the cut 0, and its Laplace transform there (Definition 5.4) sums its four cosets as geometric series: with $z = e^{-y}$,

$$\mathcal{F}_0(y) = \sum_{k\geq 0} \big(z^{6k+1} + z^{6k+2} - z^{6k+4} - z^{6k+5}\big) = \frac{z + z^2 - z^4 - z^5}{1 - z^6} = \frac{(z+z^2)(1-z^3)}{(1-z^3)(1+z^3)} = \frac{z+z^2}{1+z^3}.$$

Since $\int_{(0,\infty)} x^{-2}\,|\phi|(dx) < \infty$, the Laplace side (Theorem 5.11) at one unit exponent and the index 2 reads $\Gamma(2)\,(\mathsf{T}_{2;(1)}[x]\,\phi)((0,\infty)) = \int_0^\infty y\,\mathcal{F}_0(y)\,dy$, and

$$P_{\binom{\mathcal{E}}{2}}(\mathbf{1};1,1) = 2\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+6\mathbb{Z}} + \mathbb{I}_{2+6\mathbb{Z}} - \mathbb{I}_{4+6\mathbb{Z}} - \mathbb{I}_{5+6\mathbb{Z}}\big))((0,\infty)).$$

Cosets and dilations. $1 + 3\mathbb{Z} = (1+6\mathbb{Z}) \sqcup (4+6\mathbb{Z})$ and $2+3\mathbb{Z} = (2+6\mathbb{Z}) \sqcup (5+6\mathbb{Z})$, so

$$\phi = \mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}} + 2\big(\mathbb{I}_{2+6\mathbb{Z}} - \mathbb{I}_{4+6\mathbb{Z}}\big),$$

and the Trailmoment, an integral against its input (Definition 5.1), splits accordingly, every part converging. The dilation $\tau(x) = 2x$ carries $\mathbb{I}_{1+3\mathbb{Z}}$ to $\mathbb{I}_{2+6\mathbb{Z}}$ and $\mathbb{I}_{2+3\mathbb{Z}}$ to $\mathbb{I}_{4+6\mathbb{Z}}$, so transport at one unit exponent (Theorem 5.9 and the paragraph after it) gives

$$(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{2+6\mathbb{Z}} - \mathbb{I}_{4+6\mathbb{Z}}\big))((0,\infty)) = 2^{-2}\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}\big))((0,\infty)),$$

and

$$P_{\binom{\mathcal{E}}{2}}(\mathbf{1};1,1) = 2\,\big(1+2\cdot 2^{-2}\big)\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}\big))((0,\infty)) = 3\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}\big))((0,\infty)),$$

which is $2.343907238689458890\ldots$ to eighteen places. The receipt checks the identities of rational functions exactly, the decomposition of the input at every $m \leq 600$, the decomposition and the dilation together on the exact heads $\sum_{m\leq 6N} \phi(m) m^{-2}$, and the integral of the node form against the Trailmoment to twenty-five places.

Answer. F is convergent, with every proper wall 1. The pair $S = \{1,2\}$ is a series bunch of width one with $\gamma_S^K = \alpha_1\alpha_2/(\alpha_1+\alpha_2)$, $\gamma_S^\Psi = \alpha_1 + \alpha_2$ and $\varrho_S = 1/(u(1-u))$; it is not pinned, so it does not split off. At the root bunch S the Skeleton values are Q^{-1} and 1, $Q = u(1-u)$; the node form is $\int_0^1 Q^{-1} \Gamma_2^2 \begin{bmatrix} 1, & 1 \\ 1, & 1 \end{bmatrix} \mathsf{B}^{\{J\}}_{(1,1),1;I_J}(Q^{-1},1)\,du = \int_0^1 \frac{\log Q}{Q-1}\,du$, and its share side reads the stacked Gamma $\Gamma_0^4\big[a,\ a,\ 1-a,\ 1-a\big]$ against Q^{-a}. The period is

$$P_{\binom{[3]}{2}}(\mathbf{1};1,1) = 3\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}\big))((0,\infty)).$$

□

Exercise 51.2 (The list $\binom{[4]}{2}$ at its centre)**.** Let $\mathcal{E} = \{1,2,3,4\}$ and $F = \big(\binom{\mathcal{E}}{2}; 1, 1\big)$ at the masses $\mathbf{1}$.

(i) Find the widths and the walls of the sets of marks. Show that the sets of width one with at least two marks are the complements $R = \mathcal{E} \setminus \{e\}$ of single marks, that the cell of each is a dual pair (Definition 21.1), and that $\binom{\mathcal{E}}{2} = \binom{\{e,h\}}{1}\langle e, \mathcal{A}_R\rangle$ with the mark e left in place.

(ii) Show that $P^{\mathrm{mot}}_{\binom{[4]}{2}} = P^{\mathrm{mot}}_{\binom{[3]}{2}}$, and find the period.

(iii) Show that no pair of marks splits off.

Solution. Receipt: `ex2_dualpair_u24.py`.

(i) *Letters, widths, walls.* The complement of a pair is a pair, so $K = \Psi = e_2(\alpha)$ (Definition 7.1). A set S has $r(S) = \min(|S|, 2)$, and by Definition 7.12 its width is $c(S) = r(S) + r(\mathcal{E} \setminus S) - 2$:

$$c(\{e\}) = 1 + 2 - 2 = 1, \qquad c(\{e, f\}) = 2 + 2 - 2 = 2, \qquad c(\mathcal{E} \setminus \{e\}) = 2 + 1 - 2 = 1.$$

At the centre the walls are the widths (Proposition 8.21(ii)): 1, 2 and 1. Every proper wall is at least 1, and F is convergent (Definition 34.1).

The cells. The sets of width one with at least two marks are the triples $R = \mathcal{E} \setminus \{e\}$. The restriction $\mathcal{B}|R$ has the three pairs of R and the contraction $\mathcal{B}/\{e\}$ the three marks of R, the bases through e less e (Definition 7.5). The cell of the bunch R (Theorem 19.4) has the bases of $\mathcal{B}|R = \binom{R}{2}$ and the sets $X + g$, $X \in \mathcal{B}/\{e\} = \binom{R}{1}$:

$$\mathcal{A}_R = \mathcal{B}_R^+ = \binom{R+g}{2}, \qquad \mathcal{A}_R \setminus g = \binom{R}{2}, \qquad \mathcal{A}_R/g = \binom{R}{1}.$$

The complements in R of the pairs of R are its marks, so $\mathcal{A}_R/g$ is the dual table of $\mathcal{A}_R \setminus g$, the weights being 1: the cell is a dual pair (Definition 21.1). The cell of the room $\{e\}$ has the basis $\{e\}$ of $\mathcal{B}|\{e\}$ and the set $\varnothing + g$, $\varnothing$ being the one basis of $\mathcal{B}/R$: it is $\binom{\{e,g\}}{1}$. By Proposition 21.3(iii) the list is the gluing of the two cells, that is $\binom{\{e,h\}}{1}\langle e, \mathcal{A}_R\rangle$ with $\mathcal{A}_R$ at h (Definition 21.1). On the bases: the core basis $\{e\}$ avoids h and takes $X \in \mathcal{A}_R/g$, giving the three pairs through e; the core basis $\{h\}$ takes $X \in \mathcal{A}_R \setminus g$, giving the three pairs of R.

The letters of the substitution. For $e = 4$ and $R = \{1, 2, 3\}$ the cell has $k = K_{\mathcal{A}_R \setminus g} = e_2(\alpha_R)$ and $\psi = \Psi_{\mathcal{A}_R \setminus g} = e_1(\alpha_R)$, and Proposition 21.8(ii) gives the one hinge value $\gamma := \gamma^K_{\mathcal{A}_R} = \gamma^\Psi_{\mathcal{A}_R} = e_2(\alpha_R)/e_1(\alpha_R)$, with $\varrho_{\mathcal{A}_R} = 1$. The mark 4 left in place has $k = \alpha_4$ and $\psi = 1$, so $\pi = e_1(\alpha_R)$, and Proposition 21.8(iii), with $r_{\mathcal{H}} = 1$ and $q = 2$, reads

$$K_{\mathcal{H}}\big(\pi\alpha_4,\ e_2(\alpha_R)\big) = e_1(\alpha_R)\,\alpha_4 + e_2(\alpha_R) = e_2(\alpha) = \pi^0 K, \qquad \Psi_{\mathcal{H}}\big(\pi\alpha_4,\ e_2(\alpha_R)\big) = e_2(\alpha) = \pi^0 \Psi,$$

$K_{\mathcal{H}}$ and $\Psi_{\mathcal{H}}$ both being $x_4 + x_h$.

(ii) *The dual-pair factorization.* Corollary 34.18(a) applies with $q = 1$ at $s = t = 1$. The wall of the cell is

$$\kappa_R = m_R - (s+t)\big(r(R) - 1\big) = 3 - 2\,(2 - 1) = 1,$$

the wall of F at R found above. The pinned family has the form of $(\mathcal{B}|R; \kappa_R, s+t-\kappa_R) = \big(\binom{R}{2}; 1, 1\big)$ at the masses $\mathbf{1}_R$, and the core is $\big(\binom{\{e,h\}}{1}; 1, 1\big)$ at the masses $m_e = 1$ and $\kappa_R = 1$ at h:

$$P^{\mathrm{mot}}_{\binom{[4]}{2}} = P^{\mathrm{mot}}_{(\binom{\{e,h\}}{1};1,1)} \cdot P^{\mathrm{mot}}_{(\binom{R}{2};1,1)}.$$

The coefficients of the forms show the factorization: with γ as above,

$$\frac{\alpha^{\mathbf{1}}}{K\,\Psi} = \frac{\alpha_4\,\gamma}{(\alpha_4 + \gamma)^2} \cdot \frac{\alpha^{\mathbf{1}_R}}{e_1(\alpha_R)\,e_2(\alpha_R)},$$

the coefficient of the core at (α_4, γ) times that of $\big(\binom{R}{2}; 1, 1\big)$, an identity of rational functions.

The core. The core is a family on two marks with both letters $\alpha_e + \alpha_h$ and the walls $\kappa_{\{e\}} = \kappa_{\{h\}} = 1$; its form is $\Omega/(\alpha_e + \alpha_h)^2$, and by Lemma S18 its motivic period is its Beta corner,

$$P^{\mathrm{mot}}_{(\binom{\{e,h\}}{1};1,1)} = \Gamma_{\!2\atop 1}\!\begin{bmatrix} 1,\ 1 \\ 2 \end{bmatrix} = \frac{\Gamma(1)\,\Gamma(1)}{\Gamma(2)} = 1.$$

The pinned family $\binom{R}{2};1,1)$ at $\mathbf{1}_R$ is $\binom{[3]}{2}$ at its centre with its marks named by R; renaming the coordinates is an isomorphism of the models that maps the positive cells and the forms onto each other, so the two have one motivic period (Lemma 34.7). Hence

$$P^{\mathrm{mot}}_{\binom{[4]}{2}} = P^{\mathrm{mot}}_{\binom{[3]}{2}}, \qquad \mathrm{C}\big(P^{\mathrm{mot}}_{\binom{[4]}{2}}\big) = \mathrm{C}\big(P^{\mathrm{mot}}_{\binom{[3]}{2}}\big),$$

and by Exercise 51.1 the period is $P_{\binom{[4]}{2}}(\mathbf{1};1,1) = 3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))$.

(iii) *The pairs.* A pair has width two in the one active table, so by Corollary 28.26(i) it does not split off, and its law is carried by no point. The list is connected and its only sets with two marks on each side are the pairs, so it is prime (Theorem 23.16), and Corollary 28.26(ii) says the same. The complements of the marks do split off, as in (ii): $\mathbf{E}_{\mathcal{B}\setminus e} = e_2(\alpha_R)\,e_1(\alpha_R) = \mathbf{E}_{\mathcal{B}/e}$ (Corollary 28.26(iv)).

Answer. The marks and the triples have width one and the pairs width two, and at the centre these are the walls. Every triple $R = \mathcal{E}\setminus\{e\}$ has the cell $\binom{R+g}{2}$, a dual pair, and $\binom{\mathcal{E}}{2} = \binom{\{e,h\}}{1}\langle e, \binom{R+g}{2}\rangle$. With $\kappa_R = 1$, the core on two marks has the motivic period $\Gamma^{\,2}_{\,1}\left[\begin{matrix}1,\ 1\\ 2\end{matrix}\right] = 1$, so

$$P^{\mathrm{mot}}_{\binom{[4]}{2}} = P^{\mathrm{mot}}_{\binom{[3]}{2}}, \qquad P_{\binom{[4]}{2}}(\mathbf{1};1,1) = 3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty)).$$

No pair splits off: pairs have width two. □

Exercise 51.3 (The list $\binom{[4]}{1}$ at its full leans and at its centre)**.** Let $\mathcal{E} = \{1,2,3,4\}$ and $\mathcal{B} = \binom{\mathcal{E}}{1}$.

(i) Show that at its full leans $(s,0)$ and $(0,t)$, at positive integer masses on the balance wall with every proper wall at least 1, the periods are rational, and so are the motivic periods.

(ii) At the centre, show that $S = \{3,4\}$ has width one, find its hinge values and its law (Definition 28.5), and show that the law is not a point mass.

(iii) Read the period at the root bunch S, write it on the share side of the node form, show that it is the critical-line word $\mathrm{crit}_1[1]$ (Definition 36.3), and evaluate it.

Solution. Receipt: `ex3_critical_u14.py`.

(i) *Counts.* The bases are the four marks, so $K = e_1(\alpha)$ and $\Psi = e_3(\alpha)$ (Definition 7.1), of the rank 1 and the corank 3. A proper set S has $d(S) = 0$, a basis avoiding it, $r(S) = 1$, $d^*(S) = |S|-1$ and $c(S) = 1$ (Theorem 7.13, Definition 7.12).

The full lean $(s,0)$. The balance wall is $m_{\mathcal{E}} = s$, and the walls are $\kappa_S = m_S - s\,d(S) = m_S$ (Definition 8.1), at least 1 at positive integer masses. On the shape simplex $K(u) = |u| = 1$, so (Theorems 8.7 and 3.9)

$$P_{\mathcal{B}}(\mathbf{m};s,0) = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m-1}}\,d\omega_{\mathcal{E}}(u) = \Gamma^{\,4}_{\,1}\left[\begin{matrix}m_1,\ m_2,\ m_3,\ m_4\\ s\end{matrix}\right],$$

a positive rational number; at $(4,0)$ and the masses $\mathbf{1}$ it is $\Gamma^{\,4}_{\,1}\left[\begin{matrix}1,\ 1,\ 1,\ 1\\ 4\end{matrix}\right] = \frac{1}{6}$.

The full lean $(0,t)$. The balance wall is $m_{\mathcal{E}} = 3t$. By Corollary 34.8, with $2\sigma = t$, the family $(\mathcal{B};0,t)$ at $\mathbf{m}$ has the motivic period of $(\mathcal{B};t,0)$ at $t\mathbf{1}-\mathbf{m}$, whose total is $4t-3t = t$ and whose walls are those of $(\mathcal{B};0,t)$ at the complements, so

$$P_{\mathcal{B}}(\mathbf{m};0,t) = \Gamma^{\,4}_{\,1}\left[\begin{matrix}t-m_1,\ t-m_2,\ t-m_3,\ t-m_4\\ t\end{matrix}\right],$$

rational again; the walls of $(\mathcal{B};0,t)$ at the complements of the marks are $t-m_e\ge 1$.

The motivic periods. The set $\{3,4\}$ has width one with two marks on each side, and by Proposition 21.3(iii) the list is the gluing of its two cells: the cell of the bunch has the bases $\{3\},\{4\}$ of $\mathcal{B}|\{3,4\}$ and g, the one basis of $\mathcal{B}/\{1,2\}$ being $\varnothing$, and the cell of the room has the bases $\{1\},\{2\},\{g\}$. Both are the parallel nodes $\binom{\{3,4,g\}}{1}$ and $\binom{\{1,2,g\}}{1}$ on three marks, so the splitting of Corollary 34.18(b) has no prime node, and at either full lean P_F^{mot} is a positive rational number c. Its period is c, so $c=P_F$: the motivic periods are the rational numbers displayed.

(ii) *The centre.* At $(1,1)$ and the masses $\mathbf{1}$ every proper wall is the width 1 (Proposition 8.21(ii)), and $F=(\mathcal{B};1,1)$ is convergent. The pair $S=\{3,4\}$, with the room $S'=\{1,2\}$, has width one, and no basis contains both of its marks: it is a parallel bunch (Definition 19.1). The restriction $\mathcal{B}|S$ has the bases $\{3\},\{4\}$, and the contraction $\mathcal{B}/S'$, the bases meeting S' in one mark less that mark, has the one basis $\varnothing$ (Definition 7.5), whose complement in S is S. So (Definition 19.2, Proposition 19.3)

$$\gamma_S^K=\frac{\alpha_3+\alpha_4}{1},\qquad \gamma_S^\Psi=\frac{\alpha_3\alpha_4}{\alpha_3+\alpha_4},\qquad \varrho_S=\frac{\alpha_3\alpha_4}{(\alpha_3+\alpha_4)^2}=u(1-u)$$

on the shape $(u,1-u)$ of S, and likewise $\varrho_{S'}=u'(1-u')$ on the shape $(u',1-u')$ of S'.

The law. At width one the stations of the room are D_0 and D_1, of the excesses 0 and 1, and the seam letters are the letters of $\mathcal{B}/S'$ and of $\mathcal{B}|S$ (Proposition 28.3(ii)):

$$V_{D_0}=K_{\mathcal{B}/S'}=1,\qquad V_{D_1}=K_{\mathcal{B}|S}=\alpha_3+\alpha_4,$$
$$V_{D_0}^*=\Psi_{\mathcal{B}/S'}=\alpha_3\alpha_4,\qquad V_{D_1}^*=\Psi_{\mathcal{B}|S}=\alpha_3+\alpha_4.$$

With $d(S)=0$ and $d^*(S)=1$ the wall of the piece is $\kappa_S=2-(1\cdot 0+1\cdot 1)=1$ and its top wall is $(1+1)\cdot 1-1=1$ (Definition 28.4). Definition 28.5 gives the scale, the scale form on the shape $(u,1-u)$, and the seam vector $(\varrho^\rho_{D_0},\varrho^\rho_{D_1},\varrho^{\rho*}_{D_0},\varrho^{\rho*}_{D_1})$:

$$\gamma_S^\rho=\Big(\frac{\alpha_3+\alpha_4}{1}\Big)^{1/2}\Big(\frac{\alpha_3\alpha_4}{\alpha_3+\alpha_4}\Big)^{1/2}=(\alpha_3\alpha_4)^{1/2},$$
$$\phi_S^\rho(du)=1^{-1}\big(u+(1-u)\big)^{-1}\big(u(1-u)\big)^{-1/2}\,du=\varrho_S^{-1/2}\,du,$$
$$\boldsymbol{\varrho}_S^\rho=\Big(1,\ \frac{\alpha_3+\alpha_4}{(\alpha_3\alpha_4)^{1/2}},\ \frac{\alpha_3\alpha_4}{(\alpha_3+\alpha_4)(\alpha_3\alpha_4)^{1/2}},\ 1\Big)=\big(1,\ \varrho_S^{-1/2},\ \varrho_S^{1/2},\ 1\big).$$

The seam torus is cut out by $x_{D_0}=1$, $x^*_{D_1}=1$ and $x_{D_1}x^*_{D_0}=1$: it is the curve of the points $(1,x,x^{-1},1)$, $x>0$. The law is the image of $\varrho_S^{-1/2}du$ under $x=\varrho_S^{-1/2}=(u(1-u))^{-1/2}$. Since $u(1-u)\le\frac14$ it is carried by $x\ge 2$; each $x>2$ has the two preimages u and $1-u$, with $(1-2u)\,du=-2x^{-3}dx$ and $|1-2u|=(1-4x^{-2})^{1/2}$, so the law is

$$\frac{4\,dx}{x\,(x^2-4)^{1/2}}\quad\text{on } x>2,\qquad \text{of total mass } \int_0^1\big(u(1-u)\big)^{-1/2}du=\Gamma_{\substack{2\\1}}\!\left[\begin{matrix}\frac12,\ \frac12\\ 1\end{matrix}\right]$$

(Theorem 3.9). It has a density, so it is not a point mass. Theorem 28.25 says the same through the hinge values: $\mathbf{E}_{\mathcal{B}|S}=(\alpha_3+\alpha_4)^2$ and $\mathbf{E}_{\mathcal{B}/S'}=\alpha_3\alpha_4$ differ, ϱ_S not being constant, so S is not pinned and does not split off.

(iii) *The node form at the root bunch S.* As in Exercise 51.1(ii), a tree of bunches whose root carries S has the step $\mathcal{E}$ as a node step of the root (Definition 34.9(ii)), and the period is read by the node form there (Theorem 34.16(v)). Read $\alpha_S=\rho\,(u,1-u)$ and $\alpha_{S'}=(u',1-u')$ on the section $\alpha_1+\alpha_2=1$; then $d\alpha_3\,d\alpha_4=\rho\,d\rho\,du$ (Lemma 2.7) and

$$K(\rho u,u')=\rho+1,\qquad \Psi(\rho u,u')=\rho\,u'(1-u')+\rho^2u(1-u)=\rho\,\varrho_S\Big(\rho+\frac{\varrho_{S'}}{\varrho_S}\Big):$$

the Skeleton values (Theorem 34.16(ii)) are $\xi_K = 1$ and $\xi_\Psi = \varrho_{S'}/\varrho_S = \gamma^\Psi_{S'}/\gamma^\Psi_S$, the hinge value of the room over that of the bunch on $\Delta_S \times \Delta_{S'}$ (Theorem 34.16(iv)). By Theorem 8.7,

$$P_{\mathcal{B}}(\mathbf{1}; 1, 1) = \int_{\Delta_S \times \Delta_{S'}} \int_0^\infty \frac{d\rho}{(\rho+1)\,(\varrho_S\,\rho + \varrho_{S'})}\, d\omega_S(u)\, d\omega_{S'}(u'),$$

the node form with one term: $A_t = \varrho_S^{-1}$, $B_t = 1$, $\nu_t = 1$ and $\mathbf{w}_t = (1,1)$, so that $\mathrm{Sc}_t = \Gamma_{\,2}^{\,2}\!\begin{bmatrix} 1,\ 1 \\ 1,\ 1 \end{bmatrix} \mathsf{B}^{\{J\}}_{(1,1),\,1;\,I_J}\big(1, \varrho_{S'}/\varrho_S\big)$: the bunch and the room meet through the one Skeleton value $\varrho_{S'}/\varrho_S$.

The share side. Corollary 15.5, with $\xi_0 = 1$, $\xi_1 = \varrho_{S'}/\varrho_S$ and one share a, writes the term as

$$\begin{aligned} A_t\,\mathrm{Sc}_t &= \varrho_S^{-1}\,\frac{1}{2\pi i}\int_{(c)} \Gamma_{\,0}^{\,4}\big[a,\ a,\ 1-a,\ 1-a\big]\,\Big(\frac{\varrho_{S'}}{\varrho_S}\Big)^{a-1} da \\ &= \frac{1}{2\pi i}\int_{(c)} \Gamma_{\,0}^{\,4}\big[a,\ a,\ 1-a,\ 1-a\big]\ \varrho_S^{-a}\,\varrho_{S'}^{a-1}\, da. \end{aligned}$$

On the line $\mathrm{Re}\,a = \frac12$ the two conditions of Theorem 34.16(iv) are $\int_{\Delta_S} \varrho_S^{-1/2} d\omega_S < \infty$ and $\int_{\Delta_{S'}} \varrho_{S'}^{-1/2} d\omega_{S'} < \infty$, both $\Gamma_{\,1}^{\,2}\!\begin{bmatrix} \frac12,\ \frac12 \\ 1 \end{bmatrix}$; so the bunch and the room separate under the share integral, and by Theorem 3.9, for $0 < \mathrm{Re}\,a < 1$,

$$\int_0^1 \big(u(1-u)\big)^{-a} du = \Gamma_{\,1}^{\,2}\!\begin{bmatrix} 1-a,\ 1-a \\ 2-2a \end{bmatrix}, \qquad \int_0^1 \big(u'(1-u')\big)^{a-1} du' = \Gamma_{\,1}^{\,2}\!\begin{bmatrix} a,\ a \\ 2a \end{bmatrix}.$$

So

$$P_{\mathcal{B}}(\mathbf{1}; 1, 1) = \frac{1}{2\pi i}\int_{(1/2)} \Gamma_{\,0}^{\,4}\big[a,\ a,\ 1-a,\ 1-a\big]\ \Gamma_{\,1}^{\,2}\!\begin{bmatrix} 1-a,\ 1-a \\ 2-2a \end{bmatrix}\ \Gamma_{\,1}^{\,2}\!\begin{bmatrix} a,\ a \\ 2a \end{bmatrix}\, da :$$

the moments of the orders $-a$ and $a-1$ of the two ratios of hinge values, paired over one share. This is the one-share display of Theorem 19.13 at the level shares $a^d = b^d = a$ and $a^r = b^r = 1-a$, the bunch family and the room family having these two Beta corners as periods: the even share $\sigma^d = a$ is free and the odd share is 0.

The critical-line word. By the reflection at a and at $2a$ and the shift $\Gamma(2-2a) = (1-2a)\,\Gamma(1-2a)$ (Proposition 1.3(iii),(v)),

$$\Gamma_{\,1}^{\,2}\!\begin{bmatrix} a,\ a \\ 2a \end{bmatrix}\Gamma_{\,1}^{\,2}\!\begin{bmatrix} 1-a,\ 1-a \\ 2-2a \end{bmatrix} = \frac{\Gamma_{\,0}^{\,4}\big[a,\ a,\ 1-a,\ 1-a\big]}{(1-2a)\,\Gamma_{\,0}^{\,2}\big[2a,\ 1-2a\big]} = \frac{\pi^2}{\sin^2\pi a}\cdot\frac{\sin 2\pi a}{\pi\,(1-2a)} = \frac{2\pi\cot\pi a}{1-2a},$$

and $\partial_a \Gamma_{\,0}^{\,4}\big[a,\ a,\ 1-a,\ 1-a\big] = \partial_a\big(\pi^2/\sin^2\pi a\big) = -2\pi\cot\pi a\cdot\Gamma_{\,0}^{\,4}\big[a,\ a,\ 1-a,\ 1-a\big]$. So

$$P_{\mathcal{B}}(\mathbf{1}; 1, 1) = -\frac{1}{2\pi i}\int_{(1/2)} \frac{\partial_a \Gamma_{\,0}^{\,4}\big[a,\ a,\ 1-a,\ 1-a\big]}{1-2a}\, da = \mathrm{crit}_1[1],$$

Definition 36.3 at $k = 1$ and $R = 1$, where $-\frac{2}{(2k)!} = -1$.

Closing. The integrand is holomorphic on $0 < \mathrm{Re}\,a < 1$, regular at $a = \frac12$ (Definition 36.3), and on every vertical line of the strip it is $O(e^{-2\pi|\,\mathrm{Im}\,a|})$, since $|\sin\pi a|^2 = \sin^2\pi\,\mathrm{Re}\,a + \sinh^2\pi\,\mathrm{Im}\,a$; so

the line moves to $\operatorname{Re} a = c$ with $\frac{1}{2} < c < 1$, and an integration by parts, with no boundary terms, gives

$$P_{\mathcal{B}}(\mathbf{1};1,1) = \frac{1}{2\pi i}\int_{(c)} \Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\ g(a)\,da, \qquad g(a) = \partial_a \frac{1}{1-2a} = \frac{2}{(1-2a)^2}.$$

The function g is rational, holomorphic on $\operatorname{Re} a \geq c$, and $O(|a|^{-2})$, so the right closing of Lemma 5.16(i) applies: with $g'(a) = 8(1-2a)^{-3}$,

$$P_{\mathcal{B}}(\mathbf{1};1,1) = -\sum_{m\geq 1} g'(m) = \sum_{m\geq 1} \frac{8}{(2m-1)^3} = \sum_{m\geq 1} \Big(m - \frac{1}{2}\Big)^{-3}.$$

(Closed directly, the integrand $-\partial_a \Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]/(1-2a)$ has at $a = m \geq 1$ a pole of order three with the residue $8(1-2m)^{-3}$; the receipt checks $m \leq 6$.)

The Trailmoment. At one unit exponent and the coefficient function x the last sum is $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty))$ (Definition 5.2). The lattice relation at the step 2 (Theorem 5.12(i), $q = 2$, $b = 0$, the cut 0) assembles $\frac{1}{2}\mathbb{Z}$ from the cosets $\mathbb{Z}$ and $\frac{1}{2}+\mathbb{Z}$, the coefficient function $x/2$ multiplying the kernel by 2^3:

$$(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = 2^3\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

So $P_{\mathcal{B}}(\mathbf{1};1,1) = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, which is $\mathrm{crit}_1[1]$ of Theorem 36.8; it is $8.414398322117159997\ldots$ to eighteen places. The receipt checks the relation on the exact heads $\sum_{m\leq N}(m-\frac{1}{2})^{-3} + \sum_{m\leq N} m^{-3} = 8\sum_{k\leq 2N} k^{-3}$ and the share integral against $7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ to twenty-five places. At a full lean one letter is active with one Skeleton value, no share is free, and the period is rational, by (i); at the centre the one free share runs over the critical line and carries a Trailmoment of the odd index three.

Answer. At $(s,0)$ the period is $\Gamma_{\substack{4\\1}}\begin{bmatrix}\mathbf{m}\\ s\end{bmatrix}$ and at $(0,t)$ it is $\Gamma_{\substack{4\\1}}\begin{bmatrix}t\mathbf{1}-\mathbf{m}\\ t\end{bmatrix}$, rational, and the motivic periods are these rational numbers, the list having no prime node. At the centre $S = \{3,4\}$ is a parallel bunch of width one with $\gamma_S^K = \alpha_3+\alpha_4$, $\gamma_S^\Psi = \alpha_3\alpha_4/(\alpha_3+\alpha_4)$ and $\varrho_S = u(1-u)$; its law is $4\,dx/(x(x^2-4)^{1/2})$ on the points $(1,x,x^{-1},1)$, $x > 2$, of its seam torus, not a point mass. Read at the root bunch S,

$$\begin{aligned} P_{\binom{[4]}{1}}(\mathbf{1};1,1) &= \frac{1}{2\pi i}\int_{(1/2)} \Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\,\Gamma_{\substack{2\\1}}\begin{bmatrix}1-a,\ 1-a\\ 2-2a\end{bmatrix}\Gamma_{\substack{2\\1}}\begin{bmatrix}a,\ a\\ 2a\end{bmatrix}\,da \\ &= \mathrm{crit}_1[1] = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)). \end{aligned}$$

□

51.2 Substitutions of dual pairs

Exercise 51.4 (Substitutions on five and on eight marks)**.** Let $\mathrm{P}_5 = \binom{\{1,2,h\}}{1}\langle 1,2,\mathcal{A}\rangle$, the marks $1,2$ left in place and the cell $\mathcal{A} = \binom{\{3,4,5,g\}}{2}$ at h, and let $\mathrm{M}_8 = \binom{\{b_1,b_2,h_1,h_2\}}{2}\langle b_1,b_2,\mathcal{A}_1,\mathcal{A}_2\rangle$, the marks b_1,b_2 left in place and the cells $\mathcal{A}_i = \binom{\{x_i,y_i,z_i,g_i\}}{2}$ at h_i, with $R = \{3,4,5\}$, $R_i = \{x_i,y_i,z_i\}$ and $B = \{b_1,b_2\}$ (Definition 21.1).

(i) Find the bases of P_5, show that its cell is a dual pair and that P_5 is convergent at its centre, and find its motivic period there.

(ii) Find the rank of M_8, the widths of R_1, R_2 and B, and its letters; show that it is convergent at its centre and find its motivic period there. What becomes of the bunch B?

Solution. Receipt: `ex4_substitution.py`, with the helper `subst.py`.

(i) *The bases.* By Definition 21.1, the core bases $\{1\}$ and $\{2\}$ avoid h and take a basis of $\mathcal{A}/g = \binom{R}{1}$, and the core basis $\{h\}$ takes a basis of $\mathcal{A} \setminus g = \binom{R}{2}$: the bases of P_5 are the six pairs $\{1,x\},\{2,x\}$, $x \in R$, and the three pairs of R, all the pairs of $\{1,\dots,5\}$ but $\{1,2\}$, and its rank is $1+(2-1)=2$ (Proposition 21.3(i)). The complements in R of the pairs of R are its marks, so $\mathcal{A}/g$ is the dual of $\mathcal{A} \setminus g$: the cell is a dual pair, and by Proposition 21.8(ii) both letters read it through the one hinge value $\gamma = e_2(\alpha_R)/e_1(\alpha_R)$. By Theorem 21.4, with $K_{\mathcal{A}/g} = \Psi_{\mathcal{A}\setminus g} = e_1(\alpha_R)$,

$$K = e_1(\alpha_R)\,\big(\alpha_1+\alpha_2+\gamma\big), \qquad \Psi = e_1(\alpha_R)\,\big(\alpha_1\alpha_2+(\alpha_1+\alpha_2)\,\gamma\big).$$

Walls. R is a bunch of width one with the cell $\mathcal{A}$ (Proposition 21.3(i)); its wall at the centre is $\kappa_R = m_R-(s+t)(r(R)-1) = 3-2\cdot 1 = 1$, $r(R)=2$ (Corollary 34.18(a)), and $2\sigma-\kappa_R = 1$. Both letters read R through γ, so by Theorem 19.9(iii) P_5 has every proper wall positive exactly when its two cell families do: $(\mathcal{A};1,1)$ at the masses $(\mathbf{1}_R,1)$, which is $\binom{[4]}{2}$ at its centre, and $\big(\binom{\{1,2,h\}}{1};1,1\big)$ at the masses $(1,1,\kappa_R)=\mathbf{1}$, which is $\binom{[3]}{1}$ at its centre; both are connected, and their walls are their widths (Proposition 8.21(ii)). So every proper wall of P_5 is positive, an integer and so at least 1 (the receipt finds the values 1 and 2), and P_5 is convergent at its centre.

The motivic period. Corollary 34.18(a) with $q=1$:

$$P^{\mathrm{mot}}_{\mathrm{P}_5} = P^{\mathrm{mot}}_{(\binom{\{1,2,h\}}{1};1,1)} \cdot P^{\mathrm{mot}}_{(\binom{R}{2};1,1)},$$

the core at the masses $\mathbf{1}$ and the pinned family $(\mathcal{B}|R;\kappa_R,2-\kappa_R) = \big(\binom{R}{2};1,1\big)$ at $\mathbf{1}_R$. The core has the letters e_1 and e_2 of $\alpha_1,\alpha_2,\alpha_h$, so its form is $\Omega/(e_1e_2)$, the form of $\binom{[3]}{2}$ at its centre, whose letters are e_2 and e_1; the pinned family is $\binom{[3]}{2}$ at its centre on the marks of R. Two convergent families with one form have one motivic period (Lemma 34.7, with the renaming of the coordinates), so

$$P^{\mathrm{mot}}_{\mathrm{P}_5} = \big(P^{\mathrm{mot}}_{\binom{[3]}{2}}\big)^2,$$

the two factors being one motivic period. In coefficients,

$$\frac{\alpha^{\mathbf{1}}}{K\,\Psi} = \frac{\alpha_1\alpha_2\,\gamma}{e_1(\alpha_1,\alpha_2,\gamma)\,e_2(\alpha_1,\alpha_2,\gamma)} \cdot \frac{\alpha^{\mathbf{1}_R}}{e_1(\alpha_R)\,e_2(\alpha_R)},$$

the core at $(\alpha_1,\alpha_2,\gamma)$ times $\binom{[3]}{2}$ on R.

(ii) *Rank and widths.* By Proposition 21.3(i) M_8 is a list of rank $2+(2-1)+(2-1)=4$ on eight marks, and each R_i is a bunch of width one with the cell $\mathcal{A}_i$; from the six core bases, with three choices at each cell, it has $6\cdot 3\cdot 3 = 54$ bases. The pairs of the core meet B in 0, 1 or 2 marks, so B has the width two in the core, and by Corollary 21.11 in M_8.

Letters. Both cells are dual pairs with $k_i = e_2(\alpha_{R_i})$, $\psi_i = e_1(\alpha_{R_i})$ and the hinge values $\gamma_i = e_2(\alpha_{R_i})/e_1(\alpha_{R_i})$, and the marks left in place have $k=\alpha_b$, $\psi=1$ (Proposition 21.8(ii),(iii)). With $\pi = e_1(\alpha_{R_1})\,e_1(\alpha_{R_2})$, $r_{\mathcal{H}}=2$ and $q=4$, Proposition 21.8(iii) gives, $K_{\mathcal{H}} = \Psi_{\mathcal{H}} = e_2$ being homogeneous of degree two,

$$\pi^2\,e_2(\alpha_{b_1},\alpha_{b_2},\gamma_1,\gamma_2) = \pi\,K, \qquad \pi^2\,e_2(\alpha_{b_1},\alpha_{b_2},\gamma_1,\gamma_2) = \pi^{4-2-1}\,\Psi,$$

so $K=\Psi = e_1(\alpha_{R_1})\,e_1(\alpha_{R_2})\,e_2(\alpha_{b_1},\alpha_{b_2},\gamma_1,\gamma_2)$: M_8 is identically self-dual, its cobases being its bases.

Walls. The walls of the cells are $\kappa_{R_i} = 3-2\,(2-1) = 1$ (Corollary 34.18(a)). Theorem 19.9(iii), at R_1 and then at R_2 in the cell of the room, both letters reading each R_i through γ_i, reduces the positivity of the walls to that of the three families $(\mathcal{A}_1; 1, 1)$, $(\mathcal{A}_2; 1, 1)$ and $(\binom{\{b_1,b_2,h_1,h_2\}}{2}; 1, 1)$, all at the masses $\mathbf{1}$ (the hinges carrying $2\sigma - \kappa_{R_i} = 1$ and $\kappa_{R_i} = 1$): three copies of $\binom{[4]}{2}$ at its centre, convergent by Exercise 51.2. So M_8 is convergent at its centre; its 254 proper walls are positive integers (the receipt lists them, the least being 1).

The motivic period. Corollary 34.18(a) with $q = 2$, the pinned families $\big(\binom{R_i}{2}; 1, 1\big)$ at $\mathbf{1}_{R_i}$ and the core $\binom{\{b_1,b_2,h_1,h_2\}}{2}$ at its centre:

$$P^{\mathrm{mot}}_{\mathrm{M}_8} = P^{\mathrm{mot}}_{\binom{[4]}{2}} \cdot \big(P^{\mathrm{mot}}_{\binom{[3]}{2}}\big)^2 = \big(P^{\mathrm{mot}}_{\binom{[3]}{2}}\big)^3,$$

the second equality by Exercise 51.2(ii), every factor at its centre. In coefficients,

$$\frac{\alpha^{\mathbf{1}}}{K\,\Psi} = \frac{\alpha_{b_1}\alpha_{b_2}\,\gamma_1\gamma_2}{e_2(\alpha_{b_1},\alpha_{b_2},\gamma_1,\gamma_2)^2} \cdot \prod_{i=1}^{2} \frac{\alpha^{\mathbf{1}_{R_i}}}{e_1(\alpha_{R_i})\,e_2(\alpha_{R_i})}.$$

The bunch B. B is left in place and keeps in M_8 the width two it has in the core (Theorem 34.17(v)); by Corollary 28.26(i) it does not split off and enters the period through a law carried by no point. The cells split off around it, and B goes whole into the core $\binom{[4]}{2}$, which splits further only at the complement of a single mark, a set of width one (Exercise 51.2). Along B, $K(\tau\alpha_B, \alpha_{R_1}, \alpha_{R_2}) = \pi\, e_2(\tau\alpha_{b_1}, \tau\alpha_{b_2}, \gamma_1, \gamma_2)$, so the Skeleton values of B in M_8 are those of the pair $\{1, 2\}$ of $\binom{[4]}{2}$ at the point $(\alpha_{b_1}, \alpha_{b_2}, \gamma_1, \gamma_2)$.

Answer. P_5 has as its bases the pairs of $\{1, \dots, 5\}$ other than $\{1, 2\}$, its cell $\binom{\{3,4,5,g\}}{2}$ is a dual pair, and $P^{\mathrm{mot}}_{\mathrm{P}_5} = (P^{\mathrm{mot}}_{\binom{[3]}{2}})^2$. M_8 has the rank 4; R_1, R_2 have width one and B width two; $K = \Psi = e_1(\alpha_{R_1})e_1(\alpha_{R_2})\,e_2(\alpha_{b_1}, \alpha_{b_2}, \gamma_1, \gamma_2)$, and $P^{\mathrm{mot}}_{\mathrm{M}_8} = (P^{\mathrm{mot}}_{\binom{[3]}{2}})^3$. So

$$P_{\mathrm{P}_5}(\mathbf{1}; 1, 1) = \Big(3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))\Big)^2,$$
$$P_{\mathrm{M}_8}(\mathbf{1}; 1, 1) = \Big(3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))\Big)^3,$$

with $\mathrm{C}(P^{\mathrm{mot}}_{\mathrm{P}_5}) \subseteq \mathrm{C}(P^{\mathrm{mot}}_{\binom{[3]}{2}}) \cdot \mathrm{C}(P^{\mathrm{mot}}_{\binom{[3]}{2}})$, and likewise with three factors for M_8 (Theorem 34.17(ii)). The bunch B of width two does not split off and goes whole into the core. □

51.3 Full leans

Exercise 51.5 (Full leans at the exponents $(0, 2)$)**.** Let $F = (\mathcal{B}; 0, 2)$ be a convergent family of one list on $\mathcal{E}$, $|\mathcal{E}| \geq 3$, at the masses $\mathbf{1}$ (Definition 34.1).

(i) Show that $|\mathcal{E}| = 2r$ and that $\kappa_S = 2r(S) - |S|$. At a set R of width one with at least two marks on each side, show that the two hinge walls κ_R and $\kappa_{\mathcal{E}\setminus R}$ are 1, that both cell families have the masses $\mathbf{1}$, and that the Beta corner of the split is 1.

(ii) Show that every node of the splitting of Corollary 34.18(b) is prime and that $P^{\mathrm{mot}}_F = \prod_v P^{\mathrm{mot}}_{F_v}$, every $F_v = (\mathcal{H}_v; 0, 2)$ at the masses $\mathbf{1}$.

(iii) Let T_6 be the list of rank three on $\{1, \dots, 6\}$ whose bases are the nineteen triples other than $\{1, 2, 3\}$, let $\mathcal{T}$ be the cell T_6 on $\{1, \dots, 5\} \sqcup \{g\}$ with the mark 6 as its hinge g, let $\mathcal{T}'$ be

a copy of it on $\{1',\dots,5'\}\sqcup\{g\}$, and let $\mathrm{D}_{10}=\mathcal{T}\oplus_g\mathcal{T}'$ (Lemma 21.12). Find the motivic period of $(\mathrm{D}_{10};0,2)$ at the masses $\mathbf{1}$, and decide whether $\{1',\dots,5'\}$ splits off at the centre and at $(0,2)$.

Solution. Receipt: `ex5_fulllean.py`, with the helper `subst.py`.

(i) *Balance and walls.* At $(0,2)$ only Ψ is active, $\sigma=\frac{0+2}{2}=1$, and the imbalance is $\kappa_{\mathcal{E}}=n-2\,(n-r)=2r-n$ (Definition 8.1); on the balance wall $n=2r$. The walls are $\kappa_S=m_S-2\,d^*(S)$ with $d^*(S)=|S|-r(S)$ (Definition 7.12), so

$$\kappa_S=|S|-2\big(|S|-r(S)\big)=2r(S)-|S|.$$

The hinge walls. For a set R of width one, Proposition 8.16 gives

$$\kappa_R+\kappa_{\mathcal{E}\setminus R}=\kappa_{\mathcal{E}}+2\sigma\,c(R)=0+2\cdot1\cdot1=2,$$

and both walls are integers at least 1, F being convergent; so $\kappa_R=\kappa_{\mathcal{E}\setminus R}=1$. By Proposition 21.3(iii) the list is $\mathcal{H}\langle\mathcal{A}\rangle$, the cell of the bunch $\mathcal{A}$ on $R\sqcup\{g\}$ substituted at the marker h of the cell of the room $\mathcal{H}$, the marks of $L=\mathcal{E}\setminus R$ left in place. One letter is active, so the cell is read through the one hinge value $\gamma^{\Psi}_{\mathcal{A}}$, and Theorem 34.17 applies with $q=1$, $2\sigma=2$, $\kappa_1=\kappa_R=1$ and the loads $a=0$, $b=1$. The core family $(\mathcal{H};0,2)$ carries the masses $\mathbf{1}_L$ and $\kappa_R=1$ at h, and the cell family $(\mathcal{A};0,2)$ the masses $\mathbf{1}_R$ and $2\sigma-\kappa_R=1$ at g: both have the masses $\mathbf{1}$, and both are convergent (Theorem 34.17(i),(iii)). The pinned family G has the tables $\mathcal{A}\setminus g$ at $(0,2-1)=(0,1)$ and $\mathcal{A}/g$ at $(0,1)$, at the masses $\mathbf{1}_R$. The Beta corner of the split is

$$\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}\kappa_1,\ 2\sigma-\kappa_1\\ 2\sigma\end{bmatrix}=\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}1,\ 1\\ 2\end{bmatrix}=\frac{\Gamma(1)\,\Gamma(1)}{\Gamma(2)}=1.$$

So Theorem 34.17(ii),(iii) give $P^{\mathrm{mot}}_F=P^{\mathrm{mot}}_{(\mathcal{H};0,2)}P^{\mathrm{mot}}_G$ and $P^{\mathrm{mot}}_{(\mathcal{A};0,2)}=1\cdot P^{\mathrm{mot}}_G$, that is

$$P^{\mathrm{mot}}_F=P^{\mathrm{mot}}_{(\mathcal{H};0,2)}\,P^{\mathrm{mot}}_{(\mathcal{A};0,2)},$$

both at the masses $\mathbf{1}$: the split costs no factor.

(ii) *The nodes.* Each of the two families of (i) is again a convergent family of one list at $(0,2)$ with the masses $\mathbf{1}$, so (i) applies to it: it has an even number $2r_v$ of marks, and every further split costs no factor. A node made by a split has at least three marks, two of one side and its marker, and an even number of them, hence at least four; so no node on two or three marks occurs. A node at which the splitting stops has no set of width one with two marks on each side, and it is connected, since $\kappa_S+\kappa_{\mathcal{E}_v\setminus S}=2c(S)$ with both walls at least 1 gives $c(S)\geq1$ (Proposition 8.16, Definition 8.1); so it is prime (Theorem 23.16), as Corollary 34.18(b) says of the nodes on at least four marks. By induction on the number of marks,

$$P^{\mathrm{mot}}_F=\prod_v P^{\mathrm{mot}}_{F_v},\qquad F_v=(\mathcal{H}_v;0,2)\text{ at the masses }\mathbf{1},\qquad \mathrm{C}\big(P^{\mathrm{mot}}_F\big)\subseteq\prod_v\mathrm{C}\big(P^{\mathrm{mot}}_{F_v}\big)$$

(Lemma 34.6(iii)): the rational constant of Corollary 34.18(b) is 1, and every marker carries the mass 1. By Corollary 34.8 the same holds at $(2,0)$, whose masses $2\sigma\mathbf{1}-\mathbf{1}$ are again $\mathbf{1}$.

(iii) *The list* T_6. Every set of at most three marks other than $\{1,2,3\}$ lies in a basis, $r(\{1,2,3\})=2$, and every set of at least four marks contains a basis; $n=6=2r$, so T_6 at $(0,2)$ with the masses $\mathbf{1}$ is on its balance wall. By (i) its walls are $2r(S)-|S|$:

S	$\lvert S\rvert=1$	$\lvert S\rvert=2$	$\lvert S\rvert=3,\ S\neq\{1,2,3\}$	$S=\{1,2,3\}$	$\lvert S\rvert=4$	$\lvert S\rvert=5$
$r(S)$	1	2	3	2	3	3
κ_S	1	2	3	1	2	1

all at least 1: $(\mathrm{T}_6; 0, 2)$ is convergent. Its widths $c(S) = r(S) + r(\mathcal{E} \setminus S) - 3$ (Definition 7.12) are 1 at one mark, 2 at two marks, 3 at a triple other than $\{1, 2, 3\}$ and $\{4, 5, 6\}$ and 2 at these two: the sets with two marks on each side have widths two and three, and T_6 is connected and prime (Theorem 23.16). The mark 6 is neither a loop nor a coloop, so $\mathcal{T}$ is a cell, with the hinge minors

$$\mathcal{T} \setminus g: \text{ the nine triples of } \{1, \dots, 5\} \text{ other than } \{1, 2, 3\}, \qquad \mathcal{T}/g: \text{ the ten pairs of } \{1, \dots, 5\},$$

of the ranks 3 and 2 (Definition 7.3).

The list D_{10}. By Lemma 21.12 it is a list on ten marks of rank $3 + 3 - 1 = 5$, with the $9 \cdot 10 + 10 \cdot 9 = 180$ bases $X \sqcup Y$, $X \in \mathcal{T} \setminus g$ and $Y \in \mathcal{T}'/g$ or $X \in \mathcal{T}/g$ and $Y \in \mathcal{T}' \setminus g$, and $S = \{1', \dots, 5'\}$ is a bunch of width one with the cell of the bunch $\mathcal{T}'$ and the cell of the room $\mathcal{T}$. At $(0, 2)$ with the masses $\mathbf{1}$ it is on its balance wall, $10 = 2 \cdot 5$, and its hinge walls are (Lemma 21.12)

$$\kappa_S = m_S - 2\left(|S| - r_{\mathcal{T}'}\right) = 5 - 2\,(5 - 3) = 1, \qquad \kappa_{\mathcal{E}\setminus S} = 5 - 2\,(5 - 3) = 1,$$

adding to $2\sigma = 2$. One letter is active, so by Theorem 19.9(iii) every proper wall of D_{10} is positive exactly when both cell families are, $(\mathcal{T}'; 0, 2)$ and $(\mathcal{T}; 0, 2)$ at the masses $\mathbf{1}$, the hinge carrying $2\sigma - \kappa = 1$: two copies of $(\mathrm{T}_6; 0, 2)$ at $\mathbf{1}$, whose walls are displayed above. So $(\mathrm{D}_{10}; 0, 2)$ is convergent; the receipt finds the least of its 1022 proper walls to be 1.

The motivic period. The split at S gives the nodes $\mathcal{T}'$ and $\mathcal{T}$, copies of T_6 with the marker in place of the mark 6, both prime; by (ii), and Lemma 34.7 for the renaming of the marks,

$$P^{\mathrm{mot}}_{(\mathrm{D}_{10};0,2)} = \left(P^{\mathrm{mot}}_{(\mathrm{T}_6;0,2)}\right)^2, \qquad \mathrm{C}\big(P^{\mathrm{mot}}_{(\mathrm{D}_{10};0,2)}\big) \subseteq \mathrm{C}\big(P^{\mathrm{mot}}_{(\mathrm{T}_6;0,2)}\big) \cdot \mathrm{C}\big(P^{\mathrm{mot}}_{(\mathrm{T}_6;0,2)}\big),$$

all at the masses $\mathbf{1}$, with the Beta corner 1 and no other factor.

The set S *at the centre.* At $(1, 1)$ with the masses $\mathbf{1}$ the walls are the widths (Proposition 8.21(ii)), all positive, D_{10} being connected by (ii); so Theorem 28.25 applies. With both letters active and $|S| = 5$, S is pinned exactly when $\mathbf{E}_{\mathrm{D}_{10}|S} = \mathbf{E}_{\mathrm{D}_{10}/(\mathcal{E}\setminus S)}$, and these hinge minors are $\mathcal{T}' \setminus g$ and $\mathcal{T}'/g$ (Theorem 19.4), with 9 and 10 bases. At the gauge $\mathbf{1}$ the even letter of a list is the square of its number of bases, so the two even letters take the values 81 and 100 there and differ: at the centre S is not pinned, it does not split off, and its law is carried by no point. At $(0, 2)$ one letter is active, so S is pinned (Theorem 28.25(ii)) and splits off, with the Beta corner 1, as above.

Answer. At $(0, 2)$ with the masses $\mathbf{1}$, $|\mathcal{E}| = 2r$ and $\kappa_S = 2r(S) - |S|$; at a set of width one the two hinge walls add to 2 and are 1, the cell families keep the masses $\mathbf{1}$, and the Beta corner is $\Gamma_{\substack{2\\1}}\begin{bmatrix}1, & 1\\ & 2\end{bmatrix} = 1$. Every node of the splitting is prime, and $P^{\mathrm{mot}}_F = \prod_v P^{\mathrm{mot}}_{F_v}$ with every $F_v = (\mathcal{H}_v; 0, 2)$ at the masses $\mathbf{1}$. For $\mathrm{D}_{10} = \mathcal{T} \oplus_g \mathcal{T}'$, a list of rank five on ten marks with 180 bases,

$$P^{\mathrm{mot}}_{(\mathrm{D}_{10};0,2)} = \left(P^{\mathrm{mot}}_{(\mathrm{T}_6;0,2)}\right)^2 \quad \text{at the masses } \mathbf{1};$$

the set $\{1', \dots, 5'\}$ splits off at $(0, 2)$ and does not split off at the centre, where its hinge minors have the even letters 81 and 100 at the gauge $\mathbf{1}$. □

Chapter 52

Exercises on rays: the small centres

The exercises of this chapter read the list $\binom{[3]}{1}$ along its rays. Its centre ray is read four ways. A certificate proves its law at every strength, the low-end relation fixes the scale, and positivity with the native value of the direction brackets the value $\ell = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ to 10^{-7}. The box of its walls gives a spanning set of two insertions, the one-step matrix, a certificate supported on the stems and the law, with no closing computed, and shows why the closing relations alone do not give the law. The gauge end gives the values and their slopes to a hundred digits with no datum from the low end, and returns the low-end coefficients. The torus through the native gauge carries a second solution of the law; the torus value, the values and the torus read backwards are tied by an explicit connection, which reads the mean of $\log|K|$ on the torus and writes ℓ as three times a Trailmoment of index two of the lattice inputs of step three. A lean of three marks and the balanced directions of small size have their values in $\mathbb{Q} + \mathbb{Q}\ell$, read as values of insertions along the centre ray and reduced in its boxes. Every number displayed is exact unless it is labelled as a rendering; the receipts that recompute them are the scripts named in the solutions.

52.1 The centre ray, bracketed

Exercise 52.1 (The centre ray of $\binom{[3]}{1}$, bracketed)**.** Let F be the list $\binom{[3]}{1}$: the marks $\mathcal{E} = \{1, 2, 3\}$, the bases $\{1\}, \{2\}, \{3\}$ and unit weights, so that its letters are

$$K = \alpha_1 + \alpha_2 + \alpha_3, \qquad \Psi = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3.$$

Let $P(k) = P_F(k\mathbf{1}; k, k)$ be the values of its centre ray at the strength 1, and $\ell = P(1)$.

(i) From the closings of the ray find a candidate recurrence of the values, and prove it by a certificate.

(ii) Show that it holds at every k with $\operatorname{Re} k > 0$, and read its low-end relation.

(iii) Find a rational q with $0 < \ell - q < 10^{-6}$, and prove both inequalities.

Exact computation may be done by machine; floating point, quadrature and integer-relation searches serve only as checks.

Solution. Receipt: `ex18_centre_bracket.py`. Throughout, e_1, e_2, e_3 are the elementary symmetric polynomials of $\alpha_1, \alpha_2, \alpha_3$, so that $K = e_1$ and $\Psi = e_2$, and $D_e = \alpha_e\,\partial/\partial\alpha_e$, which is ∂_{y_e} in the coordinates $y_e = \log\alpha_e$.

Step 1: the ray. The list has rank 1. For a nonempty proper S, $r(S) = 1$ and $r^*(S) = |S| + r(\mathcal{E} \setminus S) - 1 = |S|$, so $d(S) = 0$, $d^*(S) = |S| - 1$ and the width is $c(S) = 1$ (Definition 7.12).

The family is connected at the strength 1, and its centre ray (Definition 38.13) has the origin 0, the direction $(\mathbf{1};1,1)$ and the walls $\kappa_{1,S}=c(S)=1$ for every nonempty proper S. On the chart $\alpha_3=1$ of the mark $e_0=3$, with $\alpha_1=e^{y_1}$, $\alpha_2=e^{y_2}$ and $dy=d\alpha_1\,d\alpha_2/(\alpha_1\alpha_2)$,

$$f=\frac{\alpha^{\mathbf{1}}}{K\Psi}=\frac{e_3}{e_1e_2},\qquad P(k)=\int_{\mathbb{R}^2}f^k\,dy,\qquad \ell=\int_0^\infty\int_0^\infty\frac{d\alpha_1\,d\alpha_2}{(1+\alpha_1+\alpha_2)(\alpha_1+\alpha_2+\alpha_1\alpha_2)}.$$

The measure is dy: with $d\alpha_1\,d\alpha_2$ in its place the integrand is of the order of $1/(\alpha_1+\alpha_2)$ at infinity and the integral diverges. The volume of the direction (Proposition 39.30) is $V_1=\sum_\pi\prod_{l=2}^3\kappa_{1,U_l}^{-1}=3!\cdot1=6$.

Step 2: the stride. (a) Both sides of

$$e_1e_2-9e_3=\alpha_1(\alpha_2-\alpha_3)^2+\alpha_2(\alpha_3-\alpha_1)^2+\alpha_3(\alpha_1-\alpha_2)^2$$

equal $\sum_{i\neq j}\alpha_i^2\alpha_j-6e_3$, the left one because $e_1e_2=\sum_{i\neq j}\alpha_i^2\alpha_j+3e_3$. So $0<f\le\frac19$ on the orthant, with equality exactly on the diagonal, which on the chart is the point $y=0$: the native gauge of the direction is $\alpha^\star=\mathbf{1}$ and $f(\alpha^\star)=\frac19$ (Theorem 38.14(i)). Every $P(k)$, $k>0$, converges and is positive, and since $f<\frac19$ off one point,

$$P(k+1)=\int f\cdot f^k\,dy<\tfrac19\,P(k)\qquad(k>0).$$

(b) Order the coordinates, $\alpha_{(1)}\ge\alpha_{(2)}\ge\alpha_{(3)}$. Then $\alpha_{(1)}\le e_1\le3\alpha_{(1)}$, $\alpha_{(1)}\alpha_{(2)}\le e_2\le3\alpha_{(1)}\alpha_{(2)}$ and $e_3=\alpha_{(1)}\alpha_{(2)}\alpha_{(3)}$, so $\frac19\,\alpha_{(3)}/\alpha_{(1)}\le f\le\alpha_{(3)}/\alpha_{(1)}$. On the chart $\alpha_{(3)}/\alpha_{(1)}=e^{-\varphi(y)}$ with $\varphi(y)=\max(y_1,y_2,0)-\min(y_1,y_2,0)$. The lines $y_1=y_2$, $y_1=0$, $y_2=0$ cut the plane into six cones, one for each order of the entries of $(y_1,y_2,0)$, and on each φ is the largest entry less the smallest; for real $k>0$

$$\int_{y_1\ge y_2\ge0}e^{-ky_1}dy=\int_{y_1\ge0\ge y_2}e^{-k(y_1-y_2)}dy=\int_{0\ge y_1\ge y_2}e^{ky_2}dy=\frac{1}{k^2},$$

and the three cones with y_1 and y_2 exchanged give the same, so $\int e^{-k\varphi}dy=6/k^2$ and

$$\frac{6\cdot9^{-k}}{k^2}\le P(k)\le\frac{6}{k^2}\qquad(k>0).$$

The six cones are the six orders π of Proposition 39.30, each contributing $\prod_l\kappa_{1,U_l}^{-1}=1$. On $[k_1,\infty)$ the integrands f^k are dominated by f^{k_1}, as $0<f<1$, so P is continuous on $(0,\infty)$.

Step 3: the candidate recurrence. Here $f^{-1}=e_1e_2/e_3=(\alpha_1+\alpha_2+\alpha_3)(\alpha_1^{-1}+\alpha_2^{-1}+\alpha_3^{-1})$, a Laurent polynomial, and the closings of 1 (Definition 39.26) are its constant terms,

$$x_j=\mathsf{C}_j(1)=\mathrm{CT}\left(f^{-j}\right)=\sum_{a_1+a_2+a_3=j}\Big(\frac{j!}{a_1!\,a_2!\,a_3!}\Big)^2,$$

$$(x_j)_{j\ge0}=1,\ 3,\ 15,\ 93,\ 639,\ 4653,\ 35169,\ 272835,\ 2157759,\ \ldots,$$

since the constant term pairs the monomial α^a of $(\sum_e\alpha_e)^j$ with the monomial α^{-a} of $(\sum_e\alpha_e^{-1})^j$, both with the coefficient $j!/(a_1!a_2!a_3!)$. Exact linear algebra with the ansatz $\sum_{i=0}^2p_i'(j)\,x_{j+i}=0$, $\deg p_i'\le2$, on $x_0,\ldots,x_{15}$ (14 equations in 9 unknowns) has a space of solutions of dimension one,

$$(j+2)^2x_{j+2}-(10j^2+30j+23)\,x_{j+1}+9(j+1)^2x_j=0.$$

By Corollary 39.32 a law $\sum_i p_i(k)\,\mathrm{Sh}^i[1]=0$ gives $\sum_i p_i(-j)\,x_{j-i}=0$; read back, $p_i(k)=p'_{2-i}(-k-2)$ is the candidate

$$k^2P(k)-(10k^2+10k+3)\,P(k+1)+9(k+1)^2\,P(k+2)=0. \tag{L}$$

At this point (L) is a candidate: sixteen closings do not prove their own recurrence, and Corollary 39.32 runs from a law to its closings.

Step 4: the certificate. The defects of the direction (Definition 38.1, Lemma 38.2(ii)) are

$$c_e=D_e\log f=1-R_e-R_e^*,\qquad R_e=\frac{\alpha_e}{e_1},\qquad R_e^*=\frac{\alpha_e(e_1-\alpha_e)}{e_2},$$

the occupations of e under the basis law and under the cobasis law; $\sum_e R_e=1$ and $\sum_e R_e^*=2$, so $\sum_e c_e=0$. The origin is 0, so its defects vanish, and a certificate of (L) (Proposition 38.5(i): an explicit element of the relation space $\mathcal{L}$ of Definition 38.4 whose image is the recurrence) is a choice of insertions g_1,g_2,g_3 with

$$k^2-(10k^2+10k+3)\,f+9(k+1)^2f^2=\sum_{e=1}^{3}\big(k\,c_e+D_e\big)g_e. \tag{C}$$

The ansatz: $g_e=N_e/(e_1^2e_2)$, N_e homogeneous of degree 4 with coefficients in $\mathbb{Q}+\mathbb{Q}k$, N_1 symmetric in α_2,α_3 and g_e the image of g_1 under the exchange of the marks 1 and e. Multiplied by $e_1^3e_2^2$, (C) is a polynomial identity of degree 7 in α and 2 in k, linear in the 18 coefficients of N_1: 99 equations of rank 10. One solution is

$$g_1=-\frac{\alpha_1^2\big(k\,\alpha_1(\alpha_2+\alpha_3)+(2+3k)(\alpha_2^2+\alpha_3^2)-(1-k)\,\alpha_2\alpha_3\big)}{e_1^2e_2}, \tag{G}$$

and within the ansatz the others differ from it by one insertion h added at every mark, h with a symmetric numerator, an 8-dimensional space; such an h contributes $k\,h\sum_e c_e+\sum_e D_eh=0$, by Euler's identity for h of degree 0. Each g_e is a polynomial over $K^2\Psi$, homogeneous of degree 0, so a regular function on X_F: an insertion (Definition 38.1).

Verification. Write $g_e=g_e^{(0)}+k\,g_e^{(1)}$, with $e_1^2e_2\,g_1^{(0)}=-\alpha_1^2(2\alpha_2^2-\alpha_2\alpha_3+2\alpha_3^2)$ and $e_1^2e_2\,g_1^{(1)}=-\alpha_1^2(\alpha_1\alpha_2+\alpha_1\alpha_3+3\alpha_2^2+\alpha_2\alpha_3+3\alpha_3^2)$. The powers of k split (C) into

$$\sum_e c_eg_e^{(1)}=1-10f+9f^2,\qquad \sum_e\big(D_eg_e^{(1)}+c_eg_e^{(0)}\big)=-10f+18f^2,\qquad \sum_e D_eg_e^{(0)}=-3f+9f^2.$$

Times $e_1^3e_2^2$ every sum over the marks is a symmetric polynomial of degree 7, and in the basis of the products of e_1,e_2,e_3 exact expansion gives

$$\sum_e e_1^3e_2^2\,c_eg_e^{(1)}=e_1^3e_2^2-10\,e_1^2e_2e_3+9\,e_1e_3^2,\qquad \sum_e e_1^3e_2^2\,D_eg_e^{(0)}=-3\,e_1^2e_2e_3+9\,e_1e_3^2,$$
$$\sum_e e_1^3e_2^2\,D_eg_e^{(1)}=-2\,e_1e_2^3-3\,e_1^2e_2e_3+9\,e_1e_3^2,\qquad \sum_e e_1^3e_2^2\,c_eg_e^{(0)}=2\,e_1e_2^3-7\,e_1^2e_2e_3+9\,e_1e_3^2,$$

which, with $e_1^3e_2^2f=e_1^2e_2e_3$ and $e_1^3e_2^2f^2=e_1e_3^2$, are the three right sides times $e_1^3e_2^2$. So (C) holds identically in α and k: $\sum_i p_i(k)f^i\in\mathcal{L}$, with $(p_0,p_1,p_2)=\big(k^2,\,-(10k^2+10k+3),\,9(k+1)^2\big)$. To test another certificate on one point: at $\alpha=(1,2,4)$ and $k=\frac{1}{3}$, $f=\frac{4}{49}$, the left side of (C) is $-\frac{2809}{7203}$, and (G) gives $(g_1,g_2,g_3)=\big(-\frac{85}{1029},-\frac{310}{1029},-\frac{424}{1029}\big)$; agreement at a point is a test, the identity above is the proof.

Step 5: the recurrence at every strength. The terms of g_e are the insertions $\alpha^v K^{-2}\Psi^{-1}$ for the monomials α^v of N_e, with the data $(v;2,1)$ and the walls $\kappa_U = v_U - 2d(U) - d^*(U) = v_U - |U| + 1$. The monomials of N_1 are $\alpha_1^3\alpha_2$, $\alpha_1^3\alpha_3$, $\alpha_1^2\alpha_2^2$, $\alpha_1^2\alpha_3^2$ and $\alpha_1^2\alpha_2\alpha_3$; each has two marks in its support, so $v_U \geq 1$ for every pair U, and every wall is at least 0 (the least is 0). With the walls $k\kappa_{1,U} = k$ of the ray, the condition of Proposition 38.5(i) holds at every k with $\operatorname{Re} k > 0$, where the values converge, and the certificate gives (L) there. The integration by parts behind it handles the mark $e_0 = 3$ on its own chart: by Euler's identity $D_3 = -D_1 - D_2$ on functions of degree 0, so $\sum_e D_e(g_e f^k) = \partial_{y_1}\big((g_1 - g_3)f^k\big) + \partial_{y_2}\big((g_2 - g_3)f^k\big)$; y_3 is not a coordinate of the chart. Corollary 39.34(i) gives the same as an identity of meromorphic functions of k, for the element (C) of $\mathcal{L}$.

Step 6: the native values; what positivity selects. The symbol of (L) (Corollary 38.7) is $\chi(\lambda) = 1 - 10\lambda + 9\lambda^2 = (1-\lambda)(1-9\lambda)$. By Theorem 38.17 every sign η, $\eta_3 = 1$, with $K(\eta) \neq 0$ is a native gauge of native value $K(\eta)^{-2}$, with the Hessian $2\big(\kappa_\eta(X_e, X_{e'})\big)_{e,e'\neq 3}$ of the indicators under $\mathbb{P}_\eta(\{e\}) = \eta_e/K(\eta)$: at $\eta = (1,1,1)$, $K(\eta) = 3$, the native value is $\frac{1}{9}$ and the Hessian is $2\left(\begin{smallmatrix} 2/9 & -1/9 \\ -1/9 & 2/9 \end{smallmatrix}\right)$, of determinant $\frac{4}{27}$; at $(-1,1,1)$, $(1,-1,1)$, $(-1,-1,1)$, $K(\eta) = \pm 1$, the native value is 1 and the determinant is -4 (at $(-1,1,1)$ the matrix is $2\left(\begin{smallmatrix} -2 & 1 \\ 1 & 0 \end{smallmatrix}\right)$). All four are nondegenerate, so Corollary 38.7 at $g = 1$ makes $\frac{1}{9}$ and 1 roots of χ; χ has degree 2, so they are all its roots. (The corollary runs from native values to roots; a root of a symbol need not be a native value.) The root other than $f(\alpha^\star) = \frac{1}{9}$ has the modulus $1 > \frac{1}{9}$, so Theorem 41.3(iii) applies: up to a constant factor the values are the only solution of (L) whose generating function has a radius of convergence of at least 9, and the low-end relation fixes the factor. The rational solutions of (L) grow like the native value 1 of the signs and the values decay like 9^{-k}, so a bracket from positivity shrinks by about $\frac{1}{9}$ per step of the strength.

Step 7: the low-end relation. By Step 2(b), $6 \cdot 9^{-k} \leq k^2 P(k) \leq 6$ for real $k > 0$, so $k^2 P(k) \to 6$ as $k \to 0^+$: the low-end coefficient L_0 is $V_1 = 6$ (Definition 39.33). Corollary 39.34(ii) applies to the element (C): $p_0 = k^2$ vanishes to the order $n - 1 = 2$ at 0, $[k^2]p_0 = 1$, $p_1(0) = -3$, $p_2(0) = 9$, and

$$6 - 3\ell + 9P(2) = 0, \qquad P(2) = \frac{\ell - 2}{3}. \tag{S}$$

Directly: take (L) at real $k \in (0,1)$ and let $k \to 0^+$; $k^2P(k) \to 6$, $P(k+1) \to \ell$ and $P(k+2) \to P(2)$ by the continuity of Step 2, while $10k^2 + 10k + 3 \to 3$ and $9(k+1)^2 \to 9$.

A solution that looks right goes wrong here. Reading (L) at $k = 0$ as $0 \cdot P(0) - 3P(1) + 9P(2) = 0$ treats $0 \cdot P(0)$ as 0; but $P(0) = \int dy = \infty$, and the product $k^2P(k)$ tends to $V_1 = 6$. The misreading gives $P(2) = \ell/3$, which Step 2(a) refutes at once: $P(2) < \frac{1}{9}P(1)$ would say $\frac{\ell}{3} < \frac{\ell}{9}$. Every $P(K)$ would then be ℓ times a positive rational, positivity would fix no digit of ℓ, and the bracket of Step 9 would be empty. Passing the limit under the integral, $k^2 f^k \to 0$ pointwise, fails for the same reason: there is no integrable majorant, $\int dy$ being infinite.

Step 8: the rational solutions and their signs. For integers $k \geq 1$, $9(k+1)^2 \neq 0$, so (L) gives $P(k+2)$ from $P(k)$ and $P(k+1)$. Let c and r be the solutions with $c(1) = 1$, $c(2) = \frac{1}{3}$ and $r(1) = 0$, $r(2) = -\frac{2}{3}$. From $P(1) = \ell$, (S) and induction,

$$P(K) = c(K)\,\ell + r(K) \qquad (K \geq 1).$$

Signs at every K. Put $x'_j = 9^j c(j+1)$. Then (L) at $k = j+1$, times 9^{j+1}, reads

$$(j+2)^2 x'_{j+2} = (10j^2 + 30j + 23)\,x'_{j+1} - 9(j+1)^2 x'_j, \qquad x'_0 = 1,\ x'_1 = 3.$$

If $x'_{j+1} \geq 3x'_j > 0$, then $(j+2)^2 x'_{j+2} \geq \big(10j^2 + 30j + 23 - 3(j+1)^2\big)x'_{j+1} = (7j^2 + 24j + 20)\,x'_{j+1} \geq 3(j+2)^2 x'_{j+1}$, because $7j^2 + 24j + 20 - 3(j+2)^2 = 4(j+1)(j+2) \geq 0$. So $x'_{j+1} \geq 3x'_j > 0$ for

every j, $x'_j \geq 3^j$, and

$$c(K) \geq 3^{1-K} > 0, \qquad 9c(K) \geq 3c(K-1) > c(K-1) \qquad (K \geq 2).$$

The closings satisfy the same recurrence with the same first two terms (Corollary 39.34(i) for the element (C)), so $x'_j = x_j$ and $c(K) = 9^{1-K} x_{K-1}$ at every K; the signs do not use this. In exact arithmetic:

K	$c(K)$	$r(K)$	$a_K = -r(K)/c(K)$
1	1	0	0
2	1/3	−2/3	2
3	5/27	−23/54	23/10
4	31/243	−145/486	145/62
5	71/729	−1331/5832	1331/568
6	517/6561	−242353/1312200	242353/103400
7	11723/177147	−5495507/35429400	5495507/2344600
8	10105/177147	−46422941/347208120	46422941/19805800
9	26639/531441	−2447622221/20832487200	2447622221/1044248800
10	5773279/129140163	−530455590781/5062294389600	530455590781/226312536800

The frame of c and r (Proposition 38.11, $h = 2$) satisfies $C_{k+1} = \frac{p_0(k)}{p_2(k)} C_k = \frac{k^2}{9(k+1)^2} C_k$ with $C_1 = c(1)r(2) - c(2)r(1) = -\frac{2}{3}$, so

$$c(k)\, r(k+1) - c(k+1)\, r(k) = -\frac{2}{3} \cdot \frac{9^{1-k}}{k^2} \qquad (k \geq 1).$$

Step 9: the bracket and the answer. (a) *Below, by positivity.* $P(K) = c(K)\ell + r(K) > 0$ and $c(K) > 0$ give $\ell > a_K = -r(K)/c(K)$. (b) *Above, by the native value.* $P(K) < \frac{1}{9} P(K-1)$ (Step 2(a)) is $\big(9c(K) - c(K-1)\big)\ell < r(K-1) - 9r(K)$, and $9c(K) - c(K-1) > 0$ (Step 8), so

$$\ell < b_K = \frac{r(K-1) - 9\, r(K)}{9\, c(K) - c(K-1)}.$$

At $K = 2$ this is the first bracket: $P(2) = \frac{\ell - 2}{3} > 0$ gives $\ell > 2$, and $P(2) < \frac{1}{9} P(1)$ gives $\ell < 3$. (c) *The width.* With the frame at $k = K-1$ and $c(K) = 9^{1-K} x_{K-1}$,

$$\begin{aligned} b_K - a_K &= \frac{r(K-1)\, c(K) - r(K)\, c(K-1)}{c(K)\big(9c(K) - c(K-1)\big)} = \frac{2 \cdot 9^{2-K}}{3(K-1)^2\, c(K)\big(9c(K) - c(K-1)\big)} \\ &= \frac{2 \cdot 9^{K-1}}{3(K-1)^2\, x_{K-1}(x_{K-1} - x_{K-2})}, \end{aligned}$$

which the bounds of Step 8 make at most $1/(K-1)^2$: the brackets close on one number. In exact arithmetic, with the width also as a rendering:

K	b_K	$b_K - a_K$	rendering
2	3	1	
3	19/8	3/40	$7.5 \cdot 10^{-2}$
4	61/26	3/403	$7.44 \cdot 10^{-3}$
5	3413/1456	81/103376	$7.84 \cdot 10^{-4}$
6	104539/44600	243/2882275	$8.43 \cdot 10^{-5}$
7	149014/63575	2187/238492712	$9.17 \cdot 10^{-6}$
8	909876251/388187800	19683/19613188595	$1.0036 \cdot 10^{-6}$
9	6414407843/2736630400	6561/59511099776	$1.1025 \cdot 10^{-7}$
10	232184895407/99058909600	177147/14589151085627	$1.214 \cdot 10^{-8}$

At $K = 8$ the certified width exceeds 10^{-6}; at $K = 9$ it is $\frac{6561}{59511099776} < \frac{111}{10^9}$.

Answer. $q = a_9 = \dfrac{2447622221}{1044248800}$, with

$$\frac{2447622221}{1044248800} < \ell < \frac{6414407843}{2736630400}, \qquad b_9 - a_9 = \frac{6561}{59511099776} < 1.11 \cdot 10^{-7},$$

so $0 < \ell - q < 10^{-6}$.

What does not certify. The value a_8 is also a correct answer, but the bracket at $K = 8$ is too wide to show it; the upper bound at $K = 9$ does, $\ell - a_8 < b_9 - a_8 = \frac{19683}{22574408320} < 10^{-6}$. A rate of convergence is no bound, and neither is the last step: $a_9 - a_8$ is smaller than $\ell - a_8$, since $a_9 < \ell$. The bound $P(K) \le 9^{-K}\ell$ does not follow from $f \le \frac{1}{9}$ (it would need $\int dy < \infty$); iterating Step 2(a) gives $P(K) < 9^{1-K}P(1) = 9^{1-K}\ell$, that is $\big(c(K) - 9^{1-K}\big)\ell < -r(K)$ with $c(K) - 9^{1-K} = 9^{1-K}(x_{K-1} - 1) > 0$ for $K \ge 2$: a valid and cruder upper bound, which needs $K = 10$ to certify six digits. A sign of $c(K)$ or of $9c(K) - c(K-1)$ that is not proved can reverse an inequality: here Step 8 proves both at every K.

Check (floating point, not part of the solution). On the chart the integral over α_2 is elementary, $\int_0^\infty \frac{d\alpha_2}{(\alpha_2+A)(A\alpha_2+B)} = \frac{\log(A^2/B)}{A^2-B}$ with $A = 1 + \alpha_1$, $B = \alpha_1$, so

$$\ell = \int_0^\infty \frac{\log\big((1+x)^2/x\big)}{1+x+x^2}\,dx,$$

which renders as $2.343907238689458890601562288872277069095\ldots$; then $\ell - a_9$ and $b_9 - \ell$ render as $9.6062 \cdot 10^{-8}$ and $1.4186 \cdot 10^{-8}$. Exercise 52.3 computes ℓ to a hundred digits without quadrature. □

52.2 The centre ray by its box

Exercise 52.2 (The centre of $\binom{[3]}{1}$ by its box)**.** Let F, P and ℓ be as in Exercise 52.1. Read the centre ray of F by steps (1) to (4) and (6) of the procedure of Definition 41.8, with the box $\mathfrak{B}_2(1,1)$ (Definition 39.11) modulo the group G generated by the permutations of the marks and the Cremona: give its symbols, orbits and relations, the pivots and the free orbits, the spanning set at $\mathfrak{s} = 1$, the matrix $\mathbf{B}(k)$, a certificate supported on the stems, the law of [1], $\mathbf{B}(\infty)$ and the closings of the spanning set. Show that the closing relations alone do not give the law.

Solution. Receipt: `ex19_centre_box.py`. The elimination is small enough to run once over $\mathbb{Q}(k)$, so the evaluation at random strengths of step (2) of the procedure is not needed.

Step 1: walls and symmetries. The walls of the direction are $\kappa_{1,S} = 1$ and $V_1 = 6$ (Exercise 52.1, Step 1). Every permutation β of the marks carries the bases and keeps the masses **1** and

the exponents, and the Cremona is a symmetry because the masses are $\sigma\mathbf{1}$ and $s = t$ (Proposition 39.7(ii)). So $G = S_3 \times \{1, \mathrm{cr}\}$, of order 12, acting on the symbols by $(u; a, b) \mapsto (u \circ \beta^{-1}; a, b)$ and $(u; a, b) \mapsto (-u + (a+b)\mathbf{1}; b, a)$ (Proposition 39.7(iii)).

Step 2: the box. Both letters are active, so the box with one unit of each active letter at $\mathfrak{t} = 2$ is $\mathfrak{B} = \mathfrak{B}_2(1,1)$: the symbols $(u; a, b)$ with $a, b \in \{0, 1\}$, $u_{\mathcal{E}} = a + 2b$, and wall degree $\deg_\kappa = \max_U(-\kappa_U) \le 2$, where $\kappa_U = u_U - a\,d(U) - b\,d^*(U) = u_U - b(|U| - 1)$. It has 110 symbols, 19 at the level $(0,0)$, 27 at each of $(1,0)$ and $(0,1)$ and 37 at $(1,1)$. G keeps it, the Cremona exchanging the levels $(1,0)$ and $(0,1)$, and the symbols fall into 16 orbits:

orbit of	level	$\deg_\kappa$	size		orbit of	level	$\deg_\kappa$	size	
$\alpha_1^4/(\alpha_3 K\Psi)$	$(1,1)$	2	12	pivot	$\alpha_1^3/(\alpha_2\alpha_3 K)$	$(1,0),(0,1)$	2	6	pivot
$\alpha_1^4\alpha_2/(\alpha_3^2 K\Psi)$	$(1,1)$	2	6	pivot	$\alpha_1^2\alpha_2/(\alpha_3^2 K)$	$(1,0),(0,1)$	2	12	pivot
$\alpha_1^3/(K\Psi)$	$(1,1)$	1	6	pivot	$\alpha_1^3/(\alpha_3^2 K)$	$(1,0),(0,1)$	2	12	pivot
$\alpha_1^3\alpha_2/(\alpha_3 K\Psi)$	$(1,1)$	1	6	pivot	$\alpha_1\alpha_2/(\alpha_3 K)$	$(1,0),(0,1)$	1	6	pivot
$\alpha_1^2\alpha_2/(K\Psi)$	$(1,1)$	0	6	pivot	$\alpha_1^2/(\alpha_3 K)$	$(1,0),(0,1)$	1	12	pivot
$\alpha_1\alpha_2\alpha_3/(K\Psi)$	$(1,1)$	-1	1	free	α_1/K	$(1,0),(0,1)$	0	6	pivot
$\alpha_1^2/(\alpha_2\alpha_3)$	$(0,0)$	2	6	pivot	α_1/α_3	$(0,0)$	1	6	free
α_1^2/α_3^2	$(0,0)$	2	6	pivot	1	$(0,0)$	0	1	free

The stems are the 19 Laurent symbols. The relations (Definition 39.11(iv)) are the letter identities for K of the 46 symbols with $a = 0$, those for Ψ of the 46 symbols with $b = 0$, and the mark relations of the 19 stems at the 3 marks: 149 relations, which give 22 distinct combinations of orbit averages (Definition 39.16(i)), of rank 13 over $\mathbb{Q}(k)$. With the ranking of Definition 39.16(ii) (positive levels above the Laurent orbits, then the total level, then the wall degree), the elimination makes the 13 orbits marked in the table pivots; the free orbits are those of $f = \alpha_1\alpha_2\alpha_3/(K\Psi)$, of α_1/α_3 and of 1.

Step 3: the spanning set and the matrix. The free Laurent orbits of wall degree at most $\mathfrak{s} = 1$ give the spanning set $\mathbf{b} = (1, b_1)$, with $b_1 = \frac{1}{6}\sum_{i\ne j}\alpha_i/\alpha_j$, the average of the orbit of α_1/α_3 (Definition 39.16(iii)). The first target is reduced already: $f^{-1} = (\sum_e \alpha_e)(\sum_e \alpha_e^{-1}) = 3 + 6b_1$ identically. The second, $f^{-1}b_1$, has 19 monomials, and its reduction is a combination of 1 and b_1. So the box is accepted at $\mathfrak{s} = 1$, with

$$\mathbf{B}(k) = \begin{pmatrix} 3 & 6 \\ \dfrac{2(k-1)}{k-2} & \dfrac{(k-1)(7k-11)}{(k-2)^2} \end{pmatrix},$$

and by Theorem 39.17(i) and (iii) $[f^{-1}b_i] = \sum_l \mathbf{B}_{il}(k)[b_l]$ in the module, and $\langle b_i\rangle_{k-1} = \sum_l \mathbf{B}_{il}(k)\langle b_l\rangle_k$ as meromorphic functions of k.

Step 4: the certificate. Theorem 39.17(ii) provides combinations G_e of stems with $f^{-1}b_1 - \mathbf{B}_{21} - \mathbf{B}_{22}\,b_1 = \sum_e \big(k\,c_e + D_e\big)G_e$, c_e and D_e as in Exercise 52.1. With G_1 a combination of stems symmetric under the exchange of the marks 2 and 3 and G_e its image under the exchange of 1 and e, the identity is linear in 11 unknowns, 37 equations, with a family of solutions of dimension 6; one of them is

$$G_1 = \frac{\alpha_2^2+\alpha_3^2}{6(k-2)\,\alpha_1^2} + \frac{(5k^2-6k-4)\,\alpha_2\alpha_3}{6(k+1)(k-2)^2\,\alpha_1^2} + \frac{3k-4}{6(k+1)(k-2)^2}\Big(\frac{\alpha_3^2}{\alpha_1\alpha_2}+\frac{\alpha_2^2}{\alpha_1\alpha_3}\Big) + \frac{5k-9}{6(k-2)^2}\,\frac{\alpha_2+\alpha_3}{\alpha_1} + \frac{3k-4}{6(k+1)(k-2)}\Big(\frac{\alpha_3^2}{\alpha_2^2}+\frac{\alpha_2^2}{\alpha_3^2}\Big),$$

whose terms are stems, of wall degree 2, 2, 2, 1 and 2. The identity holds as one of rational functions in α and k (exact expansion); at $\alpha = (1, 2, 4)$, $k = \frac{1}{3}$, both sides are $\frac{11903}{800}$. Its coefficients have poles at $k = 2$ and $k = -1$, and by Theorem 39.17(iii) it integrates term by term at every real $k > 2$.

Step 5: the law. By Proposition 39.6 the rows are $w_0 = (1, 0)$, $w_1(k) = w_0\mathbf{B}(k) = (3, 6)$ and

$$w_2(k) = w_1(k-1)\,\mathbf{B}(k) = \Big(\frac{3(7k-10)}{k-2},\ \frac{6(10k^2-30k+23)}{(k-2)^2}\Big),$$

and their first dependence is $q_0w_0 + q_1w_1 + q_2w_2 = 0$ with $(q_0, q_1, q_2) = \big(9(k-1)^2,\ -(10k^2-30k+23),\ (k-2)^2\big)$. Then $p_j(k) = q_{2-j}(k+2)$ gives $\big(k^2,\ -(10k^2+10k+3),\ 9(k+1)^2\big)$: the recurrence (L) of Exercise 52.1, now with no closing computed, and with a certificate assembled from those of the rows (Theorem 39.17(iv)). As a check of Proposition 39.6(iv), $\det\mathbf{B}(k) = 9(k-1)^2/(k-2)^2$ and $\det[w_0; w_1] = 6$, so $q_0 \cdot 6 = q_2 \cdot 6 \cdot \det\mathbf{B}(k)$.

The spanning set is independent: at the native gauges $(1, 1, 1)$ and $(-1, 1, 1)$ (Exercise 52.1, Step 6) the vectors of values $\beta_c = (1, b_1(\alpha_c))$ are $(1, 1)$ and $(1, -\frac{1}{3})$, which span $\mathbb{C}^2$, so by Proposition 39.38(i) $1, b_1$ are masters of the submodule $\mathfrak{N}$ they span and $\mathbf{B} = \mathbf{B}_\mathbf{b}$. By Proposition 39.6(ii) the order 2 is then the dimension of the cyclic submodule of [1], which is $\mathfrak{N}$, and (L) is the law of [1]. Finally

$$\mathbf{B}(\infty) = \begin{pmatrix} 3 & 6 \\ 2 & 7 \end{pmatrix}, \qquad \mathbf{B}(\infty)\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 9\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{B}(\infty)\begin{pmatrix} 1 \\ -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{1}{3} \end{pmatrix},$$

the reciprocals of the native values $\frac{1}{9}$ and 1, with the vectors β_c as eigenvectors (Proposition 39.38(ii)).

Step 6: checks. The closings of the spanning set are $\mathsf{C}_j(1) = x_j$ and, by the first row, $\mathsf{C}_j(b_1) = \frac{1}{6}(x_{j+1} - 3x_j) = 0, 1, 8, 60, 456, 3535, 27888, \ldots$; they satisfy $\mathsf{C}_{j+1}(\mathbf{b}) = \mathbf{B}(-j)\,\mathsf{C}_j(\mathbf{b})$ (Theorem 39.29) for $0 \le j < 12$. The low-end relation of (L) returns $V_1 = 6$ (Exercise 52.1, Step 7).

Step 7: the closing relations alone. The closing relations $(k\,c_e + D_e)(\alpha^w f^{-1})$ involve Laurent insertions only (Proposition 39.15). Let $\omega = e^{2\pi i/3}$ and $z = (1, \omega, \omega^2)$. Then $K(z) = 1+\omega+\omega^2 = 0$ and $\Psi(z) = \omega+\omega^2+\omega^3 = 0$, so $f^{-1} = K\Psi/e_3$ and every $D_ef^{-1} = (\Psi D_eK + KD_e\Psi)/e_3 - f^{-1}$ vanish at z: z is a singular point of the zero set of f^{-1} on the torus. By Proposition 39.15(i) the evaluation at z vanishes on every closing relation, and by (ii) it takes the value $p_2(k-2) = 9(k-1)^2 \ne 0$ on the element $\sum_{i=0}^{2} p_{2-i}(k-2)f^{-i}$ of $\mathcal{L}$ that the law gives: no combination of closing relations gives the law. Concretely, at the strength 9 the elimination of the closing relations of the Laurent monomials of wall degree at most $R-1$, on the invariant Laurent combinations of wall degree at most R, leaves three free orbits, 1, b_1 and $b_2 = \frac{1}{6}\sum_{i\ne j}\alpha_i^2/\alpha_j^2$, for every R from 3 to 7; the law at $k = 7$, read at the strength 9 as $49f^{-2} - 563f^{-1} + 576$, reduces for each such R to

$$504 + 248\,b_1 - 392\,b_2 \ne 0,$$

and at z, where $b_1(z) = b_2(z) = -\frac{1}{2}$, this is $576 = 9(k+1)^2$. At $k = 10$ and $k = \frac{23}{3}$ the reductions are $880 + \frac{946}{3}b_1 - \frac{2200}{3}b_2$ and $\frac{8671}{15} + \frac{1313}{5}b_1 - \frac{6877}{15}b_2$, with the values 1089 and 676 at z, again $9(k+1)^2$. The point z lies off X_F; the box supplies the relations that the closing relations lack through its units of the letters, by the letter identities and the mark relations of the occupations.

Answer. The box $\mathfrak{B}_2(1, 1)$ modulo G has 110 symbols in 16 orbits and 149 relations of rank 13 modulo G; it is accepted at $\mathfrak{s} = 1$ with the masters 1 and $b_1 = \frac{1}{6}\sum_{i\ne j}\alpha_i/\alpha_j$ and the matrix $\mathbf{B}_\mathbf{b}(k)$ above, with the certificate G_1 on the stems; the law of [1] is (L), and $\mathbf{B}_\mathbf{b}(\infty)$ has the eigenvalues 9 and 1. The closing relations alone leave the evaluation at $(1, \omega, \omega^2)$, on which the law takes the value $9(k+1)^2$. □

52.3 The centre ray from its gauge end

Exercise 52.3 (The centre of $\binom{[3]}{1}$ from its gauge end)**.** Let F, P, ℓ and the law (L) be as in Exercise 52.1, and let μ_f be the stride's law of the centre ray (Definition 40.1), with the density ϱ_f.

(i) Write the density operator of (L), its exponents at $f(\alpha^\star) = \frac{1}{9}$, and the germs there that extend by zero.

(ii) Find $\varrho_f(\frac{1}{9})$ and the Taylor coefficients of the germ of ϱ_f at $\frac{1}{9}$.

(iii) Read $P(1)$, $P'(1)$ and $P''(1)$ from the gauge end (Theorem 40.6) to sixty digits, find the low-end coefficients $\mathsf{L}_0, \mathsf{L}_1, \mathsf{L}_2$ from the faces of the sectors, and check the output against the low-end relation and its first two derivatives.

(iv) Find the arc integrals $S_j = -\int_0^{2\pi/3} \log^{j-1}\left(2\sin\frac{\vartheta}{2}\right) d\vartheta$ for $j = 1, 2$.

Solution. Receipt: `ex20_gauge_end.py`. Theorem 40.6 applies: the ray has the origin 0 and positive walls, and (L) holds at every k with $\operatorname{Re} k > 0$ (Exercise 52.1, Step 5), in particular at all large integers; $f(\alpha^\star) = \frac{1}{9}$ is a simple root of the symbol $\chi(\lambda) = (1-\lambda)(1-9\lambda)$. Write $f^\star = \frac{1}{9}$.

Step 1: the density operator. With $\theta = t\,d/dt$ and $(p_0, p_1, p_2) = \left(k^2, -(10k^2{+}10k{+}3), 9(k{+}1)^2\right)$,

$$\begin{aligned}\sum_i t^i p_i(-1-i-\theta) &= (\theta+1)^2 - t\,(10\theta^2+30\theta+23) + 9t^2(\theta+2)^2 \\ &= t^2(t-1)(9t-1)\,\partial_t^2 + t\,(45t^2-40t+3)\,\partial_t + (36t^2-23t+1),\end{aligned}$$

of order $D = 2$, with the leading coefficient $(-1)^D t^D \chi(t)$; its singular points are 0, $\frac{1}{9}$, 1 and ∞. The second symbol is $\chi_1(\lambda) = -10\lambda + 18\lambda^2$.

Step 2: the exponents at $f^\star$. In $s = f^\star - t$ the operator is $b_2\partial_s^2 + b_1\partial_s + b_0$ with

$$b_2 = \tfrac{1}{81}s(9s-1)^2(9s+8), \qquad b_1 = \tfrac{1}{81}(9s-1)(405s^2+270s-8), \qquad b_0 = \tfrac{1}{9}(324s^2+135s-10),$$

so $b_{2,0} = 0$, $b_{2,1} = \frac{8}{81} = -(f^\star)^2\chi'(f^\star)$ and $b_{1,0} = \frac{8}{81}$. Theorem 40.6(i) gives $\gamma = D - 1 - b_{1,0}/b_{2,1} = 0$, which is $-1 + \chi_1(f^\star)/(f^\star\chi'(f^\star)) = -1 + \frac{-8/9}{-8/9}$, and $\frac{n-3}{2} = 0$ as Theorem 40.6(iii) requires. The polynomials $R_m(x) = \sum_j b_{j,m-1+j}[x]_j$ are

$$R_0 = \tfrac{8}{81}x^2, \qquad R_1 = -\tfrac{1}{9}(15x^2+23x+10), \qquad R_2 = (2x+3)(3x+5), \qquad R_3 = 9(x+2)^2,$$

and $R_m = 0$ for $m \geq 4$. The exponents at $s = 0$ are 0 and 0: one solution $\phi = \sum_m e_m s^m$, $e_0 = 1$, holomorphic at $s = 0$, with $R_0(N)e_N = -\sum_{m=1}^3 R_m(N-m)e_{N-m}$, and one of the form $\phi\log s + \psi$, ψ holomorphic with $\psi(0) = 0$.

Step 3: the line. For a germ v on $s > 0$ with $(b_2\partial_s^2 + b_1\partial_s + b_0)v = 0$, integration by parts on (ϵ, ∞) shows that the distribution $\mathfrak{D}(v\mathbf{1}_{s>0})$ pairs with a test function w as

$$\lim_{\epsilon\to 0^+}\Big[b_2\,(v'w - vw') + (b_1 - b_2')\,vw\Big]_{s=\epsilon}.$$

For $v = \phi$ this is $(b_{1,0} - b_{2,1})\,\phi(0)w(0) = 0$: ϕ extends by zero (Definition 40.4). For $v = \phi\log s + \psi$ the term $b_2 v'w$ tends to $b_{2,1}w(0)$, while the terms in $\log\epsilon$ have the coefficient $(b_{1,0} - b_{2,1})w(0) = 0$, so the pairing is $\frac{8}{81}w(0)$: the germ leaves a point mass $\frac{8}{81}\delta_{f^\star}$ and does not extend by zero, and neither does any germ with a logarithmic term. So the germs that extend by zero form the line of

ϕ (Theorem 40.6(ii)), the germ of ϱ_f spans it (Theorem 40.6(iii)), and $\varrho_f = \varrho_f(f^\star)\,\phi(f^\star - t)$ near $f^\star$, the case $\gamma \in \{0, \dots, D-2\}$ of Theorem 40.6(iv). In exact arithmetic

$$(e_m)_{m\ge 0} = 1,\ \tfrac{45}{4},\ \tfrac{3645}{32},\ \tfrac{70713}{64},\ \tfrac{21500397}{2048},\ \tfrac{807258879}{8192},\ \tfrac{15034997331}{16384},\ \tfrac{139237010559}{16384},\ \dots,$$

all positive.

Step 4: the value at the native value. By Theorem 38.15, on the centre ray the Hessian of $-\log f$ at $y = 0$ is twice the covariance matrix of the indicators of the marks $1, 2$ under the uniform law on the three marks,

$$H = 2\begin{pmatrix} \frac{2}{9} & -\frac{1}{9} \\ -\frac{1}{9} & \frac{2}{9} \end{pmatrix}, \qquad \det H = \tfrac{4}{27}.$$

By Definition 38.4 there are coordinates v near $y = 0$ with $\log f = \log f^\star - \frac{1}{2}|v|^2$, here real, with $dy = J(v)\,dv$ and $J(0) = (\det H)^{-1/2}$. So $\mu_f([t, f^\star]) = \int_{|v|\le\rho} J\,dv$ with $\rho^2 = 2\log(f^\star/t)$, whose derivative in ρ is $2\pi\rho J(0) + O(\rho^2)$, and $\varrho_f(t) = -\frac{d}{dt}\mu_f([t, f^\star]) = \frac{1}{\rho t}\big(2\pi\rho J(0) + O(\rho^2)\big)$, which tends to

$$\varrho_f(f^\star) = \frac{2\pi}{f^\star\sqrt{\det H}} = 2\pi \cdot 9 \cdot \frac{3\sqrt{3}}{2} = 27\sqrt{3}\,\Gamma(\tfrac{1}{2})^2.$$

Step 5: the values and the slopes. The other singular points of the operator lie at $s = \frac{1}{9}$ ($t = 0$) and $s = -\frac{8}{9}$, so the series ϕ converges on $|s| < \frac{1}{9}$, which contains $t \in (0, f^\star]$, and $\varrho_f(t) = 27\sqrt{3}\,\Gamma(\frac{1}{2})^2 \sum_m e_m(\frac{1}{9} - t)^m$ there: this series is the continuation of the germ along $(0, f^\star)$ in Theorem 40.6(v). Its terms are positive, so by monotone convergence Theorem 40.6(v) gives, for every real $K > 0$ and $j \in \mathbb{N}$,

$$P^{(j)}(K) = 27\sqrt{3}\,\Gamma(\tfrac{1}{2})^2 \sum_{m\ge 0} e_m\,\frac{\partial^j}{\partial a^j}\Big[9^{-a-m-1}\,\Gamma\,{}_{1}^{2}\begin{bmatrix} a+1,\ m+1 \\ a+m+2 \end{bmatrix}\Big]_{a=K},$$

the Beta corner $\int_0^{1/9} t^a(\frac{1}{9} - t)^m\,dt = 9^{-a-m-1}\Gamma\,{}_{1}^{2}\begin{bmatrix} a+1,\ m+1 \\ a+m+2 \end{bmatrix}$ and its derivatives in a through the digamma function and its derivatives. At $K = 150$ the terms fall below 10^{-150} times the first after 520 terms. Differentiated j times, (L) is $\sum_{l\le j}\binom{j}{l}\sum_i p_i^{(j-l)}(k)\,P^{(l)}(k+i) = 0$, an identity of meromorphic functions, solved for $P^{(j)}(k)$ because $p_0(k) = k^2 \ne 0$ at $k \ge 1$. The run from $K = 150, 151$ down to $k = 1$, at 140 digits, agrees with the run from $K = 200, 201$ to 138 digits. The output, rendered to sixty digits:

$$\begin{aligned} P(1) &= \ell = 2.34390723868945889060156228887227706909540300963585626066\ldots, \\ P'(1) &= -8.07515226353164401026965124146737650134735782164232219222 1\ldots, \\ P''(1) &= 31.24865700624850564562564881961470385267564378216839540571\ldots, \end{aligned}$$

and $P(2) = 0.11463574622981963020052076295742568969846766987861875355 5\ldots$, $P'(2) = -0.31664753731523294093363384099033569196227349182039350377 7\ldots$ No datum of the low end enters.

Step 6: the low-end coefficients. $\mathsf{L}_0 = V_1 = 6$ (Definition 39.33), and $\mathsf{L}_1 = 0$ by Proposition 39.35, the weights being 1. For L_2 use the sector coordinates of an order π, $\alpha_{\pi(1)} = \rho$, $\alpha_{\pi(2)} = \rho q_2$, $\alpha_{\pi(3)} = \rho q_2 q_3$: then $K = \rho(1 + q_2 + q_2 q_3)$, $\Psi = \rho^2 q_2(1 + q_3 + q_2 q_3)$ and $e_3 = \rho^3 q_2^2 q_3$, so

$$f = q_2 q_3\,\varphi_\pi, \qquad \varphi_\pi = \big((1 + q_2 + q_2 q_3)(1 + q_3 + q_2 q_3)\big)^{-1}, \qquad \kappa_2 = \kappa_3 = 1,$$

the same for all six orders. In the formula of Proposition 39.35 at $i=2$: $A=\varnothing$ gives $\frac12\log^2\varphi_\pi(0)=0$; $A=\{2\}$ gives, with $m=1$, $\int_0^1(\log\varphi_\pi)_{[\{2\}]}\,dq_2/q_2=-\int_0^1\log(1+q)\,dq/q$ and, with $m=0$, nothing, $1_{[A]}$ being 0; $A=\{3\}$ gives the same as $\{2\}$; $A=\{2,3\}$ gives nothing. Expanding $\log(1+q)$,

$$\int_0^1\log(1+q)\,\frac{dq}{q}=\sum_{m\ge1}\frac{(-1)^{m-1}}{m^2}=\frac12(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),$$

so each order contributes $-(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $\mathsf{L}_2=-6\,(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))=-\pi^2=-\Gamma(\frac12)^4$, by the classical value $\sum_{m\ge1}m^{-2}=\pi^2/6$.

Step 7: the output against the low end. By Definition 39.33, $k^2P(k)=\mathsf{L}_0+\mathsf{L}_1k+\mathsf{L}_2k^2+O(k^3)$ near 0, while $P(k+1)$ and $P(k+2)$ are holomorphic there. Expanding (L) at $k=0$ to the orders 0, 1 and 2:

$$\mathsf{L}_0-3P(1)+9P(2)=0,\qquad \mathsf{L}_1-10P(1)-3P'(1)+18P(2)+9P'(2)=0,$$
$$\mathsf{L}_2-10P(1)-10P'(1)-\tfrac32P''(1)+9P(2)+18P'(2)+\tfrac92P''(2)=0.$$

The first is the low-end relation (Corollary 39.34(ii)). With $\mathsf{L}_0=6$, $\mathsf{L}_1=0$ and $\mathsf{L}_2=-\Gamma(\frac12)^4$, the three left sides, read on the output of Step 5, are below $2\cdot10^{-138}$ in modulus: the gauge end returns the low-end coefficients it did not use. The expansion at the orders 3 and 4 reads L_3 and L_4 off the output; they render as $4.80822761263837714159895264604579996 3\ldots$ and $15.48481981165258402025123201954778137 6\ldots$, and agree with $4\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $\frac{7}{60}\Gamma(\frac12)^8+\frac34\ell^2$ to 138 digits (a check).

Step 8: the arc integrals. $S_1=-\frac{2\pi}{3}=-\frac23\Gamma(\frac12)^2$. For S_2, expand $-\log\big(2\sin\frac{\vartheta}{2}\big)=\sum_{m\ge1}\frac{\cos m\vartheta}{m}$ on $(0,2\pi)$, a series that converges in $L^2(0,2\pi)$ and so may be integrated term by term over $(0,x)$:

$$-\int_0^x\log\big(2\sin\tfrac{\vartheta}{2}\big)\,d\vartheta=\sum_{m\ge1}\frac{\sin mx}{m^2}\qquad(0\le x\le2\pi).$$

At $x=\frac{2\pi}{3}$, $\sin\frac{2\pi m}{3}=\frac{\sqrt3}{2}\chi(m)$ with $\chi(m)=1,-1,0$ for $m\equiv1,2,0$ modulo 3, so

$$S_2=\frac{\sqrt3}{2}\Big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty))-(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\Big),$$

which renders as $0.67662773760643575001413503618301352396 1\ldots$; $2\sqrt3\,S_2$ agrees with the value of ℓ of Step 5 to 140 digits, and Exercise 52.4 proves $\ell=2\sqrt3\,S_2$.

Answer. The germ at $\frac19$ is $\varrho_f(t)=27\sqrt3\,\Gamma(\frac12)^2\sum_m e_m(\frac19-t)^m$, holomorphic, the line of the germs that extend by zero; it holds on all of $(0,\frac19]$. The values and slopes at 1 are the renderings above, the low-end coefficients are $\mathsf{L}_0=6$, $\mathsf{L}_1=0$, $\mathsf{L}_2=-\Gamma(\frac12)^4$, and $S_1=-\frac23\Gamma(\frac12)^2$, $S_2=\frac{\sqrt3}{2}\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty))-(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$. □

52.4 The torus at three marks

Exercise 52.4 (The torus at three marks)**.** Let F, P, ℓ, (L), the solution c and the closings x_j be as in Exercise 52.1, and S_2 as in Exercise 52.3. On the torus $\alpha_3=1$, $\alpha_1=e^{i\vartheta_1}$, $\alpha_2=e^{i\vartheta_2}$, let

$$T(k)=\frac{1}{(2\pi)^2}\int_{[0,2\pi]^2}|K|^{-2k}\,d\vartheta,$$

the torus value of the centre ray (Definition 40.11), wherever the integral converges.

(i) Show that T converges for $\operatorname{Re} k < 1$, satisfies (L), continues meromorphically, has $T(-j) = x_j$ and has a simple pole at $k = 1$ with the residue $R = -1/(\sqrt{3}\,\Gamma(\frac{1}{2})^2)$.

(ii) Show that $U(k) = 9^{1-k}T(1-k)$ satisfies (L), that $U(K) = c(K)$, and that $P(k)U(k+1) - P(k+1)U(k) = \frac{2}{3}\cdot 9^{1-k}/k^2$ for every k.

(iii) Show that $T = \dfrac{P}{2\Gamma(\frac{1}{2})^4} - \dfrac{\cot(\pi k)}{\sqrt{3}}\,U$.

(iv) Find the torus mean of $\log|K|$, and show that $\ell = 2\sqrt{3}\,S_2 = 3\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$.

Solution. Receipt: `ex21_torus.py`.

Step 1: the stride on the torus. On the torus $\alpha^{-1} = \bar\alpha$, and $\Psi(\alpha) = e_3\,K(\alpha^{-1})$, so $\Psi = e_3\bar K$ and $f = e_3/(K\Psi) = |K|^{-2}$, real and at least $\frac{1}{9}$. Put $F = |K|^2 = 3 + 2\cos\vartheta_1 + 2\cos\vartheta_2 + 2\cos(\vartheta_1 - \vartheta_2)$. Its gradient $\big(-2\sin\vartheta_1 - 2\sin(\vartheta_1-\vartheta_2),\ -2\sin\vartheta_2 + 2\sin(\vartheta_1-\vartheta_2)\big)$ vanishes exactly when $\sin\vartheta_1 = -\sin\vartheta_2 = -\sin(\vartheta_1-\vartheta_2)$: either $\vartheta_2 = -\vartheta_1$ and $\sin\vartheta_1(1+2\cos\vartheta_1) = 0$, or $\vartheta_2 = \vartheta_1+\pi$ and $\sin\vartheta_1 = 0$. So F has six critical points: $(0,0)$, the gauge $\mathbf{1}$, with $F = 9$, its maximum; $(\pi,0)$, $(0,\pi)$, (π,π), the signs, with $F = 1$; and the two zeros $z_\pm = (\frac{2\pi}{3}, \frac{4\pi}{3}), (\frac{4\pi}{3}, \frac{2\pi}{3})$ of K, where $F = Q(\delta) + O(|\delta|^3)$ in $\delta = \vartheta - \vartheta(z_\pm)$ with $Q(\delta) = \delta_1^2 - \delta_1\delta_2 + \delta_2^2$, of determinant $\frac{3}{4}$. In $u = \log f = -\log F$ the Hessian at $(0,0)$ is $H = \left(\begin{smallmatrix} 4/9 & -2/9 \\ -2/9 & 4/9 \end{smallmatrix}\right)$, of determinant $\frac{4}{27}$, the Hessian of Exercise 52.3, Step 4, read on the torus; at the signs it has the determinant -4 (Theorem 38.17(ii) read on the torus): they are nondegenerate saddles.

Step 2: convergence, the closings and the law. Near $z_\pm$, $|K|^{-2k}$ is of the size $|\delta|^{-2\operatorname{Re}k}$, integrable in two variables exactly when $\operatorname{Re}k < 1$; so T is holomorphic on $\operatorname{Re}k < 1$, with $|T(k)| \le T(\operatorname{Re}k)$. For $j \in \mathbb{N}$, $T(-j)$ is the mean of the Laurent polynomial $f^{-j} = |K|^{2j}$ over the torus, its constant term: $T(-j) = \mathrm{CT}(f^{-j}) = x_j$ (Definition 39.26). Multiply the certificate (C) of Exercise 52.1 by $f^k = \exp(k\log f)$, $\log f$ real on the torus off $z_\pm$: $\sum_e (k\,c_e + D_e)g_e\cdot f^k = \sum_e D_e(g_e f^k) = D_1H_1 + D_2H_2$ with $H_i = (g_i - g_3)f^k$ (Euler's identity, as in Step 5 there), and on the torus $D_e = -i\,\partial_{\vartheta_e}$. There $|\Psi| = |K|$, so $|g_e| \le C|K|^{-3}$, $|D_e g_e| \le C|K|^{-4}$ and $|c_e| \le C|K|^{-1}$; for $\operatorname{Re}k \le -3$ the H_i are C^1 on the torus and $\int \partial_{\vartheta_i} H_i\,d\vartheta = 0$. So (L) holds for T at $\operatorname{Re}k \le -3$, and by the identity theorem wherever $T(k)$, $T(k+1)$, $T(k+2)$ are holomorphic, $\operatorname{Re}k < -1$. Solved for $T(k+2)$, (L) continues T meromorphically to $\mathbb{C}$, with poles only at the positive integers; on $\operatorname{Re}k < 2$ the only possible pole is at 1, the argument of $T(k+2)$ at the zero $k = -1$ of $9(k+1)^2$.

Step 3: the residue at 1. Cut out the regions $Q(\delta) < \varepsilon^2$ around $z_\pm$. Off them the integrand is bounded for $\operatorname{Re}k \le \frac{3}{2}$, and on them $F^{-k} - Q^{-k} = O(|\delta|^{1-2\operatorname{Re}k})$ is integrable for $\operatorname{Re}k < \frac{3}{2}$; both give functions holomorphic near $k = 1$. And

$$\int_{Q(\delta)<\varepsilon^2} Q(\delta)^{-k}\,d\delta = \frac{2\pi}{\sqrt{3/4}}\int_0^\varepsilon r^{1-2k}\,dr = \frac{4\pi}{\sqrt{3}}\cdot\frac{\varepsilon^{2-2k}}{2-2k}.$$

Two zeros and the factor $(2\pi)^{-2}$ give $T(k) = \frac{1}{\sqrt{3}\,\pi}\cdot\frac{1}{1-k} + O(1)$ near 1: a simple pole with the residue $R = -1/(\sqrt{3}\,\pi) = -1/(\sqrt{3}\,\Gamma(\frac{1}{2})^2)$. Write $T(k) = R/(k-1) + T_0 + O(k-1)$.

Step 4: the torus read backwards. Read at $-k-1$, (L) for T is $(k+1)^2T(-k-1) - (10k^2 + 10k + 3)\,T(-k) + 9k^2T(1-k) = 0$. With $U(k) = 9^{1-k}T(1-k)$, $U(k+1) = 9^{-k}T(-k)$ and $U(k+2) = 9^{-k-1}T(-k-1)$,

$$k^2U(k) - (10k^2+10k+3)\,U(k+1) + 9(k+1)^2U(k+2)$$
$$= 9^{-k}\big(9k^2T(1-k) - (10k^2+10k+3)\,T(-k) + (k+1)^2T(-k-1)\big) = 0.$$

So U solves (L), and $U(K) = 9^{1-K}x_{K-1} = c(K)$ (Exercise 52.1, Step 8). Its poles are at $0, -1, -2, \dots$, and $U(k) = -9R/k + O(1)$ at 0. It is the solution of the type of the signs: $c(K)$ decays like the native value 1, not like 9^{-K}.

Step 5: the frame of the chart and the torus. For meromorphic solutions X, Y of (L), $W(X,Y)(k) = X(k)Y(k+1) - X(k+1)Y(k)$ satisfies $p_2(k)W(k+1) = p_0(k)W(k)$ (substitute $p_2X(k+2) = -p_0X(k) - p_1X(k+1)$ and the same for Y; Proposition 38.11 with $h = 2$), so $k^2 9^{k-1}W(X,Y)(k)$ is periodic of period 1. A periodic function h of period 1, holomorphic on the strip $|\operatorname{Re} k| \le \frac12$ with $|h(k)| \le C(1+|k|)^N$ there, is constant: $h(k) = \mathcal{H}(e^{2\pi i k})$ with $\mathcal{H}$ holomorphic on $\mathbb{C}^\times$ and $|\mathcal{H}(z)| = O(|\log|z||^N)$ at 0 and at ∞, so both singularities are removable and $\mathcal{H}$ is constant by Liouville's theorem. On the strip, P is holomorphic except for its pole at 0, $P(k) = 6/k^2 + O(1)$ ($\mathsf{L}_0 = 6$, $\mathsf{L}_1 = 0$: Exercise 52.3, Step 6), and polynomially bounded, being continued from $\operatorname{Re} k > 0$, where $|P(k)| \le P(\operatorname{Re} k)$, by (L) solved for $P(k)$; $T(k)$ and $T(-k)$ are bounded; $T(k+1)$ and $T(1-k)$ have a simple pole at 0 only and are polynomially bounded, by (L).

(a) In $k^2 9^{k-1}W(P,U)$ the pole of $P(k)$ at 0 meets the holomorphic $U(k+1)$ and the simple pole of $U(k)$ meets the holomorphic $P(k+1)$, so the product with k^2 is holomorphic at 0: the function is constant, and at $k = 1$ it is $P(1)c(2) - P(2)c(1) = \frac{\ell}{3} - \frac{\ell-2}{3} = \frac23$. So

$$P(k)U(k+1) - P(k+1)U(k) = \frac{2}{3} \cdot \frac{9^{1-k}}{k^2} \qquad \text{for every } k,$$

the frame of the chart and the torus read backwards; at the integers it is the frame of r and c of Exercise 52.1, Step 8.

(b) Let $m_1 = (2\pi)^{-2} \int \log|K| \, d\vartheta$, so that $T(-k) = 1 + 2m_1 k + O(k^2)$ and $U(k+1) = 1 + (2m_1 - \log 9)\, k + O(k^2)$. Near 0, $P(k)U(k+1) = 6/k^2 + 6(2m_1 - \log 9)/k + O(1)$ and $P(k+1)U(k) = -9R\ell/k + O(1)$, while $\frac23 \cdot 9^{1-k}/k^2 = 6/k^2 - 6\log 9/k + O(1)$. The terms in $1/k$ give $12m_1 + 9R\ell = 0$:

$$m_1 = -\tfrac34 R\,\ell = \frac{\sqrt{3}\,\ell}{4\,\Gamma(\frac12)^2}.$$

Step 6: the connection up to one constant. $W(P,U) \neq 0$, so P and U are independent solutions and Cramer's rule on $T(k), T(k+1)$ gives $T = \alpha P + \beta U$ with $\alpha = W(T,U)/W(P,U)$ and $\beta = W(P,T)/W(P,U)$, periodic of period 1 by (a). First,

$$\alpha(k) = \tfrac{k^2}{6}\, T(k)T(-k) - \tfrac{3k^2}{2}\, T(k+1)T(1-k),$$

holomorphic on the strip, the double pole $T(k+1)T(1-k) = -R^2/k^2 + O(1)$ being cancelled by k^2, and polynomially bounded: α is constant, equal to its limit at 0, $\alpha = \frac32 R^2 = \frac{1}{2\pi^2} = \frac{1}{2\Gamma(\frac12)^4}$. Next,

$$\beta(k) = \tfrac{3k^2}{2}\, 9^{k-1}\big(P(k)T(k+1) - P(k+1)T(k)\big) = \frac{R}{k} + T_0 + R\log 9 + O(k) \qquad \text{near } 0,$$

from $P(k) = 6/k^2 + O(1)$ and $T(k+1) = R/k + T_0 + O(k)$. The function $R\,\pi\cot(\pi k)$ has simple poles at the integers with the residue R, so $\beta - R\,\pi\cot(\pi k)$ is periodic, entire and polynomially bounded on the strip, $\cot(\pi k)$ being bounded away from the real axis: it is a constant $b_0 = T_0 + R\log 9$. So

$$T = \frac{P}{2\Gamma(\frac12)^4} + \Big(b_0 - \frac{\cot(\pi k)}{\sqrt{3}}\Big)U.$$

Step 7: $b_0 = 0$, by the Fourier transform at the frequency 0. Put $t = e^u$. The image ν of $d\vartheta/(2\pi)^2$ under $u = -\log F$ is carried by $[u_0, \infty)$, $u_0 = -\log 9$, and has a density $n(u)$ with the following properties, by the Morse lemma at the nondegenerate critical points of Step 1 and the regularity of the other level sets: (a) near the maximum, $u - u_0 = \frac12 |v|^2$ in coordinates v with $d\vartheta = J\,dv$, $J(0) = (\det H)^{-1/2}$, so n is continuous on $[u_0, u_0 + \delta)$ with $n(u_0) = (2\pi)^{-2}\, 2\pi J(0) = \frac{3\sqrt3}{4\pi} > 0$; in t, the density of the torus law at $\frac19$ is $9\, n(u_0) = \frac{27\sqrt3}{4\pi} = (2\pi)^{-2} \varrho_f(\frac19)$, with $\varrho_f(\frac19)$ of Exercise 52.3; (b) near each saddle, $u = v_1 v_2$ in coordinates with $d\vartheta = \tilde J\, dv$, $\tilde J(0) = \frac12$, the inverse square root of the modulus 4 of the determinant of the Hessian,, and the density of $v_1 v_2$ on $|v_1|, |v_2| < \delta$ at $s \neq 0$ is $\int_{|s|/\delta < |v_1| < \delta} \tilde J(v_1, s/v_1)\, dv_1/|v_1|$; writing $\tilde J(v) = \tilde J(0) + v_1 A(v) + v_2 B(v)$, this is $2\tilde J(0) \log(\delta^2/|s|)$ plus two integrals that converge, as $s \to 0$ from either side, to the same limits; so near $u = 0$, with the three saddles, $n(u) = -a \log|u| + b(u)$ with $a = 3 \cdot 2\tilde J(0)/(2\pi)^2 = \frac{3}{4\pi^2}$ and b continuous at 0; (c) near $z_\pm$, the nondegenerate minima of F, $n(u) = O(e^{-u})$, so $e^{cu} n(u)$ is integrable on $[u_0, \infty)$ for $0 < c < 1$.

Fix $c \in (0,1)$ and put $k = c + iy$, $y > 0$. Then $T(k) = \int e^{iyu} \psi(u)\, du$ with $\psi = e^{cu} n$; $P(k) = \int e^{iyu} \varphi(u)\, du$ with $\varphi(u) = e^{(c+1)u} \varrho_f(e^u)$, carried by $u \le u_0 < 0$; and, with $w = u_0 - u$ in $T(1-k)$, $U(k) = 9^{1-c} \int e^{iyw} \tilde\psi(w)\, dw$ with $\tilde\psi(w) = e^{(1-c)(u_0 - w)} n(u_0 - w)$, carried by $w \le 0$. And $\beta(k) = b_0 + \frac{i}{\sqrt3} + O(e^{-2\pi y})$, as $\cot(\pi k) \to -i$.

Let χ be smooth with support in $[1,2]$ and $\int \chi = 1$, $g(s) = s\chi(s)$, $\hat g(w) = \int g(s) e^{isw} ds$, rapidly decreasing, and for a function Φ of y put $M_Y[\Phi] = Y^{-1} \int \Phi(y)\, y\, \chi(y/Y)\, dy$. For $\Phi(y) = \int e^{iyu} \psi(u)\, du$ with ψ integrable, Fubini gives $M_Y[\Phi] = \int \psi(w/Y)\, \hat g(w)\, dw$. Now $\int \hat g = 2\pi g(0) = 0$, $\int_0^\infty \hat g = i \int g(s)\, s^{-1} ds = i$, $\int_{-\infty}^0 \hat g = -i$, and $\int \log|w|\, \hat g(w)\, dw = -\pi$ (with $\hat g = -i\hat\chi'$, integrate by parts and use $\mathrm{PV} \int e^{isw} dw/w = i\pi$ for $s > 0$). So, as $Y \to \infty$: if ψ is bounded near 0 with one-sided limits $\psi(0\pm)$, $M_Y[\Phi] \to i\big(\psi(0+) - \psi(0-)\big)$; if $\psi = -a \log|u| + b(u)$ near 0 with b continuous at 0, $M_Y[\Phi] \to \pi a$.

Apply M_Y to the connection along $k = c + iy$. The torus value gives πa, by (b). The values give 0, φ vanishing near 0. The last term gives $\big(b_0 + \frac{i}{\sqrt3}\big)\, 9^{1-c}\, i\big(0 - \tilde\psi(0-)\big)$, the factor $O(e^{-2\pi y})$ of β contributing nothing in the limit, with $9^{1-c}\tilde\psi(0-) = 9^{1-c} e^{(1-c)u_0} n(u_0) = n(u_0)$. With $Q = n(u_0) > 0$:

$$\pi a = \Big(b_0 + \frac{i}{\sqrt3}\Big)(-i)\, Q = \frac{Q}{\sqrt3} - i\, b_0 Q.$$

a, b_0 and Q are real, so $b_0 = 0$ (and $\pi a = Q/\sqrt3$: both are $\frac{3}{4\pi}$). Hence (iii), and $T_0 = -R \log 9 = \dfrac{2 \log 3}{\sqrt3\, \Gamma(\frac12)^2}$.

Step 8: what the connection carries. (a) At $k = 1 + j$, $j \ge 0$, P and U are holomorphic and $-\cot(\pi k)/\sqrt3$ has the residue $-1/(\sqrt3\,\pi)$, so $\mathrm{Res}_{k=1+j} T = -\frac{9^{-j} x_j}{\sqrt3\, \Gamma(\frac12)^2}$: the poles of the torus value carry its values at the negative integers. (b) Near 0, $P = 2\Gamma(\frac12)^4 \big(T + \frac{\cot(\pi k)}{\sqrt3} U\big)$ and $\frac{\cot(\pi k)}{\sqrt3} U = \frac{1}{\sqrt3 \pi k} \cdot \frac{-9R}{k} + O(\frac1k) = \frac{3}{\pi^2 k^2} + O(\frac1k)$: the low-end coefficient $\mathsf{L}_0 = 6$ is the residue of the torus at the zeros of K. (c) For real $a \ge 0$, $a \neq 1$, the mean of $\log|a + e^{i\vartheta}|$ over ϑ is $\log\max(a, 1)$ (for $a > 1$ write $\log|a + e^{i\vartheta}| = \log a + \mathrm{Re} \log(1 + e^{i\vartheta}/a)$, whose second term is a power series in $e^{i\vartheta}$ without constant term; for $a < 1$ the same with $\log|1 + a e^{-i\vartheta}|$). Integrating over ϑ_1 first, with $a = |1 + e^{i\vartheta_2}| = 2|\cos\frac{\vartheta_2}{2}|$, which exceeds 1 exactly when $|\vartheta_2| < \frac{2\pi}{3}$,

$$m_1 = \frac{1}{\pi} \int_0^{2\pi/3} \log\big(2\cos\tfrac{\vartheta}{2}\big)\, d\vartheta = -\frac{1}{\pi} \int_0^{\pi/3} \log\big(2\sin\tfrac{\varphi}{2}\big)\, d\varphi = \frac{1}{\pi} \sum_{m \ge 1} \frac{\sin(m\pi/3)}{m^2},$$

by $\vartheta = \pi - \varphi$, $\int_0^\pi \log(2\sin\frac{\varphi}{2})\,d\varphi = 0$ and the Fourier series of Exercise 52.3, Step 8. Here $\sin\frac{m\pi}{3} = \frac{\sqrt3}{2}\varepsilon(m)$ with $\varepsilon(m) = 1,1,0,-1,-1,0$ for $m \equiv 1,\dots,6$ modulo 6, and $\varepsilon(m) = \chi(m)$ for odd m, $-\chi(m)$ for even m, χ as in Exercise 52.3; $\chi(2m) = -\chi(m)$, so $\sum_m \varepsilon(m)m^{-2} = \sum_m \chi(m)m^{-2} + \frac12\sum_m \chi(m)m^{-2}$, and $m_1 = \frac{3\sqrt3}{4\pi}\sum_m \chi(m)m^{-2} = \frac{3}{2\pi}S_2$. With Step 5(b):

$$\ell = \frac{4\pi m_1}{\sqrt3} = 2\sqrt3\,S_2 = 3\Big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\Big).$$

Checks (floating point, not part of the solution). By the mean over one circle, $T(k) = \frac{2}{\pi}\int_0^2 I_k(a)\,\frac{da}{\sqrt{4-a^2}}$ with $I_k(a)$ the mean of $|a+e^{i\vartheta}|^{-2k}$, a hypergeometric series in a^2 or a^{-2}; this gives $T(-j) = x_j$, (L) on T at $k = -1.3$ and $k = -2.7+0.4i$ to 20 digits, and, with P from the gauge end, the connection of (iii) at $k = 0.2, 0.3, \dots, 0.8$ to between 17 and 23 digits, $k^2 9^{k-1}W(P,U) = \frac23$ at the same points to 20 digits, $\beta(\frac12) = b_0$ below $2\cdot10^{-23}$, and T_0 rendering as $0.40379779973879724450\ldots$; m_1 renders as $0.32306594721945051409363651072 4\ldots$

Answer. T continues to a meromorphic solution of (L) with $T(-j) = x_j$ and the residue $-1/(\sqrt3\,\Gamma(\frac12)^2)$ at 1; $U(k) = 9^{1-k}T(1-k)$ is the solution c of Exercise 52.1, with $P(k)U(k+1) - P(k+1)U(k) = \frac23\cdot 9^{1-k}/k^2$; the connection is $T = \frac{P}{2\Gamma(\frac12)^4} - \frac{\cot(\pi k)}{\sqrt3}U$; the torus mean of $\log|K|$ is $\frac{\sqrt3\,\ell}{4\Gamma(\frac12)^2}$, and $\ell = 2\sqrt3\,S_2 = 3\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$. □

52.5 A lean of three marks

Exercise 52.5 (The lean $(2\mathbf{1};4,1)$ of $\binom{[3]}{1}$)**.** Let F and ℓ be as in Exercise 52.1, and consider the ray of F with the origin 0 and the direction $(2\mathbf{1};4,1)$, with the values $P_\lambda(k) = P_F(2k\mathbf{1};4k,k)$.

(i) Find its walls, V_1, its stride and native value, and its closings.

(ii) Reduce its box $\mathfrak{B}_2(1,1)$ modulo the permutations of the marks: the spanning set at $\mathfrak{s} = 1$, the matrix, the law of [1], its symbol and its native gauges, and $\mathbf{B}(\infty)$.

(iii) Read the low-end relation, and show that $54\,P_\lambda(1) = 11 - 4\ell$.

(iv) Write $P_\lambda(1)$ on the share side, and say why the gap theorem cannot be applied to the rational coordinates of this ray.

Solution. Receipt: `ex22_lean.py`.

Step 1: the ray. By Definition 8.1 and the counts of Exercise 52.1, Step 1, the walls of the direction are $\kappa_{1,S} = m_S - 4d(S) - d^*(S) = m_S - (|S|-1)$: 2 on the marks and 3 on the pairs. Every order π has the tails U_2, a pair, and U_3, a mark, so $V_1 = 6\cdot\frac{1}{3\cdot2} = 1$ (Proposition 39.30). The stride is $f = \alpha^{2\mathbf{1}}K^{-4}\Psi^{-1} = e_3^2/(e_1^4e_2)$. The defects $c_e = 2 - 4R_e - R_e^*$ vanish at $\mathbf{1}$ $(2 - \frac43 - \frac23 = 0)$, so $\alpha^\star = \mathbf{1}$ and $f(\alpha^\star) = \frac{1}{3^4\cdot3} = \frac{1}{243}$. The constant term of $f^{-j} = e_1^{4j}\big(\sum_e \alpha^{\mathcal{E}\setminus e}\big)^j\alpha^{-2j\mathbf{1}}$ pairs the monomial $\alpha^{(j-b_1,j-b_2,j-b_3)}$ of the second factor with the monomial $\alpha^{(j+b_1,j+b_2,j+b_3)}$ of e_1^{4j}, so

$$x_j = \mathrm{CT}(f^{-j}) = (4j)!\,j!\sum_{b_1+b_2+b_3=j}\ \prod_{e=1}^{3}\frac{1}{b_e!\,(b_e+j)!},$$

$$(x_j)_{j\ge0} = 1,\ 36,\ 4620,\ 762300,\ 140216076,\ 27409661136,\ \ldots$$

Step 2: the box. The permutations of the marks are symmetries; the Cremona is not, the exponents 4 and 1 being different (Proposition 39.7(ii)). Modulo S_3 the box $\mathfrak{B}_2(1,1)$ (Definition 39.11), the box with one unit of each letter at $\mathfrak{t} = 2$, has 398 symbols in 79 orbits (17, 19, 20 and 23 at

the levels $(0,0)$, $(1,0)$, $(0,1)$, $(1,1)$) and 82 stems. Its relations are 184 letter identities for K, 178 for Ψ and 246 mark relations of stems; modulo S_3 they give 117 distinct combinations of orbit averages, of rank 76. The free orbits are those of $\alpha_1\alpha_2\alpha_3/(K\Psi)$ (wall degree $-\frac{1}{3}$), of α_1/α_3 (wall degree $\frac{1}{2}$) and of 1, so the spanning set at $\mathfrak{s}=1$ is again $(1,b_1)$, $b_1=\frac{1}{6}\sum_{i\neq j}\alpha_i/\alpha_j$, and the box is accepted there (Definition 39.16) with

$$\mathbf{B}_{11}=\frac{199k^3-420k^2+299k-72}{(k-1)^2(3k-2)},\qquad \mathbf{B}_{12}=\frac{(2k-1)(265k^2-424k+171)}{(k-1)^2(3k-2)},$$
$$\mathbf{B}_{21}=\frac{3710k^4-12349k^3+15022k^2-7997k+1584}{3(k-1)(2k-3)(3k-4)(3k-2)},$$
$$\mathbf{B}_{22}=\frac{(2k-1)(4706k^3-13323k^2+12238k-3681)}{3(k-1)(2k-3)(3k-4)(3k-2)}.$$

Certificates on the stems (Theorem 39.17(ii)): for each row, G_1 a combination of the stems symmetric under the exchange of the marks 2 and 3, G_e its images; the identity is linear in 44 unknowns with 118 equations and a family of solutions of dimension 24; a solution with 20 nonzero coefficients, rational functions of k, satisfies it exactly for each row. It is not displayed here; the receipt prints it.

Step 3: the law. The rows of Proposition 39.6 give the first dependence with

$$q_0=-972(k-1)^2(4k-3)(4k-1)(265k^2-954k+860),$$
$$q_1=-1721705k^6+13084958k^5-40594556k^4+65762882k^3$$
$$-58662047k^2+27325016k-5197140,$$
$$q_2=3(k-2)^2(3k-5)(3k-4)(265k^2-424k+171),$$

and $p_j(k)=q_{2-j}(k+2)$, divided by their common factor, is the recurrence

$$-3k^2(3k+1)(3k+2)(265k^2+636k+383)\,P_\lambda(k)$$
$$+(1721705k^6+7575502k^5+13047276k^4+11068046k^3+4766659k^2+965100k+75600)\,P_\lambda(k+1)$$
$$+972(k+1)^2(4k+5)(4k+7)(265k^2+106k+12)\,P_\lambda(k+2)=0,$$

of degree 6, with the symbol $265(243\lambda-1)(64\lambda+27)$; its reflection is satisfied by the closings of Step 1 (Corollary 39.34(i)). The native gauges: at $\mathbf{1}$ the native value $\frac{1}{243}$, and the Hessian of $\log f$ on the chart $\alpha_3=1$ has the determinant $\frac{25}{27}$; at $\alpha=(3,3,-2)$ and its permutations all three defects vanish (exact evaluation), the native value is $\frac{(-18)^2}{4^4\cdot(-3)}=-\frac{27}{64}$, and the Hessian has the determinant $-\frac{75}{2}$. Both are nondegenerate, their native values are the roots of the symbol (Corollary 38.7), and the vectors $(1,b_1(\alpha_c))$ are $(1,1)$ and $(1,-\frac{7}{18})$, which span: by Proposition 39.38(i) the spanning set is a set of masters and the recurrence is the law of [1]. Finally

$$\mathbf{B}(\infty)=\begin{pmatrix}\frac{199}{3} & \frac{530}{3}\\ \frac{1855}{27} & \frac{4706}{27}\end{pmatrix},$$

with the eigenvalues 243 and $-\frac{64}{27}$ and the eigenvectors $(1,1)$ and $(1,-\frac{7}{18})$ (Proposition 39.38(ii)).

Step 4: the low-end relation. p_0 vanishes to the order 2 at 0 with $[k^2]p_0=-3\cdot1\cdot2\cdot383=-2298$, $p_1(0)=75600$ and $p_2(0)=972\cdot5\cdot7\cdot12=408240$; with $\mathsf{L}_0=V_1=1$, Corollary 39.34(ii) gives $-2298+75600\,P_\lambda(1)+408240\,P_\lambda(2)=0$, that is

$$12600\,P_\lambda(1)+68040\,P_\lambda(2)=383.$$

Step 5: the value, read on the centre ray. At $k = 2$ the centre ray of Exercise 52.1 has the stride $f_{\mathrm{c}} = \alpha^{\mathbf{1}}/(K\Psi)$, and

$$\langle \Psi K^{-2} \rangle_2 = \int \Psi K^{-2} \, \frac{\alpha^{\mathbf{21}}}{K^2\Psi^2} \, dy = \int \frac{\alpha^{\mathbf{21}}}{K^4\Psi} \, dy = P_\lambda(1) :$$

the value of the lean is the value of the insertion $\Psi K^{-2} = \sum_e \alpha^{\mathcal{E}\setminus e} K^{-2}$ along the centre ray. In the box $\mathfrak{B}_2(2,2)$ of the centre ray modulo S_3 and the Cremona (339 symbols, 43 orbits, 499 relations of rank 40, the free orbits those of f_{c}^2, of α_1/α_3 and of 1) the orbit average of ΨK^{-2} reduces to

$$\Psi K^{-2} \equiv -\frac{2(k-1)}{3(k+1)} + \frac{(k-1)^2}{k(k+1)} \, b_1.$$

The relations of a box evaluate into $\mathcal{L}$ and their values vanish identically in k (Proposition 39.14(i)); the symmetries keep the values (Proposition 39.7(ii)); and at $k = 2$ every value involved converges. So, with $\langle b_1 \rangle_k = \frac{1}{6}\big(\langle f_{\mathrm{c}}^{-1} \rangle_k - 3\langle 1 \rangle_k\big) = \frac{1}{6}\big(P(k-1) - 3P(k)\big)$ from $f_{\mathrm{c}}^{-1} = 3 + 6b_1$, $\langle b_1 \rangle_2 = \frac{1}{6}\big(\ell - (\ell-2)\big) = \frac{1}{3}$, and

$$P_\lambda(1) = -\frac{2}{9} \, P(2) + \frac{1}{6} \cdot \frac{1}{3} = -\frac{2}{9} \cdot \frac{\ell - 2}{3} + \frac{1}{18} = \frac{11 - 4\ell}{54}.$$

Then Step 4 gives $P_\lambda(2) = \frac{2800\ell - 6551}{204120}$, and the law gives every $P_\lambda(K)$ in $\mathbb{Q} + \mathbb{Q}\ell$.

Step 6: the share side. By Proposition 8.15, $P_F(\mathbf{m}; s, t)$ is the period of the dual table, whose bases are the sets $\mathcal{E} \setminus e$, at the exponents (t, s); that table is one circuit, and Corollary 18.22 with the exponents $(1, 4)$, $\nu_e = m_e - 1 = 1$ and unit weights (its condition $m_S > |S| - 1$ holds: $2 > 0$, $4 > 1$) gives

$$P_\lambda(1) = \frac{1}{\Gamma(4)\Gamma(1)} \int_0^\infty \mathsf{G}_{1;(1,-1)}(1, y)^3 \, dy,$$

that is $\frac{1}{6} \int_0^\infty \mathsf{G}_{1;(1,-1)}(1, y)^3 \, dy$. A check by quadrature: both this integral and $\frac{11-4\ell}{54}$ render as $0.03008094528226230439988427489834 9847\ldots$

Step 7: why the gap does not apply. Write $P_\lambda(K) = c_\lambda(K)\,\ell + r_\lambda(K)$ with rationals from Steps 3 to 5, and let D_K be the least common denominator of $c_\lambda(K)$ and $r_\lambda(K)$, the least integer that clears the row. Theorem 37.2 needs $D_K P_\lambda(K) \to 0$. But the values decay no faster than 243^{-K}: as in Exercise 52.1, Step 2(b), $f \geq \frac{1}{243}\,\alpha_{(2)}\alpha_{(3)}^2/\alpha_{(1)}^3$, and on the cone of an order the exponent $3y_{(1)} - y_{(2)} - 2y_{(3)} = 3(y_{(1)} - y_{(2)}) + 2(y_{(2)} - y_{(3)})$ has the walls 3 and 2 as coefficients, so $P_\lambda(K) \geq 243^{-K} \cdot 6 \cdot \frac{1}{6K^2} = 243^{-K}/K^2$. The exact denominators satisfy $D_K > 243^K K^2$ for every K with $3 \leq K \leq 240$, so $D_K P_\lambda(K) > 1$ throughout that range; $\frac{1}{K} \log D_K$ renders as about 10.3 at $K = 240$, against $\log 243 = 5.49$. The clearing integers of this ray grow faster than its values decay, and the hypothesis of Theorem 37.2 fails on every computed row.

Answer. Walls 2 and 3, $V_1 = 1$, $f(\alpha^\star) = \frac{1}{243}$; the masters $1, b_1$, the law above with the symbol $265(243\lambda - 1)(64\lambda + 27)$ and the native gauges $\mathbf{1}$ and $(3, 3, -2)$; the low-end relation $12600P_\lambda(1) + 68040P_\lambda(2) = 383$; and $54\,P_\lambda(1) = 11 - 4\ell$, with the share side $\frac{1}{6} \int_0^\infty \mathsf{G}_{1;(1,-1)}(1, y)^3 dy$. □

52.6 The balanced directions

Exercise 52.6 (The balanced directions of $\binom{[3]}{1}$)**.** Let F and ℓ be as in Exercise 52.1. For a direction $(\mathbf{m}; s, t)$ on the balance wall put $c_e = m_e - t$.

(i) List the balanced directions with $1 \leq s, t \leq 5$, positive walls and $m_1 \geq m_2 \geq m_3$, and sort them by $S = s + t$ and the multiset $\{|c_e|\}$.

(ii) Show that $\Gamma(sk)\Gamma(tk)\,P_F(k\mathbf{m};ks,kt)$ depends on the direction only through S and $\{|c_e|\}$, so that within a class the values are in the ratio of two Beta corners.

(iii) For every direction with $|c_e|$ constant find $P_F(\mathbf{m};s,t)$ in $\mathbb{Q}+\mathbb{Q}\ell$, and its native value $f(\alpha^\star)$.

Solution. Receipt: `ex23_directions.py`.

Step 1: the directions. The balance wall is $m_{\mathcal{E}} = s+2t$, and the walls are $\kappa_S = m_S - t(|S|-1)$ (Exercise 52.5, Step 1): $m_e > 0$ and $m_e + m_{e'} > t$. There are 123 such directions. The multiset $\{|c_e|\}$, with $\sum_e c_e = s-t$, has one value on 17 of them, two on 70 and three on 36; the classes of equal S and $\{|c_e|\}$ number 54: 9, 31 and 14. The 17 directions with $|c_e|$ constant fall into nine classes:

$\lvert c_e\rvert$, S	directions $(\mathbf{m};s,t)$	$P_F(\mathbf{m};s,t)$
0, $2c'$	$(c'\mathbf{1};c',c')$, $c'=1,\dots,5$	ℓ, $\frac{\ell-2}{3}$, $\frac{10\ell-23}{54}$, $\frac{62\ell-145}{486}$, $\frac{568\ell-1331}{5832}$
1, 5	$(4,2,2;2,3)$, $(3,3,1;3,2)$	$\frac{11-4\ell}{18}$
	$(3,3,3;1,4)$, $(2,2,2;4,1)$	$\frac{11-4\ell}{54}$
1, 7	$(5,3,3;3,4)$, $(4,4,2;4,3)$	$\frac{19-8\ell}{54}$
	$(4,4,4;2,5)$, $(3,3,3;5,2)$	$\frac{19-8\ell}{108}$
1, 9	$(6,4,4;4,5)$, $(5,5,3;5,4)$	$\frac{751-320\ell}{2916}$
2, 8	$(7,3,3;3,5)$, $(5,5,1;5,3)$	$\frac{80\ell-181}{648}$

the last column being found in Step 3.

Step 2: the classes. The walls are positive, so Proposition 8.15 and Corollary 18.22 apply as in Exercise 52.5, Step 6, at the data $(k\mathbf{m};ks,kt)$, with $\nu_e = km_e - kt = kc_e$:

$$\Gamma(sk)\Gamma(tk)\,P_F(k\mathbf{m};ks,kt) = \int_0^\infty y^{tk-1}\prod_e \mathsf{G}_{kc_e;(1,-1)}(1,y)\,dy.$$

Proposition 2.3 with $A=1$ and $\mathbf{h}=\frac12\log y$ gives $\mathsf{G}_{\nu;(1,-1)}(1,y) = y^{\nu/2}\,\mathsf{G}_{\nu;(1,-1)}(\sqrt{y},\sqrt{y})$, and with $A=-1$ it gives $\mathsf{G}_{\nu;(1,-1)}(\xi_1,\xi_2) = \mathsf{G}_{-\nu;(1,-1)}(\xi_2,\xi_1)$, so at equal coefficients the Gramma is even in ν. With $\sum_e c_e = s-t$,

$$\Gamma(sk)\Gamma(tk)\,P_F(k\mathbf{m};ks,kt) = \int_0^\infty y^{\frac{Sk}{2}-1}\prod_e \mathsf{G}_{k|c_e|;(1,-1)}(\sqrt{y},\sqrt{y})\,dy,$$

which depends on S and $\{|c_e|\}$ only. So two directions d,d' of one class have

$$\frac{P_{d'}(k)}{P_d(k)} = \frac{\Gamma(sk)\Gamma(tk)}{\Gamma(s'k)\Gamma(t'k)} = \Gamma_1^2\begin{bmatrix} sk,\ tk \\ Sk\end{bmatrix}\Big/\Gamma_1^2\begin{bmatrix} s'k,\ t'k \\ Sk\end{bmatrix},$$

two Beta corners at the same total, rational at the integers; for instance $\frac{\Gamma(4k)\Gamma(k)}{\Gamma(3k)\Gamma(2k)} = 3, 7, \frac{33}{2}, 39$ at $k=1,2,3,4$ for $(4,2,2;2,3)$ against $(3,3,3;1,4)$. A class has one value span.

Step 3: the values, read on the centre ray. Along the centre ray of Exercise 52.1, with $f_{\mathrm{c}} = \alpha^{\mathbf{1}}/(K\Psi)$, the insertion $\alpha^u K^{-a}\Psi^{-b}$ has the value $\langle \alpha^u K^{-a}\Psi^{-b}\rangle_k = P_F(k\mathbf{1}+u;k+a,k+b)$ (Definition 38.1). So

$$P_F(k\mathbf{1}+2\mathbf{e}_1;k,k+1) = \langle \alpha_1^2\Psi^{-1}\rangle_k, \qquad P_F(3\mathbf{1}+4\mathbf{e}_1;3,5) = \langle \alpha_1^4\Psi^{-2}\rangle_3,$$

and $(4,2,2;2,3)$, $(5,3,3;3,4)$, $(6,4,4;4,5)$ are $\langle \alpha_1^2\Psi^{-1}\rangle_k$ at $k=2,3,4$. Reduced in the boxes of the centre ray modulo S_3 and the Cremona, as in Exercise 52.2 (the orbit averages have the same values, Proposition 39.7(ii)),

$$\alpha_1^2\Psi^{-1} \equiv -\frac{2}{3} + \frac{k-1}{k}\,b_1 \quad \text{in } \mathfrak{B}_2(1,1),$$

$$\alpha_1^4\Psi^{-2} \equiv \frac{7k+5}{9(k+1)} - \frac{(k-1)(2k^2-6k-5)}{3k(k-2)(k+1)}\, b_1 \quad \text{in } \mathfrak{B}_2(2,2),$$

and the relations of a box hold at every strength where the values converge (Proposition 39.14(i)). With $P(K) = c(K)\ell + r(K)$ (Exercise 52.1, Step 8) and $\langle b_1\rangle_k = \frac{1}{6}\big(P(k-1) - 3P(k)\big)$, so that $\langle b_1\rangle_2 = \frac{1}{3}$, $\langle b_1\rangle_3 = \frac{11-4\ell}{108}$ and $\langle b_1\rangle_4 = \frac{19-8\ell}{243}$,

$$P_F(4,2,2;2,3) = -\tfrac{2}{3}\cdot\tfrac{\ell-2}{3} + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{11-4\ell}{18}, \qquad P_F(5,3,3;3,4) = -\tfrac{2}{3}\cdot\tfrac{10\ell-23}{54} + \tfrac{2}{3}\cdot\tfrac{11-4\ell}{108} = \tfrac{19-8\ell}{54},$$
$$P_F(6,4,4;4,5) = -\tfrac{2}{3}\cdot\tfrac{62\ell-145}{486} + \tfrac{3}{4}\cdot\tfrac{19-8\ell}{243} = \tfrac{751-320\ell}{2916},$$
$$P_F(7,3,3;3,5) = \tfrac{13}{18}\cdot\tfrac{10\ell-23}{54} + \tfrac{5}{18}\cdot\tfrac{11-4\ell}{108} = \tfrac{80\ell-181}{648}.$$

The centre's class is the centre ray read at $c'k$: $P_F(c'\mathbf{1}; c', c') = P(c') = c(c')\ell + r(c')$. The other members follow by the ratios of Step 2 at $k=1$: $\frac{\Gamma(2)\Gamma(3)}{\Gamma(3)\Gamma(2)} = 1$ for $(3,3,1;3,2)$, $\frac{\Gamma(2)\Gamma(3)}{\Gamma(1)\Gamma(4)} = \frac{1}{3}$ for $(3,3,3;1,4)$ and $(2,2,2;4,1)$, 1 for $(4,4,2;4,3)$, $\frac{\Gamma(3)\Gamma(4)}{\Gamma(2)\Gamma(5)} = \frac{1}{2}$ for $(4,4,4;2,5)$ and $(3,3,3;5,2)$, and 1 for $(5,5,3;5,4)$ and $(5,5,1;5,3)$. The value of $(2,2,2;4,1)$ agrees with Exercise 52.5, where it was read on another insertion.

Step 4: the native values. The native gauge on the orthant is unique (Theorem 38.14(i)), so a symmetry of the masses fixes it: for $m_2 = m_3$ it is $(x,1,1)$, for $m_1 = m_2$ it is $(1,1,x)$, with $x > 0$ the root of the defect equation $m_e = s\,\alpha_e/K + t\,\alpha_e(K-\alpha_e)/\Psi$ at one mark. Exactly: $f(\alpha^\star) = 9^{-c'}$ on the centre's class; $\frac{1}{243}$ at $\mathbf{1}$ for $(3,3,3;1,4)$ and $(2,2,2;4,1)$, and $\frac{64}{6561}$ at $(4,1,1)$ and $(1,1,\frac{1}{4})$ for $(4,2,2;2,3)$ and $(3,3,1;3,2)$; $\frac{1}{2187}$ at $\mathbf{1}$ for $(4,4,4;2,5)$ and $(3,3,3;5,2)$, and $\frac{3125}{3779136}$ at $(\frac{5}{2},1,1)$ and $(1,1,\frac{2}{5})$; $\frac{1}{12500}$ at $(2,1,1)$ and $(1,1,\frac{1}{2})$; $\frac{823543}{553584375}$ at $(7,1,1)$ and $(1,1,\frac{1}{7})$. Within a class the native values are in the ratio of the rates of the Beta corners, $\frac{64}{27} = \frac{4^4}{3^3 2^2}$ and $\frac{3125}{1728} = \frac{5^5 2^2}{4^4 3^3}$.

Check (not part of the solution). The least recurrences of the closings $\mathrm{CT}(f^{-j})$, searched by linear algebra modulo the prime $2^{61}-1$ on 60 closings to the order 4 and the degree 24, have the order 2 on 16 of the 17 directions with $|c_e|$ constant; on the seventeenth, $(5\mathbf{1};5,5)$, the closings are those of the centre at $5j$, whose recurrence of order 2 has the degree 18, found with 70 closings. They have the order 3 on 5 directions and the order 4 on 2, all with two values of $|c_e|$, and none within these bounds on the other 100.

Answer. 123 directions in 54 classes; within a class $\Gamma(sk)\Gamma(tk)P$ is one function of k, and the values are in the ratio of two Beta corners at the same total; the 17 directions with $|c_e|$ constant have the values of the table, all in $\mathbb{Q} + \mathbb{Q}\ell$, and the native values of Step 4. □

52.7 The centre of $\binom{[4]}{1}$

The list $\binom{[4]}{1}$ has the marks $1,2,3,4$ and the four singletons as its bases, every weight 1, so that $K(\alpha) = \alpha_1+\alpha_2+\alpha_3+\alpha_4$ and $\Psi(\alpha) = \alpha^{[4]}K(\alpha^{-1})$. Its centre ray at the strength one (Definition 38.13) has the stride $f = \big(K(\alpha)K(\alpha^{-1})\big)^{-1}$, so that $f^{-1} = \sum_{e,e'} \alpha_e/\alpha_{e'}$; the walls of the direction are the widths, $\kappa_{1,S} = r(S) + r([4]\setminus S) - 1 = 1$ for every nonempty proper S, and the volume of the direction is $V_1 = \sum_\pi \prod_{l=2}^4 \kappa_{1,U_l}^{-1} = 4! = 24$ (Proposition 39.30). Its values are $P(k) = \langle 1\rangle_k = P_{\binom{[4]}{1}}(k\mathbf{1};k,k)$, and throughout this section $v = P(1)$. The five exercises close v on the half-lattice, derive the law of the values and its rows, write the row at every step in closed form, read the coefficient of the coordinate as a closing in two ways, and clear the rows.

Exercise 52.7 (The closing on the half-lattice)**.** Write $v = P(1)$ on its share side, merge the marks in pairs, and close the remaining share. Show that

$$v = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Solution. Receipt: `ex24a_halflattice.py`.

The share side. The dual list $\binom{[4]}{3}$ has the bases $[4]-e$, and replacing the table by its dual exchanges the exponents (Proposition 8.15), which at the centre are equal: the two lists have the same values. Corollary 18.22 applies to $\binom{[4]}{3}$ at $s=t=1$ with the masses $\mathbf{1}$: they lie on the balance wall, $m_{[4]}=4=3s+t$, they meet $m_S=|S|>s(|S|-1)$, and $\nu_e=m_e-s=0$. So

$$v=\frac{1}{\Gamma(1)^2}\int_{a_1+a_2+a_3+a_4=1}\prod_{e=1}^{4}\Gamma(a_e)^2,$$

a fibre integral of dimension 3.

Merging the pairs. Put $c=a_1+a_2$. Corollary 4.7 with the entries $a=b=0$, $c=d=c$ and the variable a_1 merges the pair $\{1,2\}$,

$$\frac{1}{2\pi i}\int\Gamma(a_1)^2\,\Gamma(c-a_1)^2\,da_1=\Gamma\begin{matrix}4\\1\end{matrix}\!\left[\begin{matrix}c,\ c,\ c,\ c\\ \ \ 2c\end{matrix}\right],$$

and the same with the total $1-c$ merges $\{3,4\}$. One share is left:

$$v=\frac{1}{2\pi i}\int_{(\frac12)}\Gamma\begin{matrix}4\\1\end{matrix}\!\left[\begin{matrix}c,\ c,\ c,\ c\\ \ \ 2c\end{matrix}\right]\Gamma\begin{matrix}4\\1\end{matrix}\!\left[\begin{matrix}1-c,\ 1-c,\ 1-c,\ 1-c\\ \ \ 2-2c\end{matrix}\right]dc.$$

The integrand. By the reflection and the shift (Proposition 1.3(iii),(v)), $\Gamma(c)^4\Gamma(1-c)^4=\pi^4/\sin^4\pi c$ and $\Gamma(2c)\Gamma(2-2c)=(1-2c)\,\Gamma(2c)\Gamma(1-2c)=(1-2c)\,\pi/\sin 2\pi c$, so the integrand is

$$w(c)=\frac{\pi^4}{\sin^4\pi c}\cdot\frac{\sin 2\pi c}{(1-2c)\,\pi}=\frac{2\pi^3\cos\pi c}{(1-2c)\sin^3\pi c}.$$

At $c=\frac12$ the zero of $\cos\pi c$ cancels that of $1-2c$, so the poles of w are the integers. The factor $2\pi^3\cos\pi c/\sin^3\pi c$ is $-\partial_c\Gamma\begin{matrix}4\\0\end{matrix}\!\big[c,\ c,\ 1-c,\ 1-c\big]=2\sum_{m\in\mathbb{Z}}(c+m)^{-3}$ (Lemma 5.16), of period one, and at $c=m+\varepsilon$

$$\frac{2\pi^3\cos\pi\varepsilon}{\sin^3\pi\varepsilon}=\frac{2}{\varepsilon^3}-\frac{2\pi^4}{15}\,\varepsilon+O(\varepsilon^3):$$

no terms in ε^{-2}, ε^{-1} or ε^0.

The closing. At $c=N+\frac12+iy$, $|\cos\pi c|=|\sinh\pi y|$ and $|\sin\pi c|=\cosh\pi y$, so $|w(c)|\le 4\pi^3e^{-2\pi|y|}/(2N)$: the integrals over the lines $\operatorname{Re}c=N+\frac12$ tend to zero, and so do those over the horizontal segments $\operatorname{Im}c=\pm Y$ as $Y\to\infty$. The line closes to the right (Theorem 4.5) on the triple poles at the integers $m\ge1$, with the residues

$$\operatorname*{Res}_{c=m}w=2\,[\varepsilon^2]\,\frac{1}{1-2m-2\varepsilon}=\frac{2\cdot4}{(1-2m)^3}=-\frac{8}{(2m-1)^3},$$

and closing to the right takes minus their sum:

$$v=\sum_{m\ge1}\frac{8}{(2m-1)^3}=\sum_{x\in\frac12+\mathbb{Z},\ x>0}x^{-3}=(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))$$

(Definition 5.2).

The coset relation. Theorem 5.12(i) at $q=2$ and $b=0$, with the kernel x^{-3} and the cut 0, assembles $\frac12\mathbb{Z}$ from its two cosets, the rescaled kernel being $(x/2)^{-3}=8x^{-3}$:

$$(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))+(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))=8\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

So $v = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty)) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, which is $8.41439832211715999 78\ldots$ to 19 places; the receipt evaluates the contour integral by quadrature as a check. The coset $\frac12+\mathbb{Z}$ is the doubled share $2c$ of a merged pair, the lower entry of $\Gamma_{\!4\atop 1}\left[\begin{matrix} c,\ c,\ c,\ c \\ & 2c & \end{matrix}\right]$: the word has one lower Gamma of slope 2 and one of slope -2, which pair off (Proposition 43.2), and its rational factor $(1-2c)^{-1}$ has its pole on that coset. □

Exercise 52.8 (The law, the low-end relation and the rows)**.** (i) With $\vartheta = y\frac{d}{dy}$, compute the operators $\mathbf{R}_0,\ldots,\mathbf{R}_5$ of Theorem 42.12 for $n=4$, and from them the law of the values, $\sum_{k=0}^{2} c_{4,k}(s)\,P(s+k)=0$.

(ii) Give the closings $\mathsf{C}_j(1)$ for $j\le 8$, and check the law read at $k=-j$ on them.

(iii) Give the symbol, its roots and the native gauges they belong to.

(iv) Write the low-end relation.

(v) Write the rows $P(k)=a_kv-b_k$ for $k\le 6$ (Definition 36.5), and the value span of the ray.

Solution. Receipt: `ex24b_law.py`.

(i) *One mark.* At the strength s the masses are $s\mathbf{1}$ and the exponents (s,s), and Corollary 18.22, as in Exercise 52.7, gives the share side in its first form,

$$\Gamma(s)^2P(s)=\int_0^\infty y^{s-1}\,\mathsf{g}(y)^4\,dy,\qquad \mathsf{g}(y)=\mathsf{G}_{0;(1,-1)}(1,y),\quad \operatorname{Re}s>0.$$

The Gramma g satisfies $\vartheta^2\mathsf{g}=y\,\mathsf{g}$. Write $\mathsf{g}_\nu=\mathsf{G}_{\nu;(1,-1)}(1,y)$. The ladder (Theorem 2.4(i)) in the coefficient y of the exponent -1 gives $\vartheta\mathsf{g}_\nu=-y\,\mathsf{g}_{\nu-1}$, so $\vartheta^2\mathsf{g}_0=-y\,\mathsf{g}_{-1}+y^2\mathsf{g}_{-2}$; the exchange relation (Theorem 2.4(ii)) at $\nu=-1$ reads $-\mathsf{g}_{-1}=\mathsf{g}_0-y\,\mathsf{g}_{-2}$, and substituting $y\,\mathsf{g}_{-2}=\mathsf{g}_0+\mathsf{g}_{-1}$ gives $\vartheta^2\mathsf{g}_0=y\,\mathsf{g}_0$.

The operators. Theorem 42.12(i) at $n=4$ has $\mathbf{R}_0=1$, $\mathbf{R}_1=\vartheta$ and $\mathbf{R}_{i+1}=\vartheta\mathbf{R}_i-i(5-i)\,y\,\mathbf{R}_{i-1}$, with the coefficients $i(5-i)=4,6,6,4$; an operator is written $\sum_k y^kq(\vartheta)$, and $\vartheta\,y^kq(\vartheta)=y^k(\vartheta+k)\,q(\vartheta)$. So

$$\begin{aligned}
\mathbf{R}_2&=\vartheta^2-4y,\\
\mathbf{R}_3&=\vartheta^3-2y(5\vartheta+2),\\
\mathbf{R}_4&=\vartheta^4-2y(8\vartheta^2+7\vartheta+2)+24y^2,\\
\mathbf{R}_5&=\vartheta^5-2y(2\vartheta+1)(5\vartheta^2+5\vartheta+2)+64y^2(\vartheta+1),
\end{aligned}$$

and $\mathbf{Q}_4=\mathbf{R}_5$ annihilates every product of four solutions of $\vartheta^2u=yu$, g^4 among them; the receipt checks it on the basis $u^{4-i}(\vartheta u)^i$. The leading coefficients $1,-20,64$ of $q_{4,0},q_{4,1},q_{4,2}$ are those of $\xi(\xi^2-16\eta)(\xi^2-4\eta)=\xi^5-20\eta\xi^3+64\eta^2\xi$, as Theorem 42.12(i) says.

The law. Integrated against y^{s-1}, a term $y^kq(\vartheta)$ becomes $q(-s-k)$ times the transform at $s+k$: integration by parts turns ϑ into $-s$, the boundary terms vanishing because g is $O(|\log y|)$ at 0 (Lemma 20.29(i)) and at most $\sqrt{\pi}\,y^{-1/4}e^{-2\sqrt{y}}$ for $y\ge1$ (Lemma 20.29(ii)). With $\Gamma(s+k)^2=\Gamma(s)^2(s)_k^2$ and a division by $\Gamma(s)^2s^2$ this is Theorem 42.12(ii):

$$c_{4,0}(s)=-s^3,\qquad c_{4,1}(s)=q_{4,1}(-s-1)=2(2s+1)(5s^2+5s+2),$$
$$c_{4,2}(s)=q_{4,2}(-s-2)\,(s+1)^2=-64(s+1)^3,$$

$$-s^3P(s)+2(2s+1)(5s^2+5s+2)\,P(s+1)-64(s+1)^3P(s+2)=0$$

identically in s. At $s=1$ the coefficients are -1, 72, -512.

(ii) *The closings.* By Corollary 39.32 $\mathsf{C}_j(1)$ is the constant term of $f^{-j}=K(\alpha)^jK(\alpha^{-1})^j$; the multinomial expansions of the two factors pair the monomial α^v with α^{-v}, so

$$\mathsf{C}_j(1)=\sum_{|v|=j}\binom{j}{v}^2=1,\ 4,\ 28,\ 256,\ 2716,\ 31504,\ 387136,\ 4951552,\ 65218204\qquad(j=0,\dots,8).$$

The law read at $k=-j$ (Corollary 39.32, Theorem 42.12(iii)) is $\sum_{i\le\min(2,j)}c_{4,i}(-j)\,\mathsf{C}_{j-i}(1)=0$: at $j=1$, $1\cdot4+(-4)\cdot1=0$; at $j=2$, $8\cdot28+(-72)\cdot4+64\cdot1=0$; the receipt checks it for $j\le40$, and also that $\mathbf{Q}_4$ annihilates $\mathsf{h}^4=\sum_j\mathsf{C}_j(1)\,y^j/j!^2$, $\mathsf{h}(y)=\sum_m y^m/m!^2$, through y^{40}.

(iii) *The symbol.* The coefficients of s^3 in $c_{4,0},c_{4,1},c_{4,2}$ are $-1,20,-64$, so the symbol (Corollary 38.7) is

$$\chi_4(\lambda)=-1+20\lambda-64\lambda^2=-(1-4\lambda)(1-16\lambda),$$

with the roots $\frac{1}{16}$ and $\frac14$. By Proposition 42.9(ii) the native gauges of the ray are the signs η with $K(\eta)\neq0$, of native value $K(\eta)^{-2}$: $\eta=\mathbf{1}$, with $K=4$ and the native value $\frac{1}{16}=f(\alpha^\star)$, $\alpha^\star=\mathbf{1}$ being the native gauge of the direction (Theorem 38.14); and the signs with one or three marks at -1, with $K=\pm2$ and the native value $\frac14$. The signs with two marks at -1 have $K=0$ and lie off X_F.

(iv) *The low-end relation.* By Corollary 39.34(ii), with $[s^3]c_{4,0}=-1$, $\mathsf{L}_0=V_1=24$, $c_{4,1}(0)=4$ and $c_{4,2}(0)=-64$,

$$-24+4P(1)-64P(2)=0,\qquad P(2)=\frac{P(1)-6}{16}.$$

(v) *The rows.* $P(1)=v$ and $P(2)=\frac{1}{16}v-\frac38$, and the law at $s=k\ge1$, whose last coefficient $c_{4,2}(k)$ does not vanish, gives $P(k+2)=-\big(c_{4,0}(k)P(k)+c_{4,1}(k)P(k+1)\big)/c_{4,2}(k)$; at $k=1$, $P(3)=\big(72P(2)-P(1)\big)/512=\frac{7}{1024}v-\frac{27}{512}$. The rows $(a_k;b_k)$:

k	1	2	3	4	5	6
a_k	1	$\frac{1}{16}$	$\frac{7}{1024}$	$\frac{1}{1024}$	$\frac{679}{4194304}$	$\frac{1969}{67108864}$
b_k	0	$\frac38$	$\frac{27}{512}$	$\frac{37}{4608}$	$\frac{25555}{18874368}$	$\frac{9304913}{37748736000}$

So every value lies in $\mathbb{Q}+\mathbb{Q}\,v$, $v=(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))$ (Exercise 52.7), and this is the value span of the ray (Corollary 42.15(i), with $h=2$): the order $h=2$ leaves one value free after the low-end relation. As renderings, $P(2)=0.15089989513232249986\dots$, $P(3)=0.004785926030097773422 4\dots$ and $P(4)=0.000187671971972553650171 46\dots$; the receipt checks $P(2)$ against the quadrature of $\int_0^\infty y\,\mathsf{g}(y)^4dy$. □

Exercise 52.9 (The row at every step)**.** For $n\ge0$ put $p_l=l+\frac12$ and $j=n-l$ for $0\le l\le n$,

$$c_l=\binom{2l}{l}\binom{2j}{j},\qquad S_l=\frac{c_l}{\binom{n}{l}}=\frac{(2l)!\,(2j)!}{l!\,j!\,n!},\qquad w_l=S_l^2c_l,\qquad F(\sigma)=\binom{\sigma-1}{n}^2,$$

$F[\sigma,p]=\big(F(\sigma)-F(p)\big)/(\sigma-p)$, and for a polynomial g

$$\mathcal{J}[g]=\frac{1}{2\pi i}\int_{(\frac12)}g(\sigma)\sum_{m\in\mathbb{Z}}\frac{d\sigma}{(\sigma+m)^3}.$$

Show that

$$256^n\,P(n+1)=\sum_{l=0}^{n}w_l\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty))-16^n\sum_{l=0}^{n}c_l\,\mathcal{J}\big[F[\,\cdot\,,p_l]\big],$$

evaluate $\mathcal{J}$ on the binomial polynomials $\binom{\sigma}{i}$, and read the rows of $P(2)$ and $P(3)$ off the formula.

Solution. Receipt: `ex24c_row.py`.

(a) *The word.* At the strength $k = n+1$ the share side of Exercise 52.7, with the shares of total k and the pairs merged by Corollary 4.7 at the totals σ and $k-\sigma$, is

$$P(k) = \frac{1}{\Gamma(k)^2}\,\frac{1}{2\pi i}\int \frac{\Gamma(\sigma)^4\,\Gamma(k-\sigma)^4}{\Gamma(2\sigma)\,\Gamma(2k-2\sigma)}\,d\sigma \qquad (0 < \operatorname{Re}\sigma < k).$$

With $\Gamma(k-\sigma) = (1-\sigma)_n\Gamma(1-\sigma)$ and $\Gamma(2k-2\sigma) = (2-2\sigma)_{2n}(1-2\sigma)\Gamma(1-2\sigma)$ (Proposition 1.3(iii)), the reflections of Exercise 52.7 give the factor $2\pi^3\cos\pi\sigma/\sin^3\pi\sigma = 2\sum_m(\sigma+m)^{-3}$ against $(1-\sigma)_n^4/\big((1-2\sigma)(2-2\sigma)_{2n}\big)$. Splitting $(2-2\sigma)_{2n}$ into its factors $2(1-\sigma+t)$ and $2(\frac{3}{2}-\sigma+t)$, $0 \le t < n$, gives $(1-2\sigma)(2-2\sigma)_{2n} = 2\cdot 4^n(\frac{1}{2}-\sigma)_{n+1}(1-\sigma)_n$, so

$$P(n+1) = \frac{1}{4^n\,n!^2}\,\frac{1}{2\pi i}\int \rho(\sigma)\sum_m \frac{d\sigma}{(\sigma+m)^3}, \qquad \rho(\sigma) = \frac{(1-\sigma)_n^3}{(\frac{1}{2}-\sigma)_{n+1}}.$$

The triple poles of the kernel at $1,\dots,n$ meet the triple zeros of ρ, and the simple poles of ρ at the p_l meet the zeros of the kernel, so the integrand is regular on $0 < \operatorname{Re}\sigma < k$ and the line may be $\operatorname{Re}\sigma = \frac{1}{2}$.

(b) *Partial fractions.* With $\Omega(\sigma) = \prod_{i=1}^n(\sigma-i) = n!\binom{\sigma-1}{n}$, $(1-\sigma)_n = (-1)^n\Omega$ and $(\frac{1}{2}-\sigma)_{n+1} = (-1)^{n+1}\prod_{i=0}^n(\sigma-p_i)$, so $\rho/\Omega^2 = -\Omega/\prod_i(\sigma-p_i)$, whose residue at p_l is $-\Omega(p_l)/\prod_{i\ne l}(l-i)$, with $\prod_{i\ne l}(l-i) = (-1)^j l!\,j!$ and

$$\Omega(p_l) = \prod_{t=0}^{l-1}\big(t+\tfrac{1}{2}\big)\prod_{t=1}^{j}\big(\tfrac{1}{2}-t\big) = \frac{(-1)^j(2l-1)!!\,(2j-1)!!}{2^n}.$$

As $(2l-1)!!/l! = \binom{2l}{l}/2^l$, the residue is $-c_l/4^n$:

$$\frac{\rho}{\Omega^2} = -\frac{1}{4^n}\sum_{l=0}^{n}\frac{c_l}{\sigma-p_l}.$$

Multiplying by $\Omega^2 = n!^2F$ and writing $\Omega(\sigma)^2 = \Omega(p_l)^2 + (\sigma-p_l)\,n!^2F[\sigma,p_l]$ splits ρ into a polynomial and its poles,

$$\rho = \mathcal{P} + \sum_{l=0}^{n}\frac{r_l}{\sigma-p_l}, \qquad \mathcal{P} = -\frac{n!^2}{4^n}\sum_l c_l\,F[\sigma,p_l], \qquad r_l = -\frac{c_l\,\Omega(p_l)^2}{4^n} = -\frac{n!^2\,w_l}{64^n},$$

the last because $(2l-1)!!\,(2j-1)!! = S_l\,n!/2^n$. The receipt checks the partial fractions and the r_l for $n \le 3$.

(c) *The pole terms.* The kernel is $\pi^3\cos\pi\sigma/\sin^3\pi\sigma$, with the principal part $(\sigma-i)^{-3}$ and no terms of orders $-2,-1,0$ at every integer i (Exercise 52.7), so the residue of $(\sigma-p_l)^{-1}$ times the kernel at i is $\frac{1}{2}\partial_\sigma^2(\sigma-p_l)^{-1}|_{\sigma=i} = (i-p_l)^{-3}$; at p_l the kernel vanishes. On the lines $\operatorname{Re}\sigma = N+\frac{1}{2}$ the integrand is $O(N^{-1}e^{-2\pi|\operatorname{Im}\sigma|})$, so the line closes to the right (Theorem 4.5):

$$\frac{1}{2\pi i}\int_{(\frac{1}{2})}\frac{1}{\sigma-p_l}\sum_m\frac{d\sigma}{(\sigma+m)^3} = -\sum_{i\ge 1}\frac{1}{(i-p_l)^3} = -(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((l,\infty)),$$

the terms $i \le l$, at the negative points $i-p_l$, cancelling the head $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,l])$ of the terms at the positive points (Proposition 5.3). The polynomial part gives $\mathcal{J}[\mathcal{P}]$.

(d) *Assembling.* By (a) to (c),

$$P(n+1)=\frac{1}{4^n n!^2}\Big(-\frac{n!^2}{4^n}\sum_l c_l\,\mathcal{J}\big[F[\cdot,p_l]\big]+\frac{n!^2}{64^n}\sum_l w_l\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty))\Big),$$

which is the stated formula after multiplication by 256^n. With $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty))=v-(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,l])$ the row of $P(n+1)$ is

$$a_{n+1}=\frac{1}{256^n}\sum_{l=0}^{n}w_l,\qquad b_{n+1}=\frac{1}{256^n}\Big(\sum_{l=0}^{n}w_l\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,l])+16^n\sum_{l=0}^{n}c_l\,\mathcal{J}\big[F[\cdot,p_l]\big]\Big).$$

(e) *The functional $\mathcal{J}$.* By Lemma 5.16, $\sum_m(\sigma+m)^{-3}=-\frac12\partial_\sigma\Gamma_4^{\,0}\big[\sigma,\ \sigma,\ 1-\sigma,\ 1-\sigma\big]$, and on the line the stacked Gamma is $2\pi\,p_{\mathrm{crit}}(y)$, $\sigma=\frac12+iy$, decaying like $e^{-2\pi|y|}$ (Definition 36.1). Integrating by parts,

$$\mathcal{J}[g]=\frac12\,\mathbb{E}\big[g'(X)\big],\qquad X=\tfrac12+iY,\quad Y\sim\mathbb{P}_{\mathrm{crit}}.$$

By Theorem 36.2(iii), $\mathbb{E}[e^{tX}]=te^t/(e^t-1)$ for $|t|<2\pi$, so $\mathbb{E}[X^m]=1,\frac12,\frac16,0,-\frac{1}{30},0,\frac{1}{42},0,-\frac{1}{30}$ for $m=0,\dots,8$, and at $t=\log(1+z)$, $\mathbb{E}[(1+z)^X]=(1+z)\log(1+z)/z$. Since $\sum_i\binom{\sigma}{i}z^i=(1+z)^\sigma$, whose derivative in σ is $\log(1+z)\,(1+z)^\sigma$,

$$\sum_{i\ge0}\mathcal{J}\Big[\binom{\sigma}{i}\Big]z^i=\frac12\log(1+z)\,\mathbb{E}\big[(1+z)^X\big]=\frac{(1+z)\log^2(1+z)}{2z}.$$

With $\log^2(1+z)=2\sum_{m\ge2}(-1)^m(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,m-1])\,z^m/m$, the head $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,m])=\sum_{r=1}^m\frac1r$ (Definition 5.2), the coefficients are

$$\lambda_0=0,\qquad \lambda_i=\mathcal{J}\Big[\binom{\sigma}{i}\Big]=(-1)^i\,\frac{(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,i-1])-1}{i(i+1)}\quad(i\ge1),$$

that is $0,\frac12,0,-\frac{1}{24},\frac{1}{24},-\frac{13}{360},\frac{11}{360},-\frac{29}{1120},\frac{223}{10080}$ for $i=0,\dots,8$; the receipt checks them against the moments for $i<40$. A polynomial written in the binomial basis, $g=\sum_i(\Delta^ig)(0)\binom{\sigma}{i}$, has $\mathcal{J}[g]=\sum_i(\Delta^ig)(0)\,\lambda_i$.

(f) *The first steps.* At $n=1$: $c_0=c_1=2$, $S_0=S_1=2$, $w_0=w_1=8$; $F(\sigma)=(\sigma-1)^2$, $F[\sigma,p]=\sigma+p-2$, so $F[\sigma,\frac12]=\sigma-\frac32$ and $F[\sigma,\frac32]=\sigma-\frac12$, each with $\mathcal{J}=\frac12$; with $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,1])=8$,

$$256\,P(2)=16v-8\cdot8-16\Big(2\cdot\frac12+2\cdot\frac12\Big)=16v-96,\qquad P(2)=\frac{1}{16}v-\frac38.$$

At $n=2$: $(c_0,c_1,c_2)=(6,4,6)$, $(S_0,S_1,S_2)=(6,2,6)$, $(w_0,w_1,w_2)=(216,16,216)$, and

$$\begin{aligned}F[\sigma,\tfrac12]&=\tfrac14\sigma^3-\tfrac{11}{8}\sigma^2+\tfrac{41}{16}\sigma-\tfrac{55}{32},&\mathcal{J}&=\tfrac{21}{32},\\F[\sigma,\tfrac32]&=\tfrac14\sigma^3-\tfrac98\sigma^2+\tfrac{25}{16}\sigma-\tfrac{21}{32},&\mathcal{J}&=\tfrac{9}{32},\\F[\sigma,\tfrac52]&=\tfrac14\sigma^3-\tfrac78\sigma^2+\tfrac{17}{16}\sigma-\tfrac{11}{32},&\mathcal{J}&=\tfrac{5}{32},\end{aligned}$$

for instance $\mathcal{J}\big[F[\cdot,\frac12]\big]=\frac12\mathbb{E}\big[\frac34X^2-\frac{11}{4}X+\frac{41}{16}\big]=\frac12\big(\frac18-\frac{11}{8}+\frac{41}{16}\big)=\frac{21}{32}$. With the heads $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,1])=8$ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,2])=8+\frac{8}{27}$,

$$256^2P(3)=448\,v-\Big(16\cdot8+216\cdot\frac{224}{27}\Big)-256\Big(6\cdot\frac{21}{32}+4\cdot\frac{9}{32}+6\cdot\frac{5}{32}\Big)=448\,v-1920-1536,$$

so $P(3) = \frac{7}{1024}v - \frac{27}{512}$. Both rows are those of Exercise 52.8.

So the row of $P(n+1)$ at every step is $a_{n+1} = 256^{-n}\sum_l w_l$ and b_{n+1} as in (d), with $\mathcal{J}$ read through the λ_i. For $n = 0,1,2,3$ the sums are $\sum_l w_l = 1, 16, 448, 16384$, $256^n b_{n+1} = 0, 96, 3456, \frac{1212416}{9}$ and $\sum_l c_l \mathcal{J}[F[\cdot, p_l]] = 0, 2, 6, \frac{853}{54}$; the receipt checks that the formula reproduces the rows of the law for $n \le 40$. □

Exercise 52.10 (The coefficient of the coordinate is a closing)**.** Show that $a_k = \mathsf{C}_{k-1}(1)/64^{k-1}$ for every $k \ge 1$, and give the denominator of a_k:

(i) from the law of Exercise 52.8, by Theorem 43.8;

(ii) directly from the row of Exercise 52.9, by a telescoping identity in the summation index.

Solution. Receipt: `ex24d_closing.py`.

(i) *The law is reflexive.* With $h = 2$, Definition 43.7 asks for $c_{4,2-i}(-k-1) = \gamma\lambda^i c_{4,i}(k)$, $i = 0,1,2$. Here

$$c_{4,2}(-k-1) = 64k^3 = -64\,c_{4,0}(k), \qquad c_{4,0}(-k-1) = (k+1)^3,$$
$$c_{4,1}(-k-1) = -2(2k+1)(5k^2+5k+2) = -c_{4,1}(k),$$

so $\gamma = -64$, $\gamma\lambda = -1$ gives the ratio $\lambda = \frac{1}{64}$, and $\gamma\lambda^2 c_{4,2}(k) = -\frac{64}{4096}\cdot(-64)(k+1)^3 = (k+1)^3$ checks the third. The ray has the origin 0, every letter exponent positive and positive walls, as Proposition 43.6 asks. By Theorem 43.8(i), $y_k = 64^{1-k}\mathsf{C}_{k-1}(1)$ solves the law on $k \ge 1$ with the low-end coefficient $-\big(c_{4,1}(0)y_1 + c_{4,2}(0)y_2\big) = -\big(4\cdot 1 - 64\cdot\frac{4}{64}\big) = 0$, and since $c_{4,2}(k) \ne 0$ for $k \ge 0$, Theorem 43.8(ii) gives

$$P(k) = \frac{\mathsf{C}_{k-1}(1)}{64^{k-1}}\,P(1) + r_k, \qquad a_k = \frac{\mathsf{C}_{k-1}(1)}{64^{k-1}}.$$

By Theorem 43.8(iii), with $\lambda = \frac{1}{64}$, the denominator of a_k is $64^{k-1}/\gcd\big(64^{k-1}, \mathsf{C}_{k-1}(1)\big)$:

k	1	2	3	4	5	6	7	8
$\mathsf{C}_{k-1}(1)$	1	4	28	256	2716	31504	387136	4951552
$v_2\big(\mathsf{C}_{k-1}(1)\big)$	0	2	2	8	2	4	6	9
$\log_2 \mathrm{den}(a_k)$	0	4	10	10	22	26	30	33

with v_2 the exponent of 2.

(ii) *The closings across the pairs.* Split $f^{-1} = (X+Y)(X'+Y')$ with $X = \alpha_1 + \alpha_2$, $X' = \alpha_1^{-1} + \alpha_2^{-1}$, and Y, Y' the same on $\{3,4\}$. In $(X+Y)^n(X'+Y')^n = \sum_{l,l'}\binom{n}{l}\binom{n}{l'}X^l Y^{n-l} X'^{l'} Y'^{n-l'}$ a term has a constant term in α_1, α_2 only if $l = l'$, X^l and $X'^{l'}$ being homogeneous of degrees l and $-l'$, and $\mathrm{CT}\,\big((XX')^l\big) = \sum_i \binom{l}{i}^2 = \binom{2l}{l}$. So

$$\mathsf{C}_n(1) = \sum_{l=0}^{n}\binom{n}{l}^2\binom{2l}{l}\binom{2j}{j} =: U(n), \qquad j = n-l.$$

Put $T(n) = 4^{-n}\sum_l w_l$, so that $a_{n+1} = 4^n T(n)/256^n$ by Exercise 52.9. It remains to show $T = U$.

The telescoping identities. Both T and U satisfy

$$(n+2)^3 X_{n+2} - 2(2n+3)(5n^2+15n+12)\,X_{n+1} + 64(n+1)^3 X_n = 0 \qquad (n \ge 0),$$

which is the law of the closings of Exercise 52.8(ii) at $j = n+2$: $c_{4,0}(-n-2) = (n+2)^3$, $c_{4,1}(-n-2) = -2(2n+3)(5n^2+15n+12)$, $c_{4,2}(-n-2) = 64(n+1)^3$. Write $t(n,l)$ for the

summand of either sum, $t_T(n,l) = 4^{-n}S_l^2 c_l$ and $t_U(n,l) = \binom{n}{l}^2 c_l$, zero outside $0 \le l \le n$. Their ratios are rational:

$$\frac{t_T(n,l+1)}{t_T(n,l)} = \frac{(2l+1)^3\, j}{(l+1)(2j-1)^3}, \qquad \frac{t_T(n+1,l)}{t_T(n,l)} = \frac{2(2j+1)^3}{(j+1)(n+1)^2},$$
$$\frac{t_U(n,l+1)}{t_U(n,l)} = \frac{j^3(2l+1)}{(l+1)^3(2j-1)}, \qquad \frac{t_U(n+1,l)}{t_U(n,l)} = \frac{2(n+1)^2(2j+1)}{(j+1)^3}.$$

With $G = Rt$ and the rational functions

$$R_T = \frac{4l\,(2j+1)^3\,\kappa_T}{(n+1)^2(j+1)(j+2)}, \qquad R_U = -\frac{4(n+1)^2\,l^3\,(2j+1)\,\kappa_U}{(j+1)^3(j+2)^3},$$

$$\kappa_T = 12j^3 + 10lj^2 + 2l^2j + 59j^2 + 33lj + 4l^2 + 94j + 26l + 49,$$
$$\kappa_U = 12j^3 + 10lj^2 + 2l^2j + 62j^2 + 35lj + 3l^2 + 104j + 30l + 56,$$

each summand satisfies, for $0 \le l \le n-1$,

$$64(n+1)^3 t(n,l) - 2(2n+3)(5n^2+15n+12)\,t(n+1,l) + (n+2)^3 t(n+2,l) = G(n,l+1) - G(n,l).$$

Divided by $t(n,l)$ and written with the ratios this is an identity of rational functions of (n,l); cleared of denominators it is an identity of polynomials, which the receipt verifies coefficient by coefficient for both T and U. Summing it over $0 \le l \le n-1$, with $G(n,0) = 0$ (the factor l), leaves $G(n,n)$, and the full sum over l of the left side is $G(n,n)$ plus the left side at $l = n, n+1, n+2$, where only the summands with $l \le n+1$ at $n+1$ and $l \le n+2$ at $n+2$ survive. Divided by $t(n,n)$, with $t(n+1,n)/t(n,n)$, $t(n+1,n+1)/t(n,n)$, $t(n+2,n)/t(n,n)$, $t(n+2,n+1)/t(n,n)$ and $t(n+2,n+2)/t(n,n)$ read off the ratios, this boundary sum is a rational function of n that vanishes identically (receipt). So both sums satisfy the recurrence; $T(0) = U(0) = 1$, $T(1) = \frac{1}{4}(8+8) = 4 = U(1)$, and the leading coefficient $(n+2)^3$ does not vanish, so $T = U$.

Hence $\sum_l w_l = 4^n \mathsf{C}_n(1)$ and $a_{n+1} = 4^n \mathsf{C}_n(1)/256^n = \mathsf{C}_n(1)/64^n$, as in (i). The coefficient of the coordinate v in $P(k)$ is the closing $\mathsf{C}_{k-1}(1)$ over 64^{k-1}, with no lattice sum: $64^{k-1}a_k \in \mathbb{Z}$, and the denominator of a_k is $64^{k-1}/\gcd\big(64^{k-1}, \mathsf{C}_{k-1}(1)\big)$. □

Exercise 52.11 (The clearing)**.** Show that for every $k \ge 1$ the part of the denominator of b_k prime to 2 divides d_{k-1}^3, and that $256^{k-1}d_{k-1}^3\, b_k \in \mathbb{Z}$; so $256^{k-1}d_{k-1}^3$ clears the row of $P(k)$. Compare it with the bound of the frame.

Solution. Receipt: `ex24e_clearing.py`.

Write $n = k-1$. By Exercise 52.9(d), $256^n b_{n+1}$ is the sum of the heads and the moments,

$$\sum_l w_l\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,l)) \qquad \text{and} \qquad 16^n \sum_l c_l\,\mathcal{J}\big[F[\cdot,p_l]\big];$$

and by Exercise 52.10 $64^n a_{n+1} = \mathsf{C}_n(1) \in \mathbb{Z}$. Two facts from outside the handbook are used: Legendre's formula, $v_q(m!) = \sum_{t\ge1}\lfloor m/q^t\rfloor$ for a prime q, and Kummer's theorem, that $v_q\binom{a+b}{a}$ is the number of carries when a and b are added in base q; v_q is the exponent of q.

The integers. $S_l \in \mathbb{Z}$: by Legendre's formula, $v_q(S_l) = \sum_{t\ge1}\big(\lfloor 2l/q^t\rfloor + \lfloor 2j/q^t\rfloor - \lfloor l/q^t\rfloor - \lfloor j/q^t\rfloor - \lfloor n/q^t\rfloor\big)$, and every term is nonnegative by $\lfloor 2x\rfloor + \lfloor 2y\rfloor \ge \lfloor x\rfloor + \lfloor y\rfloor + \lfloor x+y\rfloor$ (with $x = a+\alpha$, $y = b+\beta$, $0 \le \alpha,\beta < 1$, it reads $\lfloor 2\alpha\rfloor + \lfloor 2\beta\rfloor \ge \lfloor \alpha+\beta\rfloor$, true since $\alpha+\beta \ge 1$ forces $\max(\alpha,\beta) \ge \frac{1}{2}$). So $w_l = S_l^2 c_l \in \mathbb{Z}$.

Fix an odd prime q and $e \ge 1$ with $q^{e-1} \le n < q^e$, so that $v_q(d_n^3) = 3(e-1)$; it suffices that both sums have $v_q \ge -3(e-1)$.

The heads. $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,l]) = 8\sum_{i=1}^{l}(2i-1)^{-3}$, and $2l-1 < 2n < 2q^e$, so the only odd number up to $2l-1$ with $v_q \ge e$ is q^e, when $q^e \le 2l-1$. In that case, at the level q^e of Legendre's formula, $\lfloor l/q^e\rfloor = \lfloor j/q^e\rfloor = \lfloor n/q^e\rfloor = 0$ and $\lfloor 2l/q^e\rfloor = 1$, so $v_q(S_l) \ge 1$ and $v_q\binom{2l}{l} \ge 1$, the other levels contributing ≥ 0; then $v_q(w_l) \ge 3$, which pays for $(q^e)^{-3}$. Otherwise the head already has $v_q \ge -3(e-1)$.

The moments. Vandermonde's identity twice, $\binom{\sigma}{m} = \sum_t \binom{\sigma-p}{t}\binom{p}{m-t}$ and $\binom{\sigma-p-1}{t-1} = \sum_i \binom{\sigma}{i}\binom{-p-1}{t-1-i}$, with $\binom{\sigma-p}{t} = \frac{\sigma-p}{t}\binom{\sigma-p-1}{t-1}$, gives the binomial coefficients of the divided difference,

$$\big(\Delta^i F[\cdot,p]\big)(0) = \sum_m f_m \sum_{t=i+1}^{m} \frac{1}{t}\binom{p}{m-t}\binom{-p-1}{t-1-i}, \qquad F = \sum_m f_m\binom{\sigma}{m}.$$

Here $f_m = \sum_{i,i'\le n}(-1)^{i+i'}\binom{m}{i}\binom{i}{m-i'}$, from $\binom{\sigma-1}{n} = \sum_{i\le n}(-1)^{n-i}\binom{\sigma}{i}$ and $\binom{\sigma}{i}\binom{\sigma}{i'} = \sum_m \binom{m}{i}\binom{i}{m-i'}\binom{\sigma}{m}$; so $f_m \in \mathbb{Z}$, and $q \mid f_m$ for $m \ge q^e$: a nonzero term has $i < q^e$ and $m-i \le i' < q^e$, and adding i and $m-i$ in base q carries into the place q^e, so $q \mid \binom{m}{i}$ (Kummer's theorem). At $p = p_l$, a q-adic integer, the binomials of p and of $-p-1$ are q-integral. Since $t \le 2n < 2q^e$, the only t with $v_q(t) \ge e$ is $t = q^e$, where $m \ge q^e$ and $q \mid f_m$. So $v_q\big((\Delta^i F[\cdot,p_l])(0)\big) \ge 1-e$ for every i, and $\ge 2-e$ for $i \ge q^e$, where every t and m exceed q^e.

For $1 \le i \le 2n-1$: $v_q(\lambda_i) \ge 2-2e$ for $i < q^e$ and $v_q(\lambda_i) \ge 1-2e$ for $i \ge q^e$. The heads of the counting input at index one have $v_q\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,m])\big) \ge 1-e$ for $m < q^e$ and $\ge -e$ for $q^e \le m < 2q^e$, while

$$v_q\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-1])\big) \ge 2-e :$$

grouping the $r < q^e$ by $v_q(r) = s$, $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-1]) = \sum_{s<e} q^{-s}\sum_{r'<q^{e-s},\ q\nmid r'}\frac{1}{r'}$, and $\sum_{r'<q^a,\ q\nmid r'}\frac{1}{r'}$ has $v_q \ge a$, pairing r' with $q^a - r'$, $\frac{1}{r'} + \frac{1}{q^a-r'} = \frac{q^a}{r'(q^a-r')}$; each term has $v_q \ge e-2s \ge 2-e$. The integers i and $i+1$ are coprime and below $2q^e$, so $v_q\big(i(i+1)\big) \le e-1$ unless $q^e \in \{i, i+1\}$, where it is e. For $i < q^e-1$ this gives $v_q(\lambda_i) \ge (1-e)-(e-1)$. At $i = q^e-1$, $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-2]) - 1 = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-1]) - \frac{q^e}{q^e-1}$ has $v_q \ge \min(2-e,e)$, so $v_q(\lambda_i) \ge \min(2-e,e) - e \ge 2-2e$. At $i = q^e$, $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-1]) - 1$ has $v_q \ge \min(2-e,0)$, so $v_q(\lambda_i) \ge \min(2-e,0) - e \ge 1-2e$; and for $q^e < i \le 2n-1$, $v_q(\lambda_i) \ge -e-(e-1)$. In each case the product of the coefficient and λ_i has $v_q \ge 3-3e$, and 16^n and c_l are q-integral. The receipt checks every one of these estimates for $n \le 30$ and every odd prime $q \le 2n$; for instance $v_3\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,26])\big) = -1$, $v_5\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,24])\big) = 1$.

The prime 2. For $p \in \frac12+\mathbb{Z}$, $v_2\binom{p}{r} = -r - v_2(r!) \ge 1-2r$ for $r \ge 1$; the product of the two binomials in the coefficient, of total order $m-1-i \le 2n-1$, has $v_2 \ge 3-4n$; $v_2(1/t) \ge -\lfloor\log_2 2n\rfloor$; and $v_2(\lambda_i) \ge -2\lfloor\log_2 2n\rfloor$, from $v_2\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,i-1])\big) \ge -\lfloor\log_2(i-1)\rfloor$ and $v_2(i(i+1)) \le \lfloor\log_2(i+1)\rfloor$. So each term of $16^n\sum_l c_l\mathcal{J}[F[\cdot,p_l]]$ has $v_2 \ge -3\lfloor\log_2 n\rfloor = -v_2(d_n^3)$, and the heads, w_l times 8 times odd denominators, are 2-integral. Hence $256^n d_n^3 b_{n+1} \in \mathbb{Z}$.

The numbers. The receipt checks both statements for $k \le 80$. The first denominators, with their parts at 2:

k	2	3	4	5	6	7	8
$\operatorname{den}(b_k)$	2^3	2^9	$2^9\cdot 9$	$2^{21}\cdot 9$	$2^{25}\cdot 1125$	$2^{29}\cdot 125$	$2^{32}\cdot 42875$
d_{k-1}^3	1	2^3	$2^3\cdot 27$	$2^6\cdot 27$	$2^6\cdot 3375$	$2^6\cdot 3375$	$2^6\cdot 1157625$

The frame. The sequences (a_k) and (b_k) both solve the law of Exercise 52.8: (a_k) by Exercise 52.10(i), and $b_k = a_k v - P(k)$ as a difference of two solutions. They are independent,

$(a_1,a_2)=(1,\frac{1}{16})$ and $(b_1,b_2)=(0,\frac{3}{8})$. Their frame (Proposition 38.11) $C_k=a_kb_{k+1}-a_{k+1}b_k$ has $C_1=\frac{3}{8}$ and $C_{k+1}=\frac{c_{4,0}(k)}{c_{4,2}(k)}C_k=\frac{k^3}{64(k+1)^3}C_k$, so

$$C_k=\frac{3}{8\cdot 64^{k-1}k^3}.$$

By Theorem 43.3(i), every integer D_k that clears the rows of $P(k)$ and $P(k+1)$ has $D_k^2\geq|C_k|^{-1}=\frac{8}{3}\,64^{k-1}k^3$, so $\liminf_k\frac{1}{k}\log D_k\geq\log 8$. The clearing found grows like 256^k times d_{k-1}^3; its merit (Definition 43.4) is at most $-\log f(\alpha^\star)-\log 256=\log 16-\log 256=-\log 16$, so Theorem 43.5(ii) does not apply to these rows. For every clearing the frame bounds the merit by $\log 16-\log 8=\log 2$, and Corollary 43.9, with the ratio $\frac{1}{64}$, by $-\log 4+\liminf_k\frac{1}{k}\log\gcd\big(64^{k-1},\mathsf{C}_{k-1}(1)\big)$.

So the row of $P(k)$ is cleared by $256^{k-1}d_{k-1}^3$: the coefficient a_k by 64^{k-1}, the rational part b_k by 256^{k-1} at the prime 2 and by d_{k-1}^3 at the odd primes. □

52.8 The centre of $\binom{[5]}{1}$

The list $\binom{[5]}{1}$ has the marks $1,\dots,5$ and the five singletons as its bases, every weight 1. Its centre ray at the strength one has the stride $f=\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$, $K(\alpha)=\sum_{e=1}^5\alpha_e$, the walls $\kappa_{1,S}=1$ for every nonempty proper S, the volume of the direction $V_1=5!=120$, and the values $P(k)=\langle 1\rangle_k=P_{\binom{[5]}{1}}(k\mathbf{1};k,k)$. In this section P_4 denotes the values of the centre ray of $\binom{[4]}{1}$, continued in the strength, and $v=P_4(1)=7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ (Section 52.7); $\mathsf{g}(y)=\mathsf{G}_{0;(1,-1)}(1,y)$ and $\mathsf{h}(y)=\sum_{m\geq 0}y^m/m!^2$, both solutions of $\vartheta^2u=yu$, $\vartheta=y\frac{d}{dy}$ (Exercise 52.8; $\vartheta^2y^m=m^2y^m$ gives $\vartheta^2\mathsf{h}=y\,\mathsf{h}$).

Exercise 52.12 (The law, the low-end relation and the value span)**.** (i) Compute the operators $\mathbf{R}_2,\dots,\mathbf{R}_6$ of Theorem 42.12 for $n=5$ and the law of the values $\sum_{k=0}^3 c_{5,k}(s)\,P(s+k)=0$.

(ii) Give the closings $\mathsf{C}_j(1)$ for $j\leq 6$, the symbol, its roots and their native gauges.

(iii) Write the low-end relation and the value span, and write $\Gamma(s)^2P(s)$ for $s=3,4$ in 1, $P(1)$ and $P(2)$.

Solution. Receipt: `ex25a_law.py`.

(i) As in Exercise 52.8, Corollary 18.22 and Proposition 8.15 give $\Gamma(s)^2P(s)=\int_0^\infty y^{s-1}\mathsf{g}(y)^5dy$. Theorem 42.12(i) at $n=5$ has the coefficients $i(6-i)=5,8,9,8,5$ in $\mathbf{R}_{i+1}=\vartheta\mathbf{R}_i-i(6-i)\,y\,\mathbf{R}_{i-1}$:

$$\begin{aligned}
\mathbf{R}_2&=\vartheta^2-5y,\\
\mathbf{R}_3&=\vartheta^3-y(13\vartheta+5),\\
\mathbf{R}_4&=\vartheta^4-y(22\vartheta^2+18\vartheta+5)+45y^2,\\
\mathbf{R}_5&=\vartheta^5-y(30\vartheta^3+40\vartheta^2+23\vartheta+5)+y^2(149\vartheta+130),\\
\mathbf{R}_6&=\vartheta^6-y(35\vartheta^4+70\vartheta^3+63\vartheta^2+28\vartheta+5)+y^2(259\vartheta^2+518\vartheta+285)-225y^3.
\end{aligned}$$

The leading coefficients $1,-35,259,-225$ are those of $(\xi^2-25\eta)(\xi^2-9\eta)(\xi^2-\eta)$. Theorem 42.12(ii) gives

$$\begin{gathered}
c_{5,0}(s)=s^4,\qquad c_{5,1}(s)=-(35s^4+70s^3+63s^2+28s+5),\\
c_{5,2}(s)=(s+1)^2(259s^2+518s+285),\qquad c_{5,3}(s)=-225(s+1)^2(s+2)^2,
\end{gathered}$$

and the law $\sum_{k=0}^{3} c_{5,k}(s)P(s+k) = 0$ identically in s; every $c_{5,k}$ has degree 4 and $c_{5,k}(-s-k) = c_{5,k}(s)$. At $s=1$ and $s=2$ the coefficients are $(1,-201,4248,-8100)$ and $(16,-1433,21213,-32400)$.

(ii) The closings are $\mathsf{C}_j(1) = \sum_{|v|=j} \binom{j}{v}^2 = 1, 5, 45, 545, 7885, 127905, 2241225$ for $j \le 6$, and they satisfy $\sum_{i \le \min(3,j)} c_{5,i}(-j)\,\mathsf{C}_{j-i}(1) = 0$ (Corollary 39.32; the receipt checks $j \le 40$). The coefficients of s^4 give the symbol

$$\chi_5(\lambda) = 1 - 35\lambda + 259\lambda^2 - 225\lambda^3 = (1-\lambda)(1-9\lambda)(1-25\lambda),$$

with the roots $\frac{1}{25}, \frac{1}{9}, 1$. By Proposition 42.9(ii) they are the native values $K(\eta)^{-2}$ of the signs: $\eta = \mathbf{1}$, with $K=5$, the native gauge of the direction; the signs with one or four marks at -1, $K = \pm 3$; and those with two or three marks at -1, $K = \pm 1$.

(iii) *The low-end relation.* By Corollary 39.34(ii), with $[s^4]c_{5,0} = 1$, $\mathsf{L}_0 = V_1 = 120$ and $\big(c_{5,1}(0), c_{5,2}(0), c_{5,3}(0)\big) = (-5, 285, -900)$,

$$120 - 5P(1) + 285P(2) - 900P(3) = 0, \qquad P(3) = \frac{2}{15} - \frac{1}{180}P(1) + \frac{19}{60}P(2).$$

The value span. The law has order 3 and its last coefficient does not vanish at the positive integers, so every value is a rational combination of 1, $P(1)$ and $P(2)$: the value span of the ray is $\mathbb{Q} + \mathbb{Q}\,P(1) + \mathbb{Q}\,P(2)$ (Corollary 42.15(i), $h=3$). The law at $s=1$ gives $P(4) = \big(P(1) - 201P(2) + 4248P(3)\big)/8100$, so

$$\Gamma(3)^2 P(3) = \frac{8}{15} - \frac{1}{45}P(1) + \frac{19}{15}P(2),$$
$$\Gamma(4)^2 P(4) = \frac{944}{375} - \frac{113}{1125}P(1) + \frac{1907}{375}P(2).$$

As renderings, $P(1) = 39.94558801199624569894\ldots$, $P(2) = 0.2946961074182751313903\ldots$ and $P(3) = 0.00473383839491413155016 85\ldots$ (quadrature of the share side, a check; Exercise 52.15 reads them exactly from the gauge end). □

Exercise 52.13 (The exact pairing and the Hessian)**.** Show that for every integer $\sigma \ge 1$ the concomitant

$$B(\sigma) = \sum_{i=1}^{3} \sum_{j=1}^{i} c_{5,i}(\sigma - j)\,\mathsf{C}_{\sigma-j}(1)\,P(\sigma - j + i),$$

with the closings at negative indices read as 0, equals -120. Deduce $\det H$, H the Hessian of $-\log f$ at $\mathbf{1}$ on the chart $\alpha_5 = 1$, and compute H directly.

Solution. Receipt: `ex25b_pairing.py`.

The adjoint. The law is self-adjoint under the reflection of the strength: $c_{5,i}(-\sigma - i) = c_{5,i}(\sigma)$ for $i = 0, \ldots, 3$ (Exercise 52.12), so $c_{5,i}(\sigma - i) = c_{5,i}(-\sigma)$, and $\sum_i c_{5,i}(\sigma - i)\,\mathsf{C}_{\sigma-i}(1) = \sum_i c_{5,i}(-\sigma)\,\mathsf{C}_{\sigma-i}(1)$ is the law of the closings at $j = \sigma$, which vanishes for $\sigma \ge 0$; for $\sigma < 0$ every closing in it is 0. The closings solve the adjoint recurrence (Proposition 42.8(i)).

The concomitant is constant. Shifting j by one in $B(\sigma+1)$,

$$B(\sigma+1) - B(\sigma) = \mathsf{C}_\sigma(1) \sum_{i=1}^{3} c_{5,i}(\sigma)P(\sigma + i) - P(\sigma) \sum_{i=1}^{3} c_{5,i}(\sigma - i)\,\mathsf{C}_{\sigma-i}(1),$$

and by the law and the adjoint recurrence the two sums are $-c_{5,0}(\sigma)P(\sigma)$ and $-c_{5,0}(\sigma)\mathsf{C}_\sigma(1)$, so the difference vanishes (the receipt checks the identity symbolically). At $\sigma=1$ only $j=1$ survives, and by the low-end relation of Exercise 52.12

$$B(1) = -5P(1) + 285P(2) - 900P(3) = -120.$$

At $\sigma=2$, with $\mathsf{C}_1(1)=5$, the five terms are $-201\cdot 5\,P(2)$, $4248\cdot 5\,P(3)$, $285\,P(2)$, $-8100\cdot 5\,P(4)$ and $-900\,P(3)$, so $B(2) = -720P(2) + 20340P(3) - 40500P(4)$; with the rows of Exercise 52.12,

$$20340P(3) = 2712 - 113P(1) + 6441P(2), \qquad 40500P(4) = 2832 - 113P(1) + 5721P(2),$$

and $B(2)=-120$, the coefficients of $P(1)$ and $P(2)$ cancelling. The receipt checks $B(\sigma)=-120$ exactly in the value span for $\sigma\le 50$. This is Corollary 42.16 at $n=5$, with $(-1)^n n! = -120$.

The limit. Every $c_{5,i}$ has degree $4 = n-1$, $\chi_5(0)=1$, and $\lambda_\star = \frac{1}{25}$ is a simple root. By Proposition 42.8(iii),

$$V_1 \det H = -\frac{\lambda_\star \chi_5'(\lambda_\star)}{\chi_5(0)} = \prod_{c\neq\star}\Big(1-\frac{\lambda_\star}{\lambda_c}\Big):$$

$$\chi_5'\big(\tfrac{1}{25}\big) = -25\cdot\frac{24}{25}\cdot\frac{16}{25} = -\frac{384}{25}, \qquad \Big(1-\frac{9}{25}\Big)\Big(1-\frac{1}{25}\Big) = \frac{384}{625},$$

so $120\det H = \frac{384}{625}$ and $\det H = \frac{16}{3125} = \frac{2^4}{5^5}$.

Directly. By Theorem 38.15 at the centre, $H = 2\big(\kappa_{\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\neq 5}$, the covariances of the indicators under the weight law at $\mathbf{1}$, which picks one mark of five with the chance $\frac{1}{5}$: $\operatorname{Var} X_e = \frac{1}{5} - \frac{1}{25}$ and $\operatorname{Cov}(X_e, X_{e'}) = -\frac{1}{25}$. So

$$H = 2\Big(\frac{1}{5}I - \frac{1}{25}J\Big) = \frac{1}{25}\begin{pmatrix} 8 & -2 & -2 & -2 \\ -2 & 8 & -2 & -2 \\ -2 & -2 & 8 & -2 \\ -2 & -2 & -2 & 8 \end{pmatrix},$$

J the matrix of ones, with the eigenvalues $\frac{2}{5}$ on the vectors of sum zero and $\frac{2}{25}$ on $\mathbf{1}$: $\det H = \big(\frac{2}{5}\big)^3\frac{2}{25} = \frac{16}{3125}$, as Proposition 42.9(ii) gives at $\eta=\mathbf{1}$, $(2\sigma)^{n-1}/K(\mathbf{1})^n = 2^4/5^5$.

So $B(\sigma) = -120$ at every strength, the low-end relation carried to every σ, and its limit at high strength gives $\det H = 2^4/5^5$, the value of the Hessian computed directly. □

Exercise 52.14 (The lowered settings and the frame)**.** For $j=0,1,2$ the lowered setting with j marks lowered is $\xi_{5,j}(s) = \Gamma(s)^{-2}\int_0^\infty y^{s-1}\mathsf{g}(y)^{5-j}\mathsf{h}(y)^j\,dy$ (Proposition 42.14); put $x^{(j)}(s) = \Gamma(s)^2\xi_{5,j}(s)$, so that $x^{(0)}(s) = \Gamma(s)^2P(s)$.

(i) Show that the $x^{(j)}$ satisfy the law of Exercise 52.12 in this normalization, and find their low-end coefficients and their rates.

(ii) Write the frame $\Phi(s) = \det\big[x^{(i)}(s+l)\big]_{i,l=0}^{2}$ in terms of $\Phi(1)$, and find $\Phi(1)$ from the behaviour of the $x^{(j)}$ at large s.

(iii) Deduce the value of $x^{(1)}(1)\,x^{(2)}(2) - x^{(1)}(2)\,x^{(2)}(1)$.

Solution. Receipt: `ex25c_lowered.py`.

(i) *The law.* $\mathsf{g}^{5-j}\mathsf{h}^j$ is a product of five solutions of $\vartheta^2 u = yu$, so $\mathbf{Q}_5 = \mathbf{R}_6$ annihilates it (Theorem 42.12(i)), and integration against y^{s-1} turns it into the law for $x(s) = \Gamma(s)^2\xi(s)$:

$$s^6x(s) - (35s^4+70s^3+63s^2+28s+5)\,x(s+1) + (259s^2+518s+285)\,x(s+2) - 225\,x(s+3) = 0,$$

the coefficients being $q_{5,k}(-s-k)$; this is the law of Exercise 52.12 multiplied by $\Gamma(s)^2 s^2$ (Proposition 42.14(i)). The boundary terms vanish: near 0, $\mathsf{g}^{5-j}\mathsf{h}^j = O(|\log y|^{5-j})$ (Lemma 20.29(i)), and for $y \ge 1$, $\mathsf{g} \le \sqrt{\pi}\, y^{-1/4} e^{-2\sqrt{y}}$ (Lemma 20.29(ii)) and $\mathsf{h} \le e^{2\sqrt{y}}$, so the integrand decays like $e^{-2(5-2j)\sqrt{y}}$, $j < \frac{5}{2}$.

The low-end coefficients. Near $y = 0$, $\mathsf{g}(y) = -\log y + 2(\log\Gamma)'(1) + O(y|\log y|)$ (Lemma 20.29(i)) and $\mathsf{h}(y) = 1 + O(y)$, and $\int_0^1 y^{s-1}(-\log y)^{5-j} dy = \Gamma(6-j)\, s^{j-6}$; so $x^{(j)}(s) = \Gamma(6-j)\, s^{j-6}\big(1+O(s)\big)$ and $s^4 \xi_{5,j}(s) \to 120, 0, 0$ for $j = 0, 1, 2$, $\Gamma(s)^{-2}$ being $s^2(1+O(s))$. Read at $s \to 0$, the law gives

$$-5x^{(j)}(1) + 285x^{(j)}(2) - 225x^{(j)}(3) = -120,\ 0,\ 0 \qquad (j = 0, 1, 2):$$

the lowered settings with $j \ge 1$ have no low-end coefficient (Proposition 42.14(ii)).

The rates. $x^{(j)}(s+1)/\big(s^2 x^{(j)}(s)\big) \to (5-2j)^{-2} = \frac{1}{25}, \frac{1}{9}, 1$ (Proposition 42.14(iii)): the setting with j marks lowered carries the native value of the signs with j marks at -1. This also follows from the amplitudes below.

(ii) *The frame.* The end coefficients of the law in x are s^6 and -225, with no zero at the integers $s \ge 1$, so Proposition 38.11 gives $\Phi(s+1) = (-1)^3 \frac{s^6}{-225}\Phi(s) = \frac{s^6}{225}\Phi(s)$ and

$$\Phi(s) = \Phi(1)\,\Gamma(s)^6\, 225^{1-s}.$$

One mark at large y. By the torus homogeneity (Theorem 2.4(iii)) $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(\lambda,\lambda)$ with $\lambda = \sqrt{y}$, and Theorem 4.3(iii) at $t = (1,-1)$, $\boldsymbol{\xi} = (1,1)$, $\nu = \nu' = 0$ applies: $\Phi_{\boldsymbol{\xi}}(y') = e^{y'} + e^{-y'}$, $y^\star = 0$, $\Phi_{\boldsymbol{\xi}}(0) = 2$, $\sum_k w_k^\star t_k^2 = 2$, so

$$\mathsf{g}(y) = e^{-2\sqrt{y}}\Big(\frac{2\pi}{\sqrt{y}}\Big)^{\frac12} 2^{-\frac12}\big(1 + O(y^{-\frac12})\big) = \sqrt{\pi}\, y^{-\frac14} e^{-2\sqrt{y}}\big(1 + O(y^{-\frac12})\big).$$

$\mathsf{h}(y)$ is the constant term of $e^{\sqrt{y}(z+z^{-1})}$, that is $\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{2\sqrt{y}\cos\theta} d\theta$, and Laplace's method (an outside result; it applies because $\cos\theta$ has one maximum on the circle, at $\theta = 0$, where its second derivative is -1) gives $\mathsf{h}(y) = e^{2\sqrt{y}}(4\pi\sqrt{y})^{-1/2}\big(1 + O(y^{-1/2})\big)$. So, with $\rho_j = 2(5-2j) = 10, 6, 2$,

$$\mathsf{g}^{5-j}\mathsf{h}^j = \pi^{\frac{5-2j}{2}} 2^{-j}\, y^{-\frac54} e^{-\rho_j\sqrt{y}}\big(1 + O(y^{-\frac12})\big).$$

The amplitudes. With $y = u^2$, $x^{(j)}(s) = 2\int_0^\infty u^{2s-1}\mathsf{g}(u^2)^{5-j}\mathsf{h}(u^2)^j\, du$; the part $u \le u_0$ is $O(u_0^{2s})$, negligible against $\Gamma(2s)\rho_j^{-2s}$, and the rest gives

$$x^{(j)}(s) = A_j\,\Gamma\big(2s - \tfrac52\big)\,\rho_j^{-2s}\big(1 + O(s^{-1})\big), \qquad A_j = 2\pi^{\frac{5-2j}{2}} 2^{-j} \rho_j^{\frac52},$$

$A_0 = 200\sqrt{10}\,\pi^{5/2}$, $A_1 = 36\sqrt{6}\,\pi^{3/2}$, $A_2 = 2\sqrt{2}\,\pi^{1/2}$, $\prod_j A_j = \pi^{9/2}\, 120^{5/2}$. Factoring $A_i\rho_i^{-2s}$ out of the rows and $\Gamma(2s + 2l - \frac52)$ out of the columns,

$$\Phi(s) = \prod_i A_i\rho_i^{-2s} \prod_{l=0}^{2} \Gamma\big(2s + 2l - \tfrac52\big)\big(V + O(s^{-1})\big), \qquad V = \det\big[u_i^l\big] = \prod_{i<i'}(u_{i'} - u_i) = \frac{16}{16875},$$

$u_i = \rho_i^{-2} = \frac{1}{100}, \frac{1}{36}, \frac14$. Theorem 4.3(iii) for $\Gamma(\lambda + a) = \lambda^{\lambda+a}\mathsf{G}_{\lambda+a;(1)}(\lambda)$, with $\Phi_\xi(y') = e^{y'} - y'$, $y^\star = 0$ and $w^\star = 1$, gives $\Gamma(\lambda+a) = \lambda^{\lambda+a}e^{-\lambda}(2\pi/\lambda)^{1/2}\big(1 + O(\lambda^{-1})\big)$, hence $\prod_l \Gamma(2s + 2l - \frac52) = \Gamma(2s)^3(2s)^{-3/2}\big(1 + O(s^{-1})\big)$ and $\Gamma(2s) = 2^{2s-1}\pi^{-1/2}s^{1/2}\Gamma(s)^2\big(1 + O(s^{-1})\big)$, so

$$\Phi(s) = \prod_j A_j \cdot V \cdot 2^{-\frac92}\pi^{-\frac32}\,\Gamma(s)^6\Big(\frac{64}{\prod_j \rho_j^2}\Big)^s\big(1 + O(s^{-1})\big), \qquad \frac{64}{\prod_j \rho_j^2} = \frac{64}{14400} = \frac{1}{225}.$$

Comparing with $\Phi(s) = 225\,\Phi(1)\,\Gamma(s)^6\,225^{-s}$,

$$\Phi(1) = \frac{\pi^{9/2}\,120^{5/2}}{225}\cdot\frac{16}{16875}\cdot\frac{1}{2^{9/2}\pi^{3/2}} = \frac{128\sqrt{15}}{16875}\,\pi^3 = \frac{128\sqrt{15}}{16875}\,\Gamma\big(\tfrac12\big)^6,$$

which is $0.91088056479820303289039 72\ldots$ to 25 places; the receipt computes the nine settings by quadrature and reproduces it, and checks the law, the low-end coefficients and the amplitudes.

(iii) *The block.* By (i), $x^{(j)}(3) = \big(-5x^{(j)}(1) + 285x^{(j)}(2) + 120\,\delta_{j0}\big)/225$, so subtracting $-\frac{5}{225}$ times the first column and $\frac{285}{225}$ times the second from the third leaves the column $(\frac{120}{225}, 0, 0)$:

$$\Phi(1) = \frac{120}{225}\Big(x^{(1)}(1)\,x^{(2)}(2) - x^{(1)}(2)\,x^{(2)}(1)\Big), \qquad x^{(1)}(1)\,x^{(2)}(2) - x^{(1)}(2)\,x^{(2)}(1) = \frac{16\sqrt{15}}{1125}\,\Gamma\big(\tfrac12\big)^6,$$

which is $1.70790105899663068666949 5\ldots$ to 24 places. The values $P(1), P(2)$ do not enter Φ: their coefficients are minors that the vanishing low-end coefficients of the lowered settings remove. □

Exercise 52.15 (The reading from the gauge end)**.** Read $P(1)$, $P(2)$ and $P(3)$ from the gauge end (Theorem 40.6, Definition 40.10): the native gauge and the constant $g(0)$; the density operator of the law and its exponents at $f^\star = \frac{1}{25}$; the germs that extend by zero; the Taylor coefficients of the germ; the values at a large strength and the law run back; and the low-end relation read on the output.

Solution. Receipt: `ex25d_gauge.py`.

(1) *The native gauge.* $\alpha^\star = \mathbf{1}$ (Theorem 38.14), $f^\star = f(\mathbf{1}) = \frac{1}{25}$, $\det H = \frac{16}{3125}$ (Exercise 52.13) and $(\omega_0/dy)(\mathbf{1}) = 1$, the origin being 0. By Proposition 40.3(ii), with $n = 5$,

$$g(0) = \frac{1}{\Gamma(2)\sqrt{\det H}}\Big(\frac{2\pi}{f^\star}\Big)^2 = \frac{25\sqrt5}{4}\,(50\pi)^2 = 15625\sqrt5\,\Gamma\big(\tfrac12\big)^4.$$

(2) *The operator.* The density operator of the law $\mathbf{p} = (c_{5,0}, \ldots, c_{5,3})$ is $\mathfrak{D}_\mathbf{p} = \sum_i t^i c_{5,i}(-1 - i - \theta)$, $\theta = t\frac{d}{dt}$ (Definition 40.4). With $\theta^m = \sum_j S(m,j)\,t^j\partial_t^j$, $S(m,j)$ the number of partitions of a set of m elements into j nonempty parts, $\mathfrak{D}_\mathbf{p} = \sum_{j=0}^4 a_j(t)\partial_t^j$ with

$$\begin{aligned}
a_0 &= 1 - 201t + 4248t^2 - 8100t^3,\\
a_1 &= -t\,(24300t^3 - 16965t^2 + 1232t - 15),\\
a_2 &= -t^2(16650t^3 - 14271t^2 + 1358t - 25),\\
a_3 &= -2t^3(1800t^3 - 1813t^2 + 210t - 5),\\
a_4 &= -t^4(t-1)(9t-1)(25t-1) = t^4\chi_5(t).
\end{aligned}$$

In $s = \frac{1}{25} - t$, $\mathfrak{D}_\mathbf{p} = \sum_j b_j(s)\partial_s^j$ with $b_{4,0} = 0$ and $b_{4,1} = -(f^\star)^4\chi_5'(f^\star) = \frac{384}{25^5}$. The second symbol is $\chi_1(\lambda) = -70\lambda + 1036\lambda^2 - 1350\lambda^3$, and Theorem 40.6(i) gives

$$\gamma = -1 + \frac{\chi_1(f^\star)}{f^\star\chi_5'(f^\star)} = -1 + \frac{-768/625}{-384/625} = 1 = \frac{n-3}{2}, \qquad R_0(x) = \frac{384}{25^5}\,x(x-1)^2(x-2)\,:$$

the exponents at $f^\star$ are $0, 1, 2$ and $\gamma = 1$, a double root. The further polynomials $R_m(x) = \sum_j b_{j,m-3+j}[x]_j$ are

$$R_1 = -\tfrac{8}{25^4}\,x(x-1)(163x^2 - 67x + 72), \qquad R_2 = \tfrac{1}{25^3}\,x(1385x^3 + 1840x^2 + 1631x + 576),$$

$$R_3 = -\tfrac{1}{625}(180x^4 + 1290x^3 + 2129x^2 + 1619x + 476),$$
$$R_4 = -\tfrac{1}{25}(490x^4 + 2530x^3 + 5454x^2 + 5789x + 2499),$$
$$R_5 = (x+2)^2(196x^2 + 802x + 819), \qquad R_6 = 225(x+2)^2(x+3)^2,$$

and $R_m = 0$ for $m \geq 7$.

(3) *The line.* As $\gamma = 1 \in \{0, 1, 2\}$, the germ is $\varrho_f = \sum_r u_r s^r$ with $u_0 = 0$, $u_1 = g(0)$ (Theorem 40.6(iv)). The germs that extend by zero are fixed by the conditions $\mathsf{p}_r(u) = 0$, $r \leq 2$, the point masses of $\mathfrak{D}_{\mathbf{p}}$ applied to the germ cut at $f^\star$:

$$\mathsf{p}_0(u) = -\frac{384}{25^5}u_0, \qquad \mathsf{p}_1(u) = -\frac{576}{25^4}u_0, \qquad \mathsf{p}_2(u) = \frac{576}{25^3}u_0 - \frac{1344}{25^4}u_1 + \frac{768}{25^5}u_2,$$

the coefficient of u_r in p_r being $r!\, b_{4,1}(r-\gamma)$. They leave one line, $u_0 = 0$ and $u_2 = \frac{1344 \cdot 25}{768}u_1 = \frac{175}{4}u_1$ (Theorem 40.6(ii)): the germ vanishes to the first order, as the exponent $\gamma = 1$ says, and the receipt also derives the three conditions from the distributional derivatives of the cut germ.

(4) *The germ.* For $N \geq 3$, $R_0(N)u_N = -\sum_{m=1}^{\min(N,6)} R_m(N-m)\, u_{N-m}$. At $N = 3$, $R_0(3) = \frac{4608}{25^5}$, $R_1(2) = -\frac{9440}{25^4}$, $R_2(1) = \frac{5432}{25^3}$, so

$$u_3 = -\frac{25^5}{4608}\Big(-\frac{9440}{25^4}\cdot\frac{175}{4} + \frac{5432}{25^3}\Big)u_1 = \frac{48125}{32}u_1, \qquad u_4 = \frac{35703125}{768}u_1,$$

and the receipt continues to u_{400}, all rational multiples of u_1, and checks that the series solves $\mathfrak{D}_{\mathbf{p}}\varrho = 0$ to the order kept.

(5) *Large strengths.* The singular points of $\mathfrak{D}_{\mathbf{p}}$ in t are $0, \frac{1}{25}, \frac{1}{9}, 1$, so the series converges for $|s| < \frac{1}{25}$, on $0 < t \leq f^\star$, and by Proposition 40.9(ii) at $j = 0$

$$P(K) = \int_0^{f^\star} t^K \varrho_f(t)\, dt = g(0)\sum_{r\geq 1}\frac{u_r}{u_1}\,(f^\star)^{K+r+1}\,\mathrm{B}(K+1, r+1),$$

the terms falling like $\prod_{i\leq r}\frac{i}{K+1+i}$; at $K = 150$ the term $r = 400$ is $3 \cdot 10^{-136}$ of the sum. These are the values at $K = 150, 151, 152$.

(6) *The run back.* The law holds at every positive integer (Proposition 40.5), and $c_{5,0}(k) = k^4 \neq 0$, so $P(k) = -\big(c_{5,1}(k)P(k+1) + c_{5,2}(k)P(k+2) + c_{5,3}(k)P(k+3)\big)/k^4$ from $k = 149$ down to $k = 1$. In 150-digit arithmetic, with the renderings to 60 places,

$$P(1) = 39.945588011996245698944028498423995891849782924769859572366 6\ldots,$$
$$P(2) = 0.294696107418275131390303208655999730074422376946254727936899\ldots,$$
$$P(3) = 0.004733833949141315501684746638711048457734958673148110666870 36\ldots,$$

and the reading from $K = 250$ agrees to 10^{-133}.

(7) *Checks.* The low-end relation read on the output (Definition 40.10 step (7)) gives $5P(1) - 285P(2) + 900P(3) = 120.000\ldots$ to all the places computed: the output returns $V_1 = 120$, which the reading did not use. The values agree with the quadrature of Exercise 52.12 to its 22 places.

So $P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1) = 39.945588011996245698944028498 42\ldots$, read from the law, the native gauge $\mathbf{1}$ and $\det H$ alone, with the germ of the density ϱ_f exact over $\mathbb{Q}$ up to the factor $15625\sqrt{5}\,\Gamma(\frac{1}{2})^4$. $\square$

Exercise 52.16 (The value as a Trailmoment of the slope input of four marks)**.** Let $\tilde{p}_4 = -P_4'$ be the slope input of the centre ray of $\binom{[4]}{1}$ (Definition 42.1). Show that $P(1) = (\mathsf{T}_{0;(1)}[x]\,\tilde{p}_4\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \sum_{x\geq 1}\tilde{p}_4(x)$, find the recurrence of the slope input and its relation at the cut, and write $\tilde{p}_4(m)$, $m \leq 5$, in 1, v and $\tilde{p}_4(1)$.

Solution. Receipt: `ex25e_slope.py`.

The mark added. By Proposition 42.11(i), merging four marks, $\Gamma(s)^2 P(s) = \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2\Gamma(s-d)^2 P_4(d)\,dd$; at $s=1$ the kernel is $\Gamma_4^{\,0}\big[d,\ d,\ 1-d,\ 1-d\big]$, whose right closing reads the derivative of the rest at the positive integers (Lemma 5.16(i)). Proposition 42.11(ii) at $s=1$, with $W_1 = 1$, gives

$$P(1) = (\mathsf{T}_{0;(1)}[x]\,\tilde{p}_4\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \sum_{x\geq 1}\tilde{p}_4(x)\,:$$

the centre of five marks at the strength one is the slope input of four marks summed over the positive integers.

The recurrence. By Proposition 42.2(ii), $L\tilde{p}_4 = L'P_4$ for the law L of Exercise 52.8:

$$\sum_{i=0}^{2} c_{4,i}(\sigma)\,\tilde{p}_4(\sigma+i) = \sum_{i=0}^{2} c'_{4,i}(\sigma)\,P_4(\sigma+i),$$
$$c'_{4,0} = -3\sigma^2, \qquad c'_{4,1} = 6(10\sigma^2+10\sigma+3), \qquad c'_{4,2} = -192(\sigma+1)^2.$$

The cut. Proposition 42.6 at $x_0 = 0$, $j=0$, with $c_{4,0} = -\sigma^3$, so $\gamma_3 = -1$ and $\gamma_4 = 0$, and with $\ell_1 = 0$, every weight being 1:

$$\sum_{i=1}^{2}\big(c_{4,i}(0)\,\tilde{p}_4(i) - c'_{4,i}(0)\,P_4(i)\big) = \gamma_3\ell_1 + \gamma_4 V_1 = 0, \qquad 4\tilde{p}_4(1) - 64\tilde{p}_4(2) - 18P_4(1) + 192P_4(2) = 0.$$

With $P_4(1) = v$ and $P_4(2) = \frac{1}{16}v - \frac{3}{8}$ (Exercise 52.8),

$$\tilde{p}_4(2) = \frac{1}{16}\tilde{p}_4(1) - \frac{3}{32}v - \frac{9}{8}.$$

The rows. The recurrence, whose coefficient $c_{4,2}(m) = -64(m+1)^3$ does not vanish, gives every slope from the first two:

$$\tilde{p}_4(3) = \tfrac{7}{1024}\tilde{p}_4(1) - \tfrac{57}{4096}v - \tfrac{279}{2048},$$
$$\tilde{p}_4(4) = \tfrac{1}{1024}\tilde{p}_4(1) - \tfrac{73}{32768}v - \tfrac{2819}{147456},$$
$$\tilde{p}_4(5) = \tfrac{679}{4194304}\tilde{p}_4(1) - \tfrac{13081}{33554432}v - \tfrac{463499}{150994944},$$

the coefficient of $\tilde{p}_4(1)$ being a_m of Exercise 52.8, since the recurrence has the law as its homogeneous part. So every $\tilde{p}_4(m)$ lies in $\mathbb{Q} + \mathbb{Q}\,v + \mathbb{Q}\,\tilde{p}_4(1)$.

The value. The slopes have the root rate $\frac{1}{16}$ along the positive integers (Proposition 42.2(iii)). As renderings, $\tilde{p}_4(1) = -P'_4(1) = 39.381582123122165624939809345 1\ldots$ by quadrature of $-\int_0^\infty \log y\,\mathsf{g}(y)^4 dy + 2(\log\Gamma)'(1)\,v$, then $\tilde{p}_4(2) = 0.547499039997\ldots$, $\tilde{p}_4(3) = 0.0158846667692\ldots$, $\tilde{p}_4(4) = 0.00059555377139\ldots$, $\tilde{p}_4(5) = 0.0000253984887\ldots$ from the rows; the sum over $m \leq 70$ is $39.945588011996245694894402849842\ldots$, the value of Exercise 52.15 to all 36 places computed, and the steps $m \geq 2$ add $0.564005888874080074\ldots$ to the first slope. The receipt also checks Proposition 42.11(iii) at $n=4$, $5\,(\mathsf{T}_{0;(1)}[x]\,P_4\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 5\,(\mathsf{T}_{0;(1)}[x]\,(x-1)\,\tilde{p}_4\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,\tilde{p}_4\mathbb{I}_{\mathbb{Z}})((0,\infty))$.

So $P_{\binom{[5]}{1}}(\mathbf{1};1,1) = (\mathsf{T}_{0;(1)}[x]\,\tilde{p}_4\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, a Trailmoment at the index zero of the slope input of four marks, whose density lies in $\mathbb{Q} + \mathbb{Q}\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + \mathbb{Q}\,\tilde{p}_4(1)$ at every point. □

52.9 W_3 at $(\mathbf{1};0,2)$

The wheel W_3 (Proposition 39.49) has the hub, the rim vertices $1,2,3$, the spokes s_1,s_2,s_3 and the rims r_1,r_2,r_3, r_i joining i to $i+1$; its bases are the 16 spanning trees, the triples of marks other than the four triangles $\{r_1,r_2,r_3\}$, $\{s_1,s_2,r_1\}$, $\{s_2,s_3,r_2\}$, $\{s_3,s_1,r_3\}$, its rank is 3, its cobases are the complements of the bases and every weight is 1. The ray of this section has the origin 0 and the direction $(\mathbf{1};0,2)$: masses 1, exponent 0 for K and 2 for Ψ, on the balance wall, $m_{\mathcal{E}}=6=0\cdot 3+2\cdot 3$. Its stride is $f=\alpha^{\mathbf{1}}/\Psi^2$, only Ψ being active; its walls are $\kappa_{1,S}=|S|-2\big(|S|-r(S)\big)=2r(S)-|S|$, with the values $1,2,3$ on the nonempty proper S; and its volume of the direction is $V_1=\sum_\pi\prod_{l=2}^{6}\kappa_{1,U_l}^{-1}=84$. At $\mathbf{1}$ every mark lies in 8 of the 16 bases, so every dual occupation is $\frac12$, the defects $1-2\cdot\frac12$ vanish, and $\alpha^\star=\mathbf{1}$ with $f^\star=\Psi(\mathbf{1})^{-2}=\frac{1}{256}$. The values are $P(k)=\langle 1\rangle_k=P_{W_3}(k\mathbf{1};0,2k)$. Since K has the exponent 0, the constant terms of the powers of $f^{-1}=\Psi^2/\alpha^{\mathbf{1}}$ are written explicitly: $\mathrm{CT}(f^{-j})=[\alpha^{j\mathbf{1}}]\,\Psi^{2j}$.

Exercise 52.17 (The box, the law and the spectrum)**.** (i) Read the box $\mathfrak{B}_2$ with one unit of Ψ (Definition 39.11) modulo the automorphisms of W_3, find its free orbits, and show that it is accepted at $\mathfrak{s}=1$ with the spanning set $b_1=1$, $b_2=b=\frac14\sum_T\alpha^T/\alpha^{\mathcal{E}\setminus T}$, the sum over the four triangles T; give its matrix $\mathbf{B}(k)$.

(ii) Derive the law of [1] from the rows, its symbol, and show that b_1,b_2 are masters.

(iii) Show that $\mathbf{B}$ is not regular at $k=\infty$, and repair it by a change of masters.

Solution. Receipt: `ex26a_box.py` (with `w3common.py`).

(i) *Symmetries.* The 24 permutations of the four vertices of the graph induce permutations of the marks that carry the spanning trees onto themselves and keep the masses and the exponents: maps ϕ_β of the first kind of Proposition 39.7(ii), forming a group G. The level bounds are $\bar a=0$, K being inactive, and $\bar b=1$; G keeps Ψ, so it keeps the box.

The box. $\mathfrak{B}_2(0,1)$ holds the symbols of wall degree at most 2 at the levels $(0,0)$ and $(0,1)$ (Definition 39.11(iv)): 1167 Laurent symbols α^u, $u_{\mathcal{E}}=0$, which are the stems, and 2848 symbols at the level $(0,1)$, $u_{\mathcal{E}}=3$. Its relations are the letter identities of Ψ, $\alpha^u-\sum_T(u+\mathbf{1}_{\mathcal{E}\setminus T};0,1)$, of the 1167 Laurent symbols, and the mark relations of the stems at the six marks (Definition 39.11(iii)), here, with the origin 0, $m_{1,e}=1$, $s_1=0$ and $t_1=2$,

$$\mathrm{MR}_e(\alpha^u)=(k+u_e)\,\alpha^u-2k\sum_{T\not\ni e}(u+\mathbf{1}_{\mathcal{E}\setminus T};0,1),$$

7002 of them. Modulo G (Definition 39.16(i)) there are 66 Laurent orbits, of wall degrees 0 (one orbit), 1 (eight) and 2 (fifty-seven), and 145 orbits at the level $(0,1)$: 211 orbit averages, on which the averaged relations, 389 distinct ones, have the rank 200 over $\mathbb{Q}(k)$.

The reduction. The ranking puts the orbits at the level $(0,1)$ highest, then the Laurent orbits by decreasing wall degree, with the orbit of $\alpha^{r_1r_2r_3}/\alpha^{s_1s_2s_3}$ last among those of wall degree 1 (Definition 39.16(ii)). The free orbits are 11: nine at the level $(0,1)$ and the Laurent orbits of 1 and of $\alpha^{r_1r_2r_3}/\alpha^{s_1s_2s_3}$. The latter orbit has four members, $\alpha^T/\alpha^{\mathcal{E}\setminus T}$ for the four triangles T, each with its complement the star at the opposite vertex; so the spanning set at $\mathfrak{s}=1$ is $b_1=1$, $b_2=b$ (Definition 39.16(iii)). The targets $f^{-1}b_1=\Psi^2/\alpha^{\mathbf{1}}$ and $f^{-1}b_2$ involve 9 and 22 orbits.

Acceptance. The reductions of the two targets were computed by elimination at integer strengths and reconstructed over $\mathbb{Q}(k)$ (Proposition 41.10(i),(ii)); with the common denominator $Q_c=(k-1)^2(k-2)^5(k^2-k+1)$ every $Q_c\mathbf{B}_{il}$ is a polynomial of degree at most 10:

$$\mathbf{B}_{11}=-\frac{2(2k-1)(k^2-4k+2)(7k^2-8k+3)}{(k-1)^2(k^2-k+1)},\qquad \mathbf{B}_{12}=\frac{14k^2(2k-1)}{k^2-k+1},$$

$$\mathbf{B}_{21} = -\frac{2k(k-3)(k-1)(k+1)(2k-1)(7k^2-14k+8)}{(k-2)^4(k^2-k+1)},$$
$$\mathbf{B}_{22} = \frac{2k^2(k-1)(2k-1)(k^2-2)(7k^2-20k+15)}{(k-2)^5(k^2-k+1)}.$$

The proof that these are the reductions over $\mathbb{Q}(k)$: for each target x_i the row $x_i - \sum_l \mathbf{B}_{il} b_l$, times Q_c, stacked under the relation matrix $A(k)$, whose entries are affine in k (Proposition 41.9(iii)), has the rank 200 at 230 integer strengths; its minors of order 201 are polynomials of degree at most 219, so they vanish identically, the row lies in $\mathcal{R}^G_{\mathfrak{B}}$ over $\mathbb{Q}(k)$, and $\sum_l \mathbf{B}_{il} b_l$, a combination of free orbits, is the reduction (Definition 39.16(ii)). So the box is accepted at $\mathfrak{s} = 1$ with the matrix $\mathbf{B}$, the classes of b_1, b_2 span a submodule $\mathfrak{N} \ni [1]$, there is a certificate supported on the stems, and $\langle b_i \rangle_{k-1} = \sum_l \mathbf{B}_{il}(k) \langle b_l \rangle_k$ at every strength (Theorem 39.17(i)–(iii)).

(ii) *The law.* With $w_0 = (1, 0)$ and $w_i(k) = w_{i-1}(k-1)\mathbf{B}(k)$, the first dependence is $q_0 w_0 + q_1 w_1 + q_2 w_2 = 0$ with

$$q_0 = 16(k-1)(2k-3)(2k-1)(4k-5)(4k-3), \qquad q_2 = (k-2)^5,$$
$$q_1 = -2(2k-3)(65k^4 - 390k^3 + 885k^2 - 900k + 346),$$

and $p_j(k) = q_{2-j}(k+2)$ (Proposition 39.6(ii), Theorem 39.17(iv)) gives

$$k^5 P(k) - 2(2k+1)(65k^4 + 130k^3 + 105k^2 + 40k + 6)\, P(k+1) + 16(k+1)(2k+1)(2k+3)(4k+3)(4k+5)\, P(k+2) = 0$$

at every strength (Corollary 39.34(i)). Its symbol is $1 - 260\lambda + 1024\lambda^2 = (1 - 4\lambda)(1 - 256\lambda)$, and at both roots the exponent $-\chi_1(\lambda)/(\lambda\chi'(\lambda))$, with the second symbol $\chi_1(\lambda) = 10\lambda(512\lambda - 65)$, is $-\frac{5}{2} = -\frac{n-1}{2}$ (Proposition 40.7). The constant terms $\mathrm{CT}(f^{-j}) = 1, 12, 756, 78960, 10451700, 1587790512, 263964176784$ $(j \le 6)$ satisfy $\sum_i p_i(-j)\,\mathrm{CT}(f^{i-j}) = 0$ (Corollary 39.34(i)), a check.

Masters. Two native gauges: $\mathbf{1}$, with $\Psi = 16$, the native value $\frac{1}{256}$ and the determinant $\frac{3}{256}$ of the Hessian of $-\log f$ on the chart; and $\alpha_c = (1, \omega, \omega^2, \omega^2, 1, \omega)$ on $(s_1, s_2, s_3, r_1, r_2, r_3)$, $\omega = e^{2\pi i/3}$, at which every triangle and every star has the product 1, $\Psi(\alpha_c) = -2$, every dual occupation is $\frac{1}{2}$ (so $c^{(1)} = 0$), the native value is $\prod_e \alpha_{c,e}/\Psi^2 = \frac{1}{4}$, and the Hessian has the determinant $-\frac{3}{2}$. Both are nondegenerate with distinct native values, so by Corollary 38.7 at $g = 1$ every recurrence of $\langle 1 \rangle$ that follows from the relations has order at least 2: the cyclic submodule of $[1]$, inside $\mathfrak{N}$ of dimension at most 2, has dimension 2. So b_1, b_2 are masters, $\mathbf{B} = \mathbf{B_b}$, and the relation above is the law of [1].

(iii) *The spectrum.* $b(\alpha_c) = 1 = b(\mathbf{1})$, so the vectors $\beta_c = (1, 1)$ of the two native gauges are proportional with distinct native values, and $\mathbf{B}$ is not regular at $k = \infty$ (Proposition 39.38(ii)); indeed $\mathbf{B}(k)/k \to \begin{bmatrix} -28 & 28 \\ -28 & 28 \end{bmatrix}$. The change $T(k) = \begin{bmatrix} 1 & 0 \\ -k & k \end{bmatrix}$, to the elements $[1]$ and $k([b] - [1])$, gives $\tilde{\mathbf{B}}(k) = T(k-1)\mathbf{B}(k)T(k)^{-1}$ with

$$\tilde{\mathbf{B}}(\infty) = \begin{bmatrix} 88 & 28 \\ 504 & 172 \end{bmatrix}, \qquad \det\big(\lambda - \tilde{\mathbf{B}}(\infty)\big) = (\lambda - 256)(\lambda - 4),$$

the eigenvectors $(\frac{1}{6}, 1)$ and $(-\frac{1}{3}, 1)$: the reciprocals of the native values $\frac{1}{256}$ and $\frac{1}{4}$ (Proposition 39.38(iii)). As $T(k-1)T(k)^{-1} = \mathrm{diag}\big(1, \frac{k-1}{k}\big) \to I$, the eigenvalues of $\mathbf{B}(k)$ tend to 256 and 4: $\mathrm{tr}\,\mathbf{B}(k) \to 260$ and

$$\det \mathbf{B}(k) = \frac{16k^2(2k-1)^2(4k-5)(4k-3)(k^2-3k+3)}{(k-2)^5(k-1)(k^2-k+1)} \to 1024.$$

So the box gives the masters $1, b$, the matrix $\mathbf{B_b}$ above and the law of [1] of symbol $(1-4\lambda)(1-256\lambda)$; $\mathbf{B_b}$ grows like k, and the masters 1 and $k(b-1)$ repair it. □

Exercise 52.18 (The low-end relation, the brackets and the value)**.** From the law of Exercise 52.17: write the low-end relation; show $7 < P(1) < \frac{64}{7}$; give the brackets of the two conditions $P(K) > 0$ and $P(K) < f^\star P(K-1)$ at every K, their widths and their rate; and give $P(1)$.

Solution. Receipt: `ex26b_bracket.py`.

The low-end relation. By Corollary 39.34(ii), with $[k^5]p_0 = 1$, $\mathsf{L}_0 = V_1 = 84$ (Definition 39.33, Proposition 39.31(ii), which allows the exponent 0 of K), $p_1(0) = -12$ and $p_2(0) = 720$:

$$84 - 12P(1) + 720P(2) = 0, \qquad P(2) = \frac{P(1) - 7}{60},$$

and every value lies in $\mathbb{Q} + \mathbb{Q}\,P(1)$, p_2 having no zero at $k \geq 0$.

The first bracket. Proposition 41.7(ii), with $a = 1$: $-aV_1/p_1(0) = 7$ and $-aV_1/\big(p_1(0) + f^\star p_2(0)\big) = -84/(-12 + \frac{720}{256}) = -84/(-\frac{147}{16}) = \frac{64}{7}$, both positive, so $7 < P(1) < \frac{64}{7}$.

The brackets. Let c and r solve the law with $(c(1), c(2)) = (1, \frac{1}{60})$, of low-end coefficient 0, and $(r(1), r(2)) = (0, -\frac{7}{60})$, of low-end coefficient 84; then $P(K) = c(K)P(1) + r(K)$ (Proposition 43.6(ii)):

K	1	2	3	4	5	6
$c(K)$	1	$\frac{1}{60}$	$\frac{1}{900}$	$\frac{47}{400400}$	$\frac{79}{5105100}$	$\frac{2273}{977728752}$
$r(K)$	0	$-\frac{7}{60}$	$-\frac{173}{21600}$	$-\frac{73219}{86486400}$	$-\frac{140653}{1260230400}$	$-\frac{3541028833}{211189410432000}$

Positivity at K (Proposition 41.4(i) with $p = 1$) and at $K-1$ with $p = f^\star - t$, where $c(K) > 0$ and $256c(K) - c(K-1) > 0$ (true for $2 \leq K \leq 90$), give

$$a_K = -\frac{r(K)}{c(K)} < P(1) < b_K = \frac{r(K-1) - 256\,r(K)}{256\,c(K) - c(K-1)};$$

$K = 2$ is the first bracket. The widths are 2.143, $1.297 \cdot 10^{-2}$, $1.513 \cdot 10^{-4}$, $2.042 \cdot 10^{-6}$ at $K = 2, \ldots, 5$, $3.50 \cdot 10^{-19}$ at $K = 12$ and $1.161 \cdot 10^{-33}$ at $K = 20$; the brackets are nested, and the ratio of consecutive widths, 0.01524 at $K = 10$ and 0.01562 at $K = 80$, tends to $\frac{1}{64} = f^\star/\lambda'$, $\lambda' = \frac{1}{4}$ (Theorem 41.3(iv)). At $K = 12$

$$\frac{125311627268888255929639}{17374611099178960704000} < P(1) < \frac{2909077508602860430922579 47}{40334717113590600816192000}.$$

The value. The roots $\frac{1}{256}$ and $\frac{1}{4}$ are distinct, the second outside the disc $|\lambda| \leq f^\star$, and p_0, p_2 have the largest degree 5 and no zero at $k \geq 1$: by Theorem 41.3(iii) the values are the minimal solution, and by (iv), with $x^{(0)} = r$ and $x^{(1)} = c$, the approximants are $M^{(K)} = a_K$. At $K = 90$, of width $4 \cdot 10^{-160}$, as a rendering,

$$P(1) = 7.212341418957565712398428969068699944589917754042993290753 62\ldots,$$

and $P(2) = (P(1) - 7)/60$. The continued fraction of Proposition 41.7(iii), $\rho_1 = \mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(0)$ and $P(1) = -aV_1/(p_1(0) + p_2(0)\rho_1)$, gives the same number, to 20 places at depth 10 and to 39 at depth 40. □

52.10 Positivity against the gauge end

The centre ray of $\binom{[3]}{1}$ at the strength one has the stride $\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$, $K=\alpha_1+\alpha_2+\alpha_3$, $V_1=3!=6$, and the values $P(k)=P_{\binom{[3]}{1}}(k\mathbf{1};k,k)$; $\ell=P(1)$. By Theorem 42.12 at $n=3$ its law is $s^2P(s)-(10s^2+10s+3)P(s+1)+9(s+1)^2P(s+2)=0$, of symbol $(1-\lambda)(1-9\lambda)$ with the roots $f^\star=\frac{1}{9}$ and 1 (Proposition 42.9), and the low-end relation (Corollary 39.34(ii)) is $6-3\ell+9P(2)=0$. So $P(K)=c(K)\,\ell+r(K)$ with $(c(1),c(2))=(1,\frac{1}{3})$ and $(r(1),r(2))=(0,-\frac{2}{3})$; the law is reflexive with the ratio $\frac{1}{9}$, and $c(K)=9^{1-K}\mathsf{C}_{K-1}(1)$ (Theorem 43.8(ii)), $\mathsf{C}_j(1)=1,3,15,93,639,4653,\dots$.

Exercise 52.19 (All positivity conditions on the centre of $\binom{[3]}{1}$)**.** (i) Read ℓ from the gauge end.

(ii) For $N\le 22$ compare the bracket of the two conditions $P(K)>0$, $P(K)<\frac{1}{9}P(K-1)$ at $K=N+1$ with the bracket of all positivity conditions of degree at most N at the strength 1 (Proposition 41.4), both read on $P(1),\dots,P(N+1)$.

(iii) Find the rate of the second (Theorem 41.5).

Solution. Receipt: `ex27_positivity.py`.

(i) *The gauge end.* $n=3$, $f^\star=\frac{1}{9}$, $\det H=(2\sigma)^2/K(\mathbf{1})^3=\frac{4}{27}$ (Proposition 42.9(ii)), so $g(0)=\frac{2\pi}{f^\star\sqrt{\det H}}=27\sqrt{3}\,\Gamma(\frac{1}{2})^2$ (Proposition 40.3(ii)). The density operator (Definition 40.4) is $a_0+a_1\partial_t+a_2\partial_t^2$ with $a_0=1-23t+36t^2$, $a_1=t(3-40t+45t^2)$, $a_2=t^2(t-1)(9t-1)$. At $\frac{1}{9}$, $b_{2,1}=\frac{8}{81}$, $\gamma=0=\frac{n-3}{2}$, $R_0=\frac{8}{81}x^2$, $R_1=-\frac{1}{9}(15x^2+23x+10)$, $R_2=(2x+3)(3x+5)$, $R_3=9(x+2)^2$: the exponents are 0 and $\gamma=0$, and the line is the holomorphic germ, $u_0=g(0)$ and $R_0(N)u_N=-\sum_{m=1}^{3}R_m(N-m)u_{N-m}$ (Theorem 40.6(iv)), so $u_r/u_0=1,\frac{45}{4},\frac{3645}{32},\frac{70713}{64},\frac{21500397}{2048}$ for $r\le 4$. The Beta sums at $K=200,201$ with 500 terms and the law run back give, as a rendering,

$$\ell=2.34390723868945889060156228887227706909540300963585626066630\dots;$$

the reading from $K=300$ agrees to 10^{-129}, and $3P(1)-9P(2)=6.000\dots$ returns V_1.

(ii) *The two conditions.* At $K=N+1$, $a_K=-r(K)/c(K)<\ell<b_K=\big(r(K-1)-9r(K)\big)/\big(9c(K)-c(K-1)\big)$: at $K=2$, $2<\ell<3$; at $K=3$, with $c(3)=\frac{5}{27}$ and $r(3)=-\frac{23}{54}$, $\frac{23}{10}<\ell<\frac{19}{8}$.

All conditions. By Proposition 41.4(ii) the conditions of degree at most N at the strength 1 are the positive semidefiniteness of the Hankel matrices $[P(1+i+i')]$, $[P(2+i+i')]$, $[\frac{1}{9}P(1+i+i')-P(2+i+i')]$ and $[\frac{1}{9}P(2+i+i')-P(3+i+i')]$ of the sizes allowed by N. Each is affine in ℓ; the set where one is positive semidefinite is an interval, its least eigenvalue being concave in ℓ, whose ends are the roots of its determinant nearest to a point where it is positive definite, such as the gauge-end ℓ. The determinants are computed exactly as polynomials in ℓ and their real roots isolated in ball arithmetic. At $N=1$ the conditions are $\frac{1}{3}\ell-\frac{2}{3}\ge 0$ and $\frac{2}{3}-\frac{2}{9}\ell\ge 0$: $2\le\ell\le 3$. At $N=2$ the binding ones are

$$\det\begin{bmatrix}P(1)&P(2)\\P(2)&P(3)\end{bmatrix}=\frac{4\ell^2+\ell-24}{54}\ge 0,\qquad \frac{1}{9}P(2)-P(3)=\frac{19}{54}-\frac{4}{27}\ell\ge 0,$$

so $\frac{\sqrt{385}-1}{8} \le \ell \le \frac{19}{8}$, of width $\frac{20-\sqrt{385}}{8} = 0.0473\ldots$. The widths:

N	two conditions	all conditions
1	1	1
2	$7.50 \cdot 10^{-2}$	$4.73 \cdot 10^{-2}$
4	$7.84 \cdot 10^{-4}$	$6.90 \cdot 10^{-5}$
8	$1.10 \cdot 10^{-7}$	$9.52 \cdot 10^{-11}$
12	$1.64 \cdot 10^{-11}$	$1.04 \cdot 10^{-16}$
16	$2.46 \cdot 10^{-15}$	$1.03 \cdot 10^{-22}$
22	$4.58 \cdot 10^{-21}$	$9.10 \cdot 10^{-32}$

and the ratio of the widths of all conditions at N and $N-1$ is 0.0382, 0.0335, 0.0320, 0.0314, 0.0308 at $N = 4, 8, 12, 16, 22$. Every bracket contains the gauge-end ℓ.

(iii) *The rate.* The free direction is $u = c$. By Proposition 40.12(i), $\mathsf{C}_j(1) = \int t^{-j} \nu_f^{\mathbb{T}}(dt)$, the torus law (Definition 40.11) being carried by $[\frac{1}{9}, \infty)$; with $\tau = 1/(9t)$, $c(K) = \int_0^1 \tau^K \nu(\tau)\, d\tau$, $\nu(\tau)$ being τ^{-1} times the density of the image of $\nu_f^{\mathbb{T}}$, and by Proposition 40.12(iii) $\nu(\tau) \to \frac{1}{9} \cdot \frac{g(0)}{(2\pi)^2} = \frac{3\sqrt{3}}{4\pi} \neq 0$ as $\tau \uparrow 1$: the exponent $a = 0$ of Theorem 41.5 at $\lambda = 1$. With $w = 2\lambda/f^\star - 1 = 17$ the bracket of degree N has length at most $C\, N\, e^{-N g(1)}$,

$$e^{-g(1)} = \frac{1}{17 + \sqrt{288}} = 17 - 12\sqrt{2} = 0.02943\ldots,$$

against the factor $\frac{1}{9}$ per step of the two conditions (Theorem 41.3(iv)): $-\log_{10}(17 - 12\sqrt{2}) = 1.531$ digits of ℓ per value against $\log_{10} 9 = 0.954$. The ratios of the table approach $17 - 12\sqrt{2}$ from above, as the factor N allows; with the nine values $P(1), \ldots, P(9)$ all conditions give ℓ to 10^{-10}, the two conditions to 10^{-7}. □

52.11 What the closings see

Exercise 52.20 (What the closings see at small centres)**.** Take the following lists, with unit weights, on the marks $[n] = \{1, \ldots, n\}$: the lists $\binom{[n]}{1}$, $n = 3, 4, 5$, whose bases are the single marks; the list H_4 on [4], whose bases are the pairs other than $\{1, 2\}$; the wheel W_3 of Proposition 39.49, whose marks are the six edges $e_1 = 01$, $e_2 = 02$, $e_3 = 03$, $e_4 = 12$, $e_5 = 13$, $e_6 = 23$ of the complete graph on the vertices $0, 1, 2, 3$ and whose bases are its sixteen spanning trees; the list N_6 on [6], whose bases are the nineteen triples other than $\{1, 2, 3\}$; and the uniform lists $\binom{[5]}{2}$ and $\binom{[6]}{3}$. Read each on its centre ray at the strength 1 (Definition 38.13), with origin 0, direction $(\mathbf{1}; 1, 1)$ and stride $f = \alpha^{\mathbf{1}}(K\Psi)^{-1}$, and read H_4 also on the ray with origin 0 and direction $(1, 1, 3, 3; 3, 1)$. Let G be generated by the automorphisms that keep the masses and, at a centre, the Cremona, with the exchanges where they are named. For a box accepted at $\mathfrak{s}$ (Definition 39.16), with the spanning set $\mathbf{b} = (b_1, \ldots, b_{\mathfrak{m}})$, $b_1 = 1$, and the matrix $\mathbf{B}$, write $\mathbf{B} = P_0/Q_0$ with Q_0 the monic least common denominator of its entries, $D_0 = \max(\deg P_0, \deg Q_0)$, and let $\mathcal{S}_D$ be the space of the pairs (P, Q) of a matrix and a scalar of polynomials of degree at most D with $Q(-j)\, \mathsf{C}_{j+1}(\mathbf{b}) = P(-j)\, \mathsf{C}_j(\mathbf{b})$ for all large j, the closing equations of Proposition 39.45.

(i) At the centres of $\binom{[3]}{1}$, $\binom{[4]}{1}$, $\binom{[5]}{1}$, H_4, W_3, with and without its exchanges, and N_6 with its exchanges, and on the ray $(1, 1, 3, 3; 3, 1)$ of H_4, show that $\dim \mathcal{S}_D = 0, 1, 2, 3$ at $D = D_0 - 1, \ldots, D_0 + 2$, the line at D_0 being spanned by (P_0, Q_0), and find a degree up to which no element of the span of $\mathbf{b}$ with polynomial coordinates is annihilated by the closings. Work the centre of $\binom{[3]}{1}$ in full.

(ii) At the centre of N_6, with G generated by the automorphisms and the Cremona but not the exchanges, show that the closings miss a nonzero submodule of the span of the spanning set.

(iii) At the centre of $\binom{[5]}{2}$, with every symmetry of the four kinds in G: find the native gauges; show that the spanning set of the box $\mathfrak{B}_3(1,1)$ at $\mathfrak{s}=2$ is a set of masters of its span $\mathfrak{N}$; show that $\mathfrak{N}$ is the direct sum of the cyclic submodule of [1] and a line L on which the closings, the values and $A_\star$ vanish, reading $\binom{[5]}{2}$ as the contraction of $\binom{[6]}{3}$ (Proposition 39.46); find which solutions at the native gauges see L; and read the centre of $\binom{[6]}{3}$ by its box.

Solution. Every ray here has origin 0, positive letter exponents and positive walls, so Definition 39.26, Theorem 39.29, Proposition 39.30 and Theorem 39.41 apply. Three facts are used throughout.

(a) If an element $m=\sum_l p_l(k)[b_l]$ with polynomial coefficients has values that vanish identically, then $\sum_l p_l(-j)\,\mathsf{C}_j(b_l)=0$ for every $j\ge 0$, not only for large j: the b_l are Laurent insertions, so their closings are defined from $j=0$ on, and Proposition 39.30 gives $V_1\sum_l p_l(-j)\mathsf{C}_j(b_l)=\lim_{k\to -j}(k+j)^{n-1}\langle m\rangle_k=0$. This holds for the elements of $\mathcal{L}$ (Proposition 38.5(ii)) and for the elements on which the closings vanish (Theorem 39.41). The box gives $Q_0(k)\big(f^{-1}b_i-\sum_l \mathbf{B}_{il}(k)\,b_l\big)\in\mathcal{L}$ (Theorem 39.17(i)), and $\mathsf{C}_j(f^{-1}b_i)=\mathsf{C}_{j+1}(b_i)$, so

$$\mathsf{C}_{j+1}(\mathbf{b})=\mathbf{B}(-j)\,\mathsf{C}_j(\mathbf{b})\qquad\text{for every } j\ge 0 \text{ with } Q_0(-j)\neq 0.$$

(b) By (a), $(P,Q)\in\mathcal{S}_D$ exactly when every row of $Q\mathbf{B}-P$, read as an element of the span of $\mathbf{b}$, is annihilated by the closings; Q_0 times such a row is a polynomial row p with $p(-j)\cdot\mathsf{C}_j(\mathbf{b})=0$ for every $j\ge 0$, so every pair of $\mathcal{S}_D$ satisfies its equation at every $j\ge 0$ with $Q_0(-j)\neq 0$. The equations at finitely many such j cut out a space that contains $\mathcal{S}_D$, and so does that system reduced modulo a prime, whose rank is at most its rank over $\mathbb{Q}$. The pairs (cP_0,cQ_0), $\deg c\le D-D_0$, lie in $\mathcal{S}_D$. So when a computed dimension is $\max(0,D-D_0+1)$, it is $\dim\mathcal{S}_D$.

(c) Likewise the polynomial rows of degree at most e that the closings annihilate at finitely many j contain the coordinate rows, of degree at most e, of the elements of the span of $\mathbf{b}$ on which the closings vanish.

(i) The centre of $\binom{[3]}{1}$. The letters are $K=\alpha_1+\alpha_2+\alpha_3$ and $\Psi=\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3$, and

$$f^{-1}=\frac{K\Psi}{\alpha_1\alpha_2\alpha_3}=3+\sum_{i\neq j}\frac{\alpha_i}{\alpha_j}.$$

For a nonempty proper U, $d(U)=0$ and $d^*(U)=|U|-1$, so every wall of the direction is $\kappa_{1,U}=1$ and the wall degree of α^u is $\max_U(-u_U)$ (Definition 39.11(ii)). G is generated by the six bijections of [3] and the Cremona; it holds every symmetry of the four kinds, the list having no exchange (its bases have one mark, its cobases two). The box $\mathfrak{B}_2(1,1)$ has 110 symbols; modulo G its relations, taken at one symbol of each orbit, are 26 combinations of 16 orbit averages, of rank 13 over $\mathbb{Q}(k)$ (receipts `boxexact.py`, `boxes_all.py`, exact elimination over $\mathbb{Z}[k]$). The free orbits are those of 1, of α_3/α_1, of wall degree 1, and of $(\mathbf{1};1,1)$, at a positive level; so at $\mathfrak{s}=1$ the spanning set is $b_1=1$, $b_2=e_G(\alpha_3/\alpha_1)=\frac{1}{6}\sum_{i\neq j}\alpha_i/\alpha_j$, both targets reduce to it, and

$$\mathbf{B}(k)=\begin{bmatrix}3 & 6\\ \dfrac{2(k-1)}{k-2} & \dfrac{(k-1)(7k-11)}{(k-2)^2}\end{bmatrix},$$

with $Q_0=(k-2)^2$ and

$$P_0=\begin{bmatrix}3(k-2)^2 & 6(k-2)^2\\ 2(k-1)(k-2) & (k-1)(7k-11)\end{bmatrix},$$

so $D_0 = 2$; the first row is the Laurent identity $f^{-1}b_1 = 3b_1 + 6b_2$.

The native gauge of the direction, $\mathbf{1}$, has the native value $\frac{1}{9}$, and the three signs with one scale -1 have $K(\eta) = 1$, so they are native gauges of the native value 1 (Theorem 38.17(ii)); on a chart $\det(-H) = \frac{4}{27}$ at $\mathbf{1}$ and -4 at the signs (receipt `signs_span.py`), so all four are nondegenerate. Their solutions, summed over the two orbits, have distinct types and do not vanish at [1], so the span $\mathfrak{N}$ of b_1, b_2 has dimension 2 (Lemma 39.3): $\mathbf{b}$ is a set of masters and $\mathbf{B} = \mathbf{B_b}$. The rows $w_0 = (1, 0)$ and $w_1 = (3, 6)$ are independent, so $\mathfrak{N}$ is the cyclic submodule of [1] (Proposition 39.6(ii)).

The closings are the constant terms $\mathsf{C}_j(b) = \mathrm{CT}(b\, f^{-j})$; $\mathsf{C}_j(1)$ is the coefficient of $\alpha_1^j\alpha_2^j\alpha_3^j$ in $(K\Psi)^j$. By direct expansion,

j	0	1	2	3	4	5	6	7	8
$\mathsf{C}_j(b_1)$	1	3	15	93	639	4653	35169	272835	2157759
$\mathsf{C}_j(b_2)$	0	1	8	60	456	3535	27888	223209	1807760

for instance $\mathsf{C}_1(b_2) = \mathrm{CT}\left(b_2(3 + 6b_2)\right) = 6 \cdot \frac{6}{36} = 1$, the six products $\frac{\alpha_i}{\alpha_j}\frac{\alpha_j}{\alpha_i}$ in b_2^2 being the only constant ones. Q_0 has no zero at a nonpositive integer, so (a) gives $\mathsf{C}_{j+1} = \mathbf{B}(-j)\,\mathsf{C}_j$ for every $j \geq 0$: at $j = 1$ and $j = 2$,

$$\mathbf{B}(-1) = \begin{bmatrix} 3 & 6 \\ \frac{4}{3} & 4 \end{bmatrix}, \quad \mathbf{B}(-1)\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 15 \\ 8 \end{bmatrix}; \qquad \mathbf{B}(-2) = \begin{bmatrix} 3 & 6 \\ \frac{3}{2} & \frac{75}{16} \end{bmatrix}, \quad \mathbf{B}(-2)\begin{bmatrix} 15 \\ 8 \end{bmatrix} = \begin{bmatrix} 93 \\ 60 \end{bmatrix},$$

and the direct expansion agrees for every $j \leq 79$ (receipt `three_exact.py`).

The closing equations at the degree D have $5(D + 1)$ unknowns, the coefficients of the four entries of P and of Q, and two equations for each j. With $1 \leq j \leq 79$, 158 equations, exact elimination over $\mathbb{Q}$ gives the solution dimensions $0, 1, 2, 3$ at $D = 1, 2, 3, 4$; by (b) these are $\dim \mathcal{S}_D$, and the line at $D = 2$ is spanned by (P_0, Q_0). So the closings alone single out $\mathbf{B_b}$ among the matrices of degree at most 2, with no source supplied. The rows (p_1, p_2) of degree at most 24 with $p_1(-j)\mathsf{C}_j(b_1) + p_2(-j)\mathsf{C}_j(b_2) = 0$ for $1 \leq j \leq 80$, 50 unknowns and 80 equations, are only 0; by (c), no nonzero element of $\mathfrak{N}$ with polynomial coordinates of degree at most 24 is annihilated by the closings.

(i) The other rays. The same computation runs on the other boxes (receipt `blind2.py`, driven by `run_blind2.sh`): the closings at $j \leq 7$ are read on the torus as constant terms and continued by (a), and the equations are reduced modulo the prime $2^{61} - 1$; by (b) and (c) the dimensions found are upper bounds, attained here. The matrices $\mathbf{B}$ are those of the boxes built from Definition 39.11 and eliminated exactly over $\mathbb{Q}(k)$ (receipt `boxes_all.py`); at N_6 they are the rayreader's, checked against an elimination of the whole box at a random prime and strength. The spanning sets, with exponents in the order of the marks, are: $1, e_G(\alpha_4/\alpha_1)$ for $\binom{[4]}{1}$; $1, e_G(\alpha_5/\alpha_1), e_G(\alpha_5^2/\alpha_1^2)$ for $\binom{[5]}{1}$; $1, e_G\alpha^u$ with $u = (-1,0,0,1), (-1,1,-1,1), (-1,1,0,0)$ for H_4 at its centre, and $u = (0,1,-1,0), (0,2,-2,0), (0,2,-1,-1)$ on its ray $(1,1,3,3;3,1)$; $1, e_G\alpha^u$ with $u = (-1,-1,0,0,1,1), (-1,-1,0,1,0,1), (-1,-1,1,1,-1,1), (-1,-1,1,1,0,0)$ for W_3, with or without the exchanges; and for N_6 with its exchanges $1, e_G\alpha^u$ with the twelve $u = (-1,-1,0,0,1,1)$, $(-1,-1,1,-1,1,1)$, $(-1,-1,1,0,0,1)$, $(-1,0,0,0,0,1)$, $(-1,0,1,-1,0,1)$, $(-1,0,1,0,0,0)$, $(-1,-1,-1,1,1,1)$, $(-2,-2,0,0,2,2)$, $(-2,-2,0,1,1,2)$, $(-2,-2,1,-1,2,2)$, $(-2,-2,1,0,1,2)$, $(-2,-2,1,1,1,1)$. In the table, $\dim \mathcal{S}_D$ is listed at $D = D_0 - 1, \ldots, D_0 + 2$, h is the first index at which the rows $w_0, \ldots, w_h$ of [1] are dependent (Proposition 39.6(ii)), and the last but one column gives the degree up to which no nonzero row is annihilated, at the j from 10 to the bound shown.

ray	$\lvert G\rvert$	box, $\mathfrak{s}$	$\mathfrak{m}$	D_0	$\dim \mathcal{S}_D$	rows	h
$\binom{[4]}{1}$	48	$(2;1,1)$, 1	2	3	$0,1,2,3$	24, $j \le 120$	2
$\binom{[5]}{1}$	240	$(3;1,1)$, 2	3	6	$0,1,2,3$	24, $j \le 120$	3
H_4	8	$(2;1,1)$, 1	4	3	$0,1,2,3$	24, $j \le 150$	4
H_4, $(1,1,3,3;3,1)$	4	$(2;1,1)$, 1	4	10	$0,1,2,3$	24, $j \le 150$	4
W_3	48, 96	$(2;1,1)$, 1	5	9	$0,1,2,3$	40, $j \le 260$	4
N_6, exchanges	144	$(3;1,1)$, 2	13	16	$0,1,2,3$	32, $j \le 480$	11

The denominators are $Q_0 = (k-2)^3$, $(k-3)^4$, $(k-2)^3$, $k(k-1)^3(k-\frac{4}{3})^3(k-\frac{5}{3})^3$, $(k-2)^5(k-\frac{3}{2})(k^3-3k^2+\frac{3}{2}k+2)$, and for N_6 a polynomial of degree 16 with the factors $(k-4)(k-3)^5(k-2)^3(k-\frac{5}{2})$ and an irreducible sextic; only that of the ray $(1,1,3,3;3,1)$ vanishes at a nonpositive integer, at $k=0$, below the j used. At $\binom{[4]}{1}$ the native gauges $\mathbf{1}$ and the four signs with one scale -1, of the native values $\frac{1}{16}$ and $\frac{1}{4}$, and at $\binom{[5]}{1}$ the native gauges $\mathbf{1}$ and the signs with one and with two scales -1, of the native values $\frac{1}{25}$, $\frac{1}{9}$ and 1, are nondegenerate (receipt `signs_span.py`), so by Lemma 39.3 their spanning sets are sets of masters. At H_4, W_3 and N_6 the signs of the largest native value are degenerate native gauges, and the computations are read for the spanning set as it stands; then h bounds the dimension of the cyclic submodule of [1] from above. At each of these rays the closings alone single out the box's matrix among the matrices of degree at most D_0; at W_3 and at N_6 the rows of [1] become dependent before the size of the spanning set is reached, and the closings single out the whole matrix on the spanning set, not only the law that the rows of [1] give.

(ii) N_6 *without the exchanges.* Now $\lvert G\rvert = 72$, and the box $\mathfrak{B}_3(1,1)$ is accepted at $\mathfrak{s}=2$ with 15 elements: the thirteen above, here averaged over the smaller group, and $e_G\alpha^u$ for $u = (-1,0,0,-1,1,1)$ and $(-2,-2,2,-2,2,2)$. The bijection x with $x(i) = i+3$ and $x(i+3) = i$, $i \le 3$, carries the bases, the triples other than $\{1,2,3\}$, onto the complements of the bases, the triples other than $\{4,5,6\}$: it exchanges the list, and at the centre it is a symmetry of the ray (Proposition 39.7(ii)). With G_x generated by G and x, G is normal of index 2 in G_x, and for $b = e_G\alpha^{(-1,-1,1,0,0,1)}$, an element of the spanning set, $x^*b = e_G\alpha^{(-1,0,0,-1,1,1)}$, another (receipt `n_exchange.py`); so $(1-e_{G_x})b = \frac{1}{2}(b - x^*b)$, on which the closings, the values and $A_\star$ vanish (Corollary 39.43, applied to G_x). It is not 0: at the sign η with the scales -1 on $\{4,5\}$, $K(\eta) = -5$, a nondegenerate native gauge of the native value $\frac{1}{25}$ with $\det(-H) = -\frac{9216}{15625}$ on a chart, $b(\eta) = -\frac{1}{3}$ and $x^*b(\eta) = 1$, so the leading coefficient of $A_\eta(b - x^*b)$ is $-\frac{4}{3}$ times that of $A_\eta(1)$ (Theorem 38.6(i)). So the closings miss the nonzero submodule generated by $b - x^*b$, the part of the span that the exchanges negate. The computation of (i) agrees: the rows of degree at most e annihilated at $10 \le j \le 400$ have the dimension $e+1$ for $e \le 8$, so by (c) they are exactly the multiples $p(k)(b - x^*b)$, $\deg p \le e$; and the solution spaces of the closing equations at $D = 17, 18, 19, 20$ have dimensions at most $400, 431, 462, 493$, growing by $31 = 15 \cdot 2 + 1$, the slope of Proposition 39.45 for a missed submodule of dimension two.

(iii) The centre of $\binom{[5]}{2}$. Here $K = e_2$, $\Psi = e_3$, $f^{-1} = e_2e_3/(\alpha_1 \cdots \alpha_5)$, and G is generated by the bijections of [5] and the Cremona, $\lvert G\rvert = 240$: the list has no exchange, its bases having two marks and its cobases three, so this is every symmetry of the four kinds.

The native gauges. On the chart $\alpha_5 = 1$ the defects times e_2e_3 generate an ideal whose saturation by the scales and the letters is zero-dimensional and radical of degree 66 (receipt `u25.sing`, an exact Gröbner basis computation): there are 66 native gauges, all simple zeros of the defects and so nondegenerate. Five points, checked exactly to be native gauges (receipt `u25_core.py`),

and their orbits under G account for all 66; with $\zeta = e^{2\pi i/3}$ and the spanning set $\mathbf{b}$ found below,

point	native value	orbit	$(b_1, \dots, b_5)$ at the point
$\mathbf{1}$	$\frac{1}{100}$	1	$(1,1,1,1,1)$
$(1,1,1,1,-1)$	$\frac{1}{4}$	5	$(1,-\frac{3}{5},\frac{1}{5},1,\frac{1}{5})$
$(1,1,1,-1,-1)$	$\frac{1}{4}$	10	$(1,\frac{1}{5},-\frac{1}{5},1,-\frac{1}{5})$
$(\sqrt{3}-2,\sqrt{3}-2,\sqrt{3}-2,1,1)$	$-\frac{1}{8}$	20	$(1,1,-\frac{4}{5},\frac{113}{5},\frac{23}{5})$
$(1,1,\zeta,\zeta,\zeta^2)$	1	30	$(1,\frac{1}{10},-\frac{1}{5},\frac{1}{10},-\frac{1}{5})$

On the chart $\det(-H)$ is $\frac{81}{3125}$, -3, -3, $-\frac{81}{16}$ and 81 at these points. The orbits are the table's gauge, the signs with one and with two scales -1, the points with three scales $\sqrt{3}-2$ and two scales 1 or two and three, which the Cremona joins, and the points with two scales 1, two ζ and one ζ^2.

The box. The box $\mathfrak{B}_3(1,1)$ has 10322 symbols; modulo G its relations at the orbit representatives are 196 combinations of 114 orbit averages, of rank 103, eliminated exactly over $\mathbb{Q}(k)$. The free Laurent orbits of wall degree at most 2 give the spanning set

$$b_1 = 1, \qquad b_2 = e_G\alpha^{(-1,-1,0,1,1)}, \qquad b_3 = e_G\frac{\alpha_5}{\alpha_1} = \frac{1}{20}\sum_{i\neq j}\frac{\alpha_i}{\alpha_j},$$
$$b_4 = e_G\alpha^{(-2,-2,0,2,2)}, \qquad b_5 = e_G\alpha^{(-2,-2,1,1,2)},$$

of wall degrees $0,1,1,2,2$, and the box is accepted at $\mathfrak{s} = 2$. With $q(k) = 3k^3+4k^2-61k+130$, its matrix has the rows $\mathbf{B}_{1\bullet} = (10,30,60,0,0)$, the Laurent identity $f^{-1} = 10b_1+30b_2+60b_3$, and

$$\tfrac{q}{k-1}\mathbf{B}_{2\bullet} = \Big(27k^2+111k-130,\ \tfrac{2(51k^3+137k^2-756k+580)}{k-2},\ \tfrac{16(9k^3+33k^2-155k+108)}{k-2},$$
$$-3(3k^2-9k-10),\ 12(3k^2-k+2)\Big),$$
$$\tfrac{q}{k-1}\mathbf{B}_{3\bullet} = \Big(6(5k^2+8k-65),\ \tfrac{6(15k^3+6k^2-313k+440)}{k-2},\ \tfrac{4(33k^3+74k^2-971k+1340)}{k-2},$$
$$-12(k-5)(k-2),\ 12(5k^2-14k+5)\Big),$$

$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{41} = k(27k^5+245k^4-1789k^3+3709k^2-5388k+5820),$$
$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{42} = \tfrac{2(45k^7+225k^6-2310k^5+3606k^4+1139k^3-157k^2-7152k+2460)}{k-2},$$
$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{43} = \tfrac{4(216k^7-552k^6-925k^5-1703k^4+18473k^3-28953k^2+33288k-33444)}{3(k-2)},$$
$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{44} = 3k^6-283k^5+3037k^4-13753k^3+33126k^2-43566k+25020,$$
$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{45} = -4(k-2)(27k^5-373k^4+1321k^3-1507k^2-450k+1878),$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{51} = 2k(15k^4+52k^3-232k^2+153k-420),$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{52} = \tfrac{2(45k^6+111k^5-963k^4+461k^3+936k^2+478k-540)}{k-2},$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{53} = \tfrac{4(117k^6+225k^5-2281k^4+1565k^3+1564k^2-1562k+5028)}{3(k-2)},$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{54} = -2(6k^5-17k^4-185k^3+1052k^2-2368k+2280),$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{55} = 4(9k^5+32k^4-250k^3+412k^2-65k-522),$$

so $Q_0 = (k-3)^4(k-2)\,q(k)/3$ and $D_0 = 8$; it equals the rayreader's matrix.

Masters. The five orbit sums of the solutions A_c (Theorem 39.8(iii)) have the types $\frac{1}{100}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{8}, 1$. Those alone in their type do not vanish at [1]. On b_1, b_2 the two of the type $\frac{1}{4}$ have the leading coefficients $5C(1, -\frac{3}{5})$ and $10C'(1, \frac{1}{5})$, $C, C' \neq 0$, and $\det \begin{bmatrix} 1 & -\frac{3}{5} \\ 1 & \frac{1}{5} \end{bmatrix} = \frac{4}{5} \neq 0$, so they are linearly independent on $\mathfrak{N}$. By Lemma 39.3, $\dim \mathfrak{N} \geq 5$: the spanning set is a set of masters of $\mathfrak{N}$, of dimension exactly 5, and $\mathbf{B} = \mathbf{B_b}$. The five rows of values in the table have the determinant $\frac{3888}{625}$, so $\mathbf{B}(\infty)$ is diagonalizable with the eigenvalues $100, 4, 4, -8, 1$ (Proposition 39.38(i), (ii)); indeed

$$\mathbf{B}(\infty) = \begin{bmatrix} 10 & 30 & 60 & 0 & 0 \\ 9 & 34 & 48 & -3 & 12 \\ 10 & 30 & 44 & -4 & 20 \\ 9 & 30 & 96 & 1 & -36 \\ 10 & 30 & 52 & -4 & 12 \end{bmatrix}, \qquad \det(\lambda - \mathbf{B}(\infty)) = (\lambda - 100)(\lambda - 4)^2(\lambda + 8)(\lambda - 1)$$

(receipt `binf_exact.py`).

The law and the line. The rows $w_i(k) = w_{i-1}(k-1)\mathbf{B}(k)$, $w_0 = (1, 0, 0, 0, 0)$, are independent for $i \leq 3$ and dependent at $i = 4$, so the cyclic submodule $\mathfrak{N}_1$ of [1] has dimension 4 and the law of [1] (Proposition 39.6(ii)) is $\sum_{j=0}^{4} p_j(k)\,\mathrm{Sh}^j[1] = 0$ with

$$\begin{aligned} p_0 &= k^4(33k^2 + 88k + 60), \\ p_1 &= -(3201k^6 + 14938k^5 + 27699k^4 + 26214k^3 + 13598k^2 + 3736k + 440), \\ p_2 &= -4(2772k^6 + 18480k^5 + 48699k^4 + 63150k^3 + 40515k^2 + 11296k + 1140), \\ p_3 &= 16(k+2)^2(7491k^4 + 34958k^3 + 51724k^2 + 24507k + 4430), \\ p_4 &= -3200(k+2)^2(k+3)^2(33k^2 + 22k + 5). \end{aligned}$$

Put

$$\begin{aligned} m = {} & k(k-5)(k-2)(k+2)\,[b_1] - (3k^4 + 11k^3 - 71k^2 + 227k - 210)\,[b_2] \\ & - \tfrac{2}{3}(3k^4 - 11k^3 + 119k^2 - 125k - 90)\,[b_3] - (k-5)(k-2)^3\,[b_4] + (k-2)^2(5k^2 - 14k + 5)\,[b_5] \end{aligned}$$

and $\tilde{m} = \frac{k-1}{q(k)}\,m$. With $\tilde{w}$ the coordinate row of $\tilde{m}$, the identity $\tilde{w}(k-1)\mathbf{B}(k) = 4\,\tilde{w}(k)$ holds exactly, that is $\mathrm{Sh}^{-1}\,\tilde{m} = 4\tilde{m}$, so $L = \mathbb{C}(k)\tilde{m}$ is a submodule; and at $k = \frac{37}{3}$ the rows $w_0, \ldots, w_3, \tilde{w}$ have rank 5. So

$$\mathfrak{N} = \mathfrak{N}_1 \oplus L.$$

The table one mark larger. $\binom{[5]}{2}$ is the contraction at the mark 6 of $\binom{[6]}{3}$, which is connected and identically self-dual, so Proposition 39.46 applies, with G' generated by the T_β, β a bijection of [6], and the Cremona, and $G \subseteq G'$. The native gauges lift by $\alpha_6 = \Psi/K$: the five points of the table to $\alpha_6 = 1, -1, 1, 1, \zeta^2$ (receipt `u36_lift.py`). The lifts are native gauges of the centre of $\binom{[6]}{3}$, which has 66 of them, all simple (receipt `u36.sing`), in orbits of S_6 and the Cremona of $1, 15, 20, 30$ points with the native values $\frac{1}{400}, \frac{1}{16}, -\frac{1}{32}, \frac{1}{4}$, a quarter of those below them: the two kinds of signs lift into one orbit of 15, the signs of $\binom{[6]}{3}$ with two scales -1, and this is Proposition 39.48(ii) at $r = 2$, with $C_2 = 2$.

The element ℓ. Put $S = \sum_{i \neq j \leq 5} \alpha_i/\alpha_j = 20\,b_3$ and $T = \frac{K}{\Psi}\sum_i \alpha_i + \frac{\Psi}{K}\sum_i \alpha_i^{-1}$. Since S is invariant under G, the image of the stabilizer of the mark 6, and the transpositions $(e\,6)$, $e \leq 5$,

with the identity, represent its cosets, $e_{G'}[S] = \frac{1}{6}\big([S] + \sum_{e\le 5} T_{(e\,6)}[S]\big)$. By Proposition 39.46(iii), with $\frac{\Gamma(k+1)\Gamma(k-1)}{\Gamma(k)^2} = \frac{k}{k-1}$,

$$T_{(e\,6)}\Big[\frac{\alpha_i}{\alpha_j}\Big] = \Big[\frac{\alpha_i}{\alpha_j}\Big]\ (e \notin \{i,j\}), \qquad \frac{k}{k-1}\Big[\frac{\Psi}{K\alpha_j}\Big]\ (e=i), \qquad \frac{k}{k-1}\Big[\frac{K\alpha_i}{\Psi}\Big]\ (e=j).$$

Each pair (i,j) is left alone by three of the five e, and each α_j^{-1} and each α_i arises four times, so $\sum_e T_{(e\,6)}[S] = 3[S] + \frac{4k}{k-1}[T]$, $e_{G'}[S] = \frac{2}{3}[S] + \frac{2k}{3(k-1)}[T]$, and

$$\ell = 3(1-e_{G'})[S] = [S] - \frac{2k}{k-1}\,[T].$$

The symbols of T, at the levels $(0,1)$ and $(1,0)$ and of wall degrees 0 and 1, lie in the box, which reduces T to the spanning set (receipt `u25_core.py`):

$$[T] = \frac{10(k-1)}{k}\,[b_3] - \frac{10}{k\,q(k)}\,m.$$

So

$$\ell = 20[b_3] - \frac{2k}{k-1}\Big(\frac{10(k-1)}{k}[b_3] - \frac{10}{k\,q(k)}m\Big) = \frac{20}{(k-1)\,q(k)}\,m = \frac{20}{(k-1)^2}\,\tilde{m}.$$

Since $e_{G'}(1-e_{G'}) = 0$, $e_{G'}\ell = 0$, hence $e_{G'}\tilde{m} = 0$ and $L \subseteq (1-e_{G'})\mathfrak{M}^G$: the closings, the values and $A_\star$ vanish on L (Proposition 39.46(iv)). The values can be seen to vanish directly as well: by the formula for π_* in Proposition 39.46, $\pi_*(\alpha_i/\alpha_6) = \frac{k}{k-1}\frac{K}{\Psi}\alpha_i$ and $\pi_*(\alpha_6/\alpha_i) = \frac{k}{k-1}\frac{\Psi}{K}\alpha_i^{-1}$, so $\ell = \pi_*\ell'$ with $\ell' = \sum_{i\ne j\le 5}\frac{\alpha_i}{\alpha_j} - 2\sum_{i\le 5}\big(\frac{\alpha_i}{\alpha_6} + \frac{\alpha_6}{\alpha_i}\big)$, three times the part of $\sum_{i\ne j\le 5}\alpha_i/\alpha_j$ that the average over the bijections of [6] removes; the values of $\binom{[6]}{3}$ are kept by those bijections (Proposition 39.7(ii)), so $\langle \ell'\rangle'_k = 0$ and $\langle \ell\rangle_k = \Gamma^{\,2}_{\,1}\Big[\begin{matrix}k,\ k\\ 2k\end{matrix}\Big]^{-1}\langle \ell'\rangle'_k = 0$ (Proposition 39.46(i)). As a numerical check, a trapezoidal quadrature over the chart gives $|\langle \ell\rangle_k|/\langle 1\rangle_k$ equal to $1.0\cdot 10^{-13}$ at $k = \frac{13}{2}$ and $2.8\cdot 10^{-13}$ at $k = 8$, the accuracy of the quadrature (receipt `ell_check.py`). So L is a relation among the values of the five masters that the relations of the ray do not produce, m being nonzero in $\mathfrak{N}$, nor its symmetries of the four kinds, m lying in $\mathfrak{M}^G$.

What sees L. A solution V of the type $(\lambda, \rho+\mathbb{Z})$ has $\varsigma^{-1}V(\tilde{m}) = V(\mathrm{Sh}^{-1}\tilde{m}) = 4V(\tilde{m})$, which forces $V(\tilde{m}) = 0$ unless $\lambda^{-1} = 4$: the solutions at the orbits of the native values $\frac{1}{100}$, $-\frac{1}{8}$ and 1 vanish on L. At a sign, $A_c(b_l)/A_c(1) = b_l(\alpha_c) + O(1/k)$ (Theorem 38.6(i)), and the coefficients of k^4 in m are $(1,-3,-2,-1,5)$, so $A_c(m)/A_c(1)$ begins with

$$\big(1+\tfrac{9}{5}-\tfrac{2}{5}-1+1\big)k^4 = \tfrac{12}{5}k^4 \quad (\text{one scale } -1), \qquad \big(1-\tfrac{3}{5}+\tfrac{2}{5}-1-1\big)k^4 = -\tfrac{6}{5}k^4 \quad (\text{two scales } -1):$$

the solutions at both kinds of signs see L. With the solutions at the fifteen signs transported from one of them by G' as in Proposition 39.48(i), $A_c(1)$ is the same for all fifteen, $T_\gamma[1] = [1]$; their sum, invariant under G', has on m the leading coefficient $5\cdot\frac{12}{5} - 10\cdot\frac{6}{5} = 0$ times that of $A_c(1)$, as it must, while the mean over the five less the mean over the ten has $\frac{12}{5} + \frac{6}{5} = \frac{18}{5}$ and sees L.

The closings. The rows of degree at most e annihilated by the closings at $10 \le j \le 150$ have the dimension $\max(0, e-3)$ for $e \le 24$, which is the dimension of the multiples $p(k)\,w_m$, $\deg p \le e-4$, of the primitive coordinate row w_m of $3m$, of degree 4; by (c) the closings annihilate L and no element of $\mathfrak{N}$ outside L with polynomial coordinates of degree at most 24. The solution spaces of the closing equations at $D = 7, 8, 9, 10$ have dimensions at most $24, 30, 36, 42$, growing

by $6 = 5 \cdot 1 + 1$, the slope of Proposition 39.45 for a missed line; a step of more than one already shows that something is missed.

The centre of $\binom{[6]}{3}$ *by its box.* Modulo S_6 and the Cremona, of order 1440, the box $\mathfrak{B}_3(1,1)$, with 78296 symbols, has 406 relations at orbit representatives on 226 orbit averages, of rank 186; it is accepted at $\mathfrak{s} = 2$ with the spanning set 1, $e\,\alpha^{(-1,-1,-1,1,1,1)}$, $e\,\alpha^{(-1,-1,0,0,1,1)}$, $e\,\alpha^{(-2,-2,-2,2,2,2)}$, e the average over that group, and $Q_0 = (k-3)^5(k-2)^2(k+1)(k^3 - 8k^2 + \frac{71}{3}k - \frac{76}{3})$, $D_0 = 11$. The four orbits of native gauges give solutions of four distinct types, none vanishing at [1], so the spanning set is a set of masters of a submodule $\mathfrak{N}'$ of dimension 4 (Lemma 39.3), and $\mathbf{B}'(\infty)$ has the simple spectrum $400, 16, 4, -32$. Its rows of [1] are independent up to $h = 4$, so $\mathfrak{N}'$ is the cyclic submodule of [1]. Since $\pi_*[1] = [1]$ and $\pi_* \operatorname{Sh}' = \frac{k}{2(2k+1)} \operatorname{Sh} \pi_*$, $\pi_* \mathfrak{N}' = \mathfrak{N}_1$, and the law of [1] at the centre of $\binom{[6]}{3}$ is the law of $\binom{[5]}{2}$ for $\Gamma_{\!2\,1}\begin{bmatrix} k,\ k \\ 2k \end{bmatrix} \langle 1 \rangle_k$: its coefficients are proportional to $p_i(k) \prod_{l=i}^{3} \frac{k+l}{2(2k+2l+1)}$, which is the law that the box gives, exactly (receipt `u36_lift.py`). So

$$\mathfrak{N} \cap \mathfrak{M}^{G'} = \mathfrak{N}_1 = \pi_* \mathfrak{N}', \qquad \mathfrak{N} \cap (1 - e_{G'}) \mathfrak{M}^G = L,$$

$\mathfrak{N}_1$ lying in $\mathfrak{M}^{G'}$, which contains [1], and L in $(1 - e_{G'})\mathfrak{M}^G$. At the centre of $\binom{[6]}{3}$ the closing equations have the dimensions $0, 1, 2, 3$ at $D = 10, \ldots, 13$, and no nonzero row of degree at most 40 is annihilated at $10 \le j \le 200$: the closings there single out $\mathbf{B}'$.

So: at the small centres of (i) the closings alone single out the box's matrix at the degree D_0, with nothing of low degree missed; at the centre of N_6 without its exchanges they miss the part that the exchanges negate; and at the centre of $\binom{[5]}{2}$, with every symmetry of the four kinds, the span of the box is the cyclic submodule of [1], which the box of $\binom{[6]}{3}$ reads, plus the line L that the symmetric group of $\binom{[6]}{3}$ hides from the closings, the values and $A_\star$, and that the signs see. $\square$

Chapter 53

Exercises on rays: other lists

These exercises read rays of lists other than the lists $\binom{\mathcal{E}}{1}$: the list H_4 on four marks at its centre and on a ray strictly inside its walls whose native value has a conjugate on the slit, and the five-mark lists along their hinge trees. Each exercise defines its list by its marks and its bases; every weight is 1. A recurrence of the closings is proved for every index by an explicit telescoping identity, and Theorems 39.29 and 39.41, with Corollary 39.44, carry it to the values. A *certificate* of a recurrence is an explicit element of the relation space $\mathcal{L}$ (Definition 38.4) that equals the recurrence applied to an insertion, as in Proposition 38.5(i); a telescoping identity among binomial sums is not one. Numbers are exact; a decimal is a rendering of an exact value, truncated. The receipts are the scripts `ex28_h4_centre.py`, `ex29_h4_lean.py` and `ex30_trees.py`, with the shared tools in `common.py`; each prints the numbers its solution displays.

53.1 The list H_4

In this section H_4 is the list on the marks $a, b, 2, 3$ whose bases are the five pairs other than $\{a, b\}$: $\{a, 2\}, \{a, 3\}, \{b, 2\}, \{b, 3\}, \{2, 3\}$. With $A = \alpha_a + \alpha_b$ and $B = \alpha_2 + \alpha_3$ its letters are

$$K = AB + \alpha_2\alpha_3, \qquad \Psi = AB + \alpha_a\alpha_b,$$

the second from the cobases $\{b, 3\}, \{b, 2\}, \{a, 3\}, \{a, 2\}, \{a, b\}$, and $\Psi(\alpha) = \alpha_a\alpha_b\alpha_2\alpha_3\, K(\alpha^{-1})$. The pair $\{a, b\}$ is a parallel class.

Exercise 53.1 (H_4 at its centre)**.** Let $P(k) = \langle 1 \rangle_k = P_{\mathrm{H}_4}(k\mathbf{1}; k, k)$ be the values of the centre ray of H_4 at the strength one (Definition 38.13).

(i) Write the stride, the walls of the direction and V_1.

(ii) Find every native gauge of the ray with its native value, and decide which are nondegenerate.

(iii) Find the closings and a law of the values of order four, prove the law, and give its symbol.

(iv) Write $\Gamma(\sigma)^2 P(\sigma)$ on the share side of the hinge tree of H_4, close it at the integer strengths, and decide which of its two closings converges.

(v) Read the low-end relation, the principal parts of P at $k = 0$ and at $k = -1$, and the relation that the law gives at $k = -1$.

(vi) Give $P(1)$ and $P(2)$, and show that every value $P(k)$, $k \geq 1$, lies in $\mathbb{Q} + \mathbb{Q}\,P(1) + \mathbb{Q}\,P(2)$.

Solution. (i) Stride, walls, V_1. By Definition 38.13 the stride and its reciprocal are

$$f=\frac{\alpha_a\alpha_b\alpha_2\alpha_3}{K\Psi}=\frac{1}{K(\alpha)\,K(\alpha^{-1})},\qquad f^{-1}=\frac{(AB+\alpha_2\alpha_3)(AB+\alpha_a\alpha_b)}{\alpha_a\alpha_b\alpha_2\alpha_3},$$

and the walls of the direction are the widths, $\kappa_{1,S}=c(S)=r(S)+r(\mathcal{E}\setminus S)-2$ (Definition 7.12, Proposition 8.21(ii)). A single mark has rank one and its complement rank two; $\{a,b\}$ has rank one and $\{2,3\}$ rank two; the other pairs have rank two with complements of rank two; a triple has rank two and its complement rank one. So $\kappa_{1,S}=2$ for the four pairs $\{a,2\},\{a,3\},\{b,2\},\{b,3\}$, and $\kappa_{1,S}=1$ for every other nonempty proper S: all walls are positive. In $V_1=\sum_\pi\prod_{l=2}^{4}\kappa_{1,U_l}^{-1}$ (Proposition 39.30) the tails U_2 and U_4 have three marks and one mark, of wall 1, so an order contributes κ_{1,U_3}^{-1}, U_3 being its last two marks. Each of the six pairs ends four of the 24 orders, so

$$V_1=4\,(1+1)+4\cdot4\cdot\tfrac{1}{2}=16.$$

(ii) Native gauges. By Proposition 38.16 a point α of X_F is a native gauge exactly when the occupations $R_e=\alpha_e\partial_{\alpha_e}\log K$ agree at α and at α^{-1}, and its native value is $\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$. Here

$$R_a(\alpha)=\frac{\alpha_aB}{K},\quad R_a(\alpha^{-1})=\frac{\alpha_bB}{\Psi},\qquad R_2(\alpha)=\frac{\alpha_2(A+\alpha_3)}{K},\quad R_2(\alpha^{-1})=\frac{\alpha_3A+\alpha_a\alpha_b}{\Psi},$$

and R_b, R_3 follow by exchanging $a\leftrightarrow b$ and $2\leftrightarrow3$. So the native gauges are the solutions in X_F of

$$\begin{aligned}B(\alpha_a\Psi-\alpha_bK)&=0,&\qquad B(\alpha_b\Psi-\alpha_aK)&=0,\\ \alpha_2(A+\alpha_3)\Psi&=(\alpha_3A+\alpha_a\alpha_b)K,&\qquad \alpha_3(A+\alpha_2)\Psi&=(\alpha_2A+\alpha_a\alpha_b)K.\end{aligned}$$

The occupations add up to the rank 2 at α and at α^{-1}, so any three of the four equations imply the fourth.

If $B=0$, then $K=\alpha_2\alpha_3$ and $\Psi=\alpha_a\alpha_b$, and with $\alpha_3=-\alpha_2$ the third equation reads $\alpha_2A(\alpha_a\alpha_b-\alpha_2^2)=0$: either $A=0$ or $\alpha_a\alpha_b=\alpha_2^2$. If $B\neq0$, the first two equations give $(\alpha_a-\alpha_b)(\Psi+K)=0$ and $A(\Psi-K)=0$. For $A=0$ they leave $K=\alpha_2\alpha_3$, $\Psi=\alpha_a\alpha_b=-\alpha_a^2$ and $\Psi+K=0$, that is $\alpha_a^2=\alpha_2\alpha_3$, and the third equation, $\alpha_2\alpha_3\Psi=\alpha_a\alpha_bK$, holds there. For $A\neq0$ they give $\Psi=K$, that is $\alpha_a\alpha_b=\alpha_2\alpha_3$, and then $\alpha_a=\alpha_b$, since $\Psi+K=2K\neq0$; the third equation becomes $A(\alpha_2-\alpha_3)=0$, so $\alpha_2=\alpha_3$ and $\alpha_2^2=\alpha_a^2$. Up to scale the native gauges are therefore:

- the table's gauge $\mathbf{1}$, where $K=\Psi=5$: native value $\frac{1}{25}$;
- the sign $(1,1,-1,-1)$, where $K=\Psi=-3$: native value $\frac{1}{9}$;
- the three curves $\{A=0,\ \alpha_a^2=\alpha_2\alpha_3\}$, $\{B=0,\ \alpha_2^2=\alpha_a\alpha_b\}$ and $\{A=B=0\}$, on which $K=\alpha_2\alpha_3$ and $\Psi=\alpha_a\alpha_b$: native value 1.

No letter vanishes at these points of the torus, so they lie in X_F. On the chart $y_3=0$, in the coordinates y_a,y_b,y_2, the Hessian of $-\log f$ at $\mathbf{1}$ is twice the covariance matrix of the indicators under the uniform weight law on the five bases (Theorem 38.15), and at the sign it is twice the covariance matrix under the signed measure $\mathbb{P}_\eta$ with the masses $\frac13,\frac13,\frac13,\frac13,-\frac13$ on $\{a,2\},\{a,3\},\{b,2\},\{b,3\},\{2,3\}$ (Theorem 38.17(ii)):

$$H_{\mathbf{1}}=\frac{2}{25}\begin{bmatrix}6&-4&-1\\-4&6&-1\\-1&-1&6\end{bmatrix},\quad\det H_{\mathbf{1}}=\frac{32}{625};\qquad H_\eta=\frac{2}{9}\begin{bmatrix}2&-4&1\\-4&2&1\\1&1&2\end{bmatrix},\quad\det H_\eta=-\frac{32}{81}.$$

Both are nonsingular: the table's gauge and the sign are nondegenerate native gauges (Definition 38.4). The three curves are sources of dimension one (Definition 39.19), along which f is constant and the Hessian is singular.

(iii) Closings and the law. The origin is 0 and both letters have the exponent one, so the closings are $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j})$ (Definition 39.26). In

$$(K\Psi)^j = \sum_{u,v} \binom{j}{u}\binom{j}{v} (AB)^{u+v} (\alpha_2\alpha_3)^{j-u} (\alpha_a\alpha_b)^{j-v}$$

the monomial $(\alpha_a\alpha_b\alpha_2\alpha_3)^j$ takes $\alpha_a^v\alpha_b^v$ from A^{u+v} and $\alpha_2^u\alpha_3^u$ from B^{u+v}, so $u = v$ and

$$\mathsf{C}_j(1) = \sum_{u=0}^{j} F(j,u), \qquad F(j,u) = \binom{j}{u}^2 \binom{2u}{u}^2; \qquad \mathsf{C}_j(1) = 1,\ 5,\ 53,\ 761,\ 12661,\ 229705,\ \ldots$$

Exact linear algebra on the first 150 closings gives one line of recurrences $\sum_{i=0}^{4} p_i(j)\,\mathsf{C}_{j+i}(1) = 0$ with $\deg p_i \le 7$, spanned by

$$\begin{aligned}
p_0(j) &= 225(j+1)^2(j+2)^2\,w(j), \qquad w(j) = 144j^3 + 1296j^2 + 3853j + 3790,\\
p_1(j) &= -2(j+2)^2(34848j^5 + 435600j^4 + 2135762j^3 + 5129622j^2 + 6027508j + 2768385),\\
p_2(j) &= 2(21168j^7 + 381024j^6 + 2917047j^5 + 12308044j^4\\
&\qquad + 30893608j^3 + 46094457j^2 + 37814839j + 13140768),\\
p_3(j) &= -2(j+3)(2592j^6 + 42768j^5 + 289602j^4 + 1028216j^3 + 2015044j^2 + 2063287j + 861336),\\
p_4(j) &= (j+3)(j+4)^3(144j^3 + 864j^2 + 1693j + 1089);
\end{aligned}$$

no recurrence of order at most three and degree at most 30 holds on the first 200 closings. The recurrence holds for every $j \in \mathbb{N}$, by a telescoping identity. For $0 \le u \le j+4$ put

$$T(j,u) = \frac{((j+4)!)^2\,((2u)!)^2}{(u!)^6\,((j+4-u)!)^2} > 0, \qquad G(j,u) = \frac{T(j,u)\,A(j,u)}{\prod_{l=1}^{4}(j+l)^2}, \qquad G(j,j+5) = 0,$$

with $A(j,u) = u^4(j+1)^2(j+2)^2 \sum_{q=0}^{4} Q_q(j)\,u^q$ and

$$\begin{aligned}
Q_0 &= (j+4)(6912j^6 + 125568j^5 + 938496j^4 + 3695236j^3 + 8085594j^2 + 9321748j + 4422321),\\
Q_1 &= -4(5760j^6 + 106272j^5 + 809680j^4 + 3260831j^3 + 7321143j^2 + 8687769j + 4256385),\\
Q_2 &= 2(13248j^5 + 198864j^4 + 1186148j^3 + 3514356j^2 + 5173077j + 3027087),\\
Q_3 &= -4(22j + 63)\,w(j), \qquad Q_4 = 15\,w(j).
\end{aligned}$$

For $0 \le u \le j+4$, $F(j+i,u) = T(j,u)\prod_{l=i+1}^{4}\big((j+l-u)/(j+l)\big)^2$, which vanishes for $u > j+i$ as F does, and $T(j,u+1) = T(j,u)\,4(2u+1)^2(j+4-u)^2/(u+1)^4$, which vanishes at $u = j+4$. So, divided by $T(j,u)/\big((u+1)^4\prod_l (j+l)^2\big)$, the identity

$$\sum_{i=0}^{4} p_i(j)\,F(j+i,u) = G(j,u+1) - G(j,u) \qquad (0 \le u \le j+4)$$

is the identity of polynomials

$$(u+1)^4 \sum_{i=0}^{4} p_i(j) \prod_{l\le i}(j+l)^2 \prod_{l>i}(j+l-u)^2 = 4(2u+1)^2(j+4-u)^2 A(j,u+1) - (u+1)^4 A(j,u),$$

which holds in $\mathbb{Q}[j,u]$, both sides expanded. Summed over $0 \le u \le j+4$, the right side telescopes to $G(j,j+5) - G(j,0) = 0$, as $A(j,0) = 0$, and the left side is $\sum_i p_i(j)\,\mathsf{C}_{j+i}(1)$. So the closings satisfy the recurrence for every j.

The direction has the integral origin 0, both letter exponents 1 and positive walls. Written as $\sum_i c_i(-j)\,\mathsf{C}_{j-i}(1) = 0$ with $c_i(k) = -p_{4-i}(-k-4)$, the recurrence says that the closings vanish on $\sum_i c_i(k)\,\mathrm{Sh}^i[1]$ (Theorem 39.29); by Theorem 39.41 the values vanish on it as well, and the continuation of its value vanishes identically; Corollary 39.44 states the same for the recurrences of $\langle 1\rangle$. With $\langle f^i g\rangle_k = \langle g\rangle_{k+i}$ (Proposition 38.5(ii)) this is the law

$$\sum_{i=0}^{4} c_i(k)\,P(k+i) = 0,$$

an identity of meromorphic functions of k, with

$$\begin{aligned}
c_0(k) &= k^3(k+1)(144k^3+864k^2+1693k+1075),\\
c_1(k) &= -2(k+1)(2592k^6+19440k^5+56322k^4+80296k^3+60004k^2+23017k+3580),\\
c_2(k) &= 2(21168k^7+211680k^6+884919k^5+2003456k^4\\
&\qquad +2646984k^3+2031207k^2+830703k+137260),\\
c_3(k) &= -2(k+2)^2(34848k^5+261360k^4+741842k^3+984642k^2+598948k+127215),\\
c_4(k) &= 225(k+2)^2(k+3)^2(144k^3+432k^2+397k+102).
\end{aligned}$$

Its symbol (Corollary 38.7) is read off the coefficients of k^7:

$$\chi(\lambda) = 144 - 5184\lambda + 42336\lambda^2 - 69696\lambda^3 + 32400\lambda^4 = 144(25\lambda-1)(9\lambda-1)(\lambda-1)^2.$$

Its roots are the native values $\frac{1}{25}$ and $\frac{1}{9}$ of the two nondegenerate native gauges and the native value 1 of the curves, doubled; the root $\frac{1}{25} = f(\mathbf{1})$ is forced by Proposition 38.9.

(iv) The share side and its closings. A set of two marks has width one exactly for $\{a,b\}$ and $\{2,3\}$ (part (i)), so H_4 is the two-sum, at a hinge g, of the parallel cell $\mathcal{P}$ of $\{a,b\}$, with the core $\binom{\{g,a,b\}}{1}$, and the series cell $\mathcal{S}$ of $\{2,3\}$, with the core $\binom{\{g,2,3\}}{2}$: the bases $\{2,3\}$ and $\{x,y\}$, $x \in \{a,b\}$, $y \in \{2,3\}$, are those of the two-sum (Definition 23.15, Theorem 23.16). At the centre of real strength $\sigma > 0$, with the masses σ and $s = t = \sigma$, every mark is a one-mark cell of ratio 1, whose transform and dual transform are $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y)$ (Definition 23.19(1)), and Theorem 23.14 gives

$$f_{\mathcal{S}} = \mathsf{g}^2, \quad \kappa_{\mathcal{S}} = 2\sigma - \sigma = \sigma; \qquad \check{f}_{\mathcal{P}} = \mathsf{g}^2, \quad \kappa_{\mathcal{P}} = 2\sigma - \sigma = \sigma.$$

The root of two one-mark leaves at the strength z (Definition 23.19(5)), whose transforms are again g, and the Beta integral give

$$\int_0^\infty y^{z-1}\mathsf{g}(y)^2\,dy = \Gamma(z)^2 P_2(z),$$

$$P_2(z) = P_{\binom{[2]}{1}}(z\mathbf{1}; z, z) = \int_0^\infty x^{z-1}(1+x)^{-2z}\,dx = \frac{\Gamma(z)^2}{\Gamma(2z)} \qquad (z>0).$$

By Proposition 23.8(iii), with $s = t = \kappa = \sigma$, the two Mellin transforms of the laws are $\int \varrho_{\mathcal{S}}^{-z} d\phi_{\mathcal{S}} = P_2(z)$ and $\int \varrho_{\mathcal{P}}^{z-\sigma} d\phi_{\mathcal{P}} = P_2(z)$ for $z > 0$. Theorem 23.9(iv) for one table, with $\mathcal{S}$ in the place of $\mathcal{A}$ and $\mathcal{P}$ in the place of $\mathcal{A}'$, on a line $\mathrm{Re}\, z = c$ with $0 < c < \sigma$, gives

$$\Gamma(\sigma)^2 P(\sigma) = \frac{1}{2\pi i}\int_{(c)} \Gamma(z)^2\,\Gamma(\sigma-z)^2\,P_2(z)^2\,dz = \frac{1}{2\pi i}\int_{(c)} \mathcal{F}(z,\sigma)\,dz, \qquad \mathcal{F} = \frac{\Gamma(z)^6\,\Gamma(\sigma-z)^2}{\Gamma(2z)^2},$$

and both sides are holomorphic in σ on $\operatorname{Re}\sigma > c$.

Closing (Theorem 4.5) moves the line across the poles of the Gammas on one side and needs the contribution of the moved line to tend to zero. At an integer strength $k \geq 1$, by the shift and the reflection of Proposition 1.3(iii),(v),

$$\Gamma(z)\,\Gamma(k-z) = \frac{\pi\,(z-1)(z-2)\cdots(z-k+1)}{\sin\pi(z-k+1)},$$
$$\mathcal{F}(z,k) = \frac{\pi^2\,V_k(z)\,P_2(z)^2}{\sin^2\pi(z-k+1)}, \qquad V_k(z) = \prod_{j=1}^{k-1}(z-j)^2.$$

On the line $\operatorname{Re} z = k+N+\frac12$, $|\sin\pi(z-k+1)| = \cosh(\pi \operatorname{Im} z)$, $|P_2(z)| \leq \int_0^1 \big(w(1-w)\big)^{\operatorname{Re} z-1} dw \leq 4^{1-\operatorname{Re} z}$ and $V_k(z)$ is a polynomial, so the contribution of that line is $O(N^{2k-2}16^{-N})$ and tends to zero. To the right the poles are the integers $x \geq k$, double, with the principal part $(z-x)^{-2}$ of $\pi^2/\sin^2$ and no simple part (Lemma 5.16), and closing to the right gives

$$\Gamma(k)^2 P(k) = -\sum_{x\geq k} \big(V_k P_2^2\big)'(x),$$
$$P(1) = \sum_{m\geq 0} \frac{4\,(m!)^4}{((2m+1)!)^2}\big(h_{2m+1} - h_m\big) = 4 + \frac{5}{54} + \frac{47}{13500} + \frac{319}{2058000} + \cdots,$$

with $h_n = \sum_{i=1}^n i^{-1}$, since $P_2'/P_2 = 2(\log\Gamma)'(x) - 2(\log\Gamma)'(2x)$ and $(\log\Gamma)'(n+1) - (\log\Gamma)'(1) = h_n$ (Proposition 1.3(vi)). The terms are rational and decrease like 16^{-x}: the share side of H_4 closes on the lattice of the integers to its right.

To the left the poles are at $z = -m$, $m \in \mathbb{N}$, of order four: $\Gamma(z)^6$ has order six there and $\Gamma(2z)^{-2}$ a double zero. With the duplication formula $\Gamma(2z) = \pi^{-1/2}2^{2z-1}\Gamma(z)\Gamma(z+\frac12)$ and the reflection,

$$\mathcal{F}(z,\sigma) = 4\pi^3\,16^{-z}\,\frac{\cos^2\pi z}{\sin^4\pi z}\,\mathcal{B}(z)^2, \qquad \mathcal{B}(z) = \frac{\Gamma(\frac12 - z)\,\Gamma(\sigma - z)}{\Gamma(1-z)^2},$$

so the residue at $-m$ is $16^m\,\mathcal{B}(-m)^2\big(c + O(m^{-1})\big)$ with $c = \frac{2}{3\pi}\log 16\,\big(2\pi^2 - (\log 16)^2\big) = 7.0911\ldots$: it is $4\pi^{-1}16^m/3!$ times the third derivative at 0 of $16^{-\varepsilon}\mathcal{B}(\varepsilon - m)^2(\pi\varepsilon)^4\cos^2(\pi\varepsilon)\sin^{-4}(\pi\varepsilon)$, in which the derivatives of $\log\mathcal{B}$ at $-m$ are $O(m^{-1})$; and $\mathcal{B}(-m)^2$ grows like $m^{2\sigma-3}$. At $\sigma = 1$ the ratios of consecutive residues are 14.5771, 15.2453, 15.4869, 15.6114 at $m = 10, 20, 30, 40$, against $16\big((m+\frac12)/(m+1)\big)^2 = 14.5785$, 15.2472, 15.4880, 15.6121, and the residue over $16^m\mathcal{B}(-m)^2$ is 7.1273, 7.1173, 7.1104, 7.1063. The residues grow like 16^m, and the closing to the left diverges.

(v) The low end. By (iii) the law holds as an identity of meromorphic functions. Near $k = 0$ the terms $i \geq 1$ are holomorphic, c_0 vanishes to order three, $[k^3]c_0 = 1075$, and $\lim_{k\to 0} k^3 P(k) = V_1\,\mathsf{C}_0(1) = 16$ (Proposition 39.30); the value of the law at 0 is the low-end relation, by the argument of Corollary 39.34(ii), which uses of the law only that it holds as an identity of meromorphic functions:

$$17200 - 7160\,P(1) + 274520\,P(2) - 1017720\,P(3) + 826200\,P(4) = 0.$$

At $k = -1$ the coefficients c_0 and c_1 vanish, simply, while $c_2(-1) = -342$, $c_3(-1) = 4842$, $c_4(-1) = -6300$, so the value of the law at -1 is

$$\operatorname*{CT}_{k=-1}\big(c_0(k)P(k) + c_1(k)P(k+1)\big) - 342\,P(1) + 4842\,P(2) - 6300\,P(3) = 0,$$

and the first term needs the principal parts of P at -1 and at 0. The share side gives them. On the line $\operatorname{Re} z = \frac12$ the bound $|\Gamma(c'+i\tau)| \leq C_{c'}e^{-\pi|\tau|/2}(1+|\tau|)^{c'-1/2}$, with $1/\Gamma(2z) = \Gamma(1-2z)\sin(2\pi z)/\pi$,

makes $|\mathcal{F}(z,\sigma)|$ at most a constant times $(1+|\operatorname{Im} z|)^{2\operatorname{Re}\sigma}e^{-2\pi|\operatorname{Im} z|}$ for $|\operatorname{Im} z|\ge 1$, uniformly for σ in compact sets; so $J(\sigma)=\frac{1}{2\pi i}\int_{(1/2)}\mathcal{F}\,dz$ is holomorphic wherever no pole of $\Gamma(\sigma-z)^2$ lies on the line, in particular on the strips $|\operatorname{Re}\sigma|<\frac12$ and $-\frac32<\operatorname{Re}\sigma<-\frac12$. For $\operatorname{Re}\sigma>\frac12$, $\Gamma(\sigma)^2P(\sigma)=J(\sigma)$. As σ moves left the double poles $z=\sigma$ and then $z=\sigma+1$ of $\Gamma(\sigma-z)^2$ cross the line, and the identity of meromorphic functions that continues it is

$$\Gamma(\sigma)^2P(\sigma)=J(\sigma)-R_0(\sigma)\qquad(|\operatorname{Re}\sigma|<\tfrac12),$$
$$\Gamma(\sigma)^2P(\sigma)=J(\sigma)-R_0(\sigma)-R_1(\sigma)\qquad(-\tfrac32<\operatorname{Re}\sigma<-\tfrac12),$$

with the residues, by Proposition 5.14 at $p=2$,

$$R_m(\sigma)=\operatorname*{Res}_{z=\sigma+m}\mathcal{F}=\frac{\Gamma(\sigma+m)^6}{(m!)^2\,\Gamma(2\sigma+2m)^2}\Big(6(\log\Gamma)'(\sigma+m)-4(\log\Gamma)'(2\sigma+2m)-2(\log\Gamma)'(m+1)\Big).$$

$\Gamma(\sigma)^{-2}$ vanishes to order two at 0 and at -1, so J enters P there only from the order $(\sigma-\sigma_0)^2$ on, and the principal parts of P are those of $-R_0\,\Gamma^{-2}$ and of $-(R_0+R_1)\,\Gamma^{-2}$. Expanding every Gamma factor at its pole or regular point by Proposition 5.14, in $(\log\Gamma)'(1)$ and the Trailmoments $(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of the counting input, gives $R_0(x)=-16x^{-5}-32(\log\Gamma)'(1)\,x^{-4}+O(x^{-3})$ near 0 and $(R_0+R_1)(-1+x)=-80x^{-5}+O(x^{-4})$ near -1, and, the terms in $(\log\Gamma)'(1)$ cancelling,

$$P(x)=\frac{16}{x^3}-\frac{24\,(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))}{x}+O(1),$$
$$P(-1+x)=\frac{80}{x^3}-\frac{152}{x^2}-\frac{120\,(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))}{x}+O(1),$$

with $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))=\sum_{n\ge1}n^{-2}=\frac16\Gamma(\frac12)^4$ (Lemma 5.16 at $a=\frac12$). The top coefficients are $V_1\,\mathsf{C}_0(1)=16$ and $V_1\,\mathsf{C}_1(1)=80$ (Proposition 39.30), and the coefficient 0 of x^{-2} at 0 is the low-end coefficient L_1, which vanishes for a list (Proposition 39.35, every weight being 1). With

$$c_0(-1+x)=-102x-91x^2+453x^3+O(x^4),\qquad c_1(-1+x)=510x-514x^2-3056x^3+O(x^4),$$

and $T=(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, the coefficients of x^{-2} and x^{-1} in $c_0(k)P(k)+c_1(k)P(k+1)$ at $k=-1+x$ are $-102\cdot80+510\cdot16=0$ and $-102\cdot(-152)-91\cdot80-514\cdot16=0$, as the law asks, and its constant term is

$$-102\cdot(-120T)-91\cdot(-152)+453\cdot80+510\cdot(-24T)-514\cdot0-3056\cdot16=1176:$$

the multiples of $\Gamma(\frac12)^4$ cancel. Hence $-342\,P(1)+4842\,P(2)-6300\,P(3)=-1176$, that is

$$57\,P(1)-807\,P(2)+1050\,P(3)=196,$$

the second low-end relation.

(vi) Values and their span. The share side at $\sigma=1,\dots,5$, integrated on the line $\operatorname{Re}z=\frac12$ in ball arithmetic, and the closing of (iv), summed over its first 80 terms at $k=1,2,3$, agree to 80 digits and give

$$P(1)=4.0962369983378828500555532283512\ldots,$$
$$P(2)=0.047716498691513110297412080890663\ldots;$$

the law at $k=1$, the low-end relation and the second low-end relation hold at these values to 75 digits. For $k\ge1$, $c_4(k)>0$, so the law gives $P(k+4)$ as a rational combination of

$P(k),\dots,P(k+3)$; the second low-end relation gives $P(3)$, and then the low-end relation $P(4)$, in $\mathbb{Q}+\mathbb{Q}\,P(1)+\mathbb{Q}\,P(2)$. So every value lies in $\mathbb{Q}+\mathbb{Q}\,P(1)+\mathbb{Q}\,P(2)$. An integer-relation search at 290 bits finds no relation among 1, $P(1)$, $P(2)$ with coefficients below 10^{25}.

Answer. $V_1=16$. The native gauges are $\mathbf{1}$ (native value $\frac{1}{25}$, $\det H=\frac{32}{625}$), the sign $(1,1,-1,-1)$ ($\frac{1}{9}$, $\det H=-\frac{32}{81}$) and three curves of native value 1. The closings are $\sum_u \binom{j}{u}^2\binom{2u}{u}^2$; the law is the display of (iii), of symbol $144(25\lambda-1)(9\lambda-1)(\lambda-1)^2$, proved by the telescoping identity and Theorem 39.41. On the share side $\Gamma(\sigma)^2P(\sigma)=\frac{1}{2\pi i}\int\Gamma(z)^6\Gamma(\sigma-z)^2\Gamma(2z)^{-2}dz$; it closes to the right, $\Gamma(k)^2P(k)=-\sum_{x\ge k}(V_kP_2^2)'(x)$, and not to the left. The low-end relation is $17200-7160P(1)+274520P(2)-1017720P(3)+826200P(4)=0$, the second low-end relation $57P(1)-807P(2)+1050P(3)=196$, and the values lie in $\mathbb{Q}+\mathbb{Q}\,P(1)+\mathbb{Q}\,P(2)$. □

Exercise 53.2 (H_4 on a ray with a conjugate on the slit)**.** On H_4 take the masses $\mathbf{m}_1=(1,1,3,3)$ on $(a,b,2,3)$ and the letter exponents $s_1=3$, $t_1=1$, and let $P(k)=\langle1\rangle_k=P_{\mathrm{H}_4}\big(k\mathbf{m}_1;3k,k\big)$, $k\ge1$, be the values of the ray with the origin 0 and the direction $(\mathbf{m}_1;3,1)$.

(i) Show that the direction lies on the balance wall and strictly inside its walls, and compute V_1.

(ii) Find the native gauge $\alpha^\star$ of the direction on the positive orthant, $f^\star=f(\alpha^\star)$ and its conjugate over $\mathbb{Q}$, and show that the conjugate is the native value of a nondegenerate native gauge and lies on the slit $(0,f^\star)$.

(iii) Find the closings and a law of the values of order four, prove the law, and give its symbol.

(iv) Write the low-end relation, and show that Theorem 41.3(iii) does not apply.

(v) Read $P(1)$ and $P(2)$ from the gauge end (Theorem 40.6), and check them against the low-end relation.

(vi) Write $\Gamma(3k)\Gamma(k)P(k)$ on the share side, read the second low-end relation from it, and give the value span.

Solution. (i) Walls and V_1. The direction has $m_{1,\mathcal{E}}=8=3r+1\cdot r^*=3\cdot2+2$, on the balance wall. Its walls are $\kappa_{1,S}=m_{1,S}-3d(S)-d^*(S)$ (Definition 8.1), with $d(S)=|S|-r^*(S)$ and $d^*(S)=|S|-r(S)$ (Definition 7.12). A single mark has $d=d^*=0$; $\{a,b\}$ has $d=0$, $d^*=1$; $\{2,3\}$ has $d=1$, $d^*=0$; the four pairs $\{x,y\}$, $x\in\{a,b\}$, $y\in\{2,3\}$, have $d=d^*=0$; every triple has $d=d^*=1$. So

$$\kappa_{1,S}=1\quad\text{for } S\in\big\{\{a\},\{b\},\{a,b\},\{a,b,2\},\{a,b,3\}\big\},$$
$$\kappa_{1,S}=3\quad\text{for } S\in\big\{\{2\},\{3\},\{2,3\},\{a,2,3\},\{b,2,3\}\big\},$$

and $\kappa_{1,S}=4$ for the four pairs $\{x,y\}$: all positive. An order of the marks is the chain of its tails $U_2\supset U_3\supset U_4$, and $V_1=\sum\big(\kappa_{1,U_2}\kappa_{1,U_3}\kappa_{1,U_4}\big)^{-1}$ (Proposition 39.30). Below $U_2=\{a,b,y\}$ the pairs and their marks give $1\cdot(1+1)+2\cdot\frac14(1+\frac13)=\frac83$; below $U_2=\{x,2,3\}$ they give $2\cdot\frac14(1+\frac13)+\frac13(\frac13+\frac13)=\frac89$. So

$$V_1=2\cdot\frac83+2\cdot\frac13\cdot\frac89=\frac{160}{27}.$$

(ii) The native gauge and its conjugate. The exchanges $a\leftrightarrow b$ and $2\leftrightarrow3$ keep the direction, and $\alpha^\star$ is unique up to scale (Theorem 38.14(i)), so $\alpha^\star=(x,x,1,1)$ for some $x>0$. On that slice $K=4x+1$, $\Psi=x(x+4)$ and

$$f=\frac{\alpha_a\alpha_b\alpha_2^3\alpha_3^3}{K^3\Psi}=\frac{x}{(x+4)(4x+1)^3},\qquad\frac{d}{dx}\log f=\frac{-4(3x^2+8x-1)}{x(x+4)(4x+1)},$$

so $x^\star = \frac{\sqrt{19}-4}{3}$, where the gradient of $\log f$ vanishes in all four coordinates, and

$$f^\star = \frac{7643 + 1064\sqrt{19}}{1366875} = 0.0089846317\ldots, \qquad \det H = \frac{376096\sqrt{19} - 1614848}{151875} = 0.16142523\ldots,$$

H the Hessian of $-\log f$ on the chart $y_3 = 0$; $f^\star$ is a root of $1366875\lambda^2 - 15286\lambda + 27$. By Proposition 38.10 the conjugate point $\alpha' = (x', x', 1, 1)$, $x' = \frac{-\sqrt{19}-4}{3} = -2.786\ldots$, where $K = \frac{-13-4\sqrt{19}}{3}$ and $\Psi = x'(x'+4)$ do not vanish, is a nondegenerate native gauge, with

$$f(\alpha') = \frac{7643 - 1064\sqrt{19}}{1366875} = 0.0021985415\ldots,$$
$$\det H(\alpha') = \frac{-376096\sqrt{19} - 1614848}{151875} = -21.4269\ldots,$$

and $0 < f(\alpha') < f^\star$: the conjugate lies on the slit, at a gauge where the class $\{a, b\}$ is flipped.

(iii) Closings and the law. Every letter exponent of the direction is positive and the origin is 0, so $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j}) = [\alpha^{j\mathbf{m}_1}] K^{3j}\Psi^j$ (Definition 39.26). In

$$K^{3j}\Psi^j = \sum_{u,v} \binom{3j}{u}\binom{j}{v} (AB)^{4j-u-v} (\alpha_2\alpha_3)^u (\alpha_a\alpha_b)^v$$

the monomial $\alpha_a^j\alpha_b^j\alpha_2^{3j}\alpha_3^{3j}$ takes $(\alpha_a\alpha_b)^{j-v}$ from A^{4j-u-v} and $(\alpha_2\alpha_3)^{3j-u}$ from B^{4j-u-v}, so $u = 2j + v$, with the coefficient $\binom{2j-2v}{j-v}^2$; with $w = j - v$,

$$\mathsf{C}_j(1) = \sum_{w=0}^{j} F(j, w), \qquad F(j,w) = \binom{3j}{w}\binom{j}{w}\binom{2w}{w}^2;$$
$$\mathsf{C}_j(1) = 1,\ 13,\ 589,\ 37597,\ 2791949,\ \ldots$$

On the first 300 closings there is no recurrence of order one, two, three with degree at most 138, 91, 68, and none of order four with degree at most 24; at order four and degree 25 exact linear algebra on the first 170 closings gives one line, spanned by $p_0, \ldots, p_4 \in \mathbb{Z}[j]$ with coefficients of up to 36 digits, and the recurrence holds on all 300. It holds for every j. For $0 \le w \le j + 4$ put

$$T(j,w) = \frac{((2w)!)^2}{(w!)^6\,(3j+12-w)!\,(j+4-w)!},$$
$$G(j,w) = (3j)!\,j!\,T(j,w)\,A(j,w), \qquad G(j, j+5) = 0,$$

with $A \in \mathbb{Q}[j, w]$, a polynomial of 416 terms, of degree 16 in w and divisible by w^4, printed by the receipt. As $F(j+i, w) = (3j)!\,j!\,T(j,w)\prod_{l=1}^{3i}(3j+l)\prod_{l=1}^{i}(j+l)\prod_{l=3i+1}^{12}(3j+l-w)\prod_{l=i+1}^{4}(j+l-w)$ and $T(j, w+1) = T(j,w)\,4(2w+1)^2(3j+12-w)(j+4-w)/(w+1)^4$, the identity $\sum_i p_i(j)F(j+i,w) = G(j, w+1) - G(j,w)$ for $0 \le w \le j+4$ is the identity of polynomials

$$(w+1)^4 N(j,w) = 4(2w+1)^2(3j+12-w)(j+4-w)A(j,w+1) - (w+1)^4 A(j,w),$$

$N = \sum_i p_i(j)\prod_{l=1}^{3i}(3j+l)\prod_{l=1}^{i}(j+l)\prod_{l=3i+1}^{12}(3j+l-w)\prod_{l=i+1}^{4}(j+l-w)$, which holds in $\mathbb{Q}[j,w]$; summed over w it telescopes to 0, as $A(j,0) = 0$. As in Exercise 53.1(iii), Theorems 39.29 and 39.41 turn it into the law

$$\sum_{i=0}^{4} c_i(k)\,P(k+i) = 0, \qquad c_i(k) = -p_{4-i}(-k-4),$$

an identity of meromorphic functions of k. The coefficients factor as

$$c_0 = 3k^3(k+1)(3k+1)^3(3k+2)^3a_0, \qquad c_1 = -2(k+1)a_1, \qquad c_2 = 2(3k+4)(3k+5)a_2,$$
$$c_3 = -2(3k+4)(3k+5)(3k+7)(3k+8)(k+2)^2a_3,$$
$$c_4 = 151875(3k+4)(3k+5)(3k+7)(3k+8)(3k+10)(3k+11)(k+2)^2(k+3)^2a_4,$$

with $a_i \in \mathbb{Z}[k]$ of degrees $15, 24, 23, 19, 15$, printed by the receipt, and the symbol is

$$\chi(\lambda) = 168132676253771376\,(\lambda-1)^2\,(1366875\lambda^2 - 15286\lambda + 27)\,:$$

$f^\star$, its conjugate, and 1, doubled. The root 1 is the native value of the line $\{A = B = 0\}$, a source of dimension one (Definition 39.19): there $K = \alpha_2\alpha_3$ and $\Psi = \alpha_a\alpha_b$, the occupations of K and of Ψ are $(0,0,1,1)$ and $(1,1,0,0)$ on $(a,b,2,3)$, so the usage $3R + R^*$ is $\mathbf{m}_1$, and $f = 1$.

(iv) The low end and positive selection. c_0 vanishes to order three at 0 with $[k^3]c_0 = 656679698393897858976$, and with $V_1 = \frac{160}{27}$ the argument of Corollary 39.34(ii) gives the low-end relation, divided by 15360,

$$253348649071719853 + 204923553451974000\,P(1) - 272918959559705480400\,P(2)$$
$$+ 1177478827066019041200\,P(3) - 1011712684533185250000\,P(4) = 0.$$

Theorem 41.3 asks the roots of the symbol to be distinct, which the double root 1 already violates, and (iii) asks every root other than $f^\star$ to have modulus greater than $f^\star$. The conjugate $f(\alpha')$ lies in $(0, f^\star)$, so a solution of the law that grows like $f(\alpha')^k$ grows more slowly than the values, and growth does not separate it from them: the hypothesis fails. Set up anyway, the equations of Theorem 41.3(iv), with $P(1), P(2), P(3)$ free, $P(4)$ from the low-end relation and the solution required to vanish at $K, K+1, K+2$, return $P(1) = -2.6 \cdot 10^{23}$ at $K = 40$ and $-7.4 \cdot 10^{47}$ at $K = 80$.

(v) The gauge end. $f^\star$ is a simple root of χ, and the density operator $\mathfrak{D}_{\mathbf{p}}$ of the law (Theorem 40.6) has order $D = 25$; at $f^\star$ its coefficient $b_{D,0}$ vanishes and the extra exponent is $\gamma = D - 1 - b_{D-1,0}/b_{D,1} = \frac{1}{2} = \frac{n-3}{2}$, as Theorem 40.6(iii) requires. As $\gamma \notin \{0, \dots, 23\}$, the germ of the density is $\varrho_f(t) = s^{1/2}g(0)\Phi(s)$, $s = f^\star - t$, with Φ fixed by the recursion of Theorem 40.6(iv), 1800 terms computed over $\mathbb{Q}(\sqrt{19})$ in ball arithmetic, and the scale read off the Hessian at $\alpha^\star$ (the end of the stride's law at $f^\star$, Definition 40.1, with $(\omega_0/dy)(\alpha^\star) = 1$):

$$g(0) = \frac{1}{\Gamma(\frac{3}{2})\sqrt{\det H}}\Big(\frac{2\pi}{f^\star}\Big)^{3/2} = 51938.4198592508155915\dots$$

By Theorem 40.6(v) the density continues holomorphically along $(0, f^\star)$, through the root $f(\alpha')$ of the symbol, so the series of Φ converges on $[0, f^\star)$: the coefficients g_m of $g = g(0)\Phi$ have $|g_m|(f^\star)^m \le 5.2 \cdot 10^5$ for $m \le 1800$. The Beta integral gives

$$P(k) = \int_0^{f^\star} t^k \varrho_f(t)\,dt = \sum_{m \ge 0} g_m\,(f^\star)^{k+\frac{3}{2}+m}\,\mathrm{B}\big(k+1, m+\tfrac{3}{2}\big),$$

read at $k = K, \dots, K+3$, and the law, with $c_0(k) \neq 0$ at every integer $1 \le k \le 200$, runs back to $k = 1$. From $K = 80$ and from $K = 120$ the readings agree to 92 digits:

$$P(1) = 0.48830319489526888388541464396010900874354031922042\dots,$$

$$P(2) = 0.00132119179498773222717641714721510065075969610885\ldots$$

The low-end relation, which the reading does not use, returns $V_1 = \frac{160}{27}$ from these values to 191 digits.

(vi) The share side and the second low-end relation. The widths of (i) of Exercise 53.1 do not depend on the direction, so the hinge tree is the same: the series cell $\mathcal{S}$ of $\{2,3\}$, whose two marks have the masses $3k$, and the parallel cell $\mathcal{P}$ of $\{a,b\}$, with the masses k, at $s = 3k$, $t = k$. Every mark is again a one-mark cell whose transform, or dual transform, is g (Definition 23.19(1): $s = m_2 = m_3$, $t = m_a = m_b$), the hinge walls are $\kappa_{\mathcal{S}} = 6k - 3k = 3k$ and $\kappa_{\mathcal{P}} = 2k - k = k$, the walls of $\{2,3\}$ and $\{a,b\}$ (Theorem 23.14), and Proposition 23.8(iii) gives $\int \varrho_{\mathcal{S}}^{-z} d\phi_{\mathcal{S}} = P_2(z) = \int \varrho_{\mathcal{P}}^{z-k} d\phi_{\mathcal{P}}$. Theorem 23.9(iv), on a line $\operatorname{Re} z = c$, $0 < c < k$, gives

$$\Gamma(3k)\Gamma(k)\,P(k) = \frac{1}{2\pi i}\int_{(c)} \frac{\Gamma(z)^6\,\Gamma(k-z)\,\Gamma(3k-z)}{\Gamma(2z)^2}\,dz,$$

which at $k = 1$, on $c = \frac{1}{2}$ in ball arithmetic, agrees with (v) to 60 digits. At $k = -1$ the coefficients c_0 and c_1 vanish, simply, and the second low-end relation needs the principal parts of P at 0 and -1. As in Exercise 53.1(v), the integral on $\operatorname{Re} z = \frac{1}{2}$ is holomorphic near $k = 0$ and $k = -1$, $\Gamma(3k)\Gamma(k)$ has a double pole at both, and the principal parts of P come from the poles that cross the line, $z = k$ and $z = 3k$ near 0, and $z = k, k+1$ and $z = 3k, \ldots, 3k+3$ near -1, each simple, with the residues $(-1)^{m+1}\Gamma(2k-m)\,\Gamma(k+m)^6/\big(m!\,\Gamma(2k+2m)^2\big)$ at $z = k+m$ and $(-1)^{m+1}\Gamma(-2k-m)\,\Gamma(3k+m)^6/\big(m!\,\Gamma(6k+2m)^2\big)$ at $z = 3k+m$ (Proposition 1.3(iv)). Expanded by Proposition 5.14, with $T = (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$,

$$P(x) = \frac{160}{27x^3} - \frac{208\,T}{9x} + O(1), \qquad P(-1+x) = \frac{2080}{27x^3} - \frac{21118}{81x^2} - \frac{2704\,T + \frac{887}{9}}{9x} + O(1),$$

with the top coefficients $V_1\mathsf{C}_0(1)$ and $V_1\mathsf{C}_1(1) = \frac{160}{27}\cdot 13$ (Proposition 39.30). With the expansions of c_0 and c_1 at -1 to the order three, the multiples of T cancel in the constant term at $k = -1$, which is -2077616120857361856, and $(c_2, c_3, c_4)(-1) = 625190400\cdot(7005765879, -74409017489, 88989468750)$; so

$$7005765879\,P(1) - 74409017489\,P(2) + 88989468750\,P(3) = \frac{5981712159277}{1800},$$

which the values of (v) satisfy to 109 digits. For $k \geq 1$, $c_4(k) > 0$ (its real roots are below $\frac{1}{9}$), so, as in Exercise 53.1(vi), every value lies in $\mathbb{Q} + \mathbb{Q}\,P(1) + \mathbb{Q}\,P(2)$.

Answer. $V_1 = \frac{160}{27}$; $f^\star = \frac{7643+1064\sqrt{19}}{1366875}$ at $\alpha^\star = (x^\star, x^\star, 1, 1)$, $x^\star = \frac{\sqrt{19}-4}{3}$, $\det H = \frac{376096\sqrt{19}-1614848}{151875}$; its conjugate $\frac{7643-1064\sqrt{19}}{1366875}$ is the native value of the nondegenerate native gauge $(x', x', 1, 1)$, $x' = \frac{-\sqrt{19}-4}{3}$, and lies on the slit. The closings are $\sum_w \binom{3j}{w}\binom{j}{w}\binom{2w}{w}^2$, the law has order four, degree 25 and the symbol $(\lambda-1)^2(1366875\lambda^2 - 15286\lambda + 27)$ up to a constant, positive selection does not apply, and the gauge end gives $P(1) = 0.4883031948952688838854\ldots$ The second low-end relation is $7005765879P(1) - 74409017489P(2) + 88989468750P(3) = \frac{5981712159277}{1800}$, and the value span is $\mathbb{Q} + \mathbb{Q}\,P(1) + \mathbb{Q}\,P(2)$. □

53.2 The five-mark lists

In this section the marks are $1, \ldots, 5$, every weight is 1, and $\ell = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ is the value at the strength one of the centre ray of the list $\binom{[3]}{1}$, $P_3(\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)$ $P_2(\sigma) = P_{\binom{[2]}{1}}(\sigma\mathbf{1}; \sigma, \sigma) =$

$\Gamma(\sigma)^2/\Gamma(2\sigma)$, and $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y)$ as in Exercise 53.1(iv). The lists are $\binom{[5]}{1}$ and $\binom{[5]}{2}$, and three lists of rank two named by their parallel classes:

- P_5, whose bases are the nine pairs other than $\{1,2\}$: the classes $\{1,2\},\{3\},\{4\},\{5\}$;
- R_5, whose bases are the eight pairs $\{x,y\}$ with x,y in different classes of $\{1,2\},\{3,4\},\{5\}$;
- T_5, whose bases are the seven pairs $\{x,y\}$ with x,y in different classes of $\{1,2,3\},\{4\},\{5\}$.

Exercise 53.3 (The five-mark lists along their hinge trees)**.** (i) Find, for each of the five lists, the sets of width one with at least two marks on each side, and its canonical hinge tree.

(ii) Show that $P_{\mathrm{P}_5}(\sigma\mathbf{1};\sigma,\sigma) = P_3(\sigma)^2$. Prove the law of P_3, and derive from it the law of the centre ray of P_5, its symbol, its low-end relation and its value span.

(iii) Show that $q(\sigma) = P_{\binom{[4]}{2}}(\sigma\mathbf{1};\sigma,\sigma) = P_2(\sigma)P_3(\sigma)$, and find the law of q.

(iv) Write the centres of T_5 and R_5 on their share sides, close T_5 at the integer strengths, and write $P_{\mathrm{T}_5}(\mathbf{1};1,1)$ in the coordinates 1, ℓ and $\ell' = \partial_\sigma P_3(\sigma)|_{\sigma=1}$.

Solution. (i) Widths and trees. The width of S is $c(S) = r(S) + r(\mathcal{E}\setminus S) - r$ (Definition 7.12). In $\binom{[5]}{1}$ every nonempty set has rank one, so every proper S has width 1, and the list is one parallel node. In $\binom{[5]}{2}$ every set of two or three marks has rank two and so has its complement: width 2, and the list is prime (Theorem 23.16); so is $\binom{[4]}{2}$, whose pairs have width 2. In a list of rank two whose classes are the parallel classes, a set of two or three marks has width one exactly when it, or its complement, lies in one class, and width two otherwise. So the sets of width one with at least two marks on each side are $\{1,2\}$ for P_5; $\{1,2\}$ and $\{3,4\}$ for R_5; $\{1,2,3\}$ and its pairs for T_5; each with its complement. The canonical hinge trees (Definition 23.15, Theorem 23.16) are:

- $\binom{[5]}{1}$: one parallel node; $\binom{[5]}{2}$: one prime node;
- P_5: the parallel node $\binom{\{g,1,2\}}{1}$ and the prime node $\binom{\{g,3,4,5\}}{2}$;
- R_5: the series node $\binom{\{g_1,g_2,5\}}{2}$ with the parallel nodes $\binom{\{g_1,1,2\}}{1}$ and $\binom{\{g_2,3,4\}}{1}$;
- T_5: the series node $\binom{\{g,4,5\}}{2}$ with the parallel node $\binom{\{g,1,2,3\}}{1}$ of four marks.

For P_5, for instance, the bases of the two-sum are $\{x,y\}\subseteq\{3,4,5\}$, from $\{g\}$ and a pair of $\{3,4,5\}$, and $\{1,y\}$, $\{2,y\}$, $y\in\{3,4,5\}$, from $\{1\}$ or $\{2\}$ and $\{g,y\}$: the nine pairs other than $\{1,2\}$.

(ii) The doubled mark of $\binom{[4]}{2}$. The cell $\mathcal{A} = \binom{\{g,3,4,5\}}{2}$ of the bunch $\{3,4,5\}$ has the hinge minors $\mathcal{A}\setminus g = \binom{\{3,4,5\}}{2}$ and $\mathcal{A}/g = \binom{\{3,4,5\}}{1}$, dual with the identity and the same weights, so by Corollary 23.12(ii) its ratio is identically 1 and $\phi_{\mathcal{A}}(\Delta) = P_{\binom{[3]}{2}}(\sigma\mathbf{1};\kappa_{\mathcal{A}},2\sigma-\kappa_{\mathcal{A}})$, with the hinge wall $\kappa_{\mathcal{A}} = \sigma\, c(\{3,4,5\}) = \sigma(2+1-2) = \sigma$. By Corollary 23.12(i) the cell acts in the gluing as $\phi_{\mathcal{A}}(\Delta)$ times one mark e of ratio 1 and mass σ, and the parallel cell of $\{1,2\}$ glued to that mark is the list $\binom{\{1,2,e\}}{1}$. By Proposition 8.15 the dual table $\binom{[3]}{1}$ at the swapped exponents (σ,σ) has the same period, so

$$P_{\mathrm{P}_5}(\sigma\mathbf{1};\sigma,\sigma) = P_{\binom{[3]}{2}}(\sigma\mathbf{1};\sigma,\sigma)\,P_{\binom{\{1,2,e\}}{1}}(\sigma\mathbf{1};\sigma,\sigma) = P_3(\sigma)^2.$$

The law of P_3. The centre ray of $\binom{[3]}{1}$ has $f^{-1} = (\alpha_1+\alpha_2+\alpha_3)(\alpha_1^{-1}+\alpha_2^{-1}+\alpha_3^{-1})$, and its closings are (Definition 39.26)

$$\mathsf{C}_j(1) = \sum_{a+b+c=j}\Big(\frac{j!}{a!\,b!\,c!}\Big)^2 = \sum_{u=0}^{j}\binom{j}{u}^2\binom{2u}{u} = 1,\ 3,\ 15,\ 93,\ 639,\ 4653,\ \ldots,$$

grouping $a+b=u$. No recurrence of order one and degree at most 40 holds on the first 120 closings, and exact linear algebra gives

$$(j+2)^2\,\mathsf{C}_{j+2}(1)-(10j^2+30j+23)\,\mathsf{C}_{j+1}(1)+9(j+1)^2\,\mathsf{C}_j(1)=0.$$

It holds for every j: with $F(j,u)=\binom{j}{u}^2\binom{2u}{u}$, $T(j,u)=((j+2)!)^2(2u)!/\big((u!)^4((j+2-u)!)^2\big)$ for $0\le u\le j+2$, $A(j,u)=u^3(j+1)^2(3u-4j-8)$ and $G(j,u)=T(j,u)A(j,u)/((j+1)^2(j+2)^2)$, $G(j,j+3)=0$, one has $F(j+i,u)=T(j,u)\prod_{l=i+1}^{2}\big((j+l-u)/(j+l)\big)^2$ and $T(j,u+1)=T(j,u)\,2(2u+1)(j+2-u)^2/(u+1)^3$, so the identity $\sum_i p_i(j)F(j+i,u)=G(j,u+1)-G(j,u)$, $p_0=9(j+1)^2$, $p_1=-(10j^2+30j+23)$, $p_2=(j+2)^2$, is the identity of polynomials

$$(u+1)^3\sum_{i=0}^{2}p_i(j)\prod_{l\le i}(j+l)^2\prod_{l>i}(j+l-u)^2=2(2u+1)(j+2-u)^2A(j,u+1)-(u+1)^3A(j,u),$$

which holds in $\mathbb{Q}[j,u]$; summed over u it telescopes to 0. As in Exercise 53.1(iii), Theorems 39.29 and 39.41 give the law of the values, an identity of meromorphic functions,

$$k^2P_3(k)-(10k^2+10k+3)\,P_3(k+1)+9(k+1)^2\,P_3(k+2)=0,$$

of symbol $9\lambda^2-10\lambda+1=(9\lambda-1)(\lambda-1)$. The walls of the direction are all 1, so $V_1=3!=6$ (Proposition 39.30), and the low-end relation (Corollary 39.34(ii)) is $6-3P_3(1)+9P_3(2)=0$. So $P_3(1)=\ell$, $P_3(2)=\frac{\ell}{3}-\frac{2}{3}$, and the law gives

$$P_3(3)=\frac{5\ell}{27}-\frac{23}{54},\qquad P_3(4)=\frac{31\ell}{243}-\frac{145}{486},\qquad P_3(k)\in\mathbb{Q}+\mathbb{Q}\,\ell\quad(k\ge1).$$

The law of the squares. Write the law as $P_3(k+2)=a(k)P_3(k)+b(k)P_3(k+1)$; then $P_3(k+3)=a^{(1)}P_3(k)+b^{(1)}P_3(k+1)$ with $a^{(1)}=b(k+1)a(k)$ and $b^{(1)}=a(k+1)+b(k+1)b(k)$. The squares $x(k)=P_3(k)^2,\dots,x(k+3)$ are quadratic forms in $P_3(k),P_3(k+1)$, and the combination $d_0x(k)+d_1x(k+1)+d_2x(k+2)+x(k+3)$ with $d_2=-a^{(1)}b^{(1)}/(ab)$, $d_0=-d_2a^2-(a^{(1)})^2$, $d_1=-d_2b^2-(b^{(1)})^2$ vanishes identically. Cleared of denominators, the law of the centre ray of P_5 is $\sum_{i=0}^{3}d_i(k)\,x(k+i)=0$ with

$$\begin{gathered}d_0=k^4(10k^2+30k+23),\qquad d_3=-729(k+2)^4(10k^2+10k+3),\\ d_1=-(10k^2+10k+3)\,w(k),\qquad d_2=9(10k^2+30k+23)\,w(k),\\ w(k)=91k^4+364k^3+506k^2+284k+60,\end{gathered}$$

which the squares of the values above satisfy identically in ℓ at $k=1,\dots,5$. Its symbol is $-10(\lambda-1)(9\lambda-1)(81\lambda-1)$, the products of the roots $\frac19$ and 1 in pairs; $\frac{1}{81}=K(\mathbf{1})^{-2}$, $K(\mathbf{1})=9$ the number of bases. The walls of P_5 are 1 on single marks, on $\{1,2\}$ and on their complements, and 2 on the other sets of two or three marks, and $V_1=36$, the square of the 6 of $\binom{[3]}{1}$, as $P_3(k)^2\sim36k^{-4}$ at 0 requires. d_0 vanishes to order four at 0, $[k^4]d_0=23$, and the low-end relation

$$23\cdot36+d_1(0)\,x(1)+d_2(0)\,x(2)+d_3(0)\,x(3)=828-180\,x(1)+12420\,x(2)-34992\,x(3)=0$$

holds identically in ℓ at $x(i)=P_3(i)^2$. Every value $x(k)=P_3(k)^2$ lies in $\mathbb{Q}+\mathbb{Q}\,\ell+\mathbb{Q}\,\ell^2$. As a check, the procedure that reads a ray from its list by a box (Definition 41.8, Theorem 39.17), run on P_5 by the supplied reader, returns the same law and $P_{\mathrm{P}_5}(\mathbf{1};1,1)=5.4939011435808440124442576078206911958\ldots$, which is ℓ^2 to 87 digits.

(iii) $\binom{[4]}{2}$ *at its centre.* $\binom{[4]}{2}$ is identically self-dual: the complement of a pair is a pair, and the weights are 1. Corollary 20.19 at a mark, and Proposition 8.15 for the deletion $\binom{[3]}{2}$, give $q(\sigma) = \Gamma{}^{2}_{1}\!\left[\begin{smallmatrix}\sigma,\ \sigma\\ 2\sigma\end{smallmatrix}\right] P_3(\sigma) = P_2(\sigma)P_3(\sigma)$. With $P_2(\sigma{+}1)/P_2(\sigma) = \sigma/(2(2\sigma{+}1))$ (Proposition 1.3(iii)), the law of P_3 multiplied through by $\sigma P_2(\sigma)$ is the law of q:

$$\sigma^3 q(\sigma) - 2(2\sigma+1)(10\sigma^2+10\sigma+3)\,q(\sigma+1) + 36(\sigma+1)(2\sigma+1)(2\sigma+3)\,q(\sigma+2) = 0,$$

with the symbol $(4\lambda-1)(36\lambda-1)$ and $V_1 = 12 = 2\cdot 6$, the walls of $\binom{[4]}{2}$ being $1, 2, 1$ on sets of one, two and three marks.

(iv) The triangle rows. T_5 is H_4 of Section 53.1 with the class enlarged to $\{1,2,3\}$: the two-sum of the series cell $\mathcal{S}$ of $\{4,5\}$, with $f_{\mathcal{S}} = \mathsf{g}^2$ and $\kappa_{\mathcal{S}} = \sigma$, and the parallel cell $\mathcal{P}$ of $\{1,2,3\}$, with $\check{f}_{\mathcal{P}} = \mathsf{g}^3$ and $\kappa_{\mathcal{P}} = 3\sigma - 2\sigma = \sigma$ (Theorem 23.14). The root of three one-mark leaves (Definition 23.19(5)) gives $\int_0^\infty y^{z-1}\mathsf{g}^3 dy = \Gamma(z)^2 P_3(z)$, so Proposition 23.8(iii) gives $\int \varrho_{\mathcal{P}}^{z-\sigma} d\phi_{\mathcal{P}} = P_3(z)$, and $\int \varrho_{\mathcal{S}}^{-z} d\phi_{\mathcal{S}} = P_2(z)$ as in Exercise 53.1(iv). Theorem 23.9(iv) gives, for $0 < c < \sigma$,

$$\Gamma(\sigma)^2 P_{\mathrm{T}_5}(\sigma\mathbf{1};\sigma,\sigma) = \frac{1}{2\pi i}\int_{(c)} \Gamma(z)^2\,\Gamma(\sigma-z)^2\,q(z)\,dz.$$

$|q(z)| \le q(\operatorname{Re} z)$, the modulus of the integrand of a period being the integrand at the real parts, and $q(x) \le 4^{1-x}P_3(1)$ for $x \ge 1$; so the closing to the right of Exercise 53.1(iv) applies, the moved line contributing $O(N^{2s-2}4^{-N})$, and at an integer strength s

$$\Gamma(s)^2 P_{\mathrm{T}_5}(s\mathbf{1};s,s) = -\sum_{x\ge s} \big(V_s\,q\big)'(x), \qquad V_s(x) = \prod_{j=1}^{s-1}(x-j)^2.$$

At $s = 1$ this needs the slopes $q'(x) = P_2(x)\big(P_3'(x) + 2(h_{x-1} - h_{2x-1})P_3(x)\big)$. The law of P_3 holds as an identity of meromorphic functions, so its derivative in k holds as well and gives $P_3'(k+2)$ from $P_3'(k)$, $P_3'(k+1)$ and the values, starting from $P_3'(1) = \ell'$ and $P_3'(2)$. Near $k = 0$, $k^2 P_3(k) = 6 + \mathsf{L}_1 k + O(k^2)$ with $\mathsf{L}_1 = 0$ (Proposition 39.35, every weight being 1), so the derivative of the law at $k = 0$ reads $-10\ell - 3\ell' + 18P_3(2) + 9P_3'(2) = 0$:

$$P_3'(2) = \frac{4\ell}{9} + \frac{\ell'}{3} + \frac{4}{3}, \qquad P_3'(3) = \frac{26\ell}{81} + \frac{5\ell'}{27} + \frac{13}{18}, \qquad P_3'(k) \in \mathbb{Q} + \mathbb{Q}\,\ell + \mathbb{Q}\,\ell'.$$

So $P_{\mathrm{T}_5}(\mathbf{1};1,1) = -\sum_{x\ge1} q'(x) = a + b\,\ell + c\,\ell'$, where a, b, c are series of rational numbers whose terms decrease like 4^{-x}; summed to $x = 65$,

$$a = -0.46159963274916253008918\ldots, \qquad b = 2.01706417220768446716615\ldots,$$
$$c = -1.06282955174549612502149\ldots$$

With $\Gamma(\sigma)^2 P_3(\sigma) = \int_0^\infty y^{\sigma-1}\mathsf{g}(y)^3\,dy$, so that $\ell' = \int_0^\infty \log y\,\mathsf{g}^3 dy - 2(\log\Gamma)'(1)\,\ell$, quadrature gives

$$\ell = 2.34390723868945889060156228887227706909 54\ldots,$$
$$\ell' = -8.07515226353164401026965124146737650 13\ldots,$$

and

$$P_{\mathrm{T}_5}(\mathbf{1};1,1) = 12.84872214191555595731233993125353463661 9\ldots,$$

which the reading of T_5 by the box procedure (Definition 41.8) reproduces to 53 digits. For R_5 the series root has three cells, the mark 5, with the transform g, and the parallel cells of $\{1,2\}$

and $\{3,4\}$, each with the transform $f_{\mathcal{P}_2}$ whose Mellin transform is $\Gamma(w)^2 P_2(\sigma-w)$, $0<w<\sigma$ (Proposition 23.8(iii), with $\check{f}_{\mathcal{P}_2}=\mathsf{g}^2$); by Definition 23.19(5)

$$\Gamma(\sigma)^2 P_{\mathrm{R}_5}(\sigma\mathbf{1};\sigma,\sigma)=\int_0^\infty y^{\sigma-1}\,\mathsf{g}(y)\,f_{\mathcal{P}_2}(y)^2\,dy,$$

and the box procedure reads $P_{\mathrm{R}_5}(\mathbf{1};1,1)=7.865837368364021592421974697446 8210 9\ldots$

Answer. The trees are those listed in (i). $P_{\mathrm{P}_5}=P_3^2$, with the law $\sum_{i=0}^3 d_i(k)x(k+i)=0$ of symbol $-10(\lambda-1)(9\lambda-1)(81\lambda-1)$, $V_1=36$, the low-end relation $828-180x(1)+12420x(2)-34992x(3)=0$, and the value span $\mathbb{Q}+\mathbb{Q}\,\ell+\mathbb{Q}\,\ell^2$. $P_{\binom{[4]}{2}}=P_2P_3$ obeys $\sigma^3q(\sigma)-2(2\sigma+1)(10\sigma^2+10\sigma+3)q(\sigma+1)+36(\sigma+1)(2\sigma+1)(2\sigma+3)q(\sigma+2)=0$. $\Gamma(\sigma)^2P_{\mathrm{T}_5}(\sigma)=\frac{1}{2\pi i}\int\Gamma(z)^2\Gamma(\sigma-z)^2q(z)\,dz$ closes to the right, $\Gamma(s)^2P_{\mathrm{T}_5}(s)=-\sum_{x\geq s}(V_sq)'(x)$, and $P_{\mathrm{T}_5}(\mathbf{1};1,1)=a+b\,\ell+c\,\ell'=12.8487221419\ldots$ with the series a,b,c of (iv). □

53.3 Larger lists and the cost of a reading

Exercise 53.4 (The list N_6: letters and constant terms)**.** Let N_6 be the list of rank three on $\mathcal{E}=\{1,\ldots,6\}$ whose bases are the nineteen triples other than $a=\{1,2,3\}$, and put $b=\{4,5,6\}$. On a write $K_a=\alpha_1+\alpha_2+\alpha_3$, $\Psi_a=\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3$ and $\alpha^a=\alpha_1\alpha_2\alpha_3$, the letters and the monomial of the list $\binom{a}{1}$, and K_b, Ψ_b, α^b the same on b.

(i) Show that the letters of N_6 are $K=Q+\alpha^b$ and $\Psi=Q+\alpha^a$, with $Q=\Psi_aK_b+K_a\Psi_b$.

(ii) For $i,l\in\mathbb{N}$ with $2i+l=3D$, $D\in\mathbb{N}$, let $\gamma_{i,l}$ be the coefficient of $\alpha_1^D\alpha_2^D\alpha_3^D$ in $\Psi_a^i\,K_a^l$. Show that

$$\gamma_{i,l}=\sum_w\binom{i}{w}\binom{l}{w+\frac{l-i}{3}}\binom{2D-2w}{D-w}.$$

(iii) The centre ray of N_6 at the strength 1 (Definition 38.13) has the direction $(\mathbf{1};1,1)$, and the ray with origin 0 and direction $(\mathbf{1};0,2)$, the full cobasis lean, has the stride $f=\alpha^{\mathcal{E}}\Psi^{-2}$. Write the constant terms $\mathrm{CT}(f^{-j})$ of the two reciprocal strides as sums of products of two numbers $\gamma_{i,l}$, and compute them for $j\leq 4$.

Solution. (i) A triple T other than a meets a in at most two marks. If it meets a in two marks it is a pair of a and a mark of b, and these monomials α^T add up to Ψ_aK_b; if it meets a in one mark it is a mark of a and a pair of b, which add up to $K_a\Psi_b$; if it meets a in no mark it is b. So $K=\sum_{T\in\mathcal{B}}\alpha^T=\Psi_aK_b+K_a\Psi_b+\alpha^b=Q+\alpha^b$. The cobases are the complements of the bases; the complement of a triple is a triple, and the complement of a is b, so the cobases are the triples other than b, and the same count with a and b exchanged gives $\Psi=\sum_{T\in\mathcal{B}}\alpha^{\mathcal{E}\setminus T}=\Psi_bK_a+K_b\Psi_a+\alpha^a=Q+\alpha^a$.

(ii) Write $\Psi_a=\alpha_1\alpha_2+\alpha_3(\alpha_1+\alpha_2)$ and $K_a=\alpha_3+(\alpha_1+\alpha_2)$. Take w of the i factors of Ψ_a^i equal to $\alpha_1\alpha_2$ and v of the l factors of K_a^l equal to α_3:

$$\Psi_a^iK_a^l=\sum_{w,v}\binom{i}{w}\binom{l}{v}(\alpha_1\alpha_2)^w\,\alpha_3^{\,i-w+v}\,(\alpha_1+\alpha_2)^{\,i-w+l-v}.$$

The power D of α_3 asks $v=D-i+w=w+\frac{l-i}{3}$, since $D-i=\frac{2i+l}{3}-i$. Then $i-w+l-v=2i+l-D-2w=2D-2w$, and the coefficient of $\alpha_1^{D-w}\alpha_2^{D-w}$ in $(\alpha_1+\alpha_2)^{2D-2w}$ is $\binom{2D-2w}{D-w}$, which gives the formula. The first values are

$$\gamma_{0,0}=1,\quad \gamma_{1,1}=3,\quad \gamma_{3,0}=\gamma_{0,3}=6,\quad \gamma_{2,2}=15,\quad \gamma_{4,1}=\gamma_{1,4}=36,\quad \gamma_{3,3}=93;$$

for instance $\gamma_{2,2} = \binom{2}{0}\binom{2}{0}\binom{4}{2} + \binom{2}{1}\binom{2}{1}\binom{2}{1} + \binom{2}{2}\binom{2}{2}\binom{0}{0} = 6 + 8 + 1$. Each $\gamma_{i,l}$ is the constant term $\mathrm{CT}\left(\alpha^{-D\mathbf{1}}\Psi_a^i K_a^l\right)$ on $\binom{a}{1}$: the first constant term of the three-mark ray with direction $(D\mathbf{1}; l, i)$.

(iii) *The centre.* The stride is $f = \alpha^{\mathcal{E}}(K\Psi)^{-1}$ (Definition 38.13), so $\mathrm{CT}(f^{-j})$ is the coefficient of $\alpha^{j\mathbf{1}}$ in $(K\Psi)^j$; every letter exponent of the direction is 1, and this is the closing $\mathsf{C}_j(1)$ of Definition 39.26. Expand $K^j = (\Psi_a K_b + K_a \Psi_b + \alpha^b)^j$ with u_1, u_2, u_3 factors of the three kinds and $\Psi^j = (K_a\Psi_b + \Psi_a K_b + \alpha^a)^j$ with v_1, v_2, v_3. The product splits into a part on a and a part on b,

$$\Psi_a^{u_1+v_2} K_a^{u_2+v_1} (\alpha^a)^{v_3} \cdot \Psi_b^{u_2+v_1} K_b^{u_1+v_2} (\alpha^b)^{u_3},$$

and the coefficient of $\alpha^{j\mathbf{1}}$ is the product of the coefficient of $\alpha_1^{j-v_3}\alpha_2^{j-v_3}\alpha_3^{j-v_3}$ in the first factor without $(\alpha^a)^{v_3}$ and the like coefficient on b. Put $p = j - u_3 = u_1 + u_2$, $q = j - v_3 = v_1 + v_2$, $i = u_1 + v_2$ and $l = u_2 + v_1$. The degrees ask $2i + l = 3q$ and $i + 2l = 3p$, that is $i = 2q - p$ and $l = 2p - q$, and the two coefficients are $\gamma_{2q-p,\,2p-q}$ and $\gamma_{2p-q,\,2q-p}$. The multinomial weights summed over u_1, with $u_2 = p - u_1$, $v_2 = i - u_1$, $v_1 = q - i + u_1$, give

$$\sum_{u_1} \frac{j!}{u_1!\,u_2!\,u_3!} \cdot \frac{j!}{v_1!\,v_2!\,v_3!} = \binom{j}{p}\binom{j}{q}\sum_{u_1}\binom{p}{u_1}\binom{q}{i-u_1} = \binom{j}{p}\binom{j}{q}\binom{p+q}{2q-p}$$

by Vandermonde's identity. So

$$\mathrm{CT}(f^{-j}) = \sum_{p,q=0}^{j} \binom{j}{p}\binom{j}{q}\binom{p+q}{2q-p}\,\gamma_{2q-p,\,2p-q}\,\gamma_{2p-q,\,2q-p}.$$

At $j = 1$ the terms $(p, q) = (0, 0)$ and $(1, 1)$ survive, $1 + 2 \cdot 3 \cdot 3 = 19$, the number of bases. At $j = 2$ the terms $(0,0)$, $(1,1)$, $(1,2)$, $(2,1)$, $(2,2)$ are

$$1 + 2\cdot 2\cdot 2\cdot 3\cdot 3 + 2\cdot 1\cdot 1\cdot 6\cdot 6 + 1\cdot 2\cdot 1\cdot 6\cdot 6 + 1\cdot 1\cdot 6\cdot 15\cdot 15 = 1 + 72 + 72 + 72 + 1350 = 1567,$$

and at $j = 3$ the eight terms $(0,0), (1,1), (1,2), (2,1), (2,2), (2,3), (3,2), (3,3)$ are $1 + 162 + 324 + 324 + 12150 + 19440 + 19440 + 172980 = 224821$.

The full cobasis lean. Here $f^{-1} = \Psi^2/\alpha^{\mathcal{E}}$, and $\mathrm{CT}(f^{-j})$ is the coefficient of $\alpha^{j\mathbf{1}}$ in Ψ^{2j}; the direction has the letter exponent $s = 0$, so Definition 39.26 does not apply and the constant term is taken as written. With v_1, v_2, v_3 factors $K_a\Psi_b$, $\Psi_a K_b$, α^a out of $2j$ and $p = v_1 + v_2$, the part on b is $\Psi_b^{v_1} K_b^{v_2}$, of degree $3j$, so $2v_1 + v_2 = 3j$: $v_1 = 3j - p$ and $v_2 = 2p - 3j$; the part on a is $K_a^{v_1}\Psi_a^{v_2}(\alpha^a)^{v_3}$ with $v_3 = 2j - p$. So

$$\mathrm{CT}(f^{-j}) = \sum_{p} \binom{2j}{p}\binom{p}{2p-3j}\,\gamma_{2p-3j,\,3j-p}\,\gamma_{3j-p,\,2p-3j}.$$

At $j = 1$ only $p = 2$ survives: $\binom{2}{2}\binom{2}{1}\gamma_{1,1}^2 = 18$, the bases whose complement is a basis, all but b. At $j = 2$, $p = 3$ and $p = 4$ give $\binom{4}{3}\binom{3}{3}\gamma_{0,3}\gamma_{3,0} + \binom{4}{4}\binom{4}{2}\gamma_{2,2}^2 = 144 + 1350 = 1494$, and at $j = 3$, $p = 5$ and $p = 6$ give $\binom{6}{5}\binom{5}{1}\gamma_{1,4}\gamma_{4,1} + \binom{6}{6}\binom{6}{3}\gamma_{3,3}^2 = 38880 + 172980 = 211860$.

The closing values for $j \le 4$, by the two formulas and, as a check, by expanding $(K\Psi)^j$ and Ψ^{2j} with every exponent above j dropped (receipt `nsix_exact.py`), are

j	0	1	2	3	4
centre $(\mathbf{1}; 1, 1)$	1	19	1567	224821	41485807
lean $(\mathbf{1}; 0, 2)$	1	18	1494	211860	38486070

So the constant terms of N_6 are sums of products of two first constant terms of the three-mark list $\binom{[3]}{1}$ along its directions $(D\mathbf{1}; l, i)$, one factor for each of the triples a and b; the sums cost $O(j^2)$ and $O(j)$ products of numbers $\gamma_{i,l}$, each a sum of $O(j)$ terms. □

Exercise 53.5 (N_6 at its full cobasis lean)**.** Read the ray of N_6 (Exercise 53.4) with origin 0 and direction $(\mathbf{1};0,2)$, whose stride is $f=\alpha^{\mathcal{E}}\Psi^{-2}$, by the procedure of Definition 41.8:

(i) its walls and the volume V_1 of its direction;

(ii) the native gauge $\alpha^\star$ of the direction, $f^\star=f(\alpha^\star)$ and the Hessian H there;

(iii) the box with one unit of each active letter, its spanning set and its matrix $\mathbf{B}$, the matrix at infinite strength and its spectrum, and that the spanning set is a set of masters;

(iv) the law of [1], its two end coefficients, its symbol, and the strengths at which its rows line up;

(v) the low-end relation and the constant terms;

(vi) the value $\langle 1\rangle_1=P_{\mathrm{N}_6}(\mathbf{1};0,2)$.

Solution. Throughout, S is a nonempty proper subset of $\mathcal{E}$, $r(S)$ the largest trace of a basis on S, and the *type* of S is the pair $(i,j)=(|S\cap a|,|S\cap b|)$.

(i) *Walls.* By Definition 8.1 the walls of the direction are $\kappa_{1,S}=m_{1,S}-s\,d(S)-t\,d^*(S)$, with $d(S)$ and $d^*(S)$ the least traces of a basis and of a cobasis on S. Here $m_1=\mathbf{1}$, $s=0$, $t=2$, and the least trace of a cobasis is $d^*(S)=|S|-r(S)$, so

$$\kappa_{1,S}=2\,r(S)-|S|.$$

The rank of N_6 is $r(S)=\min(|S|,3)$, except $r(a)=2$, so $\kappa_{1,S}$ depends on the type alone:

(i,j)	$(1,0)$	$(0,1)$	$(2,0)$	$(1,1)$	$(0,2)$	$(3,0)$	$(2,1)$
$\kappa_{1,S}$	1	1	2	2	2	1	3
$F(i,j)$	1	1	1	1	1	3	1

(i,j)	$(1,2)$	$(0,3)$	$(3,1)$	$(2,2)$	$(1,3)$	$(3,2)$	$(2,3)$
$\kappa_{1,S}$	3	3	2	2	2	1	1
$F(i,j)$	1	1	3	2	2	12	10

Every wall is positive, the least is 1: the direction is strictly inside its walls, and counted with the sizes of the types the 62 walls are 1 thirteen times, 2 thirty times and 3 nineteen times.

The volume. By Proposition 39.30, $V_1=\sum_\pi\prod_{l=2}^{6}\kappa_{1,U_l}^{-1}$ over the orders π of $\mathcal{E}$ with their tails $U_l=\{\pi(l),\dots,\pi(6)\}$. An order is a chain of tails $U_6\subset\cdots\subset U_2$, so $V_1=\sum_e F(\mathcal{E}\setminus\{e\})$ with $F(\{e\})=\kappa_{1,\{e\}}^{-1}$ and $F(U)=\kappa_{1,U}^{-1}\sum_{e\in U}F(U\setminus\{e\})$. F depends on the type alone, $F(i,j)=\kappa_{1,(i,j)}^{-1}\big(i\,F(i-1,j)+j\,F(i,j-1)\big)$, which gives the last row of the table: for instance $F(3,0)=1^{-1}\cdot 3\,F(2,0)=3$, $F(3,1)=2^{-1}\big(3F(2,1)+F(3,0)\big)=3$, $F(3,2)=1^{-1}\big(3F(2,2)+2F(3,1)\big)=12$ and $F(2,3)=1^{-1}\big(2F(1,3)+3F(2,2)\big)=10$. So

$$V_1=3\,F(2,3)+3\,F(3,2)=30+36=66.$$

(ii) *The native gauge.* By Theorem 38.14(i), $f\le f(\alpha^\star)$ on the positive orthant with equality only at $\alpha\propto\alpha^\star$. The 36 automorphisms of N_6, the permutations of a and of b, keep the masses $\mathbf{1}$ and the exponents, so they are symmetries of the ray (Proposition 39.7(ii)); they keep f and the orthant and therefore fix $\alpha^\star$ up to scale. On the chart $\alpha_6=1$ this puts $\alpha^\star$ on the slice $\alpha=(x,x,x,1,1,1)$, where $Q=3x^2\cdot 3+3x\cdot 3$, $\Psi=x^3+9x^2+9x$ and

$$f=\frac{x^3}{x^2(x^2+9x+9)^2}=\frac{x}{(x^2+9x+9)^2},\qquad \frac{d}{dx}\log f=-\frac{3(x^2+3x-3)}{x(x^2+9x+9)}.$$

So $\alpha^\star = (x^\star, x^\star, x^\star, 1, 1, 1)$ with $x^\star = \frac{\sqrt{21}-3}{2}$, and with $x^2 = 3 - 3x$, $x^2 + 9x + 9 = 6(x + 2)$ and $f = x/(36(x + 2)^2)$ there:

$$f^\star = \frac{7\sqrt{21} - 27}{1800} = 0.0028211277026060444701\ldots$$

At a complex point fixed by the automorphisms the differential of $\log f$ in $y = \log \alpha$ is fixed by them too, so it is constant on a and on b, (u, u, u, v, v, v), and $u+v = 0$ because f is homogeneous of degree 0; it vanishes as soon as its derivative along the slice, $3u$, does. So the other root $x' = -\frac{\sqrt{21}+3}{2}$ gives a native gauge $\alpha' = (x', x', x', 1, 1, 1)$ too (Definition 38.4), at which $K = 9x'^2 + 9x' + 1$ and $\Psi = 6x'(x' + 2)$ do not vanish, of the conjugate native value $-\frac{7\sqrt{21}+27}{1800}$. The Hessian of $-\log f$ at $\log \alpha^\star$ on the chart is $H = 2\big(\kappa_{\alpha^\star}(X_e, X_{e'})\big)_{e,e' \neq 6}$, twice the covariances of the indicators under the cobasis law at $\alpha^\star$ (Theorem 38.15); computed in $\mathbb{Q}(\sqrt{21})$ (receipt `hessians.py`),

$$\det H = \frac{13049\sqrt{21} - 59339}{37500} = 0.0122408064927668332\ldots$$

(iii) *The box* (Procedure 41.8, the masters). The letter K has the exponent 0 along the ray and Ψ the exponent $2k$, so only Ψ is active, and the box with one unit of each active letter is $\mathfrak{B}_2(0, 1)$ of Definition 39.11(iv): the symbols of wall degree at most 2 at the levels $(0, 0)$ and $(0, 1)$; its stems are the Laurent symbols of wall degree at most 2. The 36 automorphisms keep it (Definition 39.11(vi)), and modulo them (Definition 39.16(i)) there are 104 orbits of stems and 329 orbits of symbols; the relations are the 104 letter identities $\alpha^u - \sum_T \alpha^{u+\mathbf{1}\varepsilon\setminus T}\Psi^{-1}$ of the stems and their $104 \cdot 6 = 624$ mark relations, 728 in all. One elimination, the orbits ranked as in Definition 39.16(ii), done modulo a prime of 61 bits at a random strength, which gives the pivots over $\mathbb{Q}(k)$ off finitely many strengths, leaves free among the Laurent orbits of wall degree at most 1 those of the seven orbit averages

$$b_1 = 1, \quad b_2 = \Big[\frac{\alpha_3\alpha_5\alpha_6}{\alpha_1\alpha_2\alpha_4}\Big], \quad b_3 = \Big[\frac{\alpha_3\alpha_6}{\alpha_1\alpha_2}\Big], \quad b_4 = \Big[\frac{\alpha_5\alpha_6}{\alpha_1\alpha_4}\Big],$$
$$b_5 = \Big[\frac{\alpha_6}{\alpha_1}\Big], \quad b_6 = \Big[\frac{\alpha_3\alpha_6}{\alpha_1\alpha_4}\Big], \quad b_7 = \Big[\frac{\alpha_2\alpha_3\alpha_6}{\alpha_1\alpha_4\alpha_5}\Big],$$

$[\,\cdot\,]$ the average over the 36 automorphisms, of wall degree 0 and 1 (Proposition 39.12(ii): the wall degree of α^u is $\max_U(-u_U)/\kappa_{1,U}$; for $b_2, \ldots, b_7$ it is 1, reached at $U = \{1\}$), and every target $f^{-1}b_i$, of wall degree at most 2, reduces to them. So the box is accepted at $\mathfrak{s} = 1$ (Definition 39.16(iii)), and by Theorem 39.17(i) $[f^{-1}b_i] = \sum_l \mathbf{B}_{il}(k)[b_l]$ in the module, the classes of $b_1, \ldots, b_7$ spanning a submodule $\mathfrak{N}$ that contains $[1]$.

The matrix (Procedure 41.8, the matrix over $\mathbb{Q}(k)$). Input: the 728 relations, with coefficients $c_0 + c_1 k$, $c_0, c_1 \in \mathbb{Z}$. Method: the targets need 292 of them, of rank 292; eliminations modulo primes of 62 bits at random strengths, 24 strengths on the first prime, every entry read as a quotient of polynomials, and Chinese remaindering with rational reconstruction over three primes. Output: the common denominator of the entries is $Q_c = (k - 2)^5(k + 1)$, and the row i of $\mathbf{B}$ is a factor ϕ_i times a row of polynomials:

i	ϕ_i	$\mathbf{B}_{il}/\phi_i$, $l = 1, \ldots, 7$
1	$\frac{6(2k-1)}{(k-2)(k+1)}$	6; $4(k-2)$; $6(k-1)$; $6(k-1)$; $8(k+1)$; -24; $-(k+1)$
2	$\frac{2(k-1)(2k-1)}{(k-2)^5(k+1)}$	$k^4 + 20k^3 - 36k^2 + 40k - 16$; $2k(k-2)(8k^2 - 19k + 10)$; $2k(12k^3 - 37k^2 + 42k - 20)$; $2k(12k^3 - 37k^2 + 42k - 20)$; $4k(k+1)(8k^2 - 12k + 9)$; $-4(k^4 + 20k^3 - 48k^2 + 36k - 8)$; $-8(k-1)^3(k+1)$

3	$\frac{2(k-1)(2k-1)}{(k-2)^4(k+1)}$	$-2k^3+33k^2-42k+16$; $(k-2)(15k^2-33k+14)$; $25k^3-75k^2+70k-16$; $19k^3-57k^2+70k-40$; $2k(k+1)(14k-11)$; $4(2k^3-27k^2+44k-20)$; $-(k-1)(k+1)(5k-6)$
4	$\frac{2(k-1)(2k-1)}{(k-2)^4(k+1)}$	the row 3 with its third and fourth entries exchanged
5	$\frac{2(k-1)(2k-1)}{(k-2)^3(k+1)}$	$24(k-1)$; $4(k-2)(3k-5)$; $2(9k^2-23k+16)$; $2(9k^2-23k+16)$; $(k+1)(33k-32)$; $4(3k^2-23k+22)$; $-(k+1)(3k-4)$
6	$\frac{2(k-1)(2k-1)}{(k-2)^4(k+1)}$	$2k^3+11k^2-26k+16$; $(k-2)(k-1)(7k-10)$; $13k^3-41k^2+40k-8$; $13k^3-41k^2+40k-8$; $(k+1)(33k^2-73k+48)$; $4(k^3-13k^2+23k-14)$; $-(k+1)(k^2-2)$
7	$\frac{2(k-1)(2k-1)}{(k-2)^5(k+1)}$	$-2k^4+23k^3-44k^2+53k-16$; $(k-2)(k-1)^2(16k-7)$; $27k^4-94k^3+99k^2-34k+22$; $27k^4-94k^3+99k^2-34k+22$; $(k+1)(8k^3+17k^2-35k+32)$; $-13k^4-39k^3+231k^2-273k+22$; $-(k+1)(17k^3-80k^2+131k-73)$

Check: **B** agrees with an elimination of the whole box at a prime and a strength drawn at random, and again at three fresh ones. Receipt: `rayreader.py N "1 1 1 1 1 1 ; 0 2"`, read by `analyze_run.py` and `matrix_rows_tex.py`; the re-check by `tests/check_run.py`.

Infinite strength. Every entry has a numerator of degree at most that of its denominator, so **B** is regular at $k=\infty$, with

$$\mathbf{B}(\infty)=\begin{pmatrix} 0 & 48 & 72 & 72 & 96 & 0 & -12\\ 4 & 64 & 96 & 96 & 128 & -16 & -32\\ -8 & 60 & 100 & 76 & 112 & 32 & -20\\ -8 & 60 & 76 & 100 & 112 & 32 & -20\\ 0 & 48 & 72 & 72 & 132 & 48 & -12\\ 8 & 28 & 52 & 52 & 132 & 16 & -4\\ -8 & 64 & 108 & 108 & 32 & -52 & -68 \end{pmatrix},$$

$$\det\big(\lambda-\mathbf{B}(\infty)\big)=(\lambda-24)^2\,(\lambda^2-324\lambda-10800)\,(\lambda^3+28\lambda^2-320\lambda+1728).$$

By Proposition 39.38(ii), at every nondegenerate native gauge α_c the vector $\beta_c=(b_l(\alpha_c))_l$ is an eigenvector of $\mathbf{B}(\infty)$ for $f(\alpha_c)^{-1}$. At $\alpha^\star$ every monomial of the orbit of α^u takes the value $(x^\star)^{u_1+u_2+u_3}$, so $\beta_\star=(1,\frac{1}{x^\star},\frac{1}{x^\star},\frac{1}{x^\star},\frac{1}{x^\star},1,x^\star)$ with $\frac{1}{x^\star}=\frac{\sqrt{21}+3}{6}$; the first row gives

$$\frac{48+72+72+96}{x^\star}-12x^\star=48(\sqrt{21}+3)-6(\sqrt{21}-3)=162+42\sqrt{21}=\frac{1800}{7\sqrt{21}-27}=\frac{1}{f^\star},$$

and the other rows the same, exactly. The eigenvalue 24 belongs to two rational native gauges: at $\alpha=(-1,\frac{3}{2},-1,1,1,1)$ and at $\alpha=(-1,-1,-1,1,-\frac{2}{3},1)$ every dual occupation is $R_e^*=\frac{1}{2}$, so every defect $1-2R_e^*$ vanishes, $f=\frac{1}{24}$, and the Hessian of $\log f$ on the chart has the determinant $\frac{625}{864}$ at both; their vectors

$$\beta=\big(1,\tfrac{17}{18},\tfrac{17}{18},-\tfrac{4}{9},-\tfrac{4}{9},-\tfrac{7}{18},\tfrac{11}{9}\big),\qquad \beta'=\big(1,\tfrac{17}{18},-\tfrac{4}{9},\tfrac{17}{18},-\tfrac{4}{9},-\tfrac{7}{18},\tfrac{11}{9}\big)$$

satisfy $\mathbf{B}(\infty)\beta=24\beta$ and $\mathbf{B}(\infty)\beta'=24\beta'$ exactly, and are independent. Receipt: `nsix_gauges24.py`.

The native gauges and the masters. Input: the critical equations $1-2R_e^*(\alpha)=0$ on the chart, saturated by the coordinates and the letters. Method: a Groebner basis over $\mathbb{Q}$, and, separately, the critical points of $\alpha^{\mathbf{a}}\Psi^{-b}$ for random complex exponents counted by monodromy and carried to $(\mathbf{1};2)$. Output: the ideal is zero-dimensional with 35 points counted with multiplicity, and the 35

native gauges found by the paths are distinct and nondegenerate, in 7 orbits of the automorphisms of sizes $1,1,3,3,9,9,9$: $\alpha^\star$, α', the two rational orbits of native value $\frac{1}{24}$, and three orbits whose native values are the roots of $1728\lambda^3 - 320\lambda^2 + 28\lambda + 1$, the real one $-0.0265230928\ldots$ and $0.1058541390\ldots \pm 0.1030231201\ldots\, i$. Check: Newton's method at 40 digits at each point, residual below 10^{-38}; the seven vectors β_c, one per orbit, have the singular values from 8.93 down to 0.358, and each is an eigenvector of $\mathbf{B}(\infty)$ to 10^{-38}. Receipts: `gb_direction.py`, `critpoints.py`, `polish.py`, `betas.py`. So the vectors β_c span $\mathbb{C}^7$, and by Proposition 39.38(i) the spanning set is a set of masters of $\mathfrak{N}$: $\dim_{\mathbb{C}(k)} \mathfrak{N} = 7$, $\mathbf{B} = \mathbf{B_b}$, and $\mathbf{B}(\infty)$ is diagonalizable with the eigenvalues $f(\alpha_c)^{-1}$, 24 twice for the two orbits of native value $\frac{1}{24}$.

(iv) *The law* (Procedure 41.8, the law). Input: $\mathbf{B}$. Method: the rows $w_0 = (1,0,\ldots,0)$, $w_i(k) = w_{i-1}(k-1)\,\mathbf{B}(k)$ of Proposition 39.6, evaluated modulo primes at consecutive strengths; their first dependence; the coefficients reconstructed over $\mathbb{Q}$ from four primes and made integral and primitive. Output: $w_0,\ldots,w_5$ are independent and $w_0,\ldots,w_6$ dependent, so $h = 6$, and the relation $\sum_{j=0}^{6} c_j(k)\langle 1\rangle_{k+j} = 0$ of Proposition 39.6(ii) has $c_j \in \mathbb{Z}[k]$ of degree 24, with coefficients of up to 31 digits. Check: the relation holds at a prime and a point drawn at random, and at three fresh ones. Since $b_1,\ldots,b_7$ are masters, Proposition 39.6(ii) makes it the law of [1] (Definition 39.1): the cyclic submodule of [1] has dimension 6, inside $\mathfrak{N}$ of dimension 7. Factored over $\mathbb{Q}$,

$$c_0 = k^5(k+1)^3(k+2)(k+3)\,r(k),$$
$$c_6 = 6998400\,(2k+1)(2k+3)(2k+5)(2k+7)(2k+9)(2k+11)(k+4)(k+5)^3\,r(k-1),$$

with r irreducible of degree 14 and integer coefficients of up to 18 digits, and the symbol (Corollary 38.7), the coefficients of k^{24},

$$\chi(\lambda) = 7276255875\,(24\lambda-1)(10800\lambda^2+324\lambda-1)(1728\lambda^3-320\lambda^2+28\lambda+1).$$

Its six roots are $\frac{1}{24}$, $f^\star$ and its conjugate (the roots of $10800\lambda^2+324\lambda-1$, $\frac{-27\pm7\sqrt{21}}{1800}$), and the three roots of the cubic: the six distinct native values of (iii), as Corollary 38.7 asks, and the reciprocals of the eigenvalues of $\mathbf{B}(\infty)$, 24 counted once.

Line-up strengths. In the notation of Proposition 39.6, $p_j = c_j$ and $q_6(k) = c_0(k-6)$, and Proposition 39.6(iii) gives

$$c_0(k-6) \;\Big|\; \det \mathbf{W}(k)_J \prod_{j=0}^{5} Q_c(k-j)^{6-j}, \qquad Q_c = (k-2)^5(k+1),$$

for every set J of six columns with $\det \mathbf{W}_J \neq 0$. The irreducible factor $r(k-6)$, of degree 14, divides no linear factor $Q_c(k-j)$, so it divides $\det \mathbf{W}(k)_J$: at its fourteen roots, none a root of a $Q_c(k-j)$, the rows $w_0,\ldots,w_5$ are dependent. So 14 of the 24 degrees of c_0 are strengths at which $\langle 1\rangle_k,\ldots,\langle 1\rangle_{k-5}$, read as linear forms in the values of the masters, line up; the other 10 are the integers $0,-1,-2,-3$.

(v) *The low end.* By Corollary 39.34(ii), c_0 vanishes to order $n-1=5$ at $k=0$, as it does, and with $\mathsf{L}_0 = V_1 = 66$ (Definition 39.33) the low-end relation $[k^5]c_0 \cdot 66 + \sum_{i=1}^{6} c_i(0)\langle 1\rangle_i = 0$, divided by 91238400, reads

$$64402904777 - 12632598108\,\langle 1\rangle_1 - 2849737197672\,\langle 1\rangle_2 + 380837271432240\,\langle 1\rangle_3$$
$$+ 14853466134400320\,\langle 1\rangle_4 - 369781615117353600\,\langle 1\rangle_5 + 3400346988339840000\,\langle 1\rangle_6 = 0.$$

By Corollary 39.34(i) the constant terms obey $\sum_{i=0}^{\min(6,j)} c_i(-j)\,\mathrm{CT}(f^{i-j}) = 0$ for every $j \in \mathbb{N}$; with the constant terms of Exercise 53.4 this holds exactly for $0 \le j \le 60$ (receipt `nsix_ctcheck.py`), a check of the law that uses neither the box nor the values.

(vi) *The value* (Procedure 41.8, the value). By Theorem 40.6 the values are the moments $\langle 1\rangle_k = \int_0^{f^\star} t^k \varrho_f(t)\,dt$ of the density ϱ_f of the stride's law (Definition 40.1); $f^\star$ is a simple root of χ, the density operator of the law has order 24, its exponent at $f^\star$ beyond $0,\dots,22$ is $\frac{n-3}{2} = \frac{3}{2}$, and the germ of ϱ_f is $(f^\star - t)^{3/2} g(f^\star - t)$ with g fixed by $g(0) = \big(2\Gamma(\frac12)^2/f^\star\big)^{5/2}\big/\big(\Gamma(\frac52)\sqrt{\det H}\big)$, the constant of the branch of Theorem 38.15. Input: the law, $f^\star$ and $\det H$. Method: the germ to 168 terms, the Beta sums $\int_0^{f^\star} t^K (f^\star - t)^{\frac32+m} dt = (f^\star)^{K+\frac52+m}\Gamma\,{}_2_1\!\left[\begin{matrix} K+1,\ \frac52+m \\ K+\frac72+m \end{matrix}\right]$ at the six strengths from $K = 57$ on, and the law run back to $k = 1$, in ball arithmetic at 1188 bits; again from $K = 82$. Output:

$$P_{\mathrm{N}_6}(\mathbf{1}; 0, 2) = \langle 1\rangle_1 = 4.75680256375822944340803728605708\ldots,$$

the two starting strengths agreeing to 50 digits. Check: the exponent read from the law at $f^\star$ is $\frac32$ to 16 digits; the reading at 700 digits of Exercise 53.7 agrees with this value, and on its values $\langle 1\rangle_1, \dots, \langle 1\rangle_6$ the low-end relation of (v), which the reading does not use, holds to 731 digits relative to its largest term. Receipt: `rayreader.py N "1 1 1 1 1 1 ; 0 2" --digits 30`.

So N_6 at $(\mathbf{1}; 0, 2)$ has the volume $V_1 = 66$, the native value $f^\star = \frac{7\sqrt{21}-27}{1800}$, seven masters with $\mathbf{B}(\infty)$ of spectrum $\{24, 24\}$, the reciprocals of the roots of $10800\lambda^2 + 324\lambda - 1$ and of $1728\lambda^3 - 320\lambda^2 + 28\lambda + 1$, a law of order 6 and degree 24, and the value $P_{\mathrm{N}_6}(\mathbf{1}; 0, 2) = 4.7568025637582294434\ldots$, with $P/V_1 = 0.0720727\ldots$ □

Exercise 53.6 (N_6 at its centre)**.** Read the centre ray of N_6 at the strength 1, with the direction $(\mathbf{1}; 1, 1)$ and the stride $f = \alpha^{\mathcal{E}}(K\Psi)^{-1}$, by the procedure of Definition 41.8: its walls and V_1; the native gauge of the direction and the signs; the box, the spanning set and its matrix at infinite strength; the law of [1], its ends, its symbol and its line-up strengths; the low-end relation, the closings and the value.

Solution. Walls and volume. With $m_1 = \mathbf{1}$ and $s = t = 1$, Definition 8.1 and $d(S) = 3 - r(\mathcal{E} \setminus S)$, $d^*(S) = |S| - r(S)$ give $\kappa_{1,S} = r(S) + r(\mathcal{E}\setminus S) - 3$, which depends on the type $(i, j) = (|S\cap a|, |S\cap b|)$ alone. With the recursion of Exercise 53.5(i):

(i,j)	$(1,0)$	$(0,1)$	$(2,0)$	$(1,1)$	$(0,2)$	$(3,0)$	$(2,1)$
$\kappa_{1,S}$	1	1	2	2	2	2	3
$F(i,j)$	1	1	1	1	1	$\frac32$	1

(i,j)	$(1,2)$	$(0,3)$	$(3,1)$	$(2,2)$	$(1,3)$	$(3,2)$	$(2,3)$
$\kappa_{1,S}$	3	2	2	2	2	1	1
$F(i,j)$	1	$\frac32$	$\frac94$	2	$\frac94$	$\frac{21}{2}$	$\frac{21}{2}$

Every wall is positive: 1 twelve times, 2 thirty-two times and 3 eighteen times, the triples a and b having the wall 2. By Proposition 39.30,

$$V_1 = 3\,F(2,3) + 3\,F(3,2) = \tfrac{63}{2} + \tfrac{63}{2} = 63.$$

The native gauge and the signs. On the centre ray $\alpha^\star = \mathbf{1}$ and $f^\star = K(\mathbf{1})^{-2} = 19^{-2} = \frac{1}{361}$ (Theorem 38.14, Definition 38.13), and $H = 2\big(\kappa_{\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\neq 6}$, the covariances under the weight law, which is uniform on the nineteen bases (Theorem 38.15): each mark of a lies in 9 bases and each mark of b in 10, and

$$H = \frac{1}{361}\begin{pmatrix} 180 & -48 & -48 & -28 & -28 \\ -48 & 180 & -48 & -28 & -28 \\ -48 & -48 & 180 & -28 & -28 \\ -28 & -28 & -28 & 180 & -48 \\ -28 & -28 & -28 & -48 & 180 \end{pmatrix}, \qquad \det H = \frac{580608}{361^3}.$$

By Theorem 38.17(ii) every sign η with $K(\eta) \neq 0$ is a native gauge of native value $K(\eta)^{-2}$. A sign is fixed by its partition of the marks into the scales 1 and -1, and the 32 signs of the chart $\eta_6 = 1$ fall into eight orbits of the automorphisms: $K(\eta) = 19$ when all scales agree; $|K(\eta)| = 5$ when two marks of a, or two marks of b, stand against the other four, two orbits of three signs; $|K(\eta)| = 3$ when one mark of a and one of b stand against the other four, nine signs; and $|K(\eta)| = 1$ on the other four orbits, one mark against five (3 and 3 signs), a against b (1), and two marks of a with one of b against the rest (9). The Hessian of $\log f$ at a sign, in the chart $\alpha_e = \eta_e e^{y_e}$, has the determinants $-\frac{580608}{361^3}$, $\frac{9216}{15625}$, $\frac{1024}{729}$ at the first three kinds and 0 at the last: the native values $\frac{1}{361}$, $\frac{1}{25}$ (two orbits) and $\frac{1}{9}$ are nondegenerate, and the native value 1 is degenerate at every sign (receipt `nsix_signs.py`). On the slice $(x, x, x, 1, 1, 1)$, fixed by the automorphisms, $f = x^2/\big((x^2+9x+9)(9x^2+9x+1)\big)$ and $\frac{d}{dx}\log f$ has the numerator $-18(x-1)(x+1)(x^2+5x+1)$; as at the lean, a critical point of the slice is a native gauge, and $x^2 + 5x + 1 = 0$ gives the two native gauges $\big(\frac{-5\pm\sqrt{21}}{2}\big)$ of native value $-\frac{1}{80}$, exchanged by the Cremona.

The box (Procedure 41.8, the masters). Both letters are active, and the symmetries are the 36 automorphisms and the Cremona (Proposition 39.7), 72 maps, which keep every box with equal level bounds. The box $\mathfrak{B}_2(1,1)$ is not accepted: at $\mathfrak{s} = 1$ eight Laurent orbits are free and seven of their targets do not reduce to them, and at $\mathfrak{s} = 0$ the target of 1 does not reduce. The box $\mathfrak{B}_3(1,1)$ has 252 orbits of stems and 1909 orbits of symbols, and 3774 relations: the letter identities of the symbols at the levels $(0,0)$, $(1,0)$ and $(0,1)$ and the $252 \cdot 6$ mark relations of the stems. It is accepted at $\mathfrak{s} = 2$, with the spanning set of the fifteen averages, over the 72 maps, of 1 and of α^u for

$$\begin{gathered} u = (-1,-1,0,0,1,1),\ (-1,-1,1,-1,1,1),\ (-1,-1,1,0,0,1),\ (-1,0,0,-1,1,1), \\ (-1,0,0,0,0,1),\ (-1,0,1,-1,0,1),\ (-1,0,1,0,0,0),\ (-1,-1,-1,1,1,1), \\ (-2,-2,0,0,2,2),\ (-2,-2,0,1,1,2),\ (-2,-2,1,-1,2,2), \\ (-2,-2,1,0,1,2),\ (-2,-2,1,1,1,1),\ (-2,-2,2,-2,2,2), \end{gathered}$$

of the wall degrees 1 (the first seven), $\frac{3}{2}$ (the eighth, at $U = a$) and 2 (Theorem 39.17(i)).

The matrix. Input: the 3774 relations. Method: as in Exercise 53.5; the targets need 1023 relations, of rank 1023; 55 strengths on the first prime; three primes. Output: $\mathbf{B} \in \mathrm{Mat}_{15}(\mathbb{Q}(k))$ with 8 vanishing entries, numerators and denominators of degree at most 17 in lowest terms, and the common denominator

$$Q_c = (2k-5)(k-2)^2(k-3)^5(k-4)\,S_9(k),$$

S_9 irreducible of degree 9, $S_9 = 6555027k^9 - 143215495k^8 + \cdots - 6615463200$. Check: $\mathbf{B}$ agrees with eliminations of the whole box at four primes and strengths drawn at random. Receipt: `rayreader.py N "1 1 1 1 1 1 ; 1 1"`, read by `analyze_run.py`. $\mathbf{B}$ is regular at $k = \infty$, and

$$\det\big(\lambda - \mathbf{B}(\infty)\big) = (\lambda - 361)\,(\lambda - 25)^2\,(\lambda - 9)\,(\lambda - 1)^8\,(\lambda + 80)\,\big(\lambda^2 + \tfrac{34}{3}\lambda - \tfrac{361}{3}\big).$$

By Proposition 39.38(ii) the vectors β_η of the nondegenerate signs and β of the slice gauge are eigenvectors: computed exactly, $\mathbf{B}(\infty)\beta_\eta = K(\eta)^2\beta_\eta$ at the signs of native values $\frac{1}{361}$, $\frac{1}{25}$ (both orbits) and $\frac{1}{9}$, and $\mathbf{B}(\infty)\beta = -80\beta$ at $\big(\frac{-5-\sqrt{21}}{2}\big)$, five independent vectors (receipt `nsix_centre_exact.py`). The roots of $\lambda^2 + \frac{34}{3}\lambda - \frac{361}{3}$ are $\frac{-17\pm14\sqrt{7}}{3}$, the reciprocals of $\frac{17\pm14\sqrt{7}}{361}$, which are native values of gauges off the signs and the slice: the monodromy count of Exercise 53.5, run at this direction, reaches eighteen nondegenerate native gauges at each (receipt `critpoints.py`).

The law. The rows of Proposition 39.6 first depend at $h = 11$, and the relation $\sum_{j=0}^{11} c_j(k)\langle 1\rangle_{k+j} = 0$ has $c_j \in \mathbb{Z}[k]$ of degree 52 with coefficients of up to 88 digits, from ten primes; it holds at random points of four primes. By Proposition 39.6(ii) this relation, of order 11 with coprime coefficients, annihilates [1], so the values satisfy it; its order is at least the dimension of the cyclic submodule of [1], with equality when the spanning set is a set of masters (Definition 39.1). Factored over $\mathbb{Q}$,

$$c_0 = 3\,(2k+1)\,k^5(k+1)^2(k+2)^2(k+3)\,S_{41}(k),$$
$$c_{11} = -4691556000\,(k+4)(k+5)(k+6)(k+7)^2(k+8)^2(k+9)^2(k+10)^2\,S_{41}(k-1),$$

with S_{41} irreducible of degree 41, and the symbol is

$$\chi(\lambda) \propto (361\lambda-1)(25\lambda-1)(9\lambda-1)(\lambda-1)^5(80\lambda+1)(361\lambda^2-34\lambda-3),$$

whose roots are the native values found: $\frac{1}{361}, \frac{1}{25}, \frac{1}{9}$, the degenerate 1 five times, $-\frac{1}{80}$ and $\frac{17\pm14\sqrt{7}}{361}$. The eigenvalues of $\mathbf{B}(\infty)$ are the reciprocals of the roots of χ, with their multiplicities, and one more 25 and three more 1.

Line-up strengths. $q_{11}(k) = c_0(k-11)$ divides $\det \mathbf{W}(k)_J \prod_{j=0}^{10} Q_c(k-j)^{11-j}$ (Proposition 39.6(iii)). The irreducible factors of Q_c have degrees 1 and 9, so $S_{41}(k-11)$ divides $\det \mathbf{W}(k)_J$: 41 of the 52 degrees of c_0 are strengths at which the rows $w_0, \dots, w_{10}$ line up, and the other 11 are the rational roots $0, -\frac{1}{2}, -1, -2, -3$.

The low end and the closings. c_0 vanishes to order 5 at 0, and the low-end relation of Corollary 39.34(ii), $[k^5]c_0 \cdot 63 + \sum_{i=1}^{11} c_i(0)\langle 1\rangle_i = 0$, has integer coefficients of up to 69 digits after division by their greatest common divisor 11377315676160 (receipt `lowend_forms.py`). The closings $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j})$ of Exercise 53.4 satisfy $\sum_{i=0}^{\min(11,j)} c_i(-j)\,\mathsf{C}_{j-i}(1) = 0$ exactly for $0 \le j \le 60$ (Corollary 39.32; receipt `nsix_ctcheck.py`).

The value. By Theorem 40.6, with $f^\star = \frac{1}{361}$ a simple root of χ, the density operator of order 52, the exponent $\frac{3}{2}$ at $f^\star$ (read from the law to 16 digits), and $g(0) = \big(2\Gamma(\frac{1}{2})^2 \cdot 361\big)^{5/2} \big/ \big(\Gamma(\frac{5}{2})\sqrt{\det H}\big)$: Beta sums from $K = 57$ and from $K = 82$ with 168 terms of the germ at 1188 bits, the law run back to $k = 1$. Output:

$$P_{\mathrm{N}_6}(\mathbf{1}; 1, 1) = \langle 1\rangle_1 = 4.57891247888240480638020373022855\ldots,$$

the two starts agreeing to 50 digits. Receipt: `rayreader.py N "1 1 1 1 1 1 ; 1 1" --digits 30`.

So the centre of N_6 has $V_1 = 63$, $f^\star = \frac{1}{361}$, a box accepted at $\mathfrak{B}_3(1,1)$ with fifteen spanning elements, the law of [1] of order 11 and degree 52, and $P_{\mathrm{N}_6}(\mathbf{1}; 1, 1) = 4.5789124788824048\ldots$, with $P/V_1 = 0.0726811\ldots$ □

Exercise 53.7 (The value span of N_6)**.** The *value span* of a ray is the $\mathbb{Q}$-span of its values $\langle 1\rangle_k$, $k \ge 1$. For the two rays of N_6 of Exercises 53.5 and 53.6:

(i) bound the value span by the law of [1] and the low-end relation;

(ii) find the integer relations of small height among 1 and the first values, read to 700 digits, by lattice reduction, and among them and the numbers $\ell = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$.

Solution. (i) *The lean.* The leading coefficient of the law, $c_6 = 6998400\,(2k+1)(2k+3)(2k+5)(2k+7)(2k+9)(2k+11)(k+4)(k+5)^3\,r(k-1)$, has its rational roots at negative numbers, and

r, irreducible of degree 14, has none; so $c_6(k) \neq 0$ for every integer $k \geq 0$. The law holds at every $k \geq 1$ (Corollary 39.34(i)), so $\langle 1\rangle_{k+6} = -c_6(k)^{-1} \sum_{i<6} c_i(k) \langle 1\rangle_{k+i}$ for $k \geq 1$, and by induction every value lies in the $\mathbb{Q}$-span of $\langle 1\rangle_1, \ldots, \langle 1\rangle_6$. The low-end relation of Exercise 53.5(v) has the constant term $64402904777 \neq 0$ and the coefficient $3400346988339840000 \neq 0$ at $\langle 1\rangle_6$, so it puts 1 in the value span and $\langle 1\rangle_6$ in $\mathbb{Q} + \mathbb{Q}\langle 1\rangle_1 + \cdots + \mathbb{Q}\langle 1\rangle_5$. So the value span of the lean is

$$\mathbb{Q} + \mathbb{Q}\langle 1\rangle_1 + \mathbb{Q}\langle 1\rangle_2 + \cdots + \mathbb{Q}\langle 1\rangle_5,$$

of dimension at most 6. *The centre.* The same with $c_{11} = -4691556000\,(k+4)(k+5)(k+6)(k+7)^2(k+8)^2(k+9)^2(k+10)^2\,S_{41}(k-1)$, nonzero at every integer $k \geq 0$, and the low-end relation of Exercise 53.6, whose constant term and coefficient of $\langle 1\rangle_{11}$ are nonzero: the value span of the centre is $\mathbb{Q} + \mathbb{Q}\langle 1\rangle_1 + \cdots + \mathbb{Q}\langle 1\rangle_{10}$, of dimension at most 11.

(ii) *The values.* Input: the laws and the native gauges of the two rays. Method: Theorem 40.6 at 700 digits, the Beta sums from the strengths 861 and 886 with 2580 terms of the germ, at 20224 bits, and the laws run back; this gives $\langle 1\rangle_1, \ldots, \langle 1\rangle_7$ at the lean and $\langle 1\rangle_1, \ldots, \langle 1\rangle_{12}$ at the centre. Check: the two starting strengths agree to 700 digits; the low-end relations, which the reading does not use, hold on the values to 731 digits relative to their largest terms; and the values agree with those of Exercises 53.5 and 53.6. Receipt: `valuespan_N.py`.

The relations. Method: for numbers $v_0 = 1, v_1, \ldots, v_N$, the Lenstra–Lenstra–Lovász reduction of the lattice spanned by the rows $\big(\mathbf{e}_i,\ \lfloor 10^{660} v_i \rfloor\big)$; a reduced row $(x, *)$ with a small residual $\sum_i x_i v_i$ is a candidate integer relation, and the first rows that are not relations show the size of the generic vectors. Output at the lean, for $1, \langle 1\rangle_1, \ldots, \langle 1\rangle_6$: three vectors of heights $3.0 \cdot 10^9$, $5.2 \cdot 10^9$ and $1.4 \cdot 10^{10}$ with residuals below 10^{-724},

$$\begin{aligned}
&15476 - 3451\langle 1\rangle_1 + 556832\langle 1\rangle_2 - 17388081\langle 1\rangle_3 \\
&\qquad + 111849150\langle 1\rangle_4 - 425617416\langle 1\rangle_5 - 2986275600\langle 1\rangle_6, \\
&84581 - 18857\langle 1\rangle_1 + 3035312\langle 1\rangle_2 - 97811235\langle 1\rangle_3 \\
&\qquad + 798867930\langle 1\rangle_4 - 5212068120\langle 1\rangle_5 + 1459458000\langle 1\rangle_6, \\
&5489069 - 1222045\langle 1\rangle_1 + 190733734\langle 1\rangle_2 - 4628283534\langle 1\rangle_3 \\
&\qquad + 13506396540\langle 1\rangle_4 - 1721929104\langle 1\rangle_5 - 44906400\langle 1\rangle_6,
\end{aligned}$$

and then vectors of height $3 \cdot 10^{157}$ and residual 10^{-503}. Check: the low-end relation, which is exact, lies in the rational span of the three (exact linear algebra; receipt `valuespan_exact.py`). Solved for the last values, the three read

$$\langle 1\rangle_4 = \tfrac{556429709}{1602852300}\langle 1\rangle_3 - \tfrac{137833049}{9617113800}\langle 1\rangle_2 + \tfrac{26497699}{288513414000}\langle 1\rangle_1 - \tfrac{170029763}{412162020000},$$

and like expressions of $\langle 1\rangle_5$ and $\langle 1\rangle_6$ with denominators of up to 20 digits. Among $1, \langle 1\rangle_1, \langle 1\rangle_2, \langle 1\rangle_3$ and ℓ the reduction returns one vector of height $1.1 \cdot 10^7$ and residual 10^{-727}, the next of height 10^{163}:

$$3906\,\langle 1\rangle_1 - 506772\,\langle 1\rangle_2 + 10837800\,\langle 1\rangle_3 = 15263 + 1047\,\ell.$$

ℓ is the value of the minor $\binom{\{1,2,3,5\}}{2} = (\mathrm{N}_6/4) \setminus 6$ of N_6 at its own full lean $(\mathbf{1}; 0, 2)$ and at its centre: the procedure reads $\binom{[4]}{2}$ at both, with laws of order 2 and degrees 3 and 3, and returns ℓ to the 32 digits printed (Exercise 53.10).

At the centre, for $1, \langle 1\rangle_1, \ldots, \langle 1\rangle_{12}$, four vectors of heights $5.4 \cdot 10^{23}$ to $1.1 \cdot 10^{24}$ with residuals below 10^{-714}, the next of height $2.5 \cdot 10^{62}$ and residual 10^{-598}; the law read at $k = 1$ and the low-end relation, both exact, lie in their rational span, and they express $\langle 1\rangle_9, \ldots, \langle 1\rangle_{12}$

through $1, \langle 1\rangle_1, \dots, \langle 1\rangle_8$. Among $1, \langle 1\rangle_1, \dots, \langle 1\rangle_8$, ℓ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ the reduction returns two vectors, of heights $1.4 \cdot 10^{25}$ and $1.6 \cdot 10^{25}$ and residuals below 10^{-710}, the next of height $2.9 \cdot 10^{67}$; their two coefficients at ℓ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ form the nonsingular matrix $\begin{pmatrix} 146909488464924911235 & -115139161649314013904 \\ -10928078403398636616 & 24473573045860222080 \end{pmatrix}$, so they write ℓ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ through the values. Here $7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ is the value of the minor $\binom{\{1,2,3,6\}}{1} = \mathrm{N}_6/\{4,5\}$ at its centre, and ℓ that of $\binom{\{1,2,3,5\}}{2}$.

So the value span of N_6 is $\mathbb{Q} + \sum_{i\le 5}\mathbb{Q}\langle 1\rangle_i$ at the lean and $\mathbb{Q} + \sum_{i\le 10}\mathbb{Q}\langle 1\rangle_i$ at the centre, of dimensions at most 6 and 11; the integer relations that lattice reduction finds at 700 digits, holding there to 724 and 714 digits, cut them to $\mathbb{Q} + \mathbb{Q}\langle 1\rangle_1 + \mathbb{Q}\langle 1\rangle_2 + \mathbb{Q}\langle 1\rangle_3$, containing ℓ, and to $\mathbb{Q} + \sum_{i\le 8}\mathbb{Q}\langle 1\rangle_i$, containing ℓ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$: at the lean the value of the minor $\binom{\{1,2,3,5\}}{2}$ at its own lean, at the centre the values of both minors on four marks at their centres, and besides them two numbers at the lean and six at the centre. □

Exercise 53.8 (V_8 at its full cobasis lean, read from its list)**.** Let V_8 be the list of rank four on $\mathcal{E} = \{1, \dots, 8\}$ with the pairs $P_1 = \{1,2\}$, $P_2 = \{3,4\}$, $P_3 = \{5,6\}$, $P_4 = \{7,8\}$, whose bases are the sixty-five sets of four marks other than the five unions $P_i \cup P_j$ with $\{i,j\} \neq \{3,4\}$. Read its ray with origin 0 and direction $(\mathbf{1}; 0, 2)$, of stride $f = \alpha^{\mathcal{E}}\Psi^{-2}$, by the procedure of Definition 41.8:

(i) the walls, V_1, the native gauge of the direction, $f^\star$ and $\det H$;

(ii) the box, its spanning set, the matrix $\mathbf{B}$ and its common denominator;

(iii) the relation of the rows and its line-up strengths;

(iv) the native gauges of the ray, and what they say about the dimension, the masters, the order of the law of [1] and $\mathbf{B}$ at infinite strength;

(v) the value $\langle 1\rangle_1 = P_{\mathrm{V}_8}(\mathbf{1}; 0, 2)$.

Solution. (i) *Walls and volume.* As for N_6 at its lean (Exercise 53.5(i)), $s = 0$, $t = 2$ and $d^*(S) = |S| - r(S)$ give $\kappa_{1,S} = 2r(S) - |S|$. Input: the 254 nonempty proper subsets of $\mathcal{E}$ and their ranks. Method: $\kappa_{1,S}$ for each, and the recursion over chains $F(U) = \kappa_{1,U}^{-1}\sum_{e\in U} F(U\setminus\{e\})$, $V_1 = \sum_e F(\mathcal{E}\setminus\{e\})$ of Exercise 53.5(i) (Proposition 39.30), in exact rationals. Output: the walls are 1 sixteen times, 2 sixty-one times, 3 one hundred and twelve times and 4 sixty-five times, all positive, and

$$V_1 = 300.$$

Check: the same program gives $V_1 = 66$ and 63 for the two rays of N_6, as computed by hand in Exercises 53.5 and 53.6. Receipt: `exact_data.py`.

The automorphisms. The non-bases are the unions of two pairs that involve P_1 or P_2, so an automorphism keeps $\{P_1, P_2\}$ and $\{P_3, P_4\}$; the 64 permutations that exchange the marks within pairs, P_1 with P_2 and P_3 with P_4 are automorphisms, and they keep the masses $\mathbf{1}$, so they are symmetries of the ray (Proposition 39.7(ii)).

The native gauge. The points fixed by the 64 automorphisms are, on the chart $\alpha_8 = 1$, the slice $\alpha = (x, x, x, x, 1, 1, 1, 1)$. A cobasis is the complement of a basis, so the cobases are the sets of four other than the complements $P_3\cup P_4$, $P_2\cup P_4$, $P_2\cup P_3$, $P_1\cup P_4$, $P_1\cup P_3$ of the non-bases. On the slice a set C of four has $\alpha^C = x^{|C\cap(P_1\cup P_2)|}$, the sets of four give $\sum_j \binom{4}{j}\binom{4}{4-j}x^j = 1+16x+36x^2+16x^3+x^4$, and the five sets left out give $1 + 4x^2$, so

$$\Psi = x\,(x^3 + 16x^2 + 32x + 16), \qquad f = \frac{x^4}{\Psi^2} = \frac{x^2}{(x^3 + 16x^2 + 32x + 16)^2},$$

$$\frac{d}{dx}\log f = \frac{-4\,(x^3+8x^2-8)}{x\,(x^3+16x^2+32x+16)}.$$

At a point fixed by the automorphisms the differential of $\log f$ in $y = \log\alpha$ is (u,u,u,u,v,v,v,v) with $u+v=0$, f being homogeneous of degree 0, and it vanishes when its derivative along the slice does, as in Exercise 53.5(ii). So the three roots of x^3+8x^2-8 give native gauges (Definition 38.4); at a root $x^3+16x^2+32x+16 = 8(x+1)(x+3)$, and the native value is

$$f = \Big(\frac{x}{8(x+1)(x+3)}\Big)^2.$$

The positive root $x^\star = 0.94566781799051109644\ldots$ gives $\alpha^\star$ (Theorem 38.14(i)) and

$$f^\star = 2.3709304548895617852\ldots\cdot 10^{-4};$$

the roots $-1.0748031540504515212\ldots$ and $-7.8708646639400595752\ldots$ give the native values $0.87033768220817625690\ldots$ and $8.6423132047649619559\ldots\cdot 10^{-4}$. The Hessian of $-\log f$ at $\log\alpha^\star$ on the chart is twice the covariances of the indicators under the cobasis law at $\alpha^\star$ (Theorem 38.15); reduced in $\mathbb{Q}(x^\star)$ (receipt `hessians.py`),

$$\det H = \frac{395719931704\,x^{\star 2} + 2812581517252\,x^\star - 3012741822955}{379727508532},$$

which is $0.0024055146373001934835\ldots$

(ii) *The box* (Procedure 41.8, the masters). Only Ψ is active. Input: the boxes $\mathfrak{B}_{\mathfrak{t}}(0,1)$ modulo the 64 automorphisms (Definition 39.11(iv), Definition 39.16(i)). Method: as in Exercise 53.5(iii). Output: $\mathfrak{B}_2(0,1)$ has 1155 orbits of stems, 4745 orbits of symbols and 10395 relations; at $\mathfrak{s}=1$ it leaves 36 free Laurent orbits of which 33 targets do not reduce to them, and at $\mathfrak{s}=0$ the target of 1 does not reduce: not accepted. $\mathfrak{B}_3(0,1)$ has 9723 orbits of stems, 32810 orbits of symbols and $87507 = 9723\cdot 9$ relations, the letter identities of the stems and their eight mark relations; one elimination leaves 60 free Laurent orbits of wall degree at most 2, every target reduces to them, and the box is accepted at $\mathfrak{s}=2$ (Theorem 39.17(i)), with a spanning set of 60 orbit averages. Receipt: `rayreader.py V "1 1 1 1 1 1 1 1 ; 0 2" --digits 30` (log `runlogs/V_lean.stdout`).

The matrix (the matrix over $\mathbb{Q}(k)$). Input: the 87507 relations. Method: the targets need 19581 of them, of rank 19581; eliminations modulo primes of 62 bits at random strengths; every entry read as a quotient of polynomials; Chinese remaindering with rational reconstruction. Output: on the first prime 178 strengths determine every entry, the largest sum of the degrees of a numerator and a denominator being 163; all the denominators divide one polynomial of degree 85, over which the numerators have degree at most 88, so each further prime needs 94 strengths; the entries become reconstructible over $\mathbb{Q}$ at seven primes, 432 bits. The common denominator is, up to a constant factor,

$$Q_c = k\,(k-1)^2(2k-3)(k-2)^8(2k-5)(k-3)^7\,S_{65}(k),$$

S_{65} irreducible of degree 65; 72 entries vanish, and in lowest terms the numerators have degree at most 83 and the denominators at most 80, so $d_{\mathbf{B}} = 83$. Check: $\mathbf{B}$ agrees with an elimination of the whole box at a prime and a strength drawn at random.

(iii) *The relation of the rows.* Input: $\mathbf{B}$. Method: the rows w_i of Proposition 39.6 modulo primes, their first dependence, reconstructed over $\mathbb{Q}$. Output (receipt as in (ii)): $w_0,\ldots,w_{23}$ are independent and w_{24} depends on them, so $h=24$, and the relation $\sum_{j=0}^{24} p_j(k)\langle 1\rangle_{k+j} = 0$ has $p_j \in \mathbb{Z}[k]$ of degree 377, with coefficients of up to 753 digits, from 63 primes, and over $\mathbb{Q}$

$$p_0 = k^7(k+1)^5(k+2)^5\prod_{j=3}^{7}(k+j)^3\prod_{j=8}^{13}(k+j)\,S_{339}(k),$$

S_{339} irreducible of degree 339, while p_{24} is a product of 38 linear factors, with roots at integers, halves and quarters from $-\frac{1}{2}$ to $-\frac{47}{2}$, and $S_{339}(k-1)$. Check: the relation holds at a prime and a point drawn at random. *Line-up strengths.* With $q_{24}(k) = p_0(k-24)$, Proposition 39.6(iii) gives

$$p_0(k-24) \;\Big|\; \det \mathbf{W}(k)_J \prod_{j=0}^{23} Q_c(k-j)^{24-j}$$

for every set J of 24 columns with $\det \mathbf{W}_J \neq 0$. The irreducible factors of Q_c have degree at most 65, so the irreducible $S_{339}(k-24)$ divides none of the $Q_c(k-j)$ and divides $\det \mathbf{W}(k)_J$: 339 of the 377 degrees of p_0 are strengths at which the rows $w_0, \ldots, w_{23}$ line up, and the other 38 are the integers $0, \ldots, -13$.

(iv) *The native gauges.* A native gauge of the ray is a point of X_F with $1 = 2R_e^*(\alpha)$ for every mark (Definition 38.4). Input: these equations, and the same with exponents $(\mathbf{a}; b)$ in place of $(\mathbf{1}; 2)$, $|\mathbf{a}| = 4b$, which are linear in the exponents. Method: as in Exercise 53.9(iii): Newton's method at random complex exponents, the solutions carried around random triangles of exponents until several turns in a row add nothing, run from two seeds; then every solution carried to $(\mathbf{1}; 2)$ along four routes, the straight segment and three broken ones through random exponents. Output: from both seeds, 839 solutions at random exponents, the count reaching 839 at the sixteenth and seventeenth turns and then staying there for four and six turns; at the direction, along every route, the same 331 nondegenerate native gauges, in 31 orbits of the 64 automorphisms, of sizes $1, 1, 1, 2, 2, 2, 2$, eight of 4, fourteen of 16 and two of 32, with 18 distinct native values:

one orbit:	$f^\star$, 0.000818456 (16), 0.000864231 (1), 0.124846403 (16), 0.870337682 (1), $-0.592832429 \pm 0.713227829\,i$ (16 each),
two orbits:	0.004487298 (4), -0.013533898 (8), -0.026238829 (8), 0.066968510 (32), $0.007758455 \pm 0.013607932\,i$ (32 each), 0.109146811 (32), $\frac{1}{8}$ (64), 0.870512702 (4),
three orbits:	0.021107646 (24), 0.416392354 (24),

the number of native gauges in brackets (renderings). The other 508 paths stop before the direction, at τ between 0.598 and $1 - 1.8 \cdot 10^{-6}$ along the segment, where the step collapses. Check: one native gauge of each orbit polished by Newton's method at 40 digits, with the Hessian determinants of modulus at least $\det H = 0.0024055\ldots$, reached at $\alpha^\star$; the three native gauges of the slice in (i) are the three orbits of size 1. Receipts: `critpoints.py` (`crit_V8_lean.json`, `crit_V8_lean_seed2.json`), `census_paths.py` (`census_V8_routes.json`), `escape_check.py` (`escape_V8.out`), `betas.py` (`betas_V8.out`).

So, by Theorem 38.6(iii), the ray has dimension at least 331; by Theorem 39.8(iii) the invariant module under the 64 automorphisms has dimension at least 31; and by Corollary 38.7 the law of [1] has order at least 18, so that with (iii) and Proposition 39.6(ii) it lies between 18 and 24. The vectors $\beta_c = (b_l(\alpha_c))_l$ of the spanning set at one native gauge of each orbit have rank 31, their singular values running from 1262 down to 0.036 (receipt `betas.py`): they do not span $\mathbb{C}^{60}$, and Proposition 39.38(i) does not decide whether the 60 are masters. At large strength the entries of $\mathbf{B}$ grow like k^3 (output, receipt as in (ii)): $\mathbf{B}$ is not regular at $k = \infty$, and Proposition 39.38(ii) does not apply to it.

(v) *The value* (Theorem 40.6). Input: the relation of (iii), $f^\star$ and $\det H$; $f^\star$ is a simple root of the symbol, and the exponent of the density at $f^\star$ is $\frac{n-3}{2} = \frac{5}{2}$. Method: as in Exercise 53.5(vi). Output (receipt as in (ii)):

$$P_{\mathrm{V}_8}(\mathbf{1}; 0, 2) = \langle 1 \rangle_1 = 7.12823937562968675472229135353893457156078898940720766490704\ldots,$$

two starting strengths agreeing to $2.4 \cdot 10^{-82}$, and $P/V_1 = 0.0237607979187656\ldots$ Check: the low-end relation of Corollary 39.34(ii), with $V_1 = 300$, holds on the values to 73 digits.

So V_8 at $(\mathbf{1}; 0, 2)$ has $V_1 = 300$, $f^\star = \big(x^\star/(8(x^\star+1)(x^\star+3))\big)^2$ with $x^{\star 3} + 8x^{\star 2} = 8$, a box accepted at $\mathfrak{B}_3(0,1)$ with 60 spanning elements, a relation of order 24 and degree 377 of which 339 degrees are line-up strengths, at least 331 native gauges in 31 orbits with 18 native values, and the value $7.1282393756296867547\ldots$ □

Exercise 53.9 (Sizing a reading: Q_6 and N_7)**.** Let Q_6 be the list of rank three on $\{1,\ldots,6\}$ whose bases are the seventeen triples other than $\{1,3,6\}$, $\{2,3,4\}$ and $\{2,5,6\}$, and N_7 the list of rank three on $\{1,\ldots,7\}$ whose bases are the thirty-four triples other than $\{1,2,5\}$. For the rays with origin 0 of Q_6 along $(1,3,2,3,1,2;2,2)$ and of N_7 along $(1,2,2,1,2,1,1;2,1)$, find what sizes their reading before any relation is written: the walls, V_1, the native gauge of the direction, the native gauges of the ray and their native values, with the bounds these give on the dimension of the ray and on the order of a law of [1]; then the size of the box with one unit of each letter at $\mathfrak{t} = 2$, and whether it is accepted.

Solution. (i) *Walls and volume* (Definition 8.1, Proposition 39.30). For Q_6, $s = t = 2$ and the masses add up to $12 = 2\cdot 3 + 2\cdot 3$, on the balance wall; the walls $\kappa_{1,S} = m_{1,S} - 2d(S) - 2d^*(S)$ of the 62 sets S are, by height and count,

$$1(4),\ 2(7),\ 3(13),\ 4(13),\ 5(12),\ 6(9),\ 7(3),\ 8(1),$$

all positive, and the recursion over chains of Exercise 53.5(i), run over the 62 sets (no automorphism of Q_6 fixes the masses), gives $V_1 = \frac{16493}{2430}$. For N_7, $s = 2$, $t = 1$ and $10 = 2\cdot 3 + 1\cdot 4$; the 126 walls are $1(7),\ 2(17),\ 3(29),\ 4(39),\ 5(29),\ 6(5)$, and $V_1 = \frac{3519}{128}$. The automorphisms of N_7 fixing the masses are the 12 permutations that exchange the marks 2 and 5 of the triple $\{1,2,5\}$ or not and permute the marks $4,6,7$ of mass 1. Receipt: `exact_data.py`.

(ii) *The native gauge of the direction* (Theorem 38.14(i)), by Newton's method on the chart: for Q_6, $\alpha^\star = (0.45796,\ 3.83890,\ 1.55835,\ 3.85580,\ 0.54507,\ 1)$, $f^\star = 8.3244647577707389\ldots \cdot 10^{-5}$ and $\det H = 0.13385450029314584\ldots$; for N_7, $\alpha^\star = (1.15660,\ 3.63195,\ 3.14019,\ 1,\ 3.63195,\ 1,\ 1)$, constant on the orbits of the automorphisms as it must be, $f^\star = 7.2002200376070769\ldots \cdot 10^{-5}$ and $\det H = 0.027073861743142843\ldots$ (renderings; receipt `hessians.py`).

(iii) *The native gauges.* A native gauge is a point of X_F at which $m_{1,e} = s\,R_e + t\,R_e^*$ for every mark (Definition 38.4). Input: these equations, and the same with the exponents $(\mathbf{a}; b_K, b_\Psi)$ in place of $(\mathbf{m}_1; s, t)$, $|\mathbf{a}| = 3b_K + (n-3)b_\Psi$, which are linear in the exponents. Method: for random exponents, a Groebner basis of the equations, multiplied by $K\Psi$, on the chart and saturated by the coordinates and the letters, modulo the prime 32003, and, separately, Newton's method at random complex exponents with the solutions carried around random triangles of exponents until several turns add nothing; then each solution carried along the segment to $(\mathbf{m}_1; s, t)$, a path that either arrives at a point of X_F or runs to its boundary; and a Groebner basis at $(\mathbf{m}_1; s, t)$ itself. Output: for Q_6, 114 solutions at random exponents by both methods; 90 of the paths arrive, at 90 distinct native gauges, all nondegenerate, with 90 distinct native values, and 24 run off; at the direction the Groebner basis has exactly 90 points counted with multiplicity, so these are all the native gauges. For N_7, 1174 solutions at random exponents; all 1174 arrive, at distinct nondegenerate native gauges, in 268 orbits of the 12 automorphisms (32 of size 1, 28 of size 2, 86 of size 3, 106 of size 6, 16 of size 12), with 83 distinct native values. Check: the native gauges of Q_6 polished at 40 digits, residuals below 10^{-37}, the Hessian determinants at least 0.13, the native values at least $1.6 \cdot 10^{-4}$ apart, and the same 90 values from two further runs with other random exponents. Receipts: `gb_count.py`, `gb_direction.py`, `critpoints.py`, `polish.py`.

So, by Theorem 38.6(iii), the dimension of the ray is at least 90 for Q_6 and at least 1174 for N_7; by Theorem 39.8(iii) the invariant module of N_7 under its 12 automorphisms has dimension at least 268; and by Corollary 38.7 every relation $\sum_j p_j(k) f^j \in \mathcal{L}$, the law of [1] among them, has order at least 90 for Q_6 and 83 for N_7, the number of distinct native values. These counts are read off the list and the direction alone.

(iv) *The box of* Q_6. Both letters are active, and the box with one unit of each is $\mathfrak{B}_2(1,1)$ (Definition 39.11(iv)). Input: the box, with no symmetry. Output: 361682 relations on 225874 symbols, the mark relations of 31785 stems and the letter identities, each of $1 + |\mathcal{B}| = 18$ terms (receipt `box_test.py Q6.json "1 3 2 3 1 2 ; 2 2" 2 1 1`). Method for the rest: the procedure, as in Exercise 53.5(iii) and (iv); receipt `rayreader.py Q6.json "1 3 2 3 1 2 ; 2 2" --digits 30`. Output: one elimination modulo a prime at a random strength leaves 114 free Laurent symbols of wall degree at most 1, and every target reduces to them, so the box is accepted at $\mathfrak{s} = 1$, and the spanning set of 114 spans a submodule that contains [1] (Theorem 39.17(i)). The targets need 78921 of the relations. Modulo the first prime **B** has a common denominator of degree 126 and numerators of degree up to 129, so it is not regular at $k = \infty$; it needs 271 strengths on that prime and 135 on each further one; and modulo that prime the rows first depend at $h = 90$, where 12288 consecutive strengths do not determine the relation, so that its degree exceeds 6100. By (iii) and Proposition 39.6(ii), $90 \le h \le 114$ whatever the spanning set, and the rows meet the lower bound.

The box of N_7. The box $\mathfrak{B}_2(1,1)$ modulo the 12 automorphisms has 71320 orbits of symbols, 7613 orbits of stems and 100719 relations of up to $1 + |\mathcal{B}| = 35$ terms. One elimination modulo $2^{61} - 1$ at a random strength has rank 63883; of the 359 Laurent orbits of wall degree at most 1, 247 are free, and their targets all lie in the box, but 246 of them do not reduce to the 247; at $\mathfrak{s} = 0$ the target of 1 does not reduce. So the box with one unit of each letter at $\mathfrak{t} = 2$ is not accepted (Definition 39.16(iii)), and Theorem 39.18 sends the search to larger boxes. By Definition 39.16(ii) the class of every invariant Laurent symbol of wall degree at most 1 is a combination of the 247 free ones, fewer than the 268 dimensions of the invariant module found in (iii). Receipt: `box_test.py N7.json "1 2 2 1 2 1 1 ; 2 1" 2 1 1`. The next boxes of the search are larger: $\mathfrak{B}_2(2,1)$ has 236600 relations on 158724 symbols, and $\mathfrak{B}_3(1,1)$ has 707989 relations of up to 35 terms (receipt for the search, `rayreader.py N7.json "1 2 2 1 2 1 1 ; 2 1" --digits 30`).

So before any relation is written, the walls give $V_1 = \frac{16493}{2430}$ and $\frac{3519}{128}$, and the critical equations give at least 90 and 1174 dimensions and laws of order at least 90 and 83; the box of Q_6 at $\mathfrak{t} = 2$ is accepted with 114 spanning elements, and the box of N_7 with one unit of each letter at $\mathfrak{t} = 2$ is not accepted. □

Exercise 53.10 (A table of rays read by the procedure)**.** Let $\binom{[n]}{1}$ be the list of rank one on $\{1, \dots, n\}$ whose bases are the single marks, $\binom{[n]}{2}$ the list of rank two whose bases are all the pairs, H_4 the list of rank two on $\{1,2,3,4\}$ whose bases are the five pairs other than $\{1,2\}$, and W_3 the wheel with three spokes (Proposition 39.49), of rank three on its six spokes and rims, whose bases are the sixteen spanning trees of its graph. Read by the procedure of Definition 41.8 the rays with origin 0 of

$$\tbinom{[3]}{1},\ \tbinom{[4]}{1},\ \tbinom{[5]}{1},\ \tbinom{[4]}{2},\ \tbinom{[5]}{2},\ \mathrm{H}_4,\ \mathrm{N}_6 \text{ along } (\mathbf{1}; 1, 1); \qquad \tbinom{[4]}{2},\ W_3,\ \mathrm{N}_6,\ \mathrm{V}_8 \text{ along } (\mathbf{1}; 0, 2);$$
$$\tbinom{[3]}{1} \text{ along } (2\mathbf{1}; 4, 1); \qquad \mathrm{H}_4 \text{ along } (1, 1, 3, 3; 3, 1),$$

N_6 and V_8 as in Exercises 53.4 and 53.8, and the ray of $\binom{[3]}{1}$ with origin $(u_0; 0, 0)$, $u_0 = (-\frac{1}{2}, 0, \frac{1}{2})$, and direction $(\mathbf{1}; 1, 1)$, whose value at $k = 1$ is $P_{\binom{[3]}{1}}(\frac{1}{2}, 1, \frac{3}{2}; 1, 1)$. For each ray give:

(i) the accepted box with its stems, symbols and relations modulo the symmetries of the ray, the bound $\mathfrak{s}$ at which it is accepted, the size $\mathfrak{m}$ of its spanning set, the largest degree $d_{\mathbf{B}}$ of

a numerator or a denominator of an entry of $\mathbf{B}$ in lowest terms, and the order h and the degree D of the relation of the rows;

(ii) whether the spanning set is a set of masters and that relation the law of [1], and the spectrum of $\mathbf{B}(\infty)$ where $\mathbf{B}$ is regular at $k = \infty$;

(iii) V_1 and the value $\langle 1 \rangle_1$, with the checks that the reading allows.

Solution. (i) *The boxes, the matrices and the rows.* Input: the list and the direction. Method: the procedure as in Exercise 53.5(iii) and (iv). The boxes $\mathfrak{B}_\mathfrak{t}$ with one unit of each active letter are tried for $\mathfrak{t} = 2, 3, \ldots$ (Definition 39.11(iv), Theorem 39.18), each reduced modulo the symmetries of the ray: the automorphisms that fix the masses and the origin, and the Cremona at a centre with origin 0 (Proposition 39.7, Definition 39.16(i)). One elimination modulo a prime of 61 bits at a random strength decides acceptance (Definition 39.16(iii)); $\mathbf{B}$ over $\mathbb{Q}(k)$ comes from eliminations of the relations that the targets need, modulo primes of 62 bits at random strengths, with rational reconstruction; and the rows w_i of Proposition 39.6 give the first dependence, reconstructed over $\mathbb{Q}$ from several primes. Output:

ray	box	stems	symbols	relations	$\mathfrak{s}$	$\mathfrak{m}$	$d_\mathbf{B}$	(h, D)
$\binom{[3]}{1}$, $(\mathbf{1}; 1, 1)$	$\mathfrak{B}_2(1,1)$	4	30	32	1	2	2	$(2, 2)$
$\binom{[4]}{1}$, $(\mathbf{1}; 1, 1)$	$\mathfrak{B}_2(1,1)$	5	42	46	1	2	3	$(2, 3)$
$\binom{[5]}{1}$, $(\mathbf{1}; 1, 1)$	$\mathfrak{B}_3(1,1)$	10	100	112	2	3	5	$(3, 4)$
$\binom{[3]}{1}$, $(2\mathbf{1}; 4, 1)$	$\mathfrak{B}_2(1,1)$	17	79	124	1	2	4	$(2, 6)$
$\binom{[3]}{1}$, $(\mathbf{1}; 1, 1)$, u_0	$\mathfrak{B}_2(1,1)$	19	110	149	1	4	3	$(3, 4)$
$\binom{[4]}{2}$, $(\mathbf{1}; 0, 2)$	$\mathfrak{B}_2(0,1)$	8	20	40	1	2	4	$(2, 3)$
$\binom{[4]}{2}$, $(\mathbf{1}; 1, 1)$	$\mathfrak{B}_2(1,1)$	7	33	66	1	2	4	$(2, 3)$
$\binom{[5]}{2}$, $(\mathbf{1}; 1, 1)$	$\mathfrak{B}_3(1,1)$	21	234	245	2	5	8	$(4, 6)$
H_4, $(\mathbf{1}; 1, 1)$	$\mathfrak{B}_2(1,1)$	17	102	180	1	4	3	$(4, 7)$
H_4, $(1, 1, 3, 3; 3, 1)$	$\mathfrak{B}_2(1,1)$	115	631	1000	1	4	9	$(4, 25)$
W_3, $(\mathbf{1}; 0, 2)$	$\mathfrak{B}_2(0,1)$	66	211	462	1	2	8	$(2, 5)$
N_6, $(\mathbf{1}; 0, 2)$	$\mathfrak{B}_2(0,1)$	104	329	728	1	7	6	$(6, 24)$
N_6, $(\mathbf{1}; 1, 1)$	$\mathfrak{B}_3(1,1)$	252	1909	3774	2	15	17	$(11, 52)$
V_8, $(\mathbf{1}; 0, 2)$	$\mathfrak{B}_3(0,1)$	9723	32810	87507	2	60	83	$(24, 377)$

At the centres of $\binom{[5]}{1}$, $\binom{[5]}{2}$ and N_6 the box $\mathfrak{B}_2(1,1)$ is not accepted: at $\mathfrak{s} = 1$ it leaves 2, 3 and 8 free Laurent orbits of which 1, 2 and 7 targets do not reduce to them, and at $\mathfrak{s} = 0$ the target of 1 does not reduce; at the lean of V_8, $\mathfrak{B}_2(0,1)$ leaves 36 free orbits at $\mathfrak{s} = 1$, of which 33 targets do not reduce, and fails at $\mathfrak{s} = 0$ the same way. Every other box in the table is the first one tried. Check: every $\mathbf{B}$ agrees with an elimination of the whole box at a prime and a strength drawn at random, and every relation of the rows holds at a random prime and point; on the thirteen rays other than V_8 both checks are repeated at three fresh primes and points. Receipts: `runs_table.sh`, `table_summary.py`, `tests/check_run.py` (`table_checkrun.out`).

(ii) *Masters and the law of* [1]. By Proposition 39.6(ii), with Theorem 39.17(iii), h is at least the dimension of the cyclic submodule of [1], with equality when the spanning set is a set of masters, and then the relation of the rows is the law of [1]; by Corollary 38.7 that dimension is at least the number of distinct native values of nondegenerate native gauges. So the relation is the law of [1] as soon as either the vectors $\beta_c = (b_l(\alpha_c))_l$ at nondegenerate native gauges span $\mathbb{C}^\mathfrak{m}$ (Proposition 39.38(i)), or h distinct native values are found; in the second case, when moreover $h = \mathfrak{m}$, the cyclic submodule of [1] fills the span of the $\mathfrak{m}$ classes, and the spanning set is a set of masters. Input: the critical equations of each direction. Method: the census of Exercise 53.9(iii),

monodromy at random exponents and paths to the direction; each native gauge polished at 40 digits with its Hessian; and the vectors β_c at one native gauge of each orbit, each b_l averaged over the symmetries of the ray. Output and conclusion:

- $\binom{[3]}{1}$, $\binom{[4]}{1}$ at their centres: the table's gauge, of native value $\frac{1}{9}$ and $\frac{1}{16}$, and the signs with one scale -1, of native value 1 and $\frac{1}{4}$ (Theorem 38.17(ii)), 4 and 5 native gauges in all, every one nondegenerate; the two vectors β_c are independent: masters, and the law of [1].
- $\binom{[3]}{1}$ along $(\mathbf{21};4,1)$: 4 native gauges, of native values $\frac{1}{243}$ and $-\frac{27}{64}$, two independent vectors: masters. The ray with origin u_0: the 4 native gauges of the centre direction, which no symmetry of the ray permutes, give four independent vectors: masters.
- $\binom{[5]}{2}$ at its centre: 66 native gauges in 6 orbits of the 120 automorphisms, of native values $\frac{1}{100}, \frac{1}{4}, -\frac{1}{8}, 1$, all nondegenerate, with vectors of rank 5: masters, and $h = 4 < \mathfrak{m}$, the cyclic submodule of [1] of dimension 4 inside $\mathfrak{N}$ of dimension 5 (Exercise 52.20).
- $\binom{[5]}{1}$ at its centre: 16 native gauges, the table's gauge ($\frac{1}{25}$), the five signs with one scale -1 ($\frac{1}{9}$) and the ten with two (1); three distinct native values and $h = \mathfrak{m} = 3$: masters and the law of [1], although the three vectors β_c lie in a plane. $\binom{[4]}{2}$ at both directions: 4 native gauges, of native values $\frac{1}{36}$ (the table's gauge) and $\frac{1}{4}$ (an orbit of three), with $h = \mathfrak{m} = 2$, and W_3 at $(\mathbf{1};0,2)$: 3 native gauges, of native values $\frac{1}{256}$ and $\frac{1}{4}$, with $h = \mathfrak{m} = 2$: masters and the law of [1], the two vectors β_c being proportional.
- N_6: Exercises 53.5 and 53.6. V_8: Exercise 53.8.
- H_4: of the 6 critical points at random exponents, two arrive at nondegenerate native gauges at each direction. At the centre they are the table's gauge and the sign $(-1,-1,1,1)$, of native values $\frac{1}{25}$ and $\frac{1}{9}$ and Hessian determinants $-\frac{32}{625}$ and $\frac{32}{81}$ on the chart; along $(1,1,3,3;3,1)$ they are the points $(x,x,1,1)$ with $3x^2+8x-1=0$, of native values $\frac{7643\pm1064\sqrt{19}}{1366875}$, the larger one, at $x=\frac{\sqrt{19}-4}{3}>0$, being $f^\star$. The other four paths meet in pairs at the points $(\mp i,\pm i,-1,1)$ at the centre and $(\mp\frac{i}{\sqrt{3}},\pm\frac{i}{\sqrt{3}},-1,1)$ along $(1,1,3,3;3,1)$, native gauges of native value 1 whose Hessian is singular (exact; receipt `h4_gauges.py`). So Corollary 38.7 bounds the dimension of the cyclic submodule of [1] below by 2 only, and the two vectors β_c span a plane: the relation of the rows is a relation of order 4 in $\mathcal{L}$, of order at least that of the law of [1] (Theorem 39.17(iii)), which lies between 2 and 4.

Infinite strength. $\mathbf{B}$ is regular at $k=\infty$ on nine of the rays, with the spectra in the table of (iii), and at every nondegenerate native gauge of the census β_c is an eigenvector of $\mathbf{B}(\infty)$ for $f(\alpha_c)^{-1}$, to 10^{-38} at 40 digits (Proposition 39.38(ii)). On $\binom{[5]}{1}$, $\binom{[4]}{2}$ and W_3 some entries of $\mathbf{B}$ have numerators of larger degree than their denominators, and Proposition 39.38(ii) says why they must: nondegenerate native gauges of distinct native values have dependent vectors β_c there, three in a plane for $\binom{[5]}{1}$ and two on a line for the others, which a regular $\mathbf{B}$ would make independent eigenvectors. For V_8 see Exercise 53.8.

(iii) *The values.* V_1 is computed from the walls by Proposition 39.30 with the recursion over chains of Exercise 53.5(i) (receipt `exact_data.py`); for the ray with origin u_0 it is that of the direction $(\mathbf{1};1,1)$. The values come from Theorem 40.6. Input: the relation of the rows, $f^\star$ and $\det H$. Method: as in Exercise 53.5(vi), asked to 60 digits. Output:

ray	V_1	eigenvalues of $\mathbf{B}(\infty)$	$\langle 1\rangle_1$
$\binom{[3]}{1}$, $(\mathbf{1};1,1)$	6	9, 1	2.3439072386894588906…
$\binom{[4]}{1}$, $(\mathbf{1};1,1)$	24	16, 4	8.4143983221171599977…
$\binom{[5]}{1}$, $(\mathbf{1};1,1)$	120	not regular	39.945588011996245698…
$\binom{[3]}{1}$, $(2\mathbf{1};4,1)$	1	243, $-\frac{64}{27}$	0.030080945282262304399…
$\binom{[3]}{1}$, $(\mathbf{1};1,1)$, u_0	6	9, 1, 1, 1	4.7773947923309302997…
$\binom{[4]}{2}$, $(\mathbf{1};0,2)$	12	not regular	2.3439072386894588906…
$\binom{[4]}{2}$, $(\mathbf{1};1,1)$	12	not regular	2.3439072386894588906…
$\binom{[5]}{2}$, $(\mathbf{1};1,1)$	30	100, 4, 4, 1, −8	4.0297141707110496073…
H_4, $(\mathbf{1};1,1)$	16	25, 9, 1, 1	4.0962369983378828500…
H_4, $(1,1,3,3;3,1)$	$\frac{160}{27}$	$\frac{7643\pm1064\sqrt{19}}{27}$, 1, 1	0.48830319489526888388…
W_3, $(\mathbf{1};0,2)$	84	not regular	7.2123414189575657123…
N_6, $(\mathbf{1};0,2)$	66	Exercise 53.5	4.7568025637582294434…
N_6, $(\mathbf{1};1,1)$	63	Exercise 53.6	4.5789124788824048063…
V_8, $(\mathbf{1};0,2)$	300	not regular	7.1282393756296867547…

Check: on every ray the two starting strengths agree to at least 82 digits. On the thirteen rays with origin 0 the low-end relation of Corollary 39.34(ii), $[k^{n-1}]p_0 \cdot V_1 + \sum_{i=1}^{h} p_i(0)\langle 1\rangle_i = 0$, which the reading does not use and which holds for the relation of the rows as well as for the law of [1], holds on the values $\langle 1\rangle_1, \dots, \langle 1\rangle_h$ to at least 89 digits relative to its largest term on the twelve read here at 60 digits, and to 73 digits on V_8 (Exercise 53.8(v)). Five values agree with numbers computed otherwise: those of $\binom{[3]}{1}$ at its centre and of $\binom{[4]}{2}$ at both directions with $\frac{1}{3}\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\frac{1}{3}+\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\frac{2}{3}+\mathbb{Z}})((0,\infty))\big)$, summed as the series of Definition 5.2, to 90 digits; those of $\binom{[4]}{1}$ at its centre and of W_3 at $(\mathbf{1};0,2)$ with $7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $6\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, to 89 digits; and that of the ray with origin u_0 with

$$\int_0^\infty 2\log\frac{(1+u^2)^2}{u^2}\,\frac{du}{1+u^2+u^4},$$

computed by quadrature at 90 digits, to 84 digits. Receipts: `table_check.py` (`table_check.out`), `table_masters.py` (`table_masters.out`).

So the procedure reads every ray of the table from its list and its direction, with boxes of up to 87507 relations; the relation of the rows is the law of [1] on all of them but the two rays of H_4 and V_8, where the census bounds the law's order below by 2 and 18; every value passes the low-end relation, and the five that can be compared with numbers computed otherwise agree with them to 84 digits or more. In particular $\binom{[4]}{2}$ has the value ℓ of $\binom{[3]}{1}$ at its centre both at $(\mathbf{1};0,2)$ and at $(\mathbf{1};1,1)$, to 90 digits. □

Chapter 54

Exercises on the content

The exercises of this chapter work the statements of Chapter 45 on the two smallest centre values met in the exercises on rays. The value ℓ of the list of three marks is placed in the content of level 3 as a combination of two trails of depth one and of two chains; the value of the half-lattice met at four marks is placed in the content of level 2, and the coset relation returns it to level 1. The cutting rule is then computed on the symbols of two trails, subset by subset, and the three parts of Theorem 45.9 are checked on them. Throughout, $\omega = e^{2\pi i/3}$, so that $\omega^3 = 1$, $\omega^2 + \omega + 1 = 0$, $\bar{\omega} = \omega^2 = \omega^{-1}$ and $\mu_3 = \{1, \omega, \omega^2\}$. Every number displayed is exact; decimal values are labelled renderings, and the receipts that recompute them are the scripts named in each solution.

54.1 Two centre values in the content

Exercise 54.1 (The values of three and four marks as trails and chains)**.** Let $\ell = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = 3\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$ (Exercise 52.4) and $v = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty))$ (Exercise 52.7).

(i) Write ℓ as a combination, with coefficients in $\mathbb{Q}(\mu_3)$, of the trails of level 3 of depth one at the index 2 and the frequencies $\frac{1}{3}$ and $\frac{2}{3}$ (Definition 45.1), and show that ℓ is a value of $\mathcal{C}_2^{(3)}$.

(ii) Write these two trails as chains of level 3 from 0 to 1 (Proposition 45.5(i)), check the first by integrating it, and write ℓ as a combination of chains.

(iii) Show that v is a value of $\mathcal{C}_3^{(2)}$ by writing it with the trails of level 2 at the index 3 and the frequencies 0 and $\frac{1}{2}$. Then use the coset relation to show that v is a value of $\mathcal{C}_3^{(1)}$, find the trail at the frequency $\frac{1}{2}$ in terms of the one at the frequency 0, and write v as a chain.

Solution. Receipt: `ct_values.py`.

(i) *The cosets as frequencies.* For an integer m, $\omega^m - \omega^{-m}$ is 0, $\omega - \omega^2$ and $\omega^2 - \omega$ for m in $3\mathbb{Z}$, $1 + 3\mathbb{Z}$ and $2 + 3\mathbb{Z}$, as $\omega^{-1} = \omega^2$. So the difference of the two indicators is

$$\mathbf{1}_{1+3\mathbb{Z}}(m) - \mathbf{1}_{2+3\mathbb{Z}}(m) = \frac{\omega^m - \omega^{-m}}{\omega - \omega^2} \qquad (m \in \mathbb{Z}).$$

At the index 2 every sum below converges absolutely, so the value of a lattice input at the cut 0 is the sum of its density times m^{-2} over the points $m > 0$ (Definition 5.2), and the sums may be split and recombined. With $\omega^{-m} = e^{2\pi i \cdot 2m/3}$,

$$(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty)) = \sum_{m\geq 1} \frac{\omega^m - \omega^{-m}}{\omega-\omega^2}\, m^{-2}$$
$$= \frac{1}{\omega-\omega^2}\Big((\mathsf{T}_{2;(1)}[x]\, e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\, e^{4\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big),$$

the lattice inputs $e^{2\pi i a x}\mathbb{I}_{\mathbb{Z}}$ at the frequencies $a = \frac{1}{3}$ and $a = \frac{2}{3}$ on the counting lattice (Definition 5.2).

The constant. $(\omega-\omega^2)^2 = \omega^2 - 2\omega^3 + \omega^4 = \omega^2 + \omega - 2 = -3$. So $\frac{1}{\omega-\omega^2} = -\frac{\omega-\omega^2}{3}$ and $\frac{3}{\omega-\omega^2} = \omega^2 - \omega$, and

$$\ell = (\omega^2-\omega)\Big((\mathsf{T}_{2;(1)}[x]\, e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\, e^{4\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big).$$

The trails. For $d=1$, $\nu_1 = 2$ and $a_1 \in \{\frac{1}{3}, \frac{2}{3}\} \subseteq \frac{1}{3}\mathbb{Z}$, the trail of level 3 of Definition 45.1 is $\big(\mathsf{T}_{2;I_1}[y]\, e^{2\pi i a_1 y}\,\mathbb{I}_{\mathbb{Z}}\big)(\{0<y\}) = \sum_{m\geq 1} e^{2\pi i a_1 m} m^{-2} = (\mathsf{T}_{2;(1)}[x]\, e^{2\pi i a_1 x}\mathbb{I}_{\mathbb{Z}})((0,\infty))$: it has the depth 1, the positive integer index 2 and a frequency in $\frac{1}{3}\mathbb{Z}$, and the condition at index one does not arise. Write T' and T'' for the trails at $\frac{1}{3}$ and $\frac{2}{3}$ in this solution. Then

$$\ell = \omega^2 T' - \omega T' - \omega^2 T'' + \omega T'',$$

the value of the combination $[\omega^2 T'] - [\omega T'] - [\omega^2 T''] + [\omega T'']$, with the rational coefficients ± 1, of the classes of the words ζT with $\zeta \in \mu_3$ and T a trail of level 3 of weight 2. So ℓ is a value of $\mathcal{C}_2^{(3)}$. Since $\bar{\omega} = \omega^2$, $T'' = \overline{T'}$, and $\ell = (\omega^2-\omega)(T' - \overline{T'}) = (-i\sqrt{3})(2i\,\mathrm{Im}\,T') = 2\sqrt{3}\,\mathrm{Im}\,T'$, with $\mathrm{Im}\,T' = \sum_{m\geq 1}\sin(\frac{2\pi m}{3})m^{-2} = \frac{\sqrt{3}}{2}\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$, the S_2 of Exercise 52.4.

Rendering: both sides of the display for ℓ are $2.3439072386894588\ldots$.

(ii) *The chains.* Proposition 45.5(i) with $d=1$ and $\nu_1 = 2$ gives $c_1 = e^{2\pi i a_1}$, the letters c_1^{-1} and one 0, and the sign $(-1)^1$. At $a_1 = \frac{1}{3}$, $c_1 = \omega$ and $c_1^{-1} = \omega^2$; at $a_1 = \frac{2}{3}$, $c_1 = \omega^2$ and $c_1^{-1} = \omega$:

$$T' = -\mathrm{I}(0;\omega^2,0;1), \qquad T'' = -\mathrm{I}(0;\omega,0;1).$$

The check. For the ends 0 and 1 the path is $\gamma(t) = t$, and for a letter b with $|b| = 1$, $b \neq 1$ (Definition 45.4),

$$\mathrm{I}(0;b,0;1) = \int_{0<t_1<t_2<1} \frac{dt_1}{t_1 - b}\,\frac{dt_2}{t_2}.$$

For $0 \leq t < 1$, $1 - t/b$ lies in the open disc of radius 1 about 1, where the principal logarithm is holomorphic; $\log(1-t/b)$ vanishes at $t=0$ and has the derivative $\frac{-1/b}{1-t/b} = \frac{1}{t-b}$, so the inner integral is $\log(1-t_2/b) = -\sum_{n\geq 1} b^{-n} t_2^n/n$. The series of absolute values, $\sum_n t_2^{n-1}/n$, integrates over $(0,1)$ to $\sum_n n^{-2} < \infty$, so the series integrates term by term, $\int_0^1 t_2^{n-1}\,dt_2 = \frac{1}{n}$:

$$\mathrm{I}(0;b,0;1) = -\sum_{n\geq 1} b^{-n} n^{-2}.$$

At $b = \omega^2$, $b^{-n} = \omega^{-2n} = \omega^n$, and $\mathrm{I}(0;\omega^2,0;1) = -\sum_n \omega^n n^{-2} = -T'$, as Proposition 45.5(i) states.

Rendering: the double integral is $0.5483113556160754\ldots - 0.6766277376064357\ldots i$, and so is $-T'$.

The value. Substituting,

$$\ell = (\omega^2-\omega)\big(-\mathrm{I}(0;\omega^2,0;1)+\mathrm{I}(0;\omega,0;1)\big) = (\omega-\omega^2)\big(\mathrm{I}(0;\omega^2,0;1)-\mathrm{I}(0;\omega,0;1)\big),$$

a combination of chains of level 3 from 0 to 1 with first letter not 0 and last letter not 1, as Proposition 45.5(iii) describes the values of $\mathcal{C}_2^{(3)}$.

(iii) *Level* 2. The positive points of $\frac{1}{2}+\mathbb{Z}$ are the $\frac{2m-1}{2}$, $m\geq 1$, so $v=\sum_{m\geq 1} 8(2m-1)^{-3}$ (Definition 5.2), absolutely convergent. The indicator of the odd integers is $\frac{1-(-1)^n}{2}=\frac{1-e^{\pi i n}}{2}$, so

$$v = 8\sum_{n\geq 1}\frac{1-e^{\pi i n}}{2}\,n^{-3} = 4\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 4\,(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

$(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty))$ are the trails of depth one at the index 3 and the frequencies 0 and $\frac{1}{2}$, both in $\frac{1}{2}\mathbb{Z}$: trails of level 2. So v is a value of $\mathcal{C}_3^{(2)}$, as Proposition 45.3(i) gives directly for the coset $b=\frac{1}{2}\in\frac{1}{2}\mathbb{Z}$ at the frequency 0, with $M=2$ and $M'=1$.

Level 1. The coset relation of Exercise 52.7, Theorem 5.12(i) at $q=2$ and $b=0$, is $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))+v = 8\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. So

$$v = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),$$

a rational multiple of a trail of level 1, and v is a value of $\mathcal{C}_3^{(1)}$. Comparing the two expressions of v, $4\,(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = 4\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$:

$$(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\tfrac{3}{4}\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

The chain. Proposition 45.5(i) with $d=1$, $\nu_1=3$ and $a_1=0$ gives $c_1=1$, the letters $1,0,0$, and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\mathrm{I}(0;1,0,0;1)$; so $v=-7\,\mathrm{I}(0;1,0,0;1)$.

Rendering: $v=8.4143983221171599\ldots$, the value of both expressions above; the ratio of the trail at the frequency $\frac{1}{2}$ to the trail at the frequency 0 renders as -0.75, and the triple integral $\mathrm{I}(0;1,0,0;1)$ as $-1.2020569031595942\ldots$.

Answer.

$$\begin{aligned}\ell &= (\omega^2-\omega)\Big((\mathsf{T}_{2;(1)}[x]\,e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,e^{4\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big)\\ &= (\omega-\omega^2)\big(\mathrm{I}(0;\omega^2,0;1)-\mathrm{I}(0;\omega,0;1)\big),\end{aligned}$$

a value of $\mathcal{C}_2^{(3)}$. $v = 4\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 4\,(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty))$ is a value of $\mathcal{C}_3^{(2)}$, and by the coset relation $v = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -7\,\mathrm{I}(0;1,0,0;1)$, a value of $\mathcal{C}_3^{(1)}$, with $(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\frac{3}{4}\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. □

54.2 The cutting rule on two trails

Exercise 54.2 (The cutting rule on $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and on the trail of index 2 at the frequency $\frac{1}{3}$)**.** Let σ_3 be the symbol of the trail $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of level 1, and σ' that of the trail $(\mathsf{T}_{2;(1)}[x]\,e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of level 3 (Definition 45.7).

(i) Compute $\Delta\sigma_3$ and $\Delta\sigma'$ (Definition 45.8), subset by subset.

(ii) Check the three parts of Theorem 45.9 on σ_3 and σ': coassociativity and the counit, the weights, and the compatibility with the reversal and the rotations.

(iii) Read the left factors of positive weight as symbols of trails (Proposition 45.5(ii)), and compute the chains of the two factors of weight one in $\Delta\sigma'$.

Solution. Receipts: `ct_cutting.py` for (i) and (ii), `ct_values.py` for the renderings in (iii).

The symbols. Both trails have depth $d = 1$. For $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\nu_1 = 3$ and $c_1 = 1$; for $(\mathsf{T}_{2;(1)}[x]\,e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\nu_1 = 2$ and $c_1 = \omega$, $c_1^{-1} = \omega^2$. So, with $x = \mathrm{I}^{\mathfrak{s}}(0;1,0,0;1)$ and $y = \mathrm{I}^{\mathfrak{s}}(0;\omega^2,0;1)$,

$$\sigma_3 = -x, \qquad \sigma' = -y.$$

Δ is $\mathbb{Q}(\mu_M)$-linear, so $\Delta\sigma_3 = -\Delta x$ and $\Delta\sigma' = -\Delta y$. In $\mathfrak{S}_M$ a run with no letters is 1, and a run of at least one letter between equal ends is 0 (Definition 45.7); these two rules decide every right factor below.

(i) *The cutting rule on x.* The entries are $a_0 = 0$, $a_1 = 1$, $a_2 = 0$, $a_3 = 0$, $a_4 = 1$, and $w = 3$. For each $S \subseteq \{1,2,3\}$, the left factor keeps the letters of S between the ends 0 and 1, and the right factors are the runs between consecutive kept positions $s_0 = 0 < s_1 < \cdots < s_{|S|+1} = 4$:

S	left factor	runs, and the term
$\varnothing$	1	$\mathrm{I}^{\mathfrak{s}}(0;1,0,0;1) = x$: the term $1 \otimes x$
$\{1\}$	$\mathrm{I}^{\mathfrak{s}}(0;1;1)$	$\mathrm{I}^{\mathfrak{s}}(0;\varnothing;1) = 1$, $\mathrm{I}^{\mathfrak{s}}(1;0,0;1) = 0$: no term
$\{2\}$	$\mathrm{I}^{\mathfrak{s}}(0;0;1)$	$\mathrm{I}^{\mathfrak{s}}(0;1;0) = 0$: no term
$\{3\}$	$\mathrm{I}^{\mathfrak{s}}(0;0;1)$	$\mathrm{I}^{\mathfrak{s}}(0;1,0;0) = 0$: no term
$\{1,2\}$	$\mathrm{I}^{\mathfrak{s}}(0;1,0;1)$	1, 1, $\mathrm{I}^{\mathfrak{s}}(0;0;1)$
$\{1,3\}$	$\mathrm{I}^{\mathfrak{s}}(0;1,0;1)$	1, $\mathrm{I}^{\mathfrak{s}}(1;0;0)$, 1
$\{2,3\}$	$\mathrm{I}^{\mathfrak{s}}(0;0,0;1)$	$\mathrm{I}^{\mathfrak{s}}(0;1;0) = 0$: no term
$\{1,2,3\}$	x	1, 1, 1, 1: the term $x \otimes 1$

With $z = \mathrm{I}^{\mathfrak{s}}(0;1,0;1)$ and $R = \mathrm{I}^{\mathfrak{s}}(0;0;1) + \mathrm{I}^{\mathfrak{s}}(1;0;0)$,

$$\Delta x = x \otimes 1 + 1 \otimes x + z \otimes R.$$

The cutting rule on y. The entries are $a_0 = 0$, $a_1 = \omega^2$, $a_2 = 0$, $a_3 = 1$, and $w = 2$:

S	left factor	runs, and the term
$\varnothing$	1	y: the term $1 \otimes y$
$\{1\}$	$\mathrm{I}^{\mathfrak{s}}(0;\omega^2;1)$	$\mathrm{I}^{\mathfrak{s}}(0;\varnothing;\omega^2) = 1$, $\mathrm{I}^{\mathfrak{s}}(\omega^2;0;1)$
$\{2\}$	$\mathrm{I}^{\mathfrak{s}}(0;0;1)$	$\mathrm{I}^{\mathfrak{s}}(0;\omega^2;0) = 0$: no term
$\{1,2\}$	y	1, 1, 1: the term $y \otimes 1$

With $A = \mathrm{I}^{\mathfrak{s}}(0;\omega^2;1)$ and $B = \mathrm{I}^{\mathfrak{s}}(\omega^2;0;1)$,

$$\Delta y = y \otimes 1 + 1 \otimes y + A \otimes B.$$

(ii) *The pieces are primitive.* A formal chain u of weight one has two subsets: $S = \varnothing$ gives $1 \otimes u$ and $S = \{1\}$ gives $u \otimes 1$, its two runs being empty. So $\Delta u = u \otimes 1 + 1 \otimes u$ for $u = \mathrm{I}^{\mathfrak{s}}(0;0;1)$, $\mathrm{I}^{\mathfrak{s}}(1;0;0)$, A and B, and $\Delta R = R \otimes 1 + 1 \otimes R$ by linearity. For z, with the entries $0,1,0,1$: $S = \{1\}$ leaves the run $\mathrm{I}^{\mathfrak{s}}(1;0;1) = 0$ and $S = \{2\}$ the run $\mathrm{I}^{\mathfrak{s}}(0;1;0) = 0$, so $\Delta z = z \otimes 1 + 1 \otimes z$ as well.

Coassociativity. Δ is an algebra homomorphism with $\Delta 1 = 1 \otimes 1$, so on y

$$(\Delta \otimes \mathrm{id})\Delta y = 1 \otimes 1 \otimes y + (y \otimes 1 + 1 \otimes y + A \otimes B) \otimes 1 + (A \otimes 1 + 1 \otimes A) \otimes B,$$

$$(\mathrm{id}\otimes\Delta)\Delta y = y\otimes 1\otimes 1 + 1\otimes(y\otimes 1 + 1\otimes y + A\otimes B) + A\otimes(B\otimes 1 + 1\otimes B),$$

and both are the sum of the six tensors $1\otimes 1\otimes y$, $1\otimes y\otimes 1$, $y\otimes 1\otimes 1$, $A\otimes B\otimes 1$, $A\otimes 1\otimes B$ and $1\otimes A\otimes B$. On x the same expansion, with z and R primitive, gives on both sides the six tensors $1\otimes 1\otimes x$, $1\otimes x\otimes 1$, $x\otimes 1\otimes 1$, $z\otimes R\otimes 1$, $z\otimes 1\otimes R$ and $1\otimes z\otimes R$.

The counit. $\varepsilon(1) = 1$ and ε vanishes on the formal chains, so $(\varepsilon\otimes\mathrm{id})\Delta x = x + \varepsilon(z)R = x$ and $(\mathrm{id}\otimes\varepsilon)\Delta x = x + z\,\varepsilon(R) = x$, and in the same way on y. This is Theorem 45.9(i) on the two symbols.

The weights. The terms of Δx have the weights $(3,0)$, $(0,3)$ and $(2,1)$, those of Δy the weights $(2,0)$, $(0,2)$ and $(1,1)$; the sums are 3 and 2, and $\Delta x - x\otimes 1 - 1\otimes x = z\otimes R$ and $\Delta y - y\otimes 1 - 1\otimes y = A\otimes B$ have both factors of positive weight: Theorem 45.9(ii).

The reversal. $\varrho y = (-1)^2\,\mathrm{I}^{\mathfrak{s}}(1;0,\omega^2;0)$. Its entries $1,0,\omega^2,0$ give: $S=\varnothing$, $1\otimes\varrho y$; $S=\{1\}$, the left factor $\mathrm{I}^{\mathfrak{s}}(1;0;0)$ and the run $\mathrm{I}^{\mathfrak{s}}(0;\omega^2;0) = 0$, no term; $S=\{2\}$, the left factor $\mathrm{I}^{\mathfrak{s}}(1;\omega^2;0)$ and the runs $\mathrm{I}^{\mathfrak{s}}(1;0;\omega^2)$ and 1; $S=\{1,2\}$, $\varrho y\otimes 1$. On the other side $\varrho A = -\mathrm{I}^{\mathfrak{s}}(1;\omega^2;0)$ and $\varrho B = -\mathrm{I}^{\mathfrak{s}}(1;0;\omega^2)$, so

$$\Delta\varrho y = \varrho y\otimes 1 + 1\otimes\varrho y + \mathrm{I}^{\mathfrak{s}}(1;\omega^2;0)\otimes\mathrm{I}^{\mathfrak{s}}(1;0;\omega^2) = (\varrho\otimes\varrho)\Delta y.$$

$\varrho x = (-1)^3\,\mathrm{I}^{\mathfrak{s}}(1;0,0,1;0)$. Of the eight subsets of the entries $1,0,0,1,0$, those with a term are $\varnothing$, $\{1,2,3\}$, $\{1,3\}$, with the left factor $\mathrm{I}^{\mathfrak{s}}(1;0,1;0)$ and the run $\mathrm{I}^{\mathfrak{s}}(0;0;1)$, and $\{2,3\}$, with the same left factor and the run $\mathrm{I}^{\mathfrak{s}}(1;0;0)$; the other four leave a run between equal ends. So $\Delta\varrho x = \varrho x\otimes 1 + 1\otimes\varrho x - \mathrm{I}^{\mathfrak{s}}(1;0,1;0)\otimes R$. On the other side $\varrho z = (-1)^2\,\mathrm{I}^{\mathfrak{s}}(1;0,1;0)$ and $\varrho R = -\mathrm{I}^{\mathfrak{s}}(1;0;0) - \mathrm{I}^{\mathfrak{s}}(0;0;1) = -R$, so $(\varrho\otimes\varrho)(z\otimes R) = -\mathrm{I}^{\mathfrak{s}}(1;0,1;0)\otimes R$, and $\Delta\varrho x = (\varrho\otimes\varrho)\Delta x$.

The rotations. ρ_λ, $\lambda\in\mu_3$, multiplies every entry by λ and fixes the entry 0. It changes neither which ends of a run are equal nor which runs are empty, so the same subsets give the terms of $\Delta\rho_\lambda y$, with every entry multiplied by λ: $\Delta\rho_\lambda y = \rho_\lambda y\otimes 1 + 1\otimes\rho_\lambda y + \mathrm{I}^{\mathfrak{s}}(0;\lambda\omega^2;\lambda)\otimes\mathrm{I}^{\mathfrak{s}}(\lambda\omega^2;0;\lambda) = (\rho_\lambda\otimes\rho_\lambda)\Delta y$. At level 1 the only rotation of x is the identity. This is Theorem 45.9(iii) on the two symbols.

(iii) *The left factor of* Δx. The chain $\mathrm{I}(0;1,0;1)$ has the first letter $1\neq 0$ and the last letter $0\neq 1$. In Proposition 45.5(ii) its one nonzero letter is at $j_1 = 1$, with $j_2 = w+1 = 3$, so $d=1$, $c_1=1$, $\nu_1 = 2$, $c_2 = 1$, and the frequency $a_1' = 0$: $\mathrm{I}(0;1,0;1) = -(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, and $z = -\sigma_2$ with σ_2 the symbol of $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. The right factor $R = \mathrm{I}^{\mathfrak{s}}(0;0;1) - \varrho\,\mathrm{I}^{\mathfrak{s}}(0;0;1)$ is a formal chain of weight one minus its reversal; the first letter of $\mathrm{I}^{\mathfrak{s}}(0;0;1)$ is its first end, so it has no realization (Definition 45.7), and $\int_0^1 dt/t$ diverges. Rendering: $\mathrm{I}(0;1,0;1) = -1.6449340668482264\ldots$.

The factors of Δy. A has the letter $\omega^2\neq 0,1$. In Proposition 45.5(ii), $d=1$, $c_1=\omega$, $\nu_1 = 1$ and $c_2 = 1$, with the frequency $a_1' = \frac{1}{3}$, $e^{2\pi i a_1'} = \omega$: $\mathrm{I}(0;\omega^2;1)$ is minus the trail of level 3 of depth one at the index 1 and the frequency $\frac{1}{3}$, $\lim_N\sum_{m\le N}\omega^m m^{-1}$, which is a trail because $e^{2\pi i/3}\neq 1$ (Definition 45.1); A is minus its symbol. On the path $\gamma(t) = t$, $t-\omega^2 = t + \frac{1}{2} + \frac{\sqrt{3}}{2}i$ has the imaginary part $\frac{\sqrt{3}}{2} > 0$, so the principal logarithm is continuous along it and

$$\mathrm{I}(0;\omega^2;1) = \int_0^1 \frac{dt}{t-\omega^2} = \log\Big(\frac{3}{2} + \frac{\sqrt{3}}{2}i\Big) - \log\Big(\frac{1}{2} + \frac{\sqrt{3}}{2}i\Big) = \Big(\frac{1}{2}\log 3 + \frac{\pi i}{6}\Big) - \frac{\pi i}{3} = \frac{1}{2}\log 3 - \frac{\pi i}{6}.$$

B has the ends ω^2 and 1, so it is a chain between two singular points with no end at 0, read as it stands. Its letter 0 is not on the open segment from ω^2 to 1 and differs from both ends, so the chain is defined and converges (Definition 45.4, Proposition 45.6(i)). On $\gamma(t) = \omega^2 + t(1-\omega^2) = -\frac{1}{2} + \frac{3}{2}t - \frac{\sqrt{3}}{2}(1-t)i$ the imaginary part is negative for $t<1$, so the argument of $\gamma(t)$ runs

continuously from $-\frac{2\pi}{3}$ to 0 and the modulus from 1 to 1:

$$\mathrm{I}(\omega^2;0;1)=\int_0^1\frac{\gamma'(t)\,dt}{\gamma(t)}=\log 1-\log\omega^2=0-\Big(-\frac{2\pi i}{3}\Big)=\frac{2\pi i}{3}.$$

Rendering: the integrals render as $0.5493061443340548\ldots-0.5235987755982988\ldots i$ and $2.0943951023931954\ldots i$.

Answer. With σ_2 the symbol of $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $\sigma_1'=-\mathrm{I}^{\mathfrak{s}}(0;\omega^2;1)$ the symbol of the trail of index 1 at the frequency $\frac{1}{3}$,

$$\Delta\sigma_3=\sigma_3\otimes 1+1\otimes\sigma_3+\sigma_2\otimes\big(\mathrm{I}^{\mathfrak{s}}(0;0;1)+\mathrm{I}^{\mathfrak{s}}(1;0;0)\big),\qquad \Delta\sigma'=\sigma'\otimes 1+1\otimes\sigma'+\sigma_1'\otimes\mathrm{I}^{\mathfrak{s}}(\omega^2;0;1).$$

Both are coassociative and counital, graded by the weight, and compatible with the reversal and the rotations, as Theorem 45.9 states. The left factors σ_2 and σ_1' are symbols of trails; the right factor $\mathrm{I}^{\mathfrak{s}}(\omega^2;0;1)$ is a chain between the singular points ω^2 and 1 with the value $\frac{2\pi i}{3}$, and $\mathrm{I}(0;\omega^2;1)=\frac{1}{2}\log 3-\frac{\pi i}{6}$. □

Aliases

The text uses the program's own names. This list gives, for a reader who arrives from the literature, the classical names under which some of its objects and results appear there. An alias is a name and nothing more: the statement of the handbook, with its hypotheses, is the one that holds, and where the classical result is narrower or wider the handbook's statement decides.

Lists, tables and their letters

In this handbook	Classical name
a table, its marks and bases (Definition 7.1)	a matroid with weights on its bases, its ground set and bases
a list (the weights all 1)	a matroid
the list $\binom{\mathcal{E}}{r}$, $\lvert\mathcal{E}\rvert = n$	the uniform matroid $U_{r,n}$
the dual table	the dual matroid
a molecule; a connected table	a connected component; a connected matroid
the width $c(S)$ of a set (Definition 7.12)	the connectivity $r(S)+r(\mathcal{E}\setminus S)-r(\mathcal{E})$ of the set
the two-sum $\mathcal{A} \oplus_g \mathcal{A}'$ of two cells (Lemma 21.2)	the 2-sum of matroids
for the list of the spanning trees of a graph: the basis polynomial K and the cobasis polynomial Ψ	the spanning-tree polynomial, and the graph polynomial, also called the first Symanzik or the Kirchhoff polynomial
the Cremona (Theorem 8.14)	the Cremona transformation $\alpha \mapsto \alpha^{-1}$
the period at the full cobasis lean $(\mathbf{1}; 0, 2)$ of the list of the spanning trees of a primitive logarithmically divergent graph	its Feynman period $\int \Omega/\Psi^2$; the volume of the direction V_1 of that ray is its Hepp bound
the list $\binom{[n]}{1}$	the cycle matroid of the banana graph with n edges
the wheel W_p (Proposition 39.49); W_3	the cycle matroid of the wheel with p spokes; W_3 is the cycle matroid of K_4
H_4	the cycle matroid of a triangle with one edge doubled
N_6	$U_{3,6}$ with one basis removed: six points of a plane, exactly three of them collinear
V_8	the Vámos matroid
the whirl list on six marks (rank three, three dependent sets of three marks meeting pairwise)	the rank-three whirl

In this handbook	Classical name
F_7, A_9, Z_9 (Chapter 30)	the Fano plane, the affine plane $AG(2,3)$, the Pappus configuration

Values

In this handbook	Classical name
$(\mathsf{T}_{m;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$	$\zeta(m)$
$\Gamma(\frac{1}{2})^2$	π
$(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))$; so $P_{\binom{[3]}{1}}(\mathbf{1};1,1) = 3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))$	$L(\chi_{-3},2)$; the value is $3L(\chi_{-3},2)$
$P_{\binom{[4]}{1}}(\mathbf{1};1,1) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; the period of W_3 at $(\mathbf{1};0,2)$, $6\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$	$7\zeta(3)$; $6\zeta(3)$
the signed sums of Trailmoments of lattice inputs at the cosets prime to $\lvert D\rvert$, weighted by the Kronecker signs	$L(\chi_D,2)$
the arc integrals of the three-mark centre (Part XI)	the log-sine integrals $\mathrm{Ls}_j(\frac{2\pi}{3})$; the second is the Clausen value $\mathrm{Cl}_2(\frac{2\pi}{3})$
the stacked Gammas at fifteenths, at twentieths	Chowla–Selberg periods of $\mathbb{Q}(\sqrt{-15})$, of $\mathbb{Q}(\sqrt{-5})$
the Gramma of one mark, $\mathsf{G}_{\nu;(1,-1)}(1,y) = 2y^{\nu/2}K_\nu(2\sqrt{y})$	the modified Bessel function K_ν
the centre values of $\binom{[n]}{1}$, $P_{\binom{[n]}{1}}(s\mathbf{1};s,s) = 2^{n+1-2s}\Gamma(s)^{-2}\int_0^\infty x^{2s-1}K_0(x)^n\,dx$	Bessel moments; at $n=3$ the equal-mass two-loop vacuum (sunrise) integral with propagator powers s in dimension $2s$

Rays

In this handbook	Classical name
the relations along a ray, reduced in a box (Definition 39.11)	integration-by-parts reduction on the parametric form, in the manner of Laporta
the strength	the parameter that plays the dimension
masters (Definition 39.1)	master integrals
the one-step matrix	the dimension-shift recurrence written as a first-order difference system
the law of [1]	the scalar recurrence obtained from a cyclic vector
a certificate (Theorem 39.17(ii))	a creative-telescoping certificate
the wall degree	the tropical degree of an insertion

In this handbook	Classical name
the search through boxes (Theorem 39.18)	Laporta's algorithm, its seeds grown until the targets reduce
the dimension of a ray (Definition 38.4)	the number of master integrals
the spectrum at infinite strength (Proposition 39.38)	the characteristic roots of a difference system (Birkhoff); a change of masters is a gauge transformation
the evaluation modulo primes in the reading procedure (Definition 41.8)	finite-field reconstruction
the closings (Definition 39.26)	the classical periods of the Laurent polynomial f^{-1}; the law read at the low end is the Picard–Fuchs operator of its pencil, read on coefficients
the closings of the centre ray of $\binom{[3]}{1}$	the sums of squares of trinomial coefficients
positive selection (Theorem 41.3)	Apéry limits; the minimal solution of Pincherle and Perron
positivity as the moment problem of the stride's law (Proposition 41.4); its rate (Theorem 41.5); what it cannot see (Proposition 41.6)	the Hausdorff moment problem, its four families of Markov and Lukács; the Green's function of the complement of a segment; the indeterminacy of a moment problem along a Stieltjes transform
the stride's law (Definition 40.1)	the pushforward of Lebesgue measure under the stride, a density of states; the values are its moments
the gauge end (Theorem 40.6)	Laplace's method with the Morse lemma at the end point, and the Frobenius method at a regular singular point; the germ is the Borel sum of the expansion at large strength
the torus value (Definition 40.11)	the zeta Mahler measure of the letter; at three marks the moments of the planar walk of three unit steps
the merit (Definition 43.4, Theorem 43.5)	Apéry's criterion
the law of the values of $\binom{\mathcal{E}}{1}$ (Theorem 42.12)	the recurrences of the Bessel moments $\int_0^\infty x^{2s-1}K_0(x)^n\,dx$

The motive and the content

In this handbook	Classical name
the motive and the motivic period of a family (Definitions 34.1 and 34.5)	motivic periods in Brown's affine model
small steps along the tree (Theorem 34.13)	Brown's small graphs principle in its affine form, here for families of lists
dual pairs and full leans (Corollary 34.18)	at $(0, 2)$, the product identity of Feynman periods at a two-vertex cut (Schnetz), as an identity of motivic periods

In this handbook	Classical name
chains (Definition 45.4); the content (Definition 45.1)	iterated integrals on $\{0\}$ and the roots of unity of order M; the multiple polylogarithms at roots of unity
the cutting rule (Definition 45.8, Theorem 45.9)	Goncharov's coaction on iterated integrals
critical primes (Definition 43.11)	primes of bad reduction

The family as inference

In this handbook	Classical name
the interaction of a split of the prior marks of a regression design (Proposition 47.27)	minus the logarithm of the generalized variance inflation factor